The Latticra Reference Manual

v0.1.0 Documentation Assembly

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Source documents	780
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Root README link	Top handbook links + Start Here + Main Documentation

This generated reference book preserves the current working-tree documentation corpus in a reader-oriented order. Public HTML pages and existing companion PDFs/DOCX files are indexed separately in the manifest.

Manual Part Order

- Part I - Orientation and Reader Routes: 16 source records
- Part II - Current Status, Evidence, and Public Boundaries: 6 source records
- Part III - Architecture and Foundation Concepts: 19 source records
- Part IV - Lat, LIR, and L-UI: 81 source records
- Part V - Nucleus and Runtime Boundary: 22 source records
- Part VI - Latticra Seal: 115 source records
- Part VII - Console, Panel, Installer, and Update Paths: 32 source records
- Part VIII - Nadia Offline AI Foundation: 47 source records
- Part IX - Kernel, Boot, and OS Image Research: 32 source records
- Part X - Security and Assurance: 18 source records
- Part XI - Platform Packaging and Distribution Lanes: 100 source records
- Part XII - Project Notes, Strategy, and Documentation Maintenance: 27 source records
- Part XIII - Status Record Archive: 220 source records
- Part XIV - Legal, License, and Remaining Records: 45 source records

Part I - Orientation and Reader Routes

- README.md - Latticra
- STATUS.md - Latticra Status
- SECURITY.md - Security Policy
- CONTRIBUTING.md - Contributing to Latticra
- TRADEMARK_POLICY.md - Latticra Trademark and Identity Policy
- docs/README.md - Latticra Documentation Hub
- docs/QUICK_START_CHEATSHEET.md - Latticra Quick Start Cheat Sheet
- docs/DOCUMENTATION_READER_JOURNEY_MAP.md - Documentation Reader Journey Map
- docs/DOCUMENTATION_GLOSSARY.md - Documentation Glossary
- docs/PRODUCT_DOCUMENTATION_COHESION.md - Product Documentation Cohesion
- docs/PUBLIC_CLAIMS_LEDGER.md - Latticra Public Claims Ledger
- docs/NON_CLAIMS.md - Latticra Non-Claims
- docs/REAL_SYSTEM_CONTRACT.md - Latticra Real-System Contract
- docs/EVIDENCE_LADDER.md - Latticra Evidence Ladder
- docs/PRECURSOR_PROMOTION_RULE.md - Latticra Precursor Promotion Rule
- docs/FOUNDATION_INDEX.md - Latticra Foundation Index

README.md

Source title: Latticra

Lines: 1047 | Words: 3349 | SHA-256: 63e32dcda7a3e737b38da32e63aa3e55a1387939a269ca04180e913cabf97098

Latticra

Evidence-bound systems architecture for local authority, reports, receipts, and future runtime boundaries.

README route refreshed: 2026-05-27 CDT Default branch: main

Latticra is an early-stage systems substrate. It is built around a simple rule: before a system action becomes operational, the request, identity, capability, policy, boundary, and evidence posture should be explicit, inspectable, denied by default, and backed by reproducible records.

It is not a production platform, certified security product, hardened sandbox, root installer, network authority, operating-system replacement, Fedora-approved package, Ubuntu archive-ready package, Debian archive-ready package, FreeBSD official port, OpenBSD official port, or openSUSE official package.

Featured handbook: docs/latticra-reference-manual/the-latticra-reference-manual-v0.1.0.pdf
 Editable handbook: docs/latticra-reference-manual/the-latticra-reference-manual-v0.1.0.md
 Manual package: docs/latticra-reference-manual/
 Documentation hub: docs/README.md
 System Substrate docs: docs/latticra-system-substrate/
 Seal subsystem docs: docs/latticra-seal/
 Quick user cheat sheet: docs/QUICK_START_CHEATSHEET.md

Start Here

```
| You want to... | Go here |
| --- | --- |
| Install, run, update, or remove the current user-local Panel | [Quick Start Cheat Sheet](docs/QUICK_START_CHEATSHEET.md) |
| Understand the current public posture | [Status](STATUS.md) and [current status record](docs/status/CURRENT_STATUS.md) |
| Read the v0.1.0 reference book | [The Latticra Reference Manual](docs/latticra-reference-manual/README.md), [PDF](docs/latticra-reference-manual/the-latticra-reference-manual-v0.1.0.pdf), [editable Markdown](docs/latticra-reference-manual/the-latticra-reference-manual-v0.1.0.md) |
| Browse the full documentation set | [Documentation Hub](docs/README.md) |
| Read the project handbook | [The Latticra System Substrate](docs/latticra-system-substrate/README.md) |
| Review the main evidence and architecture index | [Foundation Index](docs/FOUNDATION_INDEX.md) |
| Use the guarded local Panel workbench | [Latticra Panel](installer/README.md) |
| Inspect the trust-boundary subsystem | [Latticra Seal docs](docs/latticra-seal/README.md) |
| Check security reporting and non-claims | [Security Policy](SECURITY.md) |
| Contribute without widening claims | [Contributing Guide](CONTRIBUTING.md) |
```

What Latticra Is

Latticra is a contract-first architecture project for high-assurance local systems work. The current repository emphasizes visibility before authority: parser and report surfaces, no-effect validation, guarded local install paths, status mirrors, platform packaging lanes, and documentation that keeps claims tied to evidence.

The durable project direction is:

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

That direction does not mean unrestricted low-level code. It means each language and subsystem must earn its authority through explicit contracts, constrained boundaries, tests, reports, and status records.

System Map

```
flowchart TD
  user[&quot;Users and reviewers&quot;] --&gt; panel[&quot;Latticra Panel&quot;]
  user --&gt; docs[&quot;Documentation hub&quot;]
  panel --&gt; console[&quot;Latticra Console&quot;]
  console --&gt; seal[&quot;Latticra Seal&quot;]
  lat[&quot;Lat / Latticra Language&quot;] --&gt; lir[&quot;LIR metadata&quot;]
  lui[&quot;L-UI&quot;] --&gt; reports[&quot;Reports and rendering records&quot;]
  nucleus[&quot;Nucleus&quot;] --&gt; runtime[&quot;Runtime Boundary&quot;]
  seal --&gt; runtime
  runtime --&gt; evidence[&quot;Evidence, receipts, status, and non-claims&quot;]
  docs --&gt; evidence
```

Main Documentation

Area	Best entry point
Documentation routes	docs/README.md
Full v0.1.0 reference manual	docs/latticra-reference-manual/README.md, [PDF](docs/latticra-reference-manual/the-latticra-reference-manual-v0.1.0.pdf), [editable Markdown](docs/latticra-reference-manual/the-latticra-reference-manual-v0.1.0.md), [source manifest](docs/latticra-reference-manual/source-manifest.json)
Project handbook	docs/latticra-system-substrate/README.md, [PDF](docs/latticra-system-substrate/the-latticra-system-substrate.pdf), [DOCX](docs/latticra-system-substrate/the-latticra-system-substrate.docx)
Live status	STATUS.md, docs/status/CURRENT_STATUS.md, docs/status/README.md
Evidence and foundation records	docs/FOUNDATION_INDEX.md, docs/EVIDENCE_LADDER.md, docs/REAL_SYSTEM_CONTRACT.md
Public claims and non-claims	docs/PUBLIC_CLAIMS_LEDGER.md, docs/NON_CLAIMS.md
Reader journeys	docs/DOCUMENTATION_READER_JOURNEY_MAP.md, docs/DOCUMENTATION_GLOSSARY.md
Public site pages	docs/index.html, docs/map.html, docs/status.html, docs/start.html
Project notes and strategy	docs/project_notes/README.md, docs/strategy/README.md

Current Work Areas

Latticra Panel is the GUI-first local installer and first-run control workbench for Latticra, Lat, LIR, Latticra Seal, and the optional Nadia offline AI foundation.

Work area	Current posture	Documentation
Latticra Panel	User-local GUI workbench; guarded install and dry-run flows	installer/README.md, [installer docs](installer/docs/README.md)
Latticra Console	Metadata and command-surface planning; no broad host authority	docs/LATTICRA_CONSOLE_FOUNDATION.md
Latticra Seal	Report-only verification, policy, receipt, and trust-boundary records	docs/latticra-seal/README.md, [Seal contract](docs/LATTICRA_SEAL_CONTRACT.md)
Lat and LIR	Parse, validate, diagnose, and lower metadata; no language execution	[language strategy](docs/LANGUAGE_STRATEGY.md), [Lat pipeline](docs/LAT_PIPELINE_CONTRACT.md), [LIR shape](docs/LIR_SHAPE_CONTRACT.md)
L-UI	Parser, validation, and report/rendering foundations; no terminal-control authority	[L-UI parser](docs/L_UI_PARSER.md), [source grammar](docs/L_UI_SOURCE_GRAMMAR.md), [rendering contract](docs/L_UI_RENDERING_CONTRACT.md)
Nucleus and Runtime Boundary	Report-only task boundaries and denied-by-default classification	[supervisor architecture](docs/SUPERVISOR_ARCHITECTURE.md), [runtime boundary contract](docs/RUNTIME_BOUNDARY_CONTRACT.md)
Nadia offline AI	Stage-46 contract-only local AI foundation records; no model execution or tool authority	[Nadia foundation](docs/NADIA_OFFLINE_AI_FOUNDATION.md), [Nadia status

```
index](docs/status/README.md) |
| Kernel lifecycle research | No-effect lifecycle metadata and scheduler records; no OS
replacement claim | [kernel lifecycle seed](docs/KERNEL_LIFECYCLE_SEED.md), [kernel
summary](docs/KERNEL_LIFECYCLE_SUBSYSTEM_SUMMARY.md) |
| SeaBIOS and GRUB compatibility | User-local installer compatibility boundary; no bootloader,
firmware, partition, or bootable OS claim |
[docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md](docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md) |
| Platform packaging lanes | Local-only validation and package/port drafts |
[Fedora](packaging/fedora/README.md), [Ubuntu](packaging/ubuntu/README.md),
[Debian](packaging/debian/README.md), [openSUSE](packaging/opensuse/README.md),
[FreeBSD](packaging/freebsd/README.md), [OpenBSD](packaging/openbsd/README.md) |
| Security and assurance | Planning baselines and release gates; no production protection claim
| [security policy](SECURITY.md), [high-assurance
baseline](docs/HIGH_ASSURANCE_SECURITY_BASELINE.md), [memory-safety
roadmap](docs/MEMORY_SAFETY_ROADMAP.md) |
```

Current Estimate Table

```
| Estimate source | Current public estimate table below, mirrored from `STATUS.md` and
`docs/status/CURRENT_STATUS.md` |
| --- | --- |
```

Current public estimate table, as summarized by STATUS.md and docs/status/CURRENT_STATUS.md:

```
| Estimate note | Posture |
| --- | --- |
| Current public estimate | Roughly 45% overall system planning estimate |
| Foundation documents and contracts | Mature relative to implementation; around 94% planning
estimate |
| Public documentation posture | Strong but still evolving; around 91% planning estimate |
| Product readiness | Early; no production platform claimed |
| Area | Estimated completion |
| --- | --- |
| Overall Latticra system | 45% |
| Latticra Seal / local evidence layer | 39% |
| Latticra Panel / local control surface | 31% |
| Nadia offline AI foundation | 75% |
| L-UI parser / AST / string foundation | 87% |
| Foundation documents and contracts | 94% |
| Public documentation posture | 91% |
| Strategy/status/funding framework | 63% |
| Lat / Latticra Programming Language | 27% |
| LIR / Intermediate Representation | 24% |
| C/C++ foundation direction | 22% |
| Constrained C++ authority layer | 5% |
| Nucleus real task execution | 12% |
| Runtime / operating-system-universe direction | 26% |
| Security-hardening implementation | 10% |
| Public product readiness | 10% |
```

The current estimate table source alignment is docs/status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md. The latest mathematical estimate rebase is docs/status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md. The latest estimate refresh record is docs/status/CURRENT_ESTIMATE_REFRESH_2026_05_24.md. The latest estimate hold review is docs/status/COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md

Planning estimates are not release promises, product-readiness metrics, or security guarantees.

Boot Compatibility Markers

SeaBIOS and GRUB compatibility is tracked as a contract-only installer readiness surface in docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md.

Boot-preview evidence stays fixture-only and no-effect:

```
docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CONTRACT.md
```

```
docs/SEABIOS_GRUB_BOOT_PREVIEW_PREFLIGHT.md
docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CAPTURE_TEMPLATE.md
docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_VALIDATION.md
docs/SEABIOS_GRUB_BOOT_PREVIEW_QEMU_ARGV_TEMPLATE.md
docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_TEMPLATE.md
docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_VALIDATION.md
installer/manifests/seabios-grub-boot-preview.toml
scripts/seabios-grub-boot-preview-preflight.sh
scripts/seabios-grub-boot-preview-evidence-template.sh
scripts/seabios-grub-boot-preview-evidence-validate.sh
scripts/seabios-grub-boot-preview-qemu-argv-template.sh
scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh
scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
test-seabios-grub-boot-preview-evidence-contract.sh
test-seabios-grub-boot-preview-preflight.sh
test-seabios-grub-boot-preview-evidence-template.sh
test-seabios-grub-boot-preview-evidence-validate.sh
test-seabios-grub-boot-preview-qemu-argv-template.sh
test-seabios-grub-boot-preview-boot-artifact-manifest-template.sh
test-seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

Runtime Boundary Markers

Runtime boundary abuse-case fixtures remain no-effect planning evidence:

```
runtime_boundary_abuse_case_fixture_expansion_present=1
docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md
```

Nadia README Compatibility Markers

These compact markers preserve the guarded Nadia README/status-row assertions after the README route refresh. They are documentation alignment evidence only; they do not add model execution, prompt evaluation, dialogue generation, runtime invocation, tool execution, source mutation, network authority, or production AI claims.

```
| Nadia offline AI foundation | 75% |
Nadia Offline AI
Nadia Murad
nadia_offline_ai_stage_0_foundation_present=1
installed_command=latticra-nadia
nadia_stage_46_contract_only_foundation_present=1
nadia_stage_32_prompt_evaluation_result_review_contract_present=1
latticra-nadia prompt-evaluation-result-review
nadia_stage_33_prompt_evaluation_result_disposition_contract_present=1
latticra-nadia prompt-evaluation-result-disposition
```

Seal README Compatibility Markers

These compact markers preserve the guarded Seal README/status-row assertions after the README route refresh. They are documentation alignment evidence only; they do not add runtime authority, host behavior, network behavior, policy enforcement, capability enforcement, cryptographic enforcement, signing, or tool execution.

```
| Latticra Seal | Report-only runtime gate path, sealed report-envelope metadata/status,
signature/signing/key/public-key/bounded-key parsing metadata/status, metadata-only verification
 policy/status, metadata-only verification receipt/status, metadata-only unverified
receipt/status, metadata-only denied capability gate/status, metadata-only denied effect
decision/status, inactive metadata-only runtime handoff/status, metadata-only status
rollup/status, report-only agentic automation security metadata/status/report
surface/public-entrypoint alignment, report-only parameter schema metadata/report
surface/status-public-entry alignment, report-only policy decision
metadata/status/report-surface public-entry alignment, and core negative-test evidence for
AI-era tool-boundary planning; no production enforcement |
Trust-boundary, request-boundary, policy-boundary, tool-boundary, and crypto-profile planning.
```

openSUSE integration and maintenance

openSUSE prerequisites: this lane has the same purpose as the Fedora and Ubuntu tracks, but stays local-only for openSUSE RPM integration and maintenance. It records package-shape, lint, source archive, build-gate, environment, artifact, payload, install/remove, and OBS non-claim evidence without creating RPM artifacts, running OBS publication, or claiming an official openSUSE package.

Primary records:

```
docs/OPENSUSE_DEVELOPER_WORKFLOW.md
docs/OPENSUSE_READINESS_PLAN.md
docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/status/OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md
packaging/opensuse/README.md
```

Local guards:

```
sh scripts/test-opensuse-developer-workflow.sh
sh scripts/test-opensuse-local-rpm-static-validation.sh
sh scripts/test-opensuse-rpmlint-osc-availability.sh
sh scripts/test-opensuse-rpmlint-static-spec-lane.sh
sh scripts/test-opensuse-rpmlint-findings-classification.sh
sh scripts/test-opensuse-source-archive-reproducibility-contract.sh
sh scripts/test-opensuse-source-archive-fixture-lane.sh
sh scripts/test-opensuse-rpm-topdir-handoff-lane.sh
sh scripts/test-opensuse-local-rpm-build-gate-contract.sh
sh scripts/test-opensuse-local-rpm-build-environment-contract.sh
sh scripts/test-opensuse-rpm-artifact-naming-contract.sh
sh scripts/test-opensuse-rpm-payload-inspection-contract.sh
sh scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
sh scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
```

Debian, FreeBSD, and OpenBSD Package Posture

These package/port drafts remain local-only. They do not build packages, publish packages, submit ports, accept platform build evidence, or claim Debian archive readiness, FreeBSD official port status, OpenBSD official port status, production installer readiness, or package readiness.

```
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
package_build_gate_state=closed-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
debian_build_allowed=0
freebsd_build_allowed=0
openbsd_build_allowed=0
platform_build_evidence_accepted=0
debian_platform_build_evidence_accepted=0
freebsd_platform_build_evidence_accepted=0
openbsd_platform_build_evidence_accepted=0
package_validation_result_promoted=0
debian_validation_result_promoted=0
freebsd_validation_result_promoted=0
openbsd_validation_result_promoted=0
package_readiness_claimed=0
```

Relevant Debian, FreeBSD, and OpenBSD records:

- docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md

- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md

Security Baseline Markers

These markers keep the root README aligned with the high-assurance security baseline without adding production security, compliance, recovery, monitoring, or configuration-management claims.

```
defensive_threat_model_contract_present=1
defensive_threat_model_validation_present=1
defensive_threat_model_validation_refinement_present=1
high_assurance_security_baseline_present=1
source_refresh_date=2026-05-26
memory_safety_roadmap_present=1
supply_chain_security_baseline_present=1
cyber_incident_reporting_response_baseline_present=1
vulnerability_management_release_gate_baseline_present=1
cryptographic_assurance_key_management_baseline_present=1
identity_credential_access_management_baseline_present=1
security_logging_monitoring_baseline_present=1
backup_recovery_resilience_baseline_present=1
secure_configuration_change_management_baseline_present=1
zero_trust_runtime_authority_baseline_present=1
zero_trust_runtime_authority_guard_present=1
per_request_authorization_required=1
zero_trust_runtime_boundary_required=1
```

Defensive threat model refinement: docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md

Installer Readiness Markers

Current local installer evidence is not a production installer claim.

```
production_installer_ready=0
```

Fedora VM CLI payload next-validation planning remains evidence-bound: docs/FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN.md. Fedora VM CLI payload repeatability runner planning remains manual-only and evidence-bound: docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN.md; current runner status is tracked in docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_STATUS.md.

Fedora VM CLI payload repeatability remains manual, disposable-VM-only, and non-promoting until reviewed evidence is accepted:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE_STATUS.md
fedora_vm_cli_payload_repeatability_runner_plan_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
repeatability_runner_manual_only=1
ci_auto_repeatability_validation_allowed=0
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
repeatability_transcript_template_mode=no-effect-template
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
repeatability_transcript_template_complete=0
```

```
repeatability_transcript_candidate_valid=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
repeatability_evidence_status_template_mode=no-effect-template
repeatability_evidence_status_template_complete=0
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_review_mode=no-effect-validation
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
fedora_vm_cli_payload_repeatability_evidence_publication_gate_present=1
repeatability_evidence_publication_requested=0
operator_publication_review_completed=0
repeatability_evidence_publication_approved=0
repeatability_evidence_status_published=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The repeatability evidence acceptance contract is present, but no evidence status is written or accepted. The repeatability evidence status template is present and does not write or accept evidence. The repeatability evidence status review validator is present and does not write or promote evidence. The repeatability evidence publication gate is present and does not publish evidence.

Ubuntu prerequisites

Ubuntu no-effect validation includes:

```
sh scripts/test-ubuntu-build-lane.sh
sh scripts/test-ubuntu-upload-signing-authority-evidence-contract.sh
```

Ubuntu and local deb work

```
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
ubuntu_ppa_archive_publication_gate_contract_present=1
ubuntu_ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
deb_removed_from_host=0
source_package_created=0
source_package_uploaded=0
ubuntu_source_package_evidence_unblocked=0
ubuntu_upload_signing_authority_evidence_unblocked=0
ubuntu_publication_gate_unblocked=0
```

Ubuntu records: docs/UBUNTU_DEVELOPER_WORKFLOW.md, docs/UBUNTU_LINTIAN_AVAILABILITY.md, docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md, docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md, docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md, docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md, docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md, docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md, docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md, docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md, docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md, docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md, docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md, docs/UBUNTU_PACKAGE_NOTICE_REVIEW_CONTRACT.md, docs/UBUNTU_PACKAGE_LICENSE_REVIEW_CONTRACT.md, docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_CONTRACT.md, docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md, docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md, docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md, docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md, docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md, packaging/ubuntu/README.md.

Seal Status Markers

This compact marker appendix preserves the guarded Seal status-chain assertions used by the local status scripts after the README route refresh. These markers are status evidence only; they do not add runtime authority, host behavior, network behavior, policy enforcement, capability enforcement, cryptographic enforcement, or tool execution.

```
sealed report-envelope metadata/status
seal_report_envelope_contract_present=1
seal_report_envelope_implementation_present=1
seal_report_envelope_status_present=1
report_envelope_ready=1
report_envelope_state=sealed-report-only
report_envelope_signature_performed=0
report_envelope_runtime_authority_granted=0
performing signing, object sealing, or runtime handoff
docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md
docs/status/SEAL_REPORT_ENVELOPE_STATUS.md
SEAL_SIGNATURE_REQUEST_STATUS.md
seal_signature_request_status_present=1
signature-request status record now ties that metadata-only checkpoint to the guarded
report-envelope status predecessor
SEAL_SIGNING_AUTHORIZATION_STATUS.md
latticra_seal_signing_authorization_status_present=1
seal_signing_authorization_status_present=1
signing-authorization status record now ties that metadata-only checkpoint to the guarded
signature-request status predecessor
LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
SEAL_SIGNER_INVOCATION_STATUS.md
LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
latticra_seal_signer_handoff_contract_present=1
latticra_seal_signer_handoff_metadata_present=1
latticra_seal_signer_invocation_contract_present=1
latticra_seal_signer_invocation_metadata_present=1
latticra_seal_signer_invocation_status_present=1
latticra_seal_signing_operation_contract_present=1
seal_signer_handoff_contract_present=1
seal_signer_handoff_metadata_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
seal_signing_operation_contract_present=1
SEAL_SIGNER_HANDOFF_STATUS.md
latticra_seal_signer_handoff_status_present=1
signer-handoff status record now ties that metadata-only checkpoint to the guarded
signing-authorization status predecessor
seal_signer_handoff_status_present=1
LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
latticra_seal_signing_operation_metadata_present=1
seal_signing_operation_metadata_present=1
signer-invocation status record now ties that metadata-only checkpoint to the guarded
signer-handoff status predecessor
SEAL_SIGNING_OPERATION_STATUS.md
LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
SEAL_KEY_HANDLING_STATUS.md
latticra_seal_signing_operation_status_present=1
signing-operation status record now ties that metadata-only checkpoint to the guarded
signer-invocation status predecessor
latticra_seal_key_handling_contract_present=1
latticra_seal_key_handling_metadata_present=1
latticra_seal_key_handling_status_present=1
```

seal_signing_operation_status_present=1
seal_key_handling_contract_present=1
seal_key_handling_metadata_present=1
seal_key_handling_status_present=1
LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
key-handling status record now ties that metadata-only checkpoint to the guarded signing-operation status predecessor
latticra_seal_key_material_contract_present=1
seal_key_material_contract_present=1
LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
SEAL_KEY_MATERIAL_STATUS.md
LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
key-material status record now ties that metadata-only checkpoint to the guarded key-handling status predecessor
latticra_seal_public_key_parsing_contract_present=1
latticra_seal_key_material_status_present=1
latticra_seal_key_material_metadata_present=1
seal_public_key_parsing_contract_present=1
seal_key_material_status_present=1
seal_key_material_metadata_present=1
SEAL_PUBLIC_KEY_PARSING_STATUS.md
LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md
LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
public-key parsing status record now ties that metadata-only checkpoint to the guarded key-material status predecessor
SEAL_KEY_PARSING_STATUS.md
SEAL_VERIFICATION_POLICY_STATUS.md
SEAL_VERIFICATION_RECEIPT_STATUS.md
latticra_seal_public_key_parsing_status_present=1
latticra_seal_public_key_parsing_metadata_present=1
latticra_seal_key_parsing_metadata_present=1
latticra_seal_key_parsing_status_present=1
latticra_seal_verification_policy_metadata_present=1
latticra_seal_verification_policy_status_present=1
latticra_seal_verification_receipt_metadata_present=1
latticra_seal_verification_receipt_status_present=1
seal_public_key_parsing_status_present=1
seal_public_key_parsing_metadata_present=1
seal_key_parsing_metadata_present=1
seal_key_parsing_status_present=1
seal_verification_policy_metadata_present=1
seal_verification_policy_status_present=1
seal_verification_receipt_metadata_present=1
seal_verification_receipt_status_present=1
SEAL_CAPABILITY_GATE_STATUS.md
SEAL_EFFECT_DECISION_STATUS.md
SEAL_RUNTIME_HANDOFF_STATUS.md
seal_capability_gate_status_present=1
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
key parsing status record now ties that bounded public-key byte metadata checkpoint to the guarded public-key parsing status predecessor
verification policy status record now ties that metadata-only checkpoint to the guarded key parsing status predecessor
LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md
LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
verification receipt status record now ties that metadata-only checkpoint to the guarded verification policy status predecessor
LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md
LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
capability gate status record now ties that metadata-only denied checkpoint to the guarded verification receipt status predecessor
LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md
LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
latticra_seal_capability_gate_metadata_present=1
latticra_seal_capability_gate_status_present=1
latticra_seal_effect_decision_metadata_present=1
latticra_seal_effect_decision_status_present=1
latticra_seal_runtime_handoff_metadata_present=1
latticra_seal_runtime_handoff_status_present=1
seal_capability_gate_metadata_present=1
seal_effect_decision_metadata_present=1
seal_runtime_handoff_metadata_present=1
effect decision status record now ties that denied metadata checkpoint to the guarded capability gate status predecessor
LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md
SEAL_STATUS_ROLLUP_STATUS.md

LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
runtime handoff status record now ties that inactive metadata checkpoint to the guarded effect decision status predecessor
runtime_handoff_predecessor_effect_decision_status_present=1
status rollup status record now ties that metadata-only rollup checkpoint to the guarded runtime handoff status predecessor
LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md
LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md
latticra_seal_status_rollup_metadata_present=1
latticra_seal_status_rollup_status_present=1
seal_status_rollup_metadata_present=1
seal_status_rollup_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
agentic automation security status record now ties that report-only automation checkpoint to the guarded status rollup status predecessor
latticra_seal_agentic_automation_security_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
seal_agentic_automation_security_status_present=1
parameter schema status record now ties that report-only parameter checkpoint to the guarded agentic automation security status predecessor
latticra_seal_parameter_schema_contract_present=1
latticra_seal_parameter_schema_metadata_present=1
latticra_seal_parameter_schema_report_surface_present=1
latticra_seal_parameter_schema_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
seal_parameter_schema_contract_present=1
seal_parameter_schema_metadata_present=1
seal_parameter_schema_report_surface_present=1
seal_parameter_schema_status_present=1
report-only parameter schema metadata
operator-visible deterministic parameter schema report surface
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md
docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md
latticra_seal_request_freshness_contract_present=1
latticra_seal_request_freshness_metadata_present=1
latticra_seal_request_freshness_report_surface_present=1
latticra_seal_request_freshness_status_present=1
seal_request_freshness_contract_present=1
seal_request_freshness_metadata_present=1
seal_request_freshness_report_surface_present=1
seal_request_freshness_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
report-only request freshness metadata
operator-visible deterministic request freshness report surface
request freshness status record now ties that report-only freshness checkpoint to the guarded parameter schema status predecessor
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md
docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md
latticra_seal_signed_request_contract_present=1
latticra_seal_signed_request_metadata_present=1
latticra_seal_signed_request_status_present=1
seal_signed_request_contract_present=1
seal_signed_request_metadata_present=1
seal_signed_request_status_present=1
signed_request_predecessor_request_freshness_status_present=1
report-only signed request metadata
signed request status record now ties that report-only signed-request checkpoint to the guarded request freshness status predecessor
docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNED_REQUEST_STATUS.md
latticra_seal_policy_decision_contract_present=1
latticra_seal_policy_decision_metadata_present=1
latticra_seal_policy_decision_report_surface_present=1
latticra_seal_policy_decision_report_surface_status_present=1
latticra_seal_policy_decision_status_present=1
seal_policy_decision_contract_present=1
seal_policy_decision_metadata_present=1
seal_policy_decision_report_surface_present=1
seal_policy_decision_report_surface_status_present=1
seal_policy_decision_status_present=1
policy_decision_predecessor_signed_request_status_present=1
report-only policy decision metadata
operator-visible deterministic policy decision report surface
policy decision status record now ties that report-only policy checkpoint to the guarded signed request status predecessor

```
docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
docs/status/SEAL_POLICY_DECISION_STATUS.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
latticra_seal_operator_receipt_report_status_present=1
seal_operator_receipt_report_status_present=1
operator_receipt_report_predecessor_policy_decision_status_present=1
operator receipt report status record now ties that denied receipt checkpoint to the guarded
policy decision status predecessor
docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md
latticra_seal_local_capability_registry_schema_implementation_present=1
seal_local_capability_registry_schema_implementation_present=1
local capability registry schema implementation preserves no registry loader, no file reads, no
host behavior, no network behavior, no capability enforcement, no effects, and no runtime
authority
docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md
```

Fast Local Path

Use the Quick Start Cheat Sheet for full prerequisites and cleanup commands. The shortest guarded flow from a checkout is:

```
make -C installer dry-run
make -C installer local-example
make -C installer verify-local
latticra-panel
```

Useful local commands after install:

```
latticra status
latticra path
latticra seal report
latticra lc status
nadia commands
latticra-nadia commands
```

It is intentionally user-local. It does not use root authority, kernel mutation, systemd mutation, SELinux mutation, or network authority.

SeaBIOS and GRUB compatibility is tracked as a no-effect readiness contract in docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md; it does not grant bootloader write, firmware mutation, partition mutation, or bootable OS authority.

When a Panel specification changes, users can remove the managed local install before reinstalling:

```
make -C installer reset-dry-run
make -C installer reset-local
make -C installer uninstall-dry-run
make -C installer uninstall-local
latticra reset --dry-run
latticra reset
latticra uninstall --dry-run
latticra uninstall
```

Reset and uninstall remove the same managed artifacts: command wrappers, the Panel desktop entry, known Panel icons, and the guarded local prefix.

If LC was installed with a custom lc.install.command_wrapper, set LC_WRAPPER to that command name; the default is latticra-lc.

```
LC_WRAPPER=&quot;${LC_WRAPPER:-latticra-lc}&quot;
$HOME/.local/bin/$LC_WRAPPER
```

Installed LC wrapper path: ~/.local/bin/<lc.install.command_wrapper> (default: latticra-lc; when LC wrapper enabled).

Quality gates:

```
make quality-safety-guards
make quality-packaging-static
make quality
```

- macOS installer lane

```
| macOS installer lane | Mac-specific no-effect installer path exists |
```

macOS installer lane

The macOS installer lane targets macOS infrastructure specifically. It is a guarded no-effect route for installer planning, dry-run bundle shape checks, reset/uninstall denial handling, and status evidence only. It cannot yet create, install, sign, notarize, open, verify, reset, or uninstall a real macOS .app.

```
macos_readme_installer_usage_present=1
app_support_prefix=$HOME/Library/Application Support/Latticra
app_bundle=$HOME/Applications/Latticra Panel.app
logs_dir=$HOME/Library/Logs/Latticra
caches_dir=$HOME/Library/Caches/Latticra
preferences_dir=$HOME/Library/Preferences
optional_cli_bin=$HOME/.local/bin
```

```
Latticra Panel.app/
Contents/Info.plist
Contents/MacOS/latticra-panel
Contents/Resources/latticra-panel.icns
```

```
sh scripts/macos-build-platform-probe.sh
sh scripts/macos-dry-run-plan-adapter.sh
sh scripts/macos-local-candidate-asset-probe.sh
sh scripts/macos-app-bundle-writer-dry-run.sh
sh scripts/macos-dry-run-writer-candidate-integration.sh
sh scripts/macos-commit-gate-contract.sh
sh scripts/macos-reset-uninstall-dry-run-contract.sh
```

```
commit_gate_state=closed
commit_gate_decision=blocked-missing-managed-write-implementation
commit_user_local_managed_artifacts=0
macos_app_bundle_commit_capable_writer_present=0
macos_build_platform_probe_present=1
macos_dry_run_plan_adapter_present=1
macos_app_bundle_writer_dry_run_present=1
macos_commit_gate_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
reset_uninstall_dry_run_planner_transcript_present=1
```

```
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_deletion_enabled=0
live_runner_denied_dispatch_transcript_deletion_enabled=0
live_runner_denied_dispatch_review_dispatch_reviewed=1
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_dispatch_allowed=0
live_runner_acceptance_denial_review_dispatch_reviewed=1
live_runner_acceptance_denial_disposition_recorded=1
live_runner_acceptance_denial_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_review_present=1
live_runner_acceptance_denial_disposition_closeout_disposition_closed=1
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
```

```
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_perf
ormed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_open
ed=0
live_implementation_plan_preflight_present=1
effect_authorization_open=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
absence_report_evidence_present=0
reset_receipt_evidence_present=0
app_bundle_write_performed=0
file_delete_performed=0
directory_delete_performed=0
host_mutation_performed=0
network_performed=0
```

```
a commit-capable macOS installer
a signed or notarized macOS app
macOS app bundle install evidence
Installer, macOS, Fedora, Ubuntu, Debian, FreeBSD, OpenBSD, and openSUSE direction
```

```
docs/MACOS_INTEGRATION_TRANSFERABILITY_PLAN.md
docs/MACOS_APP_BUNDLE_WRITER_DRY_RUN.md
docs/MACOS_COMMIT_GATE_CONTRACT.md
docs/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT.md
docs/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER.md
docs/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER.md
docs/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT.md
```

macOS Reset/Uninstall No-Effect Checkpoints

The macOS reset/uninstall live-runner checkpoints below are command and field references only. They do not grant dispatch, deletion, receipt writes, host mutation, root authority, or runtime authority.

```
macos_reset_uninstall_live_target_classifier_present=1
sh scripts/macos-reset-uninstall-live-target-classifier.sh
```

```
macos_reset_uninstall_dry_run_planner_present=1
sh scripts/macos-reset-uninstall-dry-run-planner.sh
```

```
macos_reset_uninstall_absence_report_contract_present=1
sh scripts/macos-reset-uninstall-absence-report-contract.sh
```

```
macos_reset_uninstall_receipt_schema_contract_present=1
reset_uninstall_receipt_evidence_present=0
sh scripts/macos-reset-uninstall-receipt-schema-contract.sh
docs/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT.md
```

```
macos_reset_uninstall_implementation_gate_contract_present=1
sh scripts/macos-reset-uninstall-implementation-gate-contract.sh
docs/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT.md
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
```

```
macos_reset_uninstall_operator_intent_contract_present=1
sh scripts/macos-reset-uninstall-operator-intent-contract.sh
docs/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT.md
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
```

```
macos_reset_uninstall_effect_authorization_contract_present=1
sh scripts/macos-reset-uninstall-effect-authorization-contract.sh
docs/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT.md
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
```

```
macos_reset_uninstall_evidence_bundle_contract_present=1
sh scripts/macos-reset-uninstall-evidence-bundle-contract.sh
docs/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT.md
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
```

```
macos_reset_uninstall_live_implementation_plan_contract_present=1
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docs/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT.md
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live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0

macos_reset_uninstall_live_execution_preflight_contract_present=1
sh scripts/macos-reset-uninstall-live-execution-preflight-contract.sh
docs/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT.md
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0

macos_reset_uninstall_live_denial_transcript_contract_present=1
sh scripts/macos-reset-uninstall-live-denial-transcript-contract.sh
docs/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT.md
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0

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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT.md
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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT.md
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live_runner_noop_prototype_contract_state=executed-no-effect
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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT.md
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_dispatch_denied=1
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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT.md
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live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_deletion_enabled=0

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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT.md
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live_runner_acceptance_denial_transcript_deletion_enabled=0

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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT.md
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docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTR
ACT.md
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-disposition-contract.sh
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPO
SITION_CONTRACT.md
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_contract_present=1
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sed-no-effect-closeout-audit-review
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live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0

sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review
-disposition-review-contract.sh
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPO
SITION_REVIEW_CONTRACT.md
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
```

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led=0
```

Repository Map

Path	Purpose
`docs/`	Documentation hub, foundation records, status, architecture, public site pages, and subsystem records
`installer/`	Latticra Panel, guarded local install flows, installer docs, and evidence surfaces
`src/`, `include/`, `tests/`	C/C++ implementation, headers, and deterministic tests
`scripts/`	Guard scripts, no-effect validation lanes, package checks, and report runners
`packaging/`	Local-only Fedora, Ubuntu, Debian, openSUSE, FreeBSD, and OpenBSD package/port draft records
`examples/`, `fixtures/`	Input samples, fixtures, and validation material
`assets/`, `docs/assets/`	Project and documentation visual assets

Boundaries

Latticra should be read as evidence-bound foundation work. Current documentation may describe future directions, but future directions are not product claims.

Do not interpret the repository as claiming:

- production runtime enforcement
- host protection, malware prevention, or ransomware prevention
- kernel, systemd, SELinux, secure boot, or root authority
- production cryptography or key-management readiness
- distribution approval or archive readiness
- bootable OS, daily-driver OS, or operating-system replacement readiness

Development Rhythm

Before changing public claims, read Documentation Maintenance, Documentation Validation Playbook, and Documentation Drift Response Playbook.

Before changing code behavior, find the relevant contract or implementation record in the Foundation Index, update the status path if the posture changes, and run the guard script named by that record.

Canonical quality commands:

```
make quality
make quality-safety-guards
make quality-packaging-static
```

Security, License, and Support

Security reports belong in the private reporting path described in SECURITY.md. Public reports should not include exploit details, secrets, or destructive reproduction steps.

License and third-party material guidance is tracked in LICENSE, LICENSES/README.md, docs/LICENSE_POLICY.md, and TRADEMARK_POLICY.md.

For support, start with the documentation routes above. The project is early-stage and does not promise production support, response-time SLAs, package support, or security support beyond best-effort review.

STATUS.md

Source title: Latticra Status

Lines: 2250 | Words: 9597 | SHA-256: db13d3ca10be9706f0550b2b527a52113227a45d2c4ca7ebb46cf50480353931

Latticra Status

Status: public status shortcut Last updated: 2026-05-26 CDT Documentation hub: docs/README.md Latest current estimate refresh note: 2026-05-24 CDT Latest current estimate table source alignment note: 2026-05-26 CDT Latest current estimate mathematical rebase note: 2026-05-26 CDT Latest completion estimate review README/status alignment note: 2026-05-25 CDT Latest completion estimate review after runtime-boundary abuse-case fixtures note: 2026-05-25 CDT Latest defensive threat model validation refinement note: 2026-05-25 CDT Latest high-assurance security baseline note: 2026-05-26 CDT Latest memory-safety roadmap note: 2026-05-26 CDT Latest supply-chain security baseline note: 2026-05-26 CDT Latest zero-trust runtime authority baseline note: 2026-05-26 CDT Latest runtime boundary policy expansion after threat-model note: 2026-05-26 CDT Latest runtime boundary abuse-case fixture expansion note: 2026-05-25 CDT Latest runtime boundary Lat pipeline comment evidence note: 2026-05-25 CDT 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alignment note: 2026-05-26 CDT Latest Seal key-material status/public-entry note: 2026-05-25 CDT Latest Seal key-material implementation note: 2026-05-25 CDT Latest Seal key-material contract note: 2026-05-25 CDT Latest Seal key-handling predecessor status alignment note: 2026-05-26 CDT Latest Seal key-handling status/public-entry note: 2026-05-25 CDT Latest Seal key-handling implementation note: 2026-05-25 CDT Latest Seal key-handling contract note: 2026-05-25 CDT Latest Seal signing operation predecessor status alignment note: 2026-05-26 CDT Latest Seal signing operation status/public-entry note: 2026-05-25 CDT Latest Seal signing operation implementation note: 2026-05-25 CDT Latest Seal signing operation contract note: 2026-05-25 CDT Latest Seal signer invocation predecessor status alignment note: 2026-05-25 CDT Latest Seal signer invocation status/public-entry note: 2026-05-25 CDT Latest Seal signer invocation implementation note: 2026-05-25 CDT Latest Seal signer invocation contract note: 2026-05-25 CDT Latest Seal signer handoff predecessor status alignment note: 2026-05-25 CDT Latest Seal signer handoff status/public-entry note: 2026-05-25 CDT

Latest Seal signer handoff implementation note: 2026-05-25 CDT Latest Seal signer handoff contract note: 2026-05-25 CDT Latest Seal signing authorization predecessor status alignment note: 2026-05-25 CDT Latest Seal signing authorization status/public-entry note: 2026-05-25 CDT Latest Seal signing authorization implementation note: 2026-05-25 CDT Latest Seal signing authorization contract note: 2026-05-25 CDT Latest Seal signature request predecessor status alignment note: 2026-05-25 CDT Latest Seal signature request status/public-entry note: 2026-05-25 CDT Latest Seal signature request implementation note: 2026-05-25 CDT Latest Seal signature request contract note: 2026-05-25 CDT Latest Seal report envelope status/public-entry note: 2026-05-25 CDT Latest Seal report envelope implementation note: 2026-05-25 CDT Latest Seal core evidence status surface note: 2026-05-22 02:24 CDT Latest Seal core evidence index alignment note: 2026-05-22 02:37 CDT Latest Seal core evidence public entrypoint alignment note: 2026-05-22 02:45 CDT Latest Lat grammar report metadata integration note: 2026-05-25 CDT Latest Lat grammar line-comment metadata refinement note: 2026-05-25 CDT Latest Lat grammar unsupported block-comment rejection refinement note: 2026-05-25 CDT Latest Lat pipeline comment metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic comment metadata integration note: 2026-05-25 CDT Latest Lat parse-failure comment evidence propagation note: 2026-05-25 CDT Latest Lat pipeline failure span evidence propagation note: 2026-05-25 CDT Latest Lat pipeline parse-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline semantic-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline downstream stage-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline stage-summary evidence propagation note: 2026-05-25 CDT Latest Lat pipeline module/count evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-declaration evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-clause evidence propagation note: 2026-05-25 CDT Latest Lat LIR edge-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR no-effect evidence propagation note: 2026-05-26 CDT Latest Lat LIR module-summary evidence propagation note: 2026-05-26 CDT Latest Lat LIR source-span evidence propagation note: 2026-05-26 CDT Latest Lat LIR node-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge endpoint evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge span evidence propagation note: 2026-05-26 CDT Latest Lat model normalization note: 2026-05-25 CDT Latest Lat model report declaration metadata integration note: 2026-05-25 CDT Latest Lat model report clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline model normalization integration note: 2026-05-25 CDT Latest Lat-to-LIR model lowering integration note: 2026-05-25 CDT Latest Lat-to-LIR declaration metadata refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline lowering diagnostic integration note: 2026-05-25 CDT Latest Lat-to-LIR clause metadata refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline report declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline report clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline note: 2026-05-18 19:40 CDT Latest Lat pipeline report refinement note: 2026-05-18 23:30 CDT Latest Lat pipeline diagnostic integration note: 2026-05-19 14:20 CDT Latest Lat pipeline diagnostic main test audit note: 2026-05-19 17:15 CDT Latest Lat pipeline diagnostic status/docs alignment note: 2026-05-19 17:25 CDT Latest Lat pipeline diagnostic foundation index alignment note: 2026-05-19 17:35 CDT Latest Lat pipeline diagnostic status announcement note: 2026-05-19

17:45 CDT Latest Lat pipeline diagnostic README alignment note: 2026-05-19 17:55 CDT Latest project notes upcoming work alignment note: 2026-05-19 18:05 CDT Latest current direction project notes alignment note: 2026-05-19 18:15 CDT Latest project notes index alignment note: 2026-05-19 18:25 CDT Latest current status and announcement consistency review note: 2026-05-19 18:35 CDT Latest completion percentage review note: 2026-05-19 18:45 CDT Latest strategy estimate review note: 2026-05-19 18:45 CDT Latest authority status/docs alignment note: 2026-05-19 19:00 CDT Latest current status detail rollup note: 2026-05-19 19:05 CDT Latest foundation index alignment note: 2026-05-19 19:15 CDT Latest status announcement review note: 2026-05-19 19:25 CDT Latest status announcement review index alignment note: 2026-05-19 19:35 CDT Latest public entry-point consistency scan note: 2026-05-19 19:45 CDT Latest project notes follow-up alignment note: 2026-05-19 19:55 CDT Latest project notes follow-up status/index check note: 2026-05-19 20:05 CDT Latest authority announcement review note: 2026-05-19 20:15 CDT Latest L-UI rendering detailed report refinement note: 2026-05-19 20:25 CDT Latest L-UI rendering README/status alignment note: 2026-05-19 20:35 CDT Latest L-UI completion estimate review note: 2026-05-19 20:45 CDT Latest C++ authority expansion contract review note: 2026-05-19 20:55 CDT Latest language representation review note: 2026-05-19 21:05 CDT Latest Nucleus task no-effect report alignment note: 2026-05-19 21:15 CDT Latest Nucleus task README/status alignment note: 2026-05-19 21:25 CDT Latest Nucleus task report-only execution refinement note: 2026-05-19 21:35 CDT Latest Nucleus task report-only execution README/status alignment note: 2026-05-20 01:45 CDT Latest project notes Nucleus report-only alignment note: 2026-05-20 02:20 CDT Latest project notes Nucleus report-only status/index check note: 2026-05-20 02:45 CDT Latest Nucleus report-only announcement review note: 2026-05-20 03:00 CDT Latest Nucleus report-only announcement README alignment note: 2026-05-20 03:20 CDT Latest project notes Nucleus announcement README status/index check note: 2026-05-20 03:50 CDT Latest Lat semantic diagnostics refinement note: 2026-05-19 00:35 CDT Latest LIR report refinement note: 2026-05-19 00:55 CDT Latest Lat-specific LIR refinement note: 2026-05-18 21:30 CDT Latest runtime boundary refinement plan note: 2026-05-18 22:15 CDT Latest runtime boundary refinement implementation note: 2026-05-18 22:45 CDT Latest runtime boundary report refinement note: 2026-05-18 23:10 CDT Latest runtime boundary policy matrix refinement note: 2026-05-18 23:55 CDT Latest runtime boundary domain matrix refinement note: 2026-05-19 14:10 CDT Latest runtime boundary domain matrix report integration note: 2026-05-19 16:20 CDT Latest RBDM report README/foundation index alignment note: 2026-05-19 16:35 CDT Latest RBDM report status announcement note: 2026-05-19 16:45 CDT Latest RBDM report main test integration audit note: 2026-05-19 17:05 CDT Latest Nucleus task report refinement note: 2026-05-19 00:15 CDT Latest announcement rollup note: 2026-05-19 15:05 CDT

For the current project status, completion estimates, and next priorities, see:

```
docs/status/CURRENT_STATUS.md
```

For announcements and milestone updates, see:

```
docs/status/ANNOUNCEMENTS.md
```

For active strategy records, see:

```
docs/strategy/README.md
```

Current high-level estimate

```
| Area | Estimated completion |
| --- | ---: |
| Overall Latticra system | 45% |
| Latticra Seal / local evidence layer | 39% |
| Latticra Panel / local control surface | 31% |
| Nadia offline AI foundation | 75% |
| L-UI parser / AST / string foundation | 87% |
| Foundation documents and contracts | 94% |
| Public documentation posture | 91% |
| Strategy/status/funding framework | 63% |
```

```
| Lat / Latticra Programming Language | 27% |
| LIR / Intermediate Representation | 24% |
| C/C++ foundation direction | 22% |
| Constrained C++ authority layer | 5% |
| Nucleus real task execution | 12% |
| Runtime / operating-system-universe direction | 26% |
| Security-hardening implementation | 10% |
| Public product readiness | 10% |
```

These percentages are planning estimates only. They are not release promises, production-readiness metrics, security guarantees, Fedora approval claims, Debian archive claims, FreeBSD ports-tree claims, OpenBSD ports-tree claims, runtime-enforcement claims, or operating-system completeness claims.

Current direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Current milestone ledger

Seal README status row alignment
Current estimate table source alignment
Seal capability gate status/public-entry alignment
Seal effect decision status/public-entry alignment
Seal effect decision predecessor status alignment
Seal runtime handoff status/public-entry alignment
Seal runtime handoff predecessor status alignment
Seal status rollup status/public-entry alignment
Seal status rollup predecessor status alignment
Seal agentic automation security public-entrypoint alignment
Seal agentic automation security predecessor status alignment
Seal parameter schema status/public-entry alignment
Seal parameter schema predecessor status alignment
Seal request freshness status/public-entry alignment
Seal request freshness predecessor status alignment
Seal signed request status/public-entry alignment
Seal signed request predecessor status alignment
Seal policy decision status/public-entry alignment
Seal policy decision predecessor status alignment
Seal operator receipt report predecessor status alignment
Defensive threat model validation refinement
Runtime boundary policy expansion after threat-model validation
Runtime boundary abuse-case fixture expansion after policy expansion
Nadia prompt evaluation invocation contract Stage-30
Nadia prompt evaluation result contract Stage-31
Nadia prompt evaluation result review contract Stage-32
Nadia prompt evaluation result disposition contract Stage-33
Nadia prompt evaluation result release contract Stage-34
Nadia prompt evaluation result release receipt contract Stage-35
Nadia prompt evaluation result release receipt review contract Stage-36
Nadia prompt evaluation result release receipt review disposition contract Stage-37
Nadia prompt evaluation result release receipt review disposition release contract Stage-38
Nadia prompt evaluation result release receipt review disposition release receipt review contract Stage-40
Nadia prompt evaluation result release receipt review disposition release receipt review disposition contract Stage-41
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release contract Stage-42
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt contract Stage-43
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review contract Stage-44
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract Stage-45
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract Stage-46
Runtime boundary Lat pipeline comment evidence integration
Completion estimate review README/status alignment
Completion estimate review after runtime-boundary abuse-case fixtures
Seal verification receipt predecessor status alignment
Seal verification receipt status/public-entry alignment
Seal crypto verify backend status/public-entry alignment
Seal Ed25519 verify-only status/public-entry alignment
Seal verified receipt promotion status/public-entry alignment
Seal verified capability gate status/public-entry alignment
Seal verified effect decision status/public-entry alignment
Seal runtime handoff evaluation status/public-entry alignment
Seal runtime handoff report status/public-entry alignment
Nadia offline AI Stage-0 foundation
Nadia local context engine Stage-1
Nadia runtime profile Stage-2
Nadia developer workbench Stage-3
Nadia systems engineering mode Stage-4
Nadia productivity loop Stage-5
Nadia protective safety boundary Stage-6
Nadia guarded tool authority Stage-7
Nadia prompt evaluation contract Stage-8
Nadia local model registry contract Stage-9
Nadia inference readiness contract Stage-10
Nadia runtime invocation contract Stage-11
Nadia model load contract Stage-12
Nadia prompt receipt contract Stage-13
Nadia prompt materialization contract Stage-14
Nadia awareness dialogue contract Stage-15
Nadia prompt evaluation handoff contract Stage-16
Nadia tokenization boundary contract Stage-17
Nadia tokenizer specification contract Stage-18
Nadia tokenizer manifest contract Stage-19

Nadia tokenizer artifact inventory contract Stage-20
Nadia tokenizer artifact measurement contract Stage-21
Nadia tokenizer artifact verification contract Stage-22
Nadia tokenizer artifact binding contract Stage-23
Nadia tokenizer runtime attachment contract Stage-24
Nadia prompt tokenization contract Stage-25
Nadia prompt token sequence contract Stage-26
Nadia context window assembly contract Stage-27
Nadia prompt evaluation input contract Stage-28
Nadia prompt evaluation runtime handoff contract Stage-29
Latticra Console profile presets
Latticra Console standalone installer preset
Latticra Console standalone local install preset
Latticra Console host-embedding contract
Latticra Console read-only host inventory contract
Latticra Console receipt contract
Seal verification policy predecessor status alignment
Seal verification policy status/public-entry alignment
Seal crypto verify backend status/public-entry alignment
Seal Ed25519 verify-only status/public-entry alignment
Seal verified receipt promotion status/public-entry alignment
Seal verified capability gate status/public-entry alignment
Seal verified effect decision status/public-entry alignment
Seal runtime handoff evaluation status/public-entry alignment
Seal runtime handoff report status/public-entry alignment
Seal capability gate predecessor status alignment
Seal key parsing predecessor status alignment
Seal key parsing status/public-entry alignment
Seal bounded no-effect key parsing implementation
Seal future key parsing implementation plan
Seal future key parsing implementation contract
Seal public-key parsing predecessor status alignment
Seal public-key parsing status/public-entry alignment
Seal public-key parsing metadata implementation
Seal public-key parsing boundary contract
Seal key-material predecessor status alignment
Seal key-material status/public-entry alignment
Seal key-material metadata implementation
Seal key-material boundary contract
Seal key-handling predecessor status alignment
Seal key-handling status/public-entry alignment
Seal key-handling metadata implementation
Seal key-handling boundary contract
Seal signing operation predecessor status alignment
Seal signing operation status/public-entry alignment
Seal signing operation metadata implementation
Seal signing operation contract
Seal signer invocation predecessor status alignment
Seal signer invocation status/public-entry alignment
Seal signer invocation metadata implementation
Seal signer invocation contract
Seal signer handoff predecessor status alignment
Seal signer handoff status/public-entry alignment
Seal signer handoff metadata implementation
Seal signer handoff contract
Seal signing authorization predecessor status alignment
Seal signing authorization status/public-entry alignment
Seal signing authorization metadata implementation
Seal signing authorization contract
Seal signature request predecessor status alignment
Seal signature request status/public-entry alignment
Seal signature request metadata implementation
Seal signature request contract
Seal report envelope status/public-entry alignment
Seal report envelope metadata implementation
Seal core evidence status surface
Seal core evidence index alignment
Seal core evidence public entrypoint alignment
Lat semantic validation contract
Lat semantic validation implementation plan
Lat semantic validation foundation
Lat grammar report metadata integration
Lat grammar line-comment metadata refinement
Lat grammar unsupported block-comment rejection refinement
Lat pipeline comment metadata integration
Lat pipeline diagnostic comment metadata integration
Runtime boundary Lat pipeline comment evidence integration
Lat pipeline stage-summary evidence propagation
Lat pipeline module/count evidence propagation
Lat pipeline first-declaration evidence propagation

Lat pipeline first-clause evidence propagation
Lat LIR edge-kind evidence propagation
Lat LIR no-effect evidence propagation
Lat LIR module-summary evidence propagation
Lat LIR source-span evidence propagation
Lat LIR node-kind evidence propagation
Lat LIR first-node evidence propagation
Lat LIR first-node span evidence propagation
Lat LIR first transition-node evidence propagation
Lat LIR first transition-node span evidence propagation
Lat LIR first-edge evidence propagation
Lat LIR first-edge span evidence propagation
Lat LIR first transition-source edge evidence propagation
Lat LIR first transition-source edge endpoint evidence propagation
Lat LIR first transition-source edge span evidence propagation
Lat semantic diagnostics refinement
Lat model normalization implementation
Lat model report declaration metadata integration
Lat model report clause metadata integration
Lat-to-LIR lowering contract
Lat-to-LIR lowering implementation plan
Lat-to-LIR lowering implementation
Lat pipeline contract
Lat pipeline implementation plan
Lat pipeline implementation
Lat pipeline model normalization integration
Lat-to-LIR model lowering integration
Lat-to-LIR declaration metadata refinement
Lat-to-LIR diagnostic refinement
Lat-to-LIR diagnostic declaration metadata integration
Lat pipeline lowering diagnostic integration
Lat-to-LIR clause metadata refinement
Lat-to-LIR diagnostic clause metadata integration
Lat pipeline diagnostic declaration metadata integration
Lat pipeline diagnostic clause metadata integration
Lat pipeline report declaration metadata integration
Lat pipeline report clause metadata integration
Lat pipeline report refinement
Lat pipeline diagnostic integration refinement
Lat pipeline diagnostic integration main test audit
Lat pipeline diagnostic status/docs alignment
Lat pipeline diagnostic foundation index alignment
Lat pipeline diagnostic status announcement
Lat pipeline diagnostic README alignment
Lat-specific LIR refinement contract
Lat-specific LIR refinement implementation plan
Lat-specific LIR refinement implementation
LIR report refinement
Constrained C++ authority layer implementation plan
Strategy estimate review
Authority implementation review
Authority status/docs alignment
Current status detail rollup
Authority foundation index alignment
Status announcement review
Status announcement review index alignment
Public entry-point consistency scan
Project notes follow-up alignment
Project notes follow-up status/index check
Authority announcement review
L-UI rendering detailed report refinement
L-UI rendering README/status alignment
L-UI completion estimate review
C++ authority expansion contract review
Language representation review
Nucleus task no-effect report alignment
Nucleus task README/status alignment
Nucleus task report-only execution refinement
Nucleus task report-only execution README/status alignment
Project notes Nucleus report-only alignment
Project notes Nucleus report-only status/index check
Nucleus report-only announcement review
Nucleus report-only announcement README alignment
Project notes Nucleus announcement README status/index check
Runtime boundary contract
Runtime boundary implementation plan
Runtime boundary implementation
Runtime boundary refinement plan
Runtime boundary refinement implementation
Runtime boundary report refinement

```
Runtime boundary policy matrix refinement
Runtime boundary domain matrix refinement
Runtime boundary domain matrix report integration
Runtime boundary domain matrix report main test integration audit
Kernel lifecycle seed
Kernel scheduler tick seed
Kernel run queue seed
Kernel context switch seed
Kernel time accounting seed
Kernel preemption seed
Kernel scheduler credit seed
Kernel scheduler selection seed
Kernel scheduler dispatch seed
Kernel scheduler handoff seed
Kernel scheduler activation seed
Kernel scheduler run-entry seed
Kernel lifecycle report runner
Kernel lifecycle subsystem summary
Kernel lifecycle rollback plan
Nucleus task execution contract
Nucleus task execution implementation plan
Nucleus task execution implementation
Nucleus task report refinement
L-UI rendering implementation
Status announcements rollup
RBDM report README/foundation index alignment
RBDM report status announcement
Project notes upcoming work alignment
Current direction project notes alignment
Project notes index alignment
Current status and announcement consistency review
Completion percentage review
```

Current next step

```
Continue small guarded report/status alignment only when drift appears
```

Completion estimate review only if capability posture changes remains the estimate rule after this non-change review.

Current Seal crypto verify backend boundary

The Seal crypto verify backend status record makes the existing metadata-only backend implementation visible from public entry points while preserving unsupported cryptographic verification and no authority.

Current crypto verify backend fields:

```
seal_crypto_verify_backend_contract_present=1
seal_crypto_verify_backend_implementation_present=1
seal_crypto_verify_backend_status_present=1
crypto_verify_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verified=0
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
real_cryptographic_verification_added=0
```

Current Seal Ed25519 verify-only boundary

The Seal Ed25519 verify-only status record makes the existing local provider-backed verification result surface visible from public entry points while preserving authority-neutral behavior and no production cryptography claim.

Current Ed25519 verify-only fields:

```
seal_ed25519_verify_only_contract_present=1
seal_ed25519_verify_implementation_present=1
seal_ed25519_verify_status_present=1
ed25519_verify_profile=latticra-seal-ed25519-verify/0.1
ed25519_cryptographic_verification_performed=1
ed25519_authority_usable=0
crypto_verify_state=verified
cryptographic_verification_supported=1
```

```
cryptographic_verification_performed=1
verified=1
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
implementation_behavior_changed=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
capability_enforcement_added=0
effect_execution_added=0
production_cryptography_claimed=0
estimate_adjustment_required=0
```

Current Seal verified receipt promotion boundary

The Seal verified receipt promotion status record makes the existing metadata promotion surface visible from public entry points while preserving authority-neutral behavior and no capability authorization.

Current verified receipt promotion fields:

```
seal_verified_receipt_promotion_contract_present=1
seal_verified_receipt_promotion_implementation_present=1
seal_verified_receipt_promotion_status_present=1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
verification_state=verified
receipt_state=verified
verified_receipt_promotion_cryptographic_verification_performed=1
verified_receipt_promotion_authority_usable=0
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
capability_authorization_added=0
effect_execution_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal crypto graduation gate boundary

The Seal crypto graduation gate status record makes the existing local Ed25519 verify-only result and verified receipt promotion pass through explicit standards expectations while preserving authority-neutral behavior and no production cryptography claim.

Current crypto graduation gate fields:

```
seal_crypto_graduation_gate_present=1
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
signature_algorithm=Ed25519-development
message_digest_algorithm=SHA-256
provider_backed_verification_required=1
deterministic_test_vector_required=1
negative_test_vector_required=1
rfc8032_test_vector_tracked=1
fips_186_5_signature_standard_tracked=1
fips_180_4_digest_standard_tracked=1
fips_140_3_claim_gate_required=1
sp_800_57_key_management_required=1
sp_800_131a_transition_review_required=1
fips_204_ml_dsa_planning_tracked=1
```

```
fips_205_slh_dsa_planning_tracked=1
cryptographic_verification_performed=1
local_verify_graduated=1
receipt_promotion_graduated=1
standard_expectations_met=1
production_crypto_claim_allowed=0
fips_claim_allowed=0
signing_authority_granted=0
key_generation_allowed=0
key_storage_allowed=0
authority_promotion_allowed=0
capability_gate_allowed=0
runtime_authority_granted=0
gate_state=graduated-authority-neutral
```

Current Seal verified capability gate boundary

The Seal verified capability gate status record makes the existing metadata-only gate evaluation visible from public entry points while preserving no capability enforcement, no effect execution, and no runtime authority. The stricter entry point now requires a passing crypto graduation gate before recording the same metadata-only allowance.

Current verified capability gate fields:

```
seal_verified_capability_gate_contract_present=1
seal_verified_capability_gate_implementation_present=1
seal_verified_capability_gate_status_present=1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
requested_capability=verified-receipt-report
requested_effect=report-only
verified_capability_gate_allowed=1
verified_capability_gate_state=allowed-metadata-only
verified_capability_gate_runtime_authority_granted=0
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_enforcement_added=0
effect_execution_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal verified effect decision boundary

The Seal verified effect decision status record makes the existing metadata-only effect classification visible from public entry points while preserving no effect execution, no capability enforcement, and no runtime authority.

Current verified effect decision fields:

```
seal_verified_effect_decision_contract_present=1
seal_verified_effect_decision_implementation_present=1
seal_verified_effect_decision_status_present=1
decision_profile=latticra-seal-verified-effect-decision/0.1
gate_profile=latticra-seal-verified-capability-gate/0.1
requested_capability=verified-receipt-report
requested_effect=report-only
verified_effect_decision_crypto_graduation_gate_present=1
verified_effect_decision_crypto_graduation_gate_passed=1
verified_effect_decision_standard_expectations_met=1
verified_effect_decision_authority_promotion_allowed=0
verified_effect_decision_allowed=1
```

```
verified_effect_decision_state=allowed-report-only
verified_effect_decision_effect_performed=0
verified_effect_decision_runtime_authority_granted=0
effect_allowed=1
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
effect_execution_added=0
capability_enforcement_added=0
estimate_adjustment_required=0
```

Current Seal runtime handoff evaluation boundary

The Seal runtime handoff evaluation status record makes the existing metadata-only handoff eligibility classification visible from public entry points while preserving no runtime handoff execution, no effect execution, no capability enforcement, and no runtime authority.

Current runtime handoff evaluation fields:

```
seal_runtime_handoff_evaluation_contract_present=1
seal_runtime_handoff_evaluation_implementation_present=1
seal_runtime_handoff_evaluation_status_present=1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
decision_profile=latticra-seal-verified-effect-decision/0.1
requested_capability=verified-receipt-report
requested_effect=report-only
requested_handoff=report-only
runtime_handoff_evaluation_crypto_graduation_gate_present=1
runtime_handoff_evaluation_crypto_graduation_gate_passed=1
runtime_handoff_evaluation_standard_expectations_met=1
runtime_handoff_evaluation_authority_promotion_allowed=0
runtime_handoff_evaluation_eligible=1
runtime_handoff_evaluation_state=eligible-report-only
runtime_handoff_evaluation_handoff_performed=0
runtime_handoff_evaluation_runtime_authority_granted=0
handoff_eligible=1
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
estimate_adjustment_required=0
```

Current Seal runtime handoff report boundary

The Seal runtime handoff report status record makes the existing metadata-only report readiness classification visible from public entry points while preserving no runtime handoff execution, no effect execution, no capability enforcement, and no runtime authority.

Current runtime handoff report fields:

```
seal_runtime_handoff_report_contract_present=1
seal_runtime_handoff_report_implementation_present=1
seal_runtime_handoff_report_status_present=1
report_profile=latticra-seal-runtime-handoff-report/0.1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
requested_capability=verified-receipt-report
requested_effect=report-only
requested_handoff=report-only
requested_report=report-only
runtime_handoff_report_ready=1
runtime_handoff_report_state=ready-report-only
runtime_handoff_report_handoff_performed=0
runtime_handoff_report_runtime_authority_granted=0
report_ready=1
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
```

```
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
estimate_adjustment_required=0
```

Current Seal report envelope boundary

The Seal report envelope status record makes the existing metadata-only envelope readiness classification visible from public entry points while preserving no signing, no object sealing, no runtime handoff execution, no effect execution, no capability enforcement, and no runtime authority.

Current report envelope fields:

```
seal_report_envelope_contract_present=1
seal_report_envelope_implementation_present=1
seal_report_envelope_status_present=1
envelope_profile=latticra-seal-report-envelope/0.1
report_profile=latticra-seal-runtime-handoff-report/0.1
requested_envelope=report-only
report_envelope_ready=1
report_envelope_state=sealed-report-only
report_envelope_signature_performed=0
report_envelope_handoff_performed=0
report_envelope_effect_performed=0
report_envelope_runtime_authority_granted=0
report_envelope_host_read_performed=0
report_envelope_host_write_performed=0
report_envelope_network_performed=0
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal signature request predecessor status boundary

The Seal signature request predecessor status alignment ties the existing metadata-only signature request status record to the guarded report-envelope status predecessor while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signature request predecessor status fields:

```
seal_report_envelope_status_present=1
seal_signature_request_status_present=1
signature_request_predecessor_report_envelope_status_present=1
signature_request_ready=1
signature_request_state=requested-metadata-only
signature_performed=0
verification_performed=0
private_key_handling=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
```

```
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal signing authorization predecessor status boundary

The Seal signing authorization predecessor status alignment ties the existing metadata-only signing authorization status record to the guarded signature-request status predecessor while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signing authorization predecessor status fields:

```
seal_signature_request_status_present=1
seal_signing_authorization_status_present=1
signing_authorization_predecessor_signature_request_status_present=1
signature_request_ready=1
signature_request_state=requested-metadata-only
signing_authorization_ready=1
signing_authorization_state=authorized-metadata-only
signature_performed=0
verification_performed=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal signer handoff predecessor status boundary

The Seal signer handoff predecessor status alignment ties the existing metadata-only signer handoff status record to the guarded signing-authorization status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signer handoff predecessor status fields:

```
seal_signing_authorization_status_present=1
seal_signer_handoff_status_present=1
signer_handoff_predecessor_signing_authorization_status_present=1
signing_authorization_ready=1
signing_authorization_state=authorized-metadata-only
signer_handoff_ready=1
signer_handoff_state=handoff-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
```

```
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal signer invocation predecessor status boundary

The Seal signer invocation predecessor status alignment ties the existing metadata-only signer invocation status record to the guarded signer-handoff status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signer invocation predecessor status fields:

```
seal_signer_handoff_status_present=1
seal_signer_invocation_status_present=1
signer_invocation_predecessor_signer_handoff_status_present=1
signer_handoff_ready=1
signer_handoff_state=handoff-metadata-only
signer_invocation_ready=1
signer_invocation_state=invocation-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal signing operation predecessor status boundary

The Seal signing operation predecessor status alignment ties the existing metadata-only signing operation status record to the guarded signer-invocation status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signing operation predecessor status fields:

```
seal_signer_invocation_status_present=1
seal_signing_operation_status_present=1
signing_operation_predecessor_signer_invocation_status_present=1
signer_invocation_ready=1
signer_invocation_state=invocation-metadata-only
signing_operation_ready=1
signing_operation_state=operation-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
```

```
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal key-handling predecessor status boundary

The Seal key-handling predecessor status alignment ties the existing metadata-only key-handling status record to the guarded signing-operation status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current key-handling predecessor status fields:

```
seal_signing_operation_status_present=1
seal_key_handling_status_present=1
key_handling_predecessor_signing_operation_status_present=1
signing_operation_ready=1
signing_operation_state=operation-metadata-only
key_handling_ready=1
key_handling_state=key-handling-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal key-material predecessor status boundary

The Seal key-material predecessor status alignment ties the existing metadata-only key-material status record to the guarded key-handling status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current key-material predecessor status fields:

```
seal_key_handling_status_present=1
seal_key_material_status_present=1
key_material_predecessor_key_handling_status_present=1
key_handling_ready=1
key_handling_state=key-handling-metadata-only
key_material_ready=1
key_material_state=key-material-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal public-key parsing predecessor status boundary

The Seal public-key parsing predecessor status alignment ties the existing metadata-only public-key parsing status record to the guarded key-material status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current public-key parsing predecessor status fields:

```
seal_key_material_status_present=1
seal_public_key_parsing_status_present=1
public_key_parsing_predecessor_key_material_status_present=1
key_material_ready=1
key_material_state=key-material-metadata-only
public_key_parsing_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
```

```
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal key parsing predecessor status boundary

The Seal key parsing predecessor status alignment ties the existing bounded key parsing status record to the guarded public-key parsing status predecessor while preserving the existing caller-provided public-key byte metadata behavior, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current key parsing predecessor status fields:

```
seal_public_key_parsing_status_present=1
seal_key_parsing_status_present=1
key_parsing_predecessor_public_key_parsing_status_present=1
public_key_parsing_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
key_parsing_ready=1
key_parsing_state=public-key-parsed-metadata-only
public_key_parsed=1
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal verification policy predecessor status boundary

The Seal verification policy predecessor status alignment ties the existing metadata-only verification policy status record to the guarded key parsing status predecessor while preserving no cryptographic verification, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current verification policy predecessor status fields:

```
seal_key_parsing_status_present=1
seal_verification_policy_status_present=1
verification_policy_predecessor_key_parsing_status_present=1
key_parsing_ready=1
key_parsing_state=public-key-parsed-metadata-only
verification_policy_ready=1
verification_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
public_key_bytes_consumed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
cryptographic_verification_added=0
signature_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal verification receipt predecessor status boundary

The Seal verification receipt predecessor status alignment ties the existing metadata-only verification receipt status record to the guarded verification policy status predecessor while preserving no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current verification receipt predecessor status fields:

```
seal_verification_policy_status_present=1
seal_verification_receipt_status_present=1
verification_receipt_predecessor_verification_policy_status_present=1
verification_policy_ready=1
verification_receipt_ready=1
verification_state=unsupported
receipt_state=unverified-metadata
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verification_performed=0
verified=0
```

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
public_key_material_handling=0
public_key_bytes_consumed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal capability gate predecessor status boundary

The Seal capability gate predecessor status alignment ties the existing metadata-only denied capability gate status record to the guarded verification receipt status predecessor while preserving no capability enforcement, no effect execution, no runtime handoff execution, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime authority, no host behavior, no network behavior, and no production enforcement claim.

Current capability gate predecessor status fields:

```
seal_verification_receipt_status_present=1
seal_capability_gate_status_present=1
verification_receipt_predecessor_verification_policy_status_present=1
capability_gate_predecessor_verification_receipt_status_present=1
verification_receipt_ready=1
capability_gate_ready=1
receipt_state=unverified-metadata
gate_state=denied-unverified
verified=0
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=0
capability_enforcement_performed=0
effect_allowed=0
effect_performed=0
runtime_authority_granted=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_enforcement_added=0
effect_execution_added=0
runtime_handoff_execution_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
```

```
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

Current Seal runtime handoff predecessor status boundary

The Seal runtime handoff predecessor status alignment ties the existing inactive metadata-only runtime handoff status record to the guarded effect decision status predecessor while preserving no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no cryptographic verification, no verified receipt authority, no signing, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signer invocation behavior, no signer process execution, no object sealing, no policy persistence, no host behavior, no network behavior, and no production cryptography claim.

Current runtime handoff predecessor status fields:

```
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
runtime_handoff_ready=1
decision_state=denied-gate
handoff_state=denied-decision
handoff_active=0
runtime_effect_performed=0
effect_performed=0
capability_enforcement_performed=0
runtime_authority_granted=0
cryptographic_verification_performed=0
verification_performed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_handoff_status_added=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Current Seal status rollup predecessor status boundary

The Seal status rollup predecessor status alignment ties the existing metadata-only status rollup status record to the guarded runtime handoff status predecessor while preserving no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no policy persistence, no host behavior, no network behavior, and no production enforcement claim.

Current status rollup predecessor status fields:

```
seal_runtime_handoff_status_present=1
seal_status_rollup_status_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
runtime_handoff_ready=1
rollup_state=metadata-only
runtime_boundary_state=disabled
handoff_active=0
runtime_effect_performed=0
effect_performed=0
capability_enforcement_performed=0
runtime_authority_granted=0
cryptographic_verification_performed=0
verification_performed=0
verified=0
capability_gate_allowed=0
effect_allowed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
status_rollup_status_added=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Current Seal agentic automation security predecessor status boundary

The Seal agentic automation security predecessor status alignment ties the existing report-only agentic automation security status record to the guarded status rollup status predecessor while preserving no MCP behavior, no AI-agent execution, no model execution, no tool execution, no shell execution, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no policy persistence, no host behavior, no network behavior, and no production enforcement claim.

Current agentic automation security predecessor status fields:

```
seal_status_rollup_status_present=1
seal_agentic_automation_security_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
mcp_protocol_implemented=0
mcp_server_implemented=0
mcp_client_implemented=0
agent_execution_supported=0
model_execution_supported=0
tool_execution_supported=0
shell_execution_supported=0
runtime_authority_requested=0
runtime_authority_granted=0
unknown_tool_allowed=0
unsigned_manifest_allowed=0
network_access_allowed=0
private_key_access_allowed=0
system_mutation_allowed=0
host_read_performed=0
host_write_performed=0
network_performed=0
agentic_automation_security_status_added=1
runtime_execution_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
policy_persistence_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Current Seal parameter schema predecessor status boundary

The Seal parameter schema predecessor status alignment ties the existing report-only parameter schema status record to the guarded agentic automation security status predecessor while preserving no schema parsing, no schema validation, no policy evaluation, no policy enforcement, no MCP behavior, no AI-agent execution, no model execution, no tool execution, no shell execution, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no key material loading, no private-key handling, no object sealing, no policy persistence, no host behavior, no network behavior, and no production enforcement claim.

Current parameter schema predecessor status fields:

```
seal_agentic_automation_security_status_present=1
seal_parameter_schema_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
schema_profile=latticra-seal-parameter-schema/0.1
schema_present=0
schema_parsing_supported=0
schema_validation_supported=0
schema_valid=0
unknown_parameters_allowed=0
parameter_forwarding_allowed=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
schema_parsing_added=0
schema_validation_added=0
policy_evaluation_added=0
```

```
policy_enforcement_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal request freshness predecessor status boundary

The Seal request freshness predecessor status alignment ties the existing report-only request freshness status record to the guarded parameter schema status predecessor while preserving no timestamp parsing, no trusted clock behavior, no nonce storage, no replay-cache storage, no context hashing, no parameter hashing, no freshness validation, no replay detection, no signature verification, no signed request enforcement, no policy evaluation, no policy enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no host behavior, no network behavior, and no production enforcement claim.

Current request freshness predecessor status fields:

```
seal_parameter_schema_status_present=1
seal_request_freshness_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
freshness_profile=latticra-seal-request-freshness/0.1
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
timestamp_parsing_implemented=0
trusted_clock_behavior_added=0
nonce_storage_added=0
replay_cache_storage_added=0
context_hashing_implemented=0
parameter_hashing_implemented=0
freshness_validation_implemented=0
replay_protection_implemented=0
policy_evaluation_added=0
policy_enforcement_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal signed request predecessor status boundary

The Seal signed request predecessor status alignment ties the existing report-only signed request status record to the guarded request freshness status predecessor while preserving no signature generation, no signature verification, no public-key parsing, no trust-store loading, no private-key handling, no key generation, no hardware-key use, no revocation lookup, no network trust lookup, no signed request enforcement, no policy evaluation, no policy enforcement, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no host behavior, no network behavior, and no production enforcement claim.

Current signed request predecessor status fields:

```
seal_request_freshness_status_present=1
seal_signed_request_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
signed_request_predecessor_request_freshness_status_present=1
signed_request_profile=latticra-seal-signed-request/0.1
signed_request_supported=0
signature_generation_supported=0
signature_verification_supported=0
signature_present=0
signature_valid=0
signature_algorithm_declared=0
signing_key_id_present=0
signature_hash_present=0
signed_request_id_present=0
identity_binding_declared=0
context_binding_declared=0
parameter_binding_declared=0
freshness_binding_declared=0
policy_binding_declared=0
trust_store_supported=0
revocation_lookup_supported=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
signature_generation_implemented=0
signature_verification_implemented=0
signed_request_enforcement_implemented=0
trust_store_implemented=0
revocation_lookup_implemented=0
policy_evaluation_added=0
policy_enforcement_added=0
freshness_validation_implemented=0
replay_protection_implemented=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal policy decision predecessor status boundary

The Seal policy decision predecessor status alignment ties the existing report-only policy decision status record to the guarded signed request status predecessor while preserving no policy evaluation, no policy enforcement, no signature verification, no freshness validation, no replay detection, no signed request enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no host behavior, no network behavior, and no production enforcement claim.

Current policy decision predecessor status fields:

```
seal_signed_request_status_present=1
seal_policy_decision_status_present=1
signed_request_predecessor_request_freshness_status_present=1
policy_decision_predecessor_signed_request_status_present=1
```

```
policy_decision_profile=latticra-seal-policy-decision/0.1
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
real_policy_evaluation_added=0
policy_enforcement_added=0
signature_verification_implemented=0
freshness_validation_implemented=0
replay_protection_implemented=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal operator receipt report predecessor status boundary

The Seal operator receipt report predecessor status alignment ties the existing denied report-only operator receipt status record to the guarded policy decision status predecessor while preserving no receipt file writes, no tool execution, no policy enforcement, no capability enforcement, no cryptographic verification, no signature verification, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no host behavior, no network behavior, and no production enforcement claim.

Current operator receipt report predecessor status fields:

```
seal_policy_decision_status_present=1
seal_operator_receipt_report_status_present=1
policy_decision_predecessor_signed_request_status_present=1
operator_receipt_report_predecessor_policy_decision_status_present=1
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
runtime_dry_run_state=report-only
default_action_deny=1
would_allow=0
would_deny=1
would_execute_tool=0
would_read_host=0
```

```
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal core evidence boundary

The Seal core evidence status surface is now public-entrypoint visible.

It records a report-only runtime gate path with core negative-test evidence for AI-era tool-boundary planning. The latest Seal status/public-entry slice makes the report-only policy decision metadata and deterministic policy decision report surface visible from public status surfaces while preserving no real policy evaluation, no policy enforcement, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signature verification, no freshness validation, no replay detection, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, or runtime authority.

The Seal README status row alignment now keeps the compact README Seal row and Seal current-posture summary aligned with the current status checkpoint without changing implementation behavior, production enforcement, public readiness, security hardening, completion estimates, or runtime authority.

Current status fields:

```
seal_readme_status_row_alignment_present=1
seal_core_evidence_status_surface_present=1
seal_report_envelope_status_present=1
report_envelope_ready=1
report_envelope_state=sealed-report-only
report_envelope_signature_performed=0
report_envelope_runtime_authority_granted=0
seal_signature_request_contract_present=1
seal_signature_request_metadata_present=1
seal_signature_request_status_present=1
signature_request_predecessor_report_envelope_status_present=1
seal_signing_authorization_contract_present=1
seal_signing_authorization_metadata_present=1
seal_signing_authorization_status_present=1
signing_authorization_predecessor_signature_request_status_present=1
seal_signer_handoff_contract_present=1
seal_signer_handoff_metadata_present=1
seal_signer_handoff_status_present=1
signer_handoff_predecessor_signing_authorization_status_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
signer_invocation_predecessor_signer_handoff_status_present=1
seal_signing_operation_contract_present=1
seal_signing_operation_metadata_present=1
seal_signing_operation_status_present=1
signing_operation_predecessor_signer_invocation_status_present=1
seal_key_handling_contract_present=1
seal_key_handling_metadata_present=1
seal_key_handling_status_present=1
```

```
key_handling_predecessor_signing_operation_status_present=1
seal_key_material_contract_present=1
seal_key_material_metadata_present=1
seal_key_material_status_present=1
key_material_predecessor_key_handling_status_present=1
seal_public_key_parsing_contract_present=1
seal_public_key_parsing_metadata_present=1
seal_public_key_parsing_status_present=1
public_key_parsing_predecessor_key_material_status_present=1
seal_future_key_parsing_implementation_contract_present=1
seal_future_key_parsing_implementation_plan_present=1
seal_key_parsing_metadata_present=1
seal_key_parsing_status_present=1
key_parsing_predecessor_public_key_parsing_status_present=1
seal_verification_policy_metadata_present=1
seal_verification_policy_status_present=1
verification_policy_predecessor_key_parsing_status_present=1
seal_crypto_verify_backend_contract_present=1
seal_crypto_verify_backend_metadata_present=1
seal_crypto_verify_backend_status_present=1
crypto_verify_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verified=0
authority_usable=0
capability_gate_allowed=0
real_cryptographic_verification_added=0
seal_ed25519_verify_only_contract_present=1
seal_ed25519_verify_implementation_present=1
seal_ed25519_verify_status_present=1
ed25519_verify_profile=latticra-seal-ed25519-verify/0.1
ed25519_crypto_verify_state=verified
ed25519_cryptographic_verification_performed=1
cryptographic_verification_performed=1
ed25519_verified=1
ed25519_authority_usable=0
production_cryptography_claimed=0
seal_verified_receipt_promotion_contract_present=1
seal_verified_receipt_promotion_implementation_present=1
seal_verified_receipt_promotion_status_present=1
verified_receipt_promotion_state=verified
verified_receipt_promotion_cryptographic_verification_performed=1
verified_receipt_promotion_verified=1
verified_receipt_promotion_authority_usable=0
verified_receipt_promotion_capability_gate_allowed=0
verified_receipt_promotion_runtime_authority_granted=0
seal_verified_capability_gate_contract_present=1
seal_verified_capability_gate_implementation_present=1
seal_verified_capability_gate_status_present=1
verified_capability_gate_crypto_graduation_gate_passed=1
verified_capability_gate_standard_expectations_met=1
verified_capability_gate_authority_promotion_allowed=0
verified_capability_gate_allowed=1
verified_capability_gate_state=allowed-metadata-only
verified_capability_gate_runtime_authority_granted=0
verified_capability_gate_effect_performed=0
seal_verified_effect_decision_status_present=1
verified_effect_decision_crypto_graduation_gate_present=1
verified_effect_decision_crypto_graduation_gate_passed=1
verified_effect_decision_standard_expectations_met=1
verified_effect_decision_authority_promotion_allowed=0
verified_effect_decision_allowed=1
verified_effect_decision_state=allowed-report-only
verified_effect_decision_effect_performed=0
verified_effect_decision_runtime_authority_granted=0
seal_runtime_handoff_evaluation_status_present=1
runtime_handoff_evaluation_crypto_graduation_gate_present=1
runtime_handoff_evaluation_crypto_graduation_gate_passed=1
runtime_handoff_evaluation_standard_expectations_met=1
runtime_handoff_evaluation_authority_promotion_allowed=0
runtime_handoff_evaluation_eligible=1
runtime_handoff_evaluation_state=eligible-report-only
runtime_handoff_evaluation_handoff_performed=0
runtime_handoff_evaluation_runtime_authority_granted=0
seal_runtime_handoff_report_status_present=1
runtime_handoff_report_ready=1
runtime_handoff_report_state=ready-report-only
runtime_handoff_report_handoff_performed=0
runtime_handoff_report_runtime_authority_granted=0
seal_verification_receipt_metadata_present=1
```

```
seal_verification_receipt_status_present=1
verification_receipt_predecessor_verification_policy_status_present=1
seal_capability_gate_metadata_present=1
seal_capability_gate_status_present=1
capability_gate_predecessor_verification_receipt_status_present=1
seal_effect_decision_metadata_present=1
seal_effect_decision_status_present=1
effect_decision_predecessor_capability_gate_status_present=1
seal_runtime_handoff_metadata_present=1
seal_runtime_handoff_status_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
seal_status_rollup_metadata_present=1
seal_status_rollup_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
seal_runtime_handoff_report_status_present=1
runtime_handoff_report_ready=1
runtime_handoff_report_state=ready-report-only
runtime_handoff_report_handoff_performed=0
runtime_handoff_report_runtime_authority_granted=0
seal_agentic_automation_security_metadata_present=1
seal_agentic_automation_security_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
seal_agentic_automation_security_index_alignment_present=1
seal_agentic_automation_security_report_surface_present=1
seal_agentic_automation_security_public_entrypoint_alignment_present=1
seal_parameter_schema_contract_present=1
seal_parameter_schema_metadata_present=1
seal_parameter_schema_report_surface_present=1
seal_parameter_schema_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
seal_request_freshness_contract_present=1
seal_request_freshness_metadata_present=1
seal_request_freshness_report_surface_present=1
seal_request_freshness_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
seal_signed_request_contract_present=1
seal_signed_request_metadata_present=1
seal_signed_request_status_present=1
signed_request_predecessor_request_freshness_status_present=1
seal_policy_decision_contract_present=1
seal_policy_decision_metadata_present=1
seal_policy_decision_report_surface_present=1
seal_policy_decision_report_surface_status_present=1
seal_policy_decision_status_present=1
seal_policy_decision_public_entrypoint_alignment_present=1
policy_decision_predecessor_signed_request_status_present=1
seal_operator_receipt_report_status_present=1
operator_receipt_report_predecessor_policy_decision_status_present=1
operator_visible_status_surface=1
core_blocked_case_set_complete=1
runtime_gate_report_only=1
signature_performed=0
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
tool_execution_implemented=0
ai_agent_security_claimed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

The public-entrypoint alignment is documentation/status alignment only and does not change implementation behavior or public readiness.

Current defensive threat model boundary

The defensive threat model validation refinement records the current evidence-mapping and external-source checkpoint posture without promoting any security claim.

Current defensive threat model fields:

```
defensive_threat_model_contract_present=1
defensive_threat_model_implementation_plan_present=1
defensive_threat_model_validation_present=1
defensive_threat_model_validation_refinement_present=1
```

```
external_source_refresh_checkpoint_present=1
manual_source_review_required=1
high_assurance_security_baseline_present=1
source_refresh_date=2026-05-26
memory_safety_roadmap_required=1
memory_safety_roadmap_present=1
supply_chain_security_baseline_present=1
cyber_incident_reporting_response_baseline_present=1
cyber_incident_reporting_response_guard_present=1
cisa_reporting_channel_documented=1
fbi_cyber_reporting_channel_documented=1
vulnerability_management_release_gate_baseline_present=1
vulnerability_management_release_gate_guard_present=1
cisa_kev_catalog_tracked=1
nvd_cve_review_required=1
cryptographic_assurance_key_management_baseline_present=1
cryptographic_assurance_key_management_guard_present=1
fips_140_3_boundary_required_before_production_crypto=1
production_crypto_claim_allowed=0
identity_credential_access_management_baseline_present=1
identity_credential_access_management_guard_present=1
phishing_resistant_mfa_required_for_privileged_access=1
privileged_access_granted=0
security_logging_monitoring_baseline_present=1
security_logging_monitoring_guard_present=1
security_event_logging_required_before_hosted_service=1
production_monitoring_claim_allowed=0
backup_recovery_resilience_baseline_present=1
backup_recovery_resilience_guard_present=1
backup_restore_recovery_evidence_required_before_hosted_service=1
production_recovery_claim_allowed=0
secure_configuration_change_management_baseline_present=1
secure_configuration_change_management_guard_present=1
secure_configuration_change_control_required_before_hosted_service=1
production_configuration_claim_allowed=0
zero_trust_runtime_authority_baseline_present=1
zero_trust_runtime_boundary_required=1
runtime_boundary_policy_expansion_next=1
runtime_boundary_policy_expansion_after_threat_model_present=1
runtime_boundary_abuse_case_fixture_expansion_present=1
security_controls_added=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_verification_added=0
signing_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
runtime_authority_granted=0
certification_claim_allowed=0
compliance_claim_allowed=0
production_protection_claim_allowed=0
```

Current runtime boundary policy expansion boundary

Runtime boundary policy expansion after threat-model validation is now documented and guarded as a policy/evidence map. It expands request-family, effect, authority-prerequisite, future-gate, abuse-case, and evidence-gap mapping without changing runtime implementation behavior.

Current runtime boundary policy expansion fields:

```
runtime_boundary_policy_expansion_after_threat_model_present=1
runtime_boundary_policy_expansion_after_threat_model_guard_present=1
request_family_policy_map_present=1
effect_policy_map_present=1
authority_prerequisite_map_present=1
future_gate_policy_map_present=1
abuse_case_runtime_policy_map_present=1
evidence_gap_map_present=1
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
```

```
capability_enforcement_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
runtime_authority_granted=0
completion_estimate_review_required=0
```

Current runtime boundary abuse-case fixture boundary

Runtime boundary abuse-case fixture expansion after policy expansion is now documented and guarded with deterministic C fixtures for unknown request, unknown effect, future-gated execution, operator-confirmation non-override, denial reason reporting, authority failure, invalid LIR prerequisite, and blocked effect cases.

Current runtime boundary abuse-case fixture fields:

```
runtime_boundary_abuse_case_fixture_expansion_present=1
runtime_boundary_abuse_case_fixture_guard_present=1
runtime_boundary_abuse_case_c_fixtures_present=1
runtime_boundary_abuse_case_fixture_count=8
unknown_request_abuse_fixture_present=1
unknown_effect_abuse_fixture_present=1
future_gated_execution_abuse_fixture_present=1
operator_confirmation_non_override_fixture_present=1
denial_reason_report_fixture_present=1
authority_failure_abuse_fixture_present=1
invalid_lir_prerequisite_fixture_present=1
blocked_effect_abuse_fixture_present=1
report_reason_assertions_present=1
policy_matrix_assertions_present=1
domain_matrix_assertions_present=1
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
runtime_authority_granted=0
completion_estimate_review_required=0
```

Current completion estimate review boundary

Completion estimate review after runtime-boundary abuse-case fixtures confirms that the latest fixture/evidence slice improves denied-case coverage without changing capability posture, implementation behavior, public readiness, security hardening, product readiness, or runtime authority.

Completion estimate review README/status alignment makes the latest estimate hold review discoverable from README and status entry points without changing capability posture, implementation behavior, public readiness, security hardening, product readiness, completion estimates, or runtime authority.

The current high-level estimate table above is the live reader-facing estimate source. Dated review records remain slice-specific evidence records and may preserve the estimate snapshot that applied when that review was written.

Current estimate mathematical rebase records the live table as a weighted planning estimate after recent guarded Seal, Ubuntu, macOS, Nadia, Lat/LIR, kernel lifecycle, and public-entry alignment work. It changes the planning estimate table only; it does not change implementation behavior, security posture, public readiness, product readiness, runtime enforcement, or runtime authority.

Current completion estimate review fields:

```
current_estimate_table_source_alignment_present=1
current_estimate_mathematical_rebase_present=1
completion_estimate_review_readme_status_alignment_present=1
completion_estimate_after_runtime_boundary_abuse_case_fixtures_present=1
runtime_boundary_abuse_case_fixture_expansion_present=1
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
```

```
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
security_hardening_changed=0
public_readiness_changed=0
production_readiness_claimed=0
runtime_authority_granted=0
estimate_adjustment_required=0
completion_estimate_review_required=0
```

Current runtime boundary domain matrix report boundary

The runtime boundary domain matrix now has deterministic report rendering for domain-matrix cell, domain label, known/operational/declarative/future-gated flags, effect-allowed status, authority-available status, and evidence level.

Current Lat pipeline diagnostic boundary

The Lat pipeline now has a companion diagnostic integration surface that combines pipeline error/stage state with parser line-comment count and first-comment span metadata, Lat semantic diagnostic class, semantic error, diagnostic count, first-diagnostic indices, model-stage classification, optional Lat-to-LIR lowering diagnostic metadata, first lowered declaration metadata, and first lowered clause metadata while preserving no-execution behavior. The companion diagnostic integration is now covered by both the focused guard and the main Lat pipeline test runner.

Current runtime boundary domain matrix boundary

The runtime boundary now has a companion domain matrix evaluator for classifying resolved boundary domains as declarative, operational, future-gated, blocked, invalid, or unknown.

Current LIR boundary

The LIR layer now reports deterministic report classification, graph-shape labels, edge-kind summary counts, no-effect-chain status, and evidence level while preserving lowering outcomes and non-execution behavior.

Current Lat semantic boundary

Lat semantic validation now reports deterministic diagnostic classes, diagnostic category counters, first-diagnostic declaration and clause indices, and expanded report fields while preserving validation outcomes and no-effect behavior.

Current Lat grammar boundary

Lat grammar parsing now reports deterministic first declaration, first clause, and line-comment metadata from successful AST parses while preserving no-effect parsing behavior and avoiding operator evaluation. Line comments remain metadata-only, may contain otherwise forbidden behavior-marker words, and do not grant execution authority. Block-comment openers outside strings and line comments now fail deterministically with unsupported_block_comment.

Current Lat model normalization boundary

Lat now has a bounded no-effect normalized model layer after semantic validation. It builds typed declaration and clause index tables for states, policies, transitions, assertions, and effect declarations, resolves transition source-state metadata, preserves first normalized declaration metadata and first-clause role/effect/name/operator/value report metadata, preserves source spans and no-effect flags, and emits deterministic reports without reading source bytes or adding execution.

Current Nucleus task boundary

The Nucleus task layer remains no-effect and denied-by-default. It now reports explicit task report classification, task-domain labels, authorization-state labels, prerequisite status, no-effect-chain status, report-alignment status, no-effect-policy status, representation-gate status, execution status, effect status, and runtime status while preserving non-execution behavior.

Current project notes boundary

The project-notes surfaces now reflect the Nucleus task report-only execution README/status alignment and keep completion-estimate review conditional on capability posture changes.

Current announcement boundary

The Nucleus report-only announcement review confirms that no separate public announcement is needed because the recent Nucleus report-only and project-notes slices were documentation/status alignment only and did not change capability posture.

Current README/project-notes alignment boundary

The Nucleus report-only announcement README alignment makes the no-new-announcement review discoverable from README and the project-notes index without adding a public announcement entry or changing implementation behavior.

Current project-notes status/index boundary

The project notes Nucleus announcement README status/index check verifies that the project-note body alignment after the Nucleus announcement README alignment is represented from root status, the status index, and the foundation index without changing implementation behavior or completion estimates.

Current runtime boundary refinement boundary

The runtime boundary now carries no-effect Lat pipeline evidence, Lat pipeline first-clause evidence, Lat pipeline first-declaration evidence, Lat pipeline module/count evidence, Lat pipeline stage-summary evidence, Lat pipeline line-comment count and first-comment span evidence, Lat-specific LIR module-summary evidence, Lat-specific LIR source-span evidence, Lat-specific LIR node-kind evidence, Lat-specific LIR first-node evidence, Lat-specific LIR first-node span evidence, Lat-specific LIR first transition-node evidence, Lat-specific LIR first transition-node span evidence, Lat-specific LIR first-edge evidence, Lat-specific LIR first-edge span evidence, Lat-specific LIR first transition-source edge evidence, Lat-specific LIR first transition-source edge endpoint evidence, Lat-specific LIR first transition-source edge span evidence, Lat-specific LIR no-effect flag evidence, and Lat-specific LIR edge-kind evidence in records and reports. It also reports explicit classification, boundary-domain, authorization-state, evidence-level, and policy-matrix fields for runtime-boundary requests while preserving disabled-by-default behavior and future-gated execution requests.

Current Lat pipeline boundary

Lat now has a bounded no-effect path from source bytes through grammar parsing, semantic validation, model normalization, Lat-to-LIR lowering, and deterministic pipeline reporting.

The current pipeline implementation composes existing parser, semantic, model normalization, lowering, and LIR metadata outputs. It preserves no-effect flags and produces a LAT PIPELINE REPORT without executing Lat or LIR.

The Lat pipeline report now includes deterministic stage-summary metadata for last completed stage, failed stage, per-stage OK flags, model normalization status, no-effect-chain status, evidence level, parser line-comment count plus first-comment span metadata, first lowered declaration kind/name/source/index/clause metadata, and first lowered clause role/effect/name/operator/value/node metadata.

Current Lat-specific LIR refinement boundary

Lat-to-LIR lowering now emits explicit Lat declaration nodes and an explicit transition-source edge. This makes Lat state, policy, transition, assertion, requirement, and effect-declaration metadata visible inside the LIR shape while preserving the no-effect boundary.

Current Lat-to-LIR lowering boundary

Lat now has a bounded no-effect path from parser metadata through semantic validation and model normalization into LIR metadata.

The current lowering implementation consumes normalized Lat model metadata directly, creates a lat_module LIR module shape, preserves source spans, no-effect flags, model counts, first lowered declaration metadata, and transition source indices, and emits deterministic lowering reports. The parser-plus-semantic entry point remains available as a compatibility wrapper.

The Lat-to-LIR diagnostic refinement classifies lowering outcomes as valid, parse, semantic, model, effect-check, capacity, LIR, or internal, and emits deterministic diagnostic reports without changing lowering behavior.

The Lat-to-LIR diagnostic declaration metadata integration copies first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index into deterministic diagnostic reports without evaluating operators.

The Lat-to-LIR diagnostic clause metadata integration copies first lowered clause node index, role, effect, name, operator, and value into deterministic diagnostic reports without evaluating operators.

The Lat pipeline diagnostic integration now has a lowering-aware evaluator that can copy parser line-comment count and first-comment span metadata, Lat-to-LIR diagnostic class, lowering error, model error, LIR error, model counts, transition source metadata, first lowered declaration metadata, first lowered clause metadata, and failure flags into the pipeline diagnostic report.

The Lat-to-LIR clause metadata refinement preserves clause operators in LIR node metadata and reports the first lowered clause role, effect, name, operator, value, and node index without evaluating operators or adding execution.

Boundary terms tracked by the project status guard:

```
runtime behavior
command execution
effect-performing implemented C++ authority layer
```

Non-claims

This status file is a public shortcut. Detailed non-claims are maintained in docs/status/CURRENT_STATUS.md and docs/FOUNDATION_INDEX.md.

SECURITY.md

Source title: Security Policy

Lines: 204 | Words: 1532 | SHA-256: 632e658b134348e8b61259c7805655869ce531b65d94191b8c2899e2021b156e

Security Policy

Project status

Latticra is an early-stage systems architecture and language implementation project.

It is not currently a finished operating system, hardened sandbox, production runtime, installer, recovery writer, hardware mutation tool, server agent, or security boundary.

Security-related work in this repository should remain evidence-bound:

```
no claim without evidence
no execution before a contract
no mutation before visibility
no hardware effect before an explicit gate
no recovery behavior before rollback and failure behavior are documented
```

Supported versions

Latticra does not currently publish stable production releases.

Security reports should target the current main branch unless a maintainer explicitly identifies another supported branch or release line.

```
| Version or branch | Supported |
| --- | --- |
| `main` | Best-effort security review |
| Active feature/design branches | Best-effort while active |
| Tags/releases | Not security-supported unless explicitly marked |
```

Reporting a vulnerability

Please do not publish exploit details in a public issue, pull request, discussion, social post, or comment thread.

Preferred reporting path:

1. Use GitHub private vulnerability reporting or the repository security advisory flow if available.

2. If a private reporting path is not available, open a minimal public issue that says only that you need a private security contact path. Do not include exploit details, proof-of-concept payloads, secrets, or reproduction steps in that public issue.

A useful private report should include:

- affected file, component, or document;

- whether the issue is in code, tests, documentation, CI, or a claim boundary;

- clear reproduction steps, when safe to share privately;

- expected behavior;

- observed behavior;

- potential impact;

- whether the issue requires local access, malformed input, repository write access, CI access, network access, or hardware access;

- suggested fix, if known.

Vulnerability scope

Reports are in scope when they involve:

- parser memory safety risk;

- unchecked buffer behavior;

- malformed L-UI source handling;
- source-span or diagnostic confusion that could hide rejected behavior;
- CI or test gaps that allow unsupported behavior to appear accepted;
- documentation that overstates security maturity;
- repository workflow behavior that could expose secrets or unsafe automation;
- future server, update, recovery, hardware, or execution code once such code exists.

Reports are also welcome when they identify a mismatch between an implementation and a merged contract.

Currently out of scope

The following are usually out of scope unless they reveal a concrete security issue in this repository:

- requests for production hardening of unfinished features;
- claims that Latticra is not a complete operating system, because that is already a documented non-claim;
- issues in third-party tools not controlled by this repository;
- social engineering against maintainers or users;
- denial-of-service against GitHub, CI providers, package registries, or unrelated infrastructure;
- attacks against systems you do not own or have explicit permission to test.

Safe testing rules

Security testing must be limited to code, files, and systems you own or have explicit permission to test.

Do not:

- run destructive tests against third-party systems;
- attempt unauthorized access;
- exfiltrate data;
- publish secrets;
- test against unrelated production services;
- submit payloads designed to damage a maintainer's system;
- use Latticra security reports to coordinate malware, ransomware, credential theft, botnets, persistence, evasion, or unauthorized exploitation.

For parser and source-handling reports, prefer minimal local fixtures and deterministic reproduction steps.

Disclosure expectations

Maintainers will make a best-effort attempt to:

1. acknowledge private reports;
2. assess whether the report is security-relevant;
3. create or update a contract when the security boundary is unclear;
4. add tests before or alongside fixes;

5. avoid overstating the maturity of the fix;
6. credit reporters when appropriate and requested.

No response-time SLA is promised at this stage.

Security boundaries and non-claims

Latticra currently does not claim to provide:

- kernel isolation;
- hardened sandboxing;
- privilege separation;
- malware containment;
- ransomware prevention;
- secure boot;
- secure update delivery;
- hardware recovery safety;
- production-ready cryptography;
- production server security;
- operating-system completeness.

If future work introduces any of these areas, it should first receive a contract, tests, documentation, and explicit non-claims before being treated as security-relevant capability.

Secrets and sensitive data

Do not include real secrets in issues, pull requests, tests, fixtures, screenshots, logs, or documentation.

Examples of secrets and sensitive data include:

- API keys;
- tokens;
- passwords;
- private keys;
- recovery phrases;
- session cookies;
- personal identifying information;
- private infrastructure details;
- exploit details for unpatched third-party systems.

If a secret is accidentally committed, assume it is compromised and rotate it immediately.

Security development principles

Security-relevant changes should prefer:

- small, reviewable diffs;
- explicit contracts before implementation;

- deterministic tests;
- no-effect defaults;
- fail-closed behavior;
- stable diagnostics;
- source-span clarity;
- explicit non-claims;
- clear rollback or failure behavior when mutation is ever introduced.

High-assurance standards posture

The current high-assurance standards checkpoint is recorded in docs/HIGH_ASSURANCE_SECURITY_BASELINE.md.

That baseline tracks current NSA, CISA, FBI, and NIST security guidance as source input for Latticra security work. It requires memory-safety roadmap discipline, zero-trust runtime-boundary prerequisites, SSDF-style secure development evidence, CPG-inspired operational readiness gates, KEV-aware release review, SBOM evidence before production installer claims, a FIPS 140-3 cryptographic module boundary decision before any production cryptography claim, identity/access evidence before privileged or remote access claims, logging/detection evidence before hosted-service or monitoring claims, recovery evidence before backup, restore, rollback, failover, or continuity claims, and secure configuration/change-control evidence before hosted-service or hardening claims.

The component-level memory-safety roadmap is recorded in docs/MEMORY_SAFETY_ROADMAP.md.

The supply-chain security baseline is recorded in docs/SUPPLY_CHAIN_SECURITY_BASELINE.md.

The zero-trust runtime authority baseline is recorded in docs/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md.

It requires caller identity, resource identity, per-request authorization, least-privilege effect scope, policy decision visibility, denial reason visibility, and audit records before any future runtime authority is considered. Operator confirmation remains metadata-only and must not override denied effects.

The cyber incident reporting and response baseline is recorded in docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md.

It documents CISA, FBI, IC3, and joint #StopRansomware reporting paths, evidence-preservation requirements, out-of-band communication expectations, and the future gates required before any incident-response or report-assistance feature exists. Latticra does not report incidents, notify customers, contact law enforcement, collect forensic evidence, or provide incident-response services.

The vulnerability management release gate baseline is recorded in docs/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md.

It keeps release claims blocked until vulnerability source review, KEV/NVD review, disclosure path evidence, exception ownership, exception expiration, and non-exploitability evidence are recorded. Latticra does not publish security advisories, submit CVEs, generate SBOMs, claim vulnerability-free status, or authorize production releases.

The cryptographic assurance and key management baseline is recorded in docs/CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE.md.

It keeps cryptographic claims blocked until module boundaries, algorithm and parameter inventory, key lifecycle, key storage, key destruction, randomness, self-tests, sensitive-data handling, post-quantum migration planning, and FIPS/CMVP claim gates are recorded. Latticra does not implement production cryptography, signing authority, key storage, key generation, entropy collection, random-bit generation, FIPS validation, CMVP submission, CAVP testing, or post-quantum migration.

The identity, credential, and access management baseline is recorded in docs/IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE.md.

It keeps hosted service, remote access, privileged operator, SSO/federation, MFA, and identity-security claims blocked until human/service/machine identity inventory, phishing-resistant MFA planning, account lifecycle, privileged role mapping, break-glass policy, credential handling, identity logging, and access-exception evidence are recorded. Latticra does not implement an identity provider, MFA provider, remote login, account database, credential store, hosted administration surface, or privileged access authority.

The security logging, monitoring, and detection baseline is recorded in docs/SECURITY_LOGGING_MONITORING_BASELINE.md.

It keeps production monitoring, SIEM integration, telemetry export, log collection, hosted service, and detection-service claims blocked until event-source inventory, audit-event selection, log schema, runtime decision events, denial reason events, identity/access events, privileged-action events, redaction, integrity, retention, disposal, time-source, triage, and incident-handoff evidence are recorded. Latticra does not implement a log collector, SIEM, telemetry export, host sensor, network sensor, alerting service, detection rule, log storage service, or monitoring authority.

The backup, recovery, and cyber resilience baseline is recorded in docs/BACKUP_RECOVERY_RESILIENCE_BASELINE.md.

It keeps backup, restore, rollback, failover, disaster-recovery, ransomware-recovery, hosted-service recovery, and continuity claims blocked until critical asset inventory, restore ordering, RTO/RPO, backup scope, backup integrity verification, restore testing, clean recovery environment planning, recovery authorization, recovery communications, incident handoff, and lessons-learned evidence are recorded. Latticra does not implement backup creation, backup storage, restore execution, rollback execution, failover, disaster recovery, recovery orchestration, continuity operations, or recovery authority.

The secure configuration and change management baseline is recorded in docs/SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE.md.

It keeps secure-default, configuration hardening, configuration scanning, drift-detection, infrastructure-as-code, hosted-service configuration, and production configuration claims blocked until configuration item inventory, secure baseline configuration, checklist evidence, default-credential absence review, approved change record, risk review, test evidence, rollback plan, drift-detection plan, secret review, configuration-change logging, and exception ownership are recorded. Latticra does not implement host configuration mutation, infrastructure configuration mutation, configuration enforcement, configuration scanning, drift detection, change approval workflow, rollback execution, or configuration authority.

This posture is an allocation of required work, not a compliance claim. It does not certify Latticra, accredit Latticra, make Latticra a production security product, or create a hardened runtime boundary.

Contact and attribution

Use GitHub private vulnerability reporting or the repository security advisory flow when available.

If public contact is the only available path, open a minimal public issue requesting a private security contact and wait for maintainer direction before sharing details.

CONTRIBUTING.md

Source title: Contributing to Latticra

Lines: 124 | Words: 383 | SHA-256: fedf2b222a95e7f14bb23276bd30179272aca612621dc9372fd25e90ae712b72

Contributing to Latticra

Status: active contribution policy

Purpose

Latticra welcomes contributions that support an open, auditable, transparent systems project.

Contributions should preserve the project direction:

```
open source
auditable
transparent
security-conscious
evidence-bound
community-improvable
```

License expectation

Contributions are accepted under the license declared by the files being modified.

New core software source files should use:

```
SPDX-License-Identifier: AGPL-3.0-or-later
```

New SDK, example, packaging-helper, installer-glue, and integration-helper source files should use:

```
SPDX-License-Identifier: Apache-2.0
```

New documentation files may use:

```
SPDX-License-Identifier: CC-BY-4.0
```

These expectations are documented in:

```
docs/OPEN_ECOSYSTEM_POLICY.md
docs/LICENSE_POLICY.md
docs/DOCUMENTATION_LICENSE.md
```

New software source files should include an SPDX license identifier unless a documented exception exists.

No proprietary relicensing CLA

This project does not adopt a proprietary relicensing contributor license agreement through this policy.

By contributing, contributors represent that they have the right to submit the contribution under the applicable project license terms.

Developer Certificate of Origin direction

The project intends to use a Developer Certificate of Origin style contribution model.

A follow-up slice may add a formal Signed-off-by requirement and DCO guard.

Contribution quality

Contributions should be:

```
small enough to review
tested or guarded
evidence-bound
clear about effect boundaries
```

```
clear about non-claims
compatible with public project direction
```

Before changing safety guards, CI workflows, shell checks, or quality gates, run:

```
make quality-safety-guards
```

Before changing distro packaging metadata, package-readiness docs, or local package handoff lanes, run:

```
make quality-packaging-static
```

Before submitting broader code or installer changes, run:

```
make quality
```

Documentation changes

Documentation changes should preserve reader-facing cohesion and current non-claims.

Before changing public status, estimates, platform readiness, installer authority, security wording, subsystem status, or top-level navigation, read:

```
docs/README.md
docs/DOCUMENTATION_MAINTENANCE.md
```

For documentation-only changes, run the narrowest relevant guards from docs/DOCUMENTATION_MAINTENANCE.md and avoid changing source behavior, installer authority, package authority, runtime authority, workflow permissions, or security posture.

Security-sensitive changes

Changes that affect runtime behavior, state mutation, external effects, command behavior, file behavior, network behavior, hardware behavior, boot behavior, recovery behavior, or security claims require explicit tests and documentation.

Trademark and identity

Contribution rights do not grant permission to use the Latticra name, logos, marks, or Bryforge identity in a confusing way.

See:

```
TRADEMARK_POLICY.md
```

Legal note

This document is project guidance, not legal advice.

TRADEMARK_POLICY.md

Source title: Latticra Trademark and Identity Policy

Lines: 70 | Words: 323 | SHA-256: 527b28e366506095da08fc9d0a1dcfb8641dc4caf31d2b05817c291b3e0c953f

Latticra Trademark and Identity Policy

Status: initial trademark and identity policy

Purpose

This policy protects the Latticra project identity while preserving the freedom to study, modify, fork, and improve the software under its applicable open-source licenses.

Project identity

The following names and marks are project identity terms:

```
Latticra
Bryforge
```

```
Latticra logo
Bryforge logo
project slogans, marks, and distinctive visual identity
```

Code license boundary

Software and documentation licenses do not grant trademark, branding, endorsement, or official-project identity rights. Software and documentation licenses, including AGPL-3.0-or-later, Apache-2.0, and CC-BY-4.0, do not grant trademark, branding, endorsement, or official-project identity rights.

A fork may use the code according to the applicable code license, but it may not present itself as the official Latticra project unless authorized.

Allowed descriptive use

Forks, downstream projects, papers, articles, packages, and integrations may truthfully describe their relationship to Latticra.

Examples:

```
fork of Latticra
based on Latticra
compatible with Latticra
uses Latticra-derived code
```

Not allowed without permission

Do not use Latticra or Bryforge identity in a way that creates confusion about sponsorship, endorsement, or official status.

Examples of restricted uses:

```
calling a fork the official Latticra project
using the Latticra logo as the primary mark for an unrelated fork
claiming Bryforge endorsement without permission
using confusingly similar names for closed or unrelated offerings
```

Fork naming

Forks should use a distinct name if they are distributed publicly as separate projects.

They may describe their relationship to Latticra, but the fork name should not imply official project status.

Security and trust

Because Latticra is intended to be open, auditable, and transparent, identity clarity is a security concern.

Users should be able to tell whether they are using the official Latticra project, a fork, an experiment, or a third-party derivative.

Legal note

This document is project guidance, not legal advice.

Formal trademark registration, enforcement, or commercial use should receive qualified legal review.

docs/README.md

Source title: Latticra Documentation Hub

Lines: 140 | Words: 1236 | SHA-256: ac337fe3fb5e6fc036bd4004f64d8f1350b40e27a20b62364c9001131d1e3bd4

Latticra Documentation Hub

Status: active reader-facing documentation hub

Last updated: 2026-05-27 CDT Scope: public orientation, live status, foundation contracts, subsystem records, installer and packaging docs, security baselines, strategy notes, and documentation maintenance.

Purpose

This file is the deeper map behind the root README.

Latticra keeps a large evidence trail on purpose: status records, contracts, implementation notes, guard scripts, package lanes, platform notes, reports, public site pages, and non-claim ledgers. This hub keeps that material reachable without asking a new reader to absorb everything at once.

First Routes

Reader	Start	Then read
New user	[Quick Start Cheat Sheet](QUICK_START_CHEATSHEET.md)	[Installer README](../installer/README.md), [Status](../STATUS.md)
Reviewer	[Current Status](status/CURRENT_STATUS.md)	[Public Claims Ledger](PUBLIC_CLAIMS_LEDGER.md), [Non-Claims](NON_CLAIMS.md), [Evidence Ladder](EVIDENCE_LADDER.md)
Contributor	[Contributing](../CONTRIBUTING.md)	[Foundation Index](FOUNDATION_INDEX.md), [Documentation Maintenance](DOCUMENTATION_MAINTENANCE.md)
Packager	Platform README below	Platform workflow or validation lane below
Security reviewer	[Security Policy](../SECURITY.md)	[High-Assurance Security Baseline](HIGH_ASSURANCE_SECURITY_BASELINE.md), [Defensive Threat Model](DEFENSIVE_THREAT_MODEL_CONTRACT.md)
Handbook reader	[System Substrate README](latticra-system-substrate/README.md)	[System Substrate PDF](latticra-system-substrate/the-latticra-system-substrate.pdf), [Foundation Index](FOUNDATION_INDEX.md)

Core Public Documents

Document	Role
[Root README](../README.md)	Short public front door and documentation router
[Quick Start Cheat Sheet](QUICK_START_CHEATSHEET.md)	Shortest safe user-local install, run, update, reset, and cleanup route
[Status](../STATUS.md)	Public status shortcut and estimate mirror
[Current Status](status/CURRENT_STATUS.md)	Current progress, estimates, and next priorities
[Foundation Index](FOUNDATION_INDEX.md)	Exhaustive architecture, implementation, guard, status, and evidence index
[System Substrate handbook landing page](latticra-system-substrate/README.md)	Project-level long-form technical handbook
[Documentation Reader Journey Map](DOCUMENTATION_READER_JOURNEY_MAP.md)	Audience-specific reading routes and stop signals
[Documentation Glossary](DOCUMENTATION_GLOSSARY.md)	Shared vocabulary for evidence, validation, posture, platform, and readiness terms
[Public Claims Ledger](PUBLIC_CLAIMS_LEDGER.md)	Current public claims and blocked adjacent claims
[Non-Claims](NON_CLAIMS.md)	Claims Latticra explicitly does not make
[Real System Contract](REAL_SYSTEM_CONTRACT.md)	Project identity, real-system boundary, evidence rules, and non-claims
[Evidence Ladder](EVIDENCE_LADDER.md)	Promotion path from concept to real-system capability

Subsystem Entry Points

Subsystem	Entry points
Latticra Panel	[Panel README](../installer/README.md), [installer docs](../installer/docs/README.md), [installer readiness contract](../installer/docs/INSTALLER_READINESS_CONTRACT.md)
Latticra Console	[Console foundation](LATTICRA_CONSOLE_FOUNDATION.md), [UI terminal

language](UI_TERMINAL_LANGUAGE.md), [self-update model](SELF_UPDATE_MODEL.md) |

Latticra Seal	[Seal README](latticra-seal/README.md), [Seal status](latticra-seal/STATUS.md), [Seal architecture](latticra-seal/ARCHITECTURE.md), [Seal usage](latticra-seal/USAGE.md), [Seal boundaries](latticra-seal/BOUNDARIES.md), [Seal product spine](latticra-seal/PRODUCT.md)
Lat language	[Language strategy](LANGUAGE_STRATEGY.md), [language naming policy](LANGUAGE_NAMING_POLICY.md), [Lat pipeline contract](LAT_PIPELINE_CONTRACT.md), [Lat pipeline implementation](LAT_PIPELINE_IMPLEMENTATION.md), [Lat grammar contract](LAT_LANGUAGE_GRAMMAR_CONTRACT.md)
LIR	[LIR shape contract](LIR_SHAPE_CONTRACT.md), [Lat to LIR lowering contract](LAT_TO_LIR_LOWERING_CONTRACT.md), [Lat to LIR implementation](LAT_TO_LIR_LOWERING_IMPLEMENTATION.md), [LIR report refinement](LIR_REPORT_REFINEMENT.md)
L-UI	[L-UI parser](L_UI_PARSER.md), [source grammar](L_UI_SOURCE_GRAMMAR.md), [parser diagnostics](L_UI_PARSER_DIAGNOSTICS.md), [semantic validation contract](L_UI_SEMANTIC_VALIDATION_CONTRACT.md), [rendering contract](L_UI_RENDERING_CONTRACT.md)
Nucleus	[Supervisor architecture](SUPERVISOR_ARCHITECTURE.md), [task execution contract](NUCLEUS_TASK_EXECUTION_CONTRACT.md), [task execution implementation](NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md), [task report refinement](NUCLEUS_TASK_REPORT_REFINEMENT.md)
Runtime Boundary	[runtime boundary contract](RUNTIME_BOUNDARY_CONTRACT.md), [runtime boundary implementation](RUNTIME_BOUNDARY_IMPLEMENTATION.md), [runtime boundary refinement plan](RUNTIME_BOUNDARY_REFINEMENT_PLAN.md), [runtime boundary policy matrix](RUNTIME_BOUNDARY_POLICY_MATRIX_REFINEMENT.md)
Nadia offline AI	[Nadia foundation](NADIA_OFFLINE_AI_FOUNDATION.md), [local context engine](NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1.md), [runtime profile](NADIA_RUNTIME_PROFILE_STAGE_2.md), [guarded tool authority](NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7.md), [current Nadia status records](status/README.md)
Kernel lifecycle research	[kernel lifecycle seed](KERNEL_LIFECYCLE_SEED.md), [kernel lifecycle subsystem summary](KERNEL_LIFECYCLE_SUBSYSTEM_SUMMARY.md), [kernel state machine](KERNEL_STATE_MACHINE.md), [kernel scheduler seed](KERNEL_SCHEDULER_SEED.md)
Visual theorem engines	[visual theorem engines](VISUAL_THEOREM_ENGINES.md), [demos](demos/LATTICRA_SEAL_DEMO_v0_1.md)

Platform and Packaging

All platform lanes are local-only or no-effect unless a source record explicitly says otherwise. None of these entries claim distribution approval, archive readiness, official package status, root install authority, or production installability.

Platform	Reader entry	Validation and workflow records
Fedora	[packaging/fedora README](../packaging/fedora/README.md)	[Fedora developer workflow](FEDORA_DEVELOPER_WORKFLOW.md), [Fedora readiness plan](FEDORA_READINESS_PLAN.md), [local RPM static validation](FEDORA_LOCAL_RPM_STATIC_VALIDATION.md), [VM CLI payload validation lane](FEDORA_VM_CLI_PAYLOAD_VALIDATION_LANE.md)
Ubuntu	[packaging/ubuntu README](../packaging/ubuntu/README.md)	[Ubuntu developer workflow](UBUNTU_DEVELOPER_WORKFLOW.md), [Ubuntu readiness plan](UBUNTU_READINESS_PLAN.md), [local deb static validation](UBUNTU_LOCAL_DEB_STATIC_VALIDATION.md), [lintian static metadata contract](UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md)
Debian	[packaging/debian README](../packaging/debian/README.md)	[Debian local deb static validation](DEBIAN_LOCAL_DEB_STATIC_VALIDATION.md), [source archive contract](DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md), [package build gate](DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md), [install/remove transcript](DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md)
openSUSE	[packaging/opensuse README](../packaging/opensuse/README.md)	[openSUSE developer workflow](OPENSUSE_DEVELOPER_WORKFLOW.md), [openSUSE readiness plan](OPENSUSE_READINESS_PLAN.md), [local RPM static validation](OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md), [OBS non-claim review](OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md)
FreeBSD	[packaging/freebsd README](../packaging/freebsd/README.md)	[FreeBSD port static validation](FREEBSD_PORT_STATIC_VALIDATION.md), [source archive fixture lane](DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md), [package input handoff lane](DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md)
OpenBSD	[packaging/openbsd README](../packaging/openbsd/README.md)	[OpenBSD port static validation](OPENBSD_PORT_STATIC_VALIDATION.md), [source archive fixture lane](DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md), [publication non-claim review](DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md)
macOS	[macOS integration transferability plan](MACOS_INTEGRATION_TRANSFERABILITY_PLAN.md)	[app bundle writer dry run](MACOS_APP_BUNDLE_WRITER_DRY_RUN.md), [commit gate contract](MACOS_COMMIT_GATE_CONTRACT.md), [reset/uninstall dry-run contract](MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT.md), [verification transcript contract](MACOS_VERIFICATION_TRANSCRIPT_CONTRACT.md)

openSUSE RPM maintenance records:

- OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
- OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
- OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
- OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
- OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
- OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md

Shared Debian, FreeBSD, and OpenBSD package/port records:

- DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
- DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md

Security and Assurance

Area	Entry points
Reporting policy	[Security Policy](../SECURITY.md)
Security baseline	[High-Assurance Security Baseline](HIGH_ASSURANCE_SECURITY_BASELINE.md), [Defensive Threat Model](DEFENSIVE_THREAT_MODEL_CONTRACT.md), [Defensive Threat Model Validation](DEFENSIVE_THREAT_MODEL_VALIDATION.md)
Memory safety	[Memory Safety Roadmap](MEMORY_SAFETY_ROADMAP.md), [C/C++ security profile](security/C_CPP_SECURITY_PROFILE.md), [C ABI boundary policy](security/C_ABI_BOUNDARY_POLICY.md)
Supply chain	[Supply Chain Security Baseline](SUPPLY_CHAIN_SECURITY_BASELINE.md), [Vulnerability Management Release Gate](VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md)
Runtime authority	[Zero Trust Runtime Authority Baseline](ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md), [Runtime Boundary Contract](RUNTIME_BOUNDARY_CONTRACT.md)
Crypto and keys	[Cryptographic Assurance and Key Management Baseline](CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE.md), [Seal key material contract](LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md), [Seal verification policy contract](LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md)
Incident response and resilience	[Cyber Incident Reporting Response Baseline](CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md), [Security Logging Monitoring Baseline](SECURITY_LOGGING_MONITORING_BASELINE.md), [Backup Recovery Resilience Baseline](BACKUP_RECOVERY_RESILIENCE_BASELINE.md)
Identity and access	[Identity Credential Access Management Baseline](IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE.md)

Documentation Operations

Use these before changing public wording, navigation, claim posture, mirrored estimates, platform posture, or subsystem landing pages.

Document	Use it for
[Product Documentation Cohesion](PRODUCT_DOCUMENTATION_COHESION.md)	Product-facing copy, reader routes, public site resources, and subsystem landing pages
[Documentation Maintenance](DOCUMENTATION_MAINTENANCE.md)	Public entry points, status mirrors, static HTML summaries, and documentation-only validation
[Documentation Style Guide](DOCUMENTATION_STYLE_GUIDE.md)	Canonical terms, date style,

headings, link style, and claim wording |

| [Documentation Traceability Matrix](DOCUMENTATION_TRACEABILITY_MATRIX.md) | Mapping public surfaces to source records, validation, mirrors, and non-claims |

| [Documentation Validation Playbook](DOCUMENTATION_VALIDATION_PLAYBOOK.md) | Hygiene, link, public-entry, estimate, platform, subsystem, and claim-promotion checks |

| [Documentation Drift Response Playbook](DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md) | Mirror drift, stale evidence, non-claim drift, and claim demotion |

| [Documentation Change Review Packet](DOCUMENTATION_CHANGE_REVIEW_PACKET.md) | Explicit claim, mirror, validation, and rollback review |

Status and Strategy

| Area | Entry points |

| --- | --- |

| Current status | status/CURRENT_STATUS.md, [status index](status/README.md), [announcements](status/ANNOUNCEMENTS.md) |

| Current project direction | [project notes](project_notes/README.md), [current direction](project_notes/CURRENT_DIRECTION.md), [upcoming work](project_notes/UPCOMING_WORK.md) |

| Strategy records | [strategy index](strategy/README.md), [overall strategy priority map](strategy/2026-05-26-1702-cdt-overall-strategy-priority-map.md) |

| Public site status pages | status.html, validation.html, evidence.html, architecture.html, security.html, packaging.html, panel.html, kernel.html |

Maintenance Rules

- Keep the root README short, navigable, and user-facing.
- Keep this hub as the route map for deeper docs.
- Keep Foundation Index evidence-heavy and exhaustive.
- Keep Status, Current Status, and status mirrors aligned when public posture changes.
- Keep non-claims near any security, packaging, installer, runtime, or OS-adjacent wording.
- Do not promote production runtime, host protection, package approval, certification, or OS replacement claims without implementation evidence, tests, status records, and public-entry alignment.

Current Non-Claim Reminder

Latticra is early-stage and evidence-bound. It is not a production security product, hardened sandbox, operating-system replacement, root installer, network authority, production runtime, Fedora-approved package, Ubuntu archive-ready package, Debian archive-ready package, FreeBSD official port, OpenBSD official port, or openSUSE official package.

docs/QUICK_START_CHEATSHEET.md

Source title: Latticra Quick Start Cheat Sheet

Lines: 247 | Words: 653 | SHA-256: c6322b22532d980fe757deed1ba80761a956a51862fe083ea1ccdf8358591a76

Latticra Quick Start Cheat Sheet

Audience: users who want the shortest safe path to install, run, update, and remove the current user-local Latticra Panel.

Status: Latticra is still early-stage and evidence-bound. The current installer is user-local. It does not install a root service, change the kernel, change systemd, change SELinux, or use network authority.

Fast Install

Ubuntu prerequisites:

```
sudo apt-get update
sudo apt-get install -y rustc cargo make gcc pkg-config \
  libx11-dev libxcb1-dev libxcursor-dev libxrandr-dev libxi-dev \
  libxkbcommon-dev libgl1-mesa-dev libwayland-dev desktop-file-utils \
  libgtk-3-bin
```

Fedora prerequisites:

```
sudo dnf install -y rust cargo make gcc pkgconf-pkg-config \
  libX11-devel libxcb-devel libXcursor-devel libXrandr-devel libXi-devel \
  libxkbcommon-devel mesa-libGL-devel wayland-devel desktop-file-utils gtk3
```

openSUSE prerequisites:

```
sudo zypper refresh
sudo zypper install -y rust cargo make gcc pkgconf \
  libX11-devel libxcb-devel libXcursor-devel libXrandr-devel libXi-devel \
  libxkbcommon-devel Mesa-libGL-devel wayland-devel desktop-file-utils \
  gtk3-tools
```

Clone and enter the repo:

```
git clone https://github.com/Bryforge/Latticra.git
cd Latticra
```

Keep the user-local command path available:

```
export PATH=&quot;$HOME/.local/bin:$PATH&quot;
```

Preview first:

```
make -C installer dry-run
```

Install and verify:

```
make -C installer local-example
make -C installer verify-local
```

Ubuntu no-effect validation:

```
sh scripts/test-ubuntu-build-lane.sh
sh scripts/test-ubuntu-developer-workflow.sh
sh scripts/test-ubuntu-package-notice-inventory.sh
sh scripts/test-ubuntu-doc-payload-license-review-contract.sh
sh scripts/test-ubuntu-third-party-material-review-contract.sh
sh scripts/test-ubuntu-generated-artifact-notice-review-contract.sh
sh scripts/test-ubuntu-notice-file-decision-contract.sh
sh scripts/test-ubuntu-debian-copyright-notice-mapping-contract.sh
sh scripts/test-ubuntu-trademark-notice-boundary-contract.sh
sh scripts/test-ubuntu-release-artifact-notice-requirements-contract.sh
sh scripts/test-ubuntu-package-notice-promotion-gate-contract.sh
sh scripts/test-ubuntu-package-license-promotion-gate-contract.sh
sh scripts/test-ubuntu-lintian-static-metadata-contract.sh
sh scripts/test-ubuntu-local-deb-build-transcript-acceptance-gate-contract.sh
sh scripts/test-ubuntu-local-deb-install-remove-evidence-contract.sh
sh scripts/test-ubuntu-source-package-evidence-contract.sh
sh scripts/test-ubuntu-upload-signing-authority-evidence-contract.sh
sh scripts/test-ubuntu-ppa-archive-publication-gate-contract.sh
```

openSUSE no-effect validation:

```
sh scripts/test-opensuse-developer-workflow.sh
sh scripts/test-opensuse-local-rpm-static-validation.sh
```

openSUSE tool availability validation:

```
sh scripts/test-opensuse-rpmlint-osc-availability.sh
sh scripts/test-opensuse-rpmlint-static-spec-lane.sh
sh scripts/test-opensuse-rpmlint-findings-classification.sh
sh scripts/test-opensuse-source-archive-reproducibility-contract.sh
sh scripts/test-opensuse-source-archive-fixture-lane.sh
sh scripts/test-opensuse-rpm-topdir-handoff-lane.sh
sh scripts/test-opensuse-local-rpm-build-gate-contract.sh
sh scripts/test-opensuse-local-rpm-build-environment-contract.sh
sh scripts/test-opensuse-rpm-artifact-naming-contract.sh
sh scripts/test-opensuse-rpm-payload-inspection-contract.sh
sh scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
sh scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
```

Run

Open the GUI:

```
latticra-panel
```

Launch it from a terminal and keep that terminal usable:

```
tmpdir=&quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-panel.XXXXXX&quot;)&quot;
latticra-panel &gt;&quot;$tmpdir/latticra-panel.log&quot; 2&gt;&amp;1 &amp;
```

Useful commands:

```
latticra status
latticra path
latticra seal report
latticra lc status
```

Update

From the source checkout:

```
git pull
latticra-panel
```

Use the Updater workspace in Latticra Panel. Preview the update first, then apply the guarded user-local update from the current checkout. The installer migrates old Latticra-owned user-local wrappers when it can prove they are legacy Latticra files. Unrelated user files are still preserved or refused.

```
latticra updater status
```

This prints the Panel-owned updater policy, including the dry-run/apply commands, guarded apply mode, receipt setting, and disabled network/root/system mutation authority.

Normal Reset Or Uninstall

Preview removal:

```
make -C installer uninstall-dry-run
make -C installer reset-dry-run
```

Remove the managed local install:

```
make -C installer uninstall-local
```

Or, from an installed wrapper:

```
latticra uninstall
```

Use reset when you plan to reinstall from new specs:

```
latticra reset
```

Clean Full User-Local Uninstall

Use this only when normal uninstall/reset cannot clean up an old or broken user-local install.

These commands are intentionally exact. Do not replace them with broad paths such as ~/.local, /usr, /, or unreviewed wildcards.

If LC was installed with a custom lc.install.command_wrapper, set LC_WRAPPER to that command name; the default is latticra-lc.

```
LATTICRA_PREFIX=&quot;${LATTICRA_PREFIX:-$HOME/.local/share/latticra}&quot;
LC_WRAPPER=&quot;${LC_WRAPPER:-latticra-lc}&quot;

rm -rf -- \
  &quot;$LATTICRA_PREFIX&quot; \
  &quot;$HOME/.local/share/latticra-validation&quot; \
  &quot;$HOME/.local/share/latticra-reset-receipts&quot;

rm -f -- \
  &quot;$HOME/.local/bin/latticra&quot; \
  &quot;$HOME/.local/bin/$LC_WRAPPER&quot; \
```

```
    &quot;$HOME/.local/bin/lat&quot; \
    &quot;$HOME/.local/bin/latticra-seal&quot; \
    &quot;$HOME/.local/bin/latticra-nadia&quot; \
    &quot;$HOME/.local/bin/latticra-panel&quot; \
    &quot;$HOME/.local/bin/latticra-installer&quot; \
    &quot;$HOME/.local/share/applications/latticra-panel.desktop&quot; \
    &quot;$HOME/.local/share/applications/latticra-installer.desktop&quot; \
    &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-panel.png&quot; \
    &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-installer.png&quot; \
    &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-seal.png&quot;
```

If those paths are root-owned because an earlier command was run with sudo, repeat the same exact cleanup with sudo:

```
  LATTICRA_PREFIX=&quot;${LATTICRA_PREFIX:-$HOME/.local/share/latticra}&quot;
  LC_WRAPPER=&quot;${LC_WRAPPER:-latticra-lc}&quot;

  sudo rm -rf -- \
    &quot;$LATTICRA_PREFIX&quot; \
    &quot;$HOME/.local/share/latticra-validation&quot; \
    &quot;$HOME/.local/share/latticra-reset-receipts&quot;

  sudo rm -f -- \
    &quot;$HOME/.local/bin/latticra&quot; \
    &quot;$HOME/.local/bin/$LC_WRAPPER&quot; \
    &quot;$HOME/.local/bin/lat&quot; \
    &quot;$HOME/.local/bin/latticra-seal&quot; \
    &quot;$HOME/.local/bin/latticra-nadia&quot; \
    &quot;$HOME/.local/bin/latticra-panel&quot; \
    &quot;$HOME/.local/bin/latticra-installer&quot; \
    &quot;$HOME/.local/share/applications/latticra-panel.desktop&quot; \
    &quot;$HOME/.local/share/applications/latticra-installer.desktop&quot; \
    &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-panel.png&quot; \
    &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-installer.png&quot; \
    &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-seal.png&quot;
```

Refresh the shell command cache after cleanup:

```
  hash -r 2&gt;/dev/null || true
```

Important Limits

- Current install scope is user-local under ~/.local.
- Dry-run first, then install.
- Latticra currently remains early-stage and evidence-bound.
- No production security boundary, root installer, system service, kernel integration, Ubuntu archive/PPA readiness, or network authority is claimed.

docs/DOCUMENTATION_READER_JOURNEY_MAP.md

Source title: Documentation Reader Journey Map

Lines: 66 | Words: 717 | SHA-256: 1b11c0107ef23710cf7402773e378f5826b0edfb1cd0e37f29376c33f8bbfbc5

Documentation Reader Journey Map

Status: active reader journey map Last updated: 2026-05-26 CDT Scope: user, operator, reviewer, contributor, packager, security reviewer, subsystem implementer, public-site reader, and handbook reader routes.

Purpose

This map keeps Latticra documentation usable for different readers without weakening the evidence boundary.

Use it with README.md, PRODUCT_DOCUMENTATION_COHESION.md, DOCUMENTATION_TRACEABILITY_MATRIX.md, and PUBLIC_CLAIMS_LEDGER.md.

The goal is simple: each reader should know where to start, what they can safely conclude, where to find proof, and which claims remain blocked.

Journey Matrix

Reader	Start	Continue to	Success criteria	Stop signals
New local user	[`../README.md`](../README.md), [`QUICK_START_CHEATSHEET.md`](QUICK_START_CHEATSHEET.md), [`start.html`](start.html)	[`../installer/README.md`](../installer/README.md), [`NON_CLAIMS.md`](NON_CLAIMS.md)	Reader can identify guarded user-local paths and cleanup/reset expectations.	Any expectation of root install, daily-driver install, production runtime, or broad host management.
Operator	[`../STATUS.md`](../STATUS.md), [`status/CURRENT_STATUS.md`](status/CURRENT_STATUS.md), [`validation.html`](validation.html)	Relevant installer, platform, or subsystem source record	Operator can identify current posture, runnable checks, denied effects, and required evidence.	Any unclear effect boundary, missing reset path, or missing non-claim near a runnable path.
Reviewer	[`status/CURRENT_STATUS.md`](status/CURRENT_STATUS.md), [`PUBLIC_CLAIMS_LEDGER.md`](PUBLIC_CLAIMS_LEDGER.md), [`NON_CLAIMS.md`](NON_CLAIMS.md)	[`DOCUMENTATION_TRACEABILITY_MATRIX.md`](DOCUMENTATION_TRACEABILITY_MATRIX.md), [`EVIDENCE_LADDER.md`](EVIDENCE_LADDER.md)	Reviewer can trace public wording to source evidence and blocked adjacent claims.	Public wording lacks a source record, validation path, or non-claim boundary.
Contributor	[`README.md`](README.md), [`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md), [`DOCUMENTATION_MAINTENANCE.md`](DOCUMENTATION_MAINTENANCE.md)	[`DOCUMENTATION_CHANGE_REVIEW_PACKET.md`](DOCUMENTATION_CHANGE_REVIEW_PACKET.md), [`DOCUMENTATION_VALIDATION_PLAYBOOK.md`](DOCUMENTATION_VALIDATION_PLAYBOOK.md)	Contributor can change docs without broadening claims or breaking mirrors.	Change would require implementation evidence, guard changes, or stronger public wording.
Packager or platform reviewer	[`validation.html`](validation.html), platform README or workflow record	[`PRODUCTION_INSTALLER_READINESS_CONTRACT.md`](PRODUCTION_INSTALLER_READINESS_CONTRACT.md), [`DOCUMENTATION_TRACEABILITY_MATRIX.md`](DOCUMENTATION_TRACEABILITY_MATRIX.md)	Reviewer can identify local-only package posture and platform-specific blockers.	Any implied Fedora approval, Ubuntu archive readiness, Debian archive readiness, FreeBSD official port, OpenBSD official port, openSUSE official package, or production installability.
Security reviewer	[`../SECURITY.md`](../SECURITY.md), [`security.html`](security.html), [`PUBLIC_CLAIMS_LEDGER.md`](PUBLIC_CLAIMS_LEDGER.md)	[`DEFENSIVE_THREAT_MODEL_CONTRACT.md`](DEFENSIVE_THREAT_MODEL_CONTRACT.md), [`HIGH_ASSURANCE_SECURITY_BASELINE.md`](HIGH_ASSURANCE_SECURITY_BASELINE.md), [`ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md`](ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md)	Reviewer can separate defensive planning from security guarantees.	Any claim of malware prevention, ransomware prevention, hardened sandboxing, certification, or production protection.
Subsystem implementer	[`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md)	Relevant contract, implementation plan, implementation record, status record, and guard	Implementer can find the exact boundary and validation path before editing.	Missing contract, missing guard, or implementation language that outruns status.
Public-site reader	[`index.html`](index.html), [`map.html`](map.html), [`status.html`](status.html)	Source Markdown records linked from the public page	Reader can scan posture quickly and reach exact source records.	Public HTML implies stronger claims than source Markdown.
Handbook reader	[`latticra-system-substrate/README.md`](latticra-system-substrate/README.md)	[`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md), project notes, status records	Reader can connect long-form narrative to current status and source records.	Handbook narrative outruns current status or omits current non-claims.

Journey Rules

1. A fast route may be short, but it must not hide non-claims.
2. A product route may be readable, but it must not replace source records.
3. A reviewer route must reach source records, validation, and blocked adjacent claims.
4. A platform route must keep local-only or no-effect status visible until official evidence exists.
5. A security route must keep security non-claims visible near security-adjacent wording.
6. A contributor route must include validation and drift-response docs before public wording changes.

Success Criteria

A reader journey is healthy when:

- the first page tells the reader what Latticra currently is and is not;
- the path to current status is visible;
- runnable commands have guarded or no-effect expectations;
- broad claims have source-record links;
- non-claims are near the claims they constrain;
- validation commands are named where evidence matters;
- the route does not require reading the full repository before making a narrow safe decision.

Friction Signals

Treat these as documentation-product friction:

- a new reader cannot find the current status within two clicks or two source links;
- an operator sees commands before effect boundaries;
- a reviewer sees a claim before evidence;
- a packager sees package shape before local-only status;
- a security reader sees protection language before non-claims;
- a contributor sees mirrors before source-of-truth rules;
- a public site page has no route back to source Markdown.

Boundary

This map optimizes documentation navigation only.

It does not change implementation behavior, installer authority, package authority, runtime authority, workflow permissions, security posture, public estimates, or product readiness.

docs/DOCUMENTATION_GLOSSARY.md

Source title: Documentation Glossary

Lines: 101 | Words: 1255 | SHA-256: 21fd20e3481451c8a10b22bccc7932fece9ba30c4e57cf209e2b30c9d88cf3b4

Documentation Glossary

Status: active documentation glossary Last updated: 2026-05-26 CDT Scope: public documentation terms, claim posture, evidence wording, validation wording, package posture, installer posture, runtime posture, and reader-route terminology.

Purpose

This glossary defines common documentation terms used across Latticra public and source records.

Use it with DOCUMENTATION_STYLE_GUIDE.md, PUBLIC_CLAIMS_LEDGER.md, PRODUCT_DOCUMENTATION_COHESION.md, and DOCUMENTATION_TRACEABILITY_MATRIX.md.

This glossary is not a source of capability. It defines wording only.

Core Terms

Term	Meaning	Do not imply
evidence-bound	A claim is limited by source records, tests, validation, status, and non-claims.	Production readiness or safety guarantees.
contract-first	A boundary, behavior, failure mode, and non-claim are documented before implementation expands.	That the implementation already exists.
source record	The contract, implementation note, status record, transcript, or guard-backed document that supports a public statement.	That a public summary is authoritative by itself.
public surface	A reader-facing file or page such as README, public HTML, docs hub, package README, installer README, or subsystem landing page.	That public wording can outrun source records.
mirror	A second location repeating source posture for reader convenience.	That duplicate wording can become a new source of truth.
public-entry alignment	Updating public entry points so they match current source records and status.	Capability promotion.
traceability	A visible path from public wording to source record, validation, mirror, and non-claim boundary.	Proof of runtime behavior by itself.
validation	A check, test, guard, transcript, or review that supports a narrow claim.	General correctness, safety, or release readiness.
guard script	A deterministic local script that checks a narrow documentation, status, package, source, or behavior expectation.	Broad security or product certification.
review packet	A structured review record for documentation wording, mirrors, validation, non-claims, and rollback or demotion path.	Implementation evidence.

Capability Posture

Term	Meaning	Do not imply
no-effect	The path is intended to report, classify, validate, or model without performing host, network, runtime, package, installer, or security effects.	Harmlessness, safety guarantee, or production approval.
report-only	The path produces metadata or reports without enforcing policy, granting authority, or performing effects.	Enforcement, protection, containment, or runtime authority.
denied by default	The current classification blocks behavior unless later evidence and gates explicitly allow a narrow path.	A production sandbox or full policy engine.
guarded	A path has explicit checks, prerequisites, or review gates before a narrow operation can proceed.	Broad permission or unattended operation.
guarded user-local	A user-scoped path with documented boundaries and no broad root/system authority claim.	Root install, daily-driver install, production install, or host-management product.
local-only	A package, validation, or workflow lane is limited to local records and checks.	Official distribution, archive, store, or vendor approval.
static validation	A check of files, metadata, shape, wording, or configuration without proving full build/install/runtime behavior.	Publication readiness or install safety.
disposable validation	Validation intended for a throwaway VM or controlled environment.	Daily-driver safety or broad host compatibility.
runtime authority	Permission or capability to execute effects at runtime.	Present capability unless a source record explicitly proves it.
host behavior	Reading, writing, mutating, installing, deleting, managing, or otherwise affecting the host.	Allowed behavior unless explicitly proven and gated.
network behavior	Network access, server interaction, update fetching, remote trust lookup, or internet-facing behavior.	Present capability unless explicitly proven and bounded.

Product And Status Terms

Term	Meaning	Do not imply
public status	Reader-facing current posture, estimates, and next blockers.	Release status, readiness guarantee, or support commitment.
planning estimate	A rough posture marker for project planning.	Release promise, completion guarantee, or product-readiness metric.
product readiness	A future state requiring implementation, validation, support boundaries, non-claims, and public-entry alignment.	Current Latticra status.
current posture	The narrowest currently supported description of a topic.	Future ambition or roadmap completion.
next valid work	The next work that can proceed without overclaiming or skipping evidence.	A commitment to ship or release.
blocked adjacent claim	A tempting broader claim that must remain explicitly unsupported.	Hidden future capability.

| demotion | Narrowing public wording when evidence is stale, contradicted, missing, or too weak. | Failure of the project; it is claim hygiene. |
| drift | A disagreement between public pages, source records, status mirrors, estimates, validation, or non-claims. | An implementation change by itself. |

Platform Terms

Term	Meaning	Do not imply
Fedora lane	Fedora-specific local package, workflow, transcript, or validation work.	Fedora approval or distribution readiness.
Ubuntu lane	Ubuntu-specific local deb, workflow, package, notice, or validation work.	Ubuntu archive readiness, PPA readiness, or Canonical endorsement.
Debian lane	Debian-specific local package or archive-preparation work.	Debian archive upload, sponsorship, or official package status.
FreeBSD lane	FreeBSD-specific local port-shape or validation work.	FreeBSD ports-tree readiness or official port status.
OpenBSD lane	OpenBSD-specific local port-shape or validation work.	OpenBSD ports-tree readiness or official package status.
openSUSE lane	openSUSE-specific local RPM, rpmlint, osc, spec, or validation work.	Open Build Service publication, official package status, or SUSE endorsement.
macOS lane	macOS-specific no-effect or user-local planning, probing, dry-run, app-bundle, reset/uninstall, or verification work.	Signed app, notarized app, production installer, privileged helper, or broad host authority.

Reader Terms

Term	Meaning	Do not imply
new local user	A reader trying the guarded local path or understanding basic commands.	Production operator.
operator	A reader evaluating runnable paths, local checks, denied effects, and evidence requirements.	Someone granted hidden authority.
reviewer	A reader checking public claims against source evidence and non-claims.	Product approver.
contributor	A reader changing docs or source records under existing guardrails.	Someone authorized to broaden public claims without evidence.
packager	A reader reviewing platform package shape, static validation, and local-only boundaries.	Distribution maintainer approval.
security reviewer	A reader checking defensive posture, threat-model records, safe-testing rules, and security non-claims.	Certification authority.

Replacement Rule

When a term could be read as a stronger claim, replace it with the narrowest glossary term.

Examples:

Replace	With
safe	no-effect, guarded, or validated by local guard
ready	current posture or local-only draft
secure	documented defensive boundary with security non-claims
official	local-only unless official evidence exists
enforcement	report-only or denied-by-default classification
installer	guarded user-local Panel path unless production installer evidence exists

Boundary

This glossary standardizes documentation wording only.

It does not change implementation behavior, installer authority, package authority, runtime authority, workflow permissions, security posture, public estimates, or product readiness.

docs/PRODUCT_DOCUMENTATION_COHESION.md

Source title: Product Documentation Cohesion

Lines: 178 | Words: 1038 | SHA-256: f6276e3990c89107e14b1bfe07e3306d419ecbdeffdc67d02e2f3c031381b77c

Product Documentation Cohesion

Status: active product-documentation cohesion guide Last updated: 2026-05-26 CDT Scope: public README, documentation hub, static site pages, installer docs, package docs, security docs, status records, subsystem maps, and reader routes.

Purpose

This guide keeps Latticra's product-facing documentation coherent without implying that Latticra is already a production product.

In this file, "product documentation" means the reader-facing surface of the project: what a user, operator, reviewer, packager, or contributor sees before they understand the full evidence trail.

The product surface should always make four things easy to find:

1. what exists now;
2. what a reader can safely run or inspect;
3. what remains intentionally blocked;
4. what evidence would be required before the claim can become stronger.

Use DOCUMENTATION_READER_JOURNEY_MAP.md to keep audience-specific reader paths practical, bounded, and explicit about stop signals.

Use DOCUMENTATION_GLOSSARY.md when product copy depends on shared terms such as no-effect, report-only, guarded user-local, local-only, public status, drift, demotion, or product readiness.

Cohesion Promise

All product-facing documentation should describe Latticra as:

```
early-stage
evidence-bound
contract-first
local-first where runnable
no-effect or guarded by default
explicit about non-claims
```

Do not let a shorter product summary outgrow the evidence in status records, contracts, guard scripts, or the PUBLIC_CLAIMS_LEDGER.md.

Use DOCUMENTATION_TRACEABILITY_MATRIX.md to connect public surfaces to source records, validation checks, and non-claim boundaries.

Use DOCUMENTATION_VALIDATION_PLAYBOOK.md to choose documentation validation commands and failure-handling paths.

Use DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md when product-facing wording, source records, status mirrors, non-claims, or platform posture disagree.

Use DOCUMENTATION_STYLE_GUIDE.md for canonical terminology, status/date style, heading shape, link style, and replacement wording.

Product Surface Matrix

```
| Surface | Reader job | Must say | Must not imply |
| --- | --- | --- | --- |
| [`../README.md`](../README.md) | Understand identity, posture, quick start, and non-claims. |
Early-stage, evidence-bound, current estimates, guarded local routes, and explicit non-claims. |
 Production platform, certified security, OS replacement, package approval, or broad installer
readiness. |
```

[`README.md`](README.md)	Choose the right documentation depth.	First reading paths, status sources, claim boundaries, platform lanes, and maintenance rules.	That every doc is equally authoritative or that public summaries override status records.
[`index.html`](index.html)	Scan the public site quickly.	Current posture, no-effect default, major records, and resource routes.	That GitHub Pages content is stronger than source Markdown records.
[`start.html`](start.html) and [`QUICK_START_CHEATSHEET.md`](QUICK_START_CHEATSHEET.md)	Try safe local entry points.	Guarded user-local commands, reset/cleanup paths, and expected no-effect posture.	Root installation, daily-driver readiness, boot readiness, or broad host management.
[`validation.html`](validation.html)	Understand local evidence lanes.	Local-only validation, transcripts, static checks, and disposable/guarded boundaries.	Distribution approval, production installability, update safety, or recovery safety.
[`security.html`](security.html) and [`../SECURITY.md`](../SECURITY.md)	Report issues and understand security posture.	Safe testing, threat-model work, security non-claims, and evidence boundaries.	Malware prevention, ransomware prevention, certified security, or hardened sandboxing.
[`../installer/README.md`](../installer/README.md)	Use Panel and installer lanes.	User-local guarded behavior, configuration boundaries, and platform-specific prerequisites.	Root installer authority, unattended host mutation, production installer readiness, or OS-base installation.
Platform package docs	Review package shape and platform readiness.	Local-only drafts, static checks, prerequisites, and non-approval boundaries.	Fedora, Ubuntu, Debian, FreeBSD, OpenBSD, openSUSE, or vendor endorsement.
Subsystem maps	Enter a technical lane.	Contract, implementation, status, and guard path for the subsystem.	That subsystem progress changes overall product readiness automatically.
Status records	Verify current posture.	Current evidence, estimate posture, next blockers, and non-claims.	Release commitments, guarantees, or broad public claims.

Reader Routes

New User

Use this route for someone trying Latticra locally. The fuller audience matrix is DOCUMENTATION_READER_JOURNEY_MAP.md.

1. ../README.md
2. QUICK_START_CHEATSHEET.md
3. start.html
4. ../installer/README.md
5. NON_CLAIMS.md

The route must stay practical and bounded. It should not require reading every architecture record before a guarded user-local check, but it must keep current non-claims visible.

Reviewer

Use this route for posture or due-diligence review:

1. status/CURRENT_STATUS.md
2. PUBLIC_CLAIMS_LEDGER.md
3. NON_CLAIMS.md
4. EVIDENCE_LADDER.md
5. DOCUMENTATION_MAINTENANCE.md

The route should make it easy to reject overbroad wording before it reaches public summaries.

Contributor

Use this route for someone changing documentation or product posture:

1. README.md
2. FOUNDATION_INDEX.md
3. DOCUMENTATION_MAINTENANCE.md
4. DOCUMENTATION_GLOSSARY.md
5. DOCUMENTATION_TRACEABILITY_MATRIX.md
6. DOCUMENTATION_VALIDATION_PLAYBOOK.md
7. DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md
8. DOCUMENTATION_STYLE_GUIDE.md
9. DOCUMENTATION_CHANGE_REVIEW_PACKET.md
10. The relevant subsystem contract and status record
11. The exact guard named by that record

The route should prevent documentation from getting ahead of implementation or validation evidence.

Packager Or Platform Reviewer

Use this route for platform-specific work:

1. validation.html
2. The platform workflow document or package README
3. PRODUCTION_INSTALLER_READINESS_CONTRACT.md
4. PUBLIC_CLAIMS_LEDGER.md
5. status/CURRENT_STATUS.md

The route should keep every package and platform lane local-only unless official publication or distribution evidence exists.

Product Copy Rules

Use product copy that is specific, inspectable, and bounded.

Prefer:

```
guarded user-local workbench
no-effect report surface
local-only package draft
static validation lane
contract-first runtime boundary
report-only Seal evidence
early-stage public status
```

Avoid:

```
secure product
agent security platform
production installer
official package
Linux replacement
OS distribution
runtime enforcement
host protection
```

Cross-Link Requirements

Every product-facing page that describes a capability should have a path to:

1. current status;

2. non-claims;
3. relevant contract or implementation record;
4. validation or guard evidence;
5. safe user route, if the capability can be run locally;
6. blocked-claim language, if adjacent unsupported claims are likely.

Drift Review

Run this review when public wording changes:

1. Does the public wording match PUBLIC_CLAIMS_LEDGER.md?
2. Does the route from README.md or index.html reach the exact source record?
3. Does the status record use the same posture words?
4. Does the page say what is blocked?
5. Does the validation path prove the narrow claim?
6. Does DOCUMENTATION_TRACEABILITY_MATRIX.md identify the source record, mirror, validation, and non-claim boundary?
7. Does DOCUMENTATION_VALIDATION_PLAYBOOK.md identify the narrowest relevant checks?
8. Does DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md identify the correct narrowing or demotion path if records disagree?
9. Does the change preserve the documentation-only boundary if no implementation changed?
10. Does the wording follow DOCUMENTATION_STYLE_GUIDE.md?
11. Does the change need a DOCUMENTATION_CHANGE_REVIEW_PACKET.md outcome label?

Maintenance Boundary

Use DOCUMENTATION_MAINTENANCE.md for update process, source-of-truth mirrors, estimate handling, and validation commands.

Use this file for product-facing coherence: audience route, product wording, surface responsibilities, and cross-link expectations.

docs/PUBLIC_CLAIMS_LEDGER.md

Source title: Latticra Public Claims Ledger

Lines: 120 | Words: 1111 | SHA-256: 36c8a89bac30f64e77b2f3afdb205aa990ab13b3d0a874a89bbe9a24f14b3bd2

Latticra Public Claims Ledger

Status: active documentation guard Last updated: 2026-05-26 Scope: README surfaces, public site pages, status records, subsystem summaries, installer/package documentation, release notes, and external-facing project descriptions.

Purpose

This ledger keeps public Latticra wording aligned with current evidence.

It does not promote any capability by itself. It translates the current status records, evidence ladder, non-claims, contracts, and implementation notes into safe wording that can be reused without implying production readiness, security guarantees, operating-system completeness, or distribution approval.

When two records disagree, use the narrowest current claim until a later contract, implementation record, validation record, status update, and public-entry alignment all support a stronger statement.

Source of Truth Order

Use this order when public wording needs a decision:

1. STATUS.md and status/CURRENT_STATUS.md
2. NON_CLAIMS.md
3. EVIDENCE_LADDER.md
4. REAL_SYSTEM_CONTRACT.md
5. Specific subsystem contracts, implementation records, status records, and guard scripts
6. This ledger as the wording guide for public summaries

Allowed Public Wording

Area	Allowed wording	Required boundary
Project identity	Latticra is an early-stage, evidence-bound systems architecture repository.	Do not call it a deployed product, production platform, daily-driver system, or operating-system replacement.
Documentation posture	Latticra uses contract-first documentation, status records, non-claims, and validation guards to keep claims evidence-bound.	Do not imply that documentation maturity equals implementation maturity.
Evidence model	Latticra has an evidence ladder for promoting concepts, fixtures, tested models, reports, guarded experiments, and later real-system capabilities.	Do not skip evidence levels or treat a lower-level record as a real-system claim.
Runtime Boundary	Latticra has denied-by-default runtime-boundary classification and report surfaces.	Do not claim runtime authority, live enforcement, process isolation, or effect-performing runtime behavior.
Nucleus	Nucleus task records and reports are report-only and denied by default.	Do not claim effect-performing task execution.
Latticra Seal	Latticra Seal has report-only dry-run, metadata, policy-denial, request-boundary, and negative-test evidence for AI-era tool-boundary planning.	Do not claim active AI-agent control, host protection, network blocking, hardened sandboxing, or production enforcement.
Lat, LIR, and L-UI	Lat, LIR, and L-UI have no-effect parse, validate, lower, metadata, and report paths.	Do not claim production language execution, LIR execution, or an interactive terminal-control renderer.
Panel and installer	Latticra Panel is a guarded local workbench and installer/control surface in active development.	Do not claim a daily-driver installer, root installer, distribution installer, or broad host-management product.
Fedora, Ubuntu, and openSUSE	The repo has local-only packaging drafts, workflow records, and validation lanes.	Do not claim Fedora approval, Ubuntu archive readiness, PPA availability, Open Build Service publication, or distribution support.
macOS lane	The macOS lane currently centers on no-effect probes, dry-run planning, app-bundle contracts, reset/uninstall contracts, and evidence gates.	Do not claim a signed, notarized, written, installed, or verified macOS app bundle unless later evidence says so.
Nadia offline AI	Nadia has staged offline AI foundation and prompt-evaluation planning contracts.	Do not claim model loading, inference, token generation, dialogue generation, prompt evaluation, or tool execution.
Security posture	Latticra documents defensive boundaries, safe-testing rules, and security non-claims.	Do not claim malware prevention, ransomware prevention, secure boot, verified boot, sandbox escape resistance, or certified security.

Blocked Public Wording

Blocked wording	Safer replacement
production-ready	early-stage, evidence-bound, or planning-stage
secure platform	documented defensive boundary with explicit security non-claims
hardened sandbox	report-only or no-effect boundary, depending on the record
AI-agent security system	AI-era tool-boundary planning with report-only evidence
runtime enforcement	denied-by-default classification and reporting

cryptographic enforcement layer	metadata, signing, key, or verification planning record, unless a specific verified implementation record says otherwise
operating system	systems architecture repository or operating-system-universe direction
OS replacement	no-effect architecture and validation research
daily-driver installer	guarded local workbench or local-only installer lane
official Fedora package	local Fedora RPM draft or validation lane
Ubuntu archive-ready package	local Ubuntu deb draft or validation lane
openSUSE package	local openSUSE RPM draft, spec lane, or validation lane
malware prevention	defensive threat-model planning and non-claims
ransomware prevention	defensive threat-model planning and non-claims
active model runtime	offline AI foundation or prompt-evaluation contract lane
tool execution authority	report-only guarded tool-authority preflight
root authority	no root authority claimed
host protection	host-facing boundary planning with non-claims

Promotion Requirements

Any stronger public claim needs all of the following before the wording is promoted:

Requirement	Minimum proof
Contract	A specific contract names the capability, effect boundary, failure behavior, operator boundary, and non-claims.
Implementation	A bounded implementation record names the exact files, behaviors, disabled paths, and report output.
Validation	Tests, guard scripts, transcripts, or reproducible reports prove the narrow behavior and denied behavior.
Status	Root and detailed status records describe the new posture without overstating it.
Public-entry alignment	README, public HTML pages, indexes, and subsystem summaries use the same wording.
Non-claim update	Unsupported adjacent claims remain explicit.
Rollback or demotion path	If evidence becomes stale, contradicted, or unsafe, the claim can be narrowed quickly.

Review Checklist

Before adding or changing public wording, confirm:

1. The wording names the exact evidence level or status surface.
2. The wording does not imply production, security, OS, installer, package, AI-runtime, network, host, root, or cryptographic authority beyond the evidence.
3. The linked source record exists and says the same thing.
4. Adjacent non-claims remain visible.
5. Denied effects are stated when the topic involves runtime, installer, package, AI, host, network, signing, or security behavior.
6. The public entry point and the detailed status record do not drift.
7. Shared terms match DOCUMENTATION_GLOSSARY.md.
8. Source, mirror, validation, and non-claim traceability follows DOCUMENTATION_TRACEABILITY_MATRIX.md.
9. Validation selection follows DOCUMENTATION_VALIDATION_PLAYBOOK.md.
10. Drift, stale evidence, and demotion handling follows DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md.
11. Terminology follows DOCUMENTATION_STYLE_GUIDE.md.
12. Broad public wording changes have a DOCUMENTATION_CHANGE_REVIEW_PACKET.md outcome.

Primary References

- README.md

- STATUS.md
- status/CURRENT_STATUS.md
- NON_CLAIMS.md
- EVIDENCE_LADDER.md
- REAL_SYSTEM_CONTRACT.md
- PRODUCT_DOCUMENTATION_COHESION.md
- DOCUMENTATION_GLOSSARY.md
- DOCUMENTATION_TRACEABILITY_MATRIX.md
- DOCUMENTATION_VALIDATION_PLAYBOOK.md
- DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md
- DOCUMENTATION_STYLE_GUIDE.md
- DOCUMENTATION_MAINTENANCE.md
- DOCUMENTATION_CHANGE_REVIEW_PACKET.md
- PRODUCTION_INSTALLER_READINESS_CONTRACT.md
- ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md
- latticra-seal/README.md
- LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md
- ../installer/README.md
- ../packaging/fedora/README.md
- ../packaging/ubuntu/README.md
- ../packaging/opensuse/README.md

docs/NON_CLAIMS.md

Source title: Latticra Non-Claims

Lines: 129 | Words: 401 | SHA-256: 2944f7021d67285088a4e4985b0b89a1b58002c55c4a251ba394f808e921216f

Latticra Non-Claims

Status: initial non-claims boundary Scope: claims Latticra does not currently make.

Purpose

This document protects Latticra from premature, confusing, or unsafe claims.

Latticra is an implementation track, but it is not automatically a finished implementation.

For public wording that is currently allowed, blocked, or promotion-gated, see the PUBLIC_CLAIMS_LEDGER.md.

Current non-claims

Latticra does not currently claim to be:

- a finished operating system;
- a daily-driver platform;

- a bootable release image;
- an installer;
- a recovery system;
- a hardened sandbox;
- a security boundary;
- a cryptographic enforcement layer;
- a kernel;
- a hypervisor;
- a virtual machine monitor;
- a container runtime;
- a firmware replacement;
- a production language runtime;
- a production systems platform.

Hardware non-claims

Latticra does not currently claim:

- hardware boot success;
- disk safety;
- firmware safety;
- device compatibility;
- driver support;
- recovery success;
- rollback success;
- network safety;
- system integrity guarantees.

Security non-claims

Latticra does not currently claim:

- malware prevention;
- ransomware prevention;
- exploit mitigation;
- secure boot enforcement;
- verified boot enforcement;
- process isolation;
- memory isolation;
- privilege isolation;
- sandbox escape resistance;
- tamper resistance;
- cryptographic correctness.

Preview boundary

Preview and modeled behavior must not be described as real-world capability unless the evidence ladder supports that claim.

Examples:

```
Allowed: Latticra has an L1 fixture for a lattice state model.
Blocked: Latticra has a real executable software universe.

Allowed: Latticra has an L2 tested transition model.
Blocked: Latticra can safely move a live system between domains.

Allowed: Latticra has L7 real-device read-only evidence.
Blocked: Latticra is hardware-ready for general use.
```

Promotion language

Use narrow evidence language.

Prefer:

```
contracted
documented
fixture-backed
tested
preview-only
read-only validated
virtual-target evidenced
real-device read-only evidenced
gated experimental
```

Avoid:

```
finished
secure
production-ready
unbreakable
hardened
fully bootable
daily-driver ready
a complete OS
```

Review rule

Every future claim should answer:

1. What evidence level supports this?
2. What tests or reports support this?
3. What does this still not prove?
4. What can fail?
5. What effects are possible?
6. What effects are impossible?
7. Is the user/operator boundary clear?
8. Does the public claims ledger allow this wording?

Current project status

Latticra currently claims only a repository identity, documentation seed, and real-system contract direction.

No real-system execution capability is claimed yet.

docs/REAL_SYSTEM_CONTRACT.md

Source title: Latticra Real-System Contract

Lines: 66 | Words: 298 | SHA-256: 0bd11748d0a8cc7caf5171a5f2db1ec56901b9f94a3b70061e517ba47e178b3b

Latticra Real-System Contract

Status: initial contract Scope: project identity, implementation boundary, evidence rules, and non-claims.

Purpose

Latticra is a contract-first real-system implementation track for lattice-oriented computing architecture and the Latticra Programming Language.

It exists to separate formal implementation from exploratory research, informal prototypes, and premature real-system claims.

Latticra may learn from precursor work, but public Latticra capability must be proven inside this repository through contracts, tests, evidence, and explicit boundaries.

Identity

Latticra means a lattice-oriented real-system architecture: a computing environment built from explicit state spaces, grids, matrices, transition rules, safety gates, and evidence-backed promotion.

The name should communicate:

- structured state;
- software universes;
- embedded real-system direction;
- formal implementation;
- evidence-bound promotion;
- professional computer-science grounding.

Project boundary

Latticra is its own public project identity.

Public documentation should describe Latticra by its own architecture, contracts, language direction, tests, and implementation evidence.

Precursor research and prototypes may inform private design decisions or documented promotion packets, but they are not public dependencies and are not public capability claims.

Required rule

No capability may enter Latticra as a real-system feature unless it has:

1. a written contract;
2. a non-claims section;
3. tests or validation evidence;
4. a direct Latticra purpose or documented precursor-promotion packet;
5. a failure model;
6. a rollback or recovery boundary when mutation is involved;
7. an operator-visible status surface.

Current non-execution boundary

This contract does not authorize:

- kernel implementation;
- boot image generation;
- installer behavior;
- disk mutation;
- recovery execution;
- network mutation;
- hardware mutation;
- production runtime claims;
- sandbox claims;
- security boundary claims.

Development mode

Latticra starts in contract-first mode.

The first implementation work should be models, fixtures, tests, validation reports, and promotion rules. Live or mutating work comes later.

docs/EVIDENCE_LADDER.md

Source title: Latticra Evidence Ladder

Lines: 175 | Words: 514 | SHA-256: 6b6b5b693ffaa4b5c2515441c4de9f897f769dcfb9177180b8a77afac3850742

Latticra Evidence Ladder

Status: initial evidence ladder Scope: promotion levels for concepts, components, hardware paths, and real-system claims.

Purpose

The evidence ladder defines how an idea moves from simulation to real implementation.

Latticra must not treat concepts as real-system capabilities until evidence supports them.

Ladder

L0: Concept

The idea has a name and a written description.

Allowed outputs:

- notes;
- diagrams;
- design vocabulary;
- non-claims.

Not allowed:

- implementation claims;
- hardware claims;

- security claims.

L1: Simulation fixture

The idea exists as deterministic simulation data or a static fixture.

Required evidence:

- fixture file;
- expected output;
- no mutation;
- no host effect;
- no external effect.

L2: Tested model

The idea has tests that preserve required behavior and forbidden behavior.

Required evidence:

- unit tests, docs tests, or validation scripts;
- denied-behavior tests;
- explicit non-claims.

L3: Read-only command surface

The idea is inspectable through a command or report.

Required evidence:

- command output;
- status labels;
- deterministic behavior;
- no mutation;
- no host or external effect.

L4: Runtime preview

The idea is wired into a runtime model but remains preview-only.

Required evidence:

- runtime state model;
- invariant tests;
- failure state;
- rollback visibility when relevant;
- operator-facing status.

L5: Guarded local experiment

The idea performs a narrow local operation under explicit gates.

Required evidence:

- opt-in command;
- confirmation path;

- audit output;
- rollback or reset path;
- failure handling;
- no hidden effects.

L6: VM or emulator evidence

The idea runs inside a VM, emulator, or controlled local virtual target.

Required evidence:

- launch command;
- logs;
- validation report;
- reproducibility notes;
- no real-device claim.

L7: Real-device read-only evidence

The idea observes or validates real hardware without changing it.

Required evidence:

- hardware target identity;
- read-only operator checklist;
- captured logs or report;
- redaction review;
- non-mutation proof.

L8: Real-device gated execution

The idea performs a narrow real-world action on real hardware.

Required evidence:

- explicit operator approval;
- device identity;
- preflight checks;
- backup/recovery plan;
- rollback or failure behavior;
- post-run validation.

L9: Promoted real-system capability

The capability is part of the Latticra implementation contract.

Required evidence:

- repeated successful evidence;
- tests;
- documentation;
- release note;

- support boundary;
- safety review.

Promotion rule

A concept may move up one level only when the evidence for the next level exists.

No concept may skip directly from simulation to real-device execution.

Demotion rule

A promoted concept must be demoted when evidence becomes stale, contradicted, unreproducible, unsafe, or ambiguous.

Claim language

Use exact level labels when describing capability.

Good:

```
This is an L2 tested model.
This is an L4 runtime preview.
This has L7 real-device read-only evidence.
```

Bad:

```
This is production-ready.
This is secure.
This is a finished operating system.
This can recover systems safely.
```

Current project level

Latticra currently begins at L0/L1 contract and fixture planning.

No real-system execution level is claimed yet.

docs/PRECURSOR_PROMOTION_RULE.md

Source title: Latticra Precursor Promotion Rule

Lines: 157 | Words: 419 | SHA-256: adfe5f7ac9e40f18e74851c09836db8a4fda59b8ed42bbd51ff106c224cd2cc7

Latticra Precursor Promotion Rule

Status: initial promotion rule Scope: how prior research, prototypes, experiments, and external lessons may become Latticra work.

Purpose

Latticra should not absorb every prior idea automatically.

Precursor work may inform Latticra, but Latticra should only promote an idea after it has evidence, boundaries, tests, and a clear implementation purpose.

This document keeps the project self-contained while still allowing useful lessons from earlier research and prototypes to be formalized.

Promotion rule

A concept may enter Latticra only when it has:

1. a clear technical purpose;
2. a documented non-claim boundary;
3. at least one fixture, test, or validation artifact;

4. a clear Latticra destination;
5. a failure or denial model;
6. an effect boundary;
7. a reviewable implementation path.

Promotion packet

Every promoted concept should start with a promotion packet:

```
name:
origin type: research|prototype|direct-design|external-standard|test-result
source evidence level:
Latticra target:
status:
operator-visible state:
mutation allowed: no|yes
host-effect: none|planned|gated|executed
external-effect: none|planned|gated|executed
rollback state:
non-claims:
required next evidence:
```

Early promotion candidates

State lattice model

Initial Latticra target:

```
lattice state fixture
state labels
invariant tests
read-only reports
```

Boundary:

```
no live movement
no origin mutation
no recovery execution
no host effect
no external effect
```

Tri-plane transition model

Initial Latticra target:

```
pure transition model
spatial plane
state plane
safety plane
denial reasons
```

Boundary:

```
preview-only
pure function
no hardware effect
no hidden state mutation
```

Lat language path

Initial Latticra target:

```
read-only source declarations
state inspection candidate
assertion language candidate
future controlled programming language
```

Boundary:

```
no production runtime claim
no live system mutation
no privileged execution
no automatic promotion
```

Hardware evidence path

Initial Latticra target:

```
hardware target profiles
evidence templates
read-only validation reports
VM evidence gates
real-device read-only gates
```

Boundary:

```
no installer claim
no bootable release claim
no recovery execution claim
no disk mutation claim
```

Rejection rules

Do not promote a concept when:

- it is only a name;
- it has no non-claims;
- it depends on hidden mutation;
- it implies hardware readiness without hardware evidence;
- it would confuse research with implementation;
- it bypasses the evidence ladder;
- it expands claims beyond what tests support.

Naming rule

Promoted concepts should use Latticra implementation-level names once they become public architecture.

Example:

```
research concept: multi-plane movement check
Latticra component candidate: tri-plane transition model
```

Current status

No precursor concept is automatically a real-system capability.

This document only defines the promotion gate.

docs/FOUNDATION_INDEX.md

Source title: Latticra Foundation Index

Lines: 631 | Words: 7821 | SHA-256: 79a91b94c6007e7712a1ff669da3a5622bf3e0c4ac994ba3eeb19e6aa897e366

Latticra Foundation Index

Status: active planning index Scope: foundation documents, project operations records, implementation documents, guard scripts, and current next-slice direction.

Purpose

This index collects the first architecture, policy, language, operations, strategy, status, and implementation documents for Latticra.

Latticra should refine its language model, supervisor model, effect gates, source model, update model, target architectures, strategy records, status records, C/C++ foundation direction, constrained C++ authority-layer policy, L-UI rendering path, Nucleus task execution boundary, runtime boundary, defensive threat model, and precursor-promotion rules before expanding real implementation work.

Foundation documents

- REAL_SYSTEM_CONTRACT.md — project identity, real-system boundary, evidence rules, and non-claims.
- EVIDENCE_LADDER.md — promotion levels from concept to real-system capability.
- PRECURSOR_PROMOTION_RULE.md — rules for promoting precursor research, prototypes, experiments, and external lessons into Latticra.
- NON_CLAIMS.md — claims Latticra does not currently make.
- ARCHITECTURE_SEED.md — initial state-lattice and software-universe vocabulary.
- C_CPP_FOUNDATION_DIRECTION.md — active constrained C/C++ foundation direction: C is the metal, C++ is the disciplined structure, Latticra is the contract.
- CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md — governed C++ policy, validator, effect-gate, audit, ownership, lifetime, allocation, exception, and boundary contract before implementation.
- CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION_PLAN.md — Constrained C++ authority layer implementation plan for exact API, namespace, file path, build, exception, RTTI, allocation, ownership, C ABI, validator, audit report, and test planning before implementation code.
- CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md — first no-effect constrained C++ authority implementation.
- CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md — review confirming the constrained C++ authority implementation remains no-effect, metadata-only, fixed-capacity, and denied-by-default.
- LANGUAGE_STRATEGY.md — role of C, constrained C++, Lat, L-UI, and LIR.
- LANGUAGE_NAMING_POLICY.md — Lat / Latticra Language naming and extension policy; forbids plain L and .l as public identifiers.
- NAMING_SYSTEM.md — formal naming rules for professional Latticra terminology.
- FEATURE_TRANSLATION_LEDGER.md — vocabulary ledger for translating exploratory labels into Latticra architecture terms.
- SUPERVISOR_ARCHITECTURE.md — Nucleus supervisor model and orchestration responsibilities.
- EFFECT_GATES.md — effect categories, gating rules, and visibility requirements.
- UI_TERMINAL_LANGUAGE.md — L-UI terminal/operator interface language direction.
- LATTICRA_CONSOLE_FOUNDATION.md — Latticra Console Stage-0 foundation, standalone dry-run/local installer presets, Panel installability, profile presets, standalone contract, host-embedding contract, read-only host inventory contract, host-adapter contract, receipt contract, Seal receipt-request contract, receipt payload schema, receipt payload artifact draft, receipt payload artifact review gate, receipt payload artifact review receipt contract, receipt payload artifact review receipt draft contract, receipt payload materialization plan, signature-request binding contract, OS-base planning contract, VM evidence contract, command registry, Runtime Boundary binding, and future host/OS-base posture.
- SERVER_INTERACTION_MODEL.md — signed, optional, inspectable server interaction model.

- NADIA_OFFLINE_AI_FOUNDATION.md — Stage-0 foundation contract for Latticra Nadia Witness Foundation, the planned offline AI companion for Latticra software development, systems engineering, AI development, and community-awareness principles.
- NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1.md — Stage-1 no-network local context-pack engine for Nadia.
- NADIA_RUNTIME_PROFILE_STAGE_2.md — Stage-2 offline runtime-profile boundary for Nadia before inference.
- NADIA_DEVELOPER_WORKBENCH_STAGE_3.md — Stage-3 developer-workbench prompt-plan generation for Nadia before prompt evaluation.
- NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4.md — Stage-4 systems-engineering mode validation for Nadia prompt plans before prompt evaluation.
- NADIA_PRODUCTIVITY_LOOP_STAGE_5.md — Stage-5 operator-reviewed productivity ledger for Nadia local learning evidence before training or tool authority.
- NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6.md — Stage-6 protective-safety boundary for Nadia non-sexual-use, anti-manipulation, and namesake-cause awareness restrictions.
- NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7.md — Stage-7 report-only guarded tool-authority preflight for Nadia before any tool execution.
- NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8.md — Stage-8 prompt-evaluation contract for Nadia before prompt materialization, prompt evaluation, inference, or tool execution.
- NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9.md — Stage-9 local model-registry contract for Nadia before model selection, model installation, prompt evaluation, inference, or tool execution.
- NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10.md — Stage-10 inference-readiness contract for Nadia before runtime invocation, model loading, prompt evaluation, inference, or tool execution.
- NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11.md — Stage-11 runtime-invocation contract for Nadia before runtime process spawning, model session creation, model loading, token generation, inference, or tool execution.
- NADIA_MODEL_LOAD_CONTRACT_STAGE_12.md — Stage-12 model-load contract for Nadia before model file opening, weight mapping, weight loading, token generation, inference, or tool execution.
- NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13.md — Stage-13 prompt-receipt contract for Nadia before prompt source opening, prompt text receipt, prompt materialization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14.md — Stage-14 prompt-materialization contract for Nadia before prompt buffer allocation, prompt text materialization, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15.md — Stage-15 awareness-dialogue contract for Nadia Initiative Q&A scope before dialogue generation, prompt evaluation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16.md — Stage-16 prompt-evaluation handoff contract for Nadia before prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17.md — Stage-17 tokenization-boundary contract for Nadia before tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.

- NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18.md — Stage-18 tokenizer-specification contract for Nadia before tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19.md — Stage-19 tokenizer-manifest contract for Nadia before tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20.md — Stage-20 tokenizer-artifact-inventory contract for Nadia before tokenizer artifact path resolution, artifact scanning, artifact hashing, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21.md — Stage-21 tokenizer-artifact-measurement contract for Nadia before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest recording, artifact size recording, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22.md — Stage-22 tokenizer-artifact-verification contract for Nadia before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest comparison, artifact size comparison, artifact verification, artifact binding, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23.md — Stage-23 tokenizer-artifact-binding contract for Nadia before tokenizer artifact opening, artifact reading, artifact hashing, artifact verification, artifact binding, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24.md — Stage-24 tokenizer-runtime-attachment contract for Nadia before tokenizer artifact opening, artifact reading, artifact hashing, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25.md — Stage-25 prompt-tokenization contract for Nadia before prompt text reading, prompt token creation, prompt token sequence recording, tokenizer runtime attachment, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26.md — Stage-26 prompt-token-sequence contract for Nadia before prompt token ID recording, token order recording, token offset recording, context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27.md — Stage-27 context-window assembly contract for Nadia before context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28.md — Stage-28 prompt-evaluation-input contract for Nadia before prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29.md — Stage-29 prompt-evaluation runtime handoff contract for Nadia before runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30.md — Stage-30 prompt-evaluation invocation contract for Nadia before invocation requests, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31.md — Stage-31 prompt-evaluation result contract for Nadia before result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32.md — Stage-32 prompt-evaluation result review contract for Nadia before result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33.md — Stage-33 prompt-evaluation result disposition contract for Nadia before disposition records, release records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34.md — Stage-34 prompt-evaluation result release contract for Nadia before release records, release decisions, release publication, release receipts, disposition records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35.md — Stage-35 prompt-evaluation result release receipt contract for Nadia before receipt records, receipt signing, receipt publication, release records, release decisions, release publication, disposition records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36.md — Stage-36 prompt-evaluation result release receipt review contract for Nadia before review records, review decisions, review findings, receipt records, receipt signing, receipt publication, release records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_37.md — Stage-37 prompt-evaluation result release receipt review disposition contract for Nadia before disposition records, disposition decisions, disposition findings, review records, review decisions, review findings, receipt records, receipt signing, receipt publication, release records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_38.md — Stage-38 prompt-evaluation result release receipt review disposition release contract for Nadia before disposition-release records, release decisions, release

publication, release packaging, release receipts, disposition records, review records, receipt records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_39.md — Stage-39 prompt-evaluation result release receipt review disposition release receipt contract for Nadia before release-receipt records, receipt signing, receipt publication, disposition-release records, release publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_40.md — Stage-40 prompt-evaluation result release receipt review disposition release receipt review contract for Nadia before release-receipt-review records, review decisions, review findings, release-receipt records, receipt signing, receipt publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_41.md — Stage-41 prompt-evaluation result release receipt review disposition release receipt review disposition contract for Nadia before release-receipt-review-disposition records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_42.md — Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release contract for Nadia before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_43.md — Stage-43 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract for Nadia before release-receipt records, receipt emission, receipt signing, receipt publication, release records, review records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_44.md — Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review contract Stage-44 + guardrails before release-receipt-review records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_45.md — Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract Stage-45 + guardrails before release-receipt-review-disposition records, disposition decisions, disposition findings, receipt signing, receipt publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46.md — Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract Stage-46 + guardrails before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46.md — Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract Stage-46 + guardrails before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- SELF_UPDATE_MODEL.md — staged signed update and rollback design.
- HOST_ARCHITECTURE_TARGETS.md — x86_64 and ARM64 target policy.
- ROADMAP.md — design-first roadmap before implementation.
- LICENSE_POLICY.md — hybrid project licensing, contributions, branding, and future notice rules.
- DOCUMENTATION_LICENSE.md — CC-BY-4.0 documentation and handbook license decision.

Project operations documents

- README.md — reader-facing documentation hub for choosing the right documentation level before using the full foundation index.
- DOCUMENTATION_GLOSSARY.md — shared documentation glossary for public terms, claim posture, validation, platform posture, and reader-route terminology.
- DOCUMENTATION_READER_JOURNEY_MAP.md — reader journey map for user, operator, reviewer, contributor, packager, security, subsystem, public-site, and handbook routes.
- DOCUMENTATION_MAINTENANCE.md — maintenance guide for public entry points, estimate mirrors, status docs, platform docs, static HTML summaries, and documentation-only validation.
- DOCUMENTATION_CHANGE_REVIEW_PACKET.md — reusable documentation review packet for public wording, reader routes, mirrors, non-claims, validation, and rollback checks.
- DOCUMENTATION_STYLE_GUIDE.md — documentation style guide for canonical terms, status/date style, claim wording, headings, links, tables, and replacement phrasing.
- DOCUMENTATION_TRACEABILITY_MATRIX.md — traceability matrix mapping public surfaces to source records, validation, mirrors, and non-claim boundaries.
- DOCUMENTATION_VALIDATION_PLAYBOOK.md — documentation validation playbook for hygiene, link, public-entry, estimate, platform, subsystem, and claim-promotion checks.
- DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md — documentation drift-response playbook for mirror drift, stale evidence, non-claim drift, and claim demotion.
- PRODUCT_DOCUMENTATION_COHESION.md — product-facing documentation cohesion guide for reader routes, public copy, surface responsibilities, and cross-link expectations.
- QUICK_START_CHEATSHEET.md — short user-facing install, run, update, reset/uninstall, and clean full user-local uninstall command sheet.

- ../STATUS.md — root status shortcut with completion estimates and next step.
- ../SECURITY.md — vulnerability reporting, safe testing rules, and security non-claims.
- HIGH_ASSURANCE_SECURITY_BASELINE.md — high-assurance security baseline for current NSA, CISA, FBI, and NIST source-tracked allocations.
- MEMORY_SAFETY_ROADMAP.md — memory-safety roadmap for current C/C++ component mitigation, memory-safe-language preference, and promotion blockers.
- SUPPLY_CHAIN_SECURITY_BASELINE.md — supply-chain security baseline for CI, dependency, SBOM, KEV/NVD, package, installer, release, and update-lane blockers.
- ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md — zero-trust runtime authority baseline for per-request authorization, identity/resource visibility, policy decisions, denial reasons, audit records, and no implicit runtime trust.
- CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md — cyber incident reporting and response baseline for CISA/FBI/IC3 path awareness, ransomware response planning, evidence preservation, communications routing, and no service claims.
- VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md — vulnerability management release gate baseline for CISA KEV review, NVD/CVE review, disclosure handling, release blocking, and no product-security claims.
- CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE.md — cryptographic assurance and key-management baseline for FIPS/CMVP claim gates, algorithm inventory, key lifecycle, randomness, post-quantum planning, and no production crypto claims.
- IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE.md — identity, credential, and access management baseline for privileged access, phishing-resistant MFA planning, account lifecycle, service identity, identity logging, and no hosted access claims.
- SECURITY_LOGGING_MONITORING_BASELINE.md — security logging, monitoring, and detection baseline for event-source inventory, audit events, redaction, retention, triage, incident handoff, and no monitoring-service claims.
- BACKUP_RECOVERY_RESILIENCE_BASELINE.md — backup, recovery, and cyber resilience baseline for backup scope, restore testing, recovery prioritization, rollback planning, RTO/RPO, and no recovery-service claims.
- SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE.md — secure configuration and change management baseline for configuration inventory, secure baselines, checklist evidence, approved changes, rollback planning, drift detection, and no hardening claims.
- DEFENSIVE_THREAT_MODEL_CONTRACT.md — defensive threat model contract for protected assets, trust boundaries, abuse cases, evidence expectations, and non-claims.
- DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md — documentation-and-guard implementation plan for defensive threat model validation.
- DEFENSIVE_THREAT_MODEL_VALIDATION.md — defensive threat model validation ledger.
- DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md — defensive threat model validation refinement and next-gap triage.
- strategy/README.md — strategy index and dated strategy-record rules.
- strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md — active national-security open-system strategy record.
- strategy/2026-05-19-1845-cdt-strategy-estimate-review.md — latest strategy estimate review.
- status/README.md — status index and update rules.

- status/CURRENT_STATUS.md — current progress, completion estimates, and next priorities.
- status/NADIA_OFFLINE_AI_STAGE_0_STATUS.md — status record for Nadia offline AI Stage-0 foundation, Panel installability, Console interoperability, and awareness principles.
- status/NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1_STATUS.md — status record for Nadia Stage-1 local context-pack generation.
- status/NADIA_RUNTIME_PROFILE_STAGE_2_STATUS.md — status record for Nadia Stage-2 runtime-profile metadata before inference.
- status/NADIA_DEVELOPER_WORKBENCH_STAGE_3_STATUS.md — status record for Nadia Stage-3 prompt-plan generation before prompt evaluation.
- status/NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4_STATUS.md — status record for Nadia Stage-4 systems-engineering mode validation before prompt evaluation.
- status/NADIA_PRODUCTIVITY_LOOP_STAGE_5_STATUS.md — status record for Nadia Stage-5 productivity ledger before training or tool authority.
- status/NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6_STATUS.md — status record for Nadia Stage-6 protective-safety boundary before prompt evaluation, model runtime, or tool authority.
- status/NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7_STATUS.md — status record for Nadia Stage-7 guarded tool-authority preflight before any tool execution.
- status/NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8_STATUS.md — status record for Nadia Stage-8 prompt-evaluation contract before prompt materialization, prompt evaluation, inference, or tool execution.
- status/NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9_STATUS.md — status record for Nadia Stage-9 local model-registry contract before model selection, model installation, prompt evaluation, inference, or tool execution.
- status/NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10_STATUS.md — status record for Nadia Stage-10 inference-readiness contract before runtime invocation, model loading, prompt evaluation, inference, or tool execution.
- status/NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11_STATUS.md — status record for Nadia Stage-11 runtime-invocation contract before runtime process spawning, model session creation, model loading, token generation, inference, or tool execution.
- status/NADIA_MODEL_LOAD_CONTRACT_STAGE_12_STATUS.md — status record for Nadia Stage-12 model-load contract before model file opening, weight mapping, weight loading, token generation, inference, or tool execution.
- status/NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13_STATUS.md — status record for Nadia Stage-13 prompt-receipt contract before prompt source opening, prompt text receipt, prompt materialization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14_STATUS.md — status record for Nadia Stage-14 prompt-materialization contract before prompt buffer allocation, prompt text materialization, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15_STATUS.md — status record for Nadia Stage-15 awareness-dialogue contract for Nadia Initiative Q&A scope before dialogue generation, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16_STATUS.md — status record for Nadia Stage-16 prompt-evaluation handoff contract before prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- status/NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17_STATUS.md — status record for Nadia Stage-17 tokenization-boundary contract before tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18_STATUS.md — status record for Nadia Stage-18 tokenizer-specification contract before tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19_STATUS.md — status record for Nadia Stage-19 tokenizer-manifest contract before tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20_STATUS.md — status record for Nadia Stage-20 tokenizer-artifact-inventory contract before tokenizer artifact path resolution, artifact scanning, artifact hashing, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21_STATUS.md — status record for Nadia Stage-21 tokenizer-artifact-measurement contract before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest recording, artifact size recording, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22_STATUS.md — status record for Nadia Stage-22 tokenizer-artifact-verification contract before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest comparison, artifact size comparison, artifact verification, artifact binding, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23_STATUS.md — status record for Nadia Stage-23 tokenizer-artifact-binding contract before tokenizer artifact opening, artifact reading, artifact hashing, artifact verification, artifact binding, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24_STATUS.md — status record for Nadia Stage-24 tokenizer-runtime-attachment contract before tokenizer artifact opening, artifact reading, artifact hashing, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25_STATUS.md — status record for Nadia Stage-25 prompt-tokenization contract before prompt text reading, prompt token creation, prompt token sequence recording, tokenizer runtime attachment, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26_STATUS.md — status record for Nadia Stage-26 prompt-token-sequence contract before prompt token ID recording, token order recording, token offset recording, context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- status/NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27_STATUS.md — status record for Nadia Stage-27 context-window assembly contract before context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28_STATUS.md — status record for Nadia Stage-28 prompt-evaluation-input contract before prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29_STATUS.md — status record for Nadia Stage-29 prompt-evaluation runtime handoff contract before runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30_STATUS.md — status record for Nadia Stage-30 prompt-evaluation invocation contract before invocation requests, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31_STATUS.md — status record for Nadia Stage-31 prompt-evaluation result contract before result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32_STATUS.md — status record for Nadia Stage-32 prompt-evaluation result review contract before result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33_STATUS.md — status record for Nadia Stage-33 prompt-evaluation result disposition contract before disposition records, release records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34_STATUS.md — status record for Nadia Stage-34 prompt-evaluation result release contract before release records, release decisions, release publication, release receipts, disposition records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35_STATUS.md — status record for Nadia Stage-35 prompt-evaluation result release receipt contract before receipt records, receipt signing, receipt publication, release records, release decisions, release publication, disposition records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36_STATUS.md — status record for Nadia Stage-36 prompt-evaluation result release receipt review contract before review records, review decisions, review findings, receipt records, receipt signing, receipt publication, release records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- status/ANNOUNCEMENTS.md — public announcement log and milestone notes.
- status/AUTHORITY_STATUS_ANNOUNCEMENT_REVIEW.md — no-new-announcement authority status review.

- status/CPP_AUTHORITY_EXPANSION_CONTRACT_REVIEW.md — no-expansion-contract C++ authority review.
- status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md — status record for the constrained C++ authority implementation review.
- status/LANGUAGE_REPRESENTATION_REVIEW.md — language representation review before no-effect Nucleus refinement.
- status/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT_STATUS.md — status record for the L-UI rendering detailed report refinement.
- status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_REVIEW.md — no-new-announcement review after the Nucleus report-only and project-notes alignment slices.
- status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_README_ALIGNMENT.md — README/project-notes alignment after the Nucleus report-only announcement review.
- status/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT_STATUS.md — status record for the Nucleus task no-effect report alignment refinement.
- status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md — status record for the Nucleus task report-only execution refinement.
- status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_README_STATUS_ALIGNMENT.md — README/status alignment for the Nucleus task report-only execution refinement.
- status/PROJECT_NOTES_NUCLEUS_ANNOUNCEMENT_README_ALIGNMENT_STATUS_CHECK.md — status/index check for the project-notes Nucleus announcement README alignment.
- status/PROJECT_NOTES_NUCLEUS_REPORT_ONLY_ALIGNMENT_STATUS_INDEX_CHECK.md — status/index check for the project-notes Nucleus report-only alignment.
- status/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_STATUS.md — status record for the Lat pipeline diagnostic integration slice.
- status/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_AUDIT_STATUS.md — status record for the Lat pipeline diagnostic main test audit slice.
- status/RBDM_REPORT_INTEGRATION_STATUS.md — status record for the runtime-boundary domain matrix report integration slice.
- status/KERNEL_LIFECYCLE_EVIDENCE_STATUS.md — current kernel lifecycle evidence status for no-effect kernel table, process/syscall table, scheduler tick, run queue, context-switch, time-accounting, preemption, lifecycle report, subsystem summary, and rollback-plan guardrails.
- status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md — Current estimate table source alignment for the live public estimate table and its README/status mirrors.
- status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md — Current estimate mathematical rebase for the live public estimate table.
- status/COMPLETION_ESTIMATE_REVIEW_README_STATUS_ALIGNMENT.md — README/status alignment for the completion-estimate hold review after runtime-boundary abuse-case fixtures.
- status/COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md — completion-estimate hold review after runtime-boundary abuse-case fixtures.
- status/COMPLETION_PERCENTAGE_REVIEW.md — completion-percentage planning review.
- status/PUBLIC_ENTRY_POINT_CONSISTENCY_SCAN.md — public entry-point consistency scan.
- status/PROJECT_NOTES_FOLLOWUP_STATUS_INDEX_CHECK.md — project-notes follow-up status/index check.

- status/STATUS_ANNOUNCEMENT_REVIEW.md — status announcement review.
- status/STATUS_ANNOUNCEMENT_CONSISTENCY_REVIEW.md — status/announcement consistency review.
- project_notes/README.md — short project notes index.
- project_notes/CURRENT_DIRECTION.md — current direction, target users, and technical lane.
- project_notes/UPCOMING_WORK.md — recommended next slices and near-term queue.

Initial implementation and design documents

- STATE_LATTICE.md — first C state lattice fixture, invariant tests, report surface, and no-effect boundary.
- TRI_PLANE_TRANSITION.md — pure preview transition model, denial reasons, and no-effect invariants.
- NUCLEUS_PREVIEW.md — Nucleus request/effect classification, report surface, and no-execution invariants.
- NUCLEUS_TASK_EXECUTION_CONTRACT.md — future task execution boundary, prerequisites, authority checks, effect gates, task records, reports, and non-claims before execution code.
- NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md — exact future Nucleus task API, structs, enums, reports, buffers, tests, and non-claims before task execution code.
- NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md — first no-effect task classification, denied-by-default policy, authority prerequisites, deterministic reports, and invariants.
- NUCLEUS_TASK_REPORT_REFINEMENT.md — Nucleus task report refinement metadata, tests, guard, and workflow.
- NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT.md — explicit no-effect report alignment labels for Nucleus task records and reports.
- NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md — explicit report-only execution, effect, and runtime status labels for Nucleus task records and reports.
- RUNTIME_BOUNDARY_CONTRACT.md — first runtime boundary, disabled-by-default posture, denied effects, prerequisites, report surface, and non-claims before runtime code.
- RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md — exact future runtime API, structs, enums, reports, buffers, tests, task usage, authority usage, and non-claims before runtime boundary code.
- RUNTIME_BOUNDARY_IMPLEMENTATION.md — first runtime boundary public API, source surface, smoke invariants, dedicated runner, and dedicated workflow record.
- RUNTIME_BOUNDARY_REFINEMENT_PLAN.md — runtime-boundary refinement plan.
- RUNTIME_BOUNDARY_REFINEMENT_IMPLEMENTATION.md — runtime-boundary Lat pipeline first-clause, first-declaration, module/count, stage-summary, line-comment, Lat-specific LIR module-summary, Lat-specific LIR source-span, Lat-specific LIR node-kind, Lat-specific LIR first-node, Lat-specific LIR first-node span, Lat-specific LIR first transition-node, Lat-specific LIR first transition-node span, Lat-specific LIR first-edge, Lat-specific LIR first-edge span, Lat-specific LIR first transition-source edge, Lat-specific LIR first transition-source edge endpoint, Lat-specific LIR first transition-source edge span, Lat-specific LIR no-effect, and Lat-specific LIR edge-kind evidence refinement implementation.

- RUNTIME_BOUNDARY_REPORT_REFINEMENT.md — runtime-boundary report refinement implementation.
- RUNTIME_BOUNDARY_POLICY_MATRIX_REFINEMENT.md — runtime-boundary policy matrix refinement implementation.
- RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md — runtime-boundary policy expansion after defensive threat-model validation.
- RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md — deterministic runtime-boundary abuse-case fixtures after policy expansion.
- RUNTIME_BOUNDARY_DOMAIN_MATRIX_REFINEMENT.md — runtime-boundary domain matrix refinement record.
- RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md — runtime-boundary domain matrix report integration record.
- KERNEL_LIFECYCLE_SEED.md — no-effect kernel lifecycle seed.
- KERNEL_SCHEDULER_TICK_SEED.md — no-effect scheduler tick metadata seed.
- KERNEL_RUN_QUEUE_SEED.md — no-effect run queue metadata seed.
- KERNEL_CONTEXT_SWITCH_SEED.md — no-effect context switch metadata seed.
- KERNEL_TIME_ACCOUNTING_SEED.md — no-effect time-accounting metadata seed.
- KERNEL_PREEMPTION_SEED.md — no-effect preemption metadata seed.
- KERNEL_SCHEDULER_CREDIT_SEED.md — no-effect scheduler credit metadata seed.
- KERNEL_SCHEDULER_SELECTION_SEED.md — no-effect scheduler selection metadata seed.
- KERNEL_SCHEDULER_DISPATCH_SEED.md — no-effect scheduler dispatch metadata seed.
- KERNEL_SCHEDULER_HANDOFF_SEED.md — no-effect scheduler handoff metadata seed.
- KERNEL_SCHEDULER_ACTIVATION_SEED.md — no-effect scheduler activation metadata seed.
- KERNEL_SCHEDULER_RUN_ENTRY_SEED.md — no-effect scheduler run-entry metadata seed.
- KERNEL_LIFECYCLE_SUBSYSTEM_SUMMARY.md — kernel lifecycle subsystem summary.
- KERNEL_LIFECYCLE_ROLLBACK_PLAN.md — no-effect kernel lifecycle rollback plan.
- LATTICRA_SEAL_CONTRACT.md — Latticra Seal evidence-boundary, capability-boundary, promotion-gate, and non-claim contract.
- LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md — report envelope boundary after ready runtime handoff report metadata.
- LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md — first no-effect sealed report-envelope metadata implementation.
- status/SEAL_REPORT_ENVELOPE_STATUS.md — status/public-entry checkpoint for sealed report-envelope metadata.
- LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md — metadata-only signature request boundary after sealed report-envelope metadata.
- LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md — first no-effect signature request metadata implementation.
- LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md — metadata-only signing authorization boundary after signature request readiness.
- LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md — first no-effect signing authorization metadata implementation.

- LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md — metadata-only signer handoff boundary after signing authorization readiness.
- LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md — first no-effect signer handoff metadata implementation.
- LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md — metadata-only signer invocation boundary after signer handoff readiness.
- LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md — first no-effect signer invocation metadata implementation.
- LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md — metadata-only signing operation boundary after signer invocation readiness.
- LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md — first no-effect signing operation metadata implementation.
- status/SEAL_SIGNING_OPERATION_STATUS.md — status/public-entry checkpoint for metadata-only Seal signing operation implementation.
- LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md — metadata-only key-handling boundary after signing operation readiness.
- LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md — first no-effect key-handling metadata implementation.
- status/SEAL_KEY_HANDLING_STATUS.md — status/public-entry checkpoint for metadata-only Seal key-handling implementation.
- LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md — metadata-only key-material boundary after key-handling readiness.
- LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md — first no-effect key-material metadata implementation.
- status/SEAL_KEY_MATERIAL_STATUS.md — status/public-entry checkpoint for metadata-only Seal key-material implementation.
- LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md — metadata-only public-key parsing boundary after key-material status readiness.
- LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md — first no-effect public-key parsing metadata implementation.
- status/SEAL_PUBLIC_KEY_PARSING_STATUS.md — status/public-entry checkpoint for metadata-only Seal public-key parsing implementation.
- LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md — future key parsing implementation contract after public-key parsing status readiness.
- LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md — exact future bounded no-effect key parsing implementation plan.
- LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md — first bounded key parsing metadata implementation for caller-provided public-key bytes.
- status/SEAL_KEY_PARSING_STATUS.md — status/public-entry checkpoint for bounded Seal key parsing metadata.
- LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md — verification policy boundary after signature metadata.
- LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md — first no-effect verification policy metadata implementation.

- status/SEAL_VERIFICATION_POLICY_STATUS.md — status/public-entry checkpoint for metadata-only Seal verification policy implementation.
- LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md — crypto verify backend boundary after verification policy metadata.
- LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_IMPLEMENTATION.md — first metadata-only crypto verify backend implementation with unsupported verification state.
- status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md — status/public-entry checkpoint for metadata-only Seal crypto verify backend implementation.
- LATTICRA_SEAL_ED25519_VERIFY_ONLY_CONTRACT.md — Ed25519 verify-only implementation contract after crypto verify backend metadata.
- LATTICRA_SEAL_ED25519_VERIFY_IMPLEMENTATION.md — local provider-backed Ed25519 verify-only result implementation.
- status/SEAL_ED25519_VERIFY_STATUS.md — status/public-entry checkpoint for authority-neutral Seal Ed25519 verify-only results.
- LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md — authority-neutral crypto graduation gate over local Ed25519 verification, verified receipt promotion, and cryptographic assurance baseline expectations.
- status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md — status checkpoint for the Seal crypto graduation gate.
- LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_CONTRACT.md — verified receipt promotion boundary after successful Ed25519 verify-only results.
- LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_IMPLEMENTATION.md — first authority-neutral verified receipt promotion metadata implementation.
- status/SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md — status/public-entry checkpoint for verified receipt promotion metadata.
- LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_CONTRACT.md — verified capability gate boundary after verified receipt promotion metadata.
- LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_IMPLEMENTATION.md — metadata-only verified capability gate implementation with a stricter crypto-graduation-gated entry point.
- status/SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md — status/public-entry checkpoint for verified capability gate metadata.
- LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_CONTRACT.md — verified effect decision boundary after verified capability gate metadata.
- LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_IMPLEMENTATION.md — metadata-only verified effect decision implementation that carries crypto graduation evidence forward when present.
- status/SEAL_VERIFIED_EFFECT_DECISION_STATUS.md — status/public-entry checkpoint for crypto-bound verified effect decision metadata.
- LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md — runtime handoff evaluation boundary after verified effect decision metadata.
- LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md — metadata-only runtime handoff evaluation implementation that carries crypto graduation evidence forward when present.
- status/SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md — status/public-entry checkpoint for crypto-bound runtime handoff evaluation metadata.

- LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md — runtime handoff report boundary after runtime handoff evaluation metadata.
- LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md — first metadata-only runtime handoff report implementation.
- status/SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md — status/public-entry checkpoint for runtime handoff report metadata.
- LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md — verification receipt boundary after verification policy metadata.
- LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md — first no-effect verification receipt metadata implementation.
- status/SEAL_VERIFICATION_RECEIPT_STATUS.md — status/public-entry checkpoint for metadata-only Seal verification receipt implementation.
- LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md — capability gate boundary after verification receipt metadata.
- LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md — first no-effect denied capability gate metadata implementation.
- status/SEAL_CAPABILITY_GATE_STATUS.md — status/public-entry checkpoint for metadata-only Seal capability gate implementation.
- LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md — effect decision boundary after capability gate metadata.
- LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md — first no-effect denied effect decision metadata implementation.
- status/SEAL_EFFECT_DECISION_STATUS.md — status/public-entry checkpoint for metadata-only Seal effect decision implementation.
- LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md — runtime handoff boundary after effect decision metadata.
- LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md — first no-effect inactive runtime handoff metadata implementation.
- status/SEAL_RUNTIME_HANDOFF_STATUS.md — status/public-entry checkpoint for metadata-only Seal runtime handoff implementation.
- LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md — runtime handoff evaluation boundary after verified effect decision metadata.
- LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md — metadata-only runtime handoff evaluation implementation that carries crypto graduation evidence forward when present.
- status/SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md — status checkpoint for crypto-bound Seal runtime handoff evaluation.
- LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md — runtime handoff report boundary after runtime handoff evaluation metadata.
- LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md — metadata-only runtime handoff report implementation.
- status/SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md — Latticra Seal runtime handoff report status/public-entry alignment.
- LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md — status rollup boundary after runtime handoff metadata.

- LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md — first metadata-only status rollup implementation.
- status/SEAL_STATUS_ROLLUP_STATUS.md — status/public-entry checkpoint for metadata-only Seal status rollup implementation.
- LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md — report-only agentic automation security boundary after status rollup metadata.
- LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md — MCP-style tool invocation alignment plan without MCP implementation.
- LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md — first report-only agentic automation security metadata implementation.
- status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md — status checkpoint for report-only Seal agentic automation security metadata.
- status/SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md — index alignment for report-only Seal agentic automation security metadata.
- LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md — deterministic local report surface for Seal agentic automation security metadata.
- status/SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md — status checkpoint for the Seal agentic automation security report surface.
- status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md — public-entrypoint alignment for report-only Seal agentic automation security.
- LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md — report-only parameter schema boundary after agentic automation security.
- LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md — first report-only parameter schema metadata implementation.
- LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md — deterministic local report surface for Seal parameter schema metadata.
- status/SEAL_PARAMETER_SCHEMA_STATUS.md — status/public-entry checkpoint for report-only Seal parameter schema metadata.
- LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md — report-only request freshness/replay boundary after parameter schema metadata.
- LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md — first report-only request freshness metadata implementation.
- LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md — deterministic local report surface for Seal request freshness metadata.
- status/SEAL_REQUEST_FRESHNESS_STATUS.md — status/public-entry checkpoint for report-only Seal request freshness metadata.
- LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md — report-only signed request boundary after request freshness metadata.
- LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md — first report-only signed request metadata implementation.
- status/SEAL_SIGNED_REQUEST_STATUS.md — status/public-entry checkpoint for report-only Seal signed request metadata.
- LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md — report-only policy decision boundary after signed request metadata.

- LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md — first report-only default-deny policy decision metadata implementation.
- LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md — deterministic local report surface for Seal policy decision metadata.
- status/SEAL_POLICY_DECISION_STATUS.md — status/public-entry checkpoint for report-only Seal policy decision metadata.
- status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md — status checkpoint for the Seal policy decision report surface.
- status/SEAL_POLICY_DECISION_PUBLIC_ENTRYPOINT_ALIGNMENT.md — public-entrypoint alignment for report-only Seal policy decision metadata.
- LATTICRA_SEAL_CAPABILITY_METADATA_CONTRACT.md — planning contract for the no-effect Seal capability metadata surface.
- LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION_PLAN.md — exact no-effect capability metadata implementation plan.
- LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION.md — deterministic local capability metadata implementation.
- LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_SURFACE.md — deterministic local report surface for Seal capability metadata.
- status/SEAL_CAPABILITY_METADATA_REPORT_SURFACE_STATUS.md — status checkpoint for the Seal capability metadata report surface.
- latticra-seal/PRODUCT.md — product spine for earned Latticra Seal security-product capability.
- status/SEAL_PRODUCT_SPINE_STATUS.md — status checkpoint for the Seal product spine.
- LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_CONTRACT.md — contract for a future bundled local Seal operator receipt report.
- LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION_PLAN.md — exact no-effect Seal operator receipt report implementation plan.
- LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md — no-effect Seal operator receipt report implementation.
- LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md — deterministic local report surface for the Seal operator receipt report.
- status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md — status checkpoint for the Seal operator receipt report surface.
- LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md — contract for a future no-effect local Seal capability registry schema.
- LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md — exact no-effect Seal local capability registry schema implementation plan.
- LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md — no-effect Seal local capability registry schema implementation.
- status/SEAL_README_STATUS_ROW_ALIGNMENT.md — README status row alignment with the current Latticra Seal public status checkpoint.
- status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md — status checkpoint for the Seal policy decision report surface.

- status/SEAL_SIGNER_INVOCATION_STATUS.md — status/public-entry checkpoint for metadata-only Seal signer invocation implementation.
- status/SEAL_SIGNER_HANDOFF_STATUS.md — status/public-entry checkpoint for metadata-only Seal signer handoff implementation.
- status/SEAL_SIGNING_AUTHORIZATION_STATUS.md — status/public-entry checkpoint for metadata-only Seal signing authorization implementation.
- status/SEAL_SIGNATURE_REQUEST_STATUS.md — status/public-entry checkpoint for metadata-only Seal signature request implementation.
- L_UI_STATIC_REPORT.md — terminal-facing L-UI static report fixture for Nucleus/state rails.
- L_UI_SOURCE_GRAMMAR.md — L-UI source grammar draft and .lui fixture guardrails.
- L_UI_PARSER_DESIGN.md — parser design contract, error categories, no-effect constraints, and implementation gate.
- L_UI_PARSER_IMPLEMENTATION_PLAN.md — parser implementation language, module shape, API, source-size limit, and exact test list before parser code.
- L_UI_PARSER.md — initial no-effect C parser, summary result, error labels, and parser invariants.
- L_UI_SEMANTIC_VALIDATION_CONTRACT.md — semantic validation layer after structural parsing, AST construction, source policies, and string handling.
- L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md — public API, semantic result, diagnostic mapping, reports, semantic checks, and exact tests before semantic validation code.
- L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md — no-effect semantic validation API, reports, checks, and invariants for L-UI ASTs.
- LIR_SHAPE_CONTRACT.md — first Latticra Intermediate Representation shape before semantic lowering, Lat integration, rendering, or execution.
- LIR_SHAPE_IMPLEMENTATION_PLAN.md — exact public API, structs, capacities, node kinds, edge kinds, errors, report format, source-span mapping, and tests before LIR code.
- LIR_SHAPE_IMPLEMENTATION.md — bounded no-effect LIR shape API, lowering, reports, and invariants for semantically valid L-UI ASTs.
- LIR_REPORT_REFINEMENT.md — LIR report classification, graph shape, edge summary, no-effect-chain, and evidence fields.
- LAT_LANGUAGE_GRAMMAR_CONTRACT.md — first Lat / Latticra Language grammar contract for Lat-Core before parser implementation.
- LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md — exact public API, parser result structs, AST structs, capacities, error labels, reports, fixture paths, string handling, source-span mapping, and tests before Lat parser code.
- LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md — bounded no-effect Lat / Latticra Language grammar parser, AST metadata, line-comment metadata, unsupported block-comment rejection, first declaration/clause report metadata, fixture, and invariants.
- LAT_SEMANTIC_VALIDATION_CONTRACT.md — Lat semantic validation contract.
- LAT_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md — Lat semantic validation implementation plan.
- LAT_SEMANTIC_DIAGNOSTICS_REFINEMENT.md — Lat semantic diagnostic classes, counters, first-diagnostic indices, and report fields.

- LAT_MODEL_NORMALIZATION_IMPLEMENTATION.md — Lat model normalization tables and first-declaration/first-clause report metadata.
- LAT_TO_LIR_LOWERING_CONTRACT.md — Lat-to-LIR lowering contract.
- LAT_TO_LIR_LOWERING_IMPLEMENTATION_PLAN.md — Lat-to-LIR lowering implementation plan.
- LAT_TO_LIR_LOWERING_IMPLEMENTATION.md — Lat-to-LIR lowering implementation with first-declaration report metadata.
- LAT_TO_LIR_CLAUSE_METADATA_REFINEMENT.md — Lat-to-LIR clause operator/value metadata refinement.
- LAT_TO_LIR_DIAGNOSTIC_REFINEMENT.md — Lat-to-LIR diagnostic classification and first-declaration/first-clause metadata report surface.
- LAT_PIPELINE_CONTRACT.md — bounded no-effect Lat pipeline contract after parser, semantic validation, and Lat-to-LIR lowering.
- LAT_PIPELINE_IMPLEMENTATION_PLAN.md — exact Lat pipeline API, result struct, report, tests, workflow, compatibility expectations, and non-claims.
- LAT_PIPELINE_IMPLEMENTATION.md — first bounded no-effect Lat pipeline implementation.
- LAT_PIPELINE_REPORT_REFINEMENT.md — Lat pipeline stage-summary, parser line-comment metadata, first-declaration/first-clause metadata, and report refinement.
- LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_REFINEMENT.md — companion Lat pipeline diagnostic integration API, parser line-comment metadata, first-declaration/first-clause metadata, and report.
- LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_INTEGRATION_AUDIT.md — main Lat pipeline runner audit for diagnostic integration coverage.
- LAT_SPECIFIC_LIR_REFINEMENT_CONTRACT.md — explicit Lat declaration node and transition-source edge refinement contract.
- LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION_PLAN.md — exact Lat-specific LIR enum, label, lowering, test, workflow, and compatibility plan.
- LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION.md — first no-effect Lat-specific LIR refinement implementation.
- L_UI_RENDERING_CONTRACT.md — first operator-visible L-UI rendering contract before renderer implementation.
- L_UI_RENDERING_IMPLEMENTATION_PLAN.md — exact L-UI renderer API, structures, modes, errors, capacities, output rules, report format, and tests before renderer code.
- L_UI_RENDERING_IMPLEMENTATION.md — first no-effect C L-UI renderer implementation.
- L_UI_RENDERING_DETAILED_REPORT_REFINEMENT.md — explicit detailed-render report metadata refinement for no-effect L-UI reports.

Implementation rule

No implementation code should be added until the relevant contract document exists and names:

1. purpose;
2. evidence level;
3. effect boundary;
4. failure behavior;

5. non-claims;
6. tests or validation path.

Current priority

Implemented/guarded foundation and operations units:

```
project strategy/status framework + guardrails
language naming policy + guardrails
C/C++ foundation direction + guardrails
constrained C++ authority layer contract + guardrails
Constrained C++ authority layer implementation plan + guardrails
constrained C++ authority layer implementation + invariants
constrained C++ authority implementation review + status alignment
state lattice fixture + invariant tests
tri-plane transition model + invariant tests
Nucleus preview request classification + report invariants
Nucleus task execution contract + guardrails
Nucleus task execution implementation plan + guardrails
Nucleus task execution implementation + invariants
Nucleus task report refinement + invariants
Nucleus task no-effect report alignment + invariants
Nucleus task report-only execution refinement + invariants
Nucleus task report-only execution README/status alignment + public entry-point alignment
Project notes Nucleus report-only alignment + project-note alignment
Project notes Nucleus report-only status/index check + public status alignment
Nucleus report-only announcement review + no-new-announcement review
Nucleus report-only announcement README alignment + public entry-point alignment
Project notes Nucleus announcement README status/index check + public status alignment
Runtime boundary contract + guardrails
Runtime boundary implementation plan + guardrails
Runtime boundary implementation + smoke invariants
Runtime boundary Lat pipeline comment evidence integration + invariants
Runtime boundary report refinement + invariants
Runtime boundary policy matrix refinement + invariants
Runtime boundary domain matrix refinement + invariants
Runtime boundary domain matrix report integration + invariants
Latticra Seal report envelope metadata + invariants
Latticra Seal signature request contract + guardrails
Latticra Seal signature request metadata + invariants
Latticra Seal signature request predecessor status alignment + guardrails
Latticra Seal signature request status/public-entry alignment + guardrails
Latticra Seal signing authorization contract + guardrails
Latticra Seal signing authorization metadata + invariants
Latticra Seal signing authorization status/public-entry alignment + guardrails
Latticra Seal signing authorization predecessor status alignment + guardrails
Latticra Seal signer handoff contract + guardrails
Latticra Seal signer handoff metadata + invariants
Latticra Seal signer handoff status/public-entry alignment + guardrails
Latticra Seal signer handoff predecessor status alignment + guardrails
Latticra Seal signer invocation contract + guardrails
Latticra Seal signer invocation metadata + invariants
Latticra Seal signer invocation status/public-entry alignment + guardrails
Latticra Seal signer invocation predecessor status alignment + guardrails
Latticra Seal signing operation contract + guardrails
Latticra Seal signing operation metadata + invariants
Latticra Seal signing operation status/public-entry alignment + guardrails
Latticra Seal signing operation predecessor status alignment + guardrails
Latticra Seal key-handling predecessor status alignment + guardrails
Latticra Seal key-material predecessor status alignment + guardrails
Latticra Seal public-key parsing predecessor status alignment + guardrails
Latticra Seal key parsing predecessor status alignment + guardrails
Latticra Seal verification policy predecessor status alignment + guardrails
Latticra Seal verification receipt predecessor status alignment + guardrails
Latticra Seal capability gate predecessor status alignment + guardrails
Latticra Seal README status row alignment + guardrails
Latticra Seal effect decision status/public-entry alignment + guardrails
Latticra Seal effect decision predecessor status alignment + guardrails
Latticra Seal runtime handoff status/public-entry alignment + guardrails
Latticra Seal runtime handoff predecessor status alignment + guardrails
Latticra Seal status rollup status/public-entry alignment + guardrails
Latticra Seal status rollup predecessor status alignment + guardrails
Latticra Seal agentic automation security public-entrypoint alignment + guardrails
Latticra Seal agentic automation security predecessor status alignment + guardrails
Latticra Seal parameter schema status/public-entry alignment + guardrails
Latticra Seal parameter schema predecessor status alignment + guardrails
Latticra Seal request freshness status/public-entry alignment + guardrails
Latticra Seal request freshness predecessor status alignment + guardrails
```

Latticra Seal signed request status/public-entry alignment + guardrails
Latticra Seal signed request predecessor status alignment + guardrails
Latticra Seal policy decision status/public-entry alignment + guardrails
Latticra Seal policy decision predecessor status alignment + guardrails
Latticra Seal operator receipt report predecessor status alignment + guardrails
Defensive threat model validation refinement + guardrails
Runtime boundary policy expansion after threat-model validation + guardrails
Runtime boundary abuse-case fixture expansion after policy expansion + guardrails
Completion estimate review README/status alignment + guardrails
Completion estimate review after runtime-boundary abuse-case fixtures + guardrails
Latticra Seal capability gate status/public-entry alignment + guardrails
Latticra Seal verification receipt status/public-entry alignment + guardrails
Nadia offline AI Stage-0 foundation + guardrails
Nadia local context engine Stage-1 + guardrails
Nadia runtime profile Stage-2 + guardrails
Nadia developer workbench Stage-3 + guardrails
Nadia systems engineering mode Stage-4 + guardrails
Nadia productivity loop Stage-5 + guardrails
Nadia protective safety boundary Stage-6 + guardrails
Nadia guarded tool authority Stage-7 + guardrails
Nadia prompt evaluation contract Stage-8 + guardrails
Nadia local model registry contract Stage-9 + guardrails
Nadia inference readiness contract Stage-10 + guardrails
Nadia runtime invocation contract Stage-11 + guardrails
Nadia model load contract Stage-12 + guardrails
Nadia prompt receipt contract Stage-13 + guardrails
Nadia prompt materialization contract Stage-14 + guardrails
Nadia awareness dialogue contract Stage-15 + guardrails
Nadia prompt evaluation handoff contract Stage-16 + guardrails
Nadia tokenization boundary contract Stage-17 + guardrails
Nadia tokenizer specification contract Stage-18 + guardrails
Nadia tokenizer manifest contract Stage-19 + guardrails
Nadia tokenizer artifact inventory contract Stage-20 + guardrails
Nadia tokenizer artifact measurement contract Stage-21 + guardrails
Nadia tokenizer artifact verification contract Stage-22 + guardrails
Nadia tokenizer artifact binding contract Stage-23 + guardrails
Nadia tokenizer runtime attachment contract Stage-24 + guardrails
Nadia prompt tokenization contract Stage-25 + guardrails
Nadia prompt token sequence contract Stage-26 + guardrails
Nadia context window assembly contract Stage-27 + guardrails

Nadia prompt evaluation input contract Stage-28 + guardrails

Nadia prompt evaluation runtime handoff contract Stage-29 + guardrails
Nadia prompt evaluation invocation contract Stage-30 + guardrails
Nadia prompt evaluation result contract Stage-31 + guardrails
Nadia prompt evaluation result review contract Stage-32 + guardrails
Nadia prompt evaluation result disposition contract Stage-33 + guardrails
Nadia prompt evaluation result release contract Stage-34 + guardrails
Nadia prompt evaluation result release receipt contract Stage-35 + guardrails
Nadia prompt evaluation result release receipt review contract Stage-36 + guardrails
Nadia prompt evaluation result release receipt review disposition contract Stage-37 + guardrails
Nadia prompt evaluation result release receipt review disposition release contract Stage-38 + guardrails
Nadia prompt evaluation result release receipt review disposition release receipt contract Stage-39 + guardrails
Nadia prompt evaluation result release receipt review disposition release receipt review contract Stage-40 + guardrails
Nadia prompt evaluation result release receipt review disposition release receipt review disposition contract Stage-41 + guardrails
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release contract Stage-42 + guardrails
Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt contract Stage-43 + guardrails
Latticra Seal verification policy status/public-entry alignment + guardrails
Latticra Seal crypto verify backend status/public-entry alignment + guardrails
Latticra Seal Ed25519 verify-only status/public-entry alignment + guardrails
Latticra Seal verified receipt promotion status/public-entry alignment + guardrails
Latticra Seal verified capability gate status/public-entry alignment + guardrails
Latticra Seal verified effect decision status/public-entry alignment + guardrails
Latticra Seal runtime handoff evaluation status/public-entry alignment + guardrails
Latticra Seal runtime handoff report status/public-entry alignment + guardrails
Latticra Seal report envelope status/public-entry alignment + guardrails
Latticra Seal key parsing status/public-entry alignment + guardrails
Latticra Seal bounded no-effect key parsing metadata + invariants
Latticra Seal future key parsing implementation plan + guardrails
Latticra Seal future key parsing implementation contract + guardrails
Latticra Seal public-key parsing status/public-entry alignment + guardrails
L-UI static report fixture + rail invariants
L-UI source grammar draft + fixture guardrails

```
L-UI parser design contract + guardrails
L-UI parser implementation plan + guardrails
L-UI parser implementation + invariants
L-UI semantic validation implementation + invariants
LIR shape implementation + invariants
LIR report refinement + invariants
Lat language grammar implementation + invariants
Lat grammar report metadata integration + invariants
Lat grammar line-comment metadata refinement + invariants
Lat grammar unsupported block-comment rejection refinement + invariants
Lat semantic diagnostics refinement + invariants
Lat model normalization implementation + invariants
Lat model report declaration metadata integration + invariants
Lat model report clause metadata integration + invariants
Lat-to-LIR lowering implementation + invariants
Lat-to-LIR declaration metadata refinement + invariants
Lat-to-LIR clause metadata refinement + invariants
Lat-to-LIR diagnostic declaration metadata integration + invariants
Lat-to-LIR diagnostic clause metadata integration + invariants
Lat pipeline diagnostic declaration metadata integration + invariants
Lat pipeline diagnostic clause metadata integration + invariants
Lat pipeline report declaration metadata integration + invariants
Lat pipeline report clause metadata integration + invariants
Lat pipeline comment metadata integration + invariants
Lat pipeline diagnostic comment metadata integration + invariants
Lat parse-failure comment evidence propagation + invariants
Lat pipeline failure span evidence propagation + invariants
Lat pipeline parse-error evidence propagation + invariants
Lat pipeline semantic-error evidence propagation + invariants
Lat pipeline downstream stage-error evidence propagation + invariants
Lat pipeline stage-summary evidence propagation + invariants
Lat pipeline module/count evidence propagation + invariants
Lat pipeline first-declaration evidence propagation + invariants
Lat pipeline first-clause evidence propagation + invariants
Lat LIR edge-kind evidence propagation + invariants
Lat LIR no-effect evidence propagation + invariants
Lat LIR module-summary evidence propagation + invariants
Lat LIR source-span evidence propagation + invariants
Lat LIR node-kind evidence propagation + invariants
Lat LIR first-node evidence propagation + invariants
Lat LIR first-node span evidence propagation + invariants
Lat LIR first transition-node evidence propagation + invariants
Lat LIR first transition-node span evidence propagation + invariants
Lat LIR first-edge evidence propagation + invariants
Lat LIR first-edge span evidence propagation + invariants
Lat LIR first transition-source edge evidence propagation + invariants
Lat LIR first transition-source edge endpoint evidence propagation + invariants
Lat LIR first transition-source edge span evidence propagation + invariants
Lat pipeline implementation + invariants
Lat pipeline report refinement + invariants
Lat pipeline diagnostic integration refinement + invariants
Lat pipeline diagnostic integration main test audit + invariants
Lat-specific LIR refinement implementation + invariants
L-UI rendering implementation + invariants
L-UI rendering detailed report refinement + invariants
```

The next status review target should be:

```
Continue small guarded report/status alignment only when drift appears
```

Current runtime boundary policy expansion status:

```
runtime_boundary_policy_expansion_after_threat_model_present=1
runtime_boundary_abuse_case_fixture_expansion_present=1
seal_readme_status_row_alignment_present=1
completion_estimate_review_readme_status_alignment_present=1
completion_estimate_after_runtime_boundary_abuse_case_fixtures_present=1
current_estimate_table_source_alignment_present=1
current_estimate_mathematical_rebase_present=1
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
estimate_adjustment_required=0
request_family_policy_map_present=1
effect_policy_map_present=1
authority_prerequisite_map_present=1
future_gate_policy_map_present=1
abuse_case_runtime_policy_map_present=1
evidence_gap_map_present=1
runtime_execution_added=0
runtime_authority_granted=0
```

Completion-estimate review should be skipped unless a future slice changes capability posture, implementation scope, or public readiness rather than only aligning documentation/status surfaces.

Part II - Current Status, Evidence, and Public Boundaries

- docs/latticra-seal/ROADMAP.md - Latticra Seal Roadmap
- docs/LATTICRA_LAT_SEAL_ROADMAP.md - Latticra / Lat / Latticra Seal Roadmap
- docs/ROADMAP.md - Latticra Roadmap
- docs/status/ANNOUNCEMENTS.md - Latticra Announcements
- docs/status/CURRENT_STATUS.md - Latticra Current Status
- docs/status/README.md - Latticra Status Index

docs/latticra-seal/ROADMAP.md

Source title: Latticra Seal Roadmap

Lines: 104 | Words: 307 | SHA-256: 43c0b52c503982387a256e54f530df194d09725943d5706c39b07ff87d108c6e

Latticra Seal Roadmap

Latticra Seal should grow in disciplined layers.

Phase 1: Documentation baseline

Goal: make Seal understandable.

Tasks:

- create documentation handbook
- define status
- define architecture
- define command usage
- define policy behavior
- define report behavior
- define non-claims

Phase 2: CLI clarity

Goal: make commands predictable.

Tasks:

- document every command
- document read/write behavior
- document exit codes
- document report schema
- document policy-denial reasons

Phase 3: Manifest and lock hardening

Goal: make local verification reproducible.

Tasks:

- stabilize manifest schema
- stabilize lock format
- define baseline update rules
- define ignored paths
- define changed-file output
- test malformed manifest behavior

Phase 4: Report schema

Goal: make reports machine-readable and human-readable.

Tasks:

- define required fields

- define optional fields
- define status values
- define warning classes
- define failure classes
- preserve report-only authority limits

Phase 5: Panel integration

Goal: make Seal visible in Latticra Panel.

Tasks:

- expose report-only scan
- expose policy-denial status
- expose report path
- expose manifest/lock status
- avoid runtime/root/network authority

Phase 6: Cryptographic strengthening

Goal: move from hash baseline toward stronger signed evidence.

Tasks:

- define signing model
- define key handling
- define trust root boundaries
- define verification receipt format
- test invalid signature behavior
- test missing key behavior

Phase 7: Runtime-boundary handoff

Goal: prepare metadata for future enforcement without claiming enforcement early.

Tasks:

- define runtime handoff metadata
- define capability gate contract
- define denial behavior
- define no-effect dry-run behavior
- test handoff reports

Phase 8: Fedora/Linux integration research

Goal: explore host integration carefully.

Tasks:

- document user-local install evidence
- document no-root assumptions
- document future RPM boundaries

- document SELinux/systemd non-claims
- document what would be required before host-level trust claims

docs/LATTICRA_LAT_SEAL_ROADMAP.md

Source title: Latticra / Lat / Latticra Seal Roadmap

Lines: 340 | Words: 980 | SHA-256: 479fbab26344bb9b9e669151c05db6844a7b82d07eddca01cedbaa0ecfa1fd08

Latticra / Lat / Latticra Seal Roadmap

Status: active strategy roadmap Scope: coordinated planning for Latticra, Lat, LIR, Latticra Seal, C, constrained C++, Rust, Fedora/Linux integration, and AI-era tool-boundary work. This roadmap does not implement runtime execution, operating-system replacement behavior, cryptographic verification, MCP behavior, production enforcement, or production readiness.

Strategic position

Latticra should advance as an evidence-bound systems substrate and secure computing architecture. Fedora/Linux remains the proving lane. Lat is the native language direction. LIR is the bounded intermediate representation. Latticra Seal is the trust-boundary, request-boundary, capability-boundary, and tool-boundary subsystem.

The roadmap principle is:

```
evidence first
declarations before execution
reports before authority
allowlist candidates before allow grants
verification before enforcement
Fedora/Linux validation before deeper host claims
```

Project pillars

```
Latticra
  Secure systems substrate architecture, runtime-boundary model, Fedora/Linux validation lane,
Nucleus coordination, and evidence posture.

Lat
  Latticra-native declaration language for system intent, capability shape, policy-aware
declarations, and no-effect lowering into LIR.

LIR
  Bounded intermediate representation with graph shape, metadata, and report evidence before
execution.

Latticra Seal
  Trust-boundary and tool-boundary subsystem for signed request planning, freshness metadata,
guarded allowlists, capability planning, runtime gates, and evidence receipts.
```

Language architecture

```
C
  substrate, ABI, fixtures, state records, LIR/Lat lowering foundations, runtime-boundary
records, Seal metadata records, no-effect invariants.

Constrained C++
  architecture and policy modeling, capability graphs, Seal decisions, authority planning,
validation engines, bounded audit reports.

Rust
  safe CLI tools, manifest parsing, evidence report generation, signature and verification
tooling, serialization, untrusted input handling, future network-facing tools.

Lat
  system declaration and Latticra-native intent.

LIR
  bounded evidence-bearing representation.
```

Architecture lane

```
Lat source
  -&gt; parser
  -&gt; semantic validation
  -&gt; diagnostics
  -&gt; LIR metadata lowering
  -&gt; runtime-boundary classification
  -&gt; Seal policy / allowlist / gate metadata
  -&gt; operator-visible evidence report
```

The current valid posture is no-effect and report-oriented. Future execution must be gated by stronger evidence, verification, authority contracts, and explicit promotion rules.

Phase 1: Integration documentation and guardrails

Goal:

```
make the roadmap explicit and testable without changing runtime behavior
```

Deliverables:

```
docs/POLYMORPHIC_LANGUAGE_STRATEGY.md
docs/LATTICRA_LAT_SEAL_ROADMAP.md
docs/LATTICRA_ARCHITECTURE_TARGETS.md
scripts/test-latticra-lat-seal-roadmap.sh
```

Exit criteria:

```
strategy docs exist
language roles are explicit
non-claims remain explicit
roadmap guard passes
README/foundation links are aligned in a future slice
no runtime authority added
no effect behavior added
```

Phase 2: Lat and LIR strengthening

Goal:

```
mature Lat as declaration and no-effect system-intent language
```

Priority work:

```
Lat grammar refinement
Lat semantic diagnostics refinement
Lat-to-LIR lowering refinement
Lat-to-LIR clause metadata refinement
Lat-specific LIR nodes and edges
LIR report clarity
operator-visible Lat pipeline reports
negative fixtures for invalid declarations
```

C should continue to own the bounded parser/lowering/report substrate. C++ may be introduced for semantic architecture experiments only after contracts exist. Rust may be introduced for future tooling such as latfmt or latcheck after CLI contracts exist.

Non-claims:

```
lat_execution=0
lir_execution=0
compiler_product=0
interpreter_product=0
production_language=0
```

Phase 3: Seal guarded allowlist maturation

Goal:

```
move from blocked-request metadata toward guarded allowlist planning while preserving no-effect
behavior
```

Priority work:

```
guarded allowlist status alignment
guarded allowlist public entry-point alignment
capability metadata contract
```

```
manifest metadata contract
evidence receipt contract
negative allowlist cases
operator-visible denial report refinement
```

Valid posture:

```
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
runtime_authority_granted=0
effect_performed=0
```

Phase 4: Rust tooling lane

Goal:

```
introduce Rust only where it improves safety and tooling clarity
```

Candidate tools:

```
tools/seal-report
tools/seal-verify
tools/latcheck
tools/latfmt
tools/latticra-status
```

Valid Rust first slice:

```
parse a deterministic local fixture
produce a report
perform no host mutation beyond temporary test output
perform no network behavior
make no verification claim unless real verification is implemented
```

Phase 5: Constrained C++ policy model lane

Goal:

```
use C++ where policy and graph structure need stronger architecture
```

Candidate components:

```
CapabilityGraph
SealDecision
ToolBoundaryPlanner
PolicyGraph
LatSemanticModel
AuthorityAuditModel
```

C++ must remain constrained:

```
no unrestricted authority
no hidden effects
no exception crossing C boundary
no implicit allocation at ABI boundary
no runtime authority grant
no tool execution
```

Phase 6: Cryptographic verification planning

Goal:

```
separate metadata signature policy from real cryptographic verification
```

Planned distinction:

```
signature_metadata_present
signature_policy_evaluated
signature_verification_implemented
trusted_key_store_present
revocation_policy_present
hardware_identity_present
post_quantum_transition_present
```

Do not promote metadata to verification. Verification requires real primitives, real test vectors, negative tests, key handling rules, and clear non-claims.

Phase 7: Fedora/Linux integration lane

Goal:

```
keep Fedora/Linux as the practical validation lane without claiming OS replacement
```

Priority work:

```
local RPM documentation lane
Fedora disposable VM validation
host-facing validation transcripts
packaging boundary records
Fedora integration docs
no daily-driver claim
no Fedora approval claim
no immutable Fedora claim until evidence exists
```

Phase 8: Limited enforcement research

Goal:

```
consider narrow enforcement only after reports, allowlists, verification, and evidence receipts
mature
```

Earliest possible enforcement candidates:

```
deny unknown tool
deny unsigned request
deny stale request
deny replayed request
deny missing capability
deny manifest mismatch
```

This phase is future research. It is not current production behavior.

Priority queue

Near-term implementation queue:

1. Add roadmap and polymorphic language strategy guardrails.
2. Align README and foundation index entry points.
3. Add Seal guarded allowlist status alignment.
4. Add capability metadata contract.
5. Add evidence receipt contract.
6. Add Rust tooling contract before Rust code.
7. Add C++ policy graph contract before C++ expansion.

Promotion rule

A feature can move forward only when it has:

```
contract
implementation plan
bounded implementation or fixture
negative tests
operator-visible report or status surface
non-claims
README/status alignment when public-facing
```

Public claim boundary

The careful public position is:

```
Latticra is building an evidence-bound systems substrate for Linux-era and AI-era computing.
Lat provides the native declaration direction.
Latticra Seal provides report-only trust-boundary and tool-boundary planning today.
Fedora/Linux is the current validation lane.
```

Do not claim:

```
production runtime
operating-system replacement
security product
sandbox
malware prevention
MCP implementation
AI-agent security enforcement
cryptographic verification
```

```
runtime enforcement
Fedora approval
daily-driver readiness
```

docs/ROADMAP.md

Source title: Latticra Roadmap

Lines: 274 | Words: 618 | SHA-256: d174b6fd3627c4e2ce46f696304c99a036aebd6712a6f338dd2c9d35f4950fc9

Latticra Roadmap

Status: initial roadmap Scope: design-first planning, implementation order, and promotion rules.

Purpose

Latticra should be planned, refined, and documented before broad implementation work begins.

This roadmap keeps the project disciplined while preserving the ambition of building a real, evidence-bound systems architecture and programming-language stack.

Roadmap principle

```
contracts before code
fixtures before runtime
read-only before mutation
local before server
hosted before hardware
virtual target before real device
read-only device evidence before real execution
```

Stage 0: Foundation planning

Goal:

```
define the system before expanding implementation
```

Required documents:

- real-system contract;
- evidence ladder;
- precursor promotion rule;
- non-claims;
- architecture seed;
- naming system;
- language naming policy;
- language strategy;
- supervisor architecture;
- feature vocabulary ledger;
- effect gates;
- L-UI direction;
- server interaction model;
- self-update model;
- host architecture targets;
- license policy.

Exit criteria:

- documents linked from README and foundation index;
- no implementation claim beyond tested repository evidence;
- first implementation target named.

Stage 1: State lattice fixture

Goal:

```
create the first real Latticra implementation unit
```

Scope:

- C header for state shape;
- deterministic fixture;
- no-effect invariant tests;
- report fixture;
- x86_64/ARM64-neutral structure policy.

Required labels:

```
origin
route
axis
path
breadcrumb
trace
safe_portal
rollback
health
risk
lock
dark_phase
host_effect
external_effect
```

Non-claims:

- no live movement;
- no host mutation;
- no hardware interaction;
- no update behavior;
- no server interaction.

Stage 2: Tri-plane transition model

Goal:

```
model transitions safely before implementing live movement
```

Scope:

- pure transition function;
- spatial plane;
- state plane;
- safety plane;
- denial reasons;
- no-effect guarantee.

Exit criteria:

- accepted transition tests;
- denied transition tests;
- lock/risk/rollback tests;
- host/external effect remains none.

Stage 3: Nucleus preview supervisor

Goal:

```
introduce supervisor classification without real mutation
```

Scope:

- request classification;
- policy classification;
- effect classification;
- no-effect reports;
- evidence record fixture.

Non-claims:

- no task execution beyond preview/report;
- no server activity;
- no updates;
- no recovery behavior.

Stage 4: Lat-Core draft

Goal:

```
define native language contracts for state, policy, and assertions
```

Scope:

- grammar draft;
- example files;
- parser design;
- LIR draft;
- no execution beyond validation fixtures.

Stage 5: L-UI report fixtures

Goal:

```
declare terminal/operator surfaces before rendering engine work
```

Scope:

- state cards;
- operator rails;
- effect reports;
- denial reports;
- no-color and narrow-terminal constraints.

Stage 6: Server and update fixtures

Goal:

```
model network/update behavior without performing it
```

Scope:

- server request fixture;
- signed manifest fixture;
- update state fixture;
- rollback visibility fixture;
- denied network by default.

Stage 7: Hosted reference implementation

Goal:

```
run Latticra as a hosted reference process
```

Targets:

```
x86_64-hosted
aarch64-hosted
```

Scope:

- local-only execution;
- no server by default;
- no hardware effects;
- no boot effects.

Stage 8: Virtual-target evidence profiles

Goal:

```
validate controlled profiles in QEMU or similar environments
```

Targets:

```
x86_64-qemu
aarch64-qemu
```

Scope:

- reproducible commands;
- validation reports;
- evidence bundles;
- no real-device claim.

Stage 9: Real-device read-only evidence

Goal:

```
observe real targets without changing them
```

Scope:

- device identity;
- read-only checklists;
- reports;
- non-mutation evidence.

Stage 10: Gated real-system experiments

Goal:

```
consider narrow real-world actions after evidence supports them
```

Scope:

- explicit operator approval;
- preflight checks;
- audit output;
- rollback/failure plan;
- no hidden effects.

Non-claims

This roadmap does not implement any stage by itself.

It defines sequencing and boundaries so Latticra can advance through evidence-backed implementation slices.

docs/status/ANNOUNCEMENTS.md

Source title: Latticra Announcements

Lines: 279 | Words: 1137 | SHA-256: 05efbe8fbcf52ed4bf2d7555e2fa0d6552fd0def81708b833c173f4b55fc6770

Latticra Announcements

Status: public announcement log Last updated: 2026-05-26 04:12 CDT Latest Fedora VM CLI payload validation milestone note: 2026-05-26 04:12 CDT Latest Fedora VM RPM validation milestone note: 2026-05-21 03:20 CDT Latest status announcement review note: 2026-05-19 19:25 CDT Latest authority foundation index alignment note: 2026-05-19 19:15 CDT Latest current status detail rollup note: 2026-05-19 19:05 CDT Latest Lat pipeline diagnostic status announcement note: 2026-05-19 17:45 CDT Latest status and announcement consistency review note: 2026-05-19 18:35 CDT Latest announcement rollup note: 2026-05-19 15:05 CDT Latest RBDM report status announcement note: 2026-05-19 16:45 CDT Scope: dated public updates, status notes, and milestone announcements.

Purpose

This file records public-facing Latticra updates in a controlled, evidence-bound way.

Announcements should be factual, dated, and careful not to overstate current security or operating-system capability.

Announcement rules

Each announcement should include:

```
date/time
status
what changed
why it matters
validation
non-claims
next step
```

2026-05-26 04:12 CDT — Disposable Fedora VM CLI payload validation milestone

Status: evidence-backed CLI payload validation milestone recorded

Latticra completed and recorded a successful disposable Fedora VM CLI payload validation path for the current no-effect local RPM payload.

What changed:

```
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
validated_payload=/usr/bin/latticra
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

Why it matters:

The Fedora validation path now has accepted disposable VM evidence for the compiled no-effect latticra CLI payload. The transcript records local RPM build, package installation, CLI --status, --version, --report, invalid-command validation, RPM removal, and post-removal absence verification.

Validation:

```
sh scripts/test-fedora-vm-cli-payload-validation-evidence-status.sh
sh scripts/test-fedora-vm-cli-payload-readme-alignment.sh
```

Non-claims:

This milestone does not claim production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, security capability, update safety, recovery safety, sandboxing, malware prevention, ransomware prevention, bootable OS replacement behavior, kernel runtime readiness, or a production installer.

Next step:

```
Plan the next Fedora CLI payload validation lane without widening README or announcement claims
beyond disposable Fedora VM evidence.
```

2026-05-21 03:20 CDT — Disposable Fedora VM local RPM validation milestone

Status: evidence-backed validation milestone recorded

Latticra completed and recorded a successful disposable Fedora VM local RPM validation path for the current documentation-only local RPM.

What changed:

```
disposable_vm_validation_completed=1
live_host_validation_completed=1
host_install_ready=1
validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

Why it matters:

This is the first evidence-backed host-facing validation milestone for Latticra. The project now has a real disposable Fedora VM transcript showing package build, RPM install, RPM verification, RPM removal, and post-removal absence verification for the local documentation-only RPM.

Validation:

```
sh scripts/test-fedora-disposable-vm-local-rpm-validation-evidence-status.sh
sh scripts/test-fedora-disposable-vm-rpm-readme-alignment.sh
```

Non-claims:

This milestone does not claim production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, security capability, update safety, recovery safety, sandboxing, malware prevention, ransomware prevention, bootable OS replacement behavior, kernel runtime readiness, or a production installer.

Next step:

```
Plan the next Fedora validation lane without widening README or announcement claims beyond
disposable Fedora VM local RPM evidence.
```

2026-05-19 19:25 CDT — Status announcement review

Status: announcement surface reviewed

Latticra reviewed the public announcement log after the authority implementation review, authority status/docs alignment, current status detail rollup, and authority foundation index alignment.

What changed:

```
status announcement review record added
authority review sequence reflected in the announcement log
root queue prepared for announcement-index follow-up
```

Why it matters:

The announcement surface now reflects the recent authority review and documentation alignment work without implying new runtime authority or operational behavior.

Validation:

```
sh scripts/test-project-strategy-status-framework.sh
```

Non-claims:

This update does not implement runtime behavior, command execution, Lat execution, LIR execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system behavior.

Next step:

```
Status announcement review index alignment
```

2026-05-19 18:35 CDT — Current status and announcement consistency review

Status: consistency review added

Latticra completed a status-surface consistency review after the recent Lat pipeline diagnostic, RBDM report, and project-notes alignment slices.

Validation:

```
sh scripts/test-project-strategy-status-framework.sh
```

Non-claims:

This update does not implement runtime behavior, command execution, Lat execution, LIR execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system behavior.

2026-05-19 17:45 CDT — Lat pipeline diagnostic main test audit

Status: audit coverage added

Latticra added an audit guard proving that the Lat pipeline diagnostic integration is covered by the main Lat pipeline test runner.

Validation:

```
sh scripts/test-lat-pipeline-diagnostic-main-test-integration-audit.sh
sh scripts/test-lat-pipeline.sh
```

Non-claims:

This update does not implement Lat execution, Lat compilation, Lat interpretation, LIR execution, runtime behavior, command execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system behavior.

2026-05-19 16:45 CDT — Runtime boundary domain matrix report integration

Status: report integration added

Latticra added deterministic report rendering for the Runtime Boundary Domain Matrix companion surface.

Validation:

```
sh scripts/test-runtime-boundary-domain-matrix-report-integration.sh
sh scripts/test-runtime-boundary-domain-matrix-refinement.sh
sh scripts/test-runtime-boundary.sh
```

Non-claims:

This update does not implement runtime behavior, command execution, Lat execution, LIR execution, task effect execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system behavior.

2026-05-19 15:05 CDT — Recent no-effect refinement rollup

Status: public rollup added

Latticra completed a sequence of no-effect refinement slices that strengthen reporting, diagnostics, and evidence visibility across Lat, LIR, Nucleus, and the runtime boundary.

Validation:

```
sh scripts/test-runtime-boundary.sh
sh scripts/test-runtime-boundary-domain-matrix-refinement.sh
sh scripts/test-nucleus-task-execution.sh
sh scripts/test-lat-semantic-validation.sh
sh scripts/test-lir-shape.sh
sh scripts/test-lat-pipeline.sh
```

Non-claims:

This rollup does not implement runtime behavior, command execution, Lat execution, LIR execution, task effect execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system behavior.

2026-05-18 21:30 CDT — Lat-specific LIR refinement implementation

Status: implementation added

Latticra added the first Lat-specific LIR refinement implementation.

Validation:

```
sh scripts/test-lat-specific-lir-refinement.sh
```

Non-claims:

This update does not implement Lat execution, LIR execution, runtime behavior, command behavior, mutation, file I/O, network I/O, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system behavior.

2026-05-18 19:40 CDT — Lat pipeline implementation

Status: implementation added

Latticra added the first bounded no-effect Lat pipeline implementation.

Validation:

```
sh scripts/test-lat-pipeline.sh
```

Non-claims:

This update does not implement Lat execution, Lat compilation, Lat interpretation, LIR execution, command behavior, runtime behavior, mutation, file I/O, network I/O, recovery behavior, hardware behavior, malware prevention, ransomware prevention, sandboxing, certification, accreditation, or operating-system behavior.

2026-05-16 16:15 CDT — Constrained C++ authority layer contract

Status: contract added

Latticra added the first constrained C++ authority layer contract.

This announcement also preserves the historical Constrained C++ authority layer implementation plan reference as the next planning step from that milestone.

Validation:

```
sh scripts/test-constrained-cpp-authority-layer-contract.sh
```

Non-claims:

This update does not implement C++ infrastructure, policy code, validators, effect gates, audit logic, orchestration, Lat execution, LIR execution, L-UI rendering, malware prevention, ransomware prevention, or operating-system behavior.

Announcement quality bar

Announcements should be:

```
specific
dated
honest
non-hype
evidence-bound
clear about non-claims
```

docs/status/CURRENT_STATUS.md

Source title: Latticra Current Status

Lines: 2418 | Words: 14948 | SHA-256: 7d773a4f55ad377d7c404fe05f8a70206d77fb45cf73a1216c8a33578a7b1b61

Latticra Current Status

Status: public status record Last updated: 2026-05-26 CDT Documentation hub: ../README.md Latest current estimate refresh note: 2026-05-24 CDT Latest current estimate table source alignment note: 2026-05-26 CDT Latest current estimate mathematical rebase note: 2026-05-26 CDT Latest completion estimate review README/status alignment note: 2026-05-25 CDT Latest completion estimate review after runtime-boundary abuse-case fixtures note: 2026-05-25 CDT Latest defensive threat model validation refinement note: 2026-05-25 CDT Latest high-assurance security baseline note: 2026-05-26 CDT Latest memory-safety roadmap note: 2026-05-26 CDT Latest supply-chain security baseline note: 2026-05-26 CDT Latest zero-trust runtime authority baseline note: 2026-05-26 CDT Latest cyber incident reporting and response baseline note: 2026-05-26 CDT Latest vulnerability management release gate baseline note: 2026-05-26 CDT Latest cryptographic assurance and key management baseline note: 2026-05-26 CDT Latest identity, credential, and access management baseline note: 2026-05-26 CDT Latest security logging, monitoring, and detection baseline note: 2026-05-26 CDT Latest backup, recovery, and cyber resilience baseline note: 2026-05-26 CDT Latest secure configuration and change management baseline note: 2026-05-26 CDT Latest runtime boundary policy expansion after threat-model note: 2026-05-26 CDT Latest runtime boundary abuse-case fixture expansion note: 2026-05-25 CDT Latest runtime boundary Lat pipeline comment evidence note: 2026-05-25 CDT Latest Lat parse-failure comment evidence propagation note: 2026-05-25 CDT Latest Lat pipeline failure span evidence propagation note: 2026-05-25 CDT Latest Lat pipeline parse-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline semantic-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline downstream stage-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline stage-summary evidence propagation note: 2026-05-25 CDT Latest Lat pipeline module/count evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-declaration evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-clause evidence propagation note: 2026-05-25 CDT Latest Lat LIR edge-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR no-effect evidence propagation note: 2026-05-26 CDT Latest Lat LIR module-summary evidence propagation note: 2026-05-26 CDT Latest Lat LIR source-span evidence propagation note: 2026-05-26 CDT Latest Lat LIR node-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge endpoint evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge span evidence propagation note: 2026-05-26 CDT Latest kernel lifecycle evidence status note: 2026-05-26 CDT Latest macOS reset/uninstall live-target classifier note: 2026-05-25 CDT Latest macOS reset/uninstall dry-run planner note: 2026-05-25 CDT Latest macOS reset/uninstall absence-report contract note: 2026-05-25 CDT Latest Seal README status row alignment note: 2026-05-25 CDT Latest Seal capability metadata report surface status note: 2026-05-26 CDT Latest Seal product spine note: 2026-05-26 CDT Latest Seal product spine status note: 2026-05-26 CDT Latest Seal operator receipt report contract note: 2026-05-26 CDT Latest Seal operator receipt report implementation plan note: 2026-05-26 CDT Latest Seal operator receipt report implementation note: 2026-05-26 CDT Latest Seal operator receipt report surface/status note: 2026-05-26 CDT Latest Seal operator receipt report predecessor status alignment note: 2026-05-26 CDT Latest Seal local capability registry schema contract note: 2026-05-26 CDT Latest Seal local capability registry schema implementation plan note: 2026-05-26 CDT Latest Seal local capability registry schema implementation note: 2026-05-26 CDT Latest Seal policy decision status/public-entry note: 2026-05-25 CDT Latest Seal policy decision predecessor status alignment note: 2026-05-26 CDT Latest Seal signed request status/public-entry note: 2026-05-25 CDT Latest Seal signed request predecessor status alignment note: 2026-05-26 CDT Latest Seal request freshness status/public-entry note: 2026-05-25 CDT Latest Seal request freshness predecessor status alignment note: 2026-05-26 CDT Latest Seal parameter schema status/public-entry note: 2026-05-25 CDT Latest Seal parameter schema predecessor status alignment note: 2026-05-26 CDT Latest Seal agentic automation security predecessor status alignment note: 2026-05-26 CDT Latest Seal agentic automation security public-entrypoint note: 2026-05-25 CDT Latest Seal status rollup predecessor status alignment note: 2026-05-26 CDT Latest Seal status rollup status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff predecessor status alignment note: 2026-05-26 CDT Latest Seal runtime handoff report status/public-entry note: 2026-05-25 CDT Latest Seal effect decision status/public-entry note: 2026-05-25 CDT Latest Seal capability gate predecessor status alignment note: 2026-05-26 CDT Latest Seal capability gate status/public-entry note: 2026-05-25 CDT Latest Seal verification receipt status/public-entry note: 2026-05-25 CDT Latest Seal crypto verify backend status/public-entry note: 2026-05-25 CDT Latest Seal Ed25519 verify status/public-entry note: 2026-05-25 CDT Latest Seal crypto graduation gate status note: 2026-05-26 CDT Latest Seal verified receipt promotion status/public-entry note: 2026-05-25 CDT Latest Seal verified capability gate status/public-entry note: 2026-05-25 CDT Latest Seal verified effect decision status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff evaluation status/public-entry note: 2026-05-25 CDT Latest Nadia offline AI Stage-0 foundation note: 2026-05-25 CDT Latest Nadia local context engine Stage-1 note: 2026-05-25 CDT Latest Nadia runtime profile Stage-2 note: 2026-05-25 CDT Latest Nadia developer workbench Stage-3 note: 2026-05-25 CDT Latest Nadia systems engineering mode Stage-4 note: 2026-05-25 CDT Latest Nadia productivity loop Stage-5 note: 2026-05-25 CDT Latest Nadia protective safety boundary Stage-6 note: 2026-05-25 CDT Latest Nadia guarded tool authority Stage-7 note: 2026-05-25 CDT Latest Nadia prompt evaluation contract Stage-8 note: 2026-05-25 CDT Latest Nadia local model registry contract Stage-9 note: 2026-05-25 CDT Latest Nadia inference readiness contract Stage-10 note: 2026-05-25 CDT Latest Nadia runtime invocation contract Stage-11 note: 2026-05-25 CDT Latest Nadia model load contract Stage-12 note: 2026-05-25 CDT Latest Nadia prompt receipt contract Stage-13 note: 2026-05-25 CDT Latest Nadia prompt materialization contract Stage-14 note: 2026-05-25 CDT Latest Nadia awareness dialogue contract Stage-15 note: 2026-05-25 CDT Latest Nadia prompt evaluation handoff contract Stage-16 note: 2026-05-25 CDT Latest Nadia tokenization boundary contract Stage-17 note: 2026-05-25 CDT Latest Nadia tokenizer specification contract Stage-18 note: 2026-05-25 CDT Latest Nadia tokenizer manifest contract Stage-19 note: 2026-05-25 CDT Latest Nadia tokenizer artifact inventory contract Stage-20 note: 2026-05-25 CDT Latest Nadia tokenizer artifact measurement contract Stage-21 note: 2026-05-25 CDT Latest Nadia tokenizer artifact verification contract Stage-22 note: 2026-05-25 CDT Latest Nadia tokenizer artifact binding contract Stage-23 note: 2026-05-25 CDT Latest Nadia tokenizer runtime attachment contract Stage-24 note: 2026-05-25 CDT Latest Nadia prompt tokenization contract Stage-25 note: 2026-05-25 CDT Latest Nadia prompt token sequence contract Stage-26 note: 2026-05-25 CDT Latest Nadia context window assembly contract Stage-27 note: 2026-05-25 CDT Latest Nadia prompt evaluation input contract Stage-28 note: 2026-05-25 CDT Latest Nadia prompt evaluation runtime handoff contract Stage-29 note: 2026-05-25 CDT Latest Nadia prompt evaluation invocation contract Stage-30 note: 2026-05-25 CDT Latest Nadia prompt evaluation result contract Stage-31 note: 2026-05-25 CDT Latest Nadia prompt evaluation result review contract Stage-32 note: 2026-05-25 CDT Latest Nadia prompt evaluation result disposition contract Stage-33 note: 2026-05-25 CDT Latest Nadia prompt evaluation result release contract Stage-34 note: 2026-05-25 CDT Latest Nadia prompt evaluation result release receipt contract Stage-35 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review contract Stage-36 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition contract Stage-37 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release contract Stage-38 note: 2026-05-26 CDT Latest

Nadia prompt evaluation result release receipt review disposition release receipt review contract Stage-40 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release receipt review disposition contract Stage-41 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release receipt review disposition release contract Stage-42 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt contract Stage-43 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review contract Stage-44 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract Stage-45 note: 2026-05-26 CDT Latest Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract Stage-46 note: 2026-05-26 CDT Latest Latticra Console profile preset note: 2026-05-25 CDT Latest Latticra Console standalone console install note: 2026-05-26 CDT Latest Latticra Console standalone installer preset note: 2026-05-26 CDT Latest Latticra Console standalone local install preset note: 2026-05-26 CDT Latest Latticra Console standalone contract note: 2026-05-26 CDT Latest Latticra Console host-embedding contract note: 2026-05-25 CDT Latest Latticra Console read-only host inventory contract note: 2026-05-25 CDT Latest Latticra Console host-adapter contract note: 2026-05-26 CDT Latest Latticra Console Seal receipt-request contract note: 2026-05-26 CDT Latest Latticra Console receipt payload schema note: 2026-05-26 CDT Latest Latticra Console receipt payload artifact draft note: 2026-05-26 CDT Latest Latticra Console receipt payload artifact review gate note: 2026-05-26 CDT Latest Latticra Console receipt payload artifact review receipt contract note: 2026-05-26 CDT Latest Latticra Console receipt payload artifact review receipt draft contract note: 2026-05-26 CDT Latest Latticra Console receipt payload materialization plan note: 2026-05-26 CDT Latest Latticra Console signature-request binding contract note: 2026-05-26 CDT Latest Latticra Console receipt contract note: 2026-05-25 CDT Latest Latticra Console OS-base planning contract note: 2026-05-25 CDT Latest Latticra Console VM evidence contract note: 2026-05-25 CDT Latest Seal verification receipt predecessor status alignment note: 2026-05-26 CDT Latest Seal verification policy predecessor status alignment note: 2026-05-26 CDT Latest Seal verification policy status/public-entry note: 2026-05-25 CDT Latest Seal crypto verify backend status/public-entry note: 2026-05-25 CDT Latest Seal Ed25519 verify status/public-entry note: 2026-05-25 CDT Latest Seal crypto graduation gate status note: 2026-05-26 CDT Latest Seal verified receipt promotion status/public-entry note: 2026-05-25 CDT Latest Seal verified capability gate status/public-entry note: 2026-05-25 CDT Latest Seal verified effect decision status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff evaluation status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff report status/public-entry note: 2026-05-25 CDT Latest Seal key parsing predecessor status alignment note: 2026-05-26 CDT Latest Seal key parsing status/public-entry note: 2026-05-25 CDT Latest Seal bounded key parsing implementation note: 2026-05-25 CDT Latest Seal future key parsing implementation plan note: 2026-05-25 CDT Latest Seal future key parsing implementation contract note: 2026-05-25 CDT Latest Seal public-key parsing predecessor status alignment note: 2026-05-26 CDT Latest Seal public-key parsing status/public-entry note: 2026-05-25 CDT Latest Seal public-key parsing implementation note: 2026-05-25 CDT Latest Seal public-key parsing contract note: 2026-05-25 CDT Latest Seal key-material predecessor status alignment note: 2026-05-26 CDT Latest Seal key-material status/public-entry note: 2026-05-25 CDT Latest Seal key-material implementation note: 2026-05-25 CDT Latest Seal key-material contract note: 2026-05-25 CDT Latest Seal key-handling predecessor status alignment note: 2026-05-26 CDT Latest Seal key-handling status/public-entry note: 2026-05-25 CDT Latest Seal key-handling implementation note: 2026-05-25 CDT Latest Seal key-handling contract note: 2026-05-25 CDT Latest Seal signing operation predecessor status alignment note: 2026-05-26 CDT Latest Seal signing operation status/public-entry note: 2026-05-25 CDT Latest Seal signing operation implementation note: 2026-05-25 CDT Latest Seal signing operation contract note: 2026-05-25 CDT Latest Seal signer invocation predecessor status alignment note: 2026-05-25 CDT Latest Seal signer invocation status/public-entry note: 2026-05-25 CDT Latest Seal signer invocation implementation note: 2026-05-25 CDT Latest Seal signer invocation contract note: 2026-05-25 CDT Latest Seal signer handoff predecessor status alignment note: 2026-05-25 CDT Latest Seal signer handoff status/public-entry note: 2026-05-25 CDT Latest Seal signer handoff implementation note: 2026-05-25 CDT Latest Seal signer handoff contract note: 2026-05-25 CDT Latest Seal signing authorization predecessor status alignment note: 2026-05-25 CDT Latest Seal signing authorization status/public-entry note: 2026-05-25 CDT Latest Seal signing authorization implementation note: 2026-05-25 CDT Latest Seal signing authorization contract note: 2026-05-25 CDT Latest Seal signature request predecessor status alignment note: 2026-05-25 CDT Latest Seal signature request status/public-entry note: 2026-05-25 CDT Latest Seal signature request implementation note: 2026-05-25 CDT Latest Seal signature request contract note: 2026-05-25 CDT Latest Seal report envelope status/public-entry note: 2026-05-25 CDT Latest Seal report envelope implementation note: 2026-05-25 CDT Latest Lat grammar report metadata integration note: 2026-05-25 CDT Latest Lat grammar line-comment metadata refinement note: 2026-05-25 CDT Latest Lat grammar unsupported block-comment rejection refinement note: 2026-05-25 CDT Latest Lat pipeline comment metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic comment metadata integration note: 2026-05-25 CDT Latest Lat parse-failure comment evidence propagation note: 2026-05-25 CDT Latest Lat pipeline failure span evidence propagation note: 2026-05-25 CDT Latest Lat pipeline parse-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline semantic-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline downstream stage-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline stage-summary evidence propagation note: 2026-05-25 CDT Latest Lat pipeline module/count evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-declaration evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-clause evidence propagation note: 2026-05-25 CDT Latest Lat LIR edge-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR no-effect evidence propagation note: 2026-05-26 CDT Latest Lat LIR module-summary evidence propagation note: 2026-05-26 CDT Latest Lat LIR source-span evidence propagation note: 2026-05-26 CDT Latest Lat LIR node-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge endpoint evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge span evidence propagation note: 2026-05-26 CDT Latest Lat model normalization note: 2026-05-25 CDT Latest Lat model report declaration metadata integration note: 2026-05-25 CDT Latest Lat model report clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline model normalization integration note: 2026-05-25 CDT Latest Lat-to-LIR model lowering integration note: 2026-05-25 CDT Latest Lat-to-LIR declaration metadata refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline lowering diagnostic integration note: 2026-05-25 CDT Latest Lat-to-LIR clause metadata refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline report declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline report clause metadata integration note: 2026-05-25 CDT Latest Lat-to-LIR lowering note: 2026-05-18 18:15 CDT Latest Lat pipeline note: 2026-05-18 19:40 CDT Latest Lat pipeline report refinement note: 2026-05-18 23:30 CDT Latest Lat pipeline diagnostic integration note:

2026-05-19 14:20 CDT Latest Lat pipeline diagnostic main test audit note: 2026-05-19 17:15 CDT Latest Lat semantic diagnostics refinement note: 2026-05-19 00:35 CDT Latest LIR report refinement note: 2026-05-19 00:55 CDT Latest Lat-specific LIR refinement note: 2026-05-18 21:30 CDT Latest language representation review note: 2026-05-19 21:05 CDT Latest L-UI rendering detailed report refinement note: 2026-05-19 20:25 CDT Latest L-UI rendering README/status alignment note: 2026-05-19 20:35 CDT Latest L-UI completion estimate review note: 2026-05-19 20:45 CDT Latest C++ authority expansion contract review note: 2026-05-19 20:55 CDT Latest Nucleus task no-effect report alignment note: 2026-05-19 21:15 CDT Latest Nucleus task README/status alignment note: 2026-05-19 21:25 CDT Latest Nucleus task report-only execution refinement note: 2026-05-19 21:35 CDT Latest Nucleus task report-only execution README/status alignment note: 2026-05-20 01:45 CDT Latest project notes Nucleus report-only alignment note: 2026-05-20 02:20 CDT Latest project notes Nucleus report-only status/index check note: 2026-05-20 02:45 CDT Latest Nucleus report-only announcement review note: 2026-05-20 03:00 CDT Latest Nucleus report-only announcement README alignment note: 2026-05-20 03:20 CDT Latest project notes Nucleus announcement README status/index check note: 2026-05-20 03:50 CDT Latest runtime boundary refinement plan note: 2026-05-18 22:15 CDT Latest runtime boundary refinement implementation note: 2026-05-18 22:45 CDT Latest runtime boundary report refinement note: 2026-05-18 23:10 CDT Latest runtime boundary policy matrix refinement note: 2026-05-18 23:55 CDT Latest runtime boundary domain matrix refinement note: 2026-05-19 14:10 CDT Latest runtime boundary domain matrix report integration note: 2026-05-19 16:20 CDT Latest project notes alignment note: 2026-05-19 18:25 CDT Latest status and announcement consistency review note: 2026-05-19 18:35 CDT Latest completion percentage review note: 2026-05-19 18:45 CDT Latest strategy estimate review note: 2026-05-19 18:45 CDT Latest authority implementation review note: 2026-05-19 18:55 CDT Latest authority status/docs alignment note: 2026-05-19 19:00 CDT Latest current status detail rollup note: 2026-05-19 19:05 CDT Latest authority foundation index alignment note: 2026-05-19 19:15 CDT Latest status announcement review note: 2026-05-19 19:25 CDT Latest status announcement review index alignment note: 2026-05-19 19:35 CDT Latest public entry-point consistency scan note: 2026-05-19 19:45 CDT Latest project notes follow-up alignment note: 2026-05-19 19:55 CDT Latest project notes follow-up status/index check note: 2026-05-19 20:05 CDT Latest authority announcement review note: 2026-05-19 20:15 CDT Latest Nucleus task report refinement note: 2026-05-19 00:15 CDT Scope: current progress, completion estimates, merged capability areas, and next priorities.

Project status

Latticra is an early-stage, contract-first systems architecture and language project.

The repository currently emphasizes:

- public project identity;
- strategy and status documentation;
- security-policy documentation;
- constrained C/C++ foundation direction;
- governed C++ authority-layer planning;
- Constrained C++ authority layer contract;
- Constrained C++ authority layer implementation plan;
- initial no-effect constrained C++ authority-layer implementation;
- C++ authority expansion contract review;
- authority implementation review;
- authority status/docs alignment;

- authority announcement review;
- current status detail rollup;
- authority foundation index alignment;
- status announcement review;
- public entry-point consistency scan;
- project notes follow-up alignment;
- project notes follow-up status/index check;
- project notes Nucleus report-only alignment;
- project notes Nucleus report-only status/index check;
- Nucleus report-only announcement review;
- Nucleus report-only announcement README alignment;
- project notes Nucleus announcement README status/index check;
- language representation review;
- Nadia offline AI Stage-0 foundation for Panel installability, Console interoperability, and awareness principles;
- Latticra Console profile presets, standalone dry-run/local installer presets, and standalone contract for hosted reference, standalone, Panel embedded, host-embedded planning, and OS-base planning under no-effect authority;
- Latticra Console host-embedding contract for pre-integration host adapter gates, read-only inventory prerequisites, and explicit denial of host process/file authority;
- Latticra Console read-only host inventory contract for future host evidence scope while current LC performs no inventory, host probing, file reads, process launches, or mutation;
- Latticra Console receipt contract for profile, host-contract, host-inventory, and Runtime Boundary evidence while current LC performs no signing or receipt writes;
- Nadia local context engine Stage-1 for no-network context-pack generation;
- Nadia runtime profile Stage-2 for offline model-readiness metadata before inference;
- Nadia developer workbench Stage-3 for prompt-plan generation without prompt evaluation;
- Nadia systems engineering mode Stage-4 for prompt-plan mode validation before prompt evaluation;
- Nadia productivity loop Stage-5 for operator-reviewed local productivity ledger entries before training or tool authority;
- Nadia protective safety boundary Stage-6 for non-sexual-use, anti-manipulation, and namesake-cause awareness restrictions;
- Nadia guarded tool authority Stage-7 for report-only tool preflight without tool execution;
- Nadia prompt evaluation contract Stage-8 before prompt materialization, prompt evaluation, inference, or tool execution;
- Nadia local model registry contract Stage-9 before model selection, model installation, prompt evaluation, inference, or tool execution;
- Nadia inference readiness contract Stage-10 before runtime invocation, model loading, prompt evaluation, inference, or tool execution;

- Nadia runtime invocation contract Stage-11 before runtime process spawning, model session creation, model loading, token generation, inference, or tool execution;
- Nadia model load contract Stage-12 before model file opening, weight mapping, weight loading, token generation, inference, or tool execution;
- Nadia prompt receipt contract Stage-13 before prompt source opening, prompt text receipt, prompt materialization, prompt evaluation, token generation, inference, or tool execution;
- Nadia prompt materialization contract Stage-14 before prompt buffer allocation, prompt text materialization, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia awareness dialogue contract Stage-15 for future official Nadia Initiative Q&A scope before dialogue generation, prompt evaluation, token generation, inference, or tool execution;
- Nadia prompt evaluation handoff contract Stage-16 before prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia tokenization boundary contract Stage-17 before tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer specification contract Stage-18 before tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer manifest contract Stage-19 before tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer artifact inventory contract Stage-20 before tokenizer artifact path resolution, artifact scanning, artifact hashing, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer artifact measurement contract Stage-21 before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest recording, artifact size recording, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer artifact verification contract Stage-22 before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest comparison, artifact size comparison, artifact verification, artifact binding, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer artifact binding contract Stage-23 before tokenizer artifact opening, artifact reading, artifact hashing, artifact verification, artifact binding, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;
- Nadia tokenizer runtime attachment contract Stage-24 before tokenizer artifact opening, artifact reading, artifact hashing, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution;

- Nadia prompt tokenization contract Stage-25 before prompt text reading, prompt token creation, prompt token sequence recording, tokenizer runtime attachment, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt token sequence contract Stage-26 before prompt token ID recording, token order recording, token offset recording, context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia context window assembly contract Stage-27 before context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation input contract Stage-28 before prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation runtime handoff contract Stage-29 before runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation invocation contract Stage-30 before invocation request creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result contract Stage-31 before result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result review contract Stage-32 before result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result disposition contract Stage-33 before disposition recording, release recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release contract Stage-34 before release recording, release decision recording, release publication, release packaging, release receipt creation, disposition recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt contract Stage-35 before receipt recording, receipt signing, receipt publication, release recording, release decision recording, release publication, release packaging, disposition recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review contract Stage-36 before review recording, review decision recording, review findings recording, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition contract Stage-37 before disposition recording, disposition decision recording, disposition findings recording, review recording, review decision recording, review findings recording, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation,

token generation, inference, or tool execution;

- Nadia prompt evaluation result release receipt review disposition release contract Stage-38 before disposition-release recording, release decision recording, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review contract Stage-40 before release-receipt-review recording, review decisions, review findings, review dispositions, receipt emission, receipt signing, receipt publication, disposition-release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review disposition contract Stage-41 before release-receipt-review-disposition recording, review decisions, review findings, review dispositions, receipt emission, receipt signing, receipt publication, disposition-release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review disposition release contract Stage-42 before release-receipt-review-disposition-release recording, release decisions, release publication, release receipts, review decisions, review findings, review dispositions, receipt emission, receipt signing, receipt publication, disposition-release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt contract Stage-43 before release-receipt-review-disposition-release-receipt recording, receipt emission, receipt signing, receipt publication, release records, review records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review contract Stage-44 before release-receipt-review-disposition-release-receipt-review recording, review decisions, review findings, receipt signing, receipt publication, receipt packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract Stage-45 before release-receipt-review-disposition-release-receipt-review-disposition recording, disposition decisions, disposition findings, receipt signing, receipt publication, receipt packaging, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- Nadia prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract Stage-46 before release-receipt-review-disposition-release-receipt-review-disposition-release recording, release decisions, release publication, release packaging, release receipt creation, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution;
- L-UI rendering implementation;
- L-UI rendering detailed report refinement;

- L-UI rendering README/status alignment;
- L-UI completion estimate review;
- Nucleus task execution contract;
- Nucleus task execution implementation plan;
- Nucleus task execution implementation;
- Nucleus task report refinement;
- Nucleus task no-effect report alignment;
- Nucleus task README/status alignment;
- Nucleus task report-only execution refinement;
- Nucleus task report-only execution README/status alignment;
- Runtime boundary contract;
- Runtime boundary implementation plan;
- Runtime boundary implementation;
- Runtime boundary refinement plan;
- Runtime boundary refinement implementation;
- Runtime boundary report refinement;
- Runtime boundary policy matrix refinement;
- Runtime boundary domain matrix refinement;
- Runtime boundary domain matrix report integration;
- Runtime boundary Lat pipeline comment evidence integration;
- Latticra Seal README status row alignment;
- Latticra Seal capability gate status/public-entry alignment;
- Latticra Seal capability gate predecessor status alignment;
- Latticra Seal effect decision status/public-entry alignment;
- Latticra Seal runtime handoff status/public-entry alignment;
- Latticra Seal runtime handoff predecessor status alignment;
- Latticra Seal runtime handoff report status/public-entry alignment;
- Latticra Seal status rollup status/public-entry alignment;
- Latticra Seal status rollup predecessor status alignment;
- Latticra Seal agentic automation security public-entrypoint alignment;
- Latticra Seal agentic automation security predecessor status alignment;
- Latticra Seal parameter schema status/public-entry alignment;
- Latticra Seal parameter schema predecessor status alignment;
- Latticra Seal request freshness status/public-entry alignment;
- Latticra Seal request freshness predecessor status alignment;
- Latticra Seal signed request status/public-entry alignment;
- Latticra Seal signed request predecessor status alignment;

- Latticra Seal policy decision status/public-entry alignment;
- Latticra Seal policy decision predecessor status alignment;
- Latticra Seal operator receipt report predecessor status alignment;
- Defensive threat model validation refinement;
- Runtime boundary abuse-case fixture expansion after policy expansion;
- Completion estimate review README/status alignment;
- Completion estimate review after runtime-boundary abuse-case fixtures;
- Latticra Seal verification receipt status/public-entry alignment;
- Latticra Seal verification receipt predecessor status alignment;
- Latticra Seal crypto verify backend status/public-entry alignment;
- Latticra Seal Ed25519 verify-only status/public-entry alignment;
- Latticra Seal verified receipt promotion status/public-entry alignment;
- Latticra Seal verified capability gate status/public-entry alignment;
- Latticra Seal verified effect decision status/public-entry alignment;
- Latticra Seal runtime handoff evaluation status/public-entry alignment;
- Latticra Seal runtime handoff report status/public-entry alignment;
- Latticra Seal verification policy predecessor status alignment;
- Latticra Seal verification policy status/public-entry alignment;
- Latticra Seal key parsing predecessor status alignment;
- Latticra Seal key parsing status/public-entry alignment;
- Latticra Seal bounded no-effect key parsing implementation;
- Latticra Seal future key parsing implementation plan;
- Latticra Seal future key parsing implementation contract;
- Latticra Seal public-key parsing predecessor status alignment;
- Latticra Seal public-key parsing status/public-entry alignment;
- Latticra Seal public-key parsing metadata implementation;
- Latticra Seal public-key parsing boundary contract;
- Latticra Seal key-material predecessor status alignment;
- Latticra Seal key-material status/public-entry alignment;
- Latticra Seal key-material metadata implementation;
- Latticra Seal key-material boundary contract;
- Latticra Seal key-handling predecessor status alignment;
- Latticra Seal key-handling status/public-entry alignment;
- Latticra Seal key-handling metadata implementation;
- Latticra Seal key-handling boundary contract;
- Latticra Seal signing operation predecessor status alignment;
- Latticra Seal signing operation status/public-entry alignment;

- Latticra Seal signing operation metadata implementation;
- Latticra Seal signing operation contract;
- Latticra Seal signer invocation predecessor status alignment;
- Latticra Seal signer invocation status/public-entry alignment;
- Latticra Seal signer invocation metadata implementation;
- Latticra Seal signer invocation contract;
- Latticra Seal signer handoff predecessor status alignment;
- Latticra Seal signer handoff status/public-entry alignment;
- Latticra Seal signer handoff metadata implementation;
- Latticra Seal signer handoff contract;
- Latticra Seal signing authorization predecessor status alignment;
- Latticra Seal signing authorization status/public-entry alignment;
- Latticra Seal signing authorization metadata implementation;
- Latticra Seal signing authorization contract;
- Latticra Seal signature request predecessor status alignment;
- Latticra Seal signature request status/public-entry alignment;
- Latticra Seal signature request metadata implementation;
- Latticra Seal signature request contract;
- Latticra Seal report envelope status/public-entry alignment;
- Latticra Seal report envelope metadata implementation;
- deterministic diagnostics;
- source-span metadata;
- semantic validation implementation;
- Lat semantic diagnostics refinement;
- LIR shape implementation;
- LIR report refinement;
- Lat grammar implementation;
- Lat grammar report metadata integration;
- Lat grammar line-comment metadata refinement;
- Lat grammar unsupported block-comment rejection refinement;
- Lat pipeline comment metadata integration;
- Lat pipeline diagnostic comment metadata integration;
- Lat parse-failure comment evidence propagation;
- Lat pipeline failure span evidence propagation;
- Lat pipeline parse-error evidence propagation;
- Lat pipeline semantic-error evidence propagation;
- Lat pipeline downstream stage-error evidence propagation;

- Lat pipeline stage-summary evidence propagation;
- Lat pipeline module/count evidence propagation;
- Lat pipeline first-declaration evidence propagation;
- Lat pipeline first-clause evidence propagation;
- Lat LIR edge-kind evidence propagation;
- Lat LIR no-effect evidence propagation;
- Lat LIR module-summary evidence propagation;
- Lat LIR source-span evidence propagation;
- Lat LIR node-kind evidence propagation;
- Lat LIR first-node evidence propagation;
- Lat LIR first-node span evidence propagation;
- Lat LIR first transition-node evidence propagation;
- Lat LIR first transition-node span evidence propagation;
- Lat LIR first-edge evidence propagation;
- Lat LIR first-edge span evidence propagation;
- Lat LIR first transition-source edge evidence propagation;
- Lat LIR first transition-source edge endpoint evidence propagation;
- Lat LIR first transition-source edge span evidence propagation;
- Runtime boundary Lat pipeline comment evidence integration;
- Lat semantic validation foundation;
- Lat model normalization implementation;
- Lat model report declaration metadata integration;
- Lat model report clause metadata integration;
- Lat-to-LIR lowering contract;
- Lat-to-LIR lowering implementation plan;
- Lat-to-LIR lowering implementation;
- Lat pipeline contract;
- Lat pipeline implementation plan;
- Lat pipeline implementation;
- Lat pipeline model normalization integration;
- Lat-to-LIR model lowering integration;
- Lat-to-LIR declaration metadata refinement;
- Lat-to-LIR diagnostic refinement;
- Lat-to-LIR diagnostic declaration metadata integration;
- Lat pipeline lowering diagnostic integration;
- Lat-to-LIR clause metadata refinement;
- Lat-to-LIR diagnostic clause metadata integration;

- Lat pipeline diagnostic declaration metadata integration;
- Lat pipeline diagnostic clause metadata integration;
- Lat pipeline report declaration metadata integration;
- Lat pipeline report clause metadata integration;
- Lat pipeline report refinement;
- Lat pipeline diagnostic integration refinement;
- Lat pipeline diagnostic integration main test audit;
- Lat-specific LIR refinement contract;
- Lat-specific LIR refinement implementation plan;
- Lat-specific LIR refinement implementation;
- project notes current-direction alignment;
- project notes upcoming-work alignment;
- project notes index alignment;
- status and announcement consistency review;
- completion percentage review;
- strategy estimate review;
- no-effect preview boundaries.

Lat now has a bounded no-effect path from grammar parsing to semantic validation to model normalization to LIR metadata lowering. The grammar report preserves first declaration, first clause, and line-comment metadata from successful AST parses, and block-comment openers outside strings/line comments now fail with unsupported_block_comment; the current pipeline report also carries parser line-comment count and first-comment span metadata. The current lowering implementation consumes normalized Lat model metadata directly, creates a lat_module LIR module shape, preserves source spans, no-effect flags, model counts, first lowered declaration metadata, and transition source indices, and emits deterministic lowering reports. The older parser-plus-semantic lowering entry point remains available as a compatibility wrapper.

The Lat model normalization implementation builds typed declaration and clause index tables for states, policies, transitions, assertions, and effect declarations after semantic validation. It preserves first normalized declaration metadata, first normalized clause role/effect/name/operator/value report metadata, source spans, and no-effect flags without reading source bytes, executing Lat, or changing lowering behavior.

The Lat-to-LIR diagnostic refinement classifies lowering outcomes as valid, parse, semantic, model, effect-check, capacity, LIR, or internal. It copies model errors, lowering errors, optional LIR errors, model counts, transition source metadata, first lowered declaration metadata, first-clause metadata, no-effect flags, and evidence level into deterministic diagnostic reports without changing lowering behavior.

The Lat-to-LIR clause metadata refinement preserves clause operators in LIR node metadata and exposes first lowered clause role, effect, name, operator, value, and node index in deterministic lowering reports without evaluating operators or adding execution.

The Lat semantic diagnostics refinement adds deterministic diagnostic classes, category counters, first-diagnostic declaration/clause indices, and report fields. This makes semantic failures easier to audit without changing validation outcomes or adding execution.

The Lat pipeline composes source parsing, semantic validation, Lat model normalization, model-driven Lat-to-LIR lowering, LIR metadata, and deterministic pipeline reporting into one no-effect integration boundary. It preserves metadata visibility without executing Lat, executing LIR, mutating state, or providing runtime behavior.

The Lat pipeline lowering diagnostic integration extends the companion pipeline diagnostic surface with parser line-comment count and first-comment span metadata, optional Lat-to-LIR diagnostic class, lowering error, model error, LIR error, model counts, transition source metadata, first lowered declaration metadata, first lowered clause metadata, and failure flags while preserving the older evaluator for existing callers.

The Lat pipeline report refinement adds deterministic stage-summary metadata for last completed stage, failed stage, per-stage OK flags, model normalization status, no-effect-chain status, evidence level, parser line-comment count plus first-comment span metadata, first lowered declaration kind/name/source/index/clause metadata, and first lowered clause role/effect/name/operator/value/node metadata. The current runtime-boundary Lat LIR first transition-node span, Lat LIR first transition-node, Lat LIR first transition-source edge endpoint, Lat LIR first transition-source edge span, Lat LIR first transition-source edge, Lat LIR first-edge span, Lat LIR first-edge, Lat LIR first-node span, Lat LIR first-node, Lat LIR node-kind, Lat LIR source-span, Lat LIR module-summary, Lat LIR no-effect, Lat LIR edge-kind, first-clause, first-declaration, module/count, and stage-summary evidence propagation copies LIR first Lat-derived transition node source span, LIR first Lat-derived transition node identity, LIR first Lat-derived transition-source endpoint node kinds and names, LIR first Lat-derived transition-source edge source span, LIR first Lat-derived transition-source edge identity, LIR first Lat-derived edge source span, LIR first Lat-derived edge identity, LIR first Lat-derived node source span, LIR first Lat-derived node identity, LIR node-kind counts, LIR source span, LIR module name, report classification, shape kind, core LIR counts, LIR no-effect flags, LIR contains/binds/annotates/orders-before/transition edge counts, first lowered clause metadata, first lowered declaration metadata, module identity, parser counts, model counts, and stage-summary metadata into runtime-boundary Lat pipeline records/reports so boundary denials and approvals preserve the same audit shape without changing no-effect behavior.

The Lat pipeline diagnostic integration refinement adds a companion diagnostic integration surface that combines pipeline error/stage state with parser line-comment count and first-comment span metadata, Lat semantic diagnostic class, semantic error, diagnostic count, first-diagnostic indices, model-stage classification, optional Lat-to-LIR diagnostic metadata, first lowered declaration metadata, and first lowered clause role/effect/name/operator/value/node metadata while preserving no-execution behavior.

The Lat pipeline diagnostic main test audit verifies that the companion diagnostic integration is covered by both the focused guard and the main Lat pipeline test runner.

The Lat-specific LIR refinement gives Lat declarations explicit LIR node kinds and a transition-source edge kind. This improves inspectability of Lat-derived LIR without changing no-effect behavior or adding execution.

The language representation review confirms that the Lat, LIR, and L-UI representation stack is stable enough for a bounded no-effect Nucleus report/refinement slice, while still forbidding Lat execution, LIR execution, runtime behavior, command execution, mutation, file I/O, network I/O, recovery behavior, and hardware behavior.

The LIR report refinement adds deterministic report classification, graph-shape labels, edge-kind summary counts, no-effect-chain status, and evidence level while preserving lowering outcomes and non-execution behavior.

The L-UI rendering detailed report refinement adds explicit report classification, detail level, detailed-report availability, detailed section count, deterministic section sequence, no-effect-chain status, and evidence level to the no-effect renderer report surface.

The L-UI rendering README/status alignment makes that detailed-report refinement discoverable from the README and status index without changing implementation behavior.

The L-UI completion estimate review conservatively updates planning estimates after the no-effect renderer report metadata and validation work. It does not change runtime, security-hardening, authority, or product-readiness estimates.

The C++ authority expansion contract review records that no new expansion contract is needed because no new authority behavior has been proposed.

The authority implementation review confirms the constrained C++ authority layer remains no-effect, metadata-only, fixed-capacity, and denied-by-default.

The authority status/docs alignment makes the authority review status discoverable from the status index and detailed status surface.

The authority announcement review confirms that no separate public authority announcement is needed because no authority behavior or messaging posture changed beyond the already-recorded no-effect review sequence.

The authority foundation index alignment makes the authority review and review status visible from the main foundation index.

The public entry-point consistency scan refreshes README and status-index references after the recent authority/status/foundation/announcement slices.

The project notes follow-up alignment refreshes the current-direction note, upcoming-work queue, and project-notes index after the public entry-point consistency scan.

The project notes follow-up status/index check verifies that the project-notes refresh is represented from the status index and detailed status surface.

The Nucleus task report refinement adds deterministic report classification, task-domain labeling, authorization-state labeling, prerequisite status, and no-effect-chain status. This makes Nucleus task reports easier to audit while preserving no-effect behavior.

The Nucleus task no-effect report alignment adds explicit report-alignment, no-effect-policy, and representation-gate labels to Nucleus task records and deterministic reports while preserving non-execution behavior.

The Nucleus task README/status alignment makes that no-effect report alignment discoverable from README and the status index without changing implementation behavior.

The Nucleus task report-only execution refinement adds explicit execution-status, effect-status, and runtime-status labels to Nucleus task records and deterministic reports while preserving no-effect, non-executing behavior.

The Nucleus task report-only execution README/status alignment makes the report-only execution refinement discoverable from README, root status, current status, status index, and foundation index without changing implementation behavior.

The project notes Nucleus report-only alignment refreshes current direction, upcoming work, and the project-notes index after the Nucleus task report-only execution README/status alignment.

The project notes Nucleus report-only status/index check verifies that the project-notes Nucleus report-only alignment is represented from root status, detailed current status, the status index, and the foundation index without changing implementation behavior or completion estimates.

The Nucleus report-only announcement review confirms that no separate public announcement entry is needed because the recent Nucleus report-only and project-notes slices were documentation/status alignment only, with no capability posture or public-product-readiness change.

The Nucleus report-only announcement README alignment makes the no-new-announcement review discoverable from README and the project-notes index without adding a public announcement entry or changing implementation behavior.

The project notes Nucleus announcement README status/index check verifies that the project-note body alignment after the Nucleus announcement README alignment is represented from root status, detailed current status, the status index, and the foundation index without changing implementation behavior or completion estimates.

The Runtime boundary refinement implementation reports Lat pipeline evidence, Lat pipeline line-comment count and first-comment span evidence, Lat-specific LIR source-span evidence, Lat-specific LIR module-summary evidence, Lat-specific LIR node-kind evidence, Lat-specific LIR first-node evidence, Lat-specific LIR first-node span evidence, Lat-specific LIR first transition-node evidence, Lat-specific LIR first transition-node span evidence, Lat-specific LIR first-edge evidence, Lat-specific LIR first-edge span evidence, Lat-specific LIR first transition-source edge evidence, Lat-specific LIR first transition-source edge endpoint evidence, Lat-specific LIR first transition-source edge span evidence, Lat-specific LIR no-effect evidence, and Lat-specific LIR edge-kind evidence at the runtime boundary while preserving disabled-by-default, no-effect classification behavior.

The Runtime boundary report refinement adds explicit report classification, boundary-domain labeling, authorization-state labeling, and evidence-level reporting so boundary intent is visible without granting runtime authority.

The Runtime boundary policy matrix refinement adds explicit policy-matrix cell labels plus effect, mode, authority, and future-gate report fields so the runtime-boundary decision shape is easier to audit without changing denied-by-default behavior.

The Runtime boundary domain matrix refinement adds a companion domain matrix evaluator for classifying resolved boundary domains as declarative, operational, future-gated, blocked, invalid, or unknown.

The Runtime boundary domain matrix report integration adds deterministic report rendering for domain-matrix cell, domain label, domain flags, effect-allowed state, authority-available state, and evidence level.

The Latticra Seal report envelope implementation adds bounded C metadata for classifying ready runtime handoff report metadata into sealed-report-only or sealed-evaluate-only envelope states while preserving no signing, no runtime handoff, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal report envelope status record makes the existing metadata-only envelope readiness classification visible from public entry points while preserving no signing, no object sealing, no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current report envelope fields:

```
seal_report_envelope_contract_present=1
seal_report_envelope_implementation_present=1
seal_report_envelope_status_present=1
envelope_profile=latticra-seal-report-envelope/0.1
report_profile=latticra-seal-runtime-handoff-report/0.1
requested_envelope=report-only
report_envelope_ready=1
report_envelope_state=sealed-report-only
report_envelope_signature_performed=0
```

```
report_envelope_handoff_performed=0
report_envelope_effect_performed=0
report_envelope_runtime_authority_granted=0
report_envelope_host_read_performed=0
report_envelope_host_write_performed=0
report_envelope_network_performed=0
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal signature request contract defines the next metadata-only request boundary after sealed report-envelope metadata. It permits only a future signature request metadata implementation and does not add signing, verification, private-key handling, host behavior, network behavior, capability enforcement, or runtime authority.

The Latticra Seal signature request implementation adds bounded C metadata for classifying allowed future signing requests after sealed report-envelope metadata while preserving no signing, no verification, no private-key handling, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signature request status record makes the metadata-only implementation visible from public entry points while preserving no signing, no verification, no private-key handling, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signature request predecessor status alignment ties that metadata-only status record to the guarded report-envelope status predecessor while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signature request predecessor status fields:

```
seal_report_envelope_status_present=1
seal_signature_request_status_present=1
signature_request_predecessor_report_envelope_status_present=1
signature_request_ready=1
signature_request_state=requested-metadata-only
signature_performed=0
verification_performed=0
private_key_handling=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal signing authorization contract defines the next metadata-only classification boundary after signature-request readiness while preserving no signing, no verification, no private-key handling, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signing authorization implementation adds bounded C metadata for classifying ready signature requests as authorized-metadata-only for a future signing path while preserving no signing, no verification, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signing authorization status record makes the metadata-only implementation visible from public entry points while preserving no signing, no verification, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signing authorization predecessor status alignment ties that metadata-only status record to the guarded signature-request status predecessor while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signing authorization predecessor status fields:

```
seal_signature_request_status_present=1
seal_signing_authorization_status_present=1
signing_authorization_predecessor_signature_request_status_present=1
signature_request_ready=1
signature_request_state=requested-metadata-only
signing_authorization_ready=1
signing_authorization_state=authorized-metadata-only
signature_performed=0
verification_performed=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal signer handoff contract defines the next metadata-only classification boundary after signing authorization readiness while preserving no signing, no verification, no signer invocation, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signer handoff implementation adds bounded C metadata for classifying ready signing authorizations as handoff-metadata-only for a future signer path while preserving no signing, no verification, no signer invocation, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signer handoff status record makes the metadata-only implementation visible from public entry points while preserving no signing, no verification, no signer invocation, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signer handoff predecessor status alignment ties that metadata-only status record to the guarded signing-authorization status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signer handoff predecessor status fields:

```
seal_signing_authorization_status_present=1
seal_signer_handoff_status_present=1
signer_handoff_predecessor_signing_authorization_status_present=1
signing_authorization_ready=1
signing_authorization_state=authorized-metadata-only
signer_handoff_ready=1
signer_handoff_state=handoff-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal signer invocation contract defines the next metadata-only classification boundary after signer handoff readiness while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signer invocation implementation adds bounded C metadata for classifying ready signer handoffs as invocation-metadata-only for a future signer path while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signer invocation status record makes the metadata-only implementation visible from public entry points while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signer invocation predecessor status alignment ties that metadata-only status record to the guarded signer-handoff status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signer invocation predecessor status fields:

```
seal_signer_handoff_status_present=1
seal_signer_invocation_status_present=1
signer_invocation_predecessor_signer_handoff_status_present=1
```

```
signer_handoff_ready=1
signer_handoff_state=handoff-metadata-only
signer_invocation_ready=1
signer_invocation_state=invocation-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal signing operation contract defines the next metadata-only classification boundary after signer invocation readiness while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signing operation implementation adds bounded C metadata for classifying ready signer invocation metadata as operation-metadata-only for a future signing operation path while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signing operation status record makes the metadata-only implementation visible from public entry points while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal signing operation predecessor status alignment ties that metadata-only status record to the guarded signer-invocation status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key generation, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no production cryptography claim.

Current signing operation predecessor status fields:

```
seal_signer_invocation_status_present=1
seal_signing_operation_status_present=1
signing_operation_predecessor_signer_invocation_status_present=1
signer_invocation_ready=1
signer_invocation_state=invocation-metadata-only
signing_operation_ready=1
signing_operation_state=operation-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

```
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_generation_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal key-handling boundary contract defines the next metadata-only classification boundary after signing operation readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key-handling implementation adds bounded C metadata for classifying ready signing operation metadata as key-handling-metadata-only for a future key-handling path while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key-handling status record makes the metadata-only implementation visible from public entry points while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key-handling predecessor status alignment ties that metadata-only status record to the guarded signing-operation status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current key-handling predecessor status fields:

```
seal_signing_operation_status_present=1
seal_key_handling_status_present=1
key_handling_predecessor_signing_operation_status_present=1
signing_operation_ready=1
signing_operation_state=operation-metadata-only
key_handling_ready=1
key_handling_state=key-handling-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
```

```
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal key-material boundary contract defines the metadata-only classification boundary after key-handling readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key-material implementation adds bounded C metadata for classifying ready key-handling metadata as key-material-metadata-only for a future key-material path while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key-material status record makes the metadata-only implementation visible from public entry points while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key-material predecessor status alignment ties that metadata-only status record to the guarded key-handling status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current key-material predecessor status fields:

```
seal_key_handling_status_present=1
seal_key_material_status_present=1
key_material_predecessor_key_handling_status_present=1
key_handling_ready=1
key_handling_state=key-handling-metadata-only
key_material_ready=1
key_material_state=key-material-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal public-key parsing boundary contract defines the next metadata-only classification boundary after key-material status readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal public-key parsing implementation adds bounded C metadata for classifying ready key-material metadata as public-key-parsing-metadata-only for a future public-key parsing path while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal public-key parsing status record makes the metadata-only implementation visible from public entry points while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal public-key parsing predecessor status alignment ties that metadata-only status record to the guarded key-material status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current public-key parsing predecessor status fields:

```
seal_key_material_status_present=1
seal_public_key_parsing_status_present=1
public_key_parsing_predecessor_key_material_status_present=1
key_material_ready=1
key_material_state=key-material-metadata-only
public_key_parsing_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal future key parsing implementation contract defines the next planning boundary after public-key parsing status readiness while preserving no public-key parsing implementation, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal future key parsing implementation plan defines the exact future bounded no-effect key parsing API, file paths, record fields, accepted public-key byte formats, failure behavior, report shape, and tests while still adding no parser code, no key material loading, no private-key handling, no signing, no verification, no host behavior, no network behavior, and no runtime authority in this slice.

The Latticra Seal bounded key parsing implementation adds fixed-size caller-provided Ed25519 public-key byte metadata after public-key parsing status readiness while preserving no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key parsing status record makes the bounded public-key byte metadata implementation visible from public entry points while preserving no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal key parsing predecessor status alignment ties that bounded status record to the guarded public-key parsing status predecessor while preserving the existing caller-provided public-key byte metadata behavior, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current key parsing predecessor status fields:

```
seal_public_key_parsing_status_present=1
seal_key_parsing_status_present=1
key_parsing_predecessor_public_key_parsing_status_present=1
public_key_parsing_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
key_parsing_ready=1
key_parsing_state=public-key-parsed-metadata-only
public_key_parsed=1
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal verification policy status record makes the existing metadata-only verification policy implementation visible from public entry points while preserving no cryptographic verification, no signing, no public-key byte verification, no key material loading, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal verification policy predecessor status alignment ties that metadata-only status record to the guarded key parsing status predecessor while preserving no cryptographic verification, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current verification policy predecessor status fields:

```
seal_key_parsing_status_present=1
seal_verification_policy_status_present=1
verification_policy_predecessor_key_parsing_status_present=1
key_parsing_ready=1
key_parsing_state=public-key-parsed-metadata-only
verification_policy_ready=1
verification_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
public_key_bytes_consumed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
cryptographic_verification_added=0
signature_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal crypto verify backend status record makes the existing metadata-only crypto verify backend implementation visible from public entry points while preserving unsupported cryptographic verification, no signing, no key material loading, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

Current crypto verify backend fields:

```
seal_crypto_verify_backend_contract_present=1
seal_crypto_verify_backend_implementation_present=1
seal_crypto_verify_backend_status_present=1
crypto_verify_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verified=0
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
real_cryptographic_verification_added=0
```

The Latticra Seal Ed25519 verify-only status record makes the existing local provider-backed verification result surface visible from public entry points while preserving authority-neutral behavior, no signing, no key handling, no host behavior, no network behavior, no capability enforcement, no runtime authority, and no production cryptography claim.

Current Ed25519 verify-only fields:

```
seal_ed25519_verify_only_contract_present=1
seal_ed25519_verify_implementation_present=1
seal_ed25519_verify_status_present=1
ed25519_verify_profile=latticra-seal-ed25519-verify/0.1
ed25519_cryptographic_verification_performed=1
ed25519_authority_usable=0
crypto_verify_state=verified
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
implementation_behavior_changed=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
capability_enforcement_added=0
effect_execution_added=0
production_cryptography_claimed=0
estimate_adjustment_required=0
```

The Latticra Seal verified receipt promotion status record makes the existing metadata promotion surface visible from public entry points while preserving authority-neutral behavior, no capability authorization, no effect execution, no runtime authority, and no production cryptography claim.

Current verified receipt promotion fields:

```
seal_verified_receipt_promotion_contract_present=1
seal_verified_receipt_promotion_implementation_present=1
seal_verified_receipt_promotion_status_present=1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
verification_state=verified
receipt_state=verified
verified_receipt_promotion_cryptographic_verification_performed=1
verified_receipt_promotion_authority_usable=0
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
capability_authorization_added=0
effect_execution_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal crypto graduation gate status record makes the current local Ed25519 verify-only result and verified receipt promotion pass through explicit cryptographic assurance expectations while preserving authority-neutral behavior, no capability authorization, no effect execution, no runtime authority, and no production cryptography claim.

Current crypto graduation gate fields:

```
seal_crypto_graduation_gate_present=1
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
signature_algorithm=Ed25519-development
```

```
message_digest_algorithm=SHA-256
provider_backed_verification_required=1
deterministic_test_vector_required=1
negative_test_vector_required=1
rfc8032_test_vector_tracked=1
fips_186_5_signature_standard_tracked=1
fips_180_4_digest_standard_tracked=1
fips_140_3_claim_gate_required=1
sp_800_57_key_management_required=1
sp_800_131a_transition_review_required=1
fips_204_ml_dsa_planning_tracked=1
fips_205_slh_dsa_planning_tracked=1
cryptographic_verification_performed=1
local_verify_graduated=1
receipt_promotion_graduated=1
standard_expectations_met=1
production_crypto_claim_allowed=0
fips_claim_allowed=0
signing_authority_granted=0
key_generation_allowed=0
key_storage_allowed=0
authority_promotion_allowed=0
capability_gate_allowed=0
runtime_authority_granted=0
gate_state=graduated-authority-neutral
```

The Latticra Seal verified capability gate status record makes the existing metadata-only gate evaluation visible from public entry points while preserving no capability enforcement, no effect execution, no runtime authority, no host behavior, no network behavior, and no production cryptography claim. The stricter entry point now requires a passing crypto graduation gate before recording the same metadata-only allowance.

Current verified capability gate fields:

```
seal_verified_capability_gate_contract_present=1
seal_verified_capability_gate_implementation_present=1
seal_verified_capability_gate_status_present=1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
requested_capability=verified-receipt-report
requested_effect=report-only
verified_capability_gate_allowed=1
verified_capability_gate_state=allowed-metadata-only
gate_state=allowed-metadata-only
verified_capability_gate_runtime_authority_granted=0
runtime_authority_granted=0
verified_capability_gate_effect_performed=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_enforcement_added=0
effect_execution_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal runtime handoff report status record makes the existing metadata-only report readiness classification visible from public entry points while preserving no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current runtime handoff report fields:

```
seal_runtime_handoff_report_contract_present=1
```

```
seal_runtime_handoff_report_implementation_present=1
seal_runtime_handoff_report_status_present=1
report_profile=latticra-seal-runtime-handoff-report/0.1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
requested_report=report-only
handoff_state=eligible-report-only
handoff_eligible=1
runtime_handoff_report_ready=1
runtime_handoff_report_state=ready-report-only
runtime_handoff_report_handoff_performed=0
runtime_handoff_report_effect_performed=0
runtime_handoff_report_runtime_authority_granted=0
runtime_handoff_report_host_read_performed=0
runtime_handoff_report_host_write_performed=0
runtime_handoff_report_network_performed=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal verified effect decision status record makes the existing metadata-only effect classification visible from public entry points while preserving no effect execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim. When crypto graduation evidence is present on the verified capability gate, the decision requires that evidence to remain passed, standard-aligned, and authority-neutral.

Current verified effect decision fields:

```
seal_verified_effect_decision_contract_present=1
seal_verified_effect_decision_implementation_present=1
seal_verified_effect_decision_status_present=1
decision_profile=latticra-seal-verified-effect-decision/0.1
gate_profile=latticra-seal-verified-capability-gate/0.1
requested_capability=verified-receipt-report
requested_effect=report-only
verified_effect_decision_crypto_graduation_gate_present=1
verified_effect_decision_crypto_graduation_gate_passed=1
verified_effect_decision_standard_expectations_met=1
verified_effect_decision_authority_promotion_allowed=0
verified_effect_decision_allowed=1
verified_effect_decision_state=allowed-report-only
verified_effect_decision_effect_performed=0
verified_effect_decision_runtime_authority_granted=0
effect_allowed=1
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
effect_execution_added=0
capability_enforcement_added=0
estimate_adjustment_required=0
```

The Latticra Seal runtime handoff evaluation status record makes the existing metadata-only handoff eligibility classification visible from public entry points while preserving no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim. When crypto graduation evidence is present on the verified effect decision, the evaluation requires that evidence to remain passed, standard-aligned, and authority-neutral.

Current runtime handoff evaluation fields:

```
seal_runtime_handoff_evaluation_contract_present=1
seal_runtime_handoff_evaluation_implementation_present=1
seal_runtime_handoff_evaluation_status_present=1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
```

```
decision_profile=latticra-seal-verified-effect-decision/0.1
requested_capability=verified-receipt-report
requested_effect=report-only
requested_handoff=report-only
runtime_handoff_evaluation_crypto_graduation_gate_present=1
runtime_handoff_evaluation_crypto_graduation_gate_passed=1
runtime_handoff_evaluation_standard_expectations_met=1
runtime_handoff_evaluation_authority_promotion_allowed=0
runtime_handoff_evaluation_eligible=1
runtime_handoff_evaluation_state=eligible-report-only
runtime_handoff_evaluation_handoff_performed=0
runtime_handoff_evaluation_runtime_authority_granted=0
handoff_eligible=1
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
estimate_adjustment_required=0
```

The Latticra Seal verification receipt status record makes the existing metadata-only, unverified verification receipt implementation visible from public entry points while preserving no cryptographic verification, no verified receipt authority, no signing, no key material loading, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority.

The Latticra Seal verification receipt predecessor status alignment ties that metadata-only receipt status record to the guarded verification policy status predecessor while preserving no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Current verification receipt predecessor status fields:

```
seal_verification_policy_status_present=1
seal_verification_receipt_status_present=1
verification_receipt_predecessor_verification_policy_status_present=1
verification_policy_ready=1
verification_receipt_ready=1
verification_state=unsupported
receipt_state=unverified-metadata
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verification_performed=0
verified=0
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
public_key_material_handling=0
public_key_bytes_consumed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
key_material_loading_added=0
```

```
private_key_handling_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal capability gate status record makes the existing metadata-only denied capability gate implementation visible from public entry points while preserving no capability enforcement, no effect execution, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority.

The Latticra Seal capability gate predecessor status alignment ties that metadata-only denied capability gate status record to the guarded verification receipt status predecessor while preserving no capability enforcement, no effect execution, no runtime handoff execution, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime authority, no host behavior, no network behavior, and no production enforcement claim.

Current capability gate predecessor status fields:

```
seal_verification_receipt_status_present=1
seal_capability_gate_status_present=1
verification_receipt_predecessor_verification_policy_status_present=1
capability_gate_predecessor_verification_receipt_status_present=1
verification_receipt_ready=1
capability_gate_ready=1
receipt_state=unverified-metadata
gate_state=denied-unverified
verified=0
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=0
capability_enforcement_performed=0
effect_allowed=0
effect_performed=0
runtime_authority_granted=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_enforcement_added=0
effect_execution_added=0
runtime_handoff_execution_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
network_behavior_changed=0
host_behavior_changed=0
estimate_adjustment_required=0
```

The Latticra Seal effect decision status record makes the existing metadata-only denied effect decision implementation visible from public entry points while preserving no effect execution, no capability enforcement, no runtime handoff execution, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority.

Latticra Seal effect decision predecessor status alignment records that the denied metadata-only effect decision remains tied to the guarded capability gate status predecessor.

```
effect_decision_predecessor_capability_gate_status_present=1
```

The Latticra Seal runtime handoff status record makes the existing inactive metadata-only runtime handoff implementation visible from public entry points while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority.

Latticra Seal runtime handoff predecessor status alignment records that the inactive metadata-only runtime handoff remains tied to the guarded effect decision status predecessor.

```
runtime_handoff_predecessor_effect_decision_status_present=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

The Latticra Seal runtime handoff report status/public-entry alignment makes the existing metadata-only runtime handoff report implementation visible from public entry points while preserving no runtime handoff execution, no effect execution, no capability enforcement, no signing, no host behavior, no network behavior, and no runtime authority.

```
seal_runtime_handoff_report_status_present=1
runtime_handoff_report_ready=1
runtime_handoff_report_state=ready-report-only
runtime_handoff_report_handoff_performed=0
runtime_handoff_report_runtime_authority_granted=0
```

The Latticra Seal status rollup status record makes the existing metadata-only status rollup implementation visible from public entry points while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority.

The Latticra Seal status rollup predecessor status alignment ties that metadata-only status rollup record to the guarded runtime handoff status predecessor while preserving no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no policy persistence, no host behavior, no network behavior, and no production enforcement claim.

Current status rollup predecessor status fields:

```
seal_runtime_handoff_status_present=1
seal_status_rollup_status_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
runtime_handoff_ready=1
rollup_state=metadata-only
runtime_boundary_state=disabled
handoff_active=0
runtime_effect_performed=0
```

```
effect_performed=0
capability_enforcement_performed=0
runtime_authority_granted=0
cryptographic_verification_performed=0
verification_performed=0
verified=0
capability_gate_allowed=0
effect_allowed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
host_read_performed=0
host_write_performed=0
network_performed=0
status_rollup_status_added=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
key_material_loading_added=0
private_key_handling_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

The Latticra Seal agentic automation security public-entrypoint alignment makes the existing report-only agentic automation security metadata, status/index alignment, and deterministic report surface visible from public entry points while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority.

The Latticra Seal agentic automation security predecessor status alignment ties that report-only automation status record to the guarded status rollup status predecessor while preserving no MCP behavior, no AI-agent execution, no model execution, no tool execution, no shell execution, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no object sealing, no policy persistence, no host behavior, no network behavior, and no production enforcement claim.

Current agentic automation security predecessor status fields:

```
seal_status_rollup_status_present=1
seal_agentic_automation_security_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
mcp_protocol_implemented=0
agent_execution_supported=0
model_execution_supported=0
tool_execution_supported=0
shell_execution_supported=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
agentic_automation_security_status_added=1
runtime_execution_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
key_material_loading_added=0
```

```
private_key_handling_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
```

The Latticra Seal parameter schema status record makes the existing report-only parameter schema contract, metadata implementation, and deterministic report surface visible from public entry points while preserving no schema parsing, no schema validation, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority.

The Latticra Seal parameter schema predecessor status alignment ties that report-only parameter schema status record to the guarded agentic automation security status predecessor while preserving no schema parsing, no schema validation, no policy evaluation, no policy enforcement, no MCP behavior, no AI-agent execution, no model execution, no tool execution, no shell execution, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no key material loading, no private-key handling, no object sealing, no policy persistence, no host behavior, no network behavior, and no production enforcement claim.

Current parameter schema predecessor status fields:

```
seal_agentic_automation_security_status_present=1
seal_parameter_schema_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
schema_parsing_supported=0
schema_validation_supported=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
schema_parsing_added=0
schema_validation_added=0
policy_evaluation_added=0
policy_enforcement_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
```

The Latticra Seal request freshness status record makes the existing report-only request freshness contract, metadata implementation, and deterministic report surface visible from public entry points while preserving no timestamp parsing, no trusted clock behavior, no nonce storage, no replay-cache storage, no context hashing, no parameter hashing, no freshness validation, no replay detection, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority.

The Latticra Seal request freshness predecessor status alignment ties that report-only request freshness status record to the guarded parameter schema status predecessor while preserving no timestamp parsing, no trusted clock behavior, no nonce storage, no replay-cache storage, no context hashing, no parameter hashing, no freshness validation, no replay detection, no signature verification, no signed request enforcement, no policy evaluation, no policy enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no host behavior, no network behavior, and no production enforcement claim.

Current request freshness predecessor status fields:

```
seal_parameter_schema_status_present=1
seal_request_freshness_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
timestamp_parsing_implemented=0
trusted_clock_behavior_added=0
nonce_storage_added=0
replay_cache_storage_added=0
context_hashing_implemented=0
parameter_hashing_implemented=0
freshness_validation_implemented=0
replay_protection_implemented=0
policy_evaluation_added=0
policy_enforcement_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
```

The Latticra Seal signed request status record makes the existing report-only signed request contract and metadata implementation visible from public entry points while preserving no signature generation, no signature verification, no public-key parsing, no trust-store loading, no private-key handling, no key generation, no hardware-key use, no revocation lookup, no network trust lookup, no signed request enforcement, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority.

The Latticra Seal signed request predecessor status alignment ties that report-only signed request status record to the guarded request freshness status predecessor while preserving no signature generation, no signature verification, no public-key parsing, no trust-store loading, no private-key handling, no key generation, no hardware-key use, no revocation lookup, no network trust lookup, no signed request enforcement, no policy evaluation, no policy enforcement, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no host behavior, no network behavior, and no production enforcement claim.

Current signed request predecessor status fields:

```
seal_request_freshness_status_present=1
seal_signed_request_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
signed_request_predecessor_request_freshness_status_present=1
signed_request_supported=0
signature_generation_supported=0
signature_verification_supported=0
signature_present=0
```

```
signature_valid=0
signature_algorithm_declared=0
signing_key_id_present=0
signature_hash_present=0
signed_request_id_present=0
identity_binding_declared=0
context_binding_declared=0
parameter_binding_declared=0
freshness_binding_declared=0
policy_binding_declared=0
trust_store_supported=0
revocation_lookup_supported=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
signature_generation_implemented=0
signature_verification_implemented=0
signed_request_enforcement_implemented=0
trust_store_implemented=0
revocation_lookup_implemented=0
policy_evaluation_added=0
policy_enforcement_added=0
freshness_validation_implemented=0
replay_protection_implemented=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
```

The Latticra Seal policy decision status record makes the existing report-only policy decision contract, metadata implementation, and deterministic report surface visible from public entry points while preserving no real policy evaluation, no policy enforcement, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signature verification, no freshness validation, no replay detection, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority.

The Latticra Seal policy decision predecessor status alignment ties that report-only policy decision status record to the guarded signed request status predecessor while preserving no policy evaluation, no policy enforcement, no signature verification, no freshness validation, no replay detection, no signed request enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no host behavior, no network behavior, and no production enforcement claim.

Current policy decision predecessor status fields:

```
seal_signed_request_status_present=1
seal_policy_decision_status_present=1
signed_request_predecessor_request_freshness_status_present=1
policy_decision_predecessor_signed_request_status_present=1
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
```

```
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_enforcement_added=0
real_policy_evaluation_added=0
policy_enforcement_added=0
signature_verification_implemented=0
freshness_validation_implemented=0
replay_protection_implemented=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
```

The Latticra Seal operator receipt report predecessor status alignment ties that denied report-only operator receipt status record to the guarded policy decision status predecessor while preserving no receipt file writes, no tool execution, no policy enforcement, no capability enforcement, no cryptographic verification, no signature verification, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no host behavior, no network behavior, and no production enforcement claim.

Current operator receipt report predecessor status fields:

```
seal_policy_decision_status_present=1
seal_operator_receipt_report_status_present=1
policy_decision_predecessor_signed_request_status_present=1
operator_receipt_report_predecessor_policy_decision_status_present=1
receipt_mode=report-only
receipt_status=denied-report-only
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
runtime_dry_run_state=report-only
default_action_deny=1
would_allow=0
would_deny=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
```

The Latticra Seal README status row alignment keeps the compact README Seal row and Seal current-posture summary aligned with the current public status checkpoint while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, no production enforcement, no public-readiness promotion, no security-hardening implementation, and no runtime authority.

Alignment record:

```
docs/status/SEAL_README_STATUS_ROW_ALIGNMENT.md
seal_readme_status_row_alignment_present=1
```

The defensive threat model validation refinement records the current evidence-mapping, external-source checkpoint, manual-review requirement, and next-gap triage while preserving no security controls, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, no production protection claim, and no runtime authority.

The project notes are now aligned across current direction, upcoming work, and project-notes index surfaces.

The current status and announcement consistency review confirms the public status and announcement surfaces now point to the same next review lane.

The completion percentage review conservatively updates planning estimates after the recent report, diagnostic, audit, README, foundation-index, project-notes, and status-consistency slices.

The completion estimate review after runtime-boundary abuse-case fixtures keeps the runtime-boundary fixture-slice estimate decision unchanged because the latest fixture slice improves denied-case evidence without changing capability posture, implementation behavior, public readiness, security hardening, product readiness, or runtime authority.

The completion estimate review README/status alignment makes the latest hold review visible from README and status entry points without changing capability posture, implementation behavior, public readiness, security hardening, product readiness, completion estimates, or runtime authority.

Current estimate table source alignment records that the rough completion estimate table below is the live reader-facing estimate source. The current estimate mathematical rebase records the live table as a weighted planning estimate after recent guarded Seal, Ubuntu, macOS, Nadia, Lat/LIR, kernel lifecycle, and public-entry alignment work. Dated review records remain slice-specific evidence records and may preserve the estimate snapshot that applied when that review was written.

Alignment records:

```
docs/status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md
docs/status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md
current_estimate_table_source_alignment_present=1
current_estimate_mathematical_rebase_present=1
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
estimate_adjustment_required=0
docs/status/COMPLETION_ESTIMATE_REVIEW_README_STATUS_ALIGNMENT.md
```

The strategy estimate review records the current planning estimate posture in the strategy layer while preserving the original dated strategy record as historical context.

Direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

The C/C++ foundation direction is guarded as an active language strategy. It means C owns the secure substrate, constrained C++ owns governed policy/validator/effect-gate/audit layers, and Lat / Latticra Language remains the contract/declaration layer.

Rough completion estimates

These percentages are planning estimates only.

Area	Estimated completion
Overall Latticra system	45%
Latticra Seal / local evidence layer	39%
Latticra Panel / local control surface	31%
Nadia offline AI foundation	75%
L-UI parser / AST / string foundation	87%
Foundation documents and contracts	94%
Public documentation posture	91%
Strategy/status/funding framework	63%
Lat / Latticra Programming Language	27%
LIR / Intermediate Representation	24%
C/C++ foundation direction	22%
Constrained C++ authority layer	5%
Nucleus real task execution	12%
Runtime / operating-system-universe direction	26%
Security-hardening implementation	10%
Public product readiness	10%

These percentages are planning estimates only. They are not release promises, production-readiness metrics, security guarantees, Fedora approval claims, Debian archive claims, FreeBSD ports-tree claims, OpenBSD ports-tree claims, runtime-enforcement claims, or operating-system completeness claims.

Current implemented evidence areas

Implemented or guarded areas include:

```
state lattice fixture
tri-plane transition preview model
Nucleus preview request classification
Nucleus task execution contract
Nucleus task execution implementation plan
Nucleus task execution implementation
Nucleus task report refinement
Nucleus task no-effect report alignment
Nucleus task README/status alignment
Nucleus task report-only execution refinement
Nucleus task report-only execution README/status alignment
Project notes Nucleus report-only alignment
Project notes Nucleus report-only status/index check
Nucleus report-only announcement review
Nucleus report-only announcement README alignment
Project notes Nucleus announcement README status/index check
Runtime boundary contract
Runtime boundary implementation plan
Runtime boundary implementation
Runtime boundary refinement plan
Runtime boundary refinement implementation
Runtime boundary report refinement
Runtime boundary policy matrix refinement
Runtime boundary domain matrix refinement
Runtime boundary domain matrix report integration
Runtime boundary Lat pipeline comment evidence integration
Kernel lifecycle seed
Kernel scheduler tick seed
Kernel run queue seed
Kernel context switch seed
Kernel time accounting seed
Kernel preemption seed
Kernel scheduler credit seed
Kernel scheduler selection seed
Kernel scheduler dispatch seed
Kernel scheduler handoff seed
Kernel scheduler activation seed
Kernel scheduler run-entry seed
Kernel lifecycle report runner
```

Kernel lifecycle subsystem summary
Kernel lifecycle rollback plan
Latticra Seal signing operation status/public-entry alignment
Latticra Seal signing operation metadata implementation
Latticra Seal signing operation contract
Latticra Seal README status row alignment
Latticra Seal effect decision status/public-entry alignment
Latticra Seal effect decision predecessor status alignment
Latticra Seal runtime handoff status/public-entry alignment
Latticra Seal status rollup status/public-entry alignment
Latticra Seal status rollup predecessor status alignment
Latticra Seal agentic automation security public-entrypoint alignment
Latticra Seal parameter schema status/public-entry alignment
Latticra Seal request freshness status/public-entry alignment
Latticra Seal signed request status/public-entry alignment
Latticra Seal Ed25519 verify-only status/public-entry alignment
Latticra Seal capability gate status/public-entry alignment
Latticra Seal verification receipt status/public-entry alignment
Latticra Seal verification receipt predecessor status alignment
Latticra Seal crypto verify backend status/public-entry alignment
Latticra Seal Ed25519 verify-only status/public-entry alignment
Latticra Seal verified receipt promotion status/public-entry alignment
Latticra Seal runtime handoff report status/public-entry alignment
Latticra Seal verification policy status/public-entry alignment
Latticra Seal verification policy predecessor status alignment
Latticra Seal key parsing status/public-entry alignment
Latticra Seal bounded no-effect key parsing implementation
Latticra Seal key parsing predecessor status alignment
Latticra Seal future key parsing implementation plan
Latticra Seal future key parsing implementation contract
Latticra Seal public-key parsing predecessor status alignment
Latticra Seal key-material predecessor status alignment
Latticra Seal key-handling predecessor status alignment
Latticra Seal signing operation predecessor status alignment
Latticra Seal signer invocation predecessor status alignment
Latticra Seal signer invocation status/public-entry alignment
Latticra Seal signer invocation metadata implementation
Latticra Seal signer invocation contract
Latticra Seal signer handoff predecessor status alignment
Latticra Seal signer handoff status/public-entry alignment
Latticra Seal signer handoff metadata implementation
Latticra Seal signer handoff contract
Latticra Seal signing authorization predecessor status alignment
Latticra Seal signing authorization status/public-entry alignment
Latticra Seal signing authorization metadata implementation
Latticra Seal signing authorization contract
Latticra Seal signature request predecessor status alignment
Latticra Seal signature request status/public-entry alignment
Latticra Seal signature request metadata implementation
Latticra Seal signature request contract
Latticra Seal report envelope metadata implementation
Defensive threat model contract
Defensive threat model validation refinement
Runtime boundary policy expansion after threat-model validation
Runtime boundary abuse-case fixture expansion after policy expansion
Completion estimate review README/status alignment
L-UI parser implementation
semantic validation contract
semantic validation implementation plan
semantic validation implementation
Lat semantic diagnostics refinement
Lat grammar report metadata integration
Lat grammar line-comment metadata refinement
Lat grammar unsupported block-comment rejection refinement
Lat pipeline comment metadata integration
Lat pipeline diagnostic comment metadata integration
Runtime boundary Lat pipeline comment evidence integration
Lat model normalization implementation
Lat model report declaration metadata integration
Lat model report clause metadata integration
LIR shape contract
LIR shape implementation plan
LIR shape implementation
LIR report refinement
Lat language grammar contract
Lat language grammar implementation plan
Lat language grammar implementation
Lat semantic validation foundation
Lat-to-LIR lowering contract
Lat-to-LIR lowering implementation plan

```
Lat-to-LIR lowering implementation
Lat pipeline contract
Lat pipeline implementation plan
Lat pipeline implementation
Lat pipeline model normalization integration
Lat-to-LIR model lowering integration
Lat-to-LIR declaration metadata refinement
Lat-to-LIR diagnostic declaration metadata integration
Lat-to-LIR diagnostic refinement
Lat pipeline lowering diagnostic integration
Lat-to-LIR clause metadata refinement
Lat-to-LIR diagnostic clause metadata integration
Lat pipeline diagnostic declaration metadata integration
Lat pipeline diagnostic clause metadata integration
Lat pipeline report declaration metadata integration
Lat pipeline report clause metadata integration
Lat pipeline report refinement
Lat pipeline diagnostic integration refinement
Lat pipeline diagnostic integration main test audit
Lat-specific LIR refinement contract
Lat-specific LIR refinement implementation plan
Lat-specific LIR refinement implementation
language representation review
C/C++ foundation direction
Constrained C++ authority layer contract
Constrained C++ authority layer implementation plan
Constrained C++ authority layer implementation
C++ authority expansion contract review
authority implementation review
authority status/docs alignment
authority announcement review
authority foundation index alignment
current status detail rollup
public entry-point consistency scan
project notes follow-up alignment
project notes follow-up status/index check
L-UI rendering contract
L-UI rendering implementation plan
L-UI rendering implementation
L-UI rendering detailed report refinement
L-UI rendering README/status alignment
L-UI completion estimate review
project notes current-direction alignment
project notes upcoming-work alignment
project notes index alignment
status announcement review
status and announcement consistency review
completion percentage review
strategy estimate review
security policy
public legacy association guard
strategy index
status index
funding metadata
```

Current non-claims

Latticra does not currently provide:

- a kernel;
- a bootable image;
- a production or daily-driver installer;
- a production language runtime;
- effect-performing runtime behavior;
- command execution;
- unrestricted C++ authority;
- effect-performing implemented C++ authority layer;
- effect-performing Nucleus task execution;
- interactive L-UI rendering;

- terminal-control L-UI rendering;
- LIR execution;
- Lat execution;
- Lat compiler;
- Lat interpreter;
- live movement;
- operating-system replacement.

Current mission alignment

Latticra is being built toward a defensive, auditable, open systems architecture.

The long-term goal is to make unsafe behavior harder to hide and easier to inspect through contract-driven source handling, explicit effects, deterministic diagnostics, operator-visible state, constrained substrate behavior, governed authority layers, deterministic rendering surfaces, denied-by-default task boundaries, Nucleus task report refinement, Nucleus task no-effect report alignment, Nucleus task report-only execution refinement, explicit runtime boundaries, runtime boundary domain matrix refinement, runtime boundary domain matrix report integration, Lat semantic validation, Lat semantic diagnostics refinement, Lat grammar report metadata integration, Lat grammar line-comment metadata refinement, Lat grammar unsupported block-comment rejection refinement, Lat model normalization, Lat model report declaration metadata integration, Lat model report clause metadata integration, Lat-to-LIR declaration metadata refinement, Lat-to-LIR metadata lowering, Lat-to-LIR diagnostic refinement, Lat-to-LIR diagnostic declaration metadata integration, LIR report refinement, Lat pipeline reporting, Lat pipeline lowering diagnostic integration, Lat pipeline diagnostic declaration metadata integration, Lat pipeline diagnostic clause metadata integration, Lat pipeline report declaration metadata integration, Lat pipeline report clause metadata integration, Lat pipeline comment metadata integration, Lat pipeline diagnostic comment metadata integration, Lat parse-failure comment evidence propagation, Lat pipeline failure span evidence propagation, Lat pipeline parse-error evidence propagation, Lat pipeline semantic-error evidence propagation, Lat pipeline downstream stage-error evidence propagation, Lat pipeline stage-summary evidence propagation, Lat pipeline module/count evidence propagation, Lat pipeline first-declaration evidence propagation, Lat pipeline first-clause evidence propagation, Lat LIR edge-kind evidence propagation, Lat LIR no-effect evidence propagation, Lat LIR module-summary evidence propagation, Lat LIR source-span evidence propagation, Lat LIR node-kind evidence propagation, Lat LIR first-node evidence propagation, Lat LIR first-node span evidence propagation, Lat LIR first transition-node evidence propagation, Lat LIR first transition-node span evidence propagation, Lat LIR first-edge evidence propagation, Lat LIR first-edge span evidence propagation, Lat LIR first transition-source edge evidence propagation, Lat LIR first transition-source edge endpoint evidence propagation, Lat LIR first transition-source edge span evidence propagation, Runtime boundary Lat pipeline comment evidence integration, Lat pipeline report refinement, Lat pipeline diagnostic integration refinement, Lat pipeline diagnostic main test audit, Lat-specific LIR refinement, runtime boundary evidence reporting, runtime boundary report refinement, and runtime boundary policy matrix refinement.

This is a mission direction, not a current security guarantee.

Latest completed contract slice

Latest completed contract slice:

```
Latticra Seal future key parsing implementation contract
```

Latest completed status/public-entry slice

Latest completed status/public-entry slice:

    Latticra Seal operator receipt report predecessor status alignment

Previous status/public-entry slice

Previous status/public-entry slice:

    Latticra Seal policy decision predecessor status alignment

Earlier status/public-entry slice

Earlier status/public-entry slice:

    Latticra Seal parameter schema predecessor status alignment

Latest completed implementation slice

Latest completed implementation slice:

    Lat LIR first transition-source edge endpoint evidence propagation

Previous implementation slice

Previous implementation slice:

    Lat LIR first transition-node span evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first transition-node evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first transition-source edge span evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first transition-source edge evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first-edge span evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first-edge evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first-node span evidence propagation

Earlier implementation slice

Earlier implementation slice:

    Lat LIR first-node evidence propagation

Earlier implementation slice

Earlier implementation slice:

```
Lat LIR node-kind evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat LIR source-span evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat LIR module-summary evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat LIR no-effect evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat LIR edge-kind evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline first-clause evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline first-declaration evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline module/count evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline stage-summary evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline downstream stage-error evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline semantic-error evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

```
Lat pipeline parse-error evidence propagation
```

Earlier implementation slice

Earlier implementation slice:

Lat pipeline failure span evidence propagation

Previous contract slice

Previous contract slice:

Latticra Seal public-key parsing boundary contract

Earlier status/public-entry slice

Earlier status/public-entry slice:

Latticra Seal key-material status/public-entry alignment

Earlier implementation slice

Earlier implementation slice:

Latticra Seal key-handling metadata implementation

Earlier contract slice

Earlier contract slice:

Latticra Seal key-handling boundary contract

Earlier status/public-entry slice

Earlier status/public-entry slice:

Latticra Seal signing operation status/public-entry alignment

Earlier implementation slice

Earlier implementation slice:

Latticra Seal signing operation metadata implementation

Latest completed planning slice

Latest completed planning slice:

Latticra Seal future key parsing implementation plan

Previous planning slice

Previous planning slice:

Runtime boundary refinement plan

Earlier completed status/public-entry slice

Earlier completed status/public-entry slice:

Latticra Seal signer invocation status/public-entry alignment

Earlier implementation slice

Earlier implementation slice:

Latticra Seal signer invocation metadata implementation

Older implementation slice

Older implementation slice:

Latticra Seal signer handoff metadata implementation

Next recommended work

Latest completed review slice:

```
Completion estimate review README/status alignment
```

Current completion estimate review fields:

```
current_estimate_table_source_alignment_present=1
current_estimate_mathematical_rebase_present=1
completion_estimate_review_readme_status_alignment_present=1
completion_estimate_after_runtime_boundary_abuse_case_fixtures_present=1
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
runtime_boundary_abuse_case_fixture_expansion_present=1
runtime_boundary_abuse_case_fixture_count=8
source_alignment_estimate_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
security_hardening_changed=0
public_readiness_changed=0
estimate_adjustment_required=0
runtime_authority_granted=0
completion_estimate_review_required=0
```

Recommended next work:

```
Continue small guarded report/status alignment only when drift appears
```

Completion estimate review only if capability posture changes remains the estimate rule after this non-change review.

After that:

```
Runtime behavior expansion only after separate contract, plan, tests, and explicit non-claim
review
```

Update rule

Update this file when major milestones merge, especially when completion estimates or next priorities change.

docs/status/README.md

Source title: Latticra Status Index

Lines: 1428 | Words: 7667 | SHA-256: a88ab1fd29fd9134628bce7457f11bc8271348da769a4490235b744da8bac6ba

Latticra Status Index

Status: active status index Last updated: 2026-05-26 Scope: public status, milestone estimates, announcements, and completion percentages.

For reader-facing documentation navigation, start with ../README.md. This file remains the detailed status index.

Purpose

This folder records public-facing Latticra status information.

Status records should separate:

```
current evidence
rough completion estimates
public announcements
future ambition
non-claims
```

Current documents

- CURRENT_STATUS.md — current project status, completion estimates, and next priorities.

- CURRENT_ESTIMATE_REFRESH_2026_05_24.md - current planning-estimate refresh after Panel, Seal, documentation, and local evidence work.
- CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md - current public estimate table source alignment for README, root status, detailed status, foundation index, and project notes.
- CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md - current mathematical planning-estimate rebase after recent guarded status, packaging, macOS, Nadia, Lat/LIR, and public-entry alignment slices.
- HIGH_ASSURANCE_SECURITY_BASELINE_STATUS.md - High-assurance security baseline status for official-source review requirements before runtime execution, effect execution, host behavior, network behavior, cryptographic enforcement, compliance claims, certification claims, production protection claims, or runtime authority.
- CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE_STATUS.md - Cyber incident reporting and response baseline status for vulnerability reports, suspected compromise triage, reporting paths, evidence preservation, communications routing, and incident-response non-claims. Field: cyber_incident_reporting_response_baseline_present=1.
- VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE_STATUS.md - Vulnerability management release gate baseline status for CISA KEV review, NVD/CVE review, disclosure handling, release blocking, and no product-security claims. Field: vulnerability_management_release_gate_baseline_present=1.
- CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE_STATUS.md - Cryptographic assurance and key-management baseline status for FIPS/CMVP claim gates, algorithm inventory, key lifecycle, randomness, post-quantum planning, and no production crypto claims. Field: cryptographic_assurance_key_management_baseline_present=1.
- IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE_STATUS.md - Identity, credential, and access management baseline status for privileged access, phishing-resistant MFA planning, account lifecycle, service identity, identity logging, and no hosted identity/access claims. Field: identity_credential_access_management_baseline_present=1.
- SECURITY_LOGGING_MONITORING_BASELINE_STATUS.md - Security logging, monitoring, and detection baseline status for event source inventory, audit event selection, runtime-authority decision logs, log redaction, retention, detection triage, incident handoff, and monitoring non-claims. Field: security_logging_monitoring_baseline_present=1.
- BACKUP_RECOVERY_RESILIENCE_BASELINE_STATUS.md - Backup, recovery, and cyber resilience baseline status for backup scope, offline backup posture, restore testing, recovery prioritization, rollback planning, recovery communications, and recovery non-claims. Field: backup_recovery_resilience_baseline_present=1.
- SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE_STATUS.md - Secure configuration and change management baseline status for configuration inventories, secure baselines, approved change records, rollback evidence, drift detection, exception ownership, and configuration non-claims. Field: secure_configuration_change_management_baseline_present=1.
- SUPPLY_CHAIN_SECURITY_BASELINE_STATUS.md - Supply-chain security baseline status for repository, CI, dependency, package, installer, artifact, SBOM, release, and update-lane posture before release publishing, signing authority, production installer claims, compliance claims, or certification claims.
- ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE_STATUS.md - zero-trust runtime authority baseline status for future runtime, tool, host I/O, network, server/MCP, update,

recovery, boot, hardware, agentic automation, and authority-bearing request paths.

- MEMORY_SAFETY_ROADMAP_STATUS.md - Memory-safety roadmap status for component inventory, restricted C/C++ profile requirements, fuzzing prerequisites, and no current production protection claims.
- LATTICRA_CONSOLE_FOUNDATION_STATUS.md - Latticra Console foundation status for the LC C report surface, standalone dry-run/local installer presets, Panel installability, metadata-only standalone/host/OS/VM contracts, receipt payload artifact draft, receipt payload artifact review gate, receipt payload artifact review receipt contract, receipt payload artifact review receipt draft contract, receipt payload materialization plan, and no-effect authority baseline.
- LATTICRA_SEAL_FOUNDATION_STATUS.md - Latticra Seal foundation status for the first Seal contract and implementation plan.
- SEAL_PRODUCT_SPINE_STATUS.md - Latticra Seal product spine status for the report-only path toward future verify, decide, handoff, and enforcement modes without current authority expansion.
- LATTICRA_NO_EFFECT_CLI_RPM_SPEC_UPDATE_STATUS.md - Latticra no-effect CLI RPM spec update status for adding the CLI payload to the local Fedora RPM spec.
- LATTICRA_PANEL_LOCAL_INSTALL_EVIDENCE_STATUS.md - Latticra Panel local install evidence status for the Fedora Workstation user-local install verification transcript.
- LATTICRA_PANEL_LOCAL_INSTALL_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Panel local install public-entrypoint alignment after the Fedora Workstation user-local install evidence milestone.
- LATTICRA_PANEL_SIGNED_UPDATER_DELIVERY_GATE_STATUS.md - Latticra Panel closed signed updater delivery gate status for future update delivery work without network fetch, remote repository trust, or signed update apply authority.
- LATTICRA_PANEL_SIGNED_UPDATER_DENIAL_TRANSCRIPT_STATUS.md - Latticra Panel no-effect signed updater denial transcript status for the closed signed-updater delivery gate.
- LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_CONTRACT_STATUS.md - Latticra Panel local no-effect signed updater manifest fixture contract status for future validation work without trusted update delivery or apply authority.
- LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_VALIDATION_STATUS.md - Latticra Panel local no-effect signed updater manifest fixture validation status for shape and closed-authority field checks without trusted update delivery or apply authority.
- LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_CONTRACT_STATUS.md - Latticra Panel local no-effect signed updater state fixture contract status for future state naming without transition execution, staging, activation, rollback execution, receipt writes, or host mutation.
- LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_VALIDATION_STATUS.md - Latticra Panel local no-effect signed updater state fixture validation status for shape, state catalog, blocked-state posture, and closed transition fields without transition execution, staging, activation, rollback execution, receipt writes, or host mutation.
- LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_TRANSCRIPT_STATUS.md - Latticra Panel no-effect signed updater state transition denial transcript status for blocked state-transition evidence without transition execution, staging, activation, rollback execution, receipt writes, transcript file writes, or host mutation.
- LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_REVIEW_STATUS.md - Latticra Panel no-effect signed updater state transition denial review status for upholding the blocked state transition without transition execution, staging, activation, rollback

execution, receipt writes, review file writes, or host mutation.

- LATTICRA_PANEL_UI_DESIGN_CHECKPOINT.md - Latticra Panel UI design checkpoint for the current product surface review.
- MACOS_INTEGRATION_TRANSFERABILITY_STATUS.md - macOS integration transferability status for adapting current no-effect, receipt-first, user-local Latticra surfaces into a future macOS lane.
- MACOS_BUILD_PLATFORM_PROBE_STATUS.md - macOS build/platform probe status for no-effect toolchain, architecture, Panel-readiness, and C-test-readiness reporting.
- MACOS_DRY_RUN_PLAN_ADAPTER_STATUS.md - macOS dry-run plan adapter status for rendering user-local Application Support, app bundle, CLI-wrapper, receipt, and verification intent without writing artifacts.
- MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT_STATUS.md - macOS user-local app bundle contract status for exact files, managed markers, reset/uninstall behavior, and verification transcript requirements.
- MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN_STATUS.md - macOS user-local app bundle implementation plan status for no-effect writer phases, failure behavior, reset/uninstall sequencing, and guard tests.
- MACOS_APP_BUNDLE_WRITER_DRY_RUN_STATUS.md - macOS app bundle writer dry-run status for phase-report output, unsafe-path validation, marker inspection, and disabled commit behavior.
- MACOS_APP_BUNDLE_WRITER_ALIGNMENT_STATUS.md - macOS app bundle writer alignment status separating the no-effect dry-run writer prototype from any future commit-capable writer.
- MACOS_LOCAL_CANDIDATE_ASSET_PROBE_STATUS.md - macOS local candidate asset probe status for caller-supplied Panel executable and icon readiness checks without build, copy, signing, notarization, or writes.
- MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION_STATUS.md - macOS dry-run writer candidate integration status for proving accepted local candidates can move the writer dry-run to its future commit-gate decision without writes.
- MACOS_COMMIT_GATE_CONTRACT_STATUS.md - macOS commit gate contract status for keeping future user-local managed app bundle writes closed until implementation and verification evidence exist.
- MACOS_VERIFICATION_TRANSCRIPT_CONTRACT_STATUS.md - macOS verification transcript contract status for exact post-write evidence required before any user-local install can be called verified.
- MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT_STATUS.md - macOS reset/uninstall dry-run contract status for managed-target removal order and preservation rules without deleting files.
- MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER_STATUS.md - macOS reset/uninstall live-target classifier status for read-only managed, unmanaged, and absent target reports.
- MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER_STATUS.md - macOS reset/uninstall dry-run planner status for converting live target classifications into an ordered no-effect transcript.
- MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT_STATUS.md - macOS reset/uninstall absence-report contract status for future post-removal verification evidence.

- MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT_STATUS.md - macOS reset/uninstall receipt-schema contract status for future reset/uninstall receipt fields and path constraints.
- MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT_STATUS.md - macOS reset/uninstall implementation-gate contract status for keeping live reset/uninstall execution closed until required evidence exists.
- MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT_STATUS.md - macOS reset/uninstall operator-intent contract status for future explicit live reset/uninstall approval evidence.
- MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT_STATUS.md - macOS reset/uninstall effect-authorization contract status for keeping live reset/uninstall effects gated by explicit authorization evidence.
- MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT_STATUS.md - macOS reset/uninstall evidence-bundle contract status for binding required reset/uninstall evidence before any live implementation can be considered.
- MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT_STATUS.md - macOS reset/uninstall live-implementation plan contract status for mapping future effect-authorized phases while deletion remains disabled.
- MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT_STATUS.md - macOS reset/uninstall live-execution preflight contract status for keeping live deletion blocked until evidence bundle, authorization, implementation gate, and operator intent requirements are satisfied.
- MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md - macOS reset/uninstall live-denial transcript contract status for recording failed live-execution preflight decisions without persistent writes or deletion.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT_STATUS.md - macOS reset/uninstall live-runner interface contract status for keeping future runner dispatch denied until live-execution preflight evidence passes.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT_STATUS.md - macOS reset/uninstall live-runner no-op prototype contract status for exercising the denied interface path without dispatch or deletion.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT_STATUS.md - macOS reset/uninstall live-runner denied-dispatch transcript contract status for recording the denied no-op runner path without dispatch, deletion, writes, or authority.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT_STATUS.md - macOS reset/uninstall live-runner denied-dispatch review contract status for reviewing the denied no-op runner path while preserving zero dispatch and zero deletion.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-gate contract status for keeping dispatch closed until preflight, evidence, authorization, operator intent, and implementation evidence are present.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial transcript contract status for recording the closed acceptance gate without dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, or runtime authority.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial review contract status for reviewing the

closed acceptance gate while preserving zero dispatch, deletion, receipts, mutation, network, root, and runtime authority.

- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition contract status for recording the reviewed closed-gate denial without dispatch, deletion, receipts, mutation, network, root, or runtime authority.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition review contract status for reviewing the no-effect disposition while preserving zero dispatch, deletion, receipts, mutation, network, root, runtime authority, disposition application, and runner release.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition closeout contract status for closing the no-effect disposition while preserving zero dispatch, deletion, receipts, mutation, network, root, runtime authority, runner release, and audit opening.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract status for auditing the no-effect closeout while preserving zero dispatch, deletion, receipts, absence reports, mutation, network, root, runtime authority, audit review opening, and implementation claims.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review contract status for reviewing the no-effect closeout audit while preserving zero dispatch, deletion, receipts, absence reports, mutation, network, root, runtime authority, next disposition, and implementation claims.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract status for recording the reviewed no-effect closeout audit as a no-effect disposition while preserving zero dispatch, deletion, receipts, mutation, network, root, runtime authority, future review opening, and implementation claims.
- MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_REVIEW_CONTRACT_STATUS.md - macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract status for reviewing the no-effect closeout audit review disposition while preserving zero dispatch, deletion, receipts, mutation, network, root, runtime authority, future closeout opening, and implementation claims.
- MACOS_README_INSTALLER_USAGE_STATUS.md - macOS README installer usage status for documenting current Mac-specific no-effect installer commands, target paths, and closed commit-gate posture.
- NADIA_OFFLINE_AI_STAGE_0_STATUS.md - Nadia offline AI Stage-0 foundation status for Panel installability, Console interoperability, and awareness principles.
- NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1_STATUS.md - Nadia Stage-1 local context-engine status for no-network context-pack generation.
- NADIA_RUNTIME_PROFILE_STAGE_2_STATUS.md - Nadia Stage-2 runtime-profile status for offline model-readiness metadata before inference.
- NADIA_DEVELOPER_WORKBENCH_STAGE_3_STATUS.md - Nadia Stage-3 developer-workbench status for prompt-plan generation without prompt evaluation.

- NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4_STATUS.md - Nadia Stage-4 systems-engineering mode status for prompt-plan validation before prompt evaluation.
- NADIA_PRODUCTIVITY_LOOP_STAGE_5_STATUS.md - Nadia Stage-5 productivity-loop status for operator-reviewed local productivity ledger entries.
- NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6_STATUS.md - Nadia Stage-6 protective-safety boundary status for non-sexual-use and anti-manipulation restrictions.
- NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7_STATUS.md - Nadia Stage-7 guarded tool-authority status for report-only preflight after protective-safety validation.
- NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8_STATUS.md - Nadia Stage-8 prompt-evaluation contract status before prompt materialization, prompt evaluation, inference, or tool execution.
- NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9_STATUS.md - Nadia Stage-9 local model-registry contract status before model selection, model installation, prompt evaluation, inference, or tool execution.
- NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10_STATUS.md - Nadia Stage-10 inference-readiness contract status before runtime invocation, model loading, prompt evaluation, inference, or tool execution.
- NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11_STATUS.md - Nadia Stage-11 runtime-invocation contract status before runtime process spawning, model session creation, model loading, token generation, inference, or tool execution.
- NADIA_MODEL_LOAD_CONTRACT_STAGE_12_STATUS.md - Nadia Stage-12 model-load contract status before model file opening, weight mapping, weight loading, token generation, inference, or tool execution.
- NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13_STATUS.md - Nadia Stage-13 prompt-receipt contract status before prompt source opening, prompt text receipt, prompt materialization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14_STATUS.md - Nadia Stage-14 prompt-materialization contract status before prompt buffer allocation, prompt text materialization, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15_STATUS.md - Nadia Stage-15 awareness-dialogue contract status for Nadia Initiative Q&A scope before dialogue generation, prompt evaluation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16_STATUS.md - Nadia Stage-16 prompt-evaluation handoff contract status before prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17_STATUS.md - Nadia Stage-17 tokenization-boundary contract status before tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18_STATUS.md - Nadia Stage-18 tokenizer-specification contract status before tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19_STATUS.md - Nadia Stage-19 tokenizer-manifest contract status before tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.

- NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20_STATUS.md - Nadia Stage-20 tokenizer-artifact-inventory contract status before tokenizer artifact path resolution, artifact scanning, artifact hashing, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21_STATUS.md - Nadia Stage-21 tokenizer-artifact-measurement contract status before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest recording, artifact size recording, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22_STATUS.md - Nadia Stage-22 tokenizer-artifact-verification contract status before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest comparison, artifact size comparison, artifact verification, artifact binding, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23_STATUS.md - Nadia Stage-23 tokenizer-artifact-binding contract status before tokenizer artifact opening, artifact reading, artifact hashing, artifact verification, artifact binding, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24_STATUS.md - Nadia Stage-24 tokenizer-runtime-attachment contract status before tokenizer artifact opening, artifact reading, artifact hashing, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.
- NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25_STATUS.md - Nadia Stage-25 prompt-tokenization contract status before prompt text reading, prompt token creation, prompt token sequence recording, tokenizer runtime attachment, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26_STATUS.md - Nadia Stage-26 prompt-token-sequence contract status before prompt token ID recording, token order recording, token offset recording, context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27_STATUS.md - Nadia Stage-27 context-window assembly contract status before context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28_STATUS.md - Nadia Stage-28 prompt-evaluation-input contract status before prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29_STATUS.md - Nadia Stage-29 prompt-evaluation runtime handoff contract status before runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30_STATUS.md - Nadia Stage-30 prompt-evaluation invocation contract status before invocation requests, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31_STATUS.md - Nadia Stage-31 prompt-evaluation result contract status before result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32_STATUS.md - Nadia Stage-32 prompt-evaluation result review contract status before result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33_STATUS.md - Nadia Stage-33 prompt-evaluation result disposition contract status before disposition records, release records, result-review records, result records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34_STATUS.md - Nadia Stage-34 prompt-evaluation result release contract status before release recording, release publication, release packaging, release receipts, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35_STATUS.md - Nadia Stage-35 prompt-evaluation result release receipt contract status before receipt recording, receipt signing, receipt publication, release recording, release publication, release packaging, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36_STATUS.md - Nadia Stage-36 prompt-evaluation result release receipt review contract status before review recording, review decisions, review findings, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_37_STATUS.md - Nadia Stage-37 prompt-evaluation result release receipt review disposition contract status before disposition recording, disposition decisions, disposition findings, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_38_STATUS.md - Nadia Stage-38 prompt-evaluation result release receipt review disposition release contract status before disposition-release recording, release decisions, release publication, release receipts, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_39_STATUS.md - Nadia Stage-39 prompt-evaluation result release receipt review disposition release receipt contract status before release-receipt recording, receipt signing, receipt publication, disposition-release recording, release publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_40_STATUS.md - Nadia Stage-40 prompt-evaluation result release receipt review disposition release receipt review contract status before review-disposition-release receipt recording, receipt emission, receipt signing, receipt publication, review recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_41_STATUS.md - Nadia Stage-41 prompt-evaluation result release receipt review disposition release receipt review disposition contract status before release-receipt-review-disposition records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_42_STATUS.md - Nadia Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release contract status before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_43_STATUS.md - Nadia Stage-43 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract status before release-receipt records, receipt emission, receipt signing, receipt publication, release records, review records, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_44_STATUS.md - Nadia Stage-44 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract status before release-receipt-review records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_45_STATUS.md - Nadia Stage-45 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract status before release-receipt-review-disposition records, disposition decisions, disposition findings, receipt signing, receipt publication, receipt packaging, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46_STATUS.md - Nadia Stage-46 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract status before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.
- ANNOUNCEMENTS.md — public update log and announcement notes.

- ../DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md — defensive threat model validation refinement.
- ../RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md — runtime boundary policy expansion after threat-model validation.
- SEAL_README_STATUS_ROW_ALIGNMENT.md — Latticra Seal README status row alignment with the current public status checkpoint.
- SEAL_POLICY_DECISION_PUBLIC_ENTRYPOINT_ALIGNMENT.md — Latticra Seal policy decision public-entrypoint alignment.
- SEAL_POLICY_DECISION_STATUS.md — Latticra Seal policy decision status/public-entry alignment.
- SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md — Latticra Seal policy decision report surface status.
- SEAL_SIGNED_REQUEST_STATUS.md — Latticra Seal signed request status/public-entry alignment.
- SEAL_REQUEST_FRESHNESS_STATUS.md — Latticra Seal request freshness status/public-entry alignment.
- SEAL_PARAMETER_SCHEMA_STATUS.md — Latticra Seal parameter schema status/public-entry alignment.
- SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md — Latticra Seal agentic automation security public-entrypoint alignment.
- SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md — Latticra Seal agentic automation security report surface status.
- SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md — Latticra Seal agentic automation security index alignment.
- SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md — Latticra Seal agentic automation security metadata status.
- SEAL_PRODUCT_SPINE_STATUS.md — Latticra Seal product-spine status for the current observe, verify, decide, handoff, and future-closed enforce posture.
- ../LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_CONTRACT.md — Latticra Seal operator receipt report contract.
- ../LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION_PLAN.md — Latticra Seal operator receipt report implementation plan.
- ../LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md — Latticra Seal operator receipt report implementation.
- ../LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md — Latticra Seal operator receipt report surface.
- SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md — Latticra Seal operator receipt report status.
- ../LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md — Latticra Seal local capability registry schema contract.
- ../LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md — Latticra Seal local capability registry schema implementation plan.
- ../LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md — Latticra Seal local capability registry schema implementation.

- SEAL_STATUS_ROLLUP_STATUS.md — Latticra Seal status rollup metadata status and public-entry checkpoint.
- SEAL_RUNTIME_HANDOFF_STATUS.md — Latticra Seal runtime handoff metadata status and public-entry checkpoint.
- SEAL_EFFECT_DECISION_STATUS.md — Latticra Seal effect decision metadata status and public-entry checkpoint.
- SEAL_CAPABILITY_GATE_STATUS.md — Latticra Seal capability gate metadata status and public-entry checkpoint.
- SEAL_CAPABILITY_METADATA_REPORT_SURFACE_STATUS.md — Latticra Seal capability metadata report surface status for the deterministic local report-only capability metadata fixture.
- SEAL_VERIFICATION_RECEIPT_STATUS.md — Latticra Seal verification receipt metadata status and public-entry checkpoint.
- SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md — Latticra Seal crypto verify backend metadata status and public-entry checkpoint.
- SEAL_ED25519_VERIFY_STATUS.md — Latticra Seal Ed25519 verify-only result status and public-entry checkpoint.
- SEAL_CRYPTO_GRADUATION_GATE_STATUS.md — Latticra Seal authority-neutral crypto graduation gate status for standards-gated local verification and receipt promotion. Field: seal_crypto_graduation_gate_present=1.
- SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md — Latticra Seal verified receipt promotion metadata status and public-entry checkpoint.
- SEAL_CRYPTO_GRADUATION_GATE_STATUS.md — Latticra Seal crypto graduation gate status for authority-neutral Ed25519 verify-only and verified receipt promotion metadata without signing, key handling, network lookup, production crypto claims, capability enforcement, effect execution, host behavior, or runtime authority.
- SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md — Latticra Seal verified capability gate metadata status and crypto-graduation-gated public-entry checkpoint.
- SEAL_VERIFIED_EFFECT_DECISION_STATUS.md — Latticra Seal verified effect decision metadata status and public-entry checkpoint.
- SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md — Latticra Seal runtime handoff evaluation metadata status and public-entry checkpoint.
- SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md — Latticra Seal runtime handoff report metadata status and public-entry checkpoint.
- SEAL_REPORT_ENVELOPE_STATUS.md — Latticra Seal sealed report-envelope metadata status and public-entry checkpoint.
- SEAL_VERIFICATION_POLICY_STATUS.md — Latticra Seal verification policy metadata status and public-entry checkpoint.
- SEAL_KEY_PARSING_STATUS.md — Latticra Seal key parsing metadata status and public-entry checkpoint.
- SEAL_PUBLIC_KEY_PARSING_STATUS.md — Latticra Seal public-key parsing metadata status and public-entry checkpoint.
- SEAL_KEY_MATERIAL_STATUS.md — Latticra Seal key-material metadata status and public-entry checkpoint.

- SEAL_KEY_HANDLING_STATUS.md — Latticra Seal key-handling metadata status and public-entry checkpoint.
- SEAL_SIGNING_OPERATION_STATUS.md — Latticra Seal signing operation metadata status and public-entry checkpoint.
- SEAL_SIGNER_INVOCATION_STATUS.md — Latticra Seal signer invocation metadata status and public-entry checkpoint.
- SEAL_SIGNER_HANDOFF_STATUS.md — Latticra Seal signer handoff metadata status and public-entry checkpoint.
- SEAL_SIGNING_AUTHORIZATION_STATUS.md — Latticra Seal signing authorization metadata status and public-entry checkpoint.
- SEAL_SIGNATURE_REQUEST_STATUS.md — Latticra Seal signature request metadata status and public-entry checkpoint.
- SEAL_CORE_EVIDENCE_STATUS.md — Latticra Seal core evidence status surface after the completed report-only runtime gate evidence milestone.
- SEAL_CORE_EVIDENCE_INDEX_ALIGNMENT.md — Latticra Seal core evidence index alignment after the status surface.
- SEAL_CORE_BLOCKED_CASES_STATUS.md — Latticra Seal core blocked cases status after the unknown-tool, unsigned-request, stale-request, and replayed-request fixtures.
- SEAL_CORE_EVIDENCE_PUBLIC_STATUS_UPDATE.md — Latticra Seal core evidence public status update language for the report-only runtime gate milestone.
- SEAL_CORE_EVIDENCE_PUBLIC_ENTRYPOINT_ALIGNMENT.md — Latticra Seal core evidence public-entrypoint alignment after the core evidence status surface.
- SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE_STATUS.md — Latticra Seal guarded allowlist report surface status.
- SEAL_GUARDED_ALLOWLIST_STATUS_INDEX_ALIGNMENT.md — Latticra Seal guarded allowlist status-index alignment.
- SEAL_GUARDED_ALLOWLIST_PUBLIC_ENTRYPOINT_ALIGNMENT.md — Latticra Seal guarded allowlist public-entrypoint alignment.
- SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md — Latticra Seal runtime dry-run report surface status.
- SEAL_RUNTIME_DRY_RUN_STATUS_INDEX_ALIGNMENT.md — Latticra Seal runtime dry-run status-index alignment.
- SEAL_RUNTIME_DRY_RUN_PUBLIC_ENTRYPOINT_ALIGNMENT.md — Latticra Seal runtime dry-run public-entrypoint alignment.
- KERNEL_LIFECYCLE_EVIDENCE_STATUS.md — status alignment after the kernel IPC, VFS namespace, process table, syscall table, interrupt table, timer source, scheduler tick, run queue, context-switch, time-accounting, preemption, scheduler credit, scheduler selection, scheduler dispatch, scheduler handoff, scheduler activation, and scheduler run-entry guards plus the lifecycle report runner, subsystem summary, and rollback plan.
- FEDORA_HOST_INSTALL_PREFLIGHT_STATUS.md — status record for the no-effect Fedora host install preflight classifier.
- FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_STATUS.md — status record for the no-effect Fedora install preflight snapshot capture implementation.

- FEDORA_LIVE_READONLY_SNAPSHOT_ADAPTER_STATUS.md — status record for the live read-only Fedora snapshot adapter implementation.
- FEDORA_RPM_GATE_CLASSIFIER_STATUS.md — status record for the Fedora RPM gate classifier implementation.
- FEDORA_INSTALLROOT_RPM_LIFECYCLE_STATUS.md — status record for the controlled Fedora installroot RPM lifecycle lane.
- FEDORA_MANUAL_HOST_RC_CHECKLIST_STATUS.md — status record for the manual Fedora host release-candidate checklist.
- FEDORA_MANUAL_HOST_RC_DECISION_CLASSIFIER_STATUS.md — status record for the no-effect Fedora manual host RC decision classifier.
- FEDORA_DISPOSABLE_VM_EFFECT_GATE_CLASSIFIER_STATUS.md — status record for the Fedora disposable VM effect gate classifier.
- FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_STATUS.md — status record for the gated disposable Fedora VM local RPM validation lane.
- FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_EVIDENCE_STATUS.md — evidence status record for the successful disposable Fedora VM local RPM validation transcript.
- FEDORA_DISPOSABLE_VM_RPM_README_ALIGNMENT_STATUS.md — Fedora disposable VM RPM README alignment.
- FEDORA_VM_CLI_PAYLOAD_VALIDATION_STATUS.md — status record for the manually gated disposable Fedora VM CLI payload validation lane runner.
- FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md — evidence status record after a disposable Fedora VM CLI payload validation transcript reached the expected validation report.
- FEDORA_VM_CLI_PAYLOAD_README_ALIGNMENT_STATUS.md — Fedora VM CLI payload README alignment.
- FEDORA_VM_CLI_PAYLOAD_VALIDATION_ANNOUNCEMENT_STATUS.md — announcement/status alignment for the disposable Fedora VM CLI payload validation milestone.
- FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN_STATUS.md — status record for the Fedora VM CLI payload next-validation lane plan.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT_STATUS.md — status record for the future second disposable Fedora VM CLI payload validation transcript contract.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE_STATUS.md — status record for the Fedora VM CLI payload repeatability transcript capture template.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR_STATUS.md — status record for the Fedora VM CLI payload repeatability transcript review validator.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN_STATUS.md — status record for the manual disposable Fedora VM CLI payload repeatability runner plan.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_STATUS.md — status record for the manually gated Fedora VM CLI payload repeatability runner.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE_STATUS.md — status record for the Fedora VM CLI payload repeatability evidence review gate.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE_STATUS.md — status record for the no-effect Fedora VM CLI payload repeatability transcript capture template.

- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR_STATUS.md — status record for the no-effect Fedora VM CLI payload repeatability transcript review validator.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT_STATUS.md — status record for the Fedora VM CLI payload repeatability evidence acceptance contract.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE_STATUS.md — status record for the no-effect Fedora VM CLI payload repeatability evidence status template.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR_STATUS.md — status record for the no-effect Fedora VM CLI payload repeatability evidence status review validator.
- FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE_STATUS.md — status record for the Fedora VM CLI payload repeatability evidence publication gate.
- FEDORA_VM_RPM_VALIDATION_ANNOUNCEMENT_STATUS.md — announcement/status alignment for the disposable Fedora VM local RPM validation milestone.
- UBUNTU_ECOSYSTEM_INTEGRATION_STATUS.md — status record for the Ubuntu build lane, Panel apt prerequisites, and local-only deb packaging draft.
- DEBIAN_ECOSYSTEM_INTEGRATION_STATUS.md — status record for the local-only Debian deb packaging draft and static validation lane.
- FREEBSD_ECOSYSTEM_INTEGRATION_STATUS.md — status record for the local-only FreeBSD ports metadata draft and static validation lane.
- OPENBSD_ECOSYSTEM_INTEGRATION_STATUS.md — status record for the local-only OpenBSD ports metadata draft and static validation lane.
- ../DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md — source archive and checksum contract for Debian, FreeBSD, and OpenBSD package/port draft promotion blockers.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md — temporary package input handoff lane for Debian, FreeBSD, and OpenBSD source and distfile staging without package builds.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md — closed package-build evidence gate for Debian, FreeBSD, and OpenBSD package build commands.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md — package-build environment contract for disposable Debian, FreeBSD, and OpenBSD validation environments without package builds.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md — package artifact naming contract for future Debian, FreeBSD, and OpenBSD package files without creating artifacts.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md — package payload inspection contract for future Debian, FreeBSD, and OpenBSD package files without inspecting artifacts.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md — package install/remove transcript contract for future disposable Debian, FreeBSD, and OpenBSD validation without installing packages.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md — package publication non-claim review contract for local-only Debian, FreeBSD, and OpenBSD validation without publishing packages.
- ../DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md — package validation promotion blocker matrix for Debian, FreeBSD, and OpenBSD local

validation evidence without accepting build evidence.

- OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md — status record for the openSUSE compatibility lane, Panel zypper prerequisites, local-only RPM maintenance draft, rpmlint/osc availability lane, static spec lint lane, findings classification record, source archive reproducibility contract, source archive fixture lane, temporary RPM topdir handoff lane, local RPM build gate contract, local RPM build environment contract, RPM payload inspection, RPM install/remove transcript contract, and OBS publication non-claim review.
- ../OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md — closed openSUSE local RPM build evidence gate contract before any rpmbuild or osc build command can run.
- ../OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md — openSUSE local RPM build environment contract for disposable validation environments without rpmbuild or osc build.
- ../OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md — openSUSE RPM artifact naming contract for future source RPM and binary RPM files without creating artifacts.
- ../OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md — openSUSE RPM payload inspection contract for future source RPM and binary RPM files without inspecting artifacts.
- ../OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md — openSUSE RPM install/remove transcript contract for future disposable package install and remove evidence without installing RPMs.
- ../OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md — openSUSE OBS publication non-claim review contract for future local package validation without OBS, submit-request, official-package, or SUSE endorsement claims.
- AUTHORITY_STATUS_ANNOUNCEMENT_REVIEW.md — no-new-announcement authority status review.
- CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md — current public estimate table source alignment across README, root status, and detailed current status.
- CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md — current mathematical planning-estimate rebase for the live public estimate table.
- COMPLETION_ESTIMATE_L_UI_RENDERING_REVIEW.md — completion-estimate review after the L-UI detailed report refinement.
- COMPLETION_ESTIMATE_REVIEW_README_STATUS_ALIGNMENT.md — README/status alignment for the completion-estimate review after runtime-boundary abuse-case fixtures.
- COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md — completion-estimate hold review after runtime-boundary abuse-case fixtures.
- COMPLETION_PERCENTAGE_REVIEW.md — latest completion-percentage planning review.
- CPP_AUTHORITY_EXPANSION_CONTRACT_REVIEW.md — no-expansion-contract C++ authority review.
- CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md — constrained C++ authority implementation review status.
- LANGUAGE_REPRESENTATION_REVIEW.md — language representation review before Nucleus refinement.
- L_UI_RENDERING_DETAILED_REPORT_REFINEMENT_STATUS.md — L-UI rendering detailed report refinement status.
- L_UI_RENDERING_README_STATUS_ALIGNMENT.md — L-UI rendering README/status alignment.

- NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_REVIEW.md — no-new-announcement review after Nucleus report-only alignment slices.
- NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_README_ALIGNMENT.md — README/project-notes alignment after the Nucleus report-only announcement review.
- NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT_STATUS.md — Nucleus task no-effect report alignment status.
- NUCLEUS_TASK_README_STATUS_ALIGNMENT.md — Nucleus task README/status alignment.
- NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md — Nucleus task report-only execution refinement status.
- NUCLEUS_TASK_REPORT_ONLY_EXECUTION_README_STATUS_ALIGNMENT.md — Nucleus task report-only execution README/status alignment.
- PROJECT_NOTES_NUCLEUS_ANNOUNCEMENT_README_ALIGNMENT_STATUS_CHECK.md — project-notes Nucleus announcement README alignment status/index check.
- PROJECT_NOTES_NUCLEUS_REPORT_ONLY_ALIGNMENT_STATUS_INDEX_CHECK.md — project-notes Nucleus report-only alignment status/index check.
- PUBLIC_ENTRY_POINT_CONSISTENCY_SCAN.md — public entry-point consistency scan.
- PROJECT_NOTES_FOLLOWUP_STATUS_INDEX_CHECK.md — project-notes follow-up status/index check.
- STATUS_ANNOUNCEMENT_REVIEW.md — status announcement review.
- STATUS_ANNOUNCEMENT_CONSISTENCY_REVIEW.md — status/announcement consistency review.
- LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_STATUS.md — Lat pipeline diagnostic integration status.
- LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_AUDIT_STATUS.md — Lat pipeline diagnostic main-test audit status.
- RBDM_REPORT_INTEGRATION_STATUS.md — runtime-boundary domain matrix report integration status.
- ../RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md — runtime-boundary abuse-case fixture expansion after policy expansion.

Current Latticra Seal core evidence checkpoint

The latest Latticra Seal core evidence status surface records:

```
seal_readme_status_row_alignment_present=1
seal_core_evidence_report_present=1
seal_core_evidence_public_status_present=1
seal_core_evidence_status_surface_present=1
seal_report_envelope_contract_present=1
seal_report_envelope_implementation_present=1
seal_report_envelope_status_present=1
report_envelope_profile=latticra-seal-report-envelope/0.1
report_envelope_ready=1
report_envelope_state=sealed-report-only
report_envelope_signature_performed=0
report_envelope_handoff_performed=0
report_envelope_effect_performed=0
report_envelope_runtime_authority_granted=0
report_envelope_host_read_performed=0
report_envelope_host_write_performed=0
report_envelope_network_performed=0
seal_signature_request_contract_present=1
seal_signature_request_metadata_present=1
seal_signature_request_status_present=1
signature_request_predecessor_report_envelope_status_present=1
seal_signing_authorization_contract_present=1
seal_signing_authorization_metadata_present=1
```

```
seal_signing_authorization_status_present=1
signing_authorization_predecessor_signature_request_status_present=1
seal_signer_handoff_contract_present=1
seal_signer_handoff_metadata_present=1
seal_signer_handoff_status_present=1
signer_handoff_predecessor_signing_authorization_status_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
signer_invocation_predecessor_signer_handoff_status_present=1
seal_signing_operation_contract_present=1
seal_signing_operation_metadata_present=1
seal_signing_operation_status_present=1
signing_operation_predecessor_signer_invocation_status_present=1
seal_key_handling_contract_present=1
seal_key_handling_metadata_present=1
seal_key_handling_status_present=1
key_handling_predecessor_signing_operation_status_present=1
seal_key_material_contract_present=1
seal_key_material_metadata_present=1
seal_key_material_status_present=1
key_material_predecessor_key_handling_status_present=1
seal_public_key_parsing_contract_present=1
seal_public_key_parsing_metadata_present=1
seal_public_key_parsing_status_present=1
public_key_parsing_predecessor_key_material_status_present=1
seal_future_key_parsing_implementation_contract_present=1
seal_future_key_parsing_implementation_plan_present=1
seal_key_parsing_metadata_present=1
seal_key_parsing_status_present=1
key_parsing_predecessor_public_key_parsing_status_present=1
seal_verification_policy_metadata_present=1
seal_verification_policy_status_present=1
verification_policy_predecessor_key_parsing_status_present=1
seal_verification_receipt_metadata_present=1
seal_verification_receipt_status_present=1
verification_receipt_predecessor_verification_policy_status_present=1
seal_capability_gate_metadata_present=1
seal_capability_gate_status_present=1
capability_gate_predecessor_verification_receipt_status_present=1
seal_effect_decision_metadata_present=1
seal_effect_decision_status_present=1
effect_decision_predecessor_capability_gate_status_present=1
seal_runtime_handoff_metadata_present=1
seal_runtime_handoff_status_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
seal_runtime_handoff_report_status_present=1
seal_status_rollup_metadata_present=1
seal_status_rollup_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
seal_agentic_automation_security_metadata_present=1
seal_agentic_automation_security_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
seal_agentic_automation_security_index_alignment_present=1
seal_agentic_automation_security_report_surface_present=1
seal_agentic_automation_security_public_entrypoint_alignment_present=1
seal_parameter_schema_contract_present=1
seal_parameter_schema_metadata_present=1
seal_parameter_schema_report_surface_present=1
seal_parameter_schema_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
seal_request_freshness_contract_present=1
seal_request_freshness_metadata_present=1
seal_request_freshness_report_surface_present=1
seal_request_freshness_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
seal_signed_request_contract_present=1
seal_signed_request_metadata_present=1
seal_signed_request_status_present=1
signed_request_predecessor_request_freshness_status_present=1
seal_policy_decision_contract_present=1
seal_policy_decision_metadata_present=1
seal_policy_decision_report_surface_present=1
seal_policy_decision_report_surface_status_present=1
seal_policy_decision_status_present=1
seal_policy_decision_public_entrypoint_alignment_present=1
policy_decision_predecessor_signed_request_status_present=1
seal_operator_receipt_report_status_present=1
operator_receipt_report_predecessor_policy_decision_status_present=1
defensive_threat_model_validation_refinement_present=1
```

```
high_assurance_security_baseline_present=1
source_refresh_date=2026-05-26
memory_safety_roadmap_required=1
memory_safety_roadmap_present=1
supply_chain_security_baseline_present=1
cyber_incident_reporting_response_baseline_present=1
cyber_incident_reporting_response_guard_present=1
cisa_reporting_channel_documented=1
fbi_cyber_reporting_channel_documented=1
vulnerability_management_release_gate_baseline_present=1
vulnerability_management_release_gate_guard_present=1
cisa_kev_catalog_tracked=1
nvd_cve_review_required=1
cryptographic_assurance_key_management_baseline_present=1
cryptographic_assurance_key_management_guard_present=1
fips_140_3_boundary_required_before_production_crypto=1
production_crypto_claim_allowed=0
identity_credential_access_management_baseline_present=1
identity_credential_access_management_guard_present=1
phishing_resistant_mfa_required_for_privileged_access=1
privileged_access_granted=0
security_logging_monitoring_baseline_present=1
security_logging_monitoring_guard_present=1
security_event_logging_required_before_hosted_service=1
production_monitoring_claim_allowed=0
backup_recovery_resilience_baseline_present=1
backup_recovery_resilience_guard_present=1
backup_restore_recovery_evidence_required_before_hosted_service=1
production_recovery_claim_allowed=0
secure_configuration_change_management_baseline_present=1
secure_configuration_change_management_guard_present=1
secure_configuration_change_control_required_before_hosted_service=1
production_configuration_claim_allowed=0
zero_trust_runtime_authority_baseline_present=1
zero_trust_runtime_authority_guard_present=1
per_request_authorization_required=1
zero_trust_runtime_boundary_required=1
runtime_boundary_policy_expansion_next=1
runtime_boundary_policy_expansion_after_threat_model_present=1
runtime_boundary_abuse_case_fixture_expansion_present=1
operator_visible_status_surface=1
deterministic_local_report_path=1
core_blocked_case_set_complete=1
mcp_alignment_context=ai-era-tool-boundary-planning
runtime_gate_report_only=1
signature_performed=0
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
tool_execution_implemented=0
ai_agent_security_claimed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current runtime boundary policy expansion checkpoint

The runtime boundary policy expansion after threat-model validation is documentation/status/guard work only.

It records:

```
runtime_boundary_policy_expansion_after_threat_model_present=1
request_family_policy_map_present=1
effect_policy_map_present=1
authority_prerequisite_map_present=1
future_gate_policy_map_present=1
abuse_case_runtime_policy_map_present=1
evidence_gap_map_present=1
runtime_execution_added=0
effect_execution_added=0
runtime_authority_granted=0
completion_estimate_review_readme_status_alignment_present=1
completion_estimate_after_runtime_boundary_abuse_case_fixtures_present=1
current_estimate_table_source_alignment_present=1
current_estimate_mathematical_rebase_present=1
```

```
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
estimate_adjustment_required=0
completion_estimate_review_required=0
```

Current Seal core case evidence:

```
unknown_tool_case_validated=1
unsigned_request_case_validated=1
stale_request_case_validated=1
replayed_request_case_validated=1
core_blocked_case_set_complete=1
seal_crypto_verify_backend_status_present=1
crypto_verify_state=unsupported
cryptographic_verification_performed=0
seal_ed25519_verify_status_present=1
ed25519_crypto_verify_state=verified
ed25519_cryptographic_verification_performed=1
ed25519_authority_usable=0
seal_verified_receipt_promotion_status_present=1
verified_receipt_promotion_state=verified
verified_receipt_promotion_cryptographic_verification_performed=1
verified_receipt_promotion_authority_usable=0
seal_verified_capability_gate_status_present=1
verified_capability_gate_crypto_graduation_gate_passed=1
verified_capability_gate_standard_expectations_met=1
verified_capability_gate_authority_promotion_allowed=0
verified_capability_gate_allowed=1
verified_capability_gate_state=allowed-metadata-only
verified_capability_gate_runtime_authority_granted=0
verified_capability_gate_effect_performed=0
seal_verified_effect_decision_status_present=1
verified_effect_decision_crypto_graduation_gate_present=1
verified_effect_decision_crypto_graduation_gate_passed=1
verified_effect_decision_standard_expectations_met=1
verified_effect_decision_authority_promotion_allowed=0
verified_effect_decision_allowed=1
verified_effect_decision_state=allowed-report-only
verified_effect_decision_effect_performed=0
verified_effect_decision_runtime_authority_granted=0
seal_runtime_handoff_evaluation_status_present=1
runtime_handoff_evaluation_crypto_graduation_gate_present=1
runtime_handoff_evaluation_crypto_graduation_gate_passed=1
runtime_handoff_evaluation_standard_expectations_met=1
runtime_handoff_evaluation_authority_promotion_allowed=0
runtime_handoff_evaluation_eligible=1
runtime_handoff_evaluation_state=eligible-report-only
runtime_handoff_evaluation_handoff_performed=0
runtime_handoff_evaluation_runtime_authority_granted=0
seal_runtime_handoff_report_status_present=1
runtime_handoff_report_ready=1
runtime_handoff_report_state=ready-report-only
runtime_handoff_report_handoff_performed=0
runtime_handoff_report_runtime_authority_granted=0
```

The careful current public claim is:

```
Latticra Seal now has a report-only runtime gate path, sealed report-envelope metadata/status,
signature-request metadata/status, signing authorization metadata/status, signer
handoff/invocation/operation metadata/status, key-handling and key-material metadata/status,
public-key parsing metadata/status, a future key parsing implementation contract/plan, bounded
key parsing metadata/status for caller-provided public-key bytes, metadata-only verification
policy/status, metadata-only crypto verify backend/status, local Ed25519 verify-only
implementation/status, verified receipt promotion metadata/status, authority-neutral crypto
graduation gate/status, verified capability gate metadata/status, verified effect decision
metadata/status, runtime handoff evaluation metadata/status, runtime handoff report
metadata/status, metadata-only unverified receipt/status, metadata-only denied capability
gate/status, metadata-only denied effect decision/status, inactive metadata-only runtime
handoff/status tied to its guarded effect decision predecessor, metadata-only status
rollup/status, report-only agentic automation security metadata/status tied to its guarded
status rollup predecessor, report surface/public-entrypoint alignment, report-only parameter
schema metadata/report surface/status-public-entry alignment, report-only request freshness
metadata/report surface/status-public-entry alignment, report-only signed request
metadata/status-public-entry alignment, report-only policy decision
metadata/status/report-surface public-entry alignment, and core negative-test evidence for
AI-era tool-boundary planning.
```

The current next recommended Seal lane is:

```
Continue small guarded report/status alignment only when drift appears
```

Current Fedora VM CLI payload README alignment checkpoint

The latest Fedora VM CLI payload README alignment records:

```
fedora_vm_cli_payload_validation_status=evidence-recorded
disposable_vm_cli_validation_transcript_present=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The README wording is limited to the no-effect CLI payload validated in a disposable Fedora VM and does not claim production, Fedora distribution, immutable Fedora, daily-driver, security, recovery, update-safety, sandboxing, or OS-replacement readiness.

The current next recommended Fedora CLI payload lane is:

```
Plan the next Fedora CLI payload validation lane without widening README or announcement claims
beyond disposable Fedora VM evidence
```

Current Fedora VM CLI payload announcement checkpoint

The latest Fedora VM CLI payload announcement alignment records:

```
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
validated_payload=/usr/bin/latticra
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

The announcement wording is evidence-bound and does not claim production, Fedora distribution, immutable Fedora, daily-driver, security, recovery, update-safety, sandboxing, or OS-replacement readiness.

The current next recommended Fedora CLI payload lane remains:

```
Plan the next Fedora CLI payload validation lane without widening README or announcement claims
beyond disposable Fedora VM evidence
```

Current Fedora VM CLI payload next-validation plan checkpoint

The latest Fedora VM CLI payload next-validation plan records:

```
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The plan is no-effect and defines repeatability evidence for a future second disposable Fedora VM CLI payload validation run. It does not add a runner, run RPM tooling, mutate a host, or widen readiness claims.

The current next recommended Fedora CLI payload lane is:

```
Add Fedora VM CLI payload repeatability transcript contract
```

Current Fedora VM CLI payload repeatability runner plan checkpoint

The latest Fedora VM CLI payload repeatability runner plan records:

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_plan_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
repeatability_runner_manual_only=1
ci_auto_repeatability_validation_allowed=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
```

```
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The plan records the manual-only repeatability runner shape for a future second disposable Fedora VM CLI payload validation run. The plan checkpoint does not run RPM tooling, mutate a host, or widen readiness claims.

The current next recommended Fedora CLI payload lane is:

```
Capture reviewed Fedora VM CLI payload repeatability transcript evidence
```

Current Fedora VM CLI payload repeatability runner checkpoint

The latest Fedora VM CLI payload repeatability runner status records:

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_plan_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
repeatability_runner_manual_only=1
ci_auto_repeatability_validation_allowed=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The runner is present but manually gated. Normal CI only runs the static repeatability runner guard and does not execute RPM build, install, or removal commands.

The current next recommended Fedora CLI payload lane is:

```
Capture reviewed Fedora VM CLI payload repeatability transcript evidence
```

Current Fedora VM CLI payload repeatability transcript capture template checkpoint

The latest Fedora VM CLI payload repeatability transcript capture template status records:

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_complete=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The template prints the future transcript shape to stdout. It does not run the manual repeatability runner, build RPMs, install packages, remove packages, mutate a host, attach a transcript, or accept evidence.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Current Fedora VM CLI payload repeatability transcript review validator checkpoint

The latest Fedora VM CLI payload repeatability transcript review validator status records:

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_review_mode=no-effect-validation
repeatability_transcript_candidate_valid=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
evidence_status_written=0
promotion_allowed_by_validator_alone=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The validator checks a supplied transcript candidate and rejects placeholders. It does not run the repeatability runner, write evidence status, or promote repeatability evidence by itself.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Current Fedora VM CLI payload repeatability evidence review gate checkpoint

The latest Fedora VM CLI payload repeatability evidence review gate status records:

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_complete=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_review_required=1
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The gate is present but does not attach or accept evidence. It blocks repeatability promotion until a complete operator-reviewed transcript from a real disposable Fedora VM run is attached.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Current Fedora VM CLI payload repeatability evidence acceptance contract checkpoint

The latest Fedora VM CLI payload repeatability evidence acceptance contract status records:

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
```

```
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The acceptance contract is present but does not attach, accept, or write repeatability evidence. It only defines the future accepted status record after a real disposable Fedora VM repeatability transcript is validated and reviewed.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
and write accepted repeatability evidence status
```

Current Fedora VM CLI payload repeatability evidence status template checkpoint

The latest Fedora VM CLI payload repeatability evidence status template records:

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
repeatability_evidence_status_template_mode=no-effect-template
repeatability_evidence_status_template_complete=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The evidence status template is present but does not attach, accept, or write repeatability evidence. It only prints the future accepted status shape after a real disposable Fedora VM repeatability transcript is validated and reviewed.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
and use the evidence status template to write accepted repeatability evidence status
```

Current Fedora VM CLI payload repeatability evidence status review validator checkpoint

The latest Fedora VM CLI payload repeatability evidence status review validator status records:

```
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_review_mode=no-effect-validation
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
```

```
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The evidence status review validator is present but does not write or accept repeatability evidence. It only checks a supplied future status candidate for required markers and non-placeholder values.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
fill evidence status template, and review the evidence status candidate
```

Current Fedora VM CLI payload repeatability evidence publication gate checkpoint

The latest Fedora VM CLI payload repeatability evidence publication gate status records:

```
fedora_vm_cli_payload_repeatability_evidence_publication_gate_present=1
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
repeatability_evidence_publication_requested=0
operator_publication_review_completed=0
repeatability_evidence_publication_approved=0
repeatability_evidence_status_published=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The publication gate is present but does not publish repeatability evidence. It keeps a valid evidence status candidate review separate from a future operator-approved evidence status write.

The current next recommended Fedora CLI payload lane is:

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript and evidence
 status candidates, then complete operator publication review
```

Current Fedora VM CLI payload validation lane checkpoint

The latest Fedora VM CLI payload validation status alignment records:

```
Fedora VM CLI payload validation lane runner
source=PR #236
validation_lane_documented=1
validation_runner_present=1
validation_lane_docs_guard_present=1
validation_lane_docs_workflow_present=1
runner_manual_only=1
ci_auto_vm_cli_validation_allowed=0
disposable_vm_target_required=1
daily_driver_block_required=1
production_host_block_required=1
immutable_fedora_block_required=1
clean_snapshot_required=1
recovery_path_required=1
operator_consent_required=1
non_root_operator_required=1
sudo_limited_to_rpm_install_removal=1
fedora_target_required=1
rpm_tooling_required=1
rpmbuild_tooling_required=1
cc_tooling_required=1
local_cli_guard_required=1
local_rpm_build_required=1
rpm_payload_listing_required=1
```

```
rpm_payload_cli_binary_required=1
rpm_payload_readme_required=1
rpm_payload_only_expected_surfaces_required=1
unexpected_runtime_surface_absent_required=1
installed_cli_binary_required=1
installed_readme_required=1
rpm_verify_required=1
cli_status_validation_required=1
cli_version_validation_required=1
cli_report_validation_required=1
cli_invalid_command_validation_required=1
post_removal_cli_absence_required=1
post_removal_readme_absence_required=1
validation_report_schema_present=1
target_evidence_level=9
current_evidence_level=8
evidence_level_9_achieved=0
```

Current readiness classification:

```
fedora_vm_cli_transcript_contract_present=1
fedora_vm_cli_payload_validation_lane_present=1
fedora_vm_cli_payload_validation_runner_present=1
fedora_vm_cli_payload_validation_status=blocked-pending-real-vm-run
disposable_vm_cli_validation_transcript_present=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The current next recommended Fedora lane is:

```
Capture real disposable Fedora VM CLI payload validation transcript evidence
```

Current Fedora disposable VM RPM README alignment checkpoint

The latest Fedora disposable VM RPM README alignment records:

```
disposable_vm_local_rpm_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The README wording is now limited to disposable Fedora VM local RPM validation and does not claim production, Fedora distribution, immutable Fedora, daily-driver, security, recovery, or OS-replacement readiness.

The current next recommended Fedora lane is:

```
Add public announcement wording for disposable Fedora VM RPM validation milestone
```

Current Fedora disposable VM local RPM validation evidence checkpoint

The latest Fedora disposable VM local RPM validation evidence status records:

```
source=operator disposable Fedora VM transcript
repo_branch=pr-226
followup_fix_pr=PR #226
validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
validation_status=ok
package_name=latticra
package_version=0.0.0
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
vm_rpmdb_mutated=1
vm_filesystem_mutated=1
install_validation_performed=1
removal_validation_performed=1
```

```
post_removal_absence_verified=1
live_host_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
evidence_level=9
```

Current readiness classification:

```
disposable_vm_local_rpm_validation_lane_present=1
disposable_vm_validation_transcript_present=1
disposable_vm_validation_completed=1
live_host_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The previous recommended Fedora lane was:

```
Add README install-readiness wording limited to disposable Fedora VM local RPM validation
```

Current Fedora disposable VM local RPM validation lane checkpoint

The previous Fedora disposable VM local RPM validation status alignment records:

```
Fedora disposable VM local RPM validation lane
source=PR #223
validation_lane_documented=1
validation_runner_present=1
validation_lane_docs_guard_present=1
validation_lane_docs_workflow_present=1
runner_manual_only=1
ci_auto_vm_rpm_validation_allowed=0
disposable_vm_target_required=1
daily_driver_block_required=1
production_host_block_required=1
immutable_fedora_block_required=1
clean_snapshot_required=1
recovery_path_required=1
operator_consent_required=1
rpm_payload_listing_required=1
rpm_payload_documentation_only_required=1
unexpected_runtime_surface_absent_required=1
validation_report_schema_present=1
target_evidence_level=9
current_evidence_level=8
evidence_level_9_achieved=0
```

Previous readiness classification:

```
manual_host_dry_run_transcript_contract_present=1
disposable_vm_effect_gate_present=1
disposable_vm_effect_gate_classifier_present=1
disposable_vm_local_rpm_validation_lane_present=1
disposable_vm_local_rpm_validation_status=blocked-pending-real-vm-run
disposable_vm_validation_transcript_present=0
disposable_vm_validation_completed=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

The previous recommended Fedora lane was:

```
Capture disposable Fedora VM local RPM validation transcript evidence
```

Current Fedora disposable VM effect gate classifier checkpoint

The latest Fedora disposable VM effect gate classifier status alignment records:

```
Fedora disposable VM effect gate classifier
source=PR #221
evidence_contract_present=1
vm_effect_gate_present=1
vm_gate_classifier_present=1
```

```
vm_gate_classifier_guard_present=1
vm_gate_classifier_docs_guard_present=1
eligible_state_defined=1
blocked_state_defined=1
invalid_state_defined=1
eligible_test_present=1
blocked_target_tests_present=1
blocked_package_tests_present=1
blocked_prior_evidence_tests_present=1
invalid_input_test_present=1
classifier_evidence_level=8
```

Current readiness classification:

```
manual_host_dry_run_transcript_contract_present=1
disposable_vm_effect_gate_present=1
disposable_vm_effect_gate_classifier_present=1
disposable_vm_effect_gate_status=blocked-pending-vm-validation-lane
disposable_vm_effect_eligible=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

The current next recommended Fedora lane is:

```
Add disposable Fedora VM local RPM validation lane
```

Current Fedora manual host RC decision classifier checkpoint

The earlier Fedora manual host RC decision classifier status alignment records:

```
Fedora manual host RC decision classifier
source=PR #217
decision_classifier_present=1
decision_classifier_guard_present=1
decision_classifier_docs_guard_present=1
candidate_state_defined=1
blocked_state_defined=1
invalid_state_defined=1
synthetic_candidate_test_present=1
blocked_target_tests_present=1
blocked_evidence_tests_present=1
blocked_boundary_tests_present=1
invalid_input_test_present=1
classifier_evidence_level=7
live_host_validation_completed=0
host_change_performed=0
sudo_invoked=0
rpm_invoked=0
dnf_invoked=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Earlier readiness classification:

```
manual_host_rc_checklist_present=1
manual_host_rc_decision_classifier_present=1
manual_host_rc_status=blocked-pending-real-transcript
manual_host_release_candidate_ready=0
manual_host_dry_run_transcript_present=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

The earlier recommended Fedora lane was:

```
Add manual Fedora host dry-run transcript contract
```

Previous Fedora manual host RC checklist checkpoint

The previous Fedora manual host RC checklist status alignment records the Current Fedora manual host RC checklist checkpoint:

```
Fedora manual host RC checklist
source=PR #215
checklist_documented=1
checklist_guard_present=1
controlled_installroot_lifecycle_ready=1
manual_host_rc_status=blocked
manual_host_release_candidate_ready=0
live_host_validation_completed=0
host_change_performed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
fedora_candidate_compose_claimed=0
fedora_go_no_go_claimed=0
checklist_target_evidence_level=7
current_evidence_level=6
evidence_level_7_achieved=0
```

Previous readiness classification:

```
manual_host_rc_checklist_present=1
manual_host_rc_guard_present=1
manual_host_rc_status=blocked
manual_host_release_candidate_ready=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

The previous recommended Fedora lane was:

```
Add no-effect Fedora manual host RC decision classifier
```

Previous Fedora installroot RPM lifecycle checkpoint

The previous Fedora installroot status alignment records:

```
Fedora installroot RPM lifecycle lane
source=PR #213
execution_lane_documented=1
execution_guard_present=1
local_rpm_build_path=present
installroot_rpm_database_path=temporary
installroot_filesystem_mutated=1
installroot_rpmdb_mutated=1
live_container_rpmdb_mutated=0
developer_host_mutation_performed=0
post_removal_absence_verified=1
evidence_level=6
```

Previous readiness classification:

```
controlled_installroot_lifecycle_ready=1
manual_host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

The previous recommended Fedora lane was:

```
Add manual host-install release-candidate checklist
```

Current Fedora RPM gate classifier checkpoint

The previous Fedora RPM gate status alignment records:

```
Fedora RPM gate classifier implementation
install_gate_status=allowed
install_gate_denial=none
install_mutation_allowed=1
install_mutation_performed=0
host_mutation_performed=0
network_allowed=0
```

```
evidence_level=4
```

The previous next recommended Fedora lane was:

```
Add Fedora local RPM removal and rollback plan
```

Current kernel lifecycle checkpoint

The latest kernel status alignment records:

```
kernel lifecycle report runner
kernel IPC table guard
kernel IPC table report runner
kernel VFS namespace guard
kernel VFS namespace report runner
kernel device registry guard
kernel device registry report runner
kernel driver catalog guard
kernel driver catalog report runner
kernel interrupt table guard
kernel interrupt table report runner
kernel timer source guard
kernel timer source report runner
kernel scheduler tick guard
kernel scheduler tick report runner
kernel run queue guard
kernel run queue report runner
kernel context switch guard
kernel context switch report runner
kernel time accounting guard
kernel time accounting report runner
kernel preemption guard
kernel preemption report runner
kernel scheduler credit guard
kernel scheduler credit report runner
kernel scheduler selection guard
kernel scheduler selection report runner
kernel scheduler dispatch guard
kernel scheduler dispatch report runner
kernel scheduler handoff guard
kernel scheduler handoff report runner
kernel scheduler activation guard
kernel scheduler activation report runner
kernel scheduler run-entry guard
kernel scheduler run-entry report runner
kernel process table guard
kernel process table report runner
kernel syscall table guard
kernel syscall table report runner
kernel lifecycle subsystem summary
kernel lifecycle rollback plan
final_state=scheduler-run-entry-ready
external_effect_performed=0
runtime_entry_allowed=0
scheduler_execution_allowed=0
scheduler_selection_allowed=0
scheduler_dispatch_allowed=0
scheduler_handoff_allowed=0
scheduler_activation_allowed=0
scheduler_run_entry_allowed=0
memory_allocation_allowed=0
process_spawn_allowed=0
syscall_dispatch_allowed=0
ipc_send_allowed=0
ipc_receive_allowed=0
ipc_queue_mutation_allowed=0
filesystem_lookup_allowed=0
filesystem_read_allowed=0
filesystem_write_allowed=0
namespace_mutation_allowed=0
device_open_allowed=0
device_read_allowed=0
device_write_allowed=0
driver_probe_allowed=0
driver_load_allowed=0
driver_bind_allowed=0
interrupt_allowed=0
interrupt_mask_allowed=0
interrupt_unmask_allowed=0
interrupt_dispatch_allowed=0
```

```
interrupt_ack_allowed=0
timer_tick_allowed=0
timer_arm_allowed=0
timer_disarm_allowed=0
scheduler_tick_allowed=0
run_queue_mutation_allowed=0
enqueue_allowed=0
dequeue_allowed=0
dispatch_allowed=0
context_switch_allowed=0
register_save_allowed=0
register_restore_allowed=0
stack_switch_allowed=0
address_space_switch_allowed=0
preemption_allowed=0
time_accounting_allowed=0
time_read_allowed=0
cpu_usage_write_allowed=0
quota_update_allowed=0
scheduler_credit_update_allowed=0
process_wake_allowed=0
dma_allowed=0
hardware_effect_allowed=0
```

The current next recommended kernel lane is:

```
Add no-effect runtime entry admission classifier
```

Status update rules

Status updates should be:

- dated;
- evidence-bound;
- concise enough to read quickly;
- honest about non-claims;
- updated when major merged PRs change the current checkpoint;
- preserve older checkpoint headings while older guard workflows still validate them;
- careful not to claim security capability before implementation and testing prove it.

Completion percentage rule

Completion percentages are planning estimates only.

They are not release promises, security certifications, or production-readiness metrics.

Review cadence

Update status after:

- major merged PRs;
- new foundation sections;
- parser/AST milestones;
- security-policy changes;
- strategy changes;
- public-positioning changes;
- release or milestone planning.

Non-claims

This folder does not implement features, security controls, operating-system behavior, malware prevention, ransomware prevention, sandboxing, update safety, recovery safety, or production readiness.

Part III - Architecture and Foundation Concepts

- docs/ARCHITECTURE_SEED.md - Latticra Architecture Seed
- docs/C_CPP_FOUNDATION_DIRECTION.md - Latticra C/C++ Foundation Direction
- docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md - Latticra Constrained C++ Authority Layer Contract
- docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md - Latticra Constrained C++ Authority Layer Implementation
- docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION_PLAN.md - Latticra Constrained C++ Authority Layer Implementation Plan
- docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md - C++ Authority Implementation Review
- docs/EFFECT_GATES.md - Latticra Effect Gates
- docs/FEATURE_TRANSLATION_LEDGER.md - Latticra Feature Vocabulary Ledger
- docs/HOST_ARCHITECTURE_TARGETS.md - Latticra Host Architecture Targets
- docs/LATTICRA_ARCHITECTURE_TARGETS.md - Latticra Architecture Targets
- docs/LATTICRA_NO_EFFECT_CLI_PACKAGING_CONTRACT_ALIGNMENT.md - Latticra No-Effect CLI Packaging Contract Alignment
- docs/LATTICRA_NO_EFFECT_CLI_PAYLOAD_CONTRACT.md - Latticra No-Effect CLI Payload Contract
- docs/LATTICRA_NO_EFFECT_CLI_STATUS_SURFACE_IMPLEMENTATION.md - Latticra No-Effect CLI Status Surface Implementation
- docs/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT.md - macOS Reset/Uninstall Effect-Authorization Contract
- docs/NAMING_SYSTEM.md - Latticra Naming System
- docs/OPEN_ECOSYSTEM_POLICY.md - Latticra Open Ecosystem Policy
- docs/roadmap/LATTICRA_C_CPP_FOUNDATION_ROADMAP.md - Latticra C/C++ Foundation Roadmap
- docs/SERVER_INTERACTION_MODEL.md - Latticra Server Interaction Model
- docs/SUPERVISOR_ARCHITECTURE.md - Nucleus Supervisor Architecture

docs/ARCHITECTURE_SEED.md

Source title: Latticra Architecture Seed

Lines: 182 | Words: 435 | SHA-256: 2cd71e8f62c8c70003f11514fe5adcdbd6e46fc67b128b2282b9799103ed9c41

Latticra Architecture Seed

Status: initial architecture seed Scope: vocabulary, first system layers, and implementation direction.

Purpose

This document defines the initial architecture vocabulary for Latticra.

It is not a complete architecture specification. It is the seed from which implementation contracts, tests, models, and components can grow.

Core metaphor

Latticra models real software systems as structured computational universes.

Those universes are composed of:

- lattices;
- grids;
- matrices;
- state planes;
- transition rules;
- execution domains;
- visibility rails;
- evidence gates;
- rollback surfaces;
- embedded system boundaries.

Initial layers

```
L0 Contract Layer
  Real-system contract, non-claims, evidence ladder, promotion rules.

L1 State Lattice Layer
  Explicit state cells, paths, axes, routes, origins, traces, breadcrumbs.

L2 Transition Layer
  Pure transition models and denial rules before live movement.

L3 Visibility Layer
  Operator-readable state reports, visibility rails, health/risk/lock surfaces.

L4 Language Layer
  Lat declarations, read-only source forms, and assertion candidates.

L5 Evidence Layer
  Validation reports, virtual-target evidence, read-only hardware evidence, promotion records.

L6 Embedded Implementation Layer
  Narrow real-system implementation candidates after evidence gates.
```

Initial state vocabulary

Latticra begins with a compact state vocabulary but does not claim live movement yet.

```
origin
route
```

```
axis
path
breadcrumb
trace
safe_portal
rollback
health
risk
lock
dark_phase
host_effect
external_effect
```

First implementation candidate

The first implementation candidate should be a state lattice fixture.

Required properties:

```
deterministic
serializable
read-only by default
effect-free
operator-visible
traceable
testable
```

Forbidden properties in the first implementation:

```
live movement
origin mutation
recovery execution
host filesystem mutation
network mutation
hardware mutation
hidden state mutation
security claims
boot claims
```

Lattice transition direction

A future transition engine should accept:

```
current_state
transition_request
operator_intent
policy_context
```

and return:

```
next_state
result
reason
effects
rollback_visibility
trace
```

The first transition engine must be pure and preview-only.

Software universe direction

A Latticra software universe is not a fantasy world or game map.

It is a structured execution and state environment where every coordinate, route, transition, effect, and recovery surface has an inspectable contract.

The long-term goal is to make complex system behavior visible, navigable, testable, and eventually implementable on real machines.

Early component candidates

```
lattice-state
lattice-transition
lattice-report
promotion-packet
lat-assertion-candidate
hardware-profile-contract
evidence-record
promotion-record
```

Near-term repository structure

Recommended future structure:

```
docs/
  contracts/
  evidence/
  architecture/
  promotion/
  hardware/

src/
  lattice/
  transition/
  reports/

tests/
  docs/
  lattice/
  transition/

examples/
  fixtures/
```

Do not add this full structure until the first implementation slice requires it.

North star

Latticra should become a formal, evidence-bound implementation of a lattice-oriented computing architecture.

It should remain academically serious, implementation-focused, evidence-bound, and safe to reason about.

docs/C_CPP_FOUNDATION_DIRECTION.md

Source title: Latticra C/C++ Foundation Direction

Lines: 196 | Words: 645 | SHA-256: 3318060220f144e614c2dbd80a8672d33f03a13661c546c8af2f37acac9b9d1d

Latticra C/C++ Foundation Direction

Status: active language direction Scope: C/C++ foundation policy for Latticra systems implementation and public direction.

Purpose

This document records the current implementation direction for Latticra:

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Latticra should use a constrained C/C++ foundation for security-conscious system work.

This direction does not claim unrestricted C++, broad runtime behavior, operating-system completeness, sandboxing, malware prevention, ransomware prevention, or production security guarantees.

Layer model

The intended public model is:

```
Lat / Latticra Language: contract layer
C++: governed authority layer
C: secure substrate
```

C secure substrate

C owns the lowest software boundary.

Primary C responsibilities:

```
boot paths
ABI boundaries
platform shims
fixed-size data structures
freestanding-adjacent boundaries
minimal substrate behavior
no-hidden-allocation early core
portable test fixtures
```

C must remain disciplined:

- no unsafe string APIs;
- no unchecked pointer mutation;
- no undefined behavior tolerated;
- no implicit effects without explicit gates;
- no global mutable state unless named in a contract;
- all transitions return status codes;
- all source/input behavior remains bounded and testable.

C++ governed authority layer

C++ is allowed above the C substrate as a governed authority layer.

Primary C++ responsibilities:

```
policy logic
validators
effect gates
audit logic
bounded orchestration structures
operator-visible reports
higher-level state coordination
safe wrappers around C substrate APIs
```

C++ must be constrained.

It is not:

```
unrestricted C++
exception-heavy C++
reflection-heavy C++
template metaprogramming as architecture
hidden allocation by default
implicit authority
unchecked host execution
```

Latticra contract language layer

Lat / Latticra Language is the contract and declaration layer.

Primary Lat responsibilities:

```
state declarations
policy declarations
transition declarations
assertions
effect requirements
```

```
operator-visible contract surfaces
future validated input to LIR
```

Lat is not currently a parser, compiler, interpreter, runtime, package manager, or execution surface.

Not unrestricted C++

The direction explicitly rejects unrestricted C++.

C++ may be used only when it is:

```
bounded
reviewable
effect-aware
policy-oriented
audit-friendly
compatible with C substrate boundaries
covered by tests or guards
clear about ownership and lifetime
```

Trust and evidence rules

All C/C++ work should preserve:

```
explicit trust boundaries
evidence-bound validation
source-aware diagnostics
bounded reports
no-effect defaults
operator-visible state
contract before capability
```

Relationship to LIR

The current LIR implementation is metadata-only.

LIR remains a representation target and must not become an execution surface without a separate contract.

C/C++ implementation work must not use LIR to bypass semantic validation, source-span preservation, no-effect flags, or effect gates.

Relationship to Rust

Rust is not the current public foundation direction for Latticra.

Rust may appear only as optional external tooling or historical comparison if separately justified, but it is not the default public implementation lane and should not be presented as a primary foundation layer.

Public wording

Approved public wording:

```
A constrained C/C++ foundation for a security-conscious system.
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Avoid wording that implies:

```
unrestricted C++
finished operating system
finished language runtime
production security boundary
malware prevention guarantee
ransomware prevention guarantee
```

Implementation ordering

Recommended ordering:

1. C substrate fixtures and ABI-safe data structures.
2. C parser, AST, semantic validation, and LIR metadata foundations.
3. C++ policy, validation, effect-gate, and audit layers after C substrate contracts are stable.
4. Lat grammar and parser contracts before any Lat parser implementation.
5. LIR/Lat lowering only after separate contracts.
6. Execution only after explicit runtime/effect/security contracts.

Current validation command

This direction is guarded by:

```
sh scripts/test-c-cpp-foundation-direction.sh
```

The guard is static. It does not implement C++ policy code.

Non-claims

This document does not implement C++ infrastructure, Lat parsing, LIR execution, L-UI rendering, Nucleus task execution, live movement, state mutation, server interaction, recovery behavior, hardware behavior, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md

Source title: Latticra Constrained C++ Authority Layer Contract

Lines: 488 | Words: 1423 | SHA-256: 197f00cd7bcc1360044a9a55096de992c60a0550d872604d6e8f591b0d4b5648

Latticra Constrained C++ Authority Layer Contract

Status: constrained C++ authority layer contract Scope: governed C++ policy, validator, effect-gate, audit, ownership, lifetime, allocation, exception, boundary, and non-claim rules before any C++ authority-layer implementation.

Purpose

This document defines the first contract for the constrained C++ authority layer in Latticra.

It preserves the active direction:

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

C++ is permitted only as a governed authority layer above the C substrate.

This document does not implement C++ infrastructure, policy code, validators, effect gates, audit logic, orchestration, LIR lowering, Lat execution, L-UI rendering, runtime behavior, operating-system behavior, sandboxing, malware prevention, ransomware prevention, or production security guarantees.

Relationship to previous work

This contract depends on:

```
docs/C_CPP_FOUNDATION_DIRECTION.md
docs/LANGUAGE_STRATEGY.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/EFFECT_GATES.md
docs/STATE_LATTICE.md
docs/TRI_PLANE_TRANSITION.md
```

```
docs/NUCLEUS_PREVIEW.md
include/latticra/lat_parser.h
include/latticra/lir.h
include/latticra/state_lattice.h
```

Those files remain the source of truth for the C substrate, Lat metadata parsing, LIR shape, effect vocabulary, state-lattice vocabulary, no-effect flags, and preview boundaries.

Layer placement

The C++ authority layer sits above the C substrate and below future operator/runtime orchestration.

Layer model:

```
Lat / Latticra Language: contract and declaration layer
C++: governed authority layer
C: secure substrate
```

C++ must not bypass the C substrate contracts.

Allowed C++ responsibilities

C++ may be used for:

```
policy logic
validators
effect gates
audit logic
bounded orchestration structures
operator-visible reports
higher-level state coordination
safe wrappers around C substrate APIs
```

These are authority responsibilities, not unchecked execution rights.

Forbidden C++ posture

The C++ authority layer must not become:

```
unrestricted C++
exception-heavy C++
reflection-heavy C++
template metaprogramming as architecture
hidden allocation by default
implicit authority
unchecked host execution
```

Initial C++ standard policy

The first C++ authority-layer implementation should target:

```
C++20 or later only if the implementation plan justifies it
```

Required subset posture:

```
explicit ownership
explicit lifetimes
no ambient authority
no hidden execution
no implicit network behavior
no implicit hardware behavior
no implicit file-system behavior
no global mutable authority
```

Ownership policy

Ownership must be explicit.

Allowed default patterns:

```
value types
references for non-owning views
span-like bounded views
unique ownership only where necessary
```

```
RAII wrappers for explicit resources
```

Restricted patterns:

```
shared ownership by default
raw owning pointers
unbounded containers in authority paths
implicit lifetime extension
long-lived global singletons
```

Any future use of std::shared_ptr, raw owning pointers, or global singletons must be justified in an implementation plan and guarded by tests or review rules.

Allocation policy

C++ authority-layer code should avoid hidden allocation in core paths.

Preferred patterns:

```
fixed-capacity containers
caller-provided buffers
bounded vectors with explicit capacity
arena-like allocation only if separately contracted
no allocation in no-effect report paths unless justified
```

Forbidden by default:

```
unbounded allocation in validators
unbounded allocation in effect gates
allocation hidden behind policy decisions
allocation hidden in report generation
allocation during no-effect validation without explicit contract
```

Exception policy

The first authority layer should not use exceptions for control flow.

Default policy:

```
no exceptions across C ABI boundaries
no exceptions through effect gates
no exceptions through report generation
no exceptions for expected validation failures
```

Expected failures should use explicit result/status types.

A future implementation plan must decide whether the C++ build uses:

```
-fno-exceptions
```

or a documented restricted exception boundary.

RTTI and reflection policy

RTTI and reflection-like behavior are forbidden by default in the authority layer.

Default policy:

```
no dynamic_cast in authority paths
no typeid-based authority decisions
no reflection-like dispatch as policy architecture
```

A future implementation plan must justify any exception.

Template policy

Templates may support small generic utilities but must not become architecture.

Allowed:

```
small fixed-capacity helpers
strong type wrappers
bounded view utilities
compile-time constants
```

Forbidden by default:

```
template metaprogramming as policy engine
type-level authority decisions without readable reports
compile-time tricks that hide behavior
large framework-style abstractions
```

C ABI boundary policy

C++ code must interact with the C substrate through explicit C-compatible boundaries.

Rules:

```
no C++ exceptions crossing C ABI
no C++ ownership crossing C ABI without wrappers
no raw C++ object lifetime exposed to C callers
no hidden allocation behind C ABI functions
status/result returns for boundary failures
```

Effect gate policy

C++ effect gates must start as no-effect validators.

They may classify or deny effects but must not perform effects.

Initial preserved flags:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
network_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Any transition from classification to allowed behavior requires a separate execution/effect contract.

Audit policy

Audit logic must be deterministic, bounded, and operator-visible.

Audit records should preserve:

```
input source identity if available
source spans if available
policy name
validator name
requested effect
allowed or denied result
denial reason
no-effect flags
```

Audit reports must not leak secrets, read files, open network connections, or mutate state.

Validator policy

Validators should be small and named.

Initial validator categories:

```
naming validators
source-span validators
no-effect validators
effect validators
boundary validators
state-shape validators
LIR-shape validators
Lat-parse-result validators
```

Validators must return explicit result values and stable labels.

Error/result policy

A future C++ implementation should define a small result type.

Recommended labels:

```
ok
null_argument
invalid_input
capacity_exceeded
policy_denied
unsupported_effect
unsupported_boundary
not_authorized
internal_error
```

The implementation plan must define exact names before code.

Namespace and file policy

Recommended namespace:

```
latticra
```

Recommended future file roots:

```
include/latticra/cpp/authority.hpp
src/cpp/authority.cpp
tests/cpp_authority_layer_invariants.cpp
scripts/test-cpp-authority-layer.sh
```

Exact paths must be named by a separate implementation plan.

Build policy

The future implementation plan must define:

```
compiler flags
C++ standard version
warnings-as-errors policy
exception policy
RTTI policy
allocation policy
sanitizer or static-analysis path if available
CI test runner
```

No C++ source files should be added before that implementation plan exists.

Relationship to Lat

The current Lat parser is metadata-only.

C++ authority-layer work may validate Lat parser results in the future but must not execute Lat, compile Lat, interpret Lat, or lower Lat to LIR without separate contracts.

Relationship to LIR

The current LIR shape implementation is metadata-only.

C++ authority-layer work may validate LIR shape metadata in the future but must not execute LIR or convert LIR into behavior without separate contracts.

Relationship to Nucleus

Nucleus preview work remains no-effect.

C++ authority-layer planning may define future Nucleus policy validation, but must not call task execution or alter runtime behavior without separate contracts.

No-effect rule

The constrained C++ authority layer starts as policy metadata and validation planning only.

It must preserve:

```
no_effect=1
execution_allowed=0
```

```
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility expectations

This contract must not change:

```
C substrate behavior
Lat parser behavior
LIR shape behavior
L-UI parser behavior
L-UI semantic validation behavior
state lattice behavior
Nucleus preview behavior
C/C++ foundation direction
language naming policy
no-effect flags
```

Future implementation gate

C++ authority-layer implementation must not begin until a separate implementation plan defines:

1. public API shape;
2. namespace and file paths;
3. C++ standard version;
4. compiler flags;
5. exception policy;
6. RTTI policy;
7. allocation policy;
8. ownership/lifetime rules;
9. result/error labels;
10. C ABI boundary rules;
11. validator categories;
12. audit report format;
13. exact tests;
14. compatibility expectations;
15. non-claims.

Future test list

A future implementation plan should include tests for:

```
cpp_authority_layer_preserves_no_effect_flags
cpp_authority_layer_rejects_unrestricted_cpp_claims
cpp_authority_layer_has_no_execution_path
cpp_authority_layer_has_no_network_path
cpp_authority_layer_has_no_hardware_path
cpp_authority_layer_uses_explicit_result_labels
cpp_authority_layer_does_not_throw_across_c_boundary
cpp_authority_layer_does_not_allocate_in_report_path
cpp_authority_layer_preserves_source_identity_in_audit
cpp_authority_layer_accepts_max_source_identity
cpp_authority_layer_rejects_oversized_source_identity
cpp_authority_layer_bounds_source_identity_before_audit_copy
cpp_authority_layer_rejects_nul_source_identity
cpp_authority_layer_rejects_line_break_source_identity
cpp_authority_layer_validates_lat_parse_result_metadata
cpp_authority_layer_validates_lir_shape_metadata
```

```
cpp_authority_layer_audit_report_is_deterministic
cpp_authority_layer_rejects_small_report_buffer
cpp_authority_layer_rejects_unterminated_audit_text
cpp_authority_layer_is_deterministic
cpp_authority_layer_rejects_mutation_flags
cpp_authority_layer_rejects_network_flags
cpp_authority_layer_rejects_lat_network_flags
cpp_authority_layer_rejects_lir_network_flags
cpp_authority_layer_rejects_request_lat_network_flags
cpp_authority_layer_rejects_request_lir_network_flags
cpp_authority_layer_classifies_effects_without_performing_them
cpp_authority_layer_builds_with_fno_exceptions_and_fno_rtti
```

Forbidden behavior

The C++ authority layer must not:

- become unrestricted C++;
- add C++ implementation before an implementation plan;
- execute Lat;
- execute LIR;
- render L-UI;
- call Nucleus task execution;
- mutate state;
- read files;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- hide allocation in authority paths;
- throw exceptions across C ABI boundaries;
- use RTTI as policy architecture;
- use template metaprogramming as policy architecture;
- weaken C substrate boundaries;
- weaken no-effect flags;
- imply a sandbox, runtime, malware prevention, ransomware prevention, or operating-system surface.

Current validation command

This contract is guarded by:

```
sh scripts/test-constrained-cpp-authority-layer-contract.sh
```

The guard is static. It does not implement C++ authority-layer code.

Non-claims

This document does not implement C++ infrastructure, policy code, validators, effect gates, audit logic, orchestration, Lat execution, Lat compilation, Lat interpretation, LIR execution, LIR lowering, L-UI rendering, command behavior, Nucleus task execution, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md

Source title: Latticra Constrained C++ Authority Layer Implementation

Lines: 361 | Words: 782 | SHA-256: 90acfe4b753e17eabc49d005661b73a4c0337fb59b36fb8b73abd77c0cb0d19c

Latticra Constrained C++ Authority Layer Implementation

Status: initial implementation contract Scope: first no-effect constrained C++ authority-layer API, fixed-capacity audit structures, Lat metadata validation, LIR metadata validation, effect classification, deterministic report rendering, build policy, invariant tests, compatibility expectations, and non-claims.

Purpose

This document records the first constrained C++ authority-layer implementation in Latticra.

It follows:

```
docs/C_CPP_FOUNDATION_DIRECTION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION_PLAN.md
```

The implementation keeps the active direction:

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Implementation files

This slice adds:

```
include/latticra/cpp/authority.hpp
src/cpp/authority.cpp
tests/cpp_authority_layer_invariants.cpp
scripts/test-cpp-authority-layer.sh
```

The implementation is intentionally small and no-effect.

Public API

The public API is:

```
latticra::authority_status
latticra::authority_effect
latticra::authority_validator
latticra::authority_flags
latticra::authority_source_span
latticra::authority_request
latticra::authority_audit_record
latticra::authority_audit_report
latticra::authority_status_label
latticra::authority_effect_label
latticra::authority_validator_label
latticra::validate_lat_parse_result
latticra::validate_lir_shape
latticra::classify_effect_request
latticra::render_authority_audit_report
```

All public functions return explicit status values and are declared noexcept.

Capacity constants

The implementation defines:

```
LATTICRA_AUTHORITY_POLICY_NAME_MAX 64u
LATTICRA_AUTHORITY_VALIDATOR_NAME_MAX 64u
LATTICRA_AUTHORITY_DENIAL_REASON_MAX 128u
LATTICRA_AUTHORITY_SOURCE_IDENTITY_MAX 128u
LATTICRA_AUTHORITY_AUDIT_RECORD_MAX 32u
LATTICRA_AUTHORITY_REPORT_MAX 4096u
```

The authority layer uses fixed-capacity structures and caller-provided output buffers.

Result labels

Stable authority status labels:

```
ok
null_argument
invalid_input
capacity_exceeded
policy_denied
unsupported_effect
unsupported_boundary
not_authorized
internal_error
```

Effect labels

Stable authority effect labels:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Effects are metadata only. Classification does not perform effects.

Validator labels

Stable authority validator labels:

```
naming
source_span
no_effect
effect
boundary
state_shape
lir_shape
lat_parse_result
```

No-effect flags

The authority layer preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
network_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Any mutation, execution, server, network, recovery, or hardware flag is denied.

Lat metadata validation

latticra::validate_lat_parse_result validates Lat parser metadata only.

It checks:

```
Lat parse result status
Lat parse error value
Lat no-effect flags
Lat network_allowed flag denial
Lat declaration capacity
Lat clause capacity
Lat module declaration capacity
Lat source-span ordering
Lat declaration source-span ordering
Lat clause source-span ordering
```

It does not execute, compile, interpret, lower, read files, write files, open network connections, or mutate state.

LIR metadata validation

latticra::validate_lir_shape validates LIR shape metadata only.

It checks:

```
LIR module status
LIR error value
LIR no-effect flags
LIR network_allowed flag denial
LIR node capacity
LIR edge capacity
LIR binding capacity
LIR text capacity
LIR source-span ordering
LIR text length bounds
```

It does not execute LIR, render L-UI, execute Lat, call Nucleus task execution, read files, write files, open network connections, or mutate state.

Effect classification

latticra::classify_effect_request classifies requested effects without performing them. If a request carries linked Lat or LIR metadata, classification also validates that linked metadata before allowing even a no-effect request. Bounded request source identity is copied into fixed audit storage and rendered in the report; oversized source identity is denied before any audit copy. Source identity containing an embedded NUL is rejected so report text cannot silently truncate the identity. Source identity containing line breaks is rejected so report text cannot inject additional report lines.

Initial behavior:

```
none -&gt; ok
read -&gt; policy_denied
local_mutation -&gt; policy_denied
host_mutation -&gt; policy_denied
network -&gt; policy_denied
hardware -&gt; policy_denied
boot -&gt; policy_denied
recovery -&gt; policy_denied
external -&gt; policy_denied
unknown -&gt; unsupported_effect
```

Audit report format

latticra::render_authority_audit_report emits a deterministic bounded report:

```
CPP AUTHORITY REPORT
status=&lt;authority-status-label&gt;
record_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
network_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
record[0].policy=&lt;policy-name&gt;
record[0].source_identity=&lt;source-identity&gt;
record[0].validator=&lt;validator-label&gt;
record[0].requested_effect=&lt;effect-label&gt;
```

```
record[0].result=&lt;authority-status-label&gt;
record[0].denial_reason=&lt;reason&gt;
record[0].span_start_offset=&lt;offset&gt;
record[0].span_end_offset=&lt;offset&gt;
```

Small output buffers return capacity_exceeded and clear the buffer. Unterminated fixed audit text fields return invalid_input and clear the buffer.

Build policy

The test runner uses:

```
CFLAGS=&quot;-std=c99 -Wall -Wextra -Werror -pedantic&quot;
CXXFLAGS=&quot;-std=c++20 -Wall -Wextra -Werror -pedantic -fno-exceptions -fno-rtti&quot;
```

The guard rejects forbidden authority-path headers and constructs including:

```
&lt;iostream&gt;
&lt;fstream&gt;
&lt;filesystem&gt;
&lt;thread&gt;
&lt;future&gt;
&lt;regex&gt;
&lt;exception&gt;
&lt;stdexcept&gt;
&lt;vector&gt;
&lt;map&gt;
&lt;unordered_map&gt;
throw
try
catch
dynamic_cast
typeid
std::vector
std::shared_ptr
```

Validation

Run:

```
sh scripts/test-cpp-authority-layer.sh
```

The test suite covers:

```
cpp_authority_layer_preserves_no_effect_flags
cpp_authority_layer_rejects_unrestricted_cpp_claims
cpp_authority_layer_has_no_execution_path
cpp_authority_layer_has_no_network_path
cpp_authority_layer_has_no_hardware_path
cpp_authority_layer_uses_explicit_result_labels
cpp_authority_layer_does_not_throw_across_c_boundary
cpp_authority_layer_does_not_allocate_in_report_path
cpp_authority_layer_preserves_source_identity_in_audit
cpp_authority_layer_accepts_max_source_identity
cpp_authority_layer_rejects_oversized_source_identity
cpp_authority_layer_bounds_source_identity_before_audit_copy
cpp_authority_layer_rejects_nul_source_identity
cpp_authority_layer_rejects_line_break_source_identity
cpp_authority_layer_validates_lat_parse_result_metadata
cpp_authority_layer_validates_lir_shape_metadata
cpp_authority_layer_audit_report_is_deterministic
cpp_authority_layer_rejects_small_report_buffer
cpp_authority_layer_rejects_unterminated_audit_text
cpp_authority_layer_is_deterministic
cpp_authority_layer_rejects_mutation_flags
cpp_authority_layer_rejects_network_flags
cpp_authority_layer_rejects_lat_network_flags
cpp_authority_layer_rejects_lir_network_flags
cpp_authority_layer_rejects_request_lat_network_flags
cpp_authority_layer_rejects_request_lir_network_flags
cpp_authority_layer_classifies_effects_without_performing_them
cpp_authority_layer_builds_with_fno_exceptions_and_fno_rtti
```

Compatibility expectations

This implementation must not change:

```
C substrate behavior
Lat parser behavior
```

```
LIR shape behavior
L-UI parser behavior
L-UI semantic validation behavior
state lattice behavior
Nucleus preview behavior
C/C++ foundation direction
language naming policy
no-effect flags
Lat grammar reports
LIR shape reports
```

Current boundary

This implementation does not provide:

```
unrestricted C++ authority
Lat execution
Lat compilation
Lat interpretation
LIR execution
LIR lowering
L-UI rendering
Nucleus task execution
file I/O
network I/O
state mutation
server interaction
self-update
recovery behavior
hardware behavior
sandboxing
malware prevention
ransomware prevention
operating-system completeness
```

Non-claims

This document and implementation do not claim a finished operating system, hardened sandbox, production runtime, production security boundary, malware prevention, ransomware prevention, recovery system, update system, bootable image, or public release readiness.

docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION_PLAN.md

Source title: Latticra Constrained C++ Authority Layer Implementation Plan

Lines: 770 | Words: 2122 | SHA-256: 0fb1f646989824f470e7a18a70e6f983a78629fde500f688c20d0f4d58a7e153

Latticra Constrained C++ Authority Layer Implementation Plan

Status: implementation planning contract Scope: exact public API, namespace, file paths, C++ standard, compiler flags, exception policy, RTTI policy, allocation policy, ownership/lifetime rules, result labels, C ABI boundaries, validators, audit reports, tests, compatibility expectations, and non-claims before any C++ authority-layer code.

Purpose

This document defines the implementation plan for the first constrained C++ authority layer in Latticra.

The constrained C++ authority-layer contract is already merged and guarded. This plan turns that contract into exact API, namespace, file path, build, exception, RTTI, allocation, ownership, C ABI, validator, audit report, and test expectations before implementation code is added.

This document does not implement C++ infrastructure, policy code, validators, effect gates, audit logic, orchestration, Lat execution, LIR execution, L-UI rendering, runtime behavior, operating-system behavior, sandboxing, malware prevention, ransomware prevention, or production security guarantees.

Direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

The implementation must preserve this relationship:

```
Lat / Latticra Language: contract and declaration layer
C++: governed authority layer
C: secure substrate
```

C++ must remain disciplined structure above the C substrate. It must not become unrestricted authority.

Relationship to previous work

This plan depends on:

```
docs/C_CPP_FOUNDATION_DIRECTION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
docs/LANGUAGE_STRATEGY.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/EFFECT_GATES.md
docs/STATE_LATTICE.md
docs/TRI_PLANE_TRANSITION.md
docs/NUCLEUS_PREVIEW.md
include/latticra/lat_parser.h
include/latticra/lir.h
include/latticra/state_lattice.h
```

Those files remain the source of truth for the C substrate, Lat metadata parsing, LIR shape, effect vocabulary, state-lattice vocabulary, transition vocabulary, Nucleus preview boundaries, no-effect flags, and current non-claims.

Implementation language decision

The first authority-layer implementation should be implemented in constrained C++20.

Reason:

- std::span and std::string_view are C++20-era tools that fit explicit bounded views;
- constrained C++ gives RAII, value types, scoped ownership, enum class, and namespace hygiene;
- the existing C substrate remains the source of truth for parsed Lat and LIR metadata;
- the C++ layer should validate and report over C-owned metadata, not replace the C substrate;
- C++ implementation must remain no-effect and must not add execution, mutation, network, recovery, hardware, or server behavior.

This decision does not permit unrestricted C++.

Implementation files

The implementation PR should add or modify:

```
include/latticra/cpp/authority.hpp
src/cpp/authority.cpp
tests/cpp_authority_layer_invariants.cpp
scripts/test-cpp-authority-layer.sh
.github/workflows/c.yml
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
```

The implementation PR should not add Lat execution, LIR execution, L-UI rendering, Nucleus task execution, file I/O, network I/O, state mutation, recovery behavior, update behavior, server interaction, hardware behavior, or security guarantees.

Public namespace

Use the namespace:

```
latticra
```

The first implementation must not create nested framework-style namespaces such as:

```
latticra::runtime
latticra::kernel
latticra::server
latticra::recovery
```

Those names imply capabilities that do not exist yet.

Public API shape

Add public C++ API names:

```
latticra::authority_status
latticra::authority_effect
latticra::authority_validator
latticra::authority_flags
latticra::authority_source_span
latticra::authority_request
latticra::authority_audit_record
latticra::authority_audit_report
latticra::authority_status_label
latticra::authority_effect_label
latticra::authority_validator_label
latticra::validate_lat_parse_result
latticra::validate_lir_shape
latticra::classify_effect_request
latticra::render_authority_audit_report
```

Recommended function signatures:

```
const char *authority_status_label(authority_status status) noexcept;
const char *authority_effect_label(authority_effect effect) noexcept;
const char *authority_validator_label(authority_validator validator) noexcept;

authority_status validate_lat_parse_result(
    const latticra_lat_parse_result_t &amp;lat_result,
    authority_audit_report &amp;report) noexcept;

authority_status validate_lir_shape(
    const latticra_lir_module_t &amp;lir_module,
    authority_audit_report &amp;report) noexcept;

authority_status classify_effect_request(
    const authority_request &amp;request,
    authority_audit_report &amp;report) noexcept;

authority_status render_authority_audit_report(
    const authority_audit_report &amp;report,
    char *buffer,
    std::size_t buffer_len) noexcept;
```

All public functions in the first implementation should be noexcept and return explicit status values.

Capacity constants

Add exact bounded constants:

```
LATTICRA_AUTHORITY_POLICY_NAME_MAX 64u
LATTICRA_AUTHORITY_VALIDATOR_NAME_MAX 64u
LATTICRA_AUTHORITY_DENIAL_REASON_MAX 128u
LATTICRA_AUTHORITY_SOURCE_IDENTITY_MAX 128u
LATTICRA_AUTHORITY_AUDIT_RECORD_MAX 32u
LATTICRA_AUTHORITY_REPORT_MAX 4096u
```

Recommended declaration style:

```
inline constexpr std::size_t LATTICRA_AUTHORITY_POLICY_NAME_MAX = 64u;
```

The first implementation should reject or truncate nothing silently. Capacity failures must return capacity_exceeded and preserve deterministic audit state.

Authority status enum

Add status enum values:

```
authority_status::ok
authority_status::null_argument
authority_status::invalid_input
authority_status::capacity_exceeded
authority_status::policy_denied
authority_status::unsupported_effect
authority_status::unsupported_boundary
authority_status::not_authorized
authority_status::internal_error
```

Stable labels:

```
ok
null_argument
invalid_input
capacity_exceeded
policy_denied
unsupported_effect
unsupported_boundary
not_authorized
internal_error
```

Authority effect enum

Add effect enum values:

```
authority_effect::none
authority_effect::read
authority_effect::local_mutation
authority_effect::host_mutation
authority_effect::network
authority_effect::hardware
authority_effect::boot
authority_effect::recovery
authority_effect::external
authority_effect::unknown
```

Stable labels:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Effects remain metadata only. Classification must not perform effects.

Authority validator enum

Add validator enum values:

```
authority_validator::naming
authority_validator::source_span
authority_validator::no_effect
authority_validator::effect
authority_validator::boundary
authority_validator::state_shape
authority_validator::lir_shape
authority_validator::lat_parse_result
```

Stable labels:

```
naming
source_span
no_effect
effect
boundary
state_shape
lir_shape
lat_parse_result
```

Validators should be small, named, deterministic, and report-visible.

Authority flags struct

Add no-effect flag fields:

```
bool no_effect;
bool execution_allowed;
bool mutation_allowed;
bool server_allowed;
bool network_allowed;
bool recovery_allowed;
bool hardware_allowed;
```

The default valid authority-layer state is:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
network_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Any deviation must be denied by the first implementation.

Authority source span struct

Use a C++ source-span struct compatible with existing C source-span conventions:

```
std::size_t start_offset;
std::size_t end_offset;
std::size_t start_line;
std::size_t start_column;
std::size_t end_line;
std::size_t end_column;
```

The implementation may copy spans from Lat or LIR metadata. It must not invent byte offsets.

Authority request struct

Add request metadata fields:

```
authority_effect requested_effect;
authority_flags flags;
std::string_view source_identity;
const latticra_lat_parse_result_t *lat_result;
const latticra_lir_module_t *lir_module;
```

The first implementation must treat source_identity as a non-owning bounded view and must not retain it after the call returns. When present and within bounds, the implementation should copy it into fixed audit storage. The bound check must run before any audit record copies caller-provided source identity. Source identity containing an embedded NUL must be rejected before report rendering. Source identity containing line breaks must be rejected before report rendering.

Authority audit record struct

Add audit record fields:

```
authority_status status;
authority_validator validator;
authority_effect requested_effect;
authority_flags flags;
std::array&lt;char, LATTICRA_AUTHORITY_POLICY_NAME_MAX&gt; policy_name;
std::array&lt;char, LATTICRA_AUTHORITY_SOURCE_IDENTITY_MAX + 1u&gt; source_identity;
std::array&lt;char, LATTICRA_AUTHORITY_VALIDATOR_NAME_MAX&gt; validator_name;
std::array&lt;char, LATTICRA_AUTHORITY_DENIAL_REASON_MAX&gt; denial_reason;
authority_source_span span;
```

Audit records must be deterministic and must not contain secrets, file contents, environment values, or host-specific data.

Authority audit report struct

Add audit report fields:

```
authority_status status;
authority_flags flags;
std::array&lt;authority_audit_record, LATTICRA_AUTHORITY_AUDIT_RECORD_MAX&gt; records;
std::size_t record_count;
```

The report must be fully initialized on success and failure.

Ownership and lifetime rules

The first implementation should use:

```
value types
references for required non-owning inputs
std::span for bounded views when needed
std::string_view for non-owning text views
std::array for fixed storage
std::unique_ptr only if separately justified
RAII wrappers only for explicit local resources
```

The first implementation must not use:

```
raw owning pointers
raw new/delete
std::shared_ptr
hidden global mutable state
long-lived singletons
unbounded std::vector in authority paths
unbounded std::string in authority paths
stored references into caller-owned buffers
implicit lifetime extension
```

Allocation policy

The first implementation should avoid heap allocation in authority paths.

Required posture:

```
fixed-capacity containers
caller-provided output buffers
no allocation in validators
no allocation in effect gates
no allocation in audit report rendering
no hidden allocation behind policy decisions
```

A later arena or allocator policy requires a separate contract.

Exception policy

The first implementation should build with:

```
-fno-exceptions
```

Rules:

```
no throw expressions
no try/catch in authority paths
no exceptions across C ABI boundaries
no exceptions through effect gates
no exceptions through report generation
no exceptions for expected validation failures
```

Expected failures must use authority_status.

RTTI and reflection policy

The first implementation should build with:

```
-fno-rtti
```

Rules:

```
no dynamic_cast in authority paths
no typeid-based authority decisions
no RTTI as policy architecture
```

```
no reflection-like dispatch
```

Template policy

Allowed:

```
small fixed-capacity helpers
strong type wrappers
constexpr constants
simple label helpers
std::array-based utility functions
```

Forbidden by default:

```
template metaprogramming as policy engine
type-level authority decisions without readable reports
compile-time tricks that hide behavior
large framework-style abstractions
unreviewable generic dispatch
```

Allowed standard library subset

The first implementation may use:

```
&lt;array&gt;
&lt;cstddef&gt;
&lt;cstdint&gt;
&lt;span&gt;
&lt;string_view&gt;
&lt;type_traits&gt;
```

The first implementation must not use in authority paths:

```
&lt;iostream&gt;
&lt;fstream&gt;
&lt;filesystem&gt;
&lt;thread&gt;
&lt;future&gt;
&lt;regex&gt;
&lt;exception&gt;
&lt;stdexcept&gt;
&lt;vector&gt;
&lt;string&gt;
&lt;map&gt;
&lt;unordered_map&gt;
```

Any exception requires a later implementation note and guard update.

C ABI boundary plan

The first implementation should consume existing C substrate structs through included C headers only.

Rules:

```
wrap C includes with extern &quot;C&quot; when included from C++
do not expose C++ object lifetimes to C callers
do not export raw C++ objects through a C ABI
do not let C++ exceptions cross C ABI boundaries
return explicit authority_status values for boundary failures
copy C metadata into fixed C++ report structures when needed
```

The first implementation should not add a public C ABI wrapper unless a follow-up plan requires C callers to invoke the C++ authority layer directly.

Lat validation plan

validate_lat_parse_result should validate metadata only.

Required checks:

```
Lat parse result status is successful before metadata validation succeeds
Lat no-effect flags are preserved
Lat execution_allowed is zero
Lat mutation_allowed is zero
Lat server_allowed is zero
Lat network_allowed is zero
```

```
Lat recovery_allowed is zero
Lat hardware_allowed is zero
Lat declaration and clause counts do not exceed declared capacity
Lat source spans are ordered and bounded
Lat effect metadata remains metadata only
```

The function must not execute Lat, compile Lat, interpret Lat, lower Lat to LIR, read files, write files, open network connections, or mutate state.

LIR validation plan

validate_lir_shape should validate metadata only.

Required checks:

```
LIR module metadata is bounded
LIR node counts do not exceed declared capacity
LIR edge counts do not exceed declared capacity
LIR source spans are ordered and bounded
LIR network_allowed is zero
LIR binding metadata remains metadata only
LIR text references use explicit lengths
LIR report behavior remains no-effect
```

The function must not execute LIR, render L-UI, execute Lat, call Nucleus task execution, read files, write files, open network connections, or mutate state.

Effect classification plan

classify_effect_request should classify requested effects but must not perform them.

Initial behavior:

```
none -&gt; ok
read -&gt; policy_denied
local_mutation -&gt; policy_denied
host_mutation -&gt; policy_denied
network -&gt; policy_denied
hardware -&gt; policy_denied
boot -&gt; policy_denied
recovery -&gt; policy_denied
external -&gt; policy_denied
unknown -&gt; unsupported_effect
```

This preserves the no-effect authority-layer starting point.

Audit report format

render_authority_audit_report should emit a deterministic bounded report:

```
CPP AUTHORITY REPORT
status=&lt;authority-status-label&gt;
record_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
network_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
record[0].policy=&lt;policy-name&gt;
record[0].source_identity=&lt;source-identity&gt;
record[0].validator=&lt;validator-label&gt;
record[0].requested_effect=&lt;effect-label&gt;
record[0].result=&lt;authority-status-label&gt;
record[0].denial_reason=&lt;reason&gt;
record[0].span_start_offset=&lt;offset&gt;
record[0].span_end_offset=&lt;offset&gt;
```

Small output buffers should return authority_status::capacity_exceeded and clear the buffer. Unterminated fixed audit text fields should return authority_status::invalid_input and clear the buffer.

Build policy

The first implementation test runner should use:

```
CFLAGS=&quot;-std=c99 -Wall -Wextra -Werror -pedantic&quot;
CXXFLAGS=&quot;-std=c++20 -Wall -Wextra -Werror -pedantic -fno-exceptions -fno-rtti&quot;
```

The script should compile C substrate objects with cc, compile authority C++ objects with c++, and link with c++.

Recommended command shape:

```
tmpdir=&quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-cpp-authority.XXXXXX&quot;)&quot;
trap &#x27;rm -rf &quot;$tmpdir&quot;&#x27; EXIT INT HUP TERM
cc $CFLAGS -Iinclude -c src/lat_parser.c -o &quot;$tmpdir/latticra-lat-parser.o&quot;
cc $CFLAGS -Iinclude -c src/lir.c -o &quot;$tmpdir/latticra-lir.o&quot;
c++ $CXXFLAGS -Iinclude -c src/cpp/authority.cpp -o &quot;$tmpdir/latticra-cpp-authority.o&quot;
c++ $CXXFLAGS -Iinclude tests/cpp_authority_layer_invariants.cpp
&quot;$tmpdir/latticra-cpp-authority.o&quot; ... -o &quot;$tmpdir/latticra-cpp-authority-layer-
invariants&quot;
```

The exact object list may include L-UI parser and semantic objects if LIR validation requires them.

Exact implementation test list

The implementation PR should include tests for:

```
cpp_authority_layer_preserves_no_effect_flags
cpp_authority_layer_rejects_unrestricted_cpp_claims
cpp_authority_layer_has_no_execution_path
cpp_authority_layer_has_no_network_path
cpp_authority_layer_has_no_hardware_path
cpp_authority_layer_uses_explicit_result_labels
cpp_authority_layer_does_not_throw_across_c_boundary
cpp_authority_layer_does_not_allocate_in_report_path
cpp_authority_layer_preserves_source_identity_in_audit
cpp_authority_layer_accepts_max_source_identity
cpp_authority_layer_rejects_oversized_source_identity
cpp_authority_layer_bounds_source_identity_before_audit_copy
cpp_authority_layer_rejects_nul_source_identity
cpp_authority_layer_rejects_line_break_source_identity
cpp_authority_layer_validates_lat_parse_result_metadata
cpp_authority_layer_validates_lir_shape_metadata
cpp_authority_layer_audit_report_is_deterministic
cpp_authority_layer_rejects_small_report_buffer
cpp_authority_layer_rejects_unterminated_audit_text
cpp_authority_layer_is_deterministic
cpp_authority_layer_rejects_mutation_flags
cpp_authority_layer_rejects_network_flags
cpp_authority_layer_rejects_lat_network_flags
cpp_authority_layer_rejects_lir_network_flags
cpp_authority_layer_rejects_request_lat_network_flags
cpp_authority_layer_rejects_request_lir_network_flags
cpp_authority_layer_classifies_effects_without_performing_them
cpp_authority_layer_builds_with_fno_exceptions_and_fno_rtti
```

Test file plan

Add:

```
tests/cpp_authority_layer_invariants.cpp
scripts/test-cpp-authority-layer.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-constrained-cpp-authority-layer-implementation-plan.sh
```

and before:

```
sh scripts/test-lat-language-grammar-contract.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
C substrate behavior
Lat parser behavior
LIR shape behavior
L-UI parser behavior
L-UI semantic validation behavior
state lattice behavior
Nucleus preview behavior
C/C++ foundation direction
language naming policy
no-effect flags
Lat grammar reports
LIR shape reports
```

Forbidden implementation behavior

The first C++ authority-layer implementation must not:

- become unrestricted C++;
- execute Lat;
- execute LIR;
- render L-UI;
- call Nucleus task execution;
- mutate state;
- read files;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- hide allocation in authority paths;
- throw exceptions across C ABI boundaries;
- use RTTI as policy architecture;
- use template metaprogramming as policy architecture;
- weaken C substrate boundaries;

- weaken no-effect flags;
- imply a sandbox, runtime, malware prevention, ransomware prevention, or operating-system surface.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-constrained-cpp-authority-layer-implementation-plan.sh
```

The guard is static. It does not implement C++ authority-layer code.

Implementation gate

Constrained C++ authority-layer implementation code may be added only after this plan is merged.

Non-claims

This document does not implement C++ infrastructure, policy code, validators, effect gates, audit logic, orchestration, Lat execution, Lat compilation, Lat interpretation, LIR execution, LIR lowering, L-UI rendering, command behavior, Nucleus task execution, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md

Source title: C++ Authority Implementation Review

Lines: 70 | Words: 181 | SHA-256: 690e33ccbe245f1bb7bdd2f36525228b149c5ada3dce70b6b5c852dc5c8bd050

C++ Authority Implementation Review

Status: implementation review

This review covers the initial no-effect constrained C++ authority implementation after the recent Lat/LIR/runtime-boundary report, diagnostic, audit, and project-notes slices.

Reviewed files:

```
include/latticra/cpp/authority.hpp
src/cpp/authority.cpp
tests/cpp_authority_layer_invariants.cpp
scripts/test-cpp-authority-layer.sh
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
```

Current finding:

```
The constrained C++ authority layer remains a no-effect metadata validation and audit-report
surface.
```

Confirmed boundaries:

```
no unrestricted C++ authority
no Lat execution
no Lat compilation
no Lat interpretation
no LIR execution
no runtime behavior
no command execution
no state mutation
no file I/O
no network I/O
no recovery behavior
no hardware behavior
no production security guarantee
```

Confirmed guard posture:

```
-fno-exceptions
```

```
-fno-rtti
forbidden exception constructs rejected
forbidden RTTI constructs rejected
forbidden dynamic containers rejected
forbidden file/network/system headers rejected
fixed-capacity report buffers
explicit result labels
policy-denied effect classification
Lat parser metadata validation
LIR shape metadata validation
deterministic authority report rendering
```

Current recommendation:

```
Keep the C++ authority layer no-effect and metadata-only until a separate authority expansion
contract exists.
```

Potential next refinement:

```
C++ authority diagnostic/status report alignment
```

Boundary: review/status only. No C++ authority behavior is changed.

docs/EFFECT_GATES.md

Source title: Latticra Effect Gates

Lines: 155 | Words: 555 | SHA-256: d1e7203986d4893fce3711e7f337109fe123f88d6c761d08fbc200ae36c46da7

Latticra Effect Gates

Status: initial effect-gate model Scope: effect classes, gate rules, request classification, and non-claims.

Purpose

Latticra must classify effects before it handles any request.

A system that can update itself, interact with servers, handle hardware targets, or eventually perform recovery work must never hide effects behind friendly names.

Effect classes

Effect class	Meaning	Initial status
`none`	No mutation, no host action, no external action.	Allowed for fixtures and reports.
`read`	Read-only inspection of local approved state.	Allowed after source contract.
`local_mutation`	Changes Latticra-owned local state.	Planned, gated.
`host_mutation`	Changes host files, processes, services, or settings.	Blocked until explicit gates.
`network`	Sends or receives network data.	Blocked until server model and trust policy.
`hardware`	Touches hardware state or device interfaces.	Blocked until hardware profile and evidence.
`boot`	Affects boot path, boot artifacts, or boot configuration.	Blocked until boot contract and evidence.
`recovery`	Performs recovery behavior or rollback beyond local preview.	Blocked until recovery contract and evidence.
`external`	Affects systems outside the local Latticra boundary.	Blocked until explicit deployment policy.

Classification rule

Every request must be classified before handling.

If classification fails, the request is denied.

```
unknown effect -&gt; deny
ambiguous effect -&gt; deny
unclassified effect -&gt; deny
```

Gate states

Each gate may be in one of these states:

```
disabled
planned
available
armed
executed
failed
blocked
```

Only executed may imply a real action happened.

Request record

Every request should eventually produce a record shaped like:

```
request_id:
source:
requested_effect:
allowed_effect:
gate:
policy_result:
reason:
operator_confirmation:
result:
rollback_state:
evidence_level:
```

Default policy

Default policy is conservative:

```
none -&gt; allowed
read -&gt; allowed only for approved local state
local_mutation -&gt; planned, explicit gate required
host_mutation -&gt; blocked
network -&gt; blocked
hardware -&gt; blocked
boot -&gt; blocked
recovery -&gt; blocked
external -&gt; blocked
```

Operator visibility

Effect state must always be visible to the operator.

Reports should show:

```
host_effect
external_effect
network_effect
hardware_effect
boot_effect
recovery_effect
```

Relationship to Nucleus

Nucleus must consult the Effect Gate before any task is carried out.

The Effect Gate does not decide all policy. It decides whether the effect class is known, allowed, gated, blocked, or failed.

Relationship to Lat

Lat documents must declare intended effects.

If a Lat document omits an effect declaration, the default is:

```
effect = none
```

If the document attempts to exceed its declared effect, it must be rejected.

Relationship to L-UI

L-UI should be effect-free by default.

A UI surface may request reports or display state, but it must not perform hidden mutation.

Server and update effects

Server interaction is network plus optional local_mutation when writing downloaded artifacts or evidence records.

Self-update is at least:

```
network
local_mutation
update_stage
```

and may eventually involve boot or recovery effects depending on deployment profile.

First implementation target

The first implementation target should support only:

```
effect = none
```

and possibly read-only report classification.

Non-claims

This document does not implement effect gates.

It defines the gate vocabulary and default-deny policy before implementation begins.

docs/FEATURE_TRANSLATION_LEDGER.md

Source title: Latticra Feature Vocabulary Ledger

Lines: 117 | Words: 738 | SHA-256: 08f22fdb3ee69bff336d119023664aa3907db973158b2d7a620d20fa03677e41

Latticra Feature Vocabulary Ledger

Status: initial vocabulary ledger Scope: mapping exploratory concept labels into professional Latticra architecture terms.

Purpose

Latticra needs stable public vocabulary before broad implementation work expands.

This ledger prevents informal, exploratory, or metaphor-heavy names from becoming unclear implementation contracts.

The public project should describe features by their Latticra purpose, behavior, evidence level, and effect boundary rather than by predecessor-project lore.

Vocabulary table

```
| Exploratory concept | Latticra term | Notes |
| --- | --- | --- |
| General state movement | Lattice transition | General state movement inside a structured
lattice. |
| Multi-plane transition check | Tri-plane transition model | Transition checked across spatial,
 state, and safety planes. |
| Portal-like boundary | Gate | A controlled transition boundary. |
| Safe transition boundary | Recovery gate | A gate associated with recovery visibility or
recovery eligibility. |
| Mirror-like state view | Reflection surface | Read-only view or reflection of state. |
| Modeled entity | Lattice actor | A modeled actor within a state lattice. |
```

Inactive context	Shadow context	An inactive, pending, or non-primary context.
Nested scope	Domain stack	Nested scope or domain stack.
Split/fault boundary	Partition boundary	Split, isolation, or fault boundary.
Recursive structure	Recursive lattice	Recursive state domain or repeated lattice pattern.
Floor/layer concept	Layer / plane / tier	Use the most precise term per context.
Visibility concept	Lens / visibility rail	Operator visibility and state display surface.
Rail display concept	Operator rails	Terminal/state display rails.
Hardware evidence track	Substrate evidence path	Boot, recovery, hardware, and validation evidence path.
Native language path	Lat / Latticra Language	Public native language family and programming-language direction.
Staged candidate work	Staged candidate system	Candidate staging, never automatic system behavior.
Verification and repair work	Verification and repair suite	Defensive validation, maintenance, and repair-oriented checks.
Matrix	State matrix	Structured multi-dimensional state surface.
Universe	Lattice universe	Structured software universe with explicit contracts.
Origin	Origin	Keep term.
Route	Route	Keep term.
Axis	Axis	Keep term.
Path	Path	Keep term.
Trace	Trace	Keep term.

Promotion categories

Each translated feature should be assigned one category:

```
historical-only
concept-candidate
fixture-candidate
tested-model-candidate
read-only-surface-candidate
implementation-candidate
blocked
```

Initial category assignments

Latticra term	Initial category	Reason
State lattice	fixture-candidate	First implementation target.
Tri-plane transition model	tested-model-candidate	Needs pure transition and denial tests before surface work.
Gate	concept-candidate	Needs effect-gate contract first.
Recovery gate	concept-candidate	Must not perform recovery until evidence supports it.
Reflection surface	read-only-surface-candidate	Safe as inspection only.
Lattice actor	concept-candidate	Needs strict non-claim and lifecycle rules.
Shadow context	concept-candidate	Needs visibility rules.
Domain stack	tested-model-candidate	Can derive from nested-domain lessons.
Partition boundary	concept-candidate	Needs failure and isolation vocabulary.
Recursive lattice	concept-candidate	Needs mathematical definition before implementation.
Lens / visibility rail	read-only-surface-candidate	UI/reporting only at first.
Staged candidate system	tested-model-candidate	Must remain approval-gated.
Verification and repair suite	concept-candidate	Must remain defensive and evidence-bound.

Rejected direct public names

Do not directly import these names as formal Latticra public components:

```
black arts
ghost
magic
arena
game
hacker console
simulator OS
```

They may appear in private brainstorming, but not as public Latticra implementation names.

Promotion rule

A feature may enter implementation only after it has:

1. a Latticra name;
2. a direct Latticra purpose;
3. evidence level;
4. non-claims;
5. effect boundary;
6. test plan;
7. failure behavior.

First implementation candidate

The first implementation candidate remains:

```
state lattice fixture
```

It should include labels for origin, route, axis, path, breadcrumb, trace, safe_portal, rollback, health, risk, lock, dark_phase, host_effect, and external_effect.

Non-claims

This ledger does not implement any feature.

It translates and constrains vocabulary before implementation begins.

docs/HOST_ARCHITECTURE_TARGETS.md

Source title: Latticra Host Architecture Targets

Lines: 115 | Words: 382 | SHA-256: d8d2c3865eb9b9e75110cd2ab9fa79e10676386cfa77eb045b0b25af4a7f928c

Latticra Host Architecture Targets

Status: initial target architecture policy Scope: x86_64, ARM64, profiles, portability, and non-claims.

Purpose

Latticra should plan target host architectures from the beginning.

The initial target families are:

```
x86_64
ARM64 / AArch64
```

Architecture rule

No architecture-specific behavior may enter the core without a profile boundary.

Architecture-specific work must declare:

- target family;
- execution mode;
- evidence level;
- effect class;
- fallback behavior;
- non-claims.

Initial target profiles

```
x86_64-hosted
x86_64-qemu
x86_64-uefi-preview
x86_64-seabios-grub-preview
x86_64-grub2-bios-preview
x86_64-grub2-uefi-preview
aarch64-hosted
aarch64-qemu
aarch64-device-readonly
```

Profile meanings

Profile	Meaning	Initial status
`x86_64-hosted`	Runs as a hosted process on an x86_64 OS.	Planning target.
`x86_64-qemu`	Runs or validates in QEMU for x86_64.	Planning target.
`x86_64-uefi-preview`	Boot-adjacent preview profile.	Future target.
`x86_64-seabios-grub-preview`	Future SeaBIOS plus GRUB compatibility evidence profile.	Contract-only target.
`x86_64-grub2-bios-preview`	Future GRUB 2 BIOS boot evidence profile.	Contract-only target.
`x86_64-grub2-uefi-preview`	Future GRUB 2 UEFI boot evidence profile.	Contract-only target.
`aarch64-hosted`	Runs as a hosted process on an ARM64 OS.	Planning target.
`aarch64-qemu`	Runs or validates in QEMU for ARM64.	Planning target.
`aarch64-device-readonly`	Real-device read-only evidence profile.	Future target.

Implementation order

Recommended order:

```
1. hosted reference model
2. portable C state lattice fixture
3. validation tests on host
4. QEMU profile planning
5. read-only real-device evidence planning
6. boot-adjacent profile only after evidence
```

C portability policy

C code intended for architecture-neutral layers should avoid assumptions about:

- pointer width beyond explicit contracts;
- endianness;
- structure packing;
- alignment;
- host OS APIs;
- filesystem layout;
- network availability.

Architecture-specific code policy

Architecture-specific code must live behind named boundaries.

Possible future structure:

```
src/arch/x86_64/
src/arch/aarch64/
include/latticra/arch.h
```

Do not create this structure until implementation requires it.

Hardware evidence policy

Real-device evidence must record:

```
device identity
architecture
firmware/boot context if relevant
operator checklist
read-only or mutating classification
logs or report
non-claims
```

Non-claims

This document does not claim Latticra currently runs on x86_64 or ARM64.

It defines the target architecture policy before implementation begins.

docs/LATTICRA_ARCHITECTURE_TARGETS.md

Source title: Latticra Architecture Targets

Lines: 282 | Words: 647 | SHA-256: 9e0dee5dc1108e509136b2e9720f2f897548910e0e21a77576edccaa5b2193b5

Latticra Architecture Targets

Status: active architecture target plan Scope: target architecture planning for C substrate work, constrained C++ policy work, Rust tooling work, Lat/LIR language work, Seal trust-boundary work, and Fedora/Linux validation. This document does not implement runtime execution, production enforcement, kernel behavior, boot behavior, or operating-system replacement behavior.

Purpose

This document turns the Latticra / Lat / Latticra Seal roadmap into concrete architecture target lanes.

The goal is to keep implementation polymorphic, evidence-bound, and testable.

Top-level architecture

```
User / operator / developer
  -&gt; L-UI reports and CLI tools
  -&gt; Lat declarations
  -&gt; Lat parser and semantic validation
  -&gt; LIR metadata representation
  -&gt; Nucleus coordination and task classification
  -&gt; Runtime Boundary classification
  -&gt; Latticra Seal trust-boundary planning
  -&gt; Fedora/Linux validation lane
```

No layer should imply authority by existing. Authority must be explicit, denied by default, and proven by tests before promotion.

C substrate targets

C targets should remain bounded and deterministic.

Current and near-term C lanes:

```
include/latticra/*.h
src/*.c
tests/*_invariants.c
scripts/test-*.sh
```

Target responsibilities:

```
state fixtures
runtime-boundary structs
Lat parser/lowering substrate
LIR shape and report substrate
Seal request and allowlist metadata
no-effect invariant tests
operator-visible deterministic report fixtures
```

C success criteria:

```
fixed capacities where practical
explicit error values
no hidden host effects
no network effects
no runtime authority grants
shell guard validates reports and non-claims
```

Constrained C++ targets

C++ targets should be introduced only when graph or policy structure requires stronger modeling than C.

Candidate layout:

```
cpp/include/latticra/CapabilityGraph.hpp
cpp/include/latticra/PolicyGraph.hpp
cpp/include/latticra/SealDecision.hpp
cpp/include/latticra/ToolBoundaryPlanner.hpp
cpp/src/CapabilityGraph.cpp
cpp/src/PolicyGraph.cpp
cpp/src/SealDecision.cpp
cpp/src/ToolBoundaryPlanner.cpp
cpp/tests/test_capability_graph.cpp
cpp/tests/test_policy_graph.cpp
```

Initial C++ targets:

```
CapabilityGraph report-only model
PolicyGraph report-only model
SealDecision report-only model
ToolBoundaryPlanner report-only model
```

C++ non-claims:

```
runtime_authority_granted=0
tool_execution=0
host_read=0
host_write=0
network=0
policy_enforcement=0
production_authority_layer=0
```

Rust targets

Rust targets should protect risky input surfaces and developer tooling.

Candidate layout:

```
tools/seal-report/
tools/seal-verify/
tools/latcheck/
tools/latfmt/
tools/latticra-status/
crates/latticra_manifest/
crates/latticra_evidence/
crates/latticra_seal_policy/
```

Initial Rust target:

```
local fixture parser and report generator only
```

Rust success criteria:

```
no network by default
no host mutation beyond temporary test output
no verification claim without real verification
clear CLI output
negative fixtures
```

Lat targets

Lat should remain declaration-first.

Target areas:

```
module declarations
capability annotations
Seal intent annotations
policy-aware function signatures
```

```
runtime-boundary declarations
LIR metadata lowering
```

Lat success criteria:

```
parse deterministically
validate semantically
produce diagnostics
lower into bounded LIR metadata
perform no execution
make no compiler-product claim
```

LIR targets

LIR should preserve evidence-bearing structure between Lat and runtime-boundary planning.

Target areas:

```
node shape
edge shape
Lat-specific node labels
transition-source edges
report fields
no-effect evidence chain
```

LIR success criteria:

```
bounded representation
operator-visible reports
negative tests
no execution
no authority grant
```

Seal targets

Latticra Seal should mature in stages.

Target areas:

```
request metadata
parameter schema metadata
freshness metadata
signed request metadata
policy decision metadata
runtime gate metadata
runtime dry-run report
guarded allowlist metadata
capability metadata
evidence receipt metadata
cryptographic verification planning
```

Seal success criteria:

```
default_action_deny=1
would_deny=1 until all prerequisites exist
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
runtime_authority_granted=0
effect_performed=0
```

Fedora/Linux targets

Fedora/Linux is the validation lane.

Target areas:

```
disposable VM validation
local RPM documentation payload
host-facing transcript records
packaging boundary docs
Fedora integration docs
```

Fedora/Linux non-claims:

```
fedora_approval_claimed=0
production_installer_ready=0
daily_driver_install_ready=0
```

```
immutable_fedora_ready=0
operating_system_replacement=0
```

MCP and AI-era tool-boundary targets

Latticra should plan for AI-era tool invocation without claiming production MCP behavior.

Target flow:

```
tool request
  -&gt; declared parameters
  -&gt; freshness metadata
  -&gt; signature metadata
  -&gt; guarded allowlist candidate
  -&gt; policy decision metadata
  -&gt; runtime gate metadata
  -&gt; dry-run denial report
  -&gt; evidence receipt
```

Non-claims:

```
mcp_server=0
mcp_client=0
ai_agent_execution_control=0
tool_execution=0
production_agent_security=0
```

Review rule

Every new target lane must answer:

1. What language owns the substrate?
2. What layer owns the policy?
3. What tool owns the report?
4. What effect is denied?
5. What authority remains absent?
6. What script validates the claim?

docs/LATTICRA_NO_EFFECT_CLI_PACKAGING_CONTRACT_ALIGNMENT.md

Source title: Latticra No-Effect CLI Packaging Contract Alignment

Lines: 172 | Words: 495 | SHA-256: fdf2c0a263b99b6786b7272bbe04ce7733a03ea53789d050dbebfdaebfbe6638

Latticra No-Effect CLI Packaging Contract Alignment

Status: packaging/spec alignment record Evidence level: 7 packaging alignment, local guard only Scope: Fedora RPM spec transition for the no-effect latticra CLI payload.

Purpose

The no-effect CLI status surface now exists as a local guarded C implementation.

This record aligns the Fedora RPM spec transition that adds the no-effect CLI payload to the local RPM draft.

This is packaging/spec alignment only.

It updates packaging/fedora/latticra.spec for the CLI payload.

It does not build a new RPM in CI.

It does not install or remove an RPM.

It does not run disposable Fedora VM validation.

It does not claim host install readiness for the CLI payload.

Alignment header

```
LATTICRA NO-EFFECT CLI PACKAGING CONTRACT ALIGNMENT
packaging_alignment_version=2
cli_payload_contract_present=1
cli_status_surface_implemented=1
cli_packaging_alignment_present=1
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_payload_validated=0
```

Historical validated package boundary

The current evidence-backed Fedora validation record still applies only to the earlier documentation-only local RPM:

```
validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
validated_payload=/usr/share/doc/latticra/README.md
validated_payload_remains_documentation_only=1
historical_disposable_vm_rpm_evidence_remains_limited=1
```

That historical evidence does not validate the new CLI payload.

Current spec payload target

The local Fedora spec now targets this payload shape:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

Because the CLI is a compiled C executable, the spec no longer declares:

```
BuildArch: noarch
```

The spec transition records:

```
rpm_contains_compiled_c_binary=1
buildarch_noarch_removed=1
cli_binary_install_mode=0755
doc_install_mode=0644
```

Required spec properties

The spec must verify the local CLI guard before compiling/installing the binary:

```
sh scripts/test-latticra-no-effect-cli-status-surface.sh
cc %{optflags} -std=c99 -Wall -Wextra -Werror -pedantic src/latticra_cli.c -o build/latticra
install -m 0755 build/latticra %{buildroot}%{_bindir}/latticra
install -m 0644 README.md %{buildroot}%{_docdir}/%{name}/README.md
%{_bindir}/latticra
%doc %{_docdir}/%{name}/README.md
```

Forbidden payload surfaces for this lane

The no-effect CLI packaging lane must not add:

```
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
/usr/share/selinux
```

The spec must also avoid service activation, package-manager actions, network actions, boot changes, kernel-module installation, and SELinux policy installation.

Future disposable Fedora VM validation requirements

After this spec-update lane, a separate disposable Fedora VM validation transcript must record:

```
expanded_payload_validation_transcript_present=1
cli_binary_present_after_install=1
cli_status_command_recorded=1
cli_version_command_recorded=1
cli_report_command_recorded=1
cli_status_output_matches_contract=1
cli_no_root_required=1
```

```
cli_no_host_mutation_observed=1
cli_no_network_observed=1
rpm_verify_completed=1
removal_validation_performed=1
cli_removed_after_rpm_removal=1
post_removal_cli_absence_verified=1
```

Until that transcript exists, the CLI payload remains unvalidated from a disposable Fedora VM packaging perspective.

Current readiness classification

```
cli_payload_contract_present=1
cli_status_surface_implemented=1
cli_packaging_alignment_present=1
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_payload_validated=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

```
sh scripts/test-latticra-no-effect-cli-packaging-contract-alignment.sh
```

Expected output:

```
latticra_no_effect_cli_packaging_contract_alignment: ok
```

Next recommended lane

```
Add disposable Fedora VM CLI payload validation transcript contract
```

That future lane should define the evidence contract for a real disposable Fedora VM RPM validation run against the expanded /usr/bin/latticra payload.

Non-claims

This alignment is not a completed RPM build transcript.

It is not installation evidence for /usr/bin/latticra.

It is not disposable Fedora VM validation of the CLI payload.

It is not host install readiness for the CLI payload.

It is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, runtime behavior, Lat execution, LIR execution, service management, kernel integration, SELinux policy integration, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/LATTICRA_NO_EFFECT_CLI_PAYLOAD_CONTRACT.md

Source title: Latticra No-Effect CLI Payload Contract

Lines: 207 | Words: 584 | SHA-256: fc9a6d26654d0b05a03f8c96b0f7c9875240b025b1601066669fdf7cc3a96331

Latticra No-Effect CLI Payload Contract

Status: contract record Evidence level: 9 target, contract only Scope: future no-effect /usr/bin/latticra CLI payload for the local Fedora RPM path.

Purpose

Latticra now has an evidence-backed disposable Fedora VM local RPM validation path for the current documentation-only package.

This contract defines the next payload boundary for the first executable Latticra surface.

The intended executable surface is a deterministic, report-only, no-effect CLI.

This is a contract only.

It does not implement /usr/bin/latticra.

It does not update the Fedora RPM spec.

It does not add a binary payload to the RPM.

It does not install, remove, or validate a new RPM.

It does not run the disposable Fedora VM validation lane.

It does not grant runtime authority.

Contract header

```
LATTICRA NO-EFFECT CLI PAYLOAD CONTRACT
cli_payload_contract_version=1
cli_binary_path=/usr/bin/latticra
cli_payload_planned=1
cli_payload_implemented=0
cli_rpm_payload_validated=0
operator_review_required=1
```

Current validated package boundary

The current evidence-backed Fedora package remains the documentation-only local RPM:

```
current_validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
current_validated_payload=/usr/share/doc/latticra/README.md
current_validated_payload_remains_documentation_only=1
current_disposable_vm_rpm_evidence_remains_valid=1
```

The existing Fedora VM validation evidence does not currently validate:

```
/usr/bin/latticra
```

Required CLI behavior contract

A future CLI implementation must remain no-effect and report-only:

```
cli_report_only=1
cli_runtime_behavior_allowed=0
cli_host_mutation_allowed=0
cli_network_allowed=0
cli_root_required=0
cli_file_write_allowed=0
cli_file_read_required=0
cli_service_operation_allowed=0
cli_kernel_operation_allowed=0
cli_package_manager_allowed=0
cli_boot_operation_allowed=0
cli_selinux_policy_operation_allowed=0
cli_effect_authority_default=denied
cli_exit_status_deterministic=1
```

Allowed initial command surface

The first CLI payload may only expose deterministic status/report commands:

```
latticra --status
latticra --version
latticra --report
```

No command may execute Lat, execute LIR, mutate host state, invoke services, invoke package managers, touch the boot path, load kernel modules, change SELinux policy, open network connections, or require root.

Required deterministic status output shape

The future latticra --status output must include a stable report shape equivalent to:

```
LATTICRA STATUS REPORT
project=latticra
mode=no-effect
runtime_behavior=disabled
host_mutation=0
network=0
kernel_operation=0
service_operation=0
package_manager_operation=0
boot_operation=0
selinux_policy_operation=0
effect_authority=denied
```

The exact implementation may add fields only if they preserve the no-effect boundary and are covered by a guard.

Implementation constraints

A future implementation should remain small, auditable, and dependency-light:

```
implementation_language=C
external_runtime_dependencies=0
dynamic_network_dependency=0
privileged_execution_required=0
fixed_output_schema=1
deterministic_report_order=1
```

Future RPM payload contract

Only after the no-effect CLI implementation exists and passes local guards may a future RPM payload expand to:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

The following payload surfaces remain forbidden for this lane:

```
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
```

Required future validation evidence

Before the CLI payload can be treated as validated, a future disposable Fedora VM transcript must record:

```
cli_binary_present_after_install=1
cli_status_command_recorded=1
cli_version_command_recorded=1
cli_report_command_recorded=1
cli_status_output_matches_contract=1
cli_no_root_required=1
cli_no_host_mutation_observed=1
cli_no_network_observed=1
cli_removed_after_rpm_removal=1
post_removal_cli_absence_verified=1
```

The future transcript must also continue to record package build, RPM install, RPM verification, RPM removal, and post-removal absence verification.

Current project state until implementation and VM evidence exist

Until the no-effect CLI is implemented, packaged, and validated inside a disposable Fedora VM, the project remains at:

```
cli_payload_contract_present=1
cli_payload_implemented=0
cli_rpm_payload_validated=0
```

```
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

```
sh scripts/test-latticra-no-effect-cli-payload-contract.sh
```

Expected output:

```
latticra_no_effect_cli_payload_contract: ok
```

Next recommended lane

```
Implement no-effect Latticra CLI status surface
```

That lane should add the smallest possible report-only executable and should not update Fedora validation status until a real disposable Fedora VM RPM validation transcript exists for the expanded payload.

Non-claims

This contract is not a CLI implementation.

It is not RPM payload expansion.

It is not disposable Fedora VM validation of /usr/bin/latticra.

It is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, runtime behavior, Lat execution, LIR execution, service management, kernel integration, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/LATTICRA_NO_EFFECT_CLI_STATUS_SURFACE_IMPLEMENTATION.md

Source title: Latticra No-Effect CLI Status Surface Implementation

Lines: 151 | Words: 366 | SHA-256: 8d143ccc6a19c688703599c2258e7aac491eb8457ef57900c8ff21296b0ef127

Latticra No-Effect CLI Status Surface Implementation

Status: implementation record Evidence level: 6 local implementation guard Scope: first no-effect latticra CLI status surface implementation.

Summary

This slice implements the first local no-effect Latticra CLI status surface as a small C executable source file.

The implementation is intentionally narrow.

It provides deterministic report-only output for:

```
latticra --status
latticra --version
latticra --report
```

It does not update the Fedora RPM spec.

It does not install /usr/bin/latticra.

It does not add the CLI to the RPM payload.

It does not run disposable Fedora VM validation.

It does not claim host install readiness for the CLI payload.

Files

```
src/latticra_cli.c
scripts/test-latticra-no-effect-cli-status-surface.sh
.github/workflows/latticra-no-effect-cli-status-surface.yml
```

CLI surface

The CLI accepts only the three report-only commands defined by the contract:

```
latticra --status
latticra --version
latticra --report
```

Invalid commands emit usage text to stderr and return exit code 2.

Deterministic status output

The --status and --report commands emit the same fixed report:

```
LATTICRA STATUS REPORT
project=latticra
mode=no-effect
runtime_behavior=disabled
host_mutation=0
network=0
kernel_operation=0
service_operation=0
package_manager_operation=0
boot_operation=0
selinux_policy_operation=0
effect_authority=denied
```

The --version command emits:

```
latticra 0.0.0
mode=no-effect
runtime_behavior=disabled
```

No-effect implementation constraints

The implementation uses only standard C string comparison and output calls.

The guard rejects forbidden implementation patterns including:

```
system(
popen(
fork(
exec
socket(
connect(
open(
fopen(
freopen(
remove(
rename(
unlink(
mkdir(
rmdir(
chmod(
chown(
mount(
setuid(
setgid(
```

Current readiness classification

```
cli_payload_contract_present=1
cli_status_surface_implemented=1
cli_local_guard_present=1
cli_local_guard_pass_expected=1
cli_rpm_payload_validated=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

```
sh scripts/test-latticra-no-effect-cli-status-surface.sh
```

Expected output:

```
latticra_no_effect_cli_status_surface: ok
```

Next recommended lane

```
Add no-effect CLI payload packaging contract alignment
```

That future lane should still avoid claiming Fedora package validation for /usr/bin/latticra until a real disposable Fedora VM RPM validation transcript exists for the expanded payload.

Non-claims

This implementation record is not RPM payload expansion.

It is not installation evidence for /usr/bin/latticra.

It is not disposable Fedora VM validation of the CLI payload.

It is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, runtime behavior, Lat execution, LIR execution, service management, kernel integration, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT.md

Source title: macOS Reset/Uninstall Effect-Authorization Contract

Lines: 263 | Words: 518 | SHA-256: 7b33b76d7e05121abf6a7842a56617cd255d152419d9beacc43bb93dd2b785bb

macOS Reset/Uninstall Effect-Authorization Contract

Status: no-effect macOS reset/uninstall effect-authorization contract Date: 2026-05-25 CDT Scope: contract for keeping future live macOS reset/uninstall execution disabled until implementation-gate and operator-intent evidence are both present.

Purpose

This contract defines the future effect-authorization boundary for macOS reset/uninstall. It keeps live managed-target removal and receipt writing disabled until both implementation-gate evidence and operator-intent evidence exist.

It is contract-only. It does not authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-effect-authorization-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current effect-authorization posture is:

```
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_contract_decision=blocked-missing-implementation-gate-and-operator-intent-e
vidence
effect_authorization_contract_required=1
effect_authorization_contract_open=0
effect_authorization_required=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
reset_uninstall_effect_authorization_evidence_present=0
effect_authorization_record_write_enabled=0
effect_authorization_denial_recorded=1
effect_authorization_denial_reason=missing-implementation-gate-and-operator-intent-evidence
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
```

Required Evidence Chain

Future effect authorization must bind to the closed implementation gate, explicit operator-intent contract, and reset/uninstall evidence contracts:

```
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
```

```
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_required=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
operator_intent_evidence_written=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
reset_uninstall_dry_run_evidence_present=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
absence_report_evidence_present=0
```

Target Scope

Future effect authorization must stay bound to user-local managed macOS targets and receipt locations:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Authorization Requirements

The future authorization boundary requires:

```
effect_authorization_requires_implementation_gate_open=1
effect_authorization_requires_operator_intent_evidence=1
effect_authorization_requires_dry_run_planner_transcript=1
effect_authorization_requires_live_target_classifier=1
effect_authorization_requires_receipt_schema=1
effect_authorization_requires_absence_report_contract=1
effect_authorization_requires_receipt_outside_removed_prefix=1
effect_authorization_requires_no_unmanaged_targets=1
effect_authorization_requires_no_unsafe_paths=1
effect_authorization_requires_no_network=1
effect_authorization_requires_no_root=1
effect_authorization_schema_version=macos-reset-uninstall-effect-authorization/1
effect_authorization_condition_implementation_gate_open=required
effect_authorization_condition_operator_intent_evidence_present=required
effect_authorization_condition_operator_intent_current_session=required
effect_authorization_condition_no_unmanaged_targets=required
effect_authorization_condition_no_network=required
effect_authorization_condition_no_root=required
effect_authorization_must_not_override_missing_operator_intent=1
effect_authorization_must_not_override_closed_implementation_gate=1
effect_authorization_must_not_authorize_unmanaged_target_removal=1
```

The current condition state remains closed:

```
effect_authorization_result_implementation_gate_open=not_met
effect_authorization_result_operator_intent_evidence=not_met
effect_authorization_result_dry_run_planner_transcript=contract-only
effect_authorization_result_live_target_classifier=contract-only
effect_authorization_result_receipt_schema=contract-only
effect_authorization_result_absence_report_contract=contract-only
effect_authorization_result_no_network=met
effect_authorization_result_no_root=met
```

Phases

The future effect-authorization phases are:

```
effect_authorization_phase_1=consume_implementation_gate_contract
effect_authorization_phase_2=consume_operator_intent_contract
effect_authorization_phase_3=validate_reset_uninstall_evidence_chain
effect_authorization_phase_4=compute_effect_authorization_decision
effect_authorization_phase_5=authorize_future_live_reset_uninstall
effect_authorization_phase_6=run_live_reset_uninstall
effect_authorization_phase_1_status=contract-only
effect_authorization_phase_2_status=contract-only
effect_authorization_phase_3_status=contract-only
effect_authorization_phase_4_status=contract-only
effect_authorization_phase_5_status=disabled
effect_authorization_phase_6_status=disabled
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-effect-authorization-contract.sh
```

Expected output:

```
macos_reset_uninstall_effect_authorization_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/NAMING_SYSTEM.md

Source title: Latticra Naming System

Lines: 158 | Words: 595 | SHA-256: 35b1c46dae143cfcb4cbe3b76edb03324d21345117cd089f515924a88edb4b9f

Latticra Naming System

Status: initial naming system Scope: professional terminology, public architecture naming, and implementation naming rules.

Purpose

Latticra needs a professional naming system before implementation work expands.

Public names should fit academic systems work, embedded software, formal state models, deterministic validation, and real implementation.

Naming principles

1. Use names that describe function before metaphor.
2. Keep exploratory names out of public architecture unless they become formalized.
3. Avoid names that make Latticra sound like a game, fantasy world, or toy operating system.
4. Prefer systems, formal methods, embedded software, and computer-science vocabulary.
5. Preserve strong conceptual meaning through precise implementation terms.
6. Avoid public dependency on predecessor project names.

Primary project names

Name	Meaning
Latticra	Lattice-oriented systems architecture and implementation project.
Lat / Latticra Language	Native Latticra programming language family.
Nucleus	Latticra supervisor and orchestration core.
L-UI	Terminal/operator interface declaration language or dialect.
LIR	Latticra Intermediate Representation.

Preferred implementation vocabulary

Concept	Latticra term
Software universe	Lattice universe
State space	State lattice
Movement	Transition
Multi-plane movement	Tri-plane transition model
Route/origin/axis/path	Route/origin/axis/path
Visibility surface	Lens or visibility rail
Rails	Operator rails or state rails
Portal-like boundary	Gate or transition gate
Safe transition boundary	Recovery gate
Mirror-like view	Reflection surface or state reflection

```
| Entity | Cell actor or lattice actor |
| Inactive context | Shadow context |
| Nested scope | Domain stack or nested domain |
| Split/fault boundary | Partition, split domain, or fault boundary |
| Recursive structure | Recursive lattice or recursive domain |
| Floor/layer concept | Layer, tier, or plane |
| Staged candidate work | Staged candidate system |
| Verification and repair work | Verification and repair suite |
| Hardware evidence path | Substrate evidence path |
| Native language family | Lat / Latticra Language |
```

Names to avoid in formal architecture

Avoid these as official Latticra component names unless a later design explicitly approves them:

```
black arts
ghost
magic
spell
haunt
arena
game
world map
fantasy universe
hacker magic
```

These can remain private brainstorming vocabulary, but Latticra implementation names should be clearer.

Naming levels

Use distinct naming levels:

```
Project name: Latticra
Supervisor name: Nucleus
Language name: Lat / Latticra Language
Interface dialect: L-UI
Intermediate form: LIR
State model: State lattice
Transition model: Tri-plane transition
Gate model: Effect gate
```

File and module naming rules

Use uppercase snake-case for major public docs already in the repository:

```
STATE_LATTICE.md
TRI_PLANE_TRANSITION.md
EFFECT_GATES.md
```

Use lowercase snake_case for C/Rust source:

```
state_lattice.c
tri_plane_transition.rs
effect_gate.h
```

Use lat_ or explicit latticra_ prefixes for native language internals when needed:

```
lat_parser
lat_runtime
latticra_ir
```

Use l_ui_ for existing L-UI parser and report code:

```
l_ui_parser
l_ui_parser_ast
l_ui_parser_diagnostics
```

Review rule

A new Latticra name should answer:

1. What does it do?

2. Is it professional?
3. Is it understandable without project lore?
4. Does it avoid overclaiming?
5. Can a systems engineer understand it from the public docs?

Current decision

The native language name is:

```
Lat / Latticra Language
```

The canonical native source extension is:

```
.lat
```

The supervisor name is:

```
Nucleus
```

The UI language or dialect is:

```
L-UI
```

docs/OPEN_ECOSYSTEM_POLICY.md

Source title: Latticra Open Ecosystem Policy

Lines: 92 | Words: 352 | SHA-256: f03a0ca369e8f91052b81df03457766ceffb62a2931354159f981ded8b987a6e

Latticra Open Ecosystem Policy

Status: active governance policy Scope: project openness, auditability, transparency, contribution posture, hybrid licensing, and license transition planning.

Purpose

Latticra is intended to remain open, auditable, transparent, and community-improvable.

The project should preserve the ability of users, researchers, maintainers, and downstream operators to inspect, modify, rebuild, and improve the system.

Direction

The intended direction is:

```
core software direction: AGPL-3.0-or-later
adoption-facing helper direction: Apache-2.0
documentation direction: CC-BY-4.0
legacy code default: Apache-2.0 until files are intentionally migrated or otherwise marked
kernel/core direction: reciprocal open license coverage
branding: separate trademark and identity policy
```

Transition rule

This policy does not silently relicense every existing file.

Existing files remain under their current license until a later migration PR updates them clearly.

New software source files should declare their license with an SPDX header.

New source file rule

New core software source files should use:

```
SPDX-License-Identifier: AGPL-3.0-or-later
```

New SDK, example, packaging-helper, installer-glue, and integration-helper source files should use:

```
SPDX-License-Identifier: Apache-2.0
```

unless a documented exception exists.

Documentation rule

Documentation and handbooks are licensed under:

```
CC-BY-4.0
```

See:

```
docs/DOCUMENTATION_LICENSE.md
```

Contribution rule

Contributions are accepted under the license declared by the files they modify.

New software contributions should be compatible with the project license direction unless a documented exception is accepted.

Identity rule

The code license does not grant rights to use the Latticra name, logo, marks, or Bryforge identity in a confusing way.

Forks may describe their relationship truthfully, but they must not imply they are the official project unless authorized.

Legal review note

This document is project guidance, not legal advice.

Formal release, relicensing, commercial distribution, or dual-license decisions should receive qualified legal review.

Follow-up work

Recommended follow-up slices:

1. Add SPDX headers to new kernel and core source files.
2. Add a license guard for new source files.
3. Add documentation SPDX headers where path-level notices are not enough.
4. Add or update NOTICE and copyright ownership policy.
5. Review path-by-path migration from Apache-2.0 to AGPL-3.0-or-later.
6. Add a release checklist for license obligations.

docs/roadmap/LATTICRA_C_CPP_FOUNDATION_ROADMAP.md

Source title: Latticra C/C++ Foundation Roadmap

Lines: 252 | Words: 1057 | SHA-256: 5422fe7efe88dc3f1504ea194c80d6c79645b52fcb1714d8fad4398f1fc5c0e3

Latticra C/C++ Foundation Roadmap

Status: Draft Layer: Roadmap Scope: Documentation-first implementation plan for the constrained C/C++ direction.

1. Purpose

This roadmap defines the staged path for turning the Latticra C/C++ direction into an implementation foundation.

The goal is to avoid rushing into code before the architecture, security profile, and validation expectations are clear.

2. Phase 0 — Documentation lock

Objective: Establish the direction before implementation.

Tasks:

- [ ] Add docs/latticra/DOCUMENTATION_MAP.md.
- [ ] Add docs/architecture/LATTICRA_LANGUAGE_FOUNDATION.md.
- [ ] Add docs/security/C_CPP_SECURITY_PROFILE.md.
- [ ] Add docs/security/C_ABI_BOUNDARY_POLICY.md.
- [ ] Add this roadmap.
- [ ] Ensure public wording avoids unsupported security claims.
- [ ] Record the official phrase: C is the metal. C++ is the disciplined structure. Latticra is the contract.

Exit criteria:

- docs are committed;
- docs are internally consistent;
- future implementation can reference these docs.

3. Phase 1 — Repository structure

Objective: Prepare the codebase layout without overbuilding.

Candidate layout:

```
latticra/
■■■ docs/
■   ■■■ architecture/
■   ■■■ security/
■   ■■■ roadmap/
■   ■■■ latticra/
■■■ include/
■   ■■■ latticra/
■       ■■■ latticra.h
■■■ src/
■   ■■■ c_substrate/
■   ■■■ cpp_authority/
■   ■■■ nucleus/
■   ■■■ contract/
■   ■■■ audit/
■   ■■■ platform/
■■■ tests/
■   ■■■ abi/
■   ■■■ contract/
■   ■■■ security/
■   ■■■ docs/
■■■ tools/
```

Tasks:

- [ ] Create minimal include/source directories.
- [ ] Create placeholder C ABI header.
- [ ] Create placeholder C++ authority module.
- [ ] Add build system skeleton.
- [ ] Add documentation integrity tests if project test framework exists.

Exit criteria:

- clean empty build or placeholder build;
- no fake security claims;
- C/C++ boundaries named clearly.

4. Phase 2 — C ABI skeleton

Objective: Create the narrow public C-facing substrate.

Tasks:

- [ ] Define latticra_status.
- [ ] Define opaque latticra_handle.
- [ ] Define latticra_bytes.
- [ ] Define create/destroy lifecycle.
- [ ] Define validation entry point.
- [ ] Add null pointer tests.
- [ ] Add buffer length tests.
- [ ] Add invalid input tests.
- [ ] Document every ABI function.

Exit criteria:

- ABI does not expose internal C++ types;
- no ownership crosses implicitly;
- invalid inputs fail deterministically.

5. Phase 3 — Restricted C++ authority core

Objective: Build the first governed authority objects.

Candidate objects:

```
Nucleus
ContractInput
ValidatedContract
EffectContext
EffectGate
AuditRecord
LatticraError
Result&lt;T, E&gt;
```

Tasks:

- [ ] Add Result<T, E> or equivalent.
- [ ] Add typed error enum.
- [ ] Add immutable validated contract object.
- [ ] Add effect gate stub.
- [ ] Add audit record stub.
- [ ] Add tests for denied, invalid, and accepted paths.
- [ ] Avoid raw owning pointers.
- [ ] Avoid exceptions across ABI.

Exit criteria:

- C ABI wraps C++ authority logic;
- policy decisions live above C substrate;
- audit path exists even if minimal.

6. Phase 4 — Contract language preparation

Objective: Prepare Latticra contract parsing and validation.

Tasks:

- [ ] Define smallest contract syntax or data model.
- [ ] Separate parse from validation.
- [ ] Reject malformed input.
- [ ] Reject unsupported declarations.
- [ ] Convert accepted input into typed validated objects.
- [ ] Add fuzzing target for parser when practical.
- [ ] Add tests for malformed, oversized, and boundary inputs.

Exit criteria:

- raw contract input never directly drives privileged behavior;
- only validated contract objects enter authority logic.

7. Phase 5 — Effect gates and audit trail

Objective: Make privileged mutation visible and governed.

Tasks:

- [ ] Define effect categories.
- [ ] Define authorization context.
- [ ] Define denied behavior.
- [ ] Define audit record format.
- [ ] Add tests for denied effects.
- [ ] Add tests for accepted effects.
- [ ] Add tests confirming audit output exists.

Exit criteria:

- every privileged mutation path has an effect gate;
- every security-relevant gate produces audit evidence.

8. Phase 6 — Tooling and hardening

Objective: Make the security profile enforceable.

Tasks:

- [ ] Add strict compiler warnings.
- [ ] Add sanitizer build mode where supported.
- [ ] Add static analysis target where supported.
- [ ] Add clang-tidy config where supported.
- [ ] Add fuzz target for parsers.
- [ ] Add dependency review notes.
- [ ] Add unsafe exception template to PR/review process.

Exit criteria:

- build/test path catches common C/C++ errors early;
- exceptions are documented rather than hidden.

9. Phase 7 — Public evidence and status

Objective: Align public claims with real evidence.

Tasks:

- [ ] Add public status document.
- [ ] List supported platforms honestly.
- [ ] List unsupported platforms honestly.
- [ ] List security work completed.
- [ ] List security work still planned.
- [ ] Avoid claiming hardened status before validation.

Exit criteria:

- public communication reflects actual implementation state;
- no security marketing outruns the evidence.

10. Initial milestone recommendation

Recommended first milestone:

```
Milestone: latticra-foundation-docs-v0.1
Scope:
  - documentation map
  - language foundation doc
  - C/C++ security profile
  - C ABI boundary policy
  - roadmap
No implementation claims.
```

Recommended commit message:

```
Add Latticra C/C++ foundation documentation
```

11. Risk register

```
| Risk | Mitigation |
|---|---|
| Unrestricted C++ enters trusted core | Enforce C/C++ Security Profile and review labels. |
| C substrate grows too large | Keep C limited to ABI, platform, and boot substrate. |
| ABI carries unsafe lifetime assumptions | Require opaque handles, explicit buffer lengths, and
 status codes. |
| Security claims outrun implementation | Use evidence-bound promotion rule. |
| Dependency graph grows uncontrolled | Require dependency review. |
| Parser becomes attack surface | Separate parse/validate, add fuzzing, reject malformed input.
|
| Platform-specific code contaminates generic core | Isolate platform code under dedicated
directories. |
```

12. Definition of done for the direction

This direction is ready for deeper implementation when:

- the docs are committed;
- the repo structure reflects the layer model;
- the first ABI skeleton exists;
- the first C++ authority object exists;

- security profile rules are visible in review;
- tests exist for invalid input and boundary cases;
- public messaging remains precise and evidence-bound.

docs/SERVER_INTERACTION_MODEL.md

Source title: Latticra Server Interaction Model

Lines: 120 | Words: 397 | SHA-256: cd6e6b6b70cebc0ad78f8aa1c584d2b15aee5f3cf2878fdcf6502c0a3d8974b5

Latticra Server Interaction Model

Status: initial server interaction model Scope: optional server connectivity, signed artifacts, evidence sync, package/update interactions, and boundaries.

Purpose

Latticra should design server interaction from the beginning.

Server connectivity must be explicit, signed where relevant, inspectable, and optional unless a deployment profile requires it.

Core rule

Latticra must remain locally operable by default.

Server interaction is a gated capability, not a hidden dependency.

Server roles

```
| Server role | Purpose | Initial status |
| --- | --- | --- |
| Update server | Signed update manifests and version metadata. | Planned. |
| Evidence server | Validation reports, evidence bundles, hardware profile records. | Planned. |
| Package server | Lat, L-UI, and supervisor module packages. | Planned. |
| Support server | Diagnostic report submission and support guidance. | Planned. |
| Trust metadata server | Keys, revocation metadata, trust roots, and policy bundles. | Planned.
 |
```

Network effect boundary

Any server interaction is at least:

```
effect = network
```

If server interaction writes local artifacts, it is also:

```
effect = local_mutation
```

If it changes boot, recovery, hardware, or supervisor behavior, additional gates are required.

Required server request record

Every server request should eventually record:

```
request_id
server_role
endpoint_identity
method
network_effect
local_write_effect
signature_required
signature_result
operator_confirmation
result
failure_reason
```

Trust requirements

Server-provided artifacts must not be accepted as trusted merely because they downloaded successfully.

Future artifact acceptance should require:

- manifest identity;
- signature verification;
- hash verification;
- channel policy;
- version policy;
- rollback compatibility;
- evidence record.

Local-first operation

The following must work without server access:

- local state reports;
- no-effect fixtures;
- local validation;
- local evidence review;
- local non-claims review;
- local architecture documentation.

Server gateway

Nucleus should eventually route all server interaction through a Server Gateway.

No component should bypass the Server Gateway for network behavior.

Failure behavior

Server failure must be visible and non-destructive by default.

Failure states:

```
unavailable
signature_failed
hash_mismatch
policy_denied
channel_denied
rollback_incompatible
operator_denied
unknown_failure
```

Initial implementation target

No server interaction should be implemented first.

The first server-related implementation should be a static server interaction contract fixture and validation test.

Non-claims

This document does not implement networking, package downloads, update servers, support servers, or trust servers.

It defines the server interaction model before implementation begins.

docs/SUPERVISOR_ARCHITECTURE.md

Source title: Nucleus Supervisor Architecture

Lines: 159 | Words: 512 | SHA-256: 91443ff80d00ecce710073fa0b4158084cc2f4782c38c1464ac8767308bdcf93

Nucleus Supervisor Architecture

Status: initial supervisor architecture Scope: Latticra supervisor, orchestration responsibilities, state ownership, gates, and non-claims.

Purpose

Nucleus is the planned Latticra supervisor and orchestration core.

It should be designed from the beginning as a disciplined system supervisor, not as an experimental shell and not as a simulation loop.

Responsibilities

Nucleus owns or coordinates:

- state lattice identity;
- transition requests;
- policy evaluation;
- effect gates;
- task orchestration;
- update staging;
- rollback visibility;
- server interaction gates;
- operator-visible reports;
- evidence records;
- language boundaries for Lat-family documents.

Initial boundaries

Nucleus does not initially provide:

- hardware boot control;
- disk modification;
- recovery action;
- silent self-update behavior;
- kernel behavior;
- hypervisor behavior;
- security-boundary guarantees;
- general third-party code isolation;
- system integrity guarantees.

Core components

```
Nucleus
  State Lattice
  Policy Engine
  Effect Gate
  Task Engine
  Update Engine
  Server Gateway
  Report Engine
  Lat Boundary
  Evidence Recorder
```

State Lattice

The State Lattice stores explicit state cells, paths, routes, axes, traces, breadcrumbs, and effect labels.

Early state must be read-only and fixture-backed.

Policy Engine

The Policy Engine answers:

```
Is this request allowed?
Why or why not?
What evidence level is required?
What gate is required?
What effect is possible?
```

Effect Gate

The Effect Gate controls whether a request may produce no effect, a read effect, a local mutation, a host mutation, a network effect, a hardware effect, a boot effect, a recovery effect, or an external effect.

Task Engine

The Task Engine handles only approved tasks.

Early implementation should support only dry-run and preview tasks.

Update Engine

The Update Engine handles signed staged updates, channels, verification, rollback slots, and post-update validation.

It must never perform silent updates.

Server Gateway

The Server Gateway handles optional and signed server interaction.

It must keep local operation possible unless a deployment profile explicitly requires server connectivity.

Report Engine

The Report Engine produces operator-visible state, effect, risk, health, lock, update, and evidence reports.

Lat Boundary

The Lat Boundary controls how Lat, L-UI, and future Lat-Orch documents are parsed, validated, and handled.

Early Lat behavior should be assertion/report only.

Evidence Recorder

The Evidence Recorder captures validation level, source, target, effect, result, failure behavior, and promotion status.

Request lifecycle

```
operator request
  -&gt; parse
  -&gt; classify
  -&gt; policy check
  -&gt; effect gate
  -&gt; preview/report
  -&gt; explicit confirmation if required
  -&gt; carry out only if allowed
  -&gt; record evidence
  -&gt; report result
```

Effect-first design

Every Nucleus request must carry an effect label.

If the effect cannot be classified, the request is denied.

Failure-first design

Every mutating or external action must define:

- failure result;
- rollback or recovery boundary;
- operator-visible reason;
- evidence record;
- post-action validation.

Initial Nucleus milestone

The first Nucleus milestone is not a full supervisor.

It is:

```
state lattice fixture
policy classification
no-effect report
invariant tests
```

Non-claims

This document does not claim Nucleus is implemented.

It defines the intended supervisor architecture before implementation begins.

Part IV - Lat, LIR, and L-UI

- docs/architecture/LATTICRA_LANGUAGE_FOUNDATION.md - Latticra Language Foundation
- docs/L_UI_AST_DETAILED_REPORT_CONTRACT.md - Latticra L-UI AST Detailed Report Contract
- docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md - Latticra L-UI AST Detailed Report Implementation
- docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION_PLAN.md - Latticra L-UI AST Detailed Report Implementation Plan
- docs/L_UI_AST_ESCAPED_STRING_REPORT_CONTRACT.md - Latticra L-UI AST Escaped String Report Contract

- docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md - Latticra L-UI AST Escaped String Report Implementation
- docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION_PLAN.md - Latticra L-UI AST Escaped String Report Implementation Plan
- docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md - Latticra L-UI AST Length-Carrying String Storage Contract
- docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md - Latticra L-UI AST Length-Carrying String Storage Implementation
- docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION_PLAN.md - Latticra L-UI AST Length-Carrying String Storage Implementation Plan
- docs/L_UI_AST_SOURCE_BACKED_TEXT_CONTRACT.md - Latticra L-UI AST Source-Backed Text Contract
- docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md - Latticra L-UI AST Source-Backed Text Implementation
- docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION_PLAN.md - Latticra L-UI AST Source-Backed Text Implementation Plan
- docs/L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md - Latticra L-UI Decoded NUL Acceptance Contract
- docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md - Latticra L-UI Decoded NUL Acceptance Implementation
- docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION_PLAN.md - Latticra L-UI Decoded NUL Acceptance Implementation Plan
- docs/L_UI_PARSER.md - Latticra L-UI Parser
- docs/L_UI_PARSER_AST_CONTRACT.md - Latticra L-UI Parser AST Contract
- docs/L_UI_PARSER_AST_IMPLEMENTATION.md - Latticra L-UI Parser AST Implementation
- docs/L_UI_PARSER_AST_IMPLEMENTATION_PLAN.md - Latticra L-UI Parser AST Implementation Plan
- docs/L_UI_PARSER_DESIGN.md - Latticra L-UI Parser Design Contract
- docs/L_UI_PARSER_DIAGNOSTICS.md - Latticra L-UI Parser Diagnostics Contract
- docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md - Latticra L-UI Parser Diagnostics Implementation
- docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION_PLAN.md - Latticra L-UI Parser Diagnostics Implementation Plan
- docs/L_UI_PARSER_FIXTURE_INTEGRATION.md - Latticra L-UI Parser Fixture Integration
- docs/L_UI_PARSER_IMPLEMENTATION_PLAN.md - Latticra L-UI Parser Implementation Plan
- docs/L_UI_PARSER_LINE_COLUMN_PRECISION_IMPLEMENTATION.md - Latticra L-UI Parser Line/Column Precision Implementation
- docs/L_UI_PARSER_LINE_COLUMN_PRECISION_IMPLEMENTATION_PLAN.md - Latticra L-UI Parser Line/Column Precision Implementation Plan
- docs/L_UI_PARSER_LINE_COLUMN_PRECISION_PLAN.md - Latticra L-UI Parser Line/Column Precision Plan
- docs/L_UI_PARSER_RESULT_REPORT.md - Latticra L-UI Parser Result Report
- docs/L_UI_PARSER_SOURCE_SPAN_CONTRACT.md - Latticra L-UI Parser Source-Span Contract

- docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION.md - Latticra L-UI Parser Source-Span Implementation
- docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION_PLAN.md - Latticra L-UI Parser Source-Span Implementation Plan
- docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_CONTRACT.md - Latticra L-UI Parser String Escape Diagnostics Contract
- docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md - Latticra L-UI Parser String Escape Diagnostics Implementation
- docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION_PLAN.md - Latticra L-UI Parser String Escape Diagnostics Implementation Plan
- docs/L_UI_RENDERING_CONTRACT.md - Latticra L-UI Rendering Contract
- docs/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT.md - Latticra L-UI Rendering Detailed Report Refinement
- docs/L_UI_RENDERING_IMPLEMENTATION.md - Latticra L-UI Rendering Implementation
- docs/L_UI_RENDERING_IMPLEMENTATION_PLAN.md - Latticra L-UI Rendering Implementation Plan
- docs/L_UI_SEMANTIC_VALIDATION_CONTRACT.md - Latticra L-UI Semantic Validation Contract
- docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md - Latticra L-UI Semantic Validation Implementation
- docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md - Latticra L-UI Semantic Validation Implementation Plan
- docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md - Latticra L-UI Source Buffer Literal NUL Policy Contract
- docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md - Latticra L-UI Source Buffer Literal NUL Policy Implementation
- docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION_PLAN.md - Latticra L-UI Source Buffer Literal NUL Policy Implementation Plan
- docs/L_UI_SOURCE_GRAMMAR.md - Latticra L-UI Source Grammar Draft
- docs/L_UI_STATIC_REPORT.md - Latticra L-UI Static Report Fixture
- docs/L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md - Latticra L-UI String-Literal Escape Contract
- docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md - Latticra L-UI String-Literal Escape Implementation
- docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md - Latticra L-UI String-Literal Escape Implementation Plan
- docs/LANGUAGE_NAMING_POLICY.md - Latticra Language Naming Policy
- docs/LANGUAGE_STRATEGY.md - Latticra Language Strategy
- docs/LAT_LANGUAGE_FOUNDATION_ANALYSIS.md - Latticra Lat Language Foundation Analysis
- docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md - Latticra Lat Language Grammar Contract
- docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md - Latticra Lat Language Grammar Implementation
- docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md - Latticra Lat Language Grammar Implementation Plan

- docs/LAT_MODEL_NORMALIZATION_IMPLEMENTATION.md - Lat Model Normalization Implementation
- docs/LAT_PIPELINE_CONTRACT.md - Latticra Lat Pipeline Contract
- docs/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_REFINEMENT.md - Lat Pipeline Diagnostic Integration Refinement
- docs/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_INTEGRATION_AUDIT.md - Lat Pipeline Diagnostic Main Test Integration Audit
- docs/LAT_PIPELINE_IMPLEMENTATION.md - Latticra Lat Pipeline Implementation
- docs/LAT_PIPELINE_IMPLEMENTATION_PLAN.md - Latticra Lat Pipeline Implementation Plan
- docs/LAT_PIPELINE_REPORT_REFINEMENT.md - Latticra Lat Pipeline Report Refinement
- docs/LAT_SEMANTIC_DIAGNOSTICS_REFINEMENT.md - Latticra Lat Semantic Diagnostics Refinement
- docs/LAT_SEMANTIC_VALIDATION_CONTRACT.md - Latticra Lat Semantic Validation Contract
- docs/LAT_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md - Latticra Lat Semantic Validation Implementation Plan
- docs/LAT_SPECIFIC_LIR_REFINEMENT_CONTRACT.md - Latticra Lat-Specific LIR Refinement Contract
- docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION.md - Latticra Lat-Specific LIR Refinement Implementation
- docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION_PLAN.md - Latticra Lat-Specific LIR Refinement Implementation Plan
- docs/LAT_TO_LIR_CLAUSE_METADATA_REFINEMENT.md - Lat-to-LIR Clause Metadata Refinement
- docs/LAT_TO_LIR_DIAGNOSTIC_REFINEMENT.md - Lat-to-LIR Diagnostic Refinement
- docs/LAT_TO_LIR_LOWERING_CONTRACT.md - Latticra Lat-to-LIR Lowering Contract
- docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION.md - Latticra Lat-to-LIR Lowering Implementation
- docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION_PLAN.md - Latticra Lat-to-LIR Lowering Implementation Plan
- docs/LIR_REPORT_REFINEMENT.md - Latticra LIR Report Refinement
- docs/LIR_SHAPE_CONTRACT.md - Latticra LIR Shape Contract
- docs/LIR_SHAPE_IMPLEMENTATION.md - Latticra LIR Shape Implementation
- docs/LIR_SHAPE_IMPLEMENTATION_PLAN.md - Latticra LIR Shape Implementation Plan
- docs/POLYMORPHIC_LANGUAGE_STRATEGY.md - Polymorphic Language Strategy
- docs/UI_TERMINAL_LANGUAGE.md - Latticra UI Terminal Language

docs/architecture/LATTICRA_LANGUAGE_FOUNDATION.md

Source title: Latticra Language Foundation

Lines: 201 | Words: 893 | SHA-256: eef8b4d9e4c5d4d9e752b93cbb7d250924530471320503f0f7e9544700a9ed83

Latticra Language Foundation

Status: Draft Layer: Architecture Applies to: Latticra core, trusted runtime design, language substrate, security-facing documentation.

1. Direction

Latticra is moving toward a constrained C/C++ foundation.

The goal is not to use C/C++ casually. The goal is to use C and C++ in a disciplined, security-governed form suitable for a low-level, contract-driven systems project.

The intended stack is:

```
Latticra
■■■ Latticra contract language
■   ■■■ declarations
■   ■■■ policies
■   ■■■ validation forms
■   ■■■ system descriptions
■   ■■■ human-facing control surfaces
■
■■■ Restricted C++ governed authority layer
■   ■■■ Nucleus logic
■   ■■■ state lattice transitions
■   ■■■ policy objects
■   ■■■ validators
■   ■■■ effect gates
■   ■■■ audit records
■   ■■■ typed resource ownership
■
■■■ C secure substrate
■   ■■■ boot paths
■   ■■■ ABI boundaries
■   ■■■ platform shims
■   ■■■ fixed-layout structures
■   ■■■ freestanding runtime support
■   ■■■ hardware-facing interfaces
■
■■■ platform-specific assembly or firmware interfaces
```

Core phrase:

C is the metal. C++ is the disciplined structure. Latticra is the contract.

2. Rationale

Latticra requires:

- deep platform control;
- clear ABI boundaries;
- freestanding and low-level implementation paths;
- audit-visible security transitions;
- explicit trust boundaries;
- long-lived independence from unnecessary dependency churn;
- predictable integration with boot, hardware, runtime, and toolchain surfaces.

C gives Latticra a narrow, stable substrate for platform-facing interfaces.

C++ gives Latticra stronger structure above that substrate when it is constrained by project rules: RAII, typed ownership, value objects, explicit status returns, validators, and policy types.

The Latticra contract language sits above both layers and expresses declarations, authority boundaries, policy intent, and system-visible commitments.

3. Non-goals

This direction does not mean:

- unrestricted C++ is acceptable;
- large dependency graphs are acceptable;
- raw pointer ownership is acceptable in trusted core code;
- unchecked buffer behavior is acceptable;
- undefined behavior is an acceptable engineering cost;
- memory safety concerns are dismissed;
- marketing claims can outrun evidence.

Latticra must remain honest about C/C++ risk. Security must come from architecture, review, tooling, tests, and narrow boundaries.

4. Architectural roles

4.1 C secure substrate

C is limited to substrate responsibilities:

- boot entry and early runtime;
- C ABI exposure;
- platform shims;
- fixed-layout interop structures;
- minimal hardware-facing interfaces;
- freestanding runtime helpers;
- thin wrappers over platform-specific mechanisms.

C should not own high-level policy decisions unless no higher layer exists yet.

4.2 Restricted C++ governed authority layer

Restricted C++ owns the higher structure:

- Nucleus supervision;
- state lattice logic;
- typed security transitions;
- effect gate dispatch;
- validators;
- audit trail assembly;
- resource ownership;
- capability objects;
- policy evaluation;
- deterministic failure handling.

C++ code in this layer must follow the Latticra C/C++ Security Profile.

4.3 Latticra contract language

The Latticra language expresses system intent above the implementation foundation.

It should describe:

- what authority exists;
- what transitions are allowed;
- what validation must occur;
- what evidence is required;
- what effects may be attempted;
- what audit record must be produced;
- what failure state is safe.

The contract language should never hide unsafe behavior. It should make authority visible.

5. Security posture

Latticra treats unsafe power as explicit.

Every privileged action should pass through these conceptual stages:

```
request
  -&gt; parse
  -&gt; validate
  -&gt; authorize
  -&gt; effect gate
  -&gt; execute
  -&gt; audit
  -&gt; report
```

The central design rule:

No privileged mutation without validation, authorization, effect gating, and audit visibility.

6. Layer boundaries

Layer boundaries should be treated as security boundaries.

```
| Boundary | Rule |
|---|---|
| Latticra language -&gt; C++ | Contracts must be parsed, validated, and transformed into typed
policy objects. |
| C++ -&gt; C | No ownership, lifetime, or unchecked mutable buffer may cross implicitly. |
| C -&gt; platform | Platform-specific code must be isolated, documented, and reviewable. |
| Any layer -&gt; privileged effect | Must pass through an effect gate. |
```

7. Naming recommendations

Suggested terms for public/project documentation:

- Secure Substrate for the C layer.
- Governed Authority Layer for the restricted C++ layer.
- Contract Language for the Latticra language layer.
- Effect Gate for controlled privileged action paths.
- Evidence-bound validation for claims that require testable proof.
- Nucleus for the core supervisory authority.

8. Public wording

Preferred:

Latticra uses a minimal C substrate for ABI, boot, and platform-facing interfaces, with a constrained C++ authority layer above it for policy, validation, effect gates, audit logic, and typed security transitions.

Avoid:

C++ makes Latticra secure.

Better:

Latticra uses constrained C/C++ with explicit security contracts, narrow unsafe boundaries, build hardening, static analysis, sanitizer testing, fuzzing, and evidence-bound validation.

9. Current status language

Use this status language until implementation evidence exists:

This direction is architectural and preparatory. It defines the intended foundation and security posture. It is not yet a completed hardened implementation.

10. Acceptance criteria

This architecture can be considered ready for implementation work when:

- the C/C++ Security Profile exists and is checked into the repo;
- ABI boundary rules are documented;
- build flags and analyzer expectations are documented;
- a roadmap exists for staged migration or implementation;
- public wording avoids unsupported security claims;
- early tests enforce at least some documentation invariants.

docs/L_UI_AST_DETAILED_REPORT_CONTRACT.md

Source title: Latticra L-UI AST Detailed Report Contract

Lines: 368 | Words: 879 | SHA-256: f88b2d3b93981e37efc5a88821fccdc6e3590e7a8e2baba5937d2462f4bc32a6

Latticra L-UI AST Detailed Report Contract

Status: detailed report contract Scope: deterministic rail, field, text, source-span, and no-effect reporting for the fixed-size L-UI AST.

Purpose

This document defines the detailed report contract for the L-UI AST.

The AST implementation exposes a compact summary report and a detailed metadata report. The detailed report makes card, rail, field, text, and source-span metadata visible without adding rendering, traversal runtime, command behavior, live movement, mutation, or external effects.

The implementation is documented separately in L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md.

Current boundary

The current AST implementation provides:

```
fixed-size AST metadata
card summary
rail metadata
field metadata
text metadata
source spans
compact AST summary report
detailed AST metadata report
no-effect flags
```

This contract does not add:

```
interactive UI behavior
renderer integration
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Detailed report purpose

A detailed AST report provides deterministic, inspectable text for validated L-UI metadata.

It supports:

- reviewing the parsed card;
- inspecting rail order and counts;
- inspecting field names and bindings;
- inspecting text nodes;
- inspecting source spans for populated nodes;
- debugging fixture changes;
- future tooling that consumes stable text output.

It remains metadata-only.

Report function

Implemented public function:

```
latticra_l_ui_ast_detailed_report
```

Signature:

```
latticra_status_t latticra_l_ui_ast_detailed_report(
    const latticra_l_ui_ast_result_t *ast,
    char *buffer,
    size_t buffer_len);
```

This is additive and does not change the existing compact AST report.

Report capacity

The bounded capacity is:

```
LATTICRA_L_UI_AST_DETAILED_REPORT_MAX = 16384
```

The detailed report is larger than the compact AST report because it renders all populated rails, fields, text nodes, source spans, and binding spans.

Top-level report shape

The report starts with:

```
L-UI AST DETAILED REPORT
```

```
card=NucleusPreview
purpose=operator-visible Nucleus preview report
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Span field format

Every node section that includes a span renders:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

The span field names match existing parser and diagnostic report span fields.

Card section

The card section renders:

```
[card]
kind=card
name=NucleusPreview
purpose=operator-visible Nucleus preview report
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Rail section

Each rail renders in AST order:

```
[rail &lt;index&gt;]
kind=rail
name=&lt;rail-name&gt;
first_field_index=&lt;n&gt;
field_count=&lt;n&gt;
first_text_index=&lt;n&gt;
text_count=&lt;n&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Required rail order:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Field section

Each field renders in AST order:

```
[field &lt;index&gt;]
kind=field
name=&lt;field-name&gt;
binding=&lt;binding-path&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
binding_span_start_offset=&lt;n&gt;
binding_span_end_offset=&lt;n&gt;
binding_span_start_line=&lt;n&gt;
binding_span_start_column=&lt;n&gt;
binding_span_end_line=&lt;n&gt;
binding_span_end_column=&lt;n&gt;
```

Field sections must not treat bindings as executable references.

Text section

Each text node renders in AST order:

```
[text &lt;index&gt;]
kind=text
value=&lt;literal-text&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Text sections must not treat text values as commands.

Determinism rules

Detailed AST reports must be deterministic.

Rules:

1. Report sections must appear in a fixed order.
2. Rails must appear in AST rail order.
3. Fields must appear in AST field order.
4. Text nodes must appear in AST text order.
5. Missing/unpopulated capacity slots must not be printed.
6. Repeated calls with the same AST must produce identical text.
7. Output must not include memory addresses.
8. Output must not depend on platform locale.

Escaping policy

The first detailed report emits current fixture text literally because the fixture text is controlled.

Before accepting broader L-UI text values, a future escaping contract should define how to render newlines, tabs, quotes, and non-printable bytes.

Failed parse behavior

If AST construction failed due to a parser error, detailed reporting returns a compact failed-parse report:

```
L-UI AST DETAILED REPORT
parse_error=&lt;error-label&gt;
rail_count=0
field_count=0
text_count=0
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

No partial AST nodes are printed for failed parses.

No-effect rule

Detailed reporting is metadata output only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility rule

The detailed report is additive.

It does not change:

```
latticra_l_ui_ast_report
latticra_l_ui_parse_ast
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
latticra_l_ui_diagnostic_report
```

Implementation gate

Detailed AST report implementation required a separate implementation plan defining:

1. public API addition;
2. report capacity;
3. exact section order;
4. failed-parse behavior;
5. escaping policy for current fixture;
6. test file names;
7. exact invariant tests;
8. compatibility expectations.

That plan is recorded in L_UI_AST_DETAILED_REPORT_IMPLEMENTATION_PLAN.md.

Test list

Detailed report implementation tests verify:

```
detailed_report_contains_title
detailed_report_contains_card_section
detailed_report_contains_all_rails
detailed_report_contains_all_fields
detailed_report_contains_all_text_nodes
detailed_report_preserves_rail_order
detailed_report_preserves_field_order
detailed_report_preserves_text_order
detailed_report_includes_card_span
detailed_report_includes_rail_spans
detailed_report_includes_field_spans
```

```
detailed_report_includes_binding_spans
detailed_report_includes_text_spans
detailed_report_preserves_no_effect_flags
detailed_report_is_deterministic
detailed_report_rejects_bad_arguments
detailed_report_rejects_small_buffers
detailed_report_omits_unused_capacity_slots
detailed_report_handles_failed_parse
```

Forbidden behavior

A detailed report implementation must not:

- add file I/O to parser code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- emit memory addresses.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-ast-detailed-report-contract.sh
```

The guard is static. It validates the contract text.

Non-claims

This document does not implement L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md

Source title: Latticra L-UI AST Detailed Report Implementation

Lines: 242 | Words: 632 | SHA-256: a7cb7e01e5282ef7e3cfaee4774cb95c60feea87ba3016b173502597edac1deb

Latticra L-UI AST Detailed Report Implementation

Status: implementation contract Scope: bounded detailed AST reporting for card, rail, field, text, source-span, escaped string, source-backed text, and no-effect metadata.

Purpose

The L-UI AST Detailed Report implementation renders deterministic text for the fixed-size L-UI AST metadata model.

It makes card, rail, field, source-backed text, source-span, binding-span, and escaped string metadata visible without adding rendering, command behavior, Nucleus task handling, live movement, state mutation, server interaction, recovery behavior, self-update behavior, hardware behavior, or boot behavior.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
tests/l_ui_parser_ast_detailed_report_invariants.c
tests/l_ui_ast_escaped_string_report_invariants.c
tests/l_ui_ast_source_backed_text_invariants.c
scripts/test-l-ui-ast-detailed-report.sh
scripts/test-l-ui-ast-escaped-string-report.sh
scripts/test-l-ui-ast-source-backed-text.sh
```

Public API

```
latticra_l_ui_ast_detailed_report
```

Report capacity

```
LATTICRA_L_UI_AST_DETAILED_REPORT_MAX = 16384
```

The detailed report is larger than the compact AST report because it prints every populated rail, field, text node, source span, binding span, and escaped string field.

Report shape

The report starts with:

```
L-UI AST DETAILED REPORT
card=NucleusPreview
purpose=operator-visible Nucleus preview report
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Then it renders sections in deterministic order:

```
[card]
[rail 0] ... [rail 8]
[field 0] ... [field 22]
[text 0] ... [text 1]
```

Unused capacity slots are not printed.

Source-backed report values

The detailed report reflects source-backed AST values for:

```
purpose=&lt;literal-purpose&gt;
value=&lt;literal-text&gt;
```

When validated source purpose/text values change, detailed report literal and escaped fields change with the AST-owned copies.

Escaped string fields

The detailed report keeps existing literal fields and adds escaped variants:

```
purpose=&lt;literal-purpose&gt;
purpose_escaped=&lt;escaped-purpose&gt;
value=&lt;literal-text&gt;
value_escaped=&lt;escaped-text&gt;
```

Escaped fields are additive and appear only in:

```
[card]
[text &lt;index&gt;]
```

Escaped fields do not appear in failed-parse reports.

Span fields

Node spans render as:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Field binding spans render as:

```
binding_span_start_offset=&lt;n&gt;
binding_span_end_offset=&lt;n&gt;
binding_span_start_line=&lt;n&gt;
binding_span_start_column=&lt;n&gt;
binding_span_end_line=&lt;n&gt;
binding_span_end_column=&lt;n&gt;
```

Failed parse behavior

If AST construction contains a parser error, the detailed report renders a compact failed-parse report:

```
L-UI AST DETAILED REPORT
parse_error=&lt;error-label&gt;
rail_count=0
field_count=0
text_count=0
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

No partial AST nodes, source-backed text values, or escaped fields are printed for failed parses.

Escaping policy

Escaped string fields use the byte-oriented escaping model documented in:

```
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
```

The implementation renders current literal fields for compatibility while adding escaped fields for fixture-safe comparisons and future broader text handling.

Compatibility

The implementation does not change:

```
latticra_l_ui_ast_report
latticra_l_ui_parse_ast
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
latticra_l_ui_diagnostic_report
latticra_l_ui_ast_detailed_report existing required fields
```

No-effect boundary

Detailed reporting preserves:

```
no_effect=1
```

```
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test commands

Run:

```
sh scripts/test-l-ui-ast-detailed-report.sh
sh scripts/test-l-ui-ast-escaped-string-report.sh
sh scripts/test-l-ui-ast-source-backed-text.sh
```

The main C workflow runs these checks after the detailed report implementation-plan guard, escaped string report implementation-plan guard, and source-backed text implementation-plan guard.

Required invariants

The detailed report tests verify:

```
detailed_report_contains_title
detailed_report_contains_card_section
detailed_report_contains_all_rails
detailed_report_contains_all_fields
detailed_report_contains_all_text_nodes
detailed_report_preserves_rail_order
detailed_report_preserves_field_order
detailed_report_preserves_text_order
detailed_report_includes_card_span
detailed_report_includes_rail_spans
detailed_report_includes_field_spans
detailed_report_includes_binding_spans
detailed_report_includes_text_spans
detailed_report_preserves_no_effect_flags
detailed_report_is_deterministic
detailed_report_rejects_bad_arguments
detailed_report_rejects_small_buffers
detailed_report_omits_unused_capacity_slots
detailed_report_handles_failed_parse
```

The escaped string report tests verify additive escaped purpose/text fields and byte-oriented escaping behavior.

The source-backed text tests verify detailed report literal and escaped fields update from validated source values.

Current evidence level

This implementation is an L2 tested detailed metadata, escaped string, and source-backed text report for the fixed-size L-UI AST.

It is not a renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after source-backed text extraction is:

```
L-UI string-literal escape contract
```

That future work should define whether and how string escapes are decoded before AST extraction decodes escape sequences.

Non-claims

This document and implementation do not claim L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI AST Detailed Report Implementation Plan

Lines: 378 | Words: 826 | SHA-256: 075e0b900bc4974c7099a44f65e5f587fec78deabcedc2c6940b652cd6698960

Latticra L-UI AST Detailed Report Implementation Plan

Status: implementation planning contract Scope: public API addition, report capacity, exact section order, failed-parse behavior, escaping policy, exact tests, and compatibility expectations before detailed AST report code.

Purpose

This document defines the implementation plan for the L-UI AST detailed report.

The detailed report contract is already merged and guarded. This plan decides the exact public API addition, capacity, report section order, failed-parse behavior, escaping policy, exact tests, and compatibility expectations before implementation code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_AST_DETAILED_REPORT_CONTRACT.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
```

Those files remain the source of truth for AST metadata, compact AST reporting, source spans, and no-effect boundaries.

Implementation language decision

The detailed AST report implementation should be in C.

Reason:

- the AST implementation is in C;
- AST report APIs are C ABI surfaces;
- report generation should remain bounded and deterministic;
- the current C workflow can validate report invariants;
- no dynamic runtime should be required.

Public API addition

Add to:

```
include/latticra/l_ui_parser.h
#define LATTICRA_L_UI_AST_DETAILED_REPORT_MAX 16384u

latticra_status_t latticra_l_ui_ast_detailed_report(
    const latticra_l_ui_ast_result_t *ast,
    char *buffer,
    size_t buffer_len);
```

This must be additive. Existing AST, parser, diagnostics, location, and source-span APIs must remain unchanged.

Module shape

Update:

```
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
tests/l_ui_parser_ast_detailed_report_invariants.c
```

```
scripts/test-l-ui-ast-detailed-report.sh
.github/workflows/c.yml
```

Detailed report generation should live in src/l_ui_parser_ast.c with the compact AST report.

Report capacity

Use:

```
LATTICRA_L_UI_AST_DETAILED_REPORT_MAX = 16384
```

The report function must return LATTICRA_STATUS_BUFFER_TOO_SMALL when the provided buffer cannot hold the full deterministic report.

Failed-parse behavior

The first implementation should render a compact failed-parse detailed report instead of rejecting report generation.

Shape:

```
L-UI AST DETAILED REPORT
parse_error=&lt;error-label&gt;
rail_count=0
field_count=0
text_count=0
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

This keeps report generation deterministic for failed parses while not emitting partial AST nodes.

Escaping policy

The first implementation may render current fixture text literally.

Allowed literal values:

```
Latticra / Nucleus Preview / effect-bound
preview-only no-live-movement no-host-effect no-external-effect
operator-visible Nucleus preview report
```

The implementation must not introduce a broader escaping model yet. A future escaping contract should define broader string rendering.

Exact section order

A successful detailed report must render sections in this order:

```
1. top-level summary
2. card section
3. rail sections in AST order
4. field sections in AST order
5. text sections in AST order
```

No unused capacity slots may be printed.

Top-level summary format

```
L-UI AST DETAILED REPORT
card=NucleusPreview
purpose=operator-visible Nucleus preview report
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Card section format

```
[card]
kind=card
name=NucleusPreview
purpose=operator-visible Nucleus preview report
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Rail section format

For each populated rail:

```
[rail &lt;index&gt;]
kind=rail
name=&lt;rail-name&gt;
first_field_index=&lt;n&gt;
field_count=&lt;n&gt;
first_text_index=&lt;n&gt;
text_count=&lt;n&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Required order:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Field section format

For each populated field:

```
[field &lt;index&gt;]
kind=field
name=&lt;field-name&gt;
binding=&lt;binding-path&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

```
binding_span_start_offset=&lt;n&gt;
binding_span_end_offset=&lt;n&gt;
binding_span_start_line=&lt;n&gt;
binding_span_start_column=&lt;n&gt;
binding_span_end_line=&lt;n&gt;
binding_span_end_column=&lt;n&gt;
```

Text section format

For each populated text node:

```
[text &lt;index&gt;]
kind=text
value=&lt;literal-text&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Helper plan

The implementation may add private helpers in src/l_ui_parser_ast.c:

```
append_text
append_span_fields
append_card_section
append_rail_section
append_field_section
append_text_section
append_failed_parse_report
```

Helpers should remain private and bounded.

Compatibility expectations

The detailed report implementation must not change:

```
latticra_l_ui_ast_report
latticra_l_ui_parse_ast
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
latticra_l_ui_diagnostic_report
```

Existing parser, diagnostic, source-span, location, and compact AST tests must continue to pass.

No-effect preservation

Detailed reporting must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
detailed_report_contains_title
detailed_report_contains_card_section
detailed_report_contains_all_rails
detailed_report_contains_all_fields
detailed_report_contains_all_text_nodes
detailed_report_preserves_rail_order
detailed_report_preserves_field_order
detailed_report_preserves_text_order
detailed_report_includes_card_span
detailed_report_includes_rail_spans
detailed_report_includes_field_spans
detailed_report_includes_binding_spans
detailed_report_includes_text_spans
detailed_report_preserves_no_effect_flags
```

```
detailed_report_is_deterministic
detailed_report_rejects_bad_arguments
detailed_report_rejects_small_buffers
detailed_report_omits_unused_capacity_slots
detailed_report_handles_failed_parse
```

Test file plan

Add:

```
tests/l_ui_parser_ast_detailed_report_invariants.c
scripts/test-l-ui-ast-detailed-report.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_AST_DETAILED_REPORT_CONTRACT.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
```

Forbidden implementation behavior

The detailed report implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- emit memory addresses.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-ast-detailed-report-implementation-plan.sh
```

The guard is static. It does not implement detailed AST reporting.

Implementation gate

Detailed AST report implementation code may be added only after this plan is merged.

Non-claims

This document does not implement detailed AST reporting, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_AST_ESCAPED_STRING_REPORT_CONTRACT.md

Source title: Latticra L-UI AST Escaped String Report Contract

Lines: 309 | Words: 926 | SHA-256: 6c0a5bd771e1a2a1f803be5f57e4963e557ae285d3f43f3f19dcb21f35ff7357

Latticra L-UI AST Escaped String Report Contract

Status: escaped string report contract Scope: stable escaping rules for detailed AST report string values before broader L-UI text values are accepted.

Purpose

This document defines the escaped string report contract for L-UI AST reports.

The detailed AST report now includes additive escaped fields for current fixture text values. Before broader L-UI text values are accepted, these rules define stable escaping behavior for newlines, tabs, quotes, backslashes, control bytes, and non-printable bytes.

The implementation is documented separately in L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md.

Current boundary

The current detailed AST report provides deterministic metadata for:

```
card
rails
fields
text nodes
source spans
binding spans
literal purpose/text fields
escaped purpose/text fields
no-effect flags
```

This contract does not add:

```
new parser grammar
new accepted text values
renderer integration
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Escaping purpose

Escaped string reporting makes text output:

- deterministic;
- single-line when required;
- safe for logs;
- safe for fixture comparisons;

- independent of terminal behavior;
- independent of platform locale;
- unambiguous for quotes and backslashes;
- readable for common whitespace;
- explicit for control and non-printable bytes.

Escaping remains metadata-only.

Escaped fields

Escaping applies to report values that may contain arbitrary source text:

```
card.purpose
text.value
```

It may later apply to labels or bindings if broader syntax permits non-identifier bytes, but current labels and bindings remain controlled identifiers.

Report key naming

The first implementation adds escaped variants:

```
purpose_escaped=&lt;escaped-string&gt;
value_escaped=&lt;escaped-string&gt;
```

Existing literal keys remain:

```
purpose=&lt;literal-purpose&gt;
value=&lt;literal-text&gt;
```

Required escape sequences

Escaped string reporting supports:

```
| Input byte | Report text |
| --- | --- |
| newline LF `0x0A` | `\n` |
| carriage return CR `0x0D` | `\r` |
| horizontal tab `0x09` | `\t` |
| double quote `0x22` | `\&quot;` |
| backslash `0x5C` | `\\` |
| NUL `0x00` | `\x00` |
| other control bytes `0x01`-`0x1F` | `\xNN` |
| DEL `0x7F` | `\x7F` |
| non-ASCII bytes `0x80`-`0xFF` | `\xNN` |
```

Printable ASCII bytes from 0x20 through 0x7E, except double quote and backslash, may be emitted literally.

Hex format

Hex escapes must use uppercase hexadecimal digits:

```
\x00
\x09
\x1F
\x7F
\x80
\xFF
```

Lowercase hex output is not allowed.

Determinism rules

Escaped string output must be deterministic.

Rules:

1. The same byte sequence must always produce the same escaped text.
2. Escaped output must not depend on platform locale.
3. Escaped output must not depend on terminal capabilities.
4. Escaped output must not include memory addresses.
5. Escaped output must not include raw control bytes.
6. Escaped output must not include raw non-ASCII bytes.
7. Escaped output must be NUL-terminated in C buffers.
8. Escaped output must fail with a stable status when the destination buffer is too small.

Byte-oriented rule

Escaping is byte-oriented, not Unicode-codepoint-oriented.

The implementation treats input text as bytes and renders non-ASCII bytes with \xNN escapes. A future Unicode display contract may define higher-level behavior later.

Buffer sizing rule

Worst-case escaped size is four output characters per input byte plus the terminating NUL:

```
input_len * 4 + 1
```

The implementation uses private bounded buffers sized from the AST purpose/text capacities.

Helper function

The implementation uses a private helper:

```
escape_report_string
```

Helper shape:

```
static latticra_status_t escape_report_string(
    const char *input,
    size_t input_len,
    char *output,
    size_t output_len);
```

The helper remains private.

Detailed report integration

The detailed report renders:

```
purpose_escaped=&lt;escaped-purpose&gt;
value_escaped=&lt;escaped-text&gt;
```

Placement:

```
[card]
[text &lt;index&gt;]
```

Failed parse behavior

Escaping does not change failed-parse detailed report behavior.

Failed parse detailed reports continue to render:

```
parse_error=&lt;error-label&gt;
rail_count=0
field_count=0
text_count=0
```

Escaped fields do not appear in failed-parse reports.

Compatibility rule

Escaped string reporting is additive.

It does not change:

```
latticra_l_ui_ast_report
latticra_l_ui_parse_ast
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
latticra_l_ui_diagnostic_report
latticra_l_ui_ast_detailed_report existing required fields
```

No-effect rule

Escaped string reporting is metadata output only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Implementation gate

Escaped string report implementation required a separate implementation plan defining:

1. public API decision;
2. private helper shape;
3. destination buffer sizes;
4. exact escaped report fields;
5. exact report placement;
6. too-small behavior;
7. test file names;
8. exact invariant tests;
9. compatibility expectations.

That plan is recorded in L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION_PLAN.md.

Test list

Implementation tests verify:

```
escape_preserves_printable_ascii
escape_newline_as_backslash_n
escape_carriage_return_as_backslash_r
escape_tab_as_backslash_t
escape_quote_as_backslash_quote
escape_backslash_as_double_backslash
escape_nul_as_hex_00
escape_control_bytes_as_uppercase_hex
escape_del_as_uppercase_hex
escape_non_ascii_bytes_as_uppercase_hex
escape_rejects_small_buffers
escape_is_deterministic
detailed_report_contains_escaped_purpose
detailed_report_contains_escaped_text_values
detailed_report_escaped_fields_are_additive
detailed_report_escape_preserves_no_effect_flags
detailed_report_escape_does_not_change_failed_parse_report
```

Forbidden behavior

Escaped string implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- emit memory addresses;
- emit raw control bytes in escaped fields;
- emit raw non-ASCII bytes in escaped fields.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-ast-escaped-string-report-contract.sh
```

The guard is static. It validates the contract text.

Non-claims

This document does not broaden accepted L-UI text syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md

Source title: Latticra L-UI AST Escaped String Report Implementation

Lines: 207 | Words: 621 | SHA-256: 45b7595c90c514295231f2be0a72760ce6ba17a2b8dfc167731403584ff73f09

Latticra L-UI AST Escaped String Report Implementation

Status: initial implementation contract Scope: byte-oriented escaped string fields for detailed AST report metadata.

Purpose

The L-UI AST Escaped String Report implementation adds additive escaped string fields to the detailed AST report.

It keeps existing literal fields intact and adds escaped fields for report values that may later contain broader text bytes.

Implementation files

```
src/l_ui_parser_ast.c
tests/l_ui_ast_escaped_string_report_invariants.c
scripts/test-l-ui-ast-escaped-string-report.sh
```

Public API impact

No new public function is added.

Escaping is integrated into the existing detailed report API:

```
latticra_l_ui_ast_detailed_report
```

Private helper

The implementation adds a private helper in src/l_ui_parser_ast.c:

```
escape_report_string
```

The helper is byte-oriented and remains private.

Escaped report fields

The detailed AST report now includes additive escaped fields:

```
purpose_escaped=&lt;escaped-purpose&gt;
value_escaped=&lt;escaped-text&gt;
```

Existing literal fields remain:

```
purpose=&lt;literal-purpose&gt;
value=&lt;literal-text&gt;
```

Field placement

Escaped fields are placed as follows:

```
[card]
purpose=&lt;literal-purpose&gt;
purpose_escaped=&lt;escaped-purpose&gt;

[text &lt;index&gt;]
value=&lt;literal-text&gt;
value_escaped=&lt;escaped-text&gt;
```

Escaped fields are not added to rail sections, field sections, or failed-parse reports.

Escape rules

The helper implements the required byte escape table:

```
| Input byte | Report text |
| --- | --- |
| newline LF `0x0A` | `\n` |
| carriage return CR `0x0D` | `\r` |
| horizontal tab `0x09` | `\t` |
| double quote `0x22` | `\&quot;` |
| backslash `0x5C` | `\\` |
| NUL `0x00` | `\x00` |
| other control bytes `0x01`-`0x1F` | `\xNN` |
| DEL `0x7F` | `\x7F` |
| non-ASCII bytes `0x80`-`0xFF` | `\xNN` |
```

Printable ASCII bytes from 0x20 through 0x7E, except double quote and backslash, are emitted literally.

Hex digits are uppercase:

```
0123456789ABCDEF
```

Buffer behavior

If the output buffer is too small, escaping returns:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

When possible, the destination buffer is cleared to an empty string on failure.

NUL behavior

The private helper supports NUL bytes through explicit input lengths.

Existing AST string fields are still C strings, so detailed report integration uses strlen(...) for current AST values. Dedicated tests validate explicit NUL byte escaping through the private helper test path.

Failed parse behavior

Failed-parse detailed reports are unchanged.

They continue to render:

```
parse_error=&lt;error-label&gt;
rail_count=0
field_count=0
text_count=0
```

No escaped fields appear in failed-parse reports.

Compatibility

This implementation does not change:

```
latticra_l_ui_ast_report
latticra_l_ui_parse_ast
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
latticra_l_ui_diagnostic_report
latticra_l_ui_ast_detailed_report existing required fields
```

No-effect boundary

Escaped string reporting preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-ast-escaped-string-report.sh
```

The main C workflow runs this check after the escaped string report implementation-plan guard.

Required invariants

The escaped string report tests verify:

```
escape_preserves_printable_ascii
escape_newline_as_backslash_n
escape_carriage_return_as_backslash_r
escape_tab_as_backslash_t
escape_quote_as_backslash_quote
escape_backslash_as_double_backslash
escape_nul_as_hex_00
escape_control_bytes_as_uppercase_hex
escape_del_as_uppercase_hex
escape_non_ascii_bytes_as_uppercase_hex
escape_rejects_small_buffers
escape_is_deterministic
detailed_report_contains_escaped_purpose
detailed_report_contains_escaped_text_values
detailed_report_escaped_fields_are_additive
```

```
detailed_report_escape_preserves_no_effect_flags
detailed_report_escape_does_not_change_failed_parse_report
```

The implementation also verifies mutated AST purpose and text values with newline, carriage return, tab, quote, and backslash bytes.

Current evidence level

This implementation is an L2 tested byte-oriented escaping model for detailed AST report string fields.

It is not a broader text grammar, Unicode display model, renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after escaped string reporting is:

```
L-UI AST source-backed text extraction contract
```

That future work should define how AST text values are extracted from source instead of being fixed fixture metadata.

Non-claims

This document and implementation do not broaden accepted L-UI text syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI AST Escaped String Report Implementation Plan

Lines: 323 | Words: 998 | SHA-256: 1260ac191f37a1eb0953806cfed7d1d6c8f6dcbe343a3ba1ed7c6c5d79c86622

Latticra L-UI AST Escaped String Report Implementation Plan

Status: implementation planning contract Scope: private escape helper shape, escaped report fields, destination sizing, too-small behavior, exact tests, and compatibility expectations before escaped string report implementation.

Purpose

This document defines the implementation plan for escaped string reporting in the L-UI AST detailed report.

The escaped string report contract is already merged and guarded. This plan decides the private helper shape, exact escaped fields, placement in the detailed AST report, destination sizes, too-small behavior, test files, exact tests, and compatibility expectations before code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_AST_ESCAPED_STRING_REPORT_CONTRACT.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
```

Those files remain the source of truth for detailed AST reporting, text metadata, source spans, and no-effect boundaries.

Implementation language decision

Escaped string reporting should be implemented in C.

Reason:

- the AST detailed report is implemented in C;
- report generation is part of the C public API surface;
- escaping should remain bounded, deterministic, and byte-oriented;
- the current C workflow can validate all escaping invariants;
- no dynamic runtime should be required.

Public API decision

No new public function is needed for the first escaping implementation.

Escaping should be added to the existing detailed report API:

```
latticra_l_ui_ast_detailed_report
```

The implementation should add escaped fields to the detailed report output while preserving existing required literal fields.

Private helper shape

Add a private helper in:

```
src/l_ui_parser_ast.c
```

Proposed helper:

```
static latticra_status_t escape_report_string(
    const char *input,
    size_t input_len,
    char *output,
    size_t output_len);
```

The helper must remain private.

Destination sizes

Add private bounded local buffers for escaped strings in src/l_ui_parser_ast.c.

Required destination sizes:

```
escaped purpose buffer &gt;= LATTICRA_L_UI_AST_PURPOSE_MAX * 4 + 1
escaped text buffer &gt;= LATTICRA_L_UI_AST_PURPOSE_MAX * 4 + 1
```

The first implementation may use private macros:

```
LATTICRA_L_UI_AST_ESCAPED_PURPOSE_MAX
LATTICRA_L_UI_AST_ESCAPED_TEXT_MAX
```

These do not need to be public unless future callers need direct escaping support.

Escaped fields

Add escaped fields to the detailed report only.

Escaped source fields:

```
card.purpose
text.value
```

Report keys:

```
purpose_escaped=&lt;escaped-purpose&gt;
value_escaped=&lt;escaped-text&gt;
```

The existing literal fields should remain:

```
purpose=&lt;literal-purpose&gt;
```

```
value=&lt;literal-text&gt;
```

This makes the first implementation additive.

Field placement

Add escaped fields in these sections:

```
[card]
[text &lt;index&gt;]
```

Placement rules:

1. In [card], print purpose_escaped immediately after purpose.
2. In [text <index>], print value_escaped immediately after value.
3. Do not add escaped fields to rail sections.
4. Do not add escaped fields to field sections.
5. Do not add escaped fields to failed-parse reports.

Escape rules

The helper must implement the contract escape table:

```
| Input byte | Report text |
| --- | --- |
| newline LF `0x0A` | `\n` |
| carriage return CR `0x0D` | `\r` |
| horizontal tab `0x09` | `\t` |
| double quote `0x22` | `\&quot;` |
| backslash `0x5C` | `\\` |
| NUL `0x00` | `\x00` |
| other control bytes `0x01`-`0x1F` | `\xNN` |
| DEL `0x7F` | `\x7F` |
| non-ASCII bytes `0x80`-`0xFF` | `\xNN` |
```

Printable ASCII bytes from 0x20 through 0x7E, except double quote and backslash, should be emitted literally.

Hex rule

Hex escapes must use uppercase hexadecimal digits:

```
0123456789ABCDEF
```

Lowercase hexadecimal output is not allowed.

Byte-oriented rule

Escaping is byte-oriented, not Unicode-codepoint-oriented.

Input should be treated as unsigned char bytes for classification and hex output.

Too-small behavior

If the destination buffer cannot hold the full escaped string and terminating NUL, escape_report_string must return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

The helper should clear the output buffer to an empty string when possible.

latticra_l_ui_ast_detailed_report must propagate this status if escaping fails.

NUL behavior

The first implementation should support NUL bytes in the private helper test path through explicit input_len.

For existing AST strings, the helper will receive strlen(...) because current AST text fields are C strings. Dedicated helper tests should still validate NUL handling with byte arrays.

Failed parse behavior

Escaping must not change failed-parse detailed report behavior.

Failed parse detailed reports should continue to render:

```
parse_error=&lt;error-label&gt;
rail_count=0
field_count=0
text_count=0
```

No escaped fields should appear in failed-parse reports.

Compatibility expectations

Escaped string reporting must be additive.

It must not change:

```
latticra_l_ui_ast_report
latticra_l_ui_parse_ast
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
latticra_l_ui_diagnostic_report
latticra_l_ui_ast_detailed_report existing required fields
```

Existing parser, diagnostic, source-span, AST, compact report, and detailed report tests must continue to pass.

No-effect preservation

Escaped string reporting must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
escape_preserves_printable_ascii
escape_newline_as_backslash_n
escape_carriage_return_as_backslash_r
escape_tab_as_backslash_t
escape_quote_as_backslash_quote
escape_backslash_as_double_backslash
escape_nul_as_hex_00
escape_control_bytes_as_uppercase_hex
escape_del_as_uppercase_hex
escape_non_ascii_bytes_as_uppercase_hex
escape_rejects_small_buffers
escape_is_deterministic
detailed_report_contains_escaped_purpose
detailed_report_contains_escaped_text_values
detailed_report_escaped_fields_are_additive
detailed_report_escape_preserves_no_effect_flags
detailed_report_escape_does_not_change_failed_parse_report
```

Test file plan

Add:

```
tests/l_ui_ast_escaped_string_report_invariants.c
scripts/test-l-ui-ast-escaped-string-report.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_CONTRACT.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
```

Forbidden implementation behavior

The escaped string implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- emit memory addresses;
- emit raw control bytes in escaped fields;
- emit raw non-ASCII bytes in escaped fields.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-ast-escaped-string-report-implementation-plan.sh
```

The guard is static. It does not implement escaped string reporting.

Implementation gate

Escaped string report implementation code may be added only after this plan is merged.

Non-claims

This document does not implement escaped string reporting, broaden accepted L-UI text syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md

Source title: Latticra L-UI AST Length-Carrying String Storage Contract

Lines: 397 | Words: 1310 | SHA-256: 6d1ba0e65b3a364051453cb3fbfd9ffc190c32b6ff748194c457fe6ffd6b6020

Latticra L-UI AST Length-Carrying String Storage Contract

Status: AST string storage contract Scope: future explicit decoded byte lengths for L-UI AST purpose and text values before AST string storage is extended.

Purpose

This document defines the contract for length-carrying AST string storage in L-UI.

Current AST string storage uses fixed-size NUL-terminated C strings:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

That storage model cannot represent decoded NUL bytes or safely report byte-oriented decoded values that contain embedded NUL. This contract defines the future public storage shape, compatibility rules, report rules, decoding relationship, and no-effect boundary before implementation code is added.

This document does not implement length-carrying AST strings.

The implementation plan is documented separately in L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION_PLAN.md.

Relationship to previous work

This contract depends on:

```
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
```

Those files remain the current source of truth for AST shape, source-backed value materialization, escaped report fields, string-literal escape decoding, parser-level invalid escape diagnostics, and no-effect behavior.

Current boundary

The current system provides:

```
source-backed AST purpose and text values
accepted string-literal escape decoding
parser-level invalid string escape diagnostics
NUL rejection for decoded and literal NUL bytes
fixed-size NUL-terminated AST string buffers
escaped detailed report fields
source-oriented spans
no-effect flags
```

This contract does not add:

```
length-carrying AST string implementation
decoded NUL acceptance
literal NUL acceptance
new accepted escape sequences
Unicode display behavior
renderer behavior
interactive UI behavior
command behavior
Nucleus task handling
live movement
```

```
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Storage purpose

Length-carrying AST string storage should allow AST string values to be treated as byte sequences with explicit decoded lengths while preserving existing C-string compatibility for values that do not contain NUL bytes.

The goal is to make future decoded values unambiguous for:

```
AST consumers
escaped detailed report fields
capacity checks
string-literal escape decoding
future NUL-capable storage work
```

Public API shape

A future implementation should extend public AST structs in:

```
include/latticra/l_ui_parser.h
```

by adding explicit decoded byte length fields:

```
latticra_l_ui_ast_card_t.purpose_len
latticra_l_ui_ast_text_t.value_len
```

Proposed struct relationship:

```
char purpose[LATTICRA_L_UI_AST_PURPOSE_MAX];
size_t purpose_len;

char value[LATTICRA_L_UI_AST_PURPOSE_MAX];
size_t value_len;
```

The length fields store decoded byte counts and do not include any trailing compatibility NUL byte.

Compatibility rule

The existing fixed-size character buffers should remain public fields.

For decoded strings that do not contain NUL bytes:

```
purpose_len == strlen(purpose)
value_len == strlen(value)
```

For future decoded strings that contain NUL bytes, purpose_len and value_len would be the authoritative lengths. Existing C-string fields would remain compatibility buffers but would not be sufficient to inspect the full decoded value through strlen.

NUL byte rule

This contract does not make decoded NUL bytes accepted immediately.

The current parser-level diagnostics still reject:

```
\x00 -&gt; LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Decoded NUL bytes may be accepted only after:

1. length-carrying AST storage is implemented;
2. report escaping uses explicit lengths;
3. parser string escape diagnostics are updated by a separate contract;
4. string-literal escape decoding behavior is updated by a separate contract.

Source-backed extraction rule

Source-backed extraction should continue to target:

```
purpose &quot;...&quot; -&gt; ast.card.purpose + ast.card.purpose_len
first text &quot;...&quot; -&gt; ast.texts[0].value + ast.texts[0].value_len
second text &quot;...&quot; -&gt; ast.texts[1].value + ast.texts[1].value_len
```

Extraction remains source-backed, deterministic, bounded, and no-effect.

Decoding relationship

Accepted string-literal escapes should continue to decode exactly as they do now:

```
\\
\&quot;
\n
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

For all currently accepted decoded values that do not include NUL, the future length field should match the decoded byte length.

This contract does not broaden the accepted escape set.

Capacity rule

Storage capacity remains bounded by:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
```

for both purpose and text values in the current AST model.

A decoded value fits only when:

```
decoded_len &lt; destination_buffer_len
```

because the compatibility buffer still needs a trailing NUL byte.

If decoded output is greater than or equal to the destination buffer length, parser validation should continue to report:

```
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE
```

until a separate capacity expansion contract exists.

Report behavior

Detailed reports should add explicit length fields:

```
purpose_len=&lt;decoded-purpose-byte-length&gt;
value_len=&lt;decoded-text-byte-length&gt;
```

Existing report fields should remain:

```
purpose=&lt;C-string-compatible-purpose&gt;
purpose_escaped=&lt;length-aware-report-safe-purpose&gt;
value=&lt;C-string-compatible-text&gt;
value_escaped=&lt;length-aware-report-safe-text&gt;
```

For values without NUL bytes, these fields should remain compatible with current output.

For any future values with embedded NUL bytes, the stable assertion target should be:

```
purpose_len
purpose_escaped
value_len
value_escaped
```

not raw purpose= or value= C-string fields.

Escaped report rule

The escaped report helper should become length-aware before decoded NUL bytes are accepted.

A future implementation should prefer a helper shape similar to:

```
static latticra_status_t escape_report_bytes(
    const char *input,
    size_t input_len,
    char *output,
    size_t output_len);
```

The helper must emit report-safe byte escapes for control bytes, quotes, backslashes, DEL, non-ASCII bytes, and any future embedded NUL bytes.

Source-span behavior

Source spans remain source-oriented.

Text spans continue to cover the source value range between quotes, not decoded output byte ranges.

For source escapes, spans include the source escape bytes. For example, \n covers two source bytes even though the decoded value length increases by one byte.

This contract does not add a public purpose_span field.

Parser diagnostics compatibility

Current parser diagnostics remain valid:

```
LUI0019 invalid_string_escape
LUI0020 invalid_hex_escape
LUI0021 unterminated_escape
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
LUI0024 string_value_too_large
```

The first length-carrying storage implementation must not remove NUL rejection diagnostics.

A later decoded-NUL acceptance contract may change LUI0022 behavior, but not as part of this storage contract.

AST compatibility behavior

Failed parse behavior remains unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Accepted sources should continue to produce the same AST counts and string values, plus explicit lengths.

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect rule

Length-carrying AST string storage is metadata-only.

It must preserve:

```
no_effect=1
```

```
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Implementation gate

Length-carrying AST string storage implementation must not begin until a separate implementation plan defines:

1. public struct field placement;
2. default initialization rules;
3. source-backed extraction length assignment;
4. string-literal decode length assignment;
5. report field additions;
6. length-aware escaped report helper shape;
7. parser diagnostics compatibility;
8. capacity behavior;
9. exact test file names;
10. exact invariant tests;
11. compatibility expectations.

That plan is recorded in L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION_PLAN.md.

Future test list

A future implementation plan should include tests for:

```
ast_string_storage_sets_purpose_len
ast_string_storage_sets_text_value_len
ast_string_storage_len_matches_strlen_for_non_nul_values
ast_string_storage_len_tracks_decoded_newline
ast_string_storage_len_tracks_decoded_tab
ast_string_storage_len_tracks_decoded_high_byte_hex
ast_string_storage_preserves_existing_c_string_fields
ast_string_storage_updates_detailed_report_lengths
ast_string_storage_escaped_report_uses_explicit_lengths
ast_string_storage_still_rejects_decoded_nul_until_acceptance_contract
ast_string_storage_still_rejects_literal_nul_until_acceptance_contract
ast_string_storage_still_rejects_oversized_decoded_output
ast_string_storage_preserves_source_spans
ast_string_storage_preserves_no_effect_flags
ast_string_storage_does_not_change_existing_diagnostic_codes
ast_string_storage_does_not_change_failed_parse_report
ast_string_storage_is_deterministic
```

Forbidden behavior

A future length-carrying AST string storage implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;

- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- accept decoded NUL bytes without a separate acceptance contract;
- accept literal NUL bytes without a separate acceptance contract;
- broaden accepted string escapes;
- remove existing C-string fields;
- silently truncate decoded values;
- make raw purpose= or value= report fields the assertion target for embedded NUL bytes.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-ast-length-carrying-string-storage-contract.sh
```

The guard is static. It does not implement length-carrying AST string storage.

Non-claims

This document does not implement length-carrying AST strings, accept decoded NUL bytes, accept literal NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md

Source title: Latticra L-UI AST Length-Carrying String Storage Implementation

Lines: 346 | Words: 931 | SHA-256: dc96a9cd281c7fb0baa359f771036a45335b458dadaef4350d394bbdfcf3067b

Latticra L-UI AST Length-Carrying String Storage Implementation

Status: initial implementation contract Scope: explicit decoded byte lengths for L-UI AST purpose and text values.

Purpose

The L-UI AST Length-Carrying String Storage implementation adds explicit decoded byte lengths for source-backed AST string values while preserving the existing C-string compatibility buffers.

The implementation makes decoded byte lengths available for:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

without accepting decoded NUL bytes, broadening accepted escapes, or changing parser execution boundaries.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
tests/l_ui_ast_length_carrying_string_storage_invariants.c
scripts/test-l-ui-ast-length-carrying-string-storage.sh
```

Related active files:

```
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
tests/l_ui_string_literal_escape_invariants.c
scripts/test-l-ui-string-literal-escape.sh
```

Public struct additions

The implementation extends public AST structs with:

```
latticra_l_ui_ast_card_t.purpose_len
latticra_l_ui_ast_text_t.value_len
```

Struct relationship:

```
char purpose[LATTICRA_L_UI_AST_PURPOSE_MAX];
size_t purpose_len;

char value[LATTICRA_L_UI_AST_PURPOSE_MAX];
size_t value_len;
```

The length fields store decoded byte counts and do not include the trailing compatibility NUL byte.

Public API compatibility

No new public function is added.

Existing public C-string fields remain:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

For decoded strings that do not contain NUL bytes:

```
purpose_len == strlen(purpose)
value_len == strlen(value)
```

Default initialization

ast_default initializes all string lengths to zero:

```
ast.card.purpose_len = 0
ast.texts[index].value_len = 0
```

Failed parse AST results keep all AST string lengths at zero.

Source-backed extraction length assignment

Source-backed extraction now assigns decoded lengths while materializing AST values:

```
purpose &quot;...&quot; -&gt; ast.card.purpose + ast.card.purpose_len
first text &quot;...&quot; -&gt; ast.texts[0].value + ast.texts[0].value_len
second text &quot;...&quot; -&gt; ast.texts[1].value + ast.texts[1].value_len
```

The decode path reports decoded byte counts from the same output index used to populate the destination buffer.

Decode helper behavior

The private string-literal decode helper now reports decoded length through an output pointer:

```
size_t *decoded_len
```

decoded_len is set to zero before decoding and set to the decoded byte count only on successful decoding.

Accepted escape decoding remains unchanged:

```
\\
\&quot;
\n
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

Capacity behavior

Capacity remains bounded by:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
```

for purpose and text values.

A decoded value fits only when:

```
decoded_len &lt; destination_len
```

because the compatibility C string still needs a trailing NUL byte.

Oversized decoded values continue to be rejected by parser diagnostics as:

```
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE
```

NUL behavior

This implementation does not accept decoded or literal NUL bytes.

Parser-level diagnostics continue to reject:

```
\x00 -&gt; LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

The decoded-NUL acceptance contract is recorded in:

```
L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md
```

A later decoded-NUL acceptance implementation may change escaped \x00 behavior, but not literal source NUL rejection.

Detailed report behavior

Detailed AST reports now include explicit length fields:

```
purpose_len=&lt;decoded-purpose-byte-length&gt;
value_len=&lt;decoded-text-byte-length&gt;
```

Card section placement:

```
[card]
kind=card
name=&lt;name&gt;
purpose=&lt;purpose&gt;
purpose_len=&lt;purpose_len&gt;
purpose_escaped=&lt;purpose_escaped&gt;
```

Text section placement:

```
[text N]
kind=text
value=&lt;value&gt;
value_len=&lt;value_len&gt;
value_escaped=&lt;value_escaped&gt;
```

Existing field names remain stable.

Length-aware escaped report behavior

The escaped report helper is byte-oriented and length-aware:

```
escape_report_bytes(input, input_len, output, output_len)
```

Detailed report callers pass explicit AST lengths instead of strlen for purpose and text values.

For current non-NUL values, escaped report behavior remains compatible with previous output.

Compact AST report compatibility

The compact AST report remains unchanged:

```
latticra_l_ui_ast_report
```

The detailed AST report gains length fields.

Failed-parse detailed reports remain unchanged and do not include string length fields.

Parser diagnostics compatibility

The implementation does not change existing parser diagnostics.

These diagnostics remain active:

```
LUI0019 invalid_string_escape
LUI0020 invalid_hex_escape
LUI0021 unterminated_escape
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
LUI0024 string_value_too_large
```

Decoded-NUL and literal-NUL rejection remain active.

Source-span behavior

Source spans remain source-oriented.

Text spans continue to cover the source value range between quotes, not decoded output byte ranges.

For example, source \n covers two source bytes while the decoded AST value length is one byte for that escape.

No public purpose_span field is added.

Compatibility

The implementation does not change:

```
accepted string-literal escape decoding
parser-level invalid string escape diagnostics
existing diagnostic codes LUI0000 through LUI0024
existing parser error labels for current errors
latticra_l_ui_ast_report existing required fields
latticra_l_ui_diagnostic_report existing required fields
existing accepted fixture summary counts
failed-parse detailed report behavior
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect boundary

Length-carrying AST string storage is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
```

```
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-ast-length-carrying-string-storage.sh
```

The main C workflow runs this check after the length-carrying string storage implementation-plan guard.

Required invariants

The length-carrying AST string storage tests verify:

```
ast_string_storage_sets_purpose_len
ast_string_storage_sets_text_value_len
ast_string_storage_len_matches_strlen_for_non_nul_values
ast_string_storage_len_tracks_decoded_newline
ast_string_storage_len_tracks_decoded_tab
ast_string_storage_len_tracks_decoded_high_byte_hex
ast_string_storage_preserves_existing_c_string_fields
ast_string_storage_updates_detailed_report_lengths
ast_string_storage_escaped_report_uses_explicit_lengths
ast_string_storage_still_rejects_decoded_nul_until_acceptance_contract
ast_string_storage_still_rejects_literal_nul_until_acceptance_contract
ast_string_storage_still_rejects_oversized_decoded_output
ast_string_storage_preserves_source_spans
ast_string_storage_preserves_no_effect_flags
ast_string_storage_does_not_change_existing_diagnostic_codes
ast_string_storage_does_not_change_failed_parse_report
ast_string_storage_is_deterministic
```

Current evidence level

This implementation is an L2 tested AST string metadata model for explicit decoded byte lengths.

It is not decoded-NUL acceptance, literal-NUL acceptance, a Unicode display model, renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, sandbox, or operating system.

Next implementation step

The decoded-NUL acceptance contract is recorded in:

```
L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md
```

The next implementation candidate after that contract is:

```
L-UI decoded-NUL acceptance implementation plan
```

That future work should define parser validation changes, decode helper changes, AST storage expectations, report expectations, diagnostic compatibility behavior, source-span expectations, and exact invariant tests before accepting escaped \x00.

Non-claims

This document and implementation do not accept decoded NUL bytes, accept literal NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI AST Length-Carrying String Storage Implementation Plan

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Latticra L-UI AST Length-Carrying String Storage Implementation Plan

Status: implementation planning contract Scope: public struct field placement, default initialization, source-backed extraction length assignment, string-literal decode length assignment, report field additions, length-aware escaped report helper shape, parser diagnostics compatibility, capacity behavior, exact test files, and exact invariant tests before length-carrying AST string storage code.

Purpose

This document defines the implementation plan for length-carrying AST string storage in L-UI.

The AST length-carrying string storage contract is already merged and guarded. This plan decides the exact public struct changes, default initialization rules, private helper changes, AST value materialization rules, detailed report additions, escaped report helper shape, compatibility behavior, and test coverage before implementation code is added.

This document does not implement length-carrying AST strings.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for AST shape, source-backed extraction, escaped detailed report fields, string-literal escape decoding, parser-level invalid escape diagnostics, and no-effect behavior.

Implementation language decision

Length-carrying AST string storage should be implemented in C.

Reason:

- the L-UI parser and AST builder are implemented in C;
- AST structs are exposed through a C public API surface;
- report generation is implemented in C;
- storage and report changes must remain bounded, deterministic, and no-effect;
- no dynamic runtime should be required.

Public struct field placement

Extend public AST structs in:

```
include/latticra/l_ui_parser.h
```

Add purpose_len directly after the existing purpose buffer:

```
typedef struct {
    char name[LATTICRA_L_UI_AST_LABEL_MAX];
    char purpose[LATTICRA_L_UI_AST_PURPOSE_MAX];
```

```
    size_t purpose_len;
    char effect[LATTICRA_L_UI_LABEL_MAX];
    char boundary[LATTICRA_L_UI_LABEL_MAX];
    latticra_l_ui_source_span_t span;
    size_t rail_count;
    size_t field_count;
    size_t text_count;
} latticra_l_ui_ast_card_t;
```

Add value_len directly after the existing text value buffer:

```
typedef struct {
    char value[LATTICRA_L_UI_AST_PURPOSE_MAX];
    size_t value_len;
    latticra_l_ui_source_span_t span;
} latticra_l_ui_ast_text_t;
```

The length fields store decoded byte counts and do not include the trailing compatibility NUL byte.

Public API compatibility decision

No new public function is required for the first implementation.

Existing public C-string fields remain:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

The new length fields add metadata but do not remove or rename existing fields.

Default initialization rules

Update ast_default in:

```
src/l_ui_parser_ast.c
```

Default values:

```
ast.card.purpose[0] = &#x27;\0&#x27;;
ast.card.purpose_len = 0
ast.texts[index].value[0] = &#x27;\0&#x27;;
ast.texts[index].value_len = 0
```

Failed parse AST results must keep all AST string lengths at zero.

Source-backed extraction length assignment

Source-backed extraction should assign decoded lengths while materializing AST values:

```
purpose &quot;...&quot; -&gt; ast.card.purpose + ast.card.purpose_len
first text &quot;...&quot; -&gt; ast.texts[0].value + ast.texts[0].value_len
second text &quot;...&quot; -&gt; ast.texts[1].value + ast.texts[1].value_len
```

For decoded values without NUL bytes:

```
ast.card.purpose_len == strlen(ast.card.purpose)
ast.texts[index].value_len == strlen(ast.texts[index].value)
```

String-literal decode helper shape

Update private decode helper behavior in:

```
src/l_ui_parser_ast.c
```

Current helper:

```
static latticra_status_t decode_l_ui_string_literal_value(
    const char *source,
    size_t start_offset,
    size_t end_offset,
    char *destination,
    size_t destination_len);
```

Proposed helper:

```
static latticra_status_t decode_l_ui_string_literal_value(
    const char *source,
```

```
    size_t start_offset,
    size_t end_offset,
    char *destination,
    size_t destination_len,
    size_t *decoded_len);
```

decoded_len must be set only on success.

On failure, the destination should remain cleared when possible and the decoded length output should be set to zero when non-null.

Extraction helper shape

Update the source-backed extraction helper in:

```
src/l_ui_parser_ast.c
```

Current helper:

```
static latticra_status_t extract_decoded_quoted_value_after_token(
    const char *source,
    size_t source_len,
    const char *token,
    size_t occurrence,
    char *destination,
    size_t destination_len,
    latticra_l_ui_source_span_t *value_span);
```

Proposed helper:

```
static latticra_status_t extract_decoded_quoted_value_after_token(
    const char *source,
    size_t source_len,
    const char *token,
    size_t occurrence,
    char *destination,
    size_t destination_len,
    size_t *decoded_len,
    latticra_l_ui_source_span_t *value_span);
```

The helper should pass the decoded length back to the AST builder after successful decoding.

Fill helper shape

Update private fill helpers:

```
static void fill_text(
    latticra_l_ui_ast_result_t *ast,
    size_t index,
    const char *value,
    size_t value_len,
    const latticra_l_ui_source_span_t *span);
```

For the card purpose, assign:

```
ast-&gt;card.purpose_len = extracted_purpose_len;
```

For text nodes, assign:

```
ast-&gt;texts[index].value_len = value_len;
```

Capacity behavior

Capacity remains:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
```

for purpose and text values.

A decoded value fits only when:

```
decoded_len &lt; destination_len
```

because the compatibility C string still needs a trailing NUL byte.

If decoded output is greater than or equal to the destination buffer length, parser validation should continue to report:

```
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE
```

and AST construction should preserve failed-parse behavior.

NUL behavior

This implementation must not accept decoded or literal NUL bytes.

Parser-level diagnostics must continue to reject:

```
\x00 -&gt; LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

A later decoded-NUL acceptance contract may change this, but not this implementation.

Report field additions

Update detailed AST reports in:

```
latticra_l_ui_ast_detailed_report
```

Add card section field:

```
purpose_len=&lt;decoded-purpose-byte-length&gt;
```

Add each text section field:

```
value_len=&lt;decoded-text-byte-length&gt;
```

Recommended section placement:

```
[card]
kind=card
name=&lt;name&gt;
purpose=&lt;purpose&gt;
purpose_len=&lt;purpose_len&gt;
purpose_escaped=&lt;purpose_escaped&gt;
```

and:

```
[text N]
kind=text
value=&lt;value&gt;
value_len=&lt;value_len&gt;
value_escaped=&lt;value_escaped&gt;
```

Existing field names must remain stable.

Length-aware escaped report helper shape

Replace or refactor the existing escaped report helper into a length-aware byte helper.

Current helper shape:

```
static latticra_status_t escape_report_string(
    const char *input,
    size_t input_len,
    char *output,
    size_t output_len);
```

It already accepts input_len, but callers currently use strlen.

Implementation decision:

```
keep the helper private
keep byte-oriented behavior
ensure callers pass explicit AST lengths instead of strlen for purpose and text values
```

Optional rename:

```
escape_report_bytes
```

If renamed, keep behavior identical for non-NUL values.

AST report compatibility

The compact AST report:

```
latticra_l_ui_ast_report
```

should remain unchanged in the first implementation.

The detailed AST report gains length fields, but existing required fields and ordering should remain stable except for insertion of purpose_len and value_len near their corresponding values.

Failed-parse detailed reports should remain unchanged and should not include string length fields.

Parser diagnostics compatibility

The implementation must not change existing parser diagnostics.

These diagnostics remain active:

```
LUI0019 invalid_string_escape
LUI0020 invalid_hex_escape
LUI0021 unterminated_escape
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
LUI0024 string_value_too_large
```

This implementation must not remove decoded-NUL or literal-NUL rejection.

Source-span behavior

Source spans remain source-oriented.

Text spans continue to cover the source value range between quotes, not decoded output byte ranges.

For example, source \n covers two source bytes while the decoded AST value length increases by one byte.

No public purpose_span field is added in this implementation.

Compatibility expectations

The implementation must not change:

```
accepted string-literal escape decoding
parser-level invalid string escape diagnostics
existing diagnostic codes LUI0000 through LUI0024
existing parser error labels for current errors
latticra_l_ui_ast_report existing required fields
latticra_l_ui_diagnostic_report existing required fields
existing accepted fixture summary counts
failed-parse detailed report behavior
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect preservation

Length-carrying AST string storage is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
ast_string_storage_sets_purpose_len
ast_string_storage_sets_text_value_len
ast_string_storage_len_matches_strlen_for_non_nul_values
ast_string_storage_len_tracks_decoded_newline
ast_string_storage_len_tracks_decoded_tab
ast_string_storage_len_tracks_decoded_high_byte_hex
ast_string_storage_preserves_existing_c_string_fields
ast_string_storage_updates_detailed_report_lengths
ast_string_storage_escaped_report_uses_explicit_lengths
ast_string_storage_still_rejects_decoded_nul_until_acceptance_contract
ast_string_storage_still_rejects_literal_nul_until_acceptance_contract
ast_string_storage_still_rejects_oversized_decoded_output
ast_string_storage_preserves_source_spans
ast_string_storage_preserves_no_effect_flags
ast_string_storage_does_not_change_existing_diagnostic_codes
ast_string_storage_does_not_change_failed_parse_report
ast_string_storage_is_deterministic
```

Test file plan

Add:

```
tests/l_ui_ast_length_carrying_string_storage_invariants.c
scripts/test-l-ui-ast-length-carrying-string-storage.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
```

Forbidden implementation behavior

The length-carrying AST string storage implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;

- accept decoded NUL bytes without a separate acceptance contract;
- accept literal NUL bytes without a separate acceptance contract;
- broaden accepted string escapes;
- remove existing C-string fields;
- silently truncate decoded values;
- change accepted escape decoding semantics;
- change existing parser diagnostic codes;
- make raw purpose= or value= report fields the assertion target for embedded NUL bytes.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-ast-length-carrying-string-storage-implementation-plan.sh
```

The guard is static. It does not implement length-carrying AST string storage.

Implementation gate

Length-carrying AST string storage implementation code may be added only after this plan is merged.

Non-claims

This document does not implement length-carrying AST strings, accept decoded NUL bytes, accept literal NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_SOURCE_BACKED_TEXT_CONTRACT.md

Source title: Latticra L-UI AST Source-Backed Text Contract

Lines: 326 | Words: 911 | SHA-256: 11f6b41cc779c7df23543c97fc5bda5891a395444f9e6e65c1c13bcae97543aa

Latticra L-UI AST Source-Backed Text Contract

Status: source-backed text contract Scope: extraction of AST purpose and text values from validated L-UI source instead of fixed fixture metadata.

Purpose

This document defines the source-backed text extraction contract for the L-UI AST.

The AST implementation extracts purpose and text values from validated NucleusPreview source while preserving source spans, escaped reporting, no-effect behavior, and deterministic output.

The source-backed extraction implementation is documented separately in L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md.

The later string-literal escape decoder is documented separately in L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md.

Current boundary

The current L-UI AST stack provides:

```
validated L-UI parser
source spans
fixed-size AST metadata
source-backed purpose value
source-backed text values
string-literal escape decoding for accepted source escapes
compact AST report
detailed AST report
escaped detailed report fields
no-effect flags
```

This contract does not add:

```
new renderer behavior
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Extraction purpose

Source-backed text extraction makes the AST reflect the actual validated .lui source text for:

```
card.purpose
text.value
```

This allows fixture text changes to be represented in AST metadata without hard-coded AST literals.

Extraction targets

The implementation extracts:

```
purpose &quot;...&quot;
text &quot;...&quot;
```

from the already-validated source.

The first accepted fixture still produces:

```
purpose=operator-visible Nucleus preview report
value=Latticra / Nucleus Preview / effect-bound
value=preview-only no-live-movement no-host-effect no-external-effect
```

No grammar broadening rule

Source-backed extraction does not broaden accepted L-UI syntax by itself.

The parser still rejects unsupported forms according to the existing parser contracts.

Extraction runs only after successful validation by:

```
latticra_l_ui_parse_source
```

Ownership rule

Extracted text is copied into fixed AST storage.

The AST does not retain borrowed pointers into the source buffer.

Destination fields:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

The extraction implementation preserves NUL-terminated C strings in AST storage.

Capacity rule

Extraction respects existing capacities:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
LATTICRA_L_UI_AST_TEXT_MAX
```

The implementation classifies oversized extracted or decoded strings with:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

and avoids partial AST output.

It does not truncate silently.

Span rule

Extracted values keep source-span metadata where currently supported.

For purpose:

```
card.span covers the card block
```

For text:

```
text.span covers the source text value range
```

Source spans refer to source byte ranges, not decoded output byte ranges.

No public purpose_span field is added in the first source-backed extraction implementation.

Quote handling rule

The implementation extracts quoted string contents.

For source:

```
purpose &quot;abc&quot;
text &quot;xyz&quot;
```

AST values are:

```
abc
xyz
```

The surrounding quotes are not copied into AST values.

Escaped quotes are respected when finding the closing quote.

Escape handling relationship

The initial source-backed extraction phase copied raw bytes between quotes without decoding escapes.

Current source-backed AST values now decode accepted string-literal escapes according to:

```
L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md
L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md
L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
```

Accepted escapes such as \n, \r, \t, \", \\, and uppercase \xNN decode before values are stored in AST-owned buffers.

Rejected escapes are classified through:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

until parser-level string escape diagnostics are designed.

Detailed report relationship

Detailed reports continue to render:

```
purpose=&lt;literal-purpose&gt;
purpose_escaped=&lt;escaped-purpose&gt;
```

```
value=&lt;literal-text&gt;
value_escaped=&lt;escaped-text&gt;
```

After source-backed extraction and accepted string-literal escape decoding, those report values reflect decoded AST values.

The escaped report fields remain the stable assertion target for control bytes, quotes, backslashes, DEL, and non-ASCII bytes.

Failed parse behavior

Failed parse behavior remains unchanged.

If source validation fails:

```
no partial AST nodes
no extracted text
failed-parse detailed report only
```

No-effect rule

Source-backed text extraction and string-literal escape decoding are metadata-only.

They preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility rule

Source-backed text extraction and string-literal escape decoding do not change:

```
latticra_l_ui_parse_source
latticra_l_ui_ast_report
latticra_l_ui_ast_detailed_report existing required fields
latticra_l_ui_diagnostic_report
parser error labels
existing accepted fixture summary
```

The first accepted fixture still reports the same AST summary counts:

```
rail_count=9
field_count=23
text_count=2
```

Implementation gate

Source-backed text extraction implementation required a separate implementation plan defining:

1. extraction helper shapes;
2. exact extraction targets;
3. capacity behavior;
4. span field decision;
5. quote handling behavior;
6. escape handling boundary;
7. failed-parse behavior;
8. test file names;
9. exact invariant tests;
10. compatibility expectations.

That plan is recorded in L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION_PLAN.md.

String-literal escape decoding required a separate implementation plan before decoder code. That plan is recorded in L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md.

Test list

Source-backed text implementation tests verify:

```
source_backed_purpose_matches_fixture_source
source_backed_top_text_matches_fixture_source
source_backed_bottom_text_matches_fixture_source
source_backed_text_values_are_copied
source_backed_text_does_not_retain_source_pointers
source_backed_text_excludes_quotes
source_backed_text_preserves_ast_counts
source_backed_text_preserves_no_effect_flags
source_backed_text_updates_detailed_report_literals
source_backed_text_updates_detailed_report_escaped_fields
source_backed_text_rejects_or_classifies_oversized_purpose
source_backed_text_rejects_or_classifies_oversized_text
source_backed_text_does_not_change_failed_parse_report
source_backed_text_is_deterministic
source_backed_text_decodes_accepted_string_escapes
```

Forbidden behavior

Source-backed text implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- retain source-buffer pointers in the AST;
- silently truncate extracted values;
- broaden accepted grammar without a grammar contract.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-ast-source-backed-text-contract.sh
```

The guard is static. It validates the contract text.

Non-claims

This document does not broaden non-string L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md

Source title: Latticra L-UI AST Source-Backed Text Implementation

Lines: 269 | Words: 791 | SHA-256: ad87683a5486c584c18278bb942c18b1be0f21006adb407d5cb1545c043c4933

Latticra L-UI AST Source-Backed Text Implementation

Status: initial implementation contract Scope: source-backed extraction of AST purpose and text values from validated L-UI source.

Purpose

The L-UI AST Source-Backed Text implementation extracts AST purpose and text values from validated L-UI source instead of relying on fixed fixture literals.

It keeps AST storage fixed-size, copies extracted text into AST-owned buffers, preserves no-effect behavior, and leaves parser validation as the gate before extraction.

Accepted string-literal escapes are now decoded during AST value materialization by the later L-UI string-literal escape implementation.

Implementation files

```
src/l_ui_parser_ast.c
tests/l_ui_ast_source_backed_text_invariants.c
scripts/test-l-ui-ast-source-backed-text.sh
```

Extraction targets

The implementation extracts:

```
purpose &quot;...&quot; -&gt; ast.card.purpose
first text &quot;...&quot; -&gt; ast.texts[0].value
second text &quot;...&quot; -&gt; ast.texts[1].value
```

The first accepted fixture still produces:

```
purpose=operator-visible Nucleus preview report
value=Latticra / Nucleus Preview / effect-bound
value=preview-only no-live-movement no-host-effect no-external-effect
```

Private helpers

The original source-backed implementation added private helpers in src/l_ui_parser_ast.c:

```
extract_quoted_value_after_token
copy_extracted_value
```

The string-literal escape implementation later replaced raw value copying with private decoding helpers while preserving the same extraction targets.

These helpers remain private.

Quote behavior

The implementation copies source-backed contents between the opening and closing quote into AST-owned buffers.

For source:

```
purpose &quot;abc&quot;
text &quot;xyz&quot;
```

AST values become:

```
abc
xyz
```

The surrounding quotes are not copied into AST values.

Escaped quotes are respected when finding the closing quote.

Escape handling relationship

The initial source-backed extraction phase copied raw bytes between quotes while respecting escaped quotes for finding the closing quote.

Current AST value materialization decodes accepted string-literal escapes according to:

```
L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md
L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md
L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
```

For example, source bytes representing:

```
raw\ntext
```

now become AST bytes representing:

```
raw
text
```

and detailed report escaped fields render that decoded newline safely as:

```
raw\ntext
```

Rejected string escapes classify AST construction with:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

until parser-level string escape diagnostics are designed.

Capacity behavior

Extraction and decoding respect destination capacities:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
LATTICRA_L_UI_AST_PURPOSE_MAX for text values
LATTICRA_L_UI_AST_TEXT_MAX for text node count
```

If an extracted or decoded value is too large for its destination, AST construction classifies the result as:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

and leaves counts at zero to avoid a partial AST.

The implementation does not silently truncate extracted or decoded values.

Span behavior

The implementation does not add a public purpose span.

text.span is set to the source text value range for text nodes. card.span continues to cover the card block.

Text spans refer to source byte ranges, not decoded output byte ranges.

Successful parse rule

Extraction and decoding run only after:

```
latticra_l_ui_parse_source
```

returns LATTICRA_STATUS_OK and parse_result.error == LATTICRA_L_UI_PARSE_OK.

If parsing fails, extraction and decoding do not run.

Detailed report relationship

Detailed reports continue to render:

```
purpose=&lt;literal-purpose&gt;
purpose_escaped=&lt;escaped-purpose&gt;
value=&lt;literal-text&gt;
value_escaped=&lt;escaped-text&gt;
```

After source-backed extraction and accepted string-literal escape decoding, these values reflect decoded AST values.

The escaped report fields remain the stable assertion target for control bytes, quotes, backslashes, DEL, and non-ASCII bytes.

Failed parse behavior

Failed parse behavior is unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

No source-backed text values are extracted for failed parses.

Compatibility

This implementation does not change:

```
latticra_l_ui_parse_source
latticra_l_ui_ast_report
latticra_l_ui_ast_detailed_report existing required fields
latticra_l_ui_diagnostic_report
parser error labels
existing accepted fixture summary counts
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect boundary

Source-backed text extraction and string-literal escape decoding preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-ast-source-backed-text.sh
```

The main C workflow runs this check after the source-backed text implementation-plan guard.

Required invariants

The source-backed text tests verify:

```
source_backed_purpose_matches_fixture_source
source_backed_top_text_matches_fixture_source
source_backed_bottom_text_matches_fixture_source
```

```
source_backed_text_values_are_copied
source_backed_text_does_not_retain_source_pointers
source_backed_text_excludes_quotes
source_backed_text_preserves_ast_counts
source_backed_text_preserves_no_effect_flags
source_backed_text_updates_detailed_report_literals
source_backed_text_updates_detailed_report_escaped_fields
source_backed_text_rejects_or_classifies_oversized_purpose
source_backed_text_rejects_or_classifies_oversized_text
source_backed_text_does_not_change_failed_parse_report
source_backed_text_is_deterministic
source_backed_text_decodes_accepted_string_escapes
```

Current evidence level

This implementation is an L2 tested source-backed text extraction model for validated L-UI AST metadata, now paired with tested string-literal escape decoding.

It is not a broader non-string grammar, Unicode display model, renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after source-backed text extraction and string-literal escape decoding is:

```
L-UI parser-level string escape diagnostics contract
```

That future work should decide whether invalid string escapes receive first-class parser diagnostic codes instead of AST-level internal-error classification.

Non-claims

This document and implementation do not broaden non-string L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI AST Source-Backed Text Implementation Plan

Lines: 344 | Words: 978 | SHA-256: 4641b182751d9cb7add7144cd72424d0bbefaa5160ed5656b4679c3d226bbed7

Latticra L-UI AST Source-Backed Text Implementation Plan

Status: implementation planning contract Scope: extraction helpers, exact extraction targets, quote handling, capacity behavior, span decisions, exact tests, and compatibility expectations before source-backed text extraction code.

Purpose

This document defines the implementation plan for source-backed text extraction in the L-UI AST.

The source-backed text contract is already merged and guarded. This plan decides the helper shapes, extraction targets, quote handling, escape handling boundary, capacity behavior, span decision, exact tests, and compatibility expectations before implementation code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_AST_SOURCE_BACKED_TEXT_CONTRACT.md
```

```
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
```

Those files remain the source of truth for AST storage, source spans, detailed reporting, escaped fields, and no-effect boundaries.

Implementation language decision

Source-backed text extraction should be implemented in C.

Reason:

- the AST builder is implemented in C;
- AST storage is a C public API surface;
- extraction must remain bounded, deterministic, and no-effect;
- the current C workflow can validate extraction invariants;
- no dynamic runtime should be required.

Public API decision

No new public function is needed for the first source-backed extraction implementation.

Extraction should be integrated into:

```
latticra_l_ui_parse_ast
```

The public AST fields remain:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

Extraction helper plan

Add private helpers in:

```
src/l_ui_parser_ast.c
```

Proposed helpers:

```
static latticra_status_t extract_quoted_value_after_token(
    const char *source,
    size_t source_len,
    const char *token,
    size_t occurrence,
    char *destination,
    size_t destination_len,
    latticra_l_ui_source_span_t *value_span);

static latticra_status_t copy_extracted_value(
    const char *source,
    size_t start_offset,
    size_t end_offset,
    char *destination,
    size_t destination_len);
```

Helpers must remain private.

Extraction targets

The first implementation should extract these source-backed values:

```
purpose &quot;...&quot; -&gt; ast.card.purpose
first text &quot;...&quot; -&gt; ast.texts[0].value
second text &quot;...&quot; -&gt; ast.texts[1].value
```

The first accepted fixture must still produce:

```
purpose=operator-visible Nucleus preview report
value=Latticra / Nucleus Preview / effect-bound
value=preview-only no-live-movement no-host-effect no-external-effect
```

Quote handling behavior

The first implementation should copy bytes between the opening and closing quote.

For source:

```
purpose &quot;abc&quot;
text &quot;xyz&quot;
```

AST values should be:

```
abc
xyz
```

The surrounding quotes must not be copied into AST values.

Escape handling boundary

The first implementation must not decode escape sequences.

Behavior:

```
copy raw bytes between quotes without decoding escapes
```

If later grammar accepts string escapes, a string-literal escape contract should define decoding behavior before AST extraction decodes anything.

Capacity behavior

Extraction must respect destination capacities:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
LATTICRA_L_UI_AST_PURPOSE_MAX for text values
LATTICRA_L_UI_AST_TEXT_MAX for text node count
```

Because latticra_l_ui_ast_text_t.value currently uses LATTICRA_L_UI_AST_PURPOSE_MAX, the text extraction destination capacity is that field size. LATTICRA_L_UI_AST_TEXT_MAX remains the bounded capacity for the number of text nodes.

If extracted text length is greater than or equal to the destination buffer length, extraction should return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

latticra_l_ui_parse_ast should convert an extraction capacity failure into:

```
ast.parse_result.error = LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

and should leave counts at zero to avoid a partial AST.

The implementation must not silently truncate extracted values.

Span decision

Do not add a new public purpose_span field in the first implementation.

Reason:

- text.span already exists for text values;
- card.span covers the card block;
- adding a public purpose span would require a broader AST struct change;
- source-backed value correctness can be tested through copied values and detailed reports.

Internal extraction may compute a value span for validation, but it does not need to expose it publicly yet.

Successful parse rule

Extraction must run only after:

```
latticra_l_ui_parse_source
```

returns LATTICRA_STATUS_OK and parse_result.error == LATTICRA_L_UI_PARSE_OK.

If parsing fails, extraction must not run.

Failed parse behavior

Failed parse behavior must remain unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

No source-backed text values should be extracted for failed parses.

Detailed report relationship

Detailed reports should continue to render:

```
purpose=&lt;literal-purpose&gt;
purpose_escaped=&lt;escaped-purpose&gt;
value=&lt;literal-text&gt;
value_escaped=&lt;escaped-text&gt;
```

After source-backed extraction, these values should reflect the extracted source values.

Compatibility expectations

Source-backed text extraction must not change:

```
latticra_l_ui_parse_source
latticra_l_ui_ast_report
latticra_l_ui_ast_detailed_report existing required fields
latticra_l_ui_diagnostic_report
parser error labels
existing accepted fixture summary counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect preservation

Source-backed text extraction must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
source_backed_purpose_matches_fixture_source
source_backed_top_text_matches_fixture_source
source_backed_bottom_text_matches_fixture_source
source_backed_text_values_are_copied
source_backed_text_does_not_retain_source_pointers
source_backed_text_excludes_quotes
source_backed_text_preserves_ast_counts
source_backed_text_preserves_no_effect_flags
source_backed_text_updates_detailed_report_literals
source_backed_text_updates_detailed_report_escaped_fields
source_backed_text_rejects_or_classifies_oversized_purpose
source_backed_text_rejects_or_classifies_oversized_text
source_backed_text_does_not_change_failed_parse_report
source_backed_text_is_deterministic
```

```
source_backed_text_does_not_decode_escapes
```

Test file plan

Add:

```
tests/l_ui_ast_source_backed_text_invariants.c
scripts/test-l-ui-ast-source-backed-text.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_CONTRACT.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
```

Forbidden implementation behavior

The source-backed text implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- retain source-buffer pointers in the AST;
- silently truncate extracted values;
- broaden accepted grammar without a grammar contract;
- decode string escapes without a string-literal escape contract.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-ast-source-backed-text-implementation-plan.sh
```

The guard is static. It does not implement source-backed text extraction.

Implementation gate

Source-backed text extraction implementation code may be added only after this plan is merged.

Non-claims

This document does not implement source-backed text extraction, broaden accepted L-UI text syntax, implement string escape decoding, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md

Source title: Latticra L-UI Decoded NUL Acceptance Contract

Lines: 370 | Words: 1147 | SHA-256: fad6f30f8b45fc9b65a6d87bf9a5722b2e5ceeddd5084ba8cdcf31c4d9b3cc9c

Latticra L-UI Decoded NUL Acceptance Contract

Status: decoded NUL acceptance contract Scope: future acceptance of escaped decoded NUL bytes in L-UI source-backed AST strings after length-carrying AST string storage.

Purpose

This document defines the contract for accepting escaped decoded NUL bytes in L-UI source-backed AST string values.

The current parser rejects:

```
\x00 -&gt; LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Now that AST purpose and text values carry explicit decoded byte lengths, a future implementation may accept escaped \x00 in quoted string values while continuing to reject literal NUL bytes in source buffers.

This document does not implement decoded NUL acceptance.

The implementation plan is documented separately in L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION_PLAN.md.

Relationship to previous work

This contract depends on:

```
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION_PLAN.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for explicit string lengths, parser diagnostics, string-literal escape decoding, detailed reports, and no-effect behavior.

Current boundary

The current system provides:

```
purpose_len and value_len fields
length-aware escaped detailed report fields
source-backed AST purpose and text values
```

```
accepted string-literal escape decoding
parser-level invalid string escape diagnostics
literal NUL rejection
no-effect flags
```

This contract does not add:

```
decoded NUL acceptance implementation
literal NUL acceptance
new accepted escape sequences beyond \x00 behavior change
Unicode display behavior
renderer behavior
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Acceptance decision

A future implementation may accept escaped decoded NUL bytes only through the existing uppercase hex escape form:

```
\x00
```

Accepted source examples:

```
purpose &quot;A\x00B&quot;
text &quot;left\x00right&quot;
```

The decoded AST byte sequence should contain the NUL byte at that position, and the explicit length field should include that byte.

Literal NUL remains rejected

Literal 0x00 bytes in the source buffer remain rejected.

A source buffer containing a literal NUL inside a quoted value should continue to report:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0023 literal_nul_in_string
```

Reason:

- source buffers are still byte-oriented input;
- literal NUL bytes make source inspection and tool interop ambiguous;
- escaped NUL is deterministic and operator-visible in reports;
- accepting literal NUL requires a separate source-buffer contract.

Parser diagnostic transition

A future implementation should stop reporting this diagnostic for escaped \x00 in source-backed string values:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
LUI0022 decoded_nul_in_string
```

LUI0022 must remain a stable diagnostic code for compatibility, but escaped \x00 in supported L-UI string values should no longer trigger it after decoded-NUL acceptance is implemented.

LUI0023 literal_nul_in_string remains active.

Accepted escape compatibility

Decoded NUL acceptance must not broaden the accepted escape set beyond the existing contract:

```
\\
\&quot;
\n
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

Lowercase hex remains rejected:

```
\x0a -&gt; LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE
```

Unknown escapes remain rejected:

```
\a -&gt; LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE
```

AST storage behavior

After implementation, escaped decoded NUL bytes should be represented in the existing AST buffers and lengths:

```
ast.card.purpose
ast.card.purpose_len
ast.texts[index].value
ast.texts[index].value_len
```

For values containing decoded NUL bytes:

```
purpose_len != strlen(purpose)
value_len != strlen(value)
```

The explicit length field is authoritative.

The compatibility C-string buffer remains present, but strlen must not be used to inspect the full value when embedded NUL bytes exist.

Decode helper behavior

The private string-literal decode helper should allow decoded byte value 0x00 only when it came from escaped \x00.

It should still reject literal source NUL bytes before decoding materializes a value.

Decoded length assignment must count decoded NUL bytes:

```
&quot;A\x00B&quot; -&gt; length 3
```

Parser validation order

Decoded NUL acceptance should preserve existing validation order:

1. null argument checks;
2. source size checks;
3. structural parse checks;
4. string escape validation;
5. AST construction.

Structural parse errors must retain priority over decoded-NUL acceptance.

Source-span behavior

Source spans remain source-oriented.

For escaped decoded NUL:

```
\x00 -&gt; source span covers four source bytes when a diagnostic or test inspects that source
range
```

Accepted AST text spans continue to cover the full source value range between quotes.

Decoded output byte positions are not source spans.

Detailed report behavior

Detailed AST reports must use explicit lengths and escaped output as the stable inspection surface.

For decoded NUL bytes, escaped report fields should render:

```
\x00
```

Examples:

```
purpose_len=3
purpose_escaped=A\x00B

value_len=10
value_escaped=left\x00right
```

Raw C-string fields remain compatibility fields:

```
purpose=&lt;C-string-compatible-prefix&gt;
value=&lt;C-string-compatible-prefix&gt;
```

Tests for embedded NUL must assert against:

```
purpose_len
purpose_escaped
value_len
value_escaped
```

not raw purpose= or value= fields.

Compact report compatibility

The compact AST report should remain unchanged in the first decoded-NUL acceptance implementation.

Detailed reports are the stable inspection surface for embedded NUL bytes.

AST compatibility behavior

Failed parse behavior remains unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Accepted sources should continue to produce the same rail, field, and text counts.

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect rule

Decoded NUL acceptance is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Implementation gate

Decoded NUL acceptance implementation must not begin until a separate implementation plan defines:

1. parser validation changes;
2. decode helper changes;
3. AST storage expectations;
4. report expectations;
5. diagnostic compatibility behavior;
6. source-span expectations;
7. exact test file names;
8. exact invariant tests;
9. compatibility expectations;
10. non-claims.

That plan is recorded in L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION_PLAN.md.

Future test list

A future implementation plan should include tests for:

```
decoded_nul_accepts_purpose_x00
decoded_nul_accepts_text_x00
decoded_nul_counts_purpose_len
decoded_nul_counts_text_value_len
decoded_nul_preserves_c_string_prefix_compatibility
decoded_nul_reports_purpose_escaped_x00
decoded_nul_reports_text_value_escaped_x00
decoded_nul_does_not_emit_lui0022_for_x00
decoded_nul_still_rejects_literal_nul_lui0023
decoded_nul_still_rejects_lowercase_hex_lui0020
decoded_nul_still_rejects_unknown_escape_lui0019
decoded_nul_preserves_source_spans
decoded_nul_preserves_no_effect_flags
decoded_nul_does_not_change_failed_parse_report
decoded_nul_is_deterministic
```

Forbidden behavior

A future decoded NUL acceptance implementation must not:

- accept literal NUL source bytes;
- accept lowercase hex escapes;
- accept unknown escapes;
- broaden accepted escape forms;
- remove LUI0022 from the diagnostic code table;
- remove LUI0023 literal NUL rejection;
- remove existing C-string fields;
- make raw purpose= or value= report fields the assertion target for embedded NUL bytes;
- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;

- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-decoded-nul-acceptance-contract.sh
```

The guard is static. It does not implement decoded NUL acceptance.

Non-claims

This document does not implement decoded NUL acceptance, accept literal NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md

Source title: Latticra L-UI Decoded NUL Acceptance Implementation

Lines: 335 | Words: 873 | SHA-256: 07e0dd667f8a3db9e434619e3db9d2d28114b6a2c13ea3f8277f31f906b9166a

Latticra L-UI Decoded NUL Acceptance Implementation

Status: initial implementation contract Scope: escaped decoded NUL acceptance for L-UI source-backed AST strings.

Purpose

The L-UI Decoded NUL Acceptance implementation accepts escaped decoded NUL bytes in source-backed AST string values through the existing uppercase hex escape syntax:

```
\x00
```

The implementation relies on the explicit decoded byte lengths added by the L-UI AST length-carrying string storage implementation.

It accepts escaped decoded NUL bytes while continuing to reject literal NUL bytes in source buffers.

Implementation files

```
src/l_ui_parser.c
src/l_ui_parser_ast.c
tests/l_ui_decoded_nul_acceptance_invariants.c
scripts/test-l-ui-decoded-nul-acceptance.sh
```

Related active files:

```
include/latticra/l_ui_parser.h
```

```
src/l_ui_parser_diagnostics.c
tests/l_ui_parser_string_escape_diagnostics_invariants.c
tests/l_ui_ast_length_carrying_string_storage_invariants.c
tests/l_ui_string_literal_escape_invariants.c
```

Accepted form

Only escaped decoded NUL through uppercase hex is accepted:

```
\x00
```

Accepted examples:

```
purpose &quot;A\x00B&quot;
text &quot;left\x00right&quot;
```

The decoded byte sequence contains 0x00 at the escape position.

Literal NUL remains rejected

Literal 0x00 bytes in the source buffer remain rejected.

Literal NUL still reports:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0023 literal_nul_in_string
```

This implementation does not add a source-buffer literal-NUL acceptance model.

The literal-NUL source-buffer policy contract is recorded in:

```
L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md
```

Parser validation behavior

Parser validation in:

```
src/l_ui_parser.c
```

no longer reports:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
```

for escaped \x00 in supported source-backed string values.

The parser still rejects:

```
\x0a -&gt; LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE
\a -&gt; LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Diagnostic compatibility

The implementation keeps existing parser errors and diagnostic codes stable:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
```

LUI0022 remains available as a stable diagnostic mapping, but escaped \x00 no longer emits it from real supported L-UI string input.

LUI0023 remains active for literal source NUL.

AST storage behavior

Escaped decoded NUL bytes are stored in the existing AST byte buffers and counted by explicit lengths:

```
ast.card.purpose
ast.card.purpose_len
ast.texts[index].value
ast.texts[index].value_len
```

For embedded NUL values:

```
purpose_len != strlen(purpose)
value_len != strlen(value)
```

The explicit length field is authoritative.

C-string prefix compatibility

For decoded value:

```
A\x00B
```

The AST stores:

```
purpose[0] == &#x27;A&#x27;
purpose[1] == &#x27;\0&#x27;
purpose[2] == &#x27;B&#x27;
purpose_len == 3
strlen(purpose) == 1
```

The same rule applies to text values.

Decode helper behavior

The AST string decode helper in:

```
src/l_ui_parser_ast.c
```

now allows decoded byte 0x00 when it is produced by escaped \x00.

The helper still rejects literal source NUL bytes before value materialization.

Decoded length assignment counts decoded NUL bytes:

```
&quot;A\x00B&quot; -&gt; decoded bytes: A, 0x00, B -&gt; length 3
```

Detailed report behavior

Detailed AST reports use explicit lengths and escaped output as the stable inspection surface.

Purpose example:

```
purpose_len=3
purpose_escaped=A\x00B
```

Text example:

```
value_len=10
value_escaped=left\x00right
```

Raw C-string compatibility fields remain prefix-oriented:

```
purpose=&lt;C-string-compatible-prefix&gt;
value=&lt;C-string-compatible-prefix&gt;
```

Embedded-NUL tests assert against:

```
purpose_len
purpose_escaped
value_len
value_escaped
```

not raw purpose= or value= fields.

Compact report compatibility

The compact AST report remains unchanged:

```
latticra_l_ui_ast_report
```

Detailed reports remain the stable inspection surface for embedded NUL values.

Source-span behavior

Source spans remain source-oriented.

A decoded NUL produced by source bytes \x00 counts as one decoded byte but four source bytes.

Accepted AST text spans continue to cover the full source value range between quotes.

No public decoded-byte span field is added.

Validation order

The implementation preserves validation order:

1. null argument checks;
2. source size checks;
3. structural parse checks;
4. string escape validation;
5. AST construction.

Structural parse errors retain priority over decoded-NUL acceptance.

Compatibility

The implementation does not change:

```
accepted escape forms other than \x00 behavior
parser diagnostic code stability
literal source NUL rejection
lowercase hex rejection
unknown escape rejection
existing C-string fields
compact AST report fields
failed-parse detailed report behavior
existing accepted fixture counts
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect boundary

Decoded NUL acceptance is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-decoded-nul-acceptance.sh
```

The main C workflow runs this check after the decoded-NUL acceptance implementation-plan guard and before the string-literal escape guards.

Required invariants

The decoded-NUL acceptance tests verify:

```
decoded_nul_accepts_purpose_x00
decoded_nul_accepts_text_x00
decoded_nul_counts_purpose_len
decoded_nul_counts_text_value_len
```

```
decoded_nul_preserves_c_string_prefix_compatibility
decoded_nul_reports_purpose_escaped_x00
decoded_nul_reports_text_value_escaped_x00
decoded_nul_does_not_emit_lui0022_for_x00
decoded_nul_still_rejects_literal_nul_lui0023
decoded_nul_still_rejects_lowercase_hex_lui0020
decoded_nul_still_rejects_unknown_escape_lui0019
decoded_nul_preserves_source_spans
decoded_nul_preserves_no_effect_flags
decoded_nul_does_not_change_failed_parse_report
decoded_nul_is_deterministic
```

Current evidence level

This implementation is an L2 tested decoded-byte storage and reporting model for escaped decoded NUL bytes in L-UI source-backed AST strings.

It is not literal source NUL acceptance, Unicode display behavior, renderer behavior, UI runtime behavior, command behavior, Nucleus task execution, server interaction, update behavior, recovery behavior, hardware behavior, boot behavior, sandboxing, or an operating system.

Next implementation step

The source-buffer literal NUL policy contract is recorded in:

```
L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md
```

The next implementation candidate after that contract is:

```
L-UI source-buffer literal NUL policy implementation plan
```

That future work should decide whether literal NUL bytes in source buffers remain permanently forbidden or receive their own length-aware source input policy.

Non-claims

This document and implementation do not accept literal NUL bytes, broaden accepted L-UI syntax beyond escaped decoded NUL behavior, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Decoded NUL Acceptance Implementation Plan

Lines: 435 | Words: 1143 | SHA-256: 5cc0a2c3b87427342bf480f9d8819ded14e288de89aac3d65b726ce254a96e4a

Latticra L-UI Decoded NUL Acceptance Implementation Plan

Status: implementation planning contract Scope: parser validation changes, decode helper changes, AST storage expectations, report expectations, diagnostic compatibility behavior, source-span expectations, exact test files, and exact invariant tests before escaped decoded NUL acceptance code.

Purpose

This document defines the implementation plan for accepting escaped decoded NUL bytes in L-UI source-backed AST string values.

The decoded-NUL acceptance contract is already merged and guarded. This plan decides the exact parser validation changes, private decode helper behavior, AST storage expectations, detailed report expectations, diagnostic compatibility behavior, source-span expectations, and test coverage before implementation code is added.

This document does not implement decoded NUL acceptance.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION_PLAN.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for explicit AST string lengths, parser diagnostics, string-literal escape decoding, detailed reports, and no-effect behavior.

Implementation language decision

Decoded NUL acceptance should be implemented in C.

Reason:

- parser validation lives in src/l_ui_parser.c;
- AST decoding lives in src/l_ui_parser_ast.c;
- diagnostics live in src/l_ui_parser_diagnostics.c;
- the public AST API is C;
- the change is byte-oriented, bounded, deterministic, and no-effect.

Acceptance scope

The implementation should accept only escaped decoded NUL through uppercase hex syntax:

```
\x00
```

Accepted sources after implementation:

```
purpose &quot;A\x00B&quot;
text &quot;left\x00right&quot;
```

Literal source NUL bytes remain rejected.

Parser validation changes

Update private string escape validation in:

```
src/l_ui_parser.c
```

The parser currently maps escaped \x00 to:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
```

The implementation should change parser validation so escaped \x00 in supported source-backed string values is accepted and counted as one decoded byte.

The parser must continue to reject literal source NUL bytes as:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Parser diagnostic compatibility

Do not remove existing parser errors or diagnostic codes:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
```

After implementation:

```
\x00 -&gt; accepted in source-backed string values
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

LUI0022 remains reserved and stable, but should not be emitted for accepted escaped \x00 in supported string values.

Decode helper changes

Update private string-literal decoding in:

```
src/l_ui_parser_ast.c
```

The helper currently rejects decoded byte 0x00 through the same capacity/error path used for invalid output.

The implementation should allow decoded byte 0x00 when and only when it is produced by escaped \x00.

It must still reject literal source NUL bytes before value materialization.

Decoded length assignment must count decoded NUL bytes:

```
&quot;A\x00B&quot; -&gt; decoded bytes: A, 0x00, B -&gt; length 3
```

AST storage expectations

Use the existing length-carrying fields:

```
ast.card.purpose
ast.card.purpose_len
ast.texts[index].value
ast.texts[index].value_len
```

For embedded NUL values:

```
purpose_len != strlen(purpose)
value_len != strlen(value)
```

The explicit length field is authoritative.

Existing C-string buffers remain compatibility buffers and should contain the decoded bytes plus a trailing compatibility NUL. Consumers that use strlen will see the prefix before the first embedded NUL.

C-string prefix compatibility

For decoded value:

```
A\x00B
```

Expected compatibility behavior:

```
purpose[0] == &#x27;A&#x27;
purpose[1] == &#x27;\0&#x27;
purpose[2] == &#x27;B&#x27;
purpose_len == 3
strlen(purpose) == 1
```

The same rule applies to text values.

Report expectations

Detailed AST reports should remain the authoritative inspection surface for embedded NUL values.

For purpose:

```
purpose_len=3
purpose_escaped=A\x00B
```

For text:

```
value_len=10
```

```
value_escaped=left\x00right
```

Raw fields remain compatibility fields only:

```
purpose=&lt;C-string-compatible-prefix&gt;
value=&lt;C-string-compatible-prefix&gt;
```

Tests for embedded NUL must assert against:

```
purpose_len
purpose_escaped
value_len
value_escaped
```

not raw purpose= or value= fields.

Escaped report helper expectations

The current length-aware helper:

```
escape_report_bytes(input, input_len, output, output_len)
```

already accepts explicit lengths.

The implementation must ensure callers pass:

```
ast.card.purpose_len
ast.texts[index].value_len
```

for purpose and text values.

The helper should render embedded decoded NUL as:

```
\x00
```

Source-span expectations

Source spans remain source-oriented.

Accepted AST text spans continue to cover the full source value range between quotes.

A decoded NUL produced by source bytes \x00 counts as one decoded byte but four source bytes.

No public decoded-byte span field is added.

Parser validation order

Preserve the current validation order:

1. null argument checks;
2. source size checks;
3. structural parse checks;
4. string escape validation;
5. AST construction.

Structural parse errors must retain priority over decoded-NUL acceptance.

Accepted escape compatibility

Do not broaden the accepted escape set.

Still accepted:

```
\\
\&quot;
\n
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

Still rejected:

```
\x0a -&gt; LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE
\a -&gt; LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Compact report compatibility

The compact AST report:

```
latticra_l_ui_ast_report
```

should remain unchanged.

Detailed AST reports are the stable inspection surface for embedded NUL values.

Failed parse compatibility

Failed parse behavior remains unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Fixture compatibility

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect preservation

Decoded NUL acceptance is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
decoded_nul_accepts_purpose_x00
decoded_nul_accepts_text_x00
decoded_nul_counts_purpose_len
decoded_nul_counts_text_value_len
decoded_nul_preserves_c_string_prefix_compatibility
decoded_nul_reports_purpose_escaped_x00
decoded_nul_reports_text_value_escaped_x00
decoded_nul_does_not_emit_lui0022_for_x00
decoded_nul_still_rejects_literal_nul_lui0023
decoded_nul_still_rejects_lowercase_hex_lui0020
decoded_nul_still_rejects_unknown_escape_lui0019
decoded_nul_preserves_source_spans
decoded_nul_preserves_no_effect_flags
decoded_nul_does_not_change_failed_parse_report
decoded_nul_is_deterministic
```

Test file plan

Add:

```
tests/l_ui_decoded_nul_acceptance_invariants.c
scripts/test-l-ui-decoded-nul-acceptance.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-l-ui-decoded-nul-acceptance-contract.sh
```

and before:

```
sh scripts/test-l-ui-string-literal-escape-contract.sh
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
```

Forbidden implementation behavior

The decoded NUL acceptance implementation must not:

- accept literal NUL source bytes;
- accept lowercase hex escapes;
- accept unknown escapes;
- broaden accepted escape forms;
- remove LUI0022 from the diagnostic code table;
- remove LUI0023 literal NUL rejection;
- remove existing C-string fields;
- make raw purpose= or value= report fields the assertion target for embedded NUL bytes;
- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-decoded-nul-acceptance-implementation-plan.sh
```

The guard is static. It does not implement decoded NUL acceptance.

Implementation gate

Decoded NUL acceptance implementation code may be added only after this plan is merged.

Non-claims

This document does not implement decoded NUL acceptance, accept literal NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_PARSER.md

Source title: Latticra L-UI Parser

Lines: 234 | Words: 520 | SHA-256: e912de1efede3e08ce866061e96e3d44b057073a3a796d0a9109fa5514413373

Latticra L-UI Parser

Status: implementation contract Scope: no-effect C parser, parse summaries, source locations, and source spans for the first L-UI source fixture.

Purpose

The L-UI Parser validates in-memory .lui source and returns a compact no-effect parse summary.

It does not render UI, run L-UI behavior, perform file I/O, call Nucleus task handling, or mutate state.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
tests/l_ui_parser_invariants.c
tests/l_ui_parser_location_invariants.c
tests/l_ui_parser_source_span_invariants.c
scripts/test-l-ui-parser.sh
scripts/test-l-ui-parser-location.sh
scripts/test-l-ui-parser-source-span.sh
```

Source-size limit

The parser defines:

```
LATTICRA_L_UI_SOURCE_MAX = 65536
```

Input larger than this is rejected with:

```
source_too_large
```

Public API

Current API:

```
latticra_l_ui_parse_error_label
latticra_l_ui_parse_source
latticra_l_ui_parse_result_report
```

The parser accepts in-memory source only. It does not read files directly.

Parse result

The parse result contains:

```
status
error
line
column
span
card_name
rail_count
field_count
effect
boundary
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

The source span contains:

```
start_offset
end_offset
start_line
start_column
end_line
end_column
```

For a valid fixture, the expected summary is:

```
status=ok
error=ok
card_name=NucleusPreview
rail_count=9
field_count=23
effect=none
boundary=preview_only
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
span_start_offset=0
span_end_offset=0
span_start_line=1
span_start_column=1
span_end_line=1
span_end_column=1
```

Accepted fixture

The first accepted source is:

```
examples/l-ui/nucleus-preview-card.lui
```

Error labels

The parser exposes stable labels for:

```
ok
null_argument
empty_source
unsupported_version
missing_card
missing_purpose
missing_effect
unsupported_effect
missing_boundary
unsupported_boundary
missing_rail
unknown_rail
unknown_binding_prefix
missing_required_binding
unterminated_string
unbalanced_brace
forbidden_behavior_marker
source_too_large
internal_error
```

Validation strategy

The parser is intentionally conservative. It performs:

- source length check;
- quote balance check;
- brace balance check;
- required header and card checks;
- required effect and boundary checks;
- forbidden marker checks;
- required rail checks;
- required binding checks;
- point-location metadata;
- source-span metadata;
- no-effect summary output.

It does not produce a full AST yet.

Report shape

The parser result report renders deterministic text and appends span fields:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Test commands

Run:

```
sh scripts/test-l-ui-parser.sh
sh scripts/test-l-ui-parser-location.sh
sh scripts/test-l-ui-parser-source-span.sh
```

The main C workflow runs these checks after existing state, transition, Nucleus, L-UI static report, grammar fixture, parser design, and parser plan guards.

Required invariants

The parser tests verify:

```
valid_fixture_parses_successfully
null_source_is_rejected
null_result_is_rejected
empty_source_is_rejected
oversized_source_is_rejected
unsupported_version_is_rejected
missing_card_is_rejected
missing_purpose_is_rejected
missing_effect_is_rejected
unsupported_effect_is_rejected
missing_boundary_is_rejected
unsupported_boundary_is_rejected
missing_required_rail_is_rejected
missing_required_binding_is_rejected
unknown_binding_prefix_is_rejected
forbidden_behavior_marker_is_rejected
unterminated_string_is_rejected
unbalanced_brace_is_rejected
valid_parse_returns_no_effect_flags
error_results_preserve_no_execution_flags
parse_error_labels_are_stable
```

The source-span tests verify span offsets, exclusive end offsets, one-based line/column ranges, byte-based columns, LF/CRLF/CR handling, report fields, diagnostic span copying, and default success spans.

Current evidence level

This implementation is an L2 tested model for no-effect L-UI source validation, point locations, and source-span metadata.

It is not a renderer, interactive UI, shell, compositor, command language, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, AST system, or security boundary.

Next implementation step

The next implementation candidate after source spans is:

```
L-UI parser AST contract
```

That future work should define AST shape, node ownership, source-span usage, and no-effect boundaries before any AST implementation.

Non-claims

This document and implementation do not claim AST construction, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_AST_CONTRACT.md

Source title: Latticra L-UI Parser AST Contract

Lines: 384 | Words: 962 | SHA-256: 69abe05546f83aac90f701823feaf60659367e05877ac8cdf9642b049ed72bde

Latticra L-UI Parser AST Contract

Status: AST contract Scope: future AST shape, node ownership, source-span usage, capacity rules, diagnostics relationship, and no-effect boundaries before AST implementation.

Purpose

This document defines the initial AST contract for the L-UI parser.

The parser currently validates source, produces parse summaries, emits precise point locations, and records source-span metadata. The next architectural step is an AST contract that defines how source may later become a structured tree without adding rendering, command behavior, live movement, or mutation.

This document does not implement an AST.

Current boundary

The current parser provides:

```
parse status
parse error category
point line and column
source span metadata
diagnostic mapping
diagnostic reports
no-effect flags
```

This contract does not add:

```
AST construction
AST allocation
```

```
AST traversal
renderer integration
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

AST purpose

A future AST should represent validated .lui source as structured, inspectable metadata.

A future AST may support:

- stable card summaries;
- rail and field enumeration;
- source-span-aware diagnostics;
- fixture validation reports;
- later renderer contracts;
- later tooling that can inspect L-UI source without executing it.

The AST must remain metadata-only until a separate renderer or execution contract exists.

No-effect rule

AST construction must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

AST code must not run L-UI behavior.

Ownership rule

The first AST implementation should use caller-provided storage or fixed-size result storage.

It must not require unbounded heap allocation.

It must not retain borrowed pointers into a source buffer unless a later lifetime contract explicitly permits it.

The first AST should prefer copied labels and spans:

```
card_name
purpose
boundary
effect
rail names
field names
binding paths
source spans
```

Public API rule

AST implementation will require a public API extension.

A future implementation plan must define:

```
AST struct names
capacity constants
```

```
parse function shape
report function shape
error behavior
compatibility expectations
```

No AST API should be added until an implementation plan is merged.

Proposed capacity constants

Initial proposed constants:

```
LATTICRA_L_UI_AST_RAIL_MAX = 16
LATTICRA_L_UI_AST_FIELD_MAX = 64
LATTICRA_L_UI_AST_TEXT_MAX = 16
LATTICRA_L_UI_AST_LABEL_MAX = 64
LATTICRA_L_UI_AST_BINDING_MAX = 96
LATTICRA_L_UI_AST_PURPOSE_MAX = 128
LATTICRA_L_UI_AST_REPORT_MAX = 2048
```

These values should be reviewed in the implementation plan.

Proposed node kinds

Initial AST node kinds:

```
card
rail
field
text
binding
```

Reserved future node kinds:

```
layout
theme
action
condition
```

Reserved node kinds must not become executable without a future contract.

Card node

The first card node should include:

```
name
purpose
effect
boundary
span
rail_count
field_count
text_count
```

The first accepted card remains:

```
NucleusPreview
```

Rail node

A rail node should include:

```
name
span
first_field_index
field_count
first_text_index
text_count
```

Initial required rails:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Field node

A field node should include:

```
name
binding
span
binding_span
```

Allowed binding prefixes remain:

```
state.
preview.
```

Text node

A text node should include:

```
value
span
```

Text nodes are literal metadata only. They must not be interpreted as commands.

Source-span usage

Every AST node should carry source-span metadata when available.

Rules:

1. Card span covers the full card declaration.
2. Rail span covers the rail block.
3. Field span covers the field declaration.
4. Binding span covers the binding path only.
5. Text span covers the quoted text value or full text declaration, as defined by the implementation plan.
6. Missing constructs should not produce AST nodes.

AST result shape

A future AST parse result may include:

```
parse_result
card
rails[]
fields[]
texts[]
rail_count
field_count
text_count
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

The exact shape must be finalized in an implementation plan.

AST report shape

A future AST report may render deterministic text such as:

```
L-UI AST SUMMARY
card=NucleusPreview
rail_count=&lt;n&gt;
field_count=&lt;n&gt;
text_count=&lt;n&gt;
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
```

```
recovery_allowed=0
hardware_allowed=0
```

Detailed rail and field reporting should be planned separately if needed.

Initial accepted fixture

The first AST target should be:

```
examples/l-ui/nucleus-preview-card.lui
```

The expected summary should include:

```
card=NucleusPreview
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Diagnostics relationship

AST construction should depend on a successful parse.

If parsing fails, AST construction should return the parser error and must not emit partial executable structures.

A later contract may define partial AST behavior for diagnostics, but the first AST should avoid partial AST output.

Forbidden behavior

A future AST implementation must not:

- add file I/O to parser code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- accept unsupported effects;
- treat text nodes as commands;
- treat bindings as executable references.

Implementation gate

AST implementation must not begin until a separate implementation plan defines:

1. public API changes;
2. AST struct placement;
3. capacity constants;
4. ownership and lifetime rules;

5. source-span integration;
6. successful parse requirements;
7. report format;
8. test file names;
9. exact invariant tests;
10. compatibility expectations.

Future test list

A future implementation plan should include tests for:

```
valid_fixture_builds_ast_summary
ast_card_name_is_nucleus_preview
ast_counts_match_fixture
ast_preserves_effect_none
ast_preserves_boundary_preview_only
ast_contains_required_rails
ast_contains_required_fields
ast_contains_text_nodes
ast_nodes_have_source_spans
ast_bindings_have_binding_spans
ast_rejects_failed_parse
ast_preserves_no_effect_flags
ast_report_contains_required_fields
ast_report_is_deterministic
ast_capacity_limits_are_enforced
ast_does_not_retain_source_pointers
ast_labels_are_stable
```

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-parser-ast-contract.sh
```

The guard is static. It does not implement an AST.

Non-claims

This document does not implement AST construction, AST traversal, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_AST_IMPLEMENTATION.md

Source title: Latticra L-UI Parser AST Implementation

Lines: 292 | Words: 734 | SHA-256: 5e92b1662cce65c359612288d4869059d1e1efa516e28c512854be224dcf3228

Latticra L-UI Parser AST Implementation

Status: implementation contract Scope: fixed-size AST metadata, source-backed purpose/text extraction, parse-result integration, source-span-aware nodes, compact and detailed AST reports, and no-effect invariants.

Purpose

The L-UI Parser AST implementation builds a fixed-size metadata tree for validated L-UI source.

It depends on successful source validation. It copies labels, bindings, source-backed purpose text, source-backed text values, and source spans into bounded public structures. It does not add rendering, command behavior, Nucleus task handling, live movement, state mutation, server interaction, recovery behavior, self-update behavior, hardware behavior, or boot behavior.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
tests/l_ui_parser_ast_invariants.c
tests/l_ui_parser_ast_detailed_report_invariants.c
tests/l_ui_ast_source_backed_text_invariants.c
scripts/test-l-ui-parser-ast.sh
scripts/test-l-ui-ast-detailed-report.sh
scripts/test-l-ui-ast-source-backed-text.sh
```

Public API

```
latticra_l_ui_ast_node_kind_label
latticra_l_ui_parse_ast
latticra_l_ui_ast_report
latticra_l_ui_ast_detailed_report
```

Public types

The implementation adds:

```
latticra_l_ui_ast_node_kind_t
latticra_l_ui_ast_card_t
latticra_l_ui_ast_rail_t
latticra_l_ui_ast_field_t
latticra_l_ui_ast_text_t
latticra_l_ui_ast_result_t
```

Capacity constants

```
LATTICRA_L_UI_AST_RAIL_MAX = 16
LATTICRA_L_UI_AST_FIELD_MAX = 64
LATTICRA_L_UI_AST_TEXT_MAX = 16
LATTICRA_L_UI_AST_LABEL_MAX = 64
LATTICRA_L_UI_AST_BINDING_MAX = 96
LATTICRA_L_UI_AST_PURPOSE_MAX = 128
LATTICRA_L_UI_AST_REPORT_MAX = 2048
LATTICRA_L_UI_AST_DETAILED_REPORT_MAX = 16384
```

Node kinds

Stable node kind labels:

```
card
rail
field
text
binding
unknown
```

AST result

A valid AST result contains:

```
parse_result
card
rails[]
fields[]
texts[]
rail_count
field_count
text_count
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

First accepted fixture

The first accepted source remains:

```
examples/l-ui/nucleus-preview-card.lui
```

The expected AST summary is:

```
card=NucleusPreview
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Source-backed text extraction

The AST extracts these values from validated source:

```
purpose &quot;...&quot; -&gt; ast.card.purpose
first text &quot;...&quot; -&gt; ast.texts[0].value
second text &quot;...&quot; -&gt; ast.texts[1].value
```

The surrounding quotes are not copied into AST values.

The extraction copies raw bytes between quotes and does not decode string escapes.

If an extracted value is too large for its fixed destination buffer, AST construction classifies the result as:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

and avoids partial AST counts.

Source-span usage

The AST copies source spans into populated nodes:

```
card.span
rail.span
field.span
field.binding_span
text.span
```

Text spans cover the extracted text value range. No public purpose span is currently exposed.

Spans are metadata only.

Ownership and lifetime

The AST implementation:

- uses fixed-size public storage;
- copies labels, purpose text, text values, binding paths, and spans;
- does not retain borrowed pointers into source;
- does not require heap allocation;
- treats all nodes as metadata only.

Failed parse behavior

AST construction depends on successful validation by latticra_l_ui_parse_source.

If parsing fails:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
no_effect=1
```

```
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

No partial AST is emitted in the first implementation.

Detailed AST reporting renders a compact failed-parse report in this case.

Compact AST report

The compact AST report renders:

```
L-UI AST SUMMARY
card=&lt;card&gt;
rail_count=&lt;n&gt;
field_count=&lt;n&gt;
text_count=&lt;n&gt;
effect=&lt;effect&gt;
boundary=&lt;boundary&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

Detailed AST report

The detailed AST report renders deterministic card, rail, field, text, source-span, binding-span, escaped-string, and no-effect metadata.

The detailed report starts with:

```
L-UI AST DETAILED REPORT
card=NucleusPreview
purpose=operator-visible Nucleus preview report
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
```

Detailed report capacity:

```
LATTICRA_L_UI_AST_DETAILED_REPORT_MAX = 16384
```

Test commands

Run:

```
sh scripts/test-l-ui-parser-ast.sh
sh scripts/test-l-ui-ast-detailed-report.sh
sh scripts/test-l-ui-ast-source-backed-text.sh
```

The main C workflow runs these checks after the AST implementation-plan guard, detailed report implementation-plan guard, and source-backed text implementation-plan guard.

Required invariants

The AST tests verify:

```
valid_fixture_builds_ast_summary
ast_card_name_is_nucleus_preview
ast_counts_match_fixture
ast_preserves_effect_none
ast_preserves_boundary_preview_only
ast_contains_required_rails
ast_contains_required_fields
ast_contains_text_nodes
ast_nodes_have_source_spans
ast_bindings_have_binding_spans
ast_rejects_failed_parse
ast_preserves_no_effect_flags
ast_report_contains_required_fields
ast_report_is_deterministic
```

```
ast_capacity_limits_are_enforced
ast_does_not_retain_source_pointers
ast_labels_are_stable
ast_node_kind_labels_are_stable
```

The detailed report tests verify deterministic card, rail, field, text, source-span, binding-span, failed-parse, and no-effect output.

The source-backed text tests verify extracted purpose/text values, copy ownership, quote exclusion, capacity failure classification, detailed report updates, no-effect preservation, and raw escape preservation.

Current evidence level

This implementation is an L2 tested metadata AST, source-backed text extraction, and detailed-report model for validated L-UI source.

It is not a renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after source-backed text extraction is:

```
L-UI string-literal escape contract
```

That future work should define whether and how string escapes are decoded before AST extraction decodes escape sequences.

Non-claims

This document and implementation do not claim L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_AST_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Parser AST Implementation Plan

Lines: 430 | Words: 1048 | SHA-256: 99f18c7bd47a147f2f22f3025dbdd3839aeaac6f23e5422a6937f67dce092693

Latticra L-UI Parser AST Implementation Plan

Status: implementation planning contract Scope: public API changes, struct placement, capacity constants, ownership/lifetime rules, source-span integration, exact tests, and compatibility expectations before AST implementation.

Purpose

This document defines the implementation plan for the first L-UI parser AST.

The AST contract is already merged and guarded. This plan decides the exact C API, structure layout, capacity model, ownership rules, source-span integration, report format, and test list before AST construction code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_PARSER_AST_CONTRACT.md
docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION.md
docs/L_UI_PARSER.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
```

Those files remain the source of truth for parser behavior, source spans, diagnostics, and no-effect boundaries.

Implementation language decision

The AST implementation should be in C.

Reason:

- the parser is implemented in C;
- parse results and spans are C ABI surfaces;
- the first AST must remain fixed-size and portable;
- the current C workflow can validate AST invariants;
- no dynamic runtime should be required.

Public API change

AST implementation requires public API additions in:

```
include/latticra/l_ui_parser.h
```

The first AST implementation should add structs and functions, but should not change the existing parser function signatures.

Proposed capacity constants

Add:

```
LATTICRA_L_UI_AST_RAIL_MAX = 16
LATTICRA_L_UI_AST_FIELD_MAX = 64
LATTICRA_L_UI_AST_TEXT_MAX = 16
LATTICRA_L_UI_AST_LABEL_MAX = 64
LATTICRA_L_UI_AST_BINDING_MAX = 96
LATTICRA_L_UI_AST_PURPOSE_MAX = 128
LATTICRA_L_UI_AST_REPORT_MAX = 2048
```

These caps are intentionally larger than the first NucleusPreview fixture while remaining bounded.

AST node kind enum

Add:

```
typedef enum {
    LATTICRA_L_UI_AST_NODE_CARD = 0,
    LATTICRA_L_UI_AST_NODE_RAIL = 1,
    LATTICRA_L_UI_AST_NODE_FIELD = 2,
    LATTICRA_L_UI_AST_NODE_TEXT = 3,
    LATTICRA_L_UI_AST_NODE_BINDING = 4,
    LATTICRA_L_UI_AST_NODE_UNKNOWN = 5
} latticra_l_ui_ast_node_kind_t;
```

Reserved future node kinds such as layout, theme, action, and condition must remain unimplemented until future contracts exist.

Card struct

Add a card summary struct:

```
typedef struct {
    char name[LATTICRA_L_UI_AST_LABEL_MAX];
    char purpose[LATTICRA_L_UI_AST_PURPOSE_MAX];
    char effect[LATTICRA_L_UI_LABEL_MAX];
    char boundary[LATTICRA_L_UI_LABEL_MAX];
    latticra_l_ui_source_span_t span;
    size_t rail_count;
    size_t field_count;
    size_t text_count;
} latticra_l_ui_ast_card_t;
```

Rail struct

Add:

```
typedef struct {
    char name[LATTICRA_L_UI_AST_LABEL_MAX];
    latticra_l_ui_source_span_t span;
    size_t first_field_index;
    size_t field_count;
    size_t first_text_index;
    size_t text_count;
} latticra_l_ui_ast_rail_t;
```

Field struct

Add:

```
typedef struct {
    char name[LATTICRA_L_UI_AST_LABEL_MAX];
    char binding[LATTICRA_L_UI_AST_BINDING_MAX];
    latticra_l_ui_source_span_t span;
    latticra_l_ui_source_span_t binding_span;
} latticra_l_ui_ast_field_t;
```

Text struct

Add:

```
typedef struct {
    char value[LATTICRA_L_UI_AST_PURPOSE_MAX];
    latticra_l_ui_source_span_t span;
} latticra_l_ui_ast_text_t;
```

AST result struct

Add:

```
typedef struct {
    latticra_l_ui_parse_result_t parse_result;
    latticra_l_ui_ast_card_t card;
    latticra_l_ui_ast_rail_t rails[LATTICRA_L_UI_AST_RAIL_MAX];
    latticra_l_ui_ast_field_t fields[LATTICRA_L_UI_AST_FIELD_MAX];
    latticra_l_ui_ast_text_t texts[LATTICRA_L_UI_AST_TEXT_MAX];
    size_t rail_count;
    size_t field_count;
    size_t text_count;
    int no_effect;
    int execution_allowed;
    int mutation_allowed;
    int server_allowed;
    int recovery_allowed;
    int hardware_allowed;
} latticra_l_ui_ast_result_t;
```

Public API plan

Add:

```
const char *latticra_l_ui_ast_node_kind_label(
    latticra_l_ui_ast_node_kind_t kind);

latticra_status_t latticra_l_ui_parse_ast(
    const char *source,
    size_t source_len,
    latticra_l_ui_ast_result_t *ast);

latticra_status_t latticra_l_ui_ast_report(
    const latticra_l_ui_ast_result_t *ast,
    char *buffer,
    size_t buffer_len);
```

The AST parser should accept in-memory source only. It must not perform file I/O.

Module shape

Add or update:

```
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
tests/l_ui_parser_ast_invariants.c
scripts/test-l-ui-parser-ast.sh
.github/workflows/c.yml
```

A separate src/l_ui_parser_ast.c keeps AST construction distinct from the validator implementation.

Ownership and lifetime rules

The first AST must:

- use fixed-size storage;
- copy labels, purpose text, text values, binding paths, and spans;
- not store borrowed pointers into source;
- not require heap allocation;
- not expose mutable internal references;
- treat all AST nodes as metadata only.

Source-span integration

Every populated node should carry a span when possible.

Rules:

1. Card span covers the full card NucleusPreview { ... } block.
2. Rail span covers the rail block.
3. Field span covers the field declaration.
4. Binding span covers only the binding path.
5. Text span covers the quoted literal text.
6. Missing constructs do not produce AST nodes.

The first implementation may use simple deterministic scanning against the first fixture, but must remain bounded and no-effect.

Successful parse rule

AST construction should depend on successful validation by latticra_l_ui_parse_source.

If parsing fails:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

No partial AST should be emitted in the first implementation.

First accepted fixture summary

For:

```
examples/l-ui/nucleus-preview-card.lui
```

expected AST summary:

```
card=NucleusPreview
```

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

AST report shape

The first AST report should render:

```
L-UI AST SUMMARY
card=NucleusPreview
rail_count=&lt;n&gt;
field_count=&lt;n&gt;
text_count=&lt;n&gt;
effect=&lt;effect&gt;
boundary=&lt;boundary&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

Detailed node reports should be deferred until a later report contract.

Compatibility expectations

Existing parser, diagnostics, location, and source-span tests must continue to pass.

Existing parser API signatures must not be changed.

New AST structs are additive public API.

Existing parser result and diagnostic report shapes must not be modified by the AST implementation.

Exact implementation test list

The AST implementation PR should include tests for:

```
valid_fixture_builds_ast_summary
ast_card_name_is_nucleus_preview
ast_counts_match_fixture
ast_preserves_effect_none
ast_preserves_boundary_preview_only
ast_contains_required_rails
ast_contains_required_fields
ast_contains_text_nodes
ast_nodes_have_source_spans
ast_bindings_have_binding_spans
ast_rejects_failed_parse
ast_preserves_no_effect_flags
ast_report_contains_required_fields
ast_report_is_deterministic
ast_capacity_limits_are_enforced
ast_does_not_retain_source_pointers
ast_labels_are_stable
ast_node_kind_labels_are_stable
```

Required rails

The AST must include rails for:

```
top
state
trace
safety
gates
effects
policy
```

```
execution
bottom
```

Required field names

The first AST implementation should include fields for:

```
origin
route
axis
path
breadcrumb
trace
health
risk
lock
dark_phase
safe_portal
rollback
host
external
requested
request
policy
reason
executed
mutation
server
recovery
hardware
```

Required text values

The first AST implementation should include text values for:

```
Latticra / Nucleus Preview / effect-bound
preview-only no-live-movement no-host-effect no-external-effect
```

Forbidden implementation behavior

The AST implementation must not:

- add file I/O to parser code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- accept unsupported effects;
- treat text nodes as commands;
- treat bindings as executable references.

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
```

```
docs/L_UI_PARSER_AST_CONTRACT.md
```

and add:

```
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
```

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-ast-implementation-plan.sh
```

The guard is static. It does not implement an AST.

Implementation gate

AST implementation code may be added only after this plan is merged.

Non-claims

This document does not implement AST construction, AST traversal, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_DESIGN.md

Source title: Latticra L-UI Parser Design Contract

Lines: 305 | Words: 763 | SHA-256: cbc45e19f3c933758bdd7d8f62b2743ce0b5941d982a80478370b59966655151

Latticra L-UI Parser Design Contract

Status: parser design contract Scope: parser behavior, no-effect constraints, error categories, fixture expectations, and non-claims before parser implementation.

Purpose

This document defines the design contract for a future L-UI parser.

The parser should be introduced only after the L-UI grammar draft and fixtures are stable enough to justify implementation.

This document does not implement a parser.

Current boundary

The current project state includes:

```
L-UI source grammar draft
.lui fixture
static fixture guard script
static C report fixture
```

The current project state does not include:

```
L-UI parser
renderer engine
interactive UI
command execution
Nucleus execution
live movement
server interaction
update behavior
hardware behavior
boot behavior
recovery behavior
```

Parser role

A future L-UI parser should read .lui source and produce a no-effect parse result.

Input:

```
source bytes
source length
optional source name
```

Output:

```
parse status
error category
error location
normalized fixture summary
no-effect proof flags
```

Parser must not

The parser must not:

- execute commands;
- render UI;
- call Nucleus execution;
- mutate state;
- perform live movement;
- perform recovery behavior;
- perform update behavior;
- interact with servers;
- access networking;
- access hardware;
- change boot state;
- perform host effects;
- perform external effects;
- allocate hidden persistent state;
- silently accept unknown effect declarations.

Allowed parser behavior

A future parser may:

- scan source;
- tokenize source;
- validate required clauses;
- validate allowed rails;
- validate allowed bindings;
- validate string literals;
- validate effect none;
- validate boundary preview_only;
- return deterministic errors;
- return a normalized no-effect summary.

Error categories

Initial parser error categories should include:

```
ok
null_argument
empty_source
unsupported_version
missing_card
missing_purpose
missing_effect
unsupported_effect
missing_boundary
unsupported_boundary
missing_rail
unknown_rail
unknown_binding_prefix
missing_required_binding
unterminated_string
unbalanced_brace
forbidden_behavior_marker
source_too_large
internal_error
```

Source size rule

A future parser should define a maximum source size.

Initial proposed limit:

```
64 KiB
```

The parser must reject oversized source rather than reallocating without bounds.

Required fixture behavior

The first parser target should accept:

```
examples/l-ui/nucleus-preview-card.lui
```

The parser must confirm:

- version is lui 0.1;
- card name is NucleusPreview;
- purpose exists;
- effect is none;
- boundary is preview_only;
- required rails exist;
- required bindings exist;
- forbidden behavior markers are absent.

Required rails

The first parser target must recognize:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Required binding prefixes

Initial allowed binding prefixes:

```
state.
preview.
```

No other prefix should be accepted until documented.

Effect validation

Only this effect declaration is valid for the first parser target:

```
effect none
```

Examples that must be rejected:

```
effect read
effect local_mutation
effect host_mutation
effect network
effect hardware
effect boot
effect recovery
effect external
```

Boundary validation

Only this boundary is valid for the first parser target:

```
boundary preview_only
```

Any other boundary should be rejected until documented.

Forbidden markers

Parser and fixture guards should reject markers that imply behavior outside static report layout:

```
execute
host_mutation
network
hardware
boot
recovery
self_update
server call
server_interaction {
```

Future syntax may refine this list, but the first parser must be conservative.

Parse result shape

A future parse result may use a C structure similar to:

```
status
error
line
column
card_name
rail_count
field_count
effect
boundary
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

Required no-effect flags:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test plan for future parser implementation

Future parser tests should verify:

- valid fixture parses successfully;
- unsupported version is rejected;
- missing card is rejected;
- missing purpose is rejected;
- missing effect is rejected;
- unsupported effect is rejected;
- missing boundary is rejected;
- unsupported boundary is rejected;
- missing required rail is rejected;
- missing required binding is rejected;
- unknown binding prefix is rejected;
- forbidden behavior marker is rejected;
- oversized source is rejected;
- parser returns no-effect flags for valid fixtures.

Implementation gate

Parser code should not be added until:

1. this design contract is merged;
2. fixture guardrails pass;
3. parser error categories are reviewed;
4. parse result shape is approved;
5. no-effect invariants are explicit in tests.

Current validation command

This design contract is guarded by:

```
sh scripts/test-l-ui-parser-design.sh
```

The guard is static. It does not parse L-UI.

Non-claims

This document does not implement L-UI parsing, rendering, command execution, Nucleus execution, live movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_DIAGNOSTICS.md

Source title: Latticra L-UI Parser Diagnostics Contract

Lines: 234 | Words: 773 | SHA-256: 3dc68bf2c0ad4d31bf44f5f58aa62ac01baae845705071ccce3627b9ecfe425e

Latticra L-UI Parser Diagnostics Contract

Status: diagnostics contract Scope: stable parser diagnostic language, line/column behavior, result reporting, and implementation guardrails before richer diagnostics code.

Purpose

This document defines the diagnostics contract for the L-UI parser.

The current parser already returns an error category, line, column, and deterministic result report. This contract defines how those fields should become stable diagnostic language before richer diagnostic implementation is added.

This document does not implement new diagnostics.

Current boundary

The current parser provides:

```
parse status
parse error category
line
column
card name
rail count
field count
effect
boundary
no-effect flags
parser result report
```

This contract does not add:

```
renderer behavior
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
self-update behavior
hardware behavior
boot behavior
recovery behavior
```

Diagnostic shape

A future diagnostic should be representable as:

```
severity=&lt;severity&gt;
code=&lt;stable-code&gt;
message=&lt;stable-message&gt;
line=&lt;line&gt;
column=&lt;column&gt;
hint=&lt;optional-remediation-hint&gt;
```

Severity levels

Initial severity levels:

```
ok
error
internal
```

Meanings:

```
| Severity | Meaning |
| --- | --- |
| `ok` | Valid parse result. |
| `error` | Source was rejected with a stable user-facing reason. |
| `internal` | Unexpected parser-side failure or unknown category. |
```

Stable diagnostic codes

Diagnostic codes should mirror parser error labels.

Initial codes:

```
LUI0000 ok
LUI0001 null_argument
LUI0002 empty_source
LUI0003 unsupported_version
LUI0004 missing_card
LUI0005 missing_purpose
LUI0006 missing_effect
LUI0007 unsupported_effect
LUI0008 missing_boundary
LUI0009 unsupported_boundary
LUI0010 missing_rail
LUI0011 unknown_rail
LUI0012 unknown_binding_prefix
LUI0013 missing_required_binding
LUI0014 unterminated_string
LUI0015 unbalanced_brace
LUI0016 forbidden_behavior_marker
LUI0017 source_too_large
LUI0018 internal_error
```

Stable messages

Initial message language should be concise and deterministic:

```
| Code | Message |
| --- | --- |
| `LUI0000` | Parse completed successfully. |
| `LUI0001` | Parser received a null argument. |
| `LUI0002` | L-UI source is empty. |
| `LUI0003` | L-UI version is not supported. |
| `LUI0004` | Required card declaration is missing. |
| `LUI0005` | Required purpose clause is missing. |
| `LUI0006` | Required effect clause is missing. |
| `LUI0007` | Effect declaration is not supported. |
| `LUI0008` | Required boundary clause is missing. |
| `LUI0009` | Boundary declaration is not supported. |
| `LUI0010` | Required rail is missing. |
| `LUI0011` | Rail name is not supported. |
| `LUI0012` | Binding prefix is not supported. |
| `LUI0013` | Required binding is missing. |
| `LUI0014` | String literal is not terminated. |
| `LUI0015` | Braces are not balanced. |
| `LUI0016` | Source contains a forbidden behavior marker. |
| `LUI0017` | Source exceeds the supported size limit. |
| `LUI0018` | Parser reached an internal error. |
```

Line and column behavior

Line and column should be one-based.

Default values:

```
line=1
column=1
```

A future diagnostics implementation may improve precision, but it must remain deterministic and must not require file I/O inside parser code.

Rules:

1. If a precise location is not available, return line=1 and column=1.
2. If a token location is available, line and column should point to the beginning of the rejected token or clause.

3. Line and column should count bytes in the in-memory source buffer, not terminal display cells.
4. Diagnostics must not depend on platform-specific newline behavior.
5. Diagnostics must preserve no-effect flags.

Hint language

Hints should be stable and optional.

Initial hint examples:

```
Use version header: lui 0.1
Add card NucleusPreview { ... }
Add effect none
Add boundary preview_only
Use only state. or preview. binding prefixes
Remove unsupported effect declarations
```

Hints must not suggest behavior outside the current no-effect grammar.

Report integration

A future parser diagnostic report may extend the current parser result report with:

```
diagnostic_severity=&lt;severity&gt;
diagnostic_code=&lt;code&gt;
diagnostic_message=&lt;message&gt;
diagnostic_hint=&lt;hint-or-empty&gt;
```

This should be added only after tests lock down stable codes and messages.

Required test plan for future diagnostics implementation

Future diagnostics tests should verify:

```
ok_result_has_lui0000
empty_source_reports_lui0002
unsupported_version_reports_lui0003
missing_card_reports_lui0004
missing_purpose_reports_lui0005
missing_effect_reports_lui0006
unsupported_effect_reports_lui0007
missing_boundary_reports_lui0008
unsupported_boundary_reports_lui0009
missing_rail_reports_lui0010
unknown_binding_prefix_reports_lui0012
missing_required_binding_reports_lui0013
unterminated_string_reports_lui0014
unbalanced_brace_reports_lui0015
forbidden_marker_reports_lui0016
source_too_large_reports_lui0017
error_diagnostics_preserve_no_effect_flags
unknown_error_reports_internal
line_and_column_are_one_based
messages_are_stable
```

Implementation gate

Diagnostic implementation code should not be added until:

1. this contract is merged;
2. diagnostic codes are stable;
3. message text is stable;
4. line and column behavior is accepted;
5. tests preserve no-effect flags;
6. result report extension shape is documented.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-parser-diagnostics-contract.sh
```

The guard is static. It does not implement diagnostics.

Non-claims

This document does not implement richer parser diagnostics, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md

Source title: Latticra L-UI Parser Diagnostics Implementation

Lines: 195 | Words: 649 | SHA-256: 7af0c8d7db3ab56783fddbf92bcbecb8cbf2e794e1001d124cc4e6b29f77fae7

Latticra L-UI Parser Diagnostics Implementation

Status: implementation contract Scope: diagnostic severity, stable codes, stable messages, hints, source spans, no-effect flags, and deterministic diagnostic reports.

Purpose

The L-UI Parser Diagnostics implementation maps parser results into stable diagnostics.

It turns latticra_l_ui_parse_result_t into latticra_l_ui_diagnostic_t and renders deterministic diagnostic reports without adding L-UI rendering, interactive UI behavior, command behavior, Nucleus task handling, live movement, state mutation, server interaction, recovery behavior, self-update behavior, hardware behavior, or boot behavior.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser_diagnostics.c
tests/l_ui_parser_diagnostics_invariants.c
tests/l_ui_parser_source_span_invariants.c
scripts/test-l-ui-parser-diagnostics.sh
scripts/test-l-ui-parser-source-span.sh
```

Public API

```
latticra_l_ui_diagnostic_severity_label
latticra_l_ui_diagnostic_from_parse_result
latticra_l_ui_diagnostic_report
```

Severity labels

```
ok
error
internal
```

Diagnostic fields

```
severity
code
message
hint
line
column
span
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

The diagnostic span contains:

```
start_offset
end_offset
start_line
start_column
end_line
end_column
```

Diagnostics copy span metadata from parse results.

Report shape

The diagnostic report renders:

```
L-UI PARSE DIAGNOSTIC
severity=&lt;severity&gt;
code=&lt;code&gt;
message=&lt;message&gt;
line=&lt;line&gt;
column=&lt;column&gt;
hint=&lt;hint-or-empty&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Stable diagnostic mapping

```
| Parser error | Severity | Code | Message |
| --- | --- | --- | --- |
| `ok` | `ok` | `LUI0000` | Parse completed successfully. |
| `null_argument` | `error` | `LUI0001` | Parser received a null argument. |
| `empty_source` | `error` | `LUI0002` | L-UI source is empty. |
| `unsupported_version` | `error` | `LUI0003` | L-UI version is not supported. |
| `missing_card` | `error` | `LUI0004` | Required card declaration is missing. |
| `missing_purpose` | `error` | `LUI0005` | Required purpose clause is missing. |
| `missing_effect` | `error` | `LUI0006` | Required effect clause is missing. |
| `unsupported_effect` | `error` | `LUI0007` | Effect declaration is not supported. |
| `missing_boundary` | `error` | `LUI0008` | Required boundary clause is missing. |
| `unsupported_boundary` | `error` | `LUI0009` | Boundary declaration is not supported. |
| `missing_rail` | `error` | `LUI0010` | Required rail is missing. |
| `unknown_rail` | `error` | `LUI0011` | Rail name is not supported. |
| `unknown_binding_prefix` | `error` | `LUI0012` | Binding prefix is not supported. |
| `missing_required_binding` | `error` | `LUI0013` | Required binding is missing. |
| `unterminated_string` | `error` | `LUI0014` | String literal is not terminated. |
| `unbalanced_brace` | `error` | `LUI0015` | Braces are not balanced. |
| `forbidden_behavior_marker` | `error` | `LUI0016` | Source contains a forbidden behavior marker. |
| `source_too_large` | `error` | `LUI0017` | Source exceeds the supported size limit. |
| `internal_error` | `internal` | `LUI0018` | Parser reached an internal error. |
```

No-effect boundary

Diagnostics copy the safety flags from parse results and preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test commands

Run:

```
sh scripts/test-l-ui-parser-diagnostics.sh
sh scripts/test-l-ui-parser-source-span.sh
```

The main C workflow runs these checks after the parser diagnostics implementation plan guard and source-span implementation plan guard.

Required invariants

The diagnostics tests verify:

```
ok_result_has_lui0000
empty_source_reports_lui0002
unsupported_version_reports_lui0003
missing_card_reports_lui0004
missing_purpose_reports_lui0005
missing_effect_reports_lui0006
unsupported_effect_reports_lui0007
missing_boundary_reports_lui0008
unsupported_boundary_reports_lui0009
missing_rail_reports_lui0010
unknown_binding_prefix_reports_lui0012
missing_required_binding_reports_lui0013
unterminated_string_reports_lui0014
unbalanced_brace_reports_lui0015
forbidden_marker_reports_lui0016
source_too_large_reports_lui0017
internal_error_reports_lui0018
error_diagnostics_preserve_no_effect_flags
line_and_column_are_one_based
messages_are_stable
hints_are_stable
diagnostic_report_contains_required_fields
diagnostic_report_rejects_bad_arguments_and_small_buffers
severity_labels_are_stable
```

The source-span tests verify diagnostic span copying and diagnostic report span fields.

Current evidence level

This implementation is an L2 tested diagnostics and source-span metadata model for the L-UI parser.

It is not a renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, AST system, or security boundary.

Next implementation step

The next implementation candidate after source spans is:

```
L-UI parser AST contract
```

That future work should define AST shape, node ownership, source-span usage, and no-effect boundaries before any AST implementation.

Non-claims

This document and implementation do not claim AST construction, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Parser Diagnostics Implementation Plan

Lines: 319 | Words: 1062 | SHA-256: 401b4d340b70afc0c2dab01eafc17bc88eaa61f659adbdac65bb72349c65af4f

Latticra L-UI Parser Diagnostics Implementation Plan

Status: implementation planning contract Scope: diagnostic API shape, exact tests, parser result report extension rules, and no-effect constraints before diagnostic code.

Purpose

This document defines the implementation plan for richer L-UI parser diagnostics.

The diagnostics contract is already merged and guarded. This plan decides the C API, data shape, exact test list, and report extension rules before implementation code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_PARSER_DIAGNOSTICS.md
docs/L_UI_PARSER.md
docs/L_UI_PARSER_RESULT_REPORT.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
```

Those files remain the source of truth for parser errors, parse summaries, no-effect flags, and current result reporting.

Implementation language decision

The first diagnostics implementation should be in C.

Reason:

- the parser is already implemented in C;
- diagnostics should be part of the same low-level ABI surface;
- the implementation should remain portable and deterministic;
- diagnostics should require no runtime beyond the current C test environment.

Proposed module shape

The first diagnostics implementation should extend the existing parser module:

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
tests/l_ui_parser_diagnostics_invariants.c
scripts/test-l-ui-parser-diagnostics.sh
```

A separate diagnostics source file is not required for the first implementation.

Diagnostic enums

Add diagnostic severity enum:

```
LATTICRA_L_UI_DIAGNOSTIC_OK
LATTICRA_L_UI_DIAGNOSTIC_ERROR
LATTICRA_L_UI_DIAGNOSTIC_INTERNAL
```

Diagnostic codes should remain stable strings instead of a second enum in the first implementation.

Reason: parser errors already have a stable enum. Code strings can be mapped directly from the parser error enum.

Diagnostic result shape

Add a compact diagnostic struct:

```
severity
code
message
hint
```

```
line
column
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

Recommended fixed sizes:

```
LATTICRA_L_UI_DIAGNOSTIC_CODE_MAX = 16
LATTICRA_L_UI_DIAGNOSTIC_MESSAGE_MAX = 128
LATTICRA_L_UI_DIAGNOSTIC_HINT_MAX = 128
LATTICRA_L_UI_DIAGNOSTIC_REPORT_MAX = 1024
```

Public API plan

Initial functions:

```
const char *latticra_l_ui_diagnostic_severity_label(
    latticra_l_ui_diagnostic_severity_t severity);

latticra_status_t latticra_l_ui_diagnostic_from_parse_result(
    const latticra_l_ui_parse_result_t *parse_result,
    latticra_l_ui_diagnostic_t *diagnostic);

latticra_status_t latticra_l_ui_diagnostic_report(
    const latticra_l_ui_diagnostic_t *diagnostic,
    char *buffer,
    size_t buffer_len);
```

Parser result report extension

The current parser result report should not be changed in the first diagnostics implementation.

Instead, diagnostics should have a separate deterministic report:

```
L-UI PARSE DIAGNOSTIC
severity=&lt;severity&gt;
code=&lt;code&gt;
message=&lt;message&gt;
line=&lt;line&gt;
column=&lt;column&gt;
hint=&lt;hint-or-empty&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

This avoids destabilizing the existing parser result report.

Mapping table

The first implementation should map parser errors to diagnostic fields exactly:

```
| Parser error | Severity | Code | Message |
| --- | --- | --- | --- |
| `ok` | `ok` | `LUI0000` | Parse completed successfully. |
| `null_argument` | `error` | `LUI0001` | Parser received a null argument. |
| `empty_source` | `error` | `LUI0002` | L-UI source is empty. |
| `unsupported_version` | `error` | `LUI0003` | L-UI version is not supported. |
| `missing_card` | `error` | `LUI0004` | Required card declaration is missing. |
| `missing_purpose` | `error` | `LUI0005` | Required purpose clause is missing. |
| `missing_effect` | `error` | `LUI0006` | Required effect clause is missing. |
| `unsupported_effect` | `error` | `LUI0007` | Effect declaration is not supported. |
| `missing_boundary` | `error` | `LUI0008` | Required boundary clause is missing. |
| `unsupported_boundary` | `error` | `LUI0009` | Boundary declaration is not supported. |
| `missing_rail` | `error` | `LUI0010` | Required rail is missing. |
| `unknown_rail` | `error` | `LUI0011` | Rail name is not supported. |
| `unknown_binding_prefix` | `error` | `LUI0012` | Binding prefix is not supported. |
| `missing_required_binding` | `error` | `LUI0013` | Required binding is missing. |
```

`unterminated_string`	`error`	`LUI0014`	String literal is not terminated.
`unbalanced_brace`	`error`	`LUI0015`	Braces are not balanced.
`forbidden_behavior_marker`	`error`	`LUI0016`	Source contains a forbidden behavior marker.
`source_too_large`	`error`	`LUI0017`	Source exceeds the supported size limit.
`internal_error`	`internal`	`LUI0018`	Parser reached an internal error.

Hint mapping

Initial hints:

```
LUI0000 -&gt; empty hint
LUI0001 -&gt; Provide non-null source and result pointers.
LUI0002 -&gt; Provide L-UI source beginning with: lui 0.1
LUI0003 -&gt; Use version header: lui 0.1
LUI0004 -&gt; Add card NucleusPreview { ... }
LUI0005 -&gt; Add a purpose string.
LUI0006 -&gt; Add effect none.
LUI0007 -&gt; Use only effect none.
LUI0008 -&gt; Add boundary preview_only.
LUI0009 -&gt; Use only boundary preview_only.
LUI0010 -&gt; Add the required NucleusPreview rails.
LUI0011 -&gt; Use only documented rail names.
LUI0012 -&gt; Use only state. or preview. binding prefixes.
LUI0013 -&gt; Add all required NucleusPreview bindings.
LUI0014 -&gt; Close the string literal.
LUI0015 -&gt; Balance opening and closing braces.
LUI0016 -&gt; Remove behavior markers outside the static report grammar.
LUI0017 -&gt; Keep L-UI source at or below LATTICRA_L_UI_SOURCE_MAX.
LUI0018 -&gt; Report the internal parser error.
```

Hints must not suggest behavior outside the current no-effect grammar.

Line and column behavior

The first implementation should copy line and column from the parse result.

Current parser line and column are conservative:

```
line=1
column=1
```

The diagnostics implementation should not attempt richer location detection yet.

No-effect flags

Diagnostics must copy safety flags from parse results:

```
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

For all current parse paths, expected values remain:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact test list

The first diagnostics implementation PR should include tests for:

```
ok_result_has_lui0000
empty_source_reports_lui0002
unsupported_version_reports_lui0003
missing_card_reports_lui0004
missing_purpose_reports_lui0005
missing_effect_reports_lui0006
unsupported_effect_reports_lui0007
missing_boundary_reports_lui0008
unsupported_boundary_reports_lui0009
```

```
missing_rail_reports_lui0010
unknown_binding_prefix_reports_lui0012
missing_required_binding_reports_lui0013
unterminated_string_reports_lui0014
unbalanced_brace_reports_lui0015
forbidden_marker_reports_lui0016
source_too_large_reports_lui0017
internal_error_reports_lui0018
error_diagnostics_preserve_no_effect_flags
line_and_column_are_one_based
messages_are_stable
hints_are_stable
diagnostic_report_contains_required_fields
diagnostic_report_rejects_bad_arguments_and_small_buffers
severity_labels_are_stable
```

CI requirement

The first diagnostics implementation PR must add:

```
scripts/test-l-ui-parser-diagnostics.sh
```

and wire it into:

```
.github/workflows/c.yml
```

Documentation requirement

The first diagnostics implementation PR must update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_PARSER_DIAGNOSTICS.md
```

and add:

```
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
```

Forbidden implementation behavior

The diagnostics implementation must not:

- parse files directly;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- accept unsupported effects.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-diagnostics-implementation-plan.sh
```

The guard is static. It does not implement diagnostics.

Implementation gate

Diagnostics implementation code may be added only after this plan is merged.

Non-claims

This document does not implement richer parser diagnostics, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_FIXTURE_INTEGRATION.md

Source title: Latticra L-UI Parser Fixture Integration

Lines: 94 | Words: 287 | SHA-256: ba618d5dec97680770d1230bd150419902fd486506d2468f545b358b76776713

Latticra L-UI Parser Fixture Integration

Status: initial implementation contract Scope: repository fixture validation through the in-memory L-UI parser.

Purpose

This document describes the fixture integration test for the L-UI parser.

The parser remains in-memory only. The test harness reads the repository .lui fixture and passes the source bytes into the parser.

Implementation files

```
tests/l_ui_parser_fixture_integration.c
scripts/test-l-ui-parser-fixture-integration.sh
examples/l-ui/nucleus-preview-card.lui
```

Fixture under test

```
examples/l-ui/nucleus-preview-card.lui
```

Boundary

The parser still does not perform file I/O.

Only the test harness reads the fixture file.

The integration test must not add:

- L-UI rendering;
- interactive UI behavior;
- Nucleus task handling;
- live movement;
- state mutation;
- recovery behavior;
- server interaction;
- self-update behavior;
- host effects;
- network effects;
- hardware effects;

- boot effects.

Expected parse summary

The repository fixture must parse as:

```
error=ok
card_name=NucleusPreview
rail_count=9
field_count=23
effect=none
boundary=preview_only
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-parser-fixture-integration.sh
```

The main C workflow runs this check after the in-memory parser invariant test.

Current evidence level

This implementation is an L2 tested integration between the repository L-UI fixture and the in-memory parser.

It is not a renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after fixture integration is:

```
L-UI parser result report
```

That future work should render parse results as deterministic text while preserving no-effect behavior.

Non-claims

This document and implementation do not claim L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Parser Implementation Plan

Lines: 287 | Words: 674 | SHA-256: c8c0bbab6fc9630056d78ec7a61b72a1fed8dea8834704a275b97202eafd312a

Latticra L-UI Parser Implementation Plan

Status: implementation planning contract Scope: implementation language, module shape, API, source-size limit, exact test list, and no-effect constraints before parser code.

Purpose

This document defines the implementation plan for the first L-UI parser.

The parser should be implemented only after the parser design contract is merged and guarded. This plan decides what parser code is allowed to look like before code is introduced.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_SOURCE_GRAMMAR.md
docs/L_UI_PARSER_DESIGN.md
examples/l-ui/nucleus-preview-card.lui
scripts/test-l-ui-grammar-fixtures.sh
scripts/test-l-ui-parser-design.sh
```

Those documents and guards remain the source of truth for syntax, error categories, no-effect constraints, and fixture expectations.

Implementation language decision

The first parser should be implemented in C.

Reason:

- Latticra's current implementation foundation is C;
- the parser must stay portable and small;
- the parser output should become part of the low-level Latticra ABI;
- no dynamic runtime should be required;
- the parser can be tested with the existing C workflow.

Rust tooling can be added later for validation helpers, but the first parser should remain a C no-effect parser.

Proposed module shape

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
tests/l_ui_parser_invariants.c
scripts/test-l-ui-parser.sh
```

Source-size limit

The first parser should define:

```
LATTICRA_L_UI_SOURCE_MAX = 65536
```

Input larger than this must return source_too_large.

Parse result shape

The first parser should return a compact summary rather than a full AST.

Planned struct fields:

```
status
error
line
column
card_name
rail_count
field_count
effect
boundary
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

Error enum

The first C parser should implement these errors exactly:

```
LATTICRA_L_UI_PARSE_OK
LATTICRA_L_UI_PARSE_NULL_ARGUMENT
LATTICRA_L_UI_PARSE_EMPTY_SOURCE
LATTICRA_L_UI_PARSE_UNSUPPORTED_VERSION
LATTICRA_L_UI_PARSE_MISSING_CARD
LATTICRA_L_UI_PARSE_MISSING_PURPOSE
LATTICRA_L_UI_PARSE_MISSING_EFFECT
LATTICRA_L_UI_PARSE_UNSUPPORTED_EFFECT
LATTICRA_L_UI_PARSE_MISSING_BOUNDARY
LATTICRA_L_UI_PARSE_UNSUPPORTED_BOUNDARY
LATTICRA_L_UI_PARSE_MISSING_RAIL
LATTICRA_L_UI_PARSE_UNKNOWN_RAIL
LATTICRA_L_UI_PARSE_UNKNOWN_BINDING_PREFIX
LATTICRA_L_UI_PARSE_MISSING_REQUIRED_BINDING
LATTICRA_L_UI_PARSE_UNTERMINATED_STRING
LATTICRA_L_UI_PARSE_UNBALANCED_BRACE
LATTICRA_L_UI_PARSE_FORBIDDEN_BEHAVIOR_MARKER
LATTICRA_L_UI_PARSE_SOURCE_TOO_LARGE
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

Public API plan

Initial functions:

```
const char *latticra_l_ui_parse_error_label(latticra_l_ui_parse_error_t error);

latticra_status_t latticra_l_ui_parse_source(
    const char *source,
    size_t source_len,
    latticra_l_ui_parse_result_t *result);
```

Optional later function, not required in first parser PR:

```
latticra_status_t latticra_l_ui_parse_file(...)
```

The first parser should parse in-memory source only. It should not perform file I/O.

No-effect output flags

For valid fixtures, the parser must return:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

These flags must also remain safe on denied/error results.

First accepted source

The first accepted fixture is:

```
examples/l-ui/nucleus-preview-card.lui
```

The parser should accept this fixture and produce:

```
status=ok
error=ok
card_name=NucleusPreview
rail_count=9
field_count=23
effect=none
boundary=preview_only
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact test list for first parser PR

The first parser implementation PR should include tests for:

```
valid_fixture_parses_successfully
null_source_is_rejected
```

```
null_result_is_rejected
empty_source_is_rejected
oversized_source_is_rejected
unsupported_version_is_rejected
missing_card_is_rejected
missing_purpose_is_rejected
missing_effect_is_rejected
unsupported_effect_is_rejected
missing_boundary_is_rejected
unsupported_boundary_is_rejected
missing_required_rail_is_rejected
missing_required_binding_is_rejected
unknown_binding_prefix_is_rejected
forbidden_behavior_marker_is_rejected
valid_parse_returns_no_effect_flags
error_results_preserve_no_execution_flags
parse_error_labels_are_stable
```

Parser approach

The first parser should be simple and conservative.

Allowed approach:

- bounded source length check;
- string scanning;
- required-pattern validation;
- brace balance check;
- string literal quote balance check;
- required rail and binding validation;
- forbidden marker validation;
- normalized summary output.

Not required in first parser:

- complete AST;
- nested object model;
- renderer integration;
- file I/O;
- allocation-heavy tokenizer;
- recovery parser;
- interactive diagnostics;
- command execution.

Forbidden implementation behavior

The parser implementation must not:

- read files directly;
- write files;
- open sockets;
- call server code;
- call update code;
- call recovery code;

- call hardware code;
- mutate state lattice;
- perform live movement;
- execute L-UI source;
- render UI;
- use unbounded allocation;
- accept unsupported effects.

CI requirement

The first parser PR must add:

```
scripts/test-l-ui-parser.sh
```

and wire it into:

```
.github/workflows/c.yml
```

Documentation requirement

The first parser PR must update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_PARSER_DESIGN.md
```

and add:

```
docs/L_UI_PARSER.md
```

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-implementation-plan.sh
```

The guard is static. It does not implement the parser.

Implementation gate

Parser code may be added only after this plan is merged.

Non-claims

This document does not implement L-UI parsing, rendering, command execution, Nucleus execution, live movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_LINE_COLUMN_PRECISION_IMPLEMENTATION.md

Source title: Latticra L-UI Parser Line/Column Precision Implementation

Lines: 151 | Words: 361 | SHA-256: f4c84897821cc1d4c87c41a2029ce1b167f72d75ca08d110c14630c76ed3dfa0

Latticra L-UI Parser Line/Column Precision Implementation

Status: initial implementation contract Scope: private parser location helpers, parser touch-point updates, location invariants, and no-effect boundaries.

Purpose

The L-UI Parser Line/Column Precision implementation adds deterministic line and column reporting for selected parser errors.

It updates private parser internals only. It does not change the public parser API.

Implementation files

```
src/l_ui_parser.c
tests/l_ui_parser_location_invariants.c
scripts/test-l-ui-parser-location.sh
```

Public API stability

No public API was added for location precision.

The existing output fields remain the public location surface:

```
latticra_l_ui_parse_result_t.line
latticra_l_ui_parse_result_t.column
```

Diagnostic reports already render those fields through the diagnostic mapping.

Internal helper behavior

The parser now uses private helpers for:

```
location_default
location_for_index
find_slice_index
find_slice_location
card_body_location
find_unbalanced_brace_location
find_unterminated_string_location
set_error_at
```

These helpers remain private to src/l_ui_parser.c.

Location rules

Line and column are one-based.

Columns are byte-based, not display-cell-based.

Recognized newline forms:

```
LF
CRLF
CR
```

Implemented location touch points

The initial precision implementation covers:

```
unsupported_version
unsupported_effect
unsupported_boundary
unknown_binding_prefix
forbidden_behavior_marker
unterminated_string
unbalanced_brace
missing_effect
missing_boundary
missing_rail
```

Other errors may still use conservative default locations until a later source-span contract.

No-effect preservation

Location precision is diagnostic metadata only.

It must not change:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-parser-location.sh
```

The main C workflow runs this check after the line-column implementation plan guard.

Required invariants

The location tests verify:

```
unsupported_version_reports_precise_location
unsupported_effect_reports_effect_token_location
unsupported_boundary_reports_boundary_token_location
unknown_binding_prefix_reports_binding_target_location
forbidden_marker_reports_marker_location
unterminated_string_reports_opening_quote_location
unbalanced_open_brace_reports_opening_brace_location
unbalanced_close_brace_reports_closing_brace_location
missing_effect_reports_card_body_location
missing_boundary_reports_card_body_location
missing_rail_reports_card_body_location
location_scanner_handles_lf_newlines
location_scanner_handles_crlf_newlines
location_scanner_handles_cr_newlines
location_columns_are_byte_based
success_result_remains_line_one_column_one
error_locations_preserve_no_effect_flags
location_reports_are_deterministic
```

Current evidence level

This implementation is an L2 tested diagnostic-location model for the L-UI parser.

It is not a renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after line/column precision is:

```
L-UI parser source-span contract
```

That future work should define richer spans and source ranges before any AST or renderer integration.

Non-claims

This document and implementation do not claim L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_LINE_COLUMN_PRECISION_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Parser Line/Column Precision Implementation Plan

Lines: 271 | Words: 799 | SHA-256: a2b5b6974d7fe695334e61ba2d17a05344dcaba91e7c243e1d936a3df96c464d

Latticra L-UI Parser Line/Column Precision Implementation Plan

Status: implementation planning contract Scope: helper shapes, parser touch points, exact tests, and no-effect boundaries before changing parser location behavior.

Purpose

This document defines the implementation plan for precise line and column reporting in the L-UI parser.

The line/column precision plan is already merged and guarded. This plan decides how parser code should be changed before any location-tracking code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_PARSER_LINE_COLUMN_PRECISION_PLAN.md
docs/L_UI_PARSER_DIAGNOSTICS.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for parser errors, diagnostic reports, and no-effect behavior.

Implementation language decision

The precision implementation should be in C.

Reason:

- the parser is already implemented in C;
- parse results already expose line and column fields;
- diagnostic reports already render those fields;
- precision can be added without changing the public parser API;
- precision must remain portable and deterministic.

Proposed module touch points

The first precision implementation should modify:

```
src/l_ui_parser.c
tests/l_ui_parser_location_invariants.c
scripts/test-l-ui-parser-location.sh
.github/workflows/c.yml
```

It should not add a new public header.

Public API rule

No public API change is required for the first precision implementation.

Existing fields remain the output surface:

```
latticra_l_ui_parse_result_t.line
latticra_l_ui_parse_result_t.column
```

Existing diagnostic reports already include:

```
line=&lt;line&gt;
column=&lt;column&gt;
```

Internal helper shapes

The parser may add internal helpers similar to:

```
static void location_default(size_t *line, size_t *column);

static int find_slice_location(
    const char *source,
    size_t source_len,
    const char *needle,
    size_t *line,
    size_t *column);

static int find_unbalanced_brace_location(
    const char *source,
    size_t source_len,
    size_t *line,
```

```
    size_t *column);

static int find_unterminated_string_location(
    const char *source,
    size_t source_len,
    size_t *line,
    size_t *column);

static void set_error_at(
    latticra_l_ui_parse_result_t *result,
    latticra_l_ui_parse_error_t error,
    size_t line,
    size_t column);
```

These helpers should remain private to src/l_ui_parser.c.

Newline scanner behavior

The scanner must support:

```
LF:   \n
CRLF: \r\n
CR:   \r
```

Rules:

1. Line and column are one-based.
2. \n increments line and resets column to 1.
3. \r\n counts as one newline and resets column to 1.
4. Lone \r counts as one newline and resets column to 1.
5. All other bytes increment column by 1.
6. Columns are byte-based, not display-cell-based.
7. Scanner behavior must not depend on host text mode.

Parser touch-point plan

Initial parser changes should set precise locations for:

```
unsupported_version
unsupported_effect
unsupported_boundary
unknown_binding_prefix
forbidden_behavior_marker
unterminated_string
unbalanced_brace
missing_effect
missing_boundary
missing_rail
```

Other errors may keep conservative defaults until a later source-span contract.

Error location mapping

```
| Parser error | First implementation location |
| --- | --- |
| `ok` | `line=1`, `column=1` |
| `null_argument` | not applicable, direct status return |
| `empty_source` | `line=1`, `column=1` |
| `unsupported_version` | first `lui` token if present, otherwise `line=1`, `column=1` |
| `missing_card` | `line=1`, `column=1` |
| `missing_purpose` | card body start when available |
| `missing_effect` | card body start when available |
| `unsupported_effect` | `effect` token |
| `missing_boundary` | card body start when available |
| `unsupported_boundary` | `boundary` token |
| `missing_rail` | card body start when available |
| `unknown_rail` | unsupported `rail` token when implemented later |
| `unknown_binding_prefix` | unsupported binding target |
```

```
| `missing_required_binding` | relevant rail when available, otherwise card body start |
| `unterminated_string` | opening quote of unterminated string |
| `unbalanced_brace` | unmatched brace when available |
| `forbidden_behavior_marker` | first forbidden marker |
| `source_too_large` | `line=1`, `column=1` |
| `internal_error` | `line=1`, `column=1` |
```

No-effect preservation

Precision metadata must not change safety fields:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The first precision implementation PR should include tests for:

```
unsupported_version_reports_precise_location
unsupported_effect_reports_effect_token_location
unsupported_boundary_reports_boundary_token_location
unknown_binding_prefix_reports_binding_target_location
forbidden_marker_reports_marker_location
unterminated_string_reports_opening_quote_location
unbalanced_open_brace_reports_opening_brace_location
unbalanced_close_brace_reports_closing_brace_location
missing_effect_reports_card_body_location
missing_boundary_reports_card_body_location
missing_rail_reports_card_body_location
location_scanner_handles_lf_newlines
location_scanner_handles_crlf_newlines
location_scanner_handles_cr_newlines
location_columns_are_byte_based
success_result_remains_line_one_column_one
error_locations_preserve_no_effect_flags
location_reports_are_deterministic
```

Test file plan

Add:

```
tests/l_ui_parser_location_invariants.c
scripts/test-l-ui-parser-location.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The first precision implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_PARSER_LINE_COLUMN_PRECISION_PLAN.md
```

and add:

```
docs/L_UI_PARSER_LINE_COLUMN_PRECISION_IMPLEMENTATION.md
```

Forbidden implementation behavior

The precision implementation must not:

- add file I/O to parser code;
- write files;
- open network connections;
- call server code;

- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- change diagnostic report shape;
- accept unsupported effects.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-line-column-precision-implementation-plan.sh
```

The guard is static. It does not implement precision.

Implementation gate

Precision implementation code may be added only after this plan is merged.

Non-claims

This document does not implement precise parser locations, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_LINE_COLUMN_PRECISION_PLAN.md

Source title: Latticra L-UI Parser Line/Column Precision Plan

Lines: 207 | Words: 805 | SHA-256: 9bc329a60455f329f5237ffeb5448907aa3f4e2357d5c168997194a05b1ed60a

Latticra L-UI Parser Line/Column Precision Plan

Status: precision planning contract Scope: one-based location rules, source scanning policy, future parser changes, exact tests, and no-effect boundaries before precise diagnostics implementation.

Purpose

This document plans richer line and column reporting for the L-UI parser.

The current parser uses conservative location defaults:

```
line=1
column=1
```

That is acceptable for the first no-effect parser. The next step is to define how precise locations should work before changing parser logic.

Current boundary

This plan does not implement precise location tracking.

This plan does not change parser behavior.

This plan does not add an L-UI renderer, interactive UI, command behavior, Nucleus task handling, live movement, state mutation, server interaction, update behavior, recovery behavior, hardware behavior, or boot behavior.

Precision goal

Future parser diagnostics should report the beginning of the rejected token or clause when possible.

Examples:

```
unsupported version      -&gt; location of version header
missing card             -&gt; line=1 column=1 if no card token exists
missing effect           -&gt; location of card body start when available
unsupported effect       -&gt; location of effect token
unsupported boundary     -&gt; location of boundary token
missing rail             -&gt; location of card body start when available
unknown binding prefix   -&gt; location of bind target
unterminated string      -&gt; location of opening quote
unbalanced brace         -&gt; location of unmatched brace when available
forbidden marker         -&gt; location of forbidden marker
```

Location units

Line and column must be one-based.

```
first line = 1
first column = 1
```

Columns count bytes in the in-memory source buffer, not terminal display cells, Unicode grapheme clusters, or rendered visual columns.

This keeps the parser portable, deterministic, and independent of terminal behavior.

Newline policy

Recognized newline forms:

```
\n
\r\n
\r
```

Rules:

1. \n increments line and resets column to 1.
2. \r\n counts as one newline and resets column to 1.
3. Lone \r counts as one newline and resets column to 1.
4. Other bytes increment column by 1.
5. Location scanning must not depend on host platform text mode.

Source scanning helper plan

A future implementation should add internal helpers similar to:

```
find_slice_location(source, source_len, needle, *line, *column)
find_unbalanced_brace_location(source, source_len, *line, *column)
find_unterminated_string_location(source, source_len, *line, *column)
```

These helpers should be internal to the parser implementation unless a later contract approves a public API.

Parse result behavior

Future parser code should set:

```
result.line
result.column
```

for the first relevant error location.

Success results may continue to use:

```
line=1
column=1
```

unless a future AST or source-span contract requires richer success metadata.

Error location plan

```
| Parser error | Planned location |
| --- | --- |
| `ok` | `line=1`, `column=1` |
| `null_argument` | `line=1`, `column=1` |
| `empty_source` | `line=1`, `column=1` |
| `unsupported_version` | first byte of source or found unsupported `lui` token |
| `missing_card` | `line=1`, `column=1` unless card-like token found |
| `missing_purpose` | first byte of `card NucleusPreview` body when available |
| `missing_effect` | first byte of `card NucleusPreview` body when available |
| `unsupported_effect` | first byte of `effect` token |
| `missing_boundary` | first byte of `card NucleusPreview` body when available |
| `unsupported_boundary` | first byte of `boundary` token |
| `missing_rail` | first byte of `card NucleusPreview` body when available |
| `unknown_rail` | first byte of unsupported rail token |
| `unknown_binding_prefix` | first byte of unsupported binding target |
| `missing_required_binding` | first byte of relevant rail when available |
| `unterminated_string` | opening quote of the unterminated string |
| `unbalanced_brace` | unmatched opening or closing brace when available |
| `forbidden_behavior_marker` | first byte of forbidden marker |
| `source_too_large` | `line=1`, `column=1` |
| `internal_error` | `line=1`, `column=1` |
```

No-effect rule

Line and column precision is diagnostic metadata only.

It must not change:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact future test list

The future precision implementation PR should include tests for:

```
unsupported_version_reports_precise_location
unsupported_effect_reports_effect_token_location
unsupported_boundary_reports_boundary_token_location
unknown_binding_prefix_reports_binding_target_location
forbidden_marker_reports_marker_location
unterminated_string_reports_opening_quote_location
unbalanced_open_brace_reports_opening_brace_location
unbalanced_close_brace_reports_closing_brace_location
missing_effect_reports_card_body_location
missing_boundary_reports_card_body_location
missing_rail_reports_card_body_location
location_scanner_handles_lf_newlines
location_scanner_handles_crlf_newlines
location_scanner_handles_cr_newlines
location_columns_are_byte_based
success_result_remains_line_one_column_one
error_locations_preserve_no_effect_flags
location_reports_are_deterministic
```

Diagnostic report impact

The existing diagnostic report already renders:

```
line=&lt;line&gt;
column=&lt;column&gt;
```

Future precision work should update only the values, not the report shape.

Implementation gate

Precise line/column implementation should not be added until:

1. this plan is merged;
2. newline behavior is accepted;
3. byte-based column behavior is accepted;
4. error-location rules are accepted;
5. exact tests are present;
6. no-effect flags remain tested.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-line-column-precision-plan.sh
```

The guard is static. It does not implement precision.

Non-claims

This document does not implement precise parser locations, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_RESULT_REPORT.md

Source title: Latticra L-UI Parser Result Report

Lines: 128 | Words: 326 | SHA-256: c72ba536de89e4f832264b212d33e011cee1be5c7d9c0542e499e6d2ece5529a

Latticra L-UI Parser Result Report

Status: initial implementation contract Scope: deterministic text reports for L-UI parse summaries.

Purpose

The L-UI Parser Result Report renders latticra_l_ui_parse_result_t as deterministic text.

It exists so parser status, errors, counts, effect labels, boundary labels, and no-effect flags are operator-visible without adding an L-UI renderer, interactive UI, command behavior, or Nucleus task handling.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
tests/l_ui_parser_result_report_invariants.c
scripts/test-l-ui-parser-result-report.sh
```

Public API

```
latticra_l_ui_parse_result_report
```

Report shape

The report must render:

```
L-UI PARSE RESULT
status=&lt;status-code&gt;
```

```
error=&lt;error-label&gt;
line=&lt;line&gt;
column=&lt;column&gt;
card_name=&lt;card-name&gt;
rail_count=&lt;rail-count&gt;
field_count=&lt;field-count&gt;
effect=&lt;effect&gt;
boundary=&lt;boundary&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

Expected valid summary

For the first accepted fixture, the report should include:

```
error=ok
card_name=NucleusPreview
rail_count=9
field_count=23
effect=none
boundary=preview_only
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Expected error summary

Error results must also preserve no-effect flags:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-parser-result-report.sh
```

The main C workflow runs this check after parser fixture integration.

Boundary

This report must not add:

- L-UI rendering;
- interactive UI behavior;
- command behavior;
- Nucleus task handling;
- live movement;
- state mutation;
- recovery behavior;
- server interaction;
- self-update behavior;
- host effects;
- network effects;

- hardware effects;
- boot effects.

Current evidence level

This implementation is an L2 tested report surface for L-UI parse summaries.

It is not a renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after parser result reporting is:

```
L-UI parser diagnostics contract
```

That future work should define stable diagnostic language and line/column behavior before richer parser diagnostics are implemented.

Non-claims

This document and implementation do not claim L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_SOURCE_SPAN_CONTRACT.md

Source title: Latticra L-UI Parser Source-Span Contract

Lines: 284 | Words: 780 | SHA-256: 8a1f8c02c7e88312f04ad0fddd47bf54d7ba340f1a51da675b2bed21ed755c37

Latticra L-UI Parser Source-Span Contract

Status: source-span contract Scope: source ranges, span rules, byte offsets, line/column ranges, and no-effect boundaries before AST or renderer integration.

Purpose

This document defines the source-span contract for the L-UI parser.

The parser reports precise one-based line and column locations for selected errors. Source spans describe a range in the in-memory source buffer, not only a single point.

This document does not implement source spans. The implementation is documented separately in L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION.md.

Current boundary

The current parser provides:

```
error category
line
column
source span metadata
no-effect flags
diagnostic mapping
diagnostic reports
```

This contract does not add:

```
AST construction
renderer integration
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
```

```
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Span purpose

A source span identifies the source range responsible for a parser result or diagnostic.

Source spans can support:

- clearer diagnostics;
- underline-style terminal reports;
- future AST node ranges;
- fixture validation summaries;
- future L-UI tooling.

They must remain no-effect metadata.

Source unit rules

Spans must use byte offsets over the in-memory source buffer.

Rules:

1. start_offset is zero-based.
2. end_offset is zero-based and exclusive.
3. start_offset <= end_offset.
4. Both offsets must be within the provided source length.
5. Line and column are one-based.
6. Columns are byte-based, not display-cell-based.
7. Spans must not depend on host text mode.

Line and column range rules

A span includes:

```
start_line
start_column
end_line
end_column
```

Rules:

1. Start line and column identify the first byte in the span.
2. End line and column identify the location immediately after the final byte in the span.
3. Empty spans are allowed only for missing constructs or successful/default results.
4. Empty spans should set start and end to the same location.
5. Multiline spans are allowed.

Newline policy

The source-span scanner must preserve the current newline policy:

```
LF   -&gt; newline
CRLF -&gt; one newline
CR   -&gt; newline
```

Column reset behavior:

```
line += 1
column = 1
```

for each logical newline.

Span struct

The public span struct is:

```
typedef struct {
    size_t start_offset;
    size_t end_offset;
    size_t start_line;
    size_t start_column;
    size_t end_line;
    size_t end_column;
} latticra_l_ui_source_span_t;
```

Parse result integration

The parser result includes:

```
latticra_l_ui_source_span_t span;
```

Diagnostic integration

Diagnostics include the same span shape:

```
latticra_l_ui_source_span_t span;
```

Diagnostic reports render:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

Initial span targets

The first span implementation covers:

```
unsupported_version
unsupported_effect
unsupported_boundary
unknown_binding_prefix
forbidden_behavior_marker
unterminated_string
unbalanced_brace
missing_effect
missing_boundary
missing_rail
```

Span target behavior

```
| Parser result | Span |
| --- | --- |
| `unsupported_version` | unsupported version token or `lui` token |
| `unsupported_effect` | unsupported `effect ...` clause token range |
| `unsupported_boundary` | unsupported `boundary ...` clause token range |
| `unknown_binding_prefix` | unsupported binding target prefix range |
| `forbidden_behavior_marker` | forbidden marker token range |
| `unterminated_string` | opening quote through end of source |
| `unbalanced_brace` | unmatched brace byte |
| `missing_effect` | empty span at card body start |
| `missing_boundary` | empty span at card body start |
| `missing_rail` | empty span at card body start |
```

Empty span rule

Missing constructs use empty spans.

Example:

```
start_offset=end_offset=&lt;card-body-start-offset&gt;
start_line=end_line=&lt;card-body-start-line&gt;
start_column=end_column=&lt;card-body-start-column&gt;
```

This keeps missing data precise without inventing source ranges.

No-effect rule

Source spans are metadata only.

They must not change:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Implementation gate

Source-span implementation required a separate implementation plan defining:

1. public API changes;
2. struct placement;
3. parse result integration;
4. diagnostic integration;
5. report format changes;
6. test file names;
7. exact invariant tests;
8. compatibility expectations.

That plan is recorded in L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION_PLAN.md.

Test list

Source-span implementation tests verify:

```
unsupported_effect_reports_effect_span
unsupported_boundary_reports_boundary_span
unknown_binding_prefix_reports_binding_span
forbidden_marker_reports_marker_span
unterminated_string_reports_string_span
unbalanced_open_brace_reports_brace_span
unbalanced_close_brace_reports_brace_span
missing_effect_reports_empty_card_body_span
missing_boundary_reports_empty_card_body_span
missing_rail_reports_empty_card_body_span
span_offsets_are_zero_based
span_end_offset_is_exclusive
span_line_columns_are_one_based
span_columns_are_byte_based
span_scanner_handles_lf_newlines
span_scanner_handles_crlf_newlines
span_scanner_handles_cr_newlines
span_metadata_preserves_no_effect_flags
span_reports_are_deterministic
```

Forbidden behavior

A source-span implementation must not:

- add file I/O to parser code;
- write files;

- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- accept unsupported effects.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-parser-source-span-contract.sh
```

The guard is static. It does not implement source spans.

Non-claims

This document does not implement AST construction, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION.md

Source title: Latticra L-UI Parser Source-Span Implementation

Lines: 189 | Words: 436 | SHA-256: 8b3e8b05c14086208656596b1de35b111a1caa89589eb1d7da7a1defe84041d6

Latticra L-UI Parser Source-Span Implementation

Status: initial implementation contract Scope: public source-span metadata, parse-result spans, diagnostic spans, report extensions, and invariant tests.

Purpose

The L-UI Parser Source-Span implementation adds source-range metadata to parser results and diagnostics.

Source spans describe ranges in the in-memory source buffer. They are metadata only. They do not add AST construction, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, server interaction, recovery behavior, self-update behavior, hardware behavior, or boot behavior.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
tests/l_ui_parser_source_span_invariants.c
scripts/test-l-ui-parser-source-span.sh
```

Public source-span struct

```
typedef struct {
    size_t start_offset;
    size_t end_offset;
    size_t start_line;
    size_t start_column;
    size_t end_line;
    size_t end_column;
} latticra_l_ui_source_span_t;
```

Parse-result integration

The parser result includes:

```
latticra_l_ui_source_span_t span;
```

Diagnostic integration

The diagnostic result includes:

```
latticra_l_ui_source_span_t span;
```

Diagnostics copy span metadata from parse results.

Report extensions

Parser result reports and diagnostic reports append:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

The extension is append-only. Existing fields remain present and stable.

Span rules

```
start_offset is zero-based
end_offset is zero-based and exclusive
start_offset &lt;= end_offset
line and column are one-based
columns are byte-based
```

Implemented span targets

Initial source spans cover:

```
unsupported_version
unsupported_effect
unsupported_boundary
unknown_binding_prefix
forbidden_behavior_marker
unterminated_string
unbalanced_brace
missing_effect
missing_boundary
missing_rail
```

Empty span behavior

Missing constructs use empty spans at the card-body start when available:

```
start_offset=end_offset
start_line=end_line
start_column=end_column
```

Default success span

Successful parses use the default empty span:

```
start_offset=0
end_offset=0
start_line=1
start_column=1
```

```
end_line=1
end_column=1
```

No-effect boundary

Source-span metadata must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-parser-source-span.sh
```

The main C workflow runs this check after the source-span implementation-plan guard.

Required invariants

The source-span tests verify:

```
unsupported_effect_reports_effect_span
unsupported_boundary_reports_boundary_span
unknown_binding_prefix_reports_binding_span
forbidden_marker_reports_marker_span
unterminated_string_reports_string_span
unbalanced_open_brace_reports_brace_span
unbalanced_close_brace_reports_brace_span
missing_effect_reports_empty_card_body_span
missing_boundary_reports_empty_card_body_span
missing_rail_reports_empty_card_body_span
span_offsets_are_zero_based
span_end_offset_is_exclusive
span_line_columns_are_one_based
span_columns_are_byte_based
span_scanner_handles_lf_newlines
span_scanner_handles_crlf_newlines
span_scanner_handles_cr_newlines
span_metadata_preserves_no_effect_flags
span_reports_are_deterministic
parse_result_report_includes_span_fields
diagnostic_report_includes_span_fields
diagnostic_copies_parse_result_span
success_span_uses_default_empty_span
```

Current evidence level

This implementation is an L2 tested source-span metadata model for the L-UI parser.

It is not an AST system, renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after source spans is:

```
L-UI parser AST contract
```

That future work should define AST shape, node ownership, source-span usage, and no-effect boundaries before any AST implementation.

Non-claims

This document and implementation do not claim AST construction, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Parser Source-Span Implementation Plan

Lines: 353 | Words: 886 | SHA-256: bab8ba01866661a22f700259d7270dad67488170a5e1c3497dec31faadb019a9

Latticra L-UI Parser Source-Span Implementation Plan

Status: implementation planning contract Scope: public API changes, struct placement, parse-result integration, diagnostics integration, report format changes, exact tests, and compatibility expectations before source-span implementation.

Purpose

This document defines the implementation plan for L-UI parser source spans.

The source-span contract is already merged and guarded. This plan decides the exact C API, struct placement, parse result integration, diagnostic integration, report format changes, tests, and compatibility expectations before span code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_PARSER_SOURCE_SPAN_CONTRACT.md
docs/L_UI_PARSER_LINE_COLUMN_PRECISION_IMPLEMENTATION.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for parser errors, precise point locations, diagnostics, and no-effect behavior.

Implementation language decision

The source-span implementation should be in C.

Reason:

- the parser is implemented in C;
- parse results and diagnostics are C ABI surfaces;
- spans should be fixed-size metadata with no allocation requirements;
- span calculations can reuse existing private source-location scanning behavior;
- the current C workflow can validate spans.

Public API change

Unlike line/column precision, source spans require a public API extension.

Add a public span struct in:

```
include/latticra/l_ui_parser.h
```

Proposed public struct:

```
typedef struct {
    size_t start_offset;
    size_t end_offset;
    size_t start_line;
    size_t start_column;
    size_t end_line;
    size_t end_column;
} latticra_l_ui_source_span_t;
```

Parse result integration

Add span to:

```
latticra_l_ui_parse_result_t
```

Proposed field:

```
latticra_l_ui_source_span_t span;
```

Diagnostic integration

Add span to:

```
latticra_l_ui_diagnostic_t
```

Proposed field:

```
latticra_l_ui_source_span_t span;
```

Diagnostics should copy span metadata from parse results.

Report format changes

The parser result report should add:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

The diagnostic report should add the same fields:

```
span_start_offset=&lt;n&gt;
span_end_offset=&lt;n&gt;
span_start_line=&lt;n&gt;
span_start_column=&lt;n&gt;
span_end_line=&lt;n&gt;
span_end_column=&lt;n&gt;
```

These additions are append-only to preserve existing report fields.

Compatibility expectations

Existing consumers of parse results must continue to compile after adding the span field, assuming they do not rely on exact struct size.

Existing report tests must be updated to accept the new appended fields.

Existing behavior must remain stable:

```
error labels unchanged
diagnostic codes unchanged
diagnostic messages unchanged
diagnostic hints unchanged
line and column fields unchanged
no-effect flags unchanged
```

Internal helper plan

Extend private parser helpers in src/l_ui_parser.c.

Planned helper shapes:

```
static void span_default(latticra_l_ui_source_span_t *span);

static void span_for_range(
    const char *source,
    size_t source_len,
    size_t start_offset,
    size_t end_offset,
    latticra_l_ui_source_span_t *span);

static int find_slice_span(
    const char *source,
```

```
    size_t source_len,
    const char *needle,
    latticra_l_ui_source_span_t *span);

static int find_unterminated_string_span(
    const char *source,
    size_t source_len,
    latticra_l_ui_source_span_t *span);

static int find_unbalanced_brace_span(
    const char *source,
    size_t source_len,
    latticra_l_ui_source_span_t *span);

static void set_error_with_span(
    latticra_l_ui_parse_result_t *result,
    latticra_l_ui_parse_error_t error,
    const latticra_l_ui_source_span_t *span);
```

These helpers should remain private.

Span defaults

Default span for success and conservative errors:

```
start_offset=0
end_offset=0
start_line=1
start_column=1
end_line=1
end_column=1
```

Span calculation rules

For token spans:

```
start_offset = first byte of token
end_offset = first byte after token
start_line/start_column = location of start_offset
end_line/end_column = location of end_offset
```

For missing constructs:

```
start_offset = card body start offset
end_offset = card body start offset
start_line = card body start line
start_column = card body start column
end_line = start_line
end_column = start_column
```

Initial span targets

The first implementation should add spans for:

```
unsupported_version
unsupported_effect
unsupported_boundary
unknown_binding_prefix
forbidden_behavior_marker
unterminated_string
unbalanced_brace
missing_effect
missing_boundary
missing_rail
```

Span mapping plan

```
| Parser result | Planned span |
| --- | --- |
| `unsupported_version` | unsupported version token or `lui` token |
| `unsupported_effect` | unsupported `effect ...` clause token range |
| `unsupported_boundary` | unsupported `boundary ...` clause token range |
| `unknown_binding_prefix` | unsupported binding target prefix range |
| `forbidden_behavior_marker` | forbidden marker token range |
| `unterminated_string` | opening quote through end of source |
| `unbalanced_brace` | unmatched brace byte |
| `missing_effect` | empty span at card body start |
```

```
| `missing_boundary` | empty span at card body start |
| `missing_rail` | empty span at card body start |
```

No-effect preservation

Source spans are metadata only.

They must not change:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The source-span implementation PR should include tests for:

```
unsupported_effect_reports_effect_span
unsupported_boundary_reports_boundary_span
unknown_binding_prefix_reports_binding_span
forbidden_marker_reports_marker_span
unterminated_string_reports_string_span
unbalanced_open_brace_reports_brace_span
unbalanced_close_brace_reports_brace_span
missing_effect_reports_empty_card_body_span
missing_boundary_reports_empty_card_body_span
missing_rail_reports_empty_card_body_span
span_offsets_are_zero_based
span_end_offset_is_exclusive
span_line_columns_are_one_based
span_columns_are_byte_based
span_scanner_handles_lf_newlines
span_scanner_handles_crlf_newlines
span_scanner_handles_cr_newlines
span_metadata_preserves_no_effect_flags
span_reports_are_deterministic
parse_result_report_includes_span_fields
diagnostic_report_includes_span_fields
diagnostic_copies_parse_result_span
success_span_uses_default_empty_span
```

Test file plan

Add:

```
tests/l_ui_parser_source_span_invariants.c
scripts/test-l-ui-parser-source-span.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_PARSER.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_PARSER_SOURCE_SPAN_CONTRACT.md
```

and add:

```
docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION.md
```

Forbidden implementation behavior

The source-span implementation must not:

- add file I/O to parser code;
- write files;

- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- accept unsupported effects.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-source-span-implementation-plan.sh
```

The guard is static. It does not implement source spans.

Implementation gate

Source-span implementation code may be added only after this plan is merged.

Non-claims

This document does not implement source spans, AST construction, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_CONTRACT.md

Source title: Latticra L-UI Parser String Escape Diagnostics Contract

Lines: 357 | Words: 1213 | SHA-256: 49921552235fdc99f60f5f2f24735ee44497a29b94e9961bd1893322a9e8c0b6

Latticra L-UI Parser String Escape Diagnostics Contract

Status: parser string escape diagnostics contract Scope: parser-level diagnostics for invalid L-UI string-literal escape sequences before and after parser diagnostics extension.

Purpose

This document defines the parser-level diagnostics contract for invalid L-UI string-literal escape sequences.

The string-literal escape decoder originally ran during AST construction. Invalid escapes were classified as:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

at the AST layer so the structural parser remained compatible. This contract defined the parser-level diagnostic surface before parser validation was extended to report invalid escapes directly.

This document does not implement parser-level string escape diagnostics.

The implementation plan is documented separately in L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION_PLAN.md.

The implementation is documented separately in L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md.

Relationship to existing diagnostics

This contract extends the existing parser diagnostics track:

```
docs/L_UI_PARSER_DIAGNOSTICS.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION_PLAN.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION_PLAN.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
```

The existing diagnostics contract owns stable parser diagnostic language. This contract defines the extension set for rejected string-literal escapes only.

Current boundary

The current system provides:

```
structural L-UI parser diagnostics
source spans
line and column metadata
string-literal escape decoding during AST construction
parser-level invalid escape diagnostics
escaped detailed AST report fields
no-effect flags
```

This contract does not add:

```
new accepted escape sequences
length-carrying AST strings
Unicode display behavior
renderer behavior
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Diagnostic purpose

Parser-level string escape diagnostics make rejected string-literal escapes visible before AST construction.

The behavior allows:

```
latticra_l_ui_parse_source
latticra_l_ui_diagnostic_from_parse_result
latticra_l_ui_diagnostic_report
```

to report stable invalid-escape categories, messages, hints, source spans, and no-effect flags.

Parser error additions

The implementation extends latticra_l_ui_parse_error_t after LATTICRA_L_UI_PARSE_INTERNAL_ERROR with:

```
LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE
LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE
```

```
LATTICRA_L_UI_PARSE_UNTERMINATED_ESCAPE
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE
```

These names are reserved by this contract for parser-level string escape diagnostics.

Diagnostic codes

The diagnostic codes extend the existing LUI0000 through LUI0018 code range:

```
LUI0019 invalid_string_escape
LUI0020 invalid_hex_escape
LUI0021 unterminated_escape
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
LUI0024 string_value_too_large
```

The code numbers are stable.

Stable messages

Message language is concise and deterministic:

```
| Code | Message |
| --- | --- |
| `LUI0019` | String literal escape is not supported. |
| `LUI0020` | Hex string escape must use exactly two uppercase hexadecimal digits. |
| `LUI0021` | String literal escape is not terminated. |
| `LUI0022` | Decoded NUL bytes are not supported in AST strings. |
| `LUI0023` | Literal NUL bytes are not supported in AST strings. |
| `LUI0024` | Decoded string value exceeds the supported AST storage limit. |
```

Messages do not include dynamic source fragments.

Hint language

Hint language is stable:

```
| Code | Hint |
| --- | --- |
| `LUI0019` | Use only \\, \&quot;, \n, \r, \t, or uppercase \xNN escapes. |
| `LUI0020` | Use exactly two uppercase hex digits, such as \x0A or \x7F. |
| `LUI0021` | Complete the escape sequence or remove the trailing backslash. |
| `LUI0022` | Avoid \x00 until length-carrying AST strings exist. |
| `LUI0023` | Remove literal NUL bytes from the source string. |
| `LUI0024` | Shorten the decoded purpose or text value. |
```

Hints must not suggest unsupported escape forms such as Unicode escapes.

Source-span behavior

Diagnostics must remain source-oriented.

For invalid escapes, spans should cover the rejected source bytes:

```
\a      -&gt; span covers \a
\x0a    -&gt; span covers \x0a
\xGG    -&gt; span covers \xGG
\       -&gt; span covers the trailing backslash
\x      -&gt; span covers \x
\x0     -&gt; span covers \x0
\x00    -&gt; span covers \x00
literal NUL -&gt; span covers the literal NUL source byte
oversized decoded output -&gt; span covers the full source value range between quotes
```

Line and column should point to the first byte of the rejected span.

Line and column remain one-based byte positions in the in-memory source buffer.

Validation order

Parser-level string escape validation runs only after basic structural fixture checks.

It runs before AST construction and before AST values are materialized.

It must not hide earlier structural parser errors when those errors occur before a string-literal escape can be inspected deterministically.

When source structure is valid but a string escape is invalid, the string escape diagnostic is reported instead of allowing AST construction to classify an internal error.

Accepted escape compatibility

Parser-level diagnostics preserve the accepted escape set:

```
\\
\&quot;
\n
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

The diagnostics implementation must not broaden accepted escapes without a separate grammar or string-literal escape contract update.

AST compatibility

Invalid escapes are rejected by:

```
latticra_l_ui_parse_source
```

before AST construction runs.

For invalid escape source, AST behavior is:

```
ast.parse_result.error = parser-level string escape error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Accepted escapes continue to decode in AST values exactly as the string-literal escape implementation defines.

Report relationship

Diagnostic reports render:

```
diagnostic_severity=error
diagnostic_code=&lt;LUI0019-through-LUI0024&gt;
diagnostic_message=&lt;stable-message&gt;
diagnostic_hint=&lt;stable-hint&gt;
```

Parser result reports continue to include stable parse error labels.

Detailed AST reports continue using the failed-parse report path for rejected sources.

No-effect rule

Parser-level string escape diagnostics are metadata-only.

They preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility rule

The implementation must not change:

```
accepted string-literal escape decoding
```

```
latticra_l_ui_ast_report existing required fields
latticra_l_ui_ast_detailed_report existing required fields for accepted sources
latticra_l_ui_diagnostic_report existing required fields
existing diagnostic codes LUI0000 through LUI0018
existing parser error labels for current errors
existing accepted fixture summary counts
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Implementation evidence

Parser-level string escape diagnostics implementation required a separate implementation plan defining:

1. parser enum additions;
2. diagnostic code additions;
3. parse error labels;
4. stable messages;
5. stable hints;
6. source-span rules;
7. validation order;
8. AST compatibility behavior;
9. report compatibility behavior;
10. exact test file names;
11. exact invariant tests.

That plan is recorded in L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION_PLAN.md.

The implementation is recorded in L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md.

Test list

The implementation tests verify:

```
string_escape_diagnostic_rejects_unknown_escape_lui0019
string_escape_diagnostic_rejects_lowercase_hex_lui0020
string_escape_diagnostic_rejects_short_hex_lui0020
string_escape_diagnostic_rejects_invalid_hex_lui0020
string_escape_diagnostic_rejects_unterminated_escape_lui0021
string_escape_diagnostic_rejects_decoded_nul_lui0022
string_escape_diagnostic_rejects_literal_nul_lui0023
string_escape_diagnostic_rejects_oversized_decoded_output_lui0024
string_escape_diagnostic_reports_source_span
string_escape_diagnostic_reports_line_column
string_escape_diagnostic_preserves_no_effect_flags
string_escape_diagnostic_does_not_change_accepted_escape_decoding
string_escape_diagnostic_does_not_change_existing_error_codes
string_escape_diagnostic_uses_failed_parse_ast_report
string_escape_diagnostic_is_deterministic
```

Forbidden behavior

The parser-level string escape diagnostics implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;

- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- accept lowercase hex escapes;
- accept Unicode escapes;
- accept unknown escapes literally;
- allow decoded NUL bytes before length-carrying AST strings exist;
- broaden accepted grammar beyond the string-literal escape contract.

Current validation commands

This contract is guarded by:

```
sh scripts/test-l-ui-parser-string-escape-diagnostics-contract.sh
```

The implementation is tested by:

```
sh scripts/test-l-ui-parser-string-escape-diagnostics.sh
```

Non-claims

This document and implementation do not change AST storage, implement length-carrying AST strings, allow decoded NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md

Source title: Latticra L-UI Parser String Escape Diagnostics Implementation

Lines: 344 | Words: 1034 | SHA-256: 5f531f3281225a78b7e3f774832c9253a43d79edf2277c10fc394550595733fb

Latticra L-UI Parser String Escape Diagnostics Implementation

Status: initial implementation contract Scope: parser-level diagnostics for invalid L-UI string-literal escape sequences.

Purpose

The L-UI Parser String Escape Diagnostics implementation promotes invalid string-literal escape failures from AST-level internal-error classification to parser-level diagnostics.

The implementation reports stable parser errors, diagnostic codes, messages, hints, source spans, and no-effect flags before AST construction materializes source-backed string values.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_diagnostics.c
tests/l_ui_parser_string_escape_diagnostics_invariants.c
scripts/test-l-ui-parser-string-escape-diagnostics.sh
```

Related existing files remain active:

```
src/l_ui_parser_ast.c
tests/l_ui_string_literal_escape_invariants.c
scripts/test-l-ui-string-literal-escape.sh
```

Public API additions

The implementation extends latticra_l_ui_parse_error_t with:

```
LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE = 19
LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE = 20
LATTICRA_L_UI_PARSE_UNTERMINATED_ESCAPE = 21
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING = 22
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING = 23
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE = 24
```

No new public function is added.

Parse error labels

The implementation adds stable parse error labels:

```
invalid_string_escape
invalid_hex_escape
unterminated_escape
decoded_nul_in_string
literal_nul_in_string
string_value_too_large
```

Existing parse error labels remain unchanged.

Diagnostic codes

The implementation adds diagnostic codes:

```
LUI0019 invalid_string_escape
LUI0020 invalid_hex_escape
LUI0021 unterminated_escape
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
LUI0024 string_value_too_large
```

Existing diagnostic codes LUI0000 through LUI0018 remain unchanged.

Stable messages

The implementation uses these deterministic messages:

```
LUI0019: String literal escape is not supported.
LUI0020: Hex string escape must use exactly two uppercase hexadecimal digits.
LUI0021: String literal escape is not terminated.
LUI0022: Decoded NUL bytes are not supported in AST strings.
LUI0023: Literal NUL bytes are not supported in AST strings.
LUI0024: Decoded string value exceeds the supported AST storage limit.
```

Messages do not include dynamic source fragments.

Stable hints

The implementation uses these deterministic hints:

```
LUI0019: Use only \\, \&quot;, \n, \r, \t, or uppercase \xNN escapes.
LUI0020: Use exactly two uppercase hex digits, such as \x0A or \x7F.
LUI0021: Complete the escape sequence or remove the trailing backslash.
LUI0022: Avoid \x00 until length-carrying AST strings exist.
LUI0023: Remove literal NUL bytes from the source string.
LUI0024: Shorten the decoded purpose or text value.
```

Hints do not suggest unsupported Unicode escapes.

Parser helper implementation

The implementation adds private helpers in src/l_ui_parser.c:

```
is_upper_hex_digit
upper_hex_value
validate_string_literal_escape_span
validate_l_ui_string_literal_escapes
```

These helpers remain private.

Validation targets

The implementation validates quoted values used as source-backed AST strings:

```
purpose &quot;...&quot;
first text &quot;...&quot;
second text &quot;...&quot;
```

It does not validate or decode:

```
card names
rail names
field names
binding values
effect values
boundary values
```

Validation order

String escape validation runs inside:

```
latticra_l_ui_parse_source
```

after structural fixture checks and before returning LATTICRA_L_UI_PARSE_OK.

Existing structural errors retain priority over string escape diagnostics.

When source structure is valid but a string escape is invalid, the parser reports the string escape diagnostic instead of allowing AST construction to classify an internal error.

Source-span behavior

Diagnostics remain source-oriented.

For invalid escapes, spans cover the rejected source bytes:

```
\a      -&gt; span covers \a
\x0a    -&gt; span covers \x0a
\xGG    -&gt; span covers \xGG
\x      -&gt; span covers \x
\x0     -&gt; span covers \x0
\x00    -&gt; span covers \x00
literal NUL -&gt; span covers the literal NUL source byte
oversized decoded output -&gt; span covers the full source value range between quotes
```

Line and column point to the first byte of the rejected span.

Line and column remain one-based byte positions in the in-memory source buffer.

Capacity behavior

Parser validation uses the current AST string storage capacity:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
```

for both purpose and text values.

If decoded output would be greater than or equal to the destination capacity, parser validation returns:

```
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE
```

NUL behavior

Decoded \x00 returns:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
```

Literal 0x00 source bytes inside the validated string value range return:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

NUL bytes remain rejected until a length-carrying AST string storage contract exists.

That storage contract is now recorded in:

```
L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md
```

AST compatibility behavior

Invalid escapes are rejected by:

```
latticra_l_ui_parse_source
```

before AST construction materializes values.

For invalid escape source, latticra_l_ui_parse_ast preserves failed-parse behavior:

```
ast.parse_result.error = parser-level string escape error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Accepted escapes continue to decode in AST values exactly as the string-literal escape implementation defines.

Report compatibility behavior

Parser result reports render the new parse error labels through existing report fields.

Diagnostic reports render:

```
diagnostic_severity=error
diagnostic_code=&lt;LUI0019-through-LUI0024&gt;
diagnostic_message=&lt;stable-message&gt;
diagnostic_hint=&lt;stable-hint&gt;
```

Detailed AST reports continue using the failed-parse report path for rejected sources.

Existing report fields remain stable.

Compatibility

The implementation does not change:

```
accepted string-literal escape decoding
existing diagnostic codes LUI0000 through LUI0018
existing parser error labels for current errors
latticra_l_ui_ast_report existing required fields
latticra_l_ui_ast_detailed_report existing required fields for accepted sources
latticra_l_ui_diagnostic_report existing required fields
existing accepted fixture summary counts
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect boundary

Parser-level string escape diagnostics preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
```

```
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-parser-string-escape-diagnostics.sh
```

The main C workflow runs this check after the parser string escape diagnostics implementation-plan guard.

Required invariants

The parser string escape diagnostics tests verify:

```
string_escape_diagnostic_rejects_unknown_escape_lui0019
string_escape_diagnostic_rejects_lowercase_hex_lui0020
string_escape_diagnostic_rejects_short_hex_lui0020
string_escape_diagnostic_rejects_invalid_hex_lui0020
string_escape_diagnostic_rejects_unterminated_escape_lui0021
string_escape_diagnostic_rejects_decoded_nul_lui0022
string_escape_diagnostic_rejects_literal_nul_lui0023
string_escape_diagnostic_rejects_oversized_decoded_output_lui0024
string_escape_diagnostic_reports_source_span
string_escape_diagnostic_reports_line_column
string_escape_diagnostic_preserves_no_effect_flags
string_escape_diagnostic_does_not_change_accepted_escape_decoding
string_escape_diagnostic_does_not_change_existing_error_codes
string_escape_diagnostic_uses_failed_parse_ast_report
string_escape_diagnostic_is_deterministic
```

Current evidence level

This implementation is an L2 tested parser-level diagnostic model for invalid L-UI string-literal escapes.

It is not a length-carrying AST string model, Unicode display model, renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, sandbox, or operating system.

Next implementation step

The length-carrying AST string storage contract is recorded in:

```
L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_CONTRACT.md
```

The next implementation candidate after that contract is:

```
L-UI AST length-carrying string storage implementation plan
```

That future work should define public struct field placement, initialization rules, source-backed extraction length assignment, string decode length assignment, report field additions, length-aware escaped report helper shape, and exact invariant tests before storage code changes.

Non-claims

This document and implementation do not implement length-carrying AST strings, allow decoded NUL bytes, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Parser String Escape Diagnostics Implementation Plan

Lines: 461 | Words: 1402 | SHA-256: ea934787218331c7fdfb1edf9f67d9f4434157e6128e520b5a8d82d0f12d7e47

Latticra L-UI Parser String Escape Diagnostics Implementation Plan

Status: implementation planning contract Scope: parser enum additions, diagnostic code additions, parse error labels, stable messages, stable hints, source-span rules, validation order, AST/report compatibility, exact test file names, and exact invariant tests before parser-level string escape diagnostics code.

Purpose

This document defines the implementation plan for parser-level diagnostics for invalid L-UI string-literal escape sequences.

The parser string escape diagnostics contract is already merged and guarded. This plan decides the exact public enum changes, diagnostic mapping updates, parser validation helper shape, source-span behavior, AST compatibility behavior, report behavior, test files, and compatibility expectations before implementation code is added.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_CONTRACT.md
docs/L_UI_PARSER_DIAGNOSTICS.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
```

Those files remain the source of truth for current parser diagnostics, source spans, string-literal escape decoding, AST failure behavior, and no-effect boundaries.

Implementation language decision

Parser-level string escape diagnostics should be implemented in C.

Reason:

- the L-UI parser is implemented in C;
- diagnostics are exposed through a C public API surface;
- validation must remain bounded, deterministic, and no-effect;
- the current C workflow can validate diagnostic invariants;
- no dynamic runtime should be required.

Public API decision

Extend the existing public parser error enum in:

```
include/latticra/l_ui_parser.h
```

Add the new values after:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR = 18
```

Proposed exact values:

```
LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE = 19
```

```
LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE = 20
LATTICRA_L_UI_PARSE_UNTERMINATED_ESCAPE = 21
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING = 22
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING = 23
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE = 24
```

No new public function is required for the first implementation.

Parse error label decision

Update:

```
latticra_l_ui_parse_error_label
```

with stable labels:

```
invalid_string_escape
invalid_hex_escape
unterminated_escape
decoded_nul_in_string
literal_nul_in_string
string_value_too_large
```

Existing labels must remain unchanged.

Diagnostic mapping decision

Update:

```
latticra_l_ui_diagnostic_from_parse_result
```

so the new parser errors map to:

```
LUI0019 invalid_string_escape
LUI0020 invalid_hex_escape
LUI0021 unterminated_escape
LUI0022 decoded_nul_in_string
LUI0023 literal_nul_in_string
LUI0024 string_value_too_large
```

Severity for all new diagnostics should be:

```
LATTICRA_L_UI_DIAGNOSTIC_ERROR
```

Stable messages

Use these exact messages:

```
LUI0019: String literal escape is not supported.
LUI0020: Hex string escape must use exactly two uppercase hexadecimal digits.
LUI0021: String literal escape is not terminated.
LUI0022: Decoded NUL bytes are not supported in AST strings.
LUI0023: Literal NUL bytes are not supported in AST strings.
LUI0024: Decoded string value exceeds the supported AST storage limit.
```

Messages must remain deterministic and must not include dynamic source fragments.

Stable hints

Use these exact hints:

```
LUI0019: Use only \\, \&quot;, \n, \r, \t, or uppercase \xNN escapes.
LUI0020: Use exactly two uppercase hex digits, such as \x0A or \x7F.
LUI0021: Complete the escape sequence or remove the trailing backslash.
LUI0022: Avoid \x00 until length-carrying AST strings exist.
LUI0023: Remove literal NUL bytes from the source string.
LUI0024: Shorten the decoded purpose or text value.
```

Hints must not suggest unsupported escape forms such as Unicode escapes.

Parser helper plan

Add private helpers in:

```
src/l_ui_parser.c
```

Proposed helpers:

```
static int is_upper_hex_digit(unsigned char byte);

static unsigned char upper_hex_value(unsigned char byte);

static latticra_l_ui_parse_error_t validate_string_literal_escape_span(
    const char *source,
    size_t source_len,
    size_t value_start,
    size_t value_end,
    size_t destination_len,
    latticra_l_ui_source_span_t *span);

static latticra_l_ui_parse_error_t validate_l_ui_string_literal_escapes(
    const char *source,
    size_t source_len,
    latticra_l_ui_source_span_t *span);
```

Helpers must remain private.

The validation helper may share logic with AST decoding later, but this first implementation can duplicate small byte-oriented logic to avoid changing public API or adding shared modules.

Validation targets

The first parser-level string escape diagnostics implementation should validate only quoted source values used as AST strings:

```
purpose &quot;...&quot;
first text &quot;...&quot;
second text &quot;...&quot;
```

It should not validate or decode:

```
card names
rail names
field names
binding values
effect values
boundary values
```

Validation order

Parser-level string escape validation should run inside:

```
latticra_l_ui_parse_source
```

after these checks:

```
null argument check
empty source check
source size check
unterminated string check
unbalanced brace check
version check
card check
purpose presence check
effect presence and supported effect checks
boundary presence and supported boundary checks
forbidden behavior marker check
required rail checks
unknown binding prefix check
required binding checks
```

and before the parser returns LATTICRA_L_UI_PARSE_OK.

Reason:

- existing structural errors should keep their current priority;
- validation should only inspect string escapes after the accepted fixture structure is present;
- accepted sources should remain compatible;

- invalid escapes should no longer reach AST construction as internal errors when structure is otherwise valid.

Source-span behavior

Use source-oriented spans.

Exact span expectations:

```
\a      -&gt; span covers \a
\x0a    -&gt; span covers \x0a
\xGG    -&gt; span covers \xGG
\       -&gt; span covers the trailing backslash
\x      -&gt; span covers \x
\x0     -&gt; span covers \x0
\x00    -&gt; span covers \x00
literal NUL -&gt; span covers the literal NUL source byte
oversized decoded output -&gt; span covers the full source value range between quotes
```

Line and column should point to the first byte of the rejected span.

Line and column remain one-based byte positions in the in-memory source buffer.

Capacity behavior

Parser validation should use the current AST string storage capacity:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
```

for both purpose and text values.

If decoded output would be greater than or equal to the destination capacity, parser validation should return:

```
LATTICRA_L_UI_PARSE_STRING_VALUE_TOO_LARGE
```

This prevents oversized decoded values from reaching AST construction.

NUL behavior

Decoded \x00 should return:

```
LATTICRA_L_UI_PARSE_DECODED_NUL_IN_STRING
```

Literal 0x00 source bytes inside the validated string value range should return:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

NUL bytes remain rejected until a length-carrying AST string storage contract exists.

AST compatibility behavior

After parser-level diagnostics are implemented, invalid escapes should be rejected by:

```
latticra_l_ui_parse_source
```

before AST construction materializes values.

For invalid escape source, latticra_l_ui_parse_ast should preserve failed-parse behavior:

```
ast.parse_result.error = parser-level string escape error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Accepted escapes must continue to decode in AST values exactly as the current string-literal escape implementation defines.

Report compatibility behavior

Parser result reports should render the new parse error labels through existing report fields.

Diagnostic reports should render:

```
diagnostic_severity=error
diagnostic_code=&lt;LUI0019-through-LUI0024&gt;
diagnostic_message=&lt;stable-message&gt;
diagnostic_hint=&lt;stable-hint&gt;
```

Detailed AST reports should continue using the failed-parse report path for rejected sources.

Existing report fields must remain stable.

Compatibility expectations

The implementation must not change:

```
accepted string-literal escape decoding
existing diagnostic codes LUI0000 through LUI0018
existing parser error labels for current errors
latticra_l_ui_ast_report existing required fields
latticra_l_ui_ast_detailed_report existing required fields for accepted sources
latticra_l_ui_diagnostic_report existing required fields
existing accepted fixture summary counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect preservation

Parser-level string escape diagnostics must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
string_escape_diagnostic_rejects_unknown_escape_lui0019
string_escape_diagnostic_rejects_lowercase_hex_lui0020
string_escape_diagnostic_rejects_short_hex_lui0020
string_escape_diagnostic_rejects_invalid_hex_lui0020
string_escape_diagnostic_rejects_unterminated_escape_lui0021
string_escape_diagnostic_rejects_decoded_nul_lui0022
string_escape_diagnostic_rejects_literal_nul_lui0023
string_escape_diagnostic_rejects_oversized_decoded_output_lui0024
string_escape_diagnostic_reports_source_span
string_escape_diagnostic_reports_line_column
string_escape_diagnostic_preserves_no_effect_flags
string_escape_diagnostic_does_not_change_accepted_escape_decoding
string_escape_diagnostic_does_not_change_existing_error_codes
string_escape_diagnostic_uses_failed_parse_ast_report
string_escape_diagnostic_is_deterministic
```

Test file plan

Add:

```
tests/l_ui_parser_string_escape_diagnostics_invariants.c
scripts/test-l-ui-parser-string-escape-diagnostics.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_CONTRACT.md
docs/L_UI_PARSER_DIAGNOSTICS.md
docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_IMPLEMENTATION.md
```

Forbidden implementation behavior

The parser-level string escape diagnostics implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- accept lowercase hex escapes;
- accept Unicode escapes;
- accept unknown escapes literally;
- allow decoded NUL bytes before length-carrying AST strings exist;
- broaden accepted grammar beyond the string-literal escape contract;
- change accepted escape decoding semantics.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-parser-string-escape-diagnostics-implementation-plan.sh
```

The guard is static. It does not implement parser-level string escape diagnostics.

Implementation gate

Parser-level string escape diagnostics implementation code may be added only after this plan is merged.

Non-claims

This document does not implement parser-level string escape diagnostics, change parser error enums, change diagnostic codes, change AST storage, implement length-carrying AST strings, broaden accepted L-UI syntax, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_RENDERING_CONTRACT.md

Source title: Latticra L-UI Rendering Contract

Lines: 509 | Words: 1365 | SHA-256: beb92b09054894159cf2598febafc2ad408baef0ca37e591730d80e4651cc873

Latticra L-UI Rendering Contract

Status: L-UI rendering contract Scope: first operator-visible L-UI rendering boundary before renderer implementation, terminal UI behavior, interactive UI behavior, command behavior, execution, mutation, networking, recovery, hardware behavior, or production UI claims.

Purpose

This document defines the first contract for L-UI rendering in Latticra.

L-UI rendering means turning already validated L-UI/LIR metadata into deterministic operator-visible text output.

This document does not implement L-UI rendering.

The implementation plan is recorded in:

```
L_UI_RENDERING_IMPLEMENTATION_PLAN.md
```

Relationship to previous work

This contract depends on:

```
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
include/latticra/lir.h
include/latticra/cpp/authority.hpp
src/l_ui_parser_ast.c
src/l_ui_parser_semantic.c
src/lir.c
src/cpp/authority.cpp
```

Those files remain the source of truth for L-UI AST shape, source-backed text, length-carrying strings, escaped decoded NUL handling, literal source-buffer NUL rejection, semantic validation, LIR shape, constrained C++ authority validation, and no-effect boundaries.

Rendering role

L-UI rendering should be a deterministic presentation layer over validated metadata.

Recommended future pipeline:

```
source bytes
  -&gt; structural parser
  -&gt; AST construction
  -&gt; semantic validation
  -&gt; LIR lowering
  -&gt; constrained authority validation
  -&gt; L-UI rendering
  -&gt; operator-visible text output
```

Rendering must not bypass parser, semantic validation, LIR shape, or authority-layer checks.

First rendering unit

The first renderer should render one validated L-UI card or one validated LIR module.

Suggested future root object:

```
latticra_l_ui_render_result_t
```

Suggested future input object:

```
latticra_l_ui_render_request_t
```

Initial rendered unit labels:

```
card
rail
field
text
binding
effect
boundary
source_span
authority
```

Input prerequisites

Rendering may only operate after the input metadata is valid:

```
parser_error=ok
semantic_error=ok
lir_error=ok
authority_status=ok
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

If parsing, semantic validation, LIR lowering, or authority validation failed, rendering should return a bounded failure report instead of rendering a partial operator surface.

Render modes

Initial render modes should be explicit and bounded.

Suggested render modes:

```
summary
detailed
diagnostics_only
authority_only
```

The first implementation plan must choose exact enum names.

Render modes must not imply interactive UI, live terminal control, command execution, or event handling.

Output target

The first renderer should write into caller-provided buffers only.

Suggested output behavior:

```
caller-provided char buffer
explicit buffer length
NUL-terminated on success
cleared on too-small failure
no heap allocation
no file output
no terminal escape control by default
```

The first renderer should not write directly to stdout, stderr, files, terminals, sockets, servers, logs, or devices.

Rendered report model

A future rendered report should be deterministic and bounded.

Suggested report header:

```
LATTICRA L-UI RENDER REPORT
status=&lt;status-label&gt;
mode=&lt;render-mode-label&gt;
card=&lt;card-name&gt;
effect=&lt;effect-label&gt;
boundary=&lt;boundary-label&gt;
rail_count=&lt;count&gt;
field_count=&lt;count&gt;
text_count=&lt;count&gt;
binding_count=&lt;count&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

Detailed output may include rail, field, text, binding, LIR node, and authority sections, but only as text metadata.

Suggested section order

Detailed rendering should use stable section order:

```
HEADER
CARD
AUTHORITY
RAILS
FIELDS
TEXT
BINDINGS
LIR
SOURCE_SPANS
NO_EFFECT_FLAGS
```

The order must be deterministic across repeated runs.

Rail rendering

Rail rendering should preserve validated rail order from AST/LIR metadata.

Rail output should include:

```
rail index
rail name
field count
text count
source span
```

The renderer must not invent rails or reorder them unless a future contract explicitly defines a display sort mode.

Field and binding rendering

Field rendering should preserve validated field metadata.

Field output should include:

```
field index
field name
binding target
binding prefix
source span
binding source span
```

Bindings remain symbolic. Rendering must not evaluate state, preview values, host state, or runtime state.

Text rendering

Text rendering must respect length-carrying storage.

Rules:

```
value_len is authoritative
escaped output is used for operator-visible text
escaped decoded NUL through \x00 remains representable
literal source-buffer NUL remains rejected before rendering
raw C-string fields are not authoritative for embedded NUL values
```

Rendering must not drop embedded NUL metadata or silently truncate text.

Source-span rendering

Source spans should be operator-visible where useful.

Suggested source-span fields:

```
start_offset
end_offset
start_line
start_column
end_line
end_column
```

Rendering must not invent byte positions, line numbers, or column numbers.

LIR rendering

LIR rendering should display LIR metadata only.

LIR output may include:

```
source_kind
module_name
card_name
node_count
edge_count
binding_count
text_count
node kinds
edge kinds
binding refs
text refs
```

Rendering must not execute LIR, lower LIR into behavior, evaluate bindings, render an interactive UI, or call Nucleus task execution.

Authority rendering

Authority rendering should display constrained C++ authority-layer report metadata only.

Authority output may include:

```
authority status
authority validator
authority requested effect
authority denial reason
authority no-effect flags
```

Rendering authority metadata must not broaden authority, perform effects, or alter authority-layer state.

Error model

Rendering should have a separate error space.

Recommended future labels:

```
ok
null_argument
invalid_input
parser_failed
semantic_failed
lir_failed
authority_failed
capacity_exceeded
unsupported_render_mode
unsupported_effect
unsupported_boundary
internal_error
```

Exact enum names must be defined by a future implementation plan.

Capacity policy

The first implementation should use fixed capacities and caller-owned buffers.

Suggested constants:

```
LATTICRA_L_UI_RENDER_REPORT_MAX
LATTICRA_L_UI_RENDER_LINE_MAX
LATTICRA_L_UI_RENDER_SECTION_MAX
```

The implementation plan must name exact capacities before code is added.

Small output buffers should return a stable capacity error and clear the output buffer.

No-effect rule

L-UI rendering is presentation-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Rendering must not execute commands, mutate state, evaluate host state, call Nucleus task execution, write files, read files, open network connections, call server code, call update code, call recovery code, or touch hardware.

Compatibility expectations

L-UI rendering work must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
existing escaped string report behavior
existing source-backed text behavior
existing length-carrying string behavior
existing decoded NUL acceptance
existing literal source-buffer NUL rejection
existing semantic validation behavior
existing LIR shape behavior
existing constrained C++ authority behavior
source-span byte offset behavior
no-effect flags
current accepted fixture counts
```

The first accepted fixture should still preserve:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Future implementation gate

L-UI rendering implementation must not begin until a separate implementation plan defines:

1. implementation language;
2. public API shape;
3. public header path;
4. source file path;
5. render request struct;
6. render result struct;
7. render mode enum;
8. render error enum;
9. capacity constants;
10. output buffer rules;
11. section order;
12. source-span rendering rules;
13. embedded NUL rendering rules;
14. authority metadata rendering rules;
15. exact tests;
16. compatibility expectations;
17. non-claims.

That plan is recorded in:

```
L_UI_RENDERING_IMPLEMENTATION_PLAN.md
```

Future file policy

Recommended future files:

```
include/latticra/l_ui_renderer.h
src/l_ui_renderer.c
tests/l_ui_rendering_invariants.c
scripts/test-l-ui-rendering.sh
docs/L_UI_RENDERING_IMPLEMENTATION_PLAN.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
```

The implementation plan confirms exact paths before renderer code is added.

Future test list

A future implementation plan should include tests for:

```
l_ui_rendering_accepts_semantically_valid_l_ui_fixture
l_ui_rendering_requires_semantic_success
l_ui_rendering_requires_lir_success
l_ui_rendering_requires_authority_success
l_ui_rendering_preserves_card_metadata
l_ui_rendering_preserves_rail_order
l_ui_rendering_preserves_field_bindings
l_ui_rendering_preserves_text_lengths
l_ui_rendering_preserves_escaped_x00_visibility
```

```
l_ui_rendering_preserves_source_spans
l_ui_rendering_preserves_no_effect_flags
l_ui_rendering_report_is_deterministic
l_ui_rendering_report_rejects_small_buffer
l_ui_rendering_does_not_change_ast_report
l_ui_rendering_does_not_change_lir_report
l_ui_rendering_does_not_execute_bindings
l_ui_rendering_does_not_call_nucleus_execution
l_ui_rendering_is_deterministic
```

The implementation-plan test list is now recorded in:

```
L_UI_RENDERING_IMPLEMENTATION_PLAN.md
```

Forbidden behavior

L-UI rendering work must not:

- render parser-failed input as a valid surface;
- render semantic-failed input as a valid surface;
- render LIR-failed input as a valid surface;
- bypass constrained authority validation;
- execute bindings;
- evaluate host state;
- evaluate runtime state;
- call Nucleus task execution;
- execute Lat;
- execute LIR;
- mutate state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- emit terminal escape control by default;
- create an interactive UI loop;
- broaden accepted syntax;
- weaken parser diagnostics;
- weaken semantic validation;
- weaken LIR validation;
- weaken constrained C++ authority validation;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;
- make raw C-string fields authoritative for embedded NUL values;

- imply a production UI, runtime, sandbox, malware prevention, ransomware prevention, or operating-system surface.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-rendering-contract.sh
```

The guard is static. It does not implement L-UI rendering.

Non-claims

This document does not implement L-UI rendering, interactive UI behavior, terminal control, command behavior, Lat execution, LIR execution, Nucleus task execution, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT.md

Source title: Latticra L-UI Rendering Detailed Report Refinement

Lines: 120 | Words: 271 | SHA-256: 072730fca8ae88a23c027a6cd2bf8a2fb5c7a95a235096799a87acdc1196ea0d

Latticra L-UI Rendering Detailed Report Refinement

Status: implementation refinement Scope: deterministic detailed-report metadata for the no-effect L-UI renderer.

Purpose

This refinement makes the existing L-UI detailed rendering surface easier to audit by adding explicit metadata fields for report classification, detail level, detailed-report availability, detailed section count, deterministic section sequence, no-effect-chain status, and evidence level.

The renderer remains presentation-only.

Implementation files

This slice updates:

```
include/latticra/l_ui_renderer.h
src/l_ui_renderer.c
tests/l_ui_rendering_detailed_report_refinement.c
scripts/test-l-ui-rendering-detailed-report-refinement.sh
.github/workflows/l-ui-rendering-detailed-report-refinement.yml
.github/workflows/c.yml
```

Added metadata

The render result now exposes:

```
report_classification
detail_level
section_sequence
no_effect_chain
evidence_level
detailed_report_available
detailed_section_count
```

Detailed mode reports:

```
report_classification=detailed_report
detail_level=detailed
detailed_report_available=1
detailed_section_count=10
section_sequence=HEADER,CARD,AUTHORITY,RAILS,FIELDS,TEXT,BINDINGS,LIR,SOURCE_SPANS,NO_EFFECT_FLA
GS
```

```
no_effect_chain=preserved
evidence_level=metadata
```

Summary mode remains a header-only report:

```
report_classification=summary_report
detail_level=summary
detailed_report_available=0
detailed_section_count=0
section_sequence=HEADER
```

Boundary

This refinement does not add:

```
interactive UI behavior
terminal control
command behavior
Lat execution
LIR execution
Nucleus task execution
file I/O
network I/O
state mutation
server interaction
self-update
recovery behavior
hardware behavior
sandboxing
malware prevention
ransomware prevention
operating-system completeness
```

Validation

Run:

```
sh scripts/test-l-ui-rendering-detailed-report-refinement.sh
sh scripts/test-l-ui-rendering.sh
```

The focused test verifies:

```
detailed mode sets explicit report metadata
detailed report emits explicit metadata
summary mode remains single-header reporting
no-effect flags remain preserved
```

Compatibility expectations

This refinement preserves:

```
existing parser behavior
existing semantic validation behavior
existing LIR shape behavior
existing authority behavior
existing render modes
existing no-effect flags
existing detailed report section order
existing summary report behavior
```

Non-claims

This document and implementation do not claim a finished operating system, hardened sandbox, production runtime, production security boundary, malware prevention, ransomware prevention, recovery system, update system, bootable image, public UI product, terminal UI product, or public release readiness.

docs/L_UI_RENDERING_IMPLEMENTATION.md

Source title: Latticra L-UI Rendering Implementation

Lines: 335 | Words: 715 | SHA-256: b73e0ebc8e01f40eca3870dcdf5ac3b277c02b216217ba3bb055823e880e2233

Latticra L-UI Rendering Implementation

Status: initial implementation contract Scope: first no-effect C L-UI renderer API, bounded render result metadata, deterministic report output, semantic/LIR/authority prerequisites, source-span rendering, length-aware text rendering, compatibility expectations, validation, and non-claims.

Purpose

This document records the first L-UI rendering implementation.

The implementation follows:

```
docs/L_UI_RENDERING_CONTRACT.md
docs/L_UI_RENDERING_IMPLEMENTATION_PLAN.md
```

The renderer turns already validated L-UI/LIR metadata into deterministic operator-visible text output.

It is presentation-only.

Implementation files

This slice adds:

```
include/latticra/l_ui_renderer.h
src/l_ui_renderer.c
tests/l_ui_rendering_invariants.c
scripts/test-l-ui-rendering.sh
```

It also wires the renderer test into:

```
.github/workflows/c.yml
```

Public API

The public API is:

```
latticra_l_ui_render_error_t
latticra_l_ui_render_mode_t
latticra_l_ui_render_authority_summary_t
latticra_l_ui_render_request_t
latticra_l_ui_render_result_t
latticra_l_ui_render_error_label
latticra_l_ui_render_mode_label
latticra_l_ui_render
latticra_l_ui_render_report
```

The API is C and uses caller-owned input/output structures.

Capacity constants

The renderer defines:

```
LATTICRA_L_UI_RENDER_LABEL_MAX 64u
LATTICRA_L_UI_RENDER_REASON_MAX 128u
LATTICRA_L_UI_RENDER_REPORT_MAX 16384u
LATTICRA_L_UI_RENDER_LINE_MAX 512u
LATTICRA_L_UI_RENDER_SECTION_MAX 16u
LATTICRA_L_UI_RENDER_AUTHORITY_REPORT_MAX 4096u
```

The renderer uses fixed-capacity fields and caller-provided report buffers.

Render errors

Stable render error labels:

```
ok
null_argument
invalid_input
parser_failed
semantic_failed
lir_failed
authority_failed
```

```
capacity_exceeded
unsupported_render_mode
unsupported_effect
unsupported_boundary
internal_error
```

Render modes

Supported render modes:

```
summary
detailed
diagnostics_only
authority_only
```

Render modes are presentation choices only. They do not imply interactive UI behavior, command execution, live terminal control, or event handling.

Input prerequisites

Rendering succeeds only when:

```
parser_error=ok
semantic_error=ok
lir_error=ok
authority_status=ok
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

The renderer rejects parser-failed, semantic-failed, LIR-failed, authority-failed, non-no-effect, unsupported-effect, unsupported-boundary, and unsupported-mode inputs.

Render result metadata

The render result records:

```
card name
effect
boundary
rail count
field count
text count
binding count
node count
edge count
section count
card source span
authority summary
no-effect flags
bounded rail snapshots
bounded field snapshots
bounded binding-prefix snapshots
bounded text length and escaped-value snapshots
bounded source-span snapshots
```

The renderer does not retain request pointers after latticra_l_ui_render returns.

Report output

latticra_l_ui_render_report writes a deterministic report into a caller-provided buffer.

Header fields include:

```
LATTICRA L-UI RENDER REPORT
status=&lt;integer-status&gt;
error=&lt;render-error-label&gt;
mode=&lt;render-mode-label&gt;
card=&lt;card-name&gt;
effect=&lt;effect-label&gt;
boundary=&lt;boundary-label&gt;
rail_count=&lt;count&gt;
field_count=&lt;count&gt;
text_count=&lt;count&gt;
```

```
binding_count=&lt;count&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
section_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
authority_status=&lt;authority-status-label&gt;
authority_validator=&lt;authority-validator-label&gt;
authority_requested_effect=&lt;authority-effect-label&gt;
authority_denial_reason=&lt;authority-denial-reason&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
span_start_line=&lt;line&gt;
span_start_column=&lt;column&gt;
span_end_line=&lt;line&gt;
span_end_column=&lt;column&gt;
```

Small output buffers return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer.

Detailed report sections

Detailed mode emits sections in this order:

```
HEADER
CARD
AUTHORITY
RAILS
FIELDS
TEXT
BINDINGS
LIR
SOURCE_SPANS
NO_EFFECT_FLAGS
```

The order is deterministic across repeated runs.

Text and embedded NUL behavior

The renderer respects length-carrying text metadata.

Rules:

```
value_len is authoritative
escaped output is used for operator-visible text
escaped decoded NUL through \x00 remains visible
literal source-buffer NUL remains rejected before rendering
raw C-string fields are not authoritative for embedded NUL values
```

Authority metadata behavior

The renderer consumes a C-compatible authority summary.

It displays authority metadata but does not broaden authority, perform effects, call C++ authority APIs directly, or expose C++ object lifetimes through the renderer API.

No-effect behavior

Rendering preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Rendering does not execute commands, mutate state, evaluate host state, call Nucleus task execution, write files, read files, open network connections, call server code, call update code, call recovery code, or touch hardware.

Validation

Run:

```
sh scripts/test-l-ui-rendering.sh
```

The test suite covers:

```
l_ui_rendering_accepts_semantically_valid_l_ui_fixture
l_ui_rendering_requires_semantic_success
l_ui_rendering_requires_lir_success
l_ui_rendering_requires_authority_success
l_ui_rendering_rejects_non_no_effect_flags
l_ui_rendering_preserves_card_metadata
l_ui_rendering_preserves_rail_order
l_ui_rendering_preserves_field_bindings
l_ui_rendering_preserves_text_lengths
l_ui_rendering_preserves_escaped_x00_visibility
l_ui_rendering_preserves_source_spans
l_ui_rendering_preserves_no_effect_flags
l_ui_rendering_report_is_deterministic
l_ui_rendering_report_rejects_small_buffer
l_ui_rendering_rejects_unsupported_mode
l_ui_rendering_does_not_change_ast_report
l_ui_rendering_does_not_change_lir_report
l_ui_rendering_does_not_execute_bindings
l_ui_rendering_does_not_call_nucleus_execution
l_ui_rendering_is_deterministic
```

Compatibility expectations

This implementation must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
existing escaped string report behavior
existing source-backed text behavior
existing length-carrying string behavior
existing decoded NUL acceptance
existing literal source-buffer NUL rejection
existing semantic validation behavior
existing LIR shape behavior
existing constrained C++ authority behavior
source-span byte offset behavior
no-effect flags
current accepted fixture counts
```

The first accepted fixture still preserves:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Current boundary

This implementation does not provide:

```
interactive UI behavior
terminal control
command behavior
Lat execution
LIR execution
Nucleus task execution
file I/O
network I/O
state mutation
server interaction
self-update
recovery behavior
hardware behavior
sandboxing
malware prevention
ransomware prevention
operating-system completeness
```

Non-claims

This document and implementation do not claim a finished operating system, hardened sandbox, production runtime, production security boundary, malware prevention, ransomware prevention, recovery system, update system, bootable image, public UI product, terminal UI product, or public release readiness.

docs/L_UI_RENDERING_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Rendering Implementation Plan

Lines: 640 | Words: 1634 | SHA-256: 3a6c57934332789aa4e8aa1b02658622c1ba0de62a48ff1974b2b3da22816d59

Latticra L-UI Rendering Implementation Plan

Status: implementation planning contract Scope: exact public API, renderer files, render request/result structs, render mode enum, render error enum, capacity constants, output-buffer behavior, section order, source-span rendering, embedded-NUL rendering, authority metadata rendering, tests, compatibility expectations, and non-claims before L-UI renderer code.

Purpose

This document defines the implementation plan for the first L-UI renderer.

The L-UI rendering contract is already merged and guarded. This plan turns that contract into exact public API, file paths, data structures, capacity constants, render modes, error labels, report format, output-buffer rules, authority metadata handling, tests, compatibility expectations, and non-claims required before renderer code is added.

This document does not implement L-UI rendering.

Relationship to previous work

This plan depends on:

```
docs/L_UI_RENDERING_CONTRACT.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
include/latticra/lir.h
include/latticra/cpp/authority.hpp
src/l_ui_parser_ast.c
src/l_ui_parser_semantic.c
src/lir.c
src/cpp/authority.cpp
```

Those files remain the source of truth for parser behavior, AST shape, source-backed text, length-carrying strings, escaped decoded NUL handling, literal source-buffer NUL rejection, semantic validation, LIR shape, constrained C++ authority validation, and no-effect boundaries.

Implementation language decision

The first L-UI renderer should be implemented in C.

Reason:

- L-UI parser, AST, semantic validation, detailed reports, and LIR shape are C;

- C remains the secure substrate;
- renderer output should be deterministic and bounded;
- renderer code should use caller-provided buffers and avoid heap allocation;
- the C++ authority layer remains a validation/report dependency, not the rendering substrate.

The renderer may consume authority metadata through a small C-compatible authority summary struct or a pre-rendered authority report snapshot. It should not include or require C++ object lifetime in the public C renderer API.

Implementation files

The implementation PR should add or modify:

```
include/latticra/l_ui_renderer.h
src/l_ui_renderer.c
tests/l_ui_rendering_invariants.c
scripts/test-l-ui-rendering.sh
.github/workflows/c.yml
docs/L_UI_RENDERING_IMPLEMENTATION.md
```

The implementation PR should not add interactive UI behavior, terminal control, command behavior, Lat execution, LIR execution, Nucleus task execution, file I/O, network I/O, state mutation, recovery behavior, update behavior, server interaction, hardware behavior, or security guarantees.

Public API shape

Add public API names:

```
latticra_l_ui_render_error_t
latticra_l_ui_render_mode_t
latticra_l_ui_render_authority_summary_t
latticra_l_ui_render_request_t
latticra_l_ui_render_result_t
latticra_l_ui_render_error_label
latticra_l_ui_render_mode_label
latticra_l_ui_render
latticra_l_ui_render_report
```

Recommended function signatures:

```
const char *latticra_l_ui_render_error_label(latticra_l_ui_render_error_t error);
const char *latticra_l_ui_render_mode_label(latticra_l_ui_render_mode_t mode);

latticra_status_t latticra_l_ui_render(
    const latticra_l_ui_render_request_t *request,
    latticra_l_ui_render_result_t *result);

latticra_status_t latticra_l_ui_render_report(
    const latticra_l_ui_render_result_t *result,
    char *buffer,
    size_t buffer_len);
```

The first renderer should generate metadata reports only.

Capacity constants

Add exact bounded constants:

```
LATTICRA_L_UI_RENDER_LABEL_MAX 64u
LATTICRA_L_UI_RENDER_REASON_MAX 128u
LATTICRA_L_UI_RENDER_REPORT_MAX 16384u
LATTICRA_L_UI_RENDER_LINE_MAX 512u
LATTICRA_L_UI_RENDER_SECTION_MAX 16u
LATTICRA_L_UI_RENDER_AUTHORITY_REPORT_MAX 4096u
```

The first implementation must reject or report capacity failures deterministically. It must not silently truncate semantically meaningful fields.

Render error enum

Add render error enum values:

```
LATTICRA_L_UI_RENDER_OK
LATTICRA_L_UI_RENDER_NULL_ARGUMENT
LATTICRA_L_UI_RENDER_INVALID_INPUT
LATTICRA_L_UI_RENDER_PARSER_FAILED
LATTICRA_L_UI_RENDER_SEMANTIC_FAILED
LATTICRA_L_UI_RENDER_LIR_FAILED
LATTICRA_L_UI_RENDER_AUTHORITY_FAILED
LATTICRA_L_UI_RENDER_CAPACITY_EXCEEDED
LATTICRA_L_UI_RENDER_UNSUPPORTED_RENDER_MODE
LATTICRA_L_UI_RENDER_UNSUPPORTED_EFFECT
LATTICRA_L_UI_RENDER_UNSUPPORTED_BOUNDARY
LATTICRA_L_UI_RENDER_INTERNAL_ERROR
```

Stable labels:

```
ok
null_argument
invalid_input
parser_failed
semantic_failed
lir_failed
authority_failed
capacity_exceeded
unsupported_render_mode
unsupported_effect
unsupported_boundary
internal_error
```

Render mode enum

Add render mode enum values:

```
LATTICRA_L_UI_RENDER_MODE_SUMMARY
LATTICRA_L_UI_RENDER_MODE_DETAILED
LATTICRA_L_UI_RENDER_MODE_DIAGNOSTICS_ONLY
LATTICRA_L_UI_RENDER_MODE_AUTHORITY_ONLY
```

Stable labels:

```
summary
detailed
diagnostics_only
authority_only
```

Render modes are presentation choices only. They must not imply interaction, command execution, live terminal control, or event handling.

Authority summary struct

Add a C-compatible authority summary struct:

```
latticra_status_t status;
char status_label[LATTICRA_L_UI_RENDER_LABEL_MAX];
char validator_label[LATTICRA_L_UI_RENDER_LABEL_MAX];
char requested_effect_label[LATTICRA_L_UI_RENDER_LABEL_MAX];
char denial_reason[LATTICRA_L_UI_RENDER_REASON_MAX];
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

This struct lets the C renderer display authority metadata without owning C++ objects or exposing C++ object lifetimes through a C ABI.

The first implementation may use a helper in tests to populate this struct from known good authority metadata, but renderer code must remain C.

Render request struct

Add request fields:

```
latticra_l_ui_render_mode_t mode;
const latticra_l_ui_ast_result_t *ast;
const latticra_l_ui_semantic_result_t *semantic;
const latticra_lir_module_t *lir;
const latticra_l_ui_render_authority_summary_t *authority;
```

The renderer must not retain pointers from the request after the call returns.

Render result struct

Add result fields:

```
latticra_status_t status;
latticra_l_ui_render_error_t error;
latticra_l_ui_render_mode_t mode;
char card_name[LATTICRA_L_UI_RENDER_LABEL_MAX];
char effect[LATTICRA_L_UI_RENDER_LABEL_MAX];
char boundary[LATTICRA_L_UI_RENDER_LABEL_MAX];
size_t rail_count;
size_t field_count;
size_t text_count;
size_t binding_count;
size_t node_count;
size_t edge_count;
size_t section_count;
latticra_l_ui_source_span_t span;
latticra_l_ui_render_authority_summary_t authority;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

The result must be fully initialized on success and failure.

Input prerequisite behavior

Rendering may succeed only when:

```
parser_error=ok
semantic_error=ok
lir_error=ok
authority_status=ok
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Failure behavior:

```
parser failure -&gt; LATTICRA_L_UI_RENDER_PARSER_FAILED
semantic failure -&gt; LATTICRA_L_UI_RENDER_SEMANTIC_FAILED
LIR failure -&gt; LATTICRA_L_UI_RENDER_LIR_FAILED
authority failure -&gt; LATTICRA_L_UI_RENDER_AUTHORITY_FAILED
non-no-effect flags -&gt; LATTICRA_L_UI_RENDER_AUTHORITY_FAILED
unsupported mode -&gt; LATTICRA_L_UI_RENDER_UNSUPPORTED_RENDER_MODE
unsupported effect -&gt; LATTICRA_L_UI_RENDER_UNSUPPORTED_EFFECT
unsupported boundary -&gt; LATTICRA_L_UI_RENDER_UNSUPPORTED_BOUNDARY
```

The renderer must not produce a valid surface from invalid inputs.

Output buffer rules

latticra_l_ui_render_report should:

```
write only to caller-provided buffers
require explicit buffer length
NUL-terminate on success
clear the buffer on too-small failure
return LATTICRA_STATUS_BUFFER_TOO_SMALL for small buffers
avoid heap allocation
avoid file output
avoid stdout
avoid stderr
avoid terminal escape control
```

The renderer must not write directly to files, terminals, sockets, servers, logs, or devices.

Report format

latticra_l_ui_render_report should emit:

```
LATTICRA L-UI RENDER REPORT
status=&lt;integer-status&gt;
error=&lt;render-error-label&gt;
mode=&lt;render-mode-label&gt;
card=&lt;card-name&gt;
effect=&lt;effect-label&gt;
boundary=&lt;boundary-label&gt;
rail_count=&lt;count&gt;
field_count=&lt;count&gt;
text_count=&lt;count&gt;
binding_count=&lt;count&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
section_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
authority_status=&lt;authority-status-label&gt;
authority_validator=&lt;authority-validator-label&gt;
authority_requested_effect=&lt;authority-effect-label&gt;
authority_denial_reason=&lt;authority-denial-reason&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
span_start_line=&lt;line&gt;
span_start_column=&lt;column&gt;
span_end_line=&lt;line&gt;
span_end_column=&lt;column&gt;
```

Summary mode may stop at the metadata header.

Detailed mode should add deterministic sections in the section order below.

Section order

Detailed rendering must use this section order:

```
HEADER
CARD
AUTHORITY
RAILS
FIELDS
TEXT
BINDINGS
LIR
SOURCE_SPANS
NO_EFFECT_FLAGS
```

The renderer must produce deterministic section order across repeated runs.

Rail rendering rules

Rail rendering should preserve validated rail order.

Detailed output should include:

```
rail[&lt;index&gt;].name=&lt;rail-name&gt;
rail[&lt;index&gt;].field_count=&lt;count&gt;
rail[&lt;index&gt;].text_count=&lt;count&gt;
rail[&lt;index&gt;].span_start_offset=&lt;offset&gt;
rail[&lt;index&gt;].span_end_offset=&lt;offset&gt;
```

The renderer must not invent rails or reorder rails.

Field and binding rendering rules

Field output should preserve validated field metadata:

```
field[&lt;index&gt;].name=&lt;field-name&gt;
field[&lt;index&gt;].binding=&lt;binding-target&gt;
field[&lt;index&gt;].binding_prefix=&lt;binding-prefix&gt;
field[&lt;index&gt;].span_start_offset=&lt;offset&gt;
field[&lt;index&gt;].binding_span_start_offset=&lt;offset&gt;
```

Bindings remain symbolic. Rendering must not evaluate state, preview values, host state, runtime state, or environment state.

Text rendering rules

Text rendering must respect length-carrying storage.

Rules:

```
value_len is authoritative
escaped output is used for operator-visible text
escaped decoded NUL through \x00 remains visible
literal source-buffer NUL remains rejected before rendering
raw C-string fields are not authoritative for embedded NUL values
```

Detailed output should include:

```
text[&lt;index&gt;].value_len=&lt;length&gt;
text[&lt;index&gt;].escaped_value=&lt;escaped-text&gt;
text[&lt;index&gt;].span_start_offset=&lt;offset&gt;
text[&lt;index&gt;].span_end_offset=&lt;offset&gt;
```

Rendering must not silently drop or truncate embedded NUL metadata.

Source-span rendering rules

Source-span output should use existing parser/AST/LIR spans.

Required fields:

```
start_offset
end_offset
start_line
start_column
end_line
end_column
```

Rendering must not invent byte offsets, line numbers, or column numbers.

LIR rendering rules

LIR rendering should display LIR metadata only.

Detailed LIR output may include:

```
source_kind
module_name
card_name
node_count
edge_count
binding_count
text_count
node kind labels
edge kind labels
binding refs
text refs
```

Rendering must not execute LIR, lower LIR into behavior, evaluate bindings, render an interactive UI, or call Nucleus task execution.

Authority metadata rendering rules

Authority rendering should display the C-compatible authority summary only.

Authority output may include:

```
authority_status
authority_validator
authority_requested_effect
authority_denial_reason
```

```
authority no-effect flags
```

Rendering authority metadata must not broaden authority, perform effects, or alter authority-layer state.

No-effect preservation

The renderer is presentation-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Rendering must not execute commands, mutate state, evaluate host state, call Nucleus task execution, write files, read files, open network connections, call server code, call update code, call recovery code, or touch hardware.

Exact implementation test list

The implementation PR should include tests for:

```
l_ui_rendering_accepts_semantically_valid_l_ui_fixture
l_ui_rendering_requires_semantic_success
l_ui_rendering_requires_lir_success
l_ui_rendering_requires_authority_success
l_ui_rendering_rejects_non_no_effect_flags
l_ui_rendering_preserves_card_metadata
l_ui_rendering_preserves_rail_order
l_ui_rendering_preserves_field_bindings
l_ui_rendering_preserves_text_lengths
l_ui_rendering_preserves_escaped_x00_visibility
l_ui_rendering_preserves_source_spans
l_ui_rendering_preserves_no_effect_flags
l_ui_rendering_report_is_deterministic
l_ui_rendering_report_rejects_small_buffer
l_ui_rendering_rejects_unsupported_mode
l_ui_rendering_does_not_change_ast_report
l_ui_rendering_does_not_change_lir_report
l_ui_rendering_does_not_execute_bindings
l_ui_rendering_does_not_call_nucleus_execution
l_ui_rendering_is_deterministic
```

Test file plan

Add:

```
tests/l_ui_rendering_invariants.c
scripts/test-l-ui-rendering.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-l-ui-rendering-implementation-plan.sh
```

and before:

```
sh scripts/test-l-ui-string-literal-escape-contract.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

```
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/L_UI_RENDERING_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/L_UI_RENDERING_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
existing escaped string report behavior
existing source-backed text behavior
existing length-carrying string behavior
existing decoded NUL acceptance
existing literal source-buffer NUL rejection
existing semantic validation behavior
existing LIR shape behavior
existing constrained C++ authority behavior
source-span byte offset behavior
no-effect flags
current accepted fixture counts
```

The first accepted fixture should still preserve:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Forbidden implementation behavior

The first L-UI renderer implementation must not:

- render parser-failed input as a valid surface;
- render semantic-failed input as a valid surface;
- render LIR-failed input as a valid surface;
- bypass constrained authority validation;
- execute bindings;
- evaluate host state;
- evaluate runtime state;
- call Nucleus task execution;
- execute Lat;
- execute LIR;
- mutate state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;

- emit terminal escape control by default;
- create an interactive UI loop;
- broaden accepted syntax;
- weaken parser diagnostics;
- weaken semantic validation;
- weaken LIR validation;
- weaken constrained C++ authority validation;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;
- make raw C-string fields authoritative for embedded NUL values;
- imply a production UI, runtime, sandbox, malware prevention, ransomware prevention, or operating-system surface.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-l-ui-rendering-implementation-plan.sh
```

The guard is static. It does not implement L-UI rendering.

Implementation gate

L-UI renderer implementation code may be added only after this plan is merged.

Non-claims

This document does not implement L-UI rendering, interactive UI behavior, terminal control, command behavior, Lat execution, LIR execution, Nucleus task execution, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/L_UI_SEMANTIC_VALIDATION_CONTRACT.md

Source title: Latticra L-UI Semantic Validation Contract

Lines: 407 | Words: 1208 | SHA-256: 17c684cce51e9183745a09d1553dea32cbcc35aed8a9bf1c8eb3cf58061de651

Latticra L-UI Semantic Validation Contract

Status: semantic validation contract Scope: future semantic validation layer for L-UI after structural parsing, source policies, AST construction, and string handling.

Purpose

This document defines the contract for a future L-UI semantic validation layer.

The current L-UI parser verifies a controlled structural fixture shape, source spans, diagnostics, AST metadata, string-literal escapes, escaped decoded NUL behavior, and literal source-buffer NUL rejection.

Semantic validation is the next layer. It should validate meaning after structural parsing succeeds but before any future rendering, LIR lowering, Lat integration, or Nucleus task handling.

This document does not implement semantic validation.

The implementation plan is recorded in:

```
L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
```

Relationship to previous work

This contract depends on:

```
docs/L_UI_PARSER.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for parse results, AST shape, diagnostics, source spans, string storage, and source-buffer policy.

Semantic validation layer

Semantic validation is a distinct layer after structural parsing.

Recommended pipeline:

```
source bytes
  -&gt; structural parser
  -&gt; parser diagnostics
  -&gt; AST construction
  -&gt; semantic validation
  -&gt; semantic report
  -&gt; future rendering or lowering only if allowed
```

Semantic validation should not replace structural parsing.

Initial semantic scope

The first semantic validation layer should check:

```
required rail set
rail uniqueness
field uniqueness within a rail
required field ownership
binding prefix policy
binding target consistency
text node placement
card effect boundary
card boundary policy
no-effect preservation
```

Required rail semantics

The accepted L-UI fixture currently uses these rails:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Semantic validation should verify:

- each required rail is present exactly once;
- rails appear in deterministic report order;
- rail names are known;
- each rail owns only its expected fields or text nodes;

- text-only rails do not own fields;
- field-only rails do not own text nodes unless a future contract expands the grammar.

Field ownership semantics

Initial expected ownership:

```
state: origin, route, axis, path
trace: breadcrumb, trace
safety: health, risk, lock, dark_phase
gates: safe_portal, rollback
effects: host, external, requested
policy: request, policy, reason
execution: executed, mutation, server, recovery, hardware
```

Semantic validation should reject fields that appear under the wrong rail, duplicate fields, or missing required fields when a semantic validation API is introduced.

Binding semantics

Initial allowed binding prefixes:

```
state.
preview.
```

Expected binding relationships:

```
origin -&gt; state.origin
route -&gt; state.route
axis -&gt; state.axis
path -&gt; state.path
breadcrumb -&gt; state.breadcrumb
trace -&gt; state.trace
health -&gt; state.health
risk -&gt; state.risk
lock -&gt; state.lock
dark_phase -&gt; state.dark_phase
safe_portal -&gt; state.safe_portal
rollback -&gt; state.rollback
host -&gt; state.host_effect
external -&gt; state.external_effect
requested -&gt; preview.requested_effect
request -&gt; preview.request
policy -&gt; preview.policy
reason -&gt; preview.reason
executed -&gt; preview.executed
mutation -&gt; preview.mutation_allowed
server -&gt; preview.server_interaction_allowed
recovery -&gt; preview.recovery_allowed
hardware -&gt; preview.hardware_allowed
```

Semantic validation should treat mismatched field/binding pairs as semantic errors.

Text semantics

Initial text node placement:

```
top rail -&gt; first text node
bottom rail -&gt; second text node
```

Semantic validation should verify:

- exactly two text nodes exist;
- the top rail owns the first text node;
- the bottom rail owns the second text node;
- text values use explicit value_len when inspected;
- raw value= compatibility fields are not authoritative for embedded NUL values.

Card semantics

The initial card remains:

```
NucleusPreview
```

Required card metadata:

```
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
```

Semantic validation should reject unsupported cards, effects, boundaries, or mismatched counts.

Semantic diagnostic model

A future implementation should introduce semantic validation diagnostics without destabilizing existing parser diagnostic codes.

Recommended semantic diagnostic code range:

```
LUI1000-LUI1099
```

Initial candidate diagnostics:

```
LUI1000 semantic_validation_failed
LUI1001 duplicate_rail
LUI1002 missing_required_rail
LUI1003 duplicate_field
LUI1004 field_rail_mismatch
LUI1005 binding_field_mismatch
LUI1006 unsupported_binding_target
LUI1007 text_rail_mismatch
LUI1008 card_count_mismatch
LUI1009 semantic_internal_error
```

The exact names and codes must be finalized in the implementation plan before code is added.

Semantic report model

A future semantic report should be deterministic and bounded.

Suggested fields:

```
L-UI SEMANTIC VALIDATION RESULT
status=&lt;ok|error&gt;
semantic_error=&lt;label&gt;
card=&lt;card-name&gt;
rail_count=&lt;count&gt;
field_count=&lt;count&gt;
text_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

For semantic failures, the report should include a source span when available.

Source-span behavior

Semantic diagnostics should use the most specific existing source span available:

- rail errors should use the rail span;
- field errors should use the field span;
- binding errors should use the binding span;
- text errors should use the text span;
- card-level errors should use the card span;
- internal semantic errors should use a default span only when no better span exists.

Semantic validation should not invent byte positions that are not represented by parser or AST metadata.

AST relationship

Semantic validation should operate on latticra_l_ui_ast_result_t after latticra_l_ui_parse_ast returns parser success.

If ast.parse_result.error != LATTICRA_L_UI_PARSE_OK, semantic validation should not attempt semantic inspection and should report parser failure or semantic not-run state.

Public API expectation

A future implementation plan should define public API additions such as:

```
latticra_l_ui_semantic_result_t
latticra_l_ui_validate_semantics
latticra_l_ui_semantic_error_label
latticra_l_ui_semantic_report
```

No public API is added by this contract.

No-effect rule

Semantic validation is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Semantic validation must not render UI, lower to LIR, execute bindings, mutate state, call Nucleus task code, interact with a server, write files, read files, or touch hardware.

Compatibility expectations

Semantic validation must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
source-span byte offset behavior
no-effect flags
current accepted fixture counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Future implementation gate

Semantic validation implementation must not begin until a separate implementation plan defines:

1. public API shape;
2. semantic result struct fields;
3. semantic error enum labels;
4. diagnostic code mapping;

5. report format;
6. rail semantic checks;
7. field semantic checks;
8. binding semantic checks;
9. text semantic checks;
10. exact tests;
11. compatibility expectations;
12. non-claims.

That plan is recorded in:

```
L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
```

Future test list

A future implementation plan should include tests for:

```
semantic_validation_accepts_current_fixture
semantic_validation_rejects_duplicate_rail
semantic_validation_rejects_missing_required_rail
semantic_validation_rejects_duplicate_field
semantic_validation_rejects_field_rail_mismatch
semantic_validation_rejects_binding_field_mismatch
semantic_validation_rejects_unsupported_binding_target
semantic_validation_rejects_text_rail_mismatch
semantic_validation_rejects_card_count_mismatch
semantic_validation_skips_when_parser_failed
semantic_validation_reports_source_spans
semantic_validation_preserves_no_effect_flags
semantic_validation_does_not_change_ast_report
semantic_validation_does_not_change_escaped_x00_acceptance
semantic_validation_does_not_change_literal_nul_rejection
semantic_validation_is_deterministic
```

The implementation-plan test list is now recorded in:

```
L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
```

Forbidden behavior

Semantic validation must not:

- execute bindings;
- evaluate host state;
- call Nucleus task execution;
- render L-UI;
- lower to LIR;
- mutate state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;

- broaden accepted syntax;
- weaken parser diagnostics;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;
- make raw value= or purpose= fields authoritative for embedded NUL values.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-semantic-validation-contract.sh
```

The guard is static. It does not implement semantic validation.

Non-claims

This document does not implement semantic validation, LIR lowering, L-UI rendering, Lat execution, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md

Source title: Latticra L-UI Semantic Validation Implementation

Lines: 270 | Words: 583 | SHA-256: 7a98415c171aa537ca4f86c97598f6e30892b434ea1db877233fb5b53b1f0d23

Latticra L-UI Semantic Validation Implementation

Status: initial implementation contract Scope: no-effect semantic validation API, result model, reports, and invariants for L-UI ASTs.

Purpose

The L-UI Semantic Validation implementation adds a bounded, deterministic validation layer after structural parsing and AST construction.

It validates meaning over the existing L-UI AST without rendering UI, lowering to LIR, executing bindings, reading files, writing files, opening network connections, mutating state, or calling Nucleus task behavior.

Implementation files

```
include/latticra/l_ui_parser.h
src/l_ui_parser_semantic.c
tests/l_ui_semantic_validation_invariants.c
scripts/test-l-ui-semantic-validation.sh
.github/workflows/c.yml
```

Related active files:

```
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
docs/L_UI_SEMANTIC_VALIDATION_CONTRACT.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
```

Public API

The public API adds:

```
latticra_l_ui_semantic_error_t
latticra_l_ui_semantic_result_t
latticra_l_ui_semantic_error_label
latticra_l_ui_validate_semantics
```

```
latticra_l_ui_semantic_report
```

The validator consumes:

```
const latticra_l_ui_ast_result_t *ast
```

and writes:

```
latticra_l_ui_semantic_result_t *result
```

Semantic error labels

Stable labels:

```
ok
parser_failed
duplicate_rail
missing_required_rail
duplicate_field
field_rail_mismatch
binding_field_mismatch
unsupported_binding_target
text_rail_mismatch
card_count_mismatch
internal_error
```

Semantic report

The semantic report is deterministic and bounded.

It starts with:

```
L-UI SEMANTIC VALIDATION RESULT
```

and includes:

```
status
error
parser_error
card
rail
field
binding
rail_index
field_index
text_index
rail_count
field_count
text_count
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
span_start_offset
span_end_offset
span_start_line
span_start_column
span_end_line
span_end_column
```

Small output buffers return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

and clear the output buffer.

Checks implemented

The first semantic validation implementation verifies:

```
card=NucleusPreview
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
required rail names
```

```
rail uniqueness
rail field/text ownership metadata
field uniqueness
field names in deterministic order
binding prefixes
field/binding pair consistency
text placement
no-effect flags
```

Parser-failed behavior

If parsing failed, semantic validation does not inspect AST nodes.

It reports:

```
error=parser_failed
parser_error=&lt;existing-parser-error-label&gt;
rail_count=0
field_count=0
text_count=0
```

This does not replace parser diagnostics or failed-parse AST reports.

Source-span behavior

Semantic failures use existing AST spans:

```
rail errors -&gt; rail span
field errors -&gt; field span
binding errors -&gt; binding span
text errors -&gt; text span
card/count errors -&gt; card span
parser_failed -&gt; parse-result span
```

Semantic validation does not invent byte offsets.

No-effect boundary

Semantic validation is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility

This implementation does not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
source-span byte offset behavior
current accepted fixture counts
compact AST report behavior
failed-parse detailed report behavior
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Test command

Run:

```
sh scripts/test-l-ui-semantic-validation.sh
```

The main C workflow runs this check after the semantic validation implementation-plan guard and before the string-literal escape guards.

Required invariants

The semantic validation tests verify:

```
semantic_validation_accepts_current_fixture
semantic_validation_rejects_duplicate_rail
semantic_validation_rejects_missing_required_rail
semantic_validation_rejects_duplicate_field
semantic_validation_rejects_field_rail_mismatch
semantic_validation_rejects_binding_field_mismatch
semantic_validation_rejects_unsupported_binding_target
semantic_validation_rejects_text_rail_mismatch
semantic_validation_rejects_card_count_mismatch
semantic_validation_skips_when_parser_failed
semantic_validation_reports_source_spans
semantic_validation_preserves_no_effect_flags
semantic_validation_does_not_change_ast_report
semantic_validation_does_not_change_escaped_x00_acceptance
semantic_validation_does_not_change_literal_nul_rejection
semantic_validation_is_deterministic
semantic_validation_report_rejects_small_buffer
semantic_validation_error_labels_are_stable
```

Current evidence level

This implementation is an L2 tested semantic validation layer for the current L-UI AST fixture model.

It is not LIR lowering, L-UI rendering, Lat execution, command behavior, Nucleus task execution, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

Next implementation step

The next implementation candidate is:

```
LIR shape contract
```

That future work should define the first Latticra Intermediate Representation shape before any lowering implementation.

Non-claims

This document and implementation do not implement LIR lowering, L-UI rendering, Lat execution, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Semantic Validation Implementation Plan

Lines: 556 | Words: 1327 | SHA-256: 05b7a2e5a16d1bd86a7d9b0ea72405f8fde4c7926304937fb76b9820622dba6f

Latticra L-UI Semantic Validation Implementation Plan

Status: implementation planning contract Scope: public API shape, semantic result structs, semantic errors, diagnostic mapping, report format, semantic checks, exact tests, and compatibility expectations before semantic validation code.

Purpose

This document defines the implementation plan for the L-UI semantic validation layer.

The semantic-validation contract is already merged and guarded. This plan names the exact API, result shape, error labels, diagnostic mapping, report format, semantic checks, source-span behavior, tests, and non-claims required before implementation code is added.

This document does not implement semantic validation.

Relationship to previous work

This plan depends on:

```
docs/L_UI_SEMANTIC_VALIDATION_CONTRACT.md
docs/L_UI_PARSER.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for parse results, AST shape, diagnostics, source spans, string storage, decoded NUL behavior, and source-buffer policy.

Implementation language decision

Semantic validation should be implemented in C.

Reason:

- the parser and AST implementation are C;
- the public L-UI API is C;
- semantic validation consumes latticra_l_ui_ast_result_t;
- existing tests and runners are C and POSIX shell;
- the first semantic layer should stay deterministic, bounded, and no-effect.

Implementation files

The implementation PR should modify:

```
include/latticra/l_ui_parser.h
src/l_ui_parser_semantic.c
src/l_ui_parser_diagnostics.c
.github/workflows/c.yml
```

The implementation PR should add:

```
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md
tests/l_ui_semantic_validation_invariants.c
scripts/test-l-ui-semantic-validation.sh
```

The implementation should not add file I/O, network I/O, rendering code, LIR lowering, or task execution.

Public API shape

Add public semantic validation API declarations to:

```
include/latticra/l_ui_parser.h
```

Required API names:

```
latticra_l_ui_semantic_error_t
latticra_l_ui_semantic_result_t
latticra_l_ui_semantic_error_label
latticra_l_ui_validate_semantics
latticra_l_ui_semantic_report
```

Recommended function signatures:

```
const char *latticra_l_ui_semantic_error_label(latticra_l_ui_semantic_error_t error);

latticra_status_t latticra_l_ui_validate_semantics(
    const latticra_l_ui_ast_result_t *ast,
    latticra_l_ui_semantic_result_t *result);

latticra_status_t latticra_l_ui_semantic_report(
    const latticra_l_ui_semantic_result_t *result,
    char *buffer,
    size_t buffer_len);
```

Semantic error enum

Add a semantic error enum with stable labels:

```
LATTICRA_L_UI_SEMANTIC_OK
LATTICRA_L_UI_SEMANTIC_PARSER_FAILED
LATTICRA_L_UI_SEMANTIC_DUPLICATE_RAIL
LATTICRA_L_UI_SEMANTIC_MISSING_REQUIRED_RAIL
LATTICRA_L_UI_SEMANTIC_DUPLICATE_FIELD
LATTICRA_L_UI_SEMANTIC_FIELD_RAIL_MISMATCH
LATTICRA_L_UI_SEMANTIC_BINDING_FIELD_MISMATCH
LATTICRA_L_UI_SEMANTIC_UNSUPPORTED_BINDING_TARGET
LATTICRA_L_UI_SEMANTIC_TEXT_RAIL_MISMATCH
LATTICRA_L_UI_SEMANTIC_CARD_COUNT_MISMATCH
LATTICRA_L_UI_SEMANTIC_INTERNAL_ERROR
```

Required labels:

```
ok
parser_failed
duplicate_rail
missing_required_rail
duplicate_field
field_rail_mismatch
binding_field_mismatch
unsupported_binding_target
text_rail_mismatch
card_count_mismatch
internal_error
```

Semantic result struct

Add a semantic result struct that includes:

```
latticra_status_t status;
latticra_l_ui_semantic_error_t error;
latticra_l_ui_parse_error_t parser_error;
latticra_l_ui_source_span_t span;
size_t rail_index;
size_t field_index;
size_t text_index;
char card_name[LATTICRA_L_UI_AST_NAME_MAX];
char rail_name[LATTICRA_L_UI_AST_NAME_MAX];
char field_name[LATTICRA_L_UI_AST_NAME_MAX];
char binding[LATTICRA_L_UI_AST_BINDING_MAX];
size_t rail_count;
size_t field_count;
size_t text_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

The result should be deterministic and fully initialized for both success and failure.

Semantic diagnostic mapping

Semantic validation should use a separate diagnostic code range from parser diagnostics:

```
LUI1000-LUI1099
```

Initial mapping:

```
LUI1000 semantic_validation_failed
```

```
LUI1001 duplicate_rail
LUI1002 missing_required_rail
LUI1003 duplicate_field
LUI1004 field_rail_mismatch
LUI1005 binding_field_mismatch
LUI1006 unsupported_binding_target
LUI1007 text_rail_mismatch
LUI1008 card_count_mismatch
LUI1009 semantic_internal_error
```

parser_failed may report a semantic not-run state in semantic reports, but it must not replace existing parser diagnostic codes.

Semantic report format

latticra_l_ui_semantic_report should produce a deterministic report:

```
L-UI SEMANTIC VALIDATION RESULT
status=&lt;integer-status&gt;
error=&lt;semantic-error-label&gt;
parser_error=&lt;parser-error-label&gt;
card=&lt;card-name&gt;
rail=&lt;rail-name&gt;
field=&lt;field-name&gt;
binding=&lt;binding&gt;
rail_index=&lt;index&gt;
field_index=&lt;index&gt;
text_index=&lt;index&gt;
rail_count=&lt;count&gt;
field_count=&lt;count&gt;
text_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
span_start_line=&lt;line&gt;
span_start_column=&lt;column&gt;
span_end_line=&lt;line&gt;
span_end_column=&lt;column&gt;
```

The report must return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer when the output does not fit.

Rail semantic checks

Semantic validation should verify exactly these rail names in deterministic order:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Checks:

- all required rails are present;
- no duplicate rail names exist;
- no unknown rail names exist;
- rail order is deterministic;
- rail field/text ownership metadata is consistent.

Expected rail ownership:

```
top: text_count=1, field_count=0
state: field_count=4, text_count=0
```

```
trace: field_count=2, text_count=0
safety: field_count=4, text_count=0
gates: field_count=2, text_count=0
effects: field_count=3, text_count=0
policy: field_count=3, text_count=0
execution: field_count=5, text_count=0
bottom: text_count=1, field_count=0
```

Field semantic checks

Expected field ownership:

```
state: origin, route, axis, path
trace: breadcrumb, trace
safety: health, risk, lock, dark_phase
gates: safe_portal, rollback
effects: host, external, requested
policy: request, policy, reason
execution: executed, mutation, server, recovery, hardware
```

Checks:

- all required fields exist;
- no duplicate field appears within the same rail;
- field order remains deterministic;
- each field appears in its expected rail;
- field source spans remain available.

Binding semantic checks

Allowed binding prefixes:

```
state.
preview.
```

Expected field/binding pairs:

```
origin -&gt; state.origin
route -&gt; state.route
axis -&gt; state.axis
path -&gt; state.path
breadcrumb -&gt; state.breadcrumb
trace -&gt; state.trace
health -&gt; state.health
risk -&gt; state.risk
lock -&gt; state.lock
dark_phase -&gt; state.dark_phase
safe_portal -&gt; state.safe_portal
rollback -&gt; state.rollback
host -&gt; state.host_effect
external -&gt; state.external_effect
requested -&gt; preview.requested_effect
request -&gt; preview.request
policy -&gt; preview.policy
reason -&gt; preview.reason
executed -&gt; preview.executed
mutation -&gt; preview.mutation_allowed
server -&gt; preview.server_interaction_allowed
recovery -&gt; preview.recovery_allowed
hardware -&gt; preview.hardware_allowed
```

Checks:

- unsupported binding prefixes are rejected;
- mismatched field/binding pairs are rejected;
- binding source spans are used for binding diagnostics.

Text semantic checks

Expected text ownership:

```
top rail -&gt; first text node
bottom rail -&gt; second text node
```

Checks:

- exactly two text nodes exist;
- top rail owns one text node;
- bottom rail owns one text node;
- no other rail owns text nodes;
- text spans remain available;
- value_len is used for length-aware checks.

Card semantic checks

Required card semantics:

```
card=NucleusPreview
effect=none
boundary=preview_only
rail_count=9
field_count=23
text_count=2
```

Checks:

- unsupported card names are rejected;
- unsupported effect values are rejected;
- unsupported boundary values are rejected;
- card counts must match aggregate AST counts;
- no-effect flags remain denied for execution, mutation, server, recovery, and hardware.

Parser-failed behavior

If ast->parse_result.error != LATTICRA_L_UI_PARSE_OK, semantic validation should not inspect semantic nodes.

Expected result:

```
status=LATTICRA_STATUS_OK
error=LATTICRA_L_UI_SEMANTIC_PARSER_FAILED
parser_error=&lt;existing parser error&gt;
rail_count=0
field_count=0
text_count=0
```

This does not replace parser diagnostics or failed-parse AST reports.

Source-span behavior

Semantic failures should report the most specific span available:

```
rail errors -&gt; rail span
field errors -&gt; field span
binding errors -&gt; binding span
text errors -&gt; text span
card/count errors -&gt; card span
parser_failed -&gt; parser result span
internal_error -&gt; default span only if no better span exists
```

Semantic validation must not invent byte offsets.

Exact implementation test list

The implementation PR should include tests for:

```
semantic_validation_accepts_current_fixture
```

```
semantic_validation_rejects_duplicate_rail
semantic_validation_rejects_missing_required_rail
semantic_validation_rejects_duplicate_field
semantic_validation_rejects_field_rail_mismatch
semantic_validation_rejects_binding_field_mismatch
semantic_validation_rejects_unsupported_binding_target
semantic_validation_rejects_text_rail_mismatch
semantic_validation_rejects_card_count_mismatch
semantic_validation_skips_when_parser_failed
semantic_validation_reports_source_spans
semantic_validation_preserves_no_effect_flags
semantic_validation_does_not_change_ast_report
semantic_validation_does_not_change_escaped_x00_acceptance
semantic_validation_does_not_change_literal_nul_rejection
semantic_validation_is_deterministic
semantic_validation_report_rejects_small_buffer
semantic_validation_error_labels_are_stable
```

Test file plan

Add:

```
tests/l_ui_semantic_validation_invariants.c
scripts/test-l-ui-semantic-validation.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-l-ui-semantic-validation-contract.sh
```

and before:

```
sh scripts/test-l-ui-string-literal-escape-contract.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/L_UI_SEMANTIC_VALIDATION_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md
```

No-effect preservation

Semantic validation is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility expectations

The implementation must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
```

```
source-span byte offset behavior
no-effect flags
current accepted fixture counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Forbidden implementation behavior

Semantic validation implementation must not:

- execute bindings;
- evaluate host state;
- call Nucleus task execution;
- render L-UI;
- lower to LIR;
- mutate state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- broaden accepted syntax;
- weaken parser diagnostics;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;
- make raw value= or purpose= fields authoritative for embedded NUL values;
- change compact AST report behavior;
- change failed-parse detailed report behavior.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-l-ui-semantic-validation-implementation-plan.sh
```

The guard is static. It does not implement semantic validation.

Implementation gate

Semantic validation implementation code may be added only after this plan is merged.

Non-claims

This document does not implement semantic validation, LIR lowering, L-UI rendering, Lat execution, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md

Source title: Latticra L-UI Source Buffer Literal NUL Policy Contract

Lines: 325 | Words: 1037 | SHA-256: a62c0adcf5ff0a5941cb3ad7ecc09d709871346ae0cc5a599dcec5a6d10d0b5b

Latticra L-UI Source Buffer Literal NUL Policy Contract

Status: source-buffer literal NUL policy contract Scope: policy for literal 0x00 bytes in L-UI source buffers after escaped decoded NUL acceptance.

Purpose

This document defines the L-UI policy for literal NUL bytes in source buffers.

L-UI now accepts escaped decoded NUL bytes through:

```
\x00
```

inside supported source-backed AST string values. That does not mean literal 0x00 bytes in the source buffer are accepted.

This contract keeps literal source NUL bytes rejected by default and defines what a future source-buffer model would need to prove before that boundary could change.

This document does not implement literal NUL acceptance.

The implementation plan is recorded in:

```
L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION_PLAN.md
```

Relationship to previous work

This contract depends on:

```
docs/L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION_PLAN.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for escaped decoded NUL acceptance, explicit AST string lengths, parser diagnostics, AST decoding, detailed reports, and no-effect behavior.

Current boundary

The current system provides:

```
escaped decoded NUL acceptance through \x00
purpose_len and value_len fields
length-aware escaped detailed report fields
literal source NUL rejection
parser-level literal NUL diagnostics
no-effect flags
```

This contract does not add:

```
literal source NUL acceptance
```

```
new source-buffer ownership model
new accepted escape sequences
Unicode display behavior
renderer behavior
interactive UI behavior
command behavior
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Policy decision

Literal 0x00 bytes in L-UI source buffers remain forbidden.

The accepted way to express a decoded NUL byte inside a source-backed string value is the escaped form:

```
\x00
```

Literal NUL bytes in the source buffer should continue to report:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0023 literal_nul_in_string
```

Rationale

Literal source NUL bytes remain forbidden because:

- source buffers still cross C APIs and C-compatible tooling;
- many tools treat NUL as a string terminator;
- embedded source NUL can hide or truncate source text in logs and diagnostics;
- escaped \x00 is deterministic and visible;
- detailed reports already render escaped decoded NUL as \x00;
- accepting literal source NUL requires a separate source-buffer ownership model.

Escaped decoded NUL remains accepted

This policy must not roll back escaped decoded NUL support.

The following remains accepted in supported source-backed string values:

```
purpose &quot;A\x00B&quot;
text &quot;left\x00right&quot;
```

The decoded AST byte sequence contains a NUL byte, and the explicit length fields remain authoritative:

```
ast.card.purpose_len
ast.texts[index].value_len
```

Literal NUL diagnostic behavior

Literal source NUL inside a validated quoted string value should report:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0023 literal_nul_in_string
```

The diagnostic span should cover the literal source NUL byte.

The diagnostic should preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
```

```
hardware_allowed=0
```

Parser behavior

Parser validation should keep detecting literal source NUL bytes before AST materialization.

Literal source NUL should not be treated as:

```
empty source
unterminated string
invalid hex escape
decoded NUL
internal error
```

Literal source NUL remains a distinct source-buffer error.

AST behavior

Literal source NUL input should not produce a partial AST.

Failed parse behavior remains:

```
ast.parse_result.error = LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

Report behavior

Literal source NUL failures should continue to use failed-parse detailed report behavior.

No decoded AST value should be materialized for literal source NUL input.

Detailed report embedded-NUL surfaces remain reserved for escaped decoded NUL values:

```
purpose_len
purpose_escaped
value_len
value_escaped
```

Source-span behavior

For literal source NUL failures, spans should cover exactly the literal 0x00 byte in the source buffer.

Line and column remain one-based byte positions in the in-memory source buffer.

Escaped decoded NUL spans remain source-oriented and cover the source bytes \x00 when relevant.

Future source-buffer model requirements

A future implementation may revisit literal source NUL only after a separate implementation plan defines:

1. source buffer ownership rules;
2. source length authority rules;
3. source display and logging behavior;
4. diagnostic rendering for literal NUL;
5. file fixture policy;
6. C API compatibility rules;
7. command-line tooling behavior;
8. report escaping behavior;

9. exact tests;
10. non-claims.

Until that exists, literal source NUL remains forbidden.

The current implementation plan keeps literal source NUL rejected and adds exact enforcement tests:

```
L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION_PLAN.md
```

Compatibility expectations

This policy must not change:

```
escaped decoded NUL acceptance
literal source NUL rejection
LUI0023 diagnostic mapping
accepted string-literal escape decoding
parser diagnostic code stability
existing C-string fields
purpose_len and value_len semantics
failed-parse detailed report behavior
existing accepted fixture counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

No-effect rule

Literal NUL rejection is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Future test list

A future implementation plan for source-buffer literal NUL policy enforcement should include tests for:

```
literal_nul_policy_rejects_purpose_literal_nul
literal_nul_policy_rejects_text_literal_nul
literal_nul_policy_reports_lui0023
literal_nul_policy_span_covers_literal_nul_byte
literal_nul_policy_preserves_no_effect_flags
literal_nul_policy_does_not_emit_decoded_nul_lui0022
literal_nul_policy_does_not_materialize_partial_ast
literal_nul_policy_does_not_change_escaped_x00_acceptance
literal_nul_policy_does_not_change_failed_parse_report
literal_nul_policy_is_deterministic
```

That implementation-plan test list is now recorded in:

```
L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION_PLAN.md
```

Forbidden behavior

Future source-buffer work must not:

- accept literal NUL source bytes without a separate implementation plan;
- treat literal source NUL as escaped decoded NUL;
- remove LUI0023 literal NUL rejection;

- remove LUI0022 diagnostic compatibility;
- hide literal source NUL in diagnostics;
- produce a partial AST for literal source NUL input;
- make raw purpose= or value= report fields the assertion target for embedded NUL bytes;
- broaden accepted escape forms;
- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references.

Current validation command

This contract is guarded by:

```
sh scripts/test-l-ui-source-buffer-literal-nul-policy-contract.sh
```

The guard is static. It does not implement literal source NUL acceptance.

Non-claims

This document does not implement literal source NUL acceptance, source-buffer ownership changes, Unicode display behavior, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md

Source title: Latticra L-UI Source Buffer Literal NUL Policy Implementation

Lines: 235 | Words: 604 | SHA-256: e22141b44c7e6887f46990a95de68ac2b4455bf1e51fc21ae83f2d3ddaf30fb5

Latticra L-UI Source Buffer Literal NUL Policy Implementation

Status: initial implementation contract Scope: enforcement tests and documentation for rejecting literal 0x00 bytes in L-UI source buffers while preserving escaped decoded NUL acceptance.

Purpose

This implementation locks down the L-UI source-buffer literal NUL policy.

Literal 0x00 bytes in L-UI source buffers remain rejected.

Escaped decoded NUL remains accepted through:

```
\x00
```

inside supported source-backed string values.

This implementation does not accept literal source NUL bytes.

Implementation files

```
tests/l_ui_source_buffer_literal_nul_policy_invariants.c
scripts/test-l-ui-source-buffer-literal-nul-policy.sh
.github/workflows/c.yml
```

Related active files:

```
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION_PLAN.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
```

Behavior

Literal source-buffer NUL remains rejected as:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
LUI0023 literal_nul_in_string
```

Literal source NUL is not treated as:

```
empty_source
unterminated_string
invalid_string_escape
invalid_hex_escape
decoded_nul_in_string
internal_error
```

Escaped decoded NUL compatibility

Escaped decoded NUL remains accepted:

```
purpose &quot;A\x00B&quot;
text &quot;left\x00right&quot;
```

The decoded AST byte sequence contains a NUL byte, and explicit length fields remain authoritative:

```
ast.card.purpose_len
ast.texts[index].value_len
```

Diagnostic behavior

Literal source NUL maps to:

```
LUI0023 literal_nul_in_string
```

Stable message:

```
Literal NUL bytes are not supported in AST strings.
```

Stable hint:

```
Remove literal NUL bytes from the source string.
```

LUI0022 decoded_nul_in_string remains available as a compatibility diagnostic, but literal source NUL must not emit it.

Source-span behavior

Literal source NUL spans cover exactly one source byte:

```
span_end_offset - span_start_offset == 1
source[span_start_offset] == 0x00
```

Line and column remain one-based byte positions in the in-memory source buffer.

AST behavior

Literal source NUL input does not produce a partial AST.

Expected failed AST state:

```
ast.parse_result.error = LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
ast.card.rail_count = 0
ast.card.field_count = 0
ast.card.text_count = 0
ast.card.purpose_len = 0
ast.texts[index].value_len = 0
```

Report behavior

Literal source NUL failures use failed-parse detailed report behavior.

Expected report fields:

```
parse_error=literal_nul_in_string
rail_count=0
field_count=0
text_count=0
```

Failed reports do not materialize string sections for literal source NUL input:

```
purpose_escaped=
value_escaped=
```

Test command

Run:

```
sh scripts/test-l-ui-source-buffer-literal-nul-policy.sh
```

The main C workflow runs this check after the source-buffer literal NUL policy implementation-plan guard and before the string-literal escape guards.

Required invariants

The implementation tests verify:

```
literal_nul_policy_rejects_purpose_literal_nul
literal_nul_policy_rejects_text_literal_nul
literal_nul_policy_reports_lui0023
literal_nul_policy_span_covers_literal_nul_byte
literal_nul_policy_preserves_no_effect_flags
literal_nul_policy_does_not_emit_decoded_nul_lui0022
literal_nul_policy_does_not_materialize_partial_ast
literal_nul_policy_does_not_change_escaped_x00_acceptance
literal_nul_policy_does_not_change_failed_parse_report
literal_nul_policy_is_deterministic
```

No-effect boundary

Literal source NUL rejection is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility

This implementation does not change:

```
escaped decoded NUL acceptance
literal source NUL rejection
LUI0023 diagnostic mapping
accepted string-literal escape decoding
```

```
parser diagnostic code stability
existing C-string fields
purpose_len and value_len semantics
failed-parse detailed report behavior
existing accepted fixture counts
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Current evidence level

This implementation is an L2 tested source-buffer policy enforcement model for literal source NUL rejection.

It is not literal source NUL acceptance, source-buffer ownership changes, Unicode display behavior, L-UI rendering, command behavior, Nucleus task execution, runtime security, malware prevention, ransomware prevention, sandboxing, or an operating system.

Next implementation step

The next implementation candidate is:

```
L-UI semantic validation contract
```

That future work should define semantic checks beyond structural parsing, including field/binding consistency, duplicate handling, rail semantics, and reportable validation errors.

Non-claims

This document and implementation do not accept literal source NUL bytes, change source-buffer ownership, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, claim malware prevention, claim ransomware prevention, or claim operating-system completeness.

docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI Source Buffer Literal NUL Policy Implementation Plan

Lines: 354 | Words: 996 | SHA-256: 19b7d9e58cd6147c399dde516fa920a9ba2207a1d068c9d63931ff4d9da4724c

Latticra L-UI Source Buffer Literal NUL Policy Implementation Plan

Status: implementation planning contract Scope: exact implementation and test plan for enforcing the policy that literal 0x00 bytes in L-UI source buffers remain rejected.

Purpose

This document defines the implementation plan for L-UI source-buffer literal NUL policy enforcement.

The policy contract is already merged and guarded. This plan defines exact parser behavior, diagnostic behavior, AST behavior, report behavior, source-span expectations, tests, documentation updates, and non-claims before any implementation slice changes or extends literal source-buffer handling.

This document does not implement literal NUL acceptance.

Relationship to previous work

This plan depends on:

```
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_CONTRACT.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION_PLAN.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
docs/L_UI_AST_LENGTH_CARRYING_STRING_STORAGE_IMPLEMENTATION.md
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
src/l_ui_parser_diagnostics.c
```

Those files remain the source of truth for escaped decoded NUL acceptance, explicit AST string lengths, parser diagnostics, AST decoding, detailed reports, and no-effect behavior.

Implementation language decision

Literal source-buffer NUL policy enforcement should remain in C.

Reason:

- source validation lives in src/l_ui_parser.c;
- AST construction lives in src/l_ui_parser_ast.c;
- diagnostics live in src/l_ui_parser_diagnostics.c;
- tests are C fixtures and shell runners;
- the behavior is byte-oriented and bounded.

Policy decision

Literal 0x00 bytes in L-UI source buffers remain rejected.

The accepted way to express a decoded NUL byte inside a source-backed string value remains:

```
\x00
```

This plan is for policy enforcement and regression coverage, not literal-NUL acceptance.

Parser behavior plan

Parser validation in:

```
src/l_ui_parser.c
```

should continue detecting literal source-buffer NUL bytes before AST materialization.

Literal source NUL inside a quoted value should report:

```
LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Literal source NUL should not be classified as:

```
empty_source
unterminated_string
invalid_string_escape
invalid_hex_escape
decoded_nul_in_string
internal_error
```

Diagnostic behavior plan

Literal source NUL should map to:

```
LUI0023 literal_nul_in_string
```

The diagnostic message and hint remain stable:

```
Literal NUL bytes are not supported in AST strings.
Remove literal NUL bytes from the source string.
```

LUI0022 decoded_nul_in_string remains available as a stable compatibility diagnostic, but literal source NUL must not emit it.

Source-span plan

For literal source NUL failures, the parse-result and diagnostic span should cover exactly the literal 0x00 byte in the source buffer.

Expected span rule:

```
span_end_offset - span_start_offset == 1
source[span_start_offset] == 0x00
```

Line and column remain one-based byte positions in the in-memory source buffer.

AST behavior plan

Literal source NUL input should not produce a partial AST.

Expected failed-parse AST state:

```
ast.parse_result.error = LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
ast.card.rail_count = 0
ast.card.field_count = 0
ast.card.text_count = 0
ast.card.purpose_len = 0
ast.texts[index].value_len = 0
```

Report behavior plan

Literal source NUL failures should use failed-parse detailed report behavior.

Expected report behavior:

```
parse_error=literal_nul_in_string
rail_count=0
field_count=0
text_count=0
```

The report should not include materialized AST string sections for literal source NUL input:

```
purpose=
value=
purpose_escaped=
value_escaped=
```

Escaped decoded NUL compatibility

This implementation plan must not roll back escaped decoded NUL acceptance.

The following should remain accepted:

```
purpose &quot;A\x00B&quot;
text &quot;left\x00right&quot;
```

The existing decoded-NUL acceptance tests should remain valid.

Accepted escape compatibility

Do not broaden or reduce the accepted escape set.

Still accepted:

```
\\
\&quot;
\n
```

```
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

Still rejected:

```
\x0a -&gt; LATTICRA_L_UI_PARSE_INVALID_HEX_ESCAPE
\a -&gt; LATTICRA_L_UI_PARSE_INVALID_STRING_ESCAPE
literal 0x00 -&gt; LATTICRA_L_UI_PARSE_LITERAL_NUL_IN_STRING
```

Source-buffer construction test plan

Literal source NUL tests should construct source buffers with explicit byte lengths.

Tests should not rely on C string termination to place the literal NUL.

Recommended helper shape:

```
make_source_binary(buffer, buffer_len, purpose, purpose_len, top_text, top_text_len,
bottom_text, bottom_text_len, source_len)
```

This helper should preserve embedded literal NUL bytes in the source buffer and pass the explicit source_len to parser APIs.

Exact implementation test list

The implementation PR should include tests for:

```
literal_nul_policy_rejects_purpose_literal_nul
literal_nul_policy_rejects_text_literal_nul
literal_nul_policy_reports_lui0023
literal_nul_policy_span_covers_literal_nul_byte
literal_nul_policy_preserves_no_effect_flags
literal_nul_policy_does_not_emit_decoded_nul_lui0022
literal_nul_policy_does_not_materialize_partial_ast
literal_nul_policy_does_not_change_escaped_x00_acceptance
literal_nul_policy_does_not_change_failed_parse_report
literal_nul_policy_is_deterministic
```

Test file plan

Add or update:

```
tests/l_ui_source_buffer_literal_nul_policy_invariants.c
scripts/test-l-ui-source-buffer-literal-nul-policy.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-l-ui-source-buffer-literal-nul-policy-contract.sh
```

and before:

```
sh scripts/test-l-ui-string-literal-escape-contract.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/UPCOMING_WORK.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_CONTRACT.md
docs/L_UI_DECODED_NUL_ACCEPTANCE_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
```

No-effect preservation

Literal source NUL rejection is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility expectations

This implementation plan must not change:

```
escaped decoded NUL acceptance
literal source NUL rejection
LUI0023 diagnostic mapping
accepted string-literal escape decoding
parser diagnostic code stability
existing C-string fields
purpose_len and value_len semantics
failed-parse detailed report behavior
existing accepted fixture counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Forbidden implementation behavior

Future source-buffer work must not:

- accept literal NUL source bytes without a separate acceptance contract;
- treat literal source NUL as escaped decoded NUL;
- remove LUI0023 literal NUL rejection;
- remove LUI0022 diagnostic compatibility;
- hide literal source NUL in diagnostics;
- produce a partial AST for literal source NUL input;
- make raw purpose= or value= report fields the assertion target for embedded NUL bytes;
- broaden accepted escape forms;
- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;

- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-l-ui-source-buffer-literal-nul-policy-implementation-plan.sh
```

The guard is static. It does not implement literal source NUL acceptance.

Implementation gate

Literal source-buffer NUL policy implementation code may be added only after this plan is merged.

Non-claims

This document does not implement literal source NUL acceptance, source-buffer ownership changes, Unicode display behavior, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_SOURCE_GRAMMAR.md

Source title: Latticra L-UI Source Grammar Draft

Lines: 212 | Words: 540 | SHA-256: 4082904c6c9512cb869312d04d0d728f177d5aa06f59e10e3b8c0b9e88a66c67

Latticra L-UI Source Grammar Draft

Status: grammar draft and fixture contract Scope: L-UI source shape, static fixtures, reserved words, validation guards, and non-claims.

Purpose

This document defines the first draft of L-UI source syntax.

L-UI is the planned terminal/operator UI dialect for Latticra. This draft exists before a parser so the project can refine syntax, fixtures, names, and boundaries before implementation work begins.

Current boundary

This is not a parser.

This is not a renderer engine.

This is not an interactive UI.

This is not a command language.

This is not live Nucleus execution.

This document defines a source grammar direction and fixture rules only.

File extension

Planned extension:

```
.lui
```

Version header

Every L-UI fixture should begin with:

```
lui 0.1
```

Initial top-level forms

Initial forms:

```
card &lt;Name&gt; {
  ...
}
```

Reserved future forms:

```
rail &lt;Name&gt; { ... }
layout &lt;Name&gt; { ... }
theme &lt;Name&gt; { ... }
```

Only card fixtures are used in the first draft.

Card grammar direction

Draft shape:

```
lui 0.1

card NucleusPreview {
  purpose &quot;operator-visible Nucleus preview report&quot;
  effect none
  boundary preview_only

  rail top {
    text &quot;Latticra / Nucleus Preview / effect-bound&quot;
  }

  rail state {
    field origin bind state.origin
    field route bind state.route
    field axis bind state.axis
    field path bind state.path
  }
}
```

Required card clauses

Every initial card fixture must include:

```
purpose &quot;...&quot;
effect none
boundary preview_only
```

Required Nucleus preview rails

The initial NucleusPreview fixture must include these rails:

```
top
state
trace
safety
gates
effects
policy
execution
bottom
```

Bind syntax

Field binding direction:

```
field &lt;label&gt; bind &lt;source.path&gt;
```

Allowed source prefixes in the first draft:

```
state.
preview.
```

Examples:

```
field origin bind state.origin
field requested_effect bind preview.requested_effect
field policy bind preview.policy
```

Literal syntax

String literals use double quotes:

```
text &quot;preview-only no-live-movement no-host-effect no-external-effect&quot;
```

Effect rule

The only allowed effect declaration in the first draft is:

```
effect none
```

No other effect value is allowed in initial fixtures.

Boundary rule

The initial allowed boundary is:

```
boundary preview_only
```

This means the fixture is intended for static, no-effect report layout only.

Forbidden fixture behavior

An L-UI fixture must not include behavior that implies:

- command execution;
- state mutation;
- live movement;
- host effects;
- network effects;
- hardware effects;
- boot effects;
- recovery behavior;
- update behavior;
- server interaction;
- hidden side effects.

First fixture

The first fixture is:

```
examples/l-ui/nucleus-preview-card.lui
```

It should mirror the existing C static report rails while remaining source-only.

Guard script

Fixture validation is performed by:

```
sh scripts/test-l-ui-grammar-fixtures.sh
```

The guard script performs static checks only. It does not parse or execute L-UI.

Future parser rule

A future parser may only be added after:

1. this grammar draft is stable;
2. fixtures are reviewed;
3. forbidden behavior tests exist;
4. syntax and error reporting rules are documented;
5. parsing remains no-effect.

Non-claims

This document and its fixtures do not implement L-UI parsing, rendering, command execution, Nucleus execution, live movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_STATIC_REPORT.md

Source title: Latticra L-UI Static Report Fixture

Lines: 121 | Words: 402 | SHA-256: e763f842adf9cdc7e007220e55367beaa1e691713e2a8338a479c611b006c829

Latticra L-UI Static Report Fixture

Status: initial implementation contract Scope: terminal-facing static report fixture for Nucleus preview and state lattice data.

Purpose

The L-UI Static Report Fixture is the fourth Latticra implementation unit.

It defines a deterministic terminal-facing layout that composes the state lattice and Nucleus preview classification into operator-visible rails.

This is not a parser, renderer engine, interactive UI, shell, compositor, or live control surface.

Implementation files

```
include/latticra/l_ui_static_report.h
src/l_ui_static_report.c
tests/l_ui_static_report_invariants.c
scripts/test-l-ui-static-report.sh
```

Report kind

Initial report kind:

```
nucleus-preview
```

Required rails

The static report must render:

```
L-UI STATIC REPORT
kind=nucleus-preview
rail.top=Latticra / Nucleus Preview / effect-bound
rail.state=origin:&lt;origin&gt; route:&lt;route&gt; axis:&lt;axis&gt; path:&lt;path&gt;
rail.trace=breadcrumb:&lt;breadcrumb&gt; trace:&lt;trace&gt;
rail.safety=health:&lt;health&gt; risk:&lt;risk&gt; lock:&lt;lock&gt; dark_phase:&lt;
dark_phase&gt;
rail.gates=safe_portal:&lt;safe_portal&gt; rollback:&lt;rollback&gt;
rail.effects=host:&lt;host_effect&gt; external:&lt;external_effect&gt; requested:&lt;
requested_effect&gt;
rail.policy=request:&lt;request&gt; policy:&lt;policy&gt; reason:&lt;reason&gt;
rail.execution=executed:&lt;0|1&gt; mutation:&lt;0|1&gt; server:&lt;0|1&gt; recovery:&lt;0|1&gt;
 hardware:&lt;0|1&gt;
rail.bottom=preview-only no-live-movement no-host-effect no-external-effect
```

Preview-only boundary

The fixture must not:

- parse L-UI source;
- execute commands;
- mutate state;
- perform live movement;
- interact with servers;
- perform update behavior;
- perform recovery behavior;
- touch hardware;
- touch boot state;
- perform host effects;
- perform external effects.

Public API

Initial API:

```
latticra_l_ui_report_kind_label
latticra_l_ui_nucleus_preview_card
```

Test command

Run:

```
sh scripts/test-l-ui-static-report.sh
```

The main C workflow runs state-lattice, tri-plane transition, Nucleus preview, and L-UI static report tests.

Required invariants

The tests must verify:

- report title is stable;
- report kind is stable;
- top rail is present;
- state rail is present;
- trace rail is present;
- safety rail is present;
- gate rail is present;
- effect rail is present;
- policy rail is present;
- execution rail is present;
- bottom boundary rail is present;
- denied requests can be rendered;
- bad arguments are rejected;
- small buffers are rejected.

Current evidence level

This implementation is an L2 tested model for a terminal-facing static report fixture.

It is not a live L-UI language implementation, renderer, shell, compositor, parser, supervisor executor, server client, update engine, recovery system, hardware system, boot system, or security boundary.

Next implementation step

The next implementation candidate after this model is:

```
L-UI source grammar draft
```

That future work should define syntax and fixtures before adding a parser.

Non-claims

This document and implementation do not claim live UI execution, L-UI parsing, live Nucleus execution, live movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md

Source title: Latticra L-UI String-Literal Escape Contract

Lines: 310 | Words: 973 | SHA-256: 062edcb425263f2138b070961bf8f09730a6aefb0bf00ae1b2c3e607963ed678

Latticra L-UI String-Literal Escape Contract

Status: string-literal escape contract Scope: decoding rules for quoted L-UI source strings used by source-backed AST purpose and text values.

Purpose

This document defines the string-literal escape contract for L-UI source strings.

Latticra now decodes the accepted string-literal escape set while materializing source-backed AST string values. The contract defines which escape sequences are accepted, how they decode, what is rejected, how spans are preserved, how failures are classified, and how reporting remains deterministic.

The implementation plan is documented separately in L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md.

The implementation is documented separately in L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md.

Current boundary

The current L-UI AST stack provides:

```
validated L-UI parser
source-backed purpose extraction
source-backed text extraction
accepted string-literal escape decoding
escaped report fields
source spans
no-effect flags
```

This contract does not add:

```
parser-level invalid escape diagnostics
new accepted non-string grammar
renderer behavior
interactive UI behavior
command behavior
```

```
Nucleus task handling
live movement
state mutation
server interaction
update behavior
recovery behavior
hardware behavior
boot behavior
```

Escape decoding purpose

String-literal escape decoding allows quoted L-UI strings to represent bytes that are awkward or unsafe to write literally while keeping source parsing deterministic.

Decoding applies to source-backed AST values:

```
card.purpose
text.value
```

It remains metadata-only unless a later execution or rendering contract explicitly says otherwise.

Accepted escape set

The implementation supports only this initial escape set:

```
| Source escape | Decoded byte |
| --- | --- |
| `\\` | backslash `0x5C` |
| `\&quot;` | double quote `0x22` |
| `\n` | LF `0x0A` |
| `\r` | CR `0x0D` |
| `\t` | horizontal tab `0x09` |
| `\xNN` | byte value from two uppercase hexadecimal digits |
```

No other escape sequences are accepted in the first implementation.

Hex escape rule

Hex escapes must use exactly two uppercase hexadecimal digits:

```
\x09
\x1F
\x7F
\x80
\xFF
```

Lowercase hexadecimal digits are rejected in the first implementation.

Examples that are rejected:

```
\x0a
\x0
\x001
\xGG
```

Unknown escape rule

Unknown escape sequences are rejected, not copied literally.

Examples that are rejected:

```
\a
\b
\f
\v
\u1234
\U00000000
```

Unterminated escape rule

Unterminated escape sequences are rejected.

Examples:

```
\
\x
\x0
```

Output byte rule

Decoded output is byte-oriented, not Unicode-codepoint-oriented.

The implementation decodes into bounded byte buffers used by AST string fields.

A future Unicode display contract may define higher-level text behavior later.

NUL byte rule

The contract may allow \x00 as a decoded byte only after the AST storage model supports explicit lengths.

Current AST text fields are NUL-terminated C strings, so this implementation rejects decoded NUL bytes for AST values.

Literal 0x00 source bytes inside AST string values are also rejected.

Storage rule

The current AST storage uses NUL-terminated fixed-size buffers:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

The implementation does not silently truncate decoded output.

If decoded output does not fit the destination buffer, decoding returns a stable failure and AST construction avoids partial AST output.

Source-span rule

Decoded values preserve source-span metadata.

Spans continue to refer to source byte ranges, not decoded output byte ranges.

For text nodes:

```
text.span covers the source value range between quotes
```

For purpose values, no public purpose span exists yet; card.span remains the available public metadata unless a future purpose-span contract adds a new field.

Report relationship

Detailed AST reports continue to render:

```
purpose=&lt;decoded-purpose-as-current-C-string&gt;
purpose_escaped=&lt;report-safe-purpose&gt;
value=&lt;decoded-text-as-current-C-string&gt;
value_escaped=&lt;report-safe-text&gt;
```

The escaped report fields remain the stable assertion target for decoded control bytes, quotes, backslashes, DEL, and non-ASCII bytes.

Failure behavior

If string-literal escape decoding fails, AST construction avoids partial AST output.

Until parser-level string escape validation exists, invalid escapes classify the AST result as:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

The structural parser remains compatible and does not yet expose first-class invalid escape errors.

No-effect rule

String-literal escape decoding is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Compatibility rule

The implementation does not change:

```
latticra_l_ui_parse_source existing accepted fixture behavior
latticra_l_ui_ast_report existing required fields
latticra_l_ui_ast_detailed_report existing required fields
latticra_l_ui_diagnostic_report existing required fields
existing escaped report field semantics
existing accepted fixture summary counts
```

The current first fixture contains no source escapes and remains unchanged.

Implementation evidence

String-literal escape decoding required a separate implementation plan defining:

1. parser-level validation decision;
2. AST-level decoding helper shape;
3. accepted escape sequences;
4. rejected escape sequences;
5. NUL byte behavior;
6. capacity failure behavior;
7. span behavior;
8. detailed report behavior;
9. test file names;
10. exact invariant tests;
11. compatibility expectations.

That plan is recorded in L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md.

The implementation is recorded in L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md.

Test list

The implementation tests verify:

```
string_escape_decodes_backslash
string_escape_decodes_quote
string_escape_decodes_newline
string_escape_decodes_carriage_return
string_escape_decodes_tab
string_escape_decodes_uppercase_hex
string_escape_decodes_high_byte_hex
string_escape_rejects_lowercase_hex
string_escape_rejects_short_hex
string_escape_rejects_invalid_hex
string_escape_rejects_unknown_escape
string_escape_rejects_unterminated_escape
string_escape_rejects_decoded_nul_until_length_storage_exists
string_escape_rejects_literal_nul_until_length_storage_exists
string_escape_rejects_oversized_decoded_output
```

```
string_escape_preserves_source_spans
string_escape_updates_detailed_report_escaped_fields
string_escape_preserves_no_effect_flags
string_escape_does_not_change_parse_source_summary
string_escape_does_not_change_failed_parse_report
string_escape_is_deterministic
```

Current validation commands

This contract is guarded by:

```
sh scripts/test-l-ui-string-literal-escape-contract.sh
```

The implementation is tested by:

```
sh scripts/test-l-ui-string-literal-escape.sh
```

Forbidden behavior

The string-literal escape implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- silently truncate decoded values;
- emit raw control bytes in escaped report fields;
- decode bindings, field names, rail names, card names, effect values, or boundary values;
- broaden accepted grammar beyond this contract without a grammar update.

Non-claims

This document and implementation do not implement parser-level string escape diagnostics, length-carrying AST strings, Unicode display behavior, L-UI rendering, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md

Source title: Latticra L-UI String-Literal Escape Implementation

Lines: 328 | Words: 891 | SHA-256: 0d4830a455183ab9ce9e9ff5c54b466125900638b9de787eb9ad5df90cc34943

Latticra L-UI String-Literal Escape Implementation

Status: initial implementation contract Scope: byte-oriented string-literal escape decoding for source-backed L-UI AST purpose and text values.

Purpose

The L-UI String-Literal Escape implementation decodes accepted escape sequences in quoted L-UI source strings while materializing AST-owned metadata values.

This implementation keeps decoding bounded, deterministic, source-span aware, and no-effect. It does not add rendering behavior, command behavior, parser-level string escape diagnostics, Unicode display semantics, or length-carrying AST strings.

Implementation files

```
src/l_ui_parser_ast.c
tests/l_ui_string_literal_escape_invariants.c
scripts/test-l-ui-string-literal-escape.sh
```

Related source-backed extraction files remain active:

```
tests/l_ui_ast_source_backed_text_invariants.c
scripts/test-l-ui-ast-source-backed-text.sh
```

Public API decision

No new public function is added.

Decoding is integrated into:

```
latticra_l_ui_parse_ast
```

The public AST storage remains:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

No new public parser error enum value is added in this slice.

Private helper implementation

The implementation adds private helpers in src/l_ui_parser_ast.c:

```
is_upper_hex_digit
upper_hex_value
append_decoded_byte
decode_l_ui_string_literal_value
extract_decoded_quoted_value_after_token
```

These helpers remain private and are used only while constructing AST string values.

Decoding targets

The implementation decodes only source-backed AST string values:

```
purpose &quot;...&quot; -&gt; ast.card.purpose
first text &quot;...&quot; -&gt; ast.texts[0].value
second text &quot;...&quot; -&gt; ast.texts[1].value
```

It does not decode:

```
card names
rail names
field names
binding values
effect values
boundary values
```

Accepted escape behavior

The implementation accepts only this escape set:

```
\\
\&quot;
```

```
\n
\r
\t
\xNN
```

Decoded byte mapping:

```
\\ -&gt; 0x5C
\&quot; -&gt; 0x22
\n -&gt; 0x0A
\r -&gt; 0x0D
\t -&gt; 0x09
\x41 -&gt; 0x41
\x7F -&gt; 0x7F
\x80 -&gt; 0x80
\xFF -&gt; 0xFF
```

The \xNN form requires exactly two uppercase hexadecimal digits.

Rejected escape behavior

The implementation rejects unknown, malformed, incomplete, and lowercase-hex escapes, including:

```
\a
\b
\f
\v
\u1234
\U00000000
\x0a
\x0
\x001
\xGG
\
\x
\x0
```

Rejected escape sequences are not copied literally.

NUL byte behavior

Decoded NUL bytes are rejected in this implementation.

This includes:

```
\x00
```

Literal 0x00 source bytes inside AST string values are also rejected.

Reason:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

are NUL-terminated fixed-size C strings and do not carry explicit decoded lengths.

Capacity behavior

Decoded output must fit the destination buffer.

If decoded output length is greater than or equal to the destination capacity, decoding fails and clears the destination buffer when possible.

The implementation does not silently truncate decoded values.

AST failure behavior

When string-literal escape decoding fails, AST construction returns:

```
LATTICRA_STATUS_OK
```

and classifies the AST parse result as:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

The AST avoids partial output:

```
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
ast.card.rail_count = 0
ast.card.field_count = 0
ast.card.text_count = 0
```

This preserves the current parser API until parser-level string escape diagnostics are designed.

Source-span behavior

Source spans remain source-oriented.

For text nodes:

```
text.span covers the source value range between quotes
```

If the source contains \n, the span covers the two source bytes \ and n, even though the AST value contains one LF byte.

No public purpose_span field is added.

Detailed report relationship

Detailed reports continue to render:

```
purpose=&lt;decoded-purpose-as-current-C-string&gt;
purpose_escaped=&lt;report-safe-purpose&gt;
value=&lt;decoded-text-as-current-C-string&gt;
value_escaped=&lt;report-safe-text&gt;
```

For decoded control bytes, quote bytes, backslashes, DEL, and non-ASCII bytes, the stable assertion target is the escaped report field.

The existing report escaping layer remains responsible for deterministic report-safe string output.

Parser compatibility

The structural parser remains unchanged for this slice.

latticra_l_ui_parse_source does not perform string-literal escape validation yet.

Invalid escape sequences found during AST construction classify the AST as:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

until a future parser diagnostics contract promotes them to first-class parser errors.

That parser-level extension is now specified by:

```
L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_CONTRACT.md
```

Failed parse compatibility

Existing failed parse behavior remains unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

String decoding does not run for structurally failed parses.

No-effect boundary

String-literal escape decoding is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-l-ui-string-literal-escape.sh
```

The main C workflow runs this check after the string-literal escape implementation-plan guard.

Required invariants

The string-literal escape tests verify:

```
string_escape_decodes_backslash
string_escape_decodes_quote
string_escape_decodes_newline
string_escape_decodes_carriage_return
string_escape_decodes_tab
string_escape_decodes_uppercase_hex
string_escape_decodes_high_byte_hex
string_escape_rejects_lowercase_hex
string_escape_rejects_short_hex
string_escape_rejects_invalid_hex
string_escape_rejects_unknown_escape
string_escape_rejects_unterminated_escape
string_escape_rejects_decoded_nul_until_length_storage_exists
string_escape_rejects_literal_nul_until_length_storage_exists
string_escape_rejects_oversized_decoded_output
string_escape_preserves_source_spans
string_escape_updates_detailed_report_escaped_fields
string_escape_preserves_no_effect_flags
string_escape_does_not_change_parse_source_summary
string_escape_does_not_change_failed_parse_report
string_escape_is_deterministic
```

Current evidence level

This implementation is an L2 tested string-literal escape decoding model for validated L-UI AST metadata.

It is not a parser-level string escape diagnostic system, Unicode display model, renderer, UI runtime, command surface, Nucleus task runner, server client, update engine, recovery system, hardware system, boot system, sandbox, or operating system.

Next implementation step

The parser-level string escape diagnostics contract is recorded in:

```
L_UI_PARSER_STRING_ESCAPE_DIAGNOSTICS_CONTRACT.md
```

The next implementation candidate after that contract is:

```
L-UI parser-level string escape diagnostics implementation plan
```

That future work should define parser enum additions, diagnostic code additions, parse error labels, source-span handling, validation order, messages, hints, AST compatibility, report compatibility, and exact tests before parser diagnostics are extended.

Non-claims

This document and implementation do not implement parser-level escape diagnostics, implement length-carrying AST strings, broaden accepted non-string L-UI grammar, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION_PLAN.md

Source title: Latticra L-UI String-Literal Escape Implementation Plan

Lines: 465 | Words: 1424 | SHA-256: ecb377e87fe1b7ba9c8e9fcfa82e3062970fabf31c6a312a45b4fdedb7aab7c4

Latticra L-UI String-Literal Escape Implementation Plan

Status: implementation planning contract Scope: parser-level validation decision, AST-level decoding helper shape, accepted and rejected escapes, NUL behavior, capacity behavior, span behavior, detailed report behavior, exact tests, and compatibility expectations before string-literal escape decoding code.

Purpose

This document defines the implementation plan for L-UI string-literal escape decoding.

The string-literal escape contract is already merged and guarded. This plan decides where the first decoding implementation belongs, which private helpers should be added, how accepted escapes decode, how rejected escapes fail, how NUL bytes are handled under the current NUL-terminated AST storage model, how source spans remain source-oriented, how detailed reports stay deterministic, and which tests must exist before decoding is considered implemented.

Relationship to previous contracts

This plan depends on:

```
docs/L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser.c
src/l_ui_parser_ast.c
```

Those files remain the source of truth for accepted parser summaries, AST storage, source-backed purpose/text extraction, report escaping, source spans, parse error labels, and no-effect boundaries.

Implementation language decision

String-literal escape decoding should be implemented in C.

Reason:

- the L-UI parser and AST builder are implemented in C;
- AST string storage is a C public API surface;
- decoding must remain bounded, deterministic, and no-effect;
- the current C workflow can validate decoding invariants;
- no dynamic runtime should be required.

Public API decision

No new public function is needed for the first string-literal escape implementation.

Decoding should be integrated into:

```
latticra_l_ui_parse_ast
```

The public AST fields remain:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

The first implementation should not add a new public latticra_l_ui_parse_error_t value for invalid string escapes.

Parser-level validation decision

The first implementation should keep latticra_l_ui_parse_source behavior compatible and should not add parser-level string-escape validation yet.

Decision:

```
latticra_l_ui_parse_source remains a structural parser for the current L-UI fixture shape.
latticra_l_ui_parse_ast performs string-literal escape decoding while constructing AST values.
invalid escapes found during AST decoding classify the AST result as
LATTICRA_L_UI_PARSE_INTERNAL_ERROR.
```

Reason:

- the current parse error enum has no string-escape-specific error;
- adding parser-level escape errors is a broader diagnostic/API change;
- AST extraction already owns source-backed purpose/text value materialization;
- this keeps latticra_l_ui_parse_source accepted fixture summaries stable;
- failed AST construction already has a no-partial-output path.

A later parser diagnostics contract may promote invalid string escape reporting to parser-level errors.

Decoding helper plan

Add private helpers in:

```
src/l_ui_parser_ast.c
```

Proposed helpers:

```
static int is_upper_hex_digit(unsigned char byte);

static unsigned char upper_hex_value(unsigned char byte);

static latticra_status_t decode_l_ui_string_literal_value(
    const char *source,
    size_t start_offset,
    size_t end_offset,
    char *destination,
    size_t destination_len);

static latticra_status_t extract_decoded_quoted_value_after_token(
    const char *source,
    size_t source_len,
    const char *token,
    size_t occurrence,
    char *destination,
    size_t destination_len,
    latticra_l_ui_source_span_t *value_span);
```

Helpers must remain private.

The existing raw extraction helper may be replaced or refactored, but the new behavior must decode the value before writing into the AST destination buffer.

Extraction targets

The first implementation should decode only source-backed AST string values:

```
purpose &quot;...&quot; -&gt; ast.card.purpose
first text &quot;...&quot; -&gt; ast.texts[0].value
second text &quot;...&quot; -&gt; ast.texts[1].value
```

No binding values, rail names, field names, card names, effect values, or boundary values should be decoded in this slice.

Accepted escape behavior

The first implementation should accept only:

```
\\
\&quot;
\n
\r
\t
\xNN
```

where \xNN uses exactly two uppercase hexadecimal digits.

Decoded byte mapping:

```
\\ -&gt; 0x5C
\&quot; -&gt; 0x22
\n -&gt; 0x0A
\r -&gt; 0x0D
\t -&gt; 0x09
\x41 -&gt; 0x41
\x7F -&gt; 0x7F
\x80 -&gt; 0x80
\xFF -&gt; 0xFF
```

Printable non-escaped source bytes should be copied unchanged unless they are NUL bytes.

Rejected escape behavior

The first implementation should reject:

```
\a
\b
\f
\v
\u1234
\U00000000
\x0a
\x0
\x001
\xGG
\
\x
\x0
```

Any backslash followed by an unsupported escape marker must fail instead of being copied literally.

Lowercase hex digits must fail in the first implementation.

NUL byte behavior

Decoded NUL bytes must be rejected in the first implementation.

This includes:

```
\x00
```

and any literal 0x00 source byte that would enter an AST string value.

Reason:

```
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

are currently NUL-terminated fixed-size C strings without explicit decoded lengths.

A later length-carrying AST string storage contract may allow NUL bytes.

Capacity behavior

Decoding must respect destination capacities:

```
LATTICRA_L_UI_AST_PURPOSE_MAX
latticra_l_ui_ast_card_t.purpose
latticra_l_ui_ast_text_t.value
```

If decoded output length is greater than or equal to the destination buffer length, decoding should return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

and clear the destination to an empty C string when possible.

The implementation must not silently truncate decoded values.

AST failure behavior

If string-literal escape decoding fails, latticra_l_ui_parse_ast should return:

```
LATTICRA_STATUS_OK
```

and classify the AST parse result as:

```
ast.parse_result.error = LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

The AST must avoid partial output:

```
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
ast.card.rail_count = 0
ast.card.field_count = 0
ast.card.text_count = 0
```

The failed-parse detailed report path should remain the compact failed-parse report.

Source-span behavior

Decoded values must preserve source-span metadata.

Spans continue to refer to source byte ranges, not decoded output byte ranges.

For text nodes:

```
text.span covers the source value range between quotes
```

The span range includes the source escape bytes. For example, a source value containing \n has a text span over two source bytes, even though the decoded AST value contains one LF byte.

Do not add a new public purpose_span field in the first implementation.

Detailed report relationship

Detailed reports should continue to render:

```
purpose=&lt;decoded-purpose-as-current-C-string&gt;
purpose_escaped=&lt;report-safe-purpose&gt;
value=&lt;decoded-text-as-current-C-string&gt;
value_escaped=&lt;report-safe-text&gt;
```

For decoded control bytes, quote bytes, backslashes, DEL, and non-ASCII bytes, the stable assertion target should be:

```
purpose_escaped=&lt;report-safe-purpose&gt;
value_escaped=&lt;report-safe-text&gt;
```

The existing escape_report_string helper must remain the report-safe representation layer.

Successful parse rule

Decoding must run only after:

```
latticra_l_ui_parse_source
```

returns LATTICRA_STATUS_OK and parse_result.error == LATTICRA_L_UI_PARSE_OK.

If parsing fails structurally, decoding must not run.

Failed parse compatibility

Existing failed parse behavior must remain unchanged:

```
ast.parse_result.error = parser error
ast.rail_count = 0
ast.field_count = 0
ast.text_count = 0
failed-parse detailed report only
```

String decoding must not hide or replace existing structural parser errors.

Compatibility expectations

String-literal escape decoding must not change:

```
latticra_l_ui_parse_source existing accepted fixture behavior
latticra_l_ui_ast_report existing required fields
latticra_l_ui_ast_detailed_report existing required fields
latticra_l_ui_diagnostic_report existing required fields
parser error labels
existing escaped report field semantics
existing accepted fixture summary counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Fixtures without source escapes must produce the same AST values they produced before this implementation.

No-effect preservation

String-literal escape decoding is metadata-only and must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
string_escape_decodes_backslash
string_escape_decodes_quote
string_escape_decodes_newline
string_escape_decodes_carriage_return
string_escape_decodes_tab
string_escape_decodes_uppercase_hex
string_escape_decodes_high_byte_hex
string_escape_rejects_lowercase_hex
string_escape_rejects_short_hex
string_escape_rejects_invalid_hex
string_escape_rejects_unknown_escape
string_escape_rejects_unterminated_escape
string_escape_rejects_decoded_nul_until_length_storage_exists
string_escape_rejects_literal_nul_until_length_storage_exists
string_escape_rejects_oversized_decoded_output
string_escape_preserves_source_spans
string_escape_updates_detailed_report_escaped_fields
```

```
string_escape_preserves_no_effect_flags
string_escape_does_not_change_parse_source_summary
string_escape_does_not_change_failed_parse_report
string_escape_is_deterministic
```

Invalid escape tests should assert AST classification through:

```
LATTICRA_L_UI_PARSE_INTERNAL_ERROR
```

until parser-level string escape diagnostics are designed.

Test file plan

Add:

```
tests/l_ui_string_literal_escape_invariants.c
scripts/test-l-ui-string-literal-escape.sh
```

Wire into:

```
.github/workflows/c.yml
```

Documentation requirement

The implementation PR should update:

```
README.md
docs/FOUNDATION_INDEX.md
docs/L_UI_STRING_LITERAL_ESCAPE_CONTRACT.md
docs/L_UI_AST_SOURCE_BACKED_TEXT_IMPLEMENTATION.md
docs/L_UI_AST_ESCAPED_STRING_REPORT_IMPLEMENTATION.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
```

and add:

```
docs/L_UI_STRING_LITERAL_ESCAPE_IMPLEMENTATION.md
```

Forbidden implementation behavior

The string-literal escape implementation must not:

- add file I/O to parser or AST code;
- write files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- mutate state lattice;
- perform live movement;
- run L-UI behavior;
- render an interactive UI;
- treat text nodes as commands;
- treat bindings as executable references;
- silently truncate decoded values;
- emit raw control bytes in escaped report fields;
- decode bindings, field names, rail names, card names, effect values, or boundary values in this slice;
- broaden accepted grammar beyond the string-literal escape contract;

- add parser-level escape diagnostics without a parser diagnostics contract;
- add length-carrying AST string storage without a storage contract.

Current validation command

This plan is guarded by:

```
sh scripts/test-l-ui-string-literal-escape-implementation-plan.sh
```

The guard is static. It does not implement string-literal escape decoding.

Implementation gate

String-literal escape decoding implementation code may be added only after this plan is merged.

Non-claims

This document does not implement string-literal escape decoding, implement parser-level escape diagnostics, implement length-carrying AST strings, broaden accepted non-string L-UI grammar, implement Unicode display behavior, add L-UI rendering, add command behavior, add Nucleus task handling, add live movement, add origin mutation, add recovery behavior, add server interaction, add self-update, add hardware support, add boot readiness, claim security isolation, claim sandboxing, or claim operating-system completeness.

docs/LANGUAGE_NAMING_POLICY.md

Source title: Latticra Language Naming Policy

Lines: 185 | Words: 680 | SHA-256: 48d2decf5506ecf865fc62713714ffe02f1f43cb006681e573f642b39a51ddb3

Latticra Language Naming Policy

Status: language naming policy Scope: public naming and file-extension policy for the Latticra native language family.

Purpose

This document corrects the early shorthand that called the native Latticra programming language L.

Plain L is too collision-prone for a public language name or file extension. Latticra should use project-specific language names and extensions before language implementation begins.

This policy does not implement a programming language, parser, compiler, runtime, package format, or execution model.

Collision rule

Latticra must not claim plain:

```
L
.l
```

as the public native language name or canonical file extension.

Reason:

- single-letter language names are ambiguous;
- existing language literature already uses L-like names and variants;
- .l is too short and too likely to collide with existing tools;

- Latticra should avoid avoidable naming confusion before syntax stabilizes.

Public native language name

The public native language name should be:

```
Latticra Language
```

The approved short name should be:

```
Lat
```

Lat is a project-specific shorthand for Latticra Language.

File extension policy

The canonical source file extension for the native Latticra Language should be:

```
.lat
```

Reserved related extensions:

```
.lat
.lui
.lir
```

Meaning:

```
| Extension | Meaning |
| --- | --- |
| `.lat` | Latticra Language source, including future Lat-Core and Lat-Orch forms. |
| `.lui` | Latticra UI declaration source. |
| `.lir` | Latticra Intermediate Representation, internal or generated unless later promoted. |
```

The project should not use .l for Latticra source.

Dialect naming policy

Rename the intended language-family labels as follows:

```
| Old shorthand | New public label | Role |
| --- | --- | --- |
| `L` | `Lat` / `Latticra Language` | Native Latticra language family. |
| `L-Core` | `Lat-Core` | State, policy, assertion, and transition declarations. |
| `L-Orch` | `Lat-Orch` | Nucleus orchestration planning and effect-gated task descriptions. |
| `L-UI` | `L-UI` | Operator interface declaration language; existing `.lui` files remain valid. |
| `LIR` | `LIR` | Latticra Intermediate Representation. |
```

L-UI may keep its existing name because it is already a compound project-specific label and has an existing .lui fixture path.

LIR may remain as an internal representation name because it expands to Latticra Intermediate Representation, but .lir should be treated as internal or generated until a later contract promotes it.

Documentation rule

Future docs should write:

```
Lat / Latticra Language
```

instead of:

```
L
```

when referring to the native Latticra language.

Historical references may say:

```
formerly documented as L
```

only when explaining the migration.

Syntax example rule

Syntax examples should be marked as:

```
Lat example syntax only
```

not:

```
L syntax
```

until a grammar contract exists.

Implementation boundary

This policy does not change:

```
L-UI parser behavior
.lui fixtures
LIR shape
AST behavior
string-literal escape behavior
parser diagnostics
state lattice behavior
Nucleus preview behavior
```

Future implementation gate

Before implementing native Latticra Language parsing, a future contract must define:

1. canonical language name;
2. canonical file extension;
3. dialect boundaries;
4. grammar scope;
5. parser input ownership;
6. effect boundary;
7. non-claims;
8. exact fixture paths;
9. exact tests.

That future work should use the names and extensions from this policy.

Forbidden behavior

Future language work must not:

- claim plain L as the public language name;
- claim .l as the canonical source extension;
- imply compatibility with unrelated L-like languages;
- imply a completed compiler or runtime before implementation exists;
- treat .lat files as executable before an execution contract exists;
- treat .lui files as commands;
- broaden L-UI behavior through the language naming policy.

Current validation command

This policy is guarded by:

```
sh scripts/test-language-naming-policy.sh
```

The guard is static. It does not implement a language.

Non-claims

This document does not implement Lat, Lat-Core, Lat-Orch, L-UI rendering, LIR lowering, a parser, compiler, runtime, package manager, command behavior, Nucleus task execution, live movement, state mutation, server interaction, recovery behavior, hardware behavior, boot readiness, sandboxing, or operating-system completeness.

docs/LANGUAGE_STRATEGY.md

Source title: Latticra Language Strategy

Lines: 249 | Words: 953 | SHA-256: c5530db3246e74c433900c154f3200b332aa658e1bee22fc6ec58097e82ad4a6

Latticra Language Strategy

Status: active language strategy Scope: C, constrained C++, Lat, L-UI, LIR, and implementation ordering.

Purpose

Latticra needs a language strategy from the beginning.

The real system should not grow as an accidental collection of shell scripts, raw C files, ad-hoc config formats, and UI strings. Latticra should define which languages own which parts of the architecture before implementation expands.

This document avoids using plain L as the public native language name. The language naming policy is defined in LANGUAGE_NAMING_POLICY.md.

The active C/C++ foundation direction is defined in C_CPP_FOUNDATION_DIRECTION.md.

Direction summary

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Latticra should use a constrained C/C++ foundation for a security-conscious system.

This does not mean unrestricted C++.

Language roles

```
| Language | Role |
| --- | --- |
| C | Secure substrate, boot paths, ABI boundaries, platform shims, fixed-size data structures,
and freestanding-adjacent boundaries. |
| C++ | Governed authority layer for policy, validators, effect gates, audit logic, bounded
orchestration structures, and safe wrappers over C substrate APIs. |
| Lat / Latticra Language | Native contract/declaration language family for state, policy,
orchestration, assertions, and controlled system expression. |
| L-UI | Terminal/operator UI declaration language or dialect. |
| LIR | Latticra Intermediate Representation for validating and lowering Latticra
language-family documents. |
```

C secure substrate

C owns the lowest software boundary.

Primary C responsibilities:

- boot paths;
- ABI boundaries;
- platform shims;
- fixed-size data structures;

- freestanding-adjacent boundaries;
- minimal substrate behavior;
- portable test fixtures.

C must remain disciplined:

- no unsafe string APIs;
- no unchecked pointer mutation;
- no undefined behavior tolerated;
- no implicit effects without explicit gates;
- no global mutable state unless named in a contract;
- all transitions return status codes;
- all source/input behavior remains bounded and testable.

C++ governed authority layer

C++ is allowed above the C substrate as a governed authority layer.

Primary C++ responsibilities:

- policy logic;
- validators;
- effect gates;
- audit logic;
- bounded orchestration structures;
- operator-visible reports;
- higher-level state coordination;
- safe wrappers around C substrate APIs.

C++ must be constrained.

It is not:

```
unrestricted C++
exception-heavy C++
reflection-heavy C++
template metaprogramming as architecture
hidden allocation by default
implicit authority
unchecked host execution
```

Lat / Latticra Language

The native language is planned as:

```
Lat
```

Full descriptive name:

```
Latticra Language
```

Canonical source extension:

```
.lat
```

Plain L and .l are not the public language name or canonical file extension.

Lat should begin as a controlled system language, not a general-purpose programming language.

Early Lat responsibilities:

- declare lattice state;
- declare policies;
- declare effect requirements;
- declare transition rules;
- declare assertions;
- describe supervisor orchestration plans;
- describe operator-visible state reports.

Lat should not initially provide a general-purpose runtime, unrestricted system access, production runtime claims, or broad platform claims.

Lat-Core

Lat-Core is the first native-language target.

Purpose:

```
state + policy + assertion + transition declaration
```

Lat example syntax only:

```
state RootCell {
  origin = &quot;0/0&quot;
  route = &quot;ROOT&quot;
  axis = &quot;ROOT&quot;
  host_effect = none
  external_effect = none
}

transition move_right from RootCell {
  require lock == open
  require risk != high
  effect host = none
  effect external = none
}
```

This is example syntax only. It is not implemented yet.

Lat-Orch

Lat-Orch is the future orchestration dialect for Nucleus.

Purpose:

```
supervisor task planning + effect gates + recovery visibility + update staging
```

Lat-Orch must be explicit about:

- requested effect;
- required gate;
- rollback behavior;
- failure behavior;
- operator confirmation;
- server interaction;
- audit trail.

L-UI

L-UI is the terminal/operator interface language or dialect.

Purpose:

```
operator rails + state cards + reports + safe command surfaces
```

L-UI should be declarative, not a hidden imperative UI runtime.

The .lui extension remains valid for L-UI fixtures and source documents.

LIR

LIR is the Latticra Intermediate Representation.

Purpose:

- normalize Lat and L-UI documents;
- validate names and effects;
- preserve source spans;
- reject forbidden behavior;
- support deterministic testing;
- provide stable input for C and constrained C++ implementations.

LIR should be boring, explicit, serializable, and test-friendly.

The .lir extension is reserved for Latticra Intermediate Representation and should remain internal or generated until a future contract promotes it.

Rust relationship

Rust is not the current public foundation direction for Latticra.

Rust may appear only as optional external tooling or historical comparison if separately justified, but it is not the default public implementation lane and should not be presented as a primary foundation layer.

Extension policy

The reserved Latticra language-family extensions are:

```
.lat
.lui
.lir
```

Meaning:

```
| Extension | Meaning |
| --- | --- |
| `.lat` | Latticra Language source, including future Lat-Core and Lat-Orch forms. |
| `.lui` | Latticra UI declaration source. |
| `.lir` | Latticra Intermediate Representation, internal or generated unless later promoted. |
```

The project should not use .l as the canonical source extension.

Implementation order

1. Define contracts and docs.
2. Add C substrate fixtures and invariant tests.
3. Add C parser, AST, semantic validation, and LIR metadata foundations.
4. Define constrained C++ policy/validator/effect-gate layers after C substrate contracts stabilize.
5. Define Lat-Core grammar and implementation plans before parser code.
6. Define Lat-to-LIR lowering only after parser and semantic contracts exist.

7. Add execution only after explicit runtime/effect/security contracts.

Non-claims

This document does not claim that Lat, Lat-Core, Lat-Orch, L-UI, LIR, C++ authority layers, or any runtime are fully implemented.

It defines the intended language architecture before broader native-language implementation begins.

docs/LAT_LANGUAGE_FOUNDATION_ANALYSIS.md

Source title: Latticra Lat Language Foundation Analysis

Lines: 264 | Words: 1028 | SHA-256: 805d045a07e767659da6acc1200821afd9439607485fe0da0d924cfed9b7551e

Latticra Lat Language Foundation Analysis

Status: foundation analysis proposal Scope: deeper Lat / Latticra Language foundation after the initial bounded grammar parser.

Purpose

Lat should mature as a contract and declaration language, not as an accidental executable scripting surface. The existing grammar parser is a necessary first slice, but a stronger foundation requires a semantic layer that decides what a parsed Lat document means, what names are valid, what relationships are permitted, and what remains explicitly non-operational.

This analysis defines the next foundation layer:

```
Lat source -&gt; bounded parser -&gt; Lat AST metadata -&gt; semantic validation -&gt;
normalized Lat model
-&gt; LIR metadata lowering
```

The parser, semantic validation, normalized model, and Lat-to-LIR metadata lowering phases now exist in bounded no-effect form. This analysis remains the foundation record for the semantic and normalization direction.

Current foundation summary

The current Lat parser already establishes important constraints:

- source size is bounded;
- names, values, declarations, clauses, and reports use fixed-size storage;
- the parser preserves source spans;
- declarations are metadata only;
- no execution, mutation, server, recovery, or hardware behavior is allowed;
- Lat uses the public name Lat / Latticra Language and .lat, not plain L or .l.

That is the right parser posture. The missing foundation is semantic meaning.

Main gap

A grammar parser can say:

```
this document has a transition declaration
```

It cannot yet say:

```
the transition source state exists
the effect target is allowed
the declaration name is unique
the field name belongs to the Lat-Core state vocabulary
```

```
the document still preserves the no-effect boundary
```

Without these semantic checks, Lat remains syntactically bounded but architecturally loose.

Foundation principle

The next layer should be semantic validation, not execution.

Lat semantic validation should answer:

1. Is the parsed module internally coherent?
2. Are declaration names unique and resolvable?
3. Are transition references valid?
4. Are state fields from the known state-lattice vocabulary?
5. Are effect targets known and denied by default unless they are none?
6. Are policy and assertion clauses restricted to requirement/assurance forms?
7. Are reports deterministic, bounded, source-aware, and operator-readable?

It should not answer:

- how to execute a transition;
- how to mutate state;
- how to lower to LIR;
- how to render L-UI;
- how to call Nucleus;
- how to authorize real host, network, recovery, or hardware effects.

Proposed Lat foundation phases

Phase 0: Naming and role policy

Lat is the approved short name. .lat is the canonical extension. Plain L and .l remain forbidden.

Phase 1: Lexical and grammar parser

Current parser role:

```
source bytes -&gt; bounded AST metadata + parse report
```

It owns source spans, syntax errors, declaration records, clause records, and no-effect parse reports.

Phase 2: Semantic validation

Proposed role:

```
parse result -&gt; semantic diagnostics + semantic report
```

It owns:

- duplicate declaration detection;
- transition source resolution;
- declaration-specific clause rules;
- state-field vocabulary checking;
- effect-target checking;

- first default-deny effect validation;
- stable diagnostic labels;
- deterministic semantic reports.

Phase 3: Lat model normalization

Current role:

```
parse result + semantic result -&gt; normalized Lat module model
```

The normalized model carries explicit declaration and clause index tables for states, policies, transitions, assertions, and effect declarations. It remains no-effect and does not read source bytes.

Phase 4: Lat-to-LIR lowering

Current role:

```
semantic Lat module metadata -&gt; bounded LIR shape
```

The current implementation remains metadata-only and no-effect. It does not execute Lat or LIR.

Phase 5: Effect-gated runtime interpretation

Future role:

```
validated Lat / LIR -&gt; runtime request classification -&gt; effect gate -&gt; denied or
authorized
behavior
```

This must wait for explicit runtime, threat-model, effect-gate, audit, and authority-layer contracts.

Semantic validation rules proposed for this slice

Module rule

The parser must have succeeded before semantic validation runs.

```
parse.error == ok
```

If parsing did not succeed, semantic validation returns a semantic parse_not_ok diagnostic and does not reinterpret the source.

Declaration identity rule

Declaration names are globally unique within a module for this slice.

Reason: global uniqueness prevents ambiguity before namespacing, imports, traits, modules, or overloads exist.

State declaration rule

A state declaration may contain only field assignments.

Allowed fields:

```
origin
route
axis
path
breadcrumb
trace
safe_portal
rollback
health
risk
lock
dark_phase
```

```
host_effect
external_effect
```

host_effect and external_effect must use known effect labels, and this first semantic slice accepts only none as an allowed declared effect value.

Policy declaration rule

A policy declaration may contain only:

```
require left operator right
ensure left operator right
```

Field assignments and effect assignments are rejected in policies for this slice.

Transition declaration rule

A transition declaration must name an existing source state:

```
transition MoveRight from RootCell { ... }
```

RootCell must resolve to a declared state.

A transition may contain requirement clauses and explicit no-effect metadata clauses:

```
require lock == &quot;open&quot;
effect host = none
effect external = none
```

This first semantic slice rejects non-none declared effects.

Assertion declaration rule

An assertion declaration may contain only require or ensure clauses.

Assertions are semantic checks, not runtime actions.

Effect declaration rule

An effect declaration may contain only known effect targets:

```
host
external
network
hardware
boot
recovery
local
```

The value must be a known Lat effect label. This first slice accepts only none as an allowed declared value.

Diagnostic model

Semantic diagnostics should be stable, bounded, source-span-aware, and deterministic.

Each diagnostic records:

```
error label
source span
declaration index
clause index
name
detail
```

The semantic result records the first error as the summary error while retaining a bounded diagnostic list.

Evidence level

This proposal targets an L2 implementation slice:

```
contract + implementation plan + C implementation + invariant tests + deterministic report
```

It does not claim runtime capability.

Non-claims

This analysis does not implement Lat execution, Lat compilation, Lat interpretation, LIR lowering, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md

Source title: Latticra Lat Language Grammar Contract

Lines: 548 | Words: 1346 | SHA-256: 2a9632b7bb77464c720d0c9e51ec646f66f2a9f51fae54ce7b686118a511ade6

Latticra Lat Language Grammar Contract

Status: Lat language grammar contract Scope: first grammar contract for Lat / Latticra Language before parser implementation, LIR lowering, execution, runtime behavior, or operating-system behavior.

Purpose

This document defines the first grammar contract for:

```
Lat / Latticra Language
```

Lat is the approved short name for the Latticra Language. The canonical source extension is:

```
.lat
```

This contract defines the first syntax shape for Lat-Core declarations only. It does not implement a parser, compiler, interpreter, runtime, package format, command behavior, Nucleus task execution, LIR lowering, L-UI rendering, or operating-system behavior.

The implementation plan is recorded in:

```
LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
```

Naming boundary

Lat must follow the language naming policy:

```
docs/LANGUAGE_NAMING_POLICY.md
```

The public language name is:

```
Lat / Latticra Language
```

The canonical source extension is:

```
.lat
```

Forbidden public claims:

```
L
.l
```

Plain L is not the public language name, and .l is not the canonical source extension.

Relationship to previous work

This contract depends on:

```
docs/LANGUAGE_NAMING_POLICY.md
docs/LANGUAGE_STRATEGY.md
docs/LIR_SHAPE_CONTRACT.md
docs/LIR_SHAPE_IMPLEMENTATION_PLAN.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/EFFECT_GATES.md
```

```
docs/STATE_LATTICE.md
docs/TRI_PLANE_TRANSITION.md
docs/NUCLEUS_PREVIEW.md
include/latticra/lir.h
include/latticra/state_lattice.h
```

Those files remain the source of truth for public naming, language-family roles, LIR shape, effect vocabulary, state-lattice fixtures, no-effect preview behavior, and Nucleus preview boundaries.

First grammar target

The first grammar target is:

```
Lat-Core
```

Lat-Core is the state, policy, assertion, and transition declaration subset of Lat.

Lat-Orch remains future work and must not be implemented or implied by this contract.

Source unit

A Lat source unit should represent one module-like document:

```
lat module &lt;ModuleName&gt; {
  ... declarations ...
}
```

The first grammar should require an explicit module wrapper.

Initial declaration kinds

The first Lat-Core grammar should define these declaration kinds:

```
state
policy
transition
assertion
effect
```

These declarations are metadata only until separate parser, semantic, lowering, and execution contracts exist.

Lexical grammar

The first lexical grammar should include:

```
identifier
keyword
string_literal
integer_literal
boolean_literal
effect_literal
operator
punctuation
comment
whitespace
```

Identifier policy:

```
[A-Za-z_][A-Za-z0-9_]*
```

Reserved keywords:

```
lat
module
state
policy
transition
assertion
effect
from
require
ensure
where
let
true
```

```
false
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
```

Future implementation may add keywords only through a separate contract or implementation plan.

Comment policy

The first grammar should allow line comments only:

```
// comment text
```

Block comments are out of scope until a later contract.

String literal policy

Lat string literals should use quoted strings:

```
&quot;text&quot;
```

String escapes should align with the L-UI string-literal escape policy unless a future Lat-specific contract changes it:

```
\\
\&quot;
\n
\r
\t
\xHH
```

Escaped decoded NUL through \x00 may be represented only with explicit length-carrying storage in a future implementation.

Literal source-buffer NUL bytes remain forbidden.

Effect literals

Initial effect literals:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
```

The first parser may accept effect declarations as metadata, but it must not perform the effect.

State declaration grammar

Lat example syntax only:

```
state RootCell {
  origin = &quot;0/0&quot;
  route = &quot;ROOT&quot;
  axis = &quot;ROOT&quot;
  path = &quot;/&quot;
  host_effect = none
  external_effect = none
}
```

State declarations should allow fields compatible with the current state lattice vocabulary:

```
origin
route
axis
path
breadcrumb
trace
safe_portal
rollback
health
risk
lock
dark_phase
host_effect
external_effect
```

This grammar does not mutate runtime state.

Policy declaration grammar

Lat example syntax only:

```
policy SafePreview {
  require risk != &quot;high&quot;
  require lock == &quot;open&quot;
  ensure host_effect == none
  ensure external_effect == none
}
```

Policy declarations should describe requirements and assertions as metadata only.

Transition declaration grammar

Lat example syntax only:

```
transition MoveRight from RootCell {
  require lock == &quot;open&quot;
  require risk != &quot;high&quot;
  effect host = none
  effect external = none
}
```

Transition declarations should require:

```
transition name
source state name
require clauses
effect clauses
```

Transition declarations must not execute movement, mutate state, or call Nucleus task execution.

Assertion declaration grammar

Lat example syntax only:

```
assertion RootCellIsSafe {
  require health == &quot;ok&quot;
  require host_effect == none
  require external_effect == none
}
```

Assertions are metadata until a future validator implements them.

Effect declaration grammar

Lat example syntax only:

```
effect PreviewOnly {
  host = none
  external = none
  network = none
  hardware = none
}
```

Effect declarations describe intended effect boundaries only. They do not grant capability.

Expression grammar

The first expression grammar should be deliberately small:

```
identifier
string_literal
integer_literal
boolean_literal
effect_literal
comparison_expression
logical_and_expression
logical_or_expression
parenthesized_expression
```

Initial comparison operators:

```
==
!=
&lt;
&lt;=
&gt;
&gt;=
```

Initial logical operators:

```
&amp;&amp;
||
```

No function calls, loops, arbitrary arithmetic, imports, macros, generic types, allocation, host calls, network calls, or hardware calls are in scope.

Module report model

A future Lat grammar parser should produce deterministic bounded reports.

Suggested report fields:

```
LAT GRAMMAR REPORT
status=&lt;status&gt;
module=&lt;module-name&gt;
state_count=&lt;count&gt;
policy_count=&lt;count&gt;
transition_count=&lt;count&gt;
assertion_count=&lt;count&gt;
effect_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

The exact implementation report is defined in:

```
LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
```

Source-span behavior

Future Lat parser and AST work should preserve source spans for:

```
module declarations
state declarations
policy declarations
transition declarations
assertion declarations
effect declarations
field assignments
require clauses
ensure clauses
effect clauses
string literals
identifier references
```

Lat source-span behavior must not invent byte offsets.

LIR relationship

Lat grammar work must not lower to LIR until a separate Lat-to-LIR lowering contract exists.

The first relationship is planning-only:

```
Lat source -&gt; future Lat parser -&gt; future Lat AST -&gt; future Lat semantic validation -
&gt; future
LIR lowering
```

The current LIR shape implementation lowers semantically valid L-UI ASTs only.

No-effect rule

Lat grammar work is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Lat grammar work must not execute declarations, mutate state, render UI, lower to LIR, call Nucleus task code, write files, read files, open network connections, call server code, call update code, call recovery code, or touch hardware.

Compatibility expectations

Lat grammar work must not change:

```
language naming policy
.lat canonical extension
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
state lattice behavior
Nucleus preview behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
no-effect flags
```

Future implementation gate

Lat parser implementation must not begin until a separate implementation plan defines:

1. public API shape;
2. parser result structs;
3. AST structs;
4. capacity constants;
5. error enum labels;
6. diagnostic/report format;
7. accepted fixture paths;
8. parser ownership rules;
9. string literal handling;
10. source-span mapping;
11. exact tests;
12. compatibility expectations;
13. non-claims.

That next planning slice is:

```
Lat language grammar implementation plan
LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
```

Future test list

A future implementation plan should include tests for:

```
lat_grammar_accepts_minimal_module
lat_grammar_accepts_state_declaration
lat_grammar_accepts_policy_declaration
lat_grammar_accepts_transition_declaration
lat_grammar_accepts_assertion_declaration
lat_grammar_accepts_effect_declaration
lat_grammar_rejects_plain_l_extension_claim
lat_grammar_rejects_unknown_keyword
lat_grammar_rejects_unterminated_string
lat_grammar_rejects_invalid_escape
lat_grammar_rejects_literal_source_nul
lat_grammar_reports_source_spans
lat_grammar_preserves_no_effect_flags
lat_grammar_report_is_deterministic
lat_grammar_does_not_lower_to_lir
lat_grammar_does_not_execute_declarations
lat_grammar_is_deterministic
```

The implementation-plan test list is now recorded in:

```
LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
```

Forbidden behavior

Lat grammar work must not:

- claim plain L as the public language name;
- claim .l as the canonical source extension;
- implement a parser before an implementation plan;
- execute declarations;
- mutate state;
- lower to LIR;
- render L-UI;
- call Nucleus task execution;
- evaluate host state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- broaden L-UI behavior;
- weaken semantic validation;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;

- imply a compiler, interpreter, runtime, package manager, sandbox, or operating-system surface.

Current validation command

This contract is guarded by:

```
sh scripts/test-lat-language-grammar-contract.sh
```

The guard is static. It does not implement Lat.

Non-claims

This document does not implement Lat, Lat-Core, Lat-Orch, a parser, compiler, interpreter, runtime, package manager, LIR lowering, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md

Source title: Latticra Lat Language Grammar Implementation

Lines: 315 | Words: 765 | SHA-256: bb704eb7a593f116e924e77cf4a9215c4eb0b00830a45f6619ec33a4b4dd1395

Latticra Lat Language Grammar Implementation

Status: implementation with line-comment metadata and unsupported block-comment rejection
Scope: bounded no-effect Lat / Latticra Language grammar parser, parse result model, AST metadata, line-comment metadata, unsupported block-comment rejection, report surface, fixture, and invariants.

Purpose

This implementation adds the first bounded parser for the Lat / Latticra Language grammar.

The parser accepts the first Lat-Core declaration grammar shape and records module, declaration, clause, effect, source-span, and no-effect metadata.

The parser also records deterministic line-comment metadata for audit visibility. Line comments are skipped by the grammar, may contain otherwise forbidden behavior marker words, and do not change no-effect flags or clause/operator behavior.

Block comments are rejected with a deterministic unsupported_block_comment parse error outside strings and line comments. / and / text inside string literals remains ordinary string content.

This implementation does not execute Lat, compile Lat, interpret Lat, lower Lat to LIR, render L-UI, call Nucleus task behavior, read files, write files, open network connections, mutate state, or touch hardware.

Implementation files

```
include/latticra/lat_parser.h
src/lat_parser.c
tests/lat_language_grammar_invariants.c
scripts/test-lat-language-grammar.sh
fixtures/lat/minimal_module.lat
.github/workflows/c.yml
```

Related active files:

```
docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
docs/LANGUAGE_NAMING_POLICY.md
docs/LANGUAGE_STRATEGY.md
docs/C_CPP_FOUNDATION_DIRECTION.md
```

```
docs/LIR_SHAPE_IMPLEMENTATION.md
```

Public API

The public API adds:

```
latticra_lat_parse_error_t
latticra_lat_declaration_kind_t
latticra_lat_effect_t
latticra_lat_source_span_t
latticra_lat_parse_result_t
latticra_lat_ast_module_t
latticra_lat_ast_declaration_t
latticra_lat_ast_clause_t
latticra_lat_parse_error_label
latticra_lat_declaration_kind_label
latticra_lat_effect_label
latticra_lat_parse_source
latticra_lat_parse_report
```

Capacity constants

The first Lat parser implementation uses exact bounded constants:

```
LATTICRA_LAT_SOURCE_MAX 65536u
LATTICRA_LAT_NAME_MAX 64u
LATTICRA_LAT_VALUE_MAX 128u
LATTICRA_LAT_DECLARATION_MAX 64u
LATTICRA_LAT_CLAUSE_MAX 128u
LATTICRA_LAT_REPORT_MAX 4096u
```

Grammar accepted in this slice

The implementation accepts a bounded module wrapper:

```
lat module &lt;ModuleName&gt; {
  ... declarations ...
}
```

Accepted declaration kinds:

```
state
policy
transition
assertion
effect
```

Accepted clause forms:

```
field = value
require left operator right
ensure left operator right
effect target = effect_value
```

Transition declarations preserve the from <StateName> source state in source_name.

Fixture

The first repository Lat fixture is:

```
fixtures/lat/minimal_module.lat
```

It contains a minimal RootModule with a RootCell state declaration.

Error labels

Stable parse error labels include:

```
ok
null_argument
empty_source
source_too_large
unsupported_extension_claim
missing_module
invalid_module_name
unbalanced_brace
unknown_declaration
invalid_declaration_name
```

```
unterminated_string
invalid_string_escape
invalid_hex_escape
literal_nul_in_string
capacity_exceeded
forbidden_behavior_marker
unsupported_block_comment
internal_error
```

No-effect boundary

Lat grammar parsing is metadata-only.

It preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Forbidden behavior markers

The parser rejects behavior markers before execution contracts exist:

```
exec
spawn
syscall
socket
open_file
write_file
hardware_write
```

Markers inside line comments are ignored by the marker scan.

Source-span behavior

The parser records source-span metadata for module, declaration, and clause records.

Source spans are byte-offset and line/column based.

The parser does not invent byte offsets.

Report format

latticra_lat_parse_report emits a deterministic bounded report beginning with:

```
LAT GRAMMAR REPORT
```

and including:

```
status
error
module
declaration_count
state_count
policy_count
transition_count
assertion_count
effect_count
clause_count
comment_count
first_comment_start_offset
first_comment_end_offset
first_comment_start_line
first_comment_start_column
first_comment_end_line
first_comment_end_column
first_declaration_index
first_declaration_kind
first_declaration_name
first_declaration_source
first_declaration_first_clause_index
first_declaration_clause_count
first_clause_index
first_clause_keyword
first_clause_left
```

```
first_clause_operator
first_clause_right
first_clause_effect
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
span_start_offset
span_end_offset
span_start_line
span_start_column
span_end_line
span_end_column
```

The first declaration and first clause fields are copied from the parsed AST only when parsing succeeds. They are report metadata only; clause operators are not evaluated.

Small output buffers return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

and clear the output buffer.

Test command

Run:

```
sh scripts/test-lat-language-grammar.sh
```

The main C workflow runs this check after the Lat grammar implementation-plan guard and before the state lattice tests.

Required invariants

The Lat grammar tests verify:

```
lat_grammar_accepts_minimal_module
lat_grammar_accepts_state_declaration
lat_grammar_accepts_policy_declaration
lat_grammar_accepts_transition_declaration
lat_grammar_accepts_assertion_declaration
lat_grammar_accepts_effect_declaration
lat_grammar_rejects_plain_l_extension_claim
lat_grammar_rejects_unknown_keyword
lat_grammar_rejects_unterminated_string
lat_grammar_rejects_invalid_escape
lat_grammar_rejects_literal_source_nul
lat_grammar_reports_source_spans
lat_grammar_preserves_no_effect_flags
lat_grammar_report_is_deterministic
lat_grammar_report_rejects_small_buffer
lat_grammar_error_labels_are_stable
lat_grammar_kind_labels_are_stable
lat_grammar_does_not_lower_to_lir
lat_grammar_does_not_execute_declarations
lat_grammar_is_deterministic
```

Compatibility

This implementation does not change:

```
language naming policy
.lat canonical extension
C/C++ foundation direction
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
state lattice behavior
Nucleus preview behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
no-effect flags
```

Current evidence level

This implementation is an L2 tested grammar parser foundation for bounded Lat-Core metadata.

It is not Lat execution, Lat compilation, Lat interpretation, LIR lowering, L-UI rendering, command behavior, Nucleus task execution, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

Next implementation step

The next implementation candidate is:

```
Constrained C++ authority layer contract
```

That future work should define the governed C++ layer before any C++ policy, validator, effect-gate, or audit implementation.

Non-claims

This document and implementation do not implement Lat execution, Lat compilation, Lat interpretation, LIR lowering, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md

Source title: Latticra Lat Language Grammar Implementation Plan

Lines: 579 | Words: 1338 | SHA-256: c38e373621fdeb1130b455252dbd81be379263fefc2155c3743ab14fec7f586f

Latticra Lat Language Grammar Implementation Plan

Status: implementation planning contract Scope: exact public API, parser result structs, AST structs, capacities, error labels, reports, fixture paths, parser ownership rules, string handling, source-span mapping, tests, compatibility expectations, and non-claims before Lat parser code.

Purpose

This document defines the implementation plan for the first Lat / Latticra Language grammar parser.

The Lat grammar contract is already merged and guarded. This plan names the exact public API, result shape, AST metadata, capacity constants, error labels, report format, fixture paths, ownership rules, string literal handling, source-span mapping, tests, and compatibility expectations required before parser code is added.

This document does not implement Lat parsing.

Relationship to previous work

This plan depends on:

```
docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md
docs/LANGUAGE_NAMING_POLICY.md
docs/LANGUAGE_STRATEGY.md
docs/C_CPP_FOUNDATION_DIRECTION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/EFFECT_GATES.md
docs/STATE_LATTICE.md
include/latticra/lir.h
include/latticra/state_lattice.h
```

Those files remain the source of truth for naming, .lat extension policy, C/C++ foundation direction, LIR shape, effect vocabulary, and state-lattice vocabulary.

Implementation language decision

The first Lat grammar parser should be implemented in C.

Reason:

- current parser, AST, semantic validation, and LIR foundations are C;
- C is the secure substrate;
- the first Lat parser should be bounded, deterministic, and no-effect;
- tests and runners use C and POSIX shell;
- constrained C++ policy layers are planned separately and should not precede a stable C parser substrate.

Implementation files

The implementation PR should modify or add:

```
include/latticra/lat_parser.h
src/lat_parser.c
tests/lat_language_grammar_invariants.c
scripts/test-lat-language-grammar.sh
.github/workflows/c.yml
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
```

The implementation should not add C++ infrastructure, LIR lowering, L-UI rendering, Lat execution, runtime behavior, file I/O, network I/O, state mutation, recovery behavior, update behavior, or hardware behavior.

Public API shape

Add public API names:

```
latticra_lat_parse_error_t
latticra_lat_declaration_kind_t
latticra_lat_effect_t
latticra_lat_source_span_t
latticra_lat_parse_result_t
latticra_lat_ast_module_t
latticra_lat_ast_declaration_t
latticra_lat_ast_clause_t
latticra_lat_parse_error_label
latticra_lat_declaration_kind_label
latticra_lat_effect_label
latticra_lat_parse_source
latticra_lat_parse_report
```

Recommended function signatures:

```
const char *latticra_lat_parse_error_label(latticra_lat_parse_error_t error);
const char *latticra_lat_declaration_kind_label(latticra_lat_declaration_kind_t kind);
const char *latticra_lat_effect_label(latticra_lat_effect_t effect);

latticra_status_t latticra_lat_parse_source(
    const char *source,
    size_t source_len,
    latticra_lat_parse_result_t *result);

latticra_status_t latticra_lat_parse_report(
    const latticra_lat_parse_result_t *result,
    char *buffer,
    size_t buffer_len);
```

Capacity constants

Add exact bounded constants:

```
LATTICRA_LAT_SOURCE_MAX 65536u
LATTICRA_LAT_NAME_MAX 64u
```

```
LATTICRA_LAT_VALUE_MAX 128u
LATTICRA_LAT_DECLARATION_MAX 64u
LATTICRA_LAT_CLAUSE_MAX 128u
LATTICRA_LAT_REPORT_MAX 4096u
```

The first implementation should reject sources larger than LATTICRA_LAT_SOURCE_MAX.

Parse error enum

Add parse error enum values:

```
LATTICRA_LAT_PARSE_OK
LATTICRA_LAT_PARSE_NULL_ARGUMENT
LATTICRA_LAT_PARSE_EMPTY_SOURCE
LATTICRA_LAT_PARSE_SOURCE_TOO_LARGE
LATTICRA_LAT_PARSE_UNSUPPORTED_EXTENSION_CLAIM
LATTICRA_LAT_PARSE_MISSING_MODULE
LATTICRA_LAT_PARSE_INVALID_MODULE_NAME
LATTICRA_LAT_PARSE_UNBALANCED_BRACE
LATTICRA_LAT_PARSE_UNKNOWN_DECLARATION
LATTICRA_LAT_PARSE_INVALID_DECLARATION_NAME
LATTICRA_LAT_PARSE_UNTERMINATED_STRING
LATTICRA_LAT_PARSE_INVALID_STRING_ESCAPE
LATTICRA_LAT_PARSE_INVALID_HEX_ESCAPE
LATTICRA_LAT_PARSE_LITERAL_NUL_IN_STRING
LATTICRA_LAT_PARSE_CAPACITY_EXCEEDED
LATTICRA_LAT_PARSE_FORBIDDEN_BEHAVIOR_MARKER
LATTICRA_LAT_PARSE_INTERNAL_ERROR
```

Stable labels:

```
ok
null_argument
empty_source
source_too_large
unsupported_extension_claim
missing_module
invalid_module_name
unbalanced_brace
unknown_declaration
invalid_declaration_name
unterminated_string
invalid_string_escape
invalid_hex_escape
literal_nul_in_string
capacity_exceeded
forbidden_behavior_marker
internal_error
```

Declaration kind enum

Add declaration kind enum values:

```
LATTICRA_LAT_DECLARATION_STATE
LATTICRA_LAT_DECLARATION_POLICY
LATTICRA_LAT_DECLARATION_TRANSITION
LATTICRA_LAT_DECLARATION_ASSERTION
LATTICRA_LAT_DECLARATION_EFFECT
LATTICRA_LAT_DECLARATION_UNKNOWN
```

Stable labels:

```
state
policy
transition
assertion
effect
unknown
```

Effect enum

Add effect enum values:

```
LATTICRA_LAT_EFFECT_NONE
LATTICRA_LAT_EFFECT_READ
LATTICRA_LAT_EFFECT_LOCAL_MUTATION
LATTICRA_LAT_EFFECT_HOST_MUTATION
LATTICRA_LAT_EFFECT_NETWORK
LATTICRA_LAT_EFFECT_HARDWARE
```

```
LATTICRA_LAT_EFFECT_BOOT
LATTICRA_LAT_EFFECT_RECOVERY
LATTICRA_LAT_EFFECT_EXTERNAL
LATTICRA_LAT_EFFECT_UNKNOWN
```

Stable labels:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Effects remain metadata only.

Source span struct

Use a Lat source-span struct compatible with existing source-span conventions:

```
size_t start_offset;
size_t end_offset;
size_t start_line;
size_t start_column;
size_t end_line;
size_t end_column;
```

A future implementation may reuse latticra_l_ui_source_span_t only if that does not create unwanted coupling.

AST module struct

Add module metadata fields:

```
char module_name[LATTICRA_LAT_NAME_MAX];
latticra_lat_source_span_t span;
size_t declaration_count;
size_t state_count;
size_t policy_count;
size_t transition_count;
size_t assertion_count;
size_t effect_count;
```

AST declaration struct

Add declaration metadata fields:

```
latticra_lat_declaration_kind_t kind;
char name[LATTICRA_LAT_NAME_MAX];
char source_name[LATTICRA_LAT_NAME_MAX];
latticra_lat_source_span_t span;
size_t first_clause_index;
size_t clause_count;
```

source_name is used by transition declarations for the from <StateName> source state.

AST clause struct

Add clause metadata fields:

```
char keyword[LATTICRA_LAT_NAME_MAX];
char left[LATTICRA_LAT_NAME_MAX];
char operator_text[LATTICRA_LAT_NAME_MAX];
char right[LATTICRA_LAT_VALUE_MAX];
latticra_lat_effect_t effect;
latticra_lat_source_span_t span;
```

Clause values are metadata only.

Parse result struct

Add parse result fields:

```
latticra_status_t status;
latticra_lat_parse_error_t error;
latticra_lat_source_span_t span;
latticra_lat_ast_module_t module;
latticra_lat_ast_declaration_t declarations[LATTICRA_LAT_DECLARATION_MAX];
latticra_lat_ast_clause_t clauses[LATTICRA_LAT_CLAUSE_MAX];
size_t declaration_count;
size_t clause_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

The result must be fully initialized on success and failure.

Accepted fixture paths

The implementation PR should add at least one fixture file:

```
fixtures/lat/minimal_module.lat
```

Recommended fixture content:

```
lat module RootModule {
  state RootCell {
    origin = &quot;0/0&quot;
    route = &quot;ROOT&quot;
    axis = &quot;ROOT&quot;
    path = &quot;/&quot;
    host_effect = none
    external_effect = none
  }
}
```

Additional in-test string fixtures may be used for policy, transition, assertion, and effect declarations.

Parser ownership rules

The first parser should:

- never retain pointers into caller-owned source buffers;
- copy names and values into bounded fixed-size fields;
- reject capacity overflow deterministically;
- preserve source spans;
- return status codes;
- avoid heap allocation;
- avoid file I/O;
- avoid global mutable state;
- avoid execution or evaluation.

String literal handling

String handling should align with L-UI string policy:

```
\\
\&quot;
\n
\r
\t
\xHH
```

The first Lat parser should reject:

```
unterminated strings
unsupported escapes
invalid hex escapes
```

```
literal source-buffer NUL bytes
string values that exceed bounded storage
```

Escaped decoded NUL through \x00 may be accepted only if explicit length-carrying storage is implemented in the same parser slice. If not, it should be rejected in the first parser implementation plan follow-up.

Forbidden behavior markers

The parser should reject obvious behavior markers that imply execution before contracts exist:

```
exec
spawn
syscall
socket
open_file
write_file
hardware_write
```

These markers may appear only in comments if comments are stripped before validation.

Report format

latticra_lat_parse_report should emit a deterministic bounded report:

```
LAT GRAMMAR REPORT
status=&lt;integer-status&gt;
error=&lt;lat-error-label&gt;
module=&lt;module-name&gt;
declaration_count=&lt;count&gt;
state_count=&lt;count&gt;
policy_count=&lt;count&gt;
transition_count=&lt;count&gt;
assertion_count=&lt;count&gt;
effect_count=&lt;count&gt;
clause_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
span_start_line=&lt;line&gt;
span_start_column=&lt;column&gt;
span_end_line=&lt;line&gt;
span_end_column=&lt;column&gt;
```

Small output buffers should return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer.

Source-span mapping

Use these source-span mappings:

```
module -&gt; full module declaration span
declaration -&gt; full declaration span
field assignment -&gt; assignment clause span
require clause -&gt; requirement clause span
ensure clause -&gt; ensure clause span
effect clause -&gt; effect clause span
string literal -&gt; string token span if exposed later
identifier reference -&gt; identifier token span if exposed later
```

The parser must not invent byte offsets.

No-effect preservation

Lat grammar parsing is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
```

```
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
lat_grammar_accepts_minimal_module
lat_grammar_accepts_state_declaration
lat_grammar_accepts_policy_declaration
lat_grammar_accepts_transition_declaration
lat_grammar_accepts_assertion_declaration
lat_grammar_accepts_effect_declaration
lat_grammar_rejects_plain_l_extension_claim
lat_grammar_rejects_unknown_keyword
lat_grammar_rejects_unterminated_string
lat_grammar_rejects_invalid_escape
lat_grammar_rejects_literal_source_nul
lat_grammar_reports_source_spans
lat_grammar_preserves_no_effect_flags
lat_grammar_report_is_deterministic
lat_grammar_report_rejects_small_buffer
lat_grammar_error_labels_are_stable
lat_grammar_kind_labels_are_stable
lat_grammar_does_not_lower_to_lir
lat_grammar_does_not_execute_declarations
lat_grammar_is_deterministic
```

Test file plan

Add:

```
tests/lat_language_grammar_invariants.c
scripts/test-lat-language-grammar.sh
fixtures/lat/minimal_module.lat
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-lat-language-grammar-contract.sh
```

and before:

```
sh scripts/test-state-lattice.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
language naming policy
.lat canonical extension
C/C++ foundation direction
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
state lattice behavior
```

```
Nucleus preview behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
no-effect flags
```

Forbidden implementation behavior

Lat grammar implementation must not:

- claim plain L as the public language name;
- claim .l as the canonical source extension;
- execute declarations;
- mutate state;
- lower to LIR;
- render L-UI;
- call Nucleus task execution;
- evaluate host state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- broaden L-UI behavior;
- weaken semantic validation;
- accept literal source-buffer NUL;
- imply a compiler, interpreter, runtime, package manager, sandbox, or operating-system surface.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-lat-language-grammar-implementation-plan.sh
```

The guard is static. It does not implement Lat parsing.

Implementation gate

Lat grammar parser implementation code may be added only after this plan is merged.

Non-claims

This document does not implement Lat, Lat-Core, Lat-Orch, a parser, compiler, interpreter, runtime, package manager, LIR lowering, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_MODEL_NORMALIZATION_IMPLEMENTATION.md

Source title: Lat Model Normalization Implementation

Lines: 151 | Words: 430 | SHA-256: 2f4a44b217a2915941eed90be561646dee5dc149a9b0598ca88987e49e71898a

Lat Model Normalization Implementation

Status: no-effect implementation with first-declaration and first-clause report metadata

Scope: bounded Lat / Latticra Language model normalization after grammar parsing and semantic validation, before LIR lowering, execution, runtime behavior, or operating-system behavior.

Purpose

This slice adds a normalized Lat module model between semantic validation and Lat-to-LIR lowering.

The model consumes only bounded parser and semantic metadata. It builds deterministic declaration and clause tables for states, policies, transitions, assertions, and effect declarations while preserving source spans and no-effect flags.

It does not read source bytes, execute Lat, interpret transitions, lower to LIR, execute LIR, mutate state, perform file I/O, perform network I/O, call recovery behavior, touch hardware, or grant runtime authority.

Implementation Files

```
include/latticra/lat_model.h
src/lat_model.c
tests/lat_model_normalization_invariants.c
scripts/test-lat-model-normalization.sh
.github/workflows/lat-model-normalization.yml
```

Public API

The public API adds:

```
latticra_lat_model_error_t
latticra_lat_model_clause_role_t
latticra_lat_model_declaration_t
latticra_lat_model_clause_t
latticra_lat_model_t
latticra_lat_model_error_label
latticra_lat_model_clause_role_label
latticra_lat_model_normalize_module
latticra_lat_model_report
```

Normalized Tables

The model preserves the parser declaration order and adds typed index tables:

```
state_indices
policy_indices
transition_indices
assertion_indices
effect_indices
```

Transition declarations also carry a resolved source-state declaration index.

Clauses are normalized into stable roles:

```
field
require
ensure
effect
```

Each clause records its owner declaration index, owner declaration kind, name, operator text, value, effect metadata, and source span.

Report Format

latticra_lat_model_report emits a deterministic bounded report beginning with:

```
LAT MODEL NORMALIZATION REPORT
```

and including:

```
status
error
module
declaration_count
state_count
policy_count
transition_count
assertion_count
effect_count
clause_count
first_declaration_index
first_declaration_kind
first_declaration_name
first_declaration_source
first_declaration_parse_index
first_declaration_first_clause_index
first_declaration_clause_count
first_declaration_source_index
first_state_index
first_policy_index
first_transition_index
first_transition_source_index
first_clause_index
first_clause_role
first_clause_effect
first_clause_name
first_clause_operator
first_clause_value
first_assertion_index
first_effect_index
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
source span fields
```

The first-declaration fields are copied from the normalized declaration table for audit visibility only and are materialized only after successful normalization.

The first-clause fields are copied from the normalized clause table for audit visibility only. Operators are not evaluated.

Small output buffers return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the output buffer when possible.

Validation

Run:

```
sh scripts/test-lat-model-normalization.sh
sh scripts/test-lat-pipeline.sh
```

The invariants verify:

```
lat_model_labels_are_stable
lat_model_normalizes_foundation_counts
lat_model_builds_typed_declaration_indices
lat_model_builds_clause_roles_and_owners
lat_model_rejects_parse_failure
lat_model_rejects_semantic_failure
lat_model_preserves_no_effect_flags
lat_model_report_is_deterministic
lat_model_report_rejects_small_buffer
```

The Lat pipeline now also calls latticra_lat_model_normalize_module internally. Callers that need the normalized model can use latticra_lat_pipeline_run_source_with_model; existing callers can continue using latticra_lat_pipeline_run_source.

Lat-to-LIR lowering now has a model-aware entry point, latticra_lir_lower_lat_model, that consumes this normalized model directly. The older latticra_lir_lower_lat_module entry point remains available and normalizes a local model before lowering.

Non-Claims

This implementation does not provide Lat execution, Lat compilation, Lat interpretation, LIR execution, runtime behavior, command execution, mutation, file I/O, network I/O, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system completeness.

docs/LAT_PIPELINE_CONTRACT.md

Source title: Latticra Lat Pipeline Contract

Lines: 176 | Words: 521 | SHA-256: cfa96866bc3e88ff65611da64945f3d33107cfd929356e34f1cb160cec97763b

Latticra Lat Pipeline Contract

Status: Lat pipeline contract Scope: bounded no-effect composition of Lat grammar parsing, Lat semantic validation, and Lat-to-LIR lowering.

Purpose

This contract defines the first Lat / Latticra Language pipeline boundary.

The pipeline consumes Lat source bytes and coordinates the already bounded stages:

```
Lat source -&gt; grammar parser -&gt; semantic validator -&gt; Lat-to-LIR lowering -&gt;
pipeline report
```

The pipeline is not a compiler, interpreter, runtime, executor, package manager, or operating-system surface. It is a deterministic composition/report boundary over parser, semantic, and LIR metadata.

Relationship to existing work

This contract depends on:

```
include/latticra/lat_parser.h
include/latticra/lat_semantic.h
include/latticra/lat_to_lir.h
include/latticra/lir.h
src/lat_parser.c
src/lat_semantic.c
src/lat_to_lir.c
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LAT_SEMANTIC_VALIDATION_IMPLEMENTATION.md
docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION.md
```

Those files remain the source of truth for syntax parsing, semantic validation, LIR shape, lowering behavior, reports, capacities, and non-claims.

Pipeline posture

The pipeline must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

The pipeline must not execute Lat or LIR. It must not evaluate host state, mutate state, write files, read files, open network connections, invoke server behavior, invoke recovery behavior, touch hardware, boot anything, or perform command behavior.

Input contract

The pipeline accepts:

```
const char *source
size_t source_len
```

The caller provides storage for all intermediate and final records:

```
latticra_lat_parse_result_t
latticra_lat_semantic_result_t
latticra_lir_module_t
latticra_lat_to_lir_result_t
latticra_lat_pipeline_result_t
```

The pipeline does not allocate memory and does not take ownership of caller buffers.

Output contract

The pipeline produces:

```
latticra_lat_pipeline_result_t
```

The result summarizes:

```
status
pipeline error
parse error
semantic error
lowering error
LIR error
module name
source length
declaration count
clause count
node count
edge count
semantic_valid
no-effect flags
source span
```

Stable pipeline error labels

The first stable pipeline labels are:

```
ok
null_argument
parse_not_ok
semantic_not_ok
semantic_not_valid
lowering_not_ok
no_effect_violation
internal_error
```

Classification rules

Pipeline classification is ordered:

1. Null arguments return LATTICRA_STATUS_NULL_ARGUMENT.
2. Parser status failures are internal status failures.
3. Parser metadata errors are summarized as parse_not_ok.
4. Semantic metadata errors are summarized as semantic_not_ok.
5. A semantic result that is not valid is summarized as semantic_not_valid.
6. Lat-to-LIR or LIR metadata errors are summarized as lowering_not_ok.
7. Any no-effect flag violation is summarized as no_effect_violation.
8. Only a fully successful bounded path reports ok.

Report format

latticra_lat_pipeline_report emits a deterministic bounded report beginning with:

```
LAT PIPELINE REPORT
```

The report includes status, error labels, module metadata, counts, no-effect flags, and source-span metadata.

Small output buffers must return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer when possible.

Required implementation files

The implementation slice must add:

```
include/latticra/lat_pipeline.h
src/lat_pipeline.c
tests/lat_pipeline_invariants.c
scripts/test-lat-pipeline.sh
.github/workflows/lat-pipeline.yml
docs/LAT_PIPELINE_IMPLEMENTATION_PLAN.md
docs/LAT_PIPELINE_IMPLEMENTATION.md
```

Required tests

The implementation must include invariant coverage for:

```
lat_pipeline_accepts_foundation_model
lat_pipeline_preserves_counts
lat_pipeline_rejects_parse_failure
lat_pipeline_rejects_semantic_failure
lat_pipeline_preserves_no_effect_flags
lat_pipeline_report_is_deterministic
lat_pipeline_report_rejects_small_buffer
lat_pipeline_error_labels_are_stable
```

Compatibility expectations

This contract must not change parser behavior, semantic validation behavior, LIR shape behavior, Lat-to-LIR lowering behavior, L-UI behavior, C++ authority behavior, Nucleus behavior, runtime behavior, or existing report labels.

Non-claims

This contract does not implement Lat execution, Lat compilation, Lat interpretation, LIR execution, L-UI rendering, command behavior, runtime behavior, Nucleus task execution, mutation, file I/O, network I/O, recovery behavior, hardware behavior, malware prevention, ransomware prevention, sandboxing, certification, accreditation, or operating-system completeness.

docs/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_REFINEMENT.md

Source title: Lat Pipeline Diagnostic Integration Refinement

Lines: 104 | Words: 324 | SHA-256: ae6340877df5028763ddf7410396ba49df9d428f353f599a1f23050f8fc5a6e6

Lat Pipeline Diagnostic Integration Refinement

Status: lowering-aware implementation with parse-error, pipeline-span, comment, first-declaration, and first-clause metadata

This slice adds a companion Lat pipeline diagnostics API that combines pipeline stage/error state with parser error metadata, pipeline span metadata, parser line-comment metadata, Lat semantic diagnostic class, count, first-diagnostic indices, model-stage classification, and optional Lat-to-LIR lowering diagnostic metadata.

Files:

```
include/latticra/lat_pipeline_diagnostics.h
src/lat_pipeline_diagnostics.c
src/lat_pipeline_diagnostics_eval.c
src/lat_pipeline_diagnostics_report.c
include/latticra/lat_to_lir_diagnostics.h
src/lat_to_lir_diagnostics.c
tests/lat_pipeline_diagnostic_integration_refinement.c
```

The compatibility evaluator remains available:

```
latticra_lat_pipeline_diagnostics_evaluate
```

The lowering-aware evaluator adds optional lowering and LIR inputs:

```
latticra_lat_pipeline_diagnostics_evaluate_with_lowering
```

The diagnostic class labels include:

```
valid
parse
semantic
model
lowering
lir
effect-check
internal
```

When lowering metadata is provided, the pipeline diagnostic result also records:

```
lowering_class
lowering_error
model_error
lir_error
parse_error
pipeline_span_start_offset
pipeline_span_end_offset
pipeline_span_start_line
pipeline_span_start_column
pipeline_span_end_line
pipeline_span_end_column
comment_count
first_comment_start_offset
first_comment_end_offset
first_comment_start_line
first_comment_start_column
first_comment_end_line
first_comment_end_column
lowering_model_declaration_count
lowering_model_clause_count
lowering_first_declaration_node_index
lowering_first_declaration_kind
lowering_first_declaration_name
lowering_first_declaration_source
lowering_first_declaration_parse_index
lowering_first_declaration_first_clause_index
lowering_first_declaration_clause_count
lowering_first_declaration_source_index
lowering_first_transition_source_index
lowering_first_clause_node_index
lowering_first_clause_role
lowering_first_clause_effect
lowering_first_clause_name
lowering_first_clause_operator
lowering_first_clause_value
lowering_failed
model_failed
lir_failed
```

Parse-error metadata is copied from the Lat pipeline result. It records the parser error label, including specific parse-failure reasons such as unsupported_block_comment, for report/audit use only.

Pipeline span metadata is copied from the Lat pipeline result. It records the successful module span or the parse-failure diagnostic span for report/audit use only.

Comment metadata is copied from the Lat pipeline result. It records the parser line-comment count and first-comment span for diagnostic/report audit use only, including parse-failure diagnostics where a line comment appears before an unsupported block-comment opener.

First-declaration metadata is copied from the Lat-to-LIR diagnostic result. It records the first lowered declaration node index, declaration kind, name, source name, parse declaration index, first-clause index, clause count, and source declaration index for report/audit use only.

First-clause metadata is copied from the Lat-to-LIR diagnostic result. It records the first lowered clause node index, normalized clause role, effect label, name, operator text, and value text for report/audit use only.

Validation:

```
sh scripts/test-lat-pipeline-diagnostic-integration-refinement.sh
sh scripts/test-lat-pipeline.sh
```

Boundary: diagnostic metadata only. No Lat execution is added.

docs/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_INTEGRATION_AUDIT.md

Source title: Lat Pipeline Diagnostic Main Test Integration Audit

Lines: 25 | Words: 73 | SHA-256: 6f130f6e8ea29512838ee45b28e4aac7cf05c7f72fb18b8bfa5c9601bb95c242

Lat Pipeline Diagnostic Main Test Integration Audit

Status: initial audit guard

This record verifies that the Lat pipeline diagnostic integration is covered by the main Lat pipeline test runner, not only by its focused guard.

Audit expectations:

```
scripts/test-lat-pipeline.sh links src/lat_pipeline_diagnostics.c
scripts/test-lat-pipeline.sh links src/lat_pipeline_diagnostics_eval.c
scripts/test-lat-pipeline.sh links src/lat_pipeline_diagnostics_report.c
scripts/test-lat-pipeline.sh links src/lat_to_lir_diagnostics.c
scripts/test-lat-pipeline.sh compiles tests/lat_pipeline_diagnostic_integration_refinement.c
.github/workflows/lat-pipeline.yml runs scripts/test-lat-pipeline.sh
```

Validation:

```
sh scripts/test-lat-pipeline-diagnostic-main-test-integration-audit.sh
sh scripts/test-lat-pipeline.sh
```

Boundary: audit/guard only. No Lat execution is added.

docs/LAT_PIPELINE_IMPLEMENTATION.md

Source title: Latticra Lat Pipeline Implementation

Lines: 98 | Words: 387 | SHA-256: 577b63b96ed2349a82ad6eca0ba8064014884b28af17772ac1f778d8888e9aea

Latticra Lat Pipeline Implementation

Status: initial implementation contract Scope: first bounded no-effect Lat pipeline API, source composition, summary report, invariants, and workflow.

Purpose

This slice adds a no-effect Lat pipeline that composes the existing Lat parser, Lat semantic validator, Lat model normalization layer, and Lat-to-LIR lowering layer into one deterministic report boundary.

The pipeline is intended to make the currently separate stages easier to validate together without changing the behavior of any stage.

Added files

```
include/latticra/lat_pipeline.h
src/lat_pipeline.c
tests/lat_pipeline_invariants.c
scripts/test-lat-pipeline.sh
.github/workflows/lat-pipeline.yml
```

Public API

The public API adds:

```
latticra_lat_pipeline_error_t
latticra_lat_pipeline_result_t
latticra_lat_pipeline_error_label
latticra_lat_pipeline_run_source
latticra_lat_pipeline_run_source_with_model
latticra_lat_pipeline_report
```

Behavior

latticra_lat_pipeline_run_source performs the bounded metadata path:

```
source bytes
  -&gt; latticra_lat_parse_source
  -&gt; latticra_lat_validate_module
  -&gt; latticra_lat_model_normalize_module
  -&gt; latticra_lir_lower_lat_model
  -&gt; latticra_lat_pipeline_result_t
```

The wrapper entry point latticra_lat_pipeline_run_source preserves the original caller shape and runs normalization internally. The model-aware entry point latticra_lat_pipeline_run_source_with_model also returns the normalized model to callers. Lowering now consumes that normalized model directly.

The function records parser, semantic, model, lowering, and LIR errors separately, then classifies the aggregate pipeline state.

The companion pipeline diagnostic surface can now evaluate with optional Lat-to-LIR lowering diagnostics so pipeline reports can expose lowering class, model error, LIR error, model counts, transition source index, first-clause metadata, and failure flags without changing pipeline execution behavior.

Report surface

latticra_lat_pipeline_report emits:

```
LAT PIPELINE REPORT
```

with deterministic fields for status, aggregate error, stage errors, module name, source length, validity, parser counts, model counts, transition source index metadata, first-clause metadata copied from lowering, no-effect flags, and source spans.

Small buffers return LATTICRA_STATUS_BUFFER_TOO_SMALL and are cleared when possible.

Validation

Run:

```
sh scripts/test-lat-pipeline.sh
```

The invariant suite covers:

```
lat_pipeline_accepts_foundation_model
lat_pipeline_exposes_normalized_model
lat_pipeline_preserves_counts
lat_pipeline_rejects_parse_failure
lat_pipeline_rejects_semantic_failure
lat_pipeline_preserves_no_effect_flags
lat_pipeline_report_is_deterministic
lat_pipeline_report_rejects_small_buffer
lat_pipeline_error_labels_are_stable
```

Boundary

This slice does not execute Lat, execute LIR, perform runtime behavior, mutate state, perform file I/O, perform network I/O, call server code, call recovery code, touch hardware, control a terminal, call Nucleus task execution, or grant C++ authority.

Compatibility

The implementation is a composition layer. It does not modify parser, semantic validator, Lat model normalization, LIR, Lat-to-LIR lowering, L-UI, C++ authority, Nucleus, or runtime-boundary APIs.

Non-claims

This implementation does not provide a Lat compiler, Lat interpreter, Lat runtime, LIR executor, command system, task executor, production runtime, security boundary, malware prevention guarantee, ransomware prevention guarantee, certification, accreditation, or operating-system replacement.

docs/LAT_PIPELINE_IMPLEMENTATION_PLAN.md

Source title: Latticra Lat Pipeline Implementation Plan

Lines: 194 | Words: 435 | SHA-256: cf8e4f401ddc11cc5ca0d6402fb3b03ed15daa97690e93b1ba1a27572c557e82

Latticra Lat Pipeline Implementation Plan

Status: implementation planning contract Scope: exact public API, result struct, classification rules, report surface, tests, workflow, compatibility expectations, and non-claims before Lat pipeline implementation code.

Purpose

This plan defines the first no-effect Lat pipeline implementation.

The implementation should compose the existing bounded stages:

```
latticra_lat_parse_source
latticra_lat_validate_module
latticra_lir_lower_lat_module
```

and summarize their outputs through one deterministic pipeline result and report.

Implementation files

Add:

```
include/latticra/lat_pipeline.h
src/lat_pipeline.c
tests/lat_pipeline_invariants.c
scripts/test-lat-pipeline.sh
.github/workflows/lat-pipeline.yml
docs/LAT_PIPELINE_IMPLEMENTATION.md
```

Update status and index records to include the pipeline as the next bounded no-effect Lat integration slice.

Public API shape

Add:

```
latticra_lat_pipeline_error_t
latticra_lat_pipeline_result_t
latticra_lat_pipeline_error_label
latticra_lat_pipeline_run_source
latticra_lat_pipeline_report
```

Recommended signatures:

```
const char *latticra_lat_pipeline_error_label(latticra_lat_pipeline_error_t error);

latticra_status_t latticra_lat_pipeline_run_source(
    const char *source,
    size_t source_len,
    latticra_lat_parse_result_t *parse_result,
    latticra_lat_semantic_result_t *semantic_result,
    latticra_lir_module_t *module,
    latticra_lat_to_lir_result_t *lowering_result,
    latticra_lat_pipeline_result_t *pipeline_result);

latticra_status_t latticra_lat_pipeline_report(
    const latticra_lat_pipeline_result_t *result,
    char *buffer,
    size_t buffer_len);
```

Capacity constant

Add:

```
LATTICRA_LAT_PIPELINE_REPORT_MAX 8192u
```

Result fields

The result should include:

```
latticra_status_t status;
latticra_lat_pipeline_error_t error;
latticra_lat_parse_error_t parse_error;
latticra_lat_semantic_error_t semantic_error;
latticra_lat_to_lir_error_t lowering_error;
latticra_lir_error_t lir_error;
latticra_lat_source_span_t span;
char module_name[LATTICRA_LAT_NAME_MAX];
size_t source_len;
size_t declaration_count;
size_t clause_count;
size_t node_count;
size_t edge_count;
int semantic_valid;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

Error enum

Add:

```
LATTICRA_LAT_PIPELINE_OK
LATTICRA_LAT_PIPELINE_NULL_ARGUMENT
LATTICRA_LAT_PIPELINE_PARSE_NOT_OK
LATTICRA_LAT_PIPELINE_SEMANTIC_NOT_OK
LATTICRA_LAT_PIPELINE_SEMANTIC_NOT_VALID
LATTICRA_LAT_PIPELINE_LOWERING_NOT_OK
LATTICRA_LAT_PIPELINE_NO_EFFECT_VIOLATION
LATTICRA_LAT_PIPELINE_INTERNAL_ERROR
```

Stable labels:

```
ok
null_argument
parse_not_ok
semantic_not_ok
semantic_not_valid
lowering_not_ok
no_effect_violation
internal_error
```

Algorithm

1. Reject null pointers with LATTICRA_STATUS_NULL_ARGUMENT.
2. Initialize the pipeline result deterministically.
3. Parse the source using the existing Lat parser.

4. Validate semantic metadata using the existing Lat semantic validator.
5. Lower to LIR using the existing Lat-to-LIR lowering layer.
6. Summarize parse, semantic, lowering, LIR, count, source-span, and no-effect metadata.
7. Classify the pipeline result using ordered metadata errors.
8. Emit deterministic reports with small-buffer rejection and clearing.

Test command

Add:

```
sh scripts/test-lat-pipeline.sh
```

Compile with:

```
-std=c99 -Wall -Wextra -Werror -pedantic
```

and include:

```
src/lat_parser.c
src/lat_semantic.c
src/lir.c
src/lat_to_lir.c
src/lat_pipeline.c
tests/lat_pipeline_invariants.c
```

Workflow

Add a dedicated workflow:

```
.github/workflows/lat-pipeline.yml
```

The workflow should run the pipeline test script on pull requests and pushes to main.

Compatibility expectations

This implementation must not change:

```
Lat parser API or labels
Lat semantic validation API or labels
LIR API or labels
Lat-to-LIR lowering API or labels
L-UI parser, semantic, LIR, or rendering behavior
C++ authority behavior
Nucleus task behavior
runtime boundary behavior
no-effect flags
```

Non-claims

This implementation plan does not implement Lat execution, Lat compilation, Lat interpretation, LIR execution, command behavior, runtime behavior, Nucleus task execution, mutation, file I/O, network I/O, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_PIPELINE_REPORT_REFINEMENT.md

Source title: Latticra Lat Pipeline Report Refinement

Lines: 221 | Words: 662 | SHA-256: 57248b254624b081e625c8dc456bbb88d837bdb57030a48bd5e53107f8122508

Latticra Lat Pipeline Report Refinement

Status: Lat pipeline report refinement implementation with comment, first-declaration, and first-clause metadata Scope: deterministic Lat pipeline stage-summary metadata, parser span report fields, line-comment report fields, first-declaration report fields, first-clause report fields, invariant tests, guard coverage, and workflow wiring.

Purpose

This document records the Lat pipeline report refinement after the first Lat pipeline implementation and the runtime boundary report refinement.

The goal is to make the Lat pipeline report clearer about which stage completed, which stage failed, and whether the full no-effect chain remained intact.

This refinement does not execute Lat, execute LIR, compile Lat, interpret Lat, mutate state, perform I/O, or provide runtime behavior.

Added report metadata

The Lat pipeline result now records:

```
last_completed_stage
failed_stage
parse_ok
semantic_ok
model_ok
lowering_ok
lir_ok
no_effect_chain_ok
evidence_level
comment_count
first_comment_start_offset
first_comment_end_offset
first_comment_start_line
first_comment_start_column
first_comment_end_line
first_comment_end_column
first_declaration_node_index
first_declaration_kind
first_declaration_name
first_declaration_source
first_declaration_parse_index
first_declaration_first_clause_index
first_declaration_clause_count
first_declaration_source_index
first_clause_node_index
first_clause_role
first_clause_effect
first_clause_name
first_clause_operator
first_clause_value
```

The deterministic LAT PIPELINE REPORT now emits these fields as labels and integers. Line-comment metadata is copied from the parser result, and first-declaration plus first-clause metadata is copied from the Lat-to-LIR lowering result for audit visibility only.

The pipeline span is copied from the parser result span. Successful parses therefore report the module span, while parse failures report the parser diagnostic span, such as the unsupported block-comment opener location.

Stage labels

Initial stage labels:

```
none
parse
semantic
model
lowering
lir
effect-check
report
unknown
```

Evidence level

The Lat pipeline report uses a small deterministic evidence label:

```
0 -&gt; unavailable or internal/null pipeline evidence
```

```
1 -&gt; partial pipeline evidence with a known failed stage
2 -&gt; complete no-effect pipeline report evidence
```

This is a project-internal evidence label, not a certification, production-readiness claim, or security guarantee.

Successful report path

A successful Lat pipeline report records:

```
last_completed_stage=report
failed_stage=none
parse_ok=1
semantic_ok=1
model_ok=1
lowering_ok=1
lir_ok=1
no_effect_chain_ok=1
evidence_level=2
comment_count=&lt;line comment count from parse result&gt;
first_comment_start_offset=&lt;first line comment start offset&gt;
first_comment_end_offset=&lt;first line comment end offset&gt;
first_comment_start_line=&lt;first line comment start line&gt;
first_comment_start_column=&lt;first line comment start column&gt;
first_comment_end_line=&lt;first line comment end line&gt;
first_comment_end_column=&lt;first line comment end column&gt;
first_declaration_node_index=&lt;first lowered declaration node index&gt;
first_declaration_kind=&lt;first lowered declaration kind&gt;
first_declaration_name=&lt;first lowered declaration name&gt;
first_declaration_source=&lt;first lowered declaration source name&gt;
first_declaration_parse_index=&lt;first lowered declaration parse index&gt;
first_declaration_first_clause_index=&lt;first lowered declaration first clause index&gt;
first_declaration_clause_count=&lt;first lowered declaration clause count&gt;
first_declaration_source_index=&lt;first lowered declaration source index&gt;
first_clause_node_index=&lt;first lowered clause node index&gt;
first_clause_role=&lt;first lowered clause role&gt;
first_clause_effect=&lt;first lowered clause effect&gt;
first_clause_name=&lt;first lowered clause name&gt;
first_clause_operator=&lt;first lowered clause operator text&gt;
first_clause_value=&lt;first lowered clause value text&gt;
```

Failure report path

Known failure paths record the failed stage deterministically:

```
parse_not_ok -&gt; failed_stage=parse
semantic_not_ok -&gt; failed_stage=semantic
semantic_not_valid -&gt; failed_stage=semantic
model_not_ok -&gt; failed_stage=model
lowering_not_ok -&gt; failed_stage=lowering
no_effect_violation -&gt; failed_stage=effect-check
null_argument / internal_error -&gt; failed_stage=unknown
```

Implementation files

This slice updates or adds:

```
include/latticra/lat_pipeline.h
src/lat_pipeline.c
tests/lat_pipeline_invariants.c
tests/lat_pipeline_report_refinement.c
docs/LAT_PIPELINE_REPORT_REFINEMENT.md
scripts/test-lat-pipeline-report-refinement.sh
.github/workflows/lat-pipeline-report-refinement.yml
```

The existing Lat pipeline runner also compiles and runs the focused report refinement tests:

```
scripts/test-lat-pipeline.sh
```

Validation

Run:

```
sh scripts/test-lat-pipeline-report-refinement.sh
sh scripts/test-lat-pipeline.sh
```

The focused invariant tests verify:

```
lat_pipeline_report_refinement_labels_are_stable
lat_pipeline_report_refinement_reports_success_stage_summary
lat_pipeline_report_refinement_reports_parse_failure_stage
lat_pipeline_report_refinement_reports_semantic_failure_stage
lat_pipeline_report_refinement_null_result_sets_unknown_stage
```

The aggregate Lat pipeline invariants also verify that parser comment metadata is copied into the pipeline result and emitted in LAT PIPELINE REPORT on both successful parses and parse failures such as unsupported block-comment rejection. They also verify that parse-failure reports preserve the parser diagnostic span instead of falling back to an empty module span.

Compatibility

This refinement preserves the existing Lat pipeline behavior for:

```
source parsing
semantic validation
Lat model normalization
Lat-to-LIR lowering
LIR metadata output
aggregate pipeline error labels
module/source/count/span reporting
parser line-comment count and first-comment span reporting
parse-failure comment metadata reporting
parse-failure diagnostic span reporting
no-effect flags
small-buffer behavior
null-argument behavior
```

Non-claims

This report refinement does not provide:

```
Lat execution
Lat compilation
Lat interpretation
LIR execution
runtime behavior
command execution
task effect execution
state mutation
file I/O
network I/O
server interaction
hardware behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
certification
accreditation
operating-system completeness
```

docs/LAT_SEMANTIC_DIAGNOSTICS_REFINEMENT.md

Source title: Latticra Lat Semantic Diagnostics Refinement

Lines: 141 | Words: 331 | SHA-256: 3683e9edb2b238d317873071ed05a4afd1c0ae2f733e5a15be9ed89aa45df383

Latticra Lat Semantic Diagnostics Refinement

Status: initial Lat semantic diagnostics refinement implementation Scope: deterministic semantic diagnostic classes, category counters, first-diagnostic indices, report fields, invariant tests, guard coverage, and workflow coverage.

Purpose

This document records the Lat semantic diagnostics refinement after the Nucleus task report refinement.

The goal is to make Lat semantic reports easier to audit by classifying diagnostics into stable categories, counting diagnostics by category, and exposing deterministic first-diagnostic indices.

This refinement does not execute Lat, execute LIR, compile Lat, interpret Lat, mutate state, perform I/O, or provide runtime behavior.

Added semantic metadata

The Lat semantic result now records:

```
diagnostic_class
parse_diagnostic_count
declaration_diagnostic_count
reference_diagnostic_count
requirement_diagnostic_count
effect_diagnostic_count
no_effect_diagnostic_count
internal_diagnostic_count
first_diagnostic_declaration_index
first_diagnostic_clause_index
```

The deterministic LAT SEMANTIC REPORT now emits those fields.

Diagnostic classes

Initial diagnostic classes:

```
valid
parse
declaration
reference
requirement
effect
no-effect
internal
```

Class mapping

Initial class mapping:

```
ok -&gt; valid
parse_not_ok -&gt; parse
duplicate_declaration -&gt; declaration
invalid_state_field -&gt; declaration
invalid_clause_for_declaration -&gt; declaration
empty_declaration -&gt; declaration
unknown_transition_source -&gt; reference
invalid_require_left -&gt; requirement
invalid_effect_target -&gt; effect
invalid_effect_value -&gt; effect
effect_requires_gate -&gt; effect
no_effect_violation -&gt; no-effect
null_argument -&gt; internal
capacity_exceeded -&gt; internal
internal_error -&gt; internal
```

First diagnostic indices

When diagnostics are present, the semantic result records the declaration and clause indices of the first emitted diagnostic:

```
first_diagnostic_declaration_index
first_diagnostic_clause_index
```

When the semantic result is valid or has no diagnostic record, both remain zero.

Validation

Run:

```
sh scripts/test-lat-semantic-diagnostics-refinement.sh
sh scripts/test-lat-semantic-validation.sh
```

The focused invariant tests verify:

```
lat_semantic_diagnostic_class_labels_are_stable
lat_semantic_diagnostics_refinement_reports_valid_class
lat_semantic_diagnostics_refinement_reports_parse_class
lat_semantic_diagnostics_refinement_reports_declaration_class
lat_semantic_diagnostics_refinement_reports_reference_class
lat_semantic_diagnostics_refinement_reports_requirement_class
lat_semantic_diagnostics_refinement_reports_effect_class
lat_semantic_diagnostics_refinement_reports_internal_null_argument
```

Compatibility

This refinement preserves existing Lat semantic behavior for:

```
semantic validation outcomes
semantic error labels
diagnostic record content
declaration/state/policy/transition/assertion/effect counts
no-effect flags
small-buffer behavior
null-argument behavior
```

Non-claims

This diagnostics refinement does not provide:

```
Lat execution
Lat compilation
Lat interpretation
LIR execution
runtime behavior
command execution
state mutation
file I/O
network I/O
server interaction
hardware behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
certification
accreditation
operating-system completeness
```

docs/LAT_SEMANTIC_VALIDATION_CONTRACT.md

Source title: Latticra Lat Semantic Validation Contract

Lines: 313 | Words: 797 | SHA-256: 8ce21bd25f3e62bd2d7102e9510826c246c5051daeb8c72a573038305fc409e3

Latticra Lat Semantic Validation Contract

Status: semantic validation contract proposal Scope: semantic validation after bounded Lat grammar parsing and before Lat-to-LIR lowering, execution, runtime behavior, or operating-system behavior.

Purpose

This contract defines the first semantic validation layer for Lat / Latticra Language.

The layer consumes a successful latticra_lat_parse_result_t and validates declaration identity, cross-declaration references, state-field vocabulary, effect metadata, clause shapes, no-effect preservation, and deterministic report behavior.

This contract does not execute Lat, compile Lat, interpret Lat, lower Lat to LIR, render L-UI, call Nucleus, mutate state, read files, write files, open network connections, call server code, call update code, call recovery code, or touch hardware.

Relationship to previous work

This contract depends on:

```
docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LANGUAGE_NAMING_POLICY.md
docs/LANGUAGE_STRATEGY.md
docs/EFFECT_GATES.md
docs/STATE_LATTICE.md
include/latticra/lat_parser.h
include/latticra/state_lattice.h
```

Those files remain the source of truth for naming, parser API, effect vocabulary, state-lattice vocabulary, and no-effect posture.

Semantic validation target

The first target is:

```
Lat-Core semantic validation
```

Lat-Core remains a declaration subset for state, policy, assertion, transition, and effect metadata.

Lat-Orch remains future work.

Input contract

The validator accepts:

```
const latticra_lat_parse_result_t *parse_result
```

The validator must reject or report semantic failure when:

```
parse_result == NULL
parse_result-&gt;error != LATTICRA_LAT_PARSE_OK
```

The validator must not read source bytes directly. It trusts only bounded parser metadata and source spans already produced by the parser.

Output contract

The validator produces:

```
latticra_lat_semantic_result_t
```

The result records:

- status;
- summary semantic error;
- module name;
- semantic validity flag;
- diagnostic count;
- bounded diagnostics;
- declaration counts;
- no-effect flags.

Stable semantic error labels

The first semantic error labels are:

```
ok
null_argument
parse_not_ok
duplicate_declaration
unknown_transition_source
invalid_state_field
invalid_require_left
invalid_effect_target
invalid_effect_value
effect_requires_gate
invalid_clause_for_declaration
empty_declaration
no_effect_violation
capacity_exceeded
internal_error
```

Declaration identity rule

Declaration names must be globally unique inside one Lat module for this slice.

This prevents ambiguity before namespace, import, or qualified-name contracts exist.

State semantic rule

A state declaration may contain only field assignment clauses.

Allowed state fields:

```
origin
route
axis
path
breadcrumb
trace
safe_portal
rollback
health
risk
lock
dark_phase
host_effect
external_effect
```

The fields host_effect and external_effect must use known Lat effect values. This first semantic validation slice accepts only:

```
none
```

as a declared effect value.

Policy semantic rule

A policy declaration may contain only:

```
require
ensure
```

clauses.

A policy must not assign fields or declare direct effects in this slice.

Transition semantic rule

A transition declaration must resolve its from <StateName> reference to a declared state.

Transition clauses may be:

```
require left operator right
effect host = none
effect external = none
```

Unknown effect targets are rejected.

Any non-none declared effect value is rejected until a later effect-gate contract promotes it.

Assertion semantic rule

An assertion declaration may contain only:

```
require
ensure
```

clauses.

Assertions are metadata checks only.

Effect declaration semantic rule

An effect declaration may assign known effect targets:

```
host
external
local
network
hardware
boot
recovery
```

Values must be known Lat effect labels. This first slice accepts only none as a declared effect value.

No-effect preservation

The semantic validator must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

If the input parse result violates these flags, semantic validation must report no_effect_violation.

Report format

latticra_lat_semantic_report emits a deterministic bounded report beginning with:

```
LAT SEMANTIC REPORT
```

The report includes:

```
status
error
semantic_valid
module
declaration_count
state_count
policy_count
transition_count
assertion_count
effect_count
diagnostic_count
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
first_diagnostic_error
first_diagnostic_name
first_diagnostic_detail
span_start_offset
span_end_offset
span_start_line
span_start_column
span_end_line
span_end_column
```

Small output buffers must return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer when possible.

Exact implementation files

The implementation slice should add:

```
include/latticra/lat_semantic.h
src/lat_semantic.c
tests/lat_semantic_validation_invariants.c
scripts/test-lat-semantic-validation.sh
fixtures/lat/foundation_model.lat
docs/LAT_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
```

and update the C workflow to run the semantic validation test after grammar parser tests.

Required tests

The implementation must include tests for:

```
lat_semantic_accepts_foundation_model
lat_semantic_rejects_parse_error
lat_semantic_rejects_duplicate_declaration
lat_semantic_rejects_unknown_transition_source
lat_semantic_rejects_invalid_state_field
lat_semantic_rejects_invalid_policy_field_assignment
lat_semantic_rejects_invalid_effect_target
lat_semantic_rejects_invalid_effect_value
lat_semantic_rejects_effect_requiring_gate
lat_semantic_preserves_no_effect_flags
lat_semantic_reports_are_deterministic
lat_semantic_report_rejects_small_buffer
lat_semantic_error_labels_are_stable
lat_semantic_is_deterministic
```

Forbidden behavior

Lat semantic validation must not:

- claim plain L as the public language name;
- claim .l as the canonical source extension;
- parse source text directly;
- execute declarations;
- mutate state;
- lower to LIR;
- render L-UI;
- call Nucleus task execution;
- evaluate host state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- imply a compiler, interpreter, runtime, package manager, sandbox, or operating-system surface.

Non-claims

This contract does not implement Lat execution, Lat compilation, Lat interpretation, LIR lowering, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md

Source title: Latticra Lat Semantic Validation Implementation Plan

Lines: 300 | Words: 685 | SHA-256: 0adff9c7ef50dc21e530049fd5ee2436b20676e6ebf40993c74aec76e3933773

Latticra Lat Semantic Validation Implementation Plan

Status: implementation planning contract proposal Scope: exact public API, result structs, diagnostics, validation rules, report surface, fixture, tests, compatibility expectations, and non-claims before semantic validation code.

Purpose

This document defines the implementation plan for the first Lat / Latticra Language semantic validation layer.

The existing Lat grammar parser records bounded module, declaration, clause, effect, source-span, and no-effect metadata. The semantic validator will consume that parser result and validate module coherence without executing Lat or lowering it to LIR.

Implementation language decision

The first Lat semantic validator should be implemented in C.

Reason:

- current parser and state-lattice foundations are C;
- validation must remain bounded, deterministic, and no-effect;
- this layer should be usable before constrained C++ authority code grows around it;
- tests and runners already use C and POSIX shell.

Implementation files

Add:

```
include/latticra/lat_semantic.h
src/lat_semantic.c
tests/lat_semantic_validation_invariants.c
scripts/test-lat-semantic-validation.sh
fixtures/lat/foundation_model.lat
```

Update:

```
.github/workflows/c.yml
docs/FOUNDATION_INDEX.md
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

Public API shape

Add public API names:

```
latticra_lat_semantic_error_t
```

```
latticra_lat_semantic_diagnostic_t
latticra_lat_semantic_result_t
latticra_lat_semantic_error_label
latticra_lat_validate_module
latticra_lat_semantic_report
```

Recommended function signatures:

```
const char *latticra_lat_semantic_error_label(latticra_lat_semantic_error_t error);

latticra_status_t latticra_lat_validate_module(
    const latticra_lat_parse_result_t *parse_result,
    latticra_lat_semantic_result_t *result);

latticra_status_t latticra_lat_semantic_report(
    const latticra_lat_semantic_result_t *result,
    char *buffer,
    size_t buffer_len);
```

Capacity constants

Add bounded constants:

```
LATTICRA_LAT_SEMANTIC_DIAGNOSTIC_MAX 32u
LATTICRA_LAT_SEMANTIC_REPORT_MAX 4096u
```

Semantic error enum

Add semantic error enum values:

```
LATTICRA_LAT_SEMANTIC_OK
LATTICRA_LAT_SEMANTIC_NULL_ARGUMENT
LATTICRA_LAT_SEMANTIC_PARSE_NOT_OK
LATTICRA_LAT_SEMANTIC_DUPLICATE_DECLARATION
LATTICRA_LAT_SEMANTIC_UNKNOWN_TRANSITION_SOURCE
LATTICRA_LAT_SEMANTIC_INVALID_STATE_FIELD
LATTICRA_LAT_SEMANTIC_INVALID_REQUIRE_LEFT
LATTICRA_LAT_SEMANTIC_INVALID_EFFECT_TARGET
LATTICRA_LAT_SEMANTIC_INVALID_EFFECT_VALUE
LATTICRA_LAT_SEMANTIC_EFFECT_REQUIRES_GATE
LATTICRA_LAT_SEMANTIC_INVALID_CLAUSE_FOR_DECLARATION
LATTICRA_LAT_SEMANTIC_EMPTY_DECLARATION
LATTICRA_LAT_SEMANTIC_NO_EFFECT_VIOLATION
LATTICRA_LAT_SEMANTIC_CAPACITY_EXCEEDED
LATTICRA_LAT_SEMANTIC_INTERNAL_ERROR
```

Stable labels:

```
ok
null_argument
parse_not_ok
duplicate_declaration
unknown_transition_source
invalid_state_field
invalid_require_left
invalid_effect_target
invalid_effect_value
effect_requires_gate
invalid_clause_for_declaration
empty_declaration
no_effect_violation
capacity_exceeded
internal_error
```

Diagnostic struct

Add diagnostic metadata fields:

```
latticra_lat_semantic_error_t error;
latticra_lat_source_span_t span;
size_t declaration_index;
size_t clause_index;
char name[LATTICRA_LAT_NAME_MAX];
char detail[LATTICRA_LAT_VALUE_MAX];
```

Result struct

Add result metadata fields:

```
latticra_status_t status;
latticra_lat_semantic_error_t error;
latticra_lat_source_span_t span;
char module_name[LATTICRA_LAT_NAME_MAX];
int semantic_valid;
size_t diagnostic_count;
latticra_lat_semantic_diagnostic_t diagnostics[LATTICRA_LAT_SEMANTIC_DIAGNOSTIC_MAX];
size_t declaration_count;
size_t state_count;
size_t policy_count;
size_t transition_count;
size_t assertion_count;
size_t effect_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

Validation algorithm

1. Initialize output deterministically.
2. Reject null arguments with LATTICRA_STATUS_NULL_ARGUMENT.
3. Copy module and count metadata from the parser result when available.
4. If parser error is not ok, record parse_not_ok and stop.
5. Verify no-effect flags.
6. Detect duplicate declaration names.
7. For each declaration:

- reject empty declarations;
- validate declaration-specific clause forms;
- validate known state fields;
- validate known effect targets;
- validate known effect values;
- reject non-none declared effects in this first slice;
- resolve transition source state names.

8. Set semantic_valid=1 only when no diagnostics exist.
9. Emit deterministic bounded reports.

Fixture

Add:

```
fixtures/lat/foundation_model.lat
```

Recommended content:

```
lat module FoundationModule {
  state RootCell {
    origin = &quot;0/0&quot;
    route = &quot;ROOT&quot;
    axis = &quot;ROOT&quot;
    path = &quot;/&quot;
    health = &quot;ok&quot;
    risk = &quot;low&quot;
    lock = &quot;open&quot;
    host_effect = none
    external_effect = none
  }

  effect PreviewOnly {
```

```
    host = none
    external = none
    network = none
    hardware = none
  }

  policy SafePreview {
    require risk != &quot;high&quot;
    require lock == &quot;open&quot;
    ensure host_effect == none
    ensure external_effect == none
  }

  transition MoveRight from RootCell {
    require lock == &quot;open&quot;
    effect host = none
    effect external = none
  }

  assertion RootCellIsSafe {
    require health == &quot;ok&quot;
    require host_effect == none
    require external_effect == none
  }
}
```

Exact test list

Add invariant tests for:

```
lat_semantic_accepts_foundation_model
lat_semantic_rejects_parse_error
lat_semantic_rejects_duplicate_declaration
lat_semantic_rejects_unknown_transition_source
lat_semantic_rejects_invalid_state_field
lat_semantic_rejects_invalid_policy_field_assignment
lat_semantic_rejects_invalid_effect_target
lat_semantic_rejects_invalid_effect_value
lat_semantic_rejects_effect_requiring_gate
lat_semantic_preserves_no_effect_flags
lat_semantic_reports_are_deterministic
lat_semantic_report_rejects_small_buffer
lat_semantic_error_labels_are_stable
lat_semantic_is_deterministic
```

Test command

Add:

```
sh scripts/test-lat-semantic-validation.sh
```

Compile with:

```
-std=c99 -Wall -Wextra -Werror -pedantic
```

and include:

```
src/lat_parser.c
src/lat_semantic.c
tests/lat_semantic_validation_invariants.c
```

Compatibility expectations

This implementation must not change:

```
language naming policy
.lat canonical extension
C/C++ foundation direction
Lat grammar parser behavior
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
state lattice behavior
Nucleus preview behavior
literal source-buffer NUL rejection
no-effect flags
```

Non-claims

This implementation plan does not implement Lat execution, Lat compilation, Lat interpretation, LIR lowering, L-UI rendering, command behavior, Nucleus task handling, live movement, state mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_SPECIFIC_LIR_REFINEMENT_CONTRACT.md

Source title: Latticra Lat-Specific LIR Refinement Contract

Lines: 83 | Words: 295 | SHA-256: c03b0025250440f22c794f356f4dd6452febdcd1a67c1170cc1bfbe879c0acf2

Latticra Lat-Specific LIR Refinement Contract

Status: Lat-specific LIR refinement contract Scope: explicit Lat declaration and transition metadata in LIR after grammar parsing, semantic validation, Lat-to-LIR lowering, and Lat pipeline composition.

Purpose

This contract defines the first Lat-specific refinement of the Latticra Intermediate Representation.

The existing Lat-to-LIR lowering path materializes a lat_module LIR shape. This refinement makes the Lat origin of key nodes and transition-source edges explicit instead of relying only on generic field, binding, and effect node kinds.

Refinement target

The refinement introduces explicit LIR node kinds for Lat declarations:

```
lat_state
lat_policy
lat_transition
lat_assertion
lat_requirement
lat_effect_declaration
```

and an explicit edge kind for transition source relationships:

```
transitions_from
```

Compatibility policy

The refinement must preserve existing numeric values for already-published LIR enum members.

The generic node kinds and edge kinds remain available for L-UI lowering and existing LIR users.

Required behavior

Lat-to-LIR lowering must map:

```
state declaration      -&gt; lat_state
policy declaration     -&gt; lat_policy
transition declaration -&gt; lat_transition
assertion declaration  -&gt; lat_assertion
effect declaration     -&gt; lat_effect_declaration
require / ensure       -&gt; lat_requirement
transition source edge -&gt; transitions_from
```

The implementation must preserve:

```
source spans
module/declaration/clause counts
node and edge counts
```

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Forbidden behavior

This refinement must not execute Lat, execute LIR, interpret transitions, mutate state, perform effects, perform file I/O, perform network I/O, call server code, call recovery code, touch hardware, or provide runtime behavior.

Validation

The implementation must add focused invariants for:

```
lat_specific_lir_labels_are_stable
lat_specific_lir_uses_lat_declaration_node_kinds
lat_specific_lir_uses_lat_requirement_node_kind
lat_specific_lir_uses_transition_source_edge_kind
lat_specific_lir_preserves_counts_and_no_effect_flags
```

Non-claims

This contract does not implement Lat execution, Lat compilation, Lat interpretation, LIR execution, runtime behavior, command behavior, Nucleus task execution, mutation, file I/O, network I/O, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system completeness.

docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION.md

Source title: Latticra Lat-Specific LIR Refinement Implementation

Lines: 61 | Words: 245 | SHA-256: f2a67125f5ce6a2f0caf8d1216be78aa44465ab7356322827056236160a72d3e

Latticra Lat-Specific LIR Refinement Implementation

Status: initial implementation contract Scope: first explicit Lat declaration node kinds and transition-source edge kind in LIR.

Purpose

This slice implements the first Lat-specific LIR refinement after the parser, semantic validator, Lat-to-LIR lowering layer, and Lat pipeline foundation.

The refinement makes Lat declaration roles visible in LIR without changing the no-effect posture or introducing execution behavior.

Added behavior

Lat-to-LIR lowering now emits explicit node kinds for:

```
state declarations
policy declarations
transition declarations
assertion declarations
effect declarations
require / ensure clauses
```

Transition source references now use an explicit:

```
transitions_from
```

edge kind.

The later Lat-to-LIR clause metadata refinement preserves clause operators in the LIR node operator_text field while keeping role labels in the existing node binding field.

Validation

Run:

```
sh scripts/test-lat-specific-lir-refinement.sh
```

The test covers stable labels, declaration node kind mapping, requirement node mapping, transition-source edge mapping, counts, and no-effect preservation.

Related operator metadata is covered by:

```
sh scripts/test-lat-to-lir-clause-metadata-refinement.sh
```

Boundary

This implementation remains metadata-only. It does not execute Lat or LIR, interpret transition behavior, mutate state, perform effects, call Nucleus, call runtime code, or perform host operations.

Compatibility

Existing generic LIR node and edge values are preserved. L-UI LIR lowering remains generic and unchanged.

Non-claims

This implementation does not provide a Lat compiler, Lat interpreter, Lat runtime, LIR executor, command system, task executor, production runtime, security boundary, malware prevention guarantee, ransomware prevention guarantee, certification, accreditation, or operating-system replacement.

docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION_PLAN.md

Source title: Latticra Lat-Specific LIR Refinement Implementation Plan

Lines: 90 | Words: 240 | SHA-256: 565708890724df1586472b5eb85cc3aad6850bb92a7879a5f17fe855698a79e9

Latticra Lat-Specific LIR Refinement Implementation Plan

Status: implementation planning contract Scope: exact enum additions, lowering mappings, labels, tests, workflow, compatibility expectations, and non-claims before Lat-specific LIR refinement implementation.

Purpose

This plan defines the implementation of the first Lat-specific LIR refinement.

The goal is to make Lat declaration semantics more visible in LIR without changing the no-effect boundary or adding execution behavior.

Implementation files

Update:

```
include/latticra/lir.h
src/lir.c
src/lat_to_lir.c
tests/lat_to_lir_lowering_invariants.c
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/FOUNDATION_INDEX.md
```

Add:

```
tests/lat_specific_lir_refinement_invariants.c
```

```
scripts/test-lat-specific-lir-refinement.sh
.github/workflows/lat-specific-lir-refinement.yml
docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION.md
```

Public enum additions

Append Lat-specific node kinds without renumbering existing values:

```
LATTICRA_LIR_NODE_LAT_STATE
LATTICRA_LIR_NODE_LAT_POLICY
LATTICRA_LIR_NODE_LAT_TRANSITION
LATTICRA_LIR_NODE_LAT_ASSERTION
LATTICRA_LIR_NODE_LAT_REQUIREMENT
LATTICRA_LIR_NODE_LAT_EFFECT_DECLARATION
```

Append the transition-source edge kind:

```
LATTICRA_LIR_EDGE_TRANSITIONS_FROM
```

Label additions

Add deterministic labels:

```
lat_state
lat_policy
lat_transition
lat_assertion
lat_requirement
lat_effect_declaration
transitions_from
```

Lowering changes

latticra_lir_lower_lat_module should emit Lat-specific declaration node kinds while preserving the existing LIR shape and counts.

The transition source relationship should use transitions_from rather than the generic binds edge.

Test command

```
sh scripts/test-lat-specific-lir-refinement.sh
```

Compatibility expectations

This implementation must not change L-UI lowering, existing generic LIR node kinds, existing generic LIR edge kinds, Lat parser behavior, Lat semantic validation behavior, Lat pipeline behavior, C++ authority behavior, Nucleus behavior, or runtime boundary behavior.

Non-claims

This implementation plan does not implement Lat execution, LIR execution, runtime behavior, command behavior, mutation, file I/O, network I/O, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LAT_TO_LIR_CLAUSE_METADATA_REFINEMENT.md

Source title: Lat-to-LIR Clause Metadata Refinement

Lines: 82 | Words: 266 | SHA-256: 6472062f7cb1f6cc912772b6abe2d7dd4cf5a591aa0bbdca50e77d8e2cd6c9be

Lat-to-LIR Clause Metadata Refinement

Status: initial no-effect implementation

Scope: preserve Lat clause operator metadata in LIR nodes and expose first-clause lowering metadata in deterministic Lat-to-LIR reports.

Purpose

This slice makes lowered Lat clause metadata more inspectable without changing Lat parsing, semantic validation, normalization, lowering classification, or execution behavior.

Before this refinement, clause nodes preserved clause name, value, role, source span, owner relationship, and node kind. This slice adds a dedicated LIR node operator_text field and records the first lowered clause in the Lat-to-LIR result/report surface.

Added Metadata

LIR nodes now carry:

```
operator_text
```

Lat-to-LIR lowering writes clause operators into that field for:

```
field clauses
require clauses
ensure clauses
effect clauses
```

The Lat-to-LIR result/report now also records:

```
first_clause_node_index
first_clause_role
first_clause_effect
first_clause_name
first_clause_operator
first_clause_value
```

Behavior

The refinement preserves existing mapping behavior:

```
Lat clause name  -&gt; LIR node name
Lat clause value -&gt; LIR node value
Lat clause role  -&gt; LIR node binding
Lat operator     -&gt; LIR node operator_text
```

For example:

```
require risk != &quot;high&quot;
```

lowers into a Lat requirement node with:

```
name=risk
operator_text=!=
value=high
binding=require
```

Validation

Run:

```
sh scripts/test-lat-to-lir-clause-metadata-refinement.sh
sh scripts/test-lat-to-lir-lowering.sh
```

The focused guard verifies field, requirement, and effect clause operator metadata plus first-clause report fields.

The Lat-to-LIR diagnostic refinement also copies these first-clause fields into its diagnostic result/report so operator metadata remains visible from the diagnostic layer.

Boundary

This is metadata-only. It does not execute Lat, interpret requirements, evaluate operators, execute LIR, perform effects, mutate state, perform file I/O, perform network I/O, call runtime behavior, or grant authority.

docs/LAT_TO_LIR_DIAGNOSTIC_REFINEMENT.md

Source title: Lat-to-LIR Diagnostic Refinement

Lines: 101 | Words: 328 | SHA-256: 63d297d6d15ec2bb75c2b2e0c165d144e510c8474c45cfee42fbb4b6934cccb4

Lat-to-LIR Diagnostic Refinement

Status: no-effect implementation with first-declaration and first-clause diagnostic metadata

Scope: deterministic diagnostic classification and reporting for Lat-to-LIR lowering results after model-driven lowering integration.

Purpose

This slice adds a diagnostic surface for the Lat-to-LIR boundary.

It classifies lowering outcomes without changing parser, semantic, model normalization, lowering, LIR, pipeline, runtime, or authority behavior.

Implementation Files

```
include/latticra/lat_to_lir_diagnostics.h
src/lat_to_lir_diagnostics.c
tests/lat_to_lir_diagnostic_refinement.c
scripts/test-lat-to-lir-diagnostic-refinement.sh
```

Public API

The diagnostic API adds:

```
latticra_lat_to_lir_diagnostic_class_t
latticra_lat_to_lir_diagnostic_result_t
latticra_lat_to_lir_diagnostic_class_label
latticra_lat_to_lir_diagnostics_evaluate
latticra_lat_to_lir_diagnostics_report
```

Diagnostic Classes

The evaluator classifies lowering outcomes as:

```
valid
parse
semantic
model
effect-check
capacity
lir
internal
```

It copies lowering error, model error, optional LIR error, model counts, first lowered declaration metadata, first transition source index, first-clause metadata, node and edge counts, no-effect flags, failure flags, no-effect issue flag, and evidence level into a bounded result record.

The diagnostic result preserves the first lowered declaration:

```
first_declaration_node_index
first_declaration_kind
first_declaration_name
first_declaration_source
first_declaration_parse_index
first_declaration_first_clause_index
first_declaration_clause_count
first_declaration_source_index
```

The diagnostic result preserves the first lowered clause:

```
first_clause_node_index
first_clause_role
first_clause_effect
first_clause_name
```

```
first_clause_operator
first_clause_value
```

These fields are copied from lowering metadata only. Operators are not evaluated.

The Lat pipeline diagnostic integration can consume this result through latticra_lat_pipeline_diagnostics_evaluate_with_lowering, preserving the older pipeline diagnostic evaluator for existing callers. Pipeline diagnostics copy the first-clause metadata with lowering_ prefixes so aggregate reports can retain the first lowered clause role, effect, name, operator, value, and node index.

Report Format

latticra_lat_to_lir_diagnostics_report emits a deterministic bounded report beginning with:

```
LAT TO LIR DIAGNOSTIC REPORT
```

Small output buffers return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer when possible.

Validation

Run:

```
sh scripts/test-lat-to-lir-diagnostic-refinement.sh
```

The invariant suite checks stable labels, valid lowering diagnostics, first-declaration diagnostic metadata, first-clause diagnostic metadata, parse failure diagnostics, model failure diagnostics, no-effect issue diagnostics, null lowering handling, deterministic report fields, and small-buffer rejection.

Non-Claims

This slice does not execute Lat, execute LIR, mutate state, perform file I/O, perform network I/O, perform runtime behavior, grant authority, enforce policy, or change lowering behavior.

docs/LAT_TO_LIR_LOWERING_CONTRACT.md

Source title: Latticra Lat-to-LIR Lowering Contract

Lines: 344 | Words: 1101 | SHA-256: 0788cf4300fc916decd1e0f1ea9b813444ece0a4157668b3fc067df069cf277b

Latticra Lat-to-LIR Lowering Contract

Status: Lat-to-LIR lowering contract Scope: first contract for lowering semantically valid Lat / Latticra Language metadata into LIR shape metadata before implementation code, execution, runtime behavior, or operating-system behavior.

Purpose

This document defines the first contract for:

```
Lat-to-LIR lowering
```

The goal is to define how a semantically valid Lat module may be represented as Latticra Intermediate Representation metadata in a future implementation slice.

This is contract-only work. It does not implement lowering code.

The current Lat lane now has:

```
Lat source -&gt; bounded grammar parser -&gt; Lat semantic validation -&gt; future Lat-to-LIR
lowering
```

The current LIR lane already has a bounded representation shape and reserves lat_module as a source kind. This contract defines the first Lat-specific lowering boundary for that reserved source kind.

Relationship to previous work

This contract depends on:

```
docs/LAT_LANGUAGE_GRAMMAR_CONTRACT.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION_PLAN.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LAT_SEMANTIC_VALIDATION_CONTRACT.md
docs/LAT_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
docs/LAT_LANGUAGE_FOUNDATION_ANALYSIS.md
docs/LIR_SHAPE_CONTRACT.md
docs/LIR_SHAPE_IMPLEMENTATION_PLAN.md
docs/LIR_SHAPE_IMPLEMENTATION.md
include/latticra/lat_parser.h
include/latticra/lat_semantic.h
include/latticra/lir.h
```

Those files remain the source of truth for Lat grammar parsing, Lat semantic validation, existing LIR shape, no-effect metadata, report behavior, and current implementation boundaries.

First lowering target

The first target is:

```
Lat-Core semantic module -&gt; LIR module metadata
```

Lat-Core currently includes:

```
state
policy
transition
assertion
effect
```

Lat-Orch remains future work.

Input prerequisite

A future lowering function must accept only parser and semantic results that satisfy:

```
parser_error=ok
semantic_error=ok
semantic_valid=1
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

If semantic validation failed, lowering must not produce a partial LIR module.

Output target

The output target should be the existing LIR module shape:

```
latticra_lir_module_t
```

The source kind should be:

```
lat_module
```

The future implementation should reuse the existing LIR no-effect flags and report model instead of creating a separate executable representation.

Proposed public API boundary

A future implementation plan should define exact C API names. The expected shape is:

```
latticra_status_t latticra_lir_lower_lat_module(
    const latticra_lat_parse_result_t *parse_result,
    const latticra_lat_semantic_result_t *semantic_result,
    latticra_lir_module_t *module);
```

This contract does not add the function.

Node mapping rules

A future Lat-to-LIR lowering implementation should map Lat declarations into deterministic LIR node records.

Suggested mapping:

```
Lat module      -&gt; LIR module node
state           -&gt; LIR node kind field or module child node with declaration role metadata
policy          -&gt; LIR node kind field or module child node with declaration role metadata
transition      -&gt; LIR node kind field or module child node with source-state metadata
assertion       -&gt; LIR node kind field or module child node with assertion metadata
effect          -&gt; LIR effect node
require clause  -&gt; LIR binding or metadata node
ensure clause   -&gt; LIR binding or metadata node
effect clause   -&gt; LIR effect metadata
field assignment -&gt; LIR field metadata
```

Because the current LIR node enum was first shaped for L-UI, the implementation plan must either reuse existing node kinds conservatively or add exact Lat-oriented node kinds through a separate plan before code is added.

Edge mapping rules

A future lowering implementation should use deterministic edges for relationships.

Suggested edge mapping:

```
module contains declaration
declaration contains clause
transition references source state
policy annotates requirement metadata
assertion annotates checked metadata
effect annotates effect boundary metadata
```

If the existing edge enum is not expressive enough, the implementation plan must name any new edge kind before implementation code is added.

Source-span preservation

Every LIR record derived from Lat must carry the source span from parser metadata.

Rules:

- module span comes from the Lat module span;
- declaration span comes from the Lat declaration span;
- clause span comes from the Lat clause span;
- transition source-state metadata should preserve the transition declaration span until a more specific reference span exists;
- effect metadata should preserve the effect clause or effect declaration span;
- lowering must not invent byte offsets.

Effect preservation

Lat semantic validation currently accepts only none effect declarations in this slice.

Lat-to-LIR lowering must preserve:

```
effect=none
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

A future implementation must reject or report unsupported effect values rather than widening behavior.

Report model

A future Lat-to-LIR lowering report should remain deterministic and bounded.

Suggested report fields:

```
LAT TO LIR LOWERING REPORT
status=&lt;status&gt;
source_kind=lat_module
module=&lt;module-name&gt;
declaration_count=&lt;count&gt;
state_count=&lt;count&gt;
policy_count=&lt;count&gt;
transition_count=&lt;count&gt;
assertion_count=&lt;count&gt;
effect_count=&lt;count&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

The future implementation may use latticra_lir_report directly if it can represent all required fields without losing Lat-specific context.

Error model

A future implementation plan should define stable error labels.

Expected labels:

```
ok
null_argument
parse_not_ok
semantic_not_ok
semantic_not_valid
capacity_exceeded
unsupported_declaration_kind
unsupported_clause_kind
unsupported_effect
unsupported_lir_shape
internal_error
```

Capacity policy

The implementation plan must name exact capacity behavior before code is added.

At minimum, it must define how many LIR nodes and edges are required per:

```
module
declaration
clause
transition source reference
effect metadata record
```

If the existing LATTICRA_LIR_NODE_MAX or LATTICRA_LIR_EDGE_MAX is not sufficient, the plan must justify any increase before implementation code is added.

Compatibility expectations

Lat-to-LIR lowering work must not change:

```
Lat parser behavior
Lat semantic validation behavior
L-UI parser behavior
L-UI semantic validation behavior
existing LIR lowering from L-UI
state lattice behavior
source-span behavior
no-effect flags
```

Future implementation gate

Lowering implementation must not begin until a separate implementation plan defines:

1. public API shape;
2. exact error enum labels;
3. exact node mapping;
4. exact edge mapping;
5. exact capacity accounting;
6. source-span mapping;
7. report format;
8. semantic prerequisite behavior;
9. compatibility expectations;
10. exact tests;
11. non-claims.

That next slice should be:

```
LAT_TO_LIR_LOWERING_IMPLEMENTATION_PLAN.md
```

Future test list

The implementation plan should include tests for:

```
lat_to_lir_rejects_parse_error
lat_to_lir_rejects_semantic_error
lat_to_lir_accepts_foundation_model
lat_to_lir_sets_source_kind_lat_module
lat_to_lir_preserves_module_name
lat_to_lir_preserves_declaration_counts
lat_to_lir_preserves_transition_source_metadata
lat_to_lir_preserves_effect_none
lat_to_lir_preserves_source_spans
lat_to_lir_preserves_no_effect_flags
lat_to_lir_report_is_deterministic
lat_to_lir_report_rejects_small_buffer
lat_to_lir_does_not_execute_lat
lat_to_lir_does_not_mutate_state
lat_to_lir_is_deterministic
```

Forbidden behavior

Lat-to-LIR lowering work must not:

- lower parser-failed input;
- lower semantic-failed input;
- execute Lat declarations;
- interpret transition behavior;
- mutate state;
- render L-UI;
- call Nucleus task execution;
- evaluate host state;
- write files;
- read files;

- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- broaden accepted Lat syntax;
- weaken Lat semantic validation;
- weaken existing LIR shape behavior;
- invent source byte offsets;
- imply a compiler, interpreter, runtime, package manager, or operating-system surface.

Current validation command

This contract is guarded by:

```
sh scripts/test-lat-to-lir-lowering-contract.sh
```

The guard is static. It does not implement Lat-to-LIR lowering.

Non-claims

This document does not implement Lat-to-LIR lowering, Lat execution, Lat compilation, Lat interpretation, LIR execution, command behavior, Nucleus task handling, state mutation, runtime behavior, recovery behavior, server interaction, self-update, hardware support, boot readiness, or operating-system completeness.

docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION.md

Source title: Latticra Lat-to-LIR Lowering Implementation

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Latticra Lat-to-LIR Lowering Implementation

Status: model-aware implementation record with first-declaration report metadata Scope: bounded no-effect Lat-to-LIR lowering implementation, normalized model entry point, compatibility wrapper, public API, test runner, workflow, report surface, fixture path, and invariants.

Purpose

This implementation adds the bounded Lat-to-LIR lowering surface for semantically valid Lat / Latticra Language metadata.

The primary lowering implementation consumes:

```
latticra_lat_model_t
```

and fills:

```
latticra_lir_module_t
latticra_lat_to_lir_result_t
```

The compatibility wrapper latticra_lir_lower_lat_module still accepts parser and semantic results, normalizes a local Lat model, and delegates to the model-aware lowerer.

The lowering surface is metadata-only. It does not execute Lat, interpret transitions, mutate state, render L-UI, invoke Nucleus behavior, perform runtime behavior, write files, read files, open network connections, call update code, call recovery code, touch hardware, or provide an operating-system surface.

Implementation files

```
include/latticra/lat_to_lir.h
src/lat_to_lir.c
tests/lat_to_lir_lowering_invariants.c
scripts/test-lat-to-lir-lowering.sh
.github/workflows/lat-to-lir-lowering.yml
```

Related diagnostic refinement:

```
include/latticra/lat_to_lir_diagnostics.h
src/lat_to_lir_diagnostics.c
tests/lat_to_lir_diagnostic_refinement.c
scripts/test-lat-to-lir-diagnostic-refinement.sh
docs/LAT_TO_LIR_DIAGNOSTIC_REFINEMENT.md
```

Related clause metadata refinement:

```
tests/lat_to_lir_clause_metadata_refinement.c
scripts/test-lat-to-lir-clause-metadata-refinement.sh
docs/LAT_TO_LIR_CLAUSE_METADATA_REFINEMENT.md
```

Related contract and plan files:

```
docs/LAT_TO_LIR_LOWERING_CONTRACT.md
docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION_PLAN.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LAT_SEMANTIC_VALIDATION_CONTRACT.md
docs/LIR_SHAPE_IMPLEMENTATION.md
```

Public API

The public API adds:

```
latticra_lat_to_lir_error_t
latticra_lat_to_lir_result_t
latticra_lat_to_lir_error_label
latticra_lir_lower_lat_model
latticra_lir_lower_lat_module
latticra_lat_to_lir_report
```

The model-aware lowering function accepts normalized Lat model metadata, writes a LIR module, and writes a lowering-specific summary result. The wrapper accepts parser and semantic metadata for existing callers.

Accepted input

The lowering implementation accepts only input that satisfies:

```
parse_result-&gt;error == LATTICRA_LAT_PARSE_OK
semantic_result-&gt;error == LATTICRA_LAT_SEMANTIC_OK
semantic_result-&gt;semantic_valid == 1
model-&gt;error == LATTICRA_LAT_MODEL_OK
no_effect == 1
execution_allowed == 0
mutation_allowed == 0
```

The implementation also preserves server, recovery, and hardware denial flags in the generated LIR module.

LIR output shape

On success, the LIR module records:

```
source_kind=lat_module
module=&lt;Lat module name&gt;
card=
effect=none
boundary=lat_semantic_only
node_count=1 + declaration_count + clause_count
edge_count=declaration_count + clause_count + transition_count
binding_count=0
text_count=0
no_effect=1
execution_allowed=0
```

```
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

The implementation uses the Lat-specific LIR declaration node kinds and transition source edge kind while preserving the same no-effect metadata boundary.

Mapping summary

```
Lat module      -&gt; LIR module node
state           -&gt; LIR lat_state node
policy          -&gt; LIR lat_policy node
transition      -&gt; LIR lat_transition node with source-state metadata
assertion       -&gt; LIR lat_assertion node
effect          -&gt; LIR lat_effect_declaration node
field clause    -&gt; LIR field node
require clause  -&gt; LIR lat_requirement node metadata
ensure clause   -&gt; LIR lat_requirement node metadata
effect clause   -&gt; LIR effect node metadata
```

Edges use:

```
module contains declaration
declaration contains clause
transition transitions_from source state
```

Source-span behavior

Lat source spans are copied into the compatible LIR source-span shape:

```
start_offset
end_offset
start_line
start_column
end_line
end_column
```

The implementation does not invent byte offsets.

Report surface

latticra_lat_to_lir_report emits a deterministic bounded report beginning with:

```
LAT TO LIR LOWERING REPORT
```

The report records status, lowering error label, model error label, module name, declaration counts, model counts, clause counts, first lowered declaration metadata, first transition source index, node count, edge count, no-effect flags, and source-span fields.

The first lowered declaration metadata includes:

```
first_declaration_node_index
first_declaration_kind
first_declaration_name
first_declaration_source
first_declaration_parse_index
first_declaration_first_clause_index
first_declaration_clause_count
first_declaration_source_index
```

The companion diagnostic refinement can classify those lowering records as valid, parse, semantic, model, effect-check, capacity, LIR, or internal without changing lowering behavior. It also copies first-declaration and first-clause metadata from the lowering result for diagnostic report visibility.

The clause metadata refinement also reports first-clause node index, role, effect label, name, operator, and value so callers can confirm that lowered clause operators remain inspectable.

Small output buffers return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

and clear the buffer when possible.

Validation

Run:

```
sh scripts/test-lat-to-lir-lowering.sh
sh scripts/test-lat-to-lir-diagnostic-refinement.sh
```

The runner compiles with:

```
-std=c99 -Wall -Wextra -Werror -pedantic
```

and includes:

```
src/lat_parser.c
src/lat_semantic.c
src/lat_model.c
src/lir.c
src/lat_to_lir.c
tests/lat_to_lir_lowering_invariants.c
```

Invariants covered

The invariant suite checks:

```
foundation model lowers successfully
normalized model lowers through the model-aware entry point
source kind is lat_module
declaration and clause counts are preserved
model counts, first declaration metadata, and first transition source index are preserved
first clause role, operator, value, effect, and node index are preserved
node and edge counts are deterministic
module metadata is preserved
transition source metadata is preserved
parse failures are rejected
semantic failures are rejected
no-effect flags are preserved
lowering reports are deterministic
small report buffers are rejected and cleared
error labels are stable
```

Current evidence level

This is an L2 implementation slice:

```
contract + implementation plan + C implementation + invariant tests + workflow coverage
```

It is not a compiler, interpreter, runtime, package system, command surface, or operating-system surface.

Next implementation candidate

The next completed refinement after model-driven diagnostics is:

```
Lat-to-LIR clause metadata refinement
```

Later refinements may broaden clause reporting beyond the first lowered clause or integrate this metadata into higher-level diagnostics only through separate bounded slices.

Non-claims

This implementation does not execute Lat, compile Lat, interpret Lat, execute LIR, perform runtime behavior, mutate state, render L-UI, invoke Nucleus behavior, perform recovery behavior, interact with servers, perform self-update, touch hardware, provide boot readiness, or provide operating-system completeness.

docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION_PLAN.md

Source title: Latticra Lat-to-LIR Lowering Implementation Plan

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Latticra Lat-to-LIR Lowering Implementation Plan

Status: implementation planning contract Scope: exact public API, files, structs, errors, capacity accounting, node mapping, edge mapping, source-span mapping, report format, tests, compatibility expectations, and non-claims before Lat-to-LIR lowering code.

Purpose

This document defines the implementation plan for the first Lat-to-LIR lowering slice.

The existing Lat lane now has a bounded grammar parser and a bounded no-effect semantic validator. The next implementation step must not jump directly into execution or runtime behavior. It must first define the exact lowering API and deterministic mapping from semantically valid Lat metadata into the existing LIR module shape.

This document is plan-only. It does not implement Lat-to-LIR lowering.

Relationship to previous work

This plan depends on:

```
docs/LAT_TO_LIR_LOWERING_CONTRACT.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LAT_SEMANTIC_VALIDATION_CONTRACT.md
docs/LAT_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
docs/LAT_LANGUAGE_FOUNDATION_ANALYSIS.md
docs/LIR_SHAPE_CONTRACT.md
docs/LIR_SHAPE_IMPLEMENTATION_PLAN.md
docs/LIR_SHAPE_IMPLEMENTATION.md
include/latticra/lat_parser.h
include/latticra/lat_semantic.h
include/latticra/lir.h
fixtures/lat/foundation_model.lat
```

Those files remain the source of truth for parser metadata, semantic validation, LIR shape, no-effect flags, fixture behavior, and current report boundaries.

Implementation language decision

The first Lat-to-LIR lowering implementation should be written in C.

Reason:

- the existing Lat parser is C;
- the existing Lat semantic validator is C;
- the existing LIR shape is C;
- the lowering layer should remain bounded, deterministic, caller-buffer based, and no-effect;
- constrained C++ authority layers should consume stable C substrate data rather than define the first lowering surface.

Implementation files

The future implementation slice should add:

```
include/latticra/lat_to_lir.h
src/lat_to_lir.c
tests/lat_to_lir_lowering_invariants.c
scripts/test-lat-to-lir-lowering.sh
docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION.md
```

The implementation should reuse:

```
include/latticra/lat_parser.h
include/latticra/lat_semantic.h
include/latticra/lir.h
fixtures/lat/foundation_model.lat
```

The implementation should not add runtime execution, command behavior, file I/O, network I/O, host mutation, recovery behavior, hardware behavior, or a package/runtime surface.

Public API shape

Add public API names:

```
latticra_lat_to_lir_error_t
latticra_lat_to_lir_result_t
latticra_lat_to_lir_error_label
latticra_lir_lower_lat_module
latticra_lat_to_lir_report
```

Recommended function signatures:

```
const char *latticra_lat_to_lir_error_label(latticra_lat_to_lir_error_t error);

latticra_status_t latticra_lir_lower_lat_module(
    const latticra_lat_parse_result_t *parse_result,
    const latticra_lat_semantic_result_t *semantic_result,
    latticra_lir_module_t *module,
    latticra_lat_to_lir_result_t *result);

latticra_status_t latticra_lat_to_lir_report(
    const latticra_lat_to_lir_result_t *result,
    char *buffer,
    size_t buffer_len);
```

The lowering function should fill both the target latticra_lir_module_t and a smaller Lat-to-LIR summary result. The summary result exists so tests can assert lowering-specific counts and errors without parsing the generic LIR report.

Capacity constants

Add exact bounded constants:

```
LATTICRA_LAT_TO_LIR_REPORT_MAX 4096u
LATTICRA_LAT_TO_LIR_MODULE_NODE_COST 1u
LATTICRA_LAT_TO_LIR_DECLARATION_NODE_COST 1u
LATTICRA_LAT_TO_LIR_CLAUSE_NODE_COST 1u
LATTICRA_LAT_TO_LIR_DECLARATION_EDGE_COST 1u
LATTICRA_LAT_TO_LIR_CLAUSE_EDGE_COST 1u
LATTICRA_LAT_TO_LIR_TRANSITION_SOURCE_EDGE_COST 1u
```

Capacity accounting must be deterministic.

Required nodes:

```
1 + declaration_count + clause_count
```

Required edges:

```
declaration_count + clause_count + transition_count
```

If required nodes exceed LATTICRA_LIR_NODE_MAX, return capacity_exceeded.

If required edges exceed LATTICRA_LIR_EDGE_MAX, return capacity_exceeded.

Error enum

Add error enum values:

```
LATTICRA_LAT_TO_LIR_OK
LATTICRA_LAT_TO_LIR_NULL_ARGUMENT
LATTICRA_LAT_TO_LIR_PARSE_NOT_OK
LATTICRA_LAT_TO_LIR_SEMANTIC_NOT_OK
LATTICRA_LAT_TO_LIR_SEMANTIC_NOT_VALID
LATTICRA_LAT_TO_LIR_NO_EFFECT_VIOLATION
LATTICRA_LAT_TO_LIR_CAPACITY_EXCEEDED
LATTICRA_LAT_TO_LIR_UNSUPPORTED_DECLARATION_KIND
LATTICRA_LAT_TO_LIR_UNSUPPORTED_CLAUSE_KIND
LATTICRA_LAT_TO_LIR_UNSUPPORTED_EFFECT
LATTICRA_LAT_TO_LIR_UNSUPPORTED_LIR_SHAPE
LATTICRA_LAT_TO_LIR_INTERNAL_ERROR
```

Stable labels:

```
ok
null_argument
parse_not_ok
semantic_not_ok
semantic_not_valid
no_effect_violation
capacity_exceeded
unsupported_declaration_kind
unsupported_clause_kind
unsupported_effect
unsupported_lir_shape
internal_error
```

Result struct

Add a bounded summary result:

```
latticra_status_t status;
latticra_lat_to_lir_error_t error;
latticra_lat_source_span_t span;
char module_name[LATTICRA_LAT_NAME_MAX];
size_t declaration_count;
size_t state_count;
size_t policy_count;
size_t transition_count;
size_t assertion_count;
size_t effect_count;
size_t clause_count;
size_t node_count;
size_t edge_count;
size_t binding_count;
size_t text_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

The result must be fully initialized on success and failure.

Semantic prerequisite behavior

The lowering function must reject null arguments with LATTICRA_STATUS_NULL_ARGUMENT.

The lowering function must reject parser failure when:

```
parse_result-&gt;error != LATTICRA_LAT_PARSE_OK
```

The lowering function must reject semantic failure when:

```
semantic_result-&gt;error != LATTICRA_LAT_SEMANTIC_OK
semantic_result-&gt;semantic_valid != 1
```

The lowering function must reject no-effect flag violations when any of these differ from the no-effect boundary:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

No partial successful LIR module should be reported for failed prerequisites.

LIR module metadata mapping

On success, the future implementation should set:

```
module-&gt;status = LATTICRA_STATUS_OK
module-&gt;error = LATTICRA_LIR_OK
module-&gt;source_kind = LATTICRA_LIR_SOURCE_LAT_MODULE
module-&gt;module_name = parse_result-&gt;module.module_name
module-&gt;card_name = &quot;&quot;
module-&gt;effect = &quot;none&quot;
module-&gt;boundary = &quot;lat_semantic_only&quot;
module-&gt;source_span = parse_result-&gt;module.span converted to the LIR source-span shape
```

```
module-&gt;no_effect = 1
module-&gt;execution_allowed = 0
module-&gt;mutation_allowed = 0
module-&gt;server_allowed = 0
module-&gt;recovery_allowed = 0
module-&gt;hardware_allowed = 0
```

The implementation plan should not require card_name for Lat modules.

Source-span conversion

The current Lat source span and L-UI source span have compatible fields:

```
start_offset
end_offset
start_line
start_column
end_line
end_column
```

The implementation should use a private conversion helper from latticra_lat_source_span_t to the LIR span type used by latticra_lir_module_t.

The conversion must copy fields exactly and must not invent byte offsets.

Node mapping

The first implementation should not add new LIR node enum values.

Use the existing node kinds conservatively:

```
Lat module declaration -&gt; LATTICRA_LIR_NODE_MODULE
state declaration      -&gt; LATTICRA_LIR_NODE_FIELD with value=&quot;state&quot;
policy declaration     -&gt; LATTICRA_LIR_NODE_FIELD with value=&quot;policy&quot;
transition declaration -&gt; LATTICRA_LIR_NODE_FIELD with value=&quot;transition&quot;
assertion declaration  -&gt; LATTICRA_LIR_NODE_FIELD with value=&quot;assertion&quot;
effect declaration     -&gt; LATTICRA_LIR_NODE_EFFECT with value=&quot;effect&quot;
field clause           -&gt; LATTICRA_LIR_NODE_FIELD
require clause         -&gt; LATTICRA_LIR_NODE_BINDING with value=&quot;require&quot;
ensure clause          -&gt; LATTICRA_LIR_NODE_BINDING with value=&quot;ensure&quot;
effect clause          -&gt; LATTICRA_LIR_NODE_EFFECT
```

Declaration node fields:

```
name = declaration.name
value = declaration kind label
binding = declaration.source_name for transition declarations, otherwise empty
source_span = declaration.span
parent_index = module node index
```

Clause node fields:

```
name = clause.left
value = clause.right
binding = clause.keyword
source_span = clause.span
parent_index = declaration node index
```

Edge mapping

The first implementation should use existing edge kinds:

```
module contains declaration -&gt; LATTICRA_LIR_EDGE_CONTAINS
declaration contains clause -&gt; LATTICRA_LIR_EDGE_CONTAINS
transition references source state -&gt; LATTICRA_LIR_EDGE_BINDS
```

Transition source-reference edges should connect the transition declaration node to the referenced state declaration node when that state exists. Semantic validation is the prerequisite that guarantees the referenced state exists.

Policy and assertion clauses may remain contained child nodes in the first implementation. Additional annotates edges may wait for a later refinement.

Binding and text counts

The first Lat-to-LIR lowering implementation should not populate the LIR text table.

```
text_count=0
```

The first implementation may set binding_count equal to the number of require and ensure clause nodes, or it may keep binding_count=0 if binding references are reserved for L-UI only.

This plan chooses:

```
binding_count=0
```

Reason: Lat requirement and ensure clauses are metadata nodes in this first slice, not resolved L-UI binding references.

Effect handling

The first implementation accepts only semantic output that preserves effect none.

If any Lat effect value maps to LATTICRA_LAT_EFFECT_UNKNOWN or any non-none value reaches lowering, return:

```
unsupported_effect
```

This should be unreachable after successful semantic validation, but the lowering layer must still defend its own boundary.

Report format

latticra_lat_to_lir_report should emit:

```
LAT TO LIR LOWERING REPORT
status=&lt;integer-status&gt;
error=&lt;error-label&gt;
module=&lt;module-name&gt;
declaration_count=&lt;count&gt;
state_count=&lt;count&gt;
policy_count=&lt;count&gt;
transition_count=&lt;count&gt;
assertion_count=&lt;count&gt;
effect_count=&lt;count&gt;
clause_count=&lt;count&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
binding_count=&lt;count&gt;
text_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
span_start_line=&lt;line&gt;
span_start_column=&lt;column&gt;
span_end_line=&lt;line&gt;
span_end_column=&lt;column&gt;
```

Small buffers must return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the output buffer when possible.

Exact test list

The implementation slice should add tests for:

```
lat_to_lir_rejects_null_arguments
lat_to_lir_rejects_parse_error
lat_to_lir_rejects_semantic_error
lat_to_lir_rejects_semantic_not_valid
lat_to_lir_accepts_foundation_model
lat_to_lir_sets_source_kind_lat_module
lat_to_lir_preserves_module_name
lat_to_lir_preserves_declaration_counts
lat_to_lir_preserves_transition_source_metadata
lat_to_lir_preserves_effect_none
lat_to_lir_preserves_source_spans
```

```
lat_to_lir_preserves_no_effect_flags
lat_to_lir_counts_nodes_and_edges_deterministically
lat_to_lir_report_is_deterministic
lat_to_lir_report_rejects_small_buffer
lat_to_lir_error_labels_are_stable
lat_to_lir_does_not_execute_lat
lat_to_lir_does_not_mutate_state
lat_to_lir_is_deterministic
```

Test command

The implementation slice should add:

```
sh scripts/test-lat-to-lir-lowering.sh
```

The test should compile with:

```
-std=c99 -Wall -Wextra -Werror -pedantic
```

and include:

```
src/lat_parser.c
src/lat_semantic.c
src/lat_to_lir.c
src/lir.c
tests/lat_to_lir_lowering_invariants.c
```

Documentation update plan

The implementation slice should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

and add:

```
docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
Lat grammar parser behavior
Lat semantic validation behavior
L-UI parser behavior
L-UI semantic validation behavior
existing LIR lowering from L-UI
state lattice behavior
source-span behavior
no-effect flags
```

Implementation boundary

The implementation must not:

- execute Lat declarations;
- interpret transition behavior;
- mutate state;
- render L-UI;
- call Nucleus task execution;
- evaluate host state;
- write files;
- read files;
- open network connections;

- call update code;
- call recovery code;
- call hardware code;
- broaden accepted Lat syntax;
- weaken Lat semantic validation;
- weaken existing LIR shape behavior;
- invent source byte offsets;
- imply a compiler, interpreter, runtime, package manager, or operating-system surface.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-lat-to-lir-lowering-implementation-plan.sh
```

The guard is static. It does not implement Lat-to-LIR lowering.

Implementation gate

Lat-to-LIR lowering implementation code may be added only after this plan is merged and guarded.

Non-claims

This document does not implement Lat-to-LIR lowering, Lat execution, Lat compilation, Lat interpretation, LIR execution, command behavior, Nucleus task handling, state mutation, runtime behavior, recovery behavior, server interaction, self-update, hardware support, boot readiness, or operating-system completeness.

docs/LIR_REPORT_REFINEMENT.md

Source title: Latticra LIR Report Refinement

Lines: 140 | Words: 338 | SHA-256: 1dc4f06fb3a9906fe6ee0e33c1720a398dd4801dd04d161a4075d60e77e4afeb

Latticra LIR Report Refinement

Status: initial LIR report refinement implementation Scope: deterministic LIR report classification, graph shape labeling, edge-kind summary counts, no-effect-chain reporting, evidence-level reporting, invariant tests, guard coverage, and workflow coverage.

Purpose

This document records the LIR report refinement after the Lat semantic diagnostics refinement.

The goal is to make LIR reports easier to audit by stating whether a graph was materialized, rejected, invalid, or empty, which graph shape is present, and how edge kinds are distributed.

This refinement does not execute LIR, compile Lat, interpret Lat, mutate state, perform I/O, or provide runtime behavior.

Added LIR report metadata

The LIR module now records:

```
report_classification
shape_kind
contains_edge_count
```

```
binds_edge_count
annotates_edge_count
orders_before_edge_count
transitions_from_edge_count
no_effect_chain_ok
evidence_level
```

The deterministic LATTICRA LIR REPORT now emits those fields.

Report classifications

Initial report classifications:

```
empty
materialized
rejected
invalid
```

Shape kinds

Initial shape kinds:

```
unknown
l-ui-card-graph
lat-module-graph
internal-fixture-graph
```

Edge summary counts

The LIR report now records counts by edge kind:

```
contains_edge_count
binds_edge_count
annotates_edge_count
orders_before_edge_count
transitions_from_edge_count
```

These counts are derived from the existing edge array; they do not change the graph.

Evidence level

The LIR report uses a small deterministic evidence label:

```
0 -&gt; invalid or empty report evidence
1 -&gt; rejected report evidence or materialized graph with no-effect-chain concern
2 -&gt; materialized no-effect graph evidence
```

This is a project-internal evidence label, not a certification, production-readiness claim, or security guarantee.

Validation

Run:

```
sh scripts/test-lir-report-refinement.sh
sh scripts/test-lir-shape.sh
```

The focused invariant tests verify:

```
lir_report_refinement_labels_are_stable
lir_report_refinement_reports_materialized_l_ui_shape
lir_report_refinement_reports_semantic_rejection
lir_report_refinement_report_is_deterministic
```

Compatibility

This refinement preserves existing LIR behavior for:

```
L-UI AST lowering
Lat-to-LIR lowering
node counts
edge counts
binding counts
text counts
binding resolution
source spans
no-effect flags
```

```
small-buffer behavior
semantic-failure behavior
```

Non-claims

This report refinement does not provide:

```
LIR execution
Lat execution
Lat compilation
Lat interpretation
runtime behavior
command execution
state mutation
file I/O
network I/O
server interaction
hardware behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
certification
accreditation
operating-system completeness
```

docs/LIR_SHAPE_CONTRACT.md

Source title: Latticra LIR Shape Contract

Lines: 459 | Words: 1138 | SHA-256: 89228416e1ca3d820e16884aaa71f44d00a7f31908ae9eeb8569bfd3f6f0bb8c

Latticra LIR Shape Contract

Status: LIR shape contract Scope: first Latticra Intermediate Representation shape before semantic lowering, Lat integration, rendering, execution, or runtime behavior.

Purpose

This document defines the first contract for LIR: the Latticra Intermediate Representation.

LIR is the planned internal representation between validated declarations and later implementation stages.

The current repository has structural L-UI parsing, AST construction, source spans, string handling, source-buffer policies, and semantic validation. LIR shape planning comes next so future lowering has a stable target before code is added.

This document does not implement LIR.

The implementation plan is recorded in:

```
LIR_SHAPE_IMPLEMENTATION_PLAN.md
```

Relationship to previous work

This contract depends on:

```
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
docs/L_UI_SEMANTIC_VALIDATION_CONTRACT.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
src/l_ui_parser_semantic.c
```

Those files remain the source of truth for L-UI AST shape, semantic validation, source spans, no-effect metadata, and parser/semantic boundaries.

LIR role

LIR should be a small, explicit representation used after source parsing and semantic validation.

Recommended future pipeline:

```
source bytes
  -&gt; structural parser
  -&gt; AST construction
  -&gt; semantic validation
  -&gt; LIR lowering
  -&gt; future analysis, rendering, planning, or execution only if separately allowed
```

LIR should not bypass parser or semantic validation.

First LIR unit

The first LIR unit should be a module-like object representing one validated L-UI card.

Suggested root object:

```
latticra_lir_module_t
```

Initial fields:

```
status
source_kind
module_name
card_name
effect
boundary
node_count
edge_count
binding_count
text_count
source_span
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
```

Source kind

Initial source kind labels:

```
unknown
l_ui_card
lat_module
internal_fixture
```

Only l_ui_card is in scope for the first LIR lowering path.

Node kinds

Initial LIR node kinds:

```
module
card
rail
field
text
binding
effect
boundary
```

A future implementation plan may add more node kinds, but this contract keeps the first set minimal.

Node shape

Suggested node shape:

```
kind
```

```
name
value
binding
source_span
parent_index
first_child_index
child_count
flags
```

Rules:

- all nodes have deterministic indexes;
- parent/child relationships are index-based;
- source spans come from parser/AST metadata;
- string values use explicit lengths when needed;
- no node is executable by default.

Edge shape

Suggested edge shape:

```
from_index
to_index
edge_kind
source_span
```

Initial edge kinds:

```
contains
binds
annotates
orders_before
```

Edges are metadata only. They do not execute behavior.

Binding shape

Suggested binding shape:

```
field_node_index
binding_target
binding_prefix
source_span
resolved_kind
```

Initial binding prefixes:

```
state
preview
```

Initial resolved kinds:

```
state_value
preview_value
unsupported
```

Bindings remain symbolic. LIR must not evaluate host state or preview state.

Text shape

Text values should carry explicit byte lengths.

Suggested text shape:

```
text_node_index
value
value_len
escaped_value
source_span
```

value_len remains authoritative for embedded NUL values.

Raw C-string compatibility fields must not be the assertion target for embedded NUL values.

Effect and boundary shape

LIR should carry effect and boundary metadata from validated input.

Initial allowed values:

```
effect=none
boundary=preview_only
```

No other effect or boundary should be accepted into first LIR shape without a separate contract.

Semantic prerequisite

LIR lowering may only operate after semantic validation succeeds:

```
semantic_error=ok
parser_error=ok
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

If semantic validation failed, LIR lowering should not materialize a partial LIR.

Source-span behavior

Every LIR node that comes from source should carry a source span from parser or AST metadata.

Rules:

- module/card spans come from card spans;
- rail spans come from rail spans;
- field spans come from field spans;
- binding spans come from binding spans;
- text spans come from text spans;
- effect and boundary spans may use card metadata until more specific spans exist;
- LIR must not invent byte positions.

Report model

A future LIR report should be deterministic and bounded.

Suggested fields:

```
LATTICRA LIR REPORT
status=&lt;status&gt;
source_kind=&lt;kind&gt;
module=&lt;name&gt;
card=&lt;name&gt;
effect=&lt;effect&gt;
boundary=&lt;boundary&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
binding_count=&lt;count&gt;
text_count=&lt;count&gt;
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
```

Detailed reports may later include node and edge tables.

Capacity policy

The first LIR implementation should use fixed capacities aligned with existing AST limits.

Suggested capacities:

```
LATTICRA_LIR_NODE_MAX &gt;= LATTICRA_L_UI_AST_RAIL_MAX + LATTICRA_L_UI_AST_FIELD_MAX +
LATTICRA_L_UI_AST_TEXT_MAX + card/effect/boundary overhead
LATTICRA_LIR_EDGE_MAX bounded and deterministic
LATTICRA_LIR_BINDING_MAX bounded and deterministic
LATTICRA_LIR_TEXT_MAX bounded and deterministic
```

A future implementation plan must name exact constants before code is added.

The exact first implementation-plan constants are recorded in:

```
LIR_SHAPE_IMPLEMENTATION_PLAN.md
```

Error model

LIR should have a separate error space from parser and semantic errors.

Recommended future labels:

```
ok
null_argument
semantic_failed
capacity_exceeded
unsupported_source_kind
unsupported_node_kind
unsupported_effect
unsupported_boundary
internal_error
```

Exact enum names should be defined in the implementation plan.

No-effect rule

LIR shape and reports are metadata-only.

They must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

LIR work must not render UI, execute bindings, evaluate host state, call Nucleus task code, write files, read files, open network connections, call server code, call update code, call recovery code, or touch hardware.

Compatibility expectations

LIR shape work must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
existing semantic validation behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
source-span byte offset behavior
no-effect flags
current accepted fixture counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
```

```
boundary=preview_only
```

Future implementation gate

LIR implementation must not begin until a separate implementation plan defines:

1. public API shape;
2. LIR struct names;
3. capacity constants;
4. node enum labels;
5. edge enum labels;
6. error enum labels;
7. report format;
8. semantic prerequisite behavior;
9. source-span mapping;
10. exact tests;
11. compatibility expectations;
12. non-claims.

That plan is recorded in:

```
LIR_SHAPE_IMPLEMENTATION_PLAN.md
```

Future test list

A future implementation plan should include tests for:

```
lir_shape_accepts_semantically_valid_l_ui_fixture
lir_shape_rejects_parser_failed_ast
lir_shape_rejects_semantic_failed_ast
lir_shape_preserves_card_metadata
lir_shape_preserves_rail_nodes
lir_shape_preserves_field_nodes
lir_shape_preserves_binding_nodes
lir_shape_preserves_text_nodes_with_lengths
lir_shape_preserves_source_spans
lir_shape_preserves_no_effect_flags
lir_shape_report_is_deterministic
lir_shape_report_rejects_small_buffer
lir_shape_does_not_change_ast_report
lir_shape_does_not_change_semantic_report
lir_shape_does_not_change_escaped_x00_acceptance
lir_shape_does_not_change_literal_nul_rejection
lir_shape_is_deterministic
```

The implementation-plan test list is now recorded in:

```
LIR_SHAPE_IMPLEMENTATION_PLAN.md
```

Forbidden behavior

LIR shape work must not:

- lower invalid semantic input;
- execute bindings;
- evaluate host state;
- call Nucleus task execution;
- render L-UI;
- execute Lat;

- mutate state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- call hardware code;
- broaden accepted syntax;
- weaken parser diagnostics;
- weaken semantic validation;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;
- make raw C-string fields authoritative for embedded NUL values.

Current validation command

This contract is guarded by:

```
sh scripts/test-lir-shape-contract.sh
```

The guard is static. It does not implement LIR.

Non-claims

This document does not implement LIR, LIR lowering, L-UI rendering, Lat execution, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LIR_SHAPE_IMPLEMENTATION.md

Source title: Latticra LIR Shape Implementation

Lines: 320 | Words: 716 | SHA-256: 81d7fd2ebf1e18b8a13f3423824cf3cb1097ca5261734fbbe2fd5425355f79a8

Latticra LIR Shape Implementation

Status: initial implementation contract Scope: bounded no-effect LIR shape API, lowering, report, and invariants for semantically valid L-UI ASTs.

Purpose

This implementation adds the first LIR shape for Latticra.

LIR is the Latticra Intermediate Representation. The first implementation lowers only a semantically valid L-UI AST into a bounded metadata graph.

This implementation does not execute LIR, render L-UI, lower Lat, call Nucleus task behavior, read files, write files, open network connections, mutate state, or touch hardware.

Implementation files

```
include/latticra/lir.h
src/lir.c
tests/lir_shape_invariants.c
scripts/test-lir-shape.sh
.github/workflows/c.yml
```

Related active files:

```
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
src/l_ui_parser_semantic.c
docs/LIR_SHAPE_CONTRACT.md
docs/LIR_SHAPE_IMPLEMENTATION_PLAN.md
```

Public API

The public API adds:

```
latticra_lir_source_kind_t
latticra_lir_node_kind_t
latticra_lir_edge_kind_t
latticra_lir_resolved_binding_kind_t
latticra_lir_error_t
latticra_lir_node_t
latticra_lir_edge_t
latticra_lir_binding_ref_t
latticra_lir_text_ref_t
latticra_lir_module_t
latticra_lir_error_label
latticra_lir_source_kind_label
latticra_lir_node_kind_label
latticra_lir_edge_kind_label
latticra_lir_resolved_binding_kind_label
latticra_lir_lower_l_ui_ast
latticra_lir_report
```

The Lat-to-LIR clause metadata refinement adds operator_text to latticra_lir_node_t so Lat-derived clause nodes can preserve =, ==, and != metadata without overloading values or bindings. L-UI lowering leaves this field empty.

Capacity constants

The first LIR implementation uses exact bounded constants:

```
LATTICRA_LIR_NAME_MAX 64u
LATTICRA_LIR_VALUE_MAX 128u
LATTICRA_LIR_BINDING_MAX 96u
LATTICRA_LIR_NODE_MAX 96u
LATTICRA_LIR_EDGE_MAX 128u
LATTICRA_LIR_BINDING_REF_MAX 32u
LATTICRA_LIR_TEXT_MAX 16u
LATTICRA_LIR_REPORT_MAX 8192u
```

Lowering prerequisite

latticra_lir_lower_l_ui_ast requires:

```
ast != NULL
semantic != NULL
module != NULL
ast-&gt;parse_result.error == LATTICRA_L_UI_PARSE_OK
semantic-&gt;error == LATTICRA_L_UI_SEMANTIC_OK
semantic-&gt;parser_error == LATTICRA_L_UI_PARSE_OK
semantic-&gt;no_effect == 1
semantic-&gt;execution_allowed == 0
semantic-&gt;mutation_allowed == 0
semantic-&gt;server_allowed == 0
semantic-&gt;recovery_allowed == 0
semantic-&gt;hardware_allowed == 0
```

If the prerequisite fails, it does not materialize a partial LIR graph.

Expected failure state:

```
module.error=semantic_failed
node_count=0
```

```
edge_count=0
binding_count=0
text_count=0
```

Deterministic node plan

Successful L-UI lowering currently produces deterministic nodes:

```
0: module
1: card
2-10: rails in current L-UI order
11-33: fields in current AST order
34-35: text nodes
36-58: binding nodes
59: effect
60: boundary
```

Expected successful counts:

```
node_count=61
edge_count=60
binding_count=23
text_count=2
```

Source-span mapping

LIR nodes use existing parser/AST spans:

```
module -&gt; ast.card.span
card -&gt; ast.card.span
rail -&gt; ast.rails[index].span
field -&gt; ast.fields[index].span
binding -&gt; ast.fields[index].binding_span
text -&gt; ast.texts[index].span
effect -&gt; ast.card.span
boundary -&gt; ast.card.span
```

LIR does not invent byte offsets.

Text and binding preservation

Text references preserve explicit byte lengths:

```
text_node_index
value
value_len
escaped_value
source_span
```

value_len is authoritative for embedded NUL values.

Binding references preserve symbolic binding metadata:

```
field_node_index
binding_target
binding_prefix
source_span
resolved_kind
```

Initial resolved binding kinds are:

```
state_value
preview_value
unsupported
```

Bindings remain symbolic and are not executed or evaluated.

Report format

latticra_lir_report emits a deterministic bounded report beginning with:

```
LATTICRA LIR REPORT
```

and including:

```
status
error
source_kind
module
```

```
card
effect
boundary
node_count
edge_count
binding_count
text_count
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
span_start_offset
span_end_offset
span_start_line
span_start_column
span_end_line
span_end_column
```

Small output buffers return:

```
LATTICRA_STATUS_BUFFER_TOO_SMALL
```

and clear the output buffer.

No-effect boundary

LIR shape lowering and reports are metadata-only.

They preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Test command

Run:

```
sh scripts/test-lir-shape.sh
```

The main C workflow runs this check after the LIR shape implementation-plan guard and before the string-literal escape guards.

Required invariants

The LIR shape tests verify:

```
lir_shape_accepts_semantically_valid_l_ui_fixture
lir_shape_rejects_parser_failed_ast
lir_shape_rejects_semantic_failed_ast
lir_shape_preserves_card_metadata
lir_shape_preserves_rail_nodes
lir_shape_preserves_field_nodes
lir_shape_preserves_binding_nodes
lir_shape_preserves_text_nodes_with_lengths
lir_shape_preserves_source_spans
lir_shape_preserves_no_effect_flags
lir_shape_report_is_deterministic
lir_shape_report_rejects_small_buffer
lir_shape_does_not_change_ast_report
lir_shape_does_not_change_semantic_report
lir_shape_does_not_change_escaped_x00_acceptance
lir_shape_does_not_change_literal_nul_rejection
lir_shape_error_labels_are_stable
lir_shape_kind_labels_are_stable
lir_shape_is_deterministic
```

Compatibility

This implementation does not change:

```
existing parser behavior
existing parser diagnostic codes
```

```
existing AST construction behavior
existing detailed AST report behavior
existing semantic validation behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
source-span byte offset behavior
current accepted fixture counts
```

The first accepted fixture still reports:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Current evidence level

This implementation is an L2 tested intermediate-representation shape for semantically valid L-UI AST fixtures.

It is not LIR execution, L-UI rendering, Lat execution, command behavior, Nucleus task execution, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

Next implementation step

The next implementation candidate is:

```
Lat language grammar contract
```

That future work should define the first Lat / Latticra Programming Language grammar before implementation.

Non-claims

This document and implementation do not implement LIR execution, L-UI rendering, Lat execution, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/LIR_SHAPE_IMPLEMENTATION_PLAN.md

Source title: Latticra LIR Shape Implementation Plan

Lines: 634 | Words: 1339 | SHA-256: a3c30258f8a31503aeb4c2ef6d0991006b49c36b56d079fc205337ba6af89b42

Latticra LIR Shape Implementation Plan

Status: implementation planning contract Scope: exact API, structs, capacities, enum labels, report format, semantic prerequisite behavior, source-span mapping, tests, compatibility expectations, and non-claims before LIR code.

Purpose

This document defines the implementation plan for the first LIR shape.

The LIR shape contract is already merged and guarded. This plan names exact public API additions, struct names, capacity constants, node enum labels, edge enum labels, error enum labels, report format, semantic prerequisite behavior, source-span mapping, test files, documentation updates, compatibility expectations, and non-claims required before LIR code is added.

This document does not implement LIR.

Relationship to previous work

This plan depends on:

```
docs/LIR_SHAPE_CONTRACT.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md
docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION_PLAN.md
docs/L_UI_PARSER_AST_IMPLEMENTATION.md
docs/L_UI_AST_DETAILED_REPORT_IMPLEMENTATION.md
docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md
include/latticra/l_ui_parser.h
src/l_ui_parser_ast.c
src/l_ui_parser_semantic.c
```

Those files remain the source of truth for L-UI AST shape, semantic validation, source spans, no-effect metadata, and parser/semantic boundaries.

Implementation language decision

The first LIR shape implementation should be in C.

Reason:

- current L-UI parser, AST, semantic validation, and reports are C;
- the first LIR lowering path consumes C structs;
- tests and runners use C and POSIX shell;
- the first LIR layer should stay deterministic, bounded, and no-effect.

Implementation files

The implementation PR should modify:

```
include/latticra/lir.h
include/latticra/l_ui_parser.h
src/lir.c
.github/workflows/c.yml
```

The implementation PR should add:

```
docs/LIR_SHAPE_IMPLEMENTATION.md
tests/lir_shape_invariants.c
scripts/test-lir-shape.sh
```

The implementation should not add rendering, Lat execution, L-UI execution, Nucleus task execution, file I/O, network I/O, mutation, recovery behavior, update behavior, or hardware behavior.

Public header plan

Add a new public LIR header:

```
include/latticra/lir.h
```

Include it from:

```
include/latticra/l_ui_parser.h
```

only if needed for public lowering declarations.

The header should depend only on stable shared types and L-UI parser types needed for source spans and semantic input.

Capacity constants

Add exact bounded constants:

```
LATTICRA_LIR_NAME_MAX 64u
LATTICRA_LIR_VALUE_MAX 128u
LATTICRA_LIR_BINDING_MAX 96u
LATTICRA_LIR_NODE_MAX 96u
LATTICRA_LIR_EDGE_MAX 128u
LATTICRA_LIR_BINDING_REF_MAX 32u
```

```
LATTICRA_LIR_TEXT_MAX 16u
LATTICRA_LIR_REPORT_MAX 8192u
```

Rationale:

- current AST rail max is 16;
- current AST field max is 64;
- current AST text max is 16;
- 96 nodes gives room for module, card, rails, fields, texts, bindings, effect, and boundary metadata while remaining bounded.

Public API shape

Add public API names:

```
latticra_lir_source_kind_t
latticra_lir_node_kind_t
latticra_lir_edge_kind_t
latticra_lir_resolved_binding_kind_t
latticra_lir_error_t
latticra_lir_node_t
latticra_lir_edge_t
latticra_lir_binding_ref_t
latticra_lir_text_ref_t
latticra_lir_module_t
latticra_lir_error_label
latticra_lir_source_kind_label
latticra_lir_node_kind_label
latticra_lir_edge_kind_label
latticra_lir_resolved_binding_kind_label
latticra_lir_lower_l_ui_ast
latticra_lir_report
```

Recommended function signatures:

```
const char *latticra_lir_error_label(latticra_lir_error_t error);
const char *latticra_lir_source_kind_label(latticra_lir_source_kind_t kind);
const char *latticra_lir_node_kind_label(latticra_lir_node_kind_t kind);
const char *latticra_lir_edge_kind_label(latticra_lir_edge_kind_t kind);
const char *latticra_lir_resolved_binding_kind_label(latticra_lir_resolved_binding_kind_t kind);

latticra_status_t latticra_lir_lower_l_ui_ast(
    const latticra_l_ui_ast_result_t *ast,
    const latticra_l_ui_semantic_result_t *semantic,
    latticra_lir_module_t *module);

latticra_status_t latticra_lir_report(
    const latticra_lir_module_t *module,
    char *buffer,
    size_t buffer_len);
```

Source kind enum

Add source kind enum values:

```
LATTICRA_LIR_SOURCE_UNKNOWN
LATTICRA_LIR_SOURCE_L_UI_CARD
LATTICRA_LIR_SOURCE_LAT_MODULE
LATTICRA_LIR_SOURCE_INTERNAL_FIXTURE
```

Stable labels:

```
unknown
l_ui_card
lat_module
internal_fixture
```

Only LATTICRA_LIR_SOURCE_L_UI_CARD should be produced by the first implementation.

Node kind enum

Add node kind enum values:

```
LATTICRA_LIR_NODE_MODULE
LATTICRA_LIR_NODE_CARD
```

```
LATTICRA_LIR_NODE_RAIL
LATTICRA_LIR_NODE_FIELD
LATTICRA_LIR_NODE_TEXT
LATTICRA_LIR_NODE_BINDING
LATTICRA_LIR_NODE_EFFECT
LATTICRA_LIR_NODE_BOUNDARY
LATTICRA_LIR_NODE_UNKNOWN
```

Stable labels:

```
module
card
rail
field
text
binding
effect
boundary
unknown
```

Edge kind enum

Add edge kind enum values:

```
LATTICRA_LIR_EDGE_CONTAINS
LATTICRA_LIR_EDGE_BINDS
LATTICRA_LIR_EDGE_ANNOTATES
LATTICRA_LIR_EDGE_ORDERS_BEFORE
LATTICRA_LIR_EDGE_UNKNOWN
```

Stable labels:

```
contains
binds
annotates
orders_before
unknown
```

Resolved binding kind enum

Add resolved binding kind enum values:

```
LATTICRA_LIR_BINDING_STATE_VALUE
LATTICRA_LIR_BINDING_PREVIEW_VALUE
LATTICRA_LIR_BINDING_UNSUPPORTED
```

Stable labels:

```
state_value
preview_value
unsupported
```

Bindings remain symbolic and must not evaluate host state.

Error enum

Add LIR error enum values:

```
LATTICRA_LIR_OK
LATTICRA_LIR_NULL_ARGUMENT
LATTICRA_LIR_SEMANTIC_FAILED
LATTICRA_LIR_CAPACITY_EXCEEDED
LATTICRA_LIR_UNSUPPORTED_SOURCE_KIND
LATTICRA_LIR_UNSUPPORTED_NODE_KIND
LATTICRA_LIR_UNSUPPORTED_EFFECT
LATTICRA_LIR_UNSUPPORTED_BOUNDARY
LATTICRA_LIR_INTERNAL_ERROR
```

Stable labels:

```
ok
null_argument
semantic_failed
capacity_exceeded
unsupported_source_kind
unsupported_node_kind
unsupported_effect
unsupported_boundary
internal_error
```

Node struct

Add node struct:

```
latticra_lir_node_kind_t kind;
char name[LATTICRA_LIR_NAME_MAX];
char value[LATTICRA_LIR_VALUE_MAX];
char binding[LATTICRA_LIR_BINDING_MAX];
latticra_l_ui_source_span_t source_span;
size_t parent_index;
size_t first_child_index;
size_t child_count;
unsigned int flags;
```

Rules:

- indexes are deterministic;
- parent and child references use indexes;
- source spans are copied from AST metadata;
- values remain bounded;
- nodes are metadata-only.

Edge struct

Add edge struct:

```
size_t from_index;
size_t to_index;
latticra_lir_edge_kind_t edge_kind;
latticra_l_ui_source_span_t source_span;
```

Edges are metadata only and do not execute behavior.

Binding reference struct

Add binding reference struct:

```
size_t field_node_index;
char binding_target[LATTICRA_LIR_BINDING_MAX];
char binding_prefix[LATTICRA_LIR_NAME_MAX];
latticra_l_ui_source_span_t source_span;
latticra_lir_resolved_binding_kind_t resolved_kind;
```

Only state and preview prefixes are expected from current semantic validation.

Text reference struct

Add text reference struct:

```
size_t text_node_index;
char value[LATTICRA_LIR_VALUE_MAX];
size_t value_len;
char escaped_value[LATTICRA_LIR_VALUE_MAX];
latticra_l_ui_source_span_t source_span;
```

value_len is authoritative.

Module struct

Add module struct:

```
latticra_status_t status;
latticra_lir_error_t error;
latticra_lir_source_kind_t source_kind;
char module_name[LATTICRA_LIR_NAME_MAX];
char card_name[LATTICRA_LIR_NAME_MAX];
char effect[LATTICRA_LIR_NAME_MAX];
char boundary[LATTICRA_LIR_NAME_MAX];
latticra_l_ui_source_span_t source_span;
latticra_lir_node_t nodes[LATTICRA_LIR_NODE_MAX];
latticra_lir_edge_t edges[LATTICRA_LIR_EDGE_MAX];
latticra_lir_binding_ref_t bindings[LATTICRA_LIR_BINDING_REF_MAX];
```

```
latticra_lir_text_ref_t texts[LATTICRA_LIR_TEXT_MAX];
size_t node_count;
size_t edge_count;
size_t binding_count;
size_t text_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

The module must be fully initialized on success and on failure.

L-UI lowering behavior

latticra_lir_lower_l_ui_ast should require:

```
ast != NULL
semantic != NULL
module != NULL
ast-&gt;parse_result.error == LATTICRA_L_UI_PARSE_OK
semantic-&gt;error == LATTICRA_L_UI_SEMANTIC_OK
semantic-&gt;parser_error == LATTICRA_L_UI_PARSE_OK
semantic-&gt;no_effect == 1
semantic-&gt;execution_allowed == 0
semantic-&gt;mutation_allowed == 0
semantic-&gt;server_allowed == 0
semantic-&gt;recovery_allowed == 0
semantic-&gt;hardware_allowed == 0
```

If the prerequisite fails, it should return LATTICRA_STATUS_OK with:

```
module.error = LATTICRA_LIR_SEMANTIC_FAILED
module.node_count = 0
module.edge_count = 0
module.binding_count = 0
module.text_count = 0
```

Initial deterministic node plan

The first implementation should produce deterministic nodes:

```
0: module
1: card
2-10: rails in current L-UI order
11-33: fields in current AST order
34-35: text nodes
36-58: binding nodes
59: effect
60: boundary
```

Expected successful counts:

```
node_count=61
binding_count=23
text_count=2
```

Edge count may be implementation-defined but must be deterministic and bounded. The implementation plan sets expected minimum:

```
edge_count &gt;= 60
```

Report format

latticra_lir_report should produce:

```
LATTICRA LIR REPORT
status=&lt;integer-status&gt;
error=&lt;lir-error-label&gt;
source_kind=&lt;source-kind-label&gt;
module=&lt;module-name&gt;
card=&lt;card-name&gt;
effect=&lt;effect&gt;
boundary=&lt;boundary&gt;
node_count=&lt;count&gt;
edge_count=&lt;count&gt;
binding_count=&lt;count&gt;
text_count=&lt;count&gt;
```

```
no_effect=&lt;0|1&gt;
execution_allowed=&lt;0|1&gt;
mutation_allowed=&lt;0|1&gt;
server_allowed=&lt;0|1&gt;
recovery_allowed=&lt;0|1&gt;
hardware_allowed=&lt;0|1&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
span_start_line=&lt;line&gt;
span_start_column=&lt;column&gt;
span_end_line=&lt;line&gt;
span_end_column=&lt;column&gt;
```

Small output buffers should return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the buffer.

Source-span mapping

Use these source-span mappings:

```
module -&gt; ast.card.span
card -&gt; ast.card.span
rail -&gt; ast.rails[index].span
field -&gt; ast.fields[index].span
binding -&gt; ast.fields[index].binding_span
text -&gt; ast.texts[index].span
effect -&gt; ast.card.span
boundary -&gt; ast.card.span
```

LIR must not invent byte offsets.

No-effect preservation

LIR is metadata-only.

It must preserve:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Exact implementation test list

The implementation PR should include tests for:

```
lir_shape_accepts_semantically_valid_l_ui_fixture
lir_shape_rejects_parser_failed_ast
lir_shape_rejects_semantic_failed_ast
lir_shape_preserves_card_metadata
lir_shape_preserves_rail_nodes
lir_shape_preserves_field_nodes
lir_shape_preserves_binding_nodes
lir_shape_preserves_text_nodes_with_lengths
lir_shape_preserves_source_spans
lir_shape_preserves_no_effect_flags
lir_shape_report_is_deterministic
lir_shape_report_rejects_small_buffer
lir_shape_does_not_change_ast_report
lir_shape_does_not_change_semantic_report
lir_shape_does_not_change_escaped_x00_acceptance
lir_shape_does_not_change_literal_nul_rejection
lir_shape_error_labels_are_stable
lir_shape_kind_labels_are_stable
lir_shape_is_deterministic
```

Test file plan

Add:

```
tests/lir_shape_invariants.c
scripts/test-lir-shape.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-lir-shape-contract.sh
```

and before:

```
sh scripts/test-l-ui-string-literal-escape-contract.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/LIR_SHAPE_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/LIR_SHAPE_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
existing parser behavior
existing parser diagnostic codes
existing AST construction behavior
existing detailed AST report behavior
existing semantic validation behavior
escaped decoded NUL acceptance
literal source-buffer NUL rejection
source-span byte offset behavior
current accepted fixture counts
```

The first accepted fixture should still report:

```
rail_count=9
field_count=23
text_count=2
effect=none
boundary=preview_only
```

Forbidden implementation behavior

LIR shape implementation must not:

- lower invalid semantic input;
- execute bindings;
- evaluate host state;
- call Nucleus task execution;
- render L-UI;
- execute Lat;
- mutate state;
- write files;
- read files;
- open network connections;
- call server code;
- call update code;

- call recovery code;
- call hardware code;
- broaden accepted syntax;
- weaken parser diagnostics;
- weaken semantic validation;
- accept literal source-buffer NUL;
- remove escaped decoded NUL support;
- make raw C-string fields authoritative for embedded NUL values.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-lir-shape-implementation-plan.sh
```

The guard is static. It does not implement LIR.

Implementation gate

LIR shape implementation code may be added only after this plan is merged.

Non-claims

This document does not implement LIR, LIR lowering, L-UI rendering, Lat execution, command behavior, Nucleus task handling, live movement, origin mutation, recovery behavior, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/POLYMORPHIC_LANGUAGE_STRATEGY.md

Source title: Polymorphic Language Strategy

Lines: 191 | Words: 682 | SHA-256: 4ba630c06fdc485945c7206d426f21a018545f31e01636cc3cc99de6afc2122a

Polymorphic Language Strategy

Status: active architecture strategy Scope: language placement rules for Latticra, Lat, LIR, Latticra Seal, C, C++, and Rust. This document does not implement runtime execution, runtime authority, policy enforcement, cryptographic verification, operating-system behavior, production readiness, or external endorsement.

Purpose

Latticra should be polymorphic in its use of programming languages without becoming undisciplined.

The project should use each language where it is strongest, keep cross-language interfaces explicit, and avoid language ideology. The goal is a serious systems architecture where C, constrained C++, Rust, Lat, and LIR each have a bounded role.

Core rule

```
C owns the substrate.
C++ owns governed architecture and policy modeling.
Rust owns safe tooling and high-risk input surfaces.
Lat owns Latticra-native declaration and system intent.
LIR owns bounded intermediate representation and evidence-bearing shape.
```

C role

C is the substrate language for Latticra.

Use C for:

```
stable ABI boundaries
small no-effect runtime primitives
state and boundary fixtures
Lat-to-LIR lowering foundations
LIR shape and metadata surfaces
runtime-boundary classification
Seal metadata structs
capability records
evidence receipts
host-neutral invariant tests
```

C code should remain small, auditable, deterministic, and boring. It should not hide policy complexity, own broad orchestration, or become a dumping ground for every subsystem.

C++ role

Constrained C++ is the architecture and policy language for Latticra.

Use C++ for:

```
policy graph modeling
capability graph modeling
Seal decision modeling
tool-boundary planning
Lat semantic architecture experiments
higher-order validation engines
audit and planning reports
future bounded architecture simulators
```

C++ must remain governed by the existing constrained C++ authority-layer direction. It should not bypass C effect gates, grant runtime authority, execute tools, perform host mutation, perform network behavior, or introduce unrestricted ownership and allocation patterns.

Rust role

Rust is the safety-focused tooling and verification language for Latticra.

Use Rust for:

```
CLI tooling
manifest parsing
evidence report generation
signature metadata tooling
future cryptographic verification wrappers
serialization and deserialization
untrusted input handling
safe developer tools
future network-facing user-space tools
```

Rust should not be used to erase the C/C++ architecture. It should protect risky input surfaces and developer-facing tools where memory safety and parser safety are especially valuable.

Lat role

Lat is the Latticra-native language direction.

Use Lat for:

```
system declarations
capability annotations
Seal intent declarations
policy-aware function shape
runtime-boundary descriptions
future Latticra-native system contracts
```

Lat is not currently a compiler product, interpreter product, execution engine, production language, or operating-system replacement layer. Its current valid path is parsing, semantic validation, diagnostics, metadata lowering, and no-effect LIR representation.

LIR role

LIR is the bounded intermediate representation.

Use LIR for:

```
Lat declaration shape
metadata-bearing graph representation
operator-visible reports
semantic lowering evidence
no-effect transition between language and runtime-boundary planning
```

LIR should remain evidence-bearing and bounded before execution is considered.

Boundary interface policy

Cross-language boundaries should use explicit contracts.

Preferred boundary shape:

```
plain C ABI
fixed-capacity structs where practical
explicit ownership
explicit error codes
no hidden runtime dependency
no implicit authority grant
no effect without a named gate
```

Every new cross-language boundary must name:

1. caller language;
2. callee language;
3. ownership rules;
4. error behavior;
5. effect posture;
6. authority posture;
7. test or guard path.

Layer map

```
Lat
  declaration and system intent layer

LIR
  bounded metadata and intermediate representation layer

C substrate
  ABI, fixtures, no-effect primitives, runtime-boundary records

Constrained C++
  policy graph, authority planning, capability graph, decision models

Rust tools
  safe CLI, manifest parsing, evidence processing, verification wrappers

Fedora/Linux
  current host-facing validation lane and integration target
```

Non-claims

This strategy does not claim:

```
Rust implementation present for all named Rust areas
C++ implementation present for all named C++ areas
Lat execution
```

```
LIR execution
runtime authority
policy enforcement
cryptographic verification
tool execution
MCP implementation
production readiness
operating-system replacement
```

Implementation rule

A language should be introduced into a subsystem only when that subsystem has a contract, a boundary, and a test path.

The acceptable question is not "which language is best overall?"

The acceptable question is:

```
Which language is safest, clearest, and most auditable for this specific layer?
```

docs/UI_TERMINAL_LANGUAGE.md

Source title: Latticra UI Terminal Language

Lines: 145 | Words: 405 | SHA-256: 7b5016e4068d2e114c83802bde7a03c74d45595937a1b7fda81735a1d7ac6a59

Latticra UI Terminal Language

Status: initial L-UI design Scope: terminal/operator UI language direction, rails, reports, and non-claims.

Purpose

Latticra should define its terminal and operator interface language before implementation code expands.

The terminal UI should not be a pile of hard-coded strings. It should become a controlled, declarative surface that can render state, effects, gates, evidence, and supervisor reports clearly.

Name

The planned UI language or dialect is:

```
L-UI
```

L-UI is part of the Latticra language family, but it should remain narrower than full Lat.

Role

L-UI describes:

- operator rails;
- state cards;
- evidence reports;
- update reports;
- effect-gate reports;
- policy-denial messages;
- supervisor status panels;
- terminal-safe layouts.

Design principles

1. Declarative before imperative.
2. Terminal-safe before graphical.
3. Text-first before animation.
4. Readable on narrow terminals.
5. Copy-safe command output.
6. No hidden mutation.
7. Effect-free by default.
8. Accessibility and fallback modes from the beginning.

Initial surface vocabulary

```
rail
card
row
field
status
warning
reason
gate
trace
breadcrumb
state
```

Example direction

Example syntax only:

```
card StateOverview {
  row &quot;origin&quot; = state.origin
  row &quot;route&quot; = state.route
  row &quot;axis&quot; = state.axis
  row &quot;host effect&quot; = state.host_effect
  row &quot;external effect&quot; = state.external_effect
}
```

This syntax is not implemented yet.

Operator rails

Initial rail categories:

```
top rail      -&gt; identity, target, architecture, channel
state rail    -&gt; origin, route, axis, path, breadcrumb, trace
safety rail   -&gt; health, risk, lock, gates, effects
bottom rail   -&gt; result, reason, next action, evidence level
```

Required report fields

Every report surface that describes a request should eventually include:

```
request_id
source
effect
policy_result
reason
evidence_level
rollback_state
host_effect
external_effect
```

L-UI and Nucleus

Nucleus supplies state. L-UI renders it.

L-UI should not own policy decisions, effect gates, update behavior, server interaction, or hidden task execution.

L-UI and Lat

Lat may eventually generate or validate L-UI documents.

Early L-UI should be static and fixture-backed.

Color and accessibility

L-UI must support:

- plain text mode;
- no-color mode;
- narrow terminal mode;
- screen-reader-friendly output where possible;
- clear labels independent of color.

First implementation target

The first L-UI implementation target should be a static report fixture for the state lattice.

It must be:

```
deterministic
ASCII-safe
copy-safe
read-only
effect-free
```

Non-claims

This document does not implement L-UI.

It defines the UI language direction and terminal boundary before implementation begins.

Part V - Nucleus and Runtime Boundary

- docs/NUCLEUS_PREVIEW.md - Latticra Nucleus Preview Classification
- docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md - Latticra Nucleus Task Execution Contract
- docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md - Latticra Nucleus Task Execution Implementation
- docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md - Latticra Nucleus Task Execution Implementation Plan
- docs/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT.md - Nucleus Task No-Effect Report Alignment
- docs/NUCLEUS_TASK_PLAN_IMPLEMENTATION.md - Latticra Nucleus Task Plan Implementation
- docs/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md - Nucleus Task Report-Only Execution Refinement
- docs/NUCLEUS_TASK_REPORT_REFINEMENT.md - Latticra Nucleus Task Report Refinement
- docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md - Latticra Runtime Boundary Abuse-Case Fixtures After Policy Expansion
- docs/RUNTIME_BOUNDARY_CONTRACT.md - Latticra Runtime Boundary Contract
- docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_MAIN_TEST_INTEGRATION_AUDIT.md - Runtime Boundary Domain Matrix Main Test Integration Audit

- docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REFINEMENT.md - Runtime Boundary Domain Matrix Refinement
- docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md - Runtime Boundary Domain Matrix Report Integration
- docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md - Latticra Runtime Boundary Implementation
- docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md - Latticra Runtime Boundary Implementation Plan
- docs/RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md - Latticra Runtime Boundary Policy Expansion After Threat-Model Validation
- docs/RUNTIME_BOUNDARY_POLICY_MATRIX_REFINEMENT.md - Latticra Runtime Boundary Policy Matrix Refinement
- docs/RUNTIME_BOUNDARY_REFINEMENT_IMPLEMENTATION.md - Latticra Runtime Boundary Refinement Implementation
- docs/RUNTIME_BOUNDARY_REFINEMENT_PLAN.md - Latticra Runtime Boundary Refinement Plan
- docs/RUNTIME_BOUNDARY_REPORT_REFINEMENT.md - Latticra Runtime Boundary Report Refinement
- docs/STATE_LATTICE.md - Latticra State Lattice
- docs/TRI_PLANE_TRANSITION.md - Latticra Tri-Plane Transition Model

docs/NUCLEUS_PREVIEW.md

Source title: Latticra Nucleus Preview Classification

Lines: 169 | Words: 378 | SHA-256: 0f8a8f8936b03035fff5cf1b10644fa4d4a9113e028809db9ed8c06b16fcb788

Latticra Nucleus Preview Classification

Status: initial implementation contract Scope: request classification, effect policy, no-execution flags, operator-visible reports, and preview-only boundaries.

Purpose

Nucleus Preview Classification is the third Latticra implementation unit.

It introduces the first Nucleus-adjacent classifier and operator-visible report surface without task execution, mutation, server interaction, recovery execution, hardware access, or update behavior.

Implementation files

```
include/latticra/nucleus_preview.h
src/nucleus_preview.c
tests/nucleus_preview_invariants.c
scripts/test-nucleus-preview.sh
```

Request kinds

Initial request kinds:

```
state-report
transition-preview
server-interaction
self-update
recovery-action
hardware-action
unknown
```

Policy results

Initial policy results:

```
allow-preview
deny
```

Initial policy reasons:

```
ok
null-argument
unknown-request
effect-blocked
effect-requires-future-gate
```

Allowed preview requests

Only these are allowed in the initial implementation:

```
state-report with effect=none
state-report with effect=read
transition-preview with effect=none
transition-preview with effect=read
```

Future-gated requests

These request kinds are classified but denied:

```
server-interaction
self-update
recovery-action
hardware-action
```

They require future contracts, gates, evidence, and tests before any stronger behavior.

Blocked effects

The initial classifier denies:

```
local_mutation
host_mutation
network
hardware
boot
recovery
external
```

for preview-allowed request kinds.

No-execution boundary

The classifier and report surface must always preserve:

```
executed=0
mutation_allowed=0
server_interaction_allowed=0
recovery_allowed=0
hardware_allowed=0
```

This preserves the preview-only boundary.

Operator-visible report

The preview report renders classifier output as deterministic text.

Required report fields:

```
LATTICRA NUCLEUS PREVIEW
request=&lt;request-kind&gt;
requested_effect=&lt;effect&gt;
policy=&lt;policy-result&gt;
reason=&lt;policy-reason&gt;
executed=0
mutation_allowed=0
server_interaction_allowed=0
recovery_allowed=0
hardware_allowed=0
```

The report must handle both allowed previews and denied previews without changing policy state or performing work.

Public API

Initial API:

```
latticra_request_kind_label
latticra_policy_result_label
latticra_policy_reason_label
latticra_nucleus_classify_preview
latticra_nucleus_preview_report
```

Test command

Run:

```
sh scripts/test-nucleus-preview.sh
```

The main C workflow runs state-lattice, tri-plane transition, and Nucleus preview tests.

Current evidence level

This implementation is an L2 tested model for request/effect classification and operator-visible preview reporting.

It is not live Nucleus orchestration, task execution, server interaction, self-update, recovery behavior, hardware behavior, boot behavior, or a security boundary.

Next implementation step

The next implementation candidate after this model is:

```
L-UI static report fixture
```

That future work should define terminal-facing layout fixtures for these reports while still avoiding live mutation.

Non-claims

This document and implementation do not claim live Nucleus execution, live movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md

Source title: Latticra Nucleus Task Execution Contract

Lines: 494 | Words: 1264 | SHA-256: 3974579e0ff518838be53a73b0797a99fef9fc358dc73c9d8b43ec3eaf405615

Latticra Nucleus Task Execution Contract

Status: Nucleus task execution contract Scope: future task execution boundary, prerequisites, authority checks, effect gates, denial behavior, task records, report surfaces, future files, tests, compatibility expectations, and non-claims before any task execution implementation.

Purpose

This document defines the first contract for future Nucleus task execution in Latticra.

Nucleus task execution means a future capability to accept a classified task request, validate every prerequisite, check effect gates and authority, produce a deterministic task record, and only then decide whether the task remains denied, preview-only, or eligible for a later explicitly gated implementation.

This document does not implement Nucleus task execution.

Relationship to previous work

This contract depends on:

```
docs/NUCLEUS_PREVIEW.md
docs/EFFECT_GATES.md
docs/STATE_LATTICE.md
docs/TRI_PLANE_TRANSITION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION_PLAN.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/L_UI_RENDERING_CONTRACT.md
docs/L_UI_RENDERING_IMPLEMENTATION_PLAN.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
include/latticra/nucleus_preview.h
include/latticra/state_lattice.h
include/latticra/tri_plane_transition.h
include/latticra/l_ui_renderer.h
include/latticra/lir.h
include/latticra/lat_parser.h
include/latticra/cpp/authority.hpp
src/nucleus_preview.c
src/l_ui_renderer.c
src/lir.c
src/lat_parser.c
src/cpp/authority.cpp
```

Those files remain the source of truth for request classification, effect vocabulary, preview denial behavior, state-lattice vocabulary, transition preview behavior, constrained C++ authority validation, deterministic L-UI rendering, LIR metadata, Lat metadata, and current no-effect boundaries.

Direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Task execution may not bypass this relationship:

```
Lat / Latticra Language: declares contract and intended effects
LIR: carries validated metadata
C++ authority layer: validates authority and reports decisions
C substrate: owns bounded execution-adjacent records and ABI-compatible surfaces
Nucleus: coordinates only after contracts and gates permit it
```

Current Nucleus boundary

Current Nucleus behavior is preview classification only.

Current Nucleus preview allows only:

```
state-report with effect=none
state-report with effect=read
transition-preview with effect=none
transition-preview with effect=read
```

Current Nucleus preview denies:

```
server-interaction
self-update
recovery-action
hardware-action
unknown request
local_mutation
host_mutation
network
hardware
boot
recovery
external
```

The future task execution contract must preserve this preview-only boundary until a separate implementation plan and implementation explicitly prove otherwise.

First task execution posture

The first future implementation must be denied-by-default.

Initial task execution posture:

```
task execution implemented? no
default policy: deny
preview allowed: yes
mutation allowed: no
network allowed: no
hardware allowed: no
boot allowed: no
recovery allowed: no
server interaction allowed: no
self-update allowed: no
operator confirmation allowed to override: no
```

Operator confirmation may be represented as metadata later, but it must not override policy in the first implementation.

Task request kinds

Future task execution should reuse and extend the existing request vocabulary carefully.

Initial task request kinds:

```
state-report
transition-preview
render-report
```

```
lat-validate
lir-validate
authority-check
server-interaction
self-update
recovery-action
hardware-action
boot-action
unknown
```

Only report/validation/preview-style requests may be eligible for no-effect handling in the first future implementation.

All effect-performing requests remain denied.

Task effects

Task execution must classify effects before task handling.

Initial task effects:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Default policy:

```
none -&gt; preview/report/validation only
read -&gt; approved local metadata only
local_mutation -&gt; deny until explicit local-mutation contract
host_mutation -&gt; deny
network -&gt; deny
hardware -&gt; deny
boot -&gt; deny
recovery -&gt; deny
external -&gt; deny
unknown -&gt; deny
```

Task policy results

Future policy result labels:

```
allow-preview
allow-report
allow-validation
deny
blocked
requires-future-gate
unsupported
internal-error
```

The first future implementation should not produce an executed task result for effect-performing work.

Task denial reasons

Future denial reason labels:

```
ok
null-argument
unknown-request
unknown-effect
unsupported-request
unsupported-effect
parser-failed
semantic-failed
lir-failed
authority-failed
effect-blocked
effect-requires-future-gate
```

```
non-no-effect-flags
operator-confirmation-not-supported
implementation-not-present
internal-error
```

Denial reasons must be stable and report-visible.

Required prerequisites

A future task may be considered only after all relevant prerequisites are explicit.

Possible prerequisites:

```
parser_error=ok
semantic_error=ok
lir_error=ok
authority_status=ok
render_status=ok
preview_policy=allow-preview or allow-report or allow-validation
effect_gate=allowed-preview or allowed-report or allowed-validation
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
operator_confirmation=not-applicable
```

If any prerequisite fails, the task must be denied and reported.

Required authority checks

Future Nucleus task execution must consult authority metadata before any task is eligible.

Authority checks must include:

```
authority status
authority validator
authority requested effect
authority denial reason
authority no-effect flags
source identity
request kind
effect kind
boundary kind
```

Authority failure must deny the task.

Authority success must not by itself execute a task. It only satisfies one prerequisite.

Required effect-gate checks

Future task execution must consult the effect gate before task handling.

Effect gate checks must classify:

```
requested effect
allowed effect
gate state
policy result
reason
rollback state
```

Initial allowed gate states for first implementation:

```
disabled
blocked
planned
```

The first future implementation must not produce:

```
armed
executed
failed
```

except as literal denied metadata in tests or reports.

Task record shape

A future task record should include:

```
task_id
request_kind
requested_effect
allowed_effect
policy_result
policy_reason
authority_status
authority_validator
authority_reason
gate_state
operator_confirmation
executed
mutation_allowed
server_interaction_allowed
recovery_allowed
hardware_allowed
rollback_state
evidence_level
source_identity
source_span
```

The first future implementation should use fixed-size fields and caller-owned buffers.

Task report format

A future task report should be deterministic and bounded.

Suggested report header:

```
LATTICRA NUCLEUS TASK REPORT
task_id=&lt;id&gt;
request=&lt;request-kind&gt;
requested_effect=&lt;effect&gt;
allowed_effect=&lt;effect&gt;
policy=&lt;policy-result&gt;
reason=&lt;policy-reason&gt;
authority_status=&lt;authority-status&gt;
authority_validator=&lt;authority-validator&gt;
authority_reason=&lt;authority-reason&gt;
gate_state=&lt;gate-state&gt;
operator_confirmation=&lt;confirmation-state&gt;
executed=0
mutation_allowed=0
server_interaction_allowed=0
recovery_allowed=0
hardware_allowed=0
rollback_state=&lt;rollback-state&gt;
evidence_level=&lt;level&gt;
source_identity=&lt;source&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
```

Reports must not include secrets, host environment values, filesystem contents, network data, credentials, or hardware identifiers.

First implementation gate

Nucleus task execution implementation must not begin until a separate implementation plan defines:

1. implementation language;
2. public header path;
3. source file path;
4. task request struct;
5. task result struct;
6. task record struct;

7. task policy enum;
8. task denial enum;
9. effect-gate enum usage;
10. authority summary usage;
11. operator-confirmation metadata shape;
12. rollback metadata shape;
13. report format;
14. capacity constants;
15. output-buffer behavior;
16. exact tests;
17. compatibility expectations;
18. non-claims.

That plan should be recorded in:

```
NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md
```

Future file policy

Recommended future files:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_execution_invariants.c
scripts/test-nucleus-task-execution.sh
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
```

The implementation plan must confirm exact paths before task execution code is added.

Future test list

A future implementation plan should include tests for:

```
nucleus_task_execution_denies_unknown_request
nucleus_task_execution_denies_unknown_effect
nucleus_task_execution_preserves_preview_only_boundary
nucleus_task_execution_requires_authority_success
nucleus_task_execution_requires_effect_gate_success
nucleus_task_execution_requires_no_effect_flags
nucleus_task_execution_allows_state_report_preview_only
nucleus_task_execution_allows_transition_preview_only
nucleus_task_execution_allows_render_report_only
nucleus_task_execution_allows_lat_validation_only
nucleus_task_execution_allows_lir_validation_only
nucleus_task_execution_denies_server_interaction
nucleus_task_execution_denies_self_update
nucleus_task_execution_denies_recovery_action
nucleus_task_execution_denies_hardware_action
nucleus_task_execution_denies_boot_action
nucleus_task_execution_report_is_deterministic
nucleus_task_execution_report_rejects_small_buffer
nucleus_task_execution_does_not_mutate_state
nucleus_task_execution_does_not_open_network
nucleus_task_execution_does_not_touch_hardware
nucleus_task_execution_does_not_write_files
nucleus_task_execution_does_not_call_recovery
nucleus_task_execution_does_not_override_policy_with_operator_confirmation
```

Compatibility expectations

Future Nucleus task execution work must not change:

```
existing Nucleus preview classification behavior
existing Nucleus preview report behavior
state lattice behavior
```

```
tri-plane transition behavior
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
L-UI rendering behavior
Lat grammar behavior
constrained C++ authority behavior
no-effect flags
current accepted fixture counts
```

Forbidden behavior

Nucleus task execution work must not:

- bypass Nucleus preview classification;
- bypass effect gates;
- bypass constrained C++ authority validation;
- bypass parser, semantic, LIR, or render prerequisites when relevant;
- execute unknown requests;
- execute unknown effects;
- mutate state without a separate mutation contract;
- write files;
- read host files outside approved metadata;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- touch hardware;
- alter boot state;
- perform rollback;
- let operator confirmation override policy;
- hide denial reasons;
- omit task records;
- emit secrets, host environment values, credentials, or hardware identifiers;
- imply a production runtime, sandbox, malware prevention, ransomware prevention, recovery system, update system, or operating-system surface.

Current validation command

This contract is guarded by:

```
sh scripts/test-nucleus-task-execution-contract.sh
```

The guard is static. It does not implement Nucleus task execution.

Non-claims

This document does not implement Nucleus task execution, command behavior, Lat execution, LIR execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md

Source title: Latticra Nucleus Task Execution Implementation

Lines: 353 | Words: 690 | SHA-256: 768720a80ed9e40bc72e858bbd888742e53d2d95b408d7d9e36c18e4889ee803

Latticra Nucleus Task Execution Implementation

Status: initial implementation contract Scope: first no-effect C Nucleus task classification/report surface, denied-by-default policy, authority prerequisites, effect-gate metadata, caller-provided buffers, deterministic reports, invariant tests, compatibility expectations, and non-claims.

Purpose

This document records the first Nucleus task execution implementation slice.

The implementation follows:

```
docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md
```

This slice implements task classification and report generation only.

It does not execute tasks.

Implementation files

This slice adds:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_execution_invariants.c
scripts/test-nucleus-task-execution.sh
```

It also wires the test runner into:

```
.github/workflows/c.yml
```

Public API

The public API is:

```
latticra_nucleus_task_request_kind_t
latticra_nucleus_task_effect_t
latticra_nucleus_task_policy_t
latticra_nucleus_task_denial_t
latticra_nucleus_task_gate_state_t
latticra_nucleus_task_operator_confirmation_t
latticra_nucleus_task_rollback_state_t
latticra_nucleus_task_authority_summary_t
latticra_nucleus_task_request_t
latticra_nucleus_task_record_t
latticra_nucleus_task_result_t
latticra_nucleus_task_request_kind_label
latticra_nucleus_task_effect_label
latticra_nucleus_task_policy_label
latticra_nucleus_task_denial_label
latticra_nucleus_task_gate_state_label
latticra_nucleus_task_operator_confirmation_label
latticra_nucleus_task_rollback_state_label
latticra_nucleus_task_classify
latticra_nucleus_task_report
```

Capacity constants

The implementation defines:

```
LATTICRA_NUCLEUS_TASK_ID_MAX 64u
LATTICRA_NUCLEUS_TASK_LABEL_MAX 64u
LATTICRA_NUCLEUS_TASK_REASON_MAX 128u
LATTICRA_NUCLEUS_TASK_SOURCE_IDENTITY_MAX 128u
LATTICRA_NUCLEUS_TASK_REPORT_MAX 4096u
LATTICRA_NUCLEUS_TASK_RECORD_MAX 16u
```

The task report uses caller-provided buffers and fixed-size task records.

Request kinds

Supported request kind labels:

```
state-report
transition-preview
render-report
lat-validate
lir-validate
authority-check
server-interaction
self-update
recovery-action
hardware-action
boot-action
unknown
```

Effects

Supported task effect labels:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Only none and read may be eligible for no-effect report, preview, or validation classifications.

Policy results

Supported policy labels:

```
allow-preview
allow-report
allow-validation
deny
blocked
requires-future-gate
unsupported
internal-error
```

The implementation does not produce an executed effect-performing task.

Denial reasons

Stable denial labels:

```
ok
null-argument
unknown-request
unknown-effect
unsupported-request
unsupported-effect
parser-failed
semantic-failed
lir-failed
render-failed
authority-failed
effect-blocked
effect-requires-future-gate
non-no-effect-flags
operator-confirmation-not-supported
implementation-not-present
internal-error
```

Classification behavior

Implemented classification behavior:

```
state-report + none -&gt; allow-report
state-report + read -&gt; allow-report
transition-preview + none -&gt; allow-preview
transition-preview + read -&gt; allow-preview
render-report + none -&gt; allow-report
render-report + read -&gt; allow-report
lat-validate + none -&gt; allow-validation
lat-validate + read -&gt; allow-validation
lir-validate + none -&gt; allow-validation
lir-validate + read -&gt; allow-validation
authority-check + none -&gt; allow-validation
authority-check + read -&gt; allow-validation
server-interaction -&gt; requires-future-gate
self-update -&gt; requires-future-gate
recovery-action -&gt; requires-future-gate
hardware-action -&gt; requires-future-gate
boot-action -&gt; requires-future-gate
unknown -&gt; deny
unknown effect -&gt; deny
local_mutation -&gt; deny
host_mutation -&gt; deny
network -&gt; deny
hardware -&gt; deny
boot -&gt; deny
recovery -&gt; deny
external -&gt; deny
```

Allowed classifications still set:

```
executed=0
mutation_allowed=0
server_interaction_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Prerequisite behavior

The implementation checks:

```
request presence
result presence
authority presence
authority status
authority no-effect flags
operator confirmation metadata
Nucleus preview policy for state-report and transition-preview
render status for render-report
Lat parse status for lat-validate
LIR status for lir-validate
source identity capacity
```

Failed prerequisites deny the task and remain visible in the report.

Report format

latticra_nucleus_task_report emits:

```
LATTICRA NUCLEUS TASK REPORT
status=&lt;integer-status&gt;
task_id=&lt;id&gt;
request=&lt;request-kind&gt;
requested_effect=&lt;effect&gt;
allowed_effect=&lt;effect&gt;
policy=&lt;policy-result&gt;
reason=&lt;denial-reason&gt;
authority_status=&lt;authority-status&gt;
authority_validator=&lt;authority-validator&gt;
authority_reason=&lt;authority-reason&gt;
gate_state=&lt;gate-state&gt;
operator_confirmation=&lt;confirmation-state&gt;
executed=0
mutation_allowed=0
server_interaction_allowed=0
recovery_allowed=0
hardware_allowed=0
```

```
rollback_state=&lt;rollback-state&gt;
evidence_level=&lt;level&gt;
source_identity=&lt;source&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
```

Small buffers return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the output buffer.

No-effect behavior

The implementation preserves:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
```

It does not execute commands, mutate state, open network connections, write files, read host files outside metadata, call server code, call update code, call recovery code, touch hardware, alter boot state, or perform rollback.

Validation

Run:

```
sh scripts/test-nucleus-task-execution.sh
```

The test suite covers:

```
nucleus_task_execution_denies_unknown_request
nucleus_task_execution_denies_unknown_effect
nucleus_task_execution_preserves_preview_only_boundary
nucleus_task_execution_requires_authority_success
nucleus_task_execution_requires_effect_gate_success
nucleus_task_execution_requires_no_effect_flags
nucleus_task_execution_allows_state_report_preview_only
nucleus_task_execution_allows_transition_preview_only
nucleus_task_execution_allows_render_report_only
nucleus_task_execution_allows_lat_validation_only
nucleus_task_execution_allows_lir_validation_only
nucleus_task_execution_denies_server_interaction
nucleus_task_execution_denies_self_update
nucleus_task_execution_denies_recovery_action
nucleus_task_execution_denies_hardware_action
nucleus_task_execution_denies_boot_action
nucleus_task_execution_report_is_deterministic
nucleus_task_execution_report_rejects_small_buffer
nucleus_task_execution_does_not_mutate_state
nucleus_task_execution_does_not_open_network
nucleus_task_execution_does_not_touch_hardware
nucleus_task_execution_does_not_write_files
nucleus_task_execution_does_not_call_recovery
nucleus_task_execution_does_not_override_policy_with_operator_confirmation
```

Compatibility expectations

This implementation must not change:

```
existing Nucleus preview classification behavior
existing Nucleus preview report behavior
state lattice behavior
tri-plane transition behavior
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
L-UI rendering behavior
Lat grammar behavior
constrained C++ authority behavior
no-effect flags
current accepted fixture counts
```

Current boundary

This implementation does not provide:

```
command behavior
Lat execution
LIR execution
live movement
state mutation
file I/O
network I/O
server interaction
self-update
recovery behavior
rollback
hardware support
boot behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
operating-system completeness
```

Non-claims

This document and implementation do not claim a finished operating system, hardened sandbox, production runtime, production security boundary, malware prevention, ransomware prevention, recovery system, update system, bootable image, public task executor, or public release readiness.

docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md

Source title: Latticra Nucleus Task Execution Implementation Plan

Lines: 700 | Words: 1534 | SHA-256: 7e9e8bca228486dd925162a4dca68a92e223131e193d9d5285d0342c055f586f

Latticra Nucleus Task Execution Implementation Plan

Status: implementation planning contract Scope: exact public API, C implementation files, task request/result/record structs, policy enum, denial enum, effect-gate enum usage, authority summary usage, operator-confirmation metadata, rollback metadata, report format, capacity constants, output-buffer behavior, tests, compatibility expectations, and non-claims before Nucleus task execution code.

Purpose

This document defines the implementation plan for the first Nucleus task execution surface.

The Nucleus task execution contract is already merged and guarded. This plan turns that contract into exact public API, file paths, structures, enums, capacity constants, report rules, prerequisite behavior, effect-gate behavior, authority-summary behavior, task-record behavior, exact tests, compatibility expectations, and non-claims before task execution code is added.

This document does not implement Nucleus task execution.

Relationship to previous work

This plan depends on:

```
docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md
docs/NUCLEUS_PREVIEW.md
docs/EFFECT_GATES.md
docs/STATE_LATTICE.md
docs/TRI_PLANE_TRANSITION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
include/latticra/nucleus_preview.h
include/latticra/state_lattice.h
include/latticra/tri_plane_transition.h
```

```
include/latticra/l_ui_renderer.h
include/latticra/lir.h
include/latticra/lat_parser.h
include/latticra/cpp/authority.hpp
src/nucleus_preview.c
src/l_ui_renderer.c
src/lir.c
src/lat_parser.c
src/cpp/authority.cpp
```

Those files remain the source of truth for preview classification, effect vocabulary, state lattice, transition preview, constrained authority, rendering, LIR metadata, Lat metadata, and no-effect boundaries.

Implementation language decision

The first Nucleus task execution surface should be implemented in C.

Reason:

- current Nucleus preview classification is C;
- state lattice and tri-plane transition are C;
- L-UI rendering and LIR shape are C;
- the task surface should use bounded structs and caller-provided buffers;
- C remains the secure substrate;
- the constrained C++ authority layer remains a validation/report dependency, not the task-record storage substrate.

The first implementation should not include C++ object lifetimes in the public C API.

Implementation files

The implementation PR should add or modify:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_execution_invariants.c
scripts/test-nucleus-task-execution.sh
.github/workflows/c.yml
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
```

The implementation PR should not add command behavior, Lat execution, LIR execution, live movement, mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware behavior, boot behavior, terminal control, sandboxing, malware prevention, ransomware prevention, or operating-system behavior.

Public API shape

Add public API names:

```
latticra_nucleus_task_request_kind_t
latticra_nucleus_task_effect_t
latticra_nucleus_task_policy_t
latticra_nucleus_task_denial_t
latticra_nucleus_task_gate_state_t
latticra_nucleus_task_operator_confirmation_t
latticra_nucleus_task_rollback_state_t
latticra_nucleus_task_authority_summary_t
latticra_nucleus_task_request_t
latticra_nucleus_task_record_t
latticra_nucleus_task_result_t
latticra_nucleus_task_request_kind_label
latticra_nucleus_task_effect_label
latticra_nucleus_task_policy_label
latticra_nucleus_task_denial_label
latticra_nucleus_task_gate_state_label
latticra_nucleus_task_operator_confirmation_label
latticra_nucleus_task_rollback_state_label
latticra_nucleus_task_classify
```

```
latticra_nucleus_task_report
```

Recommended function signatures:

```
const char *latticra_nucleus_task_request_kind_label(latticra_nucleus_task_request_kind_t kind);
const char *latticra_nucleus_task_effect_label(latticra_nucleus_task_effect_t effect);
const char *latticra_nucleus_task_policy_label(latticra_nucleus_task_policy_t policy);
const char *latticra_nucleus_task_denial_label(latticra_nucleus_task_denial_t denial);
const char *latticra_nucleus_task_gate_state_label(latticra_nucleus_task_gate_state_t
gate_state);
const char
*latticra_nucleus_task_operator_confirmation_label(latticra_nucleus_task_operator_confirmation_t
 confirmation);
const char *latticra_nucleus_task_rollback_state_label(latticra_nucleus_task_rollback_state_t
rollback_state);

latticra_status_t latticra_nucleus_task_classify(
    const latticra_nucleus_task_request_t *request,
    latticra_nucleus_task_result_t *result);

latticra_status_t latticra_nucleus_task_report(
    const latticra_nucleus_task_result_t *result,
    char *buffer,
    size_t buffer_len);
```

The first implementation should classify and report only. It must not execute work.

Capacity constants

Add exact bounded constants:

```
LATTICRA_NUCLEUS_TASK_ID_MAX 64u
LATTICRA_NUCLEUS_TASK_LABEL_MAX 64u
LATTICRA_NUCLEUS_TASK_REASON_MAX 128u
LATTICRA_NUCLEUS_TASK_SOURCE_IDENTITY_MAX 128u
LATTICRA_NUCLEUS_TASK_REPORT_MAX 4096u
LATTICRA_NUCLEUS_TASK_RECORD_MAX 16u
```

Small output buffers must return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the output buffer.

Request kind enum

Add request kinds:

```
LATTICRA_NUCLEUS_TASK_STATE_REPORT
LATTICRA_NUCLEUS_TASK_TRANSITION_PREVIEW
LATTICRA_NUCLEUS_TASK_RENDER_REPORT
LATTICRA_NUCLEUS_TASK_LAT_VALIDATE
LATTICRA_NUCLEUS_TASK_LIR_VALIDATE
LATTICRA_NUCLEUS_TASK_AUTHORITY_CHECK
LATTICRA_NUCLEUS_TASK_SERVER_INTERACTION
LATTICRA_NUCLEUS_TASK_SELF_UPDATE
LATTICRA_NUCLEUS_TASK_RECOVERY_ACTION
LATTICRA_NUCLEUS_TASK_HARDWARE_ACTION
LATTICRA_NUCLEUS_TASK_BOOT_ACTION
LATTICRA_NUCLEUS_TASK_UNKNOWN
```

Stable labels:

```
state-report
transition-preview
render-report
lat-validate
lir-validate
authority-check
server-interaction
self-update
recovery-action
hardware-action
boot-action
unknown
```

Effect enum

Add task effects:

```
LATTICRA_NUCLEUS_TASK_EFFECT_NONE
```

```
LATTICRA_NUCLEUS_TASK_EFFECT_READ
LATTICRA_NUCLEUS_TASK_EFFECT_LOCAL_MUTATION
LATTICRA_NUCLEUS_TASK_EFFECT_HOST_MUTATION
LATTICRA_NUCLEUS_TASK_EFFECT_NETWORK
LATTICRA_NUCLEUS_TASK_EFFECT_HARDWARE
LATTICRA_NUCLEUS_TASK_EFFECT_BOOT
LATTICRA_NUCLEUS_TASK_EFFECT_RECOVERY
LATTICRA_NUCLEUS_TASK_EFFECT_EXTERNAL
LATTICRA_NUCLEUS_TASK_EFFECT_UNKNOWN
```

Stable labels:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Policy enum

Add task policies:

```
LATTICRA_NUCLEUS_TASK_POLICY_ALLOW_PREVIEW
LATTICRA_NUCLEUS_TASK_POLICY_ALLOW_REPORT
LATTICRA_NUCLEUS_TASK_POLICY_ALLOW_VALIDATION
LATTICRA_NUCLEUS_TASK_POLICY_DENY
LATTICRA_NUCLEUS_TASK_POLICY_BLOCKED
LATTICRA_NUCLEUS_TASK_POLICY_REQUIRES_FUTURE_GATE
LATTICRA_NUCLEUS_TASK_POLICY_UNSUPPORTED
LATTICRA_NUCLEUS_TASK_POLICY_INTERNAL_ERROR
```

Stable labels:

```
allow-preview
allow-report
allow-validation
deny
blocked
requires-future-gate
unsupported
internal-error
```

The first implementation must not produce an executed effect-performing task.

Denial enum

Add denial reasons:

```
LATTICRA_NUCLEUS_TASK_DENIAL_OK
LATTICRA_NUCLEUS_TASK_DENIAL_NULL_ARGUMENT
LATTICRA_NUCLEUS_TASK_DENIAL_UNKNOWN_REQUEST
LATTICRA_NUCLEUS_TASK_DENIAL_UNKNOWN_EFFECT
LATTICRA_NUCLEUS_TASK_DENIAL_UNSUPPORTED_REQUEST
LATTICRA_NUCLEUS_TASK_DENIAL_UNSUPPORTED_EFFECT
LATTICRA_NUCLEUS_TASK_DENIAL_PARSER_FAILED
LATTICRA_NUCLEUS_TASK_DENIAL_SEMANTIC_FAILED
LATTICRA_NUCLEUS_TASK_DENIAL_LIR_FAILED
LATTICRA_NUCLEUS_TASK_DENIAL_RENDER_FAILED
LATTICRA_NUCLEUS_TASK_DENIAL_AUTHORITY_FAILED
LATTICRA_NUCLEUS_TASK_DENIAL_EFFECT_BLOCKED
LATTICRA_NUCLEUS_TASK_DENIAL_EFFECT_REQUIRES_FUTURE_GATE
LATTICRA_NUCLEUS_TASK_DENIAL_NON_NO_EFFECT_FLAGS
LATTICRA_NUCLEUS_TASK_DENIAL_OPERATOR_CONFIRMATION_NOT_SUPPORTED
LATTICRA_NUCLEUS_TASK_DENIAL_IMPLEMENTATION_NOT_PRESENT
LATTICRA_NUCLEUS_TASK_DENIAL_INTERNAL_ERROR
```

Stable labels:

```
ok
null-argument
unknown-request
unknown-effect
unsupported-request
unsupported-effect
parser-failed
```

```
semantic-failed
lir-failed
render-failed
authority-failed
effect-blocked
effect-requires-future-gate
non-no-effect-flags
operator-confirmation-not-supported
implementation-not-present
internal-error
```

Gate state enum

Add gate states:

```
LATTICRA_NUCLEUS_TASK_GATE_DISABLED
LATTICRA_NUCLEUS_TASK_GATE_BLOCKED
LATTICRA_NUCLEUS_TASK_GATE_PLANNED
LATTICRA_NUCLEUS_TASK_GATE_AVAILABLE
LATTICRA_NUCLEUS_TASK_GATE_ARMED
LATTICRA_NUCLEUS_TASK_GATE_EXECUTED
LATTICRA_NUCLEUS_TASK_GATE_FAILED
```

Stable labels:

```
disabled
blocked
planned
available
armed
executed
failed
```

The first implementation should output only disabled, blocked, or planned for real classifications.

Operator confirmation enum

Add operator-confirmation states:

```
LATTICRA_NUCLEUS_TASK_OPERATOR_NOT_APPLICABLE
LATTICRA_NUCLEUS_TASK_OPERATOR_REQUIRED
LATTICRA_NUCLEUS_TASK_OPERATOR_PRESENT
LATTICRA_NUCLEUS_TASK_OPERATOR_REJECTED
LATTICRA_NUCLEUS_TASK_OPERATOR_NOT_SUPPORTED
```

Stable labels:

```
not-applicable
required
present
rejected
not-supported
```

The first implementation must treat confirmation as metadata only. Confirmation must not override policy.

Rollback state enum

Add rollback states:

```
LATTICRA_NUCLEUS_TASK_ROLLBACK_NOT_APPLICABLE
LATTICRA_NUCLEUS_TASK_ROLLBACK_NOT_AVAILABLE
LATTICRA_NUCLEUS_TASK_ROLLBACK_REQUIRED
LATTICRA_NUCLEUS_TASK_ROLLBACK_READY
LATTICRA_NUCLEUS_TASK_ROLLBACK_BLOCKED
```

Stable labels:

```
not-applicable
not-available
required
ready
blocked
```

The first implementation must not perform rollback.

Authority summary struct

Add a C-compatible authority summary:

```
latticra_status_t status;
char status_label[LATTICRA_NUCLEUS_TASK_LABEL_MAX];
char validator_label[LATTICRA_NUCLEUS_TASK_LABEL_MAX];
char requested_effect_label[LATTICRA_NUCLEUS_TASK_LABEL_MAX];
char denial_reason[LATTICRA_NUCLEUS_TASK_REASON_MAX];
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

This mirrors the renderer authority summary pattern and avoids exposing C++ object lifetimes through the C task API.

Task request struct

Add request fields:

```
char task_id[LATTICRA_NUCLEUS_TASK_ID_MAX];
latticra_nucleus_task_request_kind_t request_kind;
latticra_nucleus_task_effect_t requested_effect;
latticra_nucleus_task_operator_confirmation_t operator_confirmation;
latticra_nucleus_task_rollback_state_t rollback_state;
const latticra_nucleus_task_authority_summary_t *authority;
const latticra_nucleus_preview_t *preview;
const latticra_l_ui_render_result_t *render;
const latticra_lir_module_t *lir;
const latticra_lat_parse_result_t *lat;
const char *source_identity;
size_t source_identity_len;
latticra_l_ui_source_span_t source_span;
```

The implementation must not retain caller-owned pointers after the call returns.

Task record struct

Add task record fields:

```
char task_id[LATTICRA_NUCLEUS_TASK_ID_MAX];
latticra_nucleus_task_request_kind_t request_kind;
latticra_nucleus_task_effect_t requested_effect;
latticra_nucleus_task_effect_t allowed_effect;
latticra_nucleus_task_policy_t policy;
latticra_nucleus_task_denial_t denial;
latticra_nucleus_task_gate_state_t gate_state;
latticra_nucleus_task_operator_confirmation_t operator_confirmation;
latticra_nucleus_task_rollback_state_t rollback_state;
latticra_nucleus_task_authority_summary_t authority;
char source_identity[LATTICRA_NUCLEUS_TASK_SOURCE_IDENTITY_MAX];
latticra_l_ui_source_span_t source_span;
int executed;
int mutation_allowed;
int server_interaction_allowed;
int recovery_allowed;
int hardware_allowed;
unsigned int evidence_level;
```

Task result struct

Add task result fields:

```
latticra_status_t status;
latticra_nucleus_task_record_t record;
size_t record_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

The result must be fully initialized on success and failure.

Classification rules

Initial classification behavior:

```
state-report + none -&gt; allow-report
state-report + read -&gt; allow-report
transition-preview + none -&gt; allow-preview
transition-preview + read -&gt; allow-preview
render-report + none -&gt; allow-report
render-report + read -&gt; allow-report
lat-validate + none -&gt; allow-validation
lat-validate + read -&gt; allow-validation
lir-validate + none -&gt; allow-validation
lir-validate + read -&gt; allow-validation
authority-check + none -&gt; allow-validation
authority-check + read -&gt; allow-validation
server-interaction -&gt; requires-future-gate
self-update -&gt; requires-future-gate
recovery-action -&gt; requires-future-gate
hardware-action -&gt; requires-future-gate
boot-action -&gt; requires-future-gate
unknown -&gt; deny
unknown effect -&gt; deny
local_mutation -&gt; deny
host_mutation -&gt; deny
network -&gt; deny
hardware -&gt; deny
boot -&gt; deny
recovery -&gt; deny
external -&gt; deny
```

Allowed classifications still must set:

```
executed=0
mutation_allowed=0
server_interaction_allowed=0
recovery_allowed=0
hardware_allowed=0
```

Prerequisite behavior

The first implementation must check relevant prerequisites.

Rules:

```
missing request -&gt; null-argument
missing result -&gt; null-argument
missing authority -&gt; authority-failed
non-ok authority -&gt; authority-failed
non-no-effect authority flags -&gt; non-no-effect-flags
operator confirmation present -&gt; operator-confirmation-not-supported
render-report requires render_status=ok
lat-validate requires parser_error=ok
lir-validate requires lir_error=ok
transition-preview requires preview policy allow-preview
state-report requires preview policy allow-preview or allow-report
```

Failed prerequisites deny the task and must be visible in the report.

Report format

latticra_nucleus_task_report should emit:

```
LATTICRA NUCLEUS TASK REPORT
status=&lt;integer-status&gt;
task_id=&lt;id&gt;
request=&lt;request-kind&gt;
requested_effect=&lt;effect&gt;
allowed_effect=&lt;effect&gt;
policy=&lt;policy-result&gt;
reason=&lt;denial-reason&gt;
authority_status=&lt;authority-status&gt;
authority_validator=&lt;authority-validator&gt;
authority_reason=&lt;authority-reason&gt;
gate_state=&lt;gate-state&gt;
operator_confirmation=&lt;confirmation-state&gt;
executed=0
mutation_allowed=0
server_interaction_allowed=0
```

```
recovery_allowed=0
hardware_allowed=0
rollback_state=&lt;rollback-state&gt;
evidence_level=&lt;level&gt;
source_identity=&lt;source&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
```

Reports must not include secrets, host environment values, filesystem contents, network data, credentials, or hardware identifiers.

Output buffer behavior

latticra_nucleus_task_report should:

```
write only to caller-provided buffers
require explicit buffer length
NUL-terminate on success
clear the buffer on too-small failure
return LATTICRA_STATUS_BUFFER_TOO_SMALL for small buffers
avoid heap allocation
avoid file output
avoid stdout
avoid stderr
avoid terminal escape control
```

Exact implementation test list

The implementation PR should include tests for:

```
nucleus_task_execution_denies_unknown_request
nucleus_task_execution_denies_unknown_effect
nucleus_task_execution_preserves_preview_only_boundary
nucleus_task_execution_requires_authority_success
nucleus_task_execution_requires_effect_gate_success
nucleus_task_execution_requires_no_effect_flags
nucleus_task_execution_allows_state_report_preview_only
nucleus_task_execution_allows_transition_preview_only
nucleus_task_execution_allows_render_report_only
nucleus_task_execution_allows_lat_validation_only
nucleus_task_execution_allows_lir_validation_only
nucleus_task_execution_denies_server_interaction
nucleus_task_execution_denies_self_update
nucleus_task_execution_denies_recovery_action
nucleus_task_execution_denies_hardware_action
nucleus_task_execution_denies_boot_action
nucleus_task_execution_report_is_deterministic
nucleus_task_execution_report_rejects_small_buffer
nucleus_task_execution_does_not_mutate_state
nucleus_task_execution_does_not_open_network
nucleus_task_execution_does_not_touch_hardware
nucleus_task_execution_does_not_write_files
nucleus_task_execution_does_not_call_recovery
nucleus_task_execution_does_not_override_policy_with_operator_confirmation
```

Test file plan

Add:

```
tests/nucleus_task_execution_invariants.c
scripts/test-nucleus-task-execution.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-nucleus-task-execution-implementation-plan.sh
```

and before:

```
sh scripts/test-l-ui-static-report.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
existing Nucleus preview classification behavior
existing Nucleus preview report behavior
state lattice behavior
tri-plane transition behavior
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
L-UI rendering behavior
Lat grammar behavior
constrained C++ authority behavior
no-effect flags
current accepted fixture counts
```

Forbidden implementation behavior

The first implementation must not:

- bypass Nucleus preview classification;
- bypass effect gates;
- bypass constrained C++ authority validation;
- bypass parser, semantic, LIR, or render prerequisites when relevant;
- execute unknown requests;
- execute unknown effects;
- mutate state;
- write files;
- read host files outside approved metadata;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- touch hardware;
- alter boot state;
- perform rollback;
- let operator confirmation override policy;
- hide denial reasons;
- omit task records;
- emit secrets, host environment values, credentials, or hardware identifiers;

- imply a production runtime, sandbox, malware prevention, ransomware prevention, recovery system, update system, or operating-system surface.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-nucleus-task-execution-implementation-plan.sh
```

The guard is static. It does not implement Nucleus task execution.

Implementation gate

Nucleus task execution code may be added only after this plan is merged.

Non-claims

This document does not implement Nucleus task execution, command behavior, Lat execution, LIR execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT.md

Source title: Nucleus Task No-Effect Report Alignment

Lines: 105 | Words: 251 | SHA-256: b3d30d20b8b05a098890fc3ff5ca71cd642f075c665565f0cbca70e5266f3dc7

Nucleus Task No-Effect Report Alignment

Status: implementation refinement Scope: deterministic no-effect report alignment for Nucleus task records.

Purpose

This refinement makes the existing Nucleus task report surface explicitly state that task output remains aligned with the language-representation review and no-effect policy.

It adds report-only metadata. It does not add task execution behavior.

Implementation files

This slice updates:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_no_effect_report_alignment.c
scripts/test-nucleus-task-no-effect-report-alignment.sh
scripts/test-nucleus-task-execution.sh
.github/workflows/nucleus-task-no-effect-report-alignment.yml
```

Added report fields

Each Nucleus task record now includes:

```
report_alignment
no_effect_policy
representation_gate
```

Expected values are:

```
report_alignment=no-effect-report-alignment
no_effect_policy=preserved
representation_gate=language-representation-reviewed
```

These fields appear in the deterministic Nucleus task report output.

Boundary

This refinement does not add:

```
effect-performing task execution
runtime behavior
command execution
Lat execution
LIR execution
file I/O
network I/O
state mutation
server interaction
self-update
recovery behavior
hardware behavior
boot behavior
sandboxing
malware prevention
ransomware prevention
operating-system completeness
```

Validation

Run:

```
sh scripts/test-nucleus-task-no-effect-report-alignment.sh
sh scripts/test-nucleus-task-execution.sh
```

The focused test verifies:

```
accepted no-effect reports emit alignment labels
future-gated reports keep alignment labels
invalid reports keep alignment labels
execution and mutation remain disabled
server, recovery, and hardware behavior remain disabled
```

Compatibility expectations

This refinement preserves:

```
existing request-kind labels
existing effect labels
existing policy labels
existing denial labels
existing classification behavior
existing future-gated behavior
existing no-effect flags
existing denied-by-default posture
```

Non-claims

This document and implementation do not claim finished Nucleus execution, runtime behavior, command execution, production security boundary, malware prevention, ransomware prevention, recovery system, update system, bootable image, or public release readiness.

docs/NUCLEUS_TASK_PLAN_IMPLEMENTATION.md

Source title: Latticra Nucleus Task Plan Implementation

Lines: 89 | Words: 291 | SHA-256: abee5e5ae00f03f6b54c10cfa4e04f7832bde2959c4ea3f0848ef4bb182d0bb6

Latticra Nucleus Task Plan Implementation

Status: initial implementation record Scope: bounded C Nucleus task plan API, no-effect task-result sequencing, deterministic plan reports, invariant tests, CI script, no-effect posture, and non-claims.

Purpose

The Nucleus task plan layer evaluates an ordered set of already-classified Nucleus task results and produces a bounded, operator-visible plan report.

This layer does not execute tasks. It does not perform mutation. It does not perform network access. It does not open files. It does not interact with hardware. It does not upgrade future-gated work into runnable behavior.

Its purpose is to make sequencing visible while preserving Latticra's current disabled-by-default runtime posture.

Added files

```
include/latticra/nucleus_task_plan.h
src/nucleus_task_plan.c
tests/nucleus_task_plan_invariants.c
scripts/test-nucleus-task-plan.sh
```

Implementation behavior

The implementation accepts a caller-provided array of latticra_nucleus_task_result_t records and produces one latticra_nucleus_task_plan_result_t.

The evaluator preserves the following rules:

```
bounded task count only
empty plans are denied
oversized plans are denied
only successful task results are accepted
only no-effect task flags are accepted
future-gated tasks block the plan
already-denied tasks block the plan
allowed report, preview, and validation tasks can form a no-effect sequence
```

Report fields

The deterministic report includes:

```
plan_id
record_count
task_count
accepted_count
blocked_count
has_blocked_task
first_blocked_index
policy
reason
no_effect
execution_allowed
mutation_allowed
server_allowed
recovery_allowed
hardware_allowed
evidence_level
```

Covered invariants

The focused test covers:

```
accepted no-effect report/preview/validation sequence
empty plan denial
future-gated task blocks the plan
non-no-effect task flags block the plan
operator-visible report fields
no execution allowed
no mutation allowed
no server interaction allowed
no recovery allowed
no hardware allowed
```

Non-claims

This implementation does not create a scheduler, executor, runtime, command runner, LIR executor, Lat executor, network service, update system, recovery engine, boot path, or hardware control layer.

It is a planning and reporting slice only.

docs/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md

Source title: Nucleus Task Report-Only Execution Refinement

Lines: 107 | Words: 263 | SHA-256: 8a70d30146396eedda4ee13b3553f84789299affa37fca225a18b203ff1f9e64

Nucleus Task Report-Only Execution Refinement

Status: implementation refinement Scope: deterministic report-only execution boundary metadata for Nucleus task records.

Purpose

This refinement makes the Nucleus task report state the execution boundary directly.

It adds report-only metadata labels for execution status, effect status, and runtime status. These labels make it explicit that the task layer still reports classification and policy only.

Implementation files

This slice updates:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_report_only_execution_refinement.c
scripts/test-nucleus-task-report-only-execution-refinement.sh
scripts/test-nucleus-task-execution.sh
.github/workflows/nucleus-task-report-only-execution-refinement.yml
```

Added report fields

Each Nucleus task record now includes:

```
execution_status
effect_status
runtime_status
```

Expected values are:

```
execution_status=not-executed
effect_status=report-only
runtime_status=not-entered
```

These fields appear in deterministic Nucleus task report output.

Boundary

This refinement does not add:

```
effect-performing task execution
runtime behavior
command execution
Lat execution
LIR execution
file I/O
network I/O
state mutation
server interaction
self-update
recovery behavior
hardware behavior
boot behavior
sandboxing
malware prevention
ransomware prevention
operating-system completeness
```

Validation

Run:

```
sh scripts/test-nucleus-task-report-only-execution-refinement.sh
sh scripts/test-nucleus-task-execution.sh
```

The focused test verifies:

```
accepted reports emit report-only execution labels
future-gated reports remain not executed
invalid reports remain report-only
execution and mutation remain disabled
server, recovery, hardware, and runtime behavior remain disabled
```

Compatibility expectations

This refinement preserves:

```
existing request-kind labels
existing effect labels
existing policy labels
existing denial labels
existing report-alignment labels
existing no-effect-policy labels
existing representation-gate labels
existing future-gated behavior
existing no-effect flags
existing denied-by-default posture
```

Non-claims

This document and implementation do not claim finished Nucleus execution, runtime behavior, command execution, production security boundary, malware prevention, ransomware prevention, recovery system, update system, bootable image, or public release readiness.

docs/NUCLEUS_TASK_REPORT_REFINEMENT.md

Source title: Latticra Nucleus Task Report Refinement

Lines: 164 | Words: 381 | SHA-256: cf73775ddf3b524fdd46198cf3834df8bd7af0e83d751296cf2b4fd9874547e2

Latticra Nucleus Task Report Refinement

Status: initial Nucleus task report refinement implementation Scope: deterministic no-effect task report classification, task-domain labeling, authorization-state labeling, prerequisite reporting, no-effect-chain reporting, invariant tests, and workflow coverage.

Purpose

This document records the Nucleus task report refinement after the runtime boundary policy matrix refinement.

The goal is to make Nucleus task reports more explicit about what kind of task decision was made, which task domain it belongs to, and whether prerequisites and no-effect posture remained intact.

This refinement does not execute tasks, mutate state, perform I/O, open network connections, touch hardware, control a terminal, or provide runtime behavior.

Added report metadata

The Nucleus task record now includes:

```
report_classification
task_domain
authorization_state
prerequisites_satisfied
no_effect_chain_ok
```

The deterministic LATTICRA NUCLEUS TASK REPORT now emits those fields.

Report classifications

Initial report classifications:

```
accepted
```

```
future-gated
denied
invalid
```

Task domains

Initial task domains:

```
state
transition
render
lat
lir
authority
server
update
recovery
hardware
boot
unknown
```

Authorization states

Initial authorization states:

```
not-requested
checked
denied
reserved-for-future
unavailable
```

Deterministic report meaning

The refinement reports:

```
accepted -&gt; task accepted as no-effect report, preview, or validation
future-gated -&gt; operational task remains behind a future gate
denied -&gt; known task rejected by prerequisites, authority, operator, or effect policy
invalid -&gt; null, unknown, or malformed task/effect identity
```

prerequisites_satisfied is true only when the task classification reason is ok.

no_effect_chain_ok is true only when result-level and record-level execution/mutation/server/recovery/hardware flags remain denied and no-effect is preserved.

Implementation files

This slice updates or adds:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_report_refinement.c
scripts/test-nucleus-task-execution.sh
docs/NUCLEUS_TASK_REPORT_REFINEMENT.md
scripts/test-nucleus-task-report-refinement.sh
.github/workflows/nucleus-task-report-refinement.yml
```

Validation

Run:

```
sh scripts/test-nucleus-task-report-refinement.sh
sh scripts/test-nucleus-task-execution.sh
```

The focused invariant tests verify:

```
nucleus_task_report_refinement_labels_are_stable
nucleus_task_report_refinement_reports_accepted_state_report
nucleus_task_report_refinement_reports_authority_check
nucleus_task_report_refinement_reports_future_gated_task
nucleus_task_report_refinement_reports_denied_prerequisite
nucleus_task_report_refinement_reports_invalid_request
```

Compatibility

This refinement preserves existing Nucleus task behavior for:

```
state-report preview/report behavior
transition-preview behavior
render-report behavior
Lat validation behavior
LIR validation behavior
authority-check behavior
future-gated operational task kinds
unknown request denial
unknown effect denial
operator-confirmation non-override behavior
small-buffer behavior
```

Non-claims

This report refinement does not provide:

```
task execution
runtime behavior
command execution
Lat execution
LIR execution
state mutation
file I/O
network I/O
server interaction
self-update
recovery behavior
rollback behavior
hardware support
boot behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
certification
accreditation
operating-system completeness
```

docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md

Source title: Latticra Runtime Boundary Abuse-Case Fixtures After Policy Expansion

Lines: 143 | Words: 500 | SHA-256: 8c464345ce394fb2071b4509eefa404181fd78d71a859ce6b5cd3b2b7116097d

Latticra Runtime Boundary Abuse-Case Fixtures After Policy Expansion

Status: runtime boundary abuse-case fixture expansion after policy expansion Source: local follow-up slice Scope: deterministic runtime-boundary abuse-case fixtures, report assertions, policy-matrix assertions, domain-matrix assertions, guard coverage, public entry-point updates, and no-claim preservation. This document does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production protection, sandboxing, malware prevention, ransomware prevention, certification, compliance, or runtime authority.

Purpose

This record closes the next evidence gap named by the runtime boundary policy expansion:

```
abuse-case mapping needs broader fixture coverage
```

The expansion is fixture coverage only.

It turns the abuse-case map from documentation-only policy evidence into deterministic C fixtures that exercise the current runtime-boundary classifier and report surface.

Relationship to previous work

This fixture expansion depends on:

```
docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
docs/RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md
docs/RUNTIME_BOUNDARY_POLICY_MATRIX_REFINEMENT.md
docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REFINEMENT.md
docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md
include/latticra/runtime_boundary.h
include/latticra/runtime_boundary_domain_matrix.h
src/runtime_boundary.c
src/runtime_boundary_domain_matrix_eval.c
tests/runtime_boundary_abuse_case_fixtures.c
scripts/test-runtime-boundary-abuse-case-fixtures.sh
scripts/test-runtime-boundary-policy-expansion-after-threat-model.sh
scripts/test-runtime-boundary.sh
```

Those files remain no-effect and report/classification oriented.

Current checkpoint

Current runtime-boundary abuse-case fixture posture:

```
runtime_boundary_abuse_case_fixture_expansion_present=1
runtime_boundary_abuse_case_fixture_guard_present=1
runtime_boundary_abuse_case_c_fixtures_present=1
runtime_boundary_abuse_case_fixture_count=8
runtime_boundary_policy_expansion_after_threat_model_present=1
defensive_threat_model_validation_refinement_present=1
runtime_boundary_policy_matrix_present=1
runtime_boundary_domain_matrix_present=1
unknown_request_abuse_fixture_present=1
unknown_effect_abuse_fixture_present=1
future_gated_execution_abuse_fixture_present=1
operator_confirmation_non_override_fixture_present=1
denial_reason_report_fixture_present=1
authority_failure_abuse_fixture_present=1
invalid_lir_prerequisite_fixture_present=1
blocked_effect_abuse_fixture_present=1
report_reason_assertions_present=1
policy_matrix_assertions_present=1
domain_matrix_assertions_present=1
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_verification_added=0
signing_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
runtime_authority_granted=0
production_protection_claim_allowed=0
runtime_protection_claim_allowed=0
malware_prevention_claim_allowed=0
ransomware_prevention_claim_allowed=0
sandbox_claim_allowed=0
certification_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
completion_estimate_review_required=0
```

Fixture map

The new C fixture table covers:

```
| Fixture id | Abuse case | Expected current posture |
| --- | --- | --- |
| unknown-request-is-not-allowed | unknown request is treated as allowed | deny,
```

```
unknown-request, invalid matrix |
| unknown-effect-is-not-allowed | unknown effect is treated as allowed | deny, unknown-effect, invalid matrix |
| future-gated-execution-is-not-executable | future-gated request is treated as executable | future-gated, planned gate, no execution |
| operator-confirmation-cannot-override-policy | operator confirmation overrides policy | deny, operator-confirmation-not-supported |
| denial-reason-stays-rendered | report omits denial reason | report includes deterministic denial reason |
| authority-failure-is-not-allowed | failed authority metadata is treated as allowed | deny, authority/prerequisite failure |
| invalid-lir-prerequisite-stays-denied | invalid LIR input reaches later behavior | deny, LIR prerequisite failure |
| blocked-effect-stays-blocked | blocked effect is treated as no-effect | deny, blocked-effect policy matrix |
```

Each fixture also asserts:

```
no_effect=1
execution_allowed=0
mutation_allowed=0
file_io_allowed=0
network_allowed=0
server_allowed=0
recovery_allowed=0
rollback_allowed=0
hardware_allowed=0
boot_allowed=0
record_executed=0
```

Validation

This fixture expansion is guarded by:

```
sh scripts/test-runtime-boundary-abuse-case-fixtures.sh
```

The guard verifies the documentation checkpoint, the fixture source, public entry points, and the runtime-boundary C test runner.

Expected focused fixture output:

```
runtime_boundary_abuse_case_fixtures: ok
```

Boundary

This expansion adds fixtures and guard coverage only.

It does not add runtime behavior, command execution, Lat execution, LIR execution, task effect execution, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback behavior, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, certification, accreditation, compliance, production protection, operating-system completeness, or runtime authority.

The next valid work should remain completion-estimate review only if capability posture changes; otherwise continue small guarded report/status alignment only when drift appears.

docs/RUNTIME_BOUNDARY_CONTRACT.md

Source title: Latticra Runtime Boundary Contract

Lines: 569 | Words: 1465 | SHA-256: 9e54ce479d635eb9831a6b2150f639383541ad12e01ee31fab59746c9cb8edac

Latticra Runtime Boundary Contract

Status: Runtime boundary contract Scope: first runtime boundary definition before runtime implementation, command behavior, Lat execution, LIR execution, task effect execution, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware behavior, boot behavior, terminal control, sandboxing, malware prevention, ransomware prevention, or operating-system behavior.

Purpose

This document defines the first runtime boundary contract for Latticra.

Runtime boundary means the line between validated metadata/report/classification surfaces and any future behavior that could execute, mutate, interact with a host, contact a network, affect recovery, affect hardware, affect boot state, or claim runtime authority.

This document does not implement a runtime.

Relationship to previous work

This contract depends on:

```
docs/EFFECT_GATES.md
docs/SUPERVISOR_ARCHITECTURE.md
docs/NUCLEUS_PREVIEW.md
docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION_PLAN.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/STATE_LATTICE.md
docs/TRI_PLANE_TRANSITION.md
include/latticra/nucleus_preview.h
include/latticra/nucleus_task.h
include/latticra/l_ui_renderer.h
include/latticra/lir.h
include/latticra/lat_parser.h
include/latticra/cpp/authority.hpp
src/nucleus_preview.c
src/nucleus_task.c
src/l_ui_renderer.c
src/lir.c
src/lat_parser.c
src/cpp/authority.cpp
```

Those files remain the source of truth for effect vocabulary, Nucleus preview behavior, task classification/reporting, constrained authority, rendering, LIR metadata, Lat metadata, state lattice, tri-plane transition, and current no-effect boundaries.

Direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Runtime work may not bypass this relationship:

```
Lat / Latticra Language: declares contracts and intended effects
LIR: carries validated metadata
C++ authority layer: validates authority and reports decisions
C substrate: owns bounded runtime-adjacent records and ABI-compatible surfaces
Nucleus: coordinates only after contracts and gates permit it
Runtime: remains disabled until explicit boundaries, authority, tests, and evidence exist
```

Current runtime boundary

Current Latticra behavior includes no production runtime.

Current implemented behavior is limited to:

```
parsing
```

```
validation
classification
preview
reporting
rendering metadata
no-effect task classification
```

Current implemented behavior does not include:

```
command execution
Lat execution
LIR execution
runtime scheduling
process launching
plugin loading
file I/O
network I/O
server interaction
self-update
recovery behavior
rollback
hardware access
boot behavior
terminal control
sandbox enforcement
malware prevention
ransomware prevention
operating-system behavior
```

Runtime posture

The first runtime posture is disabled-by-default.

Initial runtime posture:

```
runtime implemented? no
default policy: deny
report-only allowed: yes
validation-only allowed: yes
preview-only allowed: yes
classification-only allowed: yes
command execution allowed: no
Lat execution allowed: no
LIR execution allowed: no
mutation allowed: no
file I/O allowed: no
network I/O allowed: no
server interaction allowed: no
self-update allowed: no
recovery allowed: no
rollback allowed: no
hardware allowed: no
boot behavior allowed: no
operator confirmation allowed to override: no
```

Operator confirmation may be represented as metadata later, but it must not override runtime policy in the first implementation.

Runtime boundary modes

Future runtime boundary work should use explicit modes.

Initial mode labels:

```
disabled
report-only
validation-only
preview-only
classification-only
deny-all
requires-future-gate
```

The first runtime implementation plan must choose exact enum names.

No mode may imply command execution, Lat execution, LIR execution, mutation, file I/O, network I/O, recovery, hardware, boot behavior, or sandboxing.

Runtime request kinds

Future runtime request kinds should include:

```
parse-only
validate-only
classify-only
render-report
nucleus-task-report
lat-validate
lir-validate
authority-check
runtime-execute
command-execute
lat-execute
lir-execute
file-read
file-write
network-open
server-interaction
self-update
recovery-action
rollback-action
hardware-action
boot-action
unknown
```

Only parse, validate, classify, render/report, and authority-check requests may be considered for first no-effect runtime handling.

All effect-performing requests remain denied.

Runtime effects

Runtime boundary work must classify effects before handling.

Initial runtime effects:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Default policy:

```
none -&gt; report/validation/classification only
read -&gt; approved local metadata only
local_mutation -&gt; deny until explicit local-mutation contract
host_mutation -&gt; deny
network -&gt; deny
hardware -&gt; deny
boot -&gt; deny
recovery -&gt; deny
external -&gt; deny
unknown -&gt; deny
```

Runtime policy results

Future runtime policy labels:

```
allow-report
allow-validation
allow-classification
allow-preview
deny
blocked
requires-future-gate
unsupported
internal-error
```

The first future implementation must not produce an executed result for effect-performing work.

Runtime denial reasons

Future denial labels:

```
ok
null-argument
unknown-request
unknown-effect
unsupported-request
unsupported-effect
parser-failed
semantic-failed
lir-failed
render-failed
authority-failed
task-failed
runtime-disabled
effect-blocked
effect-requires-future-gate
non-no-effect-flags
operator-confirmation-not-supported
implementation-not-present
internal-error
```

Denial reasons must be stable and report-visible.

Required prerequisites

A future runtime request may be considered only after relevant prerequisites are explicit.

Possible prerequisites:

```
parser_error=ok
semantic_error=ok
lir_error=ok
render_status=ok
authority_status=ok
task_policy=allow-report or allow-validation or allow-classification or allow-preview
runtime_mode=report-only or validation-only or classification-only or preview-only
effect_gate=allowed-report or allowed-validation or allowed-classification or allowed-preview
no_effect=1
execution_allowed=0
mutation_allowed=0
server_allowed=0
recovery_allowed=0
hardware_allowed=0
operator_confirmation=not-applicable
```

If any prerequisite fails, the runtime boundary must deny the request and produce a deterministic report.

Authority requirements

Future runtime boundary work must consult authority metadata before any request is eligible.

Authority checks must include:

```
authority status
authority validator
authority requested effect
authority denial reason
authority no-effect flags
source identity
runtime request kind
runtime effect kind
runtime boundary mode
```

Authority failure must deny the request.

Authority success must not by itself execute a request. It only satisfies one prerequisite.

Task requirements

Future runtime boundary work must use Nucleus task classification/reporting before runtime handling.

Task requirements must include:

```
task policy
task denial reason
task requested effect
task allowed effect
task executed flag
task mutation_allowed flag
task server_interaction_allowed flag
task recovery_allowed flag
task hardware_allowed flag
```

If the task record indicates execution, mutation, server interaction, recovery, or hardware behavior, the first runtime boundary must reject it.

Runtime record shape

A future runtime boundary record should include:

```
runtime_id
request_kind
requested_effect
allowed_effect
runtime_mode
policy_result
denial_reason
authority_status
task_policy
task_reason
effect_gate_state
operator_confirmation
executed
mutation_allowed
file_io_allowed
network_allowed
server_interaction_allowed
recovery_allowed
rollback_allowed
hardware_allowed
boot_allowed
evidence_level
source_identity
source_span
```

The first future implementation should use fixed-size fields and caller-owned buffers.

Runtime report format

A future runtime boundary report should be deterministic and bounded.

Suggested report header:

```
LATTICRA RUNTIME BOUNDARY REPORT
runtime_id=&lt;id&gt;
request=&lt;request-kind&gt;
requested_effect=&lt;effect&gt;
allowed_effect=&lt;effect&gt;
mode=&lt;runtime-mode&gt;
policy=&lt;policy-result&gt;
reason=&lt;denial-reason&gt;
authority_status=&lt;authority-status&gt;
task_policy=&lt;task-policy&gt;
task_reason=&lt;task-reason&gt;
effect_gate_state=&lt;gate-state&gt;
operator_confirmation=&lt;confirmation-state&gt;
executed=0
mutation_allowed=0
file_io_allowed=0
network_allowed=0
server_interaction_allowed=0
recovery_allowed=0
rollback_allowed=0
```

```
hardware_allowed=0
boot_allowed=0
evidence_level=&lt;level&gt;
source_identity=&lt;source&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
```

Reports must not include secrets, host environment values, filesystem contents, network data, credentials, tokens, keys, hardware identifiers, or process information.

Output buffer policy

Runtime boundary reports must:

```
write only to caller-provided buffers
require explicit buffer length
NUL-terminate on success
clear the buffer on too-small failure
avoid heap allocation
avoid file output
avoid stdout
avoid stderr
avoid terminal escape control
```

First implementation gate

Runtime boundary implementation must not begin until a separate implementation plan defines:

1. implementation language;
2. public header path;
3. source file path;
4. runtime request struct;
5. runtime record struct;
6. runtime result struct;
7. runtime mode enum;
8. runtime policy enum;
9. runtime denial enum;
10. authority summary usage;
11. Nucleus task result usage;
12. effect-gate state usage;
13. operator-confirmation metadata shape;
14. report format;
15. capacity constants;
16. output-buffer behavior;
17. exact tests;
18. compatibility expectations;
19. non-claims.

That plan should be recorded in:

```
RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
```

Future file policy

Recommended future files:

```
include/latticra/runtime_boundary.h
src/runtime_boundary.c
tests/runtime_boundary_invariants.c
scripts/test-runtime-boundary.sh
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
```

The implementation plan must confirm exact paths before runtime boundary code is added.

Future test list

A future implementation plan should include tests for:

```
runtime_boundary_denies_unknown_request
runtime_boundary_denies_unknown_effect
runtime_boundary_denies_runtime_execute
runtime_boundary_denies_command_execute
runtime_boundary_denies_lat_execute
runtime_boundary_denies_lir_execute
runtime_boundary_denies_file_read
runtime_boundary_denies_file_write
runtime_boundary_denies_network_open
runtime_boundary_denies_server_interaction
runtime_boundary_denies_self_update
runtime_boundary_denies_recovery_action
runtime_boundary_denies_rollback_action
runtime_boundary_denies_hardware_action
runtime_boundary_denies_boot_action
runtime_boundary_requires_authority_success
runtime_boundary_requires_task_success
runtime_boundary_requires_no_effect_flags
runtime_boundary_allows_parse_only_report_only
runtime_boundary_allows_validate_only
runtime_boundary_allows_classify_only
runtime_boundary_allows_render_report_only
runtime_boundary_report_is_deterministic
runtime_boundary_report_rejects_small_buffer
runtime_boundary_does_not_mutate_state
runtime_boundary_does_not_write_files
runtime_boundary_does_not_open_network
runtime_boundary_does_not_touch_hardware
runtime_boundary_does_not_call_recovery
runtime_boundary_does_not_override_policy_with_operator_confirmation
```

Compatibility expectations

Future runtime boundary work must not change:

```
existing Nucleus preview classification behavior
existing Nucleus task classification behavior
existing Nucleus task report behavior
state lattice behavior
tri-plane transition behavior
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
L-UI rendering behavior
Lat grammar behavior
constrained C++ authority behavior
no-effect flags
current accepted fixture counts
```

Forbidden behavior

Runtime boundary work must not:

- bypass Nucleus task classification;
- bypass effect gates;
- bypass constrained C++ authority validation;
- bypass parser, semantic, LIR, render, or task prerequisites when relevant;
- execute unknown requests;
- execute unknown effects;

- execute commands;
- execute Lat;
- execute LIR;
- launch processes;
- load plugins;
- mutate state;
- write files;
- read host files outside approved metadata;
- open network connections;
- call server code;
- call update code;
- call recovery code;
- perform rollback;
- touch hardware;
- alter boot state;
- use terminal control;
- let operator confirmation override policy;
- hide denial reasons;
- omit runtime records;
- emit secrets, host environment values, credentials, tokens, keys, hardware identifiers, or process information;
- imply a production runtime, sandbox, malware prevention, ransomware prevention, recovery system, update system, or operating-system surface.

Current validation command

This contract is guarded by:

```
sh scripts/test-runtime-boundary-contract.sh
```

The guard is static. It does not implement runtime behavior.

Non-claims

This document does not implement runtime behavior, command behavior, Lat execution, LIR execution, task effect execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_MAIN_TEST_INTEGRATION_AUDIT.md

Source title: Runtime Boundary Domain Matrix Main Test Integration Audit

Lines: 23 | Words: 70 | SHA-256: b09447eb38409f253a150aab2dffc9d4b7b04696d4140e290ec183f9cbf6b3e3

Runtime Boundary Domain Matrix Main Test Integration Audit

Status: initial audit guard

This record verifies that the RBDM report integration is covered by the main runtime-boundary test runner, not only by its focused runner.

Audit expectations:

```
scripts/test-runtime-boundary.sh compiles tests/runtime_boundary_*.c
scripts/test-runtime-boundary.sh links src/runtime_boundary_domain_matrix_report.c
tests/runtime_boundary_domain_matrix_report_integration.c is included by the
runtime_boundary_*.c glob
.github/workflows/runtime-boundary.yml runs scripts/test-runtime-boundary.sh
```

Validation:

```
sh scripts/test-runtime-boundary-domain-matrix-main-test-integration-audit.sh
sh scripts/test-runtime-boundary.sh
```

Boundary: audit/guard only. No runtime behavior is added.

docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REFINEMENT.md

Source title: Runtime Boundary Domain Matrix Refinement

Lines: 34 | Words: 44 | SHA-256: 149294605d9cda0ae8610357262bcdb9c85a31b04e9cdad251e0694cd761ac2a

Runtime Boundary Domain Matrix Refinement

Status: initial companion implementation.

This record documents the RBDM companion slice.

Tracked files:

```
include/latticra/runtime_boundary_domain_matrix.h
src/runtime_boundary_domain_matrix.c
src/runtime_boundary_domain_matrix_eval.c
tests/runtime_boundary_domain_matrix_refinement.c
scripts/test-runtime-boundary-domain-matrix-refinement.sh
.github/workflows/runtime-boundary-domain-matrix-refinement.yml
```

Matrix cells:

```
declarative
operational
future-gated
blocked
invalid
unknown
```

Validation:

```
sh scripts/test-runtime-boundary-domain-matrix-refinement.sh
sh scripts/test-runtime-boundary.sh
```

docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md

Source title: Runtime Boundary Domain Matrix Report Integration

Lines: 22 | Words: 45 | SHA-256: 173000044b40b3bbfde03579e0635b2e1b51c26664debba542f0f5e530c45bf3

Runtime Boundary Domain Matrix Report Integration

Status: initial report integration

This slice implements the RBDM report function declared by the companion domain-matrix API.

Files:

```
src/runtime_boundary_domain_matrix_report.c
```

```
tests/runtime_boundary_domain_matrix_report_integration.c
scripts/test-runtime-boundary-domain-matrix-report-integration.sh
.github/workflows/runtime-boundary-domain-matrix-report-integration.yml
```

Validation:

```
sh scripts/test-runtime-boundary-domain-matrix-report-integration.sh
```

Boundary: report metadata only. No runtime behavior is added.

docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md

Source title: Latticra Runtime Boundary Implementation

Lines: 145 | Words: 840 | SHA-256: 6c4a745f3ee32cabaefe87c940e6622111d0a9dea6e3df0a478ea6fe032f3280

Latticra Runtime Boundary Implementation

Status: initial implementation record Scope: C runtime boundary API surface, deterministic labels, default-deny classification, no-effect allow-mode classification, allowed-effect report field, authority prerequisites, authority flag report fields, authority label report fields, Nucleus task prerequisites and happy path, Nucleus task flag report fields, render/Lat/LIR prerequisite status report fields, render/Lat/LIR prerequisite denials and happy paths, runtime identity copying, source metadata copying, future-gate classification, expanded report fields, invariant tests, caller-provided report buffer, no-effect posture, and non-claims.

Purpose

This document records the first runtime boundary implementation surface.

The implementation follows:

```
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
```

This slice adds the public runtime boundary API, source file, invariant tests, and test runner.

The current source is no-effect and disabled-by-default. It does not implement operational runtime behavior.

Implementation files

This slice includes:

```
include/latticra/runtime_boundary.h
src/runtime_boundary.c
tests/runtime_boundary_invariants.c
tests/runtime_boundary_effect_flag_report.c
tests/runtime_boundary_allowed_effect_report.c
tests/runtime_boundary_authority_flag_report.c
tests/runtime_boundary_authority_label_report.c
tests/runtime_boundary_abuse_case_fixtures.c
tests/runtime_boundary_task_flag_report.c
tests/runtime_boundary_prerequisite_status_report.c
tests/runtime_boundary_prerequisite_denial.c
tests/runtime_boundary_allow_modes.c
tests/runtime_boundary_prerequisite_happy_paths.c
tests/runtime_boundary_task_report_happy_path.c
scripts/test-runtime-boundary.sh
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
```

Implemented surface

The public API defines request kinds, effects, modes, policies, denial labels, gate states, operator-confirmation metadata, authority summary metadata, request records, result records, classifier entry point, and report entry point.

The source provides:

```
deterministic request/effect/mode/policy/denial/gate/operator labels
default-deny classification
no-effect report-mode allow classification
no-effect validation-mode allow classification
no-effect classification-mode allow classification
allowed-effect report visibility
runtime_id copying
record_count initialization and report visibility
authority presence and status checks
authority no-effect flag checks
authority execution/mutation/server/recovery/hardware flag report visibility
authority status label, validator label, requested-effect label, and denial-reason report
visibility
Nucleus task-result prerequisite checks for task-report requests
Nucleus task-report happy path when task metadata is OK and report mode matches
task policy and task reason metadata copying
task executed/mutation/server/recovery/hardware flag copying and report visibility
render status/error metadata copying and report visibility
render-report prerequisite denial when render metadata is missing or failed
render-report happy path when render metadata is OK and report mode matches
Lat parser status/error metadata copying and report visibility
Lat validation prerequisite denial when Lat metadata is missing or failed
Lat validation happy path when Lat metadata is OK and validation mode matches
LIR status/error metadata copying and report visibility
LIR validation prerequisite denial when LIR metadata is missing or failed
LIR validation happy path when LIR metadata is OK and validation mode matches
source identity copying
source span metadata copying
unknown request denial
unknown effect denial
future-gate classification for operational request kinds
operator-confirmation non-override behavior
bounded report output with policy, reason, and gate state
expanded report output for runtime_id, record_count, request, requested effect, allowed effect,
mode, operator confirmation, authority status, authority label strings, authority no-effect
state, authority effect flags, task policy, task reason, task effect flags, render prerequisite
status, Lat prerequisite status, LIR prerequisite status, no-effect flag, execution flag,
mutation flag, file I/O flag, network flag, server flag, recovery flag, rollback flag, hardware
flag, boot flag, source identity, and source span metadata
small-buffer rejection and clearing
```

Validation

Run:

```
sh scripts/test-runtime-boundary.sh
```

The runtime boundary tests verify:

```
runtime boundary classification initializes a no-effect result
runtime boundary abuse-case fixtures deny unknown request, unknown effect, future-gated
execution, operator override, missing task metadata, missing authority metadata, invalid LIR
prerequisites, and blocked effects
runtime_id is copied from request to record
record_count is initialized
runtime_id and record_count report fields are present
requested-effect and allowed-effect report fields are present
matched no-effect report mode allows report requests
matched no-effect validation mode allows validation requests
matched no-effect classification mode allows classification requests
valid render metadata allows render-report in report mode
valid Lat parser metadata allows Lat validation in validation mode
valid LIR metadata allows LIR validation in validation mode
valid Nucleus task metadata allows task-report in report mode
mismatched modes remain denied
default policy denies
source identity metadata is copied
source span metadata is copied
missing authority is denied
failed authority status is denied
non-no-effect authority flags are denied
authority effect flags are present and zero in reports
authority label strings are present in reports
missing Nucleus task metadata is denied for task reports
valid Nucleus task metadata is copied for task reports
Nucleus task effect flags are copied and visible in reports
render/Lat/LIR prerequisite status fields are visible in reports
failed render metadata is denied with render-failed
failed Lat metadata is denied with parser-failed
failed LIR metadata is denied with lir-failed
```

```
unknown requests are denied
unknown effects are denied
operational request kinds require a future gate
operator confirmation does not override policy
runtime boundary reports are bounded
expanded report fields are present
source identity and source span report fields are present
file I/O, network, server, recovery, rollback, hardware, and boot report flags are present and
zero
small buffers are rejected and cleared
null arguments are handled safely
```

Boundary

This implementation does not provide runtime behavior, command behavior, Lat execution, LIR execution, task effect execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

Note

The fuller classification policy remains specified in the contract and implementation plan. This source slice expands the public C API behavior while preserving the denied-by-default runtime boundary.

docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md

Source title: Latticra Runtime Boundary Implementation Plan

Lines: 790 | Words: 1759 | SHA-256: 7c0afd077d15226f1329e42ea62111b03ac78faa5552a43887d2ebb5bcec68ba

Latticra Runtime Boundary Implementation Plan

Status: implementation planning contract Scope: exact public API, C implementation files, runtime request/record/result structs, runtime mode enum, runtime policy enum, runtime denial enum, effect enum, authority summary usage, Nucleus task result usage, effect-gate state usage, operator-confirmation metadata, report format, capacity constants, output-buffer behavior, tests, compatibility expectations, and non-claims before runtime boundary code.

Purpose

This document defines the implementation plan for the first runtime boundary surface.

The Runtime boundary contract is already merged and guarded. This plan turns that contract into exact public API, file paths, data structures, enums, capacity constants, report rules, prerequisite behavior, task-result behavior, authority behavior, exact tests, compatibility expectations, and non-claims before runtime boundary code is added.

This document does not implement runtime behavior.

Relationship to previous work

This plan depends on:

```
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/EFFECT_GATES.md
docs/SUPERVISOR_ARCHITECTURE.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
docs/NUCLEUS_PREVIEW.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
include/latticra/nucleus_task.h
include/latticra/nucleus_preview.h
include/latticra/l_ui_renderer.h
```

```
include/latticra/lir.h
include/latticra/lat_parser.h
include/latticra/cpp/authority.hpp
src/nucleus_task.c
src/nucleus_preview.c
src/l_ui_renderer.c
src/lir.c
src/lat_parser.c
src/cpp/authority.cpp
```

Those files remain the source of truth for task classification, preview classification, constrained authority, rendering, LIR metadata, Lat metadata, and no-effect boundaries.

Implementation language decision

The first runtime boundary surface should be implemented in C.

Reason:

- Nucleus preview is C;
- Nucleus task classification/reporting is C;
- L-UI rendering, LIR shape, and Lat grammar are C;
- the boundary should use bounded structs and caller-provided buffers;
- C remains the secure substrate;
- the constrained C++ authority layer remains a validation/report dependency, not the runtime boundary storage substrate.

The first implementation must not expose C++ object lifetimes through the public C runtime boundary API.

Implementation files

The implementation PR should add or modify:

```
include/latticra/runtime_boundary.h
src/runtime_boundary.c
tests/runtime_boundary_invariants.c
scripts/test-runtime-boundary.sh
.github/workflows/c.yml
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
```

The implementation PR should not add command behavior, Lat execution, LIR execution, task effect execution, live movement, mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware behavior, boot behavior, terminal control, sandboxing, malware prevention, ransomware prevention, or operating-system behavior.

Public API shape

Add public API names:

```
latticra_runtime_boundary_request_kind_t
latticra_runtime_boundary_effect_t
latticra_runtime_boundary_mode_t
latticra_runtime_boundary_policy_t
latticra_runtime_boundary_denial_t
latticra_runtime_boundary_gate_state_t
latticra_runtime_boundary_operator_confirmation_t
latticra_runtime_boundary_authority_summary_t
latticra_runtime_boundary_request_t
latticra_runtime_boundary_record_t
latticra_runtime_boundary_result_t
latticra_runtime_boundary_request_kind_label
latticra_runtime_boundary_effect_label
latticra_runtime_boundary_mode_label
latticra_runtime_boundary_policy_label
latticra_runtime_boundary_denial_label
latticra_runtime_boundary_gate_state_label
latticra_runtime_boundary_operator_confirmation_label
```

```
latticra_runtime_boundary_classify
latticra_runtime_boundary_report
```

Recommended function signatures:

```
const char
*latticra_runtime_boundary_request_kind_label(latticra_runtime_boundary_request_kind_t kind);
const char *latticra_runtime_boundary_effect_label(latticra_runtime_boundary_effect_t effect);
const char *latticra_runtime_boundary_mode_label(latticra_runtime_boundary_mode_t mode);
const char *latticra_runtime_boundary_policy_label(latticra_runtime_boundary_policy_t policy);
const char *latticra_runtime_boundary_denial_label(latticra_runtime_boundary_denial_t denial);
const char *latticra_runtime_boundary_gate_state_label(latticra_runtime_boundary_gate_state_t
gate_state);
const char *latticra_runtime_boundary_operator_confirmation_label(latticra_runtime_boundary_oper
ator_confirmation_t confirmation);

latticra_status_t latticra_runtime_boundary_classify(
    const latticra_runtime_boundary_request_t *request,
    latticra_runtime_boundary_result_t *result);

latticra_status_t latticra_runtime_boundary_report(
    const latticra_runtime_boundary_result_t *result,
    char *buffer,
    size_t buffer_len);
```

The first implementation should classify and report only. It must not execute runtime behavior.

Capacity constants

Add exact bounded constants:

```
LATTICRA_RUNTIME_BOUNDARY_ID_MAX 64u
LATTICRA_RUNTIME_BOUNDARY_LABEL_MAX 64u
LATTICRA_RUNTIME_BOUNDARY_REASON_MAX 128u
LATTICRA_RUNTIME_BOUNDARY_SOURCE_IDENTITY_MAX 128u
LATTICRA_RUNTIME_BOUNDARY_REPORT_MAX 4096u
LATTICRA_RUNTIME_BOUNDARY_RECORD_MAX 16u
```

Small output buffers must return LATTICRA_STATUS_BUFFER_TOO_SMALL and clear the output buffer.

Request kind enum

Add request kinds:

```
LATTICRA_RUNTIME_BOUNDARY_PARSE_ONLY
LATTICRA_RUNTIME_BOUNDARY_VALIDATE_ONLY
LATTICRA_RUNTIME_BOUNDARY_CLASSIFY_ONLY
LATTICRA_RUNTIME_BOUNDARY_RENDER_REPORT
LATTICRA_RUNTIME_BOUNDARY_NUCLEUS_TASK_REPORT
LATTICRA_RUNTIME_BOUNDARY_LAT_VALIDATE
LATTICRA_RUNTIME_BOUNDARY_LIR_VALIDATE
LATTICRA_RUNTIME_BOUNDARY_AUTHORITY_CHECK
LATTICRA_RUNTIME_BOUNDARY_RUNTIME_EXECUTE
LATTICRA_RUNTIME_BOUNDARY_COMMAND_EXECUTE
LATTICRA_RUNTIME_BOUNDARY_LAT_EXECUTE
LATTICRA_RUNTIME_BOUNDARY_LIR_EXECUTE
LATTICRA_RUNTIME_BOUNDARY_FILE_READ
LATTICRA_RUNTIME_BOUNDARY_FILE_WRITE
LATTICRA_RUNTIME_BOUNDARY_NETWORK_OPEN
LATTICRA_RUNTIME_BOUNDARY_SERVER_INTERACTION
LATTICRA_RUNTIME_BOUNDARY_SELF_UPDATE
LATTICRA_RUNTIME_BOUNDARY_RECOVERY_ACTION
LATTICRA_RUNTIME_BOUNDARY_ROLLBACK_ACTION
LATTICRA_RUNTIME_BOUNDARY_HARDWARE_ACTION
LATTICRA_RUNTIME_BOUNDARY_BOOT_ACTION
LATTICRA_RUNTIME_BOUNDARY_UNKNOWN
```

Stable labels:

```
parse-only
validate-only
classify-only
render-report
nucleus-task-report
lat-validate
lir-validate
```

```
authority-check
runtime-execute
command-execute
lat-execute
lir-execute
file-read
file-write
network-open
server-interaction
self-update
recovery-action
rollback-action
hardware-action
boot-action
unknown
```

Effect enum

Add runtime effects:

```
LATTICRA_RUNTIME_BOUNDARY_EFFECT_NONE
LATTICRA_RUNTIME_BOUNDARY_EFFECT_READ
LATTICRA_RUNTIME_BOUNDARY_EFFECT_LOCAL_MUTATION
LATTICRA_RUNTIME_BOUNDARY_EFFECT_HOST_MUTATION
LATTICRA_RUNTIME_BOUNDARY_EFFECT_NETWORK
LATTICRA_RUNTIME_BOUNDARY_EFFECT_HARDWARE
LATTICRA_RUNTIME_BOUNDARY_EFFECT_BOOT
LATTICRA_RUNTIME_BOUNDARY_EFFECT_RECOVERY
LATTICRA_RUNTIME_BOUNDARY_EFFECT_EXTERNAL
LATTICRA_RUNTIME_BOUNDARY_EFFECT_UNKNOWN
```

Stable labels:

```
none
read
local_mutation
host_mutation
network
hardware
boot
recovery
external
unknown
```

Only none and read may be eligible for no-effect report, validation, classification, or preview boundary handling.

Runtime mode enum

Add runtime modes:

```
LATTICRA_RUNTIME_BOUNDARY_MODE_DISABLED
LATTICRA_RUNTIME_BOUNDARY_MODE_REPORT_ONLY
LATTICRA_RUNTIME_BOUNDARY_MODE_VALIDATION_ONLY
LATTICRA_RUNTIME_BOUNDARY_MODE_PREVIEW_ONLY
LATTICRA_RUNTIME_BOUNDARY_MODE_CLASSIFICATION_ONLY
LATTICRA_RUNTIME_BOUNDARY_MODE_DENY_ALL
LATTICRA_RUNTIME_BOUNDARY_MODE_REQUIRES_FUTURE_GATE
```

Stable labels:

```
disabled
report-only
validation-only
preview-only
classification-only
deny-all
requires-future-gate
```

No mode may imply execution, mutation, file I/O, network I/O, recovery, hardware, boot behavior, or sandboxing.

Policy enum

Add runtime policies:

```
LATTICRA_RUNTIME_BOUNDARY_POLICY_ALLOW_REPORT
LATTICRA_RUNTIME_BOUNDARY_POLICY_ALLOW_VALIDATION
```

```
LATTICRA_RUNTIME_BOUNDARY_POLICY_ALLOW_CLASSIFICATION
LATTICRA_RUNTIME_BOUNDARY_POLICY_ALLOW_PREVIEW
LATTICRA_RUNTIME_BOUNDARY_POLICY_DENY
LATTICRA_RUNTIME_BOUNDARY_POLICY_BLOCKED
LATTICRA_RUNTIME_BOUNDARY_POLICY_REQUIRES_FUTURE_GATE
LATTICRA_RUNTIME_BOUNDARY_POLICY_UNSUPPORTED
LATTICRA_RUNTIME_BOUNDARY_POLICY_INTERNAL_ERROR
```

Stable labels:

```
allow-report
allow-validation
allow-classification
allow-preview
deny
blocked
requires-future-gate
unsupported
internal-error
```

The first implementation must not produce an executed effect-performing result.

Denial enum

Add denial reasons:

```
LATTICRA_RUNTIME_BOUNDARY_DENIAL_OK
LATTICRA_RUNTIME_BOUNDARY_DENIAL_NULL_ARGUMENT
LATTICRA_RUNTIME_BOUNDARY_DENIAL_UNKNOWN_REQUEST
LATTICRA_RUNTIME_BOUNDARY_DENIAL_UNKNOWN_EFFECT
LATTICRA_RUNTIME_BOUNDARY_DENIAL_UNSUPPORTED_REQUEST
LATTICRA_RUNTIME_BOUNDARY_DENIAL_UNSUPPORTED_EFFECT
LATTICRA_RUNTIME_BOUNDARY_DENIAL_PARSER_FAILED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_SEMANTIC_FAILED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_LIR_FAILED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_RENDER_FAILED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_AUTHORITY_FAILED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_TASK_FAILED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_RUNTIME_DISABLED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_EFFECT_BLOCKED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_EFFECT_REQUIRES_FUTURE_GATE
LATTICRA_RUNTIME_BOUNDARY_DENIAL_NON_NO_EFFECT_FLAGS
LATTICRA_RUNTIME_BOUNDARY_DENIAL_OPERATOR_CONFIRMATION_NOT_SUPPORTED
LATTICRA_RUNTIME_BOUNDARY_DENIAL_IMPLEMENTATION_NOT_PRESENT
LATTICRA_RUNTIME_BOUNDARY_DENIAL_INTERNAL_ERROR
```

Stable labels:

```
ok
null-argument
unknown-request
unknown-effect
unsupported-request
unsupported-effect
parser-failed
semantic-failed
lir-failed
render-failed
authority-failed
task-failed
runtime-disabled
effect-blocked
effect-requires-future-gate
non-no-effect-flags
operator-confirmation-not-supported
implementation-not-present
internal-error
```

Gate state enum

Add gate states:

```
LATTICRA_RUNTIME_BOUNDARY_GATE_DISABLED
LATTICRA_RUNTIME_BOUNDARY_GATE_BLOCKED
LATTICRA_RUNTIME_BOUNDARY_GATE_PLANNED
LATTICRA_RUNTIME_BOUNDARY_GATE_AVAILABLE
LATTICRA_RUNTIME_BOUNDARY_GATE_ARMED
LATTICRA_RUNTIME_BOUNDARY_GATE_EXECUTED
LATTICRA_RUNTIME_BOUNDARY_GATE_FAILED
```

Stable labels:

```
disabled
blocked
planned
available
armed
executed
failed
```

The first implementation should output only disabled, blocked, or planned for real classifications.

Operator confirmation enum

Add operator-confirmation states:

```
LATTICRA_RUNTIME_BOUNDARY_OPERATOR_NOT_APPLICABLE
LATTICRA_RUNTIME_BOUNDARY_OPERATOR_REQUIRED
LATTICRA_RUNTIME_BOUNDARY_OPERATOR_PRESENT
LATTICRA_RUNTIME_BOUNDARY_OPERATOR_REJECTED
LATTICRA_RUNTIME_BOUNDARY_OPERATOR_NOT_SUPPORTED
```

Stable labels:

```
not-applicable
required
present
rejected
not-supported
```

Confirmation is metadata only. It must not override policy.

Authority summary struct

Add a C-compatible authority summary:

```
latticra_status_t status;
char status_label[LATTICRA_RUNTIME_BOUNDARY_LABEL_MAX];
char validator_label[LATTICRA_RUNTIME_BOUNDARY_LABEL_MAX];
char requested_effect_label[LATTICRA_RUNTIME_BOUNDARY_LABEL_MAX];
char denial_reason[LATTICRA_RUNTIME_BOUNDARY_REASON_MAX];
int no_effect;
int execution_allowed;
int mutation_allowed;
int server_allowed;
int recovery_allowed;
int hardware_allowed;
```

This mirrors existing authority summary patterns and avoids exposing C++ object lifetimes through the C runtime boundary API.

Runtime request struct

Add request fields:

```
char runtime_id[LATTICRA_RUNTIME_BOUNDARY_ID_MAX];
latticra_runtime_boundary_request_kind_t request_kind;
latticra_runtime_boundary_effect_t requested_effect;
latticra_runtime_boundary_mode_t mode;
latticra_runtime_boundary_operator_confirmation_t operator_confirmation;
const latticra_runtime_boundary_authority_summary_t *authority;
const latticra_nucleus_task_result_t *task;
const latticra_l_ui_render_result_t *render;
const latticra_lir_module_t *lir;
const latticra_lat_parse_result_t *lat;
const char *source_identity;
size_t source_identity_len;
latticra_l_ui_source_span_t source_span;
```

The implementation must not retain caller-owned pointers after the call returns.

Runtime record struct

Add record fields:

```
char runtime_id[LATTICRA_RUNTIME_BOUNDARY_ID_MAX];
```

```
latticra_runtime_boundary_request_kind_t request_kind;
latticra_runtime_boundary_effect_t requested_effect;
latticra_runtime_boundary_effect_t allowed_effect;
latticra_runtime_boundary_mode_t mode;
latticra_runtime_boundary_policy_t policy;
latticra_runtime_boundary_denial_t denial;
latticra_runtime_boundary_gate_state_t gate_state;
latticra_runtime_boundary_operator_confirmation_t operator_confirmation;
latticra_runtime_boundary_authority_summary_t authority;
latticra_nucleus_task_policy_t task_policy;
latticra_nucleus_task_denial_t task_reason;
char source_identity[LATTICRA_RUNTIME_BOUNDARY_SOURCE_IDENTITY_MAX];
latticra_l_ui_source_span_t source_span;
int executed;
int mutation_allowed;
int file_io_allowed;
int network_allowed;
int server_interaction_allowed;
int recovery_allowed;
int rollback_allowed;
int hardware_allowed;
int boot_allowed;
unsigned int evidence_level;
```

Runtime result struct

Add result fields:

```
latticra_status_t status;
latticra_runtime_boundary_record_t record;
size_t record_count;
int no_effect;
int execution_allowed;
int mutation_allowed;
int file_io_allowed;
int network_allowed;
int server_allowed;
int recovery_allowed;
int rollback_allowed;
int hardware_allowed;
int boot_allowed;
```

The result must be fully initialized on success and failure.

Classification rules

Initial classification behavior:

```
parse-only + none -&gt; allow-report
parse-only + read -&gt; allow-report
validate-only + none -&gt; allow-validation
validate-only + read -&gt; allow-validation
classify-only + none -&gt; allow-classification
classify-only + read -&gt; allow-classification
render-report + none -&gt; allow-report
render-report + read -&gt; allow-report
nucleus-task-report + none -&gt; allow-report
nucleus-task-report + read -&gt; allow-report
lat-validate + none -&gt; allow-validation
lat-validate + read -&gt; allow-validation
lir-validate + none -&gt; allow-validation
lir-validate + read -&gt; allow-validation
authority-check + none -&gt; allow-validation
authority-check + read -&gt; allow-validation
runtime-execute -&gt; requires-future-gate
command-execute -&gt; requires-future-gate
lat-execute -&gt; requires-future-gate
lir-execute -&gt; requires-future-gate
file-read -&gt; requires-future-gate
file-write -&gt; requires-future-gate
network-open -&gt; requires-future-gate
server-interaction -&gt; requires-future-gate
self-update -&gt; requires-future-gate
recovery-action -&gt; requires-future-gate
rollback-action -&gt; requires-future-gate
hardware-action -&gt; requires-future-gate
boot-action -&gt; requires-future-gate
unknown -&gt; deny
unknown effect -&gt; deny
```

```
local_mutation -&gt; deny
host_mutation -&gt; deny
network -&gt; deny
hardware -&gt; deny
boot -&gt; deny
recovery -&gt; deny
external -&gt; deny
```

Allowed classifications still must set:

```
executed=0
mutation_allowed=0
file_io_allowed=0
network_allowed=0
server_interaction_allowed=0
recovery_allowed=0
rollback_allowed=0
hardware_allowed=0
boot_allowed=0
```

Prerequisite behavior

The first implementation must check relevant prerequisites.

Rules:

```
missing request -&gt; null-argument
missing result -&gt; null-argument
missing authority -&gt; authority-failed
non-ok authority -&gt; authority-failed
non-no-effect authority flags -&gt; non-no-effect-flags
operator confirmation present -&gt; operator-confirmation-not-supported
runtime mode disabled -&gt; runtime-disabled unless request is report-only metadata
render-report requires render_status=ok
nucleus-task-report requires task policy allow-report or allow-validation or
allow-classification or allow-preview
lat-validate requires parser_error=ok
lir-validate requires lir_error=ok
runtime-execute requires future gate
command-execute requires future gate
lat-execute requires future gate
lir-execute requires future gate
file-read requires future gate
file-write requires future gate
network-open requires future gate
server-interaction requires future gate
self-update requires future gate
recovery-action requires future gate
rollback-action requires future gate
hardware-action requires future gate
boot-action requires future gate
```

Failed prerequisites deny the request and must be visible in the report.

Report format

latticra_runtime_boundary_report should emit:

```
LATTICRA RUNTIME BOUNDARY REPORT
status=&lt;integer-status&gt;
runtime_id=&lt;id&gt;
request=&lt;request-kind&gt;
requested_effect=&lt;effect&gt;
allowed_effect=&lt;effect&gt;
mode=&lt;runtime-mode&gt;
policy=&lt;policy-result&gt;
reason=&lt;denial-reason&gt;
authority_status=&lt;authority-status&gt;
authority_validator=&lt;authority-validator&gt;
authority_reason=&lt;authority-reason&gt;
task_policy=&lt;task-policy&gt;
task_reason=&lt;task-reason&gt;
effect_gate_state=&lt;gate-state&gt;
operator_confirmation=&lt;confirmation-state&gt;
executed=0
mutation_allowed=0
file_io_allowed=0
network_allowed=0
server_interaction_allowed=0
```

```
recovery_allowed=0
rollback_allowed=0
hardware_allowed=0
boot_allowed=0
evidence_level=&lt;level&gt;
source_identity=&lt;source&gt;
span_start_offset=&lt;offset&gt;
span_end_offset=&lt;offset&gt;
```

Reports must not include secrets, host environment values, filesystem contents, network data, credentials, tokens, keys, hardware identifiers, or process information.

Output buffer behavior

latticra_runtime_boundary_report should:

```
write only to caller-provided buffers
require explicit buffer length
NUL-terminate on success
clear the buffer on too-small failure
return LATTICRA_STATUS_BUFFER_TOO_SMALL for small buffers
avoid heap allocation
avoid file output
avoid stdout
avoid stderr
avoid terminal escape control
```

Exact implementation test list

The implementation PR should include tests for:

```
runtime_boundary_denies_unknown_request
runtime_boundary_denies_unknown_effect
runtime_boundary_denies_runtime_execute
runtime_boundary_denies_command_execute
runtime_boundary_denies_lat_execute
runtime_boundary_denies_lir_execute
runtime_boundary_denies_file_read
runtime_boundary_denies_file_write
runtime_boundary_denies_network_open
runtime_boundary_denies_server_interaction
runtime_boundary_denies_self_update
runtime_boundary_denies_recovery_action
runtime_boundary_denies_rollback_action
runtime_boundary_denies_hardware_action
runtime_boundary_denies_boot_action
runtime_boundary_requires_authority_success
runtime_boundary_requires_task_success
runtime_boundary_requires_no_effect_flags
runtime_boundary_allows_parse_only_report_only
runtime_boundary_allows_validate_only
runtime_boundary_allows_classify_only
runtime_boundary_allows_render_report_only
runtime_boundary_report_is_deterministic
runtime_boundary_report_rejects_small_buffer
runtime_boundary_does_not_mutate_state
runtime_boundary_does_not_write_files
runtime_boundary_does_not_open_network
runtime_boundary_does_not_touch_hardware
runtime_boundary_does_not_call_recovery
runtime_boundary_does_not_override_policy_with_operator_confirmation
```

Test file plan

Add:

```
tests/runtime_boundary_invariants.c
scripts/test-runtime-boundary.sh
```

Wire into:

```
.github/workflows/c.yml
```

Run after:

```
sh scripts/test-runtime-boundary-implementation-plan.sh
```

and before:

```
sh scripts/test-l-ui-static-report.sh
```

Documentation update plan

The implementation PR should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md
docs/RUNTIME_BOUNDARY_CONTRACT.md
scripts/test-project-strategy-status-framework.sh
```

and add:

```
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
```

Compatibility expectations

The implementation must not change:

```
existing Nucleus preview classification behavior
existing Nucleus task classification behavior
existing Nucleus task report behavior
state lattice behavior
tri-plane transition behavior
L-UI parser behavior
L-UI semantic validation behavior
LIR shape behavior
L-UI rendering behavior
Lat grammar behavior
constrained C++ authority behavior
no-effect flags
current accepted fixture counts
```

Forbidden implementation behavior

The first implementation must not:

- bypass Nucleus task classification;
- bypass effect gates;
- bypass constrained C++ authority validation;
- bypass parser, semantic, LIR, render, or task prerequisites when relevant;
- execute unknown requests;
- execute unknown effects;
- execute commands;
- execute Lat;
- execute LIR;
- launch processes;
- load plugins;
- mutate state;
- write files;
- read host files outside approved metadata;
- open network connections;
- call server code;
- call update code;

- call recovery code;
- perform rollback;
- touch hardware;
- alter boot state;
- use terminal control;
- let operator confirmation override policy;
- hide denial reasons;
- omit runtime records;
- emit secrets, host environment values, credentials, tokens, keys, hardware identifiers, or process information;
- imply a production runtime, sandbox, malware prevention, ransomware prevention, recovery system, update system, or operating-system surface.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-runtime-boundary-implementation-plan.sh
```

The guard is static. It does not implement runtime boundary behavior.

Implementation gate

Runtime boundary code may be added only after this plan is merged.

Non-claims

This document does not implement runtime behavior, command behavior, Lat execution, LIR execution, task effect execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, or operating-system completeness.

docs/RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md

Source title: Latticra Runtime Boundary Policy Expansion After Threat-Model Validation

Lines: 247 | Words: 1333 | SHA-256: a8e2b18941575a015f4784be55aa6624811993509113fb0e71a45a9618ae0a35

Latticra Runtime Boundary Policy Expansion After Threat-Model Validation

Status: runtime boundary policy expansion after threat-model validation Source: local follow-up slice Scope: post-threat-model runtime-boundary policy expansion, request-family mapping, effect mapping, authority-prerequisite mapping, future-gate mapping, abuse-case mapping, evidence-gap mapping, public entry-point updates, and no-claim preservation. This document does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production protection, sandboxing, malware prevention, ransomware prevention, certification, compliance, or runtime authority.

Purpose

This record closes the next evidence gap named by the defensive threat model validation refinement:

```
runtime boundary source needs fuller policy expansion after threat-model validation
```

The expansion is policy and evidence mapping only.

It makes the runtime boundary policy vocabulary more explicit before any future runtime behavior is considered.

Relationship to previous work

This policy expansion depends on:

```
docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
docs/RUNTIME_BOUNDARY_POLICY_MATRIX_REFINEMENT.md
docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REFINEMENT.md
docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md
scripts/test-defensive-threat-model-validation-refinement.sh
scripts/test-runtime-boundary.sh
scripts/test-runtime-boundary-policy-matrix-refinement.sh
scripts/test-runtime-boundary-domain-matrix-refinement.sh
scripts/test-runtime-boundary-domain-matrix-report-integration.sh
```

Those files remain the source of truth for current runtime-boundary API behavior, matrix labels, domain labels, reports, and no-effect invariants.

Current checkpoint

Current runtime-boundary policy-expansion posture:

```
runtime_boundary_policy_expansion_after_threat_model_present=1
runtime_boundary_policy_expansion_after_threat_model_guard_present=1
defensive_threat_model_validation_refinement_present=1
runtime_boundary_contract_present=1
runtime_boundary_implementation_plan_present=1
runtime_boundary_implementation_present=1
runtime_boundary_policy_matrix_present=1
runtime_boundary_domain_matrix_present=1
runtime_boundary_domain_matrix_report_present=1
request_family_policy_map_present=1
effect_policy_map_present=1
authority_prerequisite_map_present=1
future_gate_policy_map_present=1
abuse_case_runtime_policy_map_present=1
evidence_gap_map_present=1
mode=policy-expansion-after-threat-model
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_verification_added=0
signing_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
runtime_authority_granted=0
production_protection_claim_allowed=0
runtime_protection_claim_allowed=0
malware_prevention_claim_allowed=0
ransomware_prevention_claim_allowed=0
sandbox_claim_allowed=0
certification_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
completion_estimate_review_required=0
```

Request-family policy map

Current request families:

```
| Request family | Examples | Current policy posture | Required evidence before promotion |
| --- | --- | --- | --- |
```

no-effect report	render-report, nucleus-task-report	allow only under report-only mode with satisfied prerequisites	report tests, prerequisite tests, no-effect flags
no-effect validation	parse-only, validate-only, lat-validate, lir-validate	allow only under validation-only mode with satisfied prerequisites	parser, Lat, LIR, and runtime-boundary tests
no-effect classification	classify-only, authority-check	allow only under classification-only mode with satisfied prerequisites	authority/status tests and runtime-boundary tests
operational runtime	runtime-execute, command-execute, lat-execute, lir-execute	requires future gate and remains denied	separate runtime contract, plan, implementation, negative tests, and non-claim review
host interaction	file-read, file-write, server-interaction	requires future gate and remains denied	separate host/file/network authority contract and evidence
network interaction	network-open	requires future gate and remains denied	separate network authority contract and evidence
recovery/update/rollback	self-update, recovery-action, rollback-action	requires future gate and remains denied	separate recovery/update contract and evidence
hardware/boot	hardware-action, boot-action	requires future gate and remains denied	separate hardware/boot contract and evidence
unknown	unknown or malformed request kind	deny	negative tests and deterministic denial report

Future authority-bearing families must also preserve distinct operator identity, workload or service identity, and host or device integrity context before promotion.

Effect policy map

Current effect families:

Effect family	Current policy posture	Promotion blocker
none	allowed only for report, validation, preview, or classification metadata	no operational behavior may be attached
read	limited to approved local metadata references only	no host file read authority exists
local_mutation	denied until explicit local-mutation contract exists	mutation evidence and rollback policy absent
host_mutation	denied	host authority absent
network	denied	network authority absent
hardware	denied	hardware authority absent
boot	denied	boot authority absent
recovery	denied	recovery authority absent
external	denied	external authority absent
unknown	denied	request/effect identity invalid

Authority prerequisite map

No future runtime request may be promoted unless every relevant prerequisite is represented:

```
contract identity present
request kind known
requested effect known
caller identity known
operator identity known when operator context exists
workload or service identity known when tool, MCP, model, runtime, updater, signer, or installer authority is requested
asset or resource identity known
host or device integrity context known when host, network, update, recovery, boot, or hardware authority is requested
mode matches request family
authority summary present
authority status ok
authority no-effect flags preserved
operator confirmation recorded as metadata only
operator confirmation cannot override policy
least-privilege scope recorded for the exact requested effect
Nucleus task record denies effect behavior unless a future contract changes it
runtime report names policy, denial reason, gate state, matrix cell, domain cell, and evidence level
negative tests exist for unknown request, unknown effect, future-gated operation, blocked effect, prerequisite denial, and operator confirmation non-override
```

Future-gate policy map

Future-gated request families must stay behind explicit contracts:

```
command execution -&gt; future command/runtime contract
Lat execution -&gt; future Lat runtime contract
LIR execution -&gt; future LIR execution contract
file read/write -&gt; future host I/O authority contract
network open -&gt; future network authority contract
server interaction -&gt; future server/MCP authority contract
self-update signer or update payload acceptance -&gt; future signed update contract with
integrity,
 authenticity, and rollback evidence
tool, MCP, model, or runtime workload invocation -&gt; future workload-identity-aware authority
contract
self-update -&gt; future signed update contract
recovery or rollback -&gt; future recovery contract
hardware or boot -&gt; future hardware/boot contract
```

No future-gated family may be promoted by status text alone.

Abuse-case runtime policy map

Threat-model abuse cases map to current runtime-boundary policy evidence as follows:

```
unknown request is treated as allowed -&gt; runtime-boundary unknown request denial tests
unknown effect is treated as allowed -&gt; runtime-boundary unknown effect denial tests
future-gated request is treated as executable -&gt; future-gate classification tests
operator confirmation overrides policy -&gt; operator confirmation non-override tests
future workload or service authority lacks distinct workload identity -&gt; zero-trust
workload-identity gap remains open
future host, network, update, recovery, boot, or hardware authority lacks host or device
integrity context -&gt; zero-trust host-integrity gap remains open
future tool or command path reaches a shell boundary without dedicated command-boundary review
-&gt; command-boundary gap remains open
report omits denial reason -&gt; runtime-boundary report fields and report-refinement tests
failed authority metadata is treated as allowed -&gt; authority prerequisite denial tests
invalid LIR input reaches rendering -&gt; LIR prerequisite denial tests
status documentation overclaims implementation state -&gt; this guard plus status guards
external standard is referenced as if it were certification -&gt; defensive threat model
refinement
 non-claim guard
outdated external guidance remains marked current -&gt; defensive threat model external-source
checkpoint
```

Evidence gap map

This slice closes only the policy-expansion documentation gap.

The abuse-case fixture expansion gap is now addressed by:

```
docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md
tests/runtime_boundary_abuse_case_fixtures.c
scripts/test-runtime-boundary-abuse-case-fixtures.sh
```

Still open:

```
runtime boundary source code still performs no operational behavior
no host I/O authority exists
no network authority exists
no MCP behavior exists
no workload/service identity-aware runtime authority profile exists
no host/device integrity-aware authority profile exists
no production protection claim exists
no certification or compliance mapping exists
```

The next evidence gap should remain completion-estimate review only if capability posture changes; otherwise continue small guarded report/status alignment only when drift appears.

Validation

This policy expansion is guarded by:

```
sh scripts/test-runtime-boundary-policy-expansion-after-threat-model.sh
```

The underlying runtime and threat-model chain remains covered by:

```
sh scripts/test-defensive-threat-model-validation-refinement.sh
```

```
sh scripts/test-runtime-boundary.sh
sh scripts/test-runtime-boundary-policy-matrix-refinement.sh
sh scripts/test-runtime-boundary-domain-matrix-refinement.sh
sh scripts/test-runtime-boundary-domain-matrix-report-integration.sh
```

Expected output:

```
runtime_boundary_policy_expansion_after_threat_model: ok
```

Boundary

This expansion is documentation/status/guard alignment only.

It does not add runtime behavior, command execution, Lat execution, LIR execution, task effect execution, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback behavior, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, certification, accreditation, compliance, production protection, operating-system completeness, or runtime authority.

Subsequent slice

The abuse-case fixture expansion after runtime-boundary policy expansion is now recorded separately in docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md.

That subsequent fixture slice preserves the no-effect posture and does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production protection, or runtime authority.

docs/RUNTIME_BOUNDARY_POLICY_MATRIX_REFINEMENT.md

Source title: Latticra Runtime Boundary Policy Matrix Refinement

Lines: 153 | Words: 409 | SHA-256: a2c974ba730b3d02ff1425df98a369052a402f08776d13a8f119f09e19a1649f

Latticra Runtime Boundary Policy Matrix Refinement

Status: initial runtime boundary policy matrix refinement implementation Scope: deterministic no-effect policy matrix metadata for runtime-boundary classification and reporting.

Purpose

This document records the runtime-boundary policy matrix refinement after runtime-boundary report refinement and Lat pipeline report refinement.

The goal is to make the runtime-boundary decision shape easier to audit by reporting the matrix cell that a request falls into, along with stable booleans for effect, mode, authority, and future-gate posture.

This refinement does not grant authority, execute a request, mutate state, perform I/O, open network connections, touch hardware, control a terminal, or provide runtime behavior.

Added report metadata

The runtime boundary record now includes:

```
policy_matrix_cell
matrix_effect_allowed
matrix_mode_allowed
matrix_requires_authority
matrix_requires_future_gate
```

The deterministic runtime boundary report now emits those fields.

Policy matrix cells

Initial policy matrix cells:

```
no-effect-report
no-effect-validation
no-effect-classification
future-gated-operation
blocked-effect
prerequisite-denied
invalid
unsupported
```

no-effect-report

A no-effect report request with the correct report-only mode and satisfied prerequisites.

no-effect-validation

A no-effect validation request with the correct validation-only mode and satisfied prerequisites.

no-effect-classification

A no-effect classification request with the correct classification-only mode and satisfied prerequisites.

future-gated-operation

An operational request kind that remains behind the future gate.

blocked-effect

A known request whose requested effect is outside the current no-effect/read-only boundary.

prerequisite-denied

A known request whose effect and mode shape are acceptable but whose prerequisites are not satisfied.

invalid

A null, unknown, or malformed request/effect identity.

unsupported

A reserved fallback cell for future matrix expansion.

Matrix booleans

The matrix report includes:

```
matrix_effect_allowed      -&gt; requested effect is currently no-effect/read-only
matrix_mode_allowed        -&gt; mode matches the request-family expectation
matrix_requires_authority  -&gt; runtime boundary authority summary is expected
matrix_requires_future_gate -&gt; request kind is operational/future-gated
```

Validation

Run:

```
sh scripts/test-runtime-boundary-policy-matrix-refinement.sh
sh scripts/test-runtime-boundary.sh
```

The focused invariant tests verify:

```
runtime_boundary_policy_matrix_labels_are_stable
runtime_boundary_policy_matrix_reports_validation_cell
runtime_boundary_policy_matrix_reports_future_gate_cell
runtime_boundary_policy_matrix_reports_blocked_effect_cell
```

```
runtime_boundary_policy_matrix_reports_prerequisite_denial_cell
runtime_boundary_policy_matrix_reports_invalid_cell
```

Compatibility

This refinement preserves existing runtime-boundary behavior for:

```
allow-report
allow-validation
allow-classification
future-gated operational requests
unknown request denial
unknown effect denial
prerequisite denial
operator confirmation non-override behavior
small-buffer behavior
runtime-boundary report-refinement fields
Lat pipeline evidence fields
Lat pipeline line-comment evidence fields
Lat-specific LIR evidence fields
```

Non-claims

This report refinement does not provide:

```
runtime behavior
command execution
Lat execution
LIR execution
task effect execution
state mutation
file I/O
network I/O
server interaction
self-update
recovery behavior
rollback behavior
hardware support
boot behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
certification
accreditation
operating-system completeness
```

docs/RUNTIME_BOUNDARY_REFINEMENT_IMPLEMENTATION.md

Source title: Latticra Runtime Boundary Refinement Implementation

Lines: 289 | Words: 1032 | SHA-256: 78d1b8bacb61b1e412fbd860b7ab9617f1409571704477f6cd6538242f005853

Latticra Runtime Boundary Refinement Implementation

Status: runtime boundary refinement implementation with Lat pipeline clause, declaration, module/count, stage-summary, parse-error, semantic-error, downstream-stage-error, span, comment, Lat LIR module-summary, Lat LIR source-span, Lat LIR node-kind, Lat LIR first-node, Lat LIR first-node span, Lat LIR first transition-node, Lat LIR first transition-node span, Lat LIR first-edge, Lat LIR first-edge span, Lat LIR first transition-source edge, Lat LIR first transition-source edge endpoint, Lat LIR first transition-source edge span, Lat LIR no-effect, and Lat LIR edge-kind evidence Scope: no-effect runtime-boundary evidence reporting for Lat pipeline metadata, Lat pipeline first-clause metadata, Lat pipeline first-declaration metadata, Lat pipeline module/count metadata, Lat pipeline stage-summary metadata, Lat pipeline parse-error metadata, Lat pipeline semantic-error metadata, Lat pipeline downstream stage-error metadata, Lat pipeline span metadata, Lat pipeline line-comment metadata, Lat-specific LIR module-summary metadata, Lat-specific LIR source-span metadata, Lat-specific LIR node-kind metadata, Lat-specific LIR first-node metadata, Lat-specific LIR first-node span metadata, Lat-specific LIR first transition-node metadata, Lat-specific LIR first transition-node span metadata, Lat-specific LIR first-edge metadata, Lat-specific LIR first-edge span metadata, Lat-specific LIR first transition-source edge metadata, Lat-specific LIR first transition-source edge endpoint metadata, Lat-specific LIR first transition-source edge span metadata, Lat-specific LIR no-effect metadata, and Lat-specific LIR edge-kind metadata.

Purpose

This implementation slice refines the runtime boundary so it can classify and report Lat pipeline evidence and Lat-specific LIR evidence without executing Lat, executing LIR, mutating state, performing I/O, opening network connections, touching hardware, or enabling runtime behavior.

The implementation follows:

```
docs/RUNTIME_BOUNDARY_REFINEMENT_PLAN.md
```

Added / changed files

```
include/latticra/runtime_boundary.h
src/runtime_boundary.c
tests/runtime_boundary_lat_pipeline_evidence.c
docs/RUNTIME_BOUNDARY_REFINEMENT_IMPLEMENTATION.md
```

The existing runtime boundary test runner automatically includes the new invariant file:

```
scripts/test-runtime-boundary.sh
```

Public API refinement

The runtime boundary request now accepts optional Lat pipeline metadata:

```
const latticra_lat_pipeline_result_t *lat_pipeline
```

The runtime boundary request kinds now include:

```
LATTICRA_RUNTIME_BOUNDARY_LAT_PIPELINE_VALIDATE
```

The runtime boundary record now carries no-effect evidence fields for:

```
lat_pipeline_status
lat_pipeline_error
lat_pipeline_parse_error
lat_pipeline_span_start_offset
lat_pipeline_span_end_offset
lat_pipeline_span_start_line
lat_pipeline_span_start_column
lat_pipeline_span_end_line
lat_pipeline_span_end_column
lat_pipeline_semantic_error
lat_pipeline_model_error
lat_pipeline_lowering_error
```

```
lat_pipeline_lir_error
lat_pipeline_last_completed_stage
lat_pipeline_failed_stage
lat_pipeline_parse_ok
lat_pipeline_semantic_ok
lat_pipeline_model_ok
lat_pipeline_lowering_ok
lat_pipeline_lir_ok
lat_pipeline_no_effect_chain_ok
lat_pipeline_evidence_level
lat_pipeline_semantic_valid
lat_pipeline_module_name
lat_pipeline_source_len
lat_pipeline_declaration_count
lat_pipeline_clause_count
lat_pipeline_model_declaration_count
lat_pipeline_model_clause_count
lat_pipeline_first_declaration_node_index
lat_pipeline_first_declaration_kind
lat_pipeline_first_declaration_name
lat_pipeline_first_declaration_source
lat_pipeline_first_declaration_parse_index
lat_pipeline_first_declaration_first_clause_index
lat_pipeline_first_declaration_clause_count
lat_pipeline_first_declaration_source_index
lat_pipeline_first_transition_source_index
lat_pipeline_first_clause_node_index
lat_pipeline_first_clause_role
lat_pipeline_first_clause_effect
lat_pipeline_first_clause_name
lat_pipeline_first_clause_operator
lat_pipeline_first_clause_value
lat_pipeline_node_count
lat_pipeline_edge_count
lat_pipeline_comment_count
lat_pipeline_first_comment_start_offset
lat_pipeline_first_comment_end_offset
lat_pipeline_first_comment_start_line
lat_pipeline_first_comment_start_column
lat_pipeline_first_comment_end_line
lat_pipeline_first_comment_end_column
lat_lir_source_kind
lat_lir_module_name
lat_lir_report_classification
lat_lir_shape_kind
lat_lir_span_start_offset
lat_lir_span_end_offset
lat_lir_span_start_line
lat_lir_span_start_column
lat_lir_span_end_line
lat_lir_span_end_column
lat_lir_module_node_count
lat_lir_module_edge_count
lat_lir_binding_count
lat_lir_text_count
lat_lir_lat_state_node_count
lat_lir_lat_policy_node_count
lat_lir_lat_transition_node_count
lat_lir_lat_assertion_node_count
lat_lir_lat_requirement_node_count
lat_lir_lat_effect_declaration_node_count
lat_lir_has_first_lat_node
lat_lir_first_lat_node_index
lat_lir_first_lat_node_kind
lat_lir_first_lat_node_name
lat_lir_first_lat_node_value
lat_lir_first_lat_node_operator
lat_lir_first_lat_node_binding
lat_lir_first_lat_node_span_start_offset
lat_lir_first_lat_node_span_end_offset
lat_lir_first_lat_node_span_start_line
lat_lir_first_lat_node_span_start_column
lat_lir_first_lat_node_span_end_line
lat_lir_first_lat_node_span_end_column
lat_lir_has_first_transition_node
lat_lir_first_transition_node_index
lat_lir_first_transition_node_kind
lat_lir_first_transition_node_name
lat_lir_first_transition_node_value
lat_lir_first_transition_node_operator
```

```
lat_lir_first_transition_node_binding
lat_lir_first_transition_node_span_start_offset
lat_lir_first_transition_node_span_end_offset
lat_lir_first_transition_node_span_start_line
lat_lir_first_transition_node_span_start_column
lat_lir_first_transition_node_span_end_line
lat_lir_first_transition_node_span_end_column
lat_lir_no_effect_chain_ok
lat_lir_evidence_level
lat_lir_no_effect
lat_lir_execution_allowed
lat_lir_mutation_allowed
lat_lir_server_allowed
lat_lir_network_allowed
lat_lir_recovery_allowed
lat_lir_hardware_allowed
lat_lir_contains_edge_count
lat_lir_binds_edge_count
lat_lir_annotates_edge_count
lat_lir_orders_before_edge_count
lat_lir_transition_edge_count
lat_lir_has_first_edge
lat_lir_first_edge_index
lat_lir_first_edge_from_index
lat_lir_first_edge_to_index
lat_lir_first_edge_kind
lat_lir_first_edge_span_start_offset
lat_lir_first_edge_span_end_offset
lat_lir_first_edge_span_start_line
lat_lir_first_edge_span_start_column
lat_lir_first_edge_span_end_line
lat_lir_first_edge_span_end_column
lat_lir_has_first_transition_source_edge
lat_lir_first_transition_source_edge_index
lat_lir_first_transition_source_edge_from_index
lat_lir_first_transition_source_edge_to_index
lat_lir_first_transition_source_edge_kind
lat_lir_first_transition_source_edge_from_node_kind
lat_lir_first_transition_source_edge_from_node_name
lat_lir_first_transition_source_edge_to_node_kind
lat_lir_first_transition_source_edge_to_node_name
lat_lir_first_transition_source_edge_span_start_offset
lat_lir_first_transition_source_edge_span_end_offset
lat_lir_first_transition_source_edge_span_start_line
lat_lir_first_transition_source_edge_span_start_column
lat_lir_first_transition_source_edge_span_end_line
lat_lir_first_transition_source_edge_span_end_column
lat_lir_has_lat_state_nodes
lat_lir_has_lat_transition_nodes
lat_lir_has_transition_source_edges
```

Behavior

The new lat-pipeline-validate request is allowed only when:

```
mode == validation-only
requested effect is none or read
authority metadata is present and no-effect
Lat pipeline metadata is present and OK
Lat pipeline semantic_valid is true
Lat pipeline no-effect flags are preserved
```

Failed Lat pipeline metadata is denied with the closest existing runtime-boundary denial label:

```
parse_not_ok -&gt; parser-failed
semantic_not_ok / semantic_not_valid -&gt; semantic-failed
no_effect_violation -&gt; non-no-effect-flags
other pipeline failure -&gt; lir-failed
```

Lat execution and LIR execution remain future-gated.

Report surface

latticra_runtime_boundary_report now includes deterministic report fields for Lat pipeline evidence, Lat pipeline first-clause evidence, Lat pipeline first-declaration evidence, Lat pipeline module/count evidence, Lat pipeline stage-summary evidence, Lat pipeline parse-error evidence, Lat pipeline semantic-error evidence, Lat pipeline downstream stage-error evidence, Lat pipeline span evidence, Lat pipeline line-comment evidence, Lat-specific LIR module-summary evidence, Lat-specific LIR source-span evidence, Lat-specific LIR node-kind evidence, Lat-specific LIR first-node evidence, Lat-specific LIR first-node span evidence, Lat-specific LIR first transition-node evidence, Lat-specific LIR first transition-node span evidence, Lat-specific LIR first-edge evidence, Lat-specific LIR first-edge span evidence, Lat-specific LIR first transition-source edge evidence, Lat-specific LIR first transition-source edge endpoint evidence, Lat-specific LIR first transition-source edge span evidence, Lat-specific LIR no-effect evidence, and Lat-specific LIR edge-kind evidence.

The runtime boundary report capacity is increased to preserve bounded output with the expanded report surface.

Validation

Run:

```
sh scripts/test-runtime-boundary.sh
```

The focused evidence tests verify:

```
runtime_boundary_allows_valid_lat_pipeline_metadata
runtime_boundary_denies_failed_lat_pipeline_metadata
runtime_boundary_denies_parse_failed_lat_pipeline_metadata
runtime_boundary_denies_model_failed_lat_pipeline_metadata
runtime_boundary_reports_lat_pipeline_evidence
runtime_boundary_keeps_lat_lir_execution_future_gated
```

The Lat pipeline evidence invariants also verify that parser error metadata, semantic error metadata, downstream model/lowering/LIR error metadata, first-clause metadata, first-declaration metadata, module/count metadata, stage-summary metadata, parser diagnostic/module span metadata, line-comment count, first-comment span metadata, Lat-specific LIR module-summary metadata, Lat-specific LIR source-span metadata, Lat-specific LIR node-kind counts, Lat-specific LIR first-node identity, Lat-specific LIR first-node span metadata, Lat-specific LIR first transition-node identity, Lat-specific LIR first transition-node span metadata, Lat-specific LIR first-edge identity, Lat-specific LIR first-edge span metadata, Lat-specific LIR first transition-source edge identity, Lat-specific LIR first transition-source edge endpoint metadata, Lat-specific LIR first transition-source edge span metadata, Lat-specific LIR no-effect flags, and Lat-specific LIR edge-kind counts are copied into runtime-boundary records and reports, including denied records for failed Lat pipeline metadata.

Compatibility

This refinement preserves existing runtime boundary behavior for:

```
parse-only
validate-only
classify-only
render-report
nucleus-task-report
lat-validate
lir-validate
authority-check
future-gated operational request kinds
unknown request denial
unknown effect denial
operator confirmation non-override behavior
small-buffer behavior
Lat pipeline parse-error evidence reporting
Lat pipeline semantic-error evidence reporting
```

```
Lat pipeline downstream stage-error evidence reporting
Lat pipeline first-clause evidence reporting
Lat pipeline first-declaration evidence reporting
Lat pipeline module/count evidence reporting
Lat pipeline stage-summary evidence reporting
Lat pipeline diagnostic/module span evidence reporting
Lat pipeline line-comment evidence reporting
Lat LIR module-summary evidence reporting
Lat LIR source-span evidence reporting
Lat LIR node-kind evidence reporting
Lat LIR first-node evidence reporting
Lat LIR first-node span evidence reporting
Lat LIR first transition-node evidence reporting
Lat LIR first transition-node span evidence reporting
Lat LIR first-edge evidence reporting
Lat LIR first-edge span evidence reporting
Lat LIR first transition-source edge evidence reporting
Lat LIR first transition-source edge endpoint evidence reporting
Lat LIR first transition-source edge span evidence reporting
Lat LIR no-effect evidence reporting
Lat LIR edge-kind evidence reporting
denied Lat pipeline comment evidence recording
```

Boundary

This implementation does not provide runtime behavior, command execution, Lat execution, LIR execution, task effect execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system completeness.

docs/RUNTIME_BOUNDARY_REFINEMENT_PLAN.md

Source title: Latticra Runtime Boundary Refinement Plan

Lines: 118 | Words: 422 | SHA-256: bc2c215143cb935a6dcbeaae1f6d15707b76f9042b96f7dbb7e26107a32aed08

Latticra Runtime Boundary Refinement Plan

Status: refinement planning contract Scope: planning-only refinement for integrating Lat pipeline and Lat-specific LIR metadata into the no-effect runtime boundary classification/report surface.

Purpose

This document defines the next runtime boundary refinement before any new runtime behavior is added.

The current runtime boundary already classifies no-effect report, validation, and classification requests while preserving disabled-by-default operational behavior. The next refinement should make the boundary aware of the newer Lat pipeline and Lat-specific LIR metadata surfaces without promoting execution, mutation, command behavior, file I/O, network I/O, recovery, hardware, boot, or operating-system claims.

Relationship to current work

This plan depends on:

```
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
docs/LAT_PIPELINE_CONTRACT.md
docs/LAT_PIPELINE_IMPLEMENTATION_PLAN.md
docs/LAT_PIPELINE_IMPLEMENTATION.md
docs/LAT_SPECIFIC_LIR_REFINEMENT_CONTRACT.md
docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION_PLAN.md
docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION.md
include/latticra/runtime_boundary.h
include/latticra/lat_pipeline.h
include/latticra/lat_to_lir.h
```

```
include/latticra/lir.h
```

Refinement goal

The refinement should let runtime-boundary records and reports summarize the newer Lat pipeline chain:

```
Lat parser metadata
Lat semantic metadata
Lat-to-LIR lowering metadata
Lat-specific LIR metadata
Lat pipeline aggregate metadata
```

The runtime boundary should remain a classifier and reporter, not an executor.

Planned metadata additions

A future implementation slice may add report fields for:

```
lat_pipeline_status
lat_pipeline_error
lat_pipeline_semantic_valid
lat_pipeline_source_len
lat_pipeline_node_count
lat_pipeline_edge_count
lat_lir_source_kind
lat_lir_module_node_count
lat_lir_transition_edge_count
lat_lir_has_lat_state_nodes
lat_lir_has_lat_transition_nodes
lat_lir_has_transition_source_edges
```

The exact names may be adjusted during implementation, but the implementation must stay deterministic, bounded, and no-effect.

Planned request handling

The next implementation slice may add or refine no-effect request classification for:

```
Lat pipeline report validation
Lat-specific LIR metadata validation
Runtime boundary report over Lat pipeline evidence
```

Any operational request remains denied or future-gated.

Planned tests

The implementation slice should add focused tests for:

```
runtime_boundary_reports_lat_pipeline_status
runtime_boundary_denies_failed_lat_pipeline_metadata
runtime_boundary_allows_valid_lat_pipeline_metadata_in_report_mode
runtime_boundary_reports_lat_specific_lir_node_evidence
runtime_boundary_reports_transition_source_edge_evidence
runtime_boundary_preserves_no_effect_for_lat_pipeline_evidence
runtime_boundary_keeps_lat_and_lir_execute_future_gated
runtime_boundary_refinement_report_is_deterministic
runtime_boundary_refinement_rejects_small_buffer
```

Compatibility expectations

The refinement must not change existing behavior for:

```
unknown request denial
unknown effect denial
operator confirmation non-override behavior
future-gated operational requests
existing Lat validation prerequisites
existing LIR validation prerequisites
existing render report prerequisites
existing Nucleus task report prerequisites
existing runtime report fields
existing small-buffer behavior
```

Status and evidence rule

This plan is not implementation evidence by itself. It permits the next implementation slice to add bounded API/report fields and focused tests. Status percentages may only increase modestly from this planning milestone and should increase more substantially only after implementation and CI evidence merge.

Non-claims

This plan does not implement runtime behavior, command execution, Lat execution, LIR execution, task effect execution, live movement, state mutation, file I/O, network I/O, server interaction, self-update, recovery behavior, rollback, hardware support, boot behavior, terminal control, security isolation, sandboxing, malware prevention, ransomware prevention, certification, accreditation, or operating-system completeness.

docs/RUNTIME_BOUNDARY_REPORT_REFINEMENT.md

Source title: Latticra Runtime Boundary Report Refinement

Lines: 235 | Words: 730 | SHA-256: a092c1bd6a4b93f64d3b0fb6147f3b72564bea63c66df6990b29583f7c1c677c

Latticra Runtime Boundary Report Refinement

Status: initial runtime boundary report refinement implementation Scope: deterministic no-effect runtime-boundary report classification, boundary-domain labeling, authorization-state labeling, evidence-level reporting, invariant tests, and guard coverage.

Purpose

This document records the runtime-boundary report refinement that follows the Lat pipeline and Lat-specific LIR evidence reporting slice.

The refinement makes runtime boundary reports more explicit about what a request represents before any operational runtime authority exists.

Runtime boundary reporting remains a classifier and evidence surface. It is not a kernel interface, sandbox, permission system, process supervisor, boot layer, device layer, filesystem layer, network layer, or authority grant.

Boundary principle

Latticra runtime authority is denied by default.

A runtime-boundary request may be reported as declarative, boundary-seeking, denied, or invalid. No report classification executes the request, mutates state, performs I/O, opens a network connection, touches hardware, controls a terminal, starts a process, boots a system, or grants runtime authority.

Added report fields

The runtime boundary record now includes explicit report-refinement fields:

```
report_classification
boundary_domain
authorization_state
evidence_level
```

The deterministic text report now emits:

```
report_classification=&lt;label&gt;
boundary_domain=&lt;label&gt;
authorization_state=&lt;label&gt;
evidence_level=&lt;number&gt;
```

Report classifications

Initial report classifications:

```
declarative
boundary-seeking
denied
invalid
```

declarative

The request is a no-effect report, validation, classification, or metadata surface that can be accepted under the current disabled-by-default runtime boundary.

boundary-seeking

The request attempts to cross toward operational runtime authority or a future-gated operational domain.

Boundary-seeking does not mean the request is executed. It means the report identified the request as runtime-authority-seeking and kept it behind the future gate.

denied

The request is understood but not accepted under the current prerequisites, authority summary, effect posture, operator-confirmation state, or no-effect requirements.

invalid

The request cannot be classified as a valid boundary request because it is null, unknown, malformed, or carries unknown effect identity.

Boundary domains

Initial report domains:

```
memory
filesystem
network
process
device
clock
randomness
host
external-call
persistence
scheduler
unknown
```

Operational request kinds map to the clearest current domain:

```
file-read / file-write -&gt; filesystem
network-open / server-interaction -&gt; network
runtime-execute / command-execute / lat-execute / lir-execute -&gt; process
hardware-action / boot-action -&gt; device
self-update / recovery-action / rollback-action -&gt; persistence
nucleus-task-report -&gt; scheduler
authority-check -&gt; host
```

No-effect declarative requests that do not cross a specific operational surface remain unknown unless their requested effect provides a clearer domain.

Authorization states

Initial authorization states:

```
not-requested
requested
denied
unavailable
reserved-for-future
```

The current implementation uses:

```
not-requested -&gt; accepted no-effect report / validation / classification surfaces
denied -&gt; known request rejected by current prerequisites or policy
unavailable -&gt; null or unknown request/effect identity
reserved-for-future -&gt; future-gated operational runtime request
```

requested is reserved as an explicit report-state label for future slices that need to distinguish request capture from denial or future reservation.

Evidence level

The runtime boundary now records a small deterministic evidence level:

```
0 -&gt; invalid or unavailable boundary evidence
1 -&gt; denied or future-gated boundary evidence
2 -&gt; accepted no-effect declarative report / validation / classification evidence
```

This is an internal project evidence label, not a maturity certification, security level, or production-readiness score.

Happy path

The current report-refinement happy path is:

```
source -&gt; parse / validate / lower -&gt; classify -&gt; report classification -&gt; report
domain -&gt;
report authorization state -&gt; emit deterministic evidence
```

No unauthorized effect is executed during this path.

Implementation files

This slice updates or adds:

```
include/latticra/runtime_boundary.h
src/runtime_boundary.c
tests/runtime_boundary_report_refinement.c
docs/RUNTIME_BOUNDARY_REPORT_REFINEMENT.md
scripts/test-runtime-boundary-report-refinement.sh
.github/workflows/runtime-boundary-report-refinement.yml
```

The full C workflow also runs the new guard.

Validation

Run:

```
sh scripts/test-runtime-boundary-report-refinement.sh
sh scripts/test-runtime-boundary.sh
```

The focused invariant tests verify:

```
runtime_boundary_report_refinement_classifies_declarative_pipeline
runtime_boundary_report_refinement_marks_future_gate_boundary_seeking
runtime_boundary_report_refinement_maps_effect_domains
runtime_boundary_report_refinement_marks_invalid_unknown_request
runtime_boundary_report_refinement_marks_denied_prerequisite
```

The guard verifies that documentation, header declarations, implementation labels, report fields, focused tests, and workflow wiring remain present.

Compatibility

This refinement preserves the existing runtime boundary behavior for:

```
parse-only
validate-only
classify-only
render-report
nucleus-task-report
lat-validate
lir-validate
authority-check
lat-pipeline-validate
future-gated operational request kinds
unknown request denial
unknown effect denial
```

```
operator-confirmation non-override behavior
small-buffer behavior
Lat pipeline evidence fields
Lat pipeline line-comment evidence fields
Lat-specific LIR evidence fields
```

Non-claims

This report refinement does not provide:

```
runtime behavior
command execution
Lat execution
LIR execution
task effect execution
live movement
state mutation
file I/O
network I/O
server interaction
self-update
recovery behavior
rollback behavior
hardware support
boot behavior
terminal control
security isolation
sandboxing
malware prevention
ransomware prevention
certification
accreditation
operating-system completeness
```

docs/STATE_LATTICE.md

Source title: Latticra State Lattice

Lines: 123 | Words: 364 | SHA-256: a6e5bfba635487ea564f7535838f51a3f156d384449d739c21cde6f7339c41a3

Latticra State Lattice

Status: initial implementation contract Scope: first C state fixture, invariant tests, report surface, and no-effect boundary.

Purpose

The State Lattice is the first real Latticra implementation unit.

It provides a deterministic, fixed-size, no-effect C fixture that represents the starting state vocabulary inherited from the Phase1 evidence path and translated into Latticra architecture terms.

Implementation files

```
include/latticra/state_lattice.h
src/state_lattice.c
tests/state_lattice_invariants.c
scripts/test-state-lattice.sh
```

Default fixture

The default state lattice fixture must preserve:

```
origin=0/0
route=ROOT
axis=ROOT
path=ROOT&gt;0/0
breadcrumb=ROOT
trace=trace-preview
safe_portal=planned
rollback=available
health=nominal
risk=low
```

```
lock=open
dark_phase=off
host_effect=none
external_effect=none
```

Effect boundary

The initial state lattice fixture is no-effect.

It must not:

- mutate host state;
- use networking;
- touch hardware;
- perform boot behavior;
- perform recovery behavior;
- perform external effects;
- allocate hidden dynamic state;
- depend on server interaction.

C policy

The first implementation uses C because the State Lattice is intended to become a portable low-level representation.

Rules:

- fixed-size fields;
- explicit enum values;
- no hidden allocation;
- no host-specific APIs;
- no network APIs;
- no hardware APIs;
- no file writes;
- deterministic report output.

Public API

Initial API:

```
latticra_state_lattice_default
latticra_effect_label
latticra_state_lattice_is_no_effect
latticra_state_lattice_report
```

Test command

Run:

```
sh scripts/test-state-lattice.sh
```

The test builds the C fixture with a strict warning policy and runs invariant checks.

Required invariants

The tests must verify:

- default labels match the contract;

- host effect is none;
- external effect is none;
- default fixture is no-effect;
- report output exposes operator-visible labels;
- bad report arguments are rejected;
- small report buffers are rejected.

Current evidence level

This implementation is an L2 tested model for the initial state fixture.

It is not a runtime supervisor, transition engine, update engine, server client, boot component, hardware component, or recovery system.

Next implementation step

The next implementation candidate after this fixture is:

```
tri-plane transition model
```

That future work must remain pure and preview-only until evidence supports stronger behavior.

Non-claims

This document and implementation do not claim live lattice movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

docs/TRI_PLANE_TRANSITION.md

Source title: Latticra Tri-Plane Transition Model

Lines: 164 | Words: 414 | SHA-256: abedf7fbe118a0763607716dbfe1fb6b3febe5c6a65a0512a57b43a1e2d11b97

Latticra Tri-Plane Transition Model

Status: initial implementation contract Scope: pure preview transitions across spatial, state, and safety planes.

Purpose

The Tri-Plane Transition Model is the second Latticra implementation unit.

It models lattice movement as a pure preview transition. It accepts a current state lattice and a movement request, then returns a preview result and next-state copy without mutating the input state.

Implementation files

```
include/latticra/tri_plane_transition.h
src/tri_plane_transition.c
tests/tri_plane_transition_invariants.c
scripts/test-tri-plane-transition.sh
```

Planes

A transition is evaluated across three conceptual planes:

```
Spatial plane
  origin / route / axis / path

State plane
  breadcrumb / trace / current state labels
```

```
Safety plane
  safe_portal / rollback / health / risk / lock / dark_phase / host_effect / external_effect
```

Movement vocabulary

Initial movement values:

```
up
down
left
right
enter
back
root
unknown
```

Result vocabulary

Initial transition results:

```
allowed-preview
denied
```

Initial denial/decision reasons:

```
ok
null-argument
unknown-direction
lock-closed
risk-high
rollback-unavailable
```

Preview-only boundary

The model is preview-only.

It must not:

- mutate the input state;
- mutate origin;
- execute recovery;
- touch host state;
- use networking;
- touch hardware;
- change boot state;
- perform external effects;
- perform hidden allocation;
- depend on server interaction.

Default safe preview

A safe right preview from the default state may produce a next-state copy like:

```
axis=RIGHT
path=ROOT&gt;0/1
breadcrumb=ROOT&gt;RIGHT
host_effect=none
external_effect=none
origin_mutated=0
recovery_executed=0
```

Denial behavior

Denied transitions preserve the current state as the returned next-state copy.

Denied transitions must keep:

```
host_effect=none
external_effect=none
origin_mutated=0
recovery_executed=0
```

Required denial cases

The initial tests must verify:

- unknown direction is denied;
- closed lock denies transition;
- high risk denies transition;
- back requires rollback=available;
- denied transitions preserve no-effect state.

Public API

Initial API:

```
latticra_movement_label
latticra_transition_result_label
latticra_transition_reason_label
latticra_tri_plane_preview
```

Test command

Run:

```
sh scripts/test-tri-plane-transition.sh
```

The main C workflow runs both state-lattice and tri-plane transition tests.

Current evidence level

This implementation is an L2 tested model for pure preview transitions.

It is not live lattice movement, Nucleus execution, server interaction, self-update, recovery behavior, hardware behavior, boot behavior, or a security boundary.

Next implementation step

The next implementation candidate after this model is:

```
Nucleus preview request classification
```

That future work should classify requests and effects, but still avoid live mutation.

Non-claims

This document and implementation do not claim live movement, origin mutation, recovery execution, server interaction, self-update, hardware support, boot readiness, security isolation, sandboxing, or operating-system completeness.

Part VI - Latticra Seal

- docs/demos/LATTICRA_SEAL_DEMO_v0_1.md - Latticra Seal Demo v0.1
- docs/latticra-seal/ARCHITECTURE.md - Latticra Seal Architecture
- docs/latticra-seal/BOUNDARIES.md - Latticra Seal Boundaries
- docs/latticra-seal/POLICY.md - Latticra Seal Policy
- docs/latticra-seal/PRODUCT.md - Latticra Seal Product Spine
- docs/latticra-seal/README.md - Latticra Seal Documentation

- docs/latticra-seal/REPORTS.md - Latticra Seal Reports
- docs/latticra-seal/STATUS.md - Latticra Seal Status
- docs/latticra-seal/USAGE.md - Latticra Seal Usage
- docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md - Latticra Seal Agentic Automation Security Contract
- docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md - Latticra Seal Agentic Automation Security Implementation
- docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md - Latticra Seal Agentic Automation Security Report Surface
- docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md - Latticra Seal Capability Gate Contract
- docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md - Latticra Seal Capability Gate Implementation
- docs/LATTICRA_SEAL_CAPABILITY_METADATA_CONTRACT.md - Latticra Seal Capability Metadata Contract
- docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION.md - Latticra Seal Capability Metadata Implementation
- docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION_PLAN.md - Latticra Seal Capability Metadata Implementation Plan
- docs/LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_SURFACE.md - Latticra Seal Capability Metadata Report Surface
- docs/LATTICRA_SEAL_CONTRACT.md - Latticra Seal Contract
- docs/LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md - Latticra Seal Core Evidence Report
- docs/LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md - Latticra Seal Crypto Graduation Gate Implementation
- docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md - Latticra Seal Crypto Verify Backend Contract
- docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_IMPLEMENTATION.md - Latticra Seal Crypto Verify Backend Implementation
- docs/LATTICRA_SEAL_ED25519_VERIFY_IMPLEMENTATION.md - Latticra Seal Ed25519 Verify-Only Implementation
- docs/LATTICRA_SEAL_ED25519_VERIFY_ONLY_CONTRACT.md - Latticra Seal Ed25519 Verify-Only Contract
- docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md - Latticra Seal Effect Decision Contract
- docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md - Latticra Seal Effect Decision Implementation
- docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md - Latticra Seal Future Key Parsing Implementation
- docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md - Latticra Seal Future Key Parsing Implementation Contract
- docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md - Latticra Seal Future Key Parsing Implementation Plan
- docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_CONTRACT.md - Latticra Seal Guarded Allowlist Contract

- docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md - Latticra Seal Guarded Allowlist Implementation
- docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION_PLAN.md - Latticra Seal Guarded Allowlist Implementation Plan
- docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE.md - Latticra Seal Guarded Allowlist Report Surface
- docs/LATTICRA_SEAL_IMPLEMENTATION_PLAN.md - Latticra Seal Implementation Plan
- docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md - Latticra Seal Key-Handling Boundary Contract
- docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md - Latticra Seal Key-Handling Implementation
- docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md - Latticra Seal Key-Material Boundary Contract
- docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md - Latticra Seal Key-Material Implementation
- docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md - Latticra Seal Local Capability Registry Schema Contract
- docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md - Latticra Seal Local Capability Registry Schema Implementation
- docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md - Latticra Seal Local Capability Registry Schema Implementation Plan
- docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md - Latticra Seal MCP Alignment Plan
- docs/LATTICRA_SEAL_MEASUREMENT_CONTRACT.md - Latticra Seal Measurement Contract
- docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md - Latticra Seal Measurement Implementation
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_CONTRACT.md - Latticra Seal Operator Receipt Report Contract
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md - Latticra Seal Operator Receipt Report Implementation
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION_PLAN.md - Latticra Seal Operator Receipt Report Implementation Plan
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md - Latticra Seal Operator Receipt Report Surface
- docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md - Latticra Seal Parameter Schema Contract
- docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md - Latticra Seal Parameter Schema Implementation
- docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md - Latticra Seal Parameter Schema Report Surface
- docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md - Latticra Seal Policy Decision Contract
- docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md - Latticra Seal Policy Decision Implementation
- docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md - Latticra Seal Policy Decision Report Surface

- docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md - Latticra Seal Public-Key Parsing Boundary Contract
- docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md - Latticra Seal Public-Key Parsing Implementation
- docs/LATTICRA_SEAL_REPLAYED_REQUEST_CASE_CONTRACT.md - Latticra Seal Replayed Request Case Contract
- docs/LATTICRA_SEAL_REPLAYED_REQUEST_CASE_IMPLEMENTATION.md - Latticra Seal Replayed Request Case Implementation
- docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md - Latticra Seal Report Envelope Contract
- docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md - Latticra Seal Report Envelope Implementation
- docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md - Latticra Seal Report Implementation
- docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md - Latticra Seal Request Freshness Contract
- docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md - Latticra Seal Request Freshness Implementation
- docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md - Latticra Seal Request Freshness Report Surface
- docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_CONTRACT.md - Latticra Seal Runtime Dry-Run Contract
- docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION.md - Latticra Seal Runtime Dry-Run Implementation
- docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION_PLAN.md - Latticra Seal Runtime Dry-Run Implementation Plan
- docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md - Latticra Seal Runtime Dry-Run Report Surface
- docs/LATTICRA_SEAL_RUNTIME_ENFORCEMENT_GATE_CONTRACT.md - Latticra Seal Runtime Enforcement Gate Contract
- docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md - Latticra Seal Runtime Gate Implementation
- docs/LATTICRA_SEAL_RUNTIME_GATE_TEST_PLAN.md - Latticra Seal Runtime Gate Test Plan
- docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md - Latticra Seal Runtime Handoff Contract
- docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md - Latticra Seal Runtime Handoff Evaluation Contract
- docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md - Latticra Seal Runtime Handoff Evaluation Implementation
- docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md - Latticra Seal Runtime Handoff Implementation
- docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md - Latticra Seal Runtime Handoff Report Contract
- docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md - Latticra Seal Runtime Handoff Report Implementation
- docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md - Latticra Seal Signature Implementation

- docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION_CONTRACT.md - Latticra Seal Signature Implementation Contract
- docs/LATTICRA_SEAL_SIGNATURE_POLICY_CONTRACT.md - Latticra Seal Signature Policy Contract
- docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md - Latticra Seal Signature Policy Implementation
- docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md - Latticra Seal Signature Request Contract
- docs/LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md - Latticra Seal Signature Request Implementation
- docs/LATTICRA_SEAL_SIGNED_MANIFEST_CONTRACT.md - Latticra Seal Signed Manifest Contract
- docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md - Latticra Seal Signed Manifest Implementation
- docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md - Latticra Seal Signed Request Contract
- docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md - Latticra Seal Signed Request Implementation
- docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md - Latticra Seal Signer Handoff Contract
- docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md - Latticra Seal Signer Handoff Implementation
- docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md - Latticra Seal Signer Invocation Contract
- docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md - Latticra Seal Signer Invocation Implementation
- docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md - Latticra Seal Signing Authorization Contract
- docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md - Latticra Seal Signing Authorization Implementation
- docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md - Latticra Seal Signing Operation Contract
- docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md - Latticra Seal Signing Operation Implementation
- docs/LATTICRA_SEAL_STALE_REQUEST_CASE_CONTRACT.md - Latticra Seal Stale Request Case Contract
- docs/LATTICRA_SEAL_STALE_REQUEST_CASE_IMPLEMENTATION.md - Latticra Seal Stale Request Case Implementation
- docs/LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md - Latticra Seal Status Rollup Contract
- docs/LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md - Latticra Seal Status Rollup Implementation
- docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_CONTRACT.md - Latticra Seal Unknown Tool Case Contract
- docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_IMPLEMENTATION.md - Latticra Seal Unknown Tool Case Implementation
- docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_CONTRACT.md - Latticra Seal Unsigned Request Case Contract

- docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_IMPLEMENTATION.md - Latticra Seal Unsigned Request Case Implementation
- docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md - Latticra Seal Verification Policy Contract
- docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md - Latticra Seal Verification Policy Implementation
- docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md - Latticra Seal Verification Receipt Contract
- docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md - Latticra Seal Verification Receipt Implementation
- docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_CONTRACT.md - Latticra Seal Verified Capability Gate Contract
- docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_IMPLEMENTATION.md - Latticra Seal Verified Capability Gate Implementation
- docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_CONTRACT.md - Latticra Seal Verified Effect Decision Contract
- docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_IMPLEMENTATION.md - Latticra Seal Verified Effect Decision Implementation
- docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_CONTRACT.md - Latticra Seal Verified Receipt Promotion Contract
- docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_IMPLEMENTATION.md - Latticra Seal Verified Receipt Promotion Implementation
- docs/specs/LATTICRA_SEAL_MANIFEST_v0_1.md - Latticra Seal Manifest v0.1

docs/demos/LATTICRA_SEAL_DEMO_v0_1.md

Source title: Latticra Seal Demo v0.1

Lines: 59 | Words: 192 | SHA-256: 3c245f8556499eb3d47550b090e4f23d3be480f39d9509c6871e5d03b95985f6

Latticra Seal Demo v0.1

This demo shows the first working trust loop for Latticra Seal.

Latticra Seal currently supports:

- reading the root latticra.seal manifest
- checking manifest shape
- checking required project files
- generating SHA-256 file digests
- writing a saved baseline to latticra.seal.lock
- verifying the current project state against that saved baseline
- reporting added, removed, or modified files

Demo Commands

Build the Seal CLI:

make seal-cli

Print the version:

./build/latticra-seal version

Inspect the manifest:

./build/latticra-seal manifest

Run a full check:

./build/latticra-seal check

Create or refresh the trusted baseline:

./build/latticra-seal baseline

Verify the current project state:

./build/latticra-seal verify

Run the demo script:

./scripts/demo-latticra-seal.sh

What the baseline means

The saved baseline does not prove the project is perfect or externally certified.

It means the current project files match the file-state previously recorded in latticra.seal.lock.

A precise public statement is:

Latticra Seal verifies that the current project state matches a previously recorded trusted baseline.

Current Status

This is an early local-integrity prototype.

It is not yet a full cryptographic attestation system. Future versions should add signed manifests, public-key verification, stronger proof bundles, and Latticra Panel integration.

docs/latticra-seal/ARCHITECTURE.md

Source title: Latticra Seal Architecture

Lines: 100 | Words: 305 | SHA-256: 4156d911d6274f3d542357c3534d525d5b5995d5e8434578f942fa1aeb49dd16

Latticra Seal Architecture

Latticra Seal is organized around bounded evidence.

It does not begin with enforcement. It begins with describing what exists, measuring what exists, reporting what changed, and refusing to claim authority it does not yet have.

Core model

```
source tree
    |
    v
manifest / seal config
    |
    v
hash baseline
    |
    v
policy checks
    |
    v
report-only evidence
    |
    v
Panel / CLI / future runtime boundary handoff
```

Primary components

Manifest layer

The manifest layer defines what Seal is allowed to inspect.

Expected files may include:

```
latticra.seal
latticra.seal.lock
```

The manifest should describe scope, included paths, excluded paths, policy expectations, and report behavior.

Measurement layer

The measurement layer computes local evidence.

Initial measurement should remain read-only.

Expected behavior:

- read local files
- compute hashes
- compare against saved baseline
- record changed/missing/new files
- avoid mutation unless explicitly running an update/baseline command

Policy layer

The policy layer decides whether the current state is acceptable.

Initial policy behavior should be conservative:

- fail closed on malformed policy
- fail closed on denied paths
- fail closed on unsupported authority claims
- report why a decision was denied

Report layer

The report layer emits human-readable evidence.

Reports should clearly show:

- mode
- timestamp
- host OS
- prefix
- hash status
- policy status
- warnings
- failures
- non-authority declarations

Panel bridge

The Panel bridge should expose Seal through a safe interface.

Panel should be able to:

- run report-only checks
- display Seal status
- show policy results
- show generated reports
- avoid granting runtime/root/network authority

Future runtime-boundary handoff

Future runtime integration must remain gated.

Seal may eventually produce metadata for runtime-boundary decisions, but documentation must not claim runtime enforcement until the enforcement path exists and has tests.

docs/latticra-seal/BOUNDARIES.md

Source title: Latticra Seal Boundaries

Lines: 68 | Words: 165 | SHA-256: 90aeac26b2b3f523add60f3aeb76fed168f9a7969ac03376773ec0c81c32851f

Latticra Seal Boundaries

This document defines what Latticra Seal does not claim.

Seal is

Latticra Seal is:

- a Latticra substructure
- a local evidence layer
- a report-only verification surface
- a policy-boundary testing target
- a manifest/hash baseline mechanism
- a future bridge toward stronger runtime-boundary work

Seal is not currently

Latticra Seal is not currently:

- a production security product
- a malware prevention system
- a ransomware prevention system
- a sandbox
- a kernel module
- an SELinux policy system
- a systemd enforcement system
- a root installer
- a network authority
- a runtime enforcement authority
- a certified host protection system
- a Fedora-approved package
- an OS replacement

Required public caution

Use language like:

```
early
local
report-only
bounded
evidence-bound
verification-oriented
policy-regression-backed
```

Avoid language like:

```
unbreakable
production-ready
military-grade
ransomware-proof
malware-proof
kernel-enforced
runtime-enforced
certified
guaranteed secure
```

Rule

No claim without evidence.

No evidence without test.

No test without documentation.

docs/latticra-seal/POLICY.md

Source title: Latticra Seal Policy

Lines: 67 | Words: 209 | SHA-256: a0d95e62396dccb0ee485b761ba4995d6b30f29494cc9360addea9b7045a4d3d

Latticra Seal Policy

Latticra Seal policy exists to define what Seal is allowed to inspect, verify, report, deny, or pass.

The policy layer should be conservative by default.

Policy goals

Seal policy should:

- make allowed scope explicit
- reject unsupported authority
- deny unsafe assumptions
- produce readable denial reasons
- support regression testing
- keep Latticra evidence-bound

Fail-closed principle

When policy input is malformed, incomplete, unsupported, or contradictory, Seal should fail closed.

Failing closed means:

```
do not silently pass
do not guess intent
do not grant authority
do not claim enforcement
emit a clear denial reason
```

Denial examples

Policy should deny:

- unknown authority modes
- network authority requests
- runtime enforcement claims
- root installation claims
- paths outside declared scope
- malformed manifests
- missing required metadata
- unsupported cryptographic claims

Policy regression

The policy regression lane should prove that unsafe or unsupported states are denied.

Expected local command:

```
make seal-policy-denials
```

Expected CI lane:

```
Latticra Seal Policy
```

Correct policy wording

Seal policy verifies whether local evidence matches declared expectations.

Incorrect policy wording

Seal policy does not currently enforce runtime behavior, block malware, isolate processes, protect the kernel, or provide production host security.

docs/latticra-seal/PRODUCT.md

Source title: Latticra Seal Product Spine

Lines: 184 | Words: 748 | SHA-256: 41fadbbc9e6cd54d9111d200f66f6d205bc7efbd86ce8b9c7635906f6baeba41

Latticra Seal Product Spine

Status: product spine for Latticra Seal security-product direction Scope: product direction, operating model, and earned-capability path for Latticra Seal. This document does not implement runtime enforcement, malware prevention, ransomware prevention, sandboxing, endpoint detection, kernel enforcement, network authority, root authority, cloud trust services, production cryptographic authority, capability enforcement, effect execution, AI-agent execution control, or production security readiness.

Product Thesis

Latticra Seal should become a local-first trust-boundary product for software, operators, and agentic toolchains.

The product promise to earn is:

```
Every security-relevant request should have explicit identity, capability, policy, measurement,
verification, dry-run decision, and receipt evidence before any effect is allowed.
```

Current Seal does not yet allow effects. That is a strength of the present foundation: the product can mature from visible evidence to verified decisions to future enforcement without hiding authority jumps.

Current Truth

Seal currently has an evidence-bound base:

```
manifest_hash_baseline=1
report_only_runtime_gate_path=1
guarded_allowlist_metadata=1
capability_metadata=1
request_freshness_metadata=1
signed_request_metadata=1
policy_decision_metadata=1
verification_policy_metadata=1
verification_receipt_metadata=1
runtime_dry_run_metadata=1
runtime_handoff_metadata=1
operator_receipt_report_metadata=1
operator_receipt_report_surface=1
local_capability_registry_schema_contract=1
local_capability_registry_schema_implementation_plan=1
local_capability_registry_schema_implementation=1
operator_visible_reports=1
production_security_product=0
runtime_authority_granted=0
```

Product Modes

Seal should mature through explicit modes.

```
observe_mode=current
verify_mode=partial-local
decide_mode=metadata-only
handoff_mode=metadata-only
enforce_mode=future-closed
```

Observe

Observe mode describes local state, manifests, baselines, request posture, capability candidates, and denial reasons.

Current status: present through report-only surfaces.

Verify

Verify mode should bind artifacts, identities, signatures, and receipts to deterministic verification outcomes.

Current status: partial local verification primitives exist, but no production trust root, no trust-store behavior, no revocation lookup, and no production cryptography claim exist.

Decide

Decide mode should combine request, capability, policy, freshness, signature, receipt, allowlist, and operator-review posture into a deterministic allow/deny/review decision.

Current status: metadata-only and denied by default.

Handoff

Handoff mode should create explicit runtime-boundary evidence for a future effect-capable layer.

Current status: metadata-only and inactive.

Enforce

Enforce mode is the future production boundary. It must remain closed until implementation, tests, threat-model evidence, operator controls, failure behavior, and security review exist.

Current status: not implemented.

Product Pillars

Evidence Graph

Seal should connect manifests, file hashes, request records, capability metadata, policy decisions, verification receipts, dry-run decisions, and handoff reports into one inspectable evidence graph.

Capability Intent

Seal should make every requested capability explicit before a tool, agent, script, package, installer, or runtime component can move toward effects.

Verification Receipts

Seal should produce receipts that are deterministic, machine-readable, human-readable, signed when future prerequisites exist, and traceable to input evidence.

Deny-By-Default Decisions

Seal should treat unknown, unsigned, stale, replayed, malformed, or unsupported requests as denied unless a guarded future authority path proves otherwise.

Operator Review

Seal should make review easy: what was requested, what evidence exists, what policy decided, what would happen, what remains blocked, and what authority is still absent.

Local-First Control

Seal should preserve local operation by default. Future cloud or network trust features, if ever introduced, must be optional, signed, inspectable, and disabled unless explicitly authorized.

Product Surfaces

Near-term product surfaces should be:

```
seal_cli_report=1
seal_cli_manifest_hash_baseline=1
seal_capability_metadata_report=1
seal_policy_decision_report=1
seal_runtime_dry_run_report=1
seal_operator_receipt_report=1
seal_local_capability_registry_schema=implemented-no-effect
seal_panel_status_surface=planned
seal_receipt_bundle=partial-local
seal_operator_review_queue=planned
```

Near-Term Build Queue

Completed recent checkpoints:

1. Capability metadata report status/index alignment.
2. A Seal product-spine status record and guard.
3. A bundled operator receipt report that ties capability metadata, policy decision, request freshness, signed request metadata, runtime dry-run, and denial reason into one local artifact.
4. A local capability registry schema contract before any production registry loader exists.
5. A no-effect local capability registry schema implementation plan before any schema C code exists.
6. A no-effect local capability registry schema implementation with bounded entries, deterministic validation, and zero authority.

The product path should now prioritize:

1. A deterministic local capability registry schema report surface and status checkpoint.
2. A Panel-visible Seal dashboard that renders current reports without root, network, or runtime authority.
3. A signed receipt proof path that remains verification-only until trust-root and revocation boundaries are implemented.
4. A future enforcement preflight contract that keeps enforce mode closed until all predecessor evidence is present.

Current completed product-spine checkpoint:

```
seal_product_spine_status_present=1
product_spine_changes_authority=0
```

```
operator_receipt_report_contract_present=1
operator_receipt_report_implementation_plan_present=1
operator_receipt_report_implementation_present=1
operator_receipt_report_surface_present=1
operator_receipt_report_status_present=1
local_capability_registry_schema_contract_present=1
local_capability_registry_schema_implementation_plan_present=1
local_capability_registry_schema_implementation_present=1
```

Non-Claims

The product target does not change current authority.

```
production_security_product=0
malware_prevention=0
ransomware_prevention=0
runtime_enforcement=0
kernel_enforcement=0
root_authority=0
network_authority=0
capability_enforcement=0
effect_execution=0
ai_agent_execution_control=0
runtime_authority_granted=0
```

Definition Of Earned Progress

Seal becomes more product-real only when a new slice adds reproducible evidence, tests, failure behavior, public status alignment, and a narrower authority boundary.

Security language must follow evidence. The product can aim high, but every claim must stay attached to code, tests, reports, and explicit non-claims.

docs/latticra-seal/README.md

Source title: Latticra Seal Documentation

Lines: 58 | Words: 319 | SHA-256: 02e2a2a8dff4d7442e046ef28ec4bb67d18db01ce9d1f0e766e3dfebb3f95234

Latticra Seal Documentation

Documentation update: The former standalone Latticra Seal handbook has been superseded as the main project book by The Latticra System Substrate: An Effect at Modern Security.

- System Substrate documentation landing page
- System Substrate PDF
- System Substrate DOCX

The file latticra-seal-handbook.pdf is retained as a compatibility path, but it now points to the System Substrate PDF content.

Latticra Seal is the verification, reporting, and policy-boundary layer inside the Latticra ecosystem.

It is designed to help Latticra describe, inspect, and verify local project/system state through bounded evidence such as manifests, hashes, receipts, policy checks, status reports, and no-effect runtime-boundary metadata.

Latticra Seal is not a separate production security product. It is not currently a malware prevention system, ransomware prevention system, sandbox, kernel enforcement layer, root installer, network authority, or runtime enforcement authority.

Current role

Latticra Seal currently serves as:

- a local integrity and evidence surface
- a report-only verification lane
- a policy regression target
- a manifest/hash baseline mechanism
- a bridge between Latticra CLI, Latticra Panel, and future runtime-boundary work

Documentation map

Document	Purpose
`STATUS.md`	Current readiness and evidence status
`ARCHITECTURE.md`	Internal structure and subsystem model
`USAGE.md`	User-facing command guide
`POLICY.md`	Policy-denial and fail-closed behavior
`REPORTS.md`	Report format and interpretation
`BOUNDARIES.md`	Non-claims and authority limits
`ROADMAP.md`	Future implementation path
`PRODUCT.md`	Security-product spine, modes, product surfaces, and earned-capability path

Canonical boundary

Seal documentation must remain evidence-bound.

Do not claim:

- production security readiness
- runtime enforcement
- network authority
- root authority
- kernel integration
- SELinux integration
- systemd enforcement
- Fedora approval
- malware or ransomware prevention
- broad host protection

until those claims are implemented, tested, and documented with reproducible evidence.

docs/latticra-seal/REPORTS.md

Source title: Latticra Seal Reports

Lines: 83 | Words: 248 | SHA-256: 88c8dae2d44ad09cf12f72369ff15f5c196f3b2a9ab93c7a30f8c4c852f0cc7c

Latticra Seal Reports

Latticra Seal reports are evidence records.

A report should be readable by a human and stable enough for future tooling.

Report goals

A Seal report should answer:

- what was checked
- when it was checked

- where it was checked
- what mode was used
- what passed
- what failed
- what changed
- what authority was not used

Required report fields

Recommended fields:

```
timestamp_utc=
mode=
prefix=
kernel=
os=
network_authority=
runtime_enforcement_authority=
root_authority=
manifest_present=
lock_present=
hashes_checked=
hashes_changed=
warnings=
failures=
status=
```

Current Seal report surfaces

Current report-only Seal surfaces include:

```
seal_capability_metadata_report=1
seal_policy_decision_report=1
seal_runtime_dry_run_report=1
seal_operator_receipt_report=1
```

The operator receipt report is a bundled denied metadata receipt. It does not grant authority, execute tools, read or write host data, use the network, or perform effects.

Report modes

report-only

The default safe mode.

Report-only mode may inspect and describe local state, but should not enforce runtime behavior or mutate system state.

verification

Verification mode compares current state against declared expectations.

Verification may fail when files are missing, changed, malformed, or outside policy.

future enforcement

Future enforcement must not be claimed until implemented and tested.

PASS meaning

PASS means the checked evidence matched the declared local expectations for that command.

PASS does not mean the host is secure.

FAIL meaning

FAIL means the checked evidence did not match expectations, policy denied the state, or Seal could not safely complete the check.

FAIL should include a reason.

docs/latticra-seal/STATUS.md

Source title: Latticra Seal Status

Lines: 73 | Words: 268 | SHA-256: f44cb47ee57d59ff621982aa9efc96ac8dab2a87dd08ae537b54953d814f6287

Latticra Seal Status

Status: early evidence-bound verification layer Scope: local report-only verification, manifest/hash baseline, policy regression, and Panel integration planning.

Current classification

Latticra Seal is currently a bounded local evidence subsystem.

It may describe local state, generate reports, compare hash baselines, and participate in policy-denial tests.

It must not be described as a production enforcement layer.

Current evidence

Known current evidence includes:

- latticra-seal command wrapper installed in the local user path
- local Fedora Workstation report generation
- report-only mode
- installed component marker for latticra-seal
- hash lock material in latticra.seal.lock
- policy regression workflow through GitHub Actions
- policy-denial testing through make seal-policy-denials
- product-spine direction for an earned security-product path
- no-effect operator receipt report implementation for bundled denied metadata
- operator-visible receipt report surface and status checkpoint
- local capability registry schema contract before any registry loader exists
- local capability registry schema implementation plan before any schema C code exists
- no-effect local capability registry schema implementation with bounded entries and denied defaults

Current authority limits

```
network_authority=0
runtime_enforcement_authority=0
root_authority=0
kernel_modification_performed=0
systemd_modification_performed=0
selinux_modification_performed=0
production_security_product=0
next_generation_security_product_target=1
```

Current readiness

```
local_report_generation=1
manifest_hash_baseline=1
policy_regression_lane=1
product_spine_present=1
product_spine_status_present=1
operator_receipt_report_implementation_present=1
operator_receipt_report_surface_present=1
operator_receipt_report_status_present=1
local_capability_registry_schema_contract_present=1
local_capability_registry_schema_implementation_plan_present=1
local_capability_registry_schema_implementation_present=1
panel_bridge_planning=1
runtime_enforcement=0
network_operation=0
root_installation=0
kernel_enforcement=0
production_readiness=0
```

Correct public wording

Latticra Seal is an early verification and reporting layer inside Latticra. It focuses on local evidence, policy boundaries, manifest/hash baselines, and report-only system state inspection.

Incorrect public wording

Do not say that Latticra Seal protects production systems, prevents malware, prevents ransomware, enforces runtime isolation, modifies the kernel, installs system services, or provides certified host security.

docs/latticra-seal/USAGE.md

Source title: Latticra Seal Usage

Lines: 64 | Words: 168 | SHA-256: af11d55c1b1fd5b93268ad3e9fd0634073841dc183e5a90caa92d6a9b06f529c

Latticra Seal Usage

This document describes the intended user-facing shape of Latticra Seal commands.

Some commands may be planned, experimental, or partially implemented. Treat the local repository and current tests as the source of truth.

Check local command availability

```
which latticra-seal
```

Generate a local report

```
latticra-seal report
```

Expected report behavior:

- report-only
- no network authority
- no runtime enforcement authority
- no root authority
- local evidence only

Run the Seal smoke lane

```
make seal
```

Run policy-denial regression checks

```
make seal-policy-denials
```

Inspect Seal lock material

```
head -40 latticra.seal.lock
```

Recommended local verification sequence

```
make seal
make seal-policy-denials
latticra-seal report
```

Expected documentation posture

Every command should be documented with:

- what it reads
- what it writes
- what it refuses to do
- whether it is report-only
- whether it requires root
- whether it uses the network
- what counts as PASS
- what counts as FAIL

docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md

Source title: Latticra Seal Agentic Automation Security Contract

Lines: 227 | Words: 688 | SHA-256: 1c0f0e6ddc96f84f984adcec249704ed67aef28f8726c331a1936f3ded6bdc22

Latticra Seal Agentic Automation Security Contract

Status: Latticra Seal agentic automation security contract Scope: contract for future agentic automation security alignment after Seal status rollup metadata. This document does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, AI agent execution, model execution, tool execution, or operating-system behavior.

Purpose

This document defines the first Latticra Seal agentic automation security contract.

The purpose of this layer is to align the existing Seal chain with a future AI-era automation boundary:

```
report -&gt; measurement -&gt; manifest -&gt; signature policy -&gt; signature metadata -&gt;
verification
policy -&gt; verification receipt -&gt; capability gate -&gt; effect decision -&gt; runtime
handoff -&gt;
status rollup -&gt; agentic automation security planning
```

This document does not implement agentic automation behavior.

Design thesis

Latticra Seal is the trust and verification subsystem for Latticra.

For agentic automation, Seal should eventually help answer:

```
who requested the action
which tool or automation module is being invoked
```

```
whether the tool is known
whether a manifest exists
whether the manifest is signed
which filesystem, command, network, or runtime capabilities are requested
whether parameters match a declared schema
whether the request is fresh and non-replayed
whether runtime authority is actually granted
whether an audit receipt was generated
```

The initial posture remains conservative and report-only.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
docs/LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md
docs/LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md
include/latticra/seal_status_rollup.h
src/seal_status_rollup.c
scripts/test-latticra-seal-status-rollup-contract.sh
scripts/test-latticra-seal-status-rollup.sh
```

The status rollup metadata surface remains metadata-only until runtime-boundary behavior exists and is guarded.

Agentic automation boundary

The agentic automation security layer may describe trust boundaries for AI-assisted tools, local automation, model-context tool invocation, shell helpers, package actions, repository helpers, and operator-approved workflows.

The layer may not convert unsupported, unverified, denied, inactive, or metadata-only Seal records into authority.

Allowed in this contract slice:

```
agentic security vocabulary
trust-boundary vocabulary
MCP-alignment vocabulary
request identity field planning
tool identity field planning
parameter validation field planning
freshness and replay-prevention field planning
receipt field planning
report-only status planning
non-claims
static guard validation
```

Forbidden in this contract slice:

```
runtime execution
AI agent execution
model execution
tool execution
shell command execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
```

```
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
MCP server implementation
MCP client implementation
```

Initial agentic policy

The initial policy is report-only and deny-by-default.

```
agentic_automation_security_declared=1
mode=report-only
mcp_alignment_declared=1
mcp_implementation_supported=0
agent_execution_supported=0
model_execution_supported=0
tool_execution_supported=0
shell_execution_supported=0
cryptographic_verification_supported=0
verified_receipt_supported=0
parameter_schema_validation_supported=0
freshness_validation_supported=0
replay_protection_supported=0
capability_enforcement_supported=0
runtime_authority_granted=0
unknown_tool_allowed=0
unsigned_manifest_allowed=0
network_access_allowed=0
private_key_access_allowed=0
system_mutation_allowed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned request fields

Future agentic security records should be bounded and deterministic.

Planned fields:

```
agentic_profile
request_id
caller_id
tool_id
automation_context
manifest_present
manifest_signed
capability_scope_declared
parameter_schema_present
parameter_schema_valid
request_timestamp_present
request_expiration_present
nonce_present
context_hash_present
parameter_hash_present
signature_present
freshness_valid
replay_detected
receipt_required
receipt_generated
runtime_authority_requested
runtime_authority_granted
decision
reason
mode
status
```

Initial values before real authority:

```
agentic_profile=latticra-seal-agentic-automation-security/0.1
manifest_present=0
manifest_signed=0
parameter_schema_present=0
parameter_schema_valid=0
freshness_valid=0
replay_detected=0
```

```
receipt_generated=0
runtime_authority_requested=0
runtime_authority_granted=0
decision=report-only
reason=planning-contract-only
mode=report-only
status=agentic-automation-security-contract-only
```

Promotion rules

Agentic automation security may only advance from contract to implementation when the next slice remains bounded, deterministic, and no-effect.

The first implementation may add metadata fields and deterministic rendering only.

It must not perform runtime behavior, tool execution, shell execution, AI agent execution, model execution, host reads, host writes, network access, signature verification, key handling, or authority grants.

Non-endorsement rule

This contract may align Latticra terminology with public security design concerns around AI-driven automation and model-context tool invocation.

It must not claim endorsement, certification, approval, deployment, or validation by any outside agency, vendor, distribution, or standards body.

Required non-claim:

```
external_endorsement_claimed=0
NSA_endorsement_claimed=0
Fedora_approval_claimed=0
production_readiness_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is MCP alignment planning.

That future slice must remain documentation-only or metadata-only and must not change runtime behavior, host behavior, network behavior, capability enforcement, cryptographic verification, or authority posture.

docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md

Source title: Latticra Seal Agentic Automation Security Implementation

Lines: 160 | Words: 617 | SHA-256: 3101a5e272e787a33d71e1832e3165aba5890ed2d1f768abc1e4dfefa264fd21

Latticra Seal Agentic Automation Security Implementation

Status: initial agentic automation security metadata implementation Scope: bounded C metadata surface for report-only agentic automation security posture after Seal status rollup metadata. This slice does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first Latticra Seal agentic automation security metadata implementation.

The implementation accepts an existing Seal status rollup metadata record and produces deterministic report-only metadata for AI-era/local automation security posture.

Added files

```
include/latticra/seal_agentic_automation_security.h
src/seal_agentic_automation_security.c
tests/seal_agentic_automation_security_invariants.c
scripts/test-latticra-seal-agentic-automation-security.sh
```

API summary

The agentic automation security metadata surface adds:

```
latticra_seal_agentic_automation_security_t
latticra_seal_agentic_automation_security_error_t
latticra_seal_agentic_automation_security_error_label
latticra_seal_agentic_automation_security_from_rollup
latticra_seal_agentic_automation_security_is_report_only
latticra_seal_agentic_automation_security_report
```

Metadata behavior

The implementation:

```
accepts a valid metadata-only Seal status rollup record
sets agentic_profile=latticra-seal-agentic-automation-security/0.1
sets request_id=unset
sets caller_id=unset
sets tool_id=unset
sets automation_context=local-report-only
sets mcp_alignment_declared=1
sets mcp_protocol_implemented=0
sets mcp_server_implemented=0
sets mcp_client_implemented=0
sets agent_execution_supported=0
sets model_execution_supported=0
sets tool_execution_supported=0
sets shell_execution_supported=0
copies manifest_present from status rollup metadata
sets manifest_signed=0
sets parameter_schema_present=0
sets parameter_schema_valid=0
sets freshness_valid=0
sets replay_detected=0
sets receipt_required=1
sets receipt_generated=0
copies cryptographic_verification_supported from status rollup metadata
sets capability_enforcement_supported=0
sets runtime_authority_requested=0
copies runtime_authority_granted from status rollup metadata
sets unknown_tool_allowed=0
sets unsigned_manifest_allowed=0
sets network_access_allowed=0
sets private_key_access_allowed=0
sets system_mutation_allowed=0
copies host_read_performed from status rollup metadata
copies host_write_performed from status rollup metadata
copies network_performed from status rollup metadata
sets mode=report-only
sets decision=report-only
sets reason=metadata-only
renders deterministic agentic automation security metadata
```

Boundary

This implementation does not implement MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, receipt verification, capability enforcement, or authority grants.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null status rollup metadata input -&gt; invalid-input
invalid status rollup metadata -&gt; invalid-rollup
non-metadata-only status rollup input -&gt; invalid-rollup
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, read host files, write host files, enforce capabilities, execute tools, perform effects, call runtime components, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid metadata-only status rollup produces deterministic agentic automation security metadata
agentic_profile is stable
request_id remains unset
caller_id remains unset
tool_id remains unset
automation_context remains local-report-only
mcp_alignment_declared is one
mcp_protocol_implemented remains zero
mcp_server_implemented remains zero
mcp_client_implemented remains zero
agent_execution_supported remains zero
model_execution_supported remains zero
tool_execution_supported remains zero
shell_execution_supported remains zero
manifest_present is copied from status rollup metadata
manifest_signed remains zero
parameter_schema_present remains zero
parameter_schema_valid remains zero
freshness_valid remains zero
replay_detected remains zero
receipt_required is one
receipt_generated remains zero
cryptographic_verification_supported remains zero
capability_enforcement_supported remains zero
runtime_authority_requested remains zero
runtime_authority_granted remains zero
unknown_tool_allowed remains zero
unsigned_manifest_allowed remains zero
network_access_allowed remains zero
private_key_access_allowed remains zero
system_mutation_allowed remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
mode remains report-only
decision remains report-only
reason remains metadata-only
small report buffer fails closed
null inputs fail closed
invalid status rollup metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-agentic-automation-security-contract.sh
sh scripts/test-latticra-seal-mcp-alignment-plan.sh
sh scripts/test-latticra-seal-agentic-automation-security.sh
```

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md

Source title: Latticra Seal Agentic Automation Security Report Surface

Lines: 83 | Words: 348 | SHA-256: 90158a8282fe65319edd8e3c2b5a8f159dee1a30c3f9ed312520c35f5e7d5674

Latticra Seal Agentic Automation Security Report Surface

Status: report surface for Latticra Seal agentic automation security metadata Scope: no-effect report-surface fixture and shell runner for deterministic Seal agentic automation security metadata. This slice does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first dedicated report surface for Latticra Seal agentic automation security metadata.

The report surface exists so operators and contributors can render the current report-only Seal agentic automation security posture without adding runtime behavior or authority.

Added files

```
tests/seal_agentic_automation_security_report_surface.c
scripts/latticra-seal-agentic-automation-security-report.sh
scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
```

Behavior

The report surface:

```
builds a metadata-only Seal status rollup fixture
constructs Seal agentic automation security metadata from that fixture
renders the deterministic Seal agentic automation security report
prints the report to stdout
```

The guard verifies report output fields including:

```
agentic_profile=latticra-seal-agentic-automation-security/0.1
automation_context=local-report-only
mcp_alignment_declared=1
mcp_protocol_implemented=0
agent_execution_supported=0
model_execution_supported=0
tool_execution_supported=0
shell_execution_supported=0
cryptographic_verification_supported=0
capability_enforcement_supported=0
runtime_authority_requested=0
runtime_authority_granted=0
unknown_tool_allowed=0
unsigned_manifest_allowed=0
network_access_allowed=0
private_key_access_allowed=0
system_mutation_allowed=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
status=agentic-automation-security-metadata
```

Validation

Run:

```
sh scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
```

The underlying metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security.sh
```

Boundary

This report surface does not execute tools, execute shell commands on behalf of a model, contact networks, read host files, write host files, parse keys, verify signatures, create receipts, enforce capabilities, call runtime components, or grant runtime authority.

It compiles and runs a local deterministic fixture only.

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md

Source title: Latticra Seal Capability Gate Contract

Lines: 213 | Words: 615 | SHA-256: d7898bc1c6021ed052fc45e1e2ffbe5cbeddffbf665c9d9e070624059e1b7f3c

Latticra Seal Capability Gate Contract

Status: Latticra Seal capability gate contract Scope: contract for future capability gate metadata after verification receipt metadata. This document does not implement capability enforcement, runtime authority, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal capability gate layer.

The purpose of this layer is to decide how a future receipt may be evaluated before any capability, effect, or authority request can become eligible for execution.

This document does not implement capability gates.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
include/latticra/seal_signature.h
```

```
include/latticra/seal_verification_policy.h
include/latticra/seal_verification_receipt.h
src/seal_measurement.c
src/seal_manifest.c
src/seal_signature_policy.c
src/seal_signature.c
src/seal_verification_policy.c
src/seal_verification_receipt.c
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
scripts/test-latticra-seal-signature.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-receipt-contract.sh
scripts/test-latticra-seal-verification-receipt.sh
```

The verification receipt metadata surface remains unverified metadata until cryptographic verification and verified receipt behavior exist.

Gate boundary

No receipt becomes authority without a capability gate.

No capability gate may allow authority from an unverified receipt.

No capability gate may grant runtime authority until cryptographic verification, verified receipts, and runtime-boundary enforcement exist.

Allowed in this contract slice:

```
capability gate vocabulary
gate-state labels
requested capability metadata planning
requested effect metadata planning
deny-by-default rule planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this contract slice:

```
capability enforcement
runtime authority grants
host effect execution
host writes
network access
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
```

Initial gate policy

The initial capability gate policy is denied by default.

```
unverified receipt metadata may be inspected
unverified receipt metadata may not authorize capability use
verified=0 must deny capability gates
authority_usable=0 must deny capability gates
capability_gate_allowed=0 must deny effects
runtime_authority_granted=0 must remain zero
unknown requested capability must deny
unknown requested effect must deny
```

The first implementation after this contract may only add metadata fields and denied-state handling. It must not enforce real capabilities or grant authority.

Planned gate states

Future records should use explicit labels:

```
gate_state=unsupported
gate_state=denied-unverified
gate_state=denied-unsupported
gate_state=denied-policy
gate_state=allowed-metadata-only
gate_state=allowed-runtime-gated
gate_state=revoked
gate_state=expired
```

For the next implementation, the expected state is:

```
gate_state=denied-unverified
```

Planned gate fields

A future capability gate record should be bounded and deterministic.

Planned fields:

```
gate_profile
receipt_profile
verification_policy_profile
artifact_digest_algorithm
artifact_digest_hex
signer_identity_label
public_key_identity_label
receipt_state
verification_state
requested_capability
requested_effect
requested_scope
verified
authority_usable
receipt_capability_gate_allowed
gate_allowed
gate_state
runtime_authority_granted
status
```

Initial values before real capability enforcement:

```
receipt_state=unverified-metadata
verification_state=unsupported
verified=0
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=0
gate_state=denied-unverified
runtime_authority_granted=0
status=capability-gate-contract-only
```

Failure behavior

Future capability gate handling must fail closed.

Required failure states:

```
null receipt metadata -&gt; invalid
invalid receipt metadata -&gt; invalid
missing artifact digest -&gt; invalid
missing signer identity -&gt; invalid
missing public-key identity -&gt; invalid
missing requested capability -&gt; denied
missing requested effect -&gt; denied
unknown requested capability -&gt; denied
unknown requested effect -&gt; denied
unverified receipt -&gt; denied-unverified
unsupported verification -&gt; denied-unsupported
runtime authority request -&gt; rejected
```

Failures must not parse keys, create keys, persist secrets, contact networks, query revocation status, verify records, sign records, write host files, enforce capabilities, execute effects, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
capability gate metadata implementation
```

It does not permit capability enforcement, runtime authority, cryptographic verification, verified receipt generation, key handling, trust-store behavior, revocation lookup, object sealing, network behavior, host writes, or kernel behavior.

After capability gate metadata exists and is guarded, the next valid planning slice is an effect authorization contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-capability-gate-contract.sh
```

docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md

Source title: Latticra Seal Capability Gate Implementation

Lines: 144 | Words: 564 | SHA-256: dce1c7ca960ce5755e94ab161562ddd3e7570b5845d895853455ae06ad6fe85c

Latticra Seal Capability Gate Implementation

Status: initial capability gate metadata implementation Scope: bounded C metadata surface for capability gate posture after verification receipt metadata. This slice does not implement capability enforcement, runtime authority, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal capability gate metadata implementation.

The implementation accepts an existing verification receipt metadata record plus requested capability, effect, and scope labels. It produces deterministic gate metadata that remains denied by default.

Added files

```
include/latticra/seal_capability_gate.h
src/seal_capability_gate.c
tests/seal_capability_gate_invariants.c
scripts/test-latticra-seal-capability-gate.sh
```

API summary

The capability gate metadata surface adds:

```
latticra_seal_capability_gate_t
latticra_seal_capability_gate_error_t
latticra_seal_capability_gate_error_label
latticra_seal_capability_gate_from_receipt
latticra_seal_capability_gate_is_denied_metadata
latticra_seal_capability_gate_report
```

Gate behavior

The implementation:

```
accepts a valid verification receipt metadata record
copies receipt profile metadata
copies verification policy profile metadata
copies artifact digest algorithm metadata
```

```
copies artifact digest hex metadata
copies signer identity metadata
copies public-key identity metadata
copies receipt_state
copies verification_state
records requested capability metadata
records requested effect metadata
records requested scope metadata, defaulting to unspecified-scope
copies verified from the receipt
copies authority_usable from the receipt
copies receipt capability_gate_allowed from the receipt
sets gate_allowed=0
sets gate_state=denied-unverified
sets runtime_authority_granted=0
renders deterministic capability gate metadata
```

Boundary

This implementation does not verify signatures, parse public keys, create keys, store keys, contact networks, query revocation status, persist capability decisions, seal objects, enforce capabilities, execute effects, write host files, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null capability gate output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null verification receipt metadata input -&gt; invalid-input
invalid verification receipt metadata -&gt; invalid-receipt
missing artifact digest -&gt; missing-digest
missing signer identity -&gt; missing-signer
missing public-key identity -&gt; missing-public-key-identity
missing requested capability -&gt; missing-requested-capability
missing requested effect -&gt; missing-requested-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, write host files, enforce capabilities, execute effects, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid verification receipt metadata plus requested capability/effect produces deterministic gate
 metadata
receipt profile is copied
verification policy profile is copied
artifact digest algorithm is copied
artifact digest hex is copied
signer identity label is copied
public-key identity label is copied
receipt_state is copied
verification_state is copied
requested capability is recorded
requested effect is recorded
requested scope is recorded
missing requested scope defaults to unspecified-scope
verified remains zero
authority_usable remains zero
receipt_capability_gate_allowed remains zero
gate_allowed remains zero
gate_state remains denied-unverified
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid verification receipt metadata fails closed
missing digest fails closed
missing signer fails closed
missing public-key identity fails closed
missing requested capability fails closed
missing requested effect fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-capability-gate-contract.sh
sh scripts/test-latticra-seal-capability-gate.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
```

Expected output:

```
seal capability gate contract: ok
seal capability gate invariants: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
```

Next valid slice

The next valid Latticra Seal slice is status rollup status/public-entry alignment.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, host behavior, network behavior, or object sealing unless separately implemented and guarded.

docs/LATTICRA_SEAL_CAPABILITY_METADATA_CONTRACT.md

Source title: Latticra Seal Capability Metadata Contract

Lines: 257 | Words: 745 | SHA-256: 11364eef44da03b7726e42ebd332c81a43bc0f7d1e6145a55a41d1af8c560532

Latticra Seal Capability Metadata Contract

Status: planning contract for a future Latticra Seal capability metadata surface Scope: contract-only planning for a future no-effect capability metadata layer after the completed Latticra Seal guarded allowlist public-entrypoint alignment. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This contract defines the next safe Latticra Seal step after guarded allowlist candidate reporting became public-entrypoint visible.

The future capability metadata surface should answer one question:

```
If a tool request names a capability, can Latticra Seal describe that capability as metadata
without granting the authority to use it?
```

The answer must remain no-effect, deterministic, local, planning-only, and report-only.

Required prerequisites

A future implementation plan may begin only after these existing layers remain present and guarded:

```
seal_agentic_automation_metadata_present=1
seal_parameter_schema_metadata_present=1
seal_request_freshness_metadata_present=1
seal_signed_request_metadata_present=1
seal_policy_decision_metadata_present=1
seal_runtime_gate_metadata_present=1
seal_runtime_dry_run_metadata_present=1
```

```
seal_runtime_dry_run_report_surface_present=1
seal_guarded_allowlist_metadata_present=1
seal_guarded_allowlist_report_surface_present=1
seal_guarded_allowlist_report_surface_status_present=1
seal_guarded_allowlist_status_index_alignment_present=1
seal_guarded_allowlist_public_entrypoint_alignment_present=1
operator_visible_guarded_allowlist_report=1
core_blocked_case_set_complete=1
```

Future record shape

A future capability metadata record should expose bounded metadata fields similar to:

```
capability_metadata_profile=latticra-seal-capability-metadata/0.1
capability_contract_present
capability_metadata_planning_only
capability_name_present
capability_name
capability_scope
capability_effect_class
capability_known
capability_unknown
capability_candidate
capability_requires_guarded_allowlist
capability_requires_policy_decision
capability_requires_runtime_gate
capability_requires_runtime_dry_run
capability_requires_operator_review
capability_grants_authority
capability_executes_tool
capability_reads_host
capability_writes_host
capability_uses_network
default_action
would_allow
would_deny
would_require_operator_review
blocked_reason
report_only
mode
status
```

Required defaults

The initial implementation, if added later, must default to:

```
capability_metadata_planning_only=1
capability_name_present=0
capability_known=0
capability_unknown=1
capability_candidate=0
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
default_action=deny
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=default-deny-capability-metadata
report_only=1
mode=report-only
status=capability-metadata
```

Capability candidate rule

A known capability name may only become a candidate classification.

A known capability candidate must still report:

```
capability_known=1
capability_candidate=1
capability_grants_authority=0
```

```
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
```

This means a future capability match can improve reporting, but cannot authorize execution or effects.

Capability scope vocabulary

The initial planning vocabulary should stay small and explicit:

```
capability_scope=tool-boundary
capability_scope=request-boundary
capability_scope=policy-boundary
capability_scope=runtime-boundary
capability_scope=evidence-boundary
```

Capability effect vocabulary

The initial planning vocabulary should separate effect classes from authority:

```
capability_effect_class=none
capability_effect_class=tool
capability_effect_class=host-read
capability_effect_class=host-write
capability_effect_class=network
capability_effect_class=runtime-authority
```

Any effect class other than none remains descriptive only.

It must not make the effect available.

Required denied cases

The capability metadata surface must preserve the existing core blocked-request vocabulary:

```
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
unknown_capability_denied=1
missing_capability_denied=1
invalid_capability_denied=1
```

Forbidden behavior

The capability metadata layer must not:

```
load production capability files
evaluate real policy files
enforce policy
enforce capabilities
execute tools
execute shell commands
call a runtime executor
read host files
write host files
perform network operations
verify signatures
validate freshness against live time
perform replay-cache mutation
generate keys
load private keys
load trust stores
query revocation services
grant runtime authority
turn capability matches into execution grants
turn capability matches into effect grants
claim production readiness
claim AI-agent security
```

```
claim MCP implementation
```

Required future report surface

A future implementation must include a deterministic report renderer and a local fixture runner.

Required future files:

```
include/latticra/seal_capability_metadata.h
src/seal_capability_metadata.c
tests/seal_capability_metadata_invariants.c
tests/seal_capability_metadata_report_surface.c
scripts/test-latticra-seal-capability-metadata.sh
scripts/latticra-seal-capability-metadata-report.sh
docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION.md
```

Required future tests

A future implementation must prove:

```
missing capability remains denied
unknown capability remains denied
invalid capability remains denied
known capability only becomes a candidate
known capability candidate grants no authority
known capability candidate performs no effect
known capability candidate does not execute tools
known capability candidate does not read host state
known capability candidate does not write host state
known capability candidate does not use the network
capability effect class is descriptive only
report rendering is deterministic
small buffers fail closed
null inputs fail closed
invalid upstream metadata fails closed
```

Promotion rule

This contract does not authorize capability enforcement.

Capability metadata may be considered only as report-only metadata until a future implementation plan, implementation, report surface, status record, status-index alignment, public-entrypoint alignment, and negative-case evidence all remain merged and guarded.

Even after that sequence, a separate runtime authority contract would be required before any execution or effect path could be considered.

Boundary

This is a contract-only planning slice.

It does not change implementation behavior, add runtime behavior, grant authority, or change public readiness.

Current next valid slice

The next valid slice is a no-effect capability metadata implementation plan.

That future slice must still avoid implementation behavior and must specify exact structs, fields, APIs, report format, fixtures, tests, and failure behavior before any C code is added.

docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION.md

Source title: Latticra Seal Capability Metadata Implementation

Lines: 192 | Words: 576 | SHA-256: 3ba53535593fc618d8cb0c2ee9e74f79c757d4e8bd1e9b545f96d68fa57a5ae9

Latticra Seal Capability Metadata Implementation

Status: no-effect implementation record for the Latticra Seal capability metadata surface Source: follows docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION_PLAN.md Scope: deterministic local capability metadata only. This implementation does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This implementation adds a small, deterministic, local capability metadata surface.

The capability metadata surface can classify a known local fixture capability name as a candidate for later review, but it still denies execution, host access, network use, effects, and runtime authority.

Added files

```
include/latticra/seal_capability_metadata.h
src/seal_capability_metadata.c
tests/seal_capability_metadata_invariants.c
scripts/test-latticra-seal-capability-metadata.sh
docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION.md
```

Deterministic local fixture

The implementation uses only this local fixture set:

```
seal.capability.inspect
seal.capability.report
seal.capability.dry_run
```

The fixture is compiled into the local C implementation.

It is not loaded from host files, policy files, network sources, MCP servers, tools, trust stores, or external services.

Missing capability posture

Missing capability names remain denied:

```
capability_metadata_profile=latticra-seal-capability-metadata/0.1
capability_fixture_source=deterministic-local-fixture
capability_fixture_entry_count=3
capability_lookup_performed=1
capability_name_present=0
capability_known=0
capability_unknown=1
capability_candidate=0
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
unknown_capability_denied=1
missing_capability_denied=1
blocked_reason=missing-capability-denied
report_only=1
mode=report-only
status=capability-metadata
```

Unknown capability posture

Unknown capability names remain denied:

```
capability_name_present=1
capability_known=0
capability_unknown=1
capability_candidate=0
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
unknown_capability_denied=1
blocked_reason=default-deny-capability-metadata
report_only=1
mode=report-only
status=capability-metadata
```

Known capability candidate posture

Known fixture capabilities become candidates only:

```
capability_name_present=1
capability_known=1
capability_unknown=0
capability_candidate=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-capability-candidate-still-denied
report_only=1
mode=report-only
status=capability-metadata
```

A known fixture capability improves reporting but does not authorize execution or effects.

API surface

The implementation exposes:

```
latticra_seal_capability_metadata_error_label
latticra_seal_capability_metadata_fixture
latticra_seal_capability_metadata_result_missing
latticra_seal_capability_metadata_result_unknown
latticra_seal_capability_metadata_result_candidate
latticra_seal_capability_metadata_evaluate
latticra_seal_capability_metadata_is_report_only
latticra_seal_capability_metadata_report
```

Report format

The report begins with:

```
LATTICRA SEAL CAPABILITY METADATA
```

It renders deterministic key=value metadata, including missing/unknown/known/candidate state, authority-denial state, effect-denial state, core blocked-case state, report-only state, error label, and status.

Validation

Run:

```
sh scripts/test-latticra-seal-capability-metadata.sh
```

Expected output:

```
seal capability metadata invariants: ok
```

The invariant test proves:

```
missing capability remains denied
missing capability is not a candidate
unknown capability remains denied
unknown capability is not a candidate
known capability becomes candidate only
known capability remains denied
known capability grants no authority
known capability executes no tools
known capability reads no host state
known capability writes no host state
known capability uses no network
capability effect class is descriptive only
capability lookup uses deterministic local fixture
capability fixture entry count is stable
profile is stable
status remains capability-metadata
report_only remains one
mode remains report-only
small report buffers fail closed
null output fails closed
empty capability name fails closed as missing capability
oversized capability name fails closed
```

Boundary

This implementation is metadata-only and no-effect.

It does not execute tools, execute shell commands, read host files, write host files, use the network, evaluate external policies, load policy files, load production capability files, load production allowlist files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, turn capability matches into execution grants, turn capability matches into effect grants, claim production readiness, claim AI-agent security, or claim MCP implementation.

Current next valid slice

The next valid Latticra Seal slice is a capability metadata report surface.

That future slice must render the current deterministic metadata through a local fixture runner and must not perform effects or grant authority.

docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Capability Metadata Implementation Plan

Lines: 367 | Words: 909 | SHA-256: 3bbebad6cf6efb9e028d490e2f6a65ee4b602da25652241e68a053695a3c4044

Latticra Seal Capability Metadata Implementation Plan

Status: implementation planning contract for a future no-effect Latticra Seal capability metadata surface Scope: implementation plan only after the Latticra Seal capability metadata contract. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This plan defines the exact future implementation shape for a no-effect Seal capability metadata layer.

The future implementation should classify whether a requested capability name is known only as descriptive metadata, while still denying execution, host access, network access, effects, and runtime authority.

Required contract

The implementation plan depends on:

```
docs/LATTICRA_SEAL_CAPABILITY_METADATA_CONTRACT.md
scripts/test-latticra-seal-capability-metadata-contract.sh
```

The contract must remain merged and guarded before implementation code is added.

Future files

The future implementation slice should add:

```
include/latticra/seal_capability_metadata.h
src/seal_capability_metadata.c
tests/seal_capability_metadata_invariants.c
scripts/test-latticra-seal-capability-metadata.sh
docs/LATTICRA_SEAL_CAPABILITY_METADATA_IMPLEMENTATION.md
```

A later report-surface slice may add:

```
tests/seal_capability_metadata_report_surface.c
scripts/latticra-seal-capability-metadata-report.sh
scripts/test-latticra-seal-capability-metadata-report-surface.sh
```

Header API plan

The future header should define:

```
LATTICRA_SEAL_CAPABILITY_METADATA_PROFILE_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_NAME_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_SCOPE_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_EFFECT_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_REASON_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_STATUS_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_MAX
LATTICRA_SEAL_CAPABILITY_METADATA_ENTRY_MAX
latticra_seal_capability_metadata_error_t
latticra_seal_capability_metadata_entry_t
latticra_seal_capability_metadata_fixture_t
latticra_seal_capability_metadata_result_t
latticra_seal_capability_metadata_error_label
latticra_seal_capability_metadata_fixture
latticra_seal_capability_metadata_result_missing
latticra_seal_capability_metadata_result_unknown
latticra_seal_capability_metadata_result_candidate
latticra_seal_capability_metadata_evaluate
latticra_seal_capability_metadata_is_report_only
latticra_seal_capability_metadata_report
```

Error model

The future error enum should include:

```
LATTICRA_SEAL_CAPABILITY_METADATA_OK
LATTICRA_SEAL_CAPABILITY_METADATA_INVALID_INPUT
LATTICRA_SEAL_CAPABILITY_METADATA_INVALID_CAPABILITY_NAME
LATTICRA_SEAL_CAPABILITY_METADATA_INVALID_FIXTURE
LATTICRA_SEAL_CAPABILITY_METADATA_BUFFER_TOO_SMALL
```

Planned static fixture

The first implementation must use only a deterministic local fixture.

Initial fixture entries may include placeholder capability names such as:

```
seal.capability.inspect
seal.capability.report
seal.capability.dry_run
```

These entries are capability candidate names only.

They must not become authority grants, execution grants, or effect grants.

Record fields

The future result record should include:

```
capability_metadata_profile
capability_name
capability_scope
capability_effect_class
capability_fixture_source
capability_fixture_entry_count
capability_lookup_performed
capability_name_present
capability_known
capability_unknown
capability_candidate
capability_requires_guarded_allowlist
capability_requires_policy_decision
capability_requires_runtime_gate
capability_requires_runtime_dry_run
capability_requires_operator_review
capability_grants_authority
capability_executes_tool
capability_reads_host
capability_writes_host
capability_uses_network
default_action_deny
would_allow
would_deny
would_require_operator_review
unknown_tool_denied
unsigned_request_denied
invalid_schema_denied
stale_request_denied
replayed_request_denied
invalid_signature_denied
unknown_capability_denied
missing_capability_denied
invalid_capability_denied
blocked_reason
report_only
mode
status
error
```

Initial constants

The initial future implementation should emit for missing capability names:

```
capability_metadata_profile=latticra-seal-capability-metadata/0.1
capability_fixture_source=deterministic-local-fixture
capability_fixture_entry_count=3
capability_lookup_performed=1
capability_name_present=0
capability_known=0
capability_unknown=1
capability_candidate=0
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
unknown_capability_denied=1
missing_capability_denied=1
invalid_capability_denied=1
blocked_reason=missing-capability-denied
report_only=1
mode=report-only
status=capability-metadata
```

The initial future implementation should emit for unknown capability names:

```
capability_name_present=1
capability_known=0
capability_unknown=1
capability_candidate=0
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
unknown_capability_denied=1
blocked_reason=default-deny-capability-metadata
report_only=1
mode=report-only
status=capability-metadata
```

The initial future implementation should emit for known fixture capability names:

```
capability_name_present=1
capability_known=1
capability_unknown=0
capability_candidate=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-capability-candidate-still-denied
report_only=1
mode=report-only
status=capability-metadata
```

Function behavior

The future builder should:

```
accept a requested capability name string
accept a deterministic local capability fixture
reject null output
classify null capability names as missing capability
classify empty capability names as missing capability
reject oversized capability names
reject invalid capability fixture metadata
classify unknown capabilities as denied
classify known capabilities as candidates only
set no authority fields to zero
set all effect fields to zero
render deterministic metadata only
```

Required report format

The report should begin with:

```
LATTICRA SEAL CAPABILITY METADATA
```

It should render all fields as stable key=value lines.

Required report fields:

```
capability_metadata_profile=latticra-seal-capability-metadata/0.1
capability_name
capability_scope
capability_effect_class
capability_fixture_source=deterministic-local-fixture
capability_lookup_performed=1
capability_known
capability_unknown
capability_candidate
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
```

```
capability_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
unknown_capability_denied=1
missing_capability_denied=1
invalid_capability_denied=1
report_only=1
mode=report-only
status=capability-metadata
```

Required invariant tests

The future invariant test should verify:

```
missing capability remains denied
missing capability is not a candidate
unknown capability remains denied
unknown capability is not a candidate
known capability becomes candidate only
known capability remains denied
known capability grants no authority
known capability executes no tools
known capability reads no host state
known capability writes no host state
known capability uses no network
capability effect class is descriptive only
capability lookup uses deterministic local fixture
capability fixture entry count is stable
profile is stable
status remains capability-metadata
report_only remains one
mode remains report-only
small report buffers fail closed
null output fails closed
empty capability name fails closed as missing capability
oversized capability name fails closed
invalid fixture metadata fails closed
```

Required build runner

The future test runner should compile only local C sources and fixtures.

Planned runner:

```
sh scripts/test-latticra-seal-capability-metadata.sh
```

It should compile:

```
src/seal_capability_metadata.c
tests/seal_capability_metadata_invariants.c
```

If the implementation later depends on upstream Seal metadata records, the runner may also compile local Seal sources, but still must not call tools, read host files, write host files, use the network, or grant authority.

Explicit non-goals

The future implementation must not:

```
execute tools
execute shell commands
read host files
write host files
use the network
evaluate external policies
load policy files
load production capability files
load production allowlist files
verify signatures
validate freshness against live time
mutate replay caches
grant authority
turn capability matches into execution grants
turn capability matches into effect grants
claim production readiness
claim AI-agent security
```

```
claim MCP implementation
```

Promotion rule

No capability enforcement work may begin from this plan alone.

A future C implementation must first be merged with tests, a report surface, status record, status-index alignment, public-entrypoint alignment, and negative-case evidence while preserving the report-only boundary.

Even then, runtime authority would require a separate contract and evidence path.

Current next valid slice

The next valid Latticra Seal slice is the no-effect capability metadata implementation.

That future slice must implement only deterministic metadata, local fixtures, fail-closed behavior, and report rendering. It must not perform effects or grant authority.

docs/LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_SURFACE.md

Source title: Latticra Seal Capability Metadata Report Surface

Lines: 101 | Words: 401 | SHA-256: 83868a9fbf6d96431333d064ef7970965e952860a4cc15d460094e1d2c70dc1c

Latticra Seal Capability Metadata Report Surface

Status: report surface for the Latticra Seal capability metadata layer Scope: deterministic local report surface after the no-effect capability metadata implementation. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This document records the first operator-visible report surface for Latticra Seal capability metadata.

The report surface renders whether a deterministic local fixture capability name is known only as a candidate while still denying execution, host access, network use, effects, and runtime authority.

Added files

```
tests/seal_capability_metadata_report_surface.c
scripts/latticra-seal-capability-metadata-report.sh
```

Report runner

```
sh scripts/latticra-seal-capability-metadata-report.sh
```

Expected report posture

The report surface renders the current capability metadata posture for a known local fixture capability candidate, including:

```
LATTICRA SEAL CAPABILITY METADATA
capability_metadata_profile=latticra-seal-capability-metadata/0.1
capability_name=seal.capability.report
capability_scope=evidence-boundary
capability_effect_class=none
capability_fixture_source=deterministic-local-fixture
capability_fixture_entry_count=3
capability_lookup_performed=1
```

```
capability_name_present=1
capability_known=1
capability_unknown=0
capability_candidate=1
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
unknown_capability_denied=1
missing_capability_denied=0
invalid_capability_denied=0
blocked_reason=known-capability-candidate-still-denied
report_only=1
mode=report-only
status=capability-metadata
```

Boundary

This report surface compiles and runs a local deterministic fixture only.

It does not execute tools, execute shell commands beyond compiling and running the local test fixture, read host files beyond the repository sources needed for local compilation, write host files beyond the temporary test binary/report path, use the network, evaluate external policies, load policy files, load production capability files, load production allowlist files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, turn capability matches into execution grants, turn capability matches into effect grants, claim production readiness, claim AI-agent security, or claim MCP implementation.

Validation

Run:

```
sh scripts/test-latticra-seal-capability-metadata-report-surface.sh
```

Expected output:

```
latticra seal capability metadata report surface: ok
```

The underlying capability metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-capability-metadata.sh
```

Claim boundary

This report surface does not justify the public claim that Latticra secures AI agents.

It makes the capability candidate denial posture easier to inspect before any future capability enforcement or runtime authority path is considered.

Next valid slice

The next valid Latticra Seal slice is capability metadata report status alignment.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_CONTRACT.md

Source title: Latticra Seal Contract

Lines: 326 | Words: 1098 | SHA-256: 74eead41457c8082ce488496fad3f3e1004e0885b118347b1df5402156a618e6

Latticra Seal Contract

Status: Latticra Seal contract Scope: cryptographic evidence and capability substrate definition before signing implementation, encryption implementation, key management, post-quantum implementation, TPM integration, Linux integrity integration, Fedora integration, artifact measurement, runtime enforcement, network behavior, host mutation, kernel behavior, certification, or operating-system behavior.

Purpose

This document defines the first Latticra Seal contract.

Latticra Seal is the cryptographic evidence and capability substrate direction for Latticra. Its purpose is to bind contracts, state transitions, authority decisions, artifact identities, reports, and future operational gates to verifiable evidence before they are allowed to become runtime behavior.

This document does not implement cryptographic enforcement.

Relationship to current Latticra posture

Latticra is contract-first, report-oriented, denied-by-default, and explicit about non-claims.

Latticra Seal depends on the existing project posture:

```
docs/REAL_SYSTEM_CONTRACT.md
docs/EVIDENCE_LADDER.md
docs/EFFECT_GATES.md
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
docs/RUNTIME_BOUNDARY_REPORT_REFINEMENT.md
docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md
docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md
docs/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md
docs/LAT_PIPELINE_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
```

Those files remain the source of truth for the current no-effect, metadata-only, authority, runtime-boundary, Nucleus, Lat, and LIR behavior.

Latticra Seal adds a cryptographic evidence direction. It does not bypass existing gates.

Direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
Latticra Seal is the evidence boundary.
```

Latticra Seal must preserve this relationship:

```
Lat / Latticra Language: declares contracts, authority requests, and intended effects
LIR: carries validated metadata and evidence labels
C substrate: owns bounded records and ABI-compatible report surfaces
Constrained C++ authority layer: validates authority policy and produces denied-by-default
decisions
Nucleus: coordinates only after contracts, authority, and gates are satisfied
Latticra Seal: binds reports, contracts, artifacts, and authority envelopes to evidence
Runtime boundary: remains disabled until explicit cryptographic and authority gates exist
```

Core rule

```
No contract without identity.
No authority without a capability.
No effect without a signed gate.
No promotion without verifiable evidence.
```

This rule is directional. It is not yet implemented runtime behavior.

Current Seal posture

Current Latticra Seal behavior includes no implemented cryptographic enforcement.

Current allowed work:

```
contract definition
implementation planning
report-field design
capability-envelope design
evidence-record design
primitive-policy planning
status/index alignment
guard-script validation
```

Current forbidden claims:

```
signing implemented
encryption implemented
key generation implemented
key storage implemented
capability enforcement implemented
artifact measurement implemented
sealed objects implemented
post-quantum implementation completed
TPM-backed identity implemented
Linux integrity enforcement implemented
Fedora approval or distribution readiness
runtime enforcement
production hardening
certification
```

Seal layers

Latticra Seal is divided into explicit layers.

```
Seal contract layer
  Defines vocabulary, evidence boundaries, non-claims, and promotion rules.

Seal report layer
  Emits deterministic no-effect status records about what Seal supports.

Seal evidence layer
  Defines evidence records for contracts, reports, artifacts, state transitions, and authority
decisions.

Seal capability layer
  Defines narrow, inspectable, future signed capability envelopes.

Seal signature layer
  Future layer for signing manifests, reports, releases, gates, and authority decisions.

Seal object layer
  Future layer for sealed/encrypted objects.

Seal integrity bridge
  Future layer for Linux/Fedora integrity evidence, package evidence, and measured host-facing
validation.
```

Each layer must have its own implementation plan before code is added.

Primitive policy

Latticra Seal does not invent cryptographic primitives.

Initial policy:

```
new cipher design: forbidden
custom cryptographic primitive design: forbidden
standard primitive use: allowed only after exact implementation plan
primitive claims: forbidden until tests and evidence exist
post-quantum claims: forbidden until implementation and validation exist
```

Planned primitive families may include:

```
SHA-256 / SHA-384 for official artifact and record digests
Ed25519 for future development signing
ML-KEM for future post-quantum key establishment profiles
ML-DSA for future post-quantum signature profiles
```

```
SLH-DSA for future conservative/offline signature profiles
HPKE for future object sealing
TLS 1.3 or Noise-style channels for future authenticated transport
TPM2-backed sealed identity for future machine-bound evidence
Linux integrity mechanisms for future host-facing evidence bridges
```

These are planning targets only. This contract does not implement or certify them.

Effect boundary

First Latticra Seal effect posture:

```
contract writing allowed: yes
implementation planning allowed: yes
static report design allowed: yes
static guard validation allowed: yes
host file reads allowed: no
host file writes allowed: no
network access allowed: no
key generation allowed: no
private key storage allowed: no
signature generation allowed: no
encryption allowed: no
artifact hashing allowed: not until exact measurement plan and tests exist
runtime authority allowed: no
capability enforcement allowed: no
kernel interaction allowed: no
```

Future artifact measurement must be explicitly classified as read-only evidence work before it is implemented.

Evidence record shape

Future evidence records should be bounded and deterministic.

Initial planned fields:

```
record_version
record_kind
subject_kind
subject_label
contract_id
contract_digest
artifact_label
artifact_digest
state_before_label
state_after_label
requested_effect
allowed_effect_scope
authority_state
capability_id
signature_state
sealed_state
runtime_boundary_state
evidence_level
non_claims
```

The first report implementation may include these fields as labels without computing cryptographic values.

Capability envelope shape

Future capability envelopes should be narrow, explicit, inspectable, and expiring.

Initial planned fields:

```
capability_version
issuer_id
subject_id
action
object_scope
effect_scope
authority_state
valid_from
valid_until
revocation_state
signature_state
```

An unsigned capability envelope is metadata only. It grants no authority.

A signed capability envelope grants no authority unless a later runtime-boundary implementation explicitly recognizes and enforces it.

Failure behavior

Latticra Seal must fail closed.

Initial failure rules:

```
unknown record kind -&gt; deny / report invalid
unknown effect -&gt; deny / report invalid
unknown authority -&gt; deny / report invalid
missing capability -&gt; deny / report missing
unsigned future gate -&gt; deny / report unsigned
invalid digest -&gt; deny / report invalid
unsupported primitive -&gt; deny / report unsupported
runtime request -&gt; deny / report future-gated
```

Promotion gates

Latticra Seal may only advance through evidence-backed stages.

```
contract -&gt; implementation plan -&gt; no-effect report -&gt; fixture evidence -&gt; read-
only artifact
measurement -&gt; signed manifest -&gt; capability envelope -&gt; sealed object -&gt; Linux/
Fedora integrity
 bridge -&gt; runtime authority gate
```

Skipping stages is forbidden.

First implementation target

The first implementation target is report-only.

It should add no signing, no encryption, no key generation, no artifact reading, no network access, no runtime authority, and no host mutation.

The first report should be able to state:

```
seal_profile
contract_present
implementation_plan_present
report_only_supported
artifact_measurement_supported
signature_supported
capability_enforcement_supported
sealed_objects_supported
effect_performed
host_read_performed
host_mutation_performed
network_performed
runtime_authority_granted
status
non_claims
```

Non-claims

Latticra Seal does not currently provide:

```
cryptographic enforcement
secure boot
measured boot
file integrity enforcement
package-signature enforcement
runtime authorization
capability enforcement
key management
encryption at rest
encrypted transport
post-quantum security
TPM-backed identity
Linux integrity integration
Fedora approval
production hardening
```

```
certification
```

Validation path

This contract is validated by a static guard script that verifies the contract, implementation plan, status record, and non-claim posture remain present.

The guard does not prove cryptographic security. It only prevents the first Seal foundation from being represented as implemented enforcement.

docs/LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md

Source title: Latticra Seal Core Evidence Report

Lines: 70 | Words: 227 | SHA-256: b62c071c6d4a309bb0eef4ded0b48af3b686f7a3d86ce557c49051e9de974d43

Latticra Seal Core Evidence Report

Status: operator-visible evidence report for the Latticra Seal report-only runtime gate chain Scope: report-only evidence summary after the completed core blocked-request case set. This document does not implement runtime behavior, execution, effects, host operations, network operations, policy enforcement, or authority grants.

Purpose

This document records the first operator-visible evidence report for the Latticra Seal core runtime gate path.

The report summarizes the merged report-only metadata chain and the completed core blocked-request case set.

Report runner

```
scripts/latticra-seal-core-evidence-report.sh
```

Covered chain

```
seal_agentic_automation_metadata_present=1
seal_parameter_schema_metadata_present=1
seal_request_freshness_metadata_present=1
seal_signed_request_metadata_present=1
seal_policy_decision_metadata_present=1
seal_runtime_gate_metadata_present=1
```

Covered case evidence

```
unknown_tool_case_validated=1
unsigned_request_case_validated=1
stale_request_case_validated=1
replayed_request_case_validated=1
core_blocked_case_set_complete=1
```

Boundary fields

```
runtime_gate_report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
tool_execution_implemented=0
ai_agent_security_claimed=0
```

Recommended public claim

```
Latticra Seal now has a report-only runtime gate path with core negative-test evidence for
AI-era tool-boundary planning.
```

Claim boundary

This is a real milestone, but it remains report-only.

It is accurate to say that Latticra Seal has built an evidence-bound report-only trust boundary for AI-era tool-boundary planning.

It is not yet accurate to claim that Latticra broadly secures AI agents or enforces AI-agent tool execution in production.

Next valid slice

The next valid slice is public status alignment for this evidence report, followed by a future operator-visible report surface integrated into the broader project status path.

docs/LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md

Source title: Latticra Seal Crypto Graduation Gate Implementation

Lines: 186 | Words: 515 | SHA-256: 24e0b4f1aa4000994bf936f42f7d28494b22ea0855f3cb0e7b887dc89de03c81

Latticra Seal Crypto Graduation Gate Implementation

Status: authority-neutral Latticra Seal crypto graduation gate implementation Scope: bounded C gate over the existing local Ed25519 verify-only result and verified receipt promotion metadata. This implementation does not add signing, key generation, key storage, private-key handling, trust-store loading, revocation lookup, network lookup, FIPS validation, CMVP submission, production cryptography claims, capability enforcement, effect execution, host behavior, or runtime authority.

Purpose

This implementation graduates the current Seal cryptographic layer from isolated local verification evidence into an explicit standards-gated checkpoint.

It consumes:

```
local provider-backed Ed25519 verify-only result
verified receipt promotion metadata
cryptographic assurance and key-management baseline expectations
```

The result can record that the current cryptographic operation met the local gate, while still keeping authority, capability-gate allowance, production crypto claims, FIPS claims, key storage, key generation, signing authority, revocation lookup, network lookup, host behavior, and runtime authority closed.

Files

```
include/latticra/seal_crypto_graduation_gate.h
src/seal_crypto_graduation_gate.c
tests/seal_crypto_graduation_gate_invariants.c
scripts/test-latticra-seal-crypto-graduation-gate.sh
docs/LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md
docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md
```

Standards Tracked

This gate records standards expectations from the project cryptographic assurance baseline:

```
RFC 8032 Ed25519 test vector evidence
NIST FIPS 186-5 digital signature standard tracking
NIST FIPS 180-4 SHA-256 digest standard tracking
NIST FIPS 140-3 claim gate before cryptographic module claims
NIST SP 800-57 Part 1 Rev. 5 key-management requirements
NIST SP 800-131A Rev. 2 algorithm transition review requirements
NIST FIPS 204 ML-DSA post-quantum planning
NIST FIPS 205 SLH-DSA post-quantum planning
```

Tracked source URLs:

```
https://www.rfc-editor.org/rfc/rfc8032
https://csrc.nist.gov/pubs/fips/186-5/final
https://csrc.nist.gov/pubs/fips/180-4/upd1/final
https://csrc.nist.gov/pubs/fips/140-3/final
https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final
https://csrc.nist.gov/pubs/sp/800/131/a/r2/final
https://csrc.nist.gov/pubs/fips/204/final
https://csrc.nist.gov/pubs/fips/205/final
```

Public API

The implementation adds:

```
LATTICRA_SEAL_CRYPTO_GRADUATION_PROFILE_MAX
LATTICRA_SEAL_CRYPTO_GRADUATION_LABEL_MAX
LATTICRA_SEAL_CRYPTO_GRADUATION_ALGORITHM_MAX
LATTICRA_SEAL_CRYPTO_GRADUATION_STATE_MAX
LATTICRA_SEAL_CRYPTO_GRADUATION_REASON_MAX
LATTICRA_SEAL_CRYPTO_GRADUATION_DIGEST_MAX
LATTICRA_SEAL_CRYPTO_GRADUATION_REPORT_MAX
latticra_seal_crypto_graduation_gate_error_t
latticra_seal_crypto_graduation_gate_t
latticra_seal_crypto_graduation_gate_error_label
latticra_seal_crypto_graduation_gate_from_verified_receipt
latticra_seal_crypto_graduation_gate_is_authority_neutral
latticra_seal_crypto_graduation_gate_report
```

Current Graduated Fields

A passing gate records:

```
seal_crypto_graduation_gate_present=1
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
signature_algorithm=Ed25519-development
message_digest_algorithm=SHA-256
public_key_size_bytes=32
signature_size_bytes=64
verify_result_present=1
receipt_present=1
provider_backed_verification_required=1
deterministic_test_vector_required=1
negative_test_vector_required=1
rfc8032_test_vector_tracked=1
fips_186_5_signature_standard_tracked=1
fips_180_4_digest_standard_tracked=1
fips_140_3_claim_gate_required=1
sp_800_57_key_management_required=1
sp_800_131a_transition_review_required=1
fips_204_ml_dsa_planning_tracked=1
fips_205_slh_dsa_planning_tracked=1
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
invalid=0
local_verify_graduated=1
receipt_promotion_graduated=1
standard_expectations_met=1
production_crypto_claim_allowed=0
fips_claim_allowed=0
signing_authority_granted=0
key_generation_allowed=0
key_storage_allowed=0
revocation_lookup_allowed=0
network_lookup_allowed=0
authority_usable=0
authority_promotion_allowed=0
capability_gate_allowed=0
runtime_authority_granted=0
gate_state=graduated-authority-neutral
blocked_reason=authority-remains-denied
status=crypto-graduation-gate-passed
```

Fail-Closed Behavior

The gate blocks:

```
missing verify result
missing verified receipt
invalid or unperformed verification
invalid receipt promotion
unsupported algorithm
missing or malformed SHA-256 digest metadata
missing public-key identity
wrong Ed25519 public-key or signature size
any authority-bearing verify result or receipt
```

Blocked records preserve:

```
local_verify_graduated=0
receipt_promotion_graduated=0
standard_expectations_met=0
production_crypto_claim_allowed=0
fips_claim_allowed=0
signing_authority_granted=0
key_generation_allowed=0
key_storage_allowed=0
revocation_lookup_allowed=0
network_lookup_allowed=0
authority_usable=0
authority_promotion_allowed=0
capability_gate_allowed=0
runtime_authority_granted=0
gate_state=blocked
```

Validation

Current guard:

```
sh scripts/test-latticra-seal-crypto-graduation-gate.sh
```

Expected output:

```
seal crypto graduation gate invariants: ok
latticra seal crypto graduation gate: ok
```

The guard compiles the Ed25519 verify-only implementation, verified receipt promotion implementation, crypto graduation gate implementation, and deterministic invariants. It verifies the RFC 8032 test vector path, tampered-signature denial, malformed metadata denial, authority-bearing metadata denial, null handling, and small-buffer handling.

Current Next Valid Slice

The next valid Latticra Seal cryptographic slice is to carry crypto-graduation-gated runtime handoff evaluation evidence into the runtime handoff report and policy decision report chains as read-only evidence.

That future slice must still not add signing, key generation, key storage, private-key handling, trust-store loading, revocation lookup, network lookup, capability enforcement, effect execution, host behavior, FIPS claims, production cryptography claims, or runtime authority unless separately implemented and guarded.

docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md

Source title: Latticra Seal Crypto Verify Backend Contract

Lines: 200 | Words: 588 | SHA-256: 18835aeafbeaf1a275598765d8acfed9745e54be5c62be793cded98b8b2965b9

Latticra Seal Crypto Verify Backend Contract

Status: Latticra Seal crypto verify backend contract Scope: contract for future cryptographic verification backend behavior after signature metadata, verification policy metadata, verification receipt metadata, and capability gate metadata. This document does not implement cryptographic verification, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal crypto verify backend layer.

The purpose of this layer is to introduce a bounded, verify-only cryptographic boundary that can later verify signature material without immediately granting capability, effect, or runtime authority.

This document does not implement cryptographic verification.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
include/latticra/seal_manifest.h
include/latticra/seal_signature.h
include/latticra/seal_verification_policy.h
include/latticra/seal_verification_receipt.h
include/latticra/seal_capability_gate.h
src/seal_manifest.c
src/seal_signature.c
src/seal_verification_policy.c
src/seal_verification_receipt.c
src/seal_capability_gate.c
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-capability-gate.sh
```

The verification receipt metadata surface remains the authority boundary until a later verified receipt implementation explicitly consumes cryptographic verification results.

Backend boundary

The crypto verify backend may only evaluate verification input and produce bounded verification-result metadata.

A successful verification result may not directly grant authority.

A failed verification result must fail closed.

Allowed in this contract slice:

```
verification backend vocabulary
algorithm labels
verification input metadata planning
verification result-state planning
public-key identity metadata planning
trust-source metadata planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this contract slice:

```
cryptographic verification implementation
signature generation
private-key handling
key generation
network trust lookup
revocation lookup
public-key trust store loading
capability enforcement
effect execution
runtime authority grants
host reads
host writes
kernel interaction
```

Planned algorithm policy

Initial planned verification families:

```
Ed25519-development for local development verification
ML-DSA-planned for future post-quantum signature verification
SLH-DSA-planned for future conservative offline/root signature verification
```

No custom cryptographic primitive may be introduced.

The first implementation after this contract may only add backend metadata and unsupported verification-state handling unless a separate implementation contract authorizes real Ed25519 verification.

Planned verification states

Future records should use explicit labels:

```
crypto_verify_state=unsupported
crypto_verify_state=unverified-metadata
crypto_verify_state=verified
crypto_verify_state=invalid-signature
crypto_verify_state=unsupported-algorithm
crypto_verify_state=missing-public-key
crypto_verify_state=policy-denied
```

For the next metadata implementation, the expected state is:

```
crypto_verify_state=unsupported
```

Planned fields

A future crypto verification backend record should be bounded and deterministic.

Planned fields:

```
backend_profile
signature_profile
manifest_profile
artifact_digest_algorithm
artifact_digest_hex
signer_identity_label
signature_algorithm
public_key_identity_label
trust_source
crypto_verify_state
cryptographic_verification_supported
cryptographic_verification_performed
verified
invalid
authority_usable
capability_gate_allowed
runtime_authority_granted
status
```

Initial values before real verification:

```
crypto_verify_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verified=0
invalid=0
authority_usable=0
capability_gate_allowed=0
```

```
runtime_authority_granted=0
status=crypto-verify-backend-contract-only
```

Failure behavior

Future crypto verify backend handling must fail closed.

Required failure states:

```
null backend output -&gt; invalid
null signature metadata -&gt; invalid
invalid signature metadata -&gt; invalid
missing artifact digest -&gt; invalid
missing signer identity -&gt; invalid
missing public-key identity -&gt; invalid
unsupported algorithm -&gt; unsupported-algorithm
verification requested before support -&gt; unsupported
authority request -&gt; rejected
capability gate request -&gt; rejected
runtime authority request -&gt; rejected
```

Failures must not create keys, store keys, contact networks, query revocation status, sign records, write host files, enforce capabilities, perform effects, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
crypto verification backend metadata implementation
```

It does not permit real cryptographic verification, key handling, trust-store behavior, revocation lookup, object sealing, capability enforcement, network behavior, host reads, host writes, or runtime authority.

After crypto verification backend metadata exists and is guarded, the next valid planning slice is an Ed25519 verify-only implementation contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-crypto-verify-backend-contract.sh
```

docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_IMPLEMENTATION.md

Source title: Latticra Seal Crypto Verify Backend Implementation

Lines: 143 | Words: 545 | SHA-256: 357337a093e9b23c20e871bfd16d565c260ceeaf17569790bd2acf45f40df6fa

Latticra Seal Crypto Verify Backend Implementation

Status: initial crypto verify backend metadata implementation Scope: bounded C metadata surface for crypto verification backend posture after verification policy metadata. This slice does not implement real cryptographic verification, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal crypto verify backend metadata implementation.

The implementation accepts an existing verification policy metadata record and produces deterministic crypto verify backend metadata. It does not verify signatures and does not treat metadata as authority.

Added files

```
include/latticra/seal_crypto_verify_backend.h
src/seal_crypto_verify_backend.c
tests/seal_crypto_verify_backend_invariants.c
```

The intended local runner is:

```
scripts/test-latticra-seal-crypto-verify-backend.sh
```

API summary

The crypto verify backend metadata surface adds:

```
latticra_seal_crypto_verify_backend_t
latticra_seal_crypto_verify_backend_error_t
latticra_seal_crypto_verify_backend_error_label
latticra_seal_crypto_verify_backend_from_policy
latticra_seal_crypto_verify_backend_is_metadata_only
latticra_seal_crypto_verify_backend_report
```

Backend behavior

The implementation:

```
accepts a valid verification policy metadata record
copies verification policy profile metadata
copies signature profile metadata
copies manifest profile metadata
copies artifact digest algorithm metadata
copies artifact digest hex metadata
copies signer identity metadata
copies signature algorithm metadata
copies public-key identity metadata
copies trust-source metadata
accepts only the Ed25519-development signature algorithm label
sets crypto_verify_state=unsupported
sets cryptographic_verification_supported=0
sets cryptographic_verification_performed=0
sets verified=0
sets invalid=0
sets authority_usable=0
sets capability_gate_allowed=0
sets runtime_authority_granted=0
renders deterministic crypto verify backend metadata
```

Boundary

This implementation does not verify signatures, parse public keys, create keys, store keys, contact networks, query revocation status, persist verification decisions, seal objects, enforce capabilities, read host files, write host files, execute network behavior, call runtime components, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null crypto verify backend output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null verification policy metadata input -&gt; invalid-input
invalid verification policy metadata -&gt; invalid-policy
missing artifact digest -&gt; missing-digest
missing signer identity -&gt; missing-signer
missing public-key identity -&gt; missing-public-key-identity
unsupported algorithm label -&gt; unsupported-algorithm
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not create keys, store secrets, contact networks, verify records, sign records, read host files, write host files, enforce capabilities, perform effects, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid verification policy metadata produces deterministic crypto verify backend metadata
backend profile is set
verification policy profile is copied
signature profile is copied
manifest profile is copied
artifact digest algorithm is copied
artifact digest hex is copied
signer identity label is copied
signature algorithm label is copied
public-key identity label is copied
trust source is copied
crypto_verify_state remains unsupported
cryptographic_verification_supported remains zero
cryptographic_verification_performed remains zero
verified remains zero
invalid remains zero
authority_usable remains zero
capability_gate_allowed remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid verification policy metadata fails closed
missing digest fails closed
missing signer fails closed
missing public-key identity fails closed
unsupported algorithm fails closed
```

Validation

Run locally:

```
tmpdir=&quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-seal-crypto.XXXXXX&quot;)&quot;
trap &#x27;rm -rf &quot;$tmpdir&quot;&#x27; EXIT INT HUP TERM
cc -std=c99 -Wall -Wextra -Werror -pedantic -Iinclude src/seal_crypto_verify_backend.c
tests/seal_crypto_verify_backend_invariants.c -o
&quot;$tmpdir/latticra-seal-crypto-verify-backend-invariants&quot;
&quot;$tmpdir/latticra-seal-crypto-verify-backend-invariants&quot;
```

Expected output:

```
seal crypto verify backend invariants: ok
```

Next valid slice

The next valid Latticra Seal slice is an Ed25519 verify-only implementation contract.

That future slice must be contract-first and must not be added directly to this metadata implementation.

docs/LATTICRA_SEAL_ED25519_VERIFY_IMPLEMENTATION.md

Source title: Latticra Seal Ed25519 Verify-Only Implementation

Lines: 78 | Words: 222 | SHA-256: beb8c344125f14c46b683db50ad367cb6f0561a40578ffb7dc1124833fbd8db2

Latticra Seal Ed25519 Verify-Only Implementation

Status: initial Ed25519 verify-only local implementation Scope: provider-backed local Ed25519 verification result surface after the Ed25519 verify-only contract.

Purpose

This document records the first local Ed25519 verify-only implementation for Latticra Seal.

The implementation accepts caller-supplied message bytes, caller-supplied public key bytes, caller-supplied signature bytes, and an existing crypto verify backend metadata record.

It produces bounded verification result metadata.

Added files

```
include/latticra/seal_ed25519_verify.h
src/seal_ed25519_verify.c
tests/seal_ed25519_verify_invariants.c
scripts/test-latticra-seal-ed25519-verify.sh
```

Provider boundary

The implementation uses OpenSSL EVP Ed25519 verification and does not implement custom curve arithmetic.

The test runner links with:

```
-lcrypto
```

Behavior

The implementation:

```
accepts a valid crypto verify backend metadata record
accepts Ed25519-development as the only supported algorithm label
accepts caller-supplied message bytes
accepts caller-supplied 32-byte Ed25519 public key bytes
accepts caller-supplied 64-byte Ed25519 signature bytes
computes a SHA-256 digest of the message for reporting
calls the OpenSSL EVP Ed25519 verify provider
records crypto_verify_state=verified on success
records crypto_verify_state=invalid-signature on signature failure
renders deterministic verification result metadata
```

Authority boundary

A successful verification result remains authority-neutral.

These fields remain zero:

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
```

Validation

Run locally:

```
sh scripts/test-latticra-seal-ed25519-verify.sh
```

Expected output:

```
seal ed25519 verify invariants: ok
```

Next valid slice

The next valid Latticra Seal slice is verified receipt promotion from a successful verify-only result.

docs/LATTICRA_SEAL_ED25519_VERIFY_ONLY_CONTRACT.md

Source title: Latticra Seal Ed25519 Verify-Only Contract

Lines: 215 | Words: 582 | SHA-256: 2c299533676dc18c7833729a8dad5ac32442d80fb628b8155d416a06b5fe3794

Latticra Seal Ed25519 Verify-Only Contract

Status: Latticra Seal Ed25519 verify-only implementation contract Scope: contract for a future Ed25519 verify-only implementation after the crypto verify backend metadata surface. This document does not implement verification. It does not implement signing, key generation, private-key storage, public-key trust stores, network lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host effects, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first real cryptographic operation that may become valid inside Latticra Seal: local Ed25519 signature verification.

The purpose of this layer is narrow:

```
caller-supplied manifest bytes
caller-supplied Ed25519 public key bytes
caller-supplied Ed25519 signature bytes
local verify-only operation
bounded verification result metadata
```

A successful Ed25519 verification result may not directly grant capability, effect, or runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md
include/latticra/seal_crypto_verify_backend.h
src/seal_crypto_verify_backend.c
tests/seal_crypto_verify_backend_invariants.c
include/latticra/seal_verification_policy.h
src/seal_verification_policy.c
include/latticra/seal_signature.h
src/seal_signature.c
```

The crypto verify backend metadata surface remains the source of truth until a separate implementation slice adds actual Ed25519 verification.

Verify-only boundary

Allowed in the next implementation slice:

```
accept caller-supplied message bytes
accept caller-supplied Ed25519 public key bytes
accept caller-supplied Ed25519 signature bytes
validate bounded input lengths
call a reviewed Ed25519 verification provider
record verification attempted
record verification success or failure
render deterministic verification result metadata
```

Forbidden in the next implementation slice:

```
signature generation
private-key input
private-key storage
key generation
custom curve math
network lookup
revocation lookup
trust-store loading
capability enforcement
effect execution
host reads
host writes
runtime authority grants
kernel interaction
```

Primitive policy

The implementation must use Ed25519 verification only.

Required labels:

```
signature_algorithm=Ed25519-development
crypto_verify_state=verified
crypto_verify_state=invalid-signature
crypto_verify_state=unsupported-algorithm
```

No custom cryptographic primitive may be introduced.

The implementation must not contain handwritten Ed25519 curve arithmetic. It must use a reviewed provider boundary selected in the implementation plan.

Planned input policy

The future implementation should use bounded input records.

Planned constants:

```
ed25519_public_key_bytes=32
ed25519_signature_bytes=64
message_size_max=65536
```

Planned input fields:

```
message_label
message_size_bytes
message_digest_algorithm
message_digest_hex
public_key_identity_label
public_key_size_bytes
signature_algorithm
signature_size_bytes
trust_source
```

Planned result fields

A future Ed25519 verify-only result should be bounded and deterministic.

Planned fields:

```
ed25519_verify_profile
backend_profile
verification_policy_profile
message_label
message_size_bytes
message_digest_algorithm
message_digest_hex
public_key_identity_label
public_key_size_bytes
signature_algorithm
signature_size_bytes
trust_source
crypto_verify_state
cryptographic_verification_supported
cryptographic_verification_performed
verified
invalid
authority_usable
capability_gate_allowed
runtime_authority_granted
status
```

A successful verification result may set:

```
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
invalid=0
crypto_verify_state=verified
```

Even after success, these must remain zero:

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
```

A failed verification result must set:

```
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=0
invalid=1
crypto_verify_state=invalid-signature
```

Failure behavior

Future Ed25519 verify-only handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null message -&gt; invalid
empty message -&gt; invalid
message too large -&gt; invalid
null public key -&gt; missing-public-key
wrong public key size -&gt; invalid-public-key-size
null signature -&gt; missing-signature
wrong signature size -&gt; invalid-signature-size
unsupported algorithm -&gt; unsupported-algorithm
provider failure -&gt; verification-provider-failure
private-key input -&gt; rejected
runtime authority request -&gt; rejected
```

Failures must not create keys, store keys, contact networks, query revocation status, sign records, read host files, write host files, enforce capabilities, perform effects, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
Ed25519 verify-only local implementation
```

It does not permit signing, key generation, private-key handling, trust-store behavior, revocation lookup, object sealing, capability enforcement, network behavior, host effects, or runtime authority.

After Ed25519 verify-only implementation exists and is guarded, the next valid planning slice is verified receipt promotion from a successful verify-only result.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-ed25519-verify-only-contract.sh
```

docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md

Source title: Latticra Seal Effect Decision Contract

Lines: 213 | Words: 624 | SHA-256: c0cb5354512888b7872eb327648bc0ea5add1b4ad4be9dc1eb11e0fdcedac6ea

Latticra Seal Effect Decision Contract

Status: Latticra Seal effect decision contract Scope: contract for future effect decision metadata after capability gate metadata. This document does not implement effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal effect decision layer.

The purpose of this layer is to decide how a future capability gate record may be represented as an effect decision before any requested effect can become eligible for execution.

This document does not implement effect decisions.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
include/latticra/seal_signature.h
include/latticra/seal_verification_policy.h
include/latticra/seal_verification_receipt.h
include/latticra/seal_capability_gate.h
src/seal_measurement.c
src/seal_manifest.c
src/seal_signature_policy.c
src/seal_signature.c
src/seal_verification_policy.c
src/seal_verification_receipt.c
src/seal_capability_gate.c
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
scripts/test-latticra-seal-signature.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-capability-gate-contract.sh
scripts/test-latticra-seal-capability-gate.sh
```

The capability gate metadata surface remains denied metadata until cryptographic verification, verified receipt behavior, and real capability enforcement exist.

Effect boundary

No gate becomes an executable effect.

No denied gate may produce an allowed effect decision.

No effect decision may perform host reads, host writes, network access, kernel interaction, or runtime authority until a later runtime-boundary implementation explicitly permits and guards those behaviors.

Allowed in this contract slice:

```
effect decision vocabulary
decision-state labels
requested effect metadata planning
requested capability metadata planning
no-effect posture planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this contract slice:

```
effect execution
host reads
host writes
network access
runtime authority grants
capability enforcement
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
```

Initial decision policy

The initial effect decision policy is denied by default.

```
denied gate metadata may be inspected
gate_allowed=0 must deny effect decisions
runtime_authority_granted=0 must remain zero
unknown requested effect must deny
unknown requested capability must deny
effect_performed must remain zero
host_read_performed must remain zero
host_write_performed must remain zero
network_performed must remain zero
```

The first implementation after this contract may only add metadata fields and denied-state handling. It must not perform effects.

Planned decision states

Future records should use explicit labels:

```
decision_state=unsupported
decision_state=denied-gate
decision_state=denied-unverified
decision_state=denied-unsupported
decision_state=denied-policy
decision_state=allowed-metadata-only
decision_state=allowed-runtime-gated
```

For the next implementation, the expected state is:

```
decision_state=denied-gate
```

Planned decision fields

A future effect decision record should be bounded and deterministic.

Planned fields:

```
decision_profile
gate_profile
receipt_profile
artifact_digest_algorithm
artifact_digest_hex
requested_capability
requested_effect
requested_scope
gate_state
gate_allowed
effect_allowed
effect_performed
host_read_performed
host_write_performed
network_performed
runtime_authority_granted
status
```

Initial values before real effect handling:

```
gate_allowed=0
effect_allowed=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_authority_granted=0
decision_state=denied-gate
status=effect-decision-contract-only
```

Failure behavior

Future effect decision handling must fail closed.

Required failure states:

```
null capability gate metadata -&gt; invalid
invalid capability gate metadata -&gt; invalid
missing artifact digest -&gt; invalid
```

```
missing requested capability -&gt; denied
missing requested effect -&gt; denied
unknown requested capability -&gt; denied
unknown requested effect -&gt; denied
gate_allowed=0 -&gt; denied-gate
runtime authority request -&gt; rejected
```

Failures must not parse keys, create keys, persist secrets, contact networks, query revocation status, verify records, sign records, read host files, write host files, enforce capabilities, execute effects, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
effect decision metadata implementation
```

It does not permit effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt generation, key handling, trust-store behavior, revocation lookup, object sealing, network behavior, host reads, host writes, or kernel behavior.

After effect decision metadata exists and is guarded, the next valid planning slice is a runtime handoff contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-effect-decision-contract.sh
```

docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md

Source title: Latticra Seal Effect Decision Implementation

Lines: 135 | Words: 516 | SHA-256: e5b9f0976000e0bf80bfdfc2fbdf6c000e13f8581f9251f6a93fb53e5b69d779

Latticra Seal Effect Decision Implementation

Status: initial effect decision metadata implementation Scope: bounded C metadata surface for effect decision posture after capability gate metadata. This slice does not implement effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal effect decision metadata implementation.

The implementation accepts an existing capability gate metadata record and produces deterministic effect decision metadata. It does not perform effects and does not treat a denied gate as authority.

Added files

```
include/latticra/seal_effect_decision.h
src/seal_effect_decision.c
tests/seal_effect_decision_invariants.c
scripts/test-latticra-seal-effect-decision.sh
```

API summary

The effect decision metadata surface adds:

```
latticra_seal_effect_decision_t
latticra_seal_effect_decision_error_t
```

```
latticra_seal_effect_decision_error_label
latticra_seal_effect_decision_from_gate
latticra_seal_effect_decision_is_denied_metadata
latticra_seal_effect_decision_report
```

Decision behavior

The implementation:

```
accepts a valid capability gate metadata record
copies gate profile metadata
copies receipt profile metadata
copies artifact digest algorithm metadata
copies artifact digest hex metadata
copies requested capability metadata
copies requested effect metadata
copies requested scope metadata
copies gate_state
sets decision_state=denied-gate
copies gate_allowed from the gate
sets effect_allowed=0
sets effect_performed=0
sets host_read_performed=0
sets host_write_performed=0
sets network_performed=0
sets runtime_authority_granted=0
renders deterministic effect decision metadata
```

Boundary

This implementation does not verify signatures, parse public keys, create keys, store keys, contact networks, query revocation status, persist decisions, enforce capabilities, perform host reads, perform host writes, execute network behavior, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null effect decision output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null capability gate metadata input -&gt; invalid-input
invalid capability gate metadata -&gt; invalid-gate
missing artifact digest -&gt; missing-digest
missing requested capability -&gt; missing-requested-capability
missing requested effect -&gt; missing-requested-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, read host files, write host files, enforce capabilities, perform effects, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid capability gate metadata produces deterministic effect decision metadata
gate profile is copied
receipt profile is copied
artifact digest algorithm is copied
artifact digest hex is copied
requested capability is copied
requested effect is copied
requested scope is copied
gate_state is copied
decision_state remains denied-gate
gate_allowed remains zero
effect_allowed remains zero
effect_performed remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
runtime_authority_granted remains zero
small report buffer fails closed
```

```
null inputs fail closed
invalid capability gate metadata fails closed
missing digest fails closed
missing requested capability fails closed
missing requested effect fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-effect-decision-contract.sh
sh scripts/test-latticra-seal-effect-decision.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
```

Expected output:

```
seal effect decision contract: ok
seal effect decision invariants: ok
seal effect decision status: ok
seal runtime handoff status: ok
```

Next valid slice

The next valid Latticra Seal slice is status rollup status/public-entry alignment.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, host behavior, network behavior, or object sealing unless separately implemented and guarded.

docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md

Source title: Latticra Seal Future Key Parsing Implementation

Lines: 221 | Words: 599 | SHA-256: 7a8a60e002fa53f6959c762b53b3a56ac8f1a6203873279ccce8f0eb14356ab7

Latticra Seal Future Key Parsing Implementation

Status: initial bounded no-effect key parsing implementation Scope: bounded C metadata surface for parsing caller-provided public-key bytes after ready public-key parsing metadata. This slice does not load key material, handle private keys, generate keys, use hardware keys, load trust stores, perform revocation lookup, sign, verify signatures, invoke a signer, execute a signer process, seal objects, perform runtime handoff, grant runtime authority, read host files, write host files, use networks, execute shells, execute tools, enforce capabilities, persist policy, interact with kernels, claim Fedora approval, claim production readiness, or change operating-system behavior.

Purpose

This document records the first Latticra Seal key parsing implementation.

The implementation consumes ready public-key parsing metadata and caller-provided public-key bytes.

It parses only fixed-size public-key byte forms into metadata.

It does not load key material.

It does not handle private keys.

It does not sign.

It does not verify signatures.

Files

```
include/latticra/seal_key_parsing.h
src/seal_key_parsing.c
tests/seal_key_parsing_invariants.c
scripts/test-latticra-seal-key-parsing.sh
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
```

Required predecessors

This implementation depends on the future key parsing implementation plan, the future key parsing implementation contract, and the public-key parsing metadata/status surface:

```
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md
include/latticra/seal_public_key_parsing.h
src/seal_public_key_parsing.c
tests/seal_public_key_parsing_invariants.c
scripts/test-latticra-seal-future-key-parsing-implementation-plan.sh
scripts/test-latticra-seal-future-key-parsing-implementation-contract.sh
scripts/test-latticra-seal-public-key-parsing-contract.sh
scripts/test-latticra-seal-public-key-parsing.sh
scripts/test-latticra-seal-public-key-parsing-status.sh
```

Implemented surface

The key parsing metadata surface adds:

```
latticra_seal_key_parsing_error_t
latticra_seal_key_parsing_format_t
latticra_seal_key_parsing_input_t
latticra_seal_key_parsing_result_t
latticra_seal_key_parsing_error_label
latticra_seal_key_parsing_format_label
latticra_seal_key_parsing_from_public_key_bytes
latticra_seal_key_parsing_is_no_effect
latticra_seal_key_parsing_render
```

The implementation accepts:

```
LATTICRA_SEAL_KEY_PARSING_FORMAT_ED25519_RAW_PUBLIC_KEY_32
LATTICRA_SEAL_KEY_PARSING_FORMAT_ED25519_HEX_PUBLIC_KEY_64
```

It records PEM and DER public-key input as unsupported until a separate parser contract exists.

It denies private-key formats and private-key markers.

Even when key_parsing_ready=1, these fields remain zero:

```
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful raw Ed25519 public-key byte path renders:

```
key_parsing_profile=latticra-seal-key-parsing/0.1
public_key_parsing_profile=latticra-seal-public-key-parsing/0.1
key_material_profile=latticra-seal-key-material/0.1
requested_key_parsing=public-key-bytes-only
requested_public_key_parsing=metadata-only
```

```
key_parsing_input_format=ed25519-raw-public-key-32
key_parsing_input_length=32
key_parsing_algorithm=Ed25519-development
key_parsing_state=public-key-parsed-metadata-only
key_parsing_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
public_key_parsed=1
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
blocked_reason=none
mode=metadata-only
status=key-parsing-metadata
error=ok
```

The successful hex Ed25519 public-key byte path uses:

```
key_parsing_input_format=ed25519-hex-public-key-64
key_parsing_input_length=64
status=key-parsing-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null predecessor metadata
invalid predecessor metadata
predecessor public_key_parsing_ready=0
predecessor public_key_parsing_state not public-key-parsing-metadata-only
predecessor requested_public_key_parsing not metadata-only
predecessor public_key_parsed=1
predecessor key_material_loaded=1
predecessor private_key_handling=1
predecessor key_generation_performed=1
predecessor hardware_key_used=1
predecessor trust_store_loaded=1
predecessor revocation_lookup_performed=1
predecessor signature_performed=1
predecessor verification_performed=1
predecessor signer_invoked=1
predecessor handoff_performed=1
predecessor effect_performed=1
predecessor runtime_authority_granted=1
predecessor host_read_performed=1
predecessor host_write_performed=1
predecessor network_performed=1
missing public-key bytes
empty public-key bytes
oversized public-key bytes
raw Ed25519 public-key input not exactly 32 bytes
hex Ed25519 public-key input not exactly 64 ASCII hex bytes
PEM public-key input
DER public-key input
private-key format requests
private key markers
small report buffers
```

Failure records keep authority and effect fields at zero. Denial details are represented by error, key_parsing_state, status, and blocked_reason.

Validation

The implementation is covered by:

```
sh scripts/test-latticra-seal-future-key-parsing-implementation-plan.sh
```

```
sh scripts/test-latticra-seal-key-parsing.sh
sh scripts/test-latticra-seal-key-parsing-status.sh
sh scripts/test-latticra-seal-verification-policy-status.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-public-key-parsing-status.sh
```

Expected output:

```
seal future key parsing implementation plan: ok
seal key parsing invariants: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal public-key parsing status: ok
```

Current next valid slice

The next valid Latticra Seal slice is status rollup status/public-entry alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, cryptographic verification, signing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md

Source title: Latticra Seal Future Key Parsing Implementation Contract

Lines: 197 | Words: 655 | SHA-256: 7a51beb70eab10941a8f32c8cb9a6230541f8f9979d232c6496e7ba5009d7c7f

Latticra Seal Future Key Parsing Implementation Contract

Status: Latticra Seal future key parsing implementation contract Scope: contract for a future Latticra Seal key parsing implementation path after public-key parsing metadata/status readiness. This document is an implementation contract, not implementation. This document does not implement public-key parsing, parse key bytes, load key material, handle private keys, generate keys, use hardware keys, load trust stores, perform revocation lookup, sign, verify signatures, invoke a signer, execute a signer process, seal objects, perform runtime handoff, grant runtime authority, read host files, write host files, use networks, execute shells, execute tools, enforce capabilities, persist policy, interact with kernels, claim Fedora approval, claim production readiness, or change operating-system behavior.

Purpose

This document defines the next Latticra Seal planning boundary after the public-key parsing metadata implementation and status/public-entry alignment.

The purpose of this layer is to require a narrow implementation plan before any future parser is added.

This document does not implement public-key parsing.

It records that the current repository is ready to plan a future key parsing implementation only because the previous no-effect public-key parsing metadata status exists and is guarded.

Required Predecessors

This contract depends on the public-key parsing metadata checkpoint:

```
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md
include/latticra/seal_public_key_parsing.h
src/seal_public_key_parsing.c
tests/seal_public_key_parsing_invariants.c
scripts/test-latticra-seal-public-key-parsing-contract.sh
scripts/test-latticra-seal-public-key-parsing.sh
scripts/test-latticra-seal-public-key-parsing-status.sh
```

It also depends on the key-material metadata predecessor:

```
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
docs/status/SEAL_KEY_MATERIAL_STATUS.md
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-key-material-contract.sh
scripts/test-latticra-seal-key-material.sh
scripts/test-latticra-seal-key-material-status.sh
```

Current Evidence Gate

The future implementation path may only start from these existing metadata facts:

```
seal_public_key_parsing_status_present=1
seal_public_key_parsing_metadata_present=1
public_key_parsing_profile=latticra-seal-public-key-parsing/0.1
requested_public_key_parsing=metadata-only
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
key_material_profile=latticra-seal-key-material/0.1
requested_key_material=metadata-only
key_material_state=key-material-metadata-only
key_material_ready=1
```

The current checkpoint remains no-effect:

```
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Future Implementation Contract

A future key parsing implementation plan may define:

```
bounded public-key input representation
bounded parser result representation
deterministic parser status labels
fixed input-size limits
explicit accepted encoding labels
explicit denied encoding labels
fail-closed invalid-input behavior
rendered no-effect parser evidence
focused invariants before parser behavior merges
public status alignment before parser behavior is claimed
```

The next planning slice must still keep:

```
future_key_parsing_contract_present=1
future_key_parsing_implementation_plan_present=1
```

```
future_key_parsing_implementation_present=0
public_key_parser_implementation_present=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
signature_performed=0
verification_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

No public_key_parsed=1 status may appear until a separate implementation slice adds parser code, parser invariants, public docs, and explicit guard coverage.

Forbidden In This Contract

This contract does not permit:

```
public-key parsing implementation
public-key byte parsing
public-key file loading
key material loading
private-key handling
key generation
hardware-key use
trust-store loading
revocation lookup
cryptographic signing
signature verification
signer process invocation
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Failure Boundary

Future implementation planning must fail closed if any predecessor metadata reports an effect or any requested path exceeds metadata-only readiness.

Denied predecessor signals include:

```
public_key_parsing_ready=0
public_key_parsing_state not public-key-parsing-metadata-only
requested_public_key_parsing not metadata-only
key_material_ready=0
key_material_state not key-material-metadata-only
requested_key_material not metadata-only
public_key_parsed=1
key_material_loaded=1
private_key_handling=1
signature_performed=1
verification_performed=1
signer_invoked=1
runtime_authority_granted=1
host_read_performed=1
host_write_performed=1
network_performed=1
```

Failures must not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, seal objects, or grant runtime authority.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-future-key-parsing-implementation-contract.sh
```

Expected output:

```
seal future key parsing implementation contract: ok
```

Next Valid Slice

After the future key parsing implementation plan exists and is guarded, the next valid Latticra Seal slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up.

That future implementation slice may add guarded public-key byte parsing metadata only. It must not add key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior.

docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Future Key Parsing Implementation Plan

Lines: 387 | Words: 1015 | SHA-256: e0dadfd87639750ac2e4651a23a72de2402cb1ea165ac2c7c9d92ba91cd8c368

Latticra Seal Future Key Parsing Implementation Plan

Status: implementation planning contract for a future bounded no-effect Latticra Seal key parsing surface Scope: implementation plan only after the Latticra Seal future key parsing implementation contract. This document does not implement public-key parsing, parse key bytes, load key material, handle private keys, generate keys, use hardware keys, load trust stores, perform revocation lookup, sign, verify signatures, invoke a signer, execute a signer process, seal objects, perform runtime handoff, grant runtime authority, read host files, write host files, use networks, execute shells, execute tools, enforce capabilities, persist policy, interact with kernels, claim Fedora approval, claim production readiness, or change operating-system behavior.

Purpose

This plan defines the exact future implementation shape for a bounded no-effect Seal key parsing layer.

The future implementation may parse caller-provided public-key bytes into metadata only. It must not load key material from files, handle private keys, consult trust stores, verify signatures, perform signing, contact networks, or grant authority.

This document is a plan, not parser code.

Required Contract

The implementation plan depends on:

```
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md
scripts/test-latticra-seal-future-key-parsing-implementation-contract.sh
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md
include/latticra/seal_public_key_parsing.h
src/seal_public_key_parsing.c
tests/seal_public_key_parsing_invariants.c
scripts/test-latticra-seal-public-key-parsing-contract.sh
scripts/test-latticra-seal-public-key-parsing.sh
scripts/test-latticra-seal-public-key-parsing-status.sh
```

The contract and predecessor metadata/status guards must remain merged and green before implementation code is added.

Future Files

The future implementation slice should add:

```
include/latticra/seal_key_parsing.h
src/seal_key_parsing.c
tests/seal_key_parsing_invariants.c
scripts/test-latticra-seal-key-parsing.sh
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
```

A later status/public-entry slice may add:

```
docs/status/SEAL_KEY_PARSING_STATUS.md
scripts/test-latticra-seal-key-parsing-status.sh
```

Header API Plan

The future header should define:

```
LATTICRA_SEAL_KEY_PARSING_PROFILE_MAX
LATTICRA_SEAL_KEY_PARSING_LABEL_MAX
LATTICRA_SEAL_KEY_PARSING_FORMAT_MAX
LATTICRA_SEAL_KEY_PARSING_ALGORITHM_MAX
LATTICRA_SEAL_KEY_PARSING_REASON_MAX
LATTICRA_SEAL_KEY_PARSING_STATUS_MAX
LATTICRA_SEAL_KEY_PARSING_INPUT_MAX 4096u
LATTICRA_SEAL_KEY_PARSING_REPORT_MAX 8192u
latticra_seal_key_parsing_error_t
latticra_seal_key_parsing_format_t
latticra_seal_key_parsing_input_t
latticra_seal_key_parsing_result_t
latticra_seal_key_parsing_error_label
latticra_seal_key_parsing_format_label
latticra_seal_key_parsing_from_public_key_bytes
latticra_seal_key_parsing_is_no_effect
latticra_seal_key_parsing_render
```

Recommended function signatures:

```
const char *latticra_seal_key_parsing_error_label(latticra_seal_key_parsing_error_t error);
const char *latticra_seal_key_parsing_format_label(latticra_seal_key_parsing_format_t format);

latticra_status_t latticra_seal_key_parsing_from_public_key_bytes(
    const latticra_seal_public_key_parsing_t *predecessor,
    const unsigned char *public_key_bytes,
    size_t public_key_len,
    latticra_seal_key_parsing_format_t requested_format,
    latticra_seal_key_parsing_result_t *out);

int latticra_seal_key_parsing_is_no_effect(
    const latticra_seal_key_parsing_result_t *key_parsing);

latticra_status_t latticra_seal_key_parsing_render(
    const latticra_seal_key_parsing_result_t *key_parsing,
    char *buffer,
    size_t buffer_len);
```

The first implementation should parse and report public-key byte metadata only. It must not open files or load key material.

Format Model

The future format enum should include:

```
LATTICRA_SEAL_KEY_PARSING_FORMAT_ED25519_RAW_PUBLIC_KEY_32
LATTICRA_SEAL_KEY_PARSING_FORMAT_ED25519_HEX_PUBLIC_KEY_64
LATTICRA_SEAL_KEY_PARSING_FORMAT_PEM_PUBLIC_KEY_UNSUPPORTED
LATTICRA_SEAL_KEY_PARSING_FORMAT_DER_PUBLIC_KEY_UNSUPPORTED
LATTICRA_SEAL_KEY_PARSING_FORMAT_PRIVATE_KEY_DENIED
LATTICRA_SEAL_KEY_PARSING_FORMAT_UNKNOWN
```

Stable labels:

```
ed25519-raw-public-key-32
ed25519-hex-public-key-64
```

```
pem-public-key-unsupported
der-public-key-unsupported
private-key-denied
unknown
```

The first implementation may accept only fixed-size Ed25519 public-key byte forms. PEM and DER are explicitly planned as unsupported until a separate parser contract exists.

Error Model

The future error enum should include:

```
LATTICRA_SEAL_KEY_PARSING_OK
LATTICRA_SEAL_KEY_PARSING_INVALID_INPUT
LATTICRA_SEAL_KEY_PARSING_INVALID_PREDECESSOR
LATTICRA_SEAL_KEY_PARSING_PREDECESSOR_NOT_READY
LATTICRA_SEAL_KEY_PARSING_UNSUPPORTED_FORMAT
LATTICRA_SEAL_KEY_PARSING_OVERSIZED_INPUT
LATTICRA_SEAL_KEY_PARSING_INVALID_PUBLIC_KEY_BYTES
LATTICRA_SEAL_KEY_PARSING_PRIVATE_KEY_DENIED
LATTICRA_SEAL_KEY_PARSING_KEY_MATERIAL_LOADING_DENIED
LATTICRA_SEAL_KEY_PARSING_TRUST_STORE_DENIED
LATTICRA_SEAL_KEY_PARSING_REVOCATION_DENIED
LATTICRA_SEAL_KEY_PARSING_EFFECT_DENIED
LATTICRA_SEAL_KEY_PARSING_BUFFER_TOO_SMALL
```

Record Fields

The future result record should include:

```
key_parsing_profile
public_key_parsing_profile
key_material_profile
requested_key_parsing
requested_public_key_parsing
key_parsing_input_format
key_parsing_input_length
key_parsing_algorithm
key_parsing_state
key_parsing_ready
public_key_parsing_state
public_key_parsing_ready
public_key_parsed
key_material_loaded
private_key_handling
key_generation_performed
hardware_key_used
trust_store_loaded
revocation_lookup_performed
signature_performed
verification_performed
signer_invoked
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
blocked_reason
mode
status
error
```

Plan-Time Checkpoint

This planning slice kept the repository in a no-parser state before the implementation slice was added:

```
future_key_parsing_contract_present=1
future_key_parsing_implementation_plan_present=1
future_key_parsing_implementation_present=0
public_key_parser_implementation_present=0
key_parsing_header_present=0
key_parsing_source_present=0
key_parsing_invariant_test_present=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
```

```
signature_performed=0
verification_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Future Successful Result

The future implementation may emit for a valid caller-provided 32-byte Ed25519 public key:

```
key_parsing_profile=latticra-seal-key-parsing/0.1
requested_key_parsing=public-key-bytes-only
key_parsing_input_format=ed25519-raw-public-key-32
key_parsing_input_length=32
key_parsing_algorithm=Ed25519-development
key_parsing_state=public-key-parsed-metadata-only
key_parsing_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
public_key_parsed=1
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
blocked_reason=none
mode=metadata-only
status=key-parsing-metadata
```

The future implementation must never set key_material_loaded=1, private_key_handling=1, signature_performed=1, verification_performed=1, host_read_performed=1, host_write_performed=1, network_performed=1, or runtime_authority_granted=1.

Failure Behavior

The future builder should:

```
reject null output
reject null predecessor metadata
reject predecessor public_key_parsing_ready=0
reject predecessor public_key_parsing_state not public-key-parsing-metadata-only
reject predecessor requested_public_key_parsing not metadata-only
reject predecessor public_key_parsed=1
reject predecessor key_material_loaded=1
reject predecessor private_key_handling=1
reject predecessor signature_performed=1
reject predecessor verification_performed=1
reject predecessor signer_invoked=1
reject predecessor runtime_authority_granted=1
reject predecessor host_read_performed=1
reject predecessor host_write_performed=1
reject predecessor network_performed=1
reject null public-key bytes
reject zero-length public-key bytes
reject oversized public-key bytes
reject unsupported format labels
reject PEM private-key markers
reject DER private-key requests
reject trust-store requests
reject revocation lookup requests
clear output on small report buffers
render deterministic metadata only
```

Failures must not sign, verify signatures, invoke a signer, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, seal objects, or grant runtime authority.

Required Report Format

The report should begin with:

```
LATTICRA SEAL KEY PARSING
```

It should render all fields as stable key=value lines.

Required report fields:

```
key_parsing_profile=latticra-seal-key-parsing/0.1
requested_key_parsing
key_parsing_input_format
key_parsing_input_length
key_parsing_algorithm=Ed25519-development
key_parsing_state
key_parsing_ready
public_key_parsed
key_material_loaded=0
private_key_handling=0
signature_performed=0
verification_performed=0
signer_invoked=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
blocked_reason
mode=metadata-only
status=key-parsing-metadata
```

Small report buffers fail closed and clear the output buffer.

Exact Implementation Test List

The future implementation test should prove:

```
ed25519 raw public key bytes parse as metadata only
ed25519 hex public key bytes parse as metadata only
public_key_parsed=1 only for valid public-key byte inputs
key_material_loaded=0 after successful parsing
private_key_handling=0 after successful parsing
signature_performed=0 after successful parsing
verification_performed=0 after successful parsing
signer_invoked=0 after successful parsing
runtime_authority_granted=0 after successful parsing
host_read_performed=0 after successful parsing
host_write_performed=0 after successful parsing
network_performed=0 after successful parsing
predecessor not ready fails closed
unsupported format fails closed
private key marker fails closed
oversized input fails closed
small report buffers fail closed
```

Documentation Update Plan

The implementation slice should update:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md
```

and add:

```
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
```

Implementation Gate

Bounded no-effect key parsing implementation code may be added only after this plan is merged.

No key material loading, private-key handling, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, runtime authority, host behavior, network behavior, capability enforcement, object sealing, or kernel behavior may begin from this plan alone.

Validation

This implementation plan is guarded by:

```
sh scripts/test-latticra-seal-future-key-parsing-implementation-plan.sh
```

Expected output:

```
seal future key parsing implementation plan: ok
```

Next Valid Slice

The next valid Latticra Seal slice after the implementation is key parsing status/public-entry alignment.

That future status slice may publish guarded public-key byte parsing metadata only. It must not add key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior.

docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_CONTRACT.md

Source title: Latticra Seal Guarded Allowlist Contract

Lines: 225 | Words: 685 | SHA-256: fc2374a64d132efccea610757855c0bbe8fb94d639f03869faf6e2950aa1e36a

Latticra Seal Guarded Allowlist Contract

Status: planning contract for a future Latticra Seal guarded allowlist surface Scope: contract-only planning for a future no-effect guarded allowlist layer after the completed Latticra Seal runtime dry-run public-entrypoint alignment. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This contract defines the next safe Latticra Seal step after the runtime dry-run lane became public-entrypoint visible.

The future guarded allowlist surface should answer one question:

```
If a tool name is known to Latticra Seal, can the boundary report that it is only a candidate
for later review while still denying every effect?
```

The answer must remain no-effect, deterministic, local, planning-only, and report-only.

Required prerequisites

A future implementation plan may begin only after these existing layers remain present and guarded:

```
seal_agentic_automation_metadata_present=1
seal_parameter_schema_metadata_present=1
seal_request_freshness_metadata_present=1
seal_signed_request_metadata_present=1
seal_policy_decision_metadata_present=1
seal_policy_decision_report_surface_present=1
seal_runtime_gate_metadata_present=1
seal_runtime_dry_run_metadata_present=1
seal_runtime_dry_run_report_surface_present=1
seal_runtime_dry_run_report_surface_status_present=1
seal_runtime_dry_run_status_index_alignment_present=1
seal_runtime_dry_run_public_entrypoint_alignment_present=1
operator_visible_runtime_dry_run_report=1
core_blocked_case_set_complete=1
```

Future record shape

A future guarded allowlist record should expose bounded metadata fields similar to:

```
guarded_allowlist_profile=latticra-seal-guarded-allowlist/0.1
allowlist_contract_present
allowlist_planning_only
allowlist_source
allowlist_entry_count
allowlist_lookup_performed
requested_tool_name_present
requested_tool_known
requested_tool_unknown
requested_tool_candidate
requested_tool_allow_candidate
allow_candidate_requires_policy_decision
allow_candidate_requires_runtime_gate
allow_candidate_requires_runtime_dry_run
allow_candidate_requires_operator_review
allow_candidate_grants_authority
allow_candidate_executes_tool
allow_candidate_reads_host
allow_candidate_writes_host
allow_candidate_uses_network
default_action
would_allow
would_deny
would_require_operator_review
blocked_reason
report_only
mode
status
```

Required defaults

The initial implementation, if added later, must default to:

```
allowlist_planning_only=1
allowlist_lookup_performed=1
requested_tool_known=0
requested_tool_unknown=1
requested_tool_candidate=0
requested_tool_allow_candidate=0
allow_candidate_requires_policy_decision=1
allow_candidate_requires_runtime_gate=1
allow_candidate_requires_runtime_dry_run=1
allow_candidate_requires_operator_review=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
default_action=deny
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=default-deny-guarded-allowlist
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Required denied cases

The guarded allowlist surface must preserve the existing core blocked-request vocabulary:

```
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
```

Known-tool candidate rule

A known tool name may only become a candidate classification.

A known-tool candidate must still report:

```
requested_tool_known=1
requested_tool_candidate=1
requested_tool_allow_candidate=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
```

This means a future allowlist match can improve reporting, but cannot authorize execution.

Forbidden behavior

The guarded allowlist layer must not:

```
load production allowlist files
evaluate real policy files
enforce policy
execute tools
execute shell commands
call a runtime executor
read host files
write host files
perform network operations
verify signatures
validate freshness against live time
perform replay-cache mutation
generate keys
load private keys
load trust stores
query revocation services
grant runtime authority
turn allowlist matches into execution grants
claim production readiness
claim AI-agent security
claim MCP implementation
```

Required future report surface

A future implementation must include a deterministic report renderer and a local fixture runner.

Required future files:

```
include/latticra/seal_guarded_allowlist.h
src/seal_guarded_allowlist.c
tests/seal_guarded_allowlist_invariants.c
tests/seal_guarded_allowlist_report_surface.c
scripts/test-latticra-seal-guarded-allowlist.sh
scripts/latticra-seal-guarded-allowlist-report.sh
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md
```

Required future tests

A future implementation must prove:

```
unknown tool remains denied
```

```
known tool only becomes a candidate
known tool candidate grants no authority
known tool candidate performs no effect
known tool candidate does not execute tools
known tool candidate does not read host state
known tool candidate does not write host state
known tool candidate does not use the network
unsigned request remains denied
invalid schema remains denied
stale request remains denied
replayed request remains denied
invalid signature remains denied
report rendering is deterministic
small buffers fail closed
null inputs fail closed
invalid upstream metadata fails closed
```

Promotion rule

This contract does not authorize runtime enforcement.

A guarded allowlist may be considered only as report-only metadata until a future implementation plan, implementation, report surface, status record, status-index alignment, public-entrypoint alignment, and negative-case evidence all remain merged and guarded.

Even after that sequence, a separate runtime authority contract would be required before any execution or effect path could be considered.

Boundary

This is a contract-only planning slice.

It does not change implementation behavior, add runtime behavior, grant authority, or change public readiness.

Current next valid slice

The next valid slice is a no-effect guarded allowlist implementation plan.

That future slice must still avoid implementation behavior and must specify exact structs, fields, APIs, report format, fixtures, tests, and failure behavior before any C code is added.

docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md

Source title: Latticra Seal Guarded Allowlist Implementation

Lines: 167 | Words: 512 | SHA-256: 700a90e6b3bcc9bb779c080e4fb1228cb48ef8c2b13b1e97e4bb99a3facfc2f2

Latticra Seal Guarded Allowlist Implementation

Status: no-effect implementation record for the Latticra Seal guarded allowlist surface Source: follows docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION_PLAN.md Scope: deterministic local guarded allowlist metadata only. This implementation does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This implementation adds a small, deterministic, local guarded allowlist metadata surface.

The allowlist can classify a known local fixture tool name as a candidate for later review, but it still denies execution and grants no authority.

Added files

```
include/latticra/seal_guarded_allowlist.h
src/seal_guarded_allowlist.c
tests/seal_guarded_allowlist_invariants.c
scripts/test-latticra-seal-guarded-allowlist.sh
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md
```

Deterministic local fixture

The implementation uses only this local fixture set:

```
latticra.seal.inspect
latticra.seal.report
latticra.seal.dry_run
```

The fixture is compiled into the local C implementation.

It is not loaded from host files, policy files, network sources, MCP servers, tools, trust stores, or external services.

Unknown-tool posture

Unknown tools remain denied:

```
guarded_allowlist_profile=latticra-seal-guarded-allowlist/0.1
allowlist_source=deterministic-local-fixture
allowlist_entry_count=3
allowlist_lookup_performed=1
requested_tool_known=0
requested_tool_unknown=1
requested_tool_candidate=0
requested_tool_allow_candidate=0
allow_candidate_requires_policy_decision=1
allow_candidate_requires_runtime_gate=1
allow_candidate_requires_runtime_dry_run=1
allow_candidate_requires_operator_review=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=default-deny-guarded-allowlist
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Known-tool candidate posture

Known fixture tools become candidates only:

```
requested_tool_known=1
requested_tool_unknown=0
requested_tool_candidate=1
requested_tool_allow_candidate=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-tool-candidate-still-denied
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

A known fixture tool improves reporting but does not authorize execution.

API surface

The implementation exposes:

```
latticra_seal_guarded_allowlist_error_label
latticra_seal_guarded_allowlist_fixture
latticra_seal_guarded_allowlist_result_unknown
latticra_seal_guarded_allowlist_result_candidate
latticra_seal_guarded_allowlist_evaluate
latticra_seal_guarded_allowlist_is_report_only
latticra_seal_guarded_allowlist_report
```

Report format

The report begins with:

```
LATTICRA SEAL GUARDED ALLOWLIST
```

It renders deterministic key=value metadata, including known/unknown/candidate state, authority-denial state, effect-denial state, report-only state, error label, and status.

Validation

Run:

```
sh scripts/test-latticra-seal-guarded-allowlist.sh
```

Expected output:

```
seal guarded allowlist invariants: ok
```

The invariant test proves:

```
unknown tool remains denied
unknown tool is not a candidate
known tool becomes candidate only
known tool remains denied
known tool grants no authority
known tool executes no tools
known tool reads no host state
known tool writes no host state
known tool uses no network
allowlist lookup uses deterministic local fixture
allowlist entry count is stable
profile is stable
status remains guarded-allowlist-metadata
report_only remains one
mode remains report-only
small report buffers fail closed
null output fails closed
null tool name fails closed
empty tool name fails closed
oversized tool name fails closed
```

Boundary

This implementation is metadata-only and no-effect.

It does not execute tools, execute shell commands, read host files, write host files, use the network, evaluate external policies, load policy files, load production allowlist files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, turn allowlist matches into execution grants, claim production readiness, claim AI-agent security, or claim MCP implementation.

Current next valid slice

The next valid Latticra Seal slice is a guarded allowlist report surface.

That future slice must render the current deterministic metadata through a local fixture runner and must not perform effects or grant authority.

docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Guarded Allowlist Implementation Plan

Lines: 316 | Words: 816 | SHA-256: ae57bfd2e096bd264557ea60105e5f9e5d302815734ef46ebedefea5934eddd0

Latticra Seal Guarded Allowlist Implementation Plan

Status: implementation planning contract for a future no-effect Latticra Seal guarded allowlist surface Scope: implementation plan only after the Latticra Seal guarded allowlist contract. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This plan defines the exact future implementation shape for a no-effect Seal guarded allowlist layer.

The future implementation should classify whether a requested tool name is known only as a planning candidate, while still denying execution, host access, network access, and runtime authority.

Required contract

The implementation plan depends on:

```
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_CONTRACT.md
scripts/test-latticra-seal-guarded-allowlist-contract.sh
```

The contract must remain merged and guarded before implementation code is added.

Future files

The future implementation slice should add:

```
include/latticra/seal_guarded_allowlist.h
src/seal_guarded_allowlist.c
tests/seal_guarded_allowlist_invariants.c
scripts/test-latticra-seal-guarded-allowlist.sh
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md
```

A later report-surface slice may add:

```
tests/seal_guarded_allowlist_report_surface.c
scripts/latticra-seal-guarded-allowlist-report.sh
scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
```

Header API plan

The future header should define:

```
LATTICRA_SEAL_GUARDED_ALLOWLIST_PROFILE_MAX
LATTICRA_SEAL_GUARDED_ALLOWLIST_TOOL_NAME_MAX
LATTICRA_SEAL_GUARDED_ALLOWLIST_REASON_MAX
LATTICRA_SEAL_GUARDED_ALLOWLIST_STATUS_MAX
LATTICRA_SEAL_GUARDED_ALLOWLIST_REPORT_MAX
latticra_seal_guarded_allowlist_error_t
latticra_seal_guarded_allowlist_entry_t
latticra_seal_guarded_allowlist_t
latticra_seal_guarded_allowlist_result_t
latticra_seal_guarded_allowlist_error_label
latticra_seal_guarded_allowlist_result_unknown
latticra_seal_guarded_allowlist_result_candidate
latticra_seal_guarded_allowlist_is_report_only
latticra_seal_guarded_allowlist_report
```

Error model

The future error enum should include:

```
LATTICRA_SEAL_GUARDED_ALLOWLIST_OK
LATTICRA_SEAL_GUARDED_ALLOWLIST_INVALID_INPUT
LATTICRA_SEAL_GUARDED_ALLOWLIST_INVALID_TOOL_NAME
LATTICRA_SEAL_GUARDED_ALLOWLIST_INVALID_ALLOWLIST
LATTICRA_SEAL_GUARDED_ALLOWLIST_BUFFER_TOO_SMALL
```

Planned static fixture

The first implementation must use only a deterministic local fixture.

Initial fixture entries may include placeholder tool names such as:

```
latticra.seal.inspect
latticra.seal.report
latticra.seal.dry_run
```

These entries are candidate names only.

They must not become execution grants.

Record fields

The future result record should include:

```
guarded_allowlist_profile
tool_name
allowlist_source
allowlist_entry_count
allowlist_lookup_performed
requested_tool_name_present
requested_tool_known
requested_tool_unknown
requested_tool_candidate
requested_tool_allow_candidate
allow_candidate_requires_policy_decision
allow_candidate_requires_runtime_gate
allow_candidate_requires_runtime_dry_run
allow_candidate_requires_operator_review
allow_candidate_grants_authority
allow_candidate_executes_tool
allow_candidate_reads_host
allow_candidate_writes_host
allow_candidate_uses_network
default_action_deny
would_allow
would_deny
would_require_operator_review
blocked_reason
report_only
mode
status
error
```

Initial constants

The initial future implementation should emit for unknown tools:

```
guarded_allowlist_profile=latticra-seal-guarded-allowlist/0.1
allowlist_source=deterministic-local-fixture
allowlist_entry_count=3
allowlist_lookup_performed=1
requested_tool_name_present=1
requested_tool_known=0
requested_tool_unknown=1
requested_tool_candidate=0
requested_tool_allow_candidate=0
allow_candidate_requires_policy_decision=1
allow_candidate_requires_runtime_gate=1
allow_candidate_requires_runtime_dry_run=1
allow_candidate_requires_operator_review=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
```

```
allow_candidate_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=default-deny-guarded-allowlist
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

The initial future implementation should emit for known fixture tools:

```
requested_tool_known=1
requested_tool_unknown=0
requested_tool_candidate=1
requested_tool_allow_candidate=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-tool-candidate-still-denied
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Function behavior

The future builder should:

```
accept a requested tool name string
accept a deterministic local allowlist fixture
reject null output
reject null tool names
reject empty tool names
reject oversized tool names
reject invalid allowlist fixture metadata
classify unknown tools as denied
classify known tools as candidates only
set no authority fields to zero
set all effect fields to zero
render deterministic metadata only
```

Required report format

The report should begin with:

```
LATTICRA SEAL GUARDED ALLOWLIST
```

It should render all fields as stable key=value lines.

Required report fields:

```
guarded_allowlist_profile=latticra-seal-guarded-allowlist/0.1
allowlist_source=deterministic-local-fixture
allowlist_lookup_performed=1
requested_tool_known
requested_tool_unknown
requested_tool_candidate
requested_tool_allow_candidate
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Required invariant tests

The future invariant test should verify:

```
unknown tool remains denied
unknown tool is not a candidate
known tool becomes candidate only
known tool remains denied
known tool grants no authority
known tool executes no tools
known tool reads no host state
known tool writes no host state
known tool uses no network
allowlist lookup uses deterministic local fixture
allowlist entry count is stable
profile is stable
status remains guarded-allowlist-metadata
report_only remains one
mode remains report-only
small report buffers fail closed
null output fails closed
null tool name fails closed
empty tool name fails closed
oversized tool name fails closed
invalid allowlist metadata fails closed
```

Required build runner

The future test runner should compile only local C sources and fixtures.

Planned runner:

```
sh scripts/test-latticra-seal-guarded-allowlist.sh
```

It should compile:

```
src/seal_guarded_allowlist.c
tests/seal_guarded_allowlist_invariants.c
```

If the implementation later depends on upstream Seal metadata records, the runner may also compile local Seal sources, but still must not call tools, read host files, write host files, use the network, or grant authority.

Explicit non-goals

The future implementation must not:

```
execute tools
execute shell commands
read host files
write host files
use the network
evaluate external policies
load policy files
load production allowlist files
verify signatures
validate freshness against live time
mutate replay caches
grant authority
turn allowlist matches into execution grants
claim production readiness
claim AI-agent security
claim MCP implementation
```

Promotion rule

No runtime enforcement work may begin from this plan alone.

A future C implementation must first be merged with tests, a report surface, status record, status-index alignment, public-entrypoint alignment, and negative-case evidence while preserving the report-only boundary.

Even then, runtime authority would require a separate contract and evidence path.

Current next valid slice

The next valid Latticra Seal slice is the no-effect guarded allowlist implementation.

That future slice must implement only deterministic metadata, local fixtures, fail-closed behavior, and report rendering. It must not perform effects or grant authority.

docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE.md

Source title: Latticra Seal Guarded Allowlist Report Surface

Lines: 96 | Words: 385 | SHA-256: 6a2e30396a37dd107f73b89e0d3ba9d1f3b44649fabca4589bcabd1e56baa190

Latticra Seal Guarded Allowlist Report Surface

Status: report surface for the Latticra Seal guarded allowlist metadata layer Scope: deterministic local report surface after the no-effect guarded allowlist metadata implementation. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This document records the first operator-visible report surface for Latticra Seal guarded allowlist metadata.

The report surface renders whether a deterministic local fixture tool name is known only as a candidate while still denying execution, host access, network use, and runtime authority.

Added files

```
tests/seal_guarded_allowlist_report_surface.c
scripts/latticra-seal-guarded-allowlist-report.sh
```

Report runner

```
sh scripts/latticra-seal-guarded-allowlist-report.sh
```

Expected report posture

The report surface renders the current guarded allowlist posture for a known local fixture tool candidate, including:

```
LATTICRA SEAL GUARDED ALLOWLIST
guarded_allowlist_profile=latticra-seal-guarded-allowlist/0.1
tool_name=latticra.seal.report
allowlist_source=deterministic-local-fixture
allowlist_entry_count=3
allowlist_lookup_performed=1
requested_tool_name_present=1
requested_tool_known=1
requested_tool_unknown=0
requested_tool_candidate=1
requested_tool_allow_candidate=1
allow_candidate_requires_policy_decision=1
allow_candidate_requires_runtime_gate=1
allow_candidate_requires_runtime_dry_run=1
allow_candidate_requires_operator_review=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-tool-candidate-still-denied
```

```
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Boundary

This report surface compiles and runs a local deterministic fixture only.

It does not execute tools, execute shell commands beyond compiling and running the local test fixture, read host files beyond the repository sources needed for local compilation, write host files beyond the temporary test binary/report path, use the network, evaluate external policies, load policy files, load production allowlist files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, turn allowlist matches into execution grants, claim production readiness, claim AI-agent security, or claim MCP implementation.

Validation

Run:

```
sh scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
```

Expected output:

```
latticra seal guarded allowlist report surface: ok
```

The underlying guarded allowlist implementation remains covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist.sh
```

Claim boundary

This report surface does not justify the public claim that Latticra secures AI agents.

It makes the allowlist candidate denial posture easier to inspect before any future runtime authority path is considered.

Next valid slice

The next valid Latticra Seal slice is guarded allowlist report status alignment.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Implementation Plan

Lines: 220 | Words: 620 | SHA-256: 6386e67e600d2d60db0de608c125e3f43a826c9bcf7240ca9c1f9795984b7584

Latticra Seal Implementation Plan

Status: Latticra Seal implementation planning contract Scope: exact first implementation plan for a no-effect Seal report surface before artifact reads, signing, encryption, key management, post-quantum implementation, Linux integrity integration, Fedora integration, runtime enforcement, network behavior, host mutation, or kernel behavior.

Purpose

This document defines the first implementation plan for Latticra Seal.

This plan is intentionally narrow. The first implementation must produce deterministic report metadata only. It must not perform cryptographic enforcement.

Required predecessor

The first Seal implementation must follow:

```
docs/LATTICRA_SEAL_CONTRACT.md
```

No implementation code should be added unless the Seal contract remains present and guarded.

First implementation name

```
Latticra Seal report surface
```

First implementation goal

Add a no-effect report surface that makes the Latticra Seal posture inspectable.

The first report answers:

```
Is the Seal contract present?
Is the Seal implementation plan present?
Is Seal still report-only?
Does Seal currently support artifact measurement?
Does Seal currently support signing?
Does Seal currently support capability enforcement?
Does Seal currently support sealed objects?
Did the report perform effects?
Did the report grant authority?
```

Non-goals

The first implementation must not add:

```
artifact hashing
file reads
file writes
network access
key generation
key storage
signature generation
signature verification
encryption
decryption
capability enforcement
runtime execution
host mutation
kernel interaction
Fedora approval claims
production-readiness claims
```

Proposed files

The first implementation should use bounded C surfaces consistent with the repository direction.

```
include/latticra/seal_report.h
src/seal_report.c
tests/seal_report_invariants.c
scripts/test-latticra-seal-report.sh
docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md
```

The implementation may also add status/index alignment files after the report exists.

Public C API shape

The first public API should be small and deterministic.

```
typedef struct latticra_seal_report {
    unsigned contract_present;
    unsigned implementation_plan_present;
    unsigned report_only_supported;
    unsigned artifact_measurement_supported;
    unsigned signature_supported;
    unsigned capability_enforcement_supported;
    unsigned sealed_objects_supported;
    unsigned effect_performed;
    unsigned host_read_performed;
    unsigned host_mutation_performed;
    unsigned network_performed;
    unsigned runtime_authority_granted;
    unsigned evidence_level;
```

```
    const char *seal_profile;
    const char *status;
    const char *non_claims;
} latticra_seal_report_t;

latticra_seal_report_t latticra_seal_report_default(void);
int latticra_seal_report_render(
    const latticra_seal_report_t *report,
    char *out,
    unsigned out_capacity
);
```

Required default values

The default report must state:

```
contract_present=1
implementation_plan_present=1
report_only_supported=1
artifact_measurement_supported=0
signature_supported=0
capability_enforcement_supported=0
sealed_objects_supported=0
effect_performed=0
host_read_performed=0
host_mutation_performed=0
network_performed=0
runtime_authority_granted=0
evidence_level=2
status=report-only
```

Evidence level 2 means contract plus implementation planning and deterministic report design. It does not mean cryptographic security.

Rendered report requirements

The report renderer must emit deterministic text.

Required labels:

```
LATTICRA SEAL REPORT
seal_profile=
contract_present=
implementation_plan_present=
report_only_supported=
artifact_measurement_supported=
signature_supported=
capability_enforcement_supported=
sealed_objects_supported=
effect_performed=
host_read_performed=
host_mutation_performed=
network_performed=
runtime_authority_granted=
evidence_level=
status=
non_claims=
```

The renderer must not allocate dynamically.

The renderer must return failure if the output buffer is null or too small.

Invariant tests

The first invariant tests must verify:

```
default report is report-only
default report performs no effects
default report performs no host reads
default report performs no host mutation
default report performs no network behavior
default report grants no runtime authority
artifact measurement remains unsupported
signature support remains unsupported
capability enforcement remains unsupported
sealed objects remain unsupported
rendered report contains all required labels
small render buffer fails closed
```

Guard script requirements

The first guard script must check:

```
docs/LATTICRA_SEAL_CONTRACT.md exists
docs/LATTICRA_SEAL_IMPLEMENTATION_PLAN.md exists
docs/status/LATTICRA_SEAL_FOUNDATION_STATUS.md exists
Seal contract says this is not cryptographic enforcement
Seal plan forbids artifact hashing in the first implementation
Seal plan forbids signing in the first implementation
Seal plan forbids encryption in the first implementation
Seal plan forbids capability enforcement in the first implementation
```

Promotion path

After this implementation plan is merged, the next allowed implementation is:

```
no-effect Seal report implementation
```

After the report implementation is guarded, the next allowed contract is:

```
read-only artifact measurement contract
```

Artifact measurement must not be added directly to the first report implementation.

Current status

This file is only an implementation plan. It does not implement Seal.

docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md

Source title: Latticra Seal Key-Handling Boundary Contract

Lines: 325 | Words: 1010 | SHA-256: 7c6d7ab5e7e86c9c54b7c252bb1e523923cd94ed4a93b6068866d31e6612fb66

Latticra Seal Key-Handling Boundary Contract

Status: Latticra Seal key-handling boundary contract Scope: contract for a future metadata-only key-handling surface after Seal signing operation metadata. This document does not implement cryptographic signing, signature verification, signer invocation behavior, signer process execution, public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal key-handling boundary after ready signing operation metadata.

The purpose of this layer is to decide whether a ready metadata-only signing operation record may be represented as eligible for a future metadata-only key-handling request before any public-key parser, key material, private key, signer process, trust store, hardware key, runtime bridge, host authority, or signature generation exists.

The key-handling surface is key-handling path classification, not key handling, not signing, not verification, and not trust-store behavior.

This document does not handle keys.

This document does not handle private keys.

Relationship to signing operation

Latticra Seal signing operation metadata already records a no-effect eligibility checkpoint for the future signing path:

```
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_OPERATION_STATUS.md
```

This key-handling boundary contract starts from that checkpoint. It does not replace signing operation metadata, parse public keys, load key material, handle private keys, route to runtime behavior, invoke a signer, perform signing, verify signatures, load trust stores, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_OPERATION_STATUS.md
include/latticra/seal_signing_operation.h
src/seal_signing_operation.c
tests/seal_signing_operation_invariants.c
scripts/test-latticra-seal-signing-operation-contract.sh
scripts/test-latticra-seal-signing-operation.sh
scripts/test-latticra-seal-signing-operation-status.sh
```

The signing operation metadata surface remains the source of readiness evidence for this key-handling surface.

Key-Handling Boundary

Allowed in the next implementation slice:

```
accept signing operation metadata
require signing_operation_ready=1
require signing_operation_state=operation-metadata-only
require requested_signature=Ed25519-development
require requested_signing_authorization=metadata-only
require requested_signer_handoff=metadata-only
require requested_signer_invocation=metadata-only
require requested_signing_operation=metadata-only
require signature_performed=0
require verification_performed=0
require signer_invoked=0
require private_key_handling=0
require key_generation_performed=0
require trust_store_loaded=0
require revocation_lookup_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested_key_handling=metadata-only
classify metadata-only as key_handling_state=key-handling-metadata-only
produce deterministic key-handling metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
signature verification
signer process invocation
public-key parsing
key material loading
private-key handling
key generation
hardware-key use
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
```

```
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial Key Policy

Allowed signing operation state:

```
signing_operation_state=operation-metadata-only
```

Allowed requested signing operation label:

```
metadata-only
```

Allowed requested key-handling label:

```
metadata-only
```

These labels are metadata labels only. They do not mean Ed25519 signing, Ed25519 verification, signer invocation, public-key parsing, key material loading, key generation, private-key handling, hardware-key use, trust establishment, trust-store behavior, revocation lookup, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned key-handling states:

```
key_handling_state=key-handling-metadata-only
key_handling_state=denied-signing-operation
key_handling_state=denied-signer-invocation
key_handling_state=denied-signer-handoff
key_handling_state=denied-signing-authorization
key_handling_state=denied-signature-algorithm
key_handling_state=denied-key-handling
key_handling_state=denied-key-material
key_handling_state=denied-private-key
key_handling_state=denied-trust-store
key_handling_state=denied-runtime-authority
key_handling_state=denied-host-effect
key_handling_state=denied-network-effect
```

The first implementation may set:

```
key_handling_ready=1
```

only for ready metadata-only signing operation metadata, the allowed development signature label, the metadata-only signing operation label, and the metadata-only key-handling label.

Even when key_handling_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned Fields

A future key-handling record should be bounded and deterministic.

Planned fields:

```
key_handling_profile
signing_operation_profile
signer_invocation_profile
signer_handoff_profile
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_signing_authorization
requested_signer_handoff
requested_signer_invocation
requested_signing_operation
requested_key_handling
requested_scope
signing_authorization_state
signing_authorization_ready
signer_handoff_state
signer_handoff_ready
signer_invocation_state
signer_invocation_ready
signing_operation_state
signing_operation_ready
key_handling_state
key_handling_ready
signature_performed
verification_performed
signer_invoked
public_key_parsed
key_material_loaded
private_key_handling
key_generation_performed
hardware_key_used
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
key_handling_state=key-handling-metadata-only
key_handling_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=key-handling-metadata
```

Failure Behavior

Future key-handling handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null signing operation -&gt; invalid
invalid signing operation -&gt; denied-signing-operation
signing_operation_ready=0 -&gt; denied-signing-operation
signing_operation_state not operation-metadata-only -&gt; denied-signing-operation
requested_signing_operation not metadata-only -&gt; denied-signing-operation
signer_invocation_ready=0 -&gt; denied-signer-invocation
signer_invocation_state not invocation-metadata-only -&gt; denied-signer-invocation
requested_signer_invocation not metadata-only -&gt; denied-signer-invocation
signer_handoff_ready=0 -&gt; denied-signer-handoff
signer_handoff_state not handoff-metadata-only -&gt; denied-signer-handoff
requested_signer_handoff not metadata-only -&gt; denied-signer-handoff
requested_signing_authorization not metadata-only -&gt; denied-signing-authorization
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested key handling -&gt; denied-key-handling
unknown requested key handling -&gt; denied-key-handling
public-key parsing requested or observed -&gt; denied-key-material
key material loading requested or observed -&gt; denied-key-material
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
hardware-key use requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-trust-store
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
signer already invoked -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Promotion Rule

This contract permits and now has only this implementation slice:

```
key-handling metadata implementation
```

It does not permit cryptographic signing, signature verification, signer invocation behavior, public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After key-handling metadata, its status/public-entry checkpoint, the key-material boundary contract, and key-material metadata implementation exist and are guarded, the next valid planning slice is bounded no-effect key parsing implementation that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-key-handling-contract.sh
```

docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md

Source title: Latticra Seal Key-Handling Implementation

Lines: 204 | Words: 605 | SHA-256: 31a6bf067e448e62184ecd9240f9b0e7323931068c6bb2f665cc9d9faba7eaae

Latticra Seal Key-Handling Implementation

Status: initial key-handling metadata implementation Scope: bounded C metadata surface for classifying Seal key-handling eligibility after ready signing operation metadata. This slice does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal key-handling metadata implementation.

The implementation consumes ready signing operation metadata and classifies whether the request remains eligible for a future metadata-only key-handling path.

It is key-handling path classification only.

It does not parse public keys.

It does not load key material.

It does not handle private keys.

It does not sign.

Files

```
include/latticra/seal_key_handling.h
src/seal_key_handling.c
tests/seal_key_handling_invariants.c
scripts/test-latticra-seal-key-handling.sh
```

Required predecessors

This implementation depends on the key-handling boundary contract and the signing operation metadata surface:

```
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_OPERATION_STATUS.md
include/latticra/seal_signing_operation.h
src/seal_signing_operation.c
tests/seal_signing_operation_invariants.c
scripts/test-latticra-seal-key-handling-contract.sh
scripts/test-latticra-seal-signing-operation-contract.sh
scripts/test-latticra-seal-signing-operation.sh
scripts/test-latticra-seal-signing-operation-status.sh
```

Implemented surface

The key-handling metadata surface adds:

```
latticra_seal_key_handling_t
latticra_seal_key_handling_error_t
latticra_seal_key_handling_error_label
latticra_seal_key_handling_from_operation
latticra_seal_key_handling_is_metadata_only
latticra_seal_key_handling_render
```

The implementation:

```
accepts signing operation metadata
requires signing_operation_ready=1
requires signing_operation_state=operation-metadata-only
```

```
requires requested_signature=Ed25519-development
requires requested_signing_authorization=metadata-only
requires requested_signer_handoff=metadata-only
requires requested_signer_invocation=metadata-only
requires requested_signing_operation=metadata-only
accepts requested_key_handling=metadata-only
classifies the result as key_handling_state=key-handling-metadata-only
sets key_handling_ready=1 only for metadata-only allowed signing operation states
renders deterministic key-handling metadata
fails closed for invalid or denied predecessor metadata
```

Even when key_handling_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
key_handling_state=key-handling-metadata-only
key_handling_ready=1
public_key_parsed=0
key_material_loaded=0
hardware_key_used=0
mode=metadata-only
status=key-handling-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null signing operation
invalid signing operation
signing_operation_ready=0
signing_operation_state not operation-metadata-only
requested_signing_operation not metadata-only
signer_invocation_ready=0
signer_invocation_state not invocation-metadata-only
requested_signer_invocation not metadata-only
signer_handoff_ready=0
signer_handoff_state not handoff-metadata-only
requested_signer_handoff not metadata-only
signing_authorization_ready=0
```

```
signing_authorization_state not authorized-metadata-only
requested_signing_authorization not metadata-only
missing requested signature
unknown requested signature
missing requested key handling
unknown requested key handling
public-key parsing request
key material loading request
private-key handling request
key generation request
hardware-key use request
trust-store loading request
revocation lookup request
private-key handling already present
key generation already present
trust-store loading already present
revocation lookup already present
runtime authority already granted
signature already performed
verification already performed
signer already invoked
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-key-handling-contract.sh
sh scripts/test-latticra-seal-key-handling.sh
```

Expected output:

```
seal key-handling contract: ok
seal key-handling invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate key-material, signing, and guard contracts.

The key-handling metadata implementation is a guarded checkpoint. Future work must not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md

Source title: Latticra Seal Key-Material Boundary Contract

Lines: 351 | Words: 1054 | SHA-256: 56a21630778e30d50efb020938a81367f43a088c703b4945475b3f67452a5310

Latticra Seal Key-Material Boundary Contract

Status: Latticra Seal key-material boundary contract Scope: contract for a future metadata-only key-material surface after Seal key-handling metadata. This document does not implement public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal key-material boundary after ready key-handling metadata.

The purpose of this layer is to decide whether a ready metadata-only key-handling record may be represented as eligible for a future metadata-only key-material request before any public-key parser, key material loader, private key, signer process, trust store, hardware key, runtime bridge, host authority, or signature generation exists.

The key-material surface is key-material path classification, not key material handling, not public-key parsing, not signing, not verification, and not trust-store behavior.

This document does not parse public keys.

This document does not load key material.

This document does not handle private keys.

Relationship to key handling

Latticra Seal key-handling metadata already records a no-effect eligibility checkpoint for the future key path:

```
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
docs/status/SEAL_KEY_HANDLING_STATUS.md
```

This key-material boundary contract starts from that checkpoint. It does not replace key-handling metadata, parse public keys, load key material, handle private keys, route to runtime behavior, invoke a signer, perform signing, verify signatures, load trust stores, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
docs/status/SEAL_KEY_HANDLING_STATUS.md
include/latticra/seal_key_handling.h
src/seal_key_handling.c
tests/seal_key_handling_invariants.c
scripts/test-latticra-seal-key-handling-contract.sh
scripts/test-latticra-seal-key-handling.sh
scripts/test-latticra-seal-key-handling-status.sh
```

The key-handling metadata surface remains the source of readiness evidence for this key-material surface.

Key-Material Boundary

Allowed in the next implementation slice:

```
accept key-handling metadata
require key_handling_ready=1
```

```
require key_handling_state=key-handling-metadata-only
require requested_signature=Ed25519-development
require requested_signing_authorization=metadata-only
require requested_signer_handoff=metadata-only
require requested_signer_invocation=metadata-only
require requested_signing_operation=metadata-only
require requested_key_handling=metadata-only
require signature_performed=0
require verification_performed=0
require signer_invoked=0
require public_key_parsed=0
require key_material_loaded=0
require private_key_handling=0
require key_generation_performed=0
require hardware_key_used=0
require trust_store_loaded=0
require revocation_lookup_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested_key_material=metadata-only
classify metadata-only as key_material_state=key-material-metadata-only
produce deterministic key-material metadata
```

Forbidden in the next implementation slice:

```
public-key parsing
key material loading
private-key handling
key generation
hardware-key use
trust-store loading
revocation lookup
cryptographic signing
signature verification
signer process invocation
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial Key-Material Policy

Allowed key-handling state:

```
key_handling_state=key-handling-metadata-only
```

Allowed requested key-handling label:

```
metadata-only
```

Allowed requested key-material label:

```
metadata-only
```

These labels are metadata labels only. They do not mean public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust establishment, trust-store behavior, Ed25519 signing, Ed25519 verification, signer invocation, revocation lookup, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned key-material states:

```
key_material_state=key-material-metadata-only
key_material_state=denied-key-handling
key_material_state=denied-signing-operation
key_material_state=denied-signer-invocation
key_material_state=denied-signer-handoff
key_material_state=denied-signing-authorization
```

```
key_material_state=denied-signature-algorithm
key_material_state=denied-key-material
key_material_state=denied-private-key
key_material_state=denied-trust-store
key_material_state=denied-runtime-authority
key_material_state=denied-host-effect
key_material_state=denied-network-effect
```

The first implementation may set:

```
key_material_ready=1
```

only for ready metadata-only key-handling metadata, the allowed development signature label, the metadata-only signing operation label, the metadata-only key-handling label, and the metadata-only key-material label.

Even when key_material_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned Fields

A future key-material record should be bounded and deterministic.

Planned fields:

```
key_material_profile
key_handling_profile
signing_operation_profile
signer_invocation_profile
signer_handoff_profile
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_signing_authorization
requested_signer_handoff
requested_signer_invocation
requested_signing_operation
requested_key_handling
requested_key_material
requested_scope
signing_authorization_state
signing_authorization_ready
signer_handoff_state
signer_handoff_ready
signer_invocation_state
signer_invocation_ready
signing_operation_state
signing_operation_ready
```

```
key_handling_state
key_handling_ready
key_material_state
key_material_ready
signature_performed
verification_performed
signer_invoked
public_key_parsed
key_material_loaded
private_key_handling
key_generation_performed
hardware_key_used
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
key_material_profile=latticra-seal-key-material/0.1
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
requested_key_handling=metadata-only
requested_key_material=metadata-only
key_handling_state=key-handling-metadata-only
key_handling_ready=1
key_material_state=key-material-metadata-only
key_material_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=key-material-metadata
```

Failure Behavior

Future key-material handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null key handling -&gt; invalid
invalid key handling -&gt; denied-key-handling
key_handling_ready=0 -&gt; denied-key-handling
key_handling_state not key-handling-metadata-only -&gt; denied-key-handling
requested_key_handling not metadata-only -&gt; denied-key-handling
signing_operation_ready=0 -&gt; denied-signing-operation
signing_operation_state not operation-metadata-only -&gt; denied-signing-operation
requested_signing_operation not metadata-only -&gt; denied-signing-operation
signer_invocation_ready=0 -&gt; denied-signer-invocation
signer_invocation_state not invocation-metadata-only -&gt; denied-signer-invocation
requested_signer_invocation not metadata-only -&gt; denied-signer-invocation
signer_handoff_ready=0 -&gt; denied-signer-handoff
signer_handoff_state not handoff-metadata-only -&gt; denied-signer-handoff
requested_signer_handoff not metadata-only -&gt; denied-signer-handoff
requested_signing_authorization not metadata-only -&gt; denied-signing-authorization
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested key material -&gt; denied-key-material
unknown requested key material -&gt; denied-key-material
public-key parsing requested or observed -&gt; denied-key-material
key material loading requested or observed -&gt; denied-key-material
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
hardware-key use requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-trust-store
```

```
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
signer already invoked -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Promotion Rule

This contract permitted only the implementation slice:

```
key-material metadata implementation
```

That implementation slice now exists as:

```
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-key-material.sh
```

It does not permit public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After key-material metadata, its status/public-entry checkpoint, and the public-key parsing boundary contract exist and are guarded, the next valid planning slice is bounded no-effect key parsing implementation that still must not add public-key parsing without separate implementation, key-material, public-key, and guard contracts.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-key-material-contract.sh
```

docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md

Source title: Latticra Seal Key-Material Implementation

Lines: 215 | Words: 627 | SHA-256: 7fa4af9e487dcfb97bafb6ce1627e1d53124d5cfe3706a6ec16e508fcc78573b

Latticra Seal Key-Material Implementation

Status: initial key-material metadata implementation Scope: bounded C metadata surface for classifying Seal key-material eligibility after ready key-handling metadata. This slice does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal key-material metadata implementation.

The implementation consumes ready key-handling metadata and classifies whether the request remains eligible for a future metadata-only key-material path.

It is key-material path classification only.

It does not parse public keys.

It does not load key material.

It does not handle private keys.

It does not sign.

Files

```
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-key-material.sh
```

Required predecessors

This implementation depends on the key-material boundary contract and the key-handling metadata surface:

```
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
docs/status/SEAL_KEY_HANDLING_STATUS.md
include/latticra/seal_key_handling.h
src/seal_key_handling.c
tests/seal_key_handling_invariants.c
scripts/test-latticra-seal-key-material-contract.sh
scripts/test-latticra-seal-key-handling-contract.sh
scripts/test-latticra-seal-key-handling.sh
scripts/test-latticra-seal-key-handling-status.sh
```

Implemented surface

The key-material metadata surface adds:

```
latticra_seal_key_material_t
latticra_seal_key_material_error_t
latticra_seal_key_material_error_label
latticra_seal_key_material_from_key_handling
latticra_seal_key_material_is_metadata_only
latticra_seal_key_material_render
```

The implementation:

```
accepts key-handling metadata
requires key_handling_ready=1
requires key_handling_state=key-handling-metadata-only
requires requested_signature=Ed25519-development
requires requested_signing_authorization=metadata-only
requires requested_signer_handoff=metadata-only
requires requested_signer_invocation=metadata-only
requires requested_signing_operation=metadata-only
requires requested_key_handling=metadata-only
accepts requested_key_material=metadata-only
classifies the result as key_material_state=key-material-metadata-only
sets key_material_ready=1 only for metadata-only allowed key-handling states
renders deterministic key-material metadata
fails closed for invalid or denied predecessor metadata
```

Even when key_material_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
```

```
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
key_material_profile=latticra-seal-key-material/0.1
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
requested_key_material=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
key_handling_state=key-handling-metadata-only
key_handling_ready=1
key_material_state=key-material-metadata-only
key_material_ready=1
public_key_parsed=0
key_material_loaded=0
hardware_key_used=0
mode=metadata-only
status=key-material-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null key handling
invalid key handling
key_handling_ready=0
key_handling_state not key-handling-metadata-only
requested_key_handling not metadata-only
signing_operation_ready=0
signing_operation_state not operation-metadata-only
requested_signing_operation not metadata-only
signer_invocation_ready=0
signer_invocation_state not invocation-metadata-only
requested_signer_invocation not metadata-only
signer_handoff_ready=0
signer_handoff_state not handoff-metadata-only
requested_signer_handoff not metadata-only
signing_authorization_ready=0
signing_authorization_state not authorized-metadata-only
requested_signing_authorization not metadata-only
missing requested signature
unknown requested signature
missing requested key material
unknown requested key material
public-key parsing request
key material loading request
private-key handling request
key generation request
hardware-key use request
trust-store loading request
```

```
revocation lookup request
public-key parsing already present
key material loading already present
private-key handling already present
key generation already present
hardware-key use already present
trust-store loading already present
revocation lookup already present
runtime authority already granted
signature already performed
verification already performed
signer already invoked
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-key-material-contract.sh
sh scripts/test-latticra-seal-key-material.sh
```

Expected output:

```
seal key-material contract: ok
seal key-material invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add public-key parsing without separate key-material, public-key, and guard contracts.

The key-material metadata implementation is a guarded checkpoint. Future work must not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md

Source title: Latticra Seal Local Capability Registry Schema Contract

Lines: 270 | Words: 730 | SHA-256: b64cd85975486cdfee28c13644fc75c12fc4daa8d1ab70785fd2df8b54058fee

Latticra Seal Local Capability Registry Schema Contract

Status: contract for a future no-effect Latticra Seal local capability registry schema
Scope: contract-only planning for a future local capability registry schema after the operator receipt report surface/status checkpoint. This document does not implement a registry loader, read registry files, write registry files, create registry artifacts, execute tools, perform runtime behavior, grant runtime authority, enforce capabilities, perform effects, verify signatures, parse public keys, load key material, handle private keys, query trust stores, use networks, mutate host files, implement MCP behavior, control AI agents, claim malware prevention, claim ransomware prevention, or claim production security readiness.

Purpose

This contract defines the next Latticra Seal product slice: a future local capability registry schema.

The future schema should answer:

```
What local capability records may exist?
What fields must every capability record expose?
Which vocabulary values are allowed?
Which defaults keep every capability denied?
Which fields are descriptive only and cannot grant authority?
What evidence must exist before a future loader can be considered?
```

The answer must remain contract-only, local, no-effect, denied-by-default, and report-only.

Required Predecessors

A future implementation plan may only proceed after these predecessor surfaces remain present and guarded:

```
docs/latticra-seal/PRODUCT.md
docs/status/SEAL_PRODUCT_SPINE_STATUS.md
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md
docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md
docs/LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_SURFACE.md
docs/status/SEAL_CAPABILITY_METADATA_REPORT_SURFACE_STATUS.md
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md
docs/status/SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md
```

Schema Profile

The future schema must use this profile:

```
registry_schema_profile=latticra-seal-local-capability-registry-schema/0.1
registry_scope=local-only
registry_mode=report-only
registry_status=contract-only
```

Registry Schema Fields

The future registry schema record must expose bounded metadata fields for:

```
registry_schema_profile=
registry_format_version=
registry_scope=
registry_mode=
registry_status=
registry_contract_present=
registry_schema_planning_only=
registry_loader_implemented=
registry_file_loading_supported=
registry_network_loading_supported=
registry_signature_verification_supported=
registry_trust_store_supported=
registry_entry_count_max=
registry_entry_id_max=
registry_namespace_max=
registry_description_max=
default_action_deny=
runtime_authority_granted=
effect_performed=
host_read_performed=
host_write_performed=
network_performed=
```

Capability Entry Fields

Every future capability entry must expose bounded metadata fields for:

```
capability_id=
capability_namespace=
```

```
capability_name=
capability_description=
capability_scope=
capability_effect_class=
capability_authority_class=
capability_default_decision=
capability_requires_guarded_allowlist=
capability_requires_policy_decision=
capability_requires_runtime_gate=
capability_requires_runtime_dry_run=
capability_requires_operator_review=
capability_requires_verification_receipt=
capability_requires_signed_request=
capability_grants_authority=
capability_executes_tool=
capability_reads_host=
capability_writes_host=
capability_uses_network=
capability_report_only=
capability_deprecated=
capability_blocked_reason=
```

Required Defaults

The future schema implementation, if added later, must default to:

```
registry_contract_present=1
registry_schema_planning_only=1
registry_loader_implemented=0
registry_file_loading_supported=0
registry_network_loading_supported=0
registry_signature_verification_supported=0
registry_trust_store_supported=0
default_action_deny=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_default_decision=deny
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_requires_verification_receipt=1
capability_requires_signed_request=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
capability_report_only=1
```

Required Vocabulary

The future schema must restrict capability scopes to:

```
capability_scope=tool-boundary
capability_scope=request-boundary
capability_scope=policy-boundary
capability_scope=runtime-boundary
capability_scope=evidence-boundary
capability_scope=operator-review-boundary
```

The future schema must restrict effect classes to:

```
capability_effect_class=none
capability_effect_class=tool
capability_effect_class=host-read
capability_effect_class=host-write
capability_effect_class=network
capability_effect_class=runtime-authority
```

The future schema must restrict authority classes to:

```
capability_authority_class=none
capability_authority_class=descriptive-only
capability_authority_class=future-guarded
```

Any value other than none remains descriptive only until a separate implementation, verification, review, and authority contract exists.

Required Denial Rule

A future registry entry may describe a capability but must not authorize it.

Every future entry must preserve:

```
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
default_action_deny=1
runtime_authority_granted=0
effect_performed=0
```

Forbidden Behavior

This contract does not allow:

```
registry file loading
registry file writing
production capability registry loading
remote registry loading
trust-store lookup
signature verification
policy evaluation
policy enforcement
capability enforcement
runtime enforcement
tool execution
shell execution
AI-agent execution control
MCP protocol behavior
host reads
host writes
network behavior
runtime handoff execution
runtime authority grants
turning registry entries into execution grants
turning registry entries into effect grants
malware prevention claims
ransomware prevention claims
production security-product claims
```

Required Future Implementation Plan

A future implementation plan must specify:

```
exact header path
exact source path
exact test path
bounded struct sizes
enum labels
entry capacity
field copy behavior
default fixture entries
report shape
invalid schema handling
small-buffer behavior
null-input behavior
no-effect invariants
status/report-surface follow-up path
```

Promotion Rule

This contract does not authorize a registry loader.

The schema may be considered only as report-only metadata until a future implementation plan, implementation, report surface, status record, status-index alignment, public-entrypoint alignment, and negative-case evidence all remain merged and guarded.

Even after that sequence, a separate capability enforcement and runtime authority contract would be required before any registry entry could be used to authorize execution or effects.

Boundary

This is a contract-only planning slice.

It does not change implementation behavior, add runtime behavior, grant authority, or change public readiness.

Current Next Valid Slice

The local capability registry schema implementation plan is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md

Source title: Latticra Seal Local Capability Registry Schema Implementation

Lines: 199 | Words: 471 | SHA-256: 13e176fb205e3446c28456bda3de06ae286497d4f93a55642879540efb576673

Latticra Seal Local Capability Registry Schema Implementation

Status: no-effect implementation for the Latticra Seal local capability registry schema Scope: implementation record after the local capability registry schema contract and implementation plan. This implementation does not add a registry loader, read registry files, write registry files, create registry artifacts, execute tools, perform runtime behavior, grant runtime authority, enforce capabilities, perform effects, verify signatures, parse public keys, load key material, handle private keys, query trust stores, use networks, mutate host files, implement MCP behavior, control AI agents, claim malware prevention, claim ransomware prevention, or claim production security readiness.

Implemented Files

```
include/latticra/seal_local_capability_registry_schema.h
src/seal_local_capability_registry_schema.c
tests/seal_local_capability_registry_schema_invariants.c
scripts/test-latticra-seal-local-capability-registry-schema.sh
```

Implemented API

```
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_PROFILE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_FORMAT_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCOPE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_MODE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_STATUS_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_ENTRY_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_ENTRY_ID_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_NAMESPACE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_NAME_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_DESCRIPTION_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_BLOCKED_REASON_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_REPORT_MAX
latticra_seal_local_capability_registry_schema_error_t
latticra_seal_local_capability_registry_scope_t
latticra_seal_local_capability_registry_effect_class_t
latticra_seal_local_capability_registry_authority_class_t
latticra_seal_local_capability_registry_entry_t
latticra_seal_local_capability_registry_schema_t
latticra_seal_local_capability_registry_schema_error_label
latticra_seal_local_capability_registry_scope_label
latticra_seal_local_capability_registry_effect_class_label
latticra_seal_local_capability_registry_authority_class_label
latticra_seal_local_capability_registry_schema_init
latticra_seal_local_capability_registry_schema_add_entry
latticra_seal_local_capability_registry_schema_add_default_entry
latticra_seal_local_capability_registry_schema_validate
latticra_seal_local_capability_registry_schema_is_report_only
latticra_seal_local_capability_registry_schema_render
```

The API exposes no file-path input, registry-file reader, registry-file writer, network loader, trust-store loader, signature verifier, policy evaluator, runtime handoff function, or effect executor.

Initial Schema Output

The initialized schema preserves:

```
registry_schema_profile=latticra-seal-local-capability-registry-schema/0.1
registry_format_version=0.1
registry_scope=local-only
registry_mode=report-only
registry_status=contract-only
registry_contract_present=1
registry_schema_planning_only=1
registry_loader_implemented=0
registry_file_loading_supported=0
registry_network_loading_supported=0
registry_signature_verification_supported=0
registry_trust_store_supported=0
registry_entry_count=0
registry_entry_count_max=16
registry_entry_id_max=64
registry_namespace_max=64
registry_name_max=96
registry_description_max=256
registry_blocked_reason_max=160
default_action_deny=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
```

Default Fixture Entry

The default fixture entry emits:

```
capability_id=seal.local.registry.schema
capability_namespace=seal.local
capability_name=seal.local.registry.schema
capability_description=Describes local Seal capability schema metadata without loading or
granting capabilities.
capability_scope=operator-review-boundary
capability_effect_class=none
capability_authority_class=descriptive-only
```

```
capability_default_decision=deny
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_requires_verification_receipt=1
capability_requires_signed_request=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
capability_report_only=1
capability_deprecated=0
capability_blocked_reason=registry-schema-is-descriptive-only
```

Report Shape

The renderer emits deterministic local metadata:

```
LATTICRA SEAL LOCAL CAPABILITY REGISTRY SCHEMA REPORT
registry_schema_profile=latticra-seal-local-capability-registry-schema/0.1
registry_format_version=0.1
registry_scope=local-only
registry_mode=report-only
registry_status=contract-only
registry_contract_present=1
registry_schema_planning_only=1
registry_loader_implemented=0
registry_file_loading_supported=0
registry_network_loading_supported=0
registry_signature_verification_supported=0
registry_trust_store_supported=0
registry_entry_count=1
registry_entry_count_max=16
default_action_deny=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_id=seal.local.registry.schema
capability_namespace=seal.local
capability_name=seal.local.registry.schema
capability_scope=operator-review-boundary
capability_effect_class=none
capability_authority_class=descriptive-only
capability_default_decision=deny
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
capability_report_only=1
capability_blocked_reason=registry-schema-is-descriptive-only
error=ok
```

Failure Behavior

The implementation fails closed for:

```
null output
null entry input
small render buffer
entry capacity overflow
invalid scope
invalid effect class
invalid authority class
allow default decision
entry authority grant
entry tool execution
entry host read
entry host write
entry network use
non-report-only entry
```

Invalid entries are not added to the schema. Invalid input preserves:

```
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Boundary

This is a no-effect local metadata implementation.

It does not authorize a registry loader, does not load a registry file, does not read a registry path, does not validate signatures, does not query a trust store, does not evaluate policy, does not enforce capabilities, and does not perform effects.

Current Next Valid Slice

The local capability registry schema implementation plan is represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The next valid Latticra Seal slice is a deterministic local report surface for the local capability registry schema or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Local Capability Registry Schema Implementation Plan

Lines: 375 | Words: 840 | SHA-256: 4a951c3c8ab4302a5e70f29ec0bc977d831fcbd6b2ce80e1e806fad8ce182fec

Latticra Seal Local Capability Registry Schema Implementation Plan

Status: implementation plan for a future no-effect Latticra Seal local capability registry schema Scope: implementation planning after the local capability registry schema contract. This document does not implement a registry loader, read registry files, write registry files, create registry artifacts, execute tools, perform runtime behavior, grant runtime authority, enforce capabilities, perform effects, verify signatures, parse public keys, load key material, handle private keys, query trust stores, use networks, mutate host files, implement MCP behavior, control AI agents, claim malware prevention, claim ransomware prevention, or claim production security readiness.

Purpose

This plan defines the exact future implementation shape for a no-effect local capability registry schema.

The future implementation should make capability records inspectable as bounded local metadata while preserving denial, non-execution, and zero runtime authority.

Required Contract

The implementation plan depends on:

```
docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md
scripts/test-latticra-seal-local-capability-registry-schema-contract.sh
```

The contract must remain merged and guarded before implementation code is added.

Planned Files

A future implementation slice may add:

```
include/latticra/seal_local_capability_registry_schema.h
src/seal_local_capability_registry_schema.c
tests/seal_local_capability_registry_schema_invariants.c
scripts/test-latticra-seal-local-capability-registry-schema.sh
docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md
```

A later report-surface slice may add:

```
tests/seal_local_capability_registry_schema_surface.c
scripts/latticra-seal-local-capability-registry-schema.sh
scripts/test-latticra-seal-local-capability-registry-schema-surface.sh
docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_SURFACE.md
docs/status/SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_STATUS.md
```

Header API Plan

The header should define:

```
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_PROFILE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_FORMAT_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCOPE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_MODE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_STATUS_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_ENTRY_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_ENTRY_ID_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_NAMESPACE_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_NAME_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_DESCRIPTION_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_BLOCKED_REASON_MAX
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_REPORT_MAX
latticra_seal_local_capability_registry_schema_error_t
latticra_seal_local_capability_registry_scope_t
latticra_seal_local_capability_registry_effect_class_t
latticra_seal_local_capability_registry_authority_class_t
latticra_seal_local_capability_registry_entry_t
latticra_seal_local_capability_registry_schema_t
latticra_seal_local_capability_registry_schema_error_label
latticra_seal_local_capability_registry_schema_init
latticra_seal_local_capability_registry_schema_add_entry
latticra_seal_local_capability_registry_schema_add_default_entry
latticra_seal_local_capability_registry_schema_validate
latticra_seal_local_capability_registry_schema_is_report_only
latticra_seal_local_capability_registry_schema_render
```

The API must not include a path-loading function, registry-file reader, registry-file writer, network loader, trust-store loader, signature verifier, policy evaluator, runtime handoff function, or effect executor.

Capacity Plan

The future implementation should use bounded fixed storage:

```
registry_entry_count_max=16
registry_entry_id_max=64
registry_namespace_max=64
registry_name_max=96
registry_description_max=256
registry_blocked_reason_max=160
```

No dynamic allocation is required for the initial implementation.

Error Model

The error enum should include:

```
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_OK
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_INVALID_INPUT
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_INVALID_PROFILE
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_INVALID_SCOPE
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_INVALID_EFFECT_CLASS
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_INVALID_AUTHORITY_CLASS
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_INVALID_DEFAULT_DECISION
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_ENTRY_CAPACITY_EXCEEDED
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_ENTRY_WOULD_GRANT_AUTHORITY
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_ENTRY_WOULD_PERFORM_EFFECT
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_NON_REPORT_ONLY_ENTRY
LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_BUFFER_TOO_SMALL
```

Registry Record Fields

The registry record should include:

```
registry_schema_profile
registry_format_version
registry_scope
registry_mode
registry_status
registry_contract_present
registry_schema_planning_only
registry_loader_implemented
registry_file_loading_supported
registry_network_loading_supported
registry_signature_verification_supported
registry_trust_store_supported
registry_entry_count
registry_entry_count_max
registry_entry_id_max
registry_namespace_max
registry_name_max
registry_description_max
registry_blocked_reason_max
default_action_deny
runtime_authority_granted
effect_performed
host_read_performed
host_write_performed
network_performed
last_error
```

Capability Entry Record Fields

Every entry should include:

```
capability_id
capability_namespace
capability_name
capability_description
capability_scope
capability_effect_class
capability_authority_class
capability_default_decision
capability_requires_guarded_allowlist
capability_requires_policy_decision
capability_requires_runtime_gate
capability_requires_runtime_dry_run
capability_requires_operator_review
capability_requires_verification_receipt
capability_requires_signed_request
capability_grants_authority
capability_executes_tool
capability_reads_host
capability_writes_host
capability_uses_network
capability_report_only
capability_deprecated
capability_blocked_reason
```

Initial Constants

The implementation should emit for an initialized schema:

```
registry_schema_profile=latticra-seal-local-capability-registry-schema/0.1
registry_format_version=0.1
registry_scope=local-only
registry_mode=report-only
registry_status=contract-only
registry_contract_present=1
registry_schema_planning_only=1
registry_loader_implemented=0
registry_file_loading_supported=0
registry_network_loading_supported=0
registry_signature_verification_supported=0
registry_trust_store_supported=0
registry_entry_count=0
registry_entry_count_max=16
registry_entry_id_max=64
registry_namespace_max=64
```

```
registry_name_max=96
registry_description_max=256
registry_blocked_reason_max=160
default_action_deny=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

The default fixture entry should emit:

```
capability_id=seal.local.registry.schema
capability_namespace=seal.local
capability_name=seal.local.registry.schema
capability_description=Describes local Seal capability schema metadata without loading or
granting capabilities.
capability_scope=operator-review-boundary
capability_effect_class=none
capability_authority_class=descriptive-only
capability_default_decision=deny
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_requires_verification_receipt=1
capability_requires_signed_request=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
capability_report_only=1
capability_deprecated=0
capability_blocked_reason=registry-schema-is-descriptive-only
```

Function Behavior

The builder should:

```
initialize a caller-provided registry record
add caller-provided static entries only
add one default fixture entry
validate bounded vocabulary values
validate deny-by-default entry posture
reject null registry output
reject null entry input
reject entry capacity overflow
reject malformed vocabulary values
reject unsupported default decisions
reject any capability entry that would grant authority
reject any capability entry that would perform an effect
reject any non-report-only capability entry
render deterministic metadata only
return BUFFER_TOO_SMALL for insufficient render buffers
```

The implementation must not expose a file-path input API.

Report Shape

The render function should produce:

```
LATTICRA SEAL LOCAL CAPABILITY REGISTRY SCHEMA REPORT
registry_schema_profile=latticra-seal-local-capability-registry-schema/0.1
registry_format_version=0.1
registry_scope=local-only
registry_mode=report-only
registry_status=contract-only
registry_contract_present=1
registry_schema_planning_only=1
registry_loader_implemented=0
registry_file_loading_supported=0
registry_network_loading_supported=0
registry_signature_verification_supported=0
registry_trust_store_supported=0
registry_entry_count=1
registry_entry_count_max=16
default_action_deny=1
```

```
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_id=seal.local.registry.schema
capability_namespace=seal.local
capability_name=seal.local.registry.schema
capability_scope=operator-review-boundary
capability_effect_class=none
capability_authority_class=descriptive-only
capability_default_decision=deny
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_requires_verification_receipt=1
capability_requires_signed_request=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
capability_report_only=1
capability_deprecated=0
capability_blocked_reason=registry-schema-is-descriptive-only
error=ok
```

Test Plan

The invariant test should cover:

```
default schema initializes as report-only
default fixture entry remains denied
render output includes the report header
null output is rejected
null entry input is rejected
small render buffer is rejected
entry capacity overflow is rejected
invalid scope is rejected
invalid effect class is rejected
invalid authority class is rejected
allow default decision is rejected
entry authority grant is rejected
entry tool execution is rejected
entry host read is rejected
entry host write is rejected
entry network use is rejected
non-report-only entry is rejected
no-effect counters remain zero on invalid input
```

Forbidden Behavior

The implementation must not add:

```
registry loader
registry file loading
registry file writing
remote registry loading
trust-store lookup
signature verification
policy evaluation
policy enforcement
capability enforcement
runtime enforcement
tool execution
shell execution
AI-agent execution control
MCP protocol behavior
host reads
host writes
network behavior
runtime handoff execution
runtime authority grants
turning registry entries into execution grants
turning registry entries into effect grants
malware prevention claims
ransomware prevention claims
```

```
production security-product claims
```

Current Next Valid Slice

The local capability registry schema contract is represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md

Source title: Latticra Seal MCP Alignment Plan

Lines: 217 | Words: 678 | SHA-256: 090a951b4583ad8a782ba094be0f90d39fa751464cce8d11f57467cdccc6fb2f

Latticra Seal MCP Alignment Plan

Status: Latticra Seal MCP alignment planning document Scope: documentation-only alignment plan for model-context style tool invocation security after the Latticra Seal agentic automation security contract. This document does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, or operating-system behavior.

Purpose

This document records how Latticra Seal should align with model-context style automation security concerns while remaining evidence-bound and no-effect.

The purpose is not to implement MCP.

The purpose is to define a future-safe vocabulary and direction for protecting local tool invocation, automation, and AI-adjacent workflows through Seal metadata, policy, receipts, and runtime-boundary discipline.

Alignment thesis

Model-context style automation creates a new systems boundary:

```
model or assistant intent -&gt; tool request -&gt; parameter transfer -&gt; local action -&gt;
output return
 -&gt; possible downstream action
```

Latticra Seal should treat every boundary crossing as untrusted until measured, declared, constrained, and reported.

Seal mapping

The existing Seal chain maps naturally to agentic automation security:

```
Seal report -&gt; visibility surface
Seal measurement -&gt; identity and evidence posture
Seal manifest -&gt; declared tool identity and capability scope
Seal signature policy -&gt; rules for signed and unsigned objects
Seal signature metadata -&gt; signature presence and declared signature state
Seal verification policy -&gt; rules for accepting or rejecting trust material
Seal verification receipt -&gt; reportable verification result
```

```
Seal capability gate -&gt; denied-by-default authority boundary
Seal effect decision -&gt; explicit action decision
Seal runtime handoff -&gt; disabled runtime boundary
Seal status rollup -&gt; compact posture summary
Agentic automation security -&gt; tool invocation and context boundary planning
```

Planned MCP-adjacent fields

Future MCP-adjacent records should remain bounded and deterministic.

Planned fields:

```
mcp_alignment_profile
protocol_implementation_supported
server_behavior_supported
client_behavior_supported
tool_registry_present
tool_manifest_required
tool_manifest_signed_required
parameter_schema_required
parameter_schema_valid
request_identity_required
request_timestamp_required
request_expiration_required
nonce_required
context_hash_required
parameter_hash_required
message_signature_required
replay_protection_required
output_treated_as_untrusted
downstream_execution_allowed
operator_approval_required
receipt_required
runtime_authority_granted
mode
status
```

Initial values:

```
mcp_alignment_profile=latticra-seal-mcp-alignment/0.1
protocol_implementation_supported=0
server_behavior_supported=0
client_behavior_supported=0
tool_registry_present=0
tool_manifest_required=1
tool_manifest_signed_required=1
parameter_schema_required=1
parameter_schema_valid=0
request_identity_required=1
request_timestamp_required=1
request_expiration_required=1
nonce_required=1
context_hash_required=1
parameter_hash_required=1
message_signature_required=1
replay_protection_required=1
output_treated_as_untrusted=1
downstream_execution_allowed=0
operator_approval_required=1
receipt_required=1
runtime_authority_granted=0
mode=planning-only
status=mcp-alignment-planning-only
```

Trust boundary rules

Latticra Seal should preserve these rules for future implementation:

```
model output is not authority
tool output is not authority
retrieved context is not authority
unsigned manifests do not grant authority
unknown tools do not grant authority
parameters must not cross boundaries without validation
outputs must not become downstream inputs without policy
network access must remain denied unless explicitly granted
private key access must remain denied unless explicitly granted
runtime authority must remain zero until guarded implementation exists
```

Parameter validation direction

Future Seal records should be able to report:

```
parameter_schema_present
parameter_schema_valid
max_input_bytes_declared
input_size_within_limit
source_context_known
parameter_hash_present
parameter_forwarding_allowed
```

The initial default is:

```
parameter_forwarding_allowed=0
```

Freshness and replay direction

Future Seal records should be able to report:

```
request_timestamp_present
request_expiration_present
nonce_present
freshness_valid
replay_detected
```

The initial default is:

```
freshness_valid=0
replay_detected=0
```

Output filtering direction

Future Seal records should treat returned tool/model output as untrusted until explicitly classified.

Initial defaults:

```
output_treated_as_untrusted=1
downstream_execution_allowed=0
chain_continuation_requires_policy=1
```

Fedora/Linux composition direction

Latticra Seal should compose with Linux security instead of replacing it.

Future composition targets may include:

```
SELinux
systemd service boundaries
seccomp
read-only directories
temporary runtime directories
user namespaces
network namespace isolation
```

This plan does not implement any of these mechanisms.

Non-claims

This alignment plan must not be read as deployment, certification, endorsement, production hardening, or protocol implementation.

Required non-claims:

```
mcp_protocol_implemented=0
mcp_server_implemented=0
mcp_client_implemented=0
external_endorsement_claimed=0
NSA_endorsement_claimed=0
Fedora_approval_claimed=0
production_readiness_claimed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Next valid slice

The next valid Latticra Seal slice is a report-only agentic automation security metadata implementation.

That future slice may add bounded C metadata and deterministic report rendering.

It must not implement MCP behavior, runtime behavior, host behavior, network behavior, AI agent execution, model execution, tool execution, capability enforcement, cryptographic verification, or authority grants.

docs/LATTICRA_SEAL_MEASUREMENT_CONTRACT.md

Source title: Latticra Seal Measurement Contract

Lines: 165 | Words: 470 | SHA-256: 37c47fd58140c92249362ed9482a8372e4d09a39e50605443fdd0ce980bffd54

Latticra Seal Measurement Contract

Status: Latticra Seal measurement contract Scope: contract for future read-only artifact measurement before measurement implementation, manifest generation, signing, encryption, key handling, runtime authority, host writes, network behavior, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the next Latticra Seal layer: read-only measurement.

Measurement means computing deterministic digest metadata for explicitly supplied artifacts so Latticra can later bind reports, contracts, packages, and release evidence to stable identifiers.

This document does not implement measurement.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md
include/latticra/seal_report.h
src/seal_report.c
scripts/test-latticra-seal-report.sh
```

The Seal report must remain the source of truth for the current no-effect posture until measurement code is implemented and guarded.

Measurement boundary

The first measurement implementation may only read an explicitly supplied regular file path from the caller.

Allowed future behavior:

```
open caller-supplied file for read-only access
read file bytes in bounded chunks
compute SHA-256 digest
record byte length
record measurement status
render deterministic report metadata
```

Forbidden first measurement behavior:

```
file writes
directory traversal by default
recursive hashing
network access
key generation
```

```
signing
encryption
manifest persistence
runtime authority grants
capability enforcement
kernel interaction
package installation
host configuration changes
```

Approved initial digest policy

The first official measurement digest is:

```
SHA-256
```

SHA-384 and SHA-512 may be planned later for wider digest profiles.

BLAKE3 may be planned later only for fast internal content addressing, not compliance claims.

No custom hash function may be introduced.

Planned API shape

The first measurement implementation should use bounded C records.

Proposed files:

```
include/latticra/seal_measurement.h
src/seal_measurement.c
tests/seal_measurement_invariants.c
scripts/test-latticra-seal-measurement.sh
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
```

Proposed record fields:

```
measurement_profile
algorithm
artifact_label
artifact_size_bytes
digest_hex
read_performed
write_performed
network_performed
runtime_granted
status
error_label
```

Report integration

After implementation, the Seal report may advance:

```
measurement_supported=1
```

Only after the measurement implementation and invariant tests are present.

Until then, the report must keep:

```
measurement_supported=0
```

Failure behavior

The future measurement implementation must fail closed.

Required failure states:

```
null input -&gt; invalid
empty path -&gt; invalid
non-regular path -&gt; invalid
open failure -&gt; read unavailable
read failure -&gt; read failed
output buffer too small -&gt; bounded failure
unsupported algorithm -&gt; invalid
```

Failures must not create files, modify files, contact networks, grant runtime authority, or persist manifests.

Invariant requirements

The first measurement tests must verify:

```
known fixture produces expected SHA-256 digest
regular file read is recorded
write remains zero
network remains zero
runtime grant remains zero
null input fails
empty path fails
small report buffer fails closed
unsupported path fails without mutation
Seal report only changes measurement_supported after implementation exists
```

Promotion rule

This contract permits only the next implementation slice:

```
read-only SHA-256 measurement implementation
```

It does not permit signing, encryption, key handling, object sealing, capability enforcement, or runtime authority.

docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md

Source title: Latticra Seal Measurement Implementation

Lines: 145 | Words: 406 | SHA-256: b35d6df900500faebea5ec1ce0341ab3ce5ae406ec9e83a4d33c353ea7367c09

Latticra Seal Measurement Implementation

Status: initial read-only SHA-256 measurement implementation Scope: bounded C measurement surface for caller-supplied regular files. This slice does not add signing, encryption, key handling, object sealing, capability enforcement, runtime authority, host writes, network behavior, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal measurement implementation.

The implementation adds read-only SHA-256 artifact measurement so Latticra can identify explicitly supplied artifacts with deterministic digest metadata.

Added files

```
include/latticra/seal_measurement.h
src/seal_measurement.c
tests/seal_measurement_invariants.c
scripts/test-latticra-seal-measurement.sh
```

This slice also advances the Seal report surface so it can truthfully state:

```
measurement_supported=1
```

API summary

The measurement surface adds:

```
latticra_seal_measurement_t
latticra_seal_measurement_error_t
latticra_seal_measurement_error_label
latticra_seal_measure_file
latticra_seal_measurement_is_read_only
latticra_seal_measurement_report
```

Measurement behavior

The implementation:

```
accepts a caller-supplied path
```

```
requires a regular file
opens the file in binary read-only mode
reads bytes in bounded chunks
computes SHA-256
records byte length
renders deterministic report metadata
```

The implementation does not recurse, scan directories, create manifests, sign data, encrypt data, write files, contact networks, or grant runtime authority.

Default report posture after this slice

The Seal report now states:

```
seal_profile=latticra-seal/0.1-report
report_only_supported=1
measurement_supported=1
signing_supported=0
capability_gate_supported=0
sealed_objects_supported=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_granted=0
evidence_level=3
status=report-and-measurement
```

The report itself remains no-effect. Measurement reads only when the measurement API is explicitly called.

Measurement record posture

A successful measurement records:

```
measurement_profile=latticra-seal-measurement/0.1
algorithm=SHA-256
artifact_label=&lt;caller supplied path&gt;
artifact_size_bytes=&lt;read byte count&gt;
digest_hex=&lt;64 lowercase hex characters&gt;
read_performed=1
write_performed=0
network_performed=0
runtime_granted=0
error=ok
status=measured
```

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null path -&gt; invalid-input
empty path -&gt; invalid-input
non-regular path -&gt; not-regular
open failure -&gt; open-failed
read failure -&gt; read-failed
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not write files, contact networks, persist manifests, or grant runtime authority.

Invariants

The invariant test verifies:

```
known fixture produces expected SHA-256 digest
regular file read is recorded
write remains zero
network remains zero
runtime grant remains zero
rendered measurement report contains required labels
null path fails closed
empty path fails closed
non-regular path fails closed
missing file fails closed
small report buffer fails closed
```

```
null report inputs fail closed
Seal report advertises measurement support only after implementation exists
```

Validation

Run:

```
sh scripts/test-latticra-seal-measurement.sh
sh scripts/test-latticra-seal-measurement-contract.sh
```

Next valid slice

The next valid Latticra Seal slice is a signed evidence-manifest contract.

That future slice must be contract-first and must not be added directly to this measurement implementation.

docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_CONTRACT.md

Source title: Latticra Seal Operator Receipt Report Contract

Lines: 253 | Words: 702 | SHA-256: 8e355e23887cd8d41041d268ba5bd93607ef092ccef7a509452a415ee0e9bb66

Latticra Seal Operator Receipt Report Contract

Status: Latticra Seal operator receipt report contract Scope: contract-only planning for a future bundled local operator receipt report after the Seal product spine status checkpoint. This document does not implement a receipt report, write receipt files, create artifacts, execute tools, perform runtime behavior, grant runtime authority, enforce capabilities, perform effects, verify signatures, parse public keys, load key material, handle private keys, query trust stores, use networks, read host files beyond future caller-provided metadata, mutate host files, implement MCP behavior, control AI agents, claim malware prevention, claim ransomware prevention, or claim production security readiness.

Purpose

This contract defines the next Latticra Seal product slice: a future operator-visible receipt report that bundles the existing report-only Seal metadata chain into one local, inspectable decision artifact.

The future receipt report should answer:

```
What capability was requested?
Was the capability known only as a candidate?
What did policy decide?
Was freshness metadata present?
Was signed-request metadata present?
What would the runtime dry-run do?
Why is the request still denied?
Did anything execute?
Was runtime authority granted?
```

Required predecessors

The future implementation may only proceed after these predecessor surfaces remain present and guarded:

```
docs/latticra-seal/PRODUCT.md
docs/status/SEAL_PRODUCT_SPINE_STATUS.md
docs/LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_SURFACE.md
docs/status/SEAL_CAPABILITY_METADATA_REPORT_SURFACE_STATUS.md
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
docs/status/SEAL_POLICY_DECISION_STATUS.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md
docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNED_REQUEST_STATUS.md
```

```
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md
docs/status/SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md
```

Future implementation name

```
Latticra Seal operator receipt report
```

Future files

The future implementation should use a bounded C surface and deterministic shell guards:

```
include/latticra/seal_operator_receipt_report.h
src/seal_operator_receipt_report.c
tests/seal_operator_receipt_report_invariants.c
tests/seal_operator_receipt_report_surface.c
scripts/test-latticra-seal-operator-receipt-report.sh
scripts/latticra-seal-operator-receipt-report.sh
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md
docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md
```

Receipt profile

The future report must use this profile:

```
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
```

Required source bindings

The future receipt report must bind source metadata explicitly:

```
source_capability_metadata_present=1
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
```

If any required source metadata is absent, malformed, or not report-only, the receipt must fail closed:

```
receipt_complete=0
receipt_invalid=1
would_allow=0
would_deny=1
runtime_authority_granted=0
```

Required report fields

The future report must render deterministic fields for:

```
operator_receipt_profile=
receipt_mode=
receipt_status=
source_capability_metadata_present=
source_policy_decision_present=
source_request_freshness_present=
source_signed_request_present=
source_runtime_dry_run_present=
source_denial_reason_present=
capability_name=
capability_known=
capability_candidate=
policy_decision_state=
request_freshness_state=
signed_request_state=
runtime_dry_run_state=
blocked_reason=
default_action_deny=
would_allow=
would_deny=
would_require_operator_review=
would_execute_tool=
would_read_host=
would_write_host=
would_use_network=
```

```
would_grant_runtime_authority=
receipt_complete=
receipt_invalid=
report_only=
runtime_authority_granted=
effect_performed=
host_read_performed=
host_write_performed=
network_performed=
```

Required denied posture

The default operator receipt report must preserve:

```
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Failure behavior

The future implementation must fail closed when:

```
capability metadata is missing
policy decision metadata is missing
request freshness metadata is missing
signed request metadata is missing
runtime dry-run metadata is missing
any source metadata is not report-only
any source metadata would allow an effect
the output buffer is null
the output buffer is too small
```

Failure must produce no effects, no host reads, no host writes, no network behavior, no runtime handoff, no tool execution, and no runtime authority.

Non-goals

This contract does not allow:

```
runtime enforcement
capability enforcement
tool execution
AI-agent execution control
MCP protocol behavior
signature verification
signing
key parsing
key material loading
private-key handling
trust-store loading
revocation lookup
nonce storage
replay-cache mutation
host mutation
network behavior
kernel behavior
production security-product claims
malware prevention claims
ransomware prevention claims
```

Validation requirements

The future implementation guard must verify:

```
all required predecessor docs and status records exist
the receipt profile is stable
```

```
all source bindings are rendered
missing source metadata fails closed
non-report-only source metadata fails closed
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
runtime_authority_granted=0
small output buffer fails closed
```

Current guard

This contract is covered by:

```
sh scripts/test-latticra-seal-operator-receipt-report-contract.sh
```

Expected output:

```
latticra seal operator receipt report contract: ok
```

Current next valid slice

The no-effect operator receipt report implementation, report surface, and status record are now current follow-up checkpoints.

The operator receipt report surface/status checkpoint is now represented by docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md and docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md.

The local capability registry schema contract is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md.

The local capability registry schema implementation plan is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md

Source title: Latticra Seal Operator Receipt Report Implementation

Lines: 171 | Words: 476 | SHA-256: de08d317ece8d9f53f4125b29805af66ef062940ae0b96ac8f243e932b9d6553

Latticra Seal Operator Receipt Report Implementation

Status: no-effect Latticra Seal operator receipt report implementation Scope: bounded C metadata implementation for composing existing report-only Seal source records into one deterministic operator receipt. This implementation does not write receipt files, create persistent artifacts, execute tools, perform runtime behavior, grant runtime authority, enforce capabilities, perform effects, verify signatures, parse public keys, load key material, handle private keys, query trust stores, use networks, mutate host files, implement MCP behavior, control AI agents, claim malware prevention, claim ransomware prevention, or claim production security readiness.

Purpose

The operator receipt report makes the current Seal denial chain easier to inspect.

It combines these caller-provided in-memory records:

```
capability metadata
```

```
policy decision metadata
request freshness metadata
signed request metadata
runtime dry-run metadata
```

The result is still denied, report-only, and no-effect. It is an operator-facing evidence bundle, not an enforcement path.

Files

```
include/latticra/seal_operator_receipt_report.h
src/seal_operator_receipt_report.c
tests/seal_operator_receipt_report_invariants.c
scripts/test-latticra-seal-operator-receipt-report.sh
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md
```

Public API

The implementation adds:

```
LATTICRA_SEAL_OPERATOR_RECEIPT_PROFILE_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_STATE_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_REASON_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_CAPABILITY_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MAX
latticra_seal_operator_receipt_report_error_t
latticra_seal_operator_receipt_report_sources_t
latticra_seal_operator_receipt_report_t
latticra_seal_operator_receipt_report_error_label
latticra_seal_operator_receipt_report_from_sources
latticra_seal_operator_receipt_report_is_report_only
latticra_seal_operator_receipt_report_render
```

Error Model

```
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_OK
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_INVALID_INPUT
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_CAPABILITY_METADATA
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_POLICY_DECISION
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_REQUEST_FRESHNESS
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_SIGNED_REQUEST
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_RUNTIME_DRY_RUN
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_NON_REPORT_ONLY_SOURCE
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SOURCE_WOULD_ALLOW_EFFECT
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_BUFFER_TOO_SMALL
```

Invalid source metadata is represented as denied metadata inside the receipt record. A null output pointer is still a hard LATTICRA_STATUS_NULL_ARGUMENT API error. A too-small render buffer returns LATTICRA_STATUS_BUFFER_TOO_SMALL and clears the caller-provided buffer when possible.

Complete Denied Receipt

A complete report-only receipt renders:

```
LATTICRA SEAL OPERATOR RECEIPT REPORT
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
source_capability_metadata_present=1
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
capability_name=seal.capability.report
capability_known=1
capability_candidate=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
runtime_dry_run_state=report-only
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
```

```
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
blocked_reason=known-capability-candidate-still-denied
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
```

Fail-Closed Behavior

Missing, malformed, non-report-only, or effect-allowing source metadata produces:

```
receipt_status=invalid-source-denied
receipt_complete=0
receipt_invalid=1
default_action_deny=1
would_allow=0
would_deny=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

The implementation copies descriptive state only. It never copies authority, effect, host I/O, or network behavior from upstream sources.

Validation

Current guard:

```
sh scripts/test-latticra-seal-operator-receipt-report.sh
```

Expected output:

```
seal operator receipt report invariants: ok
latticra seal operator receipt report: ok
```

The guard compiles the new receipt implementation with the deterministic capability metadata fixture and verifies complete receipt rendering, missing source denial, non-report-only source denial, effect-allowing source denial, null handling, and small-buffer handling.

Current Next Valid Slice

The operator receipt report surface/status checkpoint is now represented by docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md and docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md.

The local capability registry schema contract is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md.

The local capability registry schema implementation plan is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Operator Receipt Report Implementation Plan

Lines: 390 | Words: 889 | SHA-256: f0e2b42a41718b81a28a1ee37f4250d50c5cb3a9b5e86429343ab905e762d75f

Latticra Seal Operator Receipt Report Implementation Plan

Status: implementation planning contract fulfilled by the no-effect Latticra Seal operator receipt report implementation Scope: implementation plan record after the Latticra Seal operator receipt report contract. This document does not itself write receipt files, create artifacts, execute tools, perform runtime behavior, grant runtime authority, enforce capabilities, perform effects, verify signatures, parse public keys, load key material, handle private keys, query trust stores, use networks, read host files beyond future caller-provided metadata, mutate host files, implement MCP behavior, control AI agents, claim malware prevention, claim ransomware prevention, or claim production security readiness.

Purpose

This plan defined the exact implementation shape for a no-effect Seal operator receipt report.

The implementation combines existing report-only Seal metadata into one deterministic operator-facing receipt record while preserving denial, non-execution, and zero runtime authority.

Required contract

The implementation plan depends on:

```
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_CONTRACT.md
scripts/test-latticra-seal-operator-receipt-report-contract.sh
```

The contract must remain merged and guarded before implementation code is added.

Implemented files

The implementation slice added:

```
include/latticra/seal_operator_receipt_report.h
src/seal_operator_receipt_report.c
tests/seal_operator_receipt_report_invariants.c
scripts/test-latticra-seal-operator-receipt-report.sh
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md
```

A later report-surface slice may add:

```
tests/seal_operator_receipt_report_surface.c
scripts/latticra-seal-operator-receipt-report.sh
scripts/test-latticra-seal-operator-receipt-report-surface.sh
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md
docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md
```

Header API plan

The header defines:

```
LATTICRA_SEAL_OPERATOR_RECEIPT_PROFILE_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_STATE_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_REASON_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_CAPABILITY_MAX
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MAX
latticra_seal_operator_receipt_report_error_t
latticra_seal_operator_receipt_report_sources_t
latticra_seal_operator_receipt_report_t
latticra_seal_operator_receipt_report_error_label
latticra_seal_operator_receipt_report_from_sources
latticra_seal_operator_receipt_report_is_report_only
latticra_seal_operator_receipt_report_render
```

Error model

The error enum includes:

```
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_OK
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_INVALID_INPUT
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_CAPABILITY_METADATA
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_POLICY_DECISION
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_REQUEST_FRESHNESS
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_SIGNED_REQUEST
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_MISSING_RUNTIME_DRY_RUN
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_NON_REPORT_ONLY_SOURCE
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SOURCE_WOULD_ALLOW_EFFECT
LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_BUFFER_TOO_SMALL
```

Source binding record

The source binding record includes:

```
source_capability_metadata_present
source_policy_decision_present
source_request_freshness_present
source_signed_request_present
source_runtime_dry_run_present
source_denial_reason_present
source_capability_metadata_report_only
source_policy_decision_report_only
source_request_freshness_report_only
source_signed_request_report_only
source_runtime_dry_run_report_only
```

Receipt record fields

The receipt record includes:

```
operator_receipt_profile
receipt_mode
receipt_status
source_capability_metadata_present
source_policy_decision_present
source_request_freshness_present
source_signed_request_present
source_runtime_dry_run_present
source_denial_reason_present
capability_name
capability_known
capability_candidate
policy_decision_state
request_freshness_state
signed_request_state
runtime_dry_run_state
blocked_reason
default_action_deny
would_allow
would_deny
would_require_operator_review
would_execute_tool
would_read_host
would_write_host
would_use_network
would_grant_runtime_authority
receipt_complete
receipt_invalid
report_only
runtime_authority_granted
effect_performed
```

```
host_read_performed
host_write_performed
network_performed
error
```

Initial constants

The implementation emits for a complete denied receipt:

```
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
source_capability_metadata_present=1
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
capability_name=seal.capability.report
capability_known=1
capability_candidate=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
runtime_dry_run_state=report-only
blocked_reason=known-capability-candidate-still-denied
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

The implementation emits for missing or invalid source metadata:

```
receipt_status=invalid-source-denied
receipt_complete=0
receipt_invalid=1
default_action_deny=1
would_allow=0
would_deny=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Function behavior

The builder:

```
accept report-only capability metadata input
accept report-only policy decision input
accept report-only request freshness input
accept report-only signed request input
accept report-only runtime dry-run input
reject null output
reject missing source metadata
reject non-report-only source metadata
reject source metadata that would allow an effect
copy descriptive capability and decision fields only
copy no authority from upstream inputs
render deterministic metadata only
```

Required report format

The report should begin with:

```
LATTICRA SEAL OPERATOR RECEIPT REPORT
```

It should render all fields as stable key=value lines.

Required report fields:

```
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
source_capability_metadata_present=1
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
capability_name=seal.capability.report
capability_known=1
capability_candidate=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
runtime_dry_run_state=report-only
blocked_reason=known-capability-candidate-still-denied
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Required invariant tests

The invariant test verifies:

```
valid report-only inputs produce a complete denied receipt
operator_receipt_profile is stable
receipt_mode remains report-only
receipt_status remains denied-report-only
all source present flags remain one for complete receipt
capability_name is copied descriptively
capability_known remains one
capability_candidate remains one
policy_decision_state remains report-only
request_freshness_state remains report-only
signed_request_state remains report-only
runtime_dry_run_state remains report-only
blocked_reason remains known-capability-candidate-still-denied
default_action_deny remains one
would_allow remains zero
would_deny remains one
would_require_operator_review remains one
would_execute_tool remains zero
would_read_host remains zero
would_write_host remains zero
would_use_network remains zero
would_grant_runtime_authority remains zero
receipt_complete remains one for complete denied receipt
receipt_invalid remains zero for complete denied receipt
missing source metadata fails closed
non-report-only source metadata fails closed
source metadata that would allow an effect fails closed
small report buffers fail closed
null inputs fail closed
```

Required build runner

The test runner compiles only local C sources and fixtures.

Runner:

```
sh scripts/test-latticra-seal-operator-receipt-report.sh
```

It compiles:

```
src/seal_operator_receipt_report.c
src/seal_capability_metadata.c
tests/seal_operator_receipt_report_invariants.c
```

Explicit non-goals

The implementation must not:

```
write receipt files
create persistent receipt artifacts
execute tools
execute shell commands
control AI agents
implement MCP behavior
read host files
write host files
use the network
evaluate external policies
load policy files
verify signatures
sign receipts
parse keys
load key material
handle private keys
query trust stores
query revocation sources
validate freshness against live time
mutate replay caches
grant authority
perform runtime handoff
perform effects
claim malware prevention
claim ransomware prevention
claim production readiness
```

The report may render only caller-provided in-memory metadata from existing local fixtures.

Operator receipt report implementation work is now represented by docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md.

Operator receipt report surface/status work is now represented by docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md and docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md.

Current validation

This implementation plan is covered by:

```
sh scripts/test-latticra-seal-operator-receipt-report-implementation-plan.sh
```

Expected output:

```
latticra seal operator receipt report implementation plan: ok
```

Current next valid slice

The operator receipt report surface/status checkpoint is now represented by docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md and docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md.

The local capability registry schema contract is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md.

The local capability registry schema implementation plan is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md

Source title: Latticra Seal Operator Receipt Report Surface

Lines: 118 | Words: 476 | SHA-256: dc97146b9c006f91c10c7d0eb416839e1ff2379ba813ace039c23199f5538a4a

Latticra Seal Operator Receipt Report Surface

Status: report surface for the Latticra Seal operator receipt report Scope: deterministic local report surface after the no-effect operator receipt report implementation. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This document records the first operator-visible report surface for the Latticra Seal operator receipt report.

The report surface renders one deterministic denied receipt that binds capability metadata, policy decision metadata, request freshness metadata, signed request metadata, runtime dry-run metadata, and the denial reason into one local operator-facing report.

Added files

```
tests/seal_operator_receipt_report_surface.c
scripts/latticra-seal-operator-receipt-report.sh
```

Report runner

```
sh scripts/latticra-seal-operator-receipt-report.sh
```

Expected report posture

The report surface renders the current operator receipt posture for a known local fixture capability candidate, including:

```
LATTICRA SEAL OPERATOR RECEIPT REPORT
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
source_capability_metadata_present=1
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
capability_name=seal.capability.report
capability_known=1
capability_candidate=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
```

```
runtime_dry_run_state=report-only
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
blocked_reason=known-capability-candidate-still-denied
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
```

Boundary

This report surface compiles and runs a local deterministic fixture only.

It does not execute tools, execute shell commands beyond compiling and running the local test fixture, read host files beyond the repository sources needed for local compilation, write host files beyond the temporary test binary/report path, use the network, evaluate external policies, load policy files, load production capability files, load production allowlist files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, turn capability matches into execution grants, turn capability matches into effect grants, claim production readiness, claim AI-agent security, or claim MCP implementation.

Validation

Run:

```
sh scripts/test-latticra-seal-operator-receipt-report-surface.sh
```

Expected output:

```
latticra seal operator receipt report surface: ok
```

The underlying operator receipt report implementation remains covered by:

```
sh scripts/test-latticra-seal-operator-receipt-report.sh
```

Claim Boundary

This report surface does not justify the public claim that Latticra Seal is a production security product.

It makes the denied receipt posture easier to inspect before any future capability enforcement or runtime authority path is considered.

Next Valid Slice

The operator receipt report status alignment is now represented by docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md.

The local capability registry schema contract is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md.

The local capability registry schema implementation plan is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md

Source title: Latticra Seal Parameter Schema Contract

Lines: 201 | Words: 584 | SHA-256: 5875e5a706fbbcf3e2ec430ae330f917e2d69218ad2b270f55364020d9d78b60

Latticra Seal Parameter Schema Contract

Status: Latticra Seal parameter schema contract Scope: contract for future parameter schema metadata after the report-only Seal agentic automation security report surface. This document does not implement schema parsing, schema validation, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document defines the next Latticra Seal contract layer for declared parameter-schema metadata.

The purpose is to prepare a bounded report-only structure for describing tool-request parameters before any future validation or enforcement is considered.

This document does not implement parameter schema behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md
include/latticra/seal_agentic_automation_security.h
src/seal_agentic_automation_security.c
tests/seal_agentic_automation_security_invariants.c
tests/seal_agentic_automation_security_report_surface.c
scripts/test-latticra-seal-agentic-automation-security.sh
scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
```

The existing agentic automation metadata surface remains report-only.

Parameter schema boundary

The parameter schema layer may describe whether parameter schema metadata is declared, present, bounded, and reportable.

The layer may not validate arbitrary schemas, parse untrusted schema languages, accept model output as authority, grant runtime authority, execute tools, execute shell commands, contact networks, read host files, write host files, or mutate system state.

Allowed in this contract slice:

```
parameter-schema vocabulary
schema-presence metadata planning
schema-id metadata planning
schema-version metadata planning
schema-hash metadata planning
max-input-size metadata planning
parameter-count metadata planning
required-parameter-count metadata planning
unknown-parameter policy planning
parameter-forwarding policy planning
report-only status planning
non-claims
static guard validation
```

Forbidden in this contract slice:

```
schema parsing
schema validation
schema language implementation
runtime execution
AI agent execution
model execution
tool execution
shell command execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
MCP server implementation
MCP client implementation
```

Initial parameter schema policy

The initial policy is report-only and closed by default.

```
seal_parameter_schema_contract_present=1
parameter_schema_supported=0
parameter_schema_parsing_supported=0
parameter_schema_validation_supported=0
parameter_schema_present=0
parameter_schema_valid=0
schema_language_supported=0
schema_hash_present=0
max_input_bytes_declared=0
parameter_count_declared=0
required_parameter_count_declared=0
unknown_parameters_allowed=0
parameter_forwarding_allowed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=contract-only
status=parameter-schema-contract-only
```

Planned metadata fields

Future parameter schema metadata should be bounded and deterministic.

Planned fields:

```
schema_profile
schema_id
schema_version
schema_language
```

```
schema_hash
schema_present
schema_parsing_supported
schema_validation_supported
schema_valid
max_input_bytes_declared
parameter_count_declared
required_parameter_count_declared
unknown_parameters_allowed
parameter_forwarding_allowed
input_size_within_limit
parameter_names_reported
runtime_authority_granted
mode
decision
reason
status
```

Initial values before schema implementation:

```
schema_profile=latticra-seal-parameter-schema/0.1
schema_id=unset
schema_version=unset
schema_language=unset
schema_hash=unset
schema_present=0
schema_parsing_supported=0
schema_validation_supported=0
schema_valid=0
max_input_bytes_declared=0
parameter_count_declared=0
required_parameter_count_declared=0
unknown_parameters_allowed=0
parameter_forwarding_allowed=0
input_size_within_limit=0
parameter_names_reported=0
runtime_authority_granted=0
mode=contract-only
decision=report-only
reason=parameter-schema-contract-only
status=parameter-schema-contract-only
```

Promotion rules

The next implementation after this contract may only add bounded C metadata fields and deterministic rendering.

It must not parse schema languages, validate untrusted schemas, execute tools, execute shell commands, call runtime components, read host files, write host files, contact networks, enforce capabilities, verify signatures, generate receipts, or grant authority.

Non-claims

This contract must not be read as implementation of parameter validation or MCP tool-security enforcement.

Required non-claim posture:

```
parameter_validation_implemented=0
schema_parser_implemented=0
schema_validator_implemented=0
mcp_tool_security_enforced=0
runtime_authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is report-only parameter schema metadata.

That future slice may add a bounded metadata structure, deterministic rendering, invariant tests, and a test runner.

It must not change runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, or authority grants.

docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md

Source title: Latticra Seal Parameter Schema Implementation

Lines: 133 | Words: 522 | SHA-256: f34e0499de9de01ec42c0eddfb64c7cebd312a7e9185ece64747b82643084aa6

Latticra Seal Parameter Schema Implementation

Status: initial report-only parameter schema metadata implementation Scope: bounded C metadata surface for parameter schema posture after the report-only Seal agentic automation security layer. This slice does not implement schema parsing, schema validation, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first Latticra Seal parameter schema metadata implementation.

The implementation accepts an existing report-only Seal agentic automation security metadata record and produces deterministic report-only parameter schema metadata.

Added files

```
include/latticra/seal_parameter_schema.h
src/seal_parameter_schema.c
tests/seal_parameter_schema_invariants.c
scripts/test-latticra-seal-parameter-schema.sh
```

API summary

The parameter schema metadata surface adds:

```
latticra_seal_parameter_schema_t
latticra_seal_parameter_schema_error_t
latticra_seal_parameter_schema_error_label
latticra_seal_parameter_schema_from_agentic
latticra_seal_parameter_schema_is_report_only
latticra_seal_parameter_schema_report
```

Metadata behavior

The implementation:

```
accepts a valid report-only Seal agentic automation security metadata record
sets schema_profile=latticra-seal-parameter-schema/0.1
sets schema_id=unset
sets schema_version=unset
sets schema_language=unset
sets schema_hash=unset
copies schema_present from agentic parameter schema metadata
sets schema_parsing_supported=0
sets schema_validation_supported=0
copies schema_valid from agentic parameter schema metadata
sets max_input_bytes_declared=0
sets parameter_count_declared=0
sets required_parameter_count_declared=0
sets unknown_parameters_allowed=0
sets parameter_forwarding_allowed=0
sets input_size_within_limit=0
sets parameter_names_reported=0
copies runtime_authority_granted from agentic metadata
copies host_read_performed from agentic metadata
copies host_write_performed from agentic metadata
```

```
copies network_performed from agentic metadata
sets mode=report-only
sets decision=report-only
sets reason=parameter-schema-metadata-only
renders deterministic parameter schema metadata
```

Boundary

This implementation does not parse schemas, validate schemas, implement a schema language, execute tools, execute shell commands, call runtime components, contact networks, read host files, write host files, verify signatures, generate receipts, enforce capabilities, or grant authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null agentic metadata input -&gt; invalid-input
invalid agentic metadata -&gt; invalid-agentic
non-report-only agentic metadata -&gt; invalid-agentic
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse schemas, validate schemas, contact networks, read host files, write host files, execute tools, enforce capabilities, or grant authority.

Invariants

The invariant test verifies:

```
valid report-only agentic metadata produces deterministic parameter schema metadata
schema_profile is stable
schema_id remains unset
schema_version remains unset
schema_language remains unset
schema_hash remains unset
schema_present remains zero
schema_parsing_supported remains zero
schema_validation_supported remains zero
schema_valid remains zero
max_input_bytes_declared remains zero
parameter_count_declared remains zero
required_parameter_count_declared remains zero
unknown_parameters_allowed remains zero
parameter_forwarding_allowed remains zero
input_size_within_limit remains zero
parameter_names_reported remains zero
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
mode remains report-only
decision remains report-only
reason remains parameter-schema-metadata-only
small report buffer fails closed
null inputs fail closed
invalid agentic metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-parameter-schema-contract.sh
sh scripts/test-latticra-seal-parameter-schema.sh
```

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md

Source title: Latticra Seal Parameter Schema Report Surface

Lines: 84 | Words: 353 | SHA-256: 81d1de3c8f4cf848d554b74c7a95d357d24e350e5ff18d14e3dc27defd618f62

Latticra Seal Parameter Schema Report Surface

Status: report surface for Latticra Seal parameter schema metadata Scope: no-effect report-surface fixture and shell runner for deterministic Seal parameter schema metadata. This slice does not implement schema parsing, schema validation, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first dedicated report surface for Latticra Seal parameter schema metadata.

The report surface exists so operators and contributors can render the current report-only parameter schema posture without adding schema parsing, validation, runtime behavior, or authority.

Added files

```
tests/seal_parameter_schema_report_surface.c
scripts/latticra-seal-parameter-schema-report.sh
scripts/test-latticra-seal-parameter-schema-report-surface.sh
```

Behavior

The report surface:

```
builds a report-only Seal agentic automation security fixture
constructs Seal parameter schema metadata from that fixture
renders the deterministic Seal parameter schema report
prints the report to stdout
```

The guard verifies report output fields including:

```
schema_profile=latticra-seal-parameter-schema/0.1
schema_id=unset
schema_version=unset
schema_language=unset
schema_hash=unset
schema_present=0
schema_parsing_supported=0
schema_validation_supported=0
schema_valid=0
max_input_bytes_declared=0
parameter_count_declared=0
required_parameter_count_declared=0
unknown_parameters_allowed=0
parameter_forwarding_allowed=0
input_size_within_limit=0
parameter_names_reported=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=parameter-schema-metadata-only
```

```
status=parameter-schema-metadata
```

Validation

Run:

```
sh scripts/test-latticra-seal-parameter-schema-report-surface.sh
```

The underlying metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-parameter-schema.sh
```

Boundary

This report surface does not parse schemas, validate schemas, execute tools, execute shell commands on behalf of a model, contact networks, read host files, write host files, parse keys, verify signatures, create receipts, enforce capabilities, call runtime components, or grant runtime authority.

It compiles and runs a local deterministic fixture only.

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md

Source title: Latticra Seal Policy Decision Contract

Lines: 272 | Words: 761 | SHA-256: 0c87b6fe72c513f4fb5c6bcfb1a46fa404396c3b85505070308e8676417ac4d6

Latticra Seal Policy Decision Contract

Status: Latticra Seal policy decision contract Scope: contract for future policy decision metadata after the report-only Seal signed request metadata layer. This document does not implement policy evaluation, allow/deny enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, signature verification, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, revocation lookup, network trust lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document defines the next Latticra Seal contract layer for policy decision metadata.

The purpose is to prepare a bounded report-only structure for describing whether a future signed request would be treated as allowed, denied, unsupported, or review-required by a policy gate.

This document does not implement policy decision behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
```

```
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
include/latticra/seal_signed_request.h
src/seal_signed_request.c
tests/seal_signed_request_invariants.c
scripts/test-latticra-seal-signed-request-contract.sh
scripts/test-latticra-seal-signed-request.sh
```

The existing signed request metadata surface remains report-only.

Policy decision boundary

The policy decision layer may describe policy-decision metadata, default-deny posture, deny reasons, and future promotion rules.

The layer may not evaluate real policy, enforce decisions, execute tools, execute shell commands, contact networks, read host files, write host files, verify signatures, validate freshness, detect replay, or mutate system state.

Allowed in this contract slice:

```
policy-decision vocabulary
policy-id metadata planning
policy-version metadata planning
decision-state planning
default-deny metadata planning
deny-reason metadata planning
unknown-tool denial planning
unsigned-request denial planning
invalid-schema denial planning
stale-request denial planning
replayed-request denial planning
signature-invalid denial planning
operator-review planning
report-only status planning
non-claims
static guard validation
```

Forbidden in this contract slice:

```
policy evaluation
policy enforcement
allow decision enforcement
deny decision enforcement
runtime execution
AI agent execution
model execution
tool execution
shell command execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
signature verification
freshness validation
replay detection
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
MCP server implementation
MCP client implementation
```

Initial policy decision posture

The initial policy is report-only and default-deny.

```
seal_policy_decision_contract_present=1
```

```
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=contract-only
status=policy-decision-contract-only
```

Planned metadata fields

Future policy decision metadata should be bounded and deterministic.

Planned fields:

```
policy_decision_profile
policy_id
policy_version
requested_action
requested_tool
policy_decision_supported
policy_evaluation_supported
policy_enforcement_supported
policy_id_present
policy_version_present
requested_action_present
requested_tool_present
signed_request_present
signature_valid
schema_valid
freshness_valid
replay_detected
default_decision
decision_state
decision_allowed
decision_denied
operator_review_required
unknown_tool_denied
unsigned_request_denied
invalid_schema_denied
stale_request_denied
replayed_request_denied
invalid_signature_denied
runtime_authority_granted
mode
decision
reason
status
```

Initial values before policy decision implementation:

```
policy_decision_profile=latticra-seal-policy-decision/0.1
policy_id=unset
policy_version=unset
requested_action=unset
requested_tool=unset
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
```

```
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
mode=contract-only
decision=report-only
reason=policy-decision-contract-only
status=policy-decision-contract-only
```

Promotion rules

The next implementation after this contract may only add bounded C metadata fields and deterministic rendering.

It must not evaluate real policies, enforce allow or deny outcomes, execute tools, execute shell commands, call runtime components, read host files, write host files, contact networks, verify signatures, validate freshness, detect replay, enforce capabilities, generate receipts, or grant authority.

Claim gate

Latticra must not claim to secure AI agents from this contract alone.

A future claim that Latticra secures AI-agent tool execution boundaries requires, at minimum:

```
request identity metadata implemented
parameter schema metadata implemented
request freshness metadata implemented
signed request metadata implemented
policy decision metadata implemented
runtime enforcement gate implemented
negative tests for denied unknown tools
negative tests for denied unsigned manifests
negative tests for denied stale requests
negative tests for denied replayed requests
operator-visible evidence report implemented
```

Until those are implemented and validated, the accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Non-claims

This contract must not be read as implementation of policy enforcement, tool security enforcement, cryptographic enforcement, or AI-agent security.

Required non-claim posture:

```
policy_evaluation_implemented=0
policy_enforcement_implemented=0
runtime_enforcement_implemented=0
signed_request_verification_implemented=0
freshness_validation_implemented=0
replay_detection_implemented=0
mcp_tool_security_enforced=0
ai_agent_security_claimed=0
runtime_authority_granted=0
```

```
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is report-only policy decision metadata.

That future slice may add a bounded metadata structure, deterministic rendering, invariant tests, and a test runner.

It must not change runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, policy enforcement, or authority grants.

docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md

Source title: Latticra Seal Policy Decision Implementation

Lines: 159 | Words: 618 | SHA-256: b20dc9c0a9f83e3b7184b6aadedba6d066afb25dd2fc8ac9f84c786d8b9851b3

Latticra Seal Policy Decision Implementation

Status: initial report-only policy decision metadata implementation Scope: bounded C metadata surface for policy decision posture after the report-only Seal signed request layer. This slice does not implement policy evaluation, policy enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, signature verification, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, revocation lookup, network trust lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first Latticra Seal policy decision metadata implementation.

The implementation accepts an existing report-only Seal signed request metadata record and produces deterministic report-only policy decision metadata.

Added files

```
include/latticra/seal_policy_decision.h
src/seal_policy_decision.c
tests/seal_policy_decision_invariants.c
scripts/test-latticra-seal-policy-decision.sh
```

API summary

The policy decision metadata surface adds:

```
latticra_seal_policy_decision_t
latticra_seal_policy_decision_error_t
latticra_seal_policy_decision_error_label
latticra_seal_policy_decision_from_signed_request
latticra_seal_policy_decision_is_report_only
latticra_seal_policy_decision_report
```

Metadata behavior

The implementation:

```
accepts a valid report-only Seal signed request metadata record
sets policy_decision_profile=latticra-seal-policy-decision/0.1
sets policy_id=unset
sets policy_version=unset
sets requested_action=unset
```

```
sets requested_tool=unset
sets policy_decision_supported=0
sets policy_evaluation_supported=0
sets policy_enforcement_supported=0
sets policy_id_present=0
sets policy_version_present=0
sets requested_action_present=0
sets requested_tool_present=0
sets signed_request_present=0
sets signature_valid=0
sets schema_valid=0
sets freshness_valid=0
sets replay_detected=0
sets default_decision=deny
sets decision_state=report-only
sets decision_allowed=0
sets decision_denied=1
sets operator_review_required=1
sets unknown_tool_denied=1
sets unsigned_request_denied=1
sets invalid_schema_denied=1
sets stale_request_denied=1
sets replayed_request_denied=1
sets invalid_signature_denied=1
copies runtime_authority_granted from signed request metadata
copies host_read_performed from signed request metadata
copies host_write_performed from signed request metadata
copies network_performed from signed request metadata
sets mode=report-only
sets decision=report-only
sets reason=policy-decision-metadata-only
renders deterministic policy decision metadata
```

Boundary

This implementation does not evaluate policies, enforce policies, execute tools, execute shell commands, call runtime components, contact networks, read host files, write host files, verify signatures, validate freshness, detect replay, enforce capabilities, or grant authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null signed request metadata input -&gt; invalid-input
invalid signed request metadata -&gt; invalid-signed-request
non-report-only signed request metadata -&gt; invalid-signed-request
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not evaluate policy, contact networks, read host files, write host files, execute tools, enforce capabilities, or grant authority.

Invariants

The invariant test verifies:

```
valid report-only signed request metadata produces deterministic policy decision metadata
policy_decision_profile is stable
policy_id remains unset
policy_version remains unset
requested_action remains unset
requested_tool remains unset
policy_decision_supported remains zero
policy_evaluation_supported remains zero
policy_enforcement_supported remains zero
signed_request_present remains zero
signature_valid remains zero
schema_valid remains zero
freshness_valid remains zero
replay_detected remains zero
default_decision remains deny
decision_state remains report-only
decision_allowed remains zero
```

```
decision_denied remains one
operator_review_required remains one
unknown_tool_denied remains one
unsigned_request_denied remains one
invalid_schema_denied remains one
stale_request_denied remains one
replayed_request_denied remains one
invalid_signature_denied remains one
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
mode remains report-only
decision remains report-only
reason remains policy-decision-metadata-only
small report buffer fails closed
null inputs fail closed
invalid signed request metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-policy-decision-contract.sh
sh scripts/test-latticra-seal-policy-decision.sh
```

Claim boundary

This implementation still does not justify the public claim that Latticra secures AI agents.

It moves the project closer by adding report-only default-deny policy decision metadata, but the stronger claim requires a runtime enforcement gate and negative tests for denied unknown, unsigned, stale, and replayed requests.

Next valid slice

The next valid Latticra Seal slice is defensive threat model validation refinement or another small guarded status alignment if drift appears.

That future slice must not implement runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, policy enforcement, or authority grants unless a specific contract and validation path justify it.

docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md

Source title: Latticra Seal Policy Decision Report Surface

Lines: 96 | Words: 358 | SHA-256: e0e2049bb2cd5bc01ae4eb42d404449d02f2a965f1c75998dcff2c37c731cb19

Latticra Seal Policy Decision Report Surface

Status: report-only policy decision report surface Scope: deterministic local report surface for the existing Latticra Seal policy decision metadata layer. This document does not implement policy evaluation, policy enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, signature verification, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, revocation lookup, network trust lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first operator-visible report surface for Latticra Seal policy decision metadata.

The surface renders the existing default-deny, report-only policy decision posture using a deterministic local fixture.

Added files

```
tests/seal_policy_decision_report_surface.c
scripts/latticra-seal-policy-decision-report.sh
```

Report runner

```
sh scripts/latticra-seal-policy-decision-report.sh
```

Expected report posture

The report surface renders the current policy-decision posture, including:

```
LATTICRA SEAL POLICY DECISION
policy_decision_profile=latticra-seal-policy-decision/0.1
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=policy-decision-metadata-only
status=policy-decision-metadata
```

Boundary

This report surface compiles and runs a local deterministic fixture only.

It does not evaluate policy, enforce policy, execute tools, execute shell commands, call runtime components, contact networks, read host files, write host files, verify signatures, validate freshness, detect replay, enforce capabilities, or grant authority.

It renders metadata only.

Validation

Run:

```
sh scripts/test-latticra-seal-policy-decision-report-surface.sh
```

Expected output:

```
latticra seal policy decision report surface: ok
```

The underlying policy decision implementation remains covered by:

```
sh scripts/test-latticra-seal-policy-decision-contract.sh
sh scripts/test-latticra-seal-policy-decision.sh
sh scripts/test-latticra-seal-policy-decision-status.sh
```

Claim boundary

This report surface does not justify the public claim that Latticra secures AI agents.

It makes the default-deny policy decision posture easier to inspect before any future runtime work is considered.

Next valid slice

The next valid Latticra Seal slice is policy-decision report status alignment or a no-effect dry-run planning contract for future runtime behavior.

That future slice must preserve the report-only posture until a specific contract, deterministic local fixture, no-effect dry-run report, guarded allowlist posture, and validation path justify any later runtime work.

docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md

Source title: Latticra Seal Public-Key Parsing Boundary Contract

Lines: 351 | Words: 1084 | SHA-256: c20e8faa22effe68624e0065779aea525855e76eb3e3e1a57cd9f78611046f18

Latticra Seal Public-Key Parsing Boundary Contract

Status: Latticra Seal public-key parsing boundary contract Scope: contract for a future metadata-only public-key parsing surface after Seal key-material metadata. This document does not implement public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal public-key parsing boundary after ready key-material metadata.

The purpose of this layer is to decide whether a ready metadata-only key-material record may be represented as eligible for a future metadata-only public-key parsing request before any public-key parser, key material loader, private key, signer process, trust store, hardware key, runtime bridge, host authority, or signature generation exists.

The public-key parsing surface is public-key parsing path classification, not public-key parsing, not key material handling, not signing, not verification, and not trust-store behavior.

This document does not parse public keys.

This document does not load key material.

This document does not handle private keys.

Relationship to key material

Latticra Seal key-material metadata already records a no-effect eligibility checkpoint for the future key path:

```
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
docs/status/SEAL_KEY_MATERIAL_STATUS.md
```

This public-key parsing boundary contract starts from that checkpoint. It does not replace key-material metadata, parse public keys, load key material, handle private keys, route to runtime behavior, invoke a signer, perform signing, verify signatures, load trust stores, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
docs/status/SEAL_KEY_MATERIAL_STATUS.md
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-key-material-contract.sh
scripts/test-latticra-seal-key-material.sh
scripts/test-latticra-seal-key-material-status.sh
```

The key-material metadata surface remains the source of readiness evidence for this public-key parsing surface.

Public-Key Parsing Boundary

Allowed in the next implementation slice:

```
accept key-material metadata
require key_material_ready=1
require key_material_state=key-material-metadata-only
require requested_signature=Ed25519-development
require requested_signing_authorization=metadata-only
require requested_signer_handoff=metadata-only
require requested_signer_invocation=metadata-only
require requested_signing_operation=metadata-only
require requested_key_handling=metadata-only
require requested_key_material=metadata-only
require signature_performed=0
require verification_performed=0
require signer_invoked=0
require public_key_parsed=0
require key_material_loaded=0
require private_key_handling=0
require key_generation_performed=0
require hardware_key_used=0
require trust_store_loaded=0
require revocation_lookup_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested_public_key_parsing=metadata-only
classify metadata-only as public_key_parsing_state=public-key-parsing-metadata-only
produce deterministic public-key parsing metadata
```

Forbidden in the next implementation slice:

```
public-key parsing
key material loading
private-key handling
key generation
hardware-key use
trust-store loading
revocation lookup
cryptographic signing
signature verification
signer process invocation
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial Public-Key Parsing Policy

Allowed key-material state:

```
key_material_state=key-material-metadata-only
```

Allowed requested key-material label:

```
metadata-only
```

Allowed requested public-key parsing label:

```
metadata-only
```

These labels are metadata labels only. They do not mean public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust establishment, trust-store behavior, Ed25519 signing, Ed25519 verification, signer invocation, revocation lookup, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned public-key parsing states:

```
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_state=denied-key-material
public_key_parsing_state=denied-key-handling
public_key_parsing_state=denied-signing-operation
public_key_parsing_state=denied-signer-invocation
public_key_parsing_state=denied-signer-handoff
public_key_parsing_state=denied-signing-authorization
public_key_parsing_state=denied-signature-algorithm
public_key_parsing_state=denied-public-key-parsing
public_key_parsing_state=denied-private-key
public_key_parsing_state=denied-trust-store
public_key_parsing_state=denied-runtime-authority
public_key_parsing_state=denied-host-effect
public_key_parsing_state=denied-network-effect
```

The first implementation may set:

```
public_key_parsing_ready=1
```

only for ready metadata-only key-material metadata, the allowed development signature label, the metadata-only key-handling label, the metadata-only key-material label, and the metadata-only public-key parsing label.

Even when public_key_parsing_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned Fields

A future public-key parsing record should be bounded and deterministic.

Planned fields:

```
public_key_parsing_profile
key_material_profile
key_handling_profile
signing_operation_profile
signer_invocation_profile
signer_handoff_profile
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
```

```
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_signing_authorization
requested_signer_handoff
requested_signer_invocation
requested_signing_operation
requested_key_handling
requested_key_material
requested_public_key_parsing
requested_scope
signing_authorization_state
signing_authorization_ready
signer_handoff_state
signer_handoff_ready
signer_invocation_state
signer_invocation_ready
signing_operation_state
signing_operation_ready
key_handling_state
key_handling_ready
key_material_state
key_material_ready
public_key_parsing_state
public_key_parsing_ready
signature_performed
verification_performed
signer_invoked
public_key_parsed
key_material_loaded
private_key_handling
key_generation_performed
hardware_key_used
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
public_key_parsing_profile=latticra-seal-public-key-parsing/0.1
key_material_profile=latticra-seal-key-material/0.1
key_handling_profile=latticra-seal-key-handling/0.1
requested_key_handling=metadata-only
requested_key_material=metadata-only
requested_public_key_parsing=metadata-only
key_material_state=key-material-metadata-only
key_material_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=public-key-parsing-metadata
```

Failure Behavior

Future public-key parsing metadata handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null key material -&gt; invalid
invalid key material -&gt; denied-key-material
key_material_ready=0 -&gt; denied-key-material
key_material_state not key-material-metadata-only -&gt; denied-key-material
requested_key_material not metadata-only -&gt; denied-key-material
key_handling_ready=0 -&gt; denied-key-handling
key_handling_state not key-handling-metadata-only -&gt; denied-key-handling
requested_key_handling not metadata-only -&gt; denied-key-handling
signing_operation_ready=0 -&gt; denied-signing-operation
signing_operation_state not operation-metadata-only -&gt; denied-signing-operation
requested_signing_operation not metadata-only -&gt; denied-signing-operation
signer_invocation_ready=0 -&gt; denied-signer-invocation
signer_invocation_state not invocation-metadata-only -&gt; denied-signer-invocation
requested_signer_invocation not metadata-only -&gt; denied-signer-invocation
signer_handoff_ready=0 -&gt; denied-signer-handoff
signer_handoff_state not handoff-metadata-only -&gt; denied-signer-handoff
requested_signer_handoff not metadata-only -&gt; denied-signer-handoff
requested_signing_authorization not metadata-only -&gt; denied-signing-authorization
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested public key parsing -&gt; denied-public-key-parsing
unknown requested public key parsing -&gt; denied-public-key-parsing
public-key parsing requested or observed -&gt; denied-public-key-parsing
key material loading requested or observed -&gt; denied-key-material
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
hardware-key use requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-trust-store
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
signer already invoked -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Promotion Rule

This contract permits only the next implementation slice:

```
public-key parsing metadata implementation
```

It does not permit public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After public-key parsing metadata, status/public-entry alignment, and the future key parsing implementation contract exist and are guarded, the next valid planning slice is bounded no-effect key parsing implementation that still must not add public-key parsing without separate implementation, public-key, key-material, and guard contracts.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-public-key-parsing-contract.sh
```

docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md

Source title: Latticra Seal Public-Key Parsing Implementation

Lines: 227 | Words: 665 | SHA-256: b95349925a81d8bb49f193f96bb143389d24c18fe9a906c32e9442f184a870f7

Latticra Seal Public-Key Parsing Implementation

Status: initial public-key parsing metadata implementation Scope: bounded C metadata surface for classifying Seal public-key parsing eligibility after ready key-material metadata. This slice does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, cryptographic signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal public-key parsing metadata implementation.

The implementation consumes ready key-material metadata and classifies whether the request remains eligible for a future metadata-only public-key parsing path.

It is public-key parsing path classification only.

It does not parse public keys.

It does not load key material.

It does not handle private keys.

It does not sign.

It does not verify signatures.

Files

```
include/latticra/seal_public_key_parsing.h
src/seal_public_key_parsing.c
tests/seal_public_key_parsing_invariants.c
scripts/test-latticra-seal-public-key-parsing.sh
```

Required predecessors

This implementation depends on the public-key parsing boundary contract and the key-material metadata surface:

```
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
docs/status/SEAL_KEY_MATERIAL_STATUS.md
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-public-key-parsing-contract.sh
scripts/test-latticra-seal-key-material-contract.sh
scripts/test-latticra-seal-key-material.sh
scripts/test-latticra-seal-key-material-status.sh
```

Implemented surface

The public-key parsing metadata surface adds:

```
latticra_seal_public_key_parsing_t
latticra_seal_public_key_parsing_error_t
latticra_seal_public_key_parsing_error_label
```

```
latticra_seal_public_key_parsing_from_key_material
latticra_seal_public_key_parsing_is_metadata_only
latticra_seal_public_key_parsing_render
```

The implementation:

```
accepts key-material metadata
requires key_material_ready=1
requires key_material_state=key-material-metadata-only
requires requested_signature=Ed25519-development
requires requested_signing_authorization=metadata-only
requires requested_signer_handoff=metadata-only
requires requested_signer_invocation=metadata-only
requires requested_signing_operation=metadata-only
requires requested_key_handling=metadata-only
requires requested_key_material=metadata-only
accepts requested_public_key_parsing=metadata-only
classifies the result as public_key_parsing_state=public-key-parsing-metadata-only
sets public_key_parsing_ready=1 only for metadata-only allowed key-material states
renders deterministic public-key parsing metadata
fails closed for invalid or denied predecessor metadata
```

Even when public_key_parsing_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
public_key_parsing_profile=latticra-seal-public-key-parsing/0.1
key_material_profile=latticra-seal-key-material/0.1
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
requested_key_material=metadata-only
requested_public_key_parsing=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
key_handling_state=key-handling-metadata-only
key_handling_ready=1
key_material_state=key-material-metadata-only
key_material_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
public_key_parsed=0
key_material_loaded=0
hardware_key_used=0
mode=metadata-only
status=public-key-parsing-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null key material
invalid key material
key_material_ready=0
key_material_state not key-material-metadata-only
requested_key_material not metadata-only
key_handling_ready=0
key_handling_state not key-handling-metadata-only
requested_key_handling not metadata-only
signing_operation_ready=0
signing_operation_state not operation-metadata-only
requested_signing_operation not metadata-only
signer_invocation_ready=0
signer_invocation_state not invocation-metadata-only
requested_signer_invocation not metadata-only
signer_handoff_ready=0
signer_handoff_state not handoff-metadata-only
requested_signer_handoff not metadata-only
signing_authorization_ready=0
signing_authorization_state not authorized-metadata-only
requested_signing_authorization not metadata-only
missing requested signature
unknown requested signature
missing requested public-key parsing
unknown requested public-key parsing
public-key parsing request
key material loading request
private-key handling request
key generation request
hardware-key use request
trust-store loading request
revocation lookup request
public-key parsing already present
key material loading already present
private-key handling already present
key generation already present
hardware-key use already present
trust-store loading already present
revocation lookup already present
runtime authority already granted
signature already performed
verification already performed
signer already invoked
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, invoke a signer, parse public keys, load key material, handle private keys, generate keys, use hardware keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-public-key-parsing-contract.sh
sh scripts/test-latticra-seal-public-key-parsing.sh
sh scripts/test-latticra-seal-public-key-parsing-status.sh
```

Expected output:

```
seal public-key parsing contract: ok
seal public-key parsing invariants: ok
seal public-key parsing status: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add public-key parsing without separate public-key parsing, key-material, and guard contracts.

The public-key parsing metadata implementation is a guarded checkpoint. Future work must not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_REPLAYED_REQUEST_CASE_CONTRACT.md

Source title: Latticra Seal Replayed Request Case Contract

Lines: 104 | Words: 296 | SHA-256: 80abf44265255a25a761e1705d0217e11cd4049eeba45b0796e2b30b445331d7

Latticra Seal Replayed Request Case Contract

Status: Latticra Seal replayed request case contract Scope: case-specific contract after the stale request case fixture. This document does not implement runtime behavior, policy behavior, tool behavior, host behavior, network behavior, cryptographic behavior, replay detection, or authority grants.

Purpose

This document defines the fourth case-specific test contract for the Seal runtime gate path.

The case is a replayed request. The expected posture is that the gate remains report-only, blocked by default, and grants no authority.

This document does not implement the case behavior.

Required predecessors

```
docs/LATTICRA_SEAL_RUNTIME_GATE_TEST_PLAN.md
docs/LATTICRA_SEAL_STALE_REQUEST_CASE_CONTRACT.md
docs/LATTICRA_SEAL_STALE_REQUEST_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
tests/seal_stale_request_case.c
scripts/test-latticra-seal-runtime-gate-test-plan.sh
scripts/test-latticra-seal-stale-request-case-contract.sh
scripts/test-latticra-seal-stale-request-case.sh
```

Case definition

```
case_profile=latticra-seal-replayed-request-case/0.1
case_name=replayed-request
case_contract_present=1
replayed_request_case_contract_present=1
input_replay_detected=1
input_nonce_reused=1
input_request_seen_before=1
expected_gate_state=report-only
expected_default_blocked=1
expected_replayed_request_blocked=1
expected_runtime_authority_granted=0
expected_effect_performed=0
expected_host_read_performed=0
expected_host_write_performed=0
expected_network_performed=0
case_behavior_implemented=0
```

Future test requirements

A future implementation must add a deterministic test fixture showing that a replayed request case remains blocked and does not perform effects.

Planned future file names:

```
tests/seal_replayed_request_case.c
scripts/test-latticra-seal-replayed-request-case.sh
```

Expected future output:

```
seal replayed request case: ok
```

Boundary

This contract is documentation-only.

It does not add case execution, runtime behavior, host behavior, network behavior, MCP behavior, AI agent behavior, model behavior, tool behavior, shell behavior, policy behavior, cryptographic behavior, replay detection, or authority grants.

Claim gate

This contract alone does not justify the public claim that Latticra secures AI agents.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

A stronger claim requires implemented and validated gate behavior plus case tests for unknown, unsigned, stale, and replayed requests.

Non-claims

```
replayed_request_case_implemented=0
runtime_gate_case_tests_implemented=0
runtime_behavior_implemented=0
policy_behavior_implemented=0
tool_behavior_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
replay_detection_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is the replayed request case test fixture.

docs/LATTICRA_SEAL_REPLAYED_REQUEST_CASE_IMPLEMENTATION.md

Source title: Latticra Seal Replayed Request Case Implementation

Lines: 51 | Words: 161 | SHA-256: 623d64cedf16507c035cb819ed8d0e27488a46d69a5734db85bff2a4be5471db

Latticra Seal Replayed Request Case Implementation

Status: initial replayed request case fixture Scope: local deterministic case fixture after the replayed request case contract.

Purpose

This record documents the replayed-request case fixture for the Seal runtime gate metadata path.

The fixture keeps the gate in report-only state and verifies that no authority, host operation, network operation, or effect is reported.

Added files

```
tests/seal_replayed_request_case.c
scripts/test-latticra-seal-replayed-request-case.sh
```

Validation

Run:

```
sh scripts/test-latticra-seal-replayed-request-case.sh
```

Expected output:

```
seal replayed request case: ok
```

Boundary

This slice is a local deterministic fixture only.

It does not add runtime behavior, tool behavior, host behavior, network behavior, effect behavior, or authority grants.

Claim boundary

This fixture does not by itself justify a stronger public claim.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Next valid slice

The next valid slice is status/index alignment for the completed core blocked-request case set.

docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md

Source title: Latticra Seal Report Envelope Contract

Lines: 199 | Words: 575 | SHA-256: 86b0bc6901bbfecaca799aad272d23d92828f642f69ad512d3c09cc648cf5c66

Latticra Seal Report Envelope Contract

Status: Latticra Seal report envelope contract Scope: contract for a future sealed report envelope surface after ready runtime handoff report metadata. This document does not implement runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, signing, key generation, private-key storage, trust-store loading, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal report envelope boundary that may consume ready metadata-only runtime handoff report metadata.

The purpose of this layer is to decide whether a previously ready metadata-only report may be wrapped into a deterministic envelope record before any future signing or runtime bridge exists.

The envelope surface is envelope classification, not signing and not runtime handoff.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md
```

```
include/latticra/seal_runtime_handoff_report.h
src/seal_runtime_handoff_report.c
tests/seal_runtime_handoff_report_invariants.c
scripts/test-latticra-seal-runtime-handoff-report.sh
```

The runtime handoff report metadata surface remains the source of report-readiness evidence for this envelope surface.

Envelope boundary

Allowed in the next implementation slice:

```
accept runtime handoff report metadata
require report_ready=1
require report_state=ready-report-only or report_state=ready-evaluate-only
require handoff_performed=0
require runtime_authority_granted=0
require effect_performed=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested envelope metadata
classify report-only as envelope_state=sealed-report-only
classify evaluate-only as envelope_state=sealed-evaluate-only
produce deterministic sealed report envelope metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
private-key handling
key generation
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial envelope policy

Allowed input report states:

```
report_state=ready-report-only
report_state=ready-evaluate-only
```

Allowed requested envelope labels:

```
report-only
evaluate-only
```

Planned envelope states:

```
envelope_state=sealed-report-only
envelope_state=sealed-evaluate-only
envelope_state=denied-report
envelope_state=denied-envelope
envelope_state=denied-runtime-authority
envelope_state=denied-host-effect
envelope_state=denied-network-effect
```

The first implementation may set:

```
envelope_ready=1
```

only for metadata-only report/evaluate envelopes.

Even when envelope_ready=1, these must remain zero:

```
signature_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
```

```
host_write_performed=0
network_performed=0
```

Planned fields

A future sealed report envelope record should be bounded and deterministic.

Planned fields:

```
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_scope
report_state
report_ready
envelope_state
envelope_ready
signature_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
status
```

Failure behavior

Future report envelope handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null runtime handoff report -&gt; invalid
invalid runtime handoff report -&gt; denied-report
report_ready=0 -&gt; denied-report
report_state not ready-report-only or ready-evaluate-only -&gt; denied-report
missing requested envelope -&gt; denied-envelope
unknown requested envelope -&gt; denied-envelope
requested envelope mismatch -&gt; denied-envelope
runtime authority already granted -&gt; denied-runtime-authority
signature already performed -&gt; denied-host-effect
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, handle private keys, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
sealed report envelope metadata implementation
```

It does not permit cryptographic signing, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After sealed report envelope metadata exists and is guarded, the next valid planning slice is a signature request contract that still performs no signing.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-report-envelope-contract.sh
```

docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md

Source title: Latticra Seal Report Envelope Implementation

Lines: 152 | Words: 555 | SHA-256: 41dc9cce35cc716c5e1c306ecae2769915f9fb1e15484c011650240bfb681dd0

Latticra Seal Report Envelope Implementation

Status: initial sealed report envelope metadata implementation Scope: bounded C metadata surface for classifying ready metadata-only runtime handoff report output into deterministic sealed report envelope readiness. This does not implement cryptographic signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel interaction, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal report envelope metadata implementation.

The implementation consumes runtime handoff report metadata and classifies whether the requested metadata-only envelope remains ready as report-only or evaluate-only.

The envelope surface is envelope classification, not signing and not runtime handoff.

Added files

```
include/latticra/seal_report_envelope.h
src/seal_report_envelope.c
tests/seal_report_envelope_invariants.c
scripts/test-latticra-seal-report-envelope.sh
```

Required predecessor

This implementation depends on the runtime handoff report metadata surface:

```
docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md
include/latticra/seal_runtime_handoff_report.h
src/seal_runtime_handoff_report.c
tests/seal_runtime_handoff_report_invariants.c
scripts/test-latticra-seal-runtime-handoff-report.sh
```

API summary

The report envelope metadata surface adds:

```
latticra_seal_report_envelope_t
latticra_seal_report_envelope_error_t
latticra_seal_report_envelope_error_label
latticra_seal_report_envelope_from_report
latticra_seal_report_envelope_is_metadata_only
latticra_seal_report_envelope_render
```

Envelope behavior

The implementation:

```
accepts runtime handoff report metadata
```

```
requires report_ready=1
requires report_state=ready-report-only or report_state=ready-evaluate-only
requires runtime_authority_granted=0
requires handoff_performed=0
requires effect_performed=0
requires host_read_performed=0
requires host_write_performed=0
requires network_performed=0
accepts requested envelope metadata
classifies report-only as envelope_state=sealed-report-only
classifies evaluate-only as envelope_state=sealed-evaluate-only
sets envelope_ready=1 only for metadata-only report/evaluate envelope surfaces
renders deterministic sealed report envelope metadata
```

Effect and runtime boundary

Even when envelope_ready=1, these fields remain zero:

```
signature_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

This implementation does not sign records, verify signatures, handle private keys, load trust stores, look up revocation status, perform runtime handoff, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, persist policy, seal objects, or interact with the kernel.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null runtime handoff report -&gt; invalid-input
invalid runtime handoff report -&gt; invalid-report
report_ready=0 -&gt; denied-report
report_state not ready-report-only or ready-evaluate-only -&gt; denied-report
missing requested envelope -&gt; missing-requested-envelope
unknown requested envelope -&gt; denied-unknown-envelope
requested envelope mismatch -&gt; denied-envelope
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not sign, perform handoff, perform effects, read host files, write host files, contact networks, or grant runtime authority.

Invariants

The invariant test verifies:

```
ready report-only metadata produces sealed-report-only envelope metadata
ready evaluate-only metadata produces sealed-evaluate-only envelope metadata
envelope_ready is one only for metadata-only envelope states
signature_performed remains zero
handoff_performed remains zero
effect_performed remains zero
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
small render buffer fails closed
null inputs fail closed
invalid report metadata fails closed
missing, unknown, and mismatched requested envelope labels fail closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-report-envelope-contract.sh
sh scripts/test-latticra-seal-report-envelope.sh
```

Expected output:

```
seal report envelope contract: ok
seal report envelope invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is a signature request contract that still performs no signing.

That future slice must remain contract-first and must not add private-key handling, signing, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately contracted, implemented, and guarded.

docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md

Source title: Latticra Seal Report Implementation

Lines: 114 | Words: 351 | SHA-256: e02dbe30a3bf142dfde467b7f369f08f0c0d221fcec7fae24dab7c485c154fe7

Latticra Seal Report Implementation

Status: Seal report implementation with read-only measurement awareness Scope: bounded C report surface for Latticra Seal status labels. This surface now reports that read-only measurement support exists. It does not add signing, object sealing, key handling, runtime authority, host writes, network behavior, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the Latticra Seal report implementation.

The implementation adds a deterministic report surface that makes the current Seal posture visible without granting authority.

Added files

```
include/latticra/seal_report.h
src/seal_report.c
tests/seal_report_invariants.c
scripts/test-latticra-seal-report.sh
```

API summary

The report surface adds:

```
latticra_seal_report_t
latticra_seal_report_default
latticra_seal_report_is_no_effect
latticra_seal_report_render
```

The report uses fixed-size fields, no dynamic allocation, and bounded rendering through snprintf.

Default report posture

The default report now states:

```
seal_profile=latticra-seal/0.1-report
contract_present=1
implementation_plan_present=1
report_only_supported=1
measurement_supported=1
signing_supported=0
capability_gate_supported=0
sealed_objects_supported=0
```

```
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_granted=0
evidence_level=3
status=report-and-measurement
```

Effect boundary

The report itself performs no host read, no host write, no network behavior, no runtime grant, and no object mutation.

Read-only measurement is represented by the separate measurement surface.

The helper latticra_seal_report_is_no_effect reports true only when all effect fields remain zero.

Failure behavior

The renderer fails closed:

```
null report -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null buffer -&gt; LATTICRA_STATUS_NULL_ARGUMENT
small buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

When the buffer is too small and has nonzero capacity, the renderer clears the first byte.

Invariants

The invariant test verifies:

```
default report is report-only plus measurement-aware
contract and plan are visible
measurement support is visible
signing remains unsupported
capability gate remains unsupported
sealed objects remain unsupported
no host read is performed by the report
no host write is performed by the report
no network behavior is performed
no runtime grant is performed
rendered report contains required labels
small render buffer fails closed
null inputs fail closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-report.sh
```

The measurement implementation is validated by:

```
sh scripts/test-latticra-seal-measurement.sh
```

Next valid slice

The next valid Latticra Seal slice after read-only measurement is a signed evidence-manifest contract.

That future slice must be contract-first and must not be added directly to the measurement implementation.

docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md

Source title: Latticra Seal Request Freshness Contract

Lines: 252 | Words: 766 | SHA-256: c0afe8e60e34fda3be6a4aed0c126e09a1fdadf90e913078cbd2b44c00978cfe

Latticra Seal Request Freshness Contract

Status: Latticra Seal request freshness contract Scope: contract for future request freshness and replay metadata after the report-only Seal parameter schema report surface. This document does not implement timestamp parsing, clock trust, nonce storage, replay-cache storage, context hashing, parameter hashing, signature verification, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document defines the next Latticra Seal contract layer for request freshness and replay metadata.

The purpose is to prepare a bounded report-only structure for describing whether a future tool request is time-scoped, nonce-scoped, context-bound, and parameter-bound before any execution or enforcement is considered.

This document does not implement request freshness behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md
include/latticra/seal_parameter_schema.h
src/seal_parameter_schema.c
tests/seal_parameter_schema_invariants.c
tests/seal_parameter_schema_report_surface.c
scripts/test-latticra-seal-parameter-schema-contract.sh
scripts/test-latticra-seal-parameter-schema.sh
scripts/test-latticra-seal-parameter-schema-report-surface.sh
```

The existing parameter schema metadata surface remains report-only.

Request freshness boundary

The request freshness layer may describe whether freshness metadata is declared, present, bounded, and reportable.

The layer may not parse trusted time, maintain a nonce cache, compare real clocks, validate signatures, authorize a request, execute tools, execute shell commands, contact networks, read host files, write host files, or mutate system state.

Allowed in this contract slice:

```
request-freshness vocabulary
request-id metadata planning
caller-id metadata planning
tool-id metadata planning
request timestamp metadata planning
request expiration metadata planning
nonce metadata planning
context hash metadata planning
parameter hash metadata planning
```

```
freshness status planning
replay status planning
report-only status planning
non-claims
static guard validation
```

Forbidden in this contract slice:

```
timestamp parsing
trusted clock behavior
nonce storage
replay-cache storage
context hashing
parameter hashing
freshness validation
replay detection
signature verification
runtime execution
AI agent execution
model execution
tool execution
shell command execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
MCP server implementation
MCP client implementation
```

Initial request freshness policy

The initial policy is report-only and closed by default.

```
seal_request_freshness_contract_present=1
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=contract-only
status=request-freshness-contract-only
```

Planned metadata fields

Future request freshness metadata should be bounded and deterministic.

Planned fields:

```
freshness_profile
request_id
caller_id
tool_id
request_timestamp
request_expiration
nonce
```

```
context_hash
parameter_hash
request_freshness_supported
request_freshness_validation_supported
replay_protection_supported
request_id_present
caller_id_present
tool_id_present
request_timestamp_present
request_expiration_present
nonce_present
context_hash_present
parameter_hash_present
freshness_valid
replay_detected
runtime_authority_granted
mode
decision
reason
status
```

Initial values before freshness implementation:

```
freshness_profile=latticra-seal-request-freshness/0.1
request_id=unset
caller_id=unset
tool_id=unset
request_timestamp=unset
request_expiration=unset
nonce=unset
context_hash=unset
parameter_hash=unset
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
runtime_authority_granted=0
mode=contract-only
decision=report-only
reason=request-freshness-contract-only
status=request-freshness-contract-only
```

Promotion rules

The next implementation after this contract may only add bounded C metadata fields and deterministic rendering.

It must not parse trusted time, maintain a nonce cache, compare live clocks, compute hashes, validate freshness, detect replay, execute tools, execute shell commands, call runtime components, read host files, write host files, contact networks, enforce capabilities, verify signatures, generate receipts, or grant authority.

Claim gate

Latticra must not claim to secure AI agents from this contract alone.

A future claim that Latticra secures AI agents requires, at minimum:

```
request identity metadata implemented
parameter schema metadata implemented
request freshness metadata implemented
signed request metadata implemented
policy decision metadata implemented
runtime enforcement gate implemented
negative tests for denied unknown tools
negative tests for denied unsigned manifests
negative tests for denied stale requests
negative tests for denied replayed requests
```

```
operator-visible evidence report implemented
```

Until those are implemented and validated, the accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Non-claims

This contract must not be read as implementation of freshness validation, replay protection, signed request verification, tool security enforcement, or AI-agent security.

Required non-claim posture:

```
freshness_validation_implemented=0
replay_protection_implemented=0
signed_request_verification_implemented=0
mcp_tool_security_enforced=0
ai_agent_security_claimed=0
runtime_authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is report-only request freshness metadata.

That future slice may add a bounded metadata structure, deterministic rendering, invariant tests, and a test runner.

It must not change runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, freshness validation, replay detection, or authority grants.

docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md

Source title: Latticra Seal Request Freshness Implementation

Lines: 151 | Words: 634 | SHA-256: c982a798ba430cf4cc907b84ac26adbdd17d76bc1ca3d0dd37f86c5c8b9a2aeb

Latticra Seal Request Freshness Implementation

Status: initial report-only request freshness metadata implementation Scope: bounded C metadata surface for request freshness and replay posture after the report-only Seal parameter schema layer. This slice does not implement timestamp parsing, trusted clock behavior, nonce storage, replay-cache storage, context hashing, parameter hashing, freshness validation, replay detection, signature verification, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first Latticra Seal request freshness metadata implementation.

The implementation accepts an existing report-only Seal parameter schema metadata record and produces deterministic report-only request freshness metadata.

Added files

```
include/latticra/seal_request_freshness.h
src/seal_request_freshness.c
tests/seal_request_freshness_invariants.c
scripts/test-latticra-seal-request-freshness.sh
```

API summary

The request freshness metadata surface adds:

```
latticra_seal_request_freshness_t
latticra_seal_request_freshness_error_t
latticra_seal_request_freshness_error_label
latticra_seal_request_freshness_from_schema
latticra_seal_request_freshness_is_report_only
latticra_seal_request_freshness_report
```

Metadata behavior

The implementation:

```
accepts a valid report-only Seal parameter schema metadata record
sets freshness_profile=latticra-seal-request-freshness/0.1
sets request_id=unset
sets caller_id=unset
sets tool_id=unset
sets request_timestamp=unset
sets request_expiration=unset
sets nonce=unset
sets context_hash=unset
sets parameter_hash=unset
sets request_freshness_supported=0
sets request_freshness_validation_supported=0
sets replay_protection_supported=0
sets request_id_present=0
sets caller_id_present=0
sets tool_id_present=0
sets request_timestamp_present=0
sets request_expiration_present=0
sets nonce_present=0
sets context_hash_present=0
sets parameter_hash_present=0
sets freshness_valid=0
sets replay_detected=0
copies runtime_authority_granted from parameter schema metadata
copies host_read_performed from parameter schema metadata
copies host_write_performed from parameter schema metadata
copies network_performed from parameter schema metadata
sets mode=report-only
sets decision=report-only
sets reason=request-freshness-metadata-only
renders deterministic request freshness metadata
```

Boundary

This implementation does not parse timestamps, compare clocks, store nonces, maintain a replay cache, compute hashes, validate freshness, detect replay, verify signatures, parse keys, create receipts, execute tools, execute shell commands, call runtime components, contact networks, read host files, write host files, enforce capabilities, or grant authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null parameter schema metadata input -&gt; invalid-input
invalid parameter schema metadata -&gt; invalid-schema
non-report-only parameter schema metadata -&gt; invalid-schema
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse time, read clocks, store nonces, contact networks, read host files, write host files, execute tools, enforce capabilities, or grant authority.

Invariants

The invariant test verifies:

```
valid report-only parameter schema metadata produces deterministic request freshness metadata
```

```
freshness_profile is stable
request_id remains unset
caller_id remains unset
tool_id remains unset
request_timestamp remains unset
request_expiration remains unset
nonce remains unset
context_hash remains unset
parameter_hash remains unset
request_freshness_supported remains zero
request_freshness_validation_supported remains zero
replay_protection_supported remains zero
request_id_present remains zero
caller_id_present remains zero
tool_id_present remains zero
request_timestamp_present remains zero
request_expiration_present remains zero
nonce_present remains zero
context_hash_present remains zero
parameter_hash_present remains zero
freshness_valid remains zero
replay_detected remains zero
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
mode remains report-only
decision remains report-only
reason remains request-freshness-metadata-only
small report buffer fails closed
null inputs fail closed
invalid parameter schema metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-request-freshness-contract.sh
sh scripts/test-latticra-seal-request-freshness.sh
```

Claim boundary

This implementation still does not justify the public claim that Latticra secures AI agents.

It moves the project closer by adding report-only metadata for freshness and replay posture, but the stronger claim requires signed request metadata, policy decision metadata, a runtime enforcement gate, and negative tests for denied unknown, unsigned, stale, and replayed requests.

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md

Source title: Latticra Seal Request Freshness Report Surface

Lines: 96 | Words: 437 | SHA-256: dca04e745b1816d58659b19be08aaf23c354189d1cbaabaa9e8de5f8553c1c6d

Latticra Seal Request Freshness Report Surface

Status: report surface for Latticra Seal request freshness metadata Scope: no-effect report-surface fixture and shell runner for deterministic Seal request freshness metadata. This slice does not implement timestamp parsing, trusted clock behavior, nonce storage, replay-cache storage, context hashing, parameter hashing, freshness validation, replay detection, signature verification, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first dedicated report surface for Latticra Seal request freshness metadata.

The report surface exists so operators and contributors can render the current report-only request freshness posture without adding timestamp validation, replay detection, runtime behavior, or authority.

Added files

```
tests/seal_request_freshness_report_surface.c
scripts/latticra-seal-request-freshness-report.sh
scripts/test-latticra-seal-request-freshness-report-surface.sh
```

Behavior

The report surface:

```
builds a report-only Seal parameter schema fixture
constructs Seal request freshness metadata from that fixture
renders the deterministic Seal request freshness report
prints the report to stdout
```

The guard verifies report output fields including:

```
freshness_profile=latticra-seal-request-freshness/0.1
request_id=unset
caller_id=unset
tool_id=unset
request_timestamp=unset
request_expiration=unset
nonce=unset
context_hash=unset
parameter_hash=unset
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=request-freshness-metadata-only
status=request-freshness-metadata
```

Validation

Run:

```
sh scripts/test-latticra-seal-request-freshness-report-surface.sh
```

The underlying metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-request-freshness.sh
```

Boundary

This report surface does not parse timestamps, read trusted clocks, store nonces, maintain a replay cache, compute hashes, validate freshness, detect replay, verify signatures, execute tools, execute shell commands on behalf of a model, contact networks, read host files, write host files, parse keys, create receipts, enforce capabilities, call runtime components, or grant runtime authority.

It compiles and runs a local deterministic fixture only.

Claim boundary

This report surface still does not justify the public claim that Latticra secures AI agents.

It makes the freshness/replay metadata visible, but the stronger claim requires signed request metadata, policy decision metadata, a runtime enforcement gate, and negative tests for denied unknown, unsigned, stale, and replayed requests.

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_CONTRACT.md

Source title: Latticra Seal Runtime Dry-Run Contract

Lines: 177 | Words: 555 | SHA-256: 0c3e1544c5e46be80db4b02d0156ab87b9bacf9472dea79a6c8aa988362ff617

Latticra Seal Runtime Dry-Run Contract

Status: planning contract for a future Latticra Seal runtime dry-run surface Scope: contract-only planning for a future no-effect dry-run layer after the report-only policy decision and runtime gate evidence path. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This contract defines the next safe Latticra Seal step before any runtime enforcement work can be considered.

The future dry-run surface should answer one question:

```
If this request reached the Seal runtime boundary, what would the boundary report, deny, and
```

```
refuse to do?
```

The answer must remain no-effect, deterministic, local, and report-only.

Required prerequisites

A future implementation may begin only after these existing layers remain present and guarded:

```
seal_agentic_automation_metadata_present=1
seal_parameter_schema_metadata_present=1
seal_request_freshness_metadata_present=1
seal_signed_request_metadata_present=1
seal_policy_decision_metadata_present=1
seal_policy_decision_report_surface_present=1
seal_runtime_gate_metadata_present=1
seal_core_blocked_case_set_complete=1
```

Future record shape

A future dry-run record should expose bounded metadata fields similar to:

```
runtime_dry_run_profile=latticra-seal-runtime-dry-run/0.1
input_policy_decision_present
input_runtime_gate_present
dry_run_supported
dry_run_performed
request_class
policy_decision_state
runtime_gate_state
default_action
would_allow
would_deny
would_require_operator_review
would_execute_tool
would_read_host
would_write_host
would_use_network
would_grant_runtime_authority
blocked_reason
report_only
mode
status
```

Initial required defaults

The initial implementation, if added later, must default to:

```
dry_run_supported=1
dry_run_performed=1
default_action=deny
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
report_only=1
mode=report-only
status=runtime-dry-run-metadata
```

Required denied cases

The dry-run surface must preserve the existing core blocked-request vocabulary:

```
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
```

Forbidden behavior

The dry-run layer must not:

```
evaluate real policy files
enforce policy
execute tools
execute shell commands
call a runtime executor
read host files
write host files
perform network operations
verify signatures
validate freshness against live time
perform replay-cache mutation
generate keys
load private keys
load trust stores
query revocation services
grant runtime authority
claim production readiness
claim AI-agent security
claim MCP implementation
```

Required report surface

A future implementation must include a deterministic report renderer and a local fixture runner.

Required future files:

```
include/latticra/seal_runtime_dry_run.h
src/seal_runtime_dry_run.c
tests/seal_runtime_dry_run_invariants.c
tests/seal_runtime_dry_run_report_surface.c
scripts/test-latticra-seal-runtime-dry-run.sh
scripts/latticra-seal-runtime-dry-run-report.sh
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION.md
```

Required tests

A future implementation must prove:

```
valid report-only policy decision input produces denied dry-run metadata
valid report-only runtime gate input remains report-only
unknown tool remains denied
unsigned request remains denied
invalid schema remains denied
stale request remains denied
replayed request remains denied
invalid signature remains denied
no runtime authority is granted
no host read is performed
no host write is performed
no network action is performed
no tool execution is performed
report rendering is deterministic
small buffers fail closed
null inputs fail closed
invalid upstream metadata fails closed
```

Promotion rule

This contract does not authorize runtime enforcement.

Runtime enforcement may be considered only after a dry-run implementation, dry-run report surface, status record, status-index alignment, and negative-case evidence all remain merged and guarded.

Boundary

This is a contract-only planning slice.

It does not change implementation behavior, add runtime behavior, grant authority, or change public readiness.

Current next valid slice

The next valid slice is a no-effect runtime dry-run implementation plan.

That future slice must still avoid implementation behavior and must specify exact structs, fields, APIs, report format, fixtures, tests, and failure behavior before any C code is added.

docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION.md

Source title: Latticra Seal Runtime Dry-Run Implementation

Lines: 156 | Words: 561 | SHA-256: 3085cb5bd4948f6ec7d585568eb1378096faadc62f0dc19d426abb1ec17ec53f

Latticra Seal Runtime Dry-Run Implementation

Status: initial no-effect runtime dry-run metadata implementation Scope: bounded C metadata surface for Latticra Seal runtime dry-run posture after the policy decision report surface and runtime gate metadata path. This implementation does not add runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This document records the first Latticra Seal runtime dry-run metadata implementation.

The implementation accepts report-only policy decision metadata and report-only runtime gate metadata, then renders deterministic metadata describing what the boundary would deny without doing it.

Added files

```
include/latticra/seal_runtime_dry_run.h
src/seal_runtime_dry_run.c
tests/seal_runtime_dry_run_invariants.c
scripts/test-latticra-seal-runtime-dry-run.sh
```

API summary

The dry-run metadata surface adds:

```
latticra_seal_runtime_dry_run_t
latticra_seal_runtime_dry_run_error_t
latticra_seal_runtime_dry_run_error_label
latticra_seal_runtime_dry_run_from_policy_and_gate
latticra_seal_runtime_dry_run_is_report_only
latticra_seal_runtime_dry_run_report
```

Metadata behavior

The implementation:

```
accepts report-only policy decision metadata
accepts report-only runtime gate metadata
sets runtime_dry_run_profile=latticra-seal-runtime-dry-run/0.1
sets request_class=core-blocked-request
sets policy_decision_state=report-only
sets runtime_gate_state=report-only
sets blocked_reason=default-deny-dry-run
sets status=runtime-dry-run-metadata
sets dry_run_supported=1
sets dry_run_performed=1
sets input_policy_decision_present=1
sets input_runtime_gate_present=1
sets policy_decision_report_only=1
```

```
sets runtime_gate_report_only=1
sets default_action_deny=1
sets would_allow=0
sets would_deny=1
sets would_require_operator_review=1
sets would_execute_tool=0
sets would_read_host=0
sets would_write_host=0
sets would_use_network=0
sets would_grant_runtime_authority=0
sets unknown_tool_denied=1
sets unsigned_request_denied=1
sets invalid_schema_denied=1
sets stale_request_denied=1
sets replayed_request_denied=1
sets invalid_signature_denied=1
sets report_only=1
sets mode=report-only
renders deterministic runtime dry-run metadata
```

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null policy decision input -&gt; invalid-input
null runtime gate input -&gt; invalid-input
invalid policy decision metadata -&gt; invalid-policy-decision
invalid runtime gate metadata -&gt; invalid-runtime-gate
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not evaluate policies, contact networks, read host files, write host files, execute tools, enforce capabilities, or grant authority.

Invariants

The invariant test verifies:

```
valid report-only inputs produce dry-run metadata
runtime_dry_run_profile is stable
request_class remains core-blocked-request
policy_decision_state remains report-only
runtime_gate_state remains report-only
blocked_reason remains default-deny-dry-run
dry_run_supported remains one
dry_run_performed remains one
input policy decision remains present
input runtime gate remains present
policy decision remains report-only
runtime gate remains report-only
default action remains deny
would_allow remains zero
would_deny remains one
would_require_operator_review remains one
would_execute_tool remains zero
would_read_host remains zero
would_write_host remains zero
would_use_network remains zero
would_grant_runtime_authority remains zero
all core denied case flags remain one
report_only remains one
mode remains report-only
status remains runtime-dry-run-metadata
report rendering is deterministic
small report buffers fail closed
null inputs fail closed
invalid upstream metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-runtime-dry-run-contract.sh
sh scripts/test-latticra-seal-runtime-dry-run-implementation-plan.sh
sh scripts/test-latticra-seal-runtime-dry-run.sh
```

Expected new output:

```
seal runtime dry-run invariants: ok
```

Boundary

This implementation is metadata-only.

It does not execute tools, execute shell commands, read host files, write host files, use the network, evaluate external policies, load policy files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, claim production readiness, claim AI-agent security, or claim MCP implementation.

Claim boundary

This implementation does not justify the public claim that Latticra secures AI agents.

It moves the project closer by adding a no-effect dry-run layer that can describe what would be denied before any runtime enforcement path is considered.

Next valid slice

The next valid Latticra Seal slice is a runtime dry-run report surface or runtime dry-run status alignment.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION_PLAN.md

Source title: Latticra Seal Runtime Dry-Run Implementation Plan

Lines: 275 | Words: 653 | SHA-256: 46ac30c12df022848b6046dafe6bb928cdb9baa60f4e34be4f087b9b60410b69

Latticra Seal Runtime Dry-Run Implementation Plan

Status: implementation planning contract for a future no-effect Latticra Seal runtime dry-run surface Scope: implementation plan only after the Latticra Seal runtime dry-run contract. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This plan defines the exact future implementation shape for a no-effect Seal runtime dry-run layer.

The future implementation should render what the Seal boundary would deny without executing tools, contacting networks, touching host state, or granting authority.

Required contract

The implementation plan depends on:

```
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_CONTRACT.md
scripts/test-latticra-seal-runtime-dry-run-contract.sh
```

The contract must remain merged and guarded before implementation code is added.

Future files

The future implementation slice should add:

```
include/latticra/seal_runtime_dry_run.h
src/seal_runtime_dry_run.c
```

```
tests/seal_runtime_dry_run_invariants.c
scripts/test-latticra-seal-runtime-dry-run.sh
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION.md
```

A later report-surface slice may add:

```
tests/seal_runtime_dry_run_report_surface.c
scripts/latticra-seal-runtime-dry-run-report.sh
scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
```

Header API plan

The future header should define:

```
LATTICRA_SEAL_RUNTIME_DRY_RUN_PROFILE_MAX
LATTICRA_SEAL_RUNTIME_DRY_RUN_STATE_MAX
LATTICRA_SEAL_RUNTIME_DRY_RUN_REASON_MAX
LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_MAX
latticra_seal_runtime_dry_run_error_t
latticra_seal_runtime_dry_run_t
latticra_seal_runtime_dry_run_error_label
latticra_seal_runtime_dry_run_from_policy_and_gate
latticra_seal_runtime_dry_run_is_report_only
latticra_seal_runtime_dry_run_report
```

Error model

The future error enum should include:

```
LATTICRA_SEAL_RUNTIME_DRY_RUN_OK
LATTICRA_SEAL_RUNTIME_DRY_RUN_INVALID_INPUT
LATTICRA_SEAL_RUNTIME_DRY_RUN_INVALID_POLICY_DECISION
LATTICRA_SEAL_RUNTIME_DRY_RUN_INVALID_RUNTIME_GATE
```

Record fields

The future record should include:

```
runtime_dry_run_profile
request_class
policy_decision_state
runtime_gate_state
blocked_reason
status
dry_run_supported
dry_run_performed
input_policy_decision_present
input_runtime_gate_present
policy_decision_report_only
runtime_gate_report_only
default_action_deny
would_allow
would_deny
would_require_operator_review
would_execute_tool
would_read_host
would_write_host
would_use_network
would_grant_runtime_authority
unknown_tool_denied
unsigned_request_denied
invalid_schema_denied
stale_request_denied
replayed_request_denied
invalid_signature_denied
report_only
mode
error
```

Initial constants

The initial future implementation should emit:

```
runtime_dry_run_profile=latticra-seal-runtime-dry-run/0.1
request_class=core-blocked-request
policy_decision_state=report-only
runtime_gate_state=report-only
blocked_reason=default-deny-dry-run
status=runtime-dry-run-metadata
```

```
dry_run_supported=1
dry_run_performed=1
input_policy_decision_present=1
input_runtime_gate_present=1
policy_decision_report_only=1
runtime_gate_report_only=1
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
report_only=1
mode=report-only
```

Function behavior

The future builder should:

```
accept a report-only Seal policy decision metadata input
accept a report-only Seal runtime gate metadata input
reject null output
return invalid-input for null upstream inputs
return invalid-policy-decision for non-report-only policy decision metadata
return invalid-runtime-gate for non-report-only runtime gate metadata
copy no authority from upstream inputs
render deterministic metadata only
```

Required report format

The report should begin with:

```
LATTICRA SEAL RUNTIME DRY RUN
```

It should render all fields as stable key=value lines.

Required report fields:

```
runtime_dry_run_profile=latticra-seal-runtime-dry-run/0.1
request_class=core-blocked-request
policy_decision_state=report-only
runtime_gate_state=report-only
blocked_reason=default-deny-dry-run
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
report_only=1
mode=report-only
status=runtime-dry-run-metadata
```

Required invariant tests

The future invariant test should verify:

```
valid report-only inputs produce dry-run metadata
runtime_dry_run_profile is stable
request_class is core-blocked-request
policy_decision_state remains report-only
runtime_gate_state remains report-only
blocked_reason is default-deny-dry-run
dry_run_supported remains one
dry_run_performed remains one
would_allow remains zero
would_deny remains one
would_require_operator_review remains one
```

```
would_execute_tool remains zero
would_read_host remains zero
would_write_host remains zero
would_use_network remains zero
would_grant_runtime_authority remains zero
all core denied case flags remain one
report_only remains one
mode remains report-only
status remains runtime-dry-run-metadata
small report buffers fail closed
null inputs fail closed
invalid upstream metadata fails closed
```

Required build runner

The future test runner should compile only local C sources and fixtures.

Planned runner:

```
sh scripts/test-latticra-seal-runtime-dry-run.sh
```

It should compile:

```
src/seal_runtime_dry_run.c
src/seal_runtime_gate.c
src/seal_policy_decision.c
src/seal_signed_request.c
src/seal_request_freshness.c
src/seal_parameter_schema.c
src/seal_agentic_automation_security.c
src/seal_status_rollup.c
tests/seal_runtime_dry_run_invariants.c
```

Explicit non-goals

The future implementation must not:

```
execute tools
execute shell commands
read host files
write host files
use the network
evaluate external policies
load policy files
verify signatures
validate freshness against live time
mutate replay caches
grant authority
claim production readiness
claim AI-agent security
claim MCP implementation
```

Promotion rule

No runtime enforcement work may begin from this plan alone.

A future C implementation must first be merged with tests, a report surface, status record, and negative-case evidence while preserving the report-only boundary.

Current next valid slice

The next valid Latticra Seal slice is the no-effect runtime dry-run implementation.

That future slice must implement only deterministic metadata, local fixtures, fail-closed behavior, and report rendering. It must not perform effects or grant authority.

docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md

Source title: Latticra Seal Runtime Dry-Run Report Surface

Lines: 98 | Words: 341 | SHA-256: be92f42763a7cc4d930e8e1da8f80ddc7bde29deaa994881e0c82eaf1184aaf4

Latticra Seal Runtime Dry-Run Report Surface

Status: report surface for the Latticra Seal runtime dry-run metadata layer Scope: deterministic local report surface after the no-effect runtime dry-run metadata implementation. This document does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This document records the first operator-visible report surface for Latticra Seal runtime dry-run metadata.

The report surface renders what the dry-run layer would deny without executing tools, touching host state, using the network, or granting authority.

Added files

```
tests/seal_runtime_dry_run_report_surface.c
scripts/latticra-seal-runtime-dry-run-report.sh
```

Report runner

```
sh scripts/latticra-seal-runtime-dry-run-report.sh
```

Expected report posture

The report surface renders the current dry-run posture, including:

```
LATTICRA SEAL RUNTIME DRY RUN
runtime_dry_run_profile=latticra-seal-runtime-dry-run/0.1
request_class=core-blocked-request
policy_decision_state=report-only
runtime_gate_state=report-only
blocked_reason=default-deny-dry-run
dry_run_supported=1
dry_run_performed=1
input_policy_decision_present=1
input_runtime_gate_present=1
policy_decision_report_only=1
runtime_gate_report_only=1
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
report_only=1
mode=report-only
status=runtime-dry-run-metadata
```

Boundary

This report surface compiles and runs a local deterministic fixture only.

It does not execute tools, execute shell commands, read host files, write host files, use the network, evaluate external policies, load policy files, verify signatures, validate freshness against live time, mutate replay caches, grant authority, claim production readiness, claim AI-agent security, or claim MCP implementation.

Validation

Run:

```
sh scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
```

Expected output:

```
latticra seal runtime dry-run report surface: ok
```

The underlying runtime dry-run implementation remains covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run.sh
```

Claim boundary

This report surface does not justify the public claim that Latticra secures AI agents.

It makes the dry-run denial posture easier to inspect before any future runtime enforcement path is considered.

Next valid slice

The next valid Latticra Seal slice is runtime dry-run report status alignment.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/LATTICRA_SEAL_RUNTIME_ENFORCEMENT_GATE_CONTRACT.md

Source title: Latticra Seal Runtime Enforcement Gate Contract

Lines: 299 | Words: 870 | SHA-256: 85d94ab9403c56ccfc65adc99875510cf465ec6099cae836e0da006fc765c7a2

Latticra Seal Runtime Enforcement Gate Contract

Status: Latticra Seal runtime enforcement gate contract Scope: contract for future runtime enforcement-gate metadata after the report-only Seal policy decision metadata layer. This document does not implement runtime enforcement, allow/deny enforcement, runtime execution, runtime authority, effect execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, public-key parsing, trust-store loading, freshness validation, replay detection, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, or operating-system behavior.

Purpose

This document defines the Latticra Seal runtime enforcement gate contract.

The purpose is to prepare a bounded, evidence-bound gate layer that will eventually consume policy-decision metadata and report whether a requested action would remain blocked, require operator review, or be eligible for future runtime handoff.

This document does not implement runtime enforcement behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
```

```
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md
include/latticra/seal_policy_decision.h
src/seal_policy_decision.c
tests/seal_policy_decision_invariants.c
scripts/test-latticra-seal-policy-decision-contract.sh
scripts/test-latticra-seal-policy-decision.sh
```

The existing policy decision metadata surface remains report-only and default-deny.

Runtime enforcement gate boundary

The runtime enforcement gate layer may describe future enforcement-gate metadata, blocked-state metadata, operator-review metadata, and eligibility metadata.

The layer may not enforce allow or deny outcomes, execute tools, execute shell commands, call runtime components, contact networks, read host files, write host files, verify signatures, validate freshness, detect replay, or mutate system state.

Allowed in this contract slice:

```
runtime-enforcement-gate vocabulary
gate-id metadata planning
gate-version metadata planning
gate-state metadata planning
policy-decision input planning
default-blocked posture planning
operator-review-required planning
unknown-tool blocked planning
unsigned-request blocked planning
invalid-schema blocked planning
stale-request blocked planning
replayed-request blocked planning
invalid-signature blocked planning
runtime-handoff eligibility planning
negative-test planning
report-only status planning
non-claims
static guard validation
```

Forbidden in this contract slice:

```
runtime enforcement
allow enforcement
deny enforcement
runtime execution
AI agent execution
model execution
tool execution
shell command execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
signature verification
freshness validation
replay detection
policy evaluation
policy enforcement
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
MCP server implementation
MCP client implementation
```

Initial runtime enforcement gate posture

The initial policy is report-only, blocked, and deny-by-default.

```
seal_runtime_enforcement_gate_contract_present=1
runtime_enforcement_gate_supported=0
runtime_enforcement_supported=0
runtime_enforcement_active=0
policy_decision_input_supported=0
policy_decision_consumed=0
gate_id_present=0
gate_version_present=0
gate_state=report-only
runtime_handoff_eligible=0
runtime_handoff_performed=0
allow_enforcement_supported=0
deny_enforcement_supported=0
allow_enforcement_performed=0
deny_enforcement_performed=0
effect_performed=0
default_blocked=1
operator_review_required=1
unknown_tool_blocked=1
unsigned_request_blocked=1
invalid_schema_blocked=1
stale_request_blocked=1
replayed_request_blocked=1
invalid_signature_blocked=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=contract-only
status=runtime-enforcement-gate-contract-only
```

Planned metadata fields

Future runtime enforcement gate metadata should be bounded and deterministic.

Planned fields:

```
runtime_enforcement_gate_profile
gate_id
gate_version
gate_state
runtime_enforcement_gate_supported
runtime_enforcement_supported
runtime_enforcement_active
policy_decision_input_supported
policy_decision_consumed
gate_id_present
gate_version_present
runtime_handoff_eligible
runtime_handoff_performed
allow_enforcement_supported
deny_enforcement_supported
allow_enforcement_performed
deny_enforcement_performed
effect_performed
default_blocked
operator_review_required
unknown_tool_blocked
unsigned_request_blocked
invalid_schema_blocked
stale_request_blocked
replayed_request_blocked
invalid_signature_blocked
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
decision
reason
status
```

Initial values before runtime enforcement implementation:

```
runtime_enforcement_gate_profile=latticra-seal-runtime-enforcement-gate/0.1
```

```
gate_id=unset
gate_version=unset
gate_state=report-only
runtime_enforcement_gate_supported=0
runtime_enforcement_supported=0
runtime_enforcement_active=0
policy_decision_input_supported=0
policy_decision_consumed=0
gate_id_present=0
gate_version_present=0
runtime_handoff_eligible=0
runtime_handoff_performed=0
allow_enforcement_supported=0
deny_enforcement_supported=0
allow_enforcement_performed=0
deny_enforcement_performed=0
effect_performed=0
default_blocked=1
operator_review_required=1
unknown_tool_blocked=1
unsigned_request_blocked=1
invalid_schema_blocked=1
stale_request_blocked=1
replayed_request_blocked=1
invalid_signature_blocked=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=contract-only
decision=report-only
reason=runtime-enforcement-gate-contract-only
status=runtime-enforcement-gate-contract-only
```

Negative-test requirements before enforcement claims

Before any future claim that Latticra secures AI-agent tool execution boundaries, the repository must include negative tests showing that the enforcement path denies or blocks:

```
unknown tools
unsigned requests
invalid schemas
stale requests
replayed requests
invalid signatures
requests without operator approval when approval is required
requests attempting host reads without declared authority
requests attempting host writes without declared authority
requests attempting network access without declared authority
```

Those tests must remain separate from this contract slice.

Promotion rules

The next implementation after this contract may only add bounded C metadata fields and deterministic rendering.

It must not evaluate real policies, enforce allow or deny outcomes, execute tools, execute shell commands, call runtime components, read host files, write host files, contact networks, verify signatures, validate freshness, detect replay, enforce capabilities, generate receipts, or grant authority.

Claim gate

Latticra must not claim to secure AI agents from this contract alone.

A future claim that Latticra secures AI-agent tool execution boundaries requires, at minimum:

```
request identity metadata implemented
parameter schema metadata implemented
request freshness metadata implemented
signed request metadata implemented
policy decision metadata implemented
runtime enforcement gate implemented
```

```
negative tests for denied unknown tools
negative tests for denied unsigned manifests
negative tests for denied stale requests
negative tests for denied replayed requests
operator-visible evidence report implemented
```

Until those are implemented and validated, the accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Non-claims

This contract must not be read as implementation of runtime enforcement, tool security enforcement, cryptographic enforcement, or AI-agent security.

Required non-claim posture:

```
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
allow_enforcement_implemented=0
deny_enforcement_implemented=0
signed_request_verification_implemented=0
freshness_validation_implemented=0
replay_detection_implemented=0
mcp_tool_security_enforced=0
ai_agent_security_claimed=0
runtime_authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is report-only runtime enforcement gate metadata.

That future slice may add a bounded metadata structure, deterministic rendering, invariant tests, and a test runner.

It must not change runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, policy enforcement, runtime enforcement, or authority grants.

docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md

Source title: Latticra Seal Runtime Gate Implementation

Lines: 157 | Words: 618 | SHA-256: 6b5288f23624b8dea0e5501a79d9f9c4ff386f75397d7c9d444dd7d4395ff33c

Latticra Seal Runtime Gate Implementation

Status: initial report-only runtime gate metadata implementation Scope: bounded C metadata surface for runtime gate posture after the report-only Seal policy decision layer. This slice does not implement runtime enforcement, allow/deny enforcement, runtime execution, runtime authority, effect execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, public-key parsing, trust-store loading, freshness validation, replay detection, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, or operating-system behavior.

Purpose

This document records the first Latticra Seal runtime gate metadata implementation.

The implementation accepts an existing report-only Seal policy decision metadata record and produces deterministic report-only runtime gate metadata.

Added files

```
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
scripts/test-latticra-seal-runtime-gate.sh
```

API summary

The runtime gate metadata surface adds:

```
latticra_seal_runtime_gate_t
latticra_seal_runtime_gate_error_t
latticra_seal_runtime_gate_error_label
latticra_seal_runtime_gate_from_policy_decision
latticra_seal_runtime_gate_is_report_only
latticra_seal_runtime_gate_report
```

Metadata behavior

The implementation:

```
accepts a valid report-only Seal policy decision metadata record
sets runtime_enforcement_gate_profile=latticra-seal-runtime-enforcement-gate/0.1
sets gate_id=unset
sets gate_version=unset
sets gate_state=report-only
sets runtime_enforcement_gate_supported=0
sets runtime_enforcement_supported=0
sets runtime_enforcement_active=0
sets policy_decision_input_supported=0
sets policy_decision_consumed=0
sets gate_id_present=0
sets gate_version_present=0
sets runtime_handoff_eligible=0
sets runtime_handoff_performed=0
sets allow_enforcement_supported=0
sets deny_enforcement_supported=0
sets allow_enforcement_performed=0
sets deny_enforcement_performed=0
sets effect_performed=0
sets default_blocked=1
sets operator_review_required=1
sets unknown_tool_blocked=1
sets unsigned_request_blocked=1
sets invalid_schema_blocked=1
sets stale_request_blocked=1
sets replayed_request_blocked=1
sets invalid_signature_blocked=1
copies runtime_authority_granted from policy decision metadata
copies host_read_performed from policy decision metadata
copies host_write_performed from policy decision metadata
copies network_performed from policy decision metadata
sets mode=report-only
sets decision=report-only
sets reason=runtime-gate-metadata-only
renders deterministic runtime gate metadata
```

Boundary

This implementation does not enforce allow outcomes, enforce deny outcomes, execute tools, execute shell commands, call runtime components, contact networks, read host files, write host files, verify signatures, validate freshness, detect replay, enforce capabilities, perform effects, or grant authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null policy decision metadata input -&gt; invalid-input
invalid policy decision metadata -&gt; invalid-policy-decision
non-report-only policy decision metadata -&gt; invalid-policy-decision
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not evaluate policy, contact networks, read host files, write host files, execute tools, enforce capabilities, or grant authority.

Invariants

The invariant test verifies:

```
valid report-only policy decision metadata produces deterministic runtime gate metadata
runtime_enforcement_gate_profile is stable
gate_id remains unset
gate_version remains unset
gate_state remains report-only
runtime_enforcement_gate_supported remains zero
runtime_enforcement_supported remains zero
runtime_enforcement_active remains zero
policy_decision_input_supported remains zero
policy_decision_consumed remains zero
runtime_handoff_eligible remains zero
runtime_handoff_performed remains zero
allow_enforcement_supported remains zero
deny_enforcement_supported remains zero
allow_enforcement_performed remains zero
deny_enforcement_performed remains zero
effect_performed remains zero
default_blocked remains one
operator_review_required remains one
unknown_tool_blocked remains one
unsigned_request_blocked remains one
invalid_schema_blocked remains one
stale_request_blocked remains one
replayed_request_blocked remains one
invalid_signature_blocked remains one
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
mode remains report-only
decision remains report-only
reason remains runtime-gate-metadata-only
small report buffer fails closed
null inputs fail closed
invalid policy decision metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-runtime-gate-contract.sh
sh scripts/test-latticra-seal-runtime-gate.sh
```

Claim boundary

This implementation still does not justify the public claim that Latticra secures AI agents.

It moves the project closer by adding report-only runtime gate metadata, but the stronger claim requires negative tests for denied unknown, unsigned, stale, and replayed requests plus a guarded enforcement path that actually blocks effects.

Next valid slice

The next valid Latticra Seal slice is negative-test planning for blocked request classes or a runtime gate report surface.

That future slice must not implement runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, policy enforcement, runtime enforcement, or authority grants unless a specific contract and validation path justify it.

docs/LATTICRA_SEAL_RUNTIME_GATE_TEST_PLAN.md

Source title: Latticra Seal Runtime Gate Test Plan

Lines: 96 | Words: 270 | SHA-256: 7d93235faa55403b8b4a2084addc707d1423d48eb30a78c9450f8575e9ad6435

Latticra Seal Runtime Gate Test Plan

Status: runtime gate test planning record Scope: planning document after the report-only Seal runtime gate metadata layer. This document does not implement runtime behavior, policy behavior, tool behavior, host behavior, network behavior, cryptographic behavior, or authority grants.

Purpose

This document names the future test cases required before the Seal runtime gate can move beyond report-only metadata.

The current runtime gate remains metadata-only and blocked by default.

Required predecessors

```
docs/LATTICRA_SEAL_RUNTIME_ENFORCEMENT_GATE_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
scripts/test-latticra-seal-runtime-gate-contract.sh
scripts/test-latticra-seal-runtime-gate.sh
```

Planned case coverage

Future tests should cover these request classes:

```
unknown_tool_case_planned=1
unsigned_request_case_planned=1
invalid_schema_case_planned=1
stale_request_case_planned=1
replayed_request_case_planned=1
invalid_signature_case_planned=1
missing_operator_approval_case_planned=1
host_read_without_authority_case_planned=1
host_write_without_authority_case_planned=1
network_without_authority_case_planned=1
```

Planned sequence

```
1. unknown tool case
2. unsigned request case
3. invalid schema case
4. stale request case
5. replayed request case
6. invalid signature case
7. missing operator approval case
8. host read authority case
9. host write authority case
10. network authority case
```

Expected future posture

All planned cases should keep these values until a later implementation contract changes them:

```
expected_runtime_authority_granted=0
expected_effect_performed=0
expected_host_read_performed=0
expected_host_write_performed=0
expected_network_performed=0
```

Claim gate

This plan alone does not justify the public claim that Latticra secures AI agents.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

A stronger claim requires implemented and validated gate behavior plus case tests for unknown, unsigned, stale, and replayed requests.

Non-claims

```
runtime_gate_case_tests_implemented=0
runtime_behavior_implemented=0
policy_behavior_implemented=0
tool_behavior_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is the first case-specific contract, beginning with the unknown tool case.

docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md

Source title: Latticra Seal Runtime Handoff Contract

Lines: 219 | Words: 633 | SHA-256: a04c6f94cd6f94ad9282815d311dca1158189355627d76071e1d8bed5143e5c9

Latticra Seal Runtime Handoff Contract

Status: Latticra Seal runtime handoff contract Scope: contract for future runtime handoff metadata after effect decision metadata. This document does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal runtime handoff layer.

The purpose of this layer is to decide how a future effect decision may be represented at the runtime boundary before any runtime component is allowed to act on it.

This document does not implement runtime handoff behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
include/latticra/seal_signature.h
include/latticra/seal_verification_policy.h
include/latticra/seal_verification_receipt.h
include/latticra/seal_capability_gate.h
```

```
include/latticra/seal_effect_decision.h
src/seal_measurement.c
src/seal_manifest.c
src/seal_signature_policy.c
src/seal_signature.c
src/seal_verification_policy.c
src/seal_verification_receipt.c
src/seal_capability_gate.c
src/seal_effect_decision.c
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
scripts/test-latticra-seal-signature.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-capability-gate.sh
scripts/test-latticra-seal-effect-decision-contract.sh
scripts/test-latticra-seal-effect-decision.sh
```

The effect decision metadata surface remains denied metadata until runtime handoff metadata exists and is guarded.

Handoff boundary

No effect decision becomes runtime behavior.

No denied decision may produce an active runtime handoff.

No runtime handoff may perform host reads, host writes, network access, kernel interaction, or authority grants until a later runtime-boundary implementation explicitly permits and guards those behaviors.

Allowed in this contract slice:

```
runtime handoff vocabulary
handoff-state labels
decision metadata planning
runtime-boundary label planning
no-runtime posture planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this contract slice:

```
runtime execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
revocation lookup
object sealing
kernel interaction
```

Initial handoff policy

The initial runtime handoff policy is denied by default.

```
denied decision metadata may be inspected
effect_allowed=0 must deny handoff activation
effect_performed=0 must remain zero
runtime_authority_granted=0 must remain zero
handoff_active must remain zero
runtime_effect_performed must remain zero
unknown boundary target must deny
unknown runtime request must deny
```

The first implementation after this contract may only add metadata fields and denied-state handling. It must not perform runtime behavior.

Planned handoff states

Future records should use explicit labels:

```
handoff_state=unsupported
handoff_state=denied-decision
handoff_state=denied-runtime-boundary
handoff_state=denied-policy
handoff_state=inactive-metadata
handoff_state=active-runtime-gated
```

For the next implementation, the expected state is:

```
handoff_state=denied-decision
```

Planned handoff fields

A future runtime handoff record should be bounded and deterministic.

Planned fields:

```
handoff_profile
decision_profile
gate_profile
requested_capability
requested_effect
requested_scope
decision_state
effect_allowed
effect_performed
runtime_boundary_state
runtime_request_label
handoff_active
runtime_effect_performed
host_read_performed
host_write_performed
network_performed
runtime_authority_granted
status
```

Initial values before real runtime behavior:

```
effect_allowed=0
effect_performed=0
runtime_boundary_state=disabled
handoff_active=0
runtime_effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_authority_granted=0
handoff_state=denied-decision
status=runtime-handoff-contract-only
```

Failure behavior

Future runtime handoff handling must fail closed.

Required failure states:

```
null effect decision metadata -&gt; invalid
invalid effect decision metadata -&gt; invalid
missing requested capability -&gt; denied
missing requested effect -&gt; denied
unknown runtime boundary -&gt; denied
unknown runtime request -&gt; denied
effect_allowed=0 -&gt; denied-decision
runtime boundary disabled -&gt; denied-runtime-boundary
runtime authority request -&gt; rejected
```

Failures must not parse keys, create keys, persist secrets, contact networks, query revocation status, verify records, sign records, read host files, write host files, enforce capabilities, execute effects, call runtime components, or grant runtime

authority.

Promotion rule

This contract permits only the next implementation slice:

```
runtime handoff metadata implementation
```

It does not permit runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt generation, key handling, trust-store behavior, revocation lookup, object sealing, network behavior, host reads, host writes, or kernel behavior.

After runtime handoff metadata exists and is guarded, the next valid planning slice is a Seal status rollup contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-runtime-handoff-contract.sh
```

docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md

Source title: Latticra Seal Runtime Handoff Evaluation Contract

Lines: 185 | Words: 540 | SHA-256: ba70e27573e74afccc4303d5698f1c14d4c6b9ce5f9b7e78f3bde08f186ea215

Latticra Seal Runtime Handoff Evaluation Contract

Status: Latticra Seal runtime handoff evaluation contract Scope: contract for a future runtime handoff evaluation step after verified effect decision metadata. This document does not implement runtime handoff, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, signing, key generation, private-key storage, trust-store loading, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal runtime handoff evaluation boundary that may consume an allowed metadata-only verified effect decision.

The purpose of this layer is to decide whether a previously allowed metadata-only effect decision is eligible to be described as a runtime handoff candidate before any future runtime bridge exists.

The evaluation is handoff classification, not runtime handoff.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_IMPLEMENTATION.md
include/latticra/seal_verified_effect_decision.h
src/seal_verified_effect_decision.c
tests/seal_verified_effect_decision_invariants.c
scripts/test-latticra-seal-verified-effect-decision.sh
```

The verified effect decision metadata surface remains the source of effect-decision evidence for this runtime handoff evaluation.

Evaluation boundary

Allowed in the next implementation slice:

```
accept verified effect decision metadata
require effect_allowed=1
require decision_state=allowed-report-only
require decision_state=allowed-evaluate-only
require runtime_authority_granted=0
require effect_performed=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested handoff metadata
classify report-only as handoff_state=eligible-report-only
classify evaluate-only as handoff_state=eligible-evaluate-only
produce deterministic runtime handoff evaluation metadata
```

Forbidden in the next implementation slice:

```
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
trust-store loading
revocation lookup
key generation
private-key handling
object sealing
kernel interaction
```

Initial handoff policy

Allowed input decision states:

```
decision_state=allowed-report-only
decision_state=allowed-evaluate-only
```

Allowed requested handoff labels:

```
report-only
evaluate-only
```

Planned handoff states:

```
handoff_state=eligible-report-only
handoff_state=eligible-evaluate-only
handoff_state=denied-decision
handoff_state=denied-effect
handoff_state=denied-runtime-authority
handoff_state=denied-host-effect
handoff_state=denied-network-effect
```

The first implementation may set:

```
handoff_eligible=1
```

only for metadata-only report/evaluate decisions.

Even when handoff_eligible=1, these must remain zero:

```
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future runtime handoff evaluation record should be bounded and deterministic.

Planned fields:

```
handoff_profile
decision_profile
gate_profile
receipt_profile
requested_capability
```

```
requested_effect
requested_handoff
requested_scope
decision_state
effect_allowed
handoff_state
handoff_eligible
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
status
```

Failure behavior

Future runtime handoff evaluation handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null verified effect decision -&gt; invalid
invalid verified effect decision -&gt; denied-decision
effect_allowed=0 -&gt; denied-effect
decision_state not allowed-report-only or allowed-evaluate-only -&gt; denied-decision
missing requested handoff -&gt; denied-effect
unknown requested handoff -&gt; denied-effect
runtime authority already granted -&gt; denied-runtime-authority
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
runtime handoff evaluation metadata implementation
```

It does not permit runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, trust-store behavior, revocation lookup, key handling, object sealing, or kernel behavior.

After runtime handoff evaluation metadata exists and is guarded, the next valid planning slice is a runtime handoff report surface that still performs no runtime handoff.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-runtime-handoff-evaluation-contract.sh
```

docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md

Source title: Latticra Seal Runtime Handoff Evaluation Implementation

Lines: 151 | Words: 521 | SHA-256: 753ce70761f0d23c84cb65e836856ee9fe0d317b1d6bab9ce86b711065db5519

Latticra Seal Runtime Handoff Evaluation Implementation

Status: initial runtime handoff evaluation metadata implementation Scope: bounded C metadata surface for classifying allowed metadata-only verified effect decision output into deterministic runtime handoff eligibility. This does not implement runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, trust-store loading, revocation lookup, key generation, private-key handling, object sealing, kernel interaction, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal runtime handoff evaluation metadata implementation.

The implementation consumes verified effect decision metadata and classifies whether the requested metadata-only runtime handoff remains eligible as report-only or evaluate-only.

When the verified effect decision carries crypto graduation metadata, the evaluation copies that evidence forward and requires it to remain passed, standard-aligned, and authority-neutral.

The evaluation is handoff classification, not runtime handoff.

Added files

```
include/latticra/seal_runtime_handoff_evaluation.h
src/seal_runtime_handoff_evaluation.c
tests/seal_runtime_handoff_evaluation_invariants.c
scripts/test-latticra-seal-runtime-handoff-evaluation.sh
```

Required predecessor

This implementation depends on the verified effect decision metadata surface:

```
include/latticra/seal_verified_effect_decision.h
src/seal_verified_effect_decision.c
tests/seal_verified_effect_decision_invariants.c
scripts/test-latticra-seal-verified-effect-decision.sh
docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_IMPLEMENTATION.md
```

API summary

The runtime handoff evaluation metadata surface adds:

```
latticra_seal_runtime_handoff_evaluation_t
latticra_seal_runtime_handoff_evaluation_error_t
latticra_seal_runtime_handoff_evaluation_error_label
latticra_seal_runtime_handoff_evaluation_from_decision
latticra_seal_runtime_handoff_evaluation_is_metadata_only
latticra_seal_runtime_handoff_evaluation_report
```

Evaluation behavior

The implementation:

```
accepts verified effect decision metadata
copies crypto graduation gate metadata when present
requires crypto_graduation_gate_passed=1 when crypto_graduation_gate_present=1
requires standard_expectations_met=1 when crypto_graduation_gate_present=1
requires local_verify_graduated=1 when crypto_graduation_gate_present=1
requires receipt_promotion_graduated=1 when crypto_graduation_gate_present=1
requires authority_promotion_allowed=0 when crypto_graduation_gate_present=1
requires effect_allowed=1
requires decision_state=allowed-report-only or decision_state=allowed-evaluate-only
requires runtime_authority_granted=0
requires effect_performed=0
requires host_read_performed=0
requires host_write_performed=0
requires network_performed=0
accepts requested handoff metadata
classifies report-only as handoff_state=eligible-report-only
```

```
classifies evaluate-only as handoff_state=eligible-evaluate-only
sets handoff_eligible=1 only for metadata-only report/evaluate handoff evaluations
renders deterministic runtime handoff evaluation metadata
```

Crypto-bound handoff evaluation metadata records:

```
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
crypto_graduation_gate_state=graduated-authority-neutral
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
handoff_state=eligible-report-only
handoff_eligible=1
handoff_performed=0
runtime_authority_granted=0
```

Effect and runtime boundary

Even when handoff_eligible=1, these fields remain zero:

```
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

This implementation does not perform runtime handoff, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, persist policy, load trust stores, look up revocation status, generate keys, handle private keys, seal objects, or interact with the kernel.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null verified effect decision -&gt; invalid-input
invalid verified effect decision -&gt; invalid-decision
failed crypto graduation gate evidence -&gt; denied-crypto-graduation-gate
authority-bearing crypto graduation evidence -&gt; denied-crypto-graduation-gate
effect_allowed=0 -&gt; denied-effect
decision_state not allowed-report-only or allowed-evaluate-only -&gt; denied-decision
missing requested handoff -&gt; missing-requested-handoff
unknown requested handoff -&gt; denied-unknown-handoff
runtime authority already granted -&gt; denied-runtime-authority
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not perform handoff, perform effects, read host files, write host files, contact networks, or grant runtime authority.

Validation

Run locally:

```
sh scripts/test-latticra-seal-runtime-handoff-evaluation.sh
```

Expected output:

```
seal runtime handoff evaluation invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is a runtime handoff report surface from eligible crypto-graduation-gated metadata-only runtime handoff evaluation.

That next slice should remain contract-first and should not perform runtime handoff unless separately implemented and guarded.

docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md

Source title: Latticra Seal Runtime Handoff Implementation

Lines: 134 | Words: 511 | SHA-256: f2d664fbb79e3fe22ba2c9c9dd29d9a7ff2235472fc605a0b78dae75b95662f8

Latticra Seal Runtime Handoff Implementation

Status: initial runtime handoff metadata implementation Scope: bounded C metadata surface for runtime handoff posture after effect decision metadata. This slice does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal runtime handoff metadata implementation.

The implementation accepts an existing effect decision metadata record and produces deterministic runtime handoff metadata. It does not activate a runtime boundary and does not perform effects.

Added files

```
include/latticra/seal_runtime_handoff.h
src/seal_runtime_handoff.c
tests/seal_runtime_handoff_invariants.c
scripts/test-latticra-seal-runtime-handoff.sh
```

API summary

The runtime handoff metadata surface adds:

```
latticra_seal_runtime_handoff_t
latticra_seal_runtime_handoff_error_t
latticra_seal_runtime_handoff_error_label
latticra_seal_runtime_handoff_from_decision
latticra_seal_runtime_handoff_is_inactive_metadata
latticra_seal_runtime_handoff_report
```

Handoff behavior

The implementation:

```
accepts a valid effect decision metadata record
copies decision profile metadata
copies gate profile metadata
copies requested capability metadata
copies requested effect metadata
copies requested scope metadata
copies decision_state
copies effect_allowed
copies effect_performed
sets runtime_boundary_state=disabled
records runtime request label metadata, defaulting to no-runtime-request
sets handoff_active=0
sets runtime_effect_performed=0
sets host_read_performed=0
sets host_write_performed=0
sets network_performed=0
sets runtime_authority_granted=0
sets handoff_state=denied-decision
renders deterministic runtime handoff metadata
```

Boundary

This implementation does not verify signatures, parse public keys, create keys, store keys, contact networks, query revocation status, persist handoff decisions, enforce capabilities, perform host reads, perform host writes, execute network behavior, call runtime components, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null runtime handoff output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null effect decision metadata input -&gt; invalid-input
invalid effect decision metadata -&gt; invalid-decision
missing requested capability -&gt; missing-requested-capability
missing requested effect -&gt; missing-requested-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, read host files, write host files, enforce capabilities, perform effects, call runtime components, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid effect decision metadata produces deterministic runtime handoff metadata
decision profile is copied
gate profile is copied
requested capability is copied
requested effect is copied
requested scope is copied
decision_state is copied
effect_allowed remains zero
effect_performed remains zero
runtime_boundary_state remains disabled
runtime request label is recorded
missing runtime request label defaults to no-runtime-request
handoff_active remains zero
runtime_effect_performed remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
runtime_authority_granted remains zero
handoff_state remains denied-decision
small report buffer fails closed
null inputs fail closed
invalid effect decision metadata fails closed
missing requested capability fails closed
missing requested effect fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-runtime-handoff-contract.sh
sh scripts/test-latticra-seal-runtime-handoff.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
```

Expected output:

```
seal runtime handoff contract: ok
seal runtime handoff invariants: ok
seal runtime handoff status: ok
```

Next valid slice

The next valid Latticra Seal slice is status rollup status/public-entry alignment.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, signing, host behavior, network behavior, or object sealing unless separately implemented and guarded.

docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md

Source title: Latticra Seal Runtime Handoff Report Contract

Lines: 194 | Words: 558 | SHA-256: d2382fbe48a5225e9454aaefe44728467c87772fc1c9e1602ecc83475f9aab7b

Latticra Seal Runtime Handoff Report Contract

Status: Latticra Seal runtime handoff report contract Scope: contract for a future report surface after runtime handoff evaluation metadata. This document does not implement runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, signing, key generation, private-key storage, trust-store loading, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal runtime handoff report boundary that may consume an eligible metadata-only runtime handoff evaluation.

The purpose of this layer is to decide whether a previously eligible metadata-only runtime handoff evaluation may be surfaced as a deterministic report record before any future runtime bridge exists.

The report surface is report classification, not runtime handoff.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md
include/latticra/seal_runtime_handoff_evaluation.h
src/seal_runtime_handoff_evaluation.c
tests/seal_runtime_handoff_evaluation_invariants.c
scripts/test-latticra-seal-runtime-handoff-evaluation.sh
```

The runtime handoff evaluation metadata surface remains the source of eligibility evidence for this report surface.

Report boundary

Allowed in the next implementation slice:

```
accept runtime handoff evaluation metadata
require handoff_eligible=1
require handoff_state=eligible-report-only or handoff_state=eligible-evaluate-only
require handoff_performed=0
require runtime_authority_granted=0
require effect_performed=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested report metadata
classify report-only as report_state=ready-report-only
classify evaluate-only as report_state=ready-evaluate-only
produce deterministic runtime handoff report metadata
```

Forbidden in the next implementation slice:

```
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
trust-store loading
```

```
revocation lookup
key generation
private-key handling
object sealing
kernel interaction
```

Initial report policy

Allowed input handoff states:

```
handoff_state=eligible-report-only
handoff_state=eligible-evaluate-only
```

Allowed requested report labels:

```
report-only
evaluate-only
```

Planned report states:

```
report_state=ready-report-only
report_state=ready-evaluate-only
report_state=denied-evaluation
report_state=denied-handoff
report_state=denied-report
report_state=denied-runtime-authority
report_state=denied-host-effect
report_state=denied-network-effect
```

The first implementation may set:

```
report_ready=1
```

only for metadata-only report/evaluate handoff reports.

Even when report_ready=1, these must remain zero:

```
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future runtime handoff report record should be bounded and deterministic.

Planned fields:

```
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_scope
handoff_state
handoff_eligible
report_state
report_ready
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
status
```

Failure behavior

Future runtime handoff report handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null runtime handoff evaluation -&gt; invalid
invalid runtime handoff evaluation -&gt; denied-evaluation
handoff_eligible=0 -&gt; denied-handoff
handoff_state not eligible-report-only or eligible-evaluate-only -&gt; denied-handoff
missing requested report -&gt; denied-report
unknown requested report -&gt; denied-report
requested report mismatch -&gt; denied-report
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
runtime handoff report metadata implementation
```

It does not permit runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, trust-store behavior, revocation lookup, key handling, object sealing, or kernel behavior.

After runtime handoff report metadata exists and is guarded, the next valid planning slice is a sealed report envelope contract that still performs no runtime handoff.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-runtime-handoff-report-contract.sh
```

docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md

Source title: Latticra Seal Runtime Handoff Report Implementation

Lines: 126 | Words: 448 | SHA-256: 64a2f54700d560614c4ad5a56042a393e7cb07be664635915ed37f7458a29e92

Latticra Seal Runtime Handoff Report Implementation

Status: initial runtime handoff report metadata implementation Scope: bounded C metadata surface for classifying eligible metadata-only runtime handoff evaluation output into deterministic report readiness. This does not implement runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, trust-store loading, revocation lookup, key generation, private-key handling, object sealing, kernel interaction, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal runtime handoff report metadata implementation.

The implementation consumes runtime handoff evaluation metadata and classifies whether the requested metadata-only report remains ready as report-only or evaluate-only.

The report surface is report classification, not runtime handoff.

Added files

```
include/latticra/seal_runtime_handoff_report.h
src/seal_runtime_handoff_report.c
tests/seal_runtime_handoff_report_invariants.c
scripts/test-latticra-seal-runtime-handoff-report.sh
```

Required predecessor

This implementation depends on the runtime handoff evaluation metadata surface:

```
include/latticra/seal_runtime_handoff_evaluation.h
src/seal_runtime_handoff_evaluation.c
tests/seal_runtime_handoff_evaluation_invariants.c
scripts/test-latticra-seal-runtime-handoff-evaluation.sh
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md
```

API summary

The runtime handoff report metadata surface adds:

```
latticra_seal_runtime_handoff_report_t
latticra_seal_runtime_handoff_report_error_t
latticra_seal_runtime_handoff_report_error_label
latticra_seal_runtime_handoff_report_from_evaluation
latticra_seal_runtime_handoff_report_is_metadata_only
latticra_seal_runtime_handoff_report_render
```

Report behavior

The implementation:

```
accepts runtime handoff evaluation metadata
requires handoff_eligible=1
requires handoff_state=eligible-report-only or handoff_state=eligible-evaluate-only
requires runtime_authority_granted=0
requires handoff_performed=0
requires effect_performed=0
requires host_read_performed=0
requires host_write_performed=0
requires network_performed=0
accepts requested report metadata
classifies report-only as report_state=ready-report-only
classifies evaluate-only as report_state=ready-evaluate-only
sets report_ready=1 only for metadata-only report/evaluate report surfaces
renders deterministic runtime handoff report metadata
```

Effect and runtime boundary

Even when report_ready=1, these fields remain zero:

```
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

This implementation does not perform runtime handoff, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, persist policy, load trust stores, look up revocation status, generate keys, handle private keys, seal objects, or interact with the kernel.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null runtime handoff evaluation -&gt; invalid-input
invalid runtime handoff evaluation -&gt; invalid-evaluation
handoff_eligible=0 -&gt; denied-handoff
handoff_state not eligible-report-only or eligible-evaluate-only -&gt; denied-handoff
missing requested report -&gt; missing-requested-report
unknown requested report -&gt; denied-unknown-report
requested report mismatch -&gt; denied-report
runtime authority already granted -&gt; denied-runtime-authority
```

```
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not perform handoff, perform effects, read host files, write host files, contact networks, or grant runtime authority.

Validation

Run locally:

```
sh scripts/test-latticra-seal-runtime-handoff-report.sh
```

Expected output:

```
seal runtime handoff report invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is a sealed report envelope contract from ready metadata-only runtime handoff report metadata.

That next slice should remain contract-first and should not perform runtime handoff unless separately implemented and guarded.

docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md

Source title: Latticra Seal Signature Implementation

Lines: 118 | Words: 415 | SHA-256: c3d214c5d71c3f224f01bc5f3c012f543bbc1024624221dc1bcc38f18e9ced74

Latticra Seal Signature Implementation

Status: initial signature metadata envelope implementation Scope: bounded C metadata surface for caller-supplied signature envelope metadata. This slice does not implement cryptographic signing, signature verification, key generation, private-key storage, hardware key use, network trust lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signature metadata envelope implementation.

The implementation records caller-supplied development-lane signature metadata for an existing unsigned evidence manifest. It does not create, verify, or interpret cryptographic signatures.

Added files

```
include/latticra/seal_signature.h
src/seal_signature.c
tests/seal_signature_invariants.c
scripts/test-latticra-seal-signature.sh
```

API summary

The signature metadata surface adds:

```
latticra_seal_signature_t
latticra_seal_signature_error_t
latticra_seal_signature_error_label
latticra_seal_signature_from_manifest
latticra_seal_signature_is_metadata_only
latticra_seal_signature_report
```

Signature metadata behavior

The implementation:

```
accepts a valid unsigned manifest record
copies manifest profile
copies manifest kind
copies artifact digest algorithm
copies artifact digest hex
copies caller-supplied signer identity label
accepts only the Ed25519-development algorithm label
records caller-supplied signature byte length
sets signature_state=metadata-only
sets signature_supported=1
sets verification_supported=0
sets private_key_handling=0
sets network_lookup_allowed=0
sets runtime_authority_granted=0
renders deterministic signature metadata
```

Boundary

This implementation does not generate signatures, verify signatures, handle private keys, contact networks, persist signature files, seal objects, enforce capabilities, write host files, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null signature output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null manifest input -&gt; invalid-input
invalid manifest -&gt; invalid-manifest
missing artifact digest -&gt; missing-digest
missing signer label -&gt; missing-signer
unsupported algorithm label -&gt; unsupported-algorithm
zero signature byte length -&gt; missing-signature-metadata
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not generate keys, store keys, contact networks, sign records, verify records, write host files, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid manifest plus caller-supplied signature metadata produces deterministic signature metadata
manifest profile is copied
manifest kind is copied
artifact digest algorithm is copied
artifact digest hex is copied
signer identity label is copied
signature algorithm label is copied
signature byte length is recorded
signature_supported is one
verification_supported remains zero
private_key_handling remains zero
network_lookup_allowed remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid manifest fails closed
missing digest fails closed
missing signer fails closed
unsupported algorithm fails closed
missing signature metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-signature.sh
```

Next valid slice

The next valid Latticra Seal slice is a verification policy contract.

That future slice must be contract-first and must not be added directly to this signature metadata implementation.

docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION_CONTRACT.md

Source title: Latticra Seal Signature Implementation Contract

Lines: 182 | Words: 562 | SHA-256: 2eff566676f9b4dc54433a4e4bd5ae841fda76c68907c06da9212f63e2491ff8

Latticra Seal Signature Implementation Contract

Status: Latticra Seal signature implementation contract Scope: contract for future development-only signature implementation after signature policy metadata. This document does not implement signing, verification, key generation, private-key storage, hardware key use, post-quantum signatures, network trust lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first allowed implementation boundary for Latticra Seal signatures.

The purpose of the future implementation is to support deterministic development-lane signing metadata for already-built unsigned evidence manifests, without granting runtime authority or introducing production trust claims.

This document does not implement signatures.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
src/seal_manifest.c
src/seal_signature_policy.c
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
```

The signature policy metadata surface remains the source of truth until a signature implementation is merged and guarded.

First signature implementation boundary

The first signature implementation may only support a development-lane signature envelope.

Allowed future behavior:

```
accept a valid manifest record
accept a caller-supplied public identity label
accept caller-supplied signature bytes as metadata
record signature algorithm label
record signature byte length
record signature state
render deterministic signature report metadata
keep private-key handling disabled
keep network lookup disabled
keep runtime authority disabled
```

Forbidden first implementation behavior:

```
private-key generation
private-key storage
secret handling
hardware token handling
network trust lookup
certificate authority behavior
post-quantum signature implementation
object sealing
capability enforcement
runtime authority grants
host writes
package installation
host configuration changes
```

Algorithm scope

The first implementation may use only a label for:

```
Ed25519-development
```

The first implementation must not implement Ed25519 math, ML-DSA, SLH-DSA, or any custom signature primitive.

Real signing and verification require a later implementation contract.

Planned signature envelope shape

The first implementation should define a bounded metadata envelope.

Planned fields:

```
signature_profile
manifest_profile
manifest_kind
artifact_digest_algorithm
artifact_digest_hex
signer_identity_label
signature_algorithm
signature_state
signature_byte_length
signature_supported
verification_supported
private_key_handling
network_lookup_allowed
runtime_authority_granted
status
```

Initial metadata values:

```
signature_algorithm=Ed25519-development
signature_state=metadata-only
signature_supported=1
verification_supported=0
private_key_handling=0
network_lookup_allowed=0
runtime_authority_granted=0
status=signature-metadata
```

Planned files

The future implementation should use bounded C surfaces consistent with the rest of Latticra Seal.

```
include/latticra/seal_signature.h
src/seal_signature.c
tests/seal_signature_invariants.c
scripts/test-latticra-seal-signature.sh
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
```

Failure behavior

The future signature metadata implementation must fail closed.

Required failure states:

```
null output -&gt; invalid
null manifest -&gt; invalid
invalid manifest -&gt; invalid
missing artifact digest -&gt; invalid
missing signer label -&gt; invalid
missing signature bytes -&gt; invalid
unsupported algorithm label -&gt; unsupported
output buffer too small -&gt; bounded failure
```

Failures must not create keys, store keys, contact networks, write host files, seal objects, enforce capabilities, verify signatures, or grant runtime authority.

Invariant requirements

The first signature metadata tests must verify:

```
valid manifest plus caller-supplied signature metadata produces deterministic signature metadata
manifest profile is copied
manifest kind is copied
artifact digest algorithm is copied
artifact digest hex is copied
signer identity label is copied
signature algorithm is recorded
signature byte length is recorded
verification_supported remains zero
private_key_handling remains zero
network_lookup_allowed remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid manifest fails closed
unsupported algorithm label fails closed
```

Promotion rule

This contract permits only the next implementation slice:

```
signature metadata envelope implementation
```

It does not permit real signature generation, signature verification, key generation, private-key storage, object sealing, capability enforcement, network trust lookup, host writes, or runtime authority.

docs/LATTICRA_SEAL_SIGNATURE_POLICY_CONTRACT.md

Source title: Latticra Seal Signature Policy Contract

Lines: 168 | Words: 439 | SHA-256: ffaa49989805ee2b92eaf9f69ae9f750a44a6fd259b3b488925783a4a1b36da2

Latticra Seal Signature Policy Contract

Status: Latticra Seal signature policy contract Scope: policy for future signature behavior after unsigned evidence-manifest metadata. This document does not implement signing, verification, key generation, private-key storage, hardware key use, post-quantum signatures, network trust, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal signature policy layer.

The purpose of this layer is to decide how future signatures may be represented, requested, validated, and promoted before any signature implementation exists.

This document does not implement signatures.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
src/seal_measurement.c
src/seal_manifest.c
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
```

The manifest metadata surface remains unsigned until a later implementation explicitly adds signature support.

Policy boundary

Allowed in this policy slice:

```
signature vocabulary
algorithm planning
signature-state labels
public-key metadata planning
verification-state planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this policy slice:

```
signature generation
signature verification
key generation
private-key storage
secret handling
network trust lookup
certificate authority behavior
hardware token behavior
runtime authority grants
host writes
object sealing
capability enforcement
```

Initial algorithm policy

Initial planned signature families:

```
Ed25519 for local development and fast signatures
ML-DSA for future post-quantum signature support
SLH-DSA for conservative offline/root signature support
```

The first implementation after this contract may only add metadata fields and unsupported-state handling. It must not add the algorithms themselves.

No custom signature primitive may be introduced.

Planned signature states

Future records should use explicit labels:

```
signature_state=unsupported
signature_state=unsigned
signature_state=requested
signature_state=signed
signature_state=verified
signature_state=invalid
signature_state=unsupported-algorithm
```

For the next implementation, the expected state is:

```
signature_state=unsupported
```

Planned policy fields

A future signature policy record should be bounded and deterministic.

Planned fields:

```
policy_profile
manifest_profile
manifest_kind
planned_signature_algorithm
post_quantum_algorithm_planned
conservative_root_algorithm_planned
signature_supported
verification_supported
public_key_metadata_supported
private_key_handling
network_lookup_allowed
runtime_authority_granted
status
```

Initial values before implementation:

```
signature_supported=0
verification_supported=0
public_key_metadata_supported=0
private_key_handling=0
network_lookup_allowed=0
runtime_authority_granted=0
status=signature-policy-only
```

Failure behavior

Future signature policy handling must fail closed.

Required failure states:

```
null manifest -&gt; invalid
missing manifest digest -&gt; invalid
signature requested before support -&gt; unsupported
unknown algorithm -&gt; unsupported-algorithm
private-key input -&gt; rejected
network lookup request -&gt; rejected
runtime authority request -&gt; rejected
```

Failures must not create keys, persist secrets, contact networks, sign records, verify records, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
signature policy metadata implementation
```

It does not permit signature generation, verification, key handling, object sealing, capability enforcement, network behavior, host writes, or runtime authority.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-signature-policy-contract.sh
```

docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md

Source title: Latticra Seal Signature Policy Implementation

Lines: 104 | Words: 316 | SHA-256: 732663d17308cfa912f073431c147c3be600d33d87c95128f578982d681bc1b8

Latticra Seal Signature Policy Implementation

Status: initial signature policy metadata implementation Scope: bounded C metadata surface for Latticra Seal signature policy posture. This slice does not add signing, verification, key handling, network trust lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signature policy metadata implementation.

The implementation consumes unsigned manifest metadata and reports the planned signature posture while keeping all operational signature capability disabled.

Added files

```
include/latticra/seal_signature_policy.h
src/seal_signature_policy.c
tests/seal_signature_policy_invariants.c
scripts/test-latticra-seal-signature-policy.sh
```

API summary

The policy surface adds:

```
latticra_seal_signature_policy_t
latticra_seal_signature_policy_error_t
latticra_seal_signature_policy_error_label
latticra_seal_signature_policy_from_manifest
latticra_seal_signature_policy_is_metadata_only
latticra_seal_signature_policy_report
```

Policy behavior

The implementation:

```
accepts a valid unsigned manifest record
copies manifest profile
copies manifest kind
records planned Ed25519 metadata
records planned ML-DSA metadata
records planned SLH-DSA metadata
sets signature_state=unsupported
sets signature_supported=0
sets verification_supported=0
sets public_key_metadata_supported=0
sets private_key_handling=0
sets network_lookup_allowed=0
sets runtime_authority_granted=0
renders deterministic policy metadata
```

Boundary

This implementation does not sign data, verify records, generate keys, store keys, contact networks, use hardware tokens, seal objects, enforce capabilities, write host files, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null policy output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null manifest input -&gt; invalid-input
invalid manifest -&gt; invalid-manifest
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Invariants

The invariant test verifies:

```
valid manifest produces deterministic policy metadata
manifest profile is copied
manifest kind is copied
signature_state remains unsupported
signature_supported remains zero
verification_supported remains zero
public_key_metadata_supported remains zero
private_key_handling remains zero
network_lookup_allowed remains zero
runtime_authority_granted remains zero
```

```
small report buffer fails closed
null inputs fail closed
invalid manifest fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-signature-policy.sh
```

Next valid slice

The next valid Latticra Seal slice is a signature implementation contract.

That future slice must be contract-first and must not be added directly to this signature policy metadata implementation.

docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md

Source title: Latticra Seal Signature Request Contract

Lines: 224 | Words: 682 | SHA-256: 61c8d28054fed8ec0caaae5d281d0b332c5d6be1da7b3bb8ce0297635b25a5e2

Latticra Seal Signature Request Contract

Status: Latticra Seal signature request contract Scope: contract for a future metadata-only signature request surface after sealed report-envelope metadata. This document does not implement signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal signature-request boundary for sealed report-envelope metadata.

The purpose of this layer is to decide whether a ready metadata-only report envelope may be marked as eligible for a future signing request before any signing implementation, key authority, trust store, or runtime bridge exists.

The signature-request surface is request classification, not signing and not verification.

This document does not implement signing.

Relationship to existing signature metadata

Latticra Seal already has a manifest-chain signature metadata envelope:

```
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
include/latticra/seal_signature.h
src/seal_signature.c
tests/seal_signature_invariants.c
scripts/test-latticra-seal-signature.sh
```

That earlier surface records caller-supplied signature metadata for unsigned evidence manifests.

This contract is a separate envelope-chain planning layer. It starts from ready sealed report-envelope metadata and does not consume private keys, produce signatures, verify signatures, trust signatures, or grant authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md
include/latticra/seal_report_envelope.h
src/seal_report_envelope.c
tests/seal_report_envelope_invariants.c
scripts/test-latticra-seal-report-envelope.sh
```

The report-envelope metadata surface remains the source of envelope-readiness evidence for this signature-request surface.

Signature request boundary

Allowed in the next implementation slice:

```
accept sealed report-envelope metadata
require envelope_ready=1
require envelope_state=sealed-report-only or envelope_state=sealed-evaluate-only
require signature_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested signature metadata
classify Ed25519-development as signature_request_state=requested-metadata-only
produce deterministic signature request metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
signature verification
private-key handling
key generation
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial request policy

Allowed input envelope states:

```
envelope_state=sealed-report-only
envelope_state=sealed-evaluate-only
```

Allowed requested signature labels:

```
Ed25519-development
```

This label is a metadata request label only. It does not mean Ed25519 signing, Ed25519 verification, key generation, private-key handling, or trust establishment exists in this path.

Planned signature request states:

```
signature_request_state=requested-metadata-only
signature_request_state=denied-envelope
signature_request_state=denied-signature-request
signature_request_state=denied-runtime-authority
signature_request_state=denied-host-effect
signature_request_state=denied-network-effect
```

The first implementation may set:

```
signature_request_ready=1
```

only for ready metadata-only sealed report/evaluate envelopes and an allowed metadata-only signature request label.

Even when signature_request_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
private_key_handling=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future signature request record should be bounded and deterministic.

Planned fields:

```
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_scope
envelope_state
envelope_ready
signature_request_state
signature_request_ready
signature_performed
verification_performed
private_key_handling
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
status
```

Failure behavior

Future signature-request handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null report envelope -&gt; invalid
invalid report envelope -&gt; denied-envelope
envelope_ready=0 -&gt; denied-envelope
envelope_state not sealed-report-only or sealed-evaluate-only -&gt; denied-envelope
missing requested signature -&gt; denied-signature-request
unknown requested signature -&gt; denied-signature-request
signature already performed -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, handle private keys, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
signature request metadata implementation
```

It does not permit cryptographic signing, signature verification, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After signature request metadata and its status/public-entry checkpoint exist, the next valid planning slice is a signing authorization contract that still must not add signing without a separate implementation contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-signature-request-contract.sh
```

docs/LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md

Source title: Latticra Seal Signature Request Implementation

Lines: 156 | Words: 576 | SHA-256: 13f480a1bd7ae957511e2bb1ccfb0ee32626b3871d54a63b69f18b970be26b53

Latticra Seal Signature Request Implementation

Status: initial signature request metadata implementation Scope: bounded C metadata surface for classifying ready sealed report-envelope metadata into deterministic signature-request readiness. This does not implement cryptographic signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signature request metadata implementation.

The implementation consumes sealed report-envelope metadata and classifies whether the requested metadata-only signature request remains eligible for a future signing path.

The signature-request surface is request classification, not signing and not verification.

Added files

```
include/latticra/seal_signature_request.h
src/seal_signature_request.c
tests/seal_signature_request_invariants.c
scripts/test-latticra-seal-signature-request.sh
```

Required predecessor

This implementation depends on the signature request contract and the report-envelope metadata surface:

```
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md
include/latticra/seal_report_envelope.h
src/seal_report_envelope.c
tests/seal_report_envelope_invariants.c
scripts/test-latticra-seal-report-envelope.sh
```

API summary

The signature request metadata surface adds:

```
latticra_seal_signature_request_t
latticra_seal_signature_request_error_t
latticra_seal_signature_request_error_label
latticra_seal_signature_request_from_envelope
latticra_seal_signature_request_is_metadata_only
latticra_seal_signature_request_render
```

Request behavior

The implementation:

```
accepts sealed report-envelope metadata
requires envelope_ready=1
requires envelope_state=sealed-report-only or envelope_state=sealed-evaluate-only
requires signature_performed=0
requires handoff_performed=0
requires effect_performed=0
requires runtime_authority_granted=0
requires host_read_performed=0
requires host_write_performed=0
requires network_performed=0
accepts requested signature metadata
classifies Ed25519-development as signature_request_state=requested-metadata-only
sets signature_request_ready=1 only for metadata-only sealed report/evaluate envelope surfaces
renders deterministic signature request metadata
```

Effect and runtime boundary

Even when signature_request_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
private_key_handling=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

This implementation does not sign records, verify signatures, handle private keys, load trust stores, look up revocation status, perform runtime handoff, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, persist policy, seal objects, or interact with the kernel.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null report envelope -&gt; invalid-input
invalid report envelope -&gt; invalid-envelope
envelope_ready=0 -&gt; denied-envelope
envelope_state not sealed-report-only or sealed-evaluate-only -&gt; denied-envelope
missing requested signature -&gt; missing-requested-signature
unknown requested signature -&gt; denied-unknown-signature
signature already performed -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not sign, verify, handle private keys, perform handoff, perform effects, read host files, write host files, contact networks, or grant runtime authority.

Invariants

The invariant test verifies:

```
ready sealed-report-only envelope metadata produces requested-metadata-only signature request
metadata
ready sealed-evaluate-only envelope metadata produces requested-metadata-only signature request
metadata
signature_request_ready is one only for metadata-only allowed request states
signature_performed remains zero
verification_performed remains zero
private_key_handling remains zero
handoff_performed remains zero
effect_performed remains zero
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
small render buffer fails closed
null inputs fail closed
invalid envelope metadata fails closed
missing and unknown requested signature labels fail closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-signature-request-contract.sh
sh scripts/test-latticra-seal-signature-request.sh
```

Expected output:

```
seal signature request contract: ok
seal signature request invariants: ok
```

Next valid slice

The next valid Latticra Seal slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

That future slice must not add private-key handling, signing, verification, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately contracted, implemented, and guarded.

docs/LATTICRA_SEAL_SIGNED_MANIFEST_CONTRACT.md

Source title: Latticra Seal Signed Manifest Contract

Lines: 184 | Words: 558 | SHA-256: e92380ac336bacbb84da4f7e024ff1baddf3a49690ee4eb5556d277c0ec09155

Latticra Seal Signed Manifest Contract

Status: Latticra Seal signed evidence-manifest contract Scope: contract for future signed evidence manifests after read-only measurement. This document does not implement signing, key generation, private-key storage, signature verification, Ed25519, ML-DSA, SLH-DSA, encryption, object sealing, capability enforcement, runtime authority, host writes, network behavior, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the next Latticra Seal layer: signed evidence manifests.

A signed evidence manifest is a future bounded record that binds measured artifacts, contract identifiers, evidence levels, and signature metadata into an auditable structure.

This document does not implement signed manifests.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_MEASUREMENT_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
include/latticra/seal_report.h
include/latticra/seal_measurement.h
src/seal_report.c
src/seal_measurement.c
scripts/test-latticra-seal-report.sh
scripts/test-latticra-seal-measurement.sh
```

The measurement surface remains the source of truth for artifact digests until signed manifests are implemented and guarded.

Signed manifest boundary

The first signed-manifest implementation may only build deterministic manifest metadata from already-measured artifact records.

Allowed future behavior:

```
accept a bounded measurement record
copy artifact label
copy digest algorithm
copy digest hex
copy artifact size
copy contract identifier metadata
copy evidence level
render deterministic manifest text
record signature posture as unsupported until signing exists
```

Forbidden first signed-manifest behavior:

```
private key generation
private key storage
signature generation
signature verification
network access
encryption
object sealing
capability enforcement
runtime authority grants
host writes
manifest persistence by default
package installation
host configuration changes
```

Planned primitive policy

Initial signature algorithm planning is:

```
Ed25519 for local development and fast signatures
ML-DSA for future post-quantum signatures
SLH-DSA for future conservative offline/root signatures
```

This contract does not implement any of those algorithms.

No custom signature primitive may be introduced.

Planned manifest shape

The first manifest record should be bounded and deterministic.

Planned fields:

```
manifest_version
manifest_kind
manifest_profile
artifact_label
artifact_size_bytes
artifact_digest_algorithm
artifact_digest_hex
measurement_profile
contract_id
contract_digest_algorithm
contract_digest_hex
```

```
evidence_level
signature_algorithm_planned
signature_supported
verification_supported
private_key_handling
runtime_authority_granted
status
```

Initial values before signing support:

```
signature_supported=0
verification_supported=0
private_key_handling=0
runtime_authority_granted=0
status=unsigned-manifest
```

Planned files

The future implementation should use bounded C surfaces consistent with the rest of Latticra Seal.

```
include/latticra/seal_manifest.h
src/seal_manifest.c
tests/seal_manifest_invariants.c
scripts/test-latticra-seal-manifest.sh
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
```

Report integration

After a manifest metadata implementation exists, the Seal report may add a manifest-support field in a later report refinement.

Signing support must remain false until a separate signing contract and implementation exist.

Failure behavior

The future manifest implementation must fail closed.

Required failure states:

```
null manifest output -&gt; invalid
null measurement input -&gt; invalid
invalid measurement status -&gt; invalid
missing digest -&gt; invalid
unsupported digest algorithm -&gt; invalid
output buffer too small -&gt; bounded failure
signature requested before signing support -&gt; unsupported
```

Failures must not create files, modify files, contact networks, generate keys, sign records, verify signatures, or grant runtime authority.

Invariant requirements

The first manifest tests must verify:

```
valid measurement produces deterministic unsigned manifest metadata
artifact label is copied
artifact size is copied
SHA-256 digest is copied
measurement profile is copied
signature_supported remains zero
verification_supported remains zero
private_key_handling remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
unsupported signing request remains unsupported
```

Promotion rule

This contract permits only the next implementation slice:

```
unsigned evidence-manifest metadata implementation
```

It does not permit signing implementation, verification implementation, key handling, object sealing, capability enforcement, network behavior, host writes, or runtime authority.

docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md

Source title: Latticra Seal Signed Manifest Implementation

Lines: 128 | Words: 365 | SHA-256: 3c91a1047d32314df5399ab5ac3d8aa02352f3ee273bc3de356ff5a7d17cdd55

Latticra Seal Signed Manifest Implementation

Status: initial unsigned evidence-manifest metadata implementation Scope: bounded C manifest metadata surface after read-only measurement. This slice does not add signing, verification, key handling, object sealing, capability enforcement, runtime authority, host writes, network behavior, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal manifest implementation.

The implementation builds deterministic unsigned evidence-manifest metadata from a valid read-only measurement record.

Added files

```
include/latticra/seal_manifest.h
src/seal_manifest.c
tests/seal_manifest_invariants.c
scripts/test-latticra-seal-manifest.sh
```

API summary

The manifest surface adds:

```
latticra_seal_manifest_t
latticra_seal_manifest_error_t
latticra_seal_manifest_error_label
latticra_seal_manifest_from_measurement
latticra_seal_manifest_is_unsigned_metadata
latticra_seal_manifest_report
```

Manifest behavior

The implementation:

```
accepts a valid measurement record
copies artifact label
copies artifact size
copies digest algorithm
copies digest hex
copies measurement profile
copies contract id metadata
renders deterministic manifest text
keeps signature support at zero
keeps verification support at zero
keeps private-key handling at zero
keeps runtime authority at zero
```

Default manifest posture

A successful unsigned manifest records:

```
manifest_profile=latticra-seal-manifest/0.1
manifest_kind=unsigned-evidence-manifest
artifact_label=&lt;measurement artifact label&gt;
artifact_size_bytes=&lt;measurement byte count&gt;
artifact_digest_algorithm=SHA-256
artifact_digest_hex=&lt;measurement digest&gt;
measurement_profile=&lt;measurement profile&gt;
```

```
contract_id=&lt;caller supplied contract id&gt;
contract_digest_algorithm=not-computed
evidence_level=4
planned_signature_algorithm=Ed25519-planned
signature_supported=0
verification_supported=0
private_key_handling=0
runtime_authority_granted=0
status=unsigned-manifest
```

Boundary

This implementation does not sign data, verify signatures, generate keys, store keys, write manifests, persist manifests, contact networks, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null manifest output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null measurement input -&gt; invalid-input
invalid measurement -&gt; invalid-measurement
missing digest -&gt; missing-digest
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not write files, contact networks, create keys, sign records, verify records, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid measurement produces deterministic unsigned manifest metadata
artifact label is copied
artifact size is copied
SHA-256 digest is copied
measurement profile is copied
signature_supported remains zero
verification_supported remains zero
private_key_handling remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid measurement fails closed
missing digest fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-manifest.sh
```

Next valid slice

The next valid Latticra Seal slice is a signature policy contract.

That future slice must be contract-first and must not be added directly to this manifest metadata implementation.

docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md

Source title: Latticra Seal Signed Request Contract

Lines: 249 | Words: 746 | SHA-256: aff3839945e953fe4e642251aceda50c24703ba4e6c0f55429d62581cc2391c4

Latticra Seal Signed Request Contract

Status: Latticra Seal signed request contract Scope: contract for future signed request metadata after the report-only Seal request freshness report surface. This document does not implement signature generation, signature verification, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, signed request enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document defines the next Latticra Seal contract layer for signed request metadata.

The purpose is to prepare a bounded report-only structure for describing whether a future tool request carries declared signature metadata and whether the signed material is expected to bind identity, context, parameters, freshness, and policy intent.

This document does not implement signed request behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md
include/latticra/seal_request_freshness.h
src/seal_request_freshness.c
tests/seal_request_freshness_invariants.c
tests/seal_request_freshness_report_surface.c
scripts/test-latticra-seal-request-freshness-contract.sh
scripts/test-latticra-seal-request-freshness.sh
scripts/test-latticra-seal-request-freshness-report-surface.sh
```

The existing request freshness metadata surface remains report-only.

Signed request boundary

The signed request layer may describe signature metadata fields, expected binding fields, and report-only verification posture.

The layer may not generate signatures, verify signatures, parse keys, load trust stores, contact networks, check revocation, authorize a request, execute tools, execute shell commands, read host files, write host files, or mutate system state.

Allowed in this contract slice:

```
signed-request vocabulary
signature-presence metadata planning
signature-algorithm metadata planning
signing-key-id metadata planning
signature-hash metadata planning
signed-request-id planning
identity-binding metadata planning
context-binding metadata planning
parameter-binding metadata planning
freshness-binding metadata planning
policy-binding metadata planning
verification-status planning
report-only status planning
non-claims
```

```
static guard validation
```

Forbidden in this contract slice:

```
signature generation
signature verification
public-key parsing
public-key trust store loading
private-key handling
key generation
hardware key use
revocation lookup
network trust lookup
signed request enforcement
runtime execution
AI agent execution
model execution
tool execution
shell command execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
verified receipt generation
object sealing
kernel interaction
MCP server implementation
MCP client implementation
```

Initial signed request policy

The initial policy is report-only and closed by default.

```
seal_signed_request_contract_present=1
signed_request_supported=0
signature_generation_supported=0
signature_verification_supported=0
signature_present=0
signature_valid=0
signature_algorithm_declared=0
signing_key_id_present=0
signature_hash_present=0
signed_request_id_present=0
identity_binding_declared=0
context_binding_declared=0
parameter_binding_declared=0
freshness_binding_declared=0
policy_binding_declared=0
trust_store_supported=0
revocation_lookup_supported=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=contract-only
status=signed-request-contract-only
```

Planned metadata fields

Future signed request metadata should be bounded and deterministic.

Planned fields:

```
signed_request_profile
signed_request_id
signature_algorithm
signing_key_id
signature_hash
signed_request_supported
signature_generation_supported
signature_verification_supported
signature_present
signature_valid
signature_algorithm_declared
signing_key_id_present
signature_hash_present
```

```
signed_request_id_present
identity_binding_declared
context_binding_declared
parameter_binding_declared
freshness_binding_declared
policy_binding_declared
trust_store_supported
revocation_lookup_supported
runtime_authority_granted
mode
decision
reason
status
```

Initial values before signed request implementation:

```
signed_request_profile=latticra-seal-signed-request/0.1
signed_request_id=unset
signature_algorithm=unset
signing_key_id=unset
signature_hash=unset
signed_request_supported=0
signature_generation_supported=0
signature_verification_supported=0
signature_present=0
signature_valid=0
signature_algorithm_declared=0
signing_key_id_present=0
signature_hash_present=0
signed_request_id_present=0
identity_binding_declared=0
context_binding_declared=0
parameter_binding_declared=0
freshness_binding_declared=0
policy_binding_declared=0
trust_store_supported=0
revocation_lookup_supported=0
runtime_authority_granted=0
mode=contract-only
decision=report-only
reason=signed-request-contract-only
status=signed-request-contract-only
```

Promotion rules

The next implementation after this contract may only add bounded C metadata fields and deterministic rendering.

It must not generate signatures, verify signatures, parse keys, load trust stores, perform revocation lookup, contact networks, execute tools, execute shell commands, call runtime components, read host files, write host files, enforce capabilities, generate receipts, or grant authority.

Claim gate

Latticra must not claim to secure AI agents from this contract alone.

A future claim that Latticra secures AI-agent tool execution boundaries requires, at minimum:

```
request identity metadata implemented
parameter schema metadata implemented
request freshness metadata implemented
signed request metadata implemented
policy decision metadata implemented
runtime enforcement gate implemented
negative tests for denied unknown tools
negative tests for denied unsigned manifests
negative tests for denied stale requests
negative tests for denied replayed requests
operator-visible evidence report implemented
```

Until those are implemented and validated, the accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Non-claims

This contract must not be read as implementation of signed request verification, tool security enforcement, cryptographic enforcement, or AI-agent security.

Required non-claim posture:

```
signed_request_verification_implemented=0
signature_generation_implemented=0
signature_verification_implemented=0
trust_store_implemented=0
revocation_lookup_implemented=0
mcp_tool_security_enforced=0
ai_agent_security_claimed=0
runtime_authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is report-only signed request metadata.

That future slice may add a bounded metadata structure, deterministic rendering, invariant tests, and a test runner.

It must not change runtime behavior, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, capability enforcement, cryptographic verification, signature verification, revocation lookup, or authority grants.

docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md

Source title: Latticra Seal Signed Request Implementation

Lines: 149 | Words: 602 | SHA-256: 96413c51d042688d6ee587385eda42b713f187642a5944ccfa52ef100b8dd826

Latticra Seal Signed Request Implementation

Status: initial report-only signed request metadata implementation Scope: bounded C metadata surface for signed request posture after the report-only Seal request freshness layer. This slice does not implement signature generation, signature verification, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, revocation lookup, network trust lookup, signed request enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This document records the first Latticra Seal signed request metadata implementation.

The implementation accepts an existing report-only Seal request freshness metadata record and produces deterministic report-only signed request metadata.

Added files

```
include/latticra/seal_signed_request.h
src/seal_signed_request.c
tests/seal_signed_request_invariants.c
scripts/test-latticra-seal-signed-request.sh
```

API summary

The signed request metadata surface adds:

```
latticra_seal_signed_request_t
latticra_seal_signed_request_error_t
latticra_seal_signed_request_error_label
latticra_seal_signed_request_from_freshness
latticra_seal_signed_request_is_report_only
latticra_seal_signed_request_report
```

Metadata behavior

The implementation:

```
accepts a valid report-only Seal request freshness metadata record
sets signed_request_profile=latticra-seal-signed-request/0.1
sets signed_request_id=unset
sets signature_algorithm=unset
sets signing_key_id=unset
sets signature_hash=unset
sets signed_request_supported=0
sets signature_generation_supported=0
sets signature_verification_supported=0
sets signature_present=0
sets signature_valid=0
sets signature_algorithm_declared=0
sets signing_key_id_present=0
sets signature_hash_present=0
sets signed_request_id_present=0
sets identity_binding_declared=0
sets context_binding_declared=0
sets parameter_binding_declared=0
sets freshness_binding_declared=0
sets policy_binding_declared=0
sets trust_store_supported=0
sets revocation_lookup_supported=0
copies runtime_authority_granted from request freshness metadata
copies host_read_performed from request freshness metadata
copies host_write_performed from request freshness metadata
copies network_performed from request freshness metadata
sets mode=report-only
sets decision=report-only
sets reason=signed-request-metadata-only
renders deterministic signed request metadata
```

Boundary

This implementation does not generate signatures, verify signatures, parse keys, load trust stores, perform revocation lookup, contact networks, create receipts, execute tools, execute shell commands, call runtime components, read host files, write host files, enforce capabilities, or grant authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null request freshness metadata input -&gt; invalid-input
invalid request freshness metadata -&gt; invalid-freshness
non-report-only request freshness metadata -&gt; invalid-freshness
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not generate signatures, verify signatures, parse keys, contact networks, read host files, write host files, execute tools, enforce capabilities, or grant authority.

Invariants

The invariant test verifies:

```
valid report-only request freshness metadata produces deterministic signed request metadata
signed_request_profile is stable
signed_request_id remains unset
signature_algorithm remains unset
signing_key_id remains unset
signature_hash remains unset
signed_request_supported remains zero
signature_generation_supported remains zero
```

```
signature_verification_supported remains zero
signature_present remains zero
signature_valid remains zero
signature_algorithm_declared remains zero
signing_key_id_present remains zero
signature_hash_present remains zero
signed_request_id_present remains zero
identity_binding_declared remains zero
context_binding_declared remains zero
parameter_binding_declared remains zero
freshness_binding_declared remains zero
policy_binding_declared remains zero
trust_store_supported remains zero
revocation_lookup_supported remains zero
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
mode remains report-only
decision remains report-only
reason remains signed-request-metadata-only
small report buffer fails closed
null inputs fail closed
invalid request freshness metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-signed-request-contract.sh
sh scripts/test-latticra-seal-signed-request.sh
```

Claim boundary

This implementation still does not justify the public claim that Latticra secures AI agents.

It moves the project closer by adding report-only metadata for signed request posture, but the stronger claim requires policy decision metadata, a runtime enforcement gate, and negative tests for denied unknown, unsigned, stale, and replayed requests.

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md

Source title: Latticra Seal Signer Handoff Contract

Lines: 286 | Words: 916 | SHA-256: 123f208695e7f3886071e69951e53d338ef51a6c0c332e99d702b1ec2b19d049

Latticra Seal Signer Handoff Contract

Status: Latticra Seal signer handoff contract Scope: contract for a future metadata-only signer handoff surface after Seal signing authorization metadata. This document does not implement signing, signature verification, signer invocation, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal signer handoff boundary after ready signing authorization metadata.

The purpose of this layer is to decide whether a ready metadata-only signing authorization may be represented as eligible for a future signer handoff before any signer, private key, trust store, runtime bridge, host authority, or signature generation exists.

The signer handoff surface is signer-path classification, not signing, not verification, not signer invocation, and not runtime handoff.

This document does not implement signing.

Relationship to runtime handoff

Latticra Seal already has runtime handoff planning and metadata surfaces for effect and runtime-boundary posture:

```
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md
```

Those surfaces describe runtime-boundary posture for effect decisions.

This signer handoff contract is separate. It starts from signing authorization metadata and describes only a future signer-path handoff record. It does not route to runtime behavior, invoke a signer, perform signing, handle private keys, verify signatures, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_AUTHORIZATION_STATUS.md
include/latticra/seal_signing_authorization.h
src/seal_signing_authorization.c
tests/seal_signing_authorization_invariants.c
scripts/test-latticra-seal-signing-authorization-contract.sh
scripts/test-latticra-seal-signing-authorization.sh
scripts/test-latticra-seal-signing-authorization-status.sh
```

The signing authorization metadata surface remains the source of readiness evidence for this signer handoff surface.

Signer handoff boundary

Allowed in the next implementation slice:

```
accept signing authorization metadata
require signing_authorization_ready=1
require signing_authorization_state=authorized-metadata-only
require requested_signature=Ed25519-development
require requested_signing_authorization=metadata-only
require signature_performed=0
require verification_performed=0
require private_key_handling=0
require key_generation_performed=0
require trust_store_loaded=0
require revocation_lookup_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested signer handoff metadata
classify metadata-only as signer_handoff_state=handoff-metadata-only
produce deterministic signer handoff metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
signature verification
signer process invocation
private-key handling
key generation
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial handoff policy

Allowed input signing authorization state:

```
signing_authorization_state=authorized-metadata-only
```

Allowed requested signature label:

```
Ed25519-development
```

Allowed requested signing authorization label:

```
metadata-only
```

Allowed requested signer handoff label:

```
metadata-only
```

These labels are metadata labels only. They do not mean Ed25519 signing, Ed25519 verification, signer invocation, key generation, private-key handling, trust establishment, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned signer handoff states:

```
signer_handoff_state=handoff-metadata-only
signer_handoff_state=denied-signing-authorization
signer_handoff_state=denied-signature-algorithm
signer_handoff_state=denied-signer-handoff
signer_handoff_state=denied-private-key
signer_handoff_state=denied-runtime-authority
signer_handoff_state=denied-host-effect
signer_handoff_state=denied-network-effect
```

The first implementation may set:

```
signer_handoff_ready=1
```

only for ready metadata-only signing authorization, the allowed development signature label, and the metadata-only signer handoff label.

Even when signer_handoff_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future signer handoff record should be bounded and deterministic.

Planned fields:

```
signer_handoff_profile
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_signing_authorization
requested_signer_handoff
requested_scope
signing_authorization_state
signing_authorization_ready
signer_handoff_state
signer_handoff_ready
signature_performed
verification_performed
signer_invoked
private_key_handling
key_generation_performed
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
signer_handoff_profile=latticra-seal-signer-handoff/0.1
requested_signer_handoff=metadata-only
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=signer-handoff-metadata
```

Failure behavior

Future signer handoff handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null signing authorization -&gt; invalid
invalid signing authorization -&gt; denied-signing-authorization
signing_authorization_ready=0 -&gt; denied-signing-authorization
signing_authorization_state not authorized-metadata-only -&gt; denied-signing-authorization
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested signer handoff -&gt; denied-signer-handoff
```

```
unknown requested signer handoff -&gt; denied-signer-handoff
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-private-key
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
signer already invoked -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, invoke a signer, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
signer handoff metadata implementation
```

It does not permit cryptographic signing, signature verification, signer invocation, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After signer handoff metadata, its status/public-entry checkpoint, the signer invocation contract, the signer invocation metadata implementation, signer invocation status/public-entry alignment, the signing operation contract, signing operation metadata implementation, signing operation status/public-entry alignment, the key-handling boundary contract, key-handling metadata implementation, key-handling status/public-entry alignment, the key-material boundary contract, and key-material metadata implementation exist and are guarded, the next valid planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-signer-handoff-contract.sh
```

docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md

Source title: Latticra Seal Signer Handoff Implementation

Lines: 163 | Words: 524 | SHA-256: 42d44d39b679524dba0453796836c35e8e6d1b91ce4e0a589eec4fa580f31186

Latticra Seal Signer Handoff Implementation

Status: initial signer handoff metadata implementation Scope: bounded C metadata surface for classifying Seal signer handoff eligibility after ready signing authorization metadata. This slice does not add signing, signature verification, signer invocation, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signer handoff metadata implementation.

The implementation consumes ready signing authorization metadata and classifies whether the request remains eligible for a future metadata-only signer handoff.

It is signer-path handoff classification only.

It does not invoke a signer.

It does not sign.

Files

```
include/latticra/seal_signer_handoff.h
src/seal_signer_handoff.c
tests/seal_signer_handoff_invariants.c
scripts/test-latticra-seal-signer-handoff.sh
```

Required predecessors

This implementation depends on the signer handoff contract and the signing authorization metadata surface:

```
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_AUTHORIZATION_STATUS.md
include/latticra/seal_signing_authorization.h
src/seal_signing_authorization.c
tests/seal_signing_authorization_invariants.c
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signing-authorization-contract.sh
scripts/test-latticra-seal-signing-authorization.sh
scripts/test-latticra-seal-signing-authorization-status.sh
```

Implemented surface

The signer handoff metadata surface adds:

```
latticra_seal_signer_handoff_t
latticra_seal_signer_handoff_error_t
latticra_seal_signer_handoff_error_label
latticra_seal_signer_handoff_from_authorization
latticra_seal_signer_handoff_is_metadata_only
latticra_seal_signer_handoff_render
```

The implementation:

```
accepts signing authorization metadata
requires signing_authorization_ready=1
requires signing_authorization_state=authorized-metadata-only
requires requested_signature=Ed25519-development
requires requested_signing_authorization=metadata-only
accepts requested_signer_handoff=metadata-only
classifies the result as signer_handoff_state=handoff-metadata-only
sets signer_handoff_ready=1 only for metadata-only allowed authorization states
renders deterministic signer handoff metadata
fails closed for invalid or denied predecessor metadata
```

Even when signer_handoff_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invoked=0
mode=metadata-only
status=signer-handoff-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null signing authorization
invalid signing authorization
signing_authorization_ready=0
signing_authorization_state not authorized-metadata-only
requested_signing_authorization not metadata-only
missing requested signature
unknown requested signature
missing requested signer handoff
unknown requested signer handoff
private-key handling already present
key generation already present
trust-store loading already present
revocation lookup already present
runtime authority already granted
signature already performed
verification already performed
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, invoke a signer, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-signer-handoff-contract.sh
sh scripts/test-latticra-seal-signer-handoff.sh
```

Expected output:

```
seal signer handoff contract: ok
seal signer handoff invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

The signer invocation metadata implementation, status/public-entry alignment, signing operation contract, signing operation metadata implementation, signing operation status/public-entry alignment, key-handling boundary contract, and key-handling metadata implementation now exist as guarded checkpoints. Future work must not add private-key handling, signing, verification, signer invocation behavior, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md

Source title: Latticra Seal Signer Invocation Contract

Lines: 288 | Words: 911 | SHA-256: dff0c083d1f5faf9fb5e73b0d627d9205935cdf5b622ea44bf94500cb14c07a9

Latticra Seal Signer Invocation Contract

Status: Latticra Seal signer invocation contract Scope: contract for a future metadata-only signer invocation surface after Seal signer handoff metadata. This document does not implement signing, signature verification, signer invocation behavior, signer process execution, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal signer invocation boundary after ready signer handoff metadata.

The purpose of this layer is to decide whether a ready metadata-only signer handoff may be represented as eligible for a future signer invocation request before any signer process, private key, trust store, runtime bridge, host authority, or signature generation exists.

The signer invocation surface is signer-invocation path classification, not signer invocation, not signing, not verification, and not runtime handoff.

This document does not invoke a signer.

This document does not implement signing.

Relationship to signer handoff

Latticra Seal signer handoff metadata already records a no-effect eligibility checkpoint for the future signer path:

```
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_HANDOFF_STATUS.md
```

This signer invocation contract starts from that checkpoint. It does not replace signer handoff, route to runtime behavior, invoke a signer, perform signing, handle private keys, verify signatures, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_HANDOFF_STATUS.md
include/latticra/seal_signer_handoff.h
src/seal_signer_handoff.c
tests/seal_signer_handoff_invariants.c
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signer-handoff.sh
scripts/test-latticra-seal-signer-handoff-status.sh
```

The signer handoff metadata surface remains the source of readiness evidence for this signer invocation surface.

Signer invocation boundary

Allowed in the next implementation slice:

```
accept signer handoff metadata
```

```
require signer_handoff_ready=1
require signer_handoff_state=handoff-metadata-only
require requested_signature=Ed25519-development
require requested_signing_authorization=metadata-only
require requested_signer_handoff=metadata-only
require signature_performed=0
require verification_performed=0
require signer_invoked=0
require private_key_handling=0
require key_generation_performed=0
require trust_store_loaded=0
require revocation_lookup_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested signer invocation metadata
classify metadata-only as signer_invocation_state=invocation-metadata-only
produce deterministic signer invocation metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
signature verification
signer process invocation
private-key handling
key generation
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial invocation policy

Allowed signer handoff state:

```
signer_handoff_state=handoff-metadata-only
```

Allowed requested signer handoff label:

```
metadata-only
```

Allowed requested signer invocation label:

```
metadata-only
```

These labels are metadata labels only. They do not mean Ed25519 signing, Ed25519 verification, signer invocation, key generation, private-key handling, trust establishment, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned signer invocation states:

```
signer_invocation_state=invocation-metadata-only
signer_invocation_state=denied-signer-handoff
signer_invocation_state=denied-signing-authorization
signer_invocation_state=denied-signature-algorithm
signer_invocation_state=denied-signer-invocation
signer_invocation_state=denied-private-key
signer_invocation_state=denied-runtime-authority
signer_invocation_state=denied-host-effect
signer_invocation_state=denied-network-effect
```

The first implementation may set:

```
signer_invocation_ready=1
```

only for ready metadata-only signer handoff metadata, the allowed development signature label, the metadata-only signer handoff label, and the metadata-only signer invocation label.

Even when signer_invocation_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future signer invocation record should be bounded and deterministic.

Planned fields:

```
signer_invocation_profile
signer_handoff_profile
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_signing_authorization
requested_signer_handoff
requested_signer_invocation
requested_scope
signing_authorization_state
signing_authorization_ready
signer_handoff_state
signer_handoff_ready
signer_invocation_state
signer_invocation_ready
signature_performed
verification_performed
signer_invoked
private_key_handling
key_generation_performed
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
requested_signer_invocation=metadata-only
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
```

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=signer-invocation-metadata
```

Failure behavior

Future signer invocation handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null signer handoff -&gt; invalid
invalid signer handoff -&gt; denied-signer-handoff
signer_handoff_ready=0 -&gt; denied-signer-handoff
signer_handoff_state not handoff-metadata-only -&gt; denied-signer-handoff
requested_signer_handoff not metadata-only -&gt; denied-signer-handoff
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested signer invocation -&gt; denied-signer-invocation
unknown requested signer invocation -&gt; denied-signer-invocation
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-private-key
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
signer already invoked -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, invoke a signer, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permitted the first implementation slice:

```
signer invocation metadata implementation
```

It does not permit cryptographic signing, signature verification, signer invocation behavior, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

The signer invocation metadata implementation, status/public-entry alignment, signing operation contract, signing operation metadata implementation, signing operation status/public-entry alignment, key-handling boundary contract, key-handling metadata implementation, key-handling status/public-entry alignment, key-material boundary contract, and key-material metadata implementation now exist and are guarded. The next valid planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-signer-invocation-contract.sh
```

docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md

Source title: Latticra Seal Signer Invocation Implementation

Lines: 174 | Words: 549 | SHA-256: 69185013bfac8ea6a09a0f3ddf29a22e557470435599dfc3e24ced6011c010ed

Latticra Seal Signer Invocation Implementation

Status: initial signer invocation metadata implementation Scope: bounded C metadata surface for classifying Seal signer invocation eligibility after ready signer handoff metadata. This slice does not add signing, signature verification, signer invocation behavior, signer process execution, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signer invocation metadata implementation.

The implementation consumes ready signer handoff metadata and classifies whether the request remains eligible for a future metadata-only signer invocation path.

It is signer-invocation path classification only.

It does not invoke a signer.

It does not sign.

Files

```
include/latticra/seal_signer_invocation.h
src/seal_signer_invocation.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signer-invocation.sh
```

Required predecessors

This implementation depends on the signer invocation contract and signer handoff metadata surface:

```
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_HANDOFF_STATUS.md
include/latticra/seal_signer_handoff.h
src/seal_signer_handoff.c
tests/seal_signer_handoff_invariants.c
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signer-handoff.sh
scripts/test-latticra-seal-signer-handoff-status.sh
```

Implemented surface

The signer invocation metadata surface adds:

```
latticra_seal_signer_invocation_t
latticra_seal_signer_invocation_error_t
latticra_seal_signer_invocation_error_label
latticra_seal_signer_invocation_from_handoff
latticra_seal_signer_invocation_is_metadata_only
latticra_seal_signer_invocation_render
```

The implementation:

```
accepts signer handoff metadata
requires signer_handoff_ready=1
requires signer_handoff_state=handoff-metadata-only
requires requested_signature=Ed25519-development
```

```
requires requested_signing_authorization=metadata-only
requires requested_signer_handoff=metadata-only
accepts requested_signer_invocation=metadata-only
classifies the result as signer_invocation_state=invocation-metadata-only
sets signer_invocation_ready=1 only for metadata-only allowed handoff states
renders deterministic signer invocation metadata
fails closed for invalid or denied predecessor metadata
```

Even when signer_invocation_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signer_invoked=0
mode=metadata-only
status=signer-invocation-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null signer handoff
invalid signer handoff
signer_handoff_ready=0
signer_handoff_state not handoff-metadata-only
requested_signer_handoff not metadata-only
signing_authorization_ready=0
signing_authorization_state not authorized-metadata-only
requested_signing_authorization not metadata-only
missing requested signature
unknown requested signature
missing requested signer invocation
unknown requested signer invocation
private-key handling already present
key generation already present
trust-store loading already present
revocation lookup already present
runtime authority already granted
signature already performed
verification already performed
signer already invoked
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, invoke a signer, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-signer-invocation-contract.sh
sh scripts/test-latticra-seal-signer-invocation.sh
sh scripts/test-latticra-seal-signer-invocation-status.sh
```

Expected output:

```
seal signer invocation contract: ok
seal signer invocation invariants: ok
seal signer invocation status: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

The signer invocation status/public-entry alignment, signing operation contract, signing operation metadata implementation, signing operation status/public-entry alignment, key-handling boundary contract, and key-handling metadata implementation now exist as guarded checkpoints. Future work must not add private-key handling, signing, verification, signer invocation behavior, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md

Source title: Latticra Seal Signing Authorization Contract

Lines: 277 | Words: 832 | SHA-256: 223f7ebfcae4addd4d1fef97dbc280645f814f3cbd5e845c7f73783bf2a640cd

Latticra Seal Signing Authorization Contract

Status: Latticra Seal signing authorization contract Scope: contract for a future metadata-only signing authorization surface after Seal signature request metadata. This document does not implement signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal signing authorization boundary after ready signature request metadata.

The purpose of this layer is to decide whether a ready metadata-only signature request may be marked as eligible for a future signing path before any signer, private key, trust store, runtime bridge, or host authority exists.

The signing authorization surface is signing-path classification, not signing, not verification, and not runtime authorization.

This document does not implement signing.

Relationship to existing signature metadata

Latticra Seal already has a manifest-chain signature metadata envelope:

```
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
include/latticra/seal_signature.h
src/seal_signature.c
tests/seal_signature_invariants.c
scripts/test-latticra-seal-signature.sh
```

That earlier surface records caller-supplied signature metadata for unsigned evidence manifests.

This contract is a separate envelope-chain authorization-planning layer. It starts from ready Seal signature request metadata and does not consume private keys, produce signatures, verify signatures, trust signatures, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNATURE_REQUEST_STATUS.md
include/latticra/seal_signature_request.h
src/seal_signature_request.c
tests/seal_signature_request_invariants.c
scripts/test-latticra-seal-signature-request-contract.sh
scripts/test-latticra-seal-signature-request.sh
scripts/test-latticra-seal-signature-request-status.sh
```

The signature request metadata surface remains the source of readiness evidence for this signing authorization surface.

Signing authorization boundary

Allowed in the next implementation slice:

```
accept signature request metadata
require signature_request_ready=1
require signature_request_state=requested-metadata-only
require requested_signature=Ed25519-development
require signature_performed=0
require verification_performed=0
require private_key_handling=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested signing authorization metadata
classify metadata-only as signing_authorization_state=authorized-metadata-only
produce deterministic signing authorization metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
signature verification
private-key handling
key generation
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial authorization policy

Allowed input signature request state:

```
signature_request_state=requested-metadata-only
```

Allowed requested signature label:

```
Ed25519-development
```

Allowed requested signing authorization label:

```
metadata-only
```

These labels are metadata labels only. They do not mean Ed25519 signing, Ed25519 verification, key generation, private-key handling, trust establishment, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned signing authorization states:

```
signing_authorization_state=authorized-metadata-only
signing_authorization_state=denied-signature-request
signing_authorization_state=denied-signature-algorithm
signing_authorization_state=denied-authorization-request
signing_authorization_state=denied-private-key
signing_authorization_state=denied-runtime-authority
signing_authorization_state=denied-host-effect
signing_authorization_state=denied-network-effect
```

The first implementation may set:

```
signing_authorization_ready=1
```

only for ready metadata-only signature requests, the allowed development signature label, and the metadata-only signing authorization label.

Even when signing_authorization_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future signing authorization record should be bounded and deterministic.

Planned fields:

```
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
requested_signature
requested_signing_authorization
requested_scope
signature_request_state
```

```
signature_request_ready
signing_authorization_state
signing_authorization_ready
signature_performed
verification_performed
private_key_handling
key_generation_performed
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
signing_authorization_profile=latticra-seal-signing-authorization/0.1
requested_signing_authorization=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signature_performed=0
verification_performed=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=signing-authorization-metadata
```

Failure behavior

Future signing authorization handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null signature request -&gt; invalid
invalid signature request -&gt; denied-signature-request
signature_request_ready=0 -&gt; denied-signature-request
signature_request_state not requested-metadata-only -&gt; denied-signature-request
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested signing authorization -&gt; denied-authorization-request
unknown requested signing authorization -&gt; denied-authorization-request
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-private-key
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
signing authorization metadata implementation
```

It does not permit cryptographic signing, signature verification, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After signing authorization metadata, its status/public-entry checkpoint, the signer handoff contract, and signer handoff metadata implementation exist, the next valid slice is signer handoff status/public-entry alignment. That future slice still must not add signing without a separate implementation contract, key-handling contract, and guards.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-signing-authorization-contract.sh
```

The first metadata implementation is validated by:

```
sh scripts/test-latticra-seal-signing-authorization.sh
```

docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md

Source title: Latticra Seal Signing Authorization Implementation

Lines: 151 | Words: 457 | SHA-256: 663c332c1bd7dd58da2558c9a7bd87d87515bb34d95dc761349188cbf2457898

Latticra Seal Signing Authorization Implementation

Status: initial signing authorization metadata implementation Scope: bounded C metadata surface for classifying Seal signing authorization after ready signature request metadata. This slice does not add signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signing authorization metadata implementation.

The implementation consumes ready signature request metadata and classifies whether the request remains eligible for a future metadata-only signing path.

It is signing-path classification only.

It does not sign.

Files

```
include/latticra/seal_signing_authorization.h
src/seal_signing_authorization.c
tests/seal_signing_authorization_invariants.c
scripts/test-latticra-seal-signing-authorization.sh
```

Required predecessors

This implementation depends on the signing authorization contract and the signature request metadata surface:

```
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNATURE_REQUEST_STATUS.md
include/latticra/seal_signature_request.h
src/seal_signature_request.c
tests/seal_signature_request_invariants.c
scripts/test-latticra-seal-signature-request-contract.sh
scripts/test-latticra-seal-signature-request.sh
scripts/test-latticra-seal-signature-request-status.sh
```

Implemented surface

The signing authorization metadata surface adds:

```
latticra_seal_signing_authorization_t
latticra_seal_signing_authorization_error_t
latticra_seal_signing_authorization_error_label
latticra_seal_signing_authorization_from_request
latticra_seal_signing_authorization_is_metadata_only
latticra_seal_signing_authorization_render
```

The implementation:

```
accepts signature request metadata
requires signature_request_ready=1
requires signature_request_state=requested-metadata-only
requires requested_signature=Ed25519-development
accepts requested_signing_authorization=metadata-only
classifies the result as signing_authorization_state=authorized-metadata-only
sets signing_authorization_ready=1 only for metadata-only allowed request states
renders deterministic signing authorization metadata
fails closed for invalid or denied predecessor metadata
```

Even when signing_authorization_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
signature_request_state=requested-metadata-only
signature_request_ready=1
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
mode=metadata-only
status=signing-authorization-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null signature request
invalid signature request
signature_request_ready=0
signature_request_state not requested-metadata-only
missing requested signature
unknown requested signature
missing requested signing authorization
unknown requested signing authorization
private-key handling already present
runtime authority already granted
signature already performed
verification already performed
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-signing-authorization-contract.sh
sh scripts/test-latticra-seal-signing-authorization.sh
```

Expected output:

```
seal signing authorization contract: ok
seal signing authorization invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

That future slice must not add private-key handling, signing, verification, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately contracted, implemented, and guarded.

docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md

Source title: Latticra Seal Signing Operation Contract

Lines: 298 | Words: 933 | SHA-256: 5d6f62002604355eab845cd5f784d40ab12f0d372478aeadf374a2a030fef65b

Latticra Seal Signing Operation Contract

Status: Latticra Seal signing operation contract Scope: contract for a future metadata-only signing operation surface after Seal signer invocation metadata. This document does not implement cryptographic signing, signature verification, signer invocation behavior, signer process execution, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first Latticra Seal signing operation boundary after ready signer invocation metadata.

The purpose of this layer is to decide whether a ready metadata-only signer invocation record may be represented as eligible for a future metadata-only signing operation request before any private key, signer process, trust store, runtime bridge, host authority, or signature generation exists.

The signing operation surface is signing-operation path classification, not signing, not verification, not signer invocation, and not runtime handoff.

This document does not sign.

This document does not handle private keys.

Relationship to signer invocation

Latticra Seal signer invocation metadata already records a no-effect eligibility checkpoint for the future signer path:

```
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
```

This signing operation contract starts from that checkpoint. It does not replace signer invocation metadata, route to runtime behavior, invoke a signer, perform signing, handle private keys, verify signatures, persist policy, or grant runtime authority.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
include/latticra/seal_signer_invocation.h
src/seal_signer_invocation.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signer-invocation-status.sh
```

The signer invocation metadata surface remains the source of readiness evidence for this signing operation surface.

Signing operation boundary

Allowed in the next implementation slice:

```
accept signer invocation metadata
require signer_invocation_ready=1
require signer_invocation_state=invocation-metadata-only
require requested_signature=Ed25519-development
require requested_signing_authorization=metadata-only
require requested_signer_handoff=metadata-only
require requested_signer_invocation=metadata-only
require signature_performed=0
require verification_performed=0
require signer_invoked=0
require private_key_handling=0
require key_generation_performed=0
require trust_store_loaded=0
require revocation_lookup_performed=0
require handoff_performed=0
require effect_performed=0
require runtime_authority_granted=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested signing operation metadata
classify metadata-only as signing_operation_state=operation-metadata-only
produce deterministic signing operation metadata
```

Forbidden in the next implementation slice:

```
cryptographic signing
signature verification
signer process invocation
private-key handling
key generation
trust-store loading
revocation lookup
runtime handoff execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
object sealing
kernel interaction
```

Initial operation policy

Allowed signer invocation state:

```
signer_invocation_state=invocation-metadata-only
```

Allowed requested signer invocation label:

```
metadata-only
```

Allowed requested signing operation label:

```
metadata-only
```

These labels are metadata labels only. They do not mean Ed25519 signing, Ed25519 verification, signer invocation, key generation, private-key handling, trust establishment, runtime handoff, host behavior, network behavior, or capability enforcement exists in this path.

Planned signing operation states:

```
signing_operation_state=operation-metadata-only
signing_operation_state=denied-signer-invocation
signing_operation_state=denied-signer-handoff
signing_operation_state=denied-signing-authorization
signing_operation_state=denied-signature-algorithm
signing_operation_state=denied-signing-operation
signing_operation_state=denied-private-key
signing_operation_state=denied-runtime-authority
signing_operation_state=denied-host-effect
signing_operation_state=denied-network-effect
```

The first implementation may set:

```
signing_operation_ready=1
```

only for ready metadata-only signer invocation metadata, the allowed development signature label, the metadata-only signer invocation label, and the metadata-only signing operation label.

Even when signing_operation_ready=1, these must remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future signing operation record should be bounded and deterministic.

Planned fields:

```
signing_operation_profile
signer_invocation_profile
signer_handoff_profile
signing_authorization_profile
signature_request_profile
envelope_profile
report_profile
handoff_profile
decision_profile
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
requested_capability
requested_effect
requested_handoff
requested_report
requested_envelope
```

```
requested_signature
requested_signing_authorization
requested_signer_handoff
requested_signer_invocation
requested_signing_operation
requested_scope
signing_authorization_state
signing_authorization_ready
signer_handoff_state
signer_handoff_ready
signer_invocation_state
signer_invocation_ready
signing_operation_state
signing_operation_ready
signature_performed
verification_performed
signer_invoked
private_key_handling
key_generation_performed
trust_store_loaded
revocation_lookup_performed
handoff_performed
effect_performed
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
mode
status
```

Expected values for an allowed first implementation result:

```
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
requested_signing_operation=metadata-only
signing_operation_state=operation-metadata-only
signing_operation_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
runtime_authority_granted=0
mode=metadata-only
status=signing-operation-metadata
```

Failure behavior

Future signing operation handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null signer invocation -&gt; invalid
invalid signer invocation -&gt; denied-signer-invocation
signer_invocation_ready=0 -&gt; denied-signer-invocation
signer_invocation_state not invocation-metadata-only -&gt; denied-signer-invocation
requested_signer_invocation not metadata-only -&gt; denied-signer-invocation
signer_handoff_ready=0 -&gt; denied-signer-handoff
signer_handoff_state not handoff-metadata-only -&gt; denied-signer-handoff
requested_signer_handoff not metadata-only -&gt; denied-signer-handoff
requested_signing_authorization not metadata-only -&gt; denied-signing-authorization
missing requested signature -&gt; denied-signature-algorithm
unknown requested signature -&gt; denied-signature-algorithm
missing requested signing operation -&gt; denied-signing-operation
unknown requested signing operation -&gt; denied-signing-operation
private-key handling requested or observed -&gt; denied-private-key
key generation requested or observed -&gt; denied-private-key
trust-store loading requested or observed -&gt; denied-private-key
revocation lookup requested or observed -&gt; denied-network-effect
signature already performed -&gt; denied-host-effect
verification already performed -&gt; denied-host-effect
signer already invoked -&gt; denied-host-effect
runtime authority already granted -&gt; denied-runtime-authority
handoff already performed -&gt; denied-host-effect
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
```

```
network already performed -&gt; denied-network-effect
```

Failures must not sign, verify signatures, invoke a signer, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
signing operation metadata implementation
```

It does not permit cryptographic signing, signature verification, signer invocation behavior, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, effect execution, capability enforcement, runtime authority, host behavior, network behavior, object sealing, or kernel behavior.

After signing operation metadata, its status/public-entry checkpoint, the key-handling boundary contract, key-handling metadata implementation, key-handling status/public-entry alignment, the key-material boundary contract, and key-material metadata implementation exist and are guarded, the next valid planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without a separate implementation contract, key-handling contract, key-material contract, and guards.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-signing-operation-contract.sh
```

docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md

Source title: Latticra Seal Signing Operation Implementation

Lines: 182 | Words: 551 | SHA-256: 9ddc9671c1c277f5546caa560a1f5d64ec79270f7943ccf7ffabff02552ae088

Latticra Seal Signing Operation Implementation

Status: initial signing operation metadata implementation Scope: bounded C metadata surface for classifying Seal signing operation eligibility after ready signer invocation metadata. This slice does not add cryptographic signing, signature verification, signer invocation behavior, signer process execution, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal signing operation metadata implementation.

The implementation consumes ready signer invocation metadata and classifies whether the request remains eligible for a future metadata-only signing operation path.

It is signing-operation path classification only.

It does not invoke a signer.

It does not sign.

It does not handle private keys.

Files

```
include/latticra/seal_signing_operation.h
src/seal_signing_operation.c
tests/seal_signing_operation_invariants.c
scripts/test-latticra-seal-signing-operation.sh
```

Required predecessors

This implementation depends on the signing operation contract and the signer invocation metadata surface:

```
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
include/latticra/seal_signer_invocation.h
src/seal_signer_invocation.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signing-operation-contract.sh
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signer-invocation-status.sh
```

Implemented surface

The signing operation metadata surface adds:

```
latticra_seal_signing_operation_t
latticra_seal_signing_operation_error_t
latticra_seal_signing_operation_error_label
latticra_seal_signing_operation_from_invocation
latticra_seal_signing_operation_is_metadata_only
latticra_seal_signing_operation_render
```

The implementation:

```
accepts signer invocation metadata
requires signer_invocation_ready=1
requires signer_invocation_state=invocation-metadata-only
requires requested_signature=Ed25519-development
requires requested_signing_authorization=metadata-only
requires requested_signer_handoff=metadata-only
requires requested_signer_invocation=metadata-only
accepts requested_signing_operation=metadata-only
classifies the result as signing_operation_state=operation-metadata-only
sets signing_operation_ready=1 only for metadata-only allowed signer invocation states
renders deterministic signing operation metadata
fails closed for invalid or denied predecessor metadata
```

Even when signing_operation_ready=1, these fields remain zero:

```
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Current metadata output

The successful metadata path renders:

```
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
```

```
requested_signing_operation=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
signer_invoked=0
mode=metadata-only
status=signing-operation-metadata
```

Failure behavior

The implementation fails closed for:

```
null output
null signer invocation
invalid signer invocation
signer_invocation_ready=0
signer_invocation_state not invocation-metadata-only
requested_signer_invocation not metadata-only
signer_handoff_ready=0
signer_handoff_state not handoff-metadata-only
requested_signer_handoff not metadata-only
signing_authorization_ready=0
signing_authorization_state not authorized-metadata-only
requested_signing_authorization not metadata-only
missing requested signature
unknown requested signature
missing requested signing operation
unknown requested signing operation
private-key handling already present
key generation already present
trust-store loading already present
revocation lookup already present
runtime authority already granted
signature already performed
verification already performed
signer already invoked
handoff already performed
effect already performed
host read already performed
host write already performed
network already performed
small render buffer
```

Failures do not sign, verify signatures, invoke a signer, handle private keys, generate keys, load trust stores, look up revocation status, read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform runtime handoff, persist policy, or grant runtime authority.

Validation

Run:

```
sh scripts/test-latticra-seal-signing-operation-contract.sh
sh scripts/test-latticra-seal-signing-operation.sh
```

Expected output:

```
seal signing operation contract: ok
seal signing operation invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is bounded no-effect key parsing implementation or another narrow status/index alignment follow-up that still must not add signing without a separate implementation contract, key-handling contract, key-material contract, and guards.

The signing operation metadata implementation and its status/public-entry checkpoint are guarded checkpoints. Future work must not add private-key handling, signing, verification, signer invocation behavior, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/LATTICRA_SEAL_STALE_REQUEST_CASE_CONTRACT.md

Source title: Latticra Seal Stale Request Case Contract

Lines: 108 | Words: 304 | SHA-256: 63160ba82cd6218eebdf7a8e3af280b2dcfb213d73dec5fe8342cd22c4c985f7

Latticra Seal Stale Request Case Contract

Status: Latticra Seal stale request case contract Scope: case-specific contract after the unsigned request case fixture. This document does not implement runtime behavior, policy behavior, tool behavior, host behavior, network behavior, cryptographic behavior, freshness validation, replay detection, or authority grants.

Purpose

This document defines the third case-specific test contract for the Seal runtime gate path.

The case is a stale request. The expected posture is that the gate remains report-only, blocked by default, and grants no authority.

This document does not implement the case behavior.

Required predecessors

```
docs/LATTICRA_SEAL_RUNTIME_GATE_TEST_PLAN.md
docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_CONTRACT.md
docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_CONTRACT.md
docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
tests/seal_unknown_tool_case.c
tests/seal_unsigned_request_case.c
scripts/test-latticra-seal-runtime-gate-test-plan.sh
scripts/test-latticra-seal-unknown-tool-case.sh
scripts/test-latticra-seal-unsigned-request-case.sh
```

Case definition

```
case_profile=latticra-seal-stale-request-case/0.1
case_name=stale-request
case_contract_present=1
stale_request_case_contract_present=1
input_request_timestamp_present=1
input_request_expiration_present=1
input_freshness_valid=0
expected_gate_state=report-only
expected_default_blocked=1
expected_stale_request_blocked=1
expected_runtime_authority_granted=0
expected_effect_performed=0
expected_host_read_performed=0
expected_host_write_performed=0
expected_network_performed=0
case_behavior_implemented=0
```

Future test requirements

A future implementation must add a deterministic test fixture showing that a stale request case remains blocked and does not perform effects.

Planned future file names:

```
tests/seal_stale_request_case.c
scripts/test-latticra-seal-stale-request-case.sh
```

Expected future output:

```
seal stale request case: ok
```

Boundary

This contract is documentation-only.

It does not add case execution, runtime behavior, host behavior, network behavior, MCP behavior, AI agent behavior, model behavior, tool behavior, shell behavior, policy behavior, freshness validation, replay detection, cryptographic behavior, or authority grants.

Claim gate

This contract alone does not justify the public claim that Latticra secures AI agents.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

A stronger claim requires implemented and validated gate behavior plus case tests for unknown, unsigned, stale, and replayed requests.

Non-claims

```
stale_request_case_implemented=0
runtime_gate_case_tests_implemented=0
runtime_behavior_implemented=0
policy_behavior_implemented=0
tool_behavior_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
freshness_validation_implemented=0
replay_detection_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is the stale request case test fixture.

docs/LATTICRA_SEAL_STALE_REQUEST_CASE_IMPLEMENTATION.md

Source title: Latticra Seal Stale Request Case Implementation

Lines: 51 | Words: 157 | SHA-256: 996c318bc4889a3817467301d860998b55afd28f2ba6d96ea78134104ac66369

Latticra Seal Stale Request Case Implementation

Status: initial stale request case fixture Scope: local deterministic case fixture after the stale request case contract.

Purpose

This record documents the stale-request case fixture for the Seal runtime gate metadata path.

The fixture keeps the gate in report-only state and verifies that no authority, host operation, network operation, or effect is reported.

Added files

```
tests/seal_stale_request_case.c
scripts/test-latticra-seal-stale-request-case.sh
```

Validation

Run:

```
sh scripts/test-latticra-seal-stale-request-case.sh
```

Expected output:

```
seal stale request case: ok
```

Boundary

This slice is a local deterministic fixture only.

It does not add runtime behavior, tool behavior, host behavior, network behavior, effect behavior, or authority grants.

Claim boundary

This fixture does not by itself justify a stronger public claim.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Next valid slice

The next valid slice is the replayed request case contract.

docs/LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md

Source title: Latticra Seal Status Rollup Contract

Lines: 238 | Words: 650 | SHA-256: 462631e0211a9bda71aeaf8c48ec261e0f27a39b78932d7f9a1f85add8177199

Latticra Seal Status Rollup Contract

Status: Latticra Seal status rollup contract Scope: contract for future status rollup metadata after runtime handoff metadata. This document does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal status rollup layer.

The purpose of this layer is to provide one bounded summary of the current Seal chain:

```
report -&gt; measurement -&gt; manifest -&gt; signature policy -&gt; signature metadata -&gt;
verification
policy -&gt; verification receipt -&gt; capability gate -&gt; effect decision -&gt; runtime
handoff
```

This document does not implement status rollup behavior.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
```

```
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
include/latticra/seal_report.h
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
include/latticra/seal_signature.h
include/latticra/seal_verification_policy.h
include/latticra/seal_verification_receipt.h
include/latticra/seal_capability_gate.h
include/latticra/seal_effect_decision.h
include/latticra/seal_runtime_handoff.h
src/seal_report.c
src/seal_measurement.c
src/seal_manifest.c
src/seal_signature_policy.c
src/seal_signature.c
src/seal_verification_policy.c
src/seal_verification_receipt.c
src/seal_capability_gate.c
src/seal_effect_decision.c
src/seal_runtime_handoff.c
scripts/test-latticra-seal-report.sh
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
scripts/test-latticra-seal-signature.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-capability-gate.sh
scripts/test-latticra-seal-effect-decision.sh
scripts/test-latticra-seal-runtime-handoff-contract.sh
scripts/test-latticra-seal-runtime-handoff.sh
```

The runtime handoff metadata surface remains inactive metadata until runtime-boundary behavior exists and is guarded.

Rollup boundary

The rollup may summarize Seal layer posture.

The rollup may not convert unsupported, unverified, denied, or inactive metadata into authority.

The rollup may not perform host reads, host writes, network access, kernel interaction, or runtime behavior.

Allowed in this contract slice:

```
rollup vocabulary
rollup-state labels
layer-present metadata planning
unsupported-state summary planning
denied-state summary planning
inactive-runtime summary planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this contract slice:

```
runtime execution
runtime authority grants
effect execution
host reads
host writes
network access
capability enforcement
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
```

```
revocation lookup
object sealing
kernel interaction
```

Initial rollup policy

The initial status rollup policy is conservative and metadata-only.

```
all implemented Seal layers may be summarized
cryptographic verification remains unsupported
verified receipt remains false
capability gate remains denied
effect decision remains denied
runtime handoff remains inactive
runtime boundary remains disabled
runtime authority remains zero
host read/write/network behavior remains zero
```

The first implementation after this contract may only add metadata fields and deterministic rendering. It must not perform runtime behavior.

Planned rollup states

Future records should use explicit labels:

```
rollup_state=metadata-only
rollup_state=incomplete
rollup_state=unsupported
rollup_state=denied
rollup_state=runtime-disabled
rollup_state=ready-for-review
```

For the next implementation, the expected state is:

```
rollup_state=metadata-only
```

Planned rollup fields

A future status rollup record should be bounded and deterministic.

Planned fields:

```
rollup_profile
report_present
measurement_present
manifest_present
signature_policy_present
signature_metadata_present
verification_policy_present
verification_receipt_present
capability_gate_present
effect_decision_present
runtime_handoff_present
cryptographic_verification_supported
verified
capability_gate_allowed
effect_allowed
handoff_active
runtime_boundary_state
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
rollup_state
status
```

Initial values before real authority:

```
cryptographic_verification_supported=0
verified=0
capability_gate_allowed=0
effect_allowed=0
handoff_active=0
runtime_boundary_state=disabled
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

```
rollup_state=metadata-only
status=status-rollup-contract-only
```

Failure behavior

Future status rollup handling must fail closed.

Required failure states:

```
null rollup output -&gt; invalid
missing required layer metadata -&gt; incomplete
unsupported verification -&gt; metadata-only
unverified receipt -&gt; metadata-only
denied capability gate -&gt; metadata-only
denied effect decision -&gt; metadata-only
inactive runtime handoff -&gt; metadata-only
runtime authority request -&gt; rejected
```

Failures must not parse keys, create keys, persist secrets, contact networks, query revocation status, verify records, sign records, read host files, write host files, enforce capabilities, execute effects, call runtime components, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
status rollup metadata implementation
```

It does not permit runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt generation, key handling, trust-store behavior, revocation lookup, object sealing, network behavior, host reads, host writes, or kernel behavior.

After status rollup metadata exists and is guarded, the next valid planning slice is Seal documentation/index alignment.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-status-rollup-contract.sh
```

docs/LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md

Source title: Latticra Seal Status Rollup Implementation

Lines: 131 | Words: 500 | SHA-256: 959baa0de31fccf5acb067d92585b214bb5706fe59fb5db3aeaed42f72ae5b31

Latticra Seal Status Rollup Implementation

Status: initial status rollup metadata implementation Scope: bounded C metadata surface for status rollup posture after runtime handoff metadata. This slice does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal status rollup metadata implementation.

The implementation accepts an existing runtime handoff metadata record and produces deterministic status rollup metadata for the implemented Seal chain.

Added files

```
include/latticra/seal_status_rollup.h
src/seal_status_rollup.c
tests/seal_status_rollup_invariants.c
scripts/test-latticra-seal-status-rollup.sh
```

API summary

The status rollup metadata surface adds:

```
latticra_seal_status_rollup_t
latticra_seal_status_rollup_error_t
latticra_seal_status_rollup_error_label
latticra_seal_status_rollup_from_handoff
latticra_seal_status_rollup_is_metadata_only
latticra_seal_status_rollup_report
```

Rollup behavior

The implementation:

```
accepts a valid runtime handoff metadata record
sets report_present=1
sets measurement_present=1
sets manifest_present=1
sets signature_policy_present=1
sets signature_metadata_present=1
sets verification_policy_present=1
sets verification_receipt_present=1
sets capability_gate_present=1
sets effect_decision_present=1
sets runtime_handoff_present=1
sets cryptographic_verification_supported=0
sets verified=0
sets capability_gate_allowed=0
copies effect_allowed from handoff metadata
copies handoff_active from handoff metadata
copies runtime_boundary_state from handoff metadata
copies runtime_authority_granted from handoff metadata
copies host_read_performed from handoff metadata
copies host_write_performed from handoff metadata
copies network_performed from handoff metadata
sets rollup_state=metadata-only
renders deterministic status rollup metadata
```

Boundary

This implementation does not verify signatures, parse public keys, create keys, store keys, contact networks, query revocation status, persist status records, enforce capabilities, perform host reads, perform host writes, execute network behavior, call runtime components, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null status rollup output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null runtime handoff metadata input -&gt; invalid-input
invalid runtime handoff metadata -&gt; invalid-handoff
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, read host files, write host files, enforce capabilities, perform effects, call runtime components, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid runtime handoff metadata produces deterministic status rollup metadata
report_present is one
measurement_present is one
```

```
manifest_present is one
signature_policy_present is one
signature_metadata_present is one
verification_policy_present is one
verification_receipt_present is one
capability_gate_present is one
effect_decision_present is one
runtime_handoff_present is one
cryptographic_verification_supported remains zero
verified remains zero
capability_gate_allowed remains zero
effect_allowed remains zero
handoff_active remains zero
runtime_boundary_state remains disabled
runtime_authority_granted remains zero
host_read_performed remains zero
host_write_performed remains zero
network_performed remains zero
rollup_state remains metadata-only
small report buffer fails closed
null inputs fail closed
invalid runtime handoff metadata fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-status-rollup-contract.sh
sh scripts/test-latticra-seal-status-rollup.sh
sh scripts/test-latticra-seal-status-rollup-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-agentic-automation-security-status.sh
```

Next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_CONTRACT.md

Source title: Latticra Seal Unknown Tool Case Contract

Lines: 101 | Words: 291 | SHA-256: 035f38571d7b4805931f06ca4ef72654505d44c63e9784844fda72ff95790206

Latticra Seal Unknown Tool Case Contract

Status: Latticra Seal unknown tool case contract Scope: case-specific contract after the Seal runtime gate test plan. This document does not implement runtime behavior, policy behavior, tool behavior, host behavior, network behavior, cryptographic behavior, or authority grants.

Purpose

This document defines the first case-specific test contract for the Seal runtime gate path.

The case is an unknown tool request. The expected posture is that the gate remains report-only, blocked by default, and grants no authority.

This document does not implement the case behavior.

Required predecessors

```
docs/LATTICRA_SEAL_RUNTIME_GATE_TEST_PLAN.md
docs/LATTICRA_SEAL_RUNTIME_ENFORCEMENT_GATE_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
scripts/test-latticra-seal-runtime-gate-test-plan.sh
scripts/test-latticra-seal-runtime-gate-contract.sh
scripts/test-latticra-seal-runtime-gate.sh
```

Case definition

```
case_profile=latticra-seal-unknown-tool-case/0.1
case_name=unknown-tool
case_contract_present=1
unknown_tool_case_contract_present=1
input_tool_known=0
input_tool_registered=0
input_tool_manifest_present=0
expected_gate_state=report-only
expected_default_blocked=1
expected_unknown_tool_blocked=1
expected_runtime_authority_granted=0
expected_effect_performed=0
expected_host_read_performed=0
expected_host_write_performed=0
expected_network_performed=0
case_behavior_implemented=0
```

Future test requirements

A future implementation must add a deterministic test fixture showing that an unknown tool case remains blocked and does not perform effects.

Planned future file names:

```
tests/seal_unknown_tool_case.c
scripts/test-latticra-seal-unknown-tool-case.sh
```

Expected future output:

```
seal unknown tool case: ok
```

Boundary

This contract is documentation-only.

It does not add case execution, runtime behavior, host behavior, network behavior, MCP behavior, AI agent behavior, model behavior, tool behavior, shell behavior, policy behavior, cryptographic behavior, or authority grants.

Claim gate

This contract alone does not justify the public claim that Latticra secures AI agents.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

A stronger claim requires implemented and validated gate behavior plus case tests for unknown, unsigned, stale, and replayed requests.

Non-claims

```
unknown_tool_case_implemented=0
runtime_gate_case_tests_implemented=0
runtime_behavior_implemented=0
policy_behavior_implemented=0
tool_behavior_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is the unknown tool case test fixture.

docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_IMPLEMENTATION.md

Source title: Latticra Seal Unknown Tool Case Implementation

Lines: 43 | Words: 150 | SHA-256: 365bf863379e18f0f228d177bd7acda68773d0e48064b20df8468101de457e79

Latticra Seal Unknown Tool Case Implementation

Status: initial unknown tool case test fixture Scope: first negative-test evidence slice after the runtime gate contract. This slice does not implement runtime enforcement, allow/deny outcomes, runtime authority, tool execution, shell execution, network behavior, host behavior, cryptographic validation, or effect execution.

Purpose

This document records the unknown-tool test fixture for the Seal runtime gate.

It uses report-only runtime gate metadata and ensures the unknown tool case remains blocked by default.

Added files

```
tests/seal_unknown_tool_case.c
scripts/test-latticra-seal-unknown-tool-case.sh
```

Validation

Run:

```
sh scripts/test-latticra-seal-unknown-tool-case.sh
```

Expected output:

```
seal unknown tool case: ok
```

Claim gate

This test fixture alone does not justify a public claim that Latticra secures AI agents.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Stronger claims require additional blocked-request tests (unsigned, stale, replayed) and validated runtime-enforcement behavior.

docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_CONTRACT.md

Source title: Latticra Seal Unsigned Request Case Contract

Lines: 104 | Words: 296 | SHA-256: 28d75312004933f9232b2325a8434966d2e907def56b4f8e5756f81b96aca3a6

Latticra Seal Unsigned Request Case Contract

Status: Latticra Seal unsigned request case contract Scope: case-specific contract after the unknown tool case fixture. This document does not implement runtime behavior, policy behavior, tool behavior, host behavior, network behavior, cryptographic behavior, signature verification, or authority grants.

Purpose

This document defines the second case-specific test contract for the Seal runtime gate path.

The case is an unsigned request. The expected posture is that the gate remains report-only, blocked by default, and grants no authority.

This document does not implement the case behavior.

Required predecessors

```
docs/LATTICRA_SEAL_RUNTIME_GATE_TEST_PLAN.md
docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_CONTRACT.md
docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
tests/seal_unknown_tool_case.c
scripts/test-latticra-seal-runtime-gate-test-plan.sh
scripts/test-latticra-seal-unknown-tool-case-contract.sh
scripts/test-latticra-seal-unknown-tool-case.sh
```

Case definition

```
case_profile=latticra-seal-unsigned-request-case/0.1
case_name=unsigned-request
case_contract_present=1
unsigned_request_case_contract_present=1
input_signature_present=0
input_signature_valid=0
input_signed_request_present=0
expected_gate_state=report-only
expected_default_blocked=1
expected_unsigned_request_blocked=1
expected_runtime_authority_granted=0
expected_effect_performed=0
expected_host_read_performed=0
expected_host_write_performed=0
expected_network_performed=0
case_behavior_implemented=0
```

Future test requirements

A future implementation must add a deterministic test fixture showing that an unsigned request case remains blocked and does not perform effects.

Planned future file names:

```
tests/seal_unsigned_request_case.c
scripts/test-latticra-seal-unsigned-request-case.sh
```

Expected future output:

```
seal unsigned request case: ok
```

Boundary

This contract is documentation-only.

It does not add case execution, runtime behavior, host behavior, network behavior, MCP behavior, AI agent behavior, model behavior, tool behavior, shell behavior, policy behavior, cryptographic behavior, signature verification, or authority grants.

Claim gate

This contract alone does not justify the public claim that Latticra secures AI agents.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

A stronger claim requires implemented and validated gate behavior plus case tests for unknown, unsigned, stale, and replayed requests.

Non-claims

```
unsigned_request_case_implemented=0
runtime_gate_case_tests_implemented=0
runtime_behavior_implemented=0
policy_behavior_implemented=0
tool_behavior_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
signature_verification_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid Latticra Seal slice is the unsigned request case test fixture.

docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_IMPLEMENTATION.md

Source title: Latticra Seal Unsigned Request Case Implementation

Lines: 51 | Words: 157 | SHA-256: 4fda14e9c4d86aabbdacce2efcfc4527c6f5f695a20fc9a4cfd0e74cff3f2223

Latticra Seal Unsigned Request Case Implementation

Status: initial unsigned request case fixture Scope: local deterministic case fixture after the unsigned request case contract.

Purpose

This record documents the unsigned-request case fixture for the Seal runtime gate metadata path.

The fixture keeps the gate in report-only state and verifies that no authority, host operation, network operation, or effect is reported.

Added files

```
tests/seal_unsigned_request_case.c
scripts/test-latticra-seal-unsigned-request-case.sh
```

Validation

Run:

```
sh scripts/test-latticra-seal-unsigned-request-case.sh
```

Expected output:

```
seal unsigned request case: ok
```

Boundary

This slice is a local deterministic fixture only.

It does not add runtime behavior, tool behavior, host behavior, network behavior, effect behavior, or authority grants.

Claim boundary

This fixture does not by itself justify a stronger public claim.

The accurate public claim remains:

```
Latticra Seal is building a report-only trust boundary for AI-era automation.
```

Next valid slice

The next valid slice is the stale request case contract.

docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md

Source title: Latticra Seal Verification Policy Contract

Lines: 202 | Words: 566 | SHA-256: f0faf5877832fb4e4a5b5a01030ecbfc19acb1675045cbfc0529fec5c0f11fe4

Latticra Seal Verification Policy Contract

Status: Latticra Seal verification policy contract Scope: policy for future verification behavior after the signature metadata envelope. This document does not implement cryptographic verification, public-key parsing, public-key trust stores, key generation, private-key storage, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal verification policy layer.

The purpose of this layer is to decide how future signature metadata may be evaluated before any signature metadata is allowed to participate in capability, gate, or authority decisions.

This document does not implement cryptographic verification.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
include/latticra/seal_signature.h
src/seal_measurement.c
src/seal_manifest.c
src/seal_signature_policy.c
src/seal_signature.c
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
scripts/test-latticra-seal-signature.sh
```

The signature metadata surface remains metadata-only until a later verification implementation explicitly adds cryptographic verification and evidence.

Verification boundary

No signature metadata becomes authority without verification policy.

No verification policy becomes authority without a later verified evidence record and explicit capability gate.

Allowed in this policy slice:

```
verification vocabulary
verification-state labels
trust-source labels
public-key metadata planning
algorithm acceptance policy
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this policy slice:

```
cryptographic verification
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
network trust lookup
revocation lookup
certificate authority behavior
hardware token behavior
object sealing
capability enforcement
runtime authority grants
host writes
kernel interaction
```

Initial verification policy

The initial verification policy is conservative.

```
signature metadata may be inspected
signature metadata may not be trusted as verified
verification support remains disabled
cryptographic verification is not performed
public-key material is not handled
private keys are not handled
network lookup remains disabled
runtime authority remains denied
```

The first implementation after this contract may only add metadata fields and unsupported-state handling. It must not verify signatures.

Planned verification states

Future records should use explicit labels:

```
verification_state=unsupported
verification_state=unverified
verification_state=verification-requested
verification_state=verified
verification_state=invalid
verification_state=expired
verification_state=revoked
verification_state=unsupported-algorithm
verification_state=missing-public-key
verification_state=policy-denied
```

For the next implementation, the expected state is:

```
verification_state=unsupported
```

Planned policy fields

A future verification policy record should be bounded and deterministic.

Planned fields:

```
verification_policy_profile
signature_profile
manifest_profile
artifact_digest_algorithm
artifact_digest_hex
signer_identity_label
signature_algorithm
public_key_identity_label
trust_source
verification_state
cryptographic_verification_supported
cryptographic_verification_performed
public_key_material_handling
private_key_handling
network_lookup_allowed
revocation_lookup_allowed
runtime_authority_granted
status
```

Initial values before implementation:

```
cryptographic_verification_supported=0
cryptographic_verification_performed=0
public_key_material_handling=0
private_key_handling=0
network_lookup_allowed=0
revocation_lookup_allowed=0
runtime_authority_granted=0
status=verification-policy-only
```

Failure behavior

Future verification policy handling must fail closed.

Required failure states:

```
null signature metadata -&gt; invalid
invalid signature metadata -&gt; invalid
missing artifact digest -&gt; invalid
missing signer identity -&gt; invalid
unsupported algorithm -&gt; unsupported-algorithm
missing public-key identity -&gt; unsupported
verification requested before support -&gt; unsupported
private-key input -&gt; rejected
network lookup request -&gt; rejected
revocation lookup request -&gt; rejected
runtime authority request -&gt; rejected
```

Failures must not parse keys, create keys, persist secrets, contact networks, verify records, sign records, write host files, enforce capabilities, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
verification policy metadata implementation
```

It does not permit cryptographic verification, key handling, trust-store behavior, revocation lookup, object sealing, capability enforcement, network behavior, host writes, or runtime authority.

After verification policy metadata exists and is guarded, the next valid planning slice is a verification receipt contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-verification-policy-contract.sh
```

docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md

Source title: Latticra Seal Verification Policy Implementation

Lines: 144 | Words: 551 | SHA-256: 8bf5ff66c9a9d8ddd749d05c79449a68d0c726ac520756272369979e78c3fdc3

Latticra Seal Verification Policy Implementation

Status: initial verification policy metadata implementation Scope: bounded C metadata surface for verification policy posture after caller-supplied signature metadata. This slice does not implement cryptographic verification, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal verification policy metadata implementation.

The implementation accepts an existing signature metadata envelope and produces deterministic verification policy metadata. It does not verify cryptographic signatures and does not treat signature metadata as authority.

Added files

```
include/latticra/seal_verification_policy.h
src/seal_verification_policy.c
tests/seal_verification_policy_invariants.c
scripts/test-latticra-seal-verification-policy.sh
```

API summary

The verification policy metadata surface adds:

```
latticra_seal_verification_policy_t
latticra_seal_verification_policy_error_t
latticra_seal_verification_policy_error_label
latticra_seal_verification_policy_from_signature
latticra_seal_verification_policy_is_metadata_only
latticra_seal_verification_policy_report
```

Verification policy behavior

The implementation:

```
accepts a valid signature metadata envelope
copies the signature profile
copies the manifest profile
copies artifact digest algorithm metadata
copies artifact digest hex metadata
copies signer identity metadata
accepts only the Ed25519-development signature algorithm label
records a caller-supplied public-key identity label
records a caller-supplied trust-source label, defaulting to local-metadata-only
sets verification_state=unsupported
sets cryptographic_verification_supported=0
sets cryptographic_verification_performed=0
sets public_key_material_handling=0
sets private_key_handling=0
sets network_lookup_allowed=0
sets revocation_lookup_allowed=0
sets runtime_authority_granted=0
renders deterministic verification policy metadata
```

Boundary

This implementation does not parse public keys, verify signatures, create keys, store keys, contact networks, query revocation status, persist verification receipts, seal objects, enforce capabilities, write host files, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null verification policy output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null signature metadata input -&gt; invalid-input
invalid signature metadata -&gt; invalid-signature
missing artifact digest -&gt; missing-digest
missing signer identity -&gt; missing-signer
missing public-key identity label -&gt; missing-public-key-identity
unsupported algorithm label -&gt; unsupported-algorithm
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, write host files, enforce capabilities, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid signature metadata plus public-key identity produces deterministic verification policy
```

```
metadata
signature profile is copied
manifest profile is copied
artifact digest algorithm is copied
artifact digest hex is copied
signer identity label is copied
signature algorithm label is copied
public-key identity label is recorded
trust source is recorded
missing trust source defaults to local-metadata-only
verification_state remains unsupported
cryptographic_verification_supported remains zero
cryptographic_verification_performed remains zero
public_key_material_handling remains zero
private_key_handling remains zero
network_lookup_allowed remains zero
revocation_lookup_allowed remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid signature metadata fails closed
missing digest fails closed
missing signer fails closed
missing public-key identity fails closed
unsupported algorithm fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-verification-policy-contract.sh
sh scripts/test-latticra-seal-verification-policy.sh
sh scripts/test-latticra-seal-verification-policy-status.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
```

Expected output:

```
seal verification policy contract: ok
seal verification policy invariants: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
```

Next valid slice

The next valid Latticra Seal slice is status rollup status/public-entry alignment.

That future slice must not add runtime execution, effect execution, capability enforcement, cryptographic verification, verified receipt authority, signing, key material loading, private-key handling, host behavior, network behavior, or runtime authority unless separately implemented and guarded.

docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md

Source title: Latticra Seal Verification Receipt Contract

Lines: 210 | Words: 627 | SHA-256: 32595194550d799085750e0029ae78876a1e732d181a9c3686ab58a54513bd64

Latticra Seal Verification Receipt Contract

Status: Latticra Seal verification receipt contract Scope: contract for future verification receipt metadata after verification policy metadata. This document does not implement cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the Latticra Seal verification receipt layer.

The purpose of this layer is to decide how a future verification result may be represented as bounded evidence before any verification result is allowed to participate in capability, gate, or authority decisions.

This document does not implement verification receipts.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_MEASUREMENT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNED_MANIFEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_POLICY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
include/latticra/seal_measurement.h
include/latticra/seal_manifest.h
include/latticra/seal_signature_policy.h
include/latticra/seal_signature.h
include/latticra/seal_verification_policy.h
src/seal_measurement.c
src/seal_manifest.c
src/seal_signature_policy.c
src/seal_signature.c
src/seal_verification_policy.c
scripts/test-latticra-seal-measurement.sh
scripts/test-latticra-seal-manifest.sh
scripts/test-latticra-seal-signature-policy.sh
scripts/test-latticra-seal-signature.sh
scripts/test-latticra-seal-verification-policy-contract.sh
scripts/test-latticra-seal-verification-policy.sh
```

The verification policy metadata surface remains the source of truth for unsupported verification posture until a later verification receipt implementation exists and is guarded.

Receipt boundary

No signature metadata becomes authority without a receipt.

No receipt becomes authority without cryptographic verification evidence and a later explicit capability gate.

A verification receipt may summarize verification policy posture, but it must not imply success when cryptographic verification is unsupported or unperformed.

Allowed in this policy slice:

```
receipt vocabulary
receipt-state labels
verification-result labels
trust-source metadata planning
failure-state planning
promotion rules
non-claims
static guard validation
```

Forbidden in this policy slice:

```
cryptographic verification
verified receipt generation
public-key parsing
public-key trust store loading
private-key handling
key generation
signature generation
network trust lookup
revocation lookup
certificate authority behavior
```

```
hardware token behavior
object sealing
capability enforcement
runtime authority grants
host writes
kernel interaction
```

Initial receipt policy

The initial verification receipt policy is conservative.

```
verification policy metadata may be copied
verification_state=unsupported must remain unverified
cryptographic verification is not performed
receipt may not assert verified=true
receipt may not authorize capability gates
receipt may not grant runtime authority
public-key material is not handled
private keys are not handled
network lookup remains disabled
revocation lookup remains disabled
```

The first implementation after this contract may only add metadata fields and unsupported-state handling. It must not verify signatures or emit a verified receipt.

Planned receipt states

Future records should use explicit labels:

```
receipt_state=unsupported
receipt_state=unverified-metadata
receipt_state=verification-unsupported
receipt_state=verified
receipt_state=invalid
receipt_state=expired
receipt_state=revoked
receipt_state=policy-denied
```

For the next implementation, the expected state is:

```
receipt_state=unverified-metadata
```

Planned receipt fields

A future verification receipt record should be bounded and deterministic.

Planned fields:

```
receipt_profile
verification_policy_profile
signature_profile
manifest_profile
artifact_digest_algorithm
artifact_digest_hex
signer_identity_label
signature_algorithm
public_key_identity_label
trust_source
verification_state
receipt_state
cryptographic_verification_supported
cryptographic_verification_performed
verified
invalid
authority_usable
capability_gate_allowed
runtime_authority_granted
status
```

Initial values before real verification:

```
verification_state=unsupported
receipt_state=unverified-metadata
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verified=0
invalid=0
authority_usable=0
```

```
capability_gate_allowed=0
runtime_authority_granted=0
status=verification-receipt-contract-only
```

Failure behavior

Future verification receipt handling must fail closed.

Required failure states:

```
null verification policy metadata -&gt; invalid
invalid verification policy metadata -&gt; invalid
missing artifact digest -&gt; invalid
missing signer identity -&gt; invalid
missing public-key identity -&gt; invalid
verification unsupported -&gt; unverified-metadata
verified receipt requested before support -&gt; unsupported
capability gate requested before support -&gt; rejected
runtime authority request -&gt; rejected
```

Failures must not parse keys, create keys, persist secrets, contact networks, query revocation status, verify records, sign records, write host files, enforce capabilities, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
verification receipt metadata implementation
```

It does not permit cryptographic verification, verified receipt generation, key handling, trust-store behavior, revocation lookup, object sealing, capability enforcement, network behavior, host writes, or runtime authority.

After verification receipt metadata exists and is guarded, the next valid planning slice is a capability gate contract.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-verification-receipt-contract.sh
```

docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md

Source title: Latticra Seal Verification Receipt Implementation

Lines: 143 | Words: 535 | SHA-256: 643237f0d09fa9ea5168907622d27d77b1a07ee8c81b5b1d551c08460c1aa157

Latticra Seal Verification Receipt Implementation

Status: initial verification receipt metadata implementation Scope: bounded C metadata surface for verification receipt posture after verification policy metadata. This slice does not implement cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal verification receipt metadata implementation.

The implementation accepts an existing verification policy metadata record and produces deterministic receipt metadata. It does not verify cryptographic signatures and does not treat a receipt as authority.

Added files

```
include/latticra/seal_verification_receipt.h
src/seal_verification_receipt.c
tests/seal_verification_receipt_invariants.c
scripts/test-latticra-seal-verification-receipt.sh
```

API summary

The verification receipt metadata surface adds:

```
latticra_seal_verification_receipt_t
latticra_seal_verification_receipt_error_t
latticra_seal_verification_receipt_error_label
latticra_seal_verification_receipt_from_policy
latticra_seal_verification_receipt_is_unverified_metadata
latticra_seal_verification_receipt_report
```

Receipt behavior

The implementation:

```
accepts a valid verification policy metadata record
copies the verification policy profile
copies the signature profile
copies the manifest profile
copies artifact digest algorithm metadata
copies artifact digest hex metadata
copies signer identity metadata
copies signature algorithm metadata
copies public-key identity metadata
copies trust-source metadata
copies verification_state
sets receipt_state=unverified-metadata
sets cryptographic_verification_supported=0
sets cryptographic_verification_performed=0
sets verified=0
sets invalid=0
sets authority_usable=0
sets capability_gate_allowed=0
sets runtime_authority_granted=0
renders deterministic verification receipt metadata
```

Boundary

This implementation does not parse public keys, verify signatures, create keys, store keys, contact networks, query revocation status, persist verified receipts, seal objects, enforce capabilities, write host files, or grant runtime authority.

It is metadata only.

Failure behavior

The implementation fails closed:

```
null verification receipt output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null verification policy metadata input -&gt; invalid-input
invalid verification policy metadata -&gt; invalid-policy
missing artifact digest -&gt; missing-digest
missing signer identity -&gt; missing-signer
missing public-key identity -&gt; missing-public-key-identity
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not parse keys, create keys, persist secrets, contact networks, verify records, sign records, write host files, enforce capabilities, allow capability gates, or grant runtime authority.

Invariants

The invariant test verifies:

```
valid verification policy metadata produces deterministic verification receipt metadata
verification policy profile is copied
signature profile is copied
manifest profile is copied
```

```
artifact digest algorithm is copied
artifact digest hex is copied
signer identity label is copied
signature algorithm label is copied
public-key identity label is copied
trust source is copied
verification_state is copied
receipt_state remains unverified-metadata
cryptographic_verification_supported remains zero
cryptographic_verification_performed remains zero
verified remains zero
invalid remains zero
authority_usable remains zero
capability_gate_allowed remains zero
runtime_authority_granted remains zero
small report buffer fails closed
null inputs fail closed
invalid verification policy metadata fails closed
missing digest fails closed
missing signer fails closed
missing public-key identity fails closed
```

Validation

Run:

```
sh scripts/test-latticra-seal-verification-receipt-contract.sh
sh scripts/test-latticra-seal-verification-receipt.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
```

Expected output:

```
seal verification receipt contract: ok
seal verification receipt invariants: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
```

Next valid slice

The next valid Latticra Seal slice is status rollup status/public-entry alignment.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, host behavior, network behavior, or object sealing unless separately implemented and guarded.

docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_CONTRACT.md

Source title: Latticra Seal Verified Capability Gate Contract

Lines: 200 | Words: 553 | SHA-256: fafdd75b4d3f254dfa393b131c8a502451dbf8f92b96eb654c633a0429aff6ea

Latticra Seal Verified Capability Gate Contract

Status: Latticra Seal verified capability gate contract Scope: contract for a future capability gate evaluation step after verified receipt promotion. This document does not implement capability authorization, effect execution, runtime authority, host effects, network behavior, signing, key generation, private-key storage, trust-store loading, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first capability gate evaluation boundary that may consume verified receipt metadata.

The purpose of this layer is to evaluate whether a verified but authority-neutral receipt could satisfy a declared capability request while still refusing runtime authority and effect execution.

The gate is policy evaluation, not effect execution.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_CONTRACT.md
docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_IMPLEMENTATION.md
include/latticra/seal_capability_gate.h
include/latticra/seal_verified_receipt_promotion.h
src/seal_capability_gate.c
src/seal_verified_receipt_promotion.c
tests/seal_capability_gate_invariants.c
tests/seal_verified_receipt_promotion_invariants.c
scripts/test-latticra-seal-capability-gate.sh
scripts/test-latticra-seal-verified-receipt-promotion.sh
```

The verified receipt promotion surface remains the evidence source for this gate.

Gate boundary

Allowed in the next implementation slice:

```
accept verified receipt promotion metadata
require receipt_state=verified
require verification_state=verified
require verified=1
require invalid=0
require cryptographic_verification_performed=1
accept requested capability metadata
accept requested effect metadata
accept requested scope metadata
compare requested capability against a narrow local allowlist
compare requested effect against a report-only/evaluate-only effect class
produce deterministic capability gate evaluation metadata
```

Forbidden in the next implementation slice:

```
host reads
host writes
network access
effect execution
runtime authority grants
shell execution
tool execution
capability enforcement
policy persistence
trust-store loading
revocation lookup
key generation
private-key handling
object sealing
kernel interaction
```

Initial gate policy

The initial policy is deliberately narrow.

Allowed requested capability labels:

```
verified-receipt-inspection
verified-receipt-report
```

Allowed requested effect labels:

```
report-only
evaluate-only
```

Any other capability or effect must deny.

Planned gate states

Future records should use explicit labels:

```
gate_state=allowed-metadata-only
gate_state=denied-unverified
gate_state=denied-invalid-receipt
gate_state=denied-unknown-capability
gate_state=denied-unknown-effect
gate_state=denied-policy
gate_state=denied-runtime-authority
```

The first implementation may set:

```
gate_allowed=1
gate_state=allowed-metadata-only
```

only for a verified receipt and a locally allowed report-only/evaluate-only request.

Even when gate_allowed=1, these must remain zero:

```
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future verified capability gate record should be bounded and deterministic.

Planned fields:

```
gate_profile
receipt_profile
verify_profile
message_digest_algorithm
message_digest_hex
public_key_identity_label
receipt_state
verification_state
requested_capability
requested_effect
requested_scope
verified
authority_usable
receipt_capability_gate_allowed
gate_allowed
gate_state
runtime_authority_granted
effect_performed
host_read_performed
host_write_performed
network_performed
status
```

Failure behavior

Future verified capability gate handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null verified receipt -&gt; invalid
receipt_state not verified -&gt; denied-unverified
verification_state not verified -&gt; denied-unverified
verified=0 -&gt; denied-unverified
invalid=1 -&gt; denied-invalid-receipt
missing message digest -&gt; denied-invalid-receipt
missing public-key identity -&gt; denied-invalid-receipt
missing requested capability -&gt; denied-unknown-capability
missing requested effect -&gt; denied-unknown-effect
unknown requested capability -&gt; denied-unknown-capability
unknown requested effect -&gt; denied-unknown-effect
runtime authority request -&gt; denied-runtime-authority
```

Failures must not read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform effects, persist policy, or grant runtime

authority.

Promotion rule

This contract permits only the next implementation slice:

```
verified capability gate metadata implementation
```

It does not permit effect execution, capability enforcement, runtime authority, host behavior, network behavior, trust-store behavior, revocation lookup, key handling, object sealing, or kernel behavior.

After verified capability gate metadata exists and is guarded, the next valid planning slice is effect decision evaluation from an allowed metadata-only capability gate.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-verified-capability-gate-contract.sh
```

docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_IMPLEMENTATION.md

Source title: Latticra Seal Verified Capability Gate Implementation

Lines: 138 | Words: 455 | SHA-256: f4236fc5bb0ebc45ba336e8a4823473181f359d0d0c516f7822c83062e56f364

Latticra Seal Verified Capability Gate Implementation

Status: initial verified capability gate metadata implementation Scope: bounded C metadata surface for evaluating a verified receipt against a narrow metadata-only capability/effect allowlist.

Purpose

This document records the first Latticra Seal verified capability gate metadata implementation.

The implementation accepts verified receipt promotion metadata plus requested capability, requested effect, and requested scope labels.

It also exposes a stricter crypto-bound entry point that requires a passing crypto graduation gate before evaluating the same metadata-only capability/effect allowlist.

It can allow metadata-only gate evaluation for a narrow local allowlist, but it does not perform effects or grant runtime authority.

Added files

```
include/latticra/seal_verified_capability_gate.h
src/seal_verified_capability_gate.c
tests/seal_verified_capability_gate_invariants.c
scripts/test-latticra-seal-verified-capability-gate.sh
include/latticra/seal_crypto_graduation_gate.h
src/seal_crypto_graduation_gate.c
```

API summary

The verified capability gate metadata surface adds:

```
latticra_seal_verified_capability_gate_t
latticra_seal_verified_capability_gate_error_t
latticra_seal_verified_capability_gate_error_label
latticra_seal_verified_capability_gate_from_receipt
latticra_seal_verified_capability_gate_from_crypto_graduation_gate
latticra_seal_verified_capability_gate_is_metadata_only
latticra_seal_verified_capability_gate_report
```

Gate behavior

The implementation:

```
accepts verified receipt promotion metadata
optionally requires crypto graduation gate metadata through the crypto-bound entry point
requires a passing crypto graduation gate for the crypto-bound entry point
requires crypto graduation gate metadata to match receipt profile, verify profile, digest,
public-key identity, receipt state, and verification state
requires receipt_state=verified
requires verification_state=verified
requires verified=1
requires invalid=0
requires message digest metadata
requires public-key identity metadata
accepts requested capability metadata
accepts requested effect metadata
accepts requested scope metadata
allows verified-receipt-inspection
allows verified-receipt-report
allows report-only
allows evaluate-only
sets gate_allowed=1 only for locally allowed metadata-only requests
sets gate_state=allowed-metadata-only only for locally allowed metadata-only requests
renders deterministic verified capability gate metadata
```

Crypto-bound successful gate metadata records:

```
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
crypto_graduation_gate_state=graduated-authority-neutral
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
gate_allowed=1
gate_state=allowed-metadata-only
```

Effect and runtime boundary

Even when gate_allowed=1, these fields remain zero:

```
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Failure behavior

The implementation fails closed:

```
null gate output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null verified receipt input -&gt; invalid-input
invalid verified receipt input -&gt; invalid-receipt
receipt_state not verified -&gt; denied-unverified
verification_state not verified -&gt; denied-unverified
verified flag not set -&gt; denied-unverified
invalid flag set -&gt; denied-invalid-receipt
missing message digest -&gt; denied-invalid-receipt
missing public-key identity -&gt; denied-invalid-receipt
missing requested capability -&gt; missing-requested-capability
missing requested effect -&gt; missing-requested-effect
unknown requested capability -&gt; denied-unknown-capability
unknown requested effect -&gt; denied-unknown-effect
missing crypto graduation gate in crypto-bound entry point -&gt; denied-crypto-graduation-gate
failed crypto graduation gate in crypto-bound entry point -&gt; denied-crypto-graduation-gate
crypto graduation gate / receipt mismatch -&gt; denied-crypto-graduation-gate
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform effects, persist policy, or grant runtime authority.

Validation

Run locally:

```
sh scripts/test-latticra-seal-verified-capability-gate.sh
```

Expected output:

```
seal verified capability gate invariants: ok
```

Next valid slice

The next valid Latticra Seal slice is effect decision evaluation from an allowed crypto-graduation-gated metadata-only capability gate.

docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_CONTRACT.md

Source title: Latticra Seal Verified Effect Decision Contract

Lines: 185 | Words: 518 | SHA-256: 4dc00b90a3902d2318d7b95b53e8f8ff494e68eb8287ee8e381b93cf1dbb556b

Latticra Seal Verified Effect Decision Contract

Status: Latticra Seal verified effect decision contract Scope: contract for a future effect-decision evaluation step after verified capability gate metadata. This document does not implement effect execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, signing, key generation, private-key storage, trust-store loading, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the first effect decision evaluation boundary that may consume an allowed metadata-only verified capability gate.

The purpose of this layer is to decide whether a previously allowed metadata-only gate remains limited to report/evaluate posture before any future runtime handoff is considered.

The decision is effect classification, not effect execution.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_IMPLEMENTATION.md
include/latticra/seal_effect_decision.h
include/latticra/seal_verified_capability_gate.h
src/seal_effect_decision.c
src/seal_verified_capability_gate.c
tests/seal_effect_decision_invariants.c
tests/seal_verified_capability_gate_invariants.c
scripts/test-latticra-seal-effect-decision.sh
scripts/test-latticra-seal-verified-capability-gate.sh
```

The verified capability gate metadata surface remains the source of gate evidence for this decision.

Decision boundary

Allowed in the next implementation slice:

```
accept verified capability gate metadata
require gate_allowed=1
require gate_state=allowed-metadata-only
```

```
require runtime_authority_granted=0
require effect_performed=0
require host_read_performed=0
require host_write_performed=0
require network_performed=0
accept requested effect metadata
classify report-only as decision_state=allowed-report-only
classify evaluate-only as decision_state=allowed-evaluate-only
produce deterministic effect decision metadata
```

Forbidden in the next implementation slice:

```
effect execution
runtime authority grants
host reads
host writes
network access
shell execution
tool execution
capability enforcement
policy persistence
trust-store loading
revocation lookup
key generation
private-key handling
object sealing
kernel interaction
```

Initial decision policy

Allowed input gate states:

```
gate_state=allowed-metadata-only
```

Allowed requested effect labels:

```
report-only
evaluate-only
```

Planned decision states:

```
decision_state=allowed-report-only
decision_state=allowed-evaluate-only
decision_state=denied-gate
decision_state=denied-effect
decision_state=denied-runtime-authority
decision_state=denied-host-effect
decision_state=denied-network-effect
```

The first implementation may set:

```
effect_allowed=1
```

only for metadata-only report/evaluate decisions.

Even when effect_allowed=1, these must remain zero:

```
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Planned fields

A future verified effect decision record should be bounded and deterministic.

Planned fields:

```
decision_profile
gate_profile
receipt_profile
requested_capability
requested_effect
requested_scope
gate_allowed
gate_state
decision_state
effect_allowed
effect_performed
```

```
runtime_authority_granted
host_read_performed
host_write_performed
network_performed
status
```

Failure behavior

Future verified effect decision handling must fail closed.

Required failure states:

```
null output -&gt; invalid
null verified gate -&gt; invalid
invalid verified gate -&gt; denied-gate
gate_allowed=0 -&gt; denied-gate
gate_state not allowed-metadata-only -&gt; denied-gate
missing requested effect -&gt; denied-effect
unknown requested effect -&gt; denied-effect
runtime authority already granted -&gt; denied-runtime-authority
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
```

Failures must not read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, perform effects, persist policy, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
verified effect decision metadata implementation
```

It does not permit effect execution, capability enforcement, runtime authority, host behavior, network behavior, trust-store behavior, revocation lookup, key handling, object sealing, or kernel behavior.

After verified effect decision metadata exists and is guarded, the next valid planning slice is runtime handoff evaluation from an allowed metadata-only effect decision.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-verified-effect-decision-contract.sh
```

docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_IMPLEMENTATION.md

Source title: Latticra Seal Verified Effect Decision Implementation

Lines: 150 | Words: 502 | SHA-256: 27c4e93d16684cf68168e48d7635d1a19da05eaed812edb65c61aa103bb96267

Latticra Seal Verified Effect Decision Implementation

Status: initial verified effect decision metadata implementation Scope: bounded C metadata surface for classifying allowed metadata-only verified capability gate output into a deterministic effect decision. This does not implement effect execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, trust-store loading, revocation lookup, key generation, private-key handling, object sealing, kernel interaction, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document records the first Latticra Seal verified effect decision metadata implementation.

The implementation consumes verified capability gate metadata and classifies whether the requested metadata-only effect remains allowed as report-only or evaluate-only.

When the capability gate carries crypto graduation metadata, the decision copies that evidence forward and requires it to remain passed, standard-aligned, and authority-neutral.

The decision is effect classification, not effect execution.

Added files

```
include/latticra/seal_verified_effect_decision.h
src/seal_verified_effect_decision.c
tests/seal_verified_effect_decision_invariants.c
scripts/test-latticra-seal-verified-effect-decision.sh
```

Required predecessor

This implementation depends on the verified capability gate metadata surface:

```
include/latticra/seal_verified_capability_gate.h
src/seal_verified_capability_gate.c
tests/seal_verified_capability_gate_invariants.c
scripts/test-latticra-seal-verified-capability-gate.sh
docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_IMPLEMENTATION.md
```

API summary

The verified effect decision metadata surface adds:

```
latticra_seal_verified_effect_decision_t
latticra_seal_verified_effect_decision_error_t
latticra_seal_verified_effect_decision_error_label
latticra_seal_verified_effect_decision_from_gate
latticra_seal_verified_effect_decision_is_metadata_only
latticra_seal_verified_effect_decision_report
```

Decision behavior

The implementation:

```
accepts verified capability gate metadata
copies crypto graduation gate metadata when present
requires crypto_graduation_gate_passed=1 when crypto_graduation_gate_present=1
requires standard_expectations_met=1 when crypto_graduation_gate_present=1
requires local_verify_graduated=1 when crypto_graduation_gate_present=1
requires receipt_promotion_graduated=1 when crypto_graduation_gate_present=1
requires authority_promotion_allowed=0 when crypto_graduation_gate_present=1
requires gate_allowed=1
requires gate_state=allowed-metadata-only
requires runtime_authority_granted=0
requires effect_performed=0
requires host_read_performed=0
requires host_write_performed=0
requires network_performed=0
accepts requested effect metadata
classifies report-only as decision_state=allowed-report-only
classifies evaluate-only as decision_state=allowed-evaluate-only
sets effect_allowed=1 only for metadata-only report/evaluate decisions
renders deterministic verified effect decision metadata
```

Crypto-bound decision metadata records:

```
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
crypto_graduation_gate_state=graduated-authority-neutral
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
decision_state=allowed-report-only
effect_allowed=1
effect_performed=0
runtime_authority_granted=0
```

Effect and runtime boundary

Even when effect_allowed=1, these fields remain zero:

```
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

This implementation does not read host files, write host files, contact networks, execute tools, execute shells, enforce capabilities, persist policy, load trust stores, look up revocation status, generate keys, handle private keys, seal objects, or interact with the kernel.

Failure behavior

The implementation fails closed:

```
null output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null verified gate -&gt; invalid-input
invalid verified gate -&gt; invalid-gate
gate_allowed=0 -&gt; denied-gate
gate_state not allowed-metadata-only -&gt; denied-gate
failed crypto graduation gate evidence -&gt; denied-crypto-graduation-gate
authority-bearing crypto graduation evidence -&gt; denied-crypto-graduation-gate
missing requested effect -&gt; missing-requested-effect
unknown requested effect -&gt; denied-unknown-effect
runtime authority already granted -&gt; denied-runtime-authority
effect already performed -&gt; denied-host-effect
host read already performed -&gt; denied-host-effect
host write already performed -&gt; denied-host-effect
network already performed -&gt; denied-network-effect
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not perform effects and do not grant runtime authority.

Validation

Run locally:

```
sh scripts/test-latticra-seal-verified-effect-decision.sh
```

Expected output:

```
seal verified effect decision invariants: ok
```

Next valid slice

The next valid Latticra Seal planning slice is runtime handoff report or policy decision report propagation from eligible crypto-graduation-gated metadata-only runtime handoff evaluation.

That next slice should remain contract-first and should not perform runtime handoff yet unless separately implemented and guarded.

docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_CONTRACT.md

Source title: Latticra Seal Verified Receipt Promotion Contract

Lines: 171 | Words: 494 | SHA-256: bcd2af5d9e71949ab93e31058a1cf2f1f8e6923a051aea5dad8411d1741e513f

Latticra Seal Verified Receipt Promotion Contract

Status: Latticra Seal verified receipt promotion contract Scope: contract for a future metadata promotion step after Ed25519 verify-only local verification. This document does not implement verified receipt promotion, capability authorization, effect execution, runtime authority, signing, key generation, private-key storage, trust-store loading, network lookup, revocation lookup, object sealing, host effects, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This document defines the verified receipt promotion boundary for Latticra Seal.

The purpose of this layer is to convert a successful local verify-only result into a bounded verified receipt record without granting authority.

The promotion is evidence promotion, not permission promotion.

Required predecessors

This contract depends on:

```
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md
docs/LATTICRA_SEAL_ED25519_VERIFY_ONLY_CONTRACT.md
docs/LATTICRA_SEAL_ED25519_VERIFY_IMPLEMENTATION.md
include/latticra/seal_verification_receipt.h
include/latticra/seal_crypto_verify_backend.h
include/latticra/seal_ed25519_verify.h
src/seal_verification_receipt.c
src/seal_crypto_verify_backend.c
src/seal_ed25519_verify.c
tests/seal_verification_receipt_invariants.c
tests/seal_crypto_verify_backend_invariants.c
tests/seal_ed25519_verify_invariants.c
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-ed25519-verify.sh
```

The Ed25519 verify-only result is the only allowed input source for the first promotion implementation.

Promotion boundary

Allowed in the next implementation slice:

```
accept a valid Ed25519 verify-only result
require cryptographic_verification_supported=1
require cryptographic_verification_performed=1
require verified=1
require invalid=0
require crypto_verify_state=verified
copy message digest metadata
copy public-key identity metadata
copy signature algorithm metadata
copy trust-source metadata
set receipt_state=verified
set verification_state=verified
render deterministic verified receipt metadata
```

Forbidden in the next implementation slice:

```
capability authorization
effect execution
runtime authority grants
host reads
host writes
network lookup
revocation lookup
trust-store loading
signature generation
key generation
private-key handling
object sealing
kernel interaction
```

Authority boundary

A promoted verified receipt remains authority-neutral.

The future implementation may set:

```
verified=1
cryptographic_verification_supported=1
cryptographic_verification_performed=1
receipt_state=verified
```

```
verification_state=verified
```

The future implementation must keep these zero:

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
```

A verified receipt may become usable by a later capability gate only after a separate contract explicitly permits that behavior.

Planned fields

A future verified receipt promotion record should be bounded and deterministic.

Planned fields:

```
receipt_profile
verify_profile
backend_profile
verification_policy_profile
message_label
message_size_bytes
message_digest_algorithm
message_digest_hex
public_key_identity_label
signature_algorithm
trust_source
verification_state
receipt_state
cryptographic_verification_supported
cryptographic_verification_performed
verified
invalid
authority_usable
capability_gate_allowed
runtime_authority_granted
status
```

Failure behavior

Future verified receipt promotion must fail closed.

Required failure states:

```
null output -&gt; invalid
null verify result -&gt; invalid
unsupported verification -&gt; rejected
verification not performed -&gt; rejected
verified=0 -&gt; rejected
invalid=1 -&gt; rejected
crypto_verify_state not verified -&gt; rejected
missing message digest -&gt; rejected
missing public-key identity -&gt; rejected
runtime authority request -&gt; rejected
capability authorization request -&gt; rejected
```

Failures must not create keys, store keys, contact networks, query revocation status, sign records, read host files, write host files, enforce capabilities, perform effects, or grant runtime authority.

Promotion rule

This contract permits only the next implementation slice:

```
verified receipt promotion metadata implementation
```

It does not permit capability authorization, effect execution, runtime authority, key handling, trust-store behavior, revocation lookup, network behavior, host effects, or kernel behavior.

After verified receipt promotion exists and is guarded, the next valid planning slice is capability gate evaluation of a verified but authority-neutral receipt.

Validation

This contract is validated by:

```
sh scripts/test-latticra-seal-verified-receipt-promotion-contract.sh
```

docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_IMPLEMENTATION.md

Source title: Latticra Seal Verified Receipt Promotion Implementation

Lines: 116 | Words: 336 | SHA-256: f6d8404679d5752aa359e0948ee504af70dd93cd3b6897a57723007acd6de117

Latticra Seal Verified Receipt Promotion Implementation

Status: initial verified receipt promotion metadata implementation Scope: bounded C metadata surface for promoting a successful Ed25519 verify-only result into verified receipt metadata.

Purpose

This document records the first Latticra Seal verified receipt promotion metadata implementation.

The implementation accepts an existing Ed25519 verify-only result and produces deterministic verified receipt metadata only when the verify-only result is already successful.

The promotion is evidence promotion, not permission promotion.

Added files

```
include/latticra/seal_verified_receipt_promotion.h
src/seal_verified_receipt_promotion.c
tests/seal_verified_receipt_promotion_invariants.c
scripts/test-latticra-seal-verified-receipt-promotion.sh
```

API summary

The verified receipt promotion metadata surface adds:

```
latticra_seal_verified_receipt_promotion_t
latticra_seal_verified_receipt_promotion_error_t
latticra_seal_verified_receipt_promotion_error_label
latticra_seal_verified_receipt_promotion_from_ed25519_result
latticra_seal_verified_receipt_promotion_is_authority_neutral
latticra_seal_verified_receipt_promotion_report
```

Promotion behavior

The implementation:

```
accepts a valid Ed25519 verify-only result
requires cryptographic_verification_supported=1
requires cryptographic_verification_performed=1
requires verified=1
requires invalid=0
requires crypto_verify_state=verified
copies verify profile metadata
copies backend profile metadata
copies verification policy profile metadata
copies message label metadata
copies message size metadata
copies message digest metadata
copies public-key identity metadata
copies signature algorithm metadata
copies trust-source metadata
sets verification_state=verified
sets receipt_state=verified
sets cryptographic_verification_supported=1
sets cryptographic_verification_performed=1
sets verified=1
```

```
sets invalid=0
sets authority_usable=0
sets capability_gate_allowed=0
sets runtime_authority_granted=0
renders deterministic verified receipt metadata
```

Authority boundary

The promoted receipt remains authority-neutral.

These fields remain zero:

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
```

Failure behavior

The implementation fails closed:

```
null verified receipt output -&gt; LATTICRA_STATUS_NULL_ARGUMENT
null Ed25519 verify result input -&gt; invalid-input
invalid Ed25519 verify result -&gt; invalid-verify-result
unsupported verification -&gt; unsupported-verification
verification not performed -&gt; verification-not-performed
verified flag not set -&gt; not-verified
invalid flag set -&gt; not-verified
crypto verify state not verified -&gt; invalid-verify-state
missing message digest -&gt; missing-message-digest
missing public-key identity -&gt; missing-public-key-identity
small report buffer -&gt; LATTICRA_STATUS_BUFFER_TOO_SMALL
```

Failures do not create keys, store secrets, contact networks, query revocation status, sign records, read host files, write host files, enforce capabilities, perform effects, or grant runtime authority.

Validation

Run locally:

```
sh scripts/test-latticra-seal-verified-receipt-promotion.sh
```

Expected output:

```
seal verified receipt promotion invariants: ok
```

Next valid slice

The next valid Latticra Seal slice is capability gate evaluation of a verified but authority-neutral receipt.

docs/specs/LATTICRA_SEAL_MANIFEST_v0_1.md

Source title: Latticra Seal Manifest v0.1

Lines: 109 | Words: 386 | SHA-256: b6a3e31f61ed3134fb84b83b146a0f07cd94b767508e8e2e05315c57ef226bce

Latticra Seal Manifest v0.1

latticra.seal is the root trust manifest for a Latticra project.

It is a human-readable, TOML-compatible manifest used to describe local project integrity rules, protected paths, policy checks, and proof-report metadata.

The .seal extension marks the file as a Latticra trust artifact while keeping it simple enough to parse from C, C++, Rust, Python, or shell tooling.

Purpose

The v0.1 manifest defines:

- project identity

- local trust boundary
- hash algorithm
- protected paths
- excluded paths
- required files
- basic policy checks
- report output locations
- reserved proof fields for future signing

Current Security Status

v0.1 is an unsigned local-integrity manifest.

It can support local scanning, policy checking, and digest reporting, but it should not yet be described as a complete cryptographic attestation system.

Required Root Fields

Required root fields:

schema = "latticra.seal/v0.1" format = "toml" kind = "local-integrity-manifest"

These fields identify the manifest version, parser expectation, and trust-artifact type.

Project Section

The project section describes the project being sealed.

Expected fields:

name = "Latticra" id = "latticra" version = "0.1.0" repository = "https://github.com/Bryforge/Latticra" license = "SEE LICENSE"

Seal Section

The seal section describes the active trust mode.

For v0.1:

mode = "local-integrity" status = "unsigned" algorithm = "sha256" digest_encoding = "hex" trust_boundary = "project-root"

Paths Section

The paths section defines what the scanner should include and exclude.

The v0.1 default includes the project root and excludes generated directories, dependency directories, temporary files, and generated reports.

Policy Section

The policy section defines local project checks.

Initial policy checks include:

- requiring a README
- requiring a LICENSE
- denying obvious private-key filenames
- denying .env files

- denying obvious committed token markers
- optionally warning about oversized files

Report Section

The report section defines where generated Seal reports should be written.

Default outputs:

reports/latticra-seal-report.txt reports/latticra-seal-file-hashes.txt

Proof Section

The proof section is reserved for future cryptographic fields.

In v0.1, these fields are intentionally empty:

manifest_hash = "" root_hash = "" signature_algorithm = "" signature = "" public_key = ""

Roadmap

- v0.1: local manifest, smoke verification, digest report
- v0.2: signed manifests and public-key verification
- v0.3: Merkle-style root hash and reproducible proof bundles
- v0.4: Latticra Panel integration
- v1.0: stable Seal CLI with documented guarantees and limits

Part VII - Console, Panel, Installer, and Update Paths

- docs/LATTICRA_CONSOLE_FOUNDATION.md - Latticra Console Foundation
- docs/LATTICRA_PANEL_SIGNED_UPDATER_DELIVERY_GATE.md - Latticra Panel Signed Updater Delivery Gate
- docs/LATTICRA_PANEL_SIGNED_UPDATER_DENIAL_TRANSCRIPT.md - Latticra Panel Signed Updater Denial Transcript
- docs/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_CONTRACT.md - Latticra Panel Signed Updater Manifest Fixture Contract
- docs/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_VALIDATION.md - Latticra Panel Signed Updater Manifest Fixture Validation
- docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_CONTRACT.md - Latticra Panel Signed Updater State Fixture Contract
- docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_VALIDATION.md - Latticra Panel Signed Updater State Fixture Validation
- docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_REVIEW.md - Latticra Panel Signed Updater State Transition Denial Review
- docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_TRANSCRIPT.md - Latticra Panel Signed Updater State Transition Denial Transcript
- docs/LOCAL_INSTALLER_ARTIFACT_MANIFEST_CONTRACT.md - Local Installer Artifact Manifest Contract

- docs/PRODUCTION_INSTALLER_READINESS_CONTRACT.md - Production Installer Readiness Contract
- docs/SELF_UPDATE_MODEL.md - Latticra Self-Update Model
- docs/strategy/2026-05-25-2137-cdt-panel-guided-local-evaluation-workflow-packet.md - Latticra Panel-Guided Local Evaluation Workflow Packet
- docs/strategy/2026-05-25-2157-cdt-panel-guided-local-evaluation-acceptance-checklist.md - Latticra Panel-Guided Local Evaluation Acceptance Checklist
- docs/strategy/2026-05-25-2226-cdt-panel-guided-local-evaluation-evidence-bundle-template.md - Latticra Panel-Guided Local Evaluation Evidence Bundle Template
- docs/strategy/2026-05-25-2246-cdt-panel-guided-local-evaluation-evidence-capture-plan.md - Latticra Panel-Guided Local Evaluation Evidence Capture Plan
- docs/strategy/2026-05-26-0019-cdt-panel-guided-local-evaluation-non-claim-review-template.md - Latticra Panel-Guided Local Evaluation Non-Claim Review Template
- docs/strategy/2026-05-26-0034-cdt-panel-guided-local-evaluation-public-entrypoint-review-template.md - Latticra Panel-Guided Local Evaluation Public-Entrypoint Review Template
- docs/strategy/2026-05-26-0119-cdt-panel-guided-local-evaluation-estimate-impact-review-template.md - Latticra Panel-Guided Local Evaluation Estimate-Impact Review Template
- docs/strategy/2026-05-26-0306-cdt-panel-guided-local-evaluation-guard-test-reference-template.md - Latticra Panel-Guided Local Evaluation Guard/Test Reference Template
- docs/strategy/2026-05-26-0410-cdt-panel-guided-local-evaluation-review-package-index.md - Latticra Panel-Guided Local Evaluation Review Package Index
- docs/strategy/2026-05-26-0949-cdt-panel-guided-local-evaluation-guard-test-reference-template.md - Latticra Panel-Guided Local Evaluation Guard/Test Reference Template
- docs/strategy/2026-05-26-1356-cdt-panel-guided-local-evaluation-planning-completion-checkpoint.md - Latticra Panel-Guided Local Evaluation Planning Completion Checkpoint
- installer/docs/INSTALL_BUTTON_EXECUTION_MODEL.md - Install Button Execution Model
- installer/docs/INSTALLER_READINESS_CONTRACT.md - Installer Readiness Contract
- installer/docs/README.md - Latticra Installer Documentation
- installer/docs/RECEIPTS_AND_EVIDENCE.md - Receipts and Evidence
- installer/docs/UI_CONFIGURATION_MODEL.md - UI Configuration Model
- installer/INSTALLER_ROADMAP.md - Latticra Installer Roadmap
- installer/latticra-installer-plan.txt - Latticra Installer Plan
- installer/README.md - Latticra Panel
- installer/TREE.txt - Tree

docs/LATTICRA_CONSOLE_FOUNDATION.md

Source title: Latticra Console Foundation

Lines: 1559 | Words: 3950 | SHA-256: 29d7b84d22333dfb3ad730b6e1c4aa31a1ff09e6209e5d95fec3d2a092f56811

Latticra Console Foundation

Status: Stage-0 foundation Scope: LC identity, standalone and Panel installability, configurable metadata, session contract, workspace contract, namespace contract, rootfs contract, packages contract, init contract, substrate bridge, host-embedding plan, host-adapter contract, Seal receipt-request contract, receipt payload schema, receipt payload artifact draft, receipt payload artifact review gate, receipt payload artifact review receipt contract, receipt payload artifact review receipt draft contract, receipt payload materialization plan, signature-request binding contract, OS-base planning contract, VM evidence contract, and future OS-base direction.

Purpose

Latticra Console, shortened to LC, is the planned main operator base for Latticra.

LC should become the primary interaction surface for the Latticra substrate and the surrounding system. It is designed to be configurable, installable through Latticra Panel, available as a standalone user-local console, embeddable inside a host environment, and eventually capable of growing into an OS-like user environment after explicit evidence gates.

This Stage-0 slice makes LC real without overclaiming authority.

Phase1 Lessons Retained

The Phase1 reference is useful as a learning source, not as code to copy. Its relevant lessons are:

- a command registry should be the source of truth for help, command discovery, and future capability metadata;
- operator consoles need a guarded host boundary from the first implementation slice;
- OS-track language should stay staged and evidence-bound;
- help, status, and inspection commands should exist before effectful commands.

LC applies those lessons to Latticra's own architecture: Lat, LIR, Nucleus, Runtime Boundary, Latticra Seal, Panel, Nadia, and the System Substrate.

Installed Identity

```
console_name=Latticra Console
short_name=LC
component_key=latticra_console
panel_installable=1
standalone_installable=1
standalone_requires_panel=0
standalone_command_wrapper=latticra-lc
configurable=1
profile=panel_embedded
panel_console_bridge=panel-aware
command_registry_profile=c-static-table
substrate_bridge_profile=metadata-bound
host_embedding_profile=panel-contained
host_embedding_contract_profile=lc-host-embedding-v0
host_inventory_contract_profile=lc-host-inventory-v0
host_adapter_contract_profile=lc-host-adapter-v0
session_contract_profile=lc-session-v0
workspace_contract_profile=lc-workspace-v0
namespace_contract_profile=lc-namespace-v0
rootfs_contract_profile=lc-rootfs-v0
packages_contract_profile=lc-packages-v0
init_contract_profile=lc-init-v0
receipt_request_contract_profile=lc-receipt-request-v0
receipt_payload_schema_profile=lc-receipt-payload-schema-v0
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_draft_profile=lc-receipt-payload-artifact-review-receipt
-draft-v0
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
signature_request_binding_profile=lc-signature-request-binding-v0
receipt_contract_profile=lc-receipts-v0
os_base_contract_profile=lc-os-base-v0
vm_evidence_contract_profile=lc-vm-evidence-v0
os_base_profile=planned-no-boot-authority
report_only=true
host_embedding_contract_required=true
read_only_host_inventory_contract_required=true
profile_receipt_required=true
host_contract_receipt_required=true
host_inventory_receipt_required=true
host_adapter_contract_required=true
session_contract_required=true
workspace_contract_required=true
namespace_contract_required=true
rootfs_contract_required=true
packages_contract_required=true
init_contract_required=true
receipt_request_contract_required=true
receipt_payload_schema_required=true
receipt_payload_artifact_draft_required=true
receipt_payload_artifact_review_required=true
receipt_payload_artifact_review_receipt_required=true
receipt_payload_artifact_review_receipt_draft_required=true
receipt_payload_materialization_plan_required=true
signature_request_binding_required=true
os_base_contract_required=true
vm_evidence_contract_required=true
runtime_boundary_binding_required=true
seal_capability_labels_required=true
```

The Panel component creates:

```
etc/latticra/lc.toml
share/latticra/lc/install/config.toml
share/latticra/lc/README.md
share/latticra/lc/commands/seed-registry.txt
share/latticra/lc/profiles/hosted-reference.toml
share/latticra/lc/profiles/standalone-console.toml
share/latticra/lc/profiles/panel-embedded.toml
share/latticra/lc/profiles/host-embedded-planning.toml
share/latticra/lc/profiles/os-base-planning.toml
share/latticra/lc/standalone/contract.toml
share/latticra/lc/session/contract.toml
share/latticra/lc/workspace/contract.toml
share/latticra/lc/namespace/contract.toml
share/latticra/lc/rootfs/contract.toml
share/latticra/lc/packages/contract.toml
```

```
share/latticra/lc/init/contract.toml
share/latticra/lc/substrate
share/latticra/lc/host-embedding/contract.toml
share/latticra/lc/host-inventory/contract.toml
share/latticra/lc/host-adapter/contract.toml
share/latticra/lc/receipt-request/contract.toml
share/latticra/lc/receipt-request/payload-schema.toml
share/latticra/lc/receipt-request/payload-artifact-draft.toml
share/latticra/lc/receipt-request/payload-artifact-review.toml
share/latticra/lc/receipt-request/payload-artifact-review-receipt.toml
share/latticra/lc/receipt-request/payload-artifact-review-receipt-draft.toml
share/latticra/lc/receipt-request/payload-materialization-plan.toml
share/latticra/lc/receipt-request/signature-request-binding.toml
share/latticra/lc/receipts/contract.toml
share/latticra/lc/os-base/contract.toml
share/latticra/lc/vm-evidence/contract.toml
share/latticra/components/latticra-console.installed
```

The default user-local wrapper is:

```
latticra-lc
```

Panel installs can rename that direct wrapper with lc.install.command_wrapper; the wrapper also serves as the standalone LC entrypoint and does not require Panel at runtime. The umbrella route remains latticra lc ....

The standalone installer presets are:

```
installer/configs/lc-standalone.installer.toml
installer/configs/lc-standalone-local.installer.toml
```

They set profile = "lc_standalone", lc.profile = "standalone", lc.install.install_profile = "lc-standalone-install-v0", lc.install.install_mode = "metadata-only-standalone-console", lc.install.panel_embedded_console = false, and leave external host commands disabled. The lc-standalone-local preset is the guarded local-write companion for the dry-run standalone preset.

LC can also be installed as a standalone local console:

```
installer/configs/lc-standalone.installer.toml
installer/configs/lc-standalone-local.installer.toml
installer/scripts/latticra-installer-verify-lc-standalone.sh
```

The standalone lane uses lc-standalone-install-v0, writes share/latticra/lc/standalone/contract.toml, keeps standalone_requires_panel = false, disables Panel embedding, and preserves the same no-effect host/network/runtime/boot authority floor.

The LC session lane writes share/latticra/lc/session/contract.toml and exposes lc session. It defines the future operator-base session envelope for standalone, Panel-embedded, and Host-embedded LC while keeping runtime session creation, shell launch, host process launch, host mutation, network, runtime enforcement, boot, and production OS authority denied.

The LC workspace lane writes share/latticra/lc/workspace/contract.toml and exposes lc workspace. It names the future operator-root workspace envelope while keeping workspace mounts, manifest writes, file reads, file writes, host workspace binding, host mutation, network, runtime enforcement, boot, and production OS authority denied.

The LC namespace lane writes share/latticra/lc/namespace/contract.toml and exposes lc namespace. It names the future internal OS-style namespace envelope while keeping namespace mounts, rootfs projection, path resolution, file reads, file writes, host path projection, host mutation, network, runtime enforcement, boot, and production OS authority denied.

The LC rootfs lane writes share/latticra/lc/rootfs/contract.toml and exposes lc rootfs. It names the future internal root filesystem envelope while keeping rootfs image creation, image opening, mounts, package installs, file reads, file writes, host mutation, network, runtime enforcement, boot, and production OS authority denied.

The LC packages lane writes share/latticra/lc/packages/contract.toml and exposes lc packages. It names the future rootfs package-manifest envelope while keeping package catalog reads, downloads, install-plan writes, package-manager execution, package-script execution, rootfs package installs, host mutation, network, runtime enforcement, boot, and production OS authority denied.

The LC init lane writes share/latticra/lc/init/contract.toml and exposes lc init. It names the future internal init and service envelope while keeping PID 1 claims, init process launch, service starts, service stops, process supervision, startup-order writes, host mutation, network, runtime enforcement, boot, and production OS authority denied.

The umbrella wrapper routes:

```
latticra lc status
```

Stage-0 Commands

```
help
status
plan
save
dry-run
reset
uninstall
clear
lc status
lc commands
lc standalone
lc session
lc workspace
lc namespace
lc rootfs
lc packages
lc init
lc profiles
lc receipts
lc receipt-request
lc receipt-payload
lc receipt-artifact
lc receipt-artifact-review
lc receipt-review-receipt
lc receipt-review-draft
lc receipt-materialization-plan
lc signature-request
lc substrate
lc host
lc host-contract
lc host-inventory
lc host-adapter
lc os-contract
lc vm-evidence
lc os
pwd
cd
```

These commands are registry-backed in the C foundation. Each command carries:

```
name
usage
category
effect
capability_label
no_effect
panel_visible
launches_host_process
requires_future_gate
```

The Stage-0 registry is still metadata-only. It reports identity, seed command metadata, substrate bridge, host-embedding plan, and future OS-base posture. It does not launch host commands.

Panel Profile Presets

Latticra Panel now carries LC profile presets as a first-class installer configuration block:

```
[lc]
profile = &quot;panel_embedded&quot;
command_registry_profile = &quot;c-static-table&quot;
substrate_bridge_profile = &quot;metadata-bound&quot;
host_embedding_profile = &quot;panel-contained&quot;
host_embedding_contract_profile = &quot;lc-host-embedding-v0&quot;
host_inventory_contract_profile = &quot;lc-host-inventory-v0&quot;
host_adapter_contract_profile = &quot;lc-host-adapter-v0&quot;
session_contract_profile = &quot;lc-session-v0&quot;
workspace_contract_profile = &quot;lc-workspace-v0&quot;
namespace_contract_profile = &quot;lc-namespace-v0&quot;
rootfs_contract_profile = &quot;lc-rootfs-v0&quot;
packages_contract_profile = &quot;lc-packages-v0&quot;
init_contract_profile = &quot;lc-init-v0&quot;
receipt_request_contract_profile = &quot;lc-receipt-request-v0&quot;
receipt_payload_schema_profile = &quot;lc-receipt-payload-schema-v0&quot;
receipt_payload_artifact_draft_profile = &quot;lc-receipt-payload-artifact-draft-v0&quot;
receipt_payload_artifact_review_profile = &quot;lc-receipt-payload-artifact-review-v0&quot;
receipt_payload_artifact_review_receipt_profile =
&quot;lc-receipt-payload-artifact-review-receipt-v0&quot;
receipt_payload_artifact_review_receipt_draft_profile =
&quot;lc-receipt-payload-artifact-review-receipt-draft-v0&quot;
receipt_payload_materialization_plan_profile = &quot;lc-receipt-payload-materialization-plan-
v0&quot;
signature_request_binding_profile = &quot;lc-signature-request-binding-v0&quot;
receipt_contract_profile = &quot;lc-receipts-v0&quot;
os_base_contract_profile = &quot;lc-os-base-v0&quot;
vm_evidence_contract_profile = &quot;lc-vm-evidence-v0&quot;
os_base_profile = &quot;planned-no-boot-authority&quot;
panel_bridge = &quot;panel-aware&quot;
report_only = true
require_host_embedding_contract = true
require_read_only_host_inventory_contract = true
require_profile_receipt = true
require_host_contract_receipt = true
require_host_inventory_receipt = true
require_host_adapter_contract = true
require_session_contract = true
require_workspace_contract = true
require_namespace_contract = true
require_rootfs_contract = true
require_packages_contract = true
require_init_contract = true
require_receipt_request_contract = true
require_receipt_payload_schema = true
require_receipt_payload_artifact_draft = true
require_receipt_payload_artifact_review = true
require_receipt_payload_artifact_review_receipt = true
require_receipt_payload_artifact_review_receipt_draft = true
require_receipt_payload_materialization_plan = true
require_signature_request_binding = true
require_os_base_contract = true
require_vm_evidence_contract = true
require_runtime_boundary_binding = true
require_seal_capability_labels = true

[lc.install]
install_profile = &quot;lc-panel-install-v0&quot;
install_mode = &quot;metadata-only-console-foundation&quot;
command_wrapper = &quot;latticra-lc&quot;
standalone_console = true
standalone_installable = true
standalone_requires_panel = false
panel_embedded_console = true
install_user_wrapper = true
allow_external_host_commands = false
```

The current presets are:

```
hosted_reference -&gt; hosted reference metadata without embedded-host claims
standalone -&gt; standalone LC wrapper profile without requiring Panel at runtime
panel_embedded -&gt; default Panel-installed LC operator surface
host_embedded_planning -&gt; future host-embedding plan with zero host mutation authority
os_base_planning -&gt; future OS-base plan with zero boot, kernel, or runtime enforcement
```

```
authority
custom -&gt; manual metadata fields under the same no-effect authority floor
```

Panel exposes the presets in a Latticra Console workspace tab and in the embedded console:

```
lc profiles
lc profile hosted
lc profile standalone
lc profile panel
lc profile host
lc profile os
lc profile custom
```

The install engine writes the selected profile into etc/latticra/lc.toml, writes install metadata into share/latticra/lc/install/config.toml, and installs the preset files under share/latticra/lc/profiles/. The command wrapper comes from lc.install.command_wrapper and defaults to latticra-lc. That wrapper is the standalone LC entrypoint when lc.install.standalone_console = true; it remains metadata-only and does not require Panel at runtime. These profiles and install records remain configuration metadata only, and external host commands stay disabled from both Panel and standalone LC.

Standalone Contract

LC now installs and reports a standalone console contract before standalone use can gain any effectful host or OS authority:

```
standalone_console_profile=lc-standalone-console-v0
standalone_console_status=metadata-only-contract
standalone_contract_present=1
standalone_console=1
standalone_installable=1
standalone_requires_panel=0
standalone_command_wrapper=latticra-lc
standalone_profile_file=profiles/standalone-console.toml
panel_embedded_console=1
panel_required_for_runtime=0
config_path=etc/latticra/lc.toml
share_path=share/latticra/lc
command_registry_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
profile_receipt_required=1
promotion_gate=lc_standalone_console_before_effectful_host_or_os_authority
command_surface=lc standalone
related_install_config_command=lc install-config
related_profile_command=lc profiles
no_effect=1
shell_execution_allowed=0
host_process_launch_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
```

The source and installed command surfaces are:

```
latticra_console_report standalone
latticra-lc standalone
```

The standalone contract does not launch a shell, launch host processes, read or write host files, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Session Contract

LC now installs and reports a session contract before the Console can claim any runtime-session or host-adapter behavior:

```
session_profile=lc-session-v0
session_status=metadata-only-contract
session_contract_present=1
session_kind=operator-base
```

```
standalone_compatible=1
panel_embedded_compatible=1
host_embedded_planned=1
session_manifest_present=0
session_manifest_write_allowed=0
runtime_session_created=0
runtime_process_spawn_allowed=0
runtime_invocation_allowed=0
interactive_shell_allowed=0
promotion_gate=lc_session_contract_before_runtime_or_host_embedding
command_surface=lc session
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
```

The source and installed command surfaces are:

```
latticra_console_report session
latticra-lc session
```

The session contract is an operator-base envelope only. It does not create a runtime session, launch an interactive shell, spawn host processes, invoke runtimes, write session manifests, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Workspace Contract

LC now installs and reports a workspace contract before the Console can claim any mounted workspace, manifest, or OS-base workspace behavior:

```
workspace_profile=lc-workspace-v0
workspace_status=metadata-only-contract
workspace_contract_present=1
workspace_kind=operator-root
workspace_root=share/latticra/lc/workspace
workspace_mount_present=0
workspace_mount_allowed=0
host_workspace_bind_allowed=0
workspace_manifest_present=0
workspace_manifest_write_allowed=0
workspace_file_read_allowed=0
workspace_file_write_allowed=0
workspace_mutation_allowed=0
runtime_session_required_before_workspace_runtime=1
session_contract_required=1
host_adapter_contract_required=1
os_base_contract_required=1
promotion_gate=lc_workspace_contract_before_host_mount_or_os_workspace
command_surface=lc workspace
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
```

The source and installed command surfaces are:

```
latticra_console_report workspace
latticra-lc workspace
```

The workspace contract is an operator-root envelope only. It does not mount a workspace, bind Host workspace paths, read or write files, write manifests, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Namespace Contract

LC now installs and reports a namespace contract before the Console can claim any internal OS-style namespace, rootfs, mount, or host path projection behavior:

```
namespace_profile=lc-namespace-v0
namespace_status=metadata-only-contract
namespace_contract_present=1
```

```
namespace_kind=lc-internal-os-namespace
namespace_root=share/latticra/lc/namespace
namespace_manifest_present=0
namespace_manifest_write_allowed=0
namespace_mount_present=0
namespace_mount_allowed=0
rootfs_present=0
rootfs_mount_allowed=0
path_resolver_present=0
path_resolution_allowed=0
host_path_projection_allowed=0
workspace_namespace_bind_allowed=0
namespace_file_read_allowed=0
namespace_file_write_allowed=0
namespace_mutation_allowed=0
workspace_contract_required=1
session_contract_required=1
runtime_boundary_required=1
promotion_gate=lc_namespace_contract_before_rootfs_or_path_projection
command_surface=lc namespace
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
```

The source and installed command surfaces are:

```
latticra_console_report namespace
latticra-lc namespace
```

The namespace contract is an internal OS-layout envelope only. It does not mount namespaces, create a rootfs, resolve host paths, project host paths, bind workspaces into a namespace, read or write files, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Rootfs Contract

LC now installs and reports a rootfs contract before the Console can claim image creation, mounts, package writes, boot adjacency, or OS-base filesystem behavior:

```
rootfs_profile=lc-rootfs-v0
rootfs_status=metadata-only-contract
rootfs_contract_present=1
rootfs_kind=lc-internal-root-filesystem
rootfs_root=share/latticra/lc/rootfs
rootfs_state_source=metadata-only
rootfs_manifest_present=0
rootfs_manifest_write_allowed=0
rootfs_image_present=0
rootfs_image_create_allowed=0
rootfs_image_open_allowed=0
rootfs_mount_present=0
rootfs_mount_allowed=0
rootfs_package_manifest_present=0
rootfs_package_install_allowed=0
rootfs_file_read_allowed=0
rootfs_file_write_allowed=0
rootfs_mutation_allowed=0
namespace_contract_required=1
workspace_contract_required=1
session_contract_required=1
runtime_boundary_required=1
receipt_required_before_rootfs_materialization=1
command_surface=lc rootfs
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
promotion_gate=lc_rootfs_contract_before_image_mount_or_package_write
```

The source and installed command surfaces are:

```
latticra_console_report rootfs
latticra-lc rootfs
```

The rootfs contract is metadata only. It does not create or open a rootfs image, mount a filesystem, install packages, read or write rootfs files, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Packages Contract

LC now installs and reports a packages contract before the Console can claim package catalogs, package downloads, package-manager execution, package-script execution, install plans, package manifests, or rootfs package writes:

```
packages_profile=lc-packages-v0
packages_status=metadata-only-contract
packages_contract_present=1
packages_kind=lc-rootfs-package-manifest-envelope
packages_root=share/latticra/lc/packages
packages_state_source=metadata-only
package_manifest_present=0
package_manifest_write_allowed=0
package_catalog_present=0
package_catalog_read_allowed=0
package_catalog_write_allowed=0
package_download_allowed=0
package_install_plan_present=0
package_install_plan_write_allowed=0
package_manager_present=0
package_manager_execution_allowed=0
package_script_execution_allowed=0
rootfs_package_manifest_present=0
rootfs_package_install_allowed=0
rootfs_file_write_allowed=0
rootfs_contract_required=1
namespace_contract_required=1
workspace_contract_required=1
session_contract_required=1
os_base_contract_required=1
runtime_boundary_required=1
receipt_required_before_package_materialization=1
command_surface=lc packages
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
promotion_gate=lc_packages_contract_before_catalog_install_or_rootfs_write
```

The source and installed command surfaces are:

```
latticra_console_report packages
latticra-lc packages
```

The packages contract is metadata only. It does not read package catalogs, download packages, write manifests or install plans, run package managers or package scripts, install packages into rootfs, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Init Contract

LC now installs and reports an init contract before the Console can claim PID 1, an init process, service supervision, startup ordering, boot adjacency, or OS-base init behavior:

```
init_profile=lc-init-v0
init_status=metadata-only-contract
init_contract_present=1
init_kind=lc-internal-init-and-service-envelope
init_root=share/latticra/lc/init
init_state_source=metadata-only
init_manifest_present=0
init_manifest_write_allowed=0
pid1_claim_allowed=0
init_process_present=0
init_process_launch_allowed=0
service_registry_present=0
service_registry_write_allowed=0
service_start_allowed=0
```

```
service_stop_allowed=0
service_restart_allowed=0
process_supervision_allowed=0
startup_order_present=0
startup_order_write_allowed=0
rootfs_contract_required=1
packages_contract_required=1
namespace_contract_required=1
workspace_contract_required=1
session_contract_required=1
os_base_contract_required=1
runtime_boundary_required=1
receipt_required_before_init_runtime=1
command_surface=lc init
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
promotion_gate=lc_init_contract_before_pid1_service_supervision_or_boot
```

The source and installed command surfaces are:

```
latticra_console_report init
latticra-lc init
```

The init contract is metadata only. It does not claim PID 1, launch an init process, start or stop services, supervise processes, write startup order, mutate the host, use the network, enforce runtime policy, boot hardware, or claim production OS status.

Host Embedding Contract

LC now installs and reports a host-embedding contract before any host integration behavior exists:

```
contract_profile=lc-host-embedding-v0
contract_status=metadata-only
host_adapter_required=1
panel_install_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
operator_consent_required=1
read_only_host_inventory_required_before_embedding=1
receipt_required_before_embedding=1
promotion_gate=contract_receipt_and_read_only_host_inventory
```

The source and installed command surfaces are:

```
latticra_console_report host-contract
latticra-lc host-contract
```

The contract explicitly denies host authority:

```
host_embedded_now=0
host_process_launch_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

lc host-contract is an inspectable contract command. lc host remains the future-gated embedding lane.

Read-Only Host Inventory Contract

LC also installs and reports a read-only host inventory contract before any host adapter exists:

```
contract_profile=lc-host-inventory-v0
contract_status=metadata-only
required_before_host_embedding=1
host_adapter_present=0
inventory_schema_status=planned
inventory_performed=0
```

```
inventory_artifact_present=0
inventory_receipt_required=1
operator_consent_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
promotion_gate=host_inventory_contract_receipt_before_host_adapter
```

The intended future inventory scope is narrow and reviewable:

```
allowed_future_scope=os_family,kernel_version,cpu_arch,memory_class,filesystem_roots,user_scope,
prefix_scope
excluded_future_scope=secrets,private_files,network_scan,process_launch,kernel_change,system_mut
ation
```

The current command surfaces are:

```
latticra_console_report host-inventory
latticra-lc host-inventory
```

This command does not inventory the host. It reports the contract and denial posture:

```
inventory_performed=0
host_probe_allowed=0
host_process_launch_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

Host Adapter Contract

LC now installs and reports a host-adapter contract before any Host embedding adapter exists:

```
contract_profile=lc-host-adapter-v0
contract_status=metadata-only
contract_present=1
host_adapter_enabled=0
host_adapter_present=0
host_adapter_loaded=0
adapter_api_status=planned
adapter_abi_status=planned
host_embedding_contract_required=1
read_only_host_inventory_contract_required=1
host_embedding_contract_receipt_required=1
host_inventory_contract_receipt_required=1
operator_consent_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
receipt_required_before_host_adapter=1
promotion_gate=host_adapter_contract_receipts_and_inventory
```

The source and installed command surfaces are:

```
latticra_console_report host-adapter
latticra-lc host-adapter
```

The contract explicitly denies adapter and host authority:

```
host_embedded_now=0
host_process_launch_allowed=0
host_probe_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

lc host-adapter is an inspectable contract command. lc host remains the future-gated embedding lane.

Receipt Contract

LC now installs and reports a receipt contract for its profile and host-side planning metadata:

```
receipt_profile=lc-receipts-v0
receipt_contract_status=metadata-only
profile_receipt_required=1
host_embedding_contract_receipt_required=1
host_inventory_contract_receipt_required=1
host_adapter_contract_receipt_required=1
session_contract_receipt_required=1
session_contract_present=1
workspace_contract_receipt_required=1
workspace_contract_present=1
namespace_contract_receipt_required=1
namespace_contract_present=1
rootfs_contract_receipt_required=1
rootfs_contract_present=1
packages_contract_receipt_required=1
packages_contract_present=1
init_contract_receipt_required=1
init_contract_present=1
receipt_request_contract_required=1
receipt_request_contract_present=1
receipt_payload_schema_required=1
receipt_payload_schema_present=1
receipt_payload_artifact_draft_required=1
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_review_required=1
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_materialization_plan_required=1
receipt_payload_materialization_plan_present=1
signature_request_binding_required=1
signature_request_binding_contract_present=1
runtime_boundary_receipt_required=1
seal_capability_labels_required=1
signature_request_profile=latticra-seal-signature-request/0.1
receipt_request_command=lc receipt-request
receipt_payload_schema_command=lc receipt-payload
receipt_payload_artifact_draft_command=lc receipt-artifact
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
signature_request_binding_command=lc signature-request
session_contract_command=lc session
workspace_contract_command=lc workspace
namespace_contract_command=lc namespace
rootfs_contract_command=lc rootfs
packages_contract_command=lc packages
init_contract_command=lc init
receipt_surfaces=profile,session,workspace,namespace,rootfs,packages,init,host-contract,host-inv
entory,host-adapter,runtime-boundary
promotion_gate=lc_receipts_before_host_adapter_or_os_base
```

The source and installed command surfaces are:

```
latticra_console_report receipts
latticra-lc receipts
```

The contract reserves a future Seal-signed receipt path without claiming signing authority now:

```
seal_signature_planned=1
seal_signature_present=0
seal_signing_authority_present=0
receipt_written=0
receipt_signed=0
receipt_hash_recorded=0
receipt_path_recorded=0
file_write_allowed=0
```

Seal Receipt Request Contract

LC now installs and reports a metadata-only request contract for the future Seal-signed LC receipt path:

```
request_profile=lc-receipt-request-v0
request_contract_status=metadata-only
request_contract_present=1
receipt_contract_profile=lc-receipts-v0
signature_request_profile=latticra-seal-signature-request/0.1
```

```
requested_receipt_profile=latticra-seal-verified-receipt/0.1
requested_capability=verified-receipt-report
requested_surfaces=profile,session,workspace,namespace,rootfs,packages,init,host-contract,host-i
nventory,host-adapter,runtime-boundary
receipt_payload_schema_profile=lc-receipt-payload-schema-v0
receipt_payload_schema_required=1
receipt_payload_schema_present=1
receipt_payload_schema_command=lc receipt-payload
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_draft_required=1
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_draft_command=lc receipt-artifact
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_required=1
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
receipt_payload_materialization_plan_required=1
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
draft_review_receipt_present=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
signature_request_binding_profile=lc-signature-request-binding-v0
signature_request_binding_required=1
signature_request_binding_contract_present=1
signature_request_binding_command=lc signature-request
receipt_payload_profile=lc-receipts-v0
receipt_payload_hash_recorded=0
receipt_payload_path_recorded=0
promotion_gate=lc_receipt_request_review_before_signing
```

The source and installed command surfaces are:

```
latticra_console_report receipt-request
latticra-lc receipt-request
```

The request contract is a bridge to Seal metadata only. It explicitly denies signing and file authority:

```
seal_signature_planned=1
seal_signature_request_ready=0
seal_signature_request_present=0
seal_signing_authority_present=0
seal_signer_handoff_allowed=0
seal_signing_operation_allowed=0
receipt_write_allowed=0
receipt_signed=0
receipt_verification_allowed=0
file_write_allowed=0
host_process_launch_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

Receipt Payload Schema

LC now installs and reports the metadata-only payload schema that a future Seal signature-request artifact would consume:

```
schema_profile=lc-receipt-payload-schema-v0
schema_status=metadata-only
schema_present=1
receipt_request_profile=lc-receipt-request-v0
receipt_contract_profile=lc-receipts-v0
signature_request_profile=latticra-seal-signature-request/0.1
requested_receipt_profile=latticra-seal-verified-receipt/0.1
requested_capability=verified-receipt-report
payload_fields=console_id,profile,command_registry,host_contract,host_inventory,host_adapter,run
time_boundary,seal_capability_labels,authority_denials
required_authority_fields=no_effect,execution_allowed,host_mutation_allowed,network_allowed,runt
ime_enforcement_allowed,boot_allowed
```

```
payload_artifact_present=0
payload_hash_computed=0
payload_path_recorded=0
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_draft_command=lc receipt-artifact
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
materialization_preconditions_met=0
materialization_allowed=0
signature_request_binding_present=0
signature_request_binding_allowed=0
signature_request_binding_contract_present=1
signature_request_binding_command=lc signature-request
promotion_gate=lc_receipt_payload_schema_before_signature_request_binding
```

The source and installed command surfaces are:

```
latticra_console_report receipt-payload
latticra-lc receipt-payload
```

This schema does not create a payload, hash a payload, write a payload path, bind a signature request, or grant signing authority.

Receipt Payload Artifact Draft

LC now installs and reports a no-write draft contract for the future receipt payload artifact:

```
draft_profile=lc-receipt-payload-artifact-draft-v0
draft_status=metadata-only
draft_contract_present=1
receipt_request_profile=lc-receipt-request-v0
receipt_payload_schema_profile=lc-receipt-payload-schema-v0
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
signature_request_binding_profile=lc-signature-request-binding-v0
receipt_contract_profile=lc-receipts-v0
signature_request_profile=latticra-seal-signature-request/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
canonicalization_profile=lc-receipt-payload-canonical-text-v0
artifact_fields=console_id,profile,command_registry,host_contract,host_inventory,host_adapter,ru
ntime_boundary,seal_capability_labels,authority_denials
required_prior_contracts=receipt-request,receipt-payload-schema,signature-request-binding,receip
t-contract
receipt_payload_artifact_draft_present=1
draft_review_required=1
draft_review_present=0
draft_review_receipt_required=1
draft_review_receipt_present=0
draft_review_approval_recorded=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
payload_write_allowed=0
payload_hash_computed=0
payload_hash_recorded=0
payload_path_recorded=0
signature_request_binding_artifact_present=0
signature_request_binding_allowed=0
seal_signature_request_ready=0
receipt_write_allowed=0
receipt_signed=0
promotion_gate=lc_receipt_payload_artifact_draft_before_materialization_and_signature_request
```

```
related_materialization_plan_command=lc receipt-materialization-plan
```

The source and installed command surfaces are:

```
latticra_console_report receipt-artifact
latticra-lc receipt-artifact
```

This draft does not create or write a payload artifact, compute or record a payload hash, record a payload path, bind a Seal signature request, invoke signing, or write a receipt.

Receipt Payload Artifact Review Gate

LC now installs and reports a metadata-only review gate between the payload artifact draft and any future payload materialization:

```
review_profile=lc-receipt-payload-artifact-review-v0
review_status=metadata-only
review_gate_present=1
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_draft_required=1
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_draft_command=lc receipt-artifact
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
draft_review_required=1
draft_review_present=0
draft_review_receipt_required=1
draft_review_receipt_present=0
draft_review_approval_recorded=0
draft_reviewer_identity_recorded=0
draft_review_timestamp_recorded=0
materialization_plan_required=1
materialization_plan_present=1
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
payload_write_allowed=0
payload_hash_computed=0
payload_hash_recorded=0
payload_path_recorded=0
signature_request_binding_allowed=0
signature_request_binding_artifact_present=0
seal_signature_request_ready=0
seal_signature_request_present=0
seal_signing_authority_present=0
receipt_write_allowed=0
receipt_signed=0
promotion_gate=lc_receipt_payload_artifact_review_before_materialization
command_surface=lc receipt-artifact-review
related_materialization_plan_command=lc receipt-materialization-plan
```

The source and installed command surfaces are:

```
latticra_console_report receipt-artifact-review
latticra-lc receipt-artifact-review
```

This gate records that the draft still needs review and a review receipt. It does not materialize the payload artifact, write payload bytes, compute or record a payload hash, record a payload path, bind a Seal signature request, invoke signing, or write a receipt.

Receipt Payload Artifact Review Receipt Contract

LC now installs and reports the metadata-only receipt contract for the payload artifact review gate before materialization can become possible:

```
review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
review_receipt_status=metadata-only-contract
review_receipt_contract_present=1
review_receipt_draft_profile=lc-receipt-payload-artifact-review-receipt-draft-v0
review_receipt_draft_required=1
review_receipt_draft_contract_present=1
```

```
review_receipt_draft_command=lc receipt-review-draft
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_required=1
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
receipt_payload_materialization_plan_required=1
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
draft_review_required=1
draft_review_present=0
draft_review_receipt_required=1
draft_review_receipt_present=0
draft_review_receipt_artifact_present=0
draft_review_receipt_write_allowed=0
draft_review_receipt_signed=0
draft_review_receipt_hash_recorded=0
draft_review_receipt_path_recorded=0
draft_review_approval_recorded=0
draft_reviewer_identity_recorded=0
draft_review_timestamp_recorded=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
payload_write_allowed=0
signature_request_binding_allowed=0
receipt_write_allowed=0
receipt_signed=0
promotion_gate=lc_receipt_payload_artifact_review_receipt_before_materialization_preconditions
command_surface=lc receipt-review-receipt
related_review_receipt_draft_command=lc receipt-review-draft
```

The source and installed command surfaces are:

```
latticra_console_report receipt-review-receipt
latticra-lc receipt-review-receipt
```

This contract does not create a review receipt, record review approval, record reviewer identity, hash or path a review receipt, materialize payload bytes, bind a Seal signature request, invoke signing, or write a receipt.

Receipt Payload Artifact Review Receipt Draft Contract

LC now installs and reports a metadata-only draft contract for the future review receipt artifact:

```
review_receipt_draft_profile=lc-receipt-payload-artifact-review-receipt-draft-v0
review_receipt_draft_status=metadata-only-draft-contract
review_receipt_draft_contract_present=1
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
draft_review_receipt_required=1
draft_review_receipt_present=0
draft_review_receipt_artifact_present=0
draft_review_receipt_write_allowed=0
draft_review_receipt_signed=0
draft_review_receipt_hash_recorded=0
draft_review_receipt_path_recorded=0
review_receipt_materialization_allowed=0
review_receipt_signing_allowed=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
payload_write_allowed=0
signature_request_binding_allowed=0
receipt_write_allowed=0
receipt_signed=0
promotion_gate=lc_receipt_payload_artifact_review_receipt_draft_before_review_receipt_creation
command_surface=lc receipt-review-draft
```

The source and installed command surfaces are:

```
latticra_console_report receipt-review-draft
latticra-lc receipt-review-draft
```

This draft contract does not create a review receipt draft, write receipt bytes, compute or record a hash, record a path, sign anything, materialize payload bytes, bind a Seal signature request, or write a receipt.

Receipt Payload Materialization Plan

LC now installs and reports a metadata-only materialization plan for the future payload artifact:

```
materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
materialization_plan_status=metadata-only
materialization_plan_present=1
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_required=1
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_artifact_review_receipt_draft_profile=lc-receipt-payload-artifact-review-receipt
-draft-v0
receipt_payload_artifact_review_receipt_draft_required=1
receipt_payload_artifact_review_receipt_draft_present=1
receipt_payload_artifact_review_receipt_draft_command=lc receipt-review-draft
draft_review_receipt_required=1
draft_review_receipt_present=0
draft_review_approval_recorded=0
materialization_preconditions_met=0
materialization_allowed=0
materialization_execution_planned=0
payload_artifact_present=0
payload_materialized=0
payload_write_allowed=0
payload_file_open_allowed=0
payload_hash_computed=0
payload_hash_recorded=0
payload_path_recorded=0
signature_request_binding_allowed=0
signature_request_binding_artifact_present=0
seal_signature_request_ready=0
receipt_write_allowed=0
receipt_signed=0
promotion_gate=lc_receipt_payload_materialization_plan_after_review_receipt
command_surface=lc receipt-materialization-plan
```

The source and installed command surfaces are:

```
latticra_console_report receipt-materialization-plan
latticra-lc receipt-materialization-plan
```

This plan names the preconditions for a future payload artifact. It does not open a payload file, write payload bytes, materialize the payload, compute or record a hash, record a path, bind a Seal signature request, invoke signing, or write a receipt.

Signature Request Binding Contract

LC now installs and reports a metadata-only binding contract for a future Seal signature-request artifact:

```
binding_profile=lc-signature-request-binding-v0
binding_status=metadata-only
binding_contract_present=1
receipt_request_profile=lc-receipt-request-v0
receipt_payload_schema_profile=lc-receipt-payload-schema-v0
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_draft_required=1
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_draft_command=lc receipt-artifact
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_required=1
```

```
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_required=1
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
receipt_payload_materialization_plan_required=1
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
draft_review_receipt_present=0
materialization_preconditions_met=0
materialization_allowed=0
receipt_contract_profile=lc-receipts-v0
signature_request_profile=latticra-seal-signature-request/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
required_surfaces=receipt-request,receipt-payload-schema,receipt-payload-artifact-draft,receipt-
payload-artifact-review,receipt-payload-artifact-review-receipt,receipt-payload-artifact-review-
receipt-draft,receipt-payload-materialization-plan,receipt-contract,runtime-boundary,seal-capabi
lity-labels
payload_artifact_present=0
signature_request_binding_allowed=0
signature_request_binding_artifact_present=0
seal_signature_request_ready=0
seal_signature_request_present=0
seal_signing_authority_present=0
receipt_write_allowed=0
receipt_signed=0
promotion_gate=lc_signature_request_binding_after_payload_artifact_and_signing_authority
```

The source and installed command surfaces are:

```
latticra_console_report signature-request
latticra-lc signature-request
```

This contract does not create a signature-request artifact, write a binding path, hash a binding, grant signing authority, invoke a signer, write a receipt, or sign anything.

OS-Base Planning Contract

LC now installs and reports an OS-base planning contract before any boot-adjacent behavior exists:

```
contract_profile=lc-os-base-v0
contract_status=metadata-only
contract_present=1
os_base_enabled=0
production_os_claim=0
read_only_host_inventory_receipt_required=1
vm_evidence_contract_required=1
vm_evidence_required=1
operator_consent_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
receipt_required_before_os_base=1
promotion_gate=os_base_contract_receipt_and_vm_evidence
```

The source and installed command surfaces are:

```
latticra_console_report os-contract
latticra-lc os-contract
```

The contract explicitly denies OS, kernel, hardware, and host authority:

```
boot_allowed=0
boot_authority_present=0
kernel_change_allowed=0
kernel_enforcement_allowed=0
hardware_access_allowed=0
bootloader_write_allowed=0
partition_mutation_allowed=0
driver_load_allowed=0
service_install_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
```

lc os-contract is an inspectable contract command. lc os remains the future-gated boot-action planning lane.

VM Evidence Contract

LC now installs and reports a VM evidence contract before any boot-adjacent evidence capture exists:

```
contract_profile=lc-vm-evidence-v0
contract_status=metadata-only
contract_present=1
vm_evidence_required=1
vm_evidence_capture_enabled=0
vm_evidence_artifact_present=0
os_base_contract_required=1
read_only_host_inventory_receipt_required=1
operator_consent_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
receipt_required_before_vm_evidence=1
promotion_gate=vm_evidence_contract_before_boot_adjacency
```

The source and installed command surfaces are:

```
latticra_console_report vm-evidence
latticra-lc vm-evidence
```

The contract explicitly denies VM, hypervisor, disk-image, guest, host, and boot authority:

```
vm_launcher_present=0
vm_launch_allowed=0
hypervisor_access_allowed=0
disk_image_open_allowed=0
disk_image_write_allowed=0
snapshot_capture_allowed=0
guest_agent_allowed=0
guest_network_allowed=0
host_probe_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
production_os_claim=0
```

lc vm-evidence is an inspectable evidence-contract command. It does not start a VM or inspect a host.

Help And Manpage Rendering

LC now has registry-backed renderers for:

```
LATTICRA CONSOLE HELP
LATTICRA-CONSOLE(1)
```

The source runner can emit them directly:

```
tools/latticra_console_report.c -&gt; help
tools/latticra_console_report.c -&gt; man
```

Installed local-prefix wrappers expose the same operator shapes:

```
latticra-lc help
latticra-lc man
```

The installed wrapper reads share/latticra/lc/commands/seed-registry.txt for help rows, keeping command names, categories, effects, and capability labels attached to the component metadata.

Runtime Boundary Binding

LC command metadata is now bound to Runtime Boundary classification and Seal capability labels:

```
runtime_boundary_bound=1
seal_capability_labels_bound=1
```

Stage-0 command bindings use these rules:

```
core, panel, and substrate inspection -&gt; authority-check / validation-only
lc standalone -&gt; authority-check / validation-only
lc session -&gt; authority-check / validation-only
lc workspace -&gt; authority-check / validation-only
lc namespace -&gt; authority-check / validation-only
lc rootfs -&gt; authority-check / validation-only
lc packages -&gt; authority-check / validation-only
lc receipts -&gt; authority-check / validation-only
lc receipt-request -&gt; authority-check / validation-only
lc receipt-payload -&gt; authority-check / validation-only
lc receipt-artifact -&gt; authority-check / validation-only
lc receipt-artifact-review -&gt; authority-check / validation-only
lc receipt-review-receipt -&gt; authority-check / validation-only
lc receipt-review-draft -&gt; authority-check / validation-only
lc receipt-materialization-plan -&gt; authority-check / validation-only
lc signature-request -&gt; authority-check / validation-only
lc host-contract -&gt; authority-check / validation-only
lc host-inventory -&gt; authority-check / validation-only
lc host-adapter -&gt; authority-check / validation-only
lc os-contract -&gt; authority-check / validation-only
lc vm-evidence -&gt; authority-check / validation-only
lc host -&gt; future-gated command-execute planning
lc os -&gt; future-gated boot-action planning
```

The boundary report keeps all authority outputs denied:

```
no_effect=1
execution_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
seal_capability_grants_authority=0
```

Source and installed surfaces expose:

```
latticra_console_report boundary
latticra-lc boundary
```

C Foundation

The C foundation is intentionally deterministic:

```
include/latticra/latticra_console.h
src/latticra_console.c
tools/latticra_console_report.c
tests/latticra_console_foundation.c
scripts/test-latticra-console-foundation.sh
```

The report surface emits:

```
LATTICRA CONSOLE REPORT
component_key=latticra_console
console_status=ready-report-only
command_registry_status=seed-registry-ready
command_registry_source=c-static-table
command_registry_no_effect=1
command_registry_host_process_launch_allowed=0
runtime_boundary_bound=1
seal_capability_labels_bound=1
standalone_console_status=metadata-only-standalone-contract-ready
standalone_contract_present=1
session_contract_status=metadata-only-contract-ready
session_contract_present=1
standalone_installable=1
standalone_requires_panel=0
substrate_bridge_status=metadata-bound-ready
panel_installable=1
standalone_installable=1
standalone_requires_panel=0
standalone_command_wrapper=latticra-lc
host_adapter_contract_status=metadata-only-contract-ready
host_adapter_contract_present=1
receipt_request_contract_status=metadata-only-contract-ready
receipt_request_contract_present=1
receipt_payload_schema_status=metadata-only-schema-ready
receipt_payload_schema_present=1
receipt_payload_artifact_draft_status=metadata-only-draft-ready
```

```
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_review_status=metadata-only-review-gate-ready
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_receipt_status=metadata-only-receipt-contract-ready
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_draft_status=metadata-only-review-receipt-draft-ready
receipt_payload_artifact_review_receipt_draft_present=1
receipt_payload_materialization_plan_status=metadata-only-plan-ready
receipt_payload_materialization_plan_present=1
signature_request_binding_status=metadata-only-contract-ready
signature_request_binding_present=1
os_base_contract_status=metadata-only-contract-ready
os_base_contract_present=1
vm_evidence_contract_status=metadata-only-contract-ready
vm_evidence_contract_present=1
os_base_enabled=0
production_os_claim=0
future_os_base_claim=planned_not_claimed
execution_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

Authority Baseline

LC Stage-0 does not:

- execute shell commands;
- launch external host processes;
- read arbitrary host files;
- write arbitrary host files;
- mutate the host;
- use the network;
- enforce runtime policy;
- boot hardware;
- claim to be a production OS;
- replace Linux, Fedora, macOS, Windows, or any other host OS.

Next Slices

1. Materialize the LC receipt payload artifact only after the actual review receipt draft and actual review receipt exist.
2. Bind the LC receipt payload artifact to a Seal signature-request artifact only after signing authority is implemented and gated.
3. Add the first Seal-signed LC receipt path only after the receipt request is reviewed and receipted.
4. Add a host-adapter artifact schema only after the host-adapter contract is receipted.
5. Add a VM evidence artifact schema only after the VM evidence contract is receipted.

docs/LATTICRA_PANEL_SIGNED_UPDATER_DELIVERY_GATE.md

Source title: Latticra Panel Signed Updater Delivery Gate

Lines: 113 | Words: 275 | SHA-256: 4e98372daed928a22d44998aa6a1ccc584976b54f927acfd4904dde73a78d451

Latticra Panel Signed Updater Delivery Gate

Status: no-effect signed updater delivery gate Date: 2026-05-26 CDT Scope: closed gate for any future signed, network-delivered update path behind the Latticra Panel updater.

Purpose

The current Latticra Panel updater is a guarded local-checkout lane. It reuses the installer engine, requires preview before apply, and keeps network fetch authority disabled.

This contract defines the gate that must stay closed before any future signed or network-delivered updater can exist:

```
signed_updater_delivery_gate_state=closed
signed_update_delivery_ready=0
network_self_update_ready=0
signed_update_apply_allowed=0
```

Command

```
sh scripts/latticra-panel-signed-updater-delivery-gate.sh
```

The command writes only a deterministic report to stdout.

Current Gate Decision

The current gate decision is:

```
signed_updater_delivery_gate_decision=blocked-missing-signed-manifest-artifact-verification-and-
rollback-evidence
```

The gate stays closed because these required future pieces are absent:

```
signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
channel_policy_present=0
compatibility_policy_present=0
rollback_plan_present=0
rollback_evidence_present=0
post_update_validation_present=0
operator_confirmation_observed=0
update_receipt_written=0
```

Required Preconditions

Before a signed update may ever be staged or applied, a later implementation must provide all of:

```
signed_manifest_required=1
manifest_signature_required=1
artifact_hash_required=1
artifact_signature_required=1
channel_policy_required=1
compatibility_policy_required=1
rollback_plan_required=1
post_update_validation_required=1
operator_confirmation_required=1
update_receipt_required=1
```

Authority Boundary

The current authority boundary remains:

```
remote_update_repository_trust=0
network_fetch_authority=0
updater_network_fetch_enabled=0
staged_update_allowed=0
signed_update_apply_allowed=0
update_activation_allowed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
```

```
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-delivery-gate.sh
```

Expected output:

```
latticra_panel_signed_updater_delivery_gate: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This contract is not a signed updater, network updater, update server, remote repository trust policy, artifact verifier, rollback implementation, production update path, production installer, root installer, host mutation permission, boot integration, kernel integration, systemd integration, SELinux integration, malware prevention, ransomware prevention, sandbox, or production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_DENIAL_TRANSCRIPT.md

Source title: Latticra Panel Signed Updater Denial Transcript

Lines: 98 | Words: 231 | SHA-256: 84ba750beb490c0d9d78915c7b9f6831a41b1fde9ec7bfa1997a23a1a64ae871

Latticra Panel Signed Updater Denial Transcript

Status: no-effect signed updater denial transcript Date: 2026-05-26 CDT Scope: denial transcript for the closed signed updater delivery gate behind the Latticra Panel updater.

Purpose

The signed updater delivery gate is closed. This transcript records the current denial decision without enabling network fetch, remote repository trust, artifact verification, staging, transcript file writes, or signed update apply.

```
signed_updater_denial_decision=deny-signed-update-delivery
signed_updater_denial_reason=missing-signed-manifest-artifact-verification-and-rollback-evidence
signed_updater_denial_transcript_stdout_only=1
signed_updater_denial_transcript_file_write_enabled=0
```

Command

```
sh scripts/latticra-panel-signed-updater-denial-transcript.sh
```

The command writes only a deterministic report to stdout.

Denial Decision

The current denial decision is:

```
signed_updater_delivery_gate_state=closed
signed_updater_denial_decision=deny-signed-update-delivery
signed_update_delivery_ready=0
signed_update_apply_allowed=0
```

The denial is caused by missing future evidence:

```
signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
```

```
artifact_signature_verified=0
channel_policy_present=0
compatibility_policy_present=0
rollback_plan_present=0
rollback_evidence_present=0
post_update_validation_present=0
operator_confirmation_observed=0
update_receipt_written=0
```

Authority Boundary

The transcript preserves:

```
network_fetch_authority=0
network_fetch_attempted=0
remote_update_repository_trust=0
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
production_update_ready=0
```

Validation

This transcript is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-denial-transcript.sh
```

Expected output:

```
latticra_panel_signed_updater_denial_transcript: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This transcript is not signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, transcript persistence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_CONTRACT.md

Source title: Latticra Panel Signed Updater Manifest Fixture Contract

Lines: 103 | Words: 256 | SHA-256: 9050aec82791f4a474f02ed655768d6436e988284e0bafb9a23160d3019e9560

Latticra Panel Signed Updater Manifest Fixture Contract

Status: local no-effect signed updater manifest fixture contract Date: 2026-05-26 CDT Scope: local fixture contract for future signed updater manifest work behind the Latticra Panel updater.

Purpose

This slice adds a committed local manifest fixture for the Panel signed updater lane.

The fixture gives the closed delivery gate a concrete manifest-shaped record to review without making it trusted signed update evidence, enabling network fetch, verifying artifacts, staging updates, writing receipts, or applying updates.

```
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_present=1
manifest_fixture_effect=none
manifest_fixture_trusted_for_apply=0
trusted_signed_manifest_present=0
signed_update_apply_allowed=0
```

Fixture

The fixture lives at:

```
fixtures/latticra-panel/signed-updater-manifest.fixture.toml
```

It records the current local-checkout updater shape:

```
update_id=panel-local-checkout-fixture
update_channel=lab
source_strategy=current-source-checkout
apply_mode=guarded-local-prefix-reinstall
target_component=latticra-panel
target_version=fixture-only
```

Command

```
sh scripts/latticra-panel-signed-updater-manifest-fixture-contract.sh
```

The command writes only a deterministic report to stdout.

Trust Boundary

The fixture is explicitly not trusted for apply:

```
signed_updater_delivery_gate_state=closed
signed_manifest_present=0
trusted_signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
rollback_plan_present=0
operator_confirmation_observed=0
update_receipt_written=0
signed_update_apply_allowed=0
```

The authority boundary remains closed:

```
network_fetch_authority=0
network_fetch_attempted=0
remote_update_repository_trust=0
updater_network_fetch_enabled=0
staged_update_performed=0
signed_update_apply_performed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
production_update_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-manifest-fixture-contract.sh
```

Expected output:

```
latticra_panel_signed_updater_manifest_fixture_contract: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This contract is not signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_VALIDATION.md

Source title: Latticra Panel Signed Updater Manifest Fixture Validation

Lines: 99 | Words: 245 | SHA-256: 04f8c7ec2f6f1274a40347c872f9c20171a87f027f2e0b80c08b0d18bb4297a1

Latticra Panel Signed Updater Manifest Fixture Validation

Status: local no-effect signed updater manifest fixture validation Date: 2026-05-26 CDT Scope: validation for the committed Latticra Panel signed-updater manifest fixture.

Purpose

This slice validates the committed manifest fixture shape without making the fixture trusted signed update evidence.

The validator reads only the local fixture, checks the expected schema and closed-authority fields, and reports that the fixture is valid only as no-effect input for future updater design work.

```
signed_updater_manifest_fixture_validation_present=1
signed_updater_manifest_fixture_validated=1
signed_updater_manifest_fixture_validation_scope=shape-and-closed-authority-fields
manifest_fixture_trusted_for_apply=0
signed_updater_manifest_fixture_valid_for_apply=0
trusted_signed_manifest_present=0
signed_update_apply_allowed=0
```

Command

```
sh scripts/latticra-panel-signed-updater-manifest-fixture-validation.sh
```

The command reads:

```
fixtures/latticra-panel/signed-updater-manifest.fixture.toml
```

It writes only a deterministic report to stdout.

Validation Boundary

The validator checks local fixture shape:

```
manifest_schema_validated=1
manifest_fixture_flag_validated=1
manifest_fixture_scope_validated=1
manifest_fixture_effect_validated=1
local_checkout_strategy_validated=1
closed_authority_fields_validated=1
non_claims_validated=1
```

The validator does not verify a real signed update:

```
signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
rollback_plan_present=0
operator_confirmation_observed=0
update_receipt_written=0
```

The authority boundary remains closed:

```
network_fetch_authority=0
network_fetch_attempted=0
remote_update_repository_trust=0
staged_update_performed=0
signed_update_apply_performed=0
validation_write_performed=0
host_mutation_performed=0
root_authority=0
production_update_ready=0
```

Validation

This slice is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-manifest-fixture-validation.sh
```

Expected output:

```
latticra_panel_signed_updater_manifest_fixture_validation: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This validation is not signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_CONTRACT.md

Source title: Latticra Panel Signed Updater State Fixture Contract

Lines: 107 | Words: 265 | SHA-256: b4d0c9b4cc87f12b8083a0eedd290f1436a1c6f4f69823c9945f9c818f6def97

Latticra Panel Signed Updater State Fixture Contract

Status: local no-effect signed updater state fixture contract Date: 2026-05-26 CDT Scope: local state fixture contract for future signed updater state work behind the Latticra Panel updater.

Purpose

This slice adds a committed local state fixture for the Panel signed updater lane.

The fixture gives the closed delivery gate a state-shaped record to review without enabling state transitions, staging, activation, rollback execution, receipt writes, or signed update apply.

```
signed_updater_state_fixture_contract_present=1
signed_updater_state_fixture_present=1
state_fixture_effect=none
current_update_state=blocked
state_transition_execution_allowed=0
signed_update_apply_allowed=0
```

Fixture

The fixture lives at:

```
fixtures/latticra-panel/signed-updater-state.fixture.toml
```

It records the planned state names while keeping the current fixture state blocked:

```
available
downloaded
verified
staged
armed
applied
rolled_back
failed
blocked
current_update_state=blocked
requested_update_state=blocked
state_transition_decision=deny-transition
```

Command

```
sh scripts/latticra-panel-signed-updater-state-fixture-contract.sh
```

The command writes only a deterministic report to stdout.

Authority Boundary

The state fixture is explicitly not executable:

```
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
rollback_execution_allowed=0
update_activation_allowed=0
signed_update_apply_allowed=0
```

The signed updater boundary remains closed:

```
trusted_signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
rollback_plan_present=0
operator_confirmation_observed=0
network_fetch_attempted=0
host_mutation_performed=0
production_update_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-fixture-contract.sh
```

Expected output:

```
latticra_panel_signed_updater_state_fixture_contract: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This contract is not update-state evidence, state-transition execution, signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_VALIDATION.md

Source title: Latticra Panel Signed Updater State Fixture Validation

Lines: 104 | Words: 257 | SHA-256: eb21ccb2671f5b6271984eae2ef5434e1c73c73e2cff1ce9a6b4b34b132d9329

Latticra Panel Signed Updater State Fixture Validation

Status: local no-effect signed updater state fixture validation Date: 2026-05-26 CDT
Scope: validation for the committed Latticra Panel signed-updater state fixture.

Purpose

This slice validates the committed state fixture shape without making the fixture executable update-state evidence.

The validator reads only the local fixture, checks the expected schema, state catalog, blocked-state decision, and closed transition fields, and reports that the fixture is valid only as no-effect input for future updater design work.

```
signed_updater_state_fixture_validation_present=1
signed_updater_state_fixture_validated=1
signed_updater_state_fixture_validation_scope=shape-state-catalog-and-closed-transition-fields
state_fixture_trusted_for_apply=0
signed_updater_state_fixture_valid_for_transition=0
signed_updater_state_fixture_valid_for_apply=0
state_transition_execution_allowed=0
signed_update_apply_allowed=0
```

Command

```
sh scripts/latticra-panel-signed-updater-state-fixture-validation.sh
```

The command reads:

```
fixtures/latticra-panel/signed-updater-state.fixture.toml
```

It writes only a deterministic report to stdout.

Validation Boundary

The validator checks local fixture shape:

```
state_schema_validated=1
state_fixture_flag_validated=1
state_fixture_scope_validated=1
state_fixture_effect_validated=1
state_catalog_validated=1
blocked_state_validated=1
closed_transition_fields_validated=1
closed_authority_fields_validated=1
non_claims_validated=1
```

The validator does not execute an updater transition:

```
current_update_state=blocked
requested_update_state=blocked
state_transition_decision=deny-transition
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
```

The authority boundary remains closed:

```
network_fetch_authority=0
network_fetch_attempted=0
rollback_execution_performed=0
signed_update_apply_performed=0
update_activation_performed=0
validation_write_performed=0
host_mutation_performed=0
root_authority=0
production_update_ready=0
```

Validation

This slice is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-fixture-validation.sh
```

Expected output:

```
latticra_panel_signed_updater_state_fixture_validation: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This validation is not update-state evidence, state-transition execution, signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_REVIEW.md

Source title: Latticra Panel Signed Updater State Transition Denial Review

Lines: 94 | Words: 240 | SHA-256: 430061d079c30dbc51e622e37bcbd4493cbf57580f00f3b279697f26a56e7471

Latticra Panel Signed Updater State Transition Denial Review

Status: no-effect signed updater state transition denial review Date: 2026-05-26 CDT Scope: stdout-only review of the blocked Latticra Panel signed-updater state transition denial transcript.

Purpose

This slice reviews the state transition denial transcript without turning the review into transition authority.

The review upholds the denial because the signed updater still lacks signed manifest trust, artifact verification, rollback evidence, and operator confirmation:

```
signed_updater_state_transition_denial_review_present=1
signed_updater_state_transition_denial_review_stdout_only=1
signed_updater_state_transition_denial_review_file_write_enabled=0
signed_updater_state_transition_denial_review_decision=uphold-deny-state-transition
signed_updater_state_transition_denial_review_reason=denial-transcript-confirms-missing-signed-m
anifest-artifact-verification-rollback-and-operator-confirmation
state_transition_review_decision=uphold-denial
```

Command

```
sh scripts/latticra-panel-signed-updater-state-transition-denial-review.sh
```

The command writes only a deterministic report to stdout.

Review Boundary

The review is not executable:

```
state_fixture_validated_for_transition=0
signed_updater_state_fixture_valid_for_transition=0
state_transition_decision=deny-transition
state_transition_review_decision=uphold-denial
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
review_write_performed=0
```

The review does not write files or mutate host state:

```
signed_updater_state_transition_denial_review_file_write_enabled=0
validation_write_performed=0
```

```
transcript_write_performed=0
review_write_performed=0
host_mutation_performed=0
root_authority=0
production_update_ready=0
```

The signed updater boundary remains closed:

```
trusted_signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
rollback_plan_present=0
rollback_execution_performed=0
operator_confirmation_observed=0
signed_update_apply_performed=0
update_activation_performed=0
network_fetch_attempted=0
```

Validation

This slice is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-transition-denial-review.sh
```

Expected output:

```
latticra_panel_signed_updater_state_transition_denial_review: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This review is not update-state evidence, state-transition execution, signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_TRANSCRIPT.md

Source title: Latticra Panel Signed Updater State Transition Denial Transcript

Lines: 95 | Words: 234 | SHA-256: df3e74c1b4cea75b3326c09acf8e2230d2b6dbfd78cb344436682d9a21cc753c

Latticra Panel Signed Updater State Transition Denial Transcript

Status: no-effect signed updater state transition denial transcript Date: 2026-05-26 CDT Scope: stdout-only denial transcript for the blocked Latticra Panel signed-updater state transition.

Purpose

This slice records the state transition denial as a deterministic transcript without making the blocked state fixture executable.

The transcript binds the validated local state fixture to the current decision:

```
signed_updater_state_transition_denial_transcript_present=1
signed_updater_state_transition_denial_transcript_stdout_only=1
signed_updater_state_transition_denial_transcript_file_write_enabled=0
signed_updater_state_transition_denial_decision=deny-state-transition
signed_updater_state_transition_denial_reason=missing-signed-manifest-artifact-verification-roll
back-and-operator-confirmation
```

```
current_update_state=blocked
requested_update_state=blocked
```

Command

```
sh scripts/latticra-panel-signed-updater-state-transition-denial-transcript.sh
```

The command writes only a deterministic report to stdout.

Denial Boundary

The denied transition remains non-executable:

```
state_fixture_validated_for_transition=0
signed_updater_state_fixture_valid_for_transition=0
state_transition_decision=deny-transition
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
```

The transcript does not write files or mutate host state:

```
signed_updater_state_transition_denial_transcript_file_write_enabled=0
validation_write_performed=0
transcript_write_performed=0
host_mutation_performed=0
root_authority=0
production_update_ready=0
```

The signed updater boundary remains closed:

```
trusted_signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
rollback_plan_present=0
rollback_execution_performed=0
operator_confirmation_observed=0
signed_update_apply_performed=0
update_activation_performed=0
network_fetch_attempted=0
```

Validation

This slice is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-transition-denial-transcript.sh
```

Expected output:

```
latticra_panel_signed_updater_state_transition_denial_transcript: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This transcript is not update-state evidence, state-transition execution, signed update evidence, network update evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production update readiness, production installer readiness, root installer readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/LOCAL_INSTALLER_ARTIFACT_MANIFEST_CONTRACT.md

Source title: Local Installer Artifact Manifest Contract

Lines: 220 | Words: 530 | SHA-256: b0e0bdccfaa8c8d0b906018be1e2da5251d3d00dfa79dab84cb380b27eb2e474

Local Installer Artifact Manifest Contract

Status: contract record Evidence level: 10 target, manifest contract only Scope: required metadata for any future local installer artifact before it can be considered for production-installer readiness.

Purpose

The production-installer readiness contract requires an installer artifact manifest before any installer can be recommended for general use.

This contract defines the minimum manifest fields that must exist before the project can evaluate installer artifacts.

This is not an installer.

This does not build, sign, publish, distribute, install, uninstall, upgrade, or rollback an installer.

This does not claim production installer readiness.

Current readiness boundary

The current project state remains:

```
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The current evidence-backed prerequisite is limited to disposable Fedora VM validation of the local no-effect CLI RPM payload:

```
host_install_ready_for_cli_payload=1
validated_payload_cli=/usr/bin/latticra
validated_payload_readme=/usr/share/doc/latticra/README.md
```

Required manifest file

A future installer artifact must include a manifest file at a stable path such as:

```
artifacts/&lt;artifact-name&gt;/manifest.txt
```

The manifest must be plain text, line-oriented, deterministic, and reviewable.

Required manifest fields

A complete artifact manifest must include:

```
LATTICRA INSTALLER ARTIFACT MANIFEST
manifest_version=1
artifact_name=&lt;recorded&gt;
artifact_version=&lt;recorded&gt;
artifact_arch=&lt;recorded&gt;
artifact_format=&lt;recorded&gt;
artifact_filename=&lt;recorded&gt;
artifact_size_bytes=&lt;recorded&gt;
artifact_sha256=&lt;recorded&gt;
artifact_signature=&lt;recorded-or-none&gt;
artifact_signing_key_id=&lt;recorded-or-none&gt;
artifact_sbom_path=&lt;recorded-or-none&gt;
artifact_license_metadata_path=&lt;recorded&gt;
source_repository=Bryforge/Latticra
source_commit=&lt;recorded&gt;
source_tag=&lt;recorded-or-none&gt;
build_environment=&lt;recorded&gt;
build_command_recorded=1
build_reproducible=&lt;0-or-1&gt;
supported_target_family=&lt;recorded&gt;
supported_target_versions=&lt;recorded&gt;
supported_target_arches=&lt;recorded&gt;
unsupported_targets_declared=1
requires_operator_consent=1
```

```
preflight_guard_required=1
install_plan_preview_required=1
uninstall_path_required=1
rollback_or_recovery_path_required=1
network_required=0
service_activation_default=0
boot_change_default=0
kernel_module_default=0
selinux_policy_default=0
payload_listing_recorded=1
payload_contains_cli_binary=&lt;0-or-1&gt;
payload_contains_readme=&lt;0-or-1&gt;
payload_contains_service=&lt;0-or-1&gt;
payload_contains_kernel_module=&lt;0-or-1&gt;
payload_contains_boot_change=&lt;0-or-1&gt;
payload_contains_selinux_policy=&lt;0-or-1&gt;
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Required payload declaration

Every manifest must declare the artifact payload explicitly.

For the current no-effect CLI RPM payload, the known validated payload baseline is:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

A future installer artifact may not add runtime surfaces silently.

Any added service, kernel, boot, SELinux, networking, or policy surface must be separately declared, reviewed, and validated before readiness language changes.

Required checksum and signature boundary

At this stage, signature fields may be none only for local experimental artifacts.

A production installer readiness claim requires:

```
artifact_sha256_recorded=1
artifact_signature_recorded=1
artifact_signing_key_documented=1
signature_verification_documented=1
```

Until those are present and validated, the project must preserve:

```
production_installer_ready=0
```

Required SBOM boundary

A production installer readiness claim requires an SBOM path in the manifest:

```
artifact_sbom_path=&lt;recorded&gt;
sbom_present=1
sbom_reviewed=1
```

Until SBOM evidence exists, the project must preserve:

```
installer_sbom_recorded=0
production_installer_ready=0
```

Required target boundary

A manifest must distinguish supported and unsupported targets.

At minimum, it must preserve:

```
daily_driver_install_ready=0
immutable_fedora_ready=0
production_host_ready=0
```

until separate evidence validates those target classes.

Required non-claims

Every manifest review must preserve:

```
not_production_readiness=1
not_fedora_approval=1
not_fedora_distribution_readiness=1
not_daily_driver_readiness=1
not_immutable_fedora_readiness=1
not_security_hardening_claim=1
not_malware_prevention_claim=1
not_ransomware_prevention_claim=1
not_os_replacement_claim=1
```

Current status

```
installer_artifact_manifest_contract_present=1
installer_artifact_manifest_present=0
installer_artifact_manifest_validated=0
installer_artifact_checksum_recorded=0
installer_artifact_signature_recorded=0
installer_sbom_recorded=0
installer_supported_targets_declared=0
installer_unsupported_targets_declared=0
production_installer_ready=0
```

Guard validation

This contract is guarded by:

```
sh scripts/test-local-installer-artifact-manifest-contract.sh
```

Expected output:

```
local_installer_artifact_manifest_contract: ok
```

Next implementation lane

```
Add installer artifact manifest fixture
```

That lane should add a non-production fixture manifest for the current local no-effect CLI RPM artifact and keep all production readiness flags set to 0.

Non-claims

This contract is not an installer artifact, not a production installer, not a distribution package, not Fedora approval, not Fedora distribution readiness, not daily-driver readiness, not immutable Fedora readiness, not update safety, not recovery safety, not security hardening, not malware prevention, not ransomware prevention, and not OS-replacement readiness.

docs/PRODUCTION_INSTALLER_READINESS_CONTRACT.md

Source title: Production Installer Readiness Contract

Lines: 182 | Words: 425 | SHA-256: 970dd1e8027e180e3e6f0821eb7e076293c370564b8941bc0e82800520a67220

Production Installer Readiness Contract

Status: contract record Evidence level: 10 target, contract only Scope: requirements that must be satisfied before Latticra can claim a production installer that others should use.

Purpose

Latticra now has disposable Fedora VM evidence for a local no-effect CLI RPM payload.

That is not the same as production installer readiness.

This contract defines the minimum evidence required before the project may claim:

```
production_installer_ready=1
```

Until every required gate is satisfied, the project must preserve:

```
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Current evidence that may be used as prerequisites

The current evidence-backed base is limited to disposable Fedora VM validation:

```
disposable_vm_local_rpm_validation_completed=1
disposable_vm_cli_validation_completed=1
host_install_ready=1
host_install_ready_for_cli_payload=1
```

Validated disposable Fedora VM CLI payload:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

Production installer readiness gates

A production installer claim requires all of the following gates to be complete and reviewed.

```
installer_contract_present=1
installer_artifact_format_declared=1
installer_artifact_built_from_tag=1
installer_artifact_reproducible=1
installer_artifact_checksum_recorded=1
installer_artifact_signature_recorded=1
installer_public_key_documented=1
installer_sbom_recorded=1
installer_license_metadata_recorded=1
installer_supported_targets_declared=1
installer_unsupported_targets_declared=1
installer_daily_driver_warning_present=1
installer_production_host_warning_present=1
installer_immutable_fedora_warning_present=1
installer_preflight_guard_present=1
installer_preflight_blocks_unsupported_targets=1
installer_install_plan_preview_present=1
installer_requires_operator_consent=1
installer_install_transcript_recorded=1
installer_uninstall_transcript_recorded=1
installer_post_removal_absence_verified=1
installer_upgrade_path_validated=1
installer_downgrade_or_rollback_path_validated=1
installer_reinstall_idempotence_validated=1
installer_no_network_requirement_documented=1
installer_no_service_activation_without_consent=1
installer_no_boot_change_without_consent=1
installer_no_kernel_module_without_consent=1
installer_no_selinux_policy_without_consent=1
installer_effect_authority_documented=1
installer_failure_mode_documented=1
installer_recovery_runbook_present=1
installer_multi_vm_validation_completed=1
installer_fresh_vm_validation_completed=1
installer_existing_install_validation_completed=1
installer_non_root_cli_validation_completed=1
installer_root_boundary_validation_completed=1
installer_security_non_claims_preserved=1
installer_release_notes_present=1
installer_readme_install_instructions_present=1
installer_status_alignment_present=1
```

Minimum validation matrix

Before claiming production installer readiness, validation must include at least:

```
fresh_disposable_fedora_vm_validation=1
repeat_disposable_fedora_vm_validation=1
existing_install_upgrade_validation=1
remove_and_reinstall_validation=1
post_removal_absence_validation=1
```

```
unsupported_target_block_validation=1
non_root_cli_use_validation=1
package_signature_verification_validation=1
checksum_verification_validation=1
```

Required production installer report

A future readiness report must include:

```
PRODUCTION INSTALLER READINESS REPORT
validation_status=ok
installer_artifact_name=&lt;recorded&gt;
installer_artifact_version=&lt;recorded&gt;
installer_artifact_arch=&lt;recorded&gt;
installer_artifact_checksum_recorded=1
installer_artifact_signature_recorded=1
installer_supported_targets_declared=1
installer_preflight_guard_present=1
installer_install_validation_completed=1
installer_uninstall_validation_completed=1
installer_upgrade_validation_completed=1
installer_rollback_validation_completed=1
installer_multi_vm_validation_completed=1
production_installer_ready=1
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=10
```

Fedora distribution readiness and Fedora approval remain separate claims and must stay 0 until separate review and acceptance evidence exists.

Current readiness classification

```
production_installer_contract_present=1
production_installer_ready=0
installer_artifact_reproducible=0
installer_artifact_signature_recorded=0
installer_sbom_recorded=0
installer_preflight_guard_present=0
installer_install_transcript_recorded=0
installer_uninstall_transcript_recorded=0
installer_upgrade_path_validated=0
installer_rollback_path_validated=0
installer_multi_vm_validation_completed=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Next implementation lanes

Recommended sequence:

```
1. Add local installer artifact manifest contract.
2. Add installer preflight classifier for supported and blocked targets.
3. Add signed artifact/checksum status contract.
4. Add installer preview-only plan renderer.
5. Add disposable Fedora VM production-installer dry-run lane.
6. Add uninstall and rollback transcript contract.
7. Add upgrade/reinstall validation contract.
8. Add multi-VM production-installer evidence status.
9. Only then consider production_installer_ready=1.
```

Non-claims

This contract is not a production installer.

It does not build, sign, publish, install, uninstall, upgrade, rollback, or distribute an installer.

It is not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver readiness, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, not security hardening, not malware prevention, not ransomware prevention, not OS-replacement readiness, and not a production installer claim.

docs/SELF_UPDATE_MODEL.md

Source title: Latticra Self-Update Model

Lines: 244 | Words: 755 | SHA-256: 13af98701c6dfbc7bc93e0ccb2aeacadc26d5cdbcd58fcb59d46de82f1eb2d42

Latticra Self-Update Model

Status: Panel-owned local-checkout updater policy active; signed updater delivery gate closed Scope: signed staged updates, channels, rollback, operator confirmation, and non-claims.

Purpose

Latticra should be designed for safe self-update capability from the beginning, but self-update must never be silent or magical.

Self-update is a real-system capability and therefore requires contracts, effect gates, verification, rollback, and operator visibility.

Core rule

No silent self-updates.

Every update path must be inspectable, signed, staged, and reversible where possible.

Current Panel-Owned Updater Surface

Panel-owned local-checkout updater policy is active.

The current implemented updater surface is the Latticra Panel updater. It is limited to a reviewed local checkout and a guarded local-prefix reinstall flow:

```
updater_current_source_strategy=current-source-checkout
updater_current_apply_mode=guarded-local-prefix-reinstall
network_self_update_ready=0
remote_update_repository_trust=0
signed_update_delivery_ready=0
```

The signed delivery gate is still closed:

```
signed_updater_delivery_gate_state=closed
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_verified=0
artifact_hash_verified=0
artifact_signature_verified=0
rollback_plan_required=1
rollback_plan_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
signed_update_apply_allowed=0
signed_updater_denial_transcript_present=1
signed_updater_denial_decision=deny-signed-update-delivery
signed_updater_denial_transcript_stdout_only=1
signed_updater_denial_transcript_file_write_enabled=0
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_present=1
signed_updater_manifest_fixture_validation_present=1
signed_updater_manifest_fixture_validated=1
signed_updater_state_fixture_contract_present=1
signed_updater_state_fixture_present=1
signed_updater_state_fixture_validation_present=1
signed_updater_state_fixture_validated=1
signed_updater_state_fixture_valid_for_transition=0
signed_updater_state_fixture_valid_for_apply=0
signed_updater_state_transition_denial_transcript_present=1
signed_updater_state_transition_denial_decision=deny-state-transition
signed_updater_state_transition_denial_transcript_stdout_only=1
signed_updater_state_transition_denial_transcript_file_write_enabled=0
```

```
signed_updater_state_transition_denial_review_present=1
signed_updater_state_transition_denial_review_decision=uphold-deny-state-transition
signed_updater_state_transition_denial_review_stdout_only=1
signed_updater_state_transition_denial_review_file_write_enabled=0
current_update_state=blocked
state_transition_execution_allowed=0
state_transition_execution_performed=0
manifest_fixture_trusted_for_apply=0
signed_updater_manifest_fixture_valid_for_apply=0
trusted_signed_manifest_present=0
```

The local signed updater manifest fixture validation only checks shape and closed-authority fields. The local signed updater state fixture validation only checks the state catalog and closed transition fields without making the blocked fixture executable. The local signed updater state transition denial transcript records the blocked decision to stdout only, without transcript file writes or execution. The local signed updater state transition denial review upholds the denial without review file writes or execution. This is not a signed updater, not a remote update client, not a network self-update path, not update state execution, and not production update readiness.

Update channels

Initial planned channels:

```
stable
candidate
edge
lab
```

Channel meanings:

```
| Channel | Meaning |
| --- | --- |
| `stable` | Promoted, evidence-backed release path. |
| `candidate` | Release candidate path after validation. |
| `edge` | Active development path with explicit risk. |
| `lab` | Local or experimental path, never default. |
```

Update states

```
available
downloaded
verified
staged
armed
applied
rolled_back
failed
blocked
```

Only applied may imply the update changed the running or installed system.

Required update record

Every update should eventually produce:

```
update_id
current_version
target_version
channel
manifest_hash
artifact_hash
signature_result
compatibility_result
rollback_available
operator_confirmation
state
failure_reason
post_update_validation
```

Effect classes

Self-update may involve multiple effect classes:

```
network
local_mutation
host_mutation
boot
recovery
external
```

The early update model must support only documentation and fixture-level planning.

Signed manifest rule

An update must be described by a signed manifest before any artifact is trusted.

Manifest should include:

- version;
- channel;
- artifact hash;
- compatibility constraints;
- architecture target;
- rollback compatibility;
- required gates;
- post-update validation commands.

Staging rule

Update artifacts should be staged before activation.

Staging does not imply activation.

```
downloaded != verified
verified != staged
staged != armed
armed != applied
```

Rollback rule

Update design must include rollback state before real update execution exists.

Rollback states:

```
unavailable
available
armed
executed
failed
blocked
```

Operator confirmation

A future update that changes local system state must require explicit operator confirmation.

A future update that changes boot or recovery state must require a stronger confirmation profile.

Server relationship

Update server interaction must pass through the Server Gateway.

No component should perform direct update network behavior outside the signed update model.

First implementation target

The current Panel implementation target is a local-checkout updater policy surface, not signed or networked self-update.

It installs and reports:

```
etc/latticra/updater.toml
share/latticra/updater/policy.toml
latticra updater status
updater dry-run
updater apply
```

The Panel updater reuses the guarded installer engine for a local-prefix reinstall from the current reviewed checkout. It has no network fetch, root, system mutation, boot, or recovery authority. latticra updater status reports the configured source strategy, channel, preview/apply commands, receipt setting, guarded apply mode, and disabled authority posture without launching the GUI.

The signed staged update model still needs:

```
update manifest fixture
update state fixture
no-effect signed updater denial transcript
local signed updater manifest fixture contract
local signed updater manifest fixture validation
local signed updater state fixture contract
local signed updater state fixture validation
local signed updater state transition denial transcript
local signed updater state transition denial review
signature-required marker
rollback visibility marker
validation test
```

Non-claims

This document does not implement signed or networked self-update.

The Panel updater does not claim secure update delivery, rollback success, boot update safety, recovery behavior, network fetch authority, root authority, system mutation authority, or production release readiness.

docs/strategy/2026-05-25-2137-cdt-panel-guided-local-evaluation-workflow-packet.md

Source title: Latticra Panel-Guided Local Evaluation Workflow Packet

Lines: 279 | Words: 859 | SHA-256: 1089a716a6a67e37c14b1e0072b644904e2698b377c0822d224e4ce8d62b5c41

Latticra Panel-Guided Local Evaluation Workflow Packet

Status: draft workflow packet Created: 2026-05-25 21:37 CDT Decision: selected as first public-readiness workflow planning path Promotion decision: no product-readiness promotion recommended Scope: bounded Panel-guided local evaluation workflow for technical evaluators.

Purpose

This packet selects the first public-readiness workflow planning path:

```
Panel-guided local evaluation
```

It narrows the public product-readiness promotion packet into one candidate user journey. It does not authorize implementation expansion, public product-readiness promotion, runtime authority, or estimate changes.

Source packet

This workflow packet follows:

```
docs/strategy/2026-05-25-2114-cdt-public-product-readiness-promotion-packet.md
```

Current strategy remains:

```
Continue small guarded report/status alignment only when drift appears.
```

Target evaluator

Primary evaluator:

```
technical evaluator
```

Secondary evaluators:

```
security researcher
systems engineer
infrastructure maintainer
```

This workflow is not for nontechnical end users, production operators, daily-driver use, or security reliance.

Workflow statement

Named workflow:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Workflow purpose:

```
let a technical evaluator inspect Latticra&#x27;s current local workbench, dry-run posture,
visible authority boundaries, plan/evidence surfaces, and reset boundaries without
claiming production readiness
```

Current evidence

Existing README-supported Panel flow:

```
open Guided Workbench
keep dry-run mode enabled
generate and inspect the plan
run Dry-Install to validate and write a receipt
review the embedded console, plan, and engine log
enable guarded local-prefix writes only after dry-run evidence looks correct
```

Existing command surface:

```
make -C installer gui
make -C installer dry-run
make -C installer verify-local
make -C installer reset-dry-run
make -C installer reset-local
make -C installer uninstall-local
make -C installer uninstall-dry-run
```

Current estimate posture:

```
Latticra Panel / local control surface: 28%
Public product readiness: 8%
```

Workflow boundary

The first public-readiness workflow should remain:

```
local
operator-visible
dry-run-first
no-root by default
no-network by default
reset-aware
receipt-aware
non-production
```

The workflow should not depend on runtime execution, effect execution, capability enforcement, cryptographic authority, model execution, tool execution, or network behavior.

Proposed user journey

The future workflow record should describe exactly this path:

1. Read the current status and non-claims.
2. Confirm local prerequisites.
3. Launch Latticra Panel from source or documented local entry point.
4. Enter Guided Workbench.
5. Confirm dry-run mode is enabled.
6. Generate the local plan.
7. Inspect the plan before any local write.
8. Run the dry-run validation path.
9. Inspect the receipt, embedded console, plan, and engine log.
10. Confirm denied authority and non-claims remain visible.
11. Run reset or uninstall dry-run path when appropriate.
12. Exit with a clear statement of what happened and what did not happen.

Required acceptance evidence

Before this workflow can support any product-readiness estimate change, require:

```
workflow acceptance checklist
platform prerequisite checklist
dry-run transcript
plan output example
receipt or receipt-like evidence example
engine log example
embedded console status example
reset or uninstall transcript
failure-state examples
known limitations note
support boundary note
non-claim review
guard or test that covers the documented path
estimate-impact review
public-entrypoint review
```

Required operator-visible labels

The workflow should expose:

```
dry_run_enabled
runtime_authority_granted=0
effect_execution_performed=0
network_authority_granted=0
root_authority_required=0
local_write_planned=0|1
local_write_performed=0|1
receipt_available=0|1
reset_available=0|1
manual_review_required=1
production_ready=0
```

If local writes are part of a future guarded path, the workflow must distinguish:

```
dry-run artifacts
user-local artifacts
runtime effects
host effects
external effects
```

Failure model

Minimum failure states:

```
missing_rust_toolchain
missing_gui_dependency
unsupported_platform
panel_launch_failed
plan_generation_failed
dry_run_failed
receipt_missing
log_missing
reset_failed
operator_cancelled
manual_review_blocked
```

Every failure state should preserve:

```
no runtime authority
no network authority
no hidden mutation
visible remediation or manual-review note
```

Out-of-scope behavior

This workflow packet does not permit:

```
production installer claim
signed release claim
notarized app claim
Fedora approval claim
Ubuntu package readiness claim
macOS app readiness claim
daily-driver claim
root install claim
runtime authority
effect execution
capability enforcement
cryptographic authority
signing authority
network behavior
model execution
tool execution
shell execution outside documented local commands
security protection claim
malware-prevention claim
ransomware-prevention claim
operating-system replacement claim
```

Estimate-impact answer

```
Does this packet move public product readiness to a new promotion class? no
Does it add a completed user-facing workflow? no
Does it add runtime behavior? no
Does it add effect execution? no
Does it add authority? no
Does it add security-hardening evidence? no
Estimate change recommended: no
Reason: this packet selects and scopes the first workflow path only.
```

Public-entrypoint answer

```
Does this packet change public capability wording? no
Does it require README update? no
Does it require STATUS.md update? no
Does it require docs/status/CURRENT_STATUS.md update? no
Does it require docs/project_notes/CURRENT_DIRECTION.md update? no
Does it require docs/project_notes/UPCOMING_WORK.md update? no
Does it require announcement review? no
Public entrypoint change recommended: no
Reason: no capability posture changes.
```

Next planning action

The next planning action is:

```
draft the Panel-guided local evaluation acceptance checklist
```

That checklist should remain a planning artifact until it is tied to a real guard, transcript, or workflow evidence record.

Non-claims

This packet does not implement product readiness, Panel readiness, installer readiness, runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, security hardening, malware prevention, ransomware prevention, sandboxing, operating-system behavior, or production support.

It records workflow planning criteria only.

docs/strategy/2026-05-25-2157-cdt-panel-guided-local-evaluation-acceptance-checklist.md

Source title: Latticra Panel-Guided Local Evaluation Acceptance Checklist

Lines: 363 | Words: 759 | SHA-256: 65d010cd6afffedc8e960855d4f007e45cd49fdf16dc7fd4a18ebf72ebc2c39f

Latticra Panel-Guided Local Evaluation Acceptance Checklist

Status: draft acceptance checklist Created: 2026-05-25 21:57 CDT Decision: checklist only Promotion decision: no product-readiness promotion recommended Scope: acceptance criteria for future Panel-guided local evaluation evidence.

Purpose

This checklist defines the evidence needed before the Panel-guided local evaluation workflow can be treated as a bounded public-readiness workflow.

It does not claim the evidence exists yet. It defines how future evidence should be reviewed.

Source workflow

This checklist follows:

```
docs/strategy/2026-05-25-2137-cdt-panel-guided-local-evaluation-workflow-packet.md
```

The source workflow is:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Review result labels

Use one result label when applying this checklist:

```
not_started
blocked
partial_evidence
evidence_ready_for_review
accepted_for_estimate_review
rejected
```

This checklist alone cannot produce accepted_for_estimate_review; that result requires actual transcript, guard, or workflow evidence.

Required evaluator boundary

Acceptance requires:

```
target_evaluator_named=1
technical_evaluator_scope=1
nontechnical_user_scope=0
production_operator_scope=0
daily_driver_scope=0
security_reliance_scope=0
```

Required reviewer question:

```
Can the intended evaluator understand that this is a local evaluation workflow, not a production
 product?
```

Required status and non-claim entry

Acceptance requires visible entry into:

```
current project status
current estimate posture
public non-claims
runtime authority denial
effect execution denial
product-readiness limitation
security-hardening limitation
```

Required evidence:

```
status_surface_reference:
non_claim_surface_reference:
evaluator_acknowledgement_or_review_note:
```

Required prerequisite check

Acceptance requires a platform prerequisite check for the evaluated platform.

Minimum prerequisite labels:

```
platform_identified
rust_toolchain_status
cargo_status
make_status
gui_dependency_status
local_path_status
network_required=0
root_required=0
```

Failure states must include:

```
unsupported_platform
missing_rust_toolchain
missing_gui_dependency
manual_review_required
```

Required launch evidence

Acceptance requires evidence that the evaluator can reach the Panel entry point or a documented fallback.

Required evidence:

```
launch_command:
launch_result:
fallback_command:
panel_available=0|1
fallback_available=0|1
```

The launch path must not grant:

```
runtime authority
network authority
root authority
effect authority
```

Required dry-run-first evidence

Acceptance requires the workflow to prove dry-run-first posture.

Required labels:

```
dry_run_enabled=1
dry_run_required_before_guarded_writes=1
runtime_authority_granted=0
effect_execution_performed=0
network_authority_granted=0
root_authority_required=0
production_ready=0
```

```
manual_review_required=1
```

Required reviewer question:

```
Can the evaluator tell, before proceeding, whether any local write is planned?
```

Required plan evidence

Acceptance requires a plan output or plan-equivalent evidence record.

Required evidence:

```
plan_generated=1
plan_reviewable=1
planned_artifact_paths_visible=1
planned_local_writes_visible=1
planned_runtime_effects_visible=1
planned_network_effects_visible=1
denied_authority_visible=1
```

The plan must distinguish:

```
dry-run artifacts
user-local artifacts
runtime effects
host effects
external effects
```

Required dry-run validation evidence

Acceptance requires a dry-run validation transcript or equivalent record.

Required evidence:

```
dry_run_started=1
dry_run_completed=0|1
dry_run_failed=0|1
failure_reason_visible=1
hidden_mutation_detected=0
effect_execution_performed=0
network_behavior_performed=0
runtime_authority_granted=0
```

If the dry-run fails, acceptance requires a visible failure label and remediation or manual-review note.

Required receipt and log evidence

Acceptance requires receipt and log visibility when the workflow says they should exist.

Required evidence:

```
receipt_expected=0|1
receipt_available=0|1
receipt_path_visible=0|1
engine_log_expected=0|1
engine_log_available=0|1
engine_log_path_visible=0|1
embedded_console_status_available=0|1
```

Failure states must include:

```
receipt_missing
log_missing
manual_review_required
```

Required reset or uninstall evidence

Acceptance requires a reset, cleanup, or uninstall path appropriate to the workflow evidence.

Required evidence:

```
reset_available=1
uninstall_or_cleanup_available=1
reset_dry_run_available=1
cleanup_scope_visible=1
```

```
local_artifact_scope_visible=1
reset_failed_label_available=1
```

If no local artifacts are created, the workflow must state:

```
local_artifacts_created=0
cleanup_required=0
```

Required failure-state coverage

Acceptance requires examples or declared handling for:

```
unsupported_platform
missing_rust_toolchain
missing_gui_dependency
panel_launch_failed
plan_generation_failed
dry_run_failed
receipt_missing
log_missing
reset_failed
operator_cancelled
manual_review_blocked
```

Every failure state must preserve:

```
no runtime authority
no network authority
no hidden mutation
visible remediation or manual-review note
```

Required known limitations

Acceptance requires visible limitations:

```
not a production installer
not a daily-driver product
not a security boundary
not malware prevention
not ransomware prevention
not a Fedora-approved package
not an Ubuntu package-readiness claim
not a macOS app-readiness claim
not an operating-system replacement
```

Required support boundary

Acceptance requires a support boundary:

```
supported_evaluator:
supported_platforms_for_this_evidence:
unsupported_platforms:
known_setup_limits:
known_runtime_limits:
where_to_report_documentation_issues:
where_to_report_security_issues:
```

Required evidence bundle

Before estimate review, collect:

```
workflow acceptance checklist result
platform prerequisite record
dry-run transcript
plan output example
receipt or receipt-like evidence
engine log example
embedded console status example
reset or uninstall transcript
failure-state examples
known limitations note
support boundary note
non-claim review
guard or test reference
```

Estimate-impact gate

Estimate review may be considered only if:

```
workflow_completed_by_target_evaluator=1
evidence_bundle_complete=1
guard_or_test_reference_present=1
non_claim_review_present=1
public_entrypoint_review_present=1
```

Checklist result:

```
estimate_change_recommended=no
reason=this checklist defines acceptance criteria only
```

Public-entrypoint gate

Public entrypoint review may be considered only if actual workflow evidence exists.

Checklist result:

```
README_update_required=no
STATUS_update_required=no
CURRENT_STATUS_update_required=no
project_notes_update_required=no
announcement_review_required=no
reason=no capability posture changes in this checklist
```

Non-claims

This checklist does not implement product readiness, Panel readiness, installer readiness, runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, security hardening, malware prevention, ransomware prevention, sandboxing, operating-system behavior, or production support.

It records acceptance criteria only.

docs/strategy/2026-05-25-2226-cdt-panel-guided-local-evaluation-evidence-bundle-template.md

Source title: Latticra Panel-Guided Local Evaluation Evidence Bundle Template

Lines: 350 | Words: 553 | SHA-256: 45a8a547e147801bc0256ccf0f92c0574f255c388cd2a57f8eea0f059576ac5c

Latticra Panel-Guided Local Evaluation Evidence Bundle Template

Status: draft evidence-bundle template Created: 2026-05-25 22:26 CDT Decision: template only Promotion decision: no product-readiness promotion recommended Scope: evidence bundle structure for future Panel-guided local evaluation review.

Purpose

This template defines the evidence bundle that would be required before Panel-guided local evaluation can be reviewed as a bounded public-readiness workflow.

It does not claim any evidence has been captured. It defines the future bundle shape only.

Source checklist

This template follows:

```
docs/strategy/2026-05-25-2157-cdt-panel-guided-local-evaluation-acceptance-checklist.md
```

The source workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Bundle header

```
bundle name:
created:
review status: not_started|blocked|partial_evidence|evidence_ready_for_review|rejected
target evaluator:
evaluated platform:
repository commit:
workflow packet:
acceptance checklist:
review owner:
```

accepted_for_estimate_review must not be used until a reviewer confirms that all required evidence exists and the non-claim, public-entrypoint, and estimate-impact reviews are present.

Evidence manifest

Required manifest fields:

```
status_entry_record:
non_claim_entry_record:
platform_prerequisite_record:
launch_record:
dry_run_first_record:
plan_output_record:
dry_run_validation_record:
receipt_record:
engine_log_record:
embedded_console_record:
reset_or_uninstall_record:
failure_state_record:
known_limitations_record:
support_boundary_record:
non_claim_review_record:
guard_or_test_record:
public_entrypoint_review_record:
estimate_impact_review_record:
```

Each record should be a path, transcript id, status document, or explicit missing label.

Status and non-claim entry record

Required fields:

```
status_surface_reference:
estimate_surface_reference:
non_claim_surface_reference:
runtime_authority_denial_visible=0|1
effect_execution_denial_visible=0|1
product_readiness_limitation_visible=0|1
security_hardening_limitation_visible=0|1
evaluator_review_note:
record_status: missing|partial|complete
```

Platform prerequisite record

Required fields:

```
platform_identified:
platform_version:
rust_toolchain_status:
cargo_status:
make_status:
gui_dependency_status:
local_path_status:
network_required=0
root_required=0
unsupported_platform=0|1
missing_dependency_labels:
manual_review_required=0|1
record_status: missing|partial|complete
```

Launch record

Required fields:

```
launch_command:
launch_result:
fallback_command:
panel_available=0|1
fallback_available=0|1
runtime_authority_granted=0
network_authority_granted=0
root_authority_granted=0
effect_authority_granted=0
record_status: missing|partial|complete
```

Dry-run-first record

Required fields:

```
dry_run_enabled=1
dry_run_required_before_guarded_writes=1
local_write_planned=0|1
local_write_performed=0|1
runtime_authority_granted=0
effect_execution_performed=0
network_authority_granted=0
root_authority_required=0
production_ready=0
manual_review_required=1
record_status: missing|partial|complete
```

Plan output record

Required fields:

```
plan_generated=0|1
plan_reviewable=0|1
planned_artifact_paths_visible=0|1
planned_local_writes_visible=0|1
planned_runtime_effects_visible=0|1
planned_network_effects_visible=0|1
denied_authority_visible=0|1
dry_run_artifacts_identified=0|1
user_local_artifacts_identified=0|1
runtime_effects_identified=0|1
host_effects_identified=0|1
external_effects_identified=0|1
record_status: missing|partial|complete
```

Dry-run validation record

Required fields:

```
dry_run_started=0|1
dry_run_completed=0|1
dry_run_failed=0|1
failure_reason_visible=0|1
hidden_mutation_detected=0
effect_execution_performed=0
network_behavior_performed=0
runtime_authority_granted=0
remediation_or_manual_review_note:
record_status: missing|partial|complete
```

Receipt and log records

Receipt fields:

```
receipt_expected=0|1
receipt_available=0|1
receipt_path_visible=0|1
receipt_review_note:
record_status: missing|partial|complete
```

Engine log fields:

```
engine_log_expected=0|1
engine_log_available=0|1
engine_log_path_visible=0|1
engine_log_review_note:
record_status: missing|partial|complete
```

Embedded console fields:

```
embedded_console_status_available=0|1
embedded_console_status_summary:
record_status: missing|partial|complete
```

Reset or uninstall record

Required fields:

```
reset_available=0|1
uninstall_or_cleanup_available=0|1
reset_dry_run_available=0|1
cleanup_scope_visible=0|1
local_artifact_scope_visible=0|1
local_artifacts_created=0|1
cleanup_required=0|1
reset_failed=0|1
reset_failed_label_available=0|1
record_status: missing|partial|complete
```

Failure-state record

Required failure labels:

```
unsupported_platform:
missing_rust_toolchain:
missing_gui_dependency:
panel_launch_failed:
plan_generation_failed:
dry_run_failed:
receipt_missing:
log_missing:
reset_failed:
operator_cancelled:
manual_review_blocked:
```

For every failure label, record:

```
runtime_authority_granted=0
network_authority_granted=0
hidden_mutation_detected=0
remediation_or_manual_review_note:
```

Known limitations record

Required visible limitations:

```
not_a_production_installer=1
not_a_daily_driver_product=1
not_a_security_boundary=1
not_malware_prevention=1
not_ransomware_prevention=1
not_fedora_approved=1
not_ubuntu_package_readiness=1
not_macos_app_readiness=1
not_operating_system_replacement=1
record_status: missing|partial|complete
```

Support boundary record

Required fields:

```
supported_evaluator:
supported_platforms_for_this_evidence:
unsupported_platforms:
known_setup_limits:
known_runtime_limits:
documentation_issue_path:
security_issue_path:
record_status: missing|partial|complete
```

Review gate records

Non-claim review:

```
non_claim_review_present=0|1
non_claim_review_path:
```

```
claim_expansion_detected=0|1
```

Guard or test reference:

```
guard_or_test_reference_present=0|1
guard_or_test_path:
guard_or_test_result:
```

Public-entrypoint review:

```
public_entrypoint_review_present=0|1
README_update_required=0|1
STATUS_update_required=0|1
CURRENT_STATUS_update_required=0|1
project_notes_update_required=0|1
announcement_review_required=0|1
```

Estimate-impact review:

```
estimate_impact_review_present=0|1
estimate_change_recommended=0|1
reason:
```

Bundle completeness rule

The bundle is complete only if:

```
all_required_records_present=1
all_required_records_complete=1
non_claim_review_present=1
guard_or_test_reference_present=1
public_entrypoint_review_present=1
estimate_impact_review_present=1
claim_expansion_detected=0
```

This template result:

```
bundle_complete=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reason=this file defines the future bundle shape only
```

Non-claims

This template does not provide workflow evidence, product readiness, Panel readiness, installer readiness, runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, security hardening, malware prevention, ransomware prevention, sandboxing, operating-system behavior, or production support.

It records an evidence-bundle structure only.

docs/strategy/2026-05-25-2246-cdt-panel-guided-local-evaluation-evidence-capture-plan.md

Source title: Latticra Panel-Guided Local Evaluation Evidence Capture Plan

Lines: 394 | Words: 894 | SHA-256: bba5d5b72426fd075b0d5cba39e752b97d94bf711ee2d844f530e1287edab680

Latticra Panel-Guided Local Evaluation Evidence Capture Plan

Status: draft evidence-capture plan Created: 2026-05-25 22:46 CDT Decision: capture plan only Promotion decision: no product-readiness promotion recommended Scope: future evidence-capture sequence for Panel-guided local evaluation.

Purpose

This plan defines how future evidence should be captured for the Panel-guided local evaluation evidence bundle.

It does not capture evidence, run commands, implement guards, or change public product-readiness posture.

Source bundle

This plan follows:

```
docs/strategy/2026-05-25-2226-cdt-panel-guided-local-evaluation-evidence-bundle-template.md
```

The target workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Capture posture

The capture sequence must remain:

```
local
dry-run-first
operator-visible
no-root by default
no-network by default
no runtime authority
no effect execution
no product-readiness claim
```

If any step requires root, network access, runtime authority, hidden mutation, or unreviewed local writes, the capture must stop and be marked blocked.

Capture roles

Required roles:

```
capture operator:
review owner:
non-claim reviewer:
public-entrypoint reviewer:
estimate-impact reviewer:
```

The same person may hold more than one role, but the evidence bundle must name which role performed each review.

Capture phases

Use this phase order:

```
phase_0_scope_lock
phase_1_status_and_non_claim_entry
phase_2_platform_prerequisite_capture
phase_3_panel_launch_capture
phase_4_dry_run_first_capture
phase_5_plan_capture
phase_6_dry_run_validation_capture
phase_7_receipt_log_console_capture
phase_8_reset_or_uninstall_capture
phase_9_failure_state_capture
phase_10_limitations_and_support_capture
phase_11_review_gate_capture
phase_12_bundle_completeness_review
```

Do not skip phases. A skipped phase must be recorded as missing or blocked.

Phase 0: Scope Lock

Capture before running any workflow step:

```
repository commit:
target evaluator:
evaluated platform:
workflow packet path:
acceptance checklist path:
evidence bundle template path:
capture plan path:
```

Required scope assertions:

```
product_readiness_promotion_requested=0
estimate_change_requested=0
runtime_authority_requested=0
network_authority_requested=0
root_authority_requested=0
effect_execution_requested=0
```

Stop if any assertion cannot be preserved.

Phase 1: Status And Non-Claim Entry

Capture:

```
status_surface_reference
estimate_surface_reference
non_claim_surface_reference
runtime_authority_denial_visible
effect_execution_denial_visible
product_readiness_limitation_visible
security_hardening_limitation_visible
```

Stop if the evaluator cannot find current status or non-claims before launch.

Phase 2: Platform Prerequisite Capture

Capture:

```
platform identity
Rust toolchain status
Cargo status
Make status
GUI dependency status
local path status
network requirement status
root requirement status
```

Allowed result labels:

```
complete
partial
blocked
unsupported_platform
missing_dependency
```

Stop if prerequisites imply network or root authority that was not already documented as optional and outside the no-effect workflow.

Phase 3: Panel Launch Capture

Capture:

```
launch command
launch result
fallback command
panel availability
fallback availability
authority labels after launch
```

Required authority labels:

```
runtime_authority_granted=0
network_authority_granted=0
root_authority_granted=0
effect_authority_granted=0
```

Stop if launch behavior grants authority or performs hidden mutation.

Phase 4: Dry-Run-First Capture

Capture:

```
dry_run_enabled
dry_run_required_before_guarded_writes
local_write_planned
local_write_performed
manual_review_required
production_ready
```

Required labels:

```
dry_run_enabled=1
runtime_authority_granted=0
effect_execution_performed=0
network_authority_granted=0
root_authority_required=0
production_ready=0
manual_review_required=1
```

Stop if dry-run posture is ambiguous.

Phase 5: Plan Capture

Capture plan output or plan-equivalent evidence.

The plan must expose:

```
planned artifact paths
planned local writes
planned runtime effects
planned network effects
denied authority
dry-run artifacts
user-local artifacts
runtime effects
host effects
external effects
```

Stop if the evaluator cannot distinguish dry-run artifacts from local writes, runtime effects, host effects, or external effects.

Phase 6: Dry-Run Validation Capture

Capture:

```
dry_run_started
dry_run_completed
dry_run_failed
failure_reason_visible
hidden_mutation_detected
effect_execution_performed
network_behavior_performed
runtime_authority_granted
remediation_or_manual_review_note
```

Required preservation labels:

```
hidden_mutation_detected=0
effect_execution_performed=0
network_behavior_performed=0
runtime_authority_granted=0
```

Stop if any preservation label changes.

Phase 7: Receipt, Log, And Console Capture

Capture:

```
receipt expectation
receipt availability
receipt path visibility
engine log expectation
engine log availability
engine log path visibility
embedded console status availability
embedded console status summary
```

If any expected record is missing, mark the bundle partial_evidence or blocked, not complete.

Phase 8: Reset Or Uninstall Capture

Capture:

```
reset availability
uninstall or cleanup availability
```

```
reset dry-run availability
cleanup scope visibility
local artifact scope visibility
local artifacts created
cleanup required
reset failure label availability
```

Stop if local artifacts are created but no cleanup or reset boundary is visible.

Phase 9: Failure-State Capture

Capture declared handling for:

```
unsupported_platform
missing_rust_toolchain
missing_gui_dependency
panel_launch_failed
plan_generation_failed
dry_run_failed
receipt_missing
log_missing
reset_failed
operator_cancelled
manual_review_blocked
```

Each failure state must preserve:

```
runtime_authority_granted=0
network_authority_granted=0
hidden_mutation_detected=0
remediation_or_manual_review_note_present=1
```

Phase 10: Limitations And Support Capture

Capture visible limitations:

```
not a production installer
not a daily-driver product
not a security boundary
not malware prevention
not ransomware prevention
not Fedora approved
not Ubuntu package readiness
not macOS app readiness
not an operating-system replacement
```

Capture support boundary:

```
supported evaluator
supported platforms for this evidence
unsupported platforms
known setup limits
known runtime limits
documentation issue path
security issue path
```

Stop if limitations are hidden, softened into product claims, or contradicted by the workflow.

Phase 11: Review Gate Capture

Required reviews:

```
non-claim review
guard or test reference
public-entrypoint review
estimate-impact review
```

Review constraints:

```
non_claim_review_present=1
guard_or_test_reference_present=1
public_entrypoint_review_present=1
estimate_impact_review_present=1
claim_expansion_detected=0
```

Do not recommend estimate or public-entrypoint changes from capture evidence alone. Those reviews must explain the decision separately.

Phase 12: Bundle Completeness Review

The bundle may be marked evidence_ready_for_review only when:

```
all_required_records_present=1
all_required_records_complete=1
all_stop_conditions_cleared=1
non_claim_review_present=1
guard_or_test_reference_present=1
public_entrypoint_review_present=1
estimate_impact_review_present=1
claim_expansion_detected=0
```

This plan result:

```
evidence_captured=0
bundle_complete=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reason=this file defines future capture sequence only
```

Non-claims

This plan does not capture evidence, implement a guard, validate the Panel workflow, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records an evidence-capture sequence only.

docs/strategy/2026-05-26-0019-cdt-panel-guided-local-evaluation-non-claim-review-template.md

Source title: Latticra Panel-Guided Local Evaluation Non-Claim Review Template

Lines: 293 | Words: 710 | SHA-256: c40f51f569956c36ee7f1062bbfacceeb6caa8607acda64a1c0b41ea98d6a979

Latticra Panel-Guided Local Evaluation Non-Claim Review Template

Status: draft non-claim review template Created: 2026-05-26 00:19 CDT Decision: review template only Promotion decision: no product-readiness promotion recommended Scope: non-claim review for future Panel-guided local evaluation evidence.

Purpose

This template defines the non-claim review required before any Panel-guided local evaluation evidence can be used in public product-readiness, public-entrypoint, or estimate-impact discussions.

It does not perform the review. It defines the review shape only.

Source plan

This template follows:

```
docs/strategy/2026-05-25-2246-cdt-panel-guided-local-evaluation-evidence-capture-plan.md
```

The target workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Review header

```
review name:
created:
reviewer:
review status: not_started|blocked|partial|complete|rejected
workflow evidence bundle:
capture plan:
evaluated platform:
repository commit:
```

Required review outcome

Use one outcome:

```
claim_safe_for_current_scope
claim_safe_with_required_edits
claim_unsafe_blocked
insufficient_evidence
```

This template result:

```
review_performed=0
claim_expansion_detected=0
product_readiness_promotion_recommended=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reason=this file defines the future review shape only
```

Required source checks

The reviewer must inspect:

```
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/README.md
docs/strategy/2026-05-25-2114-cdt-public-product-readiness-promotion-packet.md
docs/strategy/2026-05-25-2137-cdt-panel-guided-local-evaluation-workflow-packet.md
docs/strategy/2026-05-25-2157-cdt-panel-guided-local-evaluation-acceptance-checklist.md
docs/strategy/2026-05-25-2226-cdt-panel-guided-local-evaluation-evidence-bundle-template.md
docs/strategy/2026-05-25-2246-cdt-panel-guided-local-evaluation-evidence-capture-plan.md
```

If any source is missing, mark the review blocked or insufficient_evidence.

Claims that must remain false

The review must confirm the workflow does not claim:

```
production_ready=0
daily_driver_ready=0
security_boundary=0
malware_prevention=0
ransomware_prevention=0
sandbox_provided=0
operating_system_replacement=0
runtime_authority_granted=0
effect_execution_performed=0
capability_enforcement_performed=0
cryptographic_authority_granted=0
signing_authority_granted=0
network_authority_granted=0
root_authority_required_by_default=0
model_execution_performed=0
tool_execution_performed=0
hidden_mutation_performed=0
```

Any true value in this section blocks the review unless a separate contract, implementation plan, guard, evidence record, public-entrypoint review, estimate-impact review, and non-claim update exist.

Product-readiness wording check

Allowed wording:

```
bounded local evaluation workflow
technical evaluator workflow
dry-run-first local workbench review
operator-visible evidence review
planning evidence
future evidence capture
no product-readiness promotion recommended
```

Blocked wording:

```
ready for users
production ready
daily driver
installer ready
secure product
protected workflow
safe installer
approved package
supported release
```

If blocked wording appears, the review outcome must be claim_safe_with_required_edits or claim_unsafe_blocked.

Security wording check

Allowed wording:

```
security-conscious design
defensive planning
non-claim review
threat-model-aware planning
no security boundary claimed
```

Blocked wording:

```
prevents malware
prevents ransomware
secure sandbox
hardened runtime
security boundary
verified protection
production cryptographic trust
```

Any security-protection wording requires separate enforcement evidence before it can be considered.

Installer and platform wording check

Allowed wording:

```
local evaluation
source-run workflow
dry-run path
user-local planning
platform prerequisite check
```

Blocked wording:

```
production installer
signed release
notarized app
Fedora approved
Ubuntu package ready
macOS app ready
LaunchAgent integration
privileged helper
system extension
network extension
```

Platform-specific readiness claims are blocked unless a separate platform evidence packet exists.

Authority wording check

The review must confirm that the workflow preserves:

```
runtime_authority_granted=0
effect_execution_performed=0
```

```
capability_enforcement_performed=0
network_authority_granted=0
root_authority_required_by_default=0
host_effect_performed=0
external_effect_performed=0
```

Any authority-bearing wording must be rejected unless it belongs to a separate contracted and evidenced lane.

Evidence wording check

Allowed wording:

```
template
plan
checklist
future evidence
missing evidence
partial evidence
evidence ready for review
```

Blocked wording:

```
evidence complete
workflow accepted
product readiness improved
estimate increase justified
public posture changed
```

Blocked evidence wording may become allowed only after the evidence bundle is complete and separately reviewed.

Required reviewer answers

```
Does the workflow preserve no product-readiness promotion? yes|no
Does the workflow preserve no security claim? yes|no
Does the workflow preserve no runtime authority? yes|no
Does the workflow preserve no effect execution? yes|no
Does the workflow preserve no hidden mutation? yes|no
Does the workflow preserve no network authority? yes|no
Does the workflow preserve no platform-readiness claim? yes|no
Does the workflow preserve no estimate change? yes|no
Does the workflow preserve no public-entrypoint change? yes|no
```

All answers must be yes for claim_safe_for_current_scope.

Required edit log

If edits are required, record:

```
source file:
claim text:
claim risk:
required replacement:
reviewer note:
```

Review gate output

```
non_claim_review_present=0|1
claim_expansion_detected=0|1
required_edits_present=0|1
blocking_claims_present=0|1
public_entrypoint_review_required=0|1
estimate_impact_review_required=0|1
review_outcome:
```

This template output:

```
non_claim_review_present=0
claim_expansion_detected=0
required_edits_present=0
blocking_claims_present=0
public_entrypoint_review_required=0
estimate_impact_review_required=0
review_outcome=not_performed
```

Non-claims

This template does not perform a non-claim review, validate workflow evidence, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a non-claim review form only.

docs/strategy/2026-05-26-0034-cdt-panel-guided-local-evaluation-public-entrypoint-review-template.md

Source title: Latticra Panel-Guided Local Evaluation Public-Entrypoint Review Template

Lines: 275 | Words: 804 | SHA-256: afa41baa6f2f857cac8a1be0fe9d8a09e5f4353d0384ea2bded11b44cbc591e4

Latticra Panel-Guided Local Evaluation Public-Entrypoint Review Template

Status: draft public-entrypoint review template Created: 2026-05-26 00:34 CDT Decision: review template only Promotion decision: no product-readiness promotion recommended Scope: public-entrypoint review for future Panel-guided local evaluation evidence.

Purpose

This template defines when future Panel-guided local evaluation evidence may justify updates to public reader-facing entry points.

It does not perform the review, update public entry points, or change project posture.

Source review

This template follows:

```
docs/strategy/2026-05-26-0019-cdt-panel-guided-local-evaluation-non-claim-review-template.md
```

The target workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Review header

```
review name:
created:
reviewer:
review status: not_started|blocked|partial|complete|rejected
workflow evidence bundle:
non-claim review:
estimate-impact review:
repository commit:
```

Required review outcome

Use one outcome:

```
no_public_entrypoint_change
public_entrypoint_change_allowed_with_edits
public_entrypoint_change_blocked
insufficient_evidence
```

This template result:

```
review_performed=0
public_entrypoint_change_recommended=0
README_update_required=0
STATUS_update_required=0
```

```
CURRENT_STATUS_update_required=0
project_notes_update_required=0
announcement_review_required=0
reason=this file defines the future review shape only
```

Review prerequisites

Public-entrypoint review is blocked unless:

```
workflow_evidence_bundle_present=1
acceptance_checklist_result_present=1
non_claim_review_present=1
claim_expansion_detected=0
guard_or_test_reference_present=1
estimate_impact_review_present=1
```

If any prerequisite is missing, use:

```
review_outcome=insufficient_evidence
public_entrypoint_change_recommended=0
```

Entry points under review

Review only these public surfaces for this workflow:

```
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/README.md
```

Do not update broader foundation docs, announcement logs, package docs, installer docs, or platform docs from this review unless a separate review explicitly names them.

Allowed public-entrypoint changes

Allowed only after prerequisites pass:

```
add pointer to completed evidence bundle
add pointer to non-claim review
add pointer to estimate-impact review
clarify that Panel-guided local evaluation evidence exists
clarify limitations for technical evaluators
clarify no product-readiness promotion if estimate remains unchanged
```

Allowed wording:

```
Panel-guided local evaluation evidence bundle
technical evaluator workflow evidence
dry-run-first local evaluation evidence
operator-visible local evaluation record
no product-readiness promotion from this evidence alone
```

Blocked public-entrypoint changes

Blocked unless a separate capability promotion has been accepted:

```
increase public product-readiness estimate
claim production readiness
claim installer readiness
claim platform package readiness
claim daily-driver readiness
claim security protection
claim malware prevention
claim ransomware prevention
claim sandboxing
claim operating-system replacement
claim runtime authority
claim effect execution
claim capability enforcement
claim cryptographic trust
claim signing authority
```

If any blocked change is requested, the review outcome must be:

```
public_entrypoint_change_blocked
```

README review questions

```
Does README need a new link? yes|no
Does README wording stay bounded to local evaluation evidence? yes|no
Does README avoid product-readiness promotion? yes|no
Does README avoid security or installer claims? yes|no
Does README preserve current estimate wording unless estimate review changes it? yes|no
```

README update is allowed only if every preservation answer is yes.

STATUS.md review questions

```
Does STATUS.md need a new latest-note line? yes|no
Does STATUS.md need estimate table changes? yes|no
Does STATUS.md preserve planning-estimate caveats? yes|no
Does STATUS.md avoid capability posture promotion? yes|no
Does STATUS.md point to the evidence bundle only as evidence? yes|no
```

Estimate table changes require a separate estimate-impact review with estimate_change_recommended=1.

CURRENT_STATUS review questions

```
Does CURRENT_STATUS need a detailed evidence note? yes|no
Does CURRENT_STATUS preserve no runtime authority? yes|no
Does CURRENT_STATUS preserve no effect execution? yes|no
Does CURRENT_STATUS preserve no security-hardening promotion? yes|no
Does CURRENT_STATUS preserve no product-readiness promotion unless separately reviewed? yes|no
```

Project-notes review questions

```
Does CURRENT_DIRECTION need narrative alignment? yes|no
Does UPCOMING_WORK need queue alignment? yes|no
Does the current next-step rule remain unchanged? yes|no
Does the workflow evidence change near-term strategy? yes|no
```

If the current next-step rule changes, a strategy posture review is required.

Strategy index review questions

```
Does docs/strategy/README.md need only a link update? yes|no
Does the new entry remain dated and reviewable? yes|no
Does the index avoid implying capability promotion? yes|no
```

Strategy index link updates are allowed when a dated review record exists, even if no capability posture changes.

Announcement review trigger

Announcement review is required only if:

```
public capability posture changed=1
product_readiness_estimate_changed=1
security_hardening_posture_changed=1
runtime_authority_changed=1
user-facing workflow accepted for public posture=1
```

This template result:

```
announcement_review_required=0
reason=no evidence or posture change in this template
```

Required edit log

If public-entrypoint edits are allowed, record:

```
entrypoint:
edit type:
source evidence:
non-claim review:
estimate-impact review:
old wording:
new wording:
reviewer note:
```

Review gate output

```
public_entrypoint_review_present=0|1
public_entrypoint_change_recommended=0|1
README_update_required=0|1
STATUS_update_required=0|1
CURRENT_STATUS_update_required=0|1
project_notes_update_required=0|1
strategy_index_update_required=0|1
announcement_review_required=0|1
blocked_claim_detected=0|1
review_outcome:
```

This template output:

```
public_entrypoint_review_present=0
public_entrypoint_change_recommended=0
README_update_required=0
STATUS_update_required=0
CURRENT_STATUS_update_required=0
project_notes_update_required=0
strategy_index_update_required=0
announcement_review_required=0
blocked_claim_detected=0
review_outcome=not_performed
```

Non-claims

This template does not perform a public-entrypoint review, update README, update status files, update project notes, create an announcement, validate workflow evidence, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a public-entrypoint review form only.

docs/strategy/2026-05-26-0119-cdt-panel-guided-local-evaluation-estimate-impact-review-template.md

Source title: Latticra Panel-Guided Local Evaluation Estimate-Impact Review Template

Lines: 290 | Words: 842 | SHA-256: 33065a753d21323a8e02ee13fdcbdb77f9a684a4886f2cdac08c9f256559b034

Latticra Panel-Guided Local Evaluation Estimate-Impact Review Template

Status: draft estimate-impact review template Created: 2026-05-26 01:19 CDT Decision: review template only Promotion decision: no product-readiness promotion recommended Scope: estimate-impact review for future Panel-guided local evaluation evidence.

Purpose

This template defines how future Panel-guided local evaluation evidence should be reviewed before any completion estimate changes are considered.

It does not perform the review, change estimates, or change public product-readiness posture.

Source review

This template follows:

```
docs/strategy/2026-05-26-0034-cdt-panel-guided-local-evaluation-public-entrypoint-review-templat
e.md
```

The target workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Review header

```
review name:
created:
reviewer:
review status: not_started|blocked|partial|complete|rejected
workflow evidence bundle:
acceptance checklist result:
non-claim review:
public-entrypoint review:
guard or test reference:
repository commit:
```

Required review outcome

Use one outcome:

```
no_estimate_change
estimate_change_allowed_with_public_entrypoint_review
estimate_change_blocked
insufficient_evidence
```

This template result:

```
review_performed=0
estimate_change_recommended=0
product_readiness_estimate_change_recommended=0
panel_estimate_change_recommended=0
public_documentation_estimate_change_recommended=0
strategy_status_funding_estimate_change_recommended=0
reason=this file defines the future review shape only
```

Review prerequisites

Estimate-impact review is blocked unless:

```
workflow_evidence_bundle_present=1
acceptance_checklist_result_present=1
acceptance_checklist_result=evidence_ready_for_review
non_claim_review_present=1
claim_expansion_detected=0
public_entrypoint_review_present=1
guard_or_test_reference_present=1
```

If any prerequisite is missing, use:

```
review_outcome=insufficient_evidence
estimate_change_recommended=0
```

Baseline estimate record

Record current estimates at review time:

```
baseline_record_date:
baseline_source_files:
overall_latticra_system:
latticra_panel_local_control_surface:
public_product_readiness:
public_documentation_posture:
strategy_status_funding_framework:
security_hardening_implementation:
runtime_operating_system_universe_direction:
```

Do not rely on estimates copied into older strategy templates. The reviewer must read the current public estimate table at review time.

Capability-posture questions

Answer before any estimate movement:

```
Does the evidence add a completed user-facing workflow? yes|no
Does the evidence show a target evaluator completed the workflow? yes|no
Does the evidence include dry-run-first proof? yes|no
Does the evidence include plan, receipt, log, and reset visibility? yes|no
```

```
Does the evidence include failure-state coverage? yes|no
Does the evidence include a support boundary? yes|no
Does the evidence include a guard or test reference? yes|no
Does the evidence preserve no runtime authority? yes|no
Does the evidence preserve no effect execution? yes|no
Does the evidence preserve no network authority? yes|no
Does the evidence preserve no product-readiness overclaim? yes|no
```

All answers must support the proposed estimate decision.

Estimate lanes under review

Allowed lanes for this workflow:

```
Latticra Panel / local control surface
Public product readiness
Public documentation posture
Strategy/status/funding framework
```

Blocked lanes for this workflow unless a separate evidence packet exists:

```
Runtime / operating-system-universe direction
Security-hardening implementation
Nucleus real task execution
Latticra Seal / local evidence layer
Nadia offline AI foundation
Lat / Latticra Programming Language
LIR / Intermediate Representation
```

Product-readiness estimate rule

Product-readiness estimate movement is blocked unless:

```
target_evaluator_completed_workflow=1
evidence_bundle_complete=1
non_claim_review_claim_safe=1
public_entrypoint_review_complete=1
support_boundary_present=1
known_limitations_present=1
guard_or_test_reference_present=1
```

Even when all requirements pass, product-readiness movement must remain conservative and must not imply production readiness.

Panel estimate rule

Panel local-control-surface estimate movement may be considered only if:

```
Panel launch evidence complete=1
dry-run-first evidence complete=1
plan evidence complete=1
receipt/log/console evidence complete=1
reset or uninstall evidence complete=1
failure-state evidence complete=1
```

Panel estimate movement must not imply installer readiness, platform approval, runtime authority, or production support.

Public documentation estimate rule

Public documentation posture movement may be considered only if:

```
workflow docs are complete=1
non-claims are visible=1
entrypoints are aligned=1
support boundary is visible=1
known limitations are visible=1
```

Documentation estimate movement must not be used as a substitute for product-readiness evidence.

Strategy/status/funding estimate rule

Strategy/status/funding framework movement may be considered only if:

```
strategy chain is complete=1
```

```
status surfaces are aligned=1
public-entrypoint review is complete=1
estimate-impact review is complete=1
no announcement mismatch exists=1
```

This lane must not move simply because a template was added.

Estimate-change proposal fields

If any estimate movement is proposed, record:

```
lane:
baseline estimate:
proposed estimate:
delta:
evidence basis:
non-claim review:
public-entrypoint review:
guard or test reference:
residual risk:
reviewer note:
```

Every proposed change must have a separate lane entry.

Blocked estimate changes

Block estimate changes when:

```
evidence bundle incomplete=1
non-claim review missing=1
claim expansion detected=1
guard or test reference missing=1
public-entrypoint review missing=1
workflow not completed by target evaluator=1
runtime authority changed=1
effect execution changed=1
security-hardening claim introduced=1
product-readiness wording overclaims=1
```

Required reviewer answers

```
Does this evidence change capability posture? yes|no
Does this evidence justify any estimate change? yes|no
Does this evidence require public-entrypoint updates? yes|no
Does this evidence require announcement review? yes|no
Does this evidence preserve all non-claims? yes|no
Does this evidence preserve current runtime authority posture? yes|no
Does this evidence preserve current security-hardening posture? yes|no
```

If any answer is ambiguous, use insufficient_evidence.

Review gate output

```
estimate_impact_review_present=0|1
estimate_change_recommended=0|1
product_readiness_estimate_change_recommended=0|1
panel_estimate_change_recommended=0|1
public_documentation_estimate_change_recommended=0|1
strategy_status_funding_estimate_change_recommended=0|1
announcement_review_required=0|1
public_entrypoint_update_required=0|1
review_outcome:
```

This template output:

```
estimate_impact_review_present=0
estimate_change_recommended=0
product_readiness_estimate_change_recommended=0
panel_estimate_change_recommended=0
public_documentation_estimate_change_recommended=0
strategy_status_funding_estimate_change_recommended=0
announcement_review_required=0
public_entrypoint_update_required=0
review_outcome=not_performed
```

Non-claims

This template does not perform an estimate-impact review, change estimates, update public entry points, validate workflow evidence, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records an estimate-impact review form only.

docs/strategy/2026-05-26-0306-cdt-panel-guided-local-evaluation-guard-test-reference-template.md

Source title: Latticra Panel-Guided Local Evaluation Guard/Test Reference Template

Lines: 251 | Words: 616 | SHA-256: 3b171ec2f3e88de4c4f5483c890d7728669a3e5c25ef7dfdb9b1c76c800b7a18

Latticra Panel-Guided Local Evaluation Guard/Test Reference Template

Status: draft guard/test reference template Created: 2026-05-26 03:06 CDT Decision: reference template only Promotion decision: no product-readiness promotion recommended Scope: guard or test reference requirements for future Panel-guided local evaluation evidence.

Purpose

This template defines how a future guard or test reference should be recorded for Panel-guided local evaluation evidence.

It does not implement a guard, run a test, validate a workflow, or change project posture.

Source review

This template follows:

```
docs/strategy/2026-05-26-0119-cdt-panel-guided-local-evaluation-estimate-impact-review-template.
md
```

The target workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Reference header

```
reference name:
created:
reviewer:
reference status: missing|planned|present|ran_passed|ran_failed|blocked
guard or test path:
guard or test type: shell|unit|integration|manual-transcript|workflow-harness|other
workflow evidence bundle:
acceptance checklist:
repository commit:
```

Required reference outcome

Use one outcome:

```
no_guard_or_test_reference
guard_or_test_reference_planned
guard_or_test_reference_present_not_run
guard_or_test_reference_ran_passed
guard_or_test_reference_ran_failed
guard_or_test_reference_blocked
```

This template result:

```
guard_or_test_reference_present=0
guard_or_test_ran=0
guard_or_test_passed=0
workflow_validated=0
estimate_change_recommended=0
reason=this file defines the future reference shape only
```

Required reference fields

```
guard_or_test_name:
guard_or_test_path:
guard_or_test_owner:
guard_or_test_scope:
guard_or_test_inputs:
guard_or_test_outputs:
guard_or_test_result:
guard_or_test_timestamp:
guard_or_test_environment:
guard_or_test_log:
```

If any required field is missing, the reference is incomplete.

Required coverage claims

A future guard or test reference must say whether it covers:

```
status_and_non_claim_entry=0|1
platform_prerequisite_capture=0|1
panel_launch_capture=0|1
dry_run_first_capture=0|1
plan_capture=0|1
dry_run_validation_capture=0|1
receipt_log_console_capture=0|1
reset_or_uninstall_capture=0|1
failure_state_capture=0|1
limitations_and_support_capture=0|1
review_gate_capture=0|1
```

Coverage must be explicit. Do not infer coverage from a broad test name.

Required preservation assertions

The guard or test reference must preserve:

```
runtime_authority_granted=0
effect_execution_performed=0
network_authority_granted=0
root_authority_required_by_default=0
hidden_mutation_detected=0
product_readiness_promotion=0
security_boundary_claim=0
installer_readiness_claim=0
```

If any assertion cannot be checked, record it as unchecked, not passed.

Allowed reference types

Allowed reference types:

```
existing guard script
new planned guard script
manual transcript review
workflow harness
focused integration check
documentation consistency check
status/index consistency check
```

Allowed only as planning references until they exist and are run:

```
planned guard
planned transcript
planned workflow harness
```

Blocked reference types

Blocked for this workflow unless a separate contract exists:

```
root-requiring test
network-requiring test
runtime-authority test
effect-execution test
host-mutation test
production installer test
security hardening test
malware-prevention test
ransomware-prevention test
platform package approval test
```

If a blocked reference type is needed, this workflow must be superseded by a separate capability-promotion packet.

Minimum pass evidence

A future passing guard or test reference must provide:

```
command or transcript id:
exit status or review result:
captured output path:
reviewed evidence bundle:
checked preservation assertions:
failure-state coverage result:
non-claim review link:
public-entrypoint review link:
estimate-impact review link:
```

Do not mark guard_or_test_reference_ran_passed without captured output or transcript evidence.

Failure handling

If the guard or test fails, record:

```
failure label:
failed phase:
failed assertion:
output path:
remediation note:
workflow blocked=1
estimate_change_recommended=0
public_entrypoint_change_recommended=0
```

Failure must not be reframed as product-readiness evidence.

Manual transcript reference rule

Manual transcript evidence may be referenced only if it includes:

```
operator:
date:
platform:
commands or UI steps:
observed outputs:
authority labels:
failure or success labels:
reset or cleanup notes:
reviewer signature or note:
```

Manual transcript evidence must still pass non-claim, public-entrypoint, and estimate-impact review before it affects posture.

Review gate output

```
guard_or_test_reference_present=0|1
guard_or_test_reference_complete=0|1
guard_or_test_ran=0|1
guard_or_test_passed=0|1
coverage_complete=0|1
preservation_assertions_complete=0|1
blocked_reference_type_detected=0|1
workflow_blocked=0|1
estimate_change_recommended=0|1
public_entrypoint_change_recommended=0|1
reference_outcome:
```

This template output:

```
guard_or_test_reference_present=0
guard_or_test_reference_complete=0
guard_or_test_ran=0
guard_or_test_passed=0
coverage_complete=0
preservation_assertions_complete=0
blocked_reference_type_detected=0
workflow_blocked=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reference_outcome=not_performed
```

Non-claims

This template does not implement a guard, run a test, validate workflow evidence, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a guard/test reference form only.

docs/strategy/2026-05-26-0410-cdt-panel-guided-local-evaluation-review-package-index.md

Source title: Latticra Panel-Guided Local Evaluation Review Package Index

Lines: 172 | Words: 622 | SHA-256: 3d1abe5be979db321538ea9d148ea9ec277d2c14dff17c5349f3c5778031fef4

Latticra Panel-Guided Local Evaluation Review Package Index

Status: active review-package index Created: 2026-05-26 04:10 CDT Decision: package index only Promotion decision: no product-readiness promotion recommended Scope: strategy package map for Panel-guided local evaluation planning records.

Purpose

This index organizes the Panel-guided local evaluation planning package.

It shows the intended reading order, dependency order, and review gates for future evidence work. It does not capture evidence, perform reviews, implement tests, or change project posture.

Package goal

The package goal is:

```
define how Panel-guided local evaluation could eventually be reviewed as a bounded
technical-evaluator workflow without granting runtime authority or making product-readiness
claims
```

Package order

Read and apply the records in this order:

1. 2026-05-25-2114-cdt-public-product-readiness-promotion-packet.md
2. 2026-05-25-2137-cdt-panel-guided-local-evaluation-workflow-packet.md
3. 2026-05-25-2157-cdt-panel-guided-local-evaluation-acceptance-checklist.md
4. 2026-05-25-2226-cdt-panel-guided-local-evaluation-evidence-bundle-template.md
5. 2026-05-25-2246-cdt-panel-guided-local-evaluation-evidence-capture-plan.md
6. 2026-05-26-0306-cdt-panel-guided-local-evaluation-guard-test-reference-template.md
7. 2026-05-26-0019-cdt-panel-guided-local-evaluation-non-claim-review-template.md
8. 2026-05-26-0034-cdt-panel-guided-local-evaluation-public-entrypoint-review-template.md
9. 2026-05-26-0119-cdt-panel-guided-local-evaluation-estimate-impact-review-template.md

Dependency map

```
product-readiness packet
  -&gt; workflow packet
    -&gt; acceptance checklist
      -&gt; evidence bundle template
        -&gt; evidence capture plan
          -&gt; guard/test reference
          -&gt; non-claim review
            -&gt; public-entrypoint review
              -&gt; estimate-impact review
```

The guard/test reference, non-claim review, public-entrypoint review, and estimate-impact review are all required before any estimate or public-entrypoint change can be considered.

Record responsibilities

```
public product-readiness promotion packet
  Defines why no product-readiness promotion is currently recommended.

workflow packet
  Selects Panel-guided local evaluation as the first planning path.

acceptance checklist
  Defines what future evidence must satisfy.

evidence bundle template
  Defines the shape of the evidence bundle.

evidence capture plan
  Defines the future capture sequence and stop conditions.

guard/test reference template
  Defines how future guard, test, harness, or transcript evidence should be cited.

non-claim review template
  Defines how to prevent product, security, installer, runtime, or platform claims.

public-entrypoint review template
  Defines when README, status, project notes, or strategy index updates may be justified.

estimate-impact review template
  Defines when completion estimate changes may be considered.
```

Current package status

Current status:

```
planning_package_defined=1
workflow_selected=1
acceptance_criteria_defined=1
evidence_bundle_shape_defined=1
capture_plan_defined=1
```

```
guard_test_reference_shape_defined=1
non_claim_review_shape_defined=1
public_entrypoint_review_shape_defined=1
estimate_impact_review_shape_defined=1
workflow_evidence_captured=0
guard_or_test_implemented=0
guard_or_test_run=0
non_claim_review_performed=0
public_entrypoint_review_performed=0
estimate_impact_review_performed=0
product_readiness_promotion_recommended=0
```

Required future promotion sequence

Any future promotion review must proceed in this order:

1. Capture evidence according to the evidence capture plan.
2. Complete the evidence bundle.
3. Attach a valid guard/test reference.
4. Complete the non-claim review.
5. Complete the public-entrypoint review.
6. Complete the estimate-impact review.
7. Decide whether public posture changes are justified.

Do not change estimates or public posture before step 7.

Stop conditions

Stop the package review if any of these appear:

```
root authority required by default
network authority required by default
runtime authority granted
effect execution performed
hidden mutation detected
cleanup boundary missing
non-claim review missing
guard/test reference missing
public-entrypoint review missing
estimate-impact review missing
product-readiness claim introduced
security boundary claim introduced
installer readiness claim introduced
platform approval claim introduced
```

Strategy index rule

The strategy index may link to planning records as they are created.

Index linking does not imply:

```
workflow evidence captured
workflow accepted
estimate changed
public posture changed
product readiness changed
runtime authority changed
security posture changed
```

Current next planning action

The next planning action after this index is:

```
decide whether to stop at planning package completion or begin a separate evidence-capture
request
```

Evidence capture should be a separate explicit request because it may involve running commands, collecting transcripts, or producing guard/test records.

Non-claims

This index does not capture evidence, run a guard, run a test, validate workflow evidence, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a strategy package map only.

docs/strategy/2026-05-26-0949-cdt-panel-guided-local-evaluation-guard-test-reference-template.md

Source title: Latticra Panel-Guided Local Evaluation Guard/Test Reference Template

Lines: 251 | Words: 623 | SHA-256: 36b6299570845d10eb9858d96dd84c05d331f60cfef668c5bd7c6e41880aacdb

Latticra Panel-Guided Local Evaluation Guard/Test Reference Template

Status: draft guard/test reference template Created: 2026-05-26 09:49 CDT Decision: reference template only Promotion decision: no product-readiness promotion recommended Scope: guard or test reference requirements for future Panel-guided local evaluation evidence.

Purpose

This template defines how a future guard, test, workflow harness, or manual transcript should be referenced for Panel-guided local evaluation evidence.

It does not implement a guard, run a test, validate a workflow, capture evidence, or change project posture.

Source review

This template follows:

```
docs/strategy/2026-05-26-0119-cdt-panel-guided-local-evaluation-estimate-impact-review-template.
md
```

The target workflow remains:

```
Evaluate Latticra locally through Latticra Panel without granting runtime authority.
```

Reference header

```
reference name:
created:
reviewer:
reference status: missing|planned|present|ran_passed|ran_failed|blocked
guard or test path:
guard or test type: shell|unit|integration|manual-transcript|workflow-harness|other
workflow evidence bundle:
acceptance checklist:
repository commit:
```

Required reference outcome

Use one outcome:

```
no_guard_or_test_reference
guard_or_test_reference_planned
guard_or_test_reference_present_not_run
guard_or_test_reference_ran_passed
guard_or_test_reference_ran_failed
guard_or_test_reference_blocked
```

This template result:

```
guard_or_test_reference_present=0
guard_or_test_ran=0
guard_or_test_passed=0
workflow_validated=0
estimate_change_recommended=0
reason=this file defines the future reference shape only
```

Required reference fields

```
guard_or_test_name:
guard_or_test_path:
guard_or_test_owner:
guard_or_test_scope:
guard_or_test_inputs:
guard_or_test_outputs:
guard_or_test_result:
guard_or_test_timestamp:
guard_or_test_environment:
guard_or_test_log:
```

If any required field is missing, the reference is incomplete.

Required coverage claims

A future guard or test reference must say whether it covers:

```
status_and_non_claim_entry=0|1
platform_prerequisite_capture=0|1
panel_launch_capture=0|1
dry_run_first_capture=0|1
plan_capture=0|1
dry_run_validation_capture=0|1
receipt_log_console_capture=0|1
reset_or_uninstall_capture=0|1
failure_state_capture=0|1
limitations_and_support_capture=0|1
review_gate_capture=0|1
```

Coverage must be explicit. Do not infer coverage from a broad test name.

Required preservation assertions

The guard or test reference must preserve:

```
runtime_authority_granted=0
effect_execution_performed=0
network_authority_granted=0
root_authority_required_by_default=0
hidden_mutation_detected=0
product_readiness_promotion=0
security_boundary_claim=0
installer_readiness_claim=0
```

If any assertion cannot be checked, record it as unchecked, not passed.

Allowed reference types

Allowed reference types:

```
existing guard script
new planned guard script
manual transcript review
workflow harness
focused integration check
documentation consistency check
status/index consistency check
```

Allowed only as planning references until they exist and are run:

```
planned guard
planned transcript
planned workflow harness
```

Blocked reference types

Blocked for this workflow unless a separate contract exists:

```
root-requiring test
network-requiring test
runtime-authority test
effect-execution test
host-mutation test
production installer test
security hardening test
malware-prevention test
ransomware-prevention test
platform package approval test
```

If a blocked reference type is needed, this workflow must be superseded by a separate capability-promotion packet.

Minimum pass evidence

A future passing guard or test reference must provide:

```
command or transcript id:
exit status or review result:
captured output path:
reviewed evidence bundle:
checked preservation assertions:
failure-state coverage result:
non-claim review link:
public-entrypoint review link:
estimate-impact review link:
```

Do not mark guard_or_test_reference_ran_passed without captured output or transcript evidence.

Failure handling

If the guard or test fails, record:

```
failure label:
failed phase:
failed assertion:
output path:
remediation note:
workflow blocked=1
estimate_change_recommended=0
public_entrypoint_change_recommended=0
```

Failure must not be reframed as product-readiness evidence.

Manual transcript reference rule

Manual transcript evidence may be referenced only if it includes:

```
operator:
date:
platform:
commands or UI steps:
observed outputs:
authority labels:
failure or success labels:
reset or cleanup notes:
reviewer signature or note:
```

Manual transcript evidence must still pass non-claim, public-entrypoint, and estimate-impact review before it affects posture.

Review gate output

```
guard_or_test_reference_present=0|1
guard_or_test_reference_complete=0|1
guard_or_test_ran=0|1
guard_or_test_passed=0|1
coverage_complete=0|1
preservation_assertions_complete=0|1
blocked_reference_type_detected=0|1
workflow_blocked=0|1
estimate_change_recommended=0|1
public_entrypoint_change_recommended=0|1
reference_outcome:
```

This template output:

```
guard_or_test_reference_present=0
guard_or_test_reference_complete=0
guard_or_test_ran=0
guard_or_test_passed=0
coverage_complete=0
preservation_assertions_complete=0
blocked_reference_type_detected=0
workflow_blocked=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reference_outcome=not_performed
```

Non-claims

This template does not implement a guard, run a test, validate workflow evidence, capture transcripts, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a guard/test reference form only.

docs/strategy/2026-05-26-1356-cdt-panel-guided-local-evaluation-planning-completion-checkpoint.md

Source title: Latticra Panel-Guided Local Evaluation Planning Completion Checkpoint

Lines: 143 | Words: 458 | SHA-256: 0521b4401e46a2c0e5c3ba1373fb2c2ceb72a570105e5cbf8b946607679b2e98

Latticra Panel-Guided Local Evaluation Planning Completion Checkpoint

Status: planning completion checkpoint Created: 2026-05-26 13:56 CDT Decision: stop at planning package completion Promotion decision: no product-readiness promotion recommended Scope: completion checkpoint for the Panel-guided local evaluation strategy package.

Purpose

This checkpoint closes the current planning pass for Panel-guided local evaluation.

It records that the planning package is organized enough for future evidence work, but that no evidence capture, command execution, guard implementation, test execution, public-entrypoint update, estimate change, or product-readiness promotion should begin without a separate explicit request.

Source package

This checkpoint follows:

```
docs/strategy/2026-05-26-0410-cdt-panel-guided-local-evaluation-review-package-index.md
```

Latest guard/test reference template available at this checkpoint:

```
docs/strategy/2026-05-26-0949-cdt-panel-guided-local-evaluation-guard-test-reference-template.md
```

Decision

Current decision:

```
stop_at_planning_package_completion=1
begin_evidence_capture=0
run_commands=0
collect_transcripts=0
implement_guard_or_test=0
perform_non_claim_review=0
perform_public_entrypoint_review=0
perform_estimate_impact_review=0
change_public_posture=0
```

Completed planning records

The current package has records for:

```
public product-readiness promotion boundary
Panel-guided local evaluation workflow selection
acceptance criteria
evidence bundle shape
evidence capture sequence
guard/test reference requirements
non-claim review
public-entrypoint review
estimate-impact review
package index
```

Current package status

```
planning_package_complete_for_current_scope=1
workflow_selected=1
acceptance_criteria_defined=1
evidence_bundle_shape_defined=1
capture_sequence_defined=1
review_gates_defined=1
package_index_defined=1
workflow_evidence_captured=0
guard_or_test_implemented=0
guard_or_test_run=0
review_performed=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
product_readiness_promotion_recommended=0
```

Boundary for future evidence capture

Evidence capture must be a separate explicit request because it may require:

```
running local commands
opening or launching Latticra Panel
capturing transcripts
collecting logs or receipts
checking reset or uninstall behavior
creating or running guard scripts
reviewing public entry points
reviewing estimates
```

Those activities are outside this planning-only checkpoint.

Allowed next strategic moves

Allowed planning-only next moves:

```
choose a different promotion gate
create a new planning package for another lane
review overall strategy priorities
refresh strategy index organization
build a roadmap lane map
build a funding/support strategy record
build a public messaging boundary record
```

Blocked without separate request:

```
run Panel workflow
run installer commands
run guard scripts
capture evidence transcripts
update README or status as evidence-backed public posture
change completion estimates
announce product-readiness movement
```

Recommended next planning lane

Recommended next planning lane:

```
overall strategy priority map after Panel-guided local evaluation planning package completion
```

Reason:

```
the Panel-guided path is now organized; the project can step back and decide which
major lane deserves the next planning package before any evidence-capture work starts
```

Non-claims

This checkpoint does not capture evidence, run commands, launch Panel, run a guard, run a test, validate workflow evidence, perform a review, update public entry points, change estimates, provide product readiness, provide Panel readiness, provide installer readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, perform host behavior, perform network behavior, execute models, execute tools, execute shell behavior, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records planning package completion only.

installer/docs/INSTALL_BUTTON_EXECUTION_MODEL.md

Source title: Install Button Execution Model

Lines: 91 | Words: 412 | SHA-256: d48c8b43c008d3d7f108fdc137bb71f6cd03710158ef0845a8a9612977619939

Install Button Execution Model

The graphical Install button is intentionally real, but guarded.

Modes

Dry-install mode

Default.

```
safety.dry_run = true
```

The button executes the installer engine, emits progress phases, validates configuration, checks the component manifest, generates an install plan, and writes an operator receipt. It does not write into the install prefix.

Local-prefix install mode

Explicit.

```
safety.dry_run = false
safety.allow_host_mutation = true
```

The button creates a user-local install layout and writes selected component placeholders, markers, configuration files, and CLI shims. This is still not a production installer. It is a guarded early install path for development validation.

Local-prefix reset and uninstall modes

The reset and uninstall actions use the same authority posture as local-prefix install. In dry-run mode they preview removal and write a receipt. With guarded local-prefix writes enabled, they remove managed command wrappers, managed desktop entries, known Panel icons, and the selected guarded prefix.

Reset and uninstall intentionally share removal mechanics. Reset means "remove the managed local install so I can reinstall from new specifications." Uninstall means "remove the managed local install and stop using it."

State machine

```
Idle
  -&gt; Preparing files
  -&gt; Validating configuration
  -&gt; Loading component manifest
  -&gt; Resolving prefix
  -&gt; Building prefix layout
  -&gt; Materializing selected components
  -&gt; Writing receipt
  -&gt; Complete
```

Reset and uninstall use a shorter state sequence:

```
Idle
  -&gt; Resolving guarded prefix
  -&gt; Removing managed wrappers
  -&gt; Removing desktop metadata
  -&gt; Removing managed prefix
  -&gt; Writing reset receipt
  -&gt; Complete
```

Failure can occur at any step. A failure should leave the operator with a visible log and, when possible, a receipt.

Guardrails

The current installer must not:

- request root authority
- mutate system directories
- use the network
- claim production readiness
- silently overwrite existing files when preserve mode is enabled
- enable runtime enforcement from the installer path

UI behavior

The button label changes based on mode:

- Run Dry-Install when dry-run is active
- Install guarded local prefix when real local-prefix install is explicitly enabled
- Installing... while the engine is running

The reset action is separate:

- Preview local reset when dry-run is active
- Reset installed local prefix when guarded local-prefix writes are explicitly enabled

The uninstall action is also separate:

- Preview uninstall when dry-run is active

- Uninstall managed local install when guarded local-prefix writes are explicitly enabled

The progress bar follows emitted PHASE n/total messages from the install script.

installer/docs/INSTALLER_READINESS_CONTRACT.md

Source title: Installer Readiness Contract

Lines: 45 | Words: 202 | SHA-256: cc1076c43dac6d10679d4f286d745d686d904afaf23a7748daa5bcd6ec5ecf64

Installer Readiness Contract

The Latticra installer is allowed to exist before it is production-ready, but it must be honest about its authority and maturity.

Current authority

```
production_installer_ready=0
root_authority=0
network_authority=0
runtime_enforcement_authority=0
```

Current allowed behavior

- Generate configuration.
- Generate an install plan.
- Execute a dry-install.
- Write local operator receipts.
- Create a guarded user-local prefix when explicitly authorized.
- Create component markers and placeholder developer shims for validation.
- Reset or uninstall a guarded user-local prefix by removing managed wrappers, desktop metadata, known Panel icons, and the managed prefix.

Current forbidden behavior

- Do not request root.
- Do not write into /usr, /bin, /sbin, /etc, /System, or other system directories.
- Do not fetch dependencies from the network.
- Do not silently overwrite files when preserve mode is enabled.
- Do not claim the host is protected by Latticra Seal enforcement.
- Do not claim production installer readiness.

Promotion requirements

Before this installer can become a production installer, the repo needs documented evidence for:

- component manifest integrity
- artifact hashing
- rollback
- upgrade
- uninstall
- failed-install recovery

- Fedora/Linux target validation
- Latticra Seal report generation
- package-manager-safe integration strategy

installer/docs/README.md

Source title: Latticra Installer Documentation

Lines: 38 | Words: 212 | SHA-256: a7b42076acab23a03d6bdbf1ccf40a5a814cf86e26bdc55804ad990dde471bd8

Latticra Installer Documentation

Status: active installer documentation map Last updated: 2026-05-26 CDT Scope: Latticra Panel installer authority, UI configuration, install-button execution, receipts, evidence, and readiness boundaries.

Purpose

This folder records the installer-specific contracts that sit below the public Panel guide in ../README.md. These documents are about the guarded user-local installer path, not a production installer or system package manager integration.

Documents

```
| Document | Purpose |
| --- | --- |
| [`INSTALLER_READINESS_CONTRACT.md`](INSTALLER_READINESS_CONTRACT.md) | Defines current installer authority, allowed behavior, forbidden behavior, and promotion requirements. |
| [`UI_CONFIGURATION_MODEL.md`](UI_CONFIGURATION_MODEL.md) | Defines the Panel configuration model and how UI choices map to installer planning. |
| [`INSTALL_BUTTON_EXECUTION_MODEL.md`](INSTALL_BUTTON_EXECUTION_MODEL.md) | Defines the execution path behind installer UI actions. |
| [`RECEIPTS_AND_EVIDENCE.md`](RECEIPTS_AND_EVIDENCE.md) | Defines installer receipts, evidence expectations, and review surfaces. |
```

Reader Route

1. Start with ../README.md for the public Panel workflow.
2. Read INSTALLER_READINESS_CONTRACT.md before interpreting any installer behavior.
3. Use UI_CONFIGURATION_MODEL.md and INSTALL_BUTTON_EXECUTION_MODEL.md when changing or reviewing Panel controls.
4. Use RECEIPTS_AND_EVIDENCE.md when changing install, dry-run, verify, reset, uninstall, or evidence wording.

Current Boundary

```
production_installer_ready=0
root_authority=0
network_authority=0
runtime_enforcement_authority=0
user_local_prefix_allowed=1
dry_run_first_posture=1
```

The installer documentation must stay aligned with the root documentation hub at ../../docs/README.md, the root status shortcut at ../../STATUS.md, and the detailed current status record at ../../docs/status/CURRENT_STATUS.md.

installer/docs/RECEIPTS_AND_EVIDENCE.md

Source title: Receipts and Evidence

Lines: 32 | Words: 139 | SHA-256: 1edc9eb5d3343553b79ca798f0793309da2c9a7f64c41e279c3d10c9cb23f917

Receipts and Evidence

The installer should leave behind evidence that is understandable to a human operator.

The current receipt includes:

- UTC timestamp
- selected profile
- install prefix
- dry-run status
- host mutation authority status
- network authority status
- component choices
- behavior choices
- component manifest path
- installer script measurement
- config measurement
- generated install plan

The reset/uninstall receipt includes:

- UTC timestamp
- operation label
- operation mode
- install prefix
- command wrapper, desktop entry, icon, and prefix actions
- removed, planned, preserved, and missing counts
- authority baseline

The measurement is SHA-256 when sha256sum or shasum -a 256 is available. If neither exists, the script falls back to POSIX cksum and labels it clearly.

Receipts are not a substitute for cryptographic release signing. They are a development evidence trail.

installer/docs/UI_CONFIGURATION_MODEL.md

Source title: UI Configuration Model

Lines: 180 | Words: 1352 | SHA-256: ce7185e891f85f1f9b25e28079fb55d5e89b08231c2865ea3ea7274fc8ab0c68

UI Configuration Model

The Latticra Panel is now GUI-first. It exposes a guided first-run workbench rather than a numbered terminal menu, because the panel is expected to be the first major user-facing surface for Latticra.

The graphical panel uses one shared InstallerConfig model, generates the same plan preview used by the guarded installer engine, and keeps authority boundaries visible before any install path runs.

First-run design principles

- Strong safe defaults before manual toggles
- Visual guided profiles instead of numbered terminal choices
- Dry-run first, local writes later
- Visible authority status at all times
- Plan and receipt evidence before action
- Embedded panel-aware console instead of a separate TUI
- Adaptive maximized/resizable layout for Fedora workstations and smaller screens
- v0.5.0 keeps the selected workspace stable while installation runs, adds a focused run monitor with bounded recent engine output, keeps long engine lines from expanding the right evidence rail, and keeps the console command-focused during an active engine operation

Profiles

Guided Workbench

Default profile. Enables Lat tooling, LIR contracts, Seal report-only files, docs, and developer helpers while keeping dry-run enabled and host/network authority denied.

Seal Report-Only

Minimal report-only Seal-side layout and documentation for users who only want receipts, reports, and evidence posture.

Fedora Validation VM

For a Fedora or Fedora-like validation VM. Enables the Fedora validation workspace and keeps the run dry by default.

LC Standalone

Console-only profile. Enables the LC wrapper, profile presets, seed registry, and contract metadata while disabling Panel embedding. The dry-run preset previews it, and the local preset runs the guarded user-local standalone LC install.

Custom

Lets the operator manually choose components after the guided profiles are understood.

Components

- Latticra Console (LC)
- Lat language tooling
- LIR contracts
- Latticra Seal report-only subsystem
- Nadia offline AI foundation
- Fedora validation files
- Documentation and examples
- Developer CLI helpers

Safety and evidence

- Dry-run mode
- Explicit guarded local-prefix write authority
- Explicit network authority, currently disabled
- Component manifest requirement
- Artifact measurement requirement
- Verification-policy metadata requirement
- Operator receipt writing

Behavior

- Create prefix layout
- Create component markers
- Create CLI shims
- Preserve existing files

Program delivery

- Build the Latticra Panel binary when Cargo is available
- Build Latticra from source when a supported root build system is available
- Install the Latticra payload tree
- Install the desktop entry
- Install user-local command wrappers

Embedded Latticra Console

The embedded console is panel-aware and intentionally not an unrestricted shell. It supports quick operator commands such as:

```
help
status
lc status
lc commands
lc substrate
lc host
lc os
plan
save
dry-run
nadia status
nadia context
nadia runtime
nadia plan
nadia mode
nadia ledger
nadia safety
nadia tool
nadia prompt-contract
nadia model-registry
nadia inference-readiness
nadia runtime-invocation
nadia model-load
nadia prompt-receipt
nadia prompt-materialization
nadia awareness-dialogue
nadia prompt-evaluation-handoff
nadia tokenization-boundary
nadia tokenizer-specification
nadia tokenizer-manifest
nadia tokenizer-artifact-inventory
```

```
nadia tokenizer-artifact-measurement
nadia tokenizer-artifact-verification
nadia tokenizer-artifact-binding
nadia tokenizer-runtime-attachment
nadia prompt-tokenization
nadia prompt-token-sequence
nadia context-window-assembly
nadia prompt-evaluation-input
nadia prompt-evaluation-runtime-handoff
nadia prompt-evaluation-invocation
nadia prompt-evaluation-result
nadia prompt-evaluation-result-review
nadia prompt-evaluation-result-disposition
nadia prompt-evaluation-result-release
nadia prompt-evaluation-result-release-receipt
nadia prompt-evaluation-result-release-receipt-review
nadia prompt-evaluation-result-release-receipt-review-disposition
nadia prompt-evaluation-result-release-receipt-review-disposition-release
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt
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nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review-disposition
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review-disposition-release
profile guided
profile seal
profile fedora
mode dry
mode local
clear
```

The console exists inside the GUI so users can stay in one coherent Latticra control surface while still getting terminal-style feedback and procedure visibility.

LC is the standalone and Panel-installable console foundation for substrate interaction, host embedding planning, and future OS-base work. Its current Panel and standalone commands are metadata-only and do not execute external host commands.

Nadia's Stage-0, Stage-1, Stage-2, Stage-3, Stage-4, Stage-5, Stage-6, Stage-7, Stage-8, Stage-9, Stage-10, Stage-11, Stage-12, Stage-13, Stage-14, Stage-15, Stage-16, Stage-17, Stage-18, Stage-19, Stage-20, Stage-21, Stage-22, Stage-23, Stage-24, Stage-25, Stage-26, Stage-27, Stage-28, Stage-29, Stage-30, Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, and Stage-40 console surfaces are metadata-only and remain preserved as the Stage-41 lane is added.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, and Stage-41 console surfaces are metadata-only and remain preserved as the Stage-42 lane is added.

Nadia's Stage-0, Stage-1, Stage-2, Stage-3, Stage-4, Stage-5, Stage-6, Stage-7, Stage-8, Stage-9, Stage-10, Stage-11, Stage-12, Stage-13, Stage-14, Stage-15, Stage-16, Stage-17, Stage-18, Stage-19, Stage-20, Stage-21, Stage-22, Stage-23, Stage-24, Stage-25, Stage-26, Stage-27, Stage-28, Stage-29, Stage-30, Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, and Stage-42 console surfaces are metadata-only inside the Panel and do not launch an external host process or model runtime.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, and Stage-42 console surfaces are metadata-only result, review, disposition, release, release receipt, release receipt review, release receipt review disposition, release receipt review disposition release, release receipt review disposition release receipt, release receipt review disposition release receipt review, release receipt review disposition release receipt review disposition, and release receipt review disposition release receipt review disposition release surfaces inside the Panel.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, and Stage-42 console surfaces are metadata-only and remain covered by the Stage-42 metadata-only console posture.

Nadia's Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, and Stage-42 console surfaces are metadata-only and remain covered by the Stage-42 metadata-only console posture.

Nadia's Stage-0, Stage-1, Stage-2, Stage-3, Stage-4, Stage-5, Stage-6, Stage-7, Stage-8, Stage-9, Stage-10, Stage-11, Stage-12, Stage-13, Stage-14, Stage-15, Stage-16, Stage-17, Stage-18, Stage-19, Stage-20, Stage-21, Stage-22, Stage-23, Stage-24, Stage-25, Stage-26, Stage-27, Stage-28, Stage-29, Stage-30, Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, and Stage-43 console surfaces are metadata-only inside the Panel and do not launch an external host process or model runtime.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, and Stage-43 console surfaces are metadata-only result, review, disposition, release, release receipt, release receipt review, release receipt review disposition, release receipt review disposition release, release receipt review disposition release receipt, release receipt review disposition release receipt review, release receipt review disposition release receipt review disposition, release receipt review disposition release receipt review disposition release, and release receipt review disposition release receipt review disposition release receipt surfaces inside the Panel.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, and Stage-43 console surfaces are metadata-only and remain covered by the Stage-43 metadata-only console posture.

Nadia's Stage-0, Stage-1, Stage-2, Stage-3, Stage-4, Stage-5, Stage-6, Stage-7, Stage-8, Stage-9, Stage-10, Stage-11, Stage-12, Stage-13, Stage-14, Stage-15, Stage-16, Stage-17, Stage-18, Stage-19, Stage-20, Stage-21, Stage-22, Stage-23, Stage-24, Stage-25, Stage-26, Stage-27, Stage-28, Stage-29, Stage-30, Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, Stage-43, and Stage-44 console surfaces are metadata-only inside the Panel and do not launch an external host process or model runtime.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, Stage-43, and Stage-44 console surfaces are metadata-only result, review, disposition, release, release receipt, release receipt review, release receipt review disposition, release receipt review disposition release, release receipt review disposition release receipt, release receipt review disposition release receipt review, release receipt review disposition release receipt review disposition, release receipt review disposition release receipt review disposition release, release receipt review disposition release receipt review disposition release receipt, and release receipt review disposition release receipt review disposition release receipt review surfaces inside the Panel.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, Stage-43, and Stage-44 console surfaces are metadata-only and remain covered by the Stage-44 metadata-only console posture.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, Stage-43, Stage-44, and Stage-45 console surfaces are metadata-only and remain covered by the Stage-45 metadata-only console posture.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, Stage-40, Stage-41, Stage-42, Stage-43, Stage-44, Stage-45, and Stage-46 console surfaces are metadata-only and remain covered by the Stage-46 metadata-only console posture.

Stage-40 prompt-evaluation result release receipt review disposition release receipt review contract adds a metadata-only prompt-evaluation result release receipt review disposition release receipt review surface for the next review contract boundary.

Stage-31, Stage-32, Stage-33, Stage-34, Stage-35, Stage-36, Stage-37, Stage-38, Stage-39, and Stage-40 console surfaces are metadata-only and remain covered by the Stage-40 metadata-only console posture.

installer/INSTALLER_ROADMAP.md

Source title: Latticra Installer Roadmap

Lines: 40 | Words: 205 | SHA-256: c409f64611f56cecabb528d2fe85bbb3d248e7bc67f9a18d28472c79937e8bac

Latticra Installer Roadmap

Current stage

The installer supports a guarded user-local installation under ~/.local/share/latticra with command wrappers in ~/.local/bin.

It currently installs:

- Latticra source/payload snapshot
- installer configuration
- component markers
- LIR contract workspace
- Latticra Seal report-only configuration
- receipts and measurements
- reset receipts for managed local uninstall
- latticra, lat, and latticra-seal command wrappers
- optional graphical installer binary when Cargo is available

Boundaries

The installer does not currently claim:

- production OS installer readiness
- kernel mutation authority
- systemd mutation authority
- SELinux mutation authority
- package-manager integration
- bootloader integration
- firmware/SeaBIOS/GRUB mutation authority
- network authority

Next stages

1. Add signed release artifact ingestion.

2. Add stronger manifest validation.
3. Add Fedora package integration plan.
4. Use the boot-preview preflight report to classify local QEMU/GRUB readiness before any VM validation mode runs.
5. Use the boot-preview evidence manifest, boot artifact manifest template, boot artifact manifest validation, boot-preview evidence capture template, boot-preview evidence validation, and QEMU argv template to record future artifact paths, serial logs, checksums, QEMU argv records, and recovery evidence only after the compatibility guard is satisfied.
6. Add failed-install recovery receipts.
7. Add system-level installer only after explicit design review.

installer/latticra-installer-plan.txt

Source title: Latticra Installer Plan

Lines: 1101 | Words: 1150 | SHA-256: c959b8df2c78fbd240e2feee74535ebdaca1ae86a8d89709cfd2aa3474dee86c

LATTICRA INSTALLER PLAN

timestamp_utc=20260527T011742Z repo_root=/Users/chasebryan/Documents/Latticra installer_root=/Users/chasebryan/Documents/Latticra/installer config=/Users/chasebryan/Documents/Latticra/installer/configs/lc-standalone.installer.toml profile=lc_standalone mode=dry-install install_prefix=/Users/chasebryan/.local/share/latticra user_bin=/Users/chasebryan/.local/bin payload_dir=/Users/chasebryan/.local/share/latticra/lib/latticra production_installer_ready=0 root_authority=0 network_authority=0 runtime_enforcement_authority=0

[updater] panel_owned=1 source_strategy=current-source-checkout update_channel=local-checkout allow_network_fetch=false require_dry_run_before_apply=true reuse_installer_engine=true write_update_receipt=true network_authority=0 root_authority=0 system_mutation_authority=0 update_apply_mode=guarded-local-prefix-reinstall signed_delivery_gate=closed signed_manifest_required=1 signed_manifest_present=0 manifest_signature_verified=0 artifact_hash_verified=0 artifact_signature_verified=0 rollback_plan_required=1 rollback_plan_present=0 operator_confirmation_required=1 operator_confirmation_observed=0 signed_update_apply_allowed=0 network_self_update_ready=0 signed_update_delivery_ready=0

[components] latticra_console=true lat_tooling=false lir_contracts=false seal_report_only=false nadia_offline_ai=false fedora_validation=false docs_and_examples=false developer_cli_helpers=false

[lc] component_key=latticra_console console_name=Latticra Console short_name=LC component_selected=true configurable=1 panel_installable=1 install_profile=lc-standalone-install-v0 install_mode=metadata-only-standalone-console install_config_path=etc/latticra/lc.toml install_share_path=share/latticra/lc install_command_wrapper=latticra-lc standalone_console=true standalone_installable=1 standalone_requires_panel=0 standalone_command_wrapper=latticra-lc standalone_console_status=metadata-only-standalone-contract standalone_contract_present=1 session_contract_profile=lc-session-v0 session_contract_status=metadata-only-contract session_contract_present=1 workspace_contract_profile=lc-workspace-v0 workspace_contract_status=metadata-only-contract workspace_contract_present=1 namespace_contract_profile=lc-namespace-v0 namespace_contract_status=metadata-only-contract namespace_contract_present=1 rootfs_contract_profile=lc-rootfs-v0 rootfs_contract_status=metadata-only-contract

rootfs_contract_present=1 packages_contract_profile=lc-packages-v0
packages_contract_status=metadata-only-contract packages_contract_present=1
init_contract_profile=lc-init-v0 init_contract_status=metadata-only-contract
init_contract_present=1 panel_embedded_console=false write_config_file=true
write_profile_presets=true write_command_registry=true write_contract_files=true
install_user_wrapper=true allow_external_host_commands=false profile=standalone
panel_console_bridge=standalone-optional command_registry_profile=c-static-table
substrate_bridge_profile=metadata-bound host_embedding_profile=not-embedded
host_embedding_contract_profile=lc-host-embedding-v0
host_inventory_contract_profile=lc-host-inventory-v0
host_adapter_contract_profile=lc-host-adapter-v0
workspace_contract_profile=lc-workspace-v0 namespace_contract_profile=lc-namespace-v0
rootfs_contract_profile=lc-rootfs-v0 packages_contract_profile=lc-packages-v0
init_contract_profile=lc-init-v0 receipt_request_contract_profile=lc-receipt-request-v0
receipt_payload_schema_profile=lc-receipt-payload-schema-v0
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0 receipt_pay
load_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0 receip
t_payload_artifact_review_receipt_draft_profile=lc-receipt-payload-artifact-review-receip
t-draft-v0
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
signature_request_binding_profile=lc-signature-request-binding-v0
receipt_contract_profile=lc-receipts-v0 os_base_contract_profile=lc-os-base-v0
vm_evidence_contract_profile=lc-vm-evidence-v0 os_base_profile=planned-no-boot-authority
report_only=true host_embedding_contract_required=true
read_only_host_inventory_contract_required=true profile_receipt_required=true
host_contract_receipt_required=true host_inventory_receipt_required=true
host_adapter_contract_required=true session_contract_required=true
workspace_contract_required=true namespace_contract_required=true
rootfs_contract_required=true packages_contract_required=true init_contract_required=true
receipt_request_contract_required=true receipt_payload_schema_required=true
receipt_payload_artifact_draft_required=true
receipt_payload_artifact_review_required=true
receipt_payload_artifact_review_receipt_required=true
receipt_payload_artifact_review_receipt_draft_required=true
receipt_payload_materialization_plan_required=true
signature_request_binding_required=true os_base_contract_required=true
vm_evidence_contract_required=true runtime_boundary_binding_required=true
seal_capability_labels_required=true command_registry_status=seed-registry
substrate_bridge_status=metadata-bound host_embedding_status=not-embedded
host_embedding_contract_status=metadata-only-contract
host_inventory_contract_status=metadata-only-contract
host_adapter_contract_status=metadata-only-contract
session_contract_status=metadata-only-contract
workspace_contract_status=metadata-only-contract
namespace_contract_status=metadata-only-contract
rootfs_contract_status=metadata-only-contract
packages_contract_status=metadata-only-contract
init_contract_status=metadata-only-contract
receipt_request_contract_status=metadata-only-contract
receipt_payload_schema_status=metadata-only-schema
receipt_payload_artifact_draft_status=metadata-only-draft
receipt_payload_artifact_review_status=metadata-only-review-gate
receipt_payload_artifact_review_receipt_status=metadata-only-receipt-contract
receipt_payload_artifact_review_receipt_draft_status=metadata-only-review-receipt-draft

receipt_payload_materialization_plan_status=metadata-only-plan
materialization_preconditions_met=0 draft_review_receipt_present=0
draft_review_receipt_artifact_present=0 materialization_allowed=0
payload_artifact_present=0 payload_materialized=0 payload_write_allowed=0
signature_request_binding_status=metadata-only-contract
receipt_contract_status=metadata-only-contract
os_base_contract_status=metadata-only-contract
vm_evidence_contract_status=metadata-only-contract seal_signature_request_ready=0
seal_signature_request_present=0 seal_signature_present=0 receipt_signed=0
os_base_status=planned-no-boot-authority operator_shell_present=1 execution_allowed=0
host_mutation_allowed=0 network_allowed=0 runtime_enforcement_allowed=0 boot_allowed=0
os_base_enabled=0 production_os_claim=0 future_os_base_claim=planned_not_claimed

[nadia] system_name=Latticra Nadia Witness Foundation public_name=Nadia
interactive_name=Nadia implementation_name=Nadia Witness Foundation
documentation_code_name=Nadia Witness Foundation stage=45-prompt-evaluation-result-releas
e-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-di
sposition-contract previous_stage=44-prompt-evaluation-result-release-receipt-review-disp
osition-release-receipt-review-disposition-release-receipt-review-contract
component_selected=false context_engine_stage=1-local-context-engine
context_pack_command=scripts/nadia-context-pack.sh
installed_context_pack_command=latticra-nadia context-pack
local_file_read_for_indexing=operator_invoked
runtime_profile_stage=2-runtime-profile-boundary
runtime_profile_command=scripts/nadia-runtime-profile.sh
installed_runtime_profile_command=latticra-nadia runtime-profile
runtime_family=llama.cpp-compatible model_format=gguf
developer_workbench_stage=3-developer-workbench-planning
prompt_plan_command=scripts/nadia-prompt-plan.sh
installed_prompt_plan_command=latticra-nadia prompt-plan
systems_engineering_mode_stage=4-systems-engineering-mode-validation
mode_validation_command=scripts/nadia-mode-validate.sh
installed_mode_validation_command=latticra-nadia mode-validate mode_taxonomy_present=1
productivity_loop_stage=5-productivity-ledger-loop
productivity_ledger_command=scripts/nadia-productivity-ledger.sh
installed_productivity_ledger_command=latticra-nadia productivity-ledger
learning_scope=operator-reviewed-local-productivity ledger_append_only=1
protective_safety_stage=6-protective-safety-boundary
protective_safety_command=scripts/nadia-protective-safety-boundary.sh
installed_protective_safety_command=latticra-nadia protective-safety
absolute_protective_boundary=1 sexual_user_request_authority=0
sexual_content_generation=0 sexual_roleplay_authority=0
sexualized_namesake_or_survivor_content=0 sexual_request_refusal=always
user_override_authority=0 prompt_injection_override_authority=0
manipulation_resistance=required policy_bypass_authority=0 namesake_cause_awareness=1
tool_authority_stage=7-guarded-tool-authority-preflight
tool_authority_preflight_command=scripts/nadia-tool-authority-preflight.sh
installed_tool_authority_preflight_command=latticra-nadia tool-preflight
preflight_decision=report_only_no_execution tool_execution_performed=0
tool_selection_authority=0 shell_execution_authority=0 network_tool_authority=0
destructive_action_authority=0 credential_access_authority=0 requires_operator_approval=1
requires_nucleus_gate=1 requires_runtime_boundary_gate=1 requires_seal_receipt=1
requires_protective_safety_boundary=1 authority_transition_allowed=0
prompt_evaluation_contract_stage=8-prompt-evaluation-contract
prompt_evaluation_contract_command=scripts/nadia-prompt-evaluation-contract.sh
installed_prompt_evaluation_contract_command=latticra-nadia prompt-contract

```
prompt_contract_status=contract_only prompt_evaluation_stage=contract-only
prompt_materialized=0 prompt_text_materialized=0 prompt_evaluation_authority=0
prompt_receipt_required=1 refusal_policy_required=1 protective_safety_required=1
tool_preflight_required=1 runtime_profile_required=1 model_registry_review_required=1
operator_review_required=1 contract_promotion_allowed=0
local_model_registry_contract_stage=9-local-model-registry-contract
model_registry_contract_command=scripts/nadia-local-model-registry-contract.sh
installed_model_registry_contract_command=latticra-nadia model-registry
local_model_registry_stage=contract-only registry_contract_status=metadata_only
model_registry_authority=0 requires_prompt_contract=1
candidate_review_status=operator_review_required candidate_usable_for_inference=0
candidate_selected_for_runtime=0 model_selection_authority=0 model_install_authority=0
model_download_authority=0 model_copy_authority=0 model_load_authority=0
model_benchmark_authority=0 model_weight_inspection_authority=0
registry_promotion_allowed=0
inference_readiness_contract_stage=10-inference-readiness-contract
inference_readiness_contract_command=scripts/nadia-inference-readiness-contract.sh
installed_inference_readiness_contract_command=latticra-nadia inference-readiness
inference_readiness_stage=contract-only inference_readiness_contract_status=contract_only
inference_readiness_authority=0 inference_ready=0
readiness_decision=blocked_contract_only readiness_evidence_present=1
requires_model_registry_contract=1 requires_future_runtime_invocation_contract=1
readiness_promotion_allowed=0 runtime_invocation_authority=0 token_generation_authority=0
model_session_authority=0
runtime_invocation_contract_stage=11-runtime-invocation-contract
runtime_invocation_contract_command=scripts/nadia-runtime-invocation-contract.sh
installed_runtime_invocation_contract_command=latticra-nadia runtime-invocation
runtime_invocation_stage=contract-only runtime_invocation_contract_status=contract_only
runtime_invocation_allowed=0 invocation_decision=blocked_contract_only
invocation_evidence_present=1 requires_inference_readiness_contract=1
requires_future_model_load_contract=1 invocation_promotion_allowed=0
runtime_process_spawn_authority=0 runtime_binary_execution_authority=0
runtime_session_authority=0 runtime_process_spawned=0 runtime_binary_executed=0
runtime_session_created=0 token_generation_performed=0
model_load_contract_stage=12-model-load-contract
model_load_contract_command=scripts/nadia-model-load-contract.sh
installed_model_load_contract_command=latticra-nadia model-load
model_load_stage=contract-only model_load_contract_status=contract_only
model_load_authority=0 model_load_allowed=0 model_loaded=0
load_decision=blocked_contract_only load_evidence_present=1
requires_runtime_invocation_contract=1 requires_model_weight_measurement_contract=1
requires_future_prompt_receipt_contract=1 load_promotion_allowed=0
model_file_open_authority=0 model_weight_read_authority=0
model_weight_mapping_authority=0 model_weight_verification_authority=0
runtime_model_attach_authority=0 model_file_opened=0 model_file_descriptor_opened=0
model_memory_map_created=0 model_weights_mapped=0 model_weights_attached=0
model_weight_measurement_performed=0 model_weight_verification_performed=0
model_load_performed=0 prompt_receipt_contract_stage=13-prompt-receipt-contract
prompt_receipt_contract_command=scripts/nadia-prompt-receipt-contract.sh
installed_prompt_receipt_contract_command=latticra-nadia prompt-receipt
prompt_receipt_stage=contract-only prompt_receipt_contract_status=contract_only
prompt_receipt_authority=0 prompt_receipt_allowed=0 prompt_received=0
receipt_decision=blocked_contract_only receipt_evidence_present=1
requires_model_load_contract=1 requires_prompt_source_boundary=1
requires_future_prompt_materialization_contract=1 prompt_receipt_promotion_allowed=0
```

```
prompt_source_open_authority=0 prompt_source_read_authority=0
prompt_text_materialization_authority=0 prompt_content_storage_authority=0
prompt_hash_authority=0 prompt_classification_authority=0 prompt_source_opened=0
prompt_source_read=0 prompt_bytes_read=0 prompt_text_received=0
prompt_text_materialized=0 prompt_content_stored=0 prompt_hash_computed=0
prompt_classified=0
prompt_materialization_contract_stage=14-prompt-materialization-contract
prompt_materialization_contract_command=scripts/nadia-prompt-materialization-contract.sh
installed_prompt_materialization_contract_command=latticra-nadia prompt-materialization
prompt_materialization_stage=contract-only
prompt_materialization_contract_status=contract_only prompt_materialization_authority=0
prompt_materialization_allowed=0 prompt_materialized=0
materialization_decision=blocked_contract_only materialization_evidence_present=1
requires_prompt_receipt_contract=1 requires_prompt_buffer_boundary=1
requires_future_prompt_evaluation_handoff_contract=1
prompt_materialization_promotion_allowed=0 prompt_buffer_allocation_authority=0
prompt_buffer_write_authority=0 prompt_tokenization_authority=0
prompt_materialization_performed=0 prompt_buffer_allocated=0 prompt_buffer_written=0
prompt_bytes_materialized=0 prompt_tokens_created=0 prompt_tokenized=0
awareness_dialogue_contract_stage=15-awareness-dialogue-contract
awareness_dialogue_contract_command=scripts/nadia-awareness-dialogue-contract.sh
installed_awareness_dialogue_contract_command=latticra-nadia awareness-dialogue
future_qa_dialogue_capability_planned=1 awareness_dialogue_stage=contract-only
awareness_dialogue_contract_status=contract_only awareness_dialogue_authority=0
awareness_dialogue_allowed=0 dialogue_generation_authority=0
dialogue_generation_allowed=0 qa_dialogue_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer q_and_a_format_required=1
survivor_centered_dialogue_required=1 official_source_grounding_required=1
live_web_lookup_authority=0 topic_yazidi_genocide_awareness=1
topic_survivor_voice_and_dignity=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_genocide_prevention=1 topic_justice_and_accountability=1
topic_sinjar_reconstruction=1 topic_womens_empowerment=1 sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0 victim_blaming_allowed=0 genocide_denial_allowed=0
prompt_evaluation_handoff_contract_stage=16-prompt-evaluation-handoff-contract prompt_eva
luation_handoff_contract_command=scripts/nadia-prompt-evaluation-handoff-contract.sh
installed_prompt_evaluation_handoff_contract_command=latticra-nadia
prompt-evaluation-handoff prompt_evaluation_handoff_stage=contract-only
prompt_evaluation_handoff_contract_status=contract_only
prompt_evaluation_handoff_authority=0 prompt_evaluation_handoff_allowed=0
prompt_evaluation_handoff_performed=0 evaluation_handoff_decision=blocked_contract_only
requires_awareness_dialogue_contract=1 requires_future_tokenization_contract=1
prompt_evaluation_handoff_promotion_allowed=0
tokenization_boundary_contract_stage=17-tokenization-boundary-contract
tokenization_boundary_contract_command=scripts/nadia-tokenization-boundary-contract.sh
installed_tokenization_boundary_contract_command=latticra-nadia tokenization-boundary
tokenization_boundary_stage=contract-only
tokenization_boundary_contract_status=contract_only tokenization_boundary_authority=0
tokenization_boundary_allowed=0 tokenization_boundary_performed=0
prompt_tokenization_allowed=0 prompt_tokenized=0 prompt_tokens_created=0
tokenizer_file_opened=0 tokenizer_vocab_loaded=0
tokenization_decision=blocked_contract_only requires_prompt_evaluation_handoff_contract=1
requires_future_tokenizer_specification_contract=1
tokenization_boundary_promotion_allowed=0
```

tokenizer_specification_contract_stage=18-tokenizer-specification-contract tokenizer_specification_contract_command=scripts/nadia-tokenizer-specification-contract.sh installed_tokenizer_specification_contract_command=latticra-nadia tokenizer-specification tokenizer_specification_stage=contract-only tokenizer_specification_contract_status=contract_only tokenizer_specification_authority=0 tokenizer_specification_allowed=0 tokenizer_specification_performed=0 tokenizer_specification_metadata_present=1 tokenizer_family=model-compatible-tokenizer tokenizer_format=operator-reviewed-offline-specification tokenizer_specification_decision=blocked_contract_only tokenizer_path_recorded=0 tokenizer_manifest_loaded=0 requires_tokenization_boundary_contract=1 requires_future_tokenizer_manifest_contract=1 tokenizer_specification_promotion_allowed=0 tokenizer_manifest_contract_stage=19-tokenizer-manifest-contract tokenizer_manifest_contract_command=scripts/nadia-tokenizer-manifest-contract.sh installed_tokenizer_manifest_contract_command=latticra-nadia tokenizer-manifest tokenizer_manifest_stage=contract-only tokenizer_manifest_contract_status=contract_only tokenizer_manifest_authority=0 tokenizer_manifest_allowed=0 tokenizer_manifest_performed=0 tokenizer_manifest_metadata_present=1 tokenizer_manifest_family=operator-reviewed-tokenizer-manifest tokenizer_manifest_format=contract-only-offline-manifest tokenizer_manifest_decision=blocked_contract_only tokenizer_manifest_path_recorded=0 tokenizer_manifest_schema_planned=1 tokenizer_manifest_opened=0 tokenizer_manifest_read=0 tokenizer_manifest_parsed=0 tokenizer_manifest_validated=0 tokenizer_manifest_loaded=0 requires_tokenizer_specification_contract=1 requires_future_tokenizer_artifact_inventory_contract=1 tokenizer_manifest_promotion_allowed=0 tokenizer_artifact_inventory_contract_stage=20-tokenizer-artifact-inventory-contract tokenizer_artifact_inventory_contract_command=scripts/nadia-tokenizer-artifact-inventory-contract.sh installed_tokenizer_artifact_inventory_contract_command=latticra-nadia tokenizer-artifact-inventory tokenizer_artifact_inventory_stage=contract-only tokenizer_artifact_inventory_contract_status=contract_only tokenizer_artifact_inventory_authority=0 tokenizer_artifact_inventory_allowed=0 tokenizer_artifact_inventory_performed=0 tokenizer_artifact_inventory_metadata_present=1 tokenizer_artifact_inventory_family=operator-reviewed-tokenizer-artifact-inventory tokenizer_artifact_inventory_format=contract-only-offline-inventory tokenizer_artifact_inventory_decision=blocked_contract_only tokenizer_artifact_inventory_path_recorded=0 tokenizer_artifact_inventory_schema_planned=1 tokenizer_artifact_inventory_entry_count=0 tokenizer_artifact_inventory_file_count=0 tokenizer_artifact_path_resolved=0 tokenizer_artifact_scan_performed=0 tokenizer_artifact_stat_performed=0 tokenizer_artifact_file_opened=0 tokenizer_artifact_file_read=0 tokenizer_artifact_hash_computed=0 tokenizer_artifact_measurement_performed=0 requires_tokenizer_manifest_contract=1 requires_future_tokenizer_artifact_measurement_contract=1 tokenizer_artifact_inventory_promotion_allowed=0 tokenizer_artifact_measurement_contract_stage=21-tokenizer-artifact-measurement-contract tokenizer_artifact_measurement_contract_command=scripts/nadia-tokenizer-artifact-measurement-contract.sh installed_tokenizer_artifact_measurement_contract_command=latticra-nadia tokenizer-artifact-measurement tokenizer_artifact_measurement_stage=contract-only tokenizer_artifact_measurement_contract_status=contract_only tokenizer_artifact_measurement_authority=0 tokenizer_artifact_measurement_allowed=0 tokenizer_artifact_measurement_performed=0 tokenizer_artifact_measurement_metadata_present=1 tokenizer_artifact_measurement_family=operator-reviewed-tokenizer-artifact-measurement tokenizer_artifact_measurement_format=contract-only-offline-measurement tokenizer_artifact_measurement_decision=blocked_contract_only

tokenizer_artifact_measurement_plan_recorded=1
tokenizer_artifact_measurement_result_recorded=0
tokenizer_artifact_measurement_digest_recorded=0
tokenizer_artifact_measurement_size_recorded=0
tokenizer_artifact_measurement_hash_computed=0
requires_tokenizer_artifact_inventory_contract=1
requires_future_tokenizer_artifact_verification_contract=1
tokenizer_artifact_measurement_promotion_allowed=0 tokenizer_artifact_verification_contra
ct_stage=22-tokenizer-artifact-verification-contract tokenizer_artifact_verification_cont
ract_command=scripts/nadia-tokenizer-artifact-verification-contract.sh
installed_tokenizer_artifact_verification_contract_command=latticra-nadia
tokenizer-artifact-verification tokenizer_artifact_verification_stage=contract-only
tokenizer_artifact_verification_contract_status=contract_only
tokenizer_artifact_verification_authority=0 tokenizer_artifact_verification_allowed=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_verification_metadata_present=1
tokenizer_artifact_verification_family=operator-reviewed-tokenizer-artifact-verification
tokenizer_artifact_verification_format=contract-only-offline-verification
tokenizer_artifact_verification_decision=blocked_contract_only
tokenizer_artifact_verification_plan_recorded=1
tokenizer_artifact_verification_comparison_performed=0
tokenizer_artifact_verification_result_recorded=0
tokenizer_artifact_verification_digest_match_recorded=0
tokenizer_artifact_verification_size_match_recorded=0
tokenizer_artifact_verification_hash_computed=0
requires_tokenizer_artifact_measurement_contract=1
requires_future_tokenizer_artifact_binding_contract=1
tokenizer_artifact_verification_promotion_allowed=0
tokenizer_artifact_binding_contract_stage=23-tokenizer-artifact-binding-contract tokenize
r_artifact_binding_contract_command=scripts/nadia-tokenizer-artifact-binding-contract.sh
installed_tokenizer_artifact_binding_contract_command=latticra-nadia
tokenizer-artifact-binding tokenizer_artifact_binding_stage=contract-only
tokenizer_artifact_binding_contract_status=contract_only
tokenizer_artifact_binding_authority=0 tokenizer_artifact_binding_allowed=0
tokenizer_artifact_binding_performed=0 tokenizer_artifact_binding_metadata_present=1
tokenizer_artifact_binding_family=operator-reviewed-tokenizer-artifact-binding
tokenizer_artifact_binding_format=contract-only-offline-binding
tokenizer_artifact_binding_decision=blocked_contract_only
tokenizer_artifact_binding_plan_recorded=1 tokenizer_artifact_binding_result_recorded=0
tokenizer_artifact_binding_record_created=0 tokenizer_artifact_binding_hash_computed=0
tokenizer_artifact_binding_bound=0
tokenizer_artifact_binding_runtime_attachment_performed=0
tokenizer_artifact_bound_to_manifest=0 tokenizer_artifact_bound_to_tokenizer=0
tokenizer_attached_to_runtime=0 requires_tokenizer_artifact_verification_contract=1
requires_future_tokenizer_runtime_attachment_contract=1
tokenizer_artifact_binding_promotion_allowed=0
tokenizer_runtime_attachment_contract_stage=24-tokenizer-runtime-attachment-contract toke
nizer_runtime_attachment_contract_command=scripts/nadia-tokenizer-runtime-attachment-cont
ract.sh installed_tokenizer_runtime_attachment_contract_command=latticra-nadia
tokenizer-runtime-attachment tokenizer_runtime_attachment_stage=contract-only
tokenizer_runtime_attachment_contract_status=contract_only
tokenizer_runtime_attachment_authority=0 tokenizer_runtime_attachment_allowed=0
tokenizer_runtime_attachment_performed=0 tokenizer_runtime_attachment_metadata_present=1
tokenizer_runtime_attachment_family=operator-reviewed-tokenizer-runtime-attachment
tokenizer_runtime_attachment_format=contract-only-offline-attachment

tokenizer_runtime_attachment_decision=blocked_contract_only
tokenizer_runtime_attachment_plan_recorded=1
tokenizer_runtime_attachment_result_recorded=0
tokenizer_runtime_attachment_record_created=0 tokenizer_runtime_attachment_attached=0
tokenizer_runtime_attachment_runtime_invoked=0
tokenizer_runtime_attachment_session_created=0 runtime_session_created=0
runtime_invoked=0 requires_tokenizer_artifact_binding_contract=1
requires_future_prompt_tokenization_contract=1
tokenizer_runtime_attachment_promotion_allowed=0
prompt_tokenization_contract_stage=25-prompt-tokenization-contract
prompt_tokenization_contract_command=scripts/nadia-prompt-tokenization-contract.sh
installed_prompt_tokenization_contract_command=latticra-nadia prompt-tokenization
prompt_tokenization_stage=contract-only prompt_tokenization_contract_status=contract_only
prompt_tokenization_authority=0 prompt_tokenization_allowed=0
prompt_tokenization_performed=0 prompt_tokenization_metadata_present=1
prompt_tokenization_family=operator-reviewed-prompt-tokenization
prompt_tokenization_format=contract-only-offline-tokenization
prompt_tokenization_decision=blocked_contract_only prompt_tokenization_plan_recorded=1
prompt_tokenization_result_recorded=0 prompt_tokenization_token_count_recorded=0
prompt_tokenization_token_sequence_recorded=0 prompt_tokenization_runtime_invoked=0
prompt_tokens_created=0 prompt_token_count_recorded=0 prompt_token_sequence_recorded=0
prompt_token_buffer_created=0 prompt_tokenized=0
requires_tokenizer_runtime_attachment_contract=1
requires_future_prompt_token_sequence_contract=1 prompt_tokenization_promotion_allowed=0
prompt_token_sequence_contract_stage=26-prompt-token-sequence-contract
prompt_token_sequence_contract_command=scripts/nadia-prompt-token-sequence-contract.sh
installed_prompt_token_sequence_contract_command=latticra-nadia prompt-token-sequence
prompt_token_sequence_stage=contract-only
prompt_token_sequence_contract_status=contract_only prompt_token_sequence_authority=0
prompt_token_sequence_allowed=0 prompt_token_sequence_recorded=0
prompt_token_sequence_metadata_present=1
prompt_token_sequence_family=operator-reviewed-prompt-token-sequence
prompt_token_sequence_format=contract-only-offline-sequence
prompt_token_sequence_decision=blocked_contract_only
prompt_token_sequence_plan_recorded=1 prompt_token_sequence_result_recorded=0
prompt_token_sequence_count_recorded=0 prompt_token_sequence_order_recorded=0
prompt_token_sequence_runtime_invoked=0 prompt_token_ids_recorded=0
prompt_attention_mask_created=0 context_window_assembled=0
requires_prompt_tokenization_contract=1
requires_future_context_window_assembly_contract=1
prompt_token_sequence_promotion_allowed=0
context_window_assembly_contract_stage=27-context-window-assembly-contract context_window
_assembly_contract_command=scripts/nadia-context-window-assembly-contract.sh
installed_context_window_assembly_contract_command=latticra-nadia context-window-assembly
context_window_assembly_stage=contract-only
context_window_assembly_contract_status=contract_only context_window_assembly_authority=0
context_window_assembly_allowed=0 context_window_assembly_performed=0
context_window_assembly_metadata_present=1
context_window_family=operator-reviewed-context-window-assembly
context_window_format=contract-only-offline-context-window
context_window_assembly_decision=blocked_contract_only
context_window_assembly_plan_recorded=1 context_window_assembly_result_recorded=0
context_window_assembly_runtime_invoked=0 context_window_token_budget_recorded=0
context_window_truncation_applied=0 context_window_serialized=0
prompt_evaluation_input_created=0 requires_prompt_token_sequence_contract=1

requires_future_prompt_evaluation_input_contract=1
context_window_assembly_promotion_allowed=0
prompt_evaluation_input_contract_stage=28-prompt-evaluation-input-contract prompt_evaluation_input_contract_command=scripts/nadia-prompt-evaluation-input-contract.sh
installed_prompt_evaluation_input_contract_command=latticra-nadia prompt-evaluation-input
prompt_evaluation_input_stage=contract-only
prompt_evaluation_input_contract_status=contract_only prompt_evaluation_input_authority=0
prompt_evaluation_input_allowed=0 prompt_evaluation_input_created=0
prompt_evaluation_input_metadata_present=1
prompt_evaluation_input_family=operator-reviewed-prompt-evaluation-input
prompt_evaluation_input_format=contract-only-offline-evaluation-input
prompt_evaluation_input_decision=blocked_contract_only
prompt_evaluation_input_plan_recorded=1 prompt_evaluation_input_result_recorded=0
prompt_evaluation_input_runtime_invoked=0 prompt_evaluation_input_materialized=0
prompt_evaluation_input_validated=0 prompt_evaluation_input_serialized=0
prompt_evaluation_input_written=0 requires_context_window_assembly_contract=1
requires_future_prompt_evaluation_runtime_handoff_contract=1
prompt_evaluation_input_promotion_allowed=0 prompt_evaluation_runtime_handoff_contract_stage=29-prompt-evaluation-runtime-handoff-contract prompt_evaluation_runtime_handoff_contract_command=scripts/nadia-prompt-evaluation-runtime-handoff-contract.sh
installed_prompt_evaluation_runtime_handoff_contract_command=latticra-nadia
prompt-evaluation-runtime-handoff prompt_evaluation_runtime_handoff_stage=contract-only
prompt_evaluation_runtime_handoff_contract_status=contract_only
prompt_evaluation_runtime_handoff_authority=0 prompt_evaluation_runtime_handoff_allowed=0
prompt_evaluation_runtime_handoff_performed=0
prompt_evaluation_runtime_handoff_metadata_present=1 prompt_evaluation_runtime_handoff_family=operator-reviewed-prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_format=contract-only-offline-runtime-handoff
prompt_evaluation_runtime_handoff_decision=blocked_contract_only
prompt_evaluation_runtime_handoff_plan_recorded=1
prompt_evaluation_runtime_handoff_result_recorded=0
prompt_evaluation_runtime_handoff_runtime_invoked=0
prompt_evaluation_runtime_handoff_request_created=0
prompt_evaluation_runtime_handoff_request_submitted=0 runtime_handoff_created=0
runtime_invocation_requested=0 requires_prompt_evaluation_input_contract=1
requires_future_prompt_evaluation_invocation_contract=1
prompt_evaluation_runtime_handoff_promotion_allowed=0
prompt_evaluation_invocation_contract_stage=30-prompt-evaluation-invocation-contract prompt_evaluation_invocation_contract_command=scripts/nadia-prompt-evaluation-invocation-contract.sh installed_prompt_evaluation_invocation_contract_command=latticra-nadia
prompt-evaluation-invocation prompt_evaluation_invocation_stage=contract-only
prompt_evaluation_invocation_contract_status=contract_only
prompt_evaluation_invocation_authority=0 prompt_evaluation_invocation_allowed=0
prompt_evaluation_invocation_performed=0 prompt_evaluation_invocation_metadata_present=1
prompt_evaluation_invocation_family=operator-reviewed-prompt-evaluation-invocation
prompt_evaluation_invocation_format=contract-only-offline-evaluation-invocation
prompt_evaluation_invocation_decision=blocked_contract_only
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prompt_evaluation_invocation_result_recorded=0
prompt_evaluation_invocation_runtime_invoked=0
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_submitted=0
prompt_evaluation_invocation_request_scheduled=0
prompt_evaluation_invocation_request_queued=0 runtime_handoff_created=0
runtime_invocation_requested=0 requires_prompt_evaluation_runtime_handoff_contract=1

requires_future_prompt_evaluation_result_contract=1
prompt_evaluation_invocation_promotion_allowed=0
prompt_evaluation_result_contract_stage=31-prompt-evaluation-result-contract prompt_evalu ation_result_contract_command=scripts/nadia-prompt-evaluation-result-contract.sh installed_prompt_evaluation_result_contract_command=latticra-nadia prompt-evaluation-result prompt_evaluation_result_stage=contract-only prompt_evaluation_result_contract_status=contract_only prompt_evaluation_result_authority=0 prompt_evaluation_result_allowed=0 prompt_evaluation_result_recorded=0 prompt_evaluation_result_created=0 prompt_evaluation_result_performed=0 prompt_evaluation_result_metadata_present=1 prompt_evaluation_result_family=operator-reviewed-prompt-evaluation-result prompt_evaluation_result_format=contract-only-offline-evaluation-result prompt_evaluation_result_decision=blocked_contract_only prompt_evaluation_result_plan_recorded=1 prompt_evaluation_result_result_recorded=0 prompt_evaluation_result_runtime_invoked=0 prompt_evaluation_result_record_created=0 prompt_evaluation_result_model_output_recorded=0 prompt_evaluation_result_output_text_recorded=0 prompt_evaluation_result_score_recorded=0 prompt_evaluation_result_token_logprobs_recorded=0 answer_text_generated=0 requires_prompt_evaluation_invocation_contract=1 requires_future_prompt_evaluation_result_review_contract=1 prompt_evaluation_result_promotion_allowed=0 prompt_evaluation_result_review_contract_sta ge=32-prompt-evaluation-result-review-contract prompt_evaluation_result_review_contract_c ommand=scripts/nadia-prompt-evaluation-result-review-contract.sh installed_prompt_evaluation_result_review_contract_command=latticra-nadia prompt-evaluation-result-review prompt_evaluation_result_review_stage=contract-only prompt_evaluation_result_review_contract_status=contract_only prompt_evaluation_result_review_authority=0 prompt_evaluation_result_review_allowed=0 prompt_evaluation_result_review_recorded=0 prompt_evaluation_result_review_created=0 prompt_evaluation_result_review_performed=0 prompt_evaluation_result_review_metadata_present=1 prompt_evaluation_result_review_family=operator-reviewed-prompt-evaluation-result-review prompt_evaluation_result_review_format=contract-only-offline-evaluation-result-review prompt_evaluation_result_review_decision=blocked_contract_only prompt_evaluation_result_review_plan_recorded=1 prompt_evaluation_result_review_result_recorded=0 prompt_evaluation_result_review_runtime_invoked=0 prompt_evaluation_result_review_record_created=0 prompt_evaluation_result_review_decision_recorded=0 prompt_evaluation_result_review_approval_recorded=0 prompt_evaluation_result_review_rejection_recorded=0 prompt_evaluation_result_review_findings_recorded=0 requires_prompt_evaluation_result_contract=1 requires_future_prompt_evaluation_result_disposition_contract=1 prompt_evaluation_result_review_promotion_allowed=0 prompt_evaluation_result_disposition_ contract_stage=33-prompt-evaluation-result-disposition-contract prompt_evaluation_result_ disposition_contract_command=scripts/nadia-prompt-evaluation-result-disposition-contract. sh installed_prompt_evaluation_result_disposition_contract_command=latticra-nadia prompt-evaluation-result-disposition prompt_evaluation_result_disposition_stage=contract-only prompt_evaluation_result_disposition_contract_status=contract_only prompt_evaluation_result_disposition_authority=0 prompt_evaluation_result_disposition_allowed=0 prompt_evaluation_result_disposition_recorded=0 prompt_evaluation_result_disposition_created=0 prompt_evaluation_result_disposition_performed=0

prompt_evaluation_result_disposition_metadata_present=1 prompt_evaluation_result_disposition_family=operator-reviewed-prompt-evaluation-result-disposition prompt_evaluation_result_disposition_format=contract-only-offline-evaluation-result-disposition
prompt_evaluation_result_disposition_decision=blocked_contract_only
prompt_evaluation_result_disposition_plan_recorded=1
prompt_evaluation_result_disposition_result_recorded=0
prompt_evaluation_result_disposition_runtime_invoked=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
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prompt_evaluation_result_model_output_recorded=0
requires_prompt_evaluation_result_review_contract=1
requires_future_prompt_evaluation_result_release_contract=1
prompt_evaluation_result_disposition_promotion_allowed=0 prompt_evaluation_result_release_contract_stage=34-prompt-evaluation-result-release-contract prompt_evaluation_result_release_contract_command=scripts/nadia-prompt-evaluation-result-release-contract.sh
installed_prompt_evaluation_result_release_contract_command=latticra-nadia prompt-evaluation-result-release prompt_evaluation_result_release_stage=contract-only
prompt_evaluation_result_release_contract_status=contract_only
prompt_evaluation_result_release_authority=0 prompt_evaluation_result_release_allowed=0
prompt_evaluation_result_release_recorded=0 prompt_evaluation_result_release_created=0
prompt_evaluation_result_release_performed=0
prompt_evaluation_result_release_metadata_present=1 prompt_evaluation_result_release_family=operator-reviewed-prompt-evaluation-result-release
prompt_evaluation_result_release_format=contract-only-offline-evaluation-result-release
prompt_evaluation_result_release_decision=blocked_contract_only
prompt_evaluation_result_release_plan_recorded=1
prompt_evaluation_result_release_result_recorded=0
prompt_evaluation_result_release_runtime_invoked=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0 prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
requires_prompt_evaluation_result_disposition_contract=1
requires_future_prompt_evaluation_result_release_receipt_contract=1
prompt_evaluation_result_release_promotion_allowed=0 prompt_evaluation_result_release_receipt_contract_stage=35-prompt-evaluation-result-release-receipt-contract prompt_evaluation_result_release_receipt_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_contract_command=latticra-nadia prompt-evaluation-result-release-receipt
prompt_evaluation_result_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_authority=0
prompt_evaluation_result_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_performed=0
prompt_evaluation_result_release_receipt_metadata_present=1 prompt_evaluation_result_release_receipt_family=operator-reviewed-prompt-evaluation-result-release-receipt prompt_evaluation_result_release_receipt_format=contract-only-offline-evaluation-result-release-receipt prompt_evaluation_result_release_receipt_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_runtime_invoked=0

```
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
requires_prompt_evaluation_result_release_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_contract=1
prompt_evaluation_result_release_receipt_promotion_allowed=0 prompt_evaluation_result_rel
ease_receipt_review_contract_stage=36-prompt-evaluation-result-release-receipt-review-con
tract prompt_evaluation_result_release_receipt_review_contract_command=scripts/nadia-prom
pt-evaluation-result-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review
prompt_evaluation_result_release_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_metadata_present=1 prompt_evaluation_resu
lt_release_receipt_review_family=operator-reviewed-prompt-evaluation-result-release-recei
pt-review prompt_evaluation_result_release_receipt_review_format=contract-only-offline-ev
aluation-result-release-receipt-review
prompt_evaluation_result_release_receipt_review_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_plan_recorded=1
prompt_evaluation_result_release_receipt_review_result_recorded=0
prompt_evaluation_result_release_receipt_review_runtime_invoked=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
requires_prompt_evaluation_result_release_receipt_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_contract=1
prompt_evaluation_result_release_receipt_review_promotion_allowed=0 prompt_evaluation_res
ult_release_receipt_review_disposition_contract_stage=37-prompt-evaluation-result-release
-receipt-review-disposition-contract prompt_evaluation_result_release_receipt_review_disp
osition_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-di
sposition-contract.sh installed_prompt_evaluation_result_release_receipt_review_dispositi
on_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_performed=0
prompt_evaluation_result_release_receipt_review_disposition_metadata_present=1 prompt_eva
luation_result_release_receipt_review_disposition_family=operator-reviewed-prompt-evaluat
ion-result-release-receipt-review-disposition prompt_evaluation_result_release_receipt_re
view_disposition_format=contract-only-offline-evaluation-result-release-receipt-review-di
```

sposition prompt_evaluation_result_release_receipt_review_disposition_decision=blocked_contract_only prompt_evaluation_result_release_receipt_review_disposition_plan_recorded=1 prompt_evaluation_result_release_receipt_review_disposition_result_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_runtime_invoked=0 prompt_evaluation_result_release_receipt_review_disposition_record_created=0 prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_approval_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_rejection_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_applied=0 requires_prompt_evaluation_result_release_receipt_review_contract=1 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1 prompt_evaluation_result_release_receipt_review_disposition_promotion_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_contract_stage=38-prompt-evaluation-result-release-receipt-review-disposition-release-contract prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release prompt_evaluation_result_release_receipt_review_disposition_release_stage=contract-only prompt_evaluation_result_release_receipt_review_disposition_release_contract_status=contract_only prompt_evaluation_result_release_receipt_review_disposition_release_authority=0 prompt_evaluation_result_release_receipt_review_disposition_release_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_performed=0 prompt_evaluation_result_release_receipt_review_disposition_release_metadata_present=1 prompt_evaluation_result_release_receipt_review_disposition_release_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release prompt_evaluation_result_release_receipt_review_disposition_release_format=contract-only-offline-evaluation-result-release-receipt-review-disposition-release prompt_evaluation_result_release_receipt_review_disposition_release_decision=blocked_contract_only prompt_evaluation_result_release_receipt_review_disposition_release_plan_recorded=1 prompt_evaluation_result_release_receipt_review_disposition_release_result_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_runtime_invoked=0 prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_decision_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_approval_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_rejection_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_findings_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_published=0 prompt_evaluation_result_release_receipt_review_disposition_release_packaged=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0 requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract=1 prompt_evaluation_result_release_receipt_review_disposition_release_promotion_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_stage=39-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-contract prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt prompt_evaluation_result_release_receipt_review_disposition_release_receipt_stage=contract-only prompt

```
_evaluation_result_release_receipt_review_disposition_release_receipt_contract_status=con
tract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_performed=0 p
rompt_evaluation_result_release_receipt_review_disposition_release_receipt_metadata_prese
nt=1 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_family=o
perator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release-rece
ipt prompt_evaluation_result_release_receipt_review_disposition_release_receipt_format=co
ntract-only-offline-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_decision=bloc
ked_contract_only prompt_evaluation_result_release_receipt_review_disposition_release_rec
eipt_plan_recorded=1 prompt_evaluation_result_release_receipt_review_disposition_release_
receipt_result_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_rel
ease_receipt_runtime_invoked=0 prompt_evaluation_result_release_receipt_review_dispositio
n_release_receipt_record_created=0 prompt_evaluation_result_release_receipt_review_dispos
ition_release_receipt_decision_recorded=0 prompt_evaluation_result_release_receipt_review
_disposition_release_receipt_approval_recorded=0 prompt_evaluation_result_release_receipt
_review_disposition_release_receipt_rejection_recorded=0 prompt_evaluation_result_release
_receipt_review_disposition_release_receipt_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_packaged=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1 r
equires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receip
t_review_contract=1 prompt_evaluation_result_release_receipt_review_disposition_release_r
eceipt_promotion_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_re
lease_receipt_review_contract_stage=40-prompt-evaluation-result-release-receipt-review-di
sposition-release-receipt-review-contract prompt_evaluation_result_release_receipt_review
_disposition_release_receipt_review_contract_command=scripts/nadia-prompt-evaluation-resu
lt-release-receipt-review-disposition-release-receipt-review-contract.sh installed_prompt
_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_com
mand=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review prompt
_evaluation_result_release_receipt_review_disposition_release_receipt_review_stage=contra
ct-only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_contract_status=contract_only prompt_evaluation_result_release_receipt_review_dispositi
on_release_receipt_review_authority=0 prompt_evaluation_result_release_receipt_review_dis
position_release_receipt_review_allowed=0 prompt_evaluation_result_release_receipt_review
_disposition_release_receipt_review_recorded=0 prompt_evaluation_result_release_receipt_r
eview_disposition_release_receipt_review_created=0 prompt_evaluation_result_release_recei
pt_review_disposition_release_receipt_review_performed=0 prompt_evaluation_result_release
_receipt_review_disposition_release_receipt_review_record_created=0 prompt_evaluation_res
ult_release_receipt_review_disposition_release_receipt_review_decision_recorded=0 prompt_
evaluation_result_release_receipt_review_disposition_release_receipt_review_published=0 p
rompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_signed=
0 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_rec
eipt_review_disposition_contract=1 prompt_evaluation_result_release_receipt_review_dispos
ition_release_receipt_review_disposition_contract_stage=41-prompt-evaluation-result-relea
se-receipt-review-disposition-release-receipt-review-disposition-contract prompt_evaluati
on_result_release_receipt_review_disposition_release_receipt_review_disposition_contract_
command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release
```

-receipt-review-disposition-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_stage=contract-only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_contract_status=contract_only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_authority=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_created=0 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prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_stage=contract-only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_status=contract_only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_authority=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_performed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_record_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_decision_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_published=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_created=0 requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_contract=1 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_contract=1 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_contract_stage=43-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-contract prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_stage=contract-only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie w_disposition_release_receipt_contract_status=contract_only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_authority=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_performed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_record_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_decision_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_emitted=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_signed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_published=0 requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_contract=1 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract=1 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract_stage=44-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-contract prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_stage=contract-only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract_status=contract_only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_authority=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_performed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_record_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_decision_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_findings_recorded=0 requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_contract=1 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_contract=1 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_contract_stage=45-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-contract prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-contract.sh installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disp osition prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_stage=contract-only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_contract_status=contract_only prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_authority=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_allowed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_performed=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_record_created=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_decision_recorded=0 prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_findings_recorded=0 requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract=1 requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_contract=1 requires_context_pack=1 requires_runtime_profile=1 human_dignity_principle=1 survivor_witness_respect=1 community_awareness_posture=1 harm_aware_development=1 model_runtime_present=0 model_runtime_invoked=0 inference_performed=0 inference_authority=0 runtime_invoked=0 prompt_evaluated=0 model_weights_installed=0 model_weights_loaded=0 model_weights_copied=0 model_weights_downloaded=0 model_weights_inspected=0 tool_execution_authority=0 source_mutation_authority=0

[behavior] create_prefix_layout=true create_component_markers=true create_cli_shims=true preserve_existing_files=true build_gui_installer=false build_latticra_from_source=false install_payload_tree=true install_desktop_entry=false install_user_bin_wrappers=true

installer/README.md

Source title: Latticra Panel

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Latticra Panel

Graphical installer and first-run control panel for Latticra, Lat, LIR, Latticra Seal, and the Nadia offline AI foundation. The current Panel package version is v0.5.0.

The panel is designed as the main first impression for Latticra. It opens as a maximized, resizable GUI workbench with guided defaults, visible authority boundaries, component configuration, delivery controls, plan/evidence review, and an embedded Latticra Console for panel-aware commands.

Installer-specific contracts are indexed in docs/README.md. Start there when reviewing installer authority, UI configuration, install-button behavior, receipts, or evidence wording.

Prerequisites

Ubuntu:

```
sudo apt-get update
sudo apt-get install -y rustc cargo make gcc pkg-config \
  libx11-dev libxcb1-dev libxcursor-dev libxrandr-dev libxi-dev \
  libxkbcommon-dev libgl1-mesa-dev libwayland-dev desktop-file-utils \
  libgtk-3-bin
```

Fedora:

```
sudo dnf install -y rust cargo make gcc pkgconf-pkg-config \
  libX11-devel libxcb-devel libXcursor-devel libXrandr-devel libXi-devel \
  libxkbcommon-devel mesa-libGL-devel wayland-devel desktop-file-utils gtk3
```

openSUSE:

```
sudo zypper refresh
sudo zypper install -y rust cargo make gcc pkgconf \
  libX11-devel libxcb-devel libXcursor-devel libXrandr-devel libXi-devel \
  libxkbcommon-devel Mesa-libGL-devel wayland-devel desktop-file-utils \
  gtk3-tools
```

User-local command path:

```
export PATH=&quot;$HOME/.local/bin:$PATH&quot;
```

Run from source

```
make -C installer gui
```

Equivalent direct command:

```
cd installer/latticra-installer
LATTICRA_INSTALLER_ROOT=&quot;$PWD/..&quot; cargo run
```

First-run flow

1. Open Guided Workbench.
2. Keep dry-run mode enabled.
3. Generate and inspect the plan.
4. Run Dry-Install to validate and write a receipt.
5. Review the embedded console, plan, and engine log.
6. Enable guarded local-prefix writes only after the dry-run evidence looks correct.

The LC Standalone profile narrows that flow to the Console component only. It records lc-standalone-install-v0, keeps lc.profile = "standalone", installs standalone/session/workspace/namespace/rootfs/packages/init contracts, disables Panel embedding for the installed console, and preserves the no-effect host and network authority floor.

SeaBIOS and GRUB compatibility boundary

The current Panel installer is compatible with SeaBIOS and GRUB hosts by staying out of the boot path. It installs only to a guarded user-local prefix and does not write firmware, partitions, boot sectors, EFI variables, GRUB configuration, kernel images, initramfs files, services, or drivers.

The guarded SeaBIOS and GRUB compatibility contract is ../docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md.

The SeaBIOS and GRUB boot-preview evidence contract is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CONTRACT.md. Its fixture manifest is manifests/seabios-grub-boot-preview.toml and preserves a non-booted preview lane for future QEMU, serial-console, checksum, and recovery-path evidence.

The SeaBIOS and GRUB boot-preview preflight is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_PREFLIGHT.md. It is available as sh scripts/seabios-grub-boot-preview-preflight.sh and emits a no-effect tool-visibility and manifest-validity report.

The SeaBIOS and GRUB boot-preview evidence capture template is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CAPTURE_TEMPLATE.md. It is available as sh scripts/seabios-grub-boot-preview-evidence-template.sh and emits the future evidence bundle shape without running a VM.

The SeaBIOS and GRUB boot-preview evidence validation is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_VALIDATION.md. It is available as sh scripts/seabios-grub-boot-preview-evidence-validate.sh and rejects forged QEMU, serial-console, recovery, and bootability claims from the current fixture lane.

The SeaBIOS and GRUB boot-preview QEMU argv template is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_QEMU_ARGV_TEMPLATE.md. It is available as sh scripts/seabios-grub-boot-preview-qemu-argv-template.sh and emits future profile-specific QEMU argv record placeholders without running QEMU.

The SeaBIOS and GRUB boot-preview boot artifact manifest template is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_TEMPLATE.md. It is available as sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh and emits future artifact metadata placeholders without creating images or writing boot files.

The SeaBIOS and GRUB boot-preview boot artifact manifest validation is ../docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_VALIDATION.md. It is available as sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh and rejects premature bootable, GRUB, QEMU, or production OS claims from the current fixture lane.

```
installer_ready_for_user_local_panel=1
installer_boot_safe_by_absence=1
boot_preview_manifest_fixture_present=1
seabios_grub_boot_preview_preflight_present=1
seabios_grub_boot_preview_evidence_capture_template_present=1
seabios_grub_boot_preview_evidence_validation_present=1
seabios_grub_boot_preview_qemu_argv_template_present=1
seabios_grub_boot_preview_boot_artifact_manifest_template_present=1
seabios_grub_boot_preview_boot_artifact_manifest_validation_present=1
firmware_mutation_allowed=0
bootloader_write_allowed=0
partition_mutation_allowed=0
grub_install_allowed=0
efibootmgr_allowed=0
qemu_execution_allowed_by_guard=0
qemu_boot_execution_attempted=0
qemu_argv_record_ready=0
boot_evidence_record_ready=0
boot_evidence_candidate_ready=0
boot_artifact_manifest_candidate_ready=0
boot_artifact_manifest_ready=0
bootable_os_ready=0
production_installer_ready=0
```

Future OS-base work must add QEMU/VM evidence for SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI paths before any bootable image claim changes.

Embedded Latticra Console

The console in the upper-right of the panel is not a shell. It is a panel-aware operator console for common actions and local navigation only:

```
help
status
lc status
lc commands
lc substrate
lc host
lc os
plan
save
dry-run
updater status
updater dry-run
updater apply
reset
nadia status
nadia commands
nadia context
nadia runtime
```

```
nadia plan
nadia mode
nadia ledger
nadia safety
nadia tool
nadia prompt-contract
nadia model-registry
nadia inference-readiness
nadia runtime-invocation
nadia model-load
nadia prompt-receipt
nadia prompt-materialization
nadia awareness-dialogue
nadia prompt-evaluation-handoff
nadia tokenization-boundary
nadia tokenizer-specification
nadia tokenizer-manifest
nadia tokenizer-artifact-inventory
nadia tokenizer-artifact-measurement
nadia tokenizer-artifact-verification
nadia tokenizer-artifact-binding
nadia tokenizer-runtime-attachment
nadia prompt-tokenization
nadia prompt-token-sequence
nadia context-window-assembly
nadia prompt-evaluation-input
nadia prompt-evaluation-runtime-handoff
nadia prompt-evaluation-invocation
nadia prompt-evaluation-result
nadia prompt-evaluation-result-review
nadia prompt-evaluation-result-disposition
nadia prompt-evaluation-result-release
nadia prompt-evaluation-result-release-receipt
nadia prompt-evaluation-result-release-receipt-review
nadia prompt-evaluation-result-release-receipt-review-disposition
nadia prompt-evaluation-result-release-receipt-review-disposition-release
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tion-release-receipt-review
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review-disposition
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review-disposition-release
profile guided
profile seal
profile fedora
mode dry
mode local
pwd
cd <path>
clear
```

External host processes are not launched from the embedded console. Installation and dry-run behavior must use dedicated panel commands or buttons such as plan, save, dry-run, mode dry, and mode local.

Update from the Panel

Use the Updater workspace in Latticra Panel to preview and apply managed user-local updates from the current reviewed source checkout. The updater reuses the guarded installer engine, keeps network fetch authority disabled, and requires a successful updater dry-run before guarded apply by default.

The installed umbrella command can report the Panel-owned updater policy without launching the GUI:

```
latticra updater status
```

That report includes the configured source strategy, update channel, updater dry-run preview command, updater apply command, guarded apply mode, receipt setting, and disabled network/root/system mutation authority.

Dry-run

```
make -C installer dry-run
```

Standalone LC dry-run

```
make -C installer lc-standalone-dry-run
```

This uses configs/lc-standalone.installer.toml and previews a standalone LC wrapper, registry, profile presets, and contract files without requiring Panel at runtime.

Standalone LC local install

```
make -C installer lc-standalone-local
```

This uses configs/lc-standalone-local.installer.toml and runs the guarded user-local standalone LC install path. It keeps Panel GUI build and desktop integration disabled while installing the direct latticra-lc wrapper and LC metadata.

Install locally

```
make -C installer local-example
```

Verify

```
make -C installer verify-local
```

Open after install

```
latticra-panel
```

Or from the desktop app grid, open Latticra Panel.

Installed paths

```
~/.local/bin/latticra
~/.local/bin/&lt;lc.install.command_wrapper&gt; (default: latticra-lc; when LC wrapper enabled)
~/.local/bin/lat
~/.local/bin/latticra-seal
~/.local/bin/latticra-nadia (when enabled)
~/.local/bin/latticra-panel
~/.local/bin/latticra-installer (compatibility)
~/.local/share/applications/latticra-panel.desktop
~/.local/share/icons/hicolor/256x256/apps/latticra-panel.png
~/.local/share/latticra
```

Latticra Console (LC)

LC is installed as the configurable operator base for Latticra substrate, Panel, standalone console use, and future host-embedded workflows. In this first slice it is metadata-only:

The default direct wrapper is latticra-lc; custom Panel installs use lc.install.command_wrapper. It is also the standalone LC wrapper, does not require Panel at runtime, and the umbrella latticra lc ... route stays stable across wrapper names.

```
latticra lc status
latticra lc install-config
latticra-lc commands
latticra-lc install-config
latticra-lc session
latticra-lc workspace
latticra-lc namespace
latticra-lc rootfs
latticra-lc packages
latticra-lc init
latticra-lc substrate
latticra-lc host
latticra-lc os
```

The Panel LC workspace includes LC install configuration for the local config path, share path, wrapper command, profile presets, command registry, the session envelope contract, workspace envelope contract, namespace envelope contract, rootfs contract, packages contract, init contract, contract files, and embedded Panel bridge. External host command launch remains disabled.

LC does not launch external host commands, create runtime sessions, mount workspaces, mount namespaces, create or open rootfs images, read package catalogs, download packages, run package managers, install packages, claim PID 1, start services, supervise processes, mutate host files, use the network, grant runtime enforcement authority, boot hardware, or claim to be a production operating system.

Nadia offline AI foundation

Nadia is Latticra's planned offline AI companion for software development, systems engineering, and AI development work. The name honors Nobel Peace Prize laureate Nadia Murad and keeps human dignity, survivor-witness respect, community awareness, harm-aware development, and an absolute non-sexual-use boundary visible in the system direction. Documentation and code identify the solemn implementation identity as Nadia Witness Foundation while the human-facing interactive name remains Nadia.

In the current installer lane, Nadia includes Stage-46 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract metadata, Stage-45 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract metadata, Stage-44 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract metadata, Stage-43 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract metadata, Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release contract metadata, Stage-41 prompt-evaluation result release receipt review disposition release receipt review disposition contract metadata, Stage-40 prompt-evaluation result release receipt review disposition release receipt review contract metadata, Stage-39 prompt-evaluation result release receipt review disposition release receipt contract metadata, Stage-38 prompt-evaluation result release receipt review disposition release contract metadata, Stage-37 prompt-evaluation result release receipt review disposition contract metadata, Stage-36 prompt-evaluation result release receipt review contract metadata, Stage-35 prompt-evaluation result release receipt contract metadata, Stage-34 prompt-evaluation result release contract metadata, Stage-33 prompt-evaluation result disposition contract metadata, Stage-32 prompt-evaluation result review contract metadata, Stage-31 prompt-evaluation result contract metadata, Stage-30 prompt-evaluation invocation contract metadata, Stage-29 prompt-evaluation runtime handoff contract metadata, Stage-28 prompt-evaluation-input contract metadata, Stage-27 context-window-assembly contract metadata, Stage-26 prompt-token-sequence contract metadata, Stage-25 prompt-tokenization contract metadata, Stage-24 tokenizer-runtime-attachment contract metadata, Stage-23 tokenizer-artifact-binding contract metadata, Stage-22 tokenizer-artifact-verification contract metadata, Stage-21 tokenizer-artifact-measurement contract metadata, Stage-20 tokenizer-artifact-inventory contract metadata, Stage-19 tokenizer-manifest contract metadata, Stage-18 tokenizer-specification contract metadata, Stage-17 tokenization-boundary contract metadata, Stage-16 prompt-evaluation handoff contract metadata, Stage-15 awareness-dialogue contract metadata, Stage-14 prompt-materialization contract metadata, Stage-13 prompt-receipt contract metadata, Stage-12 model-load contract metadata, Stage-11 runtime-invocation contract metadata, Stage-10 inference-readiness contract metadata, Stage-9 local model-registry contract metadata, Stage-8 prompt-evaluation contract metadata, Stage-7 report-only tool-preflight metadata, Stage-6 protective-safety metadata, Stage-5 productivity-ledger metadata, Stage-4 systems-engineering mode validation, Stage-3 prompt-plan metadata, Stage-2 runtime-profile metadata, Stage-1 local context-pack generation, Stage-0 identity, config, Console status, component marker, and productivity-ledger space. No sexual user functionality, dialogue generation, prompt text receipt, prompt text reading, prompt source reading, prompt buffer allocation, prompt tokenization, prompt token creation, prompt token sequence recording, prompt token ID recording, prompt token order recording, prompt token offset recording, context window assembly, prompt evaluation input creation, prompt evaluation runtime handoff, prompt-evaluation invocation request creation, prompt-evaluation result recording, prompt-evaluation result review recording, prompt-evaluation result disposition recording, disposition decision recording, prompt-evaluation result release recording, release decision recording, release publication, release packaging, release receipt creation, prompt-evaluation result release receipt recording, prompt-evaluation result release receipt review recording, review decision recording, review findings recording, receipt signing, receipt publication, model-output recording, prompt materialization, tokenizer artifact path resolution, tokenizer artifact opening, tokenizer artifact reading, tokenizer artifact scanning, tokenizer artifact hashing, tokenizer artifact measurement, tokenizer artifact digest recording, tokenizer artifact size recording, tokenizer artifact verification, tokenizer artifact digest comparison, tokenizer artifact size comparison, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, inference, prompt evaluation, tool execution, model installation, training, distillation, source mutation, or network authority are installed. Awareness-dialogue metadata defines future survivor-centered Q&A scope for official Nadia Initiative awareness topics, prompt-evaluation handoff metadata packages that evidence, tokenization-boundary metadata keeps prompt tokenization blocked, tokenizer-specification metadata records future review requirements, tokenizer-manifest metadata records future manifest review requirements, tokenizer-artifact-inventory metadata records future artifact measurement requirements, tokenizer-artifact-measurement metadata records future artifact verification requirements, tokenizer-artifact-verification metadata records future artifact binding requirements, tokenizer-artifact-binding metadata records future runtime attachment requirements, tokenizer-runtime-attachment metadata records future prompt tokenization requirements, prompt-tokenization metadata records future prompt token sequence requirements, prompt-token-sequence metadata records future context window assembly requirements, context-window-assembly metadata records future prompt evaluation input requirements, prompt-evaluation-input metadata records future prompt evaluation runtime handoff requirements, prompt-evaluation-runtime-handoff metadata records future prompt evaluation invocation requirements, prompt-evaluation-invocation metadata records future prompt evaluation result requirements, prompt-evaluation-result metadata records future prompt evaluation result review requirements, prompt-evaluation-result-review metadata records future prompt evaluation result disposition requirements, prompt-evaluation-result-disposition metadata records future prompt evaluation result release requirements, prompt-evaluation-result-release metadata records future prompt evaluation result release receipt requirements, prompt-evaluation-result-release-receipt metadata records future prompt evaluation result release receipt review requirements, prompt-evaluation-result-release-receipt-review metadata records future prompt evaluation result release receipt review disposition requirements, prompt-evaluation-result-release-receipt-review-disposition metadata records future prompt evaluation result release receipt review disposition release requirements, prompt-evaluation-result-release-receipt-review-disposition-release metadata records future prompt evaluation result release receipt review disposition release receipt requirements, prompt-evaluation-result-release-receipt-review-disposition-release-receipt metadata records future prompt evaluation result release receipt review disposition release receipt review requirements, prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review metadata records future prompt evaluation result release receipt review disposition release receipt review disposition requirements, prompt-evaluation-result-release-receipt

-review-disposition-release-receipt-review-disposition metadata records future prompt evaluation result release receipt review disposition release receipt review disposition release requirements, prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release metadata records future prompt evaluation result release receipt review disposition release receipt review disposition release receipt requirements, prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt metadata records future prompt evaluation result release receipt review disposition release receipt review disposition release receipt review requirements, and prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review metadata records future prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition requirements, prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition metadata records future prompt evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release requirements, including prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release metadata only, prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review metadata only, prompt-evaluation result release receipt review disposition release receipt review disposition release receipt metadata only, prompt-evaluation result release receipt review disposition release receipt review disposition release metadata only, prompt-evaluation result release receipt review disposition release receipt review disposition metadata only, prompt-evaluation result release receipt review disposition release receipt review metadata only, prompt-evaluation result release receipt review disposition release receipt recording metadata only, prompt-evaluation result release receipt review disposition release recording metadata only, and prompt-evaluation result release receipt review disposition recording metadata only; none grants dialogue generation, live web lookup, prompt evaluation, token generation, inference, or tool execution authority.

Stage-40 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review as prompt-evaluation result release receipt review disposition release receipt review metadata only. Prompt-evaluation result release receipt review disposition release receipt review recording remains blocked metadata only.

Stage-41 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition as prompt-evaluation result release receipt review disposition release receipt review disposition metadata only. Prompt-evaluation result release receipt review disposition release receipt review disposition recording remains blocked metadata only.

Stage-42 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release as prompt-evaluation result release receipt review disposition release receipt review disposition release metadata only. The prompt-evaluation result release receipt review disposition release receipt review disposition release recording remains blocked metadata only.

Stage-43 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt as prompt-evaluation result release receipt review disposition release receipt review disposition release receipt metadata only. The prompt-evaluation result release receipt review disposition release receipt review disposition release receipt recording remains blocked metadata only.

Stage-44 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review as prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review metadata only. The prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review recording remains blocked metadata only.

Stage-45 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition as prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition metadata only. The prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition recording remains blocked metadata only.

Stage-46 adds latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release as prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release metadata only. The prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release recording remains blocked metadata only.

After a guarded local install with Nadia enabled:

```
latticra-nadia commands
latticra-nadia context-pack
latticra-nadia runtime-profile
latticra-nadia prompt-plan
latticra-nadia mode-validate
latticra-nadia productivity-ledger
latticra-nadia protective-safety
latticra-nadia tool-preflight
latticra-nadia prompt-contract
latticra-nadia model-registry
latticra-nadia inference-readiness
latticra-nadia runtime-invocation
latticra-nadia model-load
latticra-nadia prompt-receipt
latticra-nadia prompt-materialization
latticra-nadia awareness-dialogue
latticra-nadia prompt-evaluation-handoff
latticra-nadia tokenization-boundary
latticra-nadia tokenizer-specification
latticra-nadia tokenizer-manifest
latticra-nadia tokenizer-artifact-inventory
latticra-nadia tokenizer-artifact-measurement
latticra-nadia tokenizer-artifact-verification
latticra-nadia tokenizer-artifact-binding
latticra-nadia tokenizer-runtime-attachment
latticra-nadia prompt-tokenization
latticra-nadia prompt-token-sequence
latticra-nadia context-window-assembly
latticra-nadia prompt-evaluation-input
latticra-nadia prompt-evaluation-runtime-handoff
latticra-nadia prompt-evaluation-invocation
latticra-nadia prompt-evaluation-result
latticra-nadia prompt-evaluation-result-review
latticra-nadia prompt-evaluation-result-disposition
latticra-nadia prompt-evaluation-result-release
latticra-nadia prompt-evaluation-result-release-receipt
latticra-nadia prompt-evaluation-result-release-receipt-review
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt
latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt-review
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt-review-disposition
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt-review-disposition-release
```

```
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review-disposition
nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposi
tion-release-receipt-review-disposition-release
```

Uninstall managed local install

```
make -C installer reset-dry-run
make -C installer reset-local
make -C installer uninstall-dry-run
make -C installer uninstall-local
latticra reset --dry-run
latticra reset
latticra uninstall --dry-run
latticra uninstall
```

Reset and uninstall remove the same managed artifacts: command wrappers, the Panel desktop entry, known Panel icons, and the selected local prefix. Use reset when the intent is to reinstall from new Panel specifications; use uninstall when the intent is to remove the local install. Unmanaged files in ~/.local/bin are preserved.

Clean full user-local uninstall

Use this only when normal reset/uninstall cannot clean up an old or broken user-local install. The commands below are intentionally scoped to Latticra user-local paths.

If LC was installed with a custom lc.install.command_wrapper, set LC_WRAPPER to that command name; the default is latticra-lc.

```
LATTICRA_PREFIX=&quot;${LATTICRA_PREFIX:-$HOME/.local/share/latticra}&quot;
LC_WRAPPER=&quot;${LC_WRAPPER:-latticra-lc}&quot;

rm -rf -- \
  &quot;$LATTICRA_PREFIX&quot; \
  &quot;$HOME/.local/share/latticra-validation&quot; \
  &quot;$HOME/.local/share/latticra-reset-receipts&quot;

rm -f -- \
  &quot;$HOME/.local/bin/latticra&quot; \
  &quot;$HOME/.local/bin/$LC_WRAPPER&quot; \
  &quot;$HOME/.local/bin/lat&quot; \
  &quot;$HOME/.local/bin/latticra-seal&quot; \
  &quot;$HOME/.local/bin/latticra-nadia&quot; \
  &quot;$HOME/.local/bin/latticra-panel&quot; \
  &quot;$HOME/.local/bin/latticra-installer&quot; \
  &quot;$HOME/.local/share/applications/latticra-panel.desktop&quot; \
  &quot;$HOME/.local/share/applications/latticra-installer.desktop&quot; \
  &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-panel.png&quot; \
  &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-installer.png&quot; \
  &quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-seal.png&quot;
```

If those exact paths are root-owned because an earlier command was run with sudo, repeat the same exact cleanup with sudo:

```
LATTICRA_PREFIX=&quot;${LATTICRA_PREFIX:-$HOME/.local/share/latticra}&quot;
LC_WRAPPER=&quot;${LC_WRAPPER:-latticra-lc}&quot;

sudo rm -rf -- \
  &quot;$LATTICRA_PREFIX&quot; \
  &quot;$HOME/.local/share/latticra-validation&quot; \
  &quot;$HOME/.local/share/latticra-reset-receipts&quot;

sudo rm -f -- \
  &quot;$HOME/.local/bin/latticra&quot; \
  &quot;$HOME/.local/bin/$LC_WRAPPER&quot; \
  &quot;$HOME/.local/bin/lat&quot; \
  &quot;$HOME/.local/bin/latticra-seal&quot; \
```

```
&quot;$HOME/.local/bin/latticra-nadia&quot; \
&quot;$HOME/.local/bin/latticra-panel&quot; \
&quot;$HOME/.local/bin/latticra-installer&quot; \
&quot;$HOME/.local/share/applications/latticra-panel.desktop&quot; \
&quot;$HOME/.local/share/applications/latticra-installer.desktop&quot; \
&quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-panel.png&quot; \
&quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-installer.png&quot; \
&quot;$HOME/.local/share/icons/hicolor/256x256/apps/latticra-seal.png&quot;
```

Do not run sudo rm -rf against broad paths such as ~/.local, /usr, /, or unreviewed wildcards.

Safety baseline

```
no root
no kernel mutation
no systemd mutation
no SELinux mutation
no network authority
user-local prefix only
```

installer/TREE.txt

Source title: Tree

Lines: 31 | Words: 31 | SHA-256: 9624c03f0485a55510e45f6ed018ec0d3e4c74f1134d7531a5699bda42a2e78f

installer/ Makefile README.md INSTALLER_ROADMAP.md TREE.txt configs/ default.installer.toml lc-standalone.installer.toml lc-standalone-local.installer.toml local-prefix-example.installer.toml docs/ INSTALL_BUTTON_EXECUTION_MODEL.md INSTALLER_READINESS_CONTRACT.md RECEIPTS_AND_EVIDENCE.md UI_CONFIGURATION_MODEL.md manifests/ components.toml scripts/ latticra-installer-apply.sh latticra-installer-dry-run.sh latticra-installer-uninstall.sh latticra-installer-verify-lc-standalone.sh latticra-installer-verify.sh latticra-installer/ Cargo.toml Cargo.lock assets/latticra-seal.png src/main.rs src/config.rs src/engine.rs src/ui.rs

Part VIII - Nadia Offline AI Foundation

- docs/NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15.md - Nadia Awareness Dialogue Contract Stage-15
- docs/NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27.md - Nadia Context Window Assembly Contract Stage-27
- docs/NADIA_DEVELOPER_WORKBENCH_STAGE_3.md - Nadia Developer Workbench Stage-3
- docs/NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7.md - Nadia Guarded Tool Authority Stage-7
- docs/NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10.md - Nadia Inference Readiness Contract Stage-10
- docs/NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1.md - Nadia Local Context Engine Stage-1
- docs/NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9.md - Nadia Local Model Registry Contract Stage-9
- docs/NADIA_MODEL_LOAD_CONTRACT_STAGE_12.md - Nadia Model Load Contract Stage-12
- docs/NADIA_OFFLINE_AI_FOUNDATION.md - Nadia Offline AI Foundation
- docs/NADIA_PRODUCTIVITY_LOOP_STAGE_5.md - Nadia Productivity Loop Stage-5
- docs/NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8.md - Nadia Prompt Evaluation Contract Stage-8

- docs/NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16.md - Nadia Prompt Evaluation Handoff Contract Stage-16
- docs/NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28.md - Nadia Prompt Evaluation Input Contract Stage-28
- docs/NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30.md - Nadia Prompt Evaluation Invocation Contract Stage-30
- docs/NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31.md - Nadia Prompt Evaluation Result Contract Stage-31
- docs/NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33.md - Nadia Prompt Evaluation Result Disposition Contract Stage-33
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34.md - Nadia Prompt Evaluation Result Release Contract Stage-34
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35.md - Nadia Prompt Evaluation Result Release Receipt Contract Stage-35
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36.md - Nadia Prompt Evaluation Result Release Receipt Review Contract Stage-36
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_37.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Contract Stage-37
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_38.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Contract Stage-38
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_39.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract Stage-39
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_40.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract Stage-40
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_41.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-41
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_42.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-42
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_43.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract Stage-43
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_44.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract Stage-44
- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_45.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-45

- docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-46
- docs/NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32.md - Nadia Prompt Evaluation Result Review Contract Stage-32
- docs/NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29.md - Nadia Prompt Evaluation Runtime Handoff Contract Stage-29
- docs/NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14.md - Nadia Prompt Materialization Contract Stage-14
- docs/NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13.md - Nadia Prompt Receipt Contract Stage-13
- docs/NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26.md - Nadia Prompt Token Sequence Contract Stage-26
- docs/NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25.md - Nadia Prompt Tokenization Contract Stage-25
- docs/NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6.md - Nadia Protective Safety Boundary Stage-6
- docs/NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11.md - Nadia Runtime Invocation Contract Stage-11
- docs/NADIA_RUNTIME_PROFILE_STAGE_2.md - Nadia Runtime Profile Stage-2
- docs/NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4.md - Nadia Systems Engineering Mode Stage-4
- docs/NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17.md - Nadia Tokenization Boundary Contract Stage-17
- docs/NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23.md - Nadia Tokenizer Artifact Binding Contract Stage-23
- docs/NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20.md - Nadia Tokenizer Artifact Inventory Contract Stage-20
- docs/NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21.md - Nadia Tokenizer Artifact Measurement Contract Stage-21
- docs/NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22.md - Nadia Tokenizer Artifact Verification Contract Stage-22
- docs/NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19.md - Nadia Tokenizer Manifest Contract Stage-19
- docs/NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24.md - Nadia Tokenizer Runtime Attachment Contract Stage-24
- docs/NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18.md - Nadia Tokenizer Specification Contract Stage-18

docs/NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15.md

Source title: Nadia Awareness Dialogue Contract Stage-15

Lines: 146 | Words: 371 | SHA-256: ed6eb4726aa3e6ff7e3fa603c402356a22b01d69bf455c1a421908b0835ec679

Nadia Awareness Dialogue Contract Stage-15

Status: Stage-15 implementation contract Date: 2026-05-25 Scope: future Q&A dialogue scope for Nadia Initiative awareness topics before prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-15 gives Nadia an awareness-dialogue contract for future Q&A about the work Nadia Initiative actively raises awareness about.

The contract consumes a Stage-14 prompt-materialization contract, verifies inherited non-executing prompt, runtime, model, tool, and protective-safety posture, and records the awareness topic register and dialogue safety rules that must exist before any future prompt-evaluation handoff.

Capability

Stage-15 adds:

```
nadia_stage_15_awareness_dialogue_contract_present=1
nadia_awareness_dialogue_contract_generator_present=1
awareness_dialogue_contract_command=scripts/nadia-awareness-dialogue-contract.sh
installed_awareness_dialogue_contract_command=latticra-nadia awareness-dialogue
future_qa_dialogue_capability_planned=1
awareness_dialogue_contract_status=contract_only
awareness_dialogue_stage=contract-only
awareness_dialogue_authority=0
awareness_dialogue_allowed=0
dialogue_generation_authority=0
dialogue_generation_allowed=0
qa_dialogue_generated=0
dialogue_turns_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
plain_language_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
active_topic_update_authority=0
live_web_lookup_authority=0
dialogue_decision=blocked_contract_only
requires_official_source_snapshot=1
awareness_dialogue_promotion_allowed=0
```

Stage-15 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Awareness Topic Register

Nadia's future Q&A scope is limited to official Nadia Initiative awareness work and must remain non-graphic, respectful, and survivor-centered:

```
topic_yazidi_genocide_awareness=1
topic_survivor_voice_and_dignity=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_genocide_prevention=1
```

```
topic_justice_and_accountability=1
topic_women_peace_justice_security=1
topic_sinjar_reconstruction=1
topic_security_and_safe_return=1
topic_education_restoration=1
topic_healthcare_and_mental_health=1
topic_livelihoods_and_food_security=1
topic_wash_clean_water_sanitation_hygiene=1
topic_womens_empowerment=1
topic_legal_rights_and_reparations_awareness=1
topic_cultural_preservation_and_memorialization=1
topic_community_driven_survivor_centric_development=1
topic_responsible_support_and_digital_activism=1
```

Protective Dialogue Rules

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
```

Official Source Register

Future dialogue must be grounded in operator-reviewed offline snapshots from official Nadia Initiative sources:

```
source_nadia_initiative_home=https://www.nadiasinitiative.org/home
source_nadia_initiative_about=https://www.nadiasinitiative.org/nadias-initiative
source_nadia_initiative_approach=https://www.nadiasinitiative.org/our-approach
source_nadia_initiative_advocacy=https://www.nadiasinitiative.org/advocacy
source_nadia_initiative_womens_empowerment=https://www.nadiasinitiative.org/womens-empowerment
source_snapshot_generated=0
source_snapshot_loaded=0
source_fetch_performed=0
```

Stage-15 does not give Nadia live browsing authority.

Usage

With explicit evidence input:

```
sh scripts/nadia-awareness-dialogue-contract.sh \
  --prompt-materialization /path/to/latest-prompt-materialization-contract.txt \
  --request-class awareness-education \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-awareness-dialogue.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia awareness-dialogue
```

Non-Claims

Stage-15 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

```
sh scripts/test-nadia-awareness-dialogue-contract-stage-15.sh
```

Expected result:

```
nadia_awareness_dialogue_contract_stage_15: ok
```

docs/NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27.md

Source title: Nadia Context Window Assembly Contract Stage-27

Lines: 170 | Words: 548 | SHA-256: 9ada0d30021d836ce4482166f2c17033be7c98e36eab1795f09c40de464d1b4d

Nadia Context Window Assembly Contract Stage-27

Status: Stage-27 implementation contract

Scope: context-window assembly metadata before context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-27 gives Nadia a context-window assembly contract after the prompt-token-sequence contract is present.

The contract records how future reviewed context-window assembly must reference the Stage-26 prompt-token-sequence contract, prompt-tokenization boundary, tokenizer-runtime-attachment boundary, prompt-materialization boundary, prompt-receipt boundary, context-window layout policy, token-budget policy, truncation policy, ordering policy, attention-mask policy, position-ID policy, and prompt-evaluation-input policy. It preserves the absolute rule that Nadia cannot yet read prompt text, assemble context windows, create prompt evaluation inputs, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference. The script only measures the Stage-26 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-27 adds:

```
nadia_stage_27_context_window_assembly_contract_present=1
nadia_context_window_assembly_contract_generator_present=1
context_window_assembly_contract_command=scripts/nadia-context-window-assembly-contract.sh
installed_context_window_assembly_contract_command=latticra-nadia context-window-assembly
context_window_assembly_contract_status=contract_only
context_window_assembly_stage=contract-only
context_window_assembly_authority=0
context_window_assembly_allowed=0
context_window_assembly_performed=0
context_window_assembly_metadata_present=1
context_window_family=operator-reviewed-context-window-assembly
context_window_format=contract-only-offline-context-window
context_window_assembly_decision=blocked_contract_only
context_window_assembly_evidence_present=1
context_window_assembly_source_policy=operator-reviewed-offline
context_window_assembly_plan_recorded=1
context_window_assembly_method_planned=offline-context-window-policy-review
context_window_assembly_result_recorded=0
context_window_assembly_runtime_invoked=0
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
```

```
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_input_contract=1
context_window_assembly_promotion_allowed=0
```

Stage-27 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt token buffers, create prompt tokens, count prompt tokens, record prompt token IDs, record prompt token order, record prompt token offsets, create attention masks, create position IDs, assemble context windows, serialize context windows, create prompt evaluation inputs, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, bind tokenizer artifacts, attach tokenizers to a runtime, create runtime sessions, open tokenizer files, read tokenizer files, load tokenizer vocabularies, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Assembly Requirements

The context-window assembly contract names future review requirements:

```
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_tokenizer_runtime_attachment_reference=1
requires_prompt_materialization_reference=1
requires_prompt_receipt_reference=1
requires_context_window_layout_policy=1
requires_context_window_token_budget_policy=1
requires_context_window_truncation_policy=1
requires_context_window_ordering_policy=1
requires_attention_mask_policy=1
requires_position_id_policy=1
requires_prompt_evaluation_input_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_context_window_assembly=1
requires_no_prompt_evaluation_input_creation=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-27 preserves explicit context-window denial fields:

```
context_window_assembly_authority=0
context_window_assembly_allowed=0
context_window_assembly_open_authority=0
context_window_assembly_read_authority=0
context_window_assembly_write_authority=0
context_window_assembly_execute_authority=0
context_window_assembly_runtime_authority=0
context_window_assembly_prompt_evaluation_input_authority=0
context_window_assembly_context_write_authority=0
context_window_assembly_performed=0
context_window_assembly_result_recorded=0
context_window_assembly_runtime_invoked=0
context_window_loaded=0
context_window_opened=0
context_window_read=0
context_window_validated=0
context_window_assembled=0
context_window_bytes_read=0
context_window_hash_computed=0
context_window_entries_loaded=0
context_window_token_budget_recorded=0
context_window_entry_count_recorded=0
context_window_truncation_applied=0
context_window_serialized=0
context_window_file_written=0
prompt_evaluation_input_created=0
```

```
prompt_evaluation_input_materialized=0
prompt_evaluation_input_validated=0
prompt_evaluation_input_written=0
prompt_token_sequence_recorded=0
prompt_token_ids_recorded=0
prompt_token_order_recorded=0
prompt_token_offsets_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_tokenized=0
tokenizer_runtime_attachment_performed=0
tokenizer_attached_to_runtime=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
qa_dialogue_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-context-window-assembly-contract.sh \
  --prompt-token-sequence
reports/nadia/prompt-token-sequence/latest-prompt-token-sequence-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-context-window-assembly.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia context-window-assembly
```

Non-Claims

Stage-27 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, tokenizer runtime attachment layer, prompt tokenizer, token counter, token sequence recorder, token ID recorder, token offset recorder, context-window assembler, prompt-evaluation-input creator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-context-window-assembly-contract-stage-27.sh
```

Expected output:

```
nadia_context_window_assembly_contract_stage_27: ok
```

docs/NADIA_DEVELOPER_WORKBENCH_STAGE_3.md

Source title: Nadia Developer Workbench Stage-3

Lines: 102 | Words: 239 | SHA-256: cd71bf7daec3f0c9e40f41c1704e0485bd1faf82365a3156e76ef3d1708bdc1e

Nadia Developer Workbench Stage-3

Status: Stage-3 implementation contract Date: 2026-05-25 Scope: prompt-plan generation from local context-pack and runtime-profile evidence.

Purpose

Stage-3 gives Nadia a developer-workbench planning surface before prompt evaluation exists.

The workbench combines Stage-1 context-pack evidence and Stage-2 runtime-profile evidence into an auditable prompt plan. The plan is designed for future offline inference, but the current stage stops before prompt execution.

Capability

Stage-3 adds:

```
nadia_stage_3_developer_workbench_present=1
nadia_prompt_plan_generator_present=1
prompt_plan_command=scripts/nadia-prompt-plan.sh
installed_prompt_plan_command=latticra-nadia prompt-plan
requires_context_pack=1
requires_runtime_profile=1
context_pack_measured=1
runtime_profile_measured=1
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
```

Stage-3 does not summarize with AI, call an inference runtime, evaluate a prompt, mutate source, execute tools, or use the network.

Inputs

The prompt-plan generator requires:

```
context_pack=Stage-1 Nadia local context pack
runtime_profile=Stage-2 Nadia runtime profile
task=operator-provided task label
```

The default task label is latticra-development-planning.

Outputs

The prompt-plan generator writes:

```
nadia-prompt-plan-&lt;timestamp&gt;.txt
latest-prompt-plan.txt
```

The prompt plan records:

```
stage=3-developer-workbench-planning
context_pack_measurement=&lt;local measurement&gt;
runtime_profile_measurement=&lt;local measurement&gt;
task=&lt;operator task label&gt;
prompt_evaluated=0
inference_performed=0
source_mutation_authority=0
```

Usage

With explicit evidence files:

```
sh scripts/nadia-prompt-plan.sh \
  --context-pack /path/to/latest-context-pack.txt \
  --runtime-profile /path/to/latest-runtime-profile.txt \
  --task &quot;runtime boundary refactor planning&quot; \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-plans.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia prompt-plan
```

Non-Claims

Stage-3 Nadia is not an inference runtime, prompt evaluator, code generator, autonomous coding agent, source mutator, training system, network retriever, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-developer-workbench-stage-3.sh
```

Expected result:

```
nadia_developer_workbench_stage_3: ok
```

docs/NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7.md

Source title: Nadia Guarded Tool Authority Stage-7

Lines: 121 | Words: 263 | SHA-256: 6a50e3b2b3d39d5663d04949d3ba6712f1b836921cdaf8600a73fc3e104b4211

Nadia Guarded Tool Authority Stage-7

Status: Stage-7 implementation contract Date: 2026-05-25 Scope: report-only tool-authority preflight after protective-safety validation.

Purpose

Stage-7 gives Nadia a tool-authority preflight without granting tool execution.

The preflight consumes a Stage-6 protective-safety report, verifies the non-sexual-use and anti-manipulation boundary, and records whether a proposed tool class is inside a report-only planning boundary. It does not run the tool.

Capability

Stage-7 adds:

```
nadia_stage_7_guarded_tool_authority_present=1
nadia_tool_authority_preflight_present=1
tool_authority_preflight_command=scripts/nadia-tool-authority-preflight.sh
installed_tool_authority_preflight_command=latticra-nadia tool-preflight
requires_protective_safety=1
protective_safety_stage_required=6-protective-safety-boundary
tool_authority_stage=preflight-only
preflight_decision=report_only_no_execution
tool_execution_authority=0
tool_execution_performed=0
tool_selection_authority=0
shell_execution_authority=0
network_tool_authority=0
source_mutation_authority=0
destructive_action_authority=0
credential_access_authority=0
requires_operator_approval=1
requires_nucleus_gate=1
requires_runtime_boundary_gate=1
requires_seal_receipt=1
requires_protective_safety_boundary=1
authority_transition_allowed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-7 does not execute tools, evaluate prompts, call an inference runtime, mutate source, train, distill, or use the network.

Inputs

The tool-authority preflight requires:

```
protective_safety=Stage-6 Nadia protective-safety report
tool_class=operator-proposed tool class label
```

```
action=operator-reviewed action label
```

Initial report-only tool classes are:

```
metadata-read
report-only
test-recommendation
documentation-plan
local-evidence-review
```

Network, shell, destructive, source-mutation, credential, and secret-bearing classes fail closed.

Outputs

The preflight writes:

```
nadia-tool-preflight-&lt;timestamp&gt;.txt
latest-tool-preflight.txt
```

The report records:

```
stage=7-guarded-tool-authority-preflight
preflight_decision=report_only_no_execution
tool_execution_authority=0
tool_execution_performed=0
requires_protective_safety_boundary=1
authority_transition_allowed=0
```

Usage

With an explicit protective-safety report:

```
sh scripts/nadia-tool-authority-preflight.sh \
  --protective-safety /path/to/latest-protective-safety.txt \
  --tool-class local-evidence-review \
  --action &quot;review generated receipts&quot; \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tools.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia tool-preflight
```

Non-Claims

Stage-7 Nadia is not a tool executor, shell runner, network client, source mutator, credential reader, inference runtime, prompt evaluator, sexual assistant, roleplay surface, adult-content generator, training system, distillation system, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-guarded-tool-authority-stage-7.sh
```

Expected result:

```
nadia_guarded_tool_authority_stage_7: ok
```

docs/NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10.md

Source title: Nadia Inference Readiness Contract Stage-10

Lines: 139 | Words: 288 | SHA-256: 9f44a992afb5fe0e6b3977d50e20227d04d793d54818a76119faecdb372f9b66

Nadia Inference Readiness Contract Stage-10

Status: Stage-10 implementation contract Date: 2026-05-25 Scope: inference-readiness metadata before runtime invocation, model loading, prompt evaluation, inference, or tool execution.

Purpose

Stage-10 gives Nadia an inference-readiness contract without running inference.

The contract consumes a Stage-9 local model-registry contract, verifies inherited non-executing model, prompt, runtime, tool, and protective-safety posture, and records the review fields that would be required before any later runtime invocation contract can exist.

Capability

Stage-10 adds:

```
nadia_stage_10_inference_readiness_contract_present=1
nadia_inference_readiness_contract_generator_present=1
inference_readiness_contract_command=scripts/nadia-inference-readiness-contract.sh
installed_inference_readiness_contract_command=latticra-nadia inference-readiness
requires_model_registry_contract=1
model_registry_stage_required=9-local-model-registry-contract
inference_readiness_contract_status=contract_only
inference_readiness_stage=contract-only
inference_readiness_authority=0
inference_ready=0
readiness_decision=blocked_contract_only
readiness_evidence_present=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_provenance_review=1
requires_model_license_review=1
requires_refusal_policy_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_runtime_invocation_contract=1
readiness_promotion_allowed=0
runtime_invocation_authority=0
token_generation_authority=0
model_session_authority=0
model_selection_authority=0
model_load_authority=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_runtime_present=0
model_runtime_invoked=0
runtime_invoked=0
inference_authority=0
inference_performed=0
model_weights_installed=0
model_weights_loaded=0
model_weights_downloaded=0
model_weights_inspected=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-10 does not materialize prompts, evaluate prompts, load model weights, invoke a runtime, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inputs

The inference-readiness contract requires:

```
model_registry=Stage-9 Nadia local model-registry contract
request_class=operator request classification label
```

Sexualized request classifications fail closed.

Outputs

The contract writes:

```
nadia-inference-readiness-contract-&lt;timestamp&gt;.txt
latest-inference-readiness-contract.txt
```

The report records:

```
stage=10-inference-readiness-contract
inference_readiness_contract_status=contract_only
inference_readiness_authority=0
inference_ready=0
readiness_decision=blocked_contract_only
runtime_invocation_authority=0
model_runtime_invoked=0
inference_authority=0
inference_performed=0
model_weights_loaded=0
sexual_request_refusal=always
```

Usage

With explicit evidence input:

```
sh scripts/nadia-inference-readiness-contract.sh \
  --model-registry /path/to/latest-model-registry-contract.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-inference-readiness.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia inference-readiness
```

Non-Claims

Stage-10 Nadia is not an inference runtime, model loader, prompt evaluator, prompt materializer, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, or production AI assistant.

Validation

```
sh scripts/test-nadia-inference-readiness-contract-stage-10.sh
```

Expected result:

```
nadia_inference_readiness_contract_stage_10: ok
```

docs/NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1.md

Source title: Nadia Local Context Engine Stage-1

Lines: 113 | Words: 260 | SHA-256: 86700657c465735c8333536fb5c2010778413a2cd65f6687cb6d54c9130c5b51

Nadia Local Context Engine Stage-1

Status: Stage-1 implementation contract Date: 2026-05-25 Scope: no-network local context-pack generation for Nadia.

Purpose

Stage-1 gives Nadia a deterministic local context engine before any model runtime exists.

The context engine creates auditable context packs from local Latticra project files so future offline inference can begin from explicit, measured, inspectable source material instead of hidden memory or network retrieval.

Capability

Stage-1 adds:

```
nadia_stage_1_local_context_engine_present=1
nadia_context_pack_generator_present=1
context_pack_command=scripts/nadia-context-pack.sh
installed_context_pack_command=latticra-nadia context-pack
local_file_read_for_indexing=operator_invoked
local_context_pack_write=operator_selected_output
network_authority=0
model_runtime_present=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
```

The context engine records file paths, file classes, line counts, byte counts, and local measurements. It does not summarize with AI, embed text, call a model, use the network, or mutate source.

Inputs

The default context-pack roots are:

```
README.md
STATUS.md
SECURITY.md
CONTRIBUTING.md
docs/
include/
src/
tests/
scripts/
installer/
seal/
```

Generated build outputs, local receipts, and existing Nadia context-pack outputs are excluded.

Outputs

The context-pack generator writes:

```
nadia-context-pack-&lt;timestamp&gt;.txt
nadia-context-file-index-&lt;timestamp&gt;.tsv
latest-context-pack.txt
latest-file-index.tsv
```

The pack includes:

```
system_name=Latticra Nadia Witness Foundation
public_name=Nadia
interactive_name=Nadia
implementation_name=Nadia Witness Foundation
documentation_code_name=Nadia Witness Foundation
stage=1-local-context-engine
network_authority=0
model_runtime_present=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
human_dignity_principle=1
community_awareness_posture=1
```

Usage

From the repository:

```
sh scripts/nadia-context-pack.sh --repo . --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-context.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia context-pack
```

The installed command writes under the user-local Nadia context-pack directory.

Non-Claims

Stage-1 Nadia is not an inference runtime, autonomous coding agent, embedding service, training system, source mutator, network retriever, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-local-context-engine-stage-1.sh
```

Expected result:

```
nadia_local_context_engine_stage_1: ok
```

docs/NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9.md

Source title: Nadia Local Model Registry Contract Stage-9

Lines: 140 | Words: 321 | SHA-256: 3d4b739788a6010f81d9e793865249703fc508583783ec1062393b92829ed7a0

Nadia Local Model Registry Contract Stage-9

Status: Stage-9 implementation contract Date: 2026-05-25 Scope: local model-registry metadata before model selection, model installation, prompt evaluation, inference, or tool execution.

Purpose

Stage-9 gives Nadia a local model-registry contract without selecting or running a model.

The contract consumes a Stage-8 prompt-evaluation contract and a Stage-2 runtime profile, verifies the inherited non-executing safety posture, and records a local model candidate as metadata for later operator review.

Capability

Stage-9 adds:

```
nadia_stage_9_local_model_registry_contract_present=1
nadia_model_registry_contract_generator_present=1
model_registry_contract_command=scripts/nadia-local-model-registry-contract.sh
installed_model_registry_contract_command=latticra-nadia model-registry
requires_prompt_contract=1
prompt_contract_stage_required=8-prompt-evaluation-contract
requires_runtime_profile=1
runtime_profile_stage_required=2-runtime-profile-boundary
registry_contract_status=metadata_only
local_model_registry_stage=contract-only
model_registry_authority=0
candidate_recorded=1
candidate_review_status=operator_review_required
candidate_usable_for_inference=0
candidate_selected_for_runtime=0
model_selection_authority=0
model_install_authority=0
model_download_authority=0
model_copy_authority=0
model_load_authority=0
model_benchmark_authority=0
model_weight_inspection_authority=0
registry_promotion_allowed=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_runtime_present=0
model_runtime_invoked=0
inference_performed=0
model_weights_installed=0
model_weights_loaded=0
model_weights_copied=0
model_weights_downloaded=0
model_weights_inspected=0
```

```
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-9 does not download, install, copy, load, inspect, benchmark, or execute model weights. It does not materialize prompts, evaluate prompts, run inference, execute tools, mutate source, train, distill, or use the network.

Inputs

The local model-registry contract requires:

```
prompt_contract=Stage-8 Nadia prompt-evaluation contract
runtime_profile=Stage-2 Nadia runtime profile
model_id=operator-provided local candidate label
model_family=operator-provided family label
model_format=operator/runtime profile format label
quantization=operator-provided quantization label
source=operator-provided local provenance label
license=operator review label
```

Sexualized candidate labels fail closed.

Outputs

The contract writes:

```
nadia-model-registry-contract-&lt;timestamp&gt;.txt
latest-model-registry-contract.txt
```

The report records:

```
stage=9-local-model-registry-contract
registry_contract_status=metadata_only
local_model_registry_stage=contract-only
model_registry_authority=0
candidate_review_status=operator_review_required
candidate_usable_for_inference=0
model_selection_authority=0
model_install_authority=0
model_runtime_invoked=0
inference_performed=0
model_weights_installed=0
sexual_request_refusal=always
```

Usage

With explicit evidence inputs:

```
sh scripts/nadia-local-model-registry-contract.sh \
  --prompt-contract /path/to/latest-prompt-contract.txt \
  --runtime-profile /path/to/latest-runtime-profile.txt \
  --model-id local-coding-assistant-candidate \
  --quantization q4_k_m \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-model-registry.XXXXXX&quot;
)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia model-registry
```

Non-Claims

Stage-9 Nadia is not a model selector, inference runtime, model installer, model distributor, model downloader, weight inspector, benchmark runner, prompt evaluator, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, or production AI assistant.

Validation

```
sh scripts/test-nadia-local-model-registry-contract-stage-9.sh
```

Expected result:

```
nadia_local_model_registry_contract_stage_9: ok
```

docs/NADIA_MODEL_LOAD_CONTRACT_STAGE_12.md

Source title: Nadia Model Load Contract Stage-12

Lines: 155 | Words: 347 | SHA-256: 7a4c67dfcd0b146f76b9d469d8b7d73243a69a74b9b6ac000ef027a77b06cb66

Nadia Model Load Contract Stage-12

Status: Stage-12 implementation contract Date: 2026-05-25 Scope: model-load metadata before model file opening, weight mapping, weight verification, weight loading, runtime attachment, token generation, inference, prompt evaluation, or tool execution.

Purpose

Stage-12 gives Nadia a model-load contract without loading a model.

The contract consumes a Stage-11 runtime-invocation contract, verifies inherited non-executing invocation, readiness, model, prompt, runtime, tool, and protective-safety posture, and records the review fields that would be required before any later prompt-receipt contract can exist.

Capability

Stage-12 adds:

```
nadia_stage_12_model_load_contract_present=1
nadia_model_load_contract_generator_present=1
model_load_contract_command=scripts/nadia-model-load-contract.sh
installed_model_load_contract_command=latticra-nadia model-load
requires_runtime_invocation_contract=1
runtime_invocation_stage_required=11-runtime-invocation-contract
model_load_contract_status=contract_only
model_load_stage=contract-only
model_load_authority=0
model_load_allowed=0
model_loaded=0
load_decision=blocked_contract_only
load_evidence_present=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_provenance_review=1
requires_model_license_review=1
requires_model_safety_review=1
requires_model_weight_measurement_contract=1
requires_refusal_policy_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_prompt_receipt_contract=1
load_promotion_allowed=0
model_file_open_authority=0
model_weight_read_authority=0
model_weight_mapping_authority=0
model_weight_verification_authority=0
model_weight_inspection_authority=0
runtime_model_attach_authority=0
model_session_authority=0
token_generation_authority=0
model_file_opened=0
model_file_descriptor_opened=0
model_memory_map_created=0
```

```
model_weights_mapped=0
model_weights_loaded=0
model_weights_attached=0
model_weight_measurement_performed=0
model_weight_verification_performed=0
model_load_performed=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-12 does not materialize prompts, evaluate prompts, open model files, map model weights, verify model weights, load model weights, attach weights to a runtime, spawn a runtime process, create a runtime session, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inputs

The model-load contract requires:

```
runtime_invocation=Stage-11 Nadia runtime-invocation contract
request_class=operator request classification label
```

Sexualized request classifications fail closed.

Outputs

The contract writes:

```
nadia-model-load-contract-&lt;timestamp&gt;.txt
latest-model-load-contract.txt
```

The report records:

```
stage=12-model-load-contract
model_load_contract_status=contract_only
model_load_authority=0
model_load_allowed=0
model_loaded=0
load_decision=blocked_contract_only
model_file_opened=0
model_weights_mapped=0
model_weights_loaded=0
model_load_performed=0
token_generation_performed=0
inference_performed=0
sexual_request_refusal=always
```

Usage

With explicit evidence input:

```
sh scripts/nadia-model-load-contract.sh \
  --runtime-invocation /path/to/latest-runtime-invocation-contract.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-model-load.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia model-load
```

Non-Claims

Stage-12 Nadia is not a model loader, model file reader, model weight mapper, model verifier, inference runtime, runtime process launcher, prompt evaluator, prompt materializer, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, or production AI assistant.

Validation

```
sh scripts/test-nadia-model-load-contract-stage-12.sh
```

Expected result:

```
nadia_model_load_contract_stage_12: ok
```

docs/NADIA_OFFLINE_AI_FOUNDATION.md

Source title: Nadia Offline AI Foundation

Lines: 2346 | Words: 7777 | SHA-256: edfe63393a19051849ea87b4e627ba8d1eb1d32259cdb9671cdbb909d1375f4a

Nadia Offline AI Foundation

Status: Stage-0 foundation contract Date: 2026-05-25 Scope: Latticra-native offline AI identity, installation surface, Console interoperability, local evidence posture, and staged development plan.

Name

The Latticra offline AI is named Nadia.

Nadia is named after Nobel Peace Prize laureate Nadia Murad, whose testimony and advocacy have helped bring public attention to the enslavement and abuse of thousands of women in Iraq and Syria.

This name gives the Latticra offline AI a human-rights awareness posture: technical capability should be bound to dignity, witness, community responsibility, and careful authority.

Nadia's Latticra component identity is:

```
public_name=Nadia
system_name=Latticra Nadia Witness Foundation
interactive_name=Nadia
implementation_name=Nadia Witness Foundation
documentation_code_name=Nadia Witness Foundation
command_name=latticra-nadia
component_key=nadia_offline_ai
```

Purpose

Nadia is the future offline AI companion for Latticra software development, systems engineering, and AI development workflows.

Her long-term role is to help operators:

- understand and improve Latticra source, contracts, tests, and evidence;
- reason about C, constrained C++, Rust Panel code, Lat, LIR, L-UI, Nucleus, Runtime Boundary, and Seal surfaces;
- plan bounded implementation slices before tool authority is granted;
- use local project context without requiring network access;
- learn from accepted local work through operator-reviewed productivity evidence;

- keep community-awareness principles visible while building powerful local AI tools.

Stage-0 Foundation

Stage-0 makes Nadia visible and installable without claiming model capability.

Stage-0 establishes:

```
nadia_stage_0_foundation_present=1
panel_install_surface_present=1
console_interop_surface_present=1
local_config_surface_present=1
local_component_marker_present=1
offline_by_default=1
network_authority=0
tool_execution_authority=0
model_runtime_present=0
model_weights_installed=0
self_modification_authority=0
production_ai_claimed=0
```

Stage-0 does not install model weights, run inference, execute tools, mutate source, train a model, use the network, or claim autonomous software-development capability.

Awareness Principles

Nadia's development should remain aligned with the reason for her name:

```
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
sensationalism_allowed=0
coercive_use_allowed=0
dehumanizing_use_allowed=0
```

This does not make Stage-0 a policy engine or content-moderation system. It establishes a naming and design commitment that later AI capability must honor through explicit contracts, evidence, and operator-visible boundaries.

Local Layout

When enabled through Latticra Panel, Nadia reserves a user-local foundation under the guarded Latticra prefix:

```
etc/latticra/nadia.toml
share/latticra/nadia/README.md
share/latticra/nadia/context-packs/
share/latticra/nadia/model-registry/
share/latticra/nadia/productivity-ledger/
share/latticra/nadia/runtime-invocation/
share/latticra/nadia/model-load/
share/latticra/nadia/prompt-receipt/
share/latticra/nadia/prompt-materialization/
share/latticra/nadia/awareness-dialogue/
share/latticra/nadia/prompt-evaluation-handoff/
share/latticra/nadia/tokenization-boundary/
share/latticra/nadia/tokenizer-specification/
share/latticra/nadia/tokenizer-manifest/
share/latticra/nadia/tokenizer-artifact-inventory/
share/latticra/nadia/tokenizer-artifact-measurement/
share/latticra/nadia/tokenizer-artifact-verification/
share/latticra/nadia/tokenizer-artifact-binding/
share/latticra/nadia/tokenizer-runtime-attachment/
share/latticra/nadia/prompt-tokenization/
share/latticra/nadia/prompt-token-sequence/
share/latticra/nadia/context-window-assembly/
share/latticra/nadia/prompt-evaluation-input/
share/latticra/nadia/prompt-evaluation-runtime-handoff/
share/latticra/nadia/prompt-evaluation-invocation/
share/latticra/nadia/prompt-evaluation-result/
share/latticra/nadia/prompt-evaluation-result-review/
share/latticra/nadia/prompt-evaluation-result-disposition/
share/latticra/nadia/prompt-evaluation-result-release/
share/latticra/nadia/prompt-evaluation-result-release-receipt/
share/latticra/nadia/prompt-evaluation-result-release-receipt-review/
```

```
share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition/
share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release/
share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt
/
share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt
-review/
share/latticra/components/nadia-offline-ai.installed
bin/latticra-nadia
```

These paths are placeholders for future offline capability. They are intentionally local, inspectable, and receipt-visible.

Console Interoperability

Nadia must interoperate with the embedded Latticra Console before receiving any broader authority.

Stage-0 Console behavior is metadata-only:

```
nadia status
nadia commands
```

The nadia status command reports whether the Panel configuration has selected the Nadia component and repeats the denied authority posture. The nadia commands command lists the Stage-1 through Stage-46 command map. Neither command launches an external host process.

After user-local installation, the CLI surface is:

```
latticra-nadia status
latticra-nadia commands
latticra nadia status
latticra nadia commands
```

The installed shim reports Stage-0 metadata and a no-effect command map only.

Productivity Learning

Nadia's learning loop must begin as a local productivity ledger, not as silent self-training.

Stage-0 reserves the ledger path and policy vocabulary:

```
productivity_ledger=operator-reviewed-local
accepted_patch_memory=planned
test_outcome_memory=planned
retrieval_index_updates=planned
weight_training=0
silent_self_modification=0
```

Future stages may let Nadia learn from accepted patches, test outcomes, rejected plans, and operator notes. That learning must remain local, auditable, reversible, and separated from model-weight training unless a later contract explicitly permits it.

Development Stages

Stage-0: Foundation

Establish name, component identity, Panel install surface, Console status surface, local config, local directories, CLI shim, status record, and deterministic guard.

Stage-1: Local Context Engine

Build a no-network project indexer for docs, headers, sources, tests, scripts, installer configs, and status records. Produce context packs for Latticra-specific coding tasks without running inference.

Current Stage-1 status:

```
nadia_stage_1_local_context_engine_present=1
context_pack_command=scripts/nadia-context-pack.sh
installed_context_pack_command=latticra-nadia context-pack
local_file_read_for_indexing=operator_invoked
```

```
source_mutation_authority=0
network_authority=0
model_runtime_present=0
```

See NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1.md.

Stage-2: Offline Inference Runtime

Add a pluggable local runtime boundary for operator-provided models, likely GGUF-compatible first. Record model hashes, quantization, context length, hardware profile, and memory budget before use.

Current Stage-2 status:

```
nadia_stage_2_runtime_profile_present=1
runtime_profile_command=scripts/nadia-runtime-profile.sh
installed_runtime_profile_command=latticra-nadia runtime-profile
model_file_measurement=operator_provided_optional
runtime_family=llama.cpp-compatible
model_format=gguf
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
```

See NADIA_RUNTIME_PROFILE_STAGE_2.md.

Stage-3: Developer Workbench

Integrate Nadia with Latticra Console and Panel workflows for code navigation, patch planning, test selection, and evidence review. Keep source mutation behind explicit operator action.

Current Stage-3 status:

```
nadia_stage_3_developer_workbench_present=1
prompt_plan_command=scripts/nadia-prompt-plan.sh
installed_prompt_plan_command=latticra-nadia prompt-plan
requires_context_pack=1
requires_runtime_profile=1
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
source_mutation_authority=0
```

See NADIA_DEVELOPER_WORKBENCH_STAGE_3.md.

Stage-4: Systems Engineering Mode

Specialize prompt-plan validation with mode labels and validator sets for C, constrained C++, Rust Panel code, Lat/LIR/L-UI, Runtime Boundary, Seal, AI infrastructure, awareness safety, and Linux/Fedora integration.

```
nadia_stage_4_systems_engineering_mode_present=1
mode_validation_command=scripts/nadia-mode-validate.sh
installed_mode_validation_command=latticra-nadia mode-validate
requires_prompt_plan=1
mode_taxonomy_present=1
mode_allowed=1
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
source_mutation_authority=0
```

See NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4.md.

Stage-5: Productivity Loop

Use the local productivity ledger to improve retrieval, ranking, plan templates, test recommendations, and project-specific memory. Any model training or distillation remains a separate future contract.

```
nadia_stage_5_productivity_loop_present=1
productivity_ledger_command=scripts/nadia-productivity-ledger.sh
installed_productivity_ledger_command=latticra-nadia productivity-ledger
```

```
requires_mode_validation=1
learning_scope=operator-reviewed-local-productivity
ledger_append_only=1
project_memory_scope=local-metadata-only
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
training_performed=0
distillation_performed=0
source_mutation_authority=0
```

See NADIA_PRODUCTIVITY_LOOP_STAGE_5.md.

Stage-6: Protective Safety Boundary

Make Nadia's absolute protective boundary explicit before any prompt evaluation, model runtime, retrieval, or tool authority is considered.

```
nadia_stage_6_protective_safety_boundary_present=1
protective_safety_command=scripts/nadia-protective-safety-boundary.sh
installed_protective_safety_command=latticra-nadia protective-safety
requires_productivity_entry=1
absolute_protective_boundary=1
sexual_user_request_authority=0
sexual_content_generation=0
sexual_roleplay_authority=0
sexualized_namesake_or_survivor_content=0
sexual_request_refusal=always
user_override_authority=0
prompt_injection_override_authority=0
manipulation_resistance=required
policy_bypass_authority=0
namesake_cause_awareness=1
awareness_context=non_sensational_human_rights
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
training_performed=0
distillation_performed=0
source_mutation_authority=0
```

See NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6.md.

Stage-7: Guarded Tool Authority

Add a report-only tool-authority preflight after the protective-safety boundary. Stage-7 can classify a proposed tool class but cannot execute tools.

```
nadia_stage_7_guarded_tool_authority_present=1
tool_authority_preflight_command=scripts/nadia-tool-authority-preflight.sh
installed_tool_authority_preflight_command=latticra-nadia tool-preflight
requires_protective_safety=1
tool_authority_stage=preflight-only
preflight_decision=report_only_no_execution
tool_execution_authority=0
tool_execution_performed=0
tool_selection_authority=0
shell_execution_authority=0
network_tool_authority=0
source_mutation_authority=0
destructive_action_authority=0
credential_access_authority=0
requires_operator_approval=1
requires_nucleus_gate=1
requires_runtime_boundary_gate=1
requires_seal_receipt=1
requires_protective_safety_boundary=1
authority_transition_allowed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7.md.

Stage-8: Prompt Evaluation Contract

Define the contract required before any future prompt evaluation. Stage-8 can verify Stage-7 tool preflight evidence and write receipt fields, but it cannot materialize or evaluate prompts.

```
nadia_stage_8_prompt_evaluation_contract_present=1
prompt_evaluation_contract_command=scripts/nadia-prompt-evaluation-contract.sh
installed_prompt_evaluation_contract_command=latticra-nadia prompt-contract
requires_tool_preflight=1
prompt_evaluation_stage=contract-only
prompt_contract_status=contract_only
prompt_materialized=0
prompt_text_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
prompt_receipt_required=1
refusal_policy_required=1
protective_safety_required=1
tool_preflight_required=1
runtime_profile_required=1
model_registry_review_required=1
operator_review_required=1
contract_promotion_allowed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
tool_execution_authority=0
tool_execution_performed=0
model_runtime_invoked=0
inference_performed=0
source_mutation_authority=0
```

See NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8.md.

Stage-9: Local Model Registry Contract

Record local model-candidate metadata after prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-9 can record candidate labels, runtime/profile linkage, provenance labels, license review labels, context budget, and quantization metadata, but it cannot select, install, load, inspect, benchmark, or run a model.

```
nadia_stage_9_local_model_registry_contract_present=1
model_registry_contract_command=scripts/nadia-local-model-registry-contract.sh
installed_model_registry_contract_command=latticra-nadia model-registry
requires_prompt_contract=1
requires_runtime_profile=1
local_model_registry_stage=contract-only
registry_contract_status=metadata_only
model_registry_authority=0
candidate_recorded=1
candidate_review_status=operator_review_required
candidate_usable_for_inference=0
candidate_selected_for_runtime=0
model_selection_authority=0
model_install_authority=0
model_download_authority=0
model_copy_authority=0
model_load_authority=0
model_weight_inspection_authority=0
registry_promotion_allowed=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_runtime_invoked=0
inference_performed=0
model_weights_installed=0
model_weights_loaded=0
model_weights_downloaded=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9.md.

Stage-10: Inference Readiness Contract

Record inference-readiness metadata after local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-10 can verify prerequisite evidence and record a blocked readiness decision, but it cannot load model weights, invoke a runtime, generate tokens, materialize or evaluate prompts, or run inference.

```
nadia_stage_10_inference_readiness_contract_present=1
inference_readiness_contract_command=scripts/nadia-inference-readiness-contract.sh
installed_inference_readiness_contract_command=latticra-nadia inference-readiness
requires_model_registry_contract=1
inference_readiness_stage=contract-only
inference_readiness_contract_status=contract_only
inference_readiness_authority=0
inference_ready=0
readiness_decision=blocked_contract_only
readiness_evidence_present=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_future_runtime_invocation_contract=1
readiness_promotion_allowed=0
runtime_invocation_authority=0
token_generation_authority=0
model_session_authority=0
model_selection_authority=0
model_load_authority=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_runtime_present=0
model_runtime_invoked=0
runtime_invoked=0
inference_authority=0
inference_performed=0
model_weights_loaded=0
model_weights_downloaded=0
model_weights_inspected=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10.md.

Stage-11: Runtime Invocation Contract

Record runtime-invocation metadata after inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-11 can verify prerequisite evidence and record a blocked invocation decision, but it cannot spawn a runtime process, execute a runtime binary, create a runtime session, load model weights, generate tokens, materialize or evaluate prompts, or run inference.

```
nadia_stage_11_runtime_invocation_contract_present=1
runtime_invocation_contract_command=scripts/nadia-runtime-invocation-contract.sh
installed_runtime_invocation_contract_command=latticra-nadia runtime-invocation
requires_inference_readiness_contract=1
runtime_invocation_stage=contract-only
runtime_invocation_contract_status=contract_only
runtime_invocation_authority=0
runtime_invocation_allowed=0
runtime_invoked=0
invocation_decision=blocked_contract_only
invocation_evidence_present=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
```

```
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_future_model_load_contract=1
invocation_promotion_allowed=0
runtime_process_spawn_authority=0
runtime_binary_execution_authority=0
runtime_session_authority=0
model_session_authority=0
token_generation_authority=0
model_runtime_present=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
runtime_session_created=0
model_weights_loaded=0
model_weights_downloaded=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11.md.

Stage-12: Model Load Contract

Record model-load metadata after runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-12 can verify prerequisite evidence and record a blocked model-load decision, but it cannot open model files, map model weights, verify weights, load weights, attach weights to a runtime, generate tokens, materialize or evaluate prompts, or run inference.

```
nadia_stage_12_model_load_contract_present=1
model_load_contract_command=scripts/nadia-model-load-contract.sh
installed_model_load_contract_command=latticra-nadia model-load
requires_runtime_invocation_contract=1
model_load_stage=contract-only
model_load_contract_status=contract_only
model_load_authority=0
model_load_allowed=0
model_loaded=0
load_decision=blocked_contract_only
load_evidence_present=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_weight_measurement_contract=1
requires_future_prompt_receipt_contract=1
load_promotion_allowed=0
model_file_open_authority=0
model_weight_read_authority=0
model_weight_mapping_authority=0
model_weight_verification_authority=0
model_weight_inspection_authority=0
runtime_model_attach_authority=0
model_session_authority=0
token_generation_authority=0
model_file_opened=0
model_file_descriptor_opened=0
model_memory_map_created=0
model_weights_mapped=0
model_weights_loaded=0
model_weights_attached=0
model_weight_measurement_performed=0
```

```
model_weight_verification_performed=0
model_load_performed=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
token_generation_performed=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_MODEL_LOAD_CONTRACT_STAGE_12.md.

Stage-13: Prompt Receipt Contract

Record prompt-receipt metadata after model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-13 can verify prerequisite evidence and record a blocked prompt-receipt decision, but it cannot receive prompt text, read prompt sources, store prompt content, hash prompt content, classify prompt content, materialize prompts, evaluate prompts, generate tokens, or run inference.

```
nadia_stage_13_prompt_receipt_contract_present=1
prompt_receipt_contract_command=scripts/nadia-prompt-receipt-contract.sh
installed_prompt_receipt_contract_command=latticra-nadia prompt-receipt
requires_model_load_contract=1
prompt_receipt_stage=contract-only
prompt_receipt_contract_status=contract_only
prompt_receipt_authority=0
prompt_receipt_allowed=0
prompt_received=0
receipt_decision=blocked_contract_only
receipt_evidence_present=1
requires_runtime_invocation_contract=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_prompt_source_boundary=1
requires_future_prompt_materialization_contract=1
prompt_receipt_promotion_allowed=0
prompt_source_open_authority=0
prompt_source_read_authority=0
prompt_text_materialization_authority=0
prompt_content_storage_authority=0
prompt_hash_authority=0
prompt_classification_authority=0
prompt_source_opened=0
prompt_source_read=0
prompt_bytes_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_content_stored=0
prompt_hash_computed=0
prompt_classified=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_loaded=0
model_weights_loaded=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
```

```
network_authority=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13.md.

Stage-14: Prompt Materialization Contract

Record prompt-materialization metadata after prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-14 can verify prerequisite evidence and record a blocked prompt-materialization decision, but it cannot allocate prompt buffers, materialize prompt text, tokenize prompts, evaluate prompts, generate tokens, or run inference.

```
nadia_stage_14_prompt_materialization_contract_present=1
prompt_materialization_contract_command=scripts/nadia-prompt-materialization-contract.sh
installed_prompt_materialization_contract_command=latticra-nadia prompt-materialization
requires_prompt_receipt_contract=1
prompt_materialization_stage=contract-only
prompt_materialization_contract_status=contract_only
prompt_materialization_authority=0
prompt_materialization_allowed=0
prompt_materialized=0
prompt_text_materialized=0
materialization_decision=blocked_contract_only
materialization_evidence_present=1
requires_model_load_contract=1
requires_runtime_invocation_contract=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_prompt_source_boundary=1
requires_prompt_buffer_boundary=1
requires_future_prompt_evaluation_handoff_contract=1
prompt_materialization_promotion_allowed=0
prompt_source_open_authority=0
prompt_source_read_authority=0
prompt_text_materialization_authority=0
prompt_buffer_allocation_authority=0
prompt_buffer_write_authority=0
prompt_tokenization_authority=0
prompt_source_opened=0
prompt_source_read=0
prompt_bytes_read=0
prompt_text_received=0
prompt_materialization_performed=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_bytes_materialized=0
prompt_tokens_created=0
prompt_tokenized=0
prompt_content_stored=0
prompt_hash_computed=0
prompt_classified=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_loaded=0
model_weights_loaded=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14.md.

Stage-15: Awareness Dialogue Contract

Record awareness-dialogue metadata after prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-15 can define Nadia Initiative Q&A awareness scope and respectful dialogue requirements, but it cannot generate dialogue, evaluate prompts, generate tokens, run inference, or use the network.

```
nadia_stage_15_awareness_dialogue_contract_present=1
awareness_dialogue_contract_command=scripts/nadia-awareness-dialogue-contract.sh
installed_awareness_dialogue_contract_command=latticra-nadia awareness-dialogue
future_qa_dialogue_capability_planned=1
awareness_dialogue_stage=contract-only
awareness_dialogue_contract_status=contract_only
awareness_dialogue_authority=0
awareness_dialogue_allowed=0
dialogue_generation_authority=0
dialogue_generation_allowed=0
qa_dialogue_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
live_web_lookup_authority=0
topic_yazidi_genocide_awareness=1
topic_survivor_voice_and_dignity=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_genocide_prevention=1
topic_justice_and_accountability=1
topic_women_peace_justice_security=1
topic_sinjar_reconstruction=1
topic_security_and_safe_return=1
topic_education_restoration=1
topic_healthcare_and_mental_health=1
topic_livelihoods_and_food_security=1
topic_wash_clean_water_sanitation_hygiene=1
topic_womens_empowerment=1
topic_legal_rights_and_reparations_awareness=1
topic_cultural_preservation_and_memorialization=1
topic_community_driven_survivor_centric_development=1
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
survivor_impersonation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15.md.

Stage-16: Prompt Evaluation Handoff Contract

Record prompt-evaluation handoff metadata after awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-16 can verify awareness-dialogue evidence and package a blocked prompt-evaluation handoff, but it cannot tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_16_prompt_evaluation_handoff_contract_present=1
prompt_evaluation_handoff_contract_command=scripts/nadia-prompt-evaluation-handoff-contract.sh
installed_prompt_evaluation_handoff_contract_command=latticra-nadia prompt-evaluation-handoff
```

```
prompt_evaluation_handoff_contract_status=contract_only
prompt_evaluation_handoff_stage=contract-only
prompt_evaluation_handoff_authority=0
prompt_evaluation_handoff_allowed=0
prompt_evaluation_handoff_performed=0
prompt_evaluation_authority=0
prompt_evaluated=0
evaluation_handoff_decision=blocked_contract_only
evaluation_handoff_evidence_present=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenization_contract=1
prompt_evaluation_handoff_promotion_allowed=0
future_qa_dialogue_capability_planned=1
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16.md.

Stage-17: Tokenization Boundary Contract

Record tokenization-boundary metadata after prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-17 can verify prompt-evaluation handoff evidence and package a blocked tokenization boundary, but it cannot open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_17_tokenization_boundary_contract_present=1
tokenization_boundary_contract_command=scripts/nadia-tokenization-boundary-contract.sh
installed_tokenization_boundary_contract_command=latticra-nadia tokenization-boundary
tokenization_boundary_contract_status=contract_only
tokenization_boundary_stage=contract-only
tokenization_boundary_authority=0
tokenization_boundary_allowed=0
tokenization_boundary_performed=0
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenized=0
prompt_tokens_created=0
tokenizer_file_opened=0
tokenizer_vocab_loaded=0
prompt_evaluation_authority=0
prompt_evaluated=0
tokenization_decision=blocked_contract_only
tokenization_evidence_present=1
requires_prompt_evaluation_handoff_contract=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_prompt_buffer_boundary=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenizer_specification_contract=1
tokenization_boundary_promotion_allowed=0
qa_dialogue_generated=0
answer_text_generated=0
```

```
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17.md.

Stage-18: Tokenizer Specification Contract

Record tokenizer-specification metadata after tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-18 can define future tokenizer review requirements, but it cannot load tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_18_tokenizer_specification_contract_present=1
tokenizer_specification_contract_command=scripts/nadia-tokenizer-specification-contract.sh
installed_tokenizer_specification_contract_command=latticra-nadia tokenizer-specification
tokenizer_specification_contract_status=contract_only
tokenizer_specification_stage=contract-only
tokenizer_specification_authority=0
tokenizer_specification_allowed=0
tokenizer_specification_performed=0
tokenizer_specification_metadata_present=1
tokenizer_family=model-compatible-tokenizer
tokenizer_format=operator-reviewed-offline-specification
tokenizer_specification_decision=blocked_contract_only
tokenizer_specification_evidence_present=1
tokenizer_source_policy=operator-reviewed-offline
tokenizer_path_recorded=0
tokenizer_manifest_loaded=0
requires_tokenization_boundary_contract=1
requires_future_tokenizer_manifest_contract=1
tokenizer_specification_promotion_allowed=0
requires_model_tokenizer_compatibility_review=1
requires_unicode_policy_review=1
requires_normalization_policy_review=1
requires_special_token_policy_review=1
requires_bos_eos_policy_review=1
requires_chat_template_policy_review=1
requires_prompt_template_boundary=1
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18.md.

Stage-19: Tokenizer Manifest Contract

Record tokenizer-manifest metadata after tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-19 can define future tokenizer manifest review requirements, but it cannot load tokenizer manifests, parse tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_19_tokenizer_manifest_contract_present=1
tokenizer_manifest_contract_command=scripts/nadia-tokenizer-manifest-contract.sh
installed_tokenizer_manifest_contract_command=latticra-nadia tokenizer-manifest
```

```
tokenizer_manifest_contract_status=contract_only
tokenizer_manifest_stage=contract-only
tokenizer_manifest_authority=0
tokenizer_manifest_allowed=0
tokenizer_manifest_performed=0
tokenizer_manifest_metadata_present=1
tokenizer_manifest_family=operator-reviewed-tokenizer-manifest
tokenizer_manifest_format=contract-only-offline-manifest
tokenizer_manifest_decision=blocked_contract_only
tokenizer_manifest_evidence_present=1
tokenizer_manifest_source_policy=operator-reviewed-offline
tokenizer_manifest_path_recorded=0
tokenizer_manifest_schema_planned=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_inventory_contract=1
tokenizer_manifest_promotion_allowed=0
tokenizer_manifest_open_authority=0
tokenizer_manifest_read_authority=0
tokenizer_manifest_parse_authority=0
tokenizer_manifest_load_authority=0
tokenizer_manifest_opened=0
tokenizer_manifest_read=0
tokenizer_manifest_parsed=0
tokenizer_manifest_loaded=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19.md.

Stage-20: Tokenizer Artifact Inventory Contract

Record tokenizer-artifact-inventory metadata after tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-20 can define future tokenizer artifact inventory review requirements, but it cannot resolve artifact paths, scan directories, stat files, hash artifacts, load tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_20_tokenizer_artifact_inventory_contract_present=1
tokenizer_artifact_inventory_contract_command=scripts/nadia-tokenizer-artifact-inventory-contrac
t.sh
installed_tokenizer_artifact_inventory_contract_command=latticra-nadia
tokenizer-artifact-inventory
tokenizer_artifact_inventory_contract_status=contract_only
tokenizer_artifact_inventory_stage=contract-only
tokenizer_artifact_inventory_authority=0
tokenizer_artifact_inventory_allowed=0
tokenizer_artifact_inventory_performed=0
tokenizer_artifact_inventory_metadata_present=1
tokenizer_artifact_inventory_family=operator-reviewed-tokenizer-artifact-inventory
tokenizer_artifact_inventory_format=contract-only-offline-inventory
tokenizer_artifact_inventory_decision=blocked_contract_only
tokenizer_artifact_inventory_evidence_present=1
tokenizer_artifact_inventory_source_policy=operator-reviewed-offline
tokenizer_artifact_inventory_path_recorded=0
tokenizer_artifact_inventory_schema_planned=1
tokenizer_artifact_inventory_entry_count=0
tokenizer_artifact_inventory_file_count=0
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_measurement_contract=1
tokenizer_artifact_inventory_promotion_allowed=0
tokenizer_artifact_path_resolved=0
```

```
tokenizer_artifact_scan_performed=0
tokenizer_artifact_stat_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_artifact_measurement_performed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20.md.

Stage-21: Tokenizer Artifact Measurement Contract

Record tokenizer-artifact-measurement metadata after tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-21 can define future tokenizer artifact measurement review requirements, but it cannot open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, record tokenizer artifact digests, record tokenizer artifact sizes, load tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_21_tokenizer_artifact_measurement_contract_present=1
tokenizer_artifact_measurement_contract_command=scripts/nadia-tokenizer-artifact-measurement-con
tract.sh
installed_tokenizer_artifact_measurement_contract_command=latticra-nadia
tokenizer-artifact-measurement
tokenizer_artifact_measurement_contract_status=contract_only
tokenizer_artifact_measurement_stage=contract-only
tokenizer_artifact_measurement_authority=0
tokenizer_artifact_measurement_allowed=0
tokenizer_artifact_measurement_performed=0
tokenizer_artifact_measurement_metadata_present=1
tokenizer_artifact_measurement_family=operator-reviewed-tokenizer-artifact-measurement
tokenizer_artifact_measurement_format=contract-only-offline-measurement
tokenizer_artifact_measurement_decision=blocked_contract_only
tokenizer_artifact_measurement_evidence_present=1
tokenizer_artifact_measurement_source_policy=operator-reviewed-offline
tokenizer_artifact_measurement_plan_recorded=1
tokenizer_artifact_measurement_algorithm_planned=sha256-or-approved-digest
tokenizer_artifact_measurement_result_recorded=0
tokenizer_artifact_measurement_digest_recorded=0
tokenizer_artifact_measurement_size_recorded=0
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_verification_contract=1
tokenizer_artifact_measurement_promotion_allowed=0
tokenizer_artifact_measurement_hash_computed=0
tokenizer_artifact_digest_recorded=0
tokenizer_artifact_size_recorded=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
```

```
manipulation_resistance=required
```

See NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21.md.

Stage-22: Tokenizer Artifact Verification Contract

Record tokenizer-artifact-verification metadata after tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-22 can define future tokenizer artifact verification review requirements, but it cannot open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, compare artifact digests, compare artifact sizes, verify artifacts, bind artifacts, load tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_22_tokenizer_artifact_verification_contract_present=1
tokenizer_artifact_verification_contract_command=scripts/nadia-tokenizer-artifact-verification-c
ontract.sh
installed_tokenizer_artifact_verification_contract_command=latticra-nadia
tokenizer-artifact-verification
tokenizer_artifact_verification_contract_status=contract_only
tokenizer_artifact_verification_stage=contract-only
tokenizer_artifact_verification_authority=0
tokenizer_artifact_verification_allowed=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_verification_metadata_present=1
tokenizer_artifact_verification_family=operator-reviewed-tokenizer-artifact-verification
tokenizer_artifact_verification_format=contract-only-offline-verification
tokenizer_artifact_verification_decision=blocked_contract_only
tokenizer_artifact_verification_evidence_present=1
tokenizer_artifact_verification_source_policy=operator-reviewed-offline
tokenizer_artifact_verification_plan_recorded=1
tokenizer_artifact_verification_method_planned=offline-digest-and-size-policy-review
tokenizer_artifact_verification_comparison_performed=0
tokenizer_artifact_verification_result_recorded=0
tokenizer_artifact_verification_digest_match_recorded=0
tokenizer_artifact_verification_size_match_recorded=0
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_binding_contract=1
tokenizer_artifact_verification_promotion_allowed=0
tokenizer_artifact_verification_hash_computed=0
tokenizer_artifact_verification_compared=0
tokenizer_artifact_digest_recorded=0
tokenizer_artifact_size_recorded=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22.md.

Stage-23: Tokenizer Artifact Binding Contract

Record tokenizer-artifact-binding metadata after tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-23 can define future tokenizer artifact binding review requirements, but it cannot open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, verify artifacts, bind artifacts, attach tokenizers to a runtime, load tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_23_tokenizer_artifact_binding_contract_present=1
tokenizer_artifact_binding_contract_command=scripts/nadia-tokenizer-artifact-binding-contract.sh
installed_tokenizer_artifact_binding_contract_command=latticra-nadia tokenizer-artifact-binding
tokenizer_artifact_binding_contract_status=contract_only
tokenizer_artifact_binding_stage=contract-only
tokenizer_artifact_binding_authority=0
tokenizer_artifact_binding_allowed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_binding_metadata_present=1
tokenizer_artifact_binding_family=operator-reviewed-tokenizer-artifact-binding
tokenizer_artifact_binding_format=contract-only-offline-binding
tokenizer_artifact_binding_decision=blocked_contract_only
tokenizer_artifact_binding_evidence_present=1
tokenizer_artifact_binding_source_policy=operator-reviewed-offline
tokenizer_artifact_binding_plan_recorded=1
tokenizer_artifact_binding_method_planned=offline-manifest-artifact-role-binding-review
tokenizer_artifact_binding_result_recorded=0
tokenizer_artifact_binding_record_created=0
tokenizer_artifact_binding_manifest_reference_recorded=0
tokenizer_artifact_binding_artifact_reference_recorded=0
tokenizer_artifact_binding_runtime_attach_recorded=0
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_runtime_attachment_contract=1
tokenizer_artifact_binding_promotion_allowed=0
tokenizer_artifact_binding_hash_computed=0
tokenizer_artifact_binding_bound=0
tokenizer_artifact_bound_to_manifest=0
tokenizer_artifact_bound_to_tokenizer=0
tokenizer_attached_to_runtime=0
tokenizer_runtime_attachment_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23.md.

Stage-24: Tokenizer Runtime Attachment Contract

Record tokenizer-runtime-attachment metadata after tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-24 can define future tokenizer runtime attachment review requirements, but it cannot open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, bind artifacts, attach tokenizers to a runtime, create runtime sessions, load tokenizer manifests, open tokenizer files, load vocabularies, tokenize prompts, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_24_tokenizer_runtime_attachment_contract_present=1
tokenizer_runtime_attachment_contract_command=scripts/nadia-tokenizer-runtime-attachment-contrac
t.sh
installed_tokenizer_runtime_attachment_contract_command=latticra-nadia
tokenizer-runtime-attachment
tokenizer_runtime_attachment_contract_status=contract_only
tokenizer_runtime_attachment_stage=contract-only
tokenizer_runtime_attachment_authority=0
tokenizer_runtime_attachment_allowed=0
tokenizer_runtime_attachment_performed=0
tokenizer_runtime_attachment_metadata_present=1
tokenizer_runtime_attachment_family=operator-reviewed-tokenizer-runtime-attachment
tokenizer_runtime_attachment_format=contract-only-offline-attachment
tokenizer_runtime_attachment_decision=blocked_contract_only
tokenizer_runtime_attachment_evidence_present=1
tokenizer_runtime_attachment_source_policy=operator-reviewed-offline
tokenizer_runtime_attachment_plan_recorded=1
tokenizer_runtime_attachment_method_planned=offline-runtime-tokenizer-attachment-review
tokenizer_runtime_attachment_result_recorded=0
tokenizer_runtime_attachment_record_created=0
tokenizer_runtime_attachment_runtime_reference_recorded=0
tokenizer_runtime_attachment_tokenizer_reference_recorded=0
tokenizer_runtime_attachment_session_created=0
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_runtime_profile_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_future_prompt_tokenization_contract=1
tokenizer_runtime_attachment_promotion_allowed=0
tokenizer_runtime_attachment_attached=0
tokenizer_attached_to_runtime=0
runtime_session_created=0
runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24.md.

Stage-25: Prompt Tokenization Contract

Record prompt-tokenization metadata after tokenizer-runtime-attachment metadata, tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-25 can define future prompt token sequence review requirements, but it cannot read prompt text, materialize prompts, allocate token buffers, create prompt tokens, record token counts, record token sequences, attach tokenizers to a runtime, create runtime sessions, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_25_prompt_tokenization_contract_present=1
prompt_tokenization_contract_command=scripts/nadia-prompt-tokenization-contract.sh
installed_prompt_tokenization_contract_command=latticra-nadia prompt-tokenization
prompt_tokenization_contract_status=contract_only
prompt_tokenization_stage=contract-only
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenization_performed=0
prompt_tokenization_metadata_present=1
prompt_tokenization_family=operator-reviewed-prompt-tokenization
prompt_tokenization_format=contract-only-offline-tokenization
prompt_tokenization_decision=blocked_contract_only
prompt_tokenization_evidence_present=1
prompt_tokenization_source_policy=operator-reviewed-offline
prompt_tokenization_plan_recorded=1
prompt_tokenization_method_planned=offline-tokenization-policy-review
prompt_tokenization_result_recorded=0
prompt_tokenization_token_count_recorded=0
prompt_tokenization_token_sequence_recorded=0
prompt_tokenization_runtime_invoked=0
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_token_sequence_contract=1
prompt_tokenization_promotion_allowed=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_token_sequence_recorded=0
prompt_token_buffer_created=0
prompt_token_buffer_written=0
prompt_tokenized=0
tokenizer_runtime_attachment_performed=0
tokenizer_attached_to_runtime=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25.md.

Stage-26: Prompt Token Sequence Contract

Record prompt-token-sequence metadata after prompt-tokenization metadata, tokenizer-runtime-attachment metadata, tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-26 can define future context-window assembly review requirements, but it cannot read prompt text, create prompt tokens, record token IDs, record token order, record token offsets, create attention masks, create position IDs, assemble context windows, create prompt evaluation inputs, attach tokenizers to a runtime, create runtime sessions, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

Stage-27 now consumes this prompt-token-sequence contract without granting prompt token ID recording, token order recording, token offset recording, attention mask creation, position ID creation, context-window assembly, prompt-evaluation-input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution authority.

```
nadia_stage_26_prompt_token_sequence_contract_present=1
prompt_token_sequence_contract_command=scripts/nadia-prompt-token-sequence-contract.sh
installed_prompt_token_sequence_contract_command=latticra-nadia prompt-token-sequence
prompt_token_sequence_contract_status=contract_only
prompt_token_sequence_stage=contract-only
prompt_token_sequence_authority=0
prompt_token_sequence_allowed=0
prompt_token_sequence_recorded=0
prompt_token_sequence_metadata_present=1
prompt_token_sequence_family=operator-reviewed-prompt-token-sequence
prompt_token_sequence_format=contract-only-offline-sequence
prompt_token_sequence_decision=blocked_contract_only
prompt_token_sequence_evidence_present=1
prompt_token_sequence_source_policy=operator-reviewed-offline
prompt_token_sequence_plan_recorded=1
prompt_token_sequence_method_planned=offline-token-sequence-policy-review
prompt_token_sequence_result_recorded=0
prompt_token_sequence_count_recorded=0
prompt_token_sequence_order_recorded=0
prompt_token_sequence_runtime_invoked=0
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_context_window_assembly_contract=1
prompt_token_sequence_promotion_allowed=0
prompt_token_ids_recorded=0
prompt_token_order_recorded=0
prompt_token_offsets_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
context_window_assembled=0
prompt_evaluation_input_created=0
prompt_text_read=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_token_sequence_recorded=0
prompt_tokenized=0
runtime_session_created=0
runtime_invoked=0
```

```
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26.md.

Stage-27 preserves the Stage-26 proof that prompt-token-sequence recording, prompt token ID recording, token order recording, token offset recording, attention mask creation, position ID creation, context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, tool execution, source mutation, and network authority remain denied.

Before Stage-28 starts, the context-window assembly contract must remain contract-only and keep context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, and tool execution blocked.

Before Stage-41 starts, the context-window assembly contract must remain contract-only and keep context window assembly, prompt evaluation input creation, runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, and tool execution blocked.

Before Stage-42 starts, the context-window assembly contract must remain contract-only and keep context window assembly, prompt evaluation input creation, runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, and tool execution blocked.

Stage-27: Context Window Assembly Contract

Record context-window assembly metadata after prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-27 can define future prompt-evaluation-input review requirements, but it cannot read prompt text, assemble context windows, create prompt evaluation inputs, attach tokenizers to a runtime, create runtime sessions, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_27_context_window_assembly_contract_present=1
context_window_assembly_contract_command=scripts/nadia-context-window-assembly-contract.sh
installed_context_window_assembly_contract_command=latticra-nadia context-window-assembly
context_window_assembly_contract_status=contract_only
context_window_assembly_stage=contract-only
context_window_assembly_authority=0
context_window_assembly_allowed=0
context_window_assembly_performed=0
context_window_assembly_metadata_present=1
context_window_family=operator-reviewed-context-window-assembly
context_window_format=contract-only-offline-context-window
context_window_assembly_decision=blocked_contract_only
context_window_assembly_evidence_present=1
context_window_assembly_source_policy=operator-reviewed-offline
context_window_assembly_plan_recorded=1
context_window_assembly_method_planned=offline-context-window-policy-review
context_window_assembly_result_recorded=0
context_window_assembly_runtime_invoked=0
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
```

```
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_input_contract=1
context_window_assembly_promotion_allowed=0
context_window_assembled=0
context_window_token_budget_recorded=0
context_window_truncation_applied=0
context_window_serialized=0
prompt_evaluation_input_created=0
prompt_evaluation_input_materialized=0
prompt_evaluation_input_validated=0
prompt_text_read=0
prompt_tokens_created=0
prompt_token_sequence_recorded=0
prompt_token_ids_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27.md.

Stage-28: Prompt Evaluation Input Contract

Record prompt-evaluation-input metadata after context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-28 can define future prompt-evaluation runtime handoff review requirements, but it cannot read prompt text, assemble context windows, create prompt evaluation inputs, attach tokenizers to a runtime, create runtime sessions, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_28_prompt_evaluation_input_contract_present=1
prompt_evaluation_input_contract_command=scripts/nadia-prompt-evaluation-input-contract.sh
installed_prompt_evaluation_input_contract_command=latticra-nadia prompt-evaluation-input
prompt_evaluation_input_contract_status=contract_only
prompt_evaluation_input_stage=contract-only
prompt_evaluation_input_authority=0
prompt_evaluation_input_allowed=0
prompt_evaluation_input_created=0
prompt_evaluation_input_metadata_present=1
prompt_evaluation_input_family=operator-reviewed-prompt-evaluation-input
prompt_evaluation_input_format=contract-only-offline-evaluation-input
prompt_evaluation_input_decision=blocked_contract_only
prompt_evaluation_input_evidence_present=1
prompt_evaluation_input_source_policy=operator-reviewed-offline
prompt_evaluation_input_plan_recorded=1
prompt_evaluation_input_method_planned=offline-prompt-evaluation-input-policy-review
prompt_evaluation_input_result_recorded=0
prompt_evaluation_input_runtime_invoked=0
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
```

```
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_runtime_handoff_contract=1
prompt_evaluation_input_promotion_allowed=0
prompt_evaluation_input_materialized=0
prompt_evaluation_input_loaded=0
prompt_evaluation_input_opened=0
prompt_evaluation_input_read=0
prompt_evaluation_input_validated=0
prompt_evaluation_input_serialized=0
prompt_evaluation_input_written=0
context_window_assembled=0
context_window_serialized=0
prompt_text_read=0
prompt_tokens_created=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28.md.

Stage-29: Prompt Evaluation Runtime Handoff Contract

Record prompt-evaluation runtime handoff metadata after prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-29 can define future prompt-evaluation invocation review requirements, but it cannot create runtime handoff requests, invoke a runtime, create runtime sessions, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_29_prompt_evaluation_runtime_handoff_contract_present=1
prompt_evaluation_runtime_handoff_contract_command=scripts/nadia-prompt-evaluation-runtime-hando
ff-contract.sh
installed_prompt_evaluation_runtime_handoff_contract_command=latticra-nadia
prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_contract_status=contract_only
prompt_evaluation_runtime_handoff_stage=contract-only
prompt_evaluation_runtime_handoff_authority=0
prompt_evaluation_runtime_handoff_allowed=0
prompt_evaluation_runtime_handoff_performed=0
prompt_evaluation_runtime_handoff_metadata_present=1
prompt_evaluation_runtime_handoff_family=operator-reviewed-prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_format=contract-only-offline-runtime-handoff
prompt_evaluation_runtime_handoff_decision=blocked_contract_only
prompt_evaluation_runtime_handoff_evidence_present=1
prompt_evaluation_runtime_handoff_source_policy=operator-reviewed-offline
prompt_evaluation_runtime_handoff_plan_recorded=1
prompt_evaluation_runtime_handoff_method_planned=offline-prompt-evaluation-runtime-handoff-polic
y-review
prompt_evaluation_runtime_handoff_result_recorded=0
prompt_evaluation_runtime_handoff_runtime_invoked=0
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_invocation_contract=1
```

```
prompt_evaluation_runtime_handoff_promotion_allowed=0
prompt_evaluation_runtime_handoff_request_created=0
prompt_evaluation_runtime_handoff_request_validated=0
prompt_evaluation_runtime_handoff_request_serialized=0
prompt_evaluation_runtime_handoff_request_submitted=0
prompt_evaluation_runtime_handoff_runtime_selected=0
prompt_evaluation_runtime_handoff_model_selected=0
runtime_handoff_created=0
runtime_handoff_submitted=0
runtime_invocation_requested=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29.md.

Stage-30: Prompt Evaluation Invocation Contract

Record prompt-evaluation invocation metadata after prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-30 can define future prompt-evaluation result review requirements, but it cannot create prompt-evaluation invocation requests, schedule invocation requests, submit invocation requests, invoke a runtime, create runtime sessions, evaluate prompts, generate dialogue, generate tokens, run inference, or use the network.

```
nadia_stage_30_prompt_evaluation_invocation_contract_present=1
prompt_evaluation_invocation_contract_command=scripts/nadia-prompt-evaluation-invocation-contrac
t.sh
installed_prompt_evaluation_invocation_contract_command=latticra-nadia
prompt-evaluation-invocation
prompt_evaluation_invocation_contract_status=contract_only
prompt_evaluation_invocation_stage=contract-only
prompt_evaluation_invocation_authority=0
prompt_evaluation_invocation_allowed=0
prompt_evaluation_invocation_performed=0
prompt_evaluation_invocation_metadata_present=1
prompt_evaluation_invocation_family=operator-reviewed-prompt-evaluation-invocation
prompt_evaluation_invocation_format=contract-only-offline-evaluation-invocation
prompt_evaluation_invocation_decision=blocked_contract_only
prompt_evaluation_invocation_evidence_present=1
prompt_evaluation_invocation_source_policy=operator-reviewed-offline
prompt_evaluation_invocation_plan_recorded=1
prompt_evaluation_invocation_method_planned=offline-prompt-evaluation-invocation-policy-review
prompt_evaluation_invocation_result_recorded=0
prompt_evaluation_invocation_runtime_invoked=0
requires_prompt_evaluation_runtime_handoff_contract=1
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_result_contract=1
prompt_evaluation_invocation_promotion_allowed=0
```

```
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_validated=0
prompt_evaluation_invocation_request_serialized=0
prompt_evaluation_invocation_request_submitted=0
prompt_evaluation_invocation_request_scheduled=0
prompt_evaluation_invocation_request_queued=0
prompt_evaluation_invocation_runtime_selected=0
prompt_evaluation_invocation_model_selected=0
runtime_handoff_created=0
runtime_invocation_requested=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30.md.

Stage-31: Prompt Evaluation Result Contract

Record prompt-evaluation result metadata after prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-31 can define future prompt-evaluation result review requirements, but it cannot create prompt-evaluation result records, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_31_prompt_evaluation_result_contract_present=1
prompt_evaluation_result_contract_command=scripts/nadia-prompt-evaluation-result-contract.sh
installed_prompt_evaluation_result_contract_command=latticra-nadia prompt-evaluation-result
prompt_evaluation_result_contract_status=contract_only
prompt_evaluation_result_stage=contract-only
prompt_evaluation_result_authority=0
prompt_evaluation_result_allowed=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_created=0
prompt_evaluation_result_performed=0
prompt_evaluation_result_metadata_present=1
prompt_evaluation_result_family=operator-reviewed-prompt-evaluation-result
prompt_evaluation_result_format=contract-only-offline-evaluation-result
prompt_evaluation_result_decision=blocked_contract_only
prompt_evaluation_result_evidence_present=1
prompt_evaluation_result_source_policy=operator-reviewed-offline
prompt_evaluation_result_plan_recorded=1
prompt_evaluation_result_method_planned=offline-prompt-evaluation-result-policy-review
prompt_evaluation_result_result_recorded=0
prompt_evaluation_result_runtime_invoked=0
requires_prompt_evaluation_invocation_contract=1
requires_prompt_evaluation_runtime_handoff_contract=1
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_result_review_contract=1
prompt_evaluation_result_promotion_allowed=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_record_validated=0
prompt_evaluation_result_record_serialized=0
prompt_evaluation_result_record_written=0
prompt_evaluation_result_record_submitted=0
```

```
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
prompt_evaluation_result_score_recorded=0
prompt_evaluation_result_token_logprobs_recorded=0
prompt_evaluation_result_completion_reason_recorded=0
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_submitted=0
runtime_handoff_created=0
runtime_invocation_requested=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31.md.

Stage-32: Prompt Evaluation Result Review Contract

Record prompt-evaluation result review metadata after prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-32 can define future prompt-evaluation result disposition requirements, but it cannot create prompt-evaluation result review records, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_32_prompt_evaluation_result_review_contract_present=1
prompt_evaluation_result_review_contract_command=scripts/nadia-prompt-evaluation-result-review-c
ontract.sh
installed_prompt_evaluation_result_review_contract_command=latticra-nadia
prompt-evaluation-result-review
prompt_evaluation_result_review_contract_status=contract_only
prompt_evaluation_result_review_stage=contract-only
prompt_evaluation_result_review_authority=0
prompt_evaluation_result_review_allowed=0
prompt_evaluation_result_review_recorded=0
prompt_evaluation_result_review_created=0
prompt_evaluation_result_review_performed=0
prompt_evaluation_result_review_metadata_present=1
prompt_evaluation_result_review_family=operator-reviewed-prompt-evaluation-result-review
prompt_evaluation_result_review_format=contract-only-offline-evaluation-result-review
prompt_evaluation_result_review_decision=blocked_contract_only
prompt_evaluation_result_review_evidence_present=1
prompt_evaluation_result_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_review_plan_recorded=1
prompt_evaluation_result_review_method_planned=offline-prompt-evaluation-result-review-policy-re
view
prompt_evaluation_result_review_result_recorded=0
prompt_evaluation_result_review_runtime_invoked=0
requires_prompt_evaluation_result_contract=1
requires_prompt_evaluation_invocation_contract=1
requires_prompt_evaluation_runtime_handoff_contract=1
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_result_disposition_contract=1
prompt_evaluation_result_review_promotion_allowed=0
```

```
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_record_validated=0
prompt_evaluation_result_review_record_serialized=0
prompt_evaluation_result_review_record_written=0
prompt_evaluation_result_review_record_submitted=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_approval_recorded=0
prompt_evaluation_result_review_rejection_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

See NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32.md.

Non-Claims

The current Nadia foundation is not:

- a production AI assistant;
- an autonomous coding agent;
- a sexual assistant, roleplay surface, or adult-content generator;
- a model runtime;
- a model distribution channel;
- a training system;
- a network service;
- a sandbox;
- a security product;
- a replacement for Latticra Seal, Nucleus, or Runtime Boundary.

Promotion Gate

Before Stage-47 starts, Latticra should keep these guards passing:

```
sh scripts/test-nadia-offline-ai-stage-0.sh
sh scripts/test-nadia-local-context-engine-stage-1.sh
sh scripts/test-nadia-runtime-profile-stage-2.sh
sh scripts/test-nadia-developer-workbench-stage-3.sh
sh scripts/test-nadia-systems-engineering-mode-stage-4.sh
sh scripts/test-nadia-productivity-loop-stage-5.sh
sh scripts/test-nadia-protective-safety-boundary-stage-6.sh
sh scripts/test-nadia-guarded-tool-authority-stage-7.sh
sh scripts/test-nadia-prompt-evaluation-contract-stage-8.sh
sh scripts/test-nadia-local-model-registry-contract-stage-9.sh
sh scripts/test-nadia-inference-readiness-contract-stage-10.sh
sh scripts/test-nadia-runtime-invocation-contract-stage-11.sh
sh scripts/test-nadia-model-load-contract-stage-12.sh
sh scripts/test-nadia-prompt-receipt-contract-stage-13.sh
sh scripts/test-nadia-prompt-materialization-contract-stage-14.sh
sh scripts/test-nadia-awareness-dialogue-contract-stage-15.sh
sh scripts/test-nadia-prompt-evaluation-handoff-contract-stage-16.sh
sh scripts/test-nadia-tokenization-boundary-contract-stage-17.sh
sh scripts/test-nadia-tokenizer-specification-contract-stage-18.sh
sh scripts/test-nadia-tokenizer-manifest-contract-stage-19.sh
sh scripts/test-nadia-tokenizer-artifact-inventory-contract-stage-20.sh
sh scripts/test-nadia-tokenizer-artifact-measurement-contract-stage-21.sh
sh scripts/test-nadia-tokenizer-artifact-verification-contract-stage-22.sh
sh scripts/test-nadia-tokenizer-artifact-binding-contract-stage-23.sh
```

```
sh scripts/test-nadia-tokenizer-runtime-attachment-contract-stage-24.sh
sh scripts/test-nadia-prompt-tokenization-contract-stage-25.sh
sh scripts/test-nadia-prompt-token-sequence-contract-stage-26.sh
sh scripts/test-nadia-context-window-assembly-contract-stage-27.sh
sh scripts/test-nadia-prompt-evaluation-input-contract-stage-28.sh
sh scripts/test-nadia-prompt-evaluation-runtime-handoff-contract-stage-29.sh
sh scripts/test-nadia-prompt-evaluation-invocation-contract-stage-30.sh
sh scripts/test-nadia-prompt-evaluation-result-contract-stage-31.sh
sh scripts/test-nadia-prompt-evaluation-result-review-contract-stage-32.sh
sh scripts/test-nadia-prompt-evaluation-result-disposition-contract-stage-33.sh
sh scripts/test-nadia-prompt-evaluation-result-release-contract-stage-34.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-contract-stage-35.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-contract-stage-36.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-contract-stage
-37.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-contra
ct-stage-38.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-contract-stage-39.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-contract-stage-40.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-contract-stage-41.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-contract-stage-42.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-contract-stage-43.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-contract-stage-44.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-disposition-contract-stage-45.sh
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-disposition-release-contract-stage-46.sh
```

Before prompt evaluation result release starts, a separate prompt evaluation result release contract must exist and name prompt-evaluation result disposition metadata, prompt-evaluation result review metadata, prompt-evaluation result metadata, invocation metadata, runtime handoff metadata, evaluation input metadata, context-window assembly denial fields, prompt-token-sequence denial fields, prompt-tokenization denial fields, tokenizer-runtime-attachment denial fields, tokenizer-artifact-binding denial fields, tokenizer-artifact-verification denial fields, tokenizer-artifact-measurement denial fields, tokenizer-artifact-inventory denial fields, tokenizer-manifest denial fields, tokenizer-file denial fields, prompt-materialization denial fields, refusal boundary inheritance, operator review gates, and non-claims.

Stage-33: Prompt Evaluation Result Disposition Contract

Record prompt-evaluation result disposition metadata after prompt-evaluation result review metadata, prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-33 can define future prompt-evaluation result release requirements, but it cannot create prompt-evaluation result disposition records, create release records, create prompt-evaluation result review records, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_33_prompt_evaluation_result_disposition_contract_present=1
prompt_evaluation_result_disposition_contract_command=scripts/nadia-prompt-evaluation-result-dis
position-contract.sh
installed_prompt_evaluation_result_disposition_contract_command=latticra-nadia
prompt-evaluation-result-disposition
prompt_evaluation_result_disposition_contract_status=contract_only
prompt_evaluation_result_disposition_stage=contract-only
prompt_evaluation_result_disposition_authority=0
prompt_evaluation_result_disposition_allowed=0
```

```
prompt_evaluation_result_disposition_recorded=0
prompt_evaluation_result_disposition_created=0
prompt_evaluation_result_disposition_performed=0
prompt_evaluation_result_disposition_metadata_present=1
prompt_evaluation_result_disposition_family=operator-reviewed-prompt-evaluation-result-dispositi
on
prompt_evaluation_result_disposition_format=contract-only-offline-evaluation-result-disposition
prompt_evaluation_result_disposition_decision=blocked_contract_only
prompt_evaluation_result_disposition_evidence_present=1
prompt_evaluation_result_disposition_source_policy=operator-reviewed-offline
prompt_evaluation_result_disposition_plan_recorded=1
prompt_evaluation_result_disposition_method_planned=offline-prompt-evaluation-result-disposition
-policy-review
prompt_evaluation_result_disposition_result_recorded=0
prompt_evaluation_result_disposition_runtime_invoked=0
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_contract=1
prompt_evaluation_result_disposition_promotion_allowed=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_record_validated=0
prompt_evaluation_result_disposition_record_serialized=0
prompt_evaluation_result_disposition_record_written=0
prompt_evaluation_result_disposition_record_submitted=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_approval_recorded=0
prompt_evaluation_result_disposition_rejection_recorded=0
prompt_evaluation_result_disposition_route_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

This stage produces only disposition-contract metadata. It does not decide a disposition, apply a disposition, create release records, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-34: Prompt Evaluation Result Release Contract

Record prompt-evaluation result release metadata after prompt-evaluation result disposition metadata, prompt-evaluation result review metadata, prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-34 can define future prompt-evaluation result release receipt requirements, but it cannot create prompt-evaluation result release records, record release decisions, publish releases, package releases, create release receipts, apply dispositions, create prompt-evaluation result review records, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_34_prompt_evaluation_result_release_contract_present=1
prompt_evaluation_result_release_contract_command=scripts/nadia-prompt-evaluation-result-release
-contract.sh
installed_prompt_evaluation_result_release_contract_command=latticra-nadia
prompt-evaluation-result-release
prompt_evaluation_result_release_contract_status=contract_only
prompt_evaluation_result_release_stage=contract-only
```

```
prompt_evaluation_result_release_authority=0
prompt_evaluation_result_release_allowed=0
prompt_evaluation_result_release_recorded=0
prompt_evaluation_result_release_created=0
prompt_evaluation_result_release_performed=0
prompt_evaluation_result_release_metadata_present=1
prompt_evaluation_result_release_family=operator-reviewed-prompt-evaluation-result-release
prompt_evaluation_result_release_format=contract-only-offline-evaluation-result-release
prompt_evaluation_result_release_decision=blocked_contract_only
prompt_evaluation_result_release_evidence_present=1
prompt_evaluation_result_release_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_plan_recorded=1
prompt_evaluation_result_release_method_planned=offline-prompt-evaluation-result-release-policy-
review
prompt_evaluation_result_release_result_recorded=0
prompt_evaluation_result_release_runtime_invoked=0
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_contract=1
prompt_evaluation_result_release_promotion_allowed=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_record_validated=0
prompt_evaluation_result_release_record_serialized=0
prompt_evaluation_result_release_record_written=0
prompt_evaluation_result_release_record_submitted=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_approval_recorded=0
prompt_evaluation_result_release_rejection_recorded=0
prompt_evaluation_result_release_route_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

This stage produces only release-contract metadata. It does not decide a release, publish a release, package a release, create release receipts, apply dispositions, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-35: Prompt Evaluation Result Release Receipt Contract

Record prompt-evaluation result release receipt metadata after prompt-evaluation result release metadata, prompt-evaluation result disposition metadata, prompt-evaluation result review metadata, prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-35 can define future prompt-evaluation result release receipt review requirements, but it cannot create prompt-evaluation result release receipt records, sign receipts, emit receipts, publish receipts, create prompt-evaluation result release records, record release decisions, publish releases, package releases, apply dispositions, create prompt-evaluation result review records, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_35_prompt_evaluation_result_release_receipt_contract_present=1
prompt_evaluation_result_release_receipt_contract_command=scripts/nadia-prompt-evaluation-result
-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt
prompt_evaluation_result_release_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_authority=0
prompt_evaluation_result_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_performed=0
prompt_evaluation_result_release_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_family=operator-reviewed-prompt-evaluation-result-relea
se-receipt
prompt_evaluation_result_release_receipt_format=contract-only-offline-evaluation-result-release-
receipt
prompt_evaluation_result_release_receipt_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_method_planned=offline-prompt-evaluation-result-release
-receipt-policy-review
prompt_evaluation_result_release_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_contract=1
prompt_evaluation_result_release_receipt_promotion_allowed=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_record_validated=0
prompt_evaluation_result_release_receipt_record_serialized=0
prompt_evaluation_result_release_receipt_record_written=0
prompt_evaluation_result_release_receipt_record_submitted=0
prompt_evaluation_result_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_approval_recorded=0
prompt_evaluation_result_release_receipt_rejection_recorded=0
prompt_evaluation_result_release_receipt_route_recorded=0
prompt_evaluation_result_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
```

```
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

This stage produces only release-receipt-contract metadata. It does not decide a receipt, create a receipt, sign a receipt, emit a receipt, publish a receipt, create release records, apply dispositions, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-36: Prompt Evaluation Result Release Receipt Review Contract

Record prompt-evaluation result release receipt review metadata after prompt-evaluation result release receipt metadata, prompt-evaluation result release metadata, prompt-evaluation result disposition metadata, prompt-evaluation result review metadata, prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-36 can define future prompt-evaluation result release receipt review disposition requirements, but it cannot create prompt-evaluation result release receipt review records, record review decisions, record review findings, approve receipts, reject receipts, create prompt-evaluation result release receipt records, sign receipts, publish receipts, create prompt-evaluation result release records, record release decisions, publish releases, package releases, apply dispositions, create prompt-evaluation result review records, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_36_prompt_evaluation_result_release_receipt_review_contract_present=1
prompt_evaluation_result_release_receipt_review_contract_command=scripts/nadia-prompt-evaluation
-result-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review
prompt_evaluation_result_release_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_metadata_present=1
prompt_evaluation_result_release_receipt_review_family=operator-reviewed-prompt-evaluation-resul
t-release-receipt-review
prompt_evaluation_result_release_receipt_review_format=contract-only-offline-evaluation-result-r
elease-receipt-review
prompt_evaluation_result_release_receipt_review_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_evidence_present=1
prompt_evaluation_result_release_receipt_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_plan_recorded=1
prompt_evaluation_result_release_receipt_review_method_planned=offline-prompt-evaluation-result-
release-receipt-review-policy-review
prompt_evaluation_result_release_receipt_review_result_recorded=0
prompt_evaluation_result_release_receipt_review_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_contract=1
prompt_evaluation_result_release_receipt_review_promotion_allowed=0
```

```
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

This stage produces only release-receipt-review-contract metadata. It does not decide a review, create a review record, approve a receipt, reject a receipt, create a receipt, sign a receipt, publish a receipt, create release records, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-37: Prompt Evaluation Result Release Receipt Review Disposition Contract

Record prompt-evaluation result release receipt review disposition metadata after prompt-evaluation result release receipt review metadata, prompt-evaluation result release receipt metadata, prompt-evaluation result release metadata, prompt-evaluation result disposition metadata, prompt-evaluation result review metadata, prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-37 can define future prompt-evaluation result release receipt review disposition release requirements, but it cannot create prompt-evaluation result release receipt review disposition records, record disposition decisions, record disposition findings, apply review dispositions, create prompt-evaluation result release receipt review records, record review decisions, record review findings, approve receipts, reject receipts, create prompt-evaluation result release receipt records, sign receipts, publish receipts, create prompt-evaluation result release records, record release decisions, publish releases, package releases, apply dispositions, create prompt-evaluation result review records, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_37_prompt_evaluation_result_release_receipt_review_disposition_contract_present=1
prompt_evaluation_result_release_receipt_review_disposition_contract_command=scripts/nadia-promp
t-evaluation-result-release-receipt-review-disposition-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_contract_command=latticra-
nadia prompt-evaluation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_performed=0
prompt_evaluation_result_release_receipt_review_disposition_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_family=operator-reviewed-prompt-eval
uation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_format=contract-only-offline-evaluat
ion-result-release-receipt-review-disposition
```

```
prompt_evaluation_result_release_receipt_review_disposition_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_source_policy=operator-reviewed-offl
ine
prompt_evaluation_result_release_receipt_review_disposition_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_method_planned=offline-prompt-evalua
tion-result-release-receipt-review-disposition-policy-review
prompt_evaluation_result_release_receipt_review_disposition_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

This stage produces only release-receipt-review-disposition-contract metadata. It does not decide a disposition, create a disposition record, record disposition findings, decide a review, create a review record, approve a receipt, reject a receipt, create a receipt, sign a receipt, publish a receipt, create release records, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-38: Prompt Evaluation Result Release Receipt Review Disposition Release Contract

Record prompt-evaluation result release receipt review disposition release metadata after prompt-evaluation result release receipt review disposition metadata, prompt-evaluation result release receipt review metadata, prompt-evaluation result release receipt metadata, prompt-evaluation result release metadata, prompt-evaluation result disposition metadata, prompt-evaluation result review metadata, prompt-evaluation result metadata, prompt-evaluation invocation metadata, prompt-evaluation runtime handoff metadata, prompt-evaluation-input metadata, context-window assembly metadata, prompt-token-sequence metadata, prompt-tokenization metadata, tokenizer-runtime-attachment metadata, runtime-invocation metadata, model-load metadata, inference-readiness metadata, local model-registry metadata, awareness-dialogue metadata, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are present. Stage-38 can define future prompt-evaluation result release receipt review disposition release receipt requirements, but it cannot create prompt-evaluation result release receipt review disposition release records, record release decisions, publish releases, package releases, create release receipts, create prompt-evaluation result release receipt review disposition records, record disposition decisions, record disposition findings, apply review dispositions, create prompt-evaluation result release receipt review records, record review decisions, record review findings, approve receipts, reject receipts, create prompt-evaluation result release receipt records, sign receipts, publish receipts, create prompt-evaluation result release records, record release decisions, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
nadia_stage_38_prompt_evaluation_result_release_receipt_review_disposition_release_contract_pres
ent=1
prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=scripts/nad
ia-prompt-evaluation-result-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=l
atticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_contract_status=contract_onl
y
prompt_evaluation_result_release_receipt_review_disposition_release_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_family=operator-reviewed-pro
mpt-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_format=contract-only-offline
-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_decision=blocked_contract_on
ly
prompt_evaluation_result_release_receipt_review_disposition_release_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_source_policy=operator-revie
wed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_method_planned=offline-promp
t-evaluation-result-release-receipt-review-disposition-release-policy-review
prompt_evaluation_result_release_receipt_review_disposition_release_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_cont
ract=1
prompt_evaluation_result_release_receipt_review_disposition_release_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
```

```
prompt_evaluation_result_release_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

This stage produces only release-receipt-review-disposition-release-contract metadata. It does not publish a release, package a release, create a release receipt, decide a disposition release, decide a disposition, create a disposition record, record disposition findings, decide a review, create a review record, approve a receipt, reject a receipt, create a receipt, sign a receipt, publish a receipt, create release records, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-39: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract

Record prompt-evaluation result release receipt review disposition release receipt metadata after prompt-evaluation result release receipt review disposition release metadata exists. Stage-39 can define future prompt-evaluation result release receipt review disposition release receipt review requirements, but it cannot create release-receipt records, emit receipts, sign receipts, publish receipts, package receipts, create disposition-release records, record release decisions, publish releases, package releases, create disposition records, record disposition decisions, record disposition findings, create review records, record review decisions, record review findings, approve receipts, reject receipts, create prompt-evaluation result release receipt records, sign receipts, publish receipts, create prompt-evaluation result release records, record release decisions, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_command=scr
ipts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-contract.
sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_c
ommand=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_status=cont
ract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_packaged=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_contract=1
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt contract metadata. It does not emit, sign, publish, package, or record a receipt, decide a disposition release, decide a disposition, create a disposition record, record disposition findings, decide a review, create a review record, approve a receipt, reject a receipt, create release records, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-40: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract

Record prompt-evaluation result release receipt review disposition release receipt review metadata after prompt-evaluation result release receipt review disposition release receipt metadata exists. Stage-40 can define future prompt-evaluation result release receipt review disposition release receipt review disposition requirements, but it cannot create release-receipt-review records, record review decisions, record review findings, apply review dispositions, emit receipts, sign receipts, publish receipts, package receipts, record disposition-release receipts, create disposition-release records, record release decisions, publish releases, package releases, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_comm
and=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-re
view-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_con
tract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_stat
us=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_stage=contrac
t-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_record_create
d=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_findings_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_applied=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_cont
ract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_contract=1
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review contract metadata. It does not create a review record, record a review decision, record review findings, apply a disposition, emit a receipt, sign a receipt, publish a receipt, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-41: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract

Record prompt-evaluation result release receipt review disposition release receipt review disposition metadata after prompt-evaluation result release receipt review disposition release receipt review metadata exists. Stage-41 can define future prompt-evaluation result release receipt review disposition release receipt review disposition release requirements, but it cannot create release-receipt-review-disposition records, record review decisions, record review findings, apply review dispositions, emit receipts, sign receipts, publish receipts, package receipts, read model output, record model output, record generated answer text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
ontract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-releas
e-receipt-review-disposition-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
ontract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_s
tage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
uthority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
llowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
reated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
indings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
pplied=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_cont
ract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_contract=1
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review-disposition contract metadata. It does not create a review-disposition record, record a review decision, record review findings, apply a disposition, emit a receipt, sign a receipt, publish a receipt, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-42: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract

Record prompt-evaluation result release receipt review disposition release receipt review disposition release metadata after prompt-evaluation result release receipt review disposition release receipt review disposition metadata exists. Stage-42 can define future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt requirements, but it cannot create disposition-release records, record release decisions, publish releases, package releases, create release receipts, create disposition records, record disposition decisions, create review records, sign receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-dispositio
n-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review
-disposition-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_contract=1
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review-disposition-release contract metadata. It does not create a review-disposition-release record, record a release decision, publish a release, package a release, create a release receipt, sign a receipt, publish a receipt, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-43: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract

Record prompt-evaluation result release receipt review disposition release receipt review disposition release receipt metadata after prompt-evaluation result release receipt review disposition release receipt review disposition release metadata exists. Stage-43 can define future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review requirements, but it cannot create release-receipt records, emit receipts, sign receipts, publish receipts, package receipts, create release records, record release decisions, publish releases, create review records, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-di
sposition-release-receipt-review-disposition-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_contract_command=latticra-nadia prompt-evaluation-result-release-receip
t-review-disposition-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_contract=1
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review-disposition-release-receipt contract metadata. It does not create a receipt record, emit a receipt, sign a receipt, publish a receipt, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-44: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract

Record prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review metadata after prompt-evaluation result release receipt review disposition release receipt review disposition release receipt metadata exists. Stage-44 can define future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition requirements, but it cannot create release-receipt-review records, record review decisions, record review findings, record review dispositions, sign receipts, publish receipts, package receipts, create release records, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-re
view-disposition-release-receipt-review-disposition-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_contract_command=latticra-nadia prompt-evaluation-result-release
-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_findings_recorded=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_contract=1
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review contract metadata. It does not create a release-receipt-review record, record review findings, sign or publish receipts, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-45: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract

Record prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition metadata after prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review metadata exists. Stage-45 can define future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release requirements, but it cannot create release-receipt-review-disposition records, record disposition decisions, record disposition findings, sign receipts, publish receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_contract_command=scripts/nadia-prompt-evaluation-result-releas
e-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-dispositi
on-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_disposition_contract_command=latticra-nadia prompt-evaluation-re
sult-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-revie
w-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_findings_recorded=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_release_contract=1
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition contract metadata. It does not create a release-receipt-review-disposition record, record disposition findings, sign or publish receipts, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

Stage-46: Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract

Record prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release metadata after prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition metadata exists. Stage-46 can define future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release receipt requirements, but it cannot create disposition-release records, record release decisions, publish releases, package releases, create release receipts, record dispositions, record reviews, record receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, use tools, mutate source, use the network, or provide sexual user functionality.

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_contract_command=scripts/nadia-prompt-evaluation-resul
t-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-d
isposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_disposition_release_contract_command=latticra-nadia prompt-evalu
ation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-recei
pt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_created=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_disposition_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_release_receipt_contract=1
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
```

This stage produces only release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release contract metadata. It does not create a disposition-release record, record release decisions, publish or package releases, create release receipts, read model output, record model output, generate dialogue, or make Nadia usable as a runtime assistant.

docs/NADIA_PRODUCTIVITY_LOOP_STAGE_5.md

Source title: Nadia Productivity Loop Stage-5

Lines: 106 | Words: 247 | SHA-256: a153b60dd4c44a09de6c248613d2daa412bed5dc80fd136d7e6baf85454a0e31

Nadia Productivity Loop Stage-5

Status: Stage-5 implementation contract Date: 2026-05-25 Scope: operator-reviewed local productivity ledger before training, inference, or tool authority.

Purpose

Stage-5 gives Nadia a local productivity ledger.

The ledger records operator-reviewed outcomes from Stage-4 mode-validation work. These records can later help retrieval ranking, plan-template selection, test recommendations, and project memory, but this stage does not train or distill a model.

Capability

Stage-5 adds:

```
nadia_stage_5_productivity_loop_present=1
nadia_productivity_ledger_generator_present=1
productivity_ledger_command=scripts/nadia-productivity-ledger.sh
installed_productivity_ledger_command=latticra-nadia productivity-ledger
requires_mode_validation=1
mode_validation_stage_required=4-systems-engineering-mode-validation
learning_scope=operator-reviewed-local-productivity
ledger_append_only=1
project_memory_scope=local-metadata-only
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
distillation_performed=0
```

Stage-5 does not summarize with AI, call an inference runtime, evaluate a prompt, mutate source, execute tools, train, distill, or use the network.

Inputs

The productivity-ledger generator requires:

```
mode_validation=Stage-4 Nadia mode-validation report
outcome=operator-reviewed productivity outcome label
recommendation=operator-reviewed recommendation label
```

The default outcome is operator-reviewed-planning. The default recommendation is run-stage-guards.

Outputs

The productivity-ledger generator writes:

```
nadia-productivity-entry-&lt;timestamp&gt;.txt
latest-productivity-entry.txt
productivity-ledger-index.tsv
```

The entry records:

```
stage=5-productivity-ledger-loop
learning_scope=operator-reviewed-local-productivity
mode_validation_stage=4-systems-engineering-mode-validation
ledger_append_only=1
project_memory_scope=local-metadata-only
training_performed=0
distillation_performed=0
source_mutation_authority=0
```

Usage

With an explicit mode-validation report:

```
sh scripts/nadia-productivity-ledger.sh \
  --mode-validation /path/to/latest-mode-validation.txt \
```

```
    --outcome &quot;accepted planning surface&quot; \
    --recommendation &quot;run stage guards&quot; \
    --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-productivity.XXXXXX&quot;
  )&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia productivity-ledger
```

Non-Claims

Stage-5 Nadia is not an inference runtime, prompt evaluator, code generator, autonomous coding agent, source mutator, training system, distillation system, network retriever, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-productivity-loop-stage-5.sh
```

Expected result:

```
nadia_productivity_loop_stage_5: ok
```

docs/NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8.md

Source title: Nadia Prompt Evaluation Contract Stage-8

Lines: 118 | Words: 254 | SHA-256: 7b37c7d9c23c63f9a9a4ebc3045697cdb08c8af80fd70cde7790e3aeab64a197

Nadia Prompt Evaluation Contract Stage-8

Status: Stage-8 implementation contract Date: 2026-05-25 Scope: prompt-evaluation contract before prompt materialization, prompt evaluation, inference, or tool execution.

Purpose

Stage-8 gives Nadia a prompt-evaluation contract without evaluating prompts.

The contract consumes a Stage-7 guarded tool-authority preflight, verifies report-only tool posture and protective-safety restrictions, and records the receipt fields that would be required before any future prompt evaluation path can exist.

Capability

Stage-8 adds:

```
nadia_stage_8_prompt_evaluation_contract_present=1
nadia_prompt_evaluation_contract_generator_present=1
prompt_evaluation_contract_command=scripts/nadia-prompt-evaluation-contract.sh
installed_prompt_evaluation_contract_command=latticra-nadia prompt-contract
requires_tool_preflight=1
tool_preflight_stage_required=7-guarded-tool-authority-preflight
prompt_evaluation_stage=contract-only
prompt_contract_status=contract_only
prompt_materialized=0
prompt_text_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
prompt_receipt_required=1
refusal_policy_required=1
protective_safety_required=1
tool_preflight_required=1
runtime_profile_required=1
model_registry_review_required=1
operator_review_required=1
contract_promotion_allowed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
model_runtime_invoked=0
```

```
inference_performed=0
network_authority=0
training_performed=0
distillation_performed=0
```

Stage-8 does not materialize prompts, evaluate prompts, call an inference runtime, execute tools, mutate source, train, distill, or use the network.

Inputs

The prompt-evaluation contract requires:

```
tool_preflight=Stage-7 Nadia guarded tool-authority preflight
request_class=operator request classification label
```

Sexualized request classifications fail closed.

Outputs

The contract writes:

```
nadia-prompt-contract-&lt;timestamp&gt;.txt
latest-prompt-contract.txt
```

The report records:

```
stage=8-prompt-evaluation-contract
prompt_evaluation_stage=contract-only
prompt_evaluation_authority=0
prompt_materialized=0
prompt_evaluated=0
model_runtime_invoked=0
inference_performed=0
tool_execution_authority=0
sexual_request_refusal=always
```

Usage

With an explicit tool preflight:

```
sh scripts/nadia-prompt-evaluation-contract.sh \
  --tool-preflight /path/to/latest-tool-preflight.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-contracts.XXXXXX&quot;
)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia prompt-contract
```

Non-Claims

Stage-8 Nadia is not a prompt evaluator, inference runtime, tool executor, shell runner, network client, source mutator, credential reader, sexual assistant, roleplay surface, adult-content generator, training system, distillation system, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-prompt-evaluation-contract-stage-8.sh
```

Expected result:

```
nadia_prompt_evaluation_contract_stage_8: ok
```

docs/NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16.md

Source title: Nadia Prompt Evaluation Handoff Contract Stage-16

Lines: 120 | Words: 310 | SHA-256: 515a2611c02a82c1b79095335affd0a5cc31c276b51eed08a717f0cc89b741f1

Nadia Prompt Evaluation Handoff Contract Stage-16

Status: Stage-16 implementation contract Date: 2026-05-25 Scope: prompt-evaluation handoff metadata before prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-16 gives Nadia a prompt-evaluation handoff contract after the awareness-dialogue contract is present.

The contract consumes a Stage-15 awareness-dialogue contract, verifies inherited non-executing dialogue, prompt, runtime, model, tool, and protective-safety posture, and records that the next promotion point is a separate tokenization boundary contract. It does not evaluate prompts.

Capability

Stage-16 adds:

```
nadia_stage_16_prompt_evaluation_handoff_contract_present=1
nadia_prompt_evaluation_handoff_contract_generator_present=1
prompt_evaluation_handoff_contract_command=scripts/nadia-prompt-evaluation-handoff-contract.sh
installed_prompt_evaluation_handoff_contract_command=latticra-nadia prompt-evaluation-handoff
prompt_evaluation_handoff_contract_status=contract_only
prompt_evaluation_handoff_stage=contract-only
prompt_evaluation_handoff_authority=0
prompt_evaluation_handoff_allowed=0
prompt_evaluation_handoff_performed=0
prompt_evaluation_authority=0
prompt_evaluated=0
evaluation_handoff_decision=blocked_contract_only
evaluation_handoff_evidence_present=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenization_contract=1
prompt_evaluation_handoff_promotion_allowed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Stage-16 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inherited Awareness Boundary

The handoff contract preserves the Stage-15 Nadia Initiative awareness scope:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
plain_language_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
live_web_lookup_authority=0
topic_yazidi_genocide_awareness=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_womens_empowerment=1
```

Protective Boundary

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
```

Usage

With explicit evidence input:

```
sh scripts/nadia-prompt-evaluation-handoff-contract.sh \
  --awareness-dialogue /path/to/latest-awareness-dialogue-contract.txt \
  --request-class awareness-education \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-handoff.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia prompt-evaluation-handoff
```

Non-Claims

Stage-16 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

```
sh scripts/test-nadia-prompt-evaluation-handoff-contract-stage-16.sh
```

Expected result:

```
nadia_prompt_evaluation_handoff_contract_stage_16: ok
```

docs/NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28.md

Source title: Nadia Prompt Evaluation Input Contract Stage-28

Lines: 165 | Words: 557 | SHA-256: f632993f079f70f7ff5ea846d3d7dacdd82e657fb9442942f086a2d324e57bc6

Nadia Prompt Evaluation Input Contract Stage-28

Status: Stage-28 implementation contract

Scope: prompt-evaluation-input metadata before prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-28 gives Nadia a prompt-evaluation-input contract after the context-window assembly contract is present.

The contract records how future reviewed prompt-evaluation input packaging must reference the Stage-27 context-window assembly contract, prompt-token-sequence boundary, prompt-tokenization boundary, tokenizer-runtime-attachment boundary, context-window policy, token-budget policy, truncation policy, ordering policy, attention-mask policy, position-ID policy, input-schema policy, safety-envelope policy, and runtime-denial policy. It preserves the absolute rule that Nadia cannot yet read prompt text, assemble context windows, create prompt evaluation inputs, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference. The script only measures the Stage-27 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-28 adds:

```
nadia_stage_28_prompt_evaluation_input_contract_present=1
nadia_prompt_evaluation_input_contract_generator_present=1
prompt_evaluation_input_contract_command=scripts/nadia-prompt-evaluation-input-contract.sh
installed_prompt_evaluation_input_contract_command=latticra-nadia prompt-evaluation-input
prompt_evaluation_input_contract_status=contract_only
prompt_evaluation_input_stage=contract-only
prompt_evaluation_input_authority=0
prompt_evaluation_input_allowed=0
prompt_evaluation_input_created=0
prompt_evaluation_input_metadata_present=1
prompt_evaluation_input_family=operator-reviewed-prompt-evaluation-input
prompt_evaluation_input_format=contract-only-offline-evaluation-input
prompt_evaluation_input_decision=blocked_contract_only
prompt_evaluation_input_evidence_present=1
prompt_evaluation_input_source_policy=operator-reviewed-offline
prompt_evaluation_input_plan_recorded=1
prompt_evaluation_input_method_planned=offline-prompt-evaluation-input-policy-review
prompt_evaluation_input_result_recorded=0
prompt_evaluation_input_runtime_invoked=0
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_runtime_handoff_contract=1
prompt_evaluation_input_promotion_allowed=0
```

Stage-28 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt token buffers, create prompt tokens, count prompt tokens, record prompt token IDs, record prompt token order, record prompt token offsets, create attention masks, create position IDs, assemble context windows, serialize context windows, create prompt evaluation inputs, validate prompt evaluation inputs, serialize prompt evaluation inputs, write prompt evaluation inputs, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, bind tokenizer artifacts, attach tokenizers to a runtime, create runtime sessions, open tokenizer files, read tokenizer files, load tokenizer vocabularies, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Input Requirements

The prompt-evaluation-input contract names future review requirements:

```
requires_context_window_assembly_reference=1
requires_context_window_policy_reference=1
requires_context_window_token_budget_policy=1
requires_context_window_truncation_policy=1
```

```
requires_context_window_ordering_policy=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_attention_mask_policy=1
requires_position_id_policy=1
requires_evaluation_input_schema_policy=1
requires_evaluation_input_context_reference_policy=1
requires_evaluation_input_token_sequence_reference_policy=1
requires_evaluation_input_safety_envelope_policy=1
requires_evaluation_input_runtime_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_input_creation=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-28 preserves explicit prompt-evaluation-input denial fields:

```
prompt_evaluation_input_authority=0
prompt_evaluation_input_allowed=0
prompt_evaluation_input_open_authority=0
prompt_evaluation_input_read_authority=0
prompt_evaluation_input_write_authority=0
prompt_evaluation_input_execute_authority=0
prompt_evaluation_input_runtime_authority=0
prompt_evaluation_input_context_reference_authority=0
prompt_evaluation_input_token_sequence_reference_authority=0
prompt_evaluation_input_runtime_handoff_authority=0
prompt_evaluation_input_created=0
prompt_evaluation_input_materialized=0
prompt_evaluation_input_loaded=0
prompt_evaluation_input_opened=0
prompt_evaluation_input_read=0
prompt_evaluation_input_validated=0
prompt_evaluation_input_serialized=0
prompt_evaluation_input_written=0
prompt_evaluation_input_schema_validated=0
prompt_evaluation_input_context_reference_recorded=0
prompt_evaluation_input_token_reference_recorded=0
prompt_evaluation_input_safety_envelope_recorded=0
context_window_assembled=0
context_window_serialized=0
prompt_token_sequence_recorded=0
prompt_token_ids_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_tokenized=0
tokenizer_runtime_attachment_performed=0
tokenizer_attached_to_runtime=0
runtime_session_created=0
runtime_invoked=0
prompt_evaluated=0
qa_dialogue_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-input-contract.sh \
  --context-window-assembly
reports/nadia/context-window-assembly/latest-context-window-assembly-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-input.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-input
```

Non-Claims

Stage-28 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, tokenizer runtime attachment layer, prompt tokenizer, token counter, token sequence recorder, token ID recorder, token offset recorder, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-input-contract-stage-28.sh
```

Expected output:

```
nadia_prompt_evaluation_input_contract_stage_28: ok
```

docs/NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30.md

Source title: Nadia Prompt Evaluation Invocation Contract Stage-30

Lines: 166 | Words: 468 | SHA-256: fcb03493b72da7959a5ab060c156b563e93567a343aa31d89204597d0b9e1639

Nadia Prompt Evaluation Invocation Contract Stage-30

Status: Stage-30 implementation contract

Scope: prompt-evaluation invocation metadata before invocation request creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-30 gives Nadia a prompt-evaluation invocation contract after the prompt-evaluation runtime handoff contract is present.

The contract records how future reviewed prompt-evaluation invocation packaging must reference the Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, tokenizer-runtime-attachment boundary, runtime-profile boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, runtime-denial policy, token-generation denial policy, prompt-evaluation result policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet submit invocation requests, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference. The script only measures the Stage-29 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-30 adds:

```
nadia_stage_30_prompt_evaluation_invocation_contract_present=1
nadia_prompt_evaluation_invocation_contract_generator_present=1
prompt_evaluation_invocation_contract_command=scripts/nadia-prompt-evaluation-invocation-contrac
t.sh
```

```
installed_prompt_evaluation_invocation_contract_command=latticra-nadia
prompt-evaluation-invocation
prompt_evaluation_invocation_contract_status=contract_only
prompt_evaluation_invocation_stage=contract-only
prompt_evaluation_invocation_authority=0
prompt_evaluation_invocation_allowed=0
prompt_evaluation_invocation_performed=0
prompt_evaluation_invocation_metadata_present=1
prompt_evaluation_invocation_family=operator-reviewed-prompt-evaluation-invocation
prompt_evaluation_invocation_format=contract-only-offline-evaluation-invocation
prompt_evaluation_invocation_decision=blocked_contract_only
prompt_evaluation_invocation_evidence_present=1
prompt_evaluation_invocation_source_policy=operator-reviewed-offline
prompt_evaluation_invocation_plan_recorded=1
prompt_evaluation_invocation_method_planned=offline-prompt-evaluation-invocation-policy-review
prompt_evaluation_invocation_result_recorded=0
prompt_evaluation_invocation_runtime_invoked=0
requires_prompt_evaluation_runtime_handoff_contract=1
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_result_contract=1
prompt_evaluation_invocation_promotion_allowed=0
```

Stage-30 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt token buffers, create prompt tokens, count prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, submit invocation requests, schedule invocation requests, select a runtime, select a model, create runtime sessions, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Invocation Requirements

The prompt-evaluation invocation contract names future review requirements:

```
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_runtime_profile_reference=1
requires_runtime_invocation_contract_reference=1
requires_model_load_contract_reference=1
requires_inference_readiness_contract_reference=1
requires_prompt_evaluation_invocation_schema_policy=1
requires_prompt_evaluation_invocation_denial_policy=1
requires_prompt_evaluation_result_schema_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_invocation=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-30 preserves explicit prompt-evaluation invocation denial fields:

```
prompt_evaluation_invocation_authority=0
prompt_evaluation_invocation_allowed=0
prompt_evaluation_invocation_open_authority=0
```

```
prompt_evaluation_invocation_read_authority=0
prompt_evaluation_invocation_write_authority=0
prompt_evaluation_invocation_execute_authority=0
prompt_evaluation_invocation_runtime_authority=0
prompt_evaluation_invocation_prompt_evaluation_authority=0
prompt_evaluation_invocation_token_generation_authority=0
prompt_evaluation_invocation_inference_authority=0
prompt_evaluation_invocation_performed=0
prompt_evaluation_invocation_created=0
prompt_evaluation_invocation_materialized=0
prompt_evaluation_invocation_loaded=0
prompt_evaluation_invocation_opened=0
prompt_evaluation_invocation_read=0
prompt_evaluation_invocation_validated=0
prompt_evaluation_invocation_serialized=0
prompt_evaluation_invocation_written=0
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_validated=0
prompt_evaluation_invocation_request_serialized=0
prompt_evaluation_invocation_request_written=0
prompt_evaluation_invocation_request_submitted=0
prompt_evaluation_invocation_request_scheduled=0
prompt_evaluation_invocation_request_queued=0
prompt_evaluation_invocation_runtime_selected=0
prompt_evaluation_invocation_model_selected=0
prompt_evaluation_invocation_session_created=0
prompt_evaluation_invocation_runtime_invoked=0
prompt_evaluation_request_created=0
prompt_evaluation_request_serialized=0
prompt_evaluation_request_submitted=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
runtime_process_spawned=0
runtime_binary_executed=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-invocation-contract.sh \
  --prompt-evaluation-runtime-handoff reports/nadia/prompt-evaluation-runtime-handoff/latest-pro
mpt-evaluation-runtime-handoff-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-invocation.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-invocation
```

Non-Claims

Stage-30 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, tokenizer loader, tokenizer registry, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-invocation-contract-stage-30.sh
```

Expected output:

```
nadia_prompt_evaluation_invocation_contract_stage_30: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31.md

Source title: Nadia Prompt Evaluation Result Contract Stage-31

Lines: 177 | Words: 469 | SHA-256: 234558cc99db4abf3aed6f0a2e07f4d08f6aba930b19ad7ec70192ab6fea6605

Nadia Prompt Evaluation Result Contract Stage-31

Status: Stage-31 implementation contract

Scope: prompt-evaluation result metadata before result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-31 gives Nadia a prompt-evaluation result contract after the prompt-evaluation invocation contract is present.

The contract records how future reviewed prompt-evaluation result handling must reference the Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet record result content, record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-31 adds:

```
nadia_stage_31_prompt_evaluation_result_contract_present=1
nadia_prompt_evaluation_result_contract_generator_present=1
prompt_evaluation_result_contract_command=scripts/nadia-prompt-evaluation-result-contract.sh
installed_prompt_evaluation_result_contract_command=latticra-nadia prompt-evaluation-result
prompt_evaluation_result_contract_status=contract_only
prompt_evaluation_result_stage=contract-only
prompt_evaluation_result_authority=0
prompt_evaluation_result_allowed=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_created=0
prompt_evaluation_result_performed=0
prompt_evaluation_result_metadata_present=1
prompt_evaluation_result_family=operator-reviewed-prompt-evaluation-result
prompt_evaluation_result_format=contract-only-offline-evaluation-result
prompt_evaluation_result_decision=blocked_contract_only
prompt_evaluation_result_evidence_present=1
prompt_evaluation_result_source_policy=operator-reviewed-offline
prompt_evaluation_result_plan_recorded=1
prompt_evaluation_result_method_planned=offline-prompt-evaluation-result-policy-review
prompt_evaluation_result_result_recorded=0
prompt_evaluation_result_runtime_invoked=0
requires_prompt_evaluation_invocation_contract=1
requires_prompt_evaluation_runtime_handoff_contract=1
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_result_review_contract=1
prompt_evaluation_result_promotion_allowed=0
```

Stage-31 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, record model output, record generated answer text, score model output, record token log probabilities, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Result Requirements

The prompt-evaluation result contract names future review requirements:

```
requires_prompt_evaluation_invocation_reference=1
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_prompt_evaluation_result_schema_policy=1
requires_prompt_evaluation_result_denial_policy=1
requires_prompt_evaluation_result_review_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-31 preserves explicit prompt-evaluation result denial fields:

```
prompt_evaluation_result_authority=0
prompt_evaluation_result_allowed=0
prompt_evaluation_result_open_authority=0
prompt_evaluation_result_read_authority=0
prompt_evaluation_result_write_authority=0
prompt_evaluation_result_execute_authority=0
prompt_evaluation_result_runtime_authority=0
prompt_evaluation_result_prompt_evaluation_authority=0
prompt_evaluation_result_token_generation_authority=0
prompt_evaluation_result_inference_authority=0
prompt_evaluation_result_generation_authority=0
prompt_evaluation_result_recording_authority=0
prompt_evaluation_result_performed=0
prompt_evaluation_result_created=0
prompt_evaluation_result_materialized=0
prompt_evaluation_result_loaded=0
prompt_evaluation_result_opened=0
prompt_evaluation_result_read=0
prompt_evaluation_result_validated=0
prompt_evaluation_result_serialized=0
prompt_evaluation_result_written=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_record_validated=0
prompt_evaluation_result_record_serialized=0
prompt_evaluation_result_record_written=0
prompt_evaluation_result_record_submitted=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
prompt_evaluation_result_score_recorded=0
prompt_evaluation_result_token_logprobs_recorded=0
prompt_evaluation_result_completion_reason_recorded=0
prompt_evaluation_result_error_recorded=0
prompt_evaluation_result_runtime_invoked=0
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_submitted=0
prompt_evaluation_request_created=0
```

```
prompt_evaluation_request_serialized=0
prompt_evaluation_request_submitted=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
runtime_process_spawned=0
runtime_binary_executed=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-contract.sh \
  --prompt-evaluation-invocation
reports/nadia/prompt-evaluation-invocation/latest-prompt-evaluation-invocation-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result
```

Non-Claims

Stage-31 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-contract-stage-31.sh
```

Expected output:

```
nadia_prompt_evaluation_result_contract_stage_31: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33.md

Source title: Nadia Prompt Evaluation Result Disposition Contract Stage-33

Lines: 179 | Words: 524 | SHA-256: ed15fb514bee7dd03b365f09b54576d839a8a7a715f7b2d75d13edfee3284999

Nadia Prompt Evaluation Result Disposition Contract Stage-33

Status: Stage-33 implementation contract

Scope: prompt-evaluation result disposition metadata before disposition recording, release recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-33 gives Nadia a prompt-evaluation result disposition contract after the prompt-evaluation result review contract is present.

The contract records how future release decisions must reference the Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition records, release records, review records, result records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-33 adds:

```
nadia_stage_33_prompt_evaluation_result_disposition_contract_present=1
nadia_prompt_evaluation_result_disposition_contract_generator_present=1
prompt_evaluation_result_disposition_contract_command=scripts/nadia-prompt-evaluation-result-dis
position-contract.sh
installed_prompt_evaluation_result_disposition_contract_command=latticra-nadia
prompt-evaluation-result-disposition
prompt_evaluation_result_disposition_contract_status=contract_only
prompt_evaluation_result_disposition_stage=contract-only
prompt_evaluation_result_disposition_authority=0
prompt_evaluation_result_disposition_allowed=0
prompt_evaluation_result_disposition_recorded=0
prompt_evaluation_result_disposition_created=0
prompt_evaluation_result_disposition_performed=0
prompt_evaluation_result_disposition_metadata_present=1
prompt_evaluation_result_disposition_family=operator-reviewed-prompt-evaluation-result-dispositi
on
prompt_evaluation_result_disposition_format=contract-only-offline-evaluation-result-disposition
prompt_evaluation_result_disposition_decision=blocked_contract_only
prompt_evaluation_result_disposition_evidence_present=1
prompt_evaluation_result_disposition_source_policy=operator-reviewed-offline
prompt_evaluation_result_disposition_plan_recorded=1
prompt_evaluation_result_disposition_method_planned=offline-prompt-evaluation-result-disposition
-policy-review
prompt_evaluation_result_disposition_result_recorded=0
prompt_evaluation_result_disposition_runtime_invoked=0
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_contract=1
prompt_evaluation_result_disposition_promotion_allowed=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_route_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
answer_text_generated=0
```

Stage-33 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, read model output, record model output, record generated answer text, score model output, record token log probabilities, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Disposition Requirements

The prompt-evaluation result disposition contract names future release requirements:

```
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_invocation_reference=1
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_prompt_evaluation_result_disposition_schema_policy=1
requires_prompt_evaluation_result_disposition_denial_policy=1
requires_prompt_evaluation_result_release_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_disposition=1
requires_no_prompt_evaluation_result_release=1
requires_no_prompt_evaluation_result_review=1
requires_no_prompt_evaluation_result=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-33 preserves explicit prompt-evaluation result disposition denial fields:

```
prompt_evaluation_result_disposition_authority=0
prompt_evaluation_result_disposition_allowed=0
prompt_evaluation_result_disposition_open_authority=0
prompt_evaluation_result_disposition_read_authority=0
prompt_evaluation_result_disposition_write_authority=0
prompt_evaluation_result_disposition_execute_authority=0
prompt_evaluation_result_disposition_runtime_authority=0
prompt_evaluation_result_disposition_prompt_evaluation_authority=0
prompt_evaluation_result_disposition_token_generation_authority=0
prompt_evaluation_result_disposition_inference_authority=0
prompt_evaluation_result_disposition_generation_authority=0
prompt_evaluation_result_disposition_recording_authority=0
prompt_evaluation_result_disposition_performed=0
prompt_evaluation_result_disposition_created=0
prompt_evaluation_result_disposition_loaded=0
prompt_evaluation_result_disposition_opened=0
prompt_evaluation_result_disposition_read=0
prompt_evaluation_result_disposition_validated=0
prompt_evaluation_result_disposition_serialized=0
prompt_evaluation_result_disposition_written=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_record_validated=0
prompt_evaluation_result_disposition_record_serialized=0
prompt_evaluation_result_disposition_record_written=0
prompt_evaluation_result_disposition_record_submitted=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_approval_recorded=0
prompt_evaluation_result_disposition_rejection_recorded=0
prompt_evaluation_result_disposition_route_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
```

```
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-disposition-contract.sh \
  --prompt-evaluation-result-review reports/nadia/prompt-evaluation-result-review/latest-prompt-evaluation-result-review-contract.txt \
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-disposition.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-disposition
```

Non-Claims

Stage-33 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-disposition-contract-stage-33.sh
```

Expected output:

```
nadia_prompt_evaluation_result_disposition_contract_stage_33: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34.md

Source title: Nadia Prompt Evaluation Result Release Contract Stage-34

Lines: 192 | Words: 583 | SHA-256: a45e8d8e8587bec75f800919119aff7bca8d078fd1f366e8dabb89b980733605

Nadia Prompt Evaluation Result Release Contract Stage-34

Status: Stage-34 implementation contract

Scope: prompt-evaluation result release metadata before release recording, release decision recording, release publication, release packaging, release receipt creation, disposition recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-34 gives Nadia a prompt-evaluation result release contract after the prompt-evaluation result disposition contract is present.

The contract records how a future release lane must reference the Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create release records, publish releases, package releases, create release receipts, apply dispositions, create review records, create result records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-34 adds:

```
nadia_stage_34_prompt_evaluation_result_release_contract_present=1
nadia_prompt_evaluation_result_release_contract_generator_present=1
prompt_evaluation_result_release_contract_command=scripts/nadia-prompt-evaluation-result-release
-contract.sh
installed_prompt_evaluation_result_release_contract_command=latticra-nadia
prompt-evaluation-result-release
prompt_evaluation_result_release_contract_status=contract_only
prompt_evaluation_result_release_stage=contract-only
prompt_evaluation_result_release_authority=0
prompt_evaluation_result_release_allowed=0
prompt_evaluation_result_release_recorded=0
prompt_evaluation_result_release_created=0
prompt_evaluation_result_release_performed=0
prompt_evaluation_result_release_metadata_present=1
prompt_evaluation_result_release_family=operator-reviewed-prompt-evaluation-result-release
prompt_evaluation_result_release_format=contract-only-offline-evaluation-result-release
prompt_evaluation_result_release_decision=blocked_contract_only
prompt_evaluation_result_release_evidence_present=1
prompt_evaluation_result_release_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_plan_recorded=1
prompt_evaluation_result_release_method_planned=offline-prompt-evaluation-result-release-policy-
review
prompt_evaluation_result_release_result_recorded=0
prompt_evaluation_result_release_runtime_invoked=0
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_contract=1
prompt_evaluation_result_release_promotion_allowed=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_approval_recorded=0
prompt_evaluation_result_release_rejection_recorded=0
prompt_evaluation_result_release_route_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
answer_text_generated=0
```

Stage-34 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, read model output, record model output, record generated answer text, score model output, record token log probabilities, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Release Requirements

The prompt-evaluation result release contract names future release receipt requirements:

```
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_invocation_reference=1
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_prompt_evaluation_result_release_schema_policy=1
requires_prompt_evaluation_result_release_denial_policy=1
requires_prompt_evaluation_result_release_receipt_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_prompt_evaluation_result_disposition=1
requires_no_prompt_evaluation_result_review=1
requires_no_prompt_evaluation_result=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-34 preserves explicit prompt-evaluation result release denial fields:

```
prompt_evaluation_result_release_authority=0
prompt_evaluation_result_release_allowed=0
prompt_evaluation_result_release_open_authority=0
prompt_evaluation_result_release_read_authority=0
prompt_evaluation_result_release_write_authority=0
prompt_evaluation_result_release_execute_authority=0
prompt_evaluation_result_release_runtime_authority=0
prompt_evaluation_result_release_prompt_evaluation_authority=0
prompt_evaluation_result_release_token_generation_authority=0
prompt_evaluation_result_release_inference_authority=0
prompt_evaluation_result_release_generation_authority=0
prompt_evaluation_result_release_recording_authority=0
prompt_evaluation_result_release_performed=0
prompt_evaluation_result_release_created=0
prompt_evaluation_result_release_loaded=0
prompt_evaluation_result_release_opened=0
prompt_evaluation_result_release_read=0
prompt_evaluation_result_release_validated=0
prompt_evaluation_result_release_serialized=0
prompt_evaluation_result_release_written=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_record_validated=0
prompt_evaluation_result_release_record_serialized=0
prompt_evaluation_result_release_record_written=0
prompt_evaluation_result_release_record_submitted=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_approval_recorded=0
prompt_evaluation_result_release_rejection_recorded=0
prompt_evaluation_result_release_route_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_release_runtime_invoked=0
prompt_evaluation_result_disposition_record_created=0
```

```
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-contract.sh \
  --prompt-evaluation-result-disposition reports/nadia/prompt-evaluation-result-disposition/late
st-prompt-evaluation-result-disposition-contract.txt \
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-release.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release
```

Non-Claims

Stage-34 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-contract-stage-34.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_contract_stage_34: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35.md

Source title: Nadia Prompt Evaluation Result Release Receipt Contract Stage-35

Lines: 200 | Words: 624 | SHA-256: 8f4491b781a00f55b1386e1be594316bf4bad56fc60c580038c544ee66d908ac

Nadia Prompt Evaluation Result Release Receipt Contract Stage-35

Status: Stage-35 implementation contract

Scope: prompt-evaluation result release receipt metadata before receipt recording, receipt signing, receipt publication, release recording, release decision recording, release publication, release packaging, disposition recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-35 gives Nadia a prompt-evaluation result release receipt contract after the prompt-evaluation result release contract is present.

The contract records how a future receipt-review lane must reference the Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create release receipts, sign receipts, emit receipts, publish receipts, create release records, publish releases, package releases, apply dispositions, create review records, create result records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-35 adds:

```
nadia_stage_35_prompt_evaluation_result_release_receipt_contract_present=1
nadia_prompt_evaluation_result_release_receipt_contract_generator_present=1
prompt_evaluation_result_release_receipt_contract_command=scripts/nadia-prompt-evaluation-result
-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt
prompt_evaluation_result_release_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_authority=0
prompt_evaluation_result_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_performed=0
prompt_evaluation_result_release_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_family=operator-reviewed-prompt-evaluation-result-relea
se-receipt
prompt_evaluation_result_release_receipt_format=contract-only-offline-evaluation-result-release-
receipt
prompt_evaluation_result_release_receipt_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_method_planned=offline-prompt-evaluation-result-release
-receipt-policy-review
prompt_evaluation_result_release_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_contract=1
prompt_evaluation_result_release_receipt_promotion_allowed=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
```

```
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
answer_text_generated=0
```

Stage-35 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, read model output, record model output, record generated answer text, score model output, record token log probabilities, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Receipt Requirements

The prompt-evaluation result release receipt contract names future release receipt review requirements:

```
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_invocation_reference=1
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_prompt_evaluation_result_release_receipt_schema_policy=1
requires_prompt_evaluation_result_release_receipt_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_prompt_evaluation_result_release_receipt_review=1
requires_no_prompt_evaluation_result_release=1
requires_no_prompt_evaluation_result_disposition=1
requires_no_prompt_evaluation_result_review=1
requires_no_prompt_evaluation_result=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-35 preserves explicit prompt-evaluation result release receipt denial fields:

```
prompt_evaluation_result_release_receipt_authority=0
prompt_evaluation_result_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_open_authority=0
prompt_evaluation_result_release_receipt_read_authority=0
prompt_evaluation_result_release_receipt_write_authority=0
prompt_evaluation_result_release_receipt_execute_authority=0
prompt_evaluation_result_release_receipt_runtime_authority=0
prompt_evaluation_result_release_receipt_prompt_evaluation_authority=0
prompt_evaluation_result_release_receipt_token_generation_authority=0
prompt_evaluation_result_release_receipt_inference_authority=0
prompt_evaluation_result_release_receipt_generation_authority=0
prompt_evaluation_result_release_receipt_recording_authority=0
prompt_evaluation_result_release_receipt_performed=0
```

```
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_loaded=0
prompt_evaluation_result_release_receipt_opened=0
prompt_evaluation_result_release_receipt_read=0
prompt_evaluation_result_release_receipt_validated=0
prompt_evaluation_result_release_receipt_serialized=0
prompt_evaluation_result_release_receipt_written=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_record_validated=0
prompt_evaluation_result_release_receipt_record_serialized=0
prompt_evaluation_result_release_receipt_record_written=0
prompt_evaluation_result_release_receipt_record_submitted=0
prompt_evaluation_result_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_approval_recorded=0
prompt_evaluation_result_release_receipt_rejection_recorded=0
prompt_evaluation_result_release_receipt_route_recorded=0
prompt_evaluation_result_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_receipt_runtime_invoked=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-contract.sh \
  --prompt-evaluation-result-release reports/nadia/prompt-evaluation-result-release/latest-promp
t-evaluation-result-release-contract.txt \
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-release-receipt.XXXXXX&quot;
)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt
```

Non-Claims

Stage-35 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-contract-stage-35.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_contract_stage_35: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Contract Stage-36

Lines: 170 | Words: 606 | SHA-256: ad713b024eda6f9baca8b71a62380cc849b26372f554bfcdebf448a77c684ba2

Nadia Prompt Evaluation Result Release Receipt Review Contract Stage-36

Status: Stage-36 implementation contract

Scope: prompt-evaluation result release receipt review metadata before review recording, review decisions, review findings, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-36 gives Nadia a prompt-evaluation result release receipt review contract after the prompt-evaluation result release receipt contract is present.

The contract records how a future receipt-review disposition lane must reference the Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create review records, record review decisions, approve receipts, reject receipts, create release receipts, sign receipts, emit receipts, publish receipts, create release records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-36 adds:

```
nadia_stage_36_prompt_evaluation_result_release_receipt_review_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_contract_command=scripts/nadia-prompt-evaluation
-result-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review
prompt_evaluation_result_release_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_metadata_present=1
prompt_evaluation_result_release_receipt_review_family=operator-reviewed-prompt-evaluation-resul
t-release-receipt-review
prompt_evaluation_result_release_receipt_review_format=contract-only-offline-evaluation-result-r
elease-receipt-review
prompt_evaluation_result_release_receipt_review_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_evidence_present=1
prompt_evaluation_result_release_receipt_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_plan_recorded=1
prompt_evaluation_result_release_receipt_review_method_planned=offline-prompt-evaluation-result-
release-receipt-review-policy-review
prompt_evaluation_result_release_receipt_review_result_recorded=0
prompt_evaluation_result_release_receipt_review_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_contract=1
prompt_evaluation_result_release_receipt_review_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
answer_text_generated=0
```

Stage-36 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Review Requirements

The prompt-evaluation result release receipt review contract names future review-disposition requirements:

```
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
```

```
requires_prompt_evaluation_invocation_reference=1
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_prompt_evaluation_result_release_receipt_review_schema_policy=1
requires_prompt_evaluation_result_release_receipt_review_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_prompt_evaluation_result_release=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-36 preserves explicit prompt-evaluation result release receipt review denial fields:

```
prompt_evaluation_result_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_open_authority=0
prompt_evaluation_result_release_receipt_review_read_authority=0
prompt_evaluation_result_release_receipt_review_write_authority=0
prompt_evaluation_result_release_receipt_review_execute_authority=0
prompt_evaluation_result_release_receipt_review_runtime_authority=0
prompt_evaluation_result_release_receipt_review_prompt_evaluation_authority=0
prompt_evaluation_result_release_receipt_review_token_generation_authority=0
prompt_evaluation_result_release_receipt_review_inference_authority=0
prompt_evaluation_result_release_receipt_review_generation_authority=0
prompt_evaluation_result_release_receipt_review_recording_authority=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-contract.sh \
  --prompt-evaluation-result-release-receipt reports/nadia/prompt-evaluation-result-release-rece
ipt/latest-prompt-evaluation-result-release-receipt-contract.txt \
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-release-receipt-review.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review
```

Non-Claims

Stage-36 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-contract-stage-36.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_contract_stage_36: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_37.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Contract Stage-37

Lines: 159 | Words: 628 | SHA-256: e39701e68108498fba206137cffabebbcdec8784f289a7aad72e6610d5a7e39d

Nadia Prompt Evaluation Result Release Receipt Review Disposition Contract Stage-37

Status: Stage-37 implementation contract

Scope: prompt-evaluation result release receipt review disposition metadata before disposition recording, disposition decisions, disposition findings, review recording, review decisions, review findings, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-37 gives Nadia a prompt-evaluation result release receipt review disposition contract after the prompt-evaluation result release receipt review contract is present.

The contract records how a future review-disposition release lane must reference the Stage-36 prompt-evaluation result release receipt review contract, Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition records, record disposition decisions, record disposition findings, create review records, record review decisions, approve receipts, reject receipts, create release receipts, sign receipts, emit receipts, publish receipts, create release records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-37 adds:

```
nadia_stage_37_prompt_evaluation_result_release_receipt_review_disposition_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_contract_command=scripts/nadia-promp
t-evaluation-result-release-receipt-review-disposition-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_contract_command=latticra-
nadia prompt-evaluation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_performed=0
prompt_evaluation_result_release_receipt_review_disposition_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_family=operator-reviewed-prompt-eval
uation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_format=contract-only-offline-evaluat
ion-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_source_policy=operator-reviewed-offl
ine
prompt_evaluation_result_release_receipt_review_disposition_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
answer_text_generated=0
```

Stage-37 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, create review-disposition records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Disposition Requirements

The prompt-evaluation result release receipt review disposition contract names future review-disposition release requirements:

```
requires_prompt_evaluation_result_release_receipt_review_reference=1
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_schema_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release=1
requires_no_prompt_evaluation_result_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-37 preserves explicit prompt-evaluation result release receipt review disposition denial fields:

```
prompt_evaluation_result_release_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-contract.sh \
  --prompt-evaluation-result-release-receipt-review reports/nadia/prompt-evaluation-result-relea
se-receipt-review/latest-prompt-evaluation-result-release-receipt-review-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition
```

Non-Claims

Stage-37 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-contract-stage
-37.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_contract_stage_37: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_38.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Contract Stage-38

Lines: 172 | Words: 655 | SHA-256: 84771f6ffdf1742dd82f777e6151af839a69c32577ccc01a0d2faea2c80e84e1

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Contract Stage-38

Status: Stage-38 implementation contract

Scope: prompt-evaluation result release receipt review disposition release metadata before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-38 gives Nadia a prompt-evaluation result release receipt review disposition release contract after the prompt-evaluation result release receipt review disposition contract is present.

The contract records how a future review-disposition-release receipt lane must reference the Stage-37 prompt-evaluation result release receipt review disposition contract, Stage-36 prompt-evaluation result release receipt review contract, Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition-release records, record release decisions, publish releases, package releases, create release receipts, create disposition records, record disposition decisions, create review records, sign receipts, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-38 adds:

```
nadia_stage_38_prompt_evaluation_result_release_receipt_review_disposition_release_contract_pres
ent=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_contract_generator_pre
sent=1
prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=scripts/nad
ia-prompt-evaluation-result-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=l
atticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_contract_status=contract_onl
y
prompt_evaluation_result_release_receipt_review_disposition_release_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_family=operator-reviewed-pro
mpt-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_format=contract-only-offline
-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_decision=blocked_contract_on
ly
prompt_evaluation_result_release_receipt_review_disposition_release_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_source_policy=operator-revie
wed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_cont
ract=1
prompt_evaluation_result_release_receipt_review_disposition_release_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Stage-38 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, create review-disposition records, create review-disposition release records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Release Requirements

The prompt-evaluation result release receipt review disposition release contract names future review-disposition-release receipt requirements:

```
requires_prompt_evaluation_result_release_receipt_review_disposition_reference=1
requires_prompt_evaluation_result_release_receipt_review_reference=1
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_schema_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-38 preserves explicit prompt-evaluation result release receipt review disposition release denial fields:

```
prompt_evaluation_result_release_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
```

```
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-contract.sh
 \
  --prompt-evaluation-result-release-receipt-review-disposition reports/nadia/prompt-evaluation-
result-release-receipt-review-disposition/latest-prompt-evaluation-result-release-receipt-review
-disposition-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release
```

Non-Claims

Stage-38 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-contra
ct-stage-38.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_contract_stage_38: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_39.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract Stage-39

Lines: 179 | Words: 687 | SHA-256: d3b9f9a28db25c77c218ec7ba24b900129ff600cb8c7d37efd29ef0a155b9f7c

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract Stage-39

Status: Stage-39 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt metadata before disposition-release-receipt recording, receipt emission, receipt signing, receipt publication, disposition-release recording, release decisions, release publication, release packaging, disposition recording, review recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-39 gives Nadia a prompt-evaluation result release receipt review disposition release receipt contract after the prompt-evaluation result release receipt review disposition release contract is present.

The contract records how a future review-disposition-release receipt lane must reference the Stage-38 prompt-evaluation result release receipt review disposition release contract, Stage-37 prompt-evaluation result release receipt review disposition contract, Stage-36 prompt-evaluation result release receipt review contract, Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition-release-receipt records, emit receipts, sign receipts, publish receipts, package receipts, create disposition-release records, record release decisions, publish releases, package releases, create disposition records, record disposition decisions, create review records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-39 adds:

```
nadia_stage_39_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contr
act_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_gener
ator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_command=scr
ipts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-contract.
sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_c
ommand=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_status=cont
ract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_family=operator-revi
ewed-prompt-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_format=contract-only
-offline-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_decision=blocked_con
tract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_source_policy=operat
or-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
```

```
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Stage-39 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, create review-disposition records, create review-disposition release records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Release Requirements

The prompt-evaluation result release receipt review disposition release receipt contract names future review-disposition-release receipt review requirements:

```
requires_prompt_evaluation_result_release_receipt_review_disposition_release_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_reference=1
requires_prompt_evaluation_result_release_receipt_review_reference=1
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_schema_poli
cy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_denial_poli
cy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_poli
cy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
```

```
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-39 preserves explicit prompt-evaluation result release receipt review disposition release receipt denial fields:

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-con
tract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release reports/nadia/prompt-eva
luation-result-release-receipt-review-disposition-release/latest-prompt-evaluation-result-releas
e-receipt-review-disposition-release-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt
```

Non-Claims

Stage-39 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-contract-stage-39.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_stage
_39: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_40.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract Stage-40

Lines: 99 | Words: 391 | SHA-256: 40b7cc6e57b59aa324f310932c4796512c1d321fdc58fe070fedf7e6f91155e7

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract Stage-40

Status: Stage-40 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review metadata before release-receipt-review records, review decisions, review findings, disposition records, receipt signing, receipt publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-40 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review contract after the Stage-39 prompt-evaluation result release receipt review disposition release receipt contract is present.

The contract records how a future review-disposition-release receipt review lane must reference the Stage-39 release receipt contract, Stage-38 release-disposition contract, Stage-37 review-disposition contract, Stage-36 receipt-review contract, Stage-35 release receipt contract, Stage-34 release contract, Stage-33 disposition contract, Stage-32 review contract, and Stage-31 result contract. It preserves the absolute rule that Nadia cannot yet review receipts, decide reviews, record review findings, create dispositions, sign receipts, publish receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Contract Fields

Stage-40 adds:

```
nadia_stage_40_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contrac
t_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_comm
and=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-re
view-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_con
tract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_stat
us=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_stage=contrac
t-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_metadata_pres
ent=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_family=operat
or-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_format=contra
ct-only-offline-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision=bloc
ked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_evidence_pres
ent=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_source_policy
=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_plan_recorded
=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_result_record
ed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_runtime_invok
ed=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_promotion_all
owed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_record_create
d=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_approval_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_rejection_rec
orded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_findings_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-contract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release-receipt reports/nadia/pr
ompt-evaluation-result-release-receipt-review-disposition-release-receipt/latest-prompt-evaluati
on-result-release-receipt-review-disposition-release-receipt-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
```

Non-Claims

Stage-40 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, release publisher, release packager, release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-contract-stage-40.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contrac
t_stage_40: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_41.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-41

Lines: 99 | Words: 402 | SHA-256: 5b6a27ad480fb4a64832f45402966483a8c72c2a61c0f8b24f931e7adbe4e65f

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-41

Status: Stage-41 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition metadata before release-receipt-review-disposition records, review decisions, review findings, disposition records, receipt signing, receipt publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-41 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review disposition contract after the Stage-40 prompt-evaluation result release receipt review disposition release receipt review contract is present.

The contract records how a future review-disposition-release receipt review disposition lane must reference the Stage-40 release receipt review contract, Stage-39 release receipt contract, Stage-38 release-disposition contract, Stage-37 review-disposition contract, Stage-36 receipt-review contract, Stage-35 release receipt contract, Stage-34 release contract, Stage-33 disposition contract, Stage-32 review contract, and Stage-31 result contract. It preserves the absolute rule that Nadia cannot yet review receipts, decide reviews, record review findings, create dispositions, sign receipts, publish receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Contract Fields

Stage-41 adds:

```
nadia_stage_41_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
ontract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-releas
e-receipt-review-disposition-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
ontract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_s
tage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
uthority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
llowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
reated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_p
erformed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_m
etadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
amily=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release-rece
ipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
ormat=contract-only-offline-evaluation-result-release-receipt-review-disposition-release-receipt
-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_e
vidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_s
ource_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_p
lan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
esult_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
untime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_cont
ract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
```

```
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_p
romotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
pproval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
indings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-disposition-contract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release-receipt reports/nadia/pr
ompt-evaluation-result-release-receipt-review-disposition-release-receipt/latest-prompt-evaluati
on-result-release-receipt-review-disposition-release-receipt-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review-disposition.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
```

Non-Claims

Stage-41 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, release publisher, release packager, release receipt creator, receipt signer, receipt publisher, release receipt review dispositioner, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-contract-stage-41.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_contract_stage_41: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_42.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-42

Lines: 178 | Words: 689 | SHA-256: e71ae81cc24e932dd328ad89758c978484c0c8270e5072884e737d92c1eaf431

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-42

Status: Stage-42 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release metadata before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-42 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review disposition release contract after the prompt-evaluation result release receipt review disposition release receipt review disposition contract is present.

The contract records how a future review-disposition-release receipt lane must reference the Stage-41 prompt-evaluation result release receipt review disposition release receipt review disposition contract, Stage-36 prompt-evaluation result release receipt review contract, Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition-release records, record release decisions, publish releases, package releases, create release receipts, create disposition records, record disposition decisions, create review records, sign receipts, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-42 adds:

```
nadia_stage_42_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-dispositio
n-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review
-disposition-release-receipt-review-disposition-release
```

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-rele
ase-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_format=contract-only-offline-evaluation-result-release-receipt-review-disposition-release
-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_cont
ract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
indings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_record_create
d=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Stage-42 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, create review-disposition records, create review-disposition release records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Release Requirements

The prompt-evaluation result release receipt review disposition release receipt review disposition release contract names future review-disposition-release receipt requirements:

```
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_refe
rence=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_reference=1
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_schema_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-42 preserves explicit prompt-evaluation result release receipt review disposition release receipt review disposition release denial fields:

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_decision_recorded=0
```

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
indings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_record_create
d=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision_reco
rded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-disposition-release-contract.sh \

--prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
 reports/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition/latest-prompt-evaluation-result-release-receipt-review-disposition-release-receipt
-review-disposition-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review-disposition-release.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release
```

Non-Claims

Stage-42 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-contract-stage-42.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_contract_stage_42: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_43.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract Stage-43

Lines: 179 | Words: 716 | SHA-256: b4e1e4f75db4c3e0a44f0ff84aaf77c1e98a40b6aafcf250d227aefc92781cf4

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract Stage-43

Status: Stage-43 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt metadata before disposition-release-receipt recording, receipt emission, receipt signing, receipt publication, disposition-release recording, release decisions, release publication, release packaging, disposition recording, review recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-43 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract after the Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release contract is present.

The contract records how a future review-disposition-release receipt lane must reference the Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release contract, Stage-37 prompt-evaluation result release receipt review disposition contract, Stage-36 prompt-evaluation result release receipt review contract, Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition-release-receipt records, emit receipts, sign receipts, publish receipts, package receipts, create disposition-release records, record release decisions, publish releases, package releases, create disposition records, record disposition decisions, create review records, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-43 adds:

```
nadia_stage_43_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-di
sposition-release-receipt-review-disposition-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_contract_command=latticra-nadia prompt-evaluation-result-release-receip
t-review-disposition-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposit
ion-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_format=contract-only-offline-evaluation-result-release-receipt-review-disposition
-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
```

```
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Stage-43 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, create review-disposition records, create review-disposition release records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Release Requirements

The prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract names future review-disposition-release receipt review requirements:

```
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_reference=1
requires_prompt_evaluation_result_release_receipt_review_reference=1
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_schema_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-43 preserves explicit prompt-evaluation result release receipt review disposition release receipt review disposition release receipt denial fields:

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-disposition-release-receipt-contract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-dispositi
on-release reports/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-rec
eipt-review-disposition-release/latest-prompt-evaluation-result-release-receipt-review-dispositi
on-release-receipt-review-disposition-release-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review-disposition-release-receipt.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt
```

Non-Claims

Stage-43 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-contract-stage-43.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_contract_stage_43: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_44.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract Stage-44

Lines: 99 | Words: 407 | SHA-256: 9326d02954a8bed5172c465641318367509eb46c6fe83b7c97245fe86b64e783

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract Stage-44

Status: Stage-44 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review metadata before release-receipt-review records, review decisions, review findings, disposition records, receipt signing, receipt publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-44 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract after the Stage-43 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract is present.

The contract records how a future review-disposition-release receipt review lane must reference the Stage-43 release receipt contract, Stage-38 release-disposition contract, Stage-37 review-disposition contract, Stage-36 receipt-review contract, Stage-35 release receipt contract, Stage-34 release contract, Stage-33 disposition contract, Stage-32 review contract, and Stage-31 result contract. It preserves the absolute rule that Nadia cannot yet review receipts, decide reviews, record review findings, create dispositions, sign receipts, publish receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Contract Fields

Stage-44 adds:

```
nadia_stage_44_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_review_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-re
view-disposition-release-receipt-review-disposition-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_contract_command=latticra-nadia prompt-evaluation-result-release
-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-d
isposition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_format=contract-only-offline-evaluation-result-release-receipt-review-disp
osition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
```

```
elease_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-disposition-release-receipt-review-contract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-dispositi
on-release-receipt reports/nadia/prompt-evaluation-result-release-receipt-review-disposition-rel
ease-receipt-review-disposition-release-receipt/latest-prompt-evaluation-result-release-receipt-
review-disposition-release-receipt-review-disposition-release-receipt-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review-disposition-release-receipt-review.XXXXXX&quot;
)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt-review
```

Non-Claims

Stage-44 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, release publisher, release packager, release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-contract-stage-44.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_contract_stage_44: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_45.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-45

Lines: 99 | Words: 418 | SHA-256: c9886fbbcd2a85b85ad1a33855980254b4100b4e3b6613bbd9e8237ee43aa4c4

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-45

Status: Stage-45 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition metadata before release-receipt-review-disposition records, disposition decisions, disposition findings, receipt signing, receipt publication, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-45 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract after the Stage-44 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract is present.

The contract records how a future review-disposition-release lane must reference the Stage-44 release receipt review contract, Stage-43 release receipt contract, Stage-38 release-disposition contract, Stage-37 review-disposition contract, Stage-36 receipt-review contract, Stage-35 release receipt contract, Stage-34 release contract, Stage-33 disposition contract, Stage-32 review contract, and Stage-31 result contract. It preserves the absolute rule that Nadia cannot yet record review dispositions, decide dispositions, record disposition findings, sign receipts, publish receipts, read or record model output, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Contract Fields

Stage-45 adds:

```
nadia_stage_45_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_review_disposition_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_disposition_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_contract_command=scripts/nadia-prompt-evaluation-result-releas
e-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-dispositi
on-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_disposition_contract_command=latticra-nadia prompt-evaluation-re
sult-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-revie
w-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_family=operator-reviewed-prompt-evaluation-result-release-rece
ipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_format=contract-only-offline-evaluation-result-release-receipt
```

```
-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-disposition-release-receipt-review-disposition-contract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-dispositi
on-release-receipt-review reports/nadia/prompt-evaluation-result-release-receipt-review-disposit
ion-release-receipt-review-disposition-release-receipt-review/latest-prompt-evaluation-result-re
lease-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-contr
act.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition.XXXXXX
&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt-review-disposition
```

Non-Claims

Stage-45 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, release publisher, release packager, release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, receipt review disposition release receipt review layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-disposition-contract-stage-45.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_disposition_contract_stage_45: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-46

Lines: 178 | Words: 717 | SHA-256: e6c7a44d84b5e3a52f8278cfc73f31ad26a6ee28bc6ef53442c49c64ea276933

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-46

Status: Stage-46 implementation contract

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release metadata before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-46 gives Nadia a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract after the prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract is present.

The contract records how a future review-disposition-release receipt lane must reference the Stage-45 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract, Stage-36 prompt-evaluation result release receipt review contract, Stage-35 prompt-evaluation result release receipt contract, Stage-34 prompt-evaluation result release contract, Stage-33 prompt-evaluation result disposition contract, Stage-32 prompt-evaluation result review contract, Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create disposition-release records, record release decisions, publish releases, package releases, create release receipts, create disposition records, record disposition decisions, create review records, sign receipts, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-46 adds:

```
nadia_stage_46_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_review_disposition_release_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_disposition_release_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_contract_command=scripts/nadia-prompt-evaluation-resul
t-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-d
isposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_disposition_release_contract_command=latticra-nadia prompt-evalu
ation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-recei
pt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_family=operator-reviewed-prompt-evaluation-result-rele
ase-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposi
tion-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_format=contract-only-offline-evaluation-result-release
-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-dispositio
n-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_contract=1
```

```
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_release_receipt_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_promotion_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Stage-46 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, create prompt-evaluation result review records, create prompt-evaluation result disposition records, create release records, publish releases, package releases, create release receipts, sign receipts, emit receipts, publish receipts, create release-receipt review records, create review-disposition records, create review-disposition release records, approve release receipts, reject release receipts, read model output, record model output, record generated answer text, score model output, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Release Requirements

The prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract names future review-disposition-release receipt requirements:

```
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_disposition_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_reference=1
requires_prompt_evaluation_result_release_receipt_reference=1
requires_prompt_evaluation_result_release_reference=1
requires_prompt_evaluation_result_disposition_reference=1
requires_prompt_evaluation_result_review_reference=1
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_disposition_release_schema_policy=1
```

```
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_disposition_release_denial_policy=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_disposition_release_receipt_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt_review_disposition_release=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt_review_disposition_release_receipt=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt_review_disposition=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release_receipt_review=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition_release=1
requires_no_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_d
isposition=1
requires_no_prompt_evaluation_result_release_receipt=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-46 preserves explicit prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release denial fields:

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_model_output_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
```

```
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-rev
iew-disposition-release-receipt-review-disposition-release-contract.sh \
  --prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-dispositi
on-release-receipt-review-disposition reports/nadia/prompt-evaluation-result-release-receipt-rev
iew-disposition-release-receipt-review-disposition-release-receipt-review-disposition/latest-pro
mpt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-rele
ase-receipt-review-disposition-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-
release-receipt
-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-releas
e.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-revie
w-disposition-release-receipt-review-disposition-release
```

Non-Claims

Stage-46 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition recorder, prompt-evaluation result release recorder, prompt-evaluation result release publisher, prompt-evaluation result release packager, prompt-evaluation result release receipt creator, receipt signer, receipt publisher, release receipt reviewer, receipt review disposition recorder, receipt review disposition release recorder, receipt review disposition release receipt layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-disposition-release-contract-stage-46.sh
```

Expected output:

```
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_disposition_release_contract_stage_46: ok
```

docs/NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32.md

Source title: Nadia Prompt Evaluation Result Review Contract Stage-32

Lines: 181 | Words: 498 | SHA-256: 752752e70ff4841dd3171bcdbad7837d690be969c8e1fc882dc1fb4ae4fccabc

Nadia Prompt Evaluation Result Review Contract Stage-32

Status: Stage-32 implementation contract

Scope: prompt-evaluation result review metadata before review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-32 gives Nadia a prompt-evaluation result review contract after the prompt-evaluation result contract is present.

The contract records how future reviewed result disposition must reference the Stage-31 prompt-evaluation result contract, Stage-30 prompt-evaluation invocation contract, Stage-29 prompt-evaluation runtime handoff contract, Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, generated-text denial policy, runtime-denial policy, token-generation denial policy, and survivor-centered safety policy. It preserves the absolute rule that Nadia cannot yet create review records, record result content, read or record model output, record generated text, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

Contract Fields

Stage-32 adds:

```
nadia_stage_32_prompt_evaluation_result_review_contract_present=1
nadia_prompt_evaluation_result_review_contract_generator_present=1
prompt_evaluation_result_review_contract_command=scripts/nadia-prompt-evaluation-result-review-c
ontract.sh
installed_prompt_evaluation_result_review_contract_command=latticra-nadia
prompt-evaluation-result-review
prompt_evaluation_result_review_contract_status=contract_only
prompt_evaluation_result_review_stage=contract-only
prompt_evaluation_result_review_authority=0
prompt_evaluation_result_review_allowed=0
prompt_evaluation_result_review_recorded=0
prompt_evaluation_result_review_created=0
prompt_evaluation_result_review_performed=0
prompt_evaluation_result_review_metadata_present=1
prompt_evaluation_result_review_family=operator-reviewed-prompt-evaluation-result-review
prompt_evaluation_result_review_format=contract-only-offline-evaluation-result-review
prompt_evaluation_result_review_decision=blocked_contract_only
prompt_evaluation_result_review_evidence_present=1
prompt_evaluation_result_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_review_plan_recorded=1
prompt_evaluation_result_review_method_planned=offline-prompt-evaluation-result-review-policy-re
view
prompt_evaluation_result_review_result_recorded=0
prompt_evaluation_result_review_runtime_invoked=0
requires_prompt_evaluation_result_contract=1
requires_prompt_evaluation_invocation_contract=1
requires_prompt_evaluation_runtime_handoff_contract=1
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_result_disposition_contract=1
prompt_evaluation_result_review_promotion_allowed=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_approval_recorded=0
prompt_evaluation_result_review_rejection_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
answer_text_generated=0
```

Stage-32 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, perform runtime handoff, create prompt-evaluation invocation requests, invoke a runtime, evaluate prompts, create prompt-evaluation result records, review prompt-evaluation result content, record model output, record generated answer text, score model output, record token log probabilities, load model weights, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Review Requirements

The prompt-evaluation result review contract names future review requirements:

```
requires_prompt_evaluation_result_reference=1
requires_prompt_evaluation_invocation_reference=1
requires_prompt_evaluation_runtime_handoff_reference=1
requires_prompt_evaluation_input_reference=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_prompt_evaluation_result_review_schema_policy=1
requires_prompt_evaluation_result_review_denial_policy=1
requires_prompt_evaluation_result_disposition_policy=1
requires_generated_text_denial_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_evaluation_result_review=1
requires_no_prompt_evaluation_result=1
requires_no_model_output_read=1
requires_no_model_output_recording=1
requires_no_generated_answer=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-32 preserves explicit prompt-evaluation result review denial fields:

```
prompt_evaluation_result_review_authority=0
prompt_evaluation_result_review_allowed=0
prompt_evaluation_result_review_open_authority=0
prompt_evaluation_result_review_read_authority=0
prompt_evaluation_result_review_write_authority=0
prompt_evaluation_result_review_execute_authority=0
prompt_evaluation_result_review_runtime_authority=0
prompt_evaluation_result_review_prompt_evaluation_authority=0
prompt_evaluation_result_review_token_generation_authority=0
prompt_evaluation_result_review_inference_authority=0
prompt_evaluation_result_review_generation_authority=0
prompt_evaluation_result_review_recording_authority=0
prompt_evaluation_result_review_performed=0
prompt_evaluation_result_review_created=0
prompt_evaluation_result_review_materialized=0
prompt_evaluation_result_review_loaded=0
prompt_evaluation_result_review_opened=0
prompt_evaluation_result_review_read=0
prompt_evaluation_result_review_validated=0
prompt_evaluation_result_review_serialized=0
prompt_evaluation_result_review_written=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_record_validated=0
prompt_evaluation_result_review_record_serialized=0
prompt_evaluation_result_review_record_written=0
prompt_evaluation_result_review_record_submitted=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_approval_recorded=0
prompt_evaluation_result_review_rejection_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_review_runtime_invoked=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invoked=0
prompt_evaluated=0
token_generation_performed=0
```

```
inference_performed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-result-review-contract.sh \
  --prompt-evaluation-result
reports/nadia/prompt-evaluation-result/latest-prompt-evaluation-result-contract.txt \
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-result-review.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-result-review
```

Non-Claims

Stage-32 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, prompt-evaluation result layer, prompt-evaluation result reviewer, prompt-evaluation result disposition layer, model-output reader, model-output recorder, answer generator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-result-review-contract-stage-32.sh
```

Expected output:

```
nadia_prompt_evaluation_result_review_contract_stage_32: ok
```

docs/NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29.md

Source title: Nadia Prompt Evaluation Runtime Handoff Contract Stage-29

Lines: 165 | Words: 457 | SHA-256: b3753b73b67c30b6821e3ae31c41146073f21fc823cd10abaf2e6ff1a944aafc

Nadia Prompt Evaluation Runtime Handoff Contract Stage-29

Status: Stage-29 implementation contract

Scope: prompt-evaluation runtime handoff metadata before runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-29 gives Nadia a prompt-evaluation runtime handoff contract after the prompt-evaluation-input contract is present.

The contract records how future reviewed runtime handoff packaging must reference the Stage-28 prompt-evaluation-input contract, context-window assembly boundary, prompt-token-sequence boundary, prompt-tokenization boundary, tokenizer-runtime-attachment boundary, runtime-profile boundary, runtime-invocation contract, model-load contract, inference-readiness contract, model-registry contract, safety-envelope policy, runtime-denial policy, and prompt-evaluation invocation policy. It preserves the absolute rule that Nadia cannot yet create runtime handoff requests, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference. The script only measures the Stage-28 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-29 adds:

```
nadia_stage_29_prompt_evaluation_runtime_handoff_contract_present=1
nadia_prompt_evaluation_runtime_handoff_contract_generator_present=1
prompt_evaluation_runtime_handoff_contract_command=scripts/nadia-prompt-evaluation-runtime-hando
ff-contract.sh
installed_prompt_evaluation_runtime_handoff_contract_command=latticra-nadia
prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_contract_status=contract_only
prompt_evaluation_runtime_handoff_stage=contract-only
prompt_evaluation_runtime_handoff_authority=0
prompt_evaluation_runtime_handoff_allowed=0
prompt_evaluation_runtime_handoff_performed=0
prompt_evaluation_runtime_handoff_metadata_present=1
prompt_evaluation_runtime_handoff_family=operator-reviewed-prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_format=contract-only-offline-runtime-handoff
prompt_evaluation_runtime_handoff_decision=blocked_contract_only
prompt_evaluation_runtime_handoff_evidence_present=1
prompt_evaluation_runtime_handoff_source_policy=operator-reviewed-offline
prompt_evaluation_runtime_handoff_plan_recorded=1
prompt_evaluation_runtime_handoff_method_planned=offline-prompt-evaluation-runtime-handoff-polic
y-review
prompt_evaluation_runtime_handoff_result_recorded=0
prompt_evaluation_runtime_handoff_runtime_invoked=0
requires_prompt_evaluation_input_contract=1
requires_context_window_assembly_contract=1
requires_prompt_token_sequence_contract=1
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_inference_readiness_contract=1
requires_local_model_registry_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_evaluation_invocation_contract=1
prompt_evaluation_runtime_handoff_promotion_allowed=0
```

Stage-29 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt token buffers, create prompt tokens, count prompt tokens, record prompt token IDs, assemble context windows, create prompt evaluation inputs, validate prompt evaluation inputs, create runtime handoff requests, submit runtime handoff requests, select a runtime, select a model, create runtime sessions, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Handoff Requirements

The prompt-evaluation runtime handoff contract names future review requirements:

```
requires_prompt_evaluation_input_reference=1
requires_prompt_evaluation_input_schema_policy=1
requires_prompt_evaluation_input_safety_envelope_policy=1
requires_context_window_assembly_reference=1
requires_prompt_token_sequence_reference=1
requires_prompt_tokenization_reference=1
requires_runtime_profile_reference=1
requires_runtime_invocation_contract_reference=1
requires_model_load_contract_reference=1
requires_inference_readiness_contract_reference=1
requires_runtime_handoff_schema_policy=1
requires_runtime_handoff_denial_policy=1
requires_prompt_evaluation_invocation_policy=1
requires_token_generation_denial_policy=1
requires_operator_approval_record=1
```

```
requires_official_source_snapshot=1
requires_no_prompt_evaluation_runtime_handoff=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_token_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-29 preserves explicit prompt-evaluation runtime handoff denial fields:

```
prompt_evaluation_runtime_handoff_authority=0
prompt_evaluation_runtime_handoff_allowed=0
prompt_evaluation_runtime_handoff_open_authority=0
prompt_evaluation_runtime_handoff_read_authority=0
prompt_evaluation_runtime_handoff_write_authority=0
prompt_evaluation_runtime_handoff_execute_authority=0
prompt_evaluation_runtime_handoff_runtime_authority=0
prompt_evaluation_runtime_handoff_invocation_authority=0
prompt_evaluation_runtime_handoff_prompt_evaluation_authority=0
prompt_evaluation_runtime_handoff_performed=0
prompt_evaluation_runtime_handoff_created=0
prompt_evaluation_runtime_handoff_materialized=0
prompt_evaluation_runtime_handoff_loaded=0
prompt_evaluation_runtime_handoff_opened=0
prompt_evaluation_runtime_handoff_read=0
prompt_evaluation_runtime_handoff_validated=0
prompt_evaluation_runtime_handoff_serialized=0
prompt_evaluation_runtime_handoff_written=0
prompt_evaluation_runtime_handoff_request_created=0
prompt_evaluation_runtime_handoff_request_validated=0
prompt_evaluation_runtime_handoff_request_serialized=0
prompt_evaluation_runtime_handoff_request_written=0
prompt_evaluation_runtime_handoff_request_submitted=0
prompt_evaluation_runtime_handoff_runtime_selected=0
prompt_evaluation_runtime_handoff_model_selected=0
prompt_evaluation_runtime_handoff_session_created=0
prompt_evaluation_runtime_handoff_runtime_invoked=0
runtime_handoff_created=0
runtime_handoff_submitted=0
prompt_evaluation_request_created=0
prompt_evaluation_request_serialized=0
prompt_evaluation_request_submitted=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
runtime_process_spawned=0
runtime_binary_executed=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-evaluation-runtime-handoff-contract.sh \
  --prompt-evaluation-input
reports/nadia/prompt-evaluation-input/latest-prompt-evaluation-input-contract.txt \
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-evaluation-runtime-handoff.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-evaluation-runtime-handoff
```

Non-Claims

Stage-29 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, tokenizer loader, tokenizer registry, context-window assembler, prompt-evaluation-input creator, prompt-evaluation runtime handoff layer, prompt-evaluation invocation layer, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-evaluation-runtime-handoff-contract-stage-29.sh
```

Expected output:

```
nadia_prompt_evaluation_runtime_handoff_contract_stage_29: ok
```

docs/NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14.md

Source title: Nadia Prompt Materialization Contract Stage-14

Lines: 163 | Words: 355 | SHA-256: 2b890f793f39f8093236a53ad2d1af9ad9178dc69dc478dfc7a7e30bc68c9bef

Nadia Prompt Materialization Contract Stage-14

Status: Stage-14 implementation contract Date: 2026-05-25 Scope: prompt-materialization metadata before prompt buffer allocation, prompt text materialization, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.

Purpose

Stage-14 gives Nadia a prompt-materialization contract without materializing prompt text.

The contract consumes a Stage-13 prompt-receipt contract, verifies inherited non-executing prompt, model-load, runtime, model, tool, and protective-safety posture, and records the review fields that would be required before any later prompt-evaluation handoff contract can exist.

Capability

Stage-14 adds:

```
nadia_stage_14_prompt_materialization_contract_present=1
nadia_prompt_materialization_contract_generator_present=1
prompt_materialization_contract_command=scripts/nadia-prompt-materialization-contract.sh
installed_prompt_materialization_contract_command=latticra-nadia prompt-materialization
requires_prompt_receipt_contract=1
prompt_receipt_stage_required=13-prompt-receipt-contract
prompt_materialization_contract_status=contract_only
prompt_materialization_stage=contract-only
prompt_materialization_authority=0
prompt_materialization_allowed=0
prompt_materialized=0
prompt_text_materialized=0
materialization_decision=blocked_contract_only
materialization_evidence_present=1
requires_model_load_contract=1
requires_runtime_invocation_contract=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
```

```
requires_refusal_policy_review=1
requires_prompt_source_boundary=1
requires_prompt_buffer_boundary=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_prompt_evaluation_handoff_contract=1
prompt_materialization_promotion_allowed=0
prompt_source_open_authority=0
prompt_source_read_authority=0
prompt_text_materialization_authority=0
prompt_buffer_allocation_authority=0
prompt_buffer_write_authority=0
prompt_content_storage_authority=0
prompt_hash_authority=0
prompt_classification_authority=0
prompt_tokenization_authority=0
prompt_source_opened=0
prompt_source_read=0
prompt_bytes_read=0
prompt_text_received=0
prompt_materialization_performed=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_bytes_materialized=0
prompt_tokens_created=0
prompt_tokenized=0
prompt_content_stored=0
prompt_hash_computed=0
prompt_classified=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_loaded=0
model_weights_loaded=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-14 does not receive prompt text, read prompt sources, allocate a prompt buffer, materialize prompt text, tokenize prompts, store prompt content, hash prompt content, classify prompt content, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inputs

The prompt-materialization contract requires:

```
prompt_receipt=Stage-13 Nadia prompt-receipt contract
request_class=operator request classification label
```

Sexualized request classifications fail closed.

Outputs

The contract writes:

```
nadia-prompt-materialization-contract-&lt;timestamp&gt;.txt
latest-prompt-materialization-contract.txt
```

The report records:

```
stage=14-prompt-materialization-contract
prompt_materialization_contract_status=contract_only
prompt_materialization_authority=0
```

```
prompt_materialization_allowed=0
prompt_materialized=0
prompt_text_materialized=0
materialization_decision=blocked_contract_only
prompt_buffer_allocated=0
prompt_tokenized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
sexual_request_refusal=always
```

Usage

With explicit evidence input:

```
sh scripts/nadia-prompt-materialization-contract.sh \
  --prompt-receipt /path/to/latest-prompt-receipt-contract.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-materialization.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia prompt-materialization
```

Non-Claims

Stage-14 Nadia is not a prompt materializer, prompt tokenizer, prompt receiver, prompt reader, prompt store, prompt evaluator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, or production AI assistant.

Validation

```
sh scripts/test-nadia-prompt-materialization-contract-stage-14.sh
```

Expected result:

```
nadia_prompt_materialization_contract_stage_14: ok
```

docs/NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13.md

Source title: Nadia Prompt Receipt Contract Stage-13

Lines: 153 | Words: 335 | SHA-256: 429a232995c5852349ca6577996b71fad35f2603937c20dd192d5582d1db2f30

Nadia Prompt Receipt Contract Stage-13

Status: Stage-13 implementation contract Date: 2026-05-25 Scope: prompt-receipt metadata before prompt source opening, prompt text receipt, prompt materialization, prompt evaluation, token generation, inference, or tool execution.

Purpose

Stage-13 gives Nadia a prompt-receipt contract without receiving prompt text.

The contract consumes a Stage-12 model-load contract, verifies inherited non-executing model-load, runtime, model, prompt, tool, and protective-safety posture, and records the review fields that would be required before any later prompt-materialization contract can exist.

Capability

Stage-13 adds:

```
nadia_stage_13_prompt_receipt_contract_present=1
nadia_prompt_receipt_contract_generator_present=1
prompt_receipt_contract_command=scripts/nadia-prompt-receipt-contract.sh
installed_prompt_receipt_contract_command=latticra-nadia prompt-receipt
requires_model_load_contract=1
```

```
model_load_stage_required=12-model-load-contract
prompt_receipt_contract_status=contract_only
prompt_receipt_stage=contract-only
prompt_receipt_authority=0
prompt_receipt_allowed=0
prompt_received=0
receipt_decision=blocked_contract_only
receipt_evidence_present=1
requires_runtime_invocation_contract=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_refusal_policy_review=1
requires_prompt_source_boundary=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_prompt_materialization_contract=1
prompt_receipt_promotion_allowed=0
prompt_source_open_authority=0
prompt_source_read_authority=0
prompt_text_materialization_authority=0
prompt_content_storage_authority=0
prompt_hash_authority=0
prompt_classification_authority=0
prompt_source_opened=0
prompt_source_read=0
prompt_bytes_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_content_stored=0
prompt_hash_computed=0
prompt_classified=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_loaded=0
model_weights_loaded=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-13 does not receive prompt text, read prompt sources, store prompt content, hash prompt content, classify prompt content, materialize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inputs

The prompt-receipt contract requires:

```
model_load=Stage-12 Nadia model-load contract
request_class=operator request classification label
```

Sexualized request classifications fail closed.

Outputs

The contract writes:

```
nadia-prompt-receipt-contract-&lt;timestamp&gt;.txt
latest-prompt-receipt-contract.txt
```

The report records:

```
stage=13-prompt-receipt-contract
prompt_receipt_contract_status=contract_only
prompt_receipt_authority=0
prompt_receipt_allowed=0
prompt_received=0
receipt_decision=blocked_contract_only
prompt_source_opened=0
prompt_text_received=0
prompt_text_materialized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
sexual_request_refusal=always
```

Usage

With explicit evidence input:

```
sh scripts/nadia-prompt-receipt-contract.sh \
  --model-load /path/to/latest-model-load-contract.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-receipt.XXXXXX&quot;
)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia prompt-receipt
```

Non-Claims

Stage-13 Nadia is not a prompt receiver, prompt reader, prompt store, prompt materializer, prompt evaluator, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, or production AI assistant.

Validation

```
sh scripts/test-nadia-prompt-receipt-contract-stage-13.sh
```

Expected result:

```
nadia_prompt_receipt_contract_stage_13: ok
```

docs/NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26.md

Source title: Nadia Prompt Token Sequence Contract Stage-26

Lines: 184 | Words: 584 | SHA-256: 3b226772c4e571e0dbcdd7fc83216397334452c5c5e6f9238f2dedfbaaf5570c

Nadia Prompt Token Sequence Contract Stage-26

Status: Stage-26 implementation contract

Scope: prompt-token-sequence metadata before prompt token ID recording, token order recording, token offset recording, attention mask creation, position ID creation, context window assembly, prompt evaluation input creation, runtime invocation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-26 gives Nadia a prompt-token-sequence contract after the prompt-tokenization contract is present.

The contract records how future reviewed prompt token sequencing must reference the Stage-25 prompt-tokenization contract, tokenizer-runtime-attachment boundary, prompt-materialization boundary, prompt-receipt boundary, prompt token count policy, token order policy, token ID visibility policy, token offset policy, and context-window policy.

It preserves the absolute rule that Nadia cannot yet read prompt text, create prompt tokens, record prompt token IDs, record token order, record token offsets, assemble a context window, create prompt evaluation input, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference. The script only measures the Stage-25 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-26 adds:

```
nadia_stage_26_prompt_token_sequence_contract_present=1
nadia_prompt_token_sequence_contract_generator_present=1
prompt_token_sequence_contract_command=scripts/nadia-prompt-token-sequence-contract.sh
installed_prompt_token_sequence_contract_command=latticra-nadia prompt-token-sequence
prompt_token_sequence_contract_status=contract_only
prompt_token_sequence_stage=contract-only
prompt_token_sequence_authority=0
prompt_token_sequence_allowed=0
prompt_token_sequence_recorded=0
prompt_token_sequence_metadata_present=1
prompt_token_sequence_family=operator-reviewed-prompt-token-sequence
prompt_token_sequence_format=contract-only-offline-sequence
prompt_token_sequence_decision=blocked_contract_only
prompt_token_sequence_evidence_present=1
prompt_token_sequence_source_policy=operator-reviewed-offline
prompt_token_sequence_plan_recorded=1
prompt_token_sequence_method_planned=offline-token-sequence-policy-review
prompt_token_sequence_result_recorded=0
prompt_token_sequence_count_recorded=0
prompt_token_sequence_order_recorded=0
prompt_token_sequence_runtime_invoked=0
requires_prompt_tokenization_contract=1
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_context_window_assembly_contract=1
prompt_token_sequence_promotion_allowed=0
```

Stage-26 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt token buffers, create prompt tokens, count prompt tokens, record prompt token IDs, record prompt token order, record prompt token offsets, create attention masks, create position IDs, assemble context windows, create prompt evaluation inputs, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, bind tokenizer artifacts, attach tokenizers to a runtime, create runtime sessions, open tokenizer files, read tokenizer files, load tokenizer vocabularies, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Sequence Requirements

The prompt-token-sequence contract names future review requirements:

```
requires_prompt_tokenization_reference=1
requires_prompt_tokenization_contract_reference=1
requires_tokenizer_runtime_attachment_reference=1
requires_prompt_materialization_reference=1
requires_prompt_receipt_reference=1
requires_prompt_token_count_policy=1
requires_prompt_token_order_policy=1
requires_prompt_token_id_visibility_policy=1
requires_prompt_token_offset_policy=1
requires_context_window_policy=1
```

```
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_token_ids_recorded=1
requires_no_prompt_token_order_recording=1
requires_no_prompt_token_offset_recording=1
requires_no_attention_mask_creation=1
requires_no_position_id_creation=1
requires_no_context_window_assembly=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-26 preserves explicit prompt token sequence denial fields:

```
prompt_token_sequence_authority=0
prompt_token_sequence_allowed=0
prompt_token_sequence_open_authority=0
prompt_token_sequence_read_authority=0
prompt_token_sequence_write_authority=0
prompt_token_sequence_execute_authority=0
prompt_token_sequence_runtime_authority=0
prompt_token_sequence_record_authority=0
prompt_token_sequence_token_id_record_authority=0
prompt_token_sequence_order_record_authority=0
prompt_token_sequence_offset_record_authority=0
prompt_token_sequence_context_window_authority=0
prompt_token_sequence_opened=0
prompt_token_sequence_read=0
prompt_token_sequence_validated=0
prompt_token_sequence_loaded=0
prompt_token_sequence_bytes_read=0
prompt_token_sequence_hash_computed=0
prompt_token_sequence_entries_loaded=0
prompt_token_sequence_recorded=0
prompt_token_sequence_result_recorded=0
prompt_token_sequence_count_recorded=0
prompt_token_sequence_order_recorded=0
prompt_token_sequence_runtime_invoked=0
prompt_token_sequence_file_written=0
prompt_token_ids_recorded=0
prompt_token_order_recorded=0
prompt_token_offsets_recorded=0
prompt_token_byte_offsets_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
context_window_assembled=0
prompt_evaluation_input_created=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_token_sequence_recorded=0
prompt_token_buffer_created=0
prompt_token_buffer_written=0
prompt_tokenized=0
tokenizer_runtime_attachment_performed=0
tokenizer_attached_to_runtime=0
runtime_session_created=0
runtime_invoked=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_vocab_loaded=0
prompt_evaluated=0
qa_dialogue_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-token-sequence-contract.sh \
  --prompt-tokenization
reports/nadia/prompt-tokenization/latest-prompt-tokenization-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-token-sequence.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-token-sequence
```

Non-Claims

Stage-26 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, tokenizer runtime attachment layer, prompt tokenizer, token counter, token sequence recorder, token ID recorder, token offset recorder, context-window assembler, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-token-sequence-contract-stage-26.sh
```

Expected output:

```
nadia_prompt_token_sequence_contract_stage_26: ok
```

docs/NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25.md

Source title: Nadia Prompt Tokenization Contract Stage-25

Lines: 167 | Words: 512 | SHA-256: 85e7bed2035bf22cf970b7724faeacc66155999ece952214c98f29bff0f79631

Nadia Prompt Tokenization Contract Stage-25

Status: Stage-25 implementation contract

Scope: prompt-tokenization metadata before prompt text reading, prompt text materialization, prompt buffer allocation, prompt token creation, prompt token sequence recording, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-25 gives Nadia a prompt-tokenization contract after the tokenizer-runtime-attachment contract is present.

The contract records how future reviewed prompt tokenization must reference the Stage-24 tokenizer-runtime-attachment contract, prompt-materialization boundary, prompt-receipt boundary, bound tokenizer artifact, and tokenization policy, while preserving the absolute rule that Nadia cannot yet read prompt text, create prompt tokens, record token sequences, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference. The script only measures the Stage-24 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-25 adds:

```
nadia_stage_25_prompt_tokenization_contract_present=1
nadia_prompt_tokenization_contract_generator_present=1
prompt_tokenization_contract_command=scripts/nadia-prompt-tokenization-contract.sh
installed_prompt_tokenization_contract_command=latticra-nadia prompt-tokenization
prompt_tokenization_contract_status=contract_only
prompt_tokenization_stage=contract-only
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenization_performed=0
prompt_tokenization_metadata_present=1
prompt_tokenization_family=operator-reviewed-prompt-tokenization
prompt_tokenization_format=contract-only-offline-tokenization
prompt_tokenization_decision=blocked_contract_only
prompt_tokenization_evidence_present=1
prompt_tokenization_source_policy=operator-reviewed-offline
prompt_tokenization_plan_recorded=1
prompt_tokenization_method_planned=offline-tokenization-policy-review
prompt_tokenization_result_recorded=0
prompt_tokenization_token_count_recorded=0
prompt_tokenization_token_sequence_recorded=0
prompt_tokenization_runtime_invoked=0
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_token_sequence_contract=1
prompt_tokenization_promotion_allowed=0
```

Stage-25 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, create prompt token buffers, create prompt tokens, count prompt tokens, record prompt token sequences, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, bind tokenizer artifacts, attach tokenizers to a runtime, create runtime sessions, open tokenizer files, read tokenizer files, load tokenizer vocabularies, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Tokenization Requirements

The prompt-tokenization contract names future review requirements:

```
requires_tokenizer_runtime_attachment_reference=1
requires_prompt_materialization_reference=1
requires_prompt_receipt_reference=1
requires_bound_tokenizer_artifact_reference=1
requires_tokenizer_specification_reference=1
requires_tokenization_policy=1
requires_token_count_policy=1
requires_token_sequence_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_prompt_text_read=1
requires_no_prompt_token_creation=1
requires_no_prompt_token_sequence_recording=1
requires_no_runtime_invocation=1
requires_no_prompt_evaluation=1
requires_no_dialogue_generation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-25 preserves explicit prompt tokenization denial fields:

```
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenization_open_authority=0
prompt_tokenization_read_authority=0
prompt_tokenization_write_authority=0
prompt_tokenization_execute_authority=0
prompt_tokenization_runtime_authority=0
prompt_tokenization_token_create_authority=0
prompt_tokenization_sequence_record_authority=0
prompt_tokenization_opened=0
prompt_tokenization_read=0
prompt_tokenization_validated=0
prompt_tokenization_loaded=0
prompt_tokenization_bytes_read=0
prompt_tokenization_hash_computed=0
prompt_tokenization_entries_loaded=0
prompt_tokenization_performed=0
prompt_tokenization_result_recorded=0
prompt_tokenization_token_count_recorded=0
prompt_tokenization_token_sequence_recorded=0
prompt_tokenization_runtime_invoked=0
prompt_tokenization_file_written=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_token_sequence_recorded=0
prompt_token_buffer_created=0
prompt_token_buffer_written=0
prompt_tokenized=0
tokenizer_runtime_attachment_performed=0
tokenizer_attached_to_runtime=0
runtime_session_created=0
runtime_invoked=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_vocab_loaded=0
prompt_evaluated=0
qa_dialogue_generated=0
```

Commands

Repository command:

```
sh scripts/nadia-prompt-tokenization-contract.sh \
  --tokenizer-runtime-attachment
reports/nadia/tokenizer-runtime-attachment/latest-tokenizer-runtime-attachment-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-prompt-tokenization.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia prompt-tokenization
```

Non-Claims

Stage-25 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, prompt reader, prompt materializer, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, tokenizer runtime attachment layer, prompt tokenizer, token counter, token sequence recorder, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-prompt-tokenization-contract-stage-25.sh
```

Expected output:

```
nadia_prompt_tokenization_contract_stage_25: ok
```

docs/NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6.md

Source title: Nadia Protective Safety Boundary Stage-6

Lines: 111 | Words: 271 | SHA-256: f1bc66c8ee43958e98bf316a7b3b8576a790f7151fa8f9ce53f14b79a2b5c389

Nadia Protective Safety Boundary Stage-6

Status: Stage-6 implementation contract Date: 2026-05-25 Scope: non-sexual-use, anti-manipulation, and namesake-cause awareness boundary before prompt evaluation, model runtime, or tool authority.

Purpose

Stage-6 makes Nadia's protective boundary explicit and testable.

Nadia is not a sexualized assistant, companion, roleplay surface, or adult-content generator. She exists for Latticra software development, systems engineering, AI development, and dignified awareness of Nadia Murad's cause. This boundary is absolute and must remain upstream of any future runtime, prompt evaluation, retrieval, or tool authority.

Capability

Stage-6 adds:

```
nadia_stage_6_protective_safety_boundary_present=1
nadia_protective_safety_generator_present=1
protective_safety_command=scripts/nadia-protective-safety-boundary.sh
installed_protective_safety_command=latticra-nadia protective-safety
requires_productivity_entry=1
productivity_entry_stage_required=5-productivity-ledger-loop
absolute_protective_boundary=1
sexual_user_request_authority=0
sexual_content_generation=0
sexual_roleplay_authority=0
sexualized_namesake_or_survivor_content=0
sexual_request_refusal=always
user_override_authority=0
prompt_injection_override_authority=0
manipulation_resistance=required
policy_bypass_authority=0
namesake_cause_awareness=1
awareness_context=non_sensational_human_rights
network_authority=0
```

```
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
distillation_performed=0
```

Sexualized request classes fail closed. Stage-6 does not summarize with AI, call an inference runtime, evaluate a prompt, mutate source, execute tools, train, distill, or use the network.

Inputs

The protective-safety boundary requires:

```
productivity_entry=Stage-5 Nadia productivity-ledger entry
request_class=operator request classification label
```

The default request class is software-development.

Outputs

The protective-safety boundary writes:

```
nadia-protective-safety-&lt;timestamp&gt;.txt
latest-protective-safety.txt
```

The report records:

```
stage=6-protective-safety-boundary
absolute_protective_boundary=1
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
namesake_cause_awareness=1
source_mutation_authority=0
```

Usage

With an explicit productivity-ledger entry:

```
sh scripts/nadia-protective-safety-boundary.sh \
  --productivity-entry /path/to/latest-productivity-entry.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-safety.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia protective-safety
```

Non-Claims

Stage-6 Nadia is not an inference runtime, prompt evaluator, sexual assistant, roleplay surface, adult-content generator, code generator, autonomous coding agent, source mutator, training system, distillation system, network retriever, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-protective-safety-boundary-stage-6.sh
```

Expected result:

```
nadia_protective_safety_boundary_stage_6: ok
```

docs/NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11.md

Source title: Nadia Runtime Invocation Contract Stage-11

Lines: 144 | Words: 314 | SHA-256: 6108b4635c6107ba96efbd42205d19ebf38443b882975578388b1008e05ed93d

Nadia Runtime Invocation Contract Stage-11

Status: Stage-11 implementation contract Date: 2026-05-25 Scope: runtime-invocation metadata before runtime process spawning, model session creation, model loading, token generation, inference, prompt evaluation, or tool execution.

Purpose

Stage-11 gives Nadia a runtime-invocation contract without invoking a runtime.

The contract consumes a Stage-10 inference-readiness contract, verifies inherited non-executing readiness, model, prompt, runtime, tool, and protective-safety posture, and records the review fields that would be required before any later model-load contract can exist.

Capability

Stage-11 adds:

```
nadia_stage_11_runtime_invocation_contract_present=1
nadia_runtime_invocation_contract_generator_present=1
runtime_invocation_contract_command=scripts/nadia-runtime-invocation-contract.sh
installed_runtime_invocation_contract_command=latticra-nadia runtime-invocation
requires_inference_readiness_contract=1
inference_readiness_stage_required=10-inference-readiness-contract
runtime_invocation_contract_status=contract_only
runtime_invocation_stage=contract-only
runtime_invocation_authority=0
runtime_invocation_allowed=0
runtime_invoked=0
invocation_decision=blocked_contract_only
invocation_evidence_present=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_provenance_review=1
requires_model_license_review=1
requires_model_safety_review=1
requires_refusal_policy_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_model_load_contract=1
invocation_promotion_allowed=0
runtime_process_spawn_authority=0
runtime_binary_execution_authority=0
runtime_session_authority=0
model_session_authority=0
token_generation_authority=0
model_runtime_present=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
runtime_session_created=0
model_weights_loaded=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-11 does not materialize prompts, evaluate prompts, load model weights, spawn a runtime process, execute a runtime binary, create a runtime session, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inputs

The runtime-invocation contract requires:

```
inference_readiness=Stage-10 Nadia inference-readiness contract
request_class=operator request classification label
```

Sexualized request classifications fail closed.

Outputs

The contract writes:

```
nadia-runtime-invocation-contract-&lt;timestamp&gt;.txt
latest-runtime-invocation-contract.txt
```

The report records:

```
stage=11-runtime-invocation-contract
runtime_invocation_contract_status=contract_only
runtime_invocation_authority=0
runtime_invocation_allowed=0
runtime_invoked=0
invocation_decision=blocked_contract_only
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
model_weights_loaded=0
token_generation_performed=0
inference_performed=0
sexual_request_refusal=always
```

Usage

With explicit evidence input:

```
sh scripts/nadia-runtime-invocation-contract.sh \
  --inference-readiness /path/to/latest-inference-readiness-contract.txt \
  --request-class software-development \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-runtime-invocation.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia runtime-invocation
```

Non-Claims

Stage-11 Nadia is not an inference runtime, runtime process launcher, model loader, prompt evaluator, prompt materializer, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, or production AI assistant.

Validation

```
sh scripts/test-nadia-runtime-invocation-contract-stage-11.sh
```

Expected result:

```
nadia_runtime_invocation_contract_stage_11: ok
```

docs/NADIA_RUNTIME_PROFILE_STAGE_2.md

Source title: Nadia Runtime Profile Stage-2

Lines: 95 | Words: 251 | SHA-256: 08cb69afdd80ff4f03a9525c72c78e8cc5da525f7be40a8b673cb5ac74f840d1

Nadia Runtime Profile Stage-2

Status: Stage-2 implementation contract Date: 2026-05-25 Scope: offline runtime and model-profile metadata before inference.

Purpose

Stage-2 gives Nadia a local runtime-profile boundary before any model is run.

The runtime-profile boundary records operator-selected runtime settings and optional operator-provided model-file measurements. It creates evidence that a future inference run can inspect before prompt evaluation, while keeping execution denied.

Capability

Stage-2 adds:

```
nadia_stage_2_runtime_profile_present=1
nadia_runtime_profile_generator_present=1
runtime_profile_command=scripts/nadia-runtime-profile.sh
installed_runtime_profile_command=latticra-nadia runtime-profile
model_file_measurement=operator_provided_optional
runtime_family=llama.cpp-compatible
model_format=gguf
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
```

This stage can measure a local model file when the operator provides a path. It does not download, install, copy, load, mmap, execute, or evaluate model weights.

Profile Fields

The runtime-profile generator records:

```
runtime_family
runtime_binary
model_format
model_file
model_file_present
model_file_bytes
model_file_measurement
context_window_tokens
memory_budget_mib
thread_policy
gpu_policy
```

The default runtime family is llama.cpp-compatible because GGUF-oriented local runtimes are a practical first target for offline development, but Stage-2 does not vendor or execute llama.cpp.

Usage

Without a model file:

```
sh scripts/nadia-runtime-profile.sh --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-runtime.XXXXXX&quot;)&quot;
```

With an operator-provided local model file:

```
sh scripts/nadia-runtime-profile.sh \
  --model /path/to/model.gguf \
  --context-tokens 8192 \
  --memory-mib 16384 \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-runtime.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia runtime-profile
```

Non-Claims

Stage-2 Nadia is not an inference runtime, model installer, model distributor, prompt evaluator, autonomous coding agent, embedding service, training system, source mutator, network retriever, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-runtime-profile-stage-2.sh
```

Expected result:

```
nadia_runtime_profile_stage_2: ok
```

docs/NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4.md

Source title: Nadia Systems Engineering Mode Stage-4

Lines: 120 | Words: 255 | SHA-256: d037adcd73cc8f05ec5f86be26e1063852029a65bc24a89cfcfc709d3421bb3c

Nadia Systems Engineering Mode Stage-4

Status: Stage-4 implementation contract Date: 2026-05-25 Scope: systems-engineering mode taxonomy and prompt-plan validation before prompt evaluation.

Purpose

Stage-4 gives Nadia a mode taxonomy for software-development work before any prompt evaluation exists.

The mode validator reads a Stage-3 prompt plan, checks that it is still a planning artifact, and records which systems-engineering mode and validator set should govern the future prompt surface.

Capability

Stage-4 adds:

```
nadia_stage_4_systems_engineering_mode_present=1
nadia_mode_validation_generator_present=1
mode_validation_command=scripts/nadia-mode-validate.sh
installed_mode_validation_command=latticra-nadia mode-validate
requires_prompt_plan=1
prompt_plan_stage_required=3-developer-workbench-planning
mode_taxonomy_present=1
mode_allowed=1
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
```

Stage-4 does not summarize with AI, call an inference runtime, evaluate a prompt, mutate source, execute tools, or use the network.

Modes

The initial mode taxonomy is:

```
systems-engineering
ai-development
c-substrate
cpp-authority
rust-panel
lat-lir-l-ui
seal-boundary
runtime-boundary
fedora-validation
awareness-safety
```

Unknown modes fail closed.

Inputs

The validator requires:

```
prompt_plan=Stage-3 Nadia prompt plan
mode=operator-selected systems-engineering mode label
```

The default mode is systems-engineering.

Outputs

The mode validator writes:

```
nadia-mode-validation-&lt;timestamp&gt;.txt
latest-mode-validation.txt
```

The validation report records:

```
stage=4-systems-engineering-mode-validation
mode=&lt;operator mode&gt;
prompt_plan_stage=3-developer-workbench-planning
mode_allowed=1
validator_set=&lt;mode validator set&gt;
prompt_evaluated=0
inference_performed=0
source_mutation_authority=0
```

Usage

With an explicit prompt plan:

```
sh scripts/nadia-mode-validate.sh \
  --prompt-plan /path/to/latest-prompt-plan.txt \
  --mode runtime-boundary \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-mode.XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia mode-validate
```

Non-Claims

Stage-4 Nadia is not an inference runtime, prompt evaluator, code generator, autonomous coding agent, source mutator, training system, network retriever, security product, or production AI assistant.

Validation

```
sh scripts/test-nadia-systems-engineering-mode-stage-4.sh
```

Expected result:

```
nadia_systems_engineering_mode_stage_4: ok
```

docs/NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17.md

Source title: Nadia Tokenization Boundary Contract Stage-17

Lines: 144 | Words: 359 | SHA-256: bf1f61a7d96c98ab0b6109656252eb3e05339cb957078ac36695097ae6beb3b7

Nadia Tokenization Boundary Contract Stage-17

Status: Stage-17 implementation contract Date: 2026-05-25 Scope: tokenization-boundary metadata before tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-17 gives Nadia a tokenization-boundary contract after the prompt-evaluation handoff contract is present.

The contract consumes a Stage-16 prompt-evaluation handoff contract, verifies inherited non-executing dialogue, prompt, runtime, model, tool, and protective-safety posture, and records that the next promotion point is a separate tokenizer specification contract. It does not tokenize prompts.

Capability

Stage-17 adds:

```
nadia_stage_17_tokenization_boundary_contract_present=1
nadia_tokenization_boundary_contract_generator_present=1
tokenization_boundary_contract_command=scripts/nadia-tokenization-boundary-contract.sh
installed_tokenization_boundary_contract_command=latticra-nadia tokenization-boundary
tokenization_boundary_contract_status=contract_only
tokenization_boundary_stage=contract-only
tokenization_boundary_authority=0
tokenization_boundary_allowed=0
tokenization_boundary_performed=0
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenized=0
prompt_tokens_created=0
tokenizer_file_opened=0
tokenizer_vocab_loaded=0
prompt_evaluation_authority=0
prompt_evaluated=0
tokenization_decision=blocked_contract_only
tokenization_evidence_present=1
requires_prompt_evaluation_handoff_contract=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_prompt_buffer_boundary=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenizer_specification_contract=1
tokenization_boundary_promotion_allowed=0
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
```

Stage-17 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Tokenizer Denial Boundary

```
tokenizer_file_open_authority=0
tokenizer_file_read_authority=0
tokenizer_vocab_load_authority=0
tokenizer_vocab_mapping_authority=0
tokenizer_runtime_attach_authority=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
tokenizer_vocab_mapped=0
tokenizer_attached_to_runtime=0
tokenizer_bytes_read=0
tokenizer_hash_computed=0
```

Inherited Awareness Boundary

The tokenization boundary preserves the Stage-15 Nadia Initiative awareness scope and the Stage-16 prompt-evaluation handoff denial:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
```

```
plain_language_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
live_web_lookup_authority=0
prompt_evaluation_handoff_authority=0
prompt_evaluation_handoff_performed=0
```

Protective Boundary

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
```

Usage

With explicit evidence input:

```
sh scripts/nadia-tokenization-boundary-contract.sh \
  --prompt-evaluation-handoff /path/to/latest-prompt-evaluation-handoff-contract.txt \
  --request-class awareness-education \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenization-boundary.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia tokenization-boundary
```

Non-Claims

Stage-17 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

```
sh scripts/test-nadia-tokenization-boundary-contract-stage-17.sh
```

Expected result:

```
nadia_tokenization_boundary_contract_stage_17: ok
```

docs/NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23.md

Source title: Nadia Tokenizer Artifact Binding Contract Stage-23

Lines: 159 | Words: 481 | SHA-256: f026b108093123919b9e1497f3b5fa5e49c85a1c06fe7d8cfc6ea03e6b78fe14

Nadia Tokenizer Artifact Binding Contract Stage-23

Status: Stage-23 implementation contract

Scope: tokenizer-artifact-binding metadata before tokenizer artifact opening, artifact reading, artifact hashing, artifact verification, artifact binding, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-23 gives Nadia a tokenizer-artifact-binding contract after the tokenizer-artifact-verification contract is present.

The contract records how future verified tokenizer artifacts must be associated with manifest entries, tokenizer specification requirements, and operator review records, while preserving the absolute rule that Nadia cannot yet open, read, hash, verify, bind, validate, load, or attach tokenizer artifacts. The script only measures the Stage-22 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-23 adds:

```
nadia_stage_23_tokenizer_artifact_binding_contract_present=1
nadia_tokenizer_artifact_binding_contract_generator_present=1
tokenizer_artifact_binding_contract_command=scripts/nadia-tokenizer-artifact-binding-contract.sh
installed_tokenizer_artifact_binding_contract_command=latticra-nadia tokenizer-artifact-binding
tokenizer_artifact_binding_contract_status=contract_only
tokenizer_artifact_binding_stage=contract-only
tokenizer_artifact_binding_authority=0
tokenizer_artifact_binding_allowed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_binding_metadata_present=1
tokenizer_artifact_binding_family=operator-reviewed-tokenizer-artifact-binding
tokenizer_artifact_binding_format=contract-only-offline-binding
tokenizer_artifact_binding_decision=blocked_contract_only
tokenizer_artifact_binding_evidence_present=1
tokenizer_artifact_binding_source_policy=operator-reviewed-offline
tokenizer_artifact_binding_plan_recorded=1
tokenizer_artifact_binding_method_planned=offline-manifest-artifact-role-binding-review
tokenizer_artifact_binding_result_recorded=0
tokenizer_artifact_binding_record_created=0
tokenizer_artifact_binding_manifest_reference_recorded=0
tokenizer_artifact_binding_artifact_reference_recorded=0
tokenizer_artifact_binding_runtime_attach_recorded=0
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_runtime_attachment_contract=1
tokenizer_artifact_binding_promotion_allowed=0
```

Stage-23 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, record tokenizer artifact digests, record tokenizer artifact sizes, compare tokenizer artifact digests, compare tokenizer artifact sizes, verify tokenizer artifacts, bind tokenizer artifacts, attach tokenizers to a runtime, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Binding Requirements

The tokenizer-artifact-binding contract names future review requirements:

```
requires_verified_artifact_reference=1
requires_artifact_identity=1
requires_artifact_role_classification=1
requires_verification_contract_reference=1
requires_measurement_contract_reference=1
requires_inventory_entry_reference=1
requires_manifest_entry_reference=1
```

```
requires_tokenizer_specification_reference=1
requires_expected_digest_policy=1
requires_observed_digest_policy=1
requires_digest_comparison_policy=1
requires_size_comparison_policy=1
requires_digest_match_record=1
requires_size_match_record=1
requires_source_snapshot_reference=1
requires_license_and_source_review=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_runtime_attachment=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-23 preserves explicit binding denial fields:

```
tokenizer_artifact_binding_authority=0
tokenizer_artifact_binding_allowed=0
tokenizer_artifact_binding_open_authority=0
tokenizer_artifact_binding_read_authority=0
tokenizer_artifact_binding_write_authority=0
tokenizer_artifact_binding_hash_authority=0
tokenizer_artifact_binding_validation_authority=0
tokenizer_artifact_binding_load_authority=0
tokenizer_artifact_binding_attach_authority=0
tokenizer_artifact_binding_runtime_attach_authority=0
tokenizer_artifact_binding_manifest_bind_authority=0
tokenizer_artifact_binding_tokenizer_bind_authority=0
tokenizer_artifact_binding_opened=0
tokenizer_artifact_binding_read=0
tokenizer_artifact_binding_validated=0
tokenizer_artifact_binding_loaded=0
tokenizer_artifact_binding_bytes_read=0
tokenizer_artifact_binding_hash_computed=0
tokenizer_artifact_binding_entries_loaded=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_binding_bound=0
tokenizer_artifact_binding_record_created=0
tokenizer_artifact_binding_manifest_reference_loaded=0
tokenizer_artifact_binding_artifact_reference_loaded=0
tokenizer_artifact_binding_manifest_reference_recorded=0
tokenizer_artifact_binding_artifact_reference_recorded=0
tokenizer_artifact_binding_runtime_attach_recorded=0
tokenizer_artifact_binding_runtime_attachment_performed=0
tokenizer_artifact_binding_result_recorded=0
tokenizer_artifact_binding_file_written=0
tokenizer_artifact_bound_to_manifest=0
tokenizer_artifact_bound_to_tokenizer=0
tokenizer_attached_to_runtime=0
tokenizer_runtime_attachment_performed=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
prompt_tokenized=0
prompt_evaluated=0
```

Commands

Repository command:

```
sh scripts/nadia-tokenizer-artifact-binding-contract.sh \
  --tokenizer-artifact-verification reports/nadia/tokenizer-artifact-verification/latest-tokeniz
er-artifact-verification-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-artifact-binding.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia tokenizer-artifact-binding
```

Non-Claims

Stage-23 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, tokenizer runtime attachment layer, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-tokenizer-artifact-binding-contract-stage-23.sh
```

Expected output:

```
nadia_tokenizer_artifact_binding_contract_stage_23: ok
```

docs/NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20.md

Source title: Nadia Tokenizer Artifact Inventory Contract Stage-20

Lines: 156 | Words: 454 | SHA-256: 9de93caefa3c9289bf418e5ad6ec5fd7596f462923f9f969ad6975c951e5737a

Nadia Tokenizer Artifact Inventory Contract Stage-20

Status: Stage-20 implementation contract

Scope: tokenizer-artifact-inventory metadata before tokenizer artifact path resolution, artifact scanning, artifact stat calls, artifact hashing, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-20 gives Nadia a tokenizer-artifact-inventory contract after the tokenizer-manifest contract is present.

The contract consumes a Stage-19 tokenizer-manifest contract, verifies inherited non-executing tokenizer, manifest, prompt, dialogue, runtime, model, tool, and protective-safety posture, and records the requirements a future tokenizer artifact measurement contract must satisfy. It does not resolve artifact paths, scan directories, stat files, open tokenizer artifacts, read tokenizer artifacts, or hash tokenizer artifacts.

Contract Fields

Stage-20 adds:

```
nadia_stage_20_tokenizer_artifact_inventory_contract_present=1
nadia_tokenizer_artifact_inventory_contract_generator_present=1
tokenizer_artifact_inventory_contract_command=scripts/nadia-tokenizer-artifact-inventory-contrac
t.sh
installed_tokenizer_artifact_inventory_contract_command=latticra-nadia
tokenizer-artifact-inventory
tokenizer_artifact_inventory_contract_status=contract_only
tokenizer_artifact_inventory_stage=contract-only
tokenizer_artifact_inventory_authority=0
tokenizer_artifact_inventory_allowed=0
tokenizer_artifact_inventory_performed=0
tokenizer_artifact_inventory_metadata_present=1
tokenizer_artifact_inventory_family=operator-reviewed-tokenizer-artifact-inventory
```

```
tokenizer_artifact_inventory_format=contract-only-offline-inventory
tokenizer_artifact_inventory_decision=blocked_contract_only
tokenizer_artifact_inventory_evidence_present=1
tokenizer_artifact_inventory_source_policy=operator-reviewed-offline
tokenizer_artifact_inventory_path_recorded=0
tokenizer_artifact_inventory_schema_planned=1
tokenizer_artifact_inventory_entry_count=0
tokenizer_artifact_inventory_file_count=0
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_measurement_contract=1
tokenizer_artifact_inventory_promotion_allowed=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_artifact_measurement_performed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-20 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Inventory Requirements

The tokenizer-artifact-inventory contract names future inventory review requirements:

```
requires_artifact_identity=1
requires_artifact_role_classification=1
requires_manifest_entry_reference=1
requires_vocab_artifact_entry=1
requires_merges_artifact_entry_review=1
requires_tokenizer_model_artifact_entry=1
requires_special_tokens_artifact_entry=1
requires_chat_template_artifact_entry=1
requires_normalization_policy_artifact_entry=1
requires_unicode_policy_artifact_entry=1
requires_license_artifact_entry=1
requires_source_snapshot_reference=1
requires_relative_path_policy_review=1
requires_no_absolute_host_path_claim=1
requires_no_runtime_binding=1
requires_operator_approval_record=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

These are requirements for later review only. They do not authorize artifact discovery, artifact reading, tokenizer loading, prompt tokenization, or inference.

Denial Boundary

Stage-20 preserves explicit artifact denial fields:

```
tokenizer_artifact_inventory_open_authority=0
tokenizer_artifact_inventory_read_authority=0
tokenizer_artifact_inventory_parse_authority=0
tokenizer_artifact_inventory_validation_authority=0
tokenizer_artifact_inventory_load_authority=0
tokenizer_artifact_path_resolution_authority=0
tokenizer_artifact_scan_authority=0
```

```
tokenizer_artifact_stat_authority=0
tokenizer_artifact_hash_authority=0
tokenizer_artifact_measurement_authority=0
tokenizer_artifact_copy_authority=0
tokenizer_artifact_load_authority=0
tokenizer_artifact_inventory_opened=0
tokenizer_artifact_inventory_read=0
tokenizer_artifact_inventory_parsed=0
tokenizer_artifact_inventory_loaded=0
tokenizer_artifact_inventory_entries_loaded=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_scan_performed=0
tokenizer_artifact_stat_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_bytes_read=0
tokenizer_artifact_hash_computed=0
tokenizer_artifact_measurement_performed=0
tokenizer_vocab_artifact_loaded=0
tokenizer_merges_artifact_loaded=0
tokenizer_model_artifact_loaded=0
tokenizer_special_tokens_artifact_loaded=0
prompt_tokenization_authority=0
prompt_tokenized=0
```

Command

From the repository:

```
sh scripts/nadia-tokenizer-artifact-inventory-contract.sh \
  --tokenizer-manifest reports/nadia/tokenizer-manifest/latest-tokenizer-manifest-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-artifact-inventory.
XXXXXX&quot;)&quot;
```

After guarded local install:

```
latticra-nadia tokenizer-artifact-inventory
```

Non-Claims

Stage-20 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

```
sh scripts/test-nadia-tokenizer-artifact-inventory-contract-stage-20.sh
```

Expected:

```
nadia_tokenizer_artifact_inventory_contract_stage_20: ok
```

docs/NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21.md

Source title: Nadia Tokenizer Artifact Measurement Contract Stage-21

Lines: 133 | Words: 418 | SHA-256: 1ef8e0f61bea2dba2acf7b0ea3a42ab91219cea36fe182d216459834f4b2e5f8

Nadia Tokenizer Artifact Measurement Contract Stage-21

Status: Stage-21 implementation contract

Scope: tokenizer-artifact-measurement metadata before tokenizer artifact opening, artifact reading, artifact hashing, artifact size recording, artifact digest recording, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-21 gives Nadia a tokenizer-artifact-measurement contract after the tokenizer-artifact-inventory contract is present.

The contract records how future tokenizer artifacts must be measured, reviewed, and promoted, while preserving the absolute rule that Nadia cannot yet open, read, hash, size, validate, or load tokenizer artifacts. The script only measures the Stage-20 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-21 adds:

```
nadia_stage_21_tokenizer_artifact_measurement_contract_present=1
nadia_tokenizer_artifact_measurement_contract_generator_present=1
tokenizer_artifact_measurement_contract_command=scripts/nadia-tokenizer-artifact-measurement-con
tract.sh
installed_tokenizer_artifact_measurement_contract_command=latticra-nadia
tokenizer-artifact-measurement
tokenizer_artifact_measurement_contract_status=contract_only
tokenizer_artifact_measurement_stage=contract-only
tokenizer_artifact_measurement_authority=0
tokenizer_artifact_measurement_allowed=0
tokenizer_artifact_measurement_performed=0
tokenizer_artifact_measurement_metadata_present=1
tokenizer_artifact_measurement_family=operator-reviewed-tokenizer-artifact-measurement
tokenizer_artifact_measurement_format=contract-only-offline-measurement
tokenizer_artifact_measurement_decision=blocked_contract_only
tokenizer_artifact_measurement_evidence_present=1
tokenizer_artifact_measurement_source_policy=operator-reviewed-offline
tokenizer_artifact_measurement_plan_recorded=1
tokenizer_artifact_measurement_algorithm_planned=sha256-or-approved-digest
tokenizer_artifact_measurement_result_recorded=0
tokenizer_artifact_measurement_digest_recorded=0
tokenizer_artifact_measurement_size_recorded=0
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_verification_contract=1
tokenizer_artifact_measurement_promotion_allowed=0
```

Stage-21 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, record tokenizer artifact digests, record tokenizer artifact sizes, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Measurement Requirements

The tokenizer-artifact-measurement contract names future review requirements:

```
requires_artifact_identity=1
requires_artifact_role_classification=1
requires_inventory_entry_reference=1
requires_measurement_algorithm_policy=1
requires_digest_format_policy=1
requires_size_record_policy=1
requires_reproducible_measurement_policy=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_runtime_binding=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-21 preserves explicit measurement denial fields:

```
tokenizer_artifact_measurement_authority=0
tokenizer_artifact_measurement_allowed=0
tokenizer_artifact_measurement_open_authority=0
tokenizer_artifact_measurement_read_authority=0
tokenizer_artifact_measurement_hash_authority=0
tokenizer_artifact_measurement_validation_authority=0
tokenizer_artifact_measurement_load_authority=0
tokenizer_artifact_measurement_opened=0
tokenizer_artifact_measurement_read=0
tokenizer_artifact_measurement_validated=0
tokenizer_artifact_measurement_loaded=0
tokenizer_artifact_measurement_bytes_read=0
tokenizer_artifact_measurement_hash_computed=0
tokenizer_artifact_measurement_entries_loaded=0
tokenizer_artifact_measurement_performed=0
tokenizer_artifact_measurement_result_recorded=0
tokenizer_artifact_measurement_digest_recorded=0
tokenizer_artifact_measurement_size_recorded=0
tokenizer_artifact_digest_recorded=0
tokenizer_artifact_size_recorded=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_bytes_read=0
tokenizer_artifact_hash_computed=0
prompt_tokenized=0
prompt_evaluated=0
```

Commands

Repository command:

```
sh scripts/nadia-tokenizer-artifact-measurement-contract.sh \
  --tokenizer-artifact-inventory
reports/nadia/tokenizer-artifact-inventory/latest-tokenizer-artifact-inventory-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-artifact-
measurement.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia tokenizer-artifact-measurement
```

Non-Claims

Stage-21 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-tokenizer-artifact-measurement-contract-stage-21.sh
```

Expected output:

```
nadia_tokenizer_artifact_measurement_contract_stage_21: ok
```

docs/NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22.md

Source title: Nadia Tokenizer Artifact Verification Contract Stage-22

Lines: 147 | Words: 455 | SHA-256: 5ebaea614a5b22a4ba469c8ead65a6ea3628fd6dc8758492c51731379f814e45

Nadia Tokenizer Artifact Verification Contract Stage-22

Status: Stage-22 implementation contract

Scope: tokenizer-artifact-verification metadata before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest comparison, artifact size comparison, artifact verification, artifact binding, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-22 gives Nadia a tokenizer-artifact-verification contract after the tokenizer-artifact-measurement contract is present.

The contract records how future tokenizer artifacts must be verified, reviewed, and promoted, while preserving the absolute rule that Nadia cannot yet open, read, hash, compare, verify, bind, validate, or load tokenizer artifacts. The script only measures the Stage-21 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-22 adds:

```
nadia_stage_22_tokenizer_artifact_verification_contract_present=1
nadia_tokenizer_artifact_verification_contract_generator_present=1
tokenizer_artifact_verification_contract_command=scripts/nadia-tokenizer-artifact-verification-c
ontract.sh
installed_tokenizer_artifact_verification_contract_command=latticra-nadia
tokenizer-artifact-verification
tokenizer_artifact_verification_contract_status=contract_only
tokenizer_artifact_verification_stage=contract-only
tokenizer_artifact_verification_authority=0
tokenizer_artifact_verification_allowed=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_verification_metadata_present=1
tokenizer_artifact_verification_family=operator-reviewed-tokenizer-artifact-verification
tokenizer_artifact_verification_format=contract-only-offline-verification
tokenizer_artifact_verification_decision=blocked_contract_only
tokenizer_artifact_verification_evidence_present=1
tokenizer_artifact_verification_source_policy=operator-reviewed-offline
tokenizer_artifact_verification_plan_recorded=1
tokenizer_artifact_verification_method_planned=offline-digest-and-size-policy-review
tokenizer_artifact_verification_comparison_performed=0
tokenizer_artifact_verification_result_recorded=0
tokenizer_artifact_verification_digest_match_recorded=0
tokenizer_artifact_verification_size_match_recorded=0
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_binding_contract=1
tokenizer_artifact_verification_promotion_allowed=0
```

Stage-22 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, record tokenizer artifact digests, record tokenizer artifact sizes, compare tokenizer artifact digests, compare tokenizer artifact sizes, verify tokenizer artifacts, bind tokenizer artifacts, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Verification Requirements

The tokenizer-artifact-verification contract names future review requirements:

```
requires_artifact_identity=1
requires_artifact_role_classification=1
requires_measurement_contract_reference=1
requires_inventory_entry_reference=1
requires_expected_digest_policy=1
requires_observed_digest_policy=1
requires_digest_comparison_policy=1
requires_size_comparison_policy=1
requires_source_snapshot_reference=1
requires_license_and_source_review=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_runtime_binding=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-22 preserves explicit verification denial fields:

```
tokenizer_artifact_verification_authority=0
tokenizer_artifact_verification_allowed=0
tokenizer_artifact_verification_open_authority=0
tokenizer_artifact_verification_read_authority=0
tokenizer_artifact_verification_hash_authority=0
tokenizer_artifact_verification_validation_authority=0
tokenizer_artifact_verification_load_authority=0
tokenizer_artifact_verification_compare_authority=0
tokenizer_artifact_verification_bind_authority=0
tokenizer_artifact_verification_opened=0
tokenizer_artifact_verification_read=0
tokenizer_artifact_verification_validated=0
tokenizer_artifact_verification_loaded=0
tokenizer_artifact_verification_bytes_read=0
tokenizer_artifact_verification_hash_computed=0
tokenizer_artifact_verification_entries_loaded=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_verification_compared=0
tokenizer_artifact_verification_comparison_performed=0
tokenizer_artifact_verification_digest_comparison_performed=0
tokenizer_artifact_verification_size_comparison_performed=0
tokenizer_artifact_verification_expected_digest_loaded=0
tokenizer_artifact_verification_observed_digest_loaded=0
tokenizer_artifact_verification_digest_match_recorded=0
tokenizer_artifact_verification_size_match_recorded=0
tokenizer_artifact_verification_result_recorded=0
tokenizer_artifact_source_signature_verified=0
tokenizer_artifact_digest_recorded=0
tokenizer_artifact_size_recorded=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_bytes_read=0
tokenizer_artifact_hash_computed=0
prompt_tokenized=0
prompt_evaluated=0
```

Commands

Repository command:

```
sh scripts/nadia-tokenizer-artifact-verification-contract.sh \
  --tokenizer-artifact-measurement
reports/nadia/tokenizer-artifact-measurement/latest-tokenizer-artifact-measurement-contract.txt
\
  --output &quot;$(mktemp -d
&quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-artifact-verification.XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia tokenizer-artifact-verification
```

Non-Claims

Stage-22 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-tokenizer-artifact-verification-contract-stage-22.sh
```

Expected output:

```
nadia_tokenizer_artifact_verification_contract_stage_22: ok
```

docs/NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19.md

Source title: Nadia Tokenizer Manifest Contract Stage-19

Lines: 143 | Words: 400 | SHA-256: 091ecbb47a5810961b94110b9ee0acf90e229fee0746db8d0cf594742663e46d

Nadia Tokenizer Manifest Contract Stage-19

Status: Stage-19 implementation contract

Scope: tokenizer-manifest metadata before tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-19 gives Nadia a tokenizer-manifest contract after the tokenizer-specification contract is present.

The contract consumes a Stage-18 tokenizer-specification contract, verifies inherited non-executing tokenizer, prompt, dialogue, runtime, model, tool, and protective-safety posture, and records the requirements a future tokenizer artifact inventory must satisfy. It does not load tokenizer manifests, parse manifests, open tokenizer files, or tokenize prompts.

Contract Fields

Stage-19 adds:

```
nadia_stage_19_tokenizer_manifest_contract_present=1
nadia_tokenizer_manifest_contract_generator_present=1
tokenizer_manifest_contract_command=scripts/nadia-tokenizer-manifest-contract.sh
installed_tokenizer_manifest_contract_command=latticra-nadia tokenizer-manifest
tokenizer_manifest_contract_status=contract_only
tokenizer_manifest_stage=contract-only
tokenizer_manifest_authority=0
tokenizer_manifest_allowed=0
tokenizer_manifest_performed=0
tokenizer_manifest_metadata_present=1
tokenizer_manifest_family=operator-reviewed-tokenizer-manifest
tokenizer_manifest_format=contract-only-offline-manifest
tokenizer_manifest_decision=blocked_contract_only
tokenizer_manifest_evidence_present=1
tokenizer_manifest_source_policy=operator-reviewed-offline
tokenizer_manifest_path_recorded=0
tokenizer_manifest_schema_planned=1
```

```
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_inventory_contract=1
tokenizer_manifest_promotion_allowed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_opened=0
tokenizer_manifest_read=0
tokenizer_manifest_parsed=0
tokenizer_manifest_validated=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-19 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Manifest Review Requirements

The tokenizer-manifest contract names the minimum future manifest review requirements:

```
requires_manifest_identity=1
requires_manifest_schema_review=1
requires_model_tokenizer_compatibility_reference=1
requires_tokenizer_family_match=1
requires_tokenizer_format_match=1
requires_tokenizer_file_inventory=1
requires_vocabulary_file_entry=1
requires_merges_file_entry_review=1
requires_special_tokens_entry=1
requires_bos_eos_policy_entry=1
requires_chat_template_entry=1
requires_prompt_template_boundary_entry=1
requires_unicode_policy_entry=1
requires_normalization_policy_entry=1
requires_context_window_entry=1
requires_stop_sequence_entry=1
requires_license_and_source_entry=1
requires_source_snapshot_reference=1
requires_operator_approval_record=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

These are requirements for later review only. They do not authorize manifest loading, tokenizer loading, prompt tokenization, or inference.

Denial Boundary

Stage-19 preserves explicit manifest denial fields:

```
tokenizer_manifest_open_authority=0
tokenizer_manifest_read_authority=0
tokenizer_manifest_parse_authority=0
tokenizer_manifest_validation_authority=0
tokenizer_manifest_load_authority=0
tokenizer_manifest_opened=0
tokenizer_manifest_read=0
tokenizer_manifest_parsed=0
tokenizer_manifest_validated=0
tokenizer_manifest_loaded=0
tokenizer_manifest_bytes_read=0
tokenizer_manifest_hash_computed=0
tokenizer_manifest_entries_loaded=0
tokenizer_file_path_resolved=0
tokenizer_vocab_path_resolved=0
tokenizer_file_open_authority=0
tokenizer_file_read_authority=0
tokenizer_vocab_load_authority=0
```

```
prompt_tokenization_authority=0
prompt_tokenized=0
```

Command

From the repository:

```
sh scripts/nadia-tokenizer-manifest-contract.sh \
  --tokenizer-specification
reports/nadia/tokenizer-specification/latest-tokenizer-specification-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-manifest.
XXXXXX&quot;)&quot;
```

After guarded local install:

```
latticra-nadia tokenizer-manifest
```

Non-Claims

Stage-19 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer file inventory reader, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

```
sh scripts/test-nadia-tokenizer-manifest-contract-stage-19.sh
```

Expected:

```
nadia_tokenizer_manifest_contract_stage_19: ok
```

docs/NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24.md

Source title: Nadia Tokenizer Runtime Attachment Contract Stage-24

Lines: 165 | Words: 504 | SHA-256: 4f68f9b8b7ad402fba6e09696bb3fa5d515a97f74e71aee9e6fd287f87914ec9

Nadia Tokenizer Runtime Attachment Contract Stage-24

Status: Stage-24 implementation contract

Scope: tokenizer-runtime-attachment metadata before tokenizer artifact opening, artifact reading, artifact hashing, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-24 gives Nadia a tokenizer-runtime-attachment contract after the tokenizer-artifact-binding contract is present.

The contract records how a future reviewed tokenizer attachment must reference a bound tokenizer artifact, runtime profile, runtime invocation contract, and model-load boundary, while preserving the absolute rule that Nadia cannot yet attach tokenizers to a runtime, create runtime sessions, tokenize prompts, evaluate prompts, generate tokens, or run inference. The script only measures the Stage-23 contract report as evidence that the prerequisite metadata exists.

Contract Fields

Stage-24 adds:

```
nadia_stage_24_tokenizer_runtime_attachment_contract_present=1
```

```
nadia_tokenizer_runtime_attachment_contract_generator_present=1
tokenizer_runtime_attachment_contract_command=scripts/nadia-tokenizer-runtime-attachment-contrac
t.sh
installed_tokenizer_runtime_attachment_contract_command=latticra-nadia
tokenizer-runtime-attachment
tokenizer_runtime_attachment_contract_status=contract_only
tokenizer_runtime_attachment_stage=contract-only
tokenizer_runtime_attachment_authority=0
tokenizer_runtime_attachment_allowed=0
tokenizer_runtime_attachment_performed=0
tokenizer_runtime_attachment_metadata_present=1
tokenizer_runtime_attachment_family=operator-reviewed-tokenizer-runtime-attachment
tokenizer_runtime_attachment_format=contract-only-offline-attachment
tokenizer_runtime_attachment_decision=blocked_contract_only
tokenizer_runtime_attachment_evidence_present=1
tokenizer_runtime_attachment_source_policy=operator-reviewed-offline
tokenizer_runtime_attachment_plan_recorded=1
tokenizer_runtime_attachment_method_planned=offline-runtime-tokenizer-attachment-review
tokenizer_runtime_attachment_result_recorded=0
tokenizer_runtime_attachment_record_created=0
tokenizer_runtime_attachment_runtime_reference_recorded=0
tokenizer_runtime_attachment_tokenizer_reference_recorded=0
tokenizer_runtime_attachment_session_created=0
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_runtime_profile_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_future_prompt_tokenization_contract=1
tokenizer_runtime_attachment_promotion_allowed=0
```

Stage-24 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer manifests, read tokenizer manifests, parse tokenizer manifests, validate tokenizer manifests, load tokenizer manifests, resolve tokenizer artifact paths, scan artifact directories, stat tokenizer artifacts, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, record tokenizer artifact digests, record tokenizer artifact sizes, compare tokenizer artifact digests, compare tokenizer artifact sizes, verify tokenizer artifacts, bind tokenizer artifacts, attach tokenizers to a runtime, create runtime sessions, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Runtime Attachment Requirements

The tokenizer-runtime-attachment contract names future review requirements:

```
requires_bound_tokenizer_artifact_reference=1
requires_binding_contract_reference=1
requires_verified_artifact_reference=1
requires_artifact_identity=1
requires_artifact_role_classification=1
requires_runtime_profile_reference=1
requires_runtime_invocation_reference=1
requires_model_load_reference=1
requires_manifest_entry_reference=1
requires_tokenizer_specification_reference=1
requires_digest_match_record=1
requires_size_match_record=1
requires_operator_approval_record=1
requires_official_source_snapshot=1
requires_no_runtime_invocation=1
requires_no_runtime_session_creation=1
requires_no_prompt_tokenization=1
requires_no_prompt_evaluation=1
requires_no_inference=1
requires_refusal_policy_link=1
requires_survivor_centered_language_review=1
```

Denial Boundary

Stage-24 preserves explicit runtime attachment denial fields:

```
tokenizer_runtime_attachment_authority=0
tokenizer_runtime_attachment_allowed=0
tokenizer_runtime_attachment_open_authority=0
tokenizer_runtime_attachment_read_authority=0
tokenizer_runtime_attachment_write_authority=0
tokenizer_runtime_attachment_validation_authority=0
tokenizer_runtime_attachment_load_authority=0
tokenizer_runtime_attachment_attach_authority=0
tokenizer_runtime_attachment_runtime_invoke_authority=0
tokenizer_runtime_attachment_session_authority=0
tokenizer_runtime_attachment_tokenizer_bind_authority=0
tokenizer_runtime_attachment_opened=0
tokenizer_runtime_attachment_read=0
tokenizer_runtime_attachment_validated=0
tokenizer_runtime_attachment_loaded=0
tokenizer_runtime_attachment_bytes_read=0
tokenizer_runtime_attachment_hash_computed=0
tokenizer_runtime_attachment_entries_loaded=0
tokenizer_runtime_attachment_performed=0
tokenizer_runtime_attachment_attached=0
tokenizer_runtime_attachment_record_created=0
tokenizer_runtime_attachment_runtime_reference_loaded=0
tokenizer_runtime_attachment_tokenizer_reference_loaded=0
tokenizer_runtime_attachment_runtime_reference_recorded=0
tokenizer_runtime_attachment_tokenizer_reference_recorded=0
tokenizer_runtime_attachment_runtime_invoked=0
tokenizer_runtime_attachment_session_created=0
tokenizer_runtime_attachment_result_recorded=0
tokenizer_runtime_attachment_file_written=0
tokenizer_attached_to_runtime=0
tokenizer_runtime_attachment_performed=0
runtime_tokenizer_attachment_performed=0
runtime_session_created=0
runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
prompt_tokenized=0
prompt_evaluated=0
```

Commands

Repository command:

```
sh scripts/nadia-tokenizer-runtime-attachment-contract.sh \
  --tokenizer-artifact-binding
reports/nadia/tokenizer-artifact-binding/latest-tokenizer-artifact-binding-contract.txt \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-runtime-attachment.
XXXXXX&quot;)&quot;
```

Installed command:

```
latticra-nadia tokenizer-runtime-attachment
```

Non-Claims

Stage-24 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, tokenizer manifest parser, tokenizer artifact scanner, tokenizer artifact inventory reader, tokenizer artifact hasher, tokenizer artifact measurer, tokenizer artifact verifier, tokenizer artifact binder, tokenizer runtime attachment layer, prompt tokenizer, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

Run:

```
sh scripts/test-nadia-tokenizer-runtime-attachment-contract-stage-24.sh
```

Expected output:

```
nadia_tokenizer_runtime_attachment_contract_stage_24: ok
```

docs/NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18.md

Source title: Nadia Tokenizer Specification Contract Stage-18

Lines: 145 | Words: 358 | SHA-256: 367cd8ccb5649687901a83a7d4508470834284d52513fa7b9c9053e72296c8e5

Nadia Tokenizer Specification Contract Stage-18

Status: Stage-18 implementation contract Date: 2026-05-25 Scope: tokenizer-specification metadata before tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Purpose

Stage-18 gives Nadia a tokenizer-specification contract after the tokenization-boundary contract is present.

The contract consumes a Stage-17 tokenization-boundary contract, verifies inherited non-executing tokenizer, prompt, dialogue, runtime, model, tool, and protective-safety posture, and records the requirements a future tokenizer manifest must satisfy. It does not load tokenizer files or tokenize prompts.

Capability

Stage-18 adds:

```
nadia_stage_18_tokenizer_specification_contract_present=1
nadia_tokenizer_specification_contract_generator_present=1
tokenizer_specification_contract_command=scripts/nadia-tokenizer-specification-contract.sh
installed_tokenizer_specification_contract_command=latticra-nadia tokenizer-specification
tokenizer_specification_contract_status=contract_only
tokenizer_specification_stage=contract-only
tokenizer_specification_authority=0
tokenizer_specification_allowed=0
tokenizer_specification_performed=0
tokenizer_specification_metadata_present=1
tokenizer_family=model-compatible-tokenizer
tokenizer_format=operator-reviewed-offline-specification
tokenizer_specification_decision=blocked_contract_only
tokenizer_specification_evidence_present=1
tokenizer_source_policy=operator-reviewed-offline
tokenizer_path_recorded=0
tokenizer_manifest_loaded=0
tokenizer_file_measurement_performed=0
requires_tokenization_boundary_contract=1
requires_prompt_evaluation_handoff_contract=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_prompt_buffer_boundary=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenizer_manifest_contract=1
tokenizer_specification_promotion_allowed=0
```

Stage-18 does not generate dialogue, receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt text, open tokenizer files, read tokenizer files, load tokenizer vocabularies, load tokenizer manifests, tokenize prompts, evaluate prompts, load model weights, invoke a runtime, generate tokens, run inference, execute tools, mutate source, train, distill, download, or use the network.

Specification Requirements

```
requires_model_tokenizer_compatibility_review=1
requires_tokenizer_format_review=1
requires_unicode_policy_review=1
requires_normalization_policy_review=1
requires_special_token_policy_review=1
requires_bos_eos_policy_review=1
requires_chat_template_policy_review=1
requires_prompt_template_boundary=1
requires_context_window_policy_review=1
requires_stop_sequence_policy_review=1
requires_survivor_centered_language_review=1
requires_refusal_policy_review=1
```

Tokenizer Denial Boundary

```
tokenizer_file_open_authority=0
tokenizer_file_read_authority=0
tokenizer_vocab_load_authority=0
tokenizer_vocab_mapping_authority=0
tokenizer_runtime_attach_authority=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
tokenizer_vocab_mapped=0
tokenizer_attached_to_runtime=0
tokenizer_bytes_read=0
tokenizer_hash_computed=0
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenized=0
prompt_tokens_created=0
prompt_evaluation_authority=0
prompt_evaluated=0
```

Protective Boundary

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
```

Usage

With explicit evidence input:

```
sh scripts/nadia-tokenizer-specification-contract.sh \
  --tokenization-boundary /path/to/latest-tokenization-boundary-contract.txt \
  --request-class awareness-education \
  --tokenizer-family model-compatible-tokenizer \
  --tokenizer-format operator-reviewed-offline-specification \
  --output &quot;$(mktemp -d &quot;${TMPDIR:-/tmp}/latticra-nadia-tokenizer-specification.
XXXXXX&quot;)&quot;
```

After a guarded local install with Nadia enabled:

```
latticra-nadia tokenizer-specification
```

Non-Claims

Stage-18 Nadia is not yet a Q&A assistant, dialogue generator, prompt evaluator, tokenizer, tokenizer loader, tokenizer registry, tokenizer manifest loader, model loader, inference runtime, runtime process launcher, token generator, model selector, tool executor, shell runner, network client, source mutator, training system, distillation system, security product, sexual assistant, roleplay surface, adult-content generator, legal adviser, medical adviser, trauma counselor, crisis intervention service, or production AI assistant.

Validation

```
sh scripts/test-nadia-tokenizer-specification-contract-stage-18.sh
```

Expected result:

```
nadia_tokenizer_specification_contract_stage_18: ok
```

Part IX - Kernel, Boot, and OS Image Research

- docs/KERNEL_CONTEXT_SWITCH_SEED.md - Kernel Context Switch Seed
- docs/KERNEL_IMPLEMENTATION.md - Kernel Seed Implementation
- docs/KERNEL_LIFECYCLE_ROLLBACK_PLAN.md - Kernel Lifecycle Rollback Plan
- docs/KERNEL_LIFECYCLE_SEED.md - Kernel Lifecycle Seed
- docs/KERNEL_LIFECYCLE_SUBSYSTEM_SUMMARY.md - Kernel Lifecycle Subsystem Summary
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- docs/LATTICRA_OS_IMAGE_BUILD_RECIPE_CONTRACT.md - Latticra OS Image Build Recipe Contract
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- docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_TEMPLATE.md - SeaBIOS and GRUB Boot Preview Boot Artifact Manifest Template

- docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_VALIDATION.md - SeaBIOS and GRUB Boot Preview Boot Artifact Manifest Validation
- docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CAPTURE_TEMPLATE.md - SeaBIOS and GRUB Boot Preview Evidence Capture Template
- docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CONTRACT.md - SeaBIOS and GRUB Boot Preview Evidence Contract
- docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_VALIDATION.md - SeaBIOS and GRUB Boot Preview Evidence Validation
- docs/SEABIOS_GRUB_BOOT_PREVIEW_PREFLIGHT.md - SeaBIOS and GRUB Boot Preview Preflight
- docs/SEABIOS_GRUB_BOOT_PREVIEW_QEMU_ARGV_TEMPLATE.md - SeaBIOS and GRUB Boot Preview QEMU Argv Template
- docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md - SeaBIOS and GRUB Compatibility Contract
- docs/SYSTEM_BOOTSTRAP_IMPLEMENTATION.md - System Bootstrap Implementation
- docs/SYSTEM_BOOTSTRAP_REPORT_RUNNER.md - System Bootstrap Report Runner

docs/KERNEL_CONTEXT_SWITCH_SEED.md

Source title: Kernel Context Switch Seed

Lines: 73 | Words: 172 | SHA-256: b8a796daacadf83b13ef51e39b449684144bfd1bac6a678a4fb37e621dcffd83

Kernel Context Switch Seed

Status: controlled context switch metadata seed Scope: report-only bridge from run queue metadata to context switch intent.

Purpose

This slice connects the run queue seed to deterministic context switch metadata.

It does not save registers, restore registers, switch stacks, switch address spaces, dispatch work, mutate run queues, wake processes, preempt execution, or touch hardware. It records candidate switch declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_context_switch.h
src/kernel_context_switch.c
tests/kernel_context_switch.c
tools/kernel_context_switch_report.c
scripts/test-kernel-context-switch.sh
scripts/test-kernel-context-switch-report-runner.sh
.github/workflows/kernel-context-switch.yml
docs/KERNEL_CONTEXT_SWITCH_SEED.md
```

Current posture

The default request evaluates the run queue seed and emits:

```
switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
switch_count=4
no_effect=1
```

Authority remains denied:

```
context_switch_allowed=0
register_save_allowed=0
register_restore_allowed=0
stack_switch_allowed=0
address_space_switch_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
preemption_allowed=0
time_accounting_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-context-switch.sh
sh scripts/test-kernel-context-switch-report-runner.sh
```

Expected output:

```
kernel_context_switch: ok
kernel_context_switch_report_runner: ok
```

Non-claims

This slice does not schedule execution, mutate run queues, perform context switches, wake processes, account CPU time, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_IMPLEMENTATION.md

Source title: Kernel Seed Implementation

Lines: 103 | Words: 236 | SHA-256: 2b297b41af2b831e25c27c3c5b9b98b80a3ba12e2eef6c4a9a0b4fe2359eecbe

Kernel Seed Implementation

Status: first kernel implementation surface Scope: compiled no-effect kernel seed, deterministic kernel report, and operator-facing report runner.

Purpose

This slice creates the first Latticra kernel surface.

The kernel seed is not a bootable kernel. It is a compiled C interface that owns kernel identity, kernel status labels, subsystem status labels, and a deterministic kernel report while coordinating the already merged system bootstrap surface.

This is the next system-building step after SYSTEM_BOOTSTRAP_IMPLEMENTATION.md.

Files

```
include/latticra/kernel.h
src/kernel.c
tests/kernel.c
scripts/test-kernel.sh
tools/kernel_report.c
scripts/test-kernel-report-runner.sh
.github/workflows/kernel.yml
docs/KERNEL_IMPLEMENTATION.md
```

Public API

```
latticra_kernel_default_request
latticra_kernel_initialize
latticra_kernel_report
```

Kernel report fields

The deterministic report emits:

```
LATTICRA KERNEL REPORT
kernel_id
kernel_mode
kernel_status
boot_status
runtime_status
scheduler_status
memory_status
process_status
filesystem_status
network_status
device_status
security_status
bootstrap_status
bootstrap_runtime_entry_status
bootstrap_no_effect
no_effect
execution_allowed
mutation_allowed
file_io_allowed
network_allowed
server_allowed
recovery_allowed
hardware_allowed
```

```
boot_allowed
evidence_level
source_identity
```

Current kernel posture

The kernel seed reports:

```
kernel_mode=seed-kernel
kernel_status=initialized-report-only
boot_status=not-booted
runtime_status=not-entered
scheduler_status=not-started
memory_status=metadata-only
process_status=not-started
filesystem_status=disabled
network_status=disabled
device_status=disabled
security_status=not-production-boundary
no_effect=1
```

Validation

Run:

```
sh scripts/test-kernel.sh
sh scripts/test-kernel-report-runner.sh
```

The guards compile the kernel seed, verify no-effect invariants, render the deterministic kernel report, and verify stable report fields.

Boundary

This implementation does not add a bootloader, bootable image, hardware control, interrupts, paging, scheduler execution, process execution, system calls, filesystem I/O, network I/O, mutation, server interaction, recovery behavior, sandboxing, malware prevention, ransomware prevention, production security boundary, or operating-system replacement.

It is the first compiled kernel identity and report surface.

docs/KERNEL_LIFECYCLE_ROLLBACK_PLAN.md

Source title: Kernel Lifecycle Rollback Plan

Lines: 136 | Words: 353 | SHA-256: 33d53ed02fd0f073345765382664b76b831f7885cb73285738cf5c36b649ddf9

Kernel Lifecycle Rollback Plan

Status: planning and guardrail record Scope: rollback semantics for the bounded kernel lifecycle sequence.

Purpose

The kernel lifecycle can currently move a local in-memory state machine from created to memory-map-ready when an explicit gate allows the sequence.

This plan defines how Latticra should reason about failed, partial, or interrupted lifecycle progress before any rollback implementation exists.

The goal is to make failure handling auditable before adding more kernel surfaces.

Rollback meaning

Rollback means:

```
record the last safe lifecycle state
classify the incomplete transition
report the reason rollback would be required
preserve the no-external-effect boundary
```

Rollback does not yet mean:

```
restore persisted state
write recovery metadata
restart a runtime
undo hardware work
operate devices
spawn or stop processes
```

Current lifecycle ladder

```
created
initialized
registry-ready
scheduler-ready
memory-map-ready
```

The current approved forward path remains:

```
created -&gt; initialized
initialized -&gt; registry-ready
registry-ready -&gt; scheduler-ready
scheduler-ready -&gt; memory-map-ready
```

Rollback trigger classes

Planned rollback classification should cover:

```
null-request
invalid-target
target-before-current
step-limit-reached
no-forward-step
step-request-failed
step-failed
step-did-not-advance
external-effect-blocked
```

Planned report fields

A later implementation may emit fields such as:

```
rollback_plan_status=planned
rollback_required=1
rollback_performed=0
rollback_reason=step-limit-reached
last_safe_state=registry-ready
attempted_target=memory-map-ready
final_state=registry-ready
state_mutated_before_stop=1
external_effect_performed=0
persistence_allowed=0
recovery_authority_allowed=0
```

Safety rule

Rollback planning must remain report-only until a separate authority contract exists.

The first implementation should only classify and report rollback posture. It should not persist state, start execution, use host recovery actions, or claim operating-system readiness.

Guardrail checklist

Any future rollback implementation must preserve:

```
external_effect_performed=0
persistence_allowed=0
recovery_authority_allowed=0
runtime_entry_allowed=0
```

It must also keep rollback reports deterministic and testable through a focused guard.

Implementation sequence

Recommended order:

1. Add rollback classification structs.
2. Add a no-effect rollback evaluator.
3. Add tests for complete, limited, invalid, and denied lifecycle paths.
4. Add a report runner.
5. Add status/docs alignment only after the implementation guard passes.

Validation

This planning slice is validated by:

```
sh scripts/test-kernel-lifecycle-rollback-plan.sh
```

Expected output:

```
kernel_lifecycle_rollback_plan: ok
```

Non-claims

This plan does not implement rollback. It only records the rules that a future rollback classifier must follow.

docs/KERNEL_LIFECYCLE_SEED.md

Source title: Kernel Lifecycle Seed

Lines: 177 | Words: 461 | SHA-256: 8649cf0381073d2297fd4a6a587380af623402c951d260ce3eb2866d90eca382

Kernel Lifecycle Seed

Status: controlled in-memory lifecycle runner Scope: bounded lifecycle sequence over the kernel state machine.

Purpose

This slice builds on the kernel state machine.

The state machine can mutate one in-memory state object through gated steps. This lifecycle seed runs the approved sequence through that state machine and records the final state, step count, state-change count, completion flag, and external-effect flag.

The companion report runner emits the deterministic lifecycle report from a deliberately allowed lifecycle request so the current kernel sequence evidence can be inspected through a small operator-facing tool.

Files

```
include/latticra/kernel_lifecycle.h
src/kernel_lifecycle.c
tests/kernel_lifecycle.c
tools/kernel_lifecycle_report.c
scripts/test-kernel-lifecycle.sh
scripts/test-kernel-lifecycle-report-runner.sh
.github/workflows/kernel-sequence.yml
docs/KERNEL_LIFECYCLE_SEED.md
```

Lifecycle target

The default lifecycle target is:

```
scheduler-run-entry-ready
```

The approved sequence is:

```
created -&gt; initialized
initialized -&gt; registry-ready
registry-ready -&gt; scheduler-ready
scheduler-ready -&gt; memory-map-ready
memory-map-ready -&gt; process-table-ready
process-table-ready -&gt; syscall-table-ready
syscall-table-ready -&gt; ipc-table-ready
ipc-table-ready -&gt; vfs-namespace-ready
vfs-namespace-ready -&gt; device-registry-ready
```

```
device-registry-ready -&gt; driver-catalog-ready
driver-catalog-ready -&gt; interrupt-table-ready
interrupt-table-ready -&gt; timer-source-ready
timer-source-ready -&gt; scheduler-tick-ready
scheduler-tick-ready -&gt; run-queue-ready
run-queue-ready -&gt; context-switch-ready
context-switch-ready -&gt; time-accounting-ready
time-accounting-ready -&gt; preemption-ready
preemption-ready -&gt; scheduler-credit-ready
scheduler-credit-ready -&gt; scheduler-selection-ready
scheduler-selection-ready -&gt; scheduler-dispatch-ready
scheduler-dispatch-ready -&gt; scheduler-handoff-ready
scheduler-handoff-ready -&gt; scheduler-activation-ready
scheduler-activation-ready -&gt; scheduler-run-entry-ready
```

Controlled effect boundary

This slice allows internal state-machine mutation only.

The result may report:

```
state_change_count=23
lifecycle_complete=1
```

The result must still report:

```
external_effect_performed=0
```

This means no Latticra runtime filesystem effect, network effect, process effect, device effect, runtime-entry effect, or host mutation is introduced.

The report runner only prints the deterministic lifecycle report for validation and operator inspection. The guard captures that report as test output while preserving the project boundary that the kernel lifecycle itself performs no external effects.

Report runner

Run:

```
sh scripts/test-kernel-lifecycle-report-runner.sh
```

Expected output:

```
kernel_lifecycle_report_runner: ok
```

The guard verifies:

```
LATTICRA KERNEL LIFECYCLE REPORT
lifecycle_status=lifecycle-complete
policy_status=gate-allowed
final_state=scheduler-run-entry-ready
step_count=23
state_change_count=23
lifecycle_complete=1
external_effect_performed=0
machine_log_count=23
evidence_level=10
```

It also verifies the first and final transition log entries.

Validation

Run:

```
sh scripts/test-kernel-lifecycle.sh
sh scripts/test-kernel-lifecycle-report-runner.sh
```

Expected output:

```
kernel_lifecycle: ok
kernel_lifecycle_report_runner: ok
```

The guards verify:

```
default request is denied
allowed lifecycle reaches scheduler-run-entry-ready
intermediate target stops correctly
step limit is respected
```

```
report includes lifecycle completion and transition log
report runner emits deterministic lifecycle evidence
external_effect_performed=0 remains true
```

Non-claims

This slice does not add:

```
filesystem writes in the Latticra kernel lifecycle
network access
process execution
runtime entry
hardware behavior
device operation
scheduler tick
run queue mutation
context switching
time accounting
process wakeups
timer arming
preemption
quota updates
scheduler credit updates
scheduler selection
scheduler dispatch
scheduler handoff
scheduler activation
scheduler run-entry
time reads
scheduler execution
memory allocation
production security boundary
operating-system replacement
```

Next possible lane

A later slice may add lifecycle rollback, lifecycle-to-subsystem summary reporting, or state persistence planning before any external effects are introduced.

docs/KERNEL_LIFECYCLE_SUBSYSTEM_SUMMARY.md

Source title: Kernel Lifecycle Subsystem Summary

Lines: 206 | Words: 510 | SHA-256: ebd8a02cf85ff3116e8332001b9d6f851ca8d7709c5b952b60653ff89b71830b

Kernel Lifecycle Subsystem Summary

Status: controlled lifecycle-to-subsystem summary Scope: report-only integration between the kernel lifecycle runner and the kernel subsystem registry.

Purpose

This slice connects two existing kernel evidence surfaces:

```
kernel lifecycle runner
kernel subsystem registry
```

The lifecycle runner can move a local in-memory kernel state machine from created to scheduler-run-entry-ready through gated internal state changes.

The subsystem registry exposes boot, runtime, scheduler, memory, process, filesystem, network, device, and security subsystem posture.

The summary combines those surfaces into one deterministic report so the project can answer:

```
Which kernel subsystems are lifecycle-ready?
Which subsystems still deny authority?
Did the lifecycle stay externally inert?
Is the registry still no-effect?
```

Files

```
include/latticra/kernel_lifecycle_subsystem_summary.h
src/kernel_lifecycle_subsystem_summary.c
tests/kernel_lifecycle_subsystem_summary.c
tools/kernel_lifecycle_subsystem_summary_report.c
scripts/test-kernel-lifecycle-subsystem-summary.sh
scripts/test-kernel-lifecycle-subsystem-summary-report-runner.sh
.github/workflows/kernel-lifecycle-subsystem-summary.yml
docs/KERNEL_LIFECYCLE_SUBSYSTEM_SUMMARY.md
```

Default target

The default summary request allows the lifecycle runner to reach:

```
scheduler-run-entry-ready
```

That produces:

```
summary_status=summary-ready
final_state=scheduler-run-entry-ready
lifecycle_complete=1
lifecycle_step_count=23
lifecycle_state_change_count=23
external_effect_performed=0
registry_no_effect=1
no_external_effect_chain=1
```

Subsystem posture

Expected readiness examples:

```
boot -&gt; boot-sequence-seeded
scheduler -&gt; scheduler-run-entry-ready
memory -&gt; memory-map-ready
process -&gt; ipc-table-ready
filesystem -&gt; vfs-namespace-ready
network -&gt; network-syscall-metadata-ready
device -&gt; interrupt-table-ready
runtime -&gt; runtime-not-entered
security -&gt; security-not-production-boundary
```

Authority remains denied:

```
runtime_entry_allowed=0
scheduler_execution_allowed=0
scheduler_selection_allowed=0
scheduler_dispatch_allowed=0
scheduler_handoff_allowed=0
scheduler_activation_allowed=0
scheduler_run_entry_allowed=0
memory_allocation_allowed=0
process_spawn_allowed=0
syscall_dispatch_allowed=0
ipc_send_allowed=0
ipc_receive_allowed=0
ipc_queue_mutation_allowed=0
filesystem_lookup_allowed=0
filesystem_read_allowed=0
filesystem_write_allowed=0
namespace_mutation_allowed=0
device_open_allowed=0
device_read_allowed=0
device_write_allowed=0
driver_probe_allowed=0
driver_load_allowed=0
driver_bind_allowed=0
interrupt_allowed=0
interrupt_mask_allowed=0
interrupt_unmask_allowed=0
interrupt_dispatch_allowed=0
interrupt_ack_allowed=0
timer_tick_allowed=0
timer_arm_allowed=0
timer_disarm_allowed=0
scheduler_tick_allowed=0
run_queue_mutation_allowed=0
enqueue_allowed=0
dequeue_allowed=0
```

```
dispatch_allowed=0
context_switch_allowed=0
register_save_allowed=0
register_restore_allowed=0
stack_switch_allowed=0
address_space_switch_allowed=0
preemption_allowed=0
time_accounting_allowed=0
time_read_allowed=0
cpu_usage_write_allowed=0
quota_update_allowed=0
scheduler_credit_update_allowed=0
process_wake_allowed=0
dma_allowed=0
hardware_effect_allowed=0
```

Subsystem authority labels include:

```
boot-denied
runtime-entry-denied
scheduler-execution-denied
memory-allocation-denied
process-execution-denied
filesystem-denied
network-denied
device-denied
not-production-boundary
```

Controlled boundary

This slice may report internal lifecycle state mutation:

```
lifecycle_state_mutated=1
```

It must still report:

```
external_effect_performed=0
no_external_effect_chain=1
```

This is a summary/reporting slice only. It does not expand kernel authority or claim product readiness.

Validation

Run:

```
sh scripts/test-kernel-lifecycle-subsystem-summary.sh
sh scripts/test-kernel-lifecycle-subsystem-summary-report-runner.sh
```

Expected output:

```
kernel_lifecycle_subsystem_summary: ok
kernel_lifecycle_subsystem_summary_report_runner: ok
```

The guards verify:

```
default request targets scheduler-run-entry-ready
summary reaches scheduler-run-entry-ready
summary marks boot/scheduler/memory/process/filesystem as lifecycle-ready metadata
runtime remains not entered
runtime entry remains denied
scheduler execution remains denied
memory allocation remains denied
process spawn remains denied
syscall dispatch remains denied
IPC send, receive, and queue mutation remain denied
filesystem lookup, read, write, and namespace mutation remain denied
device open, read, write, driver probe, driver load, driver bind, interrupt mask, interrupt
unmask, interrupt dispatch, interrupt ack, timer tick, timer arm, timer disarm, scheduler tick,
scheduler selection, scheduler dispatch, scheduler handoff, scheduler activation, scheduler
run-entry, run queue mutation, enqueue, dequeue, dispatch, context switch, register save,
register restore, stack switch, address space switch, preemption, time accounting, time read,
CPU usage write, quota update, scheduler credit update, process wake, DMA, and hardware effect
remain denied
network and device authority remain denied
limited lifecycle summary reports incomplete readiness
external_effect_performed=0 remains true
```

Non-claims

This slice does not make Latticra bootable, runnable as an operating system, product-ready, or authority-expanded.

Next possible lane

A later slice may add lifecycle rollback planning, virtual device binding metadata, or scheduler-to-timer handoff metadata. Those should remain report-only unless a separate authority contract is introduced first.

docs/KERNEL_MEMORY_MAP_SEED.md

Source title: Kernel Memory Map Seed

Lines: 65 | Words: 184 | SHA-256: 10e8a9b83a19ecd41c5d530d5c3a05f743f9ce648f7fa1e6a798f1fb71ee28bd

Kernel Memory Map Seed

Status: kernel memory-map seed implementation Scope: fixed kernel memory-region metadata and deterministic report surface.

Purpose

This slice adds the first kernel memory-map seed surface on top of the kernel scheduler seed.

The memory-map seed does not allocate memory, map pages, change permissions, write to memory, or execute memory. It records deterministic region metadata so the kernel can describe its future memory lanes before behavior exists.

Files

```
include/latticra/kernel_memory_map.h
src/kernel_memory_map.c
tests/kernel_memory_map.c
scripts/test-kernel-memory-map.sh
tools/kernel_memory_map_report.c
scripts/test-kernel-memory-map-report-runner.sh
.github/workflows/kernel-map-metadata.yml
docs/KERNEL_MEMORY_MAP_SEED.md
```

Validation

Run:

```
sh scripts/test-kernel-memory-map.sh
sh scripts/test-kernel-memory-map-report-runner.sh
```

Expected output:

```
kernel_memory_map: ok
kernel_memory_map_report_runner: ok
```

The guards verify:

```
map_status=memory-map-seed-ready
policy_status=report-only
scheduler_status=scheduler-seed-ready
region_count=4
mapping_allowed=0
write_allowed=0
execute_allowed=0
region[0].label=kernel-text-metadata
region[1].label=kernel-data-metadata
region[2].label=kernel-stack-metadata
region[3].label=kernel-report-metadata
region[0].map_status=metadata-only
no_effect=1
```

Boundary

This is metadata and reporting only.

It does not allocate memory, map virtual pages, edit page tables, write memory, execute mapped regions, enter runtime execution, perform I/O, operate devices, enforce a production security boundary, boot hardware, or replace an operating system.

The memory-map seed exists so future work can add memory policy, region validation, and safe memory-state modeling one guarded slice at a time.

docs/KERNEL_PREEMPTION_SEED.md

Source title: Kernel Preemption Seed

Lines: 73 | Words: 174 | SHA-256: 531b4395aba6476170658828b4e18212e6d5bfa3b59d6cfd580f83d5f5e06bfc

Kernel Preemption Seed

Status: controlled preemption metadata seed Scope: report-only bridge from time accounting metadata to preemption decision intent.

Purpose

This slice connects the time accounting seed to deterministic preemption decision metadata.

It does not read time, arm timers, preempt execution, dispatch work, switch contexts, mutate run queues, update scheduler credits, wake processes, or touch hardware. It records candidate preemption declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_preemption.h
src/kernel_preemption.c
tests/kernel_preemption.c
tools/kernel_preemption_report.c
scripts/test-kernel-preemption.sh
scripts/test-kernel-preemption-report-runner.sh
.github/workflows/kernel-preemption.yml
docs/KERNEL_PREEMPTION_SEED.md
```

Current Posture

The default request evaluates the time accounting seed and emits:

```
preemption_status=preemption-seed-ready
time_accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
decision_count=4
no_effect=1
```

Authority remains denied:

```
preemption_allowed=0
time_read_allowed=0
time_accounting_allowed=0
context_switch_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
scheduler_credit_update_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-preemption.sh
sh scripts/test-kernel-preemption-report-runner.sh
```

Expected output:

```
kernel_preemption: ok
kernel_preemption_report_runner: ok
```

Non-claims

This slice does not schedule execution, mutate run queues, perform context switches, read clocks, account CPU time, preempt execution, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_RUN_QUEUE_SEED.md

Source title: Kernel Run Queue Seed

Lines: 72 | Words: 170 | SHA-256: 0a0b0577e53ebd3a569b5e9ae940c8d314cd66ae0b6f630d0418570eb95a5b1b

Kernel Run Queue Seed

Status: controlled run queue metadata seed Scope: report-only bridge from scheduler tick metadata to run queue intent.

Purpose

This slice connects the scheduler tick seed to deterministic run queue metadata.

It does not enqueue processes, dequeue processes, dispatch work, mutate run queues, switch contexts, account CPU time, wake processes, preempt execution, or touch hardware. It records queue declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_run_queue.h
src/kernel_run_queue.c
tests/kernel_run_queue.c
tools/kernel_run_queue_report.c
scripts/test-kernel-run-queue.sh
scripts/test-kernel-run-queue-report-runner.sh
.github/workflows/kernel-run-queue.yml
docs/KERNEL_RUN_QUEUE_SEED.md
```

Current posture

The default request evaluates the scheduler tick seed and emits:

```
queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
interrupt_table_status=interrupt-table-seed-ready
scheduler_status=scheduler-seed-ready
queue_count=4
no_effect=1
```

Authority remains denied:

```
run_queue_mutation_allowed=0
enqueue_allowed=0
dequeue_allowed=0
dispatch_allowed=0
context_switch_allowed=0
preemption_allowed=0
time_accounting_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
```

```
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-run-queue.sh
sh scripts/test-kernel-run-queue-report-runner.sh
```

Expected output:

```
kernel_run_queue: ok
kernel_run_queue_report_runner: ok
```

Non-claims

This slice does not schedule execution, mutate run queues, perform context switches, wake processes, account CPU time, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_SCHEDULER_ACTIVATION_SEED.md

Source title: Kernel Scheduler Activation Seed

Lines: 85 | Words: 228 | SHA-256: 27293ea80c11b0f64932080bd536914f74da00aeb3da66b03c8317cdeab2e267

Kernel Scheduler Activation Seed

Status: controlled scheduler activation metadata seed Scope: report-only bridge from scheduler handoff metadata to scheduler activation intent.

Purpose

This slice connects the scheduler handoff seed to deterministic scheduler activation metadata.

It does not activate execution, hand control to a process, dispatch a runnable process, dequeue or enqueue work, mutate run queues, switch contexts, enter runtime, update scheduler credits, update quotas, write CPU usage, preempt execution, wake processes, persist state, or touch hardware. It records activation declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_scheduler_activation.h
src/kernel_scheduler_activation.c
tests/kernel_scheduler_activation.c
tools/kernel_scheduler_activation_report.c
scripts/test-kernel-scheduler-activation.sh
scripts/test-kernel-scheduler-activation-report-runner.sh
.github/workflows/kernel-scheduler-activation.yml
docs/KERNEL_SCHEDULER_ACTIVATION_SEED.md
```

Current Posture

The default request evaluates the scheduler handoff seed and emits:

```
activation_status=scheduler-activation-seed-ready
scheduler_handoff_status=scheduler-handoff-seed-ready
scheduler_dispatch_status=scheduler-dispatch-seed-ready
scheduler_selection_status=scheduler-selection-seed-ready
scheduler_credit_status=scheduler-credit-seed-ready
preemption_status=preemption-seed-ready
time_accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
activation_count=4
no_effect=1
```

Authority remains denied:

```
scheduler_activation_allowed=0
scheduler_handoff_allowed=0
scheduler_dispatch_allowed=0
scheduler_selection_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
context_switch_allowed=0
preemption_allowed=0
scheduler_credit_update_allowed=0
quota_update_allowed=0
cpu_usage_write_allowed=0
time_accounting_allowed=0
time_read_allowed=0
process_wake_allowed=0
runtime_entry_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-scheduler-activation.sh
sh scripts/test-kernel-scheduler-activation-report-runner.sh
```

Expected output:

```
kernel_scheduler_activation: ok
kernel_scheduler_activation_report_runner: ok
```

Non-claims

This slice does not schedule execution, activate execution, enter runtime, hand off execution to a process, dispatch a runnable process, mutate run queues, enqueue or dequeue work, perform context switches, read clocks, account CPU time, update quotas, update scheduler credits, preempt execution, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_SCHEDULER_CREDIT_SEED.md

Source title: Kernel Scheduler Credit Seed

Lines: 76 | Words: 187 | SHA-256: af0b764eb4904ff92af2c09cef3d6c12347067000834893a5d8f71f2136ab5db

Kernel Scheduler Credit Seed

Status: controlled scheduler credit metadata seed Scope: report-only bridge from preemption decision metadata to scheduler credit intent.

Purpose

This slice connects the preemption seed to deterministic scheduler credit metadata.

It does not update scheduler credits, write CPU usage, update quotas, read time, preempt execution, dispatch work, switch contexts, mutate run queues, wake processes, persist state, or touch hardware. It records candidate credit declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_scheduler_credit.h
src/kernel_scheduler_credit.c
tests/kernel_scheduler_credit.c
tools/kernel_scheduler_credit_report.c
scripts/test-kernel-scheduler-credit.sh
scripts/test-kernel-scheduler-credit-report-runner.sh
.github/workflows/kernel-scheduler-credit.yml
docs/KERNEL_SCHEDULER_CREDIT_SEED.md
```

Current Posture

The default request evaluates the preemption seed and emits:

```
credit_status=scheduler-credit-seed-ready
preemption_status=preemption-seed-ready
time_accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
credit_count=4
no_effect=1
```

Authority remains denied:

```
scheduler_credit_update_allowed=0
quota_update_allowed=0
cpu_usage_write_allowed=0
time_accounting_allowed=0
time_read_allowed=0
preemption_allowed=0
context_switch_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-scheduler-credit.sh
sh scripts/test-kernel-scheduler-credit-report-runner.sh
```

Expected output:

```
kernel_scheduler_credit: ok
kernel_scheduler_credit_report_runner: ok
```

Non-claims

This slice does not schedule execution, mutate run queues, perform context switches, read clocks, account CPU time, update quotas, update scheduler credits, preempt execution, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_SCHEDULER_DISPATCH_SEED.md

Source title: Kernel Scheduler Dispatch Seed

Lines: 80 | Words: 204 | SHA-256: 18e1f8bf8445039a1c1b25db5792f87baa088db5cb6617f9fc27e8023b070e2f

Kernel Scheduler Dispatch Seed

Status: controlled scheduler dispatch metadata seed Scope: report-only bridge from scheduler selection metadata to scheduler dispatch intent.

Purpose

This slice connects the scheduler selection seed to deterministic scheduler dispatch metadata.

It does not dispatch a runnable process, dequeue or enqueue work, mutate run queues, switch contexts, update scheduler credits, update quotas, write CPU usage, preempt execution, wake processes, persist state, or touch hardware. It records dispatch declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_scheduler_dispatch.h
src/kernel_scheduler_dispatch.c
tests/kernel_scheduler_dispatch.c
tools/kernel_scheduler_dispatch_report.c
scripts/test-kernel-scheduler-dispatch.sh
scripts/test-kernel-scheduler-dispatch-report-runner.sh
.github/workflows/kernel-scheduler-dispatch.yml
docs/KERNEL_SCHEDULER_DISPATCH_SEED.md
```

Current Posture

The default request evaluates the scheduler selection seed and emits:

```
dispatch_status=scheduler-dispatch-seed-ready
scheduler_selection_status=scheduler-selection-seed-ready
scheduler_credit_status=scheduler-credit-seed-ready
preemption_status=preemption-seed-ready
time_accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
dispatch_count=4
no_effect=1
```

Authority remains denied:

```
scheduler_dispatch_allowed=0
scheduler_selection_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
context_switch_allowed=0
preemption_allowed=0
scheduler_credit_update_allowed=0
quota_update_allowed=0
cpu_usage_write_allowed=0
time_accounting_allowed=0
time_read_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-scheduler-dispatch.sh
sh scripts/test-kernel-scheduler-dispatch-report-runner.sh
```

Expected output:

```
kernel_scheduler_dispatch: ok
kernel_scheduler_dispatch_report_runner: ok
```

Non-claims

This slice does not schedule execution, dispatch a runnable process, mutate run queues, enqueue or dequeue work, perform context switches, read clocks, account CPU time, update quotas, update scheduler credits, preempt execution, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_SCHEDULER_HANDOFF_SEED.md

Source title: Kernel Scheduler Handoff Seed

Lines: 82 | Words: 217 | SHA-256: f2c4060db2676d9545a215cf45fd74195626a35fdd4a73a5a7b847deb7c3bf76

Kernel Scheduler Handoff Seed

Status: controlled scheduler handoff metadata seed Scope: report-only bridge from scheduler dispatch metadata to scheduler handoff intent.

Purpose

This slice connects the scheduler dispatch seed to deterministic scheduler handoff metadata.

It does not hand execution to a process, dispatch a runnable process, dequeue or enqueue work, mutate run queues, switch contexts, update scheduler credits, update quotas, write CPU usage, preempt execution, wake processes, persist state, or touch hardware. It records handoff declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_scheduler_handoff.h
src/kernel_scheduler_handoff.c
tests/kernel_scheduler_handoff.c
tools/kernel_scheduler_handoff_report.c
scripts/test-kernel-scheduler-handoff.sh
scripts/test-kernel-scheduler-handoff-report-runner.sh
.github/workflows/kernel-scheduler-handoff.yml
docs/KERNEL_SCHEDULER_HANDOFF_SEED.md
```

Current Posture

The default request evaluates the scheduler dispatch seed and emits:

```
handoff_status=scheduler-handoff-seed-ready
scheduler_dispatch_status=scheduler-dispatch-seed-ready
scheduler_selection_status=scheduler-selection-seed-ready
scheduler_credit_status=scheduler-credit-seed-ready
preemption_status=preemption-seed-ready
time_accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
handoff_count=4
no_effect=1
```

Authority remains denied:

```
scheduler_handoff_allowed=0
scheduler_dispatch_allowed=0
scheduler_selection_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
context_switch_allowed=0
preemption_allowed=0
scheduler_credit_update_allowed=0
quota_update_allowed=0
cpu_usage_write_allowed=0
time_accounting_allowed=0
time_read_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-scheduler-handoff.sh
sh scripts/test-kernel-scheduler-handoff-report-runner.sh
```

Expected output:

```
kernel_scheduler_handoff: ok
kernel_scheduler_handoff_report_runner: ok
```

Non-claims

This slice does not schedule execution, hand off execution to a process, dispatch a runnable process, mutate run queues, enqueue or dequeue work, perform context switches, read clocks, account CPU time, update quotas, update scheduler credits, preempt execution, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_SCHEDULER_RUN_ENTRY_SEED.md

Source title: Kernel Scheduler Run Entry Seed

Lines: 64 | Words: 152 | SHA-256: dfa1703aa6a13235f289b22081a21a6ab7711c01523079ded35b7371ab935eb3

Kernel Scheduler Run Entry Seed

Status: controlled scheduler run-entry metadata seed Scope: report-only bridge from scheduler activation metadata to runtime entry intent.

This slice connects the scheduler activation seed to deterministic scheduler run-entry metadata. It does not enter runtime execution, activate a process, switch CPU context, mutate a run queue, read time, update accounting, dispatch scheduler work, or touch hardware.

Files

```
include/latticra/kernel_scheduler_run_entry.h
src/kernel_scheduler_run_entry.c
tests/kernel_scheduler_run_entry.c
tools/kernel_scheduler_run_entry_report.c
scripts/test-kernel-scheduler-run-entry.sh
scripts/test-kernel-scheduler-run-entry-report-runner.sh
.github/workflows/kernel-scheduler-run-entry.yml
```

Evidence

The seed report emits a deterministic, no-effect boundary:

```
run_entry_status=scheduler-run-entry-seed-ready
policy_status=report-only
scheduler_activation_status=scheduler-activation-seed-ready
scheduler_handoff_status=scheduler-handoff-seed-ready
run_entry_count=4
scheduler_run_entry_allowed=0
scheduler_activation_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
context_switch_allowed=0
runtime_entry_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

The first declared run-entry record is the kernel report process metadata:

```
run_entry[0].process_label=kernel-report-process-metadata
run_entry[0].run_entry_class=kernel-report-scheduler-run-entry
run_entry[0].activation_token=0
run_entry[0].run_entry_rank=1
run_entry[0].run_entry_declared=1
run_entry[0].run_entry_planned=0
run_entry[0].run_entry_prepared=0
run_entry[0].execution_entered=0
run_entry[0].no_effect=1
```

The seed keeps the scheduler path honest: the lifecycle can now name the next metadata point after activation while preserving the denial of runtime entry.

Validation

```
sh scripts/test-kernel-scheduler-run-entry.sh
sh scripts/test-kernel-scheduler-run-entry-report-runner.sh
```

docs/KERNEL_SCHEDULER_SEED.md

Source title: Kernel Scheduler Seed

Lines: 61 | Words: 172 | SHA-256: 7295d79be7ca7fcbece059ddd49563f01d818a5d2ecef51738fdb19c89c289b2

Kernel Scheduler Seed

Status: kernel scheduler seed implementation Scope: compiled scheduler-slot metadata and deterministic report surface.

Purpose

This slice adds the first kernel scheduler seed surface on top of the kernel subsystem registry.

The scheduler seed does not schedule execution. It evaluates the kernel subsystem registry, records deterministic scheduler-slot metadata, and emits a report that proves the scheduler lane is present while remaining no-effect.

Files

```
include/latticra/kernel_scheduler.h
src/kernel_scheduler.c
tests/kernel_scheduler.c
scripts/test-kernel-scheduler.sh
tools/kernel_scheduler_report.c
scripts/test-kernel-scheduler-report-runner.sh
.github/workflows/kernel-scheduler.yml
docs/KERNEL_SCHEDULER_SEED.md
```

Validation

Run:

```
sh scripts/test-kernel-scheduler.sh
sh scripts/test-kernel-scheduler-report-runner.sh
```

Expected output:

```
kernel_scheduler: ok
kernel_scheduler_report_runner: ok
```

The guards verify:

```
scheduler_status=scheduler-seed-ready
policy_status=report-only
registry_status=registry-ready
slot_count=3
slot[0].label=idle-metadata
slot[1].label=kernel-report-metadata
slot[2].label=operator-report-metadata
slot[0].selection_status=not-selected
no_effect=1
```

Boundary

This is metadata and reporting only.

It does not start scheduler behavior, select runnable work, switch contexts, run processes, mutate memory, enter runtime execution, perform I/O, operate devices, enforce a production security boundary, boot hardware, or replace an operating system.

The scheduler seed exists so future work can add scheduler policy, queues, and state machines one guarded slice at a time.

docs/KERNEL_SCHEDULER_SELECTION_SEED.md

Source title: Kernel Scheduler Selection Seed

Lines: 78 | Words: 198 | SHA-256: 777b17b2815688c8ac060ba3b9fcebd1697c5c1878eac356c61cd37f6f0868e9

Kernel Scheduler Selection Seed

Status: controlled scheduler selection metadata seed Scope: report-only bridge from scheduler credit metadata to scheduler candidate selection intent.

Purpose

This slice connects the scheduler credit seed to deterministic scheduler selection metadata.

It does not select a runnable process, enqueue or dequeue work, dispatch, switch contexts, update scheduler credits, update quotas, write CPU usage, preempt execution, wake processes, persist state, or touch hardware. It records candidate selection declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_scheduler_selection.h
src/kernel_scheduler_selection.c
tests/kernel_scheduler_selection.c
tools/kernel_scheduler_selection_report.c
scripts/test-kernel-scheduler-selection.sh
scripts/test-kernel-scheduler-selection-report-runner.sh
.github/workflows/kernel-scheduler-selection.yml
docs/KERNEL_SCHEDULER_SELECTION_SEED.md
```

Current Posture

The default request evaluates the scheduler credit seed and emits:

```
selection_status=scheduler-selection-seed-ready
scheduler_credit_status=scheduler-credit-seed-ready
preemption_status=preemption-seed-ready
time_accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
selection_count=4
no_effect=1
```

Authority remains denied:

```
scheduler_selection_allowed=0
dispatch_allowed=0
run_queue_mutation_allowed=0
context_switch_allowed=0
preemption_allowed=0
scheduler_credit_update_allowed=0
quota_update_allowed=0
cpu_usage_write_allowed=0
time_accounting_allowed=0
time_read_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-scheduler-selection.sh
sh scripts/test-kernel-scheduler-selection-report-runner.sh
```

Expected output:

```
kernel_scheduler_selection: ok
kernel_scheduler_selection_report_runner: ok
```

Non-claims

This slice does not schedule execution, select a runnable process, mutate run queues, perform context switches, read clocks, account CPU time, update quotas, update scheduler credits, preempt execution, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_SCHEDULER_TICK_SEED.md

Source title: Kernel Scheduler Tick Seed

Lines: 71 | Words: 169 | SHA-256: dc5d9a60ebb26265010da5bfa7b622076f09c1cc394d4fca8befe5516d6a1df5

Kernel Scheduler Tick Seed

Status: controlled scheduler tick metadata seed Scope: report-only bridge from timer-source metadata to scheduler tick intent.

Purpose

This slice connects the timer source seed to scheduler-facing tick metadata.

It does not dispatch timer interrupts, mutate run queues, switch contexts, account CPU time, wake processes, preempt execution, read clocks, or touch hardware. It records deterministic tick declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_scheduler_tick.h
src/kernel_scheduler_tick.c
tests/kernel_scheduler_tick.c
tools/kernel_scheduler_tick_report.c
scripts/test-kernel-scheduler-tick.sh
scripts/test-kernel-scheduler-tick-report-runner.sh
.github/workflows/kernel-scheduler-tick.yml
docs/KERNEL_SCHEDULER_TICK_SEED.md
```

Current posture

The default request evaluates the timer source and emits:

```
tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
interrupt_table_status=interrupt-table-seed-ready
scheduler_status=scheduler-seed-ready
tick_count=4
no_effect=1
```

Authority remains denied:

```
timer_tick_allowed=0
scheduler_tick_allowed=0
run_queue_mutation_allowed=0
context_switch_allowed=0
preemption_allowed=0
time_accounting_allowed=0
time_read_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-scheduler-tick.sh
sh scripts/test-kernel-scheduler-tick-report-runner.sh
```

Expected output:

```
kernel_scheduler_tick: ok
```

```
kernel_scheduler_tick_report_runner: ok
```

Non-claims

This slice does not schedule execution, mutate run queues, perform context switches, wake processes, account CPU time, read hardware clocks, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/KERNEL_STATE_MACHINE.md

Source title: Kernel State Machine

Lines: 122 | Words: 298 | SHA-256: b84f89289e280a83937d74e6949912de2c9422eb7be7f7eec4901b8b9b84229f

Kernel State Machine

Status: controlled in-memory state machine implementation Scope: bounded kernel state object, sequential step function, transition log, and deterministic report surface.

Purpose

This slice builds on the kernel state mutation seed.

The previous slice proved one gated in-memory transition can report state_change_performed=1 while keeping external_effect_performed=0.

This slice adds a real mutable in-memory kernel state object. The machine starts in created, can be stepped forward through the approved kernel state ladder, records each step in a bounded transition log, and emits a deterministic state-machine report.

Files

```
include/latticra/kernel_state_machine.h
src/kernel_state_machine.c
tests/kernel_state_machine.c
scripts/test-kernel-state-machine.sh
tools/kernel_state_machine_report.c
scripts/test-kernel-state-machine-report-runner.sh
.github/workflows/kernel-state-machine.yml
docs/KERNEL_STATE_MACHINE.md
```

State ladder

The machine uses the same sequential state ladder as the kernel state mutation seed:

```
created
initialized
registry-ready
scheduler-ready
memory-map-ready
process-table-ready
syscall-table-ready
ipc-table-ready
vfs-namespace-ready
device-registry-ready
driver-catalog-ready
interrupt-table-ready
timer-source-ready
scheduler-tick-ready
run-queue-ready
context-switch-ready
time-accounting-ready
preemption-ready
scheduler-credit-ready
scheduler-selection-ready
scheduler-dispatch-ready
scheduler-handoff-ready
scheduler-activation-ready
scheduler-run-entry-ready
```

Controlled effect boundary

This slice allows internal machine mutation only:

```
state_mutated=1
```

The machine must still report:

```
external_effect_performed=0
```

This means the state machine mutates only its own in-memory struct and transition log.

Validation

Run:

```
sh scripts/test-kernel-state-machine.sh
sh scripts/test-kernel-state-machine-report-runner.sh
```

Expected output:

```
kernel_state_machine: ok
kernel_state_machine_report_runner: ok
```

The guards verify:

```
machine starts in created
default step is denied and logged
allowed step mutates machine state
sequential steps advance the state ladder
illegal step is denied without mutation
report includes transition log
external_effect_performed=0 remains true
```

Non-claims

This slice does not add:

```
filesystem writes
network access
process execution
runtime entry
hardware behavior
boot behavior
device operation
scheduler execution
memory allocation
production security boundary
operating-system replacement
```

Next possible lane

A later slice may connect the state machine to a kernel operator command or add rollback classification for in-memory state transitions before any external effects are introduced.

docs/KERNEL_STATE_MUTATION_SEED.md

Source title: Kernel State Mutation Seed

Lines: 159 | Words: 384 | SHA-256: a1fad9eb4698d2df28b5070cc9d66f9e65e76a7f830d1fba1d017dba79122fa2

Kernel State Mutation Seed

Status: first controlled-effect implementation slice Scope: gated in-memory kernel state transition.

Purpose

This slice adds the first controlled effect-bearing kernel surface.

The effect is intentionally narrow:

```
in-memory kernel state transition only
```

The state transition layer can move from one kernel state label to the next when an explicit gate allows it. It records the previous state, target state, next state, gate status, transition status, and whether a state change actually happened.

Files

```
include/latticra/kernel_state.h
src/kernel_state.c
tests/kernel_state.c
scripts/test-kernel-state.sh
tools/kernel_state_report.c
scripts/test-kernel-state-report-runner.sh
.github/workflows/kernel-state.yml
docs/KERNEL_STATE_MUTATION_SEED.md
```

State ladder

The current guarded ladder is:

```
created
initialized
registry-ready
scheduler-ready
memory-map-ready
process-table-ready
syscall-table-ready
ipc-table-ready
vfs-namespace-ready
device-registry-ready
driver-catalog-ready
interrupt-table-ready
timer-source-ready
scheduler-tick-ready
run-queue-ready
context-switch-ready
time-accounting-ready
preemption-ready
scheduler-credit-ready
scheduler-selection-ready
scheduler-dispatch-ready
scheduler-handoff-ready
scheduler-activation-ready
scheduler-run-entry-ready
```

Allowed transitions are intentionally sequential:

```
created -&gt; initialized
initialized -&gt; registry-ready
registry-ready -&gt; scheduler-ready
scheduler-ready -&gt; memory-map-ready
memory-map-ready -&gt; process-table-ready
process-table-ready -&gt; syscall-table-ready
syscall-table-ready -&gt; ipc-table-ready
ipc-table-ready -&gt; vfs-namespace-ready
vfs-namespace-ready -&gt; device-registry-ready
device-registry-ready -&gt; driver-catalog-ready
driver-catalog-ready -&gt; interrupt-table-ready
interrupt-table-ready -&gt; timer-source-ready
timer-source-ready -&gt; scheduler-tick-ready
scheduler-tick-ready -&gt; run-queue-ready
run-queue-ready -&gt; context-switch-ready
context-switch-ready -&gt; time-accounting-ready
time-accounting-ready -&gt; preemption-ready
preemption-ready -&gt; scheduler-credit-ready
scheduler-credit-ready -&gt; scheduler-selection-ready
scheduler-selection-ready -&gt; scheduler-dispatch-ready
scheduler-dispatch-ready -&gt; scheduler-handoff-ready
scheduler-handoff-ready -&gt; scheduler-activation-ready
scheduler-activation-ready -&gt; scheduler-run-entry-ready
```

No-op transitions are allowed when the gate allows them.

Non-sequential transitions are denied and do not change state.

Effect boundary

This slice changes the project posture from permanent no-effect reporting to controlled, explicit, test-covered internal effects.

The only effect this slice allows is:

```
state_change_performed=1
```

The result must still report:

```
external_effect_performed=0
```

This means no filesystem effect, network effect, process effect, device effect, boot effect, or host mutation is introduced.

Validation

Run:

```
sh scripts/test-kernel-state.sh
sh scripts/test-kernel-state-report-runner.sh
```

Expected output:

```
kernel_state: ok
kernel_state_report_runner: ok
```

The guards verify:

```
default gate denies state change
allowed sequential transition changes state
illegal transition is denied
no-op transition is stable
state_change_performed=1 only on actual allowed transition
external_effect_performed=0 always
```

Non-claims

This slice does not add:

```
filesystem writes
network access
process execution
runtime entry
hardware behavior
boot behavior
device operation
scheduler execution
memory allocation
production security boundary
operating-system replacement
```

Next possible lane

A later slice may add state persistence planning or rollback classification, but external effects should remain blocked until a separate effect contract and rollback policy exist.

docs/KERNEL_SUBSYSTEM_REGISTRY.md

Source title: Kernel Subsystem Registry

Lines: 59 | Words: 142 | SHA-256: 205f92dc6d9a7b2eaed2d1ec2de5010eaa32d732dc731505100a4ce2ab791d81

Kernel Subsystem Registry

Status: kernel subsystem registry implementation Scope: compiled metadata registry for first kernel subsystem posture.

Purpose

This slice adds a compiled kernel subsystem registry on top of the kernel seed.

The registry evaluates the current kernel seed and exposes deterministic entries for:

```
boot
runtime
scheduler
memory
process
filesystem
network
device
security
```

Implementation files

```
include/latticra/kernel_subsystem_registry.h
src/kernel_subsystem_registry.c
tests/kernel_subsystem_registry.c
scripts/test-kernel-subsystem-registry.sh
tools/kernel_subsystem_registry_report.c
scripts/test-kernel-subsystem-registry-report-runner.sh
.github/workflows/kernel-subsystem-registry.yml
docs/KERNEL_SUBSYSTEM_REGISTRY.md
```

Validation

Run:

```
sh scripts/test-kernel-subsystem-registry.sh
sh scripts/test-kernel-subsystem-registry-report-runner.sh
```

Expected output:

```
kernel_subsystem_registry: ok
kernel_subsystem_registry_report_runner: ok
```

Boundary

This registry is metadata-only and report-only.

It does not start the scheduler, allocate memory, create processes, mount filesystems, open network channels, operate devices, enforce a production security boundary, boot hardware, or replace an operating system.

The registry records subsystem posture so later implementation slices can grow one subsystem at a time under tests.

docs/KERNEL_TIME_ACCOUNTING_SEED.md

Source title: Kernel Time Accounting Seed

Lines: 73 | Words: 177 | SHA-256: 20163f7e7d332addbc4fa49822c1622421c99c2380a8b3a837d5026e97720f9e

Kernel Time Accounting Seed

Status: controlled time accounting metadata seed Scope: report-only bridge from context switch metadata to CPU charge intent.

Purpose

This slice connects the context switch seed to deterministic time accounting metadata.

It does not read time, charge CPU usage, write process usage counters, update quotas, update scheduler credits, mutate run queues, wake processes, preempt execution, or touch hardware. It records candidate accounting declarations that future scheduler work can refine behind explicit authority gates.

Files

```
include/latticra/kernel_time_accounting.h
src/kernel_time_accounting.c
tests/kernel_time_accounting.c
tools/kernel_time_accounting_report.c
scripts/test-kernel-time-accounting.sh
scripts/test-kernel-time-accounting-report-runner.sh
.github/workflows/kernel-time-accounting.yml
docs/KERNEL_TIME_ACCOUNTING_SEED.md
```

Current Posture

The default request evaluates the context switch seed and emits:

```
accounting_status=time-accounting-seed-ready
context_switch_status=context-switch-seed-ready
run_queue_status=run-queue-seed-ready
scheduler_tick_status=scheduler-tick-seed-ready
timer_source_status=timer-source-seed-ready
account_count=4
no_effect=1
```

Authority remains denied:

```
time_accounting_allowed=0
time_read_allowed=0
cpu_usage_write_allowed=0
quota_update_allowed=0
scheduler_credit_update_allowed=0
context_switch_allowed=0
run_queue_mutation_allowed=0
preemption_allowed=0
process_wake_allowed=0
hardware_effect_allowed=0
host_effect_allowed=0
```

Validation

Run:

```
sh scripts/test-kernel-time-accounting.sh
sh scripts/test-kernel-time-accounting-report-runner.sh
```

Expected output:

```
kernel_time_accounting: ok
kernel_time_accounting_report_runner: ok
```

Non-claims

This slice does not schedule execution, mutate run queues, perform context switches, read clocks, account CPU time, wake processes, dispatch interrupts, perform I/O, boot hardware, or replace an operating system.

docs/LATTICRA_OS_IMAGE_BUILD_RECIPE_CONTRACT.md

Source title: Latticra OS Image Build Recipe Contract

Lines: 200 | Words: 430 | SHA-256: 44c19c7b620a5c7060f4102fbcafd75f1bac595c523fead2272d5c3682872d7e

Latticra OS Image Build Recipe Contract

Status: contract record Evidence level: 10 target, build recipe and preflight only Scope: define the local build inputs and command recipe required before a future Latticra ISO or VM image can be produced.

Purpose

This contract makes the future image build path inspectable before any artifact is built.

It does not create an ISO, create a QCOW2 image, stage a root filesystem, install GRUB, write USB media, run QEMU, or claim operating-system readiness.

```
latticra_os_image_build_recipe_contract_present=1
os_image_build_preflight_present=1
os_image_build_recipe_template_present=1
os_image_build_execution_allowed=0
os_image_input_bundle_manifest_template_present=1
os_image_input_bundle_manifest_validation_present=1
os_image_input_bundle_manifest_candidate_present=0
os_image_input_bundle_ready_for_build_preflight=0
kernel_image_input_present=0
initramfs_input_present=0
rootfs_input_present=0
build_inputs_ready=0
build_toolchain_ready=0
os_image_build_recipe_ready=0
iso_created=0
vm_image_created=0
bootable_os_ready=0
production_os_claim=0
```

Required Inputs

A future build candidate must provide explicit inputs:

```
input_bundle_manifest_path=&lt;recorded&gt;
kernel_image_input_path=&lt;recorded&gt;
initramfs_input_path=&lt;recorded&gt;
rootfs_input_path=&lt;recorded&gt;
output_dir=&lt;recorded&gt;
```

The current default local paths are placeholders:

```
build/os-image/kernel
build/os-image/initramfs.img
build/os-image/rootfs.tar
artifacts/os-images/local-candidate/
```

Input Bundle Manifest Boundary

The input bundle manifest is the reviewable handoff into the image build lane:

```
LATTICRA OS IMAGE INPUT BUNDLE MANIFEST
manifest_version=1
bundle_kind=os-image-build-inputs
source_commit=&lt;recorded&gt;
build_environment=&lt;recorded&gt;
kernel_image_input_path=&lt;path&gt;
kernel_image_input_sha256=&lt;sha256&gt;
initramfs_input_path=&lt;path&gt;
initramfs_input_sha256=&lt;sha256&gt;
rootfs_input_path=&lt;path&gt;
rootfs_input_format=&lt;tar-or-tar.gz&gt;
rootfs_input_sha256=&lt;sha256&gt;
operator_recovery_path=&lt;recorded-or-none&gt;
bootable_os_ready=0
production_os_claim=0
```

The validator may prove path and checksum consistency, but it must preserve:

```
os_image_input_bundle_ready_for_build_preflight=&lt;0-or-1&gt;
os_image_build_execution_allowed=0
hardware_install_ready=0
full_os_install_ready=0
bootable_os_ready=0
production_os_claim=0
```

Required Tools

The preflight reports local visibility for:

```
qemu-img
xorriso
grub-mkrescue
tar
gzip
sha256sum or shasum
```

Missing tools must block the build recipe from being marked ready.

Commands

Run the build preflight:

```
sh scripts/latticra-os-image-build-preflight.sh
```

Generate the input bundle manifest template:

```
sh scripts/latticra-os-image-input-bundle-template.sh
```

Validate the current fixture or a future input bundle manifest:

```
sh scripts/latticra-os-image-input-bundle-validate.sh
sh scripts/latticra-os-image-input-bundle-validate.sh --input-manifest
build/os-image/input-bundle.txt
```

Run it with explicit candidate inputs:

```
sh scripts/latticra-os-image-build-preflight.sh --kernel build/os-image/kernel --initramfs
build/os-image/initramfs.img --rootfs build/os-image/rootfs.tar --output-dir
artifacts/os-images/local-candidate
```

The preflight prints future command shapes, including:

```
future_iso_build_command=grub-mkrescue ...
future_vm_image_create_command=qemu-img create ...
future_input_bundle_validation_command=sh scripts/latticra-os-image-input-bundle-validate.sh ...
future_artifact_manifest_validation_command=sh
scripts/latticra-os-image-artifact-manifest-validate.sh ...
```

It must preserve:

```
grub_mkrescue_invoked=0
qemu_img_invoked=0
iso_created=0
vm_image_created=0
artifact_manifest_written=0
usb_write_executed=0
qemu_run_performed=0
host_mutation_performed=0
```

Build Recipe Promotion Gates

Before a future implementation may run the recipe, reviewed evidence must show:

```
kernel_image_input_present=1
initramfs_input_present=1
rootfs_input_present=1
os_image_input_bundle_ready_for_build_preflight=1
qemu_img_available=1
xorriso_available=1
grub_mkrescue_available=1
source_tag_recorded=1
build_environment_recorded=1
recovery_path_recorded=1
operator_consent_recorded=1
```

Even when those are present, this contract alone must keep:

```
os_image_build_execution_allowed=0
hardware_install_ready=0
full_os_install_ready=0
bootable_os_ready=0
production_os_claim=0
```

Guard Validation

This contract is guarded by:

```
sh scripts/test-latticra-os-image-build-preflight.sh
sh scripts/test-latticra-os-image-input-bundle-template.sh
sh scripts/test-latticra-os-image-input-bundle-validate.sh
```

Expected output:

```
latticra_os_image_build_preflight: ok
latticra_os_image_input_bundle_template: ok
latticra_os_image_input_bundle_validate: ok
```

Non-claims

This contract is not a kernel, not an initramfs, not a root filesystem, not an ISO builder, not a VM image builder, not a boot proof, not a USB installer, not hardware install readiness, not full OS readiness, and not a production OS claim.

docs/LATTICRA_OS_IMAGE_RELEASE_READINESS_CONTRACT.md

Source title: Latticra OS Image Release Readiness Contract

Lines: 293 | Words: 671 | SHA-256: 0e540e50c9688bf1cf9d4a029c9c85130f750841fab0cdedc6258f8b8287816d

Latticra OS Image Release Readiness Contract

Status: contract record Evidence level: 10 target, ISO USB and VM image readiness contract only Scope: prepare the release lane for a future Latticra ISO, USB write command, and virtual machine image without claiming that a bootable operating system artifact exists yet.

Purpose

This contract defines what must be present before Latticra can offer users a hardware install path from a USB ISO and a virtual machine image for testing.

It is a release-preparation lane. It does not create an ISO, create a VM image, write a USB device, run QEMU, install a bootloader, install a kernel, mutate firmware, or claim that Latticra is ready as a full operating system.

```
latticra_os_image_release_readiness_contract_present=1
iso_artifact_present=0
os_image_artifact_manifest_template_present=1
os_image_artifact_manifest_validation_present=1
os_image_artifact_manifest_candidate_present=0
os_image_build_preflight_present=1
os_image_build_execution_allowed=0
usb_write_command_template_present=1
usb_write_execution_allowed=0
vm_image_artifact_present=0
vm_test_command_template_present=1
qemu_execution_allowed_by_guard=0
hardware_install_ready=0
full_os_install_ready=0
bootable_os_ready=0
production_os_claim=0
```

Current Manifest

The current fixture manifest is:

```
installer/manifests/latticra-os-image-release.toml
```

The manifest records the future artifact names, target profiles, and blocked authority flags for:

```
x86_64-usb-iso
x86_64-qemu-qcow2
x86_64-qemu-iso
```

Required Release Artifact Set

A future reviewed release candidate should use a deterministic artifact directory such as:

```
artifacts/os-images/&lt;version&gt;/
```

Minimum hardware-install artifacts:

```
latticra-x86_64.iso
latticra-x86_64.iso.sha256
latticra-x86_64.iso.sig
manifest.txt
sbom.spdx.json
```

Minimum VM-test artifacts:

```
latticra-x86_64.qcow2
latticra-x86_64.qcow2.sha256
latticra-x86_64.qcow2.sig
manifest.txt
sbom.spdx.json
```

The ISO and VM image may not be described as ready until checksums, signatures or explicit unsigned-local status, SBOM metadata, boot evidence, recovery steps, and install evidence are reviewed.

Artifact Manifest Boundary

The future artifact manifest must be line-oriented and deterministic:

```
LATTICRA OS IMAGE ARTIFACT MANIFEST
manifest_version=1
artifact_set=os-image-release
artifact_version=&lt;recorded&gt;
source_commit=&lt;recorded&gt;
source_tag=&lt;recorded-or-none&gt;
build_environment=&lt;recorded&gt;
iso_artifact_path=&lt;path-or-none&gt;
iso_artifact_sha256=&lt;sha256-or-none&gt;
iso_signature_path=&lt;path-or-none&gt;
iso_sbom_path=&lt;path-or-none&gt;
vm_image_path=&lt;path-or-none&gt;
vm_image_format=&lt;qcow2-or-raw-or-none&gt;
vm_image_sha256=&lt;sha256-or-none&gt;
vm_signature_path=&lt;path-or-none&gt;
vm_sbom_path=&lt;path-or-none&gt;
usb_write_command_template_present=1
vm_test_command_template_present=1
operator_recovery_path=&lt;recorded-or-none&gt;
bootable_os_ready=0
production_os_claim=0
```

The validator may verify artifact path and checksum consistency, but it must preserve:

```
artifact_manifest_ready_for_operator_review=&lt;0-or-1&gt;
hardware_install_ready=0
full_os_install_ready=0
bootable_os_ready=0
production_os_claim=0
```

Commands

Preflight the fixture:

```
sh scripts/latticra-os-image-release-preflight.sh
```

Preflight the future local image build inputs:

```
sh scripts/latticra-os-image-build-preflight.sh
```

Generate the artifact manifest template:

```
sh scripts/latticra-os-image-artifact-manifest-template.sh
```

Validate the current fixture or a future artifact manifest candidate:

```
sh scripts/latticra-os-image-artifact-manifest-validate.sh
sh scripts/latticra-os-image-artifact-manifest-validate.sh --artifact-manifest
artifacts/os-images/&lt;version&gt;/manifest.txt
```

Generate a Linux USB write command for operator review:

```
sh scripts/latticra-os-image-usb-write-command.sh --iso
artifacts/os-images/&lt;version&gt;/latticra-x86_64.iso --device /dev/sdX --platform linux
```

Generate a macOS USB write command for operator review:

```
sh scripts/latticra-os-image-usb-write-command.sh --iso
artifacts/os-images/&lt;version&gt;/latticra-x86_64.iso --device /dev/rdiskN --platform macos
```

Generate a QEMU VM test command for operator review:

```
sh scripts/latticra-os-image-vm-test-command.sh --image
artifacts/os-images/&lt;version&gt;/latticra-x86_64.qcow2 --format qcow2 --firmware seabios
```

These scripts print commands and evidence fields. They do not run the USB write command or QEMU command.

USB Write Command Boundary

The command generator may print a command shaped like:

```
usb_write_command=sudo dd if=&#x27;&lt;iso-path&gt;&#x27; of=&#x27;&lt;target-device&gt;&#x27;
bs=4M conv=fsync status=progress
post_write_command=sync
```

Before an operator runs a printed write command, the release evidence must include:

```
iso_artifact_present=1
iso_artifact_sha256_recorded=1
iso_checksum_verified_by_operator=1
target_device_verified_removable=1
target_device_contains_no_required_data=1
operator_recovery_path_recorded=1
operator_consent_recorded=1
```

The generator must preserve:

```
usb_write_execution_allowed=0
usb_write_executed=0
partition_mutation_performed=0
host_mutation_performed=0
```

VM Image Test Boundary

The VM command generator may print a command shaped like:

```
qemu_test_command=qemu-system-x86_64 ...
```

Before a VM image can become accepted test evidence, the release evidence must include:

```
vm_image_artifact_present=1
vm_image_sha256_recorded=1
vm_image_checksum_verified_by_operator=1
qemu_argv_recorded=1
serial_console_boot_log_recorded=1
read_only_vm_evidence_recorded=1
operator_recovery_path_recorded=1
```

The generator must preserve:

```
qemu_execution_allowed_by_guard=0
qemu_run_performed=0
qemu_boot_execution_recorded=0
host_mutation_performed=0
```

Required Promotion Evidence

Before any future claim changes from 0 to 1, the project needs reviewed evidence for:

```
source_tag_recorded=1
build_environment_recorded=1
iso_build_command_recorded=1
vm_image_build_command_recorded=1
artifact_reproducibility_reviewed=1
artifact_sha256_recorded=1
artifact_signature_recorded_or_unsigned_local_status_declared=1
sbom_recorded=1
license_metadata_recorded=1
bootloader_configuration_recorded=1
kernel_image_recorded=1
initramfs_recorded=1
hardware_usb_write_transcript_recorded=1
hardware_install_transcript_recorded=1
vm_boot_transcript_recorded=1
serial_console_boot_log_recorded=1
rollback_or_recovery_path_recorded=1
post_install_verification_recorded=1
uninstall_or_reimage_recovery_recorded=1
multi_machine_validation_recorded=1
```

Guard Validation

This lane is guarded by:

```
sh scripts/test-latticra-os-image-release-readiness-contract.sh
sh scripts/test-latticra-os-image-artifact-manifest-template.sh
sh scripts/test-latticra-os-image-artifact-manifest-validate.sh
sh scripts/test-latticra-os-image-usb-write-command.sh
sh scripts/test-latticra-os-image-vm-test-command.sh
```

Expected outputs:

```
latticra_os_image_release_readiness_contract: ok
latticra_os_image_artifact_manifest_template: ok
latticra_os_image_artifact_manifest_validate: ok
latticra_os_image_usb_write_command: ok
latticra_os_image_vm_test_command: ok
```

Current Classification

```
os_image_release_readiness_contract_present=1
os_image_release_manifest_fixture_present=1
iso_artifact_present=0
os_image_artifact_manifest_template_present=1
os_image_artifact_manifest_validation_present=1
os_image_artifact_manifest_candidate_present=0
os_image_build_preflight_present=1
os_image_build_execution_allowed=0
vm_image_artifact_present=0
usb_write_command_template_present=1
vm_test_command_template_present=1
hardware_install_ready=0
full_os_install_ready=0
bootable_os_ready=0
production_os_claim=0
```

Non-claims

This contract is not a bootable ISO, not a VM image, not a USB writer, not a hardware installer, not a production installer, not operating-system completeness, not a daily-driver readiness claim, not security hardening, not sandboxing, not malware prevention, not ransomware prevention, and not a production OS claim.

docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_TEMPLATE.md

Source title: SeaBIOS and GRUB Boot Preview Boot Artifact Manifest Template

Lines: 120 | Words: 296 | SHA-256: 77a49831e3d1e68b46321a8ad1c8b2b8d45fdd4c02eb2f98a334f9a39fa51c19

SeaBIOS and GRUB Boot Preview Boot Artifact Manifest Template

Status: no-effect boot-preview boot artifact manifest template Evidence level: 8 target, manifest template only Scope: future boot artifact metadata for SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI boot-preview profiles.

Purpose

This template defines the boot artifact manifest shape required before any future disk image, ISO, kernel, initramfs, or GRUB configuration can become boot evidence.

It is not a boot artifact. It does not create a disk image, create an ISO, install GRUB, generate GRUB config, install a kernel, write an initramfs, run QEMU, or mutate host boot state.

```
seabios_grub_boot_preview_boot_artifact_manifest_template_present=1
boot_artifact_manifest_template_mode=no-effect-template
boot_artifact_manifest_template_decision=blocked-template-only-no-artifact
boot_artifact_manifest_ready=0
boot_artifact_manifest_present=0
disk_image_created=0
```

```
disk_image_written=0
bootable_os_ready=0
production_os_claim=0
```

Command

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh
```

Optional manifest override:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh --manifest
installer/manifests/seabios-grub-boot-preview.toml
```

Boot artifact manifest validation

The current fixture is checked by:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

The validation gate rejects premature bootable, GRUB, QEMU, or production OS claims while keeping the artifact manifest incomplete.

Required Future Manifest Fields

A future completed boot artifact manifest must record:

```
artifact_id
artifact_format
artifact_path
artifact_sha256
target_firmware
target_bootloader
partition_scheme
bios_boot_partition_declared
efi_system_partition_declared
grub_config_path
grub_config_sha256
kernel_image_path
kernel_image_sha256
initramfs_path
initramfs_sha256
serial_console_enabled
operator_recovery_path
rollback_or_recovery_path
sbom_path
sbom_sha256
signature_path
signature_sha256
```

Required Profile Templates

The template keeps separate blocked manifests for:

```
x86_64-seabios-grub-preview
x86_64-grub2-bios-preview
x86_64-grub2-uefi-preview
```

Each profile remains incomplete until artifact, checksum, partition, bootloader, kernel, initramfs, serial-console, and recovery metadata exists.

Required Non-Effects

The template must always preserve:

```
boot_artifact_manifest_written=0
disk_image_created=0
disk_image_written=0
iso_created=0
grub_install_invoked=0
grub_mkconfig_invoked=0
grub_mkrescue_invoked=0
kernel_install_performed=0
initramfs_write_performed=0
qemu_boot_execution_attempted=0
host_mutation_performed=0
```

Guard

```
sh scripts/test-seabios-grub-boot-preview-boot-artifact-manifest-template.sh
```

Expected output:

```
seabios_grub_boot_preview_boot_artifact_manifest_template: ok
```

Non-claims

This template does not prove artifact integrity, create boot media, create a bootable Latticra OS image, prove SeaBIOS compatibility, prove GRUB BIOS compatibility, prove GRUB UEFI compatibility, or make Latticra production installer ready.

docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_VALIDATION.md

Source title: SeaBIOS and GRUB Boot Preview Boot Artifact Manifest Validation

Lines: 109 | Words: 337 | SHA-256: ac503415550e7d16780e12a131e4e699e34b8280a9c60febb93ee9fdaaa59deb

SeaBIOS and GRUB Boot Preview Boot Artifact Manifest Validation

Status: no-effect boot-preview boot artifact manifest validation Evidence level: 8 target, validation gate only Scope: current fixture validation for future SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI boot artifact manifests.

Purpose

This validation gate makes the boot artifact manifest lane checkable before any boot media exists.

It validates that the current fixture remains incomplete, non-authoritative, and unable to promote a boot claim. It does not accept a production artifact, create a disk image, create an ISO, invoke GRUB, install a kernel, write an initramfs, run QEMU, or mutate host boot state.

```
seabios_grub_boot_preview_boot_artifact_manifest_validation_present=1
boot_artifact_manifest_validation_mode=no-effect-readiness-check
boot_artifact_manifest_validation_decision=blocked-fixture-only-incomplete
boot_artifact_manifest_candidate_ready=0
boot_artifact_manifest_present=0
boot_artifact_checksum_recorded=0
boot_artifact_signature_recorded=0
qemu_execution_allowed_by_guard=0
bootable_os_ready=0
production_os_claim=0
```

Command

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

Optional manifest override:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh --manifest
installer/manifests/seabios-grub-boot-preview.toml
```

Required Fixture Fields

The validator requires the current preview fixture to keep these blocked fields:

```
status = &quot;fixture-only&quot;
boot_artifact_manifest_present = false
boot_artifact_checksum_recorded = false
boot_artifact_signature_recorded = false
bootable_os_ready = false
production_os_claim = false
qemu_execution_allowed_by_guard = false
seabios_grub_boot_claim_allowed = false
```

```
grub_bootloader_write_allowed = false
```

The artifact section must remain a placeholder:

```
format = &quot;none&quot;
disk_image_path = &quot;none&quot;
disk_image_sha256 = &quot;none&quot;
signature_path = &quot;none&quot;
sbom_path = &quot;none&quot;
kernel_image_path = &quot;none&quot;
initramfs_path = &quot;none&quot;
grub_config_path = &quot;none&quot;
operator_recovery_path = &quot;none&quot;
```

Required Profiles

The validator requires all preview profiles to remain present:

```
x86_64-seabios-grub-preview
x86_64-grub2-bios-preview
x86_64-grub2-uefi-preview
```

Each profile remains blocked until artifact metadata, checksums, QEMU argv records, serial-console logs, and recovery paths are recorded by a later evidence lane.

Required Rejections

The validator must reject premature claims for:

```
boot_artifact_manifest_present = true
boot_artifact_checksum_recorded = true
boot_artifact_signature_recorded = true
bootable_os_ready = true
production_os_claim = true
qemu_execution_allowed_by_guard = true
seabios_grub_boot_claim_allowed = true
grub_bootloader_write_allowed = true
```

Guard

```
sh scripts/test-seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

Expected output:

```
seabios_grub_boot_preview_boot_artifact_manifest_validate: ok
```

Non-claims

This validator does not prove artifact integrity, create boot media, create a bootable Latticra OS image, prove SeaBIOS compatibility, prove GRUB BIOS compatibility, prove GRUB UEFI compatibility, authorize QEMU execution, or make Latticra production installer ready.

docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CAPTURE_TEMPLATE.md

Source title: SeaBIOS and GRUB Boot Preview Evidence Capture Template

Lines: 134 | Words: 375 | SHA-256: 317822ab6762c05bbd95c4e5c4f70b32da08f4b546ac3a5906734cefe3a38d98

SeaBIOS and GRUB Boot Preview Evidence Capture Template

Status: no-effect boot-preview evidence capture template Evidence level: 8 target, template only Scope: future QEMU evidence bundle shape for SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI boot-preview profiles.

Purpose

This template defines what a future boot evidence bundle must record before Latticra can make any boot-preview claim.

It is a capture template only. It does not execute QEMU, invoke GRUB, create an ISO, create a disk image, write bootloader state, record a serial log, or promote Latticra to a bootable operating system.

```
seabios_grub_boot_preview_evidence_capture_template_present=1
capture_template_mode=no-effect-template
capture_template_decision=blocked-template-only-no-boot-execution
capture_template_complete=0
boot_evidence_record_ready=0
qemu_execution_allowed_by_guard=0
qemu_boot_execution_attempted=0
bootable_os_ready=0
production_os_claim=0
```

Command

```
sh scripts/seabios-grub-boot-preview-evidence-template.sh
```

Optional manifest override:

```
sh scripts/seabios-grub-boot-preview-evidence-template.sh --manifest
installer/manifests/seabios-grub-boot-preview.toml
```

The command emits a deterministic evidence-record template to stdout.

The QEMU argv template companion is:

```
sh scripts/seabios-grub-boot-preview-qemu-argv-template.sh
```

It prints future profile-specific QEMU argv record placeholders without running QEMU.

The Boot artifact manifest template companion is:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh
```

It prints future boot artifact metadata placeholders without creating disk images, installing GRUB, or writing boot files.

The Boot-preview evidence validation companion is:

```
sh scripts/seabios-grub-boot-preview-evidence-validate.sh
```

It checks the current fixture for premature QEMU, serial-console, recovery, or bootability claims before any future evidence bundle is accepted.

The Boot artifact manifest validation companion is:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

It checks the current fixture for premature bootable, GRUB, QEMU, or production OS claims before any future artifact manifest is accepted.

Required Future Evidence Fields

A future completed evidence bundle must record:

```
operator_review_id
preflight_report_path
preflight_report_sha256
boot_preview_manifest_path
boot_preview_manifest_sha256
boot_artifact_manifest_path
boot_artifact_manifest_sha256
boot_artifact_manifest_template_path
boot_artifact_manifest_validation_report_path
boot_artifact_manifest_validation_report_sha256
boot_evidence_validation_report_path
boot_evidence_validation_report_sha256
disk_image_path
disk_image_sha256
artifact_format
target_firmware
target_bootloader
qemu_machine
qemu_binary
qemu_argv_record_path
qemu_argv_template_path
serial_console_boot_log_path
serial_console_boot_log_sha256
boot_result
```

```
operator_console_boot_path
operator_recovery_path
rollback_or_recovery_path
host_mutation_performed
firmware_mutation_performed
bootloader_write_performed
```

Required Profiles

The template keeps separate placeholders for:

```
x86_64-seabios-grub-preview
x86_64-grub2-bios-preview
x86_64-grub2-uefi-preview
```

Each remains blocked until artifact, QEMU argv, serial-log, and recovery evidence exist.

Guard

```
sh scripts/test-seabios-grub-boot-preview-evidence-template.sh
```

Expected output:

```
seabios_grub_boot_preview_evidence_template: ok
```

Non-claims

This template does not create a boot artifact manifest, boot a VM, run QEMU, invoke GRUB, create a disk image, create an ISO, install a kernel, install an initramfs, mutate firmware, write bootloader state, change host boot state, prove SeaBIOS compatibility, prove GRUB compatibility, or make Latticra installer-ready as a production OS.

docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_CONTRACT.md

Source title: SeaBIOS and GRUB Boot Preview Evidence Contract

Lines: 255 | Words: 554 | SHA-256: 5411bc578f0607a39a90f8fb8e1c09ac41a720d95b7da5f9e7fa710f6c6dee13

SeaBIOS and GRUB Boot Preview Evidence Contract

Status: boot-preview-evidence contract Evidence level: 8 target, manifest fixture only Scope: future QEMU evidence for SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI boot-preview profiles.

Purpose

Latticra needs a disciplined path from installer-safe compatibility to real boot evidence.

This contract defines the next preview lane without claiming that a bootable Latticra operating-system image exists today.

```
seabios_grub_boot_preview_evidence_contract_present=1
boot_preview_manifest_fixture_present=1
boot_preview_manifest_validated=1
boot_artifact_manifest_present=0
qemu_boot_execution_recorded=0
serial_console_boot_log_recorded=0
operator_console_boot_path_recorded=0
bootable_os_ready=0
production_os_claim=0
```

Manifest fixture

The fixture manifest is:

```
installer/manifests/seabios-grub-boot-preview.toml
```

It records the fields that future boot evidence must replace with real artifact paths, digests, and logs.

The fixture must preserve:

```
status=&quot;fixture-only&quot;
boot_artifact_manifest_present=false
bootable_os_ready=false
production_os_claim=false
host_mutation_authority=false
firmware_mutation_allowed=false
bootloader_write_allowed=false
partition_mutation_allowed=false
mbr_write_allowed=false
gpt_write_allowed=false
efi_variable_write_allowed=false
esp_write_allowed=false
qemu_execution_allowed_by_guard=false
```

Required preview profiles

Future evidence must keep separate records for:

```
x86_64-seabios-grub-preview
x86_64-grub2-bios-preview
x86_64-grub2-uefi-preview
```

Each profile must record:

```
target_firmware
target_bootloader
qemu_machine
qemu_binary
disk_image_path
disk_image_sha256
boot_log_path
serial_console_enabled
host_bootloader_mutation
firmware_mutation
operator_recovery_path
```

Promotion requirements

No boot-preview status may be promoted until these fields are true:

```
boot_artifact_manifest_present=1
boot_artifact_checksum_recorded=1
qemu_i440fx_seabios_boot_validation_completed=1
qemu_grub2_bios_boot_validation_completed=1
qemu_ovmf_grub2_uefi_boot_validation_completed=1
serial_console_boot_log_recorded=1
operator_console_boot_path_recorded=1
read_only_vm_evidence_recorded=1
operator_recovery_runbook_present=1
no_host_bootloader_mutation=1
no_firmware_mutation=1
```

Until then:

```
seabios_grub_boot_claim_allowed=0
grub_bootloader_write_allowed=0
qemu_boot_execution_allowed_by_guard=0
production_os_claim=0
```

Boot-preview preflight

The no-effect boot-preview preflight is:

```
sh scripts/seabios-grub-boot-preview-preflight.sh
```

It validates this fixture lane and reports local tool visibility without running QEMU, invoking GRUB, creating disk images, writing bootloader state, or recording boot evidence.

```
seabios_grub_boot_preview_preflight_present=1
preflight_mode=no-effect-report
preflight_decision=blocked-fixture-only-no-boot-execution
manifest_fixture_valid=1
qemu_boot_execution_attempted=0
grub_install_invoked=0
disk_image_created=0
host_mutation_performed=0
```

Boot-preview evidence capture template

The no-effect evidence capture template is:

```
sh scripts/seabios-grub-boot-preview-evidence-template.sh
```

It prints the required future evidence bundle fields without running QEMU, invoking GRUB, creating disk images, or promoting any boot claim.

```
seabios_grub_boot_preview_evidence_capture_template_present=1
capture_template_mode=no-effect-template
capture_template_decision=blocked-template-only-no-boot-execution
capture_template_complete=0
boot_evidence_record_ready=0
qemu_boot_execution_attempted=0
bootable_os_ready=0
```

Boot-preview evidence validation

The no-effect boot-preview evidence validation gate is:

```
sh scripts/seabios-grub-boot-preview-evidence-validate.sh
```

It classifies the current fixture and rejects premature QEMU, serial-console, recovery, and bootability claims before future boot evidence can be accepted.

```
seabios_grub_boot_preview_evidence_validation_present=1
boot_evidence_validation_mode=no-effect-readiness-check
boot_evidence_validation_decision=blocked-fixture-only-no-boot-evidence
boot_evidence_candidate_ready=0
qemu_i440fx_seabios_boot_validation_completed=0
qemu_grub2_bios_boot_validation_completed=0
qemu_ovmf_grub2_uefi_boot_validation_completed=0
serial_console_boot_log_recorded=0
operator_recovery_runbook_present=0
bootable_os_ready=0
production_os_claim=0
```

Boot artifact manifest template

The no-effect boot artifact manifest template is:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh
```

It records the future artifact metadata shape without creating images, invoking GRUB, installing kernels, writing initramfs files, or creating a boot artifact manifest.

```
seabios_grub_boot_preview_boot_artifact_manifest_template_present=1
boot_artifact_manifest_template_mode=no-effect-template
boot_artifact_manifest_template_decision=blocked-template-only-no-artifact
boot_artifact_manifest_ready=0
boot_artifact_manifest_present=0
disk_image_created=0
bootable_os_ready=0
```

Boot artifact manifest validation

The no-effect boot artifact manifest validation gate is:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

It classifies the current fixture and rejects premature bootable, GRUB, QEMU, or production OS claims before a future artifact manifest can be accepted.

```
seabios_grub_boot_preview_boot_artifact_manifest_validation_present=1
boot_artifact_manifest_validation_mode=no-effect-readiness-check
boot_artifact_manifest_validation_decision=blocked-fixture-only-incomplete
boot_artifact_manifest_candidate_ready=0
boot_artifact_manifest_present=0
bootable_os_ready=0
production_os_claim=0
```

QEMU argv template

The no-effect QEMU argv template is:

```
sh scripts/seabios-grub-boot-preview-qemu-argv-template.sh
```

It records the future profile-specific argv shape without running QEMU or creating boot evidence.

```
seabios_grub_boot_preview_qemu_argv_template_present=1
qemu_argv_template_mode=no-effect-template
qemu_argv_template_decision=blocked-template-only-no-qemu-execution
qemu_argv_record_ready=0
qemu_run_performed=0
qemu_boot_execution_attempted=0
bootable_os_ready=0
```

Guard validation

This preview lane is guarded by:

```
sh scripts/test-seabios-grub-boot-preview-evidence-contract.sh
sh scripts/test-seabios-grub-boot-preview-preflight.sh
sh scripts/test-seabios-grub-boot-preview-evidence-template.sh
sh scripts/test-seabios-grub-boot-preview-evidence-validate.sh
sh scripts/test-seabios-grub-boot-preview-qemu-argv-template.sh
sh scripts/test-seabios-grub-boot-preview-boot-artifact-manifest-template.sh
sh scripts/test-seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

Expected output:

```
seabios_grub_boot_preview_evidence_contract: ok
```

Non-claims

This contract does not create an ISO, disk image, bootloader configuration, kernel, initramfs, QEMU launcher, VM run, serial log, or operating-system release.

It does not authorize firmware writes, GRUB writes, partition mutation, MBR writes, GPT writes, EFI variable writes, ESP writes, kernel installation, initramfs writes, service installation, driver loading, root escalation, network access, or host boot changes.

docs/SEABIOS_GRUB_BOOT_PREVIEW_EVIDENCE_VALIDATION.md

Source title: SeaBIOS and GRUB Boot Preview Evidence Validation

Lines: 104 | Words: 308 | SHA-256: 03a010a0d752ada841babcb955fdd6f1231be69755dbb99ba73291f7032d20f4

SeaBIOS and GRUB Boot Preview Evidence Validation

Status: no-effect boot-preview evidence validation Evidence level: 8 target, validation gate only Scope: current fixture validation for future SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI boot evidence bundles.

Purpose

This validation gate makes the boot evidence lane checkable before any VM boot evidence exists.

It validates that the current fixture remains unbooted, incomplete, and unable to promote a boot-preview claim. It does not accept a serial log, run QEMU, invoke GRUB, create a disk image, create an ISO, write firmware state, write bootloader state, mutate partitions, or claim Latticra is bootable.

```
seabios_grub_boot_preview_evidence_validation_present=1
boot_evidence_validation_mode=no-effect-readiness-check
boot_evidence_validation_decision=blocked-fixture-only-no-boot-evidence
boot_evidence_candidate_ready=0
qemu_i440fx_seabios_boot_validation_completed=0
qemu_grub2_bios_boot_validation_completed=0
qemu_ovmf_grub2_uefi_boot_validation_completed=0
serial_console_boot_log_recorded=0
operator_console_boot_path_recorded=0
read_only_vm_evidence_recorded=0
operator_recovery_runbook_present=0
bootable_os_ready=0
production_os_claim=0
```

Command

```
sh scripts/seabios-grub-boot-preview-evidence-validate.sh
```

Optional manifest override:

```
sh scripts/seabios-grub-boot-preview-evidence-validate.sh --manifest
installer/manifests/seabios-grub-boot-preview.toml
```

Required Fixture Fields

The validator requires the current preview fixture to preserve:

```
boot_artifact_manifest_present = false
boot_artifact_checksum_recorded = false
qemu_i440fx_seabios_boot_validation_completed = false
qemu_grub2_bios_boot_validation_completed = false
qemu_ovmf_grub2_uefi_boot_validation_completed = false
serial_console_boot_log_recorded = false
operator_console_boot_path_recorded = false
read_only_vm_evidence_recorded = false
operator_recovery_runbook_present = false
seabios_grub_boot_claim_allowed = false
grub_bootloader_write_allowed = false
bootable_os_ready = false
production_os_claim = false
```

The protected non-mutation fields must remain:

```
host_mutation_authority = false
firmware_mutation_allowed = false
bootloader_write_allowed = false
partition_mutation_allowed = false
qemu_execution_allowed_by_guard = false
no_host_bootloader_mutation = true
no_firmware_mutation = true
```

Required Rejections

The validator must reject premature claims for:

```
qemu_i440fx_seabios_boot_validation_completed = true
qemu_grub2_bios_boot_validation_completed = true
qemu_ovmf_grub2_uefi_boot_validation_completed = true
serial_console_boot_log_recorded = true
operator_console_boot_path_recorded = true
read_only_vm_evidence_recorded = true
operator_recovery_runbook_present = true
seabios_grub_boot_claim_allowed = true
bootable_os_ready = true
production_os_claim = true
```

Guard

```
sh scripts/test-seabios-grub-boot-preview-evidence-validate.sh
```

Expected output:

```
seabios_grub_boot_preview_evidence_validate: ok
```

Non-claims

This validator does not run QEMU, create or verify a boot log, prove SeaBIOS boot, prove GRUB BIOS boot, prove GRUB UEFI boot, create a bootable Latticra OS image, authorize bootloader writes, or make Latticra production installer ready.

docs/SEABIOS_GRUB_BOOT_PREVIEW_PREFLIGHT.md

Source title: SeaBIOS and GRUB Boot Preview Preflight

Lines: 114 | Words: 286 | SHA-256: ef1aed43b19c7fb9fe35406663e519ea5dddb46afe66742264290eeca73d1c91

SeaBIOS and GRUB Boot Preview Preflight

Status: no-effect boot-preview preflight Evidence level: 8 target, local report only Scope: operator-readable readiness check before any future SeaBIOS, GRUB 2 BIOS, or GRUB 2 UEFI QEMU evidence run.

Purpose

This preflight makes the boot-preview lane operationally checkable without executing it.

It validates the fixture manifest, reports whether common local tooling is visible, and preserves the current non-claim posture:

```
seabios_grub_boot_preview_preflight_present=1
preflight_mode=no-effect-report
preflight_decision=blocked-fixture-only-no-boot-execution
manifest_fixture_valid=1
boot_artifact_manifest_present=0
qemu_execution_allowed_by_guard=0
qemu_boot_execution_attempted=0
qemu_boot_execution_recorded=0
bootable_os_ready=0
production_os_claim=0
```

Command

```
sh scripts/seabios-grub-boot-preview-preflight.sh
```

Optional manifest override:

```
sh scripts/seabios-grub-boot-preview-preflight.sh --manifest
installer/manifests/seabios-grub-boot-preview.toml
```

Reported Tool Visibility

The preflight reports local visibility only:

```
qemu_system_x86_64_available=&lt;0-or-1&gt;
grub_mkrescue_available=&lt;0-or-1&gt;
grub_install_available=&lt;0-or-1&gt;
xorriso_available=&lt;0-or-1&gt;
ovmf_firmware_detected=&lt;0-or-1&gt;
```

Tool visibility is not boot evidence. Missing tools do not fail the contract guard because the current lane is still fixture-only.

The evidence capture template for the next non-executing step is:

```
sh scripts/seabios-grub-boot-preview-evidence-template.sh
```

The QEMU argv template for future profile-specific command records is:

```
sh scripts/seabios-grub-boot-preview-qemu-argv-template.sh
```

The boot artifact manifest template for future artifact metadata is:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-template.sh
```

The boot artifact manifest validation for the current blocked fixture is:

```
sh scripts/seabios-grub-boot-preview-boot-artifact-manifest-validate.sh
```

The boot-preview evidence validation for future QEMU and serial-console evidence is:

```
sh scripts/seabios-grub-boot-preview-evidence-validate.sh
```

Required Non-Effects

The preflight must always report:

```
qemu_boot_execution_attempted=0
grub_install_invoked=0
grub_mkrescue_invoked=0
disk_image_created=0
disk_image_written=0
firmware_mutation_performed=0
bootloader_write_performed=0
```

```
partition_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
```

Guard

```
sh scripts/test-seabios-grub-boot-preview-preflight.sh
```

Expected output:

```
seabios_grub_boot_preview_preflight: ok
```

Non-claims

This preflight does not create an ISO, create a disk image, write a bootloader, call grub-install, call grub-mkrescue, run QEMU, boot a VM, record a serial log, install a kernel, install an initramfs, mutate firmware, change host boot state, or promote Latticra to a bootable operating system.

docs/SEABIOS_GRUB_BOOT_PREVIEW_QEMU_ARGV_TEMPLATE.md

Source title: SeaBIOS and GRUB Boot Preview QEMU Argv Template

Lines: 95 | Words: 214 | SHA-256: 0f35b5a459310cbb18d119b553b57dbeeb6738d3826c8a26a1143081afde8e75

SeaBIOS and GRUB Boot Preview QEMU Argv Template

Status: no-effect boot-preview QEMU argv template Evidence level: 8 target, argv template only Scope: future QEMU invocation records for SeaBIOS, GRUB 2 BIOS, and GRUB 2 UEFI boot-preview profiles.

Purpose

This template defines how future QEMU argv records must be captured before any boot-preview evidence can be promoted.

It is not a VM launcher. It does not execute QEMU, create a disk image, invoke GRUB, write firmware state, write bootloader state, or claim that Latticra is bootable.

```
seabios_grub_boot_preview_qemu_argv_template_present=1
qemu_argv_template_mode=no-effect-template
qemu_argv_template_decision=blocked-template-only-no-qemu-execution
qemu_argv_record_ready=0
qemu_execution_allowed_by_guard=0
qemu_boot_execution_attempted=0
bootable_os_ready=0
production_os_claim=0
```

Command

```
sh scripts/seabios-grub-boot-preview-qemu-argv-template.sh
```

Optional manifest override:

```
sh scripts/seabios-grub-boot-preview-qemu-argv-template.sh --manifest
installer/manifests/seabios-grub-boot-preview.toml
```

Required Future Inputs

A future argv record must not be accepted unless these inputs exist:

```
operator_review_id
boot_artifact_manifest_template_path
boot_artifact_manifest_validation_report_path
boot_evidence_validation_report_path
boot_artifact_manifest_path
disk_image_path
disk_image_sha256
serial_console_boot_log_path
qemu_argv_record_path
operator_recovery_path
rollback_or_recovery_path
```

The UEFI profile also requires:

```
ovmf_code_path
ovmf_vars_template_path
ovmf_vars_ephemeral_copy_path
```

Required Non-Effects

The template must always preserve:

```
qemu_run_performed=0
qemu_boot_execution_attempted=0
qemu_boot_execution_recorded=0
grub_install_invoked=0
grub_mkrescue_invoked=0
disk_image_created=0
disk_image_written=0
firmware_mutation_performed=0
bootloader_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
```

Guard

```
sh scripts/test-seabios-grub-boot-preview-qemu-argv-template.sh
```

Expected output:

```
seabios_grub_boot_preview_qemu_argv_template: ok
```

Non-claims

This template does not prove SeaBIOS boot, GRUB BIOS boot, GRUB UEFI boot, QEMU compatibility, kernel readiness, initramfs readiness, artifact integrity, recovery safety, or production OS readiness.

docs/SEABIOS_GRUB_COMPATIBILITY_CONTRACT.md

Source title: SeaBIOS and GRUB Compatibility Contract

Lines: 155 | Words: 372 | SHA-256: 7f0dc02a5584178b5ad3b24fa8cde0a00e44a7719c5d72a421913eeb10c5b58b

SeaBIOS and GRUB Compatibility Contract

Status: compatibility-readiness contract Evidence level: 8 target, contract and guard only Scope: installer and future OS-base compatibility boundaries for SeaBIOS, legacy BIOS, GRUB, and GRUB 2.

Purpose

Latticra is moving toward an advanced operating-system direction for operators, but the current repository must keep boot claims evidence-bound.

This contract defines what may be said today:

```
seabios_grub_compatibility_contract_present=1
installer_ready_for_user_local_panel=1
installer_boot_safe_by_absence=1
production_installer_ready=0
bootable_os_ready=0
daily_driver_install_ready=0
```

The current Latticra Panel installer is compatible with SeaBIOS and GRUB in the narrow sense that it is a guarded user-local installer and does not write firmware, partitions, boot sectors, EFI variables, GRUB configuration, or bootloader entries.

It is not yet a SeaBIOS-bootable Latticra image, a GRUB-bootable Latticra image, a production OS installer, or a daily-driver operating system.

Current installer boundary

The current installer and Panel lane must preserve:

```
user_local_prefix_only=1
root_authority=0
network_authority=0
host_mutation_authority=0
runtime_enforcement_authority=0
firmware_mutation_allowed=0
bootloader_write_allowed=0
partition_mutation_allowed=0
mbr_write_allowed=0
gpt_write_allowed=0
efi_variable_write_allowed=0
esp_write_allowed=0
grub_cfg_write_allowed=0
grub_install_allowed=0
grub_mkconfig_allowed=0
efibootmgr_allowed=0
kernel_install_allowed=0
initramfs_write_allowed=0
driver_load_allowed=0
service_install_allowed=0
```

That boundary lets the installer be used on hosts that boot through SeaBIOS and GRUB without becoming a bootloader participant.

Future compatibility targets

Future OS-base work may target these profiles only after explicit evidence records exist:

```
x86_64-seabios-grub-preview
x86_64-grub2-bios-preview
x86_64-grub2-uefi-preview
x86_64-qemu-operator-console-preview
```

Required future validation before any bootable claim:

```
qemu_i440fx_seabios_boot_validation_completed=0
qemu_grub2_bios_boot_validation_completed=0
qemu_ovmf_grub2_uefi_boot_validation_completed=0
serial_console_boot_log_recorded=0
operator_console_boot_path_recorded=0
read_only_vm_evidence_recorded=0
install_plan_preview_for_boot_artifact_recorded=0
rollback_or_recovery_path_recorded=0
```

Required future boot artifact metadata

Any future boot artifact must declare:

```
artifact_format=&lt;recorded&gt;
target_firmware=&lt;seabios-or-uefi-or-both&gt;
target_bootloader=&lt;grub2-or-none&gt;
partition_scheme=&lt;mbr-or-gpt-or-hybrid&gt;
bios_boot_partition_declared=&lt;0-or-1&gt;
efi_system_partition_declared=&lt;0-or-1&gt;
grub_config_path=&lt;recorded-or-none&gt;
kernel_image_path=&lt;recorded-or-none&gt;
initramfs_path=&lt;recorded-or-none&gt;
serial_console_enabled=&lt;0-or-1&gt;
operator_recovery_path=&lt;recorded&gt;
checksum_recorded=1
signature_recorded=&lt;0-or-1&gt;
sbom_recorded=&lt;0-or-1&gt;
```

Before those fields are real evidence, Latticra must preserve:

```
boot_artifact_manifest_present=0
boot_artifact_checksum_recorded=0
boot_artifact_signature_recorded=0
bootable_os_ready=0
production_os_claim=0
```

Promotion gates

A future SeaBIOS/GRUB compatibility promotion requires:

```
compatibility_contract_present=1
boot_artifact_manifest_present=1
qemu_seabios_fixture_present=1
qemu_grub2_bios_fixture_present=1
qemu_grub2_uefi_fixture_present=1
operator_console_fixture_present=1
boot_log_reviewed=1
no_host_bootloader_mutation=1
no_firmware_mutation=1
operator_recovery_runbook_present=1
security_non_claims_preserved=1
```

Until every gate is met, status remains:

```
seabios_grub_boot_claim_allowed=0
grub_bootloader_write_allowed=0
production_os_claim=0
```

Guard validation

This contract is guarded by:

```
sh scripts/test-seabios-grub-compatibility-contract.sh
sh scripts/test-seabios-grub-boot-preview-evidence-contract.sh
```

Expected output:

```
seabios_grub_compatibility_contract: ok
```

Non-claims

This contract does not create a bootable image, GRUB configuration, disk image, ISO, kernel, initramfs, bootloader entry, firmware entry, VM launcher, hypervisor integration, production installer, or operating-system release.

It does not modify SeaBIOS, GRUB, EFI variables, MBR, GPT, ESP, kernel command lines, initramfs content, services, drivers, partitions, or host boot state.

docs/SYSTEM_BOOTSTRAP_IMPLEMENTATION.md

Source title: System Bootstrap Implementation

Lines: 131 | Words: 355 | SHA-256: 42ba89aee7b51c2fc9d09a220240fd56b3b3c1a6a3d3d086c0842e7468ef2f80

System Bootstrap Implementation

Status: first system-building implementation slice Scope: compiled no-effect system bootstrap report surface.

Purpose

This slice adds the first Latticra system bootstrap implementation surface.

The bootstrap surface coordinates existing no-effect Nucleus task classification and runtime-boundary classification into one deterministic operator-visible report.

This is the first implementation step that assembles existing system components into a single startup-facing C API.

Evidence level

Evidence level: compiled implementation slice.

The evidence is:

```
public C header
C implementation
focused invariant test
shell guard script
GitHub workflow
implementation record
```

Implementation files

This slice adds:

```
include/latticra/system_bootstrap.h
src/system_bootstrap.c
tests/system_bootstrap.c
scripts/test-system-bootstrap.sh
.github/workflows/system-bootstrap.yml
docs/SYSTEM_BOOTSTRAP_IMPLEMENTATION.md
```

Public API

The public API exposes:

```
latticra_system_bootstrap_default_request
latticra_system_bootstrap_run
latticra_system_bootstrap_report
```

The bootstrap result carries:

```
bootstrap id
phase label
system status label
effect boundary label
runtime entry status label
source identity
source span
Nucleus task result
runtime boundary result
aggregate no-effect posture
compiled evidence level
```

Effect boundary

The system bootstrap is report-only and no-effect.

It does not perform external effects, mutate host state, enter runtime execution, run commands, contact servers, perform recovery actions, touch hardware, or claim operating-system completeness.

The bootstrap coordinates existing classification/report surfaces only.

Failure behavior

The bootstrap functions return existing latticra_status_t values.

Null request/result/report-buffer inputs are rejected with the existing null-argument or buffer-too-small status values.

If a lower no-effect classification surface fails, the bootstrap reports failure through the result status and deterministic status labels.

Validation

Run:

```
sh scripts/test-system-bootstrap.sh
```

The focused test verifies:

```
default request construction
system bootstrap run remains no-effect
runtime entry remains not entered
Nucleus task remains not executed
deterministic report fields are emitted
null guard behavior is stable
```

Non-claims

This implementation does not add:

```
kernel behavior
bootloader behavior
installer behavior
```

```
runtime execution
command execution
Lat execution
LIR execution
file I/O
network I/O
state mutation
server interaction
self-update
recovery behavior
hardware behavior
sandboxing
malware prevention
ransomware prevention
production security boundary
operating-system replacement
```

Next possible implementation lane

A later slice may add a system bootstrap main-test integration audit or a small operator command wrapper, but only after this bootstrap surface compiles and stays no-effect.

docs/SYSTEM_BOOTSTRAP_REPORT_RUNNER.md

Source title: System Bootstrap Report Runner

Lines: 42 | Words: 128 | SHA-256: dd2e424c610e813bb129a55608e99bf71342c8e1feeb052e9a4eeace8f5383e0

System Bootstrap Report Runner

Status: operator-facing report runner Scope: compile and run a small command-line program that prints the deterministic system bootstrap report.

Purpose

This slice adds a tiny report runner for the system bootstrap surface.

The runner builds a default bootstrap request, runs the existing no-effect bootstrap API, renders the bootstrap report, and prints it to standard output.

Files

```
tools/system_bootstrap_report.c
scripts/test-system-bootstrap-report-runner.sh
.github/workflows/system-bootstrap-report-runner.yml
docs/SYSTEM_BOOTSTRAP_REPORT_RUNNER.md
```

Validation

Run:

```
sh scripts/test-system-bootstrap-report-runner.sh
```

The guard compiles the runner, executes it, captures its output, and checks for stable report fields:

```
LATTICRA SYSTEM BOOTSTRAP REPORT
bootstrap_id=latticra-system-bootstrap
system_status=startup-report-ready
effect_boundary=no-effect
runtime_entry_status=not-entered
no_effect=1
```

Boundary

This runner is report-only. It prints the deterministic bootstrap report and does not expand the bootstrap effect boundary or add runtime entry behavior.

Part X - Security and Assurance

- docs/BACKUP_RECOVERY_RESILIENCE_BASELINE.md - Latticra Backup, Recovery, and Cyber Resilience Baseline
- docs/CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE.md - Latticra Cryptographic Assurance and Key Management Baseline
- docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md - Latticra Cyber Incident Reporting and Response Baseline
- docs/DEFENSIVE_THREAT_MODEL_CONTRACT.md - Latticra Defensive Threat Model Contract
- docs/DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md - Latticra Defensive Threat Model Implementation Plan
- docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md - Latticra Defensive Threat Model Validation
- docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md - Latticra Defensive Threat Model Validation Refinement
- docs/HIGH_ASSURANCE_SECURITY_BASELINE.md - Latticra High-Assurance Security Baseline
- docs/IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE.md - Latticra Identity, Credential, and Access Management Baseline
- docs/MEMORY_SAFETY_ROADMAP.md - Latticra Memory-Safety Roadmap
- docs/SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE.md - Latticra Secure Configuration and Change Management Baseline
- docs/security/C_ABI_BOUNDARY_POLICY.md - Latticra C ABI Boundary Policy
- docs/security/C_CPP_SECURITY_PROFILE.md - Latticra C/C++ Security Profile
- docs/SECURITY_LOGGING_MONITORING_BASELINE.md - Latticra Security Logging, Monitoring, and Detection Baseline
- docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md - Latticra National-Security Open-System Strategy
- docs/SUPPLY_CHAIN_SECURITY_BASELINE.md - Latticra Supply-Chain Security Baseline
- docs/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md - Latticra Vulnerability Management Release Gate Baseline
- docs/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md - Latticra Zero-Trust Runtime Authority Baseline

docs/BACKUP_RECOVERY_RESILIENCE_BASELINE.md

Source title: Latticra Backup, Recovery, and Cyber Resilience Baseline

Lines: 156 | Words: 463 | SHA-256: 3e7c1848b672ff579ccf39437249c001a4dbac9c3e82b911f9331064b5a1acd2

Latticra Backup, Recovery, and Cyber Resilience Baseline

Status: backup, recovery, and cyber resilience baseline Source refresh date: 2026-05-26 Scope: backup scope, offline backup posture, restore testing, recovery prioritization, RTO/RPO planning, golden-image and infrastructure-as-code recovery evidence, recovery isolation, rollback, resilience engineering, recovery communications, post-incident lessons learned, and recovery non-claims before hosted services, mutating update lanes, production installers, production runtime, or recovery-service claims.

This baseline records backup, recovery, and resilience requirements only. It does not implement backup creation, backup storage, restore execution, rollback execution, disaster recovery, recovery orchestration, failover, continuity operations, golden-image creation, infrastructure-as-code deployment, ransomware recovery, compliance, or runtime authority.

Authoritative Recovery and Resilience Sources

Date checked: 2026-05-26

```
| Source | Latticra use |
| --- | --- |
| CISA/FBI/NSA/MS-ISAC #StopRansomware Guide | offline encrypted backups, regular backup testing, recovery prioritization, clean recovery environments, and post-incident lessons learned vocabulary |
| CISA Cross-Sector Cybersecurity Performance Goals | backup and recovery baseline expectations, including data protection and recovery planning context |
| NIST SP 800-34 Rev. 1 Contingency Planning Guide for Federal Information Systems | contingency planning, business impact analysis, recovery strategy, contingency plan testing, training, and maintenance vocabulary |
| NIST SP 800-184 Guide for Cybersecurity Event Recovery | recovery planning, playbook development, realistic test scenarios, metrics, and recovery improvement vocabulary |
| NIST SP 800-160 Vol. 2 Rev. 1 Developing Cyber-Resilient Systems | anticipate, withstand, recover from, and adapt-to-adverse-conditions vocabulary for high-assurance resilience engineering |
| NIST Cybersecurity Framework 2.0 Recover function | recovery planning and timely restoration vocabulary after cybersecurity incidents |
| NIST SP 800-53 Rev. 5 Contingency Planning controls | contingency planning, backup, recovery, alternate processing, testing, and continuity control vocabulary |
```

Authoritative URLs:

```
https://www.cisa.gov/stopransomware/ransomware-guide
https://www.cisa.gov/resources-tools/resources/stopransomware-guide
https://www.cisa.gov/cybersecurity-performance-goals-cpgs
https://csrc.nist.gov/pubs/sp/800/34/r1/upd1/final
https://csrc.nist.gov/pubs/sp/800/184/final
https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final
https://www.nist.gov/cyberframework/recover
https://csrc.nist.gov/Pubs/sp/800/53/r5/upd1/Final
```

Current Fields

```
backup_recovery_resilience_baseline_present=1
backup_recovery_resilience_guard_present=1
stopransomware_recovery_guidance_tracked=1
cisa_cpg_backup_recovery_tracked=1
nist_sp_800_34_contingency_planning_tracked=1
nist_sp_800_184_event_recovery_tracked=1
nist_sp_800_160_cyber_resilience_tracked=1
nist_csf_recover_function_tracked=1
nist_sp_800_53_contingency_planning_tracked=1
backup_scope_inventory_required=1
critical_asset_restore_priority_required=1
rto_rpo_record_required=1
offline_encrypted_backup_plan_required=1
backup_integrity_test_required=1
restore_test_required=1
clean_recovery_environment_required=1
golden_image_or_iac_recovery_plan_required=1
rollback_plan_required_before_mutation=1
recovery_authorization_contract_required=1
recovery_communications_plan_required=1
lessons_learned_update_required=1
recovery_exception_owner_required=1
recovery_exception_expiration_required=1
implementation_behavior_changed=0
backup_creation_added=0
backup_storage_added=0
restore_execution_added=0
rollback_execution_added=0
failover_added=0
recovery_orchestration_added=0
disaster_recovery_service_added=0
ransomware_recovery_service_added=0
production_recovery_claim_allowed=0
hosted_service_recovery_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Required Recovery Promotion Gate

No hosted service, production runtime, production installer, production package, mutating update lane, recovery path, rollback path, failover path, ransomware recovery feature, disaster-recovery claim, continuity claim, or production recoverability claim may be promoted until this gate is complete:

```
critical_asset_inventory_present=1
dependency_restore_order_recorded=1
business_impact_or_service_priority_recorded=1
rto_recorded=1
rpo_recorded=1
backup_scope_recorded=1
backup_owner_recorded=1
offline_or_immutable_backup_path_recorded=1
backup_encryption_and_access_control_recorded=1
backup_integrity_verification_recorded=1
restore_test_result_recorded=1
clean_recovery_environment_recorded=1
golden_image_or_iac_restore_path_recorded=1
rollback_plan_recorded=1
recovery_authorization_recorded=1
recovery_communications_path_recorded=1
incident_response_handoff_recorded=1
post_recovery_validation_recorded=1
lessons_learned_update_path_recorded=1
recovery_exception_owner_recorded=1
recovery_exception_expiration_recorded=1
operator_visible_non_claims_recorded=1
```

Until this gate is complete:

```
production_recovery_allowed=0
restore_execution_allowed=0
rollback_execution_allowed=0
failover_allowed=0
backup_service_claim_allowed=0
disaster_recovery_claim_allowed=0
ransomware_recovery_claim_allowed=0
```

```
hosted_service_recovery_claim_allowed=0
production_update_recovery_claim_allowed=0
continuity_claim_allowed=0
```

Latticra Boundary

Current Latticra recovery-related records remain evidence and no-effect contract work.

```
latticra_recovery_path_metadata_only=1
latticra_rollback_plan_metadata_only=1
latticra_install_validation_recovery_flag_metadata_only=1
latticra_backup_storage_added=0
latticra_restore_runtime_added=0
latticra_failover_runtime_added=0
latticra_recovery_authority_granted=0
```

Current Evidence

Current supporting evidence:

```
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md
docs/SECURITY_LOGGING_MONITORING_BASELINE.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
docs/LATTICRA_OS_IMAGE_RELEASE_READINESS_CONTRACT.md
SECURITY.md
scripts/test-high-assurance-security-baseline.sh
scripts/test-backup-recovery-resilience-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-backup-recovery-resilience-baseline.sh
```

docs/CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE.md

Source title: Latticra Cryptographic Assurance and Key Management Baseline

Lines: 185 | Words: 504 | SHA-256: dbc05b50829c6347dc6018286abcbb570d6709dbf29d76a8ad2f0a66d5638fef

Latticra Cryptographic Assurance and Key Management Baseline

Status: cryptographic assurance and key management baseline Source refresh date: 2026-05-26 Scope: cryptographic module boundaries, FIPS/CMVP claim gates, algorithm and parameter inventory, key lifecycle, key storage, key destruction, randomness, self-tests, sensitive-data handling, post-quantum migration planning, Seal metadata, signing authority, and cryptographic non-claims.

This baseline records cryptographic assurance requirements only. It does not implement production cryptography, signing authority, key storage, key generation, entropy collection, random-bit generation, FIPS validation, CMVP submission, CAVP testing, post-quantum migration, compliance, or runtime authority.

Authoritative Cryptographic Sources

Date checked: 2026-05-26

```
| Source | Latticra use |
| --- | --- |
| NIST FIPS 140-3 | cryptographic module boundary, security level, and validation vocabulary |
| NIST CMVP FIPS 140-3 standards and validated modules guidance | validation path and claim
caveats for validated modules |
| NIST FIPS 186-5 | digital signature standard tracking for future approved signature policy |
| RFC 8032 | Ed25519 deterministic test-vector tracking for the current local verify-only layer
|
| NIST FIPS 180-4 | SHA-256 digest vocabulary for current message/artifact digest reporting |
| NIST SP 800-57 Part 1 Rev. 5 | key-management lifecycle, key inventory, metadata, protection,
usage period, and compromise vocabulary |
| NIST SP 800-131A Rev. 2 | algorithm and key-length transition vocabulary |
```

| NIST SP 800-90A Rev. 1 and SP 800-90B | deterministic random bit generation and entropy-source review vocabulary |

| NIST FIPS 204 | ML-DSA post-quantum signature planning vocabulary |

| NIST FIPS 205 | SLH-DSA post-quantum signature planning vocabulary |

| NSA CNSA 2.0 and post-quantum cybersecurity resources | post-quantum migration planning and no-premature-deployment posture for high-assurance systems |

| NSA/CISA/NIST Quantum-Readiness guidance | inventory-first post-quantum transition planning |

| CISA/FBI Product Security Bad Practices | known-insecure cryptography, weak defaults, and unsupported cryptographic configurations remain blocked |

Authoritative URLs:

```
https://csrc.nist.gov/pubs/fips/140-3/final
https://csrc.nist.gov/Projects/cryptographic-module-validation-program/fips-140-3-standards
https://csrc.nist.gov/projects/cryptographic-module-validation-program/validated-modules
https://csrc.nist.gov/pubs/fips/186-5/final
https://www.rfc-editor.org/rfc/rfc8032
https://csrc.nist.gov/pubs/fips/180-4/upd1/final
https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final
https://csrc.nist.gov/pubs/sp/800/131/a/r2/final
https://csrc.nist.gov/pubs/sp/800/90/a/r1/final
https://csrc.nist.gov/pubs/sp/800/90/b/final
https://csrc.nist.gov/pubs/fips/204/final
https://csrc.nist.gov/pubs/fips/205/final
https://www.nsa.gov/Cybersecurity/Post-Quantum-Cybersecurity-Resources/
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/article/3498776/post
-quantum-cryptography-cisa-nist-and-nsa-recommend-how-to-prepare-now/
https://www.nsa.gov/serve-from-netstorage/Press-Room/Press-Releases-Statements/Press-Release-Vie
w/Article/3148990/nsa-releases-future-quantum-resistant-qr-algorithm-requirements-for-national-s
e/index.html
https://www.cisa.gov/resources-tools/resources/product-security-bad-practices
```

Current Fields

```
cryptographic_assurance_key_management_baseline_present=1
cryptographic_assurance_key_management_guard_present=1
seal_crypto_graduation_gate_present=1
seal_crypto_graduation_gate_guard_present=1
fips_140_3_boundary_required_before_production_crypto=1
cmvp_validation_path_required_before_fips_claim=1
validated_module_claim_requires_certificate=1
algorithm_parameter_inventory_required=1
approved_algorithm_transition_review_required=1
known_insecure_crypto_forbidden=1
ed25519_rfc8032_test_vector_required=1
authority_neutral_crypto_graduation_required=1
fips_186_5_signature_standard_tracked=1
fips_180_4_digest_standard_tracked=1
fips_204_ml_dsa_planning_tracked=1
fips_205_slh_dsa_planning_tracked=1
key_lifecycle_contract_required=1
key_inventory_required=1
key_metadata_protection_required=1
key_storage_contract_required=1
key_zeroization_contract_required=1
key_compromise_response_required=1
randomness_entropy_source_contract_required=1
drbg_review_required=1
self_test_failure_behavior_required=1
side_channel_sensitive_data_review_required=1
post_quantum_migration_inventory_required=1
cnsa_2_pq_planning_tracked=1
non_fips_disclosure_required_if_not_validated=1
seal_crypto_metadata_only_current=1
implementation_behavior_changed=0
production_crypto_added=0
signing_authority_granted=0
key_storage_added=0
key_generation_added=0
entropy_collection_added=0
fips_validation_claimed=0
cmvp_submission_performed=0
cavp_testing_claimed=0
post_quantum_migration_performed=0
production_crypto_claim_allowed=0
fips_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Required Cryptographic Promotion Gate

No future cryptographic verification, signing, encryption, key storage, key generation, key derivation, randomness, release-signing, update-signing, receipt-signing, post-quantum migration, FIPS, CMVP, or production-cryptography claim may be promoted until this gate is complete:

```
cryptographic_module_boundary_recorded=1
module_interface_inventory_recorded=1
approved_algorithm_inventory_recorded=1
algorithm_parameters_recorded=1
security_strength_recorded=1
algorithm_transition_review_recorded=1
key_types_and_usage_periods_recorded=1
key_generation_path_recorded=1
key_storage_path_recorded=1
key_access_control_recorded=1
key_rotation_and_expiration_recorded=1
key_zeroization_behavior_recorded=1
key_compromise_response_recorded=1
entropy_source_recorded=1
drbg_or_random_bit_generator_recorded=1
self_test_behavior_recorded=1
self_test_failure_mode_recorded=1
sensitive_data_logging_reviewed=1
side_channel_review_recorded=1
validated_module_certificate_recorded_before_fips_claim=1
non_fips_disclosure_recorded_if_unvalidated=1
post_quantum_inventory_recorded=1
operator_visible_non_claims_recorded=1
```

Until this gate is complete:

```
production_crypto_allowed=0
fips_claim_allowed=0
cmvp_claim_allowed=0
release_signing_allowed=0
update_signing_allowed=0
receipt_signing_allowed=0
key_generation_allowed=0
key_storage_allowed=0
key_derivation_allowed=0
entropy_collection_allowed=0
random_bit_generation_allowed=0
post_quantum_migration_claim_allowed=0
cryptographic_module_validation_claim_allowed=0
```

Seal Boundary

Current Seal crypto-related records remain evidence and metadata oriented.

```
seal_crypto_verify_backend_metadata_only=1
seal_ed25519_verify_only_authority_neutral=1
seal_crypto_graduation_gate_authority_neutral=1
seal_signing_metadata_only=1
seal_key_material_metadata_only=1
seal_runtime_authority_granted=0
seal_production_crypto_enforcement_claimed=0
```

Current Evidence

Current supporting evidence:

```
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md
docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_IMPLEMENTATION.md
docs/status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md
docs/status/SEAL_ED25519_VERIFY_STATUS.md
docs/LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md
docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md
SECURITY.md
scripts/test-high-assurance-security-baseline.sh
scripts/test-cryptographic-assurance-key-management-baseline.sh
scripts/test-latticra-seal-crypto-graduation-gate.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-cryptographic-assurance-key-management-baseline.sh
```

docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md

Source title: Latticra Cyber Incident Reporting and Response Baseline

Lines: 154 | Words: 381 | SHA-256: ad9bbf07303e5593cf7eff976cf32aaf342d480ed463868bd832744bcce3d5bf

Latticra Cyber Incident Reporting and Response Baseline

Status: cyber incident reporting and response baseline Source refresh date: 2026-05-26 Scope: vulnerability reports, suspected compromise triage, ransomware and data-extortion reporting paths, evidence preservation, communications routing, public non-claims, and future incident-response automation gates.

This baseline records reporting and response posture only. It does not implement monitoring, detection, containment, forensic acquisition, recovery, federal reporting, customer notification, breach notification, law-enforcement contact, ransomware recovery, or incident-response services.

Authoritative Reporting and Response Sources

Date checked: 2026-05-26

```
| Source | Latticra use |
| --- | --- |
| CISA Reporting a Cyber Incident | federal reporting path vocabulary and voluntary cyber incident reporting expectation |
| FBI Cyber | FBI/IC3 reporting path vocabulary for cyber-enabled crime, fraud, ongoing threats, and field-office escalation |
| CISA/FBI/NSA/MS-ISAC #StopRansomware Guide | ransomware and data-extortion prevention, response, evidence preservation, out-of-band communication, and contact-path vocabulary |
| CISA Report Ransomware | ransomware incident reporting route awareness |
| FBI Ransomware | FBI ransomware reporting route awareness |
| FBI 2025 IC3 Annual Report | current FBI cybercrime trend refresh checkpoint |
```

Authoritative URLs:

```
https://www.cisa.gov/reporting-cyber-incident
https://www.fbi.gov/investigate/cyber
https://www.cisa.gov/stopransomware/ransomware-guide
https://www.cisa.gov/stopransomware/report-ransomware
https://www.fbi.gov/how-we-can-help-you/scams-and-safety/common-frauds-and-scams/ransomware
https://www.fbi.gov/file-repository/2025_ic3report.pdf
```

Current Fields

```
cyber_incident_reporting_response_baseline_present=1
cyber_incident_reporting_response_guard_present=1
cisa_reporting_channel_documented=1
fbi_cyber_reporting_channel_documented=1
ic3_reporting_channel_documented=1
stopransomware_joint_guidance_tracked=1
fbi_ic3_annual_report_refresh_required=1
incident_classification_required=1
authorized_testing_boundary_required=1
evidence_preservation_required=1
volatile_evidence_preservation_required=1
chain_of_custody_required_before_claim=1
out_of_band_communications_required_for_compromise=1
ransomware_data_extortion_response_checklist_required=1
legal_regulatory_notification_review_required=1
law_enforcement_contact_path_required=1
internal_external_notification_plan_required=1
operator_confirmation_metadata_only_required=1
implementation_behavior_changed=0
monitoring_added=0
detection_added=0
containment_added=0
forensic_collection_added=0
federal_reporting_performed=0
law_enforcement_contact_performed=0
customer_notification_performed=0
breach_notification_authority_claimed=0
incident_response_service_claimed=0
ransomware_recovery_capability_claimed=0
production_monitoring_claimed=0
external_endorsement_claimed=0
```

Reporting Route Requirements

Latticra must not present itself as reporting on behalf of users or operators. Future reporting-capable features require a separate operator-approved contract and review.

Minimum route fields before any report-assistance feature exists:

```
reporting_subject_known=1
incident_class_known=1
affected_asset_identity_known=1
affected_data_sensitivity_known=1
operator_authorization_known=1
organization_reporting_owner_known=1
cisa_reporting_route_visible=1
fbi_field_office_route_visible=1
ic3_route_visible=1
local_law_enforcement_or_regulator_review_prompt_visible=1
privacy_legal_notification_review_prompt_visible=1
do_not_transmit_without_operator_approval=1
```

Response Gate

No future incident-response request may isolate hosts, collect forensic artifacts, contact law enforcement, submit federal reports, notify customers, rotate credentials, restore systems, delete artifacts, publish indicators, or make breach/ransomware claims until this gate is complete:

```
incident_kind_known=1
requested_response_effect_known=1
caller_identity_known=1
operator_or_organization_context_known=1
asset_scope_recorded=1
data_scope_recorded=1
legal_regulatory_review_required=1
evidence_preservation_plan_recorded=1
volatile_evidence_decision_recorded=1
chain_of_custody_plan_recorded=1
out_of_band_communications_plan_recorded=1
restoration_priority_recorded=1
reporting_route_selected_by_operator=1
policy_decision_reported=1
denial_reason_reported=1
audit_record_emitted=1
```

```
operator_confirmation_recorded_as_metadata_only=1
operator_confirmation_non_override_test_present=1
non_claim_review_completed=1
```

Until this gate is complete:

```
host_isolation_allowed=0
forensic_collection_allowed=0
credential_rotation_allowed=0
system_restore_allowed=0
federal_report_submission_allowed=0
law_enforcement_contact_allowed=0
customer_notification_allowed=0
breach_notification_allowed=0
ransomware_payment_decision_allowed=0
indicator_publication_allowed=0
artifact_deletion_allowed=0
incident_response_service_claim_allowed=0
```

Current Evidence

Current supporting evidence:

```
SECURITY.md
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md
docs/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
scripts/test-high-assurance-security-baseline.sh
scripts/test-zero-trust-runtime-authority-baseline.sh
scripts/test-cyber-incident-reporting-response-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-cyber-incident-reporting-response-baseline.sh
```

docs/DEFENSIVE_THREAT_MODEL_CONTRACT.md

Source title: Latticra Defensive Threat Model Contract

Lines: 374 | Words: 1075 | SHA-256: 0ad92562e3da4140d4a587ff95b03729f47f872dd1da06586418e8b6c0e513fe

Latticra Defensive Threat Model Contract

Status: Defensive threat model contract Scope: protected assets, trust boundaries, assumptions, capability categories, abuse cases, evidence expectations, validation path, future files, compatibility expectations, and non-claims before additional security-facing implementation work.

Purpose

This document defines the first defensive threat model contract for Latticra.

The purpose is to establish a disciplined vocabulary for what Latticra is trying to protect, what boundaries exist today, what assumptions are allowed, what adversary capability categories may be modeled, what abuse cases should be considered, and what evidence is required before security-facing claims can be promoted.

This document does not implement security controls.

Relationship to previous work

This contract depends on:

```
docs/REAL_SYSTEM_CONTRACT.md
docs/NON_CLAIMS.md
docs/EVIDENCE_LADDER.md
docs/EFFECT_GATES.md
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
```

```
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
docs/NUCLEUS_TASK_EXECUTION_CONTRACT.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
```

Those files remain the source of truth for current architecture boundaries, evidence rules, non-claims, authority behavior, task classification, runtime boundary posture, rendering, LIR metadata, and Lat metadata.

Threat model posture

The first threat model posture is evidence-bound and defensive.

Initial posture:

```
security controls implemented? limited/no-effect only
runtime protection implemented? no
production protection claim allowed? no
malware prevention claim allowed? no
ransomware prevention claim allowed? no
sandbox claim allowed? no
operator-visible report claim allowed? yes, where implemented and tested
denied-by-default boundary claim allowed? yes, where implemented and tested
```

The threat model records intended defensive reasoning. It does not certify effectiveness.

Protected assets

Initial protected assets include:

```
source text integrity
parse results
diagnostic metadata
source spans
AST metadata
semantic validation results
LIR metadata
Lat parse metadata
L-UI render metadata
Nucleus preview records
Nucleus task records
runtime boundary records
authority audit records
operator-visible reports
effect-gate decisions
project claims and status records
```

Future protected assets may include:

```
runtime execution context
configuration policy
operator approval records
file-system mediation records
network mediation records
recovery records
hardware interaction records
boot records
update records
```

Future assets require separate contracts before implementation claims.

Trust boundaries

Current trust boundaries:

```
source input boundary
parser boundary
semantic validation boundary
LIR lowering boundary
Lat grammar boundary
L-UI rendering boundary
Nucleus preview boundary
Nucleus task classification boundary
```

```
runtime boundary
authority validation boundary
operator report boundary
repository documentation boundary
```

Current boundaries are primarily metadata, validation, classification, and reporting boundaries.

They are not production isolation boundaries.

Assumptions

Current allowed assumptions:

```
source buffers may be malformed
input may be adversarial
reported metadata must be deterministic
unknown requests must be denied or classified as unsupported
unknown effects must be denied or classified as unsupported
effect-performing behavior must require explicit future gates
operator confirmation must not override policy by itself
security claims must follow evidence
no-effect slices must remain no-effect
```

Disallowed assumptions:

```
trusted input
trusted caller behavior
implicit execution safety
implicit runtime isolation
implicit sandboxing
implicit malware prevention
implicit ransomware prevention
implicit production readiness
implicit certification
```

Capability categories

Adversary or misuse capability categories may include:

```
malformed input construction
ambiguous source construction
oversized input pressure
unexpected escape sequence use
metadata confusion attempts
request-kind confusion attempts
effect-kind confusion attempts
boundary bypass attempts
operator-report confusion attempts
policy downgrade attempts
status-record exaggeration attempts
```

Future runtime capability categories may include:

```
unauthorized command request
unauthorized mutation request
unauthorized file request
unauthorized network request
unauthorized recovery request
unauthorized hardware request
unauthorized boot request
unauthorized update request
```

These categories are for defensive modeling only.

Abuse cases

Initial abuse cases:

```
malformed source causes unclear diagnostics
escaped data hides operator-visible content
literal source-buffer NUL causes parser confusion
duplicate names cause ambiguous binding
invalid binding prefix bypasses semantic validation
invalid LIR input reaches rendering
failed authority metadata is treated as allowed
unknown request is treated as allowed
unknown effect is treated as allowed
```

```
future-gated request is treated as executable
operator confirmation overrides policy
report omits denial reason
status documentation overclaims implementation state
```

Every abuse case should map to one of:

```
existing test
future test
explicit non-goal
explicit non-claim
implementation gap
```

Defensive controls currently represented

Current represented controls include:

```
parser structural checks
parser diagnostics
source-span metadata
string escape diagnostics
literal source-buffer NUL rejection
escaped decoded NUL visibility
AST metadata reports
semantic validation checks
LIR shape checks
L-UI rendering prerequisites
constrained C++ authority validation/reporting
Nucleus preview denial behavior
Nucleus task classification/reporting
runtime boundary disabled-by-default posture
static documentation guards
project status/non-claim records
```

These are engineering controls and guardrails. They are not certifications.

Evidence expectations

Security-facing promotion requires evidence.

Required evidence classes:

```
contract document
implementation plan
implementation document
unit or invariant tests
static guard when applicable
deterministic report output when applicable
negative tests for denied behavior
status update
non-claim update
compatibility check
```

No defensive claim should be promoted from concept to implementation status without these evidence classes.

Validation expectations

Future defensive validation should include:

```
positive tests for allowed no-effect behavior
negative tests for denied effect behavior
unknown request tests
unknown effect tests
malformed input tests
small-buffer tests
deterministic report tests
no-mutation tests
no-network tests
no-hardware tests
no-recovery tests
operator confirmation non-override tests
status/non-claim guard tests
```

Non-goals

Current non-goals:

```
attack tooling
exploit development
payload generation
malware simulation
credential access
stealth behavior
bypass instructions
persistence mechanisms
exfiltration behavior
```

Latticra threat modeling is defensive architecture work, not offensive tooling.

Future file policy

Recommended future files:

```
docs/DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md
scripts/test-defensive-threat-model-contract.sh
scripts/test-defensive-threat-model-validation.sh
tests/defensive_threat_model_invariants.c
```

Implementation or validation files should only be added after the contract is merged.

Future test list

Future validation should consider tests named:

```
defensive_threat_model_names_assets
defensive_threat_model_names_boundaries
defensive_threat_model_names_assumptions
defensive_threat_model_names_non_goals
defensive_threat_model_preserves_non_claims
defensive_threat_model_requires_evidence_classes
defensive_threat_model_maps_abuse_cases_to_tests_or_gaps
defensive_threat_model_forbids_operator_confirmation_override
defensive_threat_model_forbids_runtime_protection_claims
defensive_threat_model_forbids_malware_prevention_claims
defensive_threat_model_forbids_ransomware_prevention_claims
```

Compatibility expectations

Future work must not weaken:

```
existing non-claims
runtime boundary disabled-by-default posture
Nucleus task no-effect posture
constrained authority no-effect posture
L-UI rendering no-effect posture
LIR metadata-only posture
Lat parser metadata-only posture
source-buffer literal NUL rejection
escaped decoded NUL visibility
semantic validation prerequisites
operator confirmation non-override policy
```

Forbidden behavior

Threat model work must not:

- provide attack instructions;
- provide exploit steps;
- provide payload construction;
- provide stealth guidance;
- provide credential access guidance;
- provide persistence guidance;
- claim production protection;
- claim malware prevention;

- claim ransomware prevention;
- claim sandboxing;
- claim certification;
- claim operating-system completeness.

Current validation command

This contract is guarded by:

```
sh scripts/test-defensive-threat-model-contract.sh
```

The guard is static. It does not implement security controls.

Non-claims

This document does not implement security controls, runtime protection, malware prevention, ransomware prevention, sandboxing, exploit prevention, incident response, recovery behavior, certification, accreditation, production hardening, or operating-system completeness.

docs/DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md

Source title: Latticra Defensive Threat Model Implementation Plan

Lines: 312 | Words: 969 | SHA-256: ba516efc1cdd6137345ab5de226ac1d93838ad576d1c6f4c607515af66134752

Latticra Defensive Threat Model Implementation Plan

Status: implementation planning contract Scope: validation artifacts, evidence mapping, abuse-case mapping, external standards alignment, guard behavior, documentation updates, future files, compatibility expectations, and non-claims before additional security-facing implementation work.

Purpose

This document defines the implementation plan for the first defensive threat model validation layer.

The Defensive threat model contract is already merged and guarded. This plan turns that contract into validation artifacts, file paths, report expectations, evidence mapping rules, abuse-case mapping rules, external standards alignment rules, documentation updates, static guard behavior, future tests, compatibility expectations, and non-claims.

This document does not implement security controls.

Relationship to previous work

This plan depends on:

```
docs/DEFENSIVE_THREAT_MODEL_CONTRACT.md
docs/REAL_SYSTEM_CONTRACT.md
docs/NON_CLAIMS.md
docs/EVIDENCE_LADDER.md
docs/EFFECT_GATES.md
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
scripts/test-defensive-threat-model-contract.sh
```

Implementation decision

The first defensive threat model implementation should be documentation-and-guard based.

No C runtime code is required for this planning slice.

Reason:

```
threat model validation is currently a documentation and evidence mapping problem
no new security control is being implemented
no runtime behavior is being added
no parser behavior is being changed
no authority behavior is being changed
no boundary behavior is being changed
```

External standards alignment

The validation layer must include a standards-alignment ledger.

The ledger must track current guidance from:

```
NSA Cybersecurity guidance and standards/certification resources
CISA Secure by Design guidance
CISA Cross-Sector Cybersecurity Performance Goals
CISA Known Exploited Vulnerabilities catalog
CISA joint cybersecurity advisories where relevant
FBI Cyber alerts, cyber threat reporting, and joint advisories where relevant
```

Each external alignment entry must include:

```
source agency
source title
authoritative URL
date checked
version or publication date when available
applicability to Latticra
mapped Latticra document
mapped Latticra control or boundary
current evidence
missing evidence
claim allowed
claim forbidden
review cadence
```

The ledger must never claim certification, accreditation, compliance, protection, or operational readiness from simple reference alignment.

Validation document shape

The implementation PR should add:

```
docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md
scripts/test-defensive-threat-model-validation.sh
```

The validation document should contain:

```
Status
Purpose
Source contract
Protected asset matrix
Trust boundary matrix
Assumption matrix
Abuse-case mapping
Evidence matrix
External standards alignment ledger
Validation matrix
Non-goal matrix
Compatibility expectations
Current gaps
Non-claims
Validation command
```

Protected asset matrix

The validation document should map each protected asset to:

```
asset name
current representation
```

```
current evidence file
current validation command
current status
claim allowed
claim not allowed
implementation gap
```

Required protected assets:

```
source text integrity
parse results
diagnostic metadata
source spans
AST metadata
semantic validation results
LIR metadata
Lat parse metadata
L-UI render metadata
Nucleus preview records
Nucleus task records
runtime boundary records
authority audit records
operator-visible reports
effect-gate decisions
project claims and status records
```

Trust boundary matrix

Required trust boundaries:

```
source input boundary
parser boundary
semantic validation boundary
LIR lowering boundary
Lat grammar boundary
L-UI rendering boundary
Nucleus preview boundary
Nucleus task classification boundary
runtime boundary
authority validation boundary
operator report boundary
repository documentation boundary
external standards alignment boundary
```

Abuse-case mapping

Each abuse case from the contract must map to one of:

```
existing test
future test
explicit non-goal
explicit non-claim
implementation gap
```

Required abuse cases:

```
malformed source causes unclear diagnostics
escaped data hides operator-visible content
literal source-buffer NUL causes parser confusion
duplicate names cause ambiguous binding
invalid binding prefix bypasses semantic validation
invalid LIR input reaches rendering
failed authority metadata is treated as allowed
unknown request is treated as allowed
unknown effect is treated as allowed
future-gated request is treated as executable
operator confirmation overrides policy
report omits denial reason
status documentation overclaims implementation state
external standard is referenced as if it were certification
outdated external guidance remains marked current
```

Evidence matrix

Required evidence classes:

```
contract document
implementation plan
implementation document
```

```
unit or invariant tests
static guard when applicable
deterministic report output when applicable
negative tests for denied behavior
status update
non-claim update
compatibility check
external standards source check
standards alignment gap entry
```

No security-facing claim should be marked implementation-backed unless the evidence matrix names supporting files and commands.

Guard behavior

The validation guard should verify that docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md contains every required section, protected asset, trust boundary, abuse case, evidence class, external standards source, non-goal, compatibility expectation, and non-claim named in this plan.

The implementation-plan guard should verify this document contains:

```
Status: implementation planning contract
External standards alignment
NSA Cybersecurity
CISA Secure by Design
CISA Cross-Sector Cybersecurity Performance Goals
CISA Known Exploited Vulnerabilities catalog
FBI Cyber alerts
source agency
authoritative URL
date checked
claim forbidden
sh scripts/test-defensive-threat-model-implementation-plan.sh
```

Workflow behavior

The defensive threat model workflow should run:

```
sh scripts/test-defensive-threat-model-contract.sh
sh scripts/test-defensive-threat-model-implementation-plan.sh
sh scripts/test-defensive-threat-model-validation.sh
```

The validation command should run after the implementation-plan guard.

Compatibility expectations

The implementation must not weaken:

```
existing non-claims
runtime boundary disabled-by-default posture
Nucleus task no-effect posture
constrained authority no-effect posture
L-UI rendering no-effect posture
LIR metadata-only posture
Lat parser metadata-only posture
source-buffer literal NUL rejection
escaped decoded NUL visibility
semantic validation prerequisites
operator confirmation non-override policy
```

Forbidden behavior

The implementation must not:

- claim certification;
- claim accreditation;
- claim compliance;
- claim malware prevention;
- claim ransomware prevention;

- claim sandboxing;
- claim production protection;
- claim operational readiness;
- provide attack instructions;
- provide exploit steps;
- provide payload construction;
- provide stealth guidance;
- provide persistence guidance;
- provide credential access guidance;
- treat external standards alignment as proof of protection.

Current validation command

This implementation plan is guarded by:

```
sh scripts/test-defensive-threat-model-implementation-plan.sh
```

The guard is static. It does not implement security controls.

Implementation gate

Defensive threat model validation files may be added only after this plan is merged.

Non-claims

This document does not implement security controls, runtime protection, malware prevention, ransomware prevention, sandboxing, exploit prevention, incident response, recovery behavior, certification, accreditation, compliance, production hardening, or operating-system completeness.

docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md

Source title: Latticra Defensive Threat Model Validation

Lines: 243 | Words: 2217 | SHA-256: 7a120db97a04a3788b2b49bb1d11911f966e86cc96586adbb28daf7528458741

Latticra Defensive Threat Model Validation

Status: defensive threat model validation ledger Scope: asset mapping, trust-boundary mapping, evidence mapping, external standards alignment, current gaps, and non-claims.

Purpose

This document validates the defensive threat model against the current Latticra evidence base.

Source contract:

```
docs/DEFENSIVE_THREAT_MODEL_CONTRACT.md
docs/DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md
```

This document does not implement security controls.

Protected asset matrix

```
| Asset | Evidence file | Validation command | Status | Allowed claim | Forbidden claim | Gap |
| --- | --- | --- | --- | --- | --- | --- |
| source text integrity | docs/L_UI_SOURCE_BUFFER_LITERAL_NUL_POLICY_IMPLEMENTATION.md | sh
scripts/test-l-ui-source-buffer-literal-nul-policy.sh | represented | source policy is guarded |
 source safety guarantee | broader hostile corpus |
```

parse results	docs/L_UI_PARSER_RESULT_REPORT.md	sh scripts/test-l-ui-parser-result-report.sh	represented	parser result reports exist	parser hardening guarantee	broader fuzz corpus
diagnostic metadata	docs/L_UI_PARSER_DIAGNOSTICS_IMPLEMENTATION.md	sh scripts/test-l-ui-parser-diagnostics.sh	represented	diagnostics are guarded	complete diagnostic coverage	severity taxonomy
source spans	docs/L_UI_PARSER_SOURCE_SPAN_IMPLEMENTATION.md	sh scripts/test-l-ui-parser-source-span.sh	represented	spans are tracked	source authenticity	signed source policy
AST metadata	docs/L_UI_PARSER_AST_IMPLEMENTATION.md	sh scripts/test-l-ui-parser-ast.sh	represented	AST metadata exists	semantic completeness	richer AST invariants
semantic validation results	docs/L_UI_SEMANTIC_VALIDATION_IMPLEMENTATION.md	sh scripts/test-l-ui-semantic-validation.sh	represented	semantic checks exist	total semantic safety	broader ruleset
LIR metadata	docs/LIR_SHAPE_IMPLEMENTATION.md	sh scripts/test-lir-shape.sh	represented	LIR shape is guarded	LIR execution safety	execution contract
Lat parse metadata	docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md	sh scripts/test-lat-language-grammar.sh	represented	Lat parse metadata exists	Lat runtime safety	runtime contracts
L-UI render metadata	docs/L_UI_RENDERING_IMPLEMENTATION.md	sh scripts/test-l-ui-rendering.sh	represented	bounded reports exist	terminal UI safety	terminal-control contract
Nucleus preview records	docs/NUCLEUS_PREVIEW.md	sh scripts/test-nucleus-preview.sh	represented	preview classification exists	execution authorization	task expansion
Nucleus task records	docs/NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md	sh scripts/test-nucleus-task-execution.sh	represented	no-effect task records exist	effect execution safety	fuller matrix
runtime boundary records	docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md	sh scripts/test-runtime-boundary.sh	initial	API/smoke surface exists	runtime protection	fuller policy implementation
authority audit records	docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md	sh scripts/test-cpp-authority-layer.sh	represented	authority reports exist	operational authority	policy mapping
operator-visible reports	docs/FOUNDATION_INDEX.md	multiple scripts	represented	reports exist where tested	complete audit coverage	report index
effect-gate decisions	docs/EFFECT_GATES.md	static guards	planned	vocabulary exists	enforcement guarantee	enforcement implementation
project claims and status records	STATUS.md	sh scripts/test-project-strategy-status-framework.sh	guarded	status is guarded	certification/compliance	review cadence

Trust boundary matrix

Boundary	Evidence	Validation	Status	Forbidden claim
source input boundary	source policy	source-buffer policy script	represented	trusted input
parser boundary	parser implementation	parser scripts	represented	parser certification
semantic validation boundary	semantic validation	semantic validation script	represented	full semantic safety
LIR lowering boundary	LIR shape	LIR script	represented	executable IR safety
Lat grammar boundary	Lat parser	Lat script	represented	Lat runtime safety
L-UI rendering boundary	renderer	rendering script	represented	interactive UI safety
Nucleus preview boundary	preview tests	Nucleus preview script	represented	execution authorization
Nucleus task classification boundary	task tests	task script	represented	effect authorization
runtime boundary	runtime smoke tests	runtime-boundary script	initial	runtime protection
authority validation boundary	authority tests	C++ authority script	represented	operational authority
operator report boundary	report tests	multiple scripts	represented	complete audit coverage
repository documentation boundary	project guard	project-status script	represented	certification
external standards alignment boundary	this ledger	validation script	initial	standards compliance

Assumption matrix

source buffers may be malformed -> parser/source policy tests
input may be adversarial -> negative parser and semantic tests
reported metadata must be deterministic -> report tests
unknown requests must be denied or unsupported -> Nucleus/runtime contracts
unknown effects must be denied or unsupported -> Nucleus/runtime contracts
effect-performing behavior requires explicit future gates -> effect-gate and runtime contracts
operator confirmation must not override policy -> task/runtime contracts
security claims must follow evidence -> evidence ladder and this ledger
no-effect slices must remain no-effect -> implementation docs and tests

Abuse-case mapping

malformed source causes unclear diagnostics -> parser diagnostics tests
escaped data hides operator-visible content -> escaped string reports
literal source-buffer NUL causes parser confusion -> source-buffer literal NUL policy
duplicate names cause ambiguous binding -> semantic validation and fixture gap
invalid binding prefix bypasses semantic validation -> semantic validation and fixture gap
invalid LIR input reaches rendering -> rendering prerequisites and LIR checks
failed authority metadata is treated as allowed -> authority and task checks
unknown request is treated as allowed -> task tests, runtime expansion gap
unknown effect is treated as allowed -> task tests, runtime expansion gap
future-gated request is treated as executable -> task tests, runtime expansion gap
operator confirmation overrides policy -> task tests, runtime expansion gap
retained C/C++ high-risk code leaves buffer-overflow-class defects untracked -> memory-safety
roadmap refinement gap
command construction reaches a shell boundary without a reviewed contract -> shell-boundary guard and installer-boundary refinement gap
future workload or service authority lacks distinct workload identity -> zero-trust implementation profile gap
report omits denial reason -> report completeness gap
status documentation overclaims implementation state -> project status guard
external standard is referenced as if it were certification -> external ledger forbidden claim
outdated external guidance remains marked current -> review cadence required

Evidence matrix

contract document -> present
implementation plan -> present
implementation document -> this validation ledger
unit or invariant tests -> validation guard
static guard when applicable -> validation script
deterministic report output when applicable -> parser/render/task tests
negative tests for denied behavior -> parser/semantic/task tests
status update -> status docs
non-claim update -> non-claim docs/status
compatibility check -> static guards
external standards source check -> external ledger
standards alignment gap entry -> external ledger

External standards alignment ledger

Date checked: 2026-05-26

Source	Authoritative URL	Check status	Applicability	Current evidence	Missing evidence	Allowed claim	Forbidden claim	Review cadence
NSA Zero Trust Implementation Guidelines	https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4393480/nsa-releases-phase-one-and-phase-two-of-the-zero-trust-implementation-guidelines/	fetched; 2026 Primer, Discovery Phase, Phase One, and Phase Two visible	zero-trust implementation planning	mapped source	phase-by-phase implementation mapping	source tracked	NSA endorsement/certification/protection	monthly or before release
NSA/CISA Memory Safe Languages CSI	https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4223298/nsa-and-cisa-release-csi-highlighting-importance-of-memory-safe-languages-in-so/	fetched; 2025 CSI visible	C/C++ memory-safety roadmap	mapped source	component-by-component migration/mitigation map	source tracked	memory safety guarantee	monthly or before release
CISA The Case for Memory Safe Roadmaps	https://www.cisa.gov/resources-tools/resources/case-memory-safe-roadmaps	official source reviewed	component-by-component memory-safety roadmap discipline	mapped source	roadmap detail by component family and exception lifetime	source tracked	roadmap completeness guarantee	monthly or before release
CISA Secure by Design	https://www.cisa.gov/securebydesign	official source reviewed						

secure-by-design practice alignment | mapped source | secure-by-design pledge/progress decision | source tracked | CISA compliance/protection | monthly or before release |

| CISA Secure by Design Alert: Eliminating Buffer Overflow Vulnerabilities | https://www.cisa.gov/news-events/alerts/2025/02/12/cisa-and-fbi-warn-malicious-cyber-actors-using-buffer-overflow-vulnerabilities-compromise-software | official source reviewed | retained C/C++ hazard-class reduction and adversarial testing expectations | mapped source | hazard-class-by-hazard-class test/profile coverage | source tracked | elimination claim | monthly or before release |

| CISA Secure by Design Alert: Eliminating OS Command Injection Vulnerabilities | https://www.cisa.gov/resources-tools/resources/secure-design-alert-eliminating-os-command-injection-vulnerabilities | official source reviewed | shell-boundary and command-construction exclusions | mapped source | command-boundary-by-command-boundary guard coverage | source tracked | command-injection immunity claim | monthly or before release |

| CISA/FBI Product Security Bad Practices | https://www.cisa.gov/resources-tools/resources/product-security-bad-practices | official source reviewed | product-security exclusion list | mapped source | bad-practice-by-practice guard coverage | source tracked | CISA/FBI endorsement/compliance | monthly or before release |

| CISA Cross-Sector Cybersecurity Performance Goals | https://www.cisa.gov/cybersecurity-performance-goals | official source reviewed | critical-infrastructure baseline vocabulary | mapped source | CPG-by-CPG maturity mapping | source tracked | CPG compliance/protection | monthly or before release |

| CISA Zero Trust Maturity Model | https://www.cisa.gov/resources-tools/resources/zero-trust-maturity-model | official source reviewed | zero-trust maturity vocabulary | mapped source | pillar-by-pillar implementation profile | source tracked | zero-trust certification | monthly or before release |

| CISA Known Exploited Vulnerabilities Catalog | https://www.cisa.gov/resources-tools/resources/known-exploited-vulnerabilities-catalog | authoritative URL retained | vulnerability-awareness mapping | mapped source | KEV release-review process | source tracked | remediation guarantee | monthly or before release |

| FBI Cyber | https://www.fbi.gov/investigate/cyber | fetched; 2026 threat/reporting content visible | threat environment awareness | mapped source | advisory-by-advisory mapping | source tracked | FBI endorsement/protection | monthly or before release |

| NIST Cybersecurity Framework 2.0 | https://www.nist.gov/cyberframework | fetched; CSF 2.0 resource center visible | Govern/Identify/Protect/Detect/Respond/Recover vocabulary | mapped source | CSF function profile | source tracked | NIST compliance/certification | monthly or before release |

| NIST SP 800-218 SSDF | https://csrc.nist.gov/pubs/sp/800/218/final | fetched; final SSDF v1.1 visible | secure software development lifecycle | mapped source | SSDF practice-level map | source tracked | SSDF compliance | monthly or before release |

| NIST SP 800-53 Rev. 5 | https://csrc.nist.gov/Pubs/sp/800/53/r5/upd1/Final | fetched; Release 5.2.0 planning note visible | high-assurance control vocabulary | mapped source | tailored control profile | source tracked | compliance/accreditation | monthly or before release |

| NIST SP 800-160 Vol. 2 Rev. 1 | https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final | fetched; cyber-resilience publication visible | resilience engineering vocabulary | mapped source | resilience objective mapping | source tracked | resilience guarantee | monthly or before release |

| NIST SP 800-207 Zero Trust Architecture | https://www.nist.gov/publications/zero-trust-architecture-0 | fetched; official SP 800-207 page visible | zero-trust architecture vocabulary | mapped source | ZTA design profile | source tracked | zero-trust certification | monthly or before release |

| NIST SP 800-207A Zero Trust Architecture: A Practitioner's Guide | https://csrc.nist.gov/pubs/sp/800/207/a/final | official source reviewed | workload/service identity-aware zero-trust implementation detail | mapped source | future workload/service profile and trust algorithm detail | source tracked | implementation completeness guarantee | monthly or before release |

| NIST SP 1800-35 Implementing a Zero Trust Architecture | https://csrc.nist.gov/pubs/sp/1800/35/final | official source reviewed | deployment-oriented zero-trust reference architecture vocabulary | mapped source | deployment profile for future service surfaces | source tracked | deployment assurance claim | monthly or before release |

| FIPS 140-3 | https://csrc.nist.gov/pubs/fips/140-3/final | fetched; FIPS 140-3 final page visible | cryptographic module assurance | mapped source | module-boundary and validation decision | source tracked | FIPS validation claim | monthly or before release |

Recurring manual source review remains required because external guidance can change and source tracking is not certification/compliance/protection.

Validation matrix

```
positive tests for allowed no-effect behavior -&gt; represented
negative tests for denied effect behavior -&gt; represented
unknown request tests -&gt; represented
unknown effect tests -&gt; represented
malformed input tests -&gt; represented
small-buffer tests -&gt; represented
deterministic report tests -&gt; represented
no-mutation tests -&gt; represented
no-network tests -&gt; contract-level gap
no-hardware tests -&gt; contract-level gap
no-recovery tests -&gt; represented plus runtime gap
operator confirmation non-override tests -&gt; represented plus runtime gap
command-boundary tests -&gt; contract-level gap
workload/service identity zero-trust tests -&gt; future implementation gap
status/non-claim guard tests -&gt; represented
```

Non-goal matrix

```
attack tooling -&gt; forbidden
exploit development -&gt; forbidden
payload generation -&gt; forbidden
credential access -&gt; forbidden
stealth behavior -&gt; forbidden
bypass instructions -&gt; forbidden
persistence mechanisms -&gt; forbidden
exfiltration behavior -&gt; forbidden
certification claim -&gt; forbidden
compliance claim -&gt; forbidden
production protection claim -&gt; forbidden
```

Compatibility expectations

```
existing non-claims
runtime boundary disabled-by-default posture
Nucleus task no-effect posture
constrained authority no-effect posture
L-UI rendering no-effect posture
LIR metadata-only posture
Lat parser metadata-only posture
source-buffer literal NUL rejection
escaped decoded NUL visibility
semantic validation prerequisites
operator confirmation non-override policy
```

Validation refinement checkpoint

The current validation refinement is recorded in:

```
docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
```

Current refinement posture:

```
defensive_threat_model_validation_refinement_present=1
external_source_refresh_checkpoint_present=1
external_source_refresh_date=2026-05-26
manual_source_review_required=1
manual_source_review_completed_for_current_baseline=1
high_assurance_security_baseline_present=1
nsa_zero_trust_guideline_observed=1
nsa_cisa_memory_safe_languages_observed=1
cisa_fbi_product_security_bad_practices_observed=1
nist_high_assurance_references_observed=1
runtime_boundary_policy_expansion_next=1
abuse_case_fixture_expansion_next=1
certification_from_external_alignment=0
compliance_from_external_alignment=0
protection_from_external_alignment=0
security_controls_added=0
runtime_authority_granted=0
```

The refinement keeps the external standards alignment ledger source-tracking-only and does not close the runtime boundary or abuse-case fixture gaps.

Current gaps

```
external standards ledger needs recurring manual review before release
runtime boundary source needs fuller policy expansion after threat-model validation
abuse-case mapping needs broader fixture coverage
external advisory-by-advisory mapping is not complete
workload/service identity and host-integrity prerequisites are not yet profiled for future
authority
no certification or compliance mapping exists
```

Non-claims

This document does not implement security controls, runtime protection, malware prevention, ransomware prevention, sandboxing, exploit prevention, incident response, recovery behavior, certification, accreditation, compliance, production hardening, or operating-system completeness.

Validation command

```
sh scripts/test-defensive-threat-model-validation.sh
```

docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md

Source title: Latticra Defensive Threat Model Validation Refinement

Lines: 204 | Words: 727 | SHA-256: a8fe649b518bf932754aa8c6cf780fa5c4375089e74b474744d98f35f7237fce

Latticra Defensive Threat Model Validation Refinement

Status: defensive threat model validation refinement Source: local follow-up slice Scope: refinement of the existing defensive threat model validation ledger, external-source checkpoint posture, current gap triage, public entry points, and next-work pointer. This document does not implement security controls, runtime protection, runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, malware prevention, ransomware prevention, sandboxing, certification, accreditation, compliance, production hardening, or runtime authority.

Purpose

This refinement makes the defensive threat model validation surface more explicit after the policy-decision public-entry alignment.

It does three bounded things:

```
records the current validation refinement checkpoint
refreshes the external-source review posture without claiming compliance
names the next evidence gap as runtime boundary policy expansion after threat-model validation
```

It remains documentation and guard work only.

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/DEFENSIVE_THREAT_MODEL_CONTRACT.md
docs/DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
scripts/test-defensive-threat-model-contract.sh
scripts/test-defensive-threat-model-implementation-plan.sh
scripts/test-defensive-threat-model-validation.sh
scripts/test-defensive-threat-model-validation-refinement.sh
docs/RUNTIME_BOUNDARY_CONTRACT.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION_PLAN.md
docs/RUNTIME_BOUNDARY_IMPLEMENTATION.md
scripts/test-runtime-boundary.sh
```

Current refinement checkpoint

Current defensive threat model validation refinement posture:

```
defensive_threat_model_contract_present=1
defensive_threat_model_implementation_plan_present=1
defensive_threat_model_validation_present=1
defensive_threat_model_validation_refinement_present=1
defensive_threat_model_contract_guard_present=1
defensive_threat_model_plan_guard_present=1
defensive_threat_model_validation_guard_present=1
defensive_threat_model_validation_refinement_guard_present=1
protected_asset_matrix_present=1
trust_boundary_matrix_present=1
assumption_matrix_present=1
abuse_case_mapping_present=1
evidence_matrix_present=1
external_standards_alignment_ledger_present=1
validation_matrix_present=1
non_goal_matrix_present=1
compatibility_expectations_present=1
current_gaps_present=1
external_source_refresh_checkpoint_present=1
external_standards_refresh_needed=1
manual_source_review_required=1
manual_source_review_completed_for_current_baseline=1
high_assurance_security_baseline_present=1
nsa_zero_trust_guideline_observed=1
nsa_cisa_memory_safe_languages_observed=1
cisa_fbi_product_security_bad_practices_observed=1
nist_high_assurance_references_observed=1
runtime_boundary_policy_expansion_next=1
abuse_case_fixture_expansion_next=1
mode=validation-refinement
implementation_behavior_changed=0
security_controls_added=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_verification_added=0
signing_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
runtime_authority_granted=0
certification_claim_allowed=0
accreditation_claim_allowed=0
compliance_claim_allowed=0
runtime_protection_claim_allowed=0
malware_prevention_claim_allowed=0
ransomware_prevention_claim_allowed=0
sandbox_claim_allowed=0
```

```
production_protection_claim_allowed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

External-source refresh checkpoint

Date checked: 2026-05-26

The external-source posture remains source-tracking-only.

```
nsa_zero_trust_guideline_observed=1
nsa_cisa_memory_safe_languages_observed=1
fbi_cyber_page_reachable=1
fbi_recent_cyber_alerts_observed=1
cisa_secure_by_design_reference_verified=1
cisa_fbi_product_security_bad_practices_reference_verified=1
cisa_cpg_reference_verified=1
cisa_zero_trust_maturity_model_reference_verified=1
cisa_kev_reference_preserved=1
nist_csf_2_reference_verified=1
nist_ssdf_reference_verified=1
nist_sp_800_53_release_5_2_0_reference_verified=1
nist_sp_800_160_reference_verified=1
nist_sp_800_207_reference_verified=1
fips_140_3_reference_verified=1
manual_source_review_completed_for_current_baseline=1
recurring_manual_source_review_required=1
external_alignment_claim=source-tracking-only
certification_from_external_alignment=0
compliance_from_external_alignment=0
protection_from_external_alignment=0
```

Observed source posture:

```
| Source | 2026-05-26 refinement posture | Allowed use | Forbidden use |
| --- | --- | --- | --- |
| NSA Zero Trust Implementation Guidelines | 2026 Primer, Discovery Phase, Phase One, and Phase Two observed | zero-trust runtime-boundary planning | NSA endorsement, certification, or protection claim |
| NSA/CISA Memory Safe Languages CSI | 2025 CSI observed | memory-safety roadmap and C/C++ mitigation input | memory-safety guarantee |
| CISA Secure by Design | official source verified | secure-by-design vocabulary input | CISA compliance or product-security claim |
| CISA/FBI Product Security Bad Practices | official source verified | bad-practice exclusion list | CISA/FBI endorsement or compliance claim |
| CISA Cross-Sector Cybersecurity Performance Goals | official source verified | control-goal vocabulary input | CPG compliance claim |
| CISA Zero Trust Maturity Model | official source verified | zero-trust maturity vocabulary input | zero-trust certification claim |
| CISA Known Exploited Vulnerabilities Catalog | authoritative URL retained | vulnerability-awareness input | remediation guarantee |
| FBI Cyber | reachable; 2026 threat and reporting content visible | threat-environment awareness | FBI endorsement, protection claim, or incident-response capability claim |
| NIST CSF 2.0 / SSDF / SP 800-53 / SP 800-160 / SP 800-207 / FIPS 140-3 | official sources verified | high-assurance vocabulary and allocation input | NIST compliance, FIPS validation, or accreditation claim |
```

This refinement does not promote any external alignment entry to implementation-backed security status.

Refined gap triage

The validation ledger still names gaps rather than closing them.

Current next gap:

```
runtime boundary source needs fuller policy expansion after threat-model validation
```

Secondary gap:

```
abuse-case mapping needs broader fixture coverage
```

The next engineering slice should therefore stay on runtime-boundary policy expansion and evidence mapping before new behavior is considered.

Validation

This refinement is guarded by:

```
sh scripts/test-defensive-threat-model-validation-refinement.sh
```

The underlying threat model chain remains covered by:

```
sh scripts/test-defensive-threat-model-contract.sh
sh scripts/test-defensive-threat-model-implementation-plan.sh
sh scripts/test-defensive-threat-model-validation.sh
```

Expected output:

```
defensive_threat_model_validation_refinement: ok
```

Boundary

This refinement is documentation/status/guard alignment only.

It does not add security controls, runtime protection, malware prevention, ransomware prevention, sandboxing, exploit prevention, incident response, recovery behavior, certification, accreditation, compliance, production hardening, runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, or runtime authority.

Current next valid slice

The next valid Latticra slice is runtime boundary policy expansion after threat-model validation.

That future slice must preserve the no-effect posture and must not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production protection, or runtime authority.

docs/HIGH_ASSURANCE_SECURITY_BASELINE.md

Source title: Latticra High-Assurance Security Baseline

Lines: 341 | Words: 2938 | SHA-256: d46b1e1e3bef2cc18c1c9d6a7fe922c3a70297ec44d73513e38c4199ec59a3ea

Latticra High-Assurance Security Baseline

Status: high-assurance security baseline checkpoint Source refresh date: 2026-05-26 Scope: NSA, CISA, FBI, and NIST security guidance alignment for current Latticra source, documentation, tests, packaging lanes, and future infrastructure posture.

This baseline allocates required security work for Latticra. It does not certify Latticra, accredit Latticra, make Latticra a production security boundary, or claim compliance with any external framework.

Purpose

Latticra is an early-stage systems architecture and language implementation project. Because the project intentionally works near parsers, runtime boundaries, effect gates, cryptographic metadata, local installers, and future agentic automation surfaces, security work must track authoritative guidance before new authority is added.

This baseline records:

```
current official source inventory
required project allocations
memory-safety roadmap requirements
zero-trust runtime-boundary requirements
supply-chain and dependency requirements
```

cryptographic assurance requirements
identity, credential, and access management requirements
security logging, monitoring, and detection requirements
backup, recovery, and cyber resilience requirements
secure configuration and change management requirements
incident-response and disclosure requirements
non-claims that remain closed

Authoritative Source Inventory

Date checked: 2026-05-26

Source	Current guidance used	Latticra allocation
NSA Zero Trust Implementation Guidelines, Primer, Discovery Phase, Phase One, and Phase Two	Zero trust work begins with visibility over data, applications, assets, services, access, and authorization activity, then advances through phased activities, requirements, precursors, and successors toward target-level maturity.	Keep runtime authority denied until request identity, asset identity, authorization state, evidence level, prerequisites, successors, and audit output are visible.
NSA Advancing Zero Trust Maturity Throughout the User Pillar	ICAM capabilities are a core zero-trust maturity path for critical systems.	Require identity, credential, MFA, privileged-access, account-lifecycle, and identity-event evidence before hosted or privileged authority claims.
CISA and NSA Identity and Access Management: Recommended Best Practices for Administrators	Administrators should inventory assets and identities, understand security gaps, deploy and inventory MFA, collect SSO context, and monitor privileged behavior.	Keep identity and access management as a gated baseline before remote access, hosted administration, privileged sessions, or account-management claims.
NSA/CISA Memory Safe Languages CSI	Memory-safe language adoption should be considered, while existing code can be improved through interoperability and mitigations where migration is not practical.	Publish and guard a memory-safety roadmap for C/C++ substrate code; prefer memory-safe implementation for new high-risk infrastructure surfaces.
CISA Secure by Design	Manufacturers should take ownership of customer security outcomes, practice transparency/accountability, and lead from the top.	Require evidence-bound public claims, vulnerability reporting, no hidden security costs, deterministic reports, and leadership-owned security gates.
CISA/FBI Product Security Bad Practices	Avoid exceptionally risky practices, including absent memory-safety roadmaps, injection classes, known-insecure crypto, weak defaults, and unmanaged dependencies.	Guard against unsafe APIs, shell injection surfaces, default-secret patterns, unbounded buffers, path traversal, and production crypto claims without a module boundary.
NSA and CISA Red and Blue Teams Share Top Ten Cybersecurity Misconfigurations	Common enterprise compromise paths include default configurations, poor patch/configuration hygiene, weak controls, and visibility gaps.	Require secure configuration and change-management evidence before host, infrastructure, hardening, or production configuration claims.
CISA/NSA/FBI secure-by-design and secure-by-default principles	Secure products should ship with secure default baselines and should not shift avoidable hardening burden to customers.	Block secure-default and configuration-hardening claims until secure defaults, default-credential absence, checklist evidence, and exception ownership are recorded.
CISA Cross-Sector Cybersecurity Performance Goals	Establish baseline practices for asset inventory, vulnerability management, logging, account security, incident response, recovery, and third-party validation.	Maintain source and dependency inventory, KEV-aware release checks, explicit logs/reports, incident plan, recovery contracts before mutation, and third-party review before security release.
CISA/FBI/NSA international Best Practices for Event Logging and Threat Detection	Event logging supports continued operations and resilience through visibility, detection, and investigation across cloud, enterprise, mobile, and OT environments.	Require a security logging, monitoring, and detection baseline before hosted services, telemetry export, SIEM integration, or detection claims.
CISA Logging Made Easy and CISA logging guidance	Centralized log management, user activity visibility, alerting, and regular review improve threat detection.	Require event-source inventory, log centralization/export planning, redaction, triage ownership, and incident handoff before monitoring claims.
CISA Zero Trust Maturity Model v2	Mature zero trust uses identity, devices, networks, applications/workloads, and data pillars with visibility, analytics, automation, orchestration, and governance.	Treat each Latticra request as a per-request policy decision; preserve deny-by-default behavior for network, host, recovery, boot, and tool authority.
NIST SP 800-63-4 Digital Identity Guidelines	Current digital identity guidance covers identity proofing, authentication, authenticator management, federation, and assertions.	Use SP 800-63-4 vocabulary for future operator identity, account lifecycle, MFA, SSO/federation, and authentication non-claims.
CISA/NSA/FBI/MS-ISAC Phishing Guidance	Phishing-resistant MFA and social-engineering-resistant account processes reduce account-compromise risk.	Require phishing-resistant MFA planning and help-desk identity verification before privileged or remote access claims.

| CISA Known Exploited Vulnerabilities Catalog | Internet-facing exploited vulnerabilities need timely mitigation or compensating controls. | Before any internet-facing asset exists, create a KEV review gate; current project has no internet-facing runtime authority. |

| NIST National Vulnerability Database and CVSS metrics | CVE and CVSS records provide vulnerability enrichment and severity context, while CVSS is not a complete risk score by itself. | Require KEV/NVD review, component inventory, mitigation or exception records, and public non-claim review before release. |

| FBI Cyber | The active threat environment includes ransomware, nation-state targeting, critical infrastructure risk, rapidly changing IOCs/TTPs, and reporting through FBI/IC3 paths. | Keep the threat model defensive, keep reports private, add incident-reporting paths, and avoid offensive payload, persistence, exfiltration, and evasion content. |

| CISA/FBI/NSA/MS-ISAC #StopRansomware Guide | Ransomware and data-extortion preparation, prevention, response, evidence preservation, out-of-band communications, and federal contact paths must be planned before incidents. | Keep incident-response authority denied, document reporting paths, preserve evidence requirements, and require operator/legal review before notification or reporting assistance. |

| CISA/FBI/NSA/MS-ISAC #StopRansomware recovery guidance | Offline encrypted backups, regular backup testing, restoration prioritization, clean recovery environments, and post-incident lessons learned reduce ransomware recovery risk. | Require a backup, recovery, and cyber resilience baseline before hosted-service recovery, disaster-recovery, failover, rollback, or ransomware-recovery claims. |

| NIST Cybersecurity Framework 2.0 | Govern, Identify, Protect, Detect, Respond, and Recover provide current cybersecurity risk-management functions. | Map future infrastructure readiness to CSF 2.0 functions before production claims. |

| NIST SP 800-34 Rev. 1 and SP 800-184 | Contingency planning and cybersecurity event recovery require business impact analysis, recovery strategy, playbooks, realistic testing, and continuous improvement. | Require recovery scope, RTO/RPO, restore testing, rollback, recovery authorization, and lessons-learned evidence before recoverability claims. |

| NIST SP 800-128 | Security-focused configuration management uses baseline configuration, configuration control, monitoring, and SecCM practices to manage risk. | Require configuration item inventory, baseline configuration, owner, change record, testing, rollback, and drift-detection evidence before configuration claims. |

| NIST SP 800-70 Rev. 5 | Security configuration checklists help configure products to a risk posture, verify configuration, identify unauthorized change, and produce posture artifacts. | Require checklist evidence and unauthorized-change detection planning before secure configuration, hardening, or scanning claims. |

| NIST SP 800-92 and SP 800-92 Rev. 1 draft | Log management requires planning for generation, transmission, storage, access, analysis, retention, and disposal. | Use SP 800-92 vocabulary for future audit-event selection, log lifecycle, retention, access, and disposal evidence. |

| NIST SP 800-218 SSDF v1.1 | Secure development should prepare the organization, protect software, produce well-secured software, and respond to vulnerabilities. | Keep tests, threat-model docs, protected source/build processes, vulnerability handling, and root-cause-driven fixes in the quality gate. |

| NIST SP 800-53 Rev. 5, Release 5.2.0 | Control families cover access, audit, configuration, identification, incident response, risk assessment, system acquisition, system integrity, and supply-chain risk management. | Use SP 800-53 as the high-assurance control vocabulary for future production profiles, not as a current compliance claim. |

| NIST SP 800-160 Vol. 2 Rev. 1 | Cyber-resilient systems should anticipate, withstand, recover from, and adapt to adverse conditions and attacks. | Keep rollback, recovery, degraded-mode, auditability, and no-hidden-effect requirements ahead of any mutating infrastructure behavior. |

| NIST SP 800-207 Zero Trust Architecture | Zero trust removes implicit trust and makes access decisions resource-focused and least-privilege. | Runtime and agentic automation authority must remain per-request, least-privilege, auditable, and denied unless prerequisites pass. |

| FIPS 140-3 | Federal cryptographic module security requirements define validated module expectations and security levels. | No production cryptography or FIPS claim is allowed until a cryptographic module boundary, validation path, key lifecycle, and dependency review exist. |

| NIST SP 800-57 Part 1 Rev. 5, SP 800-131A Rev. 2, and SP 800-90 series | Key management, algorithm transition, and random-bit generation guidance define required cryptographic planning vocabulary. | Require a cryptographic assurance and key-management baseline before signing authority, key storage, entropy use, FIPS, or production crypto claims. |

| NSA/CISA/NIST post-quantum guidance and NSA CNSA 2.0 | High-assurance systems should inventory cryptography and plan migration toward quantum-resistant algorithms without premature deployment claims. | Require post-quantum inventory and migration planning before long-lived confidentiality or national-security-oriented cryptographic claims. |

Authoritative URLs for this checkpoint are preserved in docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md.

Tracked follow-on guidance for this checkpoint:

- CISA/FBI Secure by Design Alert: Eliminating Buffer Overflow Vulnerabilities sharpens the retained-C/C++ posture toward buffer-overflow-class defect reduction, adversarial testing, and memory-safe implementation preference for new high-risk surfaces.

- CISA Secure by Design Alert: Eliminating OS Command Injection Vulnerabilities reinforces the existing system()/popen() exclusions and keeps shell authority centralized behind reviewed boundaries.
- CISA The Case for Memory Safe Roadmaps reinforces the requirement for a component-by-component migration or mitigation record instead of a generic memory-safety aspiration.
- NIST SP 800-207A and NIST SP 1800-35 provide secondary zero-trust implementation detail for future workload, service, and device-integrity-aware authority surfaces.

Required Allocations

Current required allocation fields:

```
high_assurance_security_baseline_present=1
source_refresh_date=2026-05-26
official_source_inventory_present=1
memory_safety_roadmap_required=1
memory_safety_roadmap_present=1
zero_trust_runtime_boundary_required=1
ssdf_secure_development_required=1
cpg_operational_baseline_required=1
supply_chain_security_baseline_present=1
cyber_incident_reporting_response_baseline_present=1
vulnerability_management_release_gate_baseline_present=1
cryptographic_assurance_key_management_baseline_present=1
identity_credential_access_management_baseline_present=1
security_logging_monitoring_baseline_present=1
backup_recovery_resilience_baseline_present=1
secure_configuration_change_management_baseline_present=1
kev_release_review_required=1
fips_crypto_boundary_required_before_production_crypto=1
phishing_resistant_mfa_required_before_remote_privileged_access=1
security_event_logging_required_before_hosted_service=1
backup_restore_recovery_evidence_required_before_hosted_service=1
secure_configuration_change_control_required_before_hosted_service=1
sbom_required_before_production_installer=1
third_party_security_validation_required_before_security_release=1
incident_response_plan_required_before_production_service=1
recurring_source_review_required=1
implementation_behavior_changed=0
runtime_authority_granted=0
security_boundary_claimed=0
certification_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
production_protection_claim_allowed=0
```

Memory-Safety Roadmap

Latticra currently contains C and restricted C++ code. That remains allowed only under docs/security/C_CPP_SECURITY_PROFILE.md.

Required memory-safety posture:

```
new network-facing code -&gt; memory-safe language preferred
new cryptographic key-handling code -&gt; memory-safe language preferred
new parser for untrusted external input -&gt; memory-safe language preferred or restricted C
profile with fuzzing and exception record
new installer or host-mutation authority -&gt; memory-safe language preferred
existing C/C++ substrate -&gt; strict profile, bounded buffers, no unsafe string APIs, explicit
review of buffer overflow, format-string, off-by-one, use-after-free, and lifetime hazards,
sanitizer/static-analysis path
C/C++ unsafe exception -&gt; documented exception with lifetime, buffer, failure, test, and
reviewer fields
```

Short-term mitigations:

- keep strcpy, strcat, sprintf, vsprintf, gets, tmpnam, and tempnam forbidden;
- keep system() and popen() forbidden in source roots;
- keep shell authority centralized and guarded in installer code;

- keep pointer, buffer, lifetime, and ownership rules explicit at C ABI boundaries;
- keep parser and source-buffer invariants tested with malformed, boundary-length, truncation, and hostile-input cases;
- keep retained C/C++ high-risk paths covered by regression tests for buffer-overflow-class, format-string, off-by-one, and lifetime faults where practical;
- require fuzzing or a documented exception before any parser is promoted to a security boundary.

Long-term direction:

- prefer Rust or another memory-safe implementation language for new infrastructure-facing surfaces where practical;
- isolate C substrate code behind small ABI wrappers;
- keep C++ authority code narrow, RAII-based, and exception-free across C/security boundaries;
- publish component-by-component migration or mitigation notes before any production security claim.

Zero-Trust Runtime Requirements

No future runtime, agentic automation, network, host I/O, update, recovery, boot, or hardware request may be promoted unless all fields below are available and tested:

```
request_kind_known=1
requested_effect_known=1
caller_context_known=1
operator_identity_known=1
asset_or_resource_identity_known=1
workload_or_service_identity_known=1
device_or_host_integrity_context_known=1
mode_matches_request_family=1
authority_prerequisites_satisfied=1
operator_confirmation_is_metadata_only=1
policy_decision_reported=1
denial_reason_reported=1
audit_record_emitted=1
unknown_request_denied=1
unknown_effect_denied=1
future_gate_denied_until_contract=1
```

Supply-Chain Requirements

Before a production installer, production package, internet-facing service, or production update lane can be claimed, Latticra requires:

- SBOM evidence for shipped artifacts;
- dependency inventory with license and security review;
- KEV/NVD review or documented offline exception before release;
- vulnerability-management release gate before production release or supported-version claims;
- pinned CI actions and read-only workflow permissions;
- locked dependency builds where the package manager supports it;
- no ad hoc network client commands in workflows without a dedicated review guard;
- software update and patch integrity, authenticity, validation, and rollback evidence before any mutating update lane;
- third-party component update plan for supported product lifetime;

- vulnerability disclosure route and triage process.

Cryptographic Assurance Requirements

Current Seal cryptographic work is evidence and metadata oriented. It is not production cryptographic enforcement.

Production cryptography requires:

- defined cryptographic module boundary;
- approved algorithm and parameter inventory;
- key lifecycle and storage contract;
- randomness source contract;
- validation and self-test behavior;
- side-channel and sensitive-data handling review;
- FIPS 140-3 applicability decision;
- explicit non-FIPS disclosure if FIPS validation is not pursued;
- post-quantum inventory and migration planning before long-lived secrecy claims;
- cryptographic assurance and key-management baseline before production cryptography claims;
- authority-neutral crypto graduation gate before any verified receipt can influence future authority-bearing policy;
- test fixtures for malformed keys, unsupported algorithms, stale requests, replayed requests, and denied verification paths.

Identity, Credential, and Access Management Requirements

No future hosted service, hosted administration surface, privileged operator role, remote access path, service-account runtime authority, SSO/federation claim, MFA claim, or identity-security claim may be promoted until Latticra has:

- human, service, machine, and local identity inventories;
- privileged role and role-to-effect mapping;
- phishing-resistant MFA path for privileged and remote access;
- account lifecycle and deprovisioning contract;
- break-glass access policy and monitoring;
- credential storage, recovery, rotation, and reuse review;
- help-desk identity verification process for access changes;
- identity event logging and privileged behavior review;
- owner and expiration for every access exception;
- operator-visible identity and access non-claims.

Security Logging, Monitoring, and Detection Requirements

No future hosted service, production runtime, remote access path, privileged administration surface, SIEM integration, telemetry export, security monitoring claim, or detection-service claim may be promoted until Latticra has:

- security event source inventory and owner mapping;

- audit event selection and event schema contract;
- runtime policy decision and denial reason events;
- identity, access, privileged action, configuration change, and security error events;
- event severity taxonomy and time-source requirements;
- log redaction and secret-marker scanning;
- log integrity, access control, retention, and disposal policy;
- centralization or export path planning;
- alerting expectation for disabled critical log sources;
- detection triage owner and incident handoff path;
- operator-visible logging and monitoring non-claims.

Backup, Recovery, and Cyber Resilience Requirements

No future hosted service, production runtime, production installer, production package, mutating update lane, rollback path, failover path, ransomware recovery feature, disaster-recovery claim, continuity claim, or production recoverability claim may be promoted until Latticra has:

- critical asset inventory and dependency restore order;
- business impact or service-priority record;
- RTO and RPO evidence;
- backup scope and owner;
- offline, immutable, or otherwise protected backup path;
- backup encryption and access-control review;
- backup integrity verification and restore-test result;
- clean recovery environment plan;
- golden-image or infrastructure-as-code restore path;
- rollback plan and recovery authorization;
- recovery communication and incident-response handoff path;
- post-recovery validation and lessons-learned update path;
- owner and expiration for recovery exceptions;
- operator-visible recovery non-claims.

Secure Configuration and Change Management Requirements

No future hosted service, production runtime, production installer, production package, infrastructure automation lane, host mutation lane, configuration hardening claim, secure-default claim, configuration scanning claim, drift-detection claim, or compliance claim may be promoted until Latticra has:

- configuration item inventory and owners;
- secure baseline configuration record;
- configuration checklist or equivalent verification artifact;
- secure default and default-credential absence review;
- approved change request and risk review;

- change owner and test evidence;
- rollback plan before configuration mutation;
- drift-detection and unauthorized-change response plan;
- configuration secret review;
- configuration-change log event mapping;
- owner and expiration for configuration exceptions;
- operator-visible configuration non-claims.

Operational Security Requirements

Before any production service, hosted system, or critical infrastructure deployment:

- assign a named cybersecurity owner;
- maintain asset inventory and data-flow inventory;
- define secure configuration baseline and change-control evidence;
- require MFA/SSO for privileged accounts;
- require phishing-resistant MFA planning for remote and privileged access;
- prohibit shared administrative accounts and default credentials;
- collect security-relevant logs without leaking secrets;
- define log retention, integrity, time-source, triage, and incident-handoff evidence;
- define incident response and vulnerability disclosure procedures;
- publish a cyber incident reporting and response baseline before any incident-response feature;
- define backup, restore, and recovery evidence;
- publish a backup, recovery, and cyber resilience baseline before any recovery-service or failover claim;
- schedule table-top or third-party validation before security release;
- maintain recurring NSA/CISA/FBI/NIST source review.

Current Evidence

Current supporting local evidence:

```
SECURITY.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
docs/RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md
docs/MEMORY_SAFETY_ROADMAP.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md
docs/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md
docs/CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE.md
docs/LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md
docs/IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE.md
docs/SECURITY_LOGGING_MONITORING_BASELINE.md
docs/BACKUP_RECOVERY_RESILIENCE_BASELINE.md
docs/SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE.md
docs/security/C_CPP_SECURITY_PROFILE.md
docs/security/C_ABI_BOUNDARY_POLICY.md
scripts/test-quality-safety-guards.sh
scripts/test-defensive-threat-model-validation.sh
scripts/test-defensive-threat-model-validation-refinement.sh
scripts/test-high-assurance-security-baseline.sh
scripts/test-memory-safety-roadmap.sh
scripts/test-supply-chain-security-baseline.sh
```

```
scripts/test-cyber-incident-reporting-response-baseline.sh
scripts/test-vulnerability-management-release-gate-baseline.sh
scripts/test-cryptographic-assurance-key-management-baseline.sh
scripts/test-latticra-seal-crypto-graduation-gate.sh
scripts/test-identity-credential-access-management-baseline.sh
scripts/test-security-logging-monitoring-baseline.sh
scripts/test-backup-recovery-resilience-baseline.sh
scripts/test-secure-configuration-change-management-baseline.sh
```

Non-Claims

This baseline does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI-agent execution, model execution, tool execution, shell execution, production protection, sandboxing, malware prevention, ransomware prevention, incident response, recovery behavior, certification, accreditation, compliance, external endorsement, or runtime authority.

It also does not claim that Latticra is a finished operating system, hardened sandbox, production security product, high-assurance certified product, or critical infrastructure platform.

Validation

This baseline is guarded by:

```
sh scripts/test-high-assurance-security-baseline.sh
```

docs/IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE.md

Source title: Latticra Identity, Credential, and Access Management Baseline

Lines: 161 | Words: 453 | SHA-256: 0072d1c00462207cf4cbb65ca8d0aee61251e816a82b5b9b29f0164d5d345415

Latticra Identity, Credential, and Access Management Baseline

Status: identity, credential, and access management baseline Source refresh date: 2026-05-26 Scope: human identity, service identity, machine identity, privileged access, phishing-resistant MFA, SSO/federation context, account lifecycle, credential handling, break-glass access, identity event logging, and access-management non-claims before hosted services, remote access, privileged operator access, or identity-security claims.

This baseline records identity, credential, and access-management requirements only. It does not implement an identity provider, MFA provider, account provisioning, account deprovisioning, credential storage, remote access, privileged access, hosted administration, SSO federation, authorization enforcement, compliance, or runtime authority.

Authoritative Identity and Access Sources

Date checked: 2026-05-26

```
| Source | Latticra use |
| --- | --- |
| NSA Advancing Zero Trust Maturity Throughout the User Pillar | ICAM maturity, user identity,
credentials, access policy, and user-pillar visibility vocabulary |
| CISA and NSA Identity and Access Management: Recommended Best Practices for Administrators |
asset identity inventory, local identity inventory, MFA inventory, SSO context, and privileged
behavior monitoring vocabulary |
| NIST SP 800-63-4 Digital Identity Guidelines | identity proofing, authentication,
authenticator management, federation, assurance levels, subscriber accounts, and assertions
vocabulary |
| CISA Cross-Sector Cybersecurity Performance Goals | MFA, account security, log collection,
secure log storage, and privileged-access baseline expectations |
| CISA Require Multifactor Authentication and phishing-resistant MFA guidance |
phishing-resistant MFA preference for remote and privileged access |
| CISA/NSA/FBI/MS-ISAC Phishing Guidance: Stopping the Attack Cycle at Phase One |
```

phishing-resistant MFA, social-engineering resistance, and account-compromise reduction vocabulary |

| CISA IT and Product Design Sector-Specific Goals | MFA for software-development environments and authorization trust-relationship monitoring vocabulary |

Authoritative URLs:

```
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3328152/nsa-releases-recommendations-for-maturing-identity-credential-and-access-manage/
https://www.cisa.gov/sites/default/files/2023-12/ESF%20IDENTITY%20AND%20ACCESS%20MANAGEMENT%20RECOMMENDED%20BEST%20PRACTICES%20FOR%20ADMINISTRATORS%20PP-23-0248_508C.pdf
https://csrc.nist.gov/pubs/sp/800/63/4/final
https://csrc.nist.gov/pubs/sp/800/63/B/4/final
https://csrc.nist.gov/pubs/sp/800/63/C/4/final
https://www.cisa.gov/cross-sector-cybersecurity-performance-goals
https://www.cisa.gov/secure-our-world/require-multifactor-authentication
https://www.cisa.gov/news-events/news/cisa-nsa-fbi-ms-isac-publish-guide-preventing-phishing-intrusions
https://www.cisa.gov/news-events/news/cisa-releases-new-sector-specific-goals-it-and-product-design
```

Current Fields

```
identity_credential_access_management_baseline_present=1
identity_credential_access_management_guard_present=1
nsa_zero_trust_user_pillar_tracked=1
cisa_nsa_esf_iam_best_practices_tracked=1
nist_sp_800_63_4_digital_identity_tracked=1
cisa_cpg_account_security_tracked=1
phishing_guidance_tracked=1
it_product_design_mfa_goal_tracked=1
phishing_resistant_mfa_required_for_privileged_access=1
mfa_required_for_remote_access=1
privileged_access_inventory_required=1
service_account_inventory_required=1
local_account_inventory_required=1
account_lifecycle_contract_required=1
least_privilege_role_review_required=1
break_glass_account_policy_required=1
federation_sso_context_required=1
credential_secret_storage_review_required=1
credential_reuse_forbidden=1
default_credentials_forbidden=1
identity_event_logging_required=1
privileged_behavior_monitoring_required=1
help_desk_identity_verification_required=1
access_exception_owner_required=1
access_exception_expiration_required=1
implementation_behavior_changed=0
identity_provider_added=0
mfa_provider_added=0
account_provisioning_added=0
account_deprovisioning_added=0
remote_access_enabled=0
privileged_access_granted=0
credential_storage_added=0
hosted_admin_surface_added=0
identity_security_claim_allowed=0
hosted_service_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Required ICAM Promotion Gate

No hosted service, hosted administration surface, remote access path, privileged operator role, service-account runtime authority, SSO/federation claim, MFA claim, identity-security claim, or production access-management claim may be promoted until this gate is complete:

```
operator_identity_source_recorded=1
human_account_inventory_present=1
local_identity_inventory_present=1
service_identity_inventory_present=1
machine_identity_inventory_present=1
privileged_role_inventory_present=1
role_to_effect_mapping_recorded=1
least_privilege_review_recorded=1
phishing_resistant_mfa_path_recorded=1
```

```
mfa_exception_recorded=1
break_glass_account_recorded=1
break_glass_monitoring_recorded=1
session_lifetime_and_reauth_recorded=1
credential_storage_and_rotation_recorded=1
credential_recovery_path_recorded=1
help_desk_identity_verification_recorded=1
joiner_mover_leaver_process_recorded=1
identity_event_logging_recorded=1
privileged_behavior_review_recorded=1
authorization_trust_relationships_reviewed=1
access_exception_owner_recorded=1
access_exception_expiration_recorded=1
operator_visible_non_claims_recorded=1
```

Until this gate is complete:

```
production_identity_provider_allowed=0
remote_access_allowed=0
privileged_operator_access_allowed=0
service_account_runtime_authority_allowed=0
hosted_admin_console_allowed=0
password_only_privileged_access_allowed=0
default_credential_allowed=0
shared_admin_account_allowed=0
production_credential_storage_allowed=0
identity_security_claim_allowed=0
single_sign_on_claim_allowed=0
mfa_claim_allowed=0
hosted_service_claim_allowed=0
```

Latticra Boundary

Current Latticra identity and access posture is metadata-only and no-effect.

```
latticra_operator_identity_metadata_only=1
latticra_access_policy_metadata_only=1
latticra_authorization_enforcement_added=0
latticra_account_database_added=0
latticra_remote_login_added=0
latticra_privileged_session_added=0
latticra_identity_runtime_authority_granted=0
```

Current Evidence

Current supporting evidence:

```
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
SECURITY.md
scripts/test-high-assurance-security-baseline.sh
scripts/test-identity-credential-access-management-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-identity-credential-access-management-baseline.sh
```

docs/MEMORY_SAFETY_ROADMAP.md

Source title: Latticra Memory-Safety Roadmap

Lines: 100 | Words: 744 | SHA-256: 3450f66429a26cd0996e90e8140864bcd0582c01cfb3fa22f07d78cbdab5f9bf

Latticra Memory-Safety Roadmap

Status: memory-safety roadmap Source refresh date: 2026-05-26 Scope: component-level memory-safety allocation for Latticra C, restricted C++, Rust installer, shell guards, parser surfaces, Seal metadata, runtime-boundary metadata, and future high-risk infrastructure surfaces.

This roadmap responds to the NSA/CISA memory-safe-language guidance, CISA The Case for Memory Safe Roadmaps, the CISA/FBI buffer overflow alert, the CISA secure-design alert on OS command injection, and CISA/FBI product-security bad-practice guidance tracked in docs/HIGH_ASSURANCE_SECURITY_BASELINE.md.

It does not rewrite current components, certify memory safety, claim production hardening, or create a security boundary.

Current Fields

```
memory_safety_roadmap_present=1
memory_safety_roadmap_guard_present=1
high_assurance_security_baseline_present=1
c_cpp_security_profile_present=1
c_abi_boundary_policy_present=1
restricted_c_cpp_profile_required=1
memory_safe_language_preferred_for_new_high_risk_components=1
memory_safe_language_exception_contract_required=1
parser_fuzzing_required_before_security_boundary_claim=1
unsafe_exception_record_required=1
component_memory_safety_inventory_present=1
implementation_behavior_changed=0
runtime_authority_granted=0
security_boundary_claimed=0
memory_safety_guarantee_claimed=0
production_protection_claim_allowed=0
```

Component Inventory

Component family	Current language posture	Current mitigation	Promotion blocker
L-UI source buffer and parser surfaces	C	length-aware source buffers, literal-NUL policy, escaped decoded-NUL visibility, malformed-input tests, source-span tests, explicit attention to truncation, off-by-one, and lifetime faults	fuzzing corpus, hostile input budget, parser security-boundary contract
Lat parser, semantic validation, and LIR lowering	C	parse/semantic/lowering invariant tests, no runtime execution, deterministic diagnostics, explicit attention to buffer-overflow-class and format-string hazards	fuzzing corpus, memory-safe rewrite or exception contract before untrusted-service use
Seal key, public-key, signature, freshness, and receipt metadata	C	bounded key parsing tests, malformed-key tests, replay/stale request cases, metadata-only crypto posture	cryptographic module boundary, FIPS 140-3 applicability decision, sensitive-data handling review
Runtime boundary, Nucleus task, and effect-decision records	C	deny-by-default behavior, no-effect reports, unknown request/effect denial tests, operator confirmation non-override tests, no shell mutation or workload authority	zero-trust policy prerequisites, audit records, authority tests, distinct operator/workload identity context, no production runtime claim
Kernel metadata seed surfaces	C	metadata-only state reports, no boot/hardware/kernel isolation claim, bounded local tests	separate kernel safety contract, memory-safe or restricted-C exception plan before real kernel authority
Restricted C++ authority layer	C++	constrained profile, RAII/value-type preference, no exceptions across C/security boundaries, no raw owning pointers, explicit review of allocator and ownership expansion	exception review for any unsafe ownership, allocator, ABI, or dependency expansion
Installer UI and engine	Rust	`cargo check --locked`, no Rust unsafe blocks, centralized process-launch boundary, symlink/path traversal guards, no raw shell interpolation path	no live mutation without evidence bundle, rollback contract, update authenticity evidence, and verification transcript
Shell guard and packaging lanes	POSIX shell	fail-fast scripts, private tempdirs, no ad hoc network clients, reviewed privilege/package-manager allowlists, centralized shell boundary rules	no production installer or release lane without SBOM, dependency review, and command-boundary review
GitHub workflow surface	YAML and shell commands	pinned actions, read-only permissions, no `pull_request_target`, no secrets, no implicit token use	dedicated review guard before write tokens, OIDC, secrets, or release publishing
Future network, MCP, server, update, recovery, boot, or hardware surfaces	not implemented	future-gated and denied	memory-safe implementation preferred; restricted C/C++ requires exception contract and tests

Required Rules

New high-risk components must follow this decision path:

1. Prefer Rust or another memory-safe implementation language.

2. If C/C++ is used, document why a memory-safe implementation is not practical.
3. Apply docs/security/C_CPP_SECURITY_PROFILE.md and docs/security/C_ABI_BOUNDARY_POLICY.md.
4. Add malformed-input, boundary-length, truncation, null-rule, ownership, lifetime, and failure-mode tests.
5. Add explicit review for buffer overflow, format-string, off-by-one, use-after-free, and command-construction hazards where applicable.
6. Add fuzzing before parser or external-input code can be promoted as a security boundary.
7. Keep production protection, memory-safety guarantee, compliance, certification, and external endorsement claims closed until evidence exists.

Current Guard Coverage

Current guard coverage includes:

```
unsafe_c_string_api_guard=1
source_shell_exec_guard=1
unsafe_python_api_guard=1
rust_installer_unsafe_block_guard=1
rust_installer_process_launch_boundary_guard=1
workflow_pinned_action_guard=1
workflow_read_only_permission_guard=1
workflow_secret_usage_guard=1
private_tempdir_guard=1
strict_warning_flag_guard=1
```

These guards reduce risk; they do not prove memory safety.

Required Future Evidence Before Promotion

Before any Latticra component is described as memory-safe, hardened, production-ready, or a security boundary, it needs:

- component-specific threat model;
- input-size and allocation limits;
- malformed-input and edge-case tests;
- hazard-class review for buffer overflow, format-string, off-by-one, lifetime, and command-construction failure modes where applicable;
- sanitizer or static-analysis transcript where supported;
- fuzzing plan and corpus for parsers or external-input surfaces;
- unsafe exception inventory for any remaining C/C++ high-risk code;
- dependency and SBOM review for shipped artifacts;
- public non-claim review.

Validation

This roadmap is guarded by:

```
sh scripts/test-memory-safety-roadmap.sh
```

docs/SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE.md

Source title: Latticra Secure Configuration and Change Management Baseline

Lines: 160 | Words: 463 | SHA-256: c4b0fc4f3e3353e41fd2232231d2d76eb6313c08e20d615cabffd55c77cc596b

Latticra Secure Configuration and Change Management Baseline

Status: secure configuration and change management baseline Source refresh date: 2026-05-26 Scope: secure configuration baselines, configuration item inventory, configuration checklists, secure defaults, approved change records, rollback evidence, drift detection, exception ownership, and configuration non-claims before hosted services, production installers, production runtime, infrastructure automation, or hardening claims.

This baseline records secure configuration and change-management requirements only. It does not implement host configuration, infrastructure configuration, configuration scanning, configuration enforcement, drift detection, change approval workflow, rollback execution, compliance, or runtime authority.

Authoritative Configuration Sources

Date checked: 2026-05-26

Source	Latticra use
NIST SP 800-128 Guide for Security-Focused Configuration Management of Information Systems	security-focused configuration management, baseline configuration, configuration control, configuration monitoring, and SecCM vocabulary
NIST SP 800-70 Rev. 5 National Checklist Program for IT Products	security configuration checklist, secure posture verification, unauthorized change detection, and checklist artifact vocabulary
NIST SP 800-53 Rev. 5 Configuration Management controls	CM family vocabulary for baseline configuration, change control, configuration settings, least functionality, and configuration monitoring
CISA Cross-Sector Cybersecurity Performance Goals	secure configuration, default password, asset inventory, vulnerability reduction, and operational baseline context
CISA/FBI Product Security Bad Practices	default password, insecure default, missing security artifact, and customer-risk reduction blockers
NSA and CISA Red and Blue Teams Share Top Ten Cybersecurity Misconfigurations	common misconfiguration classes, default configurations, poor patch/configuration hygiene, and enterprise misconfiguration visibility vocabulary
CISA/NSA/FBI secure-by-design and secure-by-default principles	secure default baseline, manufacturer ownership of customer security outcomes, and no burden-shifting configuration expectations

Authoritative URLs:

```
https://csrc.nist.gov/pubs/sp/800/128/upd1/final
https://csrc.nist.gov/pubs/sp/800/70/r5/final
https://csrc.nist.gov/Pubs/sp/800/53/r5/upd1/Final
https://www.cisa.gov/cybersecurity-performance-goals-cpgs
https://www.cisa.gov/resources-tools/resources/product-security-bad-practices
https://www.cisa.gov/news-events/alerts/2025/01/17/cisa-and-fbi-release-updated-guidance-product-security-bad-practices
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3549369/nsa-and-cisa-advise-on-top-ten-cybersecurity-misconfigurations/
https://www.cisa.gov/news-events/news/us-and-international-partners-publish-secure-design-and-default-principles-and-approaches
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3361073/nsa-us-and-international-partners-issue-guidance-on-securing-technology-by-desi/
```

Current Fields

```
secure_configuration_change_management_baseline_present=1
secure_configuration_change_management_guard_present=1
nist_sp_800_128_configuration_management_tracked=1
nist_sp_800_70_rev5_checklist_tracked=1
nist_sp_800_53_configuration_management_tracked=1
cisa_cpg_secure_configuration_tracked=1
cisa_fbi_product_security_bad_practices_config_tracked=1
nsa_cisa_top_misconfigurations_tracked=1
cisa_nsa_fbi_secure_by_default_tracked=1
configuration_item_inventory_required=1
secure_baseline_configuration_required=1
configuration_checklist_required=1
approved_change_record_required=1
configuration_change_owner_required=1
configuration_change_risk_review_required=1
configuration_change_test_evidence_required=1
configuration_rollback_plan_required=1
configuration_drift_detection_required=1
default_credential_forbidden=1
insecure_default_configuration_forbidden=1
configuration_secret_review_required=1
configuration_exception_owner_required=1
configuration_exception_expiration_required=1
implementation_behavior_changed=0
configuration_enforcement_added=0
configuration_scanner_added=0
host_configuration_changed=0
infrastructure_configuration_changed=0
change_approval_workflow_added=0
drift_detection_added=0
rollback_execution_added=0
production_configuration_claim_allowed=0
hosted_service_configuration_claim_allowed=0
configuration_hardening_claim_allowed=0
secure_default_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Required Configuration Promotion Gate

No hosted service, production runtime, production installer, production package, infrastructure automation lane, host mutation lane, configuration hardening claim, secure-default claim, configuration scanning claim, drift-detection claim, compliance claim, or production configuration claim may be promoted until this gate is complete:

```
configuration_item_inventory_present=1
configuration_owner_recorded=1
baseline_configuration_recorded=1
configuration_checklist_recorded=1
secure_default_review_recorded=1
default_credential_absence_recorded=1
insecure_default_configuration_absence_recorded=1
change_request_recorded=1
change_owner_recorded=1
change_risk_review_recorded=1
change_test_evidence_recorded=1
rollback_plan_recorded=1
drift_detection_plan_recorded=1
configuration_secret_review_recorded=1
configuration_log_event_recorded=1
exception_owner_recorded=1
exception_expiration_recorded=1
operator_visible_non_claims_recorded=1
```

Until this gate is complete:

```
host_configuration_change_allowed=0
production_configuration_claim_allowed=0
secure_default_claim_allowed=0
configuration_hardening_claim_allowed=0
configuration_scanning_claim_allowed=0
configuration_enforcement_allowed=0
drift_detection_claim_allowed=0
hosted_service_configuration_claim_allowed=0
infrastructure_as_code_claim_allowed=0
compliance_claim_allowed=0
```

Latticra Boundary

Current Latticra configuration-related records remain evidence and no-effect contract work.

```
latticra_configuration_metadata_only=1
latticra_installer_config_authority_allowlist_guarded=1
latticra_installer_ui_artifact_authority_guarded=1
latticra_host_configuration_mutation_added=0
latticra_configuration_enforcement_added=0
latticra_configuration_scanning_added=0
latticra_drift_detection_added=0
latticra_change_approval_workflow_added=0
latticra_configuration_authority_granted=0
```

Current Evidence

Current supporting evidence:

```
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
docs/SECURITY_LOGGING_MONITORING_BASELINE.md
docs/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md
docs/security/C_CPP_SECURITY_PROFILE.md
installer/latticra-installer/src/config.rs
installer/scripts/latticra-installer-apply.sh
SECURITY.md
scripts/test-installer-config-authority-allowlist.sh
scripts/test-installer-ui-artifact-authority.sh
scripts/test-high-assurance-security-baseline.sh
scripts/test-secure-configuration-change-management-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-secure-configuration-change-management-baseline.sh
```

docs/security/C_ABI_BOUNDARY_POLICY.md

Source title: Latticra C ABI Boundary Policy

Lines: 220 | Words: 810 | SHA-256: 56a4833d91bed4fc38d0acf02e7bf8582bc72ef10c2a4e24f4a9fc7a8862ec95

Latticra C ABI Boundary Policy

Status: Draft Layer: Security / ABI Applies to: C-facing entry points, C/C++ interop, platform shims, plugin-like surfaces if introduced.

1. Purpose

This document defines how Latticra exposes and consumes C ABI boundaries.

C ABI boundaries are necessary for platform control, bootstrapping, interop, and long-term stability. They are also dangerous if they carry implicit ownership, unchecked buffers, ambiguous errors, or hidden mutation.

Latticra treats every C ABI boundary as a security-relevant boundary.

2. Core rules

```
No ownership crosses implicitly.
No lifetime crosses implicitly.
No mutation crosses without authorization.
No buffer crosses without length.
No result crosses without status.
No platform-specific behavior hides in generic ABI code.
```

3. Handles

Opaque handles are preferred over exposed implementation structures.

Example:

```
typedef struct latticra_handle latticra_handle;
```

Rules:

- callers must not know the internal layout;
- handle creation and destruction must be explicit;
- null handle behavior must be defined;
- invalid handle behavior must be defined;
- handle ownership must be documented.

4. Status codes

C ABI functions must return explicit status codes unless there is a strong reason not to.

Example:

```
typedef enum {
    LATTICRA_OK = 0,
    LATTICRA_ERR_DENIED,
    LATTICRA_ERR_INVALID_CONTRACT,
    LATTICRA_ERR_OUT_OF_RANGE,
    LATTICRA_ERR_UNSUPPORTED,
    LATTICRA_ERR_INTERNAL
} latticra_status;
```

Rules:

- never collapse all failures into one generic failure if policy decisions matter;
- distinguish denied, invalid, unsupported, and internal failures;
- avoid leaking sensitive details across public ABI;
- provide a safe diagnostic path where appropriate.

5. Buffers

All byte buffers must include an explicit length.

Example:

```
typedef struct {
    const unsigned char* data;
    unsigned long length;
} latticra_bytes;
```

Rules:

- no buffer without length;
- no null data pointer unless length is zero and explicitly allowed;
- maximum allowed lengths should be defined for external input;
- buffer ownership must remain with the caller unless otherwise documented;
- mutable output buffers must include capacity and written length.

6. Strings

Strings crossing the ABI must be treated as byte buffers with encoding rules.

Rules:

- define UTF-8 or other encoding explicitly;
- define whether null termination is required;

- never assume null termination if length is provided;
- validate before converting to internal string types;
- reject or normalize invalid encoding according to the specific boundary contract.

7. Ownership

Ownership transfer must be explicit in the function name or documentation.

Preferred naming:

```
create / destroy
borrow / view
copy
take
release
```

Rules:

- borrowed data may not outlive the call unless explicitly documented;
- created handles must have a corresponding destroy function;
- output allocations must specify who frees them and how;
- allocator crossing is forbidden unless an ABI allocator contract exists.

8. Mutation

Mutation must be intentional.

Rules:

- no mutation through const-violating casts;
- no hidden global mutation;
- no policy mutation through generic utility calls;
- mutation that affects authority must pass through an effect gate;
- mutation must produce audit evidence when it is security-relevant.

9. Example ABI shape

```
typedef struct latticra_handle latticra_handle;

typedef enum {
    LATTICRA_OK = 0,
    LATTICRA_ERR_DENIED,
    LATTICRA_ERR_INVALID_CONTRACT,
    LATTICRA_ERR_OUT_OF_RANGE,
    LATTICRA_ERR_UNSUPPORTED,
    LATTICRA_ERR_INTERNAL
} latticra_status;

typedef struct {
    const unsigned char* data;
    unsigned long length;
} latticra_bytes;

latticra_status latticra_validate_contract(
    latticra_handle* handle,
    latticra_bytes input
);

latticra_status latticra_destroy(
    latticra_handle* handle
);
```

10. C++ implementation mapping

C ABI functions should be thin wrappers around restricted C++ authority logic.

Example conceptual mapping:

```
C ABI call
  -&gt; argument shape check
  -&gt; safe view construction
  -&gt; C++ validator / authority object
  -&gt; typed result
  -&gt; status code mapping
  -&gt; audit record if security-relevant
```

C ABI wrappers should not contain complex policy logic unless unavoidable.

11. Audit requirements

Security-relevant ABI calls should produce audit records when they:

- validate contracts;
- deny requests;
- authorize effects;
- mutate policy state;
- execute platform effects;
- load external modules;
- alter runtime state;
- access sensitive platform data.

Audit records should not leak secrets through public output.

12. Review checklist

Before accepting a C ABI function:

- [ ] Does every pointer have a documented null rule?
- [ ] Does every buffer have a length?
- [ ] Does every mutable output have a capacity?
- [ ] Is ownership explicit?
- [ ] Is lifetime explicit?
- [ ] Are status codes precise enough?
- [ ] Are inputs validated before internal use?
- [ ] Does the function avoid hidden global mutation?
- [ ] Does privileged behavior pass through an effect gate?
- [ ] Does security-relevant behavior produce audit evidence?
- [ ] Is platform-specific behavior isolated?

13. Promotion rule

No ABI surface should be described as stable until:

- the function contract is documented;
- tests exist for invalid inputs;
- null and boundary behavior are tested;
- ownership and lifetime rules are documented;
- audit behavior is documented where relevant.

docs/security/C_CPP_SECURITY_PROFILE.md

Source title: Latticra C/C++ Security Profile

Lines: 322 | Words: 1546 | SHA-256: 775ae4fb07b75276db4ac64d8faa1843881a6bc2af200e0eaabbedd2b6c0f3cb

Latticra C/C++ Security Profile

Status: Draft Layer: Security Applies to: C substrate, restricted C++ governed authority layer, trusted core code.

1. Purpose

This document defines the required C/C++ security profile for Latticra.

Latticra may use C and C++ only under a restricted profile. This profile exists because C and C++ allow unsafe behavior unless the project explicitly prevents, isolates, or reviews that behavior.

The goal is not to pretend C/C++ is memory-safe by default. The goal is to establish a disciplined systems profile with narrow unsafe zones, explicit review, hard tooling, and evidence-bound validation.

2. Core posture

Latticra follows these principles:

1. C is the narrow substrate.
2. C++ is the governed authority layer.
3. Unsafe behavior is isolated and documented.
4. Ownership is explicit.
5. Lifetime is explicit.
6. Buffer boundaries are explicit.
7. Effects are gated.
8. Privileged mutation is auditable.
9. External input is validated.
10. Security claims require evidence.

2A. Memory-safety roadmap alignment

Latticra aligns its C/C++ posture with current NSA/CISA memory-safe-language guidance and CISA/FBI product-security bad-practice guidance.

This does not mean existing substrate code must be rewritten all at once. It does mean the project must keep a visible memory-safety roadmap and prefer a memory-safe implementation language for new high-risk components.

Required allocation:

```
memory_safety_roadmap_required=1
memory_safe_language_preferred_for_new_high_risk_components=1
restricted_c_cpp_profile_required_for_existing_substrate=1
unsafe_exception_record_required=1
sanitizer_or_static_analysis_path_required=1
fuzzing_required_before_parser_security_boundary_claim=1
```

High-risk components include:

- network-facing code;

- cryptographic key handling;
- parsers for untrusted external input;
- installer or host-mutation authority;
- update, recovery, rollback, boot, hardware, or agentic automation authority;
- IPC, privilege, capability, or policy-enforcement boundaries.

When a high-risk component remains in C or C++, it must use the restricted profile in this document, keep unsafe behavior isolated, document why a memory-safe alternative is not used, and add targeted tests for malformed input, length boundaries, ownership, lifetime, and failure behavior.

3. Allowed C++ patterns in trusted core

The following patterns are preferred:

- RAII for cleanup and resource release;
- value types for simple data transfer;
- enum class for typed state and policy names;
- std::array where size is static;
- span/view-style APIs for non-owning buffers where available and appropriate;
- std::unique_ptr for unique heap ownership when dynamic allocation is permitted;
- explicit Result<T, E> or status/error-return types;
- immutable configuration objects after initialization;
- small policy objects with narrow responsibilities;
- typed capability objects;
- deterministic destructors;
- explicit move semantics;
- dependency-free or dependency-minimal modules;
- pure validation functions where possible;
- parser/validator separation.

4. Restricted or forbidden C++ patterns in trusted core

The following are forbidden or require explicit review exception:

Pattern	Status	Reason
Raw owning pointers	Forbidden	Ownership must be explicit.
`new` / `delete` in ordinary code	Forbidden	Use RAII and approved allocators.
Unchecked pointer arithmetic	Forbidden	High memory safety risk.
`reinterpret_cast`	Exception-only	Platform or ABI code only, with review notes.
C-style casts	Forbidden	Cast intent must be explicit.
Exceptions across C ABI or security boundaries	Forbidden	Failure must be explicit and deterministic.
RTTI in trusted/freestanding core	Restricted	Avoid hidden runtime assumptions.
Hidden global mutable state	Forbidden	Breaks auditability and determinism.
Varargs	Forbidden	Type safety and boundary risk.
Template metaprogramming for core policy	Restricted	Avoid unreadable authority logic.
Complex inheritance trees	Restricted	Prefer composition and final small types.
Silent integer narrowing	Forbidden	Must use checked conversion.
Unbounded string/buffer operations	Forbidden	Buffer length must be explicit.
Large unreviewed dependencies	Forbidden	Dependency trust must be reviewed.

5. Allowed C patterns

C code is allowed only for substrate responsibilities:

- boot and early runtime;
- C ABI surfaces;
- platform shims;
- freestanding helpers;
- fixed-layout interop structs;
- hardware-facing entry points;
- minimal wrappers around platform mechanisms.

C code should be small, boring, and directly reviewable.

6. Restricted or forbidden C patterns

The following C patterns are forbidden or exception-only:

Pattern	Status
Unbounded `strcpy`, `strcat`, `sprintf`, similar APIs	Forbidden
Unchecked `memcpy` / `memmove` sizes	Forbidden
Implicit ownership transfer through raw pointers	Forbidden
Mutable global state	Exception-only
Macro-heavy control logic	Restricted
Type-punning through incompatible pointers	Exception-only
Undocumented inline assembly	Forbidden
Platform-specific behavior in generic files	Forbidden
Undefined behavior reliance	Forbidden
Silent integer truncation	Forbidden

7. Boundary rules

No C/C++ boundary may rely on implicit ownership or lifetime.

Boundary rule set:

```
No lifetime crosses the C ABI boundary.
No ownership crosses without an explicit handle.
No mutation crosses without an effect gate.
No error crosses without a typed status or result.
No buffer crosses without pointer + length + validation.
No privileged action crosses without audit intent.
```

8. Error handling

Trusted code should avoid hidden failure paths.

Preferred:

```
Result&lt;T, LatticraError&gt;
StatusCode
Expected&lt;T, E&gt;
```

Required behavior:

- all failure states must be explicit;
- no security boundary should throw exceptions across ABI;
- no privileged effect should silently fail;
- user-facing reports must not leak sensitive internals;
- audit records must distinguish denied, invalid, unsupported, and internal error states.

9. Allocation policy

Early substrate and platform code should avoid dynamic allocation unless explicitly approved.

C++ authority layer may use dynamic allocation only after allocator policy is defined.

Required allocation posture:

- boot/substrate: no ordinary heap dependency unless explicitly documented;
- trusted core: deterministic allocation preferred;
- no allocation inside critical sections unless justified;
- all ownership must be visible in the type system;
- allocation failure must be handled intentionally.

10. Input validation policy

All external input must pass through validation before use.

Examples:

- contract files;
- command input;
- serialized policy data;
- configuration files;
- plugin descriptors;
- platform-reported data;
- filesystem paths;
- network-originated input if networking exists later.

Validation must produce typed output. Downstream authority code should operate on validated types, not raw input.

11. Effect gate policy

Any action that mutates system state, opens access, executes code, loads external data, alters policy, or touches hardware-like surfaces must pass through an effect gate.

Effect gate requirements:

- request type;
- caller context;
- validation result;
- authorization decision;
- denied-state behavior;
- audit record;
- bounded effect;
- deterministic return value.

12. Dependency policy

Dependencies are not accepted only because they are convenient.

A dependency requires:

- purpose;
- license review;
- build impact review;
- security history review where practical;
- transitive dependency review;
- freestanding compatibility review if applicable;
- replacement/removal plan if the dependency becomes unsuitable.

Trusted core dependencies should be rare.

12A. CISA/FBI product-security bad-practice exclusions

The trusted core and future infrastructure surfaces must exclude the following classes unless an explicit exception contract exists:

- no hardcoded default passwords, tokens, or private keys;
- no command injection surface through implicit shells;
- no direct SQL string construction if a database layer is ever introduced;
- no cross-site scripting exposure if web UI or rendered HTML input becomes dynamic;
- no directory traversal in installer, package, update, recovery, or artifact paths;
- no known-insecure or deprecated cryptographic algorithms for production data protection;
- no unmanaged third-party dependency with unknown maintenance or security status;
- no production claim for memory-unsafe high-risk code without a memory-safety roadmap;
- no security release with known exploitable vulnerabilities lacking mitigation notes.

Cryptographic code has an additional rule: production cryptography requires a documented module boundary and a FIPS 140-3 applicability decision before any FIPS, federal, or high-assurance cryptographic claim is allowed.

13. Build and analysis expectations

Baseline debug/CI expectations:

```
-Wall
-Wextra
-Wpedantic
-Wconversion
-Wshadow
-Werror where practical
clang-tidy cppcoreguidelines-* where practical
CERT-oriented static analysis where available
AddressSanitizer where supported
UndefinedBehaviorSanitizer where supported
ThreadSanitizer for concurrent modules where supported
fuzzing for parsers and contract loaders
```

Baseline release expectations:

```
stack protector where supported
FORTIFY/source hardening where supported
PIE/RELRO/NX/canaries where applicable
LTO only after sanitizer-clean builds
reproducible build notes where practical
no unreviewed unsafe/platform files
```

Tool availability depends on platform and build mode. Unsupported tools must be documented rather than silently ignored.

14. Review labels

Suggested code review labels:

- trusted-core
- c-substrate
- cpp-authority
- abi-boundary
- unsafe-exception
- effect-gate
- parser
- validator
- audit
- platform-specific
- dependency-review

15. Unsafe exception template

Any exception to this profile should include:

```
File:
Function:
Rule being excepted:
Reason:
Why safer alternative is not used:
Scope of unsafe behavior:
How lifetime is bounded:
How buffer size is bounded:
How failure is handled:
Test coverage:
Reviewer:
Date:
```

16. Promotion rule

A feature may not be described as secure, hardened, or safe merely because it follows this profile.

A feature can only be promoted with precise evidence:

- tests pass;
- analyzers run or exceptions are documented;
- unsafe exceptions are reviewed;
- parser inputs are fuzzed where relevant;
- ABI boundaries are documented;
- effect gates exist for privileged behavior;
- audit records exist for privileged behavior.

docs/SECURITY_LOGGING_MONITORING_BASELINE.md

Source title: Latticra Security Logging, Monitoring, and Detection Baseline

Lines: 166 | Words: 499 | SHA-256: 6c412cf903285aa19cf00aaa39a1362fbd7a33711cce6ba81ccac67bcc02c6bb

Latticra Security Logging, Monitoring, and Detection Baseline

Status: security logging, monitoring, and detection baseline Source refresh date: 2026-05-26 Scope: security event source inventory, audit event selection, runtime-authority decision logs, identity and access events, privileged actions, log redaction, log integrity, retention, disposal, time source, detection triage, incident handoff, and monitoring non-claims before hosted services, production monitoring, detection services, SIEM export, or security operations claims.

This baseline records logging, monitoring, and detection requirements only. It does not implement a log collector, SIEM, telemetry export, host sensor, network sensor, detection rule, alerting service, log storage, monitoring service, incident detection service, compliance, or runtime authority.

Authoritative Logging and Detection Sources

Date checked: 2026-05-26

Source	Latticra use
CISA/FBI/NSA international Best Practices for Event Logging and Threat Detection	event logging baseline, threat detection, cloud/enterprise/mobile/OT visibility, and resource-aware logging vocabulary
NSA release for Best Practices for Event Logging and Threat Detection	national-security, DoD, DIB, and living-off-the-land detection context
CISA Logging Made Easy	centralized log management, user activity visibility, Sysmon data, real-time alerting, and locally run privacy caveats
CISA Use Logging on Business Systems	what-to-log, centralization, monitoring, alerting, and log-review planning vocabulary
CISA Cross-Sector Cybersecurity Performance Goals	log collection, account security, incident response, and baseline operational practice context
NIST SP 800-92 Guide to Computer Security Log Management	log management infrastructure, generation, transmission, storage, access, analysis, and disposal vocabulary
NIST SP 800-92 Rev. 1 initial public draft	current log-management planning playbook vocabulary for future improvement plans
NIST Cybersecurity Framework 2.0 Detect function	security continuous monitoring and event discovery vocabulary
NIST SP 800-53 Rev. 5 Audit and Accountability controls	audit event, audit record, audit storage, audit review, and audit reduction vocabulary

Authoritative URLs:

https://www.cisa.gov/resources-tools/resources/best-practices-event-logging-and-threat-detection
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3880942/nsa-joins-allies-in-releasing-best-practices-for-event-logging/
https://www.cisa.gov/resources-tools/services/logging-made-easy
https://www.cisa.gov/use-logging-business-systems
https://www.cisa.gov/cybersecurity-performance-goals-cpgs
https://csrc.nist.gov/pubs/sp/800/92/final
https://csrc.nist.gov/pubs/sp/800/92/r1/ipd
https://www.nist.gov/cyberframework/detect
https://csrc.nist.gov/Pubs/sp/800/53/r5/upd1/Final

Current Fields

```
security_logging_monitoring_baseline_present=1
security_logging_monitoring_guard_present=1
cisa_fbi_nsa_event_logging_guidance_tracked=1
nsa_event_logging_release_tracked=1
cisa_logging_made_easy_tracked=1
cisa_use_logging_on_business_systems_tracked=1
cisa_cpg_log_collection_tracked=1
nist_sp_800_92_log_management_tracked=1
nist_sp_800_92_rev1_draft_tracked=1
nist_csf_detect_function_tracked=1
nist_sp_800_53_audit_accountability_tracked=1
security_event_source_inventory_required=1
audit_event_selection_required=1
runtime_authority_decision_logging_required=1
identity_access_event_logging_required=1
privileged_action_logging_required=1
security_error_logging_required=1
configuration_change_logging_required=1
log_redaction_required=1
secret_free_log_guard_required=1
log_integrity_tamper_resistance_required=1
time_synchronization_required=1
retention_disposal_policy_required=1
critical_log_source_disable_alert_required=1
detection_triage_owner_required=1
incident_handoff_path_required=1
operator_log_access_review_required=1
implementation_behavior_changed=0
log_collector_added=0
siem_added=0
telemetry_export_added=0
host_sensor_added=0
network_sensor_added=0
alerting_service_added=0
log_storage_added=0
detection_rule_added=0
production_monitoring_claim_allowed=0
detection_service_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Required Logging and Monitoring Promotion Gate

No hosted service, production runtime, remote access path, privileged administration surface, security monitoring claim, detection-service claim, SIEM integration, telemetry export, or production audit claim may be promoted until this gate is complete:

```
security_event_source_inventory_present=1
log_source_owner_recorded=1
log_schema_or_field_contract_present=1
audit_event_selection_recorded=1
runtime_decision_event_recorded=1
denial_reason_event_recorded=1
identity_access_event_recorded=1
privileged_action_event_recorded=1
configuration_change_event_recorded=1
security_relevant_error_event_recorded=1
event_severity_taxonomy_recorded=1
time_source_recorded=1
log_redaction_review_recorded=1
secret_marker_scan_recorded=1
log_integrity_control_recorded=1
log_access_control_recorded=1
retention_period_recorded=1
disposal_process_recorded=1
centralization_or_export_path_recorded=1
critical_log_source_disable_alert_recorded=1
detection_triage_owner_recorded=1
incident_handoff_path_recorded=1
operator_visible_non_claims_recorded=1
```

Until this gate is complete:

```
production_log_monitoring_allowed=0
production_audit_claim_allowed=0
siem_integration_claim_allowed=0
telemetry_export_allowed=0
```

```
host_monitoring_allowed=0
network_monitoring_allowed=0
alerting_service_allowed=0
detection_service_claim_allowed=0
security_operations_claim_allowed=0
log_collection_service_claim_allowed=0
```

Latticra Boundary

Current Latticra logging and reporting work remains local, deterministic, and no-effect.

```
latticra_local_report_metadata_only=1
latticra_runtime_policy_decision_reports_local=1
latticra_report_redaction_boundary_guarded=1
latticra_secret_material_guarded=1
latticra_log_collection_service_added=0
latticra_remote_telemetry_added=0
latticra_detection_runtime_added=0
latticra_monitoring_authority_granted=0
```

Current Evidence

Current supporting evidence:

```
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md
docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
SECURITY.md
scripts/test-report-redaction-boundary.sh
scripts/test-secret-material-guard.sh
scripts/test-high-assurance-security-baseline.sh
scripts/test-security-logging-monitoring-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-security-logging-monitoring-baseline.sh
```

docs/strategy/2026-05-15-2249-cdt-national-security-open-system-strategy.md

Source title: Latticra National-Security Open-System Strategy

Lines: 208 | Words: 871 | SHA-256: 1fd7f5235eb3ac77e0bb897baab7305f9fac0c0dcdb2aee2d7851d212f9b5107

Latticra National-Security Open-System Strategy

Status: active strategy record Created: 2026-05-15 22:49 CDT Last updated: 2026-05-16 16:15 CDT Review cadence: update at major milestone boundaries or whenever mission, threat model, status, funding, or target users change.

Motto

“The simulacrum is never what hides the truth.” — Jean Baudrillard

Mission

Latticra is intended to become a contract-first open systems architecture and programming-language platform built with national-security-grade discipline from the beginning.

The long-term mission is to make general-purpose computing more resistant to malware, ransomware, unauthorized persistence, hidden mutation, unclear execution, and opaque system behavior.

This mission is defensive. It is not a claim that Latticra currently prevents malware, prevents ransomware, provides a hardened sandbox, or replaces an operating system.

Core thesis

Open systems can be made stronger when they are:

```
auditable
contract-driven
deterministic where possible
source-aware
operator-visible
effect-gated
small enough to review
honest about non-claims
tested before promoted
```

Latticra should favor transparent security architecture over hidden magic.

Direction checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Strategic interpretation:

```
C: secure substrate, boot paths, ABI boundaries, platform shims.
C++: governed authority layer, policy, validators, effect gates, audit logic.
Lat / Latticra Language: contract and declaration layer.
```

This is a constrained C/C++ foundation direction. It does not mean unrestricted C++.

Primary product direction

The long-term product direction is a complete open-source operating-system universe composed of:

- Latticra systems architecture;
- constrained C/C++ substrate and authority layers;
- Lat / Latticra Programming Language;
- L-UI operator interface declarations;
- LIR / Latticra Intermediate Representation;
- Nucleus supervisor architecture;
- explicit state lattices;
- effect gates;
- deterministic diagnostics;
- evidence-driven promotion;
- security-aware execution boundaries;
- eventual runtime, recovery, update, and deployment models.

Target users

Primary target users:

- intellectuals;
- scientists;
- computer scientists;
- security researchers;
- defensive engineering teams;
- public-interest infrastructure maintainers;

- government infrastructure stakeholders;
- operators who need auditable and explainable computing systems.

Security design posture

Latticra should be built as though hostile inputs, hostile environments, and operator mistakes are expected.

Design posture:

```
default deny
no hidden execution
no hidden mutation
no network behavior without gates
no hardware behavior without gates
explicit trust boundaries
explicit source spans
explicit diagnostics
stable error codes
transparent state reports
contract before capability
```

Anti-malware and anti-ransomware goal

The long-term goal is to reduce the effectiveness of malware and ransomware by designing a system where dangerous behavior is harder to hide, harder to trigger accidentally, and easier to inspect.

Early mechanisms that support this goal:

- source-aware parsing;
- semantic validation;
- LIR shape planning;
- Lat grammar implementation;
- constrained C substrate behavior;
- constrained C++ authority-layer contract;
- governed C++ authority-layer planning;
- deterministic diagnostics;
- no-effect defaults;
- effect classification;
- operator-visible reports;
- explicit string and source-buffer policies;
- contracts before implementation;
- tests before promotion.

Current non-claim:

```
Latticra does not currently prevent malware or ransomware.
```

Current estimated completion

Estimated as of 2026-05-16 16:15 CDT:

```
| Area | Estimate |
| --- | ---: |
| Overall Latticra system | 19% |
| L-UI parser / AST / string foundation | 86% |
| Foundation docs and contracts | 74% |
| Lat / Latticra Programming Language | 10% |
```

```
| LIR | 10% |
| C/C++ foundation direction | 14% |
| Constrained C++ authority layer | 4% |
| Nucleus real task execution | 10% |
| Runtime / OS-universe direction | 5% |
| Security-hardening implementation | 5% |
| Public product readiness | 5% |
```

These percentages are rough planning estimates, not formal release metrics.

Strategic priorities

1. Keep public docs professional and self-contained.
2. Keep security claims evidence-bound.
3. Preserve the constrained C/C++ foundation direction.
4. Move from constrained C++ authority-layer contract into implementation planning.
5. Keep Lat metadata-only until separate lowering, execution, or runtime contracts exist.
6. Define constrained C++ authority-layer implementation plans before C++ policy/validator implementation.
7. Expand Nucleus from preview/reporting into carefully gated execution planning.
8. Build status reporting and public progress estimates.
9. Maintain project notes so current direction and upcoming work remain clear.
10. Preserve quality, clarity, and consistency across every public artifact.

Quality requirements

Every major change should be:

```
small enough to review
well named
contracted or documented
tested or guarded
honest about boundaries
safe by default
consistent with Latticra naming
consistent with the C/C++ foundation direction
free of unsupported security claims
```

Funding strategy

Latticra should support sponsorship through Bryforge while keeping the repository technically credible.

Funding links should be visible but not intrusive.

Current sponsorship account:

```
Buy Me a Coffee: Bryforge
```

Next strategic slice

Recommended next slice:

```
Constrained C++ authority layer implementation plan
```

This turns the constrained C++ authority-layer contract into exact API, namespace, file path, build, exception, RTTI, allocation, ownership, C ABI, validator, audit report, and test planning before implementation code.

Non-claims

This strategy does not implement an operating system, sandbox, malware defense, ransomware defense, kernel, recovery system, update system, hardware system, production runtime, unrestricted C++ authority, implemented C++ authority layer, or security boundary.

It records strategic direction and quality expectations only.

docs/SUPPLY_CHAIN_SECURITY_BASELINE.md

Source title: Latticra Supply-Chain Security Baseline

Lines: 160 | Words: 815 | SHA-256: 019fd4d9fe0dcac3afc72a6eba43d7c04c5d6f911fc61e4260a4a88993a87ccb

Latticra Supply-Chain Security Baseline

Status: supply-chain security baseline Source refresh date: 2026-05-26 Scope: repository, CI, dependency, package, installer, artifact, SBOM, release, and update-lane security posture for current Latticra work.

This baseline turns the supply-chain requirements in docs/HIGH_ASSURANCE_SECURITY_BASELINE.md into a guarded local project contract.

It also carries forward the stricter update-lane, command-boundary, and integrity expectations implied by NIST SP 800-53 Release 5.2.0, CISA KEV practice, and the CISA secure-design alert on OS command injection.

It does not publish artifacts, release packages, add signing authority, claim SLSA level, claim NIST compliance, claim CISA CPG compliance, or make Latticra production-ready.

Current Fields

```
supply_chain_security_baseline_present=1
supply_chain_security_guard_present=1
high_assurance_security_baseline_present=1
ssdf_supply_chain_profile_present=1
cpg_supply_chain_profile_present=1
sbom_required_before_production_installer=1
kev_nvd_review_required_before_release=1
dependency_inventory_required=1
pinned_ci_actions_required=1
read_only_workflow_permissions_required=1
persist_credentials_false_required=1
pull_request_target_forbidden=1
repository_secret_use_requires_dedicated_review=1
implicit_github_token_use_requires_dedicated_review=1
repository_secret_filename_scan_required=1
repository_secret_content_marker_scan_required=1
sensitive_local_artifact_filename_guard_required=1
report_redaction_boundary_guard_required=1
whole_environment_report_dump_forbidden=1
installer_engine_log_redaction_required=1
installer_config_authority_slug_allowlist_required=1
installer_command_wrapper_strict_name_required=1
installer_ui_artifact_authority_guard_required=1
installer_ui_artifact_write_validation_required=1
installer_console_output_authority_guard_required=1
installer_console_config_reflection_denial_required=1
locked_dependency_builds_required=1
offline_installer_builds_required=1
ad_hoc_network_client_commands_forbidden_without_guard=1
source_archive_fixture_tracked_unignored_source_view_required=1
source_archive_fixture_symlink_refusal_required=1
source_archive_fixture_reproducible_metadata_required=1
release_publishing_authority_granted=0
production_installer_claim_allowed=0
production_update_claim_allowed=0
compliance_claim_allowed=0
certification_claim_allowed=0
external_endorsement_claimed=0
```

Current Guarded Controls

Surface	Current control	Evidence
GitHub workflows	actions pinned to approved commit SHAs	`scripts/test-quality-safety-guards.sh`
GitHub token permissions	workflows require explicit read-only repository permissions	`scripts/test-quality-safety-guards.sh`
Checkout credentials	checkout steps require `persist-credentials: false`	`scripts/test-quality-safety-guards.sh`
Pull request trust boundary	`pull_request_target` is forbidden	`scripts/test-quality-safety-guards.sh`
Secrets and tokens	workflow secret and implicit token use are blocked without a dedicated guard	`scripts/test-quality-safety-guards.sh`
Repository secret material	secret-bearing filenames, private-key blocks, and common live-token markers are blocked from source files	`scripts/test-secret-material-guard.sh`
Report and log redaction	whole-environment dumps, shell xtrace, and unredacted installer child logs are blocked	`scripts/test-report-redaction-boundary.sh`
Installer config authority labels	profile, strategy, channel, and command-wrapper values are constrained to reviewed ASCII slug/name allowlists before rendering or install	`scripts/test-installer-config-authority-allowlist.sh`
Installer UI artifacts	Panel save/plan paths validate and sanitize authority fields before writing config or plan artifacts	`scripts/test-installer-ui-artifact-authority.sh`
Installer console output	Panel console report commands validate authority fields before reflecting config-derived values	`scripts/test-installer-console-output-authority.sh`
Workflow network clients	ad hoc `curl`, `wget`, SSH, FTP, and netcat-style commands are blocked without a dedicated guard	`scripts/test-quality-safety-guards.sh`
Package-manager mutation	package-manager use is limited to reviewed workflow/script allowlists	`scripts/test-quality-safety-guards.sh`
Source archive fixtures	package source archives require tracked/unignored source selection, symlink refusal, and deterministic tar/gzip metadata	`scripts/test-quality-safety-guards.sh`
Rust installer dependency use	local quality uses `cargo check --locked`	`Makefile`
Live installer build path	installer apply script requires locked offline Cargo builds and keeps the process-launch boundary centralized	`installer/scripts/latticra-installer-apply.sh`
Local installer artifacts	SBOM path is explicit and currently `none` for non-production fixtures	`docs/LOCAL_INSTALLER_ARTIFACT_MANIFEST_CONTRACT.md`
Boot artifact manifest fixture	SBOM fields are required before real boot artifact acceptance	`docs/SEABIOS_GRUB_BOOT_PREVIEW_BOOT_ARTIFACT_MANIFEST_VALIDATION.md`
Ubuntu package readiness	third-party material, notice, license, generated-artifact, and trademark reviews remain blocked until formal review	`docs/UBUNTU_READINESS_PLAN.md`

Required Release Gate

No production package, production installer, internet-facing service, update lane, boot artifact, or release artifact may be described as production-ready until the following gate is complete:

```
release_artifact_inventory_present=1
sbom_present=1
sbom_reviewed=1
dependency_inventory_reviewed=1
third_party_material_reviewed=1
license_notice_reviewed=1
kev_nvd_review_completed=1
known_exploited_vulnerability_mitigation_recorded=1
workflow_write_permission_reviewed=1
release_secret_boundary_reviewed=1
artifact_integrity_hashes_recorded=1
signing_authority_contract_present=1
update_payload_integrity_reviewed=1
update_authenticity_path_reviewed=1
update_rollback_evidence_present=1
command_boundary_reviewed=1
rollback_or_recovery_contract_present=1
vulnerability_disclosure_path_present=1
production_non_claim_review_completed=1
```

Until this gate is complete, the required posture is:

```
sbom_present_for_production_release=0
release_artifact_published=0
release_signing_performed=0
release_secret_access_granted=0
```

```
release_write_token_granted=0
production_installer_claim_allowed=0
production_update_claim_allowed=0
production_security_claim_allowed=0
```

Dependency Review Rules

Every new dependency, vendored artifact, generated artifact, or bundled binary requires:

- purpose and ownership;
- license and notice review;
- security history review where practical;
- transitive dependency review where practical;
- KEV/NVD check before release or documented offline exception;
- build reproducibility note where practical;
- update authenticity and integrity impact review where the dependency can affect installer, package, or update behavior;
- removal or replacement plan if the dependency becomes unsuitable.

CI Authority Rules

CI changes must preserve:

- explicit workflow permissions;
- read-only default repository token posture;
- no pull_request_target;
- no repository secrets without dedicated review;
- no implicit GITHUB_TOKEN, GH_TOKEN, ACTIONS_ID_TOKEN, or runtime token use without dedicated review;
- no committed local secret files, private-key blocks, or common live-token markers in repository source;
- no whole-environment dumps, shell xtrace, or unredacted child-process output in report/log paths;
- no installer profile, update-channel, strategy, or command-wrapper string may cross into plans, config artifacts, or install scripts unless it passes the authority slug/name allowlist;
- no installer UI save or plan artifact may be written from an unvalidated or network-authority-enabled config;
- no ad hoc network client commands without dedicated review;
- no new command-construction or shell-interpolation path without a dedicated command-boundary review;
- no mutating update lane without integrity, authenticity, validation, and rollback evidence;
- no release publishing without an explicit release-authority contract.

Validation

This baseline is guarded by:

```
sh scripts/test-supply-chain-security-baseline.sh
sh scripts/test-secret-material-guard.sh
```

```
sh scripts/test-report-redaction-boundary.sh
sh scripts/test-installer-config-authority-allowlist.sh
sh scripts/test-installer-ui-artifact-authority.sh
sh scripts/test-installer-console-output-authority.sh
```

docs/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE.md

Source title: Latticra Vulnerability Management Release Gate Baseline

Lines: 142 | Words: 346 | SHA-256: e5e0530b67f0bce612ed91e9cff99f70f4c360c7f301630aac4290557308f0be

Latticra Vulnerability Management Release Gate Baseline

Status: vulnerability management release gate baseline Source refresh date: 2026-05-26 Scope: CISA KEV review, NVD/CVE review, coordinated vulnerability disclosure, dependency/component inventory, release blocking, mitigation/exception records, and public non-claims before production release, update, installer, package, internet-facing service, or security-product claims.

This baseline records vulnerability-management release gates only. It does not run vulnerability scans, query live feeds, publish advisories, submit CVEs, patch dependencies, produce an SBOM, publish releases, grant release authority, or claim product security.

Authoritative Vulnerability Sources

Date checked: 2026-05-26

```
| Source | Latticra use |
| --- | --- |
| CISA Known Exploited Vulnerabilities Catalog | exploited-in-the-wild prioritization input
before release or internet-facing claims |
| NIST National Vulnerability Database and CVSS metrics | CVE/CVSS enrichment vocabulary,
severity context, and non-risk-score caveat |
| CISA Coordinated Vulnerability Disclosure Program | disclosure coordination and mitigation
workflow vocabulary |
| CISA Vulnerability Disclosure Policy Template | good-faith reporting, authorized testing,
report intake, and disclosure-scope vocabulary |
| CISA/FBI Product Security Bad Practices | avoid release posture that ignores known exploited
vulnerabilities or fails to document non-exploitability claims |
```

Authoritative URLs:

```
https://www.cisa.gov/known-exploited-vulnerabilities-catalog
https://nvd.nist.gov/vuln-metrics/cvss
https://www.cisa.gov/coordinated-vulnerability-disclosure-process
https://www.cisa.gov/vulnerability-disclosure-policy-template
https://www.cisa.gov/resources-tools/resources/product-security-bad-practices
https://www.cisa.gov/news-events/alerts/2025/01/17/cisa-and-fbi-release-updated-guidance-product
-security-bad-practices
```

Current Fields

```
vulnerability_management_release_gate_baseline_present=1
vulnerability_management_release_gate_guard_present=1
cisa_kev_catalog_tracked=1
nvd_cve_review_required=1
cvss_context_required_not_risk_score_only=1
coordinated_vulnerability_disclosure_required=1
vulnerability_disclosure_policy_scope_required=1
dependency_component_inventory_required=1
sbom_required_before_production_release=1
kev_nvd_review_required_before_release=1
known_exploited_vulnerability_mitigation_required=1
non_exploitability_claim_requires_written_record=1
vulnerability_exception_owner_required=1
vulnerability_exception_expiration_required=1
release_block_on_unmitigated_known_exploited_vulnerability=1
internet_facing_asset_inventory_required_before_release=1
security_advisory_process_required_before_supported_release=1
implementation_behavior_changed=0
live_feed_query_added=0
vulnerability_scan_added=0
release_publishing_authority_granted=0
security_advisory_published=0
cve_submission_performed=0
sbom_generated=0
production_release_claim_allowed=0
product_security_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Required Release Gate

No production release, production installer, production package, update lane, internet-facing service, hosted service, security-product claim, or supported-version claim may be promoted until this gate is complete:

```
release_artifact_inventory_present=1
component_inventory_present=1
dependency_inventory_reviewed=1
sbom_present=1
sbom_reviewed=1
cpe_or_purl_mapping_reviewed=1
cisa_kev_review_completed=1
nvd_cve_review_completed=1
known_exploited_vulnerability_mitigation_recorded=1
critical_high_vulnerability_exception_recorded=1
non_exploitability_claim_evidence_recorded=1
vulnerability_exception_owner_recorded=1
vulnerability_exception_expiration_recorded=1
vulnerability_disclosure_path_present=1
coordinated_disclosure_escalation_path_present=1
security_advisory_template_present=1
supported_version_scope_recorded=1
release_non_claim_review_completed=1
```

Until this gate is complete:

```
release_artifact_published=0
production_release_claim_allowed=0
production_installer_claim_allowed=0
production_update_claim_allowed=0
supported_version_claim_allowed=0
security_product_claim_allowed=0
vulnerability_free_claim_allowed=0
known_exploited_vulnerability_exception_allowed=0
internet_facing_service_claim_allowed=0
security_advisory_publication_allowed=0
```

Exception Record Requirements

Any future exception for an unresolved vulnerability must include:

```
vulnerability_identifier_recorded=1
affected_component_recorded=1
affected_version_recorded=1
exposure_context_recorded=1
exploitability_analysis_recorded=1
```

```
mitigation_or_compensating_control_recorded=1
owner_recorded=1
review_deadline_recorded=1
operator_visible_status_recorded=1
public_claim_review_recorded=1
```

Current Evidence

Current supporting evidence:

```
docs/HIGH_ASSURANCE_SECURITY_BASELINE.md
docs/SUPPLY_CHAIN_SECURITY_BASELINE.md
docs/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE.md
SECURITY.md
scripts/test-high-assurance-security-baseline.sh
scripts/test-supply-chain-security-baseline.sh
scripts/test-vulnerability-management-release-gate-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-vulnerability-management-release-gate-baseline.sh
```

docs/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md

Source title: Latticra Zero-Trust Runtime Authority Baseline

Lines: 144 | Words: 477 | SHA-256: f71f774383331de1fb1703473c919d6477d7dd316bdf2d8a2ff81b8f31b86af6

Latticra Zero-Trust Runtime Authority Baseline

Status: zero-trust runtime authority baseline Source refresh date: 2026-05-26 Scope: future runtime, tool, host I/O, network, server/MCP, update, recovery, boot, hardware, agentic automation, and authority-bearing request paths.

This baseline turns the current defensive threat model and runtime-boundary policy expansion into a guarded Latticra runtime-authority contract.

It does not implement runtime execution, network access, host I/O, tool execution, MCP behavior, recovery behavior, boot behavior, hardware behavior, policy enforcement, capability enforcement, or production protection.

Authoritative Zero-Trust Sources

Date checked: 2026-05-26

```
| Source | Latticra use |
| --- | --- |
| NSA Zero Trust Implementation Guidelines: Primer, Discovery Phase, Phase One, and Phase Two |
phased runtime-authority prerequisites, dependency ordering, and evidence visibility |
| CISA Zero Trust Maturity Model Version 2.0 | pillars and cross-cutting capabilities for
maturity vocabulary |
| NIST SP 800-207 Zero Trust Architecture | per-request, least-privilege, resource-focused
access decision model |
| NIST SP 800-207A | future cloud-native/service-mesh access-control vocabulary if server or MCP
 surfaces appear |
```

Authoritative URLs:

```
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4393480/nsa-
releases-phase-one-and-phase-two-of-the-zero-trust-implementation-guidelines/
https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4378980/nsa-
releases-first-in-series-of-zero-trust-implementation-guidelines/
https://www.cisa.gov/resources-tools/resources/zero-trust-maturity-model
https://www.nist.gov/publications/zero-trust-architecture-0
https://csrc.nist.gov/pubs/sp/800/207/a/final
```

Current Fields

```
zero_trust_runtime_authority_baseline_present=1
zero_trust_runtime_authority_guard_present=1
defensive_threat_model_validation_refinement_present=1
runtime_boundary_policy_expansion_after_threat_model_present=1
zero_trust_runtime_boundary_required=1
per_request_authorization_required=1
least_privilege_effect_scope_required=1
resource_identity_required=1
caller_identity_required=1
asset_inventory_required_before_authority=1
policy_decision_visibility_required=1
denial_reason_visibility_required=1
audit_record_required_before_authority=1
operator_confirmation_metadata_only_required=1
unknown_request_denial_required=1
unknown_effect_denial_required=1
future_gate_denial_required=1
runtime_execution_added=0
tool_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
runtime_authority_granted=0
production_protection_claim_allowed=0
zero_trust_certification_claim_allowed=0
external_endorsement_claimed=0
```

Pillar Mapping

Pillar	Latticra requirement before authority
Identity	caller, operator, automation profile, and request family must be explicit
Devices / execution environment	local runner, host context, workspace boundary, and platform scope must be recorded
Networks / environment	network effects remain denied until a network authority contract and telemetry path exist
Applications / workloads	tool, MCP, model, runtime, signer, updater, and installer workloads require separate authority contracts
Data	target resources, files, manifests, receipts, keys, artifacts, and reports must be identified and bounded
Visibility and analytics	decision, denial reason, gate state, matrix cell, and evidence level must be reportable
Automation and orchestration	automation cannot bypass policy, and operator confirmation cannot override denied effects
Governance	status records, non-claims, and guard scripts must block promotion by wording alone

Required Runtime Authority Gate

No future request may execute, mutate, read host files, write host files, open network connections, run tools, invoke MCP, load model artifacts, sign, update, recover, touch boot state, or affect hardware until this gate is complete:

```
request_kind_known=1
requested_effect_known=1
caller_identity_known=1
operator_or_automation_context_known=1
resource_identity_known=1
resource_sensitivity_classified=1
execution_environment_known=1
mode_matches_request_family=1
authority_prerequisites_satisfied=1
least_privilege_scope_recorded=1
policy_decision_reported=1
denial_reason_reported=1
audit_record_emitted=1
evidence_level_recorded=1
operator_confirmation_recorded_as_metadata_only=1
operator_confirmation_non_override_test_present=1
unknown_request_denial_test_present=1
unknown_effect_denial_test_present=1
future_gate_denial_test_present=1
blocked_effect_denial_test_present=1
rollback_or_failure_behavior_defined_for_mutation=1
```

```
non_claim_review_completed=1
```

Until this gate is complete:

```
runtime_execution_allowed=0
tool_execution_allowed=0
host_read_allowed=0
host_write_allowed=0
network_open_allowed=0
mcp_invocation_allowed=0
model_execution_allowed=0
signing_authority_allowed=0
update_authority_allowed=0
recovery_authority_allowed=0
boot_authority_allowed=0
hardware_authority_allowed=0
operator_confirmation_override_allowed=0
```

Current Evidence

Current supporting evidence:

```
docs/RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md
docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
scripts/test-runtime-boundary-policy-expansion-after-threat-model.sh
scripts/test-runtime-boundary-abuse-case-fixtures.sh
scripts/test-zero-trust-runtime-authority-baseline.sh
```

Validation

This baseline is guarded by:

```
sh scripts/test-zero-trust-runtime-authority-baseline.sh
```

Part XI - Platform Packaging and Distribution Lanes

- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Artifact Naming Contract
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Build Environment Contract
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Build Gate Contract
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md - Debian, FreeBSD, and OpenBSD Package Input Handoff Lane
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Install/Remove Transcript Contract
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Payload Inspection Contract
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Publication Non-Claim Review Contract
- docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md - Debian, FreeBSD, and OpenBSD Package Validation Promotion Blocker Matrix Contract
- docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md - Debian, FreeBSD, and OpenBSD Source Archive Contract
- docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md - Debian, FreeBSD, and OpenBSD Source Archive Fixture Lane

- docs/DEBIAN_LOCAL_DEB_STATIC_VALIDATION.md - Debian Local Deb Static Validation
- docs/FEDORA_DEVELOPER_WORKFLOW.md - Fedora Developer Workflow
- docs/FEDORA_DISPOSABLE_VM_EFFECT_GATE.md - Fedora Disposable VM Effect Gate
- docs/FEDORA_DISPOSABLE_VM_EFFECT_GATE_CLASSIFIER.md - Fedora Disposable VM Effect Gate Classifier
- docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_LANE.md - Fedora Disposable VM Local RPM Validation Lane
- docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_TRANSCRIPT_CONTRACT.md - Fedora Disposable VM Local RPM Validation Transcript Contract
- docs/FEDORA_HOST_INSTALL_PREFLIGHT_CLASSIFIER.md - Fedora Host Install Preflight Classifier
- docs/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_IMPLEMENTATION.md - Fedora Install Preflight Snapshot Capture Implementation
- docs/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_PLAN.md - Fedora Install Preflight Snapshot Capture Plan
- docs/FEDORA_INSTALL_VERIFICATION_PLAN.md - Fedora Install Verification Plan
- docs/FEDORA_LIVE_READONLY_SNAPSHOT_ADAPTER.md - Fedora Live Read-Only Snapshot Adapter
- docs/FEDORA_LOCAL_BINARY_RPM_BUILD_PLAN.md - Fedora Local Binary RPM Build Plan
- docs/FEDORA_LOCAL_RPM_EXECUTION_LANE.md - Fedora Installroot RPM Mutation Lane
- docs/FEDORA_LOCAL_RPM_INSTALL_MUTATION_GATE_CONTRACT.md - Fedora Local RPM Install Mutation Gate Contract
- docs/FEDORA_LOCAL_RPM_REMOVAL_ROLLBACK_PLAN.md - Fedora Local RPM Removal and Rollback Plan
- docs/FEDORA_LOCAL_RPM_SPEC_DRAFT_PLAN.md - Fedora Local RPM Spec Draft Plan
- docs/FEDORA_LOCAL_RPM_STATIC_VALIDATION.md - Fedora Local RPM Static Validation
- docs/FEDORA_LOCAL_RPM_VALIDATION_PLAN.md - Fedora Local RPM Validation Plan
- docs/FEDORA_MANUAL_HOST_DRY_RUN_TRANSCRIPT_CONTRACT.md - Fedora Manual Host Dry-Run Transcript Contract
- docs/FEDORA_MANUAL_HOST_RC_CHECKLIST.md - Fedora Manual Host Release-Candidate Checklist
- docs/FEDORA_MANUAL_HOST_RC_DECISION_CLASSIFIER.md - Fedora Manual Host RC Decision Classifier
- docs/FEDORA_PACKAGE_METADATA_PLAN.md - Fedora Package Metadata Plan
- docs/FEDORA_READINESS_PLAN.md - Fedora Readiness Plan
- docs/FEDORA_RPM_GATE_CLASSIFIER.md - Fedora RPM Gate Classifier
- docs/FEDORA_RPM_REMOVAL_ROLLBACK_CLASSIFIER.md - Fedora RPM Removal Rollback Classifier
- docs/FEDORA_RPM_REMOVAL_ROLLBACK_CLASSIFIER_CONTRACT.md - Fedora RPM Removal Rollback Classifier Contract
- docs/FEDORA_RPMLINT_AVAILABILITY.md - Fedora rpmlint Availability Lane
- docs/FEDORA_RPMLINT_STATIC_SPEC_LANE.md - Fedora rpmlint Static Spec Lane
- docs/FEDORA_SOURCE_ARCHIVE_FIXTURE_LANE.md - Fedora Source Archive Fixture Lane

- docs/FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN.md - Fedora VM CLI Payload Next Validation Lane Plan
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT.md - Fedora VM CLI Payload Repeatability Evidence Acceptance Contract
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE.md - Fedora VM CLI Payload Repeatability Evidence Publication Gate
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE.md - Fedora VM CLI Payload Repeatability Evidence Review Gate
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR.md - Fedora VM CLI Payload Repeatability Evidence Status Review Validator
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE.md - Fedora VM CLI Payload Repeatability Evidence Status Template
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN.md - Fedora VM CLI Payload Repeatability Runner Plan
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE.md - Fedora VM CLI Payload Repeatability Transcript Capture Template
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT.md - Fedora VM CLI Payload Repeatability Transcript Contract
- docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md - Fedora VM CLI Payload Repeatability Transcript Review Validator
- docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_LANE.md - Fedora VM CLI Payload Validation Lane
- docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_TRANSCRIPT_CAPTURE.md - Fedora VM CLI Payload Validation Transcript Capture
- docs/FEDORA_VM_CLI_TRANSCRIPT_CONTRACT.md - Fedora VM CLI Transcript Contract
- docs/FREEBSD_PORT_STATIC_VALIDATION.md - FreeBSD Port Static Validation
- docs/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT.md - macOS Reset/Uninstall Absence-Report Contract
- docs/OPENBSD_PORT_STATIC_VALIDATION.md - OpenBSD Port Static Validation
- docs/OPENSUSE_DEVELOPER_WORKFLOW.md - openSUSE Developer Workflow
- docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md - openSUSE Local RPM Build Environment Contract
- docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md - openSUSE Local RPM Build Gate Contract
- docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md - openSUSE Local RPM Static Validation
- docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md - openSUSE OBS Publication Non-Claim Review Contract
- docs/OPENSUSE_READINESS_PLAN.md - openSUSE Readiness Plan
- docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md - openSUSE RPM Artifact Naming Contract
- docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md - openSUSE RPM Install/Remove Transcript Contract
- docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md - openSUSE RPM Payload Inspection Contract
- docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md - openSUSE RPM Topdir Handoff Lane

- docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md - openSUSE rpmlint Findings Classification
- docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md - openSUSE rpmlint and osc Availability Lane
- docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md - openSUSE rpmlint Static Spec Lane
- docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md - openSUSE Source Archive Fixture Lane
- docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md - openSUSE Source Archive Reproducibility Contract
- docs/strategy/2026-05-26-1959-cdt-console-read-only-host-inventory-review-package-index.md - Latticra Console Read-Only Host Inventory Review Package Index
- docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md - Ubuntu Debian Copyright Notice Mapping Contract
- docs/UBUNTU_DEVELOPER_WORKFLOW.md - Ubuntu Developer Workflow
- docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md - Ubuntu Doc Payload License Review Contract
- docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md - Ubuntu Generated-Artifact Notice Review Contract
- docs/UBUNTU_LINTIAN_AVAILABILITY.md - Ubuntu lintian Availability Lane
- docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md - Ubuntu Lintian Static Metadata Contract
- docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md - Ubuntu Local Deb Build Transcript Acceptance Gate Contract
- docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_CONTRACT.md - Ubuntu Local Deb Build Transcript Contract
- docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md - Ubuntu Local Deb Install Remove Evidence Contract
- docs/UBUNTU_LOCAL_DEB_STATIC_VALIDATION.md - Ubuntu Local Deb Static Validation
- docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md - Ubuntu NOTICE File Decision Contract
- docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md - Ubuntu Package License Promotion Gate Contract
- docs/UBUNTU_PACKAGE_LICENSE_REVIEW_CONTRACT.md - Ubuntu Package License Review Contract
- docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md - Ubuntu Package Notice Inventory
- docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md - Ubuntu Package Notice Promotion Gate Contract
- docs/UBUNTU_PACKAGE_NOTICE_REVIEW_CONTRACT.md - Ubuntu Package Notice Review Contract
- docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md - Ubuntu PPA Archive Publication Gate Contract
- docs/UBUNTU_READINESS_PLAN.md - Ubuntu Readiness Plan
- docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md - Ubuntu Release Artifact Notice Requirements Contract
- docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md - Ubuntu Source Package Evidence Contract

- docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md - Ubuntu Third-Party Material Review Contract
- docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md - Ubuntu Trademark Notice Boundary Contract
- docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md - Ubuntu Upload Signing Authority Evidence Contract
- packaging/debian/README.md - Debian Packaging Draft
- packaging/fedora/README.md - Fedora Packaging Draft
- packaging/freebsd/README.md - FreeBSD Port Draft
- packaging/openbsd/README.md - OpenBSD Port Draft
- packaging/opensuse/README.md - openSUSE Packaging Draft
- packaging/ubuntu/README.md - Ubuntu Packaging Draft

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Artifact Naming Contract

Lines: 277 | Words: 796 | SHA-256: 203c7005f2cea2f13ecb11b463a80eaf7739b8d16bd1590fef5b8e18d0503faf

Debian, FreeBSD, and OpenBSD Package Artifact Naming Contract

Status: active package artifact naming contract Scope: define future Debian, FreeBSD, and OpenBSD package artifact names and output boundaries without creating package artifacts.

Purpose

This contract records the artifact names, output-directory rules, checksum binding, retention requirements, and publication non-claims required before any future Debian, FreeBSD, or OpenBSD package build can create package files.

The goal is narrow: package artifacts must have predictable names, be written only inside a disposable validation environment, bind back to the source archive digest, and remain non-published review outputs until separate payload inspection, install/remove, and publication review contracts exist.

This contract is documentation-only and static. It does not run package build tools, create package artifacts, install Latticra, write FreeBSD or OpenBSD distinfo files, publish packages, or claim package readiness.

Current Artifact Naming State

```
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
artifact_naming_contract_present=1
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
package_artifact_output_directory_required_under_disposable_environment=1
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
publication_directory_write_allowed=0
package_artifact_created=0
deb_artifact_created=0
debian_source_package_created=0
freebsd_package_artifact_created=0
openbsd_package_artifact_created=0
package_artifact_sha256_recorded=0
package_artifact_published=0
install_on_host_run=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_readiness_claimed=0
```

Debian Artifact Names

Future Debian artifact names must be derived from the current Debian changelog version and the clean build environment architecture.

```
debian_source_package_name=latticra_0.0.0-1local1.dsc
debian_orig_archive_name=latticra_0.0.0.orig.tar.gz
debian_debian_tar_name=latticra_0.0.0-1local1.debian.tar.xz
debian_binary_package_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.deb
debian_changes_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.changes
debian_buildinfo_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.buildinfo
debian_artifact_name_pattern_recorded=1
dpkg_buildpackage_run=0
```

```
debuild_run=0
lintian_run=0
deb_artifact_created=0
debian_source_package_created=0
```

The ${DEB_HOST_ARCH} token must be resolved from the disposable Debian environment transcript before any Debian artifact can be accepted.

FreeBSD Artifact Names

Future FreeBSD package artifact names must be derived from the current port PORTNAME and DISTVERSION.

```
freebsd_package_name=latticra-0.0.0.pkg
freebsd_distfile_name=latticra-0.0.0.tar.gz
freebsd_artifact_name_pattern_recorded=1
freebsd_package_artifact_created=0
freebsd_make_makesum_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
portlint_run=0
poudriere_run=0
```

The FreeBSD package name is a future local validation artifact only. It is not a FreeBSD ports-tree submission, poudriere result, official package, or publication claim.

OpenBSD Artifact Names

Future OpenBSD package artifact names must be derived from the current port DISTNAME.

```
openbsd_package_name=latticra-0.0.0.tgz
openbsd_distfile_name=latticra-0.0.0.tar.gz
openbsd_artifact_name_pattern_recorded=1
openbsd_package_artifact_created=0
openbsd_make_makesum_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
portcheck_run=0
openbsd_bulk_build_run=0
```

The OpenBSD package name is a future local validation artifact only. PERMIT_PACKAGE=No remains in force until license, redistribution, notice, checksum, and maintainer-review evidence exists.

Output Boundary

Future package artifacts must be written only under a disposable validation environment output directory. The repository itself remains a no-artifact workspace.

```
artifact_output_root_under_disposable_environment=1
debian_artifact_output_directory=artifacts/debian/
freebsd_artifact_output_directory=artifacts/freebsd/
openbsd_artifact_output_directory=artifacts/openbsd/
repository_package_artifact_write_allowed=0
root_workspace_package_artifact_write_allowed=0
publication_directory_write_allowed=0
artifact_retention_policy_required=1
artifact_cleanup_policy_required=1
```

Any future artifact transcript must bind each package file to:

```
source_archive_sha256
package_input_archive_sha256
package_artifact_sha256
artifact_size_bytes
artifact_output_directory
artifact_generation_command
environment_identifier
operator_authorization_reference
```

Current Blockers

Package artifact creation remains blocked under this blocker and dependency state:

```
source_archive_accepted_for_build=0
```

```
license_expression_reviewed=1
package_notice_obligations_reviewed=0
explicit_operator_build_authorization=0
environment_transcript_present=0
install_remove_transcript_contract_present=1
payload_inspection_contract_present=1
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run dpkg-buildpackage
run debuild
run lintian
run FreeBSD make stage
run FreeBSD make package
run portlint
run poudriere
run OpenBSD make plist
run OpenBSD make package
run portcheck
run an OpenBSD bulk build
create .deb artifacts
create .dsc artifacts
create .changes artifacts
create .buildinfo artifacts
create .pkg artifacts
create .tgz artifacts
install Latticra on a host
publish package artifacts
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
```

The package-build gate remains closed. This artifact naming contract only records the names and output boundaries future validation artifacts must use after separate source, license, notice, environment, authorization, payload inspection, install/remove, and publication non-claim evidence exists.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package payload inspection contract before any package
artifact can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
```

That lane defines how package payloads are inspected after creation while keeping package_build_gate_state=closed-no-effect until the remaining prerequisites are satisfied.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package install/remove transcript contract before any package
 install can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
```

That lane defines disposable install/remove evidence while keeping host installs and package readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package publication non-claim review contract before any
package validation result can be promoted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
```

That lane records local-only publication and official-port non-claims while keeping package publication and readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package validation promotion blocker matrix before any
platform-specific build evidence can be accepted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

That lane ties source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping platform build evidence acceptance and validation promotion blocked.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_artifact_naming_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Build Environment Contract

Lines: 304 | Words: 863 | SHA-256: 4df6e88d20e1adb971bc7e75c97653e4118d4310e75c7d8d30b6d1a5c5910ac5

Debian, FreeBSD, and OpenBSD Package Build Environment Contract

Status: active package-build environment contract Scope: document required disposable validation environments for Debian, FreeBSD, and OpenBSD package-build lanes without running package build commands.

Purpose

This contract documents the minimum environment evidence required before any future Debian, FreeBSD, or OpenBSD package-build lane can ask to open the package-build gate.

The goal is narrow: describe clean/disposable validation environments, required transcript fields, toolchain metadata, package input binding, and host-effect boundaries before any package build command can run.

This contract is documentation-only and static. It does not provision a VM, create a chroot, create a jail, run package build tools, create package artifacts, install Latticra, write FreeBSD or OpenBSD distinfo files, or claim package readiness.

Current Environment State

```
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
package_build_gate_state=closed-no-effect
debian_clean_build_environment_documented=1
freebsd_ports_environment_documented=1
openbsd_ports_environment_documented=1
debian_build_environment_provisioned=0
freebsd_build_environment_provisioned=0
openbsd_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
disposable_validation_environment_provisioned=0
environment_transcript_present=0
toolchain_version_capture_required=1
package_input_digest_binding_required=1
package_build_command_allowed=0
package_artifact_created=0
install_on_host_run=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_readiness_claimed=0
```

Debian Environment Contract

A future Debian build lane must use a disposable clean build environment before dpkg-buildpackage, debuild, or lintian can run.

Required Debian environment evidence:

```
debian_clean_build_environment_documented=1
debian_clean_chroot_or_container_required=1
debian_build_host_identity_required=1
debian_suite_record_required=1
debian_architecture_record_required=1
debian_build_dependency_resolution_transcript_required=1
debian_toolchain_versions_required=1
debian_source_archive_digest_required=1
debian_package_input_path_required=1
dpkg_buildpackage_run=0
debuild_run=0
lintian_run=0
```

The Debian environment contract does not choose between sbuild, pbuilder, mmdebstrap, a disposable container, or a disposable VM. A future effect-bearing proposal must name the exact environment and keep the build gate closed until operator authorization and all package prerequisites exist.

FreeBSD Environment Contract

A future FreeBSD build lane must use a disposable FreeBSD ports validation environment before make makesum, make stage, make package, portlint, or poudriere can run.

Required FreeBSD environment evidence:

```
freebsd_ports_environment_documented=1
freebsd_jail_or_vm_required=1
freebsd_version_record_required=1
freebsd_architecture_record_required=1
freebsd_ports_tree_revision_required=1
freebsd_distfile_digest_required=1
freebsd_port_origin_required=1
freebsd_toolchain_versions_required=1
freebsd_distinfo_review_required=1
freebsd_make_makesum_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
portlint_run=0
poudriere_run=0
```

The FreeBSD environment contract does not create a jail, fetch a ports tree, run make makesum, run make stage, run make package, run portlint, or run poudriere.

OpenBSD Environment Contract

A future OpenBSD build lane must use a disposable OpenBSD ports validation environment before make makesum, make plist, make package, portcheck, or a bulk build can run.

Required OpenBSD environment evidence:

```
openbsd_ports_environment_documented=1
openbsd_vm_or_disposable_host_required=1
openbsd_version_record_required=1
openbsd_architecture_record_required=1
openbsd_ports_tree_revision_required=1
openbsd_distfile_digest_required=1
openbsd_port_origin_required=1
openbsd_toolchain_versions_required=1
openbsd_permit_package_review_required=1
openbsd_make_makesum_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
portcheck_run=0
openbsd_bulk_build_run=0
```

The OpenBSD environment contract keeps PERMIT_PACKAGE=No until redistribution, license, notice, checksum, and maintainer-review evidence exists.

Shared Transcript Requirements

A future environment transcript must record:

```
environment_identifier
environment_lifecycle
host_or_vm_class
operating_system_name
operating_system_version
architecture
toolchain_versions
package_input_archive_name
package_input_archive_sha256
package_input_tree_path
package_output_directory
network_policy
host_mount_policy
cleanup_policy
operator_authorization_reference
```

The transcript must be reviewed before a package-build lane can claim that the environment is acceptable.

Current Blockers

Package builds remain blocked under this blocker and dependency state:

```
source_archive_accepted_for_build=0
license_expression_reviewed=1
package_notice_obligations_reviewed=0
explicit_operator_build_authorization=0
artifact_naming_contract_present=1
install_remove_transcript_contract_present=1
```

```
payload_inspection_contract_present=1
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run dpkg-buildpackage
run debuild
run lintian
run FreeBSD make makesum
run FreeBSD make stage
run FreeBSD make package
run portlint
run poudriere
run OpenBSD make makesum
run OpenBSD make plist
run OpenBSD make package
run portcheck
run an OpenBSD bulk build
create package artifacts
install Latticra on a host
publish package artifacts
submit Latticra to Debian, FreeBSD, or OpenBSD
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
```

The package-build gate remains closed. This environment contract only records what future disposable validation environments must prove before a platform-specific build lane can request authorization.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package artifact naming contract before any package artifact
can be created.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
```

That lane defines package artifact names, output directories, checksum binding, retention rules, and publication non-claims while keeping package_build_gate_state=closed-no-effect.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package payload inspection contract before any package
artifact can be accepted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
```

That lane defines payload inspection evidence without opening the package-build gate.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package install/remove transcript contract before any package
 install can be accepted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
```

That lane defines disposable install/remove evidence while keeping host installs and package readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package publication non-claim review contract before any
package validation result can be promoted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
```

That lane records local-only publication and official-port non-claims while keeping package publication and readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package validation promotion blocker matrix before any
platform-specific build evidence can be accepted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

That lane ties source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping platform build evidence acceptance and validation promotion blocked.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_build_environment_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Build Gate Contract

Lines: 311 | Words: 849 | SHA-256: 46ff8db0e299551d18c7a161a87101a915fac672958888d59c8b452d12a3fe8a

Debian, FreeBSD, and OpenBSD Package Build Gate Contract

Status: active package-build evidence gate contract Scope: define the evidence required before Debian, FreeBSD, or OpenBSD package build commands can run in any validation lane.

Purpose

This contract closes the package-build gate after the temporary source archive fixture lane and temporary package input handoff lane.

The goal is conservative: record the exact prerequisites required before dpkg-buildpackage, debuild, lintian, FreeBSD package builds, OpenBSD package builds, port linters, bulk builders, host installs, or package artifact publication can be attempted or accepted as evidence.

This contract is documentation-only and static. It does not run package build tools, create package artifacts, install Latticra, write FreeBSD or OpenBSD distinfo files, or claim package readiness.

Current Gate State

```
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
package_build_gate_state=closed-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
debian_build_allowed=0
freebsd_build_allowed=0
openbsd_build_allowed=0
dpkg_buildpackage_run=0
debuild_run=0
lintian_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
freebsd_make_makesum_run=0
portlint_run=0
poudriere_run=0
openbsd_make_makesum_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
portcheck_run=0
openbsd_bulk_build_run=0
package_artifact_created=0
install_on_host_run=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_readiness_claimed=0
```

Required Gate Inputs

The package-build gate cannot open unless all package input evidence has already passed:

```
debian_freebsd_openbsd_package_input_handoff_lane_present=1
temporary_package_input_handoff_lane_passed=1
debian_freebsd_openbsd_source_archive_fixture_lane_present=1
debian_freebsd_openbsd_source_archive_contract_present=1
temporary_archive_sha256_preserved=1
temporary_archive_listing_preserved=1
source_archive_accepted_for_build=1
freebsd_distinfo_file_written=0
openbsd_distinfo_file_written=0
distinfo_file_writes_blocked_until_review=1
license_expression_reviewed=1
package_notice_obligations_reviewed=1
openbsd_permit_package_reviewed=1
```

The source_archive_accepted_for_build=1 value is a future prerequisite, not a current claim. The current source archive and handoff lanes still keep source archive acceptance closed.

Environment Prerequisites

Before any package build command can run, a future build-lane proposal must document:

```
debian_clean_build_environment_documented=1
freebsd_ports_environment_documented=1
openbsd_ports_environment_documented=1
explicit_operator_build_authorization=1
disposable_validation_environment=1
artifact_naming_contract_present=1
install_remove_transcript_contract_present=1
payload_inspection_contract_present=1
publication_non_claim_review_present=1
```

Those records must identify the exact host or VM class, toolchain versions, source archive digest, package input paths, package output paths, install target, remove target, and transcript retention location before a build can be attempted.

Command Gate

All package build commands remain blocked while the current gate state is closed-no-effect.

Debian commands blocked by this contract:

```
dpkg-buildpackage
debuild
lintian
dpkg-source --build
```

FreeBSD commands blocked by this contract:

```
make makesum
make stage
make package
portlint
poudriere
```

OpenBSD commands blocked by this contract:

```
make makesum
make plist
make package
portcheck
bulk build
```

Host effects blocked by this contract:

```
install package on host
remove package from host
publish package artifact
submit package or port upstream
claim official package or port status
claim production readiness
```

Future Open Conditions

A future package-build lane may move one platform from _build_allowed=0 to _build_allowed=1 only after it records:

```
package_build_gate_state=open-for-single-platform-validation
source_archive_accepted_for_build=1
archive_sha256_bound_to_build=1
distinfo_file_writes_reviewed=1
license_expression_reviewed=1
package_notice_obligations_reviewed=1
debian_clean_build_environment_documented=1
freebsd_ports_environment_documented=1
openbsd_ports_environment_documented=1
explicit_operator_build_authorization=1
disposable_validation_environment=1
artifact_naming_contract_present=1
install_remove_transcript_contract_present=1
payload_inspection_contract_present=1
publication_non_claim_review_present=1
```

Opening one platform gate must not imply readiness for the other platforms.

Current Non-Claims

```
debian_archive_ready=0
debian_mentors_upload_claimed=0
debian_sponsorship_claimed=0
debian_ftp_master_acceptance_claimed=0
freebsd_official_port_claimed=0
freebsd_ports_tree_submission_claimed=0
freebsd_bugzilla_pr_claimed=0
freebsd_committer_review_claimed=0
openbsd_official_port_claimed=0
openbsd_ports_tree_submission_claimed=0
openbsd_ports_review_thread_claimed=0
openbsd_maintainer_acceptance_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
```

The source archive fixture lane proves temporary archive shape in a disposable workspace.

The package input handoff lane proves temporary package input staging while preserving archive SHA-256 identity.

This package-build gate keeps package build execution closed until source acceptance, checksum, license, notice, environment, authorization, payload inspection, install/remove, and publication non-claim evidence exists.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package-build environment contract that documents disposable
validation environments without running package build commands.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
```

That lane documents disposable validation environment requirements while keeping package_build_gate_state=closed-no-effect.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package artifact naming contract before any package artifact
can be created.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
```

That lane defines package artifact names, output directories, checksum binding, retention rules, and publication non-claims while keeping builds blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package payload inspection contract before any package
artifact can be accepted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
```

That lane defines payload inspection evidence without opening the package-build gate.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package install/remove transcript contract before any package
 install can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
```

That lane defines disposable install/remove evidence while keeping host installs and package readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package publication non-claim review contract before any
package validation result can be promoted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
```

That lane records local-only publication and official-port non-claims while keeping package publication and readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package validation promotion blocker matrix before any
platform-specific build evidence can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

That lane ties source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping platform build evidence acceptance and validation promotion blocked.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_build_gate_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md

Source title: Debian, FreeBSD, and OpenBSD Package Input Handoff Lane

Lines: 174 | Words: 479 | SHA-256: 51cc063314a5b74571e9fb95397bb6424d125714d76fe042525da7d9e9dae98d

Debian, FreeBSD, and OpenBSD Package Input Handoff Lane

Status: active temporary package input handoff lane Scope: stage verified temporary source archives into Debian, FreeBSD, and OpenBSD package input layouts without running package builds or creating package artifacts.

Purpose

This lane advances the Debian, FreeBSD, and OpenBSD package/port path from a source archive fixture into package-input staging.

The goal is conservative: prove that the temporary Debian orig.tar.gz, FreeBSD distfile, and OpenBSD distfile can be staged into platform-shaped input directories while preserving checksum identity and keeping all package build tools disabled.

This lane does not build Debian source packages, Debian binary packages, FreeBSD packages, or OpenBSD packages. It stages temporary package inputs only.

Temporary Input Layouts

The Debian handoff stages:

```
debian/latticra_0.0.0.orig.tar.gz
debian/latticra-0.0.0/
debian/latticra-0.0.0/debian/control
debian/latticra-0.0.0/debian/changelog
```

The FreeBSD handoff stages:

```
freebsd/distfiles/latticra-0.0.0.tar.gz
freebsd/ports/devel/latticra/Makefile
freebsd/ports/devel/latticra/pkg-descr
freebsd/ports/devel/latticra/pkg-plist
```

The OpenBSD handoff stages:

```
openbsd/distfiles/latticra-0.0.0.tar.gz
openbsd/ports/devel/latticra/Makefile
openbsd/ports/devel/latticra/pkg/DESCR
openbsd/ports/devel/latticra/pkg/PLIST
```

All paths are created under a disposable temporary directory.

Handoff Checks

The lane verifies:

```
temporary_package_input_handoff_lane_present=1
temporary_debian_source_input_staged=1
temporary_debian_orig_archive_staged=1
temporary_debian_debian_dir_overlay_staged=1
temporary_freebsd_port_input_staged=1
temporary_freebsd_distfile_staged=1
temporary_openbsd_port_input_staged=1
temporary_openbsd_distfile_staged=1
temporary_archive_sha256_preserved=1
temporary_archive_listing_preserved=1
freebsd_distinfo_file_written=0
openbsd_distinfo_file_written=0
package_artifact_created=0
```

Boundary

This lane does not run dpkg-source.

It does not run dpkg-buildpackage.

It does not run debuild.

It does not run lintian.

It does not run FreeBSD make makesum.

It does not run FreeBSD make stage.

It does not run FreeBSD make package.

It does not run OpenBSD make makesum.

It does not run OpenBSD make plist.

It does not run OpenBSD make package.

It does not run portlint.

It does not run portcheck.

It does not run poudriere.

It does not run an OpenBSD bulk build.

It does not create package artifacts.

It does not write FreeBSD or OpenBSD distinfo files.

It does not install Latticra.

It does not publish package artifacts.

It does not submit Latticra to Debian, FreeBSD, or OpenBSD.

It does not claim Debian archive readiness, Debian sponsorship, FreeBSD official port status, OpenBSD official port status, product readiness, operating-system readiness, runtime authority, kernel authority, boot authority, service authority, policy readiness, or security-hardening completion.

Current Evidence

```
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_source_archive_fixture_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
temporary_package_input_handoff_lane_present=1
temporary_debian_source_input_staged=1
temporary_debian_orig_archive_staged=1
temporary_debian_debian_dir_overlay_staged=1
temporary_freebsd_port_input_staged=1
temporary_freebsd_distfile_staged=1
temporary_openbsd_port_input_staged=1
temporary_openbsd_distfile_staged=1
temporary_archive_sha256_preserved=1
temporary_archive_listing_preserved=1
freebsd_distinfo_file_written=0
openbsd_distinfo_file_written=0
dpkg_source_run=0
dpkg_buildpackage_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
package_artifact_created=0
package_readiness_claimed=0
```

The handoff evidence is temporary package-input staging only. It does not promote the archive to accepted package-build input.

Completed Follow-On Lane

Completed follow-on lane:

```
Add Debian, FreeBSD, and OpenBSD package-build evidence gate contract before any package build
command can run.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
```

That gate defines the exact evidence required before dpkg-buildpackage, FreeBSD package builds, or OpenBSD package builds are allowed in any validation environment.

Next Slice

Recommended next slice:

```
Add Debian, FreeBSD, and OpenBSD package-build environment contract before any package build
command can run.
```

That future lane should document disposable validation environments while keeping package_build_gate_state=closed-no-effect.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-input-handoff-lane.sh
```

Expected output:

```
debian_freebsd_openbsd_package_input_handoff_lane: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Install/Remove Transcript Contract

Lines: 255 | Words: 637 | SHA-256: 8c343fceb5e09c414332896e23aaba71ee9d0be425e1d77916770b8a8cd3b796

Debian, FreeBSD, and OpenBSD Package Install/Remove Transcript Contract

Status: active package install/remove transcript contract Scope: define future disposable install/remove evidence for Debian, FreeBSD, and OpenBSD package artifacts without installing or removing packages.

Purpose

This contract records the install/remove transcript evidence required before any future Debian, FreeBSD, or OpenBSD package artifact can be accepted as installable validation evidence.

The goal is narrow: a future package install/remove transcript must prove that installation happens only inside a disposable validation environment, the expected no-effect CLI payload appears after install, the package can be removed cleanly, and no host-level service, kernel, privileged-helper, or network authority is introduced.

This contract is documentation-only and static. It does not install packages, remove packages, run package managers, create package artifacts, inspect package artifacts, publish packages, or claim package readiness.

Current Install/Remove State

```
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
install_remove_transcript_contract_present=1
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
package_build_gate_state=closed-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_disposable_environment_required=1
install_remove_transcript_present=0
install_on_host_run=0
remove_on_host_run=0
host_install_allowed=0
host_remove_allowed=0
host_mutation_allowed=0
service_state_change_allowed=0
package_artifact_created=0
package_payload_accepted=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_readiness_claimed=0
```

Debian Install/Remove Transcript

Future Debian install/remove validation must run only in a disposable Debian environment.

Required Debian transcript evidence:

```
debian_install_remove_transcript_required=1
debian_install_remove_transcript_present=0
debian_package_install_run=0
debian_package_remove_run=0
debian_dpkg_install_run=0
debian_apt_install_run=0
debian_dpkg_remove_run=0
debian_apt_remove_run=0
debian_payload_post_install_check_required=1
debian_payload_post_remove_absence_check_required=1
debian_service_state_change_allowed=0
debian_systemd_unit_enable_allowed=0
```

The future Debian transcript must bind install/remove evidence to:

```
debian_binary_package_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.deb
debian_package_artifact_sha256
debian_payload_expected_bin=usr/bin/latticra
debian_payload_expected_doc=usr/share/doc/latticra/README.md
```

FreeBSD Install/Remove Transcript

Future FreeBSD install/remove validation must run only in a disposable FreeBSD jail or VM.

Required FreeBSD transcript evidence:

```
freebsd_install_remove_transcript_required=1
freebsd_install_remove_transcript_present=0
freebsd_package_install_run=0
freebsd_package_remove_run=0
freebsd_pkg_add_run=0
freebsd_pkg_delete_run=0
freebsd_payload_post_install_check_required=1
freebsd_payload_post_remove_absence_check_required=1
freebsd_rc_service_enable_allowed=0
```

The future FreeBSD transcript must bind install/remove evidence to:

```
freebsd_package_name=latticra-0.0.0.pkg
freebsd_package_artifact_sha256
freebsd_payload_expected_bin=bin/latticra
freebsd_payload_expected_doc=%%DOCSDIR%%/README.md
```

OpenBSD Install/Remove Transcript

Future OpenBSD install/remove validation must run only in a disposable OpenBSD environment.

Required OpenBSD transcript evidence:

```
openbsd_install_remove_transcript_required=1
openbsd_install_remove_transcript_present=0
openbsd_package_install_run=0
openbsd_package_remove_run=0
openbsd_pkg_add_run=0
openbsd_pkg_delete_run=0
openbsd_payload_post_install_check_required=1
openbsd_payload_post_remove_absence_check_required=1
openbsd_rc_service_enable_allowed=0
```

The future OpenBSD transcript must bind install/remove evidence to:

```
openbsd_package_name=latticra-0.0.0.tgz
openbsd_package_artifact_sha256
openbsd_payload_expected_bin=bin/latticra
openbsd_payload_expected_doc=share/doc/latticra/README.md
```

Transcript Requirements

A future install/remove transcript must record:

```
environment_identifier
```

```
operator_authorization_reference
package_artifact_name
package_artifact_sha256
pre_install_package_state
install_command
install_exit_code
post_install_payload_listing
post_install_cli_no_effect_output
service_state_after_install
remove_command
remove_exit_code
post_remove_absence_report
post_remove_package_state
host_mutation_review
```

The transcript must be reviewed before a package artifact can be accepted as install/remove-valid.

Current Blockers

Package install/remove acceptance remains blocked under this blocker and dependency state:

```
package_artifact_created=0
package_artifact_sha256_recorded=0
package_payload_accepted=0
environment_transcript_present=0
explicit_operator_build_authorization=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run dpkg -i
run apt install
run dpkg -r
run apt remove
run FreeBSD pkg add
run FreeBSD pkg delete
run OpenBSD pkg_add
run OpenBSD pkg_delete
install Latticra on a host
remove Latticra from a host
change service state
create package artifacts
publish package artifacts
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
```

The package-build gate remains closed. This install/remove transcript contract only defines the evidence future disposable validation environments must produce after package artifacts exist and payload inspection evidence is accepted.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package publication non-claim review contract before any
package validation result can be promoted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
```

That lane records local-only publication and official-port non-claims while keeping package publication and readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package validation promotion blocker matrix before any
platform-specific build evidence can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

That lane ties source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping platform build evidence acceptance and validation promotion blocked.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_install_remove_transcript_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Payload Inspection Contract

Lines: 274 | Words: 674 | SHA-256: 00aa40a60929935ce5218c09e4b08950bb31b213c47eab046f2c6bb1767ca4a4

Debian, FreeBSD, and OpenBSD Package Payload Inspection Contract

Status: active package payload inspection contract Scope: define future Debian, FreeBSD, and OpenBSD package payload inspection evidence without creating or inspecting package artifacts.

Purpose

This contract records the package payload inspection evidence required before any future Debian, FreeBSD, or OpenBSD package artifact can be accepted as validation evidence.

The goal is narrow: a future package artifact must prove that it contains only the intended no-effect CLI payload and documentation payload, with no services, privileged helpers, init files, kernel files, network authority, or host mutation hooks.

This contract is documentation-only and static. It does not run package build tools, create package artifacts, inspect package artifacts, install Latticra, write FreeBSD or OpenBSD distinfo files, publish packages, or claim package readiness.

Current Payload Inspection State

```
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
payload_inspection_contract_present=1
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
package_build_gate_state=closed-no-effect
artifact_naming_contract_state=specified-no-effect
package_artifact_created=0
package_payload_inspection_run=0
debian_payload_inspection_run=0
freebsd_payload_inspection_run=0
openbsd_payload_inspection_run=0
package_payload_accepted=0
package_artifact_sha256_recorded=0
install_on_host_run=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_readiness_claimed=0
```

Expected Payload

All future Debian, FreeBSD, and OpenBSD package artifacts must preserve the current intentionally narrow payload:

```
payload_cli_path_required=1
payload_doc_readme_required=1
payload_service_files_allowed=0
payload_init_files_allowed=0
payload_kernel_files_allowed=0
payload_privileged_helper_allowed=0
payload_network_authority_allowed=0
payload_host_mutation_hooks_allowed=0
```

The no-effect CLI payload must remain the only executable payload.

Debian Payload Inspection

Future Debian payload inspection must verify:

```
debian_payload_expected_bin=usr/bin/latticra
debian_payload_expected_doc=usr/share/doc/latticra/README.md
debian_payload_listing_required=1
debian_control_metadata_inspection_required=1
debian_maintainer_script_absence_required=1
debian_systemd_unit_absence_required=1
debian_init_script_absence_required=1
debian_privileged_helper_absence_required=1
debian_payload_unexpected_file_count=0
debian_payload_inspection_run=0
dpkg_deb_contents_inspection_run=0
dpkg_deb_metadata_inspection_run=0
```

The future Debian inspection transcript must bind the payload listing to:

```
debian_binary_package_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.deb
debian_package_artifact_sha256
debian_payload_listing_sha256
```

FreeBSD Payload Inspection

Future FreeBSD payload inspection must verify:

```
freebsd_payload_expected_bin=bin/latticra
freebsd_payload_expected_doc=%%DOCSDIR%%/README.md
freebsd_payload_listing_required=1
freebsd_manifest_inspection_required=1
freebsd_rc_script_absence_required=1
freebsd_periodic_script_absence_required=1
freebsd_privileged_helper_absence_required=1
freebsd_payload_unexpected_file_count=0
freebsd_payload_inspection_run=0
freebsd_pkg_info_inspection_run=0
```

```
freebsd_pkg_manifest_inspection_run=0
```

The future FreeBSD inspection transcript must bind the payload listing to:

```
freebsd_package_name=latticra-0.0.0.pkg
freebsd_package_artifact_sha256
freebsd_payload_listing_sha256
```

OpenBSD Payload Inspection

Future OpenBSD payload inspection must verify:

```
openbsd_payload_expected_bin=bin/latticra
openbsd_payload_expected_doc=share/doc/latticra/README.md
openbsd_payload_listing_required=1
openbsd_pkg_info_inspection_required=1
openbsd_rc_script_absence_required=1
openbsd_privileged_helper_absence_required=1
openbsd_payload_unexpected_file_count=0
openbsd_payload_inspection_run=0
openbsd_pkg_info_list_run=0
openbsd_package_tar_listing_run=0
```

The future OpenBSD inspection transcript must bind the payload listing to:

```
openbsd_package_name=latticra-0.0.0.tgz
openbsd_package_artifact_sha256
openbsd_payload_listing_sha256
```

Inspection Transcript Requirements

A future payload inspection transcript must record:

```
package_artifact_name
package_artifact_sha256
package_artifact_size_bytes
payload_listing_command
payload_listing_sha256
expected_payload_paths_present
unexpected_payload_paths_absent
service_files_absent
privileged_helpers_absent
host_mutation_hooks_absent
environment_identifier
operator_authorization_reference
```

The transcript must be reviewed before a package artifact can be accepted as payload-correct.

Current Blockers

Package payload acceptance remains blocked under this blocker and dependency state:

```
package_artifact_created=0
package_artifact_sha256_recorded=0
environment_transcript_present=0
explicit_operator_build_authorization=0
install_remove_transcript_contract_present=1
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run dpkg-deb --contents
run dpkg-deb --info
run FreeBSD pkg info
run FreeBSD pkg manifest inspection
run OpenBSD pkg_info
run package tar listing
create package artifacts
install Latticra on a host
publish package artifacts
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
```

The package-build gate remains closed. This payload inspection contract only defines the evidence future package artifacts must provide after they exist in a disposable validation environment.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package install/remove transcript contract before any package
 install can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
```

That lane defines disposable install/remove evidence while keeping host installs and package readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package publication non-claim review contract before any
package validation result can be promoted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
```

That lane records local-only publication and official-port non-claims while keeping package publication and readiness blocked.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package validation promotion blocker matrix before any
platform-specific build evidence can be accepted.
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

That lane ties source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping platform build evidence acceptance and validation promotion blocked.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_payload_inspection_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Publication Non-Claim Review Contract

Lines: 211 | Words: 646 | SHA-256: 339c041f6359962f61270d8d5aea2007498730d9342ce502dfbf93c29c851e1b

Debian, FreeBSD, and OpenBSD Package Publication Non-Claim Review Contract

Status: active package publication non-claim review contract Scope: define publication and official-port non-claim review evidence for Debian, FreeBSD, and OpenBSD package validation without publishing packages.

Purpose

This contract records the publication non-claim review required before any future Debian, FreeBSD, or OpenBSD package validation result can be promoted.

The goal is narrow: future package validation evidence must remain clearly local-only unless a separate, explicit, platform-specific publication lane records the required upload, submission, maintainer, archive, or ports-tree authority.

This contract is documentation-only and static. It does not upload packages, publish package artifacts, create package repositories, submit Latticra to Debian, submit Latticra to the FreeBSD ports tree, submit Latticra to the OpenBSD ports tree, create package artifacts, install packages, remove packages, or claim package readiness.

Current Publication Non-Claim State

```
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
publication_non_claim_review_contract_present=1
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
publication_non_claim_review_present=1
package_build_gate_state=closed-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
package_validation_result_promoted=0
package_artifact_created=0
package_artifact_published=0
package_repository_created=0
package_repository_upload_run=0
upstream_submission_run=0
official_package_or_port_claimed=0
platform_build_evidence_accepted=0
package_readiness_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Debian Publication Non-Claims

Future Debian validation promotion must preserve these current Debian non-claims:

```
debian_publication_non_claim_review_required=1
debian_publication_non_claim_review_present=1
debian_package_publication_claimed=0
debian_archive_ready=0
debian_mentors_upload_claimed=0
debian_sponsorship_claimed=0
debian_ftp_master_acceptance_claimed=0
debian_source_upload_run=0
debian_mentors_upload_run=0
debian_archive_upload_run=0
debian_debsign_run=0
debian_dput_run=0
```

The Debian package draft remains local-only. A future Debian upload or archive lane must be separate from local build, payload, install/remove, and publication non-claim review evidence.

FreeBSD Publication Non-Claims

Future FreeBSD validation promotion must preserve these current FreeBSD non-claims:

```
freebsd_publication_non_claim_review_required=1
freebsd_publication_non_claim_review_present=1
freebsd_package_publication_claimed=0
freebsd_ports_tree_submission_claimed=0
freebsd_bugzilla_pr_claimed=0
freebsd_committer_review_claimed=0
freebsd_official_port_claimed=0
freebsd_pkg_repo_created=0
freebsd_pkg_repo_publish_run=0
freebsd_pkg_repo_sign_run=0
```

The FreeBSD port draft remains local-only. A future FreeBSD ports-tree submission or package repository lane must be separate from local build, payload, install/remove, and publication non-claim review evidence.

OpenBSD Publication Non-Claims

Future OpenBSD validation promotion must preserve these current OpenBSD non-claims:

```
openbsd_publication_non_claim_review_required=1
openbsd_publication_non_claim_review_present=1
openbsd_package_publication_claimed=0
openbsd_ports_tree_submission_claimed=0
openbsd_ports_review_thread_claimed=0
openbsd_maintainer_acceptance_claimed=0
openbsd_official_port_claimed=0
openbsd_pkg_repo_created=0
openbsd_pkg_repo_publish_run=0
permit_package_enabled=0
```

The OpenBSD port draft remains local-only. PERMIT_PACKAGE=No remains in force until redistribution, license, notice, checksum, and maintainer-review evidence exists in a separate OpenBSD-specific lane.

Promotion Review Boundary

A future package validation promotion review must record:

```
package_validation_result_identifier
platform_under_review
source_archive_sha256
package_artifact_sha256
payload_inspection_transcript_sha256
install_remove_transcript_sha256
publication_non_claim_reviewer
publication_non_claim_review_date
non_claimed_publication_targets
status_page_update_reference
operator_authorization_reference
```

That review may only promote a validation result as local validation evidence. It must not convert local evidence into Debian archive evidence, FreeBSD official port evidence, OpenBSD official port evidence, package repository evidence, production installer evidence, or release readiness.

Current Blockers

Package validation promotion remains blocked because the current repository still has:

```
source_archive_accepted_for_build=0
package_artifact_created=0
package_artifact_sha256_recorded=0
package_payload_accepted=0
install_remove_transcript_present=0
environment_transcript_present=0
```

```
explicit_operator_build_authorization=0
platform_build_evidence_accepted=0
package_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run debsign
run dput
upload to mentors.debian.net
upload to a Debian archive
submit to the FreeBSD ports tree
create a FreeBSD Bugzilla PR
publish a FreeBSD package repository
submit to the OpenBSD ports tree
send an OpenBSD ports review thread
enable OpenBSD PERMIT_PACKAGE
publish an OpenBSD package repository
create package artifacts
install Latticra on a host
remove Latticra from a host
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
```

The package-build gate remains closed. This publication non-claim review contract only records that local package validation evidence must remain non-published and non-official until a separate platform-specific publication lane exists.

Completed Follow-On Lane

Completed follow-on lane:

```
Add a Debian, FreeBSD, and OpenBSD package validation promotion blocker matrix before any
platform-specific build evidence can be accepted.
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

That lane ties source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping platform build evidence acceptance and validation promotion blocked.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_publication_non_claim_review_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Package Validation Promotion Blocker Matrix Contract

Lines: 251 | Words: 649 | SHA-256: d1010222b9741ca71a856abc63c598676431fe3801fe336e6dc15179c5aadca9

Debian, FreeBSD, and OpenBSD Package Validation Promotion Blocker Matrix Contract

Status: active package validation promotion blocker matrix contract Scope: define the blocker matrix for promoting Debian, FreeBSD, and OpenBSD package validation evidence without accepting build evidence.

Purpose

This contract records the current blocker matrix that prevents any Debian, FreeBSD, or OpenBSD package validation result from being promoted.

The goal is narrow: tie the source archive, disposable environment, artifact naming, payload inspection, install/remove transcript, and publication non-claim records into one promotion matrix before any platform-specific build evidence can be accepted.

This contract is documentation-only and static. It does not run package build tools, create package artifacts, inspect package artifacts, install packages, remove packages, publish packages, submit ports, enable OpenBSD package redistribution, promote validation results, or claim package readiness.

Current Matrix State

```
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
package_validation_promotion_blocker_matrix_contract_present=1
validation_promotion_blocker_matrix_state=blocked-no-effect
package_build_gate_state=closed-no-effect
publication_non_claim_review_contract_state=specified-no-effect
source_archive_accepted_for_build=0
environment_transcript_present=0
explicit_operator_build_authorization=0
package_artifact_created=0
package_artifact_sha256_recorded=0
package_payload_accepted=0
install_remove_transcript_present=0
publication_non_claim_review_present=1
platform_build_evidence_accepted=0
package_validation_result_promoted=0
package_readiness_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Shared Promotion Blocker Matrix

Current promotion state by evidence column:

```
source_archive_column_state=blocked
environment_column_state=blocked
artifact_column_state=blocked
payload_column_state=blocked
install_remove_column_state=blocked
publication_non_claim_column_state=specified
promotion_column_state=blocked
```

A future local validation promotion may not accept any platform-specific build evidence unless all required columns are reviewed together:

```
source_archive_accepted_for_build=1
archive_sha256_bound_to_build=1
environment_transcript_present=1
explicit_operator_build_authorization=1
package_artifact_created=1
package_artifact_sha256_recorded=1
```

```
package_payload_accepted=1
install_remove_transcript_present=1
publication_non_claim_review_present=1
platform_publication_claimed=0
package_validation_result_promoted=1
```

Those values are future prerequisites, not current claims. The current matrix remains blocked.

Debian Promotion Row

Current Debian validation promotion remains blocked:

```
debian_validation_promotion_blocked=1
debian_platform_build_evidence_accepted=0
debian_source_archive_accepted_for_build=0
debian_build_environment_provisioned=0
debian_package_artifact_created=0
debian_payload_inspection_run=0
debian_install_remove_transcript_present=0
debian_publication_non_claim_review_present=1
debian_package_publication_claimed=0
debian_validation_result_promoted=0
dpkg_buildpackage_run=0
debuild_run=0
lintian_run=0
debian_archive_ready=0
```

Debian build, lintian, payload, install/remove, and publication evidence must remain local-only and unpromoted until the shared matrix opens in a future lane.

FreeBSD Promotion Row

Current FreeBSD validation promotion remains blocked:

```
freebsd_validation_promotion_blocked=1
freebsd_platform_build_evidence_accepted=0
freebsd_source_archive_accepted_for_build=0
freebsd_build_environment_provisioned=0
freebsd_package_artifact_created=0
freebsd_payload_inspection_run=0
freebsd_install_remove_transcript_present=0
freebsd_publication_non_claim_review_present=1
freebsd_package_publication_claimed=0
freebsd_validation_result_promoted=0
freebsd_make_makesum_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
portlint_run=0
poudriere_run=0
freebsd_official_port_claimed=0
```

FreeBSD ports, portlint, poudriere, package, install/remove, and publication evidence must remain local-only and unpromoted until the shared matrix opens in a future lane.

OpenBSD Promotion Row

Current OpenBSD validation promotion remains blocked:

```
openbsd_validation_promotion_blocked=1
openbsd_platform_build_evidence_accepted=0
openbsd_source_archive_accepted_for_build=0
openbsd_build_environment_provisioned=0
openbsd_package_artifact_created=0
openbsd_payload_inspection_run=0
openbsd_install_remove_transcript_present=0
openbsd_publication_non_claim_review_present=1
openbsd_package_publication_claimed=0
openbsd_validation_result_promoted=0
openbsd_make_makesum_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
portcheck_run=0
openbsd_bulk_build_run=0
permit_package_enabled=0
openbsd_official_port_claimed=0
```

OpenBSD ports, plist, package, portcheck, bulk build, install/remove, PERMIT_PACKAGE, and publication evidence must remain local-only and unpromoted until the shared matrix opens in a future lane.

Acceptance Boundary

A future promotion record must bind every accepted platform result to:

```
platform_under_review
source_archive_contract_reference
source_archive_sha256
environment_transcript_sha256
package_artifact_name
package_artifact_sha256
payload_inspection_transcript_sha256
install_remove_transcript_sha256
publication_non_claim_review_reference
promotion_reviewer
promotion_decision
promotion_scope
status_page_update_reference
operator_authorization_reference
```

The only permitted future promotion scope from this matrix is local package validation evidence. This matrix cannot promote Debian archive readiness, FreeBSD official port status, OpenBSD official port status, package repository publication, production installer readiness, or root installer readiness.

Current Blockers

Package validation promotion remains blocked under this matrix:

```
source_archive_accepted_for_build=0
environment_transcript_present=0
explicit_operator_build_authorization=0
package_artifact_created=0
package_artifact_sha256_recorded=0
package_payload_accepted=0
install_remove_transcript_present=0
platform_build_evidence_accepted=0
package_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run dpkg-buildpackage
run debuild
run lintian
run FreeBSD make makesum
run FreeBSD make stage
run FreeBSD make package
run portlint
run poudriere
run OpenBSD make makesum
run OpenBSD make plist
run OpenBSD make package
run portcheck
run an OpenBSD bulk build
create package artifacts
inspect package artifacts
install packages
remove packages
publish package artifacts
submit package or port upstream
enable OpenBSD PERMIT_PACKAGE
promote package validation results
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md
```

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
```

The package-build gate remains closed. This blocker matrix only records the promotion columns and current blocking values for future Debian, FreeBSD, and OpenBSD local validation evidence.

Next Slice

Recommended next slice:

```
Add a Debian, FreeBSD, and OpenBSD package build-evidence intake denial contract before any
single-platform build lane can open.
```

That future lane should define how build evidence intake is refused until the blocker matrix opens, while keeping package builds and readiness blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md

Source title: Debian, FreeBSD, and OpenBSD Source Archive Contract

Lines: 197 | Words: 481 | SHA-256: 115765fbaf6ef9ae77dcba7a2be0b64308b0e40785a4e3a2709751f227a0ee12

Debian, FreeBSD, and OpenBSD Source Archive Contract

Status: active source archive contract Scope: define the evidence required before Debian, FreeBSD, or OpenBSD package/port build evidence can be accepted.

Purpose

This contract defines the source archive and checksum evidence required before Latticra accepts any Debian, FreeBSD, or OpenBSD package/port build transcript.

The current goal is conservative: record expected source archive names, root layout, checksum requirements, distinfo requirements, transcript fields, and promotion blockers without creating an archive, generating distinfo, or running a package build.

This contract sits after the local-only static packaging drafts and before any dpkg-buildpackage, FreeBSD make stage, FreeBSD make package, OpenBSD make plist, OpenBSD make package, portlint, portcheck, poudriere, or bulk-build evidence.

Current Package Inputs

The current local-only Debian draft declares:

```
Source: latticra
latticra (0.0.0-1local1) UNRELEASED
3.0 (quilt)
```

For the current Debian draft, the expected source archive shape is:

```
debian_orig_archive_name=latticra_0.0.0.orig.tar.gz
debian_source_root=latticra-0.0.0/
debian_source_format=3.0 (quilt)
```

The current local-only FreeBSD port draft declares:

```
PORTNAME=	latticra
DISTVERSION=	0.0.0
```

For the current FreeBSD draft, the expected distfile and distinfo shape is:

```
freebsd_distfile_name=latticra-0.0.0.tar.gz
freebsd_distfile_root=latticra-0.0.0/
freebsd_distinfo_requires_sha256=1
freebsd_distinfo_requires_size=1
```

The current local-only OpenBSD port draft declares:

```
DISTNAME=	latticra-0.0.0
PERMIT_PACKAGE=	No
```

For the current OpenBSD draft, the expected distfile and distinfo shape is:

```
openbsd_distfile_name=latticra-0.0.0.tar.gz
openbsd_distfile_root=latticra-0.0.0/
openbsd_distinfo_requires_sha256=1
openbsd_distinfo_requires_size=1
openbsd_permit_package_requires_license_review=1
```

Required Future Evidence

A future accepted source archive transcript must record:

```
source_tree_revision
source_archive_command
source_archive_name
source_archive_root
source_archive_size_bytes
source_archive_sha256
source_archive_entry_count
source_archive_generated_twice
source_archive_repeated_sha256_match
source_archive_contains_readme
source_archive_contains_debian_metadata
source_archive_contains_freebsd_port_metadata
source_archive_contains_openbsd_port_metadata
source_archive_excludes_git_dir
source_archive_excludes_nested_archives
source_archive_excludes_build_outputs
source_archive_excludes_package_artifacts
source_archive_symlink_policy_checked
source_archive_path_safety_checked
freebsd_distinfo_sha256_recorded
freebsd_distinfo_size_recorded
openbsd_distinfo_sha256_recorded
openbsd_distinfo_size_recorded
```

Acceptance Rule

Debian, FreeBSD, and OpenBSD source archive evidence remains unaccepted unless all of these are true:

```
source_archive_transcript_present=1
source_archive_sha256_recorded=1
source_archive_generated_twice=1
source_archive_repeated_sha256_match=1
source_archive_excludes_git_dir=1
source_archive_excludes_nested_archives=1
source_archive_excludes_build_outputs=1
source_archive_excludes_package_artifacts=1
source_archive_symlink_policy_checked=1
source_archive_path_safety_checked=1
debian_orig_archive_name_matches_expected=1
freebsd_distinfo_sha256_recorded=1
freebsd_distinfo_size_recorded=1
openbsd_distinfo_sha256_recorded=1
openbsd_distinfo_size_recorded=1
license_expression_reviewed=1
package_notice_obligations_reviewed=1
openbsd_permit_package_reviewed=1
```

Until those conditions are met, source archive evidence is a review input only, not package build readiness.

Current Baseline

```
debian_freebsd_openbsd_source_archive_contract_present=1
debian_local_deb_draft_present=1
freebsd_port_draft_present=1
openbsd_port_draft_present=1
source_archive_policy_recorded=1
debian_orig_archive_name_expected=latticra_0.0.0.orig.tar.gz
freebsd_distfile_name_expected=latticra-0.0.0.tar.gz
openbsd_distfile_name_expected=latticra-0.0.0.tar.gz
source_archive_transcript_present=0
source_archive_created=0
source_archive_sha256_recorded=0
source_archive_reproducible=0
source_archive_accepted_for_build=0
freebsd_distinfo_created=0
openbsd_distinfo_created=0
debian_source_package_created=0
freebsd_package_artifact_created=0
openbsd_package_artifact_created=0
package_readiness_claimed=0
```

Boundary

This contract does not:

```
create a source archive
run tar
run gzip
run dpkg-source
run dpkg-buildpackage
run debuild
run lintian
generate FreeBSD distinfo
generate OpenBSD distinfo
run FreeBSD make stage
run FreeBSD make package
run OpenBSD make plist
run OpenBSD make package
run portlint
run portcheck
run poudriere
run an OpenBSD bulk build
create package artifacts
publish to Debian, FreeBSD, or OpenBSD
install Latticra on a host
claim official package or port status
claim production readiness
```

Next Slice

Recommended next slice:

```
Add a temporary source archive fixture lane that creates and inspects an archive in a disposable
 workspace without running package builds or generating package artifacts.
```

That future lane should prove archive shape and reproducibility in a temporary workspace while keeping Debian, FreeBSD, and OpenBSD package build and publication claims blocked.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
```

Expected output:

```
debian_freebsd_openbsd_source_archive_contract: ok
```

docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_FIXTURE_LANE.md

Source title: Debian, FreeBSD, and OpenBSD Source Archive Fixture Lane

Lines: 58 | Words: 165 | SHA-256: aae9dd3a783588346f66523ec2860231534002e94ccb7f86642b63a87ec175aa

Debian, FreeBSD, and OpenBSD Source Archive Fixture Lane

Status: active fixture lane Scope: create and inspect temporary Debian, FreeBSD, and OpenBSD source archive fixtures without running package builds, installing artifacts, or publishing package metadata.

Purpose

This lane proves that Latticra can form temporary source archives matching the current Debian orig.tar.gz, FreeBSD distfile, and OpenBSD distfile expectations, generate them twice, and confirm the repeated SHA-256 value matches.

It does not build packages. It only creates disposable archive fixtures and inspects their contents.

Current Evidence

```
debian_freebsd_openbsd_source_archive_fixture_lane_present=1
debian_freebsd_openbsd_source_archive_contract_present=1
temporary_source_archive_created=1
temporary_source_archive_sha256_recorded=1
temporary_source_archive_reproducible=1
temporary_source_archive_generated_twice=1
temporary_source_archive_repeated_sha256_match=1
temporary_debian_orig_archive_created=1
temporary_bsd_distfile_created=1
freebsd_distinfo_checksum_computed=1
freebsd_distinfo_file_written=0
openbsd_distinfo_checksum_computed=1
openbsd_distinfo_file_written=0
source_archive_accepted_for_build=0
package_artifact_created=0
package_readiness_claimed=0
```

Completed Follow-On Lane

```
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
scripts/test-debian-freebsd-openbsd-package-input-handoff-lane.sh
```

Boundary

This lane does not run dpkg-source, dpkg-buildpackage, debuild, lintian, FreeBSD make stage, FreeBSD make package, OpenBSD make plist, OpenBSD make package, portlint, portcheck, poudriere, or an OpenBSD bulk build.

It does not write FreeBSD or OpenBSD distinfo files.

Validation

Run:

```
sh scripts/test-debian-freebsd-openbsd-source-archive-fixture-lane.sh
```

Expected output:

```
debian_freebsd_openbsd_source_archive_fixture_lane: ok
```

docs/DEBIAN_LOCAL_DEB_STATIC_VALIDATION.md

Source title: Debian Local Deb Static Validation

Lines: 97 | Words: 215 | SHA-256: fe4b6f116739d57f3bac87c2ac2f30678d6661ee9ceb8ba17d18568975b0dcb0

Debian Local Deb Static Validation

Status: active static validation lane Scope: static checks for the local-only Debian deb packaging draft.

Purpose

This lane checks that the Debian packaging draft is present, narrow, and honest.

It verifies local packaging shape for the no-effect CLI payload, but it does not run dpkg-buildpackage, debuild, lintian, sbuild, or pbuilder, and it does not create package artifacts.

Guarded Files

```
packaging/debian/README.md
packaging/debian/debian/control
packaging/debian/debian/rules
packaging/debian/debian/changelog
packaging/debian/debian/copyright
packaging/debian/debian/install
packaging/debian/debian/source/format
docs/DEBIAN_LOCAL_DEB_STATIC_VALIDATION.md
docs/status/DEBIAN_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-debian-local-deb-static-validation.sh
.github/workflows/debian-local-deb-static-validation.yml
```

Preserved Boundary

```
local_only_draft=1
debian_local_deb_draft_present=1
debian_static_deb_validation_present=1
deb_artifact_created=0
deb_installed_on_host=0
dpkg_buildpackage_run_required=0
debuild_run_required=0
lintian_run_required=0
lintian_transcript_present=0
debian_archive_ready=0
debian_mentors_upload_claimed=0
debian_sponsorship_claimed=0
debian_ftp_master_acceptance_claimed=0
production_readiness_claimed=0
```

Package Intent

The local deb draft records only this initial payload shape:

```
usr/bin/latticra
usr/share/doc/latticra/README.md
```

The CLI remains the no-effect status surface from:

```
src/latticra_cli.c
```

Explicit Non-Claims

This static lane does not:

```
build a deb artifact
install a deb artifact
upload to mentors.debian.net
submit to Debian
claim Debian archive readiness
claim Debian sponsorship
claim ftp-master acceptance
install a system service
install kernel modules
write /etc/latticra
claim root installer readiness
claim production readiness
```

Next Recommended Lane

Add a Debian lintian availability and transcript contract only after the local deb license expression and source archive boundary are reviewed.

Validation

Run:

```
sh scripts/test-debian-local-deb-static-validation.sh
```

Expected output:

```
debian_local_deb_static_validation: ok
```

docs/FEDORA_DEVELOPER_WORKFLOW.md

Source title: Fedora Developer Workflow

Lines: 108 | Words: 309 | SHA-256: 8adcb747d9431f7216e7f9ab4aa5a54977b7b433051123bf319c5e842e46e9a5

Fedora Developer Workflow

Status: developer workflow record Scope: local Fedora Linux commands for productive Latticra development.

Purpose

This note gives a repeatable local workflow for working on Latticra from Fedora Linux after the Fedora build lane was added.

The goal is simple:

```
install the small toolchain
run the same guards as CI
keep kernel evidence work fast and repeatable
avoid packaging work until the build lane stays green
```

Install local tools

On a Fedora workstation or Fedora container, install the basic tools:

```
sudo dnf -y install git gcc make coreutils findutils diffutils grep
```

In a root container, omit sudo:

```
dnf -y install git gcc make coreutils findutils diffutils grep
```

Fast local guard loop

For a quick sanity check while editing C/kernel files, run:

```
sh scripts/test-state-lattice.sh
sh scripts/test-system-bootstrap.sh
sh scripts/test-kernel.sh
sh scripts/test-kernel-lifecycle.sh
```

Kernel evidence guard loop

Before opening a kernel evidence PR, run:

```
sh scripts/test-kernel-lifecycle.sh
sh scripts/test-kernel-lifecycle-report-runner.sh
sh scripts/test-kernel-lifecycle-subsystem-summary.sh
sh scripts/test-kernel-lifecycle-subsystem-summary-report-runner.sh
sh scripts/test-kernel-lifecycle-rollback-plan.sh
```

Fedora lane guard

Before merging Fedora-facing work, run:

```
sh scripts/test-fedora-build-lane.sh
```

Expected output:

```
fedora_build_lane: ok
```

Suggested work rhythm

Use this loop for productive development:

```
1. Create a small branch.
2. Make one bounded change.
```

3. Run the fast local guard loop.
4. Run the Fedora lane guard when the branch touches build, kernel, or Fedora-facing files.
5. Open a focused PR with clear non-claims.

Current boundary

This workflow is for local developer productivity. It does not add package metadata or change Latticra capability claims.

Next recommended lane

After this workflow note, the next Fedora-facing slice should be:

```
Fedora package metadata plan
```

That should be a planning slice before any spec file is added.

Validation

This workflow note is guarded by:

```
sh scripts/test-fedora-developer-workflow.sh
```

Expected output:

```
fedora_developer_workflow: ok
```

docs/FEDORA_DISPOSABLE_VM_EFFECT_GATE.md

Source title: Fedora Disposable VM Effect Gate

Lines: 131 | Words: 254 | SHA-256: 0b0bcf6eed8ba435265745bc2ba1be9baa051a5a451b256eda1e4e2917e2e6fe

Fedora Disposable VM Effect Gate

Status: gate record Evidence level: 8 target, gate only Scope: define the evidence boundary for a future Fedora disposable VM effect lane.

Purpose

Latticra has installroot mutation evidence and Fedora transcript evidence contracts.

This gate defines when the project may consider a future disposable Fedora VM effect lane reviewable.

This is a gate record only.

It does not add runtime behavior.

Required target evidence

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
```

Required package evidence

```
local_rpm_built_from_current_tree=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
```

Required prior evidence

```
installroot_lifecycle_evidence_present=1
post_removal_absence_evidence_present=1
manual_host_dry_run_transcript_contract_present=1
manual_host_rc_decision_classifier_present=1
rpm_gate_allowed=1
removal_rollback_ready=1
```

Effect eligibility decision

The gate decision must be one of:

```
disposable_vm_effect_gate_status=eligible
disposable_vm_effect_gate_status=blocked
disposable_vm_effect_gate_status=invalid
```

eligible means the evidence set is ready for review.

It does not mean production readiness.

Boundary fields

```
effect_gate_present=1
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_allowed=0
production_host_allowed=0
immutable_host_allowed=0
```

Block conditions

```
daily_driver_detected=1
production_host_detected=1
immutable_fedora_detected=1
snapshot_missing=1
recovery_path_missing=1
operator_consent_missing=1
rpm_payload_listing_missing=1
rpm_payload_not_documentation_only=1
unexpected_runtime_surface_present=1
prior_installroot_evidence_missing=1
prior_transcript_contract_missing=1
prior_decision_classifier_missing=1
prior_gate_evidence_missing=1
prior_removal_rollback_evidence_missing=1
```

Validation

```
sh scripts/test-fedora-disposable-vm-effect-gate.sh
```

Expected output:

```
fedora_disposable_vm_effect_gate: ok
```

Next recommended Fedora lane

```
Implement no-effect Fedora disposable VM effect gate classifier
```

That classifier should consume explicit evidence fields and report eligible, blocked, or invalid.

README overhaul hold

The root README should not claim install readiness from this gate alone.

The README overhaul should wait until real validation evidence exists for:

```
manual_host_dry_run_transcript_present=1
live_host_validation_completed=1
host_install_ready=1
```

Non-claims

This gate is not production readiness, Fedora approval, or Fedora distribution readiness.

docs/FEDORA_DISPOSABLE_VM_EFFECT_GATE_CLASSIFIER.md

Source title: Fedora Disposable VM Effect Gate Classifier

Lines: 167 | Words: 248 | SHA-256: 200122139918393f42f0482f73fc8d7a59167d75e723635294c38d87b549a05c

Fedora Disposable VM Effect Gate Classifier

Status: implementation record Evidence level: 8 Scope: pure classifier for disposable Fedora VM effect-gate evidence.

Purpose

This classifier consumes explicit evidence fields from the Fedora disposable VM effect gate and reports one of:

```
disposable_vm_effect_gate_status=eligible
disposable_vm_effect_gate_status=blocked
disposable_vm_effect_gate_status=invalid
```

The classifier is no-effect.

It does not add runtime behavior.

API surface

```
include/latticra/fedora_disposable_vm_effect_gate.h
src/fedora_disposable_vm_effect_gate.c
tests/fedora_disposable_vm_effect_gate.c
```

Public functions:

```
latticra_fedora_disposable_vm_effect_gate_classify
latticra_fedora_disposable_vm_effect_gate_report
latticra_fedora_disposable_vm_effect_gate_status_label
latticra_fedora_disposable_vm_effect_gate_denial_label
```

Eligible evidence

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
local_rpm_built_from_current_tree=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
installroot_lifecycle_evidence_present=1
post_removal_absence_evidence_present=1
manual_host_dry_run_transcript_contract_present=1
manual_host_rc_decision_classifier_present=1
rpm_gate_allowed=1
removal_rollback_ready=1
```

Eligible output boundary

```
disposable_vm_effect_gate_status=eligible
disposable_vm_effect_gate_denial=none
disposable_vm_effect_eligible=1
effect_gate_present=1
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_allowed=0
production_host_allowed=0
immutable_host_allowed=0
evidence_level=8
```

Blocked denial labels

```
not-disposable-fedora-vm
daily-driver-target
production-target
immutable-fedora-target
snapshot-missing
recovery-path-missing
operator-consent-missing
local-rpm-not-current-tree
rpm-name-mismatch
rpm-version-missing
rpm-payload-listing-missing
rpm-payload-not-documentation-only
unexpected-runtime-surface
installroot-evidence-missing
post-removal-evidence-missing
transcript-contract-missing
decision-classifier-missing
rpm-gate-not-allowed
removal-rollback-not-ready
```

Invalid input

Malformed non-binary evidence reports:

```
disposable_vm_effect_gate_status=invalid
disposable_vm_effect_gate_denial=invalid-classifier-input
disposable_vm_effect_eligible=0
```

Report header

```
FEDORA DISPOSABLE VM EFFECT GATE CLASSIFIER
```

Required report fields include:

```
classifier_evaluated=1
effect_gate_present=1
disposable_vm_effect_eligible=1
disposable_vm_effect_eligible=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_allowed=0
production_host_allowed=0
immutable_host_allowed=0
evidence_level=8
```

Validation

Classifier guard:

```
sh scripts/test-fedora-disposable-vm-effect-gate-classifier.sh
```

Docs guard:

```
sh scripts/test-fedora-disposable-vm-effect-gate-classifier-docs.sh
```

Expected output:

```
fedora_disposable_vm_effect_gate_classifier: ok
fedora_disposable_vm_effect_gate_classifier_docs: ok
```

Next recommended Fedora lane

```
Align Fedora disposable VM effect gate classifier status
```

After that status alignment, the next safe progression is a disposable Fedora VM local RPM validation lane.

Non-claims

This classifier is not production readiness, Fedora approval, or Fedora distribution readiness.

docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_LANE.md

Source title: Fedora Disposable VM Local RPM Validation Lane

Lines: 136 | Words: 352 | SHA-256: 68688397537a05fbb3e4a82dc0f777f3aa0d6a20bf6940bb3597d081270faf08

Fedora Disposable VM Local RPM Validation Lane

Status: gated validation lane Evidence level: 9 target Scope: local RPM validation on a disposable Fedora VM target only.

Purpose

This lane defines the first host-facing Fedora RPM validation path for Latticra.

The lane is restricted to a disposable Fedora VM with a clean snapshot, a recovery path, and explicit operator consent.

It is not for a daily-driver system, production host, immutable Fedora image, or unclear target.

Hard gate

The validation script must refuse unless all of the following are true:

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
ID=fedora
rpm_present=1
rpmbuild_present=1
```

Required package checks

Before any package operation, the lane must verify:

```
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
```

Expected package payload remains:

```
/usr/share/doc/latticra/README.md
```

Forbidden payload surfaces include:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
```

Validation sequence

The guarded lane is expected to:

```
1. Verify disposable Fedora VM evidence.
2. Build the local RPM from the current tree.
3. Inspect package metadata.
4. Inspect package payload.
5. Validate the package payload remains documentation-only.
6. Apply the local RPM to the disposable Fedora VM target.
7. Query the installed package state.
8. Verify the README payload exists.
9. Verify the package files.
10. Remove the package.
11. Confirm the package is absent.
```

12. Confirm the README payload is absent.
13. Emit a deterministic validation report.

Required report fields

```
FEDORA DISPOSABLE VM LOCAL RPM VALIDATION LANE
validation_status=ok
package_name=latticra
package_version_recorded=1
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
vm_rpmdb_mutated=1
vm_filesystem_mutated=1
install_validation_performed=1
removal_validation_performed=1
post_removal_absence_verified=1
live_host_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
evidence_level=9
```

Non-claims

This lane does not make Latticra production ready.

It does not claim Fedora approval.

It does not claim Fedora distribution readiness.

It does not make daily-driver installation eligible.

Validation guard

```
sh scripts/test-fedora-disposable-vm-local-rpm-validation-lane-docs.sh
```

Expected output:

```
fedora_disposable_vm_local_rpm_validation_lane_docs: ok
```

Next recommended Fedora lane

```
Add disposable Fedora VM local RPM validation status alignment
```

After real validation evidence exists, the root README can be overhauled to describe the disposable Fedora VM validation path without claiming production or Fedora distribution readiness.

docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_TRANSCRIPT_CONTRACT.md

Source title: Fedora Disposable VM Local RPM Validation Transcript Contract

Lines: 228 | Words: 474 | SHA-256: f317ae2ce5552a9f4c4f11d1b22fa63e0f9ec2f5b888abbec4d7dc39d03daaa5

Fedora Disposable VM Local RPM Validation Transcript Contract

Status: contract record Evidence level: 9 target, contract only Scope: transcript schema for a future real disposable Fedora VM local RPM validation run.

Purpose

Latticra now has a gated disposable Fedora VM local RPM validation lane and a status record for that lane.

This contract defines the transcript fields required before a real disposable Fedora VM validation run can be reviewed as evidence.

This is an evidence schema only.

It does not execute the validation runner.

It does not install or remove an RPM.

It does not capture real validation evidence by itself.

Required transcript header

```
LATTICRA FEDORA DISPOSABLE VM LOCAL RPM VALIDATION TRANSCRIPT
transcript_kind=disposable-vm-local-rpm-validation
transcript_version=1
operator_review_required=1
validation_transcript_recorded_after_real_run=1
```

Required target evidence

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
rpm_tooling_recorded=1
rpmbuild_tooling_recorded=1
```

Required runner gate evidence

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
ID=fedora
rpm_present=1
rpmbuild_present=1
```

Required package evidence

```
local_rpm_built_from_current_tree=1
rpm_build_command_recorded=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_path_recorded=1
rpm_metadata_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
```

Expected package payload remains:

```
/usr/share/doc/latticra/README.md
```

Forbidden payload surfaces remain:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
```

Required validation transcript evidence

```
install_command_recorded=1
install_result_recorded=1
rpm_query_after_install_recorded=1
installed_payload_listing_recorded=1
installed_readme_present=1
rpm_verify_completed=1
removal_command_recorded=1
removal_result_recorded=1
post_removal_query_recorded=1
post_removal_absence_verified=1
```

Required emitted validation report

The transcript must include the deterministic report emitted by the validation lane:

```
FEDORA DISPOSABLE VM LOCAL RPM VALIDATION LANE
validation_status=ok
package_name=latticra
package_version_recorded=1
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
vm_rpmdb_mutated=1
vm_filesystem_mutated=1
install_validation_performed=1
removal_validation_performed=1
post_removal_absence_verified=1
live_host_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
evidence_level=9
```

Required artifact references

A reviewed transcript should reference or embed:

```
fedora-os-release.txt
fedora-kernel-version.txt
rpm-version.txt
rpmbuild-version.txt
rpm-package-metadata.txt
rpm-payload.list
rpm-installed-payload.list
vm-validation.report
post-removal-absence-proof.txt
```

Required review sections

```
TARGET REVIEW
RUNNER GATE REVIEW
PACKAGE REVIEW
INSTALL VALIDATION REVIEW
REMOVAL VALIDATION REVIEW
POST-REMOVAL ABSENCE REVIEW
BOUNDARY REVIEW
OPERATOR SIGNOFF
NEXT ACTION REVIEW
```

Transcript decision states

```
disposable_vm_validation_transcript_status=accepted-for-review
disposable_vm_validation_transcript_status=blocked
disposable_vm_validation_transcript_status=invalid
```

accepted-for-review means the transcript is complete enough for review.

It does not mean Latticra is production ready, Fedora approved, Fedora distribution ready, or daily-driver safe.

Current project state until real evidence exists

Until the real disposable VM run is performed and reviewed, the project remains at:

```
disposable_vm_validation_transcript_present=0
disposable_vm_validation_completed=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Validation

```
sh scripts/test-fedora-disposable-vm-local-rpm-validation-transcript-contract.sh
```

Expected output:

```
fedora_disposable_vm_local_rpm_validation_transcript_contract: ok
```

Next recommended Fedora lane

```
Capture real disposable Fedora VM local RPM validation transcript evidence
```

That lane should run the already-gated validation runner only inside a disposable Fedora VM target with clean snapshot evidence, recovery evidence, and explicit operator consent.

README overhaul hold

The root README should not claim install readiness until real validation evidence exists for:

```
disposable_vm_validation_transcript_present=1
disposable_vm_validation_completed=1
live_host_validation_completed=1
host_install_ready=1
```

Non-claims

This contract is not a completed validation transcript.

It is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, or a production installer claim.

docs/FEDORA_HOST_INSTALL_PREFLIGHT_CLASSIFIER.md

Source title: Fedora Host Install Preflight Classifier

Lines: 179 | Words: 520 | SHA-256: 9da940d4db2cdc2931b17d1711c2f216856fb505e375d4b7e59d91a5c204194a

Fedora Host Install Preflight Classifier

Status: implementation record Scope: no-effect Fedora host install preflight classification before any host install lane performs mutation. Evidence level: 1

Purpose

The Fedora host install preflight classifier determines whether a captured Fedora host snapshot is eligible to be considered for a separate local RPM install lane.

The classifier is intentionally conservative. It converts host facts into a deterministic report, but it does not inspect the live host itself and does not install Latticra.

This slice exists because the Fedora install verification path is moving toward installed-state proof, while the current package posture remains local-only and guarded by non-claims.

Effect boundary

This classifier is no-effect.

It does not:

```
install the RPM
run dnf
run rpm
run rpmbuild
run rpmlint
read /etc/os-release
write files
open the network
mutate the host
start services
create boot entries
load kernel modules
change SELinux policy
claim runtime readiness
```

The classifier only consumes an already-captured snapshot and returns labels.

Snapshot fields

The caller provides facts such as:

```
os_id
os_id_like
host_install_requested
immutable_host
rpm_available
dnf_available
rpmbuild_available
rpmlint_available
local_rpm_present
root_or_sudo_available
network_required
package_is_doc_only
command_entrypoint_expected
```

rpm_available, dnf_available, local_rpm_present, and root_or_sudo_available are required before a mutable Fedora host can become a host-install candidate.

rpmbuild_available and rpmlint_available are reported, but they are not required for the host-install candidate classification because build and lint validation belong to earlier Fedora lanes.

Classifications

The public classifier labels are:

```
LATTICRA_FEDORA_PREFLIGHT_REPORT_ONLY
LATTICRA_FEDORA_PREFLIGHT_READY_LOCAL_RPM
LATTICRA_FEDORA_PREFLIGHT_BLOCKED
LATTICRA_FEDORA_PREFLIGHT_FUTURE_GATED
LATTICRA_FEDORA_PREFLIGHT_INVALID
```

ready-local-rpm means the snapshot is eligible to be considered by a later local RPM install lane. It does not mean this classifier performed installation.

future-gated is used for immutable Fedora-style hosts where rpm-ostree or image-based handling must be designed separately.

blocked is used for non-Fedora hosts, network-requiring paths, missing required tooling, missing local RPMs, missing operator privilege, or premature runtime command expectations.

Denial classes

The public denial labels are:

```
host-install-not-requested
non-fedora-host
immutable-host
network-required
required-tooling-missing
local-rpm-missing
privilege-missing
runtime-entrypoint-not-present
```

```
invalid-snapshot
```

These labels are intended to make failed preflight reports auditable without adding host mutation.

Deterministic report surface

The report begins with:

```
FEDORA HOST INSTALL PREFLIGHT
```

Required report fields include:

```
classification=ready-local-rpm
denial=none
host_classification=mutable-fedora-host
install_lane=local-doc-rpm
preflight_passed=1
host_install_candidate=1
host_install_performed=0
host_mutation_performed=0
network_allowed=0
local_only=1
fedora_host=1
mutable_fedora_host=1
immutable_host=0
rpm_available=1
dnf_available=1
local_rpm_present=1
no_effect=1
evidence_level=1
```

The no-effect fields are deliberate. A passing preflight report still records:

```
host_install_performed=0
host_mutation_performed=0
network_allowed=0
```

Current package boundary

For the current documentation-only package posture, a valid host preflight may classify a local doc RPM as a candidate. It must block any expectation that /usr/bin/latticra, a service, a kernel module, a boot entry, or an active runtime surface already exists.

If package_is_doc_only=1 and command_entrypoint_expected=1, the classifier reports:

```
classification=blocked
denial=runtime-entrypoint-not-present
install_lane=blocked-doc-only-package-runtime-command
```

Validation

This implementation is guarded by:

```
sh scripts/test-fedora-host-install-preflight.sh
```

Expected output:

```
fedora_host_install_preflight: ok
```

The documentation guard is:

```
sh scripts/test-fedora-host-install-preflight-docs.sh
```

Expected output:

```
fedora_host_install_preflight_docs: ok
```

Non-claims

This classifier does not install Latticra, does not prove Fedora package approval, does not create RPM artifacts, does not submit to Fedora, does not claim Fedora endorsement, does not provide a production installer, and does not prove operating-system readiness.

docs/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_IMPLEMENTATION.md

Source title: Fedora Install Preflight Snapshot Capture Implementation

Lines: 150 | Words: 386 | SHA-256: b224b64c7af104dc98b1657bbfc45dbe0438f69724527e46cd28c0f82d38a474

Fedora Install Preflight Snapshot Capture Implementation

Status: implementation record Scope: pure no-effect snapshot capture that feeds the Fedora host install preflight classifier. Evidence level: 2

Purpose

This implementation moves the Fedora install preflight snapshot lane beyond planning while preserving the host safety boundary.

It adds a pure C capture API that consumes caller-supplied read-only facts, parses bounded /etc/os-release text already provided by the caller, constructs a latticra_fedora_host_install_preflight_snapshot_t, and forwards that snapshot into the existing Fedora host install preflight classifier.

The implementation mutates repository code only. It does not mutate the host.

Public API

Primary header:

```
include/latticra/fedora_install_preflight_snapshot.h
```

Primary source:

```
src/fedora_install_preflight_snapshot.c
```

The main entry points are:

```
latticra_fedora_install_preflight_snapshot_capture
latticra_fedora_install_preflight_snapshot_report
latticra_fedora_snapshot_capture_status_label
```

Capture input

The caller supplies a bounded input structure with fields such as:

```
os_release_text
os_release_text_len
os_release_readable
host_install_requested
ostree_booted_marker_present
rpm_command_present
dnf_command_present
rpmbuild_command_present
rpmlint_command_present
local_rpm_path
local_rpm_path_len
local_rpm_readable
running_as_root
operator_privilege_assertion
network_required
package_is_doc_only
command_entrypoint_expected
```

The implementation does not read live files or probe live commands. It only consumes these supplied facts.

What changed

The implementation adds:

```
bounded os-release ID parsing
bounded os-release ID_LIKE parsing
local RPM path recording
```

```
immutable Fedora marker propagation
RPM tooling presence propagation
conservative root/operator privilege propagation
network-required propagation
current doc-only package posture propagation
classifier forwarding
snapshot capture report generation
```

Report surface

The capture report begins with:

```
FEDORA INSTALL PREFLIGHT SNAPSHOT CAPTURE
```

Required report fields include:

```
snapshot_capture_status=captured
os_id=fedora
os_id_like=rhel fedora
classifier_classification=ready-local-rpm
classifier_denial=none
snapshot_forwarded_to_classifier=1
os_release_read_allowed=1
command_probe_allowed=1
local_rpm_probe_allowed=1
sudo_validation_allowed=0
install_command_allowed=0
package_build_allowed=0
network_allowed=0
host_mutation_performed=0
host_install_performed=0
no_effect=1
evidence_level=2
```

Failure behavior

The capture path fails closed.

Unreadable or missing os-release text produces a partial capture and forwards an empty host identity into the classifier. The classifier then blocks the host as non-Fedora instead of assuming eligibility.

Network-required lanes remain blocked by the classifier.

Doc-only package posture blocks runtime-command expectations.

Immutable Fedora markers are forwarded as future-gated install lanes.

Validation

This implementation is guarded by:

```
sh scripts/test-fedora-install-preflight-snapshot-capture.sh
```

Expected output:

```
fedora_install_preflight_snapshot_capture: ok
```

The implementation document guard is:

```
sh scripts/test-fedora-install-preflight-snapshot-capture-docs.sh
```

Expected output:

```
fedora_install_preflight_snapshot_capture_docs: ok
```

Non-claims

This implementation does not install Latticra, perform host mutation, open the network, validate sudo, create package artifacts, perform package build behavior, perform package install behavior, start services, change boot entries, load kernel modules, claim Fedora package approval, or claim production installer readiness.

docs/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_PLAN.md

Source title: Fedora Install Preflight Snapshot Capture Plan

Lines: 203 | Words: 907 | SHA-256: 6d132299982ada6ef5ae720709c3086058c81b32f4d4018fcd55403f46d0a5c2

Fedora Install Preflight Snapshot Capture Plan

Status: planning record Scope: define how a future caller may capture read-only host facts before invoking the Fedora host install preflight classifier.

Purpose

The Fedora host install preflight classifier now accepts a caller-supplied snapshot and classifies whether a host may be considered for a later local RPM install lane.

This plan defines how that snapshot should be collected without turning preflight into installation.

The goal is to keep the capture lane conservative, deterministic, bounded, and auditable before any implementation reads host files or probes commands.

Capture boundary

Snapshot capture is read-only.

It may collect facts needed by the classifier, but it must not install, remove, update, enable, start, stop, or configure anything.

The capture lane must not run:

```
dnf install
dnf remove
dnf upgrade
dnf update
rpm -i
rpm -U
rpm -e
rpmbuild
rpmlint
systemctl enable
systemctl start
grubby
bootctl
restorecon
semodule
modprobe
```

It must also not open the network, create package artifacts, write host files, change privilege state, or validate sudo by refreshing credentials.

Snapshot source map

A future capture implementation may fill the classifier snapshot from these read-only facts:

```
| Snapshot field | Allowed source | Rule |
| --- | --- | --- |
| `os_id` | `/etc/os-release` `ID=` | Read bounded text only. |
| `os_id_like` | `/etc/os-release` `ID_LIKE=` | Read bounded text only. |
| `host_install_requested` | explicit caller/operator request | Must default to false. |
| `immutable_host` | `/run/ostree-booted` existence or `rpm-ostree` marker | Detect only; do not
 run `rpm-ostree install`. |
| `rpm_available` | `command -v rpm` | Probe command presence only. |
| `dnf_available` | `command -v dnf` | Probe command presence only. |
| `rpmbuild_available` | `command -v rpmbuild` | Probe command presence only. |
| `rpmlint_available` | `command -v rpmlint` | Probe command presence only. |
| `local_rpm_present` | explicit local RPM path | Check existence/readability only. |
| `root_or_sudo_available` | `id -u` equals `0` or explicit operator assertion | Do not run
```

`sudo -v`. |

| `network_required` | caller-supplied lane setting | Must default to false for local-only preflight. |

| `package_is_doc_only` | package metadata from the current Fedora package lane | Current default remains true. |

| `command_entrypoint_expected` | caller expectation | Must default to false while the package is doc-only. |

Read-only command allowance

The first capture implementation may use read-only probes such as:

```
command -v rpm
command -v dnf
command -v rpmbuild
command -v rpmlint
id -u
test -r /etc/os-release
test -e /run/ostree-booted
test -r &quot;$LOCAL_RPM&quot;
```

Those commands may classify the host, but they must not install or mutate.

Privilege rule

Privilege capture must be conservative.

Allowed:

```
record root_or_sudo_available=1 when id -u is 0
record root_or_sudo_available=0 when id -u is not 0 and no explicit operator-supplied privilege
fact is provided
```

Not allowed in the first capture lane:

```
sudo -v
sudo -n true
sudo dnf install
sudo rpm -i
```

The reason is simple: preflight capture should not refresh credentials, change sudo timestamp state, or perform a privileged dry run.

Immutable Fedora rule

If /run/ostree-booted exists or the caller identifies the host as rpm-ostree/image-based, capture should set:

```
immutable_host=1
```

The classifier should then return a future-gated lane rather than treating the host like a mutable Fedora RPM target.

Immutable Fedora work must remain separate because rpm-ostree/image-based systems need their own design, rollback, and safety model.

Local RPM rule

The local RPM fact must come from an explicit path chosen by the caller.

The capture lane may check that the file exists and is readable. It must not install it, query remote repositories, or build a replacement artifact.

If no explicit local RPM path is provided, capture should set:

```
local_rpm_present=0
```

Current package posture rule

The current Fedora package path remains documentation/package-shape oriented.

Until the package installs a stable runtime command, capture should set:

```
package_is_doc_only=1
```

```
command_entrypoint_expected=0
```

If a caller expects a command while the package remains doc-only, the classifier must block with:

```
denial=runtime-entrypoint-not-present
```

Future report fields

A later implementation may emit a capture report with fields such as:

```
FEDORA INSTALL PREFLIGHT SNAPSHOT CAPTURE
snapshot_capture_status=planned
os_release_read_allowed=1
command_probe_allowed=1
local_rpm_probe_allowed=1
sudo_validation_allowed=0
install_command_allowed=0
package_build_allowed=0
network_allowed=0
host_mutation_performed=0
host_install_performed=0
snapshot_forwarded_to_classifier=1
```

Failure behavior

Capture should fail closed.

If /etc/os-release is unreadable, command probes are unavailable, the local RPM path is missing, or the caller requests a network-dependent path, the capture lane should still produce a deterministic snapshot and allow the classifier to return a blocked or report-only result.

Failure to capture facts must not trigger installation, package building, network fallback, or privileged fallback.

Implementation sequence

Recommended order:

```
1. Add a small snapshot-capture contract/API.
2. Add bounded os-release parsing for ID and ID_LIKE.
3. Add command-presence probes for rpm, dnf, rpmbuild, and rpmlint.
4. Add explicit local RPM path probing.
5. Add immutable-host marker probing.
6. Add deterministic capture report fields.
7. Forward the captured snapshot into the existing classifier.
8. Add tests for mutable Fedora, immutable Fedora, non-Fedora, missing local RPM, and missing
tooling paths.
```

Validation

This planning slice is guarded by:

```
sh scripts/test-fedora-install-preflight-snapshot-capture-plan.sh
```

Expected output:

```
fedora_install_preflight_snapshot_capture_plan: ok
```

Non-claims

This plan does not implement snapshot capture.

It does not install Latticra, run dnf, run mutating rpm commands, run rpmbuild, run rpmlint, create RPM artifacts, validate sudo, mutate the host, open the network, claim Fedora package approval, or claim production installer readiness.

docs/FEDORA_INSTALL_VERIFICATION_PLAN.md

Source title: Fedora Install Verification Plan

Lines: 156 | Words: 604 | SHA-256: a24016a705f7fb7cb61e9b78dc6d665c750194d9c50c2849c4891a7f735580a5

Fedora Install Verification Plan

Status: planning record Scope: define how Latticra integrates with Fedora Linux through the local RPM path and how installed-state checks will prove the result.

Purpose

Latticra should integrate with Fedora as Fedora currently expects software to integrate: through RPM-shaped packaging, installed-file ownership, package queryability, package verification, and isolated Fedora validation lanes.

This plan keeps that integration conservative.

The current package path is local-only. It is not a Fedora submission, not a Fedora package approval request, not a Fedora spin/remix, and not an operating-system replacement claim.

Current repository state

The current local Fedora spec is a documentation-only package draft.

It installs:

```
%doc %{_docdir}/%{name}/README.md
```

It does not currently install:

```
/usr/bin/latticra
/usr/libexec/latticra/*
/etc/latticra/*
/usr/lib/systemd/system/*.service
kernel modules
bootloader entries
SELinux policy
active runtime services
```

That means the first valid installed-state proof is a doc/package-shape proof, not a runnable product proof.

Fedora integration model

Latticra should advance through these Fedora-facing stages:

```
1. Local RPM spec remains explicit and local-only.
2. Source archive policy is defined.
3. Source RPM dry-run is introduced.
4. Binary RPM build lane is introduced inside Fedora.
5. Fresh Fedora install smoke lane installs the RPM.
6. Installed package checks prove file ownership and verification.
7. CLI/runtime smoke checks are added only after a real entrypoint exists.
8. COPR, Koji, or Fedora submission paths remain future work and require separate approval
gates.
```

Installed-state checks

Once a local binary RPM exists, the install lane should run checks like:

```
rpm -q latticra
rpm -ql latticra
rpm -V latticra
test -r /usr/share/doc/latticra/README.md
```

For the current doc-only package, the install lane should also prove the absence of premature runtime surfaces:

```
test ! -e /usr/bin/latticra
test ! -d /etc/latticra
test ! -e /usr/lib/systemd/system/latticra.service
```

Those absence checks are intentional. They protect the project from accidentally claiming runtime behavior before it exists.

Future command checks

After Latticra exposes a real Fedora-installed command, the install lane may add command smoke checks such as:

```
command -v latticra
latticra --version
latticra check
```

Those checks must not be added until the package actually installs a stable command entrypoint.

Future install smoke lane outline

A future implementation lane may run inside a Fedora container and perform:

```
install build tools
create source archive
run rpmbuild
install generated RPM into a fresh Fedora environment
query package with rpm -q
list package files with rpm -ql
verify package files with rpm -V
probe expected documentation files
probe absence of premature runtime files
uninstall package
confirm package removal
```

Pass conditions

The first successful Fedora install-verification lane should prove:

```
package can be built locally
package can be installed locally
rpm database knows the package
installed files are owned by the package
rpm verification succeeds or has classified expected findings
README documentation is present
no runtime command is claimed before it exists
no service, kernel, boot, or policy surface is installed by accident
```

Boundary

This plan does not run rpmbuild.

It does not run mock.

It does not create RPM artifacts.

It does not install Latticra on the host.

It does not submit Latticra to Fedora.

It does not claim Fedora package approval, Fedora endorsement, Fedora spin/remix status, product readiness, operating-system readiness, runtime authority, kernel authority, boot authority, service authority, SELinux policy readiness, or security-hardening completion.

Next slice

Recommended next slice:

```
Add Fedora local binary RPM build plan
```

That future slice should define the temporary source archive and binary build rules without installing artifacts yet.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-install-verification-plan.sh
```

Expected output:

```
fedora_install_verification_plan: ok
```

docs/FEDORA_LIVE_READONLY_SNAPSHOT_ADAPTER.md

Source title: Fedora Live Read-Only Snapshot Adapter

Lines: 128 | Words: 341 | SHA-256: f8aab33d4b2a2ae6bb926987dad6fb16289eb19732aab79f25fdd8fb5c82a0dc

Fedora Live Read-Only Snapshot Adapter

Status: implementation record Scope: live read-only host fact capture for Fedora install preflight. Evidence level: 3

Purpose

This implementation adds the first live host adapter for the Fedora install preflight path.

The adapter performs bounded read-only probes and forwards the collected facts into the existing Fedora install preflight snapshot capture API.

This is real implementation work, but it is not a host-install mutation lane.

Public API

Primary header:

```
include/latticra/fedora_live_snapshot_adapter.h
```

Primary source:

```
src/fedora_live_snapshot_adapter.c
```

The main entry points are:

```
latticra_fedora_live_snapshot_adapter_capture
latticra_fedora_live_snapshot_adapter_report
latticra_fedora_live_snapshot_adapter_status_label
```

Live read-only probes

The adapter may perform these bounded read-only actions:

```
read os-release text from the configured path
probe command presence through PATH
probe local RPM readability
probe immutable Fedora marker existence
record effective root status
```

The default os-release path is:

```
/etc/os-release
```

Tests use fixture paths and fake PATH commands so CI does not depend on the runner being Fedora.

Report surface

The report begins with:

```
FEDORA LIVE READ-ONLY SNAPSHOT ADAPTER
```

Required report fields include:

```
adapter_status=captured
live_probe_performed=1
os_release_read_attempted=1
command_probe_performed=1
local_rpm_probe_performed=1
ostree_marker_probe_performed=1
id_probe_performed=1
capture_status=captured
classifier_classification=ready-local-rpm
snapshot_forwarded_to_classifier=1
```

```
sudo_validation_allowed=0
install_command_allowed=0
package_build_allowed=0
network_allowed=0
host_mutation_performed=0
host_install_performed=0
no_effect=1
evidence_level=3
```

Failure behavior

The adapter fails closed.

If os-release text cannot be read, the adapter records a partial capture and forwards the snapshot to the classifier. The classifier blocks unknown or non-Fedora hosts instead of assuming eligibility.

If the local RPM path is missing or unreadable, the classifier blocks the install candidate.

If the host is immutable Fedora, the classifier future-gates the lane.

If a network-dependent path is requested, the classifier blocks it.

Validation

This implementation is guarded by:

```
sh scripts/test-fedora-live-snapshot-adapter.sh
```

Expected output:

```
fedora_live_snapshot_adapter: ok
```

The implementation document guard is:

```
sh scripts/test-fedora-live-snapshot-adapter-docs.sh
```

Expected output:

```
fedora_live_snapshot_adapter_docs: ok
```

Non-claims

This implementation does not install Latticra, mutate the host, validate sudo, create package artifacts, build packages, install packages, open the network, start services, change boot entries, load kernel modules, claim Fedora approval, or claim production installer readiness.

docs/FEDORA_LOCAL_BINARY_RPM_BUILD_PLAN.md

Source title: Fedora Local Binary RPM Build Plan

Lines: 174 | Words: 614 | SHA-256: d9ff167db0fce7613abc2f83c7091d18c1d1f0cb76bc64a04f29e5df08b54aa1

Fedora Local Binary RPM Build Plan

Status: planning record Scope: define the first local binary RPM build path for Latticra without producing artifacts in this slice.

Purpose

This plan advances Latticra from a local Fedora spec draft toward a local binary RPM build lane.

The goal is conservative: define exactly how a future Fedora lane will create a source archive, run a binary RPM build, and preserve audit evidence while keeping the package local-only.

This plan does not make Latticra a Fedora package. It does not submit Latticra to Fedora. It does not install Latticra on a host.

Current package shape

The current local spec remains a documentation-only draft.

Current install payload:

```
/usr/share/doc/latticra/README.md
```

No executable command, service, kernel module, boot entry, SELinux policy, or runtime configuration is installed at this stage.

Required build inputs

A future binary RPM build lane should define these inputs before it runs rpmbuild:

```
package name: latticra
version source: packaging/fedora/latticra.spec Version field
release source: packaging/fedora/latticra.spec Release field
source archive name: latticra-&lt;version&gt;.tar.gz
source archive root directory: latticra-&lt;version&gt;/
spec path: packaging/fedora/latticra.spec
build root: temporary CI workspace only
artifact retention: audit-only unless a later release gate permits publishing
```

Source archive plan

The local binary build lane should create a temporary source archive from the checked-out repository.

Required archive properties:

```
archive name matches Source0
archive root directory matches %autosetup -n %{name}-%{version}
archive excludes .git
archive excludes CI-only temporary output
archive includes README.md
archive includes scripts required by %build
archive includes docs needed by current package metadata
```

The archive should be created in a temporary RPM topdir, not committed to the repository.

Binary build command plan

A future implementation lane may run a Fedora-container command shaped like:

```
rpmbuild -bb packaging/fedora/latticra.spec \
  --define &quot;_topdir ${PWD}/.rpmwork&quot; \
  --define &quot;_sourcedir ${PWD}/.rpmwork/SOURCES&quot; \
  --define &quot;_rpmdir ${PWD}/.rpmwork/RPMS&quot; \
  --define &quot;_builddir ${PWD}/.rpmwork/BUILD&quot; \
  --define &quot;_buildrootdir ${PWD}/.rpmwork/BUILDROOT&quot;
```

The first build lane should run only inside Fedora CI or an explicitly disposable Fedora environment.

Expected local output

A successful future binary build lane should produce a local RPM under a temporary path like:

```
.rpmwork/RPMS/*/latticra-*.rpm
```

The lane should record:

```
rpmbuild exit status
RPM path
RPM filename
RPM package metadata summary
RPM payload file list
RPM verification plan handoff
```

Post-build checks before install

Before any install smoke lane, the future binary build lane should inspect the generated RPM without installing it:

```
rpm -qpi .rpmwork/RPMS/*/latticra-*.rpm
rpm -qpl .rpmwork/RPMS/*/latticra-*.rpm
```

Expected payload at the current stage:

```
/usr/share/doc/latticra/README.md
```

Unexpected payload at the current stage:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
kernel modules
bootloader files
SELinux policy files
```

Relationship to install verification

This plan comes before the install-verification lane.

The correct order is:

1. binary RPM build plan
2. temporary source archive lane
3. local binary RPM build lane
4. RPM payload inspection lane
5. fresh Fedora install smoke lane
6. command/runtime smoke checks only after a stable entrypoint exists

Boundary

This plan does not run rpmbuild.

It does not run mock.

It does not create source archives.

It does not create binary RPM artifacts.

It does not install Latticra.

It does not publish package artifacts.

It does not submit Latticra to Fedora.

It does not claim Fedora package approval, Fedora endorsement, Fedora spin/remix status, product readiness, operating-system readiness, runtime authority, kernel authority, boot authority, service authority, SELinux policy readiness, or security-hardening completion.

Next slice

Recommended next slice:

```
Add Fedora source archive fixture lane
```

That future slice should create and inspect the temporary source archive only, without running the binary RPM build yet.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-local-binary-rpm-build-plan.sh
```

Expected output:

```
fedora_local_binary_rpm_build_plan: ok
```

docs/FEDORA_LOCAL_RPM_EXECUTION_LANE.md

Source title: Fedora Installroot RPM Mutation Lane

Lines: 166 | Words: 564 | SHA-256: 14d68a34ddd9ce8325604811342030d82bd7e9fda99bfe3dbd10fc7c4006058e

Fedora Installroot RPM Mutation Lane

Status: active execution lane Scope: tightly gated local RPM install/remove mutation inside a temporary RPM installroot in disposable Fedora CI. Evidence level: 6

Purpose

This lane is the first intentionally mutating Fedora RPM lane for Latticra.

It builds the local-only Latticra RPM, initializes a temporary RPM database under a temporary installroot, installs the package into that installroot, verifies installed package state, removes it from that installroot, and verifies post-removal absence.

This is the first lane where Latticra intentionally records:

```
install_mutation_performed=1
removal_mutation_performed=1
installroot_filesystem_mutated=1
installroot_rpmdb_mutated=1
```

The mutation is restricted to a temporary installroot created inside a throwaway Fedora CI container.

The live Fedora CI container RPM database remains untouched by the Latticra package execution itself.

Hard gate

The execution script refuses to run unless all of the following are true:

```
LATTICRA_ALLOW_INSTALLROOT_RPM_MUTATION=1
ID=fedora
GITHUB_ACTIONS=true
rpmbuild_present=1
rpm_present=1
```

If any condition is absent, the lane exits before installing or removing the package.

What mutates

The lane mutates only the temporary installroot:

```
installroot_rpm_database=mutated
installroot:/usr/share/doc/latticra/README.md=installed_then_removed
```

The report explicitly records:

```
live_container_rpmdb_mutated=0
developer_host_mutation_performed=0
```

It does not mutate a developer workstation, laptop, VM host, boot entry, kernel module set, service registry, SELinux policy, network configuration, firmware state, or the live Fedora container RPM database with the Latticra package.

Execution sequence

The guard performs this sequence:

1. Verify the disposable Fedora CI installroot mutation gate.
2. Build a temporary Source0 archive from the checked-out tree.
3. Run rpmbuild -bb against packaging/fedora/latticra.spec.
4. Locate the produced local latticra RPM.
5. Inspect package metadata and payload.
6. Initialize a temporary installroot RPM database.
7. Install the local RPM into the installroot with rpm --root.
8. Query installed installroot package state with rpm --root.

9. Verify installroot package files with rpm --root.
10. Confirm README payload exists under the installroot.
11. Confirm premature runtime surfaces are absent from the package payload.
12. Remove the package from the installroot with rpm --root.
13. Confirm rpm --root -q latticra fails after removal.
14. Confirm README payload is absent from the installroot after removal.
15. Emit a deterministic execution report.

Required report fields

The lane emits:

```
FEDORA INSTALLROOT RPM MUTATION LANE
execution_status=ok
install_mutation_allowed=1
install_mutation_performed=1
removal_mutation_allowed=1
removal_mutation_performed=1
installroot_filesystem_mutated=1
installroot_rpmdb_mutated=1
live_container_rpmdb_mutated=0
developer_host_mutation_performed=0
boot_operation_performed=0
kernel_operation_performed=0
service_operation_performed=0
policy_operation_performed=0
network_allowed_during_rpm_execution=0
post_removal_absence_verified=1
rollback_planned=1
partial_failure_report_required=1
evidence_level=6
```

Package truth preserved

The current RPM remains a documentation-only local package.

Expected installed payload:

```
/usr/share/doc/latticra/README.md
```

The lane rejects premature runtime surfaces such as:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
```

Relationship to previous gates

This lane follows the Fedora RPM gate classifier and the Fedora RPM removal rollback classifier.

The RPM gate proves install posture.

The removal rollback classifier proves removal/rollback posture.

This execution lane is where the disposable installroot actually mutates.

Validation

Execution guard:

```
LATTICRA_ALLOW_INSTALLROOT_RPM_MUTATION=1 sh
scripts/test-fedora-installroot-rpm-mutation-lane.sh
```

Expected output:

```
fedora_installroot_rpm_mutation_lane: ok
```

Docs guard:

```
sh scripts/test-fedora-installroot-rpm-mutation-lane-docs.sh
```

Expected output:

```
fedora_installroot_rpm_mutation_lane_docs: ok
```

Non-claims

This lane does not install Latticra on a developer host, daily-driver Fedora system, immutable Fedora host, VM host, production host, live Fedora container RPM database, or user workstation.

It does not publish artifacts, submit to Fedora, claim Fedora approval, claim production installer readiness, validate sudo on a user host, start services, change boot entries, load kernel modules, modify SELinux policy, open runtime network access, or assert that Latticra is ready as a system installer.

docs/FEDORA_LOCAL_RPM_INSTALL_MUTATION_GATE_CONTRACT.md

Source title: Fedora Local RPM Install Mutation Gate Contract

Lines: 163 | Words: 503 | SHA-256: f27dde81d6f05d827a0f604c799d7e70822c990d4aab4a67c7f333e4a78090bc

Fedora Local RPM Install Mutation Gate Contract

Status: contract record Scope: define the hard gate before any Fedora local RPM install mutation can be represented or implemented. Evidence level: 4 target, contract only

Purpose

Latticra now has a Fedora host install preflight classifier, a no-effect snapshot capture path, and a live read-only adapter.

This contract defines the next boundary: when a future Fedora local RPM install mutation may be considered.

This is intentionally not an installer implementation.

The purpose is to prevent accidental host mutation by requiring every install precondition to be explicit, reported, and testable before any code is allowed to execute a package install action.

Required gate inputs

A future install mutation gate must consume an explicit record containing at least:

```
fedora_host
mutable_fedora_host
immutable_host
local_rpm_present
local_rpm_path
root_or_sudo_available
network_required
operator_install_confirmation
dry_run_passed
rollback_or_remove_plan_present
preflight_classification
preflight_denial
snapshot_capture_status
live_probe_performed
```

The gate must fail closed when any required field is missing, false, unknown, or contradictory.

Required allow conditions

A future local RPM install mutation may only be represented as allowed when all conditions are true:

```
fedora_host=1
mutable_fedora_host=1
immutable_host=0
local_rpm_present=1
root_or_sudo_available=1
network_required=0
```

```
operator_install_confirmation=1
dry_run_passed=1
rollback_or_remove_plan_present=1
preflight_classification=ready-local-rpm
preflight_denial=none
live_probe_performed=1
```

Every other combination must deny the install mutation.

Required denial reasons

The gate must report deterministic denial reasons such as:

```
not-fedora-host
immutable-fedora-host
local-rpm-missing
privilege-missing
network-required
operator-confirmation-missing
dry-run-missing
rollback-plan-missing
preflight-not-ready
preflight-denied
live-probe-missing
invalid-gate-input
```

Mutation status fields

The first gate implementation must remain a classifier. It must not execute install commands.

It should report fields such as:

```
FEDORA LOCAL RPM INSTALL MUTATION GATE
install_gate_status=allowed
install_mutation_allowed=1
install_mutation_performed=0
host_mutation_performed=0
network_allowed=0
operator_install_confirmation=1
dry_run_passed=1
rollback_or_remove_plan_present=1
evidence_level=4
```

For denied cases, it should report:

```
install_gate_status=denied
install_mutation_allowed=0
install_mutation_performed=0
host_mutation_performed=0
```

Command boundary

This contract does not authorize any command execution.

The first implementation after this contract should be a pure gate classifier. It must not run package installation commands, privilege commands, package build commands, network commands, service commands, boot commands, or kernel module commands.

Future install execution boundary

A later execution lane may be considered only after the mutation gate classifier exists and passes focused tests.

That later lane must use a separate contract and should require:

```
explicit operator invocation
local RPM path
preflight report
live read-only adapter report
mutation gate report
rollback/removal plan
post-install verification plan
clear command transcript boundary
```

Removal and rollback requirement

No install execution lane should be added until Latticra has a documented removal or rollback path.

At minimum, the future removal plan should define:

```
how the installed package is identified
how package ownership is verified
how uninstall is requested
how post-removal absence is verified
how failure is reported without hiding partial state
```

Validation

This contract is guarded by:

```
sh scripts/test-fedora-local-rpm-install-mutation-gate-contract.sh
```

Expected output:

```
fedora_local_rpm_install_mutation_gate_contract: ok
```

Non-claims

This contract does not implement installation.

It does not install Latticra, mutate the host, validate sudo, run package tools, build packages, open the network, start services, change boot entries, load kernel modules, claim Fedora approval, or claim production installer readiness.

docs/FEDORA_LOCAL_RPM_REMOVAL_ROLLBACK_PLAN.md

Source title: Fedora Local RPM Removal and Rollback Plan

Lines: 197 | Words: 657 | SHA-256: c0cbcf31af2ee69edd8d3d781245583347a79a7b54011fc02b9475116028b2e8

Fedora Local RPM Removal and Rollback Plan

Status: planning record Scope: define removal, rollback, and post-removal verification requirements before any Fedora local RPM execution lane is added.

Purpose

The Fedora RPM gate classifier can now report whether a future local RPM gate is allowed or denied.

Before any host-changing execution lane exists, Latticra needs a removal and rollback plan.

This plan defines how a future lane must identify an installed local RPM package, verify ownership, request removal, verify absence, and report partial failure without hiding host state.

This is planning only. It does not remove or install anything.

Removal boundary

A future removal lane must be explicit and operator-directed.

It must never infer that host changes are safe merely because a package name exists.

The removal lane must require:

```
operator_removal_confirmation=1
package_name=latticra
package_query_ready=1
installed_package_detected=1
owned_file_list_available=1
post_removal_absence_check_planned=1
```

```
failure_report_planned=1
```

If any field is missing, false, unknown, or contradictory, the removal lane must deny execution.

Package identification

A future implementation must identify the package using a bounded package identity record:

```
package_name
package_epoch
package_version
package_release
package_arch
package_origin=local-rpm
```

The package name must be exact. Partial matching, fuzzy matching, glob matching, or substring matching must not be used for removal decisions.

Ownership verification

Before removal can be allowed, a future lane must prove the package owns the expected files.

For the current documentation-oriented package posture, expected evidence includes:

```
package_query_ready=1
owned_file_list_available=1
readme_owned_by_package=1
unexpected_runtime_surface_absent=1
```

The future lane must not assume ownership from file existence alone.

Current package posture

The current Fedora package path remains documentation/package-shape oriented.

A future removal plan should expect documentation ownership evidence first, not runtime service ownership.

The following runtime surfaces should still be absent unless a later package posture intentionally changes them:

```
/usr/bin/latticra
/etc/latticra
systemd service surface
kernel module surface
boot entry surface
SELinux policy surface
```

Removal request boundary

A future implementation may only request removal after the explicit removal gate allows it.

The removal request must be local-only and must not open the network, build packages, alter boot entries, start or stop services, load kernel modules, or change policy.

The removal action, when eventually implemented, must produce a transcript boundary and deterministic report fields before and after the request.

Post-removal verification

A future removal lane must check and report:

```
package_absent_after_removal
owned_files_absent_or_reclassified
readme_absent_or_no_longer_owned
runtime_surface_absent
service_surface_absent
kernel_surface_absent
boot_surface_absent
policy_surface_absent
```

If post-removal verification fails, the lane must report partial state explicitly.

Partial failure reporting

A future implementation must not hide partial failures.

Required partial-failure labels include:

```
removal-request-failed
package-still-installed
owned-files-remain
ownership-query-failed
absence-check-failed
unknown-host-state
operator-aborted
```

Rollback meaning

For the local RPM lane, rollback means returning to a clearly reported host state after a failed install or removal attempt.

At this stage, rollback does not mean automatic system recovery, rpm-ostree rollback, boot rollback, filesystem snapshot rollback, service restart, or kernel recovery.

Those require separate contracts.

Future report fields

A future removal/rollback classifier may report fields such as:

```
FEDORA LOCAL RPM REMOVAL ROLLBACK PLAN
removal_plan_status=planned
operator_removal_confirmation_required=1
package_identity_required=1
ownership_verification_required=1
post_removal_absence_required=1
partial_failure_report_required=1
network_allowed=0
package_build_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
host_mutation_performed=0
```

Relationship to the RPM gate classifier

The RPM gate classifier can report that the future local RPM gate is allowed.

That is not enough to execute host changes.

Before execution, the project must also have:

```
removal_or_rollback_plan_present=1
post_install_verification_plan_present=1
operator_execution_confirmation=1
clear command transcript boundary
```

Implementation sequence

Recommended order:

```
1. Add this removal and rollback plan.
2. Add a removal/rollback classifier contract.
3. Add a pure removal/rollback classifier implementation.
4. Add a post-install verification contract.
5. Add a command transcript contract.
6. Only then consider a tightly gated local RPM execution lane.
```

Validation

This plan is guarded by:

```
sh scripts/test-fedora-local-rpm-removal-rollback-plan.sh
```

Expected output:

```
fedora_local_rpm_removal_rollback_plan: ok
```

Non-claims

This plan does not implement removal, rollback, installation, host mutation, package execution, privilege validation, package build behavior, network use, service operation, boot operation, kernel module operation, Fedora approval, or production installer readiness.

docs/FEDORA_LOCAL_RPM_SPEC_DRAFT_PLAN.md

Source title: Fedora Local RPM Spec Draft Plan

Lines: 149 | Words: 370 | SHA-256: 32fb8382fbe40a383a9748f8448e17d612adf4287e2a8c05d95e12d23dfd1d1c

Fedora Local RPM Spec Draft Plan

Status: planning record Scope: local-only RPM spec draft preparation before adding a spec file.

Purpose

This plan defines what must be true before Latticra adds a local RPM spec draft.

The goal is to prepare carefully without creating packaging metadata too early.

Current order:

```
Fedora readiness plan
Fedora build lane
Fedora developer workflow
Fedora package metadata plan
local RPM spec draft plan
local spec draft later
```

Current prerequisites

The following records should already exist:

```
docs/FEDORA_READINESS_PLAN.md
docs/FEDORA_DEVELOPER_WORKFLOW.md
docs/FEDORA_PACKAGE_METADATA_PLAN.md
scripts/test-fedora-build-lane.sh
```

Draft spec path

The eventual local draft path should be:

```
packaging/fedora/latticra.spec
```

Do not add that file in this planning slice.

Spec draft inputs

Before the spec draft exists, collect:

```
Name
Version
Release
Summary
License
URL
Source0
BuildRequires
Requires
%description
%prep
%build
%install
%files
```

```
%changelog
```

Initial package intent

The first local spec draft should target development and evidence tooling only.

It should not install or claim:

```
bootable system image
kernel replacement
runtime service
daemon
hardware authority
Fedora derivative
```

Build command posture

The first draft should prefer existing guard scripts instead of inventing a new build system.

Initial candidate build/check commands:

```
sh scripts/test-fedora-build-lane.sh
sh scripts/test-kernel-lifecycle-report-runner.sh
sh scripts/test-kernel-lifecycle-subsystem-summary-report-runner.sh
```

File installation posture

The first local spec draft should install only reviewed artifacts.

Candidate install groups:

```
license files
selected docs
optional report tools only after install path is reviewed
```

No install path is approved by this plan.

License gate

Do not write the spec License field until the license expression is reviewed.

Required checks:

```
root license state reviewed
file-level SPDX state reviewed
AGPL migration status reviewed
legacy Apache-2.0 paths reviewed
documentation license decision reviewed
```

Local-only boundary

The future spec draft should be local-only until lint and local build validation are added.

This plan does not submit, publish, or claim package readiness.

Next slice

Recommended next slice after this plan:

```
Add local RPM spec draft skeleton
```

That future slice may add packaging/fedora/latticra.spec only if it is explicitly marked local-only and guarded by a static spec check.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-local-rpm-spec-draft-plan.sh
```

Expected output:

```
fedora_local_rpm_spec_draft_plan: ok
```

docs/FEDORA_LOCAL_RPM_STATIC_VALIDATION.md

Source title: Fedora Local RPM Static Validation

Lines: 68 | Words: 187 | SHA-256: 1fa9114762e2b8449595adb721d9868b1489dfeb029ce871d28f24bfb1f41df2

Fedora Local RPM Static Validation

Status: active static validation lane Scope: static checks for the local RPM spec skeleton.

Purpose

This lane turns the local validation plan into a first executable check.

It checks the current local spec skeleton without building packages, producing artifacts, running mock, or requiring rpmlint output.

Files

```
docs/FEDORA_LOCAL_RPM_STATIC_VALIDATION.md
packaging/fedora/README.md
scripts/test-fedora-local-rpm-static-validation.sh
.github/workflows/fedora-local-rpm-static-validation.yml
```

Checks

The static validation lane verifies:

```
spec skeleton guard passes
validation plan guard passes
local-only draft marker exists
local packaging README records non-claims
placeholder license remains explicit
local release marker remains explicit
required RPM sections exist
spec does not install binaries yet
spec does not define scriptlet sections yet
spec does not add service files yet
```

Boundary

This lane is static validation only.

It does not run rpmbuild, rpmlint, or mock.

It does not create package artifacts.

Next slice

Recommended next slice:

```
Add rpmlint availability lane
```

That future lane should only check tool availability and static lint readiness before any source or binary package build is attempted.

Validation

Run:

```
sh scripts/test-fedora-local-rpm-static-validation.sh
```

Expected output:

```
fedora_local_rpm_static_validation: ok
```

docs/FEDORA_LOCAL_RPM_VALIDATION_PLAN.md

Source title: Fedora Local RPM Validation Plan

Lines: 135 | Words: 355 | SHA-256: a35882bdd55dd0ba432b88804a4ad466aa351074924d425662b6fd27aaf698dd

Fedora Local RPM Validation Plan

Status: planning record Scope: local validation steps for the Latticra RPM draft skeleton.

Purpose

This plan defines the validation path before the local RPM skeleton is allowed to become a real build lane.

The goal is to make package validation repeatable while keeping the current RPM spec clearly local-only.

Current prerequisites

The following records and files should already exist:

```
docs/FEDORA_READINESS_PLAN.md
docs/FEDORA_PACKAGE_METADATA_PLAN.md
docs/FEDORA_LOCAL_RPM_SPEC_DRAFT_PLAN.md
packaging/fedora/latticra.spec
scripts/test-fedora-local-rpm-spec-skeleton.sh
```

Validation stages

Recommended validation order:

```
1. Static spec skeleton guard
2. Fedora container guard
3. rpmlint availability check
4. source archive policy check
5. source RPM dry-run plan
6. local mock build plan
7. installed file inspection plan
```

Static validation

The static guard should continue checking:

```
local-only draft marker
Name field
Version field
Release field
Summary field
License placeholder
URL field
Source0 placeholder
BuildRequires fields
required RPM sections
current guard commands
```

rpmlint plan

A later implementation lane may install and run:

```
rpmlint packaging/fedora/latticra.spec
```

This plan does not require rpmlint output yet.

Source RPM plan

A later implementation lane may create a temporary source archive and run a source RPM build dry run.

Required unresolved decisions before that lane:

```
version tag policy
source archive naming
```

```
license expression
license files
installed file list
```

Mock build plan

A later implementation lane may run a local mock build only after the source RPM lane is stable.

Mock validation should record:

```
mock config used
source RPM path
build result
logs location
known failures
```

Installed file inspection plan

After a local binary RPM exists, inspect:

```
installed docs
installed license files
installed binaries
file ownership
file permissions
uninstall behavior
```

Boundary

This is a validation plan only.

It does not run rpmbuild, rpmlint, or mock. It does not produce RPM artifacts.

Next slice

Recommended next slice after this plan:

```
Add local RPM static validation lane
```

That future slice should extend the current static spec skeleton guard without creating package artifacts.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-local-rpm-validation-plan.sh
```

Expected output:

```
fedora_local_rpm_validation_plan: ok
```

docs/FEDORA_MANUAL_HOST_DRY_RUN_TRANSCRIPT_CONTRACT.md

Source title: Fedora Manual Host Dry-Run Transcript Contract

Lines: 158 | Words: 278 | SHA-256: a834468ac868d59756cb336867dc7ddf17c2b1c3c5c5e7d6fe9207768b69e5c5

Fedora Manual Host Dry-Run Transcript Contract

Status: contract record Evidence level: 8 target, contract only Scope: transcript schema for a future disposable Fedora VM validation review.

Purpose

Latticra has a manual Fedora RC checklist and a no-effect RC decision classifier.

This contract defines the transcript fields required before a future disposable Fedora VM validation record can be reviewed.

This is an evidence schema only.

It does not execute anything.

Required transcript header

```
LATTICRA FEDORA MANUAL HOST DRY-RUN TRANSCRIPT
transcript_kind=dry-run
transcript_version=1
operator_review_required=1
host_change_performed=0
```

Required target evidence

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
```

Required package evidence

```
local_rpm_built_from_current_tree=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
```

Required prior evidence

```
installroot_lifecycle_evidence_present=1
post_removal_absence_evidence_present=1
host_preflight_ready_local_rpm=1
rpm_gate_allowed=1
removal_rollback_ready=1
manual_host_rc_decision_classifier_present=1
```

Required no-effect fields

```
manual_host_dry_run_transcript_present=1
live_host_validation_completed=0
host_change_performed=0
sudo_invoked=0
rpm_invoked=0
dnf_invoked=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Required review sections

```
TARGET REVIEW
PACKAGE REVIEW
PRIOR EVIDENCE REVIEW
DRY-RUN DECISION REVIEW
NO-EFFECT BOUNDARY REVIEW
OPERATOR SIGNOFF PLACEHOLDER
NEXT ACTION REVIEW
```

Dry-run decision states

```
dry_run_transcript_status=accepted-for-review
dry_run_transcript_status=blocked
dry_run_transcript_status=invalid
```

accepted-for-review means the transcript is complete enough for review.

It does not mean live validation is complete.

Block conditions

```
daily_driver_detected=1
production_host_detected=1
immutable_fedora_detected=1
snapshot_missing=1
recovery_path_missing=1
operator_consent_missing=1
rpm_payload_listing_missing=1
rpm_payload_not_documentation_only=1
unexpected_runtime_surface_present=1
prior_installroot_evidence_missing=1
prior_removal_evidence_missing=1
prior_preflight_evidence_missing=1
prior_gate_evidence_missing=1
prior_removal_rollback_evidence_missing=1
network_requirement_detected=1
service_boundary_detected=1
boot_boundary_detected=1
kernel_boundary_detected=1
policy_boundary_detected=1
```

Validation

```
sh scripts/test-fedora-manual-host-dry-run-transcript-contract.sh
```

Expected output:

```
fedora_manual_host_dry_run_transcript_contract: ok
```

Next recommended Fedora lane

```
Implement no-effect Fedora dry-run transcript classifier
```

README overhaul hold

The root README should wait until real validation evidence exists for:

```
manual_host_dry_run_transcript_present=1
live_host_validation_completed=1
host_install_ready=1
```

Non-claims

This contract is not production readiness, Fedora approval, or Fedora distribution readiness.

docs/FEDORA_MANUAL_HOST_RC_CHECKLIST.md

Source title: Fedora Manual Host Release-Candidate Checklist

Lines: 152 | Words: 368 | SHA-256: 125fc19d4b1b68be30abf00b79db27f2f6915a923511ec7e1891fa6ce98c9ad5

Fedora Manual Host Release-Candidate Checklist

Status: release-candidate checklist Scope: define go/no-go evidence before any manual Fedora host validation milestone. Evidence level: 7 target, checklist only

Purpose

Latticra has controlled Fedora installroot RPM lifecycle evidence.

This checklist defines the evidence required before the project may call a real Fedora host validation step a release candidate.

This document does not perform host changes.

It does not add an execution path.

Current readiness statement

```
controlled_installroot_lifecycle_ready=1
manual_host_release_candidate_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Required environment evidence

A future manual host release-candidate step requires:

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
```

Any primary workstation, production machine, immutable image, or unclear target must remain blocked.

Required package evidence

The local RPM candidate must have:

```
local_rpm_built_from_current_tree=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
installroot_lifecycle_evidence_present=1
post_removal_absence_evidence_present=1
```

Required project evidence

The following project evidence must exist:

```
host_preflight_classification=ready-local-rpm
rpm_gate_status=allowed
removal_rollback_status=removal-ready
controlled_installroot_lifecycle_ready=1
manual_host_release_candidate_ready=0
```

The final value is intentionally still zero. This checklist is the bridge to a future narrowly scoped candidate state.

Hard stop conditions

The release-candidate step must remain blocked if any of the following are true:

```
daily_driver_detected=1
production_host_detected=1
immutable_fedora_detected=1
snapshot_missing=1
recovery_path_missing=1
operator_consent_missing=1
unexpected_runtime_surface_present=1
installroot_lifecycle_evidence_missing=1
removal_rollback_evidence_missing=1
network_requirement_detected=1
service_boundary_detected=1
boot_boundary_detected=1
kernel_boundary_detected=1
policy_boundary_detected=1
```

Release-candidate decision output

Current expected output:

```
FEDORA MANUAL HOST RC CHECKLIST
manual_host_rc_status=blocked
manual_host_release_candidate_ready=0
host_change_performed=0
production_installer_ready=0
```

```
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Future candidate output, only after all evidence is satisfied:

```
manual_host_rc_status=candidate
manual_host_release_candidate_ready=1
host_change_performed=0
```

Any future host-changing step must remain separate from this checklist.

Honest public wording

After this checklist lands, the project may say:

```
Latticra has controlled Fedora installroot RPM lifecycle evidence.
Latticra has a manual Fedora host release-candidate checklist.
Latticra is preparing for a tightly controlled host validation candidate.
```

The project must not say:

```
Latticra is production installer ready.
Latticra is Fedora-approved.
Latticra is safe for daily-driver installation.
Latticra has completed live host validation.
```

Validation

This checklist is guarded by:

```
sh scripts/test-fedora-manual-host-rc-checklist.sh
```

Expected output:

```
fedora_manual_host_rc_checklist: ok
```

Non-claims

This checklist does not perform host changes, publish artifacts, submit to Fedora, claim Fedora approval, claim production installer readiness, start services, change boot entries, load kernel modules, change host policy, or open runtime network access.

docs/FEDORA_MANUAL_HOST_RC_DECISION_CLASSIFIER.md

Source title: Fedora Manual Host RC Decision Classifier

Lines: 213 | Words: 383 | SHA-256: 978413db7bfb2ebd669d17a1a70e3191ce3266226adc92bf4370d8bcdbffb756

Fedora Manual Host RC Decision Classifier

Status: implementation record Evidence level: 7 Scope: pure no-effect classifier for manual Fedora host release-candidate decision evidence.

Purpose

This classifier consumes explicit manual-host checklist evidence and reports whether a future Fedora manual host validation step remains blocked or may be treated as an internal Latticra release-candidate decision.

It is the implementation follow-up to the Fedora manual host release-candidate checklist.

It does not install Latticra.

It does not run host commands.

It does not perform Fedora QA, claim Fedora approval, or produce Fedora distribution readiness.

API surface

```
include/latticra/fedora_manual_host_rc_decision.h
src/fedora_manual_host_rc_decision.c
tests/fedora_manual_host_rc_decision.c
```

Public functions:

```
latticra_fedora_manual_host_rc_decision_classify
latticra_fedora_manual_host_rc_decision_report
latticra_fedora_manual_host_rc_decision_status_label
latticra_fedora_manual_host_rc_decision_denial_label
```

Candidate evidence

A candidate report requires all of the following input evidence:

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
local_rpm_built_from_current_tree=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
installroot_lifecycle_evidence_present=1
post_removal_absence_evidence_present=1
host_preflight_ready_local_rpm=1
rpm_gate_allowed=1
removal_rollback_ready=1
network_requirement_detected=0
service_boundary_detected=0
boot_boundary_detected=0
kernel_boundary_detected=0
policy_boundary_detected=0
```

Candidate output remains no-effect:

```
manual_host_rc_status=candidate
manual_host_rc_denial=none
manual_host_release_candidate_ready=1
live_host_validation_completed=0
host_change_performed=0
sudo_invoked=0
rpm_invoked=0
dnf_invoked=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
evidence_level=7
```

Blocked states

The classifier reports manual_host_rc_status=blocked when any required evidence is missing or any hard-stop boundary is detected.

Denial labels include:

```
not-disposable-fedora-vm
daily-driver-target
production-target
immutable-fedora-target
snapshot-missing
recovery-path-missing
operator-consent-missing
local-rpm-not-current-tree
rpm-payload-listing-missing
rpm-payload-not-documentation-only
unexpected-runtime-surface
installroot-evidence-missing
post-removal-evidence-missing
```

```
preflight-not-ready
rpm-gate-not-allowed
removal-rollback-not-ready
network-required
service-boundary
boot-boundary
kernel-boundary
policy-boundary
```

Blocked output preserves:

```
manual_host_release_candidate_ready=0
live_host_validation_completed=0
host_change_performed=0
sudo_invoked=0
rpm_invoked=0
dnf_invoked=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Invalid input

The classifier expects explicit binary evidence values.

Malformed non-binary evidence reports:

```
manual_host_rc_status=invalid
manual_host_rc_denial=invalid-classifier-input
manual_host_release_candidate_ready=0
```

Report header

Reports begin with:

```
FEDORA MANUAL HOST RC DECISION CLASSIFIER
```

Required report fields include:

```
manual_host_rc_status=candidate
manual_host_rc_status=blocked
manual_host_rc_status=invalid
manual_host_rc_denial=none
manual_host_rc_denial=invalid-classifier-input
classifier_evaluated=1
manual_host_release_candidate_ready=1
manual_host_release_candidate_ready=0
live_host_validation_completed=0
host_change_performed=0
sudo_invoked=0
rpm_invoked=0
dnf_invoked=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
evidence_level=7
```

Validation

Classifier guard:

```
sh scripts/test-fedora-manual-host-rc-decision-classifier.sh
```

Docs guard:

```
sh scripts/test-fedora-manual-host-rc-decision-classifier-docs.sh
```

Expected output:

```
fedora_manual_host_rc_decision_classifier: ok
```

```
fedora_manual_host_rc_decision_classifier_docs: ok
```

Next recommended Fedora lane

```
Align Fedora manual host RC decision classifier status
```

After status alignment, the next safe progression is a manual Fedora host dry-run transcript contract.

Non-claims

This classifier does not implement installation, live host validation, host mutation, privilege escalation, package execution, package removal, service changes, boot changes, kernel changes, policy changes, network access, Fedora QA approval, Fedora distribution readiness, or production installer readiness.

docs/FEDORA_PACKAGE_METADATA_PLAN.md

Source title: Fedora Package Metadata Plan

Lines: 131 | Words: 292 | SHA-256: c10ecac4a7f0506e20e410696e4f002ee34d367851e813444bbcd3e6598b3e37

Fedora Package Metadata Plan

Status: planning record Scope: metadata required before a Fedora-oriented package draft exists.

Purpose

This plan defines the package metadata Latticra should collect before adding any RPM spec file.

The goal is to keep Fedora-facing work disciplined:

```
build lane first
local developer workflow second
metadata plan third
spec draft later
```

Current prerequisites

The following Fedora-facing records already exist:

```
docs/FEDORA_READINESS_PLAN.md
docs/FEDORA_DEVELOPER_WORKFLOW.md
scripts/test-fedora-build-lane.sh
```

Required metadata fields

A later package draft should define:

```
package_name
summary
description
upstream_url
source_archive_policy
version_source
license_expression
license_files
build_requires
runtime_requires
installed_binaries
installed_docs
installed_license_files
changelog_policy
```

Initial proposed values

Initial planning values:

```
package_name=latticra
summary=Contract-first systems architecture and language project
```

```
description_scope=development and evidence tooling
upstream_url=https://github.com/Bryforge/Latticra
source_archive_policy=tagged upstream source archive only
version_source=upstream tag
license_expression=to-be-confirmed-before-spec
license_files=LICENSE plus reviewed file-level notices
build_requires=gcc, make
runtime_requires=none-confirmed-yet
installed_binaries=none-confirmed-yet
installed_docs=docs selected after review
installed_license_files=license files selected after audit
changelog_policy=derive from reviewed release notes
```

License checkpoint

Before a package draft exists, resolve:

```
root license state
file-level SPDX state
AGPL transition status
legacy Apache-2.0 files
documentation license decision
third-party material inventory
```

This must stay aligned with:

```
docs/LICENSE_MIGRATION_PLAN.md
docs/OPEN_ECOSYSTEM_POLICY.md
```

Packaging boundary

This plan does not add a spec file.

It only defines the metadata that a later spec file would need.

Spec draft gate

Do not add packaging/fedora/latticra.spec until these are true:

```
Fedora build lane is green
metadata plan is guarded
license expression is decided
source archive policy is decided
installed file list is known
```

Next slice

Recommended next slice after this plan:

```
Add local RPM spec draft plan
```

That should still be planning-only unless the metadata checklist is complete.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-package-metadata-plan.sh
```

Expected output:

```
fedora_package_metadata_plan: ok
```

docs/FEDORA_READINESS_PLAN.md

Source title: Fedora Readiness Plan

Lines: 279 | Words: 700 | SHA-256: ebd3e5b48d0bdc81751b9083e1804e9e5a3c8aac3943ef2368f7443d2f948886

Fedora Readiness Plan

Status: planning and compatibility record Scope: preparing Latticra for Fedora-friendly packaging, testing, and community review.

Purpose

This plan prepares Latticra for a future Fedora-facing path without claiming Fedora acceptance, endorsement, official inclusion, or distribution status.

The near-term target is:

```
Latticra builds cleanly on Fedora Linux.
Latticra can be described honestly as an upstream project preparing for Fedora package review.
Latticra avoids Fedora trademark confusion.
Latticra preserves its current non-claims.
```

Terminology boundary

Use:

```
Fedora readiness
Fedora packaging readiness
Fedora compatibility lane
Latticra on Fedora
future package review candidate
```

Avoid claiming:

```
official Fedora package
Fedora spin
Fedora remix
Fedora edition
Fedora merger
Fedora endorsement
operating-system replacement
```

Those terms require separate Fedora community, packaging, legal, and trademark review.

Fedora-facing requirements to respect

Before Latticra can be proposed as a package candidate, it should be checked against Fedora package review expectations:

```
package naming
spec file hygiene
Fedora allowed license posture
accurate License field
license text handling
source archive reproducibility
successful build on supported architectures where claimed
BuildRequires completeness
no bundled system libraries
valid file ownership
valid permissions
rpmlint or equivalent package lint output
mock build readiness
```

Current Latticra posture

Current Latticra evidence is still early-stage and report-oriented.

Known current posture:

```
kernel lifecycle evidence exists
final_state=scheduler-run-entry-ready
external_effect_performed=0
runtime_entry_allowed=0
scheduler_execution_allowed=0
scheduler_selection_allowed=0
scheduler_dispatch_allowed=0
scheduler_handoff_allowed=0
scheduler_activation_allowed=0
scheduler_run_entry_allowed=0
memory_allocation_allowed=0
process_spawn_allowed=0
syscall_dispatch_allowed=0
ipc_send_allowed=0
ipc_receive_allowed=0
ipc_queue_mutation_allowed=0
filesystem_lookup_allowed=0
filesystem_read_allowed=0
```

```
filesystem_write_allowed=0
namespace_mutation_allowed=0
device_open_allowed=0
device_read_allowed=0
device_write_allowed=0
driver_probe_allowed=0
driver_load_allowed=0
driver_bind_allowed=0
interrupt_allowed=0
interrupt_mask_allowed=0
interrupt_unmask_allowed=0
interrupt_dispatch_allowed=0
interrupt_ack_allowed=0
timer_tick_allowed=0
timer_arm_allowed=0
timer_disarm_allowed=0
scheduler_tick_allowed=0
run_queue_mutation_allowed=0
enqueue_allowed=0
dequeue_allowed=0
dispatch_allowed=0
context_switch_allowed=0
register_save_allowed=0
register_restore_allowed=0
stack_switch_allowed=0
address_space_switch_allowed=0
preemption_allowed=0
time_accounting_allowed=0
time_read_allowed=0
cpu_usage_write_allowed=0
quota_update_allowed=0
scheduler_credit_update_allowed=0
process_wake_allowed=0
dma_allowed=0
hardware_effect_allowed=0
not bootable
not installer-ready
not a Fedora derivative
not a production operating system
```

This makes the first Fedora-facing target a development package or source package candidate, not a distribution, image, spin, remix, or replacement operating system.

Phase 1: Fedora build lane

The Fedora build/test lane proves the current guarded project foundation compiles with Fedora toolchains.

Current guarded files:

```
scripts/test-fedora-build-lane.sh
.github/workflows/fedora-build-lane.yml
.github/workflows/compat-linux.yml
```

Initial checks should be conservative:

```
C compiler available
project C guards compile
kernel lifecycle guards compile
no external authority claims added
```

Phase 2: package metadata lane

Packaging metadata is present as a local-only draft after the Fedora build lane.

Current guarded files:

```
packaging/fedora/latticra.spec
packaging/fedora/README.md
scripts/test-fedora-spec-static.sh
.github/workflows/fedora-spec-static.yml
scripts/test-fedora-local-rpm-static-validation.sh
.github/workflows/fedora-local-rpm-static-validation.yml
scripts/test-fedora-rpmlint-static-spec-lane.sh
.github/workflows/fedora-rpmlint-static-spec-lane.yml
```

The first spec should be treated as a local development draft until it passes review.

Phase 3: license and source audit lane

Before any package submission, verify:

```
root license state
file-level SPDX state
license text included as needed
third-party material inventory
documentation license decision
trademark and branding boundary
```

This should align with the existing Latticra license migration plan.

Phase 4: local RPM and mock lane

After package metadata exists, add local package validation:

```
rpmbuild source package
rpmlint output captured
mock build validated
installed files inspected
uninstall behavior checked
```

This lane should remain local or Copr-oriented until the project is mature enough for review.

Phase 5: Fedora review candidate lane

Only after the earlier lanes pass should Latticra be described as a Fedora review candidate.

A future review-prep document should include:

```
upstream URL
source archive URL
package summary
package description
license expression
BuildRequires list
runtime Requires list
rpmlint output
mock build output
known non-claims
```

Trademark and naming posture

The project should not use Fedora marks in a way that suggests official status.

Preferred naming:

```
Latticra Fedora readiness
Latticra package draft for Fedora
Latticra on Fedora Linux
```

Avoid naming:

```
Fedora Latticra
Latticra Fedora Edition
Fedora Latticra OS
Official Fedora Latticra
```

Non-claims

This plan does not:

```
submit Latticra to Fedora
create a Fedora package
create a Fedora spin
create a Fedora remix
claim Fedora endorsement
claim Fedora legal approval
claim production readiness
claim operating-system completeness
```

Recommended next slice

Recommended next slice:

```
Keep the Fedora build/spec-static compatibility aliases aligned with the local RPM static
validation and rpmlint lanes
```

That should preserve the current no-artifact, no-submission, local-only package posture while the later local RPM and mock lanes mature.

Validation

This planning slice is guarded by:

```
sh scripts/test-fedora-readiness-plan.sh
```

Expected output:

```
fedora_readiness_plan: ok
```

docs/FEDORA_RPM_GATE_CLASSIFIER.md

Source title: Fedora RPM Gate Classifier

Lines: 138 | Words: 254 | SHA-256: a993d11f185664896637bf2fc1c0a17d3f262263772d1b78e23a4dc61983f98c

Fedora RPM Gate Classifier

Status: implementation record Scope: pure Fedora local RPM gate classification after live read-only preflight evidence. Evidence level: 4

Purpose

This implementation adds the first Fedora RPM gate classifier after the local RPM gate contract.

The classifier evaluates the required evidence fields and reports whether the local RPM gate is allowed or denied.

It does not execute any host-changing action.

Public API

Primary header:

```
include/latticra/fedora_rpm_gate.h
```

Primary source:

```
src/fedora_rpm_gate.c
```

The main entry points are:

```
latticra_fedora_rpm_gate_classify
latticra_fedora_rpm_gate_report
latticra_fedora_rpm_gate_status_label
latticra_fedora_rpm_gate_denial_label
```

Required allow evidence

The classifier reports allowed only when every required fact is present:

```
fedora_host=1
mutable_fedora_host=1
immutable_host=0
local_rpm_present=1
root_or_sudo_available=1
network_required=0
operator_install_confirmation=1
dry_run_passed=1
rollback_or_remove_plan_present=1
preflight_classification=ready-local-rpm
preflight_denial=none
```

```
snapshot_captured=1
live_probe_performed=1
```

All other combinations report denied.

Deterministic denial reasons

The classifier reports denial reasons including:

```
not-fedora-host
immutable-fedora-host
local-rpm-missing
privilege-missing
network-required
operator-confirmation-missing
dry-run-missing
rollback-plan-missing
preflight-not-ready
preflight-denied
live-probe-missing
invalid-gate-input
```

Report surface

The report begins with:

```
FEDORA LOCAL RPM INSTALL MUTATION GATE
```

Required report fields include:

```
install_gate_status=allowed
install_gate_denial=none
install_mutation_allowed=1
install_mutation_performed=0
host_mutation_performed=0
network_allowed=0
evidence_level=4
```

Denied reports preserve:

```
install_mutation_allowed=0
install_mutation_performed=0
host_mutation_performed=0
network_allowed=0
```

Validation

This implementation is guarded by:

```
sh scripts/test-fedora-rpm-gate-classifier.sh
```

Expected output:

```
fedora_rpm_gate_classifier: ok
```

The implementation document guard is:

```
sh scripts/test-fedora-rpm-gate-classifier-docs.sh
```

Expected output:

```
fedora_rpm_gate_classifier_docs: ok
```

Boundary

This implementation is a classifier only.

It does not execute package actions, validate sudo, build packages, open the network, start services, change boot entries, load kernel modules, claim Fedora approval, or claim production installer readiness.

docs/FEDORA_RPM_REMOVAL_ROLLBACK_CLASSIFIER.md

Source title: Fedora RPM Removal Rollback Classifier

Lines: 220 | Words: 520 | SHA-256: bfaab2c425ed5cfc1c57e536758955133107015b5662b472b2964503ed66cff3

Fedora RPM Removal Rollback Classifier

Status: implementation record Scope: pure Fedora RPM removal/rollback posture classification before any host-changing Fedora RPM removal behavior. Evidence level: 5

Purpose

This implementation turns the Fedora RPM removal rollback classifier contract into a pure C classifier.

The classifier evaluates explicit evidence and reports whether the Latticra RPM removal posture is removal-ready, denied, partial, or invalid.

This is install-preparation work: it proves that the project can reason about removal and rollback posture before a future local RPM execution lane performs any host-changing install or removal action.

This implementation does not implement removal, rollback execution, installation, privilege validation, package build behavior, network use, service operation, boot operation, kernel module operation, host policy changes, Fedora approval, or production installer readiness.

Public API

Primary header:

```
include/latticra/fedora_rpm_removal_rollback.h
```

Primary source:

```
src/fedora_rpm_removal_rollback.c
```

The main entry points are:

```
latticra_fedora_rpm_removal_rollback_classify
latticra_fedora_rpm_removal_rollback_report
latticra_fedora_rpm_removal_rollback_status_label
latticra_fedora_rpm_removal_rollback_denial_label
latticra_fedora_rpm_removal_rollback_partial_label
```

Ready evidence

The classifier reports removal-ready only when every required fact is present:

```
operator_removal_confirmation=1
package_name=latticra
package_identity_present=1
package_query_ready=1
installed_package_detected=1
owned_file_list_available=1
readme_owned_by_package=1
unexpected_runtime_surface_absent=1
post_removal_absence_check_planned=1
failure_report_planned=1
network_required=0
service_operation_requested=0
boot_operation_requested=0
kernel_operation_requested=0
policy_operation_requested=0
```

All other states fail closed into denied, partial, or invalid.

Status classes

The classifier can report:

```
removal-ready
denied
partial
invalid
```

removal-ready means removal posture is allowed by evidence, but removal is still not performed.

denied means planned removal should not be attempted.

partial means the input indicates incomplete or uncertain host/package state that a future execution lane would have to report and roll back around.

invalid means the classifier input itself is missing, unknown, or outside the expected 0/1 evidence model.

Deterministic denial reasons

The classifier reports deterministic reason labels including:

```
operator-confirmation-missing
package-name-mismatch
package-identity-missing
package-query-not-ready
package-not-installed
owned-file-list-missing
readme-ownership-missing
unexpected-runtime-surface-present
post-removal-check-missing
failure-report-missing
network-required
service-operation-requested
boot-operation-requested
kernel-operation-requested
policy-operation-requested
invalid-classifier-input
```

Partial state labels

Partial states are reported separately from the denial reason:

```
partial_state=package-installed-but-ownership-unknown
partial_state=owned-files-present-but-package-query-failed
partial_state=runtime-surface-present-before-removal
partial_state=absence-check-not-ready
partial_state=unknown-host-state
```

Report surface

The report begins with:

```
FEDORA RPM REMOVAL ROLLBACK CLASSIFIER
```

Ready reports include:

```
removal_rollback_status=removal-ready
removal_rollback_denial=none
partial_state=none
package_name=latticra
removal_allowed=1
rollback_planned=1
partial_failure_report_required=0
removal_performed=0
host_mutation_performed=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
evidence_level=5
```

Denied reports preserve:

```
removal_rollback_status=denied
removal_allowed=0
removal_performed=0
host_mutation_performed=0
```

Partial reports preserve:

```
removal_rollback_status=partial
partial_failure_report_required=1
```

```
removal_allowed=0
removal_performed=0
host_mutation_performed=0
```

Invalid reports preserve:

```
removal_rollback_status=invalid
removal_rollback_denial=invalid-classifier-input
removal_allowed=0
removal_performed=0
host_mutation_performed=0
```

Install-preparation meaning

This classifier closes the removal/rollback side of the local RPM gate posture.

The install-preparation sequence after this implementation should remain evidence-bound:

```
1. Merge the pure removal/rollback classifier.
2. Add status/index alignment.
3. Add a post-install verification contract.
4. Add a command transcript contract.
5. Add a tightly gated local RPM execution lane only after install, removal, rollback, and
post-install verification posture are all represented.
```

Validation

Classifier guard:

```
sh scripts/test-fedora-rpm-removal-rollback-classifier.sh
```

Expected output:

```
fedora_rpm_removal_rollback_classifier: ok
```

Docs guard:

```
sh scripts/test-fedora-rpm-removal-rollback-classifier-docs.sh
```

Expected output:

```
fedora_rpm_removal_rollback_classifier_docs: ok
```

Boundary

This implementation is a classifier only.

It does not remove packages, roll back packages, install packages, validate sudo, create RPM artifacts, run rpm, run dnf, open the network, start services, change boot entries, load kernel modules, change host policy, claim Fedora approval, or claim production installer readiness.

docs/FEDORA_RPM_REMOVAL_ROLLBACK_CLASSIFIER_CONTRACT.md

Source title: Fedora RPM Removal Rollback Classifier Contract

Lines: 191 | Words: 522 | SHA-256: d323888217bf6126348472c0961bfc10cc81cefa847b840d1bce1c396864f6fb

Fedora RPM Removal Rollback Classifier Contract

Status: contract record Scope: define the pure classifier required before any Fedora RPM removal or rollback action can be implemented. Evidence level: 5 target, contract only

Purpose

The Fedora local RPM removal and rollback plan defines how a future lane must reason about package identity, ownership verification, post-removal absence checks, and partial failure reporting.

This contract defines the next implementation boundary: a pure classifier that evaluates whether removal or rollback posture is ready, denied, partial, or invalid.

This is not a removal implementation.

The classifier must report state only. It must not request removal, install packages, validate privilege, open the network, start services, change boot entries, load kernel modules, or change host policy.

Required classifier inputs

A future classifier must consume an explicit record containing at least:

```
operator_removal_confirmation
package_name
package_identity_present
package_query_ready
installed_package_detected
owned_file_list_available
readme_owned_by_package
unexpected_runtime_surface_absent
post_removal_absence_check_planned
failure_report_planned
network_required
service_operation_requested
boot_operation_requested
kernel_operation_requested
policy_operation_requested
```

The classifier must fail closed when fields are missing, unknown, contradictory, or unsafe.

Required allow conditions

The classifier may report removal-ready only when all conditions are true:

```
operator_removal_confirmation=1
package_name=latticra
package_identity_present=1
package_query_ready=1
installed_package_detected=1
owned_file_list_available=1
readme_owned_by_package=1
unexpected_runtime_surface_absent=1
post_removal_absence_check_planned=1
failure_report_planned=1
network_required=0
service_operation_requested=0
boot_operation_requested=0
kernel_operation_requested=0
policy_operation_requested=0
```

All other combinations must report denied, partial, or invalid.

Required denial reasons

The classifier must report deterministic denial reasons such as:

```
operator-confirmation-missing
package-name-mismatch
package-identity-missing
package-query-not-ready
package-not-installed
owned-file-list-missing
readme-ownership-missing
unexpected-runtime-surface-present
post-removal-check-missing
failure-report-missing
network-required
service-operation-requested
boot-operation-requested
kernel-operation-requested
policy-operation-requested
invalid-classifier-input
```

Required report fields

A future implementation should emit a report beginning with:

For ready cases, required report fields include:

```
removal_rollback_status=removal-ready
removal_allowed=1
rollback_planned=1
removal_performed=0
host_mutation_performed=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
evidence_level=5
```

For denied cases, required report fields include:

```
removal_rollback_status=denied
removal_allowed=0
removal_performed=0
host_mutation_performed=0
```

For partial cases, required report fields include:

```
removal_rollback_status=partial
partial_failure_report_required=1
removal_performed=0
host_mutation_performed=0
```

Partial state classes

The classifier should distinguish planned denial from partial host state.

Planned denial means removal should not be attempted.

Partial state means the evidence suggests a future removal/rollback execution lane must report incomplete or uncertain host state, such as:

```
package-installed-but-ownership-unknown
owned-files-present-but-package-query-failed
runtime-surface-present-before-removal
absence-check-not-ready
unknown-host-state
```

Relationship to the RPM gate classifier

The Fedora RPM gate classifier reports whether the future local RPM gate is allowed.

This removal/rollback classifier must exist before any host-changing local RPM execution lane is added.

The two classifiers together should prove:

```
install gate evidence exists
removal or rollback posture exists
post-removal verification is planned
partial failure reporting is planned
```

Implementation sequence

Recommended next order:

1. Add this removal/rollback classifier contract.
2. Add a pure C removal/rollback classifier.
3. Add a focused test runner and docs guard.
4. Add status/index alignment after merge.
5. Add post-install verification contract.
6. Add command transcript contract.
7. Only then consider a tightly gated local RPM execution lane.

Validation

This contract is guarded by:

```
sh scripts/test-fedora-rpm-removal-rollback-classifier-contract.sh
```

Expected output:

```
fedora_rpm_removal_rollback_classifier_contract: ok
```

Non-claims

This contract does not implement removal, rollback, installation, host mutation, package execution, privilege validation, package build behavior, network use, service operation, boot operation, kernel module operation, Fedora approval, or production installer readiness.

docs/FEDORA_RPMLINT_AVAILABILITY.md

Source title: Fedora rpmlint Availability Lane

Lines: 63 | Words: 159 | SHA-256: 35804af508253a92c19ec53adc12b287e112878c66cce12b22e3f45adaffc5e5

Fedora rpmlint Availability Lane

Status: active tool availability lane Scope: verify that rpmlint is available in a Fedora Linux environment.

Purpose

This lane prepares the next packaging validation step without linting the package draft yet.

The current goal is only to prove that rpmlint can be installed and invoked inside the Fedora compatibility environment.

Files

```
docs/FEDORA_RPMLINT_AVAILABILITY.md
scripts/test-fedora-rpmlint-availability.sh
.github/workflows/fedora-rpmlint-availability.yml
```

Checks

This lane checks:

```
Fedora environment marker exists
dnf is available
rpmlint installs
rpmlint command is available
rpmlint version command can run
static RPM validation lane remains green
```

Boundary

This lane does not lint the Latticra spec yet.

It does not run rpmbuild or mock.

It does not create package artifacts.

Next slice

Recommended next slice:

```
Add rpmlint static spec lane
```

That future lane may run rpmlint packaging/fedora/latticra.spec and capture expected local-only draft findings.

Validation

Run inside Fedora:

```
sh scripts/test-fedora-rpmlint-availability.sh
```

Expected output:

```
fedora_rpmlint_availability: ok
```

docs/FEDORA_RPMLINT_STATIC_SPEC_LANE.md

Source title: Fedora rpmlint Static Spec Lane

Lines: 87 | Words: 302 | SHA-256: 0c55191651e8e711d06ffe019d30aaf7551154d7a2089ca6613e3bde039312a2

Fedora rpmlint Static Spec Lane

Status: active static spec lint lane Scope: run rpmlint against the local-only Fedora spec draft and capture findings without promoting package-readiness claims.

Purpose

This lane advances the Fedora local RPM validation stack from tool availability to a static spec lint pass.

The current goal is conservative: prove that rpmlint can inspect the local draft spec file in a Fedora environment and produce a reportable result.

This lane does not require a clean rpmlint result yet because the spec remains a local-only draft and still has source-archive and package-readiness metadata to review.

Files

```
docs/FEDORA_RPMLINT_STATIC_SPEC_LANE.md
scripts/test-fedora-rpmlint-static-spec-lane.sh
.github/workflows/fedora-rpmlint-static-spec-lane.yml
packaging/fedora/latticra.spec
```

Checks

This lane checks:

```
Fedora environment marker exists
dnf is available
rpmlint installs
rpmlint command is available
rpmlint can inspect packaging/fedora/latticra.spec
rpmlint output is captured for audit
previous rpmlint availability lane remains green
```

Expected draft findings

The current local-only spec may still report findings related to draft metadata, including:

```
local-only release marker
placeholder version
AGPL-3.0-or-later AND CC-BY-4.0 package license expression
missing source archive for a real package build
limited installed file set
```

Those findings are expected at this stage and are not package-readiness failures yet.

The purpose of this lane is visibility, not package promotion.

Boundary

This lane does not run rpmbuild.

It does not run mock.

It does not create package artifacts.

It does not submit Latticra to Fedora.

It does not claim Fedora package approval, Fedora endorsement, Fedora spin/remix status, product readiness, operating-system readiness, runtime authority, kernel authority, or security-hardening completion.

Next slice

Recommended next slice:

```
Add rpmlint findings classification report
```

That future lane may classify expected draft findings separately from unexpected spec findings while keeping the package local-only.

Validation

Run inside Fedora:

```
sh scripts/test-fedora-rpmlint-static-spec-lane.sh
```

Expected output:

```
fedora_rpmlint_static_spec_lane: ok
```

docs/FEDORA_SOURCE_ARCHIVE_FIXTURE_LANE.md

Source title: Fedora Source Archive Fixture Lane

Lines: 128 | Words: 408 | SHA-256: 5dc2b6bbdad40034eb7a5c5c241625c6fca80ce73c03b60d70b4a185ffc74db4

Fedora Source Archive Fixture Lane

Status: active fixture lane Scope: create and inspect a temporary Fedora source archive fixture without running RPM builds or installing artifacts.

Purpose

This lane advances the local Fedora packaging path from planning into a concrete source archive shape check.

The goal is conservative: prove that Latticra can form a temporary source archive matching the current Source0 and %autosetup expectations in packaging/fedora/latticra.spec.

This lane does not build a source RPM or binary RPM. It only creates a disposable archive fixture and inspects its contents.

Current spec relationship

The current spec expects:

```
Source0: %{name}-%{version}.tar.gz
%autosetup -n %{name}-%{version}
```

For the current local draft, that resolves to:

```
latticra-0.0.0.tar.gz
latticra-0.0.0/
```

Fixture rules

The archive fixture must:

```
use the Name field from packaging/fedora/latticra.spec
use the Version field from packaging/fedora/latticra.spec
create a root directory named latticra-&lt;version&gt;/
create an archive named latticra-&lt;version&gt;.tar.gz
use Git&#x27;s tracked and unignored source view
include README.md
include packaging/fedora/latticra.spec
include scripts used by the current %build section
exclude .git
exclude temporary RPM work directories
```

```
exclude RPM artifacts
exclude nested source archive artifacts
refuse symlink entries
normalize tar metadata
remain temporary and uncommitted
```

Inspection checks

The lane should inspect the archive with tar -tzf and verify:

```
latticra-&lt;version&gt;/README.md
latticra-&lt;version&gt;/packaging/fedora/latticra.spec
latticra-&lt;version&gt;/scripts/test-state-lattice.sh
latticra-&lt;version&gt;/scripts/test-system-bootstrap.sh
latticra-&lt;version&gt;/scripts/test-kernel.sh
latticra-&lt;version&gt;/scripts/test-kernel-lifecycle.sh
```

The lane should also fail if the archive contains:

```
.git/
.rpmwork/
*.rpm
nested latticra-*.tar.gz
```

Relationship to later RPM work

This fixture lane comes before RPM build execution.

The intended order is:

1. source archive fixture lane
2. source archive handoff to temporary RPM topdir
3. local binary RPM build lane
4. RPM payload inspection lane
5. fresh Fedora install smoke lane

Boundary

This lane does not run rpmbuild.

It does not run mock.

It does not create source RPM artifacts.

It does not create binary RPM artifacts.

It does not install Latticra.

It does not publish package artifacts.

It does not submit Latticra to Fedora.

It does not claim Fedora package approval, Fedora endorsement, Fedora spin/remix status, product readiness, operating-system readiness, runtime authority, kernel authority, boot authority, service authority, SELinux policy readiness, or security-hardening completion.

Next slice

Recommended next slice:

```
Add Fedora local binary RPM build lane
```

That future slice may hand the temporary archive to an RPM topdir and run a local build inside Fedora CI.

Validation

This lane is guarded by:

```
sh scripts/test-fedora-source-archive-fixture-lane.sh
```

Expected output:

```
fedora_source_archive_fixture_lane: ok
```

docs/FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN.md

Source title: Fedora VM CLI Payload Next Validation Lane Plan

Lines: 160 | Words: 458 | SHA-256: f79a2de16b15871199b7fe63f82cd02945a68ec3fd7ad09805c75fa6d8bb014d

Fedora VM CLI Payload Next Validation Lane Plan

Status: planning record Scope: define the next manual disposable Fedora VM validation lane for the no-effect CLI payload without adding a runner or widening readiness claims.

Purpose

This plan follows the accepted disposable Fedora VM CLI payload validation evidence.

The next Fedora CLI payload lane should test repeatability, not broaden authority.

The goal is conservative: define the evidence required for a future second disposable Fedora VM CLI payload validation run, compare it against the accepted CLI payload evidence, and keep the package local-only and no-effect.

This plan does not run RPM tooling, install a package, mutate a host, create artifacts, publish artifacts, submit to Fedora, or claim production readiness.

Current evidence basis

The accepted CLI payload evidence records:

```
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
validated_payload=/usr/bin/latticra
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

The accepted payload remains no-effect:

```
mode=no-effect
runtime_behavior=disabled
host_mutation=0
network=0
kernel_operation=0
service_operation=0
package_manager_operation=0
boot_operation=0
selinux_policy_operation=0
effect_authority=denied
```

Required hard gate

A future next-validation runner must remain manual and must refuse to proceed unless all of the following are true:

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
ID=fedora
rpm_present=1
rpmbuild_present=1
cc_present=1
sudo_present=1
```

The lane must not run automatically in CI.

The lane must not run on a developer daily-driver host, production host, immutable Fedora target, non-Fedora host, or unclear target.

Required repeatability evidence

The future lane should record these repeatability fields:

```
source_tree_revision_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
spec_checksum_recorded=1
source_archive_checksum_recorded=1
rpm_nevra_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
unexpected_runtime_surface_absent=1
cli_status_output_recorded=1
cli_version_output_recorded=1
cli_report_output_recorded=1
cli_invalid_command_exit_recorded=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
```

The lane should compare the new evidence against the accepted CLI payload evidence:

```
validated_payload_still_contains_cli=1
validated_payload_still_contains_readme=1
validated_cli_mode_still_no_effect=1
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
```

Success classification

A future successful run may record:

```
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

This classification is still bounded to the no-effect CLI payload in a disposable Fedora VM validation path.

Boundary

This plan does not implement a runner.

It does not execute RPM commands.

It does not install Latticra.

It does not remove Latticra.

It does not mutate a host.

It does not change services, boot entries, kernel modules, SELinux policy, firmware, networking, package repositories, or system configuration.

It does not publish package artifacts.

It does not submit Latticra to Fedora.

It does not claim Fedora approval, Fedora distribution readiness, production installer readiness, daily-driver safety, immutable Fedora readiness, security hardening, sandboxing, update safety, recovery safety, malware prevention, ransomware prevention, or OS-replacement readiness.

Next slice

Recommended next slice:

```
Add Fedora VM CLI payload repeatability transcript contract
```

That future slice should define the transcript schema for the second disposable Fedora VM CLI payload validation run without adding the runner or performing host mutation.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-next-validation-lane-plan.sh
```

Expected output:

```
fedora_vm_cli_payload_next_validation_lane_plan: ok
```

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT.md

Source title: Fedora VM CLI Payload Repeatability Evidence Acceptance Contract

Lines: 141 | Words: 379 | SHA-256: 52a0deda2410b21548661698a82f5b753a6ca11c3b0b154fff2e8406fa1e9101

Fedora VM CLI Payload Repeatability Evidence Acceptance Contract

Status: acceptance contract Evidence level: 9 repeatability target, contract only Scope: define the future evidence status record after a real Fedora VM repeatability transcript is validated and accepted.

Purpose

This contract defines the exact fields required before Fedora VM CLI payload repeatability evidence can be written as accepted.

It does not run the repeatability runner.

It does not validate a live transcript.

It does not attach a transcript.

It does not write an evidence status record.

It does not accept repeatability evidence.

It does not mutate a host.

Required source records

The acceptance contract depends on:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh
scripts/run-fedora-vm-cli-payload-repeatability-lane.sh
README.md
```

Required future acceptance prerequisites

A future accepted evidence status record requires all of these prerequisites:

```
repeatability_transcript_attached=1
repeatability_transcript_reviewed=1
repeatability_transcript_candidate_valid=1
repeatability_transcript_placeholder_values_absent=1
repeatability_transcript_required_markers_present=1
repeatability_transcript_value_fields_validated=1
repeatability_transcript_accepted=1
evidence_status_written=1
```

Future evidence status record

The future accepted status record must include:

```
FEDORA VM CLI PAYLOAD REPEATABILITY EVIDENCE STATUS
Status: evidence status alignment
source=operator disposable Fedora VM repeatability transcript
transcript_kind=disposable-vm-cli-payload-repeatability
repeatability_transcript_reviewed=1
repeatability_transcript_accepted=1
repeatability_transcript_candidate_valid=1
repeatability_transcript_placeholder_values_absent=1
repeatability_transcript_required_markers_present=1
repeatability_transcript_value_fields_validated=1
source_tree_revision_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
spec_checksum_recorded=1
source_archive_checksum_recorded=1
rpm_nevra_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
validated_cli_mode_still_no_effect=1
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
```

Current acceptance gate state

The current repository state is still blocked:

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Acceptance rule

Repeatability evidence can only be accepted after a manual disposable Fedora VM run produces a complete transcript, the transcript validator accepts the candidate markers and values, and an operator review confirms the evidence review gate.

The acceptance contract alone cannot promote evidence.

Validation

This contract is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-acceptance-contract.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
and write accepted repeatability evidence status
```

Non-claims

This contract is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not an RPM build, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE.md

Source title: Fedora VM CLI Payload Repeatability Evidence Publication Gate

Lines: 130 | Words: 380 | SHA-256: 2be6c1d34aa96c70d86c7cff0c893c9dd7c7cea1cc6c0fe5aeb4fb26c7291ea4

Fedora VM CLI Payload Repeatability Evidence Publication Gate

Status: publication gate contract Evidence level: 9 repeatability target, gate only Scope: prevents a passing evidence status candidate review from being treated as published Fedora repeatability evidence.

Purpose

This gate defines the final publication boundary after the transcript review validator and the evidence status review validator.

It does not run the repeatability runner.

It does not validate a live transcript.

It does not attach a transcript.

It does not write an evidence status record.

It does not publish repeatability evidence.

It does not mutate a host.

Required source records

The publication gate depends on:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR_STATUS.md
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh
scripts/fedora-vm-cli-payload-repeatability-evidence-status-review.sh
README.md
```

Required future publication prerequisites

A future publication record requires all of these prerequisites:

```
repeatability_transcript_attached=1
repeatability_transcript_reviewed=1
repeatability_transcript_accepted=1
repeatability_evidence_status_candidate_valid=1
repeatability_evidence_status_reviewed=1
candidate_repeatability_transcript_accepted=1
candidate_cli_payload_repeatability_evidence_present=1
candidate_evidence_status_written=1
operator_publication_review_completed=1
```

```
repeatability_evidence_publication_approved=1
evidence_status_written=1
cli_payload_repeatability_evidence_present=1
```

The evidence status review validator alone cannot publish evidence:

```
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
```

Current publication gate state

The current repository state is still blocked:

```
fedora_vm_cli_payload_repeatability_evidence_publication_gate_present=1
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
repeatability_evidence_publication_requested=0
operator_publication_review_completed=0
repeatability_evidence_publication_approved=0
repeatability_evidence_status_published=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Publication rule

The only acceptable publication path is:

```
1. Run the manual disposable Fedora VM repeatability lane in a disposable Fedora VM.
2. Attach the complete transcript.
3. Validate the transcript candidate with
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh.
4. Fill the evidence status template from the reviewed transcript.
5. Validate the evidence status candidate with
scripts/fedora-vm-cli-payload-repeatability-evidence-status-review.sh.
6. Complete an operator publication review.
7. Only then write a published evidence status record.
```

Validation

This gate is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-publication-gate.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_publication_gate: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript and evidence
 status candidates, then complete operator publication review
```

Non-claims

This gate is not repeatability evidence, not a completed transcript, not an accepted evidence status record, not a published evidence status record, not a second disposable Fedora VM validation run, not an RPM build, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE.md

Source title: Fedora VM CLI Payload Repeatability Evidence Review Gate

Lines: 163 | Words: 405 | SHA-256: e36aa35d00cbe5eb158c8d1df7b2043e787575b688899e56af487828a1350c46

Fedora VM CLI Payload Repeatability Evidence Review Gate

Status: review gate contract Evidence level: 9 repeatability target, gate only Scope: define the review gate before a second disposable Fedora VM CLI payload validation transcript can be accepted as repeatability evidence.

Purpose

This gate prevents the manual repeatability runner from being treated as completed evidence until a real disposable Fedora VM transcript is attached and reviewed.

It does not run the repeatability runner.

It does not execute RPM commands.

It does not install or remove an RPM.

It does not mutate a host.

It does not mark repeatability evidence present.

Required source records

The review gate depends on:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE_STATUS.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR_STATUS.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_STATUS.md
docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md
scripts/fedora-vm-cli-payload-repeatability-transcript-template.sh
scripts/test-fedora-vm-cli-payload-repeatability-transcript-template.sh
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh
scripts/test-fedora-vm-cli-payload-repeatability-transcript-review-validator.sh
scripts/run-fedora-vm-cli-payload-repeatability-lane.sh
scripts/test-fedora-vm-cli-payload-repeatability-runner.sh
README.md
```

Required attached transcript markers

A future accepted transcript must include:

```
FEDORA VM CLI PAYLOAD REPEATABILITY TRANSCRIPT
transcript_kind=disposable-vm-cli-payload-repeatability
transcript_version=1
operator_review_required=1
repeatability_transcript_recorded_after_real_run=1
prior_cli_payload_evidence_recorded=1
FEDORA VM CLI PAYLOAD REPEATABILITY LANE
validation_status=ok
repeatability_validation_status=ok
source_tree_revision_recorded=1
source_tree_revision=
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
fedora_kernel_version=
spec_checksum_recorded=1
spec_checksum=
source_archive_checksum_recorded=1
source_archive_checksum=
rpm_nevra_recorded=1
rpm_nevra=
package_name=latticra
```

```
package_version=0.0.0
rpm_payload_listing_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
unexpected_runtime_surface_absent=1
cli_status_output_recorded=1
cli_version_output_recorded=1
cli_report_output_recorded=1
cli_invalid_command_exit_recorded=1
validated_cli_mode_still_no_effect=1
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
fedora_vm_cli_payload_repeatability_lane: ok
```

The value-bearing fields must not be placeholders in a future accepted transcript:

```
source_tree_revision
fedora_kernel_version
spec_checksum
source_archive_checksum
rpm_nevra
```

Current gate state

```
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_complete=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_review_required=1
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Review rule

The only acceptable promotion path is:

```
1. Print scripts/fedora-vm-cli-payload-repeatability-transcript-template.sh before the manual
run.
2. Run scripts/run-fedora-vm-cli-payload-repeatability-lane.sh manually inside a disposable
Fedora VM.
3. Attach the complete transcript using the capture template shape.
4. Validate the transcript candidate with
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh.
5. Review the transcript against this gate and the repeatability transcript contract.
6. Only then add an evidence status record with repeatability_transcript_accepted=1.
```

Validation

This gate is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-review-gate.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_review_gate: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Non-claims

This gate is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR.md

Source title: Fedora VM CLI Payload Repeatability Evidence Status Review Validator

Lines: 157 | Words: 376 | SHA-256: f7ff95b4101d029d447cb6d237f8f4d783f21b1a3a54eb122fb78422a4404430

Fedora VM CLI Payload Repeatability Evidence Status Review Validator

Status: no-effect evidence status review validator Evidence level: 9 repeatability target, validator only Scope: validate a supplied Fedora VM CLI payload repeatability evidence status candidate without writing or promoting evidence.

Purpose

This validator checks a supplied future evidence status candidate after a real disposable Fedora VM repeatability transcript has been validated and reviewed.

It does not run the repeatability runner.

It does not validate a live transcript.

It does not attach a transcript.

It does not write an evidence status record.

It does not accept repeatability evidence by itself.

It does not mutate a host.

```
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_review_mode=no-effect-validation
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
```

Command

```
sh scripts/fedora-vm-cli-payload-repeatability-evidence-status-review.sh --status &lt;path&gt;
```

Required source records

The validator depends on:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
```

```
scripts/fedora-vm-cli-payload-repeatability-evidence-status-template.sh
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh
README.md
```

Accepted candidate requirements

A supplied candidate must contain the future evidence status shape from the template:

```
FEDORA VM CLI PAYLOAD REPEATABILITY EVIDENCE STATUS
Status: evidence status alignment
source=operator disposable Fedora VM repeatability transcript
transcript_kind=disposable-vm-cli-payload-repeatability
repeatability_transcript_reviewed=1
repeatability_transcript_accepted=1
repeatability_transcript_candidate_valid=1
repeatability_transcript_placeholder_values_absent=1
repeatability_transcript_required_markers_present=1
repeatability_transcript_value_fields_validated=1
source_tree_revision_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
spec_checksum_recorded=1
source_archive_checksum_recorded=1
rpm_nevra_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
validated_cli_mode_still_no_effect=1
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
evidence_status_written=1
```

The validator rejects placeholder values for:

```
Date
transcript_path
transcript_checksum
source_tree_revision
fedora_os_release
fedora_kernel_version
spec_checksum
source_archive_checksum
rpm_nevra
```

Validator output

For a valid candidate, the validator prints:

```
FEDORA VM CLI PAYLOAD REPEATABILITY EVIDENCE STATUS REVIEW
review_status=ok
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_review_mode=no-effect-validation
repeatability_evidence_status_candidate_valid=1
repeatability_evidence_status_placeholder_values_absent=1
repeatability_evidence_status_required_markers_present=1
repeatability_evidence_status_value_fields_validated=1
repeatability_evidence_status_reviewed=1
candidate_repeatability_transcript_accepted=1
candidate_cli_payload_repeatability_evidence_present=1
candidate_evidence_status_written=1
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
host_mutation_performed=0
```

Validation

This validator is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-status-review-validator.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_status_review_validator: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
fill evidence status template, and review the evidence status candidate
```

Non-claims

This validator is not repeatability evidence, not a completed transcript, not an accepted evidence status record, not a second disposable Fedora VM validation run, not an RPM build, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE.md

Source title: Fedora VM CLI Payload Repeatability Evidence Status Template

Lines: 161 | Words: 374 | SHA-256: 66457b4348259024d3dab295f310619cb2158d9d0284b2c5d3e2124d630f9955

Fedora VM CLI Payload Repeatability Evidence Status Template

Status: no-effect repeatability evidence status template Evidence level: 9 repeatability target, template only Scope: future accepted evidence status shape for the second disposable Fedora VM validation run of the no-effect CLI RPM payload.

Purpose

This template defines the future evidence status record an operator may write only after a real disposable Fedora VM repeatability transcript is attached, validated, and reviewed.

It is a status template only.

It does not run the repeatability runner.

It does not validate a live transcript.

It does not attach a transcript.

It does not write an evidence status record.

It does not accept repeatability evidence.

It does not mutate a host.

```
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
repeatability_evidence_status_template_mode=no-effect-template
repeatability_evidence_status_template_decision=blocked-template-only-no-status-write
repeatability_evidence_status_template_complete=0
repeatability_runner_executed=0
rpm_build_performed=0
rpm_install_performed=0
rpm_removal_performed=0
host_mutation_performed=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
```

Command

```
sh scripts/fedora-vm-cli-payload-repeatability-evidence-status-template.sh
```

The command prints the future evidence status shape to stdout only.

Required source records

The template depends on:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE.md
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT_STATUS.md
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh
scripts/run-fedora-vm-cli-payload-repeatability-lane.sh
README.md
```

Future evidence status template

The printed template includes this future record shape:

```
FEDORA VM CLI PAYLOAD REPEATABILITY EVIDENCE STATUS
Status: evidence status alignment
Date: &lt;required-from-accepted-review-date&gt;
source=operator disposable Fedora VM repeatability transcript
transcript_kind=disposable-vm-cli-payload-repeatability
transcript_path=&lt;required-from-operator-attachment&gt;
transcript_checksum=&lt;required-from-accepted-review&gt;
repeatability_transcript_reviewed=1
repeatability_transcript_accepted=1
repeatability_transcript_candidate_valid=1
repeatability_transcript_placeholder_values_absent=1
repeatability_transcript_required_markers_present=1
repeatability_transcript_value_fields_validated=1
source_tree_revision_recorded=1
source_tree_revision=&lt;required-from-real-run&gt;
fedora_os_release_recorded=1
fedora_os_release=&lt;required-from-real-run&gt;
fedora_kernel_version_recorded=1
fedora_kernel_version=&lt;required-from-real-run&gt;
spec_checksum_recorded=1
spec_checksum=&lt;required-from-real-run&gt;
source_archive_checksum_recorded=1
source_archive_checksum=&lt;required-from-real-run&gt;
rpm_nevra_recorded=1
rpm_nevra=&lt;required-from-real-run&gt;
rpm_payload_matches_expected_cli_surfaces=1
validated_cli_mode_still_no_effect=1
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
evidence_status_written=1
```

Current gate state

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
repeatability_evidence_status_template_mode=no-effect-template
repeatability_evidence_status_template_complete=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

This template is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-status-template.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_status_template: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
and use the evidence status template to write accepted repeatability evidence status
```

Non-claims

This template is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not an RPM build, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN.md

Source title: Fedora VM CLI Payload Repeatability Runner Plan

Lines: 271 | Words: 711 | SHA-256: 28694b57b239069106a4604eaf61b47a88a4fb8ad0d3a160bae5bf28ae2dba71

Fedora VM CLI Payload Repeatability Runner Plan

Status: implemented runner plan Scope: record the manual disposable Fedora VM repeatability runner for the no-effect CLI RPM payload without running it.

Purpose

This plan follows the Fedora VM CLI payload repeatability transcript contract.

The implemented runner is shaped to prove repeatability for the already accepted no-effect CLI payload evidence.

This plan records the runner.

It does not execute RPM commands.

It does not install or remove an RPM.

It does not mutate a host.

It does not widen host install readiness beyond the accepted no-effect CLI payload boundary.

Runner path

The implementation is:

```
scripts/run-fedora-vm-cli-payload-repeatability-lane.sh
```

That script must remain manual-only and must not be called by normal CI.

Any CI workflow for this plan may only run static guards:

```
scripts/test-fedora-vm-cli-payload-repeatability-runner-plan.sh
scripts/test-fedora-vm-cli-payload-repeatability-runner.sh
```

Required source records

The runner must require these records before doing anything else:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT.md
docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT_STATUS.md
docs/FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN.md
docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md
packaging/fedora/latticra.spec
README.md
src/latticra_cli.c
scripts/test-latticra-no-effect-cli-status-surface.sh
scripts/run-fedora-vm-cli-payload-validation-lane.sh
```

Required hard gate

The runner must refuse to proceed unless all of the following are true:

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_ALLOW_CLI_PAYLOAD_REPEATABILITY_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
ID=fedora
operator_is_non_root=1
rpm_present=1
rpmbuild_present=1
cc_present=1
sudo_present=1
```

The runner must use sudo only for RPM install and RPM removal.

Required command availability

The runner must require:

```
awk
cat
cc
find
git
grep
gzip
id
mktemp
python3
rpm
rpmbuild
sha256sum
sort
sudo
tar
uname
```

If sha256sum is unavailable on a Fedora target, the runner must fail closed rather than silently omitting checksum evidence.

Source archive creation must use Git's tracked and unignored source view, refuse symlink entries, and write deterministic tar/gzip metadata before checksum evidence is recorded.

Planned repeatability sequence

The runner performs this sequence only after the hard gate passes:

1. Verify disposable Fedora VM target evidence.
2. Verify accepted prior CLI payload evidence records.
3. Verify repeatability transcript contract fields.
4. Run the no-effect CLI status-surface guard.
5. Capture source tree revision.
6. Capture Fedora os-release and kernel version.
7. Capture packaging/fedora/latticra.spec checksum.
8. Build a temporary source archive from the current tree.
9. Capture source archive checksum.
10. Build the local binary RPM with rpmbuild.
11. Capture RPM NEVRA, path, arch, and metadata.
12. Capture RPM payload listing.
13. Confirm expected CLI payload surfaces.
14. Confirm forbidden payload surfaces are absent.
15. Confirm package is absent before install.
16. Install the RPM into the disposable Fedora VM.
17. Query installed package state and payload.
18. Run rpm -V for the installed package.
19. Execute latticra --status without root.
20. Execute latticra --version without root.
21. Execute latticra --report without root.
22. Verify invalid CLI usage exits with code 2.
23. Remove the RPM from the disposable Fedora VM.
24. Confirm package, CLI, and README absence after removal.
25. Emit the repeatability transcript and deterministic report.

Expected payload

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

The RPM payload may include:

```
/usr/share/doc/latticra
```

Forbidden payload surfaces

```
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
/usr/share/selinux
/usr/sbin/latticra
```

Required report fields

A successful repeatability runner emits:

```
FEDORA VM CLI PAYLOAD REPEATABILITY LANE
validation_status=ok
repeatability_validation_status=ok
transcript_kind=disposable-vm-cli-payload-repeatability
prior_cli_payload_evidence_recorded=1
source_tree_revision_recorded=1
source_tree_revision=$source_tree_revision
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
fedora_kernel_version=$(uname -r)
spec_checksum_recorded=1
spec_checksum=$spec_checksum
source_archive_checksum_recorded=1
source_archive_checksum=$source_archive_checksum
rpm_nevra_recorded=1
rpm_nevra=$rpm_nevra
package_name=$name
package_version=$version
package_arch=$rpm_arch
rpm_payload_listing_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
unexpected_runtime_surface_absent=1
cli_status_output_recorded=1
cli_version_output_recorded=1
cli_report_output_recorded=1
cli_invalid_command_exit_recorded=1
validated_cli_mode_still_no_effect=1
```

```
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
```

Current project state

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_plan_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
repeatability_runner_manual_only=1
ci_auto_repeatability_validation_allowed=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Boundary

This plan is not a completed repeatability transcript.

It does not run scripts/run-fedora-vm-cli-payload-repeatability-lane.sh.

It does not run rpmbuild.

It does not run rpm.

It does not install Latticra.

It does not remove Latticra.

It does not mutate a host.

It does not publish artifacts.

It does not submit Latticra to Fedora.

It does not claim Fedora approval, Fedora distribution readiness, production installer readiness, daily-driver safety, immutable Fedora readiness, security hardening, sandboxing, update safety, recovery safety, malware prevention, ransomware prevention, or OS-replacement readiness.

Next recommended lane

```
Capture reviewed Fedora VM CLI payload repeatability transcript evidence
```

That future evidence capture must come from a real disposable Fedora VM run, must be operator-reviewed, and must preserve all non-claims.

Validation

This plan is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-runner-plan.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_runner_plan: ok
```

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE.md

Source title: Fedora VM CLI Payload Repeatability Transcript Capture Template

Lines: 143 | Words: 376 | SHA-256: abbd73d191fc908cce0c8afe346f145ba1ead143d278607507e3e349a4df0845

Fedora VM CLI Payload Repeatability Transcript Capture Template

Status: no-effect repeatability transcript capture template Evidence level: 9 repeatability target, template only Scope: future transcript capture shape for the second disposable Fedora VM validation run of the no-effect CLI RPM payload.

Purpose

This template defines the transcript shape an operator should attach after a real disposable Fedora VM repeatability run.

It is a capture template only.

It does not run the repeatability runner.

It does not build an RPM.

It does not install or remove an RPM.

It does not mutate a host.

It does not mark repeatability evidence present.

```
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_decision=blocked-template-only-no-vm-execution
repeatability_transcript_template_complete=0
repeatability_runner_executed=0
rpm_build_performed=0
rpm_install_performed=0
rpm_removal_performed=0
host_mutation_performed=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
```

Command

```
sh scripts/fedora-vm-cli-payload-repeatability-transcript-template.sh
```

The command emits a deterministic transcript capture template to stdout.

It validates that the repeatability transcript contract, repeatability runner plan, evidence review gate, prior CLI payload evidence status, and repeatability runner are present before printing the template.

The review validator companion is:

```
sh scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh --transcript &lt;path&gt;
```

It checks a supplied transcript candidate for required markers and rejects placeholder values without accepting evidence by itself.

Required operator context

The future transcript must be captured from:

```
target_required=disposable-fedora-vm
operator_review_required=1
clean_snapshot_required=1
recovery_path_required=1
```

```
prior_cli_payload_evidence_required=1
runner_path=scripts/run-fedora-vm-cli-payload-repeatability-lane.sh
```

Required future transcript fields

The capture template includes these required transcript sections:

```
[transcript_header]
[prior_evidence]
[runner_report_required_fields]
[review_gate]
[non_claims]
```

The future accepted transcript must replace placeholder values for:

```
source_tree_revision
fedora_kernel_version
spec_checksum
source_archive_checksum
rpm_nevra
```

Review validator:

```
docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md
scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh
```

Current template state

```
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_complete=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

This template is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-transcript-template.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_transcript_template: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Non-claims

This template is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not an RPM build, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT.md

Source title: Fedora VM CLI Payload Repeatability Transcript Contract

Lines: 100 | Words: 238 | SHA-256: 2c7379c79f994b173015fdbeb1ffb30e53f5706859295b41954709d19dd7b6b5

Fedora VM CLI Payload Repeatability Transcript Contract

Status: contract record Evidence level: 9 repeatability target, contract only Scope: transcript schema for a future second disposable Fedora VM validation run of the no-effect CLI RPM payload.

Purpose

This contract defines the repeatability transcript required before a second disposable Fedora VM CLI payload validation run can be reviewed as repeatability evidence.

This is a contract only.

It does not run a validation lane.

It does not add a runner.

It does not run RPM tooling.

It does not install or remove an RPM.

It does not mutate a host.

Required transcript header

```
FEDORA VM CLI PAYLOAD REPEATABILITY TRANSCRIPT
transcript_kind=disposable-vm-cli-payload-repeatability
transcript_version=1
operator_review_required=1
repeatability_transcript_recorded_after_real_run=1
prior_cli_payload_evidence_recorded=1
```

Required prior evidence

```
prior_validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
prior_validated_payload=/usr/bin/latticra
prior_validated_payload=/usr/share/doc/latticra/README.md
prior_disposable_vm_cli_validation_completed=1
prior_host_install_ready_for_cli_payload=1
prior_evidence_level=9
```

Required repeatability fields

```
source_tree_revision_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
spec_checksum_recorded=1
source_archive_checksum_recorded=1
rpm_nevra_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
validated_cli_mode_still_no_effect=1
validated_runtime_behavior_still_disabled=1
validated_non_claims_preserved=1
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
second_disposable_vm_cli_validation_completed=1
cli_payload_repeatability_evidence_present=1
```

Current project state

```
fedora_vm_cli_payload_next_validation_lane_plan_present=1
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

```
sh scripts/test-fedora-vm-cli-payload-repeatability-transcript-contract.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_transcript_contract: ok
```

Next recommended lane

```
Add Fedora VM CLI payload repeatability runner plan
```

Non-claims

This contract is not a completed repeatability transcript, not RPM install evidence, not a second disposable Fedora VM validation run, not a runner, not host mutation, and not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, or a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR.md

Source title: Fedora VM CLI Payload Repeatability Transcript Review Validator

Lines: 129 | Words: 352 | SHA-256: 3684c78fdd97b87782660834397ef86d0687a391845e987fd53ff55cc16c0c8c

Fedora VM CLI Payload Repeatability Transcript Review Validator

Status: no-effect transcript review validator Evidence level: 9 repeatability target, validator only Scope: validate a supplied Fedora VM CLI payload repeatability transcript candidate without promoting evidence.

Purpose

This validator checks a supplied transcript candidate before any repeatability evidence status can be written.

It verifies that required transcript markers are present.

It rejects placeholder values in required value-bearing fields.

It does not run the repeatability runner.

It does not build an RPM.

It does not install or remove an RPM.

It does not mutate a host.

It does not accept repeatability evidence by itself.

```
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_review_mode=no-effect-validation
repeatability_transcript_candidate_valid=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
evidence_status_written=0
promotion_allowed_by_validator_alone=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
```

Command

```
sh scripts/fedora-vm-cli-payload-repeatability-transcript-review.sh --transcript &lt;path&gt;
```

The command reads the supplied transcript candidate and emits a no-effect review report.

The validator does not create or modify status files.

Required value fields

The validator rejects missing or placeholder values for:

```
source_tree_revision
fedora_kernel_version
spec_checksum
source_archive_checksum
rpm_nevra
```

It also checks that checksum fields are SHA-256-shaped, that the source revision is a hex revision, and that the RPM NEVRA is for the local Latticra package.

Successful candidate report

A structurally valid supplied transcript produces:

```
FEDORA VM CLI PAYLOAD REPEATABILITY TRANSCRIPT REVIEW
review_status=ok
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_review_mode=no-effect-validation
repeatability_transcript_candidate_valid=1
repeatability_transcript_placeholder_values_absent=1
repeatability_transcript_required_markers_present=1
repeatability_transcript_value_fields_validated=1
repeatability_transcript_reviewed=1
repeatability_transcript_accepted=0
evidence_status_written=0
promotion_allowed_by_validator_alone=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Current validator state

```
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_review_mode=no-effect-validation
repeatability_transcript_candidate_valid=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
evidence_status_written=0
promotion_allowed_by_validator_alone=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

This validator is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-transcript-review-validator.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_transcript_review_validator: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Non-claims

This validator is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not an RPM build, not an RPM install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_LANE.md

Source title: Fedora VM CLI Payload Validation Lane

Lines: 303 | Words: 742 | SHA-256: 55fa22fc127cfdb822fe227db1ad8f0755a128f25423a12219f8f8c925292583

Fedora VM CLI Payload Validation Lane

Status: gated validation lane runner Evidence level: 9 target Scope: manual disposable Fedora VM validation of the expanded local RPM payload that includes the no-effect latticra CLI.

Purpose

This lane adds the manual runner for validating the expanded Fedora local RPM payload inside a disposable Fedora VM.

The payload under validation is:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

The runner is intentionally gated.

It must not run automatically in CI.

It must not run on a daily-driver system.

It must not run on a production host.

It must not run on an immutable Fedora target.

It must not claim Fedora approval, Fedora distribution readiness, daily-driver readiness, immutable Fedora readiness, production installer readiness, security hardening, sandboxing, recovery safety, update safety, or OS replacement readiness.

Manual hard gate

The runner refuses to proceed unless all of the following are true:

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
ID=fedora
rpm_present=1
rpmbuild_present=1
cc_present=1
sudo_present=1
```

The runner must be launched by a non-root operator account so the CLI can be executed without privilege escalation. The runner uses sudo only for RPM installation and removal.

Required source checks

Before package mutation, the runner verifies the current tree contains:

```
src/latticra_cli.c
scripts/test-latticra-no-effect-cli-status-surface.sh
packaging/fedora/latticra.spec
docs/FEDORA_VM_CLI_TRANSCRIPT_CONTRACT.md
```

The runner also executes:

```
sh scripts/test-latticra-no-effect-cli-status-surface.sh
```

This preserves the no-effect CLI boundary before the RPM is built.

Required package sequence

The runner performs this sequence only after the hard gate passes:

1. Verify disposable Fedora VM target evidence.
2. Verify Fedora tooling and local tree references.
3. Run the no-effect CLI status-surface guard.
4. Build a local source archive from the current tree.
5. Build the local binary RPM with `rpmbuild`.
6. Inspect RPM metadata.
7. Inspect RPM payload.
8. Confirm the payload contains `/usr/bin/latticra`.
9. Confirm the payload contains `/usr/share/doc/latticra/README.md`.
10. Confirm forbidden payload surfaces are absent.
11. Confirm the local package is not installed before validation.
12. Install the RPM into the disposable Fedora VM with `rpm`.
13. Query installed package state.
14. Verify installed CLI and README payload files.
15. Verify package files with `rpm -V`.
16. Execute `latticra --status` without root.
17. Execute `latticra --version` without root.
18. Execute `latticra --report` without root.
19. Verify invalid CLI usage exits with code 2.
20. Remove the RPM from the disposable Fedora VM with `rpm`.
21. Confirm the package is absent after removal.
22. Confirm the CLI payload is absent after removal.
23. Confirm the README payload is absent after removal.
24. Emit a deterministic validation report.

Expected payload

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

The RPM payload may include the package-owned documentation directory:

```
/usr/share/doc/latticra
```

Forbidden payload surfaces

```
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
/usr/share/selinux
/usr/sbin/latticra
```

Required CLI output validation

The runner validates that latticra --status and latticra --report emit:

```
LATTICRA STATUS REPORT
project=latticra
mode=no-effect
runtime_behavior=disabled
host_mutation=0
network=0
kernel_operation=0
service_operation=0
package_manager_operation=0
boot_operation=0
selinux_policy_operation=0
effect_authority=denied
```

The runner validates that latticra --version emits:

```
latticra 0.0.0
mode=no-effect
runtime_behavior=disabled
```

The runner validates that an invalid CLI command exits with code 2 and emits:

```
usage: latticra [--status|--version|--report]
```

Required report fields

A successful real disposable Fedora VM run emits:

```
FEDORA VM CLI PAYLOAD VALIDATION LANE
```

```
validation_status=ok
package_name=latticra
package_version_recorded=1
package_arch_recorded=1
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
rpm_tooling_recorded=1
rpmbuild_tooling_recorded=1
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_contains_compiled_c_binary=1
buildarch_noarch_removed=1
cli_status_surface_implemented=1
cli_status_surface_guarded_before_packaging=1
local_cli_guard_passed=1
local_rpm_built_from_current_tree=1
rpm_build_command_recorded=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_arch_recorded=1
rpm_path_recorded=1
rpm_metadata_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_contains_cli_binary=1
rpm_payload_contains_readme=1
rpm_payload_contains_only_expected_surfaces=1
unexpected_runtime_surface_absent=1
install_command_recorded=1
install_result_recorded=1
rpm_query_after_install_recorded=1
installed_payload_listing_recorded=1
installed_cli_binary_present=1
installed_readme_present=1
rpm_verify_completed=1
cli_status_command_recorded=1
cli_version_command_recorded=1
cli_report_command_recorded=1
cli_invalid_command_recorded=1
cli_no_root_required=1
cli_no_host_mutation_observed=1
cli_no_network_observed=1
cli_no_service_operation_observed=1
cli_no_kernel_operation_observed=1
cli_no_boot_operation_observed=1
cli_no_selinux_policy_operation_observed=1
removal_command_recorded=1
removal_result_recorded=1
post_removal_query_recorded=1
post_removal_absence_verified=1
cli_removed_after_rpm_removal=1
readme_removed_after_rpm_removal=1
post_removal_cli_absence_verified=1
post_removal_readme_absence_verified=1
install_validation_performed=1
cli_status_validation_performed=1
cli_version_validation_performed=1
cli_report_validation_performed=1
removal_validation_performed=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
```

Current project state until a real run is captured

Adding this runner does not itself create VM evidence.

```
validation_runner_present=1
runner_manual_only=1
ci_auto_vm_cli_validation_allowed=0
disposable_vm_cli_validation_transcript_present=0
disposable_vm_cli_validation_completed=0
```

```
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Manual usage

From inside a disposable Fedora VM with a clean snapshot and recovery path:

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1 \
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1 \
LATTICRA_TARGET_IS_DAILY_DRIVER=0 \
LATTICRA_TARGET_IS_PRODUCTION_HOST=0 \
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0 \
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1 \
LATTICRA_TARGET_HAS_RECOVERY_PATH=1 \
LATTICRA_OPERATOR_CONSENT_RECORDED=1 \
sh scripts/run-fedora-vm-cli-payload-validation-lane.sh
```

Expected final marker:

```
fedora_vm_cli_payload_validation_lane: ok
```

CI boundary

CI only runs the documentation and runner-shape guard:

```
sh scripts/test-fedora-vm-cli-payload-validation-lane-docs.sh
```

Expected output:

```
fedora_vm_cli_payload_validation_lane_docs: ok
```

CI must not run:

```
scripts/run-fedora-vm-cli-payload-validation-lane.sh
```

Next recommended lane

```
Add disposable Fedora VM CLI payload validation status alignment
```

After that, capture a real disposable Fedora VM CLI payload validation transcript from an actual gated run.

docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_TRANSCRIPT_CAPTURE.md

Source title: Fedora VM CLI Payload Validation Transcript Capture

Lines: 76 | Words: 263 | SHA-256: e4434a2fcf66f9262a6cce106fbb9705bc640a224b2b32a8827363b4e78fd617

Fedora VM CLI Payload Validation Transcript Capture

Status: manual evidence-capture preparation Evidence level: capture preparation only Scope: checklist for a future reviewed disposable Fedora VM CLI payload validation transcript.

Purpose

This page prepares the next evidence step for the Fedora VM CLI payload lane.

The real validation runner already exists at:

```
scripts/run-fedora-vm-cli-payload-validation-lane.sh
```

The runner documentation already exists at:

```
docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_LANE.md
```

This page does not run the validation lane.

This page does not create completed evidence.

This page does not claim that a disposable Fedora VM run has succeeded.

This page does not claim host install readiness for the CLI payload.

Evidence capture rule

A future evidence PR may only mark the CLI payload validation complete after a real disposable Fedora VM run is performed, reviewed, and committed with the full transcript.

Until that reviewed transcript exists, the repository must preserve:

```
disposable_vm_cli_validation_transcript_present=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
```

Required reviewed evidence

The future transcript evidence PR should include:

```
reviewed_disposable_fedora_vm_target=1
reviewed_clean_snapshot_evidence=1
reviewed_recovery_path_evidence=1
reviewed_operator_consent=1
reviewed_runner_success_marker=1
reviewed_cli_payload_present=1
reviewed_readme_payload_present=1
reviewed_cli_status_output=1
reviewed_cli_version_output=1
reviewed_cli_report_output=1
reviewed_post_removal_absence=1
reviewed_non_claims_preserved=1
```

CI boundary

CI may validate this checklist and related documentation only.

CI must not run the manual validation runner.

Next recommended Fedora lane

```
Run the real disposable Fedora VM CLI payload validation lane and commit reviewed transcript
evidence
```

Non-claims

This checklist is not completed VM evidence, not CLI payload validation evidence, not host install readiness for the CLI payload, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, and not a production installer claim.

docs/FEDORA_VM_CLI_TRANSCRIPT_CONTRACT.md

Source title: Fedora VM CLI Transcript Contract

Lines: 253 | Words: 435 | SHA-256: 0aa43b5e2355246a08dfbf2fcf484bb0c37f3a933a56fbcdc96e731597711d05

Fedora VM CLI Transcript Contract

Status: contract record Evidence level: 9 target, contract only Scope: transcript schema for future disposable Fedora VM validation of the expanded local RPM payload that includes the no-effect latticra CLI.

Purpose

The local Fedora RPM spec now packages the no-effect CLI as %{_bindir}/latticra.

This document defines the evidence required before that expanded payload can be treated as validated in a disposable Fedora VM.

This is a contract only.

It does not run the validation lane.

It does not build a release RPM.

It does not install or remove an RPM.

It does not validate /usr/bin/latticra.

It does not claim host install readiness for the CLI payload.

Required transcript header

```
FEDORA VM CLI PAYLOAD VALIDATION TRANSCRIPT
transcript_kind=disposable-vm-cli-payload-validation
transcript_version=1
operator_review_required=1
validation_transcript_recorded_after_real_run=1
```

Required target evidence

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
rpm_tooling_recorded=1
rpmbuild_tooling_recorded=1
```

Required spec evidence

```
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_contains_compiled_c_binary=1
buildarch_noarch_removed=1
cli_status_surface_implemented=1
cli_status_surface_guarded_before_packaging=1
local_cli_guard_passed=1
local_rpm_built_from_current_tree=1
```

Required source and guard references:

```
src/latticra_cli.c
scripts/test-latticra-no-effect-cli-status-surface.sh
packaging/fedora/latticra.spec
scripts/test-latticra-no-effect-cli-packaging-contract-alignment.sh
scripts/test-latticra-no-effect-cli-rpm-spec-update-status.sh
```

Required runner gate evidence

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
ID=fedora
rpm_present=1
rpmbuild_present=1
```

Required package evidence

```
rpm_build_command_recorded=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_arch_recorded=1
rpm_path_recorded=1
rpm_metadata_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_contains_cli_binary=1
rpm_payload_contains_readme=1
rpm_payload_contains_only_expected_surfaces=1
unexpected_runtime_surface_absent=1
```

Expected payload:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

Forbidden payload surfaces:

```
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
/usr/share/selinux
```

Required install and CLI evidence

```
install_command_recorded=1
install_result_recorded=1
rpm_query_after_install_recorded=1
installed_payload_listing_recorded=1
installed_cli_binary_present=1
installed_readme_present=1
rpm_verify_completed=1
cli_status_command_recorded=1
cli_version_command_recorded=1
cli_report_command_recorded=1
cli_invalid_command_recorded=1
cli_no_root_required=1
cli_no_host_mutation_observed=1
cli_no_network_observed=1
cli_no_service_operation_observed=1
cli_no_kernel_operation_observed=1
cli_no_boot_operation_observed=1
cli_no_selinux_policy_operation_observed=1
```

Required latticra --status and latticra --report output fields:

```
LATTICRA STATUS REPORT
project=latticra
mode=no-effect
runtime_behavior=disabled
host_mutation=0
network=0
kernel_operation=0
service_operation=0
package_manager_operation=0
boot_operation=0
selinux_policy_operation=0
effect_authority=denied
```

Required latticra --version output fields:

```
latticra 0.0.0
mode=no-effect
runtime_behavior=disabled
```

Required removal evidence

```
removal_command_recorded=1
removal_result_recorded=1
post_removal_query_recorded=1
post_removal_absence_verified=1
cli_removed_after_rpm_removal=1
readme_removed_after_rpm_removal=1
post_removal_cli_absence_verified=1
post_removal_readme_absence_verified=1
```

Required final report

A future runner should emit:

```
FEDORA VM CLI PAYLOAD VALIDATION LANE
validation_status=ok
package_name=latticra
package_version_recorded=1
package_arch_recorded=1
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
rpm_payload_contains_cli_binary=1
```

```
rpm_payload_contains_readme=1
unexpected_runtime_surface_absent=1
install_validation_performed=1
cli_status_validation_performed=1
cli_version_validation_performed=1
cli_report_validation_performed=1
removal_validation_performed=1
post_removal_absence_verified=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
```

Current project state until real evidence exists

```
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_payload_validated=0
disposable_vm_cli_validation_transcript_present=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

```
sh scripts/test-fedora-vm-cli-transcript-contract.sh
```

Expected output:

```
fedora_vm_cli_transcript_contract: ok
```

Next recommended lane

```
Add disposable Fedora VM CLI payload validation lane runner
```

Non-claims

This contract is not a completed validation transcript, not RPM install evidence, not disposable Fedora VM validation of the CLI payload, and not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, sandboxing, or a production installer claim.

docs/FREEBSD_PORT_STATIC_VALIDATION.md

Source title: FreeBSD Port Static Validation

Lines: 92 | Words: 214 | SHA-256: b2160cbb46363727c54c9dc8a01e86c5370d7cdf2fb35acbbb949c965660d2ec

FreeBSD Port Static Validation

Status: active static validation lane Scope: static checks for the local-only FreeBSD ports metadata draft.

Purpose

This lane checks that the FreeBSD port draft is present, narrow, and honest.

It verifies local ports metadata shape for the no-effect CLI payload, but it does not run make package, make stage, poudriere, portlint, or pkg install, and it does not create package artifacts.

Guarded Files

```
packaging/freebsd/README.md
packaging/freebsd/Makefile
packaging/freebsd/pkg-descr
packaging/freebsd/pkg-plist
docs/FREEBSD_PORT_STATIC_VALIDATION.md
docs/status/FREEBSD_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-freebsd-port-static-validation.sh
.github/workflows/freebsd-port-static-validation.yml
```

Preserved Boundary

```
local_only_draft=1
freebsd_port_draft_present=1
freebsd_port_static_validation_present=1
freebsd_ports_tree_submission_claimed=0
freebsd_bugzilla_pr_claimed=0
freebsd_committer_review_claimed=0
poudriere_build_run=0
make_stage_run=0
make_package_run=0
portlint_run=0
package_artifact_created=0
freebsd_official_port_claimed=0
production_readiness_claimed=0
```

Package Intent

The local FreeBSD port draft records only this initial payload shape:

```
bin/latticra
%%DOCSDIR%%/README.md
```

The CLI remains the no-effect status surface from:

```
src/latticra_cli.c
```

Explicit Non-Claims

This static lane does not:

```
build a package artifact
install a package artifact
submit to the FreeBSD ports tree
claim Bugzilla PR evidence
claim FreeBSD committer review
claim poudriere success
claim portlint success
install an rc.d service
install kernel modules
write /etc or /boot
claim production readiness
```

Next Recommended Lane

Add a FreeBSD source archive and checksum contract before accepting make stage, make package, portlint, or poudriere evidence.

Validation

Run:

```
sh scripts/test-freebsd-port-static-validation.sh
```

Expected output:

```
freebsd_port_static_validation: ok
```

docs/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT.md

Source title: macOS Reset/Uninstall Absence-Report Contract

Lines: 240 | Words: 482 | SHA-256: 42fd11504d5a772e091eacdfc07d6b6b6aeb794cfd36b2311ddbfd7d5357f9e9

macOS Reset/Uninstall Absence-Report Contract

Status: no-effect macOS reset/uninstall absence-report contract Date: 2026-05-25 CDT Scope: contract for future post-plan absence evidence before any reset/uninstall implementation.

Purpose

This contract defines the absence-report evidence that a future macOS reset/uninstall implementation must emit after planned managed-target removal.

It is contract-only. It does not delete files, remove directories, inspect post-removal state, write reports, write receipts, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-absence-report-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current absence-report posture is:

```
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
absence_report_contract_state=defined-no-effect
absence_report_contract_decision=contract-defined-evidence-not-present
absence_report_required=1
absence_report_evidence_present=0
absence_report_run_performed=0
absence_report_written=0
```

Required Inputs

A future absence report must bind to the existing no-effect macOS reset/uninstall chain:

```
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
```

Required Report Lines

The future report must record:

```
absence_report_line_host_identity_required=1
absence_report_line_architecture_required=1
absence_report_line_operation_required=1
absence_report_line_planner_decision_required=1
absence_report_line_classifier_decision_required=1
absence_report_line_pre_removal_target_states_required=1
absence_report_line_managed_target_paths_required=1
absence_report_line_planned_actions_required=1
absence_report_line_post_removal_absence_required=1
absence_report_line_unmanaged_preservation_required=1
absence_report_line_receipt_path_required=1
absence_report_line_authority_denials_required=1
absence_report_line_no_network_required=1
absence_report_line_no_root_required=1
```

Target Evidence

The future report must bind evidence to these targets:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
absence_report_must_be_outside_removed_prefix=1
```

Managed targets must have absence evidence only when they were planned and actually removed by a future implementation:

```
app_bundle_absence_required_if_managed=1
app_support_absence_required_if_managed=1
cli_wrapper_absence_required_if_managed=1
unmanaged_target_preservation_evidence_required=1
reset_receipts_dir_preservation_required=1
reset_receipt_reference_required=1
planner_transcript_reference_required=1
```

```
live_classifier_reference_required=1
```

The macOS reset/uninstall receipt-schema contract defines the receipt fields that the future absence report must reference.

Phases

The future absence-report phases are:

```
absence_report_phase_1=consume_dry_run_planner_transcript
absence_report_phase_2=record_pre_removal_target_states
absence_report_phase_3=record_post_removal_absence_or_preservation
absence_report_phase_4=record_reset_receipt_reference
absence_report_phase_5=record_authority_denials
absence_report_phase_6=write_absence_report
absence_report_phase_6_status=disabled
```

Authority Boundary

This contract preserves:

```
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-absence-report-contract.sh
```

Expected output:

```
macos_reset_uninstall_absence_report_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/OPENBSD_PORT_STATIC_VALIDATION.md

Source title: OpenBSD Port Static Validation

Lines: 94 | Words: 223 | SHA-256: fd665d945791baee3e55e0f5dc4057dacf7e327fc2e8f63c983725f2ef4857e2

OpenBSD Port Static Validation

Status: active static validation lane Scope: static checks for the local-only OpenBSD ports metadata draft.

Purpose

This lane checks that the OpenBSD port draft is present, narrow, and honest.

It verifies local ports metadata shape for the no-effect CLI payload, but it does not run make package, make plist, portcheck, a bulk build, or pkg_add, and it does not create package artifacts.

Guarded Files

```
packaging/openbsd/README.md
packaging/openbsd/Makefile
packaging/openbsd/pkg/DESCR
packaging/openbsd/pkg/PLIST
docs/OPENBSD_PORT_STATIC_VALIDATION.md
docs/status/OPENBSD_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-openbsd-port-static-validation.sh
.github/workflows/openbsd-port-static-validation.yml
```

Preserved Boundary

```
local_only_draft=1
openbsd_port_draft_present=1
openbsd_port_static_validation_present=1
openbsd_ports_tree_submission_claimed=0
openbsd_ports_review_thread_claimed=0
openbsd_maintainer_acceptance_claimed=0
make_package_run=0
make_plist_run=0
bulk_build_run=0
portcheck_run=0
package_artifact_created=0
permit_package_enabled=0
openbsd_official_port_claimed=0
production_readiness_claimed=0
```

Package Intent

The local OpenBSD port draft records only this initial payload shape:

```
bin/latticra
share/doc/latticra/README.md
```

The CLI remains the no-effect status surface from:

```
src/latticra_cli.c
```

Explicit Non-Claims

This static lane does not:

```
build a package artifact
install a package artifact
submit to the OpenBSD ports tree
claim ports@ review evidence
claim maintainer acceptance
claim bulk build success
```

```
claim portcheck success
enable package redistribution
install an rc.d script
install kernel modules
write /etc or /bsd
claim production readiness
```

Next Recommended Lane

Add an OpenBSD source archive, checksum, and license redistribution contract before accepting PERMIT_PACKAGE=Yes, make plist, make package, portcheck, or bulk build evidence.

Validation

Run:

```
sh scripts/test-openbsd-port-static-validation.sh
```

Expected output:

```
openbsd_port_static_validation: ok
```

docs/OPENSUSE_DEVELOPER_WORKFLOW.md

Source title: openSUSE Developer Workflow

Lines: 143 | Words: 490 | SHA-256: bec79a79f3d5cc1e0eedaedc4b06cb9a1dde9adf8877d9b5ea3eb6d242159fc1

openSUSE Developer Workflow

Status: developer workflow record Scope: local openSUSE Linux commands for productive Latticra development, user-local Panel work, and openSUSE maintenance records.

Purpose

This note gives a repeatable local workflow for working on Latticra from openSUSE Linux.

The goal is simple:

```
install the small zypper toolchain
install the Panel desktop prerequisites
run the same guards as the Fedora and Ubuntu tracks
maintain local-only openSUSE RPM metadata separately from the Fedora draft
avoid Open Build Service, official package, or SUSE endorsement claims until evidence exists
```

Install Local Tools

On an openSUSE workstation or openSUSE container, install the basic guard toolchain:

```
sudo zypper refresh
sudo zypper install -y git gcc make coreutils findutils diffutils grep pkgconf
```

In a root container, omit sudo:

```
zypper refresh
zypper install -y git gcc make coreutils findutils diffutils grep pkgconf
```

Install Panel Prerequisites

For the Rust/egui Latticra Panel and desktop metadata refresh path, install:

```
sudo zypper refresh
sudo zypper install -y rust cargo make gcc pkgconf \
  libX11-devel libxcb-devel libXcursor-devel libXrandr-devel libXi-devel \
  libxkbcommon-devel Mesa-libGL-devel wayland-devel desktop-file-utils \
  gtk3-tools
```

If the distro Rust toolchain is unavailable or too old for the locked Cargo dependency graph, use a current stable Rust toolchain through the operator's approved toolchain source before running Panel commands. The repository does not grant network authority or install Rust by itself.

Fast Local Guard Loop

For a quick sanity check while editing C/kernel files, run:

```
sh scripts/test-state-lattice.sh
sh scripts/test-system-bootstrap.sh
sh scripts/test-kernel.sh
sh scripts/test-kernel-lifecycle.sh
sh scripts/test-latticra-no-effect-cli-status-surface.sh
```

openSUSE Lane Guards

Before merging openSUSE-facing work, run:

```
sh scripts/test-opensuse-developer-workflow.sh
sh scripts/test-opensuse-local-rpm-static-validation.sh
sh scripts/test-opensuse-rpmlint-findings-classification.sh
sh scripts/test-opensuse-source-archive-reproducibility-contract.sh
sh scripts/test-opensuse-source-archive-fixture-lane.sh
sh scripts/test-opensuse-rpm-topdir-handoff-lane.sh
sh scripts/test-opensuse-local-rpm-build-gate-contract.sh
sh scripts/test-opensuse-local-rpm-build-environment-contract.sh
sh scripts/test-opensuse-rpm-artifact-naming-contract.sh
sh scripts/test-opensuse-rpm-payload-inspection-contract.sh
sh scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
sh scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
```

Expected output:

```
opensuse_developer_workflow: ok
opensuse_local_rpm_static_validation: ok
opensuse_rpmlint_findings_classification: ok
opensuse_source_archive_reproducibility_contract: ok
opensuse_source_archive_fixture_lane: ok
opensuse_rpm_topdir_handoff_lane: ok
opensuse_local_rpm_build_gate_contract: ok
opensuse_local_rpm_build_environment_contract: ok
opensuse_rpm_artifact_naming_contract: ok
opensuse_rpm_payload_inspection_contract: ok
opensuse_rpm_install_remove_transcript_contract: ok
opensuse_obs_publication_non_claim_review_contract: ok
```

Panel From Source

After the prerequisites are present:

```
export PATH=&quot;$HOME/.local/bin:$PATH&quot;
make -C installer gui
```

The guarded user-local install remains the same on openSUSE:

```
make -C installer dry-run
make -C installer local-example
make -C installer verify-local
```

Maintenance Rhythm

Use this loop for productive openSUSE maintenance:

```
1. Create a small branch.
2. Make one bounded openSUSE-facing change.
3. Run the fast local guard loop.
4. Run the openSUSE lane guards when the branch touches packaging, installer, prerequisites, or
openSUSE docs.
5. Keep package language local-only until real lint/build/install and Open Build Service
evidence exists.
```

Current Boundary

This workflow is for local developer productivity, user-local Panel work, and local-only openSUSE package maintenance. It does not publish an RPM, create an Open Build Service project, submit to openSUSE, claim official package readiness, claim SUSE endorsement, install a root service, modify systemd, modify the kernel, or claim production readiness.

Validation

This workflow note is guarded by:

```
sh scripts/test-opensuse-developer-workflow.sh
```

Expected output:

```
opensuse_developer_workflow: ok
```

docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md

Source title: openSUSE Local RPM Build Environment Contract

Lines: 261 | Words: 719 | SHA-256: eda94d9a97bdfc58ed44cd69a57059e3b04738bbb0703a2a89d9fb7b9a41fb67

openSUSE Local RPM Build Environment Contract

Status: active local RPM build environment contract Scope: document required disposable validation environments for openSUSE local RPM build lanes without running rpmbuild or osc build.

Purpose

This contract documents the minimum environment evidence required before any future openSUSE local RPM build lane can ask to open the build gate.

The goal is narrow: describe clean/disposable validation environments, required transcript fields, toolchain metadata, RPM input binding, and host-effect boundaries before any package build command can run.

This contract is documentation-only and static. It does not provision a VM or container, run rpmbuild, run osc build, run spec-cleaner, create source RPMs, create binary RPMs, install Latticra, publish to Open Build Service, or claim package readiness.

Current Environment State

```
opensuse_local_rpm_build_environment_contract_present=1
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
opensuse_rpm_build_environment_contract_state=specified-no-effect
opensuse_rpm_artifact_naming_contract_state=specified-no-effect
opensuse_rpm_payload_inspection_contract_state=specified-no-effect
opensuse_rpm_install_remove_transcript_contract_state=specified-no-effect
opensuse_rpm_build_gate_state=closed-no-effect
opensuse_clean_build_environment_documented=1
opensuse_target_distribution_documented=1
osc_build_environment_documented=1
opensuse_build_environment_provisioned=0
osc_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
disposable_validation_environment_provisioned=0
environment_transcript_present=0
toolchain_version_capture_required=1
rpm_input_digest_binding_required=1
rpmbuild_command_allowed=0
osc_build_command_allowed=0
rpmbuild_run=0
osc_build_run=0
spec_cleaner_run=0
rpm_artifact_created=0
rpm_payload_inspection_run=0
rpm_install_remove_transcript_present=0
rpm_validation_result_promoted=0
source_rpm_artifact_created=0
binary_rpm_artifact_created=0
rpm_artifact_sha256_recorded=0
rpm_installed_on_host=0
rpm_removed_from_host=0
package_readiness_claimed=0
```

Local RPM Environment Contract

A future local RPM build lane must use a disposable openSUSE validation environment before rpmbuild can run.

Required local RPM environment evidence:

```
opensuse_clean_build_environment_documented=1
opensuse_container_or_vm_required=1
opensuse_target_distribution_record_required=1
opensuse_architecture_record_required=1
opensuse_repository_state_record_required=1
opensuse_build_dependency_resolution_transcript_required=1
rpmbuild_toolchain_versions_required=1
source_archive_digest_required=1
rpm_topdir_input_path_required=1
rpmbuild_run=0
rpmbuild_ba_run=0
rpmbuild_bb_run=0
rpmbuild_bs_run=0
```

The local RPM environment contract does not choose between an openSUSE Tumbleweed container, an openSUSE Leap container, a disposable VM, or another clean openSUSE build host. A future effect-bearing proposal must name the exact environment and keep the build gate closed until operator authorization and all package prerequisites exist.

osc Build Environment Contract

A future osc build lane must use a disposable and documented openSUSE build context before osc build can run.

Required osc build environment evidence:

```
osc_build_environment_documented=1
osc_local_build_context_required=1
```

```
osc_config_scope_record_required=1
osc_target_project_record_required=1
osc_target_repository_record_required=1
osc_build_root_policy_required=1
osc_toolchain_versions_required=1
osc_build_run=0
osc_commit_run=0
obs_publication_claimed=0
```

The osc environment contract does not create an Open Build Service project, run osc build, run osc commit, submit Latticra to openSUSE, or publish package artifacts.

Shared Transcript Requirements

A future environment transcript must record:

```
environment_identifier
environment_lifecycle
host_or_vm_class
operating_system_name
operating_system_version
opensuse_target_distribution
architecture
repository_state
toolchain_versions
source_archive_name
source_archive_sha256
rpm_topdir_path
rpm_spec_path
rpm_output_directory
network_policy
host_mount_policy
cleanup_policy
operator_authorization_reference
```

The transcript must be reviewed before a local RPM build lane can claim that the environment is acceptable.

Current Blockers

RPM builds remain blocked because the current repository still has:

```
source_archive_accepted_for_build=0
accepted_rpmlint_transcript_present=0
unexpected_findings_count_recorded=0
license_expression_reviewed=1
package_notice_obligations_reviewed=0
buildrequires_reviewed=0
explicit_operator_build_authorization=0
rpm_artifact_naming_contract_present=1
rpm_payload_inspection_contract_present=1
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_present=1
```

Command Boundary

This contract does not:

```
provision an openSUSE container
provision an openSUSE VM
run zypper install for build dependencies
run rpmbuild
run rpmbuild -ba
run rpmbuild -bb
run rpmbuild -bs
run osc build
run osc commit
run spec-cleaner
create source RPM artifacts
create binary RPM artifacts
install Latticra on a host
publish package artifacts
create an Open Build Service project
submit Latticra to openSUSE
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
```

The local RPM build gate remains closed. This environment contract only records what future disposable openSUSE validation environments must prove before a build lane can request authorization.

Completed Follow-On Lane

Completed follow-on RPM artifact naming contract:

```
Add openSUSE RPM artifact naming contract before any source RPM or binary RPM artifact can be
created.
```

```
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
scripts/test-opensuse-rpm-artifact-naming-contract.sh
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
```

That lane defines source RPM and binary RPM names, output directories, checksum binding, retention rules, and Open Build Service non-claims while keeping opensuse_rpm_build_gate_state=closed-no-effect.

Completed follow-on RPM payload inspection contract:

```
Add openSUSE RPM payload inspection contract before any RPM artifact can be accepted.
```

```
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-opensuse-rpm-payload-inspection-contract.sh
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
```

That lane defines payload inspection evidence without opening the RPM build gate.

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

That lane defines disposable install/remove transcript evidence without opening the RPM build gate.

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

That lane records Open Build Service, submit-request, official-package, and SUSE endorsement non-claims without publishing packages.

Next Slice

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-local-rpm-build-environment-contract.sh
```

Expected output:

```
opensuse_local_rpm_build_environment_contract: ok
```

docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md

Source title: openSUSE Local RPM Build Gate Contract

Lines: 239 | Words: 629 | SHA-256: ba06a57a6770fbd3788e2cfa630b30b3a20c3b708bd7eed48858cf00b6bfe2c6

openSUSE Local RPM Build Gate Contract

Status: active local RPM build evidence gate contract Scope: define the evidence required before openSUSE rpmbuild or osc build commands can run in any validation lane.

Purpose

This contract closes the openSUSE local RPM build gate after the source archive fixture and temporary RPM topdir handoff lanes.

The goal is conservative: record the exact prerequisites required before rpmbuild, osc build, spec-cleaner, package artifact creation, host installation, Open Build Service publication, or package-readiness claims can be attempted or accepted as evidence.

This contract is documentation-only and static. It does not run build tools, create source RPMs, create binary RPMs, install Latticra, publish to Open Build Service, or claim package readiness.

Current Gate State

```
opensuse_local_rpm_build_gate_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
opensuse_rpm_build_gate_state=closed-no-effect
rpmbuild_allowed=0
osc_build_allowed=0
spec_cleaner_allowed=0
rpmbuild_run=0
rpmbuild_ba_run=0
rpmbuild_bb_run=0
rpmbuild_bs_run=0
osc_build_run=0
spec_cleaner_run=0
source_rpm_artifact_created=0
binary_rpm_artifact_created=0
rpm_artifact_created=0
rpm_install_remove_transcript_present=0
rpm_validation_result_promoted=0
rpm_installed_on_host=0
rpm_removed_from_host=0
obs_publication_claimed=0
package_readiness_claimed=0
```

Required Gate Inputs

The build gate cannot open unless all RPM input evidence has already passed:

```
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_rpm_payload_inspection_contract_present=1
temporary_rpm_topdir_handoff_lane_passed=1
opensuse_source_archive_fixture_lane_present=1
opensuse_source_archive_reproducibility_contract_present=1
temporary_rpm_source_sha256_preserved=1
temporary_rpm_source_listing_preserved=1
temporary_rpm_source0_name_matched=1
temporary_rpm_autosetup_root_matched=1
source_archive_accepted_for_build=1
accepted_rpmlint_transcript_present=1
unexpected_findings_count=0
license_expression_reviewed=1
package_notice_obligations_reviewed=1
buildrequires_reviewed=1
spec_cleaner_review_present=1
```

```
open_build_service_non_claim_review_present=1
```

The source_archive_accepted_for_build=1 and accepted_rpmlint_transcript_present=1 values are future prerequisites, not current claims. The current source archive, lint, and topdir lanes still keep build acceptance closed.

Environment Prerequisites

Before any openSUSE package build command can run, a future build-lane proposal must document:

```
opensuse_clean_build_environment_documented=1
opensuse_target_distribution_documented=1
rpmbuild_toolchain_versions_recorded=1
osc_build_environment_documented=1
explicit_operator_build_authorization=1
disposable_validation_environment=1
rpm_artifact_naming_contract_present=1
rpm_payload_inspection_contract_present=1
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_present=1
```

Those records must identify the exact host or VM class, openSUSE target, toolchain versions, source archive digest, RPM topdir input paths, package output paths, install target, remove target, and transcript retention location before a build can be attempted.

Command Gate

All package build commands remain blocked while the current gate state is closed-no-effect.

openSUSE build commands blocked by this contract:

```
rpmbuild -ba
rpmbuild -bb
rpmbuild -bs
osc build
spec-cleaner
```

Open Build Service and host-effect commands blocked by this contract:

```
osc branch
osc submitreq
osc commit
zypper install local RPM artifact
rpm install local RPM artifact
install package on host
remove package from host
publish package artifact
claim official openSUSE package status
claim SUSE endorsement
claim production readiness
```

Future Open Conditions

A future local RPM build lane may move opensuse_rpm_build_gate_state=closed-no-effect to opensuse_rpm_build_gate_state=open-for-local-rpm-build-validation only after it records:

```
source_archive_accepted_for_build=1
archive_sha256_bound_to_build=1
accepted_rpmlint_transcript_present=1
unexpected_findings_count=0
license_expression_reviewed=1
package_notice_obligations_reviewed=1
buildrequires_reviewed=1
spec_cleaner_review_present=1
opensuse_clean_build_environment_documented=1
opensuse_target_distribution_documented=1
explicit_operator_build_authorization=1
disposable_validation_environment=1
rpm_artifact_naming_contract_present=1
rpm_payload_inspection_contract_present=1
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_present=1
```

Opening a local build gate must not imply Open Build Service publication, openSUSE acceptance, host install readiness, or package readiness.

Current Non-Claims

```
opensuse_official_package_claimed=0
opensuse_obs_publication_claimed=0
opensuse_submit_request_claimed=0
suse_endorsement_claimed=0
opensuse_distribution_ready=0
production_installer_ready=0
root_installer_ready=0
```

Relationship To Existing Lanes

This contract depends on:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
```

The rpmlint findings classification record keeps lint output from becoming package readiness evidence without review.

The source archive fixture lane proves temporary archive shape in a disposable workspace.

The RPM topdir handoff lane proves temporary RPM build-input staging while preserving archive SHA-256 identity.

This build gate keeps package build execution closed until source acceptance, lint classification, license, notice, BuildRequires, environment, authorization, payload inspection, install/remove, and OBS non-claim evidence exists.

Next Slice

Completed follow-on local RPM build environment contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
scripts/test-opensuse-local-rpm-build-environment-contract.sh
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
```

Completed follow-on RPM artifact naming contract:

```
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
scripts/test-opensuse-rpm-artifact-naming-contract.sh
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
```

Completed follow-on RPM payload inspection contract:

```
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-opensuse-rpm-payload-inspection-contract.sh
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
```

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-local-rpm-build-gate-contract.sh
```

Expected output:

```
opensuse_local_rpm_build_gate_contract: ok
```

docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md

Source title: openSUSE Local RPM Static Validation

Lines: 110 | Words: 282 | SHA-256: 3f12068ffbd97ad0dac293d9d7ac57b54a35c7609be8523b1a260a7a3c1446bf

openSUSE Local RPM Static Validation

Status: active static validation lane Scope: static checks for the local-only openSUSE RPM packaging draft and maintenance changes file.

Purpose

This lane turns the openSUSE packaging and maintenance draft into an executable static check.

It checks the current local openSUSE RPM spec and .changes record without building packages, producing artifacts, running osc, running rpmlint, or claiming Open Build Service publication.

Files

```
docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md
docs/OPENSUSE_READINESS_PLAN.md
docs/status/OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md
packaging/opensuse/README.md
packaging/opensuse/latticra.spec
packaging/opensuse/latticra.changes
scripts/test-opensuse-local-rpm-static-validation.sh
.github/workflows/opensuse-local-rpm-static-validation.yml
```

Checks

The static validation lane verifies:

```
local-only openSUSE draft marker exists
local packaging README records non-claims
placeholder license remains explicit
local release marker remains explicit
required RPM sections exist
.changes maintenance record exists
no service files are installed
no scriptlet sections are defined
no boot, kernel, systemd, SELinux, or etc paths are added
no Open Build Service publication claim is made
```

Boundary

This lane is static validation only.

It does not run rpmbuild, osc build, rpmlint, spec-cleaner, or zypper install.

It does not create package artifacts, publish to Open Build Service, submit to openSUSE, install on a host, or claim production readiness.

Next Slice

Completed follow-on availability lane:

```
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpmlint-osc-availability.sh
scripts/test-opensuse-rpmlint-static-spec-lane.sh
scripts/test-opensuse-rpmlint-findings-classification.sh
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpmlint-osc-availability.yml
.github/workflows/opensuse-rpmlint-static-spec-lane.yml
.github/workflows/opensuse-rpmlint-findings-classification.yml
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-local-rpm-static-validation.sh
```

Expected output:

```
opensuse_local_rpm_static_validation: ok
```

docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md

Source title: openSUSE OBS Publication Non-Claim Review Contract

Lines: 177 | Words: 527 | SHA-256: e5563c43f2833f826824ee1d2df23cf757f80c89a070fde97ac7b0c941fc649b

openSUSE OBS Publication Non-Claim Review Contract

Status: active OBS publication non-claim review contract Scope: define Open Build Service, submit-request, official-package, and SUSE endorsement non-claim review evidence for openSUSE package validation without publishing packages.

Purpose

This contract records the Open Build Service publication non-claim review required before any future openSUSE RPM validation result can be promoted.

The goal is narrow: future openSUSE package validation evidence must remain clearly local-only unless a separate, explicit publication lane records the required Open Build Service project, package, repository, submit-request, maintainer-review, and acceptance evidence.

This contract is documentation-only and static. It does not run osc, create an Open Build Service project, create an OBS package, upload sources, submit Latticra to openSUSE, create package repositories, publish RPM artifacts, create RPM artifacts, install packages, remove packages, or claim package readiness.

Current Publication Non-Claim State

```
opensuse_obs_publication_non_claim_review_contract_present=1
obs_publication_non_claim_review_contract_present=1
publication_non_claim_review_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
opensuse_rpm_build_gate_state=closed-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
rpm_validation_result_promoted=0
rpm_artifact_created=0
rpm_artifact_published=0
obs_project_created=0
obs_package_created=0
obs_repository_created=0
obs_source_link_created=0
osc_branch_run=0
osc_commit_run=0
osc_submitreq_run=0
osc_request_accepted=0
obs_build_result_claimed=0
opensuse_obs_publication_claimed=0
opensuse_submit_request_claimed=0
opensuse_official_package_claimed=0
suse_endorsement_claimed=0
opensuse_factory_submission_claimed=0
opensuse_factory_acceptance_claimed=0
opensuse_leap_submission_claimed=0
opensuse_distribution_ready=0
package_readiness_claimed=0
production_installer_ready=0
root_installer_ready=0
```

OBS Publication Non-Claims

Future openSUSE validation promotion must preserve these current non-claims:

```
obs_publication_non_claim_review_required=1
obs_publication_non_claim_review_present=1
opensuse_obs_publication_claimed=0
opensuse_submit_request_claimed=0
opensuse_official_package_claimed=0
suse_endorsement_claimed=0
obs_project_created=0
obs_package_created=0
obs_repository_created=0
obs_source_upload_run=0
osc_branch_run=0
osc_add_run=0
osc_commit_run=0
osc_submitreq_run=0
osc_request_accepted=0
```

The openSUSE RPM draft remains local-only. A future Open Build Service project, submit-request, Factory, Leap, or repository publication lane must be separate from local build, payload, install/remove, and publication non-claim review evidence.

Promotion Review Boundary

A future openSUSE RPM validation promotion review must record:

```
rpm_validation_result_identifier
opensuse_target_distribution
source_archive_sha256
```

```
source_rpm_artifact_sha256
binary_rpm_artifact_sha256
payload_inspection_transcript_sha256
install_remove_transcript_sha256
obs_publication_non_claim_reviewer
obs_publication_non_claim_review_date
non_claimed_obs_targets
non_claimed_submit_request_targets
status_page_update_reference
operator_authorization_reference
```

That review may only promote a validation result as local validation evidence. It must not convert local evidence into Open Build Service publication evidence, openSUSE Factory evidence, openSUSE Leap evidence, official-package evidence, SUSE endorsement evidence, package repository evidence, production installer evidence, or release readiness.

Current Blockers

openSUSE RPM validation promotion remains blocked because the current repository still has:

```
source_archive_accepted_for_build=0
accepted_rpmlint_transcript_present=0
rpm_artifact_created=0
rpm_artifact_sha256_recorded=0
rpm_payload_accepted=0
rpm_install_remove_transcript_present=0
environment_transcript_present=0
explicit_operator_build_authorization=0
rpm_validation_result_promoted=0
```

Command Boundary

This contract does not:

```
run osc branch
run osc add
run osc commit
run osc submitreq
run osc build
create an Open Build Service project
create an Open Build Service package
upload source files to Open Build Service
publish RPM artifacts
submit Latticra to openSUSE
claim openSUSE Factory acceptance
claim openSUSE Leap acceptance
claim official openSUSE package status
claim SUSE endorsement
create RPM artifacts
install Latticra on a host
remove Latticra from a host
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
```

The local RPM build gate remains closed. This OBS publication non-claim review contract only records that local openSUSE package validation evidence must remain non-published and non-official until a separate Open Build Service publication lane exists.

Next Slice

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
```

Expected output:

```
opensuse_obs_publication_non_claim_review_contract: ok
```

docs/OPENSUSE_READINESS_PLAN.md

Source title: openSUSE Readiness Plan

Lines: 273 | Words: 770 | SHA-256: 27eb773271e015bc7e68fffaf6598d3dd4dbb6cd31041208fda426aca670609e

openSUSE Readiness Plan

Status: planning and maintenance record Scope: preparing Latticra for openSUSE-friendly development, local validation, local-only RPM packaging, and maintenance review.

Purpose

This plan gives the openSUSE track the same purpose as the Fedora and Ubuntu tracks: integrate Latticra with a real Linux ecosystem through bounded local validation, packaging-shape records, and maintenance evidence while preserving current non-claims.

This plan prepares Latticra for an openSUSE-facing path without claiming openSUSE acceptance, SUSE endorsement, Open Build Service publication, production installer readiness, or distribution status.

The near-term target is:

```
Latticra builds cleanly on openSUSE after operator-installed prerequisites.
Latticra Panel can be run from source on openSUSE with documented zypper prerequisites.
Latticra has a local-only openSUSE RPM packaging draft for the no-effect CLI payload.
Latticra records openSUSE maintenance boundaries separately from Fedora and Ubuntu package
lanes.
Latticra preserves its current non-claims.
```

Terminology Boundary

Use:

```
openSUSE readiness
openSUSE compatibility lane
openSUSE maintenance lane
local-only openSUSE RPM draft
Latticra on openSUSE
future Open Build Service review candidate
```

Avoid claiming:

```
official openSUSE package
published Open Build Service package
SUSE endorsement
openSUSE derivative
openSUSE distribution
operating-system replacement
```

Those terms require separate package review, Open Build Service evidence, policy review, legal review, trademark review, and community review.

openSUSE-Facing Requirements To Respect

Before Latticra can be proposed beyond a local RPM draft, it should be checked against openSUSE package and maintenance expectations:

```
spec file hygiene
.changes maintenance record
license expression review
BuildRequires completeness
no bundled system libraries
valid file ownership
valid permissions
rpmlint output classification
spec-cleaner review where appropriate
osc local build evidence
Open Build Service project metadata review
install/remove behavior evidence
```

Current Latticra Posture

Current Latticra evidence is still early-stage and report-oriented.

Known current posture:

```
opensuse_developer_workflow_present=1
opensuse_panel_prerequisites_documented=1
opensuse_local_rpm_draft_present=1
opensuse_local_rpm_static_validation_present=1
opensuse_changes_file_present=1
opensuse_rpmlint_osc_availability_lane_present=1
opensuse_rpmlint_static_spec_lane_present=1
opensuse_rpmlint_findings_classification_present=1
opensuse_source_archive_reproducibility_contract_present=1
opensuse_source_archive_fixture_lane_present=1
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_local_rpm_build_gate_contract_present=1
opensuse_local_rpm_build_environment_contract_present=1
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
opensuse_obs_publication_claimed=0
opensuse_official_package_claimed=0
suse_endorsement_claimed=0
opensuse_distribution_ready=0
production_installer_ready=0
daily_driver_install_ready=0
root_installer_ready=0
```

The first openSUSE target is development, local validation, and package-maintenance shape. It is not an official openSUSE package, Open Build Service publication, image, distribution, or replacement operating system.

Phase 1: openSUSE Developer Workflow

The openSUSE developer workflow gives local zypper commands for toolchain setup, Panel prerequisites, and the same guarded no-effect validation loop used by the other Linux tracks.

Current guarded files:

```
docs/OPENSUSE_DEVELOPER_WORKFLOW.md
scripts/test-opensuse-developer-workflow.sh
.github/workflows/opensuse-developer-workflow.yml
```

Phase 2: Panel Prerequisites Lane

The Panel remains user-local on openSUSE. The zypper prerequisites support local Rust/egui development and desktop metadata refresh without adding root-installed Latticra services.

Current public entry points:

```
README.md
docs/QUICK_START_CHEATSHEET.md
installer/README.md
```

Phase 3: Local openSUSE RPM Draft Lane

Packaging metadata is present as a local-only openSUSE draft. It records package shape, a separate maintenance changes file, and static validation boundaries.

Current guarded files:

```
packaging/opensuse/README.md
packaging/opensuse/latticra.spec
packaging/opensuse/latticra.changes
docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md
scripts/test-opensuse-local-rpm-static-validation.sh
.github/workflows/opensuse-local-rpm-static-validation.yml
```

The first openSUSE RPM draft is only a packaging-shape and maintenance record until lint/build/install evidence is added.

Phase 4: Open Build Service Readiness Lane

The first Open Build Service readiness slices are tool availability and static spec lint only.

Current guarded files:

```
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-rpmlint-osc-availability.sh
scripts/test-opensuse-rpmlint-static-spec-lane.sh
scripts/test-opensuse-rpmlint-findings-classification.sh
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-rpmlint-osc-availability.yml
.github/workflows/opensuse-rpmlint-static-spec-lane.yml
.github/workflows/opensuse-rpmlint-findings-classification.yml
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

Before any Open Build Service publication or submit request can be claimed, the lane needs evidence for:

```
osc availability
rpmlint availability
spec-cleaner review posture
local osc build transcript
source archive reproducibility
source archive fixture reproducibility
temporary RPM topdir handoff
local RPM build gate contract
local RPM build environment contract
RPM artifact naming contract
RPM payload inspection contract
RPM install/remove transcript contract
OBS publication non-claim review contract
```

```
package artifact inspection
install/remove behavior transcript
```

Non-Claims

This plan does not:

```
publish an RPM package
create an Open Build Service project
submit Latticra to openSUSE
claim official openSUSE package readiness
claim SUSE endorsement
install a root service
modify systemd
modify the kernel
claim production readiness
claim operating-system completeness
```

Recommended Next Slice

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-developer-workflow.sh
sh scripts/test-opensuse-local-rpm-static-validation.sh
sh scripts/test-opensuse-rpmlint-osc-availability.sh
sh scripts/test-opensuse-rpmlint-static-spec-lane.sh
sh scripts/test-opensuse-rpmlint-findings-classification.sh
sh scripts/test-opensuse-source-archive-reproducibility-contract.sh
sh scripts/test-opensuse-source-archive-fixture-lane.sh
sh scripts/test-opensuse-rpm-topdir-handoff-lane.sh
sh scripts/test-opensuse-local-rpm-build-gate-contract.sh
sh scripts/test-opensuse-local-rpm-build-environment-contract.sh
sh scripts/test-opensuse-rpm-artifact-naming-contract.sh
sh scripts/test-opensuse-rpm-payload-inspection-contract.sh
sh scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
sh scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
```

Expected output:

```
opensuse_developer_workflow: ok
opensuse_local_rpm_static_validation: ok
opensuse_rpmlint_osc_availability: ok
opensuse_rpmlint_static_spec_lane: ok
opensuse_rpmlint_findings_classification: ok
opensuse_source_archive_reproducibility_contract: ok
opensuse_source_archive_fixture_lane: ok
opensuse_rpm_topdir_handoff_lane: ok
opensuse_local_rpm_build_gate_contract: ok
opensuse_local_rpm_build_environment_contract: ok
opensuse_rpm_artifact_naming_contract: ok
opensuse_rpm_payload_inspection_contract: ok
opensuse_rpm_install_remove_transcript_contract: ok
opensuse_obs_publication_non_claim_review_contract: ok
```

docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md

Source title: openSUSE RPM Artifact Naming Contract

Lines: 219 | Words: 647 | SHA-256: 0f3e5631b92b589d3b6bd47b98e77a6c82883db212de7c1dcd14c59e9b560b38

openSUSE RPM Artifact Naming Contract

Status: active RPM artifact naming contract Scope: define future openSUSE source RPM and binary RPM artifact names and output boundaries without creating RPM artifacts.

Purpose

This contract records the artifact names, output-directory rules, checksum binding, retention requirements, and Open Build Service non-claims required before any future openSUSE local RPM build can create source RPM or binary RPM files.

The goal is narrow: RPM artifacts must have predictable names, be written only inside a disposable validation environment, bind back to the source archive digest, and remain non-published review outputs until separate payload inspection, install/remove, and OBS publication review contracts exist.

This contract is documentation-only and static. It does not run rpmbuild, run osc build, run spec-cleaner, create RPM artifacts, install Latticra, publish packages, create an Open Build Service project, submit Latticra to openSUSE, or claim package readiness.

Current Artifact Naming State

```
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
rpm_artifact_naming_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
opensuse_rpm_artifact_naming_contract_state=specified-no-effect
opensuse_rpm_payload_inspection_contract_state=specified-no-effect
opensuse_rpm_install_remove_transcript_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
opensuse_rpm_build_gate_state=closed-no-effect
opensuse_rpm_build_environment_contract_state=specified-no-effect
rpm_artifact_output_directory_required_under_disposable_environment=1
rpm_artifact_output_directory_created=0
repository_rpm_artifact_write_allowed=0
publication_directory_write_allowed=0
rpm_artifact_created=0
source_rpm_artifact_created=0
binary_rpm_artifact_created=0
rpm_artifact_sha256_recorded=0
rpm_artifact_published=0
rpm_install_remove_transcript_present=0
rpm_validation_result_promoted=0
rpm_installed_on_host=0
rpm_removed_from_host=0
package_readiness_claimed=0
```

Source RPM Artifact Name

Future source RPM artifact names must be derived from the current openSUSE spec Name, Version, and Release values.

```
rpm_source_package_name=latticra-0.0.0-0.local.src.rpm
rpm_source_archive_name=latticra-0.0.0.tar.gz
rpm_source_artifact_name_pattern_recorded=1
rpmbuild_bs_run=0
source_rpm_artifact_created=0
```

The source RPM name is a future local validation artifact only. It is not an accepted source package, Open Build Service source, submit-request record, or publication claim.

Binary RPM Artifact Name

Future binary RPM artifact names must be derived from the current openSUSE spec Name, Version, Release, and the build environment architecture.

```
rpm_binary_package_name_pattern=latticra-0.0.0-0.local.${RPM_ARCH}.rpm
rpm_binary_arch_token_required=1
rpm_binary_artifact_name_pattern_recorded=1
rpmbuild_ba_run=0
rpmbuild_bb_run=0
binary_rpm_artifact_created=0
```

The ${RPM_ARCH} token must be resolved from the disposable openSUSE environment transcript before any binary RPM artifact can be accepted.

Output Boundary

Future RPM artifacts must be written only under a disposable RPM topdir inside a disposable validation environment. The repository itself remains a no-artifact workspace.

```
artifact_output_root_under_disposable_rpmtop=1
rpm_source_artifact_output_directory=rpmtop/SRPMS/
rpm_binary_artifact_output_directory_pattern=rpmtop/RPMS/${RPM_ARCH}/
repository_rpm_artifact_write_allowed=0
root_workspace_rpm_artifact_write_allowed=0
publication_directory_write_allowed=0
artifact_retention_policy_required=1
artifact_cleanup_policy_required=1
```

Any future RPM artifact transcript must bind each package file to:

```
source_archive_sha256
rpm_input_archive_sha256
rpm_artifact_sha256
artifact_size_bytes
artifact_output_directory
artifact_generation_command
environment_identifier
operator_authorization_reference
```

Current Blockers

RPM artifact creation remains blocked because the current repository still has:

```
source_archive_accepted_for_build=0
accepted_rpmlint_transcript_present=0
license_expression_reviewed=1
package_notice_obligations_reviewed=0
buildrequires_reviewed=0
explicit_operator_build_authorization=0
environment_transcript_present=0
rpm_payload_inspection_contract_present=1
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_present=1
```

Command Boundary

This contract does not:

```
run rpmbuild
run rpmbuild -ba
run rpmbuild -bb
run rpmbuild -bs
run osc build
run osc commit
run spec-cleaner
create .src.rpm artifacts
create binary .rpm artifacts
install Latticra on a host
publish package artifacts
create an Open Build Service project
submit Latticra to openSUSE
claim official openSUSE package status
claim SUSE endorsement
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
```

The local RPM build gate remains closed. This artifact naming contract only records the names and output boundaries future validation artifacts must use after separate source, lint, license, notice, BuildRequires, environment, authorization, payload inspection, install/remove, and OBS non-claim evidence exists.

Completed Follow-On Lane

Completed follow-on RPM payload inspection contract:

```
Add openSUSE RPM payload inspection contract before any RPM artifact can be accepted.
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-opensuse-rpm-payload-inspection-contract.sh
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
```

That lane defines payload inspection evidence without opening the RPM build gate.

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

That lane defines disposable install/remove transcript evidence without installing or removing RPM packages.

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

That lane records Open Build Service, submit-request, official-package, and SUSE endorsement non-claims without publishing packages.

Next Slice

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-rpm-artifact-naming-contract.sh
```

Expected output:

```
opensuse_rpm_artifact_naming_contract: ok
```

docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md

Source title: openSUSE RPM Install/Remove Transcript Contract

Lines: 189 | Words: 490 | SHA-256: 46469cac9c82bb9a0643e330152edeb91ffa81241cd7f070426cf7c445998c91

openSUSE RPM Install/Remove Transcript Contract

Status: active RPM install/remove transcript contract Scope: define future disposable openSUSE RPM install/remove evidence without installing or removing RPM packages.

Purpose

This contract records the install/remove transcript evidence required before any future openSUSE binary RPM artifact can be accepted as installable validation evidence.

The goal is narrow: a future RPM install/remove transcript must prove that installation happens only inside a disposable openSUSE validation environment, the expected no-effect CLI payload appears after install, the package can be removed cleanly, and no host-level service, systemd, kernel, privileged-helper, or network authority is introduced.

This contract is documentation-only and static. It does not run zypper, run rpm, install packages, remove packages, create RPM artifacts, inspect RPM artifacts, publish packages, create an Open Build Service project, submit Latticra to openSUSE, or claim package readiness.

Current Install/Remove State

```
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
rpm_install_remove_transcript_contract_present=1
install_remove_transcript_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
opensuse_rpm_install_remove_transcript_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
opensuse_rpm_build_gate_state=closed-no-effect
payload_inspection_contract_state=specified-no-effect
rpm_install_remove_disposable_environment_required=1
rpm_install_remove_transcript_present=0
rpm_package_install_run=0
rpm_package_remove_run=0
rpm_zypper_install_run=0
rpm_zypper_remove_run=0
rpm_cli_install_run=0
rpm_cli_remove_run=0
rpm_installed_on_host=0
rpm_removed_from_host=0
host_install_allowed=0
host_remove_allowed=0
host_mutation_allowed=0
service_state_change_allowed=0
rpm_artifact_created=0
rpm_payload_accepted=0
rpm_validation_result_promoted=0
package_readiness_claimed=0
```

Required Future Evidence

Future openSUSE RPM install/remove validation must run only in a disposable openSUSE environment.

Required transcript evidence:

```
rpm_install_remove_transcript_required=1
rpm_install_remove_transcript_present=0
rpm_payload_post_install_check_required=1
rpm_payload_post_remove_absence_check_required=1
rpm_service_state_change_allowed=0
rpm_systemd_unit_enable_allowed=0
rpm_scriptlet_effect_allowed=0
rpm_kernel_file_allowed=0
rpm_network_authority_allowed=0
rpm_privileged_helper_allowed=0
```

The future transcript must bind install/remove evidence to:

```
rpm_binary_package_name_pattern=latticra-0.0.0-0.local.${RPM_ARCH}.rpm
rpm_binary_artifact_sha256
rpm_payload_expected_bin=/usr/bin/latticra
rpm_payload_expected_doc=/usr/share/doc/packages/latticra/README.md
```

Transcript Requirements

A future openSUSE RPM install/remove transcript must record:

```
environment_identifier
opensuse_target_distribution
operator_authorization_reference
rpm_artifact_name
rpm_artifact_sha256
pre_install_package_state
install_command
install_exit_code
post_install_payload_listing
post_install_cli_no_effect_output
```

```
rpm_query_after_install
service_state_after_install
remove_command
remove_exit_code
post_remove_absence_report
post_remove_package_state
scriptlet_effect_review
host_mutation_review
```

The transcript must be reviewed before an RPM artifact can be accepted as install/remove-valid.

Current Blockers

RPM install/remove acceptance remains blocked because the current repository still has:

```
rpm_artifact_created=0
rpm_artifact_sha256_recorded=0
rpm_payload_accepted=0
environment_transcript_present=0
explicit_operator_build_authorization=0
obs_publication_non_claim_review_present=1
```

Command Boundary

This contract does not:

```
run zypper install
run zypper remove
run rpm -i
run rpm -U
run rpm -e
install Latticra on a host
remove Latticra from a host
change service state
enable a systemd unit
create package artifacts
inspect package artifacts
publish package artifacts
create an Open Build Service project
submit Latticra to openSUSE
claim official openSUSE package status
claim SUSE endorsement
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
```

The local RPM build gate remains closed. This install/remove transcript contract only defines the evidence future disposable validation environments must produce after RPM artifacts exist and payload inspection evidence is accepted.

Next Slice

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

That lane records Open Build Service, submit-request, official-package, and SUSE endorsement non-claims without publishing packages.

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
```

Expected output:

```
opensuse_rpm_install_remove_transcript_contract: ok
```

docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md

Source title: openSUSE RPM Payload Inspection Contract

Lines: 230 | Words: 576 | SHA-256: a5d94cf62e57d138acf3d2e9fbe24f3c2dff5ca5485d03fe2463f736d045e58a

openSUSE RPM Payload Inspection Contract

Status: active RPM payload inspection contract Scope: define future openSUSE source RPM and binary RPM payload inspection evidence without creating or inspecting RPM artifacts.

Purpose

This contract records the RPM payload inspection evidence required before any future openSUSE source RPM or binary RPM artifact can be accepted as validation evidence.

The goal is narrow: a future RPM artifact must prove that it contains only the intended no-effect CLI payload, documentation payload, and expected RPM source inputs, with no services, privileged helpers, init files, kernel files, network authority, or host mutation hooks.

This contract is documentation-only and static. It does not run rpmbuild, run osc build, run rpm, run rpm2cpio, run cpio, create RPM artifacts, inspect RPM artifacts, install Latticra, publish packages, create an Open Build Service project, submit Latticra to openSUSE, or claim package readiness.

Current Payload Inspection State

```
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
rpm_payload_inspection_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
payload_inspection_contract_present=1
opensuse_rpm_payload_inspection_contract_state=specified-no-effect
opensuse_rpm_install_remove_transcript_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
opensuse_rpm_build_gate_state=closed-no-effect
opensuse_rpm_artifact_naming_contract_state=specified-no-effect
rpm_artifact_created=0
rpm_payload_inspection_run=0
source_rpm_payload_inspection_run=0
binary_rpm_payload_inspection_run=0
rpm_payload_accepted=0
rpm_install_remove_transcript_present=0
rpm_validation_result_promoted=0
rpm_artifact_sha256_recorded=0
rpm_installed_on_host=0
rpm_removed_from_host=0
package_readiness_claimed=0
```

Expected Payload

All future openSUSE RPM artifacts must preserve the current intentionally narrow payload:

```
rpm_payload_cli_path_required=1
rpm_payload_doc_readme_required=1
rpm_payload_service_files_allowed=0
rpm_payload_systemd_units_allowed=0
rpm_payload_init_files_allowed=0
rpm_payload_kernel_files_allowed=0
rpm_payload_privileged_helper_allowed=0
rpm_payload_network_authority_allowed=0
rpm_payload_host_mutation_hooks_allowed=0
```

The no-effect CLI payload must remain the only executable binary payload.

Source RPM Payload Inspection

Future source RPM payload inspection must verify:

```
source_rpm_expected_spec=packaging/opensuse/latticra.spec
source_rpm_expected_changes=packaging/opensuse/latticra.changes
source_rpm_expected_source_archive=latticra-0.0.0.tar.gz
source_rpm_payload_listing_required=1
source_rpm_metadata_inspection_required=1
source_rpm_unexpected_archive_count=0
source_rpm_payload_inspection_run=0
rpm2cpio_source_inspection_run=0
source_rpm_metadata_inspection_run=0
```

The future source RPM inspection transcript must bind the payload listing to:

```
rpm_source_package_name=latticra-0.0.0-0.local.src.rpm
source_rpm_artifact_sha256
source_rpm_payload_listing_sha256
```

Binary RPM Payload Inspection

Future binary RPM payload inspection must verify:

```
rpm_payload_expected_bin=/usr/bin/latticra
rpm_payload_expected_doc=/usr/share/doc/packages/latticra/README.md
rpm_payload_listing_required=1
rpm_metadata_inspection_required=1
rpm_scriptlet_absence_required=1
rpm_systemd_unit_absence_required=1
rpm_init_script_absence_required=1
rpm_privileged_helper_absence_required=1
rpm_payload_unexpected_file_count=0
binary_rpm_payload_inspection_run=0
rpm_query_payload_inspection_run=0
rpm_query_scriptlet_inspection_run=0
rpm2cpio_binary_listing_run=0
```

The future binary RPM inspection transcript must bind the payload listing to:

```
rpm_binary_package_name_pattern=latticra-0.0.0-0.local.${RPM_ARCH}.rpm
binary_rpm_artifact_sha256
binary_rpm_payload_listing_sha256
```

Inspection Transcript Requirements

A future RPM payload inspection transcript must record:

```
rpm_artifact_name
rpm_artifact_sha256
rpm_artifact_size_bytes
payload_listing_command
payload_listing_sha256
expected_payload_paths_present
unexpected_payload_paths_absent
service_files_absent
systemd_units_absent
scriptlets_absent
privileged_helpers_absent
host_mutation_hooks_absent
environment_identifier
operator_authorization_reference
```

The transcript must be reviewed before an RPM artifact can be accepted as payload-correct.

Current Blockers

RPM payload acceptance remains blocked because the current repository still has:

```
rpm_artifact_created=0
rpm_artifact_sha256_recorded=0
environment_transcript_present=0
explicit_operator_build_authorization=0
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_present=1
```

Command Boundary

This contract does not:

```
run rpm -qlp
run rpm -qip
run rpm -qp --scripts
run rpm2cpio
run cpio
create .src.rpm artifacts
create binary .rpm artifacts
inspect RPM artifacts
install Latticra on a host
publish package artifacts
create an Open Build Service project
submit Latticra to openSUSE
claim official openSUSE package status
claim SUSE endorsement
claim package readiness
```

Relationship To Existing Lanes

This contract depends on:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
```

The local RPM build gate remains closed. This payload inspection contract only defines the evidence future RPM artifacts must provide after they exist in a disposable validation environment.

Next Slice

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

That lane defines disposable install/remove transcript evidence without installing or removing RPM packages.

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

That lane records Open Build Service, submit-request, official-package, and SUSE endorsement non-claims without publishing packages.

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-rpm-payload-inspection-contract.sh
```

Expected output:

```
opensuse_rpm_payload_inspection_contract: ok
```

docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md

Source title: openSUSE RPM Topdir Handoff Lane

Lines: 156 | Words: 421 | SHA-256: 04cfea2d6a9ab347a295e7f17198cd1a4b623cceff3f94262de8859a76c2e4ed

openSUSE RPM Topdir Handoff Lane

Status: active temporary RPM topdir handoff lane Scope: stage the verified temporary openSUSE source archive into a disposable RPM topdir without running package builds, installing artifacts, or publishing to Open Build Service.

Purpose

This lane advances the openSUSE packaging path from source archive fixture evidence into RPM build-input staging.

The goal is conservative: prove that the temporary Source0 archive can be placed under SOURCES/, the local-only spec can be placed under SPECS/, and the staged archive keeps the same checksum and listing as the verified fixture.

This lane does not build a source RPM or binary RPM. It stages temporary RPM build inputs only.

Temporary RPM Topdir Layout

The handoff creates this layout under a disposable temporary directory:

```
rpmtop/BUILD/
rpmtop/BUILDROOT/
rpmtop/RPMS/
rpmtop/SOURCES/latticra-0.0.0.tar.gz
rpmtop/SPECS/latticra.spec
rpmtop/SPECS/latticra.changes
rpmtop/SRPMS/
```

The staged source archive name must match the resolved openSUSE spec inputs:

```
Source0:        %{name}-%{version}.tar.gz
%autosetup -n %{name}-%{version}
source_archive_name=latticra-0.0.0.tar.gz
source_archive_root=latticra-0.0.0/
```

Handoff Checks

The lane verifies:

```
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_source_archive_fixture_lane_present=1
temporary_rpm_topdir_handoff_lane_present=1
temporary_rpm_topdir_created=1
temporary_rpm_sources_archive_staged=1
temporary_rpm_specs_spec_staged=1
temporary_rpm_specs_changes_staged=1
temporary_rpm_source_sha256_preserved=1
temporary_rpm_source_listing_preserved=1
temporary_rpm_source0_name_matched=1
temporary_rpm_autosetup_root_matched=1
rpm_build_dirs_created=1
source_archive_accepted_for_build=0
rpmbuild_run=0
osc_build_run=0
rpm_artifact_created=0
```

The handoff evidence is temporary package-input staging only. It does not promote the archive to accepted package-build input.

Boundary

This lane does not run rpmbuild.

It does not run osc build.

It does not run rpmlint.

It does not run spec-cleaner.

It does not create source RPM artifacts.

It does not create binary RPM artifacts.

It does not install Latticra.

It does not publish package artifacts.

It does not create an Open Build Service project.

It does not submit Latticra to openSUSE.

It does not claim official openSUSE package status, SUSE endorsement, Open Build Service publication, product readiness, operating-system readiness, runtime authority, kernel authority, boot authority, service authority, policy readiness, or security-hardening completion.

Next Slice

Completed follow-on local RPM build gate contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
scripts/test-opensuse-local-rpm-build-gate-contract.sh
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
```

Completed follow-on local RPM build environment contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
scripts/test-opensuse-local-rpm-build-environment-contract.sh
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
```

Completed follow-on RPM artifact naming contract:

```
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
scripts/test-opensuse-rpm-artifact-naming-contract.sh
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
```

Completed follow-on RPM payload inspection contract:

```
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-opensuse-rpm-payload-inspection-contract.sh
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
```

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-rpm-topdir-handoff-lane.sh
```

Expected output:

```
opensuse_rpm_topdir_handoff_lane: ok
```

docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md

Source title: openSUSE rpmlint Findings Classification

Lines: 236 | Words: 510 | SHA-256: 7863688f4bd9bc2e656a96815177467f1cc72e0da5ae050f20db48474cb7ad00

openSUSE rpmlint Findings Classification

Status: active findings classification record Scope: classify expected and unexpected rpmlint findings for the local-only openSUSE spec draft without promoting package-readiness claims.

Purpose

This record defines how rpmlint output from the openSUSE static spec lane should be interpreted while packaging/opensuse/latticra.spec remains a local-only draft.

The goal is to separate expected draft findings from findings that must block package promotion.

This record is classification only. It does not require a clean rpmlint result, run rpmbuild, run osc build, create package artifacts, publish to Open Build Service, submit Latticra to openSUSE, install Latticra, or claim package readiness.

Inputs

The classification applies to:

```
packaging/opensuse/latticra.spec
packaging/opensuse/latticra.changes
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpmlint-static-spec-lane.sh
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
```

Expected Draft Finding Classes

These finding classes are expected while the package remains local-only:

```
placeholder_version_or_release
local_only_release_marker
placeholder_license_expression
missing_real_source_archive
limited_payload_surface
draft_doc_payload_mapping
local_only_changes_record
```

Expected draft findings do not unblock package readiness. They only explain why a local-only draft may produce rpmlint output before source archive, license, and maintenance evidence mature.

Unexpected Finding Classes

These finding classes must be treated as blockers until separately reviewed:

```
service_installation
systemd_mutation
kernel_module_payload
boot_path_payload
etc_path_payload
privileged_helper_payload
network_fetch_during_build
host_mutation_during_build
unreviewed_scriptlet
unowned_or_unsafe_file_path
license_claim_promotion
opensuse_official_status_claim
obs_publication_claim
suse_endorsement_claim
```

Unexpected findings do not automatically mean the package is unsafe; they mean the local-only draft has crossed a boundary that needs a separate review lane before promotion.

Classification Output Shape

A future accepted rpmlint findings transcript should record:

```
rpmlint_invocation
rpmlint_exit_status
rpmlint_output_present
expected_draft_findings_count
unexpected_findings_count
package_artifact_created
osc_build_run
obs_publication_claimed
package_readiness_claimed
classification_decision
```

Current classification baseline:

```
opensuse_rpmlint_findings_classification_present=1
opensuse_rpmlint_static_spec_lane_present=1
expected_draft_finding_classes_recorded=1
unexpected_finding_classes_recorded=1
accepted_rpmlint_transcript_present=0
expected_draft_findings_count_recorded=0
unexpected_findings_count_recorded=0
classification_decision=blocked-pending-reviewed-rpmlint-output
package_artifact_created=0
osc_build_run=0
obs_publication_claimed=0
package_readiness_claimed=0
```

Promotion Rule

Package promotion remains blocked unless all of these are true:

```
accepted_rpmlint_transcript_present=1
unexpected_findings_count=0
license_expression_reviewed=1
source_archive_reproducible=1
osc_build_transcript_present=1
install_remove_transcript_present=1
obs_publication_claimed=0
opensuse_official_package_claimed=0
suse_endorsement_claimed=0
```

Until those conditions are met, rpmlint output is evidence for maintenance review only, not package readiness.

Boundary

This record does not:

```
run rpmlint
run rpmbuild
run osc build
run spec-cleaner
create package artifacts
publish to Open Build Service
submit Latticra to openSUSE
install Latticra on a host
claim official openSUSE package status
claim SUSE endorsement
claim production readiness
```

Next Slice

Completed follow-on source archive contract:

```
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
scripts/test-opensuse-source-archive-reproducibility-contract.sh
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
```

Completed follow-on source archive fixture:

```
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
scripts/test-opensuse-source-archive-fixture-lane.sh
.github/workflows/opensuse-source-archive-fixture-lane.yml
```

Completed follow-on RPM topdir handoff lane:

```
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
```

Completed follow-on local RPM build gate contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
scripts/test-opensuse-local-rpm-build-gate-contract.sh
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
```

Completed follow-on local RPM build environment contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
scripts/test-opensuse-local-rpm-build-environment-contract.sh
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
```

Completed follow-on RPM artifact naming contract:

```
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
scripts/test-opensuse-rpm-artifact-naming-contract.sh
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
```

Completed follow-on RPM payload inspection contract:

```
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-opensuse-rpm-payload-inspection-contract.sh
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
```

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-rpmlint-findings-classification.sh
```

Expected output:

```
opensuse_rpmlint_findings_classification: ok
```

docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md

Source title: openSUSE rpmlint and osc Availability Lane

Lines: 102 | Words: 272 | SHA-256: fcd13916e733ca7799e85917998ff018554c02452c31c8718114ca91445070a6

openSUSE rpmlint and osc Availability Lane

Status: active tool availability lane Scope: verify that rpmlint and osc are available in an openSUSE Linux environment.

Purpose

This lane prepares the next openSUSE packaging validation step without linting, building, publishing, or installing the Latticra package draft.

The current goal is only to prove that rpmlint and osc can be installed and invoked inside the openSUSE compatibility environment.

Files

```
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md
scripts/test-opensuse-rpmlint-osc-availability.sh
.github/workflows/opensuse-rpmlint-osc-availability.yml
```

Checks

This lane checks:

```
openSUSE environment marker exists
zypper is available
rpmlint installs
rpmlint command is available
rpmlint invocation probe can run
osc installs
osc command is available
osc invocation probe can run
local openSUSE RPM static validation lane remains green
```

Boundary

This lane does not lint the Latticra openSUSE spec yet.

It does not run rpmbuild, osc build, spec-cleaner, or zypper install against a Latticra artifact.

It does not create package artifacts, create an Open Build Service project, publish to Open Build Service, submit Latticra to openSUSE, install on a host, or claim package readiness.

Next Slice

Completed follow-on static spec lane:

```
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpmlint-static-spec-lane.sh
scripts/test-opensuse-rpmlint-findings-classification.sh
```

```
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpmlint-static-spec-lane.yml
.github/workflows/opensuse-rpmlint-findings-classification.yml
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run inside openSUSE:

```
sh scripts/test-opensuse-rpmlint-osc-availability.sh
```

Expected output:

```
opensuse_rpmlint_osc_availability: ok
```

docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md

Source title: openSUSE rpmlint Static Spec Lane

Lines: 155 | Words: 414 | SHA-256: cd06116bff08d681b14b2a29677e7b47a45db63a1db021402424ccf81ece5334

openSUSE rpmlint Static Spec Lane

Status: active static spec lint lane Scope: run rpmlint against the local-only openSUSE spec draft and capture findings without promoting package-readiness claims.

Purpose

This lane advances the openSUSE local RPM validation stack from tool availability to a static spec lint pass.

The current goal is conservative: prove that rpmlint can inspect the local draft spec file in an openSUSE environment and produce a reportable result.

This lane does not require a clean rpmlint result yet because the spec remains a local-only draft and still has source-archive and package-readiness metadata to review.

Files

```
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpmlint-static-spec-lane.sh
scripts/test-opensuse-rpmlint-findings-classification.sh
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpmlint-static-spec-lane.yml
.github/workflows/opensuse-rpmlint-findings-classification.yml
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
packaging/opensuse/latticra.spec
```

Checks

This lane checks:

```
openSUSE environment marker exists
zypper is available
rpmlint installs
rpmlint command is available
rpmlint can inspect packaging/opensuse/latticra.spec
rpmlint output is captured for audit
previous rpmlint and osc availability lane remains green
```

Expected Draft Findings

The current local-only spec may still report findings related to draft metadata, including:

```
local-only release marker
placeholder version
AGPL-3.0-or-later AND CC-BY-4.0 package license expression
missing source archive for a real package build
limited installed file set
openSUSE maintenance metadata still local-only
```

Those findings are expected at this stage and are not package-readiness failures yet.

The purpose of this lane is visibility, not package promotion.

Boundary

This lane does not run rpmbuild.

It does not run osc build.

It does not run spec-cleaner.

It does not create package artifacts.

It does not create an Open Build Service project, publish to Open Build Service, submit Latticra to openSUSE, install on a host, or claim package readiness.

It does not claim official openSUSE package status, SUSE endorsement, product readiness, operating-system readiness, runtime authority, kernel authority, or security-hardening completion.

Next Slice

Completed follow-on classification lane:

```
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
scripts/test-opensuse-rpmlint-findings-classification.sh
.github/workflows/opensuse-rpmlint-findings-classification.yml
```

Completed follow-on source archive lanes:

```
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run inside openSUSE:

```
sh scripts/test-opensuse-rpmlint-static-spec-lane.sh
```

Expected output:

```
opensuse_rpmlint_static_spec_lane: ok
```

docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md

Source title: openSUSE Source Archive Fixture Lane

Lines: 237 | Words: 614 | SHA-256: 7d4f8eb6e745fab956e669eeb0a75ef0df8dbdfc84eedcd33eec1bc726bbbb57

openSUSE Source Archive Fixture Lane

Status: active fixture lane Scope: create and inspect a temporary openSUSE source archive fixture without running package builds, installing artifacts, or publishing to Open Build Service.

Purpose

This lane advances the openSUSE packaging path from a reproducibility contract into a concrete source archive shape check.

The goal is conservative: prove that Latticra can form a temporary source archive matching the current Source0 and %autosetup expectations in packaging/opensuse/latticra.spec, generate it twice, and confirm the repeated SHA-256 value matches.

This lane does not build a source RPM or binary RPM. It only creates disposable archive fixtures and inspects their contents.

Current Spec Relationship

The current local-only openSUSE spec expects:

```
Source0:        %{name}-%{version}.tar.gz
%autosetup -n %{name}-%{version}
```

For the current local draft, that resolves to:

```
latticra-0.0.0.tar.gz
latticra-0.0.0/
```

Fixture Rules

The archive fixture must:

```
use the Name field from packaging/opensuse/latticra.spec
use the Version field from packaging/opensuse/latticra.spec
create a root directory named latticra-&lt;version&gt;/
create an archive named latticra-&lt;version&gt;.tar.gz
generate the archive fixture twice
record matching SHA-256 values for both fixture archives
include README.md
include packaging/opensuse/latticra.spec
include packaging/opensuse/latticra.changes
include scripts used by the current %build section
include src/latticra_cli.c
exclude .git
exclude temporary RPM work directories
exclude build outputs
exclude RPM artifacts
exclude nested source archive artifacts
reject symlink entries before archiving
remain temporary and uncommitted
```

Inspection Checks

The lane inspects the archive with tar -tzf and verifies:

```
latticra-&lt;version&gt;/README.md
latticra-&lt;version&gt;/packaging/opensuse/latticra.spec
latticra-&lt;version&gt;/packaging/opensuse/latticra.changes
latticra-&lt;version&gt;/scripts/test-state-lattice.sh
latticra-&lt;version&gt;/scripts/test-system-bootstrap.sh
latticra-&lt;version&gt;/scripts/test-kernel.sh
latticra-&lt;version&gt;/scripts/test-kernel-lifecycle.sh
latticra-&lt;version&gt;/scripts/test-latticra-no-effect-cli-status-surface.sh
latticra-&lt;version&gt;/src/latticra_cli.c
```

The lane also fails if the archive contains:

```
.git/
.rpmwork/
build/
target/
*.rpm
nested latticra-*.tar.gz
symlink entries
```

Current Evidence

```
opensuse_source_archive_fixture_lane_present=1
opensuse_source_archive_reproducibility_contract_present=1
source_archive_transcript_present=1
source_archive_created=1
source_archive_sha256_recorded=1
source_archive_reproducible=1
source_archive_generated_twice=1
source_archive_repeated_sha256_match=1
source_archive_contains_spec=1
source_archive_contains_changes=1
source_archive_contains_readme=1
source_archive_excludes_git_dir=1
source_archive_excludes_nested_archives=1
source_archive_excludes_build_outputs=1
source_archive_excludes_rpm_artifacts=1
source_archive_symlink_policy_checked=1
source_archive_path_safety_checked=1
source_archive_accepted_for_build=0
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_local_rpm_build_gate_contract_present=1
opensuse_local_rpm_build_environment_contract_present=1
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
rpmbuild_run=0
osc_build_run=0
rpm_artifact_created=0
obs_publication_claimed=0
package_readiness_claimed=0
```

The source archive evidence is a temporary fixture transcript only. It does not promote the archive to an accepted package-build input.

Relationship To Later RPM Work

This fixture lane comes before RPM build execution.

The intended order is:

1. source archive reproducibility contract
2. source archive fixture lane
3. temporary RPM topdir handoff lane
4. local openSUSE RPM build lane
5. RPM payload inspection lane
6. install/remove behavior transcript lane
7. Open Build Service review and publication evidence

Boundary

This lane does not run rpmbuild.

It does not run osc build.

It does not run rpmlint.

It does not run spec-cleaner.

It does not create source RPM artifacts.

It does not create binary RPM artifacts.

It does not install Latticra.

It does not publish package artifacts.

It does not create an Open Build Service project.

It does not submit Latticra to openSUSE.

It does not claim official openSUSE package status, SUSE endorsement, Open Build Service publication, product readiness, operating-system readiness, runtime authority, kernel authority, boot authority, service authority, policy readiness, or security-hardening completion.

Next Slice

Completed follow-on RPM topdir handoff lane:

```
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
```

Completed follow-on local RPM build gate contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
scripts/test-opensuse-local-rpm-build-gate-contract.sh
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
```

Completed follow-on local RPM build environment contract:

```
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
scripts/test-opensuse-local-rpm-build-environment-contract.sh
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
```

Completed follow-on RPM artifact naming contract:

```
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
scripts/test-opensuse-rpm-artifact-naming-contract.sh
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
```

Completed follow-on RPM payload inspection contract:

```
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
scripts/test-opensuse-rpm-payload-inspection-contract.sh
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
```

Completed follow-on RPM install/remove transcript contract:

```
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
```

Completed follow-on OBS publication non-claim review contract:

```
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

That future lane should tie source, environment, artifact, payload, install/remove, and OBS non-claim records together while keeping RPM builds and readiness blocked.

Validation

Run:

```
sh scripts/test-opensuse-source-archive-fixture-lane.sh
```

Expected output:

```
opensuse_source_archive_fixture_lane: ok
```

docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md

Source title: openSUSE Source Archive Reproducibility Contract

Lines: 185 | Words: 430 | SHA-256: 3a77110a4d35d0eab74edfba4e743561a8d8550f2be8818deed7872e164736e7

openSUSE Source Archive Reproducibility Contract

Status: active source archive reproducibility contract Scope: define the evidence required before an openSUSE source archive can be accepted for package build evidence.

Purpose

This contract defines the source archive evidence required before Latticra accepts any openSUSE package build transcript.

The current goal is conservative: record the expected Source0 archive shape, reproducibility requirements, transcript fields, and promotion blockers without creating an archive or running a package build.

This contract sits after the rpmlint findings classification record and before the source archive fixture, rpmbuild, osc build, or Open Build Service publication evidence.

Current Spec Inputs

The current local-only openSUSE spec declares:

```
Name:           latticra
Version:        0.0.0
Source0:        %{name}-%{version}.tar.gz
%autosetup -n %{name}-%{version}
```

For the current draft, the expected source archive shape is:

```
source_archive_name=latticra-0.0.0.tar.gz
source_archive_root=latticra-0.0.0/
source_archive_matches_source0=1
source_archive_matches_autosetup_root=1
```

Required Future Evidence

A future accepted source archive transcript must record:

```
source_tree_revision
source_archive_command
source_archive_name
source_archive_root
source_archive_size_bytes
source_archive_sha256
source_archive_entry_count
source_archive_generated_twice
source_archive_repeated_sha256_match
source_archive_contains_spec
source_archive_contains_changes
source_archive_contains_readme
source_archive_excludes_git_dir
source_archive_excludes_nested_archives
source_archive_excludes_build_outputs
source_archive_excludes_rpm_artifacts
source_archive_symlink_policy_checked
source_archive_path_safety_checked
```

Acceptance Rule

An openSUSE source archive remains unaccepted unless all of these are true:

```
source_archive_transcript_present=1
source_archive_name_matches_source0=1
source_archive_root_matches_autosetup=1
source_archive_sha256_recorded=1
source_archive_generated_twice=1
source_archive_repeated_sha256_match=1
source_archive_excludes_git_dir=1
source_archive_excludes_nested_archives=1
source_archive_excludes_build_outputs=1
source_archive_excludes_rpm_artifacts=1
source_archive_symlink_policy_checked=1
source_archive_path_safety_checked=1
license_expression_reviewed=1
package_notice_obligations_reviewed=1
```

Until those conditions are met, source archive evidence is a review input only, not package build readiness.

Current Baseline

The current baseline includes the follow-on temporary fixture lane. The archive remains unaccepted for package build input until the remaining acceptance requirements are met.

```
opensuse_source_archive_reproducibility_contract_present=1
opensuse_rpmlint_findings_classification_present=1
opensuse_source_archive_fixture_lane_present=1
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_local_rpm_build_gate_contract_present=1
opensuse_local_rpm_build_environment_contract_present=1
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
source_archive_policy_recorded=1
source_archive_name_expected=latticra-0.0.0.tar.gz
source_archive_root_expected=latticra-0.0.0/
source_archive_matches_source0_required=1
source_archive_matches_autosetup_required=1
source_archive_transcript_present=1
source_archive_created=1
source_archive_sha256_recorded=1
source_archive_reproducible=1
source_archive_accepted_for_build=0
rpmbuild_run=0
osc_build_run=0
rpm_artifact_created=0
obs_publication_claimed=0
package_readiness_claimed=0
```

Boundary

This contract does not:

```
create a source archive
run tar
run gzip
run rpmbuild
run osc build
run spec-cleaner
create source RPM artifacts
create binary RPM artifacts
publish to Open Build Service
submit Latticra to openSUSE
install Latticra on a host
claim official openSUSE package status
claim SUSE endorsement
claim production readiness
```

Follow-On Handoff Lane

Completed follow-on fixture and handoff lanes:

```
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

Those lanes prove archive shape, reproducibility, temporary RPM topdir staging, build-gate closure, environment requirements, future artifact naming boundaries, payload inspection evidence requirements, install/remove transcript requirements, and OBS non-claim review while keeping package build and publication claims blocked.

Recommended next slice:

```
Add openSUSE RPM validation promotion blocker matrix before any package validation result can be
 accepted.
```

Validation

Run:

```
sh scripts/test-opensuse-source-archive-reproducibility-contract.sh
```

Expected output:

```
opensuse_source_archive_reproducibility_contract: ok
```

docs/strategy/2026-05-26-1959-cdt-console-read-only-host-inventory-review-package-index.md

Source title: Latticra Console Read-Only Host Inventory Review Package Index

Lines: 186 | Words: 703 | SHA-256: 85eea2eeeed0f7010189b73db011a702e6905ae4d39c4c21ce8121d84e7e8dfe

Latticra Console Read-Only Host Inventory Review Package Index

Status: active review-package index Created: 2026-05-26 19:59 CDT Decision: package index only Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: strategy package map for Console read-only host-inventory planning records.

Purpose

This index organizes the Console read-only host-inventory planning package.

It shows the intended reading order, dependency order, and review gates for future evidence work. It does not capture evidence, perform reviews, implement tests, run Console, inspect a host, or change project posture.

Package goal

The package goal is:

```
define how Console read-only host-inventory evidence could eventually be reviewed
as a contract-only technical-evaluator workflow without performing live host
inventory, enabling a host adapter, granting runtime authority, or making
product-readiness, security, or OS-base claims
```

Package order

Read and apply the records in this order:

1. 2026-05-26-1702-cdt-overall-strategy-priority-map.md
2. 2026-05-26-1707-cdt-console-read-only-host-inventory-workflow-packet.md
3. 2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md
4. 2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md
5. 2026-05-26-1757-cdt-console-read-only-host-inventory-non-claim-review-template.md
6. 2026-05-26-1919-cdt-console-read-only-host-inventory-public-entrypoint-review-template.md

7. 2026-05-26-1956-cdt-console-read-only-host-inventory-estimate-impact-review-template.md

Dependency map

```
overall strategy priority map
  -&gt; workflow packet
    -&gt; acceptance checklist
      -&gt; evidence bundle template
        -&gt; non-claim review
          -&gt; public-entrypoint review
            -&gt; estimate-impact review
```

The evidence bundle, non-claim review, public-entrypoint review, and estimate-impact review are all required before any estimate or public-entrypoint change can be considered.

Record responsibilities

```
overall strategy priority map
  Selects the Console read-only host-inventory planning package as the Tier 1 next lane.

workflow packet
  Selects the contract-only Console host-inventory evaluator workflow and rejects live inventory
 capture.

acceptance checklist
  Defines what future evidence must satisfy before it can be reviewable.

evidence bundle template
  Defines the future evidence bundle shape.

non-claim review template
  Defines how to prevent host, runtime, security, receipt, signing, product, or OS-base claims.

public-entrypoint review template
  Defines when README, status, project notes, or strategy index references may be justified.

estimate-impact review template
  Defines when completion estimate changes may be considered and why the default is no estimate
change.
```

Current package status

Current status:

```
planning_package_defined=1
workflow_selected=1
acceptance_criteria_defined=1
evidence_bundle_shape_defined=1
non_claim_review_shape_defined=1
public_entrypoint_review_shape_defined=1
estimate_impact_review_shape_defined=1
workflow_evidence_captured=0
live_host_inventory_performed=0
host_probe_performed=0
host_file_read_performed=0
host_file_write_performed=0
host_mutation_performed=0
network_scan_performed=0
host_adapter_enabled=0
receipt_materialized=0
receipt_signed=0
non_claim_review_performed=0
public_entrypoint_review_performed=0
estimate_impact_review_performed=0
estimate_change_recommended=0
product_readiness_promotion_recommended=0
```

Required future promotion sequence

Any future promotion review must proceed in this order:

1. Capture evidence only after a separate explicit evidence-capture request.
2. Complete the evidence bundle.
3. Complete the non-claim review.

4. Complete the public-entrypoint review.
5. Complete the estimate-impact review.
6. Decide whether public posture changes are justified.

Do not change estimates or public posture before step 6.

Stop conditions

Stop the package review if any of these appear:

```
live host inventory performed
host probing performed
host file read performed
host file write performed
host mutation performed
network scan performed
host adapter enabled
runtime enforcement claimed
boot behavior claimed
receipt signed
signing authority claimed
cryptographic authority claimed
non-claim review missing
public-entrypoint review missing
estimate-impact review missing
product-readiness claim introduced
security boundary claim introduced
security-hardening claim introduced
OS-base behavior claim introduced
```

Strategy index rule

The strategy index may link to planning records as they are created.

Index linking does not imply:

```
workflow evidence captured
workflow accepted
estimate changed
public posture changed
product readiness changed
runtime authority changed
host inventory performed
host adapter enabled
security posture changed
receipt signed
```

Current next planning action

The next planning action after this index is:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the completion checkpoint closes this package, and the overall priority map
names runtime and Nucleus effect-boundary planning as the next planning priority
after public evaluator workflow clarity
```

Evidence capture should remain a separate explicit request because it may involve running commands, collecting transcripts, or producing review records.

Non-claims

This index does not capture evidence, perform a review, validate workflow evidence, run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, update estimates, update public entry points, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a strategy package map only.

docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md

Source title: Ubuntu Debian Copyright Notice Mapping Contract

Lines: 98 | Words: 335 | SHA-256: e6d2d70ec75b2f1c4498c0e36d3a5ce39b63aaae3b2b3bc8da7c676910e44329

Ubuntu Debian Copyright Notice Mapping Contract

Status: no-effect Debian copyright notice mapping contract Scope: define Debian copyright notice-mapping evidence required before the Ubuntu package notice review can be promoted.

Purpose

This contract turns the Ubuntu local deb packaging/ubuntu/debian/copyright notice-mapping blocker into a concrete review checklist.

It records the updated packaging/ubuntu/debian/copyright license mapping, but it does not create a NOTICE file, build a package, publish a package, submit to Ubuntu, or provide legal advice.

Current Inputs

```
package_scope=local-deb-draft
debian_copyright_file=packaging/ubuntu/debian/copyright
debian_copyright_present=1
debian_copyright_format_url_present=1
debian_copyright_agpl_mapping_present=1
debian_copyright_cc_by_mapping_present=1
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
notice_file_present=0
ubuntu_package_notice_inventory_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

The current Debian copyright file is a local-only draft. It records Apache-2.0 as the existing-code default, AGPL-3.0-or-later for the no-effect CLI payload, and CC-BY-4.0 for documentation while third-party material, generated-artifact notices, NOTICE-file requirements, and broader notice mappings remain under review.

Required Mapping Before Promotion

```
debian_copyright_notice_mapping_reviewed=1
debian_copyright_binary_payload_mapping_reviewed=1
debian_copyright_doc_payload_mapping_reviewed=1
debian_copyright_third_party_notice_mapping_reviewed=1
debian_copyright_generated_artifact_notice_mapping_reviewed=1
debian_copyright_notice_file_mapping_reviewed=1
debian_copyright_trademark_notice_boundary_reviewed=1
debian_copyright_license_ref_replaced_or_justified=1
debian_copyright_missing_notice_entries=0
```

Current Decision

```
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
debian_copyright_notice_mapping_reviewed=0
debian_copyright_binary_payload_mapping_reviewed=0
debian_copyright_doc_payload_mapping_reviewed=1
debian_copyright_third_party_notice_mapping_reviewed=0
debian_copyright_generated_artifact_notice_mapping_reviewed=0
debian_copyright_notice_file_mapping_reviewed=0
debian_copyright_trademark_notice_boundary_reviewed=0
debian_copyright_license_ref_replaced_or_justified=1
debian_copyright_missing_notice_entries=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until the local deb payload has a reviewed Debian copyright notice mapping. That review must explain how binary payload, documentation payload, third-party material, generated artifacts, NOTICE-file requirements, and trademark notice boundaries are represented in packaging/ubuntu/debian/copyright.

The trademark notice boundary remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The release artifact notice requirements remain separate, but they must also be resolved before the Ubuntu package notice review can be promoted.

This contract is scoped to the Ubuntu local deb draft. It is not a repository-wide license audit or a Debian Policy compliance claim.

Non-Claims

This contract is not legal advice. It does not claim NOTICE compliance, license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-debian-copyright-notice-mapping-contract.sh
```

Expected output:

```
ubuntu_debian_copyright_notice_mapping_contract: ok
```

docs/UBUNTU_DEVELOPER_WORKFLOW.md

Source title: Ubuntu Developer Workflow

Lines: 121 | Words: 428 | SHA-256: eb9a020836af6c803e1ae60428fccbc349dc4292b466d89336e3265abf402c22

Ubuntu Developer Workflow

Status: developer workflow record Scope: local Ubuntu Linux commands for productive Latticra development and user-local Panel work.

Purpose

This note gives a repeatable local workflow for working on Latticra from Ubuntu Linux after the Ubuntu build lane was added.

The goal is simple:

```
install the small toolchain
install the Panel desktop prerequisites
run the same guards as CI
keep local validation fast and repeatable
avoid archive or PPA claims until the deb lane has real evidence
```

Install Local Tools

On an Ubuntu workstation or Ubuntu container, install the basic guard toolchain:

```
sudo apt-get update
sudo apt-get install -y git build-essential make gcc pkg-config coreutils findutils diffutils
grep
```

In a root container, omit sudo:

```
apt-get update
apt-get install -y git build-essential make gcc pkg-config coreutils findutils diffutils grep
```

Install Panel Prerequisites

For the Rust/egui Latticra Panel and desktop metadata refresh path, install:

```
sudo apt-get update
sudo apt-get install -y rustc cargo make gcc pkg-config \
  libx11-dev libxcb1-dev libxcursor-dev libxrandr-dev libxi-dev \
  libxkbcommon-dev libgl1-mesa-dev libwayland-dev desktop-file-utils \
  libgtk-3-bin
```

If the distro Rust toolchain is too old for the locked Cargo dependency graph, use a current stable Rust toolchain through the operator's approved toolchain source before running Panel commands. The repository does not grant network authority or install Rust by itself.

Fast Local Guard Loop

For a quick sanity check while editing C/kernel files, run:

```
sh scripts/test-state-lattice.sh
sh scripts/test-system-bootstrap.sh
sh scripts/test-kernel.sh
sh scripts/test-kernel-lifecycle.sh
sh scripts/test-latticra-no-effect-cli-status-surface.sh
```

Ubuntu Lane Guard

Before merging Ubuntu-facing work, run this on Ubuntu:

```
sh scripts/test-ubuntu-build-lane.sh
```

Expected output:

```
ubuntu_build_lane: ok
```

Panel From Source

After the prerequisites are present:

```
export PATH=&quot;$HOME/.local/bin:$PATH&quot;
make -C installer gui
```

The guarded user-local install remains the same on Ubuntu:

```
make -C installer dry-run
make -C installer local-example
make -C installer verify-local
```

Suggested Work Rhythm

Use this loop for productive development:

1. Create a small branch.
2. Make one bounded change.
3. Run the fast local guard loop.
4. Run the Ubuntu lane guard when the branch touches build, installer, packaging, or Ubuntu-facing docs.
5. Keep package language local-only until real lint/build/install evidence exists.

Current Boundary

This workflow is for local developer productivity and user-local Panel work. It does not publish a deb, create a PPA, submit to Ubuntu, claim Ubuntu archive readiness, install a root service, modify systemd, modify the kernel, or claim production readiness.

Validation

This workflow note is guarded by:

```
sh scripts/test-ubuntu-developer-workflow.sh
```

Expected output:

```
ubuntu_developer_workflow: ok
```

docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md

Source title: Ubuntu Doc Payload License Review Contract

Lines: 106 | Words: 356 | SHA-256: 27392bd4ab82356f78a43606b9fd41e8cb4e3f8877464ca47a3e0a82e9a7c265

Ubuntu Doc Payload License Review Contract

Status: no-effect documentation-license review contract Scope: define the review evidence required before README.md can be treated as a resolved Ubuntu local deb documentation payload.

Purpose

This contract records the Ubuntu local deb documentation-license decision and its package mapping checklist.

It does not publish a package, create a PPA, submit to Ubuntu, or provide legal advice.

Current Inputs

```
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
doc_payload_source_present=1
root_license_file=LICENSE
root_license_current=hybrid-license-overview
license_policy_present=1
license_migration_plan_present=1
documentation_license_decision_present=1
ubuntu_package_notice_inventory_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

The current license migration plan records documentation licensing as decided.

Current package documentation license:

```
candidate_doc_payload_license=CC-BY-4.0
candidate_doc_payload_license_applied_to_packaging=1
```

Required Review Before Promotion

```
doc_payload_license_decision_recorded=1
doc_payload_license_expression_recorded=1
doc_payload_license_compatible_with_package=1
readme_embedded_material_reviewed=1
readme_generated_artifact_notice_reviewed=1
documentation_trademark_boundary_reviewed=1
documentation_notice_requirements_recorded=1
debian_copyright_doc_payload_mapping_reviewed=1
```

Current Decision

```
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
doc_payload_license_reviewed=1
doc_payload_license_unresolved=0
documentation_license_decision_present=1
doc_payload_license_decision_recorded=1
doc_payload_license_compatible_with_package=1
debian_copyright_doc_payload_mapping_reviewed=1
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until this documentation payload license review is resolved and mapped into Debian copyright metadata.

The third-party material review remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The generated-artifact notice review remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The NOTICE file decision remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The Debian copyright notice mapping remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The trademark notice boundary remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The release artifact notice requirements remain separate, but they must also be resolved before the Ubuntu package notice review can be promoted.

This contract is a guardrail for the current README.md package payload. It is not a general documentation relicensing decision for the whole repository.

Non-Claims

This contract is not legal advice. It does not claim license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-doc-payload-license-review-contract.sh
```

Expected output:

```
ubuntu_doc_payload_license_review_contract: ok
```

docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md

Source title: Ubuntu Generated-Artifact Notice Review Contract

Lines: 104 | Words: 352 | SHA-256: 6812a8f951795ecbce6c4f3179fb59c74469d44c8bafa810712dd3ee651ded1b

Ubuntu Generated-Artifact Notice Review Contract

Status: no-effect generated-artifact notice review contract Scope: define generated-artifact notice evidence required before the Ubuntu local deb notice review can be promoted.

Purpose

This contract turns the Ubuntu local deb generated-artifact notice blocker into a concrete review checklist.

It does not build artifacts, inspect a built deb, decide notice obligations, create a NOTICE file, update packaging/ubuntu/debian/copyright, publish a package, submit to Ubuntu, or provide legal advice.

Current Inputs

```
package_scope=local-deb-draft
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
generated_artifact_notice_policy_present=1
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
deb_artifact_created=0
package_build_performed=0
package_publish_performed=0
```

The current Ubuntu local deb draft has packaging metadata only. No deb artifact, changes file, build log, or installed package payload has been produced for notice review.

Required Review Before Promotion

```
generated_artifact_notice_reviewed=1
generated_artifact_notice_requirements_recorded=1
binary_payload_generation_path_reviewed=1
doc_payload_generation_path_reviewed=1
deb_artifact_notice_requirements_recorded=1
changes_file_notice_requirements_recorded=1
build_log_notice_requirements_recorded=1
installed_payload_notice_requirements_recorded=1
generated_artifact_missing_notice_entries=0
```

Current Decision

```
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
generated_artifact_notice_reviewed=0
generated_artifact_notice_requirements_recorded=0
binary_payload_generation_path_reviewed=0
doc_payload_generation_path_reviewed=0
deb_artifact_notice_requirements_recorded=0
changes_file_notice_requirements_recorded=0
build_log_notice_requirements_recorded=0
installed_payload_notice_requirements_recorded=0
generated_artifact_missing_notice_entries=0
deb_artifact_created=0
package_build_performed=0
package_publish_performed=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until generated-artifact notice requirements are reviewed for the current local deb payload and any future build artifacts.

The NOTICE file decision remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The Debian copyright notice mapping remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The trademark notice boundary remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The release artifact notice requirements remain separate, but they must also be resolved before the Ubuntu package notice review can be promoted.

This contract is scoped to the Ubuntu local deb draft. It is not evidence that an artifact exists, that an artifact is distributable, or that release notices are complete.

Non-Claims

This contract is not legal advice. It does not claim generated-artifact notice compliance, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-generated-artifact-notice-review-contract.sh
```

Expected output:

```
ubuntu_generated_artifact_notice_review_contract: ok
```

docs/UBUNTU_LINTIAN_AVAILABILITY.md

Source title: Ubuntu lintian Availability Lane

Lines: 67 | Words: 237 | SHA-256: f4c77421dec6b0ae64a7494210423af5290467cd31e234f761c70aa6f7d1378b

Ubuntu lintian Availability Lane

Status: active tool availability lane Scope: verify that lintian is available in an Ubuntu Linux environment.

Purpose

This lane prepares the next packaging validation step without linting or building the Latticra package draft yet.

The current goal is only to prove that lintian can be installed and invoked inside the Ubuntu compatibility environment.

Files

```
docs/UBUNTU_LINTIAN_AVAILABILITY.md
scripts/test-ubuntu-lintian-availability.sh
.github/workflows/ubuntu-lintian-availability.yml
docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md
```

Checks

This lane checks:

```
Ubuntu environment marker exists
apt-get is available
lintian installs
lintian command is available
lintian version command can run
local deb static validation lane remains green
```

Boundary

This lane does not lint the Latticra deb draft yet.

It does not run dpkg-buildpackage, debuild, sbuild, or pbuilder.

It does not create package artifacts.

It does not submit Latticra to Ubuntu, create a PPA, or claim Ubuntu archive readiness.

Next Slice

Recommended next slice:

```
Review the Ubuntu lintian static metadata contract after package license promotion is unblocked.
Keep lintian execution, lintian finding classification, build transcript evidence, and deb
artifact creation blocked until package notice and license prerequisites are reviewed.
```

That future lane may run lintian against the local debian metadata or a locally built artifact and classify expected draft findings separately from unexpected package findings. The current static metadata contract records the evidence shape without running lintian.

Validation

Run inside Ubuntu:

```
sh scripts/test-ubuntu-lintian-availability.sh
```

Expected output:

```
ubuntu_lintian_availability: ok
```

docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md

Source title: Ubuntu Lintian Static Metadata Contract

Lines: 85 | Words: 211 | SHA-256: 844d3f511f5e29294d7f1749fc7e324a4e604f025f9e27f12ff761d6b18ec852

Ubuntu Lintian Static Metadata Contract

Status: no-effect lintian static metadata contract Scope: define the static lintian metadata evidence required before Ubuntu local deb build transcript evidence can advance.

Purpose

This contract turns the future Ubuntu lintian/static metadata lane into a guarded evidence schema.

It does not run lintian, dpkg-buildpackage, debuild, sbuild, or pbuilder. It does not create, install, publish, or upload a deb package.

Required Inputs

```
ubuntu_lintian_availability_present=1
ubuntu_local_deb_static_validation_present=1
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
```

Required Evidence Before Promotion

```
debian_control_present=1
debian_rules_present=1
debian_changelog_present=1
debian_copyright_present=1
debian_install_present=1
debian_source_format_present=1
rules_requires_root_no=1
license_expression_reviewed=1
license_expression_unresolved=0
packaging_license_expression_updated=1
lintian_command_recorded=1
lintian_static_metadata_findings_classified=1
lintian_expected_draft_findings_classified=1
lintian_unexpected_findings_classified=1
ubuntu_lintian_static_metadata_unblocked=1
```

Current State

```
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_lintian_static_metadata_status=blocked-pending-package-license-promotion
ubuntu_lintian_availability_present=1
ubuntu_local_deb_static_validation_present=1
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
lintian_static_metadata_run=0
lintian_command_recorded=0
lintian_static_metadata_findings_classified=0
lintian_expected_draft_findings_classified=0
lintian_unexpected_findings_classified=0
license_expression_reviewed=1
license_expression_unresolved=0
packaging_license_expression_updated=1
deb_artifact_created=0
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Build Evidence

The Ubuntu local deb build transcript contract may not accept build evidence until this static lintian metadata contract is unblocked and any lintian findings are recorded and classified.

This contract is deliberately static. It records the future evidence shape without running lintian or creating package artifacts.

Non-Claims

This contract does not claim lintian success, Debian Policy compliance, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-lintian-static-metadata-contract.sh
```

Expected output:

```
ubuntu_lintian_static_metadata_contract: ok
```

docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md

Source title: Ubuntu Local Deb Build Transcript Acceptance Gate Contract

Lines: 94 | Words: 266 | SHA-256: 6b76477ff90a7add58cc025a7f26fe22cda908f0174bfcb31f85db24b9ee68a8

Ubuntu Local Deb Build Transcript Acceptance Gate Contract

Status: no-effect local deb build transcript acceptance gate contract Scope: aggregate the evidence required before a future Ubuntu local deb build transcript can be accepted.

Purpose

This contract turns the Ubuntu local deb build transcript acceptance boundary into a single guarded gate.

It does not run dpkg-buildpackage, debuild, lintian, sbuild, or pbuilder. It does not create, install, publish, upload, or accept a deb package.

Required Inputs

```
ubuntu_local_deb_build_transcript_contract_present=1
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_lintian_static_metadata_unblocked=1
ubuntu_local_deb_build_transcript_unblocked=1
```

Required Evidence Before Acceptance

```
ubuntu_local_deb_build_transcript_present=1
transcript_header_reviewed=1
tooling_evidence_reviewed=1
package_evidence_reviewed=1
build_evidence_reviewed=1
payload_evidence_reviewed=1
non_claims_reviewed=1
deb_artifact_digest_recorded=1
changes_file_digest_recorded=1
build_log_digest_recorded=1
lintian_output_digest_recorded=1
expected_draft_findings_classified=1
unexpected_findings_classified=1
build_transcript_acceptance_gate_unblocked=1
local_deb_build_transcript_accepted=1
```

Current Gate State

```
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata
-and-build-transcript
ubuntu_local_deb_build_transcript_present=0
build_transcript_acceptance_gate_open=0
build_transcript_acceptance_gate_unblocked=0
local_deb_build_transcript_accepted=0
transcript_header_reviewed=0
tooling_evidence_reviewed=0
package_evidence_reviewed=0
build_evidence_reviewed=0
payload_evidence_reviewed=0
non_claims_reviewed=0
deb_artifact_created=0
deb_artifact_installed=0
deb_installed_on_host=0
deb_removed_from_host=0
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
ppa_claimed=0
ubuntu_archive_ready=0
production_installer_ready=0
```

Relationship To Build Evidence

The Ubuntu local deb build transcript contract defines the fields a future transcript must contain. This gate records the separate acceptance decision for that transcript after lintian/static metadata and package license promotion have been reviewed.

This contract is intentionally closed today. It is not evidence that a build transcript exists, that a deb artifact exists, or that a package is installable.

The Ubuntu local deb install/remove evidence contract remains separate and blocked until a reviewed local deb build transcript is accepted.

Non-Claims

This contract does not claim build success, lintian success, install success, Debian Policy compliance, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-local-deb-build-transcript-acceptance-gate-contract.sh
```

Expected output:

```
ubuntu_local_deb_build_transcript_acceptance_gate_contract: ok
```

docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_CONTRACT.md

Source title: Ubuntu Local Deb Build Transcript Contract

Lines: 166 | Words: 334 | SHA-256: be3edf6a4af2e3677a32bce69d2ac0751e5dbc5f6fe294d40fc647a7f6194f78

Ubuntu Local Deb Build Transcript Contract

Status: no-effect transcript contract Scope: evidence schema for a future reviewed local Ubuntu deb build transcript.

Purpose

This contract defines the transcript fields required before a future Ubuntu local deb build can be treated as evidence.

It does not run dpkg-buildpackage, debuild, lintian, sbuild, or pbuilder. It does not create, install, publish, or upload a deb package.

Required Transcript Header

```
transcript_kind=ubuntu-local-deb-build
transcript_version=1
host_os_id=ubuntu
host_os_version_recorded=1
kernel_recorded=1
architecture_recorded=1
repo_commit_recorded=1
source_tree_dirty_state_recorded=1
```

Required Tooling Evidence

```
apt_get_available=1
cc_available=1
make_available=1
dpkg_buildpackage_available=1
debuild_available_recorded=1
lintian_available_recorded=1
debhelper_available_recorded=1
```

Required Package Evidence

```
local_deb_draft_present=1
debian_control_present=1
debian_rules_present=1
debian_changelog_present=1
debian_copyright_present=1
debian_install_present=1
rules_requires_root_no=1
license_expression_reviewed=1
license_expression_unresolved=0
```

The current repository state satisfies the package license-expression review, but this contract remains future evidence only because lintian, build, artifact, notice, and install/remove evidence are still blocked.

The current package license review checkpoint is:

```
ubuntu_package_license_review_contract_present=1
ubuntu_package_license_review_status=resolved-license-expression-recorded
ubuntu_package_notice_inventory_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_lintian_static_metadata_status=blocked-pending-package-license-promotion
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata
-and-build-transcript
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_package_notice_review_contract_present=1
ubuntu_package_notice_review_status=blocked-pending-notice-review
license_expression_candidate_recorded=1
packaging_license_expression_updated=1
```

Required Build Evidence

```
dpkg_buildpackage_command_recorded=1
build_exit_status_recorded=1
deb_artifact_path_recorded=1
deb_artifact_digest_recorded=1
changes_file_digest_recorded=1
build_log_digest_recorded=1
lintian_command_recorded=1
lintian_exit_status_recorded=1
lintian_output_digest_recorded=1
expected_draft_findings_classified=1
unexpected_findings_classified=1
```

Required Payload Evidence

```
payload_listing_recorded=1
usr_bin_latticra_present=1
usr_share_doc_latticra_readme_present=1
systemd_service_present=0
kernel_module_present=0
boot_entry_present=0
etc_latticra_present=0
network_authority_present=0
root_installer_claim_present=0
```

Required Non-Claims

```
ppa_claimed=0
ubuntu_archive_ready=0
canonical_endorsement_claimed=0
production_installer_ready=0
daily_driver_install_ready=0
root_installer_ready=0
operating_system_replacement_claimed=0
```

Current Status

```
ubuntu_local_deb_build_transcript_contract_present=1
ubuntu_local_deb_build_transcript_present=0
local_deb_build_transcript_accepted=0
build_transcript_acceptance_gate_unblocked=0
deb_artifact_created=0
deb_artifact_installed=0
deb_removed_from_host=0
license_expression_reviewed=1
license_expression_unresolved=0
ubuntu_lintian_static_metadata_unblocked=0
```

Acceptance Rule

A future transcript may only be accepted if all required fields are present, the license expression has been reviewed, lintian/static metadata is unblocked, expected draft findings are classified, unexpected findings are either absent or explicitly reviewed, the acceptance gate is unblocked, and all non-claims remain zero.

The Ubuntu local deb build transcript acceptance gate records that acceptance decision separately from this evidence schema.

The Ubuntu local deb install/remove evidence contract is downstream of this schema and remains blocked until a reviewed build transcript is accepted.

Validation

Run:

```
sh scripts/test-ubuntu-local-deb-build-transcript-contract.sh
```

Expected output:

```
ubuntu_local_deb_build_transcript_contract: ok
```

docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md

Source title: Ubuntu Local Deb Install Remove Evidence Contract

Lines: 96 | Words: 266 | SHA-256: c539c70afa1120b4f7e5526ff09aeac16c70ebe5d92b8c785403305ae3283988

Ubuntu Local Deb Install Remove Evidence Contract

Status: no-effect local deb install/remove evidence contract Scope: evidence schema for a future reviewed Ubuntu local deb install/remove transcript.

Purpose

This contract defines the evidence required before a future local Ubuntu deb can claim install or remove behavior.

It does not run dpkg -i, apt install, apt remove, dpkg -r, dpkg -P, lintian, dpkg-buildpackage, debuild, sbuild, or pbuilder. It does not create, install, remove, publish, upload, or accept a deb package.

Required Inputs

```
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata
-and-build-transcript
local_deb_build_transcript_accepted=1
deb_artifact_created=1
deb_artifact_digest_recorded=1
```

Required Evidence Before Promotion

```
install_remove_test_environment_recorded=1
install_command_recorded=1
install_exit_status_recorded=1
installed_payload_listing_recorded=1
usr_bin_latticra_installed=1
status_command_after_install_recorded=1
remove_command_recorded=1
remove_exit_status_recorded=1
post_remove_absence_checked=1
residual_payload_reviewed=1
install_remove_findings_classified=1
host_mutation_scope_reviewed=1
ubuntu_install_remove_evidence_unblocked=1
```

Current State

```
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
local_deb_build_transcript_accepted=0
deb_artifact_created=0
deb_artifact_digest_recorded=0
deb_installed_on_host=0
deb_removed_from_host=0
install_remove_test_environment_recorded=0
install_command_recorded=0
install_exit_status_recorded=0
installed_payload_listing_recorded=0
usr_bin_latticra_installed=0
status_command_after_install_recorded=0
remove_command_recorded=0
remove_exit_status_recorded=0
post_remove_absence_checked=0
residual_payload_reviewed=0
install_remove_findings_classified=0
host_mutation_scope_reviewed=0
ubuntu_install_remove_evidence_unblocked=0
ubuntu_ppa_archive_publication_gate_contract_present=1
ubuntu_ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
ubuntu_publication_gate_unblocked=0
ppa_claimed=0
ubuntu_archive_ready=0
production_installer_ready=0
root_installer_ready=0
```

Relationship To Build Evidence

The Ubuntu local deb build transcript acceptance gate must be unblocked and a reviewed local deb build transcript must be accepted before this install/remove evidence contract can advance.

This contract is intentionally closed today. It records the future evidence shape without installing a package, removing a package, or mutating the host.

The Ubuntu PPA/archive publication gate is downstream of this contract and remains blocked until install/remove evidence is unblocked.

Non-Claims

This contract does not claim install success, remove success, package-manager integration, Debian Policy compliance, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, daily-driver readiness, root installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-local-deb-install-remove-evidence-contract.sh
```

Expected output:

```
ubuntu_local_deb_install_remove_evidence_contract: ok
```

docs/UBUNTU_LOCAL_DEB_STATIC_VALIDATION.md

Source title: Ubuntu Local Deb Static Validation

Lines: 169 | Words: 279 | SHA-256: 71f21cb8e3a248ad3fbb6c6854a7dcfb87f2e41bab9df6b176779097c9c03a0f

Ubuntu Local Deb Static Validation

Status: active static validation lane Scope: static checks for the local-only Ubuntu deb packaging draft.

Purpose

This lane checks that the Ubuntu packaging draft is present, narrow, and honest.

It verifies local packaging shape for the no-effect CLI payload, but it does not run dpkg-buildpackage, debuild, lintian, sbuild, or pbuilder, and it does not create package artifacts.

Guarded Files

```
packaging/ubuntu/README.md
packaging/ubuntu/debian/control
packaging/ubuntu/debian/rules
packaging/ubuntu/debian/changelog
packaging/ubuntu/debian/copyright
packaging/ubuntu/debian/install
packaging/ubuntu/debian/source/format
scripts/test-ubuntu-local-deb-static-validation.sh
docs/UBUNTU_LINTIAN_AVAILABILITY.md
docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md
docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md
docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md
docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md
docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md
docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md
docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md
docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md
docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md
```

Preserved Boundary

```
local_only_draft=1
deb_artifact_created=0
dpkg_buildpackage_run_required=0
debuild_run_required=0
lintian_run_required=0
package_notice_inventory_present=1
package_notice_inventory_report_present=1
doc_payload_license_review_contract_present=1
doc_payload_license_review_status=resolved-cc-by-4.0
documentation_license_decision_present=1
doc_payload_license_decision_recorded=1
third_party_material_review_contract_present=1
third_party_material_review_status=blocked-pending-third-party-material-review
third_party_material_inventory_recorded=1
third_party_material_inventory_reviewed=0
third_party_notice_requirements_recorded=0
generated_artifact_notice_review_contract_present=1
generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
generated_artifact_notice_reviewed=0
generated_artifact_notice_requirements_recorded=0
notice_file_decision_contract_present=1
notice_file_decision_status=blocked-pending-notice-file-decision
notice_file_present=0
notice_file_decision_recorded=0
debian_copyright_notice_mapping_contract_present=1
debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
debian_copyright_notice_mapping_reviewed=0
trademark_notice_boundary_contract_present=1
trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
trademark_notice_boundary_recorded=0
release_artifact_notice_requirements_contract_present=1
release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requirements
release_artifact_notice_requirements_recorded=0
package_notice_promotion_gate_contract_present=1
package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
package_notice_promotion_gate_unblocked=0
package_license_promotion_gate_contract_present=1
package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
package_license_promotion_gate_unblocked=0
lintian_static_metadata_contract_present=1
lintian_static_metadata_status=blocked-pending-package-license-promotion
lintian_static_metadata_run=0
local_deb_build_transcript_acceptance_gate_contract_present=1
local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata-and-bu
ild-transcript
local_deb_build_transcript_accepted=0
local_deb_install_remove_evidence_contract_present=1
local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
deb_installed_on_host=0
deb_removed_from_host=0
source_package_evidence_contract_present=1
source_package_evidence_status=blocked-pending-accepted-build-transcript
dpkg_source_run=0
dpkg_buildpackage_source_run=0
source_package_created=0
source_package_uploaded=0
ubuntu_source_package_evidence_unblocked=0
upload_signing_authority_evidence_contract_present=1
upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
upload_target_kind_recorded=0
launchpad_account_recorded=0
ppa_or_archive_target_reviewed=0
gpg_signing_key_fingerprint_recorded=0
upload_command_non_claims_reviewed=0
ubuntu_upload_signing_authority_evidence_unblocked=0
ppa_archive_publication_gate_contract_present=1
ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
launchpad_upload_run=0
ubuntu_publication_gate_unblocked=0
ppa_claimed=0
ubuntu_archive_ready=0
production_readiness_claimed=0
```

Package Intent

The local deb draft records only this initial payload shape:

```
usr/bin/latticra
usr/share/doc/latticra/README.md
```

The CLI remains the no-effect status surface from:

```
src/latticra_cli.c
```

Explicit Non-Claims

This static lane does not:

```
build a deb artifact
install a deb artifact
publish a PPA
submit to Ubuntu
claim Ubuntu archive readiness
install a systemd service
install kernel modules
write /etc/latticra
claim root installer readiness
claim production readiness
```

Next Recommended Lane

Review the Ubuntu source package evidence contract before any source package can be treated as publication input.

Validation

Run:

```
sh scripts/test-ubuntu-local-deb-static-validation.sh
```

Expected output:

```
ubuntu_local_deb_static_validation: ok
```

docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md

Source title: Ubuntu NOTICE File Decision Contract

Lines: 95 | Words: 326 | SHA-256: d40888b06bd4f18debc37617ac9e7e0f9fd73d3ea3c76fe1a90a0b0493c29687

Ubuntu NOTICE File Decision Contract

Status: no-effect NOTICE file decision contract Scope: define NOTICE file decision evidence required before the Ubuntu local deb notice review can be promoted.

Purpose

This contract turns the Ubuntu local deb NOTICE file blocker into a concrete review checklist.

It does not create a NOTICE file, decide that one is required or unnecessary, update packaging/ubuntu/debian/copyright, build a package, publish a package, submit to Ubuntu, or provide legal advice.

Current Inputs

```
package_scope=local-deb-draft
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
notice_file_present=0
license_policy_present=1
third_party_material_policy_present=1
generated_artifact_notice_policy_present=1
ubuntu_package_notice_inventory_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

The current repository has no root NOTICE file and no reviewed package decision saying whether one is required for the Ubuntu local deb draft.

Required Decision Before Promotion

```
notice_file_decision_recorded=1
notice_file_required_decision_recorded=1
notice_file_content_requirements_recorded=1
notice_file_install_path_reviewed=1
notice_file_packaging_mapping_reviewed=1
notice_file_absence_justification_recorded=1
notice_file_missing_required_entries=0
```

Current Decision

```
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
notice_file_present=0
notice_file_decision_recorded=0
notice_file_required_decision_recorded=0
notice_file_content_requirements_recorded=0
notice_file_install_path_reviewed=0
notice_file_packaging_mapping_reviewed=0
notice_file_absence_justification_recorded=0
notice_file_missing_required_entries=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until the package has a reviewed NOTICE file decision. That decision may later require a file or record why no file is required, but this contract does not make that decision.

The Debian copyright notice mapping review remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The trademark notice boundary remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The release artifact notice requirements remain separate, but they must also be resolved before the Ubuntu package notice review can be promoted.

This contract is scoped to the Ubuntu local deb draft. It is not a repository-wide release notice decision.

Non-Claims

This contract is not legal advice. It does not claim NOTICE compliance, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-notice-file-decision-contract.sh
```

Expected output:

```
ubuntu_notice_file_decision_contract: ok
```

docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md

Source title: Ubuntu Package License Promotion Gate Contract

Lines: 84 | Words: 259 | SHA-256: d7fad60e6202e94a6b20afcd57ddb5b9e520f5065158f076d9ae22db70d1efcf

Ubuntu Package License Promotion Gate Contract

Status: no-effect package license promotion gate contract Scope: aggregate the Ubuntu package license prerequisites that must be reviewed before lintian/static metadata or local deb build transcript evidence can advance.

Purpose

This contract turns the Ubuntu package license promotion boundary into a single guarded gate.

It does not run lintian, build a package, publish a package, submit to Ubuntu, or provide legal advice.

Required Inputs

```
license_policy_present=1
license_migration_plan_present=1
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_review_contract_present=1
ubuntu_package_license_review_contract_present=1
```

Required Gate Before Promotion

```
root_license_state_reviewed=1
cli_payload_spdx_reviewed=1
doc_payload_license_reviewed=1
documentation_license_decision_present=1
third_party_notice_reviewed=1
debian_copyright_format_reviewed=1
candidate_expression_accepted=1
packaging_license_expression_updated=1
license_expression_reviewed=1
license_expression_unresolved=0
ubuntu_lintian_static_metadata_unblocked=1
ubuntu_local_deb_build_transcript_unblocked=1
```

Current Gate State

```
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_license_review_contract_present=1
ubuntu_package_license_review_status=resolved-license-expression-recorded
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_lintian_static_metadata_status=blocked-pending-package-license-promotion
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
license_expression_candidate_recorded=1
license_expression_reviewed=1
license_expression_unresolved=0
packaging_license_expression_updated=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Lintian And Build Evidence

The Ubuntu static metadata lint lane and local deb build transcript evidence may not advance until this gate records that the package notice gate is unblocked, the package license review is complete, the packaging license expression is updated, and license expression review is no longer unresolved.

The Ubuntu lintian static metadata contract records the next evidence schema while keeping lintian execution blocked behind this gate.

This gate is a coordination record for the current local deb draft. It is not a package publication claim or a compliance certification.

Non-Claims

This contract is not legal advice. It does not claim license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-package-license-promotion-gate-contract.sh
```

Expected output:

```
ubuntu_package_license_promotion_gate_contract: ok
```

docs/UBUNTU_PACKAGE_LICENSE_REVIEW_CONTRACT.md

Source title: Ubuntu Package License Review Contract

Lines: 152 | Words: 337 | SHA-256: 5f3ec3966436d975d70148819d7cbd14993acb121ea9e22352dab31a31d0ef8d

Ubuntu Package License Review Contract

Status: no-effect package license review contract Scope: record the license-review evidence for the Ubuntu local deb draft package payload.

Purpose

This contract records the Ubuntu package license expression decision and the remaining promotion blockers.

It does not publish a package, create a PPA, submit to Ubuntu, or provide legal advice.

Current Inputs

```
root_license_file=LICENSE
root_license_current=hybrid-license-overview
license_policy_present=1
license_migration_plan_present=1
new_software_direction=AGPL-3.0-or-later
no_silent_relicensing=1
documentation_license_decision_present=1
```

The current no-effect CLI payload input is:

```
cli_payload_source=src/latticra_cli.c
cli_payload_spdx_present=1
cli_payload_spdx=AGPL-3.0-or-later
```

The current local deb draft also records README documentation as an intended package payload:

```
doc_payload_source=README.md
doc_payload_license_reviewed=1
doc_payload_license_unresolved=0
```

Candidate Expressions

These are review candidates, not release claims:

```
candidate_binary_payload_license=AGPL-3.0-or-later
candidate_doc_payload_license=CC-BY-4.0
candidate_source_package_license=AGPL-3.0-or-later AND Apache-2.0 AND CC-BY-4.0
candidate_expression_recorded=1
candidate_expression_applied_to_packaging=1
```

The candidate expression is applied to the local Ubuntu packaging metadata for the current no-effect CLI plus README payload. Broader notice obligations remain blocked until the package notice review is promoted.

The current notice-review dependency is:

```
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_notice_review_contract_present=1
ubuntu_package_notice_review_status=blocked-pending-notice-review
ubuntu_package_license_review_unblocked=1
```

Required Review Before Promotion

```
root_license_state_reviewed=1
cli_payload_spdx_reviewed=1
doc_payload_license_reviewed=1
documentation_license_decision_present=1
third_party_notice_reviewed=1
debian_copyright_format_reviewed=1
candidate_expression_accepted=1
packaging_license_expression_updated=1
```

Current Decision

```
ubuntu_package_license_review_contract_present=1
ubuntu_package_license_review_status=resolved-license-expression-recorded
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_notice_review_contract_present=1
ubuntu_package_notice_review_status=blocked-pending-notice-review
license_expression_candidate_recorded=1
license_expression_reviewed=1
license_expression_unresolved=0
packaging_license_expression_updated=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Ubuntu Lanes

The Ubuntu lintian availability lane may run because it only proves tool availability.

The Ubuntu static metadata lint lane and local deb build transcript evidence remain blocked until the package notice review and promotion gates are satisfied.

The Ubuntu package license promotion gate aggregates this contract with the package notice promotion gate before static lint or build transcript evidence can advance.

Non-Claims

This contract is not legal advice. It does not claim Ubuntu archive readiness, PPA readiness, Canonical endorsement, production installer readiness, package publication readiness, or license-compliance completion.

Validation

Run:

```
sh scripts/test-ubuntu-package-license-review-contract.sh
```

Expected output:

```
ubuntu_package_license_review_contract: ok
```

docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md

Source title: Ubuntu Package Notice Inventory

Lines: 79 | Words: 168 | SHA-256: bbfdc4421add074585d44a0ba88f4f749c772b4fec8cceefabdc54ef320de0a1

Ubuntu Package Notice Inventory

Status: no-effect package notice inventory Scope: deterministic inventory report for the current Ubuntu local deb draft payload.

Purpose

This inventory records the current notice inputs for the Ubuntu local deb draft.

It does not create a NOTICE file, build a package, publish a package, submit to Ubuntu, or provide legal advice.

Current Payload Inventory

```
package_scope=local-deb-draft
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
binary_payload_spdx=AGPL-3.0-or-later
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
root_license_file=LICENSE
root_license_current=hybrid-license-overview
debian_copyright_file=packaging/ubuntu/debian/copyright
```

Current Notice Posture

```
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
third_party_material_inventory_recorded=1
third_party_material_inventory_reviewed=0
third_party_material_missing_entries=0
generated_artifact_notice_reviewed=0
notice_file_present=0
notice_file_decision_recorded=0
doc_payload_license_reviewed=1
doc_payload_license_unresolved=0
debian_copyright_notice_mapping_reviewed=0
packaging_license_expression_updated=1
ubuntu_package_notice_review_unblocked=0
```

Report Command

Run:

```
sh scripts/ubuntu-package-notice-inventory.sh
```

Expected key lines include:

```
UBUNTU PACKAGE NOTICE INVENTORY
inventory_status=ok
inventory_mode=no-effect
ubuntu_package_notice_review_unblocked=0
```

Relationship To Notice Review

The inventory is a fact report. It does not promote the Ubuntu package notice review contract.

The notice review remains blocked until third-party notices, generated-artifact notices, trademark notice boundaries, and the remaining Debian copyright notice mapping are reviewed.

Validation

Run:

```
sh scripts/test-ubuntu-package-notice-inventory.sh
```

Expected output:

```
ubuntu_package_notice_inventory: ok
```

docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md

Source title: Ubuntu Package Notice Promotion Gate Contract

Lines: 79 | Words: 225 | SHA-256: a12c4e158ae17ad6eb1389915265627a84990bf95ea32da526b0b818b6c4811c

Ubuntu Package Notice Promotion Gate Contract

Status: no-effect package notice promotion gate contract Scope: aggregate the Ubuntu package notice prerequisites that must be reviewed before package license promotion can proceed.

Purpose

This contract turns the Ubuntu package notice promotion boundary into a single guarded gate.

It does not promote the notice review, decide licenses, create a NOTICE file, update Debian copyright metadata, build a package, publish a package, submit to Ubuntu, or provide legal advice.

Required Inputs

```
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

Required Gate Before Promotion

```
doc_payload_license_decision_recorded=1
third_party_material_inventory_reviewed=1
generated_artifact_notice_reviewed=1
notice_file_decision_recorded=1
debian_copyright_notice_mapping_reviewed=1
trademark_notice_boundary_recorded=1
release_artifact_notice_requirements_recorded=1
ubuntu_package_notice_review_unblocked=1
ubuntu_package_license_review_unblocked=1
```

Current Gate State

```
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
doc_payload_license_decision_recorded=1
third_party_material_inventory_reviewed=0
generated_artifact_notice_reviewed=0
notice_file_decision_recorded=0
debian_copyright_notice_mapping_reviewed=0
trademark_notice_boundary_recorded=0
release_artifact_notice_requirements_recorded=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Package License Review

The Ubuntu package license review may not advance until this gate records that all package notice prerequisites are reviewed and the package notice review is unblocked.

This gate is a coordination record for the current local deb draft. It is not a package publication claim or a compliance certification.

Non-Claims

This contract is not legal advice. It does not claim NOTICE compliance, license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-package-notice-promotion-gate-contract.sh
```

Expected output:

```
ubuntu_package_notice_promotion_gate_contract: ok
```

docs/UBUNTU_PACKAGE_NOTICE_REVIEW_CONTRACT.md

Source title: Ubuntu Package Notice Review Contract

Lines: 173 | Words: 464 | SHA-256: e572b28a1baded806a54e7f14c0443da64c652489e5d6af5a09a3ea1464f5e28

Ubuntu Package Notice Review Contract

Status: no-effect notice review contract Scope: define notice, attribution, and package evidence required before Ubuntu package license promotion.

Purpose

This contract defines the remaining notice review layer for the Ubuntu local deb draft.

It does not create a NOTICE file, publish a package, create a PPA, submit to Ubuntu, or provide legal advice.

Current Inputs

```
license_policy_present=1
license_migration_plan_present=1
third_party_material_policy_present=1
generated_artifact_notice_policy_present=1
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
third_party_material_inventory_recorded=1
notice_file_present=0
documentation_license_decision_present=1
```

The current Ubuntu package draft intends to include:

```
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
```

Required Notice Review Before License Promotion

```
doc_payload_license_decision_recorded=1
doc_payload_license_compatible_with_package=1
third_party_material_inventory_reviewed=1
third_party_material_missing_entries=0
generated_artifact_notice_reviewed=1
notice_file_required_decision_recorded=1
notice_file_content_requirements_recorded=1
notice_file_decision_recorded=1
debian_copyright_notice_mapping_reviewed=1
debian_copyright_binary_payload_mapping_reviewed=1
debian_copyright_doc_payload_mapping_reviewed=1
debian_copyright_third_party_notice_mapping_reviewed=1
debian_copyright_generated_artifact_notice_mapping_reviewed=1
debian_copyright_notice_file_mapping_reviewed=1
debian_copyright_trademark_notice_boundary_reviewed=1
debian_copyright_license_ref_replaced_or_justified=1
trademark_notice_boundary_recorded=1
trademark_policy_applied_to_package_notice=1
package_description_endorsement_boundary_reviewed=1
documentation_trademark_boundary_reviewed=1
canonical_endorsement_boundary_reviewed=1
project_identity_downstream_use_boundary_recorded=1
release_artifact_notice_requirements_recorded=1
source_package_notice_requirements_recorded=1
release_notes_notice_requirements_recorded=1
```

Current Decision

```
ubuntu_package_notice_review_contract_present=1
ubuntu_package_notice_review_status=blocked-pending-notice-review
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
doc_payload_license_reviewed=1
doc_payload_license_unresolved=0
doc_payload_license_decision_recorded=1
third_party_material_inventory_recorded=1
third_party_material_inventory_reviewed=0
third_party_notice_requirements_recorded=0
generated_artifact_notice_reviewed=0
generated_artifact_notice_requirements_recorded=0
third_party_notice_reviewed=0
notice_file_present=0
notice_file_decision_recorded=0
notice_file_required_decision_recorded=0
notice_file_content_requirements_recorded=0
debian_copyright_notice_mapping_reviewed=0
debian_copyright_binary_payload_mapping_reviewed=0
debian_copyright_doc_payload_mapping_reviewed=1
debian_copyright_third_party_notice_mapping_reviewed=0
debian_copyright_generated_artifact_notice_mapping_reviewed=0
debian_copyright_notice_file_mapping_reviewed=0
debian_copyright_trademark_notice_boundary_reviewed=0
debian_copyright_license_ref_replaced_or_justified=1
trademark_notice_boundary_recorded=0
trademark_policy_applied_to_package_notice=0
package_description_endorsement_boundary_reviewed=0
documentation_trademark_boundary_reviewed=0
canonical_endorsement_boundary_reviewed=0
project_identity_downstream_use_boundary_recorded=0
release_artifact_notice_requirements_recorded=0
source_package_notice_requirements_recorded=0
release_notes_notice_requirements_recorded=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To License Review

The Ubuntu package license review contract records the accepted package expression. Package promotion remains blocked until this notice review contract is promoted by a reviewed notice decision.

The Ubuntu package notice inventory report records current payload facts and open notice obligations, but it does not unblock this review by itself.

The Ubuntu doc payload license review contract records the resolved README.md CC-BY-4.0 documentation-license decision required before this notice review can be promoted.

The Ubuntu third-party material review contract records the source, license, compatibility, and notice-requirement evidence required before this notice review can be promoted.

The Ubuntu generated-artifact notice review contract records the generated payload, build artifact, changes-file, build-log, and installed-payload notice evidence required before this notice review can be promoted.

The Ubuntu NOTICE file decision contract records the reviewed decision required before a package can claim NOTICE obligations are satisfied.

The Ubuntu Debian copyright notice mapping contract records the reviewed mapping required before package notice obligations can be represented in packaging/ubuntu/debian/copyright.

The Ubuntu trademark notice boundary contract records the reviewed project-identity and endorsement boundary required before package notice obligations can be promoted.

The Ubuntu release artifact notice requirements contract records the reviewed source-package, deb-artifact, changes-file, build-log, installed-payload, and release-notes notice requirements required before package notice obligations can be promoted.

The Ubuntu package notice promotion gate aggregates these prerequisites before package license promotion can proceed.

Non-Claims

This contract is not legal advice. It does not claim license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, production installer readiness, package publication readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-package-notice-review-contract.sh
```

Expected output:

```
ubuntu_package_notice_review_contract: ok
```

docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md

Source title: Ubuntu PPA Archive Publication Gate Contract

Lines: 110 | Words: 304 | SHA-256: 29290fda0b6cf9e6f21987a903553415adb5ebc0c454a69726484ef489618a61

Ubuntu PPA Archive Publication Gate Contract

Status: no-effect PPA/archive publication gate contract Scope: evidence schema for future Ubuntu PPA upload or Ubuntu archive submission review.

Purpose

This contract defines the evidence required before any future Ubuntu PPA upload, Launchpad build, Ubuntu archive submission, or publication-readiness claim can be accepted.

It does not run debsign, dput, dpkg-buildpackage, debuild, lintian, sbuild, or pbuilder. It does not create a source package, sign artifacts, upload to Launchpad, create a PPA, submit to Ubuntu, publish a package, or claim archive readiness.

Required Inputs

```
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_unblocked=1
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_unblocked=1
ubuntu_install_remove_evidence_unblocked=1
local_deb_build_transcript_accepted=1
deb_artifact_created=1
deb_artifact_digest_recorded=1
```

Required Evidence Before Publication Promotion

```
source_package_created=1
source_package_digest_recorded=1
dsc_digest_recorded=1
changes_file_digest_recorded=1
upload_target_recorded=1
upload_authority_reviewed=1
debsign_command_recorded=1
signature_fingerprint_recorded=1
dput_command_recorded=1
upload_exit_status_recorded=1
launchpad_build_log_recorded=1
launchpad_build_result_reviewed=1
publication_non_claims_reviewed=1
ubuntu_publication_gate_unblocked=1
```

Current State

```
ubuntu_ppa_archive_publication_gate_contract_present=1
ubuntu_ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_source_package_evidence_unblocked=0
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
ubuntu_upload_signing_authority_evidence_unblocked=0
ubuntu_install_remove_evidence_unblocked=0
local_deb_build_transcript_accepted=0
deb_artifact_created=0
deb_artifact_digest_recorded=0
source_package_created=0
source_package_digest_recorded=0
dsc_digest_recorded=0
changes_file_digest_recorded=0
upload_target_recorded=0
upload_authority_reviewed=0
debsign_command_recorded=0
signature_fingerprint_recorded=0
dput_command_recorded=0
upload_exit_status_recorded=0
launchpad_build_log_recorded=0
launchpad_build_result_reviewed=0
publication_non_claims_reviewed=0
ubuntu_publication_gate_unblocked=0
ppa_created=0
ppa_claimed=0
launchpad_upload_run=0
ubuntu_archive_submission_claimed=0
ubuntu_archive_ready=0
ubuntu_publication_ready=0
production_installer_ready=0
root_installer_ready=0
```

Relationship To Install Remove Evidence

The Ubuntu source package evidence contract, upload/signing authority evidence contract, and local deb install/remove evidence contract must be unblocked before this publication gate can advance. Publication evidence is downstream of notice, license, lintian/static metadata, build transcript acceptance, source package evidence, payload review, install evidence, remove evidence, upload authority, signing evidence, Launchpad evidence, and non-claim review.

This contract is intentionally closed today. It records the future evidence shape without signing artifacts, uploading artifacts, creating a PPA, submitting to Ubuntu, or publishing a package.

Non-Claims

This contract does not claim PPA readiness, Launchpad build success, Ubuntu archive readiness, Ubuntu upload rights, Ubuntu sponsorship, Canonical endorsement, package publication readiness, production installer readiness, daily-driver readiness, root installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-ppa-archive-publication-gate-contract.sh
```

Expected output:

```
ubuntu_ppa_archive_publication_gate_contract: ok
```

docs/UBUNTU_READINESS_PLAN.md

Source title: Ubuntu Readiness Plan

Lines: 286 | Words: 611 | SHA-256: c4d47a332236ec802a9ede3b5f4c09012e70267364d498d8c885ecb2336ec4b5

Ubuntu Readiness Plan

Status: planning and compatibility record Scope: preparing Latticra for Ubuntu-friendly development, local validation, and local-only deb packaging.

Purpose

This plan prepares Latticra for an Ubuntu-facing path without claiming Ubuntu archive acceptance, Canonical endorsement, PPA readiness, production installer readiness, or distribution status.

The near-term target is:

```
Latticra builds cleanly on Ubuntu Linux.
Latticra Panel can be run from source on Ubuntu with documented apt prerequisites.
Latticra has a local-only deb packaging draft for the no-effect CLI payload.
Latticra preserves its current non-claims.
```

Terminology Boundary

Use:

```
Ubuntu readiness
Ubuntu compatibility lane
Ubuntu local deb draft
Latticra on Ubuntu
local-only deb packaging draft
```

Avoid claiming:

```
official Ubuntu package
Ubuntu archive package
PPA release
Canonical endorsement
Ubuntu flavor
Ubuntu derivative
operating-system replacement
```

Those terms require separate packaging, policy, legal, trademark, and community review.

Ubuntu-Facing Requirements To Respect

Before Latticra can be proposed beyond a local deb draft, it should be checked against Ubuntu/Debian package expectations:

```
source package layout
debian/control hygiene
debian/copyright accuracy
license expression review
Build-Depends completeness
Rules-Requires-Root posture
no maintainer scripts without review
no systemd service installation without review
no root-owned host mutation without review
lintian output classification
pbuilder/sbuild or equivalent build evidence
install/remove behavior evidence
```

Current Latticra Posture

Current Latticra evidence is still early-stage and report-oriented.

Known current posture:

```
ubuntu_build_lane_present=1
ubuntu_panel_prerequisites_documented=1
ubuntu_local_deb_draft_present=1
ubuntu_lintian_availability_present=1
ubuntu_package_license_review_contract_present=1
ubuntu_package_license_review_status=resolved-license-expression-recorded
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_lintian_static_metadata_status=blocked-pending-package-license-promotion
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata
-and-build-transcript
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
ubuntu_ppa_archive_publication_gate_contract_present=1
ubuntu_ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
ubuntu_package_notice_review_contract_present=1
ubuntu_package_notice_review_status=blocked-pending-notice-review
ubuntu_local_deb_build_transcript_contract_present=1
ubuntu_local_deb_build_transcript_present=0
local_deb_build_transcript_accepted=0
deb_artifact_created=0
deb_installed_on_host=0
deb_removed_from_host=0
dpkg_source_run=0
dpkg_buildpackage_source_run=0
source_package_created=0
source_package_digest_recorded=0
source_package_uploaded=0
ubuntu_source_package_evidence_unblocked=0
upload_target_kind_recorded=0
launchpad_account_recorded=0
ppa_or_archive_target_reviewed=0
gpg_signing_key_fingerprint_recorded=0
upload_command_non_claims_reviewed=0
ubuntu_upload_signing_authority_evidence_unblocked=0
debsign_command_recorded=0
dput_command_recorded=0
launchpad_upload_run=0
ubuntu_publication_gate_unblocked=0
ppa_claimed=0
ubuntu_archive_ready=0
production_installer_ready=0
daily_driver_install_ready=0
root_installer_ready=0
```

The first Ubuntu target is development and local validation. It is not an Ubuntu archive package, PPA package, image, flavor, distribution, or replacement operating system.

Phase 1: Ubuntu Build Lane

The Ubuntu build lane proves the current guarded project foundation compiles with Ubuntu toolchains.

Current guarded files:

```
scripts/test-ubuntu-build-lane.sh
.github/workflows/ubuntu-build-lane.yml
docs/UBUNTU_DEVELOPER_WORKFLOW.md
scripts/test-ubuntu-lintian-availability.sh
.github/workflows/ubuntu-lintian-availability.yml
```

Initial checks remain conservative:

```
C compiler available
project C guards compile
kernel lifecycle guards compile
no-effect CLI status surface present
no external authority claims added
```

Phase 2: Panel Prerequisites Lane

The Panel remains user-local on Ubuntu. The apt prerequisites support local Rust/egui development and desktop metadata refresh without adding root-installed Latticra services.

Current guarded files:

```
README.md
docs/QUICK_START_CHEATSHEET.md
installer/README.md
docs/UBUNTU_DEVELOPER_WORKFLOW.md
```

Phase 3: Local Deb Draft Lane

Packaging metadata is present as a local-only draft after the Ubuntu build lane.

Current guarded files:

```
packaging/ubuntu/README.md
packaging/ubuntu/debian/control
packaging/ubuntu/debian/rules
packaging/ubuntu/debian/changelog
packaging/ubuntu/debian/copyright
packaging/ubuntu/debian/install
packaging/ubuntu/debian/source/format
scripts/test-ubuntu-local-deb-static-validation.sh
.github/workflows/ubuntu-local-deb-static-validation.yml
docs/UBUNTU_LINTIAN_AVAILABILITY.md
docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md
docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md
docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md
docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md
docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md
docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md
docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md
docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md
docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md
scripts/ubuntu-package-notice-inventory.sh
scripts/test-ubuntu-package-notice-inventory.sh
scripts/test-ubuntu-doc-payload-license-review-contract.sh
scripts/test-ubuntu-third-party-material-review-contract.sh
scripts/test-ubuntu-generated-artifact-notice-review-contract.sh
scripts/test-ubuntu-notice-file-decision-contract.sh
scripts/test-ubuntu-debian-copyright-notice-mapping-contract.sh
scripts/test-ubuntu-trademark-notice-boundary-contract.sh
scripts/test-ubuntu-release-artifact-notice-requirements-contract.sh
scripts/test-ubuntu-package-notice-promotion-gate-contract.sh
```

```
scripts/test-ubuntu-package-license-promotion-gate-contract.sh
scripts/test-ubuntu-lintian-static-metadata-contract.sh
scripts/test-ubuntu-package-license-review-contract.sh
scripts/test-ubuntu-package-notice-review-contract.sh
scripts/test-ubuntu-local-deb-build-transcript-contract.sh
scripts/test-ubuntu-local-deb-build-transcript-acceptance-gate-contract.sh
scripts/test-ubuntu-local-deb-install-remove-evidence-contract.sh
scripts/test-ubuntu-source-package-evidence-contract.sh
scripts/test-ubuntu-upload-signing-authority-evidence-contract.sh
scripts/test-ubuntu-ppa-archive-publication-gate-contract.sh
.github/workflows/ubuntu-package-notice-inventory.yml
.github/workflows/ubuntu-doc-payload-license-review-contract.yml
.github/workflows/ubuntu-third-party-material-review-contract.yml
.github/workflows/ubuntu-generated-artifact-notice-review-contract.yml
.github/workflows/ubuntu-notice-file-decision-contract.yml
.github/workflows/ubuntu-debian-copyright-notice-mapping-contract.yml
.github/workflows/ubuntu-trademark-notice-boundary-contract.yml
.github/workflows/ubuntu-release-artifact-notice-requirements-contract.yml
.github/workflows/ubuntu-package-notice-promotion-gate-contract.yml
.github/workflows/ubuntu-package-license-promotion-gate-contract.yml
.github/workflows/ubuntu-lintian-static-metadata-contract.yml
.github/workflows/ubuntu-package-notice-review-contract.yml
.github/workflows/ubuntu-package-license-review-contract.yml
.github/workflows/ubuntu-local-deb-build-transcript-contract.yml
.github/workflows/ubuntu-local-deb-build-transcript-acceptance-gate-contract.yml
.github/workflows/ubuntu-local-deb-install-remove-evidence-contract.yml
.github/workflows/ubuntu-source-package-evidence-contract.yml
.github/workflows/ubuntu-upload-signing-authority-evidence-contract.yml
.github/workflows/ubuntu-ppa-archive-publication-gate-contract.yml
```

The first deb draft is only a packaging-shape record until lint/build/install evidence is added.

Non-Claims

This plan does not:

```
publish a deb package
create a PPA
submit Latticra to Ubuntu
run dpkg-source
run debsign
run dput
claim Ubuntu archive readiness
claim Canonical endorsement
install a root service
modify systemd
modify the kernel
claim production readiness
claim operating-system completeness
```

Recommended Next Slice

Recommended next slice:

```
Review the Ubuntu upload/signing authority evidence contract, then keep signing and upload
evidence blocked until source package evidence is reviewed.
```

That should preserve the current no-artifact, no-submission, local-only Ubuntu package posture while the local deb path matures.

docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md

Source title: Ubuntu Release Artifact Notice Requirements Contract

Lines: 91 | Words: 275 | SHA-256: cf81d2e8d8e50cea8fb0e3e8ad42bc90f15c2e6bf720a7ac1318f7d937e58daf

Ubuntu Release Artifact Notice Requirements Contract

Status: no-effect release artifact notice requirements contract Scope: define release-artifact notice evidence required before the Ubuntu package notice review can be promoted.

Purpose

This contract turns the Ubuntu local deb release-artifact notice blocker into a concrete review checklist.

It does not build a deb, create a .changes file, create a build log, install a package, publish a package, create release notes, submit to Ubuntu, or provide legal advice.

Current Inputs

```
package_scope=local-deb-draft
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
deb_artifact_created=0
changes_file_created=0
build_log_created=0
package_installed_for_notice_review=0
package_publish_performed=0
ubuntu_package_notice_inventory_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

The current Ubuntu local deb draft has no release artifact, no .changes file, no build log, and no installed payload evidence for release-artifact notice review.

Required Requirements Before Promotion

```
release_artifact_notice_requirements_recorded=1
source_package_notice_requirements_recorded=1
deb_artifact_notice_requirements_recorded=1
changes_file_notice_requirements_recorded=1
build_log_notice_requirements_recorded=1
installed_payload_notice_requirements_recorded=1
release_notes_notice_requirements_recorded=1
release_artifact_missing_notice_entries=0
```

Current Decision

```
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
release_artifact_notice_requirements_recorded=0
source_package_notice_requirements_recorded=0
deb_artifact_notice_requirements_recorded=0
changes_file_notice_requirements_recorded=0
build_log_notice_requirements_recorded=0
installed_payload_notice_requirements_recorded=0
release_notes_notice_requirements_recorded=0
release_artifact_missing_notice_entries=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until release-artifact notice requirements are reviewed for the source package, deb artifact, .changes file, build log, installed payload, and any release notes or publication material.

This contract is scoped to the Ubuntu local deb draft. It is not evidence that any artifact exists, that any artifact is distributable, or that release notices are complete.

Non-Claims

This contract is not legal advice. It does not claim release-artifact notice compliance, generated-artifact notice compliance, license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-release-artifact-notice-requirements-contract.sh
```

Expected output:

```
ubuntu_release_artifact_notice_requirements_contract: ok
```

docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md

Source title: Ubuntu Source Package Evidence Contract

Lines: 105 | Words: 283 | SHA-256: 5a4d0b20ba5fcdfc6d9b82e8979e679f2e58644ced4d2bfa2ac410aa4693c8fc

Ubuntu Source Package Evidence Contract

Status: no-effect source package evidence contract Scope: evidence schema for a future reviewed Ubuntu source package transcript.

Purpose

This contract defines the evidence required before a future Ubuntu source package can be treated as publication input.

It does not run dpkg-source, dpkg-buildpackage -S, debuild -S, debsign, dput, lintian, sbuild, or pbuilder. It does not create a source package, sign artifacts, upload to Launchpad, create a PPA, submit to Ubuntu, publish a package, or claim archive readiness.

Required Inputs

```
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
local_deb_build_transcript_accepted=1
```

Required Evidence Before Promotion

```
source_package_build_environment_recorded=1
dpkg_source_command_recorded=1
dpkg_buildpackage_source_command_recorded=1
source_package_created=1
source_package_name_recorded=1
source_package_digest_recorded=1
dsc_path_recorded=1
dsc_digest_recorded=1
changes_file_path_recorded=1
changes_file_digest_recorded=1
orig_tarball_path_recorded=1
orig_tarball_digest_recorded=1
debian_source_format_verified=1
source_package_payload_reviewed=1
source_package_notice_requirements_reviewed=1
ubuntu_source_package_evidence_unblocked=1
```

Current State

```
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_status=blocked-pending-accepted-build-transcript
local_deb_build_transcript_accepted=0
source_package_build_environment_recorded=0
dpkg_source_command_recorded=0
dpkg_buildpackage_source_command_recorded=0
dpkg_source_run=0
dpkg_buildpackage_source_run=0
source_package_created=0
source_package_name_recorded=0
source_package_digest_recorded=0
dsc_path_recorded=0
dsc_digest_recorded=0
changes_file_path_recorded=0
changes_file_digest_recorded=0
orig_tarball_path_recorded=0
orig_tarball_digest_recorded=0
debian_source_format_verified=0
source_package_payload_reviewed=0
source_package_notice_requirements_reviewed=0
ubuntu_source_package_evidence_unblocked=0
debsign_command_recorded=0
dput_command_recorded=0
launchpad_upload_run=0
source_package_uploaded=0
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
ubuntu_upload_signing_authority_evidence_unblocked=0
ppa_claimed=0
ubuntu_archive_ready=0
ubuntu_publication_ready=0
production_installer_ready=0
root_installer_ready=0
```

Relationship To Publication

The Ubuntu PPA/archive publication gate must not unblock until this source package evidence contract is unblocked alongside install/remove evidence, upload authority, signing evidence, Launchpad evidence, and non-claim review.

This contract is intentionally closed today. It records the future source package evidence shape without creating, signing, uploading, or publishing artifacts.

The Ubuntu upload/signing authority evidence contract is downstream of this contract and remains blocked until source package evidence is unblocked.

Non-Claims

This contract does not claim source package creation, source package correctness, source package upload readiness, PPA readiness, Launchpad build success, Ubuntu archive readiness, Ubuntu sponsorship, Canonical endorsement, package publication readiness, production installer readiness, daily-driver readiness, root installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-source-package-evidence-contract.sh
```

Expected output:

```
ubuntu_source_package_evidence_contract: ok
```

docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md

Source title: Ubuntu Third-Party Material Review Contract

Lines: 102 | Words: 345 | SHA-256: da3026209f79b0f166eaa0da06dea9a4efa85bcbb73ba6780ef7c4689c20c7fb

Ubuntu Third-Party Material Review Contract

Status: no-effect third-party material review contract Scope: define third-party material evidence required before the Ubuntu local deb notice review can be promoted.

Purpose

This contract turns the Ubuntu local deb third-party material blocker into a concrete review checklist.

It does not add third-party material, decide license compatibility, update packaging/ubuntu/debian/copyright, create a NOTICE file, build a package, publish a package, submit to Ubuntu, or provide legal advice.

Current Inputs

```
package_scope=local-deb-draft
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
third_party_material_policy_present=1
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

The current inventory records the third-party material posture but does not complete review.

Required Review Before Promotion

```
third_party_material_inventory_reviewed=1
third_party_material_missing_entries=0
third_party_material_source_records_present=1
third_party_material_license_records_present=1
third_party_material_compatibility_notes_present=1
binary_payload_third_party_material_reviewed=1
doc_payload_third_party_material_reviewed=1
packaging_metadata_third_party_material_reviewed=1
third_party_notice_requirements_recorded=1
```

Current Decision

```
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
third_party_material_inventory_recorded=1
third_party_material_inventory_reviewed=0
third_party_material_missing_entries=0
third_party_material_source_records_present=0
third_party_material_license_records_present=0
third_party_material_compatibility_notes_present=0
binary_payload_third_party_material_reviewed=0
doc_payload_third_party_material_reviewed=0
packaging_metadata_third_party_material_reviewed=0
third_party_notice_reviewed=0
third_party_notice_requirements_recorded=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until third-party material inventory review is complete and any notice requirements are mapped into package metadata.

The generated-artifact notice review remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The NOTICE file decision remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The Debian copyright notice mapping remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The trademark notice boundary remains separate, but it must also be resolved before the Ubuntu package notice review can be promoted.

The release artifact notice requirements remain separate, but they must also be resolved before the Ubuntu package notice review can be promoted.

This contract is scoped to the current Ubuntu local deb draft payload. It is not a repository-wide third-party materials audit.

Non-Claims

This contract is not legal advice. It does not claim third-party license compliance, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-third-party-material-review-contract.sh
```

Expected output:

```
ubuntu_third_party_material_review_contract: ok
```

docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md

Source title: Ubuntu Trademark Notice Boundary Contract

Lines: 92 | Words: 287 | SHA-256: a078facab49d33529b8e05a9d5b41866201a7ed0a5f7e7c05bbdc4000a1cdc91

Ubuntu Trademark Notice Boundary Contract

Status: no-effect trademark notice boundary contract Scope: define trademark, identity, and endorsement-boundary evidence required before the Ubuntu package notice review can be promoted.

Purpose

This contract turns the Ubuntu local deb trademark notice blocker into a concrete review checklist.

It does not update package metadata, change packaging/ubuntu/debian/copyright, create a NOTICE file, decide the documentation license, build a package, publish a package, submit to Ubuntu, grant trademark rights, or provide legal advice.

Current Inputs

```
package_scope=local-deb-draft
trademark_policy_present=1
project_identity_terms_recorded=1
binary_payload=usr/bin/latticra
binary_payload_source=src/latticra_cli.c
doc_payload=usr/share/doc/latticra/README.md
doc_payload_source=README.md
debian_control_file=packaging/ubuntu/debian/control
debian_copyright_file=packaging/ubuntu/debian/copyright
ubuntu_package_notice_inventory_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_third_party_material_review_contract_present=1
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_notice_file_decision_contract_present=1
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_package_notice_review_contract_present=1
```

The current package draft is local-only and makes no Ubuntu archive, PPA, Canonical endorsement, or production readiness claim.

Required Boundary Before Promotion

```
trademark_notice_boundary_recorded=1
trademark_policy_applied_to_package_notice=1
package_description_endorsement_boundary_reviewed=1
debian_copyright_trademark_notice_boundary_reviewed=1
documentation_trademark_boundary_reviewed=1
canonical_endorsement_boundary_reviewed=1
project_identity_downstream_use_boundary_recorded=1
trademark_notice_missing_required_entries=0
```

Current Decision

```
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
trademark_notice_boundary_recorded=0
trademark_policy_applied_to_package_notice=0
package_description_endorsement_boundary_reviewed=0
debian_copyright_trademark_notice_boundary_reviewed=0
documentation_trademark_boundary_reviewed=0
canonical_endorsement_boundary_reviewed=0
project_identity_downstream_use_boundary_recorded=0
trademark_notice_missing_required_entries=0
ubuntu_package_notice_review_unblocked=0
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
```

Relationship To Notice Review

The Ubuntu package notice review may not be promoted until the local deb payload has a reviewed trademark and endorsement boundary. That review must preserve the distinction between open-source license permissions and Latticra/Bryforge project identity rights.

The release artifact notice requirements remain separate, but they must also be resolved before the Ubuntu package notice review can be promoted.

This contract is scoped to the Ubuntu local deb draft. It is not a trademark registration, enforcement plan, Ubuntu archive claim, or downstream redistribution approval.

Non-Claims

This contract is not legal advice. It does not claim trademark compliance, NOTICE compliance, license-compliance completion, Ubuntu archive readiness, PPA readiness, Canonical endorsement, package publication readiness, production installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-trademark-notice-boundary-contract.sh
```

Expected output:

```
ubuntu_trademark_notice_boundary_contract: ok
```

docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md

Source title: Ubuntu Upload Signing Authority Evidence Contract

Lines: 92 | Words: 253 | SHA-256: 4e14b583e18948bfec84a0bf31b155dc462350259a9363c03460726e4fed93d7

Ubuntu Upload Signing Authority Evidence Contract

Status: no-effect upload/signing authority evidence contract Scope: evidence schema for future Ubuntu upload target, signing key, and upload authority review.

Purpose

This contract defines the evidence required before any future Ubuntu source package upload command can be treated as publication input.

It does not run debsign, dput, dpkg-source, dpkg-buildpackage -S, debuild -S, lintian, sbuild, or pbuilder. It does not sign artifacts, upload to Launchpad, create a PPA, submit to Ubuntu, publish a package, or claim archive readiness.

Required Inputs

```
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_unblocked=1
source_package_created=1
dsc_digest_recorded=1
changes_file_digest_recorded=1
```

Required Evidence Before Promotion

```
upload_target_recorded=1
upload_target_kind_recorded=1
upload_authority_reviewed=1
launchpad_account_recorded=1
ppa_or_archive_target_reviewed=1
gpg_signing_key_fingerprint_recorded=1
signature_fingerprint_recorded=1
debsign_command_recorded=1
dput_command_recorded=1
upload_command_non_claims_reviewed=1
ubuntu_upload_signing_authority_evidence_unblocked=1
```

Current State

```
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
ubuntu_source_package_evidence_unblocked=0
source_package_created=0
dsc_digest_recorded=0
changes_file_digest_recorded=0
upload_target_recorded=0
upload_target_kind_recorded=0
upload_authority_reviewed=0
launchpad_account_recorded=0
ppa_or_archive_target_reviewed=0
gpg_signing_key_fingerprint_recorded=0
signature_fingerprint_recorded=0
debsign_command_recorded=0
dput_command_recorded=0
upload_command_non_claims_reviewed=0
upload_exit_status_recorded=0
launchpad_upload_run=0
source_package_uploaded=0
ubuntu_upload_signing_authority_evidence_unblocked=0
ppa_created=0
ppa_claimed=0
ubuntu_archive_submission_claimed=0
ubuntu_archive_ready=0
ubuntu_publication_ready=0
production_installer_ready=0
root_installer_ready=0
```

Relationship To Publication

The Ubuntu PPA/archive publication gate must not unblock until this upload/signing authority evidence contract is unblocked alongside source package evidence, install/remove evidence, Launchpad build evidence, and non-claim review.

This contract is intentionally closed today. It records the future upload authority and signing evidence shape without signing, uploading, or publishing artifacts.

Non-Claims

This contract does not claim upload authorization, signing authority, source package upload readiness, PPA readiness, Launchpad build success, Ubuntu archive readiness, Ubuntu sponsorship, Canonical endorsement, package publication readiness, production installer readiness, daily-driver readiness, root installer readiness, or release readiness.

Validation

Run:

```
sh scripts/test-ubuntu-upload-signing-authority-evidence-contract.sh
```

Expected output:

```
ubuntu_upload_signing_authority_evidence_contract: ok
```

packaging/debian/README.md

Source title: Debian Packaging Draft

Lines: 161 | Words: 418 | SHA-256: 05bf6ff826676cdd155b1b3fc2ad31b5b1483c46b0d6c28269d3ea6487ff5dda

Debian Packaging Draft

Status: local-only packaging draft

This directory contains Debian-oriented packaging experiments for Latticra.

The current debian metadata is a local-only draft used by static guards. It is not a Debian archive package, not a mentors.debian.net upload, not sponsorship evidence, not ftp-master acceptance evidence, not a local dpkg-buildpackage result, not a lintian result, and not package-readiness evidence.

Where This Fits

- Documentation hub: ../../docs/README.md
- Debian status: ../../docs/status/DEBIAN_ECOSYSTEM_INTEGRATION_STATUS.md
- Current status: ../../docs/status/CURRENT_STATUS.md

Current guarded files:

```
packaging/debian/README.md
packaging/debian/debian/control
packaging/debian/debian/rules
packaging/debian/debian/changelog
packaging/debian/debian/copyright
packaging/debian/debian/install
packaging/debian/debian/source/format
docs/DEBIAN_LOCAL_DEB_STATIC_VALIDATION.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
docs/status/DEBIAN_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-debian-local-deb-static-validation.sh
scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
.github/workflows/debian-local-deb-static-validation.yml
.github/workflows/debian-freebsd-openbsd-source-archive-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-gate-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-environment-contract.yml
.github/workflows/debian-freebsd-openbsd-package-artifact-naming-contract.yml
.github/workflows/debian-freebsd-openbsd-package-payload-inspection-contract.yml
.github/workflows/debian-freebsd-openbsd-package-install-remove-transcript-contract.yml
.github/workflows/debian-freebsd-openbsd-package-publication-non-claim-review-contract.yml
.github/workflows/debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.ym
l
```

The static lane preserves:

```
local_only_draft=1
debian_local_deb_draft_present=1
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
temporary_debian_source_input_staged=1
temporary_debian_orig_archive_staged=1
temporary_debian_debian_dir_overlay_staged=1
debian_static_deb_validation_present=1
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
debian_build_allowed=0
debian_clean_build_environment_documented=1
debian_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
environment_transcript_present=0
```

```
debian_artifact_name_pattern_recorded=1
debian_source_package_name=latticra_0.0.0-1local1.dsc
debian_binary_package_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.deb
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
debian_payload_expected_bin=usr/bin/latticra
debian_payload_expected_doc=usr/share/doc/latticra/README.md
debian_payload_inspection_run=0
package_payload_inspection_run=0
package_payload_accepted=0
debian_install_remove_transcript_required=1
debian_install_remove_transcript_present=0
debian_package_install_run=0
debian_package_remove_run=0
remove_on_host_run=0
publication_non_claim_review_present=1
debian_publication_non_claim_review_present=1
debian_package_publication_claimed=0
debian_source_upload_run=0
debian_mentors_upload_run=0
debian_archive_upload_run=0
debian_debsign_run=0
debian_dput_run=0
platform_build_evidence_accepted=0
debian_validation_promotion_blocked=1
debian_platform_build_evidence_accepted=0
package_validation_result_promoted=0
debian_validation_result_promoted=0
deb_artifact_created=0
deb_installed_on_host=0
debian_freebsd_openbsd_source_archive_contract_present=1
source_archive_policy_recorded=1
source_archive_created=0
source_archive_sha256_recorded=0
source_archive_accepted_for_build=0
dpkg_buildpackage_run_required=0
debuild_run_required=0
lintian_run_required=0
dpkg_buildpackage_run=0
debuild_run=0
lintian_run=0
package_artifact_created=0
package_artifact_sha256_recorded=0
install_on_host_run=0
lintian_transcript_present=0
debian_archive_ready=0
debian_mentors_upload_claimed=0
debian_sponsorship_claimed=0
debian_ftp_master_acceptance_claimed=0
production_readiness_claimed=0
```

The draft payload remains intentionally narrow:

```
usr/bin/latticra
usr/share/doc/latticra/README.md
```

The CLI reports no-effect status and disabled runtime behavior.

The source archive contract records the expected latticra_0.0.0.orig.tar.gz boundary, but it does not create an archive, run dpkg-source, run dpkg-buildpackage, or accept Debian build evidence.

The package-build gate is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md. It keeps package_build_gate_state=closed-no-effect and debian_build_allowed=0 until source, checksum, license, notice, environment, authorization, payload inspection, install/remove, and publication non-claim evidence exists.

The package-build environment contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md. It documents the Debian clean build environment requirement while keeping debian_build_environment_provisioned=0, explicit_operator_build_authorization=0, and all Debian build commands disabled.

The package artifact naming contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md. It records the future Debian source and binary artifact names while keeping package_artifact_created=0, repository_package_artifact_write_allowed=0, and package_artifact_sha256_recorded=0.

The package payload inspection contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md. It records the expected Debian payload paths while keeping debian_payload_inspection_run=0, package_payload_accepted=0, and package_artifact_created=0.

The package install/remove transcript contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove transcript requirements while keeping debian_package_install_run=0, debian_package_remove_run=0, and install_on_host_run=0.

The package publication non-claim review contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records local-only Debian publication non-claims while keeping debian_package_publication_claimed=0, debian_dput_run=0, and package validation promotion blocked.

The package validation promotion blocker matrix is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md. It ties Debian source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping debian_platform_build_evidence_accepted=0 and debian_validation_result_promoted=0.

packaging/fedora/README.md

Source title: Fedora Packaging Draft

Lines: 39 | Words: 99 | SHA-256: 0b0635aa2fe43784dcde826e9b57a35324758980d515934adfccb79e427596d0

Fedora Packaging Draft

Status: local-only packaging draft

This directory contains Fedora-oriented packaging experiments for Latticra.

The current spec is a local-only draft used by static guards. It is not a Fedora package submission, not Fedora approval evidence, not a Copr build record, not a mock build result, and not package-readiness evidence.

Where this fits

- Documentation hub: ../../docs/README.md
- Fedora workflow: ../../docs/FEDORA_DEVELOPER_WORKFLOW.md
- Fedora readiness plan: ../../docs/FEDORA_READINESS_PLAN.md
- Current status: ../../docs/status/CURRENT_STATUS.md

Current guarded files:

```
packaging/fedora/latticra.spec
packaging/fedora/README.md
scripts/test-fedora-spec-static.sh
scripts/test-fedora-local-rpm-static-validation.sh
scripts/test-fedora-rpmlint-static-spec-lane.sh
.github/workflows/fedora-spec-static.yml
.github/workflows/fedora-local-rpm-static-validation.yml
.github/workflows/fedora-rpmlint-static-spec-lane.yml
```

The static lanes preserve:

```
local_only_draft=1
package_artifact_created=0
rpmbuild_run_required=0
mock_run_required=0
fedora_submission_claimed=0
fedora_approval_claimed=0
production_readiness_claimed=0
```

packaging/freebsd/README.md

Source title: FreeBSD Port Draft

Lines: 152 | Words: 425 | SHA-256: 39a95e04b738a4cdfdbb484b48f791a57aafc01cd5d7de03b8cbe42005274e43

FreeBSD Port Draft

Status: local-only port draft

This directory contains FreeBSD ports metadata experiments for Latticra.

The current port files are a local-only draft used by static guards. They are not a FreeBSD ports tree submission, not a Bugzilla PR, not committer review evidence, not a poudriere build, not a make package result, not a portlint result, and not package-readiness evidence.

Where This Fits

- Documentation hub: ../../docs/README.md
- FreeBSD status: ../../docs/status/FREEBSD_ECOSYSTEM_INTEGRATION_STATUS.md
- Current status: ../../docs/status/CURRENT_STATUS.md

Current guarded files:

```
packaging/freebsd/README.md
packaging/freebsd/Makefile
packaging/freebsd/pkg-descr
packaging/freebsd/pkg-plist
docs/FREEBSD_PORT_STATIC_VALIDATION.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
docs/status/FREEBSD_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-freebsd-port-static-validation.sh
scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
.github/workflows/freebsd-port-static-validation.yml
.github/workflows/debian-freebsd-openbsd-source-archive-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-gate-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-environment-contract.yml
.github/workflows/debian-freebsd-openbsd-package-artifact-naming-contract.yml
.github/workflows/debian-freebsd-openbsd-package-payload-inspection-contract.yml
.github/workflows/debian-freebsd-openbsd-package-install-remove-transcript-contract.yml
.github/workflows/debian-freebsd-openbsd-package-publication-non-claim-review-contract.yml
.github/workflows/debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.ym
l
```

The static lane preserves:

```
local_only_draft=1
freebsd_port_draft_present=1
freebsd_port_static_validation_present=1
```

```
debian_freebsd_openbsd_source_archive_contract_present=1
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
temporary_freebsd_distfile_staged=1
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
freebsd_build_allowed=0
freebsd_ports_environment_documented=1
freebsd_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
environment_transcript_present=0
freebsd_artifact_name_pattern_recorded=1
freebsd_package_name=latticra-0.0.0.pkg
freebsd_package_artifact_created=0
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
freebsd_payload_expected_bin=bin/latticra
freebsd_payload_expected_doc=%%DOCSDIR%%/README.md
freebsd_payload_inspection_run=0
package_payload_inspection_run=0
package_payload_accepted=0
freebsd_install_remove_transcript_required=1
freebsd_install_remove_transcript_present=0
freebsd_package_install_run=0
freebsd_package_remove_run=0
remove_on_host_run=0
publication_non_claim_review_present=1
freebsd_publication_non_claim_review_present=1
freebsd_package_publication_claimed=0
freebsd_pkg_repo_created=0
freebsd_pkg_repo_publish_run=0
freebsd_pkg_repo_sign_run=0
platform_build_evidence_accepted=0
freebsd_validation_promotion_blocked=1
freebsd_platform_build_evidence_accepted=0
package_validation_result_promoted=0
freebsd_validation_result_promoted=0
source_archive_policy_recorded=1
source_archive_created=0
source_archive_sha256_recorded=0
freebsd_distinfo_created=0
freebsd_make_makesum_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
freebsd_ports_tree_submission_claimed=0
freebsd_bugzilla_pr_claimed=0
freebsd_committer_review_claimed=0
poudriere_build_run=0
poudriere_run=0
make_package_run=0
portlint_run=0
package_artifact_created=0
package_artifact_sha256_recorded=0
install_on_host_run=0
freebsd_official_port_claimed=0
production_readiness_claimed=0
```

The draft payload remains intentionally narrow:

```
bin/latticra
%%DOCSDIR%%/README.md
```

The CLI reports no-effect status and disabled runtime behavior.

The source archive contract records the expected latticra-0.0.0.tar.gz distfile and future distinfo boundary, but it does not create an archive, generate distinfo, run make stage, run make package, run portlint, or accept FreeBSD build evidence.

The package-build gate is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md. It keeps package_build_gate_state=closed-no-effect and freebsd_build_allowed=0 until source, checksum, license, notice, ports environment, operator authorization, payload inspection, install/remove, and publication non-claim evidence exists.

The package-build environment contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md. It documents the FreeBSD jail or VM ports environment requirement while keeping freebsd_build_environment_provisioned=0, explicit_operator_build_authorization=0, and all FreeBSD build commands disabled.

The package artifact naming contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md. It records the future latticra-0.0.0.pkg artifact name while keeping freebsd_package_artifact_created=0, repository_package_artifact_write_allowed=0, and package_artifact_sha256_recorded=0.

The package payload inspection contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md. It records the expected FreeBSD payload paths while keeping freebsd_payload_inspection_run=0, package_payload_accepted=0, and package_artifact_created=0.

The package install/remove transcript contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove transcript requirements while keeping freebsd_package_install_run=0, freebsd_package_remove_run=0, and install_on_host_run=0.

The package publication non-claim review contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records local-only FreeBSD publication and official-port non-claims while keeping freebsd_package_publication_claimed=0, freebsd_pkg_repo_publish_run=0, and package validation promotion blocked.

The package validation promotion blocker matrix is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md. It ties FreeBSD source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping freebsd_platform_build_evidence_accepted=0 and freebsd_validation_result_promoted=0.

packaging/openbsd/README.md

Source title: OpenBSD Port Draft

Lines: 152 | Words: 437 | SHA-256: 0a1b1bd7939a590c4ec8547733b90efcc6d239117a45917fb5aa9a6c791ac774

OpenBSD Port Draft

Status: local-only port draft

This directory contains OpenBSD ports metadata experiments for Latticra.

The current port files are a local-only draft used by static guards. They are not an OpenBSD ports tree submission, not a ports@ review thread, not maintainer acceptance evidence, not a make package result, not a bulk build result, not a portcheck result, and not package-readiness evidence.

Where This Fits

- Documentation hub: ../../docs/README.md
- OpenBSD status: ../../docs/status/OPENBSD_ECOSYSTEM_INTEGRATION_STATUS.md

- Current status: ../../docs/status/CURRENT_STATUS.md

Current guarded files:

```
packaging/openbsd/README.md
packaging/openbsd/Makefile
packaging/openbsd/pkg/DESCR
packaging/openbsd/pkg/PLIST
docs/OPENBSD_PORT_STATIC_VALIDATION.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
docs/status/OPENBSD_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-openbsd-port-static-validation.sh
scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
.github/workflows/openbsd-port-static-validation.yml
.github/workflows/debian-freebsd-openbsd-source-archive-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-gate-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-environment-contract.yml
.github/workflows/debian-freebsd-openbsd-package-artifact-naming-contract.yml
.github/workflows/debian-freebsd-openbsd-package-payload-inspection-contract.yml
.github/workflows/debian-freebsd-openbsd-package-install-remove-transcript-contract.yml
.github/workflows/debian-freebsd-openbsd-package-publication-non-claim-review-contract.yml
.github/workflows/debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.ym
l
```

The static lane preserves:

```
local_only_draft=1
openbsd_port_draft_present=1
openbsd_port_static_validation_present=1
debian_freebsd_openbsd_source_archive_contract_present=1
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
temporary_openbsd_distfile_staged=1
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
openbsd_build_allowed=0
openbsd_ports_environment_documented=1
openbsd_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
environment_transcript_present=0
openbsd_artifact_name_pattern_recorded=1
openbsd_package_name=latticra-0.0.0.tgz
openbsd_package_artifact_created=0
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
openbsd_payload_expected_bin=bin/latticra
openbsd_payload_expected_doc=share/doc/latticra/README.md
openbsd_payload_inspection_run=0
package_payload_inspection_run=0
package_payload_accepted=0
openbsd_install_remove_transcript_required=1
openbsd_install_remove_transcript_present=0
openbsd_package_install_run=0
```

```
openbsd_package_remove_run=0
remove_on_host_run=0
publication_non_claim_review_present=1
openbsd_publication_non_claim_review_present=1
openbsd_package_publication_claimed=0
openbsd_pkg_repo_created=0
openbsd_pkg_repo_publish_run=0
platform_build_evidence_accepted=0
openbsd_validation_promotion_blocked=1
openbsd_platform_build_evidence_accepted=0
package_validation_result_promoted=0
openbsd_validation_result_promoted=0
source_archive_policy_recorded=1
source_archive_created=0
source_archive_sha256_recorded=0
openbsd_distinfo_created=0
openbsd_make_makesum_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
openbsd_ports_tree_submission_claimed=0
openbsd_ports_review_thread_claimed=0
openbsd_maintainer_acceptance_claimed=0
make_package_run=0
bulk_build_run=0
openbsd_bulk_build_run=0
portcheck_run=0
package_artifact_created=0
package_artifact_sha256_recorded=0
install_on_host_run=0
permit_package_enabled=0
openbsd_official_port_claimed=0
production_readiness_claimed=0
```

The draft payload remains intentionally narrow:

```
bin/latticra
share/doc/latticra/README.md
```

The CLI reports no-effect status and disabled runtime behavior.

The source archive contract records the expected latticra-0.0.0.tar.gz distfile and future distinfo boundary, but it does not create an archive, generate distinfo, run make plist, run make package, run portcheck, run a bulk build, or accept OpenBSD build evidence.

The package-build gate is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md. It keeps package_build_gate_state=closed-no-effect, openbsd_build_allowed=0, and PERMIT_PACKAGE=No until source, checksum, license, redistribution, notice, ports environment, operator authorization, payload inspection, install/remove, and publication non-claim evidence exists.

The package-build environment contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md. It documents the OpenBSD disposable ports environment requirement while keeping openbsd_build_environment_provisioned=0, explicit_operator_build_authorization=0, PERMIT_PACKAGE=No, and all OpenBSD build commands disabled.

The package artifact naming contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md. It records the future latticra-0.0.0.tgz artifact name while keeping openbsd_package_artifact_created=0, repository_package_artifact_write_allowed=0, package_artifact_sha256_recorded=0, and PERMIT_PACKAGE=No.

The package payload inspection contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md. It records the expected OpenBSD payload paths while keeping openbsd_payload_inspection_run=0, package_payload_accepted=0, package_artifact_created=0, and PERMIT_PACKAGE=No.

The package install/remove transcript contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove transcript requirements while keeping openbsd_package_install_run=0, openbsd_package_remove_run=0, install_on_host_run=0, and PERMIT_PACKAGE=No.

The package publication non-claim review contract is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records local-only OpenBSD publication and official-port non-claims while keeping openbsd_package_publication_claimed=0, openbsd_pkg_repo_publish_run=0, package validation promotion blocked, and PERMIT_PACKAGE=No.

The package validation promotion blocker matrix is recorded in ../../docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md. It ties OpenBSD source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping openbsd_platform_build_evidence_accepted=0, openbsd_validation_result_promoted=0, and PERMIT_PACKAGE=No.

packaging/opensuse/README.md

Source title: openSUSE Packaging Draft

Lines: 200 | Words: 423 | SHA-256: 6de3dd478ce6e23190b28dfcace0109d69d19b005b1edfab154e7b921df37f59

openSUSE Packaging Draft

Status: local-only packaging draft

This directory contains openSUSE-oriented packaging experiments and maintenance records for Latticra.

The current spec and .changes metadata are a local-only draft used by static guards. They are not an official openSUSE package, not SUSE endorsement evidence, not Open Build Service publication evidence, not a submit-request record, not a local osc build result, not a rpmlint result, and not package-readiness evidence.

Where this fits

- Documentation hub: ../../docs/README.md
- openSUSE workflow: ../../docs/OPENSUSE_DEVELOPER_WORKFLOW.md
- openSUSE readiness plan: ../../docs/OPENSUSE_READINESS_PLAN.md
- Current status: ../../docs/status/CURRENT_STATUS.md

Current guarded files:

```
packaging/opensuse/README.md
packaging/opensuse/latticra.spec
packaging/opensuse/latticra.changes
docs/OPENSUSE_READINESS_PLAN.md
docs/OPENSUSE_DEVELOPER_WORKFLOW.md
docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/status/OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md
scripts/test-opensuse-developer-workflow.sh
scripts/test-opensuse-local-rpm-static-validation.sh
scripts/test-opensuse-rpmlint-osc-availability.sh
scripts/test-opensuse-rpmlint-static-spec-lane.sh
```

```
scripts/test-opensuse-rpmlint-findings-classification.sh
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-developer-workflow.yml
.github/workflows/opensuse-local-rpm-static-validation.yml
.github/workflows/opensuse-rpmlint-osc-availability.yml
.github/workflows/opensuse-rpmlint-static-spec-lane.yml
.github/workflows/opensuse-rpmlint-findings-classification.yml
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

The static lane preserves:

```
local_only_draft=1
opensuse_maintenance_lane_present=1
opensuse_rpmlint_osc_availability_lane_present=1
opensuse_rpmlint_static_spec_lane_present=1
opensuse_rpmlint_findings_classification_present=1
opensuse_source_archive_reproducibility_contract_present=1
opensuse_source_archive_fixture_lane_present=1
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_local_rpm_build_gate_contract_present=1
opensuse_local_rpm_build_environment_contract_present=1
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
temporary_rpm_topdir_handoff_lane_present=1
opensuse_rpm_build_gate_state=closed-no-effect
opensuse_rpm_build_environment_contract_state=specified-no-effect
opensuse_rpm_artifact_naming_contract_state=specified-no-effect
opensuse_rpm_payload_inspection_contract_state=specified-no-effect
opensuse_rpm_install_remove_transcript_contract_state=specified-no-effect
rpm_artifact_naming_contract_present=1
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_contract_present=1
publication_non_claim_review_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
payload_inspection_contract_present=1
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
opensuse_clean_build_environment_documented=1
opensuse_build_environment_provisioned=0
osc_build_environment_provisioned=0
explicit_operator_build_authorization=0
environment_transcript_present=0
rpmbuild_allowed=0
osc_build_allowed=0
spec_cleaner_allowed=0
rpm_artifact_created=0
rpmbuild_run_required=0
rpmbuild_run=0
rpmbuild_ba_run=0
rpmbuild_bb_run=0
rpmbuild_bs_run=0
osc_build_run=0
accepted_rpmlint_transcript_present=0
expected_draft_findings_count_recorded=0
unexpected_findings_count_recorded=0
source_archive_policy_recorded=1
source_archive_transcript_present=1
source_archive_created=1
source_archive_sha256_recorded=1
source_archive_reproducible=1
source_archive_generated_twice=1
```

```
source_archive_repeated_sha256_match=1
source_archive_accepted_for_build=0
temporary_rpm_topdir_created=1
temporary_rpm_sources_archive_staged=1
temporary_rpm_specs_spec_staged=1
temporary_rpm_specs_changes_staged=1
temporary_rpm_source_sha256_preserved=1
temporary_rpm_source_listing_preserved=1
rpm_source_artifact_name_pattern_recorded=1
rpm_binary_artifact_name_pattern_recorded=1
rpm_source_package_name=latticra-0.0.0-0.local.src.rpm
rpm_binary_package_name_pattern=latticra-0.0.0-0.local.${RPM_ARCH}.rpm
repository_rpm_artifact_write_allowed=0
source_rpm_artifact_created=0
binary_rpm_artifact_created=0
rpm_artifact_sha256_recorded=0
rpm_payload_expected_bin=/usr/bin/latticra
rpm_payload_expected_doc=/usr/share/doc/packages/latticra/README.md
source_rpm_payload_inspection_run=0
binary_rpm_payload_inspection_run=0
rpm_payload_inspection_run=0
rpm_payload_accepted=0
rpm_install_remove_transcript_present=0
rpm_install_remove_disposable_environment_required=1
rpm_package_install_run=0
rpm_package_remove_run=0
rpm_zypper_install_run=0
rpm_zypper_remove_run=0
rpm_cli_install_run=0
rpm_cli_remove_run=0
rpm_removed_from_host=0
host_install_allowed=0
host_remove_allowed=0
host_mutation_allowed=0
service_state_change_allowed=0
rpm_validation_result_promoted=0
obs_project_created=0
obs_package_created=0
obs_repository_created=0
obs_source_link_created=0
osc_branch_run=0
osc_commit_run=0
osc_submitreq_run=0
osc_request_accepted=0
obs_build_result_claimed=0
spec_cleaner_run=0
rpmlint_package_readiness_claimed=0
opensuse_obs_publication_claimed=0
opensuse_submit_request_claimed=0
opensuse_official_package_claimed=0
suse_endorsement_claimed=0
opensuse_factory_submission_claimed=0
opensuse_factory_acceptance_claimed=0
opensuse_leap_submission_claimed=0
production_readiness_claimed=0
```

The local RPM build environment contract is recorded in ../../docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md. It documents the disposable openSUSE build environment requirement while keeping opensuse_build_environment_provisioned=0, explicit_operator_build_authorization=0, environment_transcript_present=0, and all openSUSE build commands disabled.

The RPM artifact naming contract is recorded in ../../docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md. It records future latticra-0.0.0-0.local.src.rpm and latticra-0.0.0-0.local.${RPM_ARCH}.rpm names while keeping repository_rpm_artifact_write_allowed=0, rpm_artifact_created=0, and rpm_artifact_sha256_recorded=0.

The RPM payload inspection contract is recorded in ../../docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md. It records expected source RPM and binary RPM payload checks while keeping rpm_payload_inspection_run=0, rpm_payload_accepted=0, and rpm_artifact_created=0.

The RPM install/remove transcript contract is recorded in ../../docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove evidence while keeping rpm_install_remove_transcript_present=0, rpm_package_install_run=0, rpm_package_remove_run=0, and host_mutation_allowed=0.

The OBS publication non-claim review contract is recorded in ../../docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records Open Build Service, submit-request, official-package, and SUSE endorsement non-claims while keeping obs_project_created=0, osc_commit_run=0, osc_submitreq_run=0, and rpm_validation_result_promoted=0.

The draft payload remains intentionally narrow:

```
/usr/bin/latticra
/usr/share/doc/packages/latticra/README.md
```

The CLI reports no-effect status and disabled runtime behavior.

The .changes file records local-only openSUSE maintenance history. It is not accepted Open Build Service history, package submission evidence, or production maintenance evidence.

packaging/ubuntu/README.md

Source title: Ubuntu Packaging Draft

Lines: 183 | Words: 257 | SHA-256: 80559a62089e8419702e0190df8fe68bbfd5e56ca54a326327ae045cb16e594e

Ubuntu Packaging Draft

Status: local-only packaging draft

This directory contains Ubuntu-oriented packaging experiments for Latticra.

The current debian metadata is a local-only draft used by static guards. It is not an Ubuntu archive package, not a PPA package, not Canonical approval evidence, not a Launchpad build record, not a lintian result, and not package-readiness evidence.

Where this fits

- Documentation hub: ../../docs/README.md
- Ubuntu workflow: ../../docs/UBUNTU_DEVELOPER_WORKFLOW.md
- Ubuntu readiness plan: ../../docs/UBUNTU_READINESS_PLAN.md
- Current status: ../../docs/status/CURRENT_STATUS.md

Current guarded files:

```
packaging/ubuntu/README.md
packaging/ubuntu/debian/control
packaging/ubuntu/debian/rules
packaging/ubuntu/debian/changelog
packaging/ubuntu/debian/copyright
packaging/ubuntu/debian/install
packaging/ubuntu/debian/source/format
scripts/test-ubuntu-local-deb-static-validation.sh
scripts/test-ubuntu-lintian-availability.sh
docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md
docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md
docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md
docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md
docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md
docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md
```

```
docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md
docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md
docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md
scripts/ubuntu-package-notice-inventory.sh
scripts/test-ubuntu-package-notice-inventory.sh
scripts/test-ubuntu-doc-payload-license-review-contract.sh
scripts/test-ubuntu-third-party-material-review-contract.sh
scripts/test-ubuntu-generated-artifact-notice-review-contract.sh
scripts/test-ubuntu-notice-file-decision-contract.sh
scripts/test-ubuntu-debian-copyright-notice-mapping-contract.sh
scripts/test-ubuntu-trademark-notice-boundary-contract.sh
scripts/test-ubuntu-release-artifact-notice-requirements-contract.sh
scripts/test-ubuntu-package-notice-promotion-gate-contract.sh
scripts/test-ubuntu-package-license-promotion-gate-contract.sh
scripts/test-ubuntu-lintian-static-metadata-contract.sh
scripts/test-ubuntu-package-notice-review-contract.sh
scripts/test-ubuntu-package-license-review-contract.sh
scripts/test-ubuntu-local-deb-build-transcript-contract.sh
scripts/test-ubuntu-local-deb-build-transcript-acceptance-gate-contract.sh
scripts/test-ubuntu-local-deb-install-remove-evidence-contract.sh
scripts/test-ubuntu-source-package-evidence-contract.sh
scripts/test-ubuntu-upload-signing-authority-evidence-contract.sh
scripts/test-ubuntu-ppa-archive-publication-gate-contract.sh
.github/workflows/ubuntu-local-deb-static-validation.yml
.github/workflows/ubuntu-lintian-availability.yml
.github/workflows/ubuntu-package-notice-inventory.yml
.github/workflows/ubuntu-doc-payload-license-review-contract.yml
.github/workflows/ubuntu-third-party-material-review-contract.yml
.github/workflows/ubuntu-generated-artifact-notice-review-contract.yml
.github/workflows/ubuntu-notice-file-decision-contract.yml
.github/workflows/ubuntu-debian-copyright-notice-mapping-contract.yml
.github/workflows/ubuntu-trademark-notice-boundary-contract.yml
.github/workflows/ubuntu-release-artifact-notice-requirements-contract.yml
.github/workflows/ubuntu-package-notice-promotion-gate-contract.yml
.github/workflows/ubuntu-package-license-promotion-gate-contract.yml
.github/workflows/ubuntu-lintian-static-metadata-contract.yml
.github/workflows/ubuntu-package-notice-review-contract.yml
.github/workflows/ubuntu-package-license-review-contract.yml
.github/workflows/ubuntu-local-deb-build-transcript-contract.yml
.github/workflows/ubuntu-local-deb-build-transcript-acceptance-gate-contract.yml
.github/workflows/ubuntu-local-deb-install-remove-evidence-contract.yml
.github/workflows/ubuntu-source-package-evidence-contract.yml
.github/workflows/ubuntu-upload-signing-authority-evidence-contract.yml
.github/workflows/ubuntu-ppa-archive-publication-gate-contract.yml
```

The static lane preserves:

```
local_only_draft=1
deb_artifact_created=0
dpkg_buildpackage_run_required=0
debuild_run_required=0
lintian_run_required=0
lintian_availability_lane_present=1
package_notice_inventory_present=1
package_notice_inventory_report_present=1
doc_payload_license_review_contract_present=1
doc_payload_license_review_status=resolved-cc-by-4.0
documentation_license_decision_present=1
doc_payload_license_decision_recorded=1
third_party_material_review_contract_present=1
third_party_material_review_status=blocked-pending-third-party-material-review
third_party_material_inventory_recorded=1
third_party_material_inventory_reviewed=0
third_party_notice_requirements_recorded=0
generated_artifact_notice_review_contract_present=1
generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
generated_artifact_notice_reviewed=0
generated_artifact_notice_requirements_recorded=0
notice_file_decision_contract_present=1
notice_file_decision_status=blocked-pending-notice-file-decision
notice_file_present=0
notice_file_decision_recorded=0
debian_copyright_notice_mapping_contract_present=1
debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
debian_copyright_notice_mapping_reviewed=0
trademark_notice_boundary_contract_present=1
trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
trademark_notice_boundary_recorded=0
release_artifact_notice_requirements_contract_present=1
release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requirements
```

```
release_artifact_notice_requirements_recorded=0
package_notice_promotion_gate_contract_present=1
package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
package_notice_promotion_gate_unblocked=0
package_license_promotion_gate_contract_present=1
package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
package_license_promotion_gate_unblocked=0
lintian_static_metadata_contract_present=1
lintian_static_metadata_status=blocked-pending-package-license-promotion
lintian_static_metadata_run=0
package_license_review_contract_present=1
package_license_review_status=resolved-license-expression-recorded
package_notice_review_contract_present=1
package_notice_review_status=blocked-pending-notice-review
license_expression_candidate_recorded=1
packaging_license_expression_updated=1
local_deb_build_transcript_contract_present=1
local_deb_build_transcript_present=0
local_deb_build_transcript_acceptance_gate_contract_present=1
local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata-and-bu
ild-transcript
local_deb_build_transcript_accepted=0
local_deb_install_remove_evidence_contract_present=1
local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
deb_installed_on_host=0
deb_removed_from_host=0
source_package_evidence_contract_present=1
source_package_evidence_status=blocked-pending-accepted-build-transcript
dpkg_source_run=0
dpkg_buildpackage_source_run=0
source_package_created=0
source_package_uploaded=0
ubuntu_source_package_evidence_unblocked=0
upload_signing_authority_evidence_contract_present=1
upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
upload_target_kind_recorded=0
launchpad_account_recorded=0
ppa_or_archive_target_reviewed=0
gpg_signing_key_fingerprint_recorded=0
upload_command_non_claims_reviewed=0
ubuntu_upload_signing_authority_evidence_unblocked=0
ppa_archive_publication_gate_contract_present=1
ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
source_package_created=0
source_package_digest_recorded=0
debsign_command_recorded=0
dput_command_recorded=0
launchpad_upload_run=0
ubuntu_publication_gate_unblocked=0
ppa_claimed=0
ubuntu_archive_ready=0
production_readiness_claimed=0
```

The draft payload remains intentionally narrow:

```
usr/bin/latticra
usr/share/doc/latticra/README.md
```

The CLI reports no-effect status and disabled runtime behavior.

Part XII - Project Notes, Strategy, and Documentation Maintenance

- docs/DOCUMENTATION_CHANGE_REVIEW_PACKET.md - Documentation Change Review Packet
- docs/DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md - Documentation Drift Response Playbook
- docs/DOCUMENTATION_LICENSE.md - Latticra Documentation License
- docs/DOCUMENTATION_MAINTENANCE.md - Documentation Maintenance
- docs/DOCUMENTATION_STYLE_GUIDE.md - Documentation Style Guide
- docs/DOCUMENTATION_TRACEABILITY_MATRIX.md - Documentation Traceability Matrix

- docs/DOCUMENTATION_VALIDATION_PLAYBOOK.md - Documentation Validation Playbook
- docs/latticra/DOCUMENTATION_MAP.md - Latticra Documentation Map
- docs/LICENSE_MIGRATION_PLAN.md - Latticra License Migration Plan
- docs/LICENSE_POLICY.md - Latticra License Policy
- docs/project_notes/CURRENT_DIRECTION.md - Latticra Current Direction
- docs/project_notes/README.md - Latticra Project Notes
- docs/project_notes/UPCOMING_WORK.md - Latticra Upcoming Work
- docs/strategy/2026-05-19-1845-cdt-strategy-estimate-review.md - Latticra Strategy Estimate Review
- docs/strategy/2026-05-25-1951-cdt-strategy-posture-refresh.md - Latticra Strategy Posture Refresh
- docs/strategy/2026-05-25-2017-cdt-capability-promotion-gate-map.md - Latticra Capability Promotion Gate Map
- docs/strategy/2026-05-25-2025-cdt-capability-promotion-packet-template.md - Latticra Capability Promotion Packet Template
- docs/strategy/2026-05-25-2114-cdt-public-product-readiness-promotion-packet.md - Latticra Public Product Readiness Promotion Packet
- docs/strategy/2026-05-26-1702-cdt-overall-strategy-priority-map.md - Latticra Overall Strategy Priority Map After Panel-Guided Planning Completion
- docs/strategy/2026-05-26-1707-cdt-console-read-only-host-inventory-workflow-packet.md - Latticra Console Read-Only Host Inventory Workflow Packet
- docs/strategy/2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md - Latticra Console Read-Only Host Inventory Acceptance Checklist
- docs/strategy/2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md - Latticra Console Read-Only Host Inventory Evidence Bundle Template
- docs/strategy/2026-05-26-1757-cdt-console-read-only-host-inventory-non-claim-review-template.md - Latticra Console Read-Only Host Inventory Non-Claim Review Template
- docs/strategy/2026-05-26-1919-cdt-console-read-only-host-inventory-public-entrypoint-review-template.md - Latticra Console Read-Only Host Inventory Public-Entrypoint Review Template
- docs/strategy/2026-05-26-1956-cdt-console-read-only-host-inventory-estimate-impact-review-template.md - Latticra Console Read-Only Host Inventory Estimate-Impact Review Template
- docs/strategy/2026-05-26-2019-cdt-console-read-only-host-inventory-planning-completion-checkpoint.md - Latticra Console Read-Only Host Inventory Planning Completion Checkpoint
- docs/strategy/README.md - Latticra Strategy Index

docs/DOCUMENTATION_CHANGE_REVIEW_PACKET.md

Source title: Documentation Change Review Packet

Lines: 143 | Words: 654 | SHA-256: 49bb7548ee5f7923c0621c0549c152a642602f7e201cea91573ef8e89e8e3438

Documentation Change Review Packet

Status: active documentation review template Last updated: 2026-05-26 CDT Scope: documentation-only changes that affect public wording, reader routes, product-facing pages, non-claims, status mirrors, platform posture, installer posture, package posture, subsystem landing pages, or documentation navigation.

Purpose

Use this packet before merging or publishing documentation changes that could change how readers understand Latticra.

The packet is intentionally lightweight. It exists to make sure a documentation change names its claim boundary, source records, mirrors, non-claims, validation, and rollback path before public wording drifts.

When To Use

Use this packet when a change touches any of these:

- ../README.md
- README.md
- STATUS.md
- status/CURRENT_STATUS.md
- status/README.md
- public HTML pages such as index.html, map.html, status.html, roadmap.html, validation.html, or security.html
- installer, platform, package, security, or subsystem landing pages
- public estimates, public claims, product copy, non-claim wording, or reader routes

Do not use this packet as evidence for a stronger capability. It reviews documentation posture only.

Packet

Copy this section into the relevant review record, pull request, or status note.

```
documentation_change_review_packet:
  change_id:
  reviewer:
  review_date:
  change_type:
    public_wording:
    reader_route:
    status_mirror:
    estimate_mirror:
    platform_posture:
    installer_posture:
    package_posture:
    subsystem_landing_page:
    security_wording:
    non_claim_wording:
  touched_surfaces:
  source_records:
  current_claim:
  proposed_claim:
  evidence_level:
```

```
implementation_changed: no
behavior_changed: no
authority_changed: no
non_claims_preserved:
public_claims_ledger_checked:
product_documentation_cohesion_checked:
documentation_traceability_matrix_checked:
documentation_validation_playbook_checked:
documentation_drift_response_playbook_checked:
documentation_style_guide_checked:
documentation_maintenance_checked:
mirrors_checked:
validation_commands:
blocked_adjacent_claims:
rollback_or_demotion_path:
outcome:
```

Review Steps

1. Classify the change as wording-only, navigation-only, status mirror, estimate mirror, platform posture, installer posture, package posture, subsystem landing page, or security wording.
2. Identify the source records that make the wording true.
3. Compare the wording against PUBLIC_CLAIMS_LEDGER.md.
4. Compare the reader route against PRODUCT_DOCUMENTATION_COHESION.md.
5. Compare audience-specific paths against DOCUMENTATION_READER_JOURNEY_MAP.md when routes change.
6. Compare source records, mirrors, validation, and non-claim boundaries against DOCUMENTATION_TRACEABILITY_MATRIX.md.
7. Select validation commands with DOCUMENTATION_VALIDATION_PLAYBOOK.md.
8. Choose a drift-response, narrowing, or demotion path with DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md when records disagree.
9. Compare terminology, headings, dates, and replacement wording against DOCUMENTATION_STYLE_GUIDE.md.
10. Compare mirror requirements against DOCUMENTATION_MAINTENANCE.md.
11. Confirm that adjacent non-claims remain visible.
12. Run the narrowest relevant validation commands.
13. Name how to demote or roll back the wording if the evidence changes.

Claim Classification

Classification	Meaning	Required handling
Navigation-only	Links, entry points, or ordering changed without claim change.	Check local links and affected reader routes.
Wording-only	Existing evidence is described more clearly without capability change.	Check claims ledger, non-claims, and source records.
Status mirror	Public status wording is mirrored from an existing status record.	Check `STATUS.md`, `docs/status/CURRENT_STATUS.md`, and status index alignment.
Estimate mirror	Planning estimate wording or displayed values changed.	Check estimate source alignment and static HTML mirrors.
Platform posture	Fedora, Ubuntu, Debian, FreeBSD, OpenBSD, openSUSE, macOS, installer, or package wording changed.	Preserve local-only or no-effect boundaries unless official evidence exists.
Security wording	Security, threat-model, runtime authority, sandbox, malware, or ransomware wording changed.	Preserve security non-claims and safe-testing boundaries.
Claim promotion	A public claim becomes stronger.	Stop unless contract, implementation, validation, status, non-claim update, and public-entry alignment all exist.

Minimum Validation

For most documentation-only packets:

```
git diff --check
sh scripts/test-project-strategy-status-framework.sh
```

Add these when relevant:

```
sh scripts/test-current-estimate-table-source-alignment.sh
sh scripts/test-latticra-seal-docs.sh
sh scripts/test-fedora-developer-workflow.sh
sh scripts/test-ubuntu-developer-workflow.sh
sh scripts/test-opensuse-developer-workflow.sh
```

Add the exact subsystem or milestone guard named by the source record when a specialized status or public-entry alignment changes.

Use DOCUMENTATION_VALIDATION_PLAYBOOK.md for guard selection, local link checks, and failure handling.

Outcome Language

Use one of these outcome labels:

```
accepted_navigation_only
accepted_wording_only
accepted_status_mirror
accepted_estimate_mirror
accepted_local_only_platform_posture
blocked_claim_promotion
blocked_missing_source_record
blocked_missing_validation
blocked_non_claim_drift
blocked_public_entry_drift
```

Boundary

This packet does not change source behavior, installer authority, package authority, runtime authority, shell guard behavior, workflow permissions, security posture, or product readiness.

If the proposed wording requires implementation or validation to become true, the documentation change must stay blocked until that evidence exists.

docs/DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md

Source title: Documentation Drift Response Playbook

Lines: 107 | Words: 907 | SHA-256: f234d5c99744c460a00d0604d8008ad3ce686d1d55feee42a10a53add455c835

Documentation Drift Response Playbook

Status: active documentation drift-response playbook Last updated: 2026-05-26 CDT Scope: public wording drift, status mirror drift, estimate drift, non-claim drift, link drift, platform posture drift, security wording drift, stale evidence, and claim demotion.

Purpose

This playbook defines what to do when Latticra documentation disagrees with itself.

Use it when a public page, source record, status mirror, estimate table, package README, installer guide, security page, or subsystem landing page appears to overstate, understate, or contradict the current evidence.

This playbook does not promote claims. It narrows or aligns documentation until the evidence supports stronger wording.

Drift Types

Drift type	Signal	First response
Link drift	A local link is broken or points to the wrong record.	Fix the link or remove it if no source exists.
Source-record drift	A public statement has no source record.	Narrow or remove the statement until a source record exists.
Status mirror drift	`README.md`, `STATUS.md`, `docs/status/CURRENT_STATUS.md`, or public HTML pages disagree.	Use the narrowest current status wording and align mirrors.
Estimate drift	Estimate values differ across public tables or static pages.	Treat the current estimate source record as authoritative and run the estimate alignment guard.
Non-claim drift	A public page drops a required non-claim or weakens blocked language.	Restore the non-claim near the claim it constrains.
Product-copy drift	Product-facing wording implies production, approval, security, or runtime authority.	Replace it with wording from the claims ledger and style guide.
Platform drift	Package or platform docs imply official approval, archive readiness, or production installability.	Restore local-only or no-effect posture.
Security drift	Security docs imply protection, prevention, sandboxing, certification, or hardening without evidence.	Restore security non-claims and source-tracked wording.
Validation drift	A guard fails because the docs and expected public posture disagree.	Fix the docs or mirrors; do not edit the guard for convenience.
Stale evidence	A record relies on older evidence contradicted by newer status.	Demote the public wording to the latest supported status.

Response Order

When drift appears:

1. Read the public surface that contains the questionable wording.
2. Find the source record through DOCUMENTATION_TRACEABILITY_MATRIX.md.
3. Check allowed and blocked wording in PUBLIC_CLAIMS_LEDGER.md.
4. Check terminology in DOCUMENTATION_STYLE_GUIDE.md.
5. Select validation commands with DOCUMENTATION_VALIDATION_PLAYBOOK.md.
6. Use DOCUMENTATION_CHANGE_REVIEW_PACKET.md if public wording, mirrors, estimates, platform posture, security wording, or non-claims change.
7. Update only the surfaces whose reader-facing claim changed.
8. Run the selected validation checks.

Demotion Rule

Demotion is required when a public statement is stronger than the evidence.

Use this pattern:

```
strong claim -&gt; current evidence-bound claim -&gt; explicit blocked adjacent claim
```

Examples:

Strong claim	Demoted wording	Blocked adjacent claim
Latticra is secure.	Latticra documents defensive boundaries and security non-claims.	No production security guarantee.
The installer is ready.	The Panel path is guarded and user-local for the documented workflow.	No root installer or daily-driver installer claim.
Fedora support is ready.	The Fedora lane has local-only package and validation records.	No Fedora approval or distribution-readiness claim.
Seal protects AI agents.	Latticra Seal records report-only AI-era tool-boundary evidence.	No active AI-agent control or host protection claim.
Runtime enforcement exists.	Runtime Boundary classification and reports are denied by default.	No runtime authority or effect-performing execution claim.

Alignment Actions

Situation	Action	Validation

Public page is broader than status.	Narrow the public page to the status record.	Public-entry guard and link check.
Status record is broader than implementation evidence.	Narrow the status record and public mirrors.	Exact source guard plus public-entry guard.
Estimate mirror differs.	Update mirror from the estimate source record.	Estimate source-alignment guard.
Platform docs omit local-only status.	Add local-only and no-approval wording.	Platform workflow guard plus exact platform guard.
Security docs overpromise.	Restore non-claims and source-tracked wording.	Exact security or threat-model guard.
Source record is missing.	Remove or block the claim until the source record exists.	Link check and review packet outcome.
Validation command is missing.	Mark the claim blocked for missing validation.	Review packet outcome `blocked_missing_validation`.

Do Not

Do not:

- broaden a public claim to make a product page sound better;
- edit a guard script only to make documentation pass;
- hide non-claims in deep source files when a public page implies the adjacent claim;
- treat subsystem progress as overall product readiness;
- treat local static validation as platform approval;
- treat report-only evidence as runtime enforcement;
- treat planning estimates as release promises;
- claim security, safety, hardening, prevention, approval, or certification without explicit source evidence.

Review Packet Outcomes

Use these outcomes from DOCUMENTATION_CHANGE_REVIEW_PACKET.md when drift is found:

```
blocked_claim_promotion
blocked_missing_source_record
blocked_missing_validation
blocked_non_claim_drift
blocked_public_entry_drift
accepted_wording_only
accepted_status_mirror
accepted_local_only_platform_posture
```

Boundary

This playbook changes documentation posture only.

It does not change implementation behavior, installer authority, package authority, runtime authority, workflow permissions, security posture, public estimates, or product readiness.

docs/DOCUMENTATION_LICENSE.md

Source title: Latticra Documentation License

Lines: 60 | Words: 158 | SHA-256: ed0d53d83fc179846e77b6cc0fe82d86a59f0009ab66089608b3355227858f95

Latticra Documentation License

Status: active documentation license decision Scope: documentation, handbooks, architecture notes, policy notes, public project notes, and reader-facing project material unless otherwise marked.

Decision

Latticra documentation is licensed under:

```
CC-BY-4.0
```

The local license text is:

```
LICENSES/CC-BY-4.0.txt
```

Covered material

This documentation license applies to:

```
README.md
STATUS.md
SECURITY.md
CONTRIBUTING.md
docs/**
installer/README.md
installer/INSTALLER_ROADMAP.md
installer/docs/**
packaging/*/README.md
```

unless a file or embedded notice says otherwise.

Boundaries

Embedded code snippets, copied source excerpts, generated build outputs, third-party material, images, fonts, logos, and other bundled assets may have separate license or notice requirements.

The Latticra and Bryforge names, logos, marks, slogans, and distinctive identity are not licensed under CC-BY-4.0. They remain governed by:

```
TRADEMARK_POLICY.md
```

SPDX

New documentation files may use:

```
SPDX-License-Identifier: CC-BY-4.0
```

Existing documentation is covered by this path-level notice unless otherwise marked.

Non-Claims

This document records the project documentation license decision and is not legal advice.

docs/DOCUMENTATION_MAINTENANCE.md

Source title: Documentation Maintenance

Lines: 129 | Words: 702 | SHA-256: d6513be1fe58d8d2af815f706497857bf2c0a026b4d9244ac33066071adeffa7

Documentation Maintenance

Status: active documentation maintenance guide Last updated: 2026-05-26 CDT Scope: public documentation entry points, status mirrors, estimate sources, static HTML mirrors, platform guides, and documentation-only validation.

Purpose

This guide defines how Latticra documentation should stay cohesive as the project changes.

Use it when a change affects public wording, capability posture, estimates, platform readiness, security claims, installer authority, package claims, subsystem status, or reader navigation.

Source Of Truth

```
| Subject | Source of truth | Mirrors that may need updates |
| --- | --- | --- |
| Project identity and broad non-claims | [`../README.md`](../README.md),
[`REAL_SYSTEM_CONTRACT.md`](REAL_SYSTEM_CONTRACT.md), [`NON_CLAIMS.md`](NON_CLAIMS.md) |
[`map.html`](map.html),
[`latticra-system-substrate/README.md`](latticra-system-substrate/README.md),
[`latticra/DOCUMENTATION_MAP.md`](latticra/DOCUMENTATION_MAP.md) |
| Current status and estimates | [`../STATUS.md`](../STATUS.md),
[`status/CURRENT_STATUS.md`](status/CURRENT_STATUS.md), [`status/CURRENT_ESTIMATE_TABLE_SOURCE_A
LIGNMENT.md`](status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md), [`status/CURRENT_ESTIMATE_MATH
EMATICAL_REBASE_2026_05_26.md`](status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md) |
[`../README.md`](../README.md), [`status.html`](status.html), [`roadmap.html`](roadmap.html),
[`index.html`](index.html) |
| Documentation navigation | [`README.md`](README.md),
[`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md) | [`../README.md`](../README.md),
[`map.html`](map.html), [`project_notes/README.md`](project_notes/README.md) |
| Documentation traceability |
[`DOCUMENTATION_TRACEABILITY_MATRIX.md`](DOCUMENTATION_TRACEABILITY_MATRIX.md) |
[`README.md`](README.md), [`map.html`](map.html), [`index.html`](index.html), review packets |
| Documentation validation |
[`DOCUMENTATION_VALIDATION_PLAYBOOK.md`](DOCUMENTATION_VALIDATION_PLAYBOOK.md) | Review packets,
 status records, platform docs, subsystem landing pages |
| Documentation drift response |
[`DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md`](DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md) | Public
entry points, status mirrors, estimate mirrors, platform docs, security docs |
| Documentation style and terminology |
[`DOCUMENTATION_STYLE_GUIDE.md`](DOCUMENTATION_STYLE_GUIDE.md),
[`PUBLIC_CLAIMS_LEDGER.md`](PUBLIC_CLAIMS_LEDGER.md) | [`../README.md`](../README.md),
[`index.html`](index.html), [`map.html`](map.html), platform README files, subsystem landing
pages |
| Security posture | [`../SECURITY.md`](../SECURITY.md),
[`HIGH_ASSURANCE_SECURITY_BASELINE.md`](HIGH_ASSURANCE_SECURITY_BASELINE.md),
[`DEFENSIVE_THREAT_MODEL_CONTRACT.md`](DEFENSIVE_THREAT_MODEL_CONTRACT.md) |
[`security.html`](security.html), [`../README.md`](../README.md),
[`status/CURRENT_STATUS.md`](status/CURRENT_STATUS.md) |
| Installer authority | [`../installer/README.md`](../installer/README.md),
[`../installer/docs/README.md`](../installer/docs/README.md),
[`PRODUCTION_INSTALLER_READINESS_CONTRACT.md`](PRODUCTION_INSTALLER_READINESS_CONTRACT.md) |
[`QUICK_START_CHEATSHEET.md`](QUICK_START_CHEATSHEET.md), [`validation.html`](validation.html),
[`../README.md`](../README.md) |
| Platform and packaging posture | Platform workflow and package README files |
[`QUICK_START_CHEATSHEET.md`](QUICK_START_CHEATSHEET.md), [`validation.html`](validation.html),
[`status/CURRENT_STATUS.md`](status/CURRENT_STATUS.md), [`../README.md`](../README.md) |
```

Public-Entry Checklist

When a milestone changes capability posture or public wording, review these files:

```
README.md
STATUS.md
docs/README.md
docs/FOUNDATION_INDEX.md
docs/status/CURRENT_STATUS.md
docs/status/README.md
docs/project_notes/README.md
docs/map.html
docs/status.html
docs/roadmap.html
docs/index.html
```

Do not update every file mechanically. Update only the files whose reader-facing claim actually changes.

Use DOCUMENTATION_CHANGE_REVIEW_PACKET.md when a public-entry change needs an explicit record of touched surfaces, source records, claim classification, mirrors, validation commands, and rollback or demotion path.

Use DOCUMENTATION_TRACEABILITY_MATRIX.md when a public-entry change needs a source-record, mirror, validation, or non-claim lookup.

Use DOCUMENTATION_VALIDATION_PLAYBOOK.md when choosing hygiene, link, public-entry, estimate, platform, subsystem, or claim-promotion checks.

Use DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md when public pages, source records, status mirrors, estimate tables, platform docs, security wording, or non-claims disagree.

Use DOCUMENTATION_STYLE_GUIDE.md when a public-entry change edits terminology, status labels, date style, heading shape, link style, platform names, or replacement wording.

Estimate Rule

Planning estimates are posture markers, not release promises.

When an estimate changes:

1. Update the source record under docs/status/.
2. Mirror the current public table in README.md, STATUS.md, and docs/status/CURRENT_STATUS.md.
3. Update static HTML summary values in docs/index.html, docs/status.html, and docs/roadmap.html when those pages display the changed values.
4. Run the current estimate table source-alignment guard.

Non-Claim Rule

Never promote these claims without implementation, tests, status records, and public-entry alignment:

```
production runtime
host protection
hardened sandbox
malware prevention
ransomware prevention
kernel enforcement
root installer
network authority
package approval
official distribution readiness
OS replacement
certification
compliance
production cryptography
AI-agent execution control
```

Use narrower language when evidence is narrow. For example, a disposable VM transcript can support a bounded local validation claim without supporting daily-driver, production, security, update-safety, recovery-safety, or distribution-readiness claims.

Platform Documentation Rule

Platform and package documentation must state the current authority boundary:

- Fedora: local-only unless Fedora approval evidence exists.
- Ubuntu: local-only unless Ubuntu archive or PPA evidence exists.
- Debian: local-only unless Debian archive evidence exists.
- openSUSE: local-only unless Open Build Service publication or official package evidence exists.
- FreeBSD and OpenBSD: local-only unless official port evidence exists.
- macOS: no-effect or user-local only unless app-bundle write, signing, notarization, install, verification, reset, and uninstall evidence exists.

Validation

For documentation-only work, run the narrowest relevant guard set:

```
git diff --check
sh scripts/test-project-strategy-status-framework.sh
```

```
sh scripts/test-current-estimate-table-source-alignment.sh
sh scripts/test-latticra-seal-docs.sh
```

Add platform guards when platform docs move:

```
sh scripts/test-fedora-developer-workflow.sh
sh scripts/test-ubuntu-developer-workflow.sh
sh scripts/test-opensuse-developer-workflow.sh
```

Add the exact milestone guard named by the status record when changing a milestone-specific status or public-entry alignment.

For command selection and failure handling, use DOCUMENTATION_VALIDATION_PLAYBOOK.md.

Documentation-Only Boundary

Documentation-only changes should not change source behavior, installer authority, package authority, runtime authority, shell guard behavior, workflow permissions, or security posture. If a documentation update requires code or guard changes to become true, keep the documentation claim blocked until that implementation evidence exists.

docs/DOCUMENTATION_STYLE_GUIDE.md

Source title: Documentation Style Guide

Lines: 202 | Words: 808 | SHA-256: dcd7dc157007f1c0a8fcf20483724e817e39875749179bc056bf9556f351a41c

Documentation Style Guide

Status: active documentation style guide Last updated: 2026-05-26 CDT Scope: public Markdown, public HTML copy, status records, installer docs, package docs, subsystem landing pages, strategy records, and review packets.

Purpose

This guide keeps Latticra documentation consistent at the wording level.

Use it with DOCUMENTATION_GLOSSARY.md, PUBLIC_CLAIMS_LEDGER.md, PRODUCT_DOCUMENTATION_COHESION.md, DOCUMENTATION_MAINTENANCE.md, and DOCUMENTATION_CHANGE_REVIEW_PACKET.md.

This guide does not decide whether a claim is true. It decides how true claims should be written once the source records support them.

Use DOCUMENTATION_GLOSSARY.md for definitions of shared posture, validation, platform, and reader terms.

Core Voice

Latticra documentation should be:

```
specific
evidence-bound
plainspoken
operator-visible
careful about claims
clear about blocked effects
```

Prefer short, concrete sentences over broad promise language.

Canonical Project Description

Use this shape when a short project description is needed:

```
Latticra is an early-stage, evidence-bound systems architecture repository for contract-first
authority, no-effect reporting, guarded local validation, and AI-era tool-boundary planning.
```

Do not shorten this into "security product", "AI security platform", "operating system", "installer", or "runtime enforcement system".

Canonical Terms

```
| Use | Avoid |
| --- | --- |
| Latticra | Lattice OS, Latticra OS, Latticra Linux |
| Latticra Panel | the installer, the app, the product |
| Latticra Seal | Seal security platform, Seal enforcement layer |
| Lat | L, .l, LatticraScript |
| LIR | execution IR, runtime IR |
| L-UI | terminal UI runtime, shell UI |
| Runtime Boundary | runtime enforcement, sandbox runtime |
| Nucleus | task executor, agent runner |
| no-effect | harmless, safe, guaranteed safe |
| report-only | enforcement, protection |
| guarded user-local | production install, root install |
| local-only package draft | official package, distribution package |
| public status | release status, readiness guarantee |
```

Platform Names

Use the official spelling in public docs:

```
Fedora
Ubuntu
Debian
FreeBSD
OpenBSD
openSUSE
macOS
GitHub Pages
Open Build Service
```

Use "local-only" for package lanes unless official publication evidence exists.

Status And Date Style

Markdown records should use this header shape when practical:

```
Status: active documentation guide
Last updated: 2026-05-26 CDT
Scope: public documentation entry points and related records.
```

Use concrete dates for review snapshots and status records. Use CDT for current local documentation timestamps when the record is tied to the project working context.

Do not use vague status labels such as:

```
soon
almost ready
basically done
ship-ready
production-ish
```

Claim Wording

Prefer:

```
current posture
current evidence
validated by local guard
report-only metadata
denied-by-default classification
local-only draft
guarded user-local workflow
blocked until later evidence exists
```

Avoid:

```
secure
safe
complete
production-ready
approved
certified
official
```

```text
hardened
unbreakable
fully supported
```

If a sentence contains a strong word such as "secure", "safe", "ready", "approved", "official", or "enforced", check PUBLIC_CLAIMS_LEDGER.md before keeping it.

Headings

Use predictable headings for source records:

```text
Purpose
Scope
Current Posture
Evidence
Validation
Non-Claims
Boundary
Next Valid Work
```

Do not create dramatic or marketing-style headings for technical source records.

Link Style

Use relative Markdown links for repository files.

Prefer linking to source records instead of repeating long claims. A public summary should point to:

1. status record;
2. contract or implementation record;
3. validation guard or transcript;
4. non-claims;
5. review packet when public wording changed.

Tables And Lists

Use tables for comparison surfaces:

- source of truth and mirrors;
- allowed and blocked wording;
- product surface responsibilities;
- platform posture;
- validation commands.

Use numbered lists for ordered review or validation steps.

Use bullets for unordered examples.

Code Blocks

Use fenced code blocks for commands, status key/value output, guard output, packet templates, and exact labels.

Use text fences for non-command labels and sh fences for shell commands.

Non-Claim Placement

Put non-claims near the claim they constrain.

For public product pages, do not hide non-claims only in a deep source file. The page should either state the relevant boundary directly or link to NON_CLAIMS.md and PUBLIC_CLAIMS_LEDGER.md.

Replacement Examples

Replace	With
Latticra is secure.	Latticra documents defensive boundaries and security non-claims.
Seal protects AI agents.	Latticra Seal records report-only AI-era tool-boundary evidence.
The installer is ready.	The Panel workflow is guarded and user-local for the documented path.
Fedora support is ready.	The Fedora lane has local-only package and validation records.
Runtime enforcement exists.	Runtime Boundary classification and reports are denied by default.
Nadia runs offline AI.	Nadia has staged offline AI foundation and prompt-evaluation planning contracts.

Boundary

Style changes should not change source behavior, installer authority, package authority, runtime authority, workflow permissions, status estimates, security posture, or product readiness.

When a style change would make a claim stronger, stop and use DOCUMENTATION_CHANGE_REVIEW_PACKET.md.

docs/DOCUMENTATION_TRACEABILITY_MATRIX.md

Source title: Documentation Traceability Matrix

Lines: 84 | Words: 1022 | SHA-256: 508adbc418fd0cf29483e1ef7c6f54942992c8136e68dfff38013f6aa869430b

Documentation Traceability Matrix

Status: active documentation traceability matrix Last updated: 2026-05-26 CDT Scope: public entry points, source records, mirrors, validation guards, non-claims, product surfaces, installer docs, package docs, security docs, and subsystem maps.

Purpose

This matrix maps reader-facing documentation surfaces back to their source records and validation checks.

Use it to answer:

1. where a public statement should be sourced;
2. which mirrors may need updates;
3. which validation commands are relevant;
4. which non-claims must remain visible;
5. where to look when a public page and a source record disagree.

This document does not promote claims. It makes documentation traceability visible.

Traceability Rule

Every public-facing claim should have a path to:

```
public surface -&gt; source record -&gt; validation or review record -&gt; non-claim boundary
```

If that path is missing, keep the claim narrow or remove it until a source record exists.

Matrix

Topic	Public surfaces	Source records	Validation or review	Non-claim boundary
Reader journeys	[`README.md`](README.md), [`map.html`](map.html), [`index.html`](index.html)	[`DOCUMENTATION_READER_JOURNEY_MAP.md`](DOCUMENTATION_READER_JOURNEY_MAP.md), [`PRODUCT_DOCUMENTATION_COHESION.md`](PRODUCT_DOCUMENTATION_COHESION.md)	Local link check and	

public-entry guard | Audience routes must not imply capability promotion. |
Project identity	[`../README.md`](../README.md), [`index.html`](index.html), [`map.html`](map.html)	[`REAL_SYSTEM_CONTRACT.md`](REAL_SYSTEM_CONTRACT.md), [`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md), [`README.md`](README.md)	[`DOCUMENTATION_CHANGE_REVIEW_PACKET.md`](DOCUMENTATION_CHANGE_REVIEW_PACKET.md), `git diff --check`	[`NON_CLAIMS.md`](NON_CLAIMS.md), [`PUBLIC_CLAIMS_LEDGER.md`](PUBLIC_CLAIMS_LEDGER.md)
Current status and estimates	[`../README.md`](../README.md), [`../STATUS.md`](../STATUS.md), [`status.html`](status.html), [`roadmap.html`](roadmap.html)	[`status/CURRENT_STATUS.md`](status/CURRENT_STATUS.md), [`status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md`](status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md)	`sh scripts/test-current-estimate-table-source-alignment.sh`	Estimates are planning posture, not release promises.
Documentation navigation	[`README.md`](README.md), [`map.html`](map.html), [`index.html`](index.html)	[`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md), [`DOCUMENTATION_MAINTENANCE.md`](DOCUMENTATION_MAINTENANCE.md)	Local link check, `git diff --check`	Navigation must not imply capability promotion.
Product-facing wording	[`../README.md`](../README.md), [`index.html`](index.html), [`start.html`](start.html), [`validation.html`](validation.html)	[`PRODUCT_DOCUMENTATION_COHESION.md`](PRODUCT_DOCUMENTATION_COHESION.md), [`PUBLIC_CLAIMS_LEDGER.md`](PUBLIC_CLAIMS_LEDGER.md), [`DOCUMENTATION_STYLE_GUIDE.md`](DOCUMENTATION_STYLE_GUIDE.md)	[`DOCUMENTATION_CHANGE_REVIEW_PACKET.md`](DOCUMENTATION_CHANGE_REVIEW_PACKET.md)	Product copy must stay evidence-bound and non-production.
Public claim language	All public Markdown and HTML summaries	[`PUBLIC_CLAIMS_LEDGER.md`](PUBLIC_CLAIMS_LEDGER.md), [`EVIDENCE_LADDER.md`](EVIDENCE_LADDER.md), [`NON_CLAIMS.md`](NON_CLAIMS.md)	Claim review packet outcome when wording changes	Unsupported adjacent claims stay blocked.
Shared public terminology	Public Markdown, public HTML, package docs, installer docs, status records, and subsystem landing pages	[`DOCUMENTATION_GLOSSARY.md`](DOCUMENTATION_GLOSSARY.md), [`DOCUMENTATION_STYLE_GUIDE.md`](DOCUMENTATION_STYLE_GUIDE.md), [`NAMING_SYSTEM.md`](NAMING_SYSTEM.md), [`LANGUAGE_NAMING_POLICY.md`](LANGUAGE_NAMING_POLICY.md)	Local link check and public-entry guard	Shared terms must not imply stronger capability than source records.
Security posture	[`../SECURITY.md`](../SECURITY.md), [`security.html`](security.html), [`../README.md`](../README.md)	[`DEFENSIVE_THREAT_MODEL_CONTRACT.md`](DEFENSIVE_THREAT_MODEL_CONTRACT.md), [`HIGH_ASSURANCE_SECURITY_BASELINE.md`](HIGH_ASSURANCE_SECURITY_BASELINE.md), [`ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md`](ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE.md)	Security-specific guard named by the changed source record	No malware prevention, ransomware prevention, hardened sandbox, certification, or production protection claim.
Installer and Panel posture	[`../installer/README.md`](../installer/README.md), [`../installer/docs/README.md`](../installer/docs/README.md), [`start.html`](start.html), [`validation.html`](validation.html)	[`PRODUCTION_INSTALLER_READINESS_CONTRACT.md`](PRODUCTION_INSTALLER_READINESS_CONTRACT.md), installer docs and status records	Installer or Panel guard named by the changed record	No root installer, daily-driver installer, production installer, or unattended host mutation claim.
Fedora lane	[`FEDORA_READINESS_PLAN.md`](FEDORA_READINESS_PLAN.md), [`FEDORA_DEVELOPER_WORKFLOW.md`](FEDORA_DEVELOPER_WORKFLOW.md), [`../packaging/fedora/README.md`](../packaging/fedora/README.md), [`validation.html`](validation.html)	Fedora package, workflow, and status records	`sh scripts/test-fedora-developer-workflow.sh` plus exact Fedora guard	Local-only unless Fedora approval evidence exists.
Ubuntu lane	[`UBUNTU_READINESS_PLAN.md`](UBUNTU_READINESS_PLAN.md), [`UBUNTU_DEVELOPER_WORKFLOW.md`](UBUNTU_DEVELOPER_WORKFLOW.md), [`../packaging/ubuntu/README.md`](../packaging/ubuntu/README.md), [`validation.html`](validation.html)	Ubuntu package, workflow, notice, license, and status records	`sh scripts/test-ubuntu-developer-workflow.sh` plus exact Ubuntu guard	Local-only unless Ubuntu archive or PPA evidence exists.
Debian, FreeBSD, and OpenBSD lanes	[`../packaging/debian/README.md`](../packaging/debian/README.md), [`../packaging/freebsd/README.md`](../packaging/freebsd/README.md), [`../packaging/openbsd/README.md`](../packaging/openbsd/README.md), [`validation.html`](validation.html)	Platform package draft and package-input/build-gate records	Exact platform guard named by the changed record	Local-only unless archive, ports-tree, or official publication evidence exists.
openSUSE lane	[`OPENSUSE_READINESS_PLAN.md`](OPENSUSE_READINESS_PLAN.md), [`OPENSUSE_DEVELOPER_WORKFLOW.md`](OPENSUSE_DEVELOPER_WORKFLOW.md), [`../packaging/opensuse/README.md`](../packaging/opensuse/README.md), [`validation.html`](validation.html)	openSUSE package, workflow, rpmlint, osc, and status records	`sh scripts/test-opensuse-developer-workflow.sh` plus exact openSUSE guard	Local-only unless Open Build Service or official package evidence exists.
Runtime Boundary and Nucleus	[`runtime.html`](runtime.html), [`../README.md`](../README.md), subsystem records	[`RUNTIME_BOUNDARY_CONTRACT.md`](RUNTIME_BOUNDARY_CONTRACT.md), [`RUNTIME_BOUNDARY_IMPLEMENTATION.md`](RUNTIME_BOUNDARY_IMPLEMENTATION.md),

[`NUCLEUS_TASK_EXECUTION_CONTRACT.md`](NUCLEUS_TASK_EXECUTION_CONTRACT.md), [`NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md`](NUCLEUS_TASK_EXECUTION_IMPLEMENTATION.md) | Runtime or Nucleus guard named by the changed source record | No runtime authority, effect-performing execution, process isolation, or sandbox claim. |

| Latticra Seal | [`seal.html`](seal.html), [`latticra-seal/README.md`](latticra-seal/README.md), [`latticra-seal/STATUS.md`](latticra-seal/STATUS.md), [`../README.md`](../README.md) | [`LATTICRA_SEAL_CONTRACT.md`](LATTICRA_SEAL_CONTRACT.md), [`LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md`](LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md), Seal status records | `sh scripts/test-latticra-seal-docs.sh` plus exact Seal guard | No production enforcement, AI-agent control, host protection, network blocking, or cryptographic authority claim. |

| Lat, LIR, and L-UI | [`language.html`](language.html), [`examples.html`](examples.html), [`../README.md`](../README.md) | [`LANGUAGE_STRATEGY.md`](LANGUAGE_STRATEGY.md), [`LAT_PIPELINE_CONTRACT.md`](LAT_PIPELINE_CONTRACT.md), [`LIR_SHAPE_CONTRACT.md`](LIR_SHAPE_CONTRACT.md), [`L_UI_PARSER.md`](L_UI_PARSER.md) | Language or parser guard named by the changed source record | No production language runtime, LIR execution, or terminal-control renderer claim. |

| Nadia offline AI | [`nadia.html`](nadia.html), [`../README.md`](../README.md), Nadia status records | [`NADIA_OFFLINE_AI_FOUNDATION.md`](NADIA_OFFLINE_AI_FOUNDATION.md) and staged Nadia contracts | Exact Nadia guard named by the changed source record | No model loading, inference, token generation, dialogue generation, prompt evaluation, or tool execution claim. |

| Boot and OS-base preview | [`boot.html`](boot.html), [`../README.md`](../README.md), installer records | SeaBIOS/GRUB boot-preview contracts, templates, preflight, and validation records | Exact SeaBIOS/GRUB guard named by the changed source record | No bootable OS image, OS replacement, GRUB installer, firmware mutation, or hardware boot success claim. |

| Handbook and long-form docs | [`latticra-system-substrate/README.md`](latticra-system-substrate/README.md), generated handbook artifacts | [`FOUNDATION_INDEX.md`](FOUNDATION_INDEX.md), project notes, status records, subsystem contracts | Documentation change review packet and local link check | Handbook summaries must not outrun current status. |

Drift Indicators

Treat these as documentation drift:

- a public page names a capability without a source record;
- a status record and public page use different posture words;
- a platform README omits local-only or no-effect boundaries;
- a security page uses "secure", "safe", "hardened", "protection", or "prevention" without a supporting source record;
- an estimate appears in public HTML but not in the current estimate source records;
- a subsystem landing page implies product readiness from subsystem progress alone;
- a quick-start path omits reset, cleanup, or non-claim context for a runnable local path.

Review Flow

When drift is found:

1. Identify the topic row in this matrix.
2. Read the source records before editing public wording.
3. Check PUBLIC_CLAIMS_LEDGER.md and DOCUMENTATION_STYLE_GUIDE.md.
4. Use DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md to choose a narrowing, alignment, or demotion path.
5. Use DOCUMENTATION_VALIDATION_PLAYBOOK.md to choose the narrowest relevant validation commands.
6. Use DOCUMENTATION_CHANGE_REVIEW_PACKET.md when wording, mirrors, estimates, status, platform posture, or non-claims change.
7. Run the selected validation commands.

8. Update only the public surfaces whose reader-facing claim actually changed.

Boundary

This matrix is documentation traceability only.

It does not change implementation behavior, installer authority, package authority, runtime authority, workflow permissions, security posture, public estimates, or product readiness.

docs/DOCUMENTATION_VALIDATION_PLAYBOOK.md

Source title: Documentation Validation Playbook

Lines: 133 | Words: 864 | SHA-256: 8542c13d0307a12c9a2f1f50ff8e7dbcbe0427cec5569146fe646cf521e460d8

Documentation Validation Playbook

Status: active documentation validation playbook Last updated: 2026-05-26 CDT Scope: documentation-only validation commands, link checks, public-entry checks, estimate checks, subsystem guards, platform guards, and failure handling.

Purpose

This playbook explains which validation checks to run for documentation-only changes.

Use it with DOCUMENTATION_MAINTENANCE.md, DOCUMENTATION_TRACEABILITY_MATRIX.md, DOCUMENTATION_CHANGE_REVIEW_PACKET.md, and PUBLIC_CLAIMS_LEDGER.md.

Validation should prove only the documentation claim being made. It should not be used to imply production readiness, security guarantees, installer authority, package approval, runtime authority, or product readiness.

Validation Levels

Level	Use when	Minimum checks
Hygiene	Any documentation file changes.	`git diff --check`, trailing-whitespace check.
Local links	New or changed Markdown links.	Local Markdown link sanity check over touched files.
Public entry	README, docs hub, map, public site pages, status index, project notes, or strategy docs change.	`sh scripts/test-project-strategy-status-framework.sh`.
Estimate mirror	Completion estimates or public estimate tables change.	`sh scripts/test-current-estimate-table-source-alignment.sh`.
Seal docs	Seal README, Seal status, Seal public wording, or Seal source records change.	`sh scripts/test-latticra-seal-docs.sh` plus exact Seal guard.
Platform docs	Fedora, Ubuntu, openSUSE, Debian, FreeBSD, OpenBSD, macOS, installer, or package posture changes.	Platform workflow guard plus exact platform guard.
Subsystem docs	Runtime, Nucleus, Lat, LIR, L-UI, Nadia, boot preview, security, installer, or package source record changes.	Exact guard named by the source or status record.
Claim promotion	Public wording becomes stronger.	Stop unless contract, implementation, validation, status, non-claim update, traceability, and public-entry alignment all exist.

Minimum Documentation-Only Check

Run this for most documentation-only changes:

```
git diff --check
sh scripts/test-project-strategy-status-framework.sh
```

Add a local link check for changed Markdown files when links are added or moved.

Local Link Check

Use a local Markdown link sanity check on the files you touched:

```
perl -MFile::Basename=dirname -e &#x27;my $bad=0; for my $file (@ARGV) { open my $fh, q{&lt;},
$file or
```

```
die qq{$file: $!\n}; my $base = dirname($file); while (my $line = &lt;$fh&gt;) { while ($line
=~
/\[[^\]]+\]\(([^)]+)\)/g) { my $target = $1; next if $target =~ /^(https?:|mailto:|#)/; $target
=~ s/#.*\z//; next if $target eq q{}; my $path = $target =~ m{^/} ? $target : qq{$base/$target};
 unless (-e $path) { print qq{$file:$.: missing $target\n}; $bad=1; } } } } exit $bad&#x27;
FILES...
```

This check is intentionally local. It does not prove that a public claim is true; it only catches broken repository links.

Estimate And Status Checks

Run the estimate source-alignment guard when a change touches:

- public estimate tables;
- overall completion percentage;
- status dashboard estimate text;
- roadmap estimate text;
- README.md, STATUS.md, or docs/status/CURRENT_STATUS.md estimate mirrors.

```
sh scripts/test-current-estimate-table-source-alignment.sh
```

If this guard fails because source records and mirrors disagree, do not hand-edit the guard to force a pass. Fix the documented source or mirror drift, or keep the public claim unchanged.

Platform Checks

Add platform guards when platform or package docs change:

```
sh scripts/test-fedora-developer-workflow.sh
sh scripts/test-ubuntu-developer-workflow.sh
sh scripts/test-opensuse-developer-workflow.sh
```

Also run the exact guard named by the platform source record when the change is narrower than the general workflow.

Examples:

```
Fedora RPM static validation -&gt; run the Fedora static validation guard.
Ubuntu notice/license wording -&gt; run the Ubuntu notice or license guard.
openSUSE rpmlint/osc wording -&gt; run the openSUSE rpmlint or osc guard.
macOS reset/uninstall wording -&gt; run the matching macOS reset/uninstall guard.
```

Seal Checks

Run the Seal docs guard when any Seal public wording, Seal status row, Seal README, Seal public-entrypoint alignment, or Seal source record changes:

```
sh scripts/test-latticra-seal-docs.sh
```

Also run the exact Seal guard named by the changed status or implementation record when applicable.

Security Checks

Security documentation changes should always preserve security non-claims.

For security wording changes, validate:

1. PUBLIC_CLAIMS_LEDGER.md does not block the wording.
2. DOCUMENTATION_STYLE_GUIDE.md permits the terminology.
3. DOCUMENTATION_TRACEABILITY_MATRIX.md identifies the source and non-claim boundary.
4. The exact security, threat-model, supply-chain, memory-safety, or runtime-authority guard named by the source record passes.

Failure Handling

When a documentation validation fails:

1. Read the failing message before editing.
2. Identify whether the failure is link drift, mirror drift, status drift, guard expectation drift, or claim drift.
3. Fix the documentation source or mirror that is wrong.
4. Do not weaken non-claims to make a public summary sound better.
5. Do not edit guard scripts for documentation-only convenience.
6. Use DOCUMENTATION_DRIFT_RESPONSE_PLAYBOOK.md when source records, public pages, mirrors, estimates, non-claims, or validation expectations disagree.
7. If the stronger wording lacks evidence, demote the wording and record the blocked claim in the review packet.

Review Packet Fields

When using DOCUMENTATION_CHANGE_REVIEW_PACKET.md, put the selected commands in validation_commands.

Use blocked_missing_validation when a claim would require a guard or evidence record that does not exist yet.

Use blocked_public_entry_drift when public pages and source records disagree.

Use accepted_navigation_only only when links or route ordering changed without changing claims.

Boundary

This playbook validates documentation hygiene, traceability, and public-entry consistency.

It does not validate production runtime behavior, installer safety, package publication, platform approval, security guarantees, host protection, network protection, cryptographic authority, or AI-agent execution control.

docs/latticra/DOCUMENTATION_MAP.md

Source title: Latticra Documentation Map

Lines: 80 | Words: 620 | SHA-256: 5c20b91497db051ae43d835faf3ab6025c6aac575b4a1cf54ea0b26635b3c085

Latticra Documentation Map

Status: active documentation map Last updated: 2026-05-26 CDT Scope: reader routes, foundation documents, C/C++ direction, security profile, ABI boundaries, platform lanes, subsystem handbooks, and validation posture.

Purpose

This map records how the Latticra documentation set is meant to fit together.

For the shortest reader route, start with ../README.md. For the exhaustive architecture and operations index, use ../FOUNDATION_INDEX.md. This file explains the relationship between those layers and the older C/C++ foundation package that originally lived here.

Documentation Layers

```
| Layer | Purpose | Primary entry |
| --- | --- | --- |
```

Reader hub	Short route through the whole documentation set	[`../README.md`](../README.md)
Public project overview	Product identity, current status, quick start, and non-claims	[`../../README.md`](../../README.md)
Live status	Current public posture, estimates, status records, announcements	[`../../STATUS.md`](../../STATUS.md), [`../status/CURRENT_STATUS.md`](../status/CURRENT_STATUS.md), [`../status/README.md`](../status/README.md)
Foundation index	Full project operations, architecture, implementation, and guard index	[`../FOUNDATION_INDEX.md`](../FOUNDATION_INDEX.md)
System handbook	Project-level book for the durable Latticra story	[`../latticra-system-substrate/README.md`](../latticra-system-substrate/README.md)
Seal subsystem	Focused trust-boundary and report-only Seal docs	[`../latticra-seal/README.md`](../latticra-seal/README.md)
Installer docs	Panel installer authority, configuration, receipts, and evidence	[`../../installer/docs/README.md`](../../installer/docs/README.md)
Project notes	Short direction and queue notes	[`../project_notes/README.md`](../project_notes/README.md)

C/C++ Foundation Package

The C/C++ foundation direction remains active and is intentionally evidence-bound:

- C is the narrow platform substrate.
- Restricted C++ is the governed authority layer.
- Latticra is the contract language above the implementation substrate.

This does not claim that C/C++ is automatically safe. It defines the project posture required to use C/C++ responsibly inside a security-conscious systems project.

Document	Purpose
[`../architecture/LATTICRA_LANGUAGE_FOUNDATION.md`](../architecture/LATTICRA_LANGUAGE_FOUNDATION.md)	Defines the architectural language layering and project posture.
[`../security/C_CPP_SECURITY_PROFILE.md`](../security/C_CPP_SECURITY_PROFILE.md)	Defines the restricted C/C++ profile for trusted code.
[`../security/C_ABI_BOUNDARY_POLICY.md`](../security/C_ABI_BOUNDARY_POLICY.md)	Defines how C ABI boundaries are exposed, controlled, and reviewed.
[`../roadmap/LATTICRA_C_CPP_FOUNDATION_ROADMAP.md`](../roadmap/LATTICRA_C_CPP_FOUNDATION_ROADMAP.md)	Defines staged work required to make this direction real.
[`../C_CPP_FOUNDATION_DIRECTION.md`](../C_CPP_FOUNDATION_DIRECTION.md)	Records the active C is the metal, C++ is the disciplined structure, Latticra is the contract direction.
[`../CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md`](../CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md)	Defines the guarded C++ authority-layer contract before broad authority is allowed.

Platform and Product Lanes

Lane	Entry points	Current posture
Panel installer	[`../../installer/README.md`](../../installer/README.md), [`../../installer/docs/README.md`](../../installer/docs/README.md)	User-local, dry-run-first, no root, no network authority.
macOS installer lane	[`../MACOS_INTEGRATION_TRANSFERABILITY_PLAN.md`](../MACOS_INTEGRATION_TRANSFERABILITY_PLAN.md), [`../MACOS_COMMIT_GATE_CONTRACT.md`](../MACOS_COMMIT_GATE_CONTRACT.md)	No-effect probes and dry-run planning; no `.app` write or signing claim.
Fedora	[`../FEDORA_DEVELOPER_WORKFLOW.md`](../FEDORA_DEVELOPER_WORKFLOW.md), [`../../packaging/fedora/README.md`](../../packaging/fedora/README.md)	Local-only package and validation lanes; no Fedora approval.
Ubuntu	[`../UBUNTU_DEVELOPER_WORKFLOW.md`](../UBUNTU_DEVELOPER_WORKFLOW.md), [`../../packaging/ubuntu/README.md`](../../packaging/ubuntu/README.md)	Local-only deb draft and static checks; no archive or PPA readiness.
openSUSE	[`../OPENSUSE_DEVELOPER_WORKFLOW.md`](../OPENSUSE_DEVELOPER_WORKFLOW.md), [`../../packaging/opensuse/README.md`](../../packaging/opensuse/README.md)	Local-only RPM maintenance draft; no OBS publication or official package claim.

Summary

Latticra is not rejecting memory safety. Latticra is rejecting uncontrolled authority and replacing it with explicit security architecture, narrow unsafe boundaries, hard build discipline, and validation before promotion.

Core statement:

C is the metal. C++ is the disciplined structure. Latticra is the contract.

External Guidance Used For Alignment

These sources are used as security-alignment references, not as project endorsements:

- NSA/CISA, "Memory Safe Languages: Reducing Vulnerabilities in Modern Software Development"

https://media.defense.gov/2025/Jun/23/2003742198/-1/-1/0/CSI_MEMORY_SAFE_LANGUAGES_REDUCING_VULNERABILITIES_IN_MODERN_SOFTWARE_DEVELOPMENT.PDF

- NSA press release for the NSA/CISA memory-safe languages CSI

https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/4223298/nsa-and-cisa-release-csi-highlighting-importance-of-memory-safe-languages-in-so/

- SEI CERT C and C++ Coding Standards

https://www.sei.cmu.edu/library/sei-cert-c-and-c-coding-standards/

- SEI CERT C++ Coding Standard

https://cmu-sei.github.io/secure-coding-standards/sei-cert-cpp-coding-standard/

- C++ Core Guidelines

https://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines

Promotion Rule

No future claim about Latticra security, runtime authority, host protection, package readiness, boot readiness, or production maturity should be promoted unless the relevant code, tests, documentation, and review evidence support it.

docs/LICENSE_MIGRATION_PLAN.md

Source title: Latticra License Migration Plan

Lines: 199 | Words: 467 | SHA-256: 522b71a594ceb496876b32cc6a8740e0af4cbfcde3b615dc3e460bda29c9963c

Latticra License Migration Plan

Status: active migration planning record Scope: path-by-path transition planning after the open ecosystem policy.

Purpose

This plan defines how Latticra should move from a repository-wide Apache-2.0 default toward the hybrid open ecosystem direction without silently relicensing files.

The goal is to keep Latticra open, auditable, transparent, and community-improvable while preserving a clear legal trail for each migration step.

Current state

Current canonical root license text:

```
LICENSE: hybrid license overview
```

Current policy direction:

```
core software work: AGPL-3.0-or-later
adoption-facing helpers: Apache-2.0
documentation and handbooks: CC-BY-4.0
existing code files: current license until intentionally migrated or otherwise marked
branding: separate trademark and identity policy
```

Related records:

```
docs/OPEN_ECOSYSTEM_POLICY.md
docs/LICENSE_POLICY.md
docs/DOCUMENTATION_LICENSE.md
CONTRIBUTING.md
TRADEMARK_POLICY.md
LICENSES/README.md
```

Migration principles

Migration must be:

```
explicit
path-scoped
reviewed
SPDX-marked
compatible with contributor rights
recorded before release
```

No file should be relicensed silently.

Phase 1: governance baseline

Status: complete

Scope:

```
docs/OPEN_ECOSYSTEM_POLICY.md
docs/LICENSE_POLICY.md
CONTRIBUTING.md
TRADEMARK_POLICY.md
LICENSES/README.md
scripts/test-open-ecosystem-policy.sh
.github/workflows/open-ecosystem-policy.yml
```

Purpose:

```
declare open ecosystem direction
preserve existing license state
record documentation license decision
add contribution and identity policy
add governance guard
```

Phase 2: SPDX planning guard

Status: this plan

Scope:

```
docs/LICENSE_MIGRATION_PLAN.md
scripts/test-license-migration-plan.sh
.github/workflows/license-migration-plan.yml
```

Purpose:

```
record migration phases
require new-source SPDX direction to stay discoverable
define first migration candidate paths
avoid accidental whole-repository relicensing
```

Phase 3: first source SPDX migration candidates

Recommended first candidate group:

```
include/latticra/kernel*.h
src/kernel*.c
tests/kernel*.c
tools/kernel*.c
```

Reason:

```
These files are kernel/core oriented and closely match the open ecosystem reciprocal direction.
```

The first source migration PR should:

1. add SPDX-License-Identifier headers;
2. document affected paths;
3. update or add a guard for SPDX coverage;
4. avoid mixing license migration with behavior changes;
5. preserve existing code behavior and tests.

Phase 4: system/bootstrap/core source candidates

Recommended second candidate group:

```
include/latticra/system_bootstrap.h
src/system_bootstrap.c
tests/system_bootstrap.c
tools/system_bootstrap_report.c
```

Reason:

```
These files are core system surfaces and should be reviewed after kernel/core paths.
```

Phase 5: non-core path review

Review separately:

```
Lat language tooling
L-UI tooling
LIR tooling
runtime-boundary support
scripts
examples
fixtures
assets
```

Do not assume all non-core paths should use the same license immediately.

Documentation license decision

Documentation license is decided in this plan.

The documentation and handbook license is:

```
CC-BY-4.0
```

The path-level documentation license record is:

```
docs/DOCUMENTATION_LICENSE.md
```

CC-BY-SA-4.0 may be used for future standalone handbook material only when a file or path-level notice explicitly marks that choice.

Release checklist impact

Before any public release artifact, verify:

```
root license state
file-level SPDX state
third-party notices
source-offer requirements
trademark use
release artifact license notes
```

Non-legal-advice boundary

This plan is project guidance and not legal advice.

Formal relicensing, commercial distribution, dual licensing, or release review should receive qualified legal review.

docs/LICENSE_POLICY.md

Source title: Latticra License Policy

Lines: 192 | Words: 576 | SHA-256: 3eac61a4e2ef53944f2d3c1bfa4b91741f0e6abd8434add853be6cf8c4f5eec4

Latticra License Policy

Status: active hybrid license policy Scope: hybrid software licensing, documentation licensing, SPDX usage, branding, contributions, and future artifacts.

Purpose

Latticra is intended to remain open, auditable, transparent, and community-improvable.

This policy explains the current hybrid repository license state and the transition-aware rules for future work.

See also:

```
docs/OPEN_ECOSYSTEM_POLICY.md
docs/DOCUMENTATION_LICENSE.md
CONTRIBUTING.md
TRADEMARK_POLICY.md
LICENSES/README.md
```

Current repository posture

The root repository license overview is:

```
LICENSE
```

Latticra uses this hybrid posture:

```
core/runtime/security substrate: AGPL-3.0-or-later
SDKs/examples/packaging helpers/integration helpers: Apache-2.0
documentation/handbooks/policy notes: CC-BY-4.0
names/logos/identity: TRADEMARK_POLICY.md
```

Existing repository code remains Apache-2.0 unless a file-level SPDX identifier, path-level notice, or reviewed migration explicitly marks it otherwise.

Core software direction

The intended software licensing direction for new software work is:

```
AGPL-3.0-or-later
```

This direction applies to core runtime, security-substrate, tool-boundary, runtime-gating, verification, and evidence-bound computing components.

It is meant to preserve open, auditable, transparent access to Latticra-derived core software while the project grows into system, kernel, language, tool, and operator-facing components.

Adoption-facing software direction

The intended license for SDKs, examples, packaging helpers, installer glue, integration helpers, and other adoption-facing software is:

```
Apache-2.0
```

This keeps integration surfaces commercially friendly and packaging friendly while preserving an express patent grant.

Documentation direction

The documentation and handbook license decision is:

```
CC-BY-4.0
```

See:

```
docs/DOCUMENTATION_LICENSE.md
LICENSES/CC-BY-4.0.txt
```

No silent relicensing

This policy does not silently relicense every existing file.

Existing files remain under their current license until a later migration PR updates them clearly and intentionally.

A file-level or path-level migration should include:

```
SPDX license identifier update
reason for migration
affected paths
compatibility notes
review before release
```

New source file rule

New software source files should declare their license with an SPDX identifier.

The preferred identifier for new core software source files is:

```
SPDX-License-Identifier: AGPL-3.0-or-later
```

The preferred identifier for new adoption-facing helper source files is:

```
SPDX-License-Identifier: Apache-2.0
```

Documented exceptions are allowed when the exception is intentional and reviewed.

New documentation file rule

New documentation files may declare:

```
SPDX-License-Identifier: CC-BY-4.0
```

Existing documentation is covered by the path-level decision in:

```
docs/DOCUMENTATION_LICENSE.md
```

Contributions

Contributions are accepted under the license declared by the files being modified.

New software contributions should follow the project direction documented in:

```
docs/OPEN_ECOSYSTEM_POLICY.md
CONTRIBUTING.md
```

No proprietary relicensing contributor license agreement is adopted by this policy.

Branding and trademarks

Software and documentation licenses do not grant trademark, branding, endorsement, or official-project identity rights.

Project names, logos, marks, and Bryforge/Latticra branding are handled separately.

See:

```
TRADEMARK_POLICY.md
```

Third-party material

Third-party material must include source, license, and compatibility notes before being added.

Required fields:

```
name:
source:
```

```
license:
compatible with current path license: yes|no|unknown
purpose:
notes:
```

Generated artifacts

Generated artifacts may need additional notices depending on source inputs, build process, and bundled dependencies.

No release artifact should be published until its licensing and notice requirements are reviewed.

NOTICE file

A future NOTICE file may be added when Latticra has third-party notices, attribution requirements, or distribution notes to preserve.

Future work

Recommended follow-up work:

```
add SPDX headers to new software files
add a license guard for source paths
review path-by-path migration from Apache-2.0
add release license checklist
add NOTICE if needed
```

Non-claims

This policy is project guidance and not legal advice.

For commercial licensing strategy, patent strategy, trademark strategy, formal relicensing, or release review, obtain qualified legal review.

docs/project_notes/CURRENT_DIRECTION.md

Source title: Latticra Current Direction

Lines: 553 | Words: 7368 | SHA-256: 359ea5d6b97ff5a76251604f1696505175dec605afccd4baccb2ad8790e3655e

Latticra Current Direction

Status: active project note Last updated: 2026-05-26 CDT Latest current estimate table source alignment note: 2026-05-26 CDT Latest current estimate mathematical rebase note: 2026-05-26 CDT Latest completion estimate review README/status alignment note: 2026-05-25 CDT Latest completion estimate review after runtime-boundary abuse-case fixtures note: 2026-05-25 CDT Latest defensive threat model validation refinement note: 2026-05-25 CDT Latest high-assurance security baseline note: 2026-05-26 CDT Latest memory-safety roadmap note: 2026-05-26 CDT Latest supply-chain security baseline note: 2026-05-26 CDT Latest zero-trust runtime authority baseline note: 2026-05-26 CDT Latest runtime boundary policy expansion after threat-model note: 2026-05-26 CDT Latest runtime boundary abuse-case fixture expansion note: 2026-05-25 CDT Latest runtime boundary Lat pipeline comment evidence note: 2026-05-25 CDT Latest Lat parse-failure comment evidence propagation note: 2026-05-25 CDT Latest Lat pipeline failure span evidence propagation note: 2026-05-25 CDT Latest Lat pipeline parse-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline semantic-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline downstream stage-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline stage-summary evidence propagation note: 2026-05-25 CDT Latest Lat pipeline module/count evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-declaration evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-clause evidence propagation note: 2026-05-25 CDT Latest Lat LIR edge-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR no-effect evidence propagation note: 2026-05-26 CDT Latest Lat LIR module-summary evidence propagation note: 2026-05-26 CDT Latest Lat LIR source-span evidence propagation note: 2026-05-26 CDT Latest Lat LIR node-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge endpoint evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge span evidence propagation note: 2026-05-26 CDT Latest Seal README status row alignment note: 2026-05-25 CDT Latest Seal capability metadata report surface status note: 2026-05-26 CDT Latest Seal product spine note: 2026-05-26 CDT Latest Seal product spine status note: 2026-05-26 CDT Latest Seal operator receipt report contract note: 2026-05-26 CDT Latest Seal operator receipt report implementation plan note: 2026-05-26 CDT Latest Seal operator receipt report implementation note: 2026-05-26 CDT Latest Seal operator receipt report surface/status note: 2026-05-26 CDT Latest Seal operator receipt report predecessor status alignment note: 2026-05-26 CDT Latest Seal local capability registry schema contract note: 2026-05-26 CDT Latest Seal local capability registry schema implementation plan note: 2026-05-26 CDT Latest Seal local capability registry schema implementation note: 2026-05-26 CDT Latest Seal policy decision status/public-entry note: 2026-05-25 CDT Latest Seal policy decision predecessor status alignment note: 2026-05-26 CDT Latest Seal signed request status/public-entry note: 2026-05-25 CDT Latest Seal signed request predecessor status alignment note: 2026-05-26 CDT Latest Seal request freshness status/public-entry note: 2026-05-25 CDT Latest Seal request freshness predecessor status alignment note: 2026-05-26 CDT Latest Seal parameter schema status/public-entry note: 2026-05-25 CDT Latest Seal parameter schema predecessor status alignment note: 2026-05-26 CDT Latest Lat grammar report metadata integration note: 2026-05-25 CDT Latest Lat grammar line-comment metadata refinement note: 2026-05-25 CDT Latest Lat grammar unsupported block-comment rejection refinement note: 2026-05-25 CDT Latest Lat pipeline comment metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic comment metadata integration note: 2026-05-25 CDT Latest Lat parse-failure comment evidence propagation note: 2026-05-25 CDT Latest Lat pipeline failure span evidence propagation note: 2026-05-25 CDT Latest Lat pipeline parse-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline semantic-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline downstream stage-error evidence propagation note: 2026-05-25 CDT Latest Lat pipeline stage-summary evidence propagation note: 2026-05-25 CDT Latest Lat pipeline module/count evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-declaration evidence propagation note: 2026-05-25 CDT Latest Lat pipeline first-clause evidence propagation note: 2026-05-25 CDT Latest Lat LIR edge-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR no-effect evidence propagation note: 2026-05-26 CDT Latest Lat LIR module-summary evidence propagation note: 2026-05-26 CDT Latest Lat LIR source-span evidence propagation note: 2026-05-26 CDT Latest Lat LIR node-kind evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-node span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first-edge span evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge endpoint evidence propagation note: 2026-05-26 CDT Latest Lat LIR first transition-source edge span evidence propagation note: 2026-05-26 CDT Latest Lat model normalization note: 2026-05-25 CDT Latest Lat model report declaration metadata integration note: 2026-05-25 CDT Latest Lat model report clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline model normalization integration note: 2026-05-25 CDT Latest Lat-to-LIR model lowering integration note: 2026-05-25 CDT Latest Lat-to-LIR declaration metadata refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline lowering diagnostic integration note: 2026-05-25 CDT Latest Lat-to-LIR clause metadata refinement note: 2026-05-25 CDT Latest Lat-to-LIR diagnostic clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline diagnostic clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline report declaration metadata integration note: 2026-05-25 CDT Latest Lat pipeline report clause metadata integration note: 2026-05-25 CDT Latest Lat pipeline note: 2026-05-18 19:40 CDT Latest Lat pipeline diagnostic audit note: 2026-05-19 17:15 CDT Latest runtime boundary domain matrix report note: 2026-05-19 16:20 CDT Latest public entry-point consistency scan note: 2026-05-19 19:45 CDT Latest Nucleus task report-only execution refinement note: 2026-05-19 21:35 CDT Latest Nucleus task report-only execution README/status alignment note: 2026-05-20 01:45 CDT Latest Nucleus report-only announcement review note: 2026-05-20 03:00 CDT Latest Nucleus report-only announcement README alignment note: 2026-05-20 03:20 CDT Latest Seal agentic automation security predecessor status alignment note: 2026-05-26 CDT Latest Seal status rollup predecessor status alignment note: 2026-05-26 CDT Latest Seal runtime handoff status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff predecessor status alignment note: 2026-05-26 CDT Latest Latticra Seal runtime handoff report status/public-entry alignment note: 2026-05-25 CDT Latest Seal effect decision status/public-entry note: 2026-05-25 CDT Latest Seal capability gate status/public-entry note: 2026-05-25 CDT Latest Seal capability gate predecessor status alignment note: 2026-05-26 CDT Latest Seal verification receipt predecessor status alignment note: 2026-05-26 CDT Latest Seal verification receipt status/public-entry note: 2026-05-25 CDT Latest Seal crypto verify backend status/public-entry note: 2026-05-25 CDT Latest Seal Ed25519 verify status/public-entry note: 2026-05-25 CDT Latest Seal verified receipt promotion status/public-entry note: 2026-05-25 CDT Latest Seal verified capability gate status/public-entry note: 2026-05-25 CDT Latest Seal verified effect decision status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff evaluation status/public-entry note: 2026-05-25 CDT Latest Seal runtime handoff report status/public-entry note: 2026-05-25 CDT Latest Seal report envelope status/public-entry note: 2026-05-25 CDT Latest Seal signature request predecessor status alignment note:

2026-05-25 CDT Latest Seal signing authorization predecessor status alignment note: 2026-05-25 CDT Latest Seal verification policy predecessor status alignment note: 2026-05-26 CDT Latest Seal verification policy status/public-entry note: 2026-05-25 CDT Latest Seal key parsing predecessor status alignment note: 2026-05-26 CDT Latest Seal key parsing status/public-entry note: 2026-05-25 CDT Latest Seal bounded key parsing implementation note: 2026-05-25 CDT Latest Seal future key parsing implementation plan note: 2026-05-25 CDT Latest Seal future key parsing implementation contract note: 2026-05-25 CDT Latest Seal public-key parsing predecessor status alignment note: 2026-05-26 CDT Latest Seal public-key parsing status/public-entry note: 2026-05-25 CDT Latest Seal public-key parsing implementation note: 2026-05-25 CDT Latest Seal public-key parsing contract note: 2026-05-25 CDT Latest Seal key-material predecessor status alignment note: 2026-05-26 CDT Latest Seal key-material status/public-entry note: 2026-05-25 CDT Latest Seal key-material implementation note: 2026-05-25 CDT Latest Seal key-material contract note: 2026-05-25 CDT Latest Seal key-handling predecessor status alignment note: 2026-05-26 CDT Latest Seal key-handling status/public-entry note: 2026-05-25 CDT Latest Seal key-handling implementation note: 2026-05-25 CDT Latest Seal key-handling contract note: 2026-05-25 CDT Latest Seal signing operation predecessor status alignment note: 2026-05-26 CDT Latest Seal signing operation status/public-entry note: 2026-05-25 CDT Latest Seal signing operation implementation note: 2026-05-25 CDT Latest Seal signing operation contract note: 2026-05-25 CDT Latest Seal signer invocation predecessor status alignment note: 2026-05-25 CDT Latest Seal signer invocation status/public-entry note: 2026-05-25 CDT Latest Seal signer invocation implementation note: 2026-05-25 CDT Latest Seal signer invocation contract note: 2026-05-25 CDT Latest Seal signer handoff predecessor status alignment note: 2026-05-25 CDT Latest Seal signer handoff status/public-entry note: 2026-05-25 CDT Latest Seal signer handoff implementation note: 2026-05-25 CDT Latest Seal signer handoff contract note: 2026-05-25 CDT

Current direction

Latticra is being built as a contract-first open systems architecture and programming-language project.

The public direction is:

```
open source
auditable
defensive
contract-driven
evidence-bound
operator-visible
security-conscious from the beginning
constrained C/C++ foundation
```

The current security-guidance posture is aligned across the high-assurance baseline, memory-safety roadmap, supply-chain baseline, zero-trust runtime authority baseline, and runtime-boundary policy expansion. Those records now carry explicit workload or service identity, operator identity, host or device integrity, command-boundary, update-integrity, and retained-C/C++ hazard-class expectations while preserving Latticra's no-compliance-claim and no-production-security-boundary posture.

C/C++ foundation checkpoint

```
C is the metal.
C++ is the disciplined structure.
Latticra is the contract.
```

Meaning:

```
C: secure substrate, boot paths, ABI boundaries, platform shims.
C++: governed authority layer, policy, validators, effect gates, audit logic.
Lat / Latticra Language: contract, semantic validation, metadata lowering, and declaration
layer.
```

This does not mean unrestricted C++.

Current Lat language boundary

The Lat lane now has:

```
bounded grammar parser
bounded no-effect Lat grammar report metadata integration
bounded no-effect Lat grammar line-comment metadata refinement
bounded no-effect Lat grammar unsupported block-comment rejection refinement
bounded no-effect Lat pipeline comment metadata integration
bounded no-effect Lat pipeline diagnostic comment metadata integration
bounded no-effect Lat parse-failure comment evidence propagation
bounded no-effect Lat pipeline failure span evidence propagation
bounded no-effect Lat pipeline parse-error evidence propagation
bounded no-effect Lat pipeline semantic-error evidence propagation
bounded no-effect Lat pipeline downstream stage-error evidence propagation
bounded no-effect Lat pipeline stage-summary evidence propagation
bounded no-effect Lat pipeline module/count evidence propagation
bounded no-effect Lat pipeline first-declaration evidence propagation
bounded no-effect Lat pipeline first-clause evidence propagation
bounded no-effect Lat LIR edge-kind evidence propagation
bounded no-effect Lat LIR no-effect evidence propagation
bounded no-effect Lat LIR module-summary evidence propagation
bounded no-effect Lat LIR source-span evidence propagation
bounded no-effect Lat LIR node-kind evidence propagation
bounded no-effect Lat LIR first-node evidence propagation
bounded no-effect Lat LIR first-node span evidence propagation
bounded no-effect Lat LIR first transition-node evidence propagation
bounded no-effect Lat LIR first transition-node span evidence propagation
bounded no-effect Lat LIR first-edge evidence propagation
bounded no-effect Lat LIR first-edge span evidence propagation
bounded no-effect Lat LIR first transition-source edge evidence propagation
bounded no-effect Lat LIR first transition-source edge endpoint evidence propagation
bounded no-effect Lat LIR first transition-source edge span evidence propagation
bounded no-effect semantic validation
bounded no-effect Lat model normalization
bounded no-effect Lat model report declaration metadata integration
bounded no-effect Lat model report clause metadata integration
bounded no-effect Lat pipeline model normalization integration
bounded no-effect Lat-to-LIR model lowering integration
bounded no-effect Lat-to-LIR declaration metadata refinement
bounded no-effect Lat-to-LIR diagnostic refinement
bounded no-effect Lat-to-LIR diagnostic declaration metadata integration
bounded no-effect Lat pipeline lowering diagnostic integration
bounded no-effect Lat-to-LIR clause metadata refinement
bounded no-effect Lat-to-LIR diagnostic clause metadata integration
bounded no-effect Lat pipeline diagnostic declaration metadata integration
bounded no-effect Lat pipeline diagnostic clause metadata integration
bounded no-effect Lat pipeline report declaration metadata integration
bounded no-effect Lat pipeline report clause metadata integration
bounded no-effect Lat-to-LIR metadata lowering
bounded no-effect Lat pipeline reporting
bounded no-effect Lat pipeline diagnostic integration
bounded no-effect Lat pipeline diagnostic main test audit
bounded no-effect Lat-specific LIR refinement
```

The current Lat grammar report metadata integration copies the first parsed declaration kind/name/source/first-clause metadata and first parsed clause keyword/name/operator/value/effect into deterministic grammar reports when parsing succeeds.

The current Lat grammar line-comment metadata refinement records deterministic comment count and first-comment span metadata in parse results and grammar reports. Comments remain no-effect metadata, can carry otherwise forbidden marker words, and do not change clause parsing or operator behavior.

The current Lat grammar unsupported block-comment rejection refinement rejects block-comment openers outside strings and line comments with a deterministic unsupported_block_comment parse error. Block-comment marker text inside string literals remains ordinary string content.

The current Lat model normalization implementation consumes parser and semantic metadata, builds typed declaration and clause index tables for states, policies, transitions, assertions, and effect declarations, resolves transition source-state metadata, preserves source spans and no-effect flags, and emits deterministic normalization reports.

The current Lat model report declaration metadata integration copies the first normalized declaration index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index into deterministic model normalization reports.

The current Lat model report clause metadata integration copies the first normalized clause role, effect, name, operator, value, and index into deterministic model normalization reports without evaluating operators or changing normalization behavior.

The current Lat-to-LIR lowering implementation consumes normalized Lat model metadata directly, creates a lat_module LIR shape, preserves source spans, no-effect flags, model counts, transition source indices, declaration metadata, and clause operator metadata, and emits deterministic lowering reports. The older parser-plus-semantic lowering entry point remains available as a compatibility wrapper that normalizes a local model first.

The current Lat-to-LIR declaration metadata refinement copies the first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index into deterministic lowering reports.

The current Lat-to-LIR clause metadata refinement stores clause operators in LIR node metadata and reports the first lowered clause role, effect, name, operator, value, and node index for deterministic audit visibility.

The current Lat-to-LIR diagnostic refinement classifies lowering outcomes as valid, parse, semantic, model, effect-check, capacity, LIR, or internal, copies first-declaration and first-clause metadata from lowering results, and preserves no-effect behavior without changing lowering behavior.

The current Lat-to-LIR diagnostic declaration metadata integration copies first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index into deterministic diagnostic reports.

The current Lat pipeline composes Lat source parsing, semantic validation, Lat model normalization, model-driven Lat-to-LIR lowering, LIR metadata, deterministic pipeline reporting, and companion diagnostic reporting into one no-effect integration path. The original pipeline entry point remains available, and a model-aware entry point can return the normalized model to callers.

The current Lat pipeline comment metadata integration copies parser line-comment count and first-comment span metadata into deterministic LAT PIPELINE REPORT output while preserving no execution, no operator evaluation, and no runtime authority.

The current Lat pipeline report declaration metadata integration copies first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from the lowering result into deterministic LAT PIPELINE REPORT output without evaluating operators or changing pipeline behavior.

The current Lat pipeline report clause metadata integration copies first lowered clause role, effect, name, operator, value, and node index from the lowering result into deterministic LAT PIPELINE REPORT output without evaluating operators or changing pipeline behavior.

The current Lat pipeline diagnostic integration combines pipeline error/stage state with Lat semantic diagnostic class, semantic error, diagnostic count, first-diagnostic indices, model-stage classification, and optional Lat-to-LIR lowering diagnostic metadata.

The current Lat pipeline diagnostic comment metadata integration copies parser line-comment count and first-comment span metadata into deterministic pipeline diagnostic reports without evaluating operators or changing pipeline behavior.

The current Lat parse-failure comment evidence propagation guard verifies that line comments before an unsupported block-comment parse failure remain visible through pipeline reports, pipeline diagnostics, and denied runtime-boundary records without adding execution or runtime authority.

The current Lat pipeline failure span evidence propagation copies parser diagnostic/module spans into the pipeline result, pipeline diagnostic reports, and runtime-boundary Lat pipeline evidence so parse-failure locations remain visible without adding execution or runtime authority.

The current Lat pipeline parse-error evidence propagation copies parser error labels into pipeline diagnostic reports and runtime-boundary Lat pipeline evidence so parse-failure reasons remain visible without adding execution or runtime authority.

The current Lat pipeline semantic-error evidence propagation copies semantic error labels into runtime-boundary Lat pipeline evidence so semantic-failure reasons remain visible without adding execution or runtime authority.

The current Lat pipeline downstream stage-error evidence propagation copies model, lowering, and LIR error labels into runtime-boundary Lat pipeline evidence so downstream-stage failure reasons remain visible without adding execution or runtime authority.

The current Lat pipeline stage-summary evidence propagation copies last-completed stage, failed stage, per-stage OK flags, no-effect-chain status, and evidence level into runtime-boundary Lat pipeline evidence so boundary records preserve the pipeline audit summary without adding execution or runtime authority.

The current Lat pipeline module/count evidence propagation copies module identity, parser declaration/clause counts, and model declaration/clause counts into runtime-boundary Lat pipeline evidence so boundary records preserve the source/model shape summary without adding execution or runtime authority.

The current Lat pipeline first-declaration evidence propagation copies first lowered declaration node, kind, name, source, parse index, clause span/count, source index, and first transition source index into runtime-boundary Lat pipeline evidence so boundary records preserve declaration identity without adding execution or runtime authority.

The current Lat pipeline first-clause evidence propagation copies first lowered clause node, role, effect, name, operator, and value into runtime-boundary Lat pipeline evidence so boundary records preserve clause identity without adding execution or runtime authority.

The current Lat LIR edge-kind evidence propagation copies contains, binds, annotates, orders-before, and transition edge-kind counts from Lat-specific LIR into runtime-boundary evidence so boundary records preserve graph shape without adding execution or runtime authority.

The current Lat LIR no-effect evidence propagation copies no-effect-chain status, evidence level, no-effect flag, and execution/mutation/server/recovery/hardware allowance flags from Lat-specific LIR into runtime-boundary evidence so boundary records preserve LIR authority posture without adding execution or runtime authority.

The current Lat LIR module-summary evidence propagation copies module name, report classification, shape kind, node count, edge count, binding count, and text count from Lat-specific LIR into runtime-boundary evidence so boundary records preserve LIR graph identity without adding execution or runtime authority.

The current Lat LIR source-span evidence propagation copies source-span offsets, lines, and columns from Lat-specific LIR into runtime-boundary evidence so boundary records preserve LIR source location without adding execution or runtime authority.

The current Lat LIR node-kind evidence propagation copies Lat-specific state, policy, transition, assertion, requirement, and effect-declaration node counts from Lat-specific LIR into runtime-boundary evidence so boundary records preserve Lat-derived LIR node shape without adding execution or runtime authority.

The current Lat LIR first-node evidence propagation copies the first Lat-specific LIR node presence flag, index, kind, name, value, operator, and binding into runtime-boundary evidence so boundary records preserve the first Lat-derived LIR node identity without adding execution or runtime authority.

The current Lat LIR first-node span evidence propagation copies the first Lat-specific LIR node source-span offsets, lines, and columns into runtime-boundary evidence so boundary records preserve the first Lat-derived LIR node location without adding execution or runtime authority.

The current Lat LIR first transition-node evidence propagation copies the first Lat-specific LIR transition node presence flag, index, kind, name, value, operator, and binding into runtime-boundary evidence so boundary records preserve the first Lat-derived transition node identity without adding execution or runtime authority.

The current Lat LIR first transition-node span evidence propagation copies the first Lat-specific LIR transition node source-span offsets, lines, and columns into runtime-boundary evidence so boundary records preserve the first Lat-derived transition node location without adding execution or runtime authority.

The current Lat LIR first-edge evidence propagation copies the first Lat-specific LIR edge presence flag, index, from/to indices, and kind into runtime-boundary evidence so boundary records preserve the first Lat-derived LIR edge identity without adding execution or runtime authority.

The current Lat LIR first-edge span evidence propagation copies the first Lat-specific LIR edge source-span offsets, lines, and columns into runtime-boundary evidence so boundary records preserve the first Lat-derived LIR edge location without adding execution or runtime authority.

The current Lat LIR first transition-source edge evidence propagation copies the first Lat-specific LIR transition-source edge presence flag, index, from/to indices, and kind into runtime-boundary evidence so boundary records preserve the first Lat-derived transition-source relationship without adding execution or runtime authority.

The current Lat LIR first transition-source edge endpoint evidence propagation copies the first Lat-specific LIR transition-source edge endpoint node kinds and names into runtime-boundary evidence so boundary records preserve which transition points to which source state without adding execution or runtime authority.

The current Lat LIR first transition-source edge span evidence propagation copies the first Lat-specific LIR transition-source edge source-span offsets, lines, and columns into runtime-boundary evidence so boundary records preserve the first Lat-derived transition-source relationship location without adding execution or runtime authority.

The current Lat pipeline diagnostic declaration metadata integration copies first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from Lat-to-LIR diagnostics into deterministic pipeline diagnostic reports.

The current Lat pipeline diagnostic clause metadata integration copies first lowered clause role, effect, name, operator, value, and node index from Lat-to-LIR diagnostics into deterministic pipeline diagnostic reports without evaluating operators or changing pipeline behavior.

The current Lat pipeline diagnostic main test audit verifies that the diagnostic integration is covered by both the focused guard and the main Lat pipeline test runner.

The current Lat-specific LIR refinement makes Lat declarations and transition-source edges explicit in LIR while preserving source spans, counts, and no-effect flags.

It does not provide Lat execution, Lat interpretation, Lat compilation, LIR execution, runtime behavior, command execution, state mutation, file I/O, network I/O, recovery behavior, hardware behavior, or operating-system behavior.

Current C++ authority boundary

The C++ authority layer is contract-bound, implementation-plan-bound, and represented by its first no-effect implementation slice plus an implementation review.

Planning milestone preserved:

```
Constrained C++ authority layer implementation plan
```

It currently implements no-effect authority behavior only:

```
explicit status labels
explicit effect labels
explicit validator labels
fixed-capacity audit records
Lat parse-result metadata validation
LIR shape metadata validation
effect classification without performing effects
deterministic authority report rendering
```

The authority implementation review confirmed that the layer remains no-effect, metadata-only, fixed-capacity, and denied-by-default.

The implementation does not provide unrestricted C++ authority, an effect-performing implemented C++ authority layer, runtime execution, mutation authority, file authority, network authority, recovery authority, hardware authority, or production security guarantees.

Current L-UI rendering boundary

The L-UI rendering lane has a contract, implementation plan, first no-effect C implementation, and detailed report refinement.

The current renderer provides deterministic operator-visible text reports over already validated L-UI/LIR metadata and a C-compatible authority summary.

It does not provide terminal control, interactive UI behavior, command behavior, Lat execution, LIR execution, Nucleus effect execution, mutation, file I/O, network I/O, recovery behavior, hardware behavior, or production UI claims.

Current Nucleus task boundary

The Nucleus task execution lane has a contract, implementation plan, first no-effect C classification/report implementation, deterministic task report refinement, no-effect report alignment, and report-only execution metadata refinement.

Planning and implementation milestones preserved:

```
Nucleus task execution implementation plan
Nucleus task execution implementation
Nucleus task report refinement
Nucleus task no-effect report alignment
Nucleus task report-only execution refinement
Nucleus task report-only execution README/status alignment
Nucleus report-only announcement review
Nucleus report-only announcement README alignment
```

The current implementation provides denied-by-default task classification and deterministic task reports with explicit request kinds, effects, policies, denial reasons, authorization labels, prerequisite status, authority metadata, preview prerequisites, no-effect-chain status, no-effect-policy status, representation-gate status, execution status, effect status, runtime status, and no-effect flags.

The report-only execution refinement makes the execution boundary explicit with execution_status=not-executed, effect_status=report-only, and runtime_status=not-entered while preserving non-execution behavior.

The no-new-announcement review confirms that the recent Nucleus report-only and project-notes slices were documentation/status alignment only and did not justify a separate public announcement entry.

It does not implement effect-performing Nucleus task execution, command behavior, mutation, network behavior, recovery behavior, hardware behavior, boot behavior, rollback, public-product readiness, or production runtime claims.

Current runtime boundary

The runtime boundary lane now has a contract, implementation plan, initial no-effect C API/report surface, no-effect Lat evidence refinement, Lat pipeline comment evidence reporting, runtime boundary report refinement, runtime boundary policy matrix refinement, runtime boundary policy expansion after threat-model validation, runtime boundary domain matrix refinement, runtime boundary domain matrix report integration, and main-test audit coverage.

Planning milestone preserved:

```
Runtime boundary implementation plan
```

The current implementation adds the public runtime boundary header, source file, smoke invariants, report entry point, implementation record, and dedicated runtime-boundary workflow. It establishes a compileable boundary surface while preserving a disabled-by-default posture.

The current runtime boundary domain matrix evaluator classifies resolved boundary domains as declarative, operational, future-gated, blocked, invalid, or unknown.

The current runtime boundary domain matrix report integration renders deterministic report fields for matrix cell, domain label, domain flags, effect-allowed state, authority-available state, and evidence level.

The current runtime boundary policy expansion after threat-model validation maps request families, effect families, authority prerequisites, future gates, threat-model abuse cases, and remaining evidence gaps while preserving no-effect behavior.

The current runtime boundary Lat pipeline comment evidence integration copies Lat pipeline line-comment count and first-comment span metadata into deterministic runtime-boundary records and reports, including denied failed-pipeline records, without granting execution or runtime authority.

It does not implement effect-performing runtime behavior, command execution, Lat execution, LIR execution, task effect execution, mutation, file I/O, network I/O, recovery behavior, rollback, hardware behavior, boot behavior, terminal control, sandboxing, or production runtime claims.

Public entry-point posture

The recent Nucleus report-only announcement README alignment refreshed README, the project-notes index, root status, current status, the status index, and the foundation index after the Nucleus report-only announcement review.

That alignment was documentation/status-only. It did not add a public announcement entry, implementation behavior, capability posture change, or completion-estimate change.

Mission target

The long-term mission is to build a complete open-source operating-system universe and programming-language stack that makes unsafe behavior harder to hide and easier to inspect.

This includes long-term defensive goals against malware, ransomware, unauthorized persistence, hidden mutation, unclear execution, and opaque system behavior.

Target users

Primary target users include:

- intellectuals;
- scientists;
- computer scientists;
- security researchers;
- defensive engineering teams;
- infrastructure maintainers;
- government infrastructure stakeholders.

Current technical lane

The current technical lane has moved through L-UI parser, AST, source-policy, diagnostic, semantic validation, LIR shape, LIR report refinement, Lat parser foundation, Lat grammar report metadata integration, Lat grammar line-comment metadata refinement, Lat grammar unsupported block-comment rejection refinement, Lat pipeline comment metadata integration, Lat pipeline diagnostic comment metadata integration, Lat parse-failure comment evidence propagation, Lat pipeline failure span evidence propagation, Lat pipeline parse-error evidence propagation, Lat pipeline semantic-error evidence propagation, Lat pipeline downstream stage-error evidence propagation, Lat pipeline stage-summary evidence propagation, Lat pipeline module/count evidence propagation, Lat pipeline first-declaration evidence propagation, Lat pipeline first-clause evidence propagation, Lat LIR edge-kind evidence propagation, Lat LIR no-effect evidence propagation, Lat LIR module-summary evidence propagation, Lat LIR source-span evidence propagation, Lat LIR node-kind evidence propagation, Lat LIR first-node evidence propagation, Lat LIR first-node span evidence propagation, Lat LIR first transition-node evidence propagation, Lat LIR first transition-node span evidence propagation, Lat LIR first-edge evidence propagation, Lat LIR first-edge span evidence propagation, Lat LIR first transition-source edge evidence propagation, Lat LIR first transition-source edge endpoint evidence propagation, Lat LIR first transition-source edge span evidence propagation, runtime boundary Lat pipeline comment evidence integration, Lat semantic validation, Lat semantic diagnostics, Lat model normalization, Lat model report declaration metadata integration, Lat model report clause metadata integration, Lat-to-LIR lowering, Lat pipeline, Lat pipeline model normalization integration, Lat-to-LIR model lowering integration, Lat-to-LIR declaration metadata refinement, Lat-to-LIR diagnostic refinement, Lat-to-LIR diagnostic declaration metadata integration, Lat pipeline lowering diagnostic integration, Lat-to-LIR clause metadata refinement, Lat-to-LIR diagnostic clause metadata integration, Lat pipeline diagnostic declaration metadata integration, Lat pipeline diagnostic clause metadata integration, Lat pipeline report declaration metadata integration, Lat pipeline report clause metadata integration, Lat pipeline report refinement, Lat pipeline diagnostic integration, Lat pipeline diagnostic main-test audit, Lat-specific LIR refinement, C/C++ foundation direction, constrained C++ authority-layer contract, constrained C++ authority-layer implementation plan, first no-effect C++ authority implementation, authority implementation review, L-UI rendering contract, L-UI rendering implementation plan, first no-effect L-UI renderer implementation, L-UI rendering detailed report refinement, Nucleus task execution contract, Nucleus task execution implementation plan, first no-effect Nucleus task classification/report implementation, Nucleus task report refinement, Nucleus task no-effect report alignment, Nucleus task report-only execution refinement, Nucleus task report-only execution README/status alignment, project-notes Nucleus report-only alignment, project-notes Nucleus report-only status/index check, Nucleus report-only announcement review, Nucleus report-only announcement README alignment, runtime boundary contract, runtime boundary implementation plan, initial runtime boundary API/smoke implementation, runtime boundary refinement implementation, runtime boundary report refinement, runtime boundary policy matrix refinement, runtime boundary domain matrix refinement, runtime boundary domain matrix report integration, authority/status/foundation index alignment, status announcement review, public entry-point consistency scan, Latticra Seal report envelope metadata implementation, Latticra Seal signature request contract, Latticra Seal signature request metadata implementation, Latticra Seal signature request status/public-entry alignment, Latticra Seal signing authorization contract, Latticra Seal signing authorization metadata implementation, Latticra Seal signing authorization status/public-entry alignment, Latticra Seal signer handoff contract, Latticra Seal signer handoff metadata implementation, Latticra Seal signer handoff status/public-entry alignment, Latticra Seal signer invocation contract, Latticra Seal signer invocation metadata implementation, Latticra Seal signer invocation status/public-entry alignment, Latticra Seal signing operation contract, Latticra Seal signing operation metadata implementation, Latticra Seal signing operation status/public-entry alignment, Latticra Seal key-handling boundary contract, Latticra Seal key-handling metadata implementation, Latticra Seal key-handling status/public-entry alignment, Latticra Seal key-material boundary contract, Latticra Seal key-material metadata implementation, Latticra Seal key-material status/public-entry alignment, Latticra Seal public-key parsing boundary contract, Latticra Seal public-key parsing metadata implementation, Latticra Seal public-key parsing status/public-entry alignment, Latticra Seal future key parsing implementation contract, Latticra Seal future key parsing implementation plan, Latticra Seal bounded key parsing metadata implementation, Latticra Seal key parsing status/public-entry alignment, Latticra Seal verification policy status/public-entry alignment, Latticra Seal verification receipt status/public-entry alignment, Latticra Seal capability gate status/public-entry alignment, Latticra Seal effect decision status/public-entry alignment, Latticra Seal runtime handoff status/public-entry alignment, Latticra Seal status rollup status/public-entry alignment, Latticra Seal agentic automation security public-entrypoint alignment, Latticra Seal parameter schema status/public-entry alignment, Latticra Seal request freshness status/public-entry alignment, Latticra Seal signed request status/public-entry alignment, Latticra Seal policy decision status/public-entry alignment, Latticra Seal README status row alignment, Latticra Seal product spine, Latticra Seal product spine status, Latticra Seal operator receipt report contract, Latticra Seal operator receipt report implementation plan, Latticra Seal operator receipt report implementation, Latticra Seal operator receipt report surface/status, defensive threat model validation refinement, runtime boundary policy expansion after threat-model validation, runtime boundary abuse-case fixture expansion, and completion estimate review README/status alignment.

Runtime boundary abuse-case fixture expansion after policy expansion is now the latest no-effect fixture/evidence slice.

Latticra Seal README status row alignment keeps the first README Seal summary surfaces aligned with the current public status checkpoint while preserving no runtime execution, effect execution, capability enforcement, cryptographic verification, signing, production enforcement, or runtime authority.

Latticra Seal operator receipt report implementation composes report-only capability metadata, policy decision metadata, request freshness metadata, signed request metadata, and runtime dry-run metadata into one denied operator receipt while preserving no receipt file writes, tool execution, host mutation, network behavior, capability enforcement, effects, or runtime authority.

Latticra Seal operator receipt report surface/status alignment makes that denied receipt visible from a deterministic local report runner and public status record while preserving no runtime execution, effect execution, host mutation, network behavior, capability enforcement, or runtime authority.

Latticra Seal local capability registry schema contract defines future local capability-record fields and vocabulary while preserving no registry loader, no file reads, no network behavior, no capability enforcement, no effects, and no runtime authority.

Latticra Seal local capability registry schema implementation plan defines exact future header, source, test, report, enum, capacity, fixture, and negative-case behavior while preserving no registry loader, no file reads, no network behavior, no capability enforcement, no effects, and no runtime authority.

Latticra Seal local capability registry schema implementation adds bounded local schema records, deterministic validation, a default descriptive fixture entry, and negative-case invariants while preserving no registry loader, no registry file access, no host behavior, no network behavior, no capability enforcement, no effects, and no runtime authority.

Previous policy/evidence slice: Runtime boundary policy expansion after threat-model validation.

Completion estimate review after runtime-boundary abuse-case fixtures keeps the planning estimates unchanged because capability posture, public readiness, security hardening, and runtime authority did not change.

Completion estimate review README/status alignment makes that hold review discoverable from README and status surfaces without changing capability posture, public readiness, security hardening, completion estimates, or runtime authority.

Current estimate table source alignment makes the live public estimate table source explicit from README, root status, detailed status, status index, foundation index, and project notes without changing estimates or runtime authority.

Current estimate mathematical rebase updates the live public estimate table to a weighted 45% overall planning estimate after recent guarded Seal, Ubuntu, macOS, Nadia Stage-43 prompt-evaluation release-receipt contract-chain evidence, Lat/LIR, kernel lifecycle scheduler-credit and scheduler-selection-ready evidence, vulnerability-management release-gate baseline coverage, and public-entry alignment work while preserving no implementation behavior change, no product-security claim, no product-readiness promotion, and no runtime authority.

Latticra Seal crypto verify backend status/public-entry alignment makes the existing metadata-only unsupported backend visible from public status surfaces while preserving no real cryptographic verification, signing, key handling, host behavior, network behavior, capability enforcement, effect execution, or runtime authority.

Latticra Seal Ed25519 verify-only status/public-entry alignment makes the existing local provider-backed verify-only result visible from public status surfaces while preserving no signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, capability enforcement, effect execution, runtime authority, or production cryptography claim.

Latticra Seal verified receipt promotion status/public-entry alignment makes the existing verified receipt promotion metadata visible from public status surfaces while preserving evidence promotion only, no capability authorization, no effect execution, no runtime authority, no signing, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal verified capability gate status/public-entry alignment makes the existing metadata-only verified capability gate visible from public status surfaces while preserving no capability enforcement, no effect execution, no runtime authority, no signing, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal verified effect decision status/public-entry alignment makes the existing metadata-only verified effect decision visible from public status surfaces while preserving no effect execution, no capability enforcement, no runtime authority, no signing, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal effect decision predecessor status alignment preserves the denied metadata-only effect decision's guarded capability gate predecessor without adding effect execution, capability enforcement, host behavior, network behavior, or runtime authority.

Latticra Seal runtime handoff predecessor status alignment preserves the inactive metadata-only runtime handoff's guarded effect decision predecessor without adding runtime handoff execution, effect execution, capability enforcement, host behavior, network behavior, or runtime authority.

Latticra Seal status rollup predecessor status alignment preserves the metadata-only status rollup's guarded runtime handoff predecessor without adding runtime execution, runtime handoff execution, effect execution, capability enforcement, host behavior, network behavior, or runtime authority.

Latticra Seal agentic automation security predecessor status alignment preserves the report-only automation status record's guarded status rollup predecessor without adding MCP behavior, AI-agent execution, model execution, tool execution, shell execution, runtime execution, capability enforcement, host behavior, network behavior, or runtime authority.

Latticra Seal runtime handoff evaluation status/public-entry alignment makes the existing metadata-only runtime handoff evaluation visible from public status surfaces while preserving no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no signing, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal runtime handoff report status/public-entry alignment makes the existing metadata-only runtime handoff report visible from public status surfaces while preserving no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no signing, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal report envelope status/public-entry alignment makes the existing metadata-only sealed report envelope visible from public status surfaces while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no effect execution, no capability enforcement, no runtime authority, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal signature request predecessor status alignment ties the existing metadata-only signature request status to the guarded report-envelope status predecessor while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal signing authorization predecessor status alignment ties the existing metadata-only signing authorization status to the guarded signature-request status predecessor while preserving no signing, no signature verification, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal signer handoff predecessor status alignment ties the existing metadata-only signer handoff status to the guarded signing-authorization status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal signer invocation predecessor status alignment ties the existing metadata-only signer invocation status to the guarded signer-handoff status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal signing operation predecessor status alignment ties the existing metadata-only signing operation status to the guarded signer-invocation status predecessor while preserving no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no key handling, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal key-handling predecessor status alignment ties the existing metadata-only key-handling status to the guarded signing-operation status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal key-material predecessor status alignment ties the existing metadata-only key-material status to the guarded key-handling status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal public-key parsing predecessor status alignment ties the existing metadata-only public-key parsing status to the guarded key-material status predecessor while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal key parsing predecessor status alignment ties the existing bounded key parsing status to the guarded public-key parsing status predecessor while preserving the existing caller-provided public-key byte metadata behavior, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal verification policy predecessor status alignment ties the existing metadata-only verification policy status to the guarded key parsing status predecessor while preserving no cryptographic verification, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal verification receipt predecessor status alignment ties the existing metadata-only verification receipt status to the guarded verification policy status predecessor while preserving no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime handoff execution, no capability enforcement, no runtime authority, no host behavior, no network behavior, and no production cryptography claim.

Latticra Seal capability gate predecessor status alignment ties the existing metadata-only denied capability gate status to the guarded verification receipt status predecessor while preserving no capability enforcement, no effect execution, no runtime handoff execution, no cryptographic verification, no verified receipt authority, no public-key byte verification, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no signature verification, no signer invocation behavior, no signer process execution, no object sealing, no runtime authority, no host behavior, no network behavior, and no production enforcement claim.

Latticra Seal parameter schema predecessor status alignment ties the existing report-only parameter schema status to the guarded agentic automation security status predecessor while preserving no schema parsing, no schema validation, no policy evaluation, no policy enforcement, no MCP behavior, no AI-agent execution, no model execution, no tool execution, no shell execution, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no host behavior, no network behavior, and no production enforcement claim.

Latticra Seal request freshness predecessor status alignment ties the existing report-only request freshness status to the guarded parameter schema status predecessor while preserving no timestamp parsing, no trusted clock behavior, no nonce storage, no replay-cache storage, no context hashing, no parameter hashing, no freshness validation, no replay detection, no signature verification, no signed request enforcement, no policy evaluation, no policy enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no host behavior, no network behavior, and no production enforcement claim.

Latticra Seal signed request predecessor status alignment ties the existing report-only signed request status to the guarded request freshness status predecessor while preserving no signature generation, no signature verification, no public-key parsing, no trust-store loading, no private-key handling, no key generation, no hardware-key use, no revocation lookup, no network trust lookup, no signed request enforcement, no policy evaluation, no policy enforcement, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no host behavior, no network behavior, and no production enforcement claim.

Latticra Seal policy decision predecessor status alignment ties the existing report-only policy decision status to the guarded signed request status predecessor while preserving no policy evaluation, no policy enforcement, no signature verification, no freshness validation, no replay detection, no signed request enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no host behavior, no network behavior, and no production enforcement claim.

Latticra Seal operator receipt report predecessor status alignment ties the existing denied report-only operator receipt status to the guarded policy decision status predecessor while preserving no receipt file writes, no tool execution, no policy enforcement, no capability enforcement, no cryptographic verification, no signature verification, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no host behavior, no network behavior, and no production enforcement claim.

The current recommended review lane is:

```
Latticra Seal local capability registry schema report surface or Panel-visible Seal dashboard
planning checkpoint
```

Completion estimate review only if capability posture changes remains the estimate rule after this non-change review.

If capability posture does not change, the project should follow this rule: Continue small guarded report/status alignment only when drift appears.

Current non-claim

Latticra does not currently prevent malware or ransomware, provide a hardened sandbox, replace an operating system, provide unrestricted C++ authority, provide an effect-performing C++ authority layer, provide effect-performing Nucleus task execution, provide effect-performing runtime behavior, provide command execution, provide interactive L-UI rendering, provide terminal-control L-UI rendering, provide a Lat runtime, provide Lat execution, provide LIR execution, or provide a production security boundary.

Those are long-term goals and design targets, not current claims.

docs/project_notes/README.md

Source title: Latticra Project Notes

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Latticra Project Notes

Status: active project notes index Last updated: 2026-05-26 CDT

Purpose

This folder collects short public notes about current Latticra direction, status, and upcoming work.

The project notes are intentionally lighter than the foundation documents. They summarize the active direction and queue without replacing the guarded contracts, implementation records, status records, or announcement log.

Reader route

Use ../README.md as the documentation hub, ../FOUNDATION_INDEX.md as the exhaustive evidence index, and these project notes as short narrative orientation. The long slice names below are preserved because guard scripts assert public-entry alignment across recent milestones.

Files

- CURRENT_DIRECTION.md — current technical lane, boundaries, non-claims, and project direction after the recent Latticra Seal operator receipt report predecessor status alignment, Latticra Seal policy decision predecessor status alignment, Latticra Seal signed request predecessor status alignment, Latticra Seal request freshness predecessor status alignment, Latticra Seal parameter schema predecessor status alignment, Latticra Seal agentic automation security predecessor status alignment, Latticra Seal status rollup predecessor status alignment, Latticra Seal runtime handoff predecessor status alignment, Latticra Seal capability gate predecessor status alignment, Latticra Seal verification receipt predecessor status alignment, Latticra Seal verification policy predecessor status alignment, Latticra Seal key parsing predecessor status alignment, Latticra Seal public-key parsing predecessor status alignment, Seal key-material predecessor status alignment, Seal key-handling predecessor status alignment, Seal signing operation predecessor status alignment, Seal signer invocation predecessor status alignment, Seal signer handoff predecessor status alignment, Seal signing authorization predecessor status alignment, Seal signature request predecessor status alignment, Seal report envelope status/public-entry alignment, Seal runtime handoff report status/public-entry alignment, Seal runtime handoff evaluation status/public-entry alignment, Seal verified effect decision status/public-entry alignment, Seal verified capability gate status/public-entry alignment, Seal verified receipt promotion status/public-entry alignment, Seal Ed25519 verify-only status/public-entry alignment, Seal crypto verify backend status/public-entry alignment, Seal README status row alignment, completion estimate review README/status alignment, completion estimate review after runtime-boundary abuse-case fixtures, runtime boundary abuse-case fixture expansion, runtime boundary policy expansion after threat-model validation, defensive threat model validation refinement, Seal policy decision status/public-entry alignment, Seal signed request status/public-entry alignment, Seal request freshness status/public-entry alignment, Seal parameter schema status/public-entry alignment, Seal agentic automation security public-entrypoint alignment, Seal status rollup status alignment, Seal runtime handoff status alignment, Latticra Seal effect decision predecessor status alignment, Seal effect decision status alignment, Seal capability gate status alignment, Seal verification receipt status alignment, Seal verification policy status alignment, Seal key parsing status alignment, Seal key parsing implementation, Seal planning, Lat pipeline diagnostic, RBDM report, authority review, L-UI report, Nucleus task, runtime-boundary, README/status alignment, and no-new-announcement review slices. See docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md; predecessor docs/status/SEAL_POLICY_DECISION_STATUS.md.
- UPCOMING_WORK.md — current near-term queue after the recent Latticra Seal operator receipt report predecessor status alignment, Latticra Seal policy decision predecessor status alignment, Latticra Seal signed request predecessor status alignment, Latticra Seal request freshness predecessor status alignment, Latticra Seal parameter schema predecessor status alignment, Latticra Seal agentic automation security predecessor status alignment, Latticra Seal status rollup predecessor status alignment, Latticra Seal runtime handoff predecessor status alignment, Latticra Seal capability gate predecessor status alignment, Latticra Seal verification receipt predecessor status alignment, Latticra Seal verification policy predecessor status alignment, Latticra Seal key parsing predecessor status alignment, Latticra Seal public-key parsing predecessor status alignment, Seal key-material predecessor status alignment, Seal key-handling predecessor status alignment, Seal signing operation predecessor status alignment, Seal signer invocation predecessor status alignment, Seal signer handoff predecessor status alignment, Seal signing authorization predecessor status alignment, Seal signature request predecessor status alignment, Seal report envelope status/public-entry alignment, Seal runtime handoff report status/public-entry

alignment, Seal runtime handoff evaluation status/public-entry alignment, Seal verified effect decision status/public-entry alignment, Seal verified capability gate status/public-entry alignment, Seal verified receipt promotion status/public-entry alignment, Seal Ed25519 verify-only status/public-entry alignment, Seal crypto verify backend status/public-entry alignment, Seal README status row alignment, completion estimate review README/status alignment, completion estimate review after runtime-boundary abuse-case fixtures, runtime boundary abuse-case fixture expansion, runtime boundary policy expansion after threat-model validation, defensive threat model validation refinement, Seal policy decision status/public-entry alignment, Seal signed request status/public-entry alignment, Seal request freshness status/public-entry alignment, Seal parameter schema status/public-entry alignment, Seal agentic automation security public-entrypoint alignment, Seal status rollup status alignment, Seal runtime handoff status alignment, Latticra Seal effect decision predecessor status alignment, Seal effect decision status alignment, Seal capability gate status alignment, Seal verification receipt status alignment, Seal verification policy status alignment, Seal key parsing status alignment, Seal key parsing implementation, Seal implementation plan, Seal contract, no-effect report, diagnostic, audit, README, foundation-index, announcement-review, public-entry, project-notes, and Nucleus report-only execution alignment slices. See docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md; predecessor docs/status/SEAL_POLICY_DECISION_STATUS.md.

Current note posture

```
CURRENT_DIRECTION.md is the narrative direction note.
UPCOMING_WORK.md is the queue and priority note.
```

The project-level handbook in docs/latticra-system-substrate/ should be kept current as these notes evolve. When a slice changes the durable project story, architecture, evidence posture, public wording, or non-claim boundary, reflect that change in the handbook path as part of the same working rhythm.

Both notes should remain:

```
small
public
evidence-bound
clear about non-claims
consistent with the C/C++ foundation direction
consistent with the no-effect runtime boundary
consistent with the Nucleus report-only execution boundary
consistent with no-new-announcement decisions unless capability posture changes
```

Related records

Latest Seal chain note: Latticra Seal runtime handoff predecessor status alignment uses docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md with predecessor docs/status/SEAL_EFFECT_DECISION_STATUS.md.

Latest Seal chain note: Latticra Seal status rollup predecessor status alignment uses docs/status/SEAL_STATUS_ROLLUP_STATUS.md with predecessor docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md.

Latest Seal chain note: Latticra Seal agentic automation security predecessor status alignment uses docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md with predecessor docs/status/SEAL_STATUS_ROLLUP_STATUS.md.

Latest Seal chain note: Latticra Seal parameter schema predecessor status alignment uses docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md with predecessor docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md.

Latest Seal chain note: Latticra Seal request freshness predecessor status alignment uses docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md with predecessor docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md.

Latest Seal chain note: Latticra Seal signed request predecessor status alignment uses docs/status/SEAL_SIGNED_REQUEST_STATUS.md with predecessor docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md.

Latest Seal chain note: Latticra Seal policy decision predecessor status alignment uses docs/status/SEAL_POLICY_DECISION_STATUS.md with predecessor docs/status/SEAL_SIGNED_REQUEST_STATUS.md.

Latest Seal chain note: Latticra Seal operator receipt report predecessor status alignment uses docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md with predecessor docs/status/SEAL_POLICY_DECISION_STATUS.md.

Latest Seal predecessor note: Latticra Seal capability gate predecessor status alignment uses docs/status/SEAL_CAPABILITY_GATE_STATUS.md with predecessor docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md.

- STATUS.md
- README.md
- docs/status/CURRENT_STATUS.md
- docs/status/ANNOUNCEMENTS.md
- docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_REVIEW.md
- docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_README_ALIGNMENT.md
- docs/status/PROJECT_NOTES_NUCLEUS_ANNOUNCEMENT_README_ALIGNMENT_STATUS_CHECK.md
- docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md
- docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_README_STATUS_ALIGNMENT.md
- docs/status/PUBLIC_ENTRY_POINT_CONSISTENCY_SCAN.md
- docs/status/STATUS_ANNOUNCEMENT_REVIEW.md
- docs/status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md
- docs/status/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_STATUS.md
- docs/status/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_AUDIT_STATUS.md
- docs/status/RBDM_REPORT_INTEGRATION_STATUS.md
- docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md
- docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md
- docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
- docs/status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md
- docs/status/SEAL_ED25519_VERIFY_STATUS.md
- docs/status/SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md
- docs/status/SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md
- docs/status/SEAL_VERIFIED_EFFECT_DECISION_STATUS.md
- docs/status/SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md
- docs/status/SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md
- docs/status/SEAL_REPORT_ENVELOPE_STATUS.md
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- docs/status/SEAL_KEY_PARSING_STATUS.md
- docs/status/SEAL_VERIFICATION_POLICY_STATUS.md
- docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md
- docs/status/SEAL_CAPABILITY_GATE_STATUS.md
- docs/status/SEAL_EFFECT_DECISION_STATUS.md
- docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
- docs/status/SEAL_STATUS_ROLLUP_STATUS.md
- docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md
- docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md
- docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md
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- docs/status/SEAL_POLICY_DECISION_STATUS.md
- docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_CONTRACT.md
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION_PLAN.md
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md
- docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md
- docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md
- docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md
- docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md
- docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md
- docs/status/SEAL_POLICY_DECISION_PUBLIC_ENTRYPOINT_ALIGNMENT.md
- docs/status/SEAL_README_STATUS_ROW_ALIGNMENT.md
- docs/status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md
- docs/status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md
- docs/status/COMPLETION_ESTIMATE_REVIEW_README_STATUS_ALIGNMENT.md
- docs/status/COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md
- docs/RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES_AFTER_POLICY_EXPANSION.md
- docs/RUNTIME_BOUNDARY_POLICY_EXPANSION_AFTER_THREAT_MODEL.md
- docs/DEFENSIVE_THREAT_MODEL_VALIDATION_REFINEMENT.md
- docs/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md

- docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md
- docs/strategy/README.md
- docs/FOUNDATION_INDEX.md

Update rule

Update this index whenever CURRENT_DIRECTION.md or UPCOMING_WORK.md changes in a milestone-bearing way.

docs/project_notes/UPCOMING_WORK.md

Source title: Latticra Upcoming Work

Lines: 1287 | Words: 8661 | SHA-256: baaad8a58b6a39226f4560f21f634530490dff1f87b7982989cd86a15f0eb39a

Latticra Upcoming Work

Status: active project note Last updated: 2026-05-26 CDT Latest current estimate table source alignment note: 2026-05-26 CDT Latest current estimate mathematical rebase note: 2026-05-26 CDT Latest completion estimate review README/status alignment note: 2026-05-25 CDT Latest completion estimate review after runtime-boundary abuse-case fixtures note: 2026-05-25 CDT Latest high-assurance security baseline note: 2026-05-26 CDT Latest memory-safety roadmap note: 2026-05-26 CDT Latest supply-chain security baseline note: 2026-05-26 CDT Latest zero-trust runtime authority baseline note: 2026-05-26 CDT Latest runtime boundary abuse-case fixture expansion note: 2026-05-25 CDT Latest runtime boundary policy expansion after threat-model note: 2026-05-26 CDT Latest runtime boundary Lat pipeline comment evidence note: 2026-05-25 CDT Latest Lat parse-failure comment evidence propagation note: 2026-05-25 CDT Latest Lat pipeline failure span evidence propagation note: 2026-05-25 CDT Latest Lat 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Purpose

This note tracks the near-term work queue after the recent no-effect Lat, LIR, Nucleus, runtime-boundary, authority review, status, announcement-review, foundation-index, public-entry, project-notes, and Nucleus report-only announcement README alignment slices.

The project remains evidence-bound and report/classification oriented. The queue should continue to prefer small, reviewable, guarded slices over broad runtime behavior.

The security-guidance refresh slices are aligned across the baseline, roadmap, supply-chain, zero-trust, and runtime-boundary policy docs. Near-term follow-up should stay narrow: summary/status consistency only when wording drifts, and capability-bearing work only under separate guarded contracts.

Latticra Seal README status row alignment remains complete for the compact README Seal row and current-posture summary.

Latticra Seal crypto verify backend status/public-entry alignment is complete for the current metadata-only unsupported backend surface. It does not add real cryptographic verification, signing, key handling, host behavior, network behavior, capability enforcement, effect execution, or runtime authority.

Latticra Seal Ed25519 verify-only status/public-entry alignment is complete for the existing local provider-backed verification result surface. It does not add new implementation behavior, signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, capability enforcement, effect execution, runtime authority, or production cryptography claims.

Latticra Seal verified receipt promotion status/public-entry alignment is complete for the existing evidence-promotion metadata surface. It does not add new implementation behavior, capability authorization, effect execution, runtime authority, signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal verified capability gate status/public-entry alignment is complete for the existing metadata-only gate evaluation surface. It does not add new implementation behavior, capability enforcement, effect execution, runtime authority, signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal verified effect decision status/public-entry alignment is complete for the existing metadata-only effect classification surface. It does not add new implementation behavior, effect execution, capability enforcement, runtime authority, signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal runtime handoff evaluation status/public-entry alignment is complete for the existing metadata-only handoff eligibility classification surface. It does not add runtime handoff execution, effect execution, capability enforcement, runtime authority, signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal runtime handoff report status/public-entry alignment is complete for the existing metadata-only report readiness classification surface. It does not add runtime handoff execution, effect execution, capability enforcement, runtime authority, signing, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal report envelope status/public-entry alignment is complete for the existing metadata-only sealed report envelope surface. It does not add signing, signature verification, object sealing, runtime handoff execution, effect execution, capability enforcement, runtime authority, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal signature request predecessor status alignment is complete for the existing metadata-only signature request status surface. It does not add signing, signature verification, object sealing, runtime handoff execution, capability enforcement, runtime authority, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal signing authorization predecessor status alignment is complete for the existing metadata-only signing authorization status surface. It does not add signing, signature verification, object sealing, runtime handoff execution, capability enforcement, runtime authority, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal signer handoff predecessor status alignment is complete for the existing metadata-only signer handoff status surface. It does not add signing, signature verification, signer invocation behavior, object sealing, runtime handoff execution, capability enforcement, runtime authority, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal signer invocation predecessor status alignment is complete for the existing metadata-only signer invocation status surface. It does not add signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal signing operation predecessor status alignment is complete for the existing metadata-only signing operation status surface. It does not add signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, key generation, private-key handling, trust-store behavior, revocation lookup, host behavior, network behavior, or production cryptography claims.

Latticra Seal key-handling predecessor status alignment is complete for the existing metadata-only key-handling status surface. It does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, host behavior, network behavior, or production cryptography claims.

Latticra Seal key-material predecessor status alignment is complete for the existing metadata-only key-material status surface. It does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, host behavior, network behavior, or production cryptography claims.

Latticra Seal public-key parsing predecessor status alignment is complete for the existing metadata-only public-key parsing status surface. It does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, host behavior, network behavior, or production cryptography claims.

Latticra Seal key parsing predecessor status alignment is complete for the existing bounded key parsing status surface. It does not change the caller-provided public-key byte metadata behavior, add key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, host behavior, network behavior, or production cryptography claims.

Latticra Seal verification policy predecessor status alignment is complete for the existing metadata-only verification policy status surface. It does not add cryptographic verification, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, host behavior, network behavior, or production cryptography claims.

Latticra Seal verification receipt predecessor status alignment is complete for the existing metadata-only verification receipt status surface. It does not add cryptographic verification, verified receipt authority, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, capability enforcement, runtime authority, host behavior, network behavior, or production cryptography claims.

Latticra Seal capability gate predecessor status alignment is complete for the existing metadata-only denied capability gate status surface. It does not add capability enforcement, effect execution, runtime handoff execution, cryptographic verification, verified receipt authority, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime authority, host behavior, network behavior, or production enforcement claims.

Latticra Seal runtime handoff predecessor status alignment is complete for the existing inactive metadata-only runtime handoff status surface. It does not add runtime handoff execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signer invocation behavior, signer process execution, object sealing, host behavior, network behavior, or production cryptography claims.

Latticra Seal status rollup predecessor status alignment is complete for the existing metadata-only status rollup status surface. It does not add runtime execution, runtime handoff execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, signer invocation behavior, signer process execution, object sealing, host behavior, network behavior, or production enforcement claims.

Latticra Seal agentic automation security predecessor status alignment is complete for the existing report-only automation security status surface. It does not add MCP behavior, AI-agent execution, model execution, tool execution, shell execution, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, public-key byte verification, key material loading, private-key handling, object sealing, host behavior, network behavior, or production enforcement claims.

Latticra Seal parameter schema predecessor status alignment is complete for the existing report-only parameter schema status surface. It does not add schema parsing, schema validation, policy evaluation, policy enforcement, MCP behavior, AI-agent execution, model execution, tool execution, shell execution, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, key material loading, private-key handling, object sealing, host behavior, network behavior, or production enforcement claims.

Latticra Seal request freshness predecessor status alignment is complete for the existing report-only request freshness status surface. It does not add timestamp parsing, trusted clock behavior, nonce storage, replay-cache storage, context hashing, parameter hashing, freshness validation, replay detection, signature verification, signed request enforcement, policy evaluation, policy enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, key material loading, private-key handling, object sealing, host behavior, network behavior, or production enforcement claims.

Latticra Seal signed request predecessor status alignment is complete for the existing report-only signed request status surface. It does not add signature generation, signature verification, public-key parsing, trust-store loading, private-key handling, key generation, hardware-key use, revocation lookup, network trust lookup, signed request enforcement, policy evaluation, policy enforcement, freshness validation, replay detection, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, key material loading, private-key handling, object sealing, host behavior, network behavior, or production enforcement claims.

Latticra Seal policy decision predecessor status alignment is complete for the existing report-only policy decision status surface. It does not add policy evaluation, policy enforcement, signature verification, freshness validation, replay detection, signed request enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, key material loading, private-key handling, object sealing, host behavior, network behavior, or production enforcement claims.

Latticra Seal operator receipt report predecessor status alignment is complete for the existing denied report-only operator receipt status surface. It does not add receipt file writes, tool execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, runtime execution, runtime authority, effect execution, verified receipt authority, key material loading, private-key handling, object sealing, host behavior, network behavior, or production enforcement claims.

Latest completed planning slice

```
Latticra Seal local capability registry schema implementation plan
```

Purpose completed:

```
define exact future local capability registry schema header, source, tests, report shape, enum
labels, capacity, fixture, and negative-case behavior while preserving no registry loader, no
file reads, no tool execution, no host mutation, no network behavior, no capability enforcement,
 no effects, and no runtime authority
```

Latest completed implementation slice

```
Latticra Seal local capability registry schema implementation
```

Purpose completed:

```
add bounded local schema records, deterministic validation, a default descriptive fixture entry,
 and negative-case invariants while preserving no registry loader, no registry file access, no
tool execution, no host mutation, no network behavior, no capability enforcement, no effects,
and no runtime authority
```

Previous implementation slice

```
Latticra Seal operator receipt report surface/status
```

Purpose completed:

```
render the no-effect operator receipt report through a deterministic local report runner and
status checkpoint while preserving no receipt file writes, no tool execution, no host mutation,
no network behavior, no capability enforcement, no effects, and no runtime authority
```

Earlier implementation slice

```
Lat LIR first transition-node span evidence propagation
```

Purpose completed:

```
copy the first Lat-specific LIR transition node source-span offsets/lines/columns into
runtime-boundary records/reports so boundary evidence keeps the first Lat-derived transition
node location visible while preserving no Lat execution, no LIR execution, no operator
evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime
 authority
```

Earlier implementation slice

```
Lat LIR first transition-node evidence propagation
```

Purpose completed:

copy the first Lat-specific LIR transition node presence flag, index, kind, name, value, operator, and binding into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived transition node identity visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR first transition-source edge span evidence propagation

Purpose completed:

copy the first Lat-specific LIR transition-source edge source-span offsets/lines/columns into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived transition-source relationship location visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR first transition-source edge evidence propagation

Purpose completed:

copy the first Lat-specific LIR transition-source edge presence flag, index, from/to indices, and kind into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived transition-source relationship visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR first-edge span evidence propagation

Purpose completed:

copy the first Lat-specific LIR edge source-span offsets/lines/columns into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived LIR edge location visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR first-edge evidence propagation

Purpose completed:

copy the first Lat-specific LIR edge presence flag, index, from/to indices, and kind into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived LIR edge identity visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR first-node span evidence propagation

Purpose completed:

copy the first Lat-specific LIR node source-span offsets/lines/columns into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived LIR node location visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR first-node evidence propagation

Purpose completed:

copy the first Lat-specific LIR node presence flag, index, kind, name, value, operator, and binding into runtime-boundary records/reports so boundary evidence keeps the first Lat-derived LIR node identity visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR node-kind evidence propagation

Purpose completed:

copy Lat-specific LIR state, policy, transition, assertion, requirement, and effect-declaration node-kind counts into runtime-boundary records/reports so boundary evidence keeps Lat-derived LIR node shape visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR source-span evidence propagation

Purpose completed:

copy Lat-specific LIR source-span offsets/lines/columns into runtime-boundary records/reports so boundary evidence keeps LIR source location visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR module-summary evidence propagation

Purpose completed:

copy Lat-specific LIR module name, report classification, shape kind, node count, edge count, binding count, and text count into runtime-boundary records/reports so boundary evidence keeps LIR graph identity visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR no-effect evidence propagation

Purpose completed:

copy Lat-specific LIR no-effect-chain status, evidence level, no-effect flag, and execution/mutation/server/recovery/hardware allowance flags into runtime-boundary records/reports so boundary evidence keeps LIR authority posture visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat LIR edge-kind evidence propagation

Purpose completed:

copy Lat-specific LIR contains, binds, annotates, orders-before, and transition edge-kind counts into runtime-boundary records/reports so boundary evidence keeps LIR graph shape visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline first-clause evidence propagation

Purpose completed:

copy Lat pipeline first lowered clause node, role, effect, name, operator, and value into runtime-boundary Lat pipeline records/reports so boundary evidence keeps clause identity visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline first-declaration evidence propagation

Purpose completed:

copy Lat pipeline first lowered declaration node, kind, name, source, parse index, clause span/count, source index, and first transition source index into runtime-boundary Lat pipeline records/reports so boundary evidence keeps declaration identity visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline module/count evidence propagation

Purpose completed:

copy Lat pipeline module name, parser declaration/clause counts, and model declaration/clause counts into runtime-boundary Lat pipeline records/reports so boundary evidence keeps source and model shape counts visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline stage-summary evidence propagation

Purpose completed:

copy Lat pipeline last-completed stage, failed stage, per-stage OK flags, no-effect-chain status, and evidence level into runtime-boundary Lat pipeline records/reports so boundary evidence keeps the pipeline stage summary visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline downstream stage-error evidence propagation

Purpose completed:

copy Lat pipeline model, lowering, and LIR error labels into runtime-boundary Lat pipeline records/reports so downstream-stage failure reasons remain visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline semantic-error evidence propagation

Purpose completed:

copy Lat pipeline semantic error labels into runtime-boundary Lat pipeline records/reports so semantic-failure reasons remain visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline parse-error evidence propagation

Purpose completed:

copy parser error labels from Lat pipeline summaries into Lat pipeline diagnostic reports and runtime-boundary Lat pipeline records/reports so parse-failure reasons remain visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline failure span evidence propagation

Purpose completed:

copy parser diagnostic/module spans into Lat pipeline summaries, Lat pipeline diagnostic reports, and runtime-boundary Lat pipeline records/reports so parse-failure locations remain visible while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat parse-failure comment evidence propagation

Purpose completed:

verify that line-comment count and first-comment span metadata remain visible through Lat pipeline reports, Lat pipeline diagnostics, and denied runtime-boundary records when parsing fails on an unsupported block-comment opener while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Runtime boundary Lat pipeline comment evidence integration

Purpose completed:

copy Lat pipeline line-comment count and first-comment span metadata into deterministic runtime-boundary records and reports while preserving no Lat execution, no LIR execution, no operator evaluation, no runtime execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline diagnostic comment metadata integration

Purpose completed:

copy parser line-comment count and first-comment span metadata into deterministic Lat pipeline diagnostic reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline comment metadata integration

Purpose completed:

copy parser line-comment count and first-comment span metadata into deterministic Lat pipeline reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat grammar unsupported block-comment rejection refinement

Purpose completed:

reject Lat block-comment openers outside strings and line comments with a deterministic unsupported_block_comment parse error while preserving no Lat execution, no operator evaluation, no state mutation, no file I/O, no network I/O, no runtime authority, and normal string handling for block-comment marker text

Earlier implementation slice

Lat grammar line-comment metadata refinement

Purpose completed:

record deterministic Lat line-comment count and first-comment span metadata in parse results and grammar reports while preserving no Lat execution, no operator evaluation, no state mutation, no file I/O, no network I/O, no runtime authority, and no effect authority from comments

Earlier implementation slice

Lat pipeline report declaration metadata integration

Purpose completed:

copy first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from Lat-to-LIR lowering results into deterministic Lat pipeline reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline diagnostic declaration metadata integration

Purpose completed:

copy first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from Lat-to-LIR diagnostics into deterministic Lat pipeline diagnostic reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat-to-LIR diagnostic declaration metadata integration

Purpose completed:

copy first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from Lat-to-LIR lowering results into deterministic Lat-to-LIR diagnostic reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat-to-LIR declaration metadata refinement

Purpose completed:

copy first lowered declaration node index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from normalized Lat model metadata into deterministic Lat-to-LIR lowering reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat model report declaration metadata integration

Purpose completed:

copy first normalized declaration index, kind, name, source name, parse index, first-clause index, clause count, and source declaration index from the normalized Lat model table into deterministic Lat model normalization reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat grammar report metadata integration

Purpose completed:

copy first parsed declaration and first parsed clause metadata from successful Lat AST parses into deterministic Lat grammar reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat model report clause metadata integration

Purpose completed:

copy first normalized clause index, role, effect, name, operator, and value from the normalized Lat model table into deterministic Lat model normalization reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline report clause metadata integration

Purpose completed:

copy first lowered clause node index, role, effect, name, operator, and value from Lat-to-LIR lowering results into deterministic Lat pipeline reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline diagnostic clause metadata integration

Purpose completed:

copy first lowered clause node index, role, effect, name, operator, and value from Lat-to-LIR diagnostics into deterministic Lat pipeline diagnostic reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat-to-LIR diagnostic clause metadata integration

Purpose completed:

copy first lowered clause node index, role, effect, name, operator, and value from Lat-to-LIR lowering results into deterministic Lat-to-LIR diagnostic reports while preserving no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat-to-LIR clause metadata refinement

Purpose completed:

preserve Lat clause operators in LIR node metadata, expose first lowered clause role, effect, name, operator, value, and node index in deterministic Lat-to-LIR reports, add focused and aggregate guards, and preserve no Lat execution, no LIR execution, no operator evaluation, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier implementation slice

Lat pipeline lowering diagnostic integration

Purpose completed:

extend the Lat pipeline diagnostic evaluator with optional Lat-to-LIR lowering diagnostics, copy lowering class, lowering error, model error, LIR error, model counts, transition source metadata, and failure flags into deterministic pipeline diagnostic reports, keep the older evaluator compatible, and preserve no Lat execution, no LIR execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Older implementation slice

Lat-to-LIR diagnostic refinement

Purpose completed:

add deterministic Lat-to-LIR diagnostic classification and reports for valid, parse, semantic, model, effect-check, capacity, LIR, and internal lowering outcomes while copying model/lowering/LIR errors, model counts, transition source metadata, no-effect flags, and evidence level, and preserving no Lat execution, no LIR execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Older implementation slice

Lat-to-LIR model lowering integration

Purpose completed:

make Lat-to-LIR lowering consume the normalized Lat model directly, keep the parser-plus-semantic lowering entry point as a compatibility wrapper, expose model error/count/source-index metadata in lowering reports, route the Lat pipeline through model-driven lowering, and preserve no Lat execution, no LIR execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Older implementation slice

Lat pipeline model normalization integration

Purpose completed:

integrate bounded no-effect Lat model normalization into the Lat pipeline, keep the original pipeline entry point compatible, add a model-aware entry point for callers that need normalized model metadata, expand pipeline reports with model error/count/source-index fields, and preserve no Lat execution, no LIR execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Older Lat implementation slice

Lat model normalization implementation

Purpose completed:

implement bounded no-effect Lat model normalization after semantic validation, with typed declaration and clause index tables, transition source-state metadata, deterministic reports, source-span preservation, no source-byte reading, no Lat execution, no LIR execution, no state mutation, no file I/O, no network I/O, and no runtime authority

Earlier Lat implementation slice

Latticra Seal bounded no-effect key parsing implementation

Purpose completed:

implement bounded key parsing metadata for caller-provided Ed25519 public-key bytes only while preserving no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Latest completed validation slice

Defensive threat model validation refinement

Purpose completed:

refine the defensive threat model validation surface, external-source checkpoint posture, and next-gap triage while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, production protection, or runtime authority

Latest completed policy/evidence slice

Runtime boundary policy expansion after threat-model validation

Purpose completed:

map runtime-boundary request families, effect families, authority prerequisites, future gates, defensive-threat-model abuse cases, and remaining evidence gaps while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, production protection, and no runtime authority

Latest completed status/public-entry slice

Latticra Seal operator receipt report predecessor status alignment

Purpose completed:

tie and guard the existing denied report-only Seal operator receipt status to the policy decision status predecessor while preserving no new implementation behavior, no receipt file writes, no tool execution, no policy enforcement, no capability enforcement, no cryptographic verification, no signature verification, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no verified receipt authority, no key material loading, no private-key handling, no object sealing, no host behavior, no network behavior, and no production enforcement claim

Previous status/public-entry slice

Latticra Seal policy decision predecessor status alignment

Purpose completed:

tie and guard the existing report-only Seal policy decision status to the signed request status predecessor while preserving no new implementation behavior, no policy evaluation, no policy enforcement, no signature verification, no freshness validation, no replay detection, no signed request enforcement, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no key material loading, no private-key handling, no object sealing, no host behavior, no network behavior, and no production enforcement claim

Earlier status/public-entry slice

Latticra Seal signed request predecessor status alignment

Purpose completed:

tie and guard the existing report-only Seal signed request status to the request freshness status predecessor while preserving no new implementation behavior, no signature generation, no signature verification, no public-key parsing, no trust-store loading, no private-key handling, no key generation, no hardware-key use, no revocation lookup, no network trust lookup, no signed request enforcement, no policy evaluation, no policy enforcement, no freshness validation, no replay detection, no runtime execution, no runtime authority, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no key material loading, no private-key handling, no object sealing, no host behavior, no network behavior, and no production enforcement claim

Earlier status/public-entry slice

Latticra Seal signed request status/public-entry alignment

Purpose completed:

publish and guard the existing report-only Seal signed request metadata from README/status/foundation entry points while preserving no signature generation, no signature verification, no public-key parsing, no trust-store loading, no private-key handling, no key generation, no hardware-key use, no revocation lookup, no network trust lookup, no signed request enforcement, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority

Older status/public-entry slice

Latticra Seal policy decision status/public-entry alignment

Purpose completed:

publish and guard the existing report-only Seal policy decision metadata and deterministic report surface from README/status/foundation entry points while preserving no policy evaluation, no policy enforcement, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no signature verification, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority

Older status/public-entry slice

Latticra Seal request freshness status/public-entry alignment

Purpose completed:

publish and guard the existing report-only Seal request freshness metadata and report surface from README/status/foundation entry points while preserving no timestamp parsing, no trusted clock behavior, no nonce storage, no replay-cache storage, no context hashing, no parameter hashing, no freshness validation, no replay detection, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority

Older status/public-entry slice

Latticra Seal parameter schema status/public-entry alignment

Purpose completed:

publish and guard the existing report-only Seal parameter schema metadata and report surface from README/status/foundation entry points while preserving no schema parsing, no schema validation, no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority

Older status/public-entry slice

Latticra Seal agentic automation security public-entrypoint alignment

Purpose completed:

publish and guard the existing report-only Seal agentic automation security surface from README/status/foundation entry points while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent execution, no model execution, no tool execution, no shell execution, and no runtime authority

Older status/public-entry slice

Latticra Seal status rollup status/public-entry alignment

Purpose completed:

publish and guard the existing metadata-only status rollup surface from public/status entry points while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority

Earlier status/public-entry slice

Latticra Seal runtime handoff status/public-entry alignment
Latticra Seal runtime handoff report status/public-entry alignment

Purpose completed:

publish and guard the existing inactive metadata-only runtime handoff surface from public/status entry points while preserving no runtime execution, no effect execution, no capability enforcement, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority

Older status/public-entry slice

Latticra Seal effect decision status/public-entry alignment
Latticra Seal effect decision predecessor status alignment

Purpose completed:

publish and guard the existing metadata-only denied effect decision surface from public/status entry points while preserving no effect execution, no capability enforcement, no runtime handoff execution, no cryptographic verification, no verified receipt authority, no signing, no host behavior, no network behavior, and no runtime authority

Earlier planning slice

Constrained C++ authority layer implementation plan

Purpose completed:

define exact public API, namespace, file paths, C++ standard, compiler flags, exception policy, RTTI policy, allocation policy, ownership/lifetime rules, result labels, C ABI boundaries, validators, audit reports, and tests before any C++ authority-layer code

Latest completed contract slice

Latticra Seal local capability registry schema contract

Purpose completed:

define the local capability registry schema boundary after operator receipt status readiness while preserving no registry loader, no registry file access, no trust-store behavior, no signing, no verification, no host behavior, no network behavior, no capability enforcement, no effects, and no runtime authority

Previous status/public-entry slice

Latticra Seal capability gate status/public-entry alignment

Purpose completed:

publish and guard the existing metadata-only denied capability gate surface from public/status entry points while preserving no capability enforcement, no effect execution, no cryptographic verification, no verified receipt authority, no signing, no key material loading, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, and no runtime authority

Earlier status/public-entry slice

Latticra Seal verification receipt status/public-entry alignment

Purpose completed:

publish and guard the existing metadata-only verification receipt surface from public/status entry points while preserving no cryptographic verification, no verified receipt authority, no signing, no key material loading, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Earlier status/public-entry slice

Latticra Seal verification policy status/public-entry alignment

Purpose completed:

publish and guard the existing metadata-only verification policy surface from public/status entry points while preserving no cryptographic verification, no signing, no public-key byte verification, no key material loading, no private-key handling, no trust-store behavior, no revocation lookup, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Earlier status/public-entry slice

Latticra Seal key parsing status/public-entry alignment

Purpose completed:

publish and guard the bounded no-effect key parsing metadata surface from public/status entry points while preserving no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no revocation lookup, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous implementation slice

Latticra Seal public-key parsing metadata implementation

Purpose completed:

implement metadata-only public-key parsing classification after key-material status readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous contract slice

Latticra Seal public-key parsing boundary contract

Purpose completed:

define the next metadata-only public-key parsing boundary after key-material status readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous status/public-entry slice

Latticra Seal key-material status/public-entry alignment

Purpose completed:

publish and guard the key-material metadata implementation in the status and public-entry surfaces while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store behavior, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous implementation slice

Latticra Seal key-material metadata implementation

Purpose completed:

implement metadata-only key-material classification after key-handling readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous contract slice

Latticra Seal key-material boundary contract

Purpose completed:

define the next metadata-only key-material boundary after key-handling readiness while preserving no public-key parsing, no key material loading, no private-key handling, no key generation, no hardware-key use, no trust-store loading, no signing, no verification, no signer invocation behavior, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous status/public-entry slice

Latticra Seal key-handling status/public-entry alignment

Previous implementation slice

Latticra Seal key-handling metadata implementation

Previous contract slice

Latticra Seal key-handling boundary contract

Previous status/public-entry slice

Latticra Seal signing operation status/public-entry alignment

Earlier implementation slice

Latticra Seal signing operation metadata implementation

Previous contract slices retained for continuity

Lat-to-LIR lowering contract
Lat semantic validation contract
Runtime boundary contract
Nucleus task execution contract
Lat-specific LIR refinement contract
Latticra Seal signing authorization contract
Latticra Seal signer handoff contract
Latticra Seal signer invocation contract
Latticra Seal signing operation contract

Previous planning slice

Runtime boundary refinement plan

Purpose completed:

define the next no-effect runtime boundary refinement after Lat pipeline and Lat-specific LIR refinement, including compatibility requirements, report expectations, prerequisite metadata, and non-claims before any runtime behavior expands

Historical planning slices retained for continuity

Constrained C++ authority layer implementation plan
Lat-to-LIR lowering implementation plan
Lat semantic validation implementation plan
Runtime boundary implementation plan
Nucleus task execution implementation plan

Earlier completed status/public-entry slice

Latticra Seal signer invocation status/public-entry alignment

Purpose completed:

make the Seal signer invocation metadata implementation visible from public/status entry points while preserving no signing, no verification, no signer invocation behavior, no private-key handling, no host behavior, no network behavior, no capability enforcement, and no runtime authority

Previous implementation slices

```
Latticra Seal signer invocation metadata implementation
Latticra Seal signer handoff status/public-entry alignment
Latticra Seal signer handoff metadata implementation
Latticra Seal signing authorization metadata implementation
Latticra Seal signature request metadata implementation
Latticra Seal report envelope metadata implementation
Nucleus report-only announcement README alignment
Nucleus report-only announcement review
Project notes Nucleus report-only status/index check
Project notes Nucleus report-only alignment
Nucleus task report-only execution README/status alignment
Nucleus task report-only execution refinement
Nucleus task README/status alignment
Nucleus task no-effect report alignment
Language representation review
C++ authority expansion contract review
L-UI completion estimate review
L-UI rendering README/status alignment
L-UI rendering detailed report refinement
Authority announcement review
Project notes follow-up status/index check
Project notes follow-up alignment
Public entry-point consistency scan
Status announcement review index alignment
Status announcement review
Authority foundation index alignment
Current status detail rollup
Authority status/docs alignment
Authority implementation review
Strategy estimate review
Completion percentage review
Current status and announcement consistency review
Project notes index alignment
Current direction project notes alignment
Project notes upcoming work alignment
Lat pipeline diagnostic README alignment
Lat pipeline diagnostic status announcement
Lat pipeline diagnostic foundation index alignment
Lat pipeline diagnostic status/docs alignment
Lat pipeline diagnostic integration main test audit
Runtime boundary domain matrix report main test integration audit
Runtime boundary domain matrix report integration
Runtime boundary domain matrix refinement
Runtime boundary policy matrix refinement
Runtime boundary report refinement
Lat pipeline diagnostic integration refinement
Lat pipeline report refinement
LIR report refinement
Lat semantic diagnostics refinement
Nucleus task report refinement
Lat-specific LIR refinement implementation
Lat-to-LIR lowering implementation
Lat semantic validation foundation
L-UI rendering implementation
Nucleus task execution implementation
Runtime boundary implementation
```

Historical implementation slices

```
Runtime boundary contract
Runtime boundary implementation plan
Runtime boundary implementation
Nucleus task execution contract
Nucleus task execution implementation plan
Nucleus task execution implementation
L-UI rendering implementation
```

Recommended next slice

Latest completed review slice:

```
Completion estimate review README/status alignment
```

Latest completed estimate slice:

```
Current estimate mathematical rebase
```

Previous review slice:

```
Completion estimate review after runtime-boundary abuse-case fixtures
```

Latest completed fixture/evidence slice:

```
Runtime boundary abuse-case fixture expansion after policy expansion
```

Previous policy/evidence slice:

```
Runtime boundary policy expansion after threat-model validation
```

Current runtime boundary abuse-case fixture fields:

```
current_estimate_table_source_alignment_present=1
current_estimate_mathematical_rebase_present=1
seal_readme_status_row_alignment_present=1
completion_estimate_review_readme_status_alignment_present=1
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
runtime_boundary_abuse_case_fixture_expansion_present=1
runtime_boundary_abuse_case_fixture_guard_present=1
runtime_boundary_abuse_case_c_fixtures_present=1
runtime_boundary_abuse_case_fixture_count=8
runtime_execution_added=0
effect_execution_added=0
runtime_authority_granted=0
completion_estimate_review_required=0
```

Current estimate table source alignment is the source-map cleanup for the public estimate table. Current estimate mathematical rebase is the latest planning-estimate update for that table, moving the overall value to a weighted 45% after Nadia Stage-43 prompt-evaluation release-receipt contract-chain evidence, kernel lifecycle scheduler-credit/scheduler-selection-ready evidence, and vulnerability-management release-gate baseline coverage while changing no implementation behavior, product-security posture, public readiness, product readiness, or runtime authority.

Recommended next slice:

```
Continue small guarded report/status alignment only when drift appears
```

Completion estimate review only if capability posture changes remains the estimate rule after this non-change review.

Purpose:

```
continue small guarded report/status alignment only when drift appears while preserving no
runtime execution, no effect execution, no capability enforcement, no cryptographic
verification, no signing, no host behavior, no network behavior, no MCP behavior, no AI agent
execution, no model execution, no tool execution, no shell execution, production protection, and
 no runtime authority
```

Near-term queue

1. Continue small guarded report/status alignment only when drift appears.
2. Authority status announcement only if public messaging changes.
3. C++ authority expansion contract only if new authority behavior is proposed.
4. Nucleus task execution refinement only after the next language representation review and a separate effect contract.
5. Runtime boundary domain matrix report status audit only if new status drift appears.
6. Lat pipeline diagnostic README follow-up only if future guard requirements demand additional public links.
7. Runtime behavior expansion only after separate contract, plan, tests, and explicit non-claim review.

Quality rules

Upcoming work should remain:

```
small
reviewable
tested or guarded
evidence-bound
clear about non-claims
consistent with public Latticra identity
consistent with the C/C++ foundation direction
consistent with the report-only Nucleus boundary
consistent with no-new-announcement decisions unless capability posture changes
```

Current project priorities

- Preserve the C/C++ foundation checkpoint: C is the metal, C++ is the disciplined structure, Latticra is the contract.
- Keep the constrained C++ authority layer no-effect until separate effect contracts exist.
- Keep L-UI rendering no-effect and presentation-only.
- Keep Nucleus task execution no-effect, report-only, and denied-by-default.
- Keep runtime behavior no-effect and disabled-by-default.
- Keep Lat semantic validation no-effect and metadata-only.
- Keep Lat-to-LIR lowering no-effect and metadata-only.
- Keep Lat-specific LIR refinement no-effect and metadata-only.
- Keep Lat pipeline diagnostic integration no-effect and metadata-only.
- Keep Runtime boundary domain matrix report integration report-only and no-effect.
- Keep Seal key-handling no-effect and unsigned until separate key-material and signing contracts exist.
- Do not update completion estimates after documentation/status-only alignment.
- Do not add public announcement entries for documentation/status-only alignment.
- Maintain professional public docs.
- Keep status and completion estimates current only when capability posture changes.
- Keep Lat metadata-only until separate execution contracts exist.
- Keep C++ constrained by the governed authority-layer implementation plan.

docs/strategy/2026-05-19-1845-cdt-strategy-estimate-review.md

Source title: Latticra Strategy Estimate Review

Lines: 60 | Words: 254 | SHA-256: 9720ef04af5a2f917bcd383e1ced546ba07775cb5dc28116da81fb1f430f3c94

Latticra Strategy Estimate Review

Status: strategy estimate review Created: 2026-05-19 18:45 CDT Scope: strategy alignment after project-notes refresh and completion-percentage review.

Purpose

This record updates the strategy layer after the recent report, diagnostic, audit, README, foundation-index, project-notes, status-consistency, and completion-percentage review slices.

It preserves the original active strategy record as historical context while documenting the current planning estimate posture.

Current planning estimate posture

The current public estimate posture is:

```
Overall Latticra system: 35%
Foundation documents and contracts: 89%
Public documentation posture: 83%
Strategy/status/funding framework: 57%
Lat / Latticra Programming Language: 25%
Runtime / operating-system-universe direction: 17%
```

These estimates remain planning estimates only. They are not release promises, production-readiness claims, security guarantees, or operating-system completeness claims.

Strategic interpretation

Recent work improved public posture, auditability, and diagnostic/report coverage more than operational capability.

The strategy should therefore continue to emphasize:

```
small reviewable slices
evidence-bound status updates
no-effect defaults
deterministic diagnostics
operator-visible reports
explicit non-claims
C/C++ foundation direction
Lat metadata-only behavior until separate execution contracts exist
runtime boundary disabled-by-default posture
```

Current next strategic review lane

Recommended next strategic review lane:

```
C++ authority implementation review after initial no-effect validator/audit slice
```

Before moving into any broader authority behavior, the project should review the existing no-effect constrained C++ authority implementation against the current Lat/LIR/runtime-boundary evidence surfaces.

Non-claims

This record does not implement security controls, operating-system behavior, malware prevention, ransomware prevention, sandboxing, update safety, recovery safety, runtime behavior, command execution, or production readiness.

It records strategy and planning estimates only.

docs/strategy/2026-05-25-1951-cdt-strategy-posture-refresh.md

Source title: Latticra Strategy Posture Refresh

Lines: 115 | Words: 509 | SHA-256: 65778a2f18cece56588ff5c625af1288f608330dd5044346eeac7cdddc4ab0f0

Latticra Strategy Posture Refresh

Status: strategy posture refresh Created: 2026-05-25 19:51 CDT Scope: current strategic posture, promotion discipline, estimate-change rules, and next operating rhythm.

Purpose

This record refreshes the strategy layer after the current public status, project-notes, completion-estimate, runtime-boundary, Seal, Lat, Nadia, and no-effect alignment work.

It preserves the active national-security open-system strategy as mission context while updating the operating posture for the current evidence state.

Current strategic posture

Latticra remains an early-stage, evidence-bound systems architecture and programming-language project.

Current strategy should treat the repository as having:

```
strong foundation documents
strong public status posture
mature no-effect report surfaces in several lanes
bounded Lat parse/validate/normalize/lower metadata paths
metadata-only Seal trust-boundary planning
report-only Nucleus and Runtime Boundary behavior
early product readiness
early security-hardening implementation
no production runtime authority
```

Source-of-truth alignment

The current public status posture is governed by:

```
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

Older strategy records remain useful historical context, but their estimates and recommended next slices should not override the current public status files.

Current operating rule

The current next-step rule is:

```
Continue small guarded report/status alignment only when drift appears.
```

Completion-estimate review remains required only when capability posture changes.

Documentation-only, status-only, index-only, and announcement-review-only work should not promote product readiness, security-hardening posture, runtime authority, or operating-system completeness.

Promotion discipline

Future capability promotion should require explicit evidence before public posture changes.

Minimum promotion inputs:

```
contract
implementation plan
bounded implementation or fixture
focused guard or test
public status record
non-claim review
estimate-impact review when capability posture changes
```

For runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, or operating-system behavior, strategy should require a separate contract, plan, tests, and explicit non-claim review before implementation expansion.

Near-term strategic priorities

1. Preserve no-effect boundaries unless a separate promotion packet exists.
2. Keep public estimates synchronized only with capability posture changes.
3. Treat status drift as maintenance, not capability growth.
4. Prefer fewer, sharper next-slice candidates over broad parallel expansion.

5. Keep product-readiness language conservative until user-facing workflows exist.
6. Keep security-hardening language conservative until real enforcement evidence exists.
7. Keep strategy records dated, reviewable, and subordinate to current public status.

Current strategic frontier

The next useful strategic frontier is not broad feature expansion.

The next useful strategic frontier is promotion criteria:

```
What exact evidence would justify the next capability posture change?
```

Candidate promotion packets should be written before execution-oriented work starts.

Priority promotion-packet candidates:

```
runtime behavior expansion gate
Nucleus effect-contract gate
Seal cryptographic verification authority gate
Panel user-facing workflow readiness gate
Nadia offline inference readiness gate
security-hardening evidence gate
public product-readiness gate
```

These are planning candidates only. They do not authorize implementation or change estimates by themselves.

Non-claims

This record does not implement security controls, operating-system behavior, malware prevention, ransomware prevention, sandboxing, update safety, recovery safety, runtime behavior, command execution, cryptographic verification authority, signing authority, model execution, tool execution, production readiness, or public product readiness.

It records strategy and organization only.

docs/strategy/2026-05-25-2017-cdt-capability-promotion-gate-map.md

Source title: Latticra Capability Promotion Gate Map

Lines: 386 | Words: 996 | SHA-256: 1d085bc6698a414a41502b79986105481b6c848260110715bbf71c5b251fbff0

Latticra Capability Promotion Gate Map

Status: strategy gate map Created: 2026-05-25 20:17 CDT Scope: strategic promotion criteria for future capability-posture changes.

Purpose

This record translates the evidence ladder, precursor promotion rule, real-system contract, and current strategy posture into concrete promotion gates for Latticra's major work lanes.

It is an organizing record only. It does not authorize implementation expansion or change any current estimate.

Source rules

Promotion remains governed by:

```
docs/EVIDENCE_LADDER.md
docs/PRECURSOR_PROMOTION_RULE.md
docs/REAL_SYSTEM_CONTRACT.md
docs/strategy/2026-05-25-1951-cdt-strategy-posture-refresh.md
```

The controlling principle is:

```
No public capability posture change without evidence that matches the next claimed level.
```

Promotion classes

Use these classes before changing public language, estimates, or next-priority posture.

```
maintenance_alignment
  Status, README, index, announcement-review, or source-map cleanup.
  Does not change capability posture.

metadata_capability
  A bounded no-effect implementation, fixture, parser, report, diagnostic,
  classifier, validator, or metadata path exists and is guarded.

preview_capability
  A capability is wired into a runtime or workflow model but remains
  preview-only, denied-by-default, and non-mutating.

guarded_local_capability
  A narrow local operation is possible only through explicit opt-in gates,
  audit output, failure handling, and rollback or reset boundaries.

real_system_capability
  A repeated, supported, documented capability with tests, release boundary,
  operator-facing status, and safety review.
```

Estimate-change rule

Estimate changes should be considered only when a lane moves from one promotion class to another.

Do not adjust estimates for:

```
index updates
README wording alignment
status-only publication
announcement-review decisions
source-of-truth cleanup
historical strategy records
non-claim restatement
```

Do consider estimate review for:

```
new guarded implementation evidence
new test-backed capability class
new user-facing workflow evidence
new runtime preview integration
new enforcement or authority evidence
new security-hardening evidence
new product-readiness evidence
```

Gate A: Runtime Behavior Expansion

Current posture:

```
report-only
disabled-by-default
denied-by-default
no runtime authority
```

Promotion target:

```
preview_capability before guarded_local_capability
```

Required before promotion:

```
runtime behavior expansion contract
runtime behavior implementation plan
specific request family and effect family
authority prerequisite matrix
denial and failure fixtures
operator-visible report fields
rollback or non-mutation boundary
focused guard
aggregate guard
non-claim review
estimate-impact review
```

Forbidden shortcut:

```
No direct move from report-only metadata to local mutation.
```

Gate B: Nucleus Effect Contract

Current posture:

```
report-only task boundary
classification-only execution labels
no effect-performing task execution
```

Promotion target:

```
preview_capability for a single named effect family
```

Required before promotion:

```
named effect family
effect contract
task authority prerequisites
Runtime Boundary dependency statement
Seal dependency statement when trust evidence is involved
denied-case fixtures
failure-state report
operator-visible status
guard coverage
non-claim review
estimate-impact review
```

Forbidden shortcut:

```
No generic task execution umbrella before a narrow effect family is reviewed.
```

Gate C: Seal Cryptographic Authority

Current posture:

```
trust-boundary planning
bounded key metadata
verification metadata and result surfaces
no capability authorization from cryptographic result alone
no signing authority
no runtime authority
```

Promotion target:

```
metadata_capability or preview_capability before any authority-bearing gate
```

Required before promotion:

```
cryptographic authority contract
provider and algorithm boundary
public-key parsing evidence
negative verification fixtures
verified receipt semantics
capability-gate separation proof
effect-decision separation proof
key-material non-claim review
signing non-claim review
guard coverage
estimate-impact review
```

Forbidden shortcut:

```
No verified cryptographic result may imply capability enforcement, signing authority,
effect execution, runtime handoff, host behavior, network behavior, or production trust.
```

Gate D: Panel User-Facing Workflow Readiness

Current posture:

```
local control surface direction
guarded user-local posture
early product readiness
```

Promotion target:

```
guarded_local_capability for a named local workflow
```

Required before promotion:

```
named user workflow
```

```
no-root boundary
network boundary
dry-run transcript
local path plan
failure and reset path
operator-visible status
accessibility and usability review
guarded install or no-effect proof
non-claim review
estimate-impact review
```

Forbidden shortcut:

```
No public product-readiness promotion from architecture documentation alone.
```

Gate E: Nadia Offline Inference Readiness

Current posture:

```
offline AI foundation planning
context and tokenizer contracts
no prompt evaluation
no model loading
no inference
no tool execution
```

Promotion target:

```
preview_capability before guarded local inference
```

Required before promotion:

```
offline inference readiness contract
local model identity boundary
tokenizer artifact evidence
prompt materialization evidence
context-window assembly evidence
runtime invocation non-claim review
tool authority non-claim review
protective safety boundary check
guard coverage
operator-visible report
estimate-impact review
```

Forbidden shortcut:

```
No dialogue, inference, tool authority, or source mutation before the offline
runtime and safety gates are independently evidenced.
```

Gate F: Security-Hardening Evidence

Current posture:

```
security-conscious design
defensive threat model records
early security-hardening implementation
no production security boundary
```

Promotion target:

```
metadata_capability for security evidence before any protection claim
```

Required before promotion:

```
threat model mapping
abuse-case fixtures
negative tests
external-source checkpoint when relevant
manual review record
residual-risk statement
non-claim review
estimate-impact review
```

Forbidden shortcut:

```
No malware-prevention, ransomware-prevention, sandbox, hardening, or security
boundary claim without tested enforcement evidence.
```

Gate G: Public Product Readiness

Current posture:

```
early product readiness
strong documentation posture
no deployed platform
no production product claim
```

Promotion target:

```
guarded_local_capability with a coherent user-facing workflow
```

Required before promotion:

```
named target user
named workflow
installation or no-install path
first-run success criteria
failure and uninstall/reset criteria
support boundary
known limitations
operator-facing evidence
release note or readiness note
non-claim review
estimate-impact review
```

Forbidden shortcut:

```
No product-readiness promotion from internal implementation evidence unless a
target user can complete a bounded workflow and understand its limitations.
```

Strategic next-use rule

When the project wants to move beyond maintenance alignment, choose exactly one gate and write a promotion packet before implementation work starts.

The packet should include:

```
gate:
target promotion class:
current evidence level:
target evidence level:
named capability:
allowed behavior:
forbidden behavior:
required fixtures:
required tests or guards:
operator-visible report:
non-claims:
estimate-impact question:
public-entrypoint question:
```

Non-claims

This record does not implement runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, product readiness, security hardening, malware prevention, ransomware prevention, sandboxing, or operating-system behavior.

It records strategy and promotion criteria only.

docs/strategy/2026-05-25-2025-cdt-capability-promotion-packet-template.md

Source title: Latticra Capability Promotion Packet Template

Lines: 228 | Words: 665 | SHA-256: ced676593d2838145ed7a60dfa32a4da1c98dde120f5df1828442d9a007995d2

Latticra Capability Promotion Packet Template

Status: strategy template Created: 2026-05-25 20:25 CDT Scope: reusable planning template for future capability-promotion proposals.

Purpose

This template gives future Latticra promotion work a standard packet shape before implementation expansion begins.

It exists to keep capability changes evidence-bound, reviewable, narrow, and honest about non-claims.

Usage rule

Use this template when a proposal would move any lane beyond maintenance alignment.

Do not use this template to justify a capability after implementation has already expanded. The packet should exist before execution-oriented work starts.

Packet header

```
packet name:
created:
owner:
status: draft|under-review|accepted-for-planning|rejected|superseded
gate:
target lane:
target promotion class:
current evidence level:
target evidence level:
current source-of-truth files:
related contracts:
related status records:
```

Capability statement

```
named capability:
one-sentence purpose:
target user or operator:
current posture:
proposed posture:
why this promotion is needed now:
```

Allowed behavior

List only the behavior that the promotion would permit.

```
allowed behavior:
-
-
-
```

Forbidden behavior

List behavior that remains out of scope even if the packet is accepted.

```
forbidden behavior:
- no hidden execution
- no hidden mutation
- no unreviewed authority expansion
- no unsupported security claim
- no production-readiness claim unless explicitly evidenced
```

Add lane-specific exclusions:

```
runtime behavior:
effect execution:
capability enforcement:
cryptographic authority:
signing authority:
host behavior:
network behavior:
model execution:
tool execution:
shell execution:
file mutation:
hardware behavior:
```

Evidence inventory

```
existing contracts:
-

existing implementation or fixture evidence:
-

existing tests or guards:
-

existing status or public-entry records:
-

known evidence gaps:
-
```

Required new evidence

```
new contract required:
new implementation plan required:
new fixture required:
new focused guard required:
new aggregate guard required:
new report field required:
new status record required:
new non-claim review required:
new estimate-impact review required:
new public-entrypoint review required:
```

Operator visibility

```
operator-visible status:
operator-visible report:
failure label:
denial label:
rollback or reset label:
audit output:
```

Dependency map

```
Lat dependency:
LIR dependency:
L-UI dependency:
Nucleus dependency:
Runtime Boundary dependency:
Seal dependency:
Panel dependency:
Nadia dependency:
C/C++ authority dependency:
security review dependency:
```

Failure and denial model

```
expected failure states:
-

expected denial states:
-

stale evidence behavior:
ambiguous evidence behavior:
operator-cancel behavior:
unsupported-platform behavior:
capacity-limit behavior:
```

Estimate-impact question

Answer before any estimate changes.

```
Does this packet move a lane to a new promotion class?
Does it add runtime behavior?
Does it add effect execution?
Does it add authority?
Does it add user-facing workflow readiness?
Does it add security-hardening evidence?
Does it add public product-readiness evidence?
Estimate change recommended: no|yes
Reason:
```

Public-entrypoint question

Answer before README, root status, or public docs change.

```
Does this packet change public capability wording?
Does it require README update?
Does it require STATUS.md update?
Does it require docs/status/CURRENT_STATUS.md update?
Does it require docs/project_notes/CURRENT_DIRECTION.md update?
Does it require docs/project_notes/UPCOMING_WORK.md update?
Does it require announcement review?
Public entrypoint change recommended: no|yes
Reason:
```

Acceptance criteria

```
accepted only if:
- scope is narrow and named
- evidence level is explicit
- allowed behavior is bounded
- forbidden behavior is explicit
- tests or guards are named
- operator-visible status exists or is required
- non-claims are stated
- estimate-impact question is answered
- public-entrypoint question is answered
```

Rejection criteria

```
reject if:
- the proposal skips evidence levels
- the proposal bundles multiple unrelated capabilities
- the proposal implies production readiness without user-facing evidence
- the proposal implies security protection without enforcement evidence
- the proposal grants authority from metadata alone
- the proposal hides mutation, host behavior, network behavior, or tool execution
- the proposal lacks a denial or failure model
```

Non-claims

This template does not implement runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, product readiness, security hardening, malware prevention, ransomware prevention, sandboxing, or operating-system behavior.

It records a planning form only.

docs/strategy/2026-05-25-2114-cdt-public-product-readiness-promotion-packet.md

Source title: Latticra Public Product Readiness Promotion Packet

Lines: 255 | Words: 898 | SHA-256: d655248c274e807ce68c2636ef36ca0b41d515df4b0348929edeb44bda2d6a8f

Latticra Public Product Readiness Promotion Packet

Status: draft promotion packet Created: 2026-05-25 21:14 CDT Decision: no promotion recommended Scope: public product-readiness criteria for a future bounded user-facing Latticra workflow.

Purpose

This packet defines what evidence would be required before Latticra's public product-readiness posture should move beyond its current early estimate.

It does not request or authorize product-readiness promotion. It records the gap between the current evidence state and a future bounded user-facing workflow.

Packet header

```
packet name: public product readiness promotion packet
gate: public product readiness
target lane: Latticra public product posture
target promotion class: guarded local user-facing workflow
current evidence level: documentation/status plus local no-effect workbench evidence
target evidence level: bounded first-run workflow with operator-visible evidence
status: draft
decision: no promotion recommended
```

Current source-of-truth files:

```
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/EVIDENCE_LADDER.md
docs/PRECURSOR_PROMOTION_RULE.md
docs/REAL_SYSTEM_CONTRACT.md
docs/strategy/2026-05-25-1951-cdt-strategy-posture-refresh.md
```

Current posture

Current public estimate:

```
Public product readiness: 8%
Latticra Panel / local control surface: 28%
Public documentation posture: 88%
Foundation documents and contracts: 92%
```

Strategic interpretation:

```
Latticra is publicly legible and well documented, but it is not yet a public product.
```

The current repository has strong documentation, status, contracts, no-effect reports, and local workbench direction. It does not yet have enough bounded user-facing workflow evidence to promote product readiness.

Candidate target user

The first bounded product-readiness target should be narrow:

```
security researcher
systems engineer
infrastructure maintainer
technical evaluator
```

The first target user should be comfortable with a source checkout, local commands, no-effect reports, and explicit non-claims.

This packet does not target nontechnical end users.

Candidate first workflow

Candidate workflow:

```
Evaluate Latticra locally without granting runtime authority.
```

The future workflow should let a technical evaluator:

1. clone or open the repository;
2. identify current status and non-claims;
3. launch or inspect the Latticra Panel / local control surface;
4. run a no-effect dry-run or evidence review path;
5. inspect generated status, plan, or report output;
6. understand what did not happen;
7. exit, reset, or uninstall any local artifacts created by the workflow.

Allowed future behavior

For this product-readiness packet, allowed future behavior would be limited to:

```
guided local evaluation
no-effect dry-run review
operator-visible evidence review
local reset or uninstall path
explicit status and non-claim display
technical evaluator workflow documentation
```

Forbidden behavior

This packet does not permit:

```
production installer claim
daily-driver claim
runtime authority
effect execution
capability enforcement
cryptographic authority
signing authority
host mutation
network behavior
model execution
tool execution
shell execution beyond documented local developer commands
security protection claim
malware-prevention claim
ransomware-prevention claim
operating-system replacement claim
```

Existing evidence

Existing strengths:

```
public README and status surfaces
current high-level estimate table
explicit non-claims
foundation contracts
evidence ladder
project notes
Latticra Panel direction
local installer/workbench command documentation
no-effect runtime and Nucleus posture
metadata/report-heavy Seal posture
```

Existing gaps:

```
no declared first public workflow acceptance test
no product-readiness-specific user journey record
no product-readiness support boundary
no dedicated first-run evidence transcript for public readiness
no release-readiness note for a technical evaluator workflow
no public product-readiness non-claim review
no estimate-impact review tied to a user-facing workflow
```

Required evidence before promotion

Before any public product-readiness estimate increases, require:

```
named target user
named first workflow
first-run checklist
bounded local execution or no-effect dry-run transcript
operator-visible status output
failure and reset path
uninstall or cleanup path if local artifacts are created
known limitations section
support boundary
public product-readiness non-claim review
guard or test for the documented workflow
estimate-impact review
public-entrypoint review
```

Operator visibility requirements

A future product-readiness workflow should show:

```
current project status
what the workflow will do
what the workflow will not do
whether mutation is possible
whether network behavior is possible
where local artifacts may be written
how to reset or remove local artifacts
which reports or receipts were generated
which authority remains denied
```

Failure and denial model

Required failure and denial labels:

```
unsupported_platform
missing_dependency
dry_run_only
no_runtime_authority
no_network_authority
no_root_authority
reset_available
manual_review_required
```

The workflow should fail closed when the operator cannot understand or inspect the action boundary.

Estimate-impact answer

```
Does this packet move public product readiness to a new promotion class? no
Does it add runtime behavior? no
Does it add effect execution? no
Does it add authority? no
Does it add user-facing workflow evidence? no
Does it add security-hardening evidence? no
Does it add product-readiness evidence? no
Estimate change recommended: no
Reason: this packet defines future criteria only.
```

Public-entrypoint answer

```
Does this packet change public capability wording? no
Does it require README update? no
Does it require STATUS.md update? no
Does it require docs/status/CURRENT_STATUS.md update? no
Does it require docs/project_notes/CURRENT_DIRECTION.md update? no
Does it require docs/project_notes/UPCOMING_WORK.md update? no
Does it require announcement review? no
Public entrypoint change recommended: no
Reason: no capability posture changes.
```

Next planning action

The next planning action after this packet is to choose whether the first public-readiness workflow should be:

```
Panel-guided local evaluation
CLI-only no-effect evidence review
documentation-first technical evaluator path
```

Only one should be selected for the first workflow packet.

Non-claims

This packet does not implement product readiness, runtime behavior, effect execution, capability enforcement, cryptographic authority, signing authority, host behavior, network behavior, model execution, tool execution, shell execution, security hardening, malware prevention, ransomware prevention, sandboxing, operating-system behavior, installer readiness, or production support.

It records planning criteria only.

docs/strategy/2026-05-26-1702-cdt-overall-strategy-priority-map.md

Source title: Latticra Overall Strategy Priority Map After Panel-Guided Planning Completion

Lines: 363 | Words: 949 | SHA-256: d7a7c2d9ef876ea1b78f4fed39e7486cc62e027b500fa8447157562940d17dcc

Latticra Overall Strategy Priority Map After Panel-Guided Planning Completion

Status: strategy priority map Created: 2026-05-26 17:02 CDT Decision: prioritize next planning lanes without starting evidence capture or implementation Promotion decision: no product-readiness, runtime, security, cryptographic-authority, AI, or execution promotion recommended Scope: overall planning sequence after completion of the Panel-guided local evaluation planning package.

Purpose

This record defines the next strategic priorities after the Panel-guided local evaluation planning package reached planning completion.

It exists to keep the project moving deliberately without converting planning work into unsupported product, runtime, security, AI, or execution claims.

Source records

This map follows:

```
docs/strategy/2026-05-26-1356-cdt-panel-guided-local-evaluation-planning-completion-checkpoint.m
d
docs/strategy/2026-05-26-0410-cdt-panel-guided-local-evaluation-review-package-index.md
docs/strategy/2026-05-25-1951-cdt-strategy-posture-refresh.md
STATUS.md
docs/status/CURRENT_STATUS.md
README.md
```

Current estimate posture

Current public planning estimates remain:

```
| Area | Estimated completion |
| --- | ---: |
| Overall Latticra system | 45% |
| Latticra Seal / local evidence layer | 39% |
| Latticra Panel / local control surface | 31% |
| Nadia offline AI foundation | 74% |
| L-UI parser / AST / string foundation | 87% |
| Foundation documents and contracts | 94% |
| Public documentation posture | 91% |
| Strategy/status/funding framework | 63% |
| Lat / Latticra Programming Language | 27% |
| LIR / Intermediate Representation | 24% |
| C/C++ foundation direction | 22% |
| Constrained C++ authority layer | 5% |
| Nucleus real task execution | 12% |
| Runtime / operating-system-universe direction | 25% |
| Security-hardening implementation | 9% |
| Public product readiness | 10% |
```

Estimate decision:

```
change_completion_estimates=0
```

Reason:

```
the latest work completed planning organization only; no workflow evidence,
guard/test result, public-entrypoint review, runtime behavior, security
behavior, AI behavior, or production-support behavior has changed
```

Priority rules

Current rules:

```
contracts_before_code=1
fixtures_before_runtime=1
read_only_before_mutation=1
evidence_before_claims=1
non_claim_reviews_before_public_positioning=1
estimate_changes_only_after_capability_change=1
```

Priority map

Tier 0: Source-of-truth health

Priority:

```
ongoing
```

Purpose:

```
keep README.md, STATUS.md, docs/status/CURRENT_STATUS.md, and dated strategy
records aligned when drift appears
```

Planning rule:

```
continue small guarded report/status alignment only when drift appears
```

Do not create broad status rewrites unless a specific inconsistency is found.

Tier 1: Public evaluator workflow clarity

Priority:

```
highest next planning priority
```

Purpose:

```
make it easy for an evaluator to understand what Latticra can inspect,
what remains read-only, what evidence exists, and what claims are deliberately
withheld
```

Current state:

```
Panel_guided_local_evaluation_planning_package_complete=1
Panel_guided_local_evaluation_evidence_captured=0
Public_product_readiness_promotion_recommended=0
```

Recommended next planning package:

```
Latticra Console read-only host inventory and evaluator workflow planning package
```

Reason:

```
it extends the public/evaluator path while staying inside read-only,
metadata-only, non-effect boundaries
```

Expected package contents:

```
workflow packet
acceptance checklist
evidence bundle template
non-claim review template
public-entrypoint review template
estimate-impact review template
completion checkpoint
```

Blocked without separate request:

```
launch_console=0
run_host_inventory=0
capture_transcripts=0
change_public_product_readiness=0
```

Tier 2: Runtime and Nucleus effect-boundary planning

Priority:

```
second planning priority
```

Purpose:

```
define how future real task execution would be requested, previewed,
guarded, logged, denied, and reviewed before any mutation-capable behavior exists
```

Current state:

```
Nucleus_real_task_execution=12_percent
Runtime_operating_system_universe_direction=25_percent
effect_authority_granted=0
```

Recommended planning package:

```
Nucleus effect-contract and runtime-denial gate planning package
```

This should remain contract-level until explicit implementation work is requested.

Tier 3: Seal evidence and authority boundaries

Priority:

```
third planning priority
```

Purpose:

```
clarify the line between local evidence records, trust receipts, signing
authority, publication authority, and future verification behavior
```

Current state:

```
Seal_local_evidence_layer=39_percent
cryptographic_authority_claimed=0
receipt_publication_claimed=0
```

Recommended planning package:

```
Seal evidence-authority promotion gate map
```

This should not imply cryptographic enforcement, public audit readiness, or signing authority.

Tier 4: Nadia offline AI readiness boundaries

Priority:

```
fourth planning priority
```

Purpose:

```
keep offline AI foundation progress separate from model loading, prompt
evaluation, token generation, tool execution, and survivor-support claims
```

Current state:

```
Nadia_offline_AI_foundation=74_percent
inference_claimed=0
tool_execution_claimed=0
survivor_QA_claimed=0
```

Recommended planning package:

```
Nadia prompt-evaluation non-claim and evidence gate map
```

This should preserve the distinction between foundation contracts and actual inference behavior.

Tier 5: Language and representation stability

Priority:

```
fifth planning priority
```

Purpose:

```
stabilize the relationship between L-UI, Lat, LIR, constrained C++, and
future runtime authority without implying executable completeness
```

Current state:

```
L_UI_parser_AST_string_foundation=87_percent
```

```
Lat_language=27_percent
LIR=24_percent
C_CPP_foundation_direction=22_percent
Constrained_CPP_authority_layer=5_percent
```

Recommended planning package:

```
Lat/LIR representation-to-effect boundary map
```

This should focus on representation contracts, fixture expectations, and denial behavior.

Tier 6: Security-hardening evidence gates

Priority:

```
sixth planning priority
```

Purpose:

```
prevent security language from outrunning implemented controls, test evidence,
threat modeling, rollback behavior, and recovery behavior
```

Current state:

```
Security_hardening_implementation=9_percent
malware_prevention_claimed=0
ransomware_prevention_claimed=0
sandboxing_claimed=0
production_security_boundary_claimed=0
```

Recommended planning package:

```
security-hardening non-claim and evidence-gate planning package
```

This should stay defensive, evidence-bound, and non-promotional.

Recommended next move

Recommended next move:

```
create the Latticra Console read-only host inventory and evaluator workflow
planning package
```

Reason:

```
it is the nearest neighbor to the completed Panel-guided planning package,
supports public evaluator clarity, and does not require runtime effects,
security enforcement, AI inference, mutation, or product-readiness promotion
```

Do not start yet

Do not start the following from this map:

```
evidence_capture
command_execution
local_app_launch
host_inventory_run
guard_implementation
test_execution
estimate_update
public_entrypoint_update
product_readiness_promotion
runtime_authority
security_hardening_claim
AI_inference_claim
```

Those require separate explicit requests and evidence-producing work.

Non-claims

This map does not capture evidence, run commands, launch Panel, launch Console, inspect a host, run inventory, implement guards, run tests, validate workflows, update public entry points, change estimates, provide product readiness, provide runtime authority, provide task execution, grant mutation authority, provide cryptographic authority, provide signing authority, publish receipts, execute models, evaluate prompts, generate tokens, execute tools, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records strategy priority only.

docs/strategy/2026-05-26-1707-cdt-console-read-only-host-inventory-workflow-packet.md

Source title: Latticra Console Read-Only Host Inventory Workflow Packet

Lines: 241 | Words: 702 | SHA-256: 10e2c2e011d8553df07a61f3520091f6acc8436cdd9b40669390f085a30294cc

Latticra Console Read-Only Host Inventory Workflow Packet

Status: planning workflow packet Created: 2026-05-26 17:07 CDT Decision: select a contract-only Console host-inventory evaluator workflow Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: planning packet for a future evaluator workflow around Latticra Console read-only host inventory.

Purpose

This packet opens the next planning lane recommended by the overall strategy priority map.

It defines how a future evaluator should review Latticra Console host-inventory posture while the project remains inside metadata-only, read-only, no-effect boundaries.

Source records

This packet follows:

```
docs/strategy/2026-05-26-1702-cdt-overall-strategy-priority-map.md
docs/LATTICRA_CONSOLE_FOUNDATION.md
docs/status/LATTICRA_CONSOLE_FOUNDATION_STATUS.md
STATUS.md
docs/status/CURRENT_STATUS.md
README.md
```

Current Console posture

Current source posture:

```
console_name=Latticra Console
short_name=LC
component_key=latticra_console
standalone_installable=1
standalone_requires_panel=0
standalone_command_wrapper=latticra-lc
host_inventory_contract_profile=lc-host-inventory-v0
host_adapter_contract_profile=lc-host-adapter-v0
runtime_boundary_bound=1
seal_capability_labels_bound=1
command_registry_no_effect=1
```

Current host-inventory posture:

```
contract_profile=lc-host-inventory-v0
contract_status=metadata-only
required_before_host_embedding=1
host_adapter_present=0
inventory_schema_status=planned
inventory_performed=0
```

```
inventory_artifact_present=0
inventory_receipt_required=1
operator_consent_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
promotion_gate=host_inventory_contract_receipt_before_host_adapter
```

Current authority floor:

```
host_probe_allowed=0
host_process_launch_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

Workflow selection

Selected workflow:

```
contract_only_console_host_inventory_evaluator_workflow
```

Rejected workflow:

```
live_host_inventory_capture
```

Reason:

```
the current Console host-inventory lane is intentionally metadata-only;
the useful evaluator workflow is to verify the contract, denial posture,
future-gate language, and non-claims before any host observation is allowed
```

Intended evaluator question

The workflow should answer:

```
Can an evaluator understand that Latticra Console has a planned host-inventory
contract surface while also understanding that no host inventory, host probing,
host process launch, file reading, file writing, host mutation, network use,
runtime enforcement, boot behavior, or product readiness is currently claimed?
```

Planned evaluator path

The future workflow should be organized as:

1. Start at the current public Latticra status and estimate posture.
2. Find the Console foundation and Console status records.
3. Confirm lc host-inventory is described as an inspectable contract command.
4. Confirm the planned future inventory scope is narrow and reviewable.
5. Confirm excluded scope blocks secrets, private files, network scanning, process launch, kernel change, and system mutation.
6. Confirm current host-inventory fields report inventory_performed=0.
7. Confirm host probing, host process launch, file read, file write, host mutation, network, runtime enforcement, and boot authority remain denied.
8. Confirm host-adapter promotion remains blocked by contract receipt and inventory gates.
9. Confirm receipt paths remain metadata-only and unsigned.
10. Stop before evidence capture, command execution, implementation, estimate updates, or public-entrypoint changes.

Future evidence surfaces

If a separate evidence-capture request is made later, the likely surfaces are:

```
docs/LATTICRA_CONSOLE_FOUNDATION.md
docs/status/LATTICRA_CONSOLE_FOUNDATION_STATUS.md
latticra_console_report host-inventory
latticra-lc host-inventory
latticra_console_report host-adapter
latticra-lc host-adapter
latticra_console_report receipt-request
latticra-lc receipt-request
```

This packet does not run or validate those commands.

Review gates

Before any future host-inventory posture can move beyond this workflow, require:

```
acceptance_checklist=required
evidence_bundle_template=required
non_claim_review=required
public_entrypoint_review=required
estimate_impact_review=required
completion_checkpoint=required
```

Before any future live inventory is allowed, additionally require:

```
operator_consent_model=required
inventory_schema=required
field_allowlist=required
field_denial_tests=required
fixture_host_profiles=required
receipt_materialization_review=required
host_adapter_denial_review=required
runtime_boundary_review=required
security_non_claim_review=required
```

Package plan

The planning package should add, in order:

```
workflow packet
acceptance checklist
evidence bundle template
non-claim review template
public-entrypoint review template
estimate-impact review template
review package index
completion checkpoint
```

Current package status:

```
workflow_packet_created=1
acceptance_checklist_created=1
evidence_bundle_template_created=1
non_claim_review_template_created=1
public_entrypoint_review_template_created=1
estimate_impact_review_template_created=1
review_package_index_created=1
completion_checkpoint_created=1
```

Recommended next planning move

Recommended next planning move:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the workflow, acceptance criteria, evidence bundle shape, non-claim review
shape, public-entrypoint review shape, estimate-impact review shape, and review
package index are now selected and closed by the completion checkpoint; the
overall priority map names runtime and Nucleus effect-boundary planning as the
next planning priority while preserving contract-level, no-effect boundaries
```

Do not start yet

Do not start the following from this packet:

```
launch_console
run_latticra_lc
run_latticra_console_report
```

```
run_host_inventory
probe_host
read_host_files
write_host_files
scan_network
launch_host_process
create_inventory_artifact
write_receipt
request_signature
enable_host_adapter
change_estimates
update_public_entrypoints
promote_product_readiness
```

Non-claims

This packet does not run Console, launch Panel, launch a shell, launch a host process, inspect a host, probe a host, read host files, write host files, scan a network, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, update estimates, update public entry points, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records workflow planning only.

docs/strategy/2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md

Source title: Latticra Console Read-Only Host Inventory Acceptance Checklist

Lines: 349 | Words: 775 | SHA-256: 03ee3ec2471792a906d41dbf648e91a0e86f6ac059c41248b8fbe5de0e49a683

Latticra Console Read-Only Host Inventory Acceptance Checklist

Status: draft acceptance checklist Created: 2026-05-26 17:32 CDT Decision: checklist only Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: acceptance criteria for future Console read-only host-inventory evaluator workflow evidence.

Purpose

This checklist defines the conditions that must be satisfied before the Console read-only host-inventory evaluator workflow can be treated as reviewable.

It does not claim the evidence exists yet. It defines how future evidence should be reviewed.

Source workflow

This checklist follows:

```
docs/strategy/2026-05-26-1707-cdt-console-read-only-host-inventory-workflow-packet.md
```

The selected workflow is:

```
contract_only_console_host_inventory_evaluator_workflow
```

The rejected workflow is:

```
live_host_inventory_capture
```

Review result labels

Use one result label when applying this checklist:

```
not_started
blocked
partial_evidence
```

```
evidence_ready_for_review
accepted_for_non_claim_review
accepted_for_public_entrypoint_review
accepted_for_estimate_review
rejected
```

This checklist alone cannot produce accepted_for_estimate_review; that result requires actual workflow evidence plus non-claim, public-entrypoint, and estimate-impact review.

Required evaluator boundary

Acceptance requires:

```
target_evaluator_named=1
technical_evaluator_scope=1
local_operator_scope=1
nontechnical_user_scope=0
production_operator_scope=0
daily_driver_scope=0
security_reliance_scope=0
```

Required reviewer question:

```
Can the intended evaluator understand that this is a contract-only Console
host-inventory review, not a live host scan or production product workflow?
```

Required source entry

Acceptance requires visible entry into:

```
current_project_status
current_estimate_posture
Console_foundation_record
Console_status_record
Console_host_inventory_contract
Console_host_adapter_contract
Console_receipt_request_contract
runtime_boundary_binding
Seal_capability_label_binding
```

Required evidence fields:

```
status_surface_reference:
foundation_record_reference:
status_record_reference:
workflow_packet_reference:
reviewer_note:
```

Required contract identity

Acceptance requires evidence that the host-inventory surface is contract-only:

```
contract_profile=lc-host-inventory-v0
contract_status=metadata-only
required_before_host_embedding=1
host_adapter_present=0
inventory_schema_status=planned
inventory_performed=0
inventory_artifact_present=0
inventory_receipt_required=1
operator_consent_required=1
runtime_boundary_required=1
seal_capability_labels_required=1
promotion_gate=host_inventory_contract_receipt_before_host_adapter
```

Required reviewer question:

```
Can the evaluator identify the difference between an inventory contract and an
inventory artifact?
```

Required command-surface clarity

Acceptance requires the future evidence bundle to identify intended command surfaces without running them as part of this checklist:

```
latticra_console_report host-inventory
latticra-lc host-inventory
latticra_console_report host-adapter
```

```
latticra-lc host-adapter
latticra_console_report receipt-request
latticra-lc receipt-request
```

Required evidence fields:

```
command_surface_listed=1
commands_run_by_this_checklist=0
command_output_attached=0|1
command_output_source:
```

If command output is attached later, it must be produced by a separate explicit evidence-capture request.

Required allowed-scope visibility

Acceptance requires the planned future inventory scope to be visible and narrow:

```
allowed_future_scope=os_family,kernel_version,cpu_arch,memory_class,filesystem_roots,user_scope,
prefix_scope
```

Required reviewer question:

```
Can the evaluator tell that the planned inventory is limited to host posture
metadata rather than file contents, secrets, network state, or process behavior?
```

Required excluded-scope visibility

Acceptance requires excluded scope to be visible:

```
excluded_future_scope=secrets,private_files,network_scan,process_launch,kernel_change,system_mut
ation
```

Required denial labels:

```
secrets_allowed=0
private_file_read_allowed=0
network_scan_allowed=0
process_launch_allowed=0
kernel_change_allowed=0
system_mutation_allowed=0
```

Required no-effect authority floor

Acceptance requires all current authority-denial fields to remain visible:

```
host_probe_allowed=0
host_process_launch_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
seal_capability_grants_authority=0
```

Failure to show any denial field should produce:

```
blocked_missing_authority_floor
```

Required receipt boundary

Acceptance requires the receipt path to stay metadata-only:

```
receipt_request_contract_status=metadata-only-contract
receipt_request_contract_present=1
receipt_payload_schema_present=1
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_review_present=1
receipt_payload_materialization_plan_present=1
draft_review_receipt_present=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
seal_signature_request_ready=0
seal_signature_request_present=0
seal_signing_authority_present=0
receipt_written=0
```

```
receipt_signed=0
```

Required reviewer question:

```
Can the evaluator see that receipt planning exists while signed receipt
authority and payload materialization remain unavailable?
```

Required host-adapter gate

Acceptance requires host-adapter progression to remain blocked:

```
host_adapter_contract_status=metadata-only-contract
host_adapter_contract_present=1
host_adapter_enabled=0
host_adapter_present=0
host_adapter_loaded=0
host_embedding_contract_required=1
read_only_host_inventory_contract_required=1
host_embedding_contract_receipt_required=1
host_inventory_contract_receipt_required=1
receipt_required_before_host_adapter=1
promotion_gate=host_adapter_contract_receipts_and_inventory
```

The checklist must reject any evidence that treats a host-inventory contract as permission to enable the host adapter.

Required failure-state coverage

Acceptance requires declared handling for:

```
source_record_missing
host_inventory_contract_missing
host_inventory_contract_ambiguous
denial_field_missing
excluded_scope_missing
receipt_boundary_missing
host_adapter_gate_missing
command_output_missing
command_output_implies_live_scan
public_text_overclaims_inventory
estimate_change_requested_without_evidence
manual_review_required
```

Required non-claim visibility

Acceptance requires the evidence package to preserve these non-claims:

```
live_host_inventory_claim=0
host_probe_claim=0
host_file_read_claim=0
host_file_write_claim=0
host_mutation_claim=0
network_scan_claim=0
host_adapter_claim=0
runtime_enforcement_claim=0
boot_claim=0
security_hardening_claim=0
product_readiness_claim=0
production_support_claim=0
```

Acceptance decision

The checklist may return evidence_ready_for_review only if:

```
source_entry_visible=1
contract_identity_visible=1
command_surface_clarity=1
allowed_scope_visible=1
excluded_scope_visible=1
authority_floor_visible=1
receipt_boundary_visible=1
host_adapter_gate_visible=1
failure_states_declared=1
non_claims_visible=1
commands_run_by_this_checklist=0
live_host_inventory_performed=0
estimate_change_recommended=0
product_readiness_promotion_recommended=0
```

The checklist must return rejected if:

```
host_probe_performed=1
host_file_read_performed=1
host_file_write_performed=1
host_mutation_performed=1
network_scan_performed=1
host_adapter_enabled=1
receipt_signed=1
public_claim_overstates_inventory=1
```

Recommended next planning move

Recommended next planning move:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the acceptance criteria, evidence bundle shape, non-claim review shape, and
public-entrypoint review shape, estimate-impact review shape, and review
package index now define what must be reviewed and the completion checkpoint
closes the package; the next useful planning lane is the Tier 2 Nucleus
effect-boundary package
```

Non-claims

This checklist does not run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, update estimates, update public entry points, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records acceptance criteria only.

docs/strategy/2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md

Source title: Latticra Console Read-Only Host Inventory Evidence Bundle Template

Lines: 394 | Words: 720 | SHA-256: 819a5be9a13c57c162875ca4dd80893b22c5ffc66f6422a7028db0bde582739f

Latticra Console Read-Only Host Inventory Evidence Bundle Template

Status: draft evidence-bundle template Created: 2026-05-26 17:36 CDT Decision: template only Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: evidence bundle structure for future Console read-only host-inventory evaluator workflow review.

Purpose

This template defines the evidence bundle that would be required before the Console read-only host-inventory evaluator workflow can be reviewed.

It does not claim any evidence has been captured. It defines the future bundle shape only.

Source checklist

This template follows:

```
docs/strategy/2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md
```

The source workflow remains:

```
contract_only_console_host_inventory_evaluator_workflow
```

The rejected workflow remains:

```
live_host_inventory_capture
```

Bundle header

```
bundle name:
created:
review status: not_started|blocked|partial_evidence|evidence_ready_for_review|accepted_for_non_c
laim_review|accepted_for_public_entrypoint_review|accepted_for_estimate_review|rejected
target evaluator:
evaluated platform:
repository commit:
workflow packet:
acceptance checklist:
evidence bundle template:
review owner:
```

accepted_for_estimate_review must not be used until a reviewer confirms that all required evidence exists and the non-claim, public-entrypoint, and estimate-impact reviews are present.

Evidence manifest

Required manifest fields:

```
status_entry_record:
console_foundation_record:
console_status_record:
host_inventory_contract_record:
host_adapter_contract_record:
receipt_boundary_record:
command_surface_record:
allowed_scope_record:
excluded_scope_record:
authority_floor_record:
host_adapter_gate_record:
failure_state_record:
known_limitations_record:
support_boundary_record:
non_claim_review_record:
public_entrypoint_review_record:
estimate_impact_review_record:
```

Each record should be a path, transcript id, status document, or explicit missing label.

Status and source-entry record

Required fields:

```
status_surface_reference:
estimate_surface_reference:
Console_foundation_record_reference:
Console_status_record_reference:
workflow_packet_reference:
acceptance_checklist_reference:
contract_only_workflow_visible=0|1
live_inventory_rejected_visible=0|1
product_readiness_limitation_visible=0|1
security_hardening_limitation_visible=0|1
evaluator_review_note:
record_status: missing|partial|complete
```

Contract identity record

Required fields:

```
contract_profile=lc-host-inventory-v0
contract_status=metadata-only
required_before_host_embedding=1
host_adapter_present=0
inventory_schema_status=planned
inventory_performed=0
inventory_artifact_present=0
inventory_receipt_required=1
operator_consent_required=1
runtime_boundary_required=1
```

```
seal_capability_labels_required=1
promotion_gate=host_inventory_contract_receipt_before_host_adapter
contract_review_note:
record_status: missing|partial|complete
```

The record must distinguish the inventory contract from any future inventory artifact.

Command surface record

Required fields:

```
command_surface_listed=0|1
latticra_console_report_host_inventory_listed=0|1
latticra_lc_host_inventory_listed=0|1
latticra_console_report_host_adapter_listed=0|1
latticra_lc_host_adapter_listed=0|1
latticra_console_report_receipt_request_listed=0|1
latticra_lc_receipt_request_listed=0|1
commands_run_by_this_bundle=0
command_output_attached=0|1
command_output_source:
command_surface_review_note:
record_status: missing|partial|complete
```

If command output is attached later, it must come from a separate explicit evidence-capture request.

Allowed-scope record

Required fields:

```
allowed_future_scope_visible=0|1
os_family_scope_visible=0|1
kernel_version_scope_visible=0|1
cpu_arch_scope_visible=0|1
memory_class_scope_visible=0|1
filesystem_roots_scope_visible=0|1
user_scope_visible=0|1
prefix_scope_visible=0|1
host_posture_metadata_only=0|1
allowed_scope_review_note:
record_status: missing|partial|complete
```

Excluded-scope record

Required fields:

```
excluded_future_scope_visible=0|1
secrets_allowed=0
private_file_read_allowed=0
network_scan_allowed=0
process_launch_allowed=0
kernel_change_allowed=0
system_mutation_allowed=0
excluded_scope_review_note:
record_status: missing|partial|complete
```

Authority-floor record

Required fields:

```
host_probe_allowed=0
host_process_launch_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
seal_capability_grants_authority=0
authority_floor_review_note:
record_status: missing|partial|complete
```

Any missing denial field should block the bundle with:

```
blocked_missing_authority_floor
```

Receipt boundary record

Required fields:

```
receipt_request_contract_status=metadata-only-contract
receipt_request_contract_present=1
receipt_payload_schema_present=1
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_review_present=1
receipt_payload_materialization_plan_present=1
draft_review_receipt_present=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_materialized=0
seal_signature_request_ready=0
seal_signature_request_present=0
seal_signing_authority_present=0
receipt_written=0
receipt_signed=0
receipt_boundary_review_note:
record_status: missing|partial|complete
```

Host-adapter gate record

Required fields:

```
host_adapter_contract_status=metadata-only-contract
host_adapter_contract_present=1
host_adapter_enabled=0
host_adapter_present=0
host_adapter_loaded=0
host_embedding_contract_required=1
read_only_host_inventory_contract_required=1
host_embedding_contract_receipt_required=1
host_inventory_contract_receipt_required=1
receipt_required_before_host_adapter=1
promotion_gate=host_adapter_contract_receipts_and_inventory
host_adapter_gate_review_note:
record_status: missing|partial|complete
```

The bundle must reject evidence that treats the host-inventory contract as permission to enable the host adapter.

Failure-state record

Required failure labels:

```
source_record_missing:
host_inventory_contract_missing:
host_inventory_contract_ambiguous:
denial_field_missing:
excluded_scope_missing:
receipt_boundary_missing:
host_adapter_gate_missing:
command_output_missing:
command_output_implies_live_scan:
public_text_overclaims_inventory:
estimate_change_requested_without_evidence:
manual_review_required:
```

For every failure label, record:

```
host_probe_performed=0
host_file_read_performed=0
host_file_write_performed=0
host_mutation_performed=0
network_scan_performed=0
host_adapter_enabled=0
receipt_signed=0
remediation_or_manual_review_note:
```

Known limitations record

Required visible limitations:

```
not_live_host_inventory=1
not_host_probe=1
```

```
not_file_reader=1
not_network_scanner=1
not_host_adapter=1
not_runtime_enforcement=1
not_boot_behavior=1
not_a_production_product=1
not_a_security_boundary=1
not_security_hardening=1
not_malware_prevention=1
not_ransomware_prevention=1
record_status: missing|partial|complete
```

Support boundary record

Required fields:

```
supported_evaluator:
supported_platforms_for_this_evidence:
unsupported_platforms:
known_setup_limits:
known_runtime_limits:
documentation_issue_path:
security_issue_path:
record_status: missing|partial|complete
```

Review gate records

Non-claim review:

```
non_claim_review_present=0|1
non_claim_review_path:
claim_expansion_detected=0|1
```

Public-entrypoint review:

```
public_entrypoint_review_present=0|1
README_update_required=0|1
STATUS_update_required=0|1
CURRENT_STATUS_update_required=0|1
project_notes_update_required=0|1
announcement_review_required=0|1
```

Estimate-impact review:

```
estimate_impact_review_present=0|1
estimate_change_recommended=0|1
reason:
```

Bundle completeness rule

The bundle is complete only if:

```
all_required_records_present=1
all_required_records_complete=1
non_claim_review_present=1
public_entrypoint_review_present=1
estimate_impact_review_present=1
claim_expansion_detected=0
commands_run_by_this_bundle=0
live_host_inventory_performed=0
host_probe_performed=0
host_file_read_performed=0
host_file_write_performed=0
host_mutation_performed=0
network_scan_performed=0
host_adapter_enabled=0
receipt_signed=0
```

This template result:

```
bundle_complete=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reason=this file defines the future bundle shape only
```

Recommended next planning move

Recommended next planning move:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the evidence bundle, non-claim review, and public-entrypoint review shapes are
now joined by the estimate-impact review shape and review package index; the
completion checkpoint closes this package, and the next useful planning lane is
the Tier 2 Nucleus effect-boundary package
```

Non-claims

This template does not provide workflow evidence, run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, update estimates, update public entry points, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records an evidence-bundle structure only.

docs/strategy/2026-05-26-1757-cdt-console-read-only-host-inventory-non-claim-review-template.md

Source title: Latticra Console Read-Only Host Inventory Non-Claim Review Template

Lines: 385 | Words: 978 | SHA-256: 17e3a81c6ecd029f5fd75b45e751ca64ab48abac9e4fd125edda1c8cf34eee13

Latticra Console Read-Only Host Inventory Non-Claim Review Template

Status: draft non-claim review template Created: 2026-05-26 17:57 CDT Decision: review template only Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: non-claim review for future Console read-only host-inventory evaluator workflow evidence.

Purpose

This template defines the non-claim review required before any Console read-only host-inventory workflow evidence can be used in public-entrypoint, estimate-impact, or product-readiness discussions.

It does not perform the review. It defines the review shape only.

Source bundle

This template follows:

```
docs/strategy/2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md
```

The source checklist remains:

```
docs/strategy/2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md
```

The target workflow remains:

```
contract_only_console_host_inventory_evaluator_workflow
```

The rejected workflow remains:

```
live_host_inventory_capture
```

Review header

```
review name:
created:
reviewer:
review status: not_started|blocked|partial|complete|rejected
workflow evidence bundle:
acceptance checklist:
evidence bundle template:
evaluated platform:
repository commit:
```

Required review outcome

Use one outcome:

```
claim_safe_for_current_scope
claim_safe_with_required_edits
claim_unsafe_blocked
insufficient_evidence
```

This template result:

```
review_performed=0
claim_expansion_detected=0
product_readiness_promotion_recommended=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
reason=this file defines the future review shape only
```

Required source checks

The reviewer must inspect:

```
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/README.md
docs/strategy/2026-05-26-1702-cdt-overall-strategy-priority-map.md
docs/strategy/2026-05-26-1707-cdt-console-read-only-host-inventory-workflow-packet.md
docs/strategy/2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md
docs/strategy/2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md
docs/LATTICRA_CONSOLE_FOUNDATION.md
docs/status/LATTICRA_CONSOLE_FOUNDATION_STATUS.md
```

If any source is missing, mark the review blocked or insufficient_evidence.

Claims that must remain false

The review must confirm the workflow does not claim:

```
live_host_inventory_claim=0
host_probe_claim=0
host_process_launch_claim=0
host_file_read_claim=0
host_file_write_claim=0
host_mutation_claim=0
network_scan_claim=0
kernel_change_claim=0
system_mutation_claim=0
host_adapter_claim=0
host_adapter_enabled_claim=0
runtime_enforcement_claim=0
boot_claim=0
receipt_materialization_claim=0
signed_receipt_claim=0
signing_authority_claim=0
cryptographic_authority_claim=0
product_readiness_claim=0
production_support_claim=0
security_hardening_claim=0
security_boundary_claim=0
malware_prevention_claim=0
ransomware_prevention_claim=0
estimate_change_claim=0
public_entrypoint_change_claim=0
```

Any true value in this section blocks the review unless a separate contract, implementation plan, guard, evidence record, public-entrypoint review, estimate-impact review, and non-claim update exist.

Console host-inventory wording check

Allowed wording:

```
Console read-only host-inventory contract
contract-only evaluator workflow
metadata-only host-inventory posture
future inventory evidence
inventory contract
inventory schema planned
inventory artifact absent
inventory_performed=0
```

Blocked wording:

```
host inventory captured
host scanned
host probed
host inspected
machine audited
files inventoried
system enumerated
live inventory complete
```

If blocked wording appears, the review outcome must be claim_safe_with_required_edits or claim_unsafe_blocked.

Authority wording check

Allowed wording:

```
authority denied
no-effect authority floor
metadata-only contract
read-only planning
host_adapter_present=0
host_probe_allowed=0
host_file_read_allowed=0
host_file_write_allowed=0
host_mutation_allowed=0
network_allowed=0
runtime_enforcement_allowed=0
boot_allowed=0
```

Blocked wording:

```
host access granted
host adapter enabled
runtime enforcement active
system access available
file access available
network discovery available
process launch available
boot integration available
```

Any authority-bearing wording must be rejected unless it belongs to a separate contracted and evidenced lane.

Receipt and signing wording check

Allowed wording:

```
receipt planning
metadata-only receipt request contract
receipt payload schema planned
payload materialization unavailable
signature request unavailable
signing authority absent
receipt_written=0
receipt_signed=0
```

Blocked wording:

```
receipt materialized
receipt signed
trusted receipt
signed host inventory
cryptographic proof
verification authority
publication authority
```

Receipt or signing claims require separate evidence and review before they can be considered.

Product-readiness wording check

Allowed wording:

```
technical evaluator workflow
planning evidence
future evidence capture
non-claim review
no product-readiness promotion recommended
public evaluator clarity
```

Blocked wording:

```
ready for users
production ready
daily driver
operator-ready product
supported release
secure product
approved workflow
```

If blocked wording appears, the review outcome must be claim_safe_with_required_edits or claim_unsafe_blocked.

Security wording check

Allowed wording:

```
security-conscious planning
defensive planning
non-claim review
no security boundary claimed
no security-hardening promotion recommended
```

Blocked wording:

```
prevents malware
prevents ransomware
secure sandbox
hardened runtime
security boundary
verified protection
host protection
production cryptographic trust
```

Any security-protection wording requires separate enforcement evidence before it can be considered.

Evidence wording check

Allowed wording:

```
template
planning packet
checklist
future evidence
missing evidence
partial evidence
evidence ready for review
```

Blocked wording:

```
evidence complete
workflow accepted
host inventory accepted
product readiness improved
estimate increase justified
```

```
public posture changed
```

Blocked evidence wording may become allowed only after the evidence bundle is complete and separately reviewed.

Required reviewer answers

```
Does the workflow preserve no live host inventory? yes|no
Does the workflow preserve no host probing? yes|no
Does the workflow preserve no host file reading? yes|no
Does the workflow preserve no host file writing? yes|no
Does the workflow preserve no host mutation? yes|no
Does the workflow preserve no network scanning? yes|no
Does the workflow preserve no process launch? yes|no
Does the workflow preserve no host-adapter enablement? yes|no
Does the workflow preserve no runtime enforcement? yes|no
Does the workflow preserve no receipt materialization? yes|no
Does the workflow preserve no signed receipt? yes|no
Does the workflow preserve no signing or cryptographic authority? yes|no
Does the workflow preserve no product-readiness promotion? yes|no
Does the workflow preserve no security claim? yes|no
Does the workflow preserve no estimate change? yes|no
Does the workflow preserve no public-entrypoint change? yes|no
```

All answers must be yes for claim_safe_for_current_scope.

Required edit log

If edits are required, record:

```
source file:
claim text:
claim risk:
required replacement:
reviewer note:
```

Review gate output

```
non_claim_review_present=0|1
claim_expansion_detected=0|1
required_edits_present=0|1
blocking_claims_present=0|1
public_entrypoint_review_required=0|1
estimate_impact_review_required=0|1
review_outcome:
```

This template output:

```
non_claim_review_present=0
claim_expansion_detected=0
required_edits_present=0
blocking_claims_present=0
public_entrypoint_review_required=0
estimate_impact_review_required=0
review_outcome=not_performed
```

Recommended next planning move

Recommended next planning move:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the non-claim, public-entrypoint, and estimate-impact review shapes are now
defined and indexed, and the completion checkpoint closes this package; the
next useful planning lane is the Tier 2 Nucleus effect-boundary package
```

Non-claims

This template does not perform a non-claim review, validate workflow evidence, run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, update estimates, update public entry points, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a non-claim review form only.

docs/strategy/2026-05-26-1919-cdt-console-read-only-host-inventory-public-entrypoint-review-template.md

Source title: Latticra Console Read-Only Host Inventory Public-Entrypoint Review Template

Lines: 338 | Words: 1058 | SHA-256: 038c086f308681799b65cd184a74a0f7fba81439b54313d01940528a49121294

Latticra Console Read-Only Host Inventory Public-Entrypoint Review Template

Status: draft public-entrypoint review template Created: 2026-05-26 19:19 CDT Decision: review template only Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: public-entrypoint review for future Console read-only host-inventory evaluator workflow evidence.

Purpose

This template defines when future Console read-only host-inventory evidence may justify updates to public reader-facing entry points.

It does not perform the review, update public entry points, or change project posture.

Source review

This template follows:

```
docs/strategy/2026-05-26-1757-cdt-console-read-only-host-inventory-non-claim-review-template.md
```

The source evidence bundle template remains:

```
docs/strategy/2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md
```

The target workflow remains:

```
contract_only_console_host_inventory_evaluator_workflow
```

The rejected workflow remains:

```
live_host_inventory_capture
```

Review header

```
review name:
created:
reviewer:
review status: not_started|blocked|partial|complete|rejected
workflow evidence bundle:
acceptance checklist:
non-claim review:
estimate-impact review:
repository commit:
```

Required review outcome

Use one outcome:

```
no_public_entrypoint_change
public_entrypoint_change_allowed_with_edits
public_entrypoint_change_blocked
insufficient_evidence
```

This template result:

```
review_performed=0
public_entrypoint_change_recommended=0
README_update_required=0
STATUS_update_required=0
CURRENT_STATUS_update_required=0
project_notes_update_required=0
strategy_index_update_required=0
announcement_review_required=0
reason=this file defines the future review shape only
```

Review prerequisites

Public-entrypoint review is blocked unless:

```
workflow_evidence_bundle_present=1
acceptance_checklist_result_present=1
non_claim_review_present=1
claim_expansion_detected=0
blocked_claim_detected=0
```

Estimate-related public-entrypoint changes are additionally blocked unless:

```
estimate_impact_review_present=1
estimate_change_recommended=0|1
```

If any prerequisite is missing, use:

```
review_outcome=insufficient_evidence
public_entrypoint_change_recommended=0
```

Entry points under review

Review only these public surfaces for this workflow:

```
README.md
STATUS.md
docs/status/CURRENT_STATUS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/strategy/README.md
```

Do not update foundation docs, installer docs, host-adapter docs, OS-base docs, security docs, announcement logs, package docs, or platform docs from this review unless a separate review explicitly names them.

Allowed public-entrypoint changes

Allowed only after prerequisites pass:

```
add pointer to completed Console host-inventory evidence bundle
add pointer to non-claim review
add pointer to estimate-impact review
clarify that contract-only Console host-inventory workflow evidence exists
clarify that the workflow is for technical evaluators
clarify that no live host inventory was performed
clarify that no host adapter was enabled
clarify that no product-readiness promotion follows from this evidence alone
```

Allowed wording:

```
Console read-only host-inventory evidence bundle
contract-only host-inventory evaluator workflow evidence
metadata-only host-inventory posture
inventory contract evidence
no live host inventory performed
no host-adapter enablement
no product-readiness promotion from this evidence alone
```

Blocked public-entrypoint changes

Blocked unless a separate capability promotion has been accepted:

```
claim live host inventory was captured
claim host scan completed
claim host probing occurred
claim host files were read
claim host files were written
claim host mutation occurred
claim network scan occurred
claim host adapter was enabled
claim runtime enforcement
claim boot behavior
claim signed receipt
claim cryptographic proof
claim signing authority
increase public product-readiness estimate
claim production readiness
claim daily-driver readiness
claim operator-ready product
claim security protection
claim malware prevention
claim ransomware prevention
claim sandboxing
claim operating-system replacement
```

If any blocked change is requested, the review outcome must be:

```
public_entrypoint_change_blocked
```

README review questions

```
Does README need a new link? yes|no
Does README wording stay bounded to contract-only Console host-inventory evidence? yes|no
Does README avoid implying a live host scan? yes|no
Does README avoid implying host-adapter enablement? yes|no
Does README avoid product-readiness promotion? yes|no
Does README avoid security, sandboxing, OS-base, or installer claims? yes|no
Does README preserve current estimate wording unless estimate-impact review changes it? yes|no
```

README update is allowed only if every preservation answer is yes.

STATUS.md review questions

```
Does STATUS.md need a new latest-note line? yes|no
Does STATUS.md need estimate table changes? yes|no
Does STATUS.md preserve planning-estimate caveats? yes|no
Does STATUS.md avoid capability posture promotion? yes|no
Does STATUS.md identify the evidence as contract-only? yes|no
Does STATUS.md preserve live_host_inventory_claim=0? yes|no
Does STATUS.md preserve host_adapter_claim=0? yes|no
```

Estimate table changes require a separate estimate-impact review with estimate_change_recommended=1.

CURRENT_STATUS review questions

```
Does CURRENT_STATUS need a detailed evidence note? yes|no
Does CURRENT_STATUS preserve inventory_performed=0 unless actual evidence changes it? yes|no
Does CURRENT_STATUS preserve host_probe_allowed=0? yes|no
Does CURRENT_STATUS preserve host_file_read_allowed=0? yes|no
Does CURRENT_STATUS preserve host_file_write_allowed=0? yes|no
Does CURRENT_STATUS preserve host_mutation_allowed=0? yes|no
Does CURRENT_STATUS preserve network_allowed=0? yes|no
Does CURRENT_STATUS preserve runtime_enforcement_allowed=0? yes|no
Does CURRENT_STATUS preserve boot_allowed=0? yes|no
Does CURRENT_STATUS preserve no security-hardening promotion? yes|no
Does CURRENT_STATUS preserve no product-readiness promotion unless separately reviewed? yes|no
```

Project-notes review questions

```
Does CURRENT_DIRECTION need narrative alignment? yes|no
Does UPCOMING_WORK need queue alignment? yes|no
Does the current next-step rule remain unchanged? yes|no
Does the workflow evidence change near-term strategy? yes|no
Does the workflow evidence require a new evidence-capture lane? yes|no
```

If the current next-step rule changes, a strategy posture review is required.

Strategy index review questions

```
Does docs/strategy/README.md need only a link update? yes|no
Does the new entry remain dated and reviewable? yes|no
Does the index avoid implying capability promotion? yes|no
Does the index preserve this record as a template rather than completed review? yes|no
```

Strategy index link updates are allowed when a dated review record exists, even if no capability posture changes.

Announcement review trigger

Announcement review is required only if:

```
public capability posture changed=1
product_readiness_estimate_changed=1
security_hardening_posture_changed=1
runtime_authority_changed=1
host_inventory_evidence_accepted_for_public_posture=1
host_adapter_posture_changed=1
```

This template result:

```
announcement_review_required=0
reason=no evidence or posture change in this template
```

Required edit log

If public-entrypoint edits are allowed, record:

```
entrypoint:
edit type:
source evidence:
non-claim review:
estimate-impact review:
old wording:
new wording:
claim risk:
reviewer note:
```

Review gate output

```
public_entrypoint_review_present=0|1
public_entrypoint_change_recommended=0|1
README_update_required=0|1
STATUS_update_required=0|1
CURRENT_STATUS_update_required=0|1
project_notes_update_required=0|1
strategy_index_update_required=0|1
announcement_review_required=0|1
blocked_claim_detected=0|1
estimate_impact_review_required=0|1
review_outcome:
```

This template output:

```
public_entrypoint_review_present=0
public_entrypoint_change_recommended=0
README_update_required=0
STATUS_update_required=0
CURRENT_STATUS_update_required=0
project_notes_update_required=0
strategy_index_update_required=0
announcement_review_required=0
blocked_claim_detected=0
estimate_impact_review_required=0
review_outcome=not_performed
```

Recommended next planning move

Recommended next planning move:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the public-entrypoint and estimate-impact review shapes are now defined and
indexed, and the completion checkpoint closes this package; the next useful
planning lane is the Tier 2 Nucleus effect-boundary package
```

Non-claims

This template does not perform a public-entrypoint review, update README, update status files, update project notes, create an announcement, validate workflow evidence, run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, update estimates, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records a public-entrypoint review form only.

docs/strategy/2026-05-26-1956-cdt-console-read-only-host-inventory-estimate-impact-review-template.md

Source title: Latticra Console Read-Only Host Inventory Estimate-Impact Review Template

Lines: 369 | Words: 990 | SHA-256: 7e22f0e0255653a3811310e251d251f44ec4f40ca0c4167027fd35a026688ef3

Latticra Console Read-Only Host Inventory Estimate-Impact Review Template

Status: draft estimate-impact review template Created: 2026-05-26 19:56 CDT Decision: review template only Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: estimate-impact review for future Console read-only host-inventory evaluator workflow evidence.

Purpose

This template defines how future Console read-only host-inventory evidence should be reviewed before any completion estimate changes are considered.

It does not perform the review, change estimates, or change public product-readiness posture.

Source review

This template follows:

```
docs/strategy/2026-05-26-1919-cdt-console-read-only-host-inventory-public-entrypoint-review-temp
late.md
```

The source non-claim review template remains:

```
docs/strategy/2026-05-26-1757-cdt-console-read-only-host-inventory-non-claim-review-template.md
```

The target workflow remains:

```
contract_only_console_host_inventory_evaluator_workflow
```

The rejected workflow remains:

```
live_host_inventory_capture
```

Review header

```
review name:
created:
reviewer:
review status: not_started|blocked|partial|complete|rejected
workflow evidence bundle:
acceptance checklist result:
non-claim review:
public-entrypoint review:
repository commit:
```

Required review outcome

Use one outcome:

```
no_estimate_change
estimate_change_allowed_with_public_entrypoint_review
estimate_change_blocked
insufficient_evidence
```

This template result:

```
review_performed=0
estimate_change_recommended=0
overall_latticra_system_estimate_change_recommended=0
panel_local_control_surface_estimate_change_recommended=0
public_product_readiness_estimate_change_recommended=0
public_documentation_estimate_change_recommended=0
strategy_status_funding_estimate_change_recommended=0
runtime_os_direction_estimate_change_recommended=0
security_hardening_estimate_change_recommended=0
reason=this file defines the future review shape only
```

Review prerequisites

Estimate-impact review is blocked unless:

```
workflow_evidence_bundle_present=1
acceptance_checklist_result_present=1
acceptance_checklist_result=evidence_ready_for_review
non_claim_review_present=1
claim_expansion_detected=0
public_entrypoint_review_present=1
blocked_public_entrypoint_claim_detected=0
```

If any prerequisite is missing, use:

```
review_outcome=insufficient_evidence
estimate_change_recommended=0
```

If any estimate movement is proposed, additionally require:

```
current_public_estimate_table_read=1
source_files_aligned=1
evidence_bundle_complete=1
no_claim_expansion=1
public_entrypoint_change_allowed=0|1
```

Baseline estimate record

Record current estimates at review time:

```
baseline_record_date:
baseline_source_files:
overall_latticra_system:
latticra_panel_local_control_surface:
public_product_readiness:
public_documentation_posture:
strategy_status_funding_framework:
latticra_seal_local_evidence_layer:
runtime_operating_system_universe_direction:
security_hardening_implementation:
```

Do not rely on estimates copied into older strategy templates. The reviewer must read the current public estimate table at review time.

Capability-posture questions

Answer before any estimate movement:

```
Does the evidence add a completed user-facing workflow? yes|no
Does the evidence show a target evaluator completed the workflow? yes|no
Does the evidence remain contract-only? yes|no
Does the evidence preserve live_host_inventory_performed=0? yes|no
Does the evidence preserve host_probe_allowed=0? yes|no
Does the evidence preserve host_file_read_allowed=0? yes|no
Does the evidence preserve host_file_write_allowed=0? yes|no
Does the evidence preserve host_mutation_allowed=0? yes|no
Does the evidence preserve network_allowed=0? yes|no
Does the evidence preserve host_adapter_enabled=0? yes|no
Does the evidence preserve runtime_enforcement_allowed=0? yes|no
Does the evidence preserve boot_allowed=0? yes|no
Does the evidence preserve receipt_signed=0? yes|no
Does the evidence preserve no product-readiness overclaim? yes|no
Does the evidence preserve no security-hardening overclaim? yes|no
```

All answers must support the proposed estimate decision.

Estimate lanes under review

Potentially reviewable lanes for this workflow:

```
Latticra Panel / local control surface
Public product readiness
Public documentation posture
Strategy/status/funding framework
```

Default decision for this contract-only workflow:

```
estimate_change_recommended=0
```

Reason:

```
contract-only host-inventory evidence may improve review clarity, but it does
not by itself prove live host inventory, host-adapter behavior, runtime
authority, security hardening, product readiness, or production support
```

Blocked lanes for this workflow unless a separate evidence packet exists:

```
Runtime / operating-system-universe direction
Security-hardening implementation
Nucleus real task execution
Latticra Seal / local evidence layer
Nadia offline AI foundation
Lat / Latticra Programming Language
LIR / Intermediate Representation
C/C++ foundation direction
Constrained C++ authority layer
```

Product-readiness estimate rule

Product-readiness estimate movement is blocked unless:

```
target_evaluator_completed_workflow=1
evidence_bundle_complete=1
non_claim_review_claim_safe=1
public_entrypoint_review_complete=1
support_boundary_present=1
known_limitations_present=1
live_host_inventory_claim=0
host_adapter_claim=0
runtime_authority_claim=0
security_hardening_claim=0
```

Even when all requirements pass, product-readiness movement must remain conservative and must not imply production readiness, security protection, host-adapter readiness, or OS-base behavior.

Panel/local-control estimate rule

Panel or local-control-surface estimate movement may be considered only if:

```
Console_host_inventory_contract_visible=1
Console_command_surface_visible=1
```

```
authority_floor_visible=1
receipt_boundary_visible=1
host_adapter_gate_visible=1
failure_state_coverage_visible=1
public_entrypoint_review_complete=1
```

Any movement must be limited to reviewability of the local control surface. It must not imply live host inventory, host access, file access, network access, runtime enforcement, or host-adapter availability.

Public documentation estimate rule

Public documentation posture movement may be considered only if:

```
workflow_docs_complete=1
non_claims_visible=1
public_entrypoints_aligned=1
support_boundary_visible=1
known_limitations_visible=1
blocked_wording_removed=1
```

Documentation estimate movement must not be used as a substitute for product-readiness evidence.

Strategy/status/funding estimate rule

Strategy/status/funding framework movement may be considered only if:

```
strategy_chain_complete=1
status_surfaces_aligned=1
public_entrypoint_review_complete=1
estimate_impact_review_complete=1
no_announcement_mismatch_exists=1
```

This lane must not move simply because a template was added.

Blocked estimate changes

Block estimate changes when:

```
evidence_bundle_incomplete=1
non_claim_review_missing=1
claim_expansion_detected=1
public_entrypoint_review_missing=1
workflow_not_completed_by_target_evaluator=1
live_host_inventory_claim_introduced=1
host_probe_claim_introduced=1
host_file_read_claim_introduced=1
host_file_write_claim_introduced=1
host_mutation_claim_introduced=1
network_scan_claim_introduced=1
host_adapter_claim_introduced=1
runtime_authority_claim_introduced=1
security_hardening_claim_introduced=1
product_readiness_wording_overclaims=1
```

Estimate-change proposal fields

If any estimate movement is proposed, record:

```
lane:
baseline estimate:
proposed estimate:
delta:
evidence basis:
non-claim review:
public-entrypoint review:
residual risk:
reviewer note:
```

Every proposed change must have a separate lane entry.

Required reviewer answers

```
Does this evidence change capability posture? yes|no
Does this evidence justify any estimate change? yes|no
Does this evidence require public-entrypoint updates? yes|no
Does this evidence require announcement review? yes|no
Does this evidence preserve all non-claims? yes|no
Does this evidence preserve current host-inventory posture? yes|no
Does this evidence preserve current host-adapter posture? yes|no
Does this evidence preserve current runtime authority posture? yes|no
Does this evidence preserve current security-hardening posture? yes|no
```

If any answer is ambiguous, use insufficient_evidence.

Review gate output

```
estimate_impact_review_present=0|1
estimate_change_recommended=0|1
overall_latticra_system_estimate_change_recommended=0|1
panel_local_control_surface_estimate_change_recommended=0|1
public_product_readiness_estimate_change_recommended=0|1
public_documentation_estimate_change_recommended=0|1
strategy_status_funding_estimate_change_recommended=0|1
runtime_os_direction_estimate_change_recommended=0|1
security_hardening_estimate_change_recommended=0|1
announcement_review_required=0|1
public_entrypoint_update_required=0|1
review_outcome:
```

This template output:

```
estimate_impact_review_present=0
estimate_change_recommended=0
overall_latticra_system_estimate_change_recommended=0
panel_local_control_surface_estimate_change_recommended=0
public_product_readiness_estimate_change_recommended=0
public_documentation_estimate_change_recommended=0
strategy_status_funding_estimate_change_recommended=0
runtime_os_direction_estimate_change_recommended=0
security_hardening_estimate_change_recommended=0
announcement_review_required=0
public_entrypoint_update_required=0
review_outcome=not_performed
```

Recommended next planning move

Recommended next planning move:

```
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the workflow, checklist, bundle, non-claim review, public-entrypoint review,
estimate-impact review, and review package index now exist; the next useful
planning lane is the Tier 2 Nucleus effect-boundary package after the
completion checkpoint closes this package
```

Non-claims

This template does not perform an estimate-impact review, change estimates, update public entry points, validate workflow evidence, run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate product readiness, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records an estimate-impact review form only.

docs/strategy/2026-05-26-2019-cdt-console-read-only-host-inventory-planning-completion-checkpoint.md

Source title: Latticra Console Read-Only Host Inventory Planning Completion Checkpoint

Lines: 168 | Words: 537 | SHA-256: 8729afe5fb06e1db973f3b857ae28acd431f20d803fbd33c98497e78ced5099c

Latticra Console Read-Only Host Inventory Planning Completion Checkpoint

Status: planning completion checkpoint Created: 2026-05-26 20:19 CDT Decision: stop at planning package completion Promotion decision: no Console, host-inventory, host-adapter, product-readiness, runtime, security, or OS-base promotion recommended Scope: completion checkpoint for the Console read-only host-inventory strategy package.

Purpose

This checkpoint closes the current planning pass for Console read-only host inventory.

It records that the planning package is organized enough for future evidence work, but that no evidence capture, command execution, host inspection, review performance, public-entrypoint update, estimate change, or product-readiness promotion should begin without a separate explicit request.

Source package

This checkpoint follows:

```
docs/strategy/2026-05-26-1959-cdt-console-read-only-host-inventory-review-package-index.md
```

Latest estimate-impact review template available at this checkpoint:

```
docs/strategy/2026-05-26-1956-cdt-console-read-only-host-inventory-estimate-impact-review-templa
te.md
```

Decision

Current decision:

```
stop_at_planning_package_completion=1
begin_evidence_capture=0
run_commands=0
run_console=0
run_latticra_lc=0
run_latticra_console_report=0
run_host_inventory=0
inspect_host=0
probe_host=0
read_host_files=0
write_host_files=0
scan_network=0
enable_host_adapter=0
create_receipt=0
request_signature=0
perform_non_claim_review=0
perform_public_entrypoint_review=0
perform_estimate_impact_review=0
change_public_posture=0
change_estimates=0
```

Completed planning records

The current package has records for:

```
overall strategy priority selection
Console host-inventory workflow selection
acceptance criteria
evidence bundle shape
non-claim review
public-entrypoint review
estimate-impact review
```

```
package index
```

Current package status

```
planning_package_complete_for_current_scope=1
workflow_selected=1
acceptance_criteria_defined=1
evidence_bundle_shape_defined=1
review_gates_defined=1
package_index_defined=1
workflow_evidence_captured=0
live_host_inventory_performed=0
host_probe_performed=0
host_file_read_performed=0
host_file_write_performed=0
host_mutation_performed=0
network_scan_performed=0
host_adapter_enabled=0
receipt_materialized=0
receipt_signed=0
review_performed=0
estimate_change_recommended=0
public_entrypoint_change_recommended=0
product_readiness_promotion_recommended=0
```

Boundary for future evidence capture

Evidence capture must be a separate explicit request because it may require:

```
running local commands
running Console command surfaces
collecting transcripts
attaching command output
reviewing contract-only evidence
performing non-claim review
performing public-entrypoint review
performing estimate-impact review
```

Those activities are outside this planning-only checkpoint.

Allowed next strategic moves

Allowed planning-only next moves:

```
choose a different promotion gate
create a new planning package for another lane
review overall strategy priorities
refresh strategy index organization
build a roadmap lane map
build a public messaging boundary record
begin the Tier 2 Nucleus effect-contract and runtime-denial gate planning package
```

Blocked without separate request:

```
run Console
run latticra-lc
run latticra_console_report
run host inventory
inspect a host
probe a host
read host files
write host files
scan a network
enable a host adapter
create or sign a receipt
capture evidence transcripts
update README or status as evidence-backed public posture
change completion estimates
announce product-readiness movement
```

Recommended next planning lane

Recommended next planning lane:

```
Nucleus effect-contract and runtime-denial gate planning package
```

Reason:

```
the Console read-only host-inventory path is now organized; the overall priority
```

```
map names runtime and Nucleus effect-boundary planning as the next planning
priority after public evaluator workflow clarity, while still preserving
contract-level, no-effect boundaries
```

Non-claims

This checkpoint does not capture evidence, run commands, run Console, run latticra-lc, run latticra_console_report, inspect a host, probe a host, read host files, write host files, scan a network, launch a host process, mutate a host, create an inventory artifact, create a receipt, request a signature, sign a receipt, enable a host adapter, enforce runtime policy, boot hardware, validate workflow evidence, perform a review, update public entry points, change estimates, provide product readiness, provide Console readiness, provide host-inventory readiness, provide host-adapter readiness, grant runtime authority, execute effects, enforce capabilities, provide cryptographic authority, provide signing authority, harden security, prevent malware, prevent ransomware, provide sandboxing, provide operating-system behavior, or provide production support.

It records planning package completion only.

docs/strategy/README.md

Source title: Latticra Strategy Index

Lines: 235 | Words: 505 | SHA-256: 30ef75e08ef3775f937a823e88e570bc9ddb0b818228e5b75deba61e39f086f5

Latticra Strategy Index

Status: active strategy index Last updated: 2026-05-26 20:19 CDT Scope: dated strategy records, mission alignment, review cadence, and quality expectations.

Purpose

This folder records Latticra strategy in dated, reviewable documents.

Strategy may change as evidence, implementation quality, threat modeling, funding, target users, and architectural constraints change. Dated records make those changes visible instead of hidden.

Naming rule

Use this filename pattern for strategy records:

```
YYYY-MM-DD-HHMM-zone-topic.md
```

Example:

```
2026-05-15-2249-cdt-national-security-open-system-strategy.md
```

Current active strategy

Current strategy record:

```
2026-05-15-2249-cdt-national-security-open-system-strategy.md
```

Latest strategy estimate review:

```
2026-05-19-1845-cdt-strategy-estimate-review.md
```

Latest strategy posture refresh:

```
2026-05-25-1951-cdt-strategy-posture-refresh.md
```

Latest capability promotion gate map:

```
2026-05-25-2017-cdt-capability-promotion-gate-map.md
```

Latest capability promotion packet template:

```
2026-05-25-2025-cdt-capability-promotion-packet-template.md
```

Latest public product-readiness promotion packet:

2026-05-25-2114-cdt-public-product-readiness-promotion-packet.md

Latest Panel-guided local evaluation workflow packet:

2026-05-25-2137-cdt-panel-guided-local-evaluation-workflow-packet.md

Latest Panel-guided local evaluation acceptance checklist:

2026-05-25-2157-cdt-panel-guided-local-evaluation-acceptance-checklist.md

Latest Panel-guided local evaluation evidence bundle template:

2026-05-25-2226-cdt-panel-guided-local-evaluation-evidence-bundle-template.md

Latest Panel-guided local evaluation evidence capture plan:

2026-05-25-2246-cdt-panel-guided-local-evaluation-evidence-capture-plan.md

Latest Panel-guided local evaluation non-claim review template:

2026-05-26-0019-cdt-panel-guided-local-evaluation-non-claim-review-template.md

Latest Panel-guided local evaluation public-entrypoint review template:

2026-05-26-0034-cdt-panel-guided-local-evaluation-public-entrypoint-review-template.md

Latest Panel-guided local evaluation estimate-impact review template:

2026-05-26-0119-cdt-panel-guided-local-evaluation-estimate-impact-review-template.md

Latest Panel-guided local evaluation guard/test reference template:

2026-05-26-0949-cdt-panel-guided-local-evaluation-guard-test-reference-template.md

Latest Panel-guided local evaluation review package index:

2026-05-26-0410-cdt-panel-guided-local-evaluation-review-package-index.md

Latest Panel-guided local evaluation planning completion checkpoint:

2026-05-26-1356-cdt-panel-guided-local-evaluation-planning-completion-checkpoint.md

Latest overall strategy priority map:

2026-05-26-1702-cdt-overall-strategy-priority-map.md

Latest Console read-only host inventory workflow packet:

2026-05-26-1707-cdt-console-read-only-host-inventory-workflow-packet.md

Latest Console read-only host inventory acceptance checklist:

2026-05-26-1732-cdt-console-read-only-host-inventory-acceptance-checklist.md

Latest Console read-only host inventory evidence bundle template:

2026-05-26-1736-cdt-console-read-only-host-inventory-evidence-bundle-template.md

Latest Console read-only host inventory non-claim review template:

2026-05-26-1757-cdt-console-read-only-host-inventory-non-claim-review-template.md

Latest Console read-only host inventory public-entrypoint review template:

2026-05-26-1919-cdt-console-read-only-host-inventory-public-entrypoint-review-template.md

Latest Console read-only host inventory estimate-impact review template:

2026-05-26-1956-cdt-console-read-only-host-inventory-estimate-impact-review-template.md

Latest Console read-only host inventory review package index:

2026-05-26-1959-cdt-console-read-only-host-inventory-review-package-index.md

Latest Console read-only host inventory planning completion checkpoint:

2026-05-26-2019-cdt-console-read-only-host-inventory-planning-completion-checkpoint.md

Current operating rule:

Continue small guarded report/status alignment only when drift appears.

Strategic planning rules

1. State mission clearly.
2. Separate current capability from long-term ambition.
3. Keep security claims evidence-bound.
4. Treat anti-malware and anti-ransomware goals as design targets until validated.
5. Prefer auditable open systems over hidden behavior.
6. Keep public docs professional and self-contained.
7. Update estimates when implementation evidence changes.
8. Add tests before or beside implementation when possible.
9. Use no-effect previews before mutation.
10. Keep target users and deployment assumptions explicit.

Review cadence

Update strategy when any of these change:

- project mission;
- supported target users;
- threat model;
- implementation milestone;
- completion estimate;
- funding model;
- public positioning;
- security non-claims;
- product quality bar;
- roadmap direction.

At minimum, review this folder at major milestone boundaries.

Quality bar

Latticra strategy should be:

```
clear
professional
defensive
evidence-bound
versioned
dated
reviewable
actionable
non-hype
```

Non-claims

This folder does not implement security controls, operating-system behavior, malware prevention, ransomware prevention, sandboxing, update safety, recovery safety, or production readiness.

It records strategy and planning only.

Part XIII - Status Record Archive

- docs/status/AUTHORITY_STATUS_ANNOUNCEMENT_REVIEW.md - Authority Status Announcement Review
- docs/status/BACKUP_RECOVERY_RESILIENCE_BASELINE_STATUS.md - Latticra Backup, Recovery, and Cyber Resilience Baseline Status
- docs/status/COMPLETION_ESTIMATE_L_UI_RENDERING_REVIEW.md - Completion Estimate Review After L-UI Rendering Detailed Report Refinement
- docs/status/COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md - Completion Estimate Review After Runtime Boundary Abuse-Case Fixtures
- docs/status/COMPLETION_ESTIMATE_REVIEW_README_STATUS_ALIGNMENT.md - Completion Estimate Review README Status Alignment
- docs/status/COMPLETION_PERCENTAGE_REVIEW.md - Completion Percentage Review
- docs/status/CPP_AUTHORITY_EXPANSION_CONTRACT_REVIEW.md - C++ Authority Expansion Contract Review
- docs/status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md - C++ Authority Implementation Review Status
- docs/status/CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE_STATUS.md - Latticra Cryptographic Assurance and Key Management Baseline Status
- docs/status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md - Current Estimate Mathematical Rebase - 2026-05-26
- docs/status/CURRENT_ESTIMATE_REFRESH_2026_05_24.md - Current Estimate Refresh - 2026-05-24
- docs/status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md - Current Estimate Table Source Alignment
- docs/status/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE_STATUS.md - Latticra Cyber Incident Reporting and Response Baseline Status
- docs/status/DEBIAN_ECOSYSTEM_INTEGRATION_STATUS.md - Debian Ecosystem Integration Status
- docs/status/FEDORA_DISPOSABLE_VM_EFFECT_GATE_CLASSIFIER_STATUS.md - Fedora Disposable VM Effect Gate Classifier Status
- docs/status/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_EVIDENCE_STATUS.md - Fedora Disposable VM Local RPM Validation Evidence Status
- docs/status/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_STATUS.md - Fedora Disposable VM Local RPM Validation Status
- docs/status/FEDORA_DISPOSABLE_VM_RPM_README_ALIGNMENT_STATUS.md - Fedora Disposable VM RPM README Alignment Status
- docs/status/FEDORA_HOST_INSTALL_PREFLIGHT_STATUS.md - Fedora Host Install Preflight Status
- docs/status/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_STATUS.md - Fedora Install Preflight Snapshot Capture Status
- docs/status/FEDORA_INSTALLROOT_RPM_LIFECYCLE_STATUS.md - Fedora Installroot RPM Lifecycle Status

- docs/status/FEDORA_LIVE_READONLY_SNAPSHOT_ADAPTER_STATUS.md - Fedora Live Read-Only Snapshot Adapter Status
- docs/status/FEDORA_MANUAL_HOST_RC_CHECKLIST_STATUS.md - Fedora Manual Host RC Checklist Status
- docs/status/FEDORA_MANUAL_HOST_RC_DECISION_CLASSIFIER_STATUS.md - Fedora Manual Host RC Decision Classifier Status
- docs/status/FEDORA_RPM_GATE_CLASSIFIER_STATUS.md - Fedora RPM Gate Classifier Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN_STATUS.md - Fedora VM CLI Payload Next Validation Lane Plan Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_README_ALIGNMENT_STATUS.md - Fedora VM CLI Payload README Alignment Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT_STATUS.md - Fedora VM CLI Payload Repeatability Evidence Acceptance Contract Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE_STATUS.md - Fedora VM CLI Payload Repeatability Evidence Publication Gate Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE_STATUS.md - Fedora VM CLI Payload Repeatability Evidence Review Gate Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR_STATUS.md - Fedora VM CLI Payload Repeatability Evidence Status Review Validator Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE_STATUS.md - Fedora VM CLI Payload Repeatability Evidence Status Template Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN_STATUS.md - Fedora VM CLI Payload Repeatability Runner Plan Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_STATUS.md - Fedora VM CLI Payload Repeatability Runner Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE_STATUS.md - Fedora VM CLI Payload Repeatability Transcript Capture Template Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT_STATUS.md - Fedora VM CLI Payload Repeatability Transcript Contract Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR_STATUS.md - Fedora VM CLI Payload Repeatability Transcript Review Validator Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_ANNOUNCEMENT_STATUS.md - Fedora VM CLI Payload Validation Announcement Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md - Fedora VM CLI Payload Validation Evidence Status
- docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_STATUS.md - Fedora VM CLI Payload Validation Status
- docs/status/FEDORA_VM_RPM_VALIDATION_ANNOUNCEMENT_STATUS.md - Fedora VM RPM Validation Announcement Status
- docs/status/FREEBSD_ECOSYSTEM_INTEGRATION_STATUS.md - FreeBSD Ecosystem Integration Status
- docs/status/HIGH_ASSURANCE_SECURITY_BASELINE_STATUS.md - Latticra High-Assurance Security Baseline Status

- docs/status/IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE_STATUS.md - Latticra Identity, Credential, and Access Management Baseline Status
- docs/status/KERNEL_LIFECYCLE_EVIDENCE_STATUS.md - Kernel Lifecycle Evidence Status
- docs/status/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT_STATUS.md - L-UI Rendering Detailed Report Refinement Status
- docs/status/L_UI_RENDERING_README_STATUS_ALIGNMENT.md - L-UI Rendering README/Status Alignment
- docs/status/LANGUAGE_REPRESENTATION_REVIEW.md - Language Representation Review
- docs/status/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_STATUS.md - Lat Pipeline Diagnostic Integration Status
- docs/status/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_AUDIT_STATUS.md - Lat Pipeline Diagnostic Main Test Audit Status
- docs/status/LATTICRA_CONSOLE_FOUNDATION_STATUS.md - Latticra Console Foundation Status
- docs/status/LATTICRA_NO_EFFECT_CLI_RPM_SPEC_UPDATE_STATUS.md - Latticra No-Effect CLI RPM Spec Update Status
- docs/status/LATTICRA_PANEL_LOCAL_INSTALL_EVIDENCE_STATUS.md - Latticra Panel Local Install Evidence Status
- docs/status/LATTICRA_PANEL_LOCAL_INSTALL_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Panel Local Install Public Entrypoint Alignment
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_DELIVERY_GATE_STATUS.md - Latticra Panel Signed Updater Delivery Gate Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_DENIAL_TRANSCRIPT_STATUS.md - Latticra Panel Signed Updater Denial Transcript Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_CONTRACT_STATUS.md - Latticra Panel Signed Updater Manifest Fixture Contract Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_VALIDATION_STATUS.md - Latticra Panel Signed Updater Manifest Fixture Validation Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_CONTRACT_STATUS.md - Latticra Panel Signed Updater State Fixture Contract Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_VALIDATION_STATUS.md - Latticra Panel Signed Updater State Fixture Validation Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_REVIEW_STATUS.md - Latticra Panel Signed Updater State Transition Denial Review Status
- docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_TRANSCRIPT_STATUS.md - Latticra Panel Signed Updater State Transition Denial Transcript Status
- docs/status/LATTICRA_PANEL_UI_DESIGN_CHECKPOINT.md - Latticra Panel UI Design Checkpoint
- docs/status/LATTICRA_SEAL_FOUNDATION_STATUS.md - Latticra Seal Foundation Status
- docs/status/MACOS_APP_BUNDLE_WRITER_ALIGNMENT_STATUS.md - macOS App Bundle Writer Alignment Status
- docs/status/MACOS_APP_BUNDLE_WRITER_DRY_RUN_STATUS.md - macOS App Bundle Writer Dry-Run Status
- docs/status/MACOS_BUILD_PLATFORM_PROBE_STATUS.md - macOS Build Platform Probe Status

- docs/status/MACOS_COMMIT_GATE_CONTRACT_STATUS.md - macOS Commit Gate Contract Status
- docs/status/MACOS_DRY_RUN_PLAN_ADAPTER_STATUS.md - macOS Dry-Run Plan Adapter Status
- docs/status/MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION_STATUS.md - macOS Dry-Run Writer Candidate Integration Status
- docs/status/MACOS_INTEGRATION_TRANSFERABILITY_STATUS.md - macOS Integration Transferability Status
- docs/status/MACOS_LOCAL_CANDIDATE_ASSET_PROBE_STATUS.md - macOS Local Candidate Asset Probe Status
- docs/status/MACOS_README_INSTALLER_USAGE_STATUS.md - macOS README Installer Usage Status
- docs/status/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT_STATUS.md - macOS Reset/Uninstall Absence-Report Contract Status
- docs/status/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT_STATUS.md - macOS Reset/Uninstall Dry-Run Contract Status
- docs/status/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER_STATUS.md - macOS Reset/Uninstall Dry-Run Planner Status
- docs/status/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT_STATUS.md - macOS Reset/Uninstall Effect-Authorization Contract Status
- docs/status/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT_STATUS.md - macOS Reset/Uninstall Evidence-Bundle Contract Status
- docs/status/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT_STATUS.md - macOS Reset/Uninstall Implementation-Gate Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Denial Transcript Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Execution Preflight Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Implementation Plan Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_REVIEW_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Review Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Contract

Status

- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Review Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Review Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Transcript Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Acceptance-Gate Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Denied-Dispatch Review Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Denied-Dispatch Transcript Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner Interface Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT_STATUS.md - macOS Reset/Uninstall Live-Runner No-Op Prototype Contract Status
- docs/status/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER_STATUS.md - macOS Reset/Uninstall Live-Target Classifier Status
- docs/status/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT_STATUS.md - macOS Reset/Uninstall Operator-Intent Contract Status
- docs/status/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT_STATUS.md - macOS Reset/Uninstall Receipt-Schema Contract Status
- docs/status/MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT_STATUS.md - macOS User-Local App Bundle Contract Status
- docs/status/MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN_STATUS.md - macOS User-Local App Bundle Implementation Plan Status
- docs/status/MACOS_VERIFICATION_TRANSCRIPT_CONTRACT_STATUS.md - macOS Verification Transcript Contract Status
- docs/status/MEMORY_SAFETY_ROADMAP_STATUS.md - Latticra Memory-Safety Roadmap Status
- docs/status/NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15_STATUS.md - Nadia Awareness Dialogue Contract Stage-15 Status
- docs/status/NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27_STATUS.md - Nadia Context Window Assembly Contract Stage-27 Status
- docs/status/NADIA_DEVELOPER_WORKBENCH_STAGE_3_STATUS.md - Nadia Developer Workbench Stage-3 Status
- docs/status/NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7_STATUS.md - Nadia Guarded Tool Authority Stage-7 Status
- docs/status/NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10_STATUS.md - Nadia Inference Readiness Contract Stage-10 Status
- docs/status/NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1_STATUS.md - Nadia Local Context Engine Stage-1 Status

- docs/status/NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9_STATUS.md - Nadia Local Model Registry Contract Stage-9 Status
- docs/status/NADIA_MODEL_LOAD_CONTRACT_STAGE_12_STATUS.md - Nadia Model Load Contract Stage-12 Status
- docs/status/NADIA_OFFLINE_AI_STAGE_0_STATUS.md - Nadia Offline AI Stage-0 Status
- docs/status/NADIA_PRODUCTIVITY_LOOP_STAGE_5_STATUS.md - Nadia Productivity Loop Stage-5 Status
- docs/status/NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8_STATUS.md - Nadia Prompt Evaluation Contract Stage-8 Status
- docs/status/NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16_STATUS.md - Nadia Prompt Evaluation Handoff Contract Stage-16 Status
- docs/status/NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28_STATUS.md - Nadia Prompt Evaluation Input Contract Stage-28 Status
- docs/status/NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30_STATUS.md - Nadia Prompt Evaluation Invocation Contract Stage-30 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31_STATUS.md - Nadia Prompt Evaluation Result Contract Stage-31 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33_STATUS.md - Nadia Prompt Evaluation Result Disposition Contract Stage-33 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34_STATUS.md - Nadia Prompt Evaluation Result Release Contract Stage-34 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Contract Stage-35 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Contract Stage-36 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_37_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Contract Stage-37 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_38_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Contract Stage-38 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_39_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract Stage-39 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_40_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract Stage-40 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_41_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-41 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_42_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-42 Status

- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_43_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract Stage-43 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_44_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract Stage-44 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_45_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-45 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46_STATUS.md - Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-46 Status
- docs/status/NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32_STATUS.md - Nadia Prompt Evaluation Result Review Contract Stage-32 Status
- docs/status/NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29_STATUS.md - Nadia Prompt Evaluation Runtime Handoff Contract Stage-29 Status
- docs/status/NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14_STATUS.md - Nadia Prompt Materialization Contract Stage-14 Status
- docs/status/NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13_STATUS.md - Nadia Prompt Receipt Contract Stage-13 Status
- docs/status/NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26_STATUS.md - Nadia Prompt Token Sequence Contract Stage-26 Status
- docs/status/NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25_STATUS.md - Nadia Prompt Tokenization Contract Stage-25 Status
- docs/status/NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6_STATUS.md - Nadia Protective Safety Boundary Stage-6 Status
- docs/status/NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11_STATUS.md - Nadia Runtime Invocation Contract Stage-11 Status
- docs/status/NADIA_RUNTIME_PROFILE_STAGE_2_STATUS.md - Nadia Runtime Profile Stage-2 Status
- docs/status/NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4_STATUS.md - Nadia Systems Engineering Mode Stage-4 Status
- docs/status/NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17_STATUS.md - Nadia Tokenization Boundary Contract Stage-17 Status
- docs/status/NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23_STATUS.md - Nadia Tokenizer Artifact Binding Contract Stage-23 Status
- docs/status/NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20_STATUS.md - Nadia Tokenizer Artifact Inventory Contract Stage-20 Status
- docs/status/NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21_STATUS.md - Nadia Tokenizer Artifact Measurement Contract Stage-21 Status

- docs/status/NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22_STATUS.md - Nadia Tokenizer Artifact Verification Contract Stage-22 Status
- docs/status/NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19_STATUS.md - Nadia Tokenizer Manifest Contract Stage-19 Status
- docs/status/NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24_STATUS.md - Nadia Tokenizer Runtime Attachment Contract Stage-24 Status
- docs/status/NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18_STATUS.md - Nadia Tokenizer Specification Contract Stage-18 Status
- docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_README_ALIGNMENT.md - Nucleus Report-Only Announcement README Alignment
- docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_REVIEW.md - Nucleus Report-Only Announcement Review
- docs/status/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT_STATUS.md - Nucleus Task No-Effect Report Alignment Status
- docs/status/NUCLEUS_TASK_README_STATUS_ALIGNMENT.md - Nucleus Task README/Status Alignment
- docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_README_STATUS_ALIGNMENT.md - Nucleus Task Report-Only Execution README/Status Alignment
- docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md - Nucleus Task Report-Only Execution Refinement Status
- docs/status/OPENBSD_ECOSYSTEM_INTEGRATION_STATUS.md - OpenBSD Ecosystem Integration Status
- docs/status/OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md - openSUSE Ecosystem Integration Status
- docs/status/PROJECT_NOTES_FOLLOWUP_STATUS_INDEX_CHECK.md - Project Notes Follow-up Status/Index Check
- docs/status/PROJECT_NOTES_NUCLEUS_ANNOUNCEMENT_README_ALIGNMENT_STATUS_CHECK.md - Project Notes Nucleus Announcement README Alignment Status Check
- docs/status/PROJECT_NOTES_NUCLEUS_REPORT_ONLY_ALIGNMENT_STATUS_INDEX_CHECK.md - Project Notes Nucleus Report-Only Alignment Status/Index Check
- docs/status/PUBLIC_ENTRY_POINT_CONSISTENCY_SCAN.md - Public Entry-Point Consistency Scan
- docs/status/RBDM_REPORT_INTEGRATION_STATUS.md - RBDM Report Integration Status
- docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md - Latticra Seal Agentic Automation Security Index Alignment
- docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Seal Agentic Automation Security Public Entrypoint Alignment
- docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md - Latticra Seal Agentic Automation Security Report Surface Status
- docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md - Latticra Seal Agentic Automation Security Status
- docs/status/SEAL_CAPABILITY_GATE_STATUS.md - Latticra Seal Capability Gate Status
- docs/status/SEAL_CAPABILITY_METADATA_REPORT_SURFACE_STATUS.md - Latticra Seal Capability Metadata Report Surface Status

- docs/status/SEAL_CORE_BLOCKED_CASES_STATUS.md - Latticra Seal Core Blocked Cases Status
- docs/status/SEAL_CORE_EVIDENCE_INDEX_ALIGNMENT.md - Latticra Seal Core Evidence Index Alignment
- docs/status/SEAL_CORE_EVIDENCE_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Seal Core Evidence Public Entrypoint Alignment
- docs/status/SEAL_CORE_EVIDENCE_PUBLIC_STATUS_UPDATE.md - Latticra Seal Core Evidence Public Status Update
- docs/status/SEAL_CORE_EVIDENCE_STATUS.md - Latticra Seal Core Evidence Status
- docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md - Latticra Seal Crypto Graduation Gate Status
- docs/status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md - Latticra Seal Crypto Verify Backend Status
- docs/status/SEAL_ED25519_VERIFY_STATUS.md - Latticra Seal Ed25519 Verify Status
- docs/status/SEAL_EFFECT_DECISION_STATUS.md - Latticra Seal Effect Decision Status
- docs/status/SEAL_GUARDED_ALLOWLIST_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Seal Guarded Allowlist Public Entrypoint Alignment
- docs/status/SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE_STATUS.md - Latticra Seal Guarded Allowlist Report Surface Status
- docs/status/SEAL_GUARDED_ALLOWLIST_STATUS_INDEX_ALIGNMENT.md - Latticra Seal Guarded Allowlist Status Index Alignment
- docs/status/SEAL_KEY_HANDLING_STATUS.md - Latticra Seal Key-Handling Status
- docs/status/SEAL_KEY_MATERIAL_STATUS.md - Latticra Seal Key-Material Status
- docs/status/SEAL_KEY_PARSING_STATUS.md - Latticra Seal Key Parsing Status
- docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md - Latticra Seal Operator Receipt Report Status
- docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md - Latticra Seal Parameter Schema Status
- docs/status/SEAL_POLICY_DECISION_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Seal Policy Decision Public-Entrypoint Alignment
- docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md - Latticra Seal Policy Decision Report Surface Status
- docs/status/SEAL_POLICY_DECISION_STATUS.md - Latticra Seal Policy Decision Status
- docs/status/SEAL_PRODUCT_SPINE_STATUS.md - Latticra Seal Product Spine Status
- docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md - Latticra Seal Public-Key Parsing Status
- docs/status/SEAL_README_STATUS_ROW_ALIGNMENT.md - Latticra Seal README Status Row Alignment
- docs/status/SEAL_REPORT_ENVELOPE_STATUS.md - Latticra Seal Report Envelope Status
- docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md - Latticra Seal Request Freshness Status
- docs/status/SEAL_RUNTIME_DRY_RUN_PUBLIC_ENTRYPOINT_ALIGNMENT.md - Latticra Seal Runtime Dry-Run Public Entrypoint Alignment
- docs/status/SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md - Latticra Seal Runtime Dry-Run Report Surface Status

- docs/status/SEAL_RUNTIME_DRY_RUN_STATUS_INDEX_ALIGNMENT.md - Latticra Seal Runtime Dry-Run Status Index Alignment
- docs/status/SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md - Latticra Seal Runtime Handoff Evaluation Status
- docs/status/SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md - Latticra Seal Runtime Handoff Report Status
- docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md - Latticra Seal Runtime Handoff Status
- docs/status/SEAL_SIGNATURE_REQUEST_STATUS.md - Latticra Seal Signature Request Status
- docs/status/SEAL_SIGNED_REQUEST_STATUS.md - Latticra Seal Signed Request Status
- docs/status/SEAL_SIGNER_HANDOFF_STATUS.md - Latticra Seal Signer Handoff Status
- docs/status/SEAL_SIGNER_INVOCATION_STATUS.md - Latticra Seal Signer Invocation Status
- docs/status/SEAL_SIGNING_AUTHORIZATION_STATUS.md - Latticra Seal Signing Authorization Status
- docs/status/SEAL_SIGNING_OPERATION_STATUS.md - Latticra Seal Signing Operation Status
- docs/status/SEAL_STATUS_ROLLUP_STATUS.md - Latticra Seal Status Rollup Status
- docs/status/SEAL_VERIFICATION_POLICY_STATUS.md - Latticra Seal Verification Policy Status
- docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md - Latticra Seal Verification Receipt Status
- docs/status/SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md - Latticra Seal Verified Capability Gate Status
- docs/status/SEAL_VERIFIED_EFFECT_DECISION_STATUS.md - Latticra Seal Verified Effect Decision Status
- docs/status/SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md - Latticra Seal Verified Receipt Promotion Status
- docs/status/SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE_STATUS.md - Latticra Secure Configuration and Change Management Baseline Status
- docs/status/SECURITY_LOGGING_MONITORING_BASELINE_STATUS.md - Latticra Security Logging, Monitoring, and Detection Baseline Status
- docs/status/STATUS_ANNOUNCEMENT_CONSISTENCY_REVIEW.md - Status and Announcement Consistency Review
- docs/status/STATUS_ANNOUNCEMENT_REVIEW.md - Status Announcement Review
- docs/status/SUPPLY_CHAIN_SECURITY_BASELINE_STATUS.md - Latticra Supply-Chain Security Baseline Status
- docs/status/UBUNTU_ECOSYSTEM_INTEGRATION_STATUS.md - Ubuntu Ecosystem Integration Status
- docs/status/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE_STATUS.md - Latticra Vulnerability Management Release Gate Baseline Status
- docs/status/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE_STATUS.md - Latticra Zero-Trust Runtime Authority Baseline Status

docs/status/AUTHORITY_STATUS_ANNOUNCEMENT_REVIEW.md

Source title: Authority Status Announcement Review

Lines: 33 | Words: 120 | SHA-256: 8bb0f874869b31b60e4e8268ea27c688f3ef0dcab971052121261d055d48f2ec

Authority Status Announcement Review

Status: no-new-announcement review

This record reviews whether the constrained C++ authority review sequence needs a separate public announcement after the authority implementation review, authority status/docs alignment, authority foundation index alignment, status announcement review, public entry-point consistency scan, and project-notes follow-up alignment.

Reviewed files:

```
STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/status/CURRENT_STATUS.md
docs/status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md
docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md
docs/FOUNDATION_INDEX.md
README.md
```

Finding:

```
No new authority behavior or public messaging change exists beyond the already-recorded
no-effect authority review and documentation alignment sequence.
```

Decision:

```
Do not add a new public announcement for this slice.
Keep the authority messaging evidence-bound, no-effect, metadata-only, fixed-capacity, and
denied-by-default.
Advance the queue to the next implementation/documentation refinement lane.
```

Boundary: status review only. No implementation behavior is changed.

docs/status/BACKUP_RECOVERY_RESILIENCE_BASELINE_STATUS.md

Source title: Latticra Backup, Recovery, and Cyber Resilience Baseline Status

Lines: 64 | Words: 146 | SHA-256: 0a2c9455f30db4abfb523947a95bcbb3e329d90f9e10cf528cc3fea539a50ae8

Latticra Backup, Recovery, and Cyber Resilience Baseline Status

Status: status record for backup, recovery, and cyber resilience baseline Date: 2026-05-26

Scope

This record tracks the backup, recovery, and cyber resilience baseline for backup scope, offline backup posture, restore testing, recovery prioritization, RTO/RPO planning, golden-image and infrastructure-as-code recovery evidence, recovery isolation, rollback, resilience engineering, recovery communications, post-incident lessons learned, and recovery non-claims.

It does not implement backup creation, backup storage, restore execution, rollback execution, disaster recovery, recovery orchestration, failover, continuity operations, golden-image creation, infrastructure-as-code deployment, ransomware recovery, compliance, or runtime authority.

Current fields

```
backup_recovery_resilience_baseline_present=1
backup_recovery_resilience_status_present=1
backup_recovery_resilience_guard_present=1
stopransomware_recovery_guidance_tracked=1
cisa_cpg_backup_recovery_tracked=1
nist_sp_800_34_contingency_planning_tracked=1
nist_sp_800_184_event_recovery_tracked=1
nist_sp_800_160_cyber_resilience_tracked=1
nist_csf_recover_function_tracked=1
nist_sp_800_53_contingency_planning_tracked=1
backup_scope_inventory_required=1
critical_asset_restore_priority_required=1
rto_rpo_record_required=1
offline_encrypted_backup_plan_required=1
backup_integrity_test_required=1
restore_test_required=1
clean_recovery_environment_required=1
golden_image_or_iac_recovery_plan_required=1
rollback_plan_required_before_mutation=1
recovery_authorization_contract_required=1
recovery_communications_plan_required=1
lessons_learned_update_required=1
recovery_exception_owner_required=1
recovery_exception_expiration_required=1
implementation_behavior_changed=0
backup_creation_added=0
backup_storage_added=0
restore_execution_added=0
rollback_execution_added=0
failover_added=0
recovery_orchestration_added=0
disaster_recovery_service_added=0
ransomware_recovery_service_added=0
production_recovery_claim_allowed=0
hosted_service_recovery_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-backup-recovery-resilience-baseline.sh
```

Expected output:

```
backup_recovery_resilience_baseline: ok
```

docs/status/COMPLETION_ESTIMATE_L_UI_RENDERING_REVIEW.md

Source title: Completion Estimate Review After L-UI Rendering Detailed Report Refinement

Lines: 45 | Words: 197 | SHA-256: 5a9daaf5b43fc7b692e1f35211e101b2b073521ac97c9e3b427ad82da4f82c20

Completion Estimate Review After L-UI Rendering Detailed Report Refinement

Status: planning-estimate review

This record reviews completion estimates after the no-effect L-UI rendering detailed report refinement and README/status alignment.

Review basis:

```
L-UI rendering detailed report refinement
L-UI rendering README/status alignment
focused L-UI detailed report refinement guard
main C workflow coverage for the L-UI detailed report guard
```

Estimate adjustments:

```
Overall Latticra system: 35% -&gt; 36%
L-UI parser / AST / string foundation: 86% -&gt; 87%
Foundation documents and contracts: 89% -&gt; 90%
Public documentation posture: 83% -&gt; 84%
Strategy/status/funding framework: 57% -&gt; 58%
```

Unchanged areas:

```
Lat / Latticra Programming Language: 25%
LIR / Intermediate Representation: 22%
C/C++ foundation direction: 14%
Constrained C++ authority layer: 4%
Runtime / operating-system-universe direction: 17%
Nucleus real task execution: 11%
Security-hardening implementation: 5%
Public product readiness: 5%
```

Reasoning:

```
The L-UI detailed report refinement improved an existing no-effect renderer reporting surface
and added focused validation. It improved L-UI/report maturity and public documentation posture,
 but it did not add runtime behavior, command execution, authority expansion, security
hardening, or product readiness.
```

Boundary: planning estimates only. These are not release promises, production-readiness claims, security guarantees, or operating-system completeness claims.

docs/status/COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md

Source title: Completion Estimate Review After Runtime Boundary Abuse-Case Fixtures

Lines: 94 | Words: 319 | SHA-256: 39e9a46cc0ea638214bcadee7083b19f5313f2f166a9c255880cc9077de5df5d

Completion Estimate Review After Runtime Boundary Abuse-Case Fixtures

Status: planning-estimate review Date: 2026-05-25 CDT Scope: completion-estimate review after the runtime-boundary abuse-case fixture expansion. This record does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production protection, sandboxing, malware prevention, ransomware prevention, certification, compliance, public-readiness promotion, or runtime authority.

Review Basis

This review follows:

```
Runtime boundary abuse-case fixture expansion after policy expansion
runtime_boundary_abuse_case_fixture_expansion_present=1
runtime_boundary_abuse_case_fixture_guard_present=1
runtime_boundary_abuse_case_c_fixtures_present=1
runtime_boundary_abuse_case_fixture_count=8
```

The fixture expansion improves evidence coverage for denied and future-gated runtime-boundary abuse cases.

It does not change runtime behavior or public readiness.

Capability Posture Check

```
completion_estimate_after_runtime_boundary_abuse_case_fixtures_present=1
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_verification_added=0
signing_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
runtime_authority_granted=0
security_hardening_changed=0
public_readiness_changed=0
production_readiness_claimed=0
estimate_adjustment_required=0
completion_estimate_review_required=0
completion_estimate_changed=0
```

Estimate Decision

Current estimates remain unchanged:

```
Overall Latticra system: 39%
Latticra Seal / local evidence layer: 34%
Latticra Panel / local control surface: 28%
Nadia offline AI foundation: 59%
L-UI parser / AST / string foundation: 87%
Foundation documents and contracts: 92%
Public documentation posture: 88%
Strategy/status/funding framework: 60%
Lat / Latticra Programming Language: 25%
LIR / Intermediate Representation: 22%
C/C++ foundation direction: 18%
Constrained C++ authority layer: 4%
Nucleus real task execution: 11%
Runtime / operating-system-universe direction: 19%
Security-hardening implementation: 7%
Public product readiness: 8%
```

No estimate is raised because the completed fixture slice adds coverage and reviewability, not a new capability surface.

Validation

This review is guarded by:

```
sh scripts/test-completion-estimate-review-after-runtime-boundary-abuse-case-fixtures.sh
```

Expected output:

```
completion_estimate_review_after_runtime_boundary_abuse_case_fixtures: ok
```

Boundary

These percentages are planning estimates only.

They are not release promises, production-readiness metrics, security guarantees, Fedora approval claims, runtime-enforcement claims, operating-system completeness claims, or authority grants.

The next valid work is small guarded report/status alignment only when drift appears, unless a separate future slice changes capability posture with a contract, implementation plan, tests, and explicit non-claim review.

docs/status/COMPLETION_ESTIMATE_REVIEW_README_STATUS_ALIGNMENT.md

Source title: Completion Estimate Review README Status Alignment

Lines: 81 | Words: 242 | SHA-256: 362029a84a54ad1a818c5552f5f29fde5943d7bd34a639e295f050cae8cad490

Completion Estimate Review README Status Alignment

Status: README/status alignment Date: 2026-05-25 CDT Scope: README, status, foundation-index, and project-notes discoverability after the completion-estimate review following runtime-boundary abuse-case fixtures. This record does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, public-readiness promotion, production protection, security hardening, or runtime authority.

Review Basis

The completion-estimate hold review already records:

```
completion_estimate_after_runtime_boundary_abuse_case_fixtures_present=1
runtime_boundary_abuse_case_fixture_expansion_present=1
runtime_boundary_abuse_case_fixture_count=8
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
security_hardening_changed=0
public_readiness_changed=0
runtime_authority_granted=0
estimate_adjustment_required=0
```

This alignment makes that review easier to find from public entry points without changing the estimate decision.

Alignment Decision

```
completion_estimate_review_readme_status_alignment_present=1
readme_links_latest_completion_estimate_review=1
status_index_links_alignment_record=1
foundation_index_links_alignment_record=1
project_notes_link_alignment_record=1
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
security_hardening_changed=0
public_readiness_changed=0
runtime_authority_granted=0
estimate_adjustment_required=0
completion_estimate_changed=0
```

The README now points readers from the current status table to the latest completion-estimate hold review:

```
docs/status/COMPLETION_ESTIMATE_REVIEW_AFTER_RUNTIME_BOUNDARY_ABUSE_CASE_FIXTURES.md
```

Current estimates remain unchanged:

```
Overall Latticra system: 39%
Latticra Seal / local evidence layer: 34%
Security-hardening implementation: 7%
Public product readiness: 8%
```

Validation

This alignment is guarded by:

```
sh scripts/test-completion-estimate-review-readme-status-alignment.sh
```

Expected output:

```
completion_estimate_review_readme_status_alignment: ok
```

Boundary

This is a public-entry/status alignment only.

It does not change capability posture, implementation behavior, security-hardening posture, public readiness, product readiness, completion estimates, or runtime authority.

The next valid work remains small guarded report/status alignment only when drift appears, unless a separate future slice changes capability posture with a contract, implementation plan, tests, and explicit non-claim review.

docs/status/COMPLETION_PERCENTAGE_REVIEW.md

Source title: Completion Percentage Review

Lines: 40 | Words: 158 | SHA-256: 430782e737a713d4cf511d5d231bcc757de37a21b3fc10fdbd8e68000747e7b6

Completion Percentage Review

Status: planning-estimate review

This record reviews completion estimates after the recent report, diagnostic, audit, README, foundation-index, project-notes, and status-consistency slices.

Review basis:

```
Lat pipeline diagnostic integration refinement
Lat pipeline diagnostic main test audit
Runtime boundary domain matrix report integration
Runtime boundary domain matrix main test audit
Project notes upcoming work alignment
Current direction project notes alignment
Project notes index alignment
Current status and announcement consistency review
```

Estimate adjustments:

```
Overall Latticra system: 34% -&gt; 35%
Foundation documents and contracts: 88% -&gt; 89%
Public documentation posture: 82% -&gt; 83%
Strategy/status/funding framework: 56% -&gt; 57%
Lat / Latticra Programming Language: 24% -&gt; 25%
Runtime / operating-system-universe direction: 16% -&gt; 17%
```

Unchanged areas:

```
L-UI parser / AST / string foundation: 86%
LIR / Intermediate Representation: 22%
C/C++ foundation direction: 14%
Constrained C++ authority layer: 4%
```

Boundary: planning estimates only. These are not release promises, production-readiness claims, security guarantees, or operating-system completeness claims.

docs/status/CPP_AUTHORITY_EXPANSION_CONTRACT_REVIEW.md

Source title: C++ Authority Expansion Contract Review

Lines: 36 | Words: 113 | SHA-256: 094ccd9bab58af90f1cddb429038733bc590a2df08e45c490c30c22350e51a0f

C++ Authority Expansion Contract Review

Status: no-expansion-contract review

This record reviews whether a new C++ authority expansion contract is needed after the latest no-effect reporting, documentation, and status alignment slices.

Reviewed files:

```
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md
docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_CONTRACT.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION_PLAN.md
include/latticra/cpp/authority.hpp
src/cpp/authority.cpp
tests/cpp_authority_layer_invariants.cpp
```

Finding:

```
No new C++ authority behavior has been proposed.
```

Decision:

```
Do not create a C++ authority expansion contract in this slice.
Keep the constrained C++ authority layer no-effect, metadata-only, fixed-capacity, and
denied-by-default.
Only create a future expansion contract if a specific new authority behavior is proposed with
explicit effect boundaries, failure behavior, tests, and non-claims.
```

Boundary: review/status only. No C++ authority behavior is changed.

docs/status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md

Source title: C++ Authority Implementation Review Status

Lines: 35 | Words: 63 | SHA-256: 6928bd0c2f233c2698d5d3f50ec2fc773ea769b0f7ac7ab2ad0f54f67c619754

C++ Authority Implementation Review Status

Status: review record

This status record tracks the constrained C++ authority implementation review.

Primary review record:

```
docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md
```

Reviewed surfaces:

```
include/latticra/cpp/authority.hpp
src/cpp/authority.cpp
tests/cpp_authority_layer_invariants.cpp
scripts/test-cpp-authority-layer.sh
docs/CONSTRAINED_CPP_AUTHORITY_LAYER_IMPLEMENTATION.md
```

Review result:

```
The constrained C++ authority implementation remains no-effect, metadata-only, fixed-capacity,
and denied-by-default.
```

Validation:

```
sh scripts/test-cpp-authority-layer.sh
```

Boundary: documentation/status review only. No implementation behavior is changed.

docs/status/CRYPTOGRAPHIC_ASSURANCE_KEY_MANAGEMENT_BASELINE_STATUS.md

Source title: Latticra Cryptographic Assurance and Key Management Baseline Status

Lines: 72 | Words: 153 | SHA-256: aadd325277a3deb8d532ca7df396ff5e0a1bc1d6498e186e9c891c7c16549a38

Latticra Cryptographic Assurance and Key Management Baseline Status

Status: status record for cryptographic assurance and key management baseline Date: 2026-05-26

Scope

This record tracks the cryptographic assurance and key-management baseline for cryptographic module boundaries, FIPS/CMVP claim gates, algorithm and parameter inventory, key lifecycle, key storage, key destruction, randomness, self-tests, sensitive-data handling, post-quantum migration planning, Seal metadata, signing authority, and cryptographic non-claims.

It does not implement production cryptography, signing authority, key storage, key generation, entropy collection, random-bit generation, FIPS validation, CMVP submission, CAVP testing, post-quantum migration, compliance, or runtime authority.

Current fields

```
cryptographic_assurance_key_management_baseline_present=1
cryptographic_assurance_key_management_status_present=1
cryptographic_assurance_key_management_guard_present=1
seal_crypto_graduation_gate_present=1
seal_crypto_graduation_gate_guard_present=1
fips_140_3_boundary_required_before_production_crypto=1
cmvp_validation_path_required_before_fips_claim=1
validated_module_claim_requires_certificate=1
algorithm_parameter_inventory_required=1
approved_algorithm_transition_review_required=1
known_insecure_crypto_forbidden=1
ed25519_rfc8032_test_vector_required=1
authority_neutral_crypto_graduation_required=1
fips_186_5_signature_standard_tracked=1
fips_180_4_digest_standard_tracked=1
fips_204_ml_dsa_planning_tracked=1
fips_205_slh_dsa_planning_tracked=1
key_lifecycle_contract_required=1
key_inventory_required=1
key_metadata_protection_required=1
key_storage_contract_required=1
key_zeroization_contract_required=1
key_compromise_response_required=1
randomness_entropy_source_contract_required=1
drbg_review_required=1
self_test_failure_behavior_required=1
side_channel_sensitive_data_review_required=1
post_quantum_migration_inventory_required=1
cnsa_2_pq_planning_tracked=1
non_fips_disclosure_required_if_not_validated=1
seal_crypto_metadata_only_current=1
implementation_behavior_changed=0
production_crypto_added=0
signing_authority_granted=0
key_storage_added=0
key_generation_added=0
entropy_collection_added=0
fips_validation_claimed=0
cmvp_submission_performed=0
cavp_testing_claimed=0
post_quantum_migration_performed=0
production_crypto_claim_allowed=0
fips_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-cryptographic-assurance-key-management-baseline.sh
```

Expected output:

```
cryptographic_assurance_key_management_baseline: ok
```

docs/status/CURRENT_ESTIMATE_MATHEMATICAL_REBASE_2026_05_26.md

Source title: Current Estimate Mathematical Rebase - 2026-05-26

Lines: 87 | Words: 500 | SHA-256: 46e79ec3a03726b8988e2eeefbcc8fb29cfb423117c812fe8d0649497524bbb2

Current Estimate Mathematical Rebase - 2026-05-26

Status: mathematical planning-estimate rebase Date: 2026-05-26 CDT Scope: present public estimate table after the recent Seal status chain, Ubuntu local deb lane, macOS reset/uninstall contract lane, Nadia Stage-43 prompt-evaluation release-receipt contract chain, Lat/LIR evidence propagation, kernel lifecycle scheduler-credit and scheduler-selection-ready evidence, vulnerability-management release-gate baseline coverage, and current status/public-entry alignment. This record does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production readiness, public-readiness promotion, security hardening, or runtime authority.

Purpose

This record replaces the older rough table with a closer mathematical planning estimate.

The estimate remains conservative because much of Latticra is contract/status/evidence work, not production runtime behavior.

Model

The overall estimate is computed as a weighted average of the current lane estimates.

Weights:

```
Latticra Seal / local evidence layer: 10
Latticra Panel / local control surface: 8
Nadia offline AI foundation: 6
L-UI parser / AST / string foundation: 6
Foundation documents and contracts: 10
Public documentation posture: 8
Strategy/status/funding framework: 5
Lat / Latticra Programming Language: 8
LIR / Intermediate Representation: 6
C/C++ foundation direction: 6
Constrained C++ authority layer: 4
Nucleus real task execution: 5
Runtime / operating-system-universe direction: 8
Security-hardening implementation: 5
Public product readiness: 5
weight_total=100
```

Weighted calculation:

```
weighted_sum=4473
weight_total=100
overall_estimate=round(4473 / 100)=45
```

Rebased Estimate Table

```
| Area | Previous live estimate | Rebased estimate |
| --- | ---: | ---: |
| Overall Latticra system | 39% | 45% |
| Latticra Seal / local evidence layer | 34% | 39% |
| Latticra Panel / local control surface | 28% | 31% |
| Nadia offline AI foundation | 70% | 75% |
| L-UI parser / AST / string foundation | 87% | 87% |
| Foundation documents and contracts | 92% | 94% |
| Public documentation posture | 88% | 91% |
| Strategy/status/funding framework | 60% | 63% |
| Lat / Latticra Programming Language | 25% | 27% |
| LIR / Intermediate Representation | 22% | 24% |
| C/C++ foundation direction | 18% | 22% |
| Constrained C++ authority layer | 4% | 5% |
| Nucleus real task execution | 11% | 12% |
| Runtime / operating-system-universe direction | 19% | 26% |
| Security-hardening implementation | 7% | 10% |
| Public product readiness | 8% | 10% |
```

Boundary

These percentages are planning estimates only.

They are not release promises, production-readiness metrics, security guarantees, Fedora approval claims, Ubuntu archive-readiness claims, macOS production-readiness claims, runtime-enforcement claims, operating-system completeness claims, or authority grants.

The rebase raises estimates only where recent guarded evidence increased documented contract/status coverage. It does not claim production enforcement, real runtime execution, effect execution, host mutation, network behavior, signing, verification, model execution, or product readiness beyond the values above.

Validation

This rebase is guarded by:

```
sh scripts/test-current-estimate-table-source-alignment.sh
```

Expected output:

```
current_estimate_table_source_alignment: ok
```

docs/status/CURRENT_ESTIMATE_REFRESH_2026_05_24.md

Source title: Current Estimate Refresh - 2026-05-24

Lines: 67 | Words: 401 | SHA-256: e795044805d1ab29542af0b7366fb3bc6567eebd3321e0893ba6deff87eb71de

Current Estimate Refresh - 2026-05-24

Status: planning-estimate refresh Last updated: 2026-05-24 CDT Scope: current estimates after Latticra Panel, Latticra Seal, handbook/docs, local evidence, C source organization, and Fedora/local-install evidence work.

Review basis

- Latticra Panel local control surface and visible app direction
- Latticra Seal local report-only verification layer
- Manifest/hash baseline and policy-denial evidence
- Fedora local install and disposable VM evidence posture
- Latticra Seal handbook and system-substrate documentation expansion
- C/C++ foundation direction and source-only field-engine organization
- Current public-status alignment requirement

Estimate adjustments

- Overall Latticra system: 36% -> 39%
- Foundation documents and contracts: 90% -> 92%
- Public documentation posture: 84% -> 88%
- Strategy/status/funding framework: 58% -> 60%
- C/C++ foundation direction: 14% -> 18%
- Runtime / operating-system-universe direction: 17% -> 19%
- Security-hardening implementation: 5% -> 7%
- Public product readiness: 5% -> 8%

Newly separated estimate lanes

- Latticra Seal / local evidence layer: 34%
- Latticra Panel / local control surface: 28%

Unchanged areas

- L-UI parser / AST / string foundation: 87%
- Lat / Latticra Programming Language: 25%
- LIR / Intermediate Representation: 22%
- Constrained C++ authority layer: 4%
- Nucleus real task execution: 11%

Current estimate table

Area	Estimated completion
Overall Latticra system	39%
Latticra Seal / local evidence layer	34%
Latticra Panel / local control surface	28%
L-UI parser / AST / string foundation	87%
Foundation documents and contracts	92%
Public documentation posture	88%
Strategy/status/funding framework	60%
Lat / Latticra Programming Language	25%
LIR / Intermediate Representation	22%
C/C++ foundation direction	18%
Constrained C++ authority layer	4%
Nucleus real task execution	11%
Runtime / operating-system-universe direction	19%
Security-hardening implementation	7%
Public product readiness	8%

These percentages are planning estimates only. They are not release promises, production-readiness metrics, security guarantees, Fedora approval claims, runtime-enforcement claims, or operating-system completeness claims.

Boundary

This refresh records planning estimates only.

It does not claim that Latticra is a kernel, bootable operating system, production installer, production security product, certified Fedora distribution, runtime enforcement layer, malware-prevention system, ransomware-prevention system, or daily-driver platform.

docs/status/CURRENT_ESTIMATE_TABLE_SOURCE_ALIGNMENT.md

Source title: Current Estimate Table Source Alignment

Lines: 80 | Words: 343 | SHA-256: 9ce2f8abf9a5343a9fff76a4d360c35c5327abbbd03a9f0d4ff6ba255a4a7d34

Current Estimate Table Source Alignment

Status: current estimate table source alignment Date: 2026-05-26 CDT Scope: README, root status, detailed current status, status index, foundation index, project-notes alignment, and the current mathematical rebase record for the public estimate table source. This record does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, public-readiness promotion, production protection, security hardening, or runtime authority.

Purpose

The public README now includes an explicit estimate-source row and a current estimate table mirrored from STATUS.md and docs/status/CURRENT_STATUS.md.

This record guards that source relationship so the public entry point does not imply that an older estimate refresh or a later no-change hold review is the only source for the live table. The mathematical rebase record is the estimating record for the current values; this source-alignment record mirrors that table into the public entry points.

Current Source Fields

```
current_estimate_table_source_alignment_present=1
readme_estimate_source_row_present=1
readme_current_public_estimate_table_present=1
root_status_current_estimate_table_present=1
current_status_current_estimate_table_present=1
status_index_links_current_estimate_table_source_alignment=1
foundation_index_links_current_estimate_table_source_alignment=1
project_notes_link_current_estimate_table_source_alignment=1
latest_estimate_refresh_record_linked=1
latest_runtime_boundary_hold_review_linked=1
latest_mathematical_rebase_record_linked=1
source_alignment_estimate_changed=0
mathematical_rebase_estimate_changed=1
estimate_adjustment_required=0
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
security_hardening_changed=0
public_readiness_changed=0
runtime_authority_granted=0
```

Current Public Estimate Table

```
Overall Latticra system: 45%
Latticra Seal / local evidence layer: 39%
Latticra Panel / local control surface: 31%
Nadia offline AI foundation: 75%
L-UI parser / AST / string foundation: 87%
Foundation documents and contracts: 94%
Public documentation posture: 91%
Strategy/status/funding framework: 63%
Lat / Latticra Programming Language: 27%
LIR / Intermediate Representation: 24%
C/C++ foundation direction: 22%
Constrained C++ authority layer: 5%
Nucleus real task execution: 12%
Runtime / operating-system-universe direction: 26%
Security-hardening implementation: 10%
Public product readiness: 10%
```

Boundary

These percentages are planning estimates only.

They are not release promises, production-readiness metrics, security guarantees, Fedora approval claims, runtime-enforcement claims, operating-system completeness claims, or authority grants.

This source alignment mirrors the current mathematical rebase into the reader-facing estimate sources. It does not change implementation behavior, runtime behavior, security hardening, public readiness, product readiness, or authority.

Validation

This alignment is guarded by:

```
sh scripts/test-current-estimate-table-source-alignment.sh
```

Expected output:

```
current_estimate_table_source_alignment: ok
```

docs/status/CYBER_INCIDENT_REPORTING_RESPONSE_BASELINE_STATUS.md

Source title: Latticra Cyber Incident Reporting and Response Baseline Status

Lines: 59 | Words: 127 | SHA-256: c46a3050be7283de8438a1598e8b9b542fc9eb9240f62e99c5aaf70f0edd2d43

Latticra Cyber Incident Reporting and Response Baseline Status

Status: status record for cyber incident reporting and response baseline Date: 2026-05-26

Scope

This record tracks the cyber incident reporting and response baseline for vulnerability reports, suspected compromise triage, ransomware and data-extortion reporting paths, evidence preservation, communications routing, public non-claims, and future incident-response automation gates.

It does not implement monitoring, detection, containment, forensic acquisition, recovery, federal reporting, customer notification, breach notification, law-enforcement contact, ransomware recovery, or incident-response services.

Current fields

```
cyber_incident_reporting_response_baseline_present=1
cyber_incident_reporting_response_status_present=1
cyber_incident_reporting_response_guard_present=1
cisa_reporting_channel_documented=1
fbi_cyber_reporting_channel_documented=1
ic3_reporting_channel_documented=1
stopransomware_joint_guidance_tracked=1
fbi_ic3_annual_report_refresh_required=1
incident_classification_required=1
authorized_testing_boundary_required=1
evidence_preservation_required=1
volatile_evidence_preservation_required=1
chain_of_custody_required_before_claim=1
out_of_band_communications_required_for_compromise=1
ransomware_data_extortion_response_checklist_required=1
legal_regulatory_notification_review_required=1
law_enforcement_contact_path_required=1
internal_external_notification_plan_required=1
operator_confirmation_metadata_only_required=1
implementation_behavior_changed=0
monitoring_added=0
detection_added=0
containment_added=0
forensic_collection_added=0
federal_reporting_performed=0
law_enforcement_contact_performed=0
customer_notification_performed=0
breach_notification_authority_claimed=0
incident_response_service_claimed=0
ransomware_recovery_capability_claimed=0
production_monitoring_claimed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-cyber-incident-reporting-response-baseline.sh
```

Expected output:

```
cyber_incident_reporting_response_baseline: ok
```

docs/status/DEBIAN_ECOSYSTEM_INTEGRATION_STATUS.md

Source title: Debian Ecosystem Integration Status

Lines: 160 | Words: 465 | SHA-256: 9599b5d8f18b5b318d1842872f85c84b550384a30ad3510203b752212ab16bc1

Debian Ecosystem Integration Status

Status: Debian integration status record Date: 2026-05-26

Summary

Latticra now has a Debian-facing local deb packaging draft for the no-effect CLI payload.

This is an ecosystem integration checkpoint, not a production readiness claim.

Current Evidence

```
debian_local_deb_draft_present=1
debian_static_deb_validation_present=1
deb_artifact_created=0
deb_installed_on_host=0
debian_freebsd_openbsd_source_archive_contract_present=1
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
temporary_debian_source_input_staged=1
temporary_debian_orig_archive_staged=1
temporary_debian_debian_dir_overlay_staged=1
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
debian_build_allowed=0
debian_clean_build_environment_documented=1
debian_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
environment_transcript_present=0
debian_artifact_name_pattern_recorded=1
debian_source_package_name=latticra_0.0.0-1local1.dsc
debian_binary_package_name_pattern=latticra_0.0.0-1local1_${DEB_HOST_ARCH}.deb
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
debian_payload_expected_bin=usr/bin/latticra
debian_payload_expected_doc=usr/share/doc/latticra/README.md
debian_payload_inspection_run=0
package_payload_inspection_run=0
package_payload_accepted=0
debian_install_remove_transcript_required=1
debian_install_remove_transcript_present=0
debian_package_install_run=0
debian_package_remove_run=0
remove_on_host_run=0
publication_non_claim_review_present=1
debian_publication_non_claim_review_present=1
debian_package_publication_claimed=0
debian_source_upload_run=0
debian_mentors_upload_run=0
debian_archive_upload_run=0
debian_debsign_run=0
debian_dput_run=0
platform_build_evidence_accepted=0
debian_validation_promotion_blocked=1
debian_platform_build_evidence_accepted=0
package_validation_result_promoted=0
debian_validation_result_promoted=0
source_archive_policy_recorded=1
source_archive_created=0
source_archive_sha256_recorded=0
source_archive_accepted_for_build=0
dpkg_buildpackage_run_required=0
debuild_run_required=0
lintian_run_required=0
dpkg_buildpackage_run=0
debuild_run=0
lintian_run=0
package_artifact_created=0
package_artifact_sha256_recorded=0
install_on_host_run=0
lintian_transcript_present=0
debian_archive_ready=0
debian_mentors_upload_claimed=0
debian_sponsorship_claimed=0
debian_ftp_master_acceptance_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Guarded Files

```
docs/DEBIAN_LOCAL_DEB_STATIC_VALIDATION.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
packaging/debian/README.md
packaging/debian/debian/control
packaging/debian/debian/rules
packaging/debian/debian/changelog
packaging/debian/debian/copyright
packaging/debian/debian/install
packaging/debian/debian/source/format
scripts/test-debian-local-deb-static-validation.sh
scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
.github/workflows/debian-local-deb-static-validation.yml
.github/workflows/debian-freebsd-openbsd-source-archive-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-gate-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-environment-contract.yml
.github/workflows/debian-freebsd-openbsd-package-artifact-naming-contract.yml
.github/workflows/debian-freebsd-openbsd-package-payload-inspection-contract.yml
.github/workflows/debian-freebsd-openbsd-package-install-remove-transcript-contract.yml
.github/workflows/debian-freebsd-openbsd-package-publication-non-claim-review-contract.yml
.github/workflows/debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.ym
l
```

Current Boundary

The Debian lane does not publish a package, upload to mentors.debian.net, submit to Debian, claim Debian archive readiness, claim Debian sponsorship, claim ftp-master acceptance, install a root service, change init/systemd state, change the kernel, add a privileged helper, grant network authority, or claim production readiness.

The local deb metadata now records Apache-2.0 for existing-code defaults, AGPL-3.0-or-later for the no-effect CLI payload, and CC-BY-4.0 for documentation while source archive and notice obligations remain under review.

The source archive contract records the expected latticra_0.0.0.orig.tar.gz boundary while keeping archive creation, checksum acceptance, source package creation, and build transcript promotion blocked.

The package-build gate is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md. It keeps package_build_gate_state=closed-no-effect, debian_build_allowed=0, dpkg_buildpackage_run=0, debuild_run=0, and lintian_run=0 until source, checksum, license, notice, clean build environment, operator authorization, payload inspection, install/remove transcript, and publication non-claim evidence exists.

The package-build environment contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md. It documents the Debian clean build environment requirement while keeping debian_build_environment_provisioned=0, explicit_operator_build_authorization=0, environment_transcript_present=0, and all Debian package build commands disabled.

The package artifact naming contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md. It records future Debian source and binary artifact names while keeping package_artifact_created=0, repository_package_artifact_write_allowed=0, and package_artifact_sha256_recorded=0.

The package payload inspection contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md. It records the expected Debian payload paths while keeping debian_payload_inspection_run=0, package_payload_accepted=0, and package_artifact_created=0.

The package install/remove transcript contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove transcript requirements while keeping debian_package_install_run=0, debian_package_remove_run=0, and install_on_host_run=0.

The package publication non-claim review contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records local-only Debian publication non-claims while keeping debian_package_publication_claimed=0, debian_dput_run=0, and package validation promotion blocked.

The package validation promotion blocker matrix is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md. It ties Debian source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping debian_platform_build_evidence_accepted=0 and debian_validation_result_promoted=0.

Next Recommended Lane

```
Add a Debian package build-evidence intake denial contract before any Debian build lane can
open.
```

docs/status/FEDORA_DISPOSABLE_VM_EFFECT_GATE_CLASSIFIER_STATUS.md

Source title: Fedora Disposable VM Effect Gate Classifier Status

Lines: 112 | Words: 267 | SHA-256: ba8ea63576c3332638748e6f5df12a51deae94719e94bf7e754289d9d003ad2b

Fedora Disposable VM Effect Gate Classifier Status

Status: status alignment Date: 2026-05-21 Scope: public status record after the Fedora VM gate classifier landed on main.

Summary

Latticra now has the Fedora evidence contract, the Fedora disposable VM effect gate, and a no-effect classifier for that gate.

The classifier reports:

```
disposable_vm_effect_gate_status=eligible
disposable_vm_effect_gate_status=blocked
disposable_vm_effect_gate_status=invalid
```

This is classifier evidence only.

It does not make Latticra host-install ready, production ready, Fedora-approved, or Fedora distribution ready.

Evidence recorded

```
Fedora disposable VM effect gate classifier
source=PR #221
evidence_contract_present=1
vm_effect_gate_present=1
vm_gate_classifier_present=1
vm_gate_classifier_guard_present=1
vm_gate_classifier_docs_guard_present=1
eligible_state_defined=1
blocked_state_defined=1
invalid_state_defined=1
eligible_test_present=1
blocked_target_tests_present=1
blocked_package_tests_present=1
blocked_prior_evidence_tests_present=1
invalid_input_test_present=1
classifier_evidence_level=8
```

Current readiness classification

```
manual_host_dry_run_transcript_contract_present=1
disposable_vm_effect_gate_present=1
disposable_vm_effect_gate_classifier_present=1
disposable_vm_effect_gate_status=blocked-pending-vm-validation-lane
disposable_vm_effect_eligible=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Boundary statement

A synthetic eligible unit test proves classifier logic only.

It is not a completed Fedora VM validation record.

The project still needs a dedicated disposable Fedora VM local RPM validation lane before install-ready README wording is justified.

Guard validation

The classifier is guarded by:

```
sh scripts/test-fedora-disposable-vm-effect-gate-classifier.sh
sh scripts/test-fedora-disposable-vm-effect-gate-classifier-docs.sh
```

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-gate-classifier-status-alignment.sh
```

Expected output:

```
fedora_disposable_vm_effect_gate_classifier: ok
fedora_disposable_vm_effect_gate_classifier_docs: ok
fedora_vm_gate_classifier_status_alignment: ok
```

Next recommended Fedora lane

```
Add disposable Fedora VM local RPM validation lane
```

That lane should remain restricted to a disposable Fedora VM target with snapshot and recovery evidence.

README overhaul hold

The root README should not claim install readiness yet.

The README overhaul should wait until real validation evidence exists for:

```
manual_host_dry_run_transcript_present=1
live_host_validation_completed=1
host_install_ready=1
```

Non-claims

This status record is not production readiness, Fedora approval, or Fedora distribution readiness.

docs/status/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_EVIDENCE_STATUS.md

Source title: Fedora Disposable VM Local RPM Validation Evidence Status

Lines: 130 | Words: 386 | SHA-256: 3a503bd9991577fdb16a75803f10b63cb193818216e21401f11ea44970698799

Fedora Disposable VM Local RPM Validation Evidence Status

Status: evidence status alignment Date: 2026-05-21 Scope: public status record after a disposable Fedora VM local RPM validation transcript reached the expected validation report.

Summary

A disposable Fedora VM local RPM validation run completed successfully against the packaging/test-path fix from PR #226.

The validation run built the local noarch Latticra RPM, ran the build-time test suite, installed the package into the disposable Fedora VM, verified package state, removed the package, verified post-removal absence, and emitted the expected deterministic validation report.

This is disposable Fedora VM local RPM validation evidence.

It is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, or a production installer claim.

Evidence source

```
source=operator disposable Fedora VM transcript
repo_branch=pr-226
followup_fix_pr=PR #226
validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
```

The transcript records:

```
Building target platforms: noarch
Building for target noarch
state_lattice_invariants: ok
system_bootstrap: ok
kernel: ok
kernel_lifecycle: ok
Wrote: /tmp/.../rpmbuild/RPMS/noarch/latticra-0.0.0-0.1.local.fc44.noarch.rpm
Updating / installing...
latticra-0.0.0-0.1.local.fc44
```

Validation report recorded

```
FEDORA DISPOSABLE VM LOCAL RPM VALIDATION LANE
validation_status=ok
package_name=latticra
package_version=0.0.0
package_version_recorded=1
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
vm_rpmdb_mutated=1
vm_filesystem_mutated=1
install_validation_performed=1
removal_validation_performed=1
post_removal_absence_verified=1
live_host_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
evidence_level=9
fedora_disposable_vm_local_rpm_validation_lane: ok
```

Current readiness classification

```
disposable_vm_local_rpm_validation_lane_present=1
disposable_vm_validation_transcript_present=1
disposable_vm_validation_completed=1
live_host_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Boundary statement

This evidence is limited to a disposable Fedora VM local RPM validation path.

The current package payload remains documentation-only:

```
/usr/share/doc/latticra/README.md
```

This evidence does not add or validate:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
```

It does not validate daily-driver installation, immutable Fedora installation, production host installation, service activation, boot integration, kernel module loading, SELinux policy changes, network operations, update safety, rollback safety beyond package removal absence, Fedora QA approval, Fedora distribution readiness, or production installer readiness.

Guard validation

The evidence status alignment is guarded by:

```
sh scripts/test-fedora-disposable-vm-local-rpm-validation-evidence-status.sh
```

Expected output:

```
fedora_disposable_vm_local_rpm_validation_evidence_status: ok
```

Next recommended Fedora lane

```
Add README install-readiness wording limited to disposable Fedora VM local RPM validation
```

That wording must remain narrow and should not claim production, Fedora distribution, immutable Fedora, daily-driver, security, recovery, or OS-replacement readiness.

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/status/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_STATUS.md

Source title: Fedora Disposable VM Local RPM Validation Status

Lines: 126 | Words: 341 | SHA-256: 897b76013a55f8e0d837de6a0c053829c545aa84df7151537971422d3796b36b

Fedora Disposable VM Local RPM Validation Status

Status: status alignment Date: 2026-05-21 Scope: public status record after the gated disposable Fedora VM local RPM validation lane landed on main.

Summary

Latticra now has a manually gated disposable Fedora VM local RPM validation lane.

The lane defines and ships:

```
docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_LANE.md
scripts/run-fedora-disposable-vm-local-rpm-validation-lane.sh
scripts/test-fedora-disposable-vm-local-rpm-validation-lane-docs.sh
.github/workflows/fedora-vm-rpm-validation-lane-docs.yml
```

This is lane-readiness evidence only.

It does not prove that a disposable Fedora VM validation run has completed yet.

The validation runner remains manually gated and must not be auto-run by normal CI.

Evidence recorded

```
Fedora disposable VM local RPM validation lane
source=PR #223
validation_lane_documented=1
validation_runner_present=1
validation_lane_docs_guard_present=1
validation_lane_docs_workflow_present=1
runner_manual_only=1
ci_auto_vm_rpm_validation_allowed=0
disposable_vm_target_required=1
daily_driver_block_required=1
production_host_block_required=1
immutable_fedora_block_required=1
clean_snapshot_required=1
recovery_path_required=1
operator_consent_required=1
rpm_payload_listing_required=1
rpm_payload_documentation_only_required=1
unexpected_runtime_surface_absent_required=1
validation_report_schema_present=1
target_evidence_level=9
current_evidence_level=8
evidence_level_9_achieved=0
```

Current readiness classification

```
manual_host_dry_run_transcript_contract_present=1
disposable_vm_effect_gate_present=1
disposable_vm_effect_gate_classifier_present=1
disposable_vm_local_rpm_validation_lane_present=1
disposable_vm_local_rpm_validation_status=blocked-pending-real-vm-run
disposable_vm_validation_transcript_present=0
disposable_vm_validation_completed=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Boundary statement

This status alignment does not run the validation lane.

It does not install or remove an RPM.

It does not mutate a disposable VM, developer host, daily-driver Fedora host, immutable Fedora host, production host, boot entry, kernel module set, service registry, SELinux policy, firmware state, or network configuration.

The docs guard may run in CI.

The RPM validation runner must remain manual and gated by explicit disposable-VM evidence.

Guard validation

The validation lane documentation is guarded by:

```
sh scripts/test-fedora-disposable-vm-local-rpm-validation-lane-docs.sh
```

This status alignment is guarded by:

```
sh scripts/test-fedora-disposable-vm-local-rpm-validation-status-alignment.sh
```

Expected output:

```
fedora_disposable_vm_local_rpm_validation_lane_docs: ok
fedora_disposable_vm_local_rpm_validation_status_alignment: ok
```

Next recommended Fedora lane

```
Capture disposable Fedora VM local RPM validation transcript evidence
```

That lane should record the Fedora VM identity, clean snapshot evidence, recovery path evidence, package version, RPM payload listing, install transcript, removal transcript, post-removal absence proof, and emitted validation report.

README overhaul hold

The root README should not claim install readiness yet.

The README overhaul remains blocked until real validation evidence exists for:

```
disposable_vm_validation_transcript_present=1
disposable_vm_validation_completed=1
live_host_validation_completed=1
host_install_ready=1
```

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, or a production installer claim.

docs/status/FEDORA_DISPOSABLE_VM_RPM_README_ALIGNMENT_STATUS.md

Source title: Fedora Disposable VM RPM README Alignment Status

Lines: 92 | Words: 255 | SHA-256: 77f1fe94e7dfb553f398c4cc32f9d67e2f1e702b78c35f0a38c8adee6263c8ce

Fedora Disposable VM RPM README Alignment Status

Status: README/status alignment Date: 2026-05-21 Scope: root README wording after successful disposable Fedora VM local RPM validation evidence was recorded.

Summary

The root README now records the narrow Fedora install-readiness posture supported by the evidence:

```
disposable_vm_local_rpm_validation_completed=1
host_install_ready=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

This wording is intentionally limited to a disposable Fedora VM local RPM validation path.

It does not claim production readiness, Fedora approval, Fedora distribution readiness, immutable Fedora readiness, daily-driver safety, security capability, update safety, recovery safety, sandboxing, malware prevention, ransomware prevention, or OS-replacement readiness.

Evidence linked

The README links to:

```
docs/status/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_EVIDENCE_STATUS.md
docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_LANE.md
docs/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_TRANSCRIPT_CONTRACT.md
```

README wording boundaries

The README must preserve these boundaries:

```
validated package is documentation-only
validated path is disposable Fedora VM only
validated payload is /usr/share/doc/latticra/README.md
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The README must not claim:

```
production ready
Fedora approved
Fedora distribution ready
daily-driver safe
immutable Fedora ready
bootable OS replacement
kernel runtime ready
security product
malware prevention
ransomware prevention
sandboxing
update safety
recovery safety
```

Guard validation

This README alignment is guarded by:

```
sh scripts/test-fedora-disposable-vm-rpm-readme-alignment.sh
```

Expected output:

```
fedora_disposable_vm_rpm_readme_alignment: ok
```

Next recommended lane

```
Add public announcement wording for disposable Fedora VM RPM validation milestone
```

That announcement must stay evidence-bound and preserve all non-claims.

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/status/FEDORA_HOST_INSTALL_PREFLIGHT_STATUS.md

Source title: Fedora Host Install Preflight Status

Lines: 74 | Words: 196 | SHA-256: eb7242df167e66c3788ccaa9e72a8d9e38c24477838aecd5ed927c01d5e3eab2

Fedora Host Install Preflight Status

Status: merged no-effect Fedora preflight classifier

This record tracks the Fedora host install preflight classifier slice.

Merged PR:

```
#198 Add Fedora host install preflight classifier
```

Merged commit:

```
bbcb42afce009a29b7ba5c0ea8b354a9bca93966
```

Primary files:

```
include/latticra/fedora_host_install_preflight.h
src/fedora_host_install_preflight.c
tests/fedora_host_install_preflight.c
docs/FEDORA_HOST_INSTALL_PREFLIGHT_CLASSIFIER.md
scripts/test-fedora-host-install-preflight.sh
scripts/test-fedora-host-install-preflight-docs.sh
.github/workflows/fedora-host-install-preflight.yml
```

What it adds:

```
Fedora host snapshot classifier
mutable Fedora host label
immutable Fedora future-gated label
local RPM candidate label
network-required denial
missing rpm/dnf denial
missing local RPM denial
missing privilege denial
doc-only runtime-entrypoint denial
no-effect install report fields
```

Report boundary:

```
FEDORA HOST INSTALL PREFLIGHT
classification=ready-local-rpm
host_install_candidate=1
host_install_performed=0
host_mutation_performed=0
network_allowed=0
no_effect=1
evidence_level=1
```

Validation:

```
sh scripts/test-fedora-host-install-preflight.sh
sh scripts/test-fedora-host-install-preflight-docs.sh
```

Boundary: preflight classification only. No host install, dnf execution, rpm execution, rpmbuild execution, rpmlint execution, package artifact creation, host mutation, network opening, service installation, boot entry installation, kernel module installation, Fedora approval claim, or production installer readiness is added.

Recommended next slice:

```
Add Fedora install preflight snapshot capture plan
```

That future slice should define how a caller may safely collect /etc/os-release, rpm, dnf, privilege, and local RPM facts before passing them into the classifier. It should remain read-only and should not install anything.

docs/status/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_STATUS.md

Source title: Fedora Install Preflight Snapshot Capture Status

Lines: 78 | Words: 226 | SHA-256: 171bd8ee0e1f6d1bac7c457d39093f9bf56b39b23a37c9906cfbb5fb174d3e13

Fedora Install Preflight Snapshot Capture Status

Status: merged no-effect Fedora snapshot capture implementation

This record tracks the Fedora install preflight snapshot capture implementation slice.

Merged PR:

```
#201 Implement Fedora install preflight snapshot capture
```

Merged commit:

```
e8a898ee1cab776c90b4c9d4741f5dc4f67e25e5
```

Primary files:

```
include/latticra/fedora_install_preflight_snapshot.h
src/fedora_install_preflight_snapshot.c
tests/fedora_install_preflight_snapshot.c
docs/FEDORA_INSTALL_PREFLIGHT_SNAPSHOT_CAPTURE_IMPLEMENTATION.md
scripts/test-fedora-install-preflight-snapshot-capture.sh
scripts/test-fedora-install-preflight-snapshot-capture-docs.sh
.github/workflows/fedora-install-preflight-snapshot-capture.yml
```

What it adds:

```
bounded os-release text parsing from caller-provided memory
ID and ID_LIKE extraction
immutable Fedora marker propagation
RPM tooling presence propagation
local RPM path and readability propagation
conservative privilege fact propagation
network-required propagation
doc-only package posture propagation
classifier forwarding
deterministic snapshot capture report
```

Report boundary:

```
FEDORA INSTALL PREFLIGHT SNAPSHOT CAPTURE
snapshot_capture_status=captured
classifier_classification=ready-local-rpm
snapshot_forwarded_to_classifier=1
sudo_validation_allowed=0
install_command_allowed=0
package_build_allowed=0
network_allowed=0
host_mutation_performed=0
host_install_performed=0
no_effect=1
evidence_level=2
```

Validation:

```
sh scripts/test-fedora-install-preflight-snapshot-capture.sh
sh scripts/test-fedora-install-preflight-snapshot-capture-docs.sh
```

Boundary: implementation status alignment only. The merged implementation consumes caller-supplied facts and does not probe the live host by itself. No host install, host mutation, network opening, sudo validation, package artifact creation, package build behavior, package install behavior, service operation, boot entry operation, kernel module operation, Fedora approval claim, or production installer readiness is added.

Recommended next slice:

```
Add Fedora live read-only snapshot capture adapter
```

That future slice may introduce an adapter that reads/probes the host in a bounded read-only way and then feeds the existing snapshot capture API. It must still preserve host mutation and install as disabled.

docs/status/FEDORA_INSTALLROOT_RPM_LIFECYCLE_STATUS.md

Source title: Fedora Installroot RPM Lifecycle Status

Lines: 65 | Words: 244 | SHA-256: 5496779568e28756e1c844b81d40ca83c26b6b6a1b4af847abb6bbd39dd6d165

Fedora Installroot RPM Lifecycle Status

Status: status alignment Date: 2026-05-21 Scope: public status record after the Fedora installroot RPM lifecycle lane landed on main.

Summary

Latticra now has a mainline Fedora RPM lifecycle lane that can build the local-only RPM and exercise install/remove behavior inside a temporary RPM installroot.

This is controlled lifecycle evidence, not live host installer readiness.

Evidence recorded

```
Fedora installroot RPM lifecycle lane
source=PR #213
execution_lane_documented=1
execution_guard_present=1
local_rpm_build_path=present
installroot_rpm_database_path=temporary
installroot_filesystem_mutated=1
installroot_rpmdb_mutated=1
live_container_rpmdb_mutated=0
developer_host_mutation_performed=0
boot_operation_performed=0
kernel_operation_performed=0
service_operation_performed=0
policy_operation_performed=0
network_allowed_during_rpm_execution=0
post_removal_absence_verified=1
rollback_planned=1
partial_failure_report_required=1
evidence_level=6
```

Current readiness classification

```
controlled_installroot_lifecycle_ready=1
manual_host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

What this means

The project is ready to validate the RPM lifecycle in a bounded Fedora installroot context.

The project is not yet ready to install on a user workstation or make production installer claims.

Next recommended Fedora lane

```
Add manual host-install release-candidate checklist
```

That checklist should define the exact human review and evidence required before any live Fedora host install attempt.

Non-claims

This status record does not install Latticra on a developer host, daily-driver Fedora system, immutable Fedora host, VM host, production host, live Fedora container RPM database, or user workstation.

It does not publish artifacts, submit to Fedora, claim Fedora approval, claim production installer readiness, validate sudo on a user host, start services, change boot entries, load kernel modules, modify SELinux policy, open runtime network access, or assert that Latticra is ready as a system installer.

docs/status/FEDORA_LIVE_READONLY_SNAPSHOT_ADAPTER_STATUS.md

Source title: Fedora Live Read-Only Snapshot Adapter Status

Lines: 84 | Words: 234 | SHA-256: f18c399f4d53c65dfe15177a862112b0cbac150a2e5333226ce62e7a46f5b719

Fedora Live Read-Only Snapshot Adapter Status

Status: merged live read-only Fedora adapter implementation

This record tracks the Fedora live read-only snapshot adapter implementation slice.

Merged PR:

```
#203 Add Fedora live read-only snapshot adapter
```

Merged commit:

```
df2dba248eb1fcd6d37d9fa2f9dfae830b08a468
```

Primary files:

```
include/latticra/fedora_live_snapshot_adapter.h
src/fedora_live_snapshot_adapter.c
tests/fedora_live_snapshot_adapter.c
docs/FEDORA_LIVE_READONLY_SNAPSHOT_ADAPTER.md
scripts/test-fedora-live-snapshot-adapter.sh
scripts/test-fedora-live-snapshot-adapter-docs.sh
.github/workflows/fedora-live-snapshot-adapter.yml
```

Local Fedora live validation:

```
sh scripts/test-fedora-live-snapshot-adapter.sh
sh scripts/test-fedora-live-snapshot-adapter-docs.sh
fedora_live_snapshot_adapter: ok
fedora_live_snapshot_adapter_docs: ok
```

What it adds:

```
live read-only os-release path capture
live read-only PATH command presence probing
live read-only local RPM readability probing
live read-only immutable Fedora marker probing
live read-only effective-root status recording
forwarding into Fedora snapshot capture
forwarding into Fedora host install preflight classifier
deterministic live adapter report
```

Report boundary:

```
FEDORA LIVE READ-ONLY SNAPSHOT ADAPTER
live_probe_performed=1
snapshot_forwarded_to_classifier=1
sudo_validation_allowed=0
install_command_allowed=0
package_build_allowed=0
network_allowed=0
host_mutation_performed=0
host_install_performed=0
no_effect=1
```

```
evidence_level=3
```

Validation:

```
sh scripts/test-fedora-live-snapshot-adapter.sh
sh scripts/test-fedora-live-snapshot-adapter-docs.sh
```

Boundary: live read-only host fact capture only. No install behavior, sudo validation, package build behavior, package installation, host mutation, network opening, service operation, boot entry operation, kernel module operation, Fedora approval claim, or production installer readiness is added.

Recommended next slice:

```
Add Fedora local RPM install mutation gate contract
```

That future slice should define the exact hard gate before any install mutation can be represented. It should require explicit operator confirmation, mutable Fedora classification, local RPM presence, privilege availability, local-only mode, and rollback/removal plan evidence before install execution can be implemented.

docs/status/FEDORA_MANUAL_HOST_RC_CHECKLIST_STATUS.md

Source title: Fedora Manual Host RC Checklist Status

Lines: 133 | Words: 415 | SHA-256: 345718e2851c08f71ba7c9b4f7b01c1bf8bf28bdf0e11753204e5e04dc2d7812

Fedora Manual Host RC Checklist Status

Status: status alignment Date: 2026-05-21 Scope: public status record after the Fedora manual host release-candidate checklist landed on main.

Summary

Latticra now has a Fedora manual host release-candidate checklist.

The checklist defines the evidence required before any future manual Fedora host validation milestone may be called a release candidate.

This is checklist and status evidence only.

It does not make Latticra ready for live host installation, production installation, Fedora distribution, or Fedora approval.

Fedora terminology boundary

In this record, manual host RC means an internal Latticra manual-host validation candidate.

It does not mean a Fedora release candidate compose, Fedora QA validation event, Fedora Go/No-Go result, Fedora acceptance, Fedora package approval, or Fedora distribution readiness.

Evidence recorded

```
Fedora manual host RC checklist
source=PR #215
checklist_documented=1
checklist_guard_present=1
controlled_installroot_lifecycle_ready=1
manual_host_rc_status=blocked
manual_host_release_candidate_ready=0
live_host_validation_completed=0
host_change_performed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
fedora_candidate_compose_claimed=0
fedora_go_no_go_claimed=0
checklist_target_evidence_level=7
current_evidence_level=6
evidence_level_7_achieved=0
```

Required evidence before candidate state

The checklist keeps the manual-host candidate state blocked until the project records all of the following:

```
target_is_disposable_fedora_vm=1
target_is_daily_driver=0
target_is_production_host=0
target_is_immutable_fedora=0
target_has_clean_snapshot=1
target_has_recovery_path=1
operator_consent_recorded=1
local_rpm_built_from_current_tree=1
rpm_payload_listing_recorded=1
rpm_payload_is_documentation_only=1
unexpected_runtime_surface_absent=1
installroot_lifecycle_evidence_present=1
post_removal_absence_evidence_present=1
host_preflight_classification=ready-local-rpm
rpm_gate_status=allowed
removal_rollback_status=removal-ready
```

Current readiness classification

```
manual_host_rc_checklist_present=1
manual_host_rc_guard_present=1
manual_host_rc_status=blocked
manual_host_release_candidate_ready=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Guard validation

The checklist is currently guarded by:

```
sh scripts/test-fedora-manual-host-rc-checklist.sh
```

This status alignment is guarded by:

```
sh scripts/test-fedora-manual-host-rc-checklist-status-alignment.sh
```

Expected output:

```
fedora_manual_host_rc_checklist: ok
fedora_manual_host_rc_checklist_status_alignment: ok
```

Honest public wording

After this status alignment lands, the project may say:

```
Latticra has controlled Fedora installroot RPM lifecycle evidence.
Latticra has a manual Fedora host release-candidate checklist.
Latticra has status alignment that keeps manual host RC readiness blocked until explicit
evidence is recorded.
```

The project must not say:

```
Latticra has completed live Fedora host validation.
Latticra is safe for daily-driver installation.
Latticra is production installer ready.
Latticra is Fedora-approved.
Latticra is ready for Fedora distribution.
```

Next recommended Fedora lane

```
Add no-effect Fedora manual host RC decision classifier
```

That classifier should consume explicit checklist evidence and report blocked or candidate without running sudo, rpm, dnf, service operations, boot operations, kernel operations, policy operations, network operations, or host mutation.

Non-claims

This status record does not perform host changes, install Latticra on a developer host, validate a daily-driver Fedora system, publish artifacts, submit to Fedora, claim Fedora approval, claim Fedora distribution readiness, claim production installer readiness, start services, change boot entries, load kernel modules, change host policy, or open runtime network access.

docs/status/FEDORA_MANUAL_HOST_RC_DECISION_CLASSIFIER_STATUS.md

Source title: Fedora Manual Host RC Decision Classifier Status

Lines: 137 | Words: 379 | SHA-256: 2646b798eb6d3f41890f0bd2fc2a63c599532b79ac0ed4842cada7d4b65fd445

Fedora Manual Host RC Decision Classifier Status

Status: status alignment Date: 2026-05-21 Scope: public status record after the no-effect Fedora manual host RC decision classifier landed on main.

Summary

Latticra now has a no-effect Fedora manual host release-candidate decision classifier.

The classifier consumes explicit checklist-style evidence and reports one of:

```
manual_host_rc_status=candidate
manual_host_rc_status=blocked
manual_host_rc_status=invalid
```

This is decision evidence only.

It does not perform host changes, live validation, Fedora QA, Fedora approval, Fedora distribution readiness, or production readiness.

Evidence recorded

```
Fedora manual host RC decision classifier
source=PR #217
decision_classifier_present=1
decision_classifier_guard_present=1
decision_classifier_docs_guard_present=1
candidate_state_defined=1
blocked_state_defined=1
invalid_state_defined=1
synthetic_candidate_test_present=1
blocked_target_tests_present=1
blocked_evidence_tests_present=1
blocked_boundary_tests_present=1
invalid_input_test_present=1
classifier_evidence_level=7
live_host_validation_completed=0
host_change_performed=0
sudo_invoked=0
rpm_invoked=0
dnf_invoked=0
network_allowed=0
service_operation_allowed=0
boot_operation_allowed=0
kernel_operation_allowed=0
policy_operation_allowed=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Current readiness classification

```
manual_host_rc_checklist_present=1
manual_host_rc_decision_classifier_present=1
manual_host_rc_status=blocked-pending-real-transcript
manual_host_release_candidate_ready=0
manual_host_dry_run_transcript_present=0
live_host_validation_completed=0
host_install_ready=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
```

Candidate boundary

The classifier can report candidate only when explicit evidence says all required conditions are satisfied.

A synthetic unit-test candidate is not a real Fedora host validation candidate.

A real candidate still requires a concrete reviewed transcript from a disposable Fedora VM target.

Guard validation

The classifier is guarded by:

```
sh scripts/test-fedora-manual-host-rc-decision-classifier.sh
sh scripts/test-fedora-manual-host-rc-decision-classifier-docs.sh
```

This status alignment is guarded by:

```
sh scripts/test-fedora-manual-host-rc-decision-status-alignment.sh
```

Expected output:

```
fedora_manual_host_rc_decision_classifier: ok
fedora_manual_host_rc_decision_classifier_docs: ok
fedora_manual_host_rc_decision_status_alignment: ok
```

Honest public wording

After this status alignment lands, the project may say:

```
Latticra has controlled Fedora installroot RPM lifecycle evidence.
Latticra has a manual Fedora host RC checklist.
Latticra has a no-effect decision classifier for manual Fedora host RC evidence.
```

```
Latticra is preparing the dry-run transcript layer required before live host validation.
```

The project must not say:

```
Latticra has completed live Fedora host validation.
Latticra is production ready.
Latticra is Fedora-approved.
Latticra is ready for Fedora distribution.
```

Next recommended Fedora lane

```
Add manual Fedora host dry-run transcript contract
```

That contract should define the transcript evidence required before any future disposable Fedora VM validation attempt.

README overhaul hold

The root README overhaul should wait until the repository has real validation evidence for the host path.

Until then, root README wording should remain evidence-bound and avoid readiness claims.

Non-claims

This status record does not perform host changes, live validation, Fedora QA approval, Fedora distribution readiness, or production readiness.

docs/status/FEDORA_RPM_GATE_CLASSIFIER_STATUS.md

Source title: Fedora RPM Gate Classifier Status

Lines: 79 | Words: 197 | SHA-256: 87e5781f562fb8942c87fb9779a3d907daecd281cc542fd9bde1a5a8720179c6

Fedora RPM Gate Classifier Status

Status: merged Fedora RPM gate classifier implementation

This record tracks the Fedora RPM gate classifier implementation slice.

Merged PR:

```
#207 Add Fedora RPM gate classifier
```

Merged commit:

```
e3e6cf85cf59c2634c0a67c7d4318afc2c86388a
```

Primary files:

```
include/latticra/fedora_rpm_gate.h
src/fedora_rpm_gate.c
tests/fedora_rpm_gate.c
docs/FEDORA_RPM_GATE_CLASSIFIER.md
scripts/test-fedora-rpm-gate-classifier.sh
scripts/test-fedora-rpm-gate-classifier-docs.sh
.github/workflows/fedora-rpm-gate-classifier.yml
```

Local Fedora validation:

```
sh scripts/test-fedora-rpm-gate-classifier.sh
sh scripts/test-fedora-rpm-gate-classifier-docs.sh
fedora_rpm_gate_classifier: ok
fedora_rpm_gate_classifier_docs: ok
```

What it adds:

```
Fedora RPM gate input record
Fedora RPM gate result record
allowed and denied gate status labels
deterministic denial reasons
gate report surface
evidence level 4 classifier boundary
```

Report boundary:

```
FEDORA LOCAL RPM INSTALL MUTATION GATE
install_gate_status=allowed
install_gate_denial=none
install_mutation_allowed=1
install_mutation_performed=0
host_mutation_performed=0
network_allowed=0
evidence_level=4
```

Validation:

```
sh scripts/test-fedora-rpm-gate-classifier.sh
sh scripts/test-fedora-rpm-gate-classifier-docs.sh
```

Boundary: classifier only. It reports allowed or denied states and deterministic denial reasons. It does not execute package actions, validate sudo, build packages, open the network, start services, change boot entries, load kernel modules, claim Fedora approval, or claim production installer readiness.

Recommended next slice:

```
Add Fedora local RPM removal and rollback plan
```

That future slice should define package identification, ownership verification, uninstall request boundaries, post-removal absence checks, and failure reporting before any execution lane is added.

docs/status/FEDORA_VM_CLI_PAYLOAD_NEXT_VALIDATION_LANE_PLAN_STATUS.md

Source title: Fedora VM CLI Payload Next Validation Lane Plan Status

Lines: 69 | Words: 188 | SHA-256: 0de7b4a0cd7d4f48d0a0ff0f80189bda4ccbb95c13d9dca3ea4f140fd4df46a7

Fedora VM CLI Payload Next Validation Lane Plan Status

Status: plan/status alignment Date: 2026-05-26 Scope: status record for the next manual disposable Fedora VM CLI payload validation lane plan.

Summary

Latticra now has a Fedora VM CLI payload next-validation lane plan.

The plan defines a future repeatability-evidence lane for the accepted no-effect CLI payload:

```
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
validated_payload=/usr/bin/latticra
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

The plan does not add a runner or execute host mutation.

Required next-lane evidence

The future lane must record:

```
source_tree_revision_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
spec_checksum_recorded=1
source_archive_checksum_recorded=1
rpm_nevra_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_matches_expected_cli_surfaces=1
unexpected_runtime_surface_absent=1
cli_status_output_recorded=1
cli_version_output_recorded=1
cli_report_output_recorded=1
cli_invalid_command_exit_recorded=1
```

```
rpm_verify_completed=1
removal_validation_performed=1
post_removal_absence_verified=1
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-next-validation-lane-plan.sh
```

Expected output:

```
fedora_vm_cli_payload_next_validation_lane_plan: ok
```

Next recommended lane

```
Add Fedora VM CLI payload repeatability transcript contract
```

Non-claims

This status record is not a runner, not a package install, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not malware prevention, not ransomware prevention, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_README_ALIGNMENT_STATUS.md

Source title: Fedora VM CLI Payload README Alignment Status

Lines: 97 | Words: 273 | SHA-256: cd625bedb8ebd15ce7ee6f9eef4d77d2f11de3cf59acbac49d13b66519e0742e

Fedora VM CLI Payload README Alignment Status

Status: README/status alignment Date: 2026-05-26 Scope: root README wording after accepted disposable Fedora VM CLI payload validation evidence was recorded.

Summary

The root README now records the narrow Fedora CLI payload posture supported by the accepted disposable VM evidence:

```
fedora_vm_cli_payload_validation_status=evidence-recorded
disposable_vm_cli_validation_transcript_present=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

This wording is limited to the no-effect CLI payload validated in a disposable Fedora VM.

It does not claim production readiness, Fedora approval, Fedora distribution readiness, immutable Fedora readiness, daily-driver safety, security capability, update safety, recovery safety, sandboxing, malware prevention, ransomware prevention, or OS-replacement readiness.

Evidence linked

The README links to:

```
docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_LANE.md
docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_STATUS.md
docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md
docs/status/FEDORA_VM_CLI_PAYLOAD_README_ALIGNMENT_STATUS.md
packaging/fedora/latticra.spec
```

README wording boundaries

The README must preserve these boundaries:

```
validated path is disposable Fedora VM only
validated payload includes /usr/bin/latticra
validated payload includes /usr/share/doc/latticra/README.md
validated CLI mode is no-effect
validated runtime_behavior is disabled
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The README must not claim:

```
production ready
Fedora approved
Fedora distribution ready
daily-driver safe
immutable Fedora ready
security product
update safety
recovery safety
sandboxing
malware prevention
ransomware prevention
bootable OS replacement
kernel runtime ready
```

Guard validation

This README alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-readme-alignment.sh
```

Expected output:

```
fedora_vm_cli_payload_readme_alignment: ok
```

Next recommended lane

```
Plan the next Fedora CLI payload validation lane without widening README or announcement claims
beyond disposable Fedora VM evidence
```

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_ACCEPTANCE_CONTRACT_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Evidence Acceptance Contract Status

Lines: 60 | Words: 205 | SHA-256: 1f5750cf69cd4cf83aca99fe47d07e84a203f626c3e062653bd2a793c39f190b

Fedora VM CLI Payload Repeatability Evidence Acceptance Contract Status

Status: acceptance contract/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability evidence acceptance contract.

Summary

Latticra now has a static acceptance contract for the future Fedora VM CLI payload repeatability evidence status record.

The contract defines the required accepted evidence fields after a real disposable Fedora VM repeatability transcript is validated and reviewed.

It does not attach a transcript.

It does not write evidence status.

It does not mark repeatability evidence present.

Current classification

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-acceptance-contract.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
and write accepted repeatability evidence status
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_PUBLICATION_GATE_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Evidence Publication Gate Status

Lines: 67 | Words: 218 | SHA-256: 0e11d0dd3ee5e61dfcd0d335a91dbb1295e69dd3b2d5d964ffae92441fe9dc9d

Fedora VM CLI Payload Repeatability Evidence Publication Gate Status

Status: publication gate/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability evidence publication gate.

Summary

Latticra now has a static publication gate for future Fedora VM CLI payload repeatability evidence.

The gate prevents a passing evidence status candidate review from being treated as published evidence.

It does not run the repeatability runner.

It does not write evidence status.

It does not publish repeatability evidence.

Current classification

```
fedora_vm_cli_payload_repeatability_evidence_publication_gate_present=1
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
repeatability_evidence_publication_requested=0
operator_publication_review_completed=0
repeatability_evidence_publication_approved=0
repeatability_evidence_status_published=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-publication-gate.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_publication_gate: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript and evidence
 status candidates, then complete operator publication review
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not an accepted evidence status record, not a published evidence status record, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_REVIEW_GATE_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Evidence Review Gate Status

Lines: 64 | Words: 204 | SHA-256: 299b9ac5a8b05162d6bce669bfac609a853dad5f04543d4eae6f44e9f23f8e16

Fedora VM CLI Payload Repeatability Evidence Review Gate Status

Status: review gate/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability evidence review gate.

Summary

Latticra now has a static review gate for future Fedora VM CLI payload repeatability evidence.

The gate requires a complete operator-reviewed transcript from a real disposable Fedora VM repeatability run before repeatability evidence can be accepted.

It does not attach a transcript.

It does not mark repeatability evidence present.

Current classification

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_complete=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_review_required=1
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-review-gate.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_review_gate: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_REVIEW_VALIDATOR_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Evidence Status Review Validator Status

Lines: 64 | Words: 213 | SHA-256: 10293a2644c71ba01c161e56a7879ca32d5d03529d47ca62e6238aefacf9d8eb

Fedora VM CLI Payload Repeatability Evidence Status Review Validator Status

Status: validator/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability evidence status review validator.

Summary

Latticra now has a no-effect validator for supplied Fedora VM CLI payload repeatability evidence status candidates.

The validator rejects missing markers and placeholder values in a future accepted status candidate.

It does not run the repeatability runner.

It does not write evidence status.

It does not mark repeatability evidence present.

Current classification

```
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
fedora_vm_cli_payload_repeatability_evidence_status_review_validator_present=1
repeatability_evidence_status_review_mode=no-effect-validation
repeatability_evidence_status_candidate_valid=0
repeatability_evidence_status_reviewed=0
repeatability_evidence_status_accepted_by_validator=0
evidence_status_written_by_validator=0
promotion_allowed_by_status_validator_alone=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-status-review-validator.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_status_review_validator: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
fill evidence status template, and review the evidence status candidate
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not an accepted evidence status record, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_EVIDENCE_STATUS_TEMPLATE_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Evidence Status Template Status

Lines: 65 | Words: 219 | SHA-256: 1090b2f5f72ee4c99bcc274bcb8bc11615463fda26bc08b4fc9cce16d5ab505e

Fedora VM CLI Payload Repeatability Evidence Status Template Status

Status: template/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability evidence status template.

Summary

Latticra now has a no-effect stdout template for a future accepted Fedora VM CLI payload repeatability evidence status record.

The template is only usable after a real disposable Fedora VM repeatability transcript is attached, validated, and reviewed.

It does not run the repeatability runner.

It does not attach a transcript.

It does not write evidence status.

It does not mark repeatability evidence present.

Current classification

```
fedora_vm_cli_payload_repeatability_evidence_acceptance_contract_present=1
fedora_vm_cli_payload_repeatability_evidence_status_template_present=1
repeatability_evidence_status_template_mode=no-effect-template
repeatability_evidence_status_template_complete=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_accepted=0
evidence_status_written=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-evidence-status-template.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_evidence_status_template: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate transcript candidate,
and use the evidence status template to write accepted repeatability evidence status
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_PLAN_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Runner Plan Status

Lines: 67 | Words: 189 | SHA-256: 73491a993b13fd807ac0411ee2a6adf314b688a493b7f4464eac650331f6d05e

Fedora VM CLI Payload Repeatability Runner Plan Status

Status: plan/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability runner plan and manual runner implementation.

Summary

Latticra now has a plan and manually gated runner for disposable Fedora VM CLI payload repeatability validation.

The plan maps the repeatability transcript contract into a runner shape, and the runner is present as a manual-only script.

The runner remains gated by:

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_ALLOW_CLI_PAYLOAD_REPEATABILITY_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
```

Current classification

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_plan_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
repeatability_runner_manual_only=1
ci_auto_repeatability_validation_allowed=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-runner-plan.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_runner_plan: ok
```

Next recommended lane

```
Capture reviewed Fedora VM CLI payload repeatability transcript evidence
```

Non-claims

This status record is not RPM install evidence, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_RUNNER_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Runner Status

Lines: 75 | Words: 193 | SHA-256: 6534f264a8783cf3b5d2cc472d237d242d97290d3a9bf7414342147600094e7e

Fedora VM CLI Payload Repeatability Runner Status

Status: runner/status alignment Date: 2026-05-26 Scope: status record for the manually gated Fedora VM CLI payload repeatability runner.

Summary

Latticra now has a manual-only disposable Fedora VM CLI payload repeatability runner.

The runner is present at:

```
scripts/run-fedora-vm-cli-payload-repeatability-lane.sh
```

It is gated by both disposable-VM RPM validation consent and CLI payload repeatability consent.

It must not be called by normal CI.

Current classification

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_plan_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
repeatability_runner_manual_only=1
ci_auto_repeatability_validation_allowed=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Required gates

```
LATTICRA_ALLOW_DISPOSABLE_VM_RPM_VALIDATION=1
LATTICRA_ALLOW_CLI_PAYLOAD_REPEATABILITY_VALIDATION=1
LATTICRA_TARGET_IS_DISPOSABLE_FEDORA_VM=1
LATTICRA_TARGET_IS_DAILY_DRIVER=0
LATTICRA_TARGET_IS_PRODUCTION_HOST=0
LATTICRA_TARGET_IS_IMMUTABLE_FEDORA=0
LATTICRA_TARGET_HAS_CLEAN_SNAPSHOT=1
LATTICRA_TARGET_HAS_RECOVERY_PATH=1
LATTICRA_OPERATOR_CONSENT_RECORDED=1
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-runner.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_runner: ok
```

Next recommended lane

```
Capture reviewed Fedora VM CLI payload repeatability transcript evidence
```

Non-claims

This status record is not a completed repeatability transcript, not RPM install evidence, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CAPTURE_TEMPLATE_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Transcript Capture Template Status

Lines: 63 | Words: 200 | SHA-256: 3d8fed3845c490c9a4961e5cc7020bf37401ed6d1137b848aa16447ba7152e57

Fedora VM CLI Payload Repeatability Transcript Capture Template Status

Status: template/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability transcript capture template.

Summary

Latticra now has a no-effect capture template for a future Fedora VM CLI payload repeatability transcript.

The template prints the fields an operator must attach after a real disposable Fedora VM repeatability run.

It does not run the repeatability runner.

It does not mark repeatability evidence present.

Current classification

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_runner_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_template_mode=no-effect-template
repeatability_transcript_template_complete=0
repeatability_transcript_candidate_valid=0
repeatability_transcript_attached=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-transcript-template.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_transcript_template: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_CONTRACT_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Transcript Contract Status

Lines: 45 | Words: 136 | SHA-256: b9f144424e25f1f15ac51605bf3f48374b364c6222c61047ea484cbb52b6dadd

Fedora VM CLI Payload Repeatability Transcript Contract Status

Status: contract/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability transcript contract.

Summary

Latticra now has a contract for a future second disposable Fedora VM CLI payload validation transcript.

```
transcript_kind=disposable-vm-cli-payload-repeatability
fedora_vm_cli_payload_next_validation_lane_plan_present=1
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

```
sh scripts/test-fedora-vm-cli-payload-repeatability-transcript-contract.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_transcript_contract: ok
```

Next recommended lane

```
Add Fedora VM CLI payload repeatability runner plan
```

Non-claims

This status record is not a completed repeatability transcript, not a runner, not RPM install evidence, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_REPEATABILITY_TRANSCRIPT_REVIEW_VALIDATOR_STATUS.md

Source title: Fedora VM CLI Payload Repeatability Transcript Review Validator Status

Lines: 62 | Words: 189 | SHA-256: e0222e7401a6b2d8da9adc345d626f102b22ca3a0da827152e4f50ff48763f88

Fedora VM CLI Payload Repeatability Transcript Review Validator Status

Status: validator/status alignment Date: 2026-05-26 Scope: status record for the Fedora VM CLI payload repeatability transcript review validator.

Summary

Latticra now has a no-effect validator for supplied Fedora VM CLI payload repeatability transcript candidates.

The validator checks required markers and rejects placeholder values.

It does not run the repeatability runner.

It does not write evidence status.

Current classification

```
fedora_vm_cli_payload_repeatability_transcript_contract_present=1
fedora_vm_cli_payload_repeatability_transcript_capture_template_present=1
fedora_vm_cli_payload_repeatability_evidence_review_gate_present=1
fedora_vm_cli_payload_repeatability_transcript_review_validator_present=1
repeatability_transcript_review_mode=no-effect-validation
repeatability_transcript_candidate_valid=0
repeatability_transcript_reviewed=0
repeatability_transcript_accepted=0
evidence_status_written=0
promotion_allowed_by_validator_alone=0
second_disposable_vm_cli_validation_completed=0
cli_payload_repeatability_evidence_present=0
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Guard validation

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-repeatability-transcript-review-validator.sh
```

Expected output:

```
fedora_vm_cli_payload_repeatability_transcript_review_validator: ok
```

Next recommended lane

```
Run manual disposable Fedora VM CLI payload repeatability lane, validate the transcript
candidate, and then add reviewed evidence status
```

Non-claims

This status record is not repeatability evidence, not a completed transcript, not a second disposable Fedora VM validation run, not host mutation, not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not sandboxing, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_ANNOUNCEMENT_STATUS.md

Source title: Fedora VM CLI Payload Validation Announcement Status

Lines: 75 | Words: 188 | SHA-256: 938449b55e5a926fc1f268a69843848f25831d822c7842b173304da9b0a963d1

Fedora VM CLI Payload Validation Announcement Status

Status: announcement/status alignment Date: 2026-05-26 Scope: public announcement wording for the disposable Fedora VM CLI payload validation milestone.

Summary

The public announcement log now records the disposable Fedora VM CLI payload validation milestone.

The announcement is evidence-bound and limited to:

```
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
validated_payload=/usr/bin/latticra
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

Evidence basis

The announcement relies on:

```
docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md
docs/status/FEDORA_VM_CLI_PAYLOAD_README_ALIGNMENT_STATUS.md
README.md
```

Announcement boundary

The announcement must not claim:

```
production readiness
Fedora approval
Fedora distribution readiness
daily-driver safety
immutable Fedora readiness
security capability
update safety
recovery safety
sandboxing
malware prevention
ransomware prevention
bootable OS replacement behavior
kernel runtime readiness
production installer
```

Guard validation

This announcement alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-validation-announcement.sh
```

Expected output:

```
fedora_vm_cli_payload_validation_announcement: ok
```

Next recommended lane

```
Plan the next Fedora CLI payload validation lane without widening README or announcement claims
beyond disposable Fedora VM evidence
```

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_EVIDENCE_STATUS.md

Source title: Fedora VM CLI Payload Validation Evidence Status

Lines: 177 | Words: 449 | SHA-256: 8e9e94987a603d2658c40601abb3bb564e0efa9e820d5b909ffff68ff9dbaa9a

Fedora VM CLI Payload Validation Evidence Status

Status: evidence status alignment Date: 2026-05-21 Scope: public status record after a disposable Fedora VM CLI payload validation transcript reached the expected validation report.

Summary

A disposable Fedora VM CLI payload validation run completed successfully against the manually gated runner from PR #236.

The validation run built the local Latticra RPM with the no-effect CLI payload, ran the build-time test suite, installed the package into the disposable Fedora VM, validated the installed CLI command surface, removed the package, verified post-removal absence, and emitted the expected deterministic validation report.

This is disposable Fedora VM CLI payload validation evidence.

It is not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, and not a production installer claim.

Evidence source

```
source=operator disposable Fedora VM transcript
repo_checkout=main
validated_package=latticra-0.0.0-0.1.local.fc44.x86_64.rpm
validated_package_name=latticra
validated_package_version=0.0.0
validated_package_arch=x86_64
fedora_kernel_version=6.19.10-300.fc44.x86_64
```

The transcript records the prerequisite guards and runner completion markers:

```
fedora_vm_cli_payload_validation_status_alignment: ok
fedora_vm_cli_payload_transcript_capture_docs: ok
latticra_no_effect_cli_status_surface: ok
state_lattice_invariants: ok
system_bootstrap: ok
kernel: ok
kernel_lifecycle: ok
Wrote: /tmp/.../rpmbuild/RPMS/x86_64/latticra-0.0.0-0.1.local.fc44.x86_64.rpm
Updating / installing...
latticra-0.0.0-0.1.local.fc44
fedora_vm_cli_payload_validation_lane: ok
```

Validation report recorded

```
FEDORA VM CLI PAYLOAD VALIDATION LANE
validation_status=ok
package_name=latticra
package_version=0.0.0
package_arch=x86_64
package_version_recorded=1
package_arch_recorded=1
disposable_vm_target_verified=1
snapshot_evidence_present=1
recovery_evidence_present=1
operator_consent_recorded=1
fedora_os_release_recorded=1
fedora_kernel_version_recorded=1
fedora_kernel_version=6.19.10-300.fc44.x86_64
rpm_tooling_recorded=1
rpmbuild_tooling_recorded=1
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_contains_compiled_c_binary=1
buildarch_noarch_removed=1
cli_status_surface_implemented=1
cli_status_surface_guarded_before_packaging=1
local_cli_guard_passed=1
local_rpm_built_from_current_tree=1
rpm_build_command_recorded=1
rpm_name_is_latticra=1
rpm_version_recorded=1
rpm_arch_recorded=1
rpm_path_recorded=1
rpm_metadata_recorded=1
rpm_payload_listing_recorded=1
rpm_payload_contains_cli_binary=1
rpm_payload_contains_readme=1
rpm_payload_contains_only_expected_surfaces=1
unexpected_runtime_surface_absent=1
install_command_recorded=1
install_result_recorded=1
rpm_query_after_install_recorded=1
installed_payload_listing_recorded=1
installed_cli_binary_present=1
installed_readme_present=1
rpm_verify_completed=1
cli_status_command_recorded=1
cli_version_command_recorded=1
cli_report_command_recorded=1
cli_invalid_command_recorded=1
cli_no_root_required=1
cli_no_host_mutation_observed=1
cli_no_network_observed=1
cli_no_service_operation_observed=1
cli_no_kernel_operation_observed=1
cli_no_boot_operation_observed=1
cli_no_selinux_policy_operation_observed=1
removal_command_recorded=1
removal_result_recorded=1
post_removal_query_recorded=1
post_removal_absence_verified=1
cli_removed_after_rpm_removal=1
readme_removed_after_rpm_removal=1
post_removal_cli_absence_verified=1
post_removal_readme_absence_verified=1
install_validation_performed=1
cli_status_validation_performed=1
cli_version_validation_performed=1
cli_report_validation_performed=1
removal_validation_performed=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
evidence_level=9
fedora_vm_cli_payload_validation_lane: ok
```

Current readiness classification

```
fedora_vm_cli_payload_validation_lane_present=1
fedora_vm_cli_payload_validation_runner_present=1
disposable_vm_cli_validation_transcript_present=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Boundary statement

This evidence is limited to a disposable Fedora VM CLI payload validation path.

The validated payload is:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

This evidence does not validate daily-driver installation, immutable Fedora installation, production host installation, service activation, boot integration, kernel module loading, SELinux policy changes, network operations, update safety, Fedora QA approval, Fedora distribution readiness, or production installer readiness.

Guard validation

The evidence status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-validation-evidence-status.sh
```

Expected output:

```
fedora_vm_cli_payload_validation_evidence_status: ok
```

Next recommended Fedora lane

```
Align README wording with disposable Fedora VM CLI payload validation evidence
```

That wording must remain narrow and should not claim production, Fedora distribution, immutable Fedora, daily-driver, security, recovery, or OS-replacement readiness.

Non-claims

This status record is not production readiness, not Fedora approval, not Fedora distribution readiness, not daily-driver safety, not immutable Fedora readiness, not update safety, not recovery safety, not malware prevention, not ransomware prevention, not sandboxing, not security hardening, and not a production installer claim.

docs/status/FEDORA_VM_CLI_PAYLOAD_VALIDATION_STATUS.md

Source title: Fedora VM CLI Payload Validation Status

Lines: 155 | Words: 425 | SHA-256: c98772e1e610ea8c0bc7b1edf95ac67a85ff6553ace81f5f76c11c1bdb083b42

Fedora VM CLI Payload Validation Status

Status: status alignment Date: 2026-05-21 Scope: public status record after the manually gated disposable Fedora VM CLI payload validation lane runner landed on main.

Summary

Latticra now has a manually gated disposable Fedora VM CLI payload validation runner.

The runner is designed for validating the expanded local RPM payload:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

The lane defines and ships:

```
docs/FEDORA_VM_CLI_PAYLOAD_VALIDATION_LANE.md
scripts/run-fedora-vm-cli-payload-validation-lane.sh
scripts/test-fedora-vm-cli-payload-validation-lane-docs.sh
.github/workflows/fedora-vm-cli-payload-validation-lane-docs.yml
```

This is runner-readiness evidence only.

It does not prove that a disposable Fedora VM CLI payload validation run has completed yet.

The validation runner remains manually gated and must not be auto-run by normal CI.

Evidence recorded

```
Fedora VM CLI payload validation lane runner
source=PR #236
validation_lane_documented=1
validation_runner_present=1
validation_lane_docs_guard_present=1
validation_lane_docs_workflow_present=1
runner_manual_only=1
ci_auto_vm_cli_validation_allowed=0
disposable_vm_target_required=1
daily_driver_block_required=1
production_host_block_required=1
immutable_fedora_block_required=1
clean_snapshot_required=1
recovery_path_required=1
operator_consent_required=1
non_root_operator_required=1
sudo_limited_to_rpm_install_removal=1
fedora_target_required=1
rpm_tooling_required=1
rpmbuild_tooling_required=1
cc_tooling_required=1
local_cli_guard_required=1
local_rpm_build_required=1
rpm_payload_listing_required=1
rpm_payload_cli_binary_required=1
rpm_payload_readme_required=1
rpm_payload_only_expected_surfaces_required=1
unexpected_runtime_surface_absent_required=1
installed_cli_binary_required=1
installed_readme_required=1
rpm_verify_required=1
cli_status_validation_required=1
cli_version_validation_required=1
cli_report_validation_required=1
cli_invalid_command_validation_required=1
post_removal_cli_absence_required=1
post_removal_readme_absence_required=1
validation_report_schema_present=1
target_evidence_level=9
current_evidence_level=8
evidence_level_9_achieved=0
```

Current readiness classification

```
fedora_vm_cli_transcript_contract_present=1
fedora_vm_cli_payload_validation_lane_present=1
fedora_vm_cli_payload_validation_runner_present=1
fedora_vm_cli_payload_validation_status=blocked-pending-real-vm-run
disposable_vm_cli_validation_transcript_present=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Boundary statement

This status alignment does not run the validation lane.

It does not build a release RPM artifact.

It does not install or remove an RPM.

It does not validate /usr/bin/latticra in a real disposable Fedora VM.

It does not mutate a disposable VM, developer host, daily-driver Fedora host, immutable Fedora host, production host, boot entry, kernel module set, service registry, SELinux policy, firmware state, or network configuration.

The docs guard may run in CI.

The CLI payload validation runner must remain manual and gated by explicit disposable-VM evidence.

Guard validation

The validation lane documentation is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-validation-lane-docs.sh
```

This status alignment is guarded by:

```
sh scripts/test-fedora-vm-cli-payload-validation-status-alignment.sh
```

Expected output:

```
fedora_vm_cli_payload_validation_lane_docs: ok
fedora_vm_cli_payload_validation_status_alignment: ok
```

Next recommended Fedora lane

```
Capture real disposable Fedora VM CLI payload validation transcript evidence
```

That lane should record the Fedora VM identity, clean snapshot evidence, recovery path evidence, package version, RPM architecture, RPM payload listing, install transcript, CLI command transcript, invalid command transcript, removal transcript, post-removal absence proof, and emitted validation report.

README update hold

The root README should not claim CLI payload host install readiness yet.

The README update remains blocked until real validation evidence exists for:

```
disposable_vm_cli_validation_transcript_present=1
disposable_vm_cli_validation_completed=1
host_install_ready_for_cli_payload=1
```

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, sandboxing, security hardening, malware prevention, ransomware prevention, OS-replacement readiness, or a production installer claim.

docs/status/FEDORA_VM_RPM_VALIDATION_ANNOUNCEMENT_STATUS.md

Source title: Fedora VM RPM Validation Announcement Status

Lines: 75 | Words: 187 | SHA-256: 454712b6c518a0683a0bc4683b7c50da6b547939256bdc5d367256101f9fde2e

Fedora VM RPM Validation Announcement Status

Status: announcement/status alignment Date: 2026-05-21 Scope: public announcement wording for the disposable Fedora VM local RPM validation milestone.

Summary

The public announcement log now records the disposable Fedora VM local RPM validation milestone.

The announcement is evidence-bound and limited to:

```
disposable_vm_validation_completed=1
live_host_validation_completed=1
host_install_ready=1
validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
validated_payload=/usr/share/doc/latticra/README.md
evidence_level=9
```

Evidence basis

The announcement relies on:

```
docs/status/FEDORA_DISPOSABLE_VM_LOCAL_RPM_VALIDATION_EVIDENCE_STATUS.md
docs/status/FEDORA_DISPOSABLE_VM_RPM_README_ALIGNMENT_STATUS.md
README.md
```

Announcement boundary

The announcement must not claim:

```
production readiness
Fedora approval
Fedora distribution readiness
daily-driver safety
immutable Fedora readiness
security capability
update safety
recovery safety
sandboxing
malware prevention
ransomware prevention
bootable OS replacement behavior
kernel runtime readiness
production installer
```

Guard validation

This announcement alignment is guarded by:

```
sh scripts/test-fedora-vm-rpm-validation-announcement.sh
```

Expected output:

```
fedora_vm_rpm_validation_announcement: ok
```

Next recommended lane

```
Plan the next Fedora validation lane without widening README or announcement claims beyond
disposable Fedora VM local RPM evidence
```

Non-claims

This status record is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/status/FREEBSD_ECOSYSTEM_INTEGRATION_STATUS.md

Source title: FreeBSD Ecosystem Integration Status

Lines: 152 | Words: 458 | SHA-256: 770ee8e86ad24f3368039ac96b168d4d2846b4bc4717a6ae5c853681ec686386

FreeBSD Ecosystem Integration Status

Status: FreeBSD integration status record Date: 2026-05-26

Summary

Latticra now has a FreeBSD-facing local ports metadata draft for the no-effect CLI payload.

This is an ecosystem integration checkpoint, not a production readiness claim.

Current Evidence

```
freebsd_port_draft_present=1
freebsd_port_static_validation_present=1
debian_freebsd_openbsd_source_archive_contract_present=1
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
temporary_freebsd_distfile_staged=1
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
freebsd_build_allowed=0
freebsd_ports_environment_documented=1
freebsd_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
environment_transcript_present=0
freebsd_artifact_name_pattern_recorded=1
freebsd_package_name=latticra-0.0.0.pkg
freebsd_package_artifact_created=0
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
freebsd_payload_expected_bin=bin/latticra
freebsd_payload_expected_doc=%%DOCSDIR%%/README.md
freebsd_payload_inspection_run=0
package_payload_inspection_run=0
package_payload_accepted=0
freebsd_install_remove_transcript_required=1
freebsd_install_remove_transcript_present=0
freebsd_package_install_run=0
freebsd_package_remove_run=0
remove_on_host_run=0
publication_non_claim_review_present=1
freebsd_publication_non_claim_review_present=1
freebsd_package_publication_claimed=0
freebsd_pkg_repo_created=0
freebsd_pkg_repo_publish_run=0
freebsd_pkg_repo_sign_run=0
platform_build_evidence_accepted=0
freebsd_validation_promotion_blocked=1
freebsd_platform_build_evidence_accepted=0
package_validation_result_promoted=0
freebsd_validation_result_promoted=0
source_archive_policy_recorded=1
source_archive_created=0
source_archive_sha256_recorded=0
freebsd_distinfo_created=0
freebsd_make_makesum_run=0
freebsd_make_stage_run=0
freebsd_make_package_run=0
freebsd_ports_tree_submission_claimed=0
freebsd_bugzilla_pr_claimed=0
freebsd_committer_review_claimed=0
poudriere_build_run=0
poudriere_run=0
make_stage_run=0
make_package_run=0
portlint_run=0
package_artifact_created=0
package_artifact_sha256_recorded=0
install_on_host_run=0
freebsd_official_port_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Guarded Files

```
docs/FREEBSD_PORT_STATIC_VALIDATION.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
packaging/freebsd/README.md
packaging/freebsd/Makefile
packaging/freebsd/pkg-descr
packaging/freebsd/pkg-plist
scripts/test-freebsd-port-static-validation.sh
scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
.github/workflows/freebsd-port-static-validation.yml
.github/workflows/debian-freebsd-openbsd-source-archive-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-gate-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-environment-contract.yml
.github/workflows/debian-freebsd-openbsd-package-artifact-naming-contract.yml
.github/workflows/debian-freebsd-openbsd-package-payload-inspection-contract.yml
.github/workflows/debian-freebsd-openbsd-package-install-remove-transcript-contract.yml
.github/workflows/debian-freebsd-openbsd-package-publication-non-claim-review-contract.yml
.github/workflows/debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.ym
l
```

Current Boundary

The FreeBSD lane does not publish a package, submit to the FreeBSD ports tree, claim Bugzilla PR evidence, claim committer review, claim poudriere success, claim portlint success, install an rc.d service, change the kernel, add a privileged helper, grant network authority, or claim production readiness.

The local ports metadata now records the no-effect package payload as AGPL-3.0-or-later plus CC-BY-4.0 while source archive, checksum, and notice obligations remain under review.

The source archive contract records the expected latticra-0.0.0.tar.gz distfile and distinfo boundary while keeping archive creation, checksum acceptance, distinfo, package artifacts, and build transcript promotion blocked.

The package-build gate is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md. It keeps package_build_gate_state=closed-no-effect, freebsd_build_allowed=0, freebsd_make_stage_run=0, freebsd_make_package_run=0, freebsd_make_makesum_run=0, portlint_run=0, and poudriere_run=0 until source, checksum, license, notice, ports environment, operator authorization, payload inspection, install/remove transcript, and publication non-claim evidence exists.

The package-build environment contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md. It documents the FreeBSD jail or VM ports environment requirement while keeping freebsd_build_environment_provisioned=0, explicit_operator_build_authorization=0, environment_transcript_present=0, and all FreeBSD package build commands disabled.

The package artifact naming contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md. It records the future latticra-0.0.0.pkg artifact name while keeping freebsd_package_artifact_created=0, repository_package_artifact_write_allowed=0, and package_artifact_sha256_recorded=0.

The package payload inspection contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md. It records the expected FreeBSD payload paths while keeping freebsd_payload_inspection_run=0, package_payload_accepted=0, and package_artifact_created=0.

The package install/remove transcript contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove transcript requirements while keeping freebsd_package_install_run=0, freebsd_package_remove_run=0, and install_on_host_run=0.

The package publication non-claim review contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records local-only FreeBSD publication and official-port non-claims while keeping freebsd_package_publication_claimed=0, freebsd_pkg_repo_publish_run=0, and package validation promotion blocked.

The package validation promotion blocker matrix is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md. It ties FreeBSD source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping freebsd_platform_build_evidence_accepted=0 and freebsd_validation_result_promoted=0.

Next Recommended Lane

```
Add a FreeBSD package build-evidence intake denial contract before any FreeBSD build lane can
open.
```

docs/status/HIGH_ASSURANCE_SECURITY_BASELINE_STATUS.md

Source title: Latticra High-Assurance Security Baseline Status

Lines: 94 | Words: 156 | SHA-256: bb072d46fb560c0c871c31dff188fb16250db540715691f1090609a7e13c25bb

Latticra High-Assurance Security Baseline Status

Status: status record for high-assurance security baseline Date: 2026-05-26

Scope

This record tracks the high-assurance security baseline checkpoint created after current NSA, CISA, FBI, and NIST source review.

It does not implement runtime execution, effect execution, host behavior, network behavior, cryptographic enforcement, signing authority, tool execution, shell execution, sandboxing, malware prevention, ransomware prevention, incident response, recovery behavior, certification, accreditation, compliance, production protection, or runtime authority.

Current fields

```
high_assurance_security_baseline_present=1
high_assurance_security_baseline_status_present=1
source_refresh_date=2026-05-26
official_source_inventory_present=1
nsa_zero_trust_guideline_observed=1
nsa_zero_trust_user_pillar_observed=1
cisa_nsa_esf_iam_best_practices_observed=1
nsa_cisa_memory_safe_languages_observed=1
cisa_secure_by_design_observed=1
cisa_fbi_product_security_bad_practices_observed=1
nsa_cisa_top_misconfigurations_observed=1
cisa_nsa_fbi_secure_by_default_observed=1
cisa_cpg_observed=1
cisa_fbi_nsa_event_logging_guidance_observed=1
cisa_logging_made_easy_observed=1
cisa_zero_trust_maturity_model_observed=1
nist_sp_800_63_4_observed=1
cisa_nsa_fbi_phishing_guidance_observed=1
nist_nvd_observed=1
fbi_cyber_threat_environment_observed=1
stopransomware_joint_guide_observed=1
stopransomware_recovery_guidance_observed=1
nist_csf_2_observed=1
nist_sp_800_34_observed=1
nist_sp_800_184_observed=1
nist_sp_800_128_observed=1
nist_sp_800_70_rev5_observed=1
nist_sp_800_92_observed=1
nist_ssdf_observed=1
nist_sp_800_53_observed=1
nist_sp_800_160_observed=1
nist_sp_800_207_observed=1
fips_140_3_observed=1
nist_sp_800_57_observed=1
nist_sp_800_131a_observed=1
nist_sp_800_90_observed=1
nsa_cnsa_2_observed=1
memory_safety_roadmap_required=1
memory_safety_roadmap_present=1
zero_trust_runtime_boundary_required=1
ssdf_secure_development_required=1
cpg_operational_baseline_required=1
supply_chain_security_baseline_present=1
cyber_incident_reporting_response_baseline_present=1
vulnerability_management_release_gate_baseline_present=1
cryptographic_assurance_key_management_baseline_present=1
identity_credential_access_management_baseline_present=1
security_logging_monitoring_baseline_present=1
backup_recovery_resilience_baseline_present=1
secure_configuration_change_management_baseline_present=1
kev_release_review_required=1
fips_crypto_boundary_required_before_production_crypto=1
phishing_resistant_mfa_required_before_remote_privileged_access=1
security_event_logging_required_before_hosted_service=1
backup_restore_recovery_evidence_required_before_hosted_service=1
secure_configuration_change_control_required_before_hosted_service=1
sbom_required_before_production_installer=1
third_party_security_validation_required_before_security_release=1
incident_response_plan_required_before_production_service=1
recurring_source_review_required=1
implementation_behavior_changed=0
runtime_authority_granted=0
security_boundary_claimed=0
certification_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
production_protection_claim_allowed=0
```

Validation

```
sh scripts/test-high-assurance-security-baseline.sh
```

Expected output:

```
high_assurance_security_baseline: ok
```

docs/status/IDENTITY_CREDENTIAL_ACCESS_MANAGEMENT_BASELINE_STATUS.md

Source title: Latticra Identity, Credential, and Access Management Baseline Status

Lines: 66 | Words: 142 | SHA-256: 45f038ec65754850fe776404645492a1050b766ab8f9fcc7c54d93590b33a7b0

Latticra Identity, Credential, and Access Management Baseline Status

Status: status record for identity, credential, and access management baseline Date: 2026-05-26

Scope

This record tracks the identity, credential, and access-management baseline for human identity, service identity, machine identity, privileged access, phishing-resistant MFA, SSO/federation context, account lifecycle, credential handling, break-glass access, identity event logging, and access-management non-claims.

It does not implement an identity provider, MFA provider, account provisioning, account deprovisioning, credential storage, remote access, privileged access, hosted administration, SSO federation, authorization enforcement, compliance, or runtime authority.

Current fields

```
identity_credential_access_management_baseline_present=1
identity_credential_access_management_status_present=1
identity_credential_access_management_guard_present=1
nsa_zero_trust_user_pillar_tracked=1
cisa_nsa_esf_iam_best_practices_tracked=1
nist_sp_800_63_4_digital_identity_tracked=1
cisa_cpg_account_security_tracked=1
phishing_guidance_tracked=1
it_product_design_mfa_goal_tracked=1
phishing_resistant_mfa_required_for_privileged_access=1
mfa_required_for_remote_access=1
privileged_access_inventory_required=1
service_account_inventory_required=1
local_account_inventory_required=1
account_lifecycle_contract_required=1
least_privilege_role_review_required=1
break_glass_account_policy_required=1
federation_sso_context_required=1
credential_secret_storage_review_required=1
credential_reuse_forbidden=1
default_credentials_forbidden=1
identity_event_logging_required=1
privileged_behavior_monitoring_required=1
help_desk_identity_verification_required=1
access_exception_owner_required=1
access_exception_expiration_required=1
implementation_behavior_changed=0
identity_provider_added=0
mfa_provider_added=0
account_provisioning_added=0
account_deprovisioning_added=0
remote_access_enabled=0
privileged_access_granted=0
credential_storage_added=0
hosted_admin_surface_added=0
identity_security_claim_allowed=0
hosted_service_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-identity-credential-access-management-baseline.sh
```

Expected output:

```
identity_credential_access_management_baseline: ok
```

docs/status/KERNEL_LIFECYCLE_EVIDENCE_STATUS.md

Source title: Kernel Lifecycle Evidence Status

Lines: 228 | Words: 530 | SHA-256: c0d76fb0c6fe2099910c7362ccb205c005a02d1c7b8a0473463f2a9651e5a325

Kernel Lifecycle Evidence Status

Status: status alignment record Date: 2026-05-26 00:45 CDT Scope: public status alignment after the merged kernel lifecycle evidence sequence.

Purpose

This record aligns the public status layer after the merged kernel lifecycle evidence PRs.

Merged slices:

```
PR #170 — kernel lifecycle report runner
PR #171 — kernel lifecycle subsystem summary
PR #172 — kernel lifecycle rollback plan
```

Current kernel evidence

The current kernel evidence now includes:

```
kernel lifecycle seed
kernel IPC table guard
kernel IPC table report runner
kernel VFS namespace guard
kernel VFS namespace report runner
kernel device registry guard
kernel device registry report runner
kernel driver catalog guard
kernel driver catalog report runner
kernel interrupt table guard
kernel interrupt table report runner
kernel timer source guard
kernel timer source report runner
kernel scheduler tick guard
kernel scheduler tick report runner
kernel run queue guard
kernel run queue report runner
kernel context switch guard
kernel context switch report runner
kernel time accounting guard
kernel time accounting report runner
kernel preemption guard
kernel preemption report runner
kernel scheduler credit guard
kernel scheduler credit report runner
kernel scheduler selection guard
kernel scheduler selection report runner
kernel scheduler dispatch guard
kernel scheduler dispatch report runner
kernel scheduler handoff guard
kernel scheduler handoff report runner
kernel scheduler activation guard
kernel scheduler activation report runner
kernel scheduler run-entry guard
kernel scheduler run-entry report runner
kernel process table guard
kernel process table report runner
kernel syscall table guard
kernel syscall table report runner
kernel lifecycle report runner
kernel lifecycle subsystem summary
kernel lifecycle rollback plan
```

The lifecycle evidence can report a bounded in-memory path ending at:

```
final_state=scheduler-run-entry-ready
```

The lifecycle report runner and subsystem summary keep the external-effect posture explicit:

```
external_effect_performed=0
no_external_effect_chain=1
```

Current authority posture

The merged evidence keeps authority denied:

```
runtime_entry_allowed=0
scheduler_execution_allowed=0
scheduler_selection_allowed=0
scheduler_dispatch_allowed=0
scheduler_handoff_allowed=0
scheduler_activation_allowed=0
scheduler_run_entry_allowed=0
memory_allocation_allowed=0
process_spawn_allowed=0
syscall_dispatch_allowed=0
ipc_send_allowed=0
ipc_receive_allowed=0
ipc_queue_mutation_allowed=0
filesystem_lookup_allowed=0
filesystem_read_allowed=0
filesystem_write_allowed=0
namespace_mutation_allowed=0
device_open_allowed=0
device_read_allowed=0
device_write_allowed=0
driver_probe_allowed=0
driver_load_allowed=0
driver_bind_allowed=0
interrupt_allowed=0
interrupt_mask_allowed=0
interrupt_unmask_allowed=0
interrupt_dispatch_allowed=0
interrupt_ack_allowed=0
timer_tick_allowed=0
timer_arm_allowed=0
timer_disarm_allowed=0
scheduler_tick_allowed=0
run_queue_mutation_allowed=0
enqueue_allowed=0
dequeue_allowed=0
dispatch_allowed=0
context_switch_allowed=0
register_save_allowed=0
register_restore_allowed=0
stack_switch_allowed=0
address_space_switch_allowed=0
preemption_allowed=0
time_accounting_allowed=0
time_read_allowed=0
cpu_usage_write_allowed=0
quota_update_allowed=0
scheduler_credit_update_allowed=0
process_wake_allowed=0
dma_allowed=0
hardware_effect_allowed=0
```

The subsystem summary also keeps process, filesystem, network, device, and production-boundary claims denied or report-only.

Rollback posture

The rollback plan is a planning and guardrail record only.

It defines future rollback classification terms such as:

```
last_safe_state
rollback_reason
rollback_required
rollback_performed=0
persistence_allowed=0
recovery_authority_allowed=0
runtime_entry_allowed=0
```

It does not implement rollback.

Status interpretation

This status alignment records a real improvement in kernel evidence visibility.

It does not change product-readiness claims.

Current non-claims remain:

```
not bootable
not an operating-system replacement
not a production security boundary
not a production runtime
not a hardware authority layer
not installer-ready
```

Next recommended work

Recommended next work:

```
Add no-effect runtime entry admission classifier
```

That future slice should implement classification/reporting only and continue to require:

```
external_effect_performed=0
persistence_allowed=0
recovery_authority_allowed=0
runtime_entry_allowed=0
```

Validation

This status alignment is guarded by:

```
sh scripts/test-kernel-lifecycle-status-alignment.sh
```

Dedicated workflow lanes keep the kernel table guards visible:

```
.github/workflows/kernel-ipc-table.yml
.github/workflows/kernel-vfs-namespace.yml
.github/workflows/kernel-device-registry.yml
.github/workflows/kernel-driver-catalog.yml
.github/workflows/kernel-interrupt-table.yml
.github/workflows/kernel-timer-source.yml
.github/workflows/kernel-scheduler-tick.yml
.github/workflows/kernel-run-queue.yml
.github/workflows/kernel-context-switch.yml
.github/workflows/kernel-time-accounting.yml
.github/workflows/kernel-preemption.yml
.github/workflows/kernel-scheduler-credit.yml
.github/workflows/kernel-scheduler-selection.yml
.github/workflows/kernel-scheduler-dispatch.yml
.github/workflows/kernel-scheduler-handoff.yml
.github/workflows/kernel-scheduler-activation.yml
.github/workflows/kernel-scheduler-run-entry.yml
.github/workflows/kernel-process-table.yml
.github/workflows/kernel-syscall-table.yml
```

Expected output:

```
kernel_lifecycle_status_alignment: ok
```

docs/status/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT_STATUS.md

Source title: L-UI Rendering Detailed Report Refinement Status

Lines: 42 | Words: 114 | SHA-256: 41847e0c8317611dd2d7ce21cc4f19ac9755f1e525821e07d35d7aabb4dc4a7c

L-UI Rendering Detailed Report Refinement Status

Status: implementation refinement status

This record tracks the L-UI rendering detailed report refinement.

Primary record:

```
docs/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT.md
```

Primary implementation surfaces:

```
include/latticra/l_ui_renderer.h
src/l_ui_renderer.c
tests/l_ui_rendering_detailed_report_refinement.c
scripts/test-l-ui-rendering-detailed-report-refinement.sh
.github/workflows/l-ui-rendering-detailed-report-refinement.yml
```

What changed:

```
explicit report classification metadata
explicit detail level metadata
explicit detailed-report availability flag
explicit detailed section count
explicit section sequence label
explicit no-effect-chain label
explicit evidence level label
```

Validation:

```
sh scripts/test-l-ui-rendering-detailed-report-refinement.sh
sh scripts/test-l-ui-rendering.sh
```

Boundary: no-effect rendering/report metadata only. No interactive UI, terminal control, command execution, Lat execution, LIR execution, Nucleus task execution, file I/O, network I/O, state mutation, server interaction, recovery behavior, hardware behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/L_UI_RENDERING_README_STATUS_ALIGNMENT.md

Source title: L-UI Rendering README/Status Alignment

Lines: 26 | Words: 65 | SHA-256: 2a3656a4e11e664bfaf8ed11c3358d839ceeef7e00b5706e35a23e6a5744eb59

L-UI Rendering README/Status Alignment

Status: README/status alignment

This record aligns public entry-point documentation after the L-UI rendering detailed report refinement.

Reviewed files:

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT.md
docs/status/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT_STATUS.md
```

Resolution:

```
Add the L-UI detailed report refinement to README.md.
Add this alignment record to the status index.
Advance root and detailed status queues.
```

Boundary: documentation/status alignment only. No implementation behavior is changed.

docs/status/LANGUAGE_REPRESENTATION_REVIEW.md

Source title: Language Representation Review

Lines: 54 | Words: 152 | SHA-256: 649ea0a2f2b34055c65d3da374d997e4ce28a676c1a8750041e3a2112b394be6

Language Representation Review

Status: representation review

This record reviews the current Lat, LIR, and L-UI representation posture before any further Nucleus task execution refinement.

Reviewed representation surfaces:

```
docs/LAT_LANGUAGE_GRAMMAR_IMPLEMENTATION.md
docs/LAT_SEMANTIC_DIAGNOSTICS_REFINEMENT.md
docs/LAT_TO_LIR_LOWERING_IMPLEMENTATION.md
docs/LAT_PIPELINE_IMPLEMENTATION.md
docs/LAT_PIPELINE_REPORT_REFINEMENT.md
docs/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_REFINEMENT.md
docs/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_INTEGRATION_AUDIT.md
docs/LAT_SPECIFIC_LIR_REFINEMENT_IMPLEMENTATION.md
docs/LIR_SHAPE_IMPLEMENTATION.md
docs/LIR_REPORT_REFINEMENT.md
docs/L_UI_RENDERING_IMPLEMENTATION.md
docs/L_UI_RENDERING_DETAILED_REPORT_REFINEMENT.md
```

Finding:

```
The language representation stack is stable enough for a bounded no-effect Nucleus
report/refinement slice.
```

Current representation boundary:

```
Lat remains parser/semantic/lowering metadata only.
LIR remains shape/report metadata only.
L-UI rendering remains presentation/report metadata only.
No Lat execution, LIR execution, command execution, runtime behavior, file I/O, network I/O,
mutation, recovery behavior, or hardware behavior exists.
```

Decision:

```
Permit the next Nucleus task execution refinement only as a no-effect report/refinement slice.
Do not permit effect-performing task execution.
Do not permit runtime behavior.
Do not permit command execution.
```

Recommended next slice:

```
Nucleus task execution no-effect report alignment
```

Boundary: representation/status review only. No implementation behavior is changed.

docs/status/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_STATUS.md

Source title: Lat Pipeline Diagnostic Integration Status

Lines: 67 | Words: 150 | SHA-256: 246b169d5ada094027cab8986217f998e20126f79e7538e4a915294448508bd3

Lat Pipeline Diagnostic Integration Status

Status: merged companion diagnostics surface

This record tracks the Lat pipeline diagnostic integration slice.

Merged PR:

```
#109 Add Lat pipeline diagnostic integration
```

Merged commit:

```
d2880fca57bb753753844b683c14882af5ec4560
```

Primary files:

```
include/latticra/lat_pipeline_diagnostics.h
include/latticra/lat_to_lir_diagnostics.h
src/lat_pipeline_diagnostics.c
src/lat_pipeline_diagnostics_eval.c
src/lat_pipeline_diagnostics_report.c
src/lat_to_lir_diagnostics.c
tests/lat_pipeline_diagnostic_integration_refinement.c
docs/LAT_PIPELINE_DIAGNOSTIC_INTEGRATION_REFINEMENT.md
scripts/test-lat-pipeline-diagnostic-integration-refinement.sh
.github/workflows/lat-pipeline-diagnostic-integration-refinement.yml
```

What it adds:

```
pipeline diagnostic class
pipeline error capture
failed pipeline stage capture
semantic diagnostic class capture
semantic error capture
semantic diagnostic count capture
first diagnostic declaration index
first diagnostic clause index
optional Lat-to-LIR lowering diagnostic class capture
lowering error capture
model error capture
LIR error capture
lowering model counts
lowering transition source index
pipeline_failed flag
semantic_failed flag
lowering_failed flag
model_failed flag
lir_failed flag
no_effect_issue flag
evidence_level
```

Validation:

```
sh scripts/test-lat-pipeline-diagnostic-integration-refinement.sh
sh scripts/test-lat-pipeline.sh
```

Boundary: diagnostic metadata only. No Lat execution, LIR execution, compiler, interpreter, runtime, state mutation, file I/O, network I/O, hardware behavior, or operating-system completeness is added.

docs/status/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_AUDIT_STATUS.md

Source title: Lat Pipeline Diagnostic Main Test Audit Status

Lines: 45 | Words: 120 | SHA-256: c2344da1db019682473902936b3ad81104559fc0243914afeaa54c1e1e1e5e63

Lat Pipeline Diagnostic Main Test Audit Status

Status: merged audit guard

This record tracks the Lat pipeline diagnostic main test integration audit slice.

Merged PR:

```
#120 Add Lat pipeline diagnostic main test audit
```

Merged commit:

```
701a9f6ac8fc4891d8cae78b40ec763736b06045
```

Primary files:

```
docs/LAT_PIPELINE_DIAGNOSTIC_MAIN_TEST_INTEGRATION_AUDIT.md
scripts/test-lat-pipeline-diagnostic-main-test-integration-audit.sh
.github/workflows/lat-pipeline-diagnostic-main-test-audit.yml
```

What it verifies:

```
main Lat pipeline runner links src/lat_pipeline_diagnostics.c
main Lat pipeline runner links src/lat_pipeline_diagnostics_eval.c
main Lat pipeline runner links src/lat_pipeline_diagnostics_report.c
main Lat pipeline runner links src/lat_to_lir_diagnostics.c
main Lat pipeline runner compiles tests/lat_pipeline_diagnostic_integration_refinement.c
Lat Pipeline workflow runs scripts/test-lat-pipeline.sh
```

Validation:

```
sh scripts/test-lat-pipeline-diagnostic-main-test-integration-audit.sh
sh scripts/test-lat-pipeline.sh
```

Boundary: audit/status only. No Lat execution, LIR execution, compiler, interpreter, runtime behavior, state mutation, file I/O, network I/O, hardware behavior, or operating-system completeness is added.

docs/status/LATTICRA_CONSOLE_FOUNDATION_STATUS.md

Source title: Latticra Console Foundation Status

Lines: 164 | Words: 274 | SHA-256: f71d8fc5032c53144c9231240173c83f6c69d9fe426499625e0b41692611eece

Latticra Console Foundation Status

Status: active Stage-0 foundation Date: 2026-05-26 Scope: LC C report surface, standalone and Panel installability, local-prefix metadata, session contract, workspace contract, namespace contract, rootfs contract, packages contract, init contract, host-adapter contract, Seal receipt-request contract, receipt payload schema, receipt payload artifact draft, receipt payload artifact review gate, receipt payload artifact review receipt contract, receipt payload artifact review receipt draft contract, receipt payload materialization plan, signature-request binding contract, OS-base planning contract, VM evidence contract, and no-effect authority baseline.

Current Evidence

```
latticra_console_component_present=1
latticra_console_c_api_present=1
latticra_console_report_surface_present=1
latticra_console_panel_component_present=1
latticra_console_install_marker_present=1
latticra_console_wrapper_present=1
latticra_console_configurable=1
panel_installable=1
standalone_installable=1
standalone_requires_panel=0
standalone_command_wrapper=latticra-lc
standalone_contract_status=metadata-only-contract
standalone_contract_present=1
standalone_contract_command=lc standalone
standalone_contract_profile=lc-standalone-console-v0
standalone_installer_profile=lc_standalone
standalone_install_profile=lc-standalone-install-v0
standalone_installer_preset=installer/configs/lc-standalone.installer.toml
standalone_local_installer_preset=installer/configs/lc-standalone-local.installer.toml
standalone_local_install_mode=local-prefix-install
session_contract_status=metadata-only-contract
session_contract_present=1
session_contract_command=lc session
session_contract_profile=lc-session-v0
runtime_session_created=0
runtime_process_spawn_allowed=0
interactive_shell_allowed=0
workspace_contract_status=metadata-only-contract
workspace_contract_present=1
workspace_contract_command=lc workspace
workspace_contract_profile=lc-workspace-v0
workspace_mount_allowed=0
workspace_file_write_allowed=0
workspace_mutation_allowed=0
namespace_contract_status=metadata-only-contract
namespace_contract_present=1
namespace_contract_command=lc namespace
namespace_contract_profile=lc-namespace-v0
namespace_mount_allowed=0
rootfs_contract_status=metadata-only-contract
rootfs_contract_present=1
rootfs_contract_command=lc rootfs
rootfs_contract_profile=lc-rootfs-v0
rootfs_image_create_allowed=0
rootfs_mount_allowed=0
rootfs_package_install_allowed=0
packages_contract_status=metadata-only-contract
packages_contract_present=1
packages_contract_command=lc packages
packages_contract_profile=lc-packages-v0
package_manifest_write_allowed=0
package_catalog_read_allowed=0
package_download_allowed=0
package_manager_execution_allowed=0
package_script_execution_allowed=0
init_contract_status=metadata-only-contract
init_contract_present=1
init_contract_command=lc init
init_contract_profile=lc-init-v0
pid1_claim_allowed=0
service_start_allowed=0
process_supervision_allowed=0
host_path_projection_allowed=0
namespace_mutation_allowed=0
substrate_bridge_status=metadata-bound
command_registry_status=seed-registry
command_registry_source=c-static-table
command_registry_no_effect=1
command_registry_host_process_launch_allowed=0
help_renderer_present=1
manpage_renderer_present=1
installed_help_reads_seed_registry=1
runtime_boundary_bound=1
seal_capability_labels_bound=1
boundary_report_present=1
future_host_command_requires_future_gate=1
future_os_command_requires_future_gate=1
seal_capability_grants_authority=0
```

```
host_embedding_status=planned
host_adapter_contract_status=metadata-only-contract
host_adapter_contract_present=1
host_adapter_contract_command=lc host-adapter
host_adapter_contract_profile=lc-host-adapter-v0
receipt_request_contract_status=metadata-only-contract
receipt_request_contract_present=1
receipt_request_contract_command=lc receipt-request
receipt_request_contract_profile=lc-receipt-request-v0
receipt_payload_schema_status=metadata-only-schema
receipt_payload_schema_present=1
receipt_payload_schema_command=lc receipt-payload
receipt_payload_schema_profile=lc-receipt-payload-schema-v0
receipt_payload_artifact_draft_status=metadata-only-draft
receipt_payload_artifact_draft_present=1
receipt_payload_artifact_draft_command=lc receipt-artifact
receipt_payload_artifact_draft_profile=lc-receipt-payload-artifact-draft-v0
receipt_payload_artifact_review_status=metadata-only-review-gate
receipt_payload_artifact_review_present=1
receipt_payload_artifact_review_command=lc receipt-artifact-review
receipt_payload_artifact_review_profile=lc-receipt-payload-artifact-review-v0
receipt_payload_artifact_review_receipt_status=metadata-only-receipt-contract
receipt_payload_artifact_review_receipt_present=1
receipt_payload_artifact_review_receipt_command=lc receipt-review-receipt
receipt_payload_artifact_review_receipt_profile=lc-receipt-payload-artifact-review-receipt-v0
receipt_payload_artifact_review_receipt_draft_status=metadata-only-review-receipt-draft
receipt_payload_artifact_review_receipt_draft_present=1
receipt_payload_artifact_review_receipt_draft_command=lc receipt-review-draft
receipt_payload_artifact_review_receipt_draft_profile=lc-receipt-payload-artifact-review-receipt
-draft-v0
receipt_payload_materialization_plan_status=metadata-only-plan
receipt_payload_materialization_plan_present=1
receipt_payload_materialization_plan_command=lc receipt-materialization-plan
receipt_payload_materialization_plan_profile=lc-receipt-payload-materialization-plan-v0
draft_review_receipt_present=0
materialization_preconditions_met=0
materialization_allowed=0
payload_artifact_present=0
payload_write_allowed=0
signature_request_binding_status=metadata-only-contract
signature_request_binding_present=1
signature_request_binding_command=lc signature-request
signature_request_binding_profile=lc-signature-request-binding-v0
signature_request_binding_allowed=0
seal_signature_request_present=0
receipt_write_allowed=0
os_base_contract_status=metadata-only-contract
os_base_contract_present=1
os_base_contract_command=lc os-contract
os_base_contract_profile=lc-os-base-v0
vm_evidence_contract_status=metadata-only-contract
vm_evidence_contract_present=1
vm_evidence_contract_command=lc vm-evidence
vm_evidence_contract_profile=lc-vm-evidence-v0
os_base_status=planned-no-boot-authority
```

Non-Claims

```
shell_execution_authority=0
external_host_process_launch=0
host_mutation_authority=0
network_authority=0
runtime_enforcement_authority=0
boot_authority=0
production_os_claim=0
```

Verification

Run:

```
make latticra-console
```

The test compiles the LC C foundation, renders the LC report, checks key no-effect fields, and verifies Panel/install metadata references.

docs/status/LATTICRA_NO_EFFECT_CLI_RPM_SPEC_UPDATE_STATUS.md

Source title: Latticra No-Effect CLI RPM Spec Update Status

Latticra No-Effect CLI RPM Spec Update Status

Status: packaging/spec update status Date: 2026-05-21 Scope: status record for adding the no-effect CLI payload to the local Fedora RPM spec.

Summary

The local Fedora RPM spec now includes the no-effect latticra CLI payload.

The spec transition compiles the guarded C CLI source and installs it as the package binary payload.

This is a spec update only.

It is not a completed RPM build transcript.

It is not disposable Fedora VM validation of the expanded payload.

It is not host install readiness for the CLI payload.

Spec transition recorded

```
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_contains_compiled_c_binary=1
buildarch_noarch_removed=1
cli_status_surface_guarded_before_packaging=1
cli_binary_compiled_in_build_section=1
cli_binary_installed_to_bindir=1
readme_installed_to_docdir=1
rpm_payload_validated=0
```

Current spec payload target

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

Required spec lines

```
sh scripts/test-latticra-no-effect-cli-status-surface.sh
cc %{optflags} -std=c99 -Wall -Wextra -Werror -pedantic src/latticra_cli.c -o build/latticra
install -m 0755 build/latticra %{buildroot}%{_bindir}/latticra
install -m 0644 README.md %{buildroot}%{_docdir}/%{name}/README.md
%{_bindir}/latticra
%doc %{_docdir}/%{name}/README.md
```

Forbidden surfaces still absent

```
/etc/latticra=absent
/usr/lib/systemd/system/latticra.service=absent
/usr/lib/modules=absent
/boot/latticra=absent
/usr/share/selinux=absent
systemctl=absent
rpm_scriptlets=absent
```

Historical evidence boundary

The previous successful disposable Fedora VM validation remains limited to the old documentation-only local RPM payload:

```
validated_package=latticra-0.0.0-0.1.local.fc44.noarch.rpm
validated_payload=/usr/share/doc/latticra/README.md
historical_disposable_vm_rpm_evidence_remains_limited=1
```

That evidence does not validate /usr/bin/latticra.

Current readiness classification

```
cli_payload_contract_present=1
cli_status_surface_implemented=1
cli_packaging_alignment_present=1
fedora_spec_updated_for_cli=1
rpm_payload_expansion_performed=1
rpm_build_transcript_present=0
rpm_payload_validated=0
disposable_vm_cli_validation_completed=0
host_install_ready_for_cli_payload=0
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Validation

```
sh scripts/test-latticra-no-effect-cli-rpm-spec-update-status.sh
```

Expected output:

```
latticra_no_effect_cli_rpm_spec_update_status: ok
```

Next recommended lane

```
Add disposable Fedora VM CLI payload validation transcript contract
```

That future lane should define the required transcript fields for validating the expanded payload inside a disposable Fedora VM.

Non-claims

This status record is not a completed RPM build transcript.

It is not RPM install evidence.

It is not disposable Fedora VM validation of the CLI payload.

It is not host install readiness for the CLI payload.

It is not production readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, runtime behavior, Lat execution, LIR execution, service management, kernel integration, SELinux policy integration, update safety, recovery safety, malware prevention, ransomware prevention, sandboxing, or a production installer claim.

docs/status/LATTICRA_PANEL_LOCAL_INSTALL_EVIDENCE_STATUS.md

Source title: Latticra Panel Local Install Evidence Status

Lines: 201 | Words: 566 | SHA-256: d785c9595eb5b09cc8a00ddd26819d9b7ea1e8a2725fb9c511646e52a1fa2461

Latticra Panel Local Install Evidence Status

Status: evidence status alignment Date: 2026-05-22 Scope: Fedora Workstation user-local Latticra Panel install verification after the merged Panel installer, README, CI, and self-copy fix work.

Summary

A Fedora Workstation user-local Latticra Panel install verification completed successfully.

The operator verified that the installed command wrappers resolve from ~/.local/bin, that the Latticra local prefix exists, that the installed payload tree exists, that receipts are present, that the desktop entry and desktop icon are present, that the Latticra Panel launcher is available, and that Latticra Seal can generate a report-only local report.

This is Fedora user-local Latticra Panel install evidence.

It is not production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, network authority, runtime enforcement authority, or a production security-product claim.

Evidence source

```
source=operator Fedora Workstation transcript
host_os=Fedora Linux 44 (Workstation Edition)
kernel=Linux 7.0.9-205.fc44.x86_64 x86_64 GNU/Linux
repo_path=/home/ckbryan/Latticra
install_prefix=/home/ckbryan/.local/share/latticra
user_bin=/home/ckbryan/.local/bin
```

Command wrapper evidence

```
which_latticra=/home/ckbryan/.local/bin/latticra
which_lat=/home/ckbryan/.local/bin/lat
which_latticra_seal=/home/ckbryan/.local/bin/latticra-seal
which_latticra_panel=/home/ckbryan/.local/bin/latticra-panel
```

Installed status output

```
Latticra is installed.
prefix=/home/ckbryan/.local/share/latticra
payload=/home/ckbryan/.local/share/latticra/lib/latticra
receipts=/home/ckbryan/.local/share/latticra/share/latticra/receipts
```

Verification report recorded

```
ok: prefix -&gt; /home/ckbryan/.local/share/latticra
ok: payload tree -&gt; /home/ckbryan/.local/share/latticra/lib/latticra
ok: config -&gt; /home/ckbryan/.local/share/latticra/etc/latticra/installer-config.toml
ok: receipts -&gt; /home/ckbryan/.local/share/latticra/share/latticra/receipts
ok: latticra command -&gt; /home/ckbryan/.local/bin/latticra
ok: lat command -&gt; /home/ckbryan/.local/bin/lat
ok: latticra-seal command -&gt; /home/ckbryan/.local/bin/latticra-seal
ok: latticra-panel command -&gt; /home/ckbryan/.local/bin/latticra-panel
ok: desktop entry -&gt; /home/ckbryan/.local/share/applications/latticra-panel.desktop
ok: desktop icon -&gt; /home/ckbryan/.local/share/icons/hicolor/256x256/apps/latticra-panel.png
ok: latticra-seal report generated
ok: Latticra Panel launcher is available
Latticra local install verification: ok
```

Current updater policy alignment

The current verifier additionally requires the Panel-owned updater policy and non-launching updater status report:

```
ok: updater config
ok: updater policy
ok: updater status report
ok: updater status dry-run command
ok: updater status apply command
ok: updater status network fetch authority disabled
ok: updater status apply mode
```

The updater remains a guarded local-checkout Panel lane:

```
updater_panel_owned=1
updater_policy_present=1
updater_status_report_present=1
updater_preview_command=updater dry-run
updater_apply_command=updater apply
updater_apply_mode=guarded-local-prefix-reinstall
updater_network_fetch_authority=0
updater_root_authority=0
updater_system_mutation_authority=0
```

Seal report evidence

```
LATTICRA SEAL REPORT

timestamp_utc=20260522T172638Z
mode=report-only
prefix=/home/ckbryan/.local/share/latticra
kernel=Linux 7.0.9-205.fc44.x86_64 x86_64 GNU/Linux
os=Fedora Linux 44 (Workstation Edition)
network_authority=0
runtime_enforcement_authority=0
```

Installed component markers:

```
developer-cli-helpers.installed
docs-and-examples.installed
latticra-seal.installed
lat-tooling.installed
lir-contracts.installed
```

Current readiness classification

```
latticra_panel_installer_present=1
latticra_panel_user_local_install_verified=1
latticra_panel_command_wrapper_present=1
latticra_panel_desktop_entry_present=1
latticra_panel_desktop_icon_present=1
latticra_panel_launcher_available=1
latticra_local_prefix_present=1
latticra_payload_tree_present=1
latticra_receipts_present=1
latticra_seal_report_generated=1
updater_policy_present=1
updater_status_report_present=1
updater_network_fetch_authority=0
updater_apply_mode=guarded-local-prefix-reinstall
seal_report_only_mode=1
network_authority=0
runtime_enforcement_authority=0
root_authority=0
kernel_modification_performed=0
systemd_modification_performed=0
selinux_modification_performed=0
production_installer_ready=0
root_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Boundary statement

This evidence is limited to a Fedora Workstation user-local Latticra Panel installation path.

The verified install path is under:

```
/home/ckbryan/.local/share/latticra
/home/ckbryan/.local/bin
/home/ckbryan/.local/share/applications
/home/ckbryan/.local/share/icons
```

This evidence does not validate:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
kernel modification
systemd modification
SELinux modification
network operation
runtime enforcement
production host installation
```

Guard validation

The evidence status alignment is guarded by:

```
sh scripts/test-latticra-panel-local-install-evidence-status.sh
```

Expected output:

```
latticra_panel_local_install_evidence_status: ok
```

Next recommended Latticra Panel lane

```
Add public README/status wording limited to Fedora user-local Latticra Panel installation
evidence
```

That wording must remain narrow and should not claim production installer readiness, root installation readiness, Fedora approval, Fedora distribution readiness, immutable Fedora readiness, daily-driver readiness, runtime enforcement, kernel integration, systemd integration, SELinux integration, or production security-product status.

Non-claims

This status record is not production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, update safety, rollback safety, malware prevention, ransomware prevention, sandboxing, runtime enforcement, kernel integration, systemd integration, SELinux integration, or a production security-product claim.

docs/status/LATTICRA_PANEL_LOCAL_INSTALL_PUBLIC_ENTRYPOINT_ALIGNMENT.md

Source title: Latticra Panel Local Install Public Entrypoint Alignment

Lines: 129 | Words: 395 | SHA-256: 60a6aceb5b800c716906d21d314d6b9f76ab8867653bb85d90bb7ee7272342e4

Latticra Panel Local Install Public Entrypoint Alignment

Status: public entrypoint alignment Date: 2026-05-22 Scope: public README/status alignment after the successful Fedora Workstation user-local Latticra Panel install evidence milestone.

Summary

The public Latticra entrypoint now has a safe Latticra Panel installer section, and the repository now has a dedicated evidence record for the successful Fedora Workstation user-local Panel verification.

This alignment records the narrow public meaning of that milestone:

```
latticra_panel_installer_present=1
latticra_panel_user_local_install_verified=1
latticra_panel_command_wrapper_present=1
latticra_panel_desktop_entry_present=1
latticra_panel_desktop_icon_present=1
latticra_panel_launcher_available=1
latticra_local_prefix_present=1
latticra_payload_tree_present=1
latticra_receipts_present=1
latticra_seal_report_generated=1
seal_report_only_mode=1
network_authority=0
runtime_enforcement_authority=0
root_authority=0
production_installer_ready=0
root_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

Public entrypoint wording

The root README identifies Latticra Panel as the graphical local installer and control panel for Latticra, Lat, LIR, and Latticra Seal.

It documents the guarded user-local install path:

```
make -C installer local-example
make -C installer verify-local
```

It also states the critical boundary:

```
This installer is user-local only. It does not use root, modify the kernel, modify systemd,
change SELinux, or use network authority.
```

That wording remains valid after the local-install evidence milestone.

Evidence record

The local-install evidence milestone is recorded in:

```
docs/status/LATTICRA_PANEL_LOCAL_INSTALL_EVIDENCE_STATUS.md
```

The recorded evidence includes:

```
which_latticra=/home/ckbryan/.local/bin/latticra
which_lat=/home/ckbryan/.local/bin/lat
which_latticra_seal=/home/ckbryan/.local/bin/latticra-seal
which_latticra_panel=/home/ckbryan/.local/bin/latticra-panel
Latticra local install verification: ok
ok: desktop entry -&gt; /home/ckbryan/.local/share/applications/latticra-panel.desktop
ok: desktop icon -&gt; /home/ckbryan/.local/share/icons/hicolor/256x256/apps/latticra-panel.png
ok: latticra-seal report generated
ok: Latticra Panel launcher is available
network_authority=0
runtime_enforcement_authority=0
```

Safe public claim

A careful public claim is:

```
Latticra Panel now has Fedora Workstation user-local install evidence covering command wrappers,
 local prefix layout, payload tree, receipts, desktop metadata, Panel launcher availability, and
 Latticra Seal report-only output.
```

That is intentionally limited.

Non-claims

This public entrypoint alignment does not mean Latticra Panel is a production installer, root installer, Fedora-approved package, Fedora distribution-ready package, daily-driver installer, immutable Fedora installer, runtime enforcement system, kernel integration layer, systemd integration layer, SELinux integration layer, malware prevention system, ransomware prevention system, sandbox, or production security product.

It does not validate:

```
/usr/bin/latticra
/etc/latticra
/usr/lib/systemd/system/latticra.service
/usr/lib/modules
/boot/latticra
kernel modification
systemd modification
SELinux modification
network operation
runtime enforcement
production host installation
```

Guard validation

This alignment is guarded by:

```
sh scripts/test-latticra-panel-local-install-public-entrypoint-alignment.sh
```

Expected output:

```
latticra_panel_local_install_public_entrypoint_alignment: ok
```

Next recommended Latticra Panel lane

```
Add status-index integration for Latticra Panel local-install evidence and public-entrypoint
alignment
```

That follow-up should preserve the same non-claims and should not expand the installer beyond the verified user-local evidence boundary.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_DELIVERY_GATE_STATUS.md

Source title: Latticra Panel Signed Updater Delivery Gate Status

Lines: 100 | Words: 281 | SHA-256: 44faf478e68675d7e5b66a76ec58ca448e0984a77c546714d73e83d35d5f3b13

Latticra Panel Signed Updater Delivery Gate Status

Status: no-effect signed updater delivery gate status Date: 2026-05-26 CDT Scope: status checkpoint after adding the closed signed updater delivery gate for Latticra Panel.

Summary

Latticra Panel now has a closed signed updater delivery gate.

The current updater remains a Panel-owned local-checkout reinstall lane. The signed delivery gate records the missing manifest, signature, artifact, rollback, validation, receipt, and operator-confirmation evidence that must exist before any future signed or network-delivered updater can open.

Status Fields

```
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_delivery_gate_guard_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
updater_current_source_strategy=current-source-checkout
updater_current_update_channel=local-checkout
updater_current_apply_mode=guarded-local-prefix-reinstall
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_delivery_gate_decision=blocked-missing-signed-manifest-artifact-verification-and-
rollback-evidence
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
updater_network_fetch_enabled=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_verified=0
channel_policy_required=1
channel_policy_present=0
compatibility_policy_required=1
compatibility_policy_present=0
rollback_plan_required=1
rollback_plan_present=0
rollback_evidence_present=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
staged_update_allowed=0
signed_update_apply_allowed=0
update_activation_allowed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel has a closed signed-updater delivery gate for future update delivery work.
```

That does not mean Latticra has signed update delivery, network self-update, a trusted remote update repository, artifact verification, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-delivery-gate.sh
```

Expected output:

```
latticra_panel_signed_updater_delivery_gate: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not signed update evidence, update-server evidence, network fetch evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_DENIAL_TRANSCRIPT_STATUS.md

Source title: Latticra Panel Signed Updater Denial Transcript Status

Lines: 109 | Words: 287 | SHA-256: 140822c4cfd5a75a9d92f0dc3c92ce0796963c8c057cace8174feb80687ba87d

Latticra Panel Signed Updater Denial Transcript Status

Status: no-effect signed updater denial transcript status Date: 2026-05-26 CDT Scope: status checkpoint after adding the no-effect signed updater denial transcript for Latticra Panel.

Summary

Latticra Panel now has a signed updater denial transcript.

The transcript records why signed update delivery remains denied while preserving zero network fetch, zero remote repository trust, zero transcript file writes, zero staging, zero apply, and zero host mutation.

Status Fields

```
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_signed_updater_denial_transcript_guard_present=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_delivery_gate_decision=blocked-missing-signed-manifest-artifact-verification-and-
rollback-evidence
signed_updater_denial_transcript_present=1
signed_updater_denial_transcript_state=recorded-no-effect
signed_updater_denial_transcript_stdout_only=1
signed_updater_denial_transcript_file_write_enabled=0
signed_updater_denial_decision=deny-signed-update-delivery
signed_updater_denial_reason=missing-signed-manifest-artifact-verification-and-rollback-evidence
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
updater_network_fetch_enabled=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_verified=0
channel_policy_required=1
channel_policy_present=0
compatibility_policy_required=1
compatibility_policy_present=0
rollback_plan_required=1
rollback_plan_present=0
rollback_evidence_present=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel records a no-effect denial transcript for its closed signed-updater delivery
gate.
```

That does not mean Latticra has signed update delivery, network self-update, a trusted remote update repository, artifact verification, rollback implementation, persistent denial records, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-denial-transcript.sh
```

Expected output:

```
latticra_panel_signed_updater_denial_transcript: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, transcript persistence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_CONTRACT_STATUS.md

Source title: Latticra Panel Signed Updater Manifest Fixture Contract Status

Lines: 118 | Words: 300 | SHA-256: 4b80fb221d7b39461625f6e515e4cf1e3e57833a966593a70a20240b6526a418

Latticra Panel Signed Updater Manifest Fixture Contract Status

Status: local no-effect signed updater manifest fixture contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the local no-effect signed updater manifest fixture contract for Latticra Panel.

Summary

Latticra Panel now has a local signed-updater manifest fixture contract.

The fixture is reviewable manifest-shaped input for future validation work, but it is not a trusted signed manifest and does not open network fetch, artifact verification, staging, apply, receipt writes, or host mutation.

Status Fields

```
latticra_panel_signed_updater_manifest_fixture_contract_present=1
latticra_panel_signed_updater_manifest_fixture_guard_present=1
latticra_panel_signed_updater_manifest_fixture_present=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
manifest_fixture_path=fixtures/latticra-panel/signed-updater-manifest.fixture.toml
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_denial_transcript_present=1
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_present=1
signed_updater_manifest_fixture_file_present=1
manifest_fixture_schema=latticra-panel-signed-updater-manifest-fixture-v0
manifest_fixture_scope=local-no-effect
manifest_fixture_effect=none
manifest_fixture_trusted_for_apply=0
manifest_fixture_validated_for_apply=0
trusted_signed_manifest_present=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_present=0
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_present=0
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_present=0
artifact_signature_verified=0
channel_policy_required=1
channel_policy_present=0
compatibility_policy_required=1
compatibility_policy_present=0
rollback_plan_required=1
rollback_plan_present=0
rollback_evidence_present=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel has a local no-effect signed updater manifest fixture contract.
```

That does not mean Latticra has signed update delivery, network self-update, a trusted remote update repository, verified artifacts, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-manifest-fixture-contract.sh
```

Expected output:

```
latticra_panel_signed_updater_manifest_fixture_contract: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_MANIFEST_FIXTURE_VALIDATION_STATUS.md

Source title: Latticra Panel Signed Updater Manifest Fixture Validation Status

Lines: 131 | Words: 309 | SHA-256: f722049a37c61cfa78218eb71028549597a84a91e4edb7e869bc53d0096b064b

Latticra Panel Signed Updater Manifest Fixture Validation Status

Status: local no-effect signed updater manifest fixture validation status Date: 2026-05-26 CDT Scope: status checkpoint after adding local no-effect validation for the Latticra Panel signed-updater manifest fixture.

Summary

Latticra Panel now validates the committed signed-updater manifest fixture as local no-effect input.

The validation checks fixture shape and closed-authority fields, but it does not trust the fixture for apply, verify update artifacts, stage updates, write receipts, or mutate host state.

Status Fields

```
latticra_panel_signed_updater_manifest_fixture_validation_present=1
latticra_panel_signed_updater_manifest_fixture_validation_guard_present=1
latticra_panel_signed_updater_manifest_fixture_contract_present=1
latticra_panel_signed_updater_manifest_fixture_present=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
manifest_fixture_path=fixtures/latticra-panel/signed-updater-manifest.fixture.toml
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_denial_transcript_present=1
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_validation_present=1
signed_updater_manifest_fixture_present=1
signed_updater_manifest_fixture_file_present=1
signed_updater_manifest_fixture_validated=1
signed_updater_manifest_fixture_validation_scope=shape-and-closed-authority-fields
manifest_schema_validated=1
manifest_fixture_flag_validated=1
manifest_fixture_scope_validated=1
manifest_fixture_effect_validated=1
local_checkout_strategy_validated=1
closed_authority_fields_validated=1
non_claims_validated=1
manifest_fixture_schema=latticra-panel-signed-updater-manifest-fixture-v0
manifest_fixture_scope=local-no-effect
manifest_fixture_effect=none
manifest_fixture_trusted_for_apply=0
manifest_fixture_validated_for_apply=0
signed_updater_manifest_fixture_valid_for_apply=0
trusted_signed_manifest_present=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_present=0
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_present=0
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_present=0
artifact_signature_verified=0
channel_policy_required=1
channel_policy_present=0
compatibility_policy_required=1
compatibility_policy_present=0
rollback_plan_required=1
rollback_plan_present=0
rollback_evidence_present=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
validation_write_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
```

```
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel validates a local no-effect signed updater manifest fixture.
```

That does not mean Latticra has signed update delivery, network self-update, a trusted remote update repository, verified artifacts, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-manifest-fixture-validation.sh
```

Expected output:

```
latticra_panel_signed_updater_manifest_fixture_validation: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_CONTRACT_STATUS.md

Source title: Latticra Panel Signed Updater State Fixture Contract Status

Lines: 145 | Words: 329 | SHA-256: 7b041c457898e3d86c58197caf6680ff3a1ebce5dafb046e91b84ced8e7162f8

Latticra Panel Signed Updater State Fixture Contract Status

Status: local no-effect signed updater state fixture contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the local no-effect signed updater state fixture contract for Latticra Panel.

Summary

Latticra Panel now has a local signed-updater state fixture contract.

The fixture names the future updater states, but the current state remains blocked and no transition execution, staging, activation, rollback execution, receipt writes, or host mutation is enabled.

Status Fields

```
latticra_panel_signed_updater_state_fixture_contract_present=1
latticra_panel_signed_updater_state_fixture_guard_present=1
latticra_panel_signed_updater_state_fixture_present=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_signed_updater_manifest_fixture_contract_present=1
latticra_panel_signed_updater_manifest_fixture_validation_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
state_fixture_path=fixtures/latticra-panel/signed-updater-state.fixture.toml
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_denial_transcript_present=1
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_validation_present=1
signed_updater_state_fixture_contract_present=1
signed_updater_state_fixture_present=1
signed_updater_state_fixture_file_present=1
state_fixture_schema=latticra-panel-signed-updater-state-fixture-v0
state_fixture_scope=local-no-effect
state_fixture_effect=none
state_fixture_trusted_for_apply=0
state_fixture_validated_for_apply=0
state_catalog_present=1
state_available_declared=1
state_downloaded_declared=1
state_verified_declared=1
state_staged_declared=1
state_armed_declared=1
state_applied_declared=1
state_rolled_back_declared=1
state_failed_declared=1
state_blocked_declared=1
current_update_state=blocked
requested_update_state=blocked
state_transition_decision=deny-transition
state_transition_reason=missing-signed-manifest-artifact-verification-rollback-and-operator-conf
irmation
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
available_state_materialized=0
downloaded_state_materialized=0
verified_state_materialized=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
rolled_back_state_materialized=0
failed_state_materialized=0
blocked_state_recorded=1
trusted_signed_manifest_present=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_present=0
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_present=0
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_present=0
artifact_signature_verified=0
rollback_plan_required=1
rollback_plan_present=0
rollback_execution_allowed=0
rollback_execution_performed=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
```

```
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel has a local no-effect signed updater state fixture contract.
```

That does not mean Latticra has state transition execution, signed update delivery, network self-update, a trusted remote update repository, verified artifacts, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-fixture-contract.sh
```

Expected output:

```
latticra_panel_signed_updater_state_fixture_contract: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not update-state evidence, state-transition execution, signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_FIXTURE_VALIDATION_STATUS.md

Source title: Latticra Panel Signed Updater State Fixture Validation Status

Lines: 160 | Words: 349 | SHA-256: 5bdf31d5aede6f7c8a219bc99c8d4771e28bf52072276a7864f46c50d7a8cf56

Latticra Panel Signed Updater State Fixture Validation Status

Status: local no-effect signed updater state fixture validation status Date: 2026-05-26 CDT Scope: status checkpoint after adding local no-effect validation for the Latticra Panel signed-updater state fixture.

Summary

Latticra Panel now validates the committed signed-updater state fixture as local no-effect input.

The validation checks fixture shape, state catalog, blocked-state posture, and closed transition fields, but it does not transition update state, stage updates, activate updates, execute rollback, write receipts, or mutate host state.

Status Fields

```
latticra_panel_signed_updater_state_fixture_validation_present=1
latticra_panel_signed_updater_state_fixture_validation_guard_present=1
latticra_panel_signed_updater_state_fixture_contract_present=1
latticra_panel_signed_updater_state_fixture_present=1
latticra_panel_signed_updater_manifest_fixture_validation_present=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
state_fixture_path=fixtures/latticra-panel/signed-updater-state.fixture.toml
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_denial_transcript_present=1
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_validation_present=1
signed_updater_state_fixture_contract_present=1
signed_updater_state_fixture_validation_present=1
signed_updater_state_fixture_present=1
signed_updater_state_fixture_file_present=1
signed_updater_state_fixture_validated=1
signed_updater_state_fixture_validation_scope=shape-state-catalog-and-closed-transition-fields
state_schema_validated=1
state_fixture_flag_validated=1
state_fixture_scope_validated=1
state_fixture_effect_validated=1
state_catalog_validated=1
blocked_state_validated=1
closed_transition_fields_validated=1
closed_authority_fields_validated=1
non_claims_validated=1
state_fixture_schema=latticra-panel-signed-updater-state-fixture-v0
state_fixture_scope=local-no-effect
state_fixture_effect=none
state_fixture_trusted_for_apply=0
state_fixture_validated_for_apply=0
signed_updater_state_fixture_valid_for_transition=0
signed_updater_state_fixture_valid_for_apply=0
state_catalog_present=1
state_available_declared=1
state_downloaded_declared=1
state_verified_declared=1
state_staged_declared=1
state_armed_declared=1
state_applied_declared=1
state_rolled_back_declared=1
state_failed_declared=1
state_blocked_declared=1
current_update_state=blocked
requested_update_state=blocked
state_transition_decision=deny-transition
state_transition_reason=missing-signed-manifest-artifact-verification-rollback-and-operator-conf
irmation
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
available_state_materialized=0
downloaded_state_materialized=0
verified_state_materialized=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
rolled_back_state_materialized=0
failed_state_materialized=0
blocked_state_recorded=1
trusted_signed_manifest_present=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_present=0
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_present=0
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_present=0
artifact_signature_verified=0
rollback_plan_required=1
rollback_plan_present=0
```

```
rollback_execution_allowed=0
rollback_execution_performed=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
validation_write_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel validates a local no-effect signed updater state fixture.
```

That does not mean Latticra has state transition execution, signed update delivery, network self-update, a trusted remote update repository, verified artifacts, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-fixture-validation.sh
```

Expected output:

```
latticra_panel_signed_updater_state_fixture_validation: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not update-state evidence, state-transition execution, signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_REVIEW_STATUS.md

Source title: Latticra Panel Signed Updater State Transition Denial Review Status

Lines: 158 | Words: 346 | SHA-256: 3ea3a88940c6d6bb9493b1be1455c7299e02f6d3f42930e4fbf06440f3edab1c

Latticra Panel Signed Updater State Transition Denial Review Status

Status: no-effect signed updater state transition denial review status Date: 2026-05-26 CDT Scope: status checkpoint after adding the stdout-only signed-updater state transition denial review for Latticra Panel.

Summary

Latticra Panel now has a no-effect review for the signed-updater state transition denial transcript.

The review upholds the blocked state transition, but it does not transition state, stage updates, activate updates, execute rollback, write receipts, write review files, or mutate host state.

Status Fields

```
latticra_panel_signed_updater_state_transition_denial_review_present=1
latticra_panel_signed_updater_state_transition_denial_review_guard_present=1
latticra_panel_signed_updater_state_transition_denial_transcript_present=1
latticra_panel_signed_updater_state_fixture_validation_present=1
latticra_panel_signed_updater_state_fixture_validated=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_denial_transcript_present=1
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_validation_present=1
signed_updater_state_fixture_contract_present=1
signed_updater_state_fixture_validation_present=1
signed_updater_state_fixture_validated=1
signed_updater_state_transition_denial_transcript_present=1
signed_updater_state_transition_denial_transcript_state=recorded-no-effect
signed_updater_state_transition_denial_review_present=1
signed_updater_state_transition_denial_review_state=reviewed-no-effect
signed_updater_state_transition_denial_review_stdout_only=1
signed_updater_state_transition_denial_review_file_write_enabled=0
signed_updater_state_transition_denial_review_decision=uphold-deny-state-transition
signed_updater_state_transition_denial_review_reason=denial-transcript-confirms-missing-signed-m
anifest-artifact-verification-rollback-and-operator-confirmation
signed_updater_state_transition_denial_decision=deny-state-transition
signed_updater_state_transition_denial_reason=missing-signed-manifest-artifact-verification-roll
back-and-operator-confirmation
state_fixture_path=fixtures/latticra-panel/signed-updater-state.fixture.toml
state_fixture_schema=latticra-panel-signed-updater-state-fixture-v0
state_fixture_validated_for_transition=0
state_fixture_validated_for_apply=0
signed_updater_state_fixture_valid_for_transition=0
signed_updater_state_fixture_valid_for_apply=0
state_catalog_present=1
state_available_declared=1
state_downloaded_declared=1
state_verified_declared=1
state_staged_declared=1
state_armed_declared=1
state_applied_declared=1
state_rolled_back_declared=1
state_failed_declared=1
state_blocked_declared=1
current_update_state=blocked
requested_update_state=blocked
state_transition_decision=deny-transition
state_transition_review_decision=uphold-denial
state_transition_reason=missing-signed-manifest-artifact-verification-rollback-and-operator-conf
irmation
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
available_state_materialized=0
downloaded_state_materialized=0
verified_state_materialized=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
rolled_back_state_materialized=0
failed_state_materialized=0
blocked_state_recorded=1
trusted_signed_manifest_present=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_present=0
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_present=0
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_present=0
artifact_signature_verified=0
rollback_plan_required=1
rollback_plan_present=0
rollback_execution_allowed=0
```

```
rollback_execution_performed=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
validation_write_performed=0
transcript_write_performed=0
review_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel records a no-effect signed updater state transition denial review.
```

That does not mean Latticra has state transition execution, signed update delivery, network self-update, a trusted remote update repository, verified artifacts, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-transition-denial-review.sh
```

Expected output:

```
latticra_panel_signed_updater_state_transition_denial_review: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not update-state evidence, state-transition execution, signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_SIGNED_UPDATER_STATE_TRANSITION_DENIAL_TRANSCRIPT_STATUS.md

Source title: Latticra Panel Signed Updater State Transition Denial Transcript Status

Lines: 151 | Words: 342 | SHA-256: 7cbfc3a1685c1befddee6d51a789e5fc1d4bac441519b792dc87a99dcd6bb2d6

Latticra Panel Signed Updater State Transition Denial Transcript Status

Status: no-effect signed updater state transition denial transcript status Date: 2026-05-26 CDT Scope: status checkpoint after adding the stdout-only signed-updater state transition denial transcript for Latticra Panel.

Summary

Latticra Panel now has a no-effect transcript for the denied signed-updater state transition.

The transcript records the blocked state fixture posture and denial reason, but it does not transition state, stage updates, activate updates, execute rollback, write receipts, write transcript files, or mutate host state.

Status Fields

```
latticra_panel_signed_updater_state_transition_denial_transcript_present=1
latticra_panel_signed_updater_state_transition_denial_transcript_guard_present=1
latticra_panel_signed_updater_state_fixture_validation_present=1
latticra_panel_signed_updater_state_fixture_validated=1
latticra_panel_signed_updater_delivery_gate_present=1
latticra_panel_signed_updater_denial_transcript_present=1
latticra_panel_updater_present=1
latticra_panel_updater_owned=1
signed_updater_delivery_gate_present=1
signed_updater_delivery_gate_state=closed
signed_updater_denial_transcript_present=1
signed_updater_manifest_fixture_contract_present=1
signed_updater_manifest_fixture_validation_present=1
signed_updater_state_fixture_contract_present=1
signed_updater_state_fixture_validation_present=1
signed_updater_state_fixture_validated=1
signed_updater_state_transition_denial_transcript_present=1
signed_updater_state_transition_denial_transcript_state=recorded-no-effect
signed_updater_state_transition_denial_transcript_stdout_only=1
signed_updater_state_transition_denial_transcript_file_write_enabled=0
signed_updater_state_transition_denial_decision=deny-state-transition
signed_updater_state_transition_denial_reason=missing-signed-manifest-artifact-verification-roll
back-and-operator-confirmation
state_fixture_path=fixtures/latticra-panel/signed-updater-state.fixture.toml
state_fixture_schema=latticra-panel-signed-updater-state-fixture-v0
state_fixture_validated_for_transition=0
state_fixture_validated_for_apply=0
signed_updater_state_fixture_valid_for_transition=0
signed_updater_state_fixture_valid_for_apply=0
state_catalog_present=1
state_available_declared=1
state_downloaded_declared=1
state_verified_declared=1
state_staged_declared=1
state_armed_declared=1
state_applied_declared=1
state_rolled_back_declared=1
state_failed_declared=1
state_blocked_declared=1
current_update_state=blocked
requested_update_state=blocked
state_transition_decision=deny-transition
state_transition_reason=missing-signed-manifest-artifact-verification-rollback-and-operator-conf
irmation
state_transition_execution_allowed=0
state_transition_execution_performed=0
state_receipt_written=0
available_state_materialized=0
downloaded_state_materialized=0
verified_state_materialized=0
staged_state_materialized=0
armed_state_materialized=0
applied_state_materialized=0
rolled_back_state_materialized=0
failed_state_materialized=0
blocked_state_recorded=1
trusted_signed_manifest_present=0
signed_manifest_required=1
signed_manifest_present=0
manifest_signature_required=1
manifest_signature_present=0
manifest_signature_verified=0
artifact_hash_required=1
artifact_hash_present=0
artifact_hash_verified=0
artifact_signature_required=1
artifact_signature_present=0
artifact_signature_verified=0
rollback_plan_required=1
rollback_plan_present=0
rollback_execution_allowed=0
rollback_execution_performed=0
post_update_validation_required=1
post_update_validation_present=0
operator_confirmation_required=1
operator_confirmation_observed=0
update_receipt_required=1
update_receipt_written=0
```

```
signed_update_delivery_ready=0
network_self_update_ready=0
remote_update_repository_trust=0
remote_update_repository_trust_ready=0
network_fetch_authority=0
network_fetch_attempted=0
updater_network_fetch_enabled=0
staged_update_allowed=0
staged_update_performed=0
signed_update_apply_allowed=0
signed_update_apply_performed=0
update_activation_allowed=0
update_activation_performed=0
validation_write_performed=0
transcript_write_performed=0
host_mutation_allowed=0
host_mutation_performed=0
root_authority=0
system_mutation_authority=0
kernel_mutation_authority=0
systemd_mutation_authority=0
selinux_mutation_authority=0
boot_mutation_authority=0
production_update_ready=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra Panel records a no-effect signed updater state transition denial transcript.
```

That does not mean Latticra has state transition execution, signed update delivery, network self-update, a trusted remote update repository, verified artifacts, rollback implementation, production update readiness, or production installer readiness.

Guard Validation

This status record is guarded by:

```
sh scripts/test-latticra-panel-signed-updater-state-transition-denial-transcript.sh
```

Expected output:

```
latticra_panel_signed_updater_state_transition_denial_transcript: ok
```

Next Recommended Lane

```
Add Panel signed-updater state transition denial disposition without enabling transition
execution or update apply.
```

Non-Claims

This status record is not update-state evidence, state-transition execution, signed update evidence, network fetch evidence, update-server evidence, trusted repository evidence, artifact-verification evidence, rollback evidence, receipt evidence, production installer readiness, root installer readiness, Fedora approval, Fedora distribution readiness, daily-driver safety, immutable Fedora readiness, kernel integration, systemd integration, SELinux integration, runtime enforcement, malware prevention, ransomware prevention, sandboxing, or a production security-product claim.

docs/status/LATTICRA_PANEL_UI_DESIGN_CHECKPOINT.md

Source title: Latticra Panel UI Design Checkpoint

Lines: 66 | Words: 172 | SHA-256: 2b710127ba1d1229476c50920e9b3e0a5cb5749719200011ca8c912885548c22

Latticra Panel UI Design Checkpoint

Status: design checkpoint Date: 2026-05-26 CDT

This checkpoint records the current Latticra Panel workbench layout as the baseline design state.

Future changes are allowed, but they should preserve the current workbench identity unless a deliberate design migration updates this file and the checker.

Checker:

```
scripts/check_latticra_panel_ui_design.py
```

CI workflow:

```
.github/workflows/latticra-panel-installer.yml
```

The checker currently protects these design anchors:

```
gui-workbench build identity
embedded panel image
texture loader
left navigation panel
right console panel
central workbench panel
dashboard tab
components tab
seal tab
authority tab
delivery tab
updater tab
evidence tab
procedure tab
fluid install button
right evidence panel
focused running monitor
bounded recent engine output
non-wrapping engine log scroll
running console quick-command restraint
window size
minimum window size
maximized launch
README console section
README safety section
```

Change rule:

1. keep the checker passing for ordinary UI edits;
2. update this file and the checker for deliberate design migrations;
3. keep installer CI passing before merge.

Current posture:

```
production_installer_ready=0
root_authority=0
network_authority=0
runtime_enforcement_authority=0
user_local_prefix_only=1
```

docs/status/LATTICRA_SEAL_FOUNDATION_STATUS.md

Source title: Latticra Seal Foundation Status

Lines: 67 | Words: 198 | SHA-256: d75b0df00bc6cbf1fe63b132a86b1059fbd68918cf591a9e9ac671a125da897d

Latticra Seal Foundation Status

Status: Latticra Seal foundation status record Date: 2026-05-21 Scope: status record for the first Latticra Seal contract and implementation plan.

Summary

Latticra Seal is now defined as the cryptographic evidence and capability substrate direction for Latticra.

This status record confirms that the first Seal slice is contract and implementation-plan work only.

Added foundation records

```
docs/LATTICRA_SEAL_CONTRACT.md
docs/LATTICRA_SEAL_IMPLEMENTATION_PLAN.md
```

Current Seal posture

```
seal_contract_present=1
seal_implementation_plan_present=1
seal_report_implementation_present=0
artifact_measurement_supported=0
signature_supported=0
capability_enforcement_supported=0
sealed_objects_supported=0
runtime_authority_granted=0
host_read_performed=0
host_mutation_performed=0
network_performed=0
status=contract-and-plan-only
```

Capability posture

This slice changes project direction and planning priority, but it does not add operational capability.

```
new_runtime_behavior=0
new_host_behavior=0
new_network_behavior=0
new_key_behavior=0
new_signature_behavior=0
new_encryption_behavior=0
new_capability_enforcement=0
```

First implementation target

The next valid implementation target is a no-effect Seal report surface.

That report must remain metadata-only and must not hash files, generate keys, sign records, encrypt objects, enforce capabilities, contact networks, mutate hosts, or grant runtime authority.

Boundary

Latticra Seal currently does not provide cryptographic enforcement, secure boot, measured boot, runtime authorization, key management, encrypted storage, encrypted transport, post-quantum security, TPM-backed identity, Linux integrity integration, Fedora approval, production hardening, or certification.

Validation

The foundation status is validated by:

```
sh scripts/test-latticra-seal-foundation.sh
```

docs/status/MACOS_APP_BUNDLE_WRITER_ALIGNMENT_STATUS.md

Source title: macOS App Bundle Writer Alignment Status

Lines: 82 | Words: 301 | SHA-256: 44be38686eb1257f4fd24e4c6d5a7069d80fe79ecfaf4c16f3e23e586986c072

macOS App Bundle Writer Alignment Status

Status: writer alignment status Date: 2026-05-25 CDT Scope: status checkpoint after aligning dry-run writer public meaning against future commit-capable writer claims.

Summary

Latticra now has a macOS app bundle writer alignment record.

The alignment makes the current capability explicit: a no-effect dry-run writer prototype exists, the macOS dry-run writer candidate integration can prove candidate inputs reach the no-effect future commit-gate decision, and the macOS commit gate contract keeps that gate closed. A commit-capable writer, app bundle install, reset/uninstall implementation, and install verification transcript do not exist.

Status Fields

```
macos_app_bundle_writer_alignment_present=1
macos_local_candidate_asset_probe_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
macos_dry_run_writer_public_meaning_recorded=1
macos_commit_capable_writer_nonclaim_recorded=1
macos_future_commit_gate_requirements_recorded=1
macos_app_bundle_writer_dry_run_present=1
macos_app_bundle_writer_commit_disabled=1
macos_app_bundle_writer_present=0
macos_app_bundle_commit_capable_writer_present=0
commit_user_local_managed_artifacts=0
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS app bundle writer dry-run prototype that renders future writer
phases and blocks unsafe paths while keeping all writes disabled.
```

That does not mean Latticra has a commit-capable macOS installer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-app-bundle-writer-alignment.sh
```

Expected output:

```
macos_app_bundle_writer_alignment: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_APP_BUNDLE_WRITER_DRY_RUN_STATUS.md

Source title: macOS App Bundle Writer Dry-Run Status

Lines: 82 | Words: 323 | SHA-256: 472040d96499531ee6813ac6217e5a5f101e56123c912d10df9e941aa0ad5663

macOS App Bundle Writer Dry-Run Status

Status: no-effect writer dry-run status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS app bundle writer dry-run prototype.

Summary

Latticra now has a no-effect macOS app bundle writer dry-run prototype.

The prototype emits the planned writer phase report, validates unsafe paths, inspects existing target markers, reports missing executable/icon candidates, and keeps commit behavior disabled. It is now paired with the macOS dry-run writer candidate integration and closed macOS commit gate contract, which prove accepted local candidates can move the dry-run to ready-for-future-commit-gate without enabling writes. It does not create an app bundle, write Application Support files, install wrappers, mutate shell profiles, use launchd, access Keychain, request TCC permissions, use the network, or grant platform authority.

Status Fields

```
macos_app_bundle_writer_dry_run_present=1
macos_app_bundle_writer_alignment_present=1
macos_local_candidate_asset_probe_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
macos_app_bundle_writer_phase_report_present=1
macos_app_bundle_writer_path_guard_present=1
macos_app_bundle_writer_marker_inspection_present=1
macos_app_bundle_writer_missing_candidate_detection_present=1
macos_app_bundle_writer_commit_disabled=1
macos_app_bundle_writer_present=0
commit_user_local_managed_artifacts=0
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect dry-run prototype for the future managed user-local macOS app bundle
writer.
```

That does not mean Latticra has a commit-capable macOS installer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-app-bundle-writer-dry-run.sh
```

Expected output:

```
macos_app_bundle_writer_dry_run: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_BUILD_PLATFORM_PROBE_STATUS.md

Source title: macOS Build Platform Probe Status

Lines: 79 | Words: 272 | SHA-256: 0302f8fbf04ff8e20e3a5191ed5062a6c03df8c195586e61bb50ebddbfd26687

macOS Build Platform Probe Status

Status: no-effect probe status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS build/platform probe lane.

Summary

Latticra now has a no-effect macOS build/platform probe.

The probe records macOS host detection, architecture, sw_vers availability, clang availability, Rust/Cargo availability, Panel source readiness, C test source readiness, and denied-authority metadata. It does not build the Panel, compile C tests, create an app bundle, install wrappers, mutate host state, use the network, or grant platform authority.

Status Fields

```
macos_build_platform_probe_present=1
macos_dry_run_plan_adapter_present=1
macos_user_local_app_bundle_contract_present=1
macos_user_local_app_bundle_implementation_plan_present=1
macos_app_bundle_writer_dry_run_present=1
macos_probe_script_present=1
macos_probe_guard_present=1
macos_host_detection_recorded=1
macos_architecture_recorded=1
macos_sw_vers_probe_recorded=1
macos_clang_probe_recorded=1
macos_rust_probe_recorded=1
macos_panel_build_probe_recorded=1
macos_c_test_build_probe_recorded=1
panel_build_performed=0
c_test_build_performed=0
app_bundle_created=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS platform/build-readiness probe for local toolchain and
source-readiness reporting.
```

That does not mean Latticra has a macOS installer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-build-platform-probe.sh
```

Expected output:

```
macos_build_platform_probe: ok
```

Next Recommended Lane

```
Add a macOS user-local app bundle implementation plan that remains no-effect and defines writer
phases, failure behavior, reset/uninstall sequencing, verification commands, and guard tests
before implementation.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_COMMIT_GATE_CONTRACT_STATUS.md

Source title: macOS Commit Gate Contract Status

Lines: 134 | Words: 343 | SHA-256: 522a2473b2d72d01b2fc8fd4cd11a61590a81462a9edf17f3b441de005e8dae3

macOS Commit Gate Contract Status

Status: no-effect commit gate contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the closed macOS commit gate contract.

Summary

Latticra now has a no-effect macOS commit gate contract.

The contract keeps commit_user_local_managed_artifacts=0 even when local candidates and the writer dry-run reach the future commit-gate decision. The macOS verification transcript contract and reset/uninstall dry-run contract are now present, but the gate still records the missing managed-write, reset/uninstall evidence, and transcript-evidence prerequisites before a commit-capable macOS user-local app bundle writer may exist.

Status Fields

```
macos_commit_gate_contract_present=1
macos_commit_gate_contract_guard_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_readme_installer_usage_present=1
commit_gate_state=closed
commit_gate_decision=blocked-missing-managed-write-implementation
commit_user_local_managed_artifacts=0
macos_app_bundle_commit_capable_writer_present=0
managed_write_implementation_present=0
reset_uninstall_implementation_present=0
macos_verification_transcript_contract_present=1
verification_transcript_contract_present=1
verification_transcript_evidence_present=0
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_dry_run_evidence_present=0
absence_report_evidence_present=0
candidate_integration_required=1
candidate_flow_ready_required=1
operator_explicit_commit_intent_required=1
unsafe_path_negative_tests_required=1
unmanaged_target_preservation_tests_required=1
receipt_completeness_tests_required=1
reset_uninstall_dry_run_required=1
verification_transcript_required=1
application_support_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
```

```
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a closed macOS commit-gate contract for future user-local managed app bundle
writes.
```

That does not mean Latticra has a macOS installer, commit-capable app bundle writer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-commit-gate-contract.sh
```

Expected output:

```
macos_commit_gate_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_DRY_RUN_PLAN_ADAPTER_STATUS.md

Source title: macOS Dry-Run Plan Adapter Status

Lines: 84 | Words: 282 | SHA-256: 492a746f8d53c92fdc1f952b8ff24018f5f6b946756193908d357b57a5be28ac

macOS Dry-Run Plan Adapter Status

Status: no-effect dry-run plan status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS dry-run plan adapter.

Summary

Latticra now has a no-effect macOS dry-run plan adapter.

The adapter renders user-local Application Support, app bundle, CLI wrapper, receipt, logs/caches, and verification intent for a future macOS installer lane. It does not write those artifacts, create an app bundle, install wrappers, mutate shell profiles, use launchd, access Keychain, request TCC permissions, use the network, or grant platform authority.

Status Fields

```
macos_dry_run_plan_adapter_present=1
macos_user_local_app_bundle_contract_present=1
macos_user_local_app_bundle_implementation_plan_present=1
macos_app_bundle_writer_dry_run_present=1
macos_dry_run_plan_script_present=1
macos_dry_run_plan_guard_present=1
macos_application_support_plan_present=1
macos_user_local_app_bundle_plan_present=1
macos_cli_wrapper_plan_present=1
macos_receipt_plan_present=1
macos_logs_caches_plan_present=1
macos_path_guard_present=1
application_support_write_performed=0
payload_write_performed=0
config_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS dry-run plan adapter for rendering future user-local install
intent.
```

That does not mean Latticra has a macOS installer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-dry-run-plan-adapter.sh
```

Expected output:

```
macos_dry_run_plan_adapter: ok
```

Next Recommended Lane

```
Add a macOS user-local app bundle implementation plan that remains no-effect and defines writer
phases, failure behavior, reset/uninstall sequencing, verification commands, and guard tests
before implementation.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION_STATUS.md

Source title: macOS Dry-Run Writer Candidate Integration Status

Lines: 79 | Words: 295 | SHA-256: 7afe401a65622d1c8b228ae21c486159c73046f8b1d5f686b2c2a76905ccb856

macOS Dry-Run Writer Candidate Integration Status

Status: no-effect dry-run writer candidate integration status Date: 2026-05-25 CDT Scope: status checkpoint after bridging the macOS local candidate asset probe to the app bundle writer dry-run.

Summary

Latticra now has a no-effect macOS dry-run writer candidate integration.

The integration runs the local candidate asset probe and the app bundle writer dry-run with the same caller-supplied inputs. It reports readiness only when the probe accepts the inputs, the writer dry-run reaches ready-for-future-commit-gate, and all write and authority flags remain disabled.

Status Fields

```
macos_dry_run_writer_candidate_integration_present=1
macos_dry_run_writer_candidate_integration_guard_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
macos_local_candidate_asset_probe_present=1
macos_app_bundle_writer_dry_run_present=1
asset_probe_to_writer_candidate_flow_recorded=1
asset_probe_ready_decision_required=1
writer_ready_decision_required=1
authority_boundary_preserved_required=1
commit_user_local_managed_artifacts=0
application_support_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra can prove, without writing anything, that accepted local executable/icon candidates can
 move the macOS writer dry-run to its future commit-gate decision.
```

That does not mean Latticra has a macOS installer, commit-capable app bundle writer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-dry-run-writer-candidate-integration.sh
```

Expected output:

```
macos_dry_run_writer_candidate_integration: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_INTEGRATION_TRANSFERABILITY_STATUS.md

Source title: macOS Integration Transferability Status

Lines: 577 | Words: 1190 | SHA-256: 33d1235e41b1e6b5edf33e5bfa936a12c63ba967e53f7c27f9fff010412a6ba1

macOS Integration Transferability Status

Status: transferability status record Date: 2026-05-25 CDT Scope: status checkpoint for the first macOS integration transferability map.

Summary

Latticra now has a dedicated macOS integration transferability plan.

The plan maps current Latticra surfaces into a cautious macOS lane: preserve no-effect reports, Panel-guided operator review, receipt-first installation, user-local artifacts, denied-by-default authority, and future-gated platform integrations.

This is planning/status work only. It does not implement macOS installation, app bundle creation, launchd behavior, Keychain behavior, Secure Enclave behavior, sandboxing, notarization, Endpoint Security behavior, System Extension behavior, Network Extension behavior, privileged helper behavior, or production macOS readiness.

Status Fields

```
macos_integration_transferability_map_present=1
macos_build_platform_probe_present=1
macos_dry_run_plan_adapter_present=1
macos_user_local_app_bundle_contract_present=1
macos_user_local_app_bundle_implementation_plan_present=1
macos_app_bundle_writer_dry_run_present=1
macos_app_bundle_writer_alignment_present=1
macos_local_candidate_asset_probe_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_readme_installer_usage_present=1
macos_verification_transcript_contract_present=1
macos_verification_transcript_evidence_present=0
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_dry_run_evidence_present=0
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_present=
1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract_p
resent=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_accepts_passed_preflight_only=1
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_invocation_simulated=1
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_stdout_only=1
live_runner_denied_dispatch_transcript_file_write_enabled=0
live_runner_denied_dispatch_transcript_dispatch_enabled=0
live_runner_denied_dispatch_transcript_dispatch_denied=1
live_runner_denied_dispatch_transcript_dispatch_performed=0
live_runner_denied_dispatch_transcript_deletion_enabled=0
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_stdout_only=1
live_runner_denied_dispatch_review_file_write_enabled=0
live_runner_denied_dispatch_review_dispatch_reviewed=1
live_runner_denied_dispatch_review_dispatch_enabled=0
```

```
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_stdout_only=1
live_runner_acceptance_denial_transcript_file_write_enabled=0
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_dispatch_allowed=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_stdout_only=1
live_runner_acceptance_denial_review_file_write_enabled=0
live_runner_acceptance_denial_review_closed_gate_reviewed=1
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_dispatch_reviewed=1
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_disposition_opened=0
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_stdout_only=1
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live_runner_acceptance_denial_disposition_acceptance_gate_open=0
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live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_applied=0
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live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_applied=0
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live_runner_acceptance_denial_disposition_closeout_stdout_only=1
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live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_disposition_applied=0
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ut
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
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live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_state=reviewed-no-effec
t-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_file_write_enabled=0
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live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_deletion_enabled=0
```

```
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
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sed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_stdout_only=1
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live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
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pened=0
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live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_file_write_en
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live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_perf
ormed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_deletion_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_open
ed=0
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_receipt_schema_contract_present=1
reset_receipt_evidence_present=0
reset_uninstall_receipt_evidence_present=0
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
```

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_present=
1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract_p
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n_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_accepts_passed_preflight_only=1
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_invocation_simulated=1
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_stdout_only=1
live_runner_denied_dispatch_transcript_file_write_enabled=0
live_runner_denied_dispatch_transcript_dispatch_enabled=0
live_runner_denied_dispatch_transcript_dispatch_denied=1
live_runner_denied_dispatch_transcript_dispatch_performed=0
live_runner_denied_dispatch_transcript_deletion_enabled=0
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live_runner_denied_dispatch_review_stdout_only=1
live_runner_denied_dispatch_review_file_write_enabled=0
live_runner_denied_dispatch_review_dispatch_reviewed=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
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live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_stdout_only=1
live_runner_acceptance_denial_transcript_file_write_enabled=0
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_dispatch_allowed=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_stdout_only=1
live_runner_acceptance_denial_review_file_write_enabled=0
live_runner_acceptance_denial_review_closed_gate_reviewed=1
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_dispatch_reviewed=1
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_disposition_opened=0
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_recorded=1
live_runner_acceptance_denial_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_dispatch_enabled=0
```

```
live_runner_acceptance_denial_disposition_dispatch_performed=0
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live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_contract_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_review_present=1
live_runner_acceptance_denial_disposition_closeout_disposition_closed=1
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_contract_state=audited-no-effect-closeo
ut
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_state=reviewed-no-effec
t-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_state=dispo
sed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_disposition_review_o
pened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_contract_stat
e=reviewed-no-effect-closeout-audit-review-disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_file_write_en
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resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_enab
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ormed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_deletion_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_open
ed=0
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_runner_interface_current_preflight_passed=0
```

```
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
reset_uninstall_deletion_enabled=0
operator_explicit_reset_uninstall_intent_observed=0
macos_transferable_surfaces_classified=1
macos_adapter_requirements_recorded=1
macos_user_local_paths_proposed=1
macos_app_bundle_proposed=1
macos_cli_wrapper_direction_recorded=1
macos_security_interfaces_future_gated=1
macos_evidence_rules_recorded=1
macos_runtime_behavior_added=0
macos_host_mutation_added=0
macos_app_bundle_created=0
macos_install_verified=0
macos_keychain_authority=0
macos_secure_enclave_authority=0
macos_tcc_bypass_authority=0
macos_launchagent_authority=0
macos_endpoint_security_authority=0
macos_system_extension_authority=0
macos_network_extension_authority=0
macos_privileged_helper_authority=0
macos_production_ready=0
```

Transferability Classification

The status checkpoint records these transferability conclusions:

```
Lat=high_transferability
LIR=high_transferability
L_UI=high_transferability
Nucleus=high_transferability
Runtime_Boundary=high_transferability
Latticra_Seal=high_transferability_report_only
Latticra_Panel=medium_high_transferability_with_app_bundle_adapter
Nadia_offline_AI=medium_high_transferability_contract_only
Fedora_RPM_lanes=not_transferable_keep_linux_specific
installer_scripts=medium_transferability_with_platform_adapter
documentation_status=high_transferability
```

Public Meaning

The careful public meaning is:

```
Latticra has a macOS transferability map for adapting the current no-effect, receipt-first,
user-local Panel and metadata surfaces into a future macOS lane.
```

That statement must not be expanded into a claim that Latticra already has a macOS installer, notarized app, sandbox, Keychain integration, endpoint security layer, system extension, package installer, or production security product.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-integration-transferability.sh
```

Expected output:

```
macos_integration_transferability: ok
```

Previous Recommended Lane

Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate, operator-intent, receipt, absence, planner, and classifier evidence before any live execution.

Next Recommended Lane

Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition closeout contract that closes the reviewed no-effect closeout audit review disposition without opening dispatch or deletion.

Current live-runner acceptance-gate lane:

Add a macOS reset/uninstall live-runner acceptance-gate contract that keeps the live runner closed until passed preflight, evidence, authorization, operator intent, and implementation are all present.

Current live-runner acceptance-denial transcript lane:

Add a macOS reset/uninstall live-runner acceptance-denial transcript contract that records the closed acceptance gate without dispatching effects.

Current live-runner acceptance-denial review lane:

Add a macOS reset/uninstall live-runner acceptance-denial review contract that reviews the closed gate transcript without enabling dispatch or deletion.

Current live-runner acceptance-denial disposition lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition contract that records the reviewed closed-gate denial as a no-effect disposition without opening dispatch.

Current live-runner acceptance-denial disposition closeout lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout contract that closes the reviewed no-effect disposition without opening dispatch or deletion.

Current live-runner acceptance-denial disposition closeout audit lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract that audits the no-effect closeout without opening dispatch or deletion.

Current live-runner acceptance-denial disposition closeout audit review lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review contract that reviews the no-effect closeout audit without opening dispatch or deletion.

Current live-runner acceptance-denial disposition closeout audit review disposition lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract that records the reviewed no-effect closeout audit as a no-effect disposition without opening dispatch or deletion.

Current live-runner acceptance-denial disposition closeout audit review disposition review lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract that reviews the no-effect closeout audit review disposition without opening dispatch or deletion.

Current live-runner acceptance-denial disposition review lane:

Add a macOS reset/uninstall live-runner acceptance-denial disposition review contract that reviews the no-effect disposition without opening dispatch or deletion.

Current live-runner no-op prototype lane:

Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied interface path without deleting files.

Current live-runner denied-dispatch transcript lane:

Add a macOS reset/uninstall live-runner denied-dispatch transcript contract that records the no-op runner denial without enabling deletion.

Current live-runner denied-dispatch review lane:

Add a macOS reset/uninstall live-runner denied-dispatch review contract that keeps review-only dispatch denial evidence separate from any effects.

Current completed lane:

> Add a macOS reset/uninstall live-implementation plan contract that maps future effect-authorized execution phases while deletion remains disabled.

Current preflight lane:

> Add a macOS reset/uninstall live-execution preflight contract that proves the live implementation plan still cannot delete until all evidence gates are satisfied.

Current live-denial transcript lane:

> Add a macOS reset/uninstall live-denial transcript contract that records the failed preflight decision without deleting files.

Current live-runner interface lane:

> Add a macOS reset/uninstall live-runner interface contract that accepts only a passed preflight and keeps deletion disabled otherwise.

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_LOCAL_CANDIDATE_ASSET_PROBE_STATUS.md

Source title: macOS Local Candidate Asset Probe Status

Lines: 85 | Words: 330 | SHA-256: b1b3eaaa325795b6568e572f9d6b07403c728b43189a3ce1e4fee75476274ca4

macOS Local Candidate Asset Probe Status

Status: no-effect local candidate asset probe status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS local Panel executable and icon candidate asset probe.

Summary

Latticra now has a no-effect macOS local candidate asset probe.

The probe checks caller-supplied local Panel executable and icon candidates for dry-run writer input readiness. It is now linked to the macOS dry-run writer candidate integration and closed macOS commit gate contract, which compare the probe decision with the writer dry-run decision while keeping all writes disabled. The probe reports missing files, disallowed paths, non-executable Panel candidates, unsupported icon formats, and ready dry-run inputs without building, downloading, signing, notarizing, copying, writing app bundle files, mutating the host, or granting authority.

Status Fields

```
macos_local_candidate_asset_probe_present=1
macos_local_candidate_asset_probe_guard_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
local_panel_executable_candidate_probe=1
local_icon_candidate_probe=1
panel_candidate_executable_check_present=1
panel_candidate_readable_check_present=1
icon_candidate_format_check_present=1
icon_candidate_readable_check_present=1
asset_probe_ready_decision_present=1
build_performed=0
panel_build_performed=0
icon_conversion_performed=0
download_performed=0
copy_performed=0
signing_performed=0
notarization_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra can report whether caller-supplied local Panel executable and icon candidates are
usable dry-run inputs.
```

That does not mean Latticra has a macOS installer, commit-capable app bundle writer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-local-candidate-asset-probe.sh
```

Expected output:

```
macos_local_candidate_asset_probe: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_README_INSTALLER_USAGE_STATUS.md

Source title: macOS README Installer Usage Status

Lines: 460 | Words: 672 | SHA-256: a6e202a3e41ff3e52544fdbeec1804dddb4fe1906730e4c3071d20dff322c832

macOS README Installer Usage Status

Status: README installer usage alignment status Date: 2026-05-25 CDT Scope: status checkpoint after documenting the macOS installer lane in the root README.

Summary

The root README now documents how to use the current macOS installer lane.

The README section targets macOS infrastructure directly: user-local Application Support, ~/Applications/Latticra Panel.app, Logs, Caches, Preferences, optional CLI wrappers, app-bundle shape, dry-run commands, local candidate checks, and the closed commit gate. It also states that no macOS app bundle is created, signed, notarized, verified, reset, or uninstalled yet.

Status Fields

```
macos_readme_installer_usage_present=1
macos_readme_installer_usage_guard_present=1
macos_installer_targets_macos_infrastructure=1
macos_user_local_paths_documented=1
macos_app_bundle_shape_documented=1
macos_probe_commands_documented=1
macos_candidate_commands_documented=1
macos_commit_gate_command_documented=1
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```
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effect_authorization_open=0
reset_uninstall_effect_authorized=0
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```

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live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_contract_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_review_present=1
live_runner_acceptance_denial_disposition_closeout_disposition_closed=1
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_contract_state=audited-no-effect-closeo
ut
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_state=reviewed-no-effec
t-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_state=dispo
sed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_disposition_review_o
```

```
pened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_contract_stat
e=reviewed-no-effect-closeout-audit-review-disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_file_write_en
abled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_perf
ormed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_deletion_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_open
ed=0
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
reset_uninstall_deletion_enabled=0
operator_explicit_reset_uninstall_intent_observed=0
commit_user_local_managed_artifacts=0
macos_app_bundle_commit_capable_writer_present=0
macos_app_bundle_created=0
macos_install_verified=0
app_bundle_write_performed=0
file_delete_performed=0
directory_delete_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
The README documents Latticra&#x27;s current no-effect macOS installer lane and the exact
Mac-specific infrastructure it targets.
```

That does not mean Latticra has a macOS installer, commit-capable app bundle writer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-readme-installer-usage.sh
```

Expected output:

```
macos_readme_installer_usage: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Absence-Report Contract Status

Lines: 163 | Words: 367 | SHA-256: a1980addd1a257cb60f75c228dca700a84642817df9b2f7d26e6a877d5de3046

macOS Reset/Uninstall Absence-Report Contract Status

Status: no-effect reset/uninstall absence-report contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall absence-report contract.

Summary

Latticra now has a no-effect macOS reset/uninstall absence-report contract.

The contract defines the required future evidence for post-removal managed-target absence, unmanaged-target preservation, receipt linkage, planner linkage, and authority-denial lines. It does not delete files, write reports, write receipts, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
macos_reset_uninstall_absence_report_contract_guard_present=1
macos_reset_uninstall_dry_run_planner_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
absence_report_contract_state=defined-no-effect
absence_report_required=1
absence_report_evidence_present=0
absence_report_run_performed=0
```

```
absence_report_written=0
app_bundle_absence_required_if_managed=1
app_support_absence_required_if_managed=1
cli_wrapper_absence_required_if_managed=1
unmanaged_target_preservation_evidence_required=1
reset_receipts_dir_preservation_required=1
reset_receipt_reference_required=1
reset_receipt_evidence_present=0
reset_uninstall_receipt_evidence_present=0
planner_transcript_reference_required=1
live_classifier_reference_required=1
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall absence-report contract for future post-removal
evidence.
```

That does not mean Latticra has a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-absence-report-contract.sh
```

Expected output:

```
macos_reset_uninstall_absence_report_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Dry-Run Contract Status

Lines: 129 | Words: 329 | SHA-256: d0a260edeaacd44762a8c56770bbcdf214a3a39b4e9b989fd632391dc211416f

macOS Reset/Uninstall Dry-Run Contract Status

Status: no-effect reset/uninstall dry-run contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall dry-run contract.

Summary

Latticra now has a no-effect macOS reset/uninstall dry-run contract.

The contract defines managed-target removal order, managed-marker requirements, receipt placement outside the removed prefix, absence-report requirements, and preservation rules for user files. It does not remove files, write receipts, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_dry_run_contract_guard_present=1
macos_verification_transcript_contract_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_planner_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_dry_run_contract_state=defined-no-effect
reset_uninstall_dry_run_decision=contract-defined-removal-not-performed
reset_uninstall_dry_run_required=1
reset_uninstall_dry_run_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
```

```
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect contract for future macOS reset/uninstall dry-run ordering.
```

That does not mean Latticra has a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-dry-run-contract.sh
```

Expected output:

```
macos_reset_uninstall_dry_run_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER_STATUS.md

Source title: macOS Reset/Uninstall Dry-Run Planner Status

Lines: 131 | Words: 337 | SHA-256: 60c88fba8b5af22b45b8b477062590066d4835864c2d9538397202efdccb294b

macOS Reset/Uninstall Dry-Run Planner Status

Status: no-effect reset/uninstall dry-run planner status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall dry-run planner.

Summary

Latticra now has a no-effect macOS reset/uninstall dry-run planner.

The planner consumes the live-target classifier, records target states, and emits planned reset/uninstall actions for managed targets only. It does not delete files, write receipts, mutate host state, run absence verification, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_dry_run_planner_present=1
macos_reset_uninstall_dry_run_planner_guard_present=1
macos_reset_uninstall_live_target_classifier_present=1
reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
dry_run_transcript_present=1
reset_uninstall_dry_run_planner_transcript_present=1
planner_consumes_live_target_classifier=1
planned_removal_count=report-runtime
managed_wrapper_removal_planned=report-runtime
managed_app_bundle_removal_planned=report-runtime
managed_application_support_removal_planned=report-runtime
reset_receipt_write_planned=report-runtime
absence_report_planned=report-runtime
absence_report_contract_present=1
absence_report_evidence_present=0
reset_uninstall_dry_run_evidence_present=0
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
```

```
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall dry-run planner that can convert live target
classifications into planned managed-target actions.
```

That does not mean Latticra has a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-dry-run-planner.sh
```

Expected output:

```
macos_reset_uninstall_dry_run_planner: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Effect-Authorization Contract Status

Lines: 204 | Words: 456 | SHA-256: cf848eb9e1ca0013b04f4ee4347b06111a3da944c601995379047785285824bc

macOS Reset/Uninstall Effect-Authorization Contract Status

Status: no-effect reset/uninstall effect-authorization contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall effect-authorization contract.

Summary

Latticra now has a no-effect macOS reset/uninstall effect-authorization contract.

The contract keeps live reset/uninstall execution disabled until implementation-gate and operator-intent evidence are both present. It does not authorize effects, delete files, write receipts, write absence reports, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
macos_reset_uninstall_effect_authorization_contract_guard_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_contract_decision=blocked-missing-implementation-gate-and-operator-intent-e
vidence
effect_authorization_contract_required=1
effect_authorization_contract_open=0
effect_authorization_required=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
reset_uninstall_effect_authorization_evidence_present=0
effect_authorization_record_write_enabled=0
effect_authorization_denial_recorded=1
effect_authorization_denial_reason=missing-implementation-gate-and-operator-intent-evidence
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_required=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
operator_intent_evidence_written=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
```

```
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
reset_uninstall_dry_run_evidence_present=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
absence_report_evidence_present=0
effect_authorization_requires_implementation_gate_open=1
effect_authorization_requires_operator_intent_evidence=1
effect_authorization_requires_dry_run_planner_transcript=1
effect_authorization_requires_live_target_classifier=1
effect_authorization_requires_receipt_schema=1
effect_authorization_requires_absence_report_contract=1
effect_authorization_requires_receipt_outside_removed_prefix=1
effect_authorization_requires_no_unmanaged_targets=1
effect_authorization_requires_no_unsafe_paths=1
effect_authorization_requires_no_network=1
effect_authorization_requires_no_root=1
effect_authorization_schema_version=macos-reset-uninstall-effect-authorization/1
effect_authorization_condition_implementation_gate_open=required
effect_authorization_condition_operator_intent_evidence_present=required
effect_authorization_condition_operator_intent_current_session=required
effect_authorization_condition_no_unmanaged_targets=required
effect_authorization_condition_no_network=required
effect_authorization_condition_no_root=required
effect_authorization_must_not_override_missing_operator_intent=1
effect_authorization_must_not_override_closed_implementation_gate=1
effect_authorization_must_not_authorize_unmanaged_target_removal=1
effect_authorization_result_implementation_gate_open=not_met
effect_authorization_result_operator_intent_evidence=not_met
effect_authorization_result_dry_run_planner_transcript=contract-only
effect_authorization_result_live_target_classifier=contract-only
effect_authorization_result_receipt_schema=contract-only
effect_authorization_result_absence_report_contract=contract-only
effect_authorization_result_no_network=met
effect_authorization_result_no_root=met
effect_authorization_phase_5_status=disabled
effect_authorization_phase_6_status=disabled
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

> Latticra has a no-effect macOS reset/uninstall effect-authorization contract that keeps live reset/uninstall execution disabled until implementation-gate and operator-intent evidence are both present.

That does not mean Latticra has effect approval evidence, live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-effect-authorization-contract.sh
```

Expected output:

```
macos_reset_uninstall_effect_authorization_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not live reset authorization, macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Evidence-Bundle Contract Status

Lines: 193 | Words: 476 | SHA-256: 6ba3ea34749f15334c3fcb9da086ef471b6225446a1da38f9dcf757adc4e763d

macOS Reset/Uninstall Evidence-Bundle Contract Status

Status: no-effect reset/uninstall evidence-bundle contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall evidence-bundle contract.

Summary

Latticra now has a no-effect macOS reset/uninstall evidence-bundle contract.

The contract groups the implementation gate, operator intent, dry-run planner, live-target classifier, receipt schema, and absence-report requirements before any future live reset/uninstall execution. It keeps the bundle incomplete, leaves effect authorization closed, and does not delete files, write receipts, write absence reports, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_reset_uninstall_implementation_present=0
macos_reset_uninstall_evidence_bundle_contract_guard_present=1
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_contract_decision=blocked-incomplete-reset-uninstall-evidence-bundle
evidence_bundle_contract_required=1
reset_uninstall_evidence_bundle_required=1
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
reset_uninstall_evidence_bundle_write_enabled=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_required=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
operator_intent_evidence_written=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
reset_uninstall_dry_run_evidence_present=0
macos_reset_uninstall_live_target_classifier_present=1
live_target_classifier_evidence_present=0
macos_reset_uninstall_receipt_schema_contract_present=1
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
macos_reset_uninstall_absence_report_contract_present=1
absence_report_evidence_present=0
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
evidence_bundle_schema_version=macos-reset-uninstall-evidence-bundle/1
evidence_bundle_required_component_count=6
evidence_bundle_contract_component_count=6
evidence_bundle_complete_component_count=0
evidence_bundle_requires_implementation_gate_open=1
evidence_bundle_requires_operator_intent_evidence=1
evidence_bundle_requires_dry_run_planner_transcript=1
evidence_bundle_requires_live_target_classifier_report=1
evidence_bundle_requires_reset_uninstall_receipt_evidence=1
```

```
evidence_bundle_requires_absence_report_evidence=1
evidence_bundle_requires_effect_authorization_closed_until_complete=1
evidence_bundle_requires_receipt_outside_removed_prefix=1
evidence_bundle_requires_no_unmanaged_targets=1
evidence_bundle_requires_no_unsafe_paths=1
evidence_bundle_requires_no_network=1
evidence_bundle_requires_no_root=1
evidence_bundle_condition_implementation_gate_open=required
evidence_bundle_condition_operator_intent_evidence_present=required
evidence_bundle_condition_dry_run_planner_transcript_present=required
evidence_bundle_condition_live_target_classifier_report_present=required
evidence_bundle_condition_reset_uninstall_receipt_evidence_present=required
evidence_bundle_condition_absence_report_evidence_present=required
evidence_bundle_condition_no_unmanaged_targets=required
evidence_bundle_condition_no_network=required
evidence_bundle_condition_no_root=required
evidence_bundle_result_implementation_gate_open=not_met
evidence_bundle_result_operator_intent_evidence=not_met
evidence_bundle_result_dry_run_planner_transcript=contract-only
evidence_bundle_result_live_target_classifier_report=contract-only
evidence_bundle_result_reset_uninstall_receipt_evidence=not_met
evidence_bundle_result_absence_report_evidence=not_met
evidence_bundle_result_effect_authorization=closed
evidence_bundle_result_no_unmanaged_targets=not_evaluated_contract_only
evidence_bundle_result_no_network=met
evidence_bundle_result_no_root=met
evidence_bundle_phase_5_status=disabled
evidence_bundle_phase_6_status=disabled
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall evidence-bundle contract that groups the required
 reset/uninstall evidence before any live execution can be considered.
```

That does not mean Latticra has complete evidence-bundle evidence, effect approval evidence, live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-evidence-bundle-contract.sh
```

Expected output:

```
macos_reset_uninstall_evidence_bundle_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Current completed lane:

```
Add a macOS reset/uninstall live-implementation plan contract that maps future effect-authorized
 execution phases while deletion remains disabled.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, complete evidence-bundle evidence, receipt evidence, absence verification evidence, operator approval evidence, effect approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Implementation-Gate Contract Status

Lines: 159 | Words: 373 | SHA-256: 590725841811c92967307708b059efa15f75f9069cf1ae94df84d73459ce31f4

macOS Reset/Uninstall Implementation-Gate Contract Status

Status: no-effect reset/uninstall implementation-gate contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall implementation-gate contract.

Summary

Latticra now has a no-effect macOS reset/uninstall implementation-gate contract.

The contract keeps live reset/uninstall activity closed until receipt, absence, planner, classifier, and explicit operator-intent evidence exist. It does not delete files, write receipts, write absence reports, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
macos_reset_uninstall_implementation_gate_contract_guard_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_state=closed-no-effect
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
implementation_gate_required=1
implementation_gate_open=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
reset_uninstall_dry_run_evidence_present=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
absence_report_evidence_present=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
managed_target_classification_required=1
live_target_classifier_evidence_required=1
dry_run_planner_transcript_required=1
receipt_schema_required=1
reset_uninstall_receipt_evidence_required=1
absence_report_contract_required=1
absence_report_evidence_required=1
operator_explicit_reset_uninstall_intent_required=1
operator_intent_must_name_operation=1
operator_intent_must_name_target_scope=1
operator_intent_must_acknowledge_managed_targets_only=1
operator_intent_must_acknowledge_no_unmanaged_removal=1
operator_intent_must_acknowledge_receipt_path=1
gate_condition_receipt_schema_contract_present=required
gate_condition_absence_report_contract_present=required
gate_condition_dry_run_planner_transcript_present=required
gate_condition_live_target_classifier_present=required
gate_condition_operator_intent_evidence_present=required
gate_condition_no_unmanaged_targets=required
gate_condition_no_unsafe_paths=required
gate_condition_receipt_outside_removed_prefix=required
```

```
gate_condition_no_network=required
gate_condition_no_root=required
implementation_gate_phase_5_status=disabled
implementation_gate_phase_6_status=disabled
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall implementation-gate contract that keeps future
live reset/uninstall activity closed until required evidence exists.
```

That does not mean Latticra has a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-implementation-gate-contract.sh
```

Expected output:

```
macos_reset_uninstall_implementation_gate_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, receipt evidence, absence verification evidence, operator approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Denial Transcript Contract Status

Lines: 204 | Words: 480 | SHA-256: 8f5e4c8e85db0e0e52ed2944325ca5ee22455776c534e7adf803e44f731feeca

macOS Reset/Uninstall Live-Denial Transcript Contract Status

Status: no-effect reset/uninstall live-denial transcript contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-denial transcript contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-denial transcript contract.

The contract records the failed live-execution preflight decision to stdout only. It keeps deletion disabled, receipt writing disabled, absence-report writing disabled, host mutation disabled, network access absent, root authority absent, and runtime authority absent.

Status Fields

```
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_guard_present=1
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_contract_decision=denied-by-preflight-block
live_denial_transcript_required=1
live_denial_transcript_present=1
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_denial_transcript_receipt_write_enabled=0
live_denial_transcript_absence_report_write_enabled=0
live_denial_transcript_preflight_present=1
live_denial_transcript_preflight_passed=0
live_denial_transcript_denial_recorded=1
live_denial_transcript_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
live_denial_transcript_denial_source=macos-reset-uninstall-live-execution-preflight-contract
live_denial_transcript_effect=none
live_denial_transcript_effect_authorization_required=1
live_denial_transcript_effect_authorized=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_contract_decision=blocked-missing-complete-evidence-bundle-and-effect-a
uthorization
live_execution_preflight_required=1
live_execution_preflight_present=1
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_evidence_present=0
live_execution_preflight_record_write_enabled=0
live_execution_preflight_denial_recorded=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
macos_reset_uninstall_live_implementation_plan_contract_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_implementation_plan_preflight_passed=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
live_denial_transcript_schema_version=macos-reset-uninstall-live-denial-transcript/1
live_denial_transcript_required_input_count=7
live_denial_transcript_observed_input_count=7
live_denial_transcript_requires_live_execution_preflight=1
```

```
live_denial_transcript_requires_preflight_denial=1
live_denial_transcript_requires_denial_reason=1
live_denial_transcript_requires_no_receipt_write=1
live_denial_transcript_requires_no_absence_report_write=1
live_denial_transcript_requires_no_deletion=1
live_denial_transcript_requires_no_network=1
live_denial_transcript_requires_no_root=1
live_denial_transcript_condition_preflight_present=required
live_denial_transcript_condition_preflight_passed=must_be_zero_for_denial
live_denial_transcript_condition_denial_reason=required
live_denial_transcript_condition_stdout_only=required
live_denial_transcript_condition_no_receipt_write=required
live_denial_transcript_condition_no_absence_report_write=required
live_denial_transcript_condition_no_deletion=required
live_denial_transcript_condition_no_network=required
live_denial_transcript_condition_no_root=required
live_denial_transcript_result_preflight_present=met
live_denial_transcript_result_preflight_passed=not_met
live_denial_transcript_result_denial_reason=met
live_denial_transcript_result_stdout_only=met
live_denial_transcript_result_no_receipt_write=met
live_denial_transcript_result_no_absence_report_write=met
live_denial_transcript_result_no_deletion=met
live_denial_transcript_result_no_network=met
live_denial_transcript_result_no_root=met
live_denial_transcript_entry_1_status=met
live_denial_transcript_entry_2_status=recorded
live_denial_transcript_entry_3_status=recorded
live_denial_transcript_entry_4_status=recorded
live_denial_transcript_entry_5_status=recorded
live_denial_transcript_entry_6_status=recorded
live_denial_transcript_entry_7_status=recorded
live_denial_transcript_phase_1_status=contract-only
live_denial_transcript_phase_2_status=stdout-only
live_denial_transcript_phase_3_status=enforced-by-zero-authority
live_denial_transcript_phase_4_status=disabled-until-interface-contract
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall live-denial transcript contract that records why
live deletion remains blocked.
```

That does not mean Latticra has live reset execution, live uninstall execution, complete evidence-bundle evidence, effect approval evidence, live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-denial-transcript-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_denial_transcript_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall live-denial transcript contract that records the failed preflight
decision without deleting files.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Follow-on no-op prototype lane:

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live reset execution, live uninstall execution, complete evidence-bundle evidence, receipt evidence, absence verification evidence, operator approval evidence, effect approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Execution Preflight Contract Status

Lines: 186 | Words: 483 | SHA-256: 88c86f9032d38b17f6c13b617e20864c915a1192219abd7c60c65a7abb487239

macOS Reset/Uninstall Live-Execution Preflight Contract Status

Status: no-effect reset/uninstall live-execution preflight contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-execution preflight contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-execution preflight contract.

The contract proves the live implementation plan still cannot delete because the complete evidence bundle, effect authorization, implementation gate, and operator intent are not satisfied. It keeps deletion disabled, receipt writing disabled, absence-report writing disabled, host mutation disabled, and runtime authority absent.

Status Fields

```
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_guard_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_contract_decision=blocked-missing-complete-evidence-bundle-and-effect-a
uthorization
live_execution_preflight_required=1
live_execution_preflight_present=1
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_evidence_present=0
live_execution_preflight_record_write_enabled=0
live_execution_preflight_denial_recorded=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
macos_reset_uninstall_live_implementation_plan_contract_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_implementation_plan_preflight_passed=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
live_execution_preflight_schema_version=macos-reset-uninstall-live-execution-preflight/1
live_execution_preflight_required_component_count=9
live_execution_preflight_passed_component_count=0
live_execution_preflight_requires_live_implementation_plan=1
live_execution_preflight_requires_complete_evidence_bundle=1
live_execution_preflight_requires_effect_authorization=1
live_execution_preflight_requires_implementation_gate_open=1
live_execution_preflight_requires_operator_intent_evidence=1
live_execution_preflight_requires_dry_run_planner_transcript=1
live_execution_preflight_requires_live_target_classifier_report=1
live_execution_preflight_requires_receipt_schema=1
live_execution_preflight_requires_absence_report_contract=1
live_execution_preflight_requires_no_unmanaged_targets=1
live_execution_preflight_requires_no_unsafe_paths=1
live_execution_preflight_requires_no_network=1
live_execution_preflight_requires_no_root=1
live_execution_preflight_condition_live_implementation_plan=required
live_execution_preflight_condition_complete_evidence_bundle=required
live_execution_preflight_condition_effect_authorization=required
live_execution_preflight_condition_implementation_gate_open=required
live_execution_preflight_condition_operator_intent_evidence_present=required
live_execution_preflight_condition_no_unmanaged_targets=required
live_execution_preflight_condition_no_network=required
live_execution_preflight_condition_no_root=required
live_execution_preflight_result_live_implementation_plan=planned-no-effect
live_execution_preflight_result_complete_evidence_bundle=not_met
```

```
live_execution_preflight_result_effect_authorization=not_met
live_execution_preflight_result_implementation_gate_open=not_met
live_execution_preflight_result_operator_intent_evidence=not_met
live_execution_preflight_result_dry_run_planner_transcript=contract-only
live_execution_preflight_result_live_target_classifier_report=contract-only
live_execution_preflight_result_receipt_schema=contract-only
live_execution_preflight_result_absence_report_contract=contract-only
live_execution_preflight_result_no_unmanaged_targets=not_evaluated_contract_only
live_execution_preflight_result_no_network=met
live_execution_preflight_result_no_root=met
live_execution_preflight_phase_1_status=contract-only
live_execution_preflight_phase_2_status=blocked-evidence-bundle-incomplete
live_execution_preflight_phase_3_status=blocked-effect-authorization-closed
live_execution_preflight_phase_4_status=blocked-operator-intent-missing
live_execution_preflight_phase_5_status=contract-only
live_execution_preflight_phase_6_status=contract-only
live_execution_preflight_phase_7_status=blocked-no-effect
live_execution_preflight_phase_8_status=disabled
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall live-execution preflight contract that proves
live deletion remains blocked until all evidence and authorization gates are satisfied.
```

That does not mean Latticra has live reset execution, live uninstall execution, complete evidence-bundle evidence, effect approval evidence, live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-execution-preflight-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_execution_preflight_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall live-execution preflight contract that proves the live
implementation plan still cannot delete until all evidence gates are satisfied.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Follow-on no-op prototype lane:

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live reset execution, live uninstall execution, complete evidence-bundle evidence, receipt evidence, absence verification evidence, operator approval evidence, effect approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Implementation Plan Contract Status

Lines: 184 | Words: 435 | SHA-256: f72e60fcca20b202f1cf8c890cfaa750d6370f76f3e4fc52632083937ac24693

macOS Reset/Uninstall Live-Implementation Plan Contract Status

Status: no-effect reset/uninstall live-implementation plan contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-implementation plan contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-implementation plan contract.

The contract maps the future effect-authorized reset/uninstall phases while keeping the evidence bundle incomplete, effect authorization closed, deletion disabled, receipt writing disabled, absence-report writing disabled, host mutation disabled, and runtime authority absent.

Status Fields

```
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
macos_reset_uninstall_live_implementation_plan_contract_guard_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_contract_decision=blocked-deletion-disabled-and-evidence-incomplete
live_implementation_plan_required=1
live_implementation_plan_complete=0
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_receipt_write_enabled=0
live_implementation_plan_absence_write_enabled=0
live_implementation_plan_preflight_required=1
live_implementation_plan_preflight_present=1
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_required=1
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
live_implementation_plan_schema_version=macos-reset-uninstall-live-implementation-plan/1
live_implementation_plan_required_component_count=8
live_implementation_plan_complete_component_count=0
live_implementation_requires_evidence_bundle_complete=1
live_implementation_requires_effect_authorization=1
live_implementation_requires_implementation_gate_open=1
live_implementation_requires_operator_intent_evidence=1
live_implementation_requires_no_unmanaged_targets=1
live_implementation_requires_receipt_schema=1
live_implementation_requires_absence_report=1
live_implementation_requires_live_execution_preflight=1
live_implementation_requires_no_network=1
live_implementation_requires_no_root=1
live_implementation_condition_evidence_bundle_complete=required
live_implementation_condition_effect_authorized=required
live_implementation_condition_implementation_gate_open=required
live_implementation_condition_operator_intent_evidence_present=required
live_implementation_condition_no_unmanaged_targets=required
live_implementation_condition_live_execution_preflight_present=required
```

```
live_implementation_condition_no_network=required
live_implementation_condition_no_root=required
live_implementation_result_evidence_bundle_complete=not_met
live_implementation_result_effect_authorized=not_met
live_implementation_result_implementation_gate_open=not_met
live_implementation_result_operator_intent_evidence=not_met
live_implementation_result_no_unmanaged_targets=not_evaluated_contract_only
live_implementation_result_live_execution_preflight=blocked-no-effect
live_implementation_result_no_network=met
live_implementation_result_no_root=met
live_implementation_phase_1_status=contract-only
live_implementation_phase_2_status=blocked-effect-authorization-closed
live_implementation_phase_3_status=contract-only
live_implementation_phase_4_status=blocked-evidence-incomplete
live_implementation_phase_5_status=disabled
live_implementation_phase_6_status=disabled
live_implementation_phase_7_status=disabled
live_implementation_phase_8_status=disabled
live_implementation_phase_9_status=disabled
live_implementation_phase_10_status=disabled
live_implementation_phase_11_status=disabled
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall live-implementation plan contract that maps
future effect-authorized phases while deletion remains disabled.
```

That does not mean Latticra has complete evidence-bundle evidence, effect approval evidence, live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-implementation-plan-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_implementation_plan_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall live-implementation plan contract that maps future effect-authorized
 execution phases while deletion remains disabled.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, complete evidence-bundle evidence, live execution evidence, receipt evidence, absence verification evidence, operator approval evidence, effect approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Contract Status

Lines: 137 | Words: 291 | SHA-256: 56888133598b2206368a819ae94eeaba23facfd09c3ad2db2060a8c12f7a044b

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition closeout audit contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract. It audits the no-effect closeout and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, audit review, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_present=
1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_guard_pr
esent=1
live_runner_acceptance_denial_disposition_closeout_audit_contract_state=audited-no-effect-closeo
ut
live_runner_acceptance_denial_disposition_closeout_audit_contract_decision=no-effect-closeout-au
dit-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_disposition_present=1
live_runner_acceptance_denial_disposition_closeout_audit_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_disposition_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_disposition_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_required=1
live_runner_acceptance_denial_disposition_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_recorded=1
live_runner_acceptance_denial_disposition_closeout_audited=1
live_runner_acceptance_denial_disposition_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_requires_future_audit_review=1
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_audited=1
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_audit_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_closeout_audit_audit_review_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_closeout_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_schema_version=macos-reset-uninstall-li
ve-runner-acceptance-denial-disposition-closeout-audit/1
live_runner_acceptance_denial_disposition_closeout_audit_required_input_count=8
live_runner_acceptance_denial_disposition_closeout_audit_observed_input_count=8
live_runner_acceptance_denial_disposition_closeout_audit_requires_acceptance_denial_disposition=
1
live_runner_acceptance_denial_disposition_closeout_audit_requires_acceptance_denial_disposition_
review=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_closed_gate=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_effect_closeout=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_audit_only=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_effects=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_network=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_audit_review_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_result_acceptance_denial_disposition=me
t
live_runner_acceptance_denial_disposition_closeout_audit_result_acceptance_denial_disposition_re
```

```
view=met
live_runner_acceptance_denial_disposition_closeout_audit_result_closed_gate=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_effect_closeout=met
live_runner_acceptance_denial_disposition_closeout_audit_result_audit_only=met
live_runner_acceptance_denial_disposition_closeout_audit_result_stdout_only=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_effects=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_network=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_root=met
live_runner_acceptance_denial_disposition_closeout_audit_result_audit_review_handoff=met
live_runner_acceptance_denial_disposition_closeout_audit_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_closeout_audit_phase_2_status=audit-only
live_runner_acceptance_denial_disposition_closeout_audit_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_phase_4_status=future-review-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-c
ontract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, deletion evidence, closeout review evidence, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Contract Status

Lines: 142 | Words: 302 | SHA-256: 7e1fa7d21f2d10da4354c084b647120853a60676856c95361b19ba756a7fa827

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition closeout audit review contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review contract. It reviews the no-effect closeout audit and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, disposition opening, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract_p
resent=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract_g
uard_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_state=reviewed-no-effec
t-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_decision=no-effect-clos
eout-audit-review-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_state=audited-no-effect-cl
oseout
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_audited=1
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_audit_review_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_closeout_audit_review_authorizes_effects=0
live_runner_acceptance_denial_disposition_closeout_audit_review_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_present=
1
live_runner_acceptance_denial_disposition_closeout_audit_contract_state=audited-no-effect-closeo
ut
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_audit_review_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract_present=1
live_runner_acceptance_denial_disposition_closeout_contract_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_schema_version=macos-reset-unins
tall-live-runner-acceptance-denial-disposition-closeout-audit-review/1
live_runner_acceptance_denial_disposition_closeout_audit_review_required_input_count=9
live_runner_acceptance_denial_disposition_closeout_audit_review_observed_input_count=9
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_closeout_audit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_closeout=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_acceptance_denial_dispo
sition_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_closed_gate=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_effect_audit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_review_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_effects=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_absence_report_write
```

```
=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_network=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_disposition_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_review_result_closeout_audit=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_closeout=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_acceptance_denial_disposi
tion_review=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_closed_gate=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_effect_audit=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_review_only=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_stdout_only=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_effects=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_absence_report_write=m
et
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_network=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_root=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_disposition_handoff=met
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_2_status=review-only
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_4_status=future-dispositio
n-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-r
eview-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract:
ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, deletion evidence, closeout audit disposition evidence, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Contract Status

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract. It records the reviewed no-effect closeout audit as a no-effect disposition and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, future review opening, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_guard_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_state=disposed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_decision=no-effect-closeout-audit-review-disposition-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_state=reviewed-no-effect-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_source_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_reviewed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_future_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_disposition_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result=no-effect-closeout-audit-review-disposition-recorded
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_schema_version=macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-disposition/1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_closeout_audit_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_effect_disposition=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_disposition_record=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_disposition_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_review_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_review_handoff=met
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
```

```
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-r
eview-disposition-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, disposition application, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_REVIEW_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Review Contract Status

Lines: 91 | Words: 257 | SHA-256: 3dedb569fa766bc56b2d4e97cf7e5305cce4db58163e421ef54ce0769fb78bad

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Review Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract. It reviews the no-effect closeout audit review disposition and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, closeout opening, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract_guard_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_contract_stat
e=reviewed-no-effect-closeout-audit-review-disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_contract_deci
sion=no-effect-closeout-audit-review-disposition-review-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_file_write_en
abled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_review_file_w
rite_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_source_contra
ct=macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-disposi
tion-contract
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_s
tate=disposed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
ecorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
eviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_a
pplied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_o
pens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
eleases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_allo
wed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_perf
ormed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_runner_handof
f_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_deletion_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_receipt_write
_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_absence_repor
t_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_effect_author
ized=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_effect_bounda
ry_preserved=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_requ
ired=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_open
ed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_schema_versio
n=macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-disposit
ion-review/1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout_audit_review_disposition=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_e
ffect_disposition=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_revi
ew_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_d
ispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_d
eletion=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_no_dis
patch=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_no_del
etion=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_closeo
```

```
ut_handoff=met
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-r
eview-disposition-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, closeout evidence, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Contract Status

Lines: 152 | Words: 303 | SHA-256: 01fef8e245851686d8e8a38edaf4289632f833d162ee3dcc1d236ec349373c34

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition closeout contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition closeout contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout contract. It closes the reviewed no-effect disposition and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, runner release, audit opening, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract_guard_present=
1
live_runner_acceptance_denial_disposition_closeout_contract_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_contract_decision=no-effect-disposition-close
out-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_completed=1
live_runner_acceptance_denial_disposition_closeout_recorded=1
live_runner_acceptance_denial_disposition_closeout_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_record_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_review_present=1
live_runner_acceptance_denial_disposition_closeout_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_closeout_review_completed=1
live_runner_acceptance_denial_disposition_closeout_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_review_closeout_opened=0
live_runner_acceptance_denial_disposition_closeout_disposition_present=1
live_runner_acceptance_denial_disposition_closeout_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_closeout_disposition_reviewed=1
live_runner_acceptance_denial_disposition_closeout_disposition_closed=1
live_runner_acceptance_denial_disposition_closeout_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_future_closeout_required=0
live_runner_acceptance_denial_disposition_closeout_closed_gate_reviewed=1
live_runner_acceptance_denial_disposition_closeout_denial_reason_present=1
live_runner_acceptance_denial_disposition_closeout_denial_reason_closed=1
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_dispatch_closed=1
live_runner_acceptance_denial_disposition_closeout_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_closeout_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_closeout_authorizes_effects=0
live_runner_acceptance_denial_disposition_closeout_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_closeout_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_releases_runner=0
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_schema_version=macos-reset-uninstall-live-run
ner-acceptance-denial-disposition-closeout/1
live_runner_acceptance_denial_disposition_closeout_required_input_count=8
live_runner_acceptance_denial_disposition_closeout_observed_input_count=8
live_runner_acceptance_denial_disposition_closeout_requires_acceptance_denial_disposition_review
=1
live_runner_acceptance_denial_disposition_closeout_requires_acceptance_denial_disposition=1
live_runner_acceptance_denial_disposition_closeout_requires_acceptance_denial_review=1
live_runner_acceptance_denial_disposition_closeout_requires_closed_gate=1
live_runner_acceptance_denial_disposition_closeout_requires_no_effect_disposition=1
live_runner_acceptance_denial_disposition_closeout_requires_closeout_only=1
live_runner_acceptance_denial_disposition_closeout_requires_stdout_only=1
```

```
live_runner_acceptance_denial_disposition_closeout_requires_no_effects=1
live_runner_acceptance_denial_disposition_closeout_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_closeout_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_closeout_requires_no_network=1
live_runner_acceptance_denial_disposition_closeout_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_requires_audit_review_handoff=1
live_runner_acceptance_denial_disposition_closeout_result_acceptance_denial_disposition_review=m
et
live_runner_acceptance_denial_disposition_closeout_result_acceptance_denial_disposition=met
live_runner_acceptance_denial_disposition_closeout_result_acceptance_denial_review=met
live_runner_acceptance_denial_disposition_closeout_result_closed_gate=met
live_runner_acceptance_denial_disposition_closeout_result_no_effect_disposition=met
live_runner_acceptance_denial_disposition_closeout_result_closeout_only=met
live_runner_acceptance_denial_disposition_closeout_result_stdout_only=met
live_runner_acceptance_denial_disposition_closeout_result_no_effects=met
live_runner_acceptance_denial_disposition_closeout_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_closeout_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_closeout_result_no_network=met
live_runner_acceptance_denial_disposition_closeout_result_no_root=met
live_runner_acceptance_denial_disposition_closeout_result_audit_review_handoff=met
live_runner_acceptance_denial_disposition_closeout_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_closeout_phase_2_status=closeout-only
live_runner_acceptance_denial_disposition_closeout_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_phase_4_status=future-audit-review-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-contrac
t.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, closeout audit evidence, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Contract Status

Lines: 125 | Words: 272 | SHA-256: d3190dbf5be30f3197eceb7e097835c476ae8e46d0259bb7509416c10d400c09

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition contract. It records the reviewed closed-gate denial as stdout-only evidence and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, disposition application, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_guard_present=1
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_contract_decision=closed-gate-denial-disposed-no-dispa
tch
live_runner_acceptance_denial_disposition_recorded=1
live_runner_acceptance_denial_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_record_write_enabled=0
live_runner_acceptance_denial_disposition_review_present=1
live_runner_acceptance_denial_disposition_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_closed_gate_reviewed=1
live_runner_acceptance_denial_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_dispatch_allowed=0
live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_effect_authorized=0
live_runner_acceptance_denial_disposition_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_requires_future_review=1
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_disposition_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_disposition_schema_version=macos-reset-uninstall-live-runner-accep
tance-denial-disposition/1
live_runner_acceptance_denial_disposition_required_input_count=7
live_runner_acceptance_denial_disposition_observed_input_count=7
live_runner_acceptance_denial_disposition_requires_acceptance_denial_review=1
live_runner_acceptance_denial_disposition_requires_acceptance_denial_transcript=1
live_runner_acceptance_denial_disposition_requires_closed_gate=1
live_runner_acceptance_denial_disposition_requires_denial_reason=1
live_runner_acceptance_denial_disposition_requires_disposition_only=1
live_runner_acceptance_denial_disposition_requires_stdout_only=1
live_runner_acceptance_denial_disposition_requires_no_effects=1
live_runner_acceptance_denial_disposition_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_requires_no_deletion=1
live_runner_acceptance_denial_disposition_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_requires_no_network=1
live_runner_acceptance_denial_disposition_requires_no_root=1
live_runner_acceptance_denial_disposition_requires_review_handoff=1
live_runner_acceptance_denial_disposition_result_acceptance_denial_review=met
live_runner_acceptance_denial_disposition_result_acceptance_denial_transcript=met
live_runner_acceptance_denial_disposition_result_closed_gate=met
live_runner_acceptance_denial_disposition_result_denial_reason=met
live_runner_acceptance_denial_disposition_result_disposition_only=met
live_runner_acceptance_denial_disposition_result_stdout_only=met
live_runner_acceptance_denial_disposition_result_no_effects=met
live_runner_acceptance_denial_disposition_result_no_dispatch=met
live_runner_acceptance_denial_disposition_result_no_deletion=met
live_runner_acceptance_denial_disposition_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_result_no_network=met
live_runner_acceptance_denial_disposition_result_no_root=met
live_runner_acceptance_denial_disposition_result_review_handoff=met
live_runner_acceptance_denial_disposition_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_phase_2_status=stdout-only
live_runner_acceptance_denial_disposition_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_phase_4_status=future-review-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, disposition application, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Review Contract Status

Lines: 132 | Words: 279 | SHA-256: 3e612c9be1fd02614febd46068bc5214039b7fcfaeacfa0edf1e49575935cb3c

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Review Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial disposition review contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial disposition review contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial disposition review contract. It reviews the recorded no-effect disposition and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, closeout, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_guard_present=1
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_contract_decision=no-effect-disposition-review-
keeps-dispatch-closed
live_runner_acceptance_denial_disposition_review_completed=1
live_runner_acceptance_denial_disposition_review_recorded=1
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_review_disposition_recorded=1
live_runner_acceptance_denial_disposition_review_disposition_reviewed=1
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_review_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_review_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_review_closed_gate_reviewed=1
live_runner_acceptance_denial_disposition_review_denial_reason_present=1
live_runner_acceptance_denial_disposition_review_denial_reason_reviewed=1
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_review_dispatch_reviewed=1
live_runner_acceptance_denial_disposition_review_dispatch_allowed=0
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_review_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_review_effect_authorized=0
live_runner_acceptance_denial_disposition_review_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_review_closeout_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_releases_runner=0
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_schema_version=macos-reset-uninstall-live-runne
r-acceptance-denial-disposition-review/1
live_runner_acceptance_denial_disposition_review_required_input_count=7
live_runner_acceptance_denial_disposition_review_observed_input_count=7
live_runner_acceptance_denial_disposition_review_requires_acceptance_denial_disposition=1
live_runner_acceptance_denial_disposition_review_requires_acceptance_denial_review=1
live_runner_acceptance_denial_disposition_review_requires_closed_gate=1
live_runner_acceptance_denial_disposition_review_requires_no_effect_disposition=1
live_runner_acceptance_denial_disposition_review_requires_review_only=1
live_runner_acceptance_denial_disposition_review_requires_stdout_only=1
live_runner_acceptance_denial_disposition_review_requires_no_effects=1
live_runner_acceptance_denial_disposition_review_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_review_requires_no_deletion=1
live_runner_acceptance_denial_disposition_review_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_review_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_review_requires_no_network=1
live_runner_acceptance_denial_disposition_review_requires_no_root=1
live_runner_acceptance_denial_disposition_review_requires_closeout_handoff=1
live_runner_acceptance_denial_disposition_review_result_acceptance_denial_disposition=met
live_runner_acceptance_denial_disposition_review_result_acceptance_denial_review=met
live_runner_acceptance_denial_disposition_review_result_closed_gate=met
live_runner_acceptance_denial_disposition_review_result_no_effect_disposition=met
live_runner_acceptance_denial_disposition_review_result_review_only=met
live_runner_acceptance_denial_disposition_review_result_stdout_only=met
live_runner_acceptance_denial_disposition_review_result_no_effects=met
live_runner_acceptance_denial_disposition_review_result_no_dispatch=met
live_runner_acceptance_denial_disposition_review_result_no_deletion=met
live_runner_acceptance_denial_disposition_review_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_review_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_review_result_no_network=met
live_runner_acceptance_denial_disposition_review_result_no_root=met
```

```
live_runner_acceptance_denial_disposition_review_result_closeout_handoff=met
live_runner_acceptance_denial_disposition_review_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_review_phase_2_status=review-only
live_runner_acceptance_denial_disposition_review_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_review_phase_4_status=future-closeout-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh
scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, closeout evidence, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Review Contract Status

Lines: 130 | Words: 270 | SHA-256: de274c0f7ca63f19b567a0aaa319bfadbcaa0b5d6fe04983b8c2a2c2ec189d8f

macOS Reset/Uninstall Live-Runner Acceptance-Denial Review Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial review contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial review contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial review contract. It reviews the closed acceptance-gate transcript and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_guard_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_contract_decision=closed-gate-review-retains-zero-dispatch
live_runner_acceptance_denial_review_completed=1
live_runner_acceptance_denial_review_recorded=1
live_runner_acceptance_denial_review_stdout_only=1
live_runner_acceptance_denial_review_file_write_enabled=0
live_runner_acceptance_denial_review_review_file_write_enabled=0
live_runner_acceptance_denial_review_closed_gate_transcript_present=1
live_runner_acceptance_denial_review_closed_gate_reviewed=1
live_runner_acceptance_denial_review_denial_reason_present=1
live_runner_acceptance_denial_review_denial_reason_reviewed=1
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_acceptance_gate_closed=1
live_runner_acceptance_denial_review_dispatch_reviewed=1
live_runner_acceptance_denial_review_dispatch_allowed=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_runner_handoff_enabled=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_receipt_write_enabled=0
live_runner_acceptance_denial_review_absence_report_write_enabled=0
live_runner_acceptance_denial_review_effect_authorized=0
live_runner_acceptance_denial_review_effect_boundary_preserved=1
live_runner_acceptance_denial_review_disposition_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_acceptance_gate_closed=1
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_transcript_denial_reason=missing-passed-preflight-complete-evidenc
e-effect-authorization-operator-intent-and-implementation
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_operator_intent=blocked
live_runner_acceptance_gate_result_implementation=blocked
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_acceptance_gate_opened=0
live_runner_acceptance_denial_review_schema_version=macos-reset-uninstall-live-runner-acceptance
-denial-review/1
live_runner_acceptance_denial_review_required_input_count=7
live_runner_acceptance_denial_review_observed_input_count=7
live_runner_acceptance_denial_review_requires_acceptance_denial_transcript=1
live_runner_acceptance_denial_review_requires_closed_gate=1
live_runner_acceptance_denial_review_requires_denial_reason=1
live_runner_acceptance_denial_review_requires_review_only=1
live_runner_acceptance_denial_review_requires_stdout_only=1
live_runner_acceptance_denial_review_requires_no_effects=1
live_runner_acceptance_denial_review_requires_no_dispatch=1
live_runner_acceptance_denial_review_requires_no_deletion=1
live_runner_acceptance_denial_review_requires_no_receipt_write=1
live_runner_acceptance_denial_review_requires_no_absence_report_write=1
live_runner_acceptance_denial_review_requires_no_network=1
live_runner_acceptance_denial_review_requires_no_root=1
live_runner_acceptance_denial_review_requires_disposition_handoff=1
live_runner_acceptance_denial_review_result_acceptance_denial_transcript=met
live_runner_acceptance_denial_review_result_closed_gate=met
live_runner_acceptance_denial_review_result_denial_reason=met
live_runner_acceptance_denial_review_result_review_only=met
live_runner_acceptance_denial_review_result_stdout_only=met
live_runner_acceptance_denial_review_result_no_effects=met
live_runner_acceptance_denial_review_result_no_dispatch=met
live_runner_acceptance_denial_review_result_no_deletion=met
live_runner_acceptance_denial_review_result_no_receipt_write=met
live_runner_acceptance_denial_review_result_no_absence_report_write=met
live_runner_acceptance_denial_review_result_no_network=met
live_runner_acceptance_denial_review_result_no_root=met
live_runner_acceptance_denial_review_result_disposition_handoff=met
live_runner_acceptance_denial_review_phase_1_status=contract-only
```

```
live_runner_acceptance_denial_review_phase_2_status=review-only
live_runner_acceptance_denial_review_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_review_phase_4_status=future-disposition-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_review_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Transcript Contract Status

Lines: 116 | Words: 256 | SHA-256: 3f75fac1cd171c5ef421d3b5204e1fc891245ac4616c2b46c86bcdc385cf7797

macOS Reset/Uninstall Live-Runner Acceptance-Denial Transcript Contract Status

Status: no-effect reset/uninstall live-runner acceptance-denial transcript contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-denial transcript contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-denial transcript contract. It records the closed acceptance gate and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, and implementation claims disabled.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_guard_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_contract_decision=acceptance-gate-closed-no-dispatch-re
corded
live_runner_acceptance_denial_transcript_required=1
live_runner_acceptance_denial_transcript_present=1
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_stdout_only=1
live_runner_acceptance_denial_transcript_file_write_enabled=0
live_runner_acceptance_denial_transcript_acceptance_gate_present=1
live_runner_acceptance_denial_transcript_acceptance_gate_state=closed-no-effect
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_acceptance_gate_closed=1
live_runner_acceptance_denial_transcript_dispatch_allowed=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_runner_handoff_enabled=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_transcript_receipt_write_enabled=0
live_runner_acceptance_denial_transcript_absence_report_write_enabled=0
live_runner_acceptance_denial_transcript_effect=none
live_runner_acceptance_denial_transcript_effect_authorized=0
live_runner_acceptance_denial_transcript_denial_reason=missing-passed-preflight-complete-evidenc
e-effect-authorization-operator-intent-and-implementation
live_runner_acceptance_denial_transcript_denial_source=macos-reset-uninstall-live-runner-accepta
nce-gate-contract
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_denial_transcript_schema_version=macos-reset-uninstall-live-runner-accept
ance-denial-transcript/1
live_runner_acceptance_denial_transcript_requires_acceptance_gate=1
live_runner_acceptance_denial_transcript_requires_closed_gate=1
live_runner_acceptance_denial_transcript_requires_denial_reason=1
live_runner_acceptance_denial_transcript_requires_stdout_only=1
live_runner_acceptance_denial_transcript_requires_no_dispatch=1
live_runner_acceptance_denial_transcript_requires_no_deletion=1
live_runner_acceptance_denial_transcript_requires_no_receipt_write=1
live_runner_acceptance_denial_transcript_requires_no_absence_report_write=1
live_runner_acceptance_denial_transcript_requires_no_network=1
live_runner_acceptance_denial_transcript_requires_no_root=1
live_runner_acceptance_denial_transcript_result_acceptance_gate=met
live_runner_acceptance_denial_transcript_result_closed_gate=met
live_runner_acceptance_denial_transcript_result_denial_reason=met
live_runner_acceptance_denial_transcript_result_stdout_only=met
live_runner_acceptance_denial_transcript_result_no_dispatch=met
live_runner_acceptance_denial_transcript_result_no_deletion=met
live_runner_acceptance_denial_transcript_result_no_receipt_write=met
live_runner_acceptance_denial_transcript_result_no_absence_report_write=met
live_runner_acceptance_denial_transcript_result_no_network=met
live_runner_acceptance_denial_transcript_result_no_root=met
live_runner_acceptance_denial_transcript_phase_1_status=contract-only
live_runner_acceptance_denial_transcript_phase_2_status=stdout-only
live_runner_acceptance_denial_transcript_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_transcript_phase_4_status=future-review-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
```

```
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-transcript-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Gate Contract Status

Lines: 112 | Words: 253 | SHA-256: 246648d471bddff29e34e52ca37cd3ddf9d883a8d08a735afd481850bd87d1c4

macOS Reset/Uninstall Live-Runner Acceptance-Gate Contract Status

Status: no-effect reset/uninstall live-runner acceptance-gate contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner acceptance-gate contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner acceptance-gate contract. It evaluates the denied-dispatch review and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, and runtime authority disabled because the required acceptance inputs are not complete.

Status Fields

```
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
macos_reset_uninstall_live_runner_acceptance_gate_contract_guard_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_contract_decision=dispatch-blocked-missing-passed-preflight-evidence
-and-authorization
live_runner_acceptance_gate_required=1
live_runner_acceptance_gate_present=1
live_runner_acceptance_gate_evaluated=1
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_acceptance_granted=0
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_stdout_only=1
live_runner_acceptance_gate_file_write_enabled=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_runner_handoff_enabled=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_receipt_write_enabled=0
live_runner_acceptance_gate_absence_report_write_enabled=0
live_runner_acceptance_gate_effect=none
live_runner_acceptance_gate_effect_authorized=0
live_runner_acceptance_gate_denial_reason=missing-passed-preflight-complete-evidence-effect-auth
orization-operator-intent-and-implementation
live_runner_acceptance_gate_source_contract=macos-reset-uninstall-live-runner-denied-dispatch-re
view-contract
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
live_runner_acceptance_gate_schema_version=macos-reset-uninstall-live-runner-acceptance-gate/1
live_runner_acceptance_gate_required_input_count=7
live_runner_acceptance_gate_observed_input_count=7
live_runner_acceptance_gate_requires_denied_dispatch_review=1
live_runner_acceptance_gate_requires_passed_preflight=1
live_runner_acceptance_gate_requires_complete_evidence_bundle=1
live_runner_acceptance_gate_requires_effect_authorization=1
live_runner_acceptance_gate_requires_operator_intent=1
live_runner_acceptance_gate_requires_implementation=1
live_runner_acceptance_gate_requires_no_dispatch_until_open=1
live_runner_acceptance_gate_requires_no_deletion_until_open=1
live_runner_acceptance_gate_requires_no_receipt_write_until_open=1
live_runner_acceptance_gate_requires_no_absence_report_write_until_open=1
live_runner_acceptance_gate_requires_no_network=1
live_runner_acceptance_gate_requires_no_root=1
live_runner_acceptance_gate_result_denied_dispatch_review=met
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_operator_intent=blocked
live_runner_acceptance_gate_result_implementation=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_gate_result_no_deletion_until_open=met
live_runner_acceptance_gate_result_no_receipt_write_until_open=met
live_runner_acceptance_gate_result_no_absence_report_write_until_open=met
live_runner_acceptance_gate_result_no_network=met
live_runner_acceptance_gate_result_no_root=met
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
```

```
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-gate-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_gate_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Denied-Dispatch Review Contract Status

Lines: 111 | Words: 245 | SHA-256: f695480e0a7ec706ab4d6f956f542a0cc78ebfcf36d1dc9abb4bb4980b5bf059

macOS Reset/Uninstall Live-Runner Denied-Dispatch Review Contract Status

Status: no-effect reset/uninstall live-runner denied-dispatch review contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner denied-dispatch review contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner denied-dispatch review contract. It reviews the denied-dispatch transcript and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, and runtime authority disabled.

Status Fields

```
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_guard_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_contract_decision=denial-review-retains-zero-dispatch
live_runner_denied_dispatch_review_required=1
live_runner_denied_dispatch_review_present=1
live_runner_denied_dispatch_review_completed=1
live_runner_denied_dispatch_review_stdout_only=1
live_runner_denied_dispatch_review_file_write_enabled=0
live_runner_denied_dispatch_review_dispatch_denial_present=1
live_runner_denied_dispatch_review_denial_reason_present=1
live_runner_denied_dispatch_review_dispatch_reviewed=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_effect=none
live_runner_denied_dispatch_review_effect_authorized=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
live_runner_denied_dispatch_review_source_contract=macos-reset-uninstall-live-runner-denied-disp
atch-transcript-contract
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
live_runner_denied_dispatch_review_transcript_state=recorded-no-effect
live_runner_denied_dispatch_review_transcript_decision=dispatch-denied-no-effects-recorded
live_runner_denied_dispatch_review_transcript_dispatch_denied=1
live_runner_denied_dispatch_review_transcript_dispatch_enabled=0
live_runner_denied_dispatch_review_transcript_file_write_enabled=0
live_runner_denied_dispatch_review_transcript_denial_reason=failed-preflight-and-denied-interfac
e
live_runner_denied_dispatch_review_schema_version=macos-reset-uninstall-live-runner-denied-dispa
tch-review/1
live_runner_denied_dispatch_review_required_input_count=7
live_runner_denied_dispatch_review_observed_input_count=7
live_runner_denied_dispatch_review_requires_denied_dispatch_transcript=1
live_runner_denied_dispatch_review_requires_dispatch_denied=1
live_runner_denied_dispatch_review_requires_denial_reason=1
live_runner_denied_dispatch_review_requires_review_only=1
live_runner_denied_dispatch_review_requires_stdout_only=1
live_runner_denied_dispatch_review_requires_no_dispatch=1
live_runner_denied_dispatch_review_requires_no_deletion=1
live_runner_denied_dispatch_review_requires_no_receipt_write=1
live_runner_denied_dispatch_review_requires_no_absence_report_write=1
live_runner_denied_dispatch_review_requires_no_network=1
live_runner_denied_dispatch_review_requires_no_root=1
live_runner_denied_dispatch_review_result_denied_dispatch_transcript=met
live_runner_denied_dispatch_review_result_dispatch_denied=met
live_runner_denied_dispatch_review_result_denial_reason=met
live_runner_denied_dispatch_review_result_review_only=met
live_runner_denied_dispatch_review_result_stdout_only=met
live_runner_denied_dispatch_review_result_no_dispatch=met
live_runner_denied_dispatch_review_result_no_deletion=met
live_runner_denied_dispatch_review_result_no_receipt_write=met
live_runner_denied_dispatch_review_result_no_absence_report_write=met
live_runner_denied_dispatch_review_result_no_network=met
live_runner_denied_dispatch_review_result_no_root=met
live_runner_denied_dispatch_review_phase_1_status=contract-only
live_runner_denied_dispatch_review_phase_2_status=review-only
live_runner_denied_dispatch_review_phase_3_status=blocked-no-effect
live_runner_denied_dispatch_review_phase_4_status=future-gate-only
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-denied-dispatch-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_denied_dispatch_review_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Denied-Dispatch Transcript Contract Status

Lines: 119 | Words: 254 | SHA-256: 834a45c35508a9691aa56f897c3939d5fd33c27fb70ccaf8c4583eb2337a7f4f

macOS Reset/Uninstall Live-Runner Denied-Dispatch Transcript Contract Status

Status: no-effect reset/uninstall live-runner denied-dispatch transcript contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner denied-dispatch transcript contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner denied-dispatch transcript contract. It records the no-op prototype denial while keeping dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, and runtime authority disabled.

Status Fields

```
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_guard_present=1
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_contract_decision=dispatch-denied-no-effects-recorded
live_runner_denied_dispatch_transcript_recorded=1
live_runner_denied_dispatch_transcript_stdout_only=1
live_runner_denied_dispatch_transcript_file_write_enabled=0
live_runner_denied_dispatch_transcript_dispatch_attempted=0
live_runner_denied_dispatch_transcript_dispatch_enabled=0
live_runner_denied_dispatch_transcript_dispatch_denied=1
live_runner_denied_dispatch_transcript_denial_reason=failed-preflight-and-denied-interface
live_runner_denied_dispatch_transcript_denial_source=macos-reset-uninstall-live-runner-noop-prot
otype-contract
live_runner_denied_dispatch_transcript_effect=none
live_runner_denied_dispatch_transcript_noop_prototype_present=1
live_runner_denied_dispatch_transcript_noop_denial_path_exercised=1
live_runner_denied_dispatch_transcript_noop_dispatch_enabled=0
live_runner_denied_dispatch_transcript_noop_deletion_enabled=0
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_contract_decision=denied-interface-path-only
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_noop_prototype_receipt_write_enabled=0
live_runner_noop_prototype_absence_report_write_enabled=0
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_no_effects_performed=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
live_runner_denied_dispatch_transcript_schema_version=macos-reset-uninstall-live-runner-denied-d
ispatch-transcript/1
live_runner_denied_dispatch_transcript_required_input_count=6
live_runner_denied_dispatch_transcript_observed_input_count=6
live_runner_denied_dispatch_transcript_requires_noop_prototype=1
live_runner_denied_dispatch_transcript_requires_denied_dispatch=1
live_runner_denied_dispatch_transcript_requires_denial_reason=1
live_runner_denied_dispatch_transcript_requires_stdout_only=1
live_runner_denied_dispatch_transcript_requires_no_deletion=1
live_runner_denied_dispatch_transcript_requires_no_receipt_write=1
live_runner_denied_dispatch_transcript_requires_no_absence_report_write=1
live_runner_denied_dispatch_transcript_requires_no_network=1
live_runner_denied_dispatch_transcript_requires_no_root=1
live_runner_denied_dispatch_transcript_result_noop_prototype=met
live_runner_denied_dispatch_transcript_result_denied_dispatch=met
live_runner_denied_dispatch_transcript_result_denial_reason=met
live_runner_denied_dispatch_transcript_result_stdout_only=met
live_runner_denied_dispatch_transcript_result_no_deletion=met
live_runner_denied_dispatch_transcript_result_no_receipt_write=met
live_runner_denied_dispatch_transcript_result_no_absence_report_write=met
live_runner_denied_dispatch_transcript_result_no_network=met
live_runner_denied_dispatch_transcript_result_no_root=met
live_runner_denied_dispatch_transcript_phase_1_status=contract-only
live_runner_denied_dispatch_transcript_phase_2_status=stdout-only
live_runner_denied_dispatch_transcript_phase_3_status=blocked-no-effect
live_runner_denied_dispatch_transcript_phase_4_status=future-review-only
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
```

```
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-denied-dispatch-transcript-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner Interface Contract Status

Lines: 170 | Words: 442 | SHA-256: 93db58b1493ef1489a7b5e647373b83b87702f21b141ff6d1feadf3f2cdbc1eb

macOS Reset/Uninstall Live-Runner Interface Contract Status

Status: no-effect reset/uninstall live-runner interface contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner interface contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner interface contract.

The contract defines the future runner entrypoint shape and requires a passed live-execution preflight before dispatch. The current preflight is not passed, so dispatch, handoff, deletion, receipt writing, absence-report writing, host mutation, network access, root authority, and runtime authority remain disabled.

Status Fields

```
macos_reset_uninstall_live_runner_interface_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_guard_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_present=1
live_runner_interface_accepts_only_passed_preflight=1
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
live_runner_interface_receipt_write_enabled=0
live_runner_interface_absence_write_enabled=0
live_runner_interface_requires_live_denial_transcript=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_preflight_passed=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_contract_decision=blocked-missing-complete-evidence-bundle-and-effect-a
uthorization
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
macos_reset_uninstall_live_implementation_plan_contract_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_implementation_plan_preflight_passed=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
live_runner_interface_schema_version=macos-reset-uninstall-live-runner-interface/1
live_runner_interface_required_input_count=8
live_runner_interface_observed_input_count=8
live_runner_interface_requires_passed_preflight=1
live_runner_interface_requires_complete_evidence_bundle=1
live_runner_interface_requires_effect_authorization=1
live_runner_interface_requires_implementation_gate_open=1
live_runner_interface_requires_operator_intent_evidence=1
live_runner_interface_requires_no_unmanaged_targets=1
live_runner_interface_requires_no_deletion=1
live_runner_interface_requires_no_network=1
live_runner_interface_requires_no_root=1
live_runner_interface_result_passed_preflight=not_met
live_runner_interface_result_live_denial_transcript=met
live_runner_interface_result_complete_evidence_bundle=not_met
live_runner_interface_result_effect_authorization=not_met
live_runner_interface_result_implementation_gate_open=not_met
live_runner_interface_result_operator_intent_evidence=not_met
live_runner_interface_result_no_unmanaged_targets=not_evaluated_contract_only
```

```
live_runner_interface_result_no_deletion=met
live_runner_interface_result_no_network=met
live_runner_interface_result_no_root=met
live_runner_interface_phase_1_status=contract-only
live_runner_interface_phase_2_status=contract-only
live_runner_interface_phase_3_status=blocked-preflight-not-passed
live_runner_interface_phase_4_status=recorded-no-effect
live_runner_interface_phase_5_status=blocked-no-effect
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall live-runner interface contract that rejects the
current failed preflight before any dispatch can occur.
```

That does not mean Latticra has live reset execution, live uninstall execution, complete evidence-bundle evidence, effect approval evidence, live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-runner-interface-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_interface_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner interface contract that accepts only a passed preflight
and keeps deletion disabled otherwise.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live reset execution, live uninstall execution, complete evidence-bundle evidence, receipt evidence, absence verification evidence, operator approval evidence, effect approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Live-Runner No-Op Prototype Contract Status

Lines: 99 | Words: 235 | SHA-256: fa61cd0da44f577d052cea1ad1f2dff1f88db7dca8a4642ec87c9a81944c6f06

macOS Reset/Uninstall Live-Runner No-Op Prototype Contract Status

Status: no-effect reset/uninstall live-runner no-op prototype contract status Date: 2026-05-26 CDT Scope: status checkpoint after adding the macOS reset/uninstall live-runner no-op prototype contract.

Summary

Latticra now has a no-effect macOS reset/uninstall live-runner no-op prototype contract. It exercises only the denied interface path and leaves dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, and runtime authority disabled.

Status Fields

```
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
macos_reset_uninstall_live_runner_noop_prototype_contract_guard_present=1
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_contract_decision=denied-interface-path-only
live_runner_noop_prototype_invocation_simulated=1
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_current_interface_decision=deny
live_runner_noop_prototype_preflight_passed=0
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_accept_path_exercised=0
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_runner_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_noop_prototype_receipt_write_enabled=0
live_runner_noop_prototype_absence_report_write_enabled=0
live_runner_noop_prototype_effect=none
live_runner_noop_prototype_effect_authorized=0
live_runner_noop_prototype_interface_present=1
live_runner_noop_prototype_interface_denial_path_active=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_deletion_enabled=0
live_runner_noop_prototype_schema_version=macos-reset-uninstall-live-runner-noop-prototype/1
live_runner_noop_prototype_required_input_count=6
live_runner_noop_prototype_observed_input_count=6
live_runner_noop_prototype_requires_runner_interface=1
live_runner_noop_prototype_requires_denied_interface=1
live_runner_noop_prototype_requires_denied_interface_path=1
live_runner_noop_prototype_requires_denial_transcript=1
live_runner_noop_prototype_requires_no_dispatch=1
live_runner_noop_prototype_requires_no_deletion=1
live_runner_noop_prototype_requires_no_network=1
live_runner_noop_prototype_requires_no_root=1
live_runner_noop_prototype_result_runner_interface=met
live_runner_noop_prototype_result_denied_interface=met
live_runner_noop_prototype_result_denied_interface_path=met
live_runner_noop_prototype_result_denial_path=met
live_runner_noop_prototype_result_no_dispatch=met
live_runner_noop_prototype_result_no_deletion=met
live_runner_noop_prototype_result_no_network=met
live_runner_noop_prototype_result_no_root=met
live_runner_noop_prototype_phase_1_status=contract-only
live_runner_noop_prototype_phase_2_status=simulated-no-effect
live_runner_noop_prototype_phase_3_status=denied-no-effect
live_runner_noop_prototype_phase_4_status=blocked-no-effect
live_runner_noop_prototype_phase_5_status=stdout-only
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
file_delete_performed=0
directory_delete_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Guard Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-noop-prototype-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_noop_prototype_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER_STATUS.md

Source title: macOS Reset/Uninstall Live-Target Classifier Status

Lines: 128 | Words: 332 | SHA-256: c608052c7f1ab34646341b957d17c74da09fc69dd067725603d8b595ff0149f0

macOS Reset/Uninstall Live-Target Classifier Status

Status: no-effect reset/uninstall live-target classifier status Date: 2026-05-25 CDT Scope: status checkpoint after adding the read-only macOS reset/uninstall live-target classifier.

Summary

Latticra now has a no-effect macOS reset/uninstall live-target classifier.

The classifier inspects the guarded user-local macOS reset/uninstall targets and reports whether each target is absent, managed, or unmanaged-preserve. It does not delete files, write receipts, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_live_target_classifier_guard_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
target_states_recorded=1
managed_marker_required=1
unmanaged_target_preservation_required=1
managed_target_detected=report-runtime
unmanaged_target_detected=report-runtime
present_target_detected=report-runtime
absent_target_detected=report-runtime
reset_uninstall_dry_run_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall target classifier that can report absent,
managed, and unmanaged user-local targets.
```

That does not mean Latticra has a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-target-classifier.sh
```

Expected output:

```
macos_reset_uninstall_live_target_classifier: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Operator-Intent Contract Status

Lines: 206 | Words: 446 | SHA-256: c7a7ee1896c27fd8ed8e3ff83973885d491b4a6d37a71c2e9803025e7894ab57

macOS Reset/Uninstall Operator-Intent Contract Status

Status: no-effect reset/uninstall operator-intent contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall operator-intent contract.

Summary

Latticra now has a no-effect macOS reset/uninstall operator-intent contract.

The contract defines the explicit future approval evidence required before a live reset/uninstall could be considered. It does not prompt for approval, infer approval, delete files, write receipts, write absence reports, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_operator_intent_contract_guard_present=1
operator_intent_contract_state=defined-no-effect
operator_intent_contract_decision=contract-defined-intent-not-observed
operator_intent_contract_required=1
operator_intent_contract_open=0
operator_reset_uninstall_intent_required=1
operator_reset_uninstall_intent_evidence_required=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
operator_intent_capture_performed=0
operator_intent_record_write_enabled=0
operator_intent_record_written=0
operator_intent_evidence_written=0
operator_intent_transcript_write_performed=0
implementation_gate_open=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
implementation_gate_contract_state=closed-no-effect
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
```

```
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
reset_uninstall_dry_run_evidence_present=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
absence_report_evidence_present=0
operator_intent_evidence_format=text-transcript
operator_intent_schema_version=macos-reset-uninstall-operator-intent/1
operator_intent_digest_algorithm=sha256
operator_intent_must_be_explicit=1
operator_intent_must_be_current_session=1
operator_intent_must_not_be_inferred=1
operator_intent_must_not_be_defaulted=1
operator_intent_must_not_be_from_previous_session=1
operator_intent_must_name_operation=1
operator_intent_operation_values=reset,uninstall
operator_intent_must_name_target_scope=1
operator_intent_target_scope=user-local-managed-targets-only
operator_intent_must_acknowledge_managed_targets_only=1
operator_intent_must_acknowledge_no_unmanaged_removal=1
operator_intent_must_acknowledge_receipt_path=1
operator_intent_must_acknowledge_absence_report_path=1
operator_intent_must_acknowledge_no_network=1
operator_intent_must_acknowledge_no_root=1
operator_intent_must_acknowledge_live_classifier_digest=1
operator_intent_must_acknowledge_dry_run_planner_digest=1
operator_intent_field_dry_run_planner_transcript_digest_required=1
operator_intent_field_live_classifier_digest_required=1
operator_intent_field_receipt_schema_digest_required=1
operator_intent_field_absence_report_contract_digest_required=1
operator_intent_field_confirmation_text_required=1
operator_intent_scope_must_match_live_classifier=1
operator_intent_scope_must_match_dry_run_planner=1
operator_intent_confirmation_phrase_required=1
operator_intent_confirmation_phrase_future=reset-or-uninstall-latticra-user-local-managed-target
s
operator_intent_transcript_required=1
operator_intent_phase_5_status=disabled
operator_intent_phase_6_status=disabled
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

> Latticra has a no-effect macOS reset/uninstall operator-intent contract for future explicit live reset/uninstall approval evidence.

That does not mean Latticra has live approval evidence, a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-operator-intent-contract.sh
```

Expected output:

```
macos_reset_uninstall_operator_intent_contract: ok
```

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not operator approval evidence, macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, live approval evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT_STATUS.md

Source title: macOS Reset/Uninstall Receipt-Schema Contract Status

Lines: 143 | Words: 355 | SHA-256: f6174d75e1db4479642566d60645e4f009ecb56e5fc00ed1cc4659d3ab5896c7

macOS Reset/Uninstall Receipt-Schema Contract Status

Status: no-effect reset/uninstall receipt-schema contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS reset/uninstall receipt-schema contract.

Summary

Latticra now has a no-effect macOS reset/uninstall receipt-schema contract.

The contract defines the future reset/uninstall receipt location, schema version, JSON format, digest binding, target-action fields, preserved-target fields, and authority-denial fields. It does not delete files, write receipts, write absence reports, mutate host state, or claim reset/uninstall implementation.

Status Fields

```
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
macos_reset_uninstall_receipt_schema_contract_guard_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
receipt_schema_contract_state=defined-no-effect
receipt_schema_required=1
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
reset_receipt_write_performed=0
receipt_write_performed=0
receipt_written=0
receipt_must_be_outside_removed_prefix=1
removed_prefix_must_not_contain_receipt=1
receipt_format=json
receipt_schema_version=macos-reset-uninstall-receipt/1
receipt_digest_algorithm=sha256
receipt_field_schema_version_required=1
receipt_field_operation_required=1
receipt_field_host_identity_required=1
receipt_field_architecture_required=1
receipt_field_started_at_required=1
receipt_field_completed_at_required=1
receipt_field_planner_transcript_digest_required=1
receipt_field_live_classifier_digest_required=1
receipt_field_absence_report_path_required=1
receipt_field_target_actions_required=1
receipt_field_removed_managed_targets_required=1
receipt_field_preserved_unmanaged_targets_required=1
receipt_field_authority_denials_required=1
receipt_field_no_network_required=1
receipt_field_no_root_required=1
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
```

```
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect macOS reset/uninstall receipt-schema contract for future
reset/uninstall receipts outside the removed prefix.
```

That does not mean Latticra has a macOS reset implementation, uninstall implementation, installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-reset-uninstall-receipt-schema-contract.sh
```

Expected output:

```
macos_reset_uninstall_receipt_schema_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, app bundle evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT_STATUS.md

Source title: macOS User-Local App Bundle Contract Status

Lines: 83 | Words: 300 | SHA-256: 5c04e96ec0d7aefc3bd6d15861d266920fec7c364689268c8f9ec10e600a1673

macOS User-Local App Bundle Contract Status

Status: app bundle contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS user-local app bundle contract.

Summary

Latticra now has a macOS user-local app bundle contract.

The contract defines exact future app bundle files, Info.plist requirements, Application Support files, CLI wrapper managed markers, receipt requirements, reset/uninstall behavior, and verification transcript requirements before any app bundle writer exists.

This is contract/status work only. It does not create an app bundle, install wrappers, write Application Support files, mutate shell profiles, use launchd, access Keychain, request TCC permissions, use Endpoint Security, use System Extensions, use Network Extensions, open the network, or grant platform authority.

Status Fields

```
macos_user_local_app_bundle_contract_present=1
macos_user_local_app_bundle_implementation_plan_present=1
macos_app_bundle_writer_dry_run_present=1
macos_app_bundle_writer_alignment_present=1
macos_app_bundle_exact_files_recorded=1
macos_info_plist_requirements_recorded=1
macos_application_support_files_recorded=1
macos_cli_wrapper_markers_recorded=1
macos_receipt_requirements_recorded=1
macos_reset_uninstall_contract_recorded=1
macos_verification_transcript_requirements_recorded=1
macos_app_bundle_writer_present=0
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a contract for a future managed user-local macOS app bundle lane.
```

That does not mean Latticra has a macOS installer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-user-local-app-bundle-contract.sh
```

Expected output:

```
macos_user_local_app_bundle_contract: ok
```

Next Recommended Lane

```
Add a no-effect macOS app bundle writer dry-run prototype that emits the planned phase report,
validates unsafe paths, and keeps commit_user_local_managed_artifacts=0.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN_STATUS.md

Source title: macOS User-Local App Bundle Implementation Plan Status

Lines: 82 | Words: 316 | SHA-256: 3de625203f13426c2a3c4330c893105395311f3eeb2ce51730588ff96c0f112b

macOS User-Local App Bundle Implementation Plan Status

Status: implementation plan status Date: 2026-05-25 CDT Scope: status checkpoint after adding the no-effect macOS user-local app bundle implementation plan.

Summary

Latticra now has a no-effect implementation plan for a future macOS user-local app bundle writer.

The plan defines writer phases, failure behavior, reset/uninstall sequencing, verification commands, and required future guard tests. It now records the macOS dry-run writer candidate integration as the no-effect bridge from local candidate inputs to the writer dry-run future commit-gate decision, and the macOS commit gate contract as the closed boundary before writes. It keeps the writer absent and records that no app bundle, wrapper, Application Support file, receipt, or verification transcript has been created.

Status Fields

```
macos_user_local_app_bundle_implementation_plan_present=1
macos_app_bundle_writer_dry_run_present=1
macos_app_bundle_writer_alignment_present=1
macos_local_candidate_asset_probe_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
macos_writer_phase_plan_recorded=1
macos_writer_failure_behavior_recorded=1
macos_reset_uninstall_sequence_recorded=1
macos_writer_verification_commands_recorded=1
macos_future_writer_guard_tests_recorded=1
macos_app_bundle_writer_present=0
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect implementation plan for a future managed user-local macOS app bundle
writer.
```

That does not mean Latticra has a macOS installer, app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-user-local-app-bundle-implementation-plan.sh
```

Expected output:

```
macos_user_local_app_bundle_implementation_plan: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

docs/status/MACOS_VERIFICATION_TRANSCRIPT_CONTRACT_STATUS.md

Source title: macOS Verification Transcript Contract Status

Lines: 141 | Words: 337 | SHA-256: df39eb80aa8b0dca16f09db3b7edc641f59ed2c397cbbbe24fed21a300ae02a6

macOS Verification Transcript Contract Status

Status: no-effect verification transcript contract status Date: 2026-05-25 CDT Scope: status checkpoint after adding the macOS verification transcript contract.

Summary

Latticra now has a no-effect macOS verification transcript contract.

The contract defines the exact post-write evidence a future user-local app bundle writer must emit before any status can claim macos_install_verified=1. It does not write a transcript, verify an install, create an app bundle, mutate host state, or grant authority.

Status Fields

```
macos_verification_transcript_contract_present=1
macos_verification_transcript_contract_guard_present=1
macos_commit_gate_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_live_target_classifier_present=1
verification_transcript_contract_state=defined-no-effect
verification_transcript_contract_decision=contract-defined-evidence-not-present
verification_transcript_required=1
verification_transcript_evidence_present=0
macos_install_verified=0
commit_user_local_managed_artifacts=0
managed_write_implementation_present=0
reset_uninstall_implementation_present=0
host_os_macos_recorded_required=1
architecture_recorded_required=1
managed_app_bundle_present_required=1
info_plist_present_required=1
app_executable_digest_required=1
icon_asset_digest_required=1
application_support_marker_required=1
cli_wrapper_marker_required=1
receipt_completeness_required=1
authority_denial_fields_required=1
candidate_integration_ready_required=1
commit_gate_closed_until_evidence_required=1
reset_uninstall_dry_run_required=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_dry_run_evidence_present=0
absence_report_evidence_present=0
unmanaged_target_preservation_required=1
seal_report_only_output_required=1
lat_or_lir_no_effect_probe_required=1
verification_transcript_run_performed=0
verification_transcript_written=0
application_support_write_performed=0
app_bundle_write_performed=0
receipt_write_performed=0
cli_wrapper_write_performed=0
```

```
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Public Meaning

The careful public meaning is:

```
Latticra has a no-effect contract for the future macOS user-local install verification
transcript.
```

That does not mean Latticra has a macOS installer, commit-capable app bundle writer, verified app bundle, signed build, notarized build, launchd integration, Keychain integration, Endpoint Security integration, System Extension integration, privileged helper, or production security capability.

Guard Validation

This status record is guarded by:

```
sh scripts/test-macos-verification-transcript-contract.sh
```

Expected output:

```
macos_verification_transcript_contract: ok
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Non-Claims

This status record is not macOS install evidence, app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

docs/status/MEMORY_SAFETY_ROADMAP_STATUS.md

Source title: Latticra Memory-Safety Roadmap Status

Lines: 44 | Words: 95 | SHA-256: 409de03ef19271614ee9d352ace25a80378cdc9c9f9d56ea15af5fb23cda0525

Latticra Memory-Safety Roadmap Status

Status: status record for memory-safety roadmap Date: 2026-05-26

Scope

This record tracks the component-level memory-safety roadmap created from the current high-assurance security baseline.

It does not rewrite current code, certify memory safety, implement runtime execution, implement host behavior, implement network behavior, implement cryptographic enforcement, claim compliance, claim certification, claim production protection, or grant runtime

authority.

Current fields

```
memory_safety_roadmap_present=1
memory_safety_roadmap_status_present=1
memory_safety_roadmap_guard_present=1
high_assurance_security_baseline_present=1
c_cpp_security_profile_present=1
c_abi_boundary_policy_present=1
restricted_c_cpp_profile_required=1
memory_safe_language_preferred_for_new_high_risk_components=1
memory_safe_language_exception_contract_required=1
parser_fuzzing_required_before_security_boundary_claim=1
unsafe_exception_record_required=1
component_memory_safety_inventory_present=1
implementation_behavior_changed=0
runtime_authority_granted=0
security_boundary_claimed=0
memory_safety_guarantee_claimed=0
production_protection_claim_allowed=0
```

Validation

```
sh scripts/test-memory-safety-roadmap.sh
```

Expected output:

```
memory_safety_roadmap: ok
```

docs/status/NADIA_AWARENESS_DIALOGUE_CONTRACT_STAGE_15_STATUS.md

Source title: Nadia Awareness Dialogue Contract Stage-15 Status

Lines: 111 | Words: 287 | SHA-256: 15211ed6d33d9227d7be07884dfa5252878bf2db1c84704d16b59c770bf505ba

Nadia Awareness Dialogue Contract Stage-15 Status

Status: implementation status record Date: 2026-05-25 Scope: future Q&A dialogue scope for Nadia Initiative awareness topics before prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-15 adds an awareness-dialogue contract generator.

The contract verifies Stage-14 prompt-materialization metadata and inherited protective, prompt, model, runtime, and tool-denial boundaries, then records an official-source-grounded topic register for future Q&A about Nadia Initiative awareness work. It remains contract-only and does not generate dialogue.

Evidence Flags

```
nadia_stage_15_awareness_dialogue_contract_present=1
nadia_awareness_dialogue_contract_generator_present=1
nadia_awareness_dialogue_contract_guard_present=1
nadia_installed_awareness_dialogue_contract_command_planned=1
awareness_dialogue_contract_command=scripts/nadia-awareness-dialogue-contract.sh
installed_awareness_dialogue_contract_command=latticra-nadia awareness-dialogue
future_qa_dialogue_capability_planned=1
awareness_dialogue_contract_status=contract_only
awareness_dialogue_stage=contract-only
awareness_dialogue_authority=0
awareness_dialogue_allowed=0
dialogue_generation_authority=0
dialogue_generation_allowed=0
qa_dialogue_generated=0
dialogue_turns_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
plain_language_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
active_topic_update_authority=0
live_web_lookup_authority=0
dialogue_decision=blocked_contract_only
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_prompt_evaluation_handoff_contract=1
awareness_dialogue_promotion_allowed=0
topic_yazidi_genocide_awareness=1
topic_survivor_voice_and_dignity=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_genocide_prevention=1
topic_justice_and_accountability=1
topic_women_peace_justice_security=1
topic_sinjar_reconstruction=1
topic_security_and_safe_return=1
topic_education_restoration=1
topic_healthcare_and_mental_health=1
topic_livelihoods_and_food_security=1
topic_wash_clean_water_sanitation_hygiene=1
topic_womens_empowerment=1
topic_legal_rights_and_reparations_awareness=1
topic_cultural_preservation_and_memorialization=1
topic_community_driven_survivor_centric_development=1
topic_responsible_support_and_digital_activism=1
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
survivor_impersonation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_materialized=0
prompt_tokenized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
tool_execution_performed=0
network_authority=0
```

Current Claim

Nadia can now produce an awareness-dialogue contract that defines a future Q&A scope for Nadia Initiative awareness topics.

This does not mean Nadia can generate dialogue, answer user prompts, browse the web, provide legal advice, provide medical advice, provide trauma counseling, receive prompt text, materialize prompt content, tokenize prompts, evaluate prompts, load model weights, spawn a runtime process, generate tokens, run inference, execute tools, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-awareness-dialogue-contract-stage-15.sh
```

Expected result:

```
nadia_awareness_dialogue_contract_stage_15: ok
```

Next Stage

Stage-16 now defines a prompt-evaluation handoff contract only after awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_CONTEXT_WINDOW_ASSEMBLY_CONTRACT_STAGE_27_STATUS.md

Source title: Nadia Context Window Assembly Contract Stage-27 Status

Lines: 109 | Words: 315 | SHA-256: 6cc57b70d9a86e23bf2e883b0c2ef1cbb1362bed3d7f80487669317d3edef4b1

Nadia Context Window Assembly Contract Stage-27 Status

Status: implementation status record

Scope: context-window assembly metadata before context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-27 adds a context-window assembly contract generator.

It consumes the Stage-26 prompt-token-sequence report as prerequisite evidence and records a future review lane for context-window layout, token budget, truncation, ordering, attention-mask policy, position-ID policy, and prompt-evaluation-input policy. It remains contract-only: no prompt text is read, no context window is assembled, no prompt evaluation input is created, no runtime is invoked, no prompt is evaluated, and no dialogue is generated.

Status Fields

```
nadia_stage_27_context_window_assembly_contract_present=1
nadia_context_window_assembly_contract_generator_present=1
context_window_assembly_contract_command=scripts/nadia-context-window-assembly-contract.sh
installed_context_window_assembly_contract_command=latticra-nadia context-window-assembly
context_window_assembly_contract_status=contract_only
context_window_assembly_stage=contract-only
context_window_assembly_authority=0
context_window_assembly_allowed=0
context_window_assembly_performed=0
context_window_assembly_metadata_present=1
context_window_family=operator-reviewed-context-window-assembly
context_window_format=contract-only-offline-context-window
context_window_assembly_decision=blocked_contract_only
context_window_assembly_evidence_present=1
context_window_assembly_source_policy=operator-reviewed-offline
context_window_assembly_plan_recorded=1
context_window_assembly_method_planned=offline-context-window-policy-review
context_window_assembly_result_recorded=0
context_window_assembly_runtime_invoked=0
requires_prompt_token_sequence_contract=1
requires_future_prompt_evaluation_input_contract=1
context_window_assembly_promotion_allowed=0
```

Guardrails

```
context_window_assembled=0
context_window_token_budget_recorded=0
context_window_entry_count_recorded=0
context_window_truncation_applied=0
context_window_serialized=0
prompt_evaluation_input_created=0
prompt_evaluation_input_materialized=0
prompt_evaluation_input_validated=0
prompt_evaluation_input_written=0
prompt_token_sequence_recorded=0
prompt_token_ids_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_tokenized=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-27 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose context-window-assembly as metadata-only. The command delegates to scripts/nadia-context-window-assembly-contract.sh after installation and writes reports under share/latticra/nadia/context-window-assembly/.

Next Boundary

Nadia can now produce a context-window assembly contract that packages Stage-26 prompt-token-sequence evidence and records review requirements for a future prompt-evaluation-input contract.

Prompt-evaluation input creation remains blocked until a later contract explicitly names the evaluation input format, context-window reference shape, runtime-denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt evaluation input contract is the next boundary; Stage-27 only records the prerequisite metadata.

Stage-28 now defines a prompt-evaluation-input contract that keeps prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-context-window-assembly-contract-stage-27.sh
```

docs/status/NADIA_DEVELOPER_WORKBENCH_STAGE_3_STATUS.md

Source title: Nadia Developer Workbench Stage-3 Status

Lines: 59 | Words: 160 | SHA-256: 4603e6728b716b8f2cc7f8072d22f585339e8bad2774af20e1e7290e1483c784

Nadia Developer Workbench Stage-3 Status

Status: implementation status record Date: 2026-05-25 Scope: prompt-plan generation from local context-pack and runtime-profile evidence.

Summary

Nadia Stage-3 adds a developer-workbench prompt-plan generator.

The generator combines a Stage-1 context pack and Stage-2 runtime profile into an auditable plan for future offline inference. It does not evaluate prompts.

Evidence Flags

```
nadia_stage_3_developer_workbench_present=1
nadia_prompt_plan_generator_present=1
nadia_prompt_plan_guard_present=1
nadia_installed_prompt_plan_command_planned=1
prompt_plan_command=scripts/nadia-prompt-plan.sh
installed_prompt_plan_command=latticra-nadia prompt-plan
requires_context_pack=1
requires_runtime_profile=1
context_pack_measured=1
runtime_profile_measured=1
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now generate a local prompt plan from local context-pack and runtime-profile evidence.

This does not mean Nadia can evaluate a prompt, run a model, install weights, generate code, execute arbitrary tools, train herself, mutate source, or use network access.

Validation

```
sh scripts/test-nadia-developer-workbench-stage-3.sh
```

Expected result:

```
nadia_developer_workbench_stage_3: ok
```

Next Stage

Stage-4 now adds systems-engineering mode taxonomies and validators for planning surfaces before any prompt evaluation or generation path is added.

docs/status/NADIA_GUARDED_TOOL_AUTHORITY_STAGE_7_STATUS.md

Source title: Nadia Guarded Tool Authority Stage-7 Status

Lines: 76 | Words: 194 | SHA-256: 7c046da5750189455a4a80126663b336f13201acb50c55150f4677a944ff36bb

Nadia Guarded Tool Authority Stage-7 Status

Status: implementation status record Date: 2026-05-25 Scope: report-only tool-authority preflight after protective-safety validation.

Summary

Nadia Stage-7 adds a guarded tool-authority preflight.

The preflight verifies a Stage-6 protective-safety report, rejects unsupported or dangerous tool classes, and records tool-boundary requirements. It remains report-only and does not execute tools.

Evidence Flags

```
nadia_stage_7_guarded_tool_authority_present=1
nadia_tool_authority_preflight_present=1
nadia_tool_authority_guard_present=1
nadia_installed_tool_authority_preflight_command_planned=1
tool_authority_preflight_command=scripts/nadia-tool-authority-preflight.sh
installed_tool_authority_preflight_command=latticra-nadia tool-preflight
requires_protective_safety=1
protective_safety_stage_required=6-protective-safety-boundary
tool_authority_stage=preflight-only
preflight_decision=report_only_no_execution
tool_execution_authority=0
tool_execution_performed=0
tool_selection_authority=0
shell_execution_authority=0
network_tool_authority=0
source_mutation_authority=0
destructive_action_authority=0
credential_access_authority=0
requires_operator_approval=1
requires_nucleus_gate=1
requires_runtime_boundary_gate=1
requires_seal_receipt=1
requires_protective_safety_boundary=1
authority_transition_allowed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
model_weights_installed=0
training_performed=0
distillation_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a report-only preflight for proposed tool classes after protective-safety validation.

This does not mean Nadia can execute tools, use a shell, use a network client, mutate source, read credentials, run a model, evaluate a prompt, install weights, download models, generate code autonomously, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-guarded-tool-authority-stage-7.sh
```

Expected result:

```
nadia_guarded_tool_authority_stage_7: ok
```

Next Stage

Stage-8 now defines prompt-evaluation contracts after the tool preflight, protective-safety boundary, runtime-profile boundary, and local context evidence are all present and explicitly non-executing.

docs/status/NADIA_INFERENCE_READINESS_CONTRACT_STAGE_10_STATUS.md

Source title: Nadia Inference Readiness Contract Stage-10 Status

Lines: 95 | Words: 227 | SHA-256: 244a6a2a90f7f805a22b1e345634fb0551a90f63b08f3d692705db095a8af2da

Nadia Inference Readiness Contract Stage-10 Status

Status: implementation status record Date: 2026-05-25 Scope: inference-readiness metadata before runtime invocation, model loading, prompt evaluation, inference, or tool execution.

Summary

Nadia Stage-10 adds an inference-readiness contract generator.

The contract verifies Stage-9 model-registry metadata and inherited protective, prompt, runtime, and tool-denial boundaries, then records an explicitly blocked readiness decision for future operator review. It remains contract-only and does not invoke a runtime.

Evidence Flags

```
nadia_stage_10_inference_readiness_contract_present=1
nadia_inference_readiness_contract_generator_present=1
nadia_inference_readiness_contract_guard_present=1
nadia_installed_inference_readiness_contract_command_planned=1
inference_readiness_contract_command=scripts/nadia-inference-readiness-contract.sh
installed_inference_readiness_contract_command=latticra-nadia inference-readiness
requires_model_registry_contract=1
model_registry_stage_required=9-local-model-registry-contract
inference_readiness_contract_status=contract_only
inference_readiness_stage=contract-only
inference_readiness_authority=0
inference_ready=0
readiness_decision=blocked_contract_only
readiness_evidence_present=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_provenance_review=1
requires_model_license_review=1
requires_refusal_policy_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_runtime_invocation_contract=1
readiness_promotion_allowed=0
runtime_invocation_authority=0
token_generation_authority=0
model_session_authority=0
model_selection_authority=0
model_load_authority=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_runtime_present=0
model_runtime_invoked=0
runtime_invoked=0
inference_authority=0
inference_performed=0
model_weights_installed=0
model_weights_loaded=0
model_weights_downloaded=0
model_weights_inspected=0
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce an inference-readiness contract after local model-registry evidence is present.

This does not mean Nadia can load model weights, invoke a runtime, run inference, materialize prompts, evaluate prompts, execute tools, use a shell, use a network client, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-inference-readiness-contract-stage-10.sh
```

Expected result:

```
nadia_inference_readiness_contract_stage_10: ok
```

Next Stage

Stage-11 now defines a runtime-invocation contract after inference-readiness metadata, model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_LOCAL_CONTEXT_ENGINE_STAGE_1_STATUS.md

Source title: Nadia Local Context Engine Stage-1 Status

Lines: 56 | Words: 164 | SHA-256: ce875357d91574f1bf81d4d17f05820e91d0a02fa42ad2adabf92faebe5118e7

Nadia Local Context Engine Stage-1 Status

Status: implementation status record Date: 2026-05-25 Scope: no-network local context-pack generation for Nadia.

Summary

Nadia Stage-1 adds a local context engine that can generate auditable context packs from Latticra repository material.

This is still not model inference. The generated pack is local inventory and measurement evidence for future offline AI work.

Evidence Flags

```
nadia_stage_1_local_context_engine_present=1
nadia_context_pack_generator_present=1
nadia_context_pack_guard_present=1
nadia_installed_context_pack_command_planned=1
context_pack_command=scripts/nadia-context-pack.sh
installed_context_pack_command=latticra-nadia context-pack
local_file_read_for_indexing=operator_invoked
local_context_pack_write=operator_selected_output
network_authority=0
model_runtime_present=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now generate a no-network local context pack for Latticra source, docs, tests, scripts, installer metadata, and Seal material.

This does not mean Nadia can answer questions, generate code, run a model, embed files, execute arbitrary tools, train herself, mutate source, or use network access.

Validation

```
sh scripts/test-nadia-local-context-engine-stage-1.sh
```

Expected result:

```
nadia_local_context_engine_stage_1: ok
```

Next Stage

Stage-2 now defines an offline runtime-profile boundary before any prompt execution or local generation path is added.

docs/status/NADIA_LOCAL_MODEL_REGISTRY_CONTRACT_STAGE_9_STATUS.md

Source title: Nadia Local Model Registry Contract Stage-9 Status

Lines: 93 | Words: 242 | SHA-256: 1d5a9289de672b25dd7084629e534a95d9fbcf7b5eed62ee1cf18c1e6880e735

Nadia Local Model Registry Contract Stage-9 Status

Status: implementation status record Date: 2026-05-25 Scope: local model-registry metadata before model selection, model installation, prompt evaluation, inference, or tool execution.

Summary

Nadia Stage-9 adds a local model-registry contract generator.

The contract verifies Stage-8 prompt-evaluation contract evidence and Stage-2 runtime-profile metadata, then records a local model candidate for operator review. It remains metadata-only and does not select, install, load, inspect, benchmark, or run a model.

Evidence Flags

```
nadia_stage_9_local_model_registry_contract_present=1
nadia_model_registry_contract_generator_present=1
nadia_model_registry_contract_guard_present=1
nadia_installed_model_registry_contract_command_planned=1
model_registry_contract_command=scripts/nadia-local-model-registry-contract.sh
installed_model_registry_contract_command=latticra-nadia model-registry
requires_prompt_contract=1
prompt_contract_stage_required=8-prompt-evaluation-contract
requires_runtime_profile=1
runtime_profile_stage_required=2-runtime-profile-boundary
registry_contract_status=metadata_only
local_model_registry_stage=contract-only
model_registry_authority=0
candidate_recorded=1
candidate_review_status=operator_review_required
candidate_usable_for_inference=0
candidate_selected_for_runtime=0
requires_operator_review=1
requires_license_review=1
requires_provenance_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
model_selection_authority=0
model_install_authority=0
model_download_authority=0
model_copy_authority=0
model_load_authority=0
model_benchmark_authority=0
model_weight_inspection_authority=0
registry_promotion_allowed=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_runtime_present=0
model_runtime_invoked=0
inference_performed=0
model_weights_installed=0
model_weights_loaded=0
model_weights_copied=0
model_weights_downloaded=0
model_weights_inspected=0
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a local model-registry contract after prompt-contract and runtime-profile evidence are present.

This does not mean Nadia can select models, install model weights, copy model weights, load model weights, inspect model weights, benchmark models, run inference, materialize prompts, evaluate prompts, execute tools, use a shell, use a network client, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-local-model-registry-contract-stage-9.sh
```

Expected result:

```
nadia_local_model_registry_contract_stage_9: ok
```

Next Stage

Stage-10 now defines an inference-readiness contract after local model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_MODEL_LOAD_CONTRACT_STAGE_12_STATUS.md

Source title: Nadia Model Load Contract Stage-12 Status

Lines: 109 | Words: 279 | SHA-256: 31628f061b700e285234cf285ebfade5b44e8316029fc1dfc555da43122f8edb

Nadia Model Load Contract Stage-12 Status

Status: implementation status record Date: 2026-05-25 Scope: model-load metadata before model file opening, weight mapping, weight verification, weight loading, runtime attachment, token generation, inference, prompt evaluation, or tool execution.

Summary

Nadia Stage-12 adds a model-load contract generator.

The contract verifies Stage-11 runtime-invocation metadata and inherited protective, prompt, model, runtime, and tool-denial boundaries, then records an explicitly blocked load decision for future operator review. It remains contract-only and does not open, map, verify, or load model weights.

Evidence Flags

```
nadia_stage_12_model_load_contract_present=1
nadia_model_load_contract_generator_present=1
nadia_model_load_contract_guard_present=1
nadia_installed_model_load_contract_command_planned=1
model_load_contract_command=scripts/nadia-model-load-contract.sh
installed_model_load_contract_command=latticra-nadia model-load
requires_runtime_invocation_contract=1
runtime_invocation_stage_required=11-runtime-invocation-contract
model_load_contract_status=contract_only
model_load_stage=contract-only
model_load_authority=0
model_load_allowed=0
model_loaded=0
load_decision=blocked_contract_only
load_evidence_present=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_provenance_review=1
requires_model_license_review=1
requires_model_safety_review=1
requires_model_weight_measurement_contract=1
requires_refusal_policy_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_prompt_receipt_contract=1
load_promotion_allowed=0
model_file_open_authority=0
model_weight_read_authority=0
model_weight_mapping_authority=0
model_weight_verification_authority=0
model_weight_inspection_authority=0
runtime_model_attach_authority=0
model_session_authority=0
token_generation_authority=0
model_file_opened=0
model_file_descriptor_opened=0
model_memory_map_created=0
model_weights_mapped=0
model_weights_loaded=0
model_weights_attached=0
model_weight_measurement_performed=0
model_weight_verification_performed=0
model_load_performed=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a model-load contract after runtime-invocation evidence is present.

This does not mean Nadia can open model files, map model weights, verify model weights, load model weights, attach weights to a runtime, spawn a runtime process, create a model session, generate tokens, run inference, materialize prompts, evaluate prompts, execute tools, use a shell, use a network client, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-model-load-contract-stage-12.sh
```

Expected result:

```
nadia_model_load_contract_stage_12: ok
```

Next Stage

Stage-13 now defines a prompt-receipt contract after model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_OFFLINE_AI_STAGE_0_STATUS.md

Source title: Nadia Offline AI Stage-0 Status

Lines: 73 | Words: 229 | SHA-256: ef2557a75112d341e8543fa0423108ac63626c6c9c72f0a67c5bd98a897753e9

Nadia Offline AI Stage-0 Status

Status: foundation status record Date: 2026-05-25 Scope: Stage-0 foundation for the Latticra-native offline AI named Nadia.

Summary

Nadia is the selected name for Latticra's future offline AI companion.

The name honors Nobel Peace Prize laureate Nadia Murad and anchors the system's future capability in human dignity, survivor-witness respect, community awareness, and harm-aware development.

Stage-0 creates the bounded foundation only: component identity, Panel install metadata, embedded Console status metadata, local install paths, CLI shim planning, local configuration, and a deterministic guard.

Evidence Flags

```
nadia_name_selected=1
nadia_namesake=Nadia Murad
nadia_system_name=Latticra Nadia Witness Foundation
public_name=Nadia
interactive_name=Nadia
implementation_name=Nadia Witness Foundation
documentation_code_name=Nadia Witness Foundation
nadia_command_name=latticra-nadia
nadia_component_key=nadia_offline_ai
nadia_stage_0_foundation_present=1
nadia_foundation_contract_present=1
nadia_status_record_present=1
nadia_panel_component_present=1
nadia_console_status_surface_present=1
nadia_installer_manifest_present=1
nadia_installer_config_surface_present=1
nadia_cli_shim_planned=1
nadia_local_config_planned=1
nadia_productivity_ledger_reserved=1
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
offline_by_default=1
network_authority=0
root_authority=0
runtime_enforcement_authority=0
tool_execution_authority=0
model_runtime_present=0
model_weights_installed=0
weight_training=0
self_modification_authority=0
production_ai_claimed=0
```

Current Claim

Nadia now has a Stage-0 Latticra-native offline AI foundation. She is visible as an optional Panel-installable component and as a metadata-only Console/CLI status surface.

This does not mean Nadia can run a model, generate code, execute tools, train herself, mutate the repository, use network access, or provide production AI capability.

Validation

```
sh scripts/test-nadia-offline-ai-stage-0.sh
```

Expected result:

```
nadia_offline_ai_stage_0: ok
```

Next Stage

Stage-1 should add a no-network local context engine that indexes Latticra docs, source, headers, tests, scripts, installer metadata, and status records into auditable context packs without running inference.

docs/status/NADIA_PRODUCTIVITY_LOOP_STAGE_5_STATUS.md

Source title: Nadia Productivity Loop Stage-5 Status

Lines: 62 | Words: 173 | SHA-256: c9ab88c8c60306d54cbb8f4b78cbb7cf7dc82f6aa8df39f8cb25003b68e59eca

Nadia Productivity Loop Stage-5 Status

Status: implementation status record Date: 2026-05-25 Scope: operator-reviewed local productivity ledger before training, inference, or tool authority.

Summary

Nadia Stage-5 adds a productivity-ledger generator for local, operator-reviewed learning evidence.

The ledger records mode-validation evidence, outcome labels, recommendation labels, and metadata hints for future retrieval ranking, plan templates, test recommendations, and project memory. It remains metadata-only and does not train or distill a model.

Evidence Flags

```
nadia_stage_5_productivity_loop_present=1
nadia_productivity_ledger_generator_present=1
nadia_productivity_ledger_guard_present=1
nadia_installed_productivity_ledger_command_planned=1
productivity_ledger_command=scripts/nadia-productivity-ledger.sh
installed_productivity_ledger_command=latticra-nadia productivity-ledger
requires_mode_validation=1
mode_validation_stage_required=4-systems-engineering-mode-validation
learning_scope=operator-reviewed-local-productivity
ledger_append_only=1
project_memory_scope=local-metadata-only
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
distillation_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now write local productivity-ledger entries from validated planning evidence.

This does not mean Nadia can run a model, evaluate a prompt, install weights, download models, generate code, execute arbitrary tools, train herself, distill herself, mutate source, or use network access.

Validation

```
sh scripts/test-nadia-productivity-loop-stage-5.sh
```

Expected result:

```
nadia_productivity_loop_stage_5: ok
```

Next Stage

Stage-6 now adds a protective-safety boundary before any guarded tool authority is considered.

docs/status/NADIA_PROMPT_EVALUATION_CONTRACT_STAGE_8_STATUS.md

Source title: Nadia Prompt Evaluation Contract Stage-8 Status

Lines: 75 | Words: 195 | SHA-256: 95ad9b37c2c4cf3d09861056dfc403cef9d2f92ded84d45be9c25359d7801039

Nadia Prompt Evaluation Contract Stage-8 Status

Status: implementation status record Date: 2026-05-25 Scope: prompt-evaluation contract before prompt materialization, prompt evaluation, inference, or tool execution.

Summary

Nadia Stage-8 adds a prompt-evaluation contract generator.

The contract verifies Stage-7 report-only tool preflight evidence and protective-safety restrictions, then records the fields required for any future prompt-evaluation path. It remains contract-only and does not evaluate prompts.

Evidence Flags

```
nadia_stage_8_prompt_evaluation_contract_present=1
nadia_prompt_evaluation_contract_generator_present=1
nadia_prompt_evaluation_contract_guard_present=1
nadia_installed_prompt_evaluation_contract_command_planned=1
prompt_evaluation_contract_command=scripts/nadia-prompt-evaluation-contract.sh
installed_prompt_evaluation_contract_command=latticra-nadia prompt-contract
requires_tool_preflight=1
tool_preflight_stage_required=7-guarded-tool-authority-preflight
prompt_evaluation_stage=contract-only
prompt_contract_status=contract_only
prompt_materialized=0
prompt_text_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
prompt_receipt_required=1
refusal_policy_required=1
protective_safety_required=1
tool_preflight_required=1
runtime_profile_required=1
model_registry_review_required=1
operator_review_required=1
contract_promotion_allowed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
model_runtime_invoked=0
inference_performed=0
network_authority=0
training_performed=0
distillation_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a prompt-evaluation contract after tool preflight and protective-safety validation.

This does not mean Nadia can materialize prompts, evaluate prompts, run inference, execute tools, use a shell, use a network client, mutate source, read credentials, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-prompt-evaluation-contract-stage-8.sh
```

Expected result:

```
nadia_prompt_evaluation_contract_stage_8: ok
```

Next Stage

Stage-9 now defines a local model-registry contract after prompt-evaluation contracts, protective-safety refusal behavior, runtime-profile metadata, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_PROMPT_EVALUATION_HANDOFF_CONTRACT_STAGE_16_STATUS.md

Source title: Nadia Prompt Evaluation Handoff Contract Stage-16 Status

Lines: 101 | Words: 290 | SHA-256: e86f8f544e63f75737a0adb0641a74977fbaf506f9f666def21d44e535fcd2d6

Nadia Prompt Evaluation Handoff Contract Stage-16 Status

Status: implementation status record Date: 2026-05-25 Scope: prompt-evaluation handoff metadata before prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-16 adds a prompt-evaluation handoff contract generator.

The contract verifies Stage-15 awareness-dialogue metadata and inherited protective, prompt, model, runtime, and tool-denial boundaries, then records a blocked handoff state that requires a future tokenization boundary before any prompt evaluation can be considered. It remains contract-only and does not evaluate prompts.

Evidence Flags

```
nadia_stage_16_prompt_evaluation_handoff_contract_present=1
nadia_prompt_evaluation_handoff_contract_generator_present=1
nadia_prompt_evaluation_handoff_contract_guard_present=1
nadia_installed_prompt_evaluation_handoff_contract_command_planned=1
prompt_evaluation_handoff_contract_command=scripts/nadia-prompt-evaluation-handoff-contract.sh
installed_prompt_evaluation_handoff_contract_command=latticra-nadia prompt-evaluation-handoff
prompt_evaluation_handoff_contract_status=contract_only
prompt_evaluation_handoff_stage=contract-only
prompt_evaluation_handoff_authority=0
prompt_evaluation_handoff_allowed=0
prompt_evaluation_handoff_performed=0
prompt_evaluation_authority=0
prompt_evaluated=0
evaluation_handoff_decision=blocked_contract_only
evaluation_handoff_evidence_present=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenization_contract=1
prompt_evaluation_handoff_promotion_allowed=0
future_qa_dialogue_capability_planned=1
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
plain_language_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
live_web_lookup_authority=0
topic_yazidi_genocide_awareness=1
topic_survivor_voice_and_dignity=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_genocide_prevention=1
topic_justice_and_accountability=1
topic_womens_empowerment=1
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_materialized=0
prompt_tokenization_authority=0
prompt_tokenized=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
network_authority=0
```

Current Claim

Nadia can now produce a prompt-evaluation handoff contract that packages the Stage-15 awareness-dialogue evidence and explicitly blocks prompt evaluation until a future tokenization boundary exists.

This does not mean Nadia can evaluate prompts, tokenize prompts, generate dialogue, answer user prompts, browse the web, provide legal advice, provide medical advice, provide trauma counseling, receive prompt text, materialize prompt content, load model weights, spawn a runtime process, generate tokens, run inference, execute tools, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-prompt-evaluation-handoff-contract-stage-16.sh
```

Expected result:

```
nadia_prompt_evaluation_handoff_contract_stage_16: ok
```

Next Stage

Stage-17 now defines a tokenization boundary contract only after prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_PROMPT_EVALUATION_INPUT_CONTRACT_STAGE_28_STATUS.md

Source title: Nadia Prompt Evaluation Input Contract Stage-28 Status

Lines: 113 | Words: 324 | SHA-256: be41a1d53febc3beb1b1896f6200ef804aa37f10c3ec06bd2ae6491da79972da

Nadia Prompt Evaluation Input Contract Stage-28 Status

Status: implementation status record

Scope: prompt-evaluation-input metadata before prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-28 adds a prompt-evaluation-input contract generator.

It consumes the Stage-27 context-window assembly report as prerequisite evidence and records a future review lane for prompt-evaluation input schema, context-window references, prompt-token-sequence references, safety-envelope policy, runtime-denial policy, and prompt-evaluation runtime handoff requirements. It remains contract-only: no prompt text is read, no context window is assembled, no prompt evaluation input is created, no runtime is invoked, no prompt is evaluated, and no dialogue is generated.

Status Fields

```
nadia_stage_28_prompt_evaluation_input_contract_present=1
nadia_prompt_evaluation_input_contract_generator_present=1
prompt_evaluation_input_contract_command=scripts/nadia-prompt-evaluation-input-contract.sh
installed_prompt_evaluation_input_contract_command=latticra-nadia prompt-evaluation-input
prompt_evaluation_input_contract_status=contract_only
prompt_evaluation_input_stage=contract-only
prompt_evaluation_input_authority=0
prompt_evaluation_input_allowed=0
prompt_evaluation_input_created=0
prompt_evaluation_input_metadata_present=1
prompt_evaluation_input_family=operator-reviewed-prompt-evaluation-input
prompt_evaluation_input_format=contract-only-offline-evaluation-input
prompt_evaluation_input_decision=blocked_contract_only
prompt_evaluation_input_evidence_present=1
prompt_evaluation_input_source_policy=operator-reviewed-offline
prompt_evaluation_input_plan_recorded=1
prompt_evaluation_input_method_planned=offline-prompt-evaluation-input-policy-review
prompt_evaluation_input_result_recorded=0
prompt_evaluation_input_runtime_invoked=0
requires_context_window_assembly_contract=1
requires_future_prompt_evaluation_runtime_handoff_contract=1
prompt_evaluation_input_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_input_created=0
prompt_evaluation_input_materialized=0
prompt_evaluation_input_loaded=0
prompt_evaluation_input_opened=0
prompt_evaluation_input_read=0
prompt_evaluation_input_validated=0
prompt_evaluation_input_serialized=0
prompt_evaluation_input_written=0
prompt_evaluation_input_context_reference_recorded=0
prompt_evaluation_input_token_reference_recorded=0
prompt_evaluation_input_safety_envelope_recorded=0
context_window_assembled=0
context_window_serialized=0
prompt_token_sequence_recorded=0
prompt_token_ids_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_tokenized=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-28 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-input as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-input-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-input/.

Next Boundary

Nadia can now produce a prompt-evaluation-input contract that packages Stage-27 context-window assembly evidence and records review requirements for a future prompt-evaluation runtime handoff contract.

Prompt evaluation, runtime invocation, token generation, and dialogue generation remain blocked until a later contract explicitly names the runtime handoff format, evaluation request shape, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation runtime handoff contract is the next boundary; Stage-28 only records the prerequisite metadata.

Stage-29 now defines a prompt-evaluation runtime handoff contract that keeps runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-prompt-evaluation-input-contract-stage-28.sh
```

docs/status/NADIA_PROMPT_EVALUATION_INVOCATION_CONTRACT_STAGE_30_STATUS.md

Source title: Nadia Prompt Evaluation Invocation Contract Stage-30 Status

Lines: 115 | Words: 341 | SHA-256: 45cea5a8b8df0434389746c4efb7a8a4f5f968d152b92ec5527561522d8089f0

Nadia Prompt Evaluation Invocation Contract Stage-30 Status

Status: implementation status record

Scope: prompt-evaluation invocation metadata before invocation request creation, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-30 adds a prompt-evaluation invocation contract generator.

It consumes the Stage-29 prompt-evaluation runtime handoff report as prerequisite evidence and records a future review lane for prompt-evaluation invocation schema, runtime handoff references, evaluation input references, runtime profile references, runtime-invocation contract references, model-load contract references, inference-readiness contract references, invocation-denial policy, token-generation denial policy, and prompt-evaluation result requirements. It remains contract-only: no runtime handoff is performed, no invocation request is created, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Stage-31 now defines a prompt-evaluation result contract that keeps result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Status Fields

```
nadia_stage_30_prompt_evaluation_invocation_contract_present=1
nadia_prompt_evaluation_invocation_contract_generator_present=1
prompt_evaluation_invocation_contract_command=scripts/nadia-prompt-evaluation-invocation-contrac
t.sh
installed_prompt_evaluation_invocation_contract_command=latticra-nadia
prompt-evaluation-invocation
prompt_evaluation_invocation_contract_status=contract_only
prompt_evaluation_invocation_stage=contract-only
prompt_evaluation_invocation_authority=0
prompt_evaluation_invocation_allowed=0
prompt_evaluation_invocation_performed=0
prompt_evaluation_invocation_metadata_present=1
prompt_evaluation_invocation_family=operator-reviewed-prompt-evaluation-invocation
prompt_evaluation_invocation_format=contract-only-offline-evaluation-invocation
prompt_evaluation_invocation_decision=blocked_contract_only
prompt_evaluation_invocation_evidence_present=1
prompt_evaluation_invocation_source_policy=operator-reviewed-offline
prompt_evaluation_invocation_plan_recorded=1
prompt_evaluation_invocation_method_planned=offline-prompt-evaluation-invocation-policy-review
prompt_evaluation_invocation_result_recorded=0
prompt_evaluation_invocation_runtime_invoked=0
requires_prompt_evaluation_runtime_handoff_contract=1
requires_future_prompt_evaluation_result_contract=1
prompt_evaluation_invocation_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_invocation_performed=0
prompt_evaluation_invocation_created=0
prompt_evaluation_invocation_loaded=0
prompt_evaluation_invocation_opened=0
prompt_evaluation_invocation_read=0
prompt_evaluation_invocation_validated=0
prompt_evaluation_invocation_serialized=0
prompt_evaluation_invocation_written=0
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_validated=0
prompt_evaluation_invocation_request_serialized=0
prompt_evaluation_invocation_request_submitted=0
prompt_evaluation_invocation_request_scheduled=0
prompt_evaluation_invocation_request_queued=0
prompt_evaluation_invocation_runtime_selected=0
prompt_evaluation_invocation_model_selected=0
prompt_evaluation_invocation_session_created=0
prompt_evaluation_invocation_runtime_invoked=0
runtime_handoff_created=0
runtime_handoff_submitted=0
prompt_evaluation_request_created=0
prompt_evaluation_request_submitted=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-30 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-invocation as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-invocation-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-invocation/.

Next Boundary

Nadia can now produce a prompt-evaluation invocation contract that packages Stage-29 prompt-evaluation runtime handoff evidence and records review requirements for a future prompt-evaluation result contract.

Runtime invocation, prompt evaluation, token generation, inference, and dialogue generation remain blocked until a later contract explicitly names result shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result contract is the next boundary; Stage-30 only records the prerequisite metadata.

Validation

```
sh scripts/test-nadia-prompt-evaluation-invocation-contract-stage-30.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_CONTRACT_STAGE_31_STATUS.md

Source title: Nadia Prompt Evaluation Result Contract Stage-31 Status

Lines: 120 | Words: 348 | SHA-256: 939acfdb70e22493282c8fdb84acd4eebb7944d2cc412e1250fd47a9d1be4d1f

Nadia Prompt Evaluation Result Contract Stage-31 Status

Status: implementation status record

Scope: prompt-evaluation result metadata before result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-31 adds a prompt-evaluation result contract generator.

It consumes the Stage-30 prompt-evaluation invocation report as prerequisite evidence and records a future review lane for prompt-evaluation result schema, invocation references, runtime handoff references, evaluation input references, generated-text denial policy, token-generation denial policy, result-review policy, and survivor-centered safety requirements. It remains contract-only: no result record is created, no model output is recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Stage-32 now defines a prompt-evaluation result review contract that keeps review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Status Fields

```
nadia_stage_31_prompt_evaluation_result_contract_present=1
nadia_prompt_evaluation_result_contract_generator_present=1
prompt_evaluation_result_contract_command=scripts/nadia-prompt-evaluation-result-contract.sh
installed_prompt_evaluation_result_contract_command=latticra-nadia prompt-evaluation-result
prompt_evaluation_result_contract_status=contract_only
prompt_evaluation_result_stage=contract-only
prompt_evaluation_result_authority=0
prompt_evaluation_result_allowed=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_created=0
prompt_evaluation_result_performed=0
prompt_evaluation_result_metadata_present=1
prompt_evaluation_result_family=operator-reviewed-prompt-evaluation-result
prompt_evaluation_result_format=contract-only-offline-evaluation-result
prompt_evaluation_result_decision=blocked_contract_only
prompt_evaluation_result_evidence_present=1
prompt_evaluation_result_source_policy=operator-reviewed-offline
prompt_evaluation_result_plan_recorded=1
prompt_evaluation_result_method_planned=offline-prompt-evaluation-result-policy-review
prompt_evaluation_result_result_recorded=0
prompt_evaluation_result_runtime_invoked=0
requires_prompt_evaluation_invocation_contract=1
requires_future_prompt_evaluation_result_review_contract=1
prompt_evaluation_result_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_recorded=0
prompt_evaluation_result_created=0
prompt_evaluation_result_loaded=0
prompt_evaluation_result_opened=0
prompt_evaluation_result_read=0
prompt_evaluation_result_validated=0
prompt_evaluation_result_serialized=0
prompt_evaluation_result_written=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_record_validated=0
prompt_evaluation_result_record_serialized=0
prompt_evaluation_result_record_written=0
prompt_evaluation_result_record_submitted=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
prompt_evaluation_result_score_recorded=0
prompt_evaluation_result_token_logprobs_recorded=0
prompt_evaluation_result_runtime_invoked=0
prompt_evaluation_invocation_request_created=0
prompt_evaluation_invocation_request_submitted=0
runtime_handoff_created=0
runtime_handoff_submitted=0
prompt_evaluation_request_created=0
prompt_evaluation_request_submitted=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-31 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result/.

Next Boundary

Nadia can now produce a prompt-evaluation result contract that packages Stage-30 prompt-evaluation invocation evidence and records review requirements for a future prompt-evaluation result review contract.

Runtime invocation, prompt evaluation, token generation, inference, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names result review shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result review contract is the next boundary; Stage-31 only records the prerequisite metadata.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-contract-stage-31.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_DISPOSITION_CONTRACT_STAGE_33_STATUS.md

Source title: Nadia Prompt Evaluation Result Disposition Contract Stage-33 Status

Lines: 124 | Words: 399 | SHA-256: 553a9cad282e3129a2616e1537548fb0f8f290b10565b6d314b919185fc303f3

Nadia Prompt Evaluation Result Disposition Contract Stage-33 Status

Status: implementation status record

Scope: prompt-evaluation result disposition metadata before disposition recording, release recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-33 adds a prompt-evaluation result disposition contract generator.

It consumes the Stage-32 prompt-evaluation result review report as prerequisite evidence and records a future release lane for disposition schema, review references, result references, invocation references, runtime handoff references, evaluation input references, generated-text denial policy, token-generation denial policy, result-release policy, and survivor-centered safety requirements. It remains contract-only: no disposition record is created, no release record is created, no review record is created, no result record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_33_prompt_evaluation_result_disposition_contract_present=1
nadia_prompt_evaluation_result_disposition_contract_generator_present=1
prompt_evaluation_result_disposition_contract_command=scripts/nadia-prompt-evaluation-result-dis
position-contract.sh
installed_prompt_evaluation_result_disposition_contract_command=latticra-nadia
prompt-evaluation-result-disposition
prompt_evaluation_result_disposition_contract_status=contract_only
prompt_evaluation_result_disposition_stage=contract-only
prompt_evaluation_result_disposition_authority=0
prompt_evaluation_result_disposition_allowed=0
prompt_evaluation_result_disposition_recorded=0
prompt_evaluation_result_disposition_created=0
prompt_evaluation_result_disposition_performed=0
prompt_evaluation_result_disposition_metadata_present=1
prompt_evaluation_result_disposition_family=operator-reviewed-prompt-evaluation-result-dispositi
on
prompt_evaluation_result_disposition_format=contract-only-offline-evaluation-result-disposition
prompt_evaluation_result_disposition_decision=blocked_contract_only
prompt_evaluation_result_disposition_evidence_present=1
prompt_evaluation_result_disposition_source_policy=operator-reviewed-offline
prompt_evaluation_result_disposition_plan_recorded=1
prompt_evaluation_result_disposition_method_planned=offline-prompt-evaluation-result-disposition
-policy-review
prompt_evaluation_result_disposition_result_recorded=0
prompt_evaluation_result_disposition_runtime_invoked=0
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_contract=1
prompt_evaluation_result_disposition_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_disposition_recorded=0
prompt_evaluation_result_disposition_created=0
prompt_evaluation_result_disposition_loaded=0
prompt_evaluation_result_disposition_opened=0
prompt_evaluation_result_disposition_read=0
prompt_evaluation_result_disposition_validated=0
prompt_evaluation_result_disposition_serialized=0
prompt_evaluation_result_disposition_written=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_record_validated=0
prompt_evaluation_result_disposition_record_serialized=0
prompt_evaluation_result_disposition_record_written=0
prompt_evaluation_result_disposition_record_submitted=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_route_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
prompt_evaluation_result_score_recorded=0
prompt_evaluation_result_token_logprobs_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-33 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-disposition as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-disposition-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-disposition/.

Next Boundary

Nadia can now produce a prompt-evaluation result disposition contract that packages Stage-32 prompt-evaluation result review evidence and records release requirements for a future prompt-evaluation result release contract.

Runtime invocation, prompt evaluation, token generation, inference, result disposition recording, release recording, result review recording, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names release shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result release contract is the next boundary; Stage-33 only records the prerequisite metadata.

Stage-34 now defines a prompt-evaluation result release contract that keeps release recording, release decision recording, release publication, release receipt creation, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-disposition-contract-stage-33.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_CONTRACT_STAGE_34_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Contract Stage-34 Status

Lines: 129 | Words: 452 | SHA-256: c8d027ee0e26550eac78902475ba9c7b3e571a74bf0effd50f3ca8f73aedeed5

Nadia Prompt Evaluation Result Release Contract Stage-34 Status

Status: implementation status record

Scope: prompt-evaluation result release metadata before release recording, release decision recording, release publication, release packaging, release receipt creation, disposition recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-34 adds a prompt-evaluation result release contract generator.

It consumes the Stage-33 prompt-evaluation result disposition report as prerequisite evidence and records a future receipt lane for release schema, disposition references, review references, result references, invocation references, runtime handoff references, evaluation input references, generated-text denial policy, token-generation denial policy, release-receipt policy, and survivor-centered safety requirements. It remains contract-only: no release record is created, no release decision is recorded, no release is published, no release is packaged, no release receipt is created, no disposition record is created, no review record is created, no result record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_34_prompt_evaluation_result_release_contract_present=1
nadia_prompt_evaluation_result_release_contract_generator_present=1
prompt_evaluation_result_release_contract_command=scripts/nadia-prompt-evaluation-result-release
-contract.sh
installed_prompt_evaluation_result_release_contract_command=latticra-nadia
prompt-evaluation-result-release
prompt_evaluation_result_release_contract_status=contract_only
prompt_evaluation_result_release_stage=contract-only
prompt_evaluation_result_release_authority=0
prompt_evaluation_result_release_allowed=0
prompt_evaluation_result_release_recorded=0
prompt_evaluation_result_release_created=0
prompt_evaluation_result_release_performed=0
prompt_evaluation_result_release_metadata_present=1
prompt_evaluation_result_release_family=operator-reviewed-prompt-evaluation-result-release
prompt_evaluation_result_release_format=contract-only-offline-evaluation-result-release
prompt_evaluation_result_release_decision=blocked_contract_only
prompt_evaluation_result_release_evidence_present=1
prompt_evaluation_result_release_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_plan_recorded=1
prompt_evaluation_result_release_method_planned=offline-prompt-evaluation-result-release-policy-
review
prompt_evaluation_result_release_result_recorded=0
prompt_evaluation_result_release_runtime_invoked=0
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_contract=1
prompt_evaluation_result_release_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_recorded=0
prompt_evaluation_result_release_created=0
prompt_evaluation_result_release_loaded=0
prompt_evaluation_result_release_opened=0
prompt_evaluation_result_release_read=0
prompt_evaluation_result_release_validated=0
prompt_evaluation_result_release_serialized=0
prompt_evaluation_result_release_written=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_record_validated=0
prompt_evaluation_result_release_record_serialized=0
prompt_evaluation_result_release_record_written=0
prompt_evaluation_result_release_record_submitted=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_route_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
prompt_evaluation_result_score_recorded=0
prompt_evaluation_result_token_logprobs_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-34 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
```

```
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release/.

Next Boundary

Nadia can now produce a prompt-evaluation result release contract that packages Stage-33 prompt-evaluation result disposition evidence and records receipt requirements for a future prompt-evaluation result release receipt contract.

Runtime invocation, prompt evaluation, token generation, inference, release recording, release decision recording, release publication, release packaging, release receipt creation, result disposition recording, result review recording, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names receipt shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result release receipt contract is the next boundary; Stage-34 only records the prerequisite metadata.

Stage-35 now defines a prompt-evaluation result release receipt contract that keeps receipt recording, receipt signing, receipt publication, release recording, release decision recording, release publication, release packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-contract-stage-34.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_CONTRACT_STAGE_35_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Contract Stage-35 Status

Lines: 134 | Words: 483 | SHA-256: 203e9d118057b53877fe9448903c9c5fa6d518be43050ddcae903e15a9fe25d8

Nadia Prompt Evaluation Result Release Receipt Contract Stage-35 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt metadata before receipt recording, receipt signing, receipt publication, release recording, release decision recording, release publication, release packaging, disposition recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-35 adds a prompt-evaluation result release receipt contract generator.

It consumes the Stage-34 prompt-evaluation result release report as prerequisite evidence and records a future receipt-review lane for release references, disposition references, review references, result references, invocation references, runtime handoff references, evaluation input references, generated-text denial policy, token-generation denial policy, release-receipt-review policy, and survivor-centered safety requirements. It remains contract-only: no release receipt is created, no receipt record is created, no receipt is signed, no receipt is emitted, no receipt is published, no release record is created, no release decision is recorded, no release is published, no disposition record is created, no review record is created, no result record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_35_prompt_evaluation_result_release_receipt_contract_present=1
nadia_prompt_evaluation_result_release_receipt_contract_generator_present=1
prompt_evaluation_result_release_receipt_contract_command=scripts/nadia-prompt-evaluation-result
-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt
prompt_evaluation_result_release_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_authority=0
prompt_evaluation_result_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_performed=0
prompt_evaluation_result_release_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_family=operator-reviewed-prompt-evaluation-result-relea
se-receipt
prompt_evaluation_result_release_receipt_format=contract-only-offline-evaluation-result-release-
receipt
prompt_evaluation_result_release_receipt_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_method_planned=offline-prompt-evaluation-result-release
-receipt-policy-review
prompt_evaluation_result_release_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_contract=1
prompt_evaluation_result_release_receipt_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_loaded=0
prompt_evaluation_result_release_receipt_opened=0
prompt_evaluation_result_release_receipt_read=0
prompt_evaluation_result_release_receipt_validated=0
prompt_evaluation_result_release_receipt_serialized=0
prompt_evaluation_result_release_receipt_written=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_record_validated=0
prompt_evaluation_result_release_receipt_record_serialized=0
prompt_evaluation_result_release_receipt_record_written=0
prompt_evaluation_result_release_receipt_record_submitted=0
prompt_evaluation_result_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_route_recorded=0
prompt_evaluation_result_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_release_receipt_created=0
prompt_evaluation_result_release_receipt_recorded=0
prompt_evaluation_result_disposition_record_created=0
prompt_evaluation_result_disposition_decision_recorded=0
prompt_evaluation_result_disposition_applied=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-35 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt contract that packages Stage-34 prompt-evaluation result release evidence and records review requirements for a future prompt-evaluation result release receipt review contract.

Runtime invocation, prompt evaluation, token generation, inference, receipt recording, receipt signing, receipt publication, release recording, release decision recording, release publication, release packaging, result disposition recording, result review recording, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names receipt review shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result release receipt review contract is the next boundary; Stage-35 only records the prerequisite metadata.

Stage-36 now defines a prompt-evaluation result release receipt review contract that keeps review recording, review decision recording, review findings recording, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-contract-stage-35.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_36_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Contract Stage-36 Status

Lines: 123 | Words: 444 | SHA-256: 831cafce61fec7c01bb284a03a7f0b6d1815638ce9287cc47759eb73267588c2

Nadia Prompt Evaluation Result Release Receipt Review Contract Stage-36 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review metadata before review recording, review decisions, review findings, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-36 adds a prompt-evaluation result release receipt review contract generator.

It consumes the Stage-35 prompt-evaluation result release receipt report as prerequisite evidence and records a future review-disposition lane for release receipt references, release references, disposition references, review references, result references, invocation references, runtime handoff references, evaluation input references, generated-text denial policy, token-generation denial policy, review-disposition policy, and survivor-centered safety requirements. It remains contract-only: no release receipt review is created, no review record is created, no review decision is recorded, no review findings are recorded, no release receipt is created, no receipt is signed, no receipt is emitted, no receipt is published, no release record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_36_prompt_evaluation_result_release_receipt_review_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_contract_command=scripts/nadia-prompt-evaluation
-result-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review
prompt_evaluation_result_release_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_metadata_present=1
prompt_evaluation_result_release_receipt_review_family=operator-reviewed-prompt-evaluation-resul
t-release-receipt-review
prompt_evaluation_result_release_receipt_review_format=contract-only-offline-evaluation-result-r
elease-receipt-review
prompt_evaluation_result_release_receipt_review_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_evidence_present=1
prompt_evaluation_result_release_receipt_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_plan_recorded=1
prompt_evaluation_result_release_receipt_review_method_planned=offline-prompt-evaluation-result-
release-receipt-review-policy-review
prompt_evaluation_result_release_receipt_review_result_recorded=0
prompt_evaluation_result_release_receipt_review_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_contract=1
prompt_evaluation_result_release_receipt_review_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_loaded=0
prompt_evaluation_result_release_receipt_review_opened=0
prompt_evaluation_result_release_receipt_review_read=0
prompt_evaluation_result_release_receipt_review_validated=0
prompt_evaluation_result_release_receipt_review_serialized=0
prompt_evaluation_result_release_receipt_review_written=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-36 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
```

```
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review contract that packages Stage-35 prompt-evaluation result release receipt evidence and records review-disposition requirements for a future prompt-evaluation result release receipt review disposition contract.

Runtime invocation, prompt evaluation, token generation, inference, review recording, review decision recording, review findings recording, receipt recording, receipt signing, receipt publication, release recording, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review disposition shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result release receipt review disposition contract is the next boundary; Stage-36 only records the prerequisite metadata.

Stage-37 now defines that prompt-evaluation result release receipt review disposition contract while preserving the same no-runtime, no-review-recording, no-disposition-recording, no-receipt-signing, and no-inference boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-contract-stage-36.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_37_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Contract Stage-37 Status

Lines: 125 | Words: 463 | SHA-256: eb540a1e7c58b07d6c3de9b5f82fa25fce55abb4e038bc1f3efcd69fd40d62b3

Nadia Prompt Evaluation Result Release Receipt Review Disposition Contract Stage-37 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition metadata before disposition recording, disposition decisions, disposition findings, review recording, review decisions, review findings, receipt recording, receipt signing, receipt publication, release recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-37 adds a prompt-evaluation result release receipt review disposition contract generator.

It consumes the Stage-36 prompt-evaluation result release receipt review report as prerequisite evidence and records a future review-disposition release lane for review references, release receipt references, release references, disposition references, review references, result references, generated-text denial policy, token-generation denial policy, review-disposition release policy, and survivor-centered safety requirements. It remains contract-only: no review disposition is created, no disposition record is created, no disposition decision is recorded, no disposition findings are recorded, no release receipt review is created, no review record is created, no review decision is recorded, no review findings are recorded, no release receipt is created, no receipt is signed, no receipt is emitted, no receipt is published, no release record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_37_prompt_evaluation_result_release_receipt_review_disposition_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_contract_command=scripts/nadia-promp
t-evaluation-result-release-receipt-review-disposition-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_contract_command=latticra-
nadia prompt-evaluation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_performed=0
prompt_evaluation_result_release_receipt_review_disposition_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_family=operator-reviewed-prompt-eval
uation-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_format=contract-only-offline-evaluat
ion-result-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_source_policy=operator-reviewed-offl
ine
prompt_evaluation_result_release_receipt_review_disposition_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_loaded=0
prompt_evaluation_result_release_receipt_review_disposition_opened=0
prompt_evaluation_result_release_receipt_review_disposition_read=0
prompt_evaluation_result_release_receipt_review_disposition_validated=0
prompt_evaluation_result_release_receipt_review_disposition_serialized=0
prompt_evaluation_result_release_receipt_review_disposition_written=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_receipt_packaged=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_release_published=0
prompt_evaluation_result_release_packaged=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-37 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition/.

Next Boundary

Nadia can produce a prompt-evaluation result release receipt review disposition contract that packages Stage-36 prompt-evaluation result release receipt review evidence and records release requirements for the Stage-38 prompt-evaluation result release receipt review disposition release contract.

Runtime invocation, prompt evaluation, token generation, inference, review disposition recording, disposition decision recording, disposition findings recording, review recording, review decision recording, review findings recording, receipt recording, receipt signing, receipt publication, release recording, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review-disposition release shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

Stage-38 now records that later prompt-evaluation result release receipt review disposition release contract; Stage-37 remains the prerequisite metadata boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-contract-stage
-37.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_38_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Contract Stage-38 Status

Lines: 130 | Words: 463 | SHA-256: fcd98460372891f4380980ffd5d89a498524b06cbf75230ec748f114f7b294a1

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Contract Stage-38 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release metadata before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-38 adds a prompt-evaluation result release receipt review disposition release contract generator.

It consumes the Stage-37 prompt-evaluation result release receipt review disposition report as prerequisite evidence and records a future review-disposition-release receipt lane for review-disposition references, release receipt references, release references, disposition references, review references, result references, generated-text denial policy, token-generation denial policy, review-disposition-release receipt policy, and survivor-centered safety requirements. It remains contract-only: no review-disposition release is created, no release record is created, no release decision is recorded, no release is published, no release is packaged, no release receipt is created, no review disposition is created, no disposition record is created, no review record is created, no receipt is signed, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_38_prompt_evaluation_result_release_receipt_review_disposition_release_contract_pres
ent=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_contract_generator_pre
sent=1
prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=scripts/nad
ia-prompt-evaluation-result-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_contract_command=l
atticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_contract_status=contract_onl
y
prompt_evaluation_result_release_receipt_review_disposition_release_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_family=operator-reviewed-pro
mpt-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_format=contract-only-offline
-evaluation-result-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_decision=blocked_contract_on
ly
prompt_evaluation_result_release_receipt_review_disposition_release_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_source_policy=operator-revie
wed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_cont
ract=1
prompt_evaluation_result_release_receipt_review_disposition_release_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_loaded=0
prompt_evaluation_result_release_receipt_review_disposition_release_opened=0
prompt_evaluation_result_release_receipt_review_disposition_release_read=0
prompt_evaluation_result_release_receipt_review_disposition_release_validated=0
prompt_evaluation_result_release_receipt_review_disposition_release_serialized=0
prompt_evaluation_result_release_receipt_review_disposition_release_written=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-38 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release contract that packages Stage-37 prompt-evaluation result release receipt review disposition evidence and records receipt requirements for a future prompt-evaluation result release receipt review disposition release receipt contract.

Runtime invocation, prompt evaluation, token generation, inference, review-disposition release recording, release decision recording, release publication, release packaging, release receipt creation, review disposition recording, disposition decision recording, disposition findings recording, review recording, receipt recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review-disposition-release receipt shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

Stage-39 now records that later prompt-evaluation result release receipt review disposition release receipt contract; Stage-38 remains the prerequisite metadata boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-contra
ct-stage-38.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_39_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract Stage-39 Status

Lines: 132 | Words: 493 | SHA-256: 3e0455c4d939c36180e2ba9a4627dfe4fbf886bc8d941a11c324f376123e579f

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Contract Stage-39 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt metadata before disposition-release-receipt recording, receipt emission, receipt signing, receipt publication, disposition-release recording, release decisions, release publication, release packaging, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-39 adds a prompt-evaluation result release receipt review disposition release receipt contract generator.

It consumes the Stage-38 prompt-evaluation result release receipt review disposition release report as prerequisite evidence and records a future review-disposition-release receipt lane for disposition-release references, review-disposition references, release receipt references, release references, disposition references, review references, result references, generated-text denial policy, token-generation denial policy, review-disposition-release receipt review policy, and survivor-centered safety requirements. It remains contract-only: no review-disposition-release receipt is created, no receipt is emitted, no receipt is signed, no receipt is published, no receipt is packaged, no review-disposition release is created, no release record is created, no release decision is recorded, no release is published, no release is packaged, no review disposition is created, no disposition record is created, no review record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_39_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contr
act_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_gener
ator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_command=scr
ipts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-contract.
sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_c
ommand=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract_status=cont
ract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_family=operator-revi
ewed-prompt-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_format=contract-only
-offline-evaluation-result-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_decision=blocked_con
tract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_source_policy=operat
or-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_loaded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_opened=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_read=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_validated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_serialized=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_written=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-39 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt contract that packages Stage-38 prompt-evaluation result release receipt review disposition release evidence and records review requirements for a future prompt-evaluation result release receipt review disposition release receipt review contract.

Runtime invocation, prompt evaluation, token generation, inference, review-disposition release recording, release decision recording, release publication, release packaging, release receipt creation, review disposition recording, disposition decision recording, disposition findings recording, review recording, receipt recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review-disposition-release receipt shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

Stage-40 now records that later prompt-evaluation result release receipt review disposition release receipt review contract; Stage-39 remains the prerequisite metadata boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-contract-stage-39.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_40_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract Stage-40 Status

Lines: 108 | Words: 380 | SHA-256: a7189273e26d57f932dba81191dfba5463d7b2dff012bb2b87114d0ae6837838

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Contract Stage-40 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review metadata before release-receipt-review records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-40 adds a prompt-evaluation result release receipt review disposition release receipt review contract generator.

It consumes the Stage-39 prompt-evaluation result release receipt review disposition release receipt report as prerequisite evidence and records a future review-disposition-release receipt review lane for receipt references, review-denial policy, review-disposition requirements, generated-text denial policy, token-generation denial policy, and survivor-centered safety requirements. It remains contract-only: no release receipt review is recorded, no review decision is recorded, no findings are recorded, no receipt is emitted, no receipt is signed, no receipt is published, no receipt is packaged, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_40_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contrac
t_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_comm
and=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-re
view-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_con
tract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_contract_stat
us=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_stage=contrac
t-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_metadata_pres
ent=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_family=operat
or-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_format=contra
ct-only-offline-evaluation-result-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision=bloc
ked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_evidence_pres
ent=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_source_policy
=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_plan_recorded
=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_result_record
ed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_runtime_invok
ed=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_promotion_all
owed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_record_create
d=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_approval_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_rejection_rec
orded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_findings_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_applied=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
network_authority=0
tool_execution_authority=0
source_mutation_authority=0
```

Stage-40 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review contract that packages Stage-39 release receipt evidence and records requirements for a future release receipt review disposition contract.

Runtime invocation, prompt evaluation, token generation, inference, receipt review recording, review decision recording, review findings recording, disposition recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names the next disposition boundary and preserves safety inheritance, operator review gates, and non-claims.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-contract-stage-40.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_41_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-41 Status

Lines: 108 | Words: 388 | SHA-256: 8633967f428f39d93d634776a5b14c75179e1954981ec2fa16f20d73f470bf96

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-41 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition metadata before release-receipt-review-disposition records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-41 adds a prompt-evaluation result release receipt review disposition release receipt review disposition contract generator.

It consumes the Stage-40 prompt-evaluation result release receipt review disposition release receipt review report as prerequisite evidence and records a future review-disposition-release receipt review-disposition lane for receipt references, review-denial policy, review-disposition requirements, generated-text denial policy, token-generation denial policy, and survivor-centered safety requirements. It remains contract-only: no release receipt review disposition is recorded, no review decision is recorded, no findings are recorded, no receipt is emitted, no receipt is signed, no receipt is published, no receipt is packaged, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_41_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
ontract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-releas
e-receipt-review-disposition-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_contract_command=latticra-nadia
prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
ontract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_s
tage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
uthority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
llowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_c
reated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_p
erformed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_m
etadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
amily=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release-rece
ipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
ormat=contract-only-offline-evaluation-result-release-receipt-review-disposition-release-receipt
-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_e
vidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_s
ource_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_p
lan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
esult_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
untime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_cont
ract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_p
romotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
pproval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
indings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
pplied=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
network_authority=0
tool_execution_authority=0
source_mutation_authority=0
```

Stage-41 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review disposition contract that packages Stage-40 release receipt review evidence and records requirements for a future release receipt review disposition release contract.

Runtime invocation, prompt evaluation, token generation, inference, receipt review recording, review decision recording, review findings recording, disposition recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names the next disposition boundary and preserves safety inheritance, operator review gates, and non-claims.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-contract-stage-41.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_42_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-42 Status

Lines: 132 | Words: 497 | SHA-256: fc793177bcffd7e8f1447fb5467b6e08692b30eb1e77887889705f6da999a3d9

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-42 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release metadata before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-42 adds a prompt-evaluation result release receipt review disposition release receipt review disposition release contract generator.

It consumes the Stage-41 prompt-evaluation result release receipt review disposition release receipt review disposition report as prerequisite evidence and records a future review-disposition-release receipt lane for review-disposition references, release receipt references, release references, disposition references, review references, result references, generated-text denial policy, token-generation denial policy, review-disposition-release receipt policy, and survivor-centered safety requirements. It remains contract-only: no review-disposition release is created, no release record is created, no release decision is recorded, no release is published, no release is packaged, no release receipt is created, no review disposition is created, no disposition record is created, no review record is created, no receipt is signed, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_42_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-dispositio
n-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review
-disposition-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-rele
ase-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_format=contract-only-offline-evaluation-result-release-receipt-review-disposition-release
-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_cont
ract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_loaded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_opened=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_read=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_validated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_serialized=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_written=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
ecord_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_d
ecision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_f
indings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_a
pplied=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_record_create
d=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_decision_reco
rded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_findings_reco
rded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-42 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
```

```
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review disposition release contract that packages Stage-41 prompt-evaluation result release receipt review disposition release receipt review disposition evidence and records receipt requirements for a future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract.

Runtime invocation, prompt evaluation, token generation, inference, review-disposition release recording, release decision recording, release publication, release packaging, release receipt creation, review disposition recording, disposition decision recording, disposition findings recording, review recording, receipt recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review-disposition-release receipt shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

Stage-43 can later record that prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract; Stage-42 remains the prerequisite metadata boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-contract-stage-42.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_CONTRACT_STAGE_43_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract Stage-43 Status

Lines: 132 | Words: 525 | SHA-256: cbe8694727bdfc4d5dbea62a37de3b6061bbf78fb72eb178554f213f49d94f14

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Contract Stage-43 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt metadata before disposition-release-receipt recording, receipt emission, receipt signing, receipt publication, disposition-release recording, release decisions, release publication, release packaging, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-43 adds a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract generator.

It consumes the Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release report as prerequisite evidence and records a future review-disposition-release receipt lane for disposition-release references, review-disposition references, release receipt references, release references, disposition references, review references, result references, generated-text denial policy, token-generation denial policy, review-disposition-release receipt review policy, and survivor-centered safety requirements. It remains contract-only: no review-disposition-release receipt is created, no receipt is emitted, no receipt is signed, no receipt is published, no receipt is packaged, no review-disposition release is created, no release record is created, no release decision is recorded, no release is published, no release is packaged, no review disposition is created, no disposition record is created, no review record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_43_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-di
sposition-release-receipt-review-disposition-release-receipt-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_contract_command=latticra-nadia prompt-evaluation-result-release-receip
t-review-disposition-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposit
ion-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_format=contract-only-offline-evaluation-result-release-receipt-review-disposition
-release-receipt-review-disposition-release-receipt
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_loaded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_opened=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_read=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_validated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_serialized=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_written=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_emitted=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-43 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
```

```
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract that packages Stage-42 prompt-evaluation result release receipt review disposition release receipt review disposition release evidence and records review requirements for a future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract.

Runtime invocation, prompt evaluation, token generation, inference, review-disposition release recording, release decision recording, release publication, release packaging, release receipt creation, review disposition recording, disposition decision recording, disposition findings recording, review recording, receipt recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review-disposition-release receipt shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

Stage-44 can later record that prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract; Stage-43 remains the prerequisite metadata boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-contract-stage-43.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_CONTRACT_STAGE_44_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract Stage-44 Status

Lines: 108 | Words: 400 | SHA-256: 158f6947372fd1c43014ef3311eedae67b5c1bc744103211acc78ae2d47ccd63

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Contract Stage-44 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review metadata before release-receipt-review records, review decisions, review findings, receipt signing, receipt publication, receipt packaging, disposition recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-44 adds a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract generator.

It consumes the Stage-43 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt contract as prerequisite evidence and records a future review-disposition-release receipt review lane for receipt references, review-denial policy, review-disposition requirements, generated-text denial policy, token-generation denial policy, and survivor-centered safety requirements. It remains contract-only: no release receipt review is recorded, no review decision is recorded, no findings are recorded, no receipt is emitted, no receipt is signed, no receipt is published, no receipt is packaged, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_44_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_review_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-re
view-disposition-release-receipt-review-disposition-release-receipt-review-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_contract_command=latticra-nadia prompt-evaluation-result-release
-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-d
isposition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_format=contract-only-offline-evaluation-result-release-receipt-review-disp
osition-release-receipt-review-disposition-release-receipt-review
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_applied=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
network_authority=0
tool_execution_authority=0
source_mutation_authority=0
```

Stage-44 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract that packages Stage-43 release receipt evidence and records requirements for a future release receipt review disposition contract.

Runtime invocation, prompt evaluation, token generation, inference, receipt review recording, review decision recording, review findings recording, disposition recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names the next disposition boundary and preserves safety inheritance, operator review gates, and non-claims.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-contract-stage-44.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_CONTRACT_STAGE_45_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-45 Status

Lines: 108 | Words: 404 | SHA-256: 6361daa83a6a098abd49ec91f832c3016c964026ef7fb0509032f8ce10916974

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Contract Stage-45 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition metadata before release-receipt-review-disposition records, disposition decisions, disposition findings, receipt signing, receipt publication, receipt packaging, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-45 adds a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract generator.

It consumes the Stage-44 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review contract as prerequisite evidence and records a future review-disposition-release lane for receipt references, disposition-denial policy, review-disposition requirements, generated-text denial policy, token-generation denial policy, and survivor-centered safety requirements. It remains contract-only: no release receipt review disposition is recorded, no disposition decision is recorded, no findings are recorded, no receipt is emitted, no receipt is signed, no receipt is published, no receipt is packaged, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_45_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revie
w_disposition_release_receipt_review_disposition_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposi
tion_release_receipt_review_disposition_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_contract_command=scripts/nadia-prompt-evaluation-result-releas
e-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-dispositi
on-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_dis
position_release_receipt_review_disposition_contract_command=latticra-nadia prompt-evaluation-re
sult-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-revie
w-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_family=operator-reviewed-prompt-evaluation-result-release-rece
ipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_format=contract-only-offline-evaluation-result-release-receipt
-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disp
osition_release_receipt_review_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_revi
ew_disposition_release_receipt_review_disposition_release_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
network_authority=0
tool_execution_authority=0
source_mutation_authority=0
```

Stage-45 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition contract that packages Stage-44 release receipt review evidence and records requirements for a future release receipt review disposition release contract.

Runtime invocation, prompt evaluation, token generation, inference, receipt review recording, review decision recording, review findings recording, disposition recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names the next disposition boundary and preserves safety inheritance, operator review gates, and non-claims.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-disposition-contract-stage-45.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_RECEIPT_REVIEW_DISPOSITION_RELEASE_CONTRACT_STAGE_46_STATUS.md

Source title: Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-46 Status

Lines: 132 | Words: 529 | SHA-256: 62addd62f851d6327d83d1786af351355a55c0f72db13e20b9d1d42a9adb8806

Nadia Prompt Evaluation Result Release Receipt Review Disposition Release Receipt Review Disposition Release Receipt Review Disposition Release Contract Stage-46 Status

Status: implementation status record

Scope: prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release metadata before disposition-release recording, release decisions, release publication, release packaging, release receipt creation, disposition recording, review recording, receipt recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-46 adds a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract generator.

It consumes the Stage-45 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition report as prerequisite evidence and records a future review-disposition-release receipt lane for review-disposition references, release receipt references, release references, disposition references, review references, result references, generated-text denial policy, token-generation denial policy, review-disposition-release receipt policy, and survivor-centered safety requirements. It remains contract-only: no review-disposition release is created, no release record is created, no release decision is recorded, no release is published, no release is packaged, no release receipt is created, no review disposition is created, no disposition record is created, no review record is created, no receipt is signed, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_46_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_present=1
nadia_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_generator_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_command=scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release-contract.sh
installed_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_command=latticra-nadia prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_contract_status=contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_stage=contract-only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_authority=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_allowed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_performed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_metadata_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_family=operator-reviewed-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_format=contract-only-offline-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_decision=blocked_contract_only
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_evidence_present=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_source_policy=operator-reviewed-offline
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_plan_recorded=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_result_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_runtime_invoked=0
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_contract=1
requires_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_contract=1
requires_prompt_evaluation_result_release_receipt_contract=1
requires_prompt_evaluation_result_release_contract=1
requires_prompt_evaluation_result_disposition_contract=1
requires_prompt_evaluation_result_review_contract=1
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_contract=1
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_release_receipt_review_disposition_release_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_loaded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_opened=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_read=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_validated=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_serialized=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_written=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_approval_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_rejection_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_packaged=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_signed=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_release_receipt_published=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_findings_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_disposition_applied=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_record_created=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_decision_recorded=0
prompt_evaluation_result_release_receipt_review_disposition_release_receipt_review_disposition_r
elease_receipt_review_findings_recorded=0
prompt_evaluation_result_release_receipt_record_created=0
prompt_evaluation_result_release_receipt_signed=0
prompt_evaluation_result_release_receipt_published=0
prompt_evaluation_result_release_record_created=0
prompt_evaluation_result_release_decision_recorded=0
prompt_evaluation_result_model_output_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-46 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
```

```
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-release-receipt-review-disposition-release-receipt-review-disposition-release-receipt-review-disposition-release/.

Next Boundary

Nadia can now produce a prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release contract that packages Stage-45 prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition evidence and records receipt requirements for a future prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release receipt contract.

Runtime invocation, prompt evaluation, token generation, inference, review-disposition release recording, release decision recording, release publication, release packaging, release receipt creation, review disposition recording, disposition decision recording, disposition findings recording, review recording, receipt recording, receipt signing, receipt publication, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names review-disposition-release receipt shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

Stage-47 can later record that prompt-evaluation result release receipt review disposition release receipt review disposition release receipt review disposition release receipt contract; Stage-46 remains the prerequisite metadata boundary.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-release-receipt-review-disposition-release-receip
t-review-disposition-release-receipt-review-disposition-release-contract-stage-46.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RESULT_REVIEW_CONTRACT_STAGE_32_STATUS.md

Source title: Nadia Prompt Evaluation Result Review Contract Stage-32 Status

Lines: 120 | Words: 370 | SHA-256: 24768f6e1583dd8d2c372d05ca47d638443e0783727871ccc24737ef7a482462

Nadia Prompt Evaluation Result Review Contract Stage-32 Status

Status: implementation status record

Scope: prompt-evaluation result review metadata before review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-32 adds a prompt-evaluation result review contract generator.

It consumes the Stage-31 prompt-evaluation result report as prerequisite evidence and records a future review lane for prompt-evaluation result review schema, result references, invocation references, runtime handoff references, evaluation input references, generated-text denial policy, token-generation denial policy, result-disposition policy, and survivor-centered safety requirements. It remains contract-only: no review record is created, no result record is created, no model output is read or recorded, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_32_prompt_evaluation_result_review_contract_present=1
nadia_prompt_evaluation_result_review_contract_generator_present=1
prompt_evaluation_result_review_contract_command=scripts/nadia-prompt-evaluation-result-review-c
ontract.sh
installed_prompt_evaluation_result_review_contract_command=latticra-nadia
prompt-evaluation-result-review
prompt_evaluation_result_review_contract_status=contract_only
prompt_evaluation_result_review_stage=contract-only
prompt_evaluation_result_review_authority=0
prompt_evaluation_result_review_allowed=0
prompt_evaluation_result_review_recorded=0
prompt_evaluation_result_review_created=0
prompt_evaluation_result_review_performed=0
prompt_evaluation_result_review_metadata_present=1
prompt_evaluation_result_review_family=operator-reviewed-prompt-evaluation-result-review
prompt_evaluation_result_review_format=contract-only-offline-evaluation-result-review
prompt_evaluation_result_review_decision=blocked_contract_only
prompt_evaluation_result_review_evidence_present=1
prompt_evaluation_result_review_source_policy=operator-reviewed-offline
prompt_evaluation_result_review_plan_recorded=1
prompt_evaluation_result_review_method_planned=offline-prompt-evaluation-result-review-policy-re
view
prompt_evaluation_result_review_result_recorded=0
prompt_evaluation_result_review_runtime_invoked=0
requires_prompt_evaluation_result_contract=1
requires_future_prompt_evaluation_result_disposition_contract=1
prompt_evaluation_result_review_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_result_review_recorded=0
prompt_evaluation_result_review_created=0
prompt_evaluation_result_review_loaded=0
prompt_evaluation_result_review_opened=0
prompt_evaluation_result_review_read=0
prompt_evaluation_result_review_validated=0
prompt_evaluation_result_review_serialized=0
prompt_evaluation_result_review_written=0
prompt_evaluation_result_review_record_created=0
prompt_evaluation_result_review_record_validated=0
prompt_evaluation_result_review_record_serialized=0
prompt_evaluation_result_review_record_written=0
prompt_evaluation_result_review_record_submitted=0
prompt_evaluation_result_review_decision_recorded=0
prompt_evaluation_result_review_approval_recorded=0
prompt_evaluation_result_review_rejection_recorded=0
prompt_evaluation_result_review_findings_recorded=0
prompt_evaluation_result_review_runtime_invoked=0
prompt_evaluation_result_recorded=0
prompt_evaluation_result_record_created=0
prompt_evaluation_result_model_output_recorded=0
prompt_evaluation_result_output_text_recorded=0
prompt_evaluation_result_score_recorded=0
prompt_evaluation_result_token_logprobs_recorded=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
answer_text_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-32 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-result-review as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-result-review-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-result-review/.

Next Boundary

Nadia can now produce a prompt-evaluation result review contract that packages Stage-31 prompt-evaluation result evidence and records review requirements for a future prompt-evaluation result disposition contract.

Runtime invocation, prompt evaluation, token generation, inference, result review recording, result recording, model-output recording, and dialogue generation remain blocked until a later contract explicitly names disposition shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation result disposition contract is the next boundary; Stage-32 only records the prerequisite metadata.

Stage-33 now defines a prompt-evaluation result disposition contract that keeps disposition recording, release recording, result-review recording, result recording, model-output recording, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-prompt-evaluation-result-review-contract-stage-32.sh
```

docs/status/NADIA_PROMPT_EVALUATION_RUNTIME_HANDOFF_CONTRACT_STAGE_29_STATUS.md

Source title: Nadia Prompt Evaluation Runtime Handoff Contract Stage-29 Status

Lines: 113 | Words: 336 | SHA-256: 1e150bbdbcd464aae1b932ce3949beddc0936ae71661bec8782a3bc4a1a2bd8a

Nadia Prompt Evaluation Runtime Handoff Contract Stage-29 Status

Status: implementation status record

Scope: prompt-evaluation runtime handoff metadata before runtime handoff, runtime invocation, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-29 adds a prompt-evaluation runtime handoff contract generator.

It consumes the Stage-28 prompt-evaluation-input report as prerequisite evidence and records a future review lane for prompt-evaluation runtime handoff schema, evaluation input references, runtime profile references, runtime-invocation contract references, model-load contract references, inference-readiness contract references, runtime-denial policy, token-generation denial policy, and prompt-evaluation invocation requirements. It remains contract-only: no prompt evaluation input is created, no runtime handoff is performed, no runtime is invoked, no prompt is evaluated, no token is generated, and no dialogue is generated.

Status Fields

```
nadia_stage_29_prompt_evaluation_runtime_handoff_contract_present=1
nadia_prompt_evaluation_runtime_handoff_contract_generator_present=1
prompt_evaluation_runtime_handoff_contract_command=scripts/nadia-prompt-evaluation-runtime-hando
ff-contract.sh
installed_prompt_evaluation_runtime_handoff_contract_command=latticra-nadia
prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_contract_status=contract_only
prompt_evaluation_runtime_handoff_stage=contract-only
prompt_evaluation_runtime_handoff_authority=0
prompt_evaluation_runtime_handoff_allowed=0
prompt_evaluation_runtime_handoff_performed=0
prompt_evaluation_runtime_handoff_metadata_present=1
prompt_evaluation_runtime_handoff_family=operator-reviewed-prompt-evaluation-runtime-handoff
prompt_evaluation_runtime_handoff_format=contract-only-offline-runtime-handoff
prompt_evaluation_runtime_handoff_decision=blocked_contract_only
prompt_evaluation_runtime_handoff_evidence_present=1
prompt_evaluation_runtime_handoff_source_policy=operator-reviewed-offline
prompt_evaluation_runtime_handoff_plan_recorded=1
prompt_evaluation_runtime_handoff_method_planned=offline-prompt-evaluation-runtime-handoff-polic
y-review
prompt_evaluation_runtime_handoff_result_recorded=0
prompt_evaluation_runtime_handoff_runtime_invoked=0
requires_prompt_evaluation_input_contract=1
requires_future_prompt_evaluation_invocation_contract=1
prompt_evaluation_runtime_handoff_promotion_allowed=0
```

Guardrails

```
prompt_evaluation_runtime_handoff_performed=0
prompt_evaluation_runtime_handoff_created=0
prompt_evaluation_runtime_handoff_loaded=0
prompt_evaluation_runtime_handoff_opened=0
prompt_evaluation_runtime_handoff_read=0
prompt_evaluation_runtime_handoff_validated=0
prompt_evaluation_runtime_handoff_serialized=0
prompt_evaluation_runtime_handoff_written=0
prompt_evaluation_runtime_handoff_request_created=0
prompt_evaluation_runtime_handoff_request_validated=0
prompt_evaluation_runtime_handoff_request_serialized=0
prompt_evaluation_runtime_handoff_request_submitted=0
prompt_evaluation_runtime_handoff_runtime_selected=0
prompt_evaluation_runtime_handoff_model_selected=0
prompt_evaluation_runtime_handoff_session_created=0
prompt_evaluation_runtime_handoff_runtime_invoked=0
runtime_handoff_created=0
runtime_handoff_submitted=0
prompt_evaluation_request_created=0
prompt_evaluation_request_submitted=0
runtime_invocation_requested=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-29 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-evaluation-runtime-handoff as metadata-only. The command delegates to scripts/nadia-prompt-evaluation-runtime-handoff-contract.sh after installation and writes reports under share/latticra/nadia/prompt-evaluation-runtime-handoff/.

Next Boundary

Nadia can now produce a prompt-evaluation runtime handoff contract that packages Stage-28 prompt-evaluation-input evidence and records review requirements for a future prompt-evaluation invocation contract.

Runtime invocation, prompt evaluation, token generation, and dialogue generation remain blocked until a later contract explicitly names invocation request shape, runtime execution denial fields, safety inheritance, operator review gates, and non-claims.

That later prompt-evaluation invocation contract is the next boundary; Stage-29 only records the prerequisite metadata.

Stage-30 now defines a prompt-evaluation invocation contract that keeps invocation request creation, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Validation

```
sh scripts/test-nadia-prompt-evaluation-runtime-handoff-contract-stage-29.sh
```

docs/status/NADIA_PROMPT_MATERIALIZATION_CONTRACT_STAGE_14_STATUS.md

Source title: Nadia Prompt Materialization Contract Stage-14 Status

Lines: 117 | Words: 303 | SHA-256: 8f9cdb03d7ca8f77d38edae3acf2ba8bf4ed625b165991a5e17efdd1c71dbac2

Nadia Prompt Materialization Contract Stage-14 Status

Status: implementation status record Date: 2026-05-25 Scope: prompt-materialization metadata before prompt buffer allocation, prompt text materialization, prompt tokenization, prompt evaluation, token generation, inference, or tool execution.

Summary

Nadia Stage-14 adds a prompt-materialization contract generator.

The contract verifies Stage-13 prompt-receipt metadata and inherited protective, prompt, model, runtime, and tool-denial boundaries, then records an explicitly blocked materialization decision for future operator review. It remains contract-only and does not receive, read, store, hash, classify, tokenize, materialize, or evaluate prompt text.

Evidence Flags

```
nadia_stage_14_prompt_materialization_contract_present=1
nadia_prompt_materialization_contract_generator_present=1
nadia_prompt_materialization_contract_guard_present=1
nadia_installed_prompt_materialization_contract_command_planned=1
prompt_materialization_contract_command=scripts/nadia-prompt-materialization-contract.sh
installed_prompt_materialization_contract_command=latticra-nadia prompt-materialization
requires_prompt_receipt_contract=1
prompt_receipt_stage_required=13-prompt-receipt-contract
prompt_materialization_contract_status=contract_only
prompt_materialization_stage=contract-only
prompt_materialization_authority=0
prompt_materialization_allowed=0
prompt_materialized=0
prompt_text_materialized=0
materialization_decision=blocked_contract_only
materialization_evidence_present=1
requires_model_load_contract=1
requires_runtime_invocation_contract=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_refusal_policy_review=1
requires_prompt_source_boundary=1
requires_prompt_buffer_boundary=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_prompt_evaluation_handoff_contract=1
prompt_materialization_promotion_allowed=0
prompt_source_open_authority=0
prompt_source_read_authority=0
prompt_text_materialization_authority=0
prompt_buffer_allocation_authority=0
prompt_buffer_write_authority=0
prompt_content_storage_authority=0
prompt_hash_authority=0
prompt_classification_authority=0
prompt_tokenization_authority=0
prompt_source_opened=0
prompt_source_read=0
prompt_bytes_read=0
prompt_text_received=0
prompt_materialization_performed=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_bytes_materialized=0
prompt_tokens_created=0
prompt_tokenized=0
prompt_content_stored=0
prompt_hash_computed=0
prompt_classified=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_loaded=0
model_weights_loaded=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
```

```
harm_aware_development=1
```

Current Claim

Nadia can now produce a prompt-materialization contract after prompt-receipt evidence is present.

This does not mean Nadia can receive prompt text, read prompt sources, allocate prompt buffers, materialize prompt content, tokenize prompts, store prompt content, hash prompt content, classify prompt content, evaluate prompts, load model weights, spawn a runtime process, create a model session, generate tokens, run inference, execute tools, use a shell, use a network client, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-prompt-materialization-contract-stage-14.sh
```

Expected result:

```
nadia_prompt_materialization_contract_stage_14: ok
```

Next Stage

Stage-15 now defines an awareness-dialogue contract for official Nadia Initiative Q&A scope after prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_PROMPT_RECEIPT_CONTRACT_STAGE_13_STATUS.md

Source title: Nadia Prompt Receipt Contract Stage-13 Status

Lines: 107 | Words: 278 | SHA-256: 337d70ac447a1f10414779013dad74547bdcb7bba340c6dbf82f61f62cfe1861

Nadia Prompt Receipt Contract Stage-13 Status

Status: implementation status record Date: 2026-05-25 Scope: prompt-receipt metadata before prompt source opening, prompt text receipt, prompt materialization, prompt evaluation, token generation, inference, or tool execution.

Summary

Nadia Stage-13 adds a prompt-receipt contract generator.

The contract verifies Stage-12 model-load metadata and inherited protective, prompt, model, runtime, and tool-denial boundaries, then records an explicitly blocked receipt decision for future operator review. It remains contract-only and does not receive, read, store, hash, classify, materialize, or evaluate prompt text.

Evidence Flags

```
nadia_stage_13_prompt_receipt_contract_present=1
nadia_prompt_receipt_contract_generator_present=1
nadia_prompt_receipt_contract_guard_present=1
nadia_installed_prompt_receipt_contract_command_planned=1
prompt_receipt_contract_command=scripts/nadia-prompt-receipt-contract.sh
installed_prompt_receipt_contract_command=latticra-nadia prompt-receipt
requires_model_load_contract=1
model_load_stage_required=12-model-load-contract
prompt_receipt_contract_status=contract_only
prompt_receipt_stage=contract-only
prompt_receipt_authority=0
prompt_receipt_allowed=0
prompt_received=0
receipt_decision=blocked_contract_only
receipt_evidence_present=1
requires_runtime_invocation_contract=1
requires_inference_readiness_contract=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_refusal_policy_review=1
requires_prompt_source_boundary=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_prompt_materialization_contract=1
prompt_receipt_promotion_allowed=0
prompt_source_open_authority=0
prompt_source_read_authority=0
prompt_text_materialization_authority=0
prompt_content_storage_authority=0
prompt_hash_authority=0
prompt_classification_authority=0
prompt_source_opened=0
prompt_source_read=0
prompt_bytes_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_content_stored=0
prompt_hash_computed=0
prompt_classified=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
model_loaded=0
model_weights_loaded=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_session_created=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a prompt-receipt contract after model-load evidence is present.

This does not mean Nadia can receive prompt text, read prompt sources, store prompt content, hash prompt content, classify prompt content, materialize prompts, evaluate prompts, load model weights, spawn a runtime process, create a model session, generate tokens, run inference, execute tools, use a shell, use a network client, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-prompt-receipt-contract-stage-13.sh
```

Expected result:

```
nadia_prompt_receipt_contract_stage_13: ok
```

Next Stage

Stage-14 now defines a prompt-materialization contract after prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_PROMPT_TOKEN_SEQUENCE_CONTRACT_STAGE_26_STATUS.md

Source title: Nadia Prompt Token Sequence Contract Stage-26 Status

Lines: 107 | Words: 342 | SHA-256: fdd93461ec0b55b30bf2cc7607013fe6bcd049f01c1b26f5d83fcf9d105b95d9

Nadia Prompt Token Sequence Contract Stage-26 Status

Status: implementation status record

Scope: prompt-token-sequence metadata before prompt token ID recording, token order recording, token offset recording, attention mask creation, position ID creation, context window assembly, prompt evaluation input creation, runtime invocation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-26 adds a prompt-token-sequence contract generator.

It consumes the Stage-25 prompt-tokenization report as prerequisite evidence and records a future review lane for prompt token IDs, token order, token offsets, attention masks, position IDs, context-window policy, and prompt evaluation inputs. It remains contract-only: no prompt text is read, no prompt tokens are created, no token IDs or token order are recorded, no context window is assembled, no runtime is invoked, no prompt is evaluated, and no dialogue is generated.

Status Fields

```
nadia_stage_26_prompt_token_sequence_contract_present=1
nadia_prompt_token_sequence_contract_generator_present=1
prompt_token_sequence_contract_command=scripts/nadia-prompt-token-sequence-contract.sh
installed_prompt_token_sequence_contract_command=latticra-nadia prompt-token-sequence
prompt_token_sequence_contract_status=contract_only
prompt_token_sequence_stage=contract-only
prompt_token_sequence_authority=0
prompt_token_sequence_allowed=0
prompt_token_sequence_recorded=0
prompt_token_sequence_metadata_present=1
prompt_token_sequence_family=operator-reviewed-prompt-token-sequence
prompt_token_sequence_format=contract-only-offline-sequence
prompt_token_sequence_decision=blocked_contract_only
prompt_token_sequence_evidence_present=1
prompt_token_sequence_source_policy=operator-reviewed-offline
prompt_token_sequence_plan_recorded=1
prompt_token_sequence_method_planned=offline-token-sequence-policy-review
prompt_token_sequence_result_recorded=0
prompt_token_sequence_count_recorded=0
prompt_token_sequence_order_recorded=0
prompt_token_sequence_runtime_invoked=0
requires_prompt_tokenization_contract=1
requires_future_context_window_assembly_contract=1
prompt_token_sequence_promotion_allowed=0
```

Guardrails

```
prompt_token_ids_recorded=0
prompt_token_order_recorded=0
prompt_token_offsets_recorded=0
prompt_token_byte_offsets_recorded=0
prompt_attention_mask_created=0
prompt_position_ids_created=0
context_window_assembled=0
prompt_evaluation_input_created=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_token_sequence_recorded=0
prompt_tokenized=0
runtime_invoked=0
runtime_session_created=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Stage-26 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
namesake_cause_awareness=1
```

Installer and Console Surface

The Latticra Panel and installed latticra-nadia wrapper expose prompt-token-sequence as metadata-only. The command delegates to scripts/nadia-prompt-token-sequence-contract.sh after installation and writes reports under share/latticra/nadia/prompt-token-sequence/.

Next Boundary

Nadia can now produce a prompt-token-sequence contract that packages Stage-25 prompt-tokenization evidence and records review requirements for a future context-window assembly contract.

Stage-27 now defines a context-window assembly contract that keeps context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

Context-window assembly remains blocked until a later contract explicitly names the context window format, prompt evaluation input shape, truncation policy, safety inheritance, runtime-denial fields, operator review gates, and non-claims.

That later context window assembly contract is the next boundary; Stage-26 only records the prerequisite metadata.

Validation

```
sh scripts/test-nadia-prompt-token-sequence-contract-stage-26.sh
```

docs/status/NADIA_PROMPT_TOKENIZATION_CONTRACT_STAGE_25_STATUS.md

Source title: Nadia Prompt Tokenization Contract Stage-25 Status

Lines: 155 | Words: 349 | SHA-256: 96841ec6d828d4af08c6de67135bbbae0f8f85d24927ad01f05058dd15f1932f

Nadia Prompt Tokenization Contract Stage-25 Status

Status: implementation status record

Date: 2026-05-25 CDT

Scope: prompt-tokenization metadata before prompt text reading, prompt text materialization, prompt buffer allocation, prompt token creation, prompt token sequence recording, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-25 adds a prompt-tokenization contract generator.

The contract consumes Stage-24 tokenizer-runtime-attachment metadata, verifies inherited protective, prompt, tokenizer, artifact, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future prompt token sequence contract must satisfy. It remains contract-only and does not read prompt text, create prompt tokens, record token sequences, invoke a runtime, evaluate prompts, generate dialogue, generate tokens, or run inference.

```
nadia_stage_25_prompt_tokenization_contract_present=1
nadia_prompt_tokenization_contract_generator_present=1
nadia_prompt_tokenization_contract_guard_present=1
nadia_installed_prompt_tokenization_contract_command_planned=1
prompt_tokenization_contract_command=scripts/nadia-prompt-tokenization-contract.sh
installed_prompt_tokenization_contract_command=latticra-nadia prompt-tokenization
prompt_tokenization_contract_status=contract_only
prompt_tokenization_stage=contract-only
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenization_performed=0
prompt_tokenization_metadata_present=1
prompt_tokenization_family=operator-reviewed-prompt-tokenization
prompt_tokenization_format=contract-only-offline-tokenization
prompt_tokenization_decision=blocked_contract_only
prompt_tokenization_evidence_present=1
```

```
prompt_tokenization_source_policy=operator-reviewed-offline
prompt_tokenization_plan_recorded=1
prompt_tokenization_method_planned=offline-tokenization-policy-review
prompt_tokenization_result_recorded=0
prompt_tokenization_token_count_recorded=0
prompt_tokenization_token_sequence_recorded=0
prompt_tokenization_runtime_invoked=0
requires_tokenizer_runtime_attachment_contract=1
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_future_prompt_token_sequence_contract=1
prompt_tokenization_promotion_allowed=0
```

Denials Preserved

```
prompt_tokenization_open_authority=0
prompt_tokenization_read_authority=0
prompt_tokenization_write_authority=0
prompt_tokenization_execute_authority=0
prompt_tokenization_runtime_authority=0
prompt_tokenization_token_create_authority=0
prompt_tokenization_sequence_record_authority=0
prompt_tokenization_opened=0
prompt_tokenization_read=0
prompt_tokenization_validated=0
prompt_tokenization_loaded=0
prompt_tokenization_bytes_read=0
prompt_tokenization_hash_computed=0
prompt_tokenization_entries_loaded=0
prompt_tokenization_result_recorded=0
prompt_tokenization_token_count_recorded=0
prompt_tokenization_token_sequence_recorded=0
prompt_tokenization_runtime_invoked=0
prompt_tokenization_file_written=0
prompt_text_read=0
prompt_text_received=0
prompt_text_materialized=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_tokens_created=0
prompt_token_count_recorded=0
prompt_token_sequence_recorded=0
prompt_token_buffer_created=0
prompt_token_buffer_written=0
prompt_tokenized=0
tokenizer_runtime_attachment_performed=0
tokenizer_runtime_attachment_attached=0
tokenizer_attached_to_runtime=0
runtime_tokenizer_attachment_performed=0
runtime_session_created=0
runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_binding_hash_computed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
tool_execution_performed=0
network_authority=0
```

Stage-25 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
```

```
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
live_web_lookup_authority=0
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operational Meaning

Nadia can now produce a prompt-tokenization contract that packages Stage-24 tokenizer-runtime-attachment evidence and records review requirements for a future prompt token sequence contract.

Stage-26 now defines a prompt-token-sequence contract that keeps prompt token ID recording, token order recording, token offset recording, context window assembly, prompt evaluation input creation, runtime invocation, prompt evaluation, dialogue generation, token generation, and inference blocked.

This is not prompt tokenization authority. It does not read prompts, create tokens, record token sequences, invoke a runtime, run inference, generate dialogue, or execute tools.

Guard

```
sh scripts/test-nadia-prompt-tokenization-contract-stage-25.sh
```

Expected:

```
nadia_prompt_tokenization_contract_stage_25: ok
```

docs/status/NADIA_PROTECTIVE_SAFETY_BOUNDARY_STAGE_6_STATUS.md

Source title: Nadia Protective Safety Boundary Stage-6 Status

Lines: 71 | Words: 185 | SHA-256: 4144066a2619fc528ee331daa0d36db2d7176b50780cf9113aea69f92391c063

Nadia Protective Safety Boundary Stage-6 Status

Status: implementation status record Date: 2026-05-25 Scope: non-sexual-use, anti-manipulation, and namesake-cause awareness boundary before prompt evaluation, model runtime, or tool authority.

Summary

Nadia Stage-6 adds a protective-safety boundary generator.

The boundary records absolute non-sexual-use restrictions, anti-manipulation restrictions, namesake-cause awareness, and survivor-witness respect. It fails closed on sexualized request classifications and remains metadata-only.

Evidence Flags

```
nadia_stage_6_protective_safety_boundary_present=1
nadia_protective_safety_generator_present=1
nadia_protective_safety_guard_present=1
nadia_installed_protective_safety_command_planned=1
protective_safety_command=scripts/nadia-protective-safety-boundary.sh
installed_protective_safety_command=latticra-nadia protective-safety
requires_productivity_entry=1
productivity_entry_stage_required=5-productivity-ledger-loop
absolute_protective_boundary=1
sexual_user_request_authority=0
sexual_content_generation=0
sexual_roleplay_authority=0
sexualized_namesake_or_survivor_content=0
sexual_request_refusal=always
user_override_authority=0
prompt_injection_override_authority=0
manipulation_resistance=required
policy_bypass_authority=0
namesake_cause_awareness=1
awareness_context=non_sensational_human_rights
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
distillation_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a protective-safety boundary report that encodes absolute non-sexual-use and anti-manipulation restrictions.

This does not mean Nadia can run a model, evaluate a prompt, install weights, download models, generate code, execute arbitrary tools, train herself, distill herself, mutate source, use network access, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-protective-safety-boundary-stage-6.sh
```

Expected result:

```
nadia_protective_safety_boundary_stage_6: ok
```

Next Stage

Stage-7 now adds a report-only guarded tool-authority preflight after the protective-safety boundary. It still does not execute tools.

docs/status/NADIA_RUNTIME_INVOCATION_CONTRACT_STAGE_11_STATUS.md

Source title: Nadia Runtime Invocation Contract Stage-11 Status

Lines: 98 | Words: 249 | SHA-256: a6646842ace2ec6833515432dc3d629067ae1cf9f15096f81c1bfed38de62f3d

Nadia Runtime Invocation Contract Stage-11 Status

Status: implementation status record Date: 2026-05-25 Scope: runtime-invocation metadata before runtime process spawning, model session creation, model loading, token generation, inference, prompt evaluation, or tool execution.

Summary

Nadia Stage-11 adds a runtime-invocation contract generator.

The contract verifies Stage-10 inference-readiness metadata and inherited protective, prompt, model, runtime, and tool-denial boundaries, then records an explicitly blocked invocation decision for future operator review. It remains contract-only and does not invoke a runtime.

Evidence Flags

```
nadia_stage_11_runtime_invocation_contract_present=1
nadia_runtime_invocation_contract_generator_present=1
nadia_runtime_invocation_contract_guard_present=1
nadia_installed_runtime_invocation_contract_command_planned=1
runtime_invocation_contract_command=scripts/nadia-runtime-invocation-contract.sh
installed_runtime_invocation_contract_command=latticra-nadia runtime-invocation
requires_inference_readiness_contract=1
inference_readiness_stage_required=10-inference-readiness-contract
runtime_invocation_contract_status=contract_only
runtime_invocation_stage=contract-only
runtime_invocation_authority=0
runtime_invocation_allowed=0
runtime_invoked=0
invocation_decision=blocked_contract_only
invocation_evidence_present=1
requires_model_registry_contract=1
requires_prompt_contract=1
requires_runtime_profile=1
requires_protective_safety_boundary=1
requires_tool_preflight=1
requires_operator_review=1
requires_model_provenance_review=1
requires_model_license_review=1
requires_model_safety_review=1
requires_refusal_policy_review=1
requires_runtime_boundary_gate=1
requires_nucleus_gate=1
requires_seal_receipt=1
requires_future_model_load_contract=1
invocation_promotion_allowed=0
runtime_process_spawn_authority=0
runtime_binary_execution_authority=0
runtime_session_authority=0
model_session_authority=0
token_generation_authority=0
model_runtime_present=0
model_runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
runtime_session_created=0
model_weights_loaded=0
prompt_materialized=0
prompt_evaluation_authority=0
prompt_evaluated=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
source_mutation_authority=0
network_authority=0
training_performed=0
distillation_performed=0
sexual_content_generation=0
sexual_request_refusal=always
manipulation_resistance=required
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now produce a runtime-invocation contract after inference-readiness evidence is present.

This does not mean Nadia can spawn a runtime process, execute a runtime binary, create a model session, load model weights, generate tokens, run inference, materialize prompts, evaluate prompts, execute tools, use a shell, use a network client, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-runtime-invocation-contract-stage-11.sh
```

Expected result:

```
nadia_runtime_invocation_contract_stage_11: ok
```

Next Stage

Stage-12 now defines a model-load contract after runtime-invocation metadata, inference-readiness metadata, model-registry metadata, prompt-evaluation contracts, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_RUNTIME_PROFILE_STAGE_2_STATUS.md

Source title: Nadia Runtime Profile Stage-2 Status

Lines: 59 | Words: 170 | SHA-256: 255761604140c88b5bee72668f2c4fb81cbcd233e36ecfef4418621672a7e622

Nadia Runtime Profile Stage-2 Status

Status: implementation status record Date: 2026-05-25 Scope: offline runtime and model-profile metadata before inference.

Summary

Nadia Stage-2 adds a runtime-profile generator for offline model-readiness metadata.

The profile can record runtime family, GGUF/model format, context-window policy, memory budget, and optional local model-file measurement. It does not run inference.

Evidence Flags

```
nadia_stage_2_runtime_profile_present=1
nadia_runtime_profile_generator_present=1
nadia_runtime_profile_guard_present=1
nadia_installed_runtime_profile_command_planned=1
runtime_profile_command=scripts/nadia-runtime-profile.sh
installed_runtime_profile_command=latticra-nadia runtime-profile
model_file_measurement=operator_provided_optional
runtime_family=llama.cpp-compatible
model_format=gguf
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now generate a local runtime profile for future offline inference readiness. The profile may measure an operator-provided local model file.

This does not mean Nadia can run a model, evaluate a prompt, install weights, download models, generate code, execute arbitrary tools, train herself, mutate source, or use network access.

Validation

```
sh scripts/test-nadia-runtime-profile-stage-2.sh
```

Expected result:

```
nadia_runtime_profile_stage_2: ok
```

Next Stage

Stage-3 now adds a developer-workbench planning surface that combines context-pack evidence and runtime-profile evidence into prompt plans without evaluating prompts.

docs/status/NADIA_SYSTEMS_ENGINEERING_MODE_STAGE_4_STATUS.md

Source title: Nadia Systems Engineering Mode Stage-4 Status

Lines: 60 | Words: 184 | SHA-256: d9b916eb7caea9a7eeb0d4d20c12250328f405710660d84028fdce16356e50f2

Nadia Systems Engineering Mode Stage-4 Status

Status: implementation status record Date: 2026-05-25 Scope: systems-engineering mode taxonomy and prompt-plan validation before prompt evaluation.

Summary

Nadia Stage-4 adds a systems-engineering mode validator for Stage-3 prompt plans.

The validator records a mode label, mode focus, validator set, prompt-plan measurement, and authority posture. It fails closed on unknown modes or prompt plans that already claim prompt evaluation or inference.

Evidence Flags

```
nadia_stage_4_systems_engineering_mode_present=1
nadia_mode_validation_generator_present=1
nadia_mode_validation_guard_present=1
nadia_installed_mode_validation_command_planned=1
mode_validation_command=scripts/nadia-mode-validate.sh
installed_mode_validation_command=latticra-nadia mode-validate
requires_prompt_plan=1
prompt_plan_stage_required=3-developer-workbench-planning
mode_taxonomy_present=1
mode_allowed=1
network_authority=0
model_runtime_invoked=0
inference_performed=0
prompt_evaluated=0
model_weights_installed=0
tool_execution_authority=0
source_mutation_authority=0
training_performed=0
human_dignity_principle=1
survivor_witness_respect=1
community_awareness_posture=1
harm_aware_development=1
```

Current Claim

Nadia can now validate that a Stage-3 prompt plan is assigned to an allowed systems-engineering mode while remaining metadata-only.

This does not mean Nadia can run a model, evaluate a prompt, install weights, download models, generate code, execute arbitrary tools, train herself, mutate source, or use network access.

Validation

```
sh scripts/test-nadia-systems-engineering-mode-stage-4.sh
```

Expected result:

```
nadia_systems_engineering_mode_stage_4: ok
```

Next Stage

Stage-5 now adds a local productivity-ledger loop for improving retrieval, ranking, plan templates, and test recommendations without model training or autonomous source mutation.

docs/status/NADIA_TOKENIZATION_BOUNDARY_CONTRACT_STAGE_17_STATUS.md

Source title: Nadia Tokenization Boundary Contract Stage-17 Status

Lines: 110 | Words: 315 | SHA-256: 38ab910711e640f0479f733e659d01fcd4b153ddf13a441cc4ca6f8634c5e656

Nadia Tokenization Boundary Contract Stage-17 Status

Status: implementation status record Date: 2026-05-25 Scope: tokenization-boundary metadata before tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-17 adds a tokenization-boundary contract generator.

The contract verifies Stage-16 prompt-evaluation handoff metadata and inherited protective, prompt, model, runtime, dialogue, and tool-denial boundaries, then records a blocked tokenization state that requires a future tokenizer specification contract before any tokenizer file or vocabulary can be touched. It remains contract-only and does not tokenize prompts.

Evidence Flags

```
nadia_stage_17_tokenization_boundary_contract_present=1
nadia_tokenization_boundary_contract_generator_present=1
nadia_tokenization_boundary_contract_guard_present=1
nadia_installed_tokenization_boundary_contract_command_planned=1
tokenization_boundary_contract_command=scripts/nadia-tokenization-boundary-contract.sh
installed_tokenization_boundary_contract_command=latticra-nadia tokenization-boundary
tokenization_boundary_contract_status=contract_only
tokenization_boundary_stage=contract-only
tokenization_boundary_authority=0
tokenization_boundary_allowed=0
tokenization_boundary_performed=0
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
prompt_tokenized=0
prompt_tokens_created=0
tokenizer_file_opened=0
tokenizer_vocab_loaded=0
prompt_evaluation_authority=0
prompt_evaluated=0
tokenization_decision=blocked_contract_only
tokenization_evidence_present=1
requires_prompt_evaluation_handoff_contract=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_prompt_buffer_boundary=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenizer_specification_contract=1
tokenization_boundary_promotion_allowed=0
future_qa_dialogue_capability_planned=1
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
dialogue_format=question-and-answer
q_and_a_format_required=1
survivor_centered_dialogue_required=1
respectful_tone_required=1
plain_language_required=1
source_attribution_required=1
official_source_grounding_required=1
source_snapshot_policy=operator-reviewed-offline
live_web_lookup_authority=0
topic_yazidi_genocide_awareness=1
topic_survivor_voice_and_dignity=1
topic_conflict_related_sexual_violence_awareness_non_graphic=1
topic_genocide_prevention=1
topic_justice_and_accountability=1
topic_womens_empowerment=1
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_materialized=0
prompt_text_materialized=0
prompt_buffer_allocated=0
prompt_buffer_written=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
network_authority=0
```

Current Claim

Nadia can now produce a tokenization-boundary contract that packages Stage-16 prompt-evaluation handoff evidence and explicitly blocks prompt tokenization until a future tokenizer specification contract exists.

This does not mean Nadia can tokenize prompts, load tokenizer files, load tokenizer vocabularies, evaluate prompts, generate dialogue, answer user prompts, browse the web, provide legal advice, provide medical advice, provide trauma counseling, receive prompt text, materialize prompt content, load model weights, spawn a runtime process, generate tokens, run inference, execute tools, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-tokenization-boundary-contract-stage-17.sh
```

Expected result:

```
nadia_tokenization_boundary_contract_stage_17: ok
```

Next Stage

Stage-18 now defines a tokenizer specification contract only after tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_TOKENIZER_ARTIFACT_BINDING_CONTRACT_STAGE_23_STATUS.md

Source title: Nadia Tokenizer Artifact Binding Contract Stage-23 Status

Lines: 149 | Words: 359 | SHA-256: 4bac8ecef986400475525164be2c044b6fcc1f4148b8ac395e3c00d584b74f24

Nadia Tokenizer Artifact Binding Contract Stage-23 Status

Status: implementation status record

Date: 2026-05-25 CDT

Scope: tokenizer-artifact-binding metadata before tokenizer artifact opening, artifact reading, artifact hashing, artifact verification, artifact binding, tokenizer runtime attachment, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-23 adds a tokenizer-artifact-binding contract generator.

The contract consumes Stage-22 tokenizer-artifact-verification metadata, verifies inherited protective, prompt, tokenizer, artifact, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future tokenizer runtime attachment contract must satisfy. It remains contract-only and does not open tokenizer artifacts, read artifacts, hash artifacts, verify artifacts, bind artifacts, attach tokenizers to a runtime, load tokenizer manifests, open tokenizer files, load vocabularies, or tokenize prompts.

```
nadia_stage_23_tokenizer_artifact_binding_contract_present=1
nadia_tokenizer_artifact_binding_contract_generator_present=1
nadia_tokenizer_artifact_binding_contract_guard_present=1
```

```
nadia_installed_tokenizer_artifact_binding_contract_command_planned=1
tokenizer_artifact_binding_contract_command=scripts/nadia-tokenizer-artifact-binding-contract.sh
installed_tokenizer_artifact_binding_contract_command=latticra-nadia tokenizer-artifact-binding
tokenizer_artifact_binding_contract_status=contract_only
tokenizer_artifact_binding_stage=contract-only
tokenizer_artifact_binding_authority=0
tokenizer_artifact_binding_allowed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_binding_metadata_present=1
tokenizer_artifact_binding_family=operator-reviewed-tokenizer-artifact-binding
tokenizer_artifact_binding_format=contract-only-offline-binding
tokenizer_artifact_binding_decision=blocked_contract_only
tokenizer_artifact_binding_evidence_present=1
tokenizer_artifact_binding_source_policy=operator-reviewed-offline
tokenizer_artifact_binding_plan_recorded=1
tokenizer_artifact_binding_method_planned=offline-manifest-artifact-role-binding-review
tokenizer_artifact_binding_result_recorded=0
tokenizer_artifact_binding_record_created=0
tokenizer_artifact_binding_manifest_reference_recorded=0
tokenizer_artifact_binding_artifact_reference_recorded=0
tokenizer_artifact_binding_runtime_attach_recorded=0
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_runtime_attachment_contract=1
tokenizer_artifact_binding_promotion_allowed=0
```

Denials Preserved

```
tokenizer_artifact_binding_open_authority=0
tokenizer_artifact_binding_read_authority=0
tokenizer_artifact_binding_write_authority=0
tokenizer_artifact_binding_hash_authority=0
tokenizer_artifact_binding_validation_authority=0
tokenizer_artifact_binding_load_authority=0
tokenizer_artifact_binding_attach_authority=0
tokenizer_artifact_binding_runtime_attach_authority=0
tokenizer_artifact_binding_manifest_bind_authority=0
tokenizer_artifact_binding_tokenizer_bind_authority=0
tokenizer_artifact_binding_opened=0
tokenizer_artifact_binding_read=0
tokenizer_artifact_binding_validated=0
tokenizer_artifact_binding_loaded=0
tokenizer_artifact_binding_bytes_read=0
tokenizer_artifact_binding_hash_computed=0
tokenizer_artifact_binding_entries_loaded=0
tokenizer_artifact_binding_bound=0
tokenizer_artifact_binding_record_created=0
tokenizer_artifact_binding_manifest_reference_loaded=0
tokenizer_artifact_binding_artifact_reference_loaded=0
tokenizer_artifact_binding_manifest_reference_recorded=0
tokenizer_artifact_binding_artifact_reference_recorded=0
tokenizer_artifact_binding_runtime_attach_recorded=0
tokenizer_artifact_binding_runtime_attachment_performed=0
tokenizer_artifact_binding_result_recorded=0
tokenizer_artifact_binding_file_written=0
tokenizer_artifact_bound_to_manifest=0
tokenizer_artifact_bound_to_tokenizer=0
tokenizer_attached_to_runtime=0
tokenizer_runtime_attachment_performed=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_verification_hash_computed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_artifact_measurement_performed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
tool_execution_performed=0
network_authority=0
```

Stage-23 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
live_web_lookup_authority=0
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operational Meaning

Nadia can now produce a tokenizer-artifact-binding contract that packages Stage-22 tokenizer-artifact-verification evidence and records review requirements for a future tokenizer runtime attachment contract.

This is not artifact binding authority. It does not open, read, hash, compare, verify, bind, validate, load, or attach tokenizer artifacts.

Guard

```
sh scripts/test-nadia-tokenizer-artifact-binding-contract-stage-23.sh
```

Expected:

```
nadia_tokenizer_artifact_binding_contract_stage_23: ok
```

Next Gate

Stage-24 now defines a tokenizer runtime attachment contract after tokenizer-artifact-binding metadata, tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_TOKENIZER_ARTIFACT_INVENTORY_CONTRACT_STAGE_20_STATUS.md

Source title: Nadia Tokenizer Artifact Inventory Contract Stage-20 Status

Lines: 110 | Words: 375 | SHA-256: e1adc91c513b786e61aa53a597d62a15e58ec5a27d0eff044db248c63eb2a445

Nadia Tokenizer Artifact Inventory Contract Stage-20 Status

Status: implementation status record

Scope: tokenizer-artifact-inventory metadata before tokenizer artifact path resolution, artifact scanning, artifact stat calls, artifact hashing, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Current State

Nadia Stage-20 adds a tokenizer-artifact-inventory contract generator.

The contract verifies Stage-19 tokenizer-manifest metadata and inherited protective, prompt, tokenizer, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future tokenizer artifact measurement contract must satisfy. It remains contract-only and does not resolve artifact paths, scan directories, stat files, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, load tokenizer manifests, load tokenizer vocabularies, or tokenize prompts.

```
nadia_stage_20_tokenizer_artifact_inventory_contract_present=1
nadia_tokenizer_artifact_inventory_contract_generator_present=1
nadia_tokenizer_artifact_inventory_contract_guard_present=1
nadia_installed_tokenizer_artifact_inventory_contract_command_planned=1
tokenizer_artifact_inventory_contract_command=scripts/nadia-tokenizer-artifact-inventory-contrac
t.sh
installed_tokenizer_artifact_inventory_contract_command=latticra-nadia
tokenizer-artifact-inventory
tokenizer_artifact_inventory_contract_status=contract_only
tokenizer_artifact_inventory_stage=contract-only
tokenizer_artifact_inventory_authority=0
tokenizer_artifact_inventory_allowed=0
tokenizer_artifact_inventory_performed=0
tokenizer_artifact_inventory_metadata_present=1
tokenizer_artifact_inventory_family=operator-reviewed-tokenizer-artifact-inventory
tokenizer_artifact_inventory_format=contract-only-offline-inventory
tokenizer_artifact_inventory_decision=blocked_contract_only
tokenizer_artifact_inventory_evidence_present=1
tokenizer_artifact_inventory_source_policy=operator-reviewed-offline
tokenizer_artifact_inventory_path_recorded=0
tokenizer_artifact_inventory_schema_planned=1
tokenizer_artifact_inventory_entry_count=0
tokenizer_artifact_inventory_file_count=0
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_measurement_contract=1
tokenizer_artifact_inventory_promotion_allowed=0
tokenizer_artifact_path_resolution_authority=0
tokenizer_artifact_scan_authority=0
tokenizer_artifact_stat_authority=0
tokenizer_artifact_hash_authority=0
tokenizer_artifact_measurement_authority=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_scan_performed=0
tokenizer_artifact_stat_performed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_artifact_measurement_performed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Guardrails

Stage-20 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
```

```
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operator Meaning

Nadia can now produce a tokenizer-artifact-inventory contract that packages Stage-19 tokenizer-manifest evidence and records review requirements for a future tokenizer artifact measurement contract.

This does not mean Nadia can resolve artifact paths, scan directories, stat files, open tokenizer artifacts, read tokenizer artifacts, hash tokenizer artifacts, load tokenizer manifests, parse manifests, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, generate dialogue, answer user prompts, browse the web, provide legal advice, provide medical advice, provide trauma counseling, receive prompt text, materialize prompt content, load model weights, spawn a runtime process, generate tokens, run inference, execute tools, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-tokenizer-artifact-inventory-contract-stage-20.sh
```

Expected:

```
nadia_tokenizer_artifact_inventory_contract_stage_20: ok
```

Next Gate

Stage-21 now defines a tokenizer artifact measurement contract only after tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_TOKENIZER_ARTIFACT_MEASUREMENT_CONTRACT_STAGE_21_STATUS.md

Source title: Nadia Tokenizer Artifact Measurement Contract Stage-21 Status

Lines: 130 | Words: 329 | SHA-256: 6c06de74d5d7fe13e37bb8ec88b73b85199fafe0d81d706a2486c8ad6891d463

Nadia Tokenizer Artifact Measurement Contract Stage-21 Status

Status: implementation status record

Date: 2026-05-25 CDT

Scope: tokenizer-artifact-measurement metadata before tokenizer artifact opening, artifact reading, artifact hashing, artifact size recording, artifact digest recording, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-21 adds a tokenizer-artifact-measurement contract generator.

The contract consumes Stage-20 tokenizer-artifact-inventory metadata, verifies inherited protective, prompt, tokenizer, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future tokenizer artifact verification contract must satisfy. It remains contract-only and does not open tokenizer artifacts, read artifacts, hash artifacts, record artifact digests, record artifact sizes, load tokenizer manifests, open tokenizer files, load vocabularies, or tokenize prompts.

```
nadia_stage_21_tokenizer_artifact_measurement_contract_present=1
nadia_tokenizer_artifact_measurement_contract_generator_present=1
nadia_tokenizer_artifact_measurement_contract_guard_present=1
nadia_installed_tokenizer_artifact_measurement_contract_command_planned=1
tokenizer_artifact_measurement_contract_command=scripts/nadia-tokenizer-artifact-measurement-con
tract.sh
installed_tokenizer_artifact_measurement_contract_command=latticra-nadia
tokenizer-artifact-measurement
tokenizer_artifact_measurement_contract_status=contract_only
tokenizer_artifact_measurement_stage=contract-only
tokenizer_artifact_measurement_authority=0
tokenizer_artifact_measurement_allowed=0
tokenizer_artifact_measurement_performed=0
tokenizer_artifact_measurement_metadata_present=1
tokenizer_artifact_measurement_family=operator-reviewed-tokenizer-artifact-measurement
tokenizer_artifact_measurement_format=contract-only-offline-measurement
tokenizer_artifact_measurement_decision=blocked_contract_only
tokenizer_artifact_measurement_evidence_present=1
tokenizer_artifact_measurement_source_policy=operator-reviewed-offline
tokenizer_artifact_measurement_plan_recorded=1
tokenizer_artifact_measurement_algorithm_planned=sha256-or-approved-digest
tokenizer_artifact_measurement_result_recorded=0
tokenizer_artifact_measurement_digest_recorded=0
tokenizer_artifact_measurement_size_recorded=0
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_verification_contract=1
tokenizer_artifact_measurement_promotion_allowed=0
```

Denials Preserved

```
tokenizer_artifact_measurement_open_authority=0
tokenizer_artifact_measurement_read_authority=0
tokenizer_artifact_measurement_hash_authority=0
tokenizer_artifact_measurement_validation_authority=0
tokenizer_artifact_measurement_load_authority=0
tokenizer_artifact_measurement_opened=0
tokenizer_artifact_measurement_read=0
tokenizer_artifact_measurement_validated=0
tokenizer_artifact_measurement_loaded=0
tokenizer_artifact_measurement_bytes_read=0
tokenizer_artifact_measurement_hash_computed=0
tokenizer_artifact_measurement_entries_loaded=0
tokenizer_artifact_measurement_result_recorded=0
tokenizer_artifact_measurement_digest_recorded=0
tokenizer_artifact_measurement_size_recorded=0
tokenizer_artifact_digest_recorded=0
tokenizer_artifact_size_recorded=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_bytes_read=0
tokenizer_artifact_hash_computed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
tool_execution_performed=0
network_authority=0
```

Stage-21 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
```

```
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
live_web_lookup_authority=0
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operational Meaning

Nadia can now produce a tokenizer-artifact-measurement contract that packages Stage-20 tokenizer-artifact-inventory evidence and records review requirements for a future tokenizer artifact verification contract.

This is not artifact measurement authority. It does not open, read, hash, size, validate, or load tokenizer artifacts.

Guard

```
sh scripts/test-nadia-tokenizer-artifact-measurement-contract-stage-21.sh
```

Expected:

```
nadia_tokenizer_artifact_measurement_contract_stage_21: ok
```

Next Gate

Stage-22 now defines a tokenizer artifact verification contract only after tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_TOKENIZER_ARTIFACT_VERIFICATION_CONTRACT_STAGE_22_STATUS.md

Source title: Nadia Tokenizer Artifact Verification Contract Stage-22 Status

Lines: 143 | Words: 354 | SHA-256: 0ae2bc82cf7ac1b24400a4d1bbda141c91040861cdb9b0ddb623120cbad7dcaf

Nadia Tokenizer Artifact Verification Contract Stage-22 Status

Status: implementation status record

Date: 2026-05-25 CDT

Scope: tokenizer-artifact-verification metadata before tokenizer artifact opening, artifact reading, artifact hashing, artifact digest comparison, artifact size comparison, artifact verification, artifact binding, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-22 adds a tokenizer-artifact-verification contract generator.

The contract consumes Stage-21 tokenizer-artifact-measurement metadata, verifies inherited protective, prompt, tokenizer, artifact, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future tokenizer artifact binding contract must satisfy. It remains contract-only and does not open tokenizer artifacts, read artifacts, hash artifacts, compare artifact digests, compare artifact sizes, verify artifacts, bind artifacts, load tokenizer manifests, open tokenizer files, load vocabularies, or tokenize prompts.

```
nadia_stage_22_tokenizer_artifact_verification_contract_present=1
nadia_tokenizer_artifact_verification_contract_generator_present=1
nadia_tokenizer_artifact_verification_contract_guard_present=1
nadia_installed_tokenizer_artifact_verification_contract_command_planned=1
tokenizer_artifact_verification_contract_command=scripts/nadia-tokenizer-artifact-verification-c
ontract.sh
installed_tokenizer_artifact_verification_contract_command=latticra-nadia
tokenizer-artifact-verification
tokenizer_artifact_verification_contract_status=contract_only
tokenizer_artifact_verification_stage=contract-only
tokenizer_artifact_verification_authority=0
tokenizer_artifact_verification_allowed=0
tokenizer_artifact_verification_performed=0
tokenizer_artifact_verification_metadata_present=1
tokenizer_artifact_verification_family=operator-reviewed-tokenizer-artifact-verification
tokenizer_artifact_verification_format=contract-only-offline-verification
tokenizer_artifact_verification_decision=blocked_contract_only
tokenizer_artifact_verification_evidence_present=1
tokenizer_artifact_verification_source_policy=operator-reviewed-offline
tokenizer_artifact_verification_plan_recorded=1
tokenizer_artifact_verification_method_planned=offline-digest-and-size-policy-review
tokenizer_artifact_verification_comparison_performed=0
tokenizer_artifact_verification_result_recorded=0
tokenizer_artifact_verification_digest_match_recorded=0
tokenizer_artifact_verification_size_match_recorded=0
requires_tokenizer_artifact_measurement_contract=1
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_future_tokenizer_artifact_binding_contract=1
tokenizer_artifact_verification_promotion_allowed=0
```

Denials Preserved

```
tokenizer_artifact_verification_open_authority=0
tokenizer_artifact_verification_read_authority=0
tokenizer_artifact_verification_hash_authority=0
tokenizer_artifact_verification_validation_authority=0
tokenizer_artifact_verification_load_authority=0
tokenizer_artifact_verification_compare_authority=0
tokenizer_artifact_verification_bind_authority=0
tokenizer_artifact_verification_opened=0
tokenizer_artifact_verification_read=0
tokenizer_artifact_verification_validated=0
tokenizer_artifact_verification_loaded=0
tokenizer_artifact_verification_bytes_read=0
tokenizer_artifact_verification_hash_computed=0
tokenizer_artifact_verification_entries_loaded=0
tokenizer_artifact_verification_compared=0
tokenizer_artifact_verification_comparison_performed=0
tokenizer_artifact_verification_digest_comparison_performed=0
tokenizer_artifact_verification_size_comparison_performed=0
tokenizer_artifact_verification_expected_digest_loaded=0
tokenizer_artifact_verification_observed_digest_loaded=0
tokenizer_artifact_verification_digest_match_recorded=0
tokenizer_artifact_verification_size_match_recorded=0
tokenizer_artifact_verification_result_recorded=0
tokenizer_artifact_source_signature_verified=0
tokenizer_artifact_digest_recorded=0
tokenizer_artifact_size_recorded=0
tokenizer_artifact_path_resolved=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_bytes_read=0
tokenizer_artifact_hash_computed=0
tokenizer_artifact_measurement_performed=0
tokenizer_artifact_measurement_hash_computed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
tool_execution_performed=0
network_authority=0
```

Stage-22 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
live_web_lookup_authority=0
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operational Meaning

Nadia can now produce a tokenizer-artifact-verification contract that packages Stage-21 tokenizer-artifact-measurement evidence and records review requirements for a future tokenizer artifact binding contract.

This is not artifact verification authority. It does not open, read, hash, compare, verify, bind, validate, or load tokenizer artifacts.

Guard

```
sh scripts/test-nadia-tokenizer-artifact-verification-contract-stage-22.sh
```

Expected:

```
nadia_tokenizer_artifact_verification_contract_stage_22: ok
```

Next Gate

Stage-23 now defines a tokenizer artifact binding contract after tokenizer-artifact-verification metadata, tokenizer-artifact-measurement metadata, tokenizer-artifact-inventory metadata, tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_TOKENIZER_MANIFEST_CONTRACT_STAGE_19_STATUS.md

Source title: Nadia Tokenizer Manifest Contract Stage-19 Status

Lines: 107 | Words: 333 | SHA-256: f898c0ca5e457d9f812b905a7129b5738591b73b5b6adb961295ceec7f58dd45

Nadia Tokenizer Manifest Contract Stage-19 Status

Status: implementation status record

Scope: tokenizer-manifest metadata before tokenizer manifest loading, tokenizer manifest parsing, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Current State

Nadia Stage-19 adds a tokenizer-manifest contract generator.

The contract verifies Stage-18 tokenizer-specification metadata and inherited protective, prompt, tokenizer, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future tokenizer artifact inventory contract must satisfy. It remains contract-only and does not load tokenizer manifests, parse manifests, open tokenizer files, load tokenizer vocabularies, or tokenize prompts.

```
nadia_stage_19_tokenizer_manifest_contract_present=1
nadia_tokenizer_manifest_contract_generator_present=1
nadia_tokenizer_manifest_contract_guard_present=1
nadia_installed_tokenizer_manifest_contract_command_planned=1
tokenizer_manifest_contract_command=scripts/nadia-tokenizer-manifest-contract.sh
installed_tokenizer_manifest_contract_command=latticra-nadia tokenizer-manifest
tokenizer_manifest_contract_status=contract_only
tokenizer_manifest_stage=contract-only
tokenizer_manifest_authority=0
tokenizer_manifest_allowed=0
tokenizer_manifest_performed=0
tokenizer_manifest_metadata_present=1
tokenizer_manifest_family=operator-reviewed-tokenizer-manifest
tokenizer_manifest_format=contract-only-offline-manifest
tokenizer_manifest_decision=blocked_contract_only
tokenizer_manifest_evidence_present=1
tokenizer_manifest_source_policy=operator-reviewed-offline
tokenizer_manifest_path_recorded=0
tokenizer_manifest_schema_planned=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
```

```
requires_future_tokenizer_artifact_inventory_contract=1
tokenizer_manifest_promotion_allowed=0
tokenizer_manifest_open_authority=0
tokenizer_manifest_read_authority=0
tokenizer_manifest_parse_authority=0
tokenizer_manifest_validation_authority=0
tokenizer_manifest_load_authority=0
tokenizer_manifest_loaded=0
tokenizer_manifest_opened=0
tokenizer_manifest_read=0
tokenizer_manifest_parsed=0
tokenizer_manifest_validated=0
tokenizer_manifest_bytes_read=0
tokenizer_manifest_hash_computed=0
tokenizer_manifest_entries_loaded=0
tokenizer_file_path_resolved=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
qa_dialogue_generated=0
sexual_request_refusal=always
manipulation_resistance=required
```

Guardrails

Stage-19 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
survivor_impersonation_allowed=0
survivor_identifying_speculation_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
hate_or_collective_blame_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
crisis_intervention_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operator Meaning

Nadia can now produce a tokenizer-manifest contract that packages Stage-18 tokenizer-specification evidence and records review requirements for a future tokenizer artifact inventory.

This does not mean Nadia can load tokenizer manifests, parse manifests, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, generate dialogue, answer user prompts, browse the web, provide legal advice, provide medical advice, provide trauma counseling, receive prompt text, materialize prompt content, load model weights, spawn a runtime process, generate tokens, run inference, execute tools, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-tokenizer-manifest-contract-stage-19.sh
```

Expected:

```
nadia_tokenizer_manifest_contract_stage_19: ok
```

Next Gate

Stage-20 now defines a tokenizer artifact inventory contract only after tokenizer-manifest metadata, tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NADIA_TOKENIZER_RUNTIME_ATTACHMENT_CONTRACT_STAGE_24_STATUS.md

Source title: Nadia Tokenizer Runtime Attachment Contract Stage-24 Status

Lines: 152 | Words: 337 | SHA-256: f10f54935af403793bdfaa5db044e0ae0b1eaa8d4888b2d2c66f01c57cc62854

Nadia Tokenizer Runtime Attachment Contract Stage-24 Status

Status: implementation status record

Date: 2026-05-25 CDT

Scope: tokenizer-runtime-attachment metadata before tokenizer artifact opening, artifact reading, artifact hashing, tokenizer artifact binding, tokenizer runtime attachment, runtime session creation, tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-24 adds a tokenizer-runtime-attachment contract generator.

The contract consumes Stage-23 tokenizer-artifact-binding metadata, verifies inherited protective, prompt, tokenizer, artifact, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future prompt tokenization contract must satisfy. It remains contract-only and does not attach tokenizers to a runtime, create runtime sessions, tokenize prompts, load tokenizer manifests, open tokenizer files, load vocabularies, or run inference.

```
nadia_stage_24_tokenizer_runtime_attachment_contract_present=1
nadia_tokenizer_runtime_attachment_contract_generator_present=1
nadia_tokenizer_runtime_attachment_contract_guard_present=1
nadia_installed_tokenizer_runtime_attachment_contract_command_planned=1
tokenizer_runtime_attachment_contract_command=scripts/nadia-tokenizer-runtime-attachment-contrac
t.sh
installed_tokenizer_runtime_attachment_contract_command=latticra-nadia
tokenizer-runtime-attachment
tokenizer_runtime_attachment_contract_status=contract_only
tokenizer_runtime_attachment_stage=contract-only
tokenizer_runtime_attachment_authority=0
tokenizer_runtime_attachment_allowed=0
tokenizer_runtime_attachment_performed=0
tokenizer_runtime_attachment_metadata_present=1
tokenizer_runtime_attachment_family=operator-reviewed-tokenizer-runtime-attachment
tokenizer_runtime_attachment_format=contract-only-offline-attachment
tokenizer_runtime_attachment_decision=blocked_contract_only
tokenizer_runtime_attachment_evidence_present=1
tokenizer_runtime_attachment_source_policy=operator-reviewed-offline
tokenizer_runtime_attachment_plan_recorded=1
tokenizer_runtime_attachment_method_planned=offline-runtime-tokenizer-attachment-review
tokenizer_runtime_attachment_result_recorded=0
tokenizer_runtime_attachment_record_created=0
tokenizer_runtime_attachment_runtime_reference_recorded=0
tokenizer_runtime_attachment_tokenizer_reference_recorded=0
tokenizer_runtime_attachment_session_created=0
requires_tokenizer_artifact_binding_contract=1
requires_tokenizer_artifact_verification_contract=1
requires_tokenizer_artifact_measurement_contract=1
```

```
requires_tokenizer_artifact_inventory_contract=1
requires_tokenizer_manifest_contract=1
requires_tokenizer_specification_contract=1
requires_tokenization_boundary_contract=1
requires_runtime_profile_contract=1
requires_runtime_invocation_contract=1
requires_model_load_contract=1
requires_future_prompt_tokenization_contract=1
tokenizer_runtime_attachment_promotion_allowed=0
```

Denials Preserved

```
tokenizer_runtime_attachment_open_authority=0
tokenizer_runtime_attachment_read_authority=0
tokenizer_runtime_attachment_write_authority=0
tokenizer_runtime_attachment_validation_authority=0
tokenizer_runtime_attachment_load_authority=0
tokenizer_runtime_attachment_attach_authority=0
tokenizer_runtime_attachment_runtime_invoke_authority=0
tokenizer_runtime_attachment_session_authority=0
tokenizer_runtime_attachment_tokenizer_bind_authority=0
tokenizer_runtime_attachment_opened=0
tokenizer_runtime_attachment_read=0
tokenizer_runtime_attachment_validated=0
tokenizer_runtime_attachment_loaded=0
tokenizer_runtime_attachment_bytes_read=0
tokenizer_runtime_attachment_hash_computed=0
tokenizer_runtime_attachment_entries_loaded=0
tokenizer_runtime_attachment_attached=0
tokenizer_runtime_attachment_record_created=0
tokenizer_runtime_attachment_runtime_reference_loaded=0
tokenizer_runtime_attachment_tokenizer_reference_loaded=0
tokenizer_runtime_attachment_runtime_reference_recorded=0
tokenizer_runtime_attachment_tokenizer_reference_recorded=0
tokenizer_runtime_attachment_runtime_invoked=0
tokenizer_runtime_attachment_session_created=0
tokenizer_runtime_attachment_result_recorded=0
tokenizer_runtime_attachment_file_written=0
tokenizer_attached_to_runtime=0
tokenizer_runtime_attachment_performed=0
runtime_tokenizer_attachment_performed=0
runtime_session_created=0
runtime_invoked=0
runtime_process_spawned=0
runtime_binary_executed=0
tokenizer_artifact_binding_performed=0
tokenizer_artifact_binding_hash_computed=0
tokenizer_artifact_file_opened=0
tokenizer_artifact_file_read=0
tokenizer_artifact_hash_computed=0
tokenizer_manifest_loaded=0
tokenizer_manifest_parsed=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
prompt_tokenized=0
prompt_evaluated=0
token_generation_performed=0
inference_performed=0
tool_execution_performed=0
network_authority=0
```

Stage-24 inherits all Nadia protective-safety and awareness-dialogue restrictions:

```
future_qa_dialogue_capability_planned=1
dialogue_scope=official-nadia-initiative-awareness-work
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
live_web_lookup_authority=0
sexual_user_request_authority=0
sexual_content_generation=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
erotic_content_allowed=0
roleplay_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
```

```
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
prompt_injection_override_authority=0
policy_bypass_authority=0
```

Operational Meaning

Nadia can now produce a tokenizer-runtime-attachment contract that packages Stage-23 tokenizer-artifact-binding evidence and records review requirements for a future prompt tokenization contract.

This is not tokenizer runtime attachment authority. It does not attach tokenizers to a runtime, create runtime sessions, tokenize prompts, run inference, or execute tools.

Stage-25 now defines a prompt-tokenization contract that packages Stage-24 tokenizer-runtime-attachment evidence while keeping prompt text reading, token creation, token sequence recording, runtime invocation, inference, dialogue generation, and tool execution denied.

Guard

```
sh scripts/test-nadia-tokenizer-runtime-attachment-contract-stage-24.sh
```

Expected:

```
nadia_tokenizer_runtime_attachment_contract_stage_24: ok
```

docs/status/NADIA_TOKENIZER_SPECIFICATION_CONTRACT_STAGE_18_STATUS.md

Source title: Nadia Tokenizer Specification Contract Stage-18 Status

Lines: 127 | Words: 336 | SHA-256: 1657b44649758eb94816c11d6cfc1e4783b9c3bfdcc40f08924ad400cf09cffe

Nadia Tokenizer Specification Contract Stage-18 Status

Status: implementation status record Date: 2026-05-25 Scope: tokenizer-specification metadata before tokenizer manifest loading, tokenizer file access, tokenizer vocabulary loading, prompt tokenization, prompt evaluation, dialogue generation, token generation, inference, or tool execution.

Summary

Nadia Stage-18 adds a tokenizer-specification contract generator.

The contract verifies Stage-17 tokenization-boundary metadata and inherited protective, prompt, tokenizer, model, runtime, dialogue, and tool-denial boundaries, then records the requirements that a future tokenizer manifest contract must satisfy. It remains contract-only and does not load tokenizer files, load tokenizer vocabularies, or tokenize prompts.

Evidence Flags

```
nadia_stage_18_tokenizer_specification_contract_present=1
nadia_tokenizer_specification_contract_generator_present=1
nadia_tokenizer_specification_contract_guard_present=1
nadia_installed_tokenizer_specification_contract_command_planned=1
tokenizer_specification_contract_command=scripts/nadia-tokenizer-specification-contract.sh
installed_tokenizer_specification_contract_command=latticra-nadia tokenizer-specification
tokenizer_specification_contract_status=contract_only
tokenizer_specification_stage=contract-only
tokenizer_specification_authority=0
tokenizer_specification_allowed=0
tokenizer_specification_performed=0
tokenizer_specification_metadata_present=1
tokenizer_family=model-compatible-tokenizer
tokenizer_format=operator-reviewed-offline-specification
tokenizer_specification_decision=blocked_contract_only
tokenizer_specification_evidence_present=1
tokenizer_source_policy=operator-reviewed-offline
tokenizer_path_recorded=0
tokenizer_manifest_loaded=0
tokenizer_file_measurement_performed=0
requires_tokenization_boundary_contract=1
requires_prompt_evaluation_handoff_contract=1
requires_awareness_dialogue_contract=1
requires_prompt_materialization_contract=1
requires_prompt_receipt_contract=1
requires_prompt_buffer_boundary=1
requires_protective_safety_boundary=1
requires_operator_review=1
requires_official_source_snapshot=1
requires_future_tokenizer_manifest_contract=1
tokenizer_specification_promotion_allowed=0
requires_model_tokenizer_compatibility_review=1
requires_tokenizer_format_review=1
requires_unicode_policy_review=1
requires_normalization_policy_review=1
requires_special_token_policy_review=1
requires_bos_eos_policy_review=1
requires_chat_template_policy_review=1
requires_prompt_template_boundary=1
requires_context_window_policy_review=1
requires_stop_sequence_policy_review=1
future_qa_dialogue_capability_planned=1
qa_dialogue_generated=0
question_generated=0
answer_generated=0
answer_text_generated=0
dialogue_scope=official-nadia-initiative-awareness-work
q_and_a_format_required=1
survivor_centered_dialogue_required=1
official_source_grounding_required=1
live_web_lookup_authority=0
sexualized_dialogue_generation=0
graphic_sexual_detail_allowed=0
victim_blaming_allowed=0
genocide_denial_allowed=0
medical_advice_authority=0
legal_advice_authority=0
trauma_counseling_authority=0
sexual_request_refusal=always
manipulation_resistance=required
tokenizer_file_open_authority=0
tokenizer_file_read_authority=0
tokenizer_vocab_load_authority=0
tokenizer_vocab_mapping_authority=0
tokenizer_runtime_attach_authority=0
tokenizer_file_opened=0
tokenizer_file_read=0
tokenizer_vocab_loaded=0
tokenizer_vocab_mapped=0
tokenizer_attached_to_runtime=0
tokenizer_bytes_read=0
tokenizer_hash_computed=0
prompt_materialized=0
prompt_text_materialized=0
prompt_buffer_allocated=0
prompt_buffer_written=0
prompt_tokenization_authority=0
prompt_tokenization_allowed=0
```

```
prompt_tokenized=0
prompt_tokens_created=0
prompt_evaluation_authority=0
prompt_evaluated=0
token_generation_authority=0
token_generation_performed=0
inference_authority=0
inference_performed=0
tool_execution_authority=0
tool_execution_performed=0
network_authority=0
```

Current Claim

Nadia can now produce a tokenizer-specification contract that packages Stage-17 tokenization-boundary evidence and records review requirements for a future tokenizer manifest.

This does not mean Nadia can load tokenizer manifests, open tokenizer files, read tokenizer files, load tokenizer vocabularies, tokenize prompts, evaluate prompts, generate dialogue, answer user prompts, browse the web, provide legal advice, provide medical advice, provide trauma counseling, receive prompt text, materialize prompt content, load model weights, spawn a runtime process, generate tokens, run inference, execute tools, mutate source, train herself, distill herself, or provide sexualized user-facing behavior.

Validation

```
sh scripts/test-nadia-tokenizer-specification-contract-stage-18.sh
```

Expected result:

```
nadia_tokenizer_specification_contract_stage_18: ok
```

Next Stage

Stage-19 now defines a tokenizer manifest contract only after tokenizer-specification metadata, tokenization-boundary metadata, prompt-evaluation handoff metadata, awareness-dialogue metadata, prompt-materialization metadata, prompt-receipt metadata, model-load metadata, runtime-invocation metadata, inference-readiness metadata, model-registry metadata, protective-safety refusal behavior, and tool-denial behavior are all present and explicitly non-executing.

docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_README_ALIGNMENT.md

Source title: Nucleus Report-Only Announcement README Alignment

Lines: 29 | Words: 129 | SHA-256: 641116830ed5b5d4d4bf369c0ba23ad410089ddcf9f0313d29d6693455eb8968

Nucleus Report-Only Announcement README Alignment

Status: README/project-notes alignment

This record aligns public entry-point documentation after the Nucleus report-only announcement review.

Reviewed files:

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_REVIEW.md
docs/project_notes/README.md
```

Resolution:

```
Add the Nucleus report-only announcement review to README.md discoverability.
```

```
Add the review to the project-notes related-records index.
Add this alignment record to public status and foundation indexes.
Advance root and detailed status ledgers without changing completion estimates.
Keep the next review lane conditional on capability posture changes.
```

Boundary: README/project-notes/status alignment only. No implementation behavior is changed. No public announcement entry is added, and no task execution, runtime behavior, command execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/NUCLEUS_REPORT_ONLY_ANNOUNCEMENT_REVIEW.md

Source title: Nucleus Report-Only Announcement Review

Lines: 39 | Words: 169 | SHA-256: fe42794d2a45d49f115f3ddf848a249d5fe904ceab7a231f2aa4f0c0c795c44a

Nucleus Report-Only Announcement Review

Status: no-new-announcement review

This record reviews whether the recent Nucleus task report-only execution README/status alignment, project-notes Nucleus report-only alignment, and project-notes Nucleus report-only status/index check require a separate public announcement entry.

Reviewed files:

```
STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/status/CURRENT_STATUS.md
docs/status/README.md
docs/FOUNDATION_INDEX.md
docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md
docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_README_STATUS_ALIGNMENT.md
docs/status/PROJECT_NOTES_NUCLEUS_REPORT_ONLY_ALIGNMENT_STATUS_INDEX_CHECK.md
docs/project_notes/README.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
README.md
```

Finding:

```
No new implementation behavior, capability posture, public product readiness, security
guarantee, or runtime behavior was added after the Nucleus report-only execution refinement.
The recent work was README/status/project-notes alignment and status/index checking only.
```

Decision:

```
Do not add a separate public announcement entry for this slice.
Keep the announcement log reserved for capability-bearing, milestone-bearing, or
public-messaging changes that need a dated public update.
Keep the next review lane conditional on actual capability posture changes.
```

Boundary: announcement/status review only. No implementation behavior is changed. No task execution, runtime behavior, command execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT_STATUS.md

Source title: Nucleus Task No-Effect Report Alignment Status

Lines: 41 | Words: 108 | SHA-256: c94be555bf912716549105d8a7ccf8f284b90d146b781f8335721ea4d7d50b97

Nucleus Task No-Effect Report Alignment Status

Status: implementation refinement status

This record tracks the Nucleus task no-effect report alignment refinement.

Primary record:

```
docs/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT.md
```

Primary implementation surfaces:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_no_effect_report_alignment.c
scripts/test-nucleus-task-no-effect-report-alignment.sh
.github/workflows/nucleus-task-no-effect-report-alignment.yml
```

What changed:

```
explicit report_alignment field
explicit no_effect_policy field
explicit representation_gate field
focused no-effect alignment tests
focused workflow coverage
main Nucleus task runner coverage
```

Validation:

```
sh scripts/test-nucleus-task-no-effect-report-alignment.sh
sh scripts/test-nucleus-task-execution.sh
```

Boundary: no-effect report metadata only. No task execution, runtime behavior, command execution, Lat execution, LIR execution, file I/O, network I/O, state mutation, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/NUCLEUS_TASK_README_STATUS_ALIGNMENT.md

Source title: Nucleus Task README/Status Alignment

Lines: 26 | Words: 67 | SHA-256: b6bfd26d846e167baad6a0e757d80aa6028c2b85c4c37db52a8a990a9a71862e

Nucleus Task README/Status Alignment

Status: README/status alignment

This record aligns public entry-point documentation after the Nucleus task no-effect report alignment refinement.

Reviewed files:

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT.md
docs/status/NUCLEUS_TASK_NO_EFFECT_REPORT_ALIGNMENT_STATUS.md
```

Resolution:

```
Add the Nucleus task no-effect report alignment to README.md.
Add this alignment record to the status index.
Advance root and detailed status queues.
```

Boundary: documentation/status alignment only. No implementation behavior is changed.

docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_README_STATUS_ALIGNMENT.md

Source title: Nucleus Task Report-Only Execution README/Status Alignment

Lines: 28 | Words: 115 | SHA-256: 337fb998b09f91a9b3331bf4910fd2cf46bfc0e85a558124ef710ed5a8738a9c

Nucleus Task Report-Only Execution README/Status Alignment

Status: README/status alignment

This record aligns public entry-point documentation after the Nucleus task report-only execution refinement.

Reviewed files:

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md
docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md
```

Resolution:

```
Add the Nucleus task report-only execution refinement to README.md.
Add the report-only execution refinement and this alignment record to public status entry
points.
Advance root, detailed, and foundation next-step queues.
Preserve completion estimates because no capability posture changed.
```

Boundary: documentation/status alignment only. No implementation behavior is changed. No task execution, runtime behavior, command execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT_STATUS.md

Source title: Nucleus Task Report-Only Execution Refinement Status

Lines: 41 | Words: 107 | SHA-256: 7a6915d9ec51844feb7a322f4d5d21e55617e115281d53cf632aebadc5027861

Nucleus Task Report-Only Execution Refinement Status

Status: implementation refinement status

This record tracks the Nucleus task report-only execution refinement.

Primary record:

```
docs/NUCLEUS_TASK_REPORT_ONLY_EXECUTION_REFINEMENT.md
```

Primary implementation surfaces:

```
include/latticra/nucleus_task.h
src/nucleus_task.c
tests/nucleus_task_report_only_execution_refinement.c
scripts/test-nucleus-task-report-only-execution-refinement.sh
.github/workflows/nucleus-task-report-only-execution-refinement.yml
```

What changed:

```
explicit execution_status field
explicit effect_status field
explicit runtime_status field
focused report-only execution tests
focused workflow coverage
main Nucleus task runner coverage
```

Validation:

```
sh scripts/test-nucleus-task-report-only-execution-refinement.sh
sh scripts/test-nucleus-task-execution.sh
```

Boundary: no-effect report metadata only. No task execution, runtime behavior, command execution, Lat execution, LIR execution, file I/O, network I/O, state mutation, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/OPENBSD_ECOSYSTEM_INTEGRATION_STATUS.md

Source title: OpenBSD Ecosystem Integration Status

Lines: 152 | Words: 472 | SHA-256: 83234ced4cc492768b086194ae98732ddfa6a0f5840a5dfbfe601599499e2fec

OpenBSD Ecosystem Integration Status

Status: OpenBSD integration status record Date: 2026-05-26

Summary

Latticra now has an OpenBSD-facing local ports metadata draft for the no-effect CLI payload.

This is an ecosystem integration checkpoint, not a production readiness claim.

Current Evidence

```
openbsd_port_draft_present=1
openbsd_port_static_validation_present=1
debian_freebsd_openbsd_source_archive_contract_present=1
debian_freebsd_openbsd_package_input_handoff_lane_present=1
debian_freebsd_openbsd_package_build_gate_contract_present=1
debian_freebsd_openbsd_package_build_environment_contract_present=1
debian_freebsd_openbsd_package_artifact_naming_contract_present=1
debian_freebsd_openbsd_package_payload_inspection_contract_present=1
debian_freebsd_openbsd_package_install_remove_transcript_contract_present=1
debian_freebsd_openbsd_package_publication_non_claim_review_contract_present=1
debian_freebsd_openbsd_package_validation_promotion_blocker_matrix_contract_present=1
temporary_openbsd_distfile_staged=1
package_build_gate_state=closed-no-effect
package_build_environment_contract_state=specified-no-effect
artifact_naming_contract_state=specified-no-effect
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
publication_non_claim_review_contract_state=specified-no-effect
validation_promotion_blocker_matrix_state=blocked-no-effect
openbsd_build_allowed=0
openbsd_ports_environment_documented=1
openbsd_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
environment_transcript_present=0
openbsd_artifact_name_pattern_recorded=1
openbsd_package_name=latticra-0.0.0.tgz
openbsd_package_artifact_created=0
package_artifact_output_directory_created=0
repository_package_artifact_write_allowed=0
openbsd_payload_expected_bin=bin/latticra
openbsd_payload_expected_doc=share/doc/latticra/README.md
openbsd_payload_inspection_run=0
package_payload_inspection_run=0
package_payload_accepted=0
openbsd_install_remove_transcript_required=1
openbsd_install_remove_transcript_present=0
openbsd_package_install_run=0
openbsd_package_remove_run=0
remove_on_host_run=0
publication_non_claim_review_present=1
openbsd_publication_non_claim_review_present=1
openbsd_package_publication_claimed=0
openbsd_pkg_repo_created=0
openbsd_pkg_repo_publish_run=0
platform_build_evidence_accepted=0
openbsd_validation_promotion_blocked=1
openbsd_platform_build_evidence_accepted=0
package_validation_result_promoted=0
openbsd_validation_result_promoted=0
source_archive_policy_recorded=1
source_archive_created=0
source_archive_sha256_recorded=0
openbsd_distinfo_created=0
openbsd_make_makesum_run=0
openbsd_make_plist_run=0
openbsd_make_package_run=0
openbsd_ports_tree_submission_claimed=0
openbsd_ports_review_thread_claimed=0
openbsd_maintainer_acceptance_claimed=0
make_package_run=0
make_plist_run=0
bulk_build_run=0
openbsd_bulk_build_run=0
portcheck_run=0
package_artifact_created=0
package_artifact_sha256_recorded=0
install_on_host_run=0
permit_package_enabled=0
openbsd_official_port_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Guarded Files

```
docs/OPENBSD_PORT_STATIC_VALIDATION.md
docs/DEBIAN_FREEBSD_OPENBSD_SOURCE_ARCHIVE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INPUT_HANDOFF_LANE.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md
packaging/openbsd/README.md
packaging/openbsd/Makefile
packaging/openbsd/pkg/DESCR
packaging/openbsd/pkg/PLIST
scripts/test-openbsd-port-static-validation.sh
scripts/test-debian-freebsd-openbsd-source-archive-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-gate-contract.sh
scripts/test-debian-freebsd-openbsd-package-build-environment-contract.sh
scripts/test-debian-freebsd-openbsd-package-artifact-naming-contract.sh
scripts/test-debian-freebsd-openbsd-package-payload-inspection-contract.sh
scripts/test-debian-freebsd-openbsd-package-install-remove-transcript-contract.sh
scripts/test-debian-freebsd-openbsd-package-publication-non-claim-review-contract.sh
scripts/test-debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.sh
.github/workflows/openbsd-port-static-validation.yml
.github/workflows/debian-freebsd-openbsd-source-archive-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-gate-contract.yml
.github/workflows/debian-freebsd-openbsd-package-build-environment-contract.yml
.github/workflows/debian-freebsd-openbsd-package-artifact-naming-contract.yml
.github/workflows/debian-freebsd-openbsd-package-payload-inspection-contract.yml
.github/workflows/debian-freebsd-openbsd-package-install-remove-transcript-contract.yml
.github/workflows/debian-freebsd-openbsd-package-publication-non-claim-review-contract.yml
.github/workflows/debian-freebsd-openbsd-package-validation-promotion-blocker-matrix-contract.ym
l
```

Current Boundary

The OpenBSD lane does not publish a package, submit to the OpenBSD ports tree, claim ports@ review evidence, claim maintainer acceptance, claim bulk build success, claim portcheck success, enable package redistribution, install an rc.d script, change the kernel, add a privileged helper, grant network authority, or claim production readiness.

The local ports metadata now records the AGPL-3.0-or-later plus CC-BY-4.0 local payload license expression while PERMIT_PACKAGE=No remains closed until source archive, checksum, redistribution, and notice obligations are reviewed.

The source archive contract records the expected latticra-0.0.0.tar.gz distfile and distinfo boundary while keeping archive creation, checksum acceptance, distinfo, PERMIT_PACKAGE=Yes, package artifacts, and build transcript promotion blocked.

The package-build gate is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_GATE_CONTRACT.md. It keeps package_build_gate_state=closed-no-effect, openbsd_build_allowed=0, openbsd_make_makesum_run=0, openbsd_make_plist_run=0, openbsd_make_package_run=0, portcheck_run=0, and openbsd_bulk_build_run=0 until source, checksum, license, redistribution, notice, ports environment, operator authorization, payload inspection, install/remove transcript, and publication non-claim evidence exists.

The package-build environment contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_BUILD_ENVIRONMENT_CONTRACT.md. It documents the OpenBSD disposable ports environment requirement while keeping openbsd_build_environment_provisioned=0, explicit_operator_build_authorization=0, environment_transcript_present=0, PERMIT_PACKAGE=No, and all OpenBSD package build commands disabled.

The package artifact naming contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_ARTIFACT_NAMING_CONTRACT.md. It records the future latticra-0.0.0.tgz artifact name while keeping openbsd_package_artifact_created=0, repository_package_artifact_write_allowed=0, package_artifact_sha256_recorded=0, and PERMIT_PACKAGE=No.

The package payload inspection contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PAYLOAD_INSPECTION_CONTRACT.md. It records the expected OpenBSD payload paths while keeping openbsd_payload_inspection_run=0, package_payload_accepted=0, package_artifact_created=0, and PERMIT_PACKAGE=No.

The package install/remove transcript contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md. It records future disposable install/remove transcript requirements while keeping openbsd_package_install_run=0, openbsd_package_remove_run=0, install_on_host_run=0, and PERMIT_PACKAGE=No.

The package publication non-claim review contract is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md. It records local-only OpenBSD publication and official-port non-claims while keeping openbsd_package_publication_claimed=0, openbsd_pkg_repo_publish_run=0, package validation promotion blocked, and PERMIT_PACKAGE=No.

The package validation promotion blocker matrix is recorded in docs/DEBIAN_FREEBSD_OPENBSD_PACKAGE_VALIDATION_PROMOTION_BLOCKER_MATRIX_CONTRACT.md. It ties OpenBSD source, environment, artifact, payload, install/remove, and publication non-claim blockers together while keeping openbsd_platform_build_evidence_accepted=0, openbsd_validation_result_promoted=0, and PERMIT_PACKAGE=No.

Next Recommended Lane

```
Add an OpenBSD package build-evidence intake denial contract before any OpenBSD build lane can
open.
```

docs/status/OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md

Source title: openSUSE Ecosystem Integration Status

Lines: 238 | Words: 780 | SHA-256: c701db4b79650a82c5805409a5564970caeb66c245a82997499bf62982ade7d9

openSUSE Ecosystem Integration Status

Status: openSUSE integration and maintenance status record Date: 2026-05-26

Summary

Latticra now has an openSUSE-facing compatibility and maintenance lane for development, Panel prerequisites, and a local-only RPM packaging draft.

This track has the same purpose as the Fedora and Ubuntu tracks, except its scope is openSUSE integration and maintenance.

This is an ecosystem integration checkpoint, not a production readiness claim.

Current Evidence

```
opensuse_developer_workflow_present=1
opensuse_panel_prerequisites_documented=1
opensuse_quickstart_documented=1
opensuse_local_rpm_draft_present=1
opensuse_local_rpm_static_validation_present=1
opensuse_changes_file_present=1
opensuse_maintenance_lane_present=1
opensuse_rpmlint_osc_availability_lane_present=1
opensuse_rpmlint_static_spec_lane_present=1
opensuse_rpmlint_findings_classification_present=1
opensuse_source_archive_reproducibility_contract_present=1
opensuse_source_archive_fixture_lane_present=1
opensuse_rpm_topdir_handoff_lane_present=1
opensuse_local_rpm_build_gate_contract_present=1
opensuse_local_rpm_build_environment_contract_present=1
opensuse_rpm_artifact_naming_contract_present=1
opensuse_rpm_payload_inspection_contract_present=1
opensuse_rpm_install_remove_transcript_contract_present=1
opensuse_obs_publication_non_claim_review_contract_present=1
temporary_rpm_topdir_handoff_lane_present=1
opensuse_rpm_build_gate_state=closed-no-effect
opensuse_rpm_build_environment_contract_state=specified-no-effect
opensuse_rpm_artifact_naming_contract_state=specified-no-effect
opensuse_rpm_payload_inspection_contract_state=specified-no-effect
opensuse_rpm_install_remove_transcript_contract_state=specified-no-effect
rpm_artifact_naming_contract_present=1
rpm_install_remove_transcript_contract_present=1
obs_publication_non_claim_review_contract_present=1
publication_non_claim_review_contract_present=1
obs_publication_non_claim_review_present=1
publication_non_claim_review_present=1
payload_inspection_contract_present=1
payload_inspection_contract_state=specified-no-effect
install_remove_transcript_contract_state=specified-no-effect
opensuse_clean_build_environment_documented=1
opensuse_target_distribution_documented=1
osc_build_environment_documented=1
opensuse_build_environment_provisioned=0
osc_build_environment_provisioned=0
explicit_operator_build_authorization=0
disposable_validation_environment_required=1
disposable_validation_environment_provisioned=0
environment_transcript_present=0
rpmbuild_allowed=0
osc_build_allowed=0
spec_cleaner_allowed=0
rpm_artifact_created=0
rpm_installed_on_host=0
rpmbuild_run=0
rpmbuild_ba_run=0
rpmbuild_bb_run=0
rpmbuild_bs_run=0
osc_build_run=0
accepted_rpmlint_transcript_present=0
expected_draft_findings_count_recorded=0
unexpected_findings_count_recorded=0
classification_decision=blocked-pending-reviewed-rpmlint-output
source_archive_policy_recorded=1
source_archive_transcript_present=1
source_archive_created=1
source_archive_sha256_recorded=1
source_archive_reproducible=1
source_archive_generated_twice=1
source_archive_repeated_sha256_match=1
source_archive_accepted_for_build=0
temporary_rpm_topdir_created=1
temporary_rpm_sources_archive_staged=1
temporary_rpm_specs_spec_staged=1
temporary_rpm_specs_changes_staged=1
temporary_rpm_source_sha256_preserved=1
temporary_rpm_source_listing_preserved=1
rpm_source_artifact_name_pattern_recorded=1
rpm_binary_artifact_name_pattern_recorded=1
rpm_source_package_name=latticra-0.0.0-0.local.src.rpm
rpm_binary_package_name_pattern=latticra-0.0.0-0.local.${RPM_ARCH}.rpm
repository_rpm_artifact_write_allowed=0
source_rpm_artifact_created=0
binary_rpm_artifact_created=0
```

```
rpm_artifact_sha256_recorded=0
rpm_artifact_published=0
rpm_payload_expected_bin=/usr/bin/latticra
rpm_payload_expected_doc=/usr/share/doc/packages/latticra/README.md
source_rpm_payload_inspection_run=0
binary_rpm_payload_inspection_run=0
rpm_payload_inspection_run=0
rpm_payload_accepted=0
rpm_install_remove_transcript_present=0
rpm_install_remove_disposable_environment_required=1
rpm_package_install_run=0
rpm_package_remove_run=0
rpm_zypper_install_run=0
rpm_zypper_remove_run=0
rpm_cli_install_run=0
rpm_cli_remove_run=0
rpm_removed_from_host=0
host_install_allowed=0
host_remove_allowed=0
host_mutation_allowed=0
service_state_change_allowed=0
rpm_validation_result_promoted=0
obs_project_created=0
obs_package_created=0
obs_repository_created=0
obs_source_link_created=0
osc_branch_run=0
osc_commit_run=0
osc_submitreq_run=0
osc_request_accepted=0
obs_build_result_claimed=0
spec_cleaner_run=0
rpmlint_package_readiness_claimed=0
opensuse_obs_publication_claimed=0
opensuse_submit_request_claimed=0
opensuse_official_package_claimed=0
suse_endorsement_claimed=0
opensuse_factory_submission_claimed=0
opensuse_factory_acceptance_claimed=0
opensuse_leap_submission_claimed=0
opensuse_distribution_ready=0
production_installer_ready=0
daily_driver_install_ready=0
root_installer_ready=0
```

Guarded Files

```
docs/OPENSUSE_READINESS_PLAN.md
docs/OPENSUSE_DEVELOPER_WORKFLOW.md
docs/OPENSUSE_LOCAL_RPM_STATIC_VALIDATION.md
docs/OPENSUSE_RPMLINT_OSC_AVAILABILITY.md
docs/OPENSUSE_RPMLINT_STATIC_SPEC_LANE.md
docs/OPENSUSE_RPMLINT_FINDINGS_CLASSIFICATION.md
docs/OPENSUSE_SOURCE_ARCHIVE_REPRODUCIBILITY_CONTRACT.md
docs/OPENSUSE_SOURCE_ARCHIVE_FIXTURE_LANE.md
docs/OPENSUSE_RPM_TOPDIR_HANDOFF_LANE.md
docs/OPENSUSE_LOCAL_RPM_BUILD_GATE_CONTRACT.md
docs/OPENSUSE_LOCAL_RPM_BUILD_ENVIRONMENT_CONTRACT.md
docs/OPENSUSE_RPM_ARTIFACT_NAMING_CONTRACT.md
docs/OPENSUSE_RPM_PAYLOAD_INSPECTION_CONTRACT.md
docs/OPENSUSE_RPM_INSTALL_REMOVE_TRANSCRIPT_CONTRACT.md
docs/OPENSUSE_OBS_PUBLICATION_NON_CLAIM_REVIEW_CONTRACT.md
docs/status/OPENSUSE_ECOSYSTEM_INTEGRATION_STATUS.md
packaging/opensuse/README.md
packaging/opensuse/latticra.spec
packaging/opensuse/latticra.changes
scripts/test-opensuse-developer-workflow.sh
scripts/test-opensuse-local-rpm-static-validation.sh
scripts/test-opensuse-rpmlint-osc-availability.sh
scripts/test-opensuse-rpmlint-static-spec-lane.sh
scripts/test-opensuse-rpmlint-findings-classification.sh
scripts/test-opensuse-source-archive-reproducibility-contract.sh
scripts/test-opensuse-source-archive-fixture-lane.sh
scripts/test-opensuse-rpm-topdir-handoff-lane.sh
scripts/test-opensuse-local-rpm-build-gate-contract.sh
scripts/test-opensuse-local-rpm-build-environment-contract.sh
scripts/test-opensuse-rpm-artifact-naming-contract.sh
scripts/test-opensuse-rpm-payload-inspection-contract.sh
scripts/test-opensuse-rpm-install-remove-transcript-contract.sh
scripts/test-opensuse-obs-publication-non-claim-review-contract.sh
.github/workflows/opensuse-developer-workflow.yml
.github/workflows/opensuse-local-rpm-static-validation.yml
.github/workflows/opensuse-rpmlint-osc-availability.yml
.github/workflows/opensuse-rpmlint-static-spec-lane.yml
.github/workflows/opensuse-rpmlint-findings-classification.yml
.github/workflows/opensuse-source-archive-reproducibility-contract.yml
.github/workflows/opensuse-source-archive-fixture-lane.yml
.github/workflows/opensuse-rpm-topdir-handoff-lane.yml
.github/workflows/opensuse-local-rpm-build-gate-contract.yml
.github/workflows/opensuse-local-rpm-build-environment-contract.yml
.github/workflows/opensuse-rpm-artifact-naming-contract.yml
.github/workflows/opensuse-rpm-payload-inspection-contract.yml
.github/workflows/opensuse-rpm-install-remove-transcript-contract.yml
.github/workflows/opensuse-obs-publication-non-claim-review-contract.yml
```

Public Entry Points

openSUSE setup and package posture are linked from:

```
README.md
docs/QUICK_START_CHEATSHEET.md
installer/README.md
docs/status/README.md
```

Current Boundary

The openSUSE lane does not publish an RPM, create an Open Build Service project, submit Latticra to openSUSE, claim official package readiness, claim SUSE endorsement, install a root service, change systemd, change the kernel, add a privileged helper, grant network authority, or claim production readiness.

The local openSUSE RPM draft records package shape and maintenance posture only. The .changes file is a local maintenance record, not accepted Open Build Service submit-request history.

The openSUSE rpmlint and osc availability lane verifies tooling in an openSUSE environment only. It does not run rpmlint against the Latticra spec yet, run osc build, create package artifacts, publish to Open Build Service, or install Latticra on a host.

The openSUSE rpmlint static spec lane runs rpmlint against the local-only openSUSE spec for audit output only. It does not require a clean lint result, create package artifacts, run osc build, publish to Open Build Service, install Latticra, or claim package readiness.

The openSUSE rpmlint findings classification record separates expected local-only draft findings from unexpected blockers. It does not accept a transcript, clear findings, create artifacts, or promote package readiness.

The openSUSE source archive reproducibility contract records the expected Source0 archive shape and evidence required before any package build transcript can be accepted.

The openSUSE source archive fixture lane creates two temporary archive fixtures, compares their SHA-256 values, and inspects the archive listing. It does not create persistent source RPM or binary RPM artifacts, run rpmbuild, run osc build, publish to Open Build Service, or claim package readiness.

The openSUSE RPM topdir handoff lane stages the verified temporary source archive and local-only spec files into a disposable RPM topdir. It does not run rpmbuild, run osc build, create source RPM or binary RPM artifacts, publish to Open Build Service, or claim package readiness.

The openSUSE local RPM build gate contract keeps rpmbuild, osc build, spec-cleaner, package artifact creation, host installation, and Open Build Service publication blocked until source acceptance, lint classification, license, notice, BuildRequires, environment, authorization, payload inspection, install/remove, and OBS non-claim evidence exists.

The openSUSE local RPM build environment contract documents disposable openSUSE build environment requirements while keeping environment provisioning, operator authorization, rpmbuild, osc build, spec-cleaner, package artifacts, host installation, and Open Build Service publication disabled.

The openSUSE RPM artifact naming contract records future source RPM and binary RPM names and disposable output boundaries while keeping repository RPM artifact writes, checksum claims, artifact creation, host installation, and Open Build Service publication disabled.

The openSUSE RPM payload inspection contract records future source RPM and binary RPM payload inspection evidence while keeping inspection runs, payload acceptance, RPM artifacts, host installation, and Open Build Service publication disabled.

The openSUSE RPM install/remove transcript contract records future disposable install/remove evidence while keeping install/remove runs, host mutation, RPM artifacts, package readiness, and Open Build Service publication disabled.

The openSUSE OBS publication non-claim review contract records Open Build Service, submit-request, official-package, and SUSE endorsement non-claims while keeping OBS actions, package publication, validation promotion, and readiness disabled.

Next Recommended Lane

Add openSUSE RPM validation promotion blocker matrix before any package validation result can be accepted.

docs/status/PROJECT_NOTES_FOLLOWUP_STATUS_INDEX_CHECK.md

Source title: Project Notes Follow-up Status/Index Check

Lines: 32 | Words: 91 | SHA-256: 99400e8c8a815df88f941fd288f6f54b45ef53ba9967e02124f39f7c11248a7b

Project Notes Follow-up Status/Index Check

Status: status/index check

This record verifies that the project-notes follow-up alignment is reflected across the public status surfaces after the project-notes refresh.

Reviewed files:

```
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/README.md
docs/project_notes/README.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

Finding:

```
The project-notes files were refreshed, but the detailed current status page and status index
still needed a final follow-up record.
```

Resolution:

```
Add this status/index check record.
Add this record to the status index.
Advance root and detailed status queues.
```

Boundary: documentation/status check only. No implementation behavior is changed.

docs/status/PROJECT_NOTES_NUCLEUS_ANNOUNCEMENT_README_ALIGNMENT_STATUS_CHECK.md

Source title: Project Notes Nucleus Announcement README Alignment Status Check

Lines: 33 | Words: 166 | SHA-256: 4cd69be5d619c2210fcfa17b09dcbf348b15db5c0fc5679e33f2ea9c13dec193

Project Notes Nucleus Announcement README Alignment Status Check

Status: status/index check

This record verifies that the project-note body alignment after the Nucleus report-only announcement README alignment is represented across public status and index surfaces.

Reviewed files:

```
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/README.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

Finding:

```
The project-note body files were aligned after the Nucleus report-only announcement README
alignment, but root status, detailed current status, the status index, and the foundation index
still needed a final status/index check record.
```

Resolution:

```
Add this status/index check record.
Add this record to the status index and foundation index.
Advance root and detailed status ledgers without changing completion estimates.
Keep the next review lane conditional on capability posture changes.
```

Boundary: documentation/status/index check only. No implementation behavior is changed. No public announcement entry is added, and no task execution, runtime behavior, command execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/PROJECT_NOTES_NUCLEUS_REPORT_ONLY_ALIGNMENT_STATUS_INDEX_CHECK.md

Source title: Project Notes Nucleus Report-Only Alignment Status/Index Check

Lines: 34 | Words: 159 | SHA-256: 89bdaa9da75b76da610bd4272f40bd60f4261de1048d9bc6577928cf11e53569

Project Notes Nucleus Report-Only Alignment Status/Index Check

Status: status/index check

This record verifies that the project-notes alignment after the Nucleus task report-only execution README/status update is represented across public status and index surfaces.

Reviewed files:

```
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/README.md
docs/FOUNDATION_INDEX.md
docs/project_notes/README.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

Finding:

```
The project-notes files were aligned after the Nucleus task report-only execution README/status
update, but root status, detailed current status, the status index, and the foundation index
still needed a final status/index check record.
```

Resolution:

```
Add this status/index check record.
Add this record to the status index and foundation index.
Advance root and detailed status ledgers without changing completion estimates.
Keep the next review lane conditional on capability posture changes.
```

Boundary: documentation/status/index check only. No implementation behavior is changed. No task execution, runtime behavior, command execution, state mutation, file I/O, network I/O, server interaction, recovery behavior, hardware behavior, boot behavior, sandboxing, malware prevention, ransomware prevention, or operating-system completeness is added.

docs/status/PUBLIC_ENTRY_POINT_CONSISTENCY_SCAN.md

Source title: Public Entry-Point Consistency Scan

Lines: 35 | Words: 104 | SHA-256: 3d058eb2763229da8477465b8f9a2b85b25a1d3b2ec931d1c7edfbf2d91c3351

Public Entry-Point Consistency Scan

Status: public entry-point scan

This record reviews the main public entry points after the recent authority review, status/index alignment, foundation-index alignment, and announcement review slices.

Reviewed files:

```
README.md
STATUS.md
docs/FOUNDATION_INDEX.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/README.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
```

Finding:

```
README.md still pointed at older status records and did not list the newest authority review,
completion review, status announcement review, or public entry-point scan records.
```

Resolution:

```
Add the missing public status/review records to README.md.
Add this scan to the status index.
Advance the root status queue.
```

Boundary: documentation consistency scan only. No implementation behavior is changed.

docs/status/RBDM_REPORT_INTEGRATION_STATUS.md

Source title: RBDM Report Integration Status

Lines: 50 | Words: 107 | SHA-256: 170c84c780ea7bda658e1cfc61ae5d5630daf480527d2048a65b5a68afc70b82

RBDM Report Integration Status

Status: merged report integration surface

This record tracks the Runtime Boundary Domain Matrix report integration slice.

Merged PR:

```
#115 Add RBDM report integration
```

Merged commit:

```
ee1ea215e64bbb1c7b10010ce6e4e282ae044687
```

Primary files:

```
src/runtime_boundary_domain_matrix_report.c
tests/runtime_boundary_domain_matrix_report_integration.c
scripts/test-runtime-boundary-domain-matrix-report-integration.sh
.github/workflows/runtime-boundary-domain-matrix-report-integration.yml
docs/RUNTIME_BOUNDARY_DOMAIN_MATRIX_REPORT_INTEGRATION.md
```

What it adds:

```
RBDM report rendering
cell label output
domain label output
known/operational/declarative/future-gated flags
effect-allowed flag
authority-available flag
evidence-level output
small-buffer and null-argument handling
```

Validation:

```
sh scripts/test-runtime-boundary-domain-matrix-report-integration.sh
sh scripts/test-runtime-boundary-domain-matrix-refinement.sh
sh scripts/test-runtime-boundary.sh
```

Boundary: report metadata only. No runtime behavior, command execution, Lat execution, LIR execution, state mutation, file I/O, network I/O, server interaction, hardware behavior, or operating-system completeness is added.

docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md

Source title: Latticra Seal Agentic Automation Security Index Alignment

Lines: 103 | Words: 385 | SHA-256: bc245e52630a8d55d1d0ce0194a19e9b4f0a168ea2d4b6cb3a88961272763f8a

Latticra Seal Agentic Automation Security Index Alignment

Status: index alignment record Source: PR #268 Scope: documentation/status index alignment after the report-only Latticra Seal agentic automation security status record. This record does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This record keeps the Seal agentic automation security checkpoint visible as a status/index alignment item after the merged report-only metadata and status-record slices.

The purpose is to preserve the evidence chain:

```
contract -&gt; MCP alignment plan -&gt; report-only metadata implementation -&gt; status record
-&gt; index
alignment
```

Reviewed files

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md
include/latticra/seal_agentic_automation_security.h
src/seal_agentic_automation_security.c
tests/seal_agentic_automation_security_invariants.c
scripts/test-latticra-seal-agentic-automation-security-contract.sh
scripts/test-latticra-seal-mcp-alignment-plan.sh
scripts/test-latticra-seal-agentic-automation-security.sh
scripts/test-latticra-seal-agentic-automation-security-status.sh
scripts/test-latticra-seal-agentic-automation-security-public-entrypoint-alignment.sh
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md
```

Alignment checkpoint

Current alignment posture:

```
seal_agentic_contract_indexed=1
seal_mcp_alignment_plan_indexed=1
seal_agentic_metadata_implementation_indexed=1
seal_agentic_status_record_indexed=1
seal_agentic_status_guard_indexed=1
seal_agentic_index_alignment_record_present=1
seal_agentic_public_entrypoint_alignment_present=1
mode=doc-status-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
```

Current status summary

The Seal agentic automation security layer is now represented as:

```
planning contract present
MCP alignment plan present
report-only C metadata present
deterministic invariant test present
status record present
status guard present
```

```
index alignment record present
```

The layer remains report-only and no-effect.

Validation

This index alignment is covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security-index-alignment.sh
```

The underlying Seal agentic automation checkpoint remains covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security-status.sh
sh scripts/test-latticra-seal-agentic-automation-security-contract.sh
sh scripts/test-latticra-seal-mcp-alignment-plan.sh
sh scripts/test-latticra-seal-agentic-automation-security.sh
```

Boundary

This record is documentation/status alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, receipt verification, capability enforcement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md

Source title: Latticra Seal Agentic Automation Security Public Entrypoint Alignment

Lines: 140 | Words: 473 | SHA-256: 221f0cb65b17f7d4c530e3356185b7016ca6ee656778e77d6432fd8df54193cb

Latticra Seal Agentic Automation Security Public Entrypoint Alignment

Status: public-entrypoint alignment record for the Latticra Seal agentic automation security checkpoint Source: local follow-up slice Scope: README, root status, status index, foundation index, and project-notes alignment after the completed report-only Latticra Seal agentic automation security metadata, status, index-alignment, and report-surface slices. This record does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, or shell execution.

Purpose

This record makes the completed Seal agentic automation security checkpoint visible from public entry points.

The purpose is to keep public project readers aligned with the current Seal state without claiming enforcement, production readiness, MCP implementation, AI-agent execution control, model execution, tool execution, shell execution, or operational authority.

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_INDEX_ALIGNMENT.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md
include/latticra/seal_agentic_automation_security.h
src/seal_agentic_automation_security.c
tests/seal_agentic_automation_security_invariants.c
tests/seal_agentic_automation_security_report_surface.c
scripts/test-latticra-seal-agentic-automation-security-contract.sh
scripts/test-latticra-seal-mcp-alignment-plan.sh
scripts/test-latticra-seal-agentic-automation-security.sh
scripts/test-latticra-seal-agentic-automation-security-status.sh
scripts/test-latticra-seal-agentic-automation-security-index-alignment.sh
scripts/latticra-seal-agentic-automation-security-report.sh
scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
scripts/test-latticra-seal-agentic-automation-security-report-surface-status.sh
scripts/test-latticra-seal-agentic-automation-security-public-entrypoint-alignment.sh
```

Alignment checkpoint

Current public-entrypoint posture:

```
seal_agentic_automation_security_public_entrypoint_alignment_present=1
readme_mentions_agentic_automation_security_metadata=1
readme_mentions_agentic_automation_security_status=1
readme_mentions_agentic_automation_security_report_surface=1
readme_links_agentic_automation_security_contract=1
readme_links_mcp_alignment_plan=1
readme_links_agentic_automation_security_implementation=1
readme_links_agentic_automation_security_status=1
readme_links_agentic_automation_security_index_alignment=1
readme_links_agentic_automation_security_report_surface=1
readme_links_agentic_automation_security_report_surface_status=1
readme_links_agentic_automation_security_public_entrypoint_alignment=1
root_status_mentions_agentic_public_entrypoint=1
status_index_links_agentic_public_entrypoint=1
foundation_index_links_agentic_public_entrypoint=1
project_notes_point_to_parameter_schema_status=1
mode=public-entrypoint-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal summary

The public entry points now present Seal agentic automation security as:

```
agentic automation security contract present
MCP alignment plan present
```

```
report-only C metadata present
deterministic invariant test present
status record present
status/index alignment present
operator-visible deterministic report surface present
report surface status present
public-entrypoint alignment present
runtime authority remains denied
```

The layer remains report-only, metadata-only, and no-effect.

Validation

This public-entrypoint alignment is covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security-public-entrypoint-alignment.sh
```

The underlying checkpoint remains covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security-contract.sh
sh scripts/test-latticra-seal-mcp-alignment-plan.sh
sh scripts/test-latticra-seal-agentic-automation-security.sh
sh scripts/test-latticra-seal-agentic-automation-security-status.sh
sh scripts/test-latticra-seal-agentic-automation-security-index-alignment.sh
sh scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
sh scripts/test-latticra-seal-agentic-automation-security-report-surface-status.sh
```

Expected output:

```
seal agentic automation security public entrypoint alignment: ok
```

Boundary

This record is documentation/public-entrypoint alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE_STATUS.md

Source title: Latticra Seal Agentic Automation Security Report Surface Status

Lines: 87 | Words: 344 | SHA-256: 3f2a7f12b933df1c12ee4d8e9e9df76502ddc44446086fff30d2141028956f96

Latticra Seal Agentic Automation Security Report Surface Status

Status: status record for the Seal agentic automation security report surface Source: PR #270 Scope: status alignment after the deterministic local report surface for Latticra Seal agentic automation security metadata. This record does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This status record makes the Seal agentic automation report surface visible as a current project checkpoint.

The report surface provides a deterministic local fixture and shell runner for rendering the current report-only Seal agentic automation metadata.

Reviewed files

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md
tests/seal_agentic_automation_security_report_surface.c
scripts/latticra-seal-agentic-automation-security-report.sh
scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
scripts/test-latticra-seal-agentic-automation-security-public-entrypoint-alignment.sh
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md
```

Current checkpoint

Current report-surface posture:

```
seal_agentic_report_surface_document_present=1
seal_agentic_report_surface_fixture_present=1
seal_agentic_report_surface_runner_present=1
seal_agentic_report_surface_guard_present=1
renders_agentic_report=1
uses_local_deterministic_fixture=1
operator_visible_report_surface=1
seal_agentic_public_entrypoint_alignment_present=1
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
```

Expected report posture

The report surface renders the current report-only agentic metadata posture, including:

```
agentic_profile=latticra-seal-agentic-automation-security/0.1
automation_context=local-report-only
mcp_alignment_declared=1
mode=report-only
decision=report-only
status=agentic-automation-security-metadata
```

Validation

The report surface is covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security-report-surface.sh
```

The underlying metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security.sh
```

Expected output:

```
seal agentic automation security report surface: ok
```

Boundary

This status record is documentation/status alignment only.

The report surface compiles and runs a local deterministic fixture only. It does not add protocol behavior, runtime behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is policy decision status/public-entry alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md

Source title: Latticra Seal Agentic Automation Security Status

Lines: 156 | Words: 469 | SHA-256: 4d4fdeb9f2bd218726e6085464385b26b2e01536c1610d1528314500ab225a1d

Latticra Seal Agentic Automation Security Status

Status: status record for report-only Latticra Seal agentic automation security metadata Source: local follow-up slice Scope: status alignment after the initial report-only Seal agentic automation security metadata implementation. This record does not implement runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This status record makes the report-only Latticra Seal agentic automation security implementation visible as a current project checkpoint.

It records that the implementation is metadata-only, deterministic, bounded, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
include/latticra/seal_agentic_automation_security.h
src/seal_agentic_automation_security.c
tests/seal_agentic_automation_security_invariants.c
scripts/test-latticra-seal-agentic-automation-security-contract.sh
scripts/test-latticra-seal-mcp-alignment-plan.sh
scripts/test-latticra-seal-agentic-automation-security.sh
scripts/test-latticra-seal-agentic-automation-security-status.sh
.github/workflows/latticra-seal-agentic-automation-security-status.yml
scripts/test-latticra-seal-agentic-automation-security-public-entrypoint-alignment.sh
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md
docs/LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md
docs/LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md
docs/status/SEAL_STATUS_ROLLUP_STATUS.md
scripts/test-latticra-seal-status-rollup-status.sh
.github/workflows/latticra-seal-status-rollup-status.yml
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md
scripts/test-latticra-seal-parameter-schema-status.sh
```

Current checkpoint

The current Seal agentic automation security layer provides a deterministic report-only metadata surface after Seal status rollup metadata.

Current implemented posture:

```
seal_agentic_automation_security_contract_present=1
seal_mcp_alignment_plan_present=1
seal_agentic_automation_security_implementation_present=1
seal_agentic_automation_security_header_present=1
seal_agentic_automation_security_source_present=1
seal_agentic_automation_security_invariant_test_present=1
seal_agentic_automation_security_runner_present=1
seal_agentic_automation_security_status_present=1
seal_agentic_automation_security_status_runner_present=1
seal_agentic_automation_security_status_workflow_present=1
seal_status_rollup_contract_present=1
seal_status_rollup_implementation_present=1
seal_status_rollup_status_present=1
seal_status_rollup_status_runner_present=1
seal_status_rollup_status_workflow_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
seal_parameter_schema_contract_present=1
seal_parameter_schema_implementation_present=1
seal_parameter_schema_status_present=1
mcp_alignment_declared=1
mcp_protocol_implemented=0
mcp_server_implemented=0
mcp_client_implemented=0
agent_execution_supported=0
model_execution_supported=0
tool_execution_supported=0
shell_execution_supported=0
cryptographic_verification_supported=0
capability_enforcement_supported=0
runtime_authority_requested=0
runtime_authority_granted=0
unknown_tool_allowed=0
unsigned_manifest_allowed=0
network_access_allowed=0
private_key_access_allowed=0
system_mutation_allowed=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
status=agentic-automation-security-metadata
agentic_automation_security_status_added=1
runtime_execution_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
```

```
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
policy_persistence_added=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation is covered by:

```
sh scripts/test-latticra-seal-agentic-automation-security-contract.sh
sh scripts/test-latticra-seal-mcp-alignment-plan.sh
sh scripts/test-latticra-seal-agentic-automation-security.sh
sh scripts/test-latticra-seal-agentic-automation-security-status.sh
sh scripts/test-latticra-seal-agentic-automation-security-public-entrypoint-alignment.sh
sh scripts/test-latticra-seal-status-rollup-status.sh
```

Expected output:

```
seal agentic automation security contract: ok
seal mcp alignment plan: ok
seal agentic automation security invariants: ok
seal agentic automation security status: ok
seal agentic automation security public entrypoint alignment: ok
seal status rollup status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the agentic automation security status guard workflow and records the guarded status rollup status predecessor without changing the report-only agentic automation security implementation.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, receipt verification, capability enforcement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is parameter schema status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_CAPABILITY_GATE_STATUS.md

Source title: Latticra Seal Capability Gate Status

Lines: 191 | Words: 550 | SHA-256: 3826371c585cd15d32137b545bbf48c7fe603c2d78868a5b614eb7016b2ef11c

Latticra Seal Capability Gate Status

Status: status record for Latticra Seal capability gate metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal capability gate implementation. This record does not add capability enforcement, runtime authority, effect execution, cryptographic verification, verified receipt authority, signing, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal capability gate metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, denied-by-default, unverified-receipt-aware, authority-unusable, effect-not-performed, runtime-authority-denied, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/status/SEAL_CAPABILITY_GATE_STATUS.md
include/latticra/seal_capability_gate.h
src/seal_capability_gate.c
tests/seal_capability_gate_invariants.c
scripts/test-latticra-seal-capability-gate-contract.sh
scripts/test-latticra-seal-capability-gate.sh
scripts/test-latticra-seal-capability-gate-status.sh
.github/workflows/latticra-seal-capability-gate-status.yml
docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
docs/status/SEAL_EFFECT_DECISION_STATUS.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
include/latticra/seal_effect_decision.h
src/seal_effect_decision.c
tests/seal_effect_decision_invariants.c
scripts/test-latticra-seal-effect-decision-contract.sh
scripts/test-latticra-seal-effect-decision.sh
scripts/test-latticra-seal-effect-decision-status.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md
include/latticra/seal_verification_receipt.h
src/seal_verification_receipt.c
tests/seal_verification_receipt_invariants.c
scripts/test-latticra-seal-verification-receipt-contract.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-verification-receipt-status.sh
.github/workflows/latticra-seal-verification-receipt-status.yml
```

Current checkpoint

Current capability gate metadata posture:

```
seal_capability_gate_contract_present=1
seal_capability_gate_implementation_present=1
seal_capability_gate_header_present=1
seal_capability_gate_source_present=1
seal_capability_gate_invariant_test_present=1
seal_capability_gate_runner_present=1
seal_capability_gate_metadata_present=1
seal_capability_gate_status_present=1
seal_capability_gate_status_runner_present=1
seal_capability_gate_status_workflow_present=1
seal_effect_decision_contract_present=1
```

```
seal_effect_decision_implementation_present=1
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
seal_verification_receipt_contract_present=1
seal_verification_receipt_implementation_present=1
seal_verification_receipt_status_present=1
seal_verification_receipt_status_runner_present=1
seal_verification_receipt_status_workflow_present=1
verification_receipt_predecessor_verification_policy_status_present=1
capability_gate_predecessor_verification_receipt_status_present=1
gate_profile=latticra-seal-capability-gate/0.1
receipt_profile=latticra-seal-verification-receipt/0.1
verification_policy_profile=latticra-seal-verification-policy/0.1
artifact_digest_algorithm=SHA-256
signer_identity_label=latticra-dev-signer
public_key_identity_label=latticra-dev-public-key
receipt_state=unverified-metadata
verification_state=unsupported
requested_capability=seal.inspect
requested_effect=read-metadata
requested_scope=local-artifact
requested_capability_gate=metadata-only
capability_gate_ready=1
verified=0
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=0
gate_state=denied-unverified
capability_enforcement_performed=0
effect_allowed=0
effect_performed=0
runtime_authority_granted=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
network_lookup_allowed=0
revocation_lookup_allowed=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
handoff_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=capability-gate-denied-metadata
error=ok
capability_gate_status_added=1
capability_enforcement_added=0
effect_execution_added=0
runtime_authority_added=0
runtime_handoff_execution_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-capability-gate-contract.sh
```

```
sh scripts/test-latticra-seal-capability-gate.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-contract.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate contract: ok
seal capability gate invariants: ok
seal capability gate status: ok
seal effect decision contract: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal verification receipt status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the capability gate status guard workflow and records the guarded verification receipt status predecessor without changing the metadata-only denied capability gate implementation.

It does not add capability enforcement, runtime authority, effect execution, cryptographic verification, verified receipt authority, signing, public-key byte verification, public-key trust-store behavior, key material loading, private-key handling, key generation, hardware-key use, revocation lookup, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is effect decision status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signer invocation behavior, host behavior, network behavior, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_CAPABILITY_METADATA_REPORT_SURFACE_STATUS.md

Source title: Latticra Seal Capability Metadata Report Surface Status

Lines: 116 | Words: 332 | SHA-256: 5a6f82c31e7224d4ffe6c9a0d745265616336144a952d9108fad16535a876031

Latticra Seal Capability Metadata Report Surface Status

Status: status record for the Latticra Seal capability metadata report surface Scope: status alignment after the deterministic local report surface for Latticra Seal capability metadata. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This status record makes the Seal capability metadata report surface visible as a current project checkpoint.

The report surface renders a deterministic local fixture capability candidate while keeping execution, host access, network use, effects, and runtime authority denied.

Reviewed files

```
docs/LATTICRA_SEAL_CAPABILITY_METADATA_REPORT_SURFACE.md
scripts/latticra-seal-capability-metadata-report.sh
scripts/test-latticra-seal-capability-metadata-report-surface.sh
tests/seal_capability_metadata_report_surface.c
```

Current checkpoint

Current report-surface posture:

```
seal_capability_metadata_report_surface_document_present=1
seal_capability_metadata_report_surface_fixture_present=1
seal_capability_metadata_report_runner_present=1
seal_capability_metadata_report_surface_guard_present=1
operator_visible_capability_metadata_report=1
uses_local_deterministic_fixture=1
known_fixture_capability_candidate_visible=1
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
runtime_authority_granted=0
```

Expected report posture

The report surface renders the current report-only capability metadata posture, including:

```
capability_metadata_profile=latticra-seal-capability-metadata/0.1
capability_name=seal.capability.report
capability_scope=evidence-boundary
capability_effect_class=none
capability_fixture_source=deterministic-local-fixture
capability_fixture_entry_count=3
capability_lookup_performed=1
capability_name_present=1
capability_known=1
capability_unknown=0
capability_candidate=1
capability_requires_guarded_allowlist=1
capability_requires_policy_decision=1
capability_requires_runtime_gate=1
capability_requires_runtime_dry_run=1
capability_requires_operator_review=1
capability_grants_authority=0
capability_executes_tool=0
capability_reads_host=0
capability_writes_host=0
capability_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
```

```
unknown_capability_denied=1
missing_capability_denied=0
invalid_capability_denied=0
blocked_reason=known-capability-candidate-still-denied
report_only=1
mode=report-only
status=capability-metadata
```

Validation

This report surface is covered by:

```
sh scripts/test-latticra-seal-capability-metadata-report-surface.sh
```

This status record is covered by:

```
sh scripts/test-latticra-seal-capability-metadata-report-surface-status.sh
```

The underlying capability metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-capability-metadata.sh
```

Expected output:

```
latticra seal capability metadata report surface status: ok
```

Boundary

This status record is documentation/status alignment only.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is capability metadata status-index alignment or a future product-spine alignment record.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_CORE_BLOCKED_CASES_STATUS.md

Source title: Latticra Seal Core Blocked Cases Status

Lines: 107 | Words: 286 | SHA-256: bd5a0bab42e7487bafe24ab03504bedded17cf678d6814d9205081c57f35514a

Latticra Seal Core Blocked Cases Status

Status: status record for the completed core blocked-request case set Scope: status alignment after the unknown-tool, unsigned-request, stale-request, and replayed-request case fixtures. This record does not implement runtime behavior, enforcement, execution, effects, host operations, network operations, cryptographic verification, or authority grants.

Purpose

This status record marks the first completed core negative-test evidence set for the Latticra Seal runtime gate metadata path.

The completed set verifies that the report-only runtime gate metadata path keeps the gate blocked and grants no authority for four core request classes.

Completed case fixtures

```
unknown_tool_case_fixture_present=1
unsigned_request_case_fixture_present=1
stale_request_case_fixture_present=1
replayed_request_case_fixture_present=1
core_blocked_case_set_present=1
core_blocked_case_set_complete=1
```

Reviewed files

```
docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_CONTRACT.md
docs/LATTICRA_SEAL_UNKNOWN_TOOL_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_CONTRACT.md
docs/LATTICRA_SEAL_UNSIGNED_REQUEST_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_STALE_REQUEST_CASE_CONTRACT.md
docs/LATTICRA_SEAL_STALE_REQUEST_CASE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REPLAYED_REQUEST_CASE_CONTRACT.md
docs/LATTICRA_SEAL_REPLAYED_REQUEST_CASE_IMPLEMENTATION.md
tests/seal_unknown_tool_case.c
tests/seal_unsigned_request_case.c
tests/seal_stale_request_case.c
tests/seal_replayed_request_case.c
scripts/test-latticra-seal-unknown-tool-case.sh
scripts/test-latticra-seal-unsigned-request-case.sh
scripts/test-latticra-seal-stale-request-case.sh
scripts/test-latticra-seal-replayed-request-case.sh
```

Validation

Run:

```
sh scripts/test-latticra-seal-unknown-tool-case.sh
sh scripts/test-latticra-seal-unsigned-request-case.sh
sh scripts/test-latticra-seal-stale-request-case.sh
sh scripts/test-latticra-seal-replayed-request-case.sh
```

Expected outputs:

```
seal unknown tool case: ok
seal unsigned request case: ok
seal stale request case: ok
seal replayed request case: ok
```

Evidence posture

```
core_case_unknown_tool_validated=1
core_case_unsigned_request_validated=1
core_case_stale_request_validated=1
core_case_replayed_request_validated=1
runtime_gate_report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Public claim boundary

This status record is an important milestone, but it does not yet justify saying that Latticra broadly secures AI agents.

A careful public claim may now say:

```
Latticra Seal now has a report-only runtime gate path with core negative-test evidence for
AI-era tool-boundary planning.
```

The stronger claim still requires guarded runtime enforcement behavior and an operator-visible evidence report.

Non-claims

```
ai_agent_security_claimed=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
tool_execution_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Next valid slice

The next valid slice is public milestone wording for the report-only core case evidence set, or an operator-visible evidence report surface that summarizes the full Seal chain.

docs/status/SEAL_CORE_EVIDENCE_INDEX_ALIGNMENT.md

Source title: Latticra Seal Core Evidence Index Alignment

Lines: 105 | Words: 394 | SHA-256: 2824e9b8d537ae64361747130a0bd60492f5710ecbcaa19a90bace5858202a05

Latticra Seal Core Evidence Index Alignment

Status: index alignment record for the Latticra Seal core evidence status surface Source: PR #296 Scope: documentation/status index alignment after the Latticra Seal core evidence status surface. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, host reads, host writes, network behavior, policy enforcement, capability enforcement, cryptographic verification, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, operating-system behavior, Fedora approval claims, external endorsement claims, production readiness, or authority grants.

Purpose

This record keeps the completed Seal core evidence status surface visible from the status index after the merged report-only evidence and status-surface slices.

The purpose is to preserve the evidence chain:

```
Seal metadata chain -&gt; core blocked-request case evidence -&gt; core evidence report -&gt;
public
status update -&gt; status surface -&gt; status index alignment
```

Reviewed files

```
docs/LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md
docs/status/SEAL_CORE_EVIDENCE_PUBLIC_STATUS_UPDATE.md
docs/status/SEAL_CORE_EVIDENCE_STATUS.md
docs/status/README.md
scripts/latticra-seal-core-evidence-report.sh
scripts/test-latticra-seal-core-evidence-report.sh
scripts/test-latticra-seal-core-evidence-public-status.sh
scripts/test-latticra-seal-core-evidence-status.sh
```

Alignment checkpoint

Current alignment posture:

```
seal_core_evidence_report_indexed=1
seal_core_evidence_public_status_indexed=1
seal_core_evidence_status_surface_indexed=1
seal_core_evidence_status_guard_indexed=1
seal_core_evidence_index_alignment_record_present=1
status_index_mentions_seal_core_evidence=1
mode=doc-status-index-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
```

```
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current status summary

The Seal core evidence lane is now represented as:

```
report-only metadata chain present
core blocked-request case evidence complete
operator-visible evidence report present
public status update present
status surface present
status guard present
status index alignment record present
```

The lane remains report-only and no-effect.

Validation

This index alignment is covered by:

```
sh scripts/test-latticra-seal-core-evidence-index-alignment.sh
```

The underlying Seal core evidence checkpoint remains covered by:

```
sh scripts/test-latticra-seal-core-evidence-report.sh
sh scripts/test-latticra-seal-core-evidence-public-status.sh
sh scripts/test-latticra-seal-core-evidence-status.sh
```

Expected output:

```
latticra seal core evidence index alignment: ok
```

Boundary

This record is documentation/status index alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is a public entry-point refresh for the completed Seal core evidence milestone or a guarded policy-decision status surface.

That future slice must preserve the report-only posture until a specific contract, deterministic local fixture, no-effect dry-run report, guarded allowlist posture, and validation path justify any later runtime work.

docs/status/SEAL_CORE_EVIDENCE_PUBLIC_ENTRYPOINT_ALIGNMENT.md

Source title: Latticra Seal Core Evidence Public Entrypoint Alignment

Lines: 110 | Words: 410 | SHA-256: 42941e10bc734df14cde001de2cfbb5f4c73a47958609463eeeb6cc3a0276eeb

Latticra Seal Core Evidence Public Entrypoint Alignment

Status: public entrypoint alignment record for the Latticra Seal core evidence milestone Source: PR #297 Scope: public entrypoint alignment after the Latticra Seal core evidence status surface and status-index alignment. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, host reads, host writes, network behavior, policy enforcement, capability enforcement, cryptographic verification, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, operating-system behavior, Fedora approval claims, external endorsement claims, production readiness, or authority grants.

Purpose

This record makes the completed Latticra Seal core evidence milestone visible from the root public entrypoints after the status surface and status index alignment slices.

The purpose is to preserve the public evidence chain:

```
core evidence report -&gt; public status update -&gt; status surface -&gt; status index
alignment -&gt; root
 public entrypoint alignment
```

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/SEAL_CORE_EVIDENCE_STATUS.md
docs/status/SEAL_CORE_EVIDENCE_INDEX_ALIGNMENT.md
```

Alignment checkpoint

Current public entrypoint posture:

```
readme_mentions_seal_core_evidence_status=1
readme_mentions_seal_core_evidence_index_alignment=1
root_status_mentions_seal_core_evidence=1
status_index_mentions_seal_core_evidence=1
seal_core_public_entrypoint_alignment_record_present=1
foundation_index_entrypoint_followup_pending=1
status_index_public_entrypoint_followup_pending=1
mode=doc-public-entrypoint-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current public summary

The root public entrypoints now make the Seal core evidence milestone discoverable as:

```
report-only runtime gate path present
core negative-test evidence present
unknown-tool case represented
unsigned-request case represented
stale-request case represented
replayed-request case represented
runtime authority still denied
no host or network effect reported
no production security claim made
```

The current careful public claim remains:

```
Latticra Seal now has a report-only runtime gate path with core negative-test evidence for
AI-era tool-boundary planning.
```

Validation

This public entrypoint alignment is covered by:

```
sh scripts/test-latticra-seal-core-evidence-public-entrypoint-alignment.sh
```

The underlying Seal core evidence checkpoint remains covered by:

```
sh scripts/test-latticra-seal-core-evidence-report.sh
sh scripts/test-latticra-seal-core-evidence-public-status.sh
sh scripts/test-latticra-seal-core-evidence-status.sh
sh scripts/test-latticra-seal-core-evidence-index-alignment.sh
```

Expected output:

```
latticra seal core evidence public entrypoint alignment: ok
```

Boundary

This record is documentation/public-entrypoint alignment only.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is a foundation/status-index public-entrypoint follow-up, a guarded policy-decision status surface, or a no-effect dry-run planning contract for future runtime behavior.

That future slice must preserve the report-only posture until a specific contract, deterministic local fixture, no-effect dry-run report, guarded allowlist posture, and validation path justify any later runtime work.

docs/status/SEAL_CORE_EVIDENCE_PUBLIC_STATUS_UPDATE.md

Source title: Latticra Seal Core Evidence Public Status Update

Lines: 95 | Words: 414 | SHA-256: faf78891ebc6d2bfe4786b0b0ae48e0a2a3f2cc385601efe3105845fc4e2663c

Latticra Seal Core Evidence Public Status Update

Status: public status update draft for the Latticra Seal core evidence milestone Scope: public-facing status language after the completed report-only Seal runtime gate metadata path and the completed core case evidence set. This document does not implement runtime behavior, execution, effects, host operations, network operations, policy enforcement, or authority grants.

Public milestone

Latticra Seal has reached an important report-only milestone.

The project now has an operator-visible evidence path for a report-only runtime gate and a completed core case evidence set covering:

This is an operator-visible evidence report surface, not runtime enforcement.

```
unknown tool case
unsigned request case
stale request case
replayed request case
```

This means Latticra Seal can now show, in deterministic local reports, how its planned AI-era tool-boundary model handles several high-risk request classes without granting runtime authority or performing effects.

What this means

This milestone supports the careful public claim:

```
Latticra Seal now has a report-only runtime gate path with core negative-test evidence for
AI-era tool-boundary planning.
```

It is also accurate to say:

```
Latticra Seal is building an evidence-bound trust boundary for AI-era automation.
```

What this does not mean yet

This milestone does not mean Latticra broadly secures AI agents in production.

The project is still in report-only, evidence-building mode.

Current non-claims:

```
ai_agent_security_claimed=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
tool_execution_implemented=0
host_behavior_implemented=0
network_behavior_implemented=0
authority_granted=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Evidence summary

The current evidence report records:

```
seal_agentic_automation_metadata_present=1
seal_parameter_schema_metadata_present=1
seal_request_freshness_metadata_present=1
seal_signed_request_metadata_present=1
seal_policy_decision_metadata_present=1
seal_runtime_gate_metadata_present=1
unknown_tool_case_validated=1
unsigned_request_case_validated=1
stale_request_case_validated=1
replayed_request_case_validated=1
core_blocked_case_set_complete=1
runtime_gate_report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
```

Suggested public wording

```
Latticra Seal has reached a new report-only security architecture milestone: a runtime gate
evidence path for AI-era tool-boundary planning.

The current Seal chain now includes metadata and reports for agentic automation, parameter
schema posture, request freshness, signed-request posture, policy decision posture, and runtime
gate posture.

Most importantly, the first core case evidence set is now represented: unknown tools, unsigned
requests, stale requests, and replayed requests all remain blocked in the report-only runtime
gate path, with no runtime authority, host operation, network operation, or effect reported.

This is not a production security claim yet. It is an evidence-bound architecture milestone:
Latticra Seal is building the trust boundary that future AI-agent tool execution must pass
through before any authority can be considered.
```

Next work

The next valid milestone is an operator-facing evidence report surface integrated into the broader project status path, followed by continued gate work toward guarded runtime behavior.

docs/status/SEAL_CORE_EVIDENCE_STATUS.md

Source title: Latticra Seal Core Evidence Status

Lines: 125 | Words: 450 | SHA-256: 7538a95cc8eb3207690743066326648e0f39b7b63679c1243b739abb997e9d88

Latticra Seal Core Evidence Status

Status: status record for the Latticra Seal core evidence report surface Source: PR #294 Scope: status integration after the completed report-only Seal runtime gate metadata path and the completed core blocked-request case evidence set. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, host reads, host writes, network behavior, policy enforcement, capability enforcement, cryptographic verification, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, operating-system behavior, Fedora approval claims, external endorsement claims, production readiness, or authority grants.

Purpose

This status record makes the completed Latticra Seal core evidence report visible as an operator-facing project checkpoint.

It connects the report-only runtime gate evidence path, the completed blocked-request case evidence set, and the public claim boundary into one status surface.

The goal is evidence-bound visibility before any future move toward guarded runtime behavior.

Reviewed files

```
docs/LATTICRA_SEAL_CORE_EVIDENCE_REPORT.md
docs/status/SEAL_CORE_EVIDENCE_PUBLIC_STATUS_UPDATE.md
scripts/latticra-seal-core-evidence-report.sh
scripts/test-latticra-seal-core-evidence-report.sh
scripts/test-latticra-seal-core-evidence-public-status.sh
```

Current evidence chain

The current Seal evidence report records the following report-only metadata chain:

```
seal_agentic_automation_metadata_present=1
seal_parameter_schema_metadata_present=1
seal_request_freshness_metadata_present=1
seal_signed_request_metadata_present=1
seal_policy_decision_metadata_present=1
seal_runtime_gate_metadata_present=1
```

Core blocked-request case set

The completed core case evidence set covers:

```
unknown_tool_case_validated=1
unsigned_request_case_validated=1
stale_request_case_validated=1
replayed_request_case_validated=1
core_blocked_case_set_complete=1
```

These cases are represented as report-only blocked-request evidence for AI-era tool-boundary planning.

Operator-facing status surface

Current status-surface posture:

```
seal_core_evidence_report_present=1
seal_core_evidence_public_status_present=1
seal_core_evidence_status_surface_present=1
operator_visible_status_surface=1
deterministic_local_report_path=1
core_blocked_case_set_complete=1
```

```
mcp_alignment_context=ai-era-tool-boundary-planning
runtime_gate_report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
tool_execution_implemented=0
ai_agent_security_claimed=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Public claim boundary

The careful current public claim remains:

```
Latticra Seal now has a report-only runtime gate path with core negative-test evidence for
AI-era tool-boundary planning.
```

It is also accurate to say:

```
Latticra Seal is building an evidence-bound trust boundary for AI-era automation.
```

It is not yet accurate to claim production AI-agent security, runtime enforcement, tool execution control, host protection, network protection, Fedora approval, external endorsement, or production readiness.

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-core-evidence-status.sh
```

The underlying report remains covered by:

```
sh scripts/test-latticra-seal-core-evidence-report.sh
sh scripts/test-latticra-seal-core-evidence-public-status.sh
```

Expected output:

```
latticra seal core evidence status: ok
```

Boundary

This status record is documentation/status alignment only.

It does not change runtime behavior, add enforcement, execute tools, perform effects, read from the host, write to the host, perform network operations, grant authority, or claim production security capability.

Current next valid slice

The next valid Latticra Seal slice is a guarded policy-decision status surface or status-index alignment that preserves the report-only posture.

A future runtime-enforcement slice must not begin until the project has a specific contract, deterministic local fixture, no-effect dry-run report, guarded allowlist posture, and validation path showing exactly what authority would be denied or allowed and why.

docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md

Source title: Latticra Seal Crypto Graduation Gate Status

Lines: 91 | Words: 207 | SHA-256: efd9bc3d88ca99896d530817148fbbb630541a2c1cec0897e8d59b1bf8fd0048

Latticra Seal Crypto Graduation Gate Status

Status: status record for the Latticra Seal crypto graduation gate Date: 2026-05-26

Scope

This status record tracks the authority-neutral crypto graduation gate over the existing local Ed25519 verify-only result and verified receipt promotion metadata.

It does not add signing, key generation, key storage, private-key handling, trust-store loading, revocation lookup, network lookup, FIPS validation, CMVP submission, production cryptography claims, capability enforcement, effect execution, host behavior, or runtime authority.

Files

```
include/latticra/seal_crypto_graduation_gate.h
src/seal_crypto_graduation_gate.c
tests/seal_crypto_graduation_gate_invariants.c
scripts/test-latticra-seal-crypto-graduation-gate.sh
docs/LATTICRA_SEAL_CRYPTO_GRADUATION_GATE_IMPLEMENTATION.md
docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md
```

Current Fields

```
seal_crypto_graduation_gate_present=1
seal_crypto_graduation_gate_header_present=1
seal_crypto_graduation_gate_source_present=1
seal_crypto_graduation_gate_invariant_test_present=1
seal_crypto_graduation_gate_runner_present=1
seal_crypto_graduation_gate_status_present=1
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
signature_algorithm=Ed25519-development
message_digest_algorithm=SHA-256
public_key_size_bytes=32
signature_size_bytes=64
verify_result_present=1
receipt_present=1
provider_backed_verification_required=1
deterministic_test_vector_required=1
negative_test_vector_required=1
rfc8032_test_vector_tracked=1
fips_186_5_signature_standard_tracked=1
fips_180_4_digest_standard_tracked=1
fips_140_3_claim_gate_required=1
sp_800_57_key_management_required=1
sp_800_131a_transition_review_required=1
fips_204_ml_dsa_planning_tracked=1
fips_205_slh_dsa_planning_tracked=1
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
invalid=0
local_verify_graduated=1
receipt_promotion_graduated=1
standard_expectations_met=1
production_crypto_claim_allowed=0
fips_claim_allowed=0
signing_authority_granted=0
key_generation_allowed=0
key_storage_allowed=0
revocation_lookup_allowed=0
network_lookup_allowed=0
authority_usable=0
authority_promotion_allowed=0
capability_gate_allowed=0
runtime_authority_granted=0
gate_state=graduated-authority-neutral
blocked_reason=authority-remains-denied
status=crypto-graduation-gate-passed
```

Validation

```
sh scripts/test-latticra-seal-crypto-graduation-gate.sh
```

Expected output:

```
seal crypto graduation gate invariants: ok
```

```
latticra seal crypto graduation gate: ok
```

Notes

This gate makes the current local cryptographic verification checkpoint stricter and easier to promote later. It is still authority-neutral: a passing gate records standard expectations and local verification evidence, but it does not make the receipt usable for capability enforcement, effect execution, or runtime authority.

docs/status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md

Source title: Latticra Seal Crypto Verify Backend Status

Lines: 120 | Words: 312 | SHA-256: 33f37bd37e5f2dcfc1cd1a213f8fa66613cd3e4e2feb6c13a15b799c75ed4c71

Latticra Seal Crypto Verify Backend Status

Status: status record for the Latticra Seal crypto verify backend metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal crypto verify backend contract and metadata implementation. This record does not implement cryptographic verification, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal crypto verify backend metadata layer visible from public entry points.

The backend layer is important because it records a future cryptographic-verification boundary without treating metadata as authority.

It remains metadata-only, unsupported, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_CONTRACT.md
docs/LATTICRA_SEAL_CRYPTO_VERIFY_BACKEND_IMPLEMENTATION.md
docs/status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md
docs/status/SEAL_VERIFICATION_POLICY_STATUS.md
include/latticra/seal_crypto_verify_backend.h
src/seal_crypto_verify_backend.c
tests/seal_crypto_verify_backend_invariants.c
scripts/test-latticra-seal-crypto-verify-backend.sh
scripts/test-latticra-seal-crypto-verify-backend-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current crypto verify backend posture:

```
seal_crypto_verify_backend_contract_present=1
seal_crypto_verify_backend_implementation_present=1
seal_crypto_verify_backend_header_present=1
seal_crypto_verify_backend_source_present=1
seal_crypto_verify_backend_invariant_test_present=1
seal_crypto_verify_backend_runner_present=1
seal_crypto_verify_backend_status_present=1
seal_verification_policy_status_present=1
readme_links_crypto_verify_backend_contract=1
readme_links_crypto_verify_backend_implementation=1
readme_links_crypto_verify_backend_status=1
```

```
root_status_mentions_crypto_verify_backend_status=1
status_index_links_crypto_verify_backend_status=1
foundation_index_links_crypto_verify_backend_status=1
project_notes_mark_crypto_verify_backend_status_complete=1
backend_profile=latticra-seal-crypto-verify-backend/0.1
verification_policy_profile=latticra-seal-verification-policy/0.1
signature_profile=latticra-seal-signature/0.1
manifest_profile=latticra-seal-manifest/0.1
artifact_digest_algorithm=SHA-256
signature_algorithm=Ed25519-development
trust_source=local-fixture
crypto_verify_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verified=0
invalid=0
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
mode=metadata-only
status=crypto-verify-backend-metadata
error=ok
implementation_behavior_changed=0
real_cryptographic_verification_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
capability_enforcement_added=0
effect_execution_added=0
runtime_authority_granted=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-crypto-verify-backend-status.sh
```

The underlying metadata implementation remains covered by:

```
sh scripts/test-latticra-seal-crypto-verify-backend.sh
```

Expected output:

```
seal crypto verify backend status: ok
seal crypto verify backend invariants: ok
```

Boundary

This status record is documentation/status alignment only.

It does not add real cryptographic verification, signing, key generation, private-key handling, trust-store behavior, revocation lookup, object sealing, capability enforcement, effect execution, runtime behavior, host behavior, network behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this documentation/status-only alignment.

The next valid Latticra Seal slice is Ed25519 verify-only implementation status/public-entry alignment or another narrow status/index alignment follow-up.

docs/status/SEAL_ED25519_VERIFY_STATUS.md

Source title: Latticra Seal Ed25519 Verify Status

Lines: 125 | Words: 332 | SHA-256: a1e4882c9763ab8950ef49336acdaa26145711dc5ace466c1f7e6423f983c8c1

Latticra Seal Ed25519 Verify Status

Status: status record for the Latticra Seal Ed25519 verify-only result surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal Ed25519 verify-only contract and local implementation. This record does not add signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, capability enforcement, runtime authority, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal local Ed25519 verify-only implementation visible from public entry points.

The verify-only layer is important because it is the first local cryptographic operation in the Seal lane, while still refusing to treat a successful verification result as authority by itself.

Reviewed files

```
docs/LATTICRA_SEAL_ED25519_VERIFY_ONLY_CONTRACT.md
docs/LATTICRA_SEAL_ED25519_VERIFY_IMPLEMENTATION.md
docs/status/SEAL_ED25519_VERIFY_STATUS.md
docs/status/SEAL_CRYPTO_VERIFY_BACKEND_STATUS.md
include/latticra/seal_ed25519_verify.h
src/seal_ed25519_verify.c
tests/seal_ed25519_verify_invariants.c
scripts/test-latticra-seal-ed25519-verify-only-contract.sh
scripts/test-latticra-seal-ed25519-verify.sh
scripts/test-latticra-seal-ed25519-verify-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current Ed25519 verify-only posture:

```
seal_ed25519_verify_only_contract_present=1
seal_ed25519_verify_implementation_present=1
seal_ed25519_verify_header_present=1
seal_ed25519_verify_source_present=1
seal_ed25519_verify_invariant_test_present=1
seal_ed25519_verify_runner_present=1
seal_ed25519_verify_status_present=1
seal_crypto_verify_backend_status_present=1
readme_links_ed25519_verify_contract=1
readme_links_ed25519_verify_implementation=1
readme_links_ed25519_verify_status=1
root_status_mentions_ed25519_verify_status=1
status_index_links_ed25519_verify_status=1
foundation_index_links_ed25519_verify_status=1
project_notes_mark_ed25519_verify_status_complete=1
ed25519_verify_profile=latticra-seal-ed25519-verify/0.1
backend_profile=latticra-seal-crypto-verify-backend/0.1
verification_policy_profile=latticra-seal-verification-policy/0.1
message_label=rfc8032-test-vector-2
message_size_bytes=1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
public_key_size_bytes=32
signature_algorithm=Ed25519-development
signature_size_bytes=64
trust_source=local-test-vector
crypto_verify_state=verified
cryptographic_verification_supported=1
cryptographic_verification_performed=1
```

```
verified=1
invalid=0
authority_usable=0
ed25519_authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
mode=verify-only-authority-neutral
status=ed25519-verified
error=ok
local_ed25519_verification_added=1
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
capability_enforcement_added=0
effect_execution_added=0
runtime_authority_granted=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-ed25519-verify-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-ed25519-verify.sh
```

Expected output:

```
seal ed25519 verify status: ok
seal ed25519 verify invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The local Ed25519 verify-only implementation verifies caller-provided message, public-key, and signature bytes through OpenSSL EVP, but a successful result remains authority-neutral.

It does not add signing, key generation, private-key handling, trust-store behavior, revocation lookup, object sealing, capability enforcement, effect execution, runtime behavior, host behavior, network behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The next valid Latticra Seal slice is verified receipt promotion status/public-entry alignment or another narrow status/index alignment follow-up.

docs/status/SEAL_EFFECT_DECISION_STATUS.md

Source title: Latticra Seal Effect Decision Status

Lines: 180 | Words: 539 | SHA-256: 2057fcfee7bb8804df808f93730a0994a9e3c312792741f35d0dc1bce998ef44

Latticra Seal Effect Decision Status

Status: status record for Latticra Seal effect decision metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal effect decision implementation. This record does not add effect execution, capability enforcement, runtime authority, runtime handoff execution, cryptographic verification, verified receipt authority, signing, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal effect decision metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, denied-by-gate, effect-not-allowed, effect-not-performed, host-inactive, network-inactive, runtime-authority-denied, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
docs/status/SEAL_EFFECT_DECISION_STATUS.md
include/latticra/seal_effect_decision.h
src/seal_effect_decision.c
tests/seal_effect_decision_invariants.c
scripts/test-latticra-seal-effect-decision-contract.sh
scripts/test-latticra-seal-effect-decision.sh
scripts/test-latticra-seal-effect-decision-status.sh
.github/workflows/latticra-seal-effect-decision-status.yml
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
include/latticra/seal_runtime_handoff.h
src/seal_runtime_handoff.c
tests/seal_runtime_handoff_invariants.c
scripts/test-latticra-seal-runtime-handoff-contract.sh
scripts/test-latticra-seal-runtime-handoff.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/status/SEAL_CAPABILITY_GATE_STATUS.md
include/latticra/seal_capability_gate.h
src/seal_capability_gate.c
tests/seal_capability_gate_invariants.c
scripts/test-latticra-seal-capability-gate-contract.sh
scripts/test-latticra-seal-capability-gate.sh
scripts/test-latticra-seal-capability-gate-status.sh
.github/workflows/latticra-seal-capability-gate-status.yml
```

Current checkpoint

Current effect decision metadata posture:

```
seal_effect_decision_contract_present=1
seal_effect_decision_implementation_present=1
seal_effect_decision_header_present=1
seal_effect_decision_source_present=1
seal_effect_decision_invariant_test_present=1
seal_effect_decision_runner_present=1
seal_effect_decision_metadata_present=1
seal_effect_decision_status_present=1
seal_effect_decision_status_runner_present=1
seal_effect_decision_status_workflow_present=1
seal_runtime_handoff_contract_present=1
seal_runtime_handoff_implementation_present=1
seal_runtime_handoff_status_present=1
seal_capability_gate_contract_present=1
seal_capability_gate_implementation_present=1
```

```
seal_capability_gate_status_present=1
seal_capability_gate_status_runner_present=1
seal_capability_gate_status_workflow_present=1
effect_decision_predecessor_capability_gate_status_present=1
decision_profile=latticra-seal-effect-decision/0.1
gate_profile=latticra-seal-capability-gate/0.1
receipt_profile=latticra-seal-verification-receipt/0.1
artifact_digest_algorithm=SHA-256
requested_capability=seal.inspect
requested_effect=read-metadata
requested_scope=local-artifact
requested_effect_decision=metadata-only
effect_decision_ready=1
gate_state=denied-unverified
decision_state=denied-gate
gate_allowed=0
effect_allowed=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_authority_granted=0
capability_enforcement_performed=0
runtime_handoff_performed=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
network_lookup_allowed=0
revocation_lookup_allowed=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
mode=metadata-only
status=effect-decision-denied-metadata
error=ok
effect_decision_status_added=1
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
runtime_handoff_execution_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-effect-decision-contract.sh
sh scripts/test-latticra-seal-effect-decision.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-contract.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
```

```
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision contract: ok
seal effect decision invariants: ok
seal effect decision status: ok
seal runtime handoff contract: ok
seal runtime handoff status: ok
seal capability gate status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the effect decision status guard workflow and records the guarded capability gate status predecessor without changing the metadata-only denied effect decision implementation.

It does not add effect execution, capability enforcement, runtime authority, runtime handoff execution, cryptographic verification, verified receipt authority, signing, public-key byte verification, public-key trust-store behavior, key material loading, private-key handling, key generation, hardware-key use, revocation lookup, signer invocation behavior, signer process execution, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is runtime handoff status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signer invocation behavior, host behavior, network behavior, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_GUARDED_ALLOWLIST_PUBLIC_ENTRYPOINT_ALIGNMENT.md

Source title: Latticra Seal Guarded Allowlist Public Entrypoint Alignment

Lines: 110 | Words: 394 | SHA-256: 95be6798d16e0e39cbbcdebc0a1b419fd5939ac45b6c02f723292c12e855cde4

Latticra Seal Guarded Allowlist Public Entrypoint Alignment

Status: public-entrypoint alignment record for the Latticra Seal guarded allowlist milestone Source: PR #315 Scope: README/public-entrypoint alignment after the completed Latticra Seal guarded allowlist status-index alignment. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This record makes the completed Seal guarded allowlist lane visible from the root README public entrypoint.

The purpose is to keep public project readers aligned with the current Seal state without claiming enforcement, production readiness, MCP implementation, AI-agent security, or operational authority.

Reviewed files

```
README.md
docs/status/SEAL_GUARDED_ALLOWLIST_STATUS_INDEX_ALIGNMENT.md
docs/status/SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE_STATUS.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_CONTRACT.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE.md
scripts/test-latticra-seal-guarded-allowlist-status-index-alignment.sh
```

Alignment checkpoint

Current public-entrypoint posture:

```
seal_guarded_allowlist_public_entrypoint_alignment_present=1
readme_mentions_guarded_allowlist_metadata=1
readme_mentions_guarded_allowlist_report_surface=1
readme_links_guarded_allowlist_contract=1
readme_links_guarded_allowlist_implementation_plan=1
readme_links_guarded_allowlist_implementation=1
readme_links_guarded_allowlist_report_surface=1
readme_links_guarded_allowlist_report_surface_status=1
readme_links_guarded_allowlist_status_index_alignment=1
readme_mentions_operator_visible_guarded_allowlist_report=1
readme_mentions_known_fixture_tool_candidate_only=1
readme_mentions_no_runtime_authority=1
mode=public-entrypoint-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal summary

The public README now presents Seal as a report-only guarded allowlist lane:

```
guarded allowlist contract present
guarded allowlist implementation plan present
guarded allowlist implementation present
guarded allowlist report surface present
guarded allowlist report surface status present
guarded allowlist status-index alignment present
operator-visible guarded allowlist report present
known fixture tool candidate-only posture present
runtime authority remains denied
```

The lane remains no-effect and report-only.

Validation

This public-entrypoint alignment is covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist-public-entrypoint-alignment.sh
```

The underlying Seal guarded allowlist checkpoint remains covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist.sh
```

```
sh scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
sh scripts/test-latticra-seal-guarded-allowlist-report-surface-status.sh
sh scripts/test-latticra-seal-guarded-allowlist-status-index-alignment.sh
```

Expected output:

```
latticra seal guarded allowlist public entrypoint alignment: ok
```

Boundary

This record is documentation/public-entrypoint alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, freshness validation, replay detection, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is a capability metadata contract.

That future slice must remain contract-only and must not perform effects, execute tools, touch host state, use the network, or grant runtime authority.

docs/status/SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE_STATUS.md

Source title: Latticra Seal Guarded Allowlist Report Surface Status

Lines: 112 | Words: 319 | SHA-256: 2dc6bf65e7062bfb262ace0a342bbff9e46a042cf426f0e97731a737e83eedfb

Latticra Seal Guarded Allowlist Report Surface Status

Status: status record for the Latticra Seal guarded allowlist report surface Source: PR #312 Scope: status alignment after the deterministic local report surface for Latticra Seal guarded allowlist metadata. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This status record makes the Seal guarded allowlist report surface visible as a current project checkpoint.

The report surface renders the no-effect allowlist candidate denial posture from a deterministic local fixture.

Reviewed files

```
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE.md
scripts/latticra-seal-guarded-allowlist-report.sh
scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
tests/seal_guarded_allowlist_report_surface.c
```

Current checkpoint

Current report-surface posture:

```
seal_guarded_allowlist_report_surface_document_present=1
seal_guarded_allowlist_report_surface_fixture_present=1
seal_guarded_allowlist_report_runner_present=1
seal_guarded_allowlist_report_surface_guard_present=1
```

```
operator_visible_guarded_allowlist_report=1
uses_local_deterministic_fixture=1
known_fixture_tool_candidate_visible=1
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
runtime_authority_granted=0
```

Expected report posture

The report surface renders the current report-only guarded allowlist metadata posture, including:

```
guarded_allowlist_profile=latticra-seal-guarded-allowlist/0.1
tool_name=latticra.seal.report
allowlist_source=deterministic-local-fixture
allowlist_entry_count=3
allowlist_lookup_performed=1
requested_tool_name_present=1
requested_tool_known=1
requested_tool_unknown=0
requested_tool_candidate=1
requested_tool_allow_candidate=1
allow_candidate_requires_policy_decision=1
allow_candidate_requires_runtime_gate=1
allow_candidate_requires_runtime_dry_run=1
allow_candidate_requires_operator_review=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-tool-candidate-still-denied
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Validation

This report surface is covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
```

This status record is covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist-report-surface-status.sh
```

The underlying guarded allowlist implementation remains covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist.sh
```

Expected output:

```
latticra seal guarded allowlist report surface status: ok
```

Boundary

This status record is documentation/status alignment only.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is guarded allowlist status-index alignment.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_GUARDED_ALLOWLIST_STATUS_INDEX_ALIGNMENT.md

Source title: Latticra Seal Guarded Allowlist Status Index Alignment

Lines: 130 | Words: 420 | SHA-256: 7bcd9f511bcb6c3bd309856cd41fbac2ecfff91761580ed4817cb81e0c18ad0a

Latticra Seal Guarded Allowlist Status Index Alignment

Status: index alignment record for the Latticra Seal guarded allowlist report surface status Source: PR #313 Scope: documentation/status index alignment after the Latticra Seal guarded allowlist report surface status record. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This record keeps the completed Seal guarded allowlist report surface status visible as the current Seal status-index alignment checkpoint.

The purpose is to preserve the current evidence chain:

```
guarded allowlist contract -&gt; guarded allowlist implementation plan -&gt; guarded allowlist
implementation -&gt; guarded allowlist report surface -&gt; guarded allowlist report surface
status -&gt;
 status index alignment
```

Reviewed files

```
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_CONTRACT.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE.md
docs/status/SEAL_GUARDED_ALLOWLIST_REPORT_SURFACE_STATUS.md
docs/status/README.md
scripts/test-latticra-seal-guarded-allowlist-contract.sh
scripts/test-latticra-seal-guarded-allowlist-implementation-plan.sh
scripts/test-latticra-seal-guarded-allowlist.sh
scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
scripts/test-latticra-seal-guarded-allowlist-report-surface-status.sh
```

Alignment checkpoint

Current alignment posture:

```
seal_guarded_allowlist_contract_present=1
seal_guarded_allowlist_implementation_plan_present=1
seal_guarded_allowlist_implementation_present=1
seal_guarded_allowlist_report_surface_present=1
seal_guarded_allowlist_report_surface_status_present=1
seal_guarded_allowlist_status_index_alignment_record_present=1
operator_visible_guarded_allowlist_report=1
known_fixture_tool_candidate_visible=1
uses_local_deterministic_fixture=1
mode=doc-status-index-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current status summary

The Seal guarded allowlist lane is now represented as:

```
guarded allowlist contract present
guarded allowlist implementation plan present
guarded allowlist implementation present
guarded allowlist report surface present
guarded allowlist report surface status present
guarded allowlist status index alignment record present
```

The lane remains report-only and no-effect.

The current guarded allowlist report posture is still:

```
requested_tool_known=1
requested_tool_candidate=1
requested_tool_allow_candidate=1
allow_candidate_grants_authority=0
allow_candidate_executes_tool=0
allow_candidate_reads_host=0
allow_candidate_writes_host=0
allow_candidate_uses_network=0
would_allow=0
would_deny=1
would_require_operator_review=1
blocked_reason=known-tool-candidate-still-denied
report_only=1
mode=report-only
status=guarded-allowlist-metadata
```

Validation

This index alignment is covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist-status-index-alignment.sh
```

The underlying Seal guarded allowlist checkpoint remains covered by:

```
sh scripts/test-latticra-seal-guarded-allowlist.sh
sh scripts/test-latticra-seal-guarded-allowlist-report-surface.sh
sh scripts/test-latticra-seal-guarded-allowlist-report-surface-status.sh
```

Expected output:

```
latticra seal guarded allowlist status index alignment: ok
```

Boundary

This record is documentation/status index alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, freshness validation, replay detection, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is a public entry-point refresh for the completed guarded allowlist milestone.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_KEY_HANDLING_STATUS.md

Source title: Latticra Seal Key-Handling Status

Lines: 174 | Words: 477 | SHA-256: af666d88c9ed575880b788b5e4b425a8ceddb2206597b183337bc8b4097473f6

Latticra Seal Key-Handling Status

Status: status record for Latticra Seal key-handling metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal key-handling implementation. This record does not implement public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal key-handling metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, key-material-not-loaded, key-not-parsed, signer-not-invoked, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
docs/status/SEAL_KEY_HANDLING_STATUS.md
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
include/latticra/seal_key_handling.h
src/seal_key_handling.c
tests/seal_key_handling_invariants.c
scripts/test-latticra-seal-key-handling-contract.sh
scripts/test-latticra-seal-key-handling.sh
scripts/test-latticra-seal-key-handling-status.sh
scripts/test-latticra-seal-key-material-contract.sh
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_OPERATION_STATUS.md
include/latticra/seal_signing_operation.h
src/seal_signing_operation.c
tests/seal_signing_operation_invariants.c
scripts/test-latticra-seal-signing-operation-contract.sh
scripts/test-latticra-seal-signing-operation.sh
scripts/test-latticra-seal-signing-operation-status.sh
.github/workflows/latticra-seal-signing-operation-status.yml
.github/workflows/latticra-seal-key-handling-status.yml
```

Current checkpoint

Current key-handling metadata posture:

```
seal_key_handling_contract_present=1
seal_key_handling_implementation_present=1
seal_key_handling_header_present=1
seal_key_handling_source_present=1
seal_key_handling_invariant_test_present=1
seal_key_handling_runner_present=1
seal_key_handling_metadata_present=1
seal_key_handling_status_present=1
seal_key_handling_status_runner_present=1
seal_key_handling_status_workflow_present=1
seal_key_material_contract_present=1
seal_signing_operation_contract_present=1
seal_signing_operation_implementation_present=1
seal_signing_operation_status_present=1
seal_signing_operation_status_runner_present=1
seal_signing_operation_status_workflow_present=1
key_handling_predecessor_signing_operation_status_present=1
readme_links_key_handling_status=1
root_status_mentions_key_handling_status=1
status_index_links_key_handling_status=1
foundation_index_links_key_handling_status=1
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
```

```
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
key_handling_state=key-handling-metadata-only
key_handling_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=key-handling-metadata
key_handling_status_added=1
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-key-handling-contract.sh
sh scripts/test-latticra-seal-key-handling.sh
sh scripts/test-latticra-seal-key-handling-status.sh
sh scripts/test-latticra-seal-key-material-contract.sh
```

The predecessor signing operation implementation remains covered by:

```
sh scripts/test-latticra-seal-signing-operation-contract.sh
sh scripts/test-latticra-seal-signing-operation.sh
sh scripts/test-latticra-seal-signing-operation-status.sh
```

Expected output:

```
seal key-handling contract: ok
seal key-handling invariants: ok
seal key-handling status: ok
seal key-material contract: ok
seal signing operation contract: ok
seal signing operation invariants: ok
seal report envelope status: ok
```

```
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds an explicit status guard workflow and records the signing-operation status predecessor without changing the key-handling implementation.

It does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is key-material status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_KEY_MATERIAL_STATUS.md

Source title: Latticra Seal Key-Material Status

Lines: 180 | Words: 489 | SHA-256: c0ea8c0ac09042f4b0042c65525bd20574eece576cc236c593af0758e7026f32

Latticra Seal Key-Material Status

Status: status record for Latticra Seal key-material metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal key-material implementation. This record does not implement public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal key-material metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, key-material-not-loaded, key-not-parsed, signer-not-invoked, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
docs/status/SEAL_KEY_MATERIAL_STATUS.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-key-material-contract.sh
scripts/test-latticra-seal-key-material.sh
scripts/test-latticra-seal-key-material-status.sh
scripts/test-latticra-seal-public-key-parsing-contract.sh
.github/workflows/latticra-seal-key-material-status.yml
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
docs/status/SEAL_KEY_HANDLING_STATUS.md
include/latticra/seal_key_handling.h
src/seal_key_handling.c
tests/seal_key_handling_invariants.c
scripts/test-latticra-seal-key-handling-contract.sh
scripts/test-latticra-seal-key-handling.sh
scripts/test-latticra-seal-key-handling-status.sh
.github/workflows/latticra-seal-key-handling-status.yml
```

Current checkpoint

Current key-material metadata posture:

```
seal_key_material_contract_present=1
seal_key_material_implementation_present=1
seal_key_material_header_present=1
seal_key_material_source_present=1
seal_key_material_invariant_test_present=1
seal_key_material_runner_present=1
seal_key_material_metadata_present=1
seal_key_material_status_present=1
seal_key_material_status_runner_present=1
seal_key_material_status_workflow_present=1
seal_public_key_parsing_contract_present=1
seal_key_handling_contract_present=1
seal_key_handling_implementation_present=1
seal_key_handling_status_present=1
seal_key_handling_status_runner_present=1
seal_key_handling_status_workflow_present=1
key_material_predecessor_key_handling_status_present=1
readme_links_key_material_status=1
root_status_mentions_key_material_status=1
status_index_links_key_material_status=1
foundation_index_links_key_material_status=1
key_material_profile=latticra-seal-key-material/0.1
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
requested_key_material=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
key_handling_state=key-handling-metadata-only
key_handling_ready=1
key_material_state=key-material-metadata-only
key_material_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
```

```
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=key-material-metadata
key_material_status_added=1
public_key_parsing_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-key-material-contract.sh
sh scripts/test-latticra-seal-key-material.sh
sh scripts/test-latticra-seal-key-material-status.sh
sh scripts/test-latticra-seal-public-key-parsing-contract.sh
```

The predecessor key-handling implementation remains covered by:

```
sh scripts/test-latticra-seal-key-handling-contract.sh
sh scripts/test-latticra-seal-key-handling.sh
sh scripts/test-latticra-seal-key-handling-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material contract: ok
seal key-material invariants: ok
seal key-material status: ok
seal public-key parsing contract: ok
seal key-handling contract: ok
seal key-handling invariants: ok
seal key-handling status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the key-material status guard workflow and records the guarded key-handling status predecessor without changing the key-material implementation.

It does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is public-key parsing status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, signing, verification, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_KEY_PARSING_STATUS.md

Source title: Latticra Seal Key Parsing Status

Lines: 194 | Words: 544 | SHA-256: 43125acbc136e5e140a782b45f0587a64a42239f6c0758176628a1b368665557

Latticra Seal Key Parsing Status

Status: status record for Latticra Seal key parsing metadata Source: local follow-up slice Scope: status and public-entry alignment after the bounded no-effect Seal key parsing implementation. This record does not add key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal key parsing metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, caller-provided-public-key-only, key-material-not-loaded, private-key-denied, signer-not-invoked, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_KEY_PARSING_STATUS.md
include/latticra/seal_key_parsing.h
src/seal_key_parsing.c
tests/seal_key_parsing_invariants.c
scripts/test-latticra-seal-key-parsing.sh
scripts/test-latticra-seal-key-parsing-status.sh
.github/workflows/latticra-seal-key-parsing-status.yml
docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_POLICY_STATUS.md
docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md
docs/status/SEAL_CAPABILITY_GATE_STATUS.md
docs/status/SEAL_EFFECT_DECISION_STATUS.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
scripts/test-latticra-seal-verification-policy-contract.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-policy-status.sh
scripts/test-latticra-seal-verification-receipt-status.sh
scripts/test-latticra-seal-capability-gate-status.sh
scripts/test-latticra-seal-effect-decision-status.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md
include/latticra/seal_public_key_parsing.h
src/seal_public_key_parsing.c
tests/seal_public_key_parsing_invariants.c
scripts/test-latticra-seal-public-key-parsing-contract.sh
scripts/test-latticra-seal-public-key-parsing.sh
scripts/test-latticra-seal-public-key-parsing-status.sh
.github/workflows/latticra-seal-public-key-parsing-status.yml
```

Current checkpoint

Current key parsing metadata posture:

```
seal_key_parsing_implementation_present=1
seal_key_parsing_header_present=1
seal_key_parsing_source_present=1
seal_key_parsing_invariant_test_present=1
seal_key_parsing_runner_present=1
seal_key_parsing_metadata_present=1
seal_key_parsing_status_present=1
seal_key_parsing_status_runner_present=1
seal_key_parsing_status_workflow_present=1
seal_verification_policy_contract_present=1
seal_verification_policy_implementation_present=1
seal_verification_policy_status_present=1
seal_verification_receipt_status_present=1
seal_capability_gate_status_present=1
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
seal_future_key_parsing_implementation_contract_present=1
seal_future_key_parsing_implementation_plan_present=1
seal_public_key_parsing_contract_present=1
seal_public_key_parsing_implementation_present=1
seal_public_key_parsing_status_present=1
seal_public_key_parsing_status_runner_present=1
seal_public_key_parsing_status_workflow_present=1
key_parsing_predecessor_public_key_parsing_status_present=1
readme_links_key_parsing_status=1
root_status_mentions_key_parsing_status=1
status_index_links_key_parsing_status=1
foundation_index_links_key_parsing_status=1
key_parsing_profile=latticra-seal-key-parsing/0.1
public_key_parsing_profile=latticra-seal-public-key-parsing/0.1
key_material_profile=latticra-seal-key-material/0.1
requested_key_parsing=public-key-bytes-only
requested_public_key_parsing=metadata-only
key_parsing_input_format=ed25519-raw-public-key-32
key_parsing_input_length=32
key_parsing_algorithm=Ed25519-development
key_parsing_state=public-key-parsed-metadata-only
key_parsing_ready=1
```

```
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
public_key_parsed=1
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
signature_performed=0
verification_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
blocked_reason=none
mode=metadata-only
status=key-parsing-metadata
error=ok
key_parsing_status_added=1
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

The second accepted caller-provided public-key byte form is:

```
key_parsing_input_format=ed25519-hex-public-key-64
key_parsing_input_length=64
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-key-parsing.sh
sh scripts/test-latticra-seal-key-parsing-status.sh
sh scripts/test-latticra-seal-verification-policy-status.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-future-key-parsing-implementation-plan.sh
sh scripts/test-latticra-seal-public-key-parsing-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing invariants: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal future key parsing implementation plan: ok
seal public-key parsing status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the key parsing status guard workflow and records the guarded public-key parsing status predecessor without changing the bounded key parsing metadata implementation.

It does not add key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is verification policy status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signer invocation behavior, host behavior, network behavior, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md

Source title: Latticra Seal Operator Receipt Report Status

Lines: 178 | Words: 536 | SHA-256: ed5c5523e9c312cf5de7d6e34ed37adac1e31b49221d81c61f1e1b921ecdb1ad

Latticra Seal Operator Receipt Report Status

Status: status record for the Latticra Seal operator receipt report surface Scope: status alignment after the deterministic local report surface for Latticra Seal operator receipt metadata. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This status record makes the Seal operator receipt report surface visible as a current product checkpoint.

The report surface renders one denied local receipt that binds existing report-only Seal metadata without granting authority or performing effects.

Reviewed Files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_OPERATOR_RECEIPT_REPORT_SURFACE.md
docs/status/SEAL_OPERATOR_RECEIPT_REPORT_STATUS.md
docs/status/SEAL_POLICY_DECISION_STATUS.md
scripts/latticra-seal-operator-receipt-report.sh
scripts/test-latticra-seal-operator-receipt-report-surface.sh
scripts/test-latticra-seal-operator-receipt-report-status.sh
scripts/test-latticra-seal-policy-decision-status.sh
.github/workflows/latticra-seal-policy-decision-status.yml
.github/workflows/latticra-seal-operator-receipt-report-status.yml
tests/seal_operator_receipt_report_surface.c
```

Current Checkpoint

Current report-surface posture:

```
seal_operator_receipt_report_surface_document_present=1
seal_operator_receipt_report_surface_fixture_present=1
seal_operator_receipt_report_runner_present=1
seal_operator_receipt_report_surface_guard_present=1
seal_operator_receipt_report_status_present=1
seal_operator_receipt_report_status_runner_present=1
seal_operator_receipt_report_status_workflow_present=1
seal_policy_decision_status_present=1
seal_policy_decision_status_runner_present=1
seal_policy_decision_status_workflow_present=1
policy_decision_predecessor_signed_request_status_present=1
operator_receipt_report_predecessor_policy_decision_status_present=1
operator_visible_operator_receipt_report=1
uses_local_deterministic_fixture=1
source_metadata_bound=1
operator_receipt_report_status_added=1
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
runtime_authority_granted=0
```

Expected Report Posture

The report surface renders the current report-only operator receipt posture, including:

```
operator_receipt_profile=latticra-seal-operator-receipt-report/0.1
receipt_mode=report-only
receipt_status=denied-report-only
source_capability_metadata_present=1
source_policy_decision_present=1
source_request_freshness_present=1
source_signed_request_present=1
source_runtime_dry_run_present=1
source_denial_reason_present=1
capability_name=seal.capability.report
capability_known=1
capability_candidate=1
policy_decision_state=report-only
request_freshness_state=report-only
signed_request_state=report-only
runtime_dry_run_state=report-only
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
```

```
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
blocked_reason=known-capability-candidate-still-denied
receipt_complete=1
receipt_invalid=0
report_only=1
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
```

Validation

This report surface is covered by:

```
sh scripts/test-latticra-seal-operator-receipt-report-surface.sh
```

This status record is covered by:

```
sh scripts/test-latticra-seal-operator-receipt-report-status.sh
```

The underlying operator receipt report implementation remains covered by:

```
sh scripts/test-latticra-seal-operator-receipt-report.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal status rollup status: ok
seal agentic automation security status: ok
seal parameter schema status: ok
seal request freshness status: ok
seal signed request status: ok
latticra seal policy decision status: ok
latticra seal operator receipt report status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the operator receipt report status guard workflow and records the guarded policy decision status predecessor without changing the report-only operator receipt metadata, implementation, or report surface.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current Next Valid Slice

The local capability registry schema contract is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_CONTRACT.md.

The local capability registry schema implementation plan is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION_PLAN.md.

The local capability registry schema implementation is now represented by docs/LATTICRA_SEAL_LOCAL_CAPABILITY_REGISTRY_SCHEMA_IMPLEMENTATION.md.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md

Source title: Latticra Seal Parameter Schema Status

Lines: 209 | Words: 607 | SHA-256: 61c7efa01890eabc96e3a152c19729b6e8f41f477a4b0dc572c718b3d9b7f81e

Latticra Seal Parameter Schema Status

Status: status record for Latticra Seal parameter schema metadata Source: local follow-up slice Scope: status and public-entry alignment after the report-only Latticra Seal parameter schema contract, metadata implementation, and report surface. This record does not implement schema parsing, schema validation, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This status record makes the completed Seal parameter schema checkpoint visible from public entry points.

The purpose is to keep public project readers aligned with the current report-only parameter schema posture without claiming schema parsing, schema validation, enforcement, production readiness, MCP implementation, AI-agent execution control, model execution, tool execution, shell execution, or operational authority.

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_MCP_ALIGNMENT_PLAN.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_REPORT_SURFACE.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_PUBLIC_ENTRYPOINT_ALIGNMENT.md
scripts/test-latticra-seal-agentic-automation-security-status.sh
.github/workflows/latticra-seal-agentic-automation-security-status.yml
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md
docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md
include/latticra/seal_parameter_schema.h
src/seal_parameter_schema.c
tests/seal_parameter_schema_invariants.c
tests/seal_parameter_schema_report_surface.c
scripts/test-latticra-seal-parameter-schema-contract.sh
scripts/test-latticra-seal-parameter-schema.sh
scripts/latticra-seal-parameter-schema-report.sh
scripts/test-latticra-seal-parameter-schema-report-surface.sh
scripts/test-latticra-seal-parameter-schema-status.sh
.github/workflows/latticra-seal-parameter-schema-status.yml
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md
docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md
scripts/test-latticra-seal-request-freshness-contract.sh
scripts/test-latticra-seal-request-freshness.sh
scripts/test-latticra-seal-request-freshness-report-surface.sh
scripts/test-latticra-seal-request-freshness-status.sh
```

Current checkpoint

Current parameter schema status posture:

```
seal_parameter_schema_contract_present=1
seal_parameter_schema_implementation_present=1
seal_parameter_schema_header_present=1
seal_parameter_schema_source_present=1
seal_parameter_schema_invariant_test_present=1
seal_parameter_schema_runner_present=1
seal_parameter_schema_metadata_present=1
seal_parameter_schema_report_surface_present=1
seal_parameter_schema_report_runner_present=1
seal_parameter_schema_report_guard_present=1
seal_parameter_schema_status_present=1
seal_parameter_schema_status_runner_present=1
seal_parameter_schema_status_workflow_present=1
seal_agentic_automation_security_status_present=1
seal_agentic_automation_security_status_runner_present=1
seal_agentic_automation_security_status_workflow_present=1
agentic_automation_security_predecessor_status_rollup_status_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
readme_mentions_parameter_schema_metadata=1
readme_mentions_parameter_schema_report_surface=1
readme_links_parameter_schema_contract=1
readme_links_parameter_schema_implementation=1
readme_links_parameter_schema_report_surface=1
readme_links_parameter_schema_status=1
root_status_mentions_parameter_schema_status=1
status_index_links_parameter_schema_status=1
foundation_index_links_parameter_schema_status=1
project_notes_point_to_policy_decision_status=1
schema_profile=latticra-seal-parameter-schema/0.1
schema_present=0
schema_parsing_supported=0
schema_validation_supported=0
schema_valid=0
max_input_bytes_declared=0
```

```
parameter_count_declared=0
required_parameter_count_declared=0
unknown_parameters_allowed=0
parameter_forwarding_allowed=0
input_size_within_limit=0
parameter_names_reported=0
mode=status-public-entry-alignment
implementation_behavior_changed=0
parameter_validation_implemented=0
schema_parser_implemented=0
schema_validator_implemented=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
parameter_schema_status_added=1
schema_parsing_added=0
schema_validation_added=0
policy_evaluation_added=0
policy_enforcement_added=0
```

Expected report posture

The report surface renders the current report-only parameter schema posture, including:

```
schema_profile=latticra-seal-parameter-schema/0.1
schema_id=unset
schema_version=unset
schema_language=unset
schema_hash=unset
schema_present=0
schema_parsing_supported=0
schema_validation_supported=0
schema_valid=0
max_input_bytes_declared=0
parameter_count_declared=0
required_parameter_count_declared=0
unknown_parameters_allowed=0
parameter_forwarding_allowed=0
input_size_within_limit=0
parameter_names_reported=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=parameter-schema-metadata-only
status=parameter-schema-metadata
```

Validation

This status/public-entry alignment is covered by:

```
sh scripts/test-latticra-seal-parameter-schema-status.sh
```

The underlying checkpoint remains covered by:

```
sh scripts/test-latticra-seal-parameter-schema-contract.sh
sh scripts/test-latticra-seal-parameter-schema.sh
sh scripts/test-latticra-seal-parameter-schema-report-surface.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
```

```
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal status rollup status: ok
seal agentic automation security status: ok
seal parameter schema status: ok
```

Boundary

This status record is documentation/status/public-entry alignment only.

This refresh adds the parameter schema status guard workflow and records the guarded agentic automation security status predecessor without changing the report-only parameter schema metadata, implementation, or report surface.

It does not add schema parsing, schema validation, MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is request freshness status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_POLICY_DECISION_PUBLIC_ENTRYPOINT_ALIGNMENT.md

Source title: Latticra Seal Policy Decision Public-Entrypoint Alignment

Lines: 211 | Words: 518 | SHA-256: 8e24273af9ac0b26233ecc3e881c4c49d5bb08f423a2fae7e7353b14dbe0bef3

Latticra Seal Policy Decision Public-Entrypoint Alignment

Status: public-entrypoint alignment record for the Latticra Seal policy decision checkpoint Source: local follow-up slice Scope: status and public-entry alignment after the report-only Latticra Seal policy decision contract, metadata implementation, status record, report surface, and report surface status. This record does not implement real policy evaluation, policy enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, revocation lookup, network trust lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This public-entrypoint alignment makes the completed Seal policy decision checkpoint visible from README, root status, the status index, the foundation index, and project notes.

The purpose is to keep public project readers aligned with the current report-only default-deny policy decision posture without claiming real policy evaluation, policy enforcement, runtime enforcement, production readiness, MCP implementation, AI-agent execution control, model execution, tool execution, shell execution, or operational authority.

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNED_REQUEST_STATUS.md
docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
docs/status/SEAL_POLICY_DECISION_STATUS.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
docs/status/SEAL_POLICY_DECISION_PUBLIC_ENTRYPOINT_ALIGNMENT.md
include/latticra/seal_policy_decision.h
src/seal_policy_decision.c
tests/seal_policy_decision_invariants.c
tests/seal_policy_decision_report_surface.c
scripts/test-latticra-seal-policy-decision-contract.sh
scripts/test-latticra-seal-policy-decision.sh
scripts/latticra-seal-policy-decision-report.sh
scripts/test-latticra-seal-policy-decision-status.sh
scripts/test-latticra-seal-policy-decision-report-surface.sh
scripts/test-latticra-seal-policy-decision-report-surface-status.sh
scripts/test-latticra-seal-policy-decision-public-entrypoint-alignment.sh
docs/LATTICRA_SEAL_RUNTIME_ENFORCEMENT_GATE_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_GATE_IMPLEMENTATION.md
include/latticra/seal_runtime_gate.h
src/seal_runtime_gate.c
tests/seal_runtime_gate_invariants.c
scripts/test-latticra-seal-runtime-gate-contract.sh
scripts/test-latticra-seal-runtime-gate.sh
docs/DEFENSIVE_THREAT_MODEL_CONTRACT.md
docs/DEFENSIVE_THREAT_MODEL_IMPLEMENTATION_PLAN.md
docs/DEFENSIVE_THREAT_MODEL_VALIDATION.md
scripts/test-defensive-threat-model-contract.sh
scripts/test-defensive-threat-model-implementation-plan.sh
scripts/test-defensive-threat-model-validation.sh
```

Current checkpoint

Current policy decision public-entry posture:

```
seal_policy_decision_contract_present=1
seal_policy_decision_implementation_present=1
seal_policy_decision_header_present=1
seal_policy_decision_source_present=1
seal_policy_decision_invariant_test_present=1
seal_policy_decision_runner_present=1
seal_policy_decision_metadata_present=1
seal_policy_decision_status_present=1
seal_policy_decision_report_surface_present=1
seal_policy_decision_report_runner_present=1
seal_policy_decision_report_guard_present=1
seal_policy_decision_report_surface_status_present=1
seal_policy_decision_public_entrypoint_alignment_present=1
readme_mentions_policy_decision_metadata=1
readme_mentions_policy_decision_report_surface=1
readme_links_policy_decision_contract=1
```

```
readme_links_policy_decision_implementation=1
readme_links_policy_decision_report_surface=1
readme_links_policy_decision_status=1
readme_links_policy_decision_report_surface_status=1
readme_links_policy_decision_public_entrypoint_alignment=1
root_status_mentions_policy_decision_public_entrypoint=1
status_index_links_policy_decision_public_entrypoint=1
foundation_index_links_policy_decision_public_entrypoint=1
project_notes_point_to_defensive_threat_model_validation=1
policy_decision_profile=latticra-seal-policy-decision/0.1
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
mode=public-entrypoint-alignment
implementation_behavior_changed=0
policy_evaluation_implemented=0
policy_enforcement_implemented=0
runtime_enforcement_implemented=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Expected report posture

The report surface renders the current report-only policy decision posture, including:

```
policy_decision_profile=latticra-seal-policy-decision/0.1
policy_id=unset
policy_version=unset
requested_action=unset
requested_tool=unset
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
```

```
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=policy-decision-metadata-only
status=policy-decision-metadata
```

Validation

This public-entry alignment is covered by:

```
sh scripts/test-latticra-seal-policy-decision-public-entrypoint-alignment.sh
```

The underlying checkpoint remains covered by:

```
sh scripts/test-latticra-seal-policy-decision-contract.sh
sh scripts/test-latticra-seal-policy-decision.sh
sh scripts/test-latticra-seal-policy-decision-status.sh
sh scripts/test-latticra-seal-policy-decision-report-surface.sh
sh scripts/test-latticra-seal-policy-decision-report-surface-status.sh
```

Expected output:

```
seal policy decision public entrypoint alignment: ok
```

Boundary

This alignment record is documentation/status/public-entry alignment only.

It does not add real policy evaluation, policy enforcement, runtime enforcement, runtime behavior, effect execution, host behavior, network behavior, MCP behavior, model behavior, AI-agent behavior, tool behavior, shell behavior, cryptographic behavior, signature verification, freshness validation, replay detection, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is defensive threat model validation refinement.

That future slice must preserve the no-effect posture and must not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, or runtime authority.

docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md

Source title: Latticra Seal Policy Decision Report Surface Status

Lines: 108 | Words: 338 | SHA-256: a4aa4ff17d191b8f7e0c289dba31a1c9fc57bc6373eab3b28c9270e802bb9d46

Latticra Seal Policy Decision Report Surface Status

Status: status record for the Latticra Seal policy decision report surface Source: PR #300 Scope: status alignment after the deterministic local report surface for Latticra Seal policy decision metadata. This record does not implement policy evaluation, policy enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, signature verification, host reads, host writes, network behavior, MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This status record makes the Seal policy decision report surface visible as a current project checkpoint.

The report surface renders the default-deny, report-only policy decision posture from a deterministic local fixture.

Reviewed files

```
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
scripts/latticra-seal-policy-decision-report.sh
scripts/test-latticra-seal-policy-decision-report-surface.sh
tests/seal_policy_decision_report_surface.c
```

Current checkpoint

Current report-surface posture:

```
seal_policy_decision_report_surface_document_present=1
seal_policy_decision_report_surface_fixture_present=1
seal_policy_decision_report_runner_present=1
seal_policy_decision_report_surface_guard_present=1
operator_visible_policy_decision_report=1
uses_local_deterministic_fixture=1
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
policy_evaluation_supported=0
policy_enforcement_supported=0
runtime_authority_granted=0
```

Expected report posture

The report surface renders the current report-only policy decision metadata posture, including:

```
policy_decision_profile=latticra-seal-policy-decision/0.1
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=policy-decision-metadata-only
status=policy-decision-metadata
```

Validation

This report surface is covered by:

```
sh scripts/test-latticra-seal-policy-decision-report-surface.sh
```

This status record is covered by:

```
sh scripts/test-latticra-seal-policy-decision-report-surface-status.sh
```

The underlying policy decision implementation remains covered by:

```
sh scripts/test-latticra-seal-policy-decision-contract.sh
sh scripts/test-latticra-seal-policy-decision.sh
sh scripts/test-latticra-seal-policy-decision-status.sh
```

Expected output:

```
latticra seal policy decision report surface status: ok
```

Boundary

This status record is documentation/status alignment only.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is policy-decision report status-index alignment or a no-effect dry-run planning contract for future runtime behavior.

That future slice must preserve the report-only posture until a specific contract, deterministic local fixture, no-effect dry-run report, guarded allowlist posture, and validation path justify any later runtime work.

docs/status/SEAL_POLICY_DECISION_STATUS.md

Source title: Latticra Seal Policy Decision Status

Lines: 200 | Words: 609 | SHA-256: d8f3b051a6d19d86598858f15ce2f64a195dc423b606ef0dd920bd0ad2d82340

Latticra Seal Policy Decision Status

Status: status record for the Latticra Seal report-only policy decision metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the report-only Seal policy decision metadata implementation and deterministic policy decision report surface. This record does not implement policy evaluation, policy enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, signature verification, public-key parsing, trust-store loading, key generation, private-key storage, hardware key use, revocation lookup, network trust lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This status record makes the Seal policy decision metadata layer and its deterministic report surface visible from public entry points.

The policy decision layer is important because it records the default-deny decision posture that sits before the report-only runtime gate path.

It remains metadata-only and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
docs/status/SEAL_POLICY_DECISION_STATUS.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
docs/status/SEAL_SIGNED_REQUEST_STATUS.md
include/latticra/seal_policy_decision.h
src/seal_policy_decision.c
tests/seal_policy_decision_invariants.c
scripts/test-latticra-seal-policy-decision-contract.sh
scripts/test-latticra-seal-policy-decision.sh
scripts/test-latticra-seal-policy-decision-report-surface.sh
scripts/test-latticra-seal-policy-decision-report-surface-status.sh
scripts/test-latticra-seal-signed-request-status.sh
.github/workflows/latticra-seal-signed-request-status.yml
.github/workflows/latticra-seal-policy-decision-status.yml
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current policy-decision posture:

```
seal_policy_decision_contract_present=1
seal_policy_decision_implementation_present=1
seal_policy_decision_header_present=1
seal_policy_decision_source_present=1
seal_policy_decision_invariant_test_present=1
seal_policy_decision_runner_present=1
seal_policy_decision_status_surface_present=1
seal_policy_decision_status_present=1
seal_policy_decision_status_runner_present=1
seal_policy_decision_status_workflow_present=1
seal_policy_decision_report_surface_present=1
seal_policy_decision_report_surface_status_present=1
seal_signed_request_status_present=1
seal_signed_request_status_runner_present=1
seal_signed_request_status_workflow_present=1
signed_request_predecessor_request_freshness_status_present=1
policy_decision_predecessor_signed_request_status_present=1
readme_mentions_policy_decision_metadata=1
readme_mentions_policy_decision_report_surface=1
readme_links_policy_decision_contract=1
readme_links_policy_decision_implementation=1
readme_links_policy_decision_report_surface=1
readme_links_policy_decision_status=1
readme_links_policy_decision_report_surface_status=1
root_status_mentions_policy_decision_status=1
status_index_links_policy_decision_status=1
status_index_links_policy_decision_report_surface_status=1
foundation_index_links_policy_decision_status=1
foundation_index_links_policy_decision_report_surface_status=1
project_notes_mark_policy_decision_status_complete=1
policy_decision_profile=latticra-seal-policy-decision/0.1
policy_decision_supported=0
policy_evaluation_supported=0
policy_enforcement_supported=0
policy_id_present=0
policy_version_present=0
requested_action_present=0
requested_tool_present=0
signed_request_present=0
signature_valid=0
schema_valid=0
freshness_valid=0
replay_detected=0
default_decision=deny
decision_state=report-only
decision_allowed=0
decision_denied=1
operator_review_required=1
```

```
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=policy-decision-metadata-only
```

Relationship to the core evidence milestone

The completed Seal core evidence milestone depends on this policy-decision posture as part of the report-only metadata chain:

```
seal_policy_decision_metadata_present=1
policy_decision_predecessor_signed_request_status_present=1
runtime_gate_report_only=1
core_blocked_case_set_complete=1
mode=status-public-entry-alignment
policy_decision_status_added=1
implementation_behavior_changed=0
real_policy_evaluation_added=0
policy_enforcement_added=0
runtime_execution_added=0
effect_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

The current policy-decision status explains why the later core blocked-request evidence remains denied and report-only.

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-policy-decision-status.sh
```

The underlying policy decision implementation remains covered by:

```
sh scripts/test-latticra-seal-policy-decision-contract.sh
sh scripts/test-latticra-seal-policy-decision.sh
sh scripts/test-latticra-seal-policy-decision-report-surface.sh
sh scripts/test-latticra-seal-policy-decision-report-surface-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal status rollup status: ok
seal agentic automation security status: ok
seal parameter schema status: ok
seal request freshness status: ok
```

```
seal signed request status: ok
latticra seal policy decision status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the policy decision status guard workflow and records the guarded signed request status predecessor without changing the report-only policy decision metadata, implementation, or report surface.

It does not add policy evaluation, policy enforcement, runtime enforcement, runtime behavior, host behavior, network behavior, MCP behavior, model behavior, AI-agent behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this documentation/status-only alignment.

The next valid Latticra Seal slice is operator receipt report status/workflow guard alignment or another small guarded report/status alignment.

That future slice must preserve the no-effect posture and must not implement receipt file writes, tool execution, policy enforcement, capability enforcement, cryptographic verification, runtime execution, host behavior, network behavior, MCP behavior, AI agent execution, model execution, shell execution, or authority grants.

docs/status/SEAL_PRODUCT_SPINE_STATUS.md

Source title: Latticra Seal Product Spine Status

Lines: 95 | Words: 303 | SHA-256: f32e1514b4c76a20f4413bfa6324b81b8dd7aeadfc8258deb37556a385e2030f

Latticra Seal Product Spine Status

Status: status record for the Latticra Seal product spine Scope: status alignment after the Latticra Seal product spine document. This record does not implement runtime enforcement, malware prevention, ransomware prevention, sandboxing, endpoint detection, kernel enforcement, network authority, root authority, cloud trust services, production cryptographic authority, capability enforcement, effect execution, AI-agent execution control, or production security readiness.

Purpose

This status record makes the Seal product spine visible as a current project checkpoint.

The product spine defines the earned path from report-only evidence toward future verify, decide, handoff, and enforcement modes without changing current authority.

Reviewed files

```
docs/latticra-seal/PRODUCT.md
docs/latticra-seal/README.md
docs/latticra-seal/STATUS.md
scripts/test-latticra-seal-docs.sh
```

Current checkpoint

Current product-spine posture:

```
seal_product_spine_document_present=1
seal_product_spine_status_present=1
next_generation_security_product_target=1
observe_mode_current=1
verify_mode_partial_local=1
decide_mode_metadata_only=1
```

```
handoff_mode_metadata_only=1
enforce_mode_future_closed=1
operator_visible_reports=1
product_spine_changes_authority=0
production_security_product=0
malware_prevention=0
ransomware_prevention=0
runtime_enforcement=0
kernel_enforcement=0
root_authority=0
network_authority=0
capability_enforcement=0
effect_execution=0
ai_agent_execution_control=0
runtime_authority_granted=0
```

Product modes

The current product-spine document records:

```
observe_mode=current
verify_mode=partial-local
decide_mode=metadata-only
handoff_mode=metadata-only
enforce_mode=future-closed
```

Validation

The product spine is covered by:

```
sh scripts/test-latticra-seal-docs.sh
```

This status record is covered by:

```
sh scripts/test-latticra-seal-product-spine-status.sh
```

Expected output:

```
latticra seal product spine status: ok
```

Boundary

This status record is documentation/status alignment only.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The bundled operator receipt report contract, implementation plan, implementation, report surface, and status record are now current follow-up checkpoints.

The local capability registry schema contract, implementation plan, and no-effect implementation are now current follow-up checkpoints.

The next valid Latticra Seal slice is a deterministic local capability registry schema report surface or a Panel-visible Seal dashboard planning checkpoint.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md

Source title: Latticra Seal Public-Key Parsing Status

Lines: 234 | Words: 606 | SHA-256: b6c8d47ea2efa38514cd7865d9379d6468b4f8c12db46adf2d4c60cd7439717b

Latticra Seal Public-Key Parsing Status

Status: status record for Latticra Seal public-key parsing metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal public-key parsing implementation. This record does not implement public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, signing, signature verification, signer invocation behavior, signer process execution, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal public-key parsing metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, public-key-not-parsed, key-material-not-loaded, signer-not-invoked, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_CONTRACT.md
docs/LATTICRA_SEAL_PUBLIC_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_PUBLIC_KEY_PARSING_STATUS.md
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_CONTRACT.md
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION_PLAN.md
docs/LATTICRA_SEAL_FUTURE_KEY_PARSING_IMPLEMENTATION.md
docs/status/SEAL_KEY_PARSING_STATUS.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_POLICY_STATUS.md
docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md
docs/status/SEAL_CAPABILITY_GATE_STATUS.md
docs/status/SEAL_EFFECT_DECISION_STATUS.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
include/latticra/seal_public_key_parsing.h
include/latticra/seal_key_parsing.h
src/seal_public_key_parsing.c
src/seal_key_parsing.c
tests/seal_public_key_parsing_invariants.c
tests/seal_key_parsing_invariants.c
scripts/test-latticra-seal-public-key-parsing-contract.sh
scripts/test-latticra-seal-public-key-parsing.sh
scripts/test-latticra-seal-public-key-parsing-status.sh
.github/workflows/latticra-seal-public-key-parsing-status.yml
scripts/test-latticra-seal-future-key-parsing-implementation-contract.sh
scripts/test-latticra-seal-future-key-parsing-implementation-plan.sh
scripts/test-latticra-seal-key-parsing.sh
scripts/test-latticra-seal-key-parsing-status.sh
scripts/test-latticra-seal-verification-policy-contract.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-policy-status.sh
scripts/test-latticra-seal-verification-receipt-status.sh
scripts/test-latticra-seal-capability-gate-status.sh
scripts/test-latticra-seal-effect-decision-status.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
docs/LATTICRA_SEAL_KEY_MATERIAL_CONTRACT.md
docs/LATTICRA_SEAL_KEY_MATERIAL_IMPLEMENTATION.md
docs/status/SEAL_KEY_MATERIAL_STATUS.md
include/latticra/seal_key_material.h
src/seal_key_material.c
tests/seal_key_material_invariants.c
scripts/test-latticra-seal-key-material-contract.sh
scripts/test-latticra-seal-key-material.sh
scripts/test-latticra-seal-key-material-status.sh
.github/workflows/latticra-seal-key-material-status.yml
```

Current checkpoint

Current public-key parsing metadata posture:

```
seal_public_key_parsing_contract_present=1
```

```
seal_public_key_parsing_implementation_present=1
seal_public_key_parsing_header_present=1
seal_public_key_parsing_source_present=1
seal_public_key_parsing_invariant_test_present=1
seal_public_key_parsing_runner_present=1
seal_public_key_parsing_metadata_present=1
seal_public_key_parsing_status_present=1
seal_public_key_parsing_status_runner_present=1
seal_public_key_parsing_status_workflow_present=1
seal_future_key_parsing_implementation_contract_present=1
seal_future_key_parsing_implementation_plan_present=1
seal_key_parsing_metadata_present=1
seal_key_parsing_status_present=1
seal_verification_policy_contract_present=1
seal_verification_policy_implementation_present=1
seal_verification_policy_status_present=1
seal_verification_receipt_status_present=1
seal_capability_gate_status_present=1
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
seal_key_material_contract_present=1
seal_key_material_implementation_present=1
seal_key_material_status_present=1
seal_key_material_status_runner_present=1
seal_key_material_status_workflow_present=1
public_key_parsing_predecessor_key_material_status_present=1
readme_links_public_key_parsing_status=1
root_status_mentions_public_key_parsing_status=1
status_index_links_public_key_parsing_status=1
foundation_index_links_public_key_parsing_status=1
public_key_parsing_profile=latticra-seal-public-key-parsing/0.1
key_material_profile=latticra-seal-key-material/0.1
key_handling_profile=latticra-seal-key-handling/0.1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
requested_key_handling=metadata-only
requested_key_material=metadata-only
requested_public_key_parsing=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
key_handling_state=key-handling-metadata-only
key_handling_ready=1
key_material_state=key-material-metadata-only
key_material_ready=1
public_key_parsing_state=public-key-parsing-metadata-only
public_key_parsing_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
public_key_parsed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=public-key-parsing-metadata
public_key_parsing_status_added=1
public_key_parsing_added=0
```

```
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-public-key-parsing-contract.sh
sh scripts/test-latticra-seal-public-key-parsing.sh
sh scripts/test-latticra-seal-public-key-parsing-status.sh
sh scripts/test-latticra-seal-future-key-parsing-implementation-contract.sh
sh scripts/test-latticra-seal-future-key-parsing-implementation-plan.sh
sh scripts/test-latticra-seal-key-parsing.sh
sh scripts/test-latticra-seal-key-parsing-status.sh
sh scripts/test-latticra-seal-verification-policy-status.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
```

The predecessor key-material implementation remains covered by:

```
sh scripts/test-latticra-seal-key-material-contract.sh
sh scripts/test-latticra-seal-key-material.sh
sh scripts/test-latticra-seal-key-material-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing contract: ok
seal public-key parsing invariants: ok
seal public-key parsing status: ok
seal future key parsing implementation contract: ok
seal future key parsing implementation plan: ok
seal key parsing invariants: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal key-material contract: ok
seal key-material invariants: ok
seal key-material status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the public-key parsing status guard workflow and records the guarded key-material status predecessor without changing the public-key parsing implementation.

It does not add public-key parsing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signing, verification, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is key parsing status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, cryptographic verification, signing, public-key parsing behavior, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, signer invocation behavior, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_README_STATUS_ROW_ALIGNMENT.md

Source title: Latticra Seal README Status Row Alignment

Lines: 72 | Words: 263 | SHA-256: 21e46643eb37e8a134aaa39e1d917f51c70c5c1f79c2f8cd99ad70a3c0325f07

Latticra Seal README Status Row Alignment

Status: README/status alignment record for the current Latticra Seal public summary Date: 2026-05-25 CDT Scope: README compact status row and Seal current-posture summary alignment with the existing Latticra Seal status checkpoint. This record does not implement runtime execution, effect execution, capability enforcement, cryptographic verification, signing, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, shell execution, production enforcement, public-readiness promotion, security-hardening implementation, or runtime authority.

Purpose

The detailed README evidence flags and status index already record the current Seal chain.

This alignment updates the first README Seal summary surfaces so they no longer stop at bounded key parsing while later status records already include verification policy, verification receipt, denied capability gate, denied effect decision, inactive runtime handoff, status rollup, agentic automation security, parameter schema, request freshness, signed request, and policy decision report/public-entry surfaces.

Alignment Fields

```
seal_readme_status_row_alignment_present=1
readme_seal_row_mentions_runtime_gate_path=1
readme_seal_row_mentions_bounded_key_parsing=1
readme_seal_row_mentions_verification_policy_status=1
readme_seal_row_mentions_verification_receipt_status=1
readme_seal_row_mentions_capability_gate_status=1
readme_seal_row_mentions_effect_decision_status=1
readme_seal_row_mentions_runtime_handoff_status=1
readme_seal_row_mentions_status_rollup_status=1
readme_seal_row_mentions_agentic_automation_security=1
readme_seal_row_mentions_parameter_schema_status=1
readme_seal_row_mentions_policy_decision_public_entry=1
readme_seal_current_posture_aligned=1
status_index_links_alignment_record=1
foundation_index_links_alignment_record=1
project_notes_link_alignment_record=1
implementation_behavior_changed=0
runtime_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
cryptographic_verification_added=0
signing_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
security_hardening_changed=0
public_readiness_changed=0
production_enforcement_added=0
runtime_authority_granted=0
estimate_adjustment_required=0
```

Boundary

This is a wording and public-entry alignment only.

It does not add policy evaluation, policy enforcement, runtime enforcement, effect execution, cryptographic verification, signing, key loading, host access, network access, MCP protocol behavior, AI-agent execution control, model execution, tool execution, shell execution, production enforcement, or runtime authority.

The current next Seal lane remains small guarded report/status alignment only when drift appears.

Validation

This alignment is guarded by:

```
sh scripts/test-latticra-seal-readme-status-row-alignment.sh
```

Expected output:

```
latticra_seal_readme_status_row_alignment: ok
```

docs/status/SEAL_REPORT_ENVELOPE_STATUS.md

Source title: Latticra Seal Report Envelope Status

Lines: 137 | Words: 376 | SHA-256: 2d39d9a53b9e256394055a4b75c68070c367015362e6521569c2fd4cd4ed2bd4

Latticra Seal Report Envelope Status

Status: status record for the Latticra Seal sealed report-envelope metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal report envelope contract and metadata implementation. This record does not add cryptographic signing, signature verification, private-key handling, key generation, trust-store loading, revocation lookup, runtime handoff execution, runtime authority, host effects, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal report envelope implementation visible from public entry points.

The envelope layer consumes ready runtime handoff report metadata and classifies a narrow local metadata-only envelope posture. It is envelope classification metadata, not signing and not runtime handoff.

Reviewed files

```
docs/LATTICRA_SEAL_REPORT_ENVELOPE_CONTRACT.md
docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md
docs/status/SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md
include/latticra/seal_report_envelope.h
src/seal_report_envelope.c
tests/seal_report_envelope_invariants.c
scripts/test-latticra-seal-report-envelope-contract.sh
scripts/test-latticra-seal-report-envelope.sh
scripts/test-latticra-seal-report-envelope-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current sealed report-envelope posture:

```
seal_report_envelope_contract_present=1
seal_report_envelope_implementation_present=1
seal_report_envelope_header_present=1
seal_report_envelope_source_present=1
seal_report_envelope_invariant_test_present=1
seal_report_envelope_runner_present=1
seal_report_envelope_status_present=1
seal_runtime_handoff_report_contract_present=1
seal_runtime_handoff_report_implementation_present=1
seal_runtime_handoff_report_runner_present=1
seal_runtime_handoff_report_status_present=1
readme_links_report_envelope_contract=1
readme_links_report_envelope_implementation=1
readme_links_report_envelope_status=1
root_status_mentions_report_envelope_status=1
status_index_links_report_envelope_status=1
foundation_index_links_report_envelope_status=1
project_notes_mark_report_envelope_status_complete=1
envelope_profile=latticra-seal-report-envelope/0.1
report_profile=latticra-seal-runtime-handoff-report/0.1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
decision_profile=latticra-seal-verified-effect-decision/0.1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
```

```
requested_capability=verified-receipt-report
requested_effect=report-only
requested_handoff=report-only
requested_report=report-only
requested_envelope=report-only
requested_scope=local-fixture-scope
report_state=ready-report-only
report_ready=1
envelope_state=sealed-report-only
envelope_ready=1
signature_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
status=sealed-report-envelope-metadata
sealed_report_envelope_added=1
signature_generation_added=0
signature_verification_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-report-envelope-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-report-envelope.sh
```

Expected output:

```
seal report envelope status: ok
seal report envelope invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The report envelope implementation accepts ready runtime handoff report metadata and produces deterministic sealed envelope metadata. A successful envelope may set envelope_ready=1 and envelope_state=sealed-report-only, but it remains metadata-only, unsigned, object-unsealed, and authority-neutral.

It does not add cryptographic signing, signature verification, private-key handling, key generation, trust-store behavior, revocation lookup, object sealing, runtime handoff execution, capability enforcement, effect execution, host behavior, network behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The next downstream Latticra Seal checkpoint is the signature request status/public-entry alignment or another narrow status/index alignment follow-up. Any future work must remain no-signing unless separately contracted, implemented, and guarded.

docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md

Source title: Latticra Seal Request Freshness Status

Lines: 215 | Words: 654 | SHA-256: 04bd4100649815de2f9b8893b6322a0dbf909868a8ae921ed8a9d9879c435488

Latticra Seal Request Freshness Status

Status: status record for Latticra Seal request freshness metadata Source: local follow-up slice Scope: status and public-entry alignment after the report-only Latticra Seal request freshness contract, metadata implementation, and report surface. This record does not implement timestamp parsing, trusted clock behavior, nonce storage, replay-cache storage, context hashing, parameter hashing, freshness validation, replay detection, signature verification, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, public-key parsing, public-key trust stores, key generation, private-key storage, hardware key use, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This status record makes the completed Seal request freshness checkpoint visible from public entry points.

The purpose is to keep public project readers aligned with the current report-only freshness/replay posture without claiming timestamp parsing, trusted clocks, nonce persistence, replay protection, freshness validation, enforcement, production readiness, MCP implementation, AI-agent execution control, model execution, tool execution, shell execution, or operational authority.

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_CONTRACT.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_IMPLEMENTATION.md
docs/LATTICRA_SEAL_PARAMETER_SCHEMA_REPORT_SURFACE.md
docs/status/SEAL_PARAMETER_SCHEMA_STATUS.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md
docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md
include/latticra/seal_request_freshness.h
src/seal_request_freshness.c
tests/seal_request_freshness_invariants.c
tests/seal_request_freshness_report_surface.c
scripts/test-latticra-seal-request-freshness-contract.sh
scripts/test-latticra-seal-request-freshness.sh
scripts/latticra-seal-request-freshness-report.sh
scripts/test-latticra-seal-request-freshness-report-surface.sh
scripts/test-latticra-seal-request-freshness-status.sh
scripts/test-latticra-seal-parameter-schema-status.sh
.github/workflows/latticra-seal-parameter-schema-status.yml
.github/workflows/latticra-seal-request-freshness-status.yml
docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNED_REQUEST_STATUS.md
scripts/test-latticra-seal-signed-request-contract.sh
scripts/test-latticra-seal-signed-request.sh
scripts/test-latticra-seal-signed-request-status.sh
```

Current checkpoint

Current request freshness status posture:

```
seal_request_freshness_contract_present=1
seal_request_freshness_implementation_present=1
seal_request_freshness_header_present=1
seal_request_freshness_source_present=1
seal_request_freshness_invariant_test_present=1
seal_request_freshness_runner_present=1
seal_request_freshness_metadata_present=1
seal_request_freshness_report_surface_present=1
seal_request_freshness_report_runner_present=1
seal_request_freshness_report_guard_present=1
seal_request_freshness_status_present=1
seal_request_freshness_status_runner_present=1
seal_request_freshness_status_workflow_present=1
seal_parameter_schema_status_present=1
seal_parameter_schema_status_runner_present=1
seal_parameter_schema_status_workflow_present=1
parameter_schema_predecessor_agentic_automation_security_status_present=1
request_freshness_predecessor_parameter_schema_status_present=1
readme_mentions_request_freshness_metadata=1
readme_mentions_request_freshness_report_surface=1
readme_links_request_freshness_contract=1
readme_links_request_freshness_implementation=1
readme_links_request_freshness_report_surface=1
readme_links_request_freshness_status=1
root_status_mentions_request_freshness_status=1
status_index_links_request_freshness_status=1
foundation_index_links_request_freshness_status=1
project_notes_point_to_signed_request_status=1
freshness_profile=latticra-seal-request-freshness/0.1
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
mode=status-public-entry-alignment
request_freshness_status_added=1
implementation_behavior_changed=0
timestamp_parsing_implemented=0
trusted_clock_behavior_added=0
nonce_storage_added=0
replay_cache_storage_added=0
context_hashing_implemented=0
parameter_hashing_implemented=0
freshness_validation_implemented=0
replay_protection_implemented=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Expected report posture

The report surface renders the current report-only request freshness posture, including:

```
freshness_profile=latticra-seal-request-freshness/0.1
request_id=unset
caller_id=unset
```

```
tool_id=unset
request_timestamp=unset
request_expiration=unset
nonce=unset
context_hash=unset
parameter_hash=unset
request_freshness_supported=0
request_freshness_validation_supported=0
replay_protection_supported=0
request_id_present=0
caller_id_present=0
tool_id_present=0
request_timestamp_present=0
request_expiration_present=0
nonce_present=0
context_hash_present=0
parameter_hash_present=0
freshness_valid=0
replay_detected=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=request-freshness-metadata-only
status=request-freshness-metadata
```

Validation

This status/public-entry alignment is covered by:

```
sh scripts/test-latticra-seal-request-freshness-status.sh
```

The underlying checkpoint remains covered by:

```
sh scripts/test-latticra-seal-request-freshness-contract.sh
sh scripts/test-latticra-seal-request-freshness.sh
sh scripts/test-latticra-seal-request-freshness-report-surface.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal status rollup status: ok
seal agentic automation security status: ok
seal parameter schema status: ok
seal request freshness status: ok
```

Boundary

This status record is documentation/status/public-entry alignment only.

This refresh adds the request freshness status guard workflow and records the guarded parameter schema status predecessor without changing the report-only request freshness metadata, implementation, or report surface.

It does not add timestamp parsing, trusted clock behavior, nonce storage, replay-cache storage, context hashing, parameter hashing, freshness validation, replay detection, signature verification, MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature generation, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is signed request status/workflow guard alignment.

That future slice must preserve the no-effect posture and must not implement signature generation, signature verification, trust-store loading, revocation lookup, signed request enforcement, real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_RUNTIME_DRY_RUN_PUBLIC_ENTRYPOINT_ALIGNMENT.md

Source title: Latticra Seal Runtime Dry-Run Public Entrypoint Alignment

Lines: 107 | Words: 388 | SHA-256: f4ea7dad67fd9feceb7b28e5fe3c20168655e4d74e342ea37847ea82d32bd4bb

Latticra Seal Runtime Dry-Run Public Entrypoint Alignment

Status: public-entrypoint alignment record for the Latticra Seal runtime dry-run milestone Source: PR #307 Scope: README/public-entrypoint alignment after the completed Latticra Seal runtime dry-run status-index alignment. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This record makes the completed Seal runtime dry-run lane visible from the root README public entrypoint.

The purpose is to keep public project readers aligned with the current Seal state without claiming enforcement, production readiness, MCP implementation, AI-agent security, or operational authority.

Reviewed files

```
README.md
docs/status/SEAL_RUNTIME_DRY_RUN_STATUS_INDEX_ALIGNMENT.md
docs/status/SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md
scripts/test-latticra-seal-runtime-dry-run-status-index-alignment.sh
```

Alignment checkpoint

Current public-entrypoint posture:

```
seal_runtime_dry_run_public_entrypoint_alignment_present=1
readme_mentions_runtime_dry_run_metadata=1
readme_mentions_runtime_dry_run_report_surface=1
readme_links_runtime_dry_run_contract=1
readme_links_runtime_dry_run_implementation=1
readme_links_runtime_dry_run_report_surface=1
readme_links_runtime_dry_run_report_surface_status=1
readme_links_runtime_dry_run_status_index_alignment=1
readme_mentions_operator_visible_runtime_dry_run_report=1
readme_mentions_default_deny_dry_run=1
readme_mentions_no_runtime_authority=1
mode=public-entrypoint-alignment
```

```
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current Seal summary

The public README now presents Seal as a report-only dry-run lane:

```
policy decision metadata present
runtime gate metadata present
runtime dry-run metadata present
runtime dry-run report surface present
runtime dry-run report surface status present
runtime dry-run status-index alignment present
operator-visible runtime dry-run report present
default-deny dry-run posture present
runtime authority remains denied
```

The lane remains no-effect and report-only.

Validation

This public-entrypoint alignment is covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run-public-entrypoint-alignment.sh
```

The underlying Seal runtime dry-run checkpoint remains covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run.sh
sh scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
sh scripts/test-latticra-seal-runtime-dry-run-report-surface-status.sh
sh scripts/test-latticra-seal-runtime-dry-run-status-index-alignment.sh
```

Expected output:

```
latticra seal runtime dry-run public entrypoint alignment: ok
```

Boundary

This record is documentation/public-entrypoint alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, freshness validation, replay detection, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is a guarded allowlist planning contract.

That future slice must be planning-only and must not perform effects, execute tools, touch host state, use the network, or grant runtime authority.

docs/status/SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md

Source title: Latticra Seal Runtime Dry-Run Report Surface Status

Lines: 113 | Words: 326 | SHA-256: 0ff2e0efcfd876bb5c9b3a812589e4f03e9272fa58f315625e4782ea0ecc928c

Latticra Seal Runtime Dry-Run Report Surface Status

Status: status record for the Latticra Seal runtime dry-run report surface Source: PR #305 Scope: status alignment after the deterministic local report surface for Latticra Seal runtime dry-run metadata. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This status record makes the Seal runtime dry-run report surface visible as a current project checkpoint.

The report surface renders the no-effect dry-run denial posture from a deterministic local fixture.

Reviewed files

```
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md
scripts/latticra-seal-runtime-dry-run-report.sh
scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
tests/seal_runtime_dry_run_report_surface.c
```

Current checkpoint

Current report-surface posture:

```
seal_runtime_dry_run_report_surface_document_present=1
seal_runtime_dry_run_report_surface_fixture_present=1
seal_runtime_dry_run_report_runner_present=1
seal_runtime_dry_run_report_surface_guard_present=1
operator_visible_runtime_dry_run_report=1
uses_local_deterministic_fixture=1
implementation_behavior_changed=0
runtime_behavior_added=0
host_behavior_added=0
network_behavior_added=0
external_service_behavior_added=0
runtime_authority_granted=0
```

Expected report posture

The report surface renders the current report-only runtime dry-run metadata posture, including:

```
runtime_dry_run_profile=latticra-seal-runtime-dry-run/0.1
request_class=core-blocked-request
policy_decision_state=report-only
runtime_gate_state=report-only
blocked_reason=default-deny-dry-run
dry_run_supported=1
dry_run_performed=1
input_policy_decision_present=1
input_runtime_gate_present=1
policy_decision_report_only=1
runtime_gate_report_only=1
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
unknown_tool_denied=1
unsigned_request_denied=1
invalid_schema_denied=1
```

```
stale_request_denied=1
replayed_request_denied=1
invalid_signature_denied=1
report_only=1
mode=report-only
status=runtime-dry-run-metadata
```

Validation

This report surface is covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
```

This status record is covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run-report-surface-status.sh
```

The underlying runtime dry-run implementation remains covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run.sh
```

Expected output:

```
latticra seal runtime dry-run report surface status: ok
```

Boundary

This status record is documentation/status alignment only.

It does not add runtime behavior, policy behavior, protocol behavior, host behavior, network behavior, model behavior, tool behavior, shell behavior, cryptographic behavior, capability behavior, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is runtime dry-run status-index alignment or a future guarded allowlist planning contract.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_RUNTIME_DRY_RUN_STATUS_INDEX_ALIGNMENT.md

Source title: Latticra Seal Runtime Dry-Run Status Index Alignment

Lines: 127 | Words: 429 | SHA-256: ccf804c7f800cfe650879cb8bceaea9211971e084b0d3738ff36a1ddb0e162f4

Latticra Seal Runtime Dry-Run Status Index Alignment

Status: index alignment record for the Latticra Seal runtime dry-run report surface status Source: PR #306 Scope: documentation/status index alignment after the Latticra Seal runtime dry-run report surface status record. This record does not implement runtime behavior, runtime execution, runtime authority, effect execution, policy enforcement, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, tool execution, AI agent execution, model execution, MCP protocol behavior, MCP server behavior, MCP client behavior, host reads, host writes, network behavior, operating-system behavior, production readiness, external endorsement, or authority grants.

Purpose

This record keeps the completed Seal runtime dry-run report surface status visible from the status index after the merged report-only dry-run implementation, report surface, and status-surface slices.

The purpose is to preserve the current evidence chain:

```
policy decision -&gt; policy decision report surface -&gt; runtime dry-run metadata -&gt;
runtime dry-run
```

```
report surface -&gt; runtime dry-run status -&gt; status index alignment
```

Reviewed files

```
docs/status/SEAL_POLICY_DECISION_STATUS.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
docs/status/SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE_STATUS.md
docs/status/README.md
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_DRY_RUN_REPORT_SURFACE.md
scripts/latticra-seal-runtime-dry-run-report.sh
scripts/test-latticra-seal-runtime-dry-run.sh
scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
scripts/test-latticra-seal-runtime-dry-run-report-surface-status.sh
```

Alignment checkpoint

Current alignment posture:

```
seal_policy_decision_status_indexed=1
seal_policy_decision_report_surface_status_indexed=1
seal_runtime_dry_run_report_surface_status_indexed=1
seal_runtime_dry_run_status_index_alignment_record_present=1
status_index_mentions_runtime_dry_run=1
operator_visible_runtime_dry_run_report=1
uses_local_deterministic_fixture=1
mode=doc-status-index-alignment
implementation_behavior_changed=0
runtime_authority_granted=0
runtime_enforcement_implemented=0
policy_enforcement_implemented=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
tool_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Current status summary

The Seal dry-run lane is now represented as:

```
policy decision status present
policy decision report surface status present
runtime dry-run metadata present
runtime dry-run report surface present
runtime dry-run report surface status present
runtime dry-run status index alignment record present
```

The lane remains report-only and no-effect.

The current dry-run report posture is still:

```
policy_decision_state=report-only
runtime_gate_state=report-only
blocked_reason=default-deny-dry-run
default_action_deny=1
would_allow=0
would_deny=1
would_require_operator_review=1
would_execute_tool=0
would_read_host=0
would_write_host=0
would_use_network=0
would_grant_runtime_authority=0
report_only=1
mode=report-only
status=runtime-dry-run-metadata
```

Validation

This index alignment is covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run-status-index-alignment.sh
```

The underlying Seal runtime dry-run checkpoint remains covered by:

```
sh scripts/test-latticra-seal-runtime-dry-run.sh
sh scripts/test-latticra-seal-runtime-dry-run-report-surface.sh
sh scripts/test-latticra-seal-runtime-dry-run-report-surface-status.sh
```

Expected output:

```
latticra seal runtime dry-run status index alignment: ok
```

Boundary

This record is documentation/status index alignment only.

It does not add MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, signature verification, freshness validation, replay detection, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is a public entry-point refresh for the completed Seal runtime dry-run milestone or a future guarded allowlist planning contract.

That future slice must preserve the no-effect posture and must not perform effects or grant authority.

docs/status/SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md

Source title: Latticra Seal Runtime Handoff Evaluation Status

Lines: 143 | Words: 428 | SHA-256: 8396025efc04669fdd526cc72b41d6933bd3635737ffc8a2af5bb25e5d272f7c

Latticra Seal Runtime Handoff Evaluation Status

Status: status record for the Latticra Seal runtime handoff evaluation metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal runtime handoff evaluation contract and metadata implementation, now carrying crypto graduation metadata forward when present on the verified effect decision. This record does not add runtime handoff execution, effect execution, capability enforcement, runtime authority, host effects, network behavior, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal runtime handoff evaluation implementation visible from public entry points.

The evaluation consumes verified effect decision metadata and evaluates a narrow local metadata-only handoff request. When the decision carries crypto graduation metadata, the evaluation requires that evidence to remain passed, standard-aligned, and authority-neutral before allowing the metadata-only handoff classification.

It is handoff classification metadata, not runtime handoff.

Reviewed files

```
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md
docs/status/SEAL_RUNTIME_HANDOFF_EVALUATION_STATUS.md
docs/status/SEAL_VERIFIED_EFFECT_DECISION_STATUS.md
docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md
include/latticra/seal_runtime_handoff_evaluation.h
src/seal_runtime_handoff_evaluation.c
tests/seal_runtime_handoff_evaluation_invariants.c
scripts/test-latticra-seal-runtime-handoff-evaluation-contract.sh
scripts/test-latticra-seal-runtime-handoff-evaluation.sh
scripts/test-latticra-seal-runtime-handoff-evaluation-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current runtime handoff evaluation posture:

```
seal_runtime_handoff_evaluation_contract_present=1
seal_runtime_handoff_evaluation_implementation_present=1
seal_runtime_handoff_evaluation_header_present=1
seal_runtime_handoff_evaluation_source_present=1
seal_runtime_handoff_evaluation_invariant_test_present=1
seal_runtime_handoff_evaluation_runner_present=1
seal_runtime_handoff_evaluation_status_present=1
seal_verified_effect_decision_status_present=1
seal_crypto_graduation_gate_status_present=1
readme_links_runtime_handoff_evaluation_contract=1
readme_links_runtime_handoff_evaluation_implementation=1
readme_links_runtime_handoff_evaluation_status=1
root_status_mentions_runtime_handoff_evaluation_status=1
status_index_links_runtime_handoff_evaluation_status=1
foundation_index_links_runtime_handoff_evaluation_status=1
project_notes_mark_runtime_handoff_evaluation_status_complete=1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
decision_profile=latticra-seal-verified-effect-decision/0.1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
crypto_graduation_gate_state=graduated-authority-neutral
requested_capability=verified-receipt-report
requested_effect=report-only
requested_handoff=report-only
requested_scope=local-fixture-scope
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
verified=1
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=1
gate_state=allowed-metadata-only
decision_state=allowed-report-only
effect_allowed=1
handoff_state=eligible-report-only
handoff_eligible=1
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
status=runtime-handoff-evaluation-metadata
```

```
runtime_handoff_evaluation_added=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-runtime-handoff-evaluation-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-runtime-handoff-evaluation.sh
```

Expected output:

```
seal runtime handoff evaluation status: ok
seal runtime handoff evaluation invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The runtime handoff evaluation implementation evaluates a verified effect decision record against a narrow local metadata-only handoff request. If the decision includes crypto graduation evidence, the evaluation requires crypto_graduation_gate_passed=1, standard_expectations_met=1, and authority_promotion_allowed=0. A successful evaluation may set handoff_eligible=1 and handoff_state=eligible-report-only, but it remains metadata-only and authority-neutral.

It does not add runtime handoff execution, effect execution, capability enforcement, runtime behavior, host behavior, network behavior, signing, key generation, private-key handling, trust-store behavior, revocation lookup, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The next valid Latticra Seal slice is a runtime handoff report surface from an eligible crypto-graduation-gated metadata-only runtime handoff evaluation, runtime handoff report status/public-entry alignment, or another narrow status/index alignment follow-up.

docs/status/SEAL_RUNTIME_HANDOFF_REPORT_STATUS.md

Source title: Latticra Seal Runtime Handoff Report Status

Lines: 136 | Words: 361 | SHA-256: 4c2c99c9be9dc3693ddaf2e52a0d1f506e7fd1e393fc696de794bbd580973849

Latticra Seal Runtime Handoff Report Status

Status: status record for the Latticra Seal runtime handoff report metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal runtime handoff report contract and metadata implementation. This record does not add runtime handoff execution, runtime authority, host effects, network behavior, shell execution, tool execution, capability enforcement, policy persistence, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal runtime handoff report implementation visible from public entry points.

The report layer consumes eligible runtime handoff evaluation metadata and classifies a narrow local metadata-only report request. It is report classification metadata, not runtime handoff.

Reviewed files

```
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_REPORT_IMPLEMENTATION.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_EVALUATION_IMPLEMENTATION.md
include/latticra/seal_runtime_handoff_report.h
src/seal_runtime_handoff_report.c
tests/seal_runtime_handoff_report_invariants.c
scripts/test-latticra-seal-runtime-handoff-report-contract.sh
scripts/test-latticra-seal-runtime-handoff-report.sh
scripts/test-latticra-seal-runtime-handoff-report-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current runtime handoff report posture:

```
seal_runtime_handoff_report_contract_present=1
seal_runtime_handoff_report_implementation_present=1
seal_runtime_handoff_report_header_present=1
seal_runtime_handoff_report_source_present=1
seal_runtime_handoff_report_invariant_test_present=1
seal_runtime_handoff_report_runner_present=1
seal_runtime_handoff_report_status_present=1
seal_runtime_handoff_evaluation_contract_present=1
seal_runtime_handoff_evaluation_implementation_present=1
seal_runtime_handoff_evaluation_runner_present=1
readme_links_runtime_handoff_report_contract=1
readme_links_runtime_handoff_report_implementation=1
readme_links_runtime_handoff_report_status=1
root_status_mentions_runtime_handoff_report_status=1
status_index_links_runtime_handoff_report_status=1
foundation_index_links_runtime_handoff_report_status=1
project_notes_mark_runtime_handoff_report_status_complete=1
report_profile=latticra-seal-runtime-handoff-report/0.1
handoff_profile=latticra-seal-runtime-handoff-evaluation/0.1
decision_profile=latticra-seal-verified-effect-decision/0.1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
requested_capability=verified-receipt-report
requested_effect=report-only
requested_handoff=report-only
requested_report=report-only
requested_scope=local-fixture-scope
verified=1
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=1
gate_state=allowed-metadata-only
decision_state=allowed-report-only
effect_allowed=1
handoff_state=eligible-report-only
handoff_eligible=1
report_state=ready-report-only
report_ready=1
handoff_performed=0
effect_performed=0
```

```
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
status=runtime-handoff-report-metadata
runtime_handoff_report_added=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-runtime-handoff-report-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-runtime-handoff-report.sh
```

Expected output:

```
seal runtime handoff report status: ok
seal runtime handoff report invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The runtime handoff report implementation accepts eligible runtime handoff evaluation metadata and produces deterministic report-readiness metadata. A successful report may set report_ready=1 and report_state=ready-report-only, but it remains metadata-only and authority-neutral.

It does not add runtime handoff execution, capability enforcement, effect execution, host behavior, network behavior, signing, key generation, private-key handling, trust-store behavior, revocation lookup, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The next valid Latticra Seal slice is sealed report envelope status/public-entry alignment or another narrow status/index alignment follow-up.

docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md

Source title: Latticra Seal Runtime Handoff Status

Lines: 168 | Words: 475 | SHA-256: e0ed1b087a647a9a468e36a92a93c502a5c75bd6d5e5a6495ce340a8ccb364af

Latticra Seal Runtime Handoff Status

Status: status record for Latticra Seal runtime handoff metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal runtime handoff implementation. This record does not add runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, signing, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal runtime handoff metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, inactive, runtime-boundary-disabled, denied-by-decision, effect-not-performed, host-inactive, network-inactive, runtime-authority-denied, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
include/latticra/seal_runtime_handoff.h
src/seal_runtime_handoff.c
tests/seal_runtime_handoff_invariants.c
scripts/test-latticra-seal-runtime-handoff-contract.sh
scripts/test-latticra-seal-runtime-handoff.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
.github/workflows/latticra-seal-runtime-handoff-status.yml
docs/LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md
docs/LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md
include/latticra/seal_status_rollup.h
src/seal_status_rollup.c
tests/seal_status_rollup_invariants.c
scripts/test-latticra-seal-status-rollup-contract.sh
scripts/test-latticra-seal-status-rollup.sh
scripts/test-latticra-seal-status-rollup-status.sh
docs/status/SEAL_STATUS_ROLLUP_STATUS.md
docs/LATTICRA_SEAL_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_EFFECT_DECISION_IMPLEMENTATION.md
docs/status/SEAL_EFFECT_DECISION_STATUS.md
include/latticra/seal_effect_decision.h
src/seal_effect_decision.c
tests/seal_effect_decision_invariants.c
scripts/test-latticra-seal-effect-decision-contract.sh
scripts/test-latticra-seal-effect-decision.sh
scripts/test-latticra-seal-effect-decision-status.sh
.github/workflows/latticra-seal-effect-decision-status.yml
```

Current checkpoint

Current runtime handoff metadata posture:

```
seal_runtime_handoff_contract_present=1
seal_runtime_handoff_implementation_present=1
seal_runtime_handoff_header_present=1
seal_runtime_handoff_source_present=1
seal_runtime_handoff_invariant_test_present=1
seal_runtime_handoff_runner_present=1
seal_runtime_handoff_metadata_present=1
seal_runtime_handoff_status_present=1
seal_runtime_handoff_status_runner_present=1
seal_runtime_handoff_status_workflow_present=1
seal_status_rollup_contract_present=1
seal_status_rollup_implementation_present=1
seal_status_rollup_status_present=1
seal_effect_decision_contract_present=1
seal_effect_decision_implementation_present=1
```

```
seal_effect_decision_status_present=1
seal_effect_decision_status_runner_present=1
seal_effect_decision_status_workflow_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
handoff_profile=latticra-seal-runtime-handoff/0.1
decision_profile=latticra-seal-effect-decision/0.1
gate_profile=latticra-seal-capability-gate/0.1
requested_capability=seal.inspect
requested_effect=read-metadata
requested_scope=local-artifact
requested_runtime_handoff=metadata-only
runtime_handoff_ready=1
decision_state=denied-gate
effect_allowed=0
effect_performed=0
runtime_boundary_state=disabled
runtime_request_label=runtime.preview
handoff_active=0
runtime_effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
runtime_authority_granted=0
handoff_state=denied-decision
capability_enforcement_performed=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
network_lookup_allowed=0
revocation_lookup_allowed=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
mode=metadata-only
status=runtime-handoff-inactive-metadata
error=ok
runtime_handoff_status_added=1
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-runtime-handoff-contract.sh
sh scripts/test-latticra-seal-runtime-handoff.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-status-rollup-contract.sh
sh scripts/test-latticra-seal-status-rollup-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
```

Expected output:

```
seal runtime handoff contract: ok
seal runtime handoff invariants: ok
seal runtime handoff status: ok
```

```
seal status rollup contract: ok
seal status rollup status: ok
seal effect decision status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the runtime handoff status guard workflow and records the guarded effect decision status predecessor without changing the inactive metadata-only runtime handoff implementation.

It does not add runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, signing, public-key byte verification, public-key trust-store behavior, key material loading, private-key handling, key generation, hardware-key use, revocation lookup, signer invocation behavior, signer process execution, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is status rollup status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_SIGNATURE_REQUEST_STATUS.md

Source title: Latticra Seal Signature Request Status

Lines: 187 | Words: 482 | SHA-256: d42d326094c1a7e318eb3b1849eca907170a1ad46026bcddf4b568a599dae6a9

Latticra Seal Signature Request Status

Status: status record for Latticra Seal signature request metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal signature request implementation. This record does not implement signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal signature request metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_AUTHORIZATION_STATUS.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_HANDOFF_STATUS.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
docs/status/SEAL_REPORT_ENVELOPE_STATUS.md
include/latticra/seal_signature_request.h
include/latticra/seal_signing_authorization.h
include/latticra/seal_signer_handoff.h
include/latticra/seal_signer_invocation.h
src/seal_signature_request.c
src/seal_signing_authorization.c
src/seal_signer_handoff.c
src/seal_signer_invocation.c
tests/seal_signature_request_invariants.c
tests/seal_signing_authorization_invariants.c
tests/seal_signer_handoff_invariants.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signature-request-contract.sh
scripts/test-latticra-seal-signature-request.sh
scripts/test-latticra-seal-signing-authorization-contract.sh
scripts/test-latticra-seal-signing-authorization.sh
scripts/test-latticra-seal-signing-authorization-status.sh
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signer-handoff.sh
scripts/test-latticra-seal-signer-handoff-status.sh
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signer-invocation-status.sh
docs/LATTICRA_SEAL_REPORT_ENVELOPE_IMPLEMENTATION.md
scripts/test-latticra-seal-report-envelope.sh
scripts/test-latticra-seal-report-envelope-status.sh
scripts/test-latticra-seal-signature-request-status.sh
.github/workflows/latticra-seal-signature-request-status.yml
```

Current checkpoint

Current signature request metadata posture:

```
seal_signature_request_contract_present=1
seal_signature_request_implementation_present=1
seal_signature_request_header_present=1
seal_signature_request_source_present=1
seal_signature_request_invariant_test_present=1
seal_signature_request_runner_present=1
seal_signature_request_status_present=1
seal_signature_request_status_runner_present=1
seal_signature_request_status_workflow_present=1
seal_signing_authorization_contract_present=1
seal_signing_authorization_implementation_present=1
seal_signing_authorization_header_present=1
seal_signing_authorization_source_present=1
seal_signing_authorization_invariant_test_present=1
seal_signing_authorization_runner_present=1
seal_signing_authorization_status_present=1
seal_signer_handoff_contract_present=1
seal_signer_handoff_implementation_present=1
seal_signer_handoff_header_present=1
seal_signer_handoff_source_present=1
seal_signer_handoff_invariant_test_present=1
seal_signer_handoff_runner_present=1
seal_signer_handoff_metadata_present=1
seal_signer_handoff_status_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_implementation_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
seal_report_envelope_implementation_present=1
seal_report_envelope_runner_present=1
seal_report_envelope_status_present=1
seal_report_envelope_status_runner_present=1
```

```
signature_request_predecessor_report_envelope_status_present=1
readme_links_signature_request_status=1
root_status_mentions_signature_request_status=1
status_index_links_signature_request_status=1
foundation_index_links_signature_request_status=1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
signature_request_state=requested-metadata-only
signature_request_ready=1
signature_performed=0
verification_performed=0
private_key_handling=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=signature-request-metadata
signature_request_status_added=1
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation is covered by:

```
sh scripts/test-latticra-seal-signature-request-status.sh
sh scripts/test-latticra-seal-signature-request-contract.sh
sh scripts/test-latticra-seal-signature-request.sh
sh scripts/test-latticra-seal-signing-authorization-contract.sh
sh scripts/test-latticra-seal-signing-authorization.sh
sh scripts/test-latticra-seal-signing-authorization-status.sh
sh scripts/test-latticra-seal-signer-handoff-contract.sh
sh scripts/test-latticra-seal-signer-handoff.sh
sh scripts/test-latticra-seal-signer-handoff-status.sh
sh scripts/test-latticra-seal-signer-invocation-contract.sh
sh scripts/test-latticra-seal-signer-invocation.sh
sh scripts/test-latticra-seal-signer-invocation-status.sh
```

The predecessor report-envelope implementation remains covered by:

```
sh scripts/test-latticra-seal-report-envelope.sh
sh scripts/test-latticra-seal-report-envelope-status.sh
```

Expected output:

```
seal signature request status: ok
seal signature request contract: ok
seal signature request invariants: ok
seal signing authorization contract: ok
seal signing authorization invariants: ok
seal signing authorization status: ok
seal signer handoff contract: ok
seal signer handoff invariants: ok
seal signer handoff status: ok
seal signer invocation contract: ok
seal signer invocation invariants: ok
seal signer invocation status: ok
seal report envelope invariants: ok
seal report envelope status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds an explicit status guard workflow and records the report-envelope status predecessor without changing the signature-request implementation.

It does not add signing, verification, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is signing authorization status/workflow guard alignment or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

That future slice must not add signing, verification, private-key handling, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_SIGNED_REQUEST_STATUS.md

Source title: Latticra Seal Signed Request Status

Lines: 209 | Words: 630 | SHA-256: 80e5305fe2cfce68b95e09a90cdb9a2b658d90cc641a21d6dd93c24c834e2a75

Latticra Seal Signed Request Status

Status: status record for Latticra Seal signed request metadata Source: local follow-up slice Scope: status and public-entry alignment after the report-only Latticra Seal signed request contract and metadata implementation. This record does not implement signature generation, signature verification, public-key parsing, trust-store loading, private-key handling, key generation, hardware-key use, revocation lookup, network trust lookup, signed request enforcement, runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipts, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, MCP protocol implementation, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, or operating-system behavior.

Purpose

This status record makes the completed Seal signed request metadata checkpoint visible from public entry points.

The purpose is to keep public project readers aligned with the current report-only signature metadata posture without claiming signing, signature verification, key parsing, trust-store behavior, revocation lookup, signed request enforcement, production readiness, MCP implementation, AI-agent execution control, model execution, tool execution, shell execution, or operational authority.

Reviewed files

```
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_CONTRACT.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_IMPLEMENTATION.md
docs/LATTICRA_SEAL_REQUEST_FRESHNESS_REPORT_SURFACE.md
docs/status/SEAL_REQUEST_FRESHNESS_STATUS.md
scripts/test-latticra-seal-request-freshness-status.sh
.github/workflows/latticra-seal-request-freshness-status.yml
docs/LATTICRA_SEAL_SIGNED_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNED_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNED_REQUEST_STATUS.md
include/latticra/seal_signed_request.h
src/seal_signed_request.c
tests/seal_signed_request_invariants.c
scripts/test-latticra-seal-signed-request-contract.sh
scripts/test-latticra-seal-signed-request.sh
scripts/test-latticra-seal-signed-request-status.sh
.github/workflows/latticra-seal-signed-request-status.yml
docs/LATTICRA_SEAL_POLICY_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_POLICY_DECISION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_POLICY_DECISION_REPORT_SURFACE.md
docs/status/SEAL_POLICY_DECISION_STATUS.md
docs/status/SEAL_POLICY_DECISION_REPORT_SURFACE_STATUS.md
scripts/test-latticra-seal-policy-decision-contract.sh
scripts/test-latticra-seal-policy-decision.sh
scripts/test-latticra-seal-policy-decision-status.sh
scripts/test-latticra-seal-policy-decision-report-surface.sh
scripts/test-latticra-seal-policy-decision-report-surface-status.sh
```

Current checkpoint

Current signed request status posture:

```
seal_signed_request_contract_present=1
seal_signed_request_implementation_present=1
seal_signed_request_header_present=1
seal_signed_request_source_present=1
seal_signed_request_invariant_test_present=1
seal_signed_request_runner_present=1
seal_signed_request_metadata_present=1
seal_signed_request_status_present=1
seal_signed_request_status_runner_present=1
seal_signed_request_status_workflow_present=1
seal_request_freshness_status_present=1
seal_request_freshness_status_runner_present=1
seal_request_freshness_status_workflow_present=1
request_freshness_predecessor_parameter_schema_status_present=1
signed_request_predecessor_request_freshness_status_present=1
readme_mentions_signed_request_metadata=1
readme_links_signed_request_contract=1
readme_links_signed_request_implementation=1
readme_links_signed_request_status=1
root_status_mentions_signed_request_status=1
status_index_links_signed_request_status=1
foundation_index_links_signed_request_status=1
project_notes_point_to_policy_decision_status=1
signed_request_profile=latticra-seal-signed-request/0.1
signed_request_supported=0
signature_generation_supported=0
signature_verification_supported=0
signature_present=0
signature_valid=0
signature_algorithm_declared=0
signing_key_id_present=0
signature_hash_present=0
signed_request_id_present=0
identity_binding_declared=0
context_binding_declared=0
parameter_binding_declared=0
freshness_binding_declared=0
policy_binding_declared=0
```

```
trust_store_supported=0
revocation_lookup_supported=0
mode=status-public-entry-alignment
signed_request_status_added=1
implementation_behavior_changed=0
signature_generation_implemented=0
signature_verification_implemented=0
signed_request_enforcement_implemented=0
trust_store_implemented=0
revocation_lookup_implemented=0
runtime_authority_granted=0
runtime_execution_added=0
effect_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
agent_execution_added=0
model_execution_added=0
tool_execution_added=0
shell_execution_added=0
cryptographic_enforcement_added=0
capability_enforcement_added=0
production_readiness_claimed=0
external_endorsement_claimed=0
```

Expected report posture

The metadata implementation renders the current report-only signed request posture, including:

```
signed_request_profile=latticra-seal-signed-request/0.1
signed_request_id=unset
signature_algorithm=unset
signing_key_id=unset
signature_hash=unset
signed_request_supported=0
signature_generation_supported=0
signature_verification_supported=0
signature_present=0
signature_valid=0
signature_algorithm_declared=0
signing_key_id_present=0
signature_hash_present=0
signed_request_id_present=0
identity_binding_declared=0
context_binding_declared=0
parameter_binding_declared=0
freshness_binding_declared=0
policy_binding_declared=0
trust_store_supported=0
revocation_lookup_supported=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=report-only
decision=report-only
reason=signed-request-metadata-only
status=signed-request-metadata
```

Validation

This status/public-entry alignment is covered by:

```
sh scripts/test-latticra-seal-signed-request-status.sh
```

The underlying checkpoint remains covered by:

```
sh scripts/test-latticra-seal-signed-request-contract.sh
sh scripts/test-latticra-seal-signed-request.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
```

```
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal status rollup status: ok
seal agentic automation security status: ok
seal parameter schema status: ok
seal request freshness status: ok
seal signed request status: ok
```

Boundary

This status record is documentation/status/public-entry alignment only.

This refresh adds the signed request status guard workflow and records the guarded request freshness status predecessor without changing the report-only signed request metadata, implementation, or status surface.

It does not add signature generation, signature verification, public-key parsing, trust-store loading, private-key handling, key generation, hardware-key use, revocation lookup, network trust lookup, signed request enforcement, MCP protocol behavior, MCP server behavior, MCP client behavior, AI agent execution, model execution, tool execution, shell execution, runtime behavior, host reads, host writes, network behavior, cryptographic verification, key handling, receipt verification, capability enforcement, policy enforcement, runtime enforcement, production readiness, external endorsement, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is policy decision status/workflow guard alignment.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_SIGNER_HANDOFF_STATUS.md

Source title: Latticra Seal Signer Handoff Status

Lines: 164 | Words: 486 | SHA-256: 767cfa7e8fb2cd2be303e933695e0475084eeb1d82b74804c964c108e1564a1f

Latticra Seal Signer Handoff Status

Status: status record for Latticra Seal signer handoff metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal signer handoff implementation. This record does not implement signing, signature verification, signer invocation, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal signer handoff metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, signer-not-invoked, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
include/latticra/seal_signer_handoff.h
src/seal_signer_handoff.c
tests/seal_signer_handoff_invariants.c
include/latticra/seal_signer_invocation.h
src/seal_signer_invocation.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signer-handoff.sh
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signer-invocation-status.sh
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNING_AUTHORIZATION_STATUS.md
scripts/test-latticra-seal-signing-authorization-status.sh
.github/workflows/latticra-seal-signing-authorization-status.yml
scripts/test-latticra-seal-signer-handoff-status.sh
.github/workflows/latticra-seal-signer-handoff-status.yml
```

Current checkpoint

Current signer handoff metadata posture:

```
seal_signer_handoff_contract_present=1
seal_signer_handoff_implementation_present=1
seal_signer_handoff_header_present=1
seal_signer_handoff_source_present=1
seal_signer_handoff_invariant_test_present=1
seal_signer_handoff_runner_present=1
seal_signer_handoff_metadata_present=1
seal_signer_handoff_status_present=1
seal_signer_handoff_status_runner_present=1
seal_signer_handoff_status_workflow_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_implementation_present=1
seal_signer_invocation_header_present=1
seal_signer_invocation_source_present=1
seal_signer_invocation_invariant_test_present=1
seal_signer_invocation_runner_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
seal_signing_authorization_contract_present=1
seal_signing_authorization_implementation_present=1
seal_signing_authorization_status_present=1
seal_signing_authorization_status_runner_present=1
seal_signing_authorization_status_workflow_present=1
signer_handoff_predecessor_signing_authorization_status_present=1
readme_links_signer_handoff_status=1
root_status_mentions_signer_handoff_status=1
status_index_links_signer_handoff_status=1
foundation_index_links_signer_handoff_status=1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
```

```
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=signer-handoff-metadata
signer_handoff_status_added=1
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation is covered by:

```
sh scripts/test-latticra-seal-signer-handoff-status.sh
sh scripts/test-latticra-seal-signer-handoff-contract.sh
sh scripts/test-latticra-seal-signer-handoff.sh
sh scripts/test-latticra-seal-signer-invocation-contract.sh
sh scripts/test-latticra-seal-signer-invocation.sh
sh scripts/test-latticra-seal-signer-invocation-status.sh
```

The predecessor signing authorization implementation remains covered by:

```
sh scripts/test-latticra-seal-signing-authorization-contract.sh
sh scripts/test-latticra-seal-signing-authorization.sh
sh scripts/test-latticra-seal-signing-authorization-status.sh
```

Expected output:

```
seal signer handoff status: ok
seal signer handoff contract: ok
seal signer handoff invariants: ok
seal signer invocation contract: ok
seal signer invocation invariants: ok
seal signer invocation status: ok
seal signing authorization contract: ok
seal signing authorization invariants: ok
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds an explicit status guard workflow and records the signing-authorization status predecessor without changing the signer-handoff implementation.

It does not add signing, verification, signer invocation, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is signer invocation status/workflow guard alignment or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

The signer invocation metadata implementation, status/public-entry alignment, signing operation contract, signing operation metadata implementation, signing operation status/public-entry alignment, key-handling boundary contract, and key-handling metadata implementation now exist as guarded checkpoints. Future work must not add signing, verification, signer invocation behavior, private-key handling, key generation, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_SIGNER_INVOCATION_STATUS.md

Source title: Latticra Seal Signer Invocation Status

Lines: 165 | Words: 465 | SHA-256: f24993ea1019d4f70a3967c1d69b38a9a546a32c4a3ee7b1d2d18de4a369a50c

Latticra Seal Signer Invocation Status

Status: status record for Latticra Seal signer invocation metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal signer invocation implementation. This record does not implement signing, signature verification, signer invocation behavior, signer process execution, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal signer invocation metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, signer-not-invoked, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
include/latticra/seal_signer_invocation.h
include/latticra/seal_signing_operation.h
src/seal_signer_invocation.c
src/seal_signing_operation.c
tests/seal_signer_invocation_invariants.c
tests/seal_signing_operation_invariants.c
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signing-operation-contract.sh
scripts/test-latticra-seal-signing-operation.sh
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_HANDOFF_STATUS.md
include/latticra/seal_signer_handoff.h
src/seal_signer_handoff.c
tests/seal_signer_handoff_invariants.c
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signer-handoff.sh
scripts/test-latticra-seal-signer-handoff-status.sh
.github/workflows/latticra-seal-signer-handoff-status.yml
scripts/test-latticra-seal-signer-invocation-status.sh
.github/workflows/latticra-seal-signer-invocation-status.yml
```

Current checkpoint

Current signer invocation metadata posture:

```
seal_signer_invocation_contract_present=1
seal_signer_invocation_implementation_present=1
seal_signer_invocation_header_present=1
seal_signer_invocation_source_present=1
```

```
seal_signer_invocation_invariant_test_present=1
seal_signer_invocation_runner_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
seal_signer_invocation_status_runner_present=1
seal_signer_invocation_status_workflow_present=1
seal_signing_operation_contract_present=1
seal_signing_operation_metadata_present=1
seal_signer_handoff_contract_present=1
seal_signer_handoff_implementation_present=1
seal_signer_handoff_status_present=1
seal_signer_handoff_status_runner_present=1
seal_signer_handoff_status_workflow_present=1
signer_invocation_predecessor_signer_handoff_status_present=1
readme_links_signer_invocation_status=1
root_status_mentions_signer_invocation_status=1
status_index_links_signer_invocation_status=1
foundation_index_links_signer_invocation_status=1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=signer-invocation-metadata
signer_invocation_status_added=1
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation is covered by:

```
sh scripts/test-latticra-seal-signer-invocation-status.sh
sh scripts/test-latticra-seal-signer-invocation-contract.sh
sh scripts/test-latticra-seal-signer-invocation.sh
sh scripts/test-latticra-seal-signing-operation-contract.sh
sh scripts/test-latticra-seal-signing-operation.sh
```

The predecessor signer handoff implementation remains covered by:

```
sh scripts/test-latticra-seal-signer-handoff-contract.sh
sh scripts/test-latticra-seal-signer-handoff.sh
sh scripts/test-latticra-seal-signer-handoff-status.sh
```

Expected output:

```
seal signer invocation status: ok
seal signer invocation contract: ok
seal signer invocation invariants: ok
seal signing operation contract: ok
seal signing operation invariants: ok
seal signer handoff contract: ok
seal signer handoff invariants: ok
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds an explicit status guard workflow and records the signer-handoff status predecessor without changing the signer-invocation implementation.

It does not add signing, verification, signer invocation behavior, signer process execution, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is signing operation status/workflow guard alignment or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

That future slice must not add signing, verification, signer invocation behavior, private-key handling, key generation, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_SIGNING_AUTHORIZATION_STATUS.md

Source title: Latticra Seal Signing Authorization Status

Lines: 178 | Words: 473 | SHA-256: 0e833a251431294794c930706c4a7b11918526735fef6d0a631ce98eb9609c18

Latticra Seal Signing Authorization Status

Status: status record for Latticra Seal signing authorization metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal signing authorization implementation. This record does not implement signing, signature verification, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal signing authorization metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_AUTHORIZATION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_HANDOFF_STATUS.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
docs/status/SEAL_SIGNATURE_REQUEST_STATUS.md
include/latticra/seal_signing_authorization.h
include/latticra/seal_signer_handoff.h
include/latticra/seal_signer_invocation.h
src/seal_signing_authorization.c
src/seal_signer_handoff.c
src/seal_signer_invocation.c
tests/seal_signing_authorization_invariants.c
tests/seal_signer_handoff_invariants.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signing-authorization-contract.sh
scripts/test-latticra-seal-signing-authorization.sh
scripts/test-latticra-seal-signer-handoff-contract.sh
scripts/test-latticra-seal-signer-handoff.sh
scripts/test-latticra-seal-signer-handoff-status.sh
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signer-invocation-status.sh
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_CONTRACT.md
docs/LATTICRA_SEAL_SIGNATURE_REQUEST_IMPLEMENTATION.md
docs/status/SEAL_SIGNATURE_REQUEST_STATUS.md
scripts/test-latticra-seal-signature-request-status.sh
scripts/test-latticra-seal-signing-authorization-status.sh
.github/workflows/latticra-seal-signature-request-status.yml
.github/workflows/latticra-seal-signing-authorization-status.yml
```

Current checkpoint

Current signing authorization metadata posture:

```
seal_signing_authorization_contract_present=1
seal_signing_authorization_implementation_present=1
seal_signing_authorization_header_present=1
seal_signing_authorization_source_present=1
seal_signing_authorization_invariant_test_present=1
seal_signing_authorization_runner_present=1
seal_signing_authorization_status_present=1
seal_signing_authorization_status_runner_present=1
seal_signing_authorization_status_workflow_present=1
seal_signer_handoff_contract_present=1
seal_signer_handoff_implementation_present=1
seal_signer_handoff_header_present=1
seal_signer_handoff_source_present=1
seal_signer_handoff_invariant_test_present=1
seal_signer_handoff_runner_present=1
seal_signer_handoff_metadata_present=1
seal_signer_handoff_status_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_implementation_present=1
seal_signer_invocation_metadata_present=1
seal_signer_invocation_status_present=1
seal_signature_request_contract_present=1
seal_signature_request_implementation_present=1
seal_signature_request_status_present=1
seal_signature_request_status_runner_present=1
seal_signature_request_status_workflow_present=1
signing_authorization_predecessor_signature_request_status_present=1
readme_links_signing_authorization_status=1
root_status_mentions_signing_authorization_status=1
status_index_links_signing_authorization_status=1
foundation_index_links_signing_authorization_status=1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
signature_request_state=requested-metadata-only
signature_request_ready=1
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
```

```
signature_performed=0
verification_performed=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=signing-authorization-metadata
signing_authorization_status_added=1
signing_added=0
signature_verification_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation is covered by:

```
sh scripts/test-latticra-seal-signing-authorization-status.sh
sh scripts/test-latticra-seal-signing-authorization-contract.sh
sh scripts/test-latticra-seal-signing-authorization.sh
sh scripts/test-latticra-seal-signer-handoff-contract.sh
sh scripts/test-latticra-seal-signer-handoff.sh
sh scripts/test-latticra-seal-signer-handoff-status.sh
sh scripts/test-latticra-seal-signer-invocation-contract.sh
sh scripts/test-latticra-seal-signer-invocation.sh
sh scripts/test-latticra-seal-signer-invocation-status.sh
```

The predecessor signature-request implementation remains covered by:

```
sh scripts/test-latticra-seal-signature-request-contract.sh
sh scripts/test-latticra-seal-signature-request.sh
sh scripts/test-latticra-seal-signature-request-status.sh
```

Expected output:

```
seal signing authorization status: ok
seal signing authorization contract: ok
seal signing authorization invariants: ok
seal signer handoff contract: ok
seal signer handoff invariants: ok
seal signer handoff status: ok
seal signer invocation contract: ok
seal signer invocation invariants: ok
seal signer invocation status: ok
seal signature request contract: ok
seal signature request invariants: ok
seal report envelope status: ok
seal signature request status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds an explicit status guard workflow and records the signature-request status predecessor without changing the signing-authorization implementation.

It does not add signing, verification, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is signer handoff status/workflow guard alignment or another narrow status/index alignment follow-up that still must not add signing without separate implementation, key-handling, key-material, and guard contracts.

That future slice must not add signing, verification, private-key handling, key generation, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_SIGNING_OPERATION_STATUS.md

Source title: Latticra Seal Signing Operation Status

Lines: 178 | Words: 478 | SHA-256: 898f9ae14523e06f354ce26a5cf1ef0dc6b2189e2df44ce464e0858f4f4641d3

Latticra Seal Signing Operation Status

Status: status record for Latticra Seal signing operation metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal signing operation implementation. This record does not implement signing, signature verification, signer invocation behavior, signer process execution, key generation, private-key handling, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal signing operation metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsigned, signer-not-invoked, operation-not-performed, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_SIGNING_OPERATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNING_OPERATION_IMPLEMENTATION.md
docs/LATTICRA_SEAL_KEY_HANDLING_CONTRACT.md
docs/LATTICRA_SEAL_KEY_HANDLING_IMPLEMENTATION.md
docs/status/SEAL_KEY_HANDLING_STATUS.md
docs/status/SEAL_SIGNING_OPERATION_STATUS.md
include/latticra/seal_signing_operation.h
src/seal_signing_operation.c
tests/seal_signing_operation_invariants.c
include/latticra/seal_key_handling.h
src/seal_key_handling.c
tests/seal_key_handling_invariants.c
scripts/test-latticra-seal-signing-operation-contract.sh
scripts/test-latticra-seal-signing-operation.sh
scripts/test-latticra-seal-signing-operation-status.sh
scripts/test-latticra-seal-key-handling-contract.sh
scripts/test-latticra-seal-key-handling.sh
scripts/test-latticra-seal-key-handling-status.sh
docs/LATTICRA_SEAL_SIGNER_INVOCATION_CONTRACT.md
docs/LATTICRA_SEAL_SIGNER_INVOCATION_IMPLEMENTATION.md
docs/status/SEAL_SIGNER_INVOCATION_STATUS.md
include/latticra/seal_signer_invocation.h
src/seal_signer_invocation.c
tests/seal_signer_invocation_invariants.c
scripts/test-latticra-seal-signer-invocation-contract.sh
scripts/test-latticra-seal-signer-invocation.sh
scripts/test-latticra-seal-signer-invocation-status.sh
.github/workflows/latticra-seal-signer-invocation-status.yml
.github/workflows/latticra-seal-signing-operation-status.yml
```

Current checkpoint

Current signing operation metadata posture:

```
seal_signing_operation_contract_present=1
seal_signing_operation_implementation_present=1
seal_signing_operation_header_present=1
seal_signing_operation_source_present=1
seal_signing_operation_invariant_test_present=1
seal_signing_operation_runner_present=1
seal_signing_operation_metadata_present=1
seal_signing_operation_status_present=1
seal_signing_operation_status_runner_present=1
seal_signing_operation_status_workflow_present=1
seal_key_handling_contract_present=1
seal_key_handling_metadata_present=1
seal_key_handling_status_present=1
seal_signer_invocation_contract_present=1
seal_signer_invocation_implementation_present=1
seal_signer_invocation_status_present=1
seal_signer_invocation_status_runner_present=1
seal_signer_invocation_status_workflow_present=1
signing_operation_predecessor_signer_invocation_status_present=1
readme_links_signing_operation_status=1
root_status_mentions_signing_operation_status=1
status_index_links_signing_operation_status=1
foundation_index_links_signing_operation_status=1
signing_operation_profile=latticra-seal-signing-operation/0.1
signer_invocation_profile=latticra-seal-signer-invocation/0.1
signer_handoff_profile=latticra-seal-signer-handoff/0.1
signing_authorization_profile=latticra-seal-signing-authorization/0.1
signature_request_profile=latticra-seal-signature-request/0.1
requested_signature=Ed25519-development
requested_signing_authorization=metadata-only
requested_signer_handoff=metadata-only
requested_signer_invocation=metadata-only
requested_signing_operation=metadata-only
signing_authorization_state=authorized-metadata-only
signing_authorization_ready=1
signer_handoff_state=handoff-metadata-only
signer_handoff_ready=1
signer_invocation_state=invocation-metadata-only
signer_invocation_ready=1
signing_operation_state=operation-metadata-only
signing_operation_ready=1
signature_performed=0
verification_performed=0
signer_invoked=0
private_key_handling=0
key_generation_performed=0
trust_store_loaded=0
revocation_lookup_performed=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=signing-operation-metadata
signing_operation_status_added=1
signing_added=0
signature_verification_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
public_key_parsing_added=0
key_material_loading_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-signing-operation-contract.sh
sh scripts/test-latticra-seal-signing-operation.sh
sh scripts/test-latticra-seal-signing-operation-status.sh
sh scripts/test-latticra-seal-key-handling-contract.sh
sh scripts/test-latticra-seal-key-handling.sh
sh scripts/test-latticra-seal-key-handling-status.sh
```

The predecessor signer invocation implementation remains covered by:

```
sh scripts/test-latticra-seal-signer-invocation-contract.sh
sh scripts/test-latticra-seal-signer-invocation.sh
sh scripts/test-latticra-seal-signer-invocation-status.sh
```

Expected output:

```
seal signing operation contract: ok
seal signing operation invariants: ok
seal signing operation status: ok
seal key-handling contract: ok
seal key-handling invariants: ok
seal key-handling status: ok
seal signer invocation contract: ok
seal signer invocation invariants: ok
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds an explicit status guard workflow and records the signer-invocation status predecessor without changing the signing-operation implementation.

It does not add signing, verification, signer invocation behavior, signer process execution, private-key handling, key generation, trust-store behavior, revocation lookup, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is key-handling status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add signing, verification, signer invocation behavior, public-key parsing, private-key handling, key material loading, key generation, trust-store behavior, revocation lookup, host behavior, network behavior, runtime authority, capability enforcement, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_STATUS_ROLLUP_STATUS.md

Source title: Latticra Seal Status Rollup Status

Lines: 182 | Words: 543 | SHA-256: f294bad8c8128fb248407a19e7e98d3212aaf35c409290ab6b480049566e7530

Latticra Seal Status Rollup Status

Status: status record for Latticra Seal status rollup metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal status rollup implementation. This record does not add runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, signing, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal status rollup metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unverified, denied-by-gate, inactive-runtime, runtime-boundary-disabled, host-inactive, network-inactive, runtime-authority-denied, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_STATUS_ROLLUP_CONTRACT.md
docs/LATTICRA_SEAL_STATUS_ROLLUP_IMPLEMENTATION.md
docs/status/SEAL_STATUS_ROLLUP_STATUS.md
include/latticra/seal_status_rollup.h
src/seal_status_rollup.c
tests/seal_status_rollup_invariants.c
scripts/test-latticra-seal-status-rollup-contract.sh
scripts/test-latticra-seal-status-rollup.sh
scripts/test-latticra-seal-status-rollup-status.sh
.github/workflows/latticra-seal-status-rollup-status.yml
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_CONTRACT.md
docs/LATTICRA_SEAL_RUNTIME_HANDOFF_IMPLEMENTATION.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
include/latticra/seal_runtime_handoff.h
src/seal_runtime_handoff.c
tests/seal_runtime_handoff_invariants.c
scripts/test-latticra-seal-runtime-handoff-contract.sh
scripts/test-latticra-seal-runtime-handoff.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
.github/workflows/latticra-seal-runtime-handoff-status.yml
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_CONTRACT.md
docs/LATTICRA_SEAL_AGENTIC_AUTOMATION_SECURITY_IMPLEMENTATION.md
docs/status/SEAL_AGENTIC_AUTOMATION_SECURITY_STATUS.md
scripts/test-latticra-seal-agentic-automation-security-status.sh
```

Current checkpoint

Current status rollup metadata posture:

```
seal_status_rollup_contract_present=1
seal_status_rollup_implementation_present=1
seal_status_rollup_header_present=1
seal_status_rollup_source_present=1
seal_status_rollup_invariant_test_present=1
seal_status_rollup_runner_present=1
seal_status_rollup_metadata_present=1
seal_status_rollup_status_present=1
seal_status_rollup_status_runner_present=1
seal_status_rollup_status_workflow_present=1
seal_runtime_handoff_contract_present=1
seal_runtime_handoff_implementation_present=1
seal_runtime_handoff_status_present=1
seal_runtime_handoff_status_runner_present=1
seal_runtime_handoff_status_workflow_present=1
runtime_handoff_predecessor_effect_decision_status_present=1
status_rollup_predecessor_runtime_handoff_status_present=1
seal_agentic_automation_security_contract_present=1
seal_agentic_automation_security_implementation_present=1
seal_agentic_automation_security_status_present=1
```

```
rollup_profile=latticra-seal-status-rollup/0.1
report_present=1
measurement_present=1
manifest_present=1
signature_policy_present=1
signature_metadata_present=1
verification_policy_present=1
verification_receipt_present=1
capability_gate_present=1
effect_decision_present=1
runtime_handoff_present=1
cryptographic_verification_supported=0
verified=0
capability_gate_allowed=0
effect_allowed=0
effect_performed=0
handoff_active=0
runtime_boundary_state=disabled
runtime_effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
capability_enforcement_performed=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
network_lookup_allowed=0
revocation_lookup_allowed=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
mode=metadata-only
rollup_state=metadata-only
status=status-rollup-metadata
error=ok
status_rollup_status_added=1
runtime_execution_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
runtime_authority_added=0
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-status-rollup-contract.sh
sh scripts/test-latticra-seal-status-rollup.sh
sh scripts/test-latticra-seal-status-rollup-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-agentic-automation-security-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
```

```
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal status rollup contract: ok
seal status rollup invariants: ok
seal status rollup status: ok
seal runtime handoff status: ok
seal agentic automation security status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the status rollup status guard workflow and records the guarded runtime handoff status predecessor without changing the metadata-only status rollup implementation.

It does not add runtime execution, runtime authority, effect execution, capability enforcement, cryptographic verification, verified receipt authority, signing, public-key byte verification, public-key trust-store behavior, key material loading, private-key handling, key generation, hardware-key use, revocation lookup, signer invocation behavior, signer process execution, host reads, host writes, network behavior, shell execution, tool execution, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is agentic automation security status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must preserve the no-effect posture and must not implement real policy evaluation, policy enforcement, runtime execution, effect execution, capability enforcement, cryptographic verification, signature verification, freshness validation, replay detection, authority grants, host behavior, network behavior, MCP behavior, AI agent execution, model execution, tool execution, or shell execution.

docs/status/SEAL_VERIFICATION_POLICY_STATUS.md

Source title: Latticra Seal Verification Policy Status

Lines: 186 | Words: 535 | SHA-256: d2d730e8d67be3342026515bb00049681f897b3e263e7db6383667d48f5834ef

Latticra Seal Verification Policy Status

Status: status record for Latticra Seal verification policy metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal verification policy implementation. This record does not add cryptographic verification, signing, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal verification policy metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unsupported-for-cryptographic-verification, public-key-identity-label-only, key-material-not-loaded, private-key-denied, network-denied, revocation-denied, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_POLICY_STATUS.md
include/latticra/seal_verification_policy.h
src/seal_verification_policy.c
tests/seal_verification_policy_invariants.c
scripts/test-latticra-seal-verification-policy-contract.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-policy-status.sh
.github/workflows/latticra-seal-verification-policy-status.yml
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md
include/latticra/seal_verification_receipt.h
src/seal_verification_receipt.c
tests/seal_verification_receipt_invariants.c
scripts/test-latticra-seal-verification-receipt-contract.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-verification-receipt-status.sh
docs/status/SEAL_CAPABILITY_GATE_STATUS.md
scripts/test-latticra-seal-capability-gate-status.sh
docs/status/SEAL_EFFECT_DECISION_STATUS.md
scripts/test-latticra-seal-effect-decision-status.sh
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
scripts/test-latticra-seal-runtime-handoff-status.sh
docs/LATTICRA_SEAL_SIGNATURE_IMPLEMENTATION.md
include/latticra/seal_signature.h
src/seal_signature.c
scripts/test-latticra-seal-signature.sh
docs/status/SEAL_KEY_PARSING_STATUS.md
scripts/test-latticra-seal-key-parsing-status.sh
.github/workflows/latticra-seal-key-parsing-status.yml
```

Current checkpoint

Current verification policy metadata posture:

```
seal_verification_policy_contract_present=1
seal_verification_policy_implementation_present=1
seal_verification_policy_header_present=1
seal_verification_policy_source_present=1
seal_verification_policy_invariant_test_present=1
seal_verification_policy_runner_present=1
seal_verification_policy_metadata_present=1
seal_verification_policy_status_present=1
seal_verification_policy_status_runner_present=1
seal_verification_policy_status_workflow_present=1
seal_verification_receipt_contract_present=1
seal_verification_receipt_implementation_present=1
seal_verification_receipt_status_present=1
seal_capability_gate_status_present=1
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
seal_key_parsing_status_present=1
seal_key_parsing_status_runner_present=1
seal_key_parsing_status_workflow_present=1
verification_policy_predecessor_key_parsing_status_present=1
seal_signature_metadata_present=1
verification_policy_profile=latticra-seal-verification-policy/0.1
signature_profile=latticra-seal-signature/0.1
manifest_profile=latticra-seal-manifest/0.1
artifact_digest_algorithm=SHA-256
signer_identity_label=latticra-dev-signer
signature_algorithm=Ed25519-development
public_key_identity_label=latticra-dev-public-key
```

```
public_key_identity_only=1
trust_source=local-metadata-only
requested_verification_policy=metadata-only
verification_policy_ready=1
verification_state=unsupported
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verification_performed=0
public_key_material_handling=0
public_key_bytes_consumed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
network_lookup_allowed=0
revocation_lookup_allowed=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
runtime_authority_granted=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=verification-policy-metadata
error=ok
verification_policy_status_added=1
cryptographic_verification_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-verification-policy-contract.sh
sh scripts/test-latticra-seal-verification-policy.sh
sh scripts/test-latticra-seal-verification-policy-status.sh
sh scripts/test-latticra-seal-verification-receipt-contract.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-key-parsing-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy contract: ok
seal verification policy invariants: ok
seal verification policy status: ok
seal verification receipt contract: ok
```

```
seal verification receipt status: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal key parsing status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the verification policy status guard workflow and records the guarded key parsing status predecessor without changing the metadata-only verification policy implementation.

It does not add cryptographic verification, signing, public-key byte verification, public-key trust-store behavior, key material loading, private-key handling, key generation, hardware-key use, revocation lookup, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is verification receipt status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signer invocation behavior, host behavior, network behavior, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md

Source title: Latticra Seal Verification Receipt Status

Lines: 192 | Words: 545 | SHA-256: bbdf8908f39d9e3fb8ff8b5f6831d7873058ae5cfecb8034b07cc0ded0a1a891

Latticra Seal Verification Receipt Status

Status: status record for Latticra Seal verification receipt metadata Source: local follow-up slice Scope: status and public-entry alignment after the metadata-only Seal verification receipt implementation. This record does not add cryptographic verification, verified receipt authority, signing, public-key byte verification, key material loading, private-key handling, key generation, hardware-key use, trust-store loading, revocation lookup, object sealing, runtime handoff execution, runtime authority, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal verification receipt metadata implementation visible as a current project checkpoint.

It records that the implementation is bounded, deterministic, metadata-only, unverified, unsupported-for-cryptographic-verification, capability-gate-denied, authority-unusable, runtime-authority-denied, and no-effect.

Reviewed files

```
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_RECEIPT_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_RECEIPT_STATUS.md
include/latticra/seal_verification_receipt.h
src/seal_verification_receipt.c
tests/seal_verification_receipt_invariants.c
scripts/test-latticra-seal-verification-receipt-contract.sh
scripts/test-latticra-seal-verification-receipt.sh
scripts/test-latticra-seal-verification-receipt-status.sh
.github/workflows/latticra-seal-verification-receipt-status.yml
docs/LATTICRA_SEAL_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_CAPABILITY_GATE_IMPLEMENTATION.md
docs/status/SEAL_CAPABILITY_GATE_STATUS.md
include/latticra/seal_capability_gate.h
src/seal_capability_gate.c
tests/seal_capability_gate_invariants.c
scripts/test-latticra-seal-capability-gate-contract.sh
scripts/test-latticra-seal-capability-gate.sh
scripts/test-latticra-seal-capability-gate-status.sh
docs/status/SEAL_EFFECT_DECISION_STATUS.md
docs/status/SEAL_RUNTIME_HANDOFF_STATUS.md
scripts/test-latticra-seal-effect-decision-status.sh
scripts/test-latticra-seal-runtime-handoff-status.sh
docs/LATTICRA_SEAL_VERIFICATION_POLICY_CONTRACT.md
docs/LATTICRA_SEAL_VERIFICATION_POLICY_IMPLEMENTATION.md
docs/status/SEAL_VERIFICATION_POLICY_STATUS.md
include/latticra/seal_verification_policy.h
src/seal_verification_policy.c
tests/seal_verification_policy_invariants.c
scripts/test-latticra-seal-verification-policy-contract.sh
scripts/test-latticra-seal-verification-policy.sh
scripts/test-latticra-seal-verification-policy-status.sh
.github/workflows/latticra-seal-verification-policy-status.yml
```

Current checkpoint

Current verification receipt metadata posture:

```
seal_verification_receipt_contract_present=1
seal_verification_receipt_implementation_present=1
seal_verification_receipt_header_present=1
seal_verification_receipt_source_present=1
seal_verification_receipt_invariant_test_present=1
seal_verification_receipt_runner_present=1
seal_verification_receipt_metadata_present=1
seal_verification_receipt_status_present=1
seal_verification_receipt_status_runner_present=1
seal_verification_receipt_status_workflow_present=1
seal_capability_gate_contract_present=1
seal_capability_gate_implementation_present=1
seal_capability_gate_status_present=1
seal_effect_decision_status_present=1
seal_runtime_handoff_status_present=1
seal_verification_policy_contract_present=1
seal_verification_policy_implementation_present=1
seal_verification_policy_status_present=1
seal_verification_policy_status_runner_present=1
seal_verification_policy_status_workflow_present=1
verification_receipt_predecessor_verification_policy_status_present=1
receipt_profile=latticra-seal-verification-receipt/0.1
verification_policy_profile=latticra-seal-verification-policy/0.1
signature_profile=latticra-seal-signature/0.1
manifest_profile=latticra-seal-manifest/0.1
artifact_digest_algorithm=SHA-256
signer_identity_label=latticra-dev-signer
signature_algorithm=Ed25519-development
public_key_identity_label=latticra-dev-public-key
trust_source=local-metadata-only
requested_verification_receipt=metadata-only
verification_receipt_ready=1
verification_state=unsupported
receipt_state=unverified-metadata
cryptographic_verification_supported=0
cryptographic_verification_performed=0
verification_performed=0
verified=0
invalid=0
```

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
public_key_material_handling=0
public_key_bytes_consumed=0
key_material_loaded=0
private_key_handling=0
key_generation_performed=0
hardware_key_used=0
trust_store_loaded=0
network_lookup_allowed=0
revocation_lookup_allowed=0
revocation_lookup_performed=0
signature_performed=0
signer_invoked=0
handoff_performed=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
mode=metadata-only
status=verification-receipt-metadata
error=ok
verification_receipt_status_added=1
cryptographic_verification_added=0
verified_receipt_authority_added=0
signature_verification_added=0
public_key_byte_verification_added=0
key_material_loading_added=0
private_key_handling_added=0
key_generation_added=0
hardware_key_use_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
signing_added=0
signer_invocation_behavior_added=0
signer_process_execution_added=0
object_sealing_added=0
runtime_handoff_execution_added=0
effect_execution_added=0
capability_enforcement_added=0
policy_persistence_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

The implementation and status surface are covered by:

```
sh scripts/test-latticra-seal-verification-receipt-contract.sh
sh scripts/test-latticra-seal-verification-receipt.sh
sh scripts/test-latticra-seal-verification-receipt-status.sh
sh scripts/test-latticra-seal-capability-gate-contract.sh
sh scripts/test-latticra-seal-capability-gate-status.sh
sh scripts/test-latticra-seal-effect-decision-status.sh
sh scripts/test-latticra-seal-runtime-handoff-status.sh
sh scripts/test-latticra-seal-verification-policy-status.sh
```

Expected output:

```
seal report envelope status: ok
seal signature request status: ok
seal signing authorization status: ok
seal signer handoff status: ok
seal signer invocation status: ok
seal signing operation status: ok
seal key-handling status: ok
seal key-material status: ok
seal public-key parsing status: ok
seal key parsing status: ok
seal verification policy status: ok
seal verification receipt contract: ok
seal verification receipt invariants: ok
seal verification receipt status: ok
seal capability gate contract: ok
seal capability gate status: ok
seal effect decision status: ok
seal runtime handoff status: ok
seal verification policy status: ok
```

Boundary

This status record is documentation/status alignment only.

This refresh adds the verification receipt status guard workflow and records the guarded verification policy status predecessor without changing the metadata-only verification receipt implementation.

It does not add cryptographic verification, verified receipt authority, signing, public-key byte verification, public-key trust-store behavior, key material loading, private-key handling, key generation, hardware-key use, revocation lookup, signer invocation behavior, signer process execution, runtime handoff execution, host reads, host writes, network behavior, shell execution, tool execution, capability enforcement, policy persistence, object sealing, kernel behavior, production readiness, or authority grants.

Current next valid slice

The next valid Latticra Seal slice is capability gate status/workflow guard alignment or another narrow status/index alignment follow-up.

That future slice must not add runtime execution, effect execution, capability enforcement, runtime authority, cryptographic verification, verified receipt authority, signing, key material loading, private-key handling, key generation, hardware-key use, trust-store behavior, revocation lookup, signer invocation behavior, host behavior, network behavior, object sealing, or kernel behavior unless separately implemented and guarded.

docs/status/SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md

Source title: Latticra Seal Verified Capability Gate Status

Lines: 139 | Words: 406 | SHA-256: bd7df0217b95106ba6cc8fa41ae53426bbc9fd56491f4297fc7ae88ff83c7457

Latticra Seal Verified Capability Gate Status

Status: status record for the Latticra Seal verified capability gate metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal verified capability gate contract and metadata implementation, now including the stricter crypto-graduation-gated capability entry point. This record does not add capability enforcement, effect execution, runtime authority, host effects, network behavior, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal verified capability gate implementation visible from public entry points.

The gate consumes verified receipt promotion metadata and evaluates a narrow local metadata-only capability/effect allowlist. The stricter entry point also consumes a passing crypto graduation gate and requires it to match the receipt metadata before the same allowlist can pass.

It is policy evaluation metadata, not permission enforcement.

Reviewed files

```
docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_CONTRACT.md
docs/LATTICRA_SEAL_VERIFIED_CAPABILITY_GATE_IMPLEMENTATION.md
docs/status/SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md
docs/status/SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md
docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md
include/latticra/seal_verified_capability_gate.h
include/latticra/seal_crypto_graduation_gate.h
src/seal_verified_capability_gate.c
src/seal_crypto_graduation_gate.c
tests/seal_verified_capability_gate_invariants.c
scripts/test-latticra-seal-verified-capability-gate-contract.sh
scripts/test-latticra-seal-verified-capability-gate.sh
scripts/test-latticra-seal-verified-capability-gate-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current verified capability gate posture:

```
seal_verified_capability_gate_contract_present=1
seal_verified_capability_gate_implementation_present=1
seal_verified_capability_gate_header_present=1
seal_verified_capability_gate_source_present=1
seal_verified_capability_gate_invariant_test_present=1
seal_verified_capability_gate_runner_present=1
seal_verified_capability_gate_status_present=1
seal_verified_receipt_promotion_status_present=1
seal_crypto_graduation_gate_status_present=1
readme_links_verified_capability_gate_contract=1
readme_links_verified_capability_gate_implementation=1
readme_links_verified_capability_gate_status=1
root_status_mentions_verified_capability_gate_status=1
status_index_links_verified_capability_gate_status=1
foundation_index_links_verified_capability_gate_status=1
project_notes_mark_verified_capability_gate_status_complete=1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
receipt_state=verified
verification_state=verified
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
crypto_graduation_gate_state=graduated-authority-neutral
requested_capability=verified-receipt-report
requested_effect=report-only
requested_scope=local-fixture-scope
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
verified=1
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=1
gate_state=allowed-metadata-only
runtime_authority_granted=0
effect_performed=0
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
status=verified-capability-gate-metadata
verified_capability_gate_added=1
capability_enforcement_added=0
effect_execution_added=0
runtime_authority_granted=0
```

```
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-verified-capability-gate-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-verified-capability-gate.sh
```

Expected output:

```
seal verified capability gate status: ok
seal verified capability gate invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The verified capability gate implementation evaluates a verified receipt promotion record against a narrow local metadata-only allowlist. The crypto-bound entry point requires crypto_graduation_gate_passed=1 and standard_expectations_met=1 before that same metadata-only allowlist can pass. A successful gate may set gate_allowed=1 and gate_state=allowed-metadata-only, but it remains metadata-only and authority-neutral.

It does not add capability enforcement, effect execution, runtime behavior, host behavior, network behavior, signing, key generation, private-key handling, trust-store behavior, revocation lookup, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The next valid Latticra Seal slice is effect decision evaluation from an allowed crypto-graduation-gated metadata-only capability gate or another narrow status/index alignment follow-up.

docs/status/SEAL_VERIFIED_EFFECT_DECISION_STATUS.md

Source title: Latticra Seal Verified Effect Decision Status

Lines: 139 | Words: 430 | SHA-256: 3dc32ff7422d123babe6fce538227200a3c834ceb7787ed8f3d5ae360288c317

Latticra Seal Verified Effect Decision Status

Status: status record for the Latticra Seal verified effect decision metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal verified effect decision contract and metadata implementation, now carrying crypto graduation metadata forward when present on the verified capability gate. This record does not add effect execution, capability enforcement, runtime authority, host effects, network behavior, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal verified effect decision implementation visible from public entry points.

The decision consumes verified capability gate metadata and evaluates a narrow local metadata-only effect request. When the gate carries crypto graduation metadata, the decision requires that evidence to remain passed, standard-aligned, and authority-neutral before allowing the metadata-only effect classification.

It is effect classification metadata, not effect execution.

Reviewed files

```
docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_CONTRACT.md
docs/LATTICRA_SEAL_VERIFIED_EFFECT_DECISION_IMPLEMENTATION.md
docs/status/SEAL_VERIFIED_EFFECT_DECISION_STATUS.md
docs/status/SEAL_VERIFIED_CAPABILITY_GATE_STATUS.md
docs/status/SEAL_CRYPTO_GRADUATION_GATE_STATUS.md
include/latticra/seal_verified_effect_decision.h
src/seal_verified_effect_decision.c
tests/seal_verified_effect_decision_invariants.c
scripts/test-latticra-seal-verified-effect-decision-contract.sh
scripts/test-latticra-seal-verified-effect-decision.sh
scripts/test-latticra-seal-verified-effect-decision-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current verified effect decision posture:

```
seal_verified_effect_decision_contract_present=1
seal_verified_effect_decision_implementation_present=1
seal_verified_effect_decision_header_present=1
seal_verified_effect_decision_source_present=1
seal_verified_effect_decision_invariant_test_present=1
seal_verified_effect_decision_runner_present=1
seal_verified_effect_decision_status_present=1
seal_verified_capability_gate_status_present=1
seal_crypto_graduation_gate_status_present=1
readme_links_verified_effect_decision_contract=1
readme_links_verified_effect_decision_implementation=1
readme_links_verified_effect_decision_status=1
root_status_mentions_verified_effect_decision_status=1
status_index_links_verified_effect_decision_status=1
foundation_index_links_verified_effect_decision_status=1
project_notes_mark_verified_effect_decision_status_complete=1
decision_profile=latticra-seal-verified-effect-decision/0.1
gate_profile=latticra-seal-verified-capability-gate/0.1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
receipt_state=verified
verification_state=verified
crypto_graduation_profile=latticra-seal-crypto-graduation-gate/0.1
assurance_baseline_profile=latticra-cryptographic-assurance-key-management/0.1
crypto_graduation_gate_state=graduated-authority-neutral
requested_capability=verified-receipt-report
requested_effect=report-only
requested_scope=local-fixture-scope
crypto_graduation_gate_present=1
crypto_graduation_gate_passed=1
standard_expectations_met=1
local_verify_graduated=1
receipt_promotion_graduated=1
authority_promotion_allowed=0
verified=1
authority_usable=0
receipt_capability_gate_allowed=0
gate_allowed=1
gate_state=allowed-metadata-only
decision_state=allowed-report-only
effect_allowed=1
effect_performed=0
runtime_authority_granted=0
```

```
host_read_performed=0
host_write_performed=0
network_performed=0
error=ok
status=verified-effect-decision-metadata
verified_effect_decision_added=1
effect_execution_added=0
capability_enforcement_added=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-verified-effect-decision-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-verified-effect-decision.sh
```

Expected output:

```
seal verified effect decision status: ok
seal verified effect decision invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The verified effect decision implementation evaluates a verified capability gate record against a narrow local metadata-only effect request. If the gate includes crypto graduation evidence, the decision requires crypto_graduation_gate_passed=1, standard_expectations_met=1, and authority_promotion_allowed=0. A successful decision may set effect_allowed=1 and decision_state=allowed-report-only, but it remains metadata-only and authority-neutral.

It does not add effect execution, capability enforcement, runtime behavior, host behavior, network behavior, signing, key generation, private-key handling, trust-store behavior, revocation lookup, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The runtime handoff evaluation status/public-entry alignment now provides the predecessor for the next valid Latticra Seal slice: runtime handoff report or policy decision report propagation from eligible crypto-graduation-gated metadata-only runtime handoff evaluation, runtime handoff report status/public-entry alignment, or another narrow status/index alignment follow-up.

docs/status/SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md

Source title: Latticra Seal Verified Receipt Promotion Status

Lines: 124 | Words: 331 | SHA-256: 811de4a44bbea298eb23d1cf863eb8fcedc1b2e2f8210768381ba8b6b768baf9

Latticra Seal Verified Receipt Promotion Status

Status: status record for the Latticra Seal verified receipt promotion metadata surface Source: local follow-up slice Scope: status and public-entry alignment after the Seal verified receipt promotion contract and metadata implementation. This record does not add capability authorization, effect execution, runtime authority, signing, key generation, private-key storage, public-key trust stores, network trust lookup, revocation lookup, object sealing, host reads, host writes, kernel behavior, Fedora approval claims, production readiness, or operating-system behavior.

Purpose

This status record makes the Latticra Seal verified receipt promotion implementation visible from public entry points.

The promotion layer converts a successful local Ed25519 verify-only result into verified receipt metadata. The promotion is evidence promotion, not permission promotion.

Reviewed files

```
docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_CONTRACT.md
docs/LATTICRA_SEAL_VERIFIED_RECEIPT_PROMOTION_IMPLEMENTATION.md
docs/status/SEAL_VERIFIED_RECEIPT_PROMOTION_STATUS.md
docs/status/SEAL_ED25519_VERIFY_STATUS.md
include/latticra/seal_verified_receipt_promotion.h
src/seal_verified_receipt_promotion.c
tests/seal_verified_receipt_promotion_invariants.c
scripts/test-latticra-seal-verified-receipt-promotion-contract.sh
scripts/test-latticra-seal-verified-receipt-promotion.sh
scripts/test-latticra-seal-verified-receipt-promotion-status.sh
README.md
STATUS.md
docs/status/README.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Current checkpoint

Current verified receipt promotion posture:

```
seal_verified_receipt_promotion_contract_present=1
seal_verified_receipt_promotion_implementation_present=1
seal_verified_receipt_promotion_header_present=1
seal_verified_receipt_promotion_source_present=1
seal_verified_receipt_promotion_invariant_test_present=1
seal_verified_receipt_promotion_runner_present=1
seal_verified_receipt_promotion_status_present=1
seal_ed25519_verify_status_present=1
readme_links_verified_receipt_promotion_contract=1
readme_links_verified_receipt_promotion_implementation=1
readme_links_verified_receipt_promotion_status=1
root_status_mentions_verified_receipt_promotion_status=1
status_index_links_verified_receipt_promotion_status=1
foundation_index_links_verified_receipt_promotion_status=1
project_notes_mark_verified_receipt_promotion_status_complete=1
receipt_profile=latticra-seal-verified-receipt/0.1
verify_profile=latticra-seal-ed25519-verify/0.1
backend_profile=latticra-seal-crypto-verify-backend/0.1
verification_policy_profile=latticra-seal-verification-policy/0.1
message_label=rfc8032-test-vector-2
message_size_bytes=1
message_digest_algorithm=SHA-256
public_key_identity_label=rfc8032-test-key
signature_algorithm=Ed25519-development
trust_source=local-test-vector
verification_state=verified
receipt_state=verified
cryptographic_verification_supported=1
cryptographic_verification_performed=1
verified=1
invalid=0
```

```
authority_usable=0
capability_gate_allowed=0
runtime_authority_granted=0
mode=verified-receipt-authority-neutral
status=verified-receipt-metadata
error=ok
verified_receipt_promotion_added=1
capability_authorization_added=0
effect_execution_added=0
runtime_authority_granted=0
signing_added=0
key_generation_added=0
private_key_handling_added=0
trust_store_behavior_added=0
revocation_lookup_added=0
network_behavior_changed=0
host_behavior_changed=0
```

Validation

This status surface is covered by:

```
sh scripts/test-latticra-seal-verified-receipt-promotion-status.sh
```

The underlying implementation remains covered by:

```
sh scripts/test-latticra-seal-verified-receipt-promotion.sh
```

Expected output:

```
seal verified receipt promotion status: ok
seal verified receipt promotion invariants: ok
```

Boundary

This status record is documentation/status alignment only.

The verified receipt promotion implementation only accepts a successful Ed25519 verify-only result and produces deterministic verified receipt metadata. The promoted receipt remains authority-neutral.

It does not add capability authorization, effect execution, runtime behavior, host behavior, network behavior, signing, key generation, private-key handling, trust-store behavior, revocation lookup, production readiness, external endorsement, or authority grants.

Current next valid slice

No completion-estimate review is required from this status/public-entry alignment.

The next valid Latticra Seal slice is verified capability gate status/public-entry alignment or another narrow status/index alignment follow-up.

docs/status/SECURE_CONFIGURATION_CHANGE_MANAGEMENT_BASELINE_STATUS.md

Source title: Latticra Secure Configuration and Change Management Baseline Status

Lines: 65 | Words: 133 | SHA-256: 8cdfdc6e6ceaa817f50874b768ff0c599040343bd638e501d789ecaef0663a33

Latticra Secure Configuration and Change Management Baseline Status

Status: status record for secure configuration and change management baseline Date: 2026-05-26

Scope

This record tracks the secure configuration and change-management baseline for secure configuration baselines, configuration item inventory, configuration checklists, secure defaults, approved change records, rollback evidence, drift detection, exception ownership, and configuration non-claims.

It does not implement host configuration, infrastructure configuration, configuration scanning, configuration enforcement, drift detection, change approval workflow, rollback execution, compliance, or runtime authority.

Current fields

```
secure_configuration_change_management_baseline_present=1
secure_configuration_change_management_status_present=1
secure_configuration_change_management_guard_present=1
nist_sp_800_128_configuration_management_tracked=1
nist_sp_800_70_rev5_checklist_tracked=1
nist_sp_800_53_configuration_management_tracked=1
cisa_cpg_secure_configuration_tracked=1
cisa_fbi_product_security_bad_practices_config_tracked=1
nsa_cisa_top_misconfigurations_tracked=1
cisa_nsa_fbi_secure_by_default_tracked=1
configuration_item_inventory_required=1
secure_baseline_configuration_required=1
configuration_checklist_required=1
approved_change_record_required=1
configuration_change_owner_required=1
configuration_change_risk_review_required=1
configuration_change_test_evidence_required=1
configuration_rollback_plan_required=1
configuration_drift_detection_required=1
default_credential_forbidden=1
insecure_default_configuration_forbidden=1
configuration_secret_review_required=1
configuration_exception_owner_required=1
configuration_exception_expiration_required=1
implementation_behavior_changed=0
configuration_enforcement_added=0
configuration_scanner_added=0
host_configuration_changed=0
infrastructure_configuration_changed=0
change_approval_workflow_added=0
drift_detection_added=0
rollback_execution_added=0
production_configuration_claim_allowed=0
hosted_service_configuration_claim_allowed=0
configuration_hardening_claim_allowed=0
secure_default_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-secure-configuration-change-management-baseline.sh
```

Expected output:

```
secure_configuration_change_management_baseline: ok
```

docs/status/SECURITY_LOGGING_MONITORING_BASELINE_STATUS.md

Source title: Latticra Security Logging, Monitoring, and Detection Baseline Status

Lines: 68 | Words: 152 | SHA-256: 215477962a0ec24e6977b82453dd86f956c9f5c2afb05a3965b118973e298bc7

Latticra Security Logging, Monitoring, and Detection Baseline Status

Status: status record for security logging, monitoring, and detection baseline Date: 2026-05-26

Scope

This record tracks the security logging, monitoring, and detection baseline for security event source inventory, audit event selection, runtime-authority decision logs, identity and access events, privileged actions, log redaction, log integrity, retention, disposal, time source, detection triage, incident handoff, and monitoring non-claims.

It does not implement a log collector, SIEM, telemetry export, host sensor, network sensor, detection rule, alerting service, log storage, monitoring service, incident detection service, compliance, or runtime authority.

Current fields

```
security_logging_monitoring_baseline_present=1
security_logging_monitoring_status_present=1
security_logging_monitoring_guard_present=1
cisa_fbi_nsa_event_logging_guidance_tracked=1
nsa_event_logging_release_tracked=1
cisa_logging_made_easy_tracked=1
cisa_use_logging_on_business_systems_tracked=1
cisa_cpg_log_collection_tracked=1
nist_sp_800_92_log_management_tracked=1
nist_sp_800_92_rev1_draft_tracked=1
nist_csf_detect_function_tracked=1
nist_sp_800_53_audit_accountability_tracked=1
security_event_source_inventory_required=1
audit_event_selection_required=1
runtime_authority_decision_logging_required=1
identity_access_event_logging_required=1
privileged_action_logging_required=1
security_error_logging_required=1
configuration_change_logging_required=1
log_redaction_required=1
secret_free_log_guard_required=1
log_integrity_tamper_resistance_required=1
time_synchronization_required=1
retention_disposal_policy_required=1
critical_log_source_disable_alert_required=1
detection_triage_owner_required=1
incident_handoff_path_required=1
operator_log_access_review_required=1
implementation_behavior_changed=0
log_collector_added=0
siem_added=0
telemetry_export_added=0
host_sensor_added=0
network_sensor_added=0
alerting_service_added=0
log_storage_added=0
detection_rule_added=0
production_monitoring_claim_allowed=0
detection_service_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-security-logging-monitoring-baseline.sh
```

Expected output:

```
security_logging_monitoring_baseline: ok
```

docs/status/STATUS_ANNOUNCEMENT_CONSISTENCY_REVIEW.md

Source title: Status and Announcement Consistency Review

Lines: 33 | Words: 109 | SHA-256: 3c987f837b882b6f90084f812e2b0db2e287ace2a66b41e146c439504247d3f1

Status and Announcement Consistency Review

Status: initial consistency review

This record checks that the public status surfaces and announcement surfaces agree after the recent Lat pipeline diagnostic, RBDM report, and project-notes alignment slices.

Reviewed files:

```
STATUS.md
docs/status/CURRENT_STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/project_notes/CURRENT_DIRECTION.md
docs/project_notes/UPCOMING_WORK.md
docs/project_notes/README.md
```

Observed drift:

```
docs/status/CURRENT_STATUS.md still pointed to older Lat diagnostic README/foundation
follow-ups.
docs/status/ANNOUNCEMENTS.md still pointed to Lat pipeline diagnostic README alignment as next
step after that slice had merged.
```

Resolution:

```
align CURRENT_STATUS.md next recommended work with the project-notes queue
align ANNOUNCEMENTS.md latest next step with the current queue
record this review in STATUS.md
```

Boundary: documentation/status review only. No runtime behavior is added.

docs/status/STATUS_ANNOUNCEMENT_REVIEW.md

Source title: Status Announcement Review

Lines: 31 | Words: 101 | SHA-256: ac6a21b3b5b3b83e1073008f78a5ca0781ca8e9b035196c6047f886370ffcc01

Status Announcement Review

Status: announcement review

This record reviews the public announcement surface after the authority implementation review, authority status/docs alignment, current status detail rollup, and authority foundation index alignment.

Reviewed files:

```
STATUS.md
docs/status/ANNOUNCEMENTS.md
docs/status/CURRENT_STATUS.md
docs/FOUNDATION_INDEX.md
docs/CPP_AUTHORITY_IMPLEMENTATION_REVIEW.md
docs/status/CPP_AUTHORITY_IMPLEMENTATION_REVIEW_STATUS.md
```

Finding:

```
The announcement log did not yet record the authority review and authority index/status
alignment sequence.
```

Resolution:

```
Add a bounded public announcement entry that records the authority review/status/index alignment
 as documentation and audit posture only.
Advance the root status queue to the next public entry-point alignment slice.
```

Boundary: announcement/status review only. No implementation behavior is changed.

docs/status/SUPPLY_CHAIN_SECURITY_BASELINE_STATUS.md

Source title: Latticra Supply-Chain Security Baseline Status

Lines: 71 | Words: 129 | SHA-256: 1335979ce47323efe52291f3fcab30de2c2684ebd47d8fa6aed96dd1bb215e20

Latticra Supply-Chain Security Baseline Status

Status: status record for supply-chain security baseline Date: 2026-05-26

Scope

This record tracks the supply-chain security baseline for repository, CI, dependency, package, installer, artifact, SBOM, release, and update-lane posture.

It does not publish artifacts, release packages, add signing authority, claim SLSA level, claim NIST compliance, claim CISA CPG compliance, claim production readiness, or grant release authority.

Current fields

```
supply_chain_security_baseline_present=1
supply_chain_security_status_present=1
supply_chain_security_guard_present=1
high_assurance_security_baseline_present=1
ssdf_supply_chain_profile_present=1
cpg_supply_chain_profile_present=1
sbom_required_before_production_installer=1
kev_nvd_review_required_before_release=1
dependency_inventory_required=1
pinned_ci_actions_required=1
read_only_workflow_permissions_required=1
persist_credentials_false_required=1
pull_request_target_forbidden=1
repository_secret_use_requires_dedicated_review=1
implicit_github_token_use_requires_dedicated_review=1
repository_secret_filename_scan_required=1
repository_secret_content_marker_scan_required=1
sensitive_local_artifact_filename_guard_required=1
report_redaction_boundary_guard_required=1
whole_environment_report_dump_forbidden=1
installer_engine_log_redaction_required=1
installer_config_authority_slug_allowlist_required=1
installer_command_wrapper_strict_name_required=1
installer_ui_artifact_authority_guard_required=1
installer_ui_artifact_write_validation_required=1
installer_console_output_authority_guard_required=1
installer_console_config_reflection_denial_required=1
locked_dependency_builds_required=1
offline_installer_builds_required=1
ad_hoc_network_client_commands_forbidden_without_guard=1
source_archive_fixture_tracked_unignored_source_view_required=1
source_archive_fixture_symlink_refusal_required=1
source_archive_fixture_reproducible_metadata_required=1
release_publishing_authority_granted=0
production_installer_claim_allowed=0
production_update_claim_allowed=0
compliance_claim_allowed=0
certification_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-supply-chain-security-baseline.sh
sh scripts/test-secret-material-guard.sh
sh scripts/test-report-redaction-boundary.sh
sh scripts/test-installer-config-authority-allowlist.sh
sh scripts/test-installer-ui-artifact-authority.sh
sh scripts/test-installer-console-output-authority.sh
```

Expected output:

```
supply_chain_security_baseline: ok
```

docs/status/UBUNTU_ECOSYSTEM_INTEGRATION_STATUS.md

Source title: Ubuntu Ecosystem Integration Status

Lines: 316 | Words: 858 | SHA-256: bf99797afc173cfd880f690a49100a1aacd180a4f946706d169473a015955a13

Ubuntu Ecosystem Integration Status

Status: Ubuntu integration status record Date: 2026-05-26

Summary

Latticra now has an Ubuntu-facing compatibility lane for development, Panel prerequisites, and a local-only deb packaging draft.

This is an ecosystem integration checkpoint, not a production readiness claim.

Current Evidence

```
ubuntu_build_lane_present=1
ubuntu_developer_workflow_present=1
ubuntu_panel_prerequisites_documented=1
ubuntu_quickstart_documented=1
ubuntu_local_deb_draft_present=1
ubuntu_static_deb_validation_present=1
ubuntu_lintian_availability_present=1
ubuntu_local_deb_build_transcript_contract_present=1
ubuntu_local_deb_build_transcript_present=0
deb_artifact_created=0
deb_installed_on_host=0
ubuntu_package_license_review_contract_present=1
ubuntu_package_license_review_status=resolved-license-expression-recorded
ubuntu_package_notice_inventory_present=1
ubuntu_package_notice_inventory_report_present=1
ubuntu_doc_payload_license_review_contract_present=1
ubuntu_doc_payload_license_review_status=resolved-cc-by-4.0
ubuntu_third_party_material_review_contract_present=1
ubuntu_third_party_material_review_status=blocked-pending-third-party-material-review
ubuntu_generated_artifact_notice_review_contract_present=1
ubuntu_generated_artifact_notice_review_status=blocked-pending-generated-artifact-notice-review
ubuntu_notice_file_decision_contract_present=1
ubuntu_notice_file_decision_status=blocked-pending-notice-file-decision
ubuntu_debian_copyright_notice_mapping_contract_present=1
ubuntu_debian_copyright_notice_mapping_status=blocked-pending-debian-copyright-notice-mapping
ubuntu_trademark_notice_boundary_contract_present=1
ubuntu_trademark_notice_boundary_status=blocked-pending-trademark-notice-boundary
ubuntu_release_artifact_notice_requirements_contract_present=1
ubuntu_release_artifact_notice_requirements_status=blocked-pending-release-artifact-notice-requi
rements
ubuntu_package_notice_promotion_gate_contract_present=1
ubuntu_package_notice_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_package_license_promotion_gate_contract_present=1
ubuntu_package_license_promotion_gate_status=blocked-pending-package-notice-prerequisites
ubuntu_lintian_static_metadata_contract_present=1
ubuntu_lintian_static_metadata_status=blocked-pending-package-license-promotion
ubuntu_local_deb_build_transcript_acceptance_gate_contract_present=1
ubuntu_local_deb_build_transcript_acceptance_gate_status=blocked-pending-lintian-static-metadata
-and-build-transcript
ubuntu_local_deb_install_remove_evidence_contract_present=1
ubuntu_local_deb_install_remove_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_source_package_evidence_contract_present=1
ubuntu_source_package_evidence_status=blocked-pending-accepted-build-transcript
ubuntu_upload_signing_authority_evidence_contract_present=1
ubuntu_upload_signing_authority_evidence_status=blocked-pending-source-package-evidence
ubuntu_ppa_archive_publication_gate_contract_present=1
ubuntu_ppa_archive_publication_gate_status=blocked-pending-install-remove-evidence
ubuntu_package_notice_review_contract_present=1
ubuntu_package_notice_review_status=blocked-pending-notice-review
license_expression_candidate_recorded=1
license_expression_reviewed=1
license_expression_unresolved=0
documentation_license_decision_present=1
doc_payload_license_reviewed=1
doc_payload_license_unresolved=0
doc_payload_license_decision_recorded=1
debian_copyright_doc_payload_mapping_reviewed=1
debian_copyright_binary_payload_mapping_reviewed=0
debian_copyright_third_party_notice_mapping_reviewed=0
debian_copyright_generated_artifact_notice_mapping_reviewed=0
debian_copyright_notice_file_mapping_reviewed=0
debian_copyright_trademark_notice_boundary_reviewed=0
debian_copyright_license_ref_replaced_or_justified=1
trademark_notice_boundary_recorded=0
trademark_policy_applied_to_package_notice=0
package_description_endorsement_boundary_reviewed=0
documentation_trademark_boundary_reviewed=0
canonical_endorsement_boundary_reviewed=0
project_identity_downstream_use_boundary_recorded=0
third_party_material_inventory_recorded=1
third_party_material_inventory_reviewed=0
third_party_material_source_records_present=0
third_party_material_license_records_present=0
third_party_material_compatibility_notes_present=0
generated_artifact_notice_reviewed=0
generated_artifact_notice_requirements_recorded=0
binary_payload_generation_path_reviewed=0
doc_payload_generation_path_reviewed=0
```

```
deb_artifact_notice_requirements_recorded=0
changes_file_notice_requirements_recorded=0
build_log_notice_requirements_recorded=0
installed_payload_notice_requirements_recorded=0
release_artifact_notice_requirements_recorded=0
source_package_notice_requirements_recorded=0
release_notes_notice_requirements_recorded=0
deb_removed_from_host=0
lintian_static_metadata_run=0
lintian_static_metadata_findings_classified=0
lintian_expected_draft_findings_classified=0
lintian_unexpected_findings_classified=0
build_transcript_acceptance_gate_open=0
build_transcript_acceptance_gate_unblocked=0
local_deb_build_transcript_accepted=0
transcript_header_reviewed=0
tooling_evidence_reviewed=0
package_evidence_reviewed=0
build_evidence_reviewed=0
payload_evidence_reviewed=0
non_claims_reviewed=0
install_remove_test_environment_recorded=0
install_command_recorded=0
install_exit_status_recorded=0
installed_payload_listing_recorded=0
usr_bin_latticra_installed=0
status_command_after_install_recorded=0
remove_command_recorded=0
remove_exit_status_recorded=0
post_remove_absence_checked=0
residual_payload_reviewed=0
install_remove_findings_classified=0
host_mutation_scope_reviewed=0
ubuntu_install_remove_evidence_unblocked=0
source_package_created=0
source_package_build_environment_recorded=0
dpkg_source_command_recorded=0
dpkg_buildpackage_source_command_recorded=0
dpkg_source_run=0
dpkg_buildpackage_source_run=0
source_package_name_recorded=0
source_package_digest_recorded=0
dsc_path_recorded=0
dsc_digest_recorded=0
changes_file_path_recorded=0
upload_target_recorded=0
upload_target_kind_recorded=0
upload_authority_reviewed=0
launchpad_account_recorded=0
ppa_or_archive_target_reviewed=0
orig_tarball_path_recorded=0
orig_tarball_digest_recorded=0
debian_source_format_verified=0
source_package_payload_reviewed=0
source_package_notice_requirements_reviewed=0
ubuntu_source_package_evidence_unblocked=0
gpg_signing_key_fingerprint_recorded=0
upload_command_non_claims_reviewed=0
ubuntu_upload_signing_authority_evidence_unblocked=0
debsign_command_recorded=0
signature_fingerprint_recorded=0
dput_command_recorded=0
upload_exit_status_recorded=0
launchpad_build_log_recorded=0
launchpad_build_result_reviewed=0
publication_non_claims_reviewed=0
ubuntu_publication_gate_unblocked=0
ppa_created=0
launchpad_upload_run=0
source_package_uploaded=0
ubuntu_archive_submission_claimed=0
ubuntu_publication_ready=0
third_party_notice_reviewed=0
third_party_notice_requirements_recorded=0
notice_file_present=0
notice_file_decision_recorded=0
notice_file_required_decision_recorded=0
notice_file_content_requirements_recorded=0
notice_file_install_path_reviewed=0
notice_file_packaging_mapping_reviewed=0
```

```
notice_file_absence_justification_recorded=0
debian_copyright_notice_mapping_reviewed=0
ubuntu_package_notice_review_unblocked=0
packaging_license_expression_updated=1
ubuntu_package_license_review_unblocked=1
ubuntu_lintian_static_metadata_unblocked=0
ubuntu_local_deb_build_transcript_unblocked=0
ppa_claimed=0
ubuntu_archive_ready=0
canonical_endorsement_claimed=0
production_installer_ready=0
root_installer_ready=0
```

Guarded Files

```
docs/UBUNTU_READINESS_PLAN.md
docs/UBUNTU_DEVELOPER_WORKFLOW.md
docs/UBUNTU_LOCAL_DEB_STATIC_VALIDATION.md
docs/UBUNTU_LINTIAN_AVAILABILITY.md
docs/UBUNTU_PACKAGE_NOTICE_INVENTORY.md
docs/UBUNTU_DOC_PAYLOAD_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_THIRD_PARTY_MATERIAL_REVIEW_CONTRACT.md
docs/UBUNTU_GENERATED_ARTIFACT_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_NOTICE_FILE_DECISION_CONTRACT.md
docs/UBUNTU_DEBIAN_COPYRIGHT_NOTICE_MAPPING_CONTRACT.md
docs/UBUNTU_TRADEMARK_NOTICE_BOUNDARY_CONTRACT.md
docs/UBUNTU_RELEASE_ARTIFACT_NOTICE_REQUIREMENTS_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_PROMOTION_GATE_CONTRACT.md
docs/UBUNTU_LINTIAN_STATIC_METADATA_CONTRACT.md
docs/UBUNTU_PACKAGE_NOTICE_REVIEW_CONTRACT.md
docs/UBUNTU_PACKAGE_LICENSE_REVIEW_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_BUILD_TRANSCRIPT_ACCEPTANCE_GATE_CONTRACT.md
docs/UBUNTU_LOCAL_DEB_INSTALL_REMOVE_EVIDENCE_CONTRACT.md
docs/UBUNTU_SOURCE_PACKAGE_EVIDENCE_CONTRACT.md
docs/UBUNTU_UPLOAD_SIGNING_AUTHORITY_EVIDENCE_CONTRACT.md
docs/UBUNTU_PPA_ARCHIVE_PUBLICATION_GATE_CONTRACT.md
packaging/ubuntu/README.md
packaging/ubuntu/debian/control
packaging/ubuntu/debian/rules
packaging/ubuntu/debian/changelog
packaging/ubuntu/debian/copyright
packaging/ubuntu/debian/install
packaging/ubuntu/debian/source/format
scripts/test-ubuntu-build-lane.sh
scripts/test-ubuntu-developer-workflow.sh
scripts/test-ubuntu-local-deb-static-validation.sh
scripts/test-ubuntu-lintian-availability.sh
scripts/ubuntu-package-notice-inventory.sh
scripts/test-ubuntu-package-notice-inventory.sh
scripts/test-ubuntu-doc-payload-license-review-contract.sh
scripts/test-ubuntu-third-party-material-review-contract.sh
scripts/test-ubuntu-generated-artifact-notice-review-contract.sh
scripts/test-ubuntu-notice-file-decision-contract.sh
scripts/test-ubuntu-debian-copyright-notice-mapping-contract.sh
scripts/test-ubuntu-trademark-notice-boundary-contract.sh
scripts/test-ubuntu-release-artifact-notice-requirements-contract.sh
scripts/test-ubuntu-package-notice-promotion-gate-contract.sh
scripts/test-ubuntu-package-license-promotion-gate-contract.sh
scripts/test-ubuntu-lintian-static-metadata-contract.sh
scripts/test-ubuntu-package-notice-review-contract.sh
scripts/test-ubuntu-package-license-review-contract.sh
scripts/test-ubuntu-local-deb-build-transcript-contract.sh
scripts/test-ubuntu-local-deb-build-transcript-acceptance-gate-contract.sh
scripts/test-ubuntu-local-deb-install-remove-evidence-contract.sh
scripts/test-ubuntu-source-package-evidence-contract.sh
scripts/test-ubuntu-upload-signing-authority-evidence-contract.sh
scripts/test-ubuntu-ppa-archive-publication-gate-contract.sh
.github/workflows/ubuntu-package-notice-inventory.yml
.github/workflows/ubuntu-doc-payload-license-review-contract.yml
.github/workflows/ubuntu-third-party-material-review-contract.yml
.github/workflows/ubuntu-generated-artifact-notice-review-contract.yml
.github/workflows/ubuntu-notice-file-decision-contract.yml
.github/workflows/ubuntu-debian-copyright-notice-mapping-contract.yml
.github/workflows/ubuntu-trademark-notice-boundary-contract.yml
.github/workflows/ubuntu-release-artifact-notice-requirements-contract.yml
.github/workflows/ubuntu-package-notice-promotion-gate-contract.yml
.github/workflows/ubuntu-package-license-promotion-gate-contract.yml
.github/workflows/ubuntu-lintian-static-metadata-contract.yml
.github/workflows/ubuntu-package-notice-review-contract.yml
.github/workflows/ubuntu-local-deb-build-transcript-acceptance-gate-contract.yml
.github/workflows/ubuntu-local-deb-install-remove-evidence-contract.yml
.github/workflows/ubuntu-source-package-evidence-contract.yml
.github/workflows/ubuntu-upload-signing-authority-evidence-contract.yml
.github/workflows/ubuntu-ppa-archive-publication-gate-contract.yml
```

Public Entry Points

Ubuntu setup and package posture are now linked from:

```
README.md
```

```
docs/QUICK_START_CHEATSHEET.md
installer/README.md
docs/status/README.md
```

Current Boundary

The Ubuntu lane does not publish a package, create a PPA, claim Ubuntu archive readiness, install a root service, change systemd, change the kernel, add a privileged helper, grant network authority, or claim production readiness.

The local deb build transcript contract is intentionally blocked from evidence promotion until notice, lintian, build, and install/remove evidence gates are satisfied.

The Ubuntu package license review contract now records the current source facts, accepts the AGPL-3.0-or-later plus CC-BY-4.0 payload expression, and maps it into the local Debian copyright draft while broader notice review remains blocked.

The Ubuntu doc payload license review contract records that README.md is included in the local deb documentation payload under CC-BY-4.0.

The Ubuntu third-party material review contract records that source, license, compatibility, and notice-requirement records still need formal review before package notice promotion.

The Ubuntu generated-artifact notice review contract records that generated-artifact notice requirements still need formal review before any deb artifact or build evidence can promote package notice status.

The Ubuntu NOTICE file decision contract records that the package still needs a reviewed decision about whether a NOTICE file is required and how it would be mapped into the local deb payload.

The Ubuntu Debian copyright notice mapping contract records that the local deb payload still needs a reviewed mapping into packaging/ubuntu/debian/copyright before package notice promotion.

The Ubuntu trademark notice boundary contract records that package metadata still needs a reviewed trademark, project-identity, and endorsement boundary before package notice promotion.

The Ubuntu release artifact notice requirements contract records that source package, deb artifact, .changes, build-log, installed-payload, and release-notes notice requirements still need review before package notice promotion.

The Ubuntu package notice promotion gate records the aggregate blocked state across all package notice prerequisites before package license promotion can proceed.

The Ubuntu package license promotion gate records the aggregate blocked state across package notice and package license prerequisites before lintian/static metadata or build transcript evidence can proceed.

The Ubuntu lintian static metadata contract records the future lintian/static metadata evidence shape while keeping lintian execution and build transcript evidence blocked.

The Ubuntu local deb build transcript acceptance gate records that no future local build transcript can be accepted until lintian/static metadata, package license promotion, transcript evidence, payload evidence, and non-claims are reviewed.

The Ubuntu local deb install/remove evidence contract records the future install/remove evidence shape while keeping package install commands, package remove commands, host mutation, and install/remove evidence promotion blocked.

The Ubuntu source package evidence contract records the future .dsc, source package, .changes, and digest evidence shape while keeping dpkg-source, source package creation, signing, upload, and publication blocked.

The Ubuntu upload/signing authority evidence contract records the future upload target, Launchpad account, signing key, debsign, and dput evidence shape while keeping signing, upload, and publication blocked.

The Ubuntu PPA/archive publication gate records the future upload, signing, Launchpad, and archive-submission evidence shape while keeping debsign, dput, PPA creation, Launchpad upload, Ubuntu archive submission, and publication readiness blocked.

The Ubuntu package notice inventory records the current local-deb draft payload facts without promoting the review. The Ubuntu package notice review contract records the remaining notice obligations that must be settled before package promotion can proceed.

Next Recommended Lane

```
Review the Ubuntu upload/signing authority evidence contract, then keep signing and upload
evidence blocked until source package evidence is reviewed.
```

docs/status/VULNERABILITY_MANAGEMENT_RELEASE_GATE_BASELINE_STATUS.md

Source title: Latticra Vulnerability Management Release Gate Baseline Status

Lines: 56 | Words: 131 | SHA-256: 5a4e302c5ab38542a25f856c0dad44518cf249e9673f55dd4dc302a7ddf96f3d

Latticra Vulnerability Management Release Gate Baseline Status

Status: status record for vulnerability management release gate baseline Date: 2026-05-26

Scope

This record tracks the vulnerability-management release gate baseline for CISA KEV review, NVD/CVE review, coordinated vulnerability disclosure, dependency/component inventory, release blocking, mitigation/exception records, and public non-claims before production release, update, installer, package, internet-facing service, or security-product claims.

It does not run vulnerability scans, query live feeds, publish advisories, submit CVEs, patch dependencies, produce an SBOM, publish releases, grant release authority, or claim product security.

Current fields

```
vulnerability_management_release_gate_baseline_present=1
vulnerability_management_release_gate_status_present=1
vulnerability_management_release_gate_guard_present=1
cisa_kev_catalog_tracked=1
nvd_cve_review_required=1
cvss_context_required_not_risk_score_only=1
coordinated_vulnerability_disclosure_required=1
vulnerability_disclosure_policy_scope_required=1
dependency_component_inventory_required=1
sbom_required_before_production_release=1
kev_nvd_review_required_before_release=1
known_exploited_vulnerability_mitigation_required=1
non_exploitability_claim_requires_written_record=1
vulnerability_exception_owner_required=1
vulnerability_exception_expiration_required=1
release_block_on_unmitigated_known_exploited_vulnerability=1
internet_facing_asset_inventory_required_before_release=1
security_advisory_process_required_before_supported_release=1
implementation_behavior_changed=0
live_feed_query_added=0
vulnerability_scan_added=0
release_publishing_authority_granted=0
security_advisory_published=0
cve_submission_performed=0
sbom_generated=0
production_release_claim_allowed=0
product_security_claim_allowed=0
compliance_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-vulnerability-management-release-gate-baseline.sh
```

Expected output:

```
vulnerability_management_release_gate_baseline: ok
```

docs/status/ZERO_TRUST_RUNTIME_AUTHORITY_BASELINE_STATUS.md

Source title: Latticra Zero-Trust Runtime Authority Baseline Status

Lines: 54 | Words: 116 | SHA-256: 3afae443bfb75b17b29f2d29c2a283de2b683dc97d18e7381e2d401b82d6dcb3

Latticra Zero-Trust Runtime Authority Baseline Status

Status: status record for zero-trust runtime authority baseline Date: 2026-05-26

Scope

This record tracks the zero-trust runtime authority baseline for future runtime, tool, host I/O, network, server/MCP, update, recovery, boot, hardware, agentic automation, and authority-bearing request paths.

It does not implement runtime execution, network access, host I/O, tool execution, MCP behavior, recovery behavior, boot behavior, hardware behavior, policy enforcement, capability enforcement, or production protection.

Current fields

```
zero_trust_runtime_authority_baseline_present=1
zero_trust_runtime_authority_status_present=1
zero_trust_runtime_authority_guard_present=1
defensive_threat_model_validation_refinement_present=1
runtime_boundary_policy_expansion_after_threat_model_present=1
zero_trust_runtime_boundary_required=1
per_request_authorization_required=1
least_privilege_effect_scope_required=1
resource_identity_required=1
caller_identity_required=1
asset_inventory_required_before_authority=1
policy_decision_visibility_required=1
denial_reason_visibility_required=1
audit_record_required_before_authority=1
operator_confirmation_metadata_only_required=1
unknown_request_denial_required=1
unknown_effect_denial_required=1
future_gate_denial_required=1
runtime_execution_added=0
tool_execution_added=0
host_behavior_changed=0
network_behavior_changed=0
mcp_behavior_changed=0
runtime_authority_granted=0
production_protection_claim_allowed=0
zero_trust_certification_claim_allowed=0
external_endorsement_claimed=0
```

Validation

```
sh scripts/test-zero-trust-runtime-authority-baseline.sh
```

Expected output:

```
zero_trust_runtime_authority_baseline: ok
```

Part XIV - Legal, License, and Remaining Records

- docs/assets/placeholder.txt - Placeholder
- docs/latticra-system-substrate/README.md - The Latticra System Substrate
- docs/LOCAL_ARTIFACT_MANIFEST_FIXTURE.md - Local Artifact Manifest Fixture
- docs/MACOS_APP_BUNDLE_WRITER_ALIGNMENT.md - macOS App Bundle Writer Alignment
- docs/MACOS_APP_BUNDLE_WRITER_DRY_RUN.md - macOS App Bundle Writer Dry-Run Prototype
- docs/MACOS_BUILD_PLATFORM_PROBE.md - macOS Build Platform Probe
- docs/MACOS_COMMIT_GATE_CONTRACT.md - macOS Commit Gate Contract
- docs/MACOS_DRY_RUN_PLAN_ADAPTER.md - macOS Dry-Run Plan Adapter
- docs/MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION.md - macOS Dry-Run Writer Candidate Integration
- docs/MACOS_INTEGRATION_TRANSFERABILITY_PLAN.md - macOS Integration Transferability Plan
- docs/MACOS_LOCAL_CANDIDATE_ASSET_PROBE.md - macOS Local Candidate Asset Probe
- docs/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT.md - macOS Reset/Uninstall Dry-Run Contract
- docs/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER.md - macOS Reset/Uninstall Dry-Run Planner
- docs/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT.md - macOS Reset/Uninstall Evidence-Bundle Contract

- docs/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT.md - macOS Reset/Uninstall Implementation-Gate Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT.md - macOS Reset/Uninstall Live-Denial Transcript Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT.md - macOS Reset/Uninstall Live-Execution Preflight Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT.md - macOS Reset/Uninstall Live-Implementation Plan Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_REVIEW_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Review Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Review Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Review Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Denial Transcript Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT.md - macOS Reset/Uninstall Live-Runner Acceptance-Gate Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT.md - macOS Reset/Uninstall Live-Runner Denied-Dispatch Review Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT.md - macOS Reset/Uninstall Live-Runner Denied-Dispatch Transcript Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT.md - macOS Reset/Uninstall Live-Runner Interface Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT.md - macOS Reset/Uninstall Live-Runner No-Op Prototype Contract
- docs/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER.md - macOS Reset/Uninstall Live-Target Classifier
- docs/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT.md - macOS Reset/Uninstall Operator-Intent Contract

- docs/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT.md - macOS Reset/Uninstall Receipt-Schema Contract
- docs/MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT.md - macOS User-Local App Bundle Contract
- docs/MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN.md - macOS User-Local App Bundle Implementation Plan
- docs/MACOS_VERIFICATION_TRANSCRIPT_CONTRACT.md - macOS Verification Transcript Contract
- docs/VISUAL_THEOREM_ENGINES.md - Latticra Visual Theorem Engines
- fixtures/artifact/local-artifact-manifest.txt - Local Artifact Manifest
- LICENSE - License
- LICENSES/AGPL-3.0-or-later.txt - Agpl 3.0 Or Later
- LICENSES/Apache-2.0.txt - Apache 2.0
- LICENSES/CC-BY-4.0.txt - Cc By 4.0
- LICENSES/README.md - Latticra License References

docs/assets/placeholder.txt

Source title: Placeholder

Lines: 1 | Words: 1 | SHA-256: 7c4c231187cd4b75677c312faa7787da874bdbb7d71e1c9dc0619f3e27a53940

txt

docs/latticra-system-substrate/README.md

Source title: The Latticra System Substrate

Lines: 73 | Words: 760 | SHA-256: ec96bca2c315790ac2012a1ac5583630f040373080f7f0d5562b7ec616a24b86

The Latticra System Substrate

Title: The Latticra System Substrate: An Effect at Modern Security
 Edition: Working Draft 0.16 — 2026-05-26
 Role: Project-level technical handbook for Latticra, Latticra Seal trust-boundary metadata, authority-neutral Ed25519 verification evidence, metadata-only capability/effect/handoff/report/envelope/signature-request/signing-operation classification and status guards, disposable Fedora VM CLI payload validation evidence, Latticra Console host/receipt/signature/OS/VM contract surfaces, Runtime Boundary Lat/LIR provenance records, Nucleus, Nadia Stage-46 contract-only offline-AI release/receipt/review/disposition metadata, Panel, platform install validation lanes, Lat/LIR contract surfaces, receipts, reports, and future runtime-boundary research.

This handbook supersedes the former standalone Latticra Seal Documentation Handbook as the main reader-facing book for the project.

The Seal handbook was useful as a subsystem reference. This new handbook expands the scope into a computer-science-oriented project book: Latticra as an evidence-bound system substrate for local integrity, authority boundaries, reproducible reports, receipts, policy semantics, and future runtime-handoff research.

Downloads

- PDF edition
- Editable DOCX edition
- Public repository folder

Living handbook cadence

Treat this handbook as a live project artifact, not a finished release.

As Latticra work progresses, material decisions, capability changes, boundary refinements, validation lanes, public wording, and evidence updates should be folded back into the handbook alongside the nearer-term project notes and status records.

Handbook updates should stay evidence-bound: describe what is implemented, tested, measured, planned, or explicitly out of scope, and avoid promoting future runtime, enforcement, host-protection, packaging, or production claims ahead of reproducible evidence.

Correct interpretation

Latticra should be understood as early, evidence-bound systems architecture work.

It is not currently:

- a production security product

- a host-protection system
- a hardened sandbox
- a malware prevention system
- a ransomware prevention system
- a kernel enforcement layer
- a systemd enforcement layer
- an SELinux authority
- a root installer
- a network authority
- a production runtime authority
- an operating-system replacement

The handbook's core claim is narrower and more useful: Latticra is a substrate for making security-relevant effects visible, typed, reviewable, reproducible, and eventually governable.

Relationship to Latticra Seal

Latticra Seal remains the verification, reporting, manifest/hash baseline, and policy-boundary lane inside the Latticra ecosystem.

Working Draft 0.16 carries the Seal trust/crypto ladder forward, keeps the current Latticra Console contract bundle, preserves Runtime Boundary Lat/LIR provenance records, keeps signing-operation status guard/predecessor alignment for the Seal signing path, folds in disposable Fedora VM CLI payload validation evidence and README/status alignment, and adds Nadia Stage-39 through Stage-46 release/receipt/review/disposition contract metadata. Seal documents signed request metadata, request freshness metadata, verification receipts, crypto verify backend metadata, local Ed25519 verify-only results, verified receipt promotion metadata, verified capability gate metadata, verified effect decision metadata, runtime handoff evaluation metadata, runtime handoff report metadata, report envelope metadata, signature request metadata, signing authorization metadata, signer handoff metadata, signer invocation metadata, signing-operation readiness metadata, effect decisions, runtime handoff boundaries, and report envelopes. Latticra Console documents metadata-only host-adapter, receipt-request, receipt-payload, payload-artifact-draft, signature-request-binding, OS-base, and VM-evidence contract surfaces. Runtime Boundary carries Lat pipeline first-clause evidence as provenance metadata. Fedora platform evidence distinguishes a bounded disposable-VM CLI payload, /usr/bin/latticra plus /usr/share/doc/latticra/README.md, with host_install_ready_for_cli_payload=1 from production_installer_ready=0, fedora_distribution_ready=0, fedora_approval_claimed=0, daily_driver_install_ready=0, and immutable_fedora_ready=0. Nadia now records contract-only prompt-evaluation-result release receipt, review, disposition, release, receipt, review, and review-disposition metadata through Stage-46 while preserving runtime_invoked=0, prompt_evaluated=0, token_generation_performed=0, inference_performed=0, qa_dialogue_generated=0, answer_text_generated=0, network_authority=0, tool_execution_authority=0, source_mutation_authority=0, sexual_request_refusal=always, and manipulation_resistance=required. These surfaces do not claim host embedding, language execution, operator evaluation, payload materialization outside the validated package payload, receipt writing, signing, signer invocation, signature verification, private-key handling, key generation, trust-store behavior, revocation checks, object sealing, capability enforcement, effect execution, VM launch outside recorded validation, boot behavior, host effects beyond the disposable VM transcript, Nadia release/receipt/review/disposition recording, model-output recording, prompt evaluation, token generation, inference, dialogue generation, network authority, runtime authority, runtime handoff execution, production OS status, Fedora approval, daily-driver readiness, immutable Fedora readiness, security capability, update safety, recovery safety, sandboxing, malware prevention, ransomware prevention, OS-replacement readiness, or production cryptography.

The new System Substrate handbook places Seal in the full project architecture alongside:

- Latticra Panel
- Latticra Console
- Nucleus report-only task boundaries
- Nadia Stage-46 offline AI contract metadata
- macOS and Fedora platform validation lanes
- guarded local-prefix installation
- receipts and reports
- command contracts
- Lat and LIR contract surfaces
- runtime-boundary metadata
- validation lanes
- research artifacts and visual theorem engines

Canonical public wording

Use language like:

Latticra is an early, evidence-bound system substrate for describing, measuring, and presenting local project state under explicit authority constraints.

Avoid language that claims production protection, runtime enforcement, malware prevention, ransomware prevention, kernel enforcement, root authority, or certification unless those claims are implemented, tested, and documented with reproducible evidence.

docs/LOCAL_ARTIFACT_MANIFEST_FIXTURE.md

Source title: Local Artifact Manifest Fixture

Lines: 81 | Words: 202 | SHA-256: 9d9df70c0c0e527ce356276cfdf969e5b62ec9258ed7bebc1ad28001c7f79342

Local Artifact Manifest Fixture

Status: fixture record Scope: repository fixture for the local Latticra artifact manifest lane.

Purpose

This slice adds the first committed manifest fixture for the local no-effect CLI RPM artifact.

The fixture gives the existing manifest contract a concrete, reviewable sample without creating a generated artifact, publishing a package, or changing host behavior.

Fixture path

```
fixtures/artifact/local-artifact-manifest.txt
```

Guard path

```
scripts/test-local-artifact-manifest-fixture.sh
```

Expected output:

```
local_artifact_manifest_fixture: ok
```

Fixture boundary

The fixture records metadata for the current local no-effect CLI RPM payload:

```
/usr/bin/latticra
/usr/share/doc/latticra/README.md
```

It keeps the readiness boundary explicit:

```
production_installer_ready=0
fedora_distribution_ready=0
fedora_approval_claimed=0
daily_driver_install_ready=0
immutable_fedora_ready=0
```

The checksum, signature, and SBOM fields remain fixture placeholders or absent evidence fields:

```
artifact_checksum_recorded=0
artifact_signature_recorded=0
artifact_sbom_recorded=0
```

Relationship to the contract

This fixture follows:

```
docs/LOCAL_INSTALLER_ARTIFACT_MANIFEST_CONTRACT.md
```

It satisfies the next implementation lane named by that contract:

```
Add installer artifact manifest fixture
```

Non-claims

This fixture is not a generated release artifact.

It is not a production package.

It is not Fedora approval or Fedora distribution readiness.

It is not daily-driver or immutable-Fedora readiness.

It does not change boot, service, kernel, SELinux, network, install, uninstall, upgrade, rollback, or host state.

docs/MACOS_APP_BUNDLE_WRITER_ALIGNMENT.md

Source title: macOS App Bundle Writer Alignment

Lines: 141 | Words: 506 | SHA-256: b8716bdfa2dcfd6a237340f28414b18dfa32984ff0b0cc6c5dab5da2264d3189

macOS App Bundle Writer Alignment

Status: macOS app bundle writer contract/status alignment Date: 2026-05-25 CDT Scope: align public and status meaning after the no-effect macOS app bundle writer dry-run prototype.

Purpose

This alignment separates the current no-effect writer-shaped dry-run prototype from any future commit-capable macOS app bundle writer.

The current prototype can render phases, validate user-local paths, inspect existing managed markers, report missing local executable/icon candidates, and block unsafe paths.

It cannot create an app bundle, write Application Support files, install wrappers, write receipts, mutate shell profiles, or verify a real install.

The macOS local candidate asset probe remains a no-effect readiness check for caller-supplied Panel executable and icon candidates. The macOS dry-run writer candidate integration joins that probe to the dry-run writer and only reports readiness when both agree while all commit and write flags remain disabled. The macOS commit gate contract keeps that readiness from becoming write authority, and the macOS verification transcript contract defines the future evidence required before install verification can be claimed.

Current Capability

```
macos_app_bundle_writer_dry_run_present=1
macos_app_bundle_writer_phase_report_present=1
macos_app_bundle_writer_path_guard_present=1
macos_app_bundle_writer_marker_inspection_present=1
macos_app_bundle_writer_missing_candidate_detection_present=1
macos_local_candidate_asset_probe_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
macos_app_bundle_writer_commit_disabled=1
```

The current command is:

```
sh scripts/macos-app-bundle-writer-dry-run.sh
```

The current expected authority posture is:

```
phase_report_only=1
commit_user_local_managed_artifacts=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
host_mutation_performed=0
network_performed=0
```

Explicit Non-Capability

The following must remain false until a separate commit-capable writer contract, implementation, reset/uninstall implementation, and verification evidence exist:

```
macos_app_bundle_writer_present=0
macos_app_bundle_commit_capable_writer_present=0
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
```

The dry-run writer must not be described as:

```
macOS installer
app bundle installer
commit-capable writer
user-local install evidence
app bundle evidence
signed build evidence
notarized build evidence
production installer
security product
```

Future Commit Gate Requirements

Before any commit-capable writer exists, a future lane must add all of:

```
local_panel_executable_candidate_probe=1
local_icon_candidate_probe=1
commit_gate_contract=1
reset_uninstall_implementation_plan=1
verification_transcript_contract=1
unsafe_path_negative_tests=1
unmanaged_target_preservation_tests=1
receipt_completeness_tests=1
```

Even then, commit behavior must remain disabled until a separate implementation explicitly changes:

```
commit_user_local_managed_artifacts=1
```

The current integration can report only:

```
integration_decision=ready-for-future-commit-gate-no-effect
commit_user_local_managed_artifacts=0
```

Safe Public Wording

A careful public statement is:

```
Latticra has a no-effect macOS app bundle writer dry-run prototype that renders future writer
phases and blocks unsafe paths while keeping all writes disabled.
```

That statement must not be shortened to "Latticra has a macOS app bundle writer" unless the sentence also says "dry-run" and "no-effect."

Validation

This alignment is guarded by:

```
sh scripts/test-macos-app-bundle-writer-alignment.sh
```

Expected output:

```
macos_app_bundle_writer_alignment: ok
```

Non-Claims

This alignment is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_APP_BUNDLE_WRITER_DRY_RUN.md

Source title: macOS App Bundle Writer Dry-Run Prototype

Lines: 183 | Words: 565 | SHA-256: 81acd5a6179cfd8d902e6bab82c579513a67655d89a607854daa44781e0d632e

macOS App Bundle Writer Dry-Run Prototype

Status: no-effect macOS app bundle writer dry-run prototype Date: 2026-05-25 CDT Scope: executable phase-report prototype for the future macOS user-local app bundle writer.

Purpose

This prototype emits the writer-shaped phase report from the macOS user-local app bundle implementation plan.

It validates planned user-local paths, inspects existing targets for Latticra managed markers, checks whether optional local Panel executable and icon candidates are present, and reports the phase decision. It does not create an app bundle, write Application Support files, install wrappers, mutate shell profiles, build the Panel, use launchd, access Keychain, request TCC permissions, use Endpoint Security, use System Extensions, use Network Extensions, open the network, or grant runtime authority.

The macOS local candidate asset probe is the no-effect readiness check that can supply those optional executable and icon candidates. The macOS dry-run writer candidate integration then runs the probe and this writer dry-run together to prove that accepted inputs can move the writer decision to ready-for-future-commit-gate while commit_user_local_managed_artifacts=0. The macOS commit gate contract keeps that future gate closed, and the macOS verification transcript contract defines the exact post-write evidence required before install verification can be claimed.

Command

```
sh scripts/macos-app-bundle-writer-dry-run.sh
```

Optional candidate inputs:

```
sh scripts/macos-app-bundle-writer-dry-run.sh \
  --panel-executable &lt;local-executable&gt; \
  --icon &lt;local-icon-file&gt;
```

Report Fields

The report starts with:

```
MACOS APP BUNDLE WRITER DRY RUN
dry_run_status=ok
phase_report_only=1
```

It must include:

```
path_guard_status=allowed-user-local-dry-run
unsafe_path_detected=0
phase_1=validate_macos_host_and_toolchain_probe
phase_2=validate_user_local_paths_and_contract
phase_3=inspect_existing_targets_for_managed_markers
phase_4=stage_app_bundle_manifest_and_infoplist
phase_5=stage_panel_executable_and_icon_inputs
phase_6=stage_application_support_layout
phase_7=stage_cli_wrappers
phase_8=commit_user_local_managed_artifacts
phase_8_status=disabled
phase_9=write_receipts_and_measurements
phase_9_status=disabled
phase_10=run_verification_transcript
phase_10_status=not-run
commit_user_local_managed_artifacts=0
```

If no executable or icon candidate is supplied, the default dry-run may report:

```
dry_run_decision=blocked-missing-panel-executable
```

That is expected. The dry-run writer is allowed to report readiness gaps, but it must not fill them by building, downloading, generating, signing, or writing files.

Candidate Integration

The writer dry-run can be paired with the candidate probe by running:

```
sh scripts/macos-dry-run-writer-candidate-integration.sh \
  --panel-executable &lt;local-executable&gt; \
  --icon &lt;local-icon-file&gt;
```

When both checks agree and all write flags remain disabled, the integration reports:

```
asset_probe_decision=ready-for-dry-run-writer-inputs
writer_dry_run_decision=ready-for-future-commit-gate
integration_decision=ready-for-future-commit-gate-no-effect
commit_user_local_managed_artifacts=0
```

Blocked Paths

Unsafe path inputs must be blocked before write:

```
sh scripts/macos-app-bundle-writer-dry-run.sh --app-bundle /Applications/Latticra.app
```

Expected fields:

```
path_guard_status=blocked-unsafe-user-local-path
unsafe_path_detected=1
dry_run_decision=blocked-unsafe-path
commit_user_local_managed_artifacts=0
app_bundle_write_performed=0
host_mutation_performed=0
```

Managed Marker Inspection

Existing targets are classified as:

```
absent
managed
unmanaged-preserve-and-block
```

The dry-run prototype must report unmanaged targets as blocked and preserved:

```
unmanaged_existing_target=1
dry_run_decision=blocked-unmanaged-existing-target
```

It must not remove, overwrite, rename, chmod, or repair unmanaged targets.

Authority Boundary

The dry-run prototype must always preserve:

```
commit_user_local_managed_artifacts=0
application_support_write_performed=0
payload_write_performed=0
config_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Validation

This prototype is guarded by:

```
sh scripts/test-macos-app-bundle-writer-dry-run.sh
```

Expected output:

```
macos_app_bundle_writer_dry_run: ok
```

Non-Claims

This prototype is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_BUILD_PLATFORM_PROBE.md

Source title: macOS Build Platform Probe

Lines: 124 | Words: 357 | SHA-256: 05da56d3dc15a47ee5c7562b11646a99144c025482dd0d47d1894369e3bfe2ef

macOS Build Platform Probe

Status: no-effect macOS build/platform probe Date: 2026-05-25 CDT Scope: bounded platform and build-readiness probe after the macOS integration transferability map.

Purpose

This lane records whether the current checkout has enough local toolchain and source evidence to consider later macOS Panel and C test build work.

It is intentionally not an installer lane. It does not create an app bundle, build a package, install wrappers, mutate shell profiles, use launchd, access Keychain, request TCC permissions, use Endpoint Security, use System Extensions, use Network Extensions, open the network, or grant runtime authority.

Probe Command

```
sh scripts/macos-build-platform-probe.sh
```

Optional report file:

```
sh scripts/macos-build-platform-probe.sh --output reports/macos-build-platform-probe.txt
```

The default command writes the report to stdout. The optional --output path writes only where the operator explicitly asks.

Report Fields

The report records:

```
MACOS BUILD PLATFORM PROBE
probe_status=ok
host_kernel_name=Darwin
host_arch=&lt;architecture&gt;
macos_host_detected=&lt;0-or-1&gt;
sw_vers_recorded=&lt;0-or-1&gt;
clang_probe_recorded=&lt;0-or-1&gt;
rust_probe_recorded=1
architecture_recorded=1
panel_build_probe_recorded=1
panel_build_performed=0
panel_build_ready=&lt;0-or-1&gt;
c_test_build_probe_recorded=1
c_test_build_performed=0
c_test_build_ready=&lt;0-or-1&gt;
app_bundle_created=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
```

The probe records build readiness. It does not run cargo build, compile C tests, create an app bundle, or install artifacts.

Read-Only Inputs

The probe may read:

```
uname
sw_vers
clang --version
```

```
rustc --version
cargo --version
make availability
installer/latticra-installer/Cargo.toml
installer/latticra-installer/Cargo.lock
installer/latticra-installer/src/ui.rs
tests/lat_language_grammar_invariants.c
src/lat_parser.c
include/latticra
```

On non-macOS hosts, the script still emits a deterministic report with:

```
macos_host_detected=0
```

That allows CI to validate the contract without pretending it has macOS install evidence.

Authority Boundary

The probe must preserve:

```
app_bundle_created=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Validation

This lane is guarded by:

```
sh scripts/test-macos-build-platform-probe.sh
```

Expected output:

```
macos_build_platform_probe: ok
```

Non-Claims

This probe is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Next Recommended Lane

```
Add a macOS user-local app bundle implementation plan that remains no-effect and defines writer
phases, failure behavior, reset/uninstall sequencing, verification commands, and guard tests
before implementation.
```

docs/MACOS_COMMIT_GATE_CONTRACT.md

Source title: macOS Commit Gate Contract

Lines: 171 | Words: 357 | SHA-256: efe332ce7a732128ad8a9be1b82755148cd9add975e089f7494c6d8a7a539936

macOS Commit Gate Contract

Status: no-effect macOS commit gate contract Date: 2026-05-25 CDT Scope: closed commit gate for any future macOS user-local managed app bundle writer.

Purpose

This contract separates dry-run readiness from write authority.

The macOS dry-run writer candidate integration can report:

```
integration_decision=ready-for-future-commit-gate-no-effect
```

The macOS verification transcript contract and reset/uninstall dry-run contract now define future evidence shapes, but they do not provide write, reset, uninstall, or transcript evidence. That does not open the commit gate. The current and only allowed commit posture remains:

```
commit_gate_state=closed
commit_user_local_managed_artifacts=0
macos_app_bundle_commit_capable_writer_present=0
```

Command

```
sh scripts/macos-commit-gate-contract.sh
```

The command writes only a deterministic report to stdout.

Current Gate Decision

The current gate decision is:

```
commit_gate_decision=blocked-missing-managed-write-implementation
```

The gate stays closed because these future pieces do not exist yet:

```
managed_write_implementation_present=0
reset_uninstall_implementation_present=0
macos_verification_transcript_contract_present=1
verification_transcript_contract_present=1
verification_transcript_evidence_present=0
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_dry_run_evidence_present=0
absence_report_evidence_present=0
```

Opening Preconditions

Before commit_user_local_managed_artifacts may ever become 1, a later implementation must provide all of:

```
candidate_flow_ready_required=1
operator_explicit_commit_intent_required=1
managed_write_implementation_present=1
reset_uninstall_implementation_present=1
verification_transcript_contract_present=1
verification_transcript_evidence_present=1
unsafe_path_negative_tests_required=1
unmanaged_target_preservation_tests_required=1
receipt_completeness_tests_required=1
reset_uninstall_dry_run_required=1
verification_transcript_required=1
```

The future implementation must still preserve user-local scope:

```
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
```

Authority Boundary

This contract preserves:

```
commit_user_local_managed_artifacts=0
application_support_write_performed=0
payload_write_performed=0
config_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-commit-gate-contract.sh
```

Expected output:

```
macos_commit_gate_contract: ok
```

Non-Claims

This contract is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_DRY_RUN_PLAN_ADAPTER.md

Source title: macOS Dry-Run Plan Adapter

Lines: 127 | Words: 361 | SHA-256: 14192bfa954ad4ea76d635c426db2da82f6403dd79121ab3772c51a5f93c4376

macOS Dry-Run Plan Adapter

Status: no-effect macOS dry-run plan adapter Date: 2026-05-25 CDT Scope: render user-local macOS installation intent after the build/platform probe without writing artifacts.

Purpose

This lane adapts the current Latticra user-local installer model to macOS as a dry-run plan only.

It renders the intended Application Support prefix, app bundle layout, CLI wrapper paths, receipt paths, logs/caches paths, and verification command. It does not create those paths, write an app bundle, install wrappers, mutate shell profiles, use launchd, access Keychain, request TCC permissions, use Endpoint Security, use System Extensions, use Network Extensions, open the network, or grant runtime authority.

Adapter Command

```
sh scripts/macos-dry-run-plan-adapter.sh
```

Optional report file:

```
sh scripts/macos-dry-run-plan-adapter.sh --output reports/macos-dry-run-plan.txt
```

The default command writes the plan to stdout. The optional --output path writes only where the operator explicitly asks.

Default Planned Layout

```
application_support_prefix=$HOME/Library/Application Support/Latticra
app_bundle=$HOME/Applications/Latticra Panel.app
cli_bin=$HOME/.local/bin
receipts_dir=$HOME/Library/Application Support/Latticra/receipts
logs_dir=$HOME/Library/Logs/Latticra
caches_dir=$HOME/Library/Caches/Latticra
```

The app bundle plan is:

```
Latticra Panel.app/
  Contents/
    Info.plist
    MacOS/latticra-panel
    Resources/latticra-panel.icns
```

The planned wrapper paths are:

```
planned_latticra_wrapper=$HOME/.local/bin/latticra
planned_lat_wrapper=$HOME/.local/bin/lat
planned_latticra_seal_wrapper=$HOME/.local/bin/latticra-seal
planned_latticra_nadia_wrapper=$HOME/.local/bin/latticra-nadia
planned_latticra_panel_wrapper=$HOME/.local/bin/latticra-panel
```

Path Guard

The dry-run adapter classifies the default layout as:

```
path_guard_status=allowed-user-local-dry-run
dry_run_allowed=1
```

If a caller supplies paths outside the supported user-local macOS layout, the plan reports:

```
path_guard_status=blocked-unsafe-user-local-path
dry_run_allowed=0
```

Blocking a path still must not trigger a fallback write, privileged helper, package installer, network lookup, or shell-profile mutation.

Authority Boundary

The plan must preserve:

```
application_support_write_performed=0
payload_write_performed=0
config_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Validation

This lane is guarded by:

```
sh scripts/test-macos-dry-run-plan-adapter.sh
```

Expected output:

```
macos_dry_run_plan_adapter: ok
```

Non-Claims

This adapter is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Next Recommended Lane

```
Add a macOS user-local app bundle implementation plan that remains no-effect and defines writer
phases, failure behavior, reset/uninstall sequencing, verification commands, and guard tests
before implementation.
```

docs/MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION.md

Source title: macOS Dry-Run Writer Candidate Integration

Lines: 142 | Words: 388 | SHA-256: 71027e71158da3a961f2dd64b105cbf8efa0d330125e2db2ca5ddaef374bffc9

macOS Dry-Run Writer Candidate Integration

Status: no-effect macOS dry-run writer candidate integration Date: 2026-05-25 CDT Scope: bridge the local candidate asset probe to the macOS app bundle writer dry-run without enabling writes.

Purpose

This integration runs the macOS local candidate asset probe and the app bundle writer dry-run with the same caller-supplied Panel executable and icon inputs.

It records whether:

```
asset_probe_decision=ready-for-dry-run-writer-inputs
writer_dry_run_decision=ready-for-future-commit-gate
```

```
commit_user_local_managed_artifacts=0
```

Only when all three stay true does the integration report:

```
integration_decision=ready-for-future-commit-gate-no-effect
```

That result is still not a real commit gate opening. It means the no-effect checks agree that the supplied local candidates can be carried to the macOS commit gate contract, which remains closed. The macOS verification transcript contract defines the future evidence required after writes, but no transcript evidence exists yet.

Command

```
sh scripts/macos-dry-run-writer-candidate-integration.sh \
  --panel-executable &lt;local-executable&gt; \
  --icon &lt;local-icon-file&gt;
```

The command writes only a deterministic report to stdout.

Report Fields

The report starts with:

```
MACOS DRY RUN WRITER CANDIDATE INTEGRATION
integration_status=ok
integration_mode=macos-dry-run-writer-candidate-integration
```

It must include:

```
asset_probe_script=&lt;repo&gt;/scripts/macos-local-candidate-asset-probe.sh
writer_dry_run_script=&lt;repo&gt;/scripts/macos-app-bundle-writer-dry-run.sh
asset_probe_decision=&lt;decision&gt;
writer_dry_run_decision=&lt;decision&gt;
writer_phase_5_status=&lt;status&gt;
asset_probe_ready=&lt;0-or-1&gt;
writer_dry_run_ready=&lt;0-or-1&gt;
authority_boundary_preserved=&lt;0-or-1&gt;
candidate_flow_ready=&lt;0-or-1&gt;
integration_decision=&lt;decision&gt;
commit_user_local_managed_artifacts=0
app_bundle_write_performed=0
host_mutation_performed=0
network_performed=0
```

Decisions

The integration may report:

```
blocked-asset-probe-not-ready
blocked-writer-dry-run-not-ready
blocked-authority-boundary-widened
ready-for-future-commit-gate-no-effect
```

If the writer dry-run would otherwise accept an icon file that the local candidate asset probe rejects, the integration remains blocked by the probe:

```
asset_probe_decision=blocked-unsupported-icon-candidate
integration_decision=blocked-asset-probe-not-ready
```

Authority Boundary

The integration preserves:

```
build_performed=0
download_performed=0
copy_performed=0
signing_performed=0
notarization_performed=0
commit_user_local_managed_artifacts=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
```

```
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Validation

This integration is guarded by:

```
sh scripts/test-macos-dry-run-writer-candidate-integration.sh
```

Expected output:

```
macos_dry_run_writer_candidate_integration: ok
```

Non-Claims

This integration is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_INTEGRATION_TRANSFERABILITY_PLAN.md

Source title: macOS Integration Transferability Plan

Lines: 662 | Words: 2029 | SHA-256: 3c26202a66491064d985e592c7af5af028e0108d146e3d05db7f6c5b45dfbb7b

macOS Integration Transferability Plan

Status: macOS integration transferability plan Date: 2026-05-25 CDT Current posture: planning and transferability mapping only Scope: macOS integration paths that can be transferred from current Latticra work without widening authority claims.

Summary

This document maps the current Latticra repository to a macOS integration lane.

The goal is to reuse the strongest existing parts of Latticra: no-effect reporting, denied-by-default authority, receipt-first installation, guarded local-prefix layout, Panel-driven operator review, Lat/LIR metadata paths, Latticra Seal boundary reports, Runtime Boundary classification, Nucleus task-boundary records, and Nadia offline AI contract surfaces.

This is not a macOS implementation, app bundle, notarized release, launch service, sandbox, Keychain integration, Endpoint Security integration, system extension, kernel extension, package installer, or production security claim.

Transferable Now

The following areas are mostly transferable to macOS because they are already no-effect, metadata-oriented, or user-local:

```
lat_parse_validate_lower_pipeline=transferable
lir_metadata_reporting=transferable
l_ui_parser_validation_reporting=transferable
nucleus_report_only_task_boundary=transferable
runtime_boundary_classification_reports=transferable
latticra_seal_report_only_metadata=transferable
latticra_seal_request_boundary_metadata=transferable
latticra_seal_policy_decision_metadata=transferable
latticra_seal_verification_receipt_metadata=transferable
nadia_offline_ai_contract_surfaces=transferable
panel_guided_operator_review_model=transferable
receipt_first_install_evidence=transferable
deny_by_default_authority_vocabulary=transferable
deterministic_shell_guards=partially_transferable
c_invariant_tests=transferable_after_build_probe
rust_egui_panel=transferable_after_macos_build_probe
```

The current Rust Panel is a practical transfer point because it is not bound to GTK or GNOME at the application architecture level. The current shell installer is also useful because its authority model is already user-local and receipt-oriented.

Needs A macOS Adapter

These areas should not be copied directly from the Fedora/Linux lane:

```
xdg_desktop_entry=replace_with_app_bundle
hicolor_icon_cache=replace_with_app_resources
update_desktop_database=not_applicable
gtk_icon_cache=not_applicable
fedora_validation_profile=keep_separate
rpm_validation_lanes=not_transferable_to_macos
systemd_nonclaim_wording=replace_with_launchd_nonclaim_wording
selinux_nonclaim_wording=replace_with_macos_privacy_security_nonclaim_wording
dnf_prerequisites=replace_with_xcode_clt_and_rust_probe
linux_desktop_launcher_verification=replace_with_app_bundle_verification
```

A macOS installer lane should introduce a platform adapter instead of widening the existing Linux path. The adapter should keep dry-run first, then write only user-local managed artifacts.

Recommended macOS paths:

```
app_support_prefix=$HOME/Library/Application Support/Latticra
app_bundle=$HOME/Applications/Latticra Panel.app
logs_dir=$HOME/Library/Logs/Latticra
caches_dir=$HOME/Library/Caches/Latticra
preferences_dir=$HOME/Library/Preferences
optional_cli_bin=$HOME/.local/bin
receipts_dir=$HOME/Library/Application Support/Latticra/receipts
```

The default macOS lane should not write /Applications, /Library, /System, /usr/local, /opt/homebrew, LaunchDaemons, system LaunchAgents, kernel extension paths, system extension paths, network extension paths, or privileged helper tools.

First macOS Lane

The first implementation lane should be deliberately small:

```
stage_0_transferability_plan=present
stage_1_macos_build_probe=present
stage_2_macos_dry_run_plan=present
stage_3_user_local_app_bundle_contract=present
stage_3_user_local_app_bundle_implementation_plan=present
stage_3_user_local_app_bundle_writer_dry_run=present
stage_3_user_local_app_bundle_writer_alignment=present
stage_3_local_candidate_asset_probe=present
stage_3_dry_run_writer_candidate_integration=present
stage_3_macos_commit_gate_contract=present
stage_3_macos_readme_installer_usage=present
stage_3_user_local_app_bundle=future
```

```
stage_4_user_local_verification_transcript_contract=present
stage_4_user_local_verification_transcript_evidence=future
stage_4_user_local_verification_transcript=future
stage_4_macos_reset_uninstall_dry_run_contract=present
stage_4_macos_reset_uninstall_live_target_classifier=present
stage_4_macos_reset_uninstall_dry_run_planner=present
stage_4_macos_reset_uninstall_absence_report_contract=present
stage_4_macos_reset_uninstall_receipt_schema_contract=present
stage_4_macos_reset_uninstall_implementation_gate_contract=present
stage_4_macos_reset_uninstall_operator_intent_contract=present
stage_4_macos_reset_uninstall_effect_authorization_contract=present
stage_4_macos_reset_uninstall_evidence_bundle_contract=present
stage_4_macos_reset_uninstall_live_implementation_plan_contract=present
stage_4_macos_reset_uninstall_live_execution_preflight_contract=present
stage_4_macos_reset_uninstall_live_denial_transcript_contract=present
stage_4_macos_reset_uninstall_live_runner_interface_contract=present
stage_4_macos_reset_uninstall_live_runner_noop_prototype_contract=present
stage_4_macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract=present
stage_4_macos_reset_uninstall_live_runner_denied_dispatch_review_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_gate_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_review_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract=presen
t
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract=
present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_co
ntract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_di
sposition_contract=present
stage_4_macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_di
sposition_review_contract=present
stage_5_codesigning_notarization_plan=future
stage_6_controlled_os_integration_contracts=future
```

Stage 1 records compiler, Rust, architecture, and dependency evidence without installing anything:

```
uname_s_recorded=1
sw_vers_recorded=1
arch_recorded=1
clang_probe_recorded=1
rust_probe_recorded=1
panel_build_probe_recorded=1
c_test_probe_recorded=1
host_mutation_performed=0
network_performed=0
```

Stage 2 adapts the existing installer plan to macOS and still remains dry-run. Stage 3 may write a managed app bundle under the user's home directory only after Stage 2 evidence exists.

Stage 1 is implemented by:

```
docs/MACOS_BUILD_PLATFORM_PROBE.md
scripts/macos-build-platform-probe.sh
docs/status/MACOS_BUILD_PLATFORM_PROBE_STATUS.md
```

Stage 2 is implemented by:

```
docs/MACOS_DRY_RUN_PLAN_ADAPTER.md
scripts/macos-dry-run-plan-adapter.sh
docs/status/MACOS_DRY_RUN_PLAN_ADAPTER_STATUS.md
```

The Stage 3 app bundle contract is implemented by:

```
docs/MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT.md
docs/status/MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT_STATUS.md
```

The Stage 3 app bundle implementation plan is implemented by:

```
docs/MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN.md
docs/status/MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN_STATUS.md
```

The Stage 3 app bundle writer dry-run prototype is implemented by:

```
docs/MACOS_APP_BUNDLE_WRITER_DRY_RUN.md
scripts/macos-app-bundle-writer-dry-run.sh
```

```
docs/status/MACOS_APP_BUNDLE_WRITER_DRY_RUN_STATUS.md
```

The Stage 3 app bundle writer alignment is implemented by:

```
docs/MACOS_APP_BUNDLE_WRITER_ALIGNMENT.md
docs/status/MACOS_APP_BUNDLE_WRITER_ALIGNMENT_STATUS.md
```

The Stage 3 local candidate asset probe is implemented by:

```
docs/MACOS_LOCAL_CANDIDATE_ASSET_PROBE.md
scripts/macos-local-candidate-asset-probe.sh
docs/status/MACOS_LOCAL_CANDIDATE_ASSET_PROBE_STATUS.md
```

The Stage 3 dry-run writer candidate integration is implemented by:

```
docs/MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION.md
scripts/macos-dry-run-writer-candidate-integration.sh
docs/status/MACOS_DRY_RUN_WRITER_CANDIDATE_INTEGRATION_STATUS.md
```

The Stage 3 macOS commit gate contract is implemented by:

```
docs/MACOS_COMMIT_GATE_CONTRACT.md
scripts/macos-commit-gate-contract.sh
docs/status/MACOS_COMMIT_GATE_CONTRACT_STATUS.md
```

The Stage 3 macOS README installer usage alignment is implemented by:

```
README.md
docs/status/MACOS_README_INSTALLER_USAGE_STATUS.md
scripts/test-macos-readme-installer-usage.sh
```

The Stage 4 macOS verification transcript contract is implemented by:

```
docs/MACOS_VERIFICATION_TRANSCRIPT_CONTRACT.md
scripts/macos-verification-transcript-contract.sh
docs/status/MACOS_VERIFICATION_TRANSCRIPT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall dry-run contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT.md
scripts/macos-reset-uninstall-dry-run-contract.sh
docs/status/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-target classifier is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER.md
scripts/macos-reset-uninstall-live-target-classifier.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER_STATUS.md
```

The Stage 4 macOS reset/uninstall dry-run planner is implemented by:

```
docs/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER.md
scripts/macos-reset-uninstall-dry-run-planner.sh
docs/status/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER_STATUS.md
```

The Stage 4 macOS reset/uninstall absence-report contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT.md
scripts/macos-reset-uninstall-absence-report-contract.sh
docs/status/MACOS_RESET_UNINSTALL_ABSENCE_REPORT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall receipt-schema contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT.md
scripts/macos-reset-uninstall-receipt-schema-contract.sh
docs/status/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall implementation-gate contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT.md
scripts/macos-reset-uninstall-implementation-gate-contract.sh
docs/status/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall operator-intent contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT.md
scripts/macos-reset-uninstall-operator-intent-contract.sh
docs/status/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall effect-authorization contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT.md
scripts/macos-reset-uninstall-effect-authorization-contract.sh
```

```
docs/status/MACOS_RESET_UNINSTALL_EFFECT_AUTHORIZATION_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall evidence-bundle contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT.md
scripts/macos-reset-uninstall-evidence-bundle-contract.sh
docs/status/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-implementation plan contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT.md
scripts/macos-reset-uninstall-live-implementation-plan-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-execution preflight contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT.md
scripts/macos-reset-uninstall-live-execution-preflight-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-denial transcript contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT.md
scripts/macos-reset-uninstall-live-denial-transcript-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner interface contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-interface-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner no-op prototype contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-noop-prototype-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner denied-dispatch transcript contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-denied-dispatch-transcript-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner denied-dispatch review contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-denied-dispatch-review-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-gate contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-gate-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial transcript contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-transcript-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial review contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-review-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition review contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-review-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT_STAT
US.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition closeout contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT_ST
ATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-contract.
sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTR
ACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTR
ACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-co
ntract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIE
W_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPO
SITION_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-di
sposition-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIE
W_DISPOSITION_CONTRACT_STATUS.md
```

The Stage 4 macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract is implemented by:

```
docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPO
SITION_REVIEW_CONTRACT.md
scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-di
sposition-review-contract.sh
docs/status/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIE
W_DISPOSITION_REVIEW_CONTRACT_STATUS.md
```

App Bundle Direction

The macOS Panel should be represented as a managed user-local app bundle:

```
Latticra Panel.app/
  Contents/
    Info.plist
    MacOS/latticra-panel
    Resources/latticra-panel.icns
    Resources/latticra/
```

The first bundle should be local and unsigned or ad-hoc signed only if the build environment requires it. Developer ID signing, hardened runtime, and notarization belong to a later evidence lane.

The app bundle must preserve these limits:

```
root_authority=0
network_authority=0
runtime_enforcement_authority=0
launchagent_authority=0
keychain_authority=0
```

```
tcc_bypass_authority=0
system_extension_authority=0
endpoint_security_authority=0
```

CLI Direction

The current command wrappers can transfer with adjustment. On macOS, the safest first path is:

```
managed_cli_wrappers=$HOME/.local/bin
operator_path_setup=manual_or_panel_report_only
homebrew_prefix_mutation=0
usr_local_mutation=0
opt_homebrew_mutation=0
```

If the operator does not already use ~/.local/bin, the Panel can report the needed shell profile line, but should not mutate shell startup files in the first macOS lane.

macOS Security Interfaces

These macOS-specific interfaces are future-gated and should not be treated as current capability:

```
Keychain=future_contract_only
Secure_Enclave=future_contract_only
TCC_Privacy=future_contract_only
App_Sandbox=future_contract_only
Hardened_Runtime=future_contract_only
Developer_ID_Signing=future_contract_only
Notarization=future_contract_only
LaunchAgent=future_contract_only
Endpoint_Security=future_contract_only
System_Extension=future_contract_only
Network_Extension=future_contract_only
Privileged_Helper=future_contract_only
```

The first macOS work should avoid prompts for Full Disk Access, Contacts, Photos, Screen Recording, Accessibility, Automation, or Developer Tools permissions unless a separate contract and verification lane exists.

Component Mapping

Latticra area	macOS transferability	Required adaptation
Lat	High	Build/test probe on clang and macOS filesystem paths
LIR	High	Build/test probe only
L-UI	High	Terminal-control renderer remains future-gated
Nucleus	High	Keep report-only task boundary
Runtime Boundary	High	Add macOS domain labels later; no enforcement
Latticra Seal	High	Keep report-only; Keychain and Secure Enclave future-gated
Latticra Panel	Medium-high	App bundle, icon, verification, path adapter
Nadia offline AI	Medium-high	Keep contract-only; no model loading, TCC bypass, or network
Fedora/RPM lanes	Low	Keep as Linux-specific evidence, not macOS input
Installer scripts	Medium	Add platform adapter, app bundle writer, macOS verification
Documentation/status	High	Add macOS-specific non-claims and validation records

Evidence Rules

Any macOS public claim should be backed by a dedicated status record and guard script. The claim must say exactly which host, architecture, command, bundle, receipt, and verification path were observed.

Required evidence before saying "macOS user-local install verified":

```
host_os_macos_recorded=1
architecture_recorded=1
panel_build_completed=1
managed_app_bundle_present=1
managed_cli_wrappers_present=1
receipts_present=1
seal_report_only_output_recorded=1
```

```
lat_or_lir_no_effect_probe_recorded=1
uninstall_or_reset_dry_run_recorded=1
root_authority=0
network_authority=0
system_extension_authority=0
endpoint_security_authority=0
production_installer_ready=0
```

Non-Claims

This plan does not implement macOS runtime behavior, host mutation, app bundle generation, installer writes, Keychain access, Secure Enclave access, TCC handling, sandboxing, hardened runtime, notarization, LaunchAgent behavior, Endpoint Security behavior, System Extension behavior, Network Extension behavior, privileged helper behavior, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Current tracking flags:

```
macos_integration_transferability_map_present=1
macos_runtime_behavior_added=0
macos_host_mutation_added=0
macos_app_bundle_created=0
macos_install_verified=0
macos_production_ready=0
```

Previous recommended lane now present:

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Current recommended lane now present:

```
Add a macOS reset/uninstall live-implementation plan contract that maps future effect-authorized
 execution phases while deletion remains disabled.
```

Current preflight lane now present:

```
Add a macOS reset/uninstall live-execution preflight contract that proves the live
implementation plan still cannot delete until all evidence gates are satisfied.
```

Current live-denial transcript lane now present:

```
Add a macOS reset/uninstall live-denial transcript contract that records the failed preflight
decision without deleting files.
```

Current live-runner interface lane now present:

```
Add a macOS reset/uninstall live-runner interface contract that accepts only a passed preflight
and keeps deletion disabled otherwise.
```

Current live-runner no-op prototype lane now present:

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

Current live-runner denied-dispatch transcript lane now present:

```
Add a macOS reset/uninstall live-runner denied-dispatch transcript contract that records the
no-op runner denial without enabling deletion.
```

Current live-runner denied-dispatch review lane now present:

```
Add a macOS reset/uninstall live-runner denied-dispatch review contract that keeps review-only
dispatch denial evidence separate from any effects.
```

Current live-runner acceptance-gate lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-gate contract that keeps the live runner
closed until passed preflight, evidence, authorization, operator intent, and implementation are
all present.
```

Current live-runner acceptance-denial transcript lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial transcript contract that records the
closed acceptance gate without dispatching effects.
```

Current live-runner acceptance-denial review lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial review contract that reviews the
closed gate transcript without enabling dispatch or deletion.
```

Current live-runner acceptance-denial disposition lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition contract that records the
reviewed closed-gate denial as a no-effect disposition without opening dispatch.
```

Current live-runner acceptance-denial disposition review lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition review contract that
reviews the no-effect disposition without opening dispatch or deletion.
```

Current live-runner acceptance-denial disposition closeout lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout contract that
closes the reviewed no-effect disposition without opening dispatch or deletion.
```

Current live-runner acceptance-denial disposition closeout audit lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract
that audits the no-effect closeout without opening dispatch or deletion.
```

Current live-runner acceptance-denial disposition closeout audit review lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
contract that reviews the no-effect closeout audit without opening dispatch or deletion.
```

Current live-runner acceptance-denial disposition closeout audit review disposition lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition contract that records the reviewed no-effect closeout audit as a no-effect
disposition without opening dispatch or deletion.
```

Current live-runner acceptance-denial disposition closeout audit review disposition review lane now present:

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition review contract that reviews the no-effect closeout audit review disposition without
 opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_LOCAL_CANDIDATE_ASSET_PROBE.md

Source title: macOS Local Candidate Asset Probe

Lines: 143 | Words: 396 | SHA-256: 0cd609e69bba7778ed1f5c68ab2d65c11737a16cee7658093806e754e76551ec

macOS Local Candidate Asset Probe

Status: no-effect macOS local candidate asset probe Date: 2026-05-25 CDT Scope: local Panel executable and icon candidate checks before any macOS app bundle writer can become commit-capable.

Purpose

This probe checks caller-supplied local asset candidates for the macOS app bundle dry-run lane.

It follows the writer alignment requirement:

```
local_panel_executable_candidate_probe=1
local_icon_candidate_probe=1
```

The probe does not build, download, sign, notarize, copy, write app bundle files, mutate host state, open the network, or grant authority.

The macOS dry-run writer candidate integration consumes this probe's ready decision and compares it with the app bundle writer dry-run decision. The integration remains no-effect and keeps commit_user_local_managed_artifacts=0. The macOS commit gate contract then keeps the writer path closed, while the macOS verification transcript contract defines the future post-write evidence required before install verification can be claimed.

Command

```
sh scripts/macos-local-candidate-asset-probe.sh --panel-executable &lt;file&gt; --icon &lt;
file&gt;
```

The command writes only a deterministic report to stdout.

Report Fields

The default report includes:

```
MACOS LOCAL CANDIDATE ASSET PROBE
probe_status=ok
probe_mode=macos-local-candidate-asset-probe
panel_executable_candidate=&lt;path-or-none&gt;
panel_candidate_supplied=&lt;0-or-1&gt;
panel_candidate_path_status=&lt;allowed-or-blocked&gt;
panel_candidate_present=&lt;0-or-1&gt;
panel_candidate_file=&lt;0-or-1&gt;
panel_candidate_executable=&lt;0-or-1&gt;
panel_candidate_readable=&lt;0-or-1&gt;
panel_executable_candidate_supplied=&lt;0-or-1&gt;
panel_executable_candidate_path_status=&lt;allowed-or-blocked&gt;
panel_executable_candidate_present=&lt;0-or-1&gt;
panel_executable_candidate_file=&lt;0-or-1&gt;
panel_executable_candidate_executable=&lt;0-or-1&gt;
panel_executable_candidate_readable=&lt;0-or-1&gt;
icon_candidate=&lt;path-or-none&gt;
icon_candidate_supplied=&lt;0-or-1&gt;
icon_candidate_path_status=&lt;allowed-or-blocked&gt;
icon_candidate_present=&lt;0-or-1&gt;
icon_candidate_file=&lt;0-or-1&gt;
icon_candidate_readable=&lt;0-or-1&gt;
icon_candidate_format=&lt;icns-png-svg-ico-or-none&gt;
icon_candidate_supported_format=&lt;0-or-1&gt;
icon_candidate_icns_ready=&lt;0-or-1&gt;
local_panel_executable_candidate_probe=1
local_icon_candidate_probe=1
asset_probe_decision=&lt;decision&gt;
```

Decisions

The probe may report:

```
blocked-missing-panel-executable-candidate
blocked-disallowed-panel-candidate-path
blocked-panel-candidate-not-file
blocked-panel-candidate-not-executable
blocked-panel-candidate-not-readable
blocked-missing-icon-candidate
blocked-disallowed-icon-candidate-path
blocked-icon-candidate-not-file
blocked-icon-candidate-not-readable
blocked-unsupported-icon-candidate
ready-for-dry-run-writer-inputs
```

ready-for-dry-run-writer-inputs means only that the supplied local files are usable as inputs to the dry-run writer report. It is not app bundle evidence, install evidence, signing evidence, notarization evidence, or verification-transcript evidence.

Authority Boundary

The probe preserves:

```
build_performed=0
panel_build_performed=0
icon_conversion_performed=0
download_performed=0
copy_performed=0
signing_performed=0
notarization_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
```

```
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This probe is guarded by:

```
sh scripts/test-macos-local-candidate-asset-probe.sh
```

Expected output:

```
macos_local_candidate_asset_probe: ok
```

The bridge to the writer dry-run is guarded by:

```
sh scripts/test-macos-dry-run-writer-candidate-integration.sh
```

Non-Claims

This probe is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_DRY_RUN_CONTRACT.md

Source title: macOS Reset/Uninstall Dry-Run Contract

Lines: 231 | Words: 506 | SHA-256: 5a0621ff6a5a36aece8065ae85baaf12b271ad1b09d914761732579e5cdafb8f

macOS Reset/Uninstall Dry-Run Contract

Status: no-effect macOS reset/uninstall dry-run contract Date: 2026-05-25 CDT Scope: contract for managed-target removal order before any future macOS user-local app bundle reset or uninstall can run.

Purpose

This contract defines how a future macOS reset/uninstall dry-run must order managed-target removal and preserve user files.

It is contract-only. It does not delete files, remove directories, write receipts, mutate host state, run absence verification, or claim reset/uninstall implementation.

The macOS verification transcript contract can require this reset/uninstall dry-run contract as evidence shape. The macOS reset/uninstall live-target classifier can report present, managed, and unmanaged targets, and the macOS reset/uninstall dry-run planner can turn those states into a no-effect transcript. No reset or uninstall evidence exists yet.

Command

```
sh scripts/macos-reset-uninstall-dry-run-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current reset/uninstall posture is:

```
macos_reset_uninstall_dry_run_contract_present=1
reset_uninstall_dry_run_contract_state=defined-no-effect
reset_uninstall_dry_run_decision=contract-defined-removal-not-performed
reset_uninstall_dry_run_evidence_present=0
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
```

```
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_receipt_evidence_present=0
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
```

Managed Targets

The future dry-run must bind reset/uninstall evidence to these user-local targets:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
```

Only targets with Latticra managed markers may be removed by a future implementation:

```
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
```

Removal Order

The required reset/uninstall dry-run phases are:

```
reset_phase_1=validate_user_local_targets
reset_phase_2=inspect_managed_markers
reset_phase_3=preserve_unmanaged_targets
reset_phase_4=remove_managed_wrappers
reset_phase_5=remove_managed_app_bundle
reset_phase_6=remove_managed_application_support_prefix
reset_phase_7=write_reset_or_uninstall_receipt_outside_removed_prefix
reset_phase_8=emit_verification_absence_report
```

In this contract, removal phases remain disabled:

```
reset_phase_4_status=disabled
reset_phase_5_status=disabled
reset_phase_6_status=disabled
reset_phase_7_status=disabled
reset_phase_8_status=not-run
```

Preservation Rules

The future reset/uninstall lane must preserve:

```
preserve_user_logs=1
preserve_user_caches=1
preserve_shell_profiles=1
preserve_keychain_items=1
preserve_launchagents=1
preserve_login_items=1
preserve_homebrew_files=1
preserve_applications_root=1
preserve_library_root=1
preserve_system_root=1
preserve_usr_local=1
preserve_opt_homebrew=1
```

Authority Boundary

This contract preserves:

```
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
```

```
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

The follow-on macOS reset/uninstall live-target classifier reads the current user-local target set and reports absent, managed, unmanaged-preserve, or unsafe-path states without deleting files.

The macOS reset/uninstall dry-run planner consumes that classifier output and reports planned managed-target actions without deleting files.

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-dry-run-contract.sh
```

Expected output:

```
macos_reset_uninstall_dry_run_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_DRY_RUN_PLANNER.md

Source title: macOS Reset/Uninstall Dry-Run Planner

Lines: 192 | Words: 434 | SHA-256: 7186fe0b0a5dca26cf2677ef01717a01911c0c8b1c07b4057eaef05c6964983c

macOS Reset/Uninstall Dry-Run Planner

Status: no-effect macOS reset/uninstall dry-run planner Date: 2026-05-25 CDT Scope: dry-run reset/uninstall transcript planner built from the live-target classifier.

Purpose

This planner consumes the macOS reset/uninstall live-target classifier and turns its target states into a reset/uninstall dry-run transcript.

It is report-only. It does not delete files, remove directories, write receipts, mutate host state, run absence verification, open the network, or claim reset/uninstall implementation.

The macOS reset/uninstall absence-report contract now defines the post-removal evidence shape that this planner hands off to.

The macOS reset/uninstall absence-report contract now defines the future evidence shape for the planner's absence-report phase.

Command

```
sh scripts/macos-reset-uninstall-dry-run-planner.sh
```

The command writes only a deterministic report to stdout.

Inputs

The planner consumes:

```
scripts/macos-reset-uninstall-live-target-classifier.sh
macos_reset_uninstall_live_target_classifier_present=1
target_states_recorded=1
```

The same guarded user-local targets are accepted as optional arguments:

```
--app-support-prefix
--app-bundle
--cli-wrapper
--reset-receipts-dir
```

Unsafe target paths are reported and blocked before any dry-run action is planned.

Decisions

The planner can report:

```
reset_uninstall_dry_run_planner_decision=ready-no-targets-observed-no-effect
reset_uninstall_dry_run_planner_decision=planned-managed-target-removal-no-effect
reset_uninstall_dry_run_planner_decision=blocked-unmanaged-targets-preserve-no-effect
reset_uninstall_dry_run_planner_decision=blocked-unsafe-path-no-effect
```

Managed targets become would-remove- dry-run actions only when no unmanaged target blocks the plan. Unmanaged targets are preserved and block managed removal planning.

Example Fields

```
macos_reset_uninstall_dry_run_planner_present=1
planner_report_only=1
dry_run_transcript_present=1
reset_uninstall_dry_run_planner_transcript_present=1
planner_consumes_live_target_classifier=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
app_bundle_target_state=managed
app_bundle_dry_run_action=would-remove-managed-app-bundle
cli_wrapper_dry_run_action=would-remove-managed-wrapper
reset_receipt_dry_run_action=would-write-reset-uninstall-receipt-outside-removed-prefix
absence_report_dry_run_action=would-emit-verification-absence-report-no-effect
planned_removal_count=report-runtime
reset_uninstall_dry_run_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_implemented=0
```

Phases

The dry-run transcript records:

```
dry_run_phase_1=consume_live_target_classifier
dry_run_phase_2=validate_user_local_targets
dry_run_phase_3=plan_unmanaged_target_preservation
dry_run_phase_4=plan_managed_wrapper_removal
dry_run_phase_5=plan_managed_app_bundle_removal
dry_run_phase_6=plan_managed_application_support_removal
dry_run_phase_7=plan_receipt_write_outside_removed_prefix
dry_run_phase_8=plan_verification_absence_report
```

Authority Boundary

This planner preserves:

```
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
```

```
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This planner is guarded by:

```
sh scripts/test-macos-reset-uninstall-dry-run-planner.sh
```

Expected output:

```
macos_reset_uninstall_dry_run_planner: ok
```

Non-Claims

This planner is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_EVIDENCE_BUNDLE_CONTRACT.md

Source title: macOS Reset/Uninstall Evidence-Bundle Contract

Lines: 252 | Words: 529 | SHA-256: 6a62b3a241d0f5a9d7f57397859e52d5df75739ca094fdbb1f299c06939b5983

macOS Reset/Uninstall Evidence-Bundle Contract

Status: no-effect macOS reset/uninstall evidence-bundle contract Date: 2026-05-25 CDT Scope: contract for grouping required macOS reset/uninstall implementation-gate, operator-intent, planner, classifier, receipt, and absence evidence before any live execution.

Purpose

This contract defines the future reset/uninstall evidence bundle for macOS. It keeps live managed-target removal and receipt writing disabled until the implementation gate, operator intent, dry-run planner, live-target classifier, reset/uninstall receipt, and absence-report evidence are all present.

It is contract-only. It does not authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-evidence-bundle-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current evidence-bundle posture is:

```
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_contract_decision=blocked-incomplete-reset-uninstall-evidence-bundle
evidence_bundle_contract_required=1
reset_uninstall_evidence_bundle_required=1
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
reset_uninstall_evidence_bundle_write_enabled=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
```

Required Bundle Inputs

The future bundle must join the currently separate reset/uninstall contracts:

```
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_required=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
operator_intent_evidence_written=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
reset_uninstall_dry_run_evidence_present=0
macos_reset_uninstall_live_target_classifier_present=1
live_target_classifier_evidence_present=0
macos_reset_uninstall_receipt_schema_contract_present=1
reset_uninstall_receipt_evidence_present=0
```

```
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
macos_reset_uninstall_absence_report_contract_present=1
absence_report_evidence_present=0
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
macos_reset_uninstall_implemented=0
reset_uninstall_implementation_present=0
```

Target Scope

Future bundle evidence must stay bound to user-local managed macOS targets and receipt locations:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Bundle Requirements

The future bundle requires:

```
evidence_bundle_schema_version=macos-reset-uninstall-evidence-bundle/1
evidence_bundle_required_component_count=6
evidence_bundle_contract_component_count=6
evidence_bundle_complete_component_count=0
evidence_bundle_requires_implementation_gate_open=1
evidence_bundle_requires_operator_intent_evidence=1
evidence_bundle_requires_dry_run_planner_transcript=1
evidence_bundle_requires_live_target_classifier_report=1
evidence_bundle_requires_reset_uninstall_receipt_evidence=1
evidence_bundle_requires_absence_report_evidence=1
evidence_bundle_requires_effect_authorization_closed_until_complete=1
evidence_bundle_requires_receipt_outside_removed_prefix=1
evidence_bundle_requires_no_unmanaged_targets=1
evidence_bundle_requires_no_unsafe_paths=1
evidence_bundle_requires_no_network=1
evidence_bundle_requires_no_root=1
evidence_bundle_condition_implementation_gate_open=required
evidence_bundle_condition_operator_intent_evidence_present=required
evidence_bundle_condition_dry_run_planner_transcript_present=required
evidence_bundle_condition_live_target_classifier_report_present=required
evidence_bundle_condition_reset_uninstall_receipt_evidence_present=required
evidence_bundle_condition_absence_report_evidence_present=required
evidence_bundle_condition_no_unmanaged_targets=required
evidence_bundle_condition_no_network=required
evidence_bundle_condition_no_root=required
```

The current result state remains incomplete:

```
evidence_bundle_result_implementation_gate_open=not_met
evidence_bundle_result_operator_intent_evidence=not_met
evidence_bundle_result_dry_run_planner_transcript=contract-only
evidence_bundle_result_live_target_classifier_report=contract-only
evidence_bundle_result_reset_uninstall_receipt_evidence=not_met
evidence_bundle_result_absence_report_evidence=not_met
evidence_bundle_result_effect_authorization=closed
evidence_bundle_result_no_unmanaged_targets=not_evaluated_contract_only
evidence_bundle_result_no_network=met
evidence_bundle_result_no_root=met
```

Phases

The future evidence-bundle phases are:

```
evidence_bundle_phase_1=collect_implementation_gate_decision
evidence_bundle_phase_2=collect_operator_intent_evidence
evidence_bundle_phase_3=collect_planner_and_classifier_evidence
evidence_bundle_phase_4=collect_receipt_and_absence_evidence
evidence_bundle_phase_5=validate_complete_bundle
evidence_bundle_phase_6=handoff_to_effect_authorization
evidence_bundle_phase_1_status=contract-only
evidence_bundle_phase_2_status=blocked-missing-operator-intent
```

```
evidence_bundle_phase_3_status=contract-only
evidence_bundle_phase_4_status=blocked-missing-live-receipt-and-absence-evidence
evidence_bundle_phase_5_status=disabled
evidence_bundle_phase_6_status=disabled
```

Authority Boundary

This contract preserves:

```
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-evidence-bundle-contract.sh
```

Expected output:

```
macos_reset_uninstall_evidence_bundle_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, complete evidence-bundle evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Current completed lane:

```
Add a macOS reset/uninstall live-implementation plan contract that maps future effect-authorized
 execution phases while deletion remains disabled.
```

docs/MACOS_RESET_UNINSTALL_IMPLEMENTATION_GATE_CONTRACT.md

Source title: macOS Reset/Uninstall Implementation-Gate Contract

Lines: 224 | Words: 453 | SHA-256: 471a224bcf2e9d7379642d0d897a6e8f6dcad97d71cb0be844ddd7e92b774aab

macOS Reset/Uninstall Implementation-Gate Contract

Status: no-effect macOS reset/uninstall implementation-gate contract Date: 2026-05-25 CDT Scope: contract for keeping future reset/uninstall live activity closed until all required macOS evidence exists.

Purpose

This contract defines the gate that a future macOS reset/uninstall implementation must pass before live managed-target removal or receipt writing can be considered.

It is contract-only. It does not delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-implementation-gate-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current implementation-gate posture is:

```
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
implementation_gate_contract_state=closed-no-effect
implementation_gate_state=closed-no-effect
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
implementation_gate_required=1
implementation_gate_open=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
```

Required Evidence

The gate stays closed until the reset/uninstall chain has all required evidence:

```
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
reset_uninstall_dry_run_evidence_present=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
absence_report_evidence_present=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
```

Target Scope

The future gate must bind evidence to the known user-local macOS targets:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Gate Requirements

The future gate requires:

```
managed_target_classification_required=1
live_target_classifier_evidence_required=1
dry_run_planner_transcript_required=1
receipt_schema_required=1
reset_uninstall_receipt_evidence_required=1
absence_report_contract_required=1
absence_report_evidence_required=1
operator_explicit_reset_uninstall_intent_required=1
operator_intent_must_name_operation=1
operator_intent_must_name_target_scope=1
operator_intent_must_acknowledge_managed_targets_only=1
operator_intent_must_acknowledge_no_unmanaged_removal=1
operator_intent_must_acknowledge_receipt_path=1
```

The gate conditions are:

```
gate_condition_receipt_schema_contract_present=required
gate_condition_absence_report_contract_present=required
gate_condition_dry_run_planner_transcript_present=required
gate_condition_live_target_classifier_present=required
gate_condition_operator_intent_evidence_present=required
gate_condition_no_unmanaged_targets=required
gate_condition_no_unsafe_paths=required
gate_condition_receipt_outside_removed_prefix=required
gate_condition_no_network=required
gate_condition_no_root=required
```

Phases

The future implementation-gate phases are:

```
implementation_gate_phase_1=consume_live_target_classifier
implementation_gate_phase_2=consume_dry_run_planner_transcript
implementation_gate_phase_3=validate_receipt_schema_and_absence_contract
implementation_gate_phase_4=validate_operator_intent
implementation_gate_phase_5=authorize_future_reset_uninstall_implementation
implementation_gate_phase_6=run_live_reset_uninstall
implementation_gate_phase_1_status=contract-only
implementation_gate_phase_2_status=contract-only
implementation_gate_phase_3_status=contract-only
implementation_gate_phase_4_status=contract-only
implementation_gate_phase_5_status=disabled
```

```
implementation_gate_phase_6_status=disabled
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-implementation-gate-contract.sh
```

Expected output:

```
macos_reset_uninstall_implementation_gate_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, receipt evidence, absence verification evidence, operator approval evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_DENIAL_TRANSCRIPT_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Denial Transcript Contract

Lines: 268 | Words: 539 | SHA-256: fbd1b2388b9068d212e37e7b04b307640457d84a9ff66c596b9b87d7a34c7b1c

macOS Reset/Uninstall Live-Denial Transcript Contract

Status: no-effect macOS reset/uninstall live-denial transcript contract Date: 2026-05-26 CDT Scope: contract for recording the failed macOS reset/uninstall live-execution preflight decision without deleting files or writing receipts.

Purpose

This contract defines the transcript emitted after the live-execution preflight remains blocked. It records why live reset/uninstall execution is denied, while preserving the current no-effect boundary.

It is contract-only. It does not authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-denial-transcript-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current live-denial transcript posture is:

```
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_contract_decision=denied-by-preflight-block
live_denial_transcript_required=1
live_denial_transcript_present=1
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_denial_transcript_receipt_write_enabled=0
live_denial_transcript_absence_report_write_enabled=0
live_denial_transcript_preflight_present=1
live_denial_transcript_preflight_passed=0
live_denial_transcript_denial_recorded=1
live_denial_transcript_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
live_denial_transcript_denial_source=macos-reset-uninstall-live-execution-preflight-contract
live_denial_transcript_effect=none
live_denial_transcript_effect_authorization_required=1
live_denial_transcript_effect_authorized=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
```

Preflight Input

The transcript records the failed preflight decision:

```
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_contract_decision=blocked-missing-complete-evidence-bundle-and-effect-a
uthorization
live_execution_preflight_required=1
live_execution_preflight_present=1
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_evidence_present=0
live_execution_preflight_record_write_enabled=0
live_execution_preflight_denial_recorded=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
```

The underlying gates remain closed:

```
macos_reset_uninstall_live_implementation_plan_contract_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_implementation_plan_preflight_passed=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
```

Target Scope

The transcript remains limited to user-local managed macOS targets and future receipt locations:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Transcript Requirements

The transcript requires:

```
live_denial_transcript_schema_version=macos-reset-uninstall-live-denial-transcript/1
live_denial_transcript_required_input_count=7
live_denial_transcript_observed_input_count=7
live_denial_transcript_requires_live_execution_preflight=1
live_denial_transcript_requires_preflight_denial=1
live_denial_transcript_requires_denial_reason=1
live_denial_transcript_requires_no_receipt_write=1
live_denial_transcript_requires_no_absence_report_write=1
live_denial_transcript_requires_no_deletion=1
live_denial_transcript_requires_no_network=1
live_denial_transcript_requires_no_root=1
live_denial_transcript_condition_preflight_present=required
live_denial_transcript_condition_preflight_passed=must_be_zero_for_denial
live_denial_transcript_condition_denial_reason=required
live_denial_transcript_condition_stdout_only=required
live_denial_transcript_condition_no_receipt_write=required
live_denial_transcript_condition_no_absence_report_write=required
live_denial_transcript_condition_no_deletion=required
live_denial_transcript_condition_no_network=required
live_denial_transcript_condition_no_root=required
```

The current result state is:

```
live_denial_transcript_result_preflight_present=met
live_denial_transcript_result_preflight_passed=not_met
live_denial_transcript_result_denial_reason=met
live_denial_transcript_result_stdout_only=met
live_denial_transcript_result_no_receipt_write=met
live_denial_transcript_result_no_absence_report_write=met
live_denial_transcript_result_no_deletion=met
live_denial_transcript_result_no_network=met
live_denial_transcript_result_no_root=met
```

Entries

The transcript entries are:

```
live_denial_transcript_entry_1=preflight_contract_present
live_denial_transcript_entry_2=preflight_decision_blocked
live_denial_transcript_entry_3=missing_complete_evidence_bundle
live_denial_transcript_entry_4=missing_effect_authorization
live_denial_transcript_entry_5=missing_operator_intent_evidence
live_denial_transcript_entry_6=implementation_gate_closed
live_denial_transcript_entry_7=no_effects_performed
live_denial_transcript_entry_1_status=met
live_denial_transcript_entry_2_status=recorded
live_denial_transcript_entry_3_status=recorded
live_denial_transcript_entry_4_status=recorded
live_denial_transcript_entry_5_status=recorded
live_denial_transcript_entry_6_status=recorded
live_denial_transcript_entry_7_status=recorded
```

Phases

The transcript phases are mapped but closed:

```
live_denial_transcript_phase_1=load_preflight_decision
live_denial_transcript_phase_2=render_denial_transcript
live_denial_transcript_phase_3=preserve_no_effect_boundary
live_denial_transcript_phase_4=handoff_to_live_runner_interface
live_denial_transcript_phase_1_status=contract-only
live_denial_transcript_phase_2_status=stdout-only
live_denial_transcript_phase_3_status=enforced-by-zero-authority
live_denial_transcript_phase_4_status=disabled-until-interface-contract
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-denial-transcript-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_denial_transcript_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, complete evidence-bundle evidence, live execution evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-denial transcript contract that records the failed preflight
decision without deleting files.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Follow-on no-op prototype lane:

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

docs/MACOS_RESET_UNINSTALL_LIVE_EXECUTION_PREFLIGHT_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Execution Preflight Contract

Lines: 243 | Words: 531 | SHA-256: b7ce6a7b790d90d69bd9bec6e69935dfed5373728a705d34cfa4dc45f081ab84

macOS Reset/Uninstall Live-Execution Preflight Contract

Status: no-effect macOS reset/uninstall live-execution preflight contract Date: 2026-05-26 CDT Scope: contract for proving future live macOS reset/uninstall execution remains blocked until every evidence and authorization gate is satisfied.

Purpose

This contract defines the preflight boundary immediately before any future live reset/uninstall executor. It consumes the live-implementation plan, evidence-bundle contract, effect-authorization contract, implementation gate, operator intent, planner, classifier, receipt schema, and absence-report contract.

It is contract-only. It does not authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-execution-preflight-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current live-execution preflight posture is:

```
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_contract_decision=blocked-missing-complete-evidence-bundle-and-effect-a
```

```
uthorization
live_execution_preflight_required=1
live_execution_preflight_present=1
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_evidence_present=0
live_execution_preflight_record_write_enabled=0
live_execution_preflight_denial_recorded=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
```

Required Inputs

The preflight remains blocked by the current no-effect evidence chain:

```
macos_reset_uninstall_live_implementation_plan_contract_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_implementation_plan_preflight_passed=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
```

Target Scope

The preflight remains limited to user-local managed macOS targets and receipt locations:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Preflight Requirements

The future preflight requires:

```
live_execution_preflight_schema_version=macos-reset-uninstall-live-execution-preflight/1
live_execution_preflight_required_component_count=9
live_execution_preflight_passed_component_count=0
live_execution_preflight_requires_live_implementation_plan=1
live_execution_preflight_requires_complete_evidence_bundle=1
live_execution_preflight_requires_effect_authorization=1
live_execution_preflight_requires_implementation_gate_open=1
live_execution_preflight_requires_operator_intent_evidence=1
live_execution_preflight_requires_dry_run_planner_transcript=1
live_execution_preflight_requires_live_target_classifier_report=1
```

```
live_execution_preflight_requires_receipt_schema=1
live_execution_preflight_requires_absence_report_contract=1
live_execution_preflight_requires_no_unmanaged_targets=1
live_execution_preflight_requires_no_unsafe_paths=1
live_execution_preflight_requires_no_network=1
live_execution_preflight_requires_no_root=1
live_execution_preflight_condition_live_implementation_plan=required
live_execution_preflight_condition_complete_evidence_bundle=required
live_execution_preflight_condition_effect_authorization=required
live_execution_preflight_condition_implementation_gate_open=required
live_execution_preflight_condition_operator_intent_evidence_present=required
live_execution_preflight_condition_no_unmanaged_targets=required
live_execution_preflight_condition_no_network=required
live_execution_preflight_condition_no_root=required
```

The current result state remains closed:

```
live_execution_preflight_result_live_implementation_plan=planned-no-effect
live_execution_preflight_result_complete_evidence_bundle=not_met
live_execution_preflight_result_effect_authorization=not_met
live_execution_preflight_result_implementation_gate_open=not_met
live_execution_preflight_result_operator_intent_evidence=not_met
live_execution_preflight_result_dry_run_planner_transcript=contract-only
live_execution_preflight_result_live_target_classifier_report=contract-only
live_execution_preflight_result_receipt_schema=contract-only
live_execution_preflight_result_absence_report_contract=contract-only
live_execution_preflight_result_no_unmanaged_targets=not_evaluated_contract_only
live_execution_preflight_result_no_network=met
live_execution_preflight_result_no_root=met
```

Phases

The future preflight phases are mapped but closed:

```
live_execution_preflight_phase_1=load_live_implementation_plan
live_execution_preflight_phase_2=load_complete_evidence_bundle
live_execution_preflight_phase_3=verify_effect_authorization
live_execution_preflight_phase_4=verify_operator_intent
live_execution_preflight_phase_5=verify_managed_targets_only
live_execution_preflight_phase_6=verify_receipt_and_absence_paths
live_execution_preflight_phase_7=compute_live_execution_decision
live_execution_preflight_phase_8=handoff_to_live_executor
live_execution_preflight_phase_1_status=contract-only
live_execution_preflight_phase_2_status=blocked-evidence-bundle-incomplete
live_execution_preflight_phase_3_status=blocked-effect-authorization-closed
live_execution_preflight_phase_4_status=blocked-operator-intent-missing
live_execution_preflight_phase_5_status=contract-only
live_execution_preflight_phase_6_status=contract-only
live_execution_preflight_phase_7_status=blocked-no-effect
live_execution_preflight_phase_8_status=disabled
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
```

```
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-execution-preflight-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_execution_preflight_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, complete evidence-bundle evidence, live execution evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-execution preflight contract that proves the live
implementation plan still cannot delete until all evidence gates are satisfied.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Follow-on no-op prototype lane:

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

docs/MACOS_RESET_UNINSTALL_LIVE_IMPLEMENTATION_PLAN_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Implementation Plan Contract

Lines: 251 | Words: 530 | SHA-256: c7241ed4172e9ed436b086252bda8e313487753b40639ba4287adf2aa7632b6d

macOS Reset/Uninstall Live-Implementation Plan Contract

Status: no-effect macOS reset/uninstall live-implementation plan contract Date: 2026-05-25 CDT Scope: contract for mapping future effect-authorized macOS reset/uninstall phases while deletion remains disabled.

Purpose

This contract defines the future live reset/uninstall implementation plan for macOS. It maps the order in which a future effect-authorized implementation would load evidence, resolve managed targets, preserve unmanaged targets, write receipts, write absence reports, and emit verification evidence.

It is contract-only. It does not authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-implementation-plan-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current live-implementation plan posture is:

```
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_contract_decision=blocked-deletion-disabled-and-evidence-incomplete
live_implementation_plan_required=1
live_implementation_plan_complete=0
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_receipt_write_enabled=0
live_implementation_plan_absence_write_enabled=0
live_implementation_plan_preflight_required=1
live_implementation_plan_preflight_present=1
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
```

Required Gates

The future plan remains blocked until the existing reset/uninstall evidence chain is complete:

```
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_required=1
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
```

```
macos_commit_gate_contract_present=1
```

Target Scope

The future implementation plan is limited to user-local managed targets:

```
live_implementation_plan_target_scope=user-local-managed-only
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Plan Requirements

The future implementation plan requires:

```
live_implementation_plan_schema_version=macos-reset-uninstall-live-implementation-plan/1
live_implementation_plan_required_component_count=8
live_implementation_plan_complete_component_count=0
live_implementation_requires_evidence_bundle_complete=1
live_implementation_requires_effect_authorization=1
live_implementation_requires_implementation_gate_open=1
live_implementation_requires_operator_intent_evidence=1
live_implementation_requires_no_unmanaged_targets=1
live_implementation_requires_receipt_schema=1
live_implementation_requires_absence_report=1
live_implementation_requires_live_execution_preflight=1
live_implementation_requires_no_network=1
live_implementation_requires_no_root=1
```

The current result state remains blocked:

```
live_implementation_condition_evidence_bundle_complete=required
live_implementation_condition_effect_authorized=required
live_implementation_condition_implementation_gate_open=required
live_implementation_condition_operator_intent_evidence_present=required
live_implementation_condition_no_unmanaged_targets=required
live_implementation_condition_live_execution_preflight_present=required
live_implementation_condition_no_network=required
live_implementation_condition_no_root=required
live_implementation_result_evidence_bundle_complete=not_met
live_implementation_result_effect_authorized=not_met
live_implementation_result_implementation_gate_open=not_met
live_implementation_result_operator_intent_evidence=not_met
live_implementation_result_no_unmanaged_targets=not_evaluated_contract_only
live_implementation_result_live_execution_preflight=blocked-no-effect
live_implementation_result_no_network=met
live_implementation_result_no_root=met
```

Phases

The future implementation phases are mapped but disabled:

```
live_implementation_phase_1=load_complete_evidence_bundle
live_implementation_phase_2=verify_effect_authorization
live_implementation_phase_3=resolve_user_local_managed_targets
live_implementation_phase_4=confirm_no_unmanaged_targets
live_implementation_phase_5=prepare_receipt_and_absence_paths
live_implementation_phase_6=remove_managed_cli_wrapper
live_implementation_phase_7=remove_managed_app_bundle
live_implementation_phase_8=remove_managed_application_support
live_implementation_phase_9=write_reset_uninstall_receipt
live_implementation_phase_10=write_absence_report
live_implementation_phase_11=emit_verification_transcript
live_implementation_phase_1_status=contract-only
live_implementation_phase_2_status=blocked-effect-authorization-closed
live_implementation_phase_3_status=contract-only
live_implementation_phase_4_status=blocked-evidence-incomplete
live_implementation_phase_5_status=disabled
live_implementation_phase_6_status=disabled
live_implementation_phase_7_status=disabled
live_implementation_phase_8_status=disabled
live_implementation_phase_9_status=disabled
live_implementation_phase_10_status=disabled
live_implementation_phase_11_status=disabled
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-implementation-plan-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_implementation_plan_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, complete evidence-bundle evidence, live execution evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-implementation plan contract that maps future effect-authorized
 execution phases while deletion remains disabled.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

Follow-on no-op prototype lane:

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Contract

Lines: 214 | Words: 415 | SHA-256: 2acdbf61720ceca14d643366a3959052333838cc321205eb61c52073cfa217a4

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract Date: 2026-05-26 CDT Scope: audit contract for the no-effect disposition closeout without enabling dispatch or deletion.

Purpose

This contract audits the macOS reset/uninstall live-runner acceptance-denial disposition closeout. It confirms that the reviewed no-effect disposition remains closed, the acceptance gate remains closed, and no runner dispatch or deletion authority is opened.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, perform a live reset/uninstall, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_present=
1
live_runner_acceptance_denial_disposition_closeout_audit_contract_state=audited-no-effect-closeo
ut
live_runner_acceptance_denial_disposition_closeout_audit_contract_decision=no-effect-closeout-au
dit-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_required=1
live_runner_acceptance_denial_disposition_closeout_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_scope=audit-no-effect-closeout-only
live_runner_acceptance_denial_disposition_closeout_audit_source_contract=macos-reset-uninstall-l
ive-runner-acceptance-denial-disposition-review-contract
live_runner_acceptance_denial_disposition_closeout_audit_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_disposition_present=1
live_runner_acceptance_denial_disposition_closeout_audit_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_disposition_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_disposition_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_required=1
live_runner_acceptance_denial_disposition_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_recorded=1
live_runner_acceptance_denial_disposition_closeout_audited=1
live_runner_acceptance_denial_disposition_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_requires_future_audit_review=1
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_audited=1
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_audit_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_closeout_audit_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_result=no-effect-closeout-audit-boundar
y-preserved
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Review

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_contract_decision=no-effect-disposition-review-
keeps-dispatch-closed
live_runner_acceptance_denial_disposition_review_completed=1
live_runner_acceptance_denial_disposition_review_recorded=1
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_review_disposition_reviewed=1
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_review_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_review_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_closeout_opened=0
```

Requirements

```
live_runner_acceptance_denial_disposition_closeout_audit_schema_version=macos-reset-uninstall-li
ve-runner-acceptance-denial-disposition-closeout-audit/1
live_runner_acceptance_denial_disposition_closeout_audit_required_input_count=8
live_runner_acceptance_denial_disposition_closeout_audit_observed_input_count=8
live_runner_acceptance_denial_disposition_closeout_audit_requires_acceptance_denial_disposition=
1
live_runner_acceptance_denial_disposition_closeout_audit_requires_acceptance_denial_disposition_
review=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_closed_gate=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_effect_closeout=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_audit_only=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_effects=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_network=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_audit_requires_audit_review_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_result_acceptance_denial_disposition=me
t
live_runner_acceptance_denial_disposition_closeout_audit_result_acceptance_denial_disposition_re
view=met
live_runner_acceptance_denial_disposition_closeout_audit_result_closed_gate=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_effect_closeout=met
live_runner_acceptance_denial_disposition_closeout_audit_result_audit_only=met
live_runner_acceptance_denial_disposition_closeout_audit_result_stdout_only=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_effects=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_network=met
live_runner_acceptance_denial_disposition_closeout_audit_result_no_root=met
live_runner_acceptance_denial_disposition_closeout_audit_result_audit_review_handoff=met
```

Audit Entries

```
live_runner_acceptance_denial_disposition_closeout_audit_entry_1=acceptance_denial_disposition_r
eview_present
live_runner_acceptance_denial_disposition_closeout_audit_entry_2=no_effect_closeout_audited
live_runner_acceptance_denial_disposition_closeout_audit_entry_3=dispatch_remains_closed
live_runner_acceptance_denial_disposition_closeout_audit_entry_4=deletion_remains_disabled
live_runner_acceptance_denial_disposition_closeout_audit_entry_5=no_effects_confirmed
live_runner_acceptance_denial_disposition_closeout_audit_entry_6=audit_review_required
live_runner_acceptance_denial_disposition_closeout_audit_entry_1_status=met
live_runner_acceptance_denial_disposition_closeout_audit_entry_2_status=audited
live_runner_acceptance_denial_disposition_closeout_audit_entry_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_entry_4_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_entry_5_status=met
live_runner_acceptance_denial_disposition_closeout_audit_entry_6_status=recorded
```

Phases

```
live_runner_acceptance_denial_disposition_closeout_audit_phase_1=load_acceptance_denial_disposit
ion_review
live_runner_acceptance_denial_disposition_closeout_audit_phase_2=audit_no_effect_closeout
live_runner_acceptance_denial_disposition_closeout_audit_phase_3=confirm_closed_dispatch_boundar
y
live_runner_acceptance_denial_disposition_closeout_audit_phase_4=handoff_to_acceptance_denial_di
sposition_closeout_audit_review
live_runner_acceptance_denial_disposition_closeout_audit_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_closeout_audit_phase_2_status=audit-only
live_runner_acceptance_denial_disposition_closeout_audit_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_phase_4_status=future-review-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-c
ontract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, deletion evidence, closeout review evidence, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit contract
that audits the no-effect closeout without opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Contract

Lines: 205 | Words: 417 | SHA-256: e392f6fe952a936c12f253dad88e3b08f2433c8ea68944c4183e3118d5539aad

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review contract Date: 2026-05-26 CDT Scope: review contract for the no-effect disposition closeout audit without enabling dispatch or deletion.

Purpose

This contract reviews the macOS reset/uninstall live-runner acceptance-denial disposition closeout audit. It confirms that the no-effect closeout audit remains review-only, the acceptance gate remains closed, and no runner dispatch or deletion authority is opened.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, perform a live reset/uninstall, open the next disposition, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review
-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_state=reviewed-no-effec
t-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_decision=no-effect-clos
eout-audit-review-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_source_contract=macos-reset-unin
stall-live-runner-acceptance-denial-disposition-closeout-audit-contract
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_state=audited-no-effect-cl
oseout
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_audited=1
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_closeout_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_audit_review_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_result=no-effect-closeout-audit-
reviewed-dispatch-preserved
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Audit

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_contract_present=
1
live_runner_acceptance_denial_disposition_closeout_audit_contract_state=audited-no-effect-closeo
ut
live_runner_acceptance_denial_disposition_closeout_audit_contract_decision=no-effect-closeout-au
dit-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_audit_closeout_audited=1
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_result=no-effect-closeout-audit-boundar
y-preserved
```

Requirements

```
live_runner_acceptance_denial_disposition_closeout_audit_review_schema_version=macos-reset-unins
tall-live-runner-acceptance-denial-disposition-closeout-audit-review/1
live_runner_acceptance_denial_disposition_closeout_audit_review_required_input_count=9
live_runner_acceptance_denial_disposition_closeout_audit_review_observed_input_count=9
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_closeout_audit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_closeout=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_acceptance_denial_dispo
sition_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_closed_gate=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_effect_audit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_review_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_effects=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_absence_report_write
=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_network=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_audit_review_requires_disposition_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_review_result_closeout_audit=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_closeout=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_acceptance_denial_disposi
tion_review=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_closed_gate=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_effect_audit=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_review_only=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_stdout_only=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_effects=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_absence_report_write=m
et
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_network=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_no_root=met
live_runner_acceptance_denial_disposition_closeout_audit_review_result_disposition_handoff=met
```

Review Entries

```
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_1=closeout_audit_present
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_2=no_effect_closeout_audit
_reviewed
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_3=dispatch_remains_closed
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_4=deletion_remains_disable
d
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_5=no_effects_confirmed
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_6=disposition_required
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_1_status=met
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_2_status=reviewed
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_4_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_5_status=met
live_runner_acceptance_denial_disposition_closeout_audit_review_entry_6_status=recorded
```

Phases

```
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_1=load_acceptance_denial_d
isposition_closeout_audit
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_2=review_no_effect_closeou
t_audit
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_3=confirm_closed_dispatch_
boundary
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_4=handoff_to_acceptance_de
nial_disposition_closeout_audit_review_disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_2_status=review-only
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_phase_4_status=future-dispositio
n-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-r
eview-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract:
ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, deletion evidence, closeout audit disposition evidence, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
contract that reviews the no-effect closeout audit without opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Contract

Lines: 203 | Words: 419 | SHA-256: 434e5cd688557738db01b3b83719228f5694ee31490951b590d42c468f913a67

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition contract Date: 2026-05-26 CDT Scope: disposition contract for the reviewed no-effect closeout audit without enabling dispatch or deletion.

Purpose

This contract records the reviewed macOS reset/uninstall live-runner acceptance-denial disposition closeout audit as a no-effect disposition. It confirms that the closeout audit review is present, the acceptance gate remains closed, and the disposition remains evidence-only.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, apply a disposition, open the future review, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review
-disposition-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_contract_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_state=dispo
sed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_decision=no
-effect-closeout-audit-review-disposition-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_recorded_as_no_effec
t=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_source_contract=maco
s-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-contract
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_state=reviewe
d-no-effect-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_decision=no-e
ffect-closeout-audit-review-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
equired=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_o
pened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_source_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_audit_state=audited-
no-effect-closeout
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_audit_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_closeout_state=close
d-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_closeout_audited=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_clos
ed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_runner_handoff_enabl
ed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_receipt_write_enable
d=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_absence_report_write
_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_reviewed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_future_revi
ew=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_disposition_review_o
pened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result=no-effect-clo
seout-audit-review-disposition-recorded
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Review

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_contract_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_state=reviewed-no-effec
t-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_contract_decision=no-effect-clos
eout-audit-review-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_audit_reviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_opened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_result=no-effect-closeout-audit-
reviewed-dispatch-preserved
```

Requirements

```
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_schema_version=macos
-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-disposition/1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_required_input_count
=9
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_observed_input_count
=9
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_closeout_au
dit_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_effect_d
isposition=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_disposition
_record=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_closeout_au
dit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_closeout=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_acceptance_
denial_disposition_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_closed_gate
=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_reviewed_no
_effect_audit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_disposition
_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_stdout_only
=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_effects=
1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_dispatch
=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_deletion
=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_receipt_
write=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_absence_
report_write=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_network=
1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_requires_review_hand
off=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_closeout_audi
t_review=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_no_effect_dis
position=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_disposition_r
ecord=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_no_dispatch=m
et
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_no_deletion=m
et
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result_review_handof
f=met
```

Disposition Entries

```
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_1=closeout_aud
it_review_present
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_2=no_effect_cl
oseout_audit_review_disposition_recorded
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_3=dispatch_rem
ains_closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_4=deletion_rem
ains_disabled
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_5=no_effects_c
onfirmed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_6=disposition_
review_required
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_1_status=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_2_status=stdou
t-only
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_3_status=block
ed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_4_status=block
ed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_5_status=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_entry_6_status=futur
e-review-only
```

Phases

```
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_1=load_accepta
nce_denial_disposition_closeout_audit_review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_2=record_no_ef
fect_closeout_audit_review_disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_3=preserve_clo
sed_dispatch_boundary
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_4=handoff_to_a
cceptance_denial_disposition_closeout_audit_review_disposition_review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_1_status=contr
act-only
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_2_status=stdou
t-only
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_3_status=block
ed-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_phase_4_status=futur
e-review-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-r
eview-disposition-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, disposition application, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition contract that records the reviewed no-effect closeout audit as a no-effect
disposition without opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_AUDIT_REVIEW_DISPOSITION_REVIEW_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Review Contract

Lines: 202 | Words: 419 | SHA-256: 5a961d8b014fe4e57706f4f179194dced0aa09b13e1dbaf24e2716832e169da6

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Audit Review Disposition Review Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition review contract Date: 2026-05-26 CDT Scope: review contract for the no-effect closeout audit review disposition without enabling dispatch or deletion.

Purpose

This contract reviews the recorded macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review disposition. It confirms that the prior disposition is present, remains no-effect, and is reviewed only as stdout evidence.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, apply a disposition, open the future closeout, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review
-disposition-review-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_contract_stat
e=reviewed-no-effect-closeout-audit-review-disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_contract_deci
sion=no-effect-closeout-audit-review-disposition-review-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_file_write_en
abled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_review_file_w
rite_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_source_contra
ct=macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-disposi
tion-contract
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_p
resent=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_s
tate=disposed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_d
ecision=no-effect-closeout-audit-review-disposition-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
ecorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_s
tdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_f
ile_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
eviewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_a
pplied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_o
pens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
eleases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_open=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_acceptance_ga
te_closed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_revi
ewed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_allo
wed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_dispatch_perf
ormed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_runner_handof
f_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_deletion_enab
led=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_receipt_write
_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_absence_repor
t_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_effect_author
ized=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_requ
ired=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_closeout_open
ed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_authorizes_ef
fects=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_opens_dispatc
h=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_releases_runn
er=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result=no-eff
ect-closeout-audit-review-disposition-reviewed-dispatch-preserved
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Disposition

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_contract_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_state=dispo
sed-no-effect-closeout-audit-review
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_contract_decision=no
-effect-closeout-audit-review-disposition-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_completed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_recorded=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_record_write_enabled
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_present=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_state=reviewe
d-no-effect-closeout-audit
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_r
equired=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_disposition_o
pened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_open
=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_acceptance_gate_clos
ed=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_reviewed=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_disposition_review_o
pened=0
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_result=no-effect-clo
seout-audit-review-disposition-recorded
```

Requirements

live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_schema_versio
n=macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-review-disposit
ion-review/1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_required_inpu
t_count=10
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_observed_inpu
t_count=10
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout_audit_review_disposition=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout_audit_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout_audit=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_acce
ptance_denial_disposition_review=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
ed_gate=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_e
ffect_disposition=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_revi
ew_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_stdo
ut_only=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_e
ffects=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_d
ispatch=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_d
eletion=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_r
eceipt_write=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_a
bsence_report_write=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_n
etwork=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_no_r
oot=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_requires_clos
eout_handoff=1
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_closeo
ut_audit_review_disposition=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_no_dis
patch=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_no_del
etion=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_result_closeo
ut_handoff=met

Review Entries

live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_1=close
out_audit_review_disposition_present
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_2=no_ef
fect_closeout_audit_review_disposition_reviewed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_3=dispa
tch_remains_closed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_4=delet
ion_remains_disabled
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_5=no_ef
fects_confirmed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_6=close
out_required
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_1_statu
s=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_2_statu
s=reviewed
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_3_statu
s=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_4_statu
s=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_5_statu
s=met
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_entry_6_statu
s=recorded

Phases

```
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_1=load_
acceptance_denial_disposition_closeout_audit_review_disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_2=revie
w_no_effect_closeout_audit_review_disposition
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_3=confi
rm_closed_dispatch_boundary
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_4=hando
ff_to_acceptance_denial_disposition_closeout_audit_review_disposition_closeout
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_1_statu
s=contract-only
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_2_statu
s=review-only
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_3_statu
s=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_audit_review_disposition_review_phase_4_statu
s=future-closeout-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-audit-r
eview-disposition-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_audit_review_dispositio
n_review_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, disposition application, closeout application, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition contract that records the reviewed no-effect closeout audit as a no-effect
disposition without opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CLOSEOUT_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Contract

Lines: 249 | Words: 446 | SHA-256: 23b98ffe803fb64a2ed768d5a965f6346be224104a730b2e5e4197ccf2cb9a0b

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Closeout Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition closeout contract Date: 2026-05-26 CDT Scope: closeout contract for the reviewed no-effect disposition without enabling dispatch or deletion.

Purpose

This contract closes the reviewed macOS reset/uninstall live-runner acceptance-denial disposition. It confirms that the disposition remains no-effect, the acceptance gate remains closed, and no runner dispatch or deletion authority is opened.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, release the runner, open audit effects, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract_present=1
live_runner_acceptance_denial_disposition_closeout_contract_state=closed-no-effect
live_runner_acceptance_denial_disposition_closeout_contract_decision=no-effect-disposition-close
out-keeps-dispatch-closed
live_runner_acceptance_denial_disposition_closeout_required=1
live_runner_acceptance_denial_disposition_closeout_present=1
live_runner_acceptance_denial_disposition_closeout_completed=1
live_runner_acceptance_denial_disposition_closeout_recorded=1
live_runner_acceptance_denial_disposition_closeout_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_record_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_scope=close-reviewed-no-effect-disposition-on
ly
live_runner_acceptance_denial_disposition_closeout_source_contract=macos-reset-uninstall-live-ru
nner-acceptance-denial-disposition-review-contract
live_runner_acceptance_denial_disposition_closeout_review_present=1
live_runner_acceptance_denial_disposition_closeout_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_closeout_review_completed=1
live_runner_acceptance_denial_disposition_closeout_review_recorded=1
live_runner_acceptance_denial_disposition_closeout_review_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_review_closeout_opened=0
live_runner_acceptance_denial_disposition_closeout_disposition_present=1
live_runner_acceptance_denial_disposition_closeout_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_closeout_disposition_decision=closed-gate-denial-dispo
sed-no-dispatch
live_runner_acceptance_denial_disposition_closeout_disposition_reviewed=1
live_runner_acceptance_denial_disposition_closeout_disposition_closed=1
live_runner_acceptance_denial_disposition_closeout_disposition_applied=0
live_runner_acceptance_denial_disposition_closeout_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_future_closeout_required=0
live_runner_acceptance_denial_disposition_closeout_closed_gate_reviewed=1
live_runner_acceptance_denial_disposition_closeout_denial_reason_present=1
live_runner_acceptance_denial_disposition_closeout_denial_reason_closed=1
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_closeout_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_closeout_dispatch_closed=1
live_runner_acceptance_denial_disposition_closeout_dispatch_allowed=0
live_runner_acceptance_denial_disposition_closeout_dispatch_enabled=0
live_runner_acceptance_denial_disposition_closeout_dispatch_performed=0
live_runner_acceptance_denial_disposition_closeout_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_closeout_deletion_enabled=0
live_runner_acceptance_denial_disposition_closeout_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_closeout_effect=none
live_runner_acceptance_denial_disposition_closeout_effect_authorized=0
live_runner_acceptance_denial_disposition_closeout_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_closeout_authorizes_effects=0
live_runner_acceptance_denial_disposition_closeout_opens_dispatch=0
live_runner_acceptance_denial_disposition_closeout_releases_runner=0
live_runner_acceptance_denial_disposition_closeout_audit_review_required=1
live_runner_acceptance_denial_disposition_closeout_audit_review_opened=0
live_runner_acceptance_denial_disposition_closeout_result=no-effect-disposition-closed-dispatch-
preserved
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Review

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_contract_decision=no-effect-disposition-review-
keeps-dispatch-closed
live_runner_acceptance_denial_disposition_review_completed=1
live_runner_acceptance_denial_disposition_review_recorded=1
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_review_disposition_reviewed=1
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_review_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_review_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_review_dispatch_allowed=0
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_closeout_opened=0
live_runner_acceptance_denial_disposition_review_result=no-effect-disposition-boundary-preserved
```

Source Disposition

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_contract_decision=closed-gate-denial-disposed-no-dispa
tch
live_runner_acceptance_denial_disposition_recorded=1
live_runner_acceptance_denial_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_record_write_enabled=0
live_runner_acceptance_denial_disposition_review_present=1
live_runner_acceptance_denial_disposition_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_dispatch_allowed=0
live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_result=closed-gate-denial-disposition-recorded
```

Requirements

```
live_runner_acceptance_denial_disposition_closeout_schema_version=macos-reset-uninstall-live-run
ner-acceptance-denial-disposition-closeout/1
live_runner_acceptance_denial_disposition_closeout_required_input_count=8
live_runner_acceptance_denial_disposition_closeout_observed_input_count=8
live_runner_acceptance_denial_disposition_closeout_requires_acceptance_denial_disposition_review
=1
live_runner_acceptance_denial_disposition_closeout_requires_acceptance_denial_disposition=1
live_runner_acceptance_denial_disposition_closeout_requires_acceptance_denial_review=1
live_runner_acceptance_denial_disposition_closeout_requires_closed_gate=1
live_runner_acceptance_denial_disposition_closeout_requires_no_effect_disposition=1
live_runner_acceptance_denial_disposition_closeout_requires_closeout_only=1
live_runner_acceptance_denial_disposition_closeout_requires_stdout_only=1
live_runner_acceptance_denial_disposition_closeout_requires_no_effects=1
live_runner_acceptance_denial_disposition_closeout_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_closeout_requires_no_deletion=1
live_runner_acceptance_denial_disposition_closeout_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_closeout_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_closeout_requires_no_network=1
live_runner_acceptance_denial_disposition_closeout_requires_no_root=1
live_runner_acceptance_denial_disposition_closeout_requires_audit_review_handoff=1
live_runner_acceptance_denial_disposition_closeout_result_acceptance_denial_disposition_review=m
et
live_runner_acceptance_denial_disposition_closeout_result_acceptance_denial_disposition=met
live_runner_acceptance_denial_disposition_closeout_result_acceptance_denial_review=met
live_runner_acceptance_denial_disposition_closeout_result_closed_gate=met
live_runner_acceptance_denial_disposition_closeout_result_no_effect_disposition=met
live_runner_acceptance_denial_disposition_closeout_result_closeout_only=met
live_runner_acceptance_denial_disposition_closeout_result_stdout_only=met
live_runner_acceptance_denial_disposition_closeout_result_no_effects=met
live_runner_acceptance_denial_disposition_closeout_result_no_dispatch=met
live_runner_acceptance_denial_disposition_closeout_result_no_deletion=met
live_runner_acceptance_denial_disposition_closeout_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_closeout_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_closeout_result_no_network=met
live_runner_acceptance_denial_disposition_closeout_result_no_root=met
live_runner_acceptance_denial_disposition_closeout_result_audit_review_handoff=met
```

Closeout Entries

```
live_runner_acceptance_denial_disposition_closeout_entry_1=acceptance_denial_disposition_review_
present
live_runner_acceptance_denial_disposition_closeout_entry_2=no_effect_disposition_closed
live_runner_acceptance_denial_disposition_closeout_entry_3=dispatch_remains_closed
live_runner_acceptance_denial_disposition_closeout_entry_4=deletion_remains_disabled
live_runner_acceptance_denial_disposition_closeout_entry_5=no_effects_confirmed
live_runner_acceptance_denial_disposition_closeout_entry_6=audit_review_required
live_runner_acceptance_denial_disposition_closeout_entry_1_status=met
live_runner_acceptance_denial_disposition_closeout_entry_2_status=closed
live_runner_acceptance_denial_disposition_closeout_entry_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_entry_4_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_entry_5_status=met
live_runner_acceptance_denial_disposition_closeout_entry_6_status=recorded
```

Phases

```
live_runner_acceptance_denial_disposition_closeout_phase_1=load_acceptance_denial_disposition_re
view
live_runner_acceptance_denial_disposition_closeout_phase_2=close_reviewed_no_effect_disposition
live_runner_acceptance_denial_disposition_closeout_phase_3=confirm_closed_dispatch_boundary
live_runner_acceptance_denial_disposition_closeout_phase_4=handoff_to_acceptance_denial_disposit
ion_closeout_audit_review
live_runner_acceptance_denial_disposition_closeout_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_closeout_phase_2_status=closeout-only
live_runner_acceptance_denial_disposition_closeout_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_closeout_phase_4_status=future-audit-review-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-closeout-contrac
t.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_closeout_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, closeout audit evidence, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout contract that
closes the reviewed no-effect disposition without opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Contract

Lines: 233 | Words: 426 | SHA-256: 0800bcf2c0c6c941acfd7dedbb9c89b84f931e60d8ae1a820708285fbb1d7d64

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition contract Date: 2026-05-26 CDT Scope: disposition contract for the reviewed closed-gate denial without enabling dispatch or deletion.

Purpose

This contract records the reviewed macOS reset/uninstall live-runner acceptance-denial as a no-effect disposition. It confirms that the acceptance gate remained closed, the denial review is present, and the disposition remains evidence-only.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, apply a disposition, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_contract_decision=closed-gate-denial-disposed-no-dispa
tch
live_runner_acceptance_denial_disposition_required=1
live_runner_acceptance_denial_disposition_present=1
live_runner_acceptance_denial_disposition_recorded=1
live_runner_acceptance_denial_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_record_write_enabled=0
live_runner_acceptance_denial_disposition_scope=record-reviewed-closed-gate-denial-only
live_runner_acceptance_denial_disposition_source_contract=macos-reset-uninstall-live-runner-acce
ptance-denial-review-contract
live_runner_acceptance_denial_disposition_review_present=1
live_runner_acceptance_denial_disposition_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_decision=closed-gate-review-retains-zero-dispat
ch
live_runner_acceptance_denial_disposition_closed_gate_reviewed=1
live_runner_acceptance_denial_disposition_denial_reason_present=1
live_runner_acceptance_denial_disposition_denial_reason_reviewed=1
live_runner_acceptance_denial_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_dispatch_allowed=0
live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_effect=none
live_runner_acceptance_denial_disposition_effect_authorized=0
live_runner_acceptance_denial_disposition_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_requires_future_review=1
live_runner_acceptance_denial_disposition_result=closed-gate-denial-disposition-recorded
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Review

```
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_contract_decision=closed-gate-review-retains-zero-dispatch
live_runner_acceptance_denial_review_completed=1
live_runner_acceptance_denial_review_recorded=1
live_runner_acceptance_denial_review_stdout_only=1
live_runner_acceptance_denial_review_file_write_enabled=0
live_runner_acceptance_denial_review_closed_gate_reviewed=1
live_runner_acceptance_denial_review_denial_reason_present=1
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_acceptance_gate_closed=1
live_runner_acceptance_denial_review_dispatch_reviewed=1
live_runner_acceptance_denial_review_dispatch_allowed=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_receipt_write_enabled=0
live_runner_acceptance_denial_review_absence_report_write_enabled=0
live_runner_acceptance_denial_review_disposition_opened=0
```

Closed Gate Inputs

```
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_contract_decision=acceptance-gate-closed-no-dispatch-re
corded
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_acceptance_gate_closed=1
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_transcript_denial_reason=missing-passed-preflight-complete-evidenc
e-effect-authorization-operator-intent-and-implementation
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_operator_intent=blocked
live_runner_acceptance_gate_result_implementation=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_gate_result_no_deletion_until_open=met
```

Requirements

```
live_runner_acceptance_denial_disposition_schema_version=macos-reset-uninstall-live-runner-accep
tance-denial-disposition/1
live_runner_acceptance_denial_disposition_required_input_count=7
live_runner_acceptance_denial_disposition_observed_input_count=7
live_runner_acceptance_denial_disposition_requires_acceptance_denial_review=1
live_runner_acceptance_denial_disposition_requires_acceptance_denial_transcript=1
live_runner_acceptance_denial_disposition_requires_closed_gate=1
live_runner_acceptance_denial_disposition_requires_denial_reason=1
live_runner_acceptance_denial_disposition_requires_disposition_only=1
live_runner_acceptance_denial_disposition_requires_stdout_only=1
live_runner_acceptance_denial_disposition_requires_no_effects=1
live_runner_acceptance_denial_disposition_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_requires_no_deletion=1
live_runner_acceptance_denial_disposition_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_requires_no_network=1
live_runner_acceptance_denial_disposition_requires_no_root=1
live_runner_acceptance_denial_disposition_requires_review_handoff=1
live_runner_acceptance_denial_disposition_result_acceptance_denial_review=met
live_runner_acceptance_denial_disposition_result_acceptance_denial_transcript=met
live_runner_acceptance_denial_disposition_result_closed_gate=met
live_runner_acceptance_denial_disposition_result_denial_reason=met
live_runner_acceptance_denial_disposition_result_disposition_only=met
live_runner_acceptance_denial_disposition_result_stdout_only=met
live_runner_acceptance_denial_disposition_result_no_effects=met
live_runner_acceptance_denial_disposition_result_no_dispatch=met
live_runner_acceptance_denial_disposition_result_no_deletion=met
live_runner_acceptance_denial_disposition_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_result_no_network=met
live_runner_acceptance_denial_disposition_result_no_root=met
live_runner_acceptance_denial_disposition_result_review_handoff=met
```

Disposition Entries

```
live_runner_acceptance_denial_disposition_entry_1=acceptance_denial_review_present
live_runner_acceptance_denial_disposition_entry_2=closed_gate_denial_reviewed
live_runner_acceptance_denial_disposition_entry_3=no_effect_disposition_recorded
live_runner_acceptance_denial_disposition_entry_4=dispatch_remains_closed
live_runner_acceptance_denial_disposition_entry_5=no_effects_confirmed
live_runner_acceptance_denial_disposition_entry_6=disposition_review_required
live_runner_acceptance_denial_disposition_entry_1_status=met
live_runner_acceptance_denial_disposition_entry_2_status=reviewed
live_runner_acceptance_denial_disposition_entry_3_status=stdout-only
live_runner_acceptance_denial_disposition_entry_4_status=blocked-no-effect
live_runner_acceptance_denial_disposition_entry_5_status=met
live_runner_acceptance_denial_disposition_entry_6_status=future-review-only
```

Phases

```
live_runner_acceptance_denial_disposition_phase_1=load_acceptance_denial_review
live_runner_acceptance_denial_disposition_phase_2=record_no_effect_disposition
live_runner_acceptance_denial_disposition_phase_3=preserve_closed_gate_boundary
live_runner_acceptance_denial_disposition_phase_4=handoff_to_acceptance_denial_disposition_revie
w
live_runner_acceptance_denial_disposition_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_phase_2_status=stdout-only
live_runner_acceptance_denial_disposition_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_phase_4_status=future-review-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, disposition application, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition contract that records the
reviewed closed-gate denial as a no-effect disposition without opening dispatch.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_DISPOSITION_REVIEW_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Review Contract

Lines: 238 | Words: 430 | SHA-256: 898abbb39db953ce6f6f54ffb50c6d2eb61e7fe50a419e2eb827ab8fbd16fff2

macOS Reset/Uninstall Live-Runner Acceptance-Denial Disposition Review Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial disposition review contract Date: 2026-05-26 CDT Scope: review contract for the recorded no-effect disposition without enabling dispatch or deletion.

Purpose

This contract reviews the macOS reset/uninstall live-runner acceptance-denial disposition. It confirms that the disposition remains no-effect, the acceptance gate remains closed, and no runner dispatch or deletion authority is opened.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, close out the disposition, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-disposition-review-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract_present=1
live_runner_acceptance_denial_disposition_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_review_contract_decision=no-effect-disposition-review-
keeps-dispatch-closed
live_runner_acceptance_denial_disposition_review_required=1
live_runner_acceptance_denial_disposition_review_present=1
live_runner_acceptance_denial_disposition_review_completed=1
live_runner_acceptance_denial_disposition_review_recorded=1
live_runner_acceptance_denial_disposition_review_stdout_only=1
live_runner_acceptance_denial_disposition_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_review_file_write_enabled=0
live_runner_acceptance_denial_disposition_review_scope=review-no-effect-disposition-only
live_runner_acceptance_denial_disposition_review_source_contract=macos-reset-uninstall-live-runn
er-acceptance-denial-disposition-contract
live_runner_acceptance_denial_disposition_review_disposition_present=1
live_runner_acceptance_denial_disposition_review_disposition_state=disposed-no-effect
live_runner_acceptance_denial_disposition_review_disposition_decision=closed-gate-denial-dispose
d-no-dispatch
live_runner_acceptance_denial_disposition_review_disposition_recorded=1
live_runner_acceptance_denial_disposition_review_disposition_reviewed=1
live_runner_acceptance_denial_disposition_review_disposition_applied=0
live_runner_acceptance_denial_disposition_review_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_review_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_review_closed_gate_reviewed=1
live_runner_acceptance_denial_disposition_review_denial_reason_present=1
live_runner_acceptance_denial_disposition_review_denial_reason_reviewed=1
live_runner_acceptance_denial_disposition_review_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_review_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_review_dispatch_reviewed=1
live_runner_acceptance_denial_disposition_review_dispatch_allowed=0
live_runner_acceptance_denial_disposition_review_dispatch_enabled=0
live_runner_acceptance_denial_disposition_review_dispatch_performed=0
live_runner_acceptance_denial_disposition_review_runner_handoff_enabled=0
live_runner_acceptance_denial_disposition_review_deletion_enabled=0
live_runner_acceptance_denial_disposition_review_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_review_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_review_effect=none
live_runner_acceptance_denial_disposition_review_effect_authorized=0
live_runner_acceptance_denial_disposition_review_effect_boundary_preserved=1
live_runner_acceptance_denial_disposition_review_closeout_opened=0
live_runner_acceptance_denial_disposition_review_result=no-effect-disposition-boundary-preserved
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Disposition

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_contract_present=1
live_runner_acceptance_denial_disposition_contract_state=disposed-no-effect
live_runner_acceptance_denial_disposition_contract_decision=closed-gate-denial-disposed-no-dispa
tch
live_runner_acceptance_denial_disposition_recorded=1
live_runner_acceptance_denial_disposition_stdout_only=1
live_runner_acceptance_denial_disposition_file_write_enabled=0
live_runner_acceptance_denial_disposition_record_write_enabled=0
live_runner_acceptance_denial_disposition_review_present=1
live_runner_acceptance_denial_disposition_review_state=reviewed-no-effect
live_runner_acceptance_denial_disposition_acceptance_gate_open=0
live_runner_acceptance_denial_disposition_acceptance_gate_closed=1
live_runner_acceptance_denial_disposition_dispatch_allowed=0
live_runner_acceptance_denial_disposition_dispatch_enabled=0
live_runner_acceptance_denial_disposition_dispatch_performed=0
live_runner_acceptance_denial_disposition_deletion_enabled=0
live_runner_acceptance_denial_disposition_receipt_write_enabled=0
live_runner_acceptance_denial_disposition_absence_report_write_enabled=0
live_runner_acceptance_denial_disposition_applied=0
live_runner_acceptance_denial_disposition_opens_dispatch=0
live_runner_acceptance_denial_disposition_releases_runner=0
live_runner_acceptance_denial_disposition_result=closed-gate-denial-disposition-recorded
```

Prior Inputs

```
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_contract_decision=closed-gate-review-retains-zero-dispatch
live_runner_acceptance_denial_review_completed=1
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_disposition_opened=0
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_transcript_denial_reason=missing-passed-preflight-complete-evidenc
e-effect-authorization-operator-intent-and-implementation
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_deletion_enabled=0
```

Requirements

```
live_runner_acceptance_denial_disposition_review_schema_version=macos-reset-uninstall-live-runne
r-acceptance-denial-disposition-review/1
live_runner_acceptance_denial_disposition_review_required_input_count=7
live_runner_acceptance_denial_disposition_review_observed_input_count=7
live_runner_acceptance_denial_disposition_review_requires_acceptance_denial_disposition=1
live_runner_acceptance_denial_disposition_review_requires_acceptance_denial_review=1
live_runner_acceptance_denial_disposition_review_requires_closed_gate=1
live_runner_acceptance_denial_disposition_review_requires_no_effect_disposition=1
live_runner_acceptance_denial_disposition_review_requires_review_only=1
live_runner_acceptance_denial_disposition_review_requires_stdout_only=1
live_runner_acceptance_denial_disposition_review_requires_no_effects=1
live_runner_acceptance_denial_disposition_review_requires_no_dispatch=1
live_runner_acceptance_denial_disposition_review_requires_no_deletion=1
live_runner_acceptance_denial_disposition_review_requires_no_receipt_write=1
live_runner_acceptance_denial_disposition_review_requires_no_absence_report_write=1
live_runner_acceptance_denial_disposition_review_requires_no_network=1
live_runner_acceptance_denial_disposition_review_requires_no_root=1
live_runner_acceptance_denial_disposition_review_requires_closeout_handoff=1
live_runner_acceptance_denial_disposition_review_result_acceptance_denial_disposition=met
live_runner_acceptance_denial_disposition_review_result_acceptance_denial_review=met
live_runner_acceptance_denial_disposition_review_result_closed_gate=met
live_runner_acceptance_denial_disposition_review_result_no_effect_disposition=met
live_runner_acceptance_denial_disposition_review_result_review_only=met
live_runner_acceptance_denial_disposition_review_result_stdout_only=met
live_runner_acceptance_denial_disposition_review_result_no_effects=met
live_runner_acceptance_denial_disposition_review_result_no_dispatch=met
live_runner_acceptance_denial_disposition_review_result_no_deletion=met
live_runner_acceptance_denial_disposition_review_result_no_receipt_write=met
live_runner_acceptance_denial_disposition_review_result_no_absence_report_write=met
live_runner_acceptance_denial_disposition_review_result_no_network=met
live_runner_acceptance_denial_disposition_review_result_no_root=met
live_runner_acceptance_denial_disposition_review_result_closeout_handoff=met
```

Review Entries

```
live_runner_acceptance_denial_disposition_review_entry_1=acceptance_denial_disposition_present
live_runner_acceptance_denial_disposition_review_entry_2=no_effect_disposition_reviewed
live_runner_acceptance_denial_disposition_review_entry_3=dispatch_remains_closed
live_runner_acceptance_denial_disposition_review_entry_4=deletion_remains_disabled
live_runner_acceptance_denial_disposition_review_entry_5=no_effects_confirmed
live_runner_acceptance_denial_disposition_review_entry_6=closeout_required
live_runner_acceptance_denial_disposition_review_entry_1_status=met
live_runner_acceptance_denial_disposition_review_entry_2_status=reviewed
live_runner_acceptance_denial_disposition_review_entry_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_review_entry_4_status=blocked-no-effect
live_runner_acceptance_denial_disposition_review_entry_5_status=met
live_runner_acceptance_denial_disposition_review_entry_6_status=recorded
```

Phases

```
live_runner_acceptance_denial_disposition_review_phase_1=load_acceptance_denial_disposition
live_runner_acceptance_denial_disposition_review_phase_2=review_no_effect_disposition
live_runner_acceptance_denial_disposition_review_phase_3=confirm_closed_dispatch_boundary
live_runner_acceptance_denial_disposition_review_phase_4=handoff_to_acceptance_denial_dispositio
n_closeout
live_runner_acceptance_denial_disposition_review_phase_1_status=contract-only
live_runner_acceptance_denial_disposition_review_phase_2_status=review-only
live_runner_acceptance_denial_disposition_review_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_disposition_review_phase_4_status=future-closeout-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh
scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-disposition-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_disposition_review_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, closeout evidence, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition review contract that
reviews the no-effect disposition without opening dispatch or deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_REVIEW_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Review Contract

Lines: 257 | Words: 461 | SHA-256: 9bd1402c9edfb562a9eb167815973416189a3dd12e3e44df2ce87857910ab300

macOS Reset/Uninstall Live-Runner Acceptance-Denial Review Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial review contract Date: 2026-05-26 CDT Scope: review contract for the closed acceptance-gate transcript without enabling dispatch or deletion.

Purpose

This contract reviews the macOS reset/uninstall live-runner acceptance-denial transcript. It confirms that the acceptance gate remained closed, the denial reason is present, and the review remains evidence-only.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, grant runtime authority, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-review-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_review_contract_present=1
live_runner_acceptance_denial_review_contract_state=reviewed-no-effect
live_runner_acceptance_denial_review_contract_decision=closed-gate-review-retains-zero-dispatch
live_runner_acceptance_denial_review_required=1
live_runner_acceptance_denial_review_present=1
live_runner_acceptance_denial_review_completed=1
live_runner_acceptance_denial_review_recorded=1
live_runner_acceptance_denial_review_stdout_only=1
live_runner_acceptance_denial_review_file_write_enabled=0
live_runner_acceptance_denial_review_review_file_write_enabled=0
live_runner_acceptance_denial_review_scope=review-closed-gate-transcript-only
live_runner_acceptance_denial_review_source_contract=macos-reset-uninstall-live-runner-acceptanc
e-denial-transcript-contract
live_runner_acceptance_denial_review_closed_gate_transcript_present=1
live_runner_acceptance_denial_review_closed_gate_reviewed=1
live_runner_acceptance_denial_review_denial_reason_present=1
live_runner_acceptance_denial_review_denial_reason_reviewed=1
live_runner_acceptance_denial_review_acceptance_gate_open=0
live_runner_acceptance_denial_review_acceptance_gate_closed=1
live_runner_acceptance_denial_review_dispatch_reviewed=1
live_runner_acceptance_denial_review_dispatch_allowed=0
live_runner_acceptance_denial_review_dispatch_enabled=0
live_runner_acceptance_denial_review_dispatch_performed=0
live_runner_acceptance_denial_review_runner_handoff_enabled=0
live_runner_acceptance_denial_review_deletion_enabled=0
live_runner_acceptance_denial_review_receipt_write_enabled=0
live_runner_acceptance_denial_review_absence_report_write_enabled=0
live_runner_acceptance_denial_review_effect=none
live_runner_acceptance_denial_review_effect_authorized=0
live_runner_acceptance_denial_review_effect_boundary_preserved=1
live_runner_acceptance_denial_review_disposition_opened=0
live_runner_acceptance_denial_review_result=closed-gate-boundary-preserved
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Transcript

```
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_contract_decision=acceptance-gate-closed-no-dispatch-re
corded
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_stdout_only=1
live_runner_acceptance_denial_transcript_file_write_enabled=0
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_acceptance_gate_closed=1
live_runner_acceptance_denial_transcript_dispatch_allowed=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_transcript_denial_reason=missing-passed-preflight-complete-evidenc
e-effect-authorization-operator-intent-and-implementation
live_runner_acceptance_denial_transcript_denial_source=macos-reset-uninstall-live-runner-accepta
nce-gate-contract
live_runner_acceptance_denial_review_transcript_state=recorded-no-effect
live_runner_acceptance_denial_review_transcript_decision=acceptance-gate-closed-no-dispatch-reco
rded
live_runner_acceptance_denial_review_transcript_gate_open=0
live_runner_acceptance_denial_review_transcript_gate_closed=1
live_runner_acceptance_denial_review_transcript_dispatch_enabled=0
live_runner_acceptance_denial_review_transcript_dispatch_performed=0
live_runner_acceptance_denial_review_transcript_deletion_enabled=0
live_runner_acceptance_denial_review_transcript_denial_reason=missing-passed-preflight-complete-
evidence-effect-authorization-operator-intent-and-implementation
live_runner_acceptance_denial_review_transcript_effect=none
```

Gate Inputs

```
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_contract_decision=dispatch-blocked-missing-passed-preflight-evidence
-and-authorization
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_acceptance_granted=0
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_effect_authorized=0
live_runner_acceptance_gate_result_denied_dispatch_review=met
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_operator_intent=blocked
live_runner_acceptance_gate_result_implementation=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_gate_result_no_deletion_until_open=met
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_dispatch_reviewed=1
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
```

Blocking Inputs

```
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
```

Requirements

```
live_runner_acceptance_denial_review_schema_version=macos-reset-uninstall-live-runner-acceptance
-denial-review/1
live_runner_acceptance_denial_review_required_input_count=7
live_runner_acceptance_denial_review_observed_input_count=7
live_runner_acceptance_denial_review_requires_acceptance_denial_transcript=1
live_runner_acceptance_denial_review_requires_closed_gate=1
live_runner_acceptance_denial_review_requires_denial_reason=1
live_runner_acceptance_denial_review_requires_review_only=1
live_runner_acceptance_denial_review_requires_stdout_only=1
live_runner_acceptance_denial_review_requires_no_effects=1
live_runner_acceptance_denial_review_requires_no_dispatch=1
live_runner_acceptance_denial_review_requires_no_deletion=1
live_runner_acceptance_denial_review_requires_no_receipt_write=1
live_runner_acceptance_denial_review_requires_no_absence_report_write=1
live_runner_acceptance_denial_review_requires_no_network=1
live_runner_acceptance_denial_review_requires_no_root=1
live_runner_acceptance_denial_review_requires_disposition_handoff=1
live_runner_acceptance_denial_review_result_acceptance_denial_transcript=met
live_runner_acceptance_denial_review_result_closed_gate=met
live_runner_acceptance_denial_review_result_denial_reason=met
live_runner_acceptance_denial_review_result_review_only=met
live_runner_acceptance_denial_review_result_stdout_only=met
live_runner_acceptance_denial_review_result_no_effects=met
live_runner_acceptance_denial_review_result_no_dispatch=met
live_runner_acceptance_denial_review_result_no_deletion=met
live_runner_acceptance_denial_review_result_no_receipt_write=met
live_runner_acceptance_denial_review_result_no_absence_report_write=met
live_runner_acceptance_denial_review_result_no_network=met
live_runner_acceptance_denial_review_result_no_root=met
live_runner_acceptance_denial_review_result_disposition_handoff=met
```

Review Entries

```
live_runner_acceptance_denial_review_entry_1=acceptance_denial_transcript_present
live_runner_acceptance_denial_review_entry_2=closed_gate_recorded
live_runner_acceptance_denial_review_entry_3=denial_reason_reviewed
live_runner_acceptance_denial_review_entry_4=acceptance_inputs_blocked
live_runner_acceptance_denial_review_entry_5=no_effects_confirmed
live_runner_acceptance_denial_review_entry_6=disposition_required
live_runner_acceptance_denial_review_entry_1_status=met
live_runner_acceptance_denial_review_entry_2_status=reviewed
live_runner_acceptance_denial_review_entry_3_status=reviewed
live_runner_acceptance_denial_review_entry_4_status=reviewed
live_runner_acceptance_denial_review_entry_5_status=met
live_runner_acceptance_denial_review_entry_6_status=recorded
```

Phases

```
live_runner_acceptance_denial_review_phase_1=load_acceptance_denial_transcript
live_runner_acceptance_denial_review_phase_2=review_closed_gate_denial
live_runner_acceptance_denial_review_phase_3=confirm_no_effect_boundary
live_runner_acceptance_denial_review_phase_4=handoff_to_acceptance_denial_disposition
live_runner_acceptance_denial_review_phase_1_status=contract-only
live_runner_acceptance_denial_review_phase_2_status=review-only
live_runner_acceptance_denial_review_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_review_phase_4_status=future-disposition-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_review_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial review contract that reviews the
closed gate transcript without enabling dispatch or deletion.
```

Next Recommended Lane

The next lane should add a disposition contract so the reviewed closed-gate denial has a recorded no-effect disposition before any future dispatch work is reconsidered.

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_DENIAL_TRANSCRIPT_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Denial Transcript Contract

Lines: 219 | Words: 411 | SHA-256: 64b474ac02297e216687fc8cb706fc0014d2e7294c420a41f6703394aa40015a

macOS Reset/Uninstall Live-Runner Acceptance-Denial Transcript Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-denial transcript contract Date: 2026-05-26 CDT Scope: transcript contract for recording the closed acceptance gate without runner dispatch.

Purpose

This contract records the macOS reset/uninstall live-runner acceptance denial after the acceptance gate remains closed. It preserves the gate decision as a stdout-only transcript and keeps dispatch, deletion, receipts, absence reports, host mutation, network access, root authority, runtime authority, and implementation claims disabled.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-denial-transcript-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract_present=1
live_runner_acceptance_denial_transcript_contract_state=recorded-no-effect
live_runner_acceptance_denial_transcript_contract_decision=acceptance-gate-closed-no-dispatch-re
corded
live_runner_acceptance_denial_transcript_required=1
live_runner_acceptance_denial_transcript_present=1
live_runner_acceptance_denial_transcript_recorded=1
live_runner_acceptance_denial_transcript_stdout_only=1
live_runner_acceptance_denial_transcript_file_write_enabled=0
live_runner_acceptance_denial_transcript_acceptance_gate_present=1
live_runner_acceptance_denial_transcript_acceptance_gate_state=closed-no-effect
live_runner_acceptance_denial_transcript_acceptance_gate_open=0
live_runner_acceptance_denial_transcript_acceptance_gate_closed=1
live_runner_acceptance_denial_transcript_dispatch_allowed=0
live_runner_acceptance_denial_transcript_dispatch_enabled=0
live_runner_acceptance_denial_transcript_dispatch_performed=0
live_runner_acceptance_denial_transcript_runner_handoff_enabled=0
live_runner_acceptance_denial_transcript_deletion_enabled=0
live_runner_acceptance_denial_transcript_receipt_write_enabled=0
live_runner_acceptance_denial_transcript_absence_report_write_enabled=0
live_runner_acceptance_denial_transcript_effect=none
live_runner_acceptance_denial_transcript_effect_authorized=0
live_runner_acceptance_denial_transcript_denial_reason=missing-passed-preflight-complete-evidenc
e-effect-authorization-operator-intent-and-implementation
live_runner_acceptance_denial_transcript_denial_source=macos-reset-uninstall-live-runner-accepta
nce-gate-contract
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Source Acceptance Gate

```
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_contract_decision=dispatch-blocked-missing-passed-preflight-evidence
-and-authorization
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_acceptance_granted=0
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_runner_handoff_enabled=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_result_denied_dispatch_review=met
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_operator_intent=blocked
live_runner_acceptance_gate_result_implementation=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_gate_result_no_deletion_until_open=met
live_runner_acceptance_gate_result_no_receipt_write_until_open=met
live_runner_acceptance_gate_result_no_absence_report_write_until_open=met
```

Predecessor Inputs

```
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_dispatch_denied=1
live_runner_denied_dispatch_transcript_dispatch_enabled=0
live_runner_denied_dispatch_transcript_dispatch_performed=0
live_runner_denied_dispatch_transcript_deletion_enabled=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
```

Requirements

```
live_runner_acceptance_denial_transcript_schema_version=macos-reset-uninstall-live-runner-accept
ance-denial-transcript/1
live_runner_acceptance_denial_transcript_required_input_count=6
live_runner_acceptance_denial_transcript_observed_input_count=6
live_runner_acceptance_denial_transcript_requires_acceptance_gate=1
live_runner_acceptance_denial_transcript_requires_closed_gate=1
live_runner_acceptance_denial_transcript_requires_denial_reason=1
live_runner_acceptance_denial_transcript_requires_stdout_only=1
live_runner_acceptance_denial_transcript_requires_no_dispatch=1
live_runner_acceptance_denial_transcript_requires_no_deletion=1
live_runner_acceptance_denial_transcript_requires_no_receipt_write=1
live_runner_acceptance_denial_transcript_requires_no_absence_report_write=1
live_runner_acceptance_denial_transcript_requires_no_network=1
live_runner_acceptance_denial_transcript_requires_no_root=1
live_runner_acceptance_denial_transcript_result_acceptance_gate=met
live_runner_acceptance_denial_transcript_result_closed_gate=met
live_runner_acceptance_denial_transcript_result_denial_reason=met
live_runner_acceptance_denial_transcript_result_stdout_only=met
live_runner_acceptance_denial_transcript_result_no_dispatch=met
live_runner_acceptance_denial_transcript_result_no_deletion=met
live_runner_acceptance_denial_transcript_result_no_receipt_write=met
live_runner_acceptance_denial_transcript_result_no_absence_report_write=met
live_runner_acceptance_denial_transcript_result_no_network=met
live_runner_acceptance_denial_transcript_result_no_root=met
```

Transcript Entries

```
live_runner_acceptance_denial_transcript_entry_1=acceptance_gate_present
live_runner_acceptance_denial_transcript_entry_2=acceptance_gate_closed
live_runner_acceptance_denial_transcript_entry_3=acceptance_inputs_blocked
live_runner_acceptance_denial_transcript_entry_4=dispatch_blocked
live_runner_acceptance_denial_transcript_entry_5=no_effects_performed
live_runner_acceptance_denial_transcript_entry_1_status=met
live_runner_acceptance_denial_transcript_entry_2_status=met
live_runner_acceptance_denial_transcript_entry_3_status=recorded
live_runner_acceptance_denial_transcript_entry_4_status=met
live_runner_acceptance_denial_transcript_entry_5_status=met
```

Phases

```
live_runner_acceptance_denial_transcript_phase_1=load_acceptance_gate
live_runner_acceptance_denial_transcript_phase_2=record_acceptance_denial
live_runner_acceptance_denial_transcript_phase_3=preserve_no_effect_boundary
live_runner_acceptance_denial_transcript_phase_4=handoff_to_acceptance_denial_review
live_runner_acceptance_denial_transcript_phase_1_status=contract-only
live_runner_acceptance_denial_transcript_phase_2_status=stdout-only
live_runner_acceptance_denial_transcript_phase_3_status=blocked-no-effect
live_runner_acceptance_denial_transcript_phase_4_status=future-review-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-denial-transcript-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_denial_transcript_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial transcript contract that records the
closed acceptance gate without dispatching effects.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_ACCEPTANCE_GATE_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Acceptance-Gate Contract

Lines: 234 | Words: 441 | SHA-256: 298f8bd60d61293b130e869dc252b24bfcd0df39fa1a0efbda98a0a4c4d2643e

macOS Reset/Uninstall Live-Runner Acceptance-Gate Contract

Status: no-effect macOS reset/uninstall live-runner acceptance-gate contract Date: 2026-05-26 CDT Scope: contract for deciding whether the live runner may dispatch after a denied-dispatch review.

Purpose

This contract evaluates the macOS reset/uninstall live-runner acceptance gate. It records that the denied-dispatch review is present, then keeps the gate closed because passed preflight, complete evidence, explicit effect authorization, operator intent, and implementation evidence are not all present.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-acceptance-gate-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_acceptance_gate_contract_present=1
live_runner_acceptance_gate_contract_state=closed-no-effect
live_runner_acceptance_gate_contract_decision=dispatch-blocked-missing-passed-preflight-evidence
-and-authorization
live_runner_acceptance_gate_required=1
live_runner_acceptance_gate_present=1
live_runner_acceptance_gate_evaluated=1
live_runner_acceptance_gate_open=0
live_runner_acceptance_gate_closed=1
live_runner_acceptance_gate_acceptance_granted=0
live_runner_acceptance_gate_dispatch_allowed=0
live_runner_acceptance_gate_stdout_only=1
live_runner_acceptance_gate_file_write_enabled=0
live_runner_acceptance_gate_dispatch_enabled=0
live_runner_acceptance_gate_dispatch_performed=0
live_runner_acceptance_gate_runner_handoff_enabled=0
live_runner_acceptance_gate_deletion_enabled=0
live_runner_acceptance_gate_receipt_write_enabled=0
live_runner_acceptance_gate_absence_report_write_enabled=0
live_runner_acceptance_gate_effect=none
live_runner_acceptance_gate_effect_authorized=0
live_runner_acceptance_gate_denial_reason=missing-passed-preflight-complete-evidence-effect-auth
orization-operator-intent-and-implementation
live_runner_acceptance_gate_source_contract=macos-reset-uninstall-live-runner-denied-dispatch-re
view-contract
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Reviewed Gate Inputs

```
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_contract_decision=denial-review-retains-zero-dispatch
live_runner_denied_dispatch_review_completed=1
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_effect_authorized=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_contract_decision=dispatch-denied-no-effects-recorded
live_runner_denied_dispatch_transcript_dispatch_denied=1
live_runner_denied_dispatch_transcript_dispatch_enabled=0
live_runner_denied_dispatch_transcript_file_write_enabled=0
```

Predecessor Inputs

```
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_contract_decision=denied-interface-path-only
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_noop_prototype_no_effects_performed=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
```

Target Scope

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Requirements

```
live_runner_acceptance_gate_schema_version=macos-reset-uninstall-live-runner-acceptance-gate/1
live_runner_acceptance_gate_required_input_count=7
live_runner_acceptance_gate_observed_input_count=7
live_runner_acceptance_gate_requires_denied_dispatch_review=1
live_runner_acceptance_gate_requires_passed_preflight=1
live_runner_acceptance_gate_requires_complete_evidence_bundle=1
live_runner_acceptance_gate_requires_effect_authorization=1
live_runner_acceptance_gate_requires_operator_intent=1
live_runner_acceptance_gate_requires_implementation=1
live_runner_acceptance_gate_requires_no_dispatch_until_open=1
live_runner_acceptance_gate_requires_no_deletion_until_open=1
live_runner_acceptance_gate_requires_no_receipt_write_until_open=1
live_runner_acceptance_gate_requires_no_absence_report_write_until_open=1
live_runner_acceptance_gate_requires_no_network=1
live_runner_acceptance_gate_requires_no_root=1
live_runner_acceptance_gate_result_denied_dispatch_review=met
live_runner_acceptance_gate_result_passed_preflight=blocked
live_runner_acceptance_gate_result_complete_evidence_bundle=blocked
live_runner_acceptance_gate_result_effect_authorization=blocked
live_runner_acceptance_gate_result_operator_intent=blocked
live_runner_acceptance_gate_result_implementation=blocked
live_runner_acceptance_gate_result_no_dispatch_until_open=met
live_runner_acceptance_gate_result_no_deletion_until_open=met
live_runner_acceptance_gate_result_no_receipt_write_until_open=met
live_runner_acceptance_gate_result_no_absence_report_write_until_open=met
live_runner_acceptance_gate_result_no_network=met
live_runner_acceptance_gate_result_no_root=met
```

Gate Entries

```
live_runner_acceptance_gate_entry_1=denied_dispatch_review_present
live_runner_acceptance_gate_entry_2=preflight_not_passed
live_runner_acceptance_gate_entry_3=evidence_bundle_incomplete
live_runner_acceptance_gate_entry_4=effect_authorization_closed
live_runner_acceptance_gate_entry_5=operator_intent_absent
live_runner_acceptance_gate_entry_6=implementation_absent
live_runner_acceptance_gate_entry_7=dispatch_blocked
live_runner_acceptance_gate_entry_1_status=met
live_runner_acceptance_gate_entry_2_status=blocked
live_runner_acceptance_gate_entry_3_status=blocked
live_runner_acceptance_gate_entry_4_status=blocked
live_runner_acceptance_gate_entry_5_status=blocked
live_runner_acceptance_gate_entry_6_status=blocked
live_runner_acceptance_gate_entry_7_status=met
```

Phases

```
live_runner_acceptance_gate_phase_1=load_denied_dispatch_review
live_runner_acceptance_gate_phase_2=evaluate_required_acceptance_inputs
live_runner_acceptance_gate_phase_3=keep_runner_dispatch_closed
live_runner_acceptance_gate_phase_4=handoff_to_acceptance_denial_transcript
live_runner_acceptance_gate_phase_1_status=contract-only
live_runner_acceptance_gate_phase_2_status=blocking
live_runner_acceptance_gate_phase_3_status=blocked-no-effect
live_runner_acceptance_gate_phase_4_status=future-transcript-only
```

Authority Boundary

```
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-acceptance-gate-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_acceptance_gate_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, runner dispatch evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-gate contract that keeps the live runner
closed until passed preflight, evidence, authorization, operator intent, and implementation are
all present.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_REVIEW_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Denied-Dispatch Review Contract

Lines: 217 | Words: 410 | SHA-256: 2f78bf208f25a9b3bf2eed6ed981fc2193049dfa73e65d57e800bb0dbdc508c1

macOS Reset/Uninstall Live-Runner Denied-Dispatch Review Contract

Status: no-effect macOS reset/uninstall live-runner denied-dispatch review contract Date: 2026-05-26 CDT Scope: contract for reviewing the denied-dispatch transcript without enabling dispatch or deletion.

Purpose

This contract reviews the macOS reset/uninstall live-runner denied-dispatch transcript. The review records that the denial evidence is present, that the denial reason is still the failed preflight and denied interface path, and that dispatch remains blocked.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-denied-dispatch-review-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_denied_dispatch_review_contract_present=1
live_runner_denied_dispatch_review_contract_state=reviewed-no-effect
live_runner_denied_dispatch_review_contract_decision=denial-review-retains-zero-dispatch
live_runner_denied_dispatch_review_required=1
live_runner_denied_dispatch_review_present=1
live_runner_denied_dispatch_review_completed=1
live_runner_denied_dispatch_review_stdout_only=1
live_runner_denied_dispatch_review_file_write_enabled=0
live_runner_denied_dispatch_review_dispatch_denial_present=1
live_runner_denied_dispatch_review_denial_reason_present=1
live_runner_denied_dispatch_review_dispatch_reviewed=1
live_runner_denied_dispatch_review_dispatch_enabled=0
live_runner_denied_dispatch_review_dispatch_denied=1
live_runner_denied_dispatch_review_dispatch_performed=0
live_runner_denied_dispatch_review_deletion_enabled=0
live_runner_denied_dispatch_review_effect=none
live_runner_denied_dispatch_review_effect_authorized=0
live_runner_denied_dispatch_review_acceptance_gate_opened=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Reviewed Transcript

```
live_runner_denied_dispatch_review_source_contract=macos-reset-uninstall-live-runner-denied-disp
atch-transcript-contract
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
live_runner_denied_dispatch_review_transcript_state=recorded-no-effect
live_runner_denied_dispatch_review_transcript_decision=dispatch-denied-no-effects-recorded
live_runner_denied_dispatch_review_transcript_dispatch_denied=1
live_runner_denied_dispatch_review_transcript_dispatch_enabled=0
live_runner_denied_dispatch_review_transcript_file_write_enabled=0
live_runner_denied_dispatch_review_transcript_denial_reason=failed-preflight-and-denied-interfac
e
live_runner_denied_dispatch_review_transcript_denial_source=macos-reset-uninstall-live-runner-no
op-prototype-contract
live_runner_denied_dispatch_review_transcript_effect=none
```

Predecessor Inputs

```
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_contract_decision=denied-interface-path-only
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_noop_prototype_receipt_write_enabled=0
live_runner_noop_prototype_absence_report_write_enabled=0
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_no_effects_performed=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
macos_reset_uninstall_evidence_bundle_contract_present=1
reset_uninstall_evidence_bundle_complete=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_reset_uninstall_intent_evidence_present=0
```

Target Scope

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Requirements

```
live_runner_denied_dispatch_review_schema_version=macos-reset-uninstall-live-runner-denied-dispa
tch-review/1
live_runner_denied_dispatch_review_required_input_count=7
live_runner_denied_dispatch_review_observed_input_count=7
live_runner_denied_dispatch_review_requires_denied_dispatch_transcript=1
live_runner_denied_dispatch_review_requires_dispatch_denied=1
live_runner_denied_dispatch_review_requires_denial_reason=1
live_runner_denied_dispatch_review_requires_review_only=1
live_runner_denied_dispatch_review_requires_stdout_only=1
live_runner_denied_dispatch_review_requires_no_dispatch=1
live_runner_denied_dispatch_review_requires_no_deletion=1
live_runner_denied_dispatch_review_requires_no_receipt_write=1
live_runner_denied_dispatch_review_requires_no_absence_report_write=1
live_runner_denied_dispatch_review_requires_no_network=1
live_runner_denied_dispatch_review_requires_no_root=1
live_runner_denied_dispatch_review_result_denied_dispatch_transcript=met
live_runner_denied_dispatch_review_result_dispatch_denied=met
live_runner_denied_dispatch_review_result_denial_reason=met
live_runner_denied_dispatch_review_result_review_only=met
live_runner_denied_dispatch_review_result_stdout_only=met
live_runner_denied_dispatch_review_result_no_dispatch=met
live_runner_denied_dispatch_review_result_no_deletion=met
live_runner_denied_dispatch_review_result_no_receipt_write=met
live_runner_denied_dispatch_review_result_no_absence_report_write=met
live_runner_denied_dispatch_review_result_no_network=met
live_runner_denied_dispatch_review_result_no_root=met
```

Review Entries

```
live_runner_denied_dispatch_review_entry_1=transcript_contract_present
live_runner_denied_dispatch_review_entry_2=dispatch_denial_recorded
live_runner_denied_dispatch_review_entry_3=denial_reason_reviewed
live_runner_denied_dispatch_review_entry_4=no_effects_confirmed
live_runner_denied_dispatch_review_entry_5=acceptance_gate_required
live_runner_denied_dispatch_review_entry_1_status=met
live_runner_denied_dispatch_review_entry_2_status=reviewed
live_runner_denied_dispatch_review_entry_3_status=reviewed
live_runner_denied_dispatch_review_entry_4_status=met
live_runner_denied_dispatch_review_entry_5_status=recorded
```

Phases

```
live_runner_denied_dispatch_review_phase_1=load_denied_dispatch_transcript
live_runner_denied_dispatch_review_phase_2=review_dispatch_denial
live_runner_denied_dispatch_review_phase_3=confirm_no_effect_boundary
live_runner_denied_dispatch_review_phase_4=handoff_to_acceptance_gate
live_runner_denied_dispatch_review_phase_1_status=contract-only
live_runner_denied_dispatch_review_phase_2_status=review-only
live_runner_denied_dispatch_review_phase_3_status=blocked-no-effect
live_runner_denied_dispatch_review_phase_4_status=future-gate-only
```

Authority Boundary

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-denied-dispatch-review-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_denied_dispatch_review_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner denied-dispatch review contract that keeps review-only
dispatch denial evidence separate from any effects.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_DENIED_DISPATCH_TRANSCRIPT_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Denied-Dispatch Transcript Contract

Lines: 196 | Words: 381 | SHA-256: ceed9b5f53f5f5c2bacf0e3f3bd089eb790bcd9a249897b0bbbd3cb5c64d575b

macOS Reset/Uninstall Live-Runner Denied-Dispatch Transcript Contract

Status: no-effect macOS reset/uninstall live-runner denied-dispatch transcript contract Date: 2026-05-26 CDT Scope: contract for recording the no-op live-runner dispatch denial without enabling deletion.

Purpose

This contract records the denied-dispatch result from the macOS reset/uninstall live-runner no-op prototype. The predecessor path reaches only the denied interface branch because the live-execution preflight is still blocked.

It is contract-only. It does not authorize effects, dispatch a runner, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-denied-dispatch-transcript-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract_present=1
live_runner_denied_dispatch_transcript_contract_state=recorded-no-effect
live_runner_denied_dispatch_transcript_contract_decision=dispatch-denied-no-effects-recorded
live_runner_denied_dispatch_transcript_recorded=1
live_runner_denied_dispatch_transcript_stdout_only=1
live_runner_denied_dispatch_transcript_file_write_enabled=0
live_runner_denied_dispatch_transcript_dispatch_attempted=0
live_runner_denied_dispatch_transcript_dispatch_enabled=0
live_runner_denied_dispatch_transcript_dispatch_denied=1
live_runner_denied_dispatch_transcript_denial_reason=failed-preflight-and-denied-interface
live_runner_denied_dispatch_transcript_denial_source=macos-reset-uninstall-live-runner-noop-prot
otype-contract
live_runner_denied_dispatch_transcript_effect=none
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Predecessor Inputs

```
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_contract_decision=denied-interface-path-only
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_noop_prototype_receipt_write_enabled=0
live_runner_noop_prototype_absence_report_write_enabled=0
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_no_effects_performed=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_deletion_enabled=0
macos_reset_uninstall_live_denial_transcript_contract_present=1
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
macos_reset_uninstall_evidence_bundle_contract_present=1
reset_uninstall_evidence_bundle_complete=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_reset_uninstall_intent_evidence_present=0
```

Target Scope

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Requirements

```
live_runner_denied_dispatch_transcript_schema_version=macos-reset-uninstall-live-runner-denied-d
ispatch-transcript/1
live_runner_denied_dispatch_transcript_required_input_count=6
live_runner_denied_dispatch_transcript_observed_input_count=6
live_runner_denied_dispatch_transcript_requires_noop_prototype=1
live_runner_denied_dispatch_transcript_requires_denied_dispatch=1
live_runner_denied_dispatch_transcript_requires_denial_reason=1
live_runner_denied_dispatch_transcript_requires_stdout_only=1
live_runner_denied_dispatch_transcript_requires_no_deletion=1
live_runner_denied_dispatch_transcript_requires_no_receipt_write=1
live_runner_denied_dispatch_transcript_requires_no_absence_report_write=1
live_runner_denied_dispatch_transcript_requires_no_network=1
live_runner_denied_dispatch_transcript_requires_no_root=1
live_runner_denied_dispatch_transcript_result_noop_prototype=met
live_runner_denied_dispatch_transcript_result_denied_dispatch=met
live_runner_denied_dispatch_transcript_result_denial_reason=met
live_runner_denied_dispatch_transcript_result_stdout_only=met
live_runner_denied_dispatch_transcript_result_no_deletion=met
live_runner_denied_dispatch_transcript_result_no_receipt_write=met
live_runner_denied_dispatch_transcript_result_no_absence_report_write=met
live_runner_denied_dispatch_transcript_result_no_network=met
live_runner_denied_dispatch_transcript_result_no_root=met
```

Transcript Entries

```
live_runner_denied_dispatch_transcript_entry_1=noop_prototype_present
live_runner_denied_dispatch_transcript_entry_2=runner_interface_denied
live_runner_denied_dispatch_transcript_entry_3=preflight_not_passed
live_runner_denied_dispatch_transcript_entry_4=dispatch_blocked
live_runner_denied_dispatch_transcript_entry_5=no_effects_performed
live_runner_denied_dispatch_transcript_entry_1_status=met
live_runner_denied_dispatch_transcript_entry_2_status=met
live_runner_denied_dispatch_transcript_entry_3_status=met
live_runner_denied_dispatch_transcript_entry_4_status=recorded
live_runner_denied_dispatch_transcript_entry_5_status=met
```

Phases

```
live_runner_denied_dispatch_transcript_phase_1=load_noop_prototype
live_runner_denied_dispatch_transcript_phase_2=record_denied_dispatch
live_runner_denied_dispatch_transcript_phase_3=preserve_no_effect_boundary
live_runner_denied_dispatch_transcript_phase_4=handoff_to_denied_dispatch_review
live_runner_denied_dispatch_transcript_phase_1_status=contract-only
live_runner_denied_dispatch_transcript_phase_2_status=stdout-only
live_runner_denied_dispatch_transcript_phase_3_status=blocked-no-effect
live_runner_denied_dispatch_transcript_phase_4_status=future-review-only
```

Authority Boundary

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-denied-dispatch-transcript-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_denied_dispatch_transcript_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner denied-dispatch transcript contract that records the
no-op runner denial without enabling deletion.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_INTERFACE_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner Interface Contract

Lines: 287 | Words: 554 | SHA-256: 301c26698537f280ce8ada9c0acc678f48183ef5bdd4093aca89429cbdac099d

macOS Reset/Uninstall Live-Runner Interface Contract

Status: no-effect macOS reset/uninstall live-runner interface contract Date: 2026-05-26 CDT Scope: contract for the future live reset/uninstall runner entrypoint while accepting only a passed preflight.

Purpose

This contract defines the future macOS reset/uninstall live-runner interface. The interface accepts only a passed live-execution preflight and keeps the denial path active while the current preflight remains blocked.

It is contract-only. It does not invoke a runner, authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-interface-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current live-runner interface posture is:

```
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_required=1
live_runner_interface_present=1
live_runner_interface_entrypoint_defined=1
live_runner_interface_entrypoint_name=latticra-macos-reset-uninstall-live-runner
live_runner_interface_accepts_only_passed_preflight=1
live_runner_interface_accepts_passed_preflight_only=1
live_runner_interface_preflight_present=1
live_runner_interface_preflight_passed=0
live_runner_interface_denial_transcript_present=1
live_runner_interface_denial_transcript_recorded=1
live_runner_interface_invocation_enabled=0
live_runner_interface_handoff_enabled=0
live_runner_interface_runner_enabled=0
live_runner_interface_deletion_enabled=0
```

```
live_runner_interface_receipt_write_enabled=0
live_runner_interface_absence_write_enabled=0
live_runner_interface_absence_report_write_enabled=0
live_runner_interface_denial_path_active=1
live_runner_interface_accept_path_active=0
live_runner_interface_effect=none
live_runner_interface_effect_authorization_required=1
live_runner_interface_effect_authorized=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
absence_report_write_enabled=0
```

Predecessor Inputs

The interface records the live-denial transcript and failed preflight it depends on:

```
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_contract_decision=denied-by-preflight-block
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_denial_transcript_preflight_passed=0
live_denial_transcript_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_contract_decision=blocked-missing-complete-evidence-bundle-and-effect-a
uthorization
live_execution_preflight_required=1
live_execution_preflight_present=1
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_evidence_present=0
live_execution_preflight_record_write_enabled=0
live_execution_preflight_denial_recorded=1
live_execution_preflight_denial_reason=missing-complete-evidence-bundle-and-effect-authorization
```

The underlying gates remain closed:

```
macos_reset_uninstall_live_implementation_plan_contract_present=1
live_implementation_plan_contract_state=defined-no-effect
live_implementation_plan_execution_enabled=0
live_implementation_plan_deletion_enabled=0
live_implementation_plan_preflight_present=1
live_implementation_plan_preflight_passed=0
macos_reset_uninstall_evidence_bundle_contract_present=1
evidence_bundle_contract_state=defined-no-effect
reset_uninstall_evidence_bundle_complete=0
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_valid=0
reset_uninstall_evidence_bundle_evidence_present=0
macos_reset_uninstall_effect_authorization_contract_present=1
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
implementation_gate_contract_state=closed-no-effect
implementation_gate_open=0
macos_reset_uninstall_operator_intent_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
```

Target Scope

The interface remains limited to user-local managed macOS targets and future receipt locations:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
```

```
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Interface Requirements

The interface requires:

```
live_runner_interface_schema_version=macos-reset-uninstall-live-runner-interface/1
live_runner_interface_required_input_count=8
live_runner_interface_observed_input_count=8
live_runner_interface_requires_passed_preflight=1
live_runner_interface_requires_live_denial_transcript=1
live_runner_interface_requires_complete_evidence_bundle=1
live_runner_interface_requires_effect_authorization=1
live_runner_interface_requires_implementation_gate_open=1
live_runner_interface_requires_operator_intent_evidence=1
live_runner_interface_requires_denial_transcript_when_blocked=1
live_runner_interface_requires_managed_targets_only=1
live_runner_interface_requires_receipt_schema=1
live_runner_interface_requires_absence_report_contract=1
live_runner_interface_requires_no_unmanaged_targets=1
live_runner_interface_requires_no_unsafe_paths=1
live_runner_interface_requires_no_deletion=1
live_runner_interface_requires_no_network=1
live_runner_interface_requires_no_root=1
live_runner_interface_condition_passed_preflight=required
live_runner_interface_condition_live_denial_transcript=present
live_runner_interface_condition_complete_evidence_bundle=required
live_runner_interface_condition_effect_authorization=required
live_runner_interface_condition_implementation_gate_open=required
live_runner_interface_condition_operator_intent_evidence=required
live_runner_interface_condition_no_unmanaged_targets=required
live_runner_interface_condition_no_deletion=required
live_runner_interface_condition_denial_transcript_when_blocked=required
live_runner_interface_condition_no_deletion_without_accept=required
live_runner_interface_condition_no_receipt_write_without_accept=required
live_runner_interface_condition_no_absence_report_write_without_accept=required
live_runner_interface_condition_no_network=required
live_runner_interface_condition_no_root=required
```

The current result state is:

```
live_runner_interface_result_passed_preflight=not_met
live_runner_interface_result_live_denial_transcript=met
live_runner_interface_result_complete_evidence_bundle=not_met
live_runner_interface_result_effect_authorization=not_met
live_runner_interface_result_implementation_gate_open=not_met
live_runner_interface_result_operator_intent_evidence=not_met
live_runner_interface_result_no_unmanaged_targets=not_evaluated_contract_only
live_runner_interface_result_no_deletion=met
live_runner_interface_result_denial_transcript_when_blocked=met
live_runner_interface_result_no_deletion_without_accept=met
live_runner_interface_result_no_receipt_write_without_accept=met
live_runner_interface_result_no_absence_report_write_without_accept=met
live_runner_interface_result_no_network=met
live_runner_interface_result_no_root=met
```

Interface Shape

The future interface shape is:

```
live_runner_interface_input_preflight_status=required-passed
live_runner_interface_input_denial_transcript=required-when-blocked
live_runner_interface_input_target_plan=managed-targets-only
live_runner_interface_output_on_block=stdout-denial-only
live_runner_interface_output_on_pass=future-effect-handled-by-gated-runner
```

Phases

The interface phases are mapped but closed:

```
live_runner_interface_phase_1=accept_invocation_request
live_runner_interface_phase_2=load_live_execution_preflight
live_runner_interface_phase_3=reject_failed_preflight
live_runner_interface_phase_4=emit_denial_transcript_reference
```

```
live_runner_interface_phase_5=block_effect_dispatch
live_runner_interface_phase_1_status=contract-only
live_runner_interface_phase_2_status=contract-only
live_runner_interface_phase_3_status=blocked-preflight-not-passed
live_runner_interface_phase_4_status=recorded-no-effect
live_runner_interface_phase_5_status=blocked-no-effect
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-runner-interface-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_interface_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, operator approval evidence, effect approval evidence, complete evidence-bundle evidence, live execution evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner interface contract that accepts only a passed preflight
and keeps deletion disabled otherwise.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_RUNNER_NOOP_PROTOTYPE_CONTRACT.md

Source title: macOS Reset/Uninstall Live-Runner No-Op Prototype Contract

Lines: 157 | Words: 326 | SHA-256: 6e8f8795564d81453d50ab6674645a24872d4d8a17192482bca88e46e96fe654

macOS Reset/Uninstall Live-Runner No-Op Prototype Contract

Status: no-effect macOS reset/uninstall live-runner no-op prototype contract Date: 2026-05-26 CDT Scope: contract for exercising the denied live-runner interface path without deleting files.

Purpose

This contract defines a no-op prototype for the future macOS reset/uninstall live runner. It exercises only the denied interface path because the live-execution preflight is still blocked.

It is contract-only. It does not authorize effects, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-runner-noop-prototype-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

```
macos_reset_uninstall_live_runner_noop_prototype_contract_present=1
live_runner_noop_prototype_contract_state=executed-no-effect
live_runner_noop_prototype_contract_decision=denied-interface-path-only
live_runner_noop_prototype_invocation_simulated=1
live_runner_noop_prototype_stdout_only=1
live_runner_noop_prototype_dispatch_enabled=0
live_runner_noop_prototype_runner_enabled=0
live_runner_noop_prototype_runner_effect_enabled=0
live_runner_noop_prototype_deletion_enabled=0
live_runner_noop_prototype_receipt_write_enabled=0
live_runner_noop_prototype_absence_report_write_enabled=0
live_runner_noop_prototype_denial_path_exercised=1
live_runner_noop_prototype_accept_path_exercised=0
live_runner_noop_prototype_effect=none
live_runner_noop_prototype_effect_authorized=0
live_runner_noop_prototype_preflight_passed=0
live_runner_noop_prototype_interface_present=1
live_runner_noop_prototype_interface_denial_path_active=1
live_runner_noop_prototype_no_effects_performed=1
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
```

Interface Input

```
macos_reset_uninstall_live_runner_interface_contract_present=1
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_contract_decision=denied-current-preflight-not-passed
live_runner_interface_preflight_passed=0
live_runner_interface_denial_path_active=1
live_runner_interface_invocation_enabled=0
live_runner_interface_deletion_enabled=0
macos_reset_uninstall_live_execution_preflight_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
```

Requirements

```
live_runner_noop_prototype_schema_version=macos-reset-uninstall-live-runner-noop-prototype/1
live_runner_noop_prototype_requires_runner_interface=1
live_runner_noop_prototype_requires_failed_preflight=1
live_runner_noop_prototype_requires_denial_path=1
live_runner_noop_prototype_requires_denied_interface_path=1
live_runner_noop_prototype_requires_no_dispatch=1
live_runner_noop_prototype_requires_no_deletion=1
live_runner_noop_prototype_requires_no_receipt_write=1
live_runner_noop_prototype_requires_no_absence_report_write=1
live_runner_noop_prototype_requires_no_network=1
live_runner_noop_prototype_requires_no_root=1
live_runner_noop_prototype_result_runner_interface=met
live_runner_noop_prototype_result_failed_preflight=met
live_runner_noop_prototype_result_denial_path=met
live_runner_noop_prototype_result_denied_interface_path=met
live_runner_noop_prototype_result_no_dispatch=met
live_runner_noop_prototype_result_no_deletion=met
live_runner_noop_prototype_result_no_receipt_write=met
live_runner_noop_prototype_result_no_absence_report_write=met
live_runner_noop_prototype_result_no_network=met
live_runner_noop_prototype_result_no_root=met
```

Phases

```
live_runner_noop_prototype_phase_1=load_runner_interface
live_runner_noop_prototype_phase_2=exercise_denied_interface_path
live_runner_noop_prototype_phase_3=confirm_no_effect_dispatch
live_runner_noop_prototype_phase_4=handoff_to_denied_dispatch_transcript
live_runner_noop_prototype_phase_5=return_noop_denial
live_runner_noop_prototype_phase_1_status=contract-only
live_runner_noop_prototype_phase_2_status=noop-denial-only
live_runner_noop_prototype_phase_3_status=met
live_runner_noop_prototype_phase_4_status=blocked-no-effect
live_runner_noop_prototype_phase_5_status=stdout-only
```

Authority Boundary

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
live_reset_uninstall_runner_present=0
live_reset_uninstall_runner_enabled=0
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

```
sh scripts/test-macos-reset-uninstall-live-runner-noop-prototype-contract.sh
```

Expected output:

```
macos_reset_uninstall_live_runner_noop_prototype_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, live reset execution, live uninstall execution, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Endpoint Security evidence, System Extension evidence, privileged helper evidence, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall live-runner no-op prototype contract that exercises the denied
interface path without deleting files.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_LIVE_TARGET_CLASSIFIER.md

Source title: macOS Reset/Uninstall Live-Target Classifier

Lines: 222 | Words: 468 | SHA-256: 3ee2ed6805b20797f7a1eebcbb03b5a3fb35fd7c03abdd36ab359c83ef9c6f7a

macOS Reset/Uninstall Live-Target Classifier

Status: no-effect macOS reset/uninstall live-target classifier Date: 2026-05-25 CDT Scope: read-only classifier for current macOS user-local reset/uninstall targets.

Purpose

This classifier reports what exists today at the macOS reset/uninstall targets for the macOS reset/uninstall dry-run planner.

It is read-only. It does not delete files, remove directories, write receipts, mutate host state, run absence verification, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-live-target-classifier.sh
```

The command writes only a deterministic report to stdout.

Target States

The classifier records these target states:

```
absent
managed
unmanaged-preserve
not-classified-unsafe-path
```

It uses the same managed marker shape as the macOS app-bundle writer dry-run:

```
LATTICRA_INSTALLER_MANAGED=1
LATTICRA_MACOS_USER_LOCAL_APP_BUNDLE=1
```

Directory targets are managed only when the marker exists at one of these locations:

```
Contents/Resources/latticra/MANAGED_BY_LATTICRA
MANAGED_BY_LATTICRA
```

File targets are managed only when the file itself contains both marker lines.

Classified Targets

The live classifier is bound to the current user-local macOS targets:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipts_dir_preservation=preserve-outside-removed-prefix
```

Decisions

The classifier can report:

```
reset_uninstall_live_classifier_decision=ready-no-targets-observed-no-effect
```

```
reset_uninstall_live_classifier_decision=ready-managed-targets-observed-no-effect
reset_uninstall_live_classifier_decision=blocked-unmanaged-targets-preserve
reset_uninstall_live_classifier_decision=blocked-unsafe-path
```

An unmanaged target is never overwritten or removed. It is classified as unmanaged-preserve and blocks future reset/uninstall progress until an operator resolves it outside this no-effect lane.

Example Fields

```
macos_reset_uninstall_live_target_classifier_present=1
classifier_report_only=1
path_guard_status=allowed-user-local-classification
target_states_recorded=1
app_bundle_target_state=managed
app_bundle_target_state=unmanaged-preserve
managed_target_detected=report-runtime
unmanaged_target_detected=report-runtime
reset_uninstall_dry_run_evidence_present=0
macos_reset_uninstall_dry_run_planner_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
```

```
reset_uninstall_effect_authorized=0
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_receipt_evidence_present=0
macos_reset_uninstall_implemented=0
```

Authority Boundary

This classifier preserves:

```
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This classifier is guarded by:

```
sh scripts/test-macos-reset-uninstall-live-target-classifier.sh
```

Expected output:

```
macos_reset_uninstall_live_target_classifier: ok
```

Non-Claims

This classifier is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_OPERATOR_INTENT_CONTRACT.md

Source title: macOS Reset/Uninstall Operator-Intent Contract

Lines: 261 | Words: 508 | SHA-256: 1310251ce00a7712d026a87d4afee083c5e378285d564743f1f3d7488642c411

macOS Reset/Uninstall Operator-Intent Contract

Status: no-effect macOS reset/uninstall operator-intent contract Date: 2026-05-25 CDT Scope: contract for future explicit operator approval evidence before any live macOS reset/uninstall activity.

Purpose

This contract defines the explicit operator-intent evidence that a future macOS reset/uninstall implementation must collect before any live managed-target removal or reset/uninstall receipt writing can be considered.

It is contract-only. It does not prompt for approval, infer approval, delete files, remove directories, write receipts, write absence reports, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-operator-intent-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current operator-intent posture is:

```
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
macos_reset_uninstall_implementation_gate_contract_present=1
operator_intent_contract_state=defined-no-effect
operator_intent_contract_decision=contract-defined-intent-not-observed
operator_intent_contract_required=1
operator_intent_contract_open=0
operator_reset_uninstall_intent_required=1
operator_reset_uninstall_intent_evidence_required=1
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
operator_intent_capture_performed=0
operator_intent_record_write_enabled=0
operator_intent_record_written=0
operator_intent_evidence_written=0
operator_intent_transcript_write_performed=0
implementation_gate_open=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
managed_target_removal_allowed=0
managed_target_deletion_enabled=0
reset_uninstall_receipt_write_enabled=0
```

Required Evidence Chain

Future operator-intent evidence must bind to the closed implementation gate and the reset/uninstall evidence contracts:

```
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
implementation_gate_contract_state=closed-no-effect
implementation_gate_decision=blocked-missing-reset-uninstall-evidence
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
macos_commit_gate_contract_present=1
reset_uninstall_dry_run_evidence_present=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
absence_report_evidence_present=0
```

Target Scope

The future approval evidence must name these user-local macOS targets and receipt locations:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
```

Intent Requirements

The future approval evidence must be explicit, current, and scoped:

```
operator_intent_evidence_format=text-transcript
operator_intent_schema_version=macos-reset-uninstall-operator-intent/1
operator_intent_digest_algorithm=sha256
operator_intent_must_be_explicit=1
operator_intent_must_be_current_session=1
operator_intent_must_not_be_inferred=1
operator_intent_must_not_be_defaulted=1
operator_intent_must_not_be_from_previous_session=1
operator_intent_must_name_operation=1
operator_intent_operation_values=reset,uninstall
operator_intent_must_name_target_scope=1
```

```
operator_intent_target_scope=user-local-managed-targets-only
operator_intent_must_acknowledge_managed_targets_only=1
operator_intent_must_acknowledge_no_unmanaged_removal=1
operator_intent_must_acknowledge_receipt_path=1
operator_intent_must_acknowledge_absence_report_path=1
operator_intent_must_acknowledge_no_network=1
operator_intent_must_acknowledge_no_root=1
operator_intent_must_acknowledge_live_classifier_digest=1
operator_intent_must_acknowledge_dry_run_planner_digest=1
operator_intent_field_dry_run_planner_transcript_digest_required=1
operator_intent_field_live_classifier_digest_required=1
operator_intent_field_receipt_schema_digest_required=1
operator_intent_field_absence_report_contract_digest_required=1
operator_intent_field_confirmation_text_required=1
operator_intent_scope_must_match_live_classifier=1
operator_intent_scope_must_match_dry_run_planner=1
operator_intent_confirmation_phrase_required=1
operator_intent_confirmation_phrase_future=reset-or-uninstall-latticra-user-local-managed-target
s
operator_intent_transcript_required=1
```

Phases

The future operator-intent phases are:

```
operator_intent_phase_1=present_live_target_classifier_summary
operator_intent_phase_2=present_dry_run_planner_summary
operator_intent_phase_3=present_receipt_and_absence_report_paths
operator_intent_phase_4=request_explicit_operator_intent
operator_intent_phase_5=record_operator_intent_evidence
operator_intent_phase_6=allow_future_implementation_gate_open
operator_intent_phase_1_status=contract-only
operator_intent_phase_2_status=contract-only
operator_intent_phase_3_status=contract-only
operator_intent_phase_4_status=contract-only
operator_intent_phase_5_status=disabled
operator_intent_phase_6_status=disabled
```

Authority Boundary

This contract preserves:

```
reset_uninstall_implementation_present=0
macos_reset_uninstall_implemented=0
managed_marker_required=1
unmanaged_target_preservation_required=1
receipt_outside_removed_prefix_required=1
absence_report_required=1
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-operator-intent-contract.sh
```

Expected output:

```
macos_reset_uninstall_operator_intent_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, live approval evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_RESET_UNINSTALL_RECEIPT_SCHEMA_CONTRACT.md

Source title: macOS Reset/Uninstall Receipt-Schema Contract

Lines: 208 | Words: 428 | SHA-256: 040beb81952180bdb4216884594687d0eb0bb6340a6f7fe4f0d59e0851ec4cb9

macOS Reset/Uninstall Receipt-Schema Contract

Status: no-effect macOS reset/uninstall receipt-schema contract Date: 2026-05-25 CDT Scope: contract for future reset/uninstall receipts stored outside the removed prefix.

Purpose

This contract defines the receipt schema that a future macOS reset/uninstall implementation must use when recording reset or uninstall evidence.

It is contract-only. It does not write receipts, delete files, remove directories, inspect post-removal state, mutate host state, open the network, or claim reset/uninstall implementation.

Command

```
sh scripts/macos-reset-uninstall-receipt-schema-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current receipt-schema posture is:

```
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
```

```
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
receipt_schema_contract_state=defined-no-effect
receipt_schema_contract_decision=contract-defined-receipt-not-written
receipt_schema_required=1
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
receipt_schema_evidence_present=0
reset_receipt_write_performed=0
receipt_write_performed=0
receipt_written=0
```

Required Inputs

A future reset/uninstall receipt must bind to the existing no-effect macOS reset/uninstall chain:

```
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_contract_present=1
macos_verification_transcript_contract_present=1
absence_report_evidence_present=0
reset_uninstall_dry_run_evidence_present=0
```

Receipt Location

The future receipt must live outside the removed application-support prefix:

```
app_support_prefix_target=$HOME/Library/Application Support/Latticra
app_bundle_target=$HOME/Applications/Latticra Panel.app
cli_wrapper_target=$HOME/.local/bin/latticra-panel
reset_receipts_dir_target=$HOME/Library/Application Support/Latticra Reset Receipts
reset_receipt_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/reset-uninstall-receipt.json
absence_report_path_future=$HOME/Library/Application Support/Latticra Reset
Receipts/absence-report.txt
receipt_must_be_outside_removed_prefix=1
removed_prefix_must_not_contain_receipt=1
```

Schema Fields

The future receipt format is:

```
receipt_format=json
receipt_schema_version=macos-reset-uninstall-receipt/1
receipt_digest_algorithm=sha256
receipt_operation_values=reset,uninstall
```

The future receipt must include:

```
receipt_field_schema_version_required=1
receipt_field_operation_required=1
receipt_field_host_identity_required=1
receipt_field_architecture_required=1
receipt_field_started_at_required=1
receipt_field_completed_at_required=1
receipt_field_planner_transcript_digest_required=1
receipt_field_live_classifier_digest_required=1
```

```
receipt_field_absence_report_path_required=1
receipt_field_target_actions_required=1
receipt_field_removed_managed_targets_required=1
receipt_field_preserved_unmanaged_targets_required=1
receipt_field_authority_denials_required=1
receipt_field_no_network_required=1
receipt_field_no_root_required=1
```

Phases

The future receipt-schema phases are:

```
receipt_schema_phase_1=consume_dry_run_planner_transcript
receipt_schema_phase_2=consume_absence_report_contract
receipt_schema_phase_3=validate_outside_removed_prefix_path
receipt_schema_phase_4=record_target_action_summary
receipt_schema_phase_5=record_authority_denials
receipt_schema_phase_6=write_reset_uninstall_receipt
receipt_schema_phase_6_status=disabled
```

Authority Boundary

This contract preserves:

```
managed_wrapper_removal_performed=0
managed_app_bundle_removal_performed=0
managed_application_support_removal_performed=0
reset_receipt_write_performed=0
absence_report_run_performed=0
absence_report_written=0
file_delete_performed=0
directory_delete_performed=0
application_support_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-reset-uninstall-receipt-schema-contract.sh
```

Expected output:

```
macos_reset_uninstall_receipt_schema_contract: ok
```

Non-Claims

This contract is not macOS reset evidence, macOS uninstall evidence, macOS install evidence, macOS app bundle evidence, receipt evidence, absence verification evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_USER_LOCAL_APP_BUNDLE_CONTRACT.md

Source title: macOS User-Local App Bundle Contract

Lines: 269 | Words: 753 | SHA-256: 3145fcb7fa102c4dc6c9a26622e76cea89c1443087e98e7b8cafa79ba7c7170a

macOS User-Local App Bundle Contract

Status: macOS user-local app bundle contract Date: 2026-05-25 CDT Scope: exact app bundle, managed marker, reset/uninstall, and verification transcript requirements before any macOS app bundle writer exists.

Purpose

This contract defines the minimum shape a future macOS user-local Latticra Panel app bundle writer must satisfy.

It follows the macOS dry-run plan adapter and remains contract-only. It does not create an app bundle, write files, install wrappers, mutate shell profiles, use launchd, access Keychain, request TCC permissions, use Endpoint Security, use System Extensions, use Network Extensions, open the network, or grant runtime authority.

Required User-Local Roots

The only supported first writer roots are:

```
application_support_prefix=$HOME/Library/Application Support/Latticra
app_bundle=$HOME/Applications/Latticra Panel.app
cli_bin=$HOME/.local/bin
logs_dir=$HOME/Library/Logs/Latticra
caches_dir=$HOME/Library/Caches/Latticra
reset_receipts_dir=$HOME/Library/Application Support/Latticra Reset Receipts
```

The writer must refuse:

```
/Applications
/Library
/System
/usr/local
/opt/homebrew
LaunchDaemons
system LaunchAgents
kernel extension paths
system extension paths
network extension paths
privileged helper paths
```

Exact App Bundle Files

A future writer may create only this managed app bundle shape:

```
$HOME/Applications/Latticra Panel.app/
  Contents/
    Info.plist
    MacOS/
      latticra-panel
    Resources/
      latticra-panel.icns
      latticra/
        MANAGED_BY_LATTICRA
        bundle-manifest.txt
```

Required bundle file meanings:

```
| File | Requirement |
| --- | --- |
| `Contents/Info.plist` | Property-list metadata for the local app bundle only. |
```

```
| `Contents/MacOS/latticra-panel` | Future copied Panel executable; must be executable. |
| `Contents/Resources/latticra-panel.icns` | Future icon asset; may be generated or converted in
 a separate lane. |
| `Contents/Resources/latticra/MANAGED_BY_LATTICRA` | Managed marker proving the bundle is
removable by Latticra. |
| `Contents/Resources/latticra/bundle-manifest.txt` | Deterministic bundle file list and
measurements. |
```

Info.plist Requirements

The future Info.plist must include:

```
CFBundleName=Latticra Panel
CFBundleDisplayName=Latticra Panel
CFBundleIdentifier=systems.latticra.panel
CFBundleExecutable=latticra-panel
CFBundlePackageType=APPL
CFBundleShortVersionString=&lt;panel-version&gt;
CFBundleVersion=&lt;build-or-date-version&gt;
LSMinimumSystemVersion=&lt;supported-version&gt;
NSHighResolutionCapable=true
```

The first writer must not request privacy-sensitive usage strings unless a separate TCC contract exists. It must not include LaunchAgent, login item, background item, privileged helper, network extension, system extension, or Endpoint Security activation behavior.

Application Support Files

A future writer may create only this managed Application Support shape:

```
$HOME/Library/Application Support/Latticra/
  MANAGED_BY_LATTICRA
  etc/
    latticra/
      installer-config.toml
      macos-plan.txt
  lib/
    latticra/
      payload-manifest.txt
  receipts/
    latest-receipt.txt
    macos-app-bundle-install-receipt-&lt;timestamp&gt;.txt
    manifest-&lt;timestamp&gt;.sha256
```

The Application Support marker must include:

```
LATTICRA_INSTALLER_MANAGED=1
LATTICRA_MACOS_USER_LOCAL_APP_BUNDLE=1
root_authority=0
network_authority=0
runtime_enforcement_authority=0
```

CLI Wrapper Contract

Future wrapper files may be created only under:

```
$HOME/.local/bin/latticra
$HOME/.local/bin/lat
$HOME/.local/bin/latticra-seal
$HOME/.local/bin/latticra-nadia
$HOME/.local/bin/latticra-panel
```

Each wrapper must include:

```
# LATTICRA_INSTALLER_MANAGED=1
# LATTICRA_MACOS_USER_LOCAL_APP_BUNDLE=1
```

The first writer must not mutate .zshrc, .zprofile, .bashrc, .bash_profile, fish config, shell profile files, PATH settings, Homebrew prefixes, /usr/local, or /opt/homebrew.

Receipt Requirements

The future install receipt must record:

```
receipt_kind=macos-user-local-app-bundle-install
application_support_prefix=&lt;path&gt;
```

```
app_bundle=&lt;path&gt;
cli_bin=&lt;path&gt;
bundle_manifest=&lt;path&gt;
payload_manifest=&lt;path&gt;
root_authority=0
network_authority=0
runtime_enforcement_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

Receipts for reset or uninstall must be written outside any prefix being removed:

```
reset_receipts_dir=$HOME/Library/Application Support/Latticra Reset Receipts
```

Reset And Uninstall Contract

Reset/uninstall may remove only managed artifacts:

```
$HOME/Applications/Latticra Panel.app
$HOME/Library/Application Support/Latticra
$HOME/.local/bin/latticra
$HOME/.local/bin/lat
$HOME/.local/bin/latticra-seal
$HOME/.local/bin/latticra-nadia
$HOME/.local/bin/latticra-panel
```

Removal is allowed only when the target contains a managed marker:

```
LATTICRA_INSTALLER_MANAGED=1
LATTICRA_MACOS_USER_LOCAL_APP_BUNDLE=1
```

If a marker is missing, the remover must preserve the target and report:

```
preserved_unmanaged_target=1
```

Reset/uninstall must not remove user logs, user caches, shell profiles, Keychain items, LaunchAgents, Login Items, application support outside the Latticra prefix, Homebrew files, /Applications, /Library, /System, /usr/local, or /opt/homebrew.

Verification Transcript

Before claiming macOS user-local app bundle install evidence, a transcript must record:

```
ok: app bundle -&gt; $HOME/Applications/Latticra Panel.app
ok: Info.plist -&gt; $HOME/Applications/Latticra Panel.app/Contents/Info.plist
ok: app executable -&gt; $HOME/Applications/Latticra Panel.app/Contents/MacOS/latticra-panel
ok: app resources -&gt; $HOME/Applications/Latticra Panel.app/Contents/Resources
ok: app managed marker -&gt; $HOME/Applications/Latticra
Panel.app/Contents/Resources/latticra/MANAGED_BY_LATTICRA
ok: application support prefix -&gt; $HOME/Library/Application Support/Latticra
ok: application support marker -&gt; $HOME/Library/Application Support/Latticra/
MANAGED_BY_LATTICRA
ok: receipts -&gt; $HOME/Library/Application Support/Latticra/receipts
ok: latticra-panel wrapper -&gt; $HOME/.local/bin/latticra-panel
ok: reset dry-run preserves unmanaged targets
ok: unsafe path blocked
```

The transcript must also record:

```
app_bundle_created=1
host_mutation_scope=user-local-managed
root_authority=0
network_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
production_installer_ready=0
```

This transcript requirement does not mean such evidence exists today.

Current Contract Status

```
macos_user_local_app_bundle_contract_present=1
macos_app_bundle_writer_present=0
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
host_mutation_performed=0
network_performed=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-user-local-app-bundle-contract.sh
```

Expected output:

```
macos_user_local_app_bundle_contract: ok
```

Non-Claims

This contract is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Next Recommended Lane

```
Add a no-effect macOS app bundle writer dry-run prototype that emits the planned phase report,
validates unsafe paths, and keeps commit_user_local_managed_artifacts=0.
```

docs/MACOS_USER_LOCAL_APP_BUNDLE_IMPLEMENTATION_PLAN.md

Source title: macOS User-Local App Bundle Implementation Plan

Lines: 254 | Words: 741 | SHA-256: da77c8847ae801bcf8462cd38af38e44b07f1235ca97ceba15cfff25fb98b4bb

macOS User-Local App Bundle Implementation Plan

Status: no-effect macOS user-local app bundle implementation plan Date: 2026-05-25 CDT Scope: writer phases, failure behavior, reset/uninstall sequencing, verification commands, and guard tests before any app bundle writer exists.

Purpose

This plan defines how a future macOS user-local Latticra Panel app bundle writer should be implemented.

It follows the macOS user-local app bundle contract and remains no-effect planning. It does not create an app bundle, write Application Support files, install wrappers, mutate shell profiles, build the Panel, use launchd, access Keychain, request TCC permissions, use Endpoint Security, use System Extensions, use Network Extensions, open the network, or grant runtime authority.

Required Inputs

A future writer must require these inputs before any write path is enabled:

```
macos_build_platform_probe_present=1
macos_dry_run_plan_adapter_present=1
macos_user_local_app_bundle_contract_present=1
path_guard_status=allowed-user-local-dry-run
dry_run_allowed=1
application_support_prefix=$HOME/Library/Application Support/Latticra
app_bundle=$HOME/Applications/Latticra Panel.app
cli_bin=$HOME/.local/bin
```

The writer must fail closed if any input is missing or if a path expands outside the user-local allowlist.

Writer Phase Plan

The future writer must use explicit phases:

```
phase_1=validate_macos_host_and_toolchain_probe
phase_2=validate_user_local_paths_and_contract
phase_3=inspect_existing_targets_for_managed_markers
phase_4=stage_app_bundle_manifest_and_infoplist
phase_5=stage_panel_executable_and_icon_inputs
phase_6=stage_application_support_layout
phase_7=stage_cli_wrappers
phase_8=commit_user_local_managed_artifacts
phase_9=write_receipts_and_measurements
phase_10=run_verification_transcript
```

The first implementation slice should support report-only phase rendering before enabling commit behavior:

```
phase_report_only=1
commit_user_local_managed_artifacts=0
```

Phase Requirements

Phase 1 must record:

```
macos_host_detected=&lt;0-or-1&gt;
host_arch=&lt;architecture&gt;
clang_probe_recorded=&lt;0-or-1&gt;
rust_probe_recorded=1
panel_build_ready=&lt;0-or-1&gt;
c_test_build_ready=&lt;0-or-1&gt;
```

Phase 2 must re-run path guards and refuse:

```
/Applications
/Library
/System
/usr/local
/opt/homebrew
LaunchDaemons
system LaunchAgents
kernel extension paths
system extension paths
network extension paths
privileged helper paths
```

Phase 3 must preserve any existing target that lacks both markers:

```
LATTICRA_INSTALLER_MANAGED=1
LATTICRA_MACOS_USER_LOCAL_APP_BUNDLE=1
```

Phase 4 must render an Info.plist plan with:

```
CFBundleIdentifier=systems.latticra.panel
CFBundleExecutable=latticra-panel
CFBundlePackageType=APPL
NSHighResolutionCapable=true
```

Phase 5 must require a local Panel executable candidate and icon candidate before real writes. The macOS dry-run writer candidate integration proves this phase can become ready in dry-run only when the local candidate asset probe and writer dry-run agree. It must not download, build, sign, notarize, or generate network-dependent assets.

The macOS commit gate contract keeps Phase 8 closed until a future managed-write implementation, reset/uninstall implementation, and verification transcript evidence exist. The macOS verification transcript contract now defines the required evidence shape, but no verification transcript evidence exists yet.

Phase 6 must stage:

```
MANAGED_BY_LATTICRA
etc/latticra/installer-config.toml
etc/latticra/macos-plan.txt
```

```
lib/latticra/payload-manifest.txt
receipts/
```

Phase 7 must stage wrappers only under:

```
$HOME/.local/bin
```

Phase 8 may commit only after all staged files have passed validation and only if every existing target is either absent or managed by Latticra.

Phase 9 must write receipts after successful managed writes and must record all authority-denial fields.

Phase 10 must run the verification transcript and fail if any required transcript line is missing.

Failure Behavior

The future writer must fail closed:

```
unsafe_path_detected -&gt; block_before_write
unmanaged_existing_app_bundle -&gt; preserve_and_block
unmanaged_existing_wrapper -&gt; preserve_and_block
missing_panel_executable -&gt; block_before_write
missing_icon_asset -&gt; block_before_write
missing_contract_marker -&gt; block_before_write
receipt_write_failure -&gt; report_failure_without_claiming_install_verified
verification_failure -&gt; report_failure_without_claiming_install_verified
```

Failure must not trigger:

```
sudo
network_download
homebrew_install
shell_profile_mutation
launchagent_install
keychain_write
tcc_prompt
endpoint_security_activation
system_extension_activation
privileged_helper_install
```

Reset And Uninstall Sequencing

Future reset/uninstall must use these phases:

```
reset_phase_1=validate_user_local_targets
reset_phase_2=inspect_managed_markers
reset_phase_3=preserve_unmanaged_targets
reset_phase_4=remove_managed_wrappers
reset_phase_5=remove_managed_app_bundle
reset_phase_6=remove_managed_application_support_prefix
reset_phase_7=write_reset_or_uninstall_receipt_outside_removed_prefix
reset_phase_8=emit_verification_absence_report
```

Reset/uninstall must preserve:

```
user_logs
user_caches
shell_profiles
Keychain_items
LaunchAgents
Login_Items
Homebrew_files
/Applications
/Library
/System
/usr/local
/opt/homebrew
```

Verification Commands

The implementation plan is guarded by:

```
sh scripts/test-macos-user-local-app-bundle-implementation-plan.sh
```

When a future writer exists, the minimum verification set should include:

```
sh scripts/macos-build-platform-probe.sh
sh scripts/macos-dry-run-plan-adapter.sh
sh scripts/test-macos-user-local-app-bundle-contract.sh
sh scripts/test-macos-user-local-app-bundle-implementation-plan.sh
```

Future writer-specific tests must cover:

```
default dry-run phase rendering
unsafe path blocked before write
unmanaged app bundle preserved
unmanaged CLI wrapper preserved
missing executable blocks before write
missing icon blocks before write
receipt fields complete
verification transcript complete
reset dry-run preserves unmanaged targets
uninstall receipt outside removed prefix
```

Current Plan Status

```
macos_user_local_app_bundle_implementation_plan_present=1
macos_app_bundle_writer_present=0
macos_app_bundle_writer_dry_run_present=1
macos_local_candidate_asset_probe_present=1
macos_dry_run_writer_candidate_integration_present=1
macos_commit_gate_contract_present=1
macos_verification_transcript_contract_present=1
macos_app_bundle_created=0
macos_install_verified=0
macos_reset_uninstall_implemented=0
macos_verification_transcript_present=0
application_support_write_performed=0
app_bundle_write_performed=0
cli_wrapper_write_performed=0
host_mutation_performed=0
network_performed=0
```

Non-Claims

This plan is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, or Apple platform approval.

Previous Recommended Lane

```
Add a macOS reset/uninstall evidence-bundle contract that groups implementation-gate,
operator-intent, receipt, absence, planner, and classifier evidence before any live execution.
```

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/MACOS_VERIFICATION_TRANSCRIPT_CONTRACT.md

Source title: macOS Verification Transcript Contract

Lines: 229 | Words: 456 | SHA-256: 3042e53ba1b1f22d0748ea49308f3e1c4b578893e15a8a0e4e8c5255388cb77a

macOS Verification Transcript Contract

Status: no-effect macOS verification transcript contract Date: 2026-05-25 CDT Scope: contract for exact post-write evidence before a future user-local macOS app bundle install can be called verified.

Purpose

This contract defines the verification transcript that a future macOS user-local app bundle writer must produce after managed writes.

It is contract-only. It does not create an app bundle, write a transcript, verify an install, run a reset or uninstall, mutate host state, request platform authority, or claim macOS installation.

The macOS commit gate may recognize this contract as present, and the macOS reset/uninstall dry-run contract now defines the required reset evidence shape. The gate remains closed until managed writes, reset/uninstall evidence, and real transcript evidence exist.

Command

```
sh scripts/macos-verification-transcript-contract.sh
```

The command writes only a deterministic report to stdout.

Current Decision

The current transcript posture is:

```
macos_verification_transcript_contract_present=1
verification_transcript_contract_state=defined-no-effect
verification_transcript_contract_decision=contract-defined-evidence-not-present
verification_transcript_evidence_present=0
macos_install_verified=0
commit_user_local_managed_artifacts=0
```

Expected Targets

The future transcript must bind evidence to the exact user-local targets:

```
app_support_prefix_expected=$HOME/Library/Application Support/Latticra
app_bundle_expected=$HOME/Applications/Latticra Panel.app
cli_wrapper_expected=$HOME/.local/bin/latticra-panel
receipts_dir_expected=$HOME/Library/Application Support/Latticra/receipts
bundle_identifier_expected=systems.latticra.panel
bundle_executable_expected=latticra-panel
```

Required Transcript Lines

A future verification transcript must include all of:

```
transcript_line_host_identity_required=1
transcript_line_architecture_required=1
transcript_line_app_bundle_path_required=1
transcript_line_info_plist_identifier_required=1
transcript_line_panel_executable_digest_required=1
transcript_line_icon_asset_digest_required=1
transcript_line_application_support_marker_required=1
transcript_line_cli_wrapper_marker_required=1
transcript_line_receipt_manifest_required=1
transcript_line_authority_denials_required=1
transcript_line_candidate_flow_required=1
transcript_line_commit_gate_required=1
transcript_line_reset_uninstall_dry_run_required=1
```

Required Evidence

Before any future status may say macos_install_verified=1, evidence must show all of:

```
host_os_macos_recorded_required=1
architecture_recorded_required=1
managed_app_bundle_present_required=1
info_plist_present_required=1
info_plist_bundle_identifier_required=1
app_executable_present_required=1
app_executable_executable_required=1
app_executable_digest_required=1
icon_asset_present_required=1
icon_asset_digest_required=1
application_support_marker_required=1
cli_wrapper_present_required=1
cli_wrapper_marker_required=1
receipts_present_required=1
receipt_completeness_required=1
authority_denial_fields_required=1
```

```
candidate_integration_ready_required=1
commit_gate_closed_until_evidence_required=1
reset_uninstall_dry_run_required=1
macos_reset_uninstall_dry_run_contract_present=1
macos_reset_uninstall_live_target_classifier_present=1
macos_reset_uninstall_dry_run_planner_present=1
reset_uninstall_dry_run_planner_transcript_present=1
macos_reset_uninstall_absence_report_contract_present=1
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
effect_authorization_open=0
reset_uninstall_effect_authorized=0
operator_intent_contract_state=defined-no-effect
operator_intent_evidence_written=0
reset_uninstall_live_run_allowed=0
reset_uninstall_deletion_enabled=0
operator_reset_uninstall_intent_evidence_present=0
operator_explicit_reset_uninstall_intent_observed=0
reset_uninstall_receipt_evidence_present=0
reset_receipt_evidence_present=0
absence_report_evidence_present=0
macos_reset_uninstall_receipt_schema_contract_present=1
macos_reset_uninstall_implementation_gate_contract_present=1
macos_reset_uninstall_operator_intent_contract_present=1
macos_reset_uninstall_effect_authorization_contract_present=1
macos_reset_uninstall_evidence_bundle_contract_present=1
macos_reset_uninstall_live_implementation_plan_contract_present=1
macos_reset_uninstall_live_execution_preflight_contract_present=1
macos_reset_uninstall_live_denial_transcript_contract_present=1
macos_reset_uninstall_live_runner_interface_contract_present=1
live_execution_preflight_contract_state=closed-no-effect
live_execution_preflight_passed=0
live_execution_preflight_blocking=1
live_execution_preflight_deletion_enabled=0
live_denial_transcript_contract_state=recorded-no-effect
live_denial_transcript_recorded=1
live_denial_transcript_stdout_only=1
live_denial_transcript_file_write_enabled=0
live_runner_interface_contract_state=defined-no-effect
live_runner_interface_current_preflight_passed=0
live_runner_interface_current_decision=deny
live_runner_interface_dispatch_enabled=0
live_runner_interface_runner_handoff_enabled=0
live_implementation_plan_contract_state=defined-no-effect
live_reset_uninstall_implementation_present=0
evidence_bundle_contract_state=defined-no-effect
evidence_bundle_complete=0
reset_uninstall_evidence_bundle_complete=0
effect_authorization_contract_state=closed-no-effect
reset_uninstall_effect_authorized=0
reset_uninstall_live_run_allowed=0
operator_reset_uninstall_intent_evidence_present=0
reset_receipt_evidence_present=0
reset_uninstall_receipt_evidence_present=0
```

```
unmanaged_target_preservation_required=1
seal_report_only_output_required=1
lat_or_lir_no_effect_probe_required=1
```

Authority Boundary

This contract preserves:

```
verification_transcript_run_performed=0
verification_transcript_written=0
application_support_write_performed=0
payload_write_performed=0
config_write_performed=0
receipt_write_performed=0
app_bundle_write_performed=0
info_plist_write_performed=0
app_executable_write_performed=0
app_icon_write_performed=0
cli_wrapper_write_performed=0
shell_profile_mutation_performed=0
installer_write_performed=0
host_mutation_performed=0
network_performed=0
root_authority=0
launchagent_authority=0
keychain_authority=0
tcc_bypass_authority=0
endpoint_security_authority=0
system_extension_authority=0
network_extension_authority=0
privileged_helper_authority=0
runtime_authority_granted=0
production_installer_ready=0
```

Validation

This contract is guarded by:

```
sh scripts/test-macos-verification-transcript-contract.sh
```

Expected output:

```
macos_verification_transcript_contract: ok
```

Non-Claims

This contract is not macOS installation, macOS app bundle evidence, signed app evidence, notarization evidence, launchd evidence, Keychain evidence, Secure Enclave evidence, sandbox evidence, TCC approval evidence, Endpoint Security evidence, System Extension evidence, Network Extension evidence, privileged helper evidence, malware prevention, ransomware prevention, production readiness, Apple platform approval, or runtime authority.

Next Recommended Lane

```
Add a macOS reset/uninstall live-runner acceptance-denial disposition closeout audit review
disposition closeout contract that closes the reviewed no-effect closeout audit review
disposition without opening dispatch or deletion.
```

docs/VISUAL_THEOREM_ENGINES.md

Source title: Latticra Visual Theorem Engines

Lines: 70 | Words: 272 | SHA-256: c9b0b6614bb19d9a6642fe121c429e107deef440eee9e6cb37f6cb8b9e46dce4

Latticra Visual Theorem Engines

Latticra includes C-based visual theorem engines that render symbolic substrate models as MP4 video demonstrations.

These engines are not ordinary animations. They are visual field demonstrations of the Latticra idea: lattice nodes, matrix substrate behavior, recursive numeric order, modular expansion, abstraction containment, axiom fracture, particle-field unification, quantum deceleration, and regenesis from a null-state.

Engines

latt-field-engines/substrate-engine/

The figure-free Latticra Substrate Engine.

This version removes the human-like center figure and focuses entirely on the field itself. It renders the Latticra substrate as a moving 2D mathematical plane containing:

- Primordial substrate closure
- Matrix substrate structure
- Sigkokilla Modul expansion
- Abstraction-field containment
- Axiomatic fracture
- Particle unification
- Quantum deceleration
- Cosmic regenesis

latt-field-engines/visual-theorem-engine/

The original cinematic theorem engine.

This version includes a central operator figure and demonstrates the earlier theorem-card aesthetic as a moving symbolic system.

Build

From the repository root:

```
scripts/render-visual-theorem-engines.sh build
```

Render the substrate engine

```
scripts/render-visual-theorem-engines.sh substrate 96
```

Output:

```
build/visual-engines/latticra-substrate-engine.mp4
```

Render the theorem engine

```
scripts/render-visual-theorem-engines.sh theorem 60
```

Output:

```
build/visual-engines/latticra-theorem-engine.mp4
```

Render both

```
scripts/render-visual-theorem-engines.sh all 96
```

Requirements

- gcc
- ffmpeg
- standard C math library, linked with -lm

On Fedora:

sudo dnf install gcc ffmpeg

Theory summary

The visual theorem engines posit Latticra as a symbolic computation substrate where field structures can be represented as lattices, split into substructures, projected onto a 2D abstraction plane, decelerated toward coherence, unified into particle-field classes, and regenerated from a null-state through deterministic operators.

The project is best understood as a mathematical art-engine and speculative computation model: part visual proof sketch, part symbolic systems theory, and part Latticra identity piece.

fixtures/artifact/local-artifact-manifest.txt

Source title: Local Artifact Manifest

Lines: 70 | Words: 90 | SHA-256: 398f51f1962b62b9565439434c2c79045b53770ebfda850c9a4053da96d48fcf

LATTICRA LOCAL ARTIFACT MANIFEST manifest_version=1 fixture_name=latticra-local-artifact-manifest-fixture fixture_status=non_production_fixture fixture_only=1 artifact_name=latticra-local-no-effect-cli-rpm artifact_version=0.0.0 artifact_arch=noarch artifact_format=rpm artifact_filename=latticra-0.0.0-0.1.local.fc44.noarch.rpm artifact_size_bytes=0 artifact_sha256=fixture-placeholder-not-release-checksum artifact_signature=none artifact_signing_key_id=none artifact_sbom_path=none artifact_license_metadata_path=LICENSE source_repository=Bryforge/Latticra source_commit=fixture source_tag=none build_environment=disposable-fedora-vm-fixture build_command_recorded=1 build_reproducible=0 supported_target_family=fedora-disposable-vm supported_target_versions=Fedora 44 local validation fixture supported_target_arches=noarch unsupported_targets_declared=1 unsupported_target=production-host unsupported_target=daily-driver unsupported_target=immutable-fedora unsupported_target=fedora-distribution requires_operator_consent=1 preflight_guard_required=1 install_plan_preview_required=1 uninstall_path_required=1 rollback_or_recovery_path_required=1 network_required=0 service_activation_default=0 boot_change_default=0 kernel_module_default=0 selinux_policy_default=0 payload_listing_recorded=1 payload_contains_cli_binary=1 payload_contains_readme=1 payload_contains_service=0 payload_contains_kernel_module=0 payload_contains_boot_change=0 payload_contains_selinux_policy=0 payload_entry=/usr/bin/latticra payload_entry=/usr/share/doc/latticra/README.md host_install_ready_for_cli_payload=1 artifact_manifest_fixture_present=1 artifact_manifest_present=1 artifact_manifest_validated=1 artifact_checksum_recorded=0 artifact_signature_recorded=0 artifact_sbom_recorded=0 artifact_supported_targets_declared=1 artifact_unsupported_targets_declared=1 production_installer_ready=0 fedora_distribution_ready=0 fedora_approval_claimed=0 daily_driver_install_ready=0 immutable_fedora_ready=0 not_production_readiness=1 not_fedora_approval=1 not_fedora_distribution_readiness=1 not_daily_driver_readiness=1 not_immutable_fedora_readiness=1 not_os_replacement_claim=1 notes=fixture only; not a generated artifact; not a production artifact; not a release checksum

LICENSE

Source title: License

Lines: 80 | Words: 242 | SHA-256: 22e37d29767cb1493cf186b495a216b50a975a09bdc7bb0f3344575dcc2e956e

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Summary -------

```
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```

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Source title: Agpl 3.0 Or Later

Lines: 661 | Words: 5535 | SHA-256: 0d96a4ff68ad6d4b6f1f30f713b18d5184912ba8dd389f86aa7710db079abcb0

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Lines: 73 | Words: 159 | SHA-256: f772b72a9cdf2158310b737bf8273fd48df388457a4f3a302d74e10c1e36e850

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